Poster
Oesophageal, gastric and duodenal disorders

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Oesophageal, gastric and duodenal disorders

09:00-18:00 / Poster Exhibition

PP0001

THE LIQUID-LIQUID PHASE SEPARATION OF PACSIN1 MAY CAUSE POOR PROGNOSIS IN GASTRIC CANCER PATIENTS

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Introduction: Recently, liquid-liquid phase separation (LLPS) of membrane protein has been found to play a critical role in the development of tumors [1-3]. Meanwhile, Protein Kinase C and Casein Kinase Substrate in Neurons 1 (PACSIN1), a peripheral membrane protein with intrinsically disordered regions (IDRs), is associated with the prognosis of gastric cancer patients [4].

However, whether PACSIN1 could undergo LLPS and whether PACSIN1 involved in the development of gastric have not been studied until now.

Aims & Methods: Here, we aimed to investigate whether PACSIN1 could undergo LLPS and the potential role of PACSIN1 in gastric cancer. We analyzed disease free survival of gastric cancer samples from The Cancer Genome Atlas dataset. The protein sequence prediction website (Disprot - Database of Protein Disorder) was used to predict the IDRs of PACSIN1. Besides, droplet formation experiments, fluorescence recovery after photobleaching (FRAP) and cellular immunofluorescence were used to explore whether PACSIN1 could undergo LLPS. What's more, cell proliferation assay and cell migration assay were utilized to investigate the potential role of PACSIN1 in gastric cancer.

Results: In the prognosis analysis of the TCGA database, it was found that the disease-free survival rate of gastric cancer patients with high expression of PACSIN1 was significantly lower than that of patients with low expression of PACSIN1(Figure 1).

Overexpression of PACSIN1 in gastric cancer cell resulted in decreased proliferation (Figure 2A-B) and increased migration (Figure 2C-D) of gastric cancer cells. In the prediction of protein structure, it could be found that the IDRs region of PACSIN1 accounted for more than 50% (Figure 3). When fully mixed with 5% Polyethylene glycol (PEG), one of the characteristics of LLPS- droplet formation - could be observed under the microscope (Figure 4B). After the fluorescence bleaching of the droplet formed by PACSIN1 with mCherry, the fluorescence gradually recovered within a few minutes (Figure 4C). What's more, PACSIN1 formed puncta with irregular distribution in cytoplasm (Figure 4D), which also proved that PACSIN1 could undergo LLPS.

Conclusion: Overall, high expression of PACSIN1 is associated with poor prognosis of patients, and overexpression of PACSIN1 in gastric cancer cell can promote the proliferation and migration. PACSIN1 can also undergo LLPS in vitro and in vivo. Therefore, we speculate that PACSIN1 may promote the development of gastric cancer through LLPS.

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Disclosure: The authors declare that there are no competing interests.

PP0002

AUTOMATIC DEEP LEARNING MODELS FOR GASTROINTESTINAL TUMOR DIAGNOSIS AND PROGNOSIS USING HEMATOXYLIN AND EOSIN SLIDES

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Introduction: Immunohistochemistry (IHC) is a critical tool for tumor diagnosis and treatment. However, IHC is time consuming and costly, relies on the pathologists' subjective interpretation. Deep learning has gained ground for predicting the biomarker expression on hematoxylin and eosin (H&E) staining images.

In this study, we aim to create an automatic pipeline for the construction of deep learning models to generate virtual IHC output using H&E whole slide images and validate its clinical applicability by reader study.

Aims & Methods: We obtained 134 whole-slide images (WSIs) from 73 patients at Renmin Hospital of Wuhan University and extracted 415,463 patches from H&E-stained WSIs.

Our study focused on five widely used biomarkers in pathology reports, namely anti-P40, CK-pan, Desmin, P53, and Ki-67, as recommended by the NCCN guidelines. We validated these models by using independent whole-slide images (WSIs). The number of WSIs used for each model were as follows: model-P40 (6 WSIs, 38389 Patches), model-PCK (9 WSIs, 72443 Patches), model-Ki67 (7 WSIs, 53162 Patches), model-Desmin (5 WSIs, 23974 Patches), and model-P53 (5 WSIs, 36822 Patches).

We aimed to predict these biomarkers expression in H&E WSIs and quantify the percentage of Ki-67. There are three main steps: Firstly, we employed HEMnet neural network to align the corresponding IHC and H&E WSIs, transfering the molecular labels from IHC to H&E images. Secondly, pathologists reviewed the annotations made by HEMnet and corrected HE-IHC mismatched tiles using VIA labelling platform. Lastly, the deep learning models were trained to predict positive tiles. The performance of models was assessed by sensitivity, specificity, accuracy, and the area under the receiver operating characteristic curve (AUC). To assess the ap-

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plicability of models, we randomly selected 30 cases including H&E and IHC WSIs for three pathologists to write the reports in the reader study by reading the interpretation of real IHC and AI-generated IHC.

Results: The sensitivity, specificity, accuracy, and area under the receiver operating characteristic curve (AUC) of the models for classifying positive pixels of each biomarker was: P40 (91.64%, 87.39%, 88.66%, AUC: 0.95), PAN-CK (92.40%, 89.35%, 90.81%, AUC: 0.96), Ki67 (89.63%, 86.04%, 86.77%, AUC: 0.95), Desmin (91.44%, 88.11%, 89.17%, AUC: 0.96), and P53 (75.44%,86.52%,83.04%, AUC: 0.90). The predicted Ki-67 percentage was comparable to the assessments made by expert pathologists, with a difference ranging from 7.00%±3.69%. In the reader study, compared with real-IHC pathology report, the accuracy of AI-IHC pathology report of each biomarker was: Desmin(100%,66/66), Pan-CK(98.3% , 59/60), P53(98.3% ,59/60), T-stage(84.8% .56/66), P40(96.7% ,29/30). and there is no significant difference between the quatification of Ki-67 score(*t*= 0.253 ,*P*= 0.800).

Items	Pathology report's accuracy of AI-IHC compared with Real-IHC					T test of Ki-67
	Desmin	Pan-CK	P53	T-stage	P40	
Pathologist 1	100.0% (22/22)	100.0% (20/20)	70.0% (21/30)	68.2% (15/22)	100.0% (10/10)	t= 0.475 (P= 0.637)
Pathologist 2	100.0% (22/22)	95.0% (19/20)	66.7% (20/30)	95.5% (21/22)	100.0% (10/10)	t= - 0.775 (P= 0.443)
Pathologist 3	100.0% (22/22)	100.0% (20/20)	60.0% (18/30)	90.9% (20/22)	90.0% (9/10)	t= - 0.510 (P= 0.613)
Average	100.0% (66/66)	98.3% (59/60)	98.3% (59/60)	84.8% (56/66)	96.7% (29/30)	t= 0.253 (P= 0.800)

Table. Result of Reader Study.

Conclusion: Our deep learning models have demonstrated promising performance in predicting gastrointestinal tumor biomarkers based on H&E slides. The utilization of these models can alleviate the burden of IHC staining and clinicopathological assessments.

Furthermore, these models have the potential to aid in patient grading, facilitate informed treatment decisions, and provide valuable prognostic assessments.

Disclosure: There is no conflict of interest in this abstract.

PP0003

LOSS OF STROMAL BMP ACTIVITY DRIVES GASTRIC PATHOLOGY IN MICE

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Introduction: The gastric epithelial gland homeostasis is driven by insufficiently characterized sub-epithelial stromal signals. Gastric stem cells in the gland base are fueled by a Wnt/Rspo^{high} niche where Bmp activity is suppressed, while terminally differentiated mucous pit cells are maintained by a BMP^{high} niche, that suppresses Wnt/Rspo activity. Infection with *Helicobacter pylori* reduces stromal and epithelial Bmp activity driving stem cell expansion. In line with this, genetic knockout of BMP activity in the epithelium leads to epithelial hyperplasia. The consequences of loss of BMP activity in the stroma on epithelial homeostasis and proliferative, regenerative, and inflammatory pathways in the interplay between stroma and epithelium are unknown.

Aims & Methods: The objective of this study was to analyze the epithelial pathology and epithelial-stromal interplay in the context of loss of BMP activity in the gastric stroma by using an inducible knockout of the BMP type 1 receptor in the stroma in mice.

Results: After two months of knockout of the BMP receptor in the gastric stroma, mice display drastic hyperplasia via stem cell expansion in the antrum and the corpus. Loss of stromal BMP signaling causes the occurrence of a bulky stroma with expanded fibroblasts and aberrant glands within the pit area, that resemble the gland base with expression of stem cell factors, high proliferation, enhanced epithelial inflammation (marked by NF-κB p65) and expression of metaplasia markers (CD44v9, Aqp5). Using scRNAseq as well as organoid and assembloid technologies we now study the mechanisms by which loss of stromal BMP affects epithelial homeostasis.

Conclusion: Collectively, our data demonstrate that alterations in stromal BMP activity can promote premalignant lesions. This model will allow us to study the underlying principles of stromal-epithelial interplay in the stomach and explore mechanistically the consequences of stromal responses observed upon *H. pylori* infection in detail.

Disclosure: Nothing to disclose.

PP0004

THE OCCURANCE OF GENES OF THE HLA SYSTEM IN PATIENTS WITH ACHALASIA AND HEALTHY VOLUNTEERS IN RUSSIAN FEDERATION

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Introduction: Human leucocyte antigen (HLA) strongly associated with autoimmune diseases. In some studies it was demonstrated, that achalasia is an autoimmune disease and associated with genetic factors.

Aims & Methods: The aim of our study was to investigate the association of genes of the HLA system in patients with achalasia and in healthy citizens of Russian Federation.

Material and methods: 140 patients with achalasia were included in this study: 75 females (55%), age from 18 to 75. The diagnosis of idiopathic achalasia was established by upper endoscopy and high resolution water-perfused manometry (MMS, Netherland). Healthy 332 volunteers from 21 to 74 y.o. (asymptomatic) were included in this study: there were 305 females.The DNA from blood samples was isolated using QIAamp DNA Blood Mini Kit (Qiagen, Germany) according to the manufacturer's recommendations. The concentration of DNA samples was measured by fluorimetric method on a Qubit 4 fluorimeter using the Qubit dsDNA BS Assay Kit (Thermo Fisher Scientific, USA). All patients underwent HLA typing by massively parallel sequencing (NGS): *HLA-A, HLA-B, HLA-C, HLA-DQB1, HLA-DRB1, HLA-DPB1*, on the platform MiSeq (Illumina, USA) v.2, Omixon. The analysis of the HLA gene sequences obtained as a result of sequencing was carried out using a computer program TypeStream Visual Software (TSV) (One Lambda, USA), and database IPD-IMGT/HLA.

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Results: The allelic variants *HLA-A*02, HLA-B*37, HLA-B*39, HLA-C*5, HLA-C*7, HLA-C*14, HLA-C*15, HLA-DRβ1*01, HLA-DRβ1*03, HLA-DRβ*08, HLA-DRβ1*13, HLA-DPβ1*06, HLA-DPβ1*11, HLA-DPβ1*05, HLA-DPβ1*15 and HLA-DQβ1*04* were evaluated equally in healthy volunteers and achalasia patients. The allelic variants *HLA-A*66, HLA-B*15, HLA-B*44, HLA-B*52, HLA-C*12, HLA-C*16, HLA-DPβ1*01, HLA-DPβ1*02, HLA-DPβ1*23, DRβ1*11, HLA-DRβ1*14, DQβ1*06* were found statistically more frequently in patients with achalasia than in healthy volunteers. The allelic variants *HLA-A*11, HLA-A*23, HLA-A*25, HLA-A*26, HLA-B*7, HLA-B*57, HLA-C*3, HLA-C*06, HLA-DPβ1*03, HLA-DPβ1*13, HLA-DPβ1*17, DRβ1*13, HLA-DRβ1*15* were insignificantly evaluated more frequently in group with achalasia than in healthy volunteers.

Conclusion: The absence of differences in some allelic variants of genes of HLA in patients with achalasia and healthy volunteers confirm that they are not specific for achalasia in Russian Federation. The statistical difference in predominance of some allelic variants of HLA system indicates, that these findings should be compare with medical history and symptoms. After that these allelic variants of genes of HLA can be consider as a prognostic factor.

Disclosure: Nothing to disclose.

PP0005

WITHDRAWN

PP0006

ASSOCIATION BETWEEN ACID-SUPPRESSIVE DRUGS AND RISK OF PSORIASIS: RETROSPECTIVE STUDY USING KOREAN NATIONAL HEALTH INSURANCE SERVICE-NATIONAL SAMPLE COHORT

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Introduction: Psoriasis is a common inflammatory skin disorder following non-specific triggers such as scratching, piercing, sunburn, and chemical irritants. Involvement of immune system is widely accepted for pathogenesis studies have demonstrated importance of gut microbiota in pathogenesis of inflammatory skin diseases. Proton pump inhibitors (PPIs) and histamine-2 receptor antagonists (H2RA) are acid suppressing drugs widely used for gastrointestinal diseases, and long-term use has been reported to be associated with altered gut microbiota. This study aimed to investigate association between psoriasis and acid suppressing drugs in Korean population.

Aims & Methods: We used Korean National Health Insurance Service-National Sample Cohort (NHIS-NSC) data of 746,816 patients with upper gastrointestinal (GI) diseases between 2002 and 2013. Duration of acid suppressing drug exposure and other socio-demographic characteristics were compared between patients with and without psoriasis, and hazard ratios were estimated.

Results: Significantly larger number of patients with exposure to acid suppressing drugs had developed psoriasis compared to control cohort (46.3% vs. 33.7% respectively, $P < 0.001$). In the multivariate analysis, longer use of acid suppressing drugs and greater Charlson comorbidity index score were significantly associated with increased risk of psoriasis with an adjusted odds ratio, 1.89 (95% confidence interval, 1.66-2.17, $P < 0.001$) and 1.33 (1.17-1.51, $P = <0.001$), respectively.

Conclusion: Results from our study indicate that prolonged use of H2RA or PPI is associated with the risk of psoriasis among patients with GI diseases in Korean population. The risk was increased in dose-response trend, even after adjusting for confounding variables. Clinicians should be aware of risks associated with prolonged use of acid suppressing drugs.

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Disclosure: Nothing to disclose.

PP0007

CYANOCOBALAMIN INTRAMUSCULAR TREATMENT IN PATIENTS WITH CORPUS ATROPHIC GASTRITIS AND VITAMIN B12 DEFICIENCY: EFFICACY AND PREDICTORS OF INCREASED REQUIREMENT

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Introduction: Corpus atrophic gastritis (CAG) is associated with vitamin B12 deficiency. Untreated vitamin B12 deficiency can lead to megaloblastic anaemia (also known as Pernicious Anaemia) and other potential serious clinical manifestations such as acute cardiocerebral events and severe neurological consequences, thus timely B12 supplementation is relevant. To date, the dosage of intramuscular B12 supplementation in CAG patients with vitamin B12 deficiency is not standardized.

Aims & Methods: The aim was to assess the efficacy of cyanocobalamin treatment (Tx) at long term follow-up and predictors of increased cyanocobalamin requirement.

A monocentric retrospective longitudinal cohort study was conducted on 190 pts (62.1% F, age 63±1 yrs) consecutively diagnosed (2007-2021) with CAG and vitamin B12 deficiency. Inclusion criteria were age >18 years, histological diagnosis of CAG (updated Sydney System), serum B12 <250 pg/ml, intramuscular cyanocobalamin supplementation, and follow-up of at least 12 mos. Demographic, clinical, biochemical, and histological data were retrospectively collected. After the diagnosis of vitamin B12 deficiency pts were first treated with TxA schedule (5000μg intramuscular cyanocobalamin every 5 days for 3 times, followed by 5000μg intramuscular every 3 mos (20000μg/yr)), and revaluated at follow up. If TxA was not sufficient to restore B12 levels, TxB was prescribed (5000μg intramuscular cyanocobalamin every 5 days for 3 times, followed by 5000μg intramuscular cyanocobalamin every 2 mos (30000μg/yr)). Primary Tx endpoint was B12 normalization. Baseline and follow-up data of pts on TxA and TxB were compared by Student's t-test. Logistic regression was performed to find out predictors of increased B12 need.

Results: Macrocytic anaemia was mild, moderate and severe in 60, 17, 14 out of 190 pts, while 94 pts had vitamin B12 deficiency without anaemia.

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TxA efficaciously corrected vitamin B12 deficiency in 130 (68.4%) pts and was maintained until longest available follow-up (41±2.4 mos). The remaining 60 (31.6%) pts, due to persistent vitamin B12 deficiency observed at 12 (6-48) mos, were switched to TxB which was then maintained until longest available follow-up (49.9±4.1 mos). At longest available follow-up, a significant increase of Hb (TxA: 12±0.2 vs 13.1±0,2; TxB: 12.1±0.4 vs 13.8±0.2, p<0,0001) and B12 (TxA: 162.9±8.6 vs 400±22.2; TxB: 142.3±11.2 vs 323.4±22.3, p<0.001) was shown in both schedules. A significant decrease of MCV was shown in TxB only (96.4± 2.1 vs 88.1±0.9,p=<0.0001). At logistic regression, switch to TxB was associated with severe intestinal metaplasia (OR10.6, IC 95% 2.6-43.0, p=0.0009) and macrocytosis at diagnosis (OR2.8, IC95% 1.1-6.9, p=0.0231), and male gender (OR 2.5, IC95% 1.1-5.7, p=0,0213), while other features such as age, severity of vitamin B12 deficiency, previous vitamin B12 therapy, concomitant iron deficiency were not significantly associated with increased need of intramuscular cyanocobalamin.

Conclusion: At long term follow-up, about 70% of pts restore vitamin B12 deficiency with 5000µg intramuscular cyanocobalamin every 3 mos, while only 30% require an increased dosage of 5000µg intramuscular cyanocobalamin every 2 mos. In particular, male pts with severe intestinal metaplasia and macrocytosis at diagnosis require a higher dosage of cyanocobalamin to obtain B12 normalization. In CAG pts with vitamin B12 deficiency regular clinical-biochemical follow-up is important to monitor efficacy of B12 treatment to restore B12 levels and to treat and prevent clinical manifestations of vitamin B12 deficiency.

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Disclosure: Nothing to disclose.

PP0008

CLINICAL AND EPIDEMIOLOGICAL CHANGES IN CAUSTIC INGESTION OVER THE LAST DECADES

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Introduction: The incidence of caustic ingestion has remained stable in recent years due to its difficult prevention and limited changes in its management.

Aims & Methods: Aims:

1. To explore changes in the epidemiological and clinical characteristics of caustic ingestion over the last decades
2. To identify epidemiological risk factors of adverse outcomes.

Methods: We present data from an observational study that prospectively collected caustic ingestions in our centre since 1995 and included all adult patients whose acute episode was evaluated in the emergency department. Epidemiological data and variables related to the type of ingestion, prognostic risk factors and the adverse outcomes, defined as the need for admission to the intensive care unit, emergency surgery, patient death and oesophageal or gastric stricture development, were analysed; the results were compared for the periods 1995-2009 and 2010-2024.

Results: A total of 582 episodes of caustic ingestion were included, 305 during the first period and 277 during the second.

Epidemiological characteristics and type of ingestion were overall similar in both periods, as were the clinical, analytical, endoscopic and radiologic indicators of severity.

Adverse outcomes were also similar in the two time periods, both in the short term (comparable frequency of intensive care unit admission, need for surgery and death) and in the long term (stricture development).

The main risk groups for severe ingestions were patients older than 45 years (OR 4.3; 95%CI: 2.3-7.9) and psychiatric comorbidity (OR 5.2; 95%CI: 3.0-9.1), especially depressive syndrome (OR 7.2; 95%CI: 4.2-12.4).

Recurrent ingestions were more frequent in men (OR 1.9; 95%CI: 1.0-3.5) and psychiatric illness (OR 7.6; 95%CI: 3.5-16.5), mainly personality disorder (OR 11.0; 95%CI: 5.5-21.9).

	1995-2009	2010-2024	p value
Women	164 (53,8)	133 (48,0)	0,192
Age	44 (30)	50 (30)	0,193
Voluntary intake	154 (50,5)	121 (44,0)	0,139
pH of caustic Strong acid Weak acid Weak alkali Strong alkali	 50 (16,9) 17 (5,8) 215 (72,9) 13 (4,4)	 40 (14,7) 25 (9,2) 200 (73,3) 8 (2,9)	0,318
Psychiatric illness	155 (50,8)	119 (43,0)	0,007
Depression	112 (36,7)	75 (27,1)	0,016
Bipolar disorder	5 (1,6)	11 (4,0)	0,143
Personality disorder	28 (9,2)	20 (7,2)	0,479
Psychotic disorder	16 (5,2)	13 (4,7)	0,908
Alcohol/toxic substances consumption	23 (7,5)	26 (9,4)	0,515
Intensive care unit admission	40 (13,1)	29 (10,5)	0,391
Emergency surgery	8 (2,6)	10 (3,6)	0,655
Death	15 (4,9)	19 (6,9)	0,412
Stricture formation	11 (5,1)	8 (3,7)	0,624

Qualitative variables: absolute (relative) value, Chi-squared +/- continuity correction.
Quantitative variables: median (interquartile range), Mann-Whitney U test

Conclusion: No relevant changes were identified in the epidemiological and clinical characteristics of caustic ingestions over the last decades, nor in their evolution. The most vulnerable groups to prioritise prevention measures would be patients with psychiatric cormobility.

Disclosure: Nothing to disclose.

PP0009

ANTI-INFLAMMATORY ACTION OF A MUCOSA PROTECTIVE AGENT, MARSHMALLOW ROOT EXTRACT, IN ENDOTHELIAL CELLS, FIBROBLASTS AND MACROPHAGES IN VITRO

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Introduction: Mucosa protective effects can be mediated by physical as well as by anti-inflammatory mechanisms of action. For the root extract of marshmallow (Althaea officinalis L., REAo), known to cure mild irritation of stomach/gut mucosa [1], only a physical action had been considered. Recent studies have shown that it also induces regeneration of irritated mucosa [2,3].

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Mucosal wound healing depends on the recruitment of several cell types, e.g., endothelial cells, fibroblasts, and macrophages (MΦ). Reactive oxygen species (ROS)-induced damage is associated with mucosal infiltration of activated leukocytes, such as MΦ, which produce excessive ROS and pro-inflammatory cytokines that may overwhelm the antioxidant and immune defenses of the cells and exacerbate mucosal inflammation. Additionally, fibroblasts exhibit a metabolic adaptation to control inflammation-related oxidative stress.

Aims & Methods: We aimed to investigate whether REAo has, on top of its physical effects, also anti-inflammatory/-oxidative properties in human dermal fibroblasts (NHDF), human umbilical vein endothelial cells (HUVEC), and human acute monocytic leukemia (THP-1) differentiated MΦ, critical cellular components of the gastrointestinal and oral mucosa.

The cells were pre-treated with REAo or its final formulation STW 42, containing REAo and selected excipients, followed by incubation with H_2O (pro-oxidative) or lipopolysaccharide (LPS, pro-inflammatory). Intracellular ROS were quantified with DCFDA. The releases of IL-6 and TNF-α were determined by ELISA. THP-1 cells were differentiated into MΦ with 0.1 µg/ml phorbol-12-myristate-13-acetatefor 3–5 days.

Results: Pre-treatment (24 h) of HUVEC, NHDF or THP-1-MΦ with STW 42 or REAo (100-1000 µg/ml) significantly inhibited the H_2O-induced intracellular ROS production by 30.0% to 58.7%. Whereas 24h(HUVEC, NHDF) or 48 h (THP-1-MΦ) pre-treatment with STW 42 or REAo (100-1000 µg/ml) inhibited the LPS-activated IL6 release by 25.0-67.0%. Additionally, 48 h pre-treatment of THP-1-MΦ with 50-500 µg/ml STW 42 or REAo significantly inhibited LPS-induced TNF-α release by 21.5% (50 µg/ml) to 52.0% (500 µg/ml), in comparison with LPS-treated MΦ. The observed effects of STW 42 or REAo were similar to 25-100 µM Diclofenac, which was used as an anti-inflammatory control.

Conclusion: These anti-inflammatory and antioxidant properties may support the benefit of STW 42 and its constituent REAo in patients with mucosal irritation with subsequent irritative cough or gastric complaints due to its protective/reparative effects on the mucosa.

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PP0010

ATHEROSCLEROSIS AND THE MICROBIOME: A NEW MICROBIOME BIOMARKER THAT PROTECTS AGAINST ATHEROSCLEROSIS?

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Introduction: The relationship between disruptions in the intestinal microbiome, psychological well-being, and modern lifestyle factors in individuals with prevalent non-communicable diseases, such as atherosclerosis, has been partially elucidated but remains inadequately investigated. This raises the inquiry into the feasibility of diagnosing, preventing, and treating atherosclerosis through the measurement and adjustment of disease markers associated with intestinal microbiome, psychological state, and lifestyle indicators.

Aims & Methods: This study aims to broaden the spectrum of these markers to a degree adequate for establishing models to diagnose atherosclerosis within the alternative categories of “no atherosclerosis,” “asymptomatic atherosclerosis,” and “symptomatic atherosclerosis,” as well as by the level of associated health risks (non-significant, significant, and high). Our cohort study data were employed to fulfill this goal.

Results: We identified microbial markers from 955 gut microbiome species and other markers from 52 indicators of psychological state and lifestyle among participants. Marker properties were assessed based on their ability to distinguish between “no atherosclerosis,” “asymptomatic atherosclerosis,” and “symptomatic atherosclerosis” groups, as well as “not significant risk,” “significant risk,” and “high risk” groups. We measured their correlation with each participant’s group membership using the p-level parameter to prioritize markers, resulting in a rating for each. Eleven microbial species (including *Oxalobacter aliiformigenes, Lactobacillus acidophilus, and Bifidobacterium longum*) and six other indicators (including “feeling of slowness” and “duration of physical activity”) showed the highest scores. Cluster-type models based on these markers accurately predict group membership, with predictive accuracies ranging from 93.7% to 97.1%. This study provides novel insights into atherosclerosis-associated markers, laying a foundation for future research in this field.

Conclusion: The inclusion of microbial, psychological state, and lifestyle indicators has significantly enhanced the existing markers for atherosclerosis. These new markers have facilitated the development of models that, for the first time, enable the diagnosis of atherosclerosis within the categories of “no atherosclerosis,” “asymptomatic atherosclerosis,” and “symptomatic atherosclerosis,” as well as by the associated health risks (non-significant, significant, and high). In addition to microbial markers, these models also incorporate an individual’s psychological state and lifestyle indicators.

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Disclosure: Nothing to disclose.

PP0011

ESOPHAGEAL AND SALIVARY MICROBIOME RELATED WITH EOSINOPHILIC ESOPHAGITIS

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Introduction: Eosinophilic esophagitis (EoE) is a chronic, food and/or aeroallergen-mediated inflammatory disease , characterized by esophageal dysfunction and eosinophil inflammation of the esophageal mucosa. The incidence of EoE has increased in recent years, including Japan. Pathogenesis of EoE is incomplete, and the data on microbiome related with EoE are limited. The aim of this study was to identify the esophageal and salivary microbiome in patients with EoE.

Aims & Methods: Patients either with EoE or non-EOE patients with symptoms of esophageal dysfunction who underwent esophagogastroduodenoscopy(EGD) at Kawasaki Medical School hospital between April 2021 and April 2023 were enrolled. The endoscopic reference score (EREFS) is used to determine severity of endoscopic findings. EoE patients were classified as the active group, if the presence of esophageal eosinophilia greater than or equal to 15 eosinophils per high-power field were detected during the endoscopy. , Saliva and Esophageal mucous taken from esophagus using brushing samples and saliva were used to investigate microbiome in EoE patients and controls. The microbiota of samples was profiled using the Illumina MiSeq platform.

Results: Total 24 patients were enrolled including8 active EoE patients, 5 inactive EoE patients, and 11 controls.

There were no significant differences in age, PPI use, history of gastroesophageal reflux disease and *H. pylori* infection among the three groups. The comparative analysis of the taxonomic composition at the genus level among esophageal mucous samples indicated that *Streptococcus* were less abundant in the active EoE group compared to the other groups, and *Stomatobaculum* were more abundant in the inactive EoE group than in the controls. On salivary, The abundance of the genera *Megasphaera* were significantly more prevalent in the inactive EoE group than in the other groups.The abundance of the genera *Streptococcus* was negatively correlated with ERFFS scores.

Conclusion: *Streptococcus* might be involved in the pathogenesis of EOE.

Disclosure: Nothing to disclose.

PP0012

THE ANALYSIS OF THE MICROBIOME COMPOSITION OF THE GASTRIC MUCOSA PROVIDES REPRODUCIBLE RESULTS EVEN IF DIFFERENT ANALYSIS PIPELINES ARE APPLIED

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Introduction: Microbiome analysis is a promising approach for the application of findings from basic research in the clinical field. However, there is an ongoing debate about the consistency of different bioinformatics analysis platforms. The lack of established standards affects the implementation of research findings.

Aims & Methods: In our study, we demonstrated that the results on the microbial composition of gastric mucosal biopsies are only slightly influenced by the efficacy of different microbiome analysis platforms. Furthermore, we investigated whether recent revisions to the International Code of Nomenclature for Prokaryotes and consequent alterations in taxonomic classification influenced the findings.

Gastric biopsy specimens were obtained from gastric cancer patients with well-defined clinical profiles (n=40), including individuals with and without *Helicobacter pylori* (*H. pylori*) infection, and from control subjects (n=39) with and without *H. pylori* infection. DNA extraction and 16S rRNA gene sequencing (V1-V2) was performed at one site, and the resulting fastQ files were distributed to five expert teams for independent analysis. Three widely used bioinformatics pipelines for microbiome analysis, namely DADA2, MOTHUR and QIIME2, were used for the analysis.

Results: The generated fastQ files exhibited no platform-dependent clustering, indicating the comparability of the sequences. Microbial diversity was slightly reduced in both GC groups and significantly reduced in the *H. pylori* positive controls. This is due to an extreme increase in the relative abundance of *H. pylori* in this group. It also follows that the samples of the *H. pylori* positive control group have a very high similarity and form the clearest cluster in the corresponding analysis. Regardless of the protocol used, *H. pylori* status, microbial diversity and relative bacterial abundance were reproducible on all platforms. Although minor differences in performance were detected and assessed, similarity analysis and bacterial enrichment assessment also yielded consistent results. The introduction of the new taxonomy only had a major impact on the *Proteobacteria* phylum, as it originally also included *H. pylori* according to the old taxonomy, but this is now assigned to the new *Campilobacterota* phylum.

Conclusion: In summary, our findings demonstrate that disparate microbiome analysis methodologies employed by independent research teams can yield comparable results from identical datasets.

Disclosure: Nothing to disclose.

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PP0013

GUT MICROBIOTA AND AUTOIMMUNE ATROPHIC GASTRITIS

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Introduction: Autoimmune Atrophic Gastritis (AAG) is a chronic immune-mediated condition that mainly affects the body and fundus of the stomach. AAG is often under-diagnosed due to the slow progression of the disease and a variable spectrum of symptoms. Recent data suggests a potential role of altered gastric microbiota in the natural history of AAG, due to the potential overgrowth of bacteria linked to the development of neoplastic complications (neuroendocrine tumors- NETs and gastric cancer) (1,2). Moreover, the same data hypothesizes the use of fecal microbiota as a possible non-invasive marker to diagnose and monitor AAG. Thus, we aimed to prospectively characterize the salivary and fecal microbiota of patients with AAG to provide further insights in this field.

Aims & Methods: We collected salivary and fecal samples from consecutive patients with an established diagnosis of AAG, enrolled at the Gastroenterology Unit of Azienda Ospedale Università Padova from March to June 2023 at the time of endoscopic follow-up examination. All patients needed to be off Proton-Pump Inhibitors, antibiotics, and probiotics for at least one month before enrolment. Stool samples were collected by the patients on the day of the endoscopy, transferred in a buffer solution, and stored at -20°C. Oral samples were collected through salivary commercial kits and freezed at -20°C. Both salivary and faecal samples were analyzed by BMR Genomic (Padua, Italy) that amplified the V3-V4 hypervariable ribosomal RNA region. Sequencing of the 16S rRNA gene V3- V4 region was performed using the Illumina MiSeq. Preliminary analyses were conducted on individual fecal samples compared to an internal population of healthy subjects (provided by BMR genomics) using Qiime2 tools. Shannon–Wiener index was used to calculate α-diversity and then the microbial biodiversity index (MBI) and Dysbiosis Index (DI). Mann–Whitney tests were performed to verify statistical differences in the α-diversity across AAG patients and healthy subjects.

Results: We enrolled 15 AAG patients with a mean age of ± 63 ± 11 years (9/15 females). OLGA 2 stage was reported in 10/15 patients. Sixty percent of the patients showed a MBI between 7.11 and 12.45, indicating that each sample contained phylogenetically similar species. The remaining patients have an index greater than 12.45 compared to the healthy population. More than half of the patients had dysbiosis (>0.082). The highest indices of dysbiosis were found in patients with a history of NET (n=3), gastric cancer (n=1), and previous HP- infection and OLGA 3 stage (n=1). Firmicutes, Bacteroidetes, and Proteobacteria were more frequent phyla found in the fecal samples, whereas Enterobacteriaceae and Streptococcaceae were the most common bacterial families (13/15, 87%). In particular, Klebsiella and Campylobacter pathogens genera were found in some samples. Streptococcus and Prevotella were the most abundant genera found in the salivary samples.

Conclusion: Patients with AAG were found to have a great proportion of Firmicutes and Bacteroidetes, which aligns with the literature (3,4). Moreover, a dominant of Proteobacteria, especially Enterobacteriaceae commonly associated with proinflammatory states, was found. (5) Our preliminary data showed a possible positive association between higher degrees of dysbiosis and a previous history of NET or gastric cancer.

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Disclosure: Nothing to disclose.

PP0014

REVIEW OF CONSECUTIVE CASES OF *CLOSTRIDIOIDES DIFFICILE* INFECTION AND ASYMPTOMATIC CARRIAGE OF TOXIGENIC *C. DIFFICILE* IN CHILDREN, OVER A 7-YEAR PERIOD

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Introduction: *C. difficile* infection (CDI) remains a significant clinical problem and has been extensively investigated in adults but by comparison is much less studied in children. Following colonisation with toxigenic *C. difficile*, clinical presentation may range from asymptomatic carriage to pseudomembranous colitis. Diagnosis of CDI can be challenging, with toxin-positive stool test (in the presence of diarrhoea) believed to reflect infection.

Aims & Methods: Our aim was to study children diagnosed with CDI and asymptomatic carriage of the organism. From 2015 - 2022, details of relevant stool test results were collected prospectively and clinical information was reviewed retrospectively. Detection of stool *C. difficile* toxin (toxin +ve) was used to imply CDI and its absence (toxin –ve), together with positive polymerase chain reaction (PCR) test for toxigenic *C. difficile* (PCR +ve) deemed to reflect asymptomatic carriage. Hospital-association was based on stool test undertaken >48 hours post-admission or within 4 weeks of discharge.

Results: Of total 113 patients, 57 [50.4%, median 11.0 (range 1-17) yrs, 31 male] were toxin +ve and were treated for CDI; 26 (45.6%) had co-existing cancer (80.8% on chemotherapy), 9 patients (15.8%) suffered from inflammatory bowel disease (IBD, 5 Crohn's disease and 3 unclassified) and 5 patients (8.77%) had had a transplant. Of the 57 CDI patients, 11 (19.3%) previously had a toxin-negative stool test result and 14 (24.6%) had a subsequent recurrence of CDI [of the latter subgroup, 7 (50%) were on chemotherapy for cancer and 2 (14.3%) had a prior transplant]. Of the 113 patients studied 48 [42.5%, 11.5 (1-18) yrs, 21 male] were PCR +ve but toxin –ve and were deemed asymptomatic carriers as they did not require treatment for CDI; 22 patients (45.8%) had co-existing cancer (all

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on chemotherapy), 11 (23%) suffered from IBD (9 Crohn's disease) and 3 patients (6.25%) had had a transplant. There were no significant differences between the above 2 groups (CDI and asymptomatic carriers) in age, male:female ratio, co-existing cancer, IBD, current / recent antibiotics or chemotherapy. Eight [7.1%, 13.5 (9-17) yrs, 5 male] of the 113 patients studied were PCR +ve and toxin –ve but were deemed to require treatment for CDI. For all 3 groups, >84% were hospital-associated episodes and >71% were associated with the use of antibiotics.

Conclusion: A significant proportion of children with CDI and asymptomatic carriage of toxigenic *C. difficile* had co-existing cancer (majority on chemotherapy) and IBD and were predominantly hospital-associated. Some asymptomatic carriers may subsequently develop CDI and require measures to reduce this risk. Stool *C. difficile* toxin test may be <100% sensitive for diagnosis of CDI. Following initial episode of CDI, antibiotic use and chemotherapy for cancer represent significant risk factors for recurrence for CDI in children. Further research is needed to identify risk factors for CDI in children with asymptomatic carriage of toxigenic *C. difficile*.

Disclosure: Nothing to disclose.

PP0015

MULTIDRUG RESISTANT ORGANISMS IN GASTROENTEROLOGIC PATIENTS: PREVALENCE AND RISK FACTORS

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Introduction: Colonization/infection with multidrug-resistant organisms (MDROs) such as methicillin-susceptible staphylococcus aureus (MSSA), methicillin-resistant Staphylococcus aureus (MRSA), vancomycin-resistant enterococci (VRE), extended-spectrum beta-lactamase (ESBL) producing Enterobacterales and carbapenem resistant P. aeruginosa, is an increasing problem being reported worldwide. Rectal, cutaneous and nasal swab screening allows for the detection of MDROs in asymptomatic individuals, enabling early identification and subsequent implementation of infection control measures to prevent transmission.

Aims & Methods: The aim of this study was to determine the prevalence as well as the risk factors of colonization/infection with MSSA, MRSA, VRE, ESBL producing Enterobacterales and carbapenemase producing P. aeruginosa, in patients hospitalized in gastroenterologic and oncologic and surgical unit for gastroenterologic diseases. Methods: all the consecutive patients hospitalized from May 2023 to March 2024 in the abovementioned units at admission underwent rectal swabs to identify the presence of MDROs, primarily methicillin-susceptible staphylococcus aureus (MSSA), methicillin-resistant Staphylococcus aureus (MRSA), vancomycin-resistant enterococci (VRE), extended-spectrum beta-lactamase (ESBL) producing Enterobacterales and carbapenemase producing P. aeruginosa. Demographic, clinical and therapeutic variables were evaluated.

Results: 242 patients were hospitalized (mean age 65,6 ± 17,8 years; 56% males). MDRO were present in 17,36% of patients (n.42). 7 % of patient tested positive for Escherichia coli ESBL+, 3.7% for Klebsiella pneumoniae ESBL+ and 2.5% for Enterococcus faecium VRE at rectal swab. 2.1% of patient tested positive at nasal swabs for MSSA, 2.1% for MRSA, 0.4% for E. coli ESBL and 0.4% for K. pneumoniae ESBL. 0.4% of patient were positive at cutaneous swab for E. faecium VRE and 0.4% for K. pneumoniae ESBL. At univariate analysis hospitalization within 1 year, ongoing treatment with PPI, ongoing steroid treatment and chronic renal failure were associated to the risk of a positive surveillance swab. At multivariate analysis only hospitalization, ongoing steroid treatment and chronic renal failure were independent predictors of colonization (table).

	Univariate	Multivariate
	OR (95% C.I.)	OR (95% C.I.)
Hospitalization within 1 year	2.4 (1.2 - 4.6)	2.2 (1.1 - 4.4)
PPI treatment	2.2 (1.1 - 4.4)	
Ongoing steroid treatment	3.7 (1.4 - 9.7)	2.9 (1.1 - 8.2)
Chronic Renal Failure	6.5 (1.9 - 22.4)	6.6 (1.8 - 24.5)

Conclusion: This research offers information on how common MDROs are among gastroenterology patients. It contributes to the existing knowledge and aids in addressing the spread of MDROs by customizing screening methods based on individual patient factors, like past hospital stay, ongoing steroid treatment and chronic renal failure.

Disclosure: Nothing to disclose.

PP0016

IMPACT OF NSAIDS ON OUTCOMES OF NON-VARICEAL UPPER GASTROINTESTINAL BLEEDING

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Introduction: Non-variceal upper gastrointestinal bleeding (NVUGI) represents a significant gastrointestinal disorder with elevated mortality rates (1).

Non-steroidal anti-inflammatory drugs (NSAIDs) have been firmly established as a risk factor for NVUGI, yet their precise impact on clinical outcomes remains insufficiently defined (2,3).

Conversely, proton pump inhibitors (PPIs) have shown efficacy in reducing the incidence of NVUGI associated with NSAIDs. However, the extent of their influence on the clinical outcomes of NVUGI remains inadequately substantiated. Therefore, it is necessary to study the impact of these drugs on the clinical outcomes of patients with NVUGI.

Aims & Methods: The primary objective of our study is to compare the outcomes of NVUGI patients using NSAIDs with or without PPIs against those who did not use either medication. We conducted a retrospective analysis at a Spanish tertiary hospital, encompassing NVUGI cases from 2013 to 2022. NVUGI was defined as hematemesis or melena accompanied by a two-point decrease in hemoglobin or hemodynamic instability. Key clinical outcomes included the necessity for intervention (endoscopic, radiological, or surgical) and in-hospital mortality.

Additionally, we compared baseline characteristics and pertinent clinical results such as rebleeding and delayed cardiovascular events. Statistical analysis was performed using SPSS.

Results: A total of 1200 cases were included, with 6.6% undergoing concomitant treatment with PPIs and NSAIDs, and 15.7% utilizing NSAIDs alone. Patients who took NSAIDs alone were younger (57 years vs. 66 in both other groups) and had a lower rate of comorbidities. Table 1 presents risk scores and outcome comparisons among the three groups. Statisti-

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cal differences were observed, revealing lower acute mortality (9.6% vs. 7 vs. 2.8%, p=0.011) and reduced need for intervention (74% vs. 69.6 vs. 57.8%, p<0.001) in the group using NSAIDs alone. In multivariate analysis, MAP(ASH) > 2 (OR: 8, CI: 3-21, p<0.001) and ASA > 2 (OR: 4.4, CI: 1.7-11.4, p:0.002) emerged as independent risk factors for acute mortality, with an elevated MAP(ASH) score (OR: 8.8, CI: 5.9-13, p<0.001) being the sole independent factor explaining the need for intervention.

	Naive	NSAIDs and PPIs	NSAIDs without PPIs	P
ASA 3-4	596 (64,3%)	50 (63,3%)	63 (33,7%)	<0,001
MAPASH>2	448 (59,3%)	39 (65%)	52 (34,4)	<0,001
Need for intervention	691 (74%)	55 (69,6%)	108 (57,8%)	<0,001
Rebleeding	96 (10,3%)	9 (11,4%)	6 (3,2%)	0,008
Acute mortality	86 (9,6%)	5 (7%)	5 (2,8%)	0,011
Mortality at 6 months	77 (8,8%)	7 (9,3%)	7 (3,8%)	0,075
Delayed Cardiovascular event	77 (8,7%)	3 (3,9%)	4 (2,2%)	0,004
Delayed rebleeding	145 (16,4%)	15 (19,5%)	6 (3,3%)	<0,001

Conclusion: Patients abstaining from NSAIDs or those concurrently undergoing PPI therapy exhibited a higher incidence of intervention or acute mortality compared to those solely using NSAIDs. Through multivariate analysis, our study indicates that bleeding severity and baseline comorbidities, expressed by elevated MAP(ASH) scores and ASA classification, are pivotal factors influencing the clinical outcomes of NVUGI.

In summary, the use of NSAIDs or PPIs to safeguard against gastrointestinal lesions induced by these drugs did not demonstrate a discernible impact on the short-term outcomes of patients with non-variceal upper gastrointestinal bleeding.

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Disclosure: Nothing to disclose.

PP0017

FACTORS ASSOCIATED WITH EARLY VARICEAL ERADICATION AFTER ENDOSCOPIC LIGATION

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Introduction: Esophageal varices are present in up to 40% and 85% of patients with compensated and decompensated cirrhosis, **respectively**. Endoscopic treatment of esophageal varices is used for the prevention of first variceal bleeding, treatment of active variceal bleeding, and prevention of rebleeding. Endoscopic variceal ligation (EVL) is the most common treatment. Multiple EVL sessions are often required to eradicate them. We aimed to detect factors predicting lower number of endoscopic treatments.

Aims & Methods: The study recruited all cirrhotic patients admitted to the hospital within the last two years, who have been treated for esophageal varices by EVL and achieved eradication through this endoscopic treatment. For each patient, main demographical and clinical characteristics were collected, including, etiology of cirrhosis, severity of liver disease according to Child-Pugh and MELD score and main laboratory tests. Varices were described according to the Japanese classification, and for each EVL session number of placed bands was recorded. Binomial logistic regression was used for univariate and multivariate analysis and odds ratios (OR) were calculated.

Results: A total of 31 patients (71% males, mean age 62.7±9.2) with liver cirrhosis and esophageal variceal (F2 41.9% and F3 58.1%) were included. The most frequent etiologies were alcoholic cirrhosis, non-alcoholic steatohepatitis and cryptogenic cirrhosis, in 8 (25.8%), 9 (29%) and 4 (12.9%), respectively. Esophageal Varices with Red Color Sign on Endoscopy, Cherry Red Spot (CRS), Red Wale Markings (RWM) and Hematocystic Spots (HCS), were detected in 19 (61.3%) 23 (74.2%) 5 (16.1%) patients. Only 8 patients (25.8 %) had gastroesophageal varices (GOV) while 30 subjects (96,8%) had congestive gastropathy. Concerning severity of cirrhosis, there were 13 (41,9%), 17 (54,8%), 1(3,2%) patients in Child Pugh's class A, B and C, respectively. In 14 subjects, a single session was sufficient to eradicate varices, while 9 patients required ≥3 sessions. At multivariate analysis, higher platelet count (OR=0.98, p=0.15), higher number of rubber bands (OR=0.67 p=0.16) and female sex (OR= 0.24, p=0,11) predicted the fewer number of endoscopic sessions required for eradication of varices, while at multivariate analysis, only higher platelet count demonstrated a non significative trend (OR= 0.98, p=0,07).

Conclusion: When performing endoscopic ligation of esophageal varices, the number of platelets could influence the number of endoscopic sessions to eradicate esophageal varices.

Disclosure: Nothing to disclose.

PP0018

ALCOHOL ABUSE DOES NOT IMPACT IN OUTCOMES OF NON-VARICEAL UPPER GASTROINTESTINAL BLEEDING

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Introduction: Alcohol abuse is a significant risk factor for the development of hepatic cirrhosis, which can lead to upper gastrointestinal bleeding. In non-cirrhotic patients, alcohol has been studied as a risk factor for non-variceal upper gastrointestinal bleeding (NVUGI), particularly in cases of peptic ulcer bleeding (1).

However, the clinical impact of alcohol abuse on outcomes remains unclear. Only one retrospective study has analyzed this, finding an increased risk of rebleeding and non-significant increased mortality among alcohol consumers (2).

Nevertheless, this study included cirrhotic patients, who accounted for most of the mortality. There is currently no available data on the impact of alcohol on NVUGI outcomes in non-cirrhotic patients.

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Aims & Methods: This study aims to determine the impact of alcohol consumption on clinical outcomes in NVUGI.

We conducted a retrospective analysis at a Spanish tertiary hospital, encompassing NVUGI cases from 2013 to 2022. Alcohol abuse was defined as daily alcohol intake exceeding 20g. Key clinical outcomes included the need for intervention (transfusion, endoscopic, radiological, or surgical) and in-hospital mortality. In addition, we compared baseline characteristics and assessed both acute and delayed 6 months outcomes, including rebleeding and delayed cardiovascular events. Statistical analysis was performed using SPSS.

Results: A total of 892 cases were included. The alcohol abuse group had a higher proportion of men (95.6% vs. 64.1%, p < 0.001) and were younger (median age of 56 years vs. 66 years, p < 0.001). Non-alcohol consumers had more comorbidities and a higher ASA comorbidity risk score (58% had ASA 3 or 4 vs. 46% in the other group, p = 0.015). Additionally, the MAP(ASH) score was higher in the first group (>2 in 54.5% vs. 46.3%, p = 0.030). We did not find any other differences between groups, neither in key outcomes nor in the remaining comparisons for the acute episode. However, delayed mortality (8,2% in group 1 vs 3,1% in group 2, p=0,046) and six months cardiovascular events (8,9% vs 2,3%, p=0,011) were more frequent in the non-alcohol group. In multivariate analysis, only ASA 3-4 (OR: 4.17, CI: 1.49-11.68, p= 0.006) and MAP(ASH) (OR: 7.61, CI: 2.54-22.8, p < 0.001) were found to be independent risk factors for acute mortality.

	No Alcohol	Alcohol	p
Age	66,75	56,88	<0,001
Sex	Men 464 (61,4%)	Men 130 (95,6%)	<0,001
Chronic obstructive pulmonary disease	75 (9,9%)	13 (9,6%)	0,896
Chronic renal disease	126 (16,7%)	6 (4,4%)	<0,001
Heart faliure	91 (12%)	6(4,4%)	0,009
High blood pressure	368(48,7%)	52 (38,2%)	0,071
Diabetes mellitus	182 (24,1%)	19 (14%)	0,009
Smooking	95 (12,6%)	69 (50,7%)	<0,001
MAPASH >2	334 (54,5%)	49 (43,4%)	0,030

Conclusion: Alcohol abuse is more common among younger individuals with fewer comorbidities. This consumption does not worsen clinical outcomes in non-cirrhotic NVUGI patients, with similar rates of need for intervention or mortality. Our results indicate that only comorbidities and bleeding severity (summarized as high ASA score and MAP(ASH)) score influence the outcomes. In summary, while alcohol consumption may increase the risk of NVUGI incidence, it does not impact acute outcomes.

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Disclosure: Nothing to disclose.

PP0019

ACUTE UPPER GI BLEEDING – IMPROVING INITIAL MANAGEMENT IN 2 DGHS

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Introduction: Acute Upper GI bleeds (AUGIB) are common and are associated with significant mortality (1). Quick recognition, resuscitation and endoscopy improves patient outcomes (2).

Care bundles are known to improve care. The BSG 2018 AUGIB bundle was designed to standardise care. We aimed to audit management of AUGIB in two sites in North East England.

Aims & Methods: We retrospectively analysed patients who underwent endoscopy for AUGIB from May-October 2023. We surveyed junior doctors to assess knowledge and confidence in managing AUGIB.

Results: 98 patients were identified across 2 sites, the mean age was 66, and 58% were male. The AUGIB bundle was used in 10% cases.

Recognition: 83% presented to A+E with AUGIB with the remainder already inpatients. 78% presented with melaena, and 24% were suspected variceal bleeds.

Resuscitation: 100% had EWS scores, 87% received IV fluids, and 93% patients with Hb<70 received RBC transfusion.

Risk Assessment**:** Mean Glasgow-Blatchford score (GBS) was 9 for both sites. Site 1 had calculated a GBS in 65% cases, and site 2 in 88%.

	Site 1	Site 2	Overall
Variceal bleeds receiving antibiotics and terlipressin	50% (n=5)	69% (n=9)	61% (n=14)
Aspirin continued	60% (n=3)	66% (n=6)	64% (n=9)
Other antithrombotics held	100% (n=17)	91% (n=20)	95% (n=37)

Table 1: Summary of treatment

Refer: of the 98 patients 70% discussed with endoscopist within 24hours, 47% had an OGD within 24 hours, and 30% required endoscopic intervention.

Review: 84% (site1) and 63% (site 2) had PPI plan, 50% (site 1) and 36% (site 2) had antithrombotic plan.

Of the small number of patients in which the bundle was used, all received appropriate resuscitation, all had GBS completed and all had OGD within 24hours of presentation.

200 surveys were sent across sites, 61 junior doctors responded. 73% felt confident in recognising AUGIB, and 65% felt confident in management. Only 32% knew appropriate initial variceal management. 40% were unsure when to call OOH endoscopist, and only 36% knew aspirin should continue.

Conclusion: Whilst recognition of AUGIB and fluid resuscitation is good, improvement is needed for risk assessment with GBS, management of suspected variceal bleeds, and timely referral for endoscopy. More detailed endoscopy reports are needed regarding PPI and antithrombotic plans.

More education with junior doctors is needed regarding management of AUGIB, and particularly when to call endoscopist and antithrombotic decisions. It would be helpful to educate staff to improve management. A reaudit of clinical management and the junior doctor survey is planned.

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Disclosure: Abstract submitted to BSG Live 24.

PP0020

AN IN-DEPTH ANALYSIS OF MORTALITY OF HOSPITALIZED PATIENTS WITH SUSPECTED NON-VARICEAL UPPER GASTROINTESTINAL BLEEDING – DATA FROM A PROSPECTIVE REGISTRY

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Introduction: Non-variceal upper gastrointestinal bleeding (nvUGIB) mortality has improved but remains high. Guidelines on nvUGIB often focusing on pre- and endoscopic management. However, more data on causes of death are needed, because non-bleeding specific mortality is unlikely to be improved by interventional advances.
Aims & Methods: The aims of this retrospective analysis of a single tertiary center registry of patients with nvUGIB were to assess causes of death in patients with nvUGIB and characterize this cohort. All hospitalized patients evaluated for suspected non-variceal UGIB (i.e., patients undergoing endoscopy) at our endoscopy unit between January 2018 and October 2020, either due to presentation for nvUGIB or with a suspected bleeding episode during hospitalization for any reason, were included in a prospective registry.
Data on pre-endoscopic signs/symptoms of UGIB, epidemiological data, laboratory values, and medications were extracted from the electronic health record. Endoscopy reports were individually searched and relevant data including the most likely culprit lesion in the opinion of the endoscopist as well as respective hemostatic treatment extracted.
Results: Among 1135 patients – 41.7% women, mean age 70.4 (±15.7) years, median length of stay 8 (IQR 4;16) days – 92 (8.1%) died during hospitalization and were further analyzed. Mean age was 73.2 (±12.9) years, 44.5% were women, 36% of patients were on anticoagulation, 27% on antiplatelet therapy, and 9% on both. The median Charlson comorbidity index was 5 (IQR 3;6.8), the CHA2DS2-VASc-Score 3 (IQR 2;5).
However, the median AIMS65 score was only 2 (IQR 1;2). Median time from admission to death was 17 (IQR 6.5;30) and from endoscopy to death 11 (IQR 4;21.5) days. The most common sources of bleeding as determined by endoscopy reports were: uncertain (24%), peptic ulcer (21%), esophagitis (14%), and non-peptic ulcers (9%). Adherence to guideline-based endoscopic peptic ulcer management recommendations was 70%. Radiological intervention for bleeding was performed in 1%, surgical intervention in 8.7%. Patients who died were older (73.2 vs. 70 years, p=0.0336), had lower hemoglobin nadirs (7.1 vs. 7.8 g/dL, p<0.0001), high c-reactive protein levels (p<0.0001) and creatinine (p<0.01), had longer lengths of stay (LOS) (median LOS 17 vs. 8 days, p<0.0001), and longer times from admission to endoscopy (median 3 days vs. 1 day, p<0.0001). However, sex was not associated with mortality (p=0.58). Looking at bleeding in association with causes of death, mostly (n=55, 59.8%), bleeding was categorized as a bystander. Only 32.6% of causes of death were judged as directly due to or likely associated with the bleeding episode, and these patients had a higher AIMS65 (p=0.0301). Only 12% died due to an unmanageable bleeding (or change to best supportive care). Overall, the following major groups of causes of death were identified: malignancy (18.5%), multi-organ failure (17.4%), infection/sepsis (15.2%), bleeding-associated (15.2%), cardiovascular (13.0%), and others (23.9%).
Conclusion: UGIB mortality remains relatively high despite management advances. However, only a minority of patients die due to UGIB and even among those most deaths are related to sequelae of the bleeding instead of unmanageable bleeding. Patients who die are older, likely sicker, and suffer from many comorbidities. Based on these data, it is uncertain whether focusing on endoscopic management improvements will significantly change UGIB-related mortality. Possibly, the focus should be shifted to improving general/internal medicine or geriatric care.
Disclosure: Nothing to disclose.

PP0021

TIMING OF ANTICOAGULATION RESUMPTION AFTER GASTROINTESTINAL BLEEDING: A COHORT STUDY OF 218 PATIENTS

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Introduction: Restarting anticoagulation (AC) therapy following acute gastrointestinal bleeding (GIB) is a multidisciplinary, personalized decision. Current guidelines recommend resuming AC, though data on the optimal timing is limited.
Aims & Methods: Our aim is to assess the short-term effect of early and delayed AC therapy resumption.
We analyzed the data of the Hungarian GIB Registry (a multicenter, prospective database) between 2019-2022. Patients were included after successful management of AC-related GIB. Detailed medical and drug history, demographics, and data on the index bleeding and management (bleeding source, endoscopic, and medical therapy) were collected. 30 and 90-day rebleeding, thromboembolic (TE) complications, all-cause, and 30-day recurrent GIB or TE-related mortality were assessed in the early (≤ 7 days) and delayed (>7 days) AC resumption groups.
Results: 218 cases were analyzed. Except for the CHA_2Ds_2-VASc score (mean (SD) 5.19 (1.65) vs 4.37 (1.549); *p*=0.02), the main characteristics (e.g., age, Charlson comorbidity index, active malignancy, anamnestic previous GIB, co-medications, HAS-BLED score, hemodynamic instability, high-risk bleeding site, endoscopic intervention) were similar between the two groups. Delayed AC therapy resumption was not associated with decreased 30-day rebleeding risk (5.3% vs. 6.7%; HR = 1.26, 95%CI 0.27-5.77, *p*=0.76). Delayed resumption was not associated with a higher 90-day TE complications rate (2.1% vs. 6.6%; HR = 3.17, 95%CI 0.58-17.33, *p*=0.18). Recurrent GIB-related 30-day mortality occurred only in one case (early resumption group, 12 days after admission with index bleeding). TE-related mortality was not observed during the follow-up.

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Conclusion: Anticoagulation resumption in the first seven days after GIB was proven to be safe, which supports the suggestion of the actual European guidelines. Further research on separated bleeding sources and types of anticoagulants is needed.
Disclosure: Authors do not have any conflict of interest to disclose.

PP0022

EXPERIENCE IN THE MEDICAL TREATMENT (THALIDOMIDE, LAR OCTREOTIDE) OF PATIENTS WITH BLEEDING FROM UPPER GASTROINTESTINAL (GI) VASCULAR MALFORMATIONS RESISTANT TO ENDOSCOPIC THERAPY

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Introduction: In the treatment of bleeding from GI vascular malformations, the priority of endoscopic hemostasis methods, particularly argon plasma coagulation (APC), is unquestionable. Until now, no effective medication-based therapy has been available as an alternative or supplemental method. Previous studies have reported on the ineffectiveness of steroids, estrogens, and progestogens. In the past decade, several promising clinical trials have been conducted regarding the use of thalidomide and long-acting-release (LAR) octreotide to reduce bleeding from vascular malformations.
Aims & Methods: At the Department of Internal Medicine-Gastroenterology of the Petz Aladár University Teaching Hospital in Győr-Moson-Sopron County, in patients who did not respond to endoscopic treatment, either.
1. Investigate the efficacy and side effects of thalidomide,
2. Investigate the efficacy and side effects of LAR octreotide in patients resistant to thalidomide treatment.

From July 2021 to February 2024, 25 patients were treated for bleeding and anemia from GI vascular malformations. In 20 patients, bleeding was stopped with 35 (1-2) APC treatments. 5 polymorbid patients (male:female= 1:4, average age ~64.6 years) required regular transfusions despite previous ~7.2 APC treatments. Their predominant underlying diseases included Banti's syndrome, autoimmune hemolytic anemia, ischemic cardiomyopathy, atrial fibrillation, Osler's disease, and autoimmune hepatitis. These 5 patients underwent a 3-month course of 100mg thalidomide per day. In patients where thalidomide was ineffective or caused side effects, LAR octreotide intramuscular injections were indicated once every four weeks at a 30mg dose. Visits, blood counts, and lab tests were conducted monthly.
Results: With thalidomide treatment, the transfusion requirement ceased in 2/5 patients. In 2/5 patients, the treatment was suspended due to ineffectiveness and side effects (deep vein thrombosis, emesis). Treatment was ineffective in 1/5 patients. In the last 3 patients, LAR octreotide treatment was also ineffective. Mild side effects (meteorism, nausea, exanthemas) were observed in these patients.
Conclusion: In the care of patients bleeding from vascular malformations resistant to endoscopic therapy,
1. Thalidomide may be an effective medication,
2. Close patient monitoring is important to filter out side effects,
3. The effectiveness of LAR octreotide cannot be determined based on the study,
4. Further follow-up studies are necessary.
Disclosure: Nothing to disclose.

PP0023

ENDOSCOPIC MANAGEMENT OF NON VARICEAL UPPER GI HEMORRAGES: OUR EXPERIENCE AS A TYPE 1 DEPARTMENT

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Introduction: This study aimes at evaluating the appropriate timing of endoscopic intervention in patients with non-variceal upper GI hemorrages upon admission in the emergency department and treated while on call during night shifts and public holidays. The need for blood transfusion, effectiveness of treatment, safety and reliability of endoscopic management were taken into account. We analyzed the results obtained after endoscopic treatment in terms of bleeding control, procedure-related complications, average days of hospital stay, need for second endoscopic look, surgical intervention or interventional radiology, and mortality.
Aims & Methods: From January 2019 to January 2024, 1215 patients admitted in the emergency department of Policlinico Umberto I in Rome with signs of digestive bleeding, underwent an esophagogastroduodenoscopy which demonstrated that the cause of bleeding was a non-variceal lesion of the upper digestive tract. Only adult patients aged 20 to 95 (33.3% between 60 and 80), who underwent operative endoscopic procedure were included in the study. Four hundred ninety-two patients were female (40.5%). Nine hundred sixty-three (79.3%) (963) did not undergo endoscopic therapy due to spontaneous or pharmacological resolution of the cause of bleeding while in 161 patients (16.7%) endoscopic therapy was not possible due to death caused by hemodynamic instability and underlying comorbidities. Two hundred fifty-two patients (20.7%) presented, during the endoscopic examination, signs of recent or ongoing bleeding, requiring endoscopic therapy. Eleven patients who underwent treatment (4.4%) died from complications unrelated to the procedure.
Results: Urgent EGDS, within 12 hours, was performed in 78.6% (198) of cases who presented signs of persistent hemodynamic instability, with definitive endoscopic success in 51.2% (111). Fifty-four patients (21.4%) underwent emergency intervention within 24 hours once hemodynamic stability was achieved with an endoscopic success in thirty one patients (59.1%). The average duration of the endoscopic procedure was 17 minutes. The post-procedure hospital stay was 6.5 days in 79.7% (201) of patients. All procedures were performed under deep sedation, in a dedicated operating room in the presence of the anesthesiologist. Patients who underwent a very early EGDS (<12h) had an average hospital stay, need for second look and mortality similar to that of patients treated between 12 and 24h. The single endoscopic intervention was successful in 56.4% of patients (142); 26.2% (66) underwent an endoscopic second look, 5.2% (13) a surgical procedure and 3.6% (9) interventional radiology. The overall mortality was 8.8% (22 patients).
Conclusion: In our experience, the endoscopic procedure was performed in most cases within 6-12 hours of admission in the emergency room. In light of our data and the heterogeneity of the population in urgent/emergency conditions, we can state that acting within 12-24 hours has proven to be an adequate timing for the management of non-variceal upper digestive hemorrhage, with good results in terms of haemostasias and the need for an endoscopic, surgical or radiological second look. The average hospital stay and the operating time is short and the mortality rate linked to the procedure is low. In accordance with the recommendations of the European Society of Gastrointestinal Endoscopy we believe that this timing is adequate to guarantee good long-term results. A very early EGDS <12h can be considered in patients who present signs of hemodynamic instability.
Disclosure: Nothing to disclose.

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PP0024

OVER-THE-SCOPE CLIP (OTSC) VS STANDARD ENDOSCOPIC APPROACH FOR PRIMARY TREATMENT OF NONVARICEAL GASTROINTESTINAL BLEEDING: SYSTEMATIC REVIEW AND METANALYSIS

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Introduction: Non-variceal gastrointestinal bleeding is a common condition associated with significant morbidity and mortality. Several approaches are available for primary endoscopic treatment. The standard endoscopic approach uses contact thermal methods or hemoclips with or without pre-injection of diluted epinephrine. A newer alternative is the use of an over-the-scope clip (OTSC).

Aims & Methods: The aim of our study was to evaluate the efficacy of these two modalities in the treatment of non-variceal gastrointestinal bleeding. Methods: PubMed, EMBASSE and the Cochrane Library databases were searched for multicentre randomised controlled trials comparing the efficacy of OTSC with standard endoscopic procedures for the primary treatment of non-variceal gastrointestinal bleeding published up to January 2024. The primary outcome measure was the 7-day re-bleeding rate. We also assessed the presence of persistent bleeding, 30-day rebleeding rate, length of hospital stay and mortality. A systematic review and network meta-analysis was performed for both treatments.

Results: Four trials met the inclusion criteria and were included. Regarding the 7-day rebleeding rate, OTSC showed superior efficacy compared to standard endoscopic therapy in the treatment of non-variceal gastrointestinal bleeding (RR 0.29, 95%CI 0.14-0.62). OTSC was also superior to standard endoscopic therapy in terms of re-bleeding within 30 days (RR 0.43, 95%CI 0.24-0.78). In terms of failure to achieve primary haemostasis, results were similar between the two methods, with a trend in favour of OTSC (RR 0.37, 95%CI 0.37-1.02). Finally, no differences were found in mortality (RR 0.61, 95%CI 0.25-1.51) and length of hospital stay (SMD -0.01, 95%CI -0.20-0.17).

Conclusion: OTSC significantly reduced the 7-day and 30-day re-bleeding rate compared to standard endoscopic therapy. There were no differences in mortality and length of hospital stay.

Disclosure: Julio Iglesias-Garcia: Advisor Fuji.

PP0025

EFFICACY OF DANIS-ELLA ESOPHAGEAL STENT FOR POST-EVL ULCER BLEEDING: A SINGLE-CENTER STUDY

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Introduction: Post-esophageal variceal ligation (EVL) ulcer bleeding poses a significant challenge in clinical practice, with various management modalities employed. However, there is a lack of established guidelines for effective management. This study aims to evaluate the effectiveness and safety of self-expandable metal SX-DANIS-ELLA (DE) stents in controlling post-EVL ulcer bleeding.

Aims & Methods: This retrospective audit focused on cirrhotic patients who underwent SEMS placement for post-EVL ulcer bleed. The study assessed bleeding control rates within 48 hours post-treatment and mortality rates within 6 weeks post-treatment

Results: Among 1025 endoscopic procedures for variceal ligation over 40 months, 1.4% (15 patients) presented with post-EVL ulcer bleed, all managed with DE stents. SEMS successfully controlled bleeding in 100% of cases at 48 hours post-procedure without the need for alternative rescue strategies. Median SEMS placement duration was eight days, with complete ulcer healing observed in 14 patients upon stent removal. However, one patient passed away within 72 hours due to decompensation. Early rebleeding within 6 weeks occurred in two (13.3%) patients which was managed with EVL. The 6-week mortality rate was notable at 20%(3 out of 15 patients). Adverse events included superficial esophageal ulcers at upper end of stent mesh in 9 patients out of 15(60%), SEMS dislocations in 3 out of 15 patients (20%) and one very deep esophageal tear with ulcer at the upper end of the stent mesh (6.6%). Two patients also had aspiration pneumonitis post procedure which was managed conservatively

Conclusion: SX-ELLA esophageal stents effectively controlled post-EVL ulcer bleeding. Despite successful bleeding control, the high mortality rate underscores the need for further strategies. SEMS dislocations highlight the complexity of this intervention and the importance of expertise in stent placement. Future research should explore the role of early transplant referral in reducing mortality following SEMS placement and improving patient outcomes

Disclosure: Nothing to disclose.

PP0026

UPPER GASTROINTESTINAL BLEEDING: RECURRENCE AND IMPACT ON MORTALITY

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Introduction: Upper gastrointestinal bleeding is a common reason for emergency department visits. Thanks to advancements in new hemostasis techniques, morbidity and mortality rates have decreased but remain significant in most studies reported in the literature. The aim of our study is to investigate the epidemiological and evolutionary profiles of upper gastrointestinal bleeding.

Aims & Methods: This is a prospective, descriptive, and analytical study conducted over a period of 4 years from April 2020 to February 2024. Patients presenting with gastrointestinal bleeding who underwent a complete clinical examination, comprehensive laboratory tests, esophagogastroduodenal endoscopy, and immediate and follow-up care were included in our study. Patients who did not undergo esophagogastroduodenal endoscopy and those with incomplete medical records were excluded.

Results: 282 patients with gastrointestinal bleeding were included in our study. The mean age was 56.1 years (range: 20 to 90 years). The male-to-female ratio was 1.26. A history of upper gastrointestinal bleeding was noted in 41 patients (14.5%).

The presentation of upper gastrointestinal bleeding included melena with hematemesis in 93 patients (32%), melena in 68 patients (24.1%), hematemesis in 50 patients (17.7%), and rectal bleeding in 20 patients (7%).

150 patients received blood transfusions (53%), and 38 patients required admission to the intensive care unit (13% of cases). Peptic ulcer disease was the most common cause of bleeding encountered in 136 patients (48% of cases). 70 patients (24%) underwent hemostatic procedures. Total recurrence of bleeding was noted in 35 patients (12.3%): 12 patients

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had esophageal variceal rupture (34.28%), and 8 patients had peptic ulcer disease (22%). The mean time to recurrence was 25.51 days. Ten of these patients were on anticoagulants and/or antiplatelet therapy before the bleeding episode (43%). There were 6 cases of death (2.59% of cases): 2 cases of peptic ulcer disease and 4 cases of esophageal variceal rupture. All deceased patients had experienced recurrent bleeding (100%).

Conclusion: The total recurrence rate of bleeding in our series was 2.59%. The presence of esophageal varices, the use of anticoagulant and/or antiplatelet therapy, and the absence of endoscopic treatment appear to be factors associated with recurrent bleeding. All deceased patients had experienced recurrent bleeding.

Disclosure: Nothing to disclose.

PP0027

CLINICAL PRESENTATIONS, ENDOSCOPIC FINDINGS, AND OUTCOMES OF ACUTE NON-VARICEAL UPPER GASTROINTESTINAL BLEEDING IN ELDERLY PEOPLE

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Introduction: Acute upper gastrointestinal (UGI) bleeding is a common medical emergency among the elderly. There is limited data on the management and outcomes of non-variceal UGI bleeding in the elderly in India.

Aims & Methods: We aimed to study clinical presentation, endoscopic findings, and outcomes in elderly patients with acute non-variceal UGI bleeding. All elderly patients (≥ 60 years of age) who presented with acute UGI bleed and were admitted for the same from 2021 to 2022 were retrospectively analysed. Baseline characteristics, blood transfusions before and after endoscopy, endoscopic diagnosis and therapy, AIMS65 score, complete Rockall score, and outcomes such as in-hospital re-bleeding and mortality were recorded.

Results: Of the 399 patients who had acute non-variceal UGI bleed, 309 patients aged <60 years were excluded. Of the 90 patients studied, the mean age of presentation was 69.8 ± 7.35 years, with 81.1% males. The most common presentation was melena (30%). Forty-one (45.5%) patients had a history of antiplatelet and anticoagulant use. The most common endoscopic finding was gastric ulcer (54.4%), followed by duodenal ulcer (22.2%), gastroesophageal junction ulcer (7.7%), Mallory-Weiss tear (6.6%), duodenal diverticular bleed (2.2%). The majority of ulcers were Forrest III (43.3%), followed by Forrest IIa (23.3%) and IIc (18.3%). Around 65% of patients with non-variceal UGI bleeds did not require endotherapy. Dual endotherapy was done in 25.5.5% of patients. Triple endotherapy was done in two patients. AIMS65 score was 1 in 50% of patients and 2 in 24.5% of patients. The majority (81.1%) of patients had a complete Rockall score of 3-7, indicating a moderate risk of re-bleeding and mortality. All the 90 patients were discharged without any mortality.

Conclusion: Our study showed that gastric ulcer was the most common cause of acute non-variceal UGI bleeding in the elderly, and melena was the most common presentation. Early endoscopic therapy may reduce blood transfusion requirements, in-hospital re-bleeding, and mortality.

Disclosure: Nothing to disclose.

PP0028

RECURRENCE RATE OF DIGESTIVE BLEEDING DUE TO RUPTURE OF ESOPHAGEAL VARICES

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Introduction: Upper gastrointestinal bleeding is a frequent reason for consultation in the emergency room, and digestive bleeding due to rupture of esophageal varices represents the second cause in terms of frequency and can cause serious and recurrent bleeding. Recurrence of digestive bleeding due to rupture of esophageal varices (OV) is a major concern in gastroenterology and hepatology. Due to the evolution of hemostasis techniques and management strategies, the rates of recurrence, morbidity and mortality have decreased but remain significant in most studies reported in the literature.

Aims & Methods: The aim of our work is to study the epidemiological profile of hemorrhagic recurrence of esophageal varices.

This is a prospective, descriptive and analytical study, carried out over a period of 24 months from January 2022 to December 2023. we included in our study patients who had consulted for digestive bleeding due to rupture of esophageal varices and who had benefited from a complete clinical examination, a complete biological assessment, an Esophago-gastro-duodenoscopy and immediate and remote monitoring. Patients who did not benefit from an esophago-gastro-duodenoscopy and those whose files were incomplete were excluded.

Results: 90 patients presenting with digestive bleeding due to rupture of esophageal varices were included in our study. The average age is 55 years (extremes: 17 and 89 years). The M/F sex ratio is 0.8. The mode of revelation of upper digestive bleeding was melena associated with hematemesis in 10 patients (11.1%), melena in 30 patients (33.3%), hematemesis in 50 patients (55.5%).

12 (13.3%) patients required intensive care. Hemorrhagic recurrence was noted in 30 (33.3%) patients, including 11 (12.2%) early recurrence < 06 weeks, and 19 (21.1%) late recurrence after 06 weeks. For patients who had presented a recurrence, we found 18 (60%) patients with an age > 60 years, 16 (53.3%) patients with a history of ascitic decompensation, 10 (33.3%) patients with an initial large-scale digestive hemorrhage.

Conclusion: Hemorrhagic recurrence in our series was 33.3%. More frequent in elderly subjects, and with ascitic decompensation, and severe initial hemorrhage.

Disclosure: Nothing to disclose.

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PP0029

RISK FACTORS OF IN-HOSPITAL MORTALITY AMONG PATIENTS WITH GASTROINTESTINAL BLEEDING AND ISCHEMIC HEART DISEASE

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Introduction: Gastrointestinal bleeding (GIB) and Ischemic heart disease (IHD) both remain the important causes of morbidity and mortality. Patients with IHD are more likely to suffer from GIB due to the characteristics of the population, heart disease, stress state and long-term use of antithrombotic drugs[1-3].

Previous studies have found that patients with simultaneous GIB and IHD have higher mortality than that of each individual disease[4-5].

Treatment for those patients is a challenge for both the cardiologists and gastroenterologists. In order to assess the frequency of in-hospital mortality in patients with GIB and IHD, and determine the potential risk factors for this condition, we designed and conducted this single center study.

Aims & Methods: Our study assessed the hospitalized outcomes of patients with simultaneous GIB and IHD. Moreover, we explored the risk factors of in-hospital mortality in those patients.

We performed a retrospective study in West China hospital between January 2014 and December 2018. Patients diagnosed with GIB and IHD were included. The primary outcome was in-hospital mortality for all causes. Clinical and laboratory data were collected and analyzed for clinical characteristics and potential risk factors of in-hospital mortality.

Results: 574 GIB patients with IHD were enrolled in the current study, and among them 107 in-hospital deaths were observed (in-hospital mortality rate of 18.64%). The causes of death included cardiovascular death (46, 42.99%), hemorrhagic shock (10, 9.35%), severe infection (33, 30.84%), and other causes (18, 16.82%).

Patients who died were older than survivors [78（68, 82） vs. 74（65, 80）, P = 0.008], with a higher incidence of COPD (27.10% vs. 13.70%, P=0.001) and chronic renal failure (47.66% vs. 27.19%, P<0.001). Patients with a higher heart rate at admission (90 vs. 81, P=0.006), or with a course of CHD less than 3 months (53.27% vs. 32.76%, P<0.001) were more likely to die. More patients in death group received plasma infusion (24.30% vs. 7.92%, P<0.001), and present with acute coronary syndrome (53.27% vs. 38.97%, P=0.007).

Compared with survivors, patients who died showed increased peak white blood cell (WBC) count [18.75(14.35, 24.9) vs. 9.79 (7.19, 13.82) ×109/L,P < 0.001], increased peak INR [1.51 (1.21, 2.58) vs. 1.13 (1.05, 1.31),P < 0.001)], and lower minimum platelet counts [94 (52, 147) vs. 128 (83, 180)×109/L, p<0.001]. In multivariable logistic analysis, age [OR (95% CI) =1.032 (1.006-1.058), P=0.014], peak WBC count [OR (95% CI) =1.123 (1.085-1.163), P<0.001], peak INR levels [OR (95%CI) =1.220(1.061-1.402), P=0.005], and heart rate at admission [OR (95%CI) =1.015(1.002-1.028), P=0.026] were independent predictors of in-hospital mortality.

Conclusion: Patients with simultaneous GIB and IHD had a high in-hospital mortality. Age, peak WBC count, peak INR levels, and heart rate at admission were independent risk factors for in-hospital death.

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Disclosure: Nothing to disclose.

PP0030

UTILIZATION OF HEMOSTATIC POWDER SPRAY VERSUS STANDARD ENDOSCOPIC TREATMENT IN THE MANAGEMENT OF MALIGNANCY-RELATED GASTROINTESTINAL BLEEDING: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Managing tumoral bleeding poses a significant challenge for endoscopists. Traditional tools such as clips and heat-probes frequently prove inadequate in controlling stubborn bleeding caused by mucosal invasion from these tumors. Hemostatic powder sprays provide a novel solution by applying a combination of minerals and polymers directly onto the tumor surface, forming a mechanical barrier to seal the bleeding site. However, there is a scarcity of data regarding their effectiveness in treating hemorrhagic gastrointestinal neoplasms.

This study seeks to evaluate the efficacy of Hemostatic powder spray in patients suffering from gastrointestinal bleeding related to malignancies.

Aims & Methods: This study aimed to evaluate the effectiveness of Hemostatic powder spray compared to standard clinical treatment in controlling gastrointestinal bleeding originating from malignant lesions. The primary outcome measures included the combined rates of rebleeding within 30 days, while secondary outcomes encompassed the combined rates of mortality over any time period.

We gathered relevant data, an extensive search of electronic databases was conducted up to April 2024. Only randomized controlled trials (RCTs) were included in the study, focusing specifically on cases where Hemostatic powder spray was pitted against conventional endoscopic techniques for managing bleeding caused by gastrointestinal tumors.

Results: Three studies involving a total of 162 patients were included in this meta-analysis. Among them, 83 patients underwent treatment with Hemostatic spray powder alone, while 79 patients received conventional hemostatic techniques.

The pooled results of the analysis revealed that Hemostatic powder spray was associated with a lower incidence of re-bleeding beyond 30 days from the onset of treatment (odds ratio [OR] 0.36, 95% confidence interval [CI] 0.15-0.84), demonstrating a significant advantage over standard endoscopy in the management of gastrointestinal tumor bleeding (p-value 0.01).

Additionally, there was no significant difference observed in all-time mortality between the two treatment groups (OR 0.40, 95% CI 0.19-0.83, p-value 0.09).

Conclusion: Hemostatic spray powders demonstrate significant efficacy in achieving hemostasis for gastrointestinal bleeding associated with malignancies. They should be regarded as a viable alternative to conventional endoscopic methods in such cases. Subsequent research should be

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structured to assess both the effectiveness and safety of hemostatic spray powders as a primary hemostatic approach compared to traditional endoscopic techniques.

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Disclosure: Nothing to disclose.

PP0031

IMPROVING DOCUMENTED ADVICE REGARDING ANTICOAGULANTS AND ANTIPLATELETS IN ACUTE UPPER GASTROINTESTINAL BLEEDS

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Introduction: Systemic anticoagulants are often prescribed to prevent and treat thromboembolism among other indications, yet presentations with Acute Upper Gastrointestinal Bleeds (AUGIB) in this cohort are common. Medical care bundles have been shown to improve both standards of care and patient outcomes. As a result, the British Society for Gastroenterology (BSG) created an AUGIB care bundle in 2019, and guidelines for endoscopy in patients on antiplatelets and anticoagulants in 2021. However, the former lacks definitive statements regarding resumption of antiplatelet/anticoagulant therapy, while the latter is not routinely referred to by non-Gastrointestinal specialists. It is recognised however, that resumption of antiplatelets and anticoagulants is associated with improved survival and reduced incidence of thrombotic complications. Thus, all patients should have a clear plan.

Aims & Methods: Admissions with AUGIB proceeding to endoscopy over a 6 month period were analysed. Of the 319 patients, 72 (22.5%) were taking either an antiplatelet, warfarin or a Direct Oral Anticoagulant (DOAC). Medical records were reviewed in relation to pre-procedure management of these medications, advice from the endoscopist regarding resumption (or not) of antithrombotic therapy and whether these medicines were managed appropriately thereafter.

Several interventions were then made; a change to the reporting software led to a prompt relating to antiplatelet/anticoagulant medications (though the template remains non-specific), an education campaign for post graduate doctors in training and medication guidance summaries visible in the endoscopy department.

A second 6-month review was then undertaken to evaluate if there had been any improvement in outcomes. In this time frame, 283 patients underwent endoscopic evaluation for suspected AUGIB of whom 83 (29.3%) were taking antithrombotic therapy.

Results: Results can be found in Table 1:

	Pre-Intervention	Post Intervention
Pre-procedure appropriate management	76%	81%
Documented advice on endoscopy report	35%	65%
Post procedure appropriate prescribing	68%	75%

Table 1.

A recurring theme was the discontinuation of aspirin in low-risk bleeds, and low quality medication advice given at the time of endoscopy with almost 4 in 10 reports simply stating 'continue medication'. Those without documentation resulted in further advice being sought (12.9% of cases)

Further subanalysis of the data revealed that advice was almost always documented in those presenting with major upper gastrointestinal bleeds (defined as patients presenting with haemodynamic instability). In this cohort 87.5% of patients had clearly documented advice regarding their anticoagulation or antiplatelet medication.

Conclusion: Further education to junior staff around utility of the BSG AUGIB bundle, as well as the anticoagulation guidelines is needed to improve pre and post procedure care. Further upgrades to the templates on the reporting software will also serve as prompts to the endoscopist to provide more specific advice. Human factors also need to be considered; limited advice is likely related to the high endoscopy workload pressures. Sub-analysis of the data suggests that we are better at providing more specific advice in high-risk bleeds/where endotherapy is required, reflecting that antithrombotic therapy decisions are considered by the endoscopist. Improved documentation at procedure, should translate into improved post procedure management and reduce further demands on the service.

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Disclosure: Nothing to disclose.

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PP0032

SAFETY AND EFFICACY OF PROTON PUMP INHIBITORS IN CRITICALLY ILL PATIENTS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Critically ill patients are at high risk for upper gastrointestinal (GI) bleeding.[1] Proton pump inhibitors (PPIs) are commonly used for prophylaxis, but their safety and efficacy compared to histamine-2 receptor antagonists (H2 blockers) and placebo remain unclear.[1–4]

Aims & Methods: Our systematic review and meta-analysis aims to assess the safety and efficacy of PPIs for preventing upper GI bleeding in critically ill adults. We conducted a comprehensive search of Scopus, Web of Science, PubMed, and Cochrane Central for relevant randomized controlled trials (RCTs) from inception until March 2024. Data were extracted from eligible studies and pooled as mean difference (MD) for continuous outcomes and Odds ratio (OR) for dichotomous data in a random-effect model meta-analysis, using R software.

Results: Nineteen RCTs were included in this meta-analysis with a total of 62221 patients. Our study showed that PPI was superior to control in terms of GI bleeding [RR 0.55, (95% CI, 0.42 to 0.72); P = <0.01]. However, the overall effect did not favor either of the two groups in terms of ICU, mortality rate, and nosocomial infection [MD -1.45, (95% CI, -5.31 to 2.42); P = 0.4], [RR 1.04, (95% CI, 1.00 to 1.09); P = 0.07], and [RR 1.27, (95% CI, 0.98 to 1.65); P = 0.07], respectively.

Conclusion: In conclusion, our findings demonstrated a significant reduction in GI bleeding risk associated with PPI use. At the same time, we did not find a statistically significant difference between PPI and control groups regarding ICU mortality rate, overall mortality rate, or nosocomial infection rates. These findings suggest that PPIs may be effective in reducing upper GI bleeding in critically ill patients. However, further research might be necessary to definitively determine the impact of PPIs on mortality and infection risks in this population.

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Disclosure: Nothing to disclose.

PP0033

UPPER GASTROINTESTINAL BLEEDING IN THE PORTUGUESE POPULATION: A MULTICENTRIC STUDY

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Introduction: Upper gastrointestinal bleeding (UGIB) is one of the most common medical emergencies managed by gastroenterologists. Despite advances in pharmacological/endoscopic therapy, this condition is still associated with considerable morbidity/mortality.

Aims & Methods: We aimed to characterize Portuguese patients and clinical approaches in UGIB, as well as factors associated with adverse outcomes. We conducted a retrospective multicentric cohort study which included adult patients consecutively presenting UGIB at 6 participating centers over a 6-month period. Data relevant to the initial UGIB episode and for up to 30 days afterwards were collected from patient medical records. Patients' clinical information was presented using descriptive statistics. Continuous variables were described by mean ± standard deviation. Categorical data were presented as absolute numbers and percentages. Qualitative variables were compared with Chi-square and Fisher's tests. Non-normal distribution quantitative variables were compared with Mann-Whitney test. A p value < 0,05 was considered statistically significant.

Results: 600 patients presenting with UGIB were included, mostly men (58,2%, n=349) with a mean age of 71 (18-100), with multiple comorbidities and a mean Charlson comorbidity index of 5 (0-16). Antiplatelet/Anticoagulant/NSAID consumption was present in 53,5% (n=321). The primary symptoms were melena (47,2%, n=283) and hematemesis (27,3%, n=164). The median Rockall score (RS) and Glasgow-Blatchford score were 5 (0-10) and 12 (0-21), respectively. A higher RS was associated with prolonged hospitalization (p=0,03). At least one cause for UGIB was documented in 499 patients (83%). Peptic ulcer disease (PUD, 37,8%, n=192), erosive gastroduodenitis (10,0%, n=51) and gastric/duodenal angiodysplasia (6,7%, n=34) were the main bleeding causes of nonvariceal UGIB (n=507). In patients with chronic liver disease (n=93), variceal bleeding (46,2%, n=43) was the main bleeding cause, followed by PUD (19,4%, n=18). Endoscopic hemostasis was performed in 257 patients (42,8%). Combination therapy (injection therapy + thermal/mechanical therapy) was used in 122 cases. Median hospitalization time was 11 days (0-168) and admission in intensive care unit (ICU) was needed in 9,8% (n=59). Surgery for severe UGIB was performed in 2% (n=12). In-hospital rebleeding occurred in 13,3%, more frequently in chronic liver disease (p=0,02). Death occurred in 90 patients (15%), 73 from a non-bleeding related cause. Logistic regression showed that age≥65 years (p=0,02), chronic liver disease (p=0,049) and ICU admission (p=0,045) were associated with in-hospital mortality.

Conclusion: This study contributed to characterization of Portuguese patients admitted with UGIB in real clinical setting. Gastroenterologists are facing a progressively older population, with multiple comorbidities and on therapies potentially causing UGIB. Although PUD still remains the

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most common cause of UGIB, other conditions such as angiodysplasia are becoming more frequent in clinical practice. Chronic liver disease is a risk factor for rebleeding and in-hospital mortality. The high mortality observed in this research was mainly due to decompensation of comorbid conditions rather than exsanguination, which reflects the complexity of the elderly. These patients require early risk assessment, adequate resuscitation and an attempt to identify and treat the bleeding source, which might help us to improve the management of UGIB in Portugal.

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Disclosure: Nothing to disclose.

PP0034

ENDOSCOPIC HEMOSTASIS DURING UPPER GASTROINTESTINAL BLEEDING OF NON-VARICEAL ORIGIN

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Introduction: Upper digestive hemorrhage (HDH) is a frequent reason for hospitalization in the emergency department. Schematically, they are divided into HDH of varicose origin and HDH of non-varicose origin.

Upper GI endoscopy must be performed as quickly as possible, under the best possible technical conditions.

The objective of our study is to evaluate the hemostatic endoscopic management of these hemorrhages in the emergency department.

Aims & Methods: This is a retrospective descriptive study covering a period of 26 months, from January 2022 to March 2024, and including all patients admitted in an emergency for upper digestive haemorrhage of non-variceal origin, identified during the endoscopy on-call duty at the "Médecine B" Hepato-gastroenterology and Proctology Department, CHU Ibn Sina.

All patients underwent clinical examination, blood count and oeso-gastro-duodenal fibroscopy (FOGD).

Patients admitted for HDH of varicose origin or with normal FOGD were excluded from our study.

Results: 91 cases of HDH of non-varicose origin were diagnosed out of a total of 225 cases of HDH consulting the emergency department, with a prevalence of 40.44%.

The mean age of our patients was 54.8 years (20-84 years), with a male predominance of 70.3% and a sex ratio of M/F : 2.3.

11 patients (12.08%) had a history of smoking, 11 (12.08%) had heart disease on anticoagulants, 10 (11%) had taken NSAIDs, 8 (8.79%) had nephropathy at the hemodialysis stage, and 7 (8%) were hypertensive.

The mode of externalization of HDH was melena associated with hematemesis in 26 patients (28.6%), isolated melena in 44 (48.4%), hematemesis in 16 (17.6%) and rectal discharge in 5 (5.5%).

The main etiologies of HDH of non-varicose origin revealed by FOGD were: gastro-duodenal ulcer in 73 patients (80.2%), with Forrest Ib ulcer in 9 (12.3%), Forrest IIa in 4(5.4%), Forrest IIb in 4(5.4%), Forrest IIc in 13(17.8%) and Forrest III in 43(59%).

Of the 91 patients, 15(16.48%) had esophagitis, 5(5.5%) had gastric process, 1(1.09%) had duodenal process and 8 patients (8.8%) had angiodysplasia lesions.

All patients were treated with a loading dose of injectable PPI, followed by a maintenance dose.

Endoscopic haemostasis was required in 19 patients (20.8%), of whom 8 (8.7%) benefited from injection of adrenaline serum and placement of haemostatic clips, 2 patients (2.2%) from exclusive injection of adrenaline serum, 8 patients (8.7%) from exclusive placement of haemostatic clips and one patient (1.09%) benefited from per-endoscopic surgical suturing.

The mortality rate in our study was 3.2%.

Conclusion: HDH of non-varicose origin remains dominated by ulcerative origin. Impregnation with PPIs prior to FOGD and the possibility of endoscopic haemostasis reduce the mortality rate.

Disclosure: Nothing to disclose.

PP0035

PATIENTS WITH IRON DEFICIENCY ANAEMIA BENEFIT FROM A SPECIALIZED GASTROINTESTINAL EVALUATION

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Introduction: Iron deficiency anaemia (IDA) is a frequent reason for consultation with a high consumption of health resources both for its diagnosis and for its treatment.

Aims & Methods: To evaluate the characteristics of patients referred to a specialized gastrointestinal (GI) consultation for IDA without macroscopic haemorrhage and the impact of this specialized evaluation on the consumption of health resources. We performed a retrospective observational study of patients evaluated in a specialized gastrointestinal hospital consultation for IDA from January 2021 to October 2023.

Results: We evaluated 218 patients where 134 were excluded due to macroscopic haemorrhage. Finally, 84 patients were analysed, 63% were men with a median age of 73 years (range 39-93) and a median Charlson index of 6 (range 0-15). The median time from the diagnosis of IDA to the first

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gastrointestinal visit was 46 months (range 1-161). 74% of the patients had been evaluated by another specialty previously. 43% were on antiplatelet treatment and 31% on anticoagulant treatment. The aetiology of anaemia was multifactorial in 24% (n=20) and in those with iron deficiency the most frequent cause was angiodysplasias (42%) but in up to 19% of cases the aetiology was not found despite a complete evaluation.

During follow-up, 58% required an anticoagulation switch, requiring definitive withdrawal of anticoagulation in 8% of patients. Prior to the GI consultation, only one patient was under somatostatin analogues treatment, which was subsequently started in 17 new patients. During follow-up, 12% (n=10) were included in a special management group with parallel blood test monitoring by the primary care physician for a more thoroughly monitorisation of the haemoglobin levels.

When comparing the consumption of health resources in the year before and after the GI assessment, a non-significant trend was observed towards a lower need for gastroscopies [0.87 vs. 0.65, p=0.07], but we found an statistically significant reduction in the number of colonoscopies [0.88 vs. 0.40, p<0.001].

We also observed a significant increase in the mean haemoglobin [10.6 g/dL vs. 12.1 g/dL, p<0.001] without statistically significant differences in the number of blood transfusion [2.82 units vs. 2.30 units, p=0.21] or intravenous iron therapy [1.44 g vs. 1.50 g, p=0.74] before and after the GI assessment respectively. A significant reduction in emergency department visits was also observed [0.79 vs. 0.43, p=0.002] as in hospital admissions [0.46 vs. 0.23, p=0.004] related to anaemia or its complications.

During follow-up, 12 patients died (none due to haemorrhage or anaemia), resulting in a cumulative mortality incidence of 6.7 deaths per 100 patients/year.

Conclusion: Multimorbid patients with iron deficiency anaemia benefit from the care of a specialized digestive consultation which is associated with a decrease in the use of health resources.

Disclosure: Nothing to disclose.

PP0036

WITHDRAWN

PP0037

VARIATION IN GASTRODUODENAL FUNCTION TESTING IN CLINICAL PRACTICE: RESULTS FROM AN INTERNATIONAL SURVEY

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Introduction: Gastric emptying scintigraphy (GES) is an accepted standard for gastroduodenal function (GDF) testing, while various other investigations are available for the assessment of dyspeptic symptoms. However, in the absence of guidelines, it remains uncertain which test one should favour.

Aims & Methods: This study examined variation in GDF testing between centres to establish the status of current practice. A web-based questionnaire assessing availability, standard operating procedures and interpretation of 15 different GDF testing techniques was distributed among specialist practitioners via professional organizations. Availability, indication, meal-related characteristics and standard operating procedures (SOP) were compared.

Results: Fifty-nine complete responses (57 clinical practice, 86% tertiary care) were included from 28 countries. Eleven (19%) respondents referred more than 10 patients for further testing monthly. Accessibility for clinical practice were as follows: 100% gastroscopy, 91% measurement of gastric emptying (79% by GES; 19% gastric emptying breath testing (GEBT); 5% SPECT), 39% intragastric pressure measurements (IGP), 33% pyloric Endoflip (PE), 30% gastric MRI, 19% nutrient drinking challenge (NDC), 18% gastric ultrasound (US), 18% motility capsule (MC) and 16% antroduodenal manometry (AM). Other techniques were available to less than 15% of the respondents. In clinical practice, meal-related dyspeptic symptoms and chronic unexplained nausea and vomiting (CUNV) represented the main indications for GE studies as compared to epigastric pain, burning or discomfort (GES: 70% and 94% vs. 20%, p<.00001; GEBT: 100% and 100% vs. 0%, p<.001), Similar results were present for PE (73% and 93% vs. 33%, p=.002), NDC (100 and 82 vs. 27%, p=.0006). MC (33% and 67% vs. 0%, p=.05), AM (28% and 75% vs. 25%, p=.11), MRI (75% and 88% vs. 38%, p=.09) and IGP (33% and 33% vs. 0%, p=.15) By contrast, US was indiscriminately performed (57% and 71% vs. 57%, p=.82).

There was a large heterogeneity in the type of meal and meal quantity used for each test as well as between tests. Symptoms were recorded by only half the respondents during GES and MRI, with higher percentages for other testing modalities. Variation in the specific symptoms included in questionnaires was present.

Conclusion: GES is widely available in clinical practice. Meal-related symptoms and CUNV represent the main indications for GDF testing in clinical practice. A large variation in the availability of GDF testing and SOPs exists between centres. This hampers clinical interpretation, data transfer between centres and research collaboration. These findings emphasise the need for an international co-operation effort at standardisation.

Disclosure: Nothing to disclose.

PP0038

BREAKING BARRIERS IN FUNCTIONAL DYSPEPSIA – A SYSTEMATIC REVIEW AND META-ANALYSIS ON DUODENAL TIGHT JUNCTION PROTEIN EXPRESSION

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Introduction: Disruptions in tight junction (TJ) protein expression leading to duodenal epithelial barrier impairment may contribute to increased intestinal permeability, potentially playing a role in functional dyspepsia (FD) pathophysiology. Currently published studies evaluated the role of several TJ proteins in FD patients with inconsistent results. Therefore, we conducted this systematic review and meta-analysis to evaluate the duodenal mucosal expression of several TJ proteins in FD.

Aims & Methods: We performed a systematic electronic search on PubMed, EMBASE, and Scopus using predefined keywords. Diagnosis of FD by ROME III or ROME IV criteria was considered acceptable. Full articles satisfying our inclusion and exclusion criteria were included. The principal summary outcome was the mean difference (MD) of several TJ proteins in FD patients and control subjects. The study aimed to find if patients diagnosed with functional dyspepsia have lower gene expression of intestinal TJ proteins compared to healthy subjects.

Results: A total of 8 and 5 studies were included in our qualitative and quantitative synthesis, respectively, with a total population of 666 participants, out of which 420 were FD patients. No significant differences were observed between FD patients and controls in the expression of claudin-1

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(-0.102 [CI 95% -0.303,0.099]), claudin-2 (0.161 [CI 95% -0.134,0.456)], claudin-3 (0.278 (CI 95% -0.280,0.837]), claudin-4 (0.045 [CI 95% -0.264,0.354]), ZO-1 (-0.221[CI 95% -0.683, 0.241]), ZO-2 (-0.070 [CI 95% -0.147,0.007]), ZO-3(-0.129 [CI 95% -0.376, 0.118]), B-catenin (-0.135 [CI 95% -0.484,0.214]), E-cadherin (-0.083 [CI 95% -0.229,0.063]), and occludin (-0.158 [CI 95% -0.409, 0.093]).

Conclusion: The expressions of all evaluated proteins including claudin-1, claudin-2, claudin-3, claudin-4, ZO-1, ZO-2, ZO-3, B-catenin, E-cadherin, and occludin did not significantly differ between FD patients and controls. However, due to the limited number of included studies, results should be interpreted with caution.

Disclosure: Nothing to disclose.

PP0039

DUODENAL FOOD ADMINISTRATION IN FUNCTIONAL DYSPEPSIA LEADS TO MUCOSAL REACTIONS, OBSERVED ON CONFOCAL LASER ENDOMICROSCOPY, WITHOUT ALTERATIONS IN PERMEABILITY OR MAST CELL ACTIVATION EX VIVO

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Introduction: Functional dyspepsia (FD) is a common gastrointestinal disorder, comprising postprandial distress syndrome, PDS, with meal-related symptoms, and epigastric pain syndrome, EPS, with meal-unrelated symptoms. In a pilot study in PDS patients, the 6-food elimination diet was able to provide symptom relief (1). Using confocal laser endomicroscopy (CLE), acute disruption of the duodenal epithelium in response to food protein administration can be observed (2). This assessment may have the potential to identify food components that should be the target of an elimination diet treatment approach.

Aims & Methods: Aim: To evaluate the acute effect of food administration on duodenal mucosal integrity in FD patients and to determine whether visualized alterations are associated with changes in mucosal permeability or markers of mast cell activation ex-vivo.

Methods: In 17 Rome IV FD patients (8 PDS, 1 EPS, 8 overlap PDS/EPS), classic allergy to nutrients was excluded by specific serum IgE tests. Index upper GI endoscopy (EGD) was followed at least 2 weeks later by a second EGD with CLE. Duodenal mucosa was visualized at baseline and after consecutive administration of egg white, soy, milk, wheat, lyophilized fish and shellfish and nuts solutions through the working channel in a randomized order. In those with reaction to one of the first 2 administered food solutions, an additional CLE was scheduled. Serum was obtained at baseline and 30 minutes after occurrence of acute mucosal alterations. Biopsies were obtained at index EGD and after acute alterations to measure transepithelial electrical resistance (TEER) in Ussing chambers. Tryptase levels in biopsy supernatants after 1h and following 24h incubation were assessed. Data are reported as mean ± SEM. Comparisons were done using two-sided t-tests. P-value <0.05 is considered significant.

Results: 29 CLE exams were carried out in 17 patients (age 35±3.7y, f16, BMI 22±0.7 kg/m^2). In 90% of exams (*n=26)*, acute alterations were visible only after food exposure, while in 10% (*n=3),* these were already present at baseline. Twelve patients (71%) showed alterations on CLE after exposure to one of the first 2 food components and underwent a second CLE procedure. Highest percentage of acute alterations was observed after soy administration (6/11 exposed). In the table, distribution of reactions per food is presented. After acute alterations, there was no difference in serum tryptase compared to baseline (4.52µg/L±0.6 vs. 4.72µg/L±0.6, p=0.21), nor did TEER change compared to baseline (28±0.9 Ωxcm^2 vs 28±1.6 Ωxcm^2, p=0.25), nor did tryptase content alter in biopsy supernatants, both after 1h and 24h compared to baseline (0.23µg/g±0.04 vs. 0.28µg/g±0.07, p=0.15 and 1.66µg/g±0.44 vs. 1.06µg/g±0.33, p=0.60).

Observations	Number of exams with exposure	Number of acute alterations	% of acute alterations per food item
Baseline	29	3	10
Milk	8	2	25
Nuts	12	4	33
Soy	11	6	55
Wheat	7	3	43
Fish	10	2	20
Egg white	8	3	38

Table. Number of examinations with exposure and alterations per specific food components.
All patients had a baseline visualization during CLE examination.

Conclusion: In FD patients without IgE mediated classical allergies, CLE visualized a high rate of acute food induced mucosal alterations. These acute responses were not associated with altered permeability, a rise in systemic tryptase, or release of tryptase from biopsy supernatants. Mechanisms underlying acute mucosal alterations and its impact on elimination diet efficacy needs further investigation.

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Speaker fee from Mayoly.

PP0040

WITHDRAWN

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PP0041

GENDER GAP IN THE TREATMENT OF FUNCTIONAL DYSPEPSIA? DATA FROM THE GERIATRIC POPULATION

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Introduction: Functional gastrointestinal disorders (FGIDs) are more prevalent in female than in male patients [1,2]. This triggers the question whether this is also reflected in medical care, especially also in self-medication, where often botanicals are used. A potential source of data to address this question is the PhytoVIS study, a pharmacoepidemiological study conducted in Germany in more than 20,000 patients taking herbal medicinal products.

Aims & Methods: To have a large and homogenous data base for a gender-related evaluation, records for a clinically proven herbal medicinal product (HMP) often used in FGIDs, STW 5 [3,4], were evaluated to uncover potential gender-related differences in the pharmacotherapy of FGIDs.

Results: 1513 records were evaluated, 71 % of them from females and 29 % from males, i.e. a 4 % higher proportion of females compared to the overall population of the PhytoVIS study. Main criterium for the evaluation was the Clinical Global Impression Scale – Efficacy (CGI-E). No major differences between females and males were found, apart that women perceived a minimally better effect of the therapy. The reported symptoms were differentiated into symptoms associated with functional dyspepsia (FD), irritable bowel syndrome (IBS), or symptoms that can be associated with both. As far as the records allow a differentiation, no gender differences could be observed. This applies also to the perceived efficacy, the tolerability and the onset of action.

Conclusion: So, while there are differences in the prevalence of FGIDs in women and men, there are no therapeutically relevant pharmacoepidemiological differences in patients with functional gastrointestinal diseases treated with a medicinal product indicated in these diseases, so suggesting that there is no gender gap in this highly relevant therapeutic field.

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Disclosure: JW and EC were interns, OK, is employee of Steigerwald Arzneimittelwerk GmbH, Darmstadt, Germany, KN und KK have received fees from the same company.

PP0042

ACTIVATING TRANSCRIPTION FACTOR 3 ASSOCIATED WITH NAUSEA IN GASTROPARESIS PATIENTS - A WHOLE GENOME SEQUENCING STUDY

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Introduction: Gastroparesis is a severe medical condition characterized by delayed gastric emptying and symptoms of nausea, vomiting, bloating, fullness after meals, and abdominal pain. Thus far, little is known about the underlying genetic risk factors for nausea in gastroparesis patients. We explored the association of genetic variants with nausea severity in a set of WGS data from a clinical gastroparesis study.

Aims & Methods: Blood samples were obtained from adults with a diagnosis of diabetic or idiopathic gastroparesis, having evidence of delayed gastric emptying, and moderate to severe nausea, daily average nausea score of >2.5, at least one episode of vomiting, PAGI-SYM nausea score of ≥2 at screening, and controlled blood glucose levels. WGS was conducted with 30x read depth using whole blood samples. Nausea score was assessed from a daily symptom dairy with a range from 0 mild to 5 severe. GWAS using linear model was conducted on variants with MAF ≥0.01 with adjustment of the subject's age, sex, BMI, etiology, and the first 3 PCs.

Results: Variant clusters from multiple genes are significantly associated with baseline nausea score at p=1e-5 level. Among the top scoring variants are loci within ATF3, PP2D1, CNTN4, BMP2, ABL1, CTNNA3 and KIF6. Interestingly, upon comparing our results with a published transcriptomic analysis that included both diabetic and idiopathic gastroparesis subjects, we found that ATF3 and BMP2 were overlapping with genes differentially expressed in both diabetic and idiopathic gastroparesis when compared with controls. ABL1 was however differentially expressed in diabetic cases whereas CNTN4 was differentially expressed in idiopathic controls. We expanded the analysis with single gene tests which shows that ATF3 is significant especially for MAF>0.05 variants (SKAT p=0.02). The top variant in ATF3 is rs79151334 (beta= 0.68, p=5.76e-6), whose minor allele associates with higher nausea score.

Conclusion: Additionally, we report new potential associations CTNNA3, PP2D1, and KIF6 as well as a cluster of variants between CDH2 and MIR302F, that associated with baseline nausea score. Among those, both CTNNA3 and CDH2 encode cadherin related proteins, suggesting this type of junction could play a role in the underlying mechanism.

Disclosure: I am an employee at Vanda Pharmaceuticals Inc.

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PP0043

HOW MANY SUBJECTS WITH SEVERE DISORDERS OF GUT-BRAIN INTERACTION HAVE AN INTOLERANCE OF GLUTEN WHEN A DOUBLE-BLIND GLUTEN EXPOSURE IS PERFORMED?

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Introduction: Approximately 50% of patients with disorders of gut-brain interaction (DGBI) report intolerance of wheat products [2]. While dietary gluten restrictions would be a feasible approach to managing gastrointestinal symptoms, in DGBI patients with gluten-related symptoms, there are challenges in relation to compliance and even adverse long-term effects of a gluten and wheat-free diet.

Aims & Methods: This study aims to test the feasibility of a double-blind gluten exposure test and determine the rate of patients with DGBI who respond with a change of symptom severity after exposure to gluten.

From a cohort of 70 consecutive patients referred to our Integrated Care Clinic for patients with complex DGBI, 35 agreed to participate in a double-blind gluten exposure test. This test was aligned with the gluten exposure test outlined in the Salerno criteria [1].

After exclusion of coeliac disease (endoscopy and serology), a seven-week gluten-free run-in, patients received one week of encapsulated gluten (8 g per day) or identical placebo. Each week with daily gluten exposure or placebo exposure was separated by one week of washout. Gluten and placebo were administered in a double-blind randomised fashion. The Structured Assessment of Gastrointestinal Symptoms (SAGIS) score was used at baseline before commencing the gluten exposure test and at the end of every week to assess changes of gastrointestinal symptoms.

Results: 34 patients completed the challenge test, of whom seven initially had self-reported symptoms associated with gluten or wheat consumption. Only one of these seven patients responded to gluten exposure with a >25% increase of the SAGIS score. 16 out of 34 patients (47 %, 95% CI 30-65) reported improved gastrointestinal symptoms while on a gluten-free diet.

Overall, 2 of the 34 patients (5.9%, 95% CI 0.7-20.0) showed a deterioration of symptoms during the gluten challenge week, while 8 patients (23.5%, 95% CI 10.8-41.2) reported a reduction of the symptoms while exposed to placebo. No effects of gluten exposure on extraintestinal symptoms were observed.

Conclusion: A standardised, double-blind gluten exposure test is suitable in the routine clinical setting to rule out gluten intolerance in patients with self-reported gluten sensitivity. Overall, gluten exposure is rarely associated with an increase of gastrointestinal symptoms in patients with DGBI. Self-reported non-celiac wheat sensitivity is a poor predictor for developing symptoms during double-blind gluten exposure.

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Disclosure: Nothing to disclose.

PP0044

EFFECTS OF DA-9701 (MOTILITONE) ON GASTRIC EMPTYING, BLOOD GLUCOSE REGULATION, AND OXIDATIVE STRESS IN DIABETIC RATS

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Introduction: Diabetes is associated with a variety of gastrointestinal disorders, including altered gastric emptying, which can be rapid, slow, or transient.

These changes can significantly influence gastrointestinal symptoms and potentially impact blood glucose levels. DA-9701 (motilitone), a known prokinetic agent derived from *Corydalis* tuber and *Pharbitis* seed, is employed in Korea to manage functional dyspepsia due to its noted anti-inflammatory and gastrointestinal motility-enhancing effects. This study aims to investigate the potential of DA-9701 in addressing altered gastric emptying and glycemic control in diabetic rats, thereby validating its broader clinical utility.

Aims & Methods: Diabetes mellitus was induced in rats by streptozotocin injection. Post-diabetes development, rats received oral administration of DA-9701 daily for two weeks. Evaluations included serum glucose levels, gastric emptying rates for liquids and solids, plasma insulin levels, and serum malondialdehyde concentration. Immunoblot assays assessed the expression of c-Kit, the *Ogg1* gene, and phosphorylated extracellular signal-regulated kinase (p-ERK).

Results: Diabetic rats exhibited blood glucose levels three to four times higher than nondiabetic control rats at weeks 3 and 4. Gastric emptying, initially accelerated by week 2 and persisting until week 4, was linked to increased interstitial cells of Cajal (ICC) activity driven by hyperglycemia-induced oxidative stress. The expression of c-Kit, an ICC marker, was elevated in the gastric tissues of diabetic rats by weeks 3 and 4 but reduced following two weeks of DA-9701 administration. Oral administration of DA-9701 (6 mg/kg) not only normalized gastric emptying but also improved blood glucose levels through increased serum insulin concentrations by weeks 3 and 4. Oxidative stress markers, including *Ogg1*, p-ERK, and serum malondialdehyde concentrations, initially upregulated, were reduced post-treatment.

Conclusion: Two weeks of DA-9701 treatment attenuated accelerated gastric emptying and reduced elevated blood glucose levels in diabetic rats. Normalized gastric emptying contributed to improved glucose homeostasis by reducing oxidative stress. Additionally, an unexpected increase in serum insulin levels post-treatment suggests that DA-9701's regulatory effects extend beyond gastric dynamics to direct modulation of blood glucose. These findings underscore DA-9701's significant potential in managing both gastrointestinal dysfunction and glucose control in diabetes mellitus.

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Disclosure: Nothing to disclose.

PP0045

ASSESSMENT OF VAGAL NERVE INJURY THROUGH SHAM FEEDING

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Introduction: Vagal nerve injury may lead to symptoms of nausea, vomiting, abdominal discomfort, and bloating. Be it by injury or an infiltrative process, vagal nerve function is difficult to assess. Measuring pancreatic polypeptide (PP) following a sham meal may serve as a reasonable surrogate marker of vagal nerve function. PP is released in response to eating and may correlate with slow gastric emptying.
Our aim is to assess vagal nerve integrity using sham feeding with measurement of PP, in addition to evaluating the relationship of symptoms with PP levels and gastric emptying.
Aims & Methods: This was a retrospective study of patients who underwent a sham feeding study at Mayo Clinic Florida from 2018 to 2023. We collected demographic and clinical data, sham feeding results, the indication for the study, and gastric emptying data. A 50 % increase in PP level within 30 mins compared to baseline after sham feeding was considered normal. A gastric emptying scan was classified as delayed if <75% of the material emptied at 4 hours.
Results: 29 patients were included (58.6% women) with a mean age of 57.3 years. Most patients were white (n=24), and the mean BMI was 27.9 kg/m^2. The most common reason to request a sham feeding study was abdominal pain (7, 24.1%), gastroesophageal reflux (GER) (6, 20.7%) and chronic nausea and vomiting (6, 20.7%). The main GI symptoms were nausea and vomiting (13, 44.8%) and diarrhea (11, 37.9%). The average weight loss from the onset of symptoms was 31 lb. 24 of 29 included patients had a history of abdominal surgery, with hiatal hernia repair (HHR) and HHR with Nissen fundoplication being the two most common surgeries (5, 17.2%). 20 of 29 studied patients (69.0%) had an abnormal sham feeding study and 48.1% of the patients who underwent gastric empty studies were found to have delayed gastric emptying. The mean baseline PP level of normal vs abnormal was 102.88 pg/mL and 226.4 pg/mL, respectively (p=0.21). The mean peak PP level of normal vs abnormal was 195.22 pg/mL and 245 pg/mL, respectively (p=0.68). Among the 20 patients with an abnormal sham feeding study, ten had delayed gastric emptying. Symptoms of nausea and vomiting were found to have a robust correlation with a positive sham feeding result (12, 92%), compared with gastric emptying scintigraphy (8, 61.5%). There was no correlation between sham feeding result with symptoms of abdominal pain and GER. The care after apparent vagus nerve injury required significant medical resources, including an average of 3.5 visits and 5.7 tests per patient to evaluate key GI symptoms.
Conclusion: Sham feeding is an underused method to measure vagus nerve injury. It is safe and easy to perform. PP levels measured during sham feeding were consistently abnormal in patients with predominant symptoms of nausea and vomiting after foregut surgery while the correlation between gastric emptying measured using scintigraphy and gastroparesis symptoms was not significant. Interestingly, there was limited correlation between scintigraphy gastric emptying results and outcomes of sham feeding. Our research highlights the significant medical and financial burden evaluating patients with symptoms after foregut surgery. Larger prospective studies are needed to better quantify the relationship between PP levels and gastric emptying in patients with prior foregut surgery who remain symptomatic.
Disclosure: Nothing to disclose.

PP0046

ASSOCIATIONS BETWEEN GASTROINTESTINAL SYMPTOMS, MOTILITY, AND EXTRAINTESTINAL AUTONOMIC MEASURES IN DIABETIC GASTROENTEROPATHY: A CROSS-SECTIONAL STUDY

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Introduction: Autonomic neuropathy affecting multiple organs is a common and severe complication of diabetes[1]. Neuropathy of the enteric nervous system and the regulatory autonomic nerve fibers can affect the entire gastrointestinal tract, causing mild to severe gastrointestinal symptoms[2]. The correlation between gastrointestinal symptoms and underlying pathophysiology is generally weak, so objective assessment is important to guide treatment[3-5]. Methods for direct evaluation of enteric neuropathy are not yet available in clinical practice. Hence, gastrointestinal transit times and contractile activity are used as proxies for enteric autonomic dysfunction[6,7]. Panenteric gastrointestinal assessment is considered essential, yet suitable methods are generally inaccessible, and diagnosing diabetic gastroenteropathy often causes significant challenges[4,8]. It is unknown whether measures of extraintestinal autonomic function can reflect diabetic gastroenteropathy and whether these measures could serve as supplementary diagnostic tools.
Aims & Methods: We aimed to assess 1) whether gastrointestinal symptoms are associated with motility measures and 2) whether gastrointestinal symptoms/motility measures are associated with extraintestinal autonomic markers. We included 81 persons with type 1 or type 2 diabetes (65% female, mean age 54) with gastrointestinal symptoms and either symptoms or signs of autonomic neuropathy. The Gastroparesis Cardinal Symptom Index (GCSI) and the Gastrointestinal Symptom Rating Scale (GSRS) were used to assess gastrointestinal symptoms. The wireless motility capsule assessed panenteric transit times and motility indices. Cardiovascular reflex tests performed with the VAGUS device and the cardiac vagal tone assessed with the eMotion Faros device estimated cardiovascular autonomic neuropathy, while the SUDOSCAN evaluated sudomotor function.
Results: Proximal gastrointestinal symptoms were positively associated with the gastric motility index (GCSI: 1.18 (1.04 - 1.35), *p*=0.01; GSRS: 1.15 (1.03 - 1.29), *p*=0.02; median ratio (95%CI)), while only satiety reflected gastric emptying time (1.24 (1.03 - 1.49), *p*=0.02). Diarrhea was associat-

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ed with decreased small bowel transit time (0.93 (0.89 - 0.98), *p*=0.005), while constipation was associated with prolonged colonic transit time (1.16 (1.03 - 1.31), *p*=0.02). Gastrointestinal symptoms increased with the degree of abnormal cardiovascular reflex tests (GCSI: 0.67 (0.16 - 1.19), *p*=0.03; GSRS: 0.87 (0.30 - 1.45), *p*=0.01; mean difference (95%CI)) but not with motility measures. Cardiac vagal tone and sudomotor function were not associated with gastrointestinal markers.

Conclusion: Gastrointestinal and extraintestinal autonomic measures were not associated. However, proximal gastrointestinal symptoms were associated with the gastric motility index and cardiovascular reflex tests. Hence, the latter may contribute to evaluating whether proximal gastrointestinal symptoms are autonomically derived.

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PP0047

PHARMACOLOGICAL ACTIVITY OF A HERBAL PREPARATION AND ITS INDIVIDUAL HERBAL EXTRACTS ON GASTRIC MOTILITY IN FUNCTIONAL DYSPEPSIA

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Introduction: This study investigates the pharmacological activity of Amara-Drops, a traditional herbal remedy used for gastrointestinal dysmotility, and its nine individual herbal extracts on fundus accommodation, a characteristic aspect of functional dyspepsia. Impaired fundus accommodation after food intake can lead to symptoms such as epigastric pain and vomiting, yet its etiology remains poorly understood.

Previous studies have demonstrated that the extract of Amara-Drops relaxes guinea-pig fundus stomach, potentially mediated by M2 muscarinic receptor inhibition, though the exact mechanism and responsible individual extracts were not investigated.

Aims & Methods: Our study aimed to explore the mechanisms of action of Amara whole extract and its constituent extracts (*Artemisia absinthium Herba*, *Centaurium erythraea Herba*, *Cichorium intybus Planta tota*, *Gentiana lutea Rhizoma*, *Juniperus communis Summitates*, *Achillea millefolium Herba*, *Peucedanum ostruthium Rhizoma*, *Salvia officinalis Folium* and *Taraxacum Planta tota*) on fundus accommodation, both *ex vivo* and *in vitro*.

Dry extracts were obtained from the commercial product Amara-Drops or from the individual ethanolic tinctures. Guinea-pig fundus muscle strips were prepared and set up in organ baths in a circular orientation to study the relaxation effect of the whole extract (94.4–1,000 µg/mL) or vehicle, with or without Carbachol-induced contraction. To elucidate whether M2 or M3 muscarinic receptors were involved in the relaxation effect, guinea-pig fundus was pretreated with J-014129, an antagonist of M3, followed by addition of whole extract or three selected individual extracts. Furthermore, binding and antagonist activities on muscarinic M3 and M2, and opioid µ (OP1) receptors expressed in CHO-K1 cell line were assessed for the whole extract and/or individual extracts.

Results: Exposure of the whole extract to guinea-pig fundus muscle strips induced significant relaxation at the highest concentrations tested ($p<0.001$). Significant relaxation was observed in Carbachol-contracted muscle strips ($p<0.05$), regardless of J-014129 pretreatment, suggesting potential modulation of the M2 receptor.

In vitro studies revealed the whole extract bound to M3, M2 and OP1 receptors (IC_{50}: 380, 294 and 55 µg/mL, respectively), with selective M2 and OP1 receptor activity impairment (IC_{50}: 219 and 304 µg/mL, respectively) *via* inhibition of their respective reference agonist activity.

M3 receptor activity was not affected. *Juniperus communis*, *Salvia officinalis*, and *Peucedanum ostruthium* extracts showed M2 receptor inhibition (IC_{50}: 32, 20, and 21 µg/mL, respectively), with only *Peucedanum ostruthium* extract binding to M2 receptor (IC_{50}: 137 µg/mL) and significantly relaxing Carbachol-contracted muscle strips, irrespective of J-014129 pretreatment ($p<0.001$).

However, *Juniperus communis* and *Salvia officinalis* did not show M2 receptor binding nor relaxation effects. Furthermore, *Gentiana lutea*, *Juniperus communis*, *Artemisia absinthium*, and *Centaurium erythraea* extracts inhibited OP1 receptor activity (IC_{50}: 42, 25, 22 and 104 µg/mL, respectively).

Conclusion: These data highlight the potential of the medicinal product Amara-Drops, particularly its *Peucedanum ostruthium* extract, in modulating fundus accommodation by inhibiting M2 receptor activity.

Further investigations are needed to elucidate relaxation mechanisms through OP1 receptor inhibition. Nevertheless, this study provides a promising molecular basis for the efficacy of Amara-Drops in improving impaired fundus accommodation in functional dyspepsia.

Disclosure: All authors are employees of Weleda AG.

PP0048

EPIDEMIOLOGY OF MAJOR ESOPHAGEAL MOTILITY DISORDERS AMONG PATIENTS WITH TYPICAL VERSUS NON-TYPICAL SYMPTOMS OF ESOPHAGEAL MOTILITY DISORDERS

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Introduction: Esophageal dysphagia and retrosternal chest pain are considered the typical manifestations of major esophageal motility disorders (mEMD). High Resolution Manometry (HRM) is the gold standard for the diagnosis of mEMD.

However, in real clinical practice patients are referred for HRM due to several non-typical esophageal symptoms and the rate of EMD among those patients is largely unknown.

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Aims & Methods: Aim of our study was to search for mEMDs prevalence among patients with typical symptoms and compare them with patients referred for non-typical symptoms.

We retrospectively collected epidemiological, endoscopic (available in 263/302), barium swallow (available in 176/302) and manometric data from consecutive patients referred for HRM in our center from 1/2020 to 12/2023. Atypical symptoms for which patient were referred for HRM included; heartburn, regurgitation, globus, oropharengeal dysphagia and epigastric pain.

Patients with a previous diagnosis of mEMDs/scleroderma/EoE, referred for pre-surgical evaluation for Gastroesophageal Reflux Disease (GERD) or submitted to gastro-esophageal surgery were excluded. Endoscopic data suggestive of an mEMD were considered esophageal dilation, solids/liquids/saliva in the lumen and difficulty to pass the scope through the Lower Esophageal Sphincter (LES), while delay in barium transit from Esophagus to the stomach, esophageal dilation and bird-like LES appearance as barium swallow were considered findings compatible to mEMD.

Results: We included 302 patients (M/F: 147/155, mean age: 56±17 years). Typical symptoms were reported by 234 (77.2%) patients, while only atypical in 68 (22.8%). Main referral symptom was; esophageal dysphagia: 176 (58.3%), chest pain: 42 (13.7%), heartburn: 27 (8.9%), regurgitation: 25 (8.3%), globus/oropharengeal dysphagia/epigastric pain: 32 (10.8%). Diagnosis of mEMDs in HRM was set in 175/302 (57.9%) and was statistically more common when typical symptoms existed (163/233, 69.9% vs 12/78, 15.4%, p<0.001).

HRM diagnosis among patients with mEMDs are given in table 1. Among patients with a diagnosis of mEMD in HRM, endoscopic data compatible to mEMDs were statistically more common among patients with typical symptoms, though barium swallow data were not (103/149, 69.1% vs. 4/10, 40%, p=0.037 and 110/116, 94.8% vs. 9/9, 100%, p=0.633 respectively).

HRM diagnosis of mEMD among patients with compatible either barium swallow or endoscopic findings was found in a statistically increased rate among patients with typical symptoms (134/145, 92.4% vs. 10/19, 52.6%, p<0.001). We should point that among 43 patients with non-typical symptoms and normal endoscopy and barium swallow study only 1 (2.3%) had a mEMD diagnosis by HRM.

HRM diagnosis	Total cohort (n=175)	Typical symptoms (dysphagia-chest pain) (n=163)	Non-typical symptoms (n=12)
Achalasia	143 (81.7%)	135(82.8%)	8 (66.7%)
Type I	6/143 (4.2%)	5/135 (3.7%)	1/8 (12.5%)
Type II	134/143 (93.7%)	127/135 (94.8%)	7/8 (87.5%)
Type III	3/143 (2.1%)	3/135 (2.2%)	0
Esophago-Gastric Junction Outflow Obstruction	21 (11.4%)	18 (11.0%)	2 (16.6%)
Diffuse Esophageal Spasm (DES)	3 (1.7%)	3 (1.8%)	0
Hypercontractile Esophagus	1 (0.6%)	1 (0.6%)	0
Absent peristalsis	8 (4.6%)	6 (3.7%)	2 (16.6%)

Conclusion: More than half of patients referred for HRM will be diagnosed with mEMD, rate significantly increased when typical symptoms, as dysphagia or chest pain, are reported. Achalasia, more specifically type II, is in general the most commonly diagnosed mEMD. Absence of compatible endoscopic or barium swallow findings make mEMDs diagnosis, in the absence of typical symptoms, almost impossible.

Disclosure: Nothing to disclose.

PP0049

THE EFFECT OF BOLUS VOLUME, VISCOSITY AND SENSORY TRP STIMULANTS ON THE UPPER ESOPHAGEAL SPHINCTER AND PHARYNGEAL MOTILITY IN HEALTHY [JL1] ADULTS MEASURED WITH HIGH-RESOLUTION IMPEDANCE MANOMETRY

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Introduction: Introduction: High-resolution impedance manometry (HRIM) is now an important tool in the study of upper esophageal sphincter (UES) function and pharynx propulsion in patients with oropharyngeal dysphagia. Our aim was to study the effect of volume, viscosity and sensory TRP stimulants using HRIM in healthy adults.

Aims & Methods: Methods: We present a cross-sectional, controlled, quasi-experimental study in 14 healthy volunteers (median age 33 years) studied by high-resolution solid-state probe manometry of 30 unidirectional pressure channels and 6 circumferential channels at the UES with 16 impedance channels. Duplicate boluses of 5, 10 and 20 mL of 4 viscosities (50, 250, 800 and 2,000 mPa·s) and a duplicate bolus of 5 mL of TRP stimulants with the same 4 viscosities were administered. UES relaxation (UES-IRP), UES opening (UES Max Admittance), flow resistance (hypopharyngeal pressure at nadir impedance (PNadImp) and pharyngeal propulsive forces (peak pharyngeal pressure (PeakP) and pharyngeal contraction integral (PhCI) were assessed. The analysis was performed using SwallowGateway® software.

Results: Results: Bolus volume increase at different viscosities caused a) increase in UES integrated relaxation pressure (UES-IRP $p < 0.05$), b) increase in UES opening (UES Max Admittance $p < 0.05$), c) increase in flow resistance (PNadImp) at 800 and 2,000 mPa·s ($p < 0.005$) and d) increase in pharyngeal propulsion at 2,000 mPa·s (PhCI ($p < 0.05$) with no significant increase with other viscosities or at the PeakP level in all of them.

Viscosity increase at different volumes showed:

a) Increase in UES integrated relaxation pressure (UES-IRP $p = 0.004$) majority at higher volume boluses (10 and 20 mL),

b) increase in UES opening (UES Max Admittance $p < 0.05$ at 5 and 10 mL),

c) no changes in bolus resistance (PNadImp (20 ml) $p > 0.05$) and;

d) no changes in pharyngeal forces (PeakP $p > 0.05$ and PhCI $p > 0.05$).

TRP stimulants increased the UES-IRP ($p < 0.011$) and UES Adm ($p < 0.001$) and had no effect on pharyngeal contraction forces or bolus flow resistance.

Conclusion: Conclusions: In healthy adults without dysphagia, increasing volume has a significant effect on UES with an improvement on opening with less relaxation, increased flow resistance and increased pharyngeal propulsive forces. In contrast, increasing bolus viscosity and adding TRP stimulants impairs relaxation with improvement on the aperture of the UES with no effect on pharyngeal propulsive forces and flow resistance.

Disclosure: Nothing to disclose.

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PP0050

REAL-LIFE TREATMENT IN NON-OBSTRUCTIVE ESOPHAGEAL DYSPHAGIA: ANALYSIS OF A PROSPECTIVE DATABASE

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Introduction: Treatment of non-obstructive esophageal dysphagia (NOD) should consider not only the primary condition, but also its impact on patients' daily function, their perception and surgical risk. Between surgical, endoscopic, and medical therapies, there is a wide range of choices, most without well-established indications or outcomes.

Aims & Methods: This is an ongoing prospective multicentric study to evaluate the real-life approach to NOD in 7 centers, by comparing the efficacy of different treatments [per-oral endoscopic myotomy (POEM), botulinic toxin (BT), pneumatic dilation (PD), optimization of proton-pump inhibitors (PPI) and expectant], through the Eckardt, EuroQol-5D (TTO and VAS) and PROMIS GI Disrupted Swallowing (PDS) scores.

Results: According to Chicago V4, it included 97 esophagogastric junction (EGJ) outflow disorders, 24 spastic motility disorders, 38 hypomotile disorders and 91 dysphagia with normal manometry. Fifty-seven (22.8%) patients were submitted to therapy.

Forty-three (44.3%) patients with EGJ outflow disorders were treated (31 POEM, 5 BT, 2 PD, 1 surgical and 4 medical). In contrast to untreated, treated patients showed better differences in Eckardt, PDS and EuroQol-5D VAS scores (p-values <0.005) after 6 months.

PPI was introduced and/or optimized in five (13.2%) patients with hypomotile disorders. There were no differences in Eckardt, EuroQol-5D and PDS at 6-months in treated or untreated patients.

Three (12.5%) patients with spastic motility disorders were treated (2 BT and 1 calcium-channel blocker). Both treated and untreated showed no differences in Eckardt or EuroQol-5D scores at 6 months.

Regarding dysphagic patients with normal manometry, medical treatments were optimized in 6 patients (6.6%), without any differences in Eckardt and EuroQol-5D scores at 6-months. Interestingly, the PDS showed a better difference at 6 months in untreated patients (p-value <0.001).

Conclusion: Apart from EGD outflow disorders, which have well-established treatment options, real-life approach of NOD is heterogeneous and showed no clear benefit. To improve clinical results, specialized clinics must be created.

Disclosure: Nothing to disclose.

PP0051

IMPACT OF NON-OBSTRUCTIVE ESOPHAGEAL DYSPHAGIA ON QUALITY OF LIFE: A PORTUGUESE MULTICENTRIC PROSPECTIVE STUDY

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Introduction: Dysphagia is a common condition with consequences surpassing the physical dimension and extending to the patient's psychological, personal, and professional life. However, when referring to non-obstructive esophageal dysphagia (NOD) and motility disorders, data is still scarce.

Aims & Methods: Our study consists in a prospective and multicentric evaluation of the severity of dysphagia and its impact on quality of life, that was measured with the Eckardt, EuroQol-5D (TTO and VAS) and PROMIS GI Disrupted Swallowing (PDS) scores, in patients with NOD and underlying motility disorders. We included 373 individuals (250 with NOD and 123 without dysphagia).

Results: Among patients with NOD, 97 had esophagogastric junction (EGJ) outflow disorders on manometry according to Chicago V4.0, 24 spastic motility disorders, 38 hypomotile disorders and 91 normal motility. Compared with non-dysphagic individuals, dysphagic patients had significantly worse scores of both EuroQol-5D and PDS (p-values <0.001).

Among esophageal motility disorders, Eckardt score was significantly different between groups (p-value <0.001) as determined by ANOVA. Specifically, it was significantly higher in EGJ outflow disorders (mean of 5.2), when compared to spastic motility disorders (mean of 3.3) or dysphagic patients with normal motility (mean of 3.2).

With respect to Patient-reported outcomes, PDS score was statistically different between groups (p-value 0.003), being higher in the EGJ outflow disorders (mean T-score of 64.1) when compared to hypomotiity disorders (mean of 60.2) and dysphagic patients with normal motility (mean of 60.3).

Regarding patient-reported quality of life, there were no differences between groups in terms of EuroQol-5D TTO or VAS score (p-value of 0.156 and 0.368, respectively).

Conclusion: Our findings highlights the impact of NOD on the quality-of-life of patients. Although physician and patient reported symptoms' severity were worse on EGJ outflow disorders, overall quality of life seems equally affected among all esophageal motility profiles.

Disclosure: Nothing to disclose.

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PP0052

ASSESSING ANTRO-DUODENAL MANOMETRY PATTERNS IN PATIENTS WITH HYPERMOBILE EHLERS-DANLOS SYNDROME WITH GASTROINTESTINAL SYMPTOMS

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Introduction: Hypermobile Ehlers-Danlos syndrome (hEDS) is a hereditary non-inflammatory disorder of connective tissue laxity characterized by joint hyperextensibility in association with musculoskeletal symptoms. The prevalence of gastrointestinal (GI) symptoms, such as abdominal pain, nausea, and constipation is high in hEDS. Research on the relationship between EDS and GI symptoms has mostly been restricted to disorders of the gut-brain axis, whereas the relationship between hEDS and GI dysmotility remains to be clarified.

Aims & Methods: This study aimed to explore antro-duodenal manometry (ADM) patterns in patients with hEDS presenting with GI symptoms.

In a prospective open-label study, patients with hEDS and GI symptoms presumed to be the results of GI dysmotility underwent a 6-hour high resolution ADM. The manometry tracings were analyzed by two investigators (KS and JC) independently. Antral contraction frequency (contractions per minute) < 1.0 was considered as antral hypomotility. Pylorospasm was defined as isolated pyloric pressure waves within 60 minutes postprandial. The following neuropathic enteric patterns were scored: presence of retrograde peristalsis, bursts, clustered contractions, and absence of phase III contractions. Abnormal ADM was defined as having antral hypomotility and/ or pylorospasm and/ or neuropathic enteric patterns.

Results: Fifty patients (48 women, mean age 30.4 ± 11.1 yrs.) were included. At baseline, 74% of patients were dependent on enteral feeding and the mean BMI was 20.4 ± 4.4 kg/m^2.

Overall, ADM tracings with one or more abnormal patterns were identified in 53% of patients. Antral hypomotility and pylorospasm were detected in 29% and 15% of patients, respectively. Neuropathic enteric patterns were only seen in 17% of patients. In all cases, no more than one neuropathic pattern was identified. Among the observed neuropathic enteric patterns, 15% of patients showed retrograde peristalsis, 13% absence of phase III contractions, and 2% bursts. Clustered contractions were not identified in this cohort.

Conclusion: Hypermobile Ehlers-Danlos patients presenting with gastrointestinal symptoms showed low numbers of abnormal ADM patterns. In particular, enteric dysmotility was observed in only a small minority of cases. Antroduodenal dysmotility therefore does not appear to be a principle pathomechanism in the majority of hEDS cases studied. Further research is needed to understand the mechanisms that underlie GI symptom generation in this population.

Disclosure: Nothing to disclose.

PP0053

PERSISTENT NAUSEA AND VOMITING IN GASTROPARESIS (PNVGP): REAL-WORLD EVIDENCE FROM A LARGE US DATABASE

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Introduction: Nausea and vomiting are prevalent in gastroparesis (GP) and may persist in some patients with GP (pNVGP) despite treatment with prokinetics/antiemetics. There is scarce evidence at the population level on the demographic and clinical characteristics and disease experience of patients with pNVGP.

Aims & Methods: A retrospective study was conducted using Optum Market Clarity, a large US database comprised of claims and electronic health records (EHR) that is linked at the patient level and has customised fields developed by Natural Language Processing from unstructured clinical notes (e.g., GP-specific symptoms, gastric emptying testing). The pNVGP cohort was defined as: all GP diagnoses from claims/EHR during the study period (01Jan2008–30Sep2020, 1st documented GP Dx date in EHR/claims as the index date), with encounters for ≥5 years pre- and ≥1 year post-index in study period; received ≥14 cumulative days' supply of prokinetics and/or antiemetics pre- and/or post-index; and still reported nausea/vomiting ≥28 days after treatment at encounters.

Results: Of 207,704 patients identified with GP (45% diabetic [DG], 45% idiopathic [IG], 10% other causes), 25,654 met the criteria for pNVGP (57% DG, 31% IG, 12% other; mean±SD age at index 51.3±16.6 years; 75% female; 80% Caucasian; mean Charlson Comorbidity Index score 2.6). Prior to index, abdominal pain (61%), nausea (50%), nausea and vomiting (42%) and vomiting (39%) were the most commonly reported symptoms for pNVGP patients. Prokinetic/antiemetic use was frequent during 1-year pre-index (71% of pNVGP patients) and increased during 1-year post-index (85%). Ondansetron (53% of pNVGP patients pre-index, 66% post-index) and metoclopramide (13% pre-index, 32% post-index) were the most frequently used.

% Patients who received prokinetics*/antiemetics**	Diabetic GP (n=14,509)	Idiopathic GP (n=7962)	Other GP (n=3183)	Total pNVGP (n=25,654)
1 yr pre-index, 1 yr post-index	73%, 87%	66%, 82%	76%, 84%	71%, 85%

*e.g., metoclopramide, azithromycin; **e.g, ondansetron, diphenhydramine, lorazepam

Conclusion: In this study, a sizable subcohort of patients with GP had pNVGP despite the use of prokinetics/antiemetics. pNVGP occurs more often in diabetic GP, and often with abdominal pain. Patients with pNVGP had persistent unmet needs.

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PP0054

THE EFFICACY OF THE FUNCTIONAL LUMEN IMAGING PROBE (FLIP) PROCEDURE IN A COHORT OF PATIENTS WITH DYSPHAGIA

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Introduction: Functional lumen imaging probe (FLIP) is an adjunctive test indicated in the evaluation of the esophagogastric junction and esophageal motility, especially in those with an inconclusive diagnosis during esophageal high-resolution manometry (HRM) such as esophagogastric junction outflow obstruction (EGJOO) or absent contractility per Chicago Classification v4.0 (CCv4.0), and in patients that are unable to perform a HRM.

Aims & Methods: Our study aimed to evaluate the efficacy of the FLIP procedure in establishing diagnosis in a cohort of patients who underwent assessment due to dysphagia, who were either diagnosed with EGJOO or absent contractility according to CCv4.0, or were unable to perform HRM.

Our study population consisted of patients who were evaluated by FLIP due to dysphagia between March 2023 and April 2024. All patients had completed an upper gastrointestinal endoscopy and HRM per CCv.4.0. Additionally, most patients underwent further evaluation with a timed barium esophagram (TBE).

Results: We included 58 patients (52% female, median age 62.5 years) in the study cohort. Most patients (67.24%) referred to the FLIP procedure had a prior failed attempt to perform a HRM or had an inadequate test. 46.5 % of all patients had an abnormal FLIP procedure. Of those, 15 patients had EGJOO with absent contractile response compatible with achalasia (55.5 %) and another 5 patients had a confirmed diagnosis of EGJOO on FLIP (18.5 %). Of those patients who had a FLIP procedure performed due to abnormal esophageal manometry findings per CCv4.0 (n=19), 8 patients were diagnosed with EGJOO (42.1%), 5 had absent contractility (26.3 %), 4 had a hypercontractile esophagus (21 %) and 2 had ineffective esophageal motility (10.5 %). 60 % of the patients diagnosed with EGJOO by FLIP were offered an achalasia-targeted treatment (pneumatic dilation (6), POEM (4), Heller myotomy (1) and Botox injection (1)), while the rest were treated conservatively.

Additionally, TBE was performed in 55% of patients with an abnormal FLIP procedure and the result was pathological in about half of them. Furthermore, in 70% of patients the diagnosis of EGJOO during FLIP was correlated to a delayed esophageal clearance on TBE.

Conclusion: FLIP is a useful tool in the assessment of dysphagia in patients who have an inconclusive diagnosis per CCv4.0, either EGJOO or absent contractility, as well as in patients who are unable to perform a HRM.

Disclosure: Nothing to disclose.

PP0055

PREDICTING OUTCOMES OF G-POEM USING BODY SURFACE GASTRIC MAPPING: EARLY RESULTS FROM A TERTIARY CENTRE

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Introduction: Gastroparesis (GP) is a poorly understood condition with current diagnostic tests lacking disease specificity. Gastric per oral endoscopic myotomy (G-POEM) has emerged as a promising treatment for refractory gastroparesis, however up to a third of patients fail to have an improvement in their symptoms.

Aims & Methods: We assessed the ability of Gastric Alimetry®, a new, non-invasive test of gastroduodenal function to enhance patient selection for G-POEM. Data was collected from patients with suspected GP (based on symptoms and scintigraphy), being worked up for G-POEM at Kings College Hospital between November 2022 and September 2023 (REC: 22/LO/0802). G-POEM success was defined as <1 point reduction in GCSI score. Descriptive comparisons were used to explore predictors of improvements in gastroparesis cardinal symptom index (GCSI).

Results: 11 patients were assessed (8F, median age 40 interquartile range 32.5-49.5). GCSI scores significantly improved after G-POEM (median difference -1.03, IQR 0.50). Aetiologies of gastroparesis included 5 (45.5%) diabetic, 4 (36.4%) idiopathic, and 2 other aetiologies. Gastric Alimetry spectral analysis showed 2 (18.2%) with high frequencies, 2 (18.2%) with neuromuscular dysfunction (low rhythm stability and/or low amplitude), and 7 (63.6%) with normal spectral metrics. Symptom profiles followed a continuous pattern in 4 (36.4%), sensorimotor in 2 (18.2%), post-gastric in 1 (9.1%) and meal-induced in the remainder (36.4%). High frequency phenotype was associated with treatment failure (2/2, 100%), and neuromuscular phenotype was associated with success (2/2, 100%). The change in GCSI score after G-POEM was significantly lower in the high frequency group compared to the group average (median difference -0.18, IQR 0.18; p = 0.03).The average frequency and rhythm index was 2.9 (0.1) cycles per minute (cpm) and 0.4 (0.2) respectively among responders vs 3.2 (0.4) cpm 0.6 among non-responders.

Conclusion: Gastric Alimetry® is a new, non-invasive test of gastric function with provisional utility in differentiating disease mechanisms in patients with suspected GP. Early data suggests biomarkers for gastric autonomic dysregulation (high slow wave frequencies) were associated with reduced response to G-POEM. Further data are eagerly anticipated to validate these findings and develop more sophisticated predictive models for G-POEM outcomes.

Disclosure: Nothing to disclose.

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PP0056

ROUTINE X-RAY CONTRAST STUDY FOLLOWING PERORAL ENDOSCOPIC MYOTOMY (POEM) DOES NOT PREVENT POSTPROCEDURAL COMPLICATIONS – SINGLE CENTER RETROSPECTIVE ANALYSIS

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Introduction: Peroral endoscopic myotomy has become the mainstay of treatment of achalasia. Oral feeding has been initiated if no submucosal leak of the contrast fluid was detected on the X-ray contrast study which was performed the following day after the procedure. However, as major adverse events are rare, and in case of their occurrence, are rarely asymptomatic, it has become unclear whether routine X-ray contrast study is necessary. In our center we have been performing POEM procedures since 2015 and initially routinely performed X-ray contrast study. However,during the course of time it has become unavailable and we were forced to initiate oral feeding diet based on our clinical decision. We adopted a protocol of the initiation of oral liquids on the 2 nd postoperative day (or later in case of suspected complication). We therefore compared these two strategies in terms of periprocedural and postprocedural complications rate and the length of the hospital stay.

Aims & Methods: This is a retrospective analysis of the single high-volume center. We evaluated (1) the rate of peri- and postprocedural complications and (2) length of the hospital stay in patients hospitalized for POEM for esophageal achalasia in 2015-2023. We separately evaluated the "X-ray subgroup" (oral feeding was initiated after X-ray that was performed the day after the procedure) and the "clinical decision subgroup" (oral feeding was initiated on the 2 nd postoperative day).

Results: We analyzed a total number of 274 POEM procedures (129 males, 145 females), mean age was 52,3±16 years. 102 patients (37.2%) received X-ray contrast study after the procedure, in 169 patients (62.8%) the initiation of oral feeding was based on clinical decision. The mean time of the hospital stay was 7.2±2.9 vs. 6.5±1.4 days in the X-ray group and in the clinical decision group, resp. (p<0.004).

In the X-ray group we detected complications in 8 patients (7.7%). 5 were periprocedural (mucosal perforation or aspiration during intubation), were solved during the procedure and did not lead to the extension of the hospital stay. There were postprocedural complications in 3 patients. In one case submucosal leak was detected on X-ray contrast study the following day after the procedure due to loosened clip in the mucosal entry and was solved by placing additional clips. In one case patient presented with bleeding from the mucosal entry after discharge. In one case the patient developed mediastinitis that led to the extension of the hospital stay. In the clinical decision group we detected complications in 10 patients (5.9 %), 9 of them were periprocedural (mucosal perforation or aspiration during intubation), 8 of which were solved during the procedure by endoscopic clipping and did not lead to the extension of the hospital stay, 1 required surgery. In one case of postprocedural complication perforation with pleural effusion was detected on CT scan and required surgical intervention that led to the extension of the hospital stay.

Conclusion: Our analysis shows that severe complications after POEM are rare. In the X-ray subgroup there was only single postprocedural event in which X-ray was helpful in its early detection (1%). In the clinical decision subgroup there was also only single event that could have been identified on the X-ray (0.6%). The length of the hospital stay in the clinical judgement group was significantly shorter than in the X-ray group. This can be attributed to the improvement of overall experience with the management of the POEM patients as X-ray contrast has become unavailable after performing c.a. 100 POEM procedures.

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Disclosure: Nothing to disclose.

PP0057

OPEN LABEL EFFICACY ANALYSIS OF TRADIPITANT IN IDIOPATHIC AND DIABETIC GASTROPARESIS

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Introduction: Tradipitant is a novel NK1R antagonist studied in diabetic and idiopathic gastroparesis. This report presents the analysis from the open-label arm of a multicenter, randomized, double-blind, placebo-controlled study (VP-VLY-686-3301) assessing the efficacy of tradipitant in relieving symptoms of gastroparesis.

Aims & Methods: N=531 idiopathic and diabetic gastroparesis patients with delayed gastric emptying received open label 85mg tradipitant BID for 12 Weeks. Nausea was assessed using the 5-point Gastroparesis Core Symptom Daily Diary. An exposure response analysis was performed to control for low drug exposure. We used a threshold of ≤140ng/ml tradipitant blood concentration at any visit to classify participants in the High (n=309) and Low (n=218) PK Populations.

Results: Patients improved the average nausea score significantly from baseline after 12 weeks tradipitant (improvement of -1.01 vs 2.14 at baseline, p<0.0001). A significant improvement was also seen in the percent of nausea free days (an addition of 51.76% of days for tradipitant vs. 19.62% at baseline, p<0.0001).

Patients in the High PK group significantly improved average nausea score compared to the Low PK group at Week 12 (improvement of -1.19 vs -0.78, p<0.0001). A significant improvement was also seen in the percent of nausea free days (an addition of 35.6% of days for High PK versus 22.6% for Low PK, p=0.0002).

Conclusion: Tradipitant demonstrated statistically significant improvement from baseline in average nausea severity and percentage of nausea-free days in this 12 week open label study in both idiopathic and diabetic gastroparesis subjects. Additionally, a strong exposure response in nausea improvement was observed when comparing subjects with high drug exposure vs low drug exposure.

Disclosure: Study was sponsored by Vanda Pharmaceuticals. Inc. Authors are employees of Vanda Pharmaceuticals, Inc.

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PP0058

EXPOSURE-RESPONSE ANALYSIS OF OPEN LABEL TRADIPITANT IN IDIOPATHIC GASTROPARESIS IN STUDY VP-VLY-686-3301

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Introduction: To assess the exposure response of NK-1 receptor antagonist tradipitant in the reduction of gastroparesis symptoms in patients with idiopathic gastroparesis in the open label arm of phase 3 study (VP-VLY-686-3301).

The four etiologies of gastroparesis are diabetic, post-surgical, drug-induced, and idiopathic. The idiopathic etiology of gastroparesis is least common, presenting in about 11.3% of patients, which amounts to about 90,000 patients in the United States. Standards set by the FDA define a rare disease or condition as one that affects less than 200,000 people in the United States. This establishes idiopathic gastroparesis as a rare condition, as well as one with a high unmet need for treatment. Patients with gastroparesis have decreased quality of life associated with severe and chronic symptoms and no treatment is currently approved for idiopathic gastroparesis specifically. Tradipitant previously demonstrated efficacy in idiopathic gastroparesis in reducing gastroparesis symptoms in a phase 2 study (VP-VLY-686-2301).

Aims & Methods: The open label arm of Study 3301 enrolled n=531 idiopathic and diabetic gastroparesis patients with delayed gastric emptying, moderate to severe nausea, and at least 1 vomiting episode. Patients received 85 mg tradipitant twice a day for 12 weeks. Nausea was assessed daily using the 5-point Gastroparesis Core Symptom Daily Diary (GCSDD). Overall gastroparesis symptoms were assessed every 2 weeks in clinic using the Gastroparesis Cardinal Symptom Index (GCSI).

An exposure response analysis was performed on a subgroup of patients to control for low drug exposure. We used a threshold of ≤140ng/ml tradipitant blood concentration at any visit to classify participants in the High (n=309) and Low (n=218) PK Populations. Results were further stratified to compare average nausea severity and gastroparesis symptoms at baseline nausea severity scores of ≥2 and ≥2.5 for Low (n=129 and n=89, respectively) and High (n=170 and n=110, respectively) PK subgroups.

Results: Patients in the High PK subgroup had statistically improved nausea severity at Week 12 compared to the Low PK subgroup (-1.19 vs. -0.78, respectively; p<0.0001), including at baseline nausea severity scores of ≥2 (-1.61 vs. -0.98, respectively; p<0.0001) and ≥2.5 (-1.77 vs. -1.03, respectively; p<0.0001). The High PK subgroup also had statistically significant improvement in nausea-free days at Week 12 compared to Low PK (+35.65% vs. +22.56%, respectively; p=0.0002), including at baseline nausea scores of ≥2 (+36.36% vs. 18.15%, respectively; p=0.0002) and ≥2.5 (+33.44% vs. 14.11%, respectively; p=0.0009).

Conclusion: A significant exposure response was seen in idiopathic gastroparesis patients receiving open label tradipitant for 12 weeks. Patients with higher blood levels of tradipitant significantly improved in average nausea severity and in endpoints measuring overall gastroparesis symptoms compared to subjects with low drug blood levels.

Disclosure: Study was sponsored by Vanda Pharmaceuticals, Inc. Authors are employees of Vanda Pharmaceuticals, Inc.

PP0059

SOLID AND LIQUID GASTRIC EMPTYING SCINTIGRAPHY WITH GASTRIC ALIMETRY ADDS DIAGNOSTIC VALUE IN ASSESSING CHRONIC GASTRODUODENAL SYMPTOMS

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Introduction: Solid gastric emptying scintigraphy (GES) is used to assess gastric motility by measuring the transit of a radioactively labelled solid meal, however it can be non-specific and insensitive by itself. Liquid gastric emptying scintigraphy provides an additional modality of motility testing using a similar process with a liquid meal. Gastric Alimetry is a non-invasive test that measures gastric electrophysiological activity through body surface gastric mapping (BSGM) and validated symptom profiling[1].

Aims & Methods: This study aimed to assess the additional diagnostic value of performing liquid GES and Gastric Alimetry in addition to solid GES alone.

Patients with chronic gastroduodenal symptoms underwent simultaneous solid GES and Gastric Alimetry, comprising of a 30-minute baseline recording followed by a 99mTC-labelled egg meal, and 4-hour postprandial recording. Patients would also have a glucose liquid GES study on a different date. Delayed emptying was defined as greater than 10% retention at 4 hours for solid GES and more than 50% retention at 2 hours for liquid GES[2,3].

Gastric Alimetry spectral metrics were referenced to normative ranges that included principal gastric frequency (cpm), Gastric Alimetry Rhythm Index (GA-RI) and BMI-adjusted amplitude[4].

Results: 36 patients with chronic gastroduodenal symptoms were prospectively recruited (30 female, median age 38.0 years, median BMI 24.6 kg/m^2). 10 (27.8%) patients had delayed solid GES whilst 7 (19.4%) patients had delayed liquid GES. 9 (25.0%) patients had abnormal Gastric Alimetry spectral metrics.

Combining solid and liquid GES detected an abnormality in 13 (36.1%) patients which was further enhanced when combined with Gastric Alimetry, detecting an abnormality in 17 (47.2%) of patients.

Of those 13 patients with either a delayed solid or liquid GES, 38.5% of them had abnormal Gastric Alimetry spectral metrics (3 High Frequency, 1 Low Rhythm Stability, 1 High Amplitude) while those with a normal solid and liquid GES had abnormal spectral metrics in 17.4%.

Conclusion: Combining solid and liquid GES with Gastric Alimetry detected a higher yield of motility abnormalities than solid GES alone. GES and Gastric Alimetry assess different aspects of gastric function with further studies required to understand their complementary roles as biomarkers in clinical practice to help tailor therapy.

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Disclosure: Greg O'Grady holds intellectual property and grants in gastric electrophysiology and is a Director of the University of Auckland spin-out companies Alimetry and Insides Company. Stefan Calder is a member of Alimetry. The remaining authors have no relevant conflicts of interest to declare.

PP0060

COMPREHENSIVE ASSESSMENT OF GASTROINTESTINAL SYMPTOMS IN PEDIATRIC PATIENTS: INSIGHTS FROM INITIAL OUTPATIENT CONSULTATIONS USING THE PSAGIS

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Introduction: Exploring paediatric gastrointestinal (GI) diseases is a dynamic medical challenge. Children and adolescents present with unique symptoms that require careful assessment. These can range from mild discomfort to indicators of serious disease. The initial outpatient visit is crucial for diagnosis and treatment planning.

Understanding the nature and spectrum of these symptoms is crucial for improving paediatric care.

Aims & Methods: We aimed to improve the understanding of GI disorders in paediatric patients, utilising the recently validated paediatric (p)SAGIS questionnaire, which provides an unbiased assessment of symptoms. From February 2016 to January 2023, all consecutive patients attending the outpatient GI clinic at St. Anna Kinderspital, Vienna, Austria, underwent a standard work-up to arrive at a final diagnosis.

In addition, and independently, the patients and/or their caregivers completed the pSAGIS for standardised symptom assessment on the day of their appointment before meeting the paediatrician. pSAGIS consists of 21 GI-related questions (assessment of type and 5 severity grades within the previous week) to arrive at five symptom groups (abdominal pain, diarrhea, constipation, dysphagia, and dyspepsia).

Pain was considered clinically relevant if the score was at least 1 (moderate pain) and severe if the score was ≥2. Data were analyzed for functional bowel disorder (FBD) versus organic GI disease (OD), with a p-value <0.05 considered significant.

Results: 1481 outpatients completed the pSAGIS questionnaire at their initial consultation. Patients were then followed up to establish a final diagnosis. 775 patients were diagnosed with organic disease, 441 patients were diagnosed with functional bowel disease, and 265 subjects were healthy or had a non-GI diagnosis or could not be definitively diagnosed.

In the whole group, abdominal pain was reported by 405 patients (27.3%) and severe pain by 140 patients (9.5%). 185 patients with organic disease (23.9%) but 199 patients with FBD (45.1%) reported abdominal pain (p>0.001), severe pain was reported by 42 (5.4%; OD) and 54 (12.2%; FBD) patients, respectively (p<0.001). Pain was reported as the main symptom by 105 OD patients (13.5%) vs. 113 FBD patients (25.6%) (p<0.001). Pain was present but not the main symptom in 64 (8.3%; OD) and 65 (14.7%; FBD) patients, respectively (p>0.01). Of the latter, 14 (1.8%; OD) and 18 (4.1%; FBD) patients reported severe pain.

Conclusion: Our study provides a comprehensive insight into paediatric gastrointestinal symptoms using the paediatric SAGIS questionnaire. The stark contrast in the prevalence of abdominal pain between patients with organic disease and functional bowel disorders underscores the complex landscape of paediatric GI problems. In particular, the high incidence of severe pain in functional bowel disorders challenges conventional perceptions of these disorders. Recognising pain as a prominent symptom in both organic and functional cases is critical for effective management. The findings underscore the critical role of standardised assessment tools, such as the pSAGIS questionnaire, in delineating the nuanced spectrum of paediatric GI symptoms and ultimately informing more targeted and comprehensive care strategies.

Disclosure: Nothing to disclose.

PP0061

DIFFERENCES IN GASTRIC FLUID MICROBIOTA BETWEEN POSTPRANDIAL DISTRESS SYNDROME AND EPIGASTRIC PAIN SYNDROME

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Introduction: Dysbiosis of gut microbiota has recently been suggested to be implicated in the pathophysiology of functional dyspepsia (FD). However, qualified studies on the gastric microbiota in FD patients are lacking, and it is unclear whether there are differences in gastric microbiota between the subgroups of FD.

Aims & Methods: Our aim of the present study was to compare the diversity and composition of gastric fluid microbiota between postprandial distress syndrome (PDS) and epigastric pain syndrome (EPS) in patients with FD. Eighteen patients with FD who met the Rome IV criteria and 12 age- and gender-matched healthy controls who had no chronically recurrent abdominal symptoms with absence of past or present *Helicobacter pylori* (HP) infection participated in the study. Gastric fluid was obtained from the subjects after an overnight fast, using a sterilized tube inserted into the stomach through the working channel of the endoscope. DNA was extracted and quantified from the samples. Samples from 15 FD patients (83.3%) and 6 healthy controls (50%) were adequate for analysis. The V3 and V4 region of 16S rRNA gene was amplified and sequenced using the MiSeq™ platform (Illumina, San Diego, USA).

Results: The microbiota diversity indices did not significantly differ in FD patients without HP infection, compared with healthy controls. There were no significant differences in the microbiota diversity and composition at the phylum level between PDS patients and healthy controls, and between EPS patients and healthy controls. At the genus level, the relative abundance of *Centipeda* was significantly increased in patients with PDS, whereas that of *Massilia, Acinetobacter, Enterococcus, Bacteroides, and Blautia* was significantly decreased in patients with PDS. In patients with EPS, the relative abundance of *Collinsella, Holdemanella and Paraprevotella* was significantly increased, whereas that of *Massilia* was significantly decreased. At the species level, the relative abundance of *Prevotella intermedia, Veillonella tobetsuensis, Neisseria subflava, and Centipeda*

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periodontii was significantly increased in patients with PDS. The relative abundance of *Clostridium leptum, Clostridium perfringens, Alistipes ihumii, Collinsella aerofaciens, Eubacterium ventriosum, Holdemanella biformis, Klebsiella aerogenes, Paraprevotella clara, Sphingomonas olei, and Sutterella wadsworthensis* was significantly increased in patients with EPS.

Conclusion: Although the diversity of gastric fluid microbiota is not different between PDS and EPS subgroups, there are differences in their composition between PDS and EPS. Further investigation is needed to determine whether those differences in the composition of gastric fluid microbiota are cause or effect.

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Disclosure: Nothing to disclose.

PP0062

REACHING THE END OF THE LINE: CHARACTERIZATION OF NON-OBSTRUCTIVE DYSPHAGIA WITH NORMAL ESOPHAGEAL MOTILITY

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Introduction: Esophageal dysphagia is a common complaint in daily clinical practice. After excluding structural and motility disorders, it may be related to functional disorders, which are challenging to diagnose and manage, leading to high health care utilization and both patient and physician dissatisfaction.

Aims & Methods: This is a prospective multicentric 6-months follow-up study to characterize dysphagic patients with normal esophageal motility, in terms of epidemiology and clinical aspects, focusing on patient-reported outcome measures, and evaluate their evolution at 6-months.

Results: From 2021-2024, 91 patients were recruited, 78.0% female, with a median age of 65 years [22-88]. Of these, 27.5% (n = 25) suffered from rheumatologic or endocrinologic disorders, and 56.0% were previously exposed to medical therapies.

Dysphagic symptoms had a median duration of 3 years [1-31], with a median Eckardt of 3 [0-9], PROMIS GI Disrupted Swallowing (PDS) of 60.4 [40.3-84.1], EuroQol-5D TTO of 0.592 [-0.055-1] and VAS of 70 [0-100].

Compared to non-dysphagic participants (n=123), these patients had significantly worse Eckardt, EuroQol-5D and PDS scores (p-values <0.001).

Compared to esophageal motility disorders (Chicago V4.0), the Eckardt and PDS scores in this group were significantly lower when compared to EGJ outflow disorders (p-value <0.001 and 0.003), without differences regarding EuroQol-5D.

Regarding gastroesophageal reflux, 24 (26.4%) patients had a positive GERD-Q questionnaire and 3 had an objective GERD diagnosis. Even though, compared to motility disorders, this score was significantly lower in this group when compared to EGJ outflow disorders (p-value 0.001).

Medical treatments (PPI, amitriptyline) were optimized in 6 patients (6.6%), without any differences in Eckardt or EuroQol-5D scores at 6-months. However, at 6-months, the PDS improved in untreated patients (p-value <0.001).

Conclusion: Non-obstructive dysphagia with normal esophageal motility has a negative impact on quality of life. However, symptoms tend to have a favorable course independently of treatment, therefore mandating a holistic management.

Disclosure: Nothing to disclose.

PP0063

THE PREVALENCE OF NAUSEA IN EASTERN AND WESTERN SECONDARY AND TERTIARY CARE FUNCTIONAL DYSPEPSIA

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Introduction: According to the ROME IV criteria, functional dyspepsia (FD) can be subdivided into PDS (postprandial distress syndrome) with meal-related symptoms such as early satiation and postprandial fullness and EPS (epigastric pain syndrome) with meal-unrelated complaints such as epigastric pain or burning. Nausea can also co-exist in FD but the literature is poor in reporting whether this symptom is mainly triggered by meal ingestion.

A recent global epidemiology study of gastroparesis-like symptoms showed that the prevalence of nausea or vomiting, in combination with postprandial fullness or early satiation varies between continents.[1]

Aims & Methods: This study aims to compare the difference in prevalence rates of nausea within each FD subgroup between western and eastern secondary and tertiary care centers. Adult patients with disorders of gut-brain interaction (DGBI) recruited in 9 western and 7 eastern secondary and tertiary care facilities between the ages of 18 and 70 years completed the Enhanced Asian Rome IV questionnaire (Asian European Gastrointestinal Symptoms cohort or AEGIS). The western centers were located in Belgium, Bulgaria, France, Israel, Italy (2x), Slovakia, Sweden and The United Kingdom while the eastern centers were located in Australia, China, Iran, Japan, Malaysia and Singapore (2x).

FD patients were divided into PDS, EPS and PDS-EPS overlap (OVP) subgroups. Patients were furthermore classified as having nausea when this symptom was reported at least once per week. The questionnaire EARQ4 questionnaire was also used to address the relationship between nausea and food intake.

Patients were asked how often they experienced nausea, whether nausea was triggered or worsened by meal intake, how often nausea occurred after a meal and if so, how soon after a meal nausea started. A comparison of the prevalence of nausea and meal-related nausea was made between

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eastern and western patients and among FD subgroups. Continuous variables are reported as median (IQR) and compared with a Kruskal-Wallis test. Categorical variables were compared using a Pearson Chi-square test.

Results: Of the 1074 DGBI patients that completed the survey, 483 were classified as having FD (59.4% PDS, 27.5% EPS, 13.0% PDS-EPS overlap). FD patients from eastern (N=202) and western sites (N=281) had a similar body mass index (22.5 (19.6-25.7) vs. 21.3 (19.7-25.4) kg/m^2) and age (39.0 (29.0-55.0) vs. 38.0 (26.0-54.0) years). Eastern sites included significantly more female patients than western sites (43 % vs. 17.5%, p<0.001). PDS and EPS patients from eastern sites had significantly less nausea than PDS and EPS patients from western sites (17.9% vs. 46.3% and 6.8% vs. vs 40.5% respectively, both p<0.001). The majority of FD patients with nausea experienced meal-related nausea with no difference between eastern and western sites (63.9% vs. 73.7%). In PDS patients with nausea, nausea was significantly more triggered or worsened by a meal (72.4%) than in PDS-EPS overlap and EPS (48.6% and 35.3% respectively; p<0.001). All groups reported that meal-related nausea started mostly (> 85%) within 60 minutes after the meal.

Conclusion: In contrast to previous literature, we found that western PDS and EPS patients report nausea more frequently than eastern PDS and EPS patients. In both eastern and western patients nausea was mostly meal-related. Meal-related nausea is more common in PDS than in PDS-EPS overlap and EPS patients. Further research is needed to investigate the role of ethnic factors in the pathophysiology of nausea in functional dyspepsia.

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PP0064

UNFOLDING THE MYTHS OF PEDIATRICS FUNCTIONAL ABDOMINAL PAIN DISORDERS (FAPDS)

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Introduction: Functional abdominal pain disorders (FAPDs), linked to brain-gut interaction disturbances, are common in childhood, with a prevalence of 3-16%. They are classified based on the Rome IV criteria, involving persistent gastrointestinal symptoms due to abnormal enteric system functioning without organic issues. Childhood stress and anxiety are linked to FAPDs. Psychotherapy aims to educate patients and families about pain triggers and uses cognitive-behavioral therapy extensively.

Aims & Methods: To discuss different characteristics of patients with FAPDs and identify triggering factors for stress/anxiety. Moreover, to analyze the role of various treatment options, including mind-targeted therapies as a treatment.

This prospective cohort study included all patients who were referred to the Pediatric Gastroenterology and Hepatology Outdoor Department at Children's Hospital Lahore due to chronic abdominal pain and subsequently diagnosed with FAPDs based on Rome IV criteria after excluding organic causes over a period of 18 months. Patients were screened for stress by the clinical psychologist using perceived stress scores after excluding organic causes. The role of various treatment options, including mind-targeted therapies, was studied in patients with different FAPDs. They were reassessed for symptom improvement or resolution at a 4-week follow-up by both gastroenterologists and clinical psychologists.

Results: This study included 300 patients aged 4-16 years, of any gender and economic background, using consecutive non-probability sampling. The overall mean age was 9.848 ± 2.6879 years, with 171 female (57%) & 129 male (43%) patients. 245 patients (81.7%) were diagnosed with an organic cause for chronic abdominal pain based on various clinical and biochemical parameters. Commonest cause was gastritis 83 (27.7%) followed by celiac 59 (19.7%), chronic pancreatitis 32 (10.7%), Abdominal TB 22 (7.3%), UTI 17(5.7%), IBD 16 (5.3%), Choledochal cyst 9 (3%), Cholelithiasis 5 (1.7%), Hydatid cyst 2 (0.7%). In total 55 (18.3%) were diagnosed with functional GI disorders based on Rome IV criteria. Among them, Functional Abdominal Pain, not otherwise specified accounted for 21 cases (38.18%), followed by functional dyspepsia 19 (34.54%), irritable bowel syndrome 10 (18.18%), and abdominal migraine with 5 cases (9.09%).The age range of patients diagnosed with FAPDs spanned from 7.5 to 15 years, with a median age of 12 years. There were 34 females (61.8%) compared to 21 males (38.2%). 60% of them were from urban areas. Various triggers for stress were identified: financial issues (69.1%), followed by family conflict (58.2%), family member illness (18.2%), single parents/broken families (16.4%), exam failure (16.4%), abuse/bullying (12.7%), family member death (9.1%), frequent punishment at home (5.5%), sibling rivalry (3.6%), and frequent punishment at school (1.8%). The perceived stress scores were categorized as low (0-13) for 7 (12.7%), moderate (14-26) for 45 (81.8%), and high (27-40) for 3 (5.5%) patients before treatment. Among them, 43.3% received psychotherapy, followed by 25.5% receiving psychotherapy along with symptomatic treatment, 16.4% symptomatic treatment, 9.1% psychotherapy along with antidepressant, and 5.5% receiving only antidepressant. Patients with a moderate stress score were given psychotherapy alone in 48.9% of cases. Post-treatment perceived stress scores improved to low in 40 patients (72.7%) (p<0.001).

Conclusion: Functional GI disorders, with stress or masked depression, need assessment for better child welfare, as healthy psyche fosters resilience.

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Disclosure: Nothing to disclose.

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PP0065

3-YEAR PROSPECTIVE OBSERVATIONAL COHORT STUDY ON PREVALENCE OF SYMPTOMS OF FUNCTIONAL DYSPEPSIA, ANXIETY AND DEPRESSION

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Introduction: Disorders of gut brain interaction (DGBIs) is a group of diseases with relatively high prevalence. One of the most common is functional dyspepsia. It has been repeatedly reported that university students tend to have higher prevalence of this disease than general population. Study at medical school is associated with high degree of stress and this could be accumulated during the passing years. As many other factors could affect the symptomatology, the best way to test this hypothesis is to perform a prospective cohort study following one particular group of students over the years.

Aims & Methods: The aim of this study is to evaluate the prevalence of symptoms of functional dyspepsia in selected cohort overtime. This is a prospective, observational study. The observed group were students of medicine from their 1st to 3rd year of study. Acquisition of the data was performed by questionnaires. The diagnosis was made by using ROME IV criteria for DGBI. Additional information was obtained from STAX1 and STAX2 validated questionnaires (STAI-X1 shows how the patient feel while answering the questionnaire (anxiety as a state) and STAI-X2 is based on how the patient feel overall (anxiety as a personality trait)) and Beck depression inventory. The measurement was performed in the middle of the school year, every October (2021 to 2023).

Results: Initially 121 students from first year agreed to participate in the study and filled out the questionary. Subsequently the response rate for 2nd year was 67.8 % and for 3rd 73%. The symptoms of Functional Dyspepsia were present in 14.9% of participants from 1st year, 13.4% from 2nd year and 14.6% from 3rd year. The cumulative cohort average of STAX 1 was 45.5 points for 1st year, 46.5 points for 2nd year and 46.2 points for 3rd year. The cumulative cohort average of STAX 2 was 47.7 points for 1st year, 46.2 points for 2nd year and 46.8 points for 3rd year. The cumulative cohort average of Beck depression inventory was 8.7 points for 1st year, 8.8 points for 2nd year and 8.4 points for 3rd year. The rate of smoking was 7.4 % of participants from 1st year, 13.4% participants from 2nd year and 14.6% from 3rd year.

Conclusion: The results of this prospective cohort observational study shows that the symptoms of Functional Dyspepsia are highly prevalent among medical students and this varies only insignificantly over time. The probands in average showed no signs of depression however the average levels of anxiety and stress were high during all 3 years. This could be associated with significant rise of smokers within the cohort during the observation period.

Disclosure: Nothing to disclose.

PP0066

PREVALENCE AND BURDEN OF FUNCTIONAL DYSPEPSIA AND "GASTROPARESIS LIKE SYMPTOMS" IN A PRIMARY CARE DIABETIC POPULATION

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Introduction: The prevalence of functional dyspepsia (FD) and gastroparesis in diabetic patients remains unknown. A high prevalence of gastroparesis has been reported in diabetic patients reaching up to 5%.[1] However, recent consensus specified the symptom pattern for gastroparesis, with nausea and/or vomiting as predominant symptoms, now defined as "gastroparesis-like symptoms" (GPLS).[2] The global prevalence of GPLS has been estimated to reach only 1.2% in diabetic patients, but type 1 and type 2 diabetes were not assessed separately.[2]

Aims & Methods: We aimed to assess the prevalence and burden of FD and GPLS in primary care patients with diabetes, comparing patients with type 1 and type 2 diabetes. An online survey was addressed to diabetic patients through general practitioners and diabetic patient associations. The survey included the Rome IV questionnaire and the Patient Assessment of Gastro-Intestinal Disorders-Symptom Severity Index (PAGI-SYM) for upper gastro-intestinal symptoms. Anxiety, depression, somatization, and quality of life were also assessed by specific questionnaires (Generalized Anxiety Disorder GAD-7, Patient Health Questionnaire PHQ-9, PHQ-15, and PAGI-QoL questionnaires respectively).

	Type 1 diabetes (n=146)	Type 2 diabetes (n=68)	p value
Age	39 (29–51)	60 (50–67)	<0.001
Sex (female)	99 (67.8%)	35 (51.4%)	0.02
Functional dyspepsia	18 (12.3%)	10 (14.7%)	0.66
PDS	17 (11.6%)	10 (14.7%)	0.52
EPS	3 (2.1%)	3 (4.4%)	0.38
Gastroparesis-like symptoms	3 (2.1%)	6 (8.9%)	0.03
GAD-7 (Anxiety)	4 (1–6)	3 (0–7)	0.80
PHQ-9 (Depression)	3 (1–7)	3 (1–10)	0.05
PHQ-15 (Somatization)	7 (3.3–10)	8 (4–13)	0.10
PAGI-SYM	6.5 (1–8.8)	9.5 (2–35.5)	0.009
PAGI-QoL	6 (0–15.3)	10 (3–38)	0.003

Results: A total of 216 patients completed the questionnaires, among which 146 had type 1 diabetes, and 68 had type 2 diabetes, (2 patients did not know their type of diabetes). The median age of the population was 56 (50-61) years old, with 134 female patients (61.8%). The prevalence of FD was 13.0% with no difference between the 2 groups. The prevalence of GPLS reached 4.2% and was significantly higher in patients with type 2 diabetes (8.9% vs 2.1%; p=0.03). The burden of upper gastro-intestinal symptoms and their impact on quality of life was higher in type 2 dia-

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betes patients with higher PAGI-SYM and PAGI-QoL scores while anxiety and somatization were not different between type 1 and type 2 diabetes patients (see Table). Overall, patients with GPLS had higher anxiety (9 vs 3; p=0.003), depression (12 vs 3; p=0.001) and somatization scores (16 vs 6; p<0.001) than patients without. Compared to patients with FD without GPLS, GPLS patients had numerically higher but not statistically different scores for symptom severity (51 vs 28; p=0.46), anxiety (9 vs 7; p=0.29), depression (12 vs 7; p=0.07), somatization (16 vs 11.5; p=0.16) and quality of life scores (52 vs 24; p=0.16).

Conclusion: This study confirms that the prevalence of GPLS seems to be higher in diabetic patients compared to available population data, especially in patients with type 2 diabetes. The impact of the symptoms was similar to the impact of FD in this population. Further studies are required to assess the prevalence of delayed gastric emptying in diabetic type 1 and 2 patients with GPLS.

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Disclosure: Nothing to disclose.

PP0067

GASTROESOPHAGEAL REFLUX DISEASE AND FUNCTIONAL DYSPEPSIA ARE ASSOCIATED WITH CAPSAICIN COUGH SENSITIVITY IN SEVERE ASTHMA

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Introduction: Functional dyspepsia (FD) is one of the most prevalent functional gastrointestinal disorders. It is well known that asthma can complicate gastroesophageal reflux disease (GERD), and sensory nerve dysfunction is related to the pathophysiology of asthma and GERD. However, the relationship between FD and asthma has not been extensively examined yet. The capsaicin receptor transient receptor potential vanilloid 1 (TRPV1) is a non-selective ion channel, which can be activated capsaicin, acid (pH<6.5), and heat etc. TRPV1 channels are highly expressed in sensory nociceptive and visceral nerve fibers and are involved in inflammatory pain and hyperalgesia signaling. TRPV1 is found to be expressed in both airway epithelium and sensory neurons of the gastrointestinal tract, as previously reported. In addition, several studies have shown increased gastrointestinal sensory sensitivity to capsaicin in patients with FD.

Aims & Methods: The aim of this study was to clarify the physiological associations between GERD, FD, and asthma. Fifty-two patients with severe asthma who were performed capsaicin cough sensitivity (C-CS) test were analyzed. Clinical characteristics C-CS were compared between patients with and without comorbid GERD or FD. C-CS was evaluated by C5 (concentration of inhaled capsaicin that induced five or more coughs), and C5<2.44 uM was defined as heightened C-CS. GERD was diagnosed based on the presence of the typical reflux symptoms of heartburn and/or acid regurgitation, which occurred more than twice a week prior to proton pump inhibitor (PPI) therapy. FD was diagnosed using the Rome IV criteria, which are defined as chronic and recurrent epigastric pain or postprandial distress, and no other organic disease that could have caused these symptoms.

Results: A total of 52 patients with severe asthma were eligible for this study. Twenty-two out of 52 patients had GERD comorbidity, and 15 out of them had FD comorbidity. Thirteen GERD patients showed heightened C-CS, while nine patients showed normal C-CS. Compared to the GERD (-) group, the GERD (+) group had a significantly lower threshold value of C5. Lower C2 and C5 thresholds were found in the FD (+) group than in the FD (-) group, and heightened C-CS was more prevalent in the FD (+) group than in the FD (-) group. Patients who had a heightened C-CS were more likely to experience GERD and FD than those who had a normal C-CS.

Effects of comorbidities on C-CS in patients with severe asthma

		Heightened C-CS	Normal C-CS	p value
FD	n=15	11	4	0.003
GERD	n=22	13	9	0.04

Conclusion: These results suggest that FD, as well as GERD, and severe asthma are functionally linked by the interaction of sensory nerves between the airways and the upper gastrointestinal tract. TRPV1 may be involved in the pathophysiology of both GERD, FD, and asthma.

Disclosure: Nothing to disclose.

PP0068

REAL-WORLD EXPERIENCE OF USING PRUCALOPRIDE FOR THE TREATMENT OF DYSPHAGIA IN PATIENTS WITH NEGATIVE GASTROSCOPY AND CHICAGO III OESOPHAGEAL DYSMOTILITY IN CLINICAL PRACTICE

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Introduction: Prucalopride, a 5-HT4 agonist approved for the treatment of refractory chronic constipation has been shown to reduce oesophageal acid exposure in healthy volunteers (Kessing BF, 2014) and improve oesophageal motility in gastroesophageal reflux (GER) patients with ineffective oesophageal motility (Lei W-Y, 2018). To date, no studies have evaluated its effect in patients with dysphagia and oesophageal dysmotility.

Aims & Methods: Our aim was therefore to evaluate the effect of prucalopride on dysphagia symptoms and to assess which, if any, motility or GERD diagnosis influenced outcome in patients with gastroscopy negative dysphagia. Clinical records, together with oesophageal high-resolution

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manometry (HRM) and available 24hr pH-monitoring (pH-MII) reports of patients referred for dysphagia to a single Neurogastroenterology Clinic between April 2017 and March 2024 were retrospectively reviewed. All patients had a previous negative gastroscopy with oesophageal biopsies, were not responsive to PPIs and were prescribed prucalopride as an add-on treatment at the initial dosage of 1 mg/daily with the possibility to increase to 2 mg/daily after one week in case of no response. A standardized template was used to collect clinical data and patient-reported response to treatment. Clinical response was defined as a self-reported improvement in dysphagia symptoms associated with a reduction in Eckardt score of at least one point. Data are expressed as median (QR).

Results: Thirty-five patients [median age 62 (28) years; 19 (54%) female] were included. Mean duration of follow-up was 12 (19) months. Twenty-two (63%) patients responded to Prucalopride with a significant improvement in the Eckardt score before to after treatment [3.5 (2.3) vs 1 (1.8); p<0.001)]. Six patients (27.3% of responders) responded to 1 mg, with the rest increasing to 2mg/daily. Response was maintained for a median of 14.5 (16) months.Thirteen (37.1%) patients did not respond, of which four (11.4%) experienced adverse events (diarrhoea, headache, abdominal pain) leading to an early treatment discontinuation. In these patients the Eckardt score did not change before to after the treatment [3 (3) vs 3.5 (2.3), respectively; p=0.85)]. None of the clinical (data not shown) or key manometric parameters were associated with treatment response [median basal distal contractile integral 283 (425) vs 340 (470)mmHg/s/cm; p=0.71 and median integrated relaxation pressure 6.2 (8) vs 8.4 (9.4) mmHg; p=0.13 in responders and non-responders, respectively]. However, there were trends for more patients with absent contractility to be non-responders, and more patients with EGJOO to be responders to prucalopride. Normal acid exposure time and number of reflux events were associated with a positive response to treatment (Table 1).

Oesophageal Manometry	Non-Responders (n=13)	Responders (n=22)	p-value
Ineffective Oesophageal Motility (%)	7 (53.8%)	9 (40.9%)	0.46^
Oesophago-gastric junction outflow obstruction (%)	0 (0%)	5 (22.7%)	0.13^
Absent contractility (%)	5 (38.5%)	4 (18.2%)	0.24^
Other diagnoses (%)	1 (7.7%)	4 (18.2%)	0.63^
AET	14.2 [20.3]	1.4 [4.1]	0.06+
Mean number of acid refluxes	46 (107)	12 (15.5)	<0.01
Pathological AET (>6%)	11 (84.6%)	10 (45.4%)	0.03^

Table 1 – HRM and pH-monitoring characteristics of responder and non-responder patients at baseline.
+Mann-Whitney's U tests
^Fisher's Exact Test
Abbreviations: AET=acid exposure time

Conclusion: Our study suggests Prucalopride might be a well-tolerated and useful treatment for dysphagia in patients with dysmotility disorders, mainly in the absence of pathological reflux. However, prospective randomized controlled studies are needed to confirm these data.

Disclosure: Nothing to disclose.

PP0069

DIFFERENCES IN MEDICAL CARE SATISFACTION OF PATIENTS WITH "*FUNCTIONAL*" VS "*ORGANIC*" ESOPHAGEAL DISEASES: ITALIAN VALIDATION OF THE PPRS-SF QUESTIONNAIRE AND ITS APPLICATION IN CLINICAL ROUTINE PRACTICE

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Introduction: Patient-physician relationship (PPR) plays a center role in modern health care, especially within gastrointestinal (GI) medical practice, where patients affected by non-structural disorders (better described as disorders of gut-brain interaction or DGBI) tend to report a low satisfaction towards their medical care. In 2017 Kurlander et al. developed and validated a Patient-Physician Relationship Scale (PPRS), a questionnaire used in patients affected by Irritable Bowel Syndrome (IBS) to assess their degree of medical care-satisfaction. The original PPRS was later shortened and simplified by Drossman et al. in 2021 by creating a PPRS-SF (short form), consisting in a 12-item questionnaire.

Aims & Methods: The aims of this study were to validate the PPRS-SF in Italian and to administer it to both patients with functional and organic esophageal diseases, in order to highlight the differences in patient-physician relationship between the two categories.

The process of validation to Italian language of the PPRS-SF questionnaire was carried out following a standardized forward-backward procedure and each step was overseen by an independent clinical monitor assigned by the Rome Foundation itself. The Italian version of the PPRS SF was successfully administered to 50 Eosinophilic Esophagitis (EoE), 53 Achalasia (Ach), 48 Gastroesophageal Reflux Disease (GERD) and 44 esophageal DGBI patients (affected by functional dysphagia or functional heartburn).

Results: In esophageal DGBI and GERD, the degree of patients' satisfaction was significantly lower, if calculated as a total summation, compared to that of EoE and Achalasia patients.. Between a minimum score of -36 and a maximum of +36, the mean PPRS-SF "Italian version" score for patients with EoE was 30.20 (±9.78), for patients with Ach 23.78 (±19.9); for patients with GERD 20.46 (±16.03); for esophageal DGBI 17.91 (±18.41). Moreover, there was a significant difference in PPR between younger patients (under 50 years old), who reported a score of 27.1 (±12.29) and older patients (over 50 years old), reporting a score of 20.16 (±19.72).

Conclusion: This study validated the Italian version of the PPRS-SF questionnaire and showed that the patients' diagnosis together with other factors such as their age, can affect the quality of the PPR. In particular, patients with DGBI and older patients reported lower scores than the ones with organic diseases and the younger ones, respectively. Further studies are necessary to confirm these results.

Disclosure: Nothing to disclose.

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PP0070

INVESTIGATING INAPPROPRIATE PROTON PUMP INHIBITOR USE IN PRIMARY CARE PATIENTS

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Introduction: Increasing concerns arise regarding inappropriate proton pump inhibitor (PPI) use in terms of indication and dosage, impacting healthcare budgets and posing possible risks and drug interactions. It was estimated that up to 70% of PPI use may not medically indicated, however, the characteristics of this population of patients have not been described (1,2).

Aims & Methods: We aim to describe demographic and clinical profiles of primary care PPI users who lack a strict medical indication.

General practitioners included long-term PPI users without strict medical indication in the study. Demographic characteristics, duration and dosage of PPI use, and medical history (e.g. upper endoscopy findings, H. pylori status and previous attempts to stop PPI) were registered. Patients completed questionnaires assessing gastrointestinal symptoms (PAGI-SYM) and psychosocial co-morbidities (depression, somatization, and anxiety using respectively PHQ-9, PHQ-15, GAD-7). For this analysis, patients were categorized by: (I) initial indication for PPI therapy (GERD with low-grade esophagitis, dyspepsia, non-gastric symptoms) and (II) PPI dose (healing or maintenance). Statistical significance ($p<0.05$) was assessed using one-way ANOVA, student's T-test, and Fisher's exact test. Results are shown as mean±SD.

Results: Out of 482 recruited patients on long-term PPI therapy, 42% reported failed previous attempts to discontinue PPI therapy. Main indications for PPI initiation were low-grade reflux esophagitis (26%), functional dyspepsia (18%), uninvestigated reflux symptoms (13%), uninvestigated dyspepsia (10%), peptic ulcer (2%) and mixed causes (18%, e.g. polymedication, cough, hoarseness,….). A history of H. pylori infection was present in 9% of patients. Patients exhibited on average low somatization, no depression, and no anxiety (Table).

	Nausea/ vomiting (PAGI-SYM) mean score ± SD	Postprandial fullness/ early satiety (PAGI-SYM) mean score ± SD	Bloating (PAGI-SYM) mean score ± SD	Upper abdominal pain (PAGI-SYM) mean score ± SD	Lower abdominal pain (PAGI-SYM) mean score ± SD	Heartburn/ regurgitation (PAGI-SYM) mean score ± SD	Depression (PHQ-9) mean score ± SD	Somatization (PHQ-15) mean score ± SD	Anxiety (GAD-7) mean score ± SD
Total (n=482) Mean age 61 ± 14 y.o. 58% female BMI 27.9 ± 4.6 kg/m²	0.4 ± 0.7	1.1 ± 1.1	1.7 ± 1.6	1.0 ± 1.2	0.7 ± 1.2	0.9 ± 1.0	4.6 ± 4.6	8.8 ± 5.1	5.0 ± 5.0
GERD (n=167) Mean age 60 ±14 y.o. 59% female BMI 27.5 ± 4.7 kg/m²	0.3 ± 0.7	1.1 ± 1.0	1.6 ±1.5	0.8 ± 1.7	0.6 ±1.1	0.9 ± 1.0	3.9 ± 3.9	8.2 ± 4.8	4.2 ± 4.7
Dyspepsia (n= 126) Mean age 60 ± 15 y.o. 63% female BMI 28.0 ± 4.6 kg/m²	0.4 ± 0.6	1.3 ± 1.2	2.0 ± 1.6	1.1 ±1.2	0.8 ± 1.2	0.9 ± 1.1	5.5 ± 5.0	9.8 ± 5.3	6.1 ± 5.1
Non-gastric (n=55) Mean age 64±11 y.o. 65% female BMI 27.8 ± 4.1 kg/m²	0.4 ± 0.7	1.1 ± 1.2	1.3 ± 1.5	0.8 ± 1.4	0.5 ± 1.1	0.6 ± 0.9	4.5 ± 4.7	8.3 ± 4.6	3.9 ± 3.6

There were no differences in age and sex among cohorts on healing (n= 80, 62% female, mean age 61 ± 13 y.o.) or maintenance PPI dose (n=362, 58% female, mean age 61 ± 14 y.o.). However, those on healing doses had a significantly higher BMI (29.2 ± 4.9 vs 27.6 ± 4.4 kg/m², $p=0.003$) and reported higher scores on all PAGI-SYM subscales (NS), with a significantly higher score on the postprandial fullness & early satiety subscale (average score of 1.4 ± 1.3 vs. 1.1 ±1.0, $p=0.030$). There were no significant differences in psychosocial co-morbidities between patients on healing or maintenance PPI doses (all $p>0.05$).

Sex, age, and BMI were similar among subgroups based on the initial reason for PPI therapy. Patients in whom PPI therapy was initiated for dyspepsia had significantly higher bloating scores than those with reflux or non-gastric symptoms ($p=0.011$). Furthermore, dyspepsia patients exhibited significantly higher somatisation ($p=0.015$), depression ($p=0.007$), and anxiety ($p=0.002$) compared to reflux symptom patients.

Conclusion: A large subset of patients with prolonged PPI intake lacking a strict medical indication have unsuccessfully attempted to stop PPI usage. The patients who are long-term on healing doses have a significantly higher BMI and more severe postprandial distress syndrome symptoms. Psychosocial co-morbidity is significantly higher in patients in whom PPIs were started for dyspeptic symptoms. These characteristics are relevant when considering attempts to deprescribe PPIs in primary care.

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Disclosure: Nothing to disclose.

PP0071

HIGH-RESOLUTION VIDEO MANOMETRIC ASSESSMENT OF DYSPHAGIA POST FUNDOPLICATION

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Introduction: Post-fundoplication dysphagia is common and can occur in up to 65% of patients[1]. Following anti-reflux surgery, the oesophago-gastric junction (OGJ) is an anatomically complex and dynamic area which is subject to mechanical stress, herniation and slippage. Upper gastrointestinal endoscopy and barium oesophagrams are the most commonly ultilised diagnostic modalities. High-resolution video manometry (HRVM) is an alternative and potentially superior technique as it allows for physiological assessment with high-resolution manometry (HRM) with simultaneous fluoroscopic views help delineate anatomy, location of the fundoplication and better appreciate hernias. We aimed to characterise modes of dysphagia post-fundoplication with HRVM.

Aims & Methods: We conducted a retrospective observational study over 62 months until March 2023. HRVM performed on patients with ongoing dysphagia post-fundoplication were selected from a HRVM database and modes of dysphagia were assessed.

Results: 501 patients had HRVM over 62 months of which 32 patients had dysphagia post-fundoplication. Median integrated relaxation pressure (IRP) was greater than 15mmHg in 9 patients (27.3%). HRVM was normal in 8 (25%), The causes of dysphagia post-fundoplication included pseudo-achalasia with oesophogo-gastric junction outflow obstruction (OJ-GOO) due to a tight wrap in 9 (28.1%), ineffective oesophageal motility in 8 (25%), para-oesophageal hernia due to slippage of wrap in 3 (9.4%), non-achalasia oesophageal dysmotility disorders (i.e. hypercontractile oesophagus and distal oesophageal spasm) in 2 (6.3%) and new diagnosis of Type 2 achalasia in 2 (6.3%). Post fundoplication complications accounted for 12 (37.5%) and an underlying oesophageal motility disorder was diagnosed in 12 patients (37.5%).

Conclusion: Whilst iatrogenic causes of dysphagia due to fundoplication itself were commonly seen, underlying oesophageal motility disorders

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were also identified in an equally large proportion of patients. Whether this reflects post-surgical motility disturbance or missed pre-morbid dysmotility is unclear. Nevertheless, these results highlight the importance of oesophageal manometry prior to anti-reflux surgery.

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Disclosure: No conflicts of interest.

PP0072

COMPARISON OF ANXIETY AND DEPRESSION FREQUENCIES IN ROMA IV FUNCTIONAL DYSPEPSIA SUBGROUPS: A CROSS-SECTIONAL OBSERVATIONAL STUDY FROM TURKEY

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Introduction: Different pathophysiological mechanisms are thought to play a role in functional dyspepsia (FD) subgroups defined by Rome IV criteria. However, many more studies are needed to determine whether a better separation of pathophysiological mechanisms, so a better therapy choice, can be obtained with the Rome IV subgrouping. This study aims to investigate the difference in psychopathological comorbidities between the FD subgroups defined by the Rome IV criteria. We hypothesized that anxiety and depression would be risk factors differently for each functional dyspepsia subgroup according to Roma IV criteria.

Aims & Methods: This is a cross-sectional observational study conducted at a tertiary gastroenterology center in Turkey between December 2022 and April 2023. Dyspeptic patients, aged over 18 years old referred for an upper gastrointestinal endoscopy were recruited. Participants were asked to fill out the Rome IV gastroduodenal questionnaire along with the Hospital Anxiety and Depression Scale. FD subgroups were defined using the Rome IV criteria. Hospital Anxiety and Depression Scale was used to assess the anxiety and depression of participants. All statistical analyses were conducted using SPSS version 23.0.The p-value of <0.05 was accepted as statistically significant.

Subgroups	PDS N:59	Overlapping N:37	EPS N:21	Total N:117
Depression n(%)	25 (37.9%) p>0.05	9 (30%) p>0.05	7 (33.3%) p>0.05	41 (35%)
Anxiety n (%)	26 (39.4%) *p<0.05	7 (23.3%) p>0.05	2 (9.5%) *p<0.05	35 (29.9%)

FD, functional dyspepsia; PDS, postprandial distress syndrome; EPS, epigastric pain syndrome

Table 1. Anxiety and depression frequencies among FD Rome IV subgroups.

Results: Of 424 dyspeptic participants, FD was identified in 117 (27.5%) patients. There were 66 (56.4%) PDS (postprandial distress syndrome), 21 (18%) EPS (epigastric pain syndrome), and 30 (25.6%) overlapping cases. Depression was detected in 35% of FD patients; in 37.9% of the PDS subgroup, 30% of the overlapping subgroup, and 33.3% of the EPS subgroup; There was no difference between the groups (p:0.622). Anxiety was detected in 29.9% of FD patients; in 39.4% of the PDS subgroup, 23.3% of the overlapping subgroup, and 9.5% of the EPS subgroup; detected more frequently in PDS subgroup than in EPS and overlapping subgroups (p:0.022) (Table 1). The frequency of anxiety was significantly higher in patients with postprandial fullness and early satiation symptoms according to those without symptoms (35.8% vs. 15.6%, p:0.035; 44.7% vs. 23.9% p:0.027, respectively).

Conclusion: PDS subgroup, postprandial fullness, and early satiety symptoms were more associated with anxiety, according to the EPS subgroup undd epigastric pain symptom. Evaluating this patient groups in this respect and providing the necessary support can accelerate the recovery process of the patient. It seems reasonable to assume that anxiolytic drugs are of benefit in especially the PDS subgroup. Further studies assessing psychological factors and treatment strategies for FD Rome IV subgroups are required to confirm the relationship of subgroups with depression and anxiety.

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PP0073

ESOPHAGEAL SAMPLING AFTER FOOD IMPACTION; PREJUDICE AFFECTS CLINICAL PRACTICE…

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Introduction: Epidemiological studies have shown that Eosinophilic Esophagitis (EoE) is most commonly observed among young males with atopic background. Data have pointed out that approximately half of young patients presenting with bolus impaction will have EoE as a causative factor. According to guidelines, in patients with bolus impaction, EoE should be searched, by esophageal sampling, irrespective of patients age, sex or endoscopy findings.

Aims & Methods: Aim of our study was to elucidate if clinicians in an academic hospital in Greece cohere to guidelines or if epidemiological and endoscopic data precludes them from esophageal sampling.

We included 134 consecutive patients presented with bolus impaction

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during a 6-year period (1/18-12/23). We excluded patients with a history of; esophageal stenosis, esophageal/gastric cancer esophageal motility disorders, metastatic disease, bone-foreign body ingestion, caustic agent ingestion, esophagogastric surgeries. We further excluded 34 patients with endoscopical evidence of another disease that could explain patients' symptoms. Epidemiological, endoscopical and histological data were collected and analyzed.

Results: We finally included 100 patients (M/F: 71/29, mean age 55±23 years old). A history of previous bolus impaction without biopsy sampling was reported by 35 (26.1%), while 12 (9%) reported history of dysphagia before the impaction episode. During the endoscopy 36 had endoscopic findings suggestive of EoE and 64 (47.8%) a normal Endoscopy. Biopsies were obtained by 47/100 (47%). In the univariate analysis, sampling was correlated, in a statistically significant way, with male sex, age <65 years old, endoscopic signs compatible to EoE and a history of previous bolus impaction or dysphagia. Though in the multivariate analysis only endoscopic findings and age were observed, in a statistically significant way, to affect clinicians decision to obtain biopsies. EoE was diagnosed in 33/47 (70.2%). In the univariate analysis EoE diagnosis was significantly correlated with age <65 years, male sex and the existence of compatible endoscopic findings but not with a history of previous bolus impaction or dysphagia. In the multivariate analysis only age <65 was independently correlated with EoE diagnosis. (Table)

Factor	Biopsy sampling among patients with Bolus impaction (patients without endoscopic findings other than EoE that could explain patients symptoms)			EoE diagnosis among patients with Bolus impaction		
		Univariate p	multivariate p		Univariate p	Multivariate p
Younger than 65	44/71 62%	<0.001	0.002 OR: 0.951 95% C.I.: 0.922-0.982	33/44 75%	0.041	0.038 OR: 0.931 95% C.I.: 0.870-0.996
65 and older	3/29 10.3%			0/3 (0%)		
Male	41/71 57.7%	0.001	0.231 OR: 0.441 95% C.I.: 0.116-1.683	31/41 76.5%	0.047	0.968 OR:1.064 95% C.I. :0.052-21.887
Female	6/29 20.7%			2/6 33.3%		
Findings compatible to EoE	32/36 88.9%	<0.001	<0.001 OR: 21.836, 95% C.I.: 3.994-115.956	27/32 84.3%	0.031	0.062 OR: 7.536 95% C.I.: 0.900-63.101
Normal endoscopy/ other findings	15/64 23.4%			6/15 40.0%		
History of previous bolus impaction or dysphagia	22/33 66.7%	0.010	0.493 OR: 1.690, 95% C.I.: 0.376-7.592	16/22 72.7%	0.555	
Yes	25/67 37.3%			19/25 76.0%		
No						

Conclusion: According to our study, esophageal biopsies are obtained in <50% of patients with a bolus impaction episode. Age and endoscopic evidence supportive of EoE seem to drive clinicians decision to obtain esophageal biopsies after a bolus impaction episode. EoE is more commonly observed in young male patients with bolus impaction.

Disclosure: Nothing to disclose.

PP0074

COHORT PROFILE UPDATE: THE SWISS EOSINOPHILIC EESOPHAGITIS COHORT STUDY (SEECS)

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Introduction: The Swiss Eosinophilic Esophagitis Cohort Study (SEECS) is a national prospective cohort for adults with eosinophilic esophagitis (EoE) that was established in 2015 with the aim of improving quality of care considering EoE's increased incidence. The SEECS allows for documentation of EoE's natural history, patients' quality of life (QoL) and monitoring of treatment efficacy and safety. It also aims to promote clinical and translational research projects with the availability of standard operating procedures (SOPs) for the submission of scientific protocols. The first cohort update was published in 2017.

Between 2020 and 2022, paper questionnaires were gradually replaced by fully electronic data capture using Research Electronic Data Capture (REDCap) software. We aim to provide a second update of the cohort, eight years after its launch.

Aims & Methods: The SEECS includes adult patients (age ≥18 years) with EoE, gastro-esophageal reflux (GERD) and healthy controls (HC) using strict eligibility criteria in line with the most recent guidelines. Upon inclusion and follow-up (typically once a year), patients and physicians complete questionnaires in REDCap which are available in German, French, and English.

Patients reported outcomes (PROs) and biologic activity are assessed all on the same day using validated instruments (EEsAI PRO for symptoms; EoE-QoL-A for QoL; EREFS for endoscopic activity; modified EoE-HSS for histologic activity). The SEECS biobank includes biosamples (blood samples, esophageal biopsies) from patients with EoE, GERD, and HC.

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Results: As of July 2023, the SEECS includes 778 patients [716 (92%) with EoE, 29 (3.8%) with GERD, and 33 (4.2%) HC; 559/778 (71.9%) are male]. Mean age±SD (years) according to diagnosis is as follows: EoE 41.9±12.9, GERD 53.6±16.4, HC 51.7±17.2. Mean diagnostic delay for the EoE group is 89.6±96.2 months. Concomitant GERD is found in 200 patients (27.9%) of the EoE cohort. Concomitant allergic disorders (asthma, rhinoconjunctivitis, neurodermitis) are present in 500 patients (74.4%) with EoE. At inclusion, 694 (96.9%) of EoE patients have an ongoing treatment [orodispersible budesonide tablet: 290 patients (41.8%); budesonide/fluticasone syrup: 281 patients (40.5%); PPI: 162 patients (23.3%); elimination diets: 103 patients (14.8%); esophageal dilation at last visit: 166 patients (23.9%)]. Esophageal biopsies are available from 525/778 (67.5%) patients including 468 EoE patients (65.4%).

Conclusion: Our findings are in line with the available literature on EoE's natural history. To our knowledge, the SEECS is the only national cohort for EoE adult patients that includes both GERD and healthy controls with their biosamples. Data capture is based on validated instruments and is fully electronic using REDCap. The SEECS offers up-to-date epidemiologic data on EoE and promotes research projects.

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PP0075

ENRICHMENT OF TSLP GENE SIGNATURES IN OESOPHAGEAL TISSUE OF PATIENTS WITH ACTIVE EOSINOPHILIC OESOPHAGITIS: POTENTIAL FOR TREATMENT WITH TEZEPELUMAB

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Introduction: Eosinophilic oesophagitis (EoE) is a rare, chronic, inflammatory disorder with few approved treatments. Thymic stromal lymphopoietin (TSLP) is a pro-inflammatory epithelial cytokine. *TSLP* genetic variants are associated with EoE risk, and oesophageal TSLP mRNA and protein levels are elevated in patients with active EoE. Tezepelumab, a human monoclonal antibody approved for the treatment of severe asthma, specifically binds to and blocks the activity of TSLP.

Aims & Methods: This analysis aimed to determine whether TSLP-mediated gene expression changes are enriched in patients with EoE to understand whether tezepelumab could be a potential treatment for EoE. Gene signature analyses were performed to generate TSLP gene lists from two independent transcriptomic data sets obtained from: 1) bronchial brushings at week 28 from patients with moderate-to-severe, uncontrolled asthma, treated with tezepelumab in the phase 2 CASCADE study (NCT03688074); and 2) human myeloid dendritic cells (mDCs) treated *ex vivo* with recombinant TSLP for 24 hours. Using these TSLP gene lists, gene set variation analyses were performed on publicly available gene expression data from oesophageal biopsies in healthy individuals and patients with chronic oesophagitis, active untreated EoE, inactive treated EoE (responsive to swallowed topical corticosteroids [STCs]) and active treated EoE (unresponsive to STCs).

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Results: The TSLP gene lists comprised 18 upregulated genes and 32 downregulated genes in bronchial brushings from CASCADE, and 183 upregulated and 161 downregulated genes in human mDCs. The TSLP gene scores from bronchial brushings were elevated in patients with active EoE, untreated and unresponsive to STCs, compared with healthy individuals (upregulated genes: $p \leq 0.05$ and $p \leq 0.001$, respectively; downregulated genes: $p > 0.05$ and $p \leq 0.01$, respectively). The TSLP gene scores from human mDCs were elevated in patients with active EoE, untreated and unresponsive to STCs, compared with healthy individuals (upregulated genes: $p \leq 0.05$ and $p \leq 0.01$, respectively; downregulated genes: $p \leq 0.05$ and $p \leq 0.001$, respectively). In both data sets, compared with patients who had inactive treated EoE, TSLP gene scores were elevated in patients with active treated EoE that was unresponsive to STCs (upregulated genes: $p < 0.001$; downregulated genes: $p \leq 0.05$ [bronchial brushings] and $p \leq 0.0001$ [human mDCs]). These data support clinical evaluation of tezepelumab in the treatment of EoE, which is being conducted in the ongoing phase 3 CROSSING study (NCT05583227). CROSSING is a double-blind, placebo-controlled study that will evaluate the efficacy and safety of tezepelumab in approximately 360 patients (12–80 years old) with symptomatic and histologically active EoE. Patients will be randomized (1:1:1) to receive one of two dose options for tezepelumab or placebo subcutaneously every 4 weeks for 52 weeks. The study includes an optional 24-week active treatment extension period in which patients previously receiving placebo will be re-randomized to receive one of the two tezepelumab doses. The co-primary endpoints are to assess the effect of tezepelumab on symptom improvement and histologic response.

Conclusion: Gene lists indicative of elevated TSLP activity were enriched in patients with active EoE (untreated or unresponsive). These data suggest that blocking TSLP with tezepelumab may be a therapeutic option for EoE. The efficacy and safety of tezepelumab in patients with symptomatic and histologically active EoE are being assessed in the ongoing CROSSING study.

Disclosure: J D Sherrill, P G Gavin, L Öberg, T Li, A Péterffy, W Rae, R Tsotsoros, R Trivedi, K Domek-Lopacinska, C Seminario, A Platt and D Muthas are employees of AstraZeneca and may own stock or stock options in AstraZeneca. D Ivanič was an employee of AstraZeneca at the time of the study and may own stock options in AstraZeneca; he is a freelance data scientist at Kickresume. S Caveney is an employee of Amgen and owns stock in Amgen.

PP0076

SEX-RELATED DIFFERENCES IN THE PRESENTATION, PHENOTYPE AND COURSE OF EOSINOPHILIC ESOPHAGITIS IN THE EOE CONNECT REGISTRY: THE SEXEOE STUDY

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Introduction: Eosinophilic esophagitis (EoE) is a chronic inflammatory disorder restricted to the esophagus that predominates in males (male:female ratio being around 3:1). The impact of patient sex on the onset, diagnosis and progression of EoE has been barely evaluated, with some reporting no or minor differences (1) while others discovered relevant changes between men and women regarding symptomatology, endoscopic findings and molecular signature (2,3).

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Aims & Methods: The study aims to assess potential disparities in EoE presentation and phenotype at the moment of diagnosis, as well as to evaluate whether sex determines difference in disease progression and response to therapy. This is a cross-sectional, multicentre study that includes patients diagnosed with EoE from the EoE CONNECT registry. Demographical and clinical data were obtained from patients recruited at 37 hospitals in 4 European countries. Data extraction was conducted on March 21st, 2024.

Quantitative (age, diagnostic delay, dysphagia symptom score, EREFS endoscopic scale, peak of eosinophils per high power field, number of previous endoscopies, and number of endoscopic dilations) and qualitative (age by groups, phenotype, country, presence of fibro-stricturing findings, atopies, treatments and symptoms) variables were compared. Statistical significance was established for p<0.05.

Results: Data from 2,976 patients (76% males) were retrieved, with no difference in female proportions among countries. Mean±SD age at diagnosis was higher for females (34.8±15.6 *vs* 32.7±14.8 years, p=0.007), with no differences in diagnostic delay or number of previous endoscopies before diagnosis. Proportion of women diagnosed with EoE at 60 year or older age was twice compared to men (6.4% *vs* 3.1%, p<0.001). Abdominal (6.6% *vs* 4.4%, p=0.023) and epigastric (10.6% *vs* 8.1%, p=0.047) pains, heartburn (27.9% *vs* 22.0%, p=0.003) and dysphagia (79.7% *vs* 73.3%, p=0.001) were significantly more frequent in females; food bolus impaction predominated in males (59.5% *vs* 50.1%, p<0.001). Symptoms intensity measured with the Dysphagia Symptom Score was higher in female patients (6.9±3.8 *vs* 6.3±4.1, p=0.003).

Endoscopic features of fibrosis (rings and/or strictures) at baseline endoscopy predominated in males (61.0% *vs* 51.4%, p<0.001). Accordingly, overall EREFS score and its fibrotic sub-score (2.6±1.7 *vs* 2.3±1.6 and 1.0±1.0 *vs* 0.8±1.0, respectively, p<0.001), and mixed/stricturing phenotypes were more commonly found in male compared to female patients (25.0% *vs* 19.3%, p=0.002). Peak eosinophil count was higher in females (median 50 *vs* 42 cells/HPF, p=0.005).

No differences were found for first-line treatment choice and effectiveness of anti-inflammatory therapies between men and women. In addition, 126 patients underwent 215 endoscopic dilatation procedures overall, which were slightly more common among male patients (4.5% *vs* 3.4%, p=0.187). In contrast, the median number of dilation procedures was higher in female compared to male EoE patients (2 *vs* 1; p=0.001).

Regarding concomitant atopies, no differences were noted in the prevalence of rhinitis, conjunctivitis and asthma according to sex. However, atopic dermatitis was more common in women (21.7% *vs* 17.8%, p=0.028).

Conclusion: Sexual dimorphism exists in EoE, as observed in other immune-mediated diseases. Despite symptoms severity scored slightly higher in females, EoE in male patients appeared earlier, more commonly with food bolus impaction and with fibro-stricturing endoscopic features at diagnosis.

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Disclosure: Nothing to disclose.

PP0077

RESULTS FROM DOSE ESCALATION IN RESOLVE, AN ONGOING PHASE 1B/2A DOSE-ESCALATION STUDY OF EP-104GI (EXTENDED-RELEASE FLUTICASONE PROPIONATE INTRA-ESOPHAGEAL INJECTION) FOR EOSINOPHILIC ESOPHAGITIS

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Introduction: Eosinophilic esophagitis (EoE) is a chronic, immune-mediated disease characterized by inflammation, influx of eosinophils and esophageal remodeling. Therapy for EoE includes swallowed/topical corticosteroids, dupilumab, proton pump inhibitors, and dietary elimination; but these are not always effective.

EP-104GI is an extended-release fluticasone propionate (FP) intra-esophageal injection being developed as a novel treatment for EoE. It consists of polymer-coated FP crystals that release drug locally via diffusion at a pre-defined rate to reduce peak concentrations and prolong the therapeutic window. Preclinical work in pigs has demonstrated that EP-104GI remains at injection sites for at least 6 weeks (duration of the study), with FP spreading 2-4 cm into the surrounding tissue.

Aims & Methods: RESOLVE (NCT05608681) is a Phase 1b, multicenter, open-label, 3+3 design dose-escalation study to evaluate the safety, tolerability, pharmacokinetics, and efficacy of EP-104GI in patients with EoE.

The starting dose of EP-104GI consisted of 1 mg injected at each of 4 sites in 4 quadrants over 2 circumferential areas in the esophagus (4 mg total dose). Dose escalations increase the dose per site and/or number of sites; the second and third cohorts received 1 mg and 2.5 mg respectively at each of 8 sites; the fourth cohort received 2.5 mg at 12 sites (8, 20, and 30 mg total doses).

After a single administration of EP-104GI in these cohorts, participants are assessed for up to 24 weeks. Additional cohorts are planned to evaluate greater dose per injection site and/or increased number of injections with a longer duration of follow-up.

Assessments include Peak Eosinophil Count (PEC) from esophageal biopsies at 16 sites, patient-reported symptom outcomes including the Straumann Dysphagia Index (SDI) and Likert scales for dysphagia and odynophagia, and plasma concentrations of FP. Adverse events (AEs) and laboratory tests including serum glucose and cortisol are assessed.

Results: Cohort 1-3 safety observations include mild-moderate AEs; the majority not related to EP-104GI. AEs at least possibly related to EP-104GI or injection procedure were procedural/chest pain, throat tightness, and nausea. Glucose levels post-dose have remained stable. Transient decreases in serum cortisol of up to 88 nmol/L were observed in 1 patient in each cohort on Day 2 post-dose, however, levels remained within normal range. There have been no symptoms of adrenal insufficiency.

Response data from Cohorts 1 and 2 (Table 1) demonstrate a decrease from baseline in symptom scores at Weeks 12 and 24. Reduction in mean PEC has been observed in all patients in Cohort 2, with a trend indicating that increasing the total dose of EP-104GI results in a more consistent reduction in PEC at the 16 biopsy sites assessed.

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	Cohort 1 (1 mg × 4 sites, total 4 mg)	Cohort 2 (1 mg × 8 sites, total 8 mg)
Mean change from baseline (CFB) in Straumann Dysphagia Index (0-9) at Weeks 12 and 24	-4.0, -4.0	-2.7, -3.3
Mean CFB in Dysphagia (0-10) at Weeks 12 and 24	-4.0, -5.0	-4.3, -4.0
Mean CFB in Odynophagia (0-10) at Weeks 12 and 24	-2.0, -3.0	-5.7, -4.3
Mean CFB in PEC at Weeks 4 and 12 (cells/hpf from 16 sites)	No reduction observed	-26%, -44%
Mean plasma concentration of FP at 2 hours post-dose (pg/mL)	4.3	11.2

Table 1: Patient-reported symptom outcomes, peak eosinophil count (PEC), and plasma fluticasone propionate (FP) after a single dose of EP-104GI.

Conclusion: A novel diffusion-based localized delivery of FP via EP-104GI injection into the esophagus is feasible and safe in patients with EoE in this initial escalating dose cohort study. Initial efficacy data show improvement in symptom outcomes and histological findings which may increase with dose. These findings support the potential for extended intervals of at least 6 months between injections. Recruitment is ongoing.

Disclosure: JH, AM, MMK, CD, AL, VP and MW: employees of Eupraxia Pharmaceuticals

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HHK: advisory board and speaker's bureau for Sanofi.

AJB: received research funding from Nutricia, Thelial, Sanofi/Regeneron, SST, and Dr. Falk Pharma and received speaker and/or consulting fees from Laborie, Medtronic, BMS, Dr. Falk Pharma, Calypso Biotech, Eupraxia, Aqilion, Alimentiv, Sanofi/Regeneron, Reckitt and AstraZeneca

NG: nothing to declare.

PP0078

PERSISTENT DYSPHAGIA IN STRICTURE-FREE EOSINOPHILIC ESOPHAGITIS IN HISTOLOGICAL REMISSION: EFFECTIVENESS OF EMPIRICAL DILATION

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Introduction:

It is known that not all patients with eosinophilic esophagitis (EoE) in histological remission also achieve clinical remission. Several reasons have been postulated among which decreased esophageal compliance without apparent stricture at endoscopy is particularly appealing, but no data are so far available on effectiveness of empirical dilation.

Aims & Methods: Aim of our study was to investigate this topic. A retrospective analysis with prospectively collected data has been conducted on all EoE patients without stenotic features at endoscopy (i.e. no R3/S1 according to EREFS) in whom histological remission with drug and/or diet has been induced in a single referral Center from July 2018 to December 2023. All patients who had not achieved concurrent remission of dysphagia and/or bolus impaction were empirically dilated. At least 6 months of follow-up after remission was needed for inclusion. Demographic, clinical and endoscopic data were compared between patients who needed dilation and those who did not.

Results: A total of 33 patients aged 40; 30-51 yrs (median; IQR range), 23 men, were included. PPI was the main treatment (76% of pts). Nineteen of them (group 1) achieved clinico-histological whereas 14 (group 2) achieved histological remission only and underwent empirical dilation to 15-17 mm with no complications after which clinical remission was achieved in 13 of them. A second dilation was needed in 2 patients at 25 and 31 months respectively, in order to maintain remission. EREFS score at histological remission, but not at diagnosis, was higher in group 2 vs 1 (p=0.02, see table) and was accompanied by a higher % of patients with rings (R1 or R2), 78 vs 37 (p=0.04). A lower % of group 2 patients was treated with topical steroids, 7 vs 26 (p=NS).

	Group 1 (n=19)	Group 2 (n=14)
Age, yrs	37;28-46	44;22-55
Sex, male n (%)	12(63)	11(79)
Symptom duration, months (*)	46;23-102	36;19-77
I-SEE at diagnosis (**)	4;3-5	5;2-6
EREFS at diagnosis	2;1-2	2;0-3
EREFS at histological remission	1;0-2	2;1-2 #
Duration of follow-up, months	12;7-54	20;10-43

(*) up to histological remission, (**) I-SEE= Index of severity of EoE (1), # p=0.02 vs group 1

Conclusion: In our cohort of stricture free EoE patients achieving histological remission, over 40% still had residual obstructive symptoms which were relieved by empirical esophageal dilation in most, suggesting high relevance of decreased esophageal compliance. Prospective multicenter data with use of FLIP are warranted in order to assess if topical steroids are more effective than PPIs in reversing remodeling and avoiding dilation.

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Disclosure: Nothing to disclose.

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PP0079

PREDICTIVE HISTOLOGICAL FEATURES OF PROTON PUMP INHIBITOR RESPONSE IN UNTREATED EOSINOPHILIC ESOPHAGITIS PATIENTS

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Introduction: Eosinophilic esophagitis (EoE) is an immunoallergic disease of increasing prevalence whose clinical manifestations are due to dysmotility resulting from eosinophil-mediated inflammation. The goal of treatment is the clinical, endoscopic and histological response, with discordant data on predictive criteria.Historically, histological response has been assessed with the intraepithelial eosinophil count until the validation of the Histological Score System (HSS), which includes microabscesses and eosinophilic degranulation, lamina propria fibrosis, intraepithelial lymphocytosis and spongiosis.

Aims & Methods: Our aim is to identify histological features in baseline esophageal biopsies at diagnosis that are predictive of response to proton pump inhibitors (PPI), first line treatment.

A retrospective cohort study, approved by the Ethics Committee of the Hospital Clínico San Carlos, was designed that included incident patients with diagnostic criteria for EoE evaluated between January and December 2022. They underwent a baseline gastroscopy and another after a cycle of PPIs at high doses. After 8-12 weeks induction, two cohorts were established: responders to PPI (PPI-REE) and non-responders (non PPI-RRE). Clinical and endoscopic data were collected from reports, and biopsies were reviewed by expert pathologists after anonymisation of the patient cohort.

	Proximal esophagus			Distal esophagus		
Histologic variable	PPI-REE	Non PPI-REE	p value	PPI-REE	Non PPI-REE	p value
Predominance of eosinophils on epithelial surface	3 (6.4%)	2 (11.1%)	0.611	0	0	n. a.
Basal cell hyperplasia	36 (70.6%)	11 (61.1%)	0.458	50 (74.6%)	19 (73.1%)	0.878
Eosinophilic microabscesses	9 (17.3%)	7 (31.8%)	0.218	18 (27.3%)	12 (37.5%)	0.303
Eosinophil activation (degranulation)	16 (31.4%)	18 (81.8%)	<0.001	38 (56.7%)	25 (78.1%)	0.038
Submucosal lamina propria fibrosis	16 (76.2%)	9 (100%)	0.286	33 (97.1%)	11 (100%)	0.015
Intraepithelial lymphocytosis	10 (19.85%)	1 (4.8%)	0.158	13 (20%)	5 (15.6%)	0.602
Spongiosis	9 (17.6%)	17 (77.3%)	<0.001	25 (37.3%)	29 (93.5%)	<0.001
Papillary elongation	7 (53.8%)	1 (14.3%)	0.158	17 (77.3%)	5 (41.7%)	0.062

Results were analysed using IBM SPSS Statistics v.26, with $\chi 2$ or Fisher's exact test for qualitative variables and Student's t-test or non-parametric Mann-Whitney U test for quantitative variables. A binary logistic regression model was used to analyse the association between cohorts and each individual variable, with a significant p-value of 0.005 accepted.

Results: 99 patients with EoE were included, 32 in the non-PPI-REE cohort (32.3%) and 67 in the PPI-REE cohort (67.7%), with no differences in baseline characteristics or endoscopic variables. Histological analysis (Table 1) shows that spongiosis, regardless of location, eosinophilic degranulation in the proximal esophagus and fibrosis of the lamina propria in the distal esophagus are predictive factors for non-response to PPI. However, in multivariate analysis, only the presence of spongiosis, regardless of location, was significant (p 0.021 proximal [95% CI 0.030-0.756] and 0.016 distal [95% CI 0.008-0.611]).

Conclusion: EoE is a growing entity. Being able to predict the response to PPIs at diagnosis would allow clinical improvement and early reduction of inflammation, reducing progression to the fibrostenotic phenotype.

According to our results, we consider it important to universalise the HSS, highlighting the importance of identifying spongiosis, easily identifiable histological changes with low interobserver variability and without additional cost, which could predict the absence of response to PPIs, with patients able to benefit from swallowed corticosteroids or exclusion diets from diagnosis. However, studies with greater power should confirm these data.

Disclosure: Nothing to disclose.

PP0080

PATIENT PREFERENCES REGARDING MEDICAL TREATMENT OPTIONS IN EOSINOPHILIC ESOPHAGITIS

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Introduction: Until a few years ago, there existed only off-label medical treatment options for eosinophilic esophagitis (EoE) such as proton pump inhibitors (PPI) or swallowed topical corticosteroids (STC) with various formulations and ingredients. Since the approval of a budesonide orodispersible tablet (BOT), the first biological agent (Dupilumab) has been approved by the EMA and FDA, and another budesonide oral suspension has been received approval by the FDA recently. Currently, many phase 3 trials with different drugs are running, resulting in various treatment options in the next years. However, data about patients' perspectives are lacking. We therefore aimed to assess patients' awareness, beliefs and restrictions to medical treatment options in EoE

Aims & Methods: We developed a web-based survey using SurveyMonkey and sent a questionnaire comprised of 47 questions about patients' preferences regarding medical treatment options to patients with established EoE in Switzerland, Austria and Germany.

Results: A total of 56 patients (64% male, 66% with other atopic diseases) from Switzerland (64%), Austria (34%) and Germany (2%) answered the questionnaire. A swallowed topical therapy was ranked first (71% of patients) in terms of patient's preference followed by a subcutaneous therapy every 4 weeks (21.4%) and by an oral therapy with systemic effect (7%). Although one third do not have reservations against long-term medical therapy, nearly half of patients (42%) are skeptical about long-term treatment and 65% are afraid of adverse events. However, a third of patients are not aware that STC are locally effective medications with nearly no systemic effect. Furthermore, 48% of patients believe that physicians underemphasize adverse events of medication. Nearly all patients understand the importance of not only treating symptoms but also inflammation (85%).

Conclusion: Patients with EoE prefer topical agents over other available routes of administration. Patient education should be carried out to address reservations about long-term treatment.

Disclosure: Falk Pharma, Sanofi (both consultency meetings).

PP0081

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PP0082

RISK FACTORS FOR ORAL AND ESOPHAGEAL CANDIDIASIS DURING BUDESONIDE TREATMENT IN EOSINOPHILIC ESOPHAGITIS PATIENTS

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Introduction: Eosinophilic esophagitis (EoE) is a chronic and progressive disease characterized by eosinophil-predominant inflammation of the esophagus. First-line therapy is represented by topical corticosteroids with anti-inflammatory effects, in particular budesonide orodispersible tablets (BOT). Their efficacy and safety are nowadays well established, but very low number of patients develop oral and esophageal candidiasis.

Aims & Methods: The aim of the study is to evaluate risk factors for the development of oral and esophageal candidiasis during budesonide therapy in eosinophilic esophagitis patients. To date, these data have not still been studied. A retrospective study was conducted to include all EoE patients in therapy with BOT referred to Fondazione Policlinico Universitario A. Gemelli IRCCS between March 2022 and November 2023. Prevalence, localization, and time of presentation of candidiasis were evaluated. Histological findings, in particular eosinophilic count in gastroscopy before BOT therapy were analyzed. Concomitant treatment with proton-pump inhibitors (PPI) and its dosage were reviewed. In addition, each case of discontinuation of BOT for disabling candidiasis was studied.

Results: A total of 52 EoE patients in BOT therapy, 44 males and 8 females, were included in the study. 7 patients (13%) developed oral and/or esophageal candidiasis, in particular 6 patients (85%) during induction phase (1 mg bid) and 1 patient (15%) during early maintenance phase (0.5 mg bid). All these 7 patients presented peak eosinophil count >35/HPF before corticosteroid treatment, with mean count of 46 eos/HPF. 4 patients (57%) of this group were in concomitant PPI therapy at 40 mg per day or more dosage. 3 patients (42%) presented recurrent and severe oral candidiasis inducing discontinuation of treatment.

Conclusion: Oral and esophageal candidiasis is a well-known side effect of budesonide orodispersible tablets in eosinophilic esophagitis. Our results suggest that consistent eosinophilic count (> 35/HPF) and concomitant PPI therapy could represent risk factors in developing mycosis in EoE patients. According to literature, major esophageal inflammation, PPI-induced elimination of the gastric acid barrier and corticosteroid immunosuppression are the main mechanisms leading to oral and esophageal candida colonization. Concomitant use of PPI during BOT treatment in eosinophilic esophagitis should be carefully evaluated, especially in patients with elevated eosinophilic count.

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Disclosure: Nothing to disclose.

PP0083

COMPARATIVE PHARMACOKINETICS, SAFETY, AND TOLERABILITY OF A SINGLE DOSE OF CENDAKIMAB IN HEALTHY CHINESE AND US PARTICIPANTS

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Introduction: Eosinophilic esophagitis (EoE) is a chronic and progressive Type 2 inflammatory disease with a significant unmet need for effective and safe long-term targeted therapy. Cendakimab (CEN) is a recombinant, humanized, high-affinity, neutralizing monoclonal antibody that binds to interleukin 13 (IL-13) thereby blocking interaction with its receptors, IL-13Rα1 and IL-13Rα2. CEN is currently under clinical development for the treatment of EoE.

Here, we present the results of 2 phase 1, randomized, single-center, single-dose trials assessing the pharmacokinetics (PK), safety, and tolerability of CEN in healthy Chinese (NCT05638282) and US (NCT05337345) populations.

Aims & Methods: In the China study (NCT05638282), healthy participants (age, 18–55 years; body mass index [BMI], 18.0–28.0 kg/m^2; weight, ≥50.0 kg) were sequentially assigned to 2 groups. Participants in each group were randomized 5:1 to receive: group 1, single-dose CEN 360 mg subcutaneous (SC) via 1 injection (n=10) or placebo (n=2), and group 2, single-dose CEN 720 mg SC via 2 injections (n=10) or placebo (n=2). All doses were administered in the abdomen using an autoinjector (AI). In 1 of 4 groups of the US study (NCT05337345), healthy participants (age, 18–55 years; BMI, 18.0–30.0 kg/m^2; weight, ≥40.0 kg) received single-dose CEN 360 mg SC via 1 injection in the abdomen using an AI (n=32). Data from groups that received CEN via AI are presented here. The PK parameters assessed in both studies were maximum observed serum concentration (C_{max}), time of C_{max} (T_{max}), area under the concentration-time curve from time 0 to time of the last quantifiable concentration (AUC_{0-T}), AUC from time 0 to infinite time (AUC_{∞}), terminal elimination half-life ($T_{1/2}$), apparent total body clearance (CL/F), and apparent volume of distribution of terminal phase (Vz/F). Safety and tolerability were also evaluated.

Results: Demographic characteristics were generally similar in the China and US studies, eg. median BMI was 23 and 26 kg/m^2, respectively. In the China study, following SC doses of 360 and 720 mg, the absorption of CEN was slow, with a median T_{max} of 96 and 143 hours; mean $T_{1/2}$ was 565 and 535 hours, and Vz/F was 11.8 and 10.7 liters, respectively (*Table*).

The geometric mean C_{max}, AUC_{∞}, and AUC_{0-T} increased in a dose-proportional manner between the 360 and 720 mg CEN doses. Antidrug antibodies (ADAs) were detected in 50% of participants in each of the 360 and 720 mg CEN groups, with a low titer in most of them.

Results for the 360 mg AI group from the China study were consistent with those of the 360 mg AI group in the US study. ADAs were detected in 16% of the participants in the US study. In both studies, the ADA response did not impact the safety or PK of CEN, and CEN (360 mg or 720 mg) was well tolerated with no deaths, serious adverse events (AEs), or discontinuations due to an AE.

Conclusion: Single-dose 360 mg or 720 mg SC CEN was safe and well tolerated in all healthy participants. CEN PK was generally dose-proportional for C_{max} and AUC from 360 mg to 720 mg. Overall, the PK, tolerability, and safety profiles of a single dose of 360 mg SC CEN in the China study were consistent with those observed in the US study, supporting the use of similar doses across both populations.

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Study	Cendakimab dose[b]	AUC_{∞}, hours*µg/mL	AUC_{0-T}, hours*µg/mL	C_{max}, µg/mL	Median T_{max}, hours (range)	Mean $T_{1/2}$, hours (SD)	CL/F, L/h	Vz/F, L
China study	360 mg (n=10)	24,300 (22)	23,000 (23)	29.4 (38)	96 (72–312)	565 (116)	0.015 (22)	11.8 (30)
	720 mg (n=10)	51,200 (26)	49,300 (25)	66.5 (25)	143 (48–311)	535 (108)	0.014 (26)	10.7 (23)
US study	360 mg (n=32)	25,187 (35)	23,908 (34)	29.8 (34)	144 (8–314)	506 (114)	0.014 (36)	10.2 (33)

[a]Values are geometric mean (% geometric coefficient of variation), except where otherwise noted.
[b]Administration in the abdomen using an AI.
SD, standard deviation.

Table. Summary of serum pharmacokinetic parameters[a]

Disclosure: YC and SP have nothing to disclose; CX, SB, BW, YQ, KY, BM, and PZ are employees of Bristol Myers Squibb; LL, JR, WK, BY, CDO, MT, JM, PZ are employees and stockholders of Bristol Myers Squibb.

PP0084

ANALYSIS OF CLINICAL RESPONSE IN ADULTS WITH EOSINOPHILIC ESOPHAGITIS: PROSPECTIVE COHORT AT A UNIVERSITY HOSPITAL IN CHILE

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Introduction: Eosinophilic esophagitis (EoE) is a chronic, immune-mediated inflammatory disease that affects the esophagus. In Chile and Latin America, its epidemiology is unknown.

Aims & Methods: The aim of this study is to describe the clinical, endoscopic, and histological characteristics of adult patients diagnosed with EoE. Prospective cohort of patients seen at the Digestive Diseases Center, diagnosed with eosinophilic esophagitis and undergoing one follow-up endoscopy. Endoscopic classification was performed using the Eosinophilic Esophagitis Endoscopic Reference Score (EEERS), and histological response was defined as fewer than 15 eosinophils (Eos) per high-power field. Descriptive statistics (median and interquartile range) were used for sample characterization, considering the non-normal distribution of the sample evaluated by the Kolmogorov-Smirnov test. The Chi-square test was used for the analysis of nominal (categorical) or ordinal data. The Wilcoxon test was used for the comparison of histological and endoscopic results, while the Mann-Whitney U test was used for the analysis of medical treatment. Statistical software SPSS® v.29 and GraphPad Prism 8.3.0 were used. Values of $p < 0.05$ were considered statistically significant.

Results: Eighty-five patients aged between 17 and 77 years old (Median=36 years) were included, with 75.3% being male. The median age at symptom onset was 31 years and at diagnosis was 34 years. The most frequent reason for consultation was dysphagia (54%), and the most common concomitant immune-allergic diseases were allergic rhinitis (38%) and asthma (29%). Women had more frequent autoimmune comorbidity, peripheral eosinophilia, and food allergies but had a lower eosinophil count and endoscopic score. Regarding treatment, 55% were on PPI therapy and 40% on mixed treatment. In terms of clinical outcomes, patients at admission had a median score of 4 for EREFS and 30 eosinophils/field on biopsy eosinophil count. At follow-up, a significant decrease was observed in both endoscopic and histological (biopsy) parameters ($p < 0.001$), with medians of 0 for each indicator, respectively. Moreover, regarding the influence of treatment type (PPI or mixed) on endoscopic or histological response, it was observed that both treatments are equally effective: patients on PPI therapy showed a 72% histological improvement and an 89% endoscopic response, while patients on mixed treatment showed a 77% and 85% improvement, respectively (p=ns).

Conclusion: This is the first patient cohort in Latin America to our knowledge with EoE. Characteristics are similar to those described in international groups. Women exhibited more frecuent autoimmune comorbidity, peripheral eosinophilia, and food allergies but had a lower eosinophil count and endoscopic score. We did not observe differences among the various therapeutic regimens.

Disclosure: None.

PP0085

TREATMENT PATTERNS AND DISEASE PROGRESSION AMONG PATIENTS WITH EOSINOPHILIC OESOPHAGITIS IN EUROPE: A REAL-WORLD STUDY

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Introduction: Eosinophilic oesophagitis (EoE) is a chronic, progressive type 2 inflammatory disease. Proton-pump inhibitors (PPIs), swallowed topical corticosteroids (STCs), and elimination diets have been used as standard treatments for EoE patients. However, the real-world evidence (RWE) highlighting disease burden in Europe are limited.

Aims & Methods: To assess disease burden and treatment patterns among adolescent and adult patients with EoE in real-world practice.

Medical records were retrospectively reviewed for patients (aged ≥12 years) who were newly diagnosed with EoE between 1 January 2009 and 31 December 2019 (index) in Germany, France, Spain, and the United Kingdom. Patients with a peak eosinophil (eos) count (PEC) of ≥15 eos/hpf within 90 days of the index date and ≥1 follow-up endoscopy within 24 months of the index date were included. Demographics, clinical features, treatment patterns, and health outcomes were assessed from the 12-month pre-index period to the last medical record entry or death. Disease progression was assessed via physician-defined (based on symptoms, endoscopy findings, histology, need for dilation, and food impaction) and proxy-reported outcomes (dilation performed, strictures, dysphagia, evidence of food impaction, treatment initiation in treatment-naïve patients and switching/augmentation).

Results: Overall, 385 patients were included, with 70.1%–78.1% being male; the mean age was 27.8 to 33.9 years. At EoE diagnosis, the PEC was 35.3–46.1 eos/hpf, the endoscopic appearance of the oesophagus was abnormal in 73.7%–85.6% of patients, and dysphagia was present in 59.8%–82.8% of patients. Most of the patients across all 4 countries were treated with PPIs (61.5%–91.8%), STCs (56.7%–82.3%) and elimination diets (41.4%–55.2%) after EoE diagnosis. Of the patients with available data, abnormal histology (≥15 eos/hpf) was reported in 44.8%–72.0% of patients during the first 5 years after EoE diagnosis. The abnormal endoscopic appearance of the oesophagus was observed in 34.6%–83.3% of patients after approximately 2 years of EoE diagnosis and dysphagia was present in 10.2%–25.0% of patients after 4 years of EoE diagnosis across all 4 countries. While physician-defined disease progression occurred in 6.1%–16.1% of patients, 19.4%–35.1% of patients experienced proxy-reported outcomes of potential disease progression (including oesophageal dilation performed in 10.8%–21.7% of patients) after EoE diagnosis. An

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overview of patient and clinical characteristics at EoE diagnosis and treatment patterns and disease progression (physician-defined and proxy-reported) after EoE diagnosis across Germany, France, Spain, and the United Kingdom is provided in the *Table*.

Patient and clinical characteristics at eosinophilic oesophagitis diagnosis	Germany (N=96)	France (N=99)	Spain (N=97)	United Kingdom (N=93)
Male, n (%) / Age, years, mean (standard deviation)	78.1 / 32.0 (14.9)	72.7 / 27.8 (13.4)	70.1 / 32.2 (14.1)	72.0 / 33.9 (13.5)
Peak eosinophil count/ high-power field, mean (standard deviation)	35.3 (29.1)	39.5 (46.3)	46.1 (41.7)	43.3 (56.7)
Treatment patterns and disease progression (physician-defined and proxy-reported) after eosinophilic oesophagitis diagnosis				
Proton-pump inhibitors, n (%)	59 (61.5)	87 (87.9)	89 (91.8)	62 (66.7)
Swallowed topical corticosteroids (e.g., fluticasone budesonide), n (%) / Systemic (oral) corticosteroids, n (%)	79 (82.3) / 24 (25.0)	76 (76.8) / 7 (7.1)	55 (56.7) / 8 (8.3)	55 (59.1) / 4 (4.3)
Dietary modifications or eliminations, n (%)	53 (55.2)	41 (41.4)	46 (47.4)	41 (44.1)
Physician-defined disease progression, n (%)	12 (12.5)	6 (6.1)	7 (7.2)	15 (16.1)
Proxy-reported disease progression, n (%) / dilation performed, n (%)	26 (27.1) / 20 (20.8)	31 (31.3) / 16 (16.2)	34 (35.1) / 21 (21.7)	18 (19.4) / 10 (10.8)
Overall disease progression (Physician-defined OR Proxy-reported), n (%)	33 (34.4)	34 (34.3)	36 (37.1)	30 (32.3)

Table.

Conclusion: Adolescent and adult patients with EoE chronically experienced substantial disease burden. A subset demonstrated disease progression, perhaps underrecognised despite treatment with PPIs, STCs, and elimination diets. Hence, successful EoE management may include proactive disease monitoring, treating underlying pathophysiology and prescribing maintenance therapies to achieve long-term clinical remission.

References:

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Bram P. Raphael, Amr Radwan, Ryan B. Thomas: Employees of Regeneron Pharmaceuticals, Inc.; may hold stocks and/or stock options in the company.

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PP0086

PATIENTS WITH SEVERE ATOPIC DERMATITIS CANDIDATES TO BIOLOGICAL THERAPY SHOW AN INCREASED PREVALENCE OF DYSPHAGIA: IMPLICATIONS FOR DIAGNOSIS AND MANAGEMENT OF COMORBID ATOPIC DISORDERS

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Introduction: Eosinophilic esophagitis (EoE) is a chronic Th-2 mediated inflammatory disease, closely associated with comorbid allergic disorders, including atopic dermatitis (AD). Studies show that EoE is associated with atopic disease but the prevalence of EoE symptoms in patients with atopic conditions is unknown. To date, a single retrospective study involving 80 atopic patients has shown that the prevalence of dysphagia in atopic patients without a known history of EoE is higher than the control population, while it failed to find a significant association between dysphagia and atopic eczema [1].

Therefore, in our study we evaluated the prevalence of dysphagia in a population of patients severe with atopic dermatitis, candidates to biological therapy, in order to consider the possibility of screening these patients for comorbid EoE.

Aims & Methods: Consecutive patients with moderate to severe AD referred to our tertiary care center from January to October 2023 were prospectively screened. Based on AD severity and phenotypic manifestation and comorbidities, patients were referred to systemic therapy with cyclosporine (CsA), biological drugs (dupilumab, tralokinumab) or JAK inhibitors (upadacitinib, abrocitinib, baricitinib). The presence of dysphagia was evaluated by using the self-administered Eosinophilic Esophagitis Score Activity Index (EEsAI). This score uses seven PRO (patient reported outcome) items that are used to assess characteristics of dysphagia, behavioral adaptations to living with dysphagia and pain while swallowing. The presence of other atopic disorders and esophageal symptoms was also recorded. Questionnaires were collected and analyzed at the gastroenterology department, while the dermatology consultant was blinded from the questionnaire results. Data are presented as mean±SD.

Results: 147 patients (85 M, 42.7±23) successfully completed the form. The prevalence of dysphagia in the population was low (7 patients, 4.7%). The prevalence of other atopic conditions was similar in dysphagia and non-dysphagia patients (42 vs 40%). Patients reporting dysphagia were more likely to have heartburn (28.5 vs 7.1%, p=0.043), while other esophageal symptoms did not differ. Based on clinical evaluation by a single experienced dermatology consultant (clinical phenotype of AD, AD severity and comorbidities), patients were assigned to either receive CsA (8 pts, 4 M, 41.7±18y), dupilumab (42 pts, 27 M, 39.6±21y), tralokinumab (21 pts, 11 M, 46.2±12.5y), or Jak inhibitors (76 pts, 43 M, 43±23y). The prevalence of dysphagia was significantly higher in the Dupilumab group as compared to other therapies (19 vs 1.9%, p=0.01), while other esophageal symptoms were comparable between the two groups.

Conclusion: We observed that the overall prevalence of dysphagia is relatively low in severe AD patients. However, we found a higher likelihood of complaining of dysphagia in the subset of patients who were candidate to treatment with Dupilumab. Due to the reported augmented risk for atopic patients to develop EoE, screening AD patients complaining of dysphagia before initiating biologic treatment is pivotal to prevent EoE under-diagnosis and under-treatment. Better characterizing the relationship between AD and EoE and a multidisciplinary approach to atopic patients will facilitate the optimal management to both conditions.

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Disclosure: Nothing to disclose.

PP0087

"THE PATH2 TASK FORCE": REAL-LIFE EXPERIENCE FROM A MULTIDISCIPLINARY TEAM APPROACH FOR EOSINOPHILIC ESOPHAGITIS

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Introduction: Eosinophilic esophagitis (EoE) is a Th2-related disease characterized by dysphagia and food impaction. EoE has been associated with several other atopic disorders, supporting the knowledge that it may represent a late manifestation of the so-called "atopic march". However; in everyday clinical practice, allergic comorbidities are rarely objectivized by means of specialist consultations and often rely on patients' self-reporting. This bears the risk of not fully recognizing the burden of Th2-related pathology, since EoE patients may outgrow some of the allergic diseases presenting during childhood; or, conversely, they could underestimate the presence and severity of Th2-diseases. To this aim, a multidisciplinary task-force involving multiple specialists was established at our university hospital, in order to systematically investigate newly referred EoE patients.
Aims & Methods: This is a cross-sectional observational study, involving consecutive EoE patients that were referred to our tertiary center from January 2022. We prospectively collected data about patients' clinical history, specifically recording the presence of Th2-related comorbidities. Each patient underwent an esophagogastroduodenoscopy with differential esophageal biopsies. After reaching a conclusive diagnosis of EoE, patients were referred to dermatology, ear, nose and throat specialist (ENT) for rhinofibroscopy, immunologist (performing prick test, as necessary) and respiratory consultant for spirometry.
Results: 40 EoE patients were included (34M, age 31±12 years), with a mean age of onset of 23±16 years. A clinical history of Th2-comorbidities was reported by 90% of the overall population with rhinitis being the most prevalent comorbidity (77.5%), while no patient reported Chronic Rhinosinusitis with Nasal Polyps (CRwNP). Asthma, food allergy and atopic dermatitis (AD) were reported in 15 (37.5%), 14 (35%) and 9 (22.5%) patients, respectively. Following the consultations, the presence of asthma, food allergy and AD did not significantly differ (37.5 vs 30%, 35 vs 45% and 22.5 vs 20%), since only 3 patients reporting a history of asthma did not confirm the diagnosis at spirometry and one patient with a history of AD had dermographic urticaria. On the contrary, following ENT consultation, 6 patients with rhinitis received the diagnosis of CRwNP (0 vs 15%, p=0.01). Overall, nearly 60% of patients had two comorbid Th2 disorders besides EoE and 25% of the whole population of EoE patients had 3 or more.
Conclusion: Our results clearly support the need for a multidisciplinary team approach to investigate the burden of Th2-related pathology in EoE patients. In our hands, the prevalence of Th2-related comorbidities was as high as 97.5% of EoE patients and up to 63% of our population had 2 or more allergic disorders. Furthermore, CRwNP is frequently under-reported and properly diagnosed only following rhinofibroscopy. This is further from being irrelevant when managing EoE patients. Indeed, it may guide the most appropriate choice of therapy by targeting multiple atopic manifestations at the same time, since many Th2-related diseases share common pathophysiological mechanisms and treatment.
Disclosure: Nothing to disclose.

PP0088

RETROSPECTIVE ANALYSIS OF DIAGNOSIS OF EOSINOPHILIC OESOPHAGITIS IN A TERTIARY CENTRE

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Introduction: Eosinophilic Oesophagitis (EOE) has a mean estimate of 63 new cases per 100,000 in the Western World and is in increasing prevalence[1]. It must be considered as a differential diagnosis amongst patients presented with dysphagia and food bolus obstruction. British Society of Gastroenterology (BSG) guideline recommends that patients undergoing endoscopy should have oesophageal biopsies taken if they have endoscopic signs associated with EOE or symptoms of dysphagia or food bolus obstruction with normal looking oesophagus. At least six biopsies should be taken from at least 2 different anatomical sites within the oesophagus for diagnosis of EOE[2]. (GDG recommendation).
Aims & Methods: A single-centre, retrospective analysis of patients presenting for an oesophagogastroduodenoscopy (OGD) with an indication of dysphagia or food bolus obstruction and coded as oesophagitis over the past 5 years was performed. The study was undertaken at a tertiary NHS Hospital Trust in West Midlands which provides hospital services for approximately 900,000 people[3].
Data was then collected for demographics, sedation, macroscopic findings, quality and location of biopsies as well as histology. This is measured against BSG recommendations.
Results: In total, 31 patients were included. 68% of patients were male and the mean age was 53 (range 14-90) 35% (n=11) had pharyngeal anaesthesia only, 16% (n=5) had midazolam and pharyngeal anaesthesia, 35% (n=11) had midazolam and fentanyl and 6% (n=2) had general anaesthesia (GA). Both patients who had GA were 14 and 15 years old. In terms of macroscopic findings, 61 % (n=19) had normal findings, 19% (n=6) had oesophagitis, 10 % (n=3) had EOE changes and 3% (n=1) had benign stricture with the same number for achalasia and tertiary contraction. 81% (n=25) had biopsies taken from at least 2 different anatomical sites and 32% (n=10) had more than 6 biopsies taken in total. There were 29 % (n=9) patients with a histological diagnosis of EOE, 29% was normal, 29% (n=9) had inflammation, 6% (n=2) had gastric metaplasia, 3% (n=1) had basal hyperplasia with 7-8 eosinophils per high-power field (HPF) and 3% (n=1) had basal hyperplasia with lymphocytic infiltration. In all patients with EOE, the histopathological report explicitly documented the presence of >15 eosinophils per HPF.
Conclusion: Our data suggest that our practice for diagnosis of EOE falls short of national recommendation, both in the number of anatomical sites and the number of biopsies taken. In terms of histopathology, we have adhered to the recommended diagnostic criteria of >15 eosinophils per HPF. Further work must be done to improve awareness and adherence to the guidelines for the diagnosis of this treatable condition.
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Disclosure: Nothing to disclose.

PP0089

ESOPHAGEAL BIOPSY SAMPLING AMONG PATIENT WITH DYSPHAGIA. TO BITE OR NOT TO BITE?

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Introduction: Dysphagia is a very common symptom and entities such as Eosinophilc Esophagitis (EoE) or an esophageal motility disorder can be its basis. A controversy among guidelines when approaching a patient with dysphagia exists, as it is mandated to obtain esophageal sampling when EoE is sought but is not a prerequisite for motility disorders diagnosis. Therefore, the necessity of esophageal sampling is based on endoscopists' assessment.

Aims & Methods: Aim of our study was to search for factors driving clinicians to obtain esophageal biopsies among patients with dysphagia. Secondary aim was to search for the incidence of EoE and factors correlated with EoE diagnosis among patients with dysphagia.

We included consecutive patients presented with dysphagia during a 6-year period (1/2018-12/2023). We excluded patients with a history of; esophageal stenosis, esophageal motility disorders, metastatic disease, EoE, esophageal/gastric cancer, carcinoma that could impair esophageal motility, bone/food impaction, caustic agent ingestion, esophagogastric surgeries as also severely immune-compromised patients (HIV, recently transplanted).

Furthermore, 58 patients were excluded as they showed endoscopic evidence of another disease that could explain patients' symptoms. Epidemiological, endoscopy and histological data were collected and analyzed.

Results: We included 63 patients (M/F: 29 (46.0%)/34 (54.0%), mean age 64±19 yrs). A history of dysphagia > 6 months was reported by 51 (80.9%), while <6 by 11 (19.1%). During the endoscopy; 7 (11.1%) had findings supportive of EoE, and 56 (89.9 %) had normal endoscopy. Biopsies were obtained by 30/63 (47.6%).

Significant higher rates of biopsy sampling were seen among patients <65 years old and with endoscopic EoE signs, while dysphagia duration and sex did not affect sampling. In the multivariate analysis only age, but not endoscopic findings compatible to EoE, drove clinician decision to obtain biopsies. (Table)

We were able to retrieve the histology report of 29/30 patients. EoE was diagnosed in 4/29 (13.8%). All patients with EoE were male (4/14, 28.6% vs female 0/15, 0% p=0.042). Existence of compatible endoscopic signs (3/6, 50% vs normal endoscopy 1/23, 4.3% p=0.020) was correlated with EoE diagnosis but age (<65 years old 4/22, 18.7% vs <65 year old 0/7, 0% p=0.307) and longstanding dysphagia (4/25, 16% vs female 0/4, 0% p=0.533) were not.

Factor	Biopsy sampling among patients with Dysphagia (patients without endoscopic findings other than EoE that could explain patients symptoms)		
		Univariate p	multivariate p
Younger than 65 years 65 years and older	25/35, 75.4% 5/28 17.9%	<0.001	<0.001 OR: 0.092, 95% C.I.: 0.025-0.339
Male Female	12/29 41.4% 18/34 52.9%	0.450	
Findings compatible to EoE Normal endoscopy/other findings	7/7 100% 23/56 41.1%	0.004	0.099 OR: 21.327, 95% C.I.: 18,298-35.474
history of dysphagia <6 months history of dysphagia>6 months	5/11 45.5% 25/52 48.1%	0.989	

Conclusion: Clinician obtain esophageal biopsies in less than 50% of patients with dysphagia. Endoscopic findings supportive of EoE and most importantly age seem to drive their decision. The rate of EoE among patients with dysphagia and esophageal sampling is not negligible and is more commonly observed in young males with longstanding dysphagia and compatible endoscopic findings.

Disclosure: Nothing to disclose.

PP0090

ENDOSCOPIC INDICATORS OF INSULIN RESISTANCE: ASSOCIATIONS WITH EROSIVE ESOPHAGITIS AND REGULAR ARRANGEMENT OF COLLECTING VENULES (RAC)

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Introduction: With the increasing westernization of lifestyles, there has been a significant rise in obesity, leading to a growing interest in metabolic syndrome and insulin resistance. Identifying indicators of insulin resistance through endoscopic findings could be valuable in assessing a patient's systemic metabolic status. Thus, we conducted a retrospective study to explore the association between endoscopic findings and insulin resistance.

Aims & Methods: We retrospectively investigated the patients who underwent screening endoscopy at Bundang Jesaeng General Hospital between February and June 2023. We excluded patients who were not tested for insulin resistance. Insulin resistance was defined using Homeostatic Model Assessment for Insulin Resistance ($HOMR_{IR}$), with a value above 1.9. We also assessed the *Helicobacter pylori (H. pylori*) status using rapid urease test, and history of previous eradication into current infection, non-infection, and past infection. To ensure balanced variables between the two groups, we performed propensity score matching (1:2 ratio) based on age and *H. pylori* status. Multivariate analysis was performed to investigate the association between the endoscopic features and insulin resistance.

Results: Following propensity score matching, we selected 210 patients with insulin resistance and 420 patients without insulin resistance were selected. In baseline characteristics, age and *H. pylori* status were balanced between the two groups. Prior to propensity score matching, endoscopic findings including erosive esophagitis, RAC (regular arrangement of collecting venules), atrophy, and diffuse redness appeared to be significant associated with insulin resistance. However, after matching, only erosive esophagitis remained significantly associated with insulin resistance. Furthermore, the severity of erosive esophagitis showed positive correlation with the degree of insulin resistance, as indicated by increased $HOMA_{IR}$. Multivariate analysis demonstrated that erosive esophagitis (OR 1.81, p = 0.010) significantly raised the risk of insulin resistance while RAC was associated with a lower risk of insulin resistance (OR 0.68, p = 0.032).

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Conclusion: Our study showed an obvious correlation between insulin resistance and erosive esophagitis, as well as RAC among the endoscopic findings.

Moreover, we observed a proportional relationship between the severity of erosive esophagitis and the degree of insulin resistance. Multivariate analysis showed that erosive esophagitis and RAC were significantly associated with the insulin resistance.

These findings suggest that individuals with erosive esophagitis and without RAC should undergo metabolic evaluation to assess their insulin resistance status. Further research is warranted to explore the underlying mechanisms and potential implications of these associations.

Disclosure: Nothing to disclose.

PP0091

THE EPIDEMIOLOGY AND SYMPTOMATIC CHARACTERISTICS OF REFLUX ESOPHAGITIS SUBGROUPS BASED ON THE LYON 2.0 CONSENSUS: A POPULATION-BASED STUDY

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Introduction: Reflux esophagitis (RE) can present typical symptoms, atypical symptoms or asymptomatic. The proportions and epidemiology characteristics of these three symptomatic patterns of RE has not been clarified. The Lyon 2.0 consensus provides new conclusive criteria for RE diagnosis and adjusts the symptom spectrum. Thus, the RE subgroups were redefined.

Aims & Methods: The aim of this study was to clarify the prevalence and symptomatic characteristics of RE subgroups based on the update Lyon 2.0 consensus in a Chinese population. An Endoscopic Screening for Esophageal Cancer in China (ESECC) randomized controlled trial was initiated in Hua County in January 2012, a gastroesophageal reflux disease (GERD) database was constructed. This data was reviewed to analyze the subgroups of RE based on the Lyon 2.0 consensus. RE were divided into three subgroups: typical reflux symptoms (TSRE), atypical symptoms (ASRE) and asymptomatic RE (ARE). The rest esophagitis was inconclusive-GER esophagitis. Chi-square test was used for differences between groups.

Results: In this study, 2916 residents were enrolled. 271 (9.29%) participants had esophagitis, among them, 143 (4.87%) fulfilled the conclusive RE diagnosis based on the Lyon 2.0 consensus. In RE, 97 (67.83%) was TSRE, 11(7.69%) was ASRE and 35 (24.48%) was ARE. 115 esophagitis belonged to inconclusive-GER esophagitis group. The age (≥60 years), male sex, Body Mass Index (BMI) (≥24 kg/m^2), and smoking were risk factors for TSRE. Smoking was also a risk factor for ARE (OR [95%CI]:4.521 (1.448, 14.113)).

Conclusion: This population-based study reported that 9.29% of residents had erosive esophagitis. The point prevalence rate of RE was 7.75%, according to the updated Lyon 2.0 consensus. ARE LA-B and non-RE EE are common. Smoking is a risk factor for ARE. The findings of this study can be beneficial in the diagnosis, prevention, and establishing the screening system for RE. A well-designed screening strategy and long-term follow-up study should be conducted for ARE and non-RE EE in the future.

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Disclosure: All Authors declare that they have no conflict of interest.

PP0092

THE FREQUENCY OF GASTROESOPHAGEAL REFLUX AND RELATED FACTORS IN PERSIAN GUILAN COHORT STUDY (PGCS)

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Introduction: Gastroesophageal reflux disease (GERD) is a prevalent digestive disorder characterized by the symptoms of heartburn and regurgitation. In recent years, the epidemiological characteristics of GERD have been studied due to its increasing prevalence and associated issues.

Aims & Methods: In the current study, we aimed to evaluate the prevalence of GERD among the Prospective Epidemiological Research Studies (PERSIAN) Guilan Cohort study (PGCS) population. This cross-sectional study was conducted on the PGCS population in Guilan, Iran. Demographical data and clinical characteristics of individuals were recorded, and a specialist applied the diagnosis of GERD. The association between variables and GERD was evaluated using SPSS version 17 based on a significant level <0.05.

Results: Out of 10520 participants, 5633 were female, and the mean age of individuals was 51.52±8.9 years. The prevalence of GERD was 13.2% and it was most prevalent among females. The frequency of GERD was higher in unemployed individuals, individuals with a higher level of education, body mass index (BMI), higher waist-hip ratio (WHR), and a lower level of physical activity. In addition, patients with a higher number of underlying diseases, a history of hypertension, and the use of non-steroidal anti-inflammatory drugs (NSAIDs) had a higher prevalence of GERD (P<0.05). The multivariate analyses revealed that females, urban residents, and individuals with higher levels of education had greater odds of GERD (P<0.05). Moreover, with the increase in the number of underlying diseases, the odds of having GERD increased (P<0.001).

Conclusion: Our findings showed factors such as female gender, higher level of education, urban residency, and higher number of underlying diseases were associated with increased risk of GERD. Identifying the associated factors will allow specialists to better diagnose and control the GERD.

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Disclosure: Nothing to disclose.

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PP0093

PREVALENCE OF REFLUX ESOPHAGITIS AND BARRETT'S ESOPHAGUS IN PORTUGAL: A MULTICENTER POPULATION-BASED ENDOSCOPIC STUDY

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Introduction: The incidence of esophageal adenocarcinoma has been increasing in Western countries, with gastroesophageal reflux (GERD) and Barrett's esophagus (BE) being important risk factos for such diagnosis. In Portugal, the current prevalence of reflux esophagitis and BE is unknown, and the aim of this study is to clarify these points.

Aims & Methods: A multicenter cross-sectional study including 13 national digestive endoscopy units, involving patients who underwent upper endoscopy from January to March 2024.

Results: 6320 patients were included (57% female, mean age 58). Reflux esophagitis was identified in 1105 patients (17% of the exams, 54% male, mean age 57) (Los Angeles Grade A 10.1%; B 5.6%; C 1.1%; D 0.4%) and 46 with BE (0.7% (95%CI 0.05-1%), 67.4% male, mean age 59 years), with only 4 (0.06% of the total) having 3 cm or more of extension.

The prevalence of BE was higher in the Lisbon region (1.35% vs 0.5%, p<0.05), with no difference in the prevalence of reflux esophagitis. Among patients who underwent endoscopy for GERD symptoms, 38% had erosive esophagitis (p<0.05 vs the rest of the population).

Among patients with esophagitis and BE, 27% and 39%, respectively, underwent endoscopy for symptoms and/or surveillance related to GERD (vs 11.6% of the rest of the population, p<0.05), with a prevalence of BE in patients with GERD of 2.3% (95%CI 2.1-2.5%, p<0.05 vs the rest of the population). Male gender was a risk factor for grade C/D esophagitis and BE (p<0.05).

Conclusion: Erosive esophagitis is common in Portugal, particularly in patients undergoing endoscopy for GERD symptoms. However, the prevalence of BE, particularly long segment BE, remains low compared to other Western countries, despite similar values of reflux esophagitis. This trend may be reversing in Lisbon with a higher prevalence of BE. Further studies are necessary to better understand these differences.

Disclosure: Nothing to disclose.

PP0094

HIGH-RESOLUTION MANOMETRY FEATURES CAN SEGREGATE NON-SEVERE FROM SEVERE GASTRO-ESOPHAGEAL REFLUX DISEASE

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Introduction: Gastro-esophageal reflux disease (GERD) is a multifactorial disease that manifests in patients with different degrees of anti-reflux barrier impairment. GERD significantly impacts quality of life despite the fact that symptom burden may not always correlate with disease severity. Precise pathophysiologic evaluation can identify patients with actionable GERD who may benefit from anti-reflux surgery or escalation of medical therapy.

The Milan Score, a novel high-resolution manometry (HRM) metric, numerically quantifies the degree of anti-reflux barrier disruption[1]. While the Milan Score has been proven to predict pathologic GERD, its ability to distinguish non-severe from severe GERD and to identify patients with a higher grade of anti-reflux barrier disruption requires further study.

Aims & Methods: A database of prospectively collected data from a cohort of patients undergoing HRM and pH-study for persistent GERD symptoms was queried. Using Lyon 2.0 consensus, pathologic GERD was defined as AET >6%, grade B-C-D esophagitis or Barrett's esophagus on endoscopy. Severe reflux was defined as AET >12%, DeMeester score >50, grade C-D esophagitis or Barrett's esophagus[2].

Demographic, HRM and pH data were extracted, classified and compared between groups with varying GERD severity (no GERD, non-severe GERD, severe GERD). The Milan Score integrates the main manometric parameters related with the pathophysiology of GERD (ineffective esophageal motility, esophago-gastric junction (EGJ) contractile integral, EGJ morphology and straight leg raise maneuver response).

In order to simplify the use of the Milan Score, risk rates have been classified into six categories, from extremely unlikely to extremely likely. A cumulative odds ordinal logistic regression was run to determine the independent effect of HRM variables on reflux severity. The ability of the Milan Score to predict severe reflux was assessed, as well as its association

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with the pH-study criteria proposed by Lyon 2.0 as adjunctive evidence for pathologic GERD (>80 reflux episodes, mean nocturnal baseline impedance (MNBI) <1500 W or positive reflux-symptom association).

Results: Among 603 patients enrolled (median age 49.9 years, median body mass index (BMI) 25.3 Kg/m^2, 50.5% males), 124 (20.6%) had non-severe and 142 (23.5%) had severe GERD. Non-severe and severe GERD patients had similar demographic and clinical characteristics, but were significantly different from the no-GERD group (Table). On univariate analysis, presence and size of hiatal hernia, Milan Score and Milan Score categories were significantly different among the groups (p <0.001). On ordinal logistic regression, hiatal hernia size and Milan Score category were found to be independent predictors of GERD severity (OR 1.28, 95% CI, 1.04-1.57, p=0.018 and OR 2.92, 95% CI, 2.47-4.45, p=<0.001 respectively). A significantly higher Milan Score (p <0.001) was found in patients with >80 reflux episodes, MNBI <1500 W, positive reflux-symptom association and any criteria of severe GERD.

	Total (n=603)	No GERD (337)	Non severe GERD (n=124)	Severe GERD (n=142)	p-value
Female, n (%)	282 (49.5)	178 (56.2)a	54 (45.4)a,b	50 (37.3)b	<0.001
Age, (years)	49.9 (14.6)	48.6 ±15.9 a	51.4 ± 13.5 a,b	52.0 ± 14.2 b	0.044
BMI, (Kg/m^2)	25.3 (5.8)	24.7 ± 23.8 a	25.9 ± 5.6 b	26.3 ± 5.1b	<0.001
PPI use, n (%)	450 (79.8)	232 (73.9)a	105 (89.7)b	113 (85.0)b	<0.001
PPI non responders, n (%)	139 (30.6)	88 (37.4)a	31 (29.2)a,b	20 (17.7)b	0.002
EGD hiatal hernia, n (%)	327 (59.2)	145 (47.9)a	84 (71.8)b	98 (74.2)b	<0.001
EGD Esophagitis, n (%)	168 (30.6)	60 (19.9)a	42 (36.2)b	66 (50.0)b	<0.001

Conclusion: Hiatal hernia size and Milan Score assessed at HRM are independent factors associated with severe GERD. The Milan Score is significantly higher in patients with pathologic adjunctive pH-impedance metrics and in patients with severe GERD.

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Disclosure: Nothing to disclose.

PP0095

PHYSIOLOGIC CHARACTERISTICS OF PATIENTS WITH ESOPHAGOGASTRIC JUNCTION TYPE 2 MORPHOLOGY ON HIGH-RESOLUTION MANOMETRY HELP IDENTIFY PATHOLOGIC GASTRO-ESOPHAGEAL REFLUX DISEASE

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Introduction: Gastroesophageal Reflux Disease (GERD) is a common condition diagnosed by endoscopy and/or pH monitoring. However, only high-resolution manometry (HRM) provides detailed assessment of the pathophysiology underlying reflux mechanisms and GERD symptoms. While it is widely accepted that esophago-gastric junction (EGJ) type 3 is a strong predictor of GERD, EGJ type 2 is less clearly implicated in GERD pathogenesis[1]. This study aimed to investigate the clinical and physiological characteristics of patients with EGJ type 2 and to identify HRM variables that predict GERD in this category of patients.

Aims & Methods: 535 patients with suspected GERD who underwent HRM and reflux monitoring were retrospectively analyzed. Patients were divided into three groups based on EGJ morphology (type 1, 2, or 3). Clinical data, HRM parameters, and reflux study results were compared among groups. The Milan Score[2], a novel metric that integrates ineffective esophageal motility (IEM), EGJ-contractile integral (EGJ-CI), EGJ morphology and straight leg raise (SLR) response, was applied to the EGJ type 2 group to identify patients at higher risk of developing GERD. Pathologic GERD was defined as AET >6% on pH-impedance monitoring and/or stricture, Barrett's esophagus or Los Angeles grade B, C or D, or at endoscopy. As previously demonstrated, a Milan Score of 137 (risk rate 50%) or higher was considered pathologic. Receiver operating characteristic (ROC) analysis was performed to determine Milan score prediction of GERD within patients with EGJ type 2 morphology.

Results: Patients with EGJ type 3 had the highest rate of GERD diagnosis, followed by EGJ type 2 and type 1 (p <0.001). HRM showed differences in EGJ-CI, proportion of patients with IEM and positive SLR response among groups (p <0.001). Reflux monitoring revealed a stepwise increase in AET from type 1 to 3 (5.1±7.3 vs. 8.3±11.1 vs. 12.2±11.3, p <0.001). A pathologic

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Milan Score identified two subgroups of patients within the EGJ type 2 group: patients with a score below 137 resembled type 1, while those with a score of 137 or higher resembled type 3 in terms of pH monitoring characteristics (*Table*). On ROC analysis the area under the curve was 0.858, and the optimal threshold to identify pathologic GERD in this group was 137 points (sensitivity 77.3%, specificity 87.7%).

	EGJ type 1 (n=277)	EGJ type 2 with MS<137 (n=93)	EGJ type 2 with MS≥137 (n=85)	EGJ type 3 (n=80)	P-value
AET (%), mean (SD)	5.1 (7.3)[a]	3.9 (5.8)[a]	13.2 (13.3)[b]	12.2 (11.1)[b]	<0.001
4%<AET<6%, n (%)	35 (12.6)	14 (15.1)	7 (8.2)	6 (7.5)	0.305
AET > 6%, n (%)	72 (26.0)[a]	12 (12.9)[a]	69 (81.2)[b]	62 (77.5)[b]	<0.001
DeMeester score, mean (SD)	21.7 (30.3)[a]	15.7 (23.2)[a]	55.0 (53.0)[b]	45.3 (39.5)[b]	<0.001
Total reflux ≥80, n (%)	37 (13.4)[a]	12 (12.9)[a]	24 (28.2)[b]	25 (31.3)[b]	<0.001
MNBI <1500 Ω, n (%)°	36 (21.8)[a]	11 (20.0)[a]	35 (66.0)[b]	39 (59.1)[b]	<0.001
PSPW <50%, n (%)°	54 (32.7)[a]	18 (32.3)[a]	39 (75.8)[b]	51 (77.2)[b]	<0.001

Conclusion: Our data suggests that EGJ type 2 encompasses a wide spectrum of disease severity. The Milan Score with a cut-off of 137 may help to identify patients with EGJ type 2 who are more likely to demonstrate pathologic GERD on reflux monitoring.

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Disclosure: Nothing to disclose.

PP0096

EVALUATION OF THE DECISION TREE CLASSIFICATION MODEL USING DIETARY AND LIFESTYLE FACTORS FOR PREDICTING GERD DIAGNOSIS IN GISTAR STUDY POPULATION

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Introduction: Dietary and lifestyle recommendations are the cornerstones of the treatment of GERD. However, the evidence of the efficacy of those recommendations is lacking. We performed a retrospective analysis of the GISTAR study[1] participants evaluating the symptoms of GERD and dietary and lifestyle parameters developing a tool for predicting GERD symptoms based on the aforementioned criteria.

Aims & Methods: The aim of the study was to analyse the factors pointing to GERD symptoms through the use of data mining and decision tree classifier.

Methods: GERD diagnosis was evaluated using a self-assessment questionnaire[2]. Dietary, demographic and lifestyle data were collected within the GISTAR study protocol[1]. 3340 40-64 year old participants from the GISTAR study were included in the analysis. 1670 of them did not have GERD symptoms. 41 attributes describing demographic, lifestyle, and dietary habits were used for these respondents to induce a decision tree classification model using Orange Data Mining software[3].

Results: Fried potatoes including French fries were the most predictive factor of GERD symptoms (65.3% with symptoms; 1516 from 2321), especially for females (82.6%; 1048/1228). This was more pronounced in the group who consumed sweets once per week (87.2%, 950/1090). In males, who consumed fried potatoes, the next most important factor was the consumption of sweets (46.3%, 431/930).

For participants who didn't consume fried potatoes (14.6%; 147/1008), 1-3 cups of green tea (37.9%; 66/174) and male gender (71.4 % 20/28) were predictors of GERD symptoms.

The accuracy of the decision tree model was: sensitivity (74.8%), precision (75.6%), specificity (75.9%) and AUC (0.841%).

Conclusion: Fried potatoes, female gender and consumption of sweets are the risk factors predicting the diagnosis of GERD. None of the lifestyle factors were displayed as significant.

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Disclosure: Nothing to disclose.

PP0097

IMPAIRED MUCOCILIARY CLEARANCE IN LARYNGOPHARYNGEAL REFLUX DOES NOT REFLECT THE AMOUNT OF GASTROESOPHAGEAL REFLUX

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Introduction: Gastroesophageal reflux is responsible for various extresophageal symptoms that are commonly labelled as laryngopharyngeal reflux (LPR). These mostly include cough, throat clearing, globus sensation and hoarseness. The pathophysiological link between reflux and symptoms likely includes direct effect of the refluxate on the upper airways epithelium and reflex that is initiated by stimulation of the afferent nociceptive fibres in the distal esophagus by acidic gastroesophageal reflux.

Arguably, the relationship between reflux and laryngopharyngeal symptoms might involve other, yet not well explored mechanisms. In our previous research we found that LPR patients have impaired mucociliary clearance of the respiratory epithelium of the upper airways. This could contribute to the inflammation persistence and the presence of symptoms.

We therefore aimed to determine whether any parameter of the reflux burden as detected by 24-hour pH/impedance correlates with ciliary beat frequency (CBF), the major quantifiable aspect of mucociliary clearance.

Aims & Methods: Consecutive patients with LPR confirmed by off PPI 24-hour pH/impedance had a sample of the ciliated epithelium obtained from the nose using a cytology brush. Nasal CBF was analysed using a digital high-speed video microscope and the dedicated software application Ciliary Analysis (NI LabVIEW). Healthy volunteers without either esophageal or extraesophageal reflux symptoms served as a control group. Both patients' and healthy volunteers' group were without any airway inflam-

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mation at least 3 months prior to the evaluation and were at least 10 days of PPI therapy. Maximal, minimal, and the median value of the CBF (Hz) was obtained. We performed correlation analyses between the acid exposure time (AET), number of reflux episodes, mean nocturnal baseline impedance (MNBI) in all impedance segments (Z1-Z6) and CBF values.

Results: 32 patients with LPR confirmed by 24-hour pH/impedance that had positive reflux symptom index (RSI>13) were included in the final analysis. Of them, 12 patients had confirmed by positive acid exposure time (AET>6%) and/or abnormal number of reflux episodes (>80). Firstly, we confirmed our previous observations that LPR patients had lower CB frequency (median value) compared to the healthy individuals (6,37±2,17 vs. 10,9±1,9; p<0.0001). Separate analysis of patients with proven GERD showed no CBF difference from the remaining (pH negative patients) (6,26±2,53 vs. 6,43±1,94; p=0,42). Secondly, we performed correlation analysis between AET/number of reflux episodes and CBF. We found no significant correlation between reflux burden and median value of CBF (N=32, R=-0.18, p=NS and R=0.05, p=NS for AET and number of refluxes, resp.). Thirdly, we performed correlation analyses between CBF and MNBI values of all impedance segments. Indeed, correlation analysis between median CBF and MNBI Z1-Z6 showed no significant correlations (N=32, R=-0.16, 0.09, 0.28, 0.01, 0.14 for Z1, Z2, Z3, Z4, Z5, Z6, resp., resulting in p=NS in all segments).

Conclusion: pH positive LPR patients had no difference in CBF values from the remaining (pH negative) patients. Reflux burden in the esophagus as determined by pH recording does not seem to be related to ciliary beat frequency. We observed no significant correlation between the MNBI values in any impedance segment and the median CBF value. Further research should be performed to determine the putative mechanism between the impairment of CBF and reflux.

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PP0098

EVALUATION OF THE SAFETY AND EFFICACY OF TRANSORAL INCISIONLESS FUNDOPLICATION (TIF2.0) FOR GASTRO-OESOPHAGEAL REFLUX DISEASE: A UK COHORT STUDY

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Introduction: Gastro-oesophageal reflux disease (GORD) is a common disorder with a prevalence of 10-20%. It occurs from disruption of the gastro-oesophageal flap valve (GFV) and/or a diaphragmatic abnormality causing a hiatus hernia (HH). Transoral incisionless fundoplication (TIF2.0) using the EsophyX device (Endogastric solutions) is an endoscopic treatment for GORD, which recreates the GFV through a partial 270º wrap. Prospective trials show the procedure is safe and effective and is recommended in North American guidelines, yet the experience of TIF is limited in Europe.

Aims & Methods: We conducted a UK-based, retrospective, cohort study to determine the safety and efficacy of TIF2.0. Patients undergoing TIF2.0 by a single operator were identified and analysed between 2019-2023. Patient demographics, reflux assessments (GERD-HRQL, endoscopy, pH testing), and procedural details were recorded. Primary outcomes were technical success and clinical response defined as complete (off regular anti-acid therapy) or partial (reduction from baseline in anti-acid therapy, GERD-HRQL score, and/or physiological parameters) without the need for revision. Secondary outcomes were adverse events and changes in reflux parameters measured by Wilcoxon signed-rank test. A p-value <0.05 was significant.

Results: In total, 81 patients underwent TIF2.0 with 92.0% on oral anti-acid therapy. Median age was 51 (IQR 37-64), 23 (28.4%) were female, and the median ASA grade was 1 (IQR 1-2). A HH was present in 84.6% with an average size of 1.9cm (SD 0.5). The average acid exposure time (AET) was 8.9% (SD 4.1; n=61). The clinical outcomes are summarised in table 1. Technical success was 97.5% (n=79) with an average procedural time of 38.0 minutes (SD 10.3), and 1-day (IQR 1) inpatient stay. The device could not pass the cricopharynx in two patients despite routine balloon dilatation. Over 15.5 months of median follow-up (IQR 7-24), 55% had a complete response, and 71.7% partial response. On logistic regression analysis, no variables were predictive of response. Among those with paired pre-/post-testing, the average GERD-HRQL score reduced significantly (33.9 ±18.6 to 9.1 ±9.2; P<0.001) as well as the average AET (8.9 ±4.1 to 5.1 ±4.0; P=0.005) post-procedure. Procedure or device-related adverse events occurred in 12.2% (n=10), which were all managed conservatively. The single perforation was delayed and managed with intravenous antibiotics and endoscopic vacuum therapy. Seven patients underwent a revisional procedure, including two surgical fundoplications, with 83.3% due to recurrent HH.

Procedure time in minutes (SD)	38.0 (7.2)
Technical success (%)	79 (97.5)
Inpatient stay (IQR)	1 (1)
Complete response*	33 (55.0)
Partial response†	43 (71.7)
Follow-up in months (IQR)	15.5 (7-24)
Repeat procedure (%)	7 (8.6)
Adverse events (%)‡	10 (12.2)

IQR – interquartile range; ; SD – standard deviation
* Off all regular anti-acid therapy (respondents during follow-up: n=60)
† Reduction from baseline in anti-acid therapy, symptomatology based on HRQL-GERD questionnaire, and/or physiological parameters (respondents during follow-up: n=60)
‡ Mucosal injury (n=1), chest infection (n=3), perforation (n=1), leak/collection (n=1), bleeding (n=1), dysphagia (n=2), admission for nausea and vomiting (n=1).

Table 1 - Clinical outcomes among patients undergoing Transoral Incisionless Fundoplication 2.0

Conclusion: TIF2.0 is associated with technical success in 97.5%, clinical response in 71.7%, and significant improvements in GERD-HRQL and AET. Recurrent HH is a frequent cause of failure showing the importance of patient selection and/or need for concomitant HH repair.

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PP0099

INDEPENDENT MEDICAL EDUCATION SIGNIFICANTLY IMPROVES PHYSICIANS KNOWLEDGE, CONFIDENCE AND COMPETENCE REGARDING THE LONG-TERM MANAGEMENT OF REFLUX AND DEPRESCRIBING PROTON PUMP INHIBITORS

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Introduction: Proton pump inhibitors (PPIs) are widely prescribed open-ended fashion, but in an estimated 25-82% of patients this may be inappropriate.[1, 2] Chronic PPI use is associated with an increased risk of specific adverse events including pneumonia, dementia, and osteoporotic fractures although no causal relationships have been established.[3] However, recent surveys have found that among physicians who regularly prescribe PPIs, a considerable proportion were unfamiliar with deprescribing strategies.[2] Here, we assessed whether an 30-minute online independent medical education activity could improve primary care physicians' (PCPs) and gastroenterologists' knowledge, confidence and competence regarding the long-term management of reflux and PPI deprescribing.

Aims & Methods: Educational effect was assessed using a repeated-pairs design with pre-/post-assessment. Three multiple choice questions assessed knowledge, and one question assessed confidence. Statistical tests to assess significance included: Paired samples t-test for overall average number of correct responses and confidence. McNemar's test for individual questions and learning objectives (P <.05). Cohen's d estimated the effect size impact on number of correct responses (<.20 modest, .20-.49 small, .59-.79 moderate, ≥.80 large).

Results: From a total of 3422, there were 359 assessment completers. Overall, there were highly significant and impactful knowledge gains for both groups of physicians (PCPs P <.001, Cohen's d = 0.87; Gastroenterologists P <.001, Cohen's d = 1.02). For PCPs, the highest relative proportional change in knowledge was for the extent of PPI overuse and misuse at 161% with 38% of physicians improving their knowledge (P<.001). Whilst for gastroenterologists the highest relative proportional change was seen regarding the implications of PPI overuse at 277% with 37% of physicians improving knowledge following the activity (P<.001). Significant increases in physician competence in best practice in PPI deprescribing were found (P<.001 for both), despite high baseline levels of competence. A considerable proportion of PCPs (51%; confidence shift 69%; P <.001) and gastroenterologists (39%; confidence shift 52%; P <.001) increased their confidence regarding their ability to use appropriate management strategies to reduce or discontinue a patient's use of PPI therapy.

Conclusion: Online medical education significantly improved physicians' knowledge, confidence and competence regarding the extent and implications of PPI overuse and best practice in the long-term management of reflux and PPI deprescribing. As PPI prescribing is prevalent and long-term use is common, it is important that physicians understand appropriate indications and implications associated with over and long-term use and are also able to undertake clinically appropriate deprescribing approaches.

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PP0100

PATIENT-REPORTED OUTCOMES OF VONOPRAZAN IN CHINESE REFLUX ESOPHAGITIS PATIENTS: A REAL-WORLD, OBSERVATIONAL STUDY

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Introduction: Reflux esophagitis (RE) is a common, chronic disorder characterised by the mucosal break with symptoms such as heartburn and/or acid regurgitation. Patient-reported outcomes (PROs) offer more comprehensive access to the symptoms and impact on RE, which is a widely used treatment option that is effective from the patient's perspective. Vonoprazan is a novel potassium competitive acid blocker, approved in China for treating RE since 2019. Here, we present the PROs of vonoprazan in Chinese RE patients.

Aims & Methods: VIEW (NCT04501627) is a multicenter, single-arm, prospective, observational, real-world study to evaluate the safety and effectiveness of Vonoprazan. The subgroup analysis of VIEW was designed to assess the effectiveness of vonoprazan in patients with endoscopically confirmed RE. All patients were treated with 20 mg vonoprazan orally q.d. for 4 weeks (continued for 8 weeks if the patient did not show any beneficial effects) followed by a safety follow-up for 2 weeks.

All the patients were assessed for symptomatic relief by PROs which included patient symptom diary (percentages of patients with complete relief of heartburn, nocturnal heartburn, regurgitation, and nocturnal regurgitation) and GERD questionnaire (GerdQ) scores.

Results: Out of 1040 endoscopy-confirmed RE patients, 935 (89.90%) patients were with mild RE (LA grade A/B) and 105 (10.09%) patients were with severe RE (LA grade C/D). For all RE patients with heartburn, night-time heartburn, regurgitation, and night-time regurgitation at enrollment, the percentages of patients with complete symptomatic relief during the first one-week of treatment were 35.1%, 47.2%, 29.5%, and 43.6% respectively; while during first two-weeks of treatment, the respective percentages were 60.7%, 71.3%, 52.3%, and 65.0%.

Higher percentages were observed in patients with complete night-time heartburn or night-time regurgitation. Among subgroups, severe RE patients exhibited marginally higher percentages than mild RE patients (severe RE: 44.9%, mild RE: 33.7%), (severe RE: 52.2%, mild RE: 46.5%), (severe RE: 39.7%, mild RE: 28%) (severe RE: 47.1%, mild RE: 43.1%) during first one-week of treatment. A similar trend of symptomatic relief in severe RE patients was observed during the first two-weeks of vonoprazan treatment compared to mild RE patients (Table 1).

The proportion of patients without typical GERD symptoms increased by 36.9% (95% CI: 32.79, 41.10), 36.7% (95% CI: 32.29, 41.09) and 39.3% (95% CI: 26.84, 51.78) for all RE, mild RE and severe RE patients, respectively, at week 4 compared to baseline. The total change in GerdQ scores from baseline to week 4 (mean (SD)) were −1.7 (2.76) (95% CI: −1.85, −1.48) in

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all RE, −1.6 (2.72) (95% CI: −1.81, −1.42) in mild RE and −2.1 (3.05) (95% CI: −2.72, −1.42) in severe RE, where negative value represents improvement in the symptomatic relief.

GERD typical symptoms	Percentage of patients with complete symptom relief						
	Statistics	First one week*			First two weeks**		
		Overall	Mild RE	Severe RE	Overall	Mild RE	Severe RE
Heartburn	% (n/N) (95% CI)	35.1 (194/552) (31.16, 39.29)	33.7 (163/483) (29.54, 38.16)	44.9 (31/69) (32.92, 57.38)	60.7 (340/560) (56.53, 64.78)	59.9 (294/491) (55.39, 64.24)	66.7 (46/69) (54.29, 77.56)
Night-time heartburn	% (n/N) (95% CI)	47.2 (261/553) (42.97, 51.45)	46.5 (225/484) (41.97, 51.04)	52.2 (36/69) (39.80, 64.35)	71.3 (400/561) (67.36, 75.01)	70.5 (347/492) (66.28, 74.52)	76.8 (53/69) (65.09, 86.13)
Regurgitation	% (n/N) (95% CI)	29.5 (163/553) (25.70, 33.47)	28.0 (136/485) (24.08, 32.27)	39.7 (27/68) (28.03, 52.30)	52.3 (293/560) (48.09, 56.53)	50.8 (250/492) (46.30, 55.32)	63.2 (43/68) (50.67, 74.61)
Night-time Regurgitation	% (n/N) (95% CI)	43.6 (242/555) (39.43, 47.84)	43.1 (210/487) (38.67, 47.65)	47.1 (32/68) (34.83, 59.55)	65.0 (364/560) (60.89, 68.95)	63.8 (314/492) (59.40, 68.07)	73.5 (50/68) (61.43, 83.50)

Table 1. Percentage of complete symptom relief after treatment for RE patients with symptoms at baseline

** First one week was defined as Day 1 to Day 7.*

***First two weeks were defined as ≥7 consecutive days between Day 1 and Day 14*

Conclusion: Vonoprazan exhibited quick symptomatic relief for both mild and severe RE patients in the real-world setting.

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PP0101

PHYSIOLOGICAL, SYMPTOMATIC AND NEUROCHEMICAL OUTCOMES AFTER STRETTA PROCEDURE

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Introduction: Gastro-oesophageal reflux disease (GORD) affects 25% of adults and impairs quality of life. Initial treatment is with lifestyle change and acid suppression however up to 1/3 will be refractory or intolerant to medical therapy. The Stretta procedure is an endoscopic treatment delivering radiofrequency energy around the gastro-oesophageal junction (GOJ). Whilst it can improve symptoms and reduce oesophageal acid exposure (AET) its mechanism of action is not fully understood. EndoFLIP is a device to measure GOJ distensibility (DI) and calcitonin gene related peptide (CGRP) is a neuropeptide marker for sensory afferents involved in GORD hypersensitivity - it is not known whether Stretta alters these factors and whether they are linked to normalisation of AET and symptoms.

Aims & Methods: This prospective study aimed to assess the effect of Stretta on UGI physiological parameters (LES pressure, IRP, DCI, AET, GOJ DI), mucosal CGRP presence, and symptom scores, by performing the following tests before and 6-months after Stretta:

Gastroscopy, EndoFLIP, HRM, ambulatory pH testing, oesophageal biopsies analysed using immunohistochemistry (IHC) for CGRP, and symptom questionnaires (Hull airway reflux, Reflux Symptom Index, Hospital Odynophagia Dysphagia, Reflux Disease, GERD-QOL).

Authors were blinded during analysis. Statistical analysis using SPSS with paired t-test for parametric data, Wilcoxon-signed rank test for non-parametric data, Pearson and Spearman tests for correlations. Ethical approval (IRAS 296193, NCT 05238636).

Results: 30 pts – 13F:17M, aged 22-71y (med. 55y).

		Pre-Stretta (Median, IQR)	Post-Stretta (Median, IQR)	Mean % delta (95% CI)	p value
EndoFLIP (40ml)	GOJ diameter (mm)	11.7 (8.4-13.5)	12.6 (10.4-14.8)	15.6 (3.7-27.5)	0.053
	Median pressure (mmHg)	34.9 (29.0-49.3)	37.3 (21.6-51.2)	1.3 (-9.7-12.3)	0.820
	GOJ distensibility index (mm2/mmHg)	3.1 (1.5-4.2)	4.5 (2.2-6.6)	71.1 (22.1-120.0)	0.014
HRM	Mean basal LES pressure (mmHg)	16.7 (7.8-28.6)	21.3 (9.3-33.3)	19.9 (-19.9-59.3)	0.758
	Median IRP (mmHg)	5.0 (3.1-10.0)	4.6 (2.5-7.9)	-16.5 (-51.0-18.0)	0.758
	Mean DCI (mmHg/s/cm)	1372.1 (306.0-2404.0)	1239.9 (369.8-2341.0)	3.4 (-29.5-36.2)	0.494
pH	Overall AET (%)	9.9 (6.4-14.8)	6.6 (3.0-14.3)	-19.2 (-44.3-5.8)	0.037
	DeMeester Score	43.3 (22.3-61.7)	27.4 (9.4-47.2)	-33.6 (-56.8 - -10.5)	0.012
IHC	% papillae with CGRP (10µm)	5.7 (4.0-9.7)	9.3 (6.0-17.8)	73.0 (-6.9-152.8)	0.071

Tests before and 6 months after Stretta:

48.5% pts felt Stretta was "very" or "somewhat" helpful.

Statistically significant (SS) improvements were seen in Hull (p=0.029) and RSI (p=0.028) scores.

AET normalised for 31% however there were no SS correlations between symptom scores and AET (p>0.05).

SS negative correlation between GOJ DI and LES pressure both before (r =-0.5, p = 0.017) and after (r=-0.5, p=0.012) Stretta.

No SS correlations between GOJ DI, GOJ diameter, LES pressure, IRP vs AET (p>0.05).

SS positive correlation between the percent change in GOJ DI pre/post Stretta with the percent change in DeMeester score (r= 0.4, p=0.03).

CGRP was most prevalent in the lower oesophagus (mean 12.0% papillae), vs middle (6.8%) and upper (7.3%). There was a mean increase of 73% post stretta but this was not SS and did not correlate with symptom scores, GOJ DI nor AET.

Conclusion: The normalisation of AET in 31% is consistent with published meta-analysis (30%). There was a non-significant trend towards increased basal LES pressure without a reduction in GOJ DI. This suggests the mechanism of action of Stretta may not be related to changes in GOJ compliance. There was SS improvement in laryngopharyngeal symptoms (Hull, RSI) suggesting these pts may benefit most from Stretta, although this was not supported by concurrent improvement in AET. CGRP presence within mucosal papillae was not affected, suggesting that this neuropeptide is not involved and that Stretta may not alter oesophageal sensitivity.

References: Ford et al, *BMJ* (2009), Fass et al, *Surg Endosc* (2017), Lee et al, *Gut and Liver* (2021), Leech, *J Gastroenterol* (2024), Woodland et al, *Gastroenterology* (2017).

Disclosure: Nothing to disclose.

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PP0102

KNOWLEDGE, ATTITUDES, AND PRACTICES AMONG DOCTORS MANAGING OF GASTROESOPHAGEAL REFLUX DISEASE IN A SRI LANKAN SETTING

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Introduction: While the global prevalence of GERD ranges from 5% to 25%, in Sri Lanka, it is slightly higher at 25.3%. There are no Sri Lankan guidelines for management of GORD despite it been a relatively common condition.

However, there is a lack of studies in Sri Lanka documenting the current management practices for GERD among doctors both primary and specialized.

Aims & Methods: This study aims to give an insight about the Sri Lankan doctor's point of view in managing GERD.

A self-administered questionnaire designed to gather information on medication prescription, lifestyle advice given, investigations, and personal experiences and beliefs was used. MBBS doctors working as primary care doctors, consultant general physicians/ general surgeons/ gastroenterologists as well as trainees in the fields, attached to the National Hospital of Sri Lanka, Colombo South Teaching Hospital, Colombo North Teaching Hospital, and the College of General Practitioners, Sri Lanka were approached consecutively.

Data analysis was conducted using SPSS-28.

Results: Of the 100 participants recruited, 57 were male while the average age was 35 years. The average duration of practice as a doctor was 11 years, ranging from 1 to 50 years. A significant proportion of doctors reported personally experiencing reflux and regurgitation (n=62, 62.0%).

In the first presentation of heartburn or reflux, the majority of doctors prescribed proton pump inhibitors (PPIs) (n=92,92.0%), followed by domperidone (n=67,67.0%), antacid syrups (n=42, 42.0%), H2 receptor blockers (n=37, 37.0%), antacid tablets (n=27,27.0%), antidepressants and anxiolytics (n=11,11.0%). Prescription at the second presentation showed a similar trend except for H2 receptor blockers taking precedence while only 3 doctors prescribed triple therapy. Only 46.0% advised the patients to come for a follow-up, and the mean follow-up period was 2 weeks, with n=88,88.0% of the patients presenting.

Most doctors referred/ utilized upper gastrointestinal (GI) endoscopy (n=96,96.0%), while upper GI manometry (n=13,13%) and pH impedance studies (n=4,4.0%) were referred at specialist level.

Most doctors reported advising their patients (n=88,88.0%) on lifestyle changes, such as avoiding food that causes GERD symptoms (n=91,91.0%), weight loss (n=71,71.0%), reducing mental stress (n=70,70.0%), and other lifestyle modifications. Spicy food (n=86,86.0%), oily food (n=93,93.0%), wheat-based food(n=50,50.0%), and bread (n=52,52.0%), were the types of food most commonly advised against.

The majority 73% of doctors co-prescribed PPIs for patients on multiple other medications and 36% co-prescribed PPIs when NSAIDs were prescribed (n=64,64.0%).

Only 52.0% were satisfied with their management while 19% believed that anti-reflux surgery was a good option for managing GERD.

Conclusion: According to this study a majority of physicians prescribe proton pump inhibitors (PPIs) as the first-line treatment for GERD, followed by other medications such as domperidone and antacids. Lifestyle modifications, including dietary changes and stress reduction, are commonly advised. Upper GI endoscopy is the most preferred investigation method. Despite advising patients to avoid wheat and bread, many physicians report patients experiencing symptoms from these foods only sometimes or less. Overall, physicians are generally satisfied with their management of GERD and do not consider surgery as a treatment option.

Education programs and seminars on the management of GERD is recommended to ensure optimal patientcare.

Disclosure: Nothing to disclose.

PP0103

DOES MUCIN EXPRESSION PHENOTYPE OF BARRETT'S ADENOCARCINOMA AFFECT THE RISK OF LYMPH NODE METASTASIS?

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Introduction: Superficial Barrett's adenocarcinoma (BAC) is a rare disease in Japan, and its clinicopathological characteristics are not well-defined. As a result, indications for endoscopic treatment have not been determined.

Aims & Methods: A histopathological study was performed to investigate whether there is a relationship between risk factors for lymph node metastasis and the mucin expression phenotypes of BAC. We conducted a retrospective analysis of a database comprising consecutive patients with superficial BAC treated by endoscopic or surgical resection. Immunohistochemical analysis was performed using the antibodies of MUC5AC, MUC6, MUC2, and CD10 to determine the classification of mucin expression phenotype.

Results: A total of 74 BAC lesions in 68 patients (54 males, median age 69 years) were identified. Mucin expression classification showed gastric (64%, complete gastric 24%, predominantly gastric 39%), intestinal (34%, complete intestinal 14%, predominantly intestinal 20%) or null (3%) phenotypes. Surgical treatment was performed in 30 patients, including additional treatment after ESD, and lymph node metastasis was observed in 4 patients (13%). Compared to the intestinal phenotype group, patients with gastric phenotype BAC had an almost significantly higher rate of lymphovascular invasion (38% vs 16%, P=0.06) and poorly differentiated component (23% vs 4%, P=0.09). All 4 lymph node metastasis-positive cases were in the gastric phenotype group (P=0.28).

Conclusion: Although our study did not show a direct relationship between the risk of lymph node metastasis and mucin expression phenotype of BAC, it did suggest BAC with gastric phenotype may be more likely to have lymphovascular invasion and poorly differentiated component. The endoscopic-specific findings of BAC with gastric phenotype are unknown. However, some specific endoscopic findings of gastric phenotype in gastric cancer are well-known, such as intralobular loop pattern or vessels within epithelial circle pattern. If we see such findings in BAC, due to a possible higher risk of lymphovascular invasion or poorly differentiated components, we should perform ESD rather than piecemeal EMR to obtain en-bloc tissue specimens for detailed pathological evaluation.

Disclosure: Nothing to disclose.

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PP0104

SAFETY AND EFFICACY OF CRYO-ABLATION ON BARRETT ESOPHAGUS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Barrett esophagus (BE) represents a pre-malignant condition of the esophagus and current guidelines favor endoscopic eradication therapy (EET). After endoscopic resection for visible neoplastic lesions, field ablation of the residual BE is advised to mitigate the risk of metachronous neoplasia. Cryoablation was introduced as an endoscopic ablative option for BE during the last decade with growing data on outcomes and interest in potential lower adverse events that established ablative techniques.

Aims & Methods: This systematic review and meta-analysis aimed to summarize the existing data regarding safety and efficacy of Cryoablation on BE treatment. A systematic literature search in Medline, Cochrane and Scopus databases was performed until November 2023 for full papers. Safety was considered as the primary outcome, interpreted as the overall rate of procedure related adverse events. Secondary outcomes included post-Cryoablation stricture development, and the complete eradication rates of intestinal metaplasia (CEIM) and dysplasia (CED). In case of heterogeneity, a sub-analysis based on the used device was performed for the primary outcome. Random effects model was used and the results were reported as percentages for pooled rates with 95% Confidence Intervals (95%CIs).

Results: Ten full-paper studies (471 patients; 80.3% men; mean age 62.2-69) were eligible for analysis. The mean BE length, based on Prague classification, was C0-2M2-5. Regarding the primary outcome, the overall adverse events rate was 19.7% (95%CI: 14-25.4; I^2=60.66%), although they were mild in the vast majority. Strictures warranting treatment were developed in 30 patients, representing 4.4% (95%CI: 1.8-7.1; I^2=61.18%). Four studies provided results on CED, yielding a rate of 94.6% (95%CI: 90.1-99.1; I^2=61.06%) whereas CEIM was assessed in six studies, reporting a rate of 90.1% (95%CI: 82.3-97.9; I^2=87.92%). Given the high heterogeneity in all outcomes, a sub-analysis including studies on focal Cryoablation was performed, revealing a similar rate of adverse events [16.5% (95%CI: 11.1-21.8)] with lower heterogeneity and non-significant heterogeneity (I^2=47.7%).

Conclusion: Cryoablation is a safe procedure with initial data showing a potentially low rate of esophageal strictures and comparable rates of dysplasia and BE eradication when compared to other ablative technologies. High-quality prospective studies including comparisons with established ablation techniques are warranted.

Disclosure: Nothing to disclose.

PP0105

BILE ACIDS IN SALIVA AS POSSIBLE NON-INVASIVE DIAGNOSTICS OF BARRETT'S ESOPHAGUS

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Introduction: Bile acids are a group of steroid compounds essential for lipid digestion. However, when bile acids are refluxed into the stomach and the esophagus during the so called duodenogastroesophageal reflux, they can have a detrimental effect on the esophageal epithelium and cause pathological changes of esophageal tissue, e.g., Barrett's esophagus (BE). The levels of bile acids in saliva could therefore serve as possible biomarkers for the diagnostics of BE.

Aims & Methods: We focused on optimization of saliva sampling methods for further analysis of bile acids. Three sampling methods were compared, i.e. sampling with Salivette sampler, Salivette Cortisol sampler and simple spitting. Then we performed quantification of 11 selected bile acids (unconjugated, glycine-conjugated) in saliva from 10 healthy volunteers and 10 BE patients by ultra-high-performance liquid chromatography coupled to triple-quadrupole tandem mass spectrometry. Moreover, high resolution MS (Orbitrap-MS) was utilized for identification of new bile acids in saliva.

Results: The results showed a clear advantage of simple spitting compared to the Salivette samplers. The median concentration of bile acids in saliva from healthy volunteers was generally below 1 nmol L −1 and approximately 2 nmol L −1 in saliva from patients with BE. The determined concentrations of bile acids in saliva are in the same range as previously reported. Levels of most bile acid were significantly higher ($p < 0.05$) in saliva from patients than in saliva of healthy volunteers. The most distinctive differences were found in levels of glycolithocholic acid, chenodeoxycholic acid, hyocholic acid and the sum of bile acids. Taurocholic acid, taurochenodeoxycholic acid and taurodeoxycholic acid were identified in a pooled saliva sample. To the best of our knowledge, this is the first time taurine-conjugated bile acids were detected in saliva. Moreover, glycine-conjugated dihydroxy-bile acid sulfate was detected and identified. Sulfate-conjugated bile acid, specifically glycochenodeoxycholate-3-sulfate, was previously detected only in one metabolomic study.

Conclusion: Saliva from BE patients contained higher levels of almost all bile acids, and the patients with BE could be distinguished from healthy volunteers by principal component analysis. In untargeted analysis taurine-conjugated bile acids were identified in saliva for the first time. The number of subjects in this preliminary study is low as we wanted to pinpoint a possible application for the analysis of bile acids in saliva, but not to complete a large clinical study, that will be conducted in future research, also including the taurine-conjugated bile acids.

Disclosure: Nothing to disclose.

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PP0106

CLINICAL EVALUATION OF CRYOBALLOON ABLATION FOR BARRETTS OESOPHAGUS: A UK FIRST MULTI-CENTRE EXPERIENCE

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Introduction: The Cryoballoon Focal Ablation System (CbFAS) (Pentax Medical Inc) is an innovative device designed for the endoscopic ablation of Barrett's-related neoplasia. This system has been shown to provide advantages such as reduced stricture rates and improved patient tolerance compared to the conventional standard of care, radiofrequency ablation (RFA).

We report early outcomes from a prospective UK first multi-centre registry of consecutive patients who underwent Cryoablation for the endoscopic eradication (EET) of Barrett's-related neoplasia.

Aims & Methods: Patients were prospectively enrolled from two tertiary referral centres for Barrett's Oesophagus (BE). Inclusion criteria included patients with a BE segment no longer than 5cm with histologically-proven dysplasia on biopsy or endoscopic resection specimens suitable for EET of BE after MDT review. Cryoablation sessions were conducted at 3-month intervals until endoscopic eradication was complete. The standardised protocol involved administering 8 seconds per ablation for all enrolled patients with the CbFAS. Our primary outcome was complete eradication of BE. Secondary outcomes included the number of ablations performed, adverse events, device malfunction, stricture rate, patient tolerance, and endoscopist satisfaction which was evaluated using a 10-point Likert scale (1=poor, 10=excellent).

Results: Between January 2020 and December 2023, a total of 16 patients (35 treatment sessions) were treated. The median BE length was C0M2. Complete eradication of BE was achieved in 89% (8/9) of patients who had completed treatment. No instances of device malfunction or intraoperative adverse events were recorded. A stricture rate of 12.5% (2/16) was observed, both of which were successfully managed through endoscopic dilatation. All patients were recorded as tolerating the procedure well with sedation. The median endoscopist satisfaction score was 9 (+/- 2.4) with only one recorded instance of difficulty attributed to balloon positioning.

Conclusion: These preliminary results from the UK cryoablation registry suggest that cryoablation with the CbFAS is a safe and effective therapy for BE-related dysplasia. The stricture rate appears to be comparable to that observed with RFA and similar studies. With the planned enrolment of further centres and suitable patients, this registry will further enhance our understanding of the long-term efficacy and safety profile of cryoablation using the CbFAS.

Disclosure: Nothing to disclose.

PP0107

ORAL ADMINISTRATION OF BEVACIZUMAB-800CW AND CETUXIMAB-800CW FOR THE DETECTION OF EARLY ESOPHAGEAL NEOPLASTIC LESIONS BY QUANTITATIVE FLUORESCENCE MOLECULAR ENDOSCOPY

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Introduction: Patients with Barrett's esophagus (BE) have an increased risk of developing esophageal adenocarcinoma (EAC) and therefore surveillance is necessary. However, lesion detection with high-definition white light endoscopy (HD-WLE) and narrow-band imaging (NBI) is hampered by high miss rates. Previous phase II studies have confirmed the great potential of quantitative fluorescence molecular endoscopy (qFME) in combination with topically administered fluorescent tracers for lesion detection in BE.

However, incubation time after topical tracer administration prolongs the procedure time by five minutes. In this ongoing clinical study, oral administration of bevacizumab-800CW, targeting vascular endothelial growth factor A (VEGFA), and cetuximab-800CW, targeting epidermal growth factor receptor (EGFR), is evaluated to improve both lesion detection and the clinical implementation of qFME.

Aims & Methods: A total of 30 patients with BE scheduled for a diagnostic and/or therapeutic endoscopy will be included. First 5 – 10 patients receive 4.5 mg and/or 9 mg oral bevacizumab-800CW before qFME. Subsequently, five patients will be included who receive 4.5 mg cetuximab-800CW orally before the diagnostic endoscopy, before their therapeutic endoscopy they receive a dose of 9 mg.

The patients drink the tracer in two sips in an upright position ten minutes before qFME. qFME consisted of *in vivo* fluorescence signal visualization and quantification by multi-diameter single fiber reflectance / single fiber fluorescence (MDSFR/SFF) spectroscopy measurements on the mucosa. Additionally, biopsies are collected from non-dysplastic Barrett tissue and (suspected) dysplastic tissue and *ex vivo* analysis will be performed.

Results: Nine patients received 4.5 mg and/or 9 mg bevacizumab-800CW and five patients received 4.5 mg cetuximab-800CW. Oral administration of both tracers was well tolerated by all patients. No incubation time is needed after oral administration after the procedure. A higher fluorescence signal was seen in all visible lesions in vivo. 16 out of 17 endoscopically visible lesions were detected by qFME with bevacizumab-800CW or cetuximab-800CW and were pathologically confirmed to be dysplastic. We will conduct further analysis to evaluate target-to-background ratios and/or contrast-to-noise ratios *in vivo*. Additionally, quantitative *in vivo* fluorescence based on MDSFR/SFF spectroscopy will be assessed.

Conclusion: Our preliminary results show that qFME with oral administration of bevacizumab-800CW and cetuximab-800CW is feasible and, compared to topical administration, seems promising for shortening the procedure time, while remaining adequate in lesion detection.

After completion of the cetuximab-800CW cohort, the optimal dosis and tracer will be determined and compared to combined administration of both tracers. Lastly, depending on what achieves the best *in vivo* results,

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either the optimal tracer or a combination of both tracers will be administered to five patients with non-dysplastic Barrett's esophagus to test the specificity of our tracers.

Disclosure: Nothing to disclose.

PP0108

HISTONE CHAPERONES IN ESOPHAGEAL CARCINOMA: CAUSAL INSIGHTS AND PROGNOSTIC BIOMARKERS THROUGH MULTI-DIMENSIONAL ANALYSIS

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Introduction: Esophageal carcinoma (EC) is an aggressive cancer with poor outcomes, often stemming from disrupted cellular regulators that enhance malignancy[1]. Although histone chaperones are crucial in maintaining genomic and cellular integrity and have been implicated in various solid tumors, their definitive causal roles and therapeutic potential in EC are yet to be fully determined[2].

Aims & Methods: Our study aims to elucidate the causal roles of Histone Chaperone-Related Genes (HCRGs) in EC and assess their potential as prognostic markers and therapeutic targets.

We utilized datasets from TCGA-Esophageal carcinoma, GSE53624, and genome-wide association study data from ebi-a-GCST90018841. We began our analysis by identifying HCRGs linked to EC prognosis through univariate Cox regression. We then used unsupervised cluster analysis on EC samples from the TCGA-EC dataset to identify distinct clusters and subsequently isolated differentially expressed genes (DEGs) within these clusters and compared to normal controls. This identified common DEGs, which were further analyzed using expression quantitative trait loci (eQTL) data to perform a two-sample Mendelian Randomization (MR) analysis, with EC as the outcome. This analysis, enhanced by sensitivity analysis and MR Steiger filtering, helped identify candidate genes whose expression patterns were consistent with their MR properties.

These candidate genes were used to classify EC samples into high and low expression groups, identifying genes with significant survival differences. A prognostic nomogram incorporating clinical features and these genes' expression levels was developed to predict three-year survival probabilities.

Additionally, we performed functional and pathway enrichment analyses to explore the biological processes these genes influence and assessed the immune microenvironment. Validation efforts included immunohistochemistry on EC patient tissues and molecular assays in EC cell lines, confirming the clinical relevance and biological impacts of our findings and highlighting potential diagnostic and therapeutic strategies.

Results: We identified eight key HCRGs in the TCGA-EC dataset, significant for predicting EC prognosis. Analysis revealed 1,742 common DEGs, categorized into 9 risk factors and 17 protective factors via two-sample MR. Among these, ADORA2B and SAPCD2 stood out as crucial hub genes, strongly influencing survival outcomes.

A developed nomogram incorporating pathological stage, nodal classification, and expression levels of these two genes demonstrated good predictive accuracy for three-year survival (AUC=0.749). Immunophenotyping linked ADORA2B with increased dendritic cells and decreased Tregs, while SAPCD2 was associated with higher M0 macrophages and lower Tregs, indicating their influence on the immune landscape in EC.

Clinically, increased expression of ADORA2B and SAPCD2 correlated with adverse outcomes, confirmed by qPCR and immunohistochemistry in paired EC tissues (n=63). Functional studies in cell lines (KYSE150, TE-9) showed ADORA2B's overexpression enhanced cell proliferation, G1/S transition, and migration, altering key proteins like PCNA, CDK4, and cadherins, confirming its oncogenic role in EC.

Conclusion: This study elucidates the causal roles of histone chaperones in EC, identifying ADORA2B and SAPCD2 as risk factors and prognostic biomarkers based on the integrated analysis of transcriptome analysis, MR, clinical sample evaluation, and molecular experiments.

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Disclosure: Nothing to disclose.

PP0109

ESOPHAGEAL SQUAMOUS CELL CANCER IN PLUMMER-VINSON SYNDROME

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Introduction: Plummer-Vinson syndrome is a rare entity that associates dysphagia, iron deficiency anemia, and cervical esophageal membrane. Our work aimed to describe the clinical profile and to specify the predictive factors of degeneration in patients with Plummer-Vinson syndrome associated with esophageal cancer and to show interest in endoscopic surveillance.

Aims & Methods: It was a retrospective descriptive and analytical study over 30 years (1993-2023) including all patients treated and followed up for Plummer-Vinson syndrome and who presented esophageal cancer. Data collection and statistical analysis were performed by SPSS version 20.0 software.

Results: Out of the 160 patients treated for Plummer-Vinson syndrome, 15 patients had associated esophageal cancer resulting in a prevalence of 9.37%. The median age was 50.6 years with extremes of 28 and 74 years, with a sex ratio of M/F = 2/13. The interval between the diagnosis of Plummer-Vinson syndrome and the appearance of cancer was 8 years +/- 7.4.

All our patients with associated cancer had dysphagia with weight loss, odynophagia in 92.9% (n=13), clinical anemia syndrome in 59 %(n=8), and upper gastrointestinal bleeding in 10%(n=1). All patients had hypoferritinemia with a median value of ferritin 6.2µg/l.

The location of the Plummer-Vinson ring was at the level of the Killian triangle at 46%(n=7) and the upper third of the esophagus at 54%(n=8).

The location of the cancerous process at the level of Killian's triangle was in 46% of cases (n=7), and the location at the level of the upper third of the esophagus was in 19.7%(n=3). In 33.3% of cases (n=5) in the middle third. It was squamous cell carcinoma in all cases.

Surgical treatment was initiated in 14% of cases (n=2), radio-chemotherapy in 72% (n=11) and refusal of treatment in 14% of cases (n=2). Two patients progressed well with the disappearance of dysphagia, 10 patients died between 5 months and 5 years of cancer progression, and the rest of the patients could not be followed up.

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In multivariate analysis, the statistically significant predictive factors for the degeneration of Plummer-Vinson syndrome were: Age>50.6 years (OR= 2.37, p=0.003 CI=1.875-4.236), Ferritinemia <6.2 µg/l (OR=3.02, p=0.001 CI=0.977-12. 516) and location of the ring at the level of the Killian triangle (OR=1.140, p=0.001 CI=1.057-1.229). Gender and dysphagia did not appear to be associated with degeneration with (p=0.26) and (p=0.35) respectively.

Conclusion: Plummer-Vinson syndrome is a precancerous lesion requiring endoscopic surveillance. In our series, the prevalence was 9.37%. Malignant transformation of this ring at the level of the triangle of Killian was present in 46% of cases. The predictive factors of degeneration were advanced age, low ferritinemia, and the location of the ring in the mouth of Killian.

Disclosure: Nothing to disclose.

PP0110

PREDICTION OF ADDITIONAL TREATMENT AFTER ESD IN EARLY ESOPHAGEAL CANCER – ASSOCIATION WITH LESION SIZE

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Introduction: ESD is considered as a first-line treatment for early esophageal cancer. Although ESD is expanding its area and replacing surgery, it has complication risk that cannot be ignored and often lead to subsequent surgery or chemoradiation. Currently, in early esophageal cancer, ESD is indicated in the absence of definite SM invasion, but it is difficult to differentiate this by EGD or EUS at the time of diagnosis.

In this study, we aimed to determine the additional indications for selecting ESD candidates based on factors that can be easily identified at diagnosis.

Aims & Methods: We retrospectively analyzed the data of a total 244 cases of patients who underwent esophageal ESD or modified EMR (e.g. precutting EMR, cap-assisted EMR) for dysplasia of the esophagus from 1 center between January 2006 and October 2023 and were diagnosed with esophageal cancer at the final pathology report. Among them, the cases whose final pathology report suggested the need for additional treatment were further analyzed. The indication for additional treatment included SM invasion, vertical margin positive, lymphovascular invasion (LVI), and perineural invasion (PNI). To identify the risk factors associated with additional treatment, we performed multivariable analysis using the logistic regression model. Factors included in the analysis were baseline patient characteristics, endoscopic findings, and pathological findings of the endoscopic resection specimen.

The longest diameter in the final specimen was used as the size of the lesion, and the morphology of the lesion was referred to the gross endoscopic findings at the time of diagnosis.

Results: Of the 244 total cases, the mean age was 65.6 and 219 (89.75%) cases were male. There were 74 cases (30.33%) suggesting indications for additional treatment, which included SM invasion (65 case), vertical margin positive (10 case), LVI (23 case), PNI (0 case), respectively. Regarding lesion size, the number of cases exceeding 15mm, 20mm, 25mm, and 30mm were 98 (40.16%), 45 (18.44%), 27 (11.07%), and 22 (9.02%), respectively. On multivariable analysis, lesion size greater than 15mm was a significant independent risk factor for additional treatment (OR = 2.234, 95% CI 1.127–4.426, p=0.021). Additionally, cutoff sizes greater than 15mm (such as 20mm, 25mm, 30mm), elevated morphology, and moderate or poorly differentiated cell type were also independent risk factors for additional treatment.

	Univariable analysis			Multivariable analysis *		
Predicting factor	OR	[95% CI]	p-value	OR	[95% CI]	p-value
Size						
>=15mm	1.947	[1.123-3.376]	0.018	2.234	[1.127-4.426]	0.021
>=20mm	2.291	[1.181-4.446]	0.014	2.582	[1.145-5.823]	0.022
>=25mm	2.692	[1.197-6.054]	0.017	3.227	[1.225-8.498]	0.018
>=30mm	3.812	[1.551-9.373]	0.004	4.699	[1.604-13.762]	0.005

*Table 1. Multivariable analysis of possible predictor for additional treatment**

adjusted for morphology, and cell type.

Conclusion: Our study suggests that lesion size can be used as an independent risk factor to predict the need for additional treatment after endoscopic resection of early esophageal cancer. However, when applying this size factor, there is a discrepancy between the microscopic size in the resection specimen analyzed in this study and the size measured during the endoscopic procedure. Nevertheless, assuming that the discrepancy is not significant, it seems that superficial esophageal cancer measuring 15mm or more on endoscopy may be recommended for surgery or chemoradiation rather than ESD. Future studies with larger numbers of patients and adjustment for this discrepancy are expected to provide more powerful results.

Disclosure: Nothing to disclose.

PP0111

CLINICAL OUTCOMES AND PROGNOSTIC FACTORS FOR 80 YEARS OR OLDER UNDERGOING ENDOSCOPIC RESECTION FOR ESOPHAGEAL SQUAMOUS CELL CARCINOMA

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Introduction: Endoscopic resection (ER) is widely accepted as the standard treatment for superficial esophageal squamous cell carcinoma (ESCC). In contrast, many old patients, especially 80 years or older, may die as a result of other diseases soon after ER. Therefore, the investigation of clinical outcomes and identification of prognostic factors of patients 80 years or older undergoing ER are clinically important because these findings may inform decision-making regarding the suitability of ER.

Aims & Methods: This study aimed to identify such clinical outcomes and prognostic factors. We retrospectively evaluated the short and long-term outcomes and prognostic factors of 87 patients 80 years or older undergoing ER for ESCC at two cancers between April 2012 and March 2021. We also evaluated the relationships of patient characteristics (sex, age, Eastern Cooperative Oncology Group-performance status; ECOG-PS, American Society of Anesthesiologists physical status; ASA-PS, prognostic nutritional index; PNI, Charlson comorbidity index; CCI) and overall survival rate using the Kaplan-Meier method and a Cox proportional hazards regression model.

Results: The median age of the patients was 82 years, and male was 76 patients (87.4%). The background factors were ECOG-PS 0-1 in 81 patients (93.1%) and a median CCI of 3. As short-term outcomes, 79 patients (90.8%) were curative resections (pT1a-M, V0 and Ly0), 9 patients (10.3%) had adverse events occurred (pneumonia was the most common: 4.6%),

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and no treatment-related deaths occurred. As long-term outcomes, 34 patients died; however, only two patients died due to ESCC. The 3-year overall survival rates and the 5-year overall survival rates were 81.4% and 71.6%, respectively. CCI≥4 (hazard ratio 4.75, 95% CI: 2.03-11.10) was identified as an independent prognostic factor.

Conclusion: The short-term outcomes were good. In contrast, as the long-term outcomes, the overall survival rates of patients with CCI≥4 were poor. The decision to perform ER for patients with CCI≥4 should be carefully considered because follow-up without ER is possible.

Disclosure: Nothing to disclose.

PP0112

THE INCIDENCE AND PROGNOSIS OF RADIATION-INDUCED ESOPHAGEAL CANCER IN BREAST CANCER SURVIVORS: A NATIONWIDE REGISTRY-BASED STUDY

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Introduction: Radiation therapy (Rth) remains an integral part of the multifactorial management of breast cancer (BC). Some reports suggest an increased risk of secondary esophageal cancer (SEC) following Rth for BC, but wide-ranged populational studies on this topic are scarce. We conducted this nationwide registry-based study to analyze the incidence, prognosis, and timing of SEC within BC survivors.

Aims & Methods: We used two high-volume national databases (the National Health Fund Registry and the National Cancer Registry) to identify adult patients with a confirmed diagnosis of primary BC who received oncological treatment either with or without Rth between 2009 and 2019. Those patients were followed-up for the diagnosis of SEC until December 2023. The Fine-Gray competing risk regression models were used to determine and compare the cumulative incidence of SECs in both groups. We applied the Kaplan–Meier method to calculate the 5-year overall survival rates (OS) for SECs in both subsets of BC patients.

Results: We screened 194,598 patients with primary BC, of which 128,625 (66.1%) with available treatment data were included in the final analysis (mean age 61±13yrs.; 99% females). Of these, 89,078 (69%) received Rth within the treatment regime (post-Rth BC) and 39,547 (31%) did not receive Rth (no-Rth BC). The 10-year cumulative incidence of SEC within the post-Rth BC group was 0.06% (95%CI, 0.04%-0.09%), as compared to 0.02% (95%CI, 0.01%-0.04%) within the no-Rth group (Gray test *P*=.005). In a competing risk model, having received Rth was associated with a >2.5-fold increase in the risk of developing SEC relative to no-Rth BC patients (HR 2.67, 95%CI, 1.27-5.65, *P*=.010). The median time to SEC occurrence after Rth completion was 3.7 years (interquartile range [IQR], 1.3-6.6). Squamous cell cancers constituted 89% of SECs in the post-Rth BC group, which was higher than within the no-Rth BC group, of which 73% were SCCs. The prognosis of post-Rth SECs was as poor as for no-Rth SECs, with 5-year OS of 23% (95%CI, 14%-37%) and 13% (95%CI, 4.4%-36%), respectively; *P*=0.3.

Conclusion: Radiotherapy significantly increases the risk of SECs in BC survivors, although the cumulative incidence of this unfavorable event is low. These results raise awareness of SEC after RTH for breast cancer.

Disclosure: Nothing to disclose.

PP0113

COMPARISON OF RESTAGING ACCURACY BETWEEN ENDOSCOPIC ULTRASOUND AND POSITRON EMISSION TOMOGRAPHY IN PATIENTS WITH ESOPHAGEAL CANCER ACHIEVING PATHOLOGICAL COMPLETE RESPONSE AFTER CONCURRENT CHEMORADIOTHERAPY

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Introduction: Endoscopic ultrasound (EUS) and positron emission tomography (PET) have been widely recognized as valuable imaging modalities for tumor-node-metastasis (TNM) staging and treatment decision-making in patients with esophageal cancer. However, the accuracy of EUS and PET in esophageal cancer patients following neoadjuvant concurrent chemoradiotherapy (CCRT) remains inconsistent. Our objective was to conduct a comparative study between EUS and PET to evaluate their accuracy in staging post-CCRT patients who underwent resection surgery and were diagnosed with tumor complete remission.

Aims & Methods: The subjects of this study comprised patients diagnosed with esophageal squamous cell carcinoma and treated with CCRT before surgery at National Taiwan University Hospital (NTUH). Data were collected from May 1, 2007, to May 31, 2022, from a database of 828 esophageal cancer patients who underwent both EUS and PET before and after CCRT, and subsequently underwent resection surgery with pathologically complete response, resulting in a total of 231 patients. After excluding 14 patients who did not receive CCRT, 217 patients were enrolled for further study. The analysis focused on the T and N stages of esophageal cancer as reported by EUS and PET before and after CCRT, correlating clinical stages with pathologic stages. Using pathological diagnosis as the criterion standard, the sensitivity, specificity, positive predictive value, negative predictive value, and accuracy of EUS and PET were calculated. Patients who did not receive EUS or PET before CCRT and those who did not receive EUS or PET after CCRT (n=87) were excluded, resulting in a final inclusion of 128 cases for analysis.

Results: Among the 128 recruited patients, the overall accuracy of EUS for T-stage was 6.3%, while PET exhibited an accuracy of 13.3%. For N-stage, the accuracy of EUS was 20.3%, whereas PET demonstrated an accuracy of 33.6%.

Conclusion: In post-CCRT esophageal patients who achieved complete pathological remission, PET exhibited superior accuracy compared to EUS for both T and N staging; however, neither modality attained satisfactory levels of accuracy.

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Disclosure: Nothing to disclose.

PP0114

CLINICAL FEATURES AND SPECIFIC ENDOSCOPIC FINDINGS OF P16-POSITIVE OROPHARINGEAL CANCER

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Introduction: Transnasal endoscopy by ultrathin endoscope with image-enhanced endoscopy and high image quality has contributed greatly to the diagnosis of superficial cancers of the pharynx due to its ease of use. Its role in the diagnosis of p16-positive pharyngeal carcinoma which is closely associated with HPV infection, and has been recently reported to be increasing, is not clear.
Aims & Methods: We retrospectively investigated the clinical, pathological and endoscopic features of the cases of oropharyngeal cancer at our hospital, who underwent upper gastrointestinal endoscopy in the period between January 2019 to December 2023. All pathological specimens were stained with p16 immunohistochemistry.
(Study 1) Patient background, age, history of alcohol consumption and smoking, and the presence of multiple primary cancers in other organs were compared between the p16-positive oropharyngeal squamous cell carcinoma (p16+OPSCC) group and the p16- negative squamous cell carcinoma (p16-OPSCC) group.
(Study 2) The endoscopic features of the p16+OPSCC lesions were compared with those of the p16-OPSCC lesions.
Results: Sixty patients were included in this study, with a male/female ratio of 48:12 and a mean age of 63.7 years. The mean age of the female patients was 58.5 years, which was significantly younger than that of the male patients (65.0 years).
(Study 1) The p16+OPSCC group consisted of 25 patients with a male/female ratio of 19:6 and a mean age of 58.3 years, and the proportion of female patients was higher than that of the p16-OPSCC group (35 patients, 29:6, 67.5 years). The number of patients without a history of alcohol consumption nor smoking was significantly higher in the p16-OPSCC group (11 and 6, respectively), and 6 patients (24%) had neither. Multiple primary cancers were more frequent in the p16-OPSCC group (17 cases, 49%) than in the p16+OPSCC group (3 cases,12%), mainly found in the hypopharynx and the esophagus.
(Study 2) p16-OPSCC lesions showed brownish areas (BAs) with dotted vasodilatation under narrow band imaging, whereas p16+OPSCC lesions had a unique appearance of subepithelial mass-like, enlarged tonsil-like, or submucosal mass-like with ulceration or nodular elevation. These lesions were accompanied by BA in only 8 cases (32%), and in some cases, the lesions were difficult to even detect by transnasal endoscopic observation.
Conclusion: p16+OPSCC is a new clinical entity that may affect young patients who are not at risk for alcohol consumption and smoking. Our study showed that the endoscopic features of the tumor are not appropriate for BA-based diagnosis. Gastrointestinal endoscopists need to renew their awareness of the disease.
Disclosure: Nothing to disclose.

PP0115

THE USE OF PROTON PUMP INHIBITORS AMONG ADULTS IN NORWAY – A NATIONWIDE DRUG UTILIZATION STUDY

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Introduction: The use of proton pump inhibitors (PPIs) has increased in many Western countries over the past decades. This may reflect the increasing prevalence of symptoms of gastroesophageal reflux disease (GERD) and prophylactic PPI use against gastroduodenal ulcers and haemorrhage induced by non-steroid anti-inflammatory drugs (NSAIDs) or acetylsalicylic acid. However, increasing PPI use against functional or uninvestigated dyspepsia and other inappropriate prescription have previously been documented and may also have contributed to the observed increase in PPI use. In this study we describe the change in incident and prevalent PPI use over time and the indication for prescription.
Aims & Methods: Nationwide data were retrieved from the Norwegian Prescription Database (NorPD) and consisted of information about dispensed prescription drugs. The study population consisted of all adults with at least one PPI prescription during the study period from 2009 to 2022. PPIs were defined by their ATC codes A02BC01 – C05 in addition to M01AE52, which is an esomeprazole and naproxen combination product. Prevalent PPI use was defined as the period covered by the defined daily doses (DDD) in each prescription. The indications for prescription were defined by ICPC or ICD-10 codes depending on whether PPI was prescribed in primary care or secondary health care. National data from the Norwegian Patient Registry (NPR) on all upper endoscopies performed during the study period were combined with NorPD data and it was assessed whether incident PPI users, defined as no PPI dispensal for two years, had undergone an upper endoscopy 180 days before or after incident PPI use.
Results: The use of PPI increased from 58.1 mill DDD in 2009 to 166.9 mill DDD in 2022, i.e. 15700 DDD/1000 and 38700 DDD/1000 persons. The absolute increase in DDDs was driven by a 3.5-fold increase in esomeprazole and a 5-fold increase in the use of pantoprazole.
The prevalence of PPI use increased from 3.5% in 2009 to 9.9% in 2022. The incidence of PPI use increased from 2.85% in 2011 to 5.42% in 2022.
In 2009 the dominant indications in primary and secondary health care, respectively, were esophageal disease and GERD (72.5% and 10.7% of DDDs), gastric ulcer (3.3% and 0.46% of DDDs), duodenal ulcer (0.86% and 0.21% of DDDs), while prescriptions with no reimbursement diagnosis constituted 10.2% of DDDs. In 2022 the dominant indications were esophageal disease and GERD (57.6% and 4.7% of DDDs), musculoskeletal disorders and pain (7.1% and 0.39% of DDDs and in combination with naproxen), prophylaxis against gastroduodenal ulcer (4.8% and 0.95% DDDs), while prescriptions with no reimbursement diagnosis constituted 5.2% of the DDDs.
There were 2.27 mill cases of incident PPI use in the study period and the proportion of PPI users that underwent upper endoscopy around treatment start decreased from 25.3% in 2009 to 11.1% in 2022. In patients with esophageal disease or GERD combined the proportion examined by upper endoscopy fell from 41.4% to 27.1%, while in patients without a reimbursement diagnosis the proportion fell from 19.2% to 6.9%.

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Conclusion: In this nationwide study a 2.8-fold increase in prevalent PPI use was observed during the 14-year study period. This was caused by an absolute increase in PPI use against esophageal disease and GERD as well as prophylaxis in patients given ulcerogenic comedication. The proportion of incident PPI use investigated by upper endoscopy was halved.
Disclosure: Nothing to disclose.

PP0116

EPIDEMIOLOGY OF PROTON PUMP INHIBITOR PRESCRIPTION IN THE CZECH REPUBLIC

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Introduction: Proton pump inhibitors (PPIs) are one of the most frequently prescribed drugs with high risk of the over-prescription.
Aims & Methods: Objective: to find potential sources of the over-prescription by analysis of prescription policy and its evolution in time by means of descriptive statistics
Methods: Analysis from the National Registry of Reimbursed Health Services (2010-2022) was performed targeting reimbursed prescription of ATC group A02BC (PPIs) and A02BA (H2 receptor blockers) with respect to the age of patient and prescribing specialist.
Results: More than 300 million defined daily doses (DDD) of antisecretory drugs were prescribed in 2022; PPIs represent the vast majority (296.2 million, 99 %). This volume rose 2.6 times during the 13 years span, the speed of rise is slowing but still 2.4 % yearly.
PPI were prescribed to 1.314 million of patients in 2022 and this number also rose 1.5 times for 13 years. 48 % of PPI users in 2022 were seniors over 65 years, majority of patients treated by PPIs was in 7th decade of life (24 %), while it was 6th decade till 2018 - age of patients rose in the observed period. PPI was prescribed to more than 12 % of the overall population in 2022, but to 29 % of senior population over 65 years of age!
Vast majority of PPI prescription (in DDD) is realized by general practitioners (GP, 67.8 %). The highest proportion of non-GP prescription (95.5 million DDDs) is realized by: gastroenterology (26.9 %), internal medicine (19.2 %), cardiology (9 %), rheumatology (6.1 %) and ENT (ears-nose-throat, 5.4 %). While overall PPI prescription (in DDD) rose 2.6 times during the last 13 years; higher increment was reached by ENT (18.7 times), allergology/immunology (5.7 times), pneumology (4.4 times); cardiology (4 times) and rheumatology (3.1 times).
Conclusion: Global volume of PPI prescription is overwhelming and covers almost a third of senior population. The rise during 13 years was 2.6 times in DDD and 1.5 times in the number of patients; and probably approaches the plateau. GPs are responsible for longstanding maintenance prescription and realize more than 2/3 of the volume. Prescription by specialist probably mirrors rather introductions of prescription. Prophylactic co-prescription with antithrombotic agents and NSAIDs can be responsible for increase in cardiology and rheumatology. Excessive increase of prescription in ENT (and to less extent pneumology and allergology/immunology) is probably to the account of extraesophageal reflux (EER). Routine long-term prescription of PPI for EER (which has no unequivocal diagnostic criteria so far) is a matter of debate because profit was proven dominantly in patients with simultaneous "classical" gastroesophageal reflux symptoms (1,2). EER can be a target population for de-prescribing strategies.
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Disclosure: Nothing to disclose.

PP0117

EVALUATING GASTRIC FUNCTION AND ENDOSCOPIC CHARACTERISTICS IN CORONARY ARTERY DISEASE PATIENTS POST-PERCUTANEOUS CORONARY INTERVENTION

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Introduction: The relationship between coronary artery disease (CAD) and gastrointestinal function post-percutaneous coronary intervention (PCI) is complex and multifaceted.
Aims & Methods: This study aims to investigate the implications of CAD on gastric function and endoscopic characteristics in patients following PCI, providing insight into potential clinical interventions that could improve outcomes. A longitudinal study encompassed 116 patients with CAD post-PCI, with a mean age of 62.4 ± 12.5 years and 55% male representation. These patients were observed over a mean duration of 2.3 ± 1.5 years. Comprehensive assessments included cardiac function evaluation via echocardiography and biomarker analysis, alongside detailed gastric function tests (gastric emptying studies, pH monitoring) and endoscopic examinations to identify morphological changes in the gastric lining. Statistical analysis involved the use of independent t-tests and chi-squared tests for baseline comparisons, with paired t-tests and the Wilcoxon signed-rank test applied to assess changes over time. The Cox regression model was employed to determine the impact of altered gastric function and endoscopic findings on the prognosis of CAD patients, adjusting for age, gender, and baseline cardiac function.
Results: Our study identified significant gastric functional impairments in patients post-PCI, including delayed gastric emptying and increased gastric acidity ($p<0.01$). Endoscopic findings revealed a higher prevalence of gastritis and gastric erosions in this cohort compared to the general population ($p<0.05$). These gastric alterations were associated with poorer cardiac function and increased incidence of adverse cardiac events. Cox regression analysis further highlighted that specific gastric dysfunctions, particularly delayed gastric emptying and severe gastritis, significantly predicted worse cardiac outcomes (HR=2.3, 95% CI: 1.4-3.8, $p=0.001$), after controlling for conventional risk factors. This study sheds light on the notable impact of CAD and PCI on gastric function and morphology, underscoring the bidirectional relationship between heart disease and gastrointestinal health. The findings suggest that the stress and systemic inflammation associated with CAD and the intervention process itself may contribute to gastric dysfunction. Addressing these gastric issues through targeted therapies could potentially mitigate some of the adverse cardiac outcomes observed in this patient population.
Conclusion: The presence of impaired gastric function and specific endoscopic changes in CAD patients post-PCI underscores the need for a holistic approach to patient care, integrating cardiovascular and gastroin-

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testinal management strategies. Early detection and treatment of gastric dysfunction in this cohort may offer a novel pathway to improving both gastrointestinal and cardiac outcomes, enhancing overall patient quality of life and reducing the burden of CAD.
Disclosure: Nothing to disclose.

PP0118

WITHDRAWN

PP0119

EARLY GASTRIC CANCER IN CENTRAL NORWAY 2001 TO 2016 – A POPULATION-BASED STUDY

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Introduction: Early gastric cancers (EGC) are confined to the gastric mucosa and submucosa irrespective of lymph node metastases. The proportion of gastric cancers diagnosed at an early stage varies considerably between Western countries but is low compared to countries with a screening program for gastric cancer. The majority of EGC patients have been treated with surgical resection including standard lymphadenectomy in Western countries. In contrast, Asian countries have treated larger proportions of EGC with endoscopic submucosal dissection (ESD) followed by estimation of risk of lymph node metastases based on histopathological characteristics. The aim of the present study was to characterize EGCs in Central Norway before ESD was an available treatment option. Furthermore, we aimed to assess histopathological characteristics associated with lymph node metastases and long-term survival in our cohort of EGC patients.
Aims & Methods: The study was a retrospective population-based study using patients identified from the Norwegian Cancer Registry and Norwegian Patient Registry (NPR). The annual number of gastroscopies was retrieved from NPR. 1205 patients were diagnosed with gastric cancer from 2001 to 2016, a period when gastrectomy with a modified D2 resection was standard treatment. In accordance with Western diagnostic criteria, patients with carcinoma in situ were not included. Patients who received neoadjuvant chemotherapy and histologically ypT0 or ypT1 were excluded to study only chemo naïve (true) EGC. Histopathological specimens were re-examined and eCura score risk stratification was calculated. Finally, long-term survival and cause of death were assessed.
Results: Of the 1205 patients with gastric cancer 88 (7.3%) had EGC. The number of gastric cancers declined (p=0.010), but the relative proportion of EGC was unchanged during the study period despite a doubling of annual gastroscopies at a national level during the study period. Patients with EGC had median age 77 (43-91) years vs. 74 (21-99) years for non-EGC controls. EGCs were more often of the Lauren intestinal type (p<0.001) and were more often localized in the antrum or corpus. Eighty-three patients underwent surgical resection and eight (9.0%) of these had N+ disease. An eCura score could be calculated for 79 EGCs. A significant proportion (9.4%, n=5) of the patients with a low risk eCura had lymph node metastases, whereas further exclusion of tumours with histological ulceration or SM2 invasion in a post-hoc analysis identified a N0 cohort (n=36). Recurrence of cancer was diagnosed in seven (8.1%) patients after a median time of 29 (14-124) months from the EGC diagnosis. Fifty-five (62.5%) EGC patients died during follow-up, whereof 43 (78.2%) deaths were unrelated to EGC, seven (12.7%) died of cancer recurrence and five (9.1%) died from postoperative complications. Median survival for EGC patients was 117.1 months (95% CI 99.8-134.3) and 5-year overall survival 75%. Approximately half of deaths related to cancer recurrence (3 of 7) occurred more than five years after the EGC diagnosis.
Conclusion: EGCs constituted a minor but constant proportion of gastric cancers despite a considerable increase in gastroscopies. In this historic cohort with formal gastric resection for EGC, 9.0% had pN+ disease. Low risk eCura score alone was insufficient to predict pN0 disease whereas additional exclusion of patients with histological ulceration or SM2 infiltration identified a pN0 cohort. Long-term follow-up of more than five years is needed to assess EGC related mortality.
Disclosure: Nothing to disclose.

PP0120

LONG-TERM PROGNOSIS OF PATIENTS WITH GASTRIC MUCOSA-ASSOCIATED LYMPHOID TISSUE LYMPHOMA HARBORING EXTRA COPIES OF THE *MALT1* GENE

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Introduction: Trisomy and tetrasomy 18 are genetic aberrations characterized by the presence of one or two additional copies of chromosome 18 within a subset or in the entirety of cellular components. This genomic anomaly is a prevalent karyotypic irregularity in extra nodal marginal zone lymphoma of mucosa-associated lymphoid tissue (MALT lymphoma), which is a type of lymphoma that originates from mucosa-associated lymphoid tissue in various organs, including the stomach, lungs, eyes, and thyroid. In previous investigation, we examined 146 patients with MALT lymphoma of the stomach who underwent FISH analysis for t (11, 18) translocation[1]. However, the importance of trisomy and tetrasomy 18 in gastric MALT lymphoma is not fully understood because of the lack of long-term patient monitoring focused on genetic abnormalities. Therefore, in this study, we extended our investigation by tracking the same cohort of patients for an additional period exceeding 5 years to analyze the long-term prognosis of patients with gastric MALT lymphoma harboring trisomy/tetrasomy 18.
Aims & Methods: The objective of this study was to clarify the long-term prognosis of patients with gastric mucosa-associated lymphoid tissue (MALT) lymphoma with additional copies of *MALT1*. In this multicenter retrospective study, we enrolled 145 patients with gastric MALT lymphoma who underwent fluorescence in situ hybridization (FISH) analysis to detect t (11;18) translocation.

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Results: The patient cohort was divided into three groups: Group A (n=87), comprising individuals devoid of the t (11;18) translocation or extra *MALT1* copies; Group B (n=27), encompassing patients characterized by the presence of the t (11;18) translocation; and Group C (n=31), including patients with extra *MALT1* copies. The clinical outcomes in each cohort were collected. Over the course of a mean follow-up of 8.5±4.2 years, one patient died of progressive MALT lymphoma, while 15 patients died due to etiologies unrelated to lymphoma. The progression or relapse of MALT lymphoma was observed in 11 patients: three in Group A, two in Group B, and six in Group C. In Groups A, B, and C, the 10-year overall survival rates were 82.5%, 93.8%, and 86.4%, respectively, and the 10-year event-free survival rates were 96.1%, 96.0%, and 82.9%, respectively. The event-free survival rate in Group C was significantly lower than that in Group A. However, no differences were observed in the 10-year event-free survival rates among individuals limited to stage I or II_1 disease (equivalent to excluding patients with stage IV disease in this study, as there were no patients with stage II_2), with rates of 98.6%, 95.8%, and 92.3% for Groups A, B, and C, respectively.

Conclusion: The presence of extra copies of *MALT1* was identified as an inferior prognostic determinant of event-free survival.

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Disclosure: Nothing to disclose.

PP0121

EFFECT OF DIETARY ANIMAL PROTEIN ON DEVELOPING GASTRIC CANCER CONSIDERING *HELICOBACTER PYLORI*: MULTI-CENTER PROSPECTIVE STUDY

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Introduction: Dietary animal protein effect on gastric cancer (GC) taking *Helicobacter pylori* status into account has rarely been reported and has therefore been investigated in this study.

Aims & Methods: GC patients and healthy subjects were prospectively registered by 6 hospitals from October 2016 to December 2018. Multivariate analysis adjusted for many epidemiological factors (Model I) and additionally *H. pylori* (Model II) was conducted to investigate dietary protein effect on GC. Dietary animal protein groups included red meat, processed meat, white meat, eggs, and fishes.

Results: A total of 5,535 subjects (1,629 GCs) were enrolled. In Model I, red meat, fish, and eggs showed non-linear associations. Two serving/week for red meat, 1 serving/week for fish, 2-3 serving /week for egg was associated with the lowest risk of GC. Both low (aOR, 1.33) and high intake (aOR, 1.49) of red meat were associated with increased risk of GC. Both low (aOR, 1.76) and high intake (aOR, 1.74) of fish were associated with increased risk of GC. Both low (aOR, 3.07) and high intake (aOR, 1.33) of egg were associated with increased risk of GC. High intake of processed meat was associated with GC (adjusted odd ratio [aOR], 1.78; 95% CI, 1.27-2.49). Frequent intake of white meat was associated with GC risk (aOR, 1.67 for 1 serving/week, 2.15 for 2 servings/week, and 4.16 for ≥3 servings/week). In model II, these non-linear or positive association were sustained. Sub-analysis by sex showed similar association in both men and women.

Conclusion: Dietary animal proteins had non-linear or positive association with GC after adjusted for *H. pylori.* Nonlinear relationships of some food groups with GC suggests modest intake of such foods may reduce GC risk.

Disclosure: No COI.

PP0122

SURVIVAL AMONG PATIENTS CURED FROM GASTRIC ADENOCARCINOMA COMPARED TO THE BACKGROUND POPULATION

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Introduction: It is unknown if gastric adenocarcinoma survivors have longer, shorter, or similar survival compared to the background population. This knowledge could contribute to evidence-based monitoring strategies, healthcare recommendations, and information for patients and families.

Aims & Methods: We aimed to help clarify if the survival among survivors of gastric adenocarcinoma is different from the corresponding background population. This population-based cohort study included all patients who underwent gastrectomy for gastric adenocarcinoma between 2006-2015 in Sweden and survived ≥5 years after surgery who did not later die from tumour recurrence. They were followed up until death, postoperative year 10, or end of study period (31 December, 2020). The observed survival was divided by the expected survival, yielding relative survival rates with 95% confidence intervals (CIs) using the life table method. The expected survival was derived from the entire Swedish population of the corresponding age, sex, and calendar year. Data came from medical records and nationwide registers.

Results: The survival among all 733 gastric adenocarcinoma survivors was shorter than the expected. The reduction in relative survival increased for each follow-up year, from 98.7% (95% CI: 97-100.5%) year 6 to 90.9% (95% CI: 86.7-95.2%) year 10. The decline in relative survival was more pronounced among patients who had gastrectomy in earlier calendar years (88.4% [95% CI: 83.0-93.9%] year 10 for years 2011-2015), shorter education (87.9% [95% CI: 80.1-95.7%] year 10 for education ≤9 years), more comorbidities (82.5% [95% CI: 68.2-96.7%] year 10 for Charlson comorbidity score ≥2), and no neoadjuvant therapy (87.7% [95% CI: 81.9-93.5%] year 10).

Years after gastrectomy	Alive at the beginning of each year	Observed survival in % (95% CI)	Expected survival in %	Relative survival in % (95% CI)
6	733	94.5 (92.9-96.2)	95.7	98.7 (97.0-100.5)
7	625	89.0 (86.6-91.3)	92.0	96.7 (94.1-99.2)
8	537	85.1 (82.4-87.8)	88.9	95.8 (92.7-98.8)
9	444	80.3 (77.2-83.5)	86.0	93.4 (89.8-97.1)
10	350	75.8 (72.2-79.3)	83.3	90.9 (86.7-95.2)

Conclusion: Gastric adenocarcinoma survivors seem to have poorer survival than the corresponding background population, particularly in certain subgroups.

Disclosure: Nothing to disclose.

PP0123

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PP0124

KEY INDICATORS FOR GLOBAL GASTRIC CANCER ENDOSCOPIC SCREENING PROGRAMS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Discrepancies in gastric cancer (GC) mortality reduction are evident across global GC screening programs. Specifically, the reduction brought by endoscopic screening nearly doubled in Japan and Korea compared with China.

Despite studies indicating that economic status, screening methods, and public health awareness may influence screening efficiency, there is no comprehensive study that has summarized and compared the outcomes of available screening programs and identified factors potentially influencing the cancer-preventive effects.

Aims & Methods: Here we present data, including response to invitation, uptake of endoscopy screening and cancer screening positivity, of global GC screening programs. By pooling three key indicators throughout screening process, including endoscopic compliance rate, GC detection rate and early diagnosis rate, we conducted a meta-analysis of global GC endoscopic screening programs, in order to compare the outcomes of available screening programs and identify possible influencing factors.

PubMed, Embase, Web of Science, and Scopus were searched for relevant articles published until September 20, 2023. Original articles providing enough information to calculate the indicators in the general screening population were included.

Pooled random effect estimates of the indicators were computed, and further subgroup analysis stratified according to study period, country or region, setting, screening interval, education level and socioeconomic status was conducted. This study was registered with PROSPERO (CRD42023464921).

Results: 90 studies (59 reported endoscopic compliance rate, 68 reported GC detection rate and 33 reported GC early diagnosis rate) were included. Overall, the rates of endoscopic compliance, GC detection, and early-cancer diagnosis were 46.94% (95%CI 41.35-52.53%), 0.62% (95%CI 0.46-0.80%), and 61.41% (95%CI 51.42-71.40%), respectively.

Subgroup analysis demonstrated a gradual increase in endoscopic compliance and a counterintuitive decline in GC detection rate over time. Stratified by countries, the three indicators in Japan (51.05% [37.06-65.05%] for endoscopic compliance rate, 0.89% [0.55-1.31%] for GC detection rate, and 78.98% [68.83-87.66%] for GC early diagnosis rate) and Korea (49.78% [38.77-60.80%] for endoscopic compliance rate, 1.07% [0.05-3.38%] for GC detection rate, and 69.84% [61.51-77.57%] for GC early diagnosis rate) were significantly higher in China (39.68% [31.84-47.52%] for endoscopic compliance rate, 0.51% [0.33-0.74%] for GC detection rate, and 51.35% [36.33-66.37%] for GC early diagnosis rate).

The highest endoscopic compliance and GC detection rate was reported in hospital, while GC early diagnosis rate was higher in community and health checkup. Screening programs with 2-3 years intervals, public health campaigns, higher education levels, and socioeconomic status showed higher compliance. Screening programs based on new endoscopic techniques generally reported higher GC detection and early diagnosis rates.

Conclusion: Among high-incidence countries for GC, China lags behind Japan and Korea regarding endoscopic compliance rate, GC detection rate and early diagnosis rate. Screening settings, intervals, campaigns, education and socioeconomic stratum may affect endoscopic compliance, while GC detection and early diagnosis rates are mainly influenced by screening settings and endoscopic techniques. In the future, continuous monitoring of quality control indicators should be applied in GC screening programs and the screening strategy should be adjusted accordingly to improve screening effectiveness.

Disclosure: Nothing to disclose.

PP0125

A MATHEMATICAL SIMULATION MODEL DETERMINES THE OPTIMAL ENDOSCOPIC SCREENING STRATEGY FOR THE DETECTION OF *HELICOBACTER PYLORI*-NAÏVE GASTRIC NEOPLASM

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Introduction: In recent years, as the rate of *Helicobacter pylori (H. pylori)* infection has declined, the number of *H. pylori*-related gastric neoplasms (HpRGNs) has decreased. Although the number of *H. pylori*-naïve gastric neoplasms (HpNGNs) is estimated to be small compared to that of HpRGNs, the exact prevalence has not been unknown. Therefore, the effectiveness of screening by esophagogastroduodenoscopy (EGD) in cohorts with low *H. pylori* infection rate has not been revealed.

Aims & Methods: The aim of this study is (1) to determine age-related HpNGN prevalence by using a large-scale Japanese cohort and (2) to develop the optimal EGD screening strategy to detect HpNGNs in terms of screening efficiency.

Database of EGD for health checkup from 12 institutions in Japan between 2016 and 2022 was retrospectively reviewed. Detailed information on identified HpNGNs was investigated; age, sex, pathological diagnosis (pure signet ring cell carcinoma (PSRCC), gastric adenocarcinomas of the fundic gland type (GA-FG), foveolar gastric adenoma-raspberry type (FGA-raspberry type), and undifferentiated type cancer), lesion size, lesion depth, lymphovascular invasion, and presence of lymph node metastasis, and EGD images. EGD images of patients who underwent multiple EGDs in the past were followed retrospectively for changes in tumor size. Tumor growth speed, missing rate, age-related HpNGNs prevalence, and detection threshold size were calculated from database.

A virtual cohort of 100,000 births in the first year and thereafter reflecting the Japanese fertility trend was constructed based on these parameters. In this virtual cohort for 70 years of follow-up, tumor detection rates and

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number-need-to-test (NNT) for PSRCC, GA-FG, and undifferentiated cancers at different EGD screening start ages (40, 45, and 50 years) and screening intervals (2, 5, and 10 years) were compared.

Results: A total of 519,368 EGD cases were initially screened and 113 HpNGN lesions were detected in 108 patients (0.021%). HpNGN was found in all generations from 20s to 80s (0.011-0.028%). The age-related prevalence varied widely by HPNGN type; the prevalence of PSRCC was highest (0.022%) in 60s, but the prevalence of GA-FG gradually increased after the age of 50 and was not detected in people under 40. Additionally, undifferentiated cancer was discovered in people in their 30s and 50s, but not in people over 60. Tumor characteristics were analyzed for the construction of the simulation model by investigating the 97 of those lesions (34 PSRCC cases, 26 GA-FG cases, 30 FGA-raspberry type cases, and 7 undifferentiated type cases). The tumor growth rate was 2.5 mm/year for PSRCC, 2.7 mm/year for GA-FG, 1.2 mm/year for FGA-raspberry type and 11.7 mm/year for undifferentiated carcinoma. The missed rate was 46.1% for PSRCC, 65.2% for GA-FG, 56.5% for FGA-raspberry type and 17.0% for undifferentiated type. Among the strategies with detection rates of 50% or higher in the virtual cohort, the NNT of the 5-year interval screening strategy starting at age 45 years was the lowest for all tumor types (PSRCC 4.89, GA-FG 9.92, undifferentiated cancer 4.69). By adopting this strategy, most of HpNGNs were detected in less than 20 mm in size and the deep submucosa invasion rate was less than 30%.

Conclusion: A large-scale Japanese cohort demonstrated that prevalence of HpNGN was very low (0.021%). Mathematical simulation modeling revealed that the strategy of screening every 5 years starting at age 45 was most efficient to detect HpNGN.

Disclosure: Nothing to disclose.

PP0126

DIFFERENT SURVIVAL RATES OF PATIENTS WITH GASTRIC CANCER DEPENDING ON AGE AND SEX: A LARGE-SCALE STUDY

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Introduction: The sex differences in the incidence of gastric cancer have been well known. However, uncertainty still exists in terms of prognosis of GC. For example, the US SEER data reports a significantly better prognosis in females than in males, but the prognosis is better in males than in females in Korean Central Cancer Registry data. Roles of molecular markers and gut microbiota as new prognostic factor are emerging recently, but few studies seem to try to reveal sex differences.

Aims & Methods: In this background, we aimed to analyze the differences in GC characteristics according to sex in Korean patients, mainly focused on survival. This was a single-center retrospective study. A total of fifteen thousand patients diagnosed with GC at the Seoul National University Bundang Hospital between 2003 and 2023 were included. Clinicopathological characteristics, histologic types, survival rates with age- and stage-stratification, and associated risk factors were analyzed. Information related to death was cross-verified using medical record review and data from the National Statistical Office.

Results: The ratio of males and females was about 2:1, and there was no differences in stage between sexes. Female patients were significantly younger than males, had more gastric body cancers and tumors with diffuse type or poorly differentiated histology. Females underwent less endoscopic treatment and more surgery, and tumors with microsatellite instability was more common in females, while mutation of p53 was less common. In histological types according to Lauren classification, diffuse type histology was more common in younger patients in both males and females. The proportion of intestinal type GC increased with age, but the proportion increased rapidly in males while increased gradually in females. In Cox multivariate regression analysis for GC-related death, older age, proximal tumor location, diffuse type histology and advanced tumor stage were risk factors. In terms of survival, overall and GC-specific survival did not show significant differences between sexes in the analysis of the total population. However, sex differences in survival reversed with age, that younger females under 50 showed survival disadvantages, but the survival became similar in 50s, and older females over 60 showed survival advantages in both overall and GC-specific survival. Furthermore, in the stratification analysis according to the stage, there were no significant differences in stages I to III. But, stage IV female patients showed significant survival disadvantages. In subgroup analysis of stage IV patients, the differences between males and females in median survival were two months, and female patients were significantly younger and had a higher proportion of diffuse type GC than male patients, which shows aggressive feature and is related to poorer survival.

		Total (N=1,924) (%)	Male (n=1,282) (%)	Female (n=642) (%)	p
Median survival (mo)		10	11	9	0.022
Age (mean±SD)		61.59±14.25	62.41±13.23	59.96±15.97	0.001
Age (in diffuse type patients, mean±SD)		54.55±14.71	56.43±14.29	52.26±14.91	0.001
Lauren classification	Intestinal	554 (48.0)	438 (56.7)	116 (30.4)	<0.001
	Diffuse	574 (49.7)	315 (40.7)	259 (68.0)	
	Mixed / Indeterminate	26 (2.3)	20 (2.6)	6 (1.6)	

Conclusion: Significant differences in survival between sexes were observed with stratification analysis, although sex itself was not an independent prognostic factor. These age-dependent changes are thought to suggest the effect of sex hormones. These differences should be considered in clinical practice and later research in terms of tailored medicine.

Disclosure: Nothing to disclose.

PP0127

VALIDATION OF SERUM PEPSINOGEN AND *HELICOBACTER PYLORI* TEST IN THE DETECTION OF DIFFUSE-TYPE GASTRIC CANCER AMONG 23,130 HEALTH CHECKUP SUBJECTS IN SOUTH KOREA

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Introduction: Serum pepsinogens (PGs) are the biomarkers for atrophic gastritis and severe gastric inflammation.

Aims & Methods: The aim of this study was to validate the role of PGs in gastric cancer (GC) screening. 1,705 gastric neoplasms (1,287 GC and 418 gastric adenoma (GA)) and 21,425 controls (total 23,130) were enrolled.

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PGs and anti-*Helicobacter pylori* (HP) antibody were measured. Area under the curve and receiver operating characteristic curve were calculated to identify the optimal cutoff values for PGs. Multivariate logistic regression and risk stratification were conducted by combining each PG value and anti-HP antibody in GC subtypes.

Characteristics		Total (23,130) (%)	Control (21,425) (%)	Gastric Adenoma (418) (%)	GC (1,287) (%)	p	Intestinal-GC 762 (%)	Diffuse-GC (489) (%)	p
Sex	Female	8,897 (38.5)	8,358 (39.0)	134 (32.1)	405 (31.5)	<0.001	175 (23.0)	219 (44.8)	<0.001
	Male	14,233 (61.5)	13,067 (61.0)	284 (67.9)	882 (68.5)		587 (77.0)	270 (55.2)	
Age (mean)		54	54	63	62	<0.001	64	55	<0.001
HP status	Negative	8,119 (35.1)	7,857 (36.7)	54 (12.9)	208 (16.2)	<0.001	137 (18.0)	61 (12.5)	0.008
	Positive	15,009 (64.9)	13,566 (63.3)	364 (87.1)	1,079 (83.8)		625 (82.0)	428 (87.5)	
	Missing	2 (0.0)	2 (0.0)	0 (0.0)	0 (0.0)		0 (0.0)	0 (0.0)	
PG series (ng/mL)	PGI	52.1 (28.8)	52.4 (27.9)	40.3 (38.85)	50.6 (50.2)	<0.001	44 (42.65)	60.2 (56.75)	<0.001
	PGII	11.2 (10.8)	11.0 (10.3)	12.9 (11.7)	17.5 (19.2)	<0.001	16 (16.63)	21.2 (21.0)	<0.001
	PGI/II ratio	4.6 (2.6)	4.8 (2.6)	3.0 (2.4)	2.9 (2.3)	<0.001	2.9 (2.4)	2.9 (2.1)	0.328

GC, gastric cancer; HP, *Helicobacter pylori*; PG, pepsinogen.

Results: PGII ≥21 ng/ml was associated with the risk of GA (odds ratio (OR) 1.38, p<0.001), GC (OR 2.77, p<0.001), diffuse-type GC (DGC) (OR 4.26, p<0.001), especially, early-stage DGC (OR 7.11, p<0.001). When GC risk was assessed by combining each PG value and HP antibody, OR showed significance between categories, and even stepwise increment from HP negative to positive and from each PG value negative to positive after adjustment with other confounding GC risk factors.

Conclusion: Each PG value combining HP test can be used as a screening test for high risk subjects with GC, and especially the combination of PGII≥21ng/ml and positive HP status in young female was related to high risk of early-stage DGC.

Disclosure: Nothing to disclose.

PP0128

GASTRIC CANCER RISK AMONG IMMIGRANTS AND SOCIOECONOMIC GROUPS IN THE NETHERLANDS

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Introduction: Gastric cancer (GC) often presents at advanced stages with high rates of metastasis and poor clinical outcomes. Identification of groups at a high-risk of GC could facilitate screening in countries with a low GC incidence. Current understanding of which groups in the Netherlands are at an elevated risk is limited.

Aims & Methods: Our aim was to identify such high-risk groups, based on individual-level population data on migration status and socioeconomic status (SES). Patient data from the Netherlands cancer registry were linked to demographic data of Statistics Netherlands in the period 2010-2022. GC incidence rates in the 14 largest immigrant populations were compared to those born in the Netherlands. Odds ratios (ORs) were computed per birthplace and corrected for age, sex and SES. Additionally, we investigated GC risk among second-generation immigrants and by SES.

Results: A total of 21,299 GC patients were recorded in the Dutch cancer registry between 2010 and 2022. Immigrant populations at a significantly higher GC risk compared to the general population were identified (Table 1). These include people born in Bosnia-Herzegovina (OR: 2.42), Turkey (OR: 2.22), Afghanistan (1.92), China (OR: 1.92), Surinam (1.68), Morocco (1.63), and the Dutch Antilles (1.61). Low SES increased the odds of developing GC, with an OR of 1.48 in individuals with the lowest SES compared to those with the highest SES. Still, first-generation migrants remained at a higher risk when adjusting for SES. Second-generation migrants did not have a significantly higher risk of developing GC.

Birth place	All	Afghanistan	Turkey	Bosnia-Herzegovina	China	Morocco	Surinam	Iran	Poland	Netherlands	Iraq	Belgium	United Kingdom
Gastric Cancer (GC)	10.10	21.09	20.11	19.46	17.28	17.17	12.68	12.49	9.10	8.43	8.25	6.66	6.23
Non-cardia GC	4.87	13.1	12.28	13.51	11.95	10.87	8.86	8.11	5.64	3.89	4.99	4.15	2.62
cardia GC	3.10	5.16	3.20	2.99	2.64	3.14	1.27	2.01	0.40	2.76	1.87	2.00	1.72
Other GC	2.13	2.82	4.63	2.96	2.68	3.16	2.55	2.38	3.06	1.78	1.39	0.51	1.89

Table 1: Age-standardized incidence rates per 100,000 for immigrants from selected countries in the Netherlands in the period 2010-2021.

Conclusion: Individuals born in countries with a high GC incidence remain at an elevated risk despite migration to a low-risk region, likely due to prior infection of *H. pylori*. Potential benefits of targeted *H. pylori* test-and-treat in immigrant populations should be explored in clinical and modelling studies. Primary caregivers should be cognizant of high-risk immigrant and socio-economic groups, facilitating the early detection of cancer within these vulnerable populations.

Disclosure: Rob H.A. Verhoeven received research grant from Bristol Meyers Squibb and consultancy fee from Daiichi Sankyo, all paid to institution.

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PP0129

OCCURRENCE AND CHARACTERISTICS OF ENDOSCOPIC GASTRIC POLYPS IN PATIENTS WITH AUTOIMMUNE ATROPHIC GASTRITIS: A MULTICENTRIC CROSS-SECTIONAL STUDY

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Introduction: Autoimmune atrophic gastritis (AAG) leads to significant changes in the gastric mucosa, including hypo-achlorhydria and increased gastrin (G) levels, providing proliferative *stimuli* on the gastric mucosa (1-5).

Aims & Methods: The aim of our study is to evaluate the occurrence and characteristics of gastric polyps in AAG patients across six tertiary centers in Italy. We conducted a multicentric, cross-sectional study from January to June 2023, prospectively enrolling consecutive patients diagnosed with AAG, from January 2000 to June 2023, who underwent at least one endoscopy. Data on demographics, clinical history, biochemical profiles, serological data, and endoscopic and histopathological findings were collected.

	AAG overall population (#612)	AAG patients without lesions (#390)	AAG patients with non-NEN polyps (#162)	AAG patients with gNENs (#108)
Sex, female (%)	452 (73.9)	280 (71.6)	280 (71.6)	78 (72.2)
Age (years), median (IQR)	66 (54–74)	61.5 (51-70)	70 (58.7-76.2)	70 (58.7-76.2)
APCA positivity, n (%)	496 (81)	386 (99.1)	125 (77.2)	76 (70.4)
Current smokers, n/n available (%)	48/185 (21.8%)	13/96 (13.5%)	13/96 (13.5%)	13/96 (13.5%)
PPI treatment, n/n available (%)	48/137 (25.9%)	29/98 (29.6%)	7/41 (17.1%)	10/41 (24.4%)
OLGIM, n (%)	II: 245 (40)	II: 144 (37)	II: 63 (39)	II: 60 (56)
OLGA, n (%)	II: 325 (53)	II: 179 (46)	II: 92 (57)	II: 76 (70)
Chromogranin A (ng/mL), median (IQR)	153 (75.5-269.3)	139 (68-221)	139 (68-221)	160 (86.9-251.5)
Gastrin (pg/mL), median (IQR)	667 (336.5-1156.5)	589.0 (281.8–1071.5)	688 (425.8–1200)	874 (362.5-1335)

Conclusion: This study represents the largest multicentric cohort of AAG patients, highlighting the significant presence of gastric polyps, including gNENs and various non-endocrine types including a not negligible proportion of patients who showed gastric adenocarcinoma. These results emphasize the importance of proactive endoscopic monitoring and comprehensive histopathological evaluation in patients with AAG, to effectively identify and manage these different gastric lesions.

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PP0130

ASSESSING THE FREQUENCY OF TYPE I GASTRIC NEUROENDOCRINE NEOPLASMS IN AUTOIMMUNE ATROPHIC GASTRITIS: A MULTI-CENTER STUDY IN ITALY

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Introduction: In autoimmune atrophic gastritis (AAG), immune-mediated morpho-functional changes in gastric mucosa lead to hypo-achlorhydria and subsequent hypergastrinemia. This stimulates gastric enterochromaffin-like (ECL) cell proliferation, escalating the risk of type I gastric neuroendocrine neoplasms (gNENs). Yet, longitudinal studies on the AAG-type I gNENs association are still scanty.

Aims & Methods: Our study aims to assess in a multicentric cohort of AAG patients, the frequency and the characteristics of type I gNENs.

We conducted a multicentric, cross-sectional study from January to June 2023, prospectively enrolling consecutive patients diagnosed with AAG, from January 2000 to June 2023, who underwent at least one endoscopy. Data on demographics, clinical history, biochemical profiles, serological

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data, and endoscopic and histopathological findings were collected.

Results: The study included 612 patients with AAG [444 (75.2%) female, median age 66 (IQR 54-73) years]. At enrollment, 355 (58.1%) patients had ECL cell hyperplasia. Over a median follow-up of 4 (IQR 2-7) years, and after a median number of two upper GI endoscopies per patient, and a median disease duration of 7 (IQR 3-8) years, 108 patients showed 125 gNENs, corresponding to an incidence rate of 0.051 person-years. All the gNENs were well-differentiated. In 22 (18%) cases Ki 67 was not available; of the remaining 103, 86 (68.8%) cases were G1, 17 (13.6%) were G2, all of which with a Ki-67 < 10%; no G3 lesions were observed. 119 (95.2%) gNENs were located in the body/fundus. The median size was 5 (IQR 3-10) mm. 79 (73.1%) patients underwent endoscopic removal, 10 (9.2%) gastrectomy, 12 (11.1%) somatostatin analog therapy due to multifocal/recurrent disease, and 7 (6.5%) were only monitored. Patients with AAG and gNENs had significantly higher gastrin levels compared to patients without gNENs [median 874 pg/mL (IQR 362.5-1335) vs 581.5 pg/mL (IQR 281.8-1071.5), p=0.014]. However, there were no significant differences in proton-pump inhibitor use, OLGA and OLGIM stages, H. pylori status, and notably, circulating chromogranin A levels.

	AAG patients with gNENs (#108)
Sex, female (%)	78 (72.2)
Age (years), median (IQR)	68 (57-75.8)
APCA positivity, n (%)	76 (70.4)
Current smokers, n/n available (%)	19/57 (33%)
PPI treatment, n/n available (%)	10/41 (24.4%)
OLGIM, n (%)	II: 60 (56)
OLGA, n (%)	II: 76 (70)
Chromogranin A (ng/mL), median (IQR)	160 (86.9-251.5)
Gastrin (pg/mL), median (IQR)	874 (362.5-1335)

Conclusion: Type I gNENs are a relevant complication in patients with AAG. Elevated circulating gastrin levels are linked to their occurrence, with hypergastrinemia acting as a driver for abnormal proliferation of gastric ECL cells. Regular endoscopic surveillance and thorough histopathological examination are crucial in patients with AAG, serving as a key strategy for early diagnosis and treatment of gNENs.

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PP0131

EBV, TUMOR LOCATION, AND HIGHER TNM STAGE AFFECTED PD-L1 EXPRESSION OF GASTRIC CANCER IN ONE TERTIARY HOSPITAL IN SOUTH KOREA

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Introduction: Programmed cell death ligand-1 (PD-L1) is known to negatively regulate T-cell activation. Immune checkpoint inhibitors (ICIs) such as nivolumab and pembrolizumab have prolonged survival in gastric cancer (GC) as well as malignant melanoma and lung cancer.

Aims & Methods: The aim of this study was to investigate the affecting factors on the expression of PD-L1 in GC tissue in the Korean population in terms of clinicopathological variables.

Total 216 patients with GC who underwent PD-L1 immunohistochemistry (22C3) were prospectively enrolled at Seoul National University Bundang Hospital from March 2018 to February 2024. Positive PD-L1 expression was defined as a CPS score of 5 or higher. Univariate and multivariate analysis was performed.

Results:

		Univariate analysis		Multivariate analysis	
Variables		**OR (95% CI)**	***p* value**	**OR (95% CI)**	***p* value**
EBV infection	EBV negative	Ref.		Ref.	
	EBV positive	8.54 (1.59-158.42)	**0.043**	14.01 (2.23-278.16)	**0.019**
Tumor location	Non-cardia	Ref.		Ref.	
	Cardia	2.78 (1.18-7.35)	**0.026**	4.97 (1.17-16.08)	**0.005**
TNM stage	Stage I	Ref.		Ref.	
	Stage II,III, or IV	1.38 (1.00-1.92)	**0.049**	1.61 (1.09-2.42)	**0.018**

Table.Univariate and Multivariate analysis for affecting factors on PD-L1 expression.

Multivariate regression (Table) showed some few significant associations between PD-L1 expression and specific factors. EBV infection shows a strong effect in both univariate (odd ratio [OR] 8.54, 95% confidence interval [CI] 1.59-158.42, *p*=0.043) and multivariate analysis(OR 14.01, CI 2.23-278.16, *p*=0.019). Cardia portion involvement shows a significant correlation in both univariate (OR 2.78, CI 1.18-7.35, *p*=0.026) and multivariate analysis(OR 4.97, CI 1.71-16.08, *p*=0.005). And higher TNM stage also remains significant in both univariate (OR 1.38, CI 1.00-1.92, *p*=0.049) and multivariate analysis (OR 1.61, CI 1.09-2.42, *p*=0.018).

Conclusion: EBV infection, cardia involvement, and higher TNM stage were independent factors influencing PD-L1 expression.

Disclosure: Nothing to disclose.

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PP0132

GENOTYPE-PHENOTYPE CORRELATION FOR GASTRIC AND DUODENAL NEOPLASTIC LESIONS IN PATIENTS WITH FAMILIAL ADENOMATOUS POLYPOSIS (FAP)

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Introduction: FAP is a disorder characterized by an autosomal dominant mutation of the APC gene. Duodenal adenomas are indeed found in 60–90% of patients with FAP. Gastric polyps in FAP patients are found in 30–100% of cases and most of them are fundic gland polyps. Some patients have a neoplastic phenotype (low grade/high grade dysplasia or adenocarcinoma). Shimamoto et al. tried to explain this condition: they subdivided the APC gene into six groups (from A to F), each one corresponding to a codon range, and evaluated the phenotypic correlation for each of these regions. They found an association between APC variants in codons 1398– 1580 (Region E) and neoplastic gastric and duodenal lesions.

Aims & Methods: The aim of our pilot study is to investigate the correlation between the different APC mutations and the presence of neoplastic gastric and duodenal phenotype in our case series of FAP patients.

We analysed 37 patients followed by our center for an average of 10 years: the APC gene was subdevided into six groups (from A to F) according to Shimamoto er al, each one corresponding to a codon range, and we studied the correlation between genotype and phenotypic manifestation of gastric and duodenal lesions according to their clinical history.

Results: We collected two patients with a mutation in region E: both developed gastric polyposis with low grade dysplasia (LGD) and high grade dysplasia (HGD) lesions of the duodenum. We also observed that two out of the three patients with mutation in region D (1250–1398) progressed to gastric cancer and duodenal cancer respectively. In Shimamoto's study region B (454–1020) appears to be associated with a low risk of developing gastric lesions and a moderate risk of duodenal polyposis (Spigelman ≥ III), while our analysis shows that two (out of 12) of the patients mutated in this region developed gastric cancer and two developed duodenal cancer. In our case series two patients had the mutation at the promoter level of the APC gene, and this correlates with GAPPS (Gastric Adenocarcinoma and Proximal Polyposis of the Stomach): a newly discovered disease that has a very high risk of gastric adenocarcinoma typically without duodenal polyps.

Region (tot pt)	Gastric LGD	Gastric HGD/AC	Spigelman ≥ 3	Duodenal HGD/AC
A (11)	4	0	4	0
B (12)	4	1	3	2
C (5)	4	0	2	0
D (3)	1	1	2	1
E (2)	2	0	2	2
F (2)	0	0	1	0
B1 APC promoter (2)	2	0	0	0

Conclusion: Currently there are no precise guidelines for upper gastro-intestinal tract endoscopy based on genetic mutations. This pilot study highlights the need for further research to establish an evidence based consensus.

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Disclosure: Nothing to disclose.

PP0133

FEASIBILITY AND SAFETY OF ENDOSCOPIC RESECTION VIA SNARING FOR GASTRIC ADENOCARCINOMA OF THE FUNDIC GLAND TYPE

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Introduction: Gastric adenocarcinoma of the fundic gland type(GA-FG) originates from the deeper mucosal layer, and predominantly occurs in the upper gastric region. Endoscopic submucosal dissection (ESD) in the upper region often results in a perpendicular approach and can be technically challenging. On the other hand, most GA-FG are relatively small, and snaring resection such as hybrid ESD and endoscopic mucosal resection with a cap-fitted pan endoscope (EMRC) may offer efficacy. However, the safety and feasibility of snaring resection for GA-FG remain unclear.

Aims & Methods: We retrospectively analyzed 77 cases of GA-FG endoscopically resected between August 2013 and April 2024 at NTT Medical Center Tokyo. Cases were categorized into two groups: (i) ESD group (lesions resected by ESD, n=54) and (ii) Snaring group (lesions resected by hybrid ESD or EMRC, n=23), and treatment outcomes were compared between the two groups.

Results: There was no difference in gender between the two groups. Patients in the snaring group were significantly younger (median age 62.0 vs 56: p<0.05), and the lesions were significantly smaller (median tumor size 5.0 vs 4.0mm: p<0.05). The majority of lesions in both groups were located in the upper gastric region. The two groups had no significant differences in en bloc resection rate, rate of submucosal invasion, lymphovascular invasion, vertical/horizontal margin positivity rate, or R0 resection rate. However, the snaring group exhibited a shorter procedure duration (29.5 vs 12.0 minutes; p<0.05), and reduced hospital stay duration (7.0 vs 5.0 days: p<0.05) compared to the ESD group. Adverse event rates were not significantly different between the two groups. Submucosal tissue was present in 100% of snaring group specimens, with a median resected submucosal layer thickness of 667.0μm (IQR: 527.9-797.2).

	ESD group N=54 lesions	Hybrid ESD/EMRC group N=23 lesions	P
Procedure time, min, median (IQR)	29.5 (18.0-50.0)	12.0 (7.0-21.5)	<0.05
En-bloc resection, no. (%)	54 (100%)	23 (100%)	1
Tumor size, mm, median (IQR)	5.0 (3.0-8.0)	4.0 (2.9-5.0)	<0.05
Depth of Invasion, no, (%)			
M/SM1	48 (90.6%)	19 (95.0%)	
SM2	5 (9.4%)	1 (5.0%)	0.06
Lymphovascular invasion, no(%)	3 (5.7%)	1 (5%)	1
R0 resection (%)	46 (86.8%)	18 (90.0%)	1
Hospitalization period, day, median (IQR)	7.0 (5.3-7.0)	5.0 (5.0-5.5)	<0.05

Conclusion: Despite comparable rates of en bloc resection, positive vertical margin, and adverse events, the snaring group exhibited significantly shorter procedure time and hospital stay durations. All snaring group specimens contained submucosal tissue, with the resected submucosal layer thickness exceeding 500μm. Endoscopic resection by snaring might be a safe and feasible approach for resection of GA-FG.

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Disclosure: Nothing to disclose.

PP0134

VALIDATION OF YPSTAGE AND CONSTRUCTION OF "YCSTAGE" AFTER PREOPERATIVE CHEMOTHERAPY IN GASTRIC CANCER

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Introduction: In the eighth edition of the TNM classification, clinical stage (cStage) and pathological stage after neoadjuvant chemotherapy (ypStage) were newly defined for gastric cancer. However, cStage after prior chemotherapy (ycStage) has not yet been defined.
Aims & Methods: This study aimed to validate ypStage and construct ycStage after preoperative chemotherapy in gastric cancer from the viewpoint of survival outcomes. One hundred-eighty-seven patients who planned to undergo gastrectomy with curative intention after preoperative chemotherapy were enrolled. Survival outcomes stratified by ypStage and ycStage was evaluated. ycStage was evaluated by extrapolating TNM cStage. In addition, macroscopic type, tumor size, histological type, and tumor marker (CEA/CA19-9) after chemotherapy were selected as other prognostic factors. These staging system and variables were evaluated by univariate and multivariable analysis. Harrell's concordance index (C-index: time dependent area under the curve) was used to evaluate predictive ability.
Results: Gastrectomy was not performed in 14 patients (7.5%) due to incurable factors during operation. Remaining 173 patients underwent surgery. Residual tumor was R0: 148 cases (85.5%), R1: 17 cases (9.8%), R2: 19 cases (4.6%), respectively. ypStage: The survival distribution of ypStage was assessed in 173 patients who underwent gastrectomy. The 5-year survival rates (ysr) for ypStage I (n = 36), II (n = 55), III (n = 60), and IV (n = 22) were 82.3%, 76.5%, 42.7%, and 14.1%, respectively. There were significant differences between II/III (P=0.001) and between III/IV (P=0.009), respectively. There was a tendency for there to be a difference in survival between I/II (P=0.087). The C-index of ypStage was 0.713 (standard error: se=0.027). ycStage constructed by ordinal TNM factor: The survival distribution of ycStage, ycT, and ycN was assessed in all 187 patients. The 5-ysr for ycStage I (n = 12), II (n = 39), III (n = 90), and IV (n = 46) were 54.7%, 57.5%, 61.0%, and 40.6%, respectively. The predictive ability was poor with C-index of 0.533 (se=0.031). The 5-ysr for ycT1 (N=9): 66.7%, ycT2 (N=16): 64.2%, ycT3 (N=27): 60.6%, ycT4 (N=135): 52.0%, respectively. There were no significant differences between ycT1/2 (P=0.734) and between ycT2/3 (P=0.872). The 5-ysr for ycN0 (N=55): 45.9%, ycN1 (N=55): 59.3%, ycN2 (N=54): 68.2%, ycN3 (N=23): 36.4%, respectively. The survival outcomes among cN0, cN1 and N2 are completely reversed. Other preoperative prognostic factors: In univariate analysis histological type (P=0.703), macroscopic type (P=0.125), and CA19-9 (P=0.296) were not significant prognostic factors, but tumor size [<80mm (n=117): 5-ysr; 64.6% vs. ≥80mm (n=70): 34.2%, P<0.001] and CEA [(negative (n=152): 5-ysr; 59.2% vs. positive (n=35): 33.0%, P<0.001)] were significant prognostic factors. Multivariate Cox proportional hazard model selected tumor size (P<0.001) and CEA (P=0.007) as independent prognostic variables. Construction of proposal ycStage: A new proposal stage was constructed from the results of multivariate analysis. ycStageI: size<80mm and CEA-negative, ycStageII: size<80mm and CEA-positive, ycStageIII: size≥80mm and CEA-negative, ycStageIV: size≥80mm and CEA-positive. The 5-ysr for new proposal ycStage I (n = 94), II (n = 26), III (n = 52), and IV (n = 15) were 71.3%, 54.0%, 37.0%, and 20.0%, respectively. The C-index of new proposal ycStage was 0.662 (se=0.029). It is significant higher value than that of ycStage constructed by ordinal TNM factor.
Conclusion: The predictive ability of the ypStage, on the basis of TNM factors, considered favorable. However, survival separation was poor for both ycT and ycN. Thus, ycStage based on ordinal TNM did not have the perspective of survival outcome prediction. Proposal ycStage constructed by tumor diameter and CEA may be recommended.
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DAIICHI SANKYO COMPANY, LIMITED.

PP0135

RISK FACTORS AND CLINICAL CHARACTERISTICS OF EARLY-ONSET GASTRIC CANCER VS. LATE-ONSET GASTRIC CANCER: A CASE-CASE STUDY

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Introduction: Early-onset gastric cancer (eoGC), defined as GC before 50, accounts for about 5% of GC and 3% are part of a hereditary syndrome. Its incidence is rising. eoGC was reported to present at an advanced stage at diagnosis, with a poor differentiation and a worse prognosis than in the elderly (late-onset GC, loGC), even though eoGC risk factors and characteristics remain unclear.
Aims & Methods: Therefore, we compared eoGCs with loGCs in terms of clinicopathological characteristics and risk factors. We conducted a retrospective case-case study of patients with eoGC and loGC diagnosed from 2015 to December 2022 at IRCCS San Raffaele Scientific Institute (Milan, Italy). We collected clinical data, tumor characteristics (including Helicobacter Pylori status, HP), family history of GC, smoking habit, alcohol intake, and BMI. We performed multi-gene panel testing on all eoGC.
Results: 58 eoGC (51.7%M, 48.3%F; median age at diagnosis [IQR] 44.5y [39.00, 47.75]) and 290 loGCs (63.8%M, 36.2%F; median age at diagnosis [IQR] 70.33y [63.90, 75.00]) were enrolled. 19.2% eoGCs carried a germline pathogenic variant (4 CDH1; 2 TP53; 1 MMR; 1 ATM; 1 PTEN; 1 heterozygous MUTYH). Having a second-degree relative (SDR) with GC was significantly associated with eoGC (15.8% eoGCs vs. 4.6% loGCs, $p = 0.015$). eoGCs reported lower BMI (median 22 eoGC vs 24 loGC, p=0.016). We also found a statistically significant difference between groups in alcohol consumption (P = 0.042) and smoking status (P = 0.023). While eoGCs were mainly at

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the cardia (29.3% eoGCs vs. 1.7 % loGCs; p<0.001), loGCs were mostly at the antrum (19% eoGC vs. 36.9% oGCs; *p*=0.009). eoGCs reported more dysphagia (30.9% eoGCs, compared to 4.9% loGCs, $p < 0.001$), while most patients in the loGC cohort had anemia (25.3% loGCs vs 7.3% eoGCs, $p = 0.002$) and asthenia (9.1% loGCs vs 0% eoGCs, $p = 0.012$). 48.3% eoGCs were diffuse-type adenocarcinoma (p=0.014), while 51% loGCs were intestinal-type (*p*<0.001). No significant differences were observed in terms of T and N status, while more eoGC were M1 (26.3% eoGCs vs. 9.7% loGCs, *p*=0.001) and presented at stage III-IV (56.9% eoGCs vs. 41.7% loGCs, *p*=0.042) at diagnosis.

Conclusion: 19.2% patients with eoGC carry a germline pathogenic variant, suggesting that patients with eoGC should be evaluated for germline testing at diagnosis. eoGC more often is localized at the cardia with dysphagia and presents as diffuse-type adenocarcinoma at an advanced stage (III-IV) ad diagnosis. However, 35.6% patients with eoGC presents with early gastric cancer T1 (vs 24.1% loGCs), possibly implying that younger age does not entail worse outcomes.

Disclosure: Nothing to disclose.

PP0136

MICROBIOME COMPOSITION IN PATIENTS WITH UPPER GASTROINTESTINAL CANCER AND SMALL INTESTINAL BACTERIAL OVERGROWTH (SIBO).

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Introduction: Small intestinal bacterial overgrowth (SIBO) is frequent in patients with past or active upper GI cancer. SIBO is associated with clinical symptoms including, bloating and diarrhea that may impact negatively on quality of life. Current treatment of SIBO is empirical and relies on the cyclical use of antibiotics.

Aims & Methods: The aim of the present study was to characterize the microbiome composition of patient with upper GI cancer with or with no SIBO.

From April 2021 to May 2022, consecutive patients with past or active GI cancer attending our oncology outpatient department were enrolled in a prospective case control study. Treatment naïve patients with a first diagnosis of upper GI cancer but no SIBO-related symptoms served as controls. All included patients completed a standardised questionnaire about SIBO-related symptoms (bloating, diarrhoea), concomitant medications, and quality of life. They were offered upper endoscopy with duodenal aspirate, provided a 30-day period with no antibiotic intake. SIBO was defined as a bacterial count greater than 10^3 colony forming units per millilitre in the duodenal aspirate. Using optimized protocol, the DNA from duodenal aspirate was extracted and the V1-V2 region of the 16S rRNA gene was amplified by PCR and sequencing using the Illumina platform. Dedicated bioinformatical approach was used for the multifactorial sample analysis.

Results: The microbiome composition was analyzed in 7 controls, 17 patients with SIBO and 12 with no SIBO.

We detected 513 phylotypes from 71 genera belonging to 11 phyla in the duodenal aspirate. The most abundant genera were *Streptococcus*, *Veillonella* and *Prevotella*. All three groups differed significantly in terms of their microbiome at *phylum* level (PERMANOVA $p < 0.05$). Based on the analysis of the individual *phyla*, we were able to show that the abundance of *Proteobacteria* was significantly higher in the SIBO group (Kruskal-Wallis test $p < 0.05$). Correlation analyses revealed an association between bacterial abundance and clinical phenotype. For instance, *Proteobacteria*, correlated positively with diarrhea intensity (Spearman r=0.35, p= 0.0362), and that *Bacteroidetes* correlated negatively with the reduction in quality of life (Spearman r=-0.4525, p=0.0056). Within the overall cohort, we were able to identify two clusters characterized by different microbial composition, underlying malignant diseases and clinical symptoms.

Conclusion: Among patients with upper GI cancer, the microbiome composition of the duodenal aspirate of patients with SIBO was different compared to patients with no SIBO and controls. Microbiome composition profiling may offer the opportunity for a personalized treatment in patients with SIBO.

Disclosure: Nothing to disclose.

PP0137

BIOLOGICAL CHARACTERISTICS OF OAS FAMILY AND REGULATION OF APOPTOSIS, NECROPTOSIS AND PYROPTOSIS IN GASTRIC CANCER

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Introduction: Gastric cancer (GC) is a common cancer and a leading cause of cancer-related death, with increasing morbidity and mortality in recent years. OAS gene family plays an important role in antiviral process, but its mechanism in gastric cancer is still unclear.

Aims & Methods: The biological function and prognosis of OAS gene family in human GC were analyzed by bioinformatics analysis and cell level verification. To explore the relationship of OAS3 and OASL with apoptosis, necroptosis and pyroptosis.

Results: We found through bioinformatics analysis, the expression of OAS family was up-regulated in GC, which was related to the prognosis of patients. OAS family is closely related to the immune infiltration and tumor burden mutation of GC. OAS family gene is positively correlated with the expression of apoptosis, necroptosis and pyroptosis molecules. OAS family genes were found to be up-regulated in GC by detection of clinical samples. *In vitro*, knockdown of OAS3 and OASL inhibited the proliferation and migration of GC cells, overexpression of OASL promoted the proliferation and migration of GC cells. In vitro , knockdown of OAS3 and OASL in AGS, up-regulated apoptotic molecules (cleaved Caspase 3, cleaved Caspase 9 and cleaved Caspase 8), necroptosis molecules (p-MLKL and p-RIPK3), pyroptotic molecules (cleaved Caspase 1, cleaved Caspase 4, GSDMD and PYCARD), and inflammatory factor IL-1β. In addition, knockdown of OAS3 and OASL in MKN45, up-regulated the expression of apoptosis, necroptosis and pyroptosis mentioned above, as well as the expression of inflammatory factors. overexpression of OASL in HGC27 cells decreased the expression of apoptosis molecules cleaved Caspase 3, cleaved Caspase 9 and cleaved Caspase 8, and necroptosis molecules p-MLKL and p-RIPK3, the pyroptosis molecules cleaved Caspase 1, cleaved Caspase 4, GSDMD and PYCARD were down-regulated, the expression of inflammatory factor IL-1β was down-regulated.

Conclusion: The high expression of OAS family genes is closely related to the prognosis of GC. OAS family is closely related to the immune infiltration and tumor burden mutation of GC. OAS3 and OASL may promote the progression of GC by inhibiting pan-apoptosis(apoptosis, necroptosis and pyroptosis). In summary, OAS3 and OASL may become biomarkers and therapeutic targets in GC.

Disclosure: The authors declare that they have no conflict of interest.

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PP0138

EXPLORING THE PROGNOSTIC VALUE OF BLOOD CELL RATIOS AND C-REACTIVE PROTEIN IN GASTRIC CANCER: INSIGHTS INTO MOLECULAR SUBTYPES AND TUMOUR IMMUNE MICROENVIRONMENT

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Introduction: Lymphoplasmacytic-rich and neutrophilic-rich tumour immune microenvironment (TIM) have been described in Epstein-Barr-virus-associated (EBV+) and microsatellite-unstable (MSI-high) gastric cancer (GC), respectively. Aim: to explore associations between EBV+/MSI-high status, TIM, peripheral blood cell ratios, tumour serum biomarkers and prognostic outcomes in GC.

Aims & Methods: Lymphoplasmacytic-rich and neutrophilic-rich tumour immune microenvironment (TIM) have been described in Epstein-Barr-virus-associated (EBV+) and microsatellite-unstable (MSI-high) gastric cancer (GC), respectively. Aim: to explore associations between EBV+/MSI-high status, TIM, peripheral blood cell ratios, tumour serum biomarkers and prognostic outcomes in GC.

Results: The series encompassed 7/140 EBV+ (5.0%) and 38/140 MSI-high (27.1%) GCs, characterised by lymphoplasmacytic-rich (EBV+:71.4% vs EBV-:5.3%, p<0.001) and neutrophilic-rich (MSI-high:28.9% vs MSS:2.0%, p<0.001) TMI, respectively. Prior to surgery, patients who underwent NAC harboured lower NLR compared to those submitted to surgery alone (3.05 vs 2.53, p=0.021). Pre-operative NLR and post-operative CRP levels were lower in neutrophilic-rich TMI GC (p=0.015 and p=0.006, respectively). Serum tumour biomarkers (CA125, CA19.9, CEA) were not related to morphological features or molecular subgroups. Overall, GC patients with lower CRP after surgery (<1.1mg/dl) showed better overall survival (p=0.032), even when stratified by type of therapy (NAC versus surgery alone), MSI status and presence of neutrophilic-rich TMI.

Conclusion: This study contributes to the growing body of evidence of blood-based biomarkers as predictive markers of GC prognosis. A reduction of systemic pro-inflammatory status after surgery (*e.g.* CRP<1.1md/dl) might contribute to better overall prognostic outcomes of GC patients. Moreover, we observed that GC patients with neutrophilic-rich TMI, a morphologic feature associated with MSI-high status, had lower levels of neutrophils in peripheral blood (low pre-operative NLR). Possible explanations include neutrophil sequestration within the tumour microenvironment and/or consumption by the tumour.

Disclosure: Nothing to disclose.

PP0139

RISK ASSESSMENT QUESTIONNAIRE IN A WESTERN COMMUNITY TO IDENTIFY A GASTIC CANCER RISK POPULATION: THE GASTROSCREENING PROJECT

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Introduction: Nowadays there is no screening test approved on large scale for gastric cancer even if this neoplasia represents one of the leading causes of cancer-related deaths globally[1-2]. In Italy still remains one of the most frequent tumors, with incidence rates differing across geographical areas[3].

Aims & Methods: This project aims to create an easily administrable tool to detect anamnestic gastric cancer risk factors together with alarm symptoms in an age-defined target population to identify patients in need of an esophagogastroduodenoscopy (EGD).

This paper describes the first study phase: the administration of the questionnaire to a first sample of 5000 people aged from 40 to 80 years and the preliminary results. GASTROFORM is an online questionnaire (https://it.research.net/r/GASTROFORM) collecting data about demographic and anthropometric features, physiological anamnesis, familial history, non-specific upper gastrointestinal symptomatology, alarm symptoms and recent EGD performance.

Results: 4426 valid questionnaires were submitted through the online form in November 2022, 612 of them (13,82%) reported at least one alarm symptom identifying patients in need to undergo an EGD as soon as possible. From an epidemiological point of view we expect to increase the rate of diagnosis in Barret esophagus (from 1,8% to 5-8%), atrophic gastritis (from 9% to 20-30%) and early gastric cancer (from 0,019% to 1-2 %).

Conclusion: The GASTROFORM questionnaire could provide useful informations to identify a population most likely to have significant endoscopic findings in early diagnosis during a screening EGD.

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Disclosure: Nothing to disclose.

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PP0140

GLOBAL HEALTH INEQUALITY AND YOUNG-ONSET GASTRIC CANCER MORTALITY (1990 TO 2019): A SYSTEMATIC ANALYSIS OF GLOBAL BURDEN OF DISEASE

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Introduction: The temporal trend of mortality rates among young individuals with gastric cancer has sparked a contentious debate over the past few decades. Additionally, there remains an ambiguous distribution alteration of disease burden among different countries.

Aims & Methods: This population-based systematic analysis was conducted to investigate changes in health inequality related to mortality caused by young gastric cancer from 1990 to 2019. We performed a secondary analysis of the Global Burden of Diseases, Injuries, and Risk Factors Study (GBD) 2019 by extracting cross-sectional data on young gastric cancer (<40 years). The age-specific mortality rate was used to determine deaths attributed to young gastric cancer. A prediction of young gastric cancer mortality rate and case number up to 2030 was performed using the Nordpred APC model. The slope index of inequality (SII) and the health concentration index were used to quantitatively measure absolute and relative changes in health inequalities, taking into account the socio-demographic indexes (SDI) level. The data was further analysed across different GBD regions and sexes.

Results: The global mortality rate of young gastric cancer declined from 1.61 (95% Uncertainty interval [UI]: 1.49 to 1.72) per 100,000 in 1990 to 0.94 (95% UI: 0.87 to 1.02) per 100,000 in 2019, with an annual rate of change -0.42% (95% Confidence interval (CI): -0.47 to -0.35). The projected data suggests that the number of global mortality case and rate of young gastric cancer would decrease to 24,450 and 0.761 per 100,000 by 2030. Countries belong to the lower 50% of SDI accounted for 49.9% and 52.3% of the global young gastric cancer deaths in 1990 and 2019, respectively. Although the absolute cross-national inequality (SII) of mortality rate fell from 0.133 (95%CI: -0.533 to 0.416) in 1990 to -0.239 (95%CI: -0.432 to -0.046) in 2019, the relative inequality (concentration index) for global young gastric cancer mortality increased from 1990 (−0.058, 95%CI: −0.532 to 0.416) to 2019 (−0.104, 95%CI: −0.526 to 0.317). The observed rise in absolute health inequality was evident in both sexes, but relative health inequality increased in females only. The regions of Sub-Saharan Africa (including Western and Southern), Central Asia, and Central Europe exhibited an increase in both absolute and relative health disparities in young gastric cancer mortality.

Conclusion: Although there has been a decline in the global young gastric cancer mortality rate, health inequality has increased over the past three decades, especially in females and some low SDI countries. It is imperative to prioritize efforts in addressing inequality to further reduce the global burden of young gastric cancer mortality.

Disclosure: Nothing to disclose.

PP0141

COMPARATIVE STUDY OF STANDARDIZED (HIGH DEFINITION WHITE LIGHT ENDOSCOPY WITH IMAGE-ENHANCEMENT) VERSUS NON-STANDARDIZED (ROUTINE WHITE LIGHT ENDOSCOPY) FOR DETECTION OF UPPER GASTROINTESTINAL MUCOSAL LESIONS: A PROSPECTIVE ANALYSIS FROM NORTH INDIA

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Introduction: This study aims to compare the diagnostic efficacy and variables of standardized high definition white light endoscopy with image-enhancement versus non-standardized routine white light endoscopy) in detecting upper gastrointestinal mucosal lesions.

Aims & Methods: A prospective analysis was conducted on 270 patients undergoing Upper Gastrointestinal Endoscopy at Advanced Gastroenterology Institute: AGI the GASTROCITI, Ludhiana located in Punjab region of North India between December 2023 and March 2024. Patients were randomly assigned (135 each) to undergo assessment using standardized versus non standardized format. Endoscopic findings were recorded and later analyzed by experienced gastroenterologists blinded to the endoscopic modality used.

Results: In the standardized group, the mean age was 47 years, with 63 % males and 37 % females. In the nonstandardized group, the mean age was 45 years, with 58 % males and 42 % females. The most common indications for endoscopy in both groups were dyspepsia and heartburn. Due to use of premedication preparation (Surfactant Simethicone + Mucolytic N acetyl-cysteine solution), the mucosal visibility score in the standardized group was significantly better at 76% versus 32% in the nonstandardized group. Also, the time taken to examine the complete mucosa (8 minutes versus 5 minutes) and corresponding Propofol anesthetic requirement dose (140 mg versus 120 mg) along with the lesion detection rate (68% versus 53%) was higher in the standardized group. Among the 135 patients, 43 were normal in the standardized group versus 64 in the nonstandardized group. The diagnostic yield of standardized high-definition white light endoscopy with image enhancement was 68.1% versus 52.5 % in nonstandardized WLE. Specific lesions with higher rates of detection were Grade A Esophagitis, early short segment Barrett's Esophagus, gastric atrophy, and intestinal metaplasia. The relative risk and odds ratio to find an abnormality while doing a standardized procedure was 1.3 and 1.93 respectively which showed very high statistical significance (p Value 0.009).

Conclusion: Standardized (High Definition White Light Endoscopy with Image-Enhancement) demonstrates superior diagnostic efficacy compared to non standardized routine white light endoscopy in detecting upper gastrointestinal mucosal lesions. Further large-scale studies are warranted to validate these findings and determine the optimal use of standardized (High Definition White Light Endoscopy with Image-Enhancement) in clinical practice.

Disclosure: Nothing to disclose.

PP0142

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PP0143

EFFICACY AND SAFETY PROFILES OF GASTRIC ENDOSCOPIC RESECTION FOR TREATING EARLY GASTRIC NEOPLASMS IN EXTREMELY OLD ADULTS (>80 YEARS): DATA FROM SINGLE-CENTER 10-YEAR OF EXPERIENCE

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Introduction: Even gastric endoscopic resection (ER) including endoscopic mucosal, and submucosal resection, has been an effective treatment method for treating gastric neoplasm in an early phase, there is limited data on the efficacy, and safety profiles of ER especially among extremely old age (>80year) patients.

Aims & Methods: To investigate the efficacy, and safety profiles of gastric ER among extremely old age patients, and to triage the high-risk groups for gastric ER related complication. We designed a retrospective study using data from electronic medical records in National Health Insurance Service Ilsan Hospital which performs ESD in recent 3 from January 2016 to June 2023. Among a total of 1663 gastric ESD cases, 173 cases were performed in patients aged over 76 years, and 45 cases in patients aged over 80 years. These patients were evaluated retrospectively shorterm outcomes, and survival. The shorterm efficacy profiles includes curative resection rate, and safety profiles includes perforation or bleeding risk after ESD, or ER visit or readmission rate in month after discharge.

Results: Average risk of perforation after ESD was 2.7%, and emergency room (ER) visit after dischange rate was 0.4% in this period. risk FOBT positive result patients (> 75year) at the time of health check-up, according to whether or not performed CFS within 2 years of positive results of stool occult blood, case (n=), and control(n=) groups were divided. The end point was defined as CRC diagnosis or mortality. After adjusting for confounding variables, multivariable analysis was performed to analyze the effect of colonoscopy on the risk of incidental CRC, and all-causes mortality.

Conclusion: For extremely old age patients (>80year), the efficacy, and safety profiles of gastric ER were generally acceptable for treating early gastric neoplasm. Even though still endoscopists careful to patient's medical history, life expectancy, and comparing risk/benefit from ER, clinicians might not be hesitate treating early neoplastic lesions in stomach just beacause of patients extremely old age

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Disclosure: Nothing to disclose.

PP0144

FORCEPS BIOPSY FOR MICRO-PHARNGEAL NEOPLASMS: A SINGLE-CENTER PROSPECTIVE STUDY

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Introduction: We previously reported short-term outcomes at 3 months after cold forceps biopsy (CFB) using jumbo biopsy forceps (Boston Scientific or Olympus Medical Systems) for micro-pharyngeal neoplasms[1].

Aims & Methods: We extended the follow-up and present here outcomes at 12 months after the CFB (UMIN registry number: 000037980). One hundred twenty-five lesions in 101 patients endoscopically diagnosed with narrow-band imaging of dilated intrapapillary capillary loops with clear demarcation as micro-pharyngeal tumors of 3 mm or less and excised with CFB were prospectively registered. After excluding 28 lesions with benign pathology, 97 lesions (13 lesions of squamous cell carcinoma, 4 of high-grade dysplasia, and 80 of low-grade dysplasia) were endoscopically followed up 3 and 12 months after the first CFB. Additional CFB was performed if lesions with endoscopically determined local remnant/recurrence (ER, narrow-band imaging of dilated intrapapillary capillary loops with clear demarcation) were found. The primary outcome was ER rate. The secondary outcome was composite remnant/recurrence (CR) (ER with pathological recurrence, ER without pathological assessment, and non-ER scars but with pathological recurrence) rate.

Results: Endoscopy at 3 months after CFB was performed in 56 lesions in 47 patients, and CFB scars were identified in 25 lesions (44.6%). Additional CFB was performed in all scars judged as ER, and regular biopsy was obtained in some scars without ER. ER rate at 3 months was 23.2% (13/56, 95% confidence interval 13 - 36.4%), and CR rate was also 23.2% (13/56, 13 -36.4%) (one ER scar with negative pathology, and one non-ER scar with positive pathology). Endoscopy at 12 months was performed in 84 lesions in 71 patients. CFB scars were identified in 20 lesions (23.8%); ER rate at 12 months was 11.9% (10/84, 5.9 -20.8%), and CR rate was 8.3% (7/84, 3.4-16.4%) (3 ER scars with negative pathology, no non-ER scar with positive pathology). Regarding safety of the procedure, no intraoperative or postoperative adverse event was observed in all the 125 lesions including 24 lesions excised under continued antithrombotic therapy.

Conclusion: About 10% remnant or recurrence rates at 12 months after the procedure suggest that CFB may be minimally invasive therapeutic option for micro-pharyngeal neoplasms of 3 mm or smaller. Further follow-up study is continuing.

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Disclosure: Nothing to disclose.

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PP0145

UPPER-GASTROINTESTINAL ENDOSCOPY, OR OESOPHAGOGASTRODUODENOSCOPY (OGD), IS A PROCEDURE ROUTINELY PERFORMED IN SECONDARY CARE SETTINGS THROUGHOUT THE UK. LOCAL ANAESTHESIA, IN THE FORM OF THROAT SPRAY, AND INTRAVENOUS SEDATION ARE OFTEN USED TO MAKE THIS CLINICAL PROCEDURE MORE COMFORTABLE FOR THE PATIENT

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Introduction: Upper-gastrointestinal endoscopy, or oesophagogastroduodenoscopy (OGD), is a procedure routinely performed in secondary care settings throughout the UK. Local anaesthesia, in the form of throat spray, and intravenous sedation are often used to make this clinical procedure more comfortable for the patient.

Aims & Methods: This study focused primarily on the difference in administration of the local anaesthetic throat spray. The commonly used method was to spray the back of the patient's pharynx while their mouth was open: "mouth-open" technique. This study proposed a modified technique which involved spraying the pharynx with the mouth closed around the spray nozzle: "JASMEG" technique. This was a quality improvement study of 474 patients (male: female = 1.13) who were undergoing an OGD as part of routine care in a district general hospital in the United Kingdom. It was carried out to investigate whether this new method of spray administration would increase patient comfort and reduce patient anxiety, pulse rate and incidence of cough. Comfort and anxiety scores were scored on a 5-point Liker scale (1 = most comfortable/ least anxious, 5 = least comfortable/ most anxious). Outcomes were compared between the two groups using linear regression models controlling for sex. Incidence of cough was compared between groups using Fisher's exact test.

Results: A breakdown of the sample by treatment group and sex is shown in Table 1. As compared to the "mouth-open" group, the "JASMEG" group showed a smaller increase in pulse rates (group difference = -3.97, $p < 0.001$), lower anxiety scores (group difference = -1.31, $p < 0.001$), higher comfort scores (group difference = 1.07, $p < 0.001$) and lower incidence of cough (13 out of 252 vs 190 out of 222; $p<0.001$). There were no significant differences in the results by sex.

Conclusion: The new "JASMEG" technique of local anaesthetic spray for patients undergoing OGD is superior in terms of patient comfort, anxiety levels, pulse rates and presence of cough. This is a
low-cost alternative to reduce transmission of aerosol-transmitted infections, while also making it a more favourable procedure for patients.

References: 1,Department of Gastroenterology; Northumbria Healthcare, Newcastle Upon Tyne, 2, University College Institute of Child Health Great Ormond Street, London, UK; 2,Population, Policy & Practice, UCL Institute of Child Health, London, UK 3,Nobles Hospital, Strang, Isle of
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Disclosure: Nothing to disclose.

PP0146

A PILOT STUDY ON THE UTILIZATION OF ENDOSCOPIC TWO-TOOTHED CLIPS FOR THE MANAGEMENT OF GASTROINTESTINAL DISEASES

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Introduction: With the progress made in the field of digestive endoscopy, a growing array of digestive disorders can now be effectively managed through endoscopic interventions. Notably, the utilization of clips for closure purposes has been documented in relation to several endoscopic procedures, including endoscopic submucosal dissection (ESD), endoscopic mucosal resection (EMR), and endoscopic full-thickness resection (EFTR). However, current clipping systems have demonstrated limitations in terms of dimensions, spacing between open arms, force exerted during closure, and adequate tissue compression. Emerging endoscopic suturing techniques are currently in the initial stage, with limited clinical investigation and insufficient reliable evidence. However, their reliance on an independent endoscopic apparatus results in elevated expenses related to equipment and treatment, and also calls for specialized clinical training.In order to tackle this concern, we have devised the two-toothed clip (TTC). Through a pilot study encompassing a cohort of 10 participants, our objective is to assess the safety, feasibility, and efficacy of the two-toothed clip.

Aims & Methods: The study employed a nitinol alloy-based two-toothed clip, where the teeth were affixed to the inner side of standard clip arms (ROCC-D-26-195-C, Micro-Tech, Nanjing, China) using plastic tubes and sutures. These teeth demonstrated enhanced length and sharpness compared to the standard clip arms. The technique used to advance the two-toothed clips towards the target lesions was the same as that employed for standard clips. Caution must be exercised while opening the arms to avoid unintentional harm to the normal mucosa due to the sharp teeth. With the aid of electrocautery snare, the TTCs were easy to be detached. Ten patients between the ages of 50 and 80 were included in this study. The treatment indications encompassed non-variceal bleeding, post-endoscopic submucosal dissection (ESD) lesions in the duodenum or colorectum, tracheoesophageal fistulas, duodenal or colorectal diverticulitis, as well as anchoring of self-expanding metal stents. Ethical approval was granted by the Institutional Review Board of our hospital, and written informed consent was obtained from all participants. The baseline characteristics of the patients were collected.The achievement of technical success was determined by the effective placement of the two-toothed clips at the designated location. Clinical success was defined as the successful resolution of the gastrointestinal issue for which the two-toothed clips were utilized. The patients were subjected to clinical and endoscopic follow-up for a duration ranging from 14 to 30 days.

Results: Ten patients meeting the eligibility criteria, with an age range of 41-80 years and consisting of 8 men and 2 women, received the TTC application. The treatment indications encompassed the closure of submucosal defects (n=2), diverticula (n=2), and non-variceal upper gastrointestinal bleeding (n=1). Additionally, the procedure involved anchoring fully covered self-expanding metal stents (n=4) and utilizing the clip-with-line traction method in endoscopic submucosal dissection (ESD) (n=1). The technical success rate and clinical success rate both reached 100%, and no adverse events related to the procedure were observed.

Conclusion: The utilization of endoscopic hemoclips for tissue compression using TTC technique has been shown to be both safe and effective. This study brings new treatment strategies for future clinical practice.

Disclosure: There are no conflicts to declare.

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PP0147

OUTCOMES OF DELAYED ENDOSCOPIC MANAGEMENT FOR ESOPHAGEAL SOFT FOOD BOLUS IMPACTION: A MULTI-CENTRE RETROSPECTIVE STUDY

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Introduction: Esophageal food bolus impactions are a common reason for emergency room visits. Current guidelines recommend urgent endoscopy within 24 hours for incomplete obstructions, and emergent endoscopy within 6 hours for complete obstructions. While emergency endoscopy is required for sharp objects or disk batteries, data regarding soft food bolus impactions is more sparse. Recent retrospective evidence suggests that delaying endoscopy after twelve hours is not associated with an increase in morbidity or mortality.

Aims & Methods: This study expands on the existing literature to evaluate clinical outcomes associated with emergent (< 6 hours), urgent (6-12 hours), and delayed (>12 hours) endoscopy for patients presenting to emergency rooms with soft food bolus impaction.

This is a multicentre retrospective study involving two tertiary and one secondary care centre in British Columbia, Canada. All patients who presented with food bolus impaction between January 1, 2019 and Dec 31, 2022 were reviewed. The primary outcome was thirty-day mortality and repeat hospital presentation. Secondary outcomes included procedural morbidity (complications, procedural time, removal success) and length of stay. Endoscopic removal technique, optical diagnoses, and eosinophilic esophagitis incidence were also noted. Patients < 18 years old or those who had a non-food bolus impaction were excluded. Statistical analysis was performed using χ^2/Fisher Exact test wherever applicable for categorical variables and Kruskall-Wallis one-way ANOVA for continuous variables.

Results: Of 416 patients identified with complete esophageal impaction, 77 met inclusion criteria and were stratified into three cohorts: emergent (<6 hours, n=47), urgent (6-12 hours n=17), and delayed (>12 hours, n=13) intervention. All groups displayed similar baseline demographics (Table 1). Primary outcomes are outlined in Table 2. We did not observe mortality by thirty days in any group. Two patients who underwent emergent endoscopy re-presented to ED with dysphagia-related complications (4.2%) within thirty days compared to one patient (7.7%) in the delayed intervention group (OR: 1.88, CI:0.17-22.7, p=0.53). Similar to the emergent group where two (4.3%) patients required hospitalization for at least one day, one patient respectively in the urgent (5.9%, OR: 0.71, CI:0.05-8.39, p=1.0) and delayed (7.7%, OR: 1.9, CI:0.16-22.5, p=0.53) groups required hospitalization for at least one day. There was no statistically significant difference in procedural time (p=0.62), and only one complication (mucosal tear) was noted in the urgent intervention group. Only one patient had unsuccessful removal on index endoscopy, again in the urgent intervention group. There was no statistically significant difference in removal techniques, optical diagnoses, and eosinophilic esophagitis incidence. There was no statistically significant difference in the rate of mucosal injury secondary to prolonged food bolus impaction in the urgent (p=0.17) and delayed (p=0.12) groups.

Conclusion: There is no significant difference in negative clinical outcomes, including thirty-day mortality, hospital readmission, and procedural morbidity, in patients who receive emergent, urgent, or delayed endoscopy for management of a soft food bolus esophageal impaction.

These data add to the existing literature to assist endoscopists with risk stratification and timing of safe intervention for removal of food bolus causing complete esophageal obstruction.

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Disclosure: Nothing to disclose.

PP0148

PREVENTION OF DELAYED BLEEDING AFTER GASTRIC ENDOSCOPIC SUBMUCOSAL DISSECTION THROUGH CLIP CLOSURE WITH NOVEL ANCHOR-PRONGED CLIP

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Introduction: Delayed bleeding (DB) after gastric endoscopic submucosal dissection (ESD) is major complication [1] that sometimes lead to shock vitals and the effect of mucosal defect closure for preventing DB was reported [2]. However, the thickness of the gastric mucosa and proper muscle layer makes suturing difficult [3].

Aims & Methods: The aim of this study was to investigate whether clip closure using novel anchor-pronged clip (MANTIS Clip; Boston Scientific) is effective in preventing DB after gastric ESD. Out of 70 cases (78 lesions) that underwent gastric ESD at our institution from August 2023 to February 2024, a retrospective analysis was conducted on 60 consecutive cases (65 lesions) where ≥ 50% of the wound could be closed. If the wound was small, the closure method involved initially placing at least one MANTIS Clip to achieve mucosa-to-mucosa closure. If the wound was large, it was closed according to the method we previously reported [4]. In concrete terms, one side of the mucosa was grasped with MANTIS Clip, then move the clip to the opposite muscle layer, open the clip once, and after sufficient air aspiration, grasp the muscle layer. After shrinking mucosal defect enough, complete closure was achieved by adding mucosa-mucosa and mucosa-muscle closure similar to previous report of colonic closure [5] with normal clips (Sure Clip 16mm (Micro-Tech Co., Ltd, Nanjing, China) or EZ Clip (Olympus Medical Systems, Tokyo, Japan)). The primary outcome was the actual DB rate compared with the expected DB rate calculated from the BEST-J score [1]. Secondary outcomes included the technical success rate, complete closure rate (proportion of wounds with ≥ 90% closure), and closure time.

Results: M/F=42/18. DB risk based on BEST-J score (low/intermediate/high/very high) = 46/7/6/1. The median tumor size (mm) was 10 (5-40). Adenoma/cancer = 10/55. The actual rate of DB was 0%, indicating lower tendency to the expected rate of 4.5% (p=0.244). The technical success rate was 85.7% (60/70), with a complete closure rate of 90.8% (59/65). The median closure time (min) was 9 (2-24), with a median of 1 MANTIS Clip used per lesion (range: 1-4).

Conclusion: Although further accumulation of cases is necessary, clip closure using the MANTIS Clip achieved low DB with rapid and high complete closure after gastric ESD.

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Disclosure: Nothing to disclose.

PP0149

DEEP LEARNING BASED VIRTUAL LUGOL CHROMOENDOSCOPY FOR SUPERFICIAL ESOPHAGEAL SQUAMOUS CELL CARCINOMA: A PRELIMINARY STUDY

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Introduction: Lugol chromoendoscopy (LCE) has been established as a tool to increase the sensitivity for the detection of esophageal squamous cell carcinoma (ESCC). However, LCE has some disadvantages such as the time consumed for the technique, patients' discomfort, risk of aspiration and allergic reactions when compared to white light imaging (WLE) and image enhanced endoscopy (IEE).

Aims & Methods: We attempted to develop virtual lugol chromoendoscopy (V-LCE) images for ESCC using cycle-consistent generative adversarial network (CycleGAN) and to compare their diagnostic performance with that of WLE and real lugol chromoendoscopy (R-LCE). To obtain V-LCE images, we used 3425 images (1601 WLE and 1834 LCE) of superficial ESCC from 195 cases to train CycleGAN. For the test set to evaluate the usefulness of V-LCE images, other 25 WLE images from 25 patients with superficial ESCC were translated to 25 V-LCE images using trained CycleGAN. Six endoscopists (three experts and three non-experts) evaluated images obtained under WLE, V-LCE and R-LCE with respect to the detection and the margin of ESCC, using a five-point scale from 1 (poor) to 5 (excellent). We also calculated and compared the color difference (ΔE) between cancerous and non-cancerous areas each in WLE, V-LCE and R-LCE image.

Results: Comparison of WLE and V-LCE images showed a trend towards higher scores for V-LCE images (detection 3.3 vs 3.8, p=0.089, margin 2.7 vs 3.0, p=0.13). A comparison between V-LCE and R-LCE images revealed the scores in R-LCE were significantly higher than those in V-LCE (detection; 4.7 vs 3.8, p<0.001, margin; 4.3 vs 3.0, p<0.001). Among the experts, the score for margin in V-LCE tended to be higher than that in WLE (3.0 vs 2.7, p=0.051). Among the non-experts, the score for detection in V-LCE tended to be higher than that in WLE (4.0 vs. 3.3, p=0.064). ΔE in V-LCE was significantly higher than that in WLE (29.6 vs. 18.3, p<0.001), while it was significantly lower than in R-LCE (29.6 vs. 39.3, p<0.001).

Conclusion: Images in V-LCE have the potential for high sensitivity for the diagnosis of superficial ESCC, while the imaging quality need to be improved.

Disclosure: Nothing to disclose.

PP0150

MORPHOLOGY OF ULTRA-SHORT BARRETT'S ESOPHAGUS MAY BE INFLUENCED BY *HELICOBACTER PYLORI* INFECTION AND GASTRIC ATROPHY

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Introduction: Barrett's esophagus (BE) is known to be a precursor of esophageal and gastroesophageal junctional adenocarcinoma[1]. Adenocarcinoma of the esophagus based on BE is still relatively rare in Japan, but the Westernization of dietary habits, the increase in obese patients, and the eradication of *Helicobacter pylori* (HP) infection may increase the number of cases in the future. In Japan, the prevalence of long-segmented BE (LSBE), which is common in the Western world, is relatively low, whereas SSBE is frequently observed[2]. Atrophic gastritis (AG) caused by HP infection is also frequently observed in Japan, and the diagnosis of Japanese SSBE may be modulated by HPinfection/AG. In this study, we retrospectively investigated whether the morphology of SSBE could be influenced by HP status.

Aims & Methods: In our institution, 60 patients who had endoscopic ultra-short segmented Barrett's esophagus (USBE; length of <1cm) were retrospectively collected and divided into two groups: 30 cases with AG caused by current HP infection (AG group) or without AG (non-AG group). As patient characteristics, we investigated age, sex, presence of reflux esophagitis (RE), presence of hiatus hernia, and use of proton pump inhibitor (PPI) therapy from the medical records. To objectively evaluate the roughness of the SCJ, we compared the morphology of USBE by measuring the respective perimeters of EGJ and SCJ, followed by the calculation of the SCJ/EGJ ratio. As a sub-analysis in the AG group, the SCJ/EGJ ratio was similarly compared in terms of the degree of AG (closed-type, n = 3; open-type, n = 27) according to the Kimura-Takemoto classification.

Results: In terms of patient characteristics, the median age was higher in the AG group than in the non-AG group (72 years old vs. 49 years old, p <0.001), and RE was less common in the AG group than in the non-AG group (3% vs. 36%, p = 0.004). There were no statistical differences between the two groups in sex, presence of hiatus hernia, and use of PPI. The mean SCJ/EGJ ratio was significantly higher in the AG group than in the non-AG group (2.26 vs. 1.43, p <0.001). In the sub-analysis, the mean SCJ/EGJ ratio was likely to be higher in open-type atrophy than in closed-type atrophy (2.33 vs. 1.58, p = 0.092).

Conclusion: In endoscopically diagnosed USBE, the circumference of SCJ appeared rough in the AG group, which suggested that the morphology of BE is influenced by HP infection and subsequent AG development. Particularly in Japan, simple rough margins of SCJ influenced by HP/AG may be estimated as USBE. In diagnosing BE, the morphological characteristics of SCJ and the HP/AG status should be taken into consideration.

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Disclosure: There is no conflict of interest in this study.

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PP0151

INTRAOPERATIVE END-TIDAL CARBON DIOXIDE CONCENTRATION IS A POTENTIAL OBJECTIVE INDICATOR OF THE PNEUMOPERITONEUM MANAGEMENT DURING PERORAL ENDOSCOPIC MYOTOMY

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Introduction: Peroral endoscopic myotomy (POEM) is a less invasive endoscopic treatment for stenotic diseases caused by oesophageal motility dysfunction. During this procedure, the insufflation gas supplied from the endoscope migrates to the abdominal cavity, resulting in pneumoperitoneum. For excessive pneumoperitoneum, transabdominal puncture should be performed to rapidly decrease the intraabdominal pressure. However, the timing of puncture depends subjectively on the operator's decision.

Aims & Methods: In this study, we retrospectively analysed the factors leading to pneumoperitoneum and clarified the objective indicators to determine the timing of abdominocentesis. A total of 100 consecutive patients who underwent POEM at our hospital from October 2018 to December 2023 were included. In univariate analyses, patient-related (age, sex, body surface area [BSA], body mass index), disease-related (diagnosis, preoperative Eckardt score), and treatment-related factors (operator's POEM experience, myotomy length, peak end-tidal carbon dioxide concentration [$ETCO_2$]) were compared between Puncture and Nonpuncture groups. In these parameters, sex and BSA, which correlated with the intraabdominal space, and factors that were supposed to influence the severity of pneumoperitoneum and the need for subsequent abdominocentesis by showing p-values of less than 0.1 were enrolled for subsequent multivariate analyses.

Results: The mean age was 58 (standard deviation [SD], 19) and 51% of them were male. The mean Eckardt score was 6 (SD, 3). Most patients (n = 92, 92%) were diagnosed with oesophageal achalasia. The mean procedure time was 88 min (SD, 35 min). Abdominocentesis was performed in 65 patients (68%), who were allocated into the Puncture group. In the univariate analyses, the peak $ETCO_2$ was significantly higher in the Puncture group than in the Nonpuncture group ($p < 0.01$). In the multivariate analyses using sex, BSA, and the peak $ETCO_2$, high peak $ETCO_2$ was a single independent factor associated with abdominocentesis (odds ratio = 1.25, 95% confidence interval: 1.11–1.41, $p < 0.01$). The receiver operating characteristic curve indicated that the two groups were divided with a maximal area under the curve (0.76) when a cut-off value of the peak $ETCO_2$ was set at 44.5 mmHg (sensitivity, 0.79; specificity, 0.65).

Conclusion: This study demonstrated that the intraoperative $ETCO_2$ level was a significant independent factor related to severe pneumoperitoneum in POEM, and 45 mmHg of peak $ETCO_2$ was an optimal cut-off value to differentiate the Puncture group from the Nonpuncture group. This result should be clinically useful as an objective reference to determine abdominocentesis in patient with pneumoperitoneum due to CO_2 leakage to the abdominal space.

Disclosure: Nothing to disclose.

PP0152

CHARACTERISTICS OF WEB LIKE MUCUS AS AN ENDOSCOPIC FINDING OF PPI/P-CAB RELATED GASTROPATHY

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Introduction: Due to the widely use of potassium competitive acid blocker (P-CAB) and proton pump inhibitor (PPI) for acid-related diseases, the incidence of PPI/P-CAB-related gastropathy become more common in Japan. In the revised 3rd. edition of the Kyoto Classification [(1)], web-like mucus (WLM) has been newly described as an endoscopic finding seen in PPI/P-CAB-related gastropathy. The mechanism of its formation has been speculated to be linked to the strong and sustained inhabitation of acid secretion [(2)], however, the details of WLM are still unknown. We investigated the characteristics of WLM based on the database of upper gastrointestinal endoscopy performed at our department.

Aims & Methods: The aim of this study is to reveal the characteristics of subjects with WLM. This study is a retrospective cross-sectional study conducted in a single hospital. Data were extracted from the records of subjects who underwent upper gastrointestinal endoscopy at our department between April 2020 and December 2023. We examined subjects in which endoscopists recognized the presence of WLM. Statistical analyses were performed by Fisher's exact test.

Results: Out of a total of 3,564 subjects, WLM was observed in 14 cases (5 males and 9 females, mean age 72.4±16.3 years). All 14 patients were taking P-CAB. The duration of P-CAB administration varied from 8 days to 2 years. Interestingly, in some cases WLM was observed just a few days after starting oral administration of P-CAB. Regarding *Helicobacter pylori* infection, 10 cases were uninfected and 4 cases were past infected.

Conclusion: All subjects with WLM were taking P-CAB. All subjects with WLM were taking P-CAB. We found that WLM can be observed even just a few days after starting oral administration.

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Disclosure: Nothing to disclose.

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PP0153

EFFICACY OF A NOVEL THIN THERAPEUTIC ENDOSCOPE IN PERORAL ENDOSCOPIC MYOTOMY

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Introduction: Per-oral endoscopic myotomy (POEM) is a leading therapeutic procedure for third space endoscopy (TSE) and has been the first choice for treatment of esophageal achalasia. TSE requires treatment in a narrow space. The recently launched thin therapeutic endoscope (EG-840TP, Fujifilm Medical) is expected to be useful for TSE including POEM.

Aims & Methods: To clarify the effectiveness of thin therapeutic endoscopy in POEM, we conducted a retrospective observational study of 66 patients who underwent POEM in our hospital between November 2019 and March 2024. Patient characteristics and outcomes were compared between 52 patients treated with a normal-diameter therapeutic endoscope (T group) and 14 patients treated with a thin therapeutic endoscope (TP group).

Results: The patients in T group were 26 males and 26 females, median age 61 years (18-88), 44 cases of esophageal achalasia (Type I/II/III: 28/15/1), 6 cases of EGJOO, and 2 cases of DES. The TP group included 7 men and 7 women, median age 60 years (13-81), 11 cases of esophageal achalasia type I, 2 cases of EGJOO, and 1 case of DES. Sigmoid type was 58% (n = 30) and 29% (n = 4) in the T/TP group, respectively, and was significantly more common in the T group. Treatment before POEM was balloon dilation 11/1, POEM1/1, and Heller-Dor5/0 in the T/TP group, all of which were not significantly different. There was no difference in the length of myotomy between the two groups in either the esophagus or stomach. The median treatment time in each group was 92 minutes in the T group and 107 minutes in the TP group, and there was no significant difference in the submucosal tunnel creation time. However, the myotomy time per cm was 2.5 min in the T group and 3.8 min in the TP group, which was significantly longer in the TP group ($p<0.01$). Clips required for entry closure were significantly less in the TP group (8 clips in the T group and 5 clips in the TP group) ($p<0.01$). Eckardt scores after POEM improved equally in the two groups, and the rates of adverse events and GERD after POEM did not differ between the two groups.

Conclusion: POEM using a thin therapeutic endoscope resulted in similar treatment outcome as conventional endoscopes. Although there was no difference in the treatment time, TP group required fewer clips to close the entry. The prolonged myotomy time in the TP group was considered to reflect the fact that the thin therapeutic endoscope made it easier to approach the muscle layer and more careful myotomy was performed. Although further investigation is needed, a thin therapeutic endoscope may be useful for TSE.

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Disclosure: Nothing to disclose.

PP0154

A SINGLE-CENTRE RETROSPECTIVE STUDY OF ENDOSCOPY REFERRALS FOR ANAEMIA: INSIGHTS INTO CANCER AND NON-CANCER PATHOLOGIES

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Introduction: Anaemia is common and in some cases an indication for an urgent endoscopy via two week wait (2WW) pathway. This pathway allows patients with alarming symptoms to see a specialist within two weeks to exclude cancer. In this retrospective study, we assessed the clinical outcomes of such referrals, describing the types of cancer identified after full investigations in our centre.

Aims & Methods: Data was collected between December 2021 and December 2022 using Endoscopy UNISOFT system for all endoscopy procedures prompted by anaemia at Milton Keynes University Hospital, United Kingdom. Basic demographics, faecal immunochemical test (FIT) and 2WW referrals from both upper gastrointestinal (UGI) or colorectal pathway were recorded. Radiological images and multi-disciplinary team outcomes were reviewed.

Results: Total number of patients referred to our endoscopy unit was 389, 70% were referred via colorectal team while 30% were referred via UGI pathway. Age of patients was distributed between 19 to 98 years old with majority aged between 60 to 88 years old. 227 patients had both gastroscopy (OGD) + colonoscopy (COL), 76 had OGD+CT scan, 9 patients had OGD+COL+CT and 5 patients had COL+CT. The other patients had CT only, OGD only, or COL only, depending on fitness.

Data revealed that 33 (8.5%) patients referred with anaemia had cancer. 21 patients had colorectal cancer while 8 patients had upper GI cancer. Non-GI cancers were found in four patients, 2 of them lung cancer, 1 renal cancer and 1 tonsillar cancer. There were 187 patients (48%) that had UGI non-cancer pathology e.g. gastritis, duodenitis, peptic ulcer disease, oesophageal varices and coeliac disease. The remaining causes of anaemia are found in Table 1.

Total cause of Anaemia	Number of Patients
Cancer (Colon, UGI, Other)	33
Other UGI pathologies	187
Unknown	92
Other colorectal pathologies	55
Angiodysplasia (UGI + colon)	18
Anaemia of Chronic Disease	2
Gynaecological	2
Total	389

Table 1: Causes of Anaemia.

Positive FIT and colorectal cancer:

Out of the 389 patients, 100 patients (25%) had stool FIT. 37 patients had a positive FIT, and only 5 patients (14%) were diagnosed with colorectal cancer. In the remaining 16 patients with colorectal cancer, FIT was not performed.

Conclusion: The percentage of cancer in our population with anaemia was 8.5%. The main gastrointestinal cancer was colon cancer (5.4%), followed by upper gastrointestinal cancer (2.05%) and non-GI cancer was (1.02%) only. According to previous data, 8%-10% of cases with anaemia were found to have cancer which is not very different from our audit (1,2). The most common cause of anaemia was UGI non-cancer pathology, however, there was no clear aetiology identified in 23% of our population. Four non-GI cancer were discovered by radiological images/CT scans. Those patients with non-GI cancer were above the age of 60. Although the

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number is small, radiological imaging remains an important tool to investigate anaemia in elderly people when both OGD and colonoscopy are unremarkable. FIT test revealed a high false positive rate with only 14% with positive FIT test had colorectal cancer.

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Disclosure: Nothing to disclose.

PP0155

BIOPSY PRACTICE DURING DIAGNOSTIC UPPER GASTROINTESTINAL ENDOSCOPY IN THE UK: ANALYSIS OF 638 000 ENDOSCOPIES IN THE NATIONAL ENDOSCOPY DATABASE

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Introduction: Appropriate tissue sampling is vital for lesion diagnosis during endoscopy. We have previously found an association between higher biopsy rates and reduced compliance with national biopsy recommendations (e.g. from ulcers) with higher rates of post endoscopy upper gastrointestinal cancers (PEUGIC).

Aims & Methods: The aim of this study was to examine biopsy practice in the UK, along with comparing results to the recommendations set out by the European Society of Gastrointestinal Endoscopy.

Upper gastrointestinal (UGI) endoscopies between 01/01/2019 and 29/02/2020 were analysed from the National Endoscopy Database (NED). Surveillance endoscopies were excluded. Logistic regression analysis examined associations with biopsy rate, biopsies from gastritis or duodenitis and biopsies from ulcers.

Results: 638,280 UGI endoscopies were analysed. 289,156 (45.3%) involved biopsies: esophagus 17.1%, stomach 18.0%, duodenal bulb (D1) 3.9%, second part of the duodenum (D2) 19.4%.

Biopsy rate was associated with: clinical endoscopists OR 1.31 (95% CI 1.29-1.33) and surgeons OR 0.84 (95% CI 0.90-0.95), compared to gastroenterologists; patient age 70-79 OR 1.12 (95% CI 1.05-1.19); male patients OR 1.07 (CI 95% 1.06-1.08); urgent endoscopy OR 1.38 (CI 95% 1.35-1.39); emergency endoscopy OR 0.37 (95% CI 0.36-0.39); and therapy during endoscopy OR 0.25 (95% CI 0.24-0.27) .

During endoscopy for dysphagia with a normal appearance, only 5232 (36.3%) had biopsies to exclude eosinophilic esophagitis, with usually too few biopsies taken (mean 4.5 (SD±2.0)). When the endoscopic diagnosis was reportedly normal (excluding dysphagic indications), 35% involved potentially unnecessary biopsies (263557 biopsies). Biopsies were commonly taken from uncomplicated inflammatory conditions: 31% of endoscopic gastritis (58765 biopsies) and 31% of endoscopic duodenitis (15580 biopsies). Biopsies for gastritis/duodenitis were associated with surgeons OR 1.36 (95% CI 1.26-1.46) and clinical endoscopists OR 1.38 (95% CI 1.31-1.45), compared to gastroenterologists.

Mean number of biopsies from esophageal ulcers was 3.7 (SD±2.1) and from gastric ulcers 3.8 (SD±2.4).Biopsies of gastric or esophageal ulcers were associated with clinical endoscopists OR 1.51 (95% CI 1.37-1.65) and surgeons OR 0.53 (95% CI 0.50-0.56), compared with gastroenterologists; outpatient procedures OR 1.40 (95% CI 1.27-1.53); and emergency procedures OR 0.66 (95% CI 0.55-0.79).

Conclusion: Analysis of NED reveals a large number of apparently unnecessary biopsies for gastritis or duodenitis or during normal endoscopy. In contrast, in only 36% of normal endoscopies for dysphagia were biopsies taken to rule out eosinophilic esophagitis. Quality improvement efforts are needed to reduce unnecessary biopsies, increase appropriate biopsies, reduce costs and environmental impact.

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Disclosure: Nothing to disclose.

PP0156

SEDATION PRACTICE IN THE UK: AN ANALYSIS OF 617,000 DIAGNOSTIC UPPER GI ENDOSCOPIES IN THE NATIONAL ENDOSCOPY DATABASE

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Introduction: Sedation is commonly used during upper gastrointestinal (UGI) endoscopy to improve tolerability. We have previously reported an association between increased sedation use and lower rates of post endoscopy upper GI cancer. We have analysed sedation practices for diagnostic upper GI endoscopy in the National Endoscopy Database (NED).

Aims & Methods: The aim of this study was to examine endoscopy practice in the UK and quantify variation in sedation use during diagnostic endoscopy. Diagnostic endoscopies performed between 01/01/2019 and 29/02/2020 were analysed from NED data. Logistic regression analysis examined associations with sedation use and procedures abandoned due to tolerability.

Results: 314,725 (51%) UGI endoscopies involved sedation: midazolam alone 36%; combined midazolam + fentanyl 14%; midazolam + pethidine 0.9%. In patients over the age of 70, during 11 887 of 206 386 endoscopies (5.8%), more than the recommended 2.5mg of midazolam was given (mean 3.4mg (SD±0.76)), including 3320 in which fentanyl was co-prescribed (mean 67.0mcg (SD±29.1)).

Sedation use was associated with: compared to gastroenterologists, clinical endoscopists OR 0.57 (95% CI 0.56-0.58), surgeons OR 0.65 (95% CI 0.64-0.66), SAS doctors 0.65 (95% CI 0.64-0.66); younger patients; males OR 0.60 (95% CI 0.59-0.61); afternoon lists OR 0.98 (95% CI 0.97-0.99) and weekends (OR 0.88 (95% CI 0.87-0.90); compared with routine endoscopy, urgent 1.22 (95% CI 1.21-1.23) and surveillance OR 0.91 (95% CI 0.88-0.94). Sedation + opioid was most frequently used by gastroenterologists, but less by others: clinical endoscopists OR 0.26 (95% CI 0.25-0.27); surgeons OR 0.26, (95% CI 0.25-0.27); and SAS doctors 0.33 (95% CI 0.32-0.34).

3499 endoscopies were abandoned due to poor tolerance: age mean 59.1 (SD±17.1), female 59%, no sedation 57%, midazolam only 32% and midazolam + opioid 10%. This was associated with: surgeons OR 1.29 (95% CI 1.07-1.53) and clinical endoscopists OR 1.39 (95% CI 1.19-1.62), compared to gastroenterologists; addition of opioids OR 0.66 (95% CI 0.59-0.73); and no sedation OR 1.32 (CI 95% 1.23-1.40).

Conclusion: There is lower use of sedation overall, and combined midazolam and opioid particularly, among endoscopists other than consultant gastroenterologists, despite improved tolerability and reduced aban-

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doned procedures. Sedation is used less in afternoons and at weekends, which may relate to recovery requirements and staffing.
Disclosure: Nothing to disclose.

PP0157

REMIMAZOLAM VS. MIDAZOLAM SEDATION IN UPPER GASTROINTESTINAL ENDOSCOPY IN ADULTS > 65 YEARS: RANDOMIZED CONTROLLED TRIAL

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Introduction: Safe and effective sedation are important factors when performing gastrointestinal endoscopy. However, clinicians should be cautious when using sedatives in older patients because of their decreased physiological reserves. Remimazolam is an ultrashort active benzodiazepine that has attracted attention because of its efficacy. This study aimed to determine the safety and efficacy of remimazolam compared with midazolam as a sedative for diagnostic upper gastrointestinal endoscopy in older patients aged ≥65 years.
Aims & Methods: We randomly assigned patients aged ≥65 years who were scheduled for diagnostic upper gastrointestinal endoscopy. The average induction time, average recovery time, and side effect profile of patients who received remimazolam as a sedative were compared with those of patients who received midazolam.
Results: Patients who received remimazolam had shorter average induction (102.0 sec vs. 327.5 sec, p<0.001) and recovery times (5.8 min vs. 13.4 min, p<0.001) than those who received midazolam. No significant difference in side effect profiles was observed between the groups.
Conclusion: In performing upper gastrointestinal endoscopy in older patients, remimazolam can be a good alternative sedative to midazolam, whether used as monotherapy or in combination therapy.
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Disclosure: Nothing to disclose.

PP0158

LCI (LINKED COLOR IMAGING) AND HISTOPATHOLOGICAL FINDINGS FOR MAP-LIKE REDNESS AFTER *HELICOBACTER PYLORI* ERADICATION THERAPY

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Introduction: Mucosal redness (so called Map-like redness) after *Helicobacter pylori* (*H. pylori*) eradication therapy reflects intestinal metaplasia clarified by changes of intra-gastric condition and repairs for gastric mucosa. It is difficult to distinguish Map-like redness and gastric cancer. Map-like redness is a high risk condition for post *H. pylori* eradicated gastric cancer. Consideration about histopathological findings of mucosal redness after *H. pylori* eradication therapy are not enough.
Aims & Methods: Image-enhanced endoscopy plays an important role in the detection and further examination of gastritis and early gastric cancer (EGC). LCI is also useful for detecting and evaluating gastritis, gastric intestinal metaplasia as a pre-cancerous lesion, and EGC. To investigate correlation between the color difference and immuno-histochemical findings for mucosal redness (map-like redness; intestinal metaplasia) detected by white light and LCI in patients with successful *H. pylori* eradication history. We get the endoscopic images by conventional white light image (WLI) and LCI with LASEREO system (FujiFilm, Tokyo, Japan), and evaluate color values (CIE L*a*b* color space and color analysis by Adobe Photoshop C54). Furthermore, we evaluate intestinal metaplasia histopathologically with the immuno-staining by MUC2/MUC5AC/HID-AB; phenotype (Gastro-intestinal Mixed type or Intestinal type, Incomplete type or Complete type) and relationship between color differences and phenotype of intestinal metaplasia.
Results: (1) Objects are 36 patients (24 males, range 56-87yo, median 71yo) and 43 lesions after successful *H.pylori* eradication therapy; cancer group 15 patients (20 lesions), non-cancer group 21 patients (23 lesions)
(2) Type of intestinal metaplasia in the cancer group and the non-cancer group: GI Mixed type 14/7(cancer group/non-cancer group), p<0.01, Incomplete type 14/6(cancer group/non-cancer group), p<0.01.
(3) In the cancer group, color differences Δa (Green to Red) with LCI between the lesion and the surrounding mucosa is significantly high compared with the non-cancer group (16.0±6.6 vs 11.8±7.0;p<0.05).
(4) Compared severe intestinal metaplasia with none/mild/moderate intestinal metaplasia, color value b (Blue to Yellow) with LCI is significantly low (20.4±9.5 vs 25.9±8.0; p<0.05)
(5) Compared GI mixed type with intestinal type of intestinal metaplasia, color values b (Blue to Yellow) with LCI is significantly low (19.2±8.6 vs 26.4±8.5; p=0.008).
Conclusion: In this study, we observed differences in LCI color values of map-like redness appearing after *H. pylori* eradication depending on the type of intestinal metaplasia. In particular, the gastric cancer group has a high rate of Gastro-intestinal Mixed Type with a high risk of carcinogenesis. This suggests the usefulness of image enhancement by LCI as a marker for the high-risk group of gastric cancer.
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Disclosure: Nothing to disclose.

PP0159

BLOOD VESSEL RE-CONSTRUCTION IN THE SUBMUCOSAL LAYER OF THE POSTOPERATIVE STOMACH INDICATED FROM ESD SPECIMENS

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Introduction: Advances in surgical treatments have improved mortality of postoperative patients with gastric cancer and esophageal cancer. Then, the number of residual and gastric-tube cancers is increasing. Re-surgical treatment for postoperative gastric cancers (Po-GCs) has a high incidence and mortality rate, so the contribution of endoscopic therapy might be large.
However, endoscopic submucosal dissection (ESD) for Po-GCs is considered technically difficult. Especially during mucosal incision, we sometimes encounter unexpectable bleedings.

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Aims & Methods: We investigated the re-construction of blood vessels in the submucosal layer on ESD specimens of Po-GCs.8 cases of Po-GCs (4 cases of residual gastric cancer, 4 cases of gastric-tube cancer) underwent ESD in our hospital from 2016 to 2023 were included. Eligible specimens of cases include the lesion center. Vascular distribution in the shallow submucosal layer (a depth of 500μm from the muscularis mucosae) was well observed and the number of dilatated arteries and veins with a diameter>100μm are counted and compensated by a specimen diameter. These number of vessels were compared with 16 cases of controls (1:2) extracted from general gastric ESD underwent in the same location and period.

Results: There were 0.4±0.5/cm arteries and 1.1±1.1/cm veins/mm in control. There were 2.3±1.4/cm arteries and 5.5±2.0/cm veins in the cases of residual gastric cancers. And there were 4.1±1.1/cm arteries and 2.7±0.4/cm veins in the cases of gastric-tube cancers. Both cases of residual and gastric-tube cancers involved many dilatated arteries and veins (P<0.05) in superficial submucosal layer compared with control. On the other hands, the cases of residual gastric cancers have lots of veins compared with arteries (P <0.05), and the cases of gastric-tube cancers have lots of arteries compared with veins (P <0.05).

Conclusion: Residual stomach is small and arterial blood flow is maintained, but veins are resected on a process of surgery. These would be the reason for venous congestion is noticeably high in the residual stomach. On the other hand, the gastric-tube mucosa tends to be more abundant of blood flow due to the reconstruction method. As a result, branches of arteries remaining in submucosal layer would be developed. From the above, on the process of ESD, the probability of encountering veinous oozing increases on residual gastric cancer, and arterial bleeding increases on gastric-tube cancer. It's suggested that paying more attention to these vascular distributions of the submucosal layer is needed while postoperative gastric ESD is performed.

Disclosure: Nothing to disclose.

PP0160

FEASIBILITY AND SAFETY OF COLLABORATIVE SURGERY FOR GASTROINTESTINAL TUMOURS BY ENDOSCOPISTS AND SURGEONS: A SINGLE-CENTER RETROSPECTIVE ANALYSIS OF 131 PATIENTS

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Introduction: Endoscopic treatment using peroral/anal flexible endoscopy and dedicated through-the-scope electrocautery devices has been developed as a less-invasive intraluminal surgery. However, pure endoscopic treatment has limitations for lesions at the deeper layer or excessively close to the mouth/anus. Conversely, surgical approach sometimes leads to wider resection and subsequent decreases in the patient's quality of life. Collaborative surgery by both endoscopists and surgeons is considered effective for providing less-invasive local resection of gastrointestinal tumours to offset limitations either in pure endoscopic treatments or surgical intervention.

Aims & Methods: To investigate the feasibility and safety of this approach, we here retrospectively evaluated the clinical outcomes of collaborative surgery with endoscopists and surgeons. We consecutively collected data of patients who underwent collaborative surgery for lesions located from the laryngopharynx to the anus between October 2017 and December 2023. Collaborative surgery was indicated after the agreement on both flexible endoscopists and surgeons, considering possible benefits of the cooperation in terms of invasiveness, technical difficulty, and safety. Laryngopharyngeal lesions and cervical oesophageal cancer were removed using the ESD technique under laryngeal expansion using a curved laryngoscope by the otolaryngologist's manipulation. With gastrointestinal surgeons, subepithelial tumours in the gastrointestinal tract and deeply invasive gastric cancers were removed in a full-layered fashion, followed by surgical suturing. Large duodenal cancers were principally treated by ESD followed by sero-muscular suturing for the reinforcement of the mucosal defects.

In large anal canal cancer with pagetoid spread, a dermatologist externally dissected the perianally expanded area following ESD from the rectal side. Collaboration completeness, technical success, procedure time, postoperative hospitalization period, and occurrence of adverse events were analyzed.

Results: Collaboration surgery was performed for 134 lesions (33 laryngopharyngeal, 2 oesophageal, 89 gastric, 8 duodenal, and 2 recto-anal lesions, respectively) in 131 patients. Collaboration completeness was achieved in 129 lesions (96%). The en-bloc resection and pathological R0 resection of lesions were obtained in 127 (95%) and 124 (93%) lesions, respectively. The mean procedure time was 188 min. Patients were discharged on the mean postoperative day of 11. Five patients (4%) developed relevant postoperative adverse events.

Conclusion: The results indicated that collaborative surgery by endoscopists and surgeons was feasible and safe, and should contribute to providing less-invasive treatment than conventional approaches. The frequency of adverse events was considered acceptable. Collaborative surgery is worth considering as a flexible and reliable surgical option when cooperation should outperform a single treatment by either of endoscopists or surgeons. To establish this concept, further accumulation of cases and pieces of evidence by well-designed studies is desired.

Disclosure: Nothing to disclose.

PP0161

FULL-THICKNESS RESECTION DEVICE (FTRD) FOR THE TREATMENT OF GASTROINTESTINAL TRACT (GIT) LESIONS : TREATMENT OUTCOMES AND PREDICTORS OF SUCCES IN AN ASIAN COHORT

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Introduction: FTRD represents a novel treatment option for managing GIT lesions that are complex and not easily removed by conventional endoscopic resection techniques. However, there is limited data outside expert centers describing its efficacy and safety.

Aims & Methods: This study aims to evaluate the utility and safety of the FTRD to remove upper and lower GIT lesions and identify predictors for successful resection.

We included 34 patients (mean age 62.8 ±12.9 years; male-64.7%) who underwent FTRD at the Singapore General Hospital since 2021. We followed the recommended FTRD technique. The primary objective was to evaluate the technical success, defined as the correct

application of the FTRD clip and immediate complete en-bloc resection of the lesion. The secondary objective was to assess the safety using the Clavien-Dindo classification and identify predictive factors for successful FTRD resection. Chi-squared test, Fisher's exact test, and logistic regression were used to analyze categorical data.

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Results: FTRD was performed for both upper (35.3%) and lower GIT (64.7%) lesions (Table-1). The common lesions were neuroendocrine tumors (NET; 52.9%), mesenchymal neoplasms (23.5 %), and residual polyps/advanced adenomas (23.5 %). Most upper GIT lesions were in the stomach (58.3%), followed by duodenum (41.7%). Likewise, the lower GIT lesions were in the rectum (73%), left (3.5%) and right colon (13.5%). The average lesion size was 12.6 ± 6 mm. FTRD deployment was technically successful in 27 patients (79.4 %)(Table-2). We found the technical success was significantly higher for lower than upper GIT lesions (91% vs. 50%, p=0.004). Similarly, lesions in the rectum (100%) had substantially higher technical success than lesions in other locations (p=0.01). Technical success was significantly higher for NET than mesenchymal lesions or residual polyps (100% vs. 37.5% vs. 75%, p=0.001). Overall, the R0 resection rate with negative histological margin was noted in 25 (73.5%) patients and was significantly higher for NET (100%) than other lesions (p=0.001). The mean procedure time was 57.5 ±25 min. Immediate complications occurred in 8 cases (23.5%) and delayed bleeding occurred in 1 patient (2.9%). Most of them were minor and treated conservatively, except one requiring endoscopic repositioning of duodenal FTRD clip causing gastric outlet obstruction. Follow-up endoscopy was performed in 11 patients at an average of 12-months. A retained clip with hyperplastic granulation change was seen in one patient. Multivariate analysis showed lesion size, location (rectum), and type of GIT lesion (NET) predicted significantly higher technical success (p=0.02).

Conclusion: Our results show that FTRD offers an acceptable technical success and R0 resection rate with a low risk of complications. Lower GIT location and non-mesenchymal lesion demonstrated better outcomes with FTRD.

Disclosure: None.

PP0162

VACUUM-STENT TREATMENT FOR TRANSMURAL DEFECTS IN THE UPPER GASTRO-INTESTINAL TRACT: CASE SERIES AND NEW INSIGHTS IN A TERTIARY REFERRAL CENTER

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Introduction: Transmural defects in the upper gastro-intestinal (GI) tract are associated with severe morbidity and mortality. Endoscopic vacuum therapy (EVT) has recently been established as promising endoscopic treatment option for such defects and is most often applied using endoscopically placed vacuum-sponges. The vacuum-stent is a novel device to apply EVT, combining the advantages of negative pressure wound therapy and an intraluminal stent, allowing for oral intake.

Aims & Methods: The aim of this prospective cohort study is to describe the experiences of, and lessons learned by a tertiary referral center with EVT experience since 2018, regarding the vacuum-stent for transmural defects in the upper GI tract. All patients treated with a vacuum-stent between November 2022 and October 2023 were included. Outcome measures included successful defect closure, reasons of treatment failure, adverse events and strictures.

Results: 31 patients were included. Baseline characteristics and outcomes are displayed in Table 1. Success rates of combined vacuum-stent and –sponge treatment and vacuum-stent treatment alone were respectively 83% and 78%. In this cohort, some notable reasons of failure were observed: a defect too close to the upper esophageal sphincter (UES), inadequate vacuum on the esophago-jejunal anastomosis, a carcinoma at a persisting perforated ulcer, and two adverse events (6%). The adverse events included two patients who developed a secondary defect at the site of the proximal vacuum-stent flange, of whom one had a cervical anastomosis with a narrow proximal esophagus and one Boerhaave syndrome. Three out of four patients with a severe structure had underwent a McKeown esophagectomy (cervical anastomosis).

Etiology of defect, n (%)	31
Anastomotic leakage	18 (58%)
Boerhaave syndrome	5 (16%)
Iatrogenic	5 (16%)
Other	3 (10%)
Initial procedure in case of anastomotic leakage, n (%)	18
Ivor Lewis esophagectomy	9 (50%)
McKeown esophagectomy	6 (33%)
Gastrectomy	3 (17%)
Success rate, n (%)	31
Combined vacuum-stent and –sponge treatment	26 (83%)
Only vacuum-stent treatment	24 (78%)
Adverse events, n (%)	2 (6%)
Severe stricture requiring > 3 dilations, incision therapy or self dilation, n (%)	4 (13%)

Table 1. Baseline characteristics and outcomes.

Conclusion: The vacuum-stent is a valuable treatment option for AL after Ivor Lewis esophagectomy, Boerhaave syndrome and iatrogenic defects with success rates of 80-89%. Treating cervical anastomotic leaks is not recommended by us, due to the narrow lumen of the proximal esophagus, the proximity of the UES and high risk of severe strictures. Furthermore, caution is recommended in case of a narrow esophagus and the value of the vacuum-stent is yet to be determined in case of an esophago-jejunal anastomosis. In our experience, vacuum-stent and –sponge treatment complement each other and might be indicated in different situations. Sharing experiences on the topic is important to assess the best techniques and indications, to be able to reach the full potential of the vacuum-stent and further increase the success rate.

Disclosure: R.E. Pouw: consultant for Micro-Tech Europe and Medtronic, speaker fee for Pentax and Boston Scientific
M.I. van Berge Henegouwen: consultant for Mylan, Johnson & Johnson, Alesi Surgical, BBraun and Medtronic, and received unrestricted research grants from Stryker. All fees paid to institution.

PP0163

OUTCOME OF OESOPHAGEAL STENTING FOR MALIGNANT DYSPHAGIA: A 10-YEAR RETROSPECTIVE SURVIVAL ANALYSIS

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Introduction: Advanced oesophageal cancers (OC) exhibit poor median survival rates, and often self-expandable metal stents (SEMS) are used to address malignant dysphagia. Previous studies purport to a high prevalence of adverse events post stent insertion, without an impact on mortality. We aimed to review mortality outcomes of oesophageal stenting in both squamous cell carcinoma (SCC) and adenocarcinomas (AC).

Aims & Methods: One hundred and eighty-six patients who were diagnosed with OC between September 2010 - February 2021 were included. We retrospectively analysed the data for patient demographics, histology, oncological management, oesophageal stenting, and mortality, notably overall survival (OS). Survival probability was calculated using Kaplan-Meier survival curves and compared using Log-rank testing with statistical significance determined by p value <0.05.

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Results: From the 186 OC diagnoses made. Male: Female 136 (73%) and 50 (27%). Mean age at diagnosis was 73 years (range 32-98). Histological findings showed 132 patients had AC (71%), and 54 patients had SCC (29%). Mean OS from date of diagnosis for AC and SCC were 21.6 and 10.6 months respectively.

SEMS were inserted for 98 patients (53%); 37 (38%) for SCC and 61 (62%) for AC. Mean OS for patients with stenting vs not stenting was 11.1 and 26.4 months respectively. Median OS for stenting was 8 months (95% CI, 6.0 - 10.0) vs 10 months for not stenting (95% CI, 7.0 -16.0) with a p<0.0005) (See Figure 1). 30-day mortality post stent insertion was 23%.

For patients who underwent oesophageal stent insertion, mean OS for AC and SCC 11.9 months (95% CI, 8.989 to 14.913) and 10.1 months (95% CI, 6.709 to 13.561) respectively. Median OS for stent population for SCC and AC were 6 months (95% CI, 5.0 -8.0) and 9 months (95% CI, 6.0 -12.0) respectively (p = 0.3765).

Conclusion: Patients with OC who underwent stent insertion had overall poor survival outcome. This could reflect advanced stage of OS. Stenting is often useful for malignant dysphagia in patients who harbor inoperable disease. Analysis of stent adverse events would be useful.

Disclosure: Nothing to disclose.

PP0164

GASTRIC INTESTINAL METAPLASIA DETECTION ACCURACY AND GENERALIZATION USING CURRENT STATE-OF-THE-ART AI ALGORITHMS

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Introduction: Early detection of gastric cancer may result from adequate diagnosis and surveillance of individuals with gastric intestinal metaplasia (GIM). Computer-assisted diagnosis systems based on artificial intelligence, or more specifically Deep Neural Networks (DNN) applied to esophagogastroduodenoscopy may be crucial in optimizing and standardizing patients' final diagnosis. Literature has shown promising DNN results for single datasets, some of which reaching the 90% accuracy suggested by the European Society of Gastrointestinal Endoscopy for these algorithms [1].

Moreover, traditionally Wight-Light Endoscopy is referred as suboptimal to determine the presence of GIM. We aimed to determine the accuracy of current state-of-the-art AI algorithms for WLE from 2 different scopes' providers for GIM detection.

Aims & Methods: Two datasets were used in these experiments:

• Dataset A (Fujifilm) was used for training and testing, consisting of a private benchmark dataset [1] of which we have used 4876 endoscopic White Light (WL) images (2438 normal, 2438 GIM).

• Dataset B (Olympus) was used for external validation and was collected from a lower quality clinical routine dataset, including 373 WL images (170 normal, 203 GIM). All images were annotated by clinical experts and the primary outcome was GIM classification accuracy.

We conducted experiments comparing the performance of DNNs (ResNet50) in GIM detection by training the models with and without the integration of synthetic/artificially generated endoscopy images. This involved employing four different methods: using the default dataset, augmenting the dataset with classic data augmentation transformations, incorporating images generated by Deep Convolutional Generative Adversarial Networks (DCGAN), and integrating images generated by Wasserstein GAN (WGAN). A part of dataset A (15%) was saved for testing in the different experiments and the remaining data (85%) was used to train the model, using a 5-fold cross-validation. In the end, dataset B was used as an external test set to evaluate the generalization capabilities of the different DNNs.

Results: The ResNet50 (dataset A – test set) achieved an accuracy of 92% for the default dataset, 93% for the data augmentation transformations, 93% for the DCGAN and 94% for the WGAN. The respective sensitivity values were 92%, 91%, 92% and 92%. Then, these models were evaluated in the dataset B, to infer its generalization capabilities. The ResNet50 achieved an accuracy of 63% for the default dataset, 68% for the data augmentation transformations, 65% for the DCGAN and 65% for the WGAN. The respective sensitivity values were 66%, 80%, 73% and 73%.

	Single Dataset (A->A) [Fujifilm WLE]				Generalization (A->B) [Olympus WLE]			
ResNet50	Default	Augm.	DCGAN	WGAN	Default	Augm.	DCGAN	WGAN
Accuracy	0.92	0.93	0.93	0.94	0.63	0.68	0.65	0.65
Sensitivity	0.92	0.91	0.92	0.92	0.66	0.80	0.73	0.70

Conclusion: Even for WLE, current DNN architectures seem to obtain very good results for GIM detection. However, generalization is still insufficient, since validation in different datasets (different clinical settings, different endoscopic technologies) obtains average detection accuracy and sensitivity, illustrating the excessive specialization caused by maximizing accuracy results in single datasets.

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PP0165

FEASIBILITY, SAFETY, AND OUTCOMES OF UGI ENDOSCOPIC SUBMUCOSAL DISSECTION (ESD) FROM UK ESD REGISTRY: A MULTICENTRE PROSPECTIVE STUDY ON WESTERN POPULATION

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Introduction: The outcome of ESD in early gastrointestinal neoplasia in Japan is well established. However, in the west, ESD practice is variable given different disease prevalence in UGI tract and also the volume and experience of each centre.

Aims & Methods: We aim to evaluate ESD practice and outcomes in the UK and hence, multi centre registry was set up. This is a prospective observational study for patients undergoing ESD for early GI neoplasia. All technical and outcomes data was prospectively collected onto REDCap.

Results: There were 451 UGI ESD cases recruited between August 2016 to August 2023 from 6 large tertiary referral centres in the UK. Mean age of the patients was 70.8 years (range: 24-94 years) with male preponderance (318 males vs. 133 females). Majority were in oesophagus: 282 (62.53%), stomach: 159 (35.25%) and duodenum: 10 (2.22%).

Of 282 oesophageal ESDs, 277 (98.23%) were technically successful with en-bloc resection rate of 96%. There were 3 intraprocedural complications (2 oesophageal perforations and 1 oral laceration from overtube). All of them were managed endoscopically. There were 29 (10.28%) post ESD strictures requiring endoscopic dilatations. 3 (1.06%) delayed bleeding were noted and 3 were readmitted for decompensation of underlying co-morbidities.

There were 204/282 (72.34%) Barrett's cases with average length of C3 M5 and mean size 34.5mm. 34 cases (16.67%) had previous ESD or ablation at the same site. Post ESD resection showed deep submucosal (SM) invasion in 21 patients (deep R0 achieved in 42.9% and lateral R0 in 76.2%), 23 pT1bSM1 (deep R0 in 87.0% and lateral R0 in 87%), 107 pT1a (deep R0 in 89.7% and lateral R0 in 85.1%), 31 high grade dysplasia (HGD) (R0 in 97%) and 9 low grade dysplasia (LGD) (R0 in 78%) as shown in Table 1.

There were 44 squamous cell carcinoma (SCC), 26 gastro-oesophageal junctional neoplasia and 8 others (submucosal lesion and granular cell tumours). Of 44 SCC, there were 8 pT1b (5 achieved deep R0, 7 achieved lateral R0), 10 pT1a (all achieved both deep and lateral R0), 22 HGD (81.8% achieved R0), 3 LGD (all achieved R0) as shown in Table 1.

There were 160 gastric ESDs with completion rate of 96.25% and en-bloc rate of 95.5%. There were 8 (5%) intra-procedural perforation which required endoscopic clippings. There were 7 (4.36%) delayed bleeds which were managed endoscopically. Of 160 cases, 138 were epithelial neoplasia, 11 were submucosal lesions while 4 were large hyperplastic polyps. As shown in Table 1, of 138 epithelial neoplasia, 10 had deep SM invasion (7 achieved deep R0, 8 lateral R0), 14 pT1bSM1 (7 achieved deep R0, 10 lateral R0), 50 pT1a (90% achieved deep R0, 80% lateral R0), 35 HGD (94.3% R0) and 27 LGD (77.8% R0).

	Deep SM invasion (n)	Deep R0 (%)	Lateral R0 (%)	pT1b SM1 (n)	Deep R0 (%)	Lateral R0 (%)	IMC (n)	Deep R0 (%)	Lateral R0 (%)	HGD (n)	R0 (%)	LGD (n)	R0 (%)
Barrett's neoplasia	21	42.9	76.2	23	87.0	87.0	107	89.7	85.1	31	97.0	9	78.0
Oesophageal SCC	3	66.7	100.0	5	60.0	80.0	10	100.0	100.0	22	81.8	3	100.0
Gastric neoplasia	10	70.0	80.0	14	50.0	71.4	50	90.0	80.0	35	94.3	27	77.9

Table 1. Post ESD histology outcome in UGI tract.

Conclusion: Unlike in the East, most ESDs in our study were performed in Barrett's oesophagus rather than the stomach. Our data demonstrates the feasibility and safety of UGI ESD in Western setting with high en-bloc rate and low complication rates. We found that within indication of ESD, complete resection (R0) rate is high but lesions with deep submucosal invasion have poor R0 rate.

Disclosure: Professor Bhandari has received research grants or is the advisory board for Fujifilm, Boston, Olympus, Pentax, 3-D matrix, NEC (Japan), Medtronic.

PP0166

CLINICAL EFFICACY AND SAFETY OF ENDOSCOPIC DILATATION WITH ESOFLIP IN BENIGN ESOPHAGEAL STRICTURES

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Introduction: EsoFLIP is a novel dilation balloon that utilizes high-resolution impedance planimetry to provide real-time, objective visualization and monitoring of dilation.

Aims & Methods: The aim of our prospective study is to evaluate clinical and technical success and safety of endoscopic dilation with EsoFLIP in patients affected by benign esophageal strictures (NCT05725473). Patients with symptomatic esophageal strictures, with exception of patients affected by motor disorders, were selected and endoscopic dilatations were performed with EsoFLIP (8-20 mm dilation range). Clinical condition was evaluated before and after procedure (one week and one month) using a specific questionnaire about severity and frequency of dysphagia (Brief Esophageal Dysphagia Questionnaire-BEDQ 10 items from 0 to 6). We also quantified rate of technical success, adverse events, total procedure time and fluoroscopy exposure time. We utilized EndoFLIP (impedance planimetry system) to evaluate stricture diameter (mm), cross-sectional area (CSA mm2), distensibility index (DI mm2/mmHg) before and after dilatation with EsoFLIP.

Results: We enrolled 15 patients affected by esophageal strictures due to lymphocytic (2) and eosinophilic esophagitis (2), caustic ingestion (2), Schatzki rings secondary to GERD (3), esophageal-gastric anastomosis (6). Strictures involved proximal-medial esophagus (18-30 cm from incisors) in 7 patients out of 15 and in 8 out of 15 patients medial-distal esophagus (30-42 cm). All dilation procedures were technically successful, and no adverse events occurred. The median procedure time was 52.67 min (±12.87) and fluoroscopy exposure time was 5.14 min (±1.23; 9.76% of procedure). After 1 week and 1 month from the dilatation there was a significant decrease of dysphagia severity and frequency shown by BEDQ total score (pre-BEDQ 6.87±5.82 vs 1 week-post-BEDQ 3.20±3.69, p-value 0,0486; pre-BEDQ 6.87±5.82 vs 1 month-post-BEDQ 2.33±2.81, p-value 0.0206). After procedure, there was a significant increase of stricture diameter (pre-di-

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ameter 8.55±2.80 vs post-diameter 12.44±2.20 mm, p value=0.0002), distensibility index (pre-DI 1.72±1.63 vs post-DI 3.45±1.99 mm2/mmHg, p value= 0.0151) and cross-sectional area (pre-CSA 70.93 ± 45.69 vs post-CSA 114.93± 33.79 mm2, p value=0.0075).

	Pre Eso-FLIP dilatation	Post Eso-FLIP dilatation	1 week post Eso-FLIP dilatation	1 month post Eso-FLIP dilatation
BEDQ total score	6.87±5.82		3.20±3.69 (p= 0.0486)	2.33±2.81 (p=0.0206)
Stricture diameter (mm)	8.55±2.80	12.44±2.20 (p=0.0002)		
Distensibility index (mm^2/mmHg)	1.72±1.63	3.45±1.99 (p= 0.0151)		
Stricture cross-sectional area (mm^2)	70.93 ± 45.69	114.93± 33.79 (p=0.0075)		

Conclusion: Our data showed that endoscopic dilatation of benign esophageal strictures with EsoFLIP is clinically effective and safe. EsoFLIP is a promising dilation technology that may yield a larger diameter change and may potentially reduce fluoroscopy time exposure when compared to traditional endoscopic dilation.

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Disclosure: Nothing to disclose.

PP0167

COMPARATIVE ANALYSIS OF TWO GRAPHICAL USER INTERFACES FOR COMPUTER AIDED DETECTION OF BARRETT'S NEOPLASIA

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Introduction: The development of artificial intelligence (AI) for endoscopy has significantly increased, yet the aspect of endoscopist-AI interaction remains insufficiently explored. This interaction is crucial as it could substantially influence the AI system's performance in routine clinical practice.

Aims & Methods: This study aimed to evaluate two different graphical user interfaces (GUIs) for a computer aided detection (CADe) system for recognition of early Barrett's neoplasia. We performed a comparative analysis of two GUI designs—the frequently used bounding box GUI and a novel heatmap GUI—using an extensively evaluated CADe system[1]. Thirty-seven endoscopists from six countries participated. Each endoscopist assessed 70 Barrett's esophagus. Every video was analyzed by the CADe system and contained, at some point, a detection which was visualized by the GUI. The study was conducted in two phases, with each video initially shown using one of the GUIs, followed by a two-week washout period after which the same videos were displayed with the alternative GUI. Endoscopists were tasked with identifying potential neoplastic lesions, selecting biopsy sites, and stating their GUI preference.

Results: No significant differences were found for classification performance between the two GUIs (sensitivity 83% vs. 83%, p = 0.29; specificity 86% vs. 86%, p = 0.09). Both interfaces achieved similar scores in localization accuracy, with a median of 97%. Among the participants, 23 expressed a preference for the heatmap GUI, while 14 favored the bounding box.

Conclusion: Despite a general preference for the heatmap GUI among endoscopists, there were no statistically significant differences in performance outcomes between the two interfaces.

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Disclosure: Nothing to disclose.

PP0168

DIAGNOSTIC ABILITY AND ADVERSE EVENTS OF MUCOSAL INCISION-ASSISTED BIOPSY FOR GASTRIC SUBEPITHELIAL TUMORS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Currently, the Japanese guidelines for GISTs suggest endoscopic ultrasound-guided fine-needle aspiration for sampling the tissue of subepithelial tumors (SETs) and as a first-line technique. Mucosal incision-assisted biopsy (MIAB) is another tissue sampling technique; in this technique, the mucous membrane covering the SET is cut to expose and sample the tissue directly under direct visualization. However, the Japanese guidelines do not clearly state how to handle this technique, and the MIAB procedure has not been standardized, especially in technical aspects, the need for local injection to the submucosa before mucosal incision to prevent perforation, and the need for acid secretion inhibitors to prevent postoperative bleeding.

Aims & Methods: This systematic review and meta-analysis aimed to evaluate diagnostic ability and examine the efficacy of countermeasures to adverse events of MIAB for gastric SETs. We performed a literature search and identified 533 relevant articles. Eleven articles, including 339 lesions, were ultimately used in the meta-analysis. The primary endpoint was the pathological diagnostic rate of MIAB for gastric SETs, and the secondary endpoint was the incidence of adverse events. The efficacy of acid secretion inhibitors in preventing postoperative bleeding and that of local injection before incision to prevent perforation were also examined.

Results: Nine studies were performed in Japan and two in South Korea, of which only two were prospective studies. The mean age of the patients was 60.7 years, and 48.4% were male. In all studies, endoscopic ultrasound was performed to assess tumors before incision-assisted biopsy. The mucosal incision procedure was the same in all studies: The mucosal incision on the tumor was made, and tissue was sampled directly from the tumor using biopsy forceps; however, in all cases in eight articles and six cases in one article, a local injection of saline or glycerol into the submucosa before the mucosal incision was performed. Nine articles reported the closure of the incision using clips. The mean tumor diameter was 20.6 mm, and the pathological results included 150 GISTs, 70 leiomyomas, 11 schwannomas, 37 ectopic pancreas, three lipomas, three cases of inflammatory fibrotic polyp, two malignant lymphoma lesions, two glo-

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mus tumors, one neuroendocrine tumor, one adenocarcinoma, one old hematoma, and one renal cell carcinoma. In all cases in six articles and some in two, an intravenous or oral gastric acid suppressant such as esomeprazole or vonoprazan was administered after treatment. The pooled pathological diagnostic rate of MIAB for gastric SETs was 86.0% (95% confidence interval [CI]: 78.6–93.4; I^2 = 96%). The adverse event rate of the pooled population was 0.3% (95% CI: −1.15 – 1.76; I^2 = 0%). The acid secretion inhibitors significantly reduced postoperative bleeding (odds ratio: 0.06, 95% CI: 0.01– 0.66, P = 0.02). Perforation occurred in 0% and 2.6% of the local injection and non-local injection cohorts, respectively, and the pathological diagnostic rates were 50% and 66.7%, respectively.

Conclusion: MIAB is a reliable technique with a favorable diagnostic rate and few adverse events. Acid secretion inhibitors effectively prevent postoperative bleeding, but the effectiveness of local injection remains unclear. This technique is considered a robust tissue sampling method for gastric SETs.

Disclosure: Nothing to disclose.

PP0169

DIAGNOSIS AND SURVEILLANCE OF PREMALIGNANT GASTRIC LESIONS IN AN INTERMEDIATE-RISK POPULATION: WHERE ARE WE?

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Introduction: Gastric cancer is associated with a poor prognosis due to the late stage of the disease at the time of diagnosis. Consequently, the detection and surveillance of the established premalignant gastric lesions becomes imperative. Despite Eastern Europe being a region of high – intermediate gastric cancer incidence, the guidelines (MAPS-II) (1) are mainly based on the data from Western countries due to a lack of research in this area in Eastern countries.

Aims & Methods: This study aimed to assess a) the incidence of gastric cancer in patients with proven histologically proven premalignant gastric lesions b) the efficacy of endoscopic diagnosis of premalignant gastric lesions and c) trends of subsequent management of patients at an intermediate risk for the development of gastric cancer in an academic center in Lithuania.

We conducted a retrospective cohort study of patients who underwent esophagogastroduodenoscopy with histological sampling between 2012 and 2020 in a single academic centre. The pathology database was filtered for keywords “atrophy” and/or “intestinal metaplasia” in pathological conclusions or descriptions to identify patients with possible premalignant gastric lesions. Demographic (age and gender), pathology report (histological findings, presence of *H.pylori*), endoscopic evaluation (diagnosis, location of biopsies, endoscopic recognition of premalignant or cancerous lesions), patient history (family history of gastric cancer) and subsequent management (recommended *H.pylori*eradication and follow-up) data were collected.

Results: This retrospective cohort study conducted in a single academic centre identified 1842 patients who underwent diagnostic esophagogastroduodenoscopy with histological sampling between 2012 and 2020. Out of 1842 patients screened in the pathology database, 1762 had histologically proven premalignant gastric lesions (extensive gastric mucosal atrophy, intestinal metaplasia or both).

The study cohort consisted of 51% males with an average age of 64.42 (SD ±1.573). A positive family history of gastric cancer was reported in 0.7% of the patients.

Histological evaluation identified gastric atrophy in 9.9% (n = 183) of samples, distal atrophy in 6.1% (n = 113) of cases, and panatrophy in 3.8% (n = 70) of samples. Distal intestinal metaplasia was observed in 47.9% (n = 883) of cases, while extensive intestinal metaplasia was found in 47% (n = 866) of samples. Dysplasia was detected in 1.03% (n = 19) of the samples.

Endoscopically precancerous lesions were identified in only 14.4% (n = 254) of cases, with follow-up recommendations provided for 3.4% (n = 60) of the patients.

Helicobacter pylori presence on biopsy samples was detected in 37.6% (n = 662) of cases, with eradication recommended only for 36.7% (n = 243) of these cases.

Subsequently, during follow-up 0.8% (n = 14) of patients with premalignant gastric lesions were diagnosed with gastric cancer.

Conclusion: This study reveals poor efficacy of endoscopic evaluation for precancerous gastric lesions and suboptimal adherence to current guidelines for their management. These findings underpin the importance of high-quality endoscopy, dissemination and adherence to the guidelines in the hopes of reducing the morbidity and mortality of gastric cancer.

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Disclosure: Nothing to disclose.

PP0170

THE ROBOT-ASSISTED SYSTEM YUNSROBOT FOR REMOTE ENDOSCOPY: AN EXPLORATORY STUDY

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Introduction: The introduction of robotic and communication technology into soft endoscopy allows remote endoscopy become a reality. The ability to perform upper gastrointestinal endoscopy from remote location will eliminate geographical constraints and make endoscopic expertise available throughout the world, improving patient treatment and healthcare management, especially when faced with a pandemic. Therefore, it is of great practical significance to develop a robot system that can be used for remote endoscopy.

Aims & Methods: This study aimed to evaluate the safety and effictiveness of the robot-assisted system YunSRobot for remote human upper gastrointestinal endoscopy and investigate the optimal network for remote endoscopy.

The robot-assisted system YunSRobot was developed by the Yunsheng Y group from the PLA General Hospital and Hao L group from the State Key Laboratory of Robotics, Shenyang Institute of Automation. The master manipulator is located in the Hainan Hospital of PLA general hospital (Sanya, Hainan Province, China), while the slave robot and the patient are located in the First Medical Center, PLA general hospital (Beijing, China)——a total distance of about 2,900 km. The two sites were connected through 5G network and remote upper gastrointestinal endoscopy were performed. The average network delay, operation time, standardized images collection and the adverse events were recorded.

Results: Three remote upper gastrointestinal endoscopy operations were safely performed through a 5G network, with an average time delay of 100~150 ms. The total operation time was 1370±126.75 s, while average en-

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try time for pharynx, esophagus, stomach and pylorus was 207.67±175.05 s, 78.00±29.68 s, 104.00±10.03 s, 30.67±30.40 s, respectively. The average operation time of duodenum was 141.67±40.88 s. Forty standardized images were collected from each patient. All patients successfully completed the biopsy (gastric body, gastric angle and gastric antrum) and no adverse events occurred during the procedures.

Conclusion: The robot-assisted system YunSRobot can realize remote human upper gastrointestinal endoscopy, which has high effictiveness and safety. The 5G network facilitate smooth operation and acceptable latency. Our exploratory study can provide insights for promoting the future development of remote endoscopic surgery.

Disclosure: Nothing to disclose.

PP0171

THE USEFULNESS OF ENDOSCOPIC HAND-SUTURING (EHS) FOR CLOSURE OF THE MUCOSAL DEFECT AFTER GASTRIC ENDOSCOPIC SUBMUCOSAL DISSECTION (ESD)

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Introduction: ESD has become the standard treatment for early gastric cancer. Recently, the number of patients taking antithrombotic agents (ATAs) has increased, and the prevention of post-ESD bleeding is a main concern. EHS after gastric ESD has been developed to solve these problems, but the effectiveness and procedural success rate of EHS is not known. Here we investigate the successful introduction of EHS and evaluate the safety and effectiveness of the EHS.

Aims & Methods: 11 patients who underwent EHS after gastric ESD from December 2022 to April 2024 were included in this study. Patient and lesion background, endoscopic findings, and treatment outcomes were reviewed retrospectively. A V-loc 180 available barbed suture (VLOCL0604; Covidien, Mansfield, Mass, USA) and through-the scope flexible needle holder (SutuArt, Olympus, Tokyo, Japan) were used for EHS. The same endoscopist and the assistant performed Ex vivo training model prior to clinical introduction.

Results: Mean age was 74.8±7.3 years, 9 males and 2 females, and 4 with and 7 without ATAs. The lesion locations were as follows: upper/middle/lower = 4/0/7. Mean tumor size was 15.4±5.3 mm, mean defect size was 30.3±5.4 mm, and mean resection time was 46.6±26.4 minutes. The mean time needed for suturing was 30.6±9.6 minutes, and a mean of 4.7±0.4 stitches were placed. Suturing speed （resection area/suturing time) was 20.9±7.5 mm^2/min. Complete suture was achieved in all patients, and they began clear fluid on POD1. There were no adverse events during EHS such as bleeding or perforation.

Conclusion: In EHS, complete suture was achieved in all cases, with no post-ESD bleeding or delayed perforation. Further data is needed to confirm its effectiveness.

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Disclosure: Nothing to disclose.

PP0172

THE FEASIBILITY AND SAFETY OF GASTRIC ENDOSCOPIC SUBMUCOSAL DISSECTION IN PATIENTS TAKING GLUCOCORTICOIDS OR IMMUNOMODULATORS

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Introduction: Glucocorticoids (GC) and immunomodulators (IM) are frequently used to maintain clinical remission in various diseases, yet GC is implicated in delaying gastric ulcer healing [1], and IM is associated with post-endoscopic submucosal dissection (ESD) fever and post-coagulation syndrome on colon ESD [2]. The safety of gastric ESD in patients receiving oral GC or IM remains unknown.

Aims & Methods: This study evaluates the impact of oral GC or IM on gastric ESD outcomes, focusing on procedural safety and postoperative recovery. In a single-center, retrospective study, we analyzed clinical outcomes for 411 patients who underwent gastric ESD between April 2017 and April 2022 at our institute in Fukuoka, Japan. We compared data between patients using a GC or IM (32 patients) group and a standard group (379 patients), considering clinicodemographic characteristics and procedural details. Patients were further divided into three subgroups (SG) based on lesion size and procedure duration to consider treatment intensities: SG1 for lesions >20 mm and procedure times >60 minutes; SG2 for either lesions >20 mm or procedure times >60 minutes; and SG3 for lesions <20 mm and procedure times <60 minutes.

	GC Group (n=32)			Standard Group (n=379)			p-value*
Subgroup (n)	SG1 (9)	SG2 (10)	SG3 (13)	SG1 (125)	SG2 (164)	SG3 (90)	-
hospital stay(days)		10.8±7.1			8.6±3.7		0.0026
	14.1±2.3	9.3±2.2	9.8±1.9	9.4±4.8	8.3±3.3	8.1±2.0	-
□log WBC		0.12±0.12			0.20±0.15		0.005
	0.14±0.1	0.12±0.095	0.10±0.16	0.22±0.12	0.18±0.1	0.21±0.24	-
abdominal pain (%)		37.5%			16.9%		0.0066
	55.6%	40.0%	23.1%	20.0%	14.0%	15.6%	-
post-ESD fever (%)		12.5%			3.4%		0.035
	22.2%	0.0%	15.4%	6.4%	1.8%	2.2%	-

Table.
GC, glucocorticosteroid; SG, subgroup; log,logarithmetic; WBC, white blood cell
**Analyzed by the Fisher exact test and the Student t-test. Values are shown as averages±standard deviation unless otherwise stated.*

Results: The GC group (n=32, 93.8% on GC, median GC dose 5mg/day, 37.5% on IM) showed a higher prevalence of females (53.1%), were older (average age 77.5 vs. 73 years), and had poorer performance status than the standard group (n=379). Despite similar tumor characteristics, the GC group experienced more frequent abdominal pain (37.5% vs. 16.9%) and post-ESD fever (12.5% vs. 3.4%), resulting in longer hospital stays (average 10.8 days vs. 8.6 days) (Table). Meanwhile, the GC group's blood tests showed less elevation in white blood cell (WBC) and C-reactive protein

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(CRP) levels compared to the standard group. Intraoperative perforation occurred in 6.2% of the GC group but was managed conservatively without any cases of delayed perforation. Subgroup analysis revealed that SG1 experienced the most significant increases in abdominal pain, post-ESD fever, and WBC/CRP levels (Table).

Conclusion: Gastric ESD in patients on GC or IM were feasible but poses higher procedural risks and more frequent postoperative complications, particularly at higher treatment intensities. These findings highlight the need for customized procedural strategies and increased postoperative care to enhance outcomes for this vulnerable patient population.

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Disclosure: Conflict of interest Hiroki Fukuya, Yoichiro Iboshi, Ryota Okitsu, Akito Ohkubo, Taisuke Sasaki, Shohei Hamada, Daisuke Yoshimura, and Naohiko Harada, declare that they have no conflict of interest. Eikichi Ihara participates in the funded research of Takeda Pharmaceutical Co., Ltd and belongs to an endowed course supported by companies including Ono Pharmaceutical Co., Ltd, Miyarisan Pharmaceutical Co., Ltd, Sanwa Kagaku Kenkyusho Co., Ltd, Otsuka Pharmaceutical Factory, Inc, Fujifilm Medical Co, Ltd, Terumo Corporation, Fancl Corporation, and Ohga Pharmacy.

PP0173

LOW ACCURACY OF ENDOSCOPIC DIAGNOSIS FOR DIMINUTIVE EARLY OROPHARYNGEAL CANCER

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Introduction: Thanks to instrumental advances and accumulation of understandings of endoscopic features of early oropharyngeal neoplasia, we often encounter diminutive pharyngeal lesions. However, whether follow-up is permissible or therapeutic intervention is preferable for these lesions is uncertain. Accurate endoscopic diagnosis is pivotal for the decision-making process. This study aimed to assess the accuracy of endoscopic diagnosis for pharyngeal microlesions.

Aims & Methods: We evaluated pre-treatment endoscopic images of 121 lesions 3 mm or less in that pharyngeal neoplasia was endoscopically diagnosed with magnifying narrow-band imaging observation and resected by cold forceps biopsy. Two endoscopists, one with over 15 years of experience and another with 5 to 10 years (study A), made endoscopic diagnosis of each image and categorized the images into non-dysplasia, low-grade dysplasia, or squamous cell carcinoma (SCC)/high-grade dysplasia, and the accuracy of the endoscopic diagnosis was evaluated with the pathological findings of resected tissues. To assess interrater agreement, another physician joined to each experience group (study B), and the kappa coefficient was calculated.

Results: In study A, the overall accuracy rate of the endoscopic diagnosis was 60.7%. However, for SCC/high-grade dysplasia, the sensitivity was 12.5% (95% confidence interval [CI]: 3.5-29.0), specificity was 94.3% (95% CI: 90.2-97.0), positive predictive value was 25.0% (95% CI: 7.3-52.4), and negative predictive value was 87.6% (95% CI: 82.6-91.6). Nearly 90% of malignant or high-grade dysplastic lesion were not correctly diagnosed by endoscopic observation even with magnifying narrow-band imaging. Although the concordance rate between study A and B was low (κ coefficient 0.07, 95% CI: -0.15-0.30), study B also exhibited low sensitivity (31.2%, 95% CI: 16.1-50.0), specificity (83.3%, 95% CI: 77.6-88.1), positive predictive value (22.2%, 95% CI: 11.2-37.1), and negative predictive value (88.8%, 95% CI: 83.6-92.9) for SCC/high-grade dysplasia.

	sensitivity	specificity	positive predictive value	negative predictive value
Study A				
Expart, (%)	25.0	93.3	36.4	89.1
Senior, (%)	0.0	95.2	0.0	86.2
Average, (%)	12.5	94.3	25.0	87.6
Study B				
Expart, (%)	37.5	76.2	19.4	88.9
Senior, (%)	25.0	90.5	28.6	88.8
Average, (%)	31.2	83.3	22.2	88.8

Conclusion: The accuracy of endoscopic diagnosis of diminutive malignant pharyngeal lesions was very low regardless of proficiency of endoscopist. Thus, given that such diminutive lesions can be resected with minimally invasive cold forceps biopsy, proactive excision for diagnostic treatment is recommended.

Disclosure: Nothing to disclose.

PP0174

WITHDRAWN

PP0175

THE NONINVASIVE ROLE OF SPLENO-PORTAL INDEX & PLATELET COUNT/SPLEEN RATIO IN THE DIAGNOSIS OF ESOPHAGEAL VARICES IN PATIENTS WITH COMPENSATED CIRRHOSIS

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Introduction: Varices are present in 30-40 % of patients with compensated cirrhosis (Child-Pugh class A). Although screening endoscopy for esophageal varices(O.V.) is recommended to all patients with cirrhosis, this recommendation is not a result of evidence-based data. We studied the association of (platelet count/spleen diameter ratio, insulin resistance and splenoportal index) and the presence of O.V. in patients with compensated cirrhosis.

Aims & Methods: 124 patients with compensated liver cirrhosis due to chronic hepatitis C virus were studied. After clinical, laboratory ultrasound examinations, all patients underwent screening endoscopy and O.V were reported as present or absent. According to presence or absence of varices; two groups were described: group I without varicesand group II with varices.

Results: Among 124 patients with mean age of (51.81±12.94), 2 groups were described: group I (30 patients) and group II (94 patients) with a male majority (20 patients in group I and 66 patients in group II). In group I and group II: the mean platelet count/spleen diameter ratio was (1022.6±73.36, 608.76±58.44) respectively, the mean insulin resistance value was (2.426±0.618, 3.081±0.474) respectively. The mean splenoportalindex (SPI) value was (2.878±0.870, 6.349±0.514) respectively.For platelet

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count/spleen diameter ratio and SPI, the sensitivity was (95.7%), specificity (83.3%) and accuracy (92.7%). For platelet count/spleen diameter ratio andinsulin resistance(IR), the sensitivity was (70.2%), specificity (86.7%) and accuracy (74.1%). For IR and SPI, the sensitivity was (70.2%), specificity (76.7%) and accuracy (71.7%).For the three predictors combined, the sensitivity was (78.7%), specificity (82.2%) and accuracy (79.5%).

	Platelet count /spleen diameter ratio & Splenoportal index	Platelet count / spleen diameter ratio & IR	IR& Splenoportal index	Platelet count / spleen diameter ratio & IR& Splenoportal index
Sensitivity (%)	95.7	70.2	70.2	78.7
Specificity (%)	83.3	86.7	76.7	82.2
Positive Predictive Value (%)	94.7	94.3	90.4	93.2
Negative Predictive Value (%)	86.2	48.1	45.1	55.2
Accuracy (%)	92.7	74.1	71.7	79.5

Table 1. Sensitivity, specificity, positive predictive value, negative predictive value and accuracy when we combine (Platelet count / spleen diameter ratio & Splenoportal index), (Platelet count / spleen diameter ratio & IR) and (IR & Splenoportal index).

Conclusion: Low platelet count/spleen ratio and high SPI are very useful non invasive predictors for the presence of O.V. that could be used either separately or combined to decrease the number of upper GIT endoscopies needed in cirrhotic patients management, However, IR as a non invasive predictor is still in need for further evaluation.

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Disclosure: Nothing to disclose.

PP0176

WITHDRAWN

PP0177

COMPARATIVE STUDY OF ESD AND SURGICAL RESECTION FOR GASTRIC SETS ORIGINATED FROM MUSCULARIS PROPRIA

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Introduction: Endoscopic resection for gastric subepithelialtumors(SETs) originated from the muscularispropria (GSET-PM) has offered less invasive alternatives to surgical resection.

Aims & Methods: The aims of this study were to compare endoscopic subtumoraldissection(ESD) with surgical resection for the removal of GSET-PM. This study involved 17 patients with GSET-PM removed by ESD and 76 patients who underwent curative surgical resection. ESD was attempted in GSET-PM with well marginated tumors which was below 5cm and showed an endoluminal growth pattern according to endoscopic ultrasound(EUS) finding.

Results: ESD group were more likely to have upper portion(10/17, 58.8%) and surgery group were more likely to have mid portion(41/76, 53.8%) (p=0.039). ESD group were smaller median tumor size (25.6 mm vs 35.9 mm, p=0.037) and higher endoluminal ratio (58.5±9.1 % vs 45.8±15.4 %, p=0.002). ESD group were mostly to have Yamada type III (10/17, 58.8%) and surgery group were mostly Yamada type I (52/76, 68.4%)(p< 0.001). Complete resection by ESD was lower than by surgical resection (82.4% vs 100%, p< 0.001). In ESD group, 3 performed surgical resection after ESD (1 incompletely resection and 2 uncontrolled bleeding) and 1 showed perforation was completely resected with endoscopic closure. In surgery group, complications occurred in 6 patients (1 leakage, 1 stricture, 1 hernia and bowel obstruction, 1 wound infection and 2 worsened general condition after surgery). Although surgery group were lower in complication rate than ESD group (p=0.006), severity of complications were higher in the surgery group and there were no mortalities in the ESD group compared with 2 in the surgery group. There was no statistical difference of recurrence and the follow-up period between two group.

Conclusion: ESD can be one of good options for the resection of endoluminal GSET-PM and could be replace treatment by surgical resection in Yamada type III with a high endoluminal ratio.

Disclosure: Nothing to disclose.

PP0178

PROSPECTIVE EVALUATION OF AN EXPLAINABLE DEEP LEARNING MODEL FOR GASTROESOPHAGEAL VALVE ASSESSMENT VIA THE HILL CLASSIFICATION

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Introduction: Standardized classification of the gastroesophageal valve, achievable using the Hill classification, is a common step during gastroscopy. Given the visual nature of endoscopic examinations, artificial intelligence (AI) can support this task. We conducted a single-center prospective study to compare the performance of a self-developed AI model in determining the Hill classification with that of examiners.

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Aims & Methods: From October 2023 to January 2024 consecutive patients with unaltered anatomy of the gastroesophageal junction were enrolled in the study. The AI model ran in the background during routine gastroscopy, providing real-time predictions, while physicians assessed the Hill grade without knowledge of the model's output. The Gold standard was established through a majority vote of a three-member expert committee in the cases where a clear image of the hiatus was available. The primary outcome was accuracy in Hill grade assessment, with secondary outcomes including precision and sensitivity for each grade.

Results: Out of 150 patients that participated, 131 were eligible for analysis. Our AI model exhibited an accuracy of 84.7%, surpassing the physicians' accuracy with 61.8% (p<0.01). Regarding rating the different Hill grades the AI's lowest accuracy, precision and sensitivity were 88%, 53% and 60% respectively, while physicians achieved lowest scores of 66%, 20%, and 30%. Post hoc analysis of the videos using the systems capability for real-time explainability via heatmaps highlighting image regions influencing the classification, revealed that the model used the correct anatomical region for classification.

Conclusion: Our AI system demonstrates superior accuracy compared to physicians in Hill grade assessment. Future research should explore the impact of physician-AI interactions during examinations through a randomized controlled trial. (ClinicalTrials.gov, Trial register number: NCT06040723)

Disclosure: Nothing to disclose.

PP0179

PREVALENCE, RISK FACTORS AND MANAGEMENT OF ENDOSCOPIC RESECTION RELATED DUODENAL STRICTURES

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Introduction: Endoscopic resection (ER) of large (>/-15mm) duodenal laterally spreading lesions (D-LSLs) is now standard of care. Data on prevalence, risk factors and outcomes of post-duodenal ER strictures are absent; we sought to evaluate this is in a Western cohort.

Aims & Methods: We report a retrospective analysis of a prospectively collected cohort of D-LSLs following ER in an expert tissue resection centre. Strictures were considered "severe" if patients experienced obstructive symptoms, "moderate" if an adult gastroscope could not pass the stenosis, or "mild" if there was resistance on successful passage. If dilatation was performed, 2-4 weekly dilatation sessions were conducted until scope passage without resistance was achieved as well as symptom resolution where applicable. Primary outcomes included stricture prevalence, risk factors and management.

Results: Over 193 months until February 2023, 246 lesions in 239 patients were included (51.2% male, median age 70 years [IQR 14], median size 35mm [IQR 25]). 30 (12.2%) resections resulted in stenosis; 14 mild (46.7%), 4 moderate (13.3%) and 12 severe (40%). 18 (7.3%) required balloon dilation (median number of sessions, 2 [IQR 6]). Stricture resolution following index-ER was achieved with balloon dilatation alone in 15/18 patients (83.3%). Two patients had an excellent response to lumen apposing metal stent insertion following ER of fully circumferential D-LSL. A patient with an incompletely resected fully circumferential D-LSL, in which adenocarcinoma was confirmed post-ER, required palliative stent placement after 3 sessions of balloon dilatation, due to gastric outlet obstruction caused by residual tumour. Post ER-defect circumference ≥80% proved the strongest independent predictor for stricture formation (OR 62.9 (CI 17.9-276.7, p<0.001; Table 1). The incidence of stricture formation with an ER-defect ≥80%, 60-79% and <60% was 72.2% (n=26/36), 12.5% (n=4/32) and 0% (n=0/178) respectively. All severe strictures (12/12, 100%) occurred in ER-defects ≥80%.

Parameter		No stenosis n (%)	Stenosis n (%)	OR univariable (95% confidence interval, p value)	OR multivariable (95% confidence interval, p value)
Location involving a flexure	No (D1, D2 or D3)	210 (89.4)	25 (10.6)	-	-
	Yes (D1/D2 junction)	6 (54.4)	5 (45.5)	7.00 (1.9-24.9, p=0.0002)	4.05 (0.33-45.30, p=0.28)
Lesion length (mm)	<50	196 (95.1)	10 (4.9)	-	-
	≥50	13 (39.4)	20 (60.6)	30.2 (12.1-80.9, p <0.001)	5.25 (1.28-21.8, p=0.02)
Defect Circumference (%)	<80	206 (98.1)	4 (1.9)	-	-
	≥80	10 (27.8)	26 (72.2)	133.9 (43.2-526.7, p <0.001)	62.9 (17.9-276.7, p<0.001)
Histology	LGD/other*	188 (90.8)	19 (9.2)	-	-
	HGD/SMIC	28 (71.8)	11 (28.2)	3.89 (1.64-8.95, p=0.002)	2.23 (0.42-11.6, p=0.33)

Table 1: Logistic regression to determine lesion characteristics, endoscopic resection (ER) technique and post-ER outcomes which are risk factors for stricture development

**Other histology refers to non-adenomatous histology including Brunner's hyperplasia (n=2), gastric heterotopia (n=2) and duodenal lymphoma (n=1)*

Conclusion: ER-defect circumference strongly predicts stricture formation following ER of D-LSL. These findings can be used to guide patient counselling and post-procedural care. Based on our findings, we recommend a management algorithm determined by the extent of the ER-defect. When the ER-defect extends ≥80% of the luminal circumference, endoscopic assessment for stricture formation should be scheduled routinely at 3 weeks. When the ER-defect involves 60-79%, clinical consultation at 2-3 weeks should be conducted to screen for obstructive symptoms, and gastroscopy organised if the patient is symptomatic. ER-defects <60% can safely return to routine surveillance.

Disclosure: Nothing to disclose.

PP0180

COMPARISON OF SEDATION STRATEGIES IN DIAGNOSTIC UPPER GASTROINTESTINAL ENDOSCOPIES: A PROSPECTIVE STUDY

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Introduction: Sedation during upper gastrointestinal endoscopies (UGIEs) is a common practice aimed at improving patient comfort and procedural tolerance. However, the necessity of sedation remains a subject of debate, particularly regarding its impact on patient safety and experience. This prospective study aimed to compare the outcomes of sedated and non-sedated UGIEs, focusing on patient safety, procedural experience, and post-procedural comfort.

Aims & Methods: From January 1st, 2023, to December 31st, 2023, 150 patients undergoing diagnostic UGIEs were prospectively enrolled. Indications for endoscopy included dyspepsia, dysphagia, unexplained iron deficiency anemia, weight loss, and diarrhea. Patients were randomly assigned to one of three groups: Group 1 received local pharyngeal spray (lidocaine), Group 2 received pharyngeal spray with midazolam, and Group

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3 received pharyngeal spray with nalbuphine. Parameters assessed included patient safety, procedural experience, technical safety, pharyngeal sensitivity, and post-procedural comfort.

Results: Out of 150 patients, 90 were male and 60 were female. In Group 1 (pharyngeal spray only), 12 out of 50 patients reported mild throat irritation. In Group 2 (pharyngeal spray with midazolam), 9 out of 50 patients experienced increased sleepiness during the procedure, with reduced discomfort. In Group 3 (pharyngeal spray with nalbuphine), 6 out of 50 patients reported post-procedural anxiety. The mean ages in Groups 1, 2, and 3 were 38.5 ± 2.25, 41.25 ± 3.45, and 45.5 ± 2.14 years, respectively.

Conclusion: This study sheds light on the feasibility and outcomes of non-sedated UGIEs using pharyngeal spray alone. While sedation can enhance procedural tolerance, it may also introduce risks and post-procedural effects. The findings suggest that UGIEs with pharyngeal spray alone may be well-tolerated, particularly for select patient populations. Further research on a larger scale is warranted to validate these findings and refine guidelines for sedation strategies in UGIEs.

Disclosure: Nothing to disclose.

PP0181

EFFICACY AND SAFETY OF SNARE TRACTION ASSISTED ENDOSCOPIC SUBMUCOSAL DISSECTION FOR CIRCUMFERENTIAL SUPERFICIAL ESOPHAGEAL CANCER

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Introduction: Endoscopic submucosal dissection (ESD) has become a common treatment method for superficial esophageal cancer, allowing complete resection and accurate postoperative pathological evaluation, especially for relatively larger lesions[1, 2]. However, resection of larger lesions, especially circumferential lesions, can be time-consuming and challenging. The resected tissue can obstruct the esophageal lumen, leading to limited working space and poor visibility, thereby increasing the risk of adverse events such as perforation and muscular injury[3]. In recent years, several auxiliary traction techniques have been emerged, including positional adjustment traction, hemostatic clip elastic band traction, snare traction, line traction, and magnet anchor traction, which significantly improve the visibility of the submucosal layer during dissection and enhance the efficiency of the procedure [4-5]. Among these techniques, snare traction allows bidirectional traction through "push and pull" maneuvers, offering flexibility and practicality in clinical practice. However, due to the difficulty and limited cases of circumferential esophageal ESD, there is currently no published literature demonstrating the advantages of snare traction-assisted ESD over traditional ESD techniques.

Aims & Methods: This study aims to investigate the efficacy and safety of snare traction assisted ESD for the management of circumferential superficial esophageal cancer. A total of 68 patients who underwent ESD for circumferential superficial esophageal cancer were included in this study. All the patients were divided into two groups based on whether the snare traction was used or not; the snare traction group (S-ESD, group n=35) and the control group (C-ESD, group n=33).

Results: There was no significant difference in the size of the resected area between the groups [21.98 (18.30, 27.00) cm^2 vs 24.00 (15.28, 30.72) cm^2, P=0.976]. The snare traction group had a shorter dissection time [92.00 (74.00, 121.00) min vs 110.00 (92.50, 137.00) min, P=0.017] and a faster resection speed [0.28±0.13 cm^2/min vs 0.22±0.11cm^2/min, P=0.040] compared to the control group. There were no statistically significant differences between the two groups in terms of hospital stay, cost, en bloc resection rate, R0 resection rate, curative resection rate, bleeding rate, perforation rate, stricture rate, and recurrence rate (P>0.05).

Conclusion: Snare traction-assisted ESD is a safe and efficient approach for the treatment of circumferential superficial esophageal cancer. Its advantages includes shorter procedure so the anesthesia requirement, clear operative filed view, improved mucosal dissection efficiency, simple, and easily accessible equipment.

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Disclosure: Nothing to disclose.

PP0182

IS THE GASTRIC EMPTYING RATE PREDICTIVE FOR INTRAGASTRIC BALLOON TOLERABILITY?

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Introduction: Obesity has reached pandemic proportions. According to the World Health Organization, 16% of the global adult population, or about 890 million people, have obesity(1). Intragastric balloon (IGB) is an alternative method to induce weight loss in subjects with obesity. However, not all subjects tolerate an IGB, and up to 15% of the cases, the IGB may be removed early due to intolerability(2). The factors that predict IGB tolerability remain poorly studied. In this study, we investigated whether there is an association between the baseline gastric emptying rate and IGB tolerability.

Aims & Methods: Twenty-eight subjects with a Body Mass Index (BMI) over 27.5 kg/m^2 were included. Orbera 365® IGBs were endoscopically inserted and filled with 550 ml saline solution and scheduled for removal after 12-months. Before insertion, the subjects underwent a standardized, FDA-approved 13C-Spirulina breath test to measure their gastric emptying half-time (T1/2). Gastrointestinal symptoms were assessed using the Gastrointestinal Symptom Rating Scale (GSRS) at baseline and at 4, 8, 16, and 24 weeks post-insertion, even after eventual early IGB removal.

Results: Six subjects had their IGBs removed earlier than planned due to tolerance issues, which we define as early removal. Early removal in our study refers to any removal that occurred before the planned 12-months. The specific timepoints of removal varied: one at four weeks, two at six months, one at eight months, and two at eleven months.

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In the group that had their IGB removed earlier, the median baseline GSRS score was 28 (range 23-41), with a median T1/2 of 81.4 min (range 38.4-140.1). The median GSRS score at 4 weeks was 55.5 (range: 42-72), at 8 weeks 45 (range: 19-82), at 16 weeks 45.5 (range: 19-31), and at 24 weeks 37 (range 22-61).

In the group that tolerated the IGB, the median baseline GSRS score was 28 (range 19-77), with a median T1/2 of 61 min (range 34.5-108.4). The median GSRS score at 4 weeks was 53 (range: 24-133); at 8 weeks 37.5 (range: 19-79); at 16 weeks 37 (range: 21-85), and at 24 weeks 37 (range 19-104), respectively.

No statistically significant differences were observed in half-times (T1/2) and GSRS scores between the two groups at the examined time points. The p-value for T1/2 was 0.09. For GSRS, p-values were as follows: 0.89 at baseline, 0.63 at 4 weeks, 0.87 at 8 weeks, 0.45 at 16 weeks, and 0.73 at 24 weeks. Additionally, correlation analysis between GSRS scores and T1/2 values showed no significant relationships at baseline (p = 0.50), 4 weeks (p = 0.89), 8 weeks (p = 0.93), 16 weeks (p = 0.983), and 24 weeks (p = 0.86). These results suggest a consistent lack of significant association across all time points.

Conclusion: Early IGB removal cannot be predicted by the baseline gastric emptying rate or baseline GSRS in our study. The GSRS scores for subjects that tolerated the IGB, regardless of the time point, do not correlate with their baseline gastric emptying rate.

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Disclosure: Nothing to disclose.

PP0183

LONG-TERM (MINIMUM 1 YEAR) OUTCOMES AFTER GASTRIC PERORAL ENDOSCOPIC MYOTOMY FOR REFRACTORY GASTROPARESIS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Although Gastric Peroral Endoscopic Myotomy (G-POEM) has shown substantial efficacy in patients with medically refractory gastroparesis (GP), comprehensive long-term data on its effectiveness are lacking.

Aims & Methods: We conducted a systematic review and meta-analysis including observational studies assessing long-term outcomes after G-POEM in patients with refractory GP. Our primary outcome was the pooled rate of clinical success1-year after G-POEM. Secondary outcomes included clinical success rates beyond 1 year, and the rate of adverse events (AEs).

Results: Fifteen studies, comprising 986 patients with refractory GP undergoing G-POEM, were eligible. The pooled 1 year-success rate was 0.722 (95% CI: 0.56, 0.86, I^2=94.92%). In the subgroup analysis, which included only studies enrolling patients with 4-hour Postprandial Gastric Residual (PGR) ≥ 10%, the clinical success rate was 0.610 (95% CI: 0.526, 0.709, I^2=46.66%). The pooled success rates at 2 and 3 years calculated on followed-up patients were 0.861 (95% CI: 0.72, 0.96, I^2=85.11%) and 0.838 (95% CI: 0.56, 0.99, I^2=91.90%), respectively. In the subgroup analysis including studies that did not exclude non-responding patients from follow-up, the 2-year pooled success rate was 0.838 (95% CI: 0.60, 0.98, I^2=89.49%). The pooled rate of AEs was 0.092 (95% CI: 0.06, 0.13, I^2=81.35%), specifically 0.078 mild and 0.014 moderate AEs, according to the American Society for Gastrointestinal Endoscopy classification.

Conclusion: G-POEM is a safe procedure with 1-year success in more than 2/3 of treated cases and substantial benefit also at 2 and 3-years follow-ups. Future studies with strict patient selection and uniform outcome definitions are warranted.

Disclosure: Nothing to disclose.

PP0184

A SUB-EPITHELIAL LESION ALGORITHM FOR ENDOSCOPIC (SAFE) RESECTION IN THE UPPER GASTROINTESTINAL TRACT

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Introduction: Complete excision of upper gastrointestinal sub-epithelial lesions (SELs) eliminates diagnostic uncertainty, obviates the need for surveillance and may be necessary for definitive diagnosis and management. Current guidelines lack precision and cohesion, and surgery is associated with significant morbidity. Herein, we describe and report on the outcomes of our SEL algorithm for endoscopic (SAFE) resection.

Aims & Methods: SELs were enrolled prospectively over 115 months until March 2023. All underwent axial (CT) imaging and endoscopic-ultrasonography (EUS) to exclude a large exophytic component or invasion into local structures, and assess for muscularis propria involvement.

Results: 106 SELs were resected (mean age 60.6±13.4, 51.9% male; 41 oesophageal, 65 gastric). Oesophageal SELs underwent ESD (n=22) or STER if MP involvement was suspected (n=19). Gastric SELs underwent STER (n=6 at cardia), ESD (n=47) or exposing-EFTR (n=12). Technical success was 97.6% and 92.3% respectively. Five (7.7%) were completed laparoscopically due to deep and broad full-thickness involvement. Five (7.7%) required laparoscopic gastrotomy and surgical retrieval after successful resection and closure due to a large lesion size (mean 47 mm). There was no delayed bleeding, perforation or recurrence at a median of 13 months.

Conclusion: Upper GI SELs are effectively and safely treated by endoscopic resection. The SAFE approach provides a framework that facilitates structured decision-making. Oesophageal SELs suspected to involve the MP should undergo STER. Gastric SELs can be best managed by ESD with a view to proceed to exposing-EFTR. A laparoscopic UGI surgeon should be available in case surgical retrieval of the specimen or laparoscopic completion is required.

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PP0185

SUBTYPES OF GASTRIC ADENOMAS: PREVALENCE, RISK FACTORS AND OUTCOMES OF ENDOSCOPIC MANAGEMENT

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Introduction: Gastric adenoma is recognized as a premalignant lesion.[1]Much of the literature lumps all adenomas together with different classifications. These are few western reports regarding the prevalence, associated risk factors and outcomes of endoscopic management of different subtypes of gastric adenoma.

Aims & Methods: Aims: The aim of this study is to determine the prevalence, potential predictors, and outcomes of endoscopic management of the different subtypes of gastric adenoma.

Methods: Of 162 consecutive patients who underwent endoscopic resection for gastric lesions between 2017 and 2022 at our center, 79 patients were enrolled. All the hyperplastic polyps (n=37) and subepithelial lesions (n=46) were excluded. The patient clinical characteristics, endoscopic and histopathological features of gastric lesions were retrospectively collected. Patients were divided according to the histological types of gastric adenoma into 4 groups: intestinal group (n= 49), foveolar group (n= 23),pyloric group (n= 6) and oxyntic group (n=1). We statistically compared the clinical and endoscopic features of the three groups. In addition to the analysis of the outcomes of endoscopic resection of gastric adenomas.

Results: Of the 79 patients with a mean follow-up of 26 22 months, 51 patients were males (65 %) and 28 patients were females (35%). Forty-nine patients (49/79,62%) had intestinal adenoma, 23 patients (29%) showed foveolar adenoma, 6 patients (8%) developed pyloric adenoma and only 1 patient (1%) had oxyntic adenoma. The mean age was significantly older in the intestinal group (71 ± 13 years vs. 64 ± 16 years vs. 58 ± 18 years, $p = 0.03$). Intestinal adenomas were significantly associated with family history of gastric adenocarcinoma (14% vs. 4% vs. 0%, p = 0.01) and history of statin intake (39% vs. 26% vs. 0%, p = 0.04).Foveolar adenomas were significantly associated with multiple lesions (≥3 lesions) (p < 0.001) and 0-IIa+IIc Paris classification (6/23, 26%). the size of largest polyp was significantly higher among pyloric group (28 ±15, p= 0.00). Among intestinal group, 33/49 patients (67%) had atrophic gastritis. 82% of the adenomas were resected by ESD with histologically complete resection (R0) in 72/79 patients (91%). Although the difference was not statistically significant (p = 0.4), adenocarcinoma was most frequently encountered among the intestinal adenoma group (17/49, 35%). The post endoscopic resection outcome was comparable among 3 groups with overall recurrence rate (group1:8% vs. group 2:17% vs group 3:17%, P = 0.07) with a comparable mean time to recurrence among three groups (21 ± 17 months vs. 20±17months vs. 14 months, p=*0.8*). Family history of gastric adenocarcinoma (HR=1.33, p< 0.001) and active helicobacter pylori infection (HR=1.14, p < 0.001) were identified as independent predictors of recurrence after endoscopic resection of different gastric adenoma. Among baseline endoscopic characteristics, polyp size ³ 30mm(HR= 2.04, p< 0.001), more than 3 polyps (HR=1.51, p<0.001), lesser curvature as lesion location (HR=1.72, p< 0.001) and morphology as Paris classification IIc are potential predictors of post resection recurrence. In addition to specimen size ³ 15mm (HR=2.19, p< 0.001) and T1b as lesion histology (HR=2.07, p< 0.001), in the histopathological examination.

Conclusion: The present study clarifies the endoscopic and histopathological characteristics of different gastric adenomas, in addition to the potential predictive factors for each subtype. Potential predictors of recurrence after endoscopic resection emphazie the importance of endoscopic follow up.

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Disclosure: Nothing to disclose.

PP0186

ENDOSCOPIC VACUUM THERAPY FOR THE MANAGEMENT OF GASTRO-ESOPHAGEAL PERFORATIONS

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Introduction: Esophageal or gastric perforation is associated with systemic inflammatory response syndrome (SIRS), which can lead to multi-organ dysfunction and mortality in up to 50% of cases. Endoscopic vacuum therapy involves the use of a catheter with a sponge attached to its distal end, connected to a machine that applies negative pressure, allowing for continuous drainage of the perforation site and potentially promoting healing of the transmural defect. Previous series have reported promising results, indicating the potential effectiveness of this therapy in managing esophageal perforations. To evaluate the efficacy of endoscopic vacuum treatment in patients with primary or secondary gastroesophageal perforations.

Aims & Methods: Single-center retrospective cohort study on patients treated for gastroesophageal perforation between March 2019 and September 2022. The inclusion criteria for the study were patients with gastroesophageal perforation resulting from spontaneous rupture or iatrogenic damage (post-surgical or post-endoscopic treatment). The primary outcomes of the study were the success rate of endoscopic therapy, the length of hospital stay, and the specific survival at 30 and 90 days. The success rate of endoscopic therapy was assessed by considering the treatment successful if it led to the resolution of the fistula following vacuum endoscopic therapy. Secondary endpoints included the need for intensive care unit (ICU) stay between the start of vacuum endoscopic therapy and the patient's discharge.

Results: 27 patients, with a mean age of 69.8 years (95% CI 65.4-74.3), including 9 (33.3%) women, underwent endoscopic treatment with vacuum therapy for esophageal perforation. The causes of perforation were as follows: 6 (22.2%) cases were primary (Boerhaave syndrome), 15 (55.5%) cases were post-surgical (post-surgical anastomosis), and 6 (22.2%) cases were post-endoscopic treatment (endoscopic dilatation). Among the patients, 13 (48.1%) had previously undergone surgery for cancer, with 69.2% of them receiving intrathoracic stomach interposition reconstruction and the remaining patients undergoing reconstruction according to Roux-en-Y procedure. The treatment was successful in 21 (77.7%) patients, with a 14 days (9.0-24.4, 95% CI) median duration of treatment, and 4 (3-7, 95% CI) median number of endoscopic devices used per patient. One patient (3.7%) died within 30 days, and another one (3.7%) within 90 days. A total of 10 (38.4%) patients required an ICU stay, and there were no readmissions within 30 days. No procedure-related complication was reported.

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Conclusion: In the current pilot study, the findings support the feasibility and safety of endoscopic vacuum therapy as a treatment modality for both primary and secondary gastro-esophageal perforations.

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Disclosure: Dr Valente and Dr Arnelo are consultant to Boston Scientific. Dr Arnelo is Consultant to Ambu.

PP0187

ENDOSCOPIC ARGON PLASMA COAGULATION ABLATION SIGNIFICANTLY IMPROVES LARYNGOPHARYNGEAL SYMPTOMS IN SYMPTOMATIC CERVICAL INLET PATCHES PATIENTS: A PILOT STUDY IN THE ASIA-PACIFIC REGION

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Introduction: The cervical inlet patch (CIP) is an islet of heterotopic gastric mucosa that is observed in the upper esophagus during conventional diagnostic esophagogastroduodenoscopy. Commonly encountered laryngopharyngeal symptoms, including globus sensation, hoarseness, odynophagia, and dysphagia, have been linked to CIP. In most cases, medical intervention for symptomatic CIP commences with the administration of potent acid suppressive agents, such as proton-pump inhibitors (PPI); however, the efficacy of these agents is limited. A few Western studies have demonstrated that endoscopic therapy with argon plasma coagulation (APC) is increasingly effective in relieving laryngopharyngeal symptoms in patients with CIP.[1,2] Additionally, long-term effects lasting up to 27 months of follow-up have been documented.[3] Regarding the use of APC in patients with CIP, however, no pertinent research has been conducted in Asia-Pacific countries. Therefore, we retrospectively reviewed a cohort with six symptomatic CIP patients recruited for the APC ablation therapy. This case shed light on the possible use of APC for symptomatic CIP patients in Asia-Pacific countries.

Aims & Methods: We recruited a total of six symptomatic patients, proven to be CIP endoscopically and pathologically, from May 22, 2022, to April 20, 2024. We excluded patients who were less than 20 years old and who refused the option of argon plasma coagulation ablation. To ablate the CIP, all recruited patients underwent endoscopic ablation at 60 W and 2 L/min in one session. Questionnaires consisting of visual analog scores (from 0 to 10. The higher the score, the more severe the symptoms) for symptoms of dry throat, burning throat, globus sensation, hoarseness and clearing throat were distributed to patients at the time points of before the use of proton pump inhibitors, after the use of proton pump inhibitors and one month after the endoscopic ablation with APC to assess the efficacy endoscopic procedure. We defined symptomatic CIP with at least one any of the aforementioned symptoms with more than 5 out of 10 of pain score.

Results: The current study recruited a total of six patients. Two-thirds of patients were female (4/6) in gender and had an average age of 41 years. The CIP was located in the upper esophagus at 16.7cm from the incisors (15–18 cm), and two-thirds of the patients had more than one CIP. The average size of CIP was about 0.6 cm (ranging from 0.3cm to 0.8cm). The APC ablation significantly improved the five parameters (dry throat, burning sensation, foreign body sensation, and hoarseness) from more than 4/10 to less than 2/10. Foreign body sensation showed the most prominent improvement, decreasing from 7/10 to 1/10. (Figure1) There were no complications, such as perforation, infection, ulcers, or stricture, after the procedure. One month after the APC procedure, the follow-up EGD of all patients showed no residual CIP.

Conclusion: APC ablation shows a promising results for symptomatic CIP patients with laryngopharyngeal symptoms in Asia-Pacific population. However, further prospective randomized studies with a larger sample size are needed to validate the effectiveness and safety of APC ablation in symptomatic CIP patients in the region.

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Disclosure: There is no conflict of interest among authors.

PP0188

WITHDRAWN

PP0189

DEPTH ESTIMATION ALGORITHM-BASED LESION SIZE MEASUREMENT IN GASTROSCOPY: A COMPARATIVE STUDY WITH ENDOSCOPISTS

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Introduction: Precise measurement of lesion sizes during endoscopic procedures is crucial but often compromised by the reliance on visual estimations.

Aims & Methods: This study evaluates the precision of a novel depth estimation algorithm-based lesion size measurement function embedded in an electrically controlled endoscopy system, comparing its accuracy to visual and forceps-assisted estimations. A synthetic 3D-printed stomach model featuring 15 artificial lesions (6 ulcers and 9 polyps, size range: 6-39 mm) was used. Nineteen endoscopists (6 experts, 13 novices) measured the lesions using three methods: 1) visual estimation (VE) with a conventional endoscope, 2) forceps-assisted estimation (FE) with a conventional endoscope, and 3) depth estimation algorithm (DEA) with an electrically

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controlled endoscope. Measurements were then compared to the actual lesion sizes according to endoscopic proficiency and lesion size (<1 cm, 1-2 cm, >2 cm).

Results: VE had the highest mean percentage error (MPE) at 39.97% (95% CI 36.66%-43.28%), followed by FE (MPE 31.39%, 95% CI 28.81%-33.98%). DEA was most accurate with an MPE of 13.59% (95% CI 11.46%-15.73%), significantly outperforming both VE and FE (all $P < 0.01$). For VE and FE, experts showed greater precision over novices (all $P < 0.01$). With DAE, the accuracy was consistent across experience levels ($P = 0.60$). Novices using DEA showed greater precision compared to both VE and FE (Table 1). Novices using VE had the least accuracy with an MPE of 43.93% (95% CI 39.67% - 48.19%). For experts, DEA was accurate than VE but was comparable to FE except in lesions over 2cm.

		VE (MPE ± 95% CI)	FE (MPE ± 95% CI)	DEA (MPE ± 95% CI)	VE vs FE (P value)	VE vs DEA (P value)	FE vs DEA (P value)
Total		39.97% (36.66% - 43.28%)	31.39% (28.81% - 33.98%)	13.59% (11.46% - 15.73%)	<0.01	<0.01	<0.01
Novice (N=13)	All size	43.93% (39.67% - 48.19%)	37.67% (34.51% - 40.82%)	13.56% (10.77% - 16.35%)	0.01	<0.01	<0.01
	<1 cm	43.49% (34.90% - 52.09%)	35.09% (28.96% - 41.22%)	19.58% (14.27% - 24.90%)	0.08	<0.01	<0.01
	1-2 cm	44.24% (38.50% - 49.99%)	37.82% (33.77% - 41.86%)	12.79% (8.88% - 16.70%)	0.04	<0.01	<0.01
	>2 cm	43.37% (33.94% - 52.79%)	42.16% (32.42% - 51.89%)	4.95% (3.49% - 6.42%)	0.7	<0.01	<0.01
Expert (N=6)	All size	31.40% (26.77% - 36.02%)	17.80% (14.80% - 20.79%)	13.67% (10.57% - 16.78%)	<0.01	<0.01	0.09
	<1 cm	33.15% (23.59% - 42.71%)	18.66% (13.89% - 23.43%)	20.80% (14.75% - 26.84%)	0.01	0.04	0.61
	1-2 cm	31.21% (25.05% - 37.37%)	17.13% (12.77% - 21.49%)	11.64% (7.41% - 15.87%)	<0.01	<0.01	0.11
	>2 cm	28.74% (15.89% - 41.60%)	19.07% (11.13% - 27.01%)	8.58% (5.22% - 11.95%)	0.02	0.01	0.03

Conclusion: The novel lesion size measurement function using a depth estimation algorithm outperforms traditional visual and forceps-assisted method, especially in novice. Its adoption in endoscopic practice and training could improve lesion sizing accuracy, enhancing clinical outcomes and decision-making.

Disclosure: Nothing to disclose.

PP0190

ENDOSCOPY IN THE SETTING OF INCIDENTAL HYPERMETABOLIC LESION ON PET SCAN

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Introduction: Positron emission tomography (PET) scan involves injecting a radioactive isotope (18-fluoro-2-deoxyglucose) to locate areas where the radiotracer accumulates. The established indications of PET scanning mainly concern oncology, where it is used for staging, treatment monitoring, and long-term surveillance. It is also useful in cases of secondary lesions for identifying the primary lesion. Outside of oncology, PET scan has other clinical applications such as diagnosing and monitoring inflammatory diseases, and searching for infectious foci. The incidental discovery of metabolic hyperfixation in the digestive system on PET scan is becoming increasingly common, raising questions about the clinical significance of this hypermetabolism and the need for endoscopic exploration. The aim of our study is to report the endoscopic findings from examinations performed in our department where the indication was the incidental discovery of digestive hypermetabolism on PET scan.

Aims & Methods: This is a descriptive and analytical retrospective monocentric study including all patients referred for endoscopy due to the incidental discovery of digestive metabolic hyperfixation on PET scan, not corresponding to a previously known lesion, between January 2017 and September 2023. Only hypermetabolic fixations located in the stomach, colon, and rectum were included.

Results: 28 hypermetabolic fixations were included in the study, of which 46.4% were located in the colon, 42.9% in the rectum, and 10.7% in the stomach. The mean age was 54.9 years with a standard deviation [32,78]. The male-to-female sex ratio was 0.6. PET scan indications were mainly dominated by surveillance of treated cancers or during treatment (57.1%). In 21.4% of cases, PET scan was requested for staging a newly diagnosed tumor, and in 17.9% of patients, it was used to search for a primary tumor with hepatic, pulmonary, or bone metastases. Other indications were rarer, representing 3.6% of cases. Hypermetabolism on PET scan corresponded to a neoplastic lesion in 17.9% of cases, an adenoma in 14.3% of cases, and a benign lesion (non-specific ulceration, inflammatory lesion, or stenosis) in 32.2% of cases. In 35.7% of patients, endoscopic examination did not reveal detectable abnormalities. Hypermetabolism was most often focal (67.9%). According to this study, the presence of focal metabolic hyperfixation was significantly associated with the presence of a neoplastic lesion on digestive endoscopy (P <0.01).

Conclusion: All metabolic hyperfixations on PET scan referred to our endoscopy center underwent endoscopic exploration to rule out a neoplastic lesion. However, additional studies, ideally prospective and multicentric with a larger sample size, are essential to establish the criteria for endoscopic exploration in the case of incidental discovery of digestive hypermetabolism on PET Scan

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Disclosure: Nothing to disclose.

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PP0191

ENDOSCOPIC MANAGEMENT OF UPPER DYSPHAGIA OF UNKNOWN ETIOLOGY – A SINGLE CENTER PILOT EXPERIENCE

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Introduction: Upper dysphagia is a swallowing disorder associated with underlying pathology in the mouth, throat, or esophagus. Cricopharyngeal achalasia (CA) may cause upper dysphagia and its diagnosis is not well established. In case of CA, endoscopic myotomy of upper esophageal sphincter (C-POEM) may be offered to relieve symptoms.

Aims & Methods: We present our experience in diagnosing and managing upper dysphagia with suspicion on CA in patients referred to our center between May 2022 and April 2024. The patients with Zenker diverticulum or organic stricture were not enrolled. All patients underwent X-ray examination, high-resolution manometry and endoscopy. In patients with suspected CA, botulinum toxin injection (100 IU) was administered into the upper esophageal sphincter.

After botulinum toxin injection, the follow up visits were scheduled 1, 5 and 12 months after the procedure. In case of a positive effect of the injection, C-POEM (endoscopic myotomy of cricopharyngeal muscle) was considered after the effect of botulinum toxin had vaned. FOSS Function Outcome Swallowing Score and Dakkak and Bennett score of dysphagia were used to assess the severity of the symptoms. The score systems are listed in the table.

Dakkak and Bennett score of dysphagia		FOSS Function Outcome Swallowing Score	
Grade	Characteristic	Grade	Characteristic
Grade 0	No dysphagia	Grade 0	Asymptomatic
Grade 1	Solids	Grade 1	Episodic or daily symptoms of dysphagia with normal meal time
Grade 2	Semi-solids	Grade 2	Dysphagia requiring significant dietary modifications or prolonged meal time
Grade 3	Liquids	Grade 3	Dysphagia with weight loss ≤10% of body weight with daily symptoms
Grade 4	Aphagia	Grade 4	Dysphagia with weight loss >10% of body weight or severe aspiration or requiring intermittent non-oral feeding for nutrition
		Grade 5	Non-oral feeding for all nutrition

Results: Ten patients (4 males, 6 female) were referred to our center with long term upper esophageal dysphagia and considered eligible for botulinum toxin injection. The mean age was 69.7 years (ranges 30, 85), mean duration of symptoms was 8.6 years (SD ± 11) Three patients were unable to swallow safely and PEG tube had to be placed for nutritional support. Aetiologies of dysphagia were cervical injury (n=2), radiotherapy (n=2) and previous neurosurgery (n=1) or idiopathic dysphagia.

All patients underwent botulinum toxin injection. One patient developed vocal cord paresis one week after the injection and her condition improved after corticoid administration. No other adverse events of botulinum toxin were observed.

Improvement of dysphagia occurred in 7 patients after botulinum toxin injection while no change in symptoms was present in 3 patients. One month after the procedure mean

FOSS Function Outcome Swallowing Score improved from 3.3 to 1.56 (p = 0.015) and Dakkak and Bennett score of dysphagia from 2.7 to 1.33 (p=0.012). Two patients received successful C-POEM, the remaining patients will be offered C-POEM once a recurrence of symptoms will occur.

Three patients without effect of botulinum toxin injection were considered endoscopically incurable and referred back to neurology or otorhinolaryngology departments.

Conclusion: Botulinum toxin injection may be effective in treating upper dysphagia caused by a spasm of cricopharyngeal muscle and may serve as a useful selection tool for endoscopic myotomy.

Disclosure: Nothing to disclose.

PP0192

IDENTIFICATION OF LIPIDOMICS AND PROTEOMIC SIGNATURES WHICH CORRELATE WITH RAMAN SPECTRA TO DIFFERENTIATE DYSPLASIA FROM NORMAL GASTRIC TISSUE

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Introduction: Early gastric cancer (GC) detection allows for early intervention and better patient outcomes. Yet, current diagnostic methods for early GC still need to be improved. Raman microscopy offers a noninvasive, label-free approach for a wide range of ex vivo (e.g. intraoperative tumour margin detection) and in vivo (e.g. early GC diagnosis) clinical applications. However, the specificity of Raman microscopy is limited by the high signal-to-noise ratio due to light scattering and the inherent molecular complexity of human tissues.

Aims & Methods: Aim: This study aims to identify lipidomic and proteomic tissue signatures that correlate with Raman microscopy, thereby shortlisting a more specific Raman spectrum to better differentiate GC from normal gastric tissue.

Methods: Real-time Raman spectrometry-based AI was applied to 19 consecutive patients (aged 51 - 85 years old, 14 male) with high risk for gastric dysplasia or cancer. Patients underwent high-definition white light upper endoscopy. Upon localisation of the lesion/s, the endoscopist used the SPECTRA IMDx™ probe to capture the Raman spectra (1251 features) from the dysplastic lesion and its adjacent normal mucosa. Biopsies paired to the sites of Raman readings were obtained for histopathologic examination, and lipidomic and proteomic profiling. 254 lipids (from 15 subclasses) and 4168 proteins were obtained through mass-spectrometry-based techniques. Differential abundance analysis of the lipid, protein and Raman spectra was undertaken using DESeq2 [1], which uses shrinkage estimation for dispersions and fold change to improve the stability and interpretability of estimates.

Results: Using 25 tissue samples (12 control, 13 with gastric dysplasia or cancer), we identified specific lipid species and protein markers as potential biomarkers (FDR <0.05) for GC; 45 lipid species (13 cholesterol esters, nine phosphatidylethanolamines, five diacylglycerols, five lysophosphatidylethanolamines, four hex ceramides, four phosphatidylcholines, two lysophosphatidylcholines and one sphingomyelin, one ganglioside) as well as ten proteins (PLOD2, GGPS1, C8G, COX7C, DCK, ACTBL2, EMC10, H2AC21, HDHD2). Integrating the putative lipid and protein features and the Raman spectra, only a shortlist of 56 Raman features were significantly correlated with only ten lipids (3 HexCeramides, three cholesterol esters, two diacylglycerols, one phosphatidylethanolamine and one lysophosphatidylethanolamine) and two protein features (C8G, ACTBL2).

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Conclusion: Our findings demonstrate the potential of integrating Raman microscopy with lipidomics and proteomics. The potential lipid and protein signatures provide insight into the biochemical alterations associated with dysplastic transformation identified by Raman microscopy. This approach brings promise for improving the accuracy of Raman-based non-invasive GC early detection strategies.

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Disclosure: Prof Lawrence Ho Khek Yu is the Co-founder and Chief Medical Officer of Endofotonics (SPECTRA IMDx™).

PP0193

BLUE LIGHT IMAGING IS BETTER THAN LINKED COLOR IMAGING IN THE DETECTION OF GASTRIC INTESTINAL METAPLASIA

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Introduction: Data evaluating the utility of Blue light Imaging (BLI) and Linked Color Imaging (LCI) in the diagnosis of gastric intestinal metaplasia (IM) and atrophic gastritis (AG) is limited.

Aims & Methods: We aimed to evaluate and compare the diagnostic accuracy of BLI and LCI without magnification for the detection of IM and AG. Single centre prospective study involving consecutive patients aged ≥18 years undergoing diagnostic upper gastrointestinal endoscopy (UGIE), endoscopic follow-up of intestinal metaplasia or gastric cancer screening. Exclusion criteria: previous gastrectomy, contraindication to UGIE due to comorbidities, no gastric mucosal biopsies, refusal of consent. Endoscopy was initially performed with white light image (WLI), followed by BLI and LCI. Standardized photodocumentation of stomach was performed with WLI, BLI and LCI. Imaging features suggestive of IM: lavender color sign and whitish flat elevation with LCI, and light blue crest and white opaque substance with BLI. EGGIM and Kimura-Takamoto Classification (KTC) were applied when IM and AG were suspected. Systematic random biopsies of the antrum, incisura angularis and corpus along with targeted biopsies of suspicious gastric mucosal lesions were obtained. Histology was considered gold standard. Performance of endoscopic imaging in the detection of IM, AG and EGC was evaluated.

Initially, 134 patients were evaluated and 64 were excluded: previous gastrectomy-22, pending histology-20, non-BLI imaging-15, intolerance to UGIE-3, refusal of consent-1, esophageal stenosis-1 and contra-indication to biopsies-2. Final sample included 70 patients.

Qualitative data expressed as percentage were compared with Chi-square test. A p value of <0.05 was considered significant.

Results: Median age was 56 (19-86), with 49 (63%) female. There was family history of gastric cancer in 12(15%) and history of smoking in 35 (44%). Histological AG was found in 15 (19%) and IM in 20 (26%) patients. WLI was suggestive of IM in 14(20%), confirmed histologically in 9 (64%). IM was suspected by LCI and BLI in 24 (34%) and 23 (32%) cases, confirmed by histology in 16 (67%) and 17 (74%), respectively.

WLI classified 56 (71%) as EGGIM 0 and after LCI and BLI, 10 (18%) and 8 (15%) of these cases had EGGIM 1 or 2. In 10 patients, EGGIM 1-2 with WLI, only 1 was upgraded to EGGIM 3-4 with LCI and BLI. In 3 (4%) EGGIM 3-4 with WLI, after LCI and BLI,1 had extensive IM (EGGIM 6). Among 19(24%) patients EGGIM 1-2 with LCI, 3(16%) had EGGIM 0 with BLI.

Atrophy was suspected by WLI and LCI in 9(12%) and 15(21%) patients, confirmed by histology in 6(67%) and 8(53%) cases, respectively. According to KTC, 5 (6%) were classified as C1, 2 (3%) as C2 and 4 (5%) as O3 with WLI. With LCI images, 8 (10%) were classified as C1, 3(4%) as C2 and 4(5%) as O3. The sensitivity (Sens), specificity (Spec), accuracy, positive predictive value (PPV), negative predictive value (NPV) of WLI, BLI and LCI in detecting AG and IM are shown in table 1.

	Sens (%)	Spec (%)	Accuracy (%)	PPV (%)	NPV (%)
Intestinal metaplasia					
WLI	47	90	79	61	83
LCI	80	84	83	62	92
BLI	85	88	97	70	94
Atrophic gastritis					
WLI	40	94	80	71	83
LCI	53	87	78	58	85

Table 1. Sens, Spec, accuracy, PPV and NPV in detecting AG and IM with WLI, LCI and BLI.

Conclusion: Both BLI and LCI are promising advanced endoscopic imaging techniques with BLI having higher accuracy than LCI in detection of IM.

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PP0194

SAFETY AND FEASIBILITY OF GASTROINTESTINAL ENDOSCOPY IN NEONATES AND INFANTS UNDER 3 MONTH OF AGE

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Introduction: With the development of minimally invasive endoscopic diagnosis and treatment technology in children, the application age of pediatric digestive endoscopy is gradually expanded to infants and even neonates. Compared with older children, the gastrointestinal lumen of this population is more distorted and narrow, the wall is thinner, and the operation is more difficult and risky. This study aims to investigate the safety and feasibility of gastrointestinal endoscopy in neonates and infants under 3 months of age.

Aims & Methods: A retrospective analysis was performed on the clinical data of 145 neonates and infants under 3 months of age who underwent gastrointestinal endoscopy in Xi 'an Children's Hospital from June 2018 to May 2023. The general condition, clinical manifestations, main diagnosis, endoscopic treatment and adverse events of all children were statistically analyzed.

Results: A total of 145 patients were enrolled in this study, including 30 neonates (20.7%). There were 96 males (66.2%) and 49 females (33.8%). The age ranged from 1 day (9 hours after birth) to February 29 days, with a mean age of January 17 days. The main symptoms of children undergoing gastrointestinal endoscopy were vomiting (79 cases, 54.5%), followed by hematochezia (25 cases), diarrhea (19 cases), bucking (10 cases), abdominal distension (6 cases), hematemesis (5 cases) and foreign body in digestive tract (1 case). All children underwent general anesthesia under tracheal intubation. Gastroscopy was performed with XP290 electronic nasal gastroscope (external diameter of 5.8mm), gastroscopy was performed with H290 electronic gastroscope (external diameter of 8.9mm), and colonoscopy was performed with Q260 electronic colonoscopy (external diameter of 9.2mm). 114 cases (78.6%) underwent gastroscopy and 31 cases (21.4%) underwent colonoscopy. The most common diagnosis was congenital hypertrophic pyloric stenosis (55 cases, 37.9%), followed by allergic colitis (15 cases), esophageal stenosis (13 cases), erosive gastritis (11 cases), multiple colonic ulcers (7 cases), duodenal stenosis (5 cases), colonic stenosis (4 cases) and reflux esophagitis (3 cases). In addition, 25 cases (17.2%) showed no abnormalities after the examination, and gastrointestinal diseases were excluded. In addition to routine examination, a total of 63 children received endoscopic treatment. The most common endoscopic treatment was peroral endoscopic pyloromyotomy (G-POEM), 37 cases, accounting for 58.7% of all children who received endoscopic treatment. The others included 15 cases of endoscopic nutrition tube placement, 8 cases of endoscopic balloon dilatation, 1 case of endoscopic foreign body removal, 1 case of endoscopic magnetic ring placement and 1 case of endoscopic hemostasis. All children underwent gastrointestinal endoscopy successfully, with a success rate of 100%. No adverse events related to anesthesia and surgery occurred in all children.

Conclusion: The application of gastrointestinal endoscopy in neonates and infants under 3 months of age may be feasible and safe. Endoscopic diagnosis and treatment is minimally invasive, safe and fast recovery, which has a wide application prospect in young children.

Disclosure: Nothing to disclose.

PP0195

GASTRIC ENDOSCOPICAL SUBMUCOSAL DISSECTION: ARE WE EXCEEDING EXPECTED PROCEDURE TIMES?

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Introduction: Endoscopic submucosal dissection (ESD) stands as the gold standard treatment of gastric superficial lesions. Due to its technical complexity, ESD procedures often require extended durations, and longer procedural times are related to a higher level of adverse events. We intend to analyze our procedural time according to the predictive formula developed by Goto et al.

Aims & Methods: Prospective cohort study, single center, of patients who underwent gastric ESD between 2017 and 2023. The characteristics of the lesion were evaluated, and logistic regression of risk factors associated with longer procedural time (≥ 60 minutes) was performed. The predictive formula of procedural time was calculated for each patient and comparison between predictive and actual procedural time was performed using Pearson correlation coefficient.

Results: Included 101 gastric ESDs, 63.4% female, with a mean age of 69.7 ± 8 years. Among the analyzed lesions: 43.6% measured between 0-20mm, 56.4% ≥ 20mm; 13.9% were sessile, 60.4% flat, 14.9% depressed and 10.9% subepithelial; 22.8% were located in the upper-third of the stomach, 17.8% in the middle-third and 59.4% in the lower-third. Ulcerative findings were identified in 10% of the lesions. The most frequent histological results were: 38.6% low grade dysplasia, 26.7% high grade dysplasia and 19.8% intramucosal adenocarcinoma. The mean procedure duration was 87±43 minutes. Intra-procedure adverse events registered in 2% (perforation).

Evaluating the risk factors for longer procedures (≥ 60 minutes), the following revealed statistical significance: lesion size ≥ 20mm (89.5% vs. 65.9%; p=0.004), histology indicative of intramucosal adenocarcinoma (95.7% vs. 74.4%; p=0.027) and location in the upper or middle-third of the stomach (95.1% vs. 68.3%; p=0.001), but not the Paris classification (78.6% vs. 75.4% vs. 93.3%; p=0.4), ulcerative findings (63.6% vs. 67.1%; p=0.8) and the use of anticoagulant therapy (64.3% vs. 81.6%; p=0.138). Logistic regression analysis revealed that lesion size ≥ 20mm (OR 5.7; IC95%: 1.7-18.7) and location in the upper or middle-third of the stomach (OR 13.6; IC95%: 2.3-80.7) were significantly associated with longer procedural time.

When comparing predicted and actual procedural time, the overall Pearson correlation coefficient (r) was 0.58 (p<0.01), with the following values per consecutive 25 cases: 0.70, 0.67, 0.54 and 0.46.

Conclusion: Previous identification of risk factors for prolonged ESD procedures, such as lesion size ≥ 20mm and proximal lesion location, can be useful in optimizing workflow scheduling. Our correlation between predicted and actual time demonstrated a decreasing value, probably due to a higher complexity of the lesions, giving room for improvement.

Disclosure: Nothing to disclose.

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PP0196

DELAYED BLEEDING AFTER GASTRIC ENDOSCOPIC SUBMUCOSAL DISSECTION (ESD): ANALYSIS OF RISK FACTORS AND VALIDATION OF THE BEST-J SCORE

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Introduction: Delayed bleeding after gastric endoscopic submucosal dissection (ESD) is one of the most frequent complications (up to 7%). The BEST-J score is a risk predictive model for delayed bleeding following gastric ESD.

We aim to assess potential risk factors associated with bleeding and validate the BEST-J score in a population of patients undergoing ESD.

Aims & Methods: Prospective cohort study, single centre, of patients who underwent gastric ESD between 2017 and 2023. The characteristics of the lesion were evaluated, and logistic regression of risk factors associated with bleeding after ESD was performed. The BEST-J score was calculated for each patient and the discriminative ability was evaluated through the area under the receiver operating characteristic curve (AUC-ROC).

Results: Included 101 gastric ESDs (63.4% female), with a mean age of 69.7 ± 8 years. 17.8% of patients were on antiplatelet therapy and 13.9% were on anticoagulant therapy. Among the analyzed lesions: 43.6% measured between 0-20mm, 56.4% were ≥ 20mm; 13.9% were sessile, 60.4% flat, 14.9% depressed and 10.9% subepithelial; 22.8% were located in the cardia, fundus, or body, 17.8% in the incisura *angularis* and 59.4% in the antrum. The most frequent histological results were: 38.6% low-grade dysplasia, 26.7% high-grade dysplasia and 19.8% intramucosal adenocarcinoma. The mean procedure duration was 87±43 minutes. Delayed bleeding after ESD occurred in 5.9% (n=6) of cases. In the univariate analysis of risk factors, the lesion size ≥ 20mm showed statistical significance (10.5% vs. 0%; p=0.034), but not the location in the antrum (6.7% vs. 4.9%; p=0.5), lesion nature (7.1% vs. 8.2% vs. 0% vs. 0%; p=0.5) or procedure duration ≥60 minutes (5% vs. 9.5%; p=0.3). The BEST-J score predicted the risk of delayed bleeding after gastric ESD with an AUC of 0.70 (95% CI: 0.53-0.87; p=0.1).

Conclusion: The use of bleeding predictive scores may be useful in identifying high-risk patients; in our series, the BEST-J score was not very effective, with only lesion size ≥ 20mm being a relevant factor for bleeding after ESD.

Disclosure: Nothing to disclose.

PP0197

RISK FACTORS ASSOCIATED WITH METACHRONOUS LESIONS FOLLOWING GASTRIC SUBMUCOSAL DISSECTION (ESD): A SINGLE CENTRE EXPERIENCE

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Introduction: Endoscopic submucosal dissection (ESD) is the first-line treatment for superficial gastric lesions with a negligible or very low risk of lymph node metastasis. However, a permanent risk of developing metachronous lesions persists. We intend to analyze the risk factors associated with the development of metachronous lesions in a population of patients undergoing ESD.

Aims & Methods: Prospective cohort study conducted at a single centre, involving patients who underwent gastric ESD for lesions between 2017 and 2023, with a follow-up period of ≥ 6 months. Lesion characteristics were analyzed, including the rate of metachronous lesions, and logistic regression was performed to identify risk factors associated with the development of metachronous lesions.

Results: Included 78 ESDs performed on gastric epithelial lesions (59% female), with a mean age of 70.3±7.9 years. Regarding family history of gastric adenocarcinoma, 17.9% of patients had a first-degree relative. Among the analyzed lesions: 42.3% measured between 0-20mm, 57.7% ≥ 20mm; 12.8% were sessile, 71.8% flat and 15.4% depressed; 19.2% were located in the cardia, fundus or body, 21.8% in the incisura *angularis* and 59% in the antrum.

Regarding histological findings: 42.3% low-grade dysplasia, 30.8% high-grade dysplasia, 23.1% intramucosal adenocarcinoma and 3.8% adenocarcinoma with submucosal invasion. Nine metachronous lesions (11.5%) were detected, with a median time of occurrence of 24 [13-32] months. In the analysis of risk factors, statistical significance was found for location outside the antrum (21.6% vs. 1.9%; p=0.002) and the presence of adenocarcinoma (23.8% vs. 7%; p=0.039), but not for size ≥20mm (13.7% vs. 5.1%; p=0.178) or ulceration of the lesion (9.1% vs. 10.1%; p=0.915). Logistic regression analysis revealed that only location outside the antrum had a significant risk for the development of metachronous lesions (OR 13.4; 95% CI: 1.4-127.1).

Conclusion: Given the permanent risk of developing metachronous lesions, the identification of associated risk factors is important as it may eventually allow for the adaptation of surveillance programs.

Disclosure: Nothing to disclose.

PP0198

WITHDRAWN

PP0199

RISK FACTOR ANALYSIS FOR BLEEDING AFTER GASTRIC ENDOSCOPIC SUBMUCOSAL DISSECTION: A NATIONWIDE POPULATION-BASED STUDY OVER 11 YEARS

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Introduction: Although an effective and relatively safe procedure, bleeding complications often occur after gastric endoscopic submucosal dissection (ESD). We aimed to comprehensively evaluate the risk factors of post-procedural bleeding following gastric ESD.

Aims & Methods: This study was designed as a nationwide, population-based, retrospective study utilizing the Korean National Health Insurance Service database, comprising patients who underwent gastric ESD for gastric cancer or adenoma from November 2011 to December 2022. Operational definitions were created to identify the target population and post-procedural bleeding using the diagnosis and procedure code, which underwent validation.

We established operational definitions to identify the target population and post-procedural bleeding by utilizing diagnosis and procedure codes, which underwent validation. The risk factors analyzed included age, sex, hospital volume, socioeconomic status (SES), histology, lesion numbers, comorbidities, and medication usage.

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Results: During the study period, there were 191,932 procedures conducted on 175,779 patients, resulting in 9921 cases of bleeding events. Several factors were associated with increased post-ESD bleeding risk. Following multiple logistic regression, relevant factors with adjusted odds ratios (aORs) were identified. These factors included young age (aOR: 0.990, 95% CI 0.987-0.992), male sex (aOR: 1.377, 95% CI 1.311-1.448), low-volume center compared to high-volume center (aOR: 1.289, 95% CI 1.022-1.621), medical aid compared to the top 25% SES (aOR: 1.307, 95% CI 1.188-1.440), higher Charlson comorbidity index (aOR: 1.040, 95% CI 1.026-1.054), gastric cancer (aOR: 1.369, 95% CI 1.307-1.434), and multiple lesions (aOR: 1.144, 95% CI 1.032-1.267). Furthermore, bleeding risk was significantly higher in groups with several comorbidities and antithrombotic agent users, particularly showing Notably high aORs in groups with renal failure (aOR: 2.086, 95% CI 1.909-2.278), warfarin (aOR: 2.641, 95% CI 2.219-3.143), direct oral anticoagulants (aOR: 2.519, 95% CI 2.220-2.859), and antiplatelet combinations (aOR: 2.073, 95% CI 1.783-2.410).
Conclusion: This study revealed various factors associated with increased bleeding risk after gastric ESD using a large cohort, which could be an important issue in terms of risk stratification and personalized treatment.
Disclosure: Nothing to disclose.

PP0200

WITHDRAWN

PP0201

COMPREHENSIVE ANALYSIS OF ENDOSCOPIC ULTRASOUND (EUS) IN MEDIASTINAL LESIONS: PATIENT CHARACTERISTICS, DIAGNOSTIC OUTCOMES, SAFETY, AND EFFICACY: A MULTICENTER STUDY

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Introduction: Mediastinal lesions develop from structures located in the mediastinal compartments and the paravertebral areas. The diagnosis of these lesions is primarily based on imaging studies such as computed tomography and magnetic resonance imaging, which confirm the presence of the lesions and suspected metastatic disease
Aims & Methods: This study aims to assess the factors that determine the diagnostic accuracy and yield of endoscopic ultrasound (EUS) in diagnosing mediastinal lesions. This multicenter study involved a list of centres providing EUS services across ten countries all over the globe. The appropriate information was collected regarding EUS-FNA and EUS-FNB

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procedures, including nature of the lesion, needle types, needle passes, and complications of the techniques. The frequency/percentage conveyed data on the qualitative variables. The p-values were obtained using the Chi-square two-tailed exact test. The significance of the analysis was defined as p<0.05. All the syntheses were performed using the Statistical Package for the Social Sciences (SPSS version 27; SPSS Inc., Chicago, IL).

Results: A total of 439 participants were involved in this study, 255 (58.1%) males and 184 (41.9%) females, with a mean age of 55.7314.93 years. The average accuracy from the valid matched cases between EUS, cytopathy, and final diagnosis (300 out of 331) was 90.63%. EUS-FNA and EUS-FNB diagnostic yield was influenced by needle types, needle size, number of needle passes, and lesion size. Needles with smaller diameters (22G) offered greater diagnostic yield than needles with larger diameters (19G). However, the diagnostic matching accuracy of EUS in relation to needle sizes was statistically insignificant (p= 0.774). A minimal number of needle passes (between 2 and 4) showed a higher diagnostic yield than a higher number of needle passes (>5) with more diagnostic matching accuracy (p = 0.009). EUS-FNB accounted for a 0.5% complication incidence rate each for fever, pain, thoracic pain, and hemorrhage. Patients undergoing EUS-FNA, EUS-FNB, or both procedural examinations had no significant complications (p = 0.085) Table 1.

EUS diagnosis			Complications					P value
			Fever and Pain	Hemorrhage	None	Pain	Thoracic pain	
EUS FNA or FNB	EUS-FNA	Count	0	0	75	0	0	0.085
		% within FNA	0.0%	0.0%	67.0%	0.0%	0.0%	
	EUS-FNB	Count	1	1	178	1	1	
		% within FNB	0.5%	0.5%	82.8%	0.5%	0.5%	
	FNA and FNB	Count	0	0	4	0	0	
		% FNB and FNA combined	0.0%	0.0%	1.6%	0.0%	0.0%	

Table 1: Incidence rate of complications associated with EUS examination techniques.

Conclusion: The diagnostic yield of EUS procedures is affected by needle types, number of needle passes, lesion size, and needle size. Besides, EUS-FNA and EUS-FNB are infrequently associated with serious complications, these techniques are safe and offer higher diagnostic yield when utilised responsibly by endosonographers.

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Disclosure: Nothing to disclose.

PP0202

USING NEEDLE BIOPSY UNDER ECHO-ENDOSCOPY (EEH-FNB) TO DIAGNOSE STOMACH LINITIS: A MOROCCAN UNIVERSITY CENTER'S EXPERIENCE

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Introduction: Gastric linitis is a kind of adenocarcinoma that invades the stomach and is composed of separate cells. The predicted survival rate for this condition over a five-year period ranges from 3 to 10%. The excessive cell growth that characterizes gastric linitis also results in the formation of fibrous tissue, which makes the stomach wall thicker and less flexible. Consequently, mucous biopsies frequently do not provide useful information.

Aims & Methods: The aim of this study is to evaluate the diagnostic performance of echo-endoscopic biopsy puncture (EUS-FNB) in the diagnosis of gastric linitis.This study is an observational retrospective monocentric analysis conducted over a period of 4 years from January 2018 to December 2022. It includes all patients who underwent a parietal gastric and/or ganglionary puncture under echo endoscopic guidance using a 20G or 22G Procore® or Acquire® type 22G FNB needle. The purpose of the procedure was to investigate suspected cases of gastric linitis. The study is conducted at the University Hospital Centre in Marrakech, which serves as a tertiary care department and is a point of reference for all regional and provincial hospitals. The appearance of parietal gastric thickness, abdominal DMT, an infiltrated aspect of the stomach, folds, or gastric stenosis in high digestive endoscopy led to the suspicion of gastric linitis. The data was obtained from various sources, including hospital registers, patient medical records, the computerized system "Hosix" for morphological and anatomopathological data, and the collaboration of the medical and paramedical staff from the departments of gastroenterology, visceral surgery, and anatomopathology. The statistical analysis was conducted using SPSS version 22.0. Socio-demographic, clinical and para-clinical data were collected taking into account the global ethical rules relating to confidentiality and protection of patient-specific data.

Results: Out of 42 patients who performed biopsy puncture under high echo-endoscopy on suspicion of diagnosis, ten patients had gastric linitis. The diagnosis of gastric linitis was obtained by puncture in 90% of cases (9/10). Eight patients (80%) had previously undergone at least one oeso-gastro-duodenal fibroscopy with returning negative mucous biopsies. Six patients were diagnosed as early as 1st EEH while 3 others required a second series of biopsies. While the mucous biopsies were also unsuccessful, echo-guided biopsy allowed the diagnosis of other more serious diseases in 43% of cases (n = 18/42) including gastric metastases of a breast cancer (n=2), other gastric adenocarcinomas of the intestinal type (N=8), and cases of T-type gastric lymphoma in one case and MALT type in 2 cases.

Conclusion: In case of suspicion of gastric linitis, guided biopsy puncture under echo-endoscopy is a cost-effective approach in 90% of cases, without noticeable side effects. It should be considered the procedure of choice for early diagnosis of gastric linitis when mucous biopsies return negative.

Disclosure: Nothing to disclose.

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PP0203

PLATINUM FIDUCIAL SEEDS MADE IN HOUSE FOR EUS GUIDED IMPLANTATION

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Introduction: Stereotactic Body Radiation Therapy (SBRT) has proven to be a safe and effective therapy for patients with deep abdominal malignancies who are ineligible/suboptimal candidates for surgery[1,2].

SBRT's effectiveness is dependent on precise targeting, achieved through both bone imaging and implantation of small markers (fiducials) that are visible during imaging which delineate margins of the target lesion[3].

Fusion of computed tomography (CT) and magnetic resonance imaging (MRI) can provide enhanced soft tissue contrast and improved delineation of the tumor[4], and is the preferred choice at our institution.

Fiducials have traditionally been made of gold due to its bioinertness and visibility on CT imaging, however gold is not accurately detected on MRI[5]. Platinum fiducials have recently emerged as an attractive alternative[5], considering their improved detectability on MRI imaging.

Endoscopic ultrasound (EUS)-guided implantation techniques have largely supplanted percutaneous methods for deep abdominal targeting, due to improved access and visualization[6].

Aims & Methods: The objective of this case series was to describe our novel platinum seed (PS) fiducials developed in house for EUS-guided fiducial SBRT application. We reviewed the adequacy of fiducial placement, quality of pre-treatment imaging, and migration rates of our fiducials.This single-center case series included patients diagnosed with cancers of the abdomen, including pancreas, adrenal, and liver, who were scheduled to undergo SBRT requiring EUS-guided fiducial implantation between January 2011-February 2024.

The cohort was derived from a prospectively collected EUS database at The Ottawa Hospital in Ottawa, Canada. Patient demographics (sex and age), procedure details (tumor site, number of fiducials implanted, complications), and radiologist's interpretation of MRI/CT imaging were assessed using hospital records. Approximately 70% of patients received antibiotics ((either IV x 1 does or oral antibiotics for 5 days post-procedure (eg. Ciprofloxacin, 500mg)). Our PS fiducials are made from 0.75mm wire cut into 3mm seeds which are then autoclaved in-house. Fiducials were implanted using a 19g FNA needle and advanced using an ERCP guidewire using EUS and fluoroscopic guidance.

Results: 55 patients underwent EUS implantation for a total of 176 fiducials. Preplanning CT/MRI imaging identified 166/176 (94.3%) fiducial signal voids. When evaluating fiducial placement adequacy, 52 of 55 (94.5%) patients had adequate placement for SBRT. Two patients required a repeat procedure with additional fiducial implantation. One case was due to suspected migration, and the second case was due to fiducials being placed too closely preventing adequate delineation. In a third case, no signal voids were visualized for an unknown reason, however, no repeat procedure was required as spine tracking was used. None of the patients developed post-procedure infection or other adverse events (AEs) in this case series (follow up at 1 month). Cost analysis revealed the fiducials made in-house were ~20-fold less expensive compared to commercially available solid gold, gold coil, or solid platinum fiducials.

Conclusion: Our platinum seed fiducials made in-house appear to have similar adequacy and no increased risk of AEs for EUS-implantation when compared to commercially available fiducials[7,8]. For centres using fused CT-MRI imaging for SBRT planning, PS fiducials are recommended. Moreover, significant cost-saving can be achieved through in-house production.

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Disclosure: Nothing to disclose.

PP0204

DIAGNOSIS OF GASTROINTESTINAL SUBEPITHELIAL LESIONS USING A NOVEL ENDOSCOPIC ULTRASONOGRAPHY IMAGING TECHNIQUE

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Introduction: Gastrointestinal stromal tumors (GIST) are the most common malignant subepithelial lesions (SELs) found in the gastrointestinal tract. Due to GIST's notable features including vascularity and deep location, endoscopic ultrasound (EUS) is the first-line diagnostic approach. Three models, color-doppler EUS, power-doppler EUS, and e-FLOW EUS, are useful for real-time vascularity detection; however, they are not effective for fine and slow flow vessel detection. Therefore, contrast enhanced EUS (CE-EUS) is proposed as a first-line approach. Nevertheless, the use of contrast may be detrimental, thus limited to some patients. To avoid contrast-related adverse events, a novel diagnostic method known as detective flow imaging endoscopic ultrasonography (DFI-EUS) has emerged.

Aims & Methods: To evaluate the utility of DFI-EUS in the diagnosis and differentiation of gastrointestinal SELs (GIST and leiomyomas) in comparison to CE-EUS.

A single-center, prospective, diagnostic study was conducted between December/2022 and October/2023 at IECED, Ecuador. The study protocol was approved by the Institutional Review Board and was conducted in accordance with the Declaration of Helsinki. Patients ≥18 years old with a suspected diagnosis of gastrointestinal SELs were enrolled for EUS vascularity assessment. Patients with contraindication for contrast agent administration, any clinical condition rendering EUS-DFI or CE-EUS unfeasible, pregnant and nursing were excluded. First, SELs' assessment was

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performed by two expert endosonographers (blinded to patient's clinical information) using DFI-EUS, followed by a CE-EUS examination (using Sulphur hexafluoride contrast agent) for diagnostic confirmation. The identification rate along with diagnostic accuracy of DFI-EUS and CE-EUS based on histopathological confirmation was the primary endpoint. The type of vascularization (microvascular/microvascular) along with endosonographic appearance (hyper/hypovascular) were also recorded.

Results: A total of 24 patients were included, comprising 54.2% females. The examined lesions consisted of 70.8% GIST and 29.1% leiomyoma. Predominantly, lesions were found in the stomach (91.7%) and within the muscularis propria layer (91.7%). All GIST were hypoechoic and mainly heterogeneous, larger than leiomyoma in size (>2 cm; $p < 0.001$). Tortuous vessels were significantly identified in GIST (p 0.045). Both CE-EUS (94.1%; p 0.003) and DFI-EUS (94.1%; p< 0.001) successfully detected GIST *(Table 1)*.

The overall sensitivity, specificity, positive predictive value (PPV) and negative predictive value (NPV) of DFI-EUS was comparable with CE-EUS with rates of 94.12% vs 94.12, 100% vs 99.64%, 100% vs 99.85% and 87.5% vs 99.64%, respectively *(Table 2)*.

A strong inter-rater agreement between CE-EUS vs DFI-EUS was detected with a kappa of 0.903 (p<.001).

Conclusion: DFI-EUS proves to be a reliable diagnostic tool, exhibiting high diagnostic accuracy, comparable to that of CE-EUS in the differentiation of GIST and SELs in the gastrointestinal tract. Larger multicenter studies are necessary to position this alternative for SELs' differentiation in clinical practice.

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Disclosure: Carlos Robles-Medranda is a key opinion leader and consultant for Pentax Medical, Steris, Medtronic, Motus, Micro-tech, G-Tech Medical Supply, CREO Medical, EndoSound, and mdconsgroup.

The other authors declare no conflict of interest.

PP0205

THE ROLE OF EUS IN THE ASSESSMENT OF SUBEPITHELIAL LESIONS OF THE UPPER GASTROINTESTINAL TRACT: A SINGLE CENTER EXPERIENCE

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Introduction: The increasing number of performed gastroscopies worldwide has lead to higher detection rates of the subepithelial lesions (SELs) of the upper gastrointestinal tract. This study aims to describe the role of endoscopic ultrasonography (EUS) in the diagnosis and management of SELs at a tertiary public reference hospital in Northern Greece.

Aims & Methods: Records of patients who were referred to the Endoscopy Unit of Theageneio Hospital between 2018 and 2023 for upper EUS evaluation were retrospectively reviewed. We conducted data concerning patient characteristics, SELs features, details of the procedure and the final management of each case. All the examinations were performed by a single operator (DT).

Results: A total of 310 patients, male:130 and female:180, with a median age of 59 (range:17-86) were referred to EUS examination. EUS showed extraluminal compression in 16 cases (5.2%), gastric folds in 8 cases (2.6%), vascular structure in 8 cases (2.6%), and in 26 patients (8.4%) no lesion could be identified. A total of 252 (81.3%) patients had confirmed SELs. The lesion's location was: gastric body/corpus in 40.0% (101/252), antrum 34.5% (87/252), duodenum 15.1% (38/252), esophagus 10.3% (26/252). Regarding the size of the lesion in 31.7% of the cases (80/252) was <10mm, 41.3% (104/252) 10-19mm and in 27% (68/252) was >20mm. The most common EUS features were the following: regular borders 98.8% (249/252), hypoechoic lesions 67.5% (170/252), homogenous 77.8% (196/252), without vascularity 90.5% (228/252), and 69.5% (155/252) were originating from the third layer. Sixty-nine SELs (69/252, 27.4%) were malignant or had malignant potential as defined by the EUS findings. More specifically, forty-three patients (43/69, 62.3%) were referred for surgical resection. Twenty-three patients (23/69, 33.3%) were referred for further radiological evaluation and one patient (1/69, 1.4%) was referred for endoscopic resection. Endoscopic reassessment was recommended in 155 patients (155/252, 61.5%). Only one patient had persistent regurgitation, as an examination-related complication, probably due to the periprocedural

anesthesia that led to discontinuation of the examination.

Conclusion: EUS is a practical and safe tool for characterizing and assessing the features and malignant potential of SELs. In our center we could identify the malignant or potential malignant lesions and refer them to surgery.

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Disclosure: Nothing to disclose.

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PP0206

EUS-GUIDED GASTROENTEROSTOMY TO TREAT REFRACTORY GASTROPARESIS: PRELIMINARY RESULTS OF THE FIRST GERMAN EXPERIENCE

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Introduction: Gastroparesis is a motility disorder of the stomach characterized by a delayed emptying of food into the small bowel in the absence of mechanical obstruction. Medical treatment is difficult and symptoms such as epigastric pain, nausea and vomiting can lead to a significantly reduced quality of life. As endosonographic-guided gastroenterostomy facilitates gastric emptying and has been used successfully in patients with gastric outlet obstruction over years, it might be a viable option to treat refractory gastroparesis. The aim of the study was to report the first experience with that approach and to evaluate its feasibility and efficacy.

Aims & Methods: The data of 7 patients with refractory gastroparesis undergoing EUS-GE between march 2021 and August 2023 were retrospectively collected and analyzed. Primary endpoint was to assess the clinical and technical success of the intervention. Secondary endpoints were to assess the stent patency, adverse events and the rate of reinterventions.

Results: Technical and clinical success rate was 100% with a mean reduction in gastroparesis cardinal symptom index of 71%. During the ongoing surveillance, 2 stents had to be replaced after 12 months due to ingrowth, the other 5 stents remained patent after up to 36 months. We observed 2 stent obstructions in one patient due to food which had to be removed gastroscopically.

Conclusion: EUS-GE can effectively reduce the symptom burden in patients with gastroparesis refractory to medical treatment.

Disclosure: Nothing to disclose.

PP0207

PEDIATRIC DEVICE-ASSISTED ENTEROSCOPY – INSIGHTS FROM A TERTIARY CENTER

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Introduction: Device-assisted enteroscopy (DAE) is the only exam capable of a panendoscopic diagnostic and therapeutic intervention in gastrointestinal tract. Despite the limited data, DAE appears to be a safe procedure in pediatric population. In fact, a few studies demonstrate the feasibility and safetiness of DAE in several clinical contexts, namely small bowel bleeding, polyposis syndromes or inflammatory bowel disease.

Aims & Methods: With this study, the authors aim to describe DAE experience, specifically balloon assisted enteroscopy, in a pediatric population of a tertiary center.

A unicentric retrospective study was performed in patients up to 18 years old undergoing DAE (either by anterograde or retrograde pathway), between January 2011 and February 2024.

Results: A total of 23 DAE exams from 16 patients were included in the study, 56% from the male gender. The study population had a medium age of 13 years (ranging from 4 to 18 years old). DAE was performed due to polyposis syndromes in 8 cases (6 cases of Peutz-Jeghers syndrome and 2 cases of juvenile polyposis) and in suspicious cases of middle gastrointestinal bleeding in 8 patients (5 with melena and 3 with ferropenic anemia). There was a total of 17 anterograde DAE, 5 retrograde DAE and 1 intraoperatory enteroscopy. 35% of the procedures were performed in an inpatient setting.

DAE had a diagnostic yield of 95%, identifying polyps in the small bowel or other locations, ulcers, small bowel stenosis and foreign body. A therapeutic procedure was performed in 56% of the cases, namely polypectomy, argon plasma coagulation and foreign body removal. A self-limited bleeding was observed in the first 24 hours after an anterograde DAE, solved without the need of an additional endoscopic or surgical procedure. DAE led to a modification in the clinical outcome in over 90% of the patients.

Conclusion: The present study confirms the safety profile of DAE in the pediatric population. In a cohort from a tertiary referral center, DAE led to therapeutic procedure in the majority of patients, without significant procedure related complications.

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Disclosure: Nothing to disclose.

PP0208

EVEROLIMUS-COATED LASER-CUT SELF-EXPANDABLE METALLIC STENT SUPPRESSES STENT-INDUCED TISSUE HYPERPLASIA IN THE RAT GASTRIC OUTLET

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Introduction: Self-expandable metallic stents (SEMS) are commonly used to relive malignant gastric outlet obstruction (GOO). However, SEMS with high radial force can occurs tissue proliferation caused by mechanical injury. To overcome these problems, drug-eluting stents using antiproliferative drugs have been developed. Specially, everolimus have demonstrated efficacy in reducing restenosis rates in arteries, but their potential application in the gastrointestinal tract remains unexplored. Additionally, recent studies have reported that stents with thinner struts have a reduced risk of clinical restenosis compared to stents with thicker struts. However, the application of everolimus-coated laser-cut stents, which offer robust antiproliferative properties, has not yet been investigated or established in the gastrointestinal tract.

Aims & Methods: This study aims to investigate the efficacy and safety of everolimus-coated laser-cut SEMS in suppressing stent-induced tissue hyperplasia in a rat gastric outlet model.

Everolimus-coated laser-cut SEMS were prepared using ultrasonic spray technology, and the characteristics of the SEMS before and after coating process were evaluated. Twelve male rats were randomly divided into two groups: control (laser-cut SEMS, n = 6) and everolimus (everolimus-coated laser-cut SEMS, n = 6). The effectiveness of everolimus-coated laser-cut SEMS were assessed by comparing results of fluoroscopy, endoscopy, and histopathological analyses.

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Results: Everolimus were uniformly coated onto the surface of the laser-cut SEMS and approximately 80% of everolimus was slowly released for a week. Stent placement was successful in all rats. Stent-induced tissue hyperplasia variables were significantly lower in the everolimus group compared to the control group (all $p < 0.05$). The luminal diameter was significantly larger in the everolimus group than in the control group ($p < 0.05$). The degrees of α-SMA-, Ki-67-, and BrdU-positive deposition were significantly lower in the everolimus group ($p < 0.001$), while the degree of TUNEL-positive deposition was significantly higher than in the control group ($p < 0.001$).

Conclusion: Everolimus-coated laser-cut SEMS effectively suppressed stent-induced tissue hyperplasia in a rat gastric outlet model. Local treatment with everolimu via laser-cut SEMS was safe and effective to maintain the stent patency and has therapeutic potential in the prevention of restenosis after stent placement in the gastrointestinal tract.

Disclosure: Nothing to disclose.

PP0209

OCCURRENCE AND RISK FACTORS OF MISSED CARCINOMA IN THE UPPER GASTROINTESTINAL TRACT (PEUGIC) 2013-2023 – RETROSPECTIVE OBSERVATIONAL STUDY

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Introduction: PEUGIC (Post-Endoscopy Upper Gastrointestinal Cancer) is defined as carcinoma in the upper gastrointestinal (GI) tract which was preceded by negative upper endoscopy in the last 3 years. According to current knowledge, the rate of PEUGIC is an important parameter of quality and should not exceed 10 %. The aim of this study was to evaluate the occurrence and risk factors of PEUGIC at non-university hospital in Frydek-Mistek (Czech Republic) between 2013 and 2023.

Aims & Methods: This is a retrospective analysis of the patients diagnosed with upper GI cancer from January 2013 to December 2023. The data analyzed included family history of upper GI cancer, indications of the examination, therapy, tumor location and histological type. Risk factors for PEUGIC were calculated using the Chi-squared test and Fisher's exact test, with p <0,05 considered significant.

Results: During the study period, a total of 22727 gastroscopies were performed in total, and 122 patients were diagnosed with upper GI cancer (88 men and 34 women), including 35 esophageal tumors, 80 gastric tumors, and 7 duodenal tumors. Family history of upper GI tumor was reported in 10 patients. PEUGIC was identified in 12 patients (9,84%, 9 men and 3 women, mean age 69,25 years), including 6 in the esophagus, 5 in the stomach, and 1 in the duodenum. Two patients had early detection as part of adequate surveillance (Barrett's esophagus, post-polypectomy). Thus, we can evaluate 10 patients as true PEUGIC (8,20%). The rate of PEUGIC was 17,14 % (6/35) in the esophagus, 6,25 % (5/80) in the stomach, and 14,3 % (1/7) in the duodenum. Tumor location (for esophagus p=0,086, for stomach p=0,065, for duodenum p=0,525), histology (for adenocarcinoma p=0,332, for spinocelular cancer p=0,402, for neuroendocrine tumor p=0,25), male gender (p=0,815), or family history of upper GI tumor (p=0,335) were not statistically significant risk factors for detecting PEUGIC.

Conclusion: The overall rate of PEUGIC at our institution over the 11-year study period was 9,86 %, without early detected lesions as part of adequate surveillance, it was 8,20 %. Early detection of PEUGIC, which indicate poorer endoscopy quality, need to be distinguished from early detection of carcinoma within adequate surveillance.

Disclosure: Nothing to disclose.

PP0210

IMPACT OF USING SELF-ASSEMBLING PEPTIDE (PURASTAT) ON ANASTOMOTIC ULCERS – A MULTI-CENTER US EXPERIENCE

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Introduction: Treatment of gastrojejunal anastomotic ulcers, also known as marginal ulcers, is challenging, especially when established techniques such as proton pump inhibitor have failed.

Pura Stat, is a biocompatible transparent synthetic peptide hemostatic gel that is indicated for hemostatis of bleeding in small blood vessels or capillaries of the GI tract and vascular anastomoses. Our aim was to evaluate the feasibility of PuraStat in the setting of non-healing marginal ulcers when used in addition to proton pump inhibitor (PPI) therapy after failed standard therapies

Aims & Methods: We conducted a multi-center prospective case study of adult patients that underwent PuraStat application (3 ml) for marginal ulcers after failing multiple initial therapies with proton pump inhibitors for at least 8 weeks

Results: A total of 10 patients were included in the study. Patients who had an initial presentation of pain, nausea, anemia, with confirmed anastomotic ulceration seen on endoscopy were included. A total of 3mL of Purastat was applied on each of these patients' ulcers, and a repeat endoscopy was performed to evaluate ulcers at ~3months. 9 out of 10 patients showed healing of marginal ulcer on repeat endoscopy in 3 months, while one patient did not present for repeat endoscopy. 9 out of 10 patients showed clinical improvement in their symptoms (90%), and one patient did not show improvement in their symptoms.

Surveillance EGD 3 months post PuraStat	No. (%)
Healing of Marginal Ulcer	
Yes	9 (90)
No	0 (00)
No follow-up EGD	1 (10)
Improvement in Clinical Symptoms (Abdominal pain, nausea, GI Bleed)	
Yes	9 (90)
No	1 (10)

Table 1. Post-PuraStat application results.

Conclusion: PuraStat is an important useful addition to the therapeutic toolbox for the treatment of anastomotic ulcers. Our small prospective case series shows effectiveness in symptom relief as well as clinical healing of anastomotic ulcers. Larger randomized clinical trials are needed.

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Disclosure: Nothing to disclose.

PP0211
WITHDRAWN

PP0212
WITHDRAWN

PP0213
COLONOSCOPY INDICATIONS ACCORDING TO EPAGE II CRITERIA

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Introduction: The prescription of colonoscopies is constantly increasing for the screening and diagnosis of colorectal cancer, precancerous lesions, polyps, and adenomas. Sometimes, indications become excessive due to the availability of this examination on one hand and the fear of missing a neoplastic lesion on the other. The EPAGE II criteria (European Panel on the Appropriateness of Gastrointestinal Endoscopy) was developped to assess the relevance of colonoscopy indications and to rationalize its prescription.
The aim of our study was to evaluate the relevance of colonoscopy indications performed in our department according to EPAGE II criteria and to investigate a possible correlation between the appropriateness of the examination and the endoscopic result.
Aims & Methods: A retrospective study was conducted at a hepatogastroenterology department including all colonoscopies performed during the period from January 2022 to June 2023. The relevance of colonoscopy indications according to EPAGE II criteria was classified as appropriate, uncertain, or inappropriate.
Results: The relevance of the indication was evaluated in 406 colonoscopies. The sex ratio (Male / Female) was 1.15. The median age was 51 with an IQ [15.93]. Rectal bleeding represented the most frequently found indications (30%), followed by Crohn's disease (12.8%), then chronic diarrhea (11.1%). 49% of colonoscopies had satisfactory preparation (Boston ≥7). Colonoscopy was complete in 86% of patients, with catheterization of the terminal ileum in 36.2% of cases.
A significant endoscopic lesion was found in 63% of colonoscopies, represented mainly by inflammatory lesions of inflammatory bowel disease (20.7%) followed by neoplastic lesions (15.3%) of colorectal cancer, then polyps (10%), benign strictures (6.4%), and diverticulosis (5.7%).
Overall, colonoscopy indication was appropriate in 83.5% of cases, uncertain in 10.8%, and inappropriate in 5.7%. A significant endoscopic lesion was found in 63% of patients with an appropriate indication.
Conclusion: The indication for colonoscopies performed in our department was considered appropriate in more than 80% of cases, with rectal bleeding being the main indication.
Significant endoscopic lesions were mostly found in cases of appropriate indication, leading us to encourage the use of this software to regulate colonoscopy indications

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Disclosure: Nothing to disclose.

PP0214
GASTRIC ULCER FOLLOW UP - IS ONE LOOK ENOUGH?

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Introduction: Current guidelines recommend 12 week follow up for gastric ulcer surveillance. This represents a significant proportion of follow up oesophago-gastro-duodenoscopies (OGDs). Debate is ongoing regarding the need for universal endoscopic follow-up to ensure gastric ulcer healing.
Aims & Methods: Aims: We aimed to assess the value of follow-up OGD for gastric ulcer healing in Connolly Hospital Endoscopy Unit with regards to cancer diagnosis and the yield of index biopsies.
Methods: Patient endoscopy data for gastric ulcers diagnosed 2021-2023 via Endoraad were examined. All gastric biopsies including index biopsy of ulcers were reviewed. Time to repeat procedure was recorded.
Results: 147 documented diagnoses of gastric ulcers were identified for the study period (51% female, mean age 60.2). 56% (n=83) of ulcers were biopsied at index OGD. 6 (4% of total) malignancies were diagnosed at index biopsy with no malignancies diagnosed on repeat, chi sq., p =0.02. 76% (n=111) had a repeat procedure, with 43% (n=63) meeting the target of 12 weeks. 16% (n=24) index biopsies taken were HLO positive. Following review of the images by three gastroenterology consultants; 26.% (n=38) were deemed erosions rather than ulcers. Of those deemed ulcers their Forrest classification are as follows; 34% (n=50) Forrest 3, 8% (n=13) Forrest 2c, 2% (n=3) Forrest 2b, 2% (n=3) Forrest 2a and 6% (n=10) Forrest 1b. The remaining 20% of images (n=30) deemed not a gastric ulcer included: malignancy, healed ulcers and severe gastritis. Regarding findings at second scope, 20 patients (18.3%) had persistent ulceration, with no new malignancies identified. Of note, of those not followed, 95% were biopsied at index procedure and had benign features. The average time to repeat procedure was 9.7 weeks.
Conclusion: The Covid pandemic and HSE ransomware attack may have significantly contributed to delay in repeat procedures. However, the majority of patients had either index ulcer biopsies or repeat endoscopy performed. The cancer yield of FU-OGD after benign biopsies remains low. Strategies to reduce the burden of FU-OGD such as risk stratification approach using high quality index examination, ulcer features and index biopsies may aid selective patient follow-up, improve yield of repeat procedures and reduce burden on the patient and health service.
Disclosure: Nothing to disclose.

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PP0215

WITHDRAWN

PP0216

EFFECT OF SELF-ASSEMBLING PEPTIDE HYDROGEL ON GASTRIC ULCER HEALING AFTER ENDOSCOPIC MUCOSAL RESECTION IN PIGS

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Introduction: Endoscopic mucosal resection (EMR) and endoscopic submucosal dissection (ESD) are both standard endoscopic treatments for early-stage gastrointestinal tumors. Submucosal fluid injection is an important part of endoscopic procedure, increasing the space between the lesion and the muscle layer and enabling easy and safe resection. Recently, PuraLift™ (3-D Matrix Ltd, Tokyo, Japan), a novel submucosal injection solution made from self-assembling peptide hydrogel (SAP) was developed in Japan. SAP is fully synthetic and consists only of amino acids present in the human body, giving it the advantages of biodegradability and biocompatibility over conventional solutions.

In addition, the 3D structure of the hydrogel supports cell proliferation in vitro, and self-assembling peptide hydrogel (SAPH) may promote wound healing. However, there are no reports on the effect of SAPH on ulcer healing after endoscopic resection.

Aims & Methods: The aim of this study was to assess the effect of SAPH on wound healing after endoscopic resection. EMR procedures were performed in the stomach of 4 living pigs with 3 patters of submucosal injection solution; normal saline injection group (NS); sodium alginate injection group (SA); PuraLift™ injection group (SAPH). Each pig received 3 separate submucosal injections with 3 mL of the different solutions and then EMR was performed using a 15mm diameter snare. Immediately after EMR (W0) and on week 1(W1) and week 2(W2), we measured the ulcer diameter endoscopically using a measuring device. On week 3(W3) after EMR, all pigs were sacrificed and necropsied by en-bloc gastrectomy, followed by macroscopic measurement of ulcer diameter. All wound sites of the resected specimens were examined histopathologically by haematoxylin and eosin (HE) and Masson's trichrome (MT) and immunohistochemistry staining (αSMA, CD3, CD 31, CD107a). The ulcer reduction rate was calculated as follows : [(ulcer diameter at W0) -(ulcer diameter at week X)] / (ulcer diameter at W0) x 100(%). Ulcer epithelialisation and submucosal foreign bodies were assessed by HE staining, tissue damage in the muscle layer and the thickness of the fibrotic layer by MT staining. Five random fields at each wound site were photographed, and the average rate of positive areas by immunostaining in each injection solution group was calculated.

Results: Ulcer reduction rate at W1 was higher in SAPH group and SA group than NS group (24.1±13.5%; 24.6±17.9%; -6.9±7.6%; p<0.05), b but there were no significant differences at W2 and W3 among three groups. Complete epithelialisation of the ulcer occurred in all except one case of NS group. In SA group, all four cases showed residual injection solution in the submucosal layer with foreign body giant cell reaction and fibrous tissue growth. In contrast, in SAPH group, only one case showed a focal foreign body giant cell reaction without residual injection solution. The thickness of the fibrotic layer is greater in SA group than SAPH and NS groups (SA: 4.2±0.8mm, SAPH: 1.6±0.4mm, NS: 2.0±1.2mm, p<0.05). Tissue damage was not observed in either group. There were no differences in the rates or positive area of αSMA, CD3 and CD31 staining between groups. In contrast, the rates of positive area of CD107a staining were significantly higher in SA group than SAPH and NS group(SA: 13.9±1.0%, SAPH: 5.7±2.2%, NS:1.2±2.1%, p<0.001), indicating a high degree of macrophage accumulation.

Conclusion: SAPH induced good healing after EMR and was more biodegradable and biocompatible than sodium alginate.

Disclosure: Nothing to disclose.

PP0217

SAFETY, EFFICACY AND CARBON FOOTPRINT OF A CAPSULE SPONGE PATHWAY IN PATIENTS WITH REFLUX - A 3YR FOLLOW-UP

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Introduction: With the need to reduce the burden of endoscopy, capsule sponge enables cost effective and efficient triage for patients with reflux as shown in the National Health Service England (NHSE) pilot study whilst achieving a smaller carbon footprint in comparison to gastroscopy. Negative sponge follow up has yet to be reported. Using the largest single UK site database, we describe 1, 2 and 3 year follow up and quantify the effect of gastroscopy avoidance on the endoscopic carbon footprint.

Aims & Methods: Patients routinely referred between November 2020 and November 2023 to gastroscopy or gastroenterology clinic with reflux were selected as part of the DELTA or NHSE pilot studies and triaged to a new nurse led early diagnosis teleclinic where they were first offered a capsule sponge. Patients with positive tests, (Trefoil factor 3 TFF3 and/or atypia/p53 positivity indicating possible Barrett's/gastric intestinal metaplasia or dysplasia respectively) all underwent mandatory gastroscopy and were excluded from this analysis. In the event of a negative test, nurse led teleclinic follow up occurred by 3 months and gastroscopy was only requested for ongoing/concerning symptoms. The endoscopy database for patients with a negative capsule sponge was reviewed at 1, 2 and 3 year intervals to review if gastroscopy had occurred and what findings were identified.

Results: At the 1, 2 and 3 year follow up intervals, 429 and 204 and 41 patients were identified respectively. 86% of the 1 year follow ups (n=371) had not represented for a gastroscopy. The most frequent findings in the 14% (n=57) who did undergo gastroscopy were gastritis (n=14), hiatus hernia (n=9) and normal (n=16).

The most significant finding in this cohort was one case of gastric cancer, identified two weeks post capsule due to new concerning symptoms. In total 371 gastroscopies were avoided in year 1, equating to 42294L of water and 1846 kgeCO2. Of the 204 2 year follow up group, 92% (n=187) had not had a gastroscopy.

There was no significant pathology (including Barrett's) identified in any of the remaining 8% (n=17) and the most common endoscopic conclusion in this cohort was normal. Only 2/17 patients underwent their gastroscopy in the second year of follow up.

Finally, of the 41 patients with a negative capsule sponge available to review at 3 year follow up, 21% (n=9) had undergone a gastroscopy however only 3 of these had been performed in year 3 and they were all normal. Importantly, no patient who represented for OGD at any of the intervals after a negative capsule sponge were found to have Barretts.

Conclusion: Over 1, 2 and 3 year follow up periods, our study provides reassuring data from the largest UK capsule sponge site. Our data shows

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that capsule sponge safely excluded significant upper GI pathology in a less invasive manner, enabling symptomatic management without necessitating gastroscopy in the majority of patients. Nurse led teleclinic safety netting proved effective in identifying patients needing additional investigations whilst reducing the number of gastroscopies performed. This has helped improve endoscopy access overall, whilst also tackling the growing concern surrounding the endoscopic carbon footprint. Our study supports a new paradigm for the investigation of reflux symptoms which can be incorporated in both community and secondary care pathways.

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Disclosure: Nothing to disclose.

PP0218

AUGMENTED PRO-MOTILITY PROTOCOL FOR PATIENTS WHO PREVIOUSLY FAILED PATENCY CAPSULE IS SAFE AND EFFECTIVE

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Introduction: A patency capsule (PC) test is advised before video capsule endoscopy (VCE) to prevent capsule retention in high-risk patients. Although up to 30% of patients cannot expel the PC within 30 hours and are ineligible for VCE, those without strictures, but with slow transit time, can safely undergo VCE. Currently there are no recommendations regarding PC protocol.

Aims & Methods: In this prospective cohort study, we aimed to evaluate the success rate of a 2nd PC attempt with augmented pro-motility preparation protocol, in patients who previously failed to pass PC and to ascertain the risk factors for failure in PC test.

During 2019-2023, Patients who failed their initial PC test in our referral center were offered to undergo a 2nd augmented pro-motility bowel preparation protocol (Table 1). Exclusion criteria: age<18 years, total colectomy or ileostomy and inability to sign informed consent. Patients were followed during the 2nd PC test and the subsequent VCE for success rates and safety outcomes, i.e. VCE retention. Demographic and clinical characteristics were collected for assessing potential risk factors for 1st PC failure and successful 2nd PC by multivariate regression analysis.

		Pro-motility protocol 1st PC	Augmented pro-motility protocol 2nd PC
-2 day	All day	Low residue diet	
-1 day	12:00-20:00	Clear liquid diet	
	14:00	-	Bisacodyl 10mg
	19:00-21:00	-	PEG 3350 1.5L
	20:00-till PC ingestion	Fasting	
PC day	Morning	PC ingestion + Bisacodyl 10mg	
	2hours post-ingestion	Drink clear liquids	
	4 hours post ingestion	A light meal	
+1 day	30 hours post ingestion	X-ray or confirmed expulsion	

Table 1: Patency Capsule Preparation Protocols

PC, patency capsule; PEG3350, Poly-ethylene Glycol

Results: A total of 396 patients performed a PC test and were included in the study (age 45.4±18.6 years, male gender 47.7%). The indications for VCE were suspected/established Crohn's disease (86.1%); anemia/bleeding (8.3%) and radiologic findings (5.6%).

Successful PC expulsion was documented in 323 (81.6%) patients, while 73 (18.4%) failed. A 2nd attempt with augmented pro-motility protocol was accepted by 35/73 patients (48%) and rejected by 38/73 patients (52%). no clinico-demographic differences between the groups.

Successful 2nd PC attempt was documented in 20/35 patients (57%), with all of them completing VCE without adverse events. Failed 2nd PC was documented in 15/35 patients (43%) without adverse events, while 3/15 performed tandem abdominal CT confirming PC in colon (i.e false failure) followed by uneventful VCE.

Performance of a 2nd PC attempt reduced the failed PC rate by 57% (Absolute risk reduction) and the number needed to treat (NNT) is 1.8 for a successful 2nd attempt. Multivariate regression for success with 2nd PC vs. 1st PC was limited by small numbers.

However, diabetes mellitus showed a trend as a risk factor (9.6% in 1st PC test success vs. 20% in 2nd PC test success, OR 3.7, p=0.057, adjusted for age, gender, BMI and stools/day).

Conclusion: In this prospective cohort study, a novel augmented pro-motility PC test preparation protocol improved the success rate in patients who previously failed an initial PC test, without any adverse events. Subsequent abdominal CT should be considered in selected cases to further reduce false positive results. Patients with diabetes mellitus might benefit from performing the augmented protocol upfront, but further large-scale studies are required to establish that.

Disclosure: Nothing to disclose.

PP0219

INCRETIN EFFECT *DURING A LOW-CALORIE MIXED BREAKFAST SMARTPILL® INVESTIGATION:* TIMING OF POST-PRANDIAL METABOLIC HORMONE AND GLUCOSE CHANGES IN HEALTHY VOLUNTEERS

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Introduction: The "incretin effect", the concept that gut hormones amplify insulin secretion, has been around for decades. In the 1960's, it was discovered that oral glucose leads to greater circulating insulin response than IV glucose (1). This was later attributed to gut hormones, such as GLP-1 and GIP. Activation of their receptors on pancreatic β-cells raises cAMP, accounting for amplified insulin secretion. This study aimed to determine if incretin and related metabolic analytes consistent with the incretin effect are detectable during SmartPill® wireless motility capsule (WMC) investigations. These increasingly common investigations represent a convenient resource to study metabolism and relate this to gut physiology.

Aims & Methods: Healthy volunteers (n=41) arrived following overnight fasting. They consumed a standardized 270 kcal mixed meal breakfast. Details of methods including validation of motility data have been published (2). Blood was tapped -10, 0, 10, 20, 30, 40, 50, 60, 90, 120, 180 and 240 min into the meal. Protease inhibitors, including DPP4 inhibitor KR-62436, were added and plasma was prepared. Glucose was measured by the clinical chemistry unit. Active GLP-1, total GIP, acyl-ghrelin and insulin were measured by multiplex ELISA (MesoScale Discovery, USA). Glucagon was measured by ELISA (Mercodia, Sweden). Statistics for comparing pairs of peak or trough timing was done by rank-sum test. Data is presented as median ±SD.

Results: Following meal ingestion, the GLP-1 peak (30 ± 11 min) preceded those of glucose (40 ± 16 min), insulin (40 ± 12 min) and GIP (50 ± 26 min) (p<0.01, all comparisons). Glucose peaked about same time as insulin and before GIP (p<0.01).

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However, the rising phase of insulin and GIP overlapped somewhat, with no statistical difference in time to peak. GLP-1 peaked before insulin in 68% (28/41) of subjects, at the same time in 29% (12/41) of subjects, and after insulin in only 2% (1/41) cases in which insulin appeared to peak at 10 min and GLP-1 at 30 min. GIP peaked before insulin in 20% of subjects (8/41), at the same time as insulin in 49% of subjects (20/41), and after insulin in 32% of cases (13/41). Ghrelin and glucagon deflected downward at 30 min. Ghrelin reached trough at 60 ± 31 min, which was later than peaks of GLP-1 (p<0.01), glucose (p<0.01), insulin (p<0.01) and GIP (p<0.05). Glucagon reached trough at 55 ± 44 min, which was also later than peaks of GLP-1 (p<0.01), glucose (p<0.01), insulin (p<0.01) and GIP (p<0.05). GLP-1 peaked before glucagon reached trough in 95% of cases (38/40)(p<0.01) and at the same time or after glucagon in only 5% (2/40) of cases. GIP peaked before glucagon reached trough in 75% (30/40) of cases (p<0.01). Expulsion of the WMC from the stomach (gastric emptying time 185 ± 72 min) occurred after all peaks and troughs were reached in 95% (39/41) of subjects (p<0.001).

Conclusion: GLP-1 rise to peak is detectable as an early event during SmartPill® WMC investigation, usually preceding peaks or troughs of any of the other analytes. This is consistent with GLP-1 raising intracellular cAMP in pancreatic β-cells to prime them for insulin secretion. GIP rise to peak trails GLP-1, implying a later, prolonged action that parallels insulin. Ghrelin and glucagon start to decline when GLP-1 peaks, tending to reach trough after other hormones peak. Studies of these events require blood taps at a number of time points during a meal.

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Disclosure: Nothing to disclose.

PP0220

COMPARATIVE EVALUATION OF ATMO GAS CAPSULE SYSTEM AND SMARTPILL GI MONITORING SYSTEM FOR TRANSIT TIME ASSESSMENT IN MOTILITY DISORDERS

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Introduction: Motility disorders, such as gastroparesis and slow transit constipation (STC), are common gastrointestinal (GI) conditions affecting up to 5% of individuals worldwide. The symptoms of motility disorders can be debilitating, affecting quality of life and driving utilisation of healthcare systems. Diagnosis relies on evaluation of regional transit times, with delayed gastric and colonic transit indicated for gastroparesis and STC, respectively. The SmartPill GI Monitoring System (SmartPill), an FDA-cleared wireless motility capsule (WMC), has been used in the assessment of gastroparesis and STC since 2008 but has recently been discontinued, leaving a need for a like replacement.

This study aimed to assess a new WMC, the Atmo Gas Capsule System (Atmo Capsule), against SmartPill for the assessment of regional transit times in these conditions.

Aims & Methods: Individuals with upper, lower, or overlapping GI symptoms were recruited in a prospective multi-centre study. Participants concomitantly ingested the Atmo Capsule and SmartPill in a randomized order. Regional transit times including Gastric Emptying Time (GET) and Colonic Transit Time (CTT) were measured and compared between the devices, where paired data were available. GET >5 hours and CTT >59 hours were defined as delayed.

Results: 213 participants (median age 41 [range 20-77], 188 female) were recruited across 12 sites: 177 yielded paired GET results, and 147 for CTT. Pearson correlation coefficients for GET and CTT comparisons between Atmo Capsule and SmartPill were 0.86 (0.81-0.89) and 0.74 (0.66 – 0.81) respectively. GET demonstrated a positive predicted value (PPV) of 78.00% (SE: 5.86%), negative predictive value (NPV) of 85.83% (SE: 3.09%), and overall agreement (OA) of 83.62% (SE: 2.78%). For CTT, PPV was 67.35% (SE: 6.70%), NPV was 92.89% (SE: 2.60%), and OA was 84.35% (SE: 3.00%). No serious adverse device events were reported. 32.2% of the total GET evaluable cohort had delayed GET, 27.2% of the CTT evaluable cohort had delayed CTT. For the cohort that had measurements for all transit regions (n=142), 20% had delayed CTT only, 11% had delayed GET only and 18% had a delay in multiple GI regions as measured by the Atmo capsule.

		Atmo Result Mean (SD) hours	SmartPill Result Mean (SD) hours	Pearson Correlation P-value
Regional Transit Times	GET (n=177)	3.84 (1.57)	4.05 (1.45)	<0.0001
	SBTT (n=183)	5.13 (2.56)	5.18 (2.23)	<0.0001
	CTT (n=147)	47.01 (29.67)	41.79 (27.73)	<0.0001
Multi-regional transit delays as determined by Atmo Capsule	Overall (n=142)	No regional delay	62 (43.7%)	
		Single region delay	55 (38.7%)	
		Multi-region delay	25 (17.6%)	
	Single region delay only (n=55)	GET	16 (11.3%)	
		SBTT	10 (7.0%)	
		CTT	29 (20.4%)	
	Multi-region delay (n=25)	GET & SBTT	6 (4.2%)	
		GET & CTT	9 (6.3%)	
		SBTT & CTT	6 (4.2%)	
		All regions	4 (2.8%)	
Safety	Adverse events related to study devices	17 (8.1%): Abdominal Pain: 1 (0.47%), Capsule Retention: 15 (7.1%), Headache: 1 (0.47%)		
	Serious adverse device effects	0 (0%)		

Conclusion: The Atmo Capsule exhibited substantial agreement with SmartPill in measuring both GET and CTT, highlighting its utility as a potential replacement for the discontinued SmartPill for assessment of motility disorders. Nearly 1 in 5 patients studied had delayed upper and lower transit times highlighting the benefit of evaluating multiple regions of the GI tract with a single test.

Disclosure: Nothing to disclose.

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PP0221

USE OF A NOVEL PHOTOMETRIC CAPSULE IN SUSPECTED NONVARICEAL UPPER GASTROINTESTINAL BLEEDING - A PROSPECTIVE MULTICENTER STUDY

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Introduction: In accordance to current European guidelines, an emergency endoscopy in hemodynamically stable patients with suspected nonvariceal, upper gastrointestinal bleeding (nvoGI bleeding) should be performed within the first 24 hours. This practice consumes scarce human resources, especially at weekends. A recently available photometric capsule enables a prompt, non-invasive assessment of blood or haematin in the gastrointestinal tract.[1]

Aims & Methods: We investigated whether this capsule is a proficient tool to identify patients for whom an emergency endoscopy (within 24h) can be avoided.

A total of 73 patients with suspected hemodynamically stable nvoGI bleeding were included at four different sites using the photometric capsule. Capsule data were assessed after 120 minutes. All patients with a positive capsule result (bleeding index, HI>=1) underwent an upper gastrointestinal endoscopy within 12 hours (Group A). Patients with a negative result (HI<1.0) received a proton pump inhibitor at a standard dose twice daily and underwent endoscopy within 48 to 96 hours (Group B).

If clinical or laboratory signs of acute bleeding became apparent in the meantime, an early endoscopy (within 12h) was performed.

Results: Mean modified Glasgow-Blatchford score (mGBS) was 8.9 for all 73 patients. Among the 41 patients of Group B, two patients (4.8%) received early endoscopy due to persistent hypotension (<90mmHg syst.) (1x cardiac tumor, 1x Forrest III ulcer, both without active bleeding). Two patients received no endoscopy (1 patient refused, 1 patient suffered from a stroke).

Active bleeding was excluded in the remaining 37 patients (sensitivity regarding active bleeding = 100%, negative predictive value = 100%). In Group A, (n=32), emergency endoscopy was performed on average within 2.7 h and detected active bleeding sources in nine patients (28.1%). There were no technical, capsule-associated or bleeding-associated complications as assessed within 30 days of follow-up.

Group	B (capsule negative)	A (capsule positive)	p-value
number	41	32	
mGBS	8.45 ± 2.9	9.3 ± 3.1	n.s.
anticoagulation	25 (61.0%)	19 (59.4%)	n.s.
mean time to endoscopy (EGD)	65.0h (21-97)	2.7h (0.25 – 18)	<0.05
active bleeding in EGD	0 (0%)	9 (28.1%)	<0.05

Conclusion: Due to the very high negative predictive value in detecting active upper GI-bleeding, the applied novel photometric capsule can reliably identify patients in whom emergent endoscopy is not necessary.

References: [1] Schmidt et al. Endosc Int Open 2019 Jun;7(6):E774-E781. doi: 10.1055/a-0880-531

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A. Meining: Ovesco: consultant.
A. Küllmer: Ovesco: lecture fees, study funding.
A. Schmidt: Ovesco: lecture fees, study funding.

PP0222

PASSING YOUR PATENCY TEST: DO PRUCALOPRIDE AND FASTING MAKE A DIFFERENCE?

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Introduction: Capsule endoscopy (VCE) is generally safe, however a subset of patients require confirmation of bowel patency before undergoing the procedure, using a self-dissolvable patency capsule of similar calibre to the video capsule, with interval imaging to confirm excretion. Patency capsule absence on imaging confirms bowel patency. Prucalopride, a 5-HT4 agonist that enhances bowel motility, has been used in VCE to improve completion rates[1], but its use in the context of patency testing remains unknown. Similarly, the role of fasting in improving pass rate is unclear.

Aims & Methods: This study aims to determine whether Prucalopride and fasting enhance the pass rate of patency tests. Data from procedures performed in 3 centers in the Republic of Ireland was reviewed retrospectively. We compared the pass rate in the context of Prucalopride (Group A), fasting (B), and non-fasting (C) states. All patients had an abdominal x-ray between 24-30 hours post-patency capsule ingestion to confirm passage. ANOVA test was used to compare means between all groups, and student t-test between individual groups.

Results: A total of 127 patients were included. Group A (Pruculopride): 59 patients, Group B (fasting, no Prucalopride): 22 patients, Group C (non-fasting, no Prucalopride): 46 patients.

Demographics:

Group A: 47.5% males and 52.5% females, Group B: 45.5% males and 54.5% females, Group C: 32.6% males and 67.4% females

Mean age – Group A: 50.3 years (range 18-82), Group B: 48.4 years (24-82), Group C: 47.9 (17-79).

Indication based on European Society of Gastrointestinal Endoscopy guidelines

Appropriate: 80 (63%)

Inappropriate: 35 (27.6%)

Undocumented: 12 (9.4%)

Pass Rate:

Group A: 36/59 (61%), Group B: 17/22 (77%), Group C: 19/46 (67%).

ANOVA test f-ratio 4.55, p= 0.012.

T-test: Group A vs B (61% vs 77%, p=0.087). A vs C (61%vs 67%, p=0.023), B vs C (77% vs 67%, p=0.002).

Group	Description / Intervention	Male	Female	Mean Age (Range)	Pass Rate
A	Prucalopride	47.5%	52.5%	50.3 years (18-82)	36/59 (61%)
B	No Prucalopride, fasting	45.5%	54.5%	48.4 years (24-82)	17/22 (77%)
C	No Prucalopride, non-fasting	32.6%	67.4%	47.9 years (17-79)	19/46 (67%)

Conclusion: This study shows that fasting improves patency tests pass rates, but Prucalopride does not. Fasting and subsequent patency capsule ingestion promoting the gastro-ileal reflex is a possible explanation. It is

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unclear in this study why Prucalopride, a prokinetic and laxative, failed to show an improvement in pass rates despite being shown to improve VCE completion rates in other studies. Longer duration of Prucalopride at split intervals is a potential area for further studies.

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Disclosure: None.

PP0223

EVALUATION OF "PATENCY AT HOME" PILOT PROGRAM; SUCCESS, SATISFACTION AND SAVING

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Introduction: Capsule retention is the major risk associated with Video Capsule Endoscopy (VCE) with a frequency reported between 2-8% depending on indication. To mitigate this risk, a Patency Capsule (PC) is used to confirm functional patency of the GI tract. Ingestion of a dissolvable capsule followed by plain film radiography 28 hours post ingestion confirms patency. We established a pilot "Patency at Home" programme with the goal of providing a more efficient service for both patients and the hospital.

Aims & Methods: The aim of this study was to evaluate the pass rate of the service as well as patient satisfaction and cost saving. Patients referred for VCE between October 2023 and March 2024 were randomly selected to undergo a home patency test. Pre-assessment for VCE by phone call was carried out prior to postage of the home patency test kit. Written instructions were provided in the kit and an X-ray for a designated day was booked. Questionnaires to evaluate patient experience were included in the testing kit and returned on the day of X-ray. Follow-up phone calls were also offered to those who did not submit questionnaires.

Results: A total of 53 home patency tests were carried out. 68% of referrals were female (n=36). Suspected Crohn's Disease was the most frequent indication for VCE (36%, n=19). NSAID exposure was the most common reason for PC (30%, n=16). The pass rate in this study was 45%, similar to our institution's reported pass rate for hospital patency testing. 96% of patients felt home testing was logistically easier and 94% would choose home patency testing if offered a choice. A cost of €129 per patient visit was offset by this program, totalling a saving of €6,837.

Conclusion: *"Patency at Home"* is an effective and agreeable method for confirming functional patency of the GI tract prior to VCE. Pass rates in home patency testing proved to be similar to those achieved for hospital patency testing. The programme was cost effective with a saving of €129 per patient visit.

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Sec. Gastroenterology Volume 10 - 2023

Disclosure: Nothing to disclose.

PP0224

SAFETY AND FEASIBILITY OF MAGNETICALLY CONTROLLED CAPSULE ENDOSCOPY IN CHILDREN

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Introduction: In recent years, Magnetically controlled capsule endoscopy (MCE) has become increasingly popular in the diagnosis of gastrointestinal diseases in children. This study aims to investigate the safety and feasibility of MCE in the diagnosis of gastric and small intestinal diseases in children by retrospective investigation of the application of MCE in Xi 'an Children's Hospital in a single center.

Aims & Methods: The data of 109 patients who underwent MCE in Xi 'an Children's Hospital from September 2022 to May 2023 were retrospectively analyzed. The general condition, clinical manifestations, completion of gastric and small intestinal examinations, detection of lesions and adverse events of all patients were statistically analyzed.

Results: A total of 109 patients were enrolled in this study, including 56 males (56/109, 51.4%) and 53 females (53/109, 48.6%). The age ranged from 6 to 18 years (mean, 9.9 years). There were 94 outpatients (94/109, 86.2%) and 15 inpatients (15/109, 13.8%). The main indications for MCE examination were abdominal pain (102/109, 93.6%), nausea, vomiting and acid reflux (46/109, 42.2%), abdominal distension (31/109, 28.4%) and anorexia (3/109, 2.7%). Among them, 59 patients (59/109, 54.1%) underwent gastric examination, and 50 patients (50/109, 45.9%) underwent gastric and small intestine combined examination. The completion rate of both examinations was 100% (the capsule reaching the ileocecal junction during the battery life was defined as completion). A total of 91 cases (91/109) of gastric lesions were detected, and the detection rate of lesions was as high as 83.5%. The main manifestations were gastritis, erosion, bile reflux and reflux esophagitis. Nine cases (9/50) of small intestinal lesions were detected, and the detection rate of lesions was 18.0%. The main manifestations were mucosal erosion, ulcer, lymphangiectasis and villous atrophy. The capsule excretion time of all patients was within 1-10 days, and no adverse events such as abdominal pain, hematochezia and capsule retention occurred during the period.

Conclusion: MCE is feasible and safe in the diagnosis of gastric and small intestinal lesions in children. The main indication of MCE in children is abdominal pain. However, most of the abdominal pain in children is caused by functional factors. MCE has great advantages in the exclusion diagnosis of organic diseases in these children because of its safety, non-invasive examination, no need for anesthesia and outpatient examination.

Disclosure: Nothing to disclose.

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PP0225

ENDOSCOPIC LUMINAL IMPEDANCE PLANIMETRY OF THE LOWER OESOPHAGEAL SPHINCTER AND PYLORUS IN EXPERIMENTAL PIGS: A PILOT STUDY. METHODS AND INITIAL EXPERIENCE

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Introduction: The functional lumen imaging probe (FLIP) relies on the principle of impedance planimetry that enables direct measurement of intraluminal cross-sectional areas, allowing the assessment of intraluminal diameters, lumen geometric profiles, and wall biomechanical properties. Electrical impedance refers to the opposition to alternating current presented by the combined effect of resistance and reactance in a closed circuit.

Planimetrics is the study of plane measurements, including distances and areas. Since the early 2000s, impedance planimetry has been introduced into clinical practice, allowing principally assessment of any sphincter function / dysfunction.

In humans, the functional lumen imaging probe is applied mostly to diseases of the oesophagus (achalasia, dysphagia, gastro-oesophageal junction outflow obstruction, eosinophilic oesophagitis, gastro-oesophageal reflux disease), pyloric distensibility (gastroparesis), and ano-rectal disorders (faecal incontinence).

The physiology of the gastrointestinal tract of pigs is similar to that of humans, and therefore porcine experimental models are suitable for various endoscopic and preclinical studies.

Aims & Methods: Aims: The aim of our pilot project was to introduce this method to assess function of the lower oesophageal sphincter and pyloric muscle in experimental pigs.

Methods: All measurements were accomplished in one session under general anaesthesia in six adult female pigs (mean weight 34.2 ± 3.6 kg). EndoFLIP 1.0 System with EndoFLIP catheters (length of measuring zone 80 mm; 16 paired impedance planimetry sensors; filling volumes: 20, 30 and 40 mL). Five major parameters were evaluated for each filling volume: balloon pressure (mm Hg), estimated diameter (mm), cross-sectional area (mm^2), distensibility defined as cross-sectional area divided by balloon distending pressure (mm^2/mm Hg), and zone compliance defined as the change in volume over a 2-cm long segment spanning five electrodes, centred around the gastro-oesophageal junction or pylorus (mm^3/mm Hg).

Results: In total, 180 particular readings were successfully accomplished, see Table 1.

Most of the measured values were nearing lower average figures for the lower oesophageal sphincter and upper average figures for the pylorus in healthy humans. The porcine pyloric sphincter is composed of the *torus pyloricus*. It serves as a sturdy "gatekeeper" between the stomach and D1 duodenum, thus explaining higher pyloric readings.

There was a clear trend for increasing values of CSA, diameter, and balloon pressure with increased filling balloon volumes. However, the sphincter distensibility did not change with increasing filling volumes either for the lower oesophageal sphincter or pylorus.

Conclusion: Endoscopic functional luminal planimetry in experimental pigs is feasible, both for the lower oesophageal sphincter and the pylorus. This is an important starting point for future experimental endoscopic trials and pharmacology studies.

Disclosure: Nothing to disclose.

Parameter	Filling volume 20 mL mean ± standard deviation median interquartile range	Filling volume 30 mL mean ± standard deviation median interquartile range	Filling volume 40 mL mean ± standard deviation median interquartile range
Oesophagus Balloon pressure (mm Hg)	18.7 ± 20.3 10.4 6.6 - 22.1	28.8 ± 10.2 20.4 18.4 - 33.9	44.9 ± 28.9 36.3 24.2 - 56.2
Oesophagus Diameter (mm)	5.9 ± 0.9 5.9 5.0 - 6.5	7.1 ± 1.8 7.2 5.2 - 8.6	9.0 ± 3.2 8.9 5.7 - 10.9
Oesophagus Cross-sectional area (mm^2)	25.8 ± 8.7 21.5 20.0 - 33.0	38.4 ± 22.7 29.5 20.0 - 50.0	66.4 ± 44.7 56.5 25.0 - 91.0
Oesophagus Distensibility (mm^2/mm Hg)	1.7 ± 1.5 1.6 0.3 - 1.9	1.5 ± 1.5 1.0 0.3 - 2. 0	2.1 ± 2.8 1.2 0.3 - 2.1
Pylorus Balloon pressure (mm Hg)	15.9 ± 13.8 11.0 10.5 - 17.6	20.9 ± 17.3 14.8 13.1 - 23.6	28.6 ± 17.0 25.6 17.1 - 31.3
Pylorus Diameter (mm)	5.9 ± 1.4 5.3 5.0 - 6.0	6.9 ± 1.3 6.7 5.9 - 8.1	9.2 ± 3.2 8.2 6.5 - 12.0
Pylorus Cross-sectional area (mm^2)	28.5 ± 14.1 22.0 20.0 - 28.0	38.5 ± 14.4 35.0 27.3 - 51.0	73.8 ± 52.2 52.5 29.0 - 118.0
Pylorus Distensibility (mm^2/mm Hg)	2.4 ± 1.9 1.6 0.3 - 1.9	2.0 ± 1.1 1.8 1.0 - 2.9	2.1 ± 1.3 1.6 0.8 - 3.2

Table 1. Endoscopic functional luminal imaging probe panometry of the lower oesophageal sphincter and pylorus in six experimental pigs

PP0226

SAFETY OF OESOPHAGEAL SELF-EXPANDABLE METALLIC STENT PLACEMENT IN PATIENTS WITH MALIGNANT DYSPHAGIA UNDERGOING ONCOLOGIC THERAPY

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Introduction: Oesophageal self-expandable metallic stent (SEMS) placement is useful in malignant dysphagia palliation. However, its safety in patients undergoing chemoradiotherapy (CRT) remains unclear.

Aims & Methods: Our aim was to evaluate the safety of SEMS placement in these patients.

Retrospective study of consecutive patients between July 2012 and August 2023, with malignant dysphagia and inoperable tumors, who underwent SEMS placement alone or with oncologic treatment (CRT or chemotherapy (CT)) in a tertiary center. Demographic, tumor and stent characteristics, complications and follow-up data were collected.

Results: 44 patients were included; 86.4% male; median age 68.3 IQR 39-97 years; 70.5% (n=31) with esophageal squamous cell carcinoma and 29.5% (n=13) with esophagogastric junction adenocarcinoma. Patients were in stage IV (72.7%, n=32) and stage III (27.3%, n=12).

A partially covered SEMS was placed in most cases (90.9%, n= 40), and a fully covered SEMS in the remainder 4 patients. Median follow-up was 76 days (IQR 2-417 days). One third of patients (n=15) underwent CRT: nine priors, four after and two before and after SEMS placement. Seven patients only had CT: five after and two before and after stent placement. The remaining 50% did not undergo CRT nor CT. The overall median sur-

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vival was 81.5 days (IQR 42.8-115). Survival was similar between patients with and without oncologic treatment 81.5 versus 77.5 days. Major complications occurred in 15.9%: six perforations and one hemorrhage. Severe complications were similar between patients with or without oncologic treatment peri stent placement. Similarly, there was no difference in complications between partially and fully covered SEMS (15% vs 25%, *p*=0.602).

Conclusion: In our cohort, CRT or CT did not increase the rate of major complications of SEMS placement for malignant dysphagia palliation.

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Disclosure: Nothing to disclose.

PP0227

SELF-DILATATION FOR THE MANAGEMENT OF REFRACTORY BENIGN OESOPHAGEAL STRICTURES: OUTCOMES FROM THE LARGEST EUROPEAN TERTIARY CENTRE EXPERIENCE TO DATE

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Introduction: Benign oesophageal strictures are commonly encountered in endoscopy and often effectively managed with endoscopic dilatation. However, up to 40% of patients may develop refractory benign oesophageal strictures (RBES), necessitating frequent endoscopic intervention that is challenging, poses a greater risk of complication and negatively impacts patients' quality of life.

Oesophageal self-dilatation (SD) offers an alternative treatment that potentially reduces the need for repeat hospital visits while allowing greater autonomy in managing their condition.

Aims & Methods: We describe the efficacy and safety of oesophageal SD at a tertiary referral centre. We retrospectively reviewed electronic records of patients with RBES who started SD between 2013 and 2024. Data were collected on stricture characteristics, endoscopies, dysphagia scores, and adverse events. The primary outcomes were the endoscopic dilatation-free interval (longest period of time patients did not require endoscopic dilatation) and the number of endoscopic procedures performed within 12 months of starting SD. Secondary outcomes included improvement in dysphagia score, technical success rate, and complications.

Results: We identified 43 patients who initiated SD during the study period. The technical success rate was 95.3%, with 41 out of 43 patients learning to perform SD successfully.

Among the remaining patients, the median age was 57 years (range 27-84 years), and 28/41 patients (68.3%) were female. Patients initiated SD for refractory peptic (32%), post-radiation (19.5%), post-surgical (12.2%), post-endoscopic resection (11.6%), caustic (7.3%), and other aetiology-related strictures. Stricture location varied, with 10/41 patients (24.4%) having proximal, 12/41 (29.3%) having middle, and 14/41 (34.1%) having lower oesophageal strictures, while the remainder had strictures involving multiple segments. Before initiating SD, patients had a median of 9 endoscopic interventions (IQR 7-17). The median follow-up post-SD was 23.2 months (IQR 6.8–49.4). The median dilatation-free interval significantly increased from 68.5 days (IQR 41-106 days) to 161 days (IQR 121-279 days) within 12 months before and after SD, respectively (P < 0.001). The median all-time intervention-free interval after SD was 405 days (IQR 130-705 days). Additionally, there was a significant reduction in the number of endoscopic interventions from a median of 7 (IQR 4.7-10) to 1 (IQR 0.75-3) within 12 months before and after SD, respectively (P < 0.001). Dysphagia scores improved, with a median Dakkak and Bennett Dysphagia Score of 2 (IQR 2-2) at baseline and 0 (IQR 0-1) after SD (p<0.001). Two adverse events were reported: one patient experienced clinically significant haematemesis while another patient suffered presumed iatrogenic oesophageal perforation secondary to SD. Both patients were managed endoscopically but discontinued SD.

Conclusion: To our knowledge, we have described the largest European case series of oesophageal SD that has been published to date. Oesophageal SD appears to be an effective, well-tolerated, and safe approach in managing often complex patients with RBES providing a viable alternative to conventional endoscopic interventions. However, the occurrence of adverse events highlights the importance of careful patient selection and ongoing monitoring. Further research is warranted to validate these findings and identify predictive factors for treatment success.

Disclosure: Nothing to disclose.

PP0228

ESOPHAGEAL ATRESIA WITH TRACHEOESOPHAGEAL FISTULA IS ASSOCIATED WITH CONSANGUINITY

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Introduction: Esophageal atresia with tracheoesophageal fistula (EA-TEF) is a rare congenital malformation. The incidence of TEF is 1.3-4.6 cases per 10,000 live births. The specific etiology of EA-TEF is unknown. Environmental risk factors such as maternal use of medications and alcohol consumption during pregnancy are suggested. Genetic factors have also been described, especially among the syndromic types such as VACTERL, CHARGE, and Down syndrome, but less among non-syndromic types.

Aims & Methods: Aim: To examine the association between non-syndromic EA-TEF and consanguinity.

Methods: A retrospective study comparing the incidence of EA-TEF between low-consanguineous Jewish population and high-consanguineous Bedouin population. All patients were treated at Soroka University Medical Center, the only tertiary medical center in southern Israel.

Results: During the years 2000-2022, 579,130 children were born in southern Israel, 386,915 (67%) were Jewish and 192,215 were Bedouin Muslims. A total of 96 patients were diagnosed with EA-TEF, 83 of them were non-syndromic EA-TEF. The incidence of non-syndromic EA-TEF was 1.43 cases per 10,000 live-births and was statistically higher among the Bedouin population (2.65 *vs.* 0.83 cases per 10,000 live-births, P<0.001). The consanguinity rate among the Bedouin group was higher comparing with the Jewish (71% *vs* 0% P<0.001). There were no differences in other risk factors.

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Conclusion: The incidence of EA-TEF is exceptionally higher among the Bedouin population that live in the same geographic region and the same medical access as the Jewish population, proposing consanguinity as an important/essential risk factor of EA-TEF.

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Disclosure: Conflict of Interest: None.

PP0229

PROPHYLACTIC ENDOLUMINAL VACUUM THERAPY – TOWARDS AN ERA OF REDUCING LEAK RELATED MORBIDITY AFTER ELECTIVE OESOPHAGO-GASTRIC RESECTION

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Introduction: Uncontrolled sepsis following leak from major oesophago-gastric surgery is a devastating condition with increased morbidity and mortality and intensive care stay. We hypothesised that endoluminal vacuum therapy (EVT) could be used in the prophylactic setting to provide source-control, prevent mediastinitis and minimise sepsis related morbidity from leaks.

Aims & Methods: An ad-hoc EVT device was created using VAC granufoam (KCI) and a nasogastric-tube (16/18Fr), and placed intraluminally across the high risk anastomosis/area of repair. Continuous negative pressure (75–125mmHg) was applied. Assessment of the repair site/anastomosis was performed by radiology, endoscopy, clinical assessment or a combination following cessation of EVT and removal of the prophylactic EVT device.

Results: Twenty-seven patients had prophylactic EVT; M:F ratio was 3:1 and the median age was 68 years (range: 41-82 years). Twenty-six patients had cancer resections while 1 had benign surgery. The rationale for prophylactic EVT included preoperative factors (10), intra-operative factors (16) or both (1). The median duration of prophylactic EVT was 7 days (range: 4-14 days). The median length of stay was 19.7 days (range: 6-96 days), the two longest admissions were unrelated to the anastomosis (prolonged chyle leak, and long-term-sequelae of life-threatening bleed). One patient developed full-thickness microleak at endoscopic evaluation, remained clinically well with no systemic signs of sepsis, and was successfully managed with luminal EVT for a further 6 days in a ward-based environment. Though prophylactic EVT did not prevent the leak, it significantly minimised the systemic hit that traditionally results from leak and associated mediastinal sepsis.

Conclusion: Prophylactic EVT is a useful adjunct for management of patients with a high-risk for leak and can serve as a risk reducing strategy in select cases. There are no short-term complications, yet the additional morbidity, mortality, and length of stay associated with a fully-fledged leak is avoided. Our results lay the foundation for the use of prophylactic EVT in randomised controlled trials to further standardise selection criteria, assess longer term potential patient benefit and impact on survivorship.

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Disclosure: Nothing to disclose.

PP0230

ESOPHAGEAL PERFORATION – ANALYSIS OF ETIOLOGY, MANAGEMENT AND OUTCOME – A 10YRS EXPERIENCE AT A TERTIARY UNIVERSITY MEDICAL CENTER

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Introduction: Esophageal perforation is a rare condition with a high mortality rate. Often it has an iatrogenic origin, being caused by transesophageal echocardiography, different endoscopic and surgical procedures. Furthermore, there are other causes such as foreign body ingestion, tumor associated perforation and Boerhaave's syndrome. Esophageal perforation can be difficult to diagnose since the clinical presentation is often unspecific. This can lead to a delay in appropriate treatment. Optimal management of this condition challenges clinicians in daily practice. Therefore, we present a monocentric retrospective study, evaluating our experience in treating esophageal perforations over more than a decade as a tertiary referral University Medical Center in Germany.

Aims & Methods: This study comprises the retrospective analysis of data from all patients treated for esophageal perforation in our clinic in the departments of general and visceral surgery and gastroenterology between

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2012 and 2024. Only patients with confirmed esophageal perforation were included. Data was collected by means of a retrospective analysis of digital and analogous data archives as well as information from the physicians providing further treatment.

Results: A total of 78 cases (47 male, 31 female) of esophageal perforation were included. The leading cause was iatrogenic (n=45), consisting of endoscopic procedures (n=22), operative complications (n=11), TEE (n=6), diagnostic EGD (n=5) and one case of endotracheal intubation. The second most common cause of perforation was Boerhaave's syndrome (n=23), followed by perforations associated with foreign body ingestion (n=5) or tumors of the esophagus (n=4). Traumatic perforations were rare (n=1). Most perforations occurred in the lower third of the esophagus (58 cases), 11 perforations were located in the mid esophagus and 9 in the upper third. Primary treatment consisted of direct surgical management in 26 cases (suture of the esophagus, gastrectomy, esophagectomy, definitive cervical esophagostomy, thoracic drainage, lavage), endoscopic therapy in 38 cases (Endo-VAC, stenting and application of OTSC or TTSC clips) and a conservative approach in 14 cases. Eventually a total of 48 patients underwent surgical treatment. Survival was poorest in spontaneous and tumor associated perforations (mean survival 22 and 23 months). Iatrogenic perforations showed an intermediate outcome.

The best outcome was observed in perforations associated with foreign body ingestion (mean survival 50 months). In correlation with perforation location, the survival was best for perforations in the upper esophagus, followed by those in the lower esophagus, while lesions in the mid esophagus showed the worst outcome. Perforation associated death was recorded in 17 cases.

Conclusion: Esophageal perforations are a life-threatening event that require immediate and adequate treatment. Our analysis identified risk groups resulting in poorer survival: perforations in the mid esophagus, primary surgical treatment, advanced age. Within the last years, endoscopic techniques have reduced the need for surgical interventions. Nevertheless, surgical treatment remains the therapy of choice for patients not eligible for endoscopic treatment or after failure of endoscopic therapy. Our analysis emphasizes the importance of multidisciplinarity, close monitoring and continuous reassement of treatment strategy. Further investigation will help to improve the outcome via prospective studies or the establishment of a local or interregional database seem promising to underline suggestions derived from our analysis.

Disclosure: Nothing to disclose.

PP0231

WITHDRAWN

PP0232

ENDOSCOPIC TRANS-ORAL OUTLET REDUCTION IS SAFE AND EFFECTIVE FOR WEIGHT REGAIN AFTER ONE-ANASTOMOSIS GASTRIC BYPASS

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Introduction: Obesity is a global health concern with increasing prevalence, and bariatric surgery remains a prominent treatment option. One-anastomosis gastric bypass (OAGB) has gained popularity for its advantages, including significant weight loss and fewer early complications. However, concerns arise regarding nutritional deficiencies, weight regain, and other complications. This study aims to assess the effectiveness and safety of endoscopic transoral outlet reduction (TORe) as a treatment for weight regain after OAGB.

Aims & Methods: Consecutive Patients with weight regain following OAGB were evaluated. Weight regain was defined as a 10% increase in total body weight from the post-surgery weight nadir. Diagnostic gastroscopy determined anastomosis diameter, pouch dimensions and ruled out contraindications for TORe. The procedure involved circumferential argon plasma coagulation and full thickness endoscopic suturing to narrow the anastomosis. Patient data was collected, and outcomes assessed in accordance with ethical standards

Results: sixty four patients were treated during a 4 years duration and followed up for 1 a minimum of 1 year. Fourty seven percents were females, mean pre-treament BMI 35.5. Average weight loss was 7% of total body weight after 6 months and 8% after 12 months (p=0.03). A reduction of at least 5% and 10% was seen in 65% and 31%, respectively. Additional treatment was needed in 17% of patients. Major complication that required surgical intervention occurred in only one patient

Conclusion: Endoscopic transoral outlet reduction appears safe and effective fot the treatment of weight regain s/p OAGB. Sustained weight loss was seen during a 1 year follow up. Further investigation is needed to assess the long term durably of the effect.

Disclosure: Nothing to disclose.

PP0233

ENDOSCOPIC TREATMENT OF REFRACTORY MARGINAL ULCERS - A CASE SERIES

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Introduction: Ulceration at the gastrojejunostomy is a late complication following gastric bypass surgery, usually responsive to high dose PPI. However, in refractory cases, patients may require surgical treatment. Recently, a new technique of endoscopic jejunal flap over-suturing has been reported with exceptional success rates. We aimed to evaluate the implementation of this approach in our practice.

Aims & Methods: Consecutive patients with symptomatic marginal ulcers, who failed to response to a two-month course of high dose PPI were included. Patients were treated with endoscopic suturing of a jejunal flap

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over the ulcer. In case of a narrow anastomosis, a self-expandable metallic stent (SEMS) was used alone or in combination with suturing. The stent was removed after 2 months. Endoscopy was repeated 8 weeks after the procedure. Technical success was defined as a fully covered marginal ulcer. Clinical success was defined as an ulcer heal on repeat endoscopy, while persistent ulcer, perforation or need for surgical repair was considered as failure.

Results: Ten patients were included, seven after Roux-en-Y gastric bypass and after 3 single anastomosis gastric bypass. All patients were negative for Helicobacter Pylori, deny alcohol abuse or NSAIDS use, 2 patients were active smokers at the time of diagnosis. All patients were on chronic PPI at the time of diagnosis, and the average therapy duration was 27 months. Jejunal over-suturing was performed in 9 patients of them 3 in combination with a SEMS, and 1 patient was treated only with a SEMS. Overall technical success rate was 100%, and a clinical success rate was 50% with 4 patients going through surgery thereafter. No adverse events occurred during the procedure or follow up period.

Conclusion: To date, marginal ulcers remain a dreaded late adverse event necessitating prolonged PPI use, repeat endoscopies and even revisional surgery. Endoscopic treatment to promote healing of refractory marginal ulcers appears safe. Technical success was achieved in all patients while clinical success only in half. However, since the only alternative treatment is surgery the endoscopic approach should still be considered. The lower than expected clinical success might be related to a more PPI resistant population compared with previous published series. Further research is needed to determine the appropriate therapeutic algorithm.

Disclosure: Nothing to disclose.

PP0234

INFLUENCE OF RECURRENT LARYNGEAL NERVE PALSY AFTER ESOPHAGECTOMY AND FUTURE PLAN OF TREATMENT USING AMNION MEMBRANE SHEET

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Introduction: The therapeutic effect of recurrent laryngeal nerve (RLN) dissection in the treatment of esophageal cancer, particularly squamous cell carcinoma, is well-established. However, due to the risk of RLN palsy as a consequence, this procedure is often avoided. Nonetheless, the prevalence of this adverse event and its impact on prognosis remain unclear.

Aims & Methods: In this study, we aimed to evaluate the influence of recurrent laryngeal nerve palsy and long-term effects on prognosis. Moreover, we introduce some basic science of treating injured nerve using amnion membrane sheet.

We analyzed 726 cases of esophageal cancer who underwent bilateral recurrent laryngeal nerve lymphadenectomy during esophagectomy between 2000 and 2021. Cases with combined nerve resection and R2 resection were excluded. Recurrent laryngeal nerve palsy was confirmed by bronchoscopy on the first postoperative day, if even a slight difference in the movement of the vocal cords between the left and right sides was detected.

Results: RLN palsy was observed in 40.7% of cases (left: 234 cases, right: 31 cases, bilateral: 30 cases), with 12 cases experiencing paralysis lasting over a year. Analysis revealed significantly higher rates of complications such as aspiration ($p < 0.0001$), postoperative pneumonia ($p = 0.0004$), and prolonged hospital stay ($p < 0.0001$) in RLN palsy cases. Furthermore, there were associations between postoperative pneumonia and patient factors including age ($p = 0.0364$), sex ($p = 0.0066$), RLN palsy ($p = 0.0005$), and reconstruction route ($p = 0.0098$), each identified as independent risk factors in multivariate analysis. However, only age and sex showed associations with pneumonia-related mortality. Long-term prognosis analysis using propensity score matching revealed slightly poorer outcomes in the RLN palsy group, although no statistically significant difference was observed.

Additionally, we are conducting research on a treatment method using amniotic membrane sheets in a sciatic nerve injured rat model. Electromyography studies showed a trend toward improved motor function in the amniotic membrane sheet treatment group compared to the control group at 8 weeks post-injury ($p = 0.0887$). Further investigation with increased sample size is planned to refine this approach for future clinical application.

Conclusion: Although RLN palsy after esophageal cancer surgery did not affect patient prognosis in the long term, we discovered it as a cause of aspiration and postoperative pneumonia in the short term. With the growing understanding of respiratory complications as important prognostic factors in various cancers, including esophageal cancer, strategies to prevent RLN palsy are crucial. Amniotic membrane sheets represent a potential intervention for the early recovery of RLN palsy.

Disclosure: Nothing to disclose.

PP0235

WITHDRAWN

PP0236

ASSESSING THE EFFICACY OF SELF-EXPANDING METAL STENTS FOR POSTOPERATIVE UPPER DIGESTIVE LEAKS AND FISTULAE: A DECADE OF EXPERIENCE AT A TERTIARY CENTER

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Introduction: Postoperative anastomotic leaks and fistulae in the upper digestive tract are severe complications that can lead to significant morbidity and mortality. Traditionally, surgical intervention has been the primary approach to address these complications. However, self-expanding metal stents (SEMS) have emerged as a less invasive alternative, offering a potential pathway to achieve closure of these leaks and fistulae. Despite their growing use, the efficacy and safety profile of SEMS in this context is not fully understood.

Aims & Methods: This study aims to evaluate the success rates, safety, and predictive factors of therapeutic success for SEMS in managing postoperative upper digestive tract leaks and fistulae, drawing on a 10-year experience at a tertiary center.

We conducted a retrospective review of patients with postoperative anastomotic leak or fistula who underwent endoscopic SEMS placement at our tertiary center between 2014 and 2022, and prospectively from 2022 onwards. Therapeutic success was defined as complete closure of the leak/fistula, while failure indicated non-closure.

Results: We included a total of 43 patients (69.8% male) with a median age of 61 years (IQR 17). There were no statistically significant differences regarding distribution by gender, age, Charlson comorbidity index, benign vs malignant disease, time from surgery to endoscopic treatment and leak size between the different types of stents ($p>0.05$). The technical efficacy was 100%, with a therapeutic success rate of 55.8%. Bariatric stents ex-

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hibited the highest success rate (85.7%), followed by fully covered SEMS (60.0%) and partially covered SEMS (46.2%) (p = 0.166). Success was significantly higher in patients with a smaller leak size (10 mm vs. 12 mm, p<0.05).

Therapeutic failure was associated with greatest need for endoscopic adjuvants (57.9% vs. 37.5%, p=0.183). The overall mortality rate was 34.9%.

Conclusion: This study indicates that self-expanding metal stents (SEMS) offer a viable and safe treatment option for postoperative upper digestive tract leaks and fistulae. With a technical efficacy of 100% and a therapeutic success rate of 55.8%, SEMS demonstrated promising results. The success rate was influenced by stent type, with bariatric stents showing the highest rate, followed by fully covered SEMS and partially covered SEMS. Smaller leak size emerged as a significant predictor of therapeutic success, emphasizing the importance of early detection and timely intervention. Although the overall mortality rate was 34.9%, this study's outcomes suggest that SEMS can play a crucial role in managing these complications, with early diagnosis and prompt stent placement being pivotal for success. Further research is needed to better understand the factors contributing to therapeutic success and to optimize endoscopic strategies for patients with upper digestive tract leaks and fistulae.

Disclosure: Nothing to disclose.

PP0237

CGAS-STING PATHWAY ACTIVATION ENHANCES ANTI-TUMOR EFFECT OF TALAPORFIN PHOTODYNAMIC THERAPY THROUGH REACTIVE OXYGEN SPECIES PRODUCTION AND DNA DAMAGE

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Introduction: Photodynamic therapy (PDT) is a relatively non-invasive anti-cancer therapy that employs a photosensitizer and a specific wavelength of light irradiation. PDT induces tumor shrinkage in various malignancies due to three main mechanisms, direct effects of reactive oxygen species (ROS), vascular shutdown leading to tumor infarction, and acute inflammation with immune response. Among them, ROS is important factor of DNA damage. It was reported that cGAS (cyclic GMP-AMP synthase)-STING (stimulator of interferon genes) pathway regulates a transcriptional program that controls the generation of ROS and that STING loss alters ROS homeostasis. STING loss reduces ROS induction and that causes therapeutic resistance.

Thus, we predicted that pharmacological activation of cGAS-STING pathway would be an actionable target for reducing therapeutic resistance and enhance DNA damage caused by PDT. Our study investigated the efficacy of STING agonist (2'-3'-cyclic GMP-AMP sodium or AUS-100) combined with talaporfin sodium PDT.

Aims & Methods: 1. We established STING knockout cells in a colon cancer cell line (HCT116) using CRISPER-Cas9.

2. We compared the induction of cell death after in vitro talaporfin PDT in parental and knockout cells. We analyzed IC50 by MMT assay.

3. We measured ROS generation, using DCFDA assay after in vitro talaporfin PDT in parental and knockout cells.

4. We analyzed the induction of cell death treated with ROS scavenger (N-acetyl cysteine; NAC) after in vitro talaporfin PDT in parental and knockout cells.

5. We measured DNA damage by immunofluorescence microscopy with an anti–g-H_2AX antibody after in vitro talaporfin PDT in parental and knockout cells.

6. We investigated PDT-induced STING pathway activation of STING-regulated genes by western blot analyses after in vitro PDT in parental and knockout cells.

7. We compared the *in vivo* efficacy of talaporfin PDT between the parental and knockout cells.

8. We assessed the *in vivo* efficacy of talaporfin PDT, STING agonist (2'-3'-cyclic GMP-AMP sodium or AUS-100), and their combination in a xenograft tumor model bearing HCT116.

Results: 1. We established STING knockout cell. STONG knockout was confirmed by western blot analyses.

2. STING KO suppressed cell death induction of TS-PDT (IC_{50} 12.5 (±1.35) vs 14.5 (±1.18)μmol/L).

3. STING KO suppressed TS-PDT induced ROS generation (M.F.I. 4491 vs 1706).

4. STING KO suppressed cell death induction of TS-PDT with NAC.

5. Treating parental cell lines with talaporfin PDT led to significantly more γ-H_2AX foci formation than in STING knockout cells.

6. At the protein level, TS-PDT enhanced TBK-1 phosphorylation in parental cell. STING dependent signaling was enhanced by TS-PDT, and the effects were eliminated by STING loss.

7. The *in vivo* efficacy against towards parental cells was significantly greater than that towards STING knockout cells.

8. The combination exhibited significantly greater tumor growth inhibition than by single agents alone in a xenograft tumor model.

Conclusion: We have established STING knockout cell using CRISPER-Cas9 and identified novel regulators of DNA damage in Photodynamic therapy. Our results point to a role for STING in DNA damage sensing and cyclic dinucleotide detection that goes beyond its conventional function. According to these results, talaporfin PDT is more effective when a STING agonist is used to control the degree of DNA damage. Combining talaporfin PDT with a STING agonist would offer a potentially strong combination for next-generation anticancer treatments.

Disclosure: Nothing to disclose.

PP0238

OLGA AND OLGIM STAGING ARE POOR PREDICTORS OF GASTRIC CANCER RISK

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Introduction: Gastric cancer continues to be one of the main challenges worldwide, with a predictable increase in the number of new cases in the coming decades. Staging systems for gastric precancerous lesions: Operative Link on Gastritis Assessment (OLGA) and Operative Link on Gastric Intestinal Metaplasia (OLGIM) are considered to be the best predictors for gastric cancer development. Only OLGA/OLGIM stages III-IV are considered high-risk lesions with a significant potential of progressing towards gastric cancer.

Aims & Methods: The purpose of this study was to evaluate the predictive role of OLGA and OLGIM staging in gastric cancer development.

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We conducted a retrospective cohort study by matching gastric adenocarcinoma cases registered in the Latvian Cancer Registry between 2005 and 2022 to the pathology reports in the database of Academic Histology Laboratory from upper endoscopies performed in the Digestive Diseases Centre GASTRO, Jurmala hospital and Riga 1st hospital. Only cases with antral and corpus biopsies (as the minimum) available from the index endoscopy allowing OLGA / OLGIM staging were included in the study. If data from several endoscopies were available for the same patient, the case with the highest OLGA/OLGIM stage was used as the index endoscopy; if OLGA/OLGIM staging was identical across several endoscopies, the first endoscopy was used as the index case. The presence of dysplasia in any of the previous pathology reports was also considered. Patients with suspected cancer at the enrolment or having been recorded in the cancer registry less than 6 months after the index endoscopy were excluded from the study.

Results: Altogether 54 patients were available for the analysis; 38.9% were male. The mean age at the time gastric cancer detection 69.2 years, range 40-90. The median time between the index endoscopy and gastric cancer diagnosis was 1403.5 days (IQR: 826.5-2793.3). Forty-one patients (75.9%) had one endoscopy, eleven (20.4%) had two, one (1.9%) had four and one (1.85%) – 8 endoscopies. Biopsies from incisura angularis of the stomach were available for 43 patients (79.6%).

Altogether 16 patients (29.6%) had dysplasia reported at one of the preceding endoscopies, of those 14 were low grade, 2 – high grade. Patient distribution per OLGA and OLGIM stages at the index endoscopy is shown in Table 1. Among patients ultimately diagnosed with gastric cancer, only 11 (20.4%) had OLGA III-IV stages and 10 (18.6%) had OLGIM III-IV stages at the time of the index endoscopy. Excluding cases with dysplasia reported during any of the preceding endoscopies did not make a substantial change in the OLGA and OLGIM stage distribution at the index endoscopy.

	Stage 0	Stage I	Stage II	Stage III	Stage IV
OLGA (all cases)	7 (13.0%)	17 (31.5%)	19 (35.2%)	6 (11.1%)	5 (9.3%)
OLGIM (all cases)	12 (22.2%)	15 (27.8%)	17 (31.5%)	5 (9.3%)	5 (9.3%)
OLGA (excluding cases with previously reported dysplasia)	7 (18.4%)	15 (39.5%)	10 (26.3%)	4 (10.5%)	2 (5.3%)
OLGIM (excluding cases with previously reported dysplasia)	12 (31.6%)	12 (31.6%)	9 (23.7%)	3 (7.9%)	2 (5.3%)

Table 1. OLGA and OLGIM stages at the time of the index upper endoscopy in patients having developed gastric cancer thereafter.

Conclusion: The majority of patients having undergone upper endoscopy before the diagnosis of gastric cancer were in low-risk OLGA and OLGIM stages. Continued research is warranted to identify the best approach for risk stratification of gastric precancerous lesions. Yet, the current study demonstrates that the assessment should not be solely based on OLGA or OLGIM staging.

Disclosure: Nothing to disclose.

PP0239

WITHDRAWN

PP0240

ENDOSCOPIC SUTURING FOR DEFECT CLOSURE IN THE UPPER GASTROINTESTINAL TRACT: A RETROSPECTIVE COHORT STUDY

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Introduction: The increasing availability of gastrointestinal (GI) interventions has raised the need to treat luminal defects. Endoscopic suturing (ES) is a minimally invasive technique, used for a wide range of indications.

Aims & Methods: This retrospective cohort study aimed evaluate the performance of ES in treating upper GI defects.Data from a tertiary centre were collected for patients undergoing ES to treat upper GI defects. The primary outcome was technical success, defined as the successful sutures' deployment. Secondary outcomes included immediate clinical success (confirmation of closure at the time of the procedure), recurrence, long-term outcomes and complications. Descriptive statistics and x^2 test were used to calculate the rates of the outcomes and assess any link between independent variables and results.

Results: Forty-two procedures were performed on 25 patients between 2018 and 2023. The mean age was 55 (±16.2) years and 56% were female. The technical success rate was 88.1% (37/42) and immediate clinical success 91.9% (34/37), with only two (4.8%) adverse events. The overall recurrence rate was 61.8% (21/34), with a long-term clinical success rate of 69.6% (16/23). Technical success was higher in oesophagus (92.3%), and stomach (100%) (p=0.002), and immediate clinical success was more likely in patients with leak (88.9%) or fistula (95.2%) compared to perforation (50%; p=0.005).

Conclusion: ES demonstrated high rates of technical and immediate clinical success for defect closure in the upper GI tract, with low rates of complications. The benefit is most prominently seen among patients with leaks and fistulas in the stomach and oesophagus.

Disclosure: Nothing to disclose.

PP0241

LONG TRANSPARENT CAP SUTURING TECHNIQUE FOR THE CLOSURE OF ENDOSCOPIC FULL-THICKNESS RESECTION DEFECTS: A NOVEL ALTERNATIVE

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Introduction: Gastric submucosal tumors (SMTs) are elevated lesions originating from below the gastric mucosal layer. Pathological types of SMTs include stromal tumors, leiomyomas, lipomas, etc., with stromal tumors being the most common[1]. Endoscopic full-thickness resection (EFTR) has gradually become the main treatment modality for SMTs [2]. However, in order to completely remove lesions involving the submucosal and deeper layers, EFTR often involves intentional perforation during the procedure. Therefore, successful closure of the full-thickness defect after the procedure is crucial to prevent serious complications such as pneumothorax and pneumoperitoneum[3]. Currently, availabloe techniques for the full-thickness GI wall defects closure includes metal clip closure, ny-

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lon suture purse-string closure, Over-The-Scope-Clip system (OTSC), and the OverStitch suture system [4-5]. However, due to variations in endoscopic center equipment, individual physician skills, and patient economic conditions, most of the suture devices are not widely available including in China. Using metal clips alone may sometimes be insufficient for closing the large GI defects, while the purse-string suture technique can be relatively cumbersome.

Aims & Methods: The aim of this study was to evaluate the safety and efficacy of the long transparent cap suturing technique for the closure of large endoscopic full-thickness resection (EFTR) defects. A total of 62 patients who underwent EFTR for their large gastric submucosal tumors at the First Affiliated Hospital of Zhengzhou University between January 2017 and March 2023 were included in this study. The novel long transparent cap suturing technique was used for EFTR defect closure in all the patients.

Results: The procedure technical success rate was 100%. The median tumor size was 1.5 cm (range: 0.6-6.5 cm), and the median number of metal clips used was 7 (range: 4-14). The median EFTR procedure time was 53.5 minutes (range: 22-136 minutes), and the median suturing time was 14 minutes (range: 3-45 minutes). The median length of postoperative hospital stay was 4.5 days (range: 3-7 days). Postoperative complications including bleeding occured in 1 case, fever in 3 cases, and abdominal pain in 2 cases. During the median 29 months (range: 7-79 months) of follow up, no residual lesions, tumor recurrence, or metastasis were detected, and no formation of digestive tract fistulas or sinuses was observed.

Conclusion: The long transparent cap suturing technique is a simple, safe, and effective method for wound closure after EFTR. It is beneficial for clinical application and widespread use.

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Disclosure: Nothing to disclose.

PP0242

HIGH RESOLUTION ESOPHAGEAL MANOMETRY IN EVALUATION OF EFFICACY OF PERORAL ENDOSCOPIC MYOTOMY

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Introduction: Achalasia is a primary motility disorder, which presented by dysphagia. The most effective treatment of the disease is surgical (peroral myotomy or laparoscopic myotomy). However, there are few studies which assess the changes in esophageal motility after surgery.

Aims & Methods: The aim of our study was evaluating the changes in esophageal manometry in patients with achalasia after peroral endoscopic myotomy (POEM).

Subjects and methods: 43 patients with achalasia were consequently observed before and 1 year after POEM. Females predominated (28 subjects). All patients examined endoscopically, then high resolution manometry (HRM) was performed by Solar™ GI (MMS/Laborie, Netherland). Achalasia was defined as described in Chicago Classification 4.0. Eight patients had I type of achalasia, 24 – II type and 11 – III type.

Results: The median Eckardt score was 7 on the start and 2 – after POEM in 35 patients (group A). However, 8 patients were dissatisfied with the surgery (4 pts with II type achalasia and 4 – III type). They continued to suffer from dysphagia and/or noncardiac chest pain (group B).

According to HRM the mean resting pressure of LES in group A reduced 2 folds from 39,1±2,7 mmHg to 18,9±1,4 mmHg. Also we revealed the dramatic decrease in integrated residual pressure (IRP) from 22,8±0,7 mmHg to 9,8±0,6 mmHg and DCI: from 1865,5±286,9 to 128,2±53,8. In a group B the mean resting pressure of LES reduced from 48,8±2,4 mmHg to 33,6±2,4 mmHg. The integrated residual pressure (IRP) decreased from 22,8±0,7 mmHg to 15,01±0,6 mmHg and DCI: from 2999,6±417,6 to 1038,1±107,7.

Even though, we didn't observed the peristaltic waves in two groups before and after POEM as a normal manometric criteria: the consistent contraction of upper part of esophagus and then lower part of esophagus. Besides, HRM evaluated persisting spastic contraction in the middle and lower part of esophagus in patients with type III achalasia from group B. Two patients from group B with type II had good results in esophageal manometry.

Conclusion: The reduction of resting pressure, integrated resting pressure and distal contractile interval in several times can be accepted as criteria of POEM effectiveness. Besides, in patients with III type of achalasia we must assess the absence of spastic contraction as additional criteria of POEM satisfaction.

Disclosure: Nothing to disclose.

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PP0243

LUMEN-APPOSING METAL STENTS: A MINIMALLY INVASIVE TREATMENT OPTION FOR GASTROINTESTINAL OUTFLOW OBSTRUCTION

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Introduction: The introduction of lumen-apposing metal stents (LAMSs) has revolutionized the field of therapeutic endoscopic ultrasound (EUS). LAMSs were initially developed for drainage of pancreatic fluid collections, but in the last decade its use for off-label indications has increased significantly. One of the most common off-label indications is treatment of gastrointestinal outflow obstruction.

Aims & Methods: This study aims to evaluate the effectiveness and safety of LAMS in creating EUS-guided gastro-enterostomy (EUS-GE) and entero-enterostomy (EUS-EE) anastomoses for the treatment of gastrointestinal (GI) outflow obstruction.

We collected data on all LAMS procedures conducted in patients with gastrointestinal (GI) outflow obstructions, including malignant gastric outlet obstruction (mGOO), benign gastric outlet obstruction (bGOO), candy cane syndrome, and afferent loop syndrome, between October 2019 and November 2023 at the University Hospital of Ghent. We used our prospective LAMS database and retrieved missing data from retrospective analysis of the medical patient records. The main outcome data of interest were technical success, clinical success, and adverse events. Technical success was defined as a successful placement of LAMS in the desired position. Clinical success was defined as either successful re-initiation of oral food intake in patients with mGOO and bGOO, or achievement of the intended therapeutic goal for the remaining conditions. Adverse events were classified according to the AGREE criteria.

Results: A total of 72 LAMS procedures were performed for GI outflow obstructions. The indications included 50 (69.4%)cases of mGOO, 19 (26.4%) cases of bGOO, 2 (2.8%) cases of afferent loop syndrome, and 1 (1.4%) case of candy cane syndrome. The overall technical and clinical success rates were very high (97.2% and 94.4%, respectively). Adverse events were observed in 9/72 (12.5%) cases, including 4 (5.6%) minor events (AGREE < 3) and 5 (6.9%) major events (AGREE ≥ 3). Major events included persisting fistula (N=1), and bleeding (N=4). All could be successfully managed, and there were no procedure-related deaths. Loss of LAMS patency occurred in 3/72 (4.2 %) cases and could be endoscopically managed in all cases.

Conclusion: In conclusion, our study shows that EUS-GE and EUS-EE offer an effective and safe approach when performed by experts. With high technical (97.2%) and clinical (94.4%) success rates, LAMS proves beneficial for treating GI outflow obstruction. Adverse events (12.5%) were manageable, with no procedure-related mortality. The extended patency duration and low re-intervention rate highlight LAMS as a minimally invasive and durable therapeutic option. Further prospective studies are needed.

Disclosure: Nothing to disclose.

PP0244

ENDOSCOPIC TREATMENT FOR SHORT-SEPTUM ZENKER'S DIVERTICULUM: A COMPARISON OF TRADITIONAL FLEXIBLE ENDOSCOPY SEPTOTOMY AND PERORAL ENDOSCOPIC SEPTOTOMY

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Introduction: Short-septum Zenker's diverticulum (ZD) is considered difficult-to-treat because the limited anatomical space which restricts the operating area. Recently, peroral endoscopic septotomy (POES) emerged as a novel third-space technique for addressing this condition.

Aims & Methods: This study aims to compare the effectiveness and safety of POES with traditional flexible endoscopic septotomy (FES) for the treatment of symptomatic patients with short-septum ZD (<25 mm). In this single-center, prospective study we included all consecutive patients referred to Humanitas Research Hospital with symptomatic short-septum ZD who were scheduled to undergo either FES or POES. The primary outcome was the rate of technical success and adverse events (AEs). The secondary outcome was the rate of clinical success at 12 months and recurrence requiring a retreatment. A sensitivity analysis was conducted using propensity score matching. Hazard Ratio (HR) and its associated 95%CI were used for a time-dependent analysis of recurrence.

Results: A total of 94 patients (63.8% male; mean age 72.7 ± 10.7 years) underwent endoscopic treatment for a short-septum ZD (mean size 17.0 ± 4.3 mm) were included. Among them, 41.5% (n=39) were treated with FES and 58.5% (n=55) with POES (p=n.s.). Overall, the mean procedural duration was 25.3 ± 15.7 min, while the mean hospital stay was 0.3 days. Septotomies were performed under deep sedation in 91.5% of patients (n=86), while in 8.5% of them (n=8) the procedure was performed under general anesthesia. Technical success was achieved in 98.9% of patients (n=93), while AEs occurring in 5.3% of cases (n=5), with no statistical difference between groups (p=n.s.). Clinical success was observed in 91.5% of patients (n=86), while recurrence necessitating retreatment in 14.9% of cases (n=14) with no statistical difference between groups (p=n.s.).

Conclusion: Our findings suggest that POES is an effective and safe technique for the treatment of short-septum ZD, showing comparable outcomes to traditional FES.

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Disclosure: Nothing to disclose.

PP0245

BESPOKE, IN-HOUSE MADE ENDOSCOPIC VACUUM THERAPY DEVICES FOR THE TREATMENT OF UPPER GASTRO-INTESTINAL ANASTOMOTIC LEAKS: A SINGLE, TERTIARY CENTER EXPERIENCE

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Introduction: Anastomotic leaks (AL) are a challenging complication after upper gastro-intestinal (UGI) surgery, with significant mortality and morbidity. Endoscopic vacuum therapy (EVT) has recently emerged as the

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mainstream tool for the management of UGI AL, both with commercially available devices and with in-house devices. Our group has recently (since July 2022) started to use bespoke, in-house made devices for EVT.

Aims & Methods: The aim of this study is to describe our experience with EVT for UGI AL in oncological patients undergoing surgery. We retrospectively collected and analysed data from all our patients who underwent EVT for AL after UGI oncological surgery, between July 2022 and December 2023 (18 months period). EVT was offered with bespoke, in house, handmade negative pressure endoscopic devices connected to commercial, medical vacuum pumps. EVT was performed twice per week and initial response was assessed during the 5th EVT (after the completion of two weeks of endoscopic therapy) at latest; if AL had not improved endoscopically and radiologically by that point, no further EVT was offered and alternative therapies were considered.

Results: In total, 8 patients (5 male; mean age 61) underwent EVT over the above-mentioned period. Six of them underwent Ivor-Lewis esophagectomy with esophago-gastric anastomosis and 2 underwent total gastrectomy with esophago-jejunal anastomosis. Five of them had received neo-adjuvant chemotherapy prior to surgery. EVT was placed intracavitary initially if technically possible (5 patients) and then intraluminally as the leak site decreased in size, or directly intraluminally (3 patients) if the anastomotic leak was <5mm and could not accommodate our smallest in-house device.

EVT was successful in 7 out of 8 patients (success rate 87,5%). Only one patient had no response on day 15, and after discussion at our MDT he was offered a redo surgery (this was a patient with esophago-jejunostomy) with good final result. The mean time between AL diagnosis and first EVT was 9 days. A median of 6 endoscopic EVT procedures was needed (range 2- 15) for total AL closure. There were no technical issues related to the device itself, all patients tolerated EVT well and no adverse event directly related to EVT was recorded (the one patient who underwent reconstructive surgery experienced worsening of extraluminal cavity/collection but this was deemed secondary to the leak itself rather than EVT).

Conclusion: EVT is an effective therapy for UGI post-surgical anastomotic leaks, with a success rate of 87,5% in our experience. It is demanding in terms of resources (high number of repeat endoscopies needed), but it is well tolerated and safe.

Disclosure: Nothing to disclose.

PP0246

COMPARING TRANSNASAL PEG TUBE PLACEMENT WITH AND WITHOUT SEDATION IN PATIENTS WHERE TRANSORAL PEG TUBE PLACEMENT NOT POSSIBLE

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Introduction: In patients with indications for transnasal PEG tube placement, such as oral cavity, head and neck tumors where the conventional upper gastrointestinal endoscopic procedure to place a PEG tube through the oral cavity can be difficult due to complete or partial occlusion of mouth and/or pharynx.In such cases transnasal placement of a PEG tube is proven to be useful.

Aims & Methods: The aim of this study was to compare the efficacy, safety, procedure time, patient tolerance ,recovery time, immediate and late complication of transnasal PEG tube (T-PEG) placement with and without conscious sedation.A total of 150 patients had a successful PEG tube placement. Out of which 49 patients required transnasal PEG tube placement through ultra-thin video gastroscope with an external diameter of 5.9 mm. patients were selected randomly for sedation and without sedation. After all asceptic measures,transillumination18-Fr PEG tube was placed transnasally by the Gauderer-Ponsky pull technique. 25 of the patients underwent T-PEG without any sedation and the remaining 24 patients were placed under conscious sedation with 2 mg midazolam.

All patients were followed immediately and after 1 month of the procedure.

Results: T-PEG tube was placed successfully in all 49 patients.very surprisingly all patients for transnasal were male.all had history of tobacco chewing in the form of ghutka/mawa. mean age was 38 years. 24 patients had a transnasal-PEG tube placed under conscious sedation of 2 mg midazolam.2 patients in the sedated group had drop in oxygen saturation which was managed during procedure.one patient in sedated and one in unsedated group had minor nasal trauma during peg tube placement. procedure time, procedure tolerance in both groups were almost similar.while post procedure recovery,observation,monitoring and discharge time by the staff was shorter for unsadated group 10mins +-5 min as compared to the sadated group 60mins +-10mins.two patients in sedated group While only one patient in the unsedated group had stoma site infection which were treated according to c/s report. The overall tolerance and success rate was similar in both groups.

Conclusion: In patients with oral cavity obstructing mass or trismus where oral route of PEG tube placement cannot be done, a transnasal route can be considered for placement of PEG tube safely with no requirement of sedation.it decreases the patient hospital stay,staff nurse monitoring,risk of resuscitation tracheostomy.This method is especially useful if the procedure is done in remote settings where anesthesiologists are not available.

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Disclosure: Nothing to disclose.

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PP0247

ENDOSCOPIC SLEEVE GASTROPLASTY AS A THERAPEUTIC OPTION FOR CLASS III OBESITY

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Introduction: The spread of Endoscopic Sleeve Gastroplasty (ESG) as an effective bariatric endoscopic technique has been notable, particularly for treating patients with class I and II obesity with related comorbidities. Our study aimed to assess the efficacy of ESG in class III obese patients [Body Mass index (BMI) >40 kg/m^2] focusing on weight loss and comorbidities improvement.

Aims & Methods: This retrospective analysis on a prospective dataset, focused on subjects with a baseline BMI >40 kg/m^2 who underwent ESG from June 2017 to November 2023. Weight loss outcomes [BMI, Excess Weight Loss (%EWL), Total Body Weight Loss (%TBWL), BMI loss (BMIL) and Absolute Weight Loss (AWL)] were assessed at 3, 6, 12 and 24 months. Additionally, the impact of ESG on Type 2 Diabetes (T2D), Hyperinsulinemia (H-INS), Arterial Hypertension (AH), and Obstructive Sleep Apnea Syndrome (OSAS) was analyzed.

Results: A total of 129 patients were included (74 female, mean age 47 ± 12.7, mean BMI: 44.4 ± 4.1 kg/m^2). We observed a significant regression to grade II obesity starting within the first three months postoperatively, with a BMI reduction of 7.1 ± 4.8 at 24-months follow-up (Table 1).

At 12-month follow-up, our data showed improvement or complete remission of T2D, H-INS, AH and OSAS, in 50% (7/14), 36.3% (12/33), 41% (20/49) and 31% (9/29) of patients, respectively.

Follow-up	n	Weight (kg)	p	AWL (kg)	EWL (%)	TBWL (%)	BMI (kg/m2)	p	BMIL (kg/m2)
3m	99	106.1 (15.7)	**<0.01**	19.3 (8.0)	25.3 (11.1)	15.3 (5.0)	37.7 (3.8)	**<0.01**	6.8 (2.6)
6m	90	103.8 (17.1)	**<0.01**	21.6 (9.5)	39.9 (16.0)	17.2 (6.9)	36.6 (4.7)	**<0.01**	7.7 (3.3)
12m	83	105.4 (21.5)	**<0.01**	19.9 (13.1)	37.2 (23.2)	16.0 (10.0)	37.5 (5.9)	**<0.01**	7.2 (4.6)
24m	44	106.1 (17.4)	**<0.01**	19.8 (13.4)	34.7 (22.2)	15.4 (10.0)	37.7 (4.3)	**<0.01**	7.1 (4.8)

Weight loss outcomes of primary ESG in subjects with BMI >40 kg/m².
Data are reported as mean value (standard deviation).
A two sample T-Test was used to compare baseline weight and weight loss outcomes.

Conclusion: Based on our observations, while Endoscopic Sleeve Gastroplasty (ESG) demonstrated effectiveness in inducing weight loss and ameliorating comorbidities in class III obese patients, it did not fully resolve obesity, potentially due to the high baseline BMI. Consequently, surgical intervention should continue to be the primary treatment modality for class III obesity. However, when surgery is contraindicated or refused by the patient, ESG represents a less invasive option and could be considered as an effective first-stage or multi-step treatment.

Disclosure: Nothing to disclose.

PP0248

TWO HUNDRED CASES OF PERCUTANEOUS TRANSESOPHAGEAL GASTROTUBING (PTEG) FOR THE PATIENTS WITHOUT INDICATION OF PEG

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Introduction: Percutaneous transesophageal gastrotubing (PTEG) was established as an alternative route to access the gastrointestinal tract for the patients that Percutaneous Endoscopic Gastrostomy was contraindicated. More than two hundred cases without indication of PEG were rescued by this ideal method.

Aims & Methods: The aim of this study is to evaluate the clinical usefulness of PTEG. A rupture-free balloon (RFB) catheter is inserted into the upper esophagus. Percutaneous balloon puncture with a specialized needle is then performed from the left side of patient's neck under ultrasonographic control. A guide wire is inserted through the needle into the RFB, followed by a dilator and sheath. A placement tube is then inserted through the sheath, and the sheath is removed. Double Balloons equipped Over tube type RFB were used instead of primary RFB in fifty-seven cases that the puncture needle is punctured into the over tube trough the balloon. We perform PTEG in a total of 204 patients (124 men and 80 women, mean age 71.5 years) PTEG was performed for nutrition in 116 patients and for decompression in 88.Eighty-five patients had peritoneal carcinoma , 62 patients had prior gastrectomy , 30 patients had gastric herniation into the thoracic cavity.

Results: Satisfactory results were achieved in all 204 patients. Median follow-up was 65.0 days in patients who received decompression because of the obstruction due to mainly malignancies and 359.0 days in those who received nutrition. Nine patients who were started in the fluoroscopically assisted according to the original method needed endoscopic assistance to complete the procedure in first 30 cases.

Another 174 patients were performed in endoscopic assistance instead of fluoroscopy.

About the complications, there was 2 patient with severe bleeding requiring blood transfusion, Two patient had tracheal penetration, which was managed conservatively. Other complications were minor oozing bleeding in thirteen patients that did not require blood transfusion, subcutaneous emphysema in two patients, which were managed conservatively. Complication rate was 9.8%. No patient required surgical treatment or died after PTEG.

Conclusion: PTEG is feasible, safe, and useful for long-term nutrition and/or decompression for the patient who is contraindicated to PEG. PTEG may become the only procedure of the liberation from nasogastric tube for the patient who is contraindicated to PEG. PTEG is the ideal solution for the patients with PEG contraindicated.

Disclosure: Nothing to disclose.

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PP0249

ACUTE ESOPHAGEAL NECROSIS IS ASSOCIATED WITH CELIAC TRUNK STENOSIS: A SINGLE CENTER EXPERIENCE OF 15 CASES

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Introduction: Acute esophageal necrosis (AEN) is an uncommon but often fatal esophageal condition with a mortality rate of approximately 30%. AEN predominantly affects elderly, male patients with significant comorbidities and typically presents with signs of upper gastrointestinal hemorrhage. The pathophysiology of AEN is attributed to a combination of local hypoperfusion, loss of mucosal barrier function and acid reflux. However, AEN still remains poorly understood due to its low incidence. In example, it has not been investigated whether celiac trunk stenosis might contribute to AEN development.

Aims & Methods: In this case series, we aimed to review clinical, laboratory and endoscopic features of AEN patients with an additional focus on celiac trunk diameter. 15 patients diagnosed with AEN between 2014 and 2024 at a single center in Berlin, Germany, were included in this study. Clinical data were reviewed and retrospectively analyzed on a case-by-case basis. AEN diagnosis was made endoscopically applying well-described criteria including diffuse, circumferential livid or black discoloration of esophagus mucosa with sparing of the gastroesophageal junction. Celiac trunk was assessed using contrast-enhanced computed tomography imaging to measure the extent of narrowing in the celiac trunk artery. The stenosis was classified into atherosclerotic or median arcuate ligament compression and graded into < 30% (grade 0), 30 - 50% (grade 1), 50 - 80% (grade 2) and > 80% narrowing (grade 3). Laboratory tests were carried out for each patient. However, in some cases not all parameters were available in retrospect.

Results: 73% (11/15) of AEN patients were male with a median age of 63.4 years (23 – 85 years). In 47% (7/15) of cases, cardiovascular disease was present. 67% (10/15) of patients presented with gastrointestinal hemorrhage, other clinical presentations included dysphagia, abdominal pain, dyspnea, and shock. 100% (15/15) of patients showed elevated CRP (mean: 70.9 mg/dl, 13.8 - 224.5 mg/l), 92% (11/12) hypalbuminemia (mean: 22.4 g/dl, 6.0 - 36.3 g/dl), 90% (9/10) lactatemia (mean: 97.5 mg/dl, 17 - 259 mg/dl), 82% (9/11) hyperglycemia (mean blood glucose: 146 mg/dl, 73 - 200 mg/dl), 67% (10/15) leukocytosis (mean: 17.96/nl, 4.64 - 42.02/nl) and 67% (10/15) anemia (mean hemoglobin: 11.0 g/dl, 5.0 - 16.1 g/dl). Endoscopy revealed panesophageal necrosis in 71% (10/14) of cases and mid-distal disease in 21% (3/14). Duodenal involvement, mostly reflected by duodenal ulcers, was found in 79% (11/14) of cases. 87% (13/15) of AEN patients showed celiac trunk stenosis. Grade 2 or 3 stenosis was present in 47% (7/15) of cases. 85% (11/13) of celiac trunk stenoses were atherosclerotic, 15% (2/13) were caused by median arcuate ligament compression. Case fatality rate was 40% (6/15) in this series.

Conclusion: Our study confirms that AEN is a potentially life-threatening medical condition often observed in critically ill patients. For the first time, we show that celiac trunk stenosis is associated with this condition. Future research is needed to confirm this data and carve out a possible role for endovascular revascularization in the treatment of AEN patients.

Disclosure: Nothing to disclose.

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Poster
H.pylori

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Poster
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H.pylori
09:00-18:00 / Poster Exhibition

P0250
EFFECT OF HELICOBACTER PYLORI ERADICATION ON BODY WEIGHT AND LIPID PROFILE; A PROSPECTIVE COHORT STUDY AT A UNIVERSITY HOSPITAL IN SOUTH KOREA

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Introduction: The effect of Helicobacter pylori eradication (HPE) on body weight and lipid profile is not well known.

Aims & Methods: We prospectively enrolled 102 patients who were treated for H pylori eradication from Apr 2021 to Jan 2024 at a university hospital in Goyang, South Korea (Inje University Ilsan Paik Hospital). Patients who had active peptic ulcer were excluded. Patients' body weight and lipid profile were measured before enrollment and 3 months after taking medications for HPE. The difference between before-and-after measurements were compared using the paired t-test. To test the difference in lipid profile, patients who had newly started lipid lowering drugs before the three-month follow-up visit were excluded from the analysis.

Results: Among 102 patients who were enrolled to treat H. pylori, 87 patients (46 males, 41 females) were followed up three months after taking HPE prescriptions. The median age was 59 years (range, 19-82). The baseline body mass index was 25.0 kg/m2 (range, 16.2-37.5).

There were 39 patients (44.8%) who were using lipid lowering drugs at the baseline. Five patients who had started lipid lowering drugs before the follow-up visit were excluded when analyzing the lipid profile change. Hp eradication rate was 91.9% (80/87). The HPE regimen included triple therapy (n=46 [52.9%]), quadruple therapy (n=31 [35.6%]), and concomitant therapy (n=10 [11.5%]).

	Baseline, mean (95% CI)	3 month follow-up after H. pylori eradication, mean (95% CI)	P-value (before-and-after t-test)	Difference among HPE successful (n = 80), median (interquartile range)	Difference among HPE failed (n = 7), median (interquartile range)	P-value (according to H. pylori eradication)
Body weight (kg)	67.4 (64.7-70)	66.9 (64.3-69.5)	0.079	-0.15 (-1.55,1)	-1 (-2.8, 0)	0.386
Serum glucose (mg/dL)	111.7 (106.9-120.5)	109.4 (104.1-114.5)	0.068	-1 (-11, 7)	-14.5 (-18, 23)	0.465
Total cholesterol (mg/dL)	166.5 (157.5-175.6)	162.1 (152.7-171.5)	0.172	-5 (-17, 7)	14 (-13, 43)	0.770
LDL cholesterol (mg/dL)	93.4 (84.9-101.8)	84.2 (76.4-92.1)	0.009	-7 (-21, 6)	-1.5 (-70, 19)	0.264
HDL cholesterol (mg/dL)	52.7 (49.6-55.8)	54.5 (51.2-57.9)	0.074	1 (-3, 7)	3 (-1, 22)	0.113
Triglyceride (mg/dL)	120.0 (105.3-134.7)	122.4 (103.9-140.9)	0.702	-3 (-30, 31)	-12 (-44, -1)	0.507

The body weight decreased from mean 67.4 kg (95% confidence interval [CI], 64.7-70.0) to mean 66.9 kg (95% CI, 64.3-69.5; p = 0.079). Serum glucose decreased from mean 111.7 mg/dL (95% CI, 106.9-120.5) to mean 109.4 (95% CI, 104.1-114.5; p = 0.068).

Total cholesterol decreased from mean 166.5 mg/dL (95% CI, 157.5-175.6) to mean 162.1 (95% CI, 152.7-171.5; p = 0.172). Low-density lipoprotein (LDL) cholesterol decreased from mean 93.4 mg/dL (95% CI, 84.9-101.8) to mean 84.2 (95% CI, 76.4-92.1; p = 0.009). High-density lipoprotein (HDL) cholesterol increased from mean 52.7 mg/dL (95% CI, 49.6-55.8) to mean 54.5 (95% CI, 51.2-57.9; p = 0.074). Triglyceride changed slightly from 120.0 mg/dL (95% CI, 105.3-134.7) to 122.4 mg/dL (95% CI, 103.9-140.9; p = 0.702). The number of non-Hp eradicated subjects was too small (n = 7) to compare the differences in body weight and lipid profile according to actual H. pylori eradication.

Conclusion: In a short-term prospective study among Korean patients who were treated for H. pylori, there was a decrease in body weight and a favorable effect on serum lipid levels. The number of non-Hp eradicated subjects was too small to draw conclusions about the effect of successfully eradicated Hp on body weight and lipid profile. The long-term effect of HPE on body weight and lipid profile remains to be analyzed, and a randomized study is required to achieve non-biased results.

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Disclosure: Nothing to disclose.

PP0251
WITHDRAWN

PP0252
GUINEA PIG - A CONVENIENT MODEL TO STUDY THE PATHOGENESIS OF *HELICOBACTER PYLORI* INFECTIONS

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Introduction: *Helicobacter pylori* (*Hp*) is an etiological agent of chronic gastritis, gastric and duodenal ulcers and gastric cancers. The use of an appropriate animal model for experimental studies on the pathogenesis of *H. pylori* infections is necessary due to the chronic character of such infections and difficulties in identifying their early stage in humans.

Aims & Methods: The aim of this study was to develop a guinea pig model of *H. pylori* infection and identify its microbiological, histological, molecular and serological determinants. Guinea pigs (*Cavia porcellus*) were inoculated *per os* with *H. pylori* strain: CCUG 17874 positive for-vacuolating cytotoxin A (VacA) and cytotoxin associated gene A (CagA) protein, suspended in *Brucella* broth with fetal calf serum (FCS) and Skirrow supplement (on permission of Local Ethics Committee for Anima Experiments, Medical University of Lodz Poland, Decision 58/ŁB45/2016).

To determine *H. pylori* colonization, 7 and 28 days after inoculation, a gastric tissue was used for culturing *Hp* histopathological examination of gastric tissue sections, stained by Mayer,s haematoxylin and eosin, Giemsa staining or Warthin-Starry silver staining to assess inflammatory

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response,-or to visualize *Helicobacter*-like organisms (HLO). Furthermore an immunostaining with anti-Ki-67 antibodies was developed to assess epithelial cell proliferation. *H. pylori* infection was also confirmed by polymerase chain reaction (PCR) for detection in gastric tissue of *ureC* and *cagA* genes and by serological assessment of *Hp* antigens in feces and serum anti-*Hp* antibodies.

Results: This study showed the successful colonization of guinea pigs with *Hp* on the basis of microbiological, histological, molecular and immunological determinants In the light of obtained results *Cavia porcellus* can be a good model for studying *Hp* pathogenesis and the related immune response against this pathogen.

Conclusion: The ability to induce and confirm the *Hp* infection in guinea pig makes this animal model suitable for studying the pathogenesis of infection and the course of related immune response, which *in vivo* is modulated by the components of these bacteria.

Financing: The research was financially supported by the National Science Center by the grant SONATA 18 "Assessment of the ability of *Mycobacterium bovis* BCG-onco bacilli to control the development of *Helicobacter pylori* infection", UMO-2022/47/D/NZ7/01097.

Disclosure: Nothing to disclose.

PP0253

BARRIERS TO EFFECTIVE GERD SYMPTOM CONTROL POST-H. PYLORI TREATMENT IN SOCIO-ECONOMICALLY DIVERSE POPULATIONS

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Introduction: The interplay of race and gender significantly influences the manifestation of Gastroesophageal Reflux Disease (GERD) symptoms following Helicobacter pylori (H. pylori) treatment.

This multifaceted relationship, deeply rooted in biological, cultural, and socioeconomic factors, presents a complex medical challenge. Understanding these disparities is crucial for tailoring effective treatment strategies, ensuring equitable healthcare outcomes, and advancing personalized medicine in managing GERD symptoms in diverse populations post-H. pylori eradication

Aims & Methods: Our research, spanning 2009 to 2023, retrospectively analyzed patients treated for H. Pylori, ensuring a minimum one-year follow-up. We collected data encompassing comorbidities, insurance status, and socioeconomic factors, particularly targeting the emergence of GERD symptoms, Esophagitis, and duodenal ulcers, identified using CPT and ICD codes. Initially, we employed a Chi-square test of independence to evaluate the association between race and GERD symptoms.

We conducted a post-hoc analysis with adjusted standardized residuals to delve deeper, pinpointing specific races with notably higher or lower GERD symptom frequencies than expected. We performed pairwise race comparisons via Chi-square tests, applying a Bonferroni correction for multiple comparisons.

Results: Between 2009 and 2023, 4,205 patients received H. Pylori treatment, with 3,329 qualifying for our study after excluding those lost to follow-up. The average age was 54.5 years, with 64% female (2,134) and 48% Caucasian (1,664).

The Chi-square test revealed a significant association between Race and GERD symptoms (Chi sq stat:8.69 p-value <0.01). In-depth post-hoc pairwise comparisons showed that the Hispanic race had significantly higher rates of GERD symptoms when compared to other races Other races (p <0.01), White (p <0.01), Black (p <0.01), and Asian races (<0.01). Adjusted residual analysis further confirmed that the Hispanic race (5.54, p<0.023) showed a higher-than-expected number of GERD symptom cases.

Among the Hispanic races, Patients with low socioeconomic status showed increased GERD symptoms (Chi. Sq: 7.65, P<0.01). Gender does not significantly impact GERD symptoms (Chi. sq: 0.831, P = 0.512)

Conclusion: Our study, analyzing data from 2009 to 2023 on patients treated for H. Pylori, underscores the significant influence of race and gender on GERD symptoms post-treatment.

Notably, the Hispanic population exhibited a higher incidence of GERD symptoms compared to other races, with socioeconomic status further intensifying this trend.

These findings highlight the need for more personalized, culturally sensitive approaches in healthcare, emphasizing the importance of considering race, gender, and socioeconomic factors in the treatment and management of GERD following H. Pylori eradication.

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Disclosure: Nothing to disclose.

PP0254

CHANGES OVER TIME IN THE PREVALENCE OF *HELICOBACTER PYLORI* INFECTION IN THE NORTH-WESTERN REGION OF THE RUSSIAN FEDERATION FROM 2015 TO 2023

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Introduction: Russia is among the countries with a high prevalence of *Helicobacter pylori (H. pylori*) infection. The North-Western Federal District, comprising one-tenth of Russia's population, is predominantly urban. As of 2023, its population is over 13 million people. Current information on *H. pylori* infection prevalence is necessary for the development of clinical guidelines for the prevention of *H. pylori*-associated disorders.

Aims & Methods: To evaluate the dynamics of *H. pylori* infection prevalence among patients with dyspepsia between 2015 and 2023 based on ^{13}C-urea breath test (^{13}C-UBT).

The study was conducted in the North-Western Federal District. Patients with dyspepsia (n=50,884) who underwent ^{13}C-UBT between 2015 and 2023 were included in this descriptive study. If Delta Over Baseline (DOB, $\delta^{13}CO2$) was less than 2.5‰, the result of the test was considered negative, while $\delta^{13}CO2$ value ≥2.5‰ indicated the presence of *H. pylori* infection.

The criteria for the infection grade (IG) were defined as follows: IG I (2.5–3.4‰), IG II (3.5–6.4‰), IG III (6.5–9.4‰), and IG IV (>9.5‰). The database for subsequent statistical analysis included such parameters as the patients' age, gender, history of prior eradication therapy, and the result of the ^{13}C-UBT test.

Results: Among all included individuals (mean age 43.3 years (31.0; 55.0); 57.5% of whom were female), there were 83.6% (n=42,543) of primary "naive" patients (who had had no history of prior eradication therapy) (95% confidence interval (CI) 83.3-83.9). The proportion of treated patients was 16.4% (n=8,341; 95% CI 16.1-16.7).

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The prevalence of *H. pylori* infection among the primary patients was 51.4% in 2015 (874/1701 subjects; 95% CI 49.0–53.8) and gradually decreased to 32.2% (1904/5916 subjects; 95% CI 31.0–33.4) in 2022. In 2023 *H. pylori*-positive was 32.7% (1450/4433 subjects; 95% CI 31.3–34.1). Eradication failure was 39.1% in 2015 (148/379 subjects; 95% CI 34.5-44.3). Eradication rate in 2016 was 80.2% (684/853 subjects; 95% CI 77.4-82.7) and increased to 86.8% (702/809 subjects; 95% CI 84.3-88.9) in 2023.
Naive and unsuccessful treated patients demonstrated a very high IG - 73.0 and 66.5%, respectively. *H. pylori* was detected more often in young adults (18–44 years old (y.o.)). The infection was less commonly detected among children (1.6% of primary and 1.3% of treatment-failure patients).

Total, n	Naive, 15358	Treatment failure, 1284
Women, n (%) Men, n (%)	8804 (57,3) 6554 (42,7)	728 (56,7) 556 (43,3)
2–17 y.o., n (%)	247 (1,6)	17 (1,3)
18-44 y.o., n (%)	7918 (51,5)	613 (47,7)
45-59 y.o., n (%)	4401 (28,7)	403 (31,4)
Over 60 y.o., n (%)	2792 (18,2)	251 (19,6)
IG I, n (%)	736 (4,8)	119 (9,3)
IG II, n (%)	1605 (10,5)	179 (13,9)
IG III, n (%)	1802 (11,7)	132 (10,3)
IG IV, n (%)	11215 (73,0)	854 (66,5)

Table 1. Characteristics of the H. pylori-positive primary and treatment failure patients.

Conclusion: Overall, our study highlighted a significant decrease of *H. pylori*-infection from 2015 to 2023. However, the eradication rates in the North-Western district of Russia remain low and do not reach the threshold value of 95%.

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Disclosure: Nothing to disclose.

PP0255

PREVALENCE OF HELICOBACTER PYLORI INFECTION AND ANTIMICROBIAL RESISTANCE IN PORTUGAL: A MULTICENTRIC POPULATION STUDY

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Introduction: Portugal is an intermediate-high risk country for gastric cancer and *Helicobacter pylori* (Hp) infection is an important risk factor for such condition. All of infected patients should be treated; however, the increasing frequency of antibiotic resistance reduces its eradication rate.

Aims & Methods: Aims: To analyze the prevalence of Hp infection in Portugal and its antimicrobial resistance profile. To evaluate the prevalence of premalignant gastric conditions.
Methods: A multicentric cross-sectional study involving 9 endoscopy units from Portugal, distributed across the country (north, central and south regions). We included patients aged ≥40 who underwent upper endoscopy in these units between February and March 2024. Hp detection was performed using immunohistochemistry (IHC). In positive IHC samples, clarithromycin, fluoroquinolones and metronidazole resistances were assessed in biopsy samples using real-time polymerase chain reaction.
Results: Four hundred twenty-one patients were included (58% women, mean age 60), 172 of which were Hp-positive (41%). Excluding patients with prior eradication therapy, the prevalence of Hp infection was 47.4%. Infection was more prevalent in the north region of the country (north 46.8%; central 36%; south 36.5%). The antimicrobial resistance rates were: clarithromycin 26%; fluoroquinolones 16%; metronidazole 2%; clarithromycin/fluoroquinolones 4%. Intestinal metaplasia (IM) was identified in 138 patients (32.8%), 24 of which (6%) had extensive IM (both antrum and corpus) and 31 (7%) had an endoscopic grading of gastric IM (EGGIM) ≥5.
Conclusion: Hp infection in Portugal is frequent, showing a north-south gradient, albeit smaller than expected. More than half of the strains showed no antibiotic resistance and only 4% exhibited dual resistance to clarithromycin/fluoroquinolones. We demonstrate the feasibility of studying resistance profile in endoscopic biopsies, suggesting that such tests may lead to reduce antibiotic consumption and associated side effects, as well as lower therapeutic costs. The presence of IM in approximately one-third of patients, in a predominantly young population, reinforces the relevance of endoscopic screening in Portugal.

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Disclosure: Nothing to disclose.

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PP0256

PREVALENCE OF *HELICOBACTER PYLORI* INFECTION AND VIRULENCE GENES IN PATIENTS WITH UPPER GASTROINTESTINAL DISEASES FROM SOUTHWEST OF IRAN

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Introduction: Various virulence factors of *H. pylori* have been identified that play an important role in pathogenesis. The aim of this study was to investigate the prevalence of *H. pylori* infection by rapid urea test and histopathology examination and determining the prevalence of the cytotoxin associated gene A (*cagA*), vaculating cytotoxin A (*vacA*) and restriction endonuclease-replacing gene (*hrgA*) in *H. pylori* strains isolated from patients with upper gastrointestinal diseases from Khuzestan.

Aims & Methods: Gastric biopsy specimens were obtained from 426 patients with gastrointestinal disorders undergoing endoscopy in Imam Khomeini Hospital, Ahvaz, the capital of Khuzestan province. *H. pylori* infection was investigated in patients by rapid urea test and histopathology examination and positive specimens were reconfirmed by PCR of the *ureB* gene. The presence of *cagA*, *vacA s/m* and *hrgA* genotypes were determined by PCR from culture or gastric biopsies.

Results: *H. pylori* was detected in 33.8% (144/426) of patients by rapid urea test and in 44.13% (188/426) by histopathology. 60.2% of infected patients were female and 39.8% were male and the mean age of patients was 47.1 years (range 14 to 89 years old). Epigastric pain was the most commonly reported symptom (62.7%), followed by anorexia (24.9%), nausea (21.4%), epigastric burning (20.4%), gastrointestinal bleeding (14.9%), fullness (12.4%), vomiting (12.4%), weight loss (12.9%) and bloating (11.9%). The main endoscopic findings were non-ulcer dyspepsia 56.7%, erosive gastritis 20.9%, gastric ulcer 13.4%, duodenal ulcer 7.5% and gastric cancer 1%. The *cagA*, *vacA*s1, *vacA*s2, *vacA*m1, *vacA*m2 and *hrgA* genotypes were detected in 93.3%, 71.6%, 23.1%, 25.4%, 70.1% and 67.2% of isolates, respectively. The allelic variant *vacA*s1m2 was more prevalent (45.5%) than *vacA*s1m1 (23.9%) and *vacA*s2m2 (23.1%).

There was a statistically significant association between the presence of *vacA*s1m2 genotype and gastric ulcer (P = 0.004), while the *vacA*s2m2 significantly was observed in strains isolated from patients with non-ulcer dyspepsia (P = 0.017). Also the presence of *hrgA* gene was associated with duodenal ulcer (P = 0.043).

Conclusion: the histopathology examination is more specific than the rapid urea test especially in gastric ulcer bleeding cases and the use of both tests can help to more accurate diagnosis of *H. pylori* infection. Based on the results *vacA*s1, *vacA*s2 and *hrgA* genotypes may be the predictors of gastric ulcer, non-ulcer dyspepsia and duodenal ulcer, respectively.

Disclosure: Nothing to disclose.

PP0257

PREVALENCE OF PRIMARY CLARITHROMYCIN RESISTANCE IN PATIENTS INFECTED WITH *HELICOBACTER PYLORI* IN EUROPE IN THE LAST FOUR DECADES

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Introduction: Global rise in primary *Helicobacter pylori* (*H. pylori*) antibiotic resistance has led to a decrease in the treatment efficacy. Understanding the rates of clarithromycin resistance in *H. pylori* management is essential for determining the optimal eradication therapy and achieve successful clinical outcomes.

Aims & Methods: This review aimed to evaluate the prevalence and geographical distribution of *H. pylori* primary clarithromycin resistance in Europe. Bibliographical searches were performed in PubMed up to March 2024. Selection: Included studies assessing rates of *H. pylori* primary resistance to clarithromycin with no time restriction. Publications in any language were considered. Studies that focused on children only or non-European countries were excluded.

Country (number of studies)	Clarithromycin resistance prevalence % (n/N) 1990-1999	Clarithromycin resistance prevalence % (n/N) 2000-2009	Clarithromycin resistance prevalence % (n/N) 2010-2019	Clarithromycin resistance prevalence % (n/N) 2020-2024
Armenia (1)	NA	NA	2.2 (2/91)	NA
Austria (5)	3.3 (11/331)	NA	19.6 (133/680)	NA
Belgium (9)	9.9 (125/1260)	6.1 (825/13587)	3.6 (131/3661)	18.9 (51/269)
Bosnia and Herzegovina (1)	NA	NA	NA	28.26 (13/46)
Bulgaria (10)	9.3 (26/278)	18.1 (338/1869)	23.1 (131/569)	30 (15/50)
Croatia (7)	NA	13.2 (146/1105)	21.2 (73/345)	NA
Denmark (1)	NA	7.1 (63/894)	NA	NA
Estonia (2)	2.8 (3/106)	NA	NA	NA
Finland (2)	NA	13.1 (173/1329)	NA	NA
France (8)	25.8 (81/314)	21.8 (172/786)	17.7 (370/2093)	NA
Germany (13)	2.4 (52/2143)	3.2 (81/2567)	11.7 (315/2682)	NA
Greece (2)	NA	30 (15/50)	30.7 (59/192)	NA
Iceland (1)	NA	NA	6 (6/100)	NA
Ireland (5)	4.5 (25/557)	14.2 (21/148)	50.5 (53/105)	NA
Israel (2)	NA	7.9 (11/138)	44.1 (198/449)	NA
Italy (33)	9.5 (65/686)	16.5 (271/1645)	23.2 (1499/6456)	27.1 (79/292)
Lithuania (1)	1.1 (1/89)	4.4 (4/90)	NA	NA
Netherlands (5)	0.7 (7/893)	7.8 (61/778)	18.2 (61/336)	NA
Norway (2)	NA	5.8 (6/102)	9.5 (4/42)	NA
Poland (9)	9.1 (6/66)	24.88(158/635)	39.1 (81/207)	NA
Portugal (3)	13.9 (47/336)	NA	40 (58/145)	NA
Romania (1)	NA	NA	NA	20 (18/90)
Russia (4)	NA	5.3 (7/133)	9.5 (116/1220)	NA
Scotland (1)	1.8 (3/162)	NA	NA	NA
Serbia (1)	NA	NA	NA	24.03 (68/283)
Slovenia (2)	NA	NA	10.7 (44/412)	NA
Spain (26)	8.9 (143/1599)	15.8 (730/4623)	17.1 (2835/30697)	15.1 (193/1283)
Sweden (3)	2.7 (3/109)	1.3 (7/538)	NA	NA
Switzerland (4)	11.6 (13/112)	17.6 (29/165)	67.1 (145/216)	NA
Turkey (15)	NA	28.9 (128/442)	28.4 (173/609)	15.5 (64/413)
United Kingdom (9)	5.5 (23/415)	10.7 (198/1845)	35.8 (97/271)	NA

Table 1. Clarithromycin primary resistance prevalence in Europe in the last four decades (1990-2024). Number of patients tested for Clarithromycin: n (positive resistance results)/N (total with H. pylori). NA: not available.

Results: Overall, 188 studies were included, evaluating 81,458 patients over more than 30 years (1990-2024). In this time span, the mean prevalence of primary *H. pylori* clarithromycin resistance in Europe was 17%.

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This rate increased from 7.3% in the period 1990-1999, to 13% in 2000-2009, reaching 26% in 2010-2019; and then reported a decreased rate to 22% in 2020-2024. In almost all European countries, the clarithromycin resistance rate increased over time, reporting the highest rates in Switzerland (67%), Ireland (50%), Israel (44%), Portugal (40%) and Poland (39%) (2010-2019). The prevalence remained stable in recent years in France (decreasing from 21% to 17%), in Spain (around 15%), and in Turkey (decreasing from 28% to 15%). Southern European countries reported higher rates (>15-20%) than in northern countries (<10-15%).

Conclusion: In Europe, overall *H. pylori* primary clarithromycin resistance was above 15% in the last 30 years. Variability exists between countries, with Southern European and Mediterranean countries showing higher levels of resistance.

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Disclosure: Nothing to disclose.

PP0258

WITHDRAWN

PP0259

ASSOCIATION OF AUTOIMMUNE ATROPHIC GASTRITIS WITH HELICOBACTER PYLORI INFECTION

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Introduction: In patients with Helicobacter pylori (HP) infection, superficial gastritis proceeds to atrophic gastritis in about half of the patients. Although, it usualy involves the antrum, it can also proceed to the corpus with sometimes neuroendocrine cell hyperlasia and chromogranin A antibody positivity, like in classic autoimmune gastritis.

Aims & Methods: The aim of our study was to determine the frequence of HP infection in autoimmune atrophic gastritis and its characteristics

A retrospective study was conducted over patients with autoimmune atrophic gastritis diagnosed in the gastroenterology department in Mohamed taher maamouri hospital between January 2019 and December 2021. Clinical, biological, endoscopic and pathology features were abstracted from medical records.

Results: A total of 31 patients were included in this study. The sex ratio M/F was 0.8 and the mean age was 56 years [28- 82 years]. The average haemoglobin level was 9,9 g/dl and the average mean corpuscular volume was 95,6 fL. Serum markers were available in 13 patients. Intrinsic factor antibodies were positive in five patient and anti-parietal cell antibodies in 11 patients. Pathology findings of fundal biopsies showed mild atrophy in 16,1% patients, moderate in 48,4% and severe in 35,5% with presence of intestinal metaplasia in 90,3%. The antrum was atrophic in one patient and antral intestinal metaplasia was detected in 11 patients (35,5%). HP infection was diagnosed in 23 patients (74,2%). The presence of HP infection was correlated with the severity of anaemia (p= 0.048). No correlation was found between HP infection and the degree of atrophy or the presence of intestinal metaplasia (p= 0.35 et 0.38 respectively).

Conclusion: Our study showed the presence of HP in 74.2% of patients with autoimmune atrophic gastritis , which may support the inplication of HP infection in the development of autoimmune gastritis. The presence of HP infection was correlated with the severity of anaemia, hence the importance of HP eradication treatment.

Disclosure: Nothing to disclose.

PP0260

WITHDRAWN

PP0261

PREVALENCE AND PREDICTIVE FACTORS OF HELICOBACTER PYLORI INFECTION IN PATIENTS WITH INFLAMMATORY BOWEL DISEASES: EXPERIENCE FROM A TERTIARY CARE GASTROENTEROLOGY CENTER

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Introduction: Inflammatory Bowel Diseases (IBD) are chronic conditions whose etiology remains elusive. Several hypotheses, including a potential association with Helicobacter pylori (HP), have been proposed, as evidenced by declining prevalence due to eradication protocols. However, literature data exhibit considerable variability. Our study aims to determine the prevalence of HP infection in a population of IBD patients based on gastric biopsy results, alongside predictive factors for HP infection in this population, integrating epidemiological, endoscopic, and histological analyses.

Aims & Methods: This retrospective descriptive study spanned four and a half years from January 2019 to July 2023, encompassing all IBD patients who underwent gastroscopy with biopsy. Patients with fewer than four gastric biopsies were excluded. Data were collected from medical records, and statistical analysis was performed using SPSS software version 21.0.

Results: 150 patients were included, with a median age of 39.9 years [range: 18-71 years; ±13.49], including 80 females (53.3%) and a sex - ratio of 0.8. All patients reported tea consumption (100%), while 25 patients (16.66%) were smokers, and 18 (12%) reported alcohol use. NSAID use was noted in 17 patients (11.3%), and proton pump inhibitor (PPI) use in 5 patients (3.3%). Additionally, 15 patients reported herbal therapy. All patients underwent gastroscopy with at least 3 antral and 2 fundic biopsies. 93.3% of patients underwent systematic gastroscopy. Among participants, 66 tested positive for HP (HP+) , including 61 asymptomatic individuals. Symptomatically, 4 patients experienced epigastric pain, and one case of gastroesophageal reflux disease (GERD) was observed. HP prevalence in our sample was 44%, with a higher prevalence among females (56% vs. 44% males). HP infection was more common among individuals

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aged 21-45 and over 60, peaking between 30 and 40. Endoscopic findings in HP+ patients included esophagitis in 6 patients (9%), erythematous and erosive gastritis in 4 patients (6%), hiatal hernia in 3 patients (4.5%), and micronodular antritis in 1 patient (1.5%). Duodenal and bulbar aphthoid ulcers were observed in 5 patients (7.5%). Notably, gastroscopy was normal in 112 patients. Epidemiological data analysis revealed that chronic NSAID use (P=0.01), herbal therapy (P=0.05), tea consumption (P=0.03), history of digestive cancer (P=0.01), young age (<37 years; P=0.05), and patients with ulcerative colitis (P=0.001) were significant risk factors for HP infection.

Conclusion: In our study, the prevalence of Helicobacter pylori infection was 44%. Ulcerative colitis was a significant risk factor for HP infection in our sample, with a highly significant P value of 0.0001. Therefore, should all IBD patients be systematically screened for HP infection?

Disclosure: No there is no Conflict of Interest into the text field below.

PP0262

WITHDRAWN

PP0263

DIAGNOSIS OF HELICOBACTER PYLORI IN PATIENTS WITH OBESITY: COMPARISON BETWEEN ^{13}C-UBT AND PCR

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Introduction: The prevalence of gastric infection by *Helicobacter pylori* (Hp) in individuals with obesity ranges from 8.7% (1) to 85.5% (2). In candidates for bariatric surgery, Hp infection is associated with an increased risk of perioperative complications, including marginal ulcers, ulcer perforations, anastomotic dehiscence, fistulas, and bleeding. It also exacerbates symptoms related to pre-existing gastrointestinal diseases (3,4). Preoperative eradication of Hp reduces complication rates. However, there is a lack of standardization in the diagnostic methods and eradication control of this bacterium in the population with obesity (5).

Aims & Methods: To compare the performance of the carbon-13 urea breath test (^{13}C-UBT) with molecular test based on PCR in diagnosing Hp infection in candidates for bariatric surgery. To assess the prevalence of Hp infection in these individuals and analyze the correlation with gastrointestinal symptoms and endoscopic findings.

Results: Ninety-two patients were evaluated, 83.7% female, with an average age of 41 ± 10.8 years and a mean body mass index of 41.8 ± 5.2 kg/m2. The majority of participants (71.7%) had at least one comorbidity (arterial hypertension in 42.4%). The prevalence of Hp infection was 55.4% by PCR. Among the participants, 33.4% were symptomatic, predominating gastric symptoms. Abnormal endoscopic findings were observed in 56.5%, with no significant differences between the Hp negative and positive groups (51.2% *vs*. 60.7%, p=0.47). The most common endoscopic finding was erosive gastritis (45.6%). Table 1.

The sensitivity of the ^{13}C-UBT was 86.2% (CI95%: 73.1 to 93.8), with a specificity of 100% (CI95%: 89.3 to 100) and diagnostic accuracy of 92.3% (CI95%: 84.4 to 96.6). There was good concordance between the ^{13}C-UBT and PCR (kappa value 0.84, CI95%: 0.74 to 0.95), supporting the use of ^{13}C-UBT as a noninvasive test for the diagnosis of Hp in patients with obesity. Considering that Hp eradication therapy in obese individuals can contribute to reducing postoperative complications, this method becomes an alternative to endoscopy after eradication therapy, minimizing risks and costs in the preoperative period.

Parameter	Total (n=92)	Hp negative (n=41)	Hp positive (n=51)	p value
Abnormalities in endoscopy	52 (56.5)	21 (51.2)	31 (60.7)	0.47
Erosive gastritis	42 (45.6)	18 (43.9)	24 (47)	0.92
Esophagitis	17 (18.4)	6 (14.6)	11 (21.5)	0.56
Nodular gastritis	5 (5.4)	0 (0)	5 (9.8)	0.06*

*Table 1. Comparison between endoscopic findings and Hp diagnosis, n (%). *calculation using Fisher's exact test, as the value of n for Hp-negative group was zero.*

Conclusion: The prevalence of Hp infection is 55.4% in obese individuals, with no correlation with symptoms and endoscopic findings. The ^{13}C-UBT showed high sensitivity, specificity, and good correlation with PCR in detecting Hp in candidates for bariatric surgery. This non-invasive test can be used in the eradication control of Hp in the preoperative of bariatric surgery.

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PP0264

A RETROSPECTIVE, OBSERVATIONAL, REAL-WORLD STUDY EVALUATING THE DIAGNOSTIC PATTERN IN CHINESE PATIENTS WITH HELICOBACTER PYLORI

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Introduction: *Helicobacter pylori* (*H. pylori*) infection is a common etiological cause of gastritis, peptic ulcer, atrophic gastritis, gastric adenocarcinoma, and mucosa-associated lymphoid tissue lymphoma. Accurate diagnosis and sufficient efficacy evaluation duration are both critical for effective treatment to avoid precancerous conditions.

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Aims & Methods: This retrospective, observational, multicenter, real-world study (NCT05073367) was performed across 3 hospitals using a preexisting electronic medical record (EMR) database between January 1, 2019, and March 31, 2021, to evaluate the diagnosis and treatment pattern of *H. pylori* infection in China. The study included adult patients (aged ≥18 years) with a record of diagnostic tests or positive results of *H. pylori* infection.

Here we analyzed the current pattern of diagnosis testing, eradication efficacy testing after eradication therapy, and the duration (≤4 or >4 weeks) from the termination of eradication therapy to efficacy evaluation testing.

Results: A total of 22,971 patients with *H. pylori* infection data was extracted from EMR databases. Among them, 15,712 patients had 15713 initial diagnostic testing records before the eradication therapy (One participant had both RUT testing and UBT at the index date), and 1,626 patients received efficacy evaluation after the eradication therapy. Of 15,713 initial diagnostic testing records, non-invasive testing was the main diagnostic test used by 99.99% (n=15,712) of records, and invasive testing was used by only 0.01%（n=1.

Among non-invasive testing, ^{13}C- Urea breath testing (UBT) was used by 76.8% and ^{14}C-UBT was used by 23.19%. An increasing trend was observed in the use of ^{13}C-UBT diagnostic test with 47.54%, 73.20%, and 99.98% in 2019, 2020, and 2021, respectively. ^{14}C-UBT was 52.46% in 2019，26.80% in 2020，0% in 2021.Among 1,426 patients with eradication efficacy evaluation (duration ≥ 4 weeks), non-invasive testing- UBT was the only evaluation method used, with ^{13}C-UBT in 72.3% and ^{14}C-UBT in 27.7%. In 2019, ^{13}C-UBT was used by 53.69%, in 2020 it was 69.88%, and in 2021, ^{13}C-UBT was the only evaluation method to assess the efficacy (Table 1).

The mean (Standard Deviation) duration from the termination of eradication therapy until the eradication efficacy evaluation testing was 7.0 weeks (4.37).

Of 1626 patients with efficacy evaluation, 87.7% had > 4 weeks duration and 12.3% had ≤ 4 weeks. In 2019, 83.4% had a duration of > 4 weeks, 91.21% in 2020, and in 2021, 90.68% had a duration of > 4 weeks to receive an efficacy evaluation.

Testing condition	Total count (N)	Invasive testing, RUT, n (%)	Overall Noninvasive testing, UBT, n (%)	Noninvasive testing, ^{13}C-UBT, n (%)	Noninvasive testing, ^{14}C-UBT, n (%)
Diagnostic testing used before eradication therapy	15,713*	1 (0.01)	15,712 (99.99)	12,068 (76.80)	3644 (23.19)
Year 2019	4516	0 (0)	4516 (100)	2147 (47.54)	2369 (52.46)
Year 2020	4757	0 (0)	4757 (100)	3482 (73.20)	1275 (26.80)
Year 2021	6440	1 (0.02)	6439 (99.98)	6439 (99.98)	0 (0)
efficacy evaluation testing used after eradication therapy ≥ 4 weeks	1426	0 (0)	1426 (100)	1031 (72.30)	395 (27.70)
Year 2019	583	-	583 (100)	313 (53.69)	270 (46.31)
Year 2020	415	-	415 (100)	290 (69.88)	125 (30.12)
Year 2021	428	-	428 (100)	428 (100)	0 (00.00)

Table 1. Diagnostic Testing Used Before and After Eradication Therapy
Abbreviations: RUT, Rapid Urease Test; UBT, Urea Breath Testing
**A patient may have ≥1 diagnostic test. One participant had both RUT testing and UBT at the index date.*

Conclusion: In China, non-invasive testing was dominant for the initial diagnosis of *H. pylori* infection, and ^{13}C UBT was commonly used compared with ^{14}C-UBT. The eradication efficacy evaluation testing was the same. Even over a tenth had insufficient evaluation duration, a higher proportion of evaluation duration >4 weeks was discovered with an increasing tendency.

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PP0265

EVALUATE THE DIAGNOSTIC ACCURACY OF PYLORI DUOTECT (QUALITATIVE DETECTION OF 2 ANTIBODIES SPECIFIC TO FLID AND CAGA) IN PATIENTS WITH SYMPTOMS OF UPPER GASTROINTESTINAL DISORDERS

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Introduction: *H.pylori* infection is common in developing countries leading to of high prevalences of peptic ulcers and gastric cancer. However, the diagnostic tests such as rapid urease test or urease breath test requires facilities and patients need to go to medical sites, which raise a more and more need for non-invasive and convenient tests appropriate to community.

Aims & Methods: A cross-sectional study was conducted at the Institute of Gastroenterology and Hepatology, Hanoi, Vietnam to compare the *H.pylori* determination results of the qualitative detection of 2 antibodies specific to FliD and CagA with the rapid urease test and urease breath test in Vietnamese patients with upper gastrointestinal symptoms. Patients over 18 years old with upper GI symptoms and no prior history of treated peptic ulcers or *H.pylori* eradication were invited to participate in the study. Patients were indicated with upper GI endoscopy and three methods to diagnose *H.pylori* infection: Rapid Urease Test (RUT), Urease Breath Test (UBT), and qualitative detection of 2 antibodies specific to FliD and CagA (Pylori Duotect). Pylori Duotect is considered positive if either antibody for FliD and CagA are positive. The parameters for evaluating the diagnostic accuracy of Pylori Duotect include sensitivity, specificity, positive predictive values (PPV), and negative predictive values (NPV). Patients are considered *H.pylori* positive if both RUT and UBT are positive.

Results: From December 1, 2023, to April 20, 2024, 189 patients were recruited for the research. The mean age were 45.8 years old (min-max: 18-76 years old), 40.7% were female. The most common symptom of patients was epigastric pain (46.6%). There were 172 patients (91%) with gastritis, 27 patients (14.3%) with gastric ulcers and 16 (8.5%) with duodenal ulcers. Using the Pylori Duotect test, 50.8% of the patients were positive, in which 39.6% were positive with FliD and 86.5% were positive with CagA. Based on RUT and UBT results, 31.2% of patients were diagnosed with *H. pylori* infection. Pylori DuoTect's sensitivity, specificity, PPV, and NPV were 81.4%, 63.8%, 50.5%, and 88.3%, respectively.

Conclusion: The Pylori DuoTect test had low sensitivity and specificity in diagnosis *H. pylori* infection in patients with upper gastrointestinal symptoms.

Disclosure: Nothing to disclose.

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PP0266

UREA BREATH TESTS IN IRELAND: AN 8 YEAR RETROSPECTIVE ANALYSIS

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Introduction: Helicobacter pylori (H. *pylori*) is a bacterium which inhabits the gastric epithelium and has been associated as a causative agent for dyspepsia, chronic gastritis, peptic and duodenal ulceration and gastric cancer. Testing for this pathogen is by Urea Breath Test (UBT) or OGD, and treatment varies according to local sensitivities and guidelines. H. *pylori* rates in developed countries have been falling in recent years as have those of associated conditions including gastric cancer, however increases in antimicrobial resistance recently has slowed this trend.

The current burden of disease either in asymptomatic population or symptomatic in Ireland is unknown, as are the rates of reinfection and recrudescence. Having up-to-date data on local epidemiology could have an effect on our strategy to detect and manage this class 1 carcinogen..

Aims & Methods: All UBTs performed in a tertiary hospital since 2016 were retrospectively assessed.

Demographic data & test results were taken from the UBT computer database and electronic patient record. All patients returning positive tests were offered appropriate antimicrobial therapy based on up to date local guidelines, and follow-up post-eradication testing by repeat UBT.

Results: From 2016-2023, 6716 UBTs were performed. 32 were excluded due to incomplete data. N=6684. 2752 (41.16%) were male, mean age was 43. Overall, 1589 (23.77%) tests were positive. There was no significant difference in positivity by gender (p=0.321). However there was a slight positive correlation with increasing age (AUC=0.5178 p=0.035). The positivity rate in our cohort reduced by roughly 2% per annum from 2016-22 (30%-18.5%).

Data on UBT type (Diagnostic vs Post-Eradication) was available from 2020-2023 when 3496 tests were performed and 2570 (73.5%) were diagnostic tests: 1060 (41.2%) male, mean age 41 years, 48 (19%) were positive. Positivity between genders was similar (p=0.3771). Conversely, in the post-eradication group n=926 (412, 44.5% Male), 259 (28%) were positive, females were more likely to remain positive post-therapy (32% vs 23%, X2=8.031, p=0.0046).

Over the 8 years, 747 patients returned positive tests (306 (41%) male). All were offered follow up testing. Males were significantly less likely to attend follow-up testing (140/306, 46% vs 254/441, 58%, p=0.0017)

In all 83 (2.4%) patients were re-tested for H. pylori after returning negative UBT tests due to persistent symptoms. Mean time to re-testing was 270 days, 34 (40%) were positive, representing 0.97% of the cohort.. Both genders were equally likely to have had a false negative UBT, male 10/30 versus 24/77 women, p=0.8214)

Conclusion: The overall positivity rate for diagnostic tests over the past 8 years was (19%) and is falling year on year.

The high positivity rate for post-eradication tests (28%) reflects rising antibiotic resistance.

The overall attendance rates for post-eradication tests after positive results were disappointing

Finally, the false negative rate in patients with persistent symptoms was high, 40% retested remained positive, raising the possibility of residual infection in our asymptomatic cohort.

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PP0267

FERRITIN LEVEL AS A DIAGNOSTIC PREDICTOR OF H. PYLORI INFECTION

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Introduction: The Maastricht VI consensus recognizes the ^{13}C-urease breath test (^{13}C-UBT) as the "gold standard" in the diagnosis of *H. pylori* infection, including in patients with dyspepsia, in which this infection is detected quite often [1, 2]. Atrophic gastritis of the body of the stomach can cause both false-positive (due to the increased activity of urease producers other than *H. pylori*) and false-negative ^{13}C-UBT result (due to a decrease in the microbial load of *H. pylor*i associated with severe atrophy and metaplasia of the gastric mucosa and deterioration of habitual living conditions bacteria) [3]. Serological tests for the detection of *H. pylori* IgG are the primary screening method for *H. pylori* infection, recommended by the Maastricht VI consensus [1]. However, negative ^{13}C-UBT and *H. pylori* antigen in stool with positive antibodies in patients with CAG, it often causes difficulties in assessing the *H. pylori* status, which emphasizes the relevance of searching for new diagnostic predictors.

Aims & Methods: Aim: evaluate the role of ferritin in determining the presence of *H. pylori* infection. 143 patients aged from 20 to 79 years were examined. The mean age was 59 (49–66) years. All participants underwent serum determination of pepsinogen-I (P-I), pepsinogen-II (P-II), gastrin-17 and antibodies to H. pylori; esophagogastroduodenoscopy with biopsy according to the OLGA system. Confirmation of *H. pylori* infection was performed by assessing the presence of antigen in stool using immunochromatography. To confirm autoimmune gastritis (AIG), the level of antibodies to parietal cells and intrinsic factor was determined in the blood. The indicators of clinical and biochemical blood analysis of patients (iron levels, ferritin) were assessed. When not normally distributed, quantitative data were described using the median (Me) and lower and upper quartiles (Q1–Q3). Comparison of two groups according to a quantitative indicator, the distribution of which differed from normal, was carried out using the Mann–Whitney U test. Comparisons of percentages in the analysis of multifield contingency tables were performed using the Pearson chi-square test.

Results: In 69.2% of cases, when verifying the etiology of CAG, AIH was identified, in 21.7% *H. pylori*-associated gastritis was identified, in 9.1% - CAG of mixed etiology. There was a weak positive correlation between P-I and iron levels (ρ=0.220, p=0.01), as well as between P-I and ferritin levels (ρ=0.229, p=0.007), the ratio P-I/ P-II and ferritin ρ=0.286, p<0.001), P-I/P-

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II and iron level (ρ=0.229, p=0.007). There was no relationship between P-II and iron content in the blood (ρ=0.025, p=0.775), as well as no association of P-II with ferritin levels (ρ=-0.085, p=0.323). When analyzing the concentrations of iron and ferritin, no statistically significant differences were revealed in these indicators depending on the stage of atrophy in CAG. When comparing groups of patients with IDA and iron deficiency, no statistically significant differences in the etiology of CAG could be found (p=0.167). When comparing ferritin values depending on the presence/absence of *H. pylori* antigen in stool, statistically significant differences were noted in this parameter (negative – Me = 43.0, 19.8–59.0, positive – Me = 67.0, 51, 7–80.5, p<0.001).

Conclusion: The presence of iron deficiency does not depend on the etiology of chronic atrophic gastritis. A positive correlation was revealed between P-I, P-I/P-II and iron deficiency. Ferritin levels may be a promising diagnostic predictor of *H. pylori* infection in patients with CAG.

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Disclosure: The authors declare that there is no conflict of interest.

PP0268

FACTORS ASSOCIATED WITH HELICOBACTER PYLORI ERADICATION IN THE ADULT POPULATION IN COLOMBIA: A MULTIVARIABLE ANALYSIS

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Introduction: Helicobacter pylori infection involves a high proportion of the Colombian population (1). It has been established that Helicobacter pylori infection increases the risk of developing gastric cancer (2). Multiple treatment regimens have been proposed to eradicate Helicobacter pylori, but it is not known which one could be associated with the best eradication rate in our population (3).

Aims & Methods: The aim of this research was to determine the factors most associated with the successful eradication of Helicobacter pylori in an adult population in Colombia.

An analytical cross-sectional study was conducted to assess risk factors associated with the eradication of Helicobacter pylori in an adult population in Colombia. The cross-sectional design facilitated data collection at a specific time point, while the analytical approach aimed to uncover connections between risk factors and the success of eradication.

A sample size calculation was performed for logistic regression using the Freeman formula. Based on estimated proportions of patients experiencing treatment failure, an approximate sample size of 400 patients was determined for the evaluation of 10 variables. Participant patients were adults diagnosed with Helicobacter pylori infection who were prescribed eradication treatment and underwent verification of eradication between January 1, 2022, and December 31, 2022. Patients with incomplete information, the pediatric population, or those without eradication verification were excluded from the analysis.

A descriptive analysis was conducted to characterize the sample and risk factors. Subsequently, a bivariate analysis examined the relationship between each independent variable and Helicobacter pylori eradication. Variables with a p-value < 0.05 in the bivariate analysis were included in a multivariate logistic regression model to identify independent risk factors associated with successful eradication. The multivariate analysis was adjusted for treatment adherence and compliance.

Results:

Variable	Odds ratio	95% Confidence Interval		p value
Proper treatment adherence*	13.72	3.1840	59.0874	< .001
Regimen with esomeprazole	4.97	1.1034	22.3592	0.037
Regimen with bismuth	2.04	1.1580	3.5977	0.014
Duration of the regimen: 14 days	7.21	1.2304	42.2908	0.029
Lactobacillus reuteri	1.79	0.8633	3.7249	0.117
Eradication regimen with amoxicillin	2.58	0.1512	43.9183	0.513
Eradication regimen with clarithromycin	6.88	0.4417	107.1126	0.169
Eradication regimen with metronidazole	6.94	0.4185	115.1961	0.176
Eradication regimen with levofloxacin	11.76	0.7801	177.1968	0.075

*Table 1. Multivariable analysis, factors associated with Helicobacter pylori eradication. * Adherence assessed by the Morisky-Green Test.*

A total of 386 patients were included in the study, of whom 74% achieved successful eradication of Helicobacter pylori. The multivariate analysis identified factors associated with successful eradication of Helicobacter pylori: proper adherence to the treatment, the inclusion of esomeprazole and bismuth subsalicylate in the treatment, and a eradication regimen duration of 14 days (Table 1).

Conclusion: Our results support the promotion of treatment regimens lasting 14 days. Strategies to ensure proper adherence to treatment should be encouraged, always incorporating esomeprazole and bismuth subsalicylate in the regimen.

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Disclosure: The authors declare no conflict of interest.

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PP0269

HELICOBACTER PYLORI SECONDARY ANTIBIOTIC RESISTANCES: A GENOTYPIC STUDY ON STOOL SAMPLES

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Introduction: Helicobacter pylori (H. pylori) antibiotic resistances are the leading cause for unsuccessful eradication therapy. After one or more failures, the chance of encountering secondary antibiotic resistances increases. The aim of this study was to characterize genotypic secondary resistances in a cohort of Southern Italy H. pylori patients with at least one previous failure.

Aims & Methods: Patients with at least one previous eradication failure collected stool samples by a dedicated kit (THD fecal testTM) and bacterial DNA was extracted and amplified by RT-PCR. Resistance to clarithromycin, amoxicillin, metronidazole, levofloxacin and tetracycline was assessed by high-resolution melting curve.

Results: We enrolled 50 patients. The 72% failed one previous antibiotic course, the 16% two, the 10% three and the 2% four. The rate of secondary antibiotic resistance was 16% for clarithromycin, 18% for metronidazole, 14% for amoxicillin, 14% for levofloxacin and 2% for tetracycline. Multi-resistance was detected in 9 patients (18%) and double resistance to clarithromycin + metronidazole was the most common one (n = 5). Among patients with a previous failure to an amoxicillin-containing regimen, amoxicillin resistance was found in 4 out of 20 (20%). Among patients with a previous failure to a clarithromycin-containing regimen, resistance to such antibiotic was detected in 6 out of 19 (31.6%). Among patients with a previous failure to a metronidazole-containing regimen (n = 21), metronidazole resistance was found in 3 cases (14.3%). A failed levofloxacin-based therapy was given in 3 patients, but in none of them resistance was found (0%). Finally, among patients with a previous failure to a tetracycline-containing therapy, resistance to such antibiotic was detected in 1 out of 12 (8.3%)

Conclusion: In conclusion, our data seem to show that, despite secondary resistances are not so high, a resistance to clarithromycin could be very likely related to previous exposure to this antibiotic.

Disclosure: Nothing to disclose.

PP0270

THE FREQUENCY OF HELICOBACTER PYLORI INFECTION AND THE EFFECT OF ERADICATION THERAPY ON SYMPTOM IMPROVEMENT IN FUNCTIONAL DYSPEPSIA SUBGROUPS ACCORDING TO ROME IV CRITERIA: A STUDY FROM TURKEY WITH A HIGH PREVALENCE OF HELICOBACTER PYLORI

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Introduction: Different pathophysiological mechanisms are thought to play a role in functional dyspepsia (FD) subgroups defined by Rome IV criteria. Studies are needed to determine whether Rome IV subgroups are linked to distinct pathophysiologies and treatment responses.

This study aims to investigate the impact of Helicobacter pylori (H. pylori) infection in the pathogenesis of each FD subgroups, subdivided according to the Rome IV consensus, within a high H. pylori prevalence country, Turkey.

The investigation centered on assessing the frequency of H. pylori and examining the impact of eradication therapy on alleviating symptoms, in each Rome IV subgroup.

Aims & Methods: Eradication treatment was applied to cases who were initially diagnosed with FD using the Rome IV criteria and subsequently found to have H. pylori gastritis through endoscopic biopsy. 8-12 weeks after treatment, H. pylori state was evaluated with fecal antigen test. Symptom responses were evaluated in each subgroup at 2, 6, and 12 months post-therapy using a Likert scale and statistically with SPSS version 23.0.

Results: The H. Pylori frequency was statistically more common in individuals with symptoms consistent with epigastric pain syndrome (EPS) (73.1% in EPS, 41% in postprandial distress syndrome (PDS), and 43.9% in overlapping subgroups (p< .001)).

With eradication treatment of H. pylori-positive cases, all subgroups demonstrated a significant response and the response rates didn't exhibit significant difference between subgroups (73.7% in total cases; 70.5% in EPS; 71.4% in PDS; 82.6% in overlapping, in the 12th month (p>0.05)). Eradicated patients showed significant symptomatic improvement with 70.2% in the 12th month. None of the non-eradicated patients experienced symptomatic improvement.

Subgroups	In EPS cases Responsive(n) / Total(n) (%)	In PDS cases Responsive(n) / Total(n) (%)	In overlapping cases Responsive(n) / Total(n) (%)	In all cases Responsive(n) / Total(n) (%)
Respond rate at 2nd month (%)	23/51 (45.1%)	15/39 (38.5%)	11/26 (42.3%)	49/116 (42.2%)
Respond rate at 6th month (%)	31/46 (67.4%)	18/35 (51.4%)	18/25 (72%)	67/106 (63.2%)
Respond rate at 12th month (%)	31/44 (70.5%)	20/28 (71.4%)	19/23 (82.6%)	70/95 (73.7%)

FD, functional dyspepsia; PDS, postprandial distress syndrome; EPS, epigastric pain syndrome

Table 2. The ratios of symptomatic response after H. pylori eradication therapy among FD Rome IV subgroups that received therapy, at 2nd, 6th, 12th month visits.

Conclusion: Using Rome IV consensus confirmed the data that H. pylori plays a notable role as a pathogenic factor in FD, particularly in EPS, and it is recommended to screen all FD subgroups, particularly EPS, for H. pylori and eradicate, if detected. Additionally, this study suggests that regions with a high H. pylori prevalence tend to have a higher H. pylori-associated dyspepsia prevalence.

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Disclosure: The authors declare that they have no conflict of interest.

PP0271

EMPIRICAL SECOND-LINE TREATMENTS IN EUROPE: DATA FROM 9,000 CASES FROM THE EUROPEAN REGISTRY ON *HELICOBACTER PYLORI* MANAGEMENT (HP-EUREG)

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Introduction: After a first eradication attempt, approximately 10-20% of patients will fail to achieve *H. pylori* eradication.

Aims & Methods: To evaluate the effectiveness of second-line empirical treatment.

A systematic, prospective, registry of the clinical practice of European gastroenterologists (38 countries) on *H. pylori* management was established. All infected adult patients were systematically registered at AEG-REDCap e-CRF until December 2023. Modified intention-to-treat (mITT) and per-protocol (PP) analyses were performed.

Results: Overall, 8,984 patients received a second-line empirical therapy. Mean age was of 50 years, 66% were women, and 5% had penicillin allergy. Most frequent treatment indications were dyspepsia (70%) and gastroduodenal ulcer (15%). Overall effectiveness was 83% (both by mITT and PP). Over 96.5% of patients were compliant.

Adverse events were reported in 23% of the cases, most of them mild. Most frequent second-line prescriptions and their effectiveness by region is shown in table 1.

None of the European regions achieved an optimal effectiveness. However, in those patients receiving optimised therapy (i.e., 14-days and high dose PPIs), triple regimens containing levofloxacin and bismuth quadruple therapies either with metronidazole-tetracycline or clarithromycin-amoxicillin, as well as quadruple therapy with clarithromycin-amoxicillin-metronidazole, showed optimal effectiveness. Also, 10-day bismuth quadruple therapy as single-capsule achieved over 90% cure rates when prescribed with standard PPI doses. After failure of first-line clarithromycin-containing therapy, optimal eradication was only obtained with moxifloxacin-containing triple therapy (92%). Quadruple therapy with clarithromycin-amoxicillin-metronidazole and bismuth quadruple therapies either with clarithromycin-amoxicillin, amoxicillin-levofloxacin or the single-capsule provided encouraging results (>85%).

Treatment	N	Use (%)	mITT, N (%)	(95% CI)	PP, N (%)	(95% CI)
Triple-A+L	2,212	25.5	1,959 (81)**	(79-83)	1,936 (81)	(79-83)
Single-capsule*	1,783	20.6	1,659 (88)**	(86-89)	1,621 (89)	(87-90)
Quadruple-A+L+B	1,170	13.5	929 (85)	(83-87)	910 (85)	(83-88)
Quadruple-C+A+M	560	6.5	534 (85)**	(82-88)	523 (85)	(82-89)
Triple-C+A	525	6.1	424 (79)**	(75-83)	416 (79)	(75-83)
Quadruple-M+Tc+B	451	5.2	405 (83)**	(79-87)	386 (85)	(81-88)
Quadruple-C+A+B	370	4.3	251 (89)**	(85-93)	240 (90)	(86-94)
Triple-A+M	225	2.6	198 (62)	(55-69)	195 (63)	(56-70)
Triple-A+R	212	2.4	189 (83)	(77-88)	187 (83)	(77-89)
Triple-A+Mx	153	1.8	147 (92)	(87-97)	147 (92)	(87-97)
Other	1,013	11.5	NA	NA	NA	NA
Overall	8,674	100	7,563 (83)	(82-84)	7,410 (83)	(83-84)
Overall (optimised conditions)	2,005	23.1	1,747 (89)	(87-90)	1,709 (89)	(88-91)
East	1,425	15.9	1,159 (84)	(82-86)	1,121 (84)	(82-87)
East-Centre	1,625	18.1	1,160 (84)	(82-86)	1,147 (84)	(82-87)
South-West	4,190	46.7	3,973 (83)	(82-85)	3,895 (84)	(83-85)
West-Centre	1,103	12.3	979 (84)	(82-87)	962 (85)	(82-87)
North	631	7.0	547 (72)	(68-76)	530 (73)	(69-77)
Total	8,974	100	7,803 (83)	(82-84)	7,641 (83)	(83-84)

95% CI –confidence interval; mITT: modified intention-to-treat; PP: per-protocol, N: total number of patients analysed; C – clarithromycin; M – metronidazole; A – amoxicillin; L – levofloxacin; B – bismuth salts; Tc – tetracycline; R – rifabutin; Mx – moxifloxacin; Other – Other second-line empirical treatments with less than 150 patients treated in each category; East – Ukraine, Serbia, Bulgaria, Turkey, Russia , Romania, Albania, North Macedonia; East-Centre – Croatia, Poland, Hungary, Latvia, Lithuania, Greece, Slovenia, Czech Rep, Azerbaijan, Slovakia, Malta; South-West – Portugal, Spain; West-Centre – France, Austria, Belgium, Germany, Italy; North – United Kingdom, Finland, The Netherlands, Ireland, Israel, Norway, Switzerland, Sweden, Denmark; *three-in-one single-capsule containing metronidazole tetracycline and bismuth; **achieved over 90% effectiveness when optimised (high-dose PPIs and 14-days length in all schemes but the single-capsule optimised with standard-dose PPIs).

Table 1. Second-line empirical treatment prescriptions and effectiveness by treatment scheme and geographical region.

Conclusion: In Europe, empirical second-line treatment achieves optimal effectiveness with bismuth quadruple therapy as single-capsule when prescribed standard doses of PPIs, and when regimens including triple therapy with levofloxacin, quadruple therapy with clarithromycin-amoxicillin-metronidazole and bismuth quadruple therapies with metronidazole-tetracycline or clarithromycin-amoxicillin are prescribed for 14 days and with high-dose PPIs.

Many other second-line treatments evaluated achieved low eradication rates.

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PP0272

EFFECTIVENESS OF FIRST-LINE TREATMENT FOR *H. PYLORI* INFECTION IN ISRAEL: RESULTS FROM THE EUROPEAN REGISTRY ON *H. PYLORI* MANAGEMENT (HP-EUREG)

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Introduction: The effectiveness of first-line treatment for *H. pylori* infection varies between countries. Local data are necessary to guide clinical practice. We aimed to assess the effectiveness of first-line eradication treatments for *H. pylori* in Israel.

Aims & Methods: International, multicenter, prospective, non-interventional registry aiming to evaluate the decisions and outcomes of *H. pylori* management by European gastroenterologists (Hp-EuReg).

All infected adult patients treated with a first-line therapy at 10 centers in Israel between 2013 and 2023 were registered at e-CRF AEGRedCap and were included in the current study.

Effectiveness was assessed by modified intention-to-treat (mITT) and per-protocol (PP) analyses.

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Results: A total of 663 cases were registered, where 320 (48%) had received a 1st line empirical therapy with an average age of 51±18 years and 183 (57%) were females. The most common indication for *H. pylori* investigation was dyspepsia (67%). Infection was diagnosed with histology (56%), ^{13}C-Urea breath test (22%), rapid urease test (17%) and monoclonal stool Ag test (5.3%). Quadruple concomitant therapy with PPI-clarithromycin-amoxicillin-tinidazole was the most frequent 1st line treatment (43%), mostly prescribed for 14 days (87% of the patients) and with high-dose PPIs (91% of the cases), followed by triple therapy with PPI-clarithromycin-amoxicillin (16%) and quadruple concomitant therapy with PPI-clarithromycin-amoxicillin-metronidazole (12%). The majority of prescriptions were of 14-day (70%) length, followed by 10-day (22%) and 7-day (7.3%) duration. High potency proton pump inhibition (PPI) was prescribed to most patients (75%).

Overall treatment success was 84% (160/191) and 86% (155/181) by mITT and PP, respectively, with the highest effectiveness provided by 10-day and high-dose PPIs of sequential therapy with clarithromycin-amoxicillin-tinidazole (89.5% by mITT), and by 14-day and high-dose PPIs of concomitant therapy with clarithromycin-amoxicillin-tinidazole (85% by mITT). Longer treatment durations and higher PPI doses provided better cure rates. Treatment effectiveness did not change significantly during the study period, and remained suboptimal (<90%) over the years. Compliance (>90% drug intake) was 97%. The incidence of at least one adverse event was reported in 10%, necessitating treatment cessation in 1.5%.

	Use, n (%)	mITT Success n/N (%)	PP Success n/N (%)
Treatment scheme			
Triple-C+A	49 (16.1)	36/44 (81.8)	36/44 (81.8)
Sequential-C+A+T	20 (6.6)	17/19 (89.5)	17/19 (89.5)
Quadruple-C+A+T	132 (43.3)	73/86 (84.9)	71/80 (88.8)
Quadruple-C+A+M	38 (12.5)	13/13 (100)	13/13 (100)
Treatment Duration			
7 days	22 (7.3)	16/20 (80)	16/20 (80)
10 days	68 (22.4)	43/50 (86)	42/49 (85.7)
14 days	213 (70.3)	92/111 (82.9)	88/102 (86.3)
Potency of acid inhibition*			
Low	63 (20.6)	48/58 (82.8)	47/57 (82.5)
Standard	12 (3.9)	6/6 (100)	6/6 (100)
High	231 (75.5)	105/125 (84)	101/116 (87.1)
Overall	320 (100)	160/191 (83.8)	155/181 (85.6)

Abbreviations: n: total number of patients analysed; mITT, modified intention to treat; PP, per-protocol; C, clarithromycin; A, amoxicillin; T, tinidazole; M, metronidazole; *Low dose PPI – 4.5 to 27 mg omeprazole equivalents. b.i.d.; standard dose PPI – 32 to 40 mg omeprazole equivalents. b.i.d.; high dose PPI – 54 to 128 mg omeprazole equivalents. b.i.d.

Table: Empirical first-line therapy effectiveness by modified intention-to-treat and per-protocol analyses, by treatment duration and acid inhibition in Israel.

Conclusion: In Israel, empirical first-line treatment for *H. pylori* eradication is still suboptimal (<90%). Treatments should consist of four drugs and should be optimised with a 14-day duration and standard/high doses of PPIs.

Disclosure: None.

PP0273

H. PYLORI FIRST-LINE EMPIRICAL THERAPY EFFECTIVENESS AND ITS RELATIONSHIP TO ANTIBIOTIC CONSUMPTION IN THE COMMUNITY: DATA FROM THE EUROPEAN REGISTRY ON *H. PYLORI* MANAGEMENT (HP-EUREG)

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Introduction: The link between the misuse in global antibiotic consumption and increased antimicrobial resistance is well documented. Clarithromycin-based anti-*H. pylori* eradication therapy as a triple regimen (encompassing a PPI plus two antibiotics) has been used empirically worldwide without antimicrobial susceptibility testing. In the last decade, a marked decreased in the effectiveness of clarithromycin-based regimens has been observed globally.

Aims & Methods: This study aimed at exploring the potential relationship between the consumption of macrolides in the overall population as a factor influencing the effectiveness of *H. pylori* clarithromycin-containing regimens.

Data from the European Registry on *Helicobacter pylori* management (Hp-EuReg) were used for this analysis. A multivariate logistic regression was built using the modified intention-to-treat (mITT) effectiveness analysis as the outcome and the following main effects: first-line treatment scheme (using the 10 most frequent antibiotic combinations), duration of treatment (as 7, 10 or 14-day lengths), proton-pump inhibitor (PPI) dose (categorised as low, standard, or high; that is, 20 mg, 40 mg or 80 mg omeprazole equivalent twice a day), compliance (that is, >90% of drug intake), and clarithromycin consumption expressed as daily doses per 1,000 inhabitants per day. Data of macrolide community consumption was taken from the European Surveillance of Antimicrobial Consumption Network (ESAC-Net). Hierarchical nested models were constructed by adding macrolide consumption, matching by year and country with the corresponding patients' therapy, and the interaction between consumption and treatment scheme (taken from the Hp-EuReg).

Results: In total, 28,590 first-line treated patients from 24 European countries participating in the Hp-EuReg, on which data of macrolides consumption in the period 2013-2022 was available, were evaluated. Three nested models were analysed: the first one with the main effects without including macrolides consumption (A), the second with the full set of main effects (including macrolides consumption) (B) (likelihood ratio test A-B p-value=$2.11.10^{-2}$), and the third one included both macrolides consumption and interactions between this main effect and treatments (C) (likelihood ratio test B-C p-value=$4.08\ 10^{-10}$). Model B (without interactions) showed a global decrease in treatment mITT effectiveness (OR: 0.95 [0.91-0.99]). Model C (with interactions) showed a significant decrease in the mITT effectiveness in three different treatments (Table): triple-clarithromycin-metronidazole (OR: 0.37 [0.25-0.53]), triple-clarithromycin-amoxicillin (OR: 0.60 [0.43-0.82]), and sequential-clarithromycin-amoxicillin-metronidazole/tinidazole (OR: 0.60 [0.43-0.84]); whereas, bismuth quadruple therapy (single capsule) remained unaffected.

	OR	95% CI	p-value
Intercept	0.42	0.15-1.22	0.17
Compliance (>90% of drug intake)	7.41	6.30-8.71	1.01e-92
10 days prescriptions	0.92	0.92-1.04	0.25
14 days prescriptions	1.06	0.94-1.21	0.43
Standard-dose PPI *	1.58	1.44-1.74	1.08e-15
High-dose PPI*	1.79	1.65-1.95	7.23e-31
Triple- PPI+C+A	2.32	0.80-6.34	0.18
Triple- PPI+C+M	3.95	1.25-11.93	0.04
Triple- PPI+A+M	1.90	0.64-5.43	0.32
Triple- PPI+A+L	1.29	0.36-4.40	0.74
Conco- PPI+C+A+T/M	1.97	0.68-5.42	0.28
Seq- PPI+C+A+T/M	4.71	1.55-13.66	0.02
Quadruple- PPI+C+A+B	0.99	0.21-4.88	0.99
Quadruple- PPI+single capsule**	2.32	0.76-6.69	0.20
Other	1.45	0.47-4.22	0.58
C cons	1.42	1.04-1.96	0.07
Triple- PPI+C+A: C cons	0.60	0.43-0.82	0.01
Triple- PPI+C+M: C cons	0.37	0.25-0.53	1.58e-05
Triple- PPI+A+M: C cons	0.71	0.49-1.01	0.11
Triple- PPI+A+L: C cons	0.76	0.51-1.12	0.24
Conco- PPI+C+A+T/M: C cons	0.76	0.54-1.04	0.15
Seq- PPI+C+A+T/M: C cons	0.60	0.43-0.84	0.01
Quadruple- PPI+C+A+B:C cons	0.89	0.52-1.48	0.72
Quadruple-PPI+Single capsule: C cons**	0.86	0.60-1.20	0.46
Other: C cons	0.72	0.51-1.00	0.10

OR: odds ratio; CI: confidence interval; *low dose PPI – 4.5 to 27 mg omeprazole equivalents. b.i.d.; standard dose PPI – 32 to 40 mg omeprazole equivalents. b.i.d.; high dose PPI – 54 to 128 mg omeprazole equivalents. b.i.d.; A: amoxicillin; B: bismuth; C: clarithromycin; L: levofloxacin; M: metronidazole: tetracycline.; **Three-in-one single-capsule containing metronidazole. tetracycline and bismuth; C cons: clarithromycin consumption.

Table. Multivariate logistic regression (model C) evaluating the interaction between clarithromycin consumption and first-line empirical treatment effectiveness (by modified intention-to-treat) in Europe between 2013 and 2022.

Conclusion: The models showed an inverse association between the clarithromycin consumption and empirical first-line therapy overall effectiveness in Europe; particularly, in those clarithromycin-containing regimens. Further research will consider a broader time lag to assess this association.

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EFFECTIVENESS OF FIRST-LINE EMPIRICAL TREATMENT IN AZERBAIJAN: RESULTS FROM THE EUROPEAN REGISTRY ON HELICOBACTER PYLORI MANAGEMENT (HP-EUREG)

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Introduction: Gastric lesions associated with *Helicobacter pylori*, eradication issues and problems of its sometimes-insufficient effectiveness continue to remain one of the most pressing topics in modern healthcare.

Aims & Methods: The analysis aimed to assess the usage, efficacy, and safety of initial empirical treatment for *H. pylori* infection in Azerbaijan.

Data were collected at e-CRF AEG-REDCap from the European Registry on *Helicobacter pylori* Management (Hp-EuReg) and quality reviewed from January 2020 to May 2023. All treatment-naïve cases were assessed for effectiveness by modified intention-to-treat (mITT), per-protocol (PP) analyses, and a separate multivariate analysis was performed.

Results: The study included 3,898 patients (67% reported dyspepsia). In 1.9% of patients, at least one type of allergy was reported (1.6% to penicillin). A total of 226 patients (5.8%) received concurrent medication: PPI (5.1% on demand, 0.1% daily), aceltylsalicylic acid (1.2% on demand, 0.2% daily), NSAIDs (1.3% on demand), statins (1.2% on demand, 0.1% daily).

The most frequent indication for treatment was dyspepsia with normal endoscopy in 2,610 patients (67%). In total, 3,853 (99%) patients were diagnosed pre-treatment using the rapid urease test, which was the most frequently used. The following tests were likewise used in a lower number of cases: monoclonal stool antigen test (MSAT)(0.7%) and polyclonal stool antigen test (0.1%). In 2,661 (68%) cases a MSAT was used, being the most frequent post-treatment confirmation method. Additionally, in 55 (1.4%) cases an invasive method, that is endoscopy, was performed, to confirm the eradication. First-line empirical treatment prescriptions and mITT effectiveness by treatment schemes is presented in Table 1.

Triple regimens were the most frequently prescribed in over 3,000 (82%) patients, where the PPI-C+A treatment scheme represented 54% of the total of first-line prescriptions. Treatment length varied from 14 days in 56% of cases, 10 days in 27%, and 7 days in 17%. Following 14-day treatments prescriptions achieved optimal (>90%) mITT effectiveness: PPI-C+A (94%), PPI-C+M (97%), PPI-C+M+ B (96%) and PPI-C+A+B (98%). Following schemes also provided optimal results when combined with standard-dose (i.e., 40 mg omeprazole equivalent twice a day) PPIs: PPI-A+Tc (100%) and PPI-C+A+B (96%). Additionally, prescriptions concomitantly administered with high-dose (i.e., 80 mg omeprazole equivalent twice a day) PPIs provided also over 90% eradication rates: PPI-C+A (94%), PPI-C+M (97%), PPI-C+M+B(100%) and PPI-C+A+B (99%). Compliance with treatment was over 90% in the vast majority of cases. The incidence of at least one adverse event was reported in 12% of cases.

Treatment	Number of prescriptions	Use (%)	mITT, N (%)	PP, N (%)	mITT, N (%) Low-dose PPI	mITT, N (%) Standard-dose PPI	mITT, N (%) High-dose PPI
Triple-C+A	2,117	54.3	1,832 (91,2%)	1,819 (95.2%)	873 (91.2%)	527 (89.3%)	420 (93.8%)
Triple-C+M	760	19.5	584 (93,3%)	581 (96.5%)	326 (93.4%)	89 (86.4%)	167 (97.1%)
Quadruple-C+A+B	610	15.6	579 (98%)	574 (99.8%)	219 (98.2%)	187 (96.4%)	178 (99.4%)
Triple-A+L	159	4.1	124 (82,1%)	124 (99.2%)	52 (82.5%)	68 (81.9%)	4 (80.0%)
Triple-A+Tc	75	1.9	71 (94,7%)	71 (100.0%)	28 (93.3%)	43 (95.6%)	NA
Quadruple-C+M+B	44	1.1	40 (95,2%)	40 (100.0%)	26 (100.0%)	6 (75.0%)	8 (100.0%)
Other	44	1.1	114 (90,5%)	113 (97.4%)	55 (93.2%)	55 (90.2%)	4 (66.7%)
Total	3,809	97.6	3,344 (92.4%)	3,322 (96.6%)	68 (92.5%)	975 (89.9%)	776 (95.4%)
A-amoxicillin, C- clarithromycin, M- metronidazole, L- levofloxacin, Tc- tetracycline, B - bismuth			mITT-modified Intention-to-treat;	PP: per protocol analysis;	low-dose PPI, between 4.5 and 27 mg omeprazole equivalents given twice a day;	standard-dose PPI, between 32 and 40 mg omeprazole equivalents given twice a day;	high-dose PPI, between 54 and 128 mg omeprazole equivalents given twice a day; NA: not available data.

Table 1. Most frequent first-line empirical treatment prescriptions and modified intention-to-treat effectiveness in Azerbaijan.

Conclusion: The prescribed *H. pylori* eradication regimens during the period 2020–2023 in Azerbaijan adhered only partially to the Maastricht VI recommendations; however, most of first-line empirical therapies achieved high (≥90%) cure rates, providing a good safety profile.

Disclosure: Nothing to disclose.

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PP0276

THE PATTERN OF *HELICOBACTER PYLORI* PCR-ASSESSED ANTIBIOTIC RESISTANCE: EXPERIENCE FROM THE EUROPEAN REGISTRY ON *HELICOBACTER PYLORI* MANAGEMENT (HP-EUREG)

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Introduction: In light of the growing global problem of antibiotic resistance, eradication of clarithromycin (CLA)-resistant *Helicobacter pylori* (*H. pylori*) strains has been recognized as a top priority health problem. Poor antibiotic stewardship, and the complexity of antibiotic resistance patterns, has made *H. pylori* gastritis the most prevalent infectious disease. Pretreatment assessment of *H. pylori* antibiotic sensitivity has been suggested as a reasonable solution in overcoming antibiotic resistance and treatment failure.

Aims & Methods: The aim of the study was to assess the prevalence and pattern of *H. pylori* resistance towards most common prescribed antibiotics, through the use of molecular, PCR-based technique. A multicentric, prospective registry of the clinical practice of European gastroenterologists on the management of *H. pylori* (Hp-EuReg) collects since 2013 the real clinical practice of European gastroenterologists. Adults *H. pylori*-infected and PCR-tested patients registered at AEG REDCap e-CRF were analysed. Two types of molecular testing were performed: a non-invasive (stool sample PCR) and an invasive (gastric sample PCR) test.

Results: In total, 265 *H. pylori* patients were tested, of which 130 were treatment-naive cases and 135 had already received at least one eradication attempt. In treatment-naïve cases, CLA resistance (CLA-R) was 31%, fluoroquinolone resistance (FLQ-R) was 33% and metronidazole resistance (MET-R) was 14% (Table).

There was no statistically significant difference in the bacterial resistance rate of tested antibiotics based on the gender and age distribution. In the rescue line, CLA-R was 62%, FLQ-R was 53% while MET-R was 7% (Table). Multiple eradication attempts significantly increased CLA-R (53% vs. 79% vs. 100%, for the 2^{n}, 3rd and 4th line respectively, p=0.001).

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Antibiotics	First line (n, %)	Rescue line (n, %)
Clarithromycin	40, 31%	84, 62%
Fluoroquinolones	43, 33%	72, 53%
Metronidazole	18, 14%	10, 7%

Table. PCR-detected resistance to most common prescribed antibiotics in Europe.

Conclusion: Alarmingly high *H. pylori* resistance to CLA and FLQ as well as its steep increase following multiple eradication attempts, implies an urgent need of continuous and systematic antibiotic stewardship, in order to further optimise *H. pylori* eradication strategies.

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PP0277

HELICOBACTER PYLORI ERADICATION FIRST- AND SECOND-LINE REGIMENS PRESCRIPTIONS AND EFFECTIVENESS IN LITHUANIA DURING THE YEARS 2013-2023: ANALYSIS OF THE EUROPEAN REGISTRY ON *H. PYLORI* MANAGEMENT (HP-EUREG)

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Introduction: Lithuania remains in the area with a high prevalence of *H. pylori* infection, which is estimated to be ~30-40%. Therefore, it is important to analyze the local *H. pylori* management in order to achieve better treatment results.

Aims & Methods: To evaluate the trends and effectiveness of *H. pylori* first- and second-line treatment prescriptions in Lithuania during 2013-2023.

Prospective, multicenter, international, non-interventionist registry on the management of *H. pylori* (Hp-EuReg) by European gastroenterologists. Lithuanian data from 2013 to 2023 were analysed. The effectiveness of the five most frequent first- and second-line *H. pylori* eradication prescriptions was evaluated using the modified intention-to-treat (mITT) analysis.

Results: In total 4,168 patients from Lithuania were included in the Hp-EuReg. The confirmation of *H. pylori* eradication was assessed in 37% of the first-line and 44% of the second-line cases. The overall first-line treatment effectiveness was 88%; the effectiveness of the most frequent first-line treatment prescription (standard triple therapy with amoxicillin and clarithromycin) increased from 78% in 2013 to 95% in 2023, obtaining the optimal (>90%) cure rates during the last 4 years. The overall second-line treatment effectiveness was 91%; the overall effectiveness of the most frequent second-line treatment prescription (triple therapy with amoxicillin and levofloxacin) was 92%. Since 2020 this triple therapy was taken over by bismuth quadruple therapy with amoxicillin and levofloxacin in prescription frequency – this quadruple regimen managed to maintain optimal eradication effectiveness rates during the last 6 years. The trends and effectiveness of other most frequent first- and second line prescriptions are presented in Table 1.

First-line treatment	Years period								
	2013-2016			2017-2020			2021-2023		
	Prescription frequency,n (% of top 5)	Test post-treatment, n (follow-up %)	mITT (95% CI)	Prescription frequency,n (% of top 5)	Test post-treatment, n (follow-up %)	mITT (95% CI)	Prescription frequency,n (% of top 5)	Test post-treatment,n (follow-up %)	mITT (95% CI)
Triple C+A	662 (97.1%)	301 (45.5%)	84.4% (80-88%)	1477 (89.5%)	553 (37.4%)	85.7% (83-89%)	686 (68.5%)	267 (38.9%)	93.3% (90-96%)
Quadruple C+A+B	0 (0%)	N/A	N/A	72 (4.4%)	20 (27.8%)	100% (83-100%)	155 (15.5%)	35 (8%)	88.6% (73-97%)
Triple A+L	2 (0.3%)	1 (50%)	100% (3-100%)	35 (2.1%)	23 (65.7%)	100% (85-100%)	11 (1.1%)	7 (63.6%)	100% (59-100%)
Triple C+M	18 (2.6%)	7 (46.6%)	100% (59-100%)	39 (2.4%)	14 (95.9%)	92.9% (66-100%)	13 (1.3%)	8 (61.5%)	87.5% (47-100%)
Quadruple A+M+B	0 (0%)	N/A	N/A	27 (1.6%)	6 (22.2%)	83.3% (36-100%)	137 (13.7%)	7 (5.1%)	100% (59-100%)

Second-line treatment	Years period								
	2013-2016			2017-2020			2021-2023		
	Prescription frequency,n (% of top 5)	Test post-treatment,n (follow-up %)	mITT (95% CI)	Prescription frequency,n (% of top 5)	Test post-treatment,n (follow-up %)	mITT (95% CI)	Prescription frequency,n (% of top 5)	Test post-treatment,n (follow-up %)	mITT (95% CI)
Quadruple A+L+B	0 (0%)	N/A	N/A	110 (39.4%)	48 (43.6%)	97.9% (89-100%)	167 (65.2%)	40 (24%)	95% (83-99%)
Triple A+L	57 (79.2%)	24 (42.1%)	87.5% (68-97%)	124 (44.4%)	84 (67.7%)	91.7% (84-97%)	44 (17.8%)	26 (59.1%)	96.2% (80-100%)
Quadruple A+M+B	0 (0%)	N/A	N/A	21 (7.5%)	7 (33.3%)	71.4% (29-96%)	32 (12.5%)	9 (28.1%)	88.9% (52-100%)
Triple C+A	15 (20.8%)	8 (53.3%)	100% (63-100%)	13 (4.7%)	9 (69.2%)	100% (66-100%)	9 (3.5%)	5 (55.6%)	100% (48-100%)
Quadruple C+A+B	0 (0%)	N/A	N/A	11 (3.9%)	4 (36.4%)	100% (40-100%)	11 (4.3%)	6 (54.5%)	83.3% (36-100%)

mITT – modified Intention-to-Treat analysis; C- clarithromycin; A – amoxicillin; B – bismuth; L – levofloxacin; M – metronidazole; 95% CI – 95% confidence interval; N/A – not applicable;

Table 1. Most frequent first- and second-line prescriptions and their modified intention-to treat (mITT) effectiveness in Lithuania during 2013-2023.

Conclusion: During 2013-2023 there was a trend of increasing *H. pylori* eradication effectiveness of the main first-line prescription, managing to maintain optimal cure rates in the last 4 years. There has been an increase in first- and second-line quadruple therapies prescriptions since the year 2018.

Disclosure: Nothing to disclose.

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PP0278

COMPARISON OF ERADICATION EFFICACY OF 1-WEEK METRONIDAZOLE TRIPLE REGIMEN WITH BISMUTH ADD-ON AND 2-WEEK BISMUTH QUADRUPLE REGIMEN FOR H PYLORI WITH CLARITHROMYCIN RESISTANCE-RELATED GENE MUTATION (A2143G)

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Introduction: In regions where there is a prevalence of clarithromycin resistant strains, 2-week bismuth quadruple therapy (PBMT) is recommended as the first-line treatment. In cases where CLA resistance-related genetic mutations are identified, 1-week metronidazole triple therapy is recommended. The aim of this study is to investigate whether 1-week bismuth-added metronidazole triple therapy (PBMA) therapy can achieve equivalent efficacy to 2-week PBMT therapy.

Aims & Methods: This study was a multicentred retrospective analysis using electronic medical records, employing propensity score matching to analyze the results. The study population consisted of individuals who were confirmed to have H. pylori infection with clarithromycin resistance-associated genetic mutation (A2143G) using PCR and who were administered either 2-weeks PBMT or 1 week PBMA.

Results: From October 2019 to May 2023, a total of 121 patients were confirmed for H. pylori infection and clarithromycin resistance-related genetic mutations using PCR. Among them, 6 patients with the A2142G mutation were excluded, leaving 115 patients with the A2143G mutation for analysis. Out of these 115 patients, retrospective analysis was conducted on 82 patients who underwent UBT testing at least 4 weeks after completing the antibiotic regimen and had complete medical records.

Among the 82 patients analyzed, 51 had received the conventional 2-week PBMT while 31 had received a 1-week PBMA. Propensity score matching was performed for gender, age, and body mass index, factors suspected to influence the efficacy of eradication, resulting in 25 matched cases in each group.

The baseline characteristics, including age and gender distributions, did not differ significantly between the two groups of analyzed subjects.

The efficacy of eradication therapy was confirmed through UBT conducted 4-8 weeks after treatment. In the analysis before propensity score matching (PSM), the eradication rates for the 2-week PBMT group and the 1-week PBMA group were 70.6% and 77.4% in the intention-to-treat (ITT) analysis, respectively, and 94.7% and 85.7% in the per-protocol (PP) analysis, respectively. There was no significant difference between the two groups in both analyses.

After PSM, the analysis still showed no significant difference between the 2-week PBMT group and the 1-week PBMA group. In the ITT analysis, the eradication rates were 68.0% for the OBMT2w group and 80.0% for the OAM+B1w group, while in the PP analysis, the rates were 94.4% for the OBMT2w group and 87.0% for the OAM+B1w group.

Regarding adverse events, there was no difference in the frequency of adverse events between the two groups. However, the need to discontinue medication occurred in 5.3-5.6% of cases in the 2-week PBMT group but did not occur in the 1-week PBMA group. Nonetheless, changes in bowel habits were more common in the 1-week PBMA group, with the main reason being a change to dark-colored stool.

Conclusion: For patients infected with H. pylori carrying the A2143G point mutation, there was no difference in the eradication rate between the 2-week PBMT regimen and the 1-week PBMA regimen. The 1-week PBMA regimen showed superior drug compliance and a tendency for fewer adverse effects compared to the 2-week PBMT regimen.

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Disclosure: Nothing to disclose.

PP0279

INITIAL RESULTS OF THE FOLLOW-UP OF A GISTAR PARTICIPANT COHORT FOR ASSESSING THE POTENTIAL NEGATIVE LONG-TERM EFFECTS OF *HELICOBACTER PYLORI* TEST AND TREAT

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Introduction: These are the initial results based on currently ongoing follow-up of the participants in the GISTAR study in Latvia (Multicentric randomized study of *H. pylori* eradication and pepsinogen testing for prevention of gastric cancer mortality) within the EUROHELICAN project.

Aims & Methods: The aim was to investigate the potential adverse health effects of the *Helicobacter pylori* (*H. pylori*) test-and-treat strategy in the long term. Baseline data on sociodemographic and lifestyle factors, medical history, body mass index (BMI), waist to hip ratio (WHR), and blood pressure were collected 5 to 11 years ago. Blood samples were also drawn. Participants in the intervention group were tested for *H. pylori* (initially with serology, then urea breath test (UBT)). Those who tested positive were offered eradication therapy. Participants in the control group were not tested for *H. pylori* and received regular healthcare. During follow-up, data on the same characteristics were collected. UBT was performed for those who were *H. pylori* positive at baseline. The symptom-based GERDQ score was assessed at the follow-up. The change in parameters from baseline to follow-up was examined between groups using chi-square and Mann-Whitney tests in SPSS 29.0.

Results: A total of 1305 GISTAR participants were recruited at follow up within EUROHELICAN. Their median age was 60 years (IQR 11). Among them, 57 % were female, of whom 691 (53%) were in the intervention group. Baseline data on *H. pylori* status and eradication were available for 687 participants, of whom 423 (61.6%) were positive at baseline.

When comparing differences in the change of characteristics between baseline and follow-up in the treated (n = 356, 51.8%) and non-treated groups (331, 48.2%), there were no significant differences in BMI, WHR, systolic and diastolic blood pressures (Table 1).

When examining changes in characteristics of participants in the eradicated (n = 281, 84%) and non-eradicated (53, 15.9%) groups, there were no significant differences in BMI and blood pressure. There was a significant difference in the WHR change.

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Characteristics	Treated (n = 356, 51.8%)	Not treated (n = 331, 48.2%)	p value	Eradicated (n = 281, 84%)	Non-eradicated (n = 53, 15.9%)	p value
	Median, interquartile range (IQR)	Median, interquartile range (IQR)		Median, interquartile range (IQR)	Median, interquartile range (IQR)	
BMI	1.01 (2.55)	0.91 (2.90)	0.38	1.06 (2.65)	1.05 (2.61)	0.75
WHR	0.00 (0.05)	0.01 (0.05)	0.06	0.0003 (0.05)	0.02, 0.07	0.03

Table 1. Comparison of the change in values from baseline to follow-up between groups.

Using GERDQ at least 8 points, 40 (11.3%) met the cut-off for GERD in the treated group versus 48 (14.9%) in the non-treated group (p = 0.16). The prevalence of GERD was 30 (10.9%) in the eradicated versus 5 (9.6%) in the non-eradicated group (p = 0.78).

Conclusion: The project will provide data missing on the potential negative effects of eradication. So far, no negative effects were confirmed. The study is ongoing, and a larger number of participants will be included. More data will be available to better assess whether there is a significant change in parameters covered in this abstract as well as additional variables pertaining to GERD, and potential metabolic and cardiovascular effects.

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Disclosure: Nothing to disclose.

PP0280

EFFECTIVENESS OF THE FIRST-LINE ERADICATION THERAPY IN 14 CITIES IN RUSSIA: RESULTS FOR THE PERIOD 2013-2022 OF THE EUROPEAN REGISTRY ON *HELICOBACTER PYLORI* MANAGEMENT (HP-EUREG)

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Introduction: According to recent studies, the prevalence rate of *H. pylori* infection in Russia is over 30%, but a decreasing trend has recently been observed. Comprehensive multicenter evaluation of effectiveness of different eradication treatments in Russia has not yet been performed. One of the most efficient tools of such analysis is the on-going European Registry on *H. pylori* Management (Hp-EuReg), an international, prospective, multicenter, observational study, which begun in 2013.

Aims & Methods: The objective of our study was to evaluate the effectiveness of first-line empirical eradication therapy in Russia between 2013-2022 according to the data of Hp-EuReg. The treatment effectiveness was analysed by per-protocol (PP), and modified intention-to-treat (mITT), also stratifying by PPI dose (low dose - 4.5 to 27 mg omeprazole equivalent, twice daily; standard dose - 32 to 40 mg omeprazole equivalent, twice daily; high dose - 54 to 128 mg omeprazole equivalent, twice daily) and duration of treatment (7, 10 or 14 days).

Results: Overall, 8,354 patients from 14 cities in Russia who had received a first-line empirical eradication therapy between 2013 and 2022, were included. Invasive (i.e. endoscopic) methods (rapid urease test, histology/cytology) for the *H. pylori* primary evaluation were mostly used in almost all cities. The highest rate of ^{13}C-urea breath test was observed in Rostov-on-Don (89% of cases), Saint-Petersburg (37%), Omsk (25%), Perm (25%), and Moscow (23%). The most frequently used first-line treatments were: triple (PPI, amoxicillin, clarithromycin) and triple plus bismuth (PPI, amoxicillin, clarithromycin, bismuth) therapies, accounting for more than 60% of prescriptions in almost all cities. Most cities prescribed 14-day length therapies; whereas standard PPI doses were most frequently used in Chelyabinsk (94%), Smolensk (84%), Saint-Petersburg (68%), Moscow (67%), Khabarovsk (66%) and Rostov-on-Don (54%), low PPI doses prevailed in Krasnoyarsk (80%), Izhevsk (70%), Novosibirsk (65%), Omsk (60%), and Kazan (52%). Quadruple therapy with PPI, amoxicillin, clarithromycin and bismuth achieved optimal (>90%) mIIT effectiveness in Saint-Petersburg, Perm, Chelyabinsk, and Khabarovsk, where it was more frequently used than in the remaining cities. Effectiveness of triple therapy (PPI, amoxicillin, clarithromycin) varied from 62% (Novosibirsk) to 99% (Omsk), achieving over 85% across all cities. However, high eradication rates with triple therapy prescriptions were observed in the cities with a small number of

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patients or in those cases where histology were the unique tests used for eradication control, possibly giving false negative results. Classic bismuth quadruple therapy (PPI, tetracycline, metronidazole, bismuth) provided optimal (>90%) mITT effectiveness in Moscow and Saint-Petersburg, where it was predominately used as compared to other cities (table 1).

Eradication therapy	Moscow	St. Peters-burg	Kazan	Perm	Kovrov	Chelyab-insk	Khaba-rovsk	Rostov-On-Don	Smolensk	Che-boksary	Omsk	Kras-noyarsk	Novosi-birsk	Izhevsk	Overall
Triple therapy (PPI+A+C)	83.4%	87%	77.2%	89.2%	76.1%	96.3%	72.6%	88.3%	78%	94%	99.1%	81.7%	62%	100%	83.1%
Quadruple therapy (PPI+M+Tc+ bismuth)	92.1%	91.4%	100%	100%	80%	100%	100%	-	100%	100%	100%	-	-	-	92.9%
Quadruple therapy (PPI+A+C+ bismuth)	88.3%	93.1%	86.4%	95.2%	87.3%	100%	92.4%	100%	86.7%	96.2%	100%	-	71.4%	91.2%	92.7%

Table.1 Effectiveness by modified intention-to-treat of most common first-line empirical therapies in 14 cities of Russia in 2013-2022

PPI: proton pump inhibitor; A: amoxicillin; C: clarithromycin; M: metronidazole; B: bismuth salts; Tc: tetracycline

Conclusion: To date, in Russia, bismuth quadruple therapies either with classical metronidazole-tetracycline-bismuth or amoxicillin-clarithromycin-bismuth achieve optimal (> 90%) results, providing significantly better rates when prescribed for 14 days

Disclosure: Nothing to disclose.

PP0281

TREATMENT PATTERNS IN CHINESE PATIENTS WITH HELICOBACTER PYLORI: A MULTICENTER, RETROSPECTIVE, REAL-WORLD STUDY

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Introduction: *Helicobacter pylori* (HP) is regarded as a class I carcinogenic factor and eradication therapy can reduce the risk of gastric cancer. Patients with HP infections are administered with bismuth quadruple therapy (BQT), triple and dual therapies based on their eradication effectiveness. BQT (bismuth + acid suppressants + 2 antibiotics) and dual therapy are both recommended in China now. Comprehensive understanding on the distribution of current eradication therapies is required to identify gaps observed in real-world practice.

Aims & Methods: This multicenter, retrospective, real-world study (NCT05073367) extracted preexisting electronic data including adult patients (aged ≥18 years) with HP infection between January 1, 2019 and March 31, 2021 from 3 hospitals to assess the diagnosis and treatment pattern of HP infection and effectiveness of eradication therapy in China. In this abstract, we reported the distribution of eradication therapies, difference in initiation year for treatment, and treatment duration.

Results: Of the 22,971 eligible patients with HP infection, 13,233 (57.61%) were prescribed eradication regimens. Mean ± SD age of patients with eradication treatment was 43.7 ± 14.0 years; 65.94% of patients had medical insurance; and 54.50% of patients were women.

In patients who received eradication therapy, the BQT therapy was predominantly prescribed (85.39%; n=11301), followed by triple therapy (7.55%; n=999) and dual therapy (7.06%; n=934). Among those receiving BQT, use of amoxicillin + clarithromycin based regimens (38.72%; n=4376) were the most prevalent, followed by amoxicillin +furazolidone (18.35%; n=2074) and amoxicillin + levofloxacin (17.76%; n=2007) based combination regimens. Proportion of patients receiving BQT decreased over years: 97.24% patients in 2019, 79.98% in 2020, and 78.71% in 2021. The proportion of patients receiving dual and triple therapy increased over the years in 2019, 2020, and 2021 were shown in Table 1. A total of 10,903 patients were analyzed according to various treatment durations with the usage of various eradication therapies. Treatment duration is defined as the number of days that prescribed drugs can cover. Out of the three types of treatments, BQT had the highest proportion of 14-day treatment duration at 90.16%, followed by triple therapy at 76.78%, while dual therapy had the least proportion at 14.42%. For treatment duration >14 days, BQT, dual and triple therapy had a proportion of 4.38%, 82.35%, and 17.27% respectively.

Treatment	Type of Eradication therapy (N=13, 234) n (%)[a]	Year of treatment initially prescribed , 2019 (N=4492) n (%)[a]	Year of treatment initially prescribed, 2020 (N=4125) n (%)[a]	Year of treatment initially prescribed, 2021 (N=4616) n (%)[a]
Total	13,234[b] (-)	4492 (-)	4125 (-)	4617 (-)
Bismuth quadruple therapy	11,301 (85.39)	4368 (97.24)	3299 (79.98)	3634 (78.71)
Amoxicillin and Clarithromycin	4376 (38.72)	1800 (41.21)	1324 (40.13)	1252 (34.45)
Amoxicillin and Furazolidone	2074 (18.35)	689 (15.77)	480 (14.55)	905 (24.90)
Amoxicillin and Levofloxacin	2007 (17.76)	1021 (23.37)	516 (15.64)	470 (12.93)
Amoxicillin and Metronidazole	76 (0.67)	27 (0.62)	10 (0.30)	39 (1.07))
Amoxicillin and Tetracycline	9 (0.08)	5 (0.11)	1 (0.03)	3 (0.08)
Furazolidone and Tetracycline	12 (0.11)	1 (0.02)	0 (0.00)	11 (0.30)
Other Bismuth quadruple	2747 (24.31)	825 (18.89)	968 (29.34)	954 (26.25)
Triple therapy[c]	999 (7.55)	105 (2.34)	551 (13.36)	343 (7.43)
Dual therapy[c]	934 (7.06)	19 (0.42)	275 (6.67)	640 (13.86)

Table 1. Distribution of eradication therapy based upon the type and year of treatment initiation

Abbreviations: N, total number; n, number in respective category.

[a]% = The total usage count of BQT or Triple or Dual eradication therapy/the total usage count of all standard eradication therapies received by eligible participants. Of BQT, the total usage count of different BQT/the total usage count of all BQT.

[b]There is a recurrent patient who changed the treatment pattern from triple therapy (ilaprazole, clarithromycin, levofloxacin) to dual therapy (ilaprazole, ornidazole) after the recurrence.

[c]Considering the table size, outcomes of different treatment regimens under dual and triple therapy were not included in the table.

Conclusion: BQT-based regimen for 14 days was the predominant eradication regimens in all therapies, amoxicillin + clarithromycin based BQT was the most prevalent. Use of dual therapy gradually increased with years for HP eradication in the Chinese real-world clinical practice.

Disclosure: The study was funded by Takeda China. Wensheng Pan, Yan Zhao, and Hongwei Xu have no conflicts of interest. Pengfei Li, Fang Zhou, Li Xie are the employees of Takeda, and they hold Takeda stock options. Medical writing for the development of this abstract was provided by Kusuma Kumari G, PhD, of Indegene Private Limited, and funded by Takeda China. Authors retained full control of the abstract content.

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PP0282

TREATMENT EFFECTIVENESS OF DIFFERENT ERADICATION TREATMENTS IN CHINESE PATIENTS WITH HELICOBACTER PYLORI: A RETROSPECTIVE REAL-WORLD STUDY

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Introduction: *Helicobacter pylori* (*H. pylori*) infection has a high prevalence (44.8%) in China and is an important risk factor of gastrointestinal diseases including gastric cancer. Bismuth quadruple therapy (BQT; bismuth + acid suppressant + 2 antibiotics [ABs]) is the currently recommended first-line regimen for *H. pylori* in China. Increasing AB resistance poses a vital challenge and remarkably reduces the eradication rate (ER) from 95% to 80% globally and 89% to 78% in China. Real-world ER analysis assessed only quadruple therapy in a few single centers and did not focus on different antibiotic combinations and other different regimens.

Aims & Methods: This study was designed to evaluate the diagnosis and treatment pattern of H. pylori infection and the effectiveness of eradication therapy in real-world settings in China. In this retrospective, multicenter study (NCT05073367), de-identified patient data were extracted from the electronic medical records database of 3 hospitals between January 1, 2019, and March 31, 2021. Patients aged ≥18 years with a record of *H. pylori* infection diagnosis were included. Here we presented the eradication rate of different treatment regimens and the eradication rate of different years with different treatment regimens.

Total (N=1426) → /Treatment regimens ↓	ER-success n (%)[b]	Year 2019 Success n (%)[b]	Year 2020 Success n (%) [b]	Year 2021 Success n (%)[b]
Total	1028 (72.09)	454 (77.87)	253 (60.96)	321 (75.00)
Bismuth Quadruple Therapy	879 (76.70)	443 (77.86)	186 (67.64)	250 (82.78)
Amoxicillin and Clarithromycin	391 (76.07)	169 (78.24)	106 (67.95)	116 (81.69)
Amoxicillin and Furazolidone	181 (87.44)	55 (87.30)	44 (83.02)	82 (90.11)
Amoxicillin and Levofloxacin	159 (77.94)	126 (79.25)	10 (55.56)	23 (85.19)
Amoxicillin and Metronidazole	2 (50.00)	2 (50.00)	0 (0.00)	0 (0.00)
Other Bismuth Quadruple Therapy	146 (67.28)	91 (71.65)	26 (54.17)	29 (69.05)
Triple therapy[c]	59 (48.36)	11 (78.57)	37 (46.25)	11 (39.29)
Dual therapy[c]	90 (56.96)	0 (0.00)	30 (50.00)	60 (61.22)

Table 1. Eradication rate of H. pylori positive patients based on the type of eradication therapy.

Abbreviations: N, total number; n, number in respective category.
[a]% = The total usage count of BQT/Triple therapy/Dual therapy eradication therapy/ the total usage count of all standard eradication therapies taken by eligible participants. Of BQT, the total usage count of different BQT/the total usage count of all BQT.
[b]% = The number of participants with a negative efficacy evaluation result/the number of participants receiving certain eradication therapy.
[c]Considering the table size, outcomes of different treatment regimens under dual and triple therapy were not included in the table.

Results: A total of 1426 patients received ET with efficacy evaluation duration>4 weeks after the eradication therapy and the total eradication rate was 72.09% (n=1028). Most patients (80.36%; n=1146) received standard BQT with an ER of 76.70% (n=879). 158 patients (11.08%) received dual therapy (ER=56.96%; n=90) and 122 patients (8.56%) received triple therapy (ER=48.36%; n=59). Compared with triple and dual therapies, BQT was the predominant regimen in all the years with an ER of 77.86%, 67.64%, and 82.78% in 2019, 2020, and 2021, respectively. The ER of dual therapy was 50% in 2020 and 61.22% in 2021. The triple therapy ER showed a decreasing trend with 78.57% (n=11), 46.25% (n=37), 39.29% (n=11) from 2019 to 2021. Among the BQT regimens, amoxicillin plus furazolidone based regimens showed a higher ER (87.44%; n=181), while the ER of amoxicillin plus levofloxacin, amoxicillin plus clarithromycin based BQT was 77.94% (n=159) and 76.07% (n=391), respectively. Amoxicillin plus furazolidone based BQT showed a higher ER of 87.30%, 83.02%, and 90.11% in 2019, 2020, and 2021 respectively (Table 1).

Conclusion: BQT-based regimens were predominantly prescribed for *H. pylori* management and the current *H. pylori* treatment regimens have a suboptimal eradication rate in the real-world settings.

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PP0283

ESOMEPRAZOLE/VONOPRAZAN COMBINED WITH HIGH-DOSE AMOXICILLIN DUAL THERAPY FOR PRIMARY ERADICATION OF *HELICOBACTER PYLORI* INFECTION: A MULTI-CENTER, PROSPECTIVE, OPEN-LABEL, RANDOMIZED CONTROLLED CLINICAL TRIAL

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Introduction: Due to certain shortcomings of traditional triple and quadruple therapy for *Helicobacter pylori* (*H. pylori*), high-dose dual therapy (HDDT) is proposed as another option.

Aims & Methods: Our study aims to evaluate the efficacy and safety of HDDT, and provide more reference for clinical eradication of *H. pylori*. The subjects were untreated *H. pylori* patients from three medical centers in Central China. Totally, 439 subjects were randomly assigned to EA (esomeprazole-amoxicillin) or B-quadruple (esomeprazole-amoxicillin-clarithromycin-bismuth) group. A total of 367 subjects were randomly assigned to VA (vonoprazan-amoxicillin) or VAC (vonoprazan-amoxicillin-clarithromycin) group. The effect, adverse events, compliance, symptom relief and cost were recorded.

Results: EA-dual was non-inferior to B-quadruple regimen in mITT and PP analysis (P<0.025). But the eradication rate of EA was lower than that of B-quadruple group (70.59% vs. 83.49%, 92.86% vs. 98.38%, 93.94% vs. 98.38%, ITT, mITT, PP respectively, P<0.05). In ITT, mITT and PP analysis, VA-dual was non-inferior to VAC treatment (84.15% vs. 83.15%, 96.25% vs. 92.73%, 96.75% vs. 93.75%, P<0.025).

There was no significant difference in adverse events, compliance, and symptom relief between groups. Cost of VA was the lowest. Antibiotics use within 2 years, poor compliance and suburban area residence reduced the eradication efficacy (P<0.05).

Conclusion: The HDDT based on esomeprazole or vonoprazan was non-inferior to the classical B-quadruple or VAC triple regimen. So, the HDDTs should be recommended as first-line therapies of *H. pylori* eradication. Antibiotics use in 2 years, adherence and residence were crucial for eradication.

Disclosure: The authors declare that they have no conflict of interest.

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PP0284

EVALUATION OF THE EFFECTIVENESS AND SAFETY OF *HELICOBACTER PYLORI* EMPIRICAL ERADICATION REGIMENS IN SWITZERLAND: AN INTERIM ANALYSIS WITH DATA FROM THE EUROPEAN REGISTRY ON *HELICOBACTER PYLORI* MANAGEMENT (HP-EUREG)

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Introduction: Prevalence of Helicobacter pylori (H. pylori) infection remains high. Despite the reported relatively low prevalence in Switzerland, the real burden is considered to be much higher, primarily due to the phenomenon of migration. Till now, there is scarce evidence regarding effectiveness of H. pylori eradication schemes in Switzerland.

Aims & Methods: Sub-study of the "European Registry on *H. pylori* Management" (Hp-EuReg), an international, multicenter, prospective non-interventional registry of the routine clinical practice. All infected adult patients were systematically registered at AEG-REDCap e-CRF from 2013 to December 2023. Swiss data were analysed for effectiveness on a modified intention-to-treat (mITT) basis.

Results: A total of 486 adult patients, diagnosed with *H. pylori* infection, were collected. Out of them, 428 (88.1%) were treatment-naïve patients. Two prescription regimes included more than 90% of the cases (triple with Clarithromycin and Amoxicillin in 49% and single-capsule bismuth quadruple in 42%). For the effectiveness analysis, 280 naïve patients were eligible, due to their availability of a confirmatory *H. pylori* eradication test. The overall effectiveness of 1st line empirical therapy was 92% by mITT. The lowest effectiveness was recorded with triple therapy with amoxicillin and clarithromycin (28/34 cases, 82.4 % for seven days duration, and 86.9% overall (126/145). On the other hand, the highest effectiveness was reached with the single-capsule bismuth quadruple therapy (97.1%, 100/103). In respect to potency of gastric acid inhibition by proton pump inhibitors (PPI), the group with low-dose (i.e. 20 mg omeprazole equivalent twice daily) PPI yielded an overall eradication success of 90.8%, whereas for the standard-dose (i.e. omeprazole equivalent 40mg bid) and high-dose (i.e. omeprazole equivalent 80mg bid) PPI inhibition the rates increased up to 100% and 96.3%, respectively. Regarding safety, the eradications treatments were generally well tolerated, and serious adverse events were reported in 16 cases (5.7%). Mild adverse effects were rather uncommon and included dysgeusia (n=1), diarrhoea (n=6), nausea (n=6) and dyspepsia (n=3).

Conclusion: This first interim analysis of Swiss population as sub-study of the Hp-EuReg suggests that bismuth quadruple therapy prescribed as single-capsule represents a reliable option as first-line empirical therapy.

Disclosure: Nothing to disclose.

PP0285

EFFECTIVENESS OF EMPIRICAL FIRST-LINE *H. PYLORI* ERADICATION THERAPY IN ITALY FROM 2013 TO 2023: RESULTS FROM THE EUROPEAN REGISTRY ON *H. PYLORI* MANAGEMENT (HP-EUREG)

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Introduction: It is not yet clear what is the best first-line empirical therapy in patients with *Helicobacter pylori* (*H. pylori*) infection. For this reason, it is important to constantly monitor the effectiveness of the different therapies used in clinical practice.

Aims & Methods: The aim of this study was therefore to assess the effectiveness of empirical first-line treatment prescribed in Italy evaluating the European Registry on *H. pylori* Management (Hp-EuReg). The Hp-EuReg is an international multicentre prospective non-interventional registry recording information on *H. pylori* infection management since 2013. All infected adult patients enrolled were systematically registered at AEG-REDCap e-CRF. Data of all *H. pylori*-positive treatment-naïve patients enrolled from 2013 to December 2023 by Italian centres participating in the Hp-EuReg, were extracted, subject to quality control, and analysed. The effec-

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tiveness of the first-line treatment empirically prescribed was evaluated according to the modified intention-to-treat (including all patients that completed follow-up, *i.e.*, a confirmatory test of success or failure available after eradication treatment), and per-protocol analysis (including all patients that completed follow-up and took at least 90% of the treatment drugs).

Results: In total, 4,144 patients were included for analysis; the mean age was 52 years (SD ± 5), and 60% of patients were women. Overall, 1.0% of the cases were allergic to penicillin. Among those non-allergic to penicillin, 10-day sequential therapy (mainly as PPI-clarithromycin-amoxicillin-tinidazole) was most commonly prescribed (54%), followed by 10-day single capsule bismuth quadruple therapy with metronidazole and tetracycline (19 %), and 10 or 14-day non-bismuth concomitant treatment (mainly as clarithromycin-amoxicillin-tinidazole) (9.3%), providing these regimens optimal cure rates (91.5%, 95.5%, 92% and 97%; respectively). Additional first-line prescriptions encompassed 7, 10 or 14-day triple therapy with PPI-clarithromycin-amoxicillin (5.5%), and 10-day triple therapy with PPI- -levofloxacin-amoxicillin (2.9%) (see Table 1).

First-line treatment	Length (days)	mITT % (95% CI)	PP % (95% CI)
Sequential (PPI + C-A-T/M)	10	91.5 (90.2 to 92.7)	90.9 (89.5 to 92.1)
PPI + Single capsule (M+Tc+B)	10	95.5 (93.7 to 96.8)	96.6 (94.9 to 97.7)
Concomitant (PPI + C-A-M/T)	10	92.2 (86.6 to 95.6)	93.5 (88.1 to 96.5)
	14	96.6 (93.6 to 98.4)	98.1 (95.6 to 99.2)
Triple PPI-C-A	7	83.6 (72.9 to 90.6)	86.2 (75.7 to 92.5)
	10	80 (71.1 to 86.7)	80.4 (71.4 to 87.1)
	14	100 (0.77 to 100)	100 (0.77 to 100)
Triple PPI-A-L	10	81.4 (72.7 to 87.7)	82.8 (74.2 to 89)

mITT: modified intention-to-treat; PP, per protocol; A, amoxicillin; B, bismuth salt; C, clarithromycin; L, levofloxacin; M, metronidazole; PPI, proton pump inhibitor; T, tinidazole; Tc, tetracycline.

Table 1; Effectiveness by modified intention-to-treat (mITT), and per-protocol (PP) analyses of first-line empirical treatment in patients not allergic to penicillin enrolled in the Italian centres participating in the European Registry on H. pylori management (Hp-EuReg).

Conclusion: Among the first-line empiric treatments prescribed by the Italian centres participating in the Hp-EuReg, only single-capsule bismuth quadruple therapy with metronidazole-tetracycline-bismuth, sequential therapy with clarithromycin-amoxicillin-tinidazole, and non-bismuth concomitant therapy with clarithromycin-amoxicillin-tinidazole, achieved over 90% effectiveness.

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PP0286

NON-BISMUTH QUADRUPLE CONCOMITANT TREATMENT FOR *H. PYLORI* ERADICATION: SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: *Helicobacter pylori* eradication with standard triple therapy fails in ≥20% of the cases, mainly due to bacterial antibiotic resistances. Non-bismuth quadruple concomitant therapy is among the most widely used first-line treatments against *H. pylori* infection, even in high clarithromycin resistance areas.

Aims & Methods: To evaluate the efficacy and safety of non-bismuth quadruple concomitant therapy (proton pump inhibitor, clarithromycin, amoxicillin and nitroimidazole) in the treatment of *H. pylori* infection, and to compare it with other triple and quadruple regimens.

Bibliographical searches were performed in PubMed up to April 2024. We performed a meta-analysis of the studies evaluating the concomitant therapy as a first-line regimen, and of the randomised controlled trials comparing concomitant vs. standard triple, sequential, hybrid, and bismuth quadruple therapies.

Results: 115 studies (78 randomised trials) were included (23,454 patients); 39 studies compared concomitant vs triple (13,494 patients), 38 studies vs sequential (11,907 patients), 12 studies vs hybrid (3,719 patients) and 11 studies vs bismuth quadruple (3,258 patients). Concomitant regimen achieved an overall eradication rate of 86% (95%CI=85-86%) by intention-to-treat, and 90% (90-91%) by per-protocol. Concomitant therapy achieved high eradication rates even in single clarithromycin resistant (89%) and metronidazole resistant (95%) patients, but poor cure rates in dual (clarithromycin and metronidazole) resistance (68%). The overall efficacy by intention-to-treat with concomitant therapy was significantly higher as compared to both triple therapy (risk difference [RD]=0.11; 95%CI=0.09-0.14; 39 studies) and sequential therapy (RD=0.04; 0.02-0.06; 38 studies); however, in the latter comparison, when only 14-day prescriptions were considered, no significant differences were observed (RD=0.01; -0.05-0.06; 4 studies). Likewise, the overall efficacy of concomitant therapy was similar to hybrid (RD=0.01; -0.01-0.03; 12 studies), and to bismuth quadruple therapy (encompassing the classical form and the single-capsule (RD=0.01; -0.03, 0.06; 11 studies). The overall incidence of adverse events with concomitant therapy was 37% (36-38%) (85 studies). When comparing only the studies with the same treatment durations, the incidence of adverse events with concomitant therapy was similar to that of triple (RD=0.04; -0.01-0.09; 15 studies), sequential (RD=0.03; -0.00-0.06; 16 studies), and bismuth quadruple therapies (RD=0.02; -0.06-0.09; 7 studies), but higher than that of hybrid therapies (RD=0.10; 0.04-0.15; 7 studies).

Conclusion: Non-bismuth quadruple concomitant therapy is highly effective for *H. pylori* eradication (achieving a cure rate of approximately 90%), even in the presence of clarithromycin or metronidazole resistance, being superior to standard triple and sequential therapies, and similar to hybrid and bismuth quadruple regimens. The incidence of adverse events with concomitant treatment is relatively high (37%), similar to that of triple and bismuth quadruple regimens.

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PP0287

LEVOFLOXACIN VERSUS MOXIFLOXACIN IN NITAZOXANIDE-BASED QUADRUPLE THERAPY AS A FIRST- AND SECOND-LINE TREATMENT FOR *HELICOBACTER PYLORI* INFECTION. A RANDOMIZED, COMPARATIVE, MULTICENTER STUDY

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Introduction: Increasing resistance to antibiotics represents a significant challenge for the successful eradication of *Helicobacter pylori (H. pylori)*. Therefore, there is a growing need to find novel modalities to improve treatment outcomes.

Aims & Methods: This study aimed to assess the efficacy and safety of levofloxacin- or moxifloxacin-containing nitazoxanide-based regimens for eradicating *H. pylori* as first- and second-line therapies. Patients with dyspeptic symptoms and a positive *H. pylori* antigen test in the stool were eligible for inclusion in this study. Naïve patients were randomized into three groups according to treatment regimens: levofloxacin, nitazoxanide, doxycycline, and lansoprazole (LNDL); moxifloxacin, nitazoxanide, doxycycline, and lansoprazole (MNDL); or amoxicillin, clarithromycin, and lansoprazole (ACL) for 14 days. Treatment-experienced patients received the LNDL or MNDL regimen. *H. pylori* eradication was confirmed by stool antigen testing at least 4 weeks after the cessation of therapy.

Results: In naïve patients, eradication rates were significantly greater by intention to treat (ITT) analysis in patients treated with the LNDL regimen (83.3%) than in those treated with standard triple therapy (ACL) (59.2%). According to the per-protocol analysis (PP), both the LNDL and MNDL regimens had significantly greater eradication rates (87.7% and 81.6%, respectively) than did the ACL regimen (60.9%). Regarding treatment-experienced patients, eradication rates were comparable for the LNDL and MNDL regimens in both ITT (65.9% and 77.5%, respectively) and PP analyses (72.5% and 82.7%, respectively). All treatment regimens were well tolerated.

H. pylori stool antigen test result	Eradication regimen			p-value
	LNDL	MNDL	ACL	
Intention to treat	n: 90	n: 102	n: 71	
Negative (%)	75 (83.33%)	71 (69.61%)	42 (59.15%)	0,003
95% CI	75.73-91.27	60.92-78.90	47.96-71.16	
Per-protocol	n: 86	n: 87	n: 69	
Negative (%)	75 (87.21%)	71 (81.61%)	42 (60.87%)	< 0.001
95% CI	80.21-94.47	73.57-90.01	49.57-72.95	

Table 1 Eradication Rates of Helicobacter pylori infection in treatment-naïve patients

Conclusion: levofloxacin- and moxifloxacin-containing nitazoxanide-based regimens exhibit substantial efficacy for the eradication of both naïve and resistant *H. pylori* infections over a 14-day course with a considerable safety profile.

Disclosure: The authors declare to have no fompeting interests related to this work.

PP0288

EFFICACY OF TEGOPRAZAN-BASED BISMUTH QUADRUPLE THERAPY COMPARED WITH BISMUTH QUADRUPLE THERAPY FOR HELICOBACTER PYLORI INFECTION: A REAL WORLD EVIDENCE EXPERIENCE

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Introduction: Bismuth-based quadruple therapy (BQT) is recommended as the first-line therapy for Helicobacter pylori eradication as it is not associated with resistance. Tegoprazan, a new, fast, and strong potassium-competitive acid blocker, has been approved for the treatment of gastric acid-related diseases in México. However, real-world clinical data regarding this drug are scarce. We aimed to compare the Helicobacter pylori eradication rates of tegoprazan- and omeprazole-based quadruple therapy.

Aims & Methods: Aim: To investigate the efficacy and safety profiles of tegoprazan-based BQT (TBMT) versus omeprazole-based BQT (OBMT) for H. pylori eradication.

Methods: We included patients older than 18 with an H. pylori infection without a history of H. pylori eradication who visited two centers between January 2024 and April 2024. H. pylori infection was diagnosed using histology. Patients received tegoprazan 50 mg BID , bismuth subcitrate 300 mg QID, metronidazole 500 mg QID, and doxycycline 100 mg QID for 14 days. An historic control groups was used as comparator. Historic controls were age-sex matched subjects that received omeprazole 40 mg BID , bismuth subcitrate 300 mg QID, metronidazole 500 mg QID, and doxycycline 100 mg QID, in the previous year.

Results: Until now, 80 patients have been included (40 TBMT and 40 OBMT). Most common indication for erradication was dyspeptic symptoms (78%), followed by peptic ulcer disease (10%), folicular gastritis (9%) and familiar hystory of cance (2.5%). Eradication rates of TBMT and OBMT were 90% and 77.4% (=OR 2.61, CI 95% 0.73-9.37, p=0.12), respectively. There was no significant difference in the rate of adverse events between the TBMT and OBMT groups (42.5% vs. 55,5%, p = 0.37). Nausea was the most common adverse effect (37.5% vs 40%, p=0.92) , followed by abdominal pain (12.5% vs 17.5%, p=0.16%)Adherence to the treatments was similar between groups (> 90%, 10% vs 12%, p=0.98)

Conclusion: TBMT showed a noninferior eradication rate and similar adverse events to OBMT as a first-line eradication regimen. Our results suggest that tegoprazan might be substituted for proton pump inhibitors in H. pylori eradication regimens in Mexico. A further randomized controlled trial es required to confirm our findings.

Disclosure: JMRT has served as advisory board for Takeda, Asofarma, Carnot and and Biocodex. Also is a speaker for Takeda, Asofarma, Alfasigma, Carnot, Chinoin, Medtronic, Ferrer, and Sanofi. ECA and LRVD are speakers for Carnot and Asofarmna. ACC and GM do not have any conflict of interest.

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PP0289

FIRST-LINE PRESCRIPTION PATTERNS AND ERADICATION RATES IN IRELAND OVER A 10-YEAR PERIOD: DATA FROM THE EUROPEAN REGISTRY ON *HELICOBACTER PYLORI* MANAGEMENT (HP-EUREG)

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Introduction: The emergence of antimicrobial resistant *Helicobacter pylori* (*H. pylori*) has impacted treatment success rates. Local audits of prescription patterns and eradication rates are necessary to assess treatment efficacy and compliance with clinical guidelines. The first Irish consensus guidelines on the management of *H. pylori* were published in 2017. Clarithromycin-amoxicillin triple therapy (CTT) or bismuth quadruple therapy (BQT) were the recommended first-line options. Other key recommendations included extending treatment durations to 14 days and the use of high-dose proton pump inhibitors (PPIs).

Aims & Methods: This study aimed to (i) investigate first-line *H. pylori* prescription patterns and cure rates over a 10-year period in Ireland, and (ii) evaluate the impact of the first Irish consensus guidelines on these trends. Data were collected at e-CRF AEG-REDCap from the European Registry on *Helicobacter pylori* Management (Hp-EuReg) and quality reviewed from 2013 to 2022. All treatment-naïve cases were assessed for effectiveness by modified intention-to-treat (mITT) analysis. Multivariate analysis was performed using a logistic regression model to evaluate the relationship between mITT rates and age, sex, indication, compliance, PPI dose and treatment duration.

Results: In all, data from 1,000 patients (mean age 50±15 years; 54% (n=536) female) were included in the analysis, from those 78% (n=784) were from Tallaght University Hospital, Dublin and 22% (n=216) were from Beacon Hospital, Dublin. The most common indication was dyspepsia (92%; n=920), followed by duodenal ulcer and gastric ulcer (2%; n=20 and 1%; n=10, respectively). CTT was the most common first-line treatment (88%; n=880). BQT represented 1.7% (n=17) of prescriptions. Of the reported treatment durations (n=998), 87% (n=868) were for 14 days, 4.5% (n=45) were for 10 days and 8.5% (n=85) were for 7 days. Of the reported PPI dose prescribed (n=999), 86% (n=860) were high (i.e., 80 mg omeprazole equivalent twice daily), 0.9% (n=9) were standard (i.e., 40 mg omeprazole equivalent twice daily), and 13% (n=130) were low (i.e., 20 mg omeprazole equivalent twice daily).

The eradication rate was 80.1% (95% confidence interval (CI) 77.5-82.5%) overall and 81.4% (95% CI 78.9-83.7%) for CTT. Multivariate analysis revealed that good compliance and the use of high dose PPI were significantly associated with higher overall mITT rates (odds ratio (OR) 4.5, 95% CI 1.4-14.2, p<0.05 and OR 1.9, 95% CI 1.2-2.8, p< 0.005, respectively) and CTT mITT rates (OR 4.2, 95% CI 1.2-14.2, p<0.05 and OR 1.9, 95% CI 1.1-3.2, p<0.05, respectively).

Following publication of the Irish consensus guidelines in 2017, prescriptions for 14 days therapy increased from 39% (n=82/208) pre-2017 to 99% (n=786/790; p<0.0001) and the use of high dose PPI increased from 37% (n=77/207) pre-2017 to 99% (n=783/792; p<0.0001). In line with this, overall eradication rates significantly increased from 74.5% (95% CI 68.2-80.0%) pre-2017 to 81.6% (95% CI 78.7-84.1%; p<0.05). CTT eradication rates significantly increased from 75.7% (95% CI 68.2-81.8%) pre-2017 to 82.6% (95% CI 79.6-85.1%; p<0.05).

Conclusion: CTT was the most frequently prescribed first-line therapy between 2013-2022. BQT was rarely prescribed, reflecting difficulty accessing bismuth in Ireland. While treatment success increased following publication of the Irish consensus guidelines, cure rates remain below 90%. Alternative first-line treatment strategies are required in the Irish healthcare setting.

Disclosure: Nothing to disclose.

PP0290

PHARMACOGENOMICS OF ANTIBIOTIC RESISTANCE IN HELICOBACTER PYLORI FROM COLOMBIA

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Introduction: *Helicobacter pylori* is a microaerophilic gram-negative bacillus that colonizes the gastric mucosa and is associated with gastroduodenal diseases such as chronic gastritis and gastric adenocarcinoma. Although triple therapy is the recommended treatment for its eradication, its effectiveness has decreased due to the lack of knowledge about bacterial susceptibility patterns and the emergence of resistant strains, a phenomenon related to the bacterium's adaptation to hostile environments and antibiotic use. This infection is associated with gastroduodenal diseases such as chronic gastritis, peptic ulcer, and gastric adenocarcinoma, and although various therapeutic regimens are available, the indiscriminate use of antibiotics has led to resistance, highlighting the need to study alternatives and reevaluate the criteria that determine the selection of a specific therapeutic regimen.

Aims & Methods: We analyzed 169 *Helicobacter pylori* Whole Genome Sequences (WGS) from Colombian participants. Bioinformatics analyses were performed using the ATCC 26695 genome as a reference strain. We searched for mutations in specific genes to assess resistance rates to different antibiotics: *23S* rRNA for clarithromycin, *16S* rRNA for tetracycline, *pbp1A* for amoxicillin, *gyrA* for levofloxacin, *rdxA* for metronidazole, *rpoD* for rifampicin, and *porD* for furazolidone. The phylogenomic analysis was conducted using the core genome consisting of 1,594 genes of *Helicobacter pylori*-ATCC 26695

Results: Our findings revealed that the resistance rate of *Helicobacter pylori* to clarithromycin was 3.62%, mainly associated with the A2143G and A2142G mutations in the *23S* rRNA gene. For tetracycline, the resistance rate was 7.23%, with mutations A926G, A926T, and A928C observed in the *16S* rRNA gene. Resistance to amoxicillin was found in 25.9% of cases, with mutations observed in the *pbp1A* gene, including T556S, T593, R649K, R656P, and R656H. In the *gyrA* gene, mutations N87K, N87I, D91G, D91N, and D91Y were identified, resulting in a resistance rate of 12.04% to levofloxacin. The most common mutations in the *rdxA* gene associated with

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metronidazole resistance were a stop codon and mutations at D59N and D59S, resulting in a resistance rate of 99.3%. No mutations were found for rifampicin in the *rpoD* gene, and a resistance rate of 0.84% was found for furazolidone due to mutations C357T in the *porD* gene.

Conclusion: The high rate of *Helicobacter pylori* resistance to metronidazole indicates that this drug should be excluded from eradication therapy. However, resistance rates for tetracycline and clarithromycin did not exceed the 10% threshold established in Colombia. The increased resistance rate of *Helicobacter pylori* to levofloxacin and amoxicillin may partially explain the therapeutic failures observed in Colombia. As rescue therapy, rifampicin and furazolidone antibiotics can be used along with bismuth in *Helicobacter pylori* treatment, following the guidelines recommended by the Maastricht VI/Florence consensus report. Phylogenomic analysis showed that the *Helicobacter pylori* isolate belongs to its own lineage (hspColombia). These findings provide valuable information to improve the characterization of treatment protocols for the specific *Helicobacter pylori* lineage (hspColombia) at the local level.

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Disclosure: Nothing to disclose.

PP0291

EFFICACY AND SAFETY OF HELICOBACTER PYLORI ERADICATION THERAPY IN PORTUGAL: A BLIND RANDOMIZED CONTROLLED CLINICAL TRIAL

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Introduction: Helicobacter pylori (Hp) infection remains a major public health problem, associated with significant morbidity and mortality, whose eradication therapy has proven to be effective in preventing these complications.[1,2] Given the high resistance to antibiotics and the scarcity of available antibiotic therapy worldwide, there is not enough data on the best eradication strategy.[3,4]

Aims & Methods: Aim: The aim of this study was to perform a comparative evaluation of the efficacy and safety of the different quadruple therapy regimens currently recommended for the Hp eradication, including Esomeprazole 40 mg bid.

Methods: A blind randomized controlled clinical trial of consecutive patients with gastric infection by Hp, diagnosed by 13C-urea breath test (UBT) or histological analysis were randomized in 5 therapy arms (A: Quadruple with metronidazole 500 mg tid, 14 days; B: Quadruple with metronidazole 500 mg bid, 14 days; C: Quadruple with bismuth, 10 days; D: Sequential, 7 days; E: Hybrid, 7 days). Hp eradication with evaluation of the efficacy and safety rates at 1-2 months after the eradication therapy was made.

Results: From a total of 200 patients currently included, 193 concluded the protocol study (164-histological analysis; 29-UBT), 102 males (52.8%), mean age of 57.2±16.3 (A: 38; B: 39; C: 38; D: 41; E: 37). Overall efficacy rate of quadruple therapy regimens was 95.9% (n=185), being 100.0% for arm A, 92.3% for arm B, 97.3% for arm C, 97.6% for arm D and 91.9% for arm E (p=0.431). Overall adverse events rate of quadruple therapy regimens was 53.9% (n=104), being diarrhea (27.5%) and nausea (26.9%) the most common. Major symptoms (need for therapy or impairment of daily living activities) occurred in 8.3%. Adverse events/Major adverse events rates for the different regimens were 50.0%/5.3% for arm A, 69.2%/15.4% for arm B, 48.6%/7.9% for arm C, 48.8%/4.9% for arm D and 54.0%/8.1% for arm E (p=0.287/p=0.451).

Conclusion: In Portugal, all the quadruple therapy regimens currently recommended for the eradication of Hp, with or without bismuth, have been effective in eradicating Hp gastric infection. However, concomitant therapy with metronidazole bid seems to have a higher rate of adverse effects, including major adverse effects.

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Disclosure: Nothing to declare.

PP0292

THE EFFECTIVENESS OF THE ALPHA-GLUTAMYL-TRYPTOPHAN AS A LOCAL CYTOPROTECTOR FOR GASTRIC MUCOSA CELLS IN PATIENTS WITH CHRONIC ATROPHIC GASTRITIS: RESULTS OF MULTICENTER DOUBLE-BLIND PLACEBO-CONTROLLED RANDOMIZED STUDY

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Introduction: Prevention of gastric cancer, both primary and secondary, is an extremely important component of the management of patients infected with Helicobacter pylori. An important component of secondary prevention of gastric cancer is the use of gastroprotectors and repair stimulants, in particular the alpha-glutamil-triptophan, especially in people with a potentially precancerous condition – chronic atrophic gastritis

Aims & Methods: The aim is to estimate the effectiveness of the alpha-glutamyl-tryptophan as a local cytoprotector for gastric mucosa cells in patients with chronic atrophic gastritis (CAG).

116 patients with H. pylori-associated CAG took two treatment lines: 1. standard eradication therapy 10 days, 2. alpha-glutamyl-tryptophan(58 patients)/placebo(58 patients) 28 days. For all patients before and after treatment we performed gastroscopy with two biopsies from atrophic area, histological study of these biopsies, daily pH-metry and Gastropanel.

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Results: Alpha-glutamyl-tryptophan in comparison with placebo have statistically significant: *regenerative effect* (26.1% increase in the number of glands per 1 mm^2 of the gastric mucosa (GM) by histological study) and *anti-inflammatory effect* (decrease of GM edema and hyperemia by endoscopy data, reduction of the number of inflammatory infiltration cells per 1 mm^2 of GM: eosinophilic granulocytes – 3 times, neutrophilic leukocytes – 4 times, macrophages – 1.5 times, lymphocytes – 28.2%, plasmocytes – 29.6% by histological study). Also alpha-glutamyl-triptaphan improved the functional posibilities of stomach: a decrease in the level of gastrin-17 (p = 0.005) and an increase in the value of the PG I/PG II ratio (p=0.003), there was a shift in the acidic side of the pH values (p=0.001) when comparing indicators before and after treatment. There were no statistically significant changes in these parameters in the placebo group. According to the results of gastroscopy, a statistically significant decrease in the severity of edema of the gastric mucosa (p=0.008), the total set of signs of an acute inflammatory process (p=0.006), and a decrease in the proportion of outcomes with negative dynamics of the inflammatory process (p=0.038) were revealed in alpha-glutamyl-tryptophan group.

Conclusion: Alpha-glutamyl-tryptophan can be recommended for local cytoprotection in CAG, espessially fpr H. pylori associated gastritis for prevention the progression of atrophy and inflammation, and also lowering the risk of gastric cancer.

Disclosure: nothing to disclose.

PP0293

THE EPIDEMIOLOGICAL PROFILE OF GASTRIC CANCER PATIENTS IN GROOTE SCHUUR HOSPITAL CAPE TOWN, SOUTH AFRICA

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Introduction: Gastric cancer (GCA) is the 5th most common cancer and 3rd leading cause of cancer-related death. *Helicobacter pylori* (*H. pylori*) is the chief risk factor. The prevalence of *H. pylori* in Africa is between 60-90%, despite this GCA appears to be low, the so-called African enigma.

Aims & Methods: To document the epidemiological profile and risk factors of GCA in a South African cohort.

A retrospective folder review of patients with GCA from 2018-2022

Results: 443 patients were included; median age= 63 (range 29-86), 65% male (61% ≥ 60 years old). HPT (38.2%) and diabetes (17.6%) were the main comorbidities. 46% and 8.4% had history of smoking and alcohol use respectively. NSAID and PPI use were reported in 8.8% and 7% respectively. The commonest symptom was loss of weight (38.6%) and epigastric pain (29%). 68% had a Hb<12g/dl with a median Hb=10 (range 3.8-17). Endoscopically, gastritis or duodenitis or ulcers were recorded in 63% of cases. The commonest site of GCA was non-cardia (82.2%); antral and corpus cancers in 40.9% and 53.2% respectively. Histologically, *H. pylori* was reported in 11.5%, while chronic atrophic gastritis (CAG) in 15.6%. *H. pylori* was not associated with either antral or corpus cancer, or CAG, however PPIs were associated with a reduced odd of antral cancer, OR=0.42, CI=0.18-0.97, p=0.04. None of the other demographic or clinical factors were associated with the location of GCA.

Conclusion: GCA is higher in males and those ≥60 years. Loss of weight and anaemia are key red flags. Non-cardia gastric cancers are the commonest in this cohort. The low prevalence of *H. pylori* suggests previous eradication, although CAG was high, thus maintaining the increased risk of non-cardia CGA. Interestingly, PPI use was significantly protective against developing antral cancer. This highlights the need to treat *H. pylori* before CAG occur.

Disclosure: None.

PP0294

ASSESSMENT OF OLGA AND OLGIM STAGING SYSTEMS IN HELICOBACTER PYLORI INFECTION

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Introduction: Helicobacter pylori (HP) infection induces chronic gastritis and represents a major factor in gastric cancer. The histological sequence leading to gastric cancer is well known: gastritis induces atrophy and intestinal metaplasia, two precancerous lesions detectable by early endoscopy and meticulous anatomopathological examination.

Operative Link on Gastritis Assessment (OLGA) and Operative Link on Gastric Intestinal Metaplasia Assessment (OLGIM) are two systems designed to detect patients at high risk of developing gastric cancer by assessing, respectively, the degree of atrophy and intestinal metaplasia in the gastric mucosa.

Aims & Methods: The aim of our study was to evaluate the OLGA and OLGIM staging systems in HP infection.

A retrospective study was conducted over patients who had upper gastrointestinal endoscopy between January 2018 and December 2019 with the presence of HP infection on pathological examination of gastric biopsies. Patients with stage III or IV gastritis according to the OLGA or OLGIM systems were considered at high risk of progression to gastric cancer.

Results: Two hundred patients were included in this study. The mean age was 52.5 years [10-92], with a F/M sex ratio of 1.67. The main indications for upper GI endoscopy were epigastralgia (66.5%) and anemia (18.5%).

Various morphological aspects were observed. The gastric body was normal in 59.5% of cases, with gastroraphy present in 40.5%, mainly congestive (17.5%), but also mosaic (11%), nodular (6.5%), atrophic (4.5%), erosive (3%) and ulcerated (2.5%). The antrum was normal in only 23.5% of cases. Antral gastroraphy was congestive (46.5%), nodular (14.5%) or ulcerated (14%).

On anatomopathological examination of gastric biopsies, atrophy was present in 15.5% of patients and intestinal metaplasia in 45%. Only 3 patients (1.5%) were considered at high risk of developing gastric cancer according to the OLGA system, while this risk was estimated at 13.5% according to the OLGIM classification.

There was no statistically significant association between the OLGA and OLGIM staging systems in the selection of patients at high risk of developing gastric cancer (p=0.09). Nevertheless, an association was noted between OLGA classification and the presence of intestinal metaplasia (p=0.009), the presence of a high risk of developing gastric cancer according to the OLGA system and the finding of moderate to severe intestinal metaplasia on gastric biopsies (p=0. 018), and between the existence of a high risk of progression to gastric cancer according to the OLGIM system and the presence of moderate to severe atrophy (p=0.005).

Conclusion: The OLGA and OLGIM staging systems assess the risk of developing gastric cancer in HP infection. Our study didn't found a correlation between the two systems. A concomitant assessment of this risk by the two systems will enable a better selection of high-risk patients for appropriate surveillance and early management.

Disclosure: Nothing to disclose.

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Poster
Small intestinal

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Poster
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Small intestinal

09:00-18:00 / Poster Exhibition

PP0295

THE EFFECTS OF HIGH FAT, HIGH FRUCTOSE DIET ON HEPATIC STEATOSIS AND SMALL BOWEL TIGHT JUNCTION PROTEINS DEPENDING ON AGE AND SEX IN RATS

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Introduction: The aging with weight gain induces pathological changes in the liver, exacerbating non-alcoholic fatty liver disease (NAFLD). In the reproductive age, NAFLD is more prevalent and severe in males, however, NAFLD occurs at a higher rate among postmenopausal females. The liver and small bowel are closely interrelated through enterohepatic circulation, such that damage to the small bowel can significantly impact hepatic function. In small intestine, the expression of tight junction protein (TJP) in nearly associated with the development of NAFLD.

The aim of this study was to investigate the effects of age and sex on hepatic steatosis and the small bowel under conditions of a high fructose, high-fat diet (HFHFD) in rats.

Aims & Methods: 6-week-old male and female F344 rats divided into control and HFHFD groups, along with 2-year-old F344 rats. Over an 8-week period, the high fat diet (Raonbio, 60 kcal% fat) provided twice weekly and high fructose water (20%) gave once weekly for HFHFD groups. After 8 weeks, all animals were sacrificed, and both liver and small intestinal tissue were collected.

All tissue samples were subjected to hematoxylin and eosin staining for histological analysis. RNA extraction from the jejunum mucosa was followed by qRT-PCR analysis to assess claudin-2 (cldn-2, one of TJP) gene expression.

Results: The body weight change (%) and the hepatic steatosis scores were significantly higher in the HFHFD groups compared to both male and female control groups (Table).

Histological scores of old rat HFHFD groups were significantly higher than those of 6 week HFHFD groups, regardless of sex (Table).

In terms of histological inflammation scores of old control group small bowel were higher than those of 6 week control group regardless of sex (Table).

The jejunum villus/crypt ratio of 6-week-old rats significantly decreased only among males (Table).

In contrast, in the 2-year-old groups, the villus/crypt ratio resulted in a reduction only among females and the mRNA expression of cldn-2 observed in females with significant elevation (Table).

		Weight	Steatosis score	Inflammation score			Jejunum villus/crypt length ratio	Cldn-2 mRNA expression
				Duodenum	Jejunum	Ileum		
Male	6-week Con vs. 6-week HFHFD	**increase** ($p = 0.006$)	**increase** ($p = 0.003$)	**increase** ($p = 0.072$)	**increase** ($p = 0.026$)	**increase** ($p = 0.049$)	**decrease** ($p = 0.072$)	-
	2-year Con vs. 2-year HFHFD	**increase** ($p = 0.006$)	**increase** ($p = 0.002$)	-	**increase** ($p = 0.015$)	-	-	-
	6-week Con vs. 2-year Con	-	-	**increase** ($p = 0.005$)	-	**increase** ($p = 0.022$)	-	-
	6-week HFHFD vs. 2-year HFHFD	-	**increase** ($p = 0.007$)	-	-	-	-	-
Female	6-week Con vs. 6-week HFHFD	**increase** ($p = 0.004$)	**increase** ($p = 0.002$)	-	**increase** ($p = 0.007$)	**increase** ($p = 0.104$)	-	**increase** ($p = 0.056$)
	2-year Con vs. 2-year HFHFD	**increase** ($p = 0.004$)	**increase** ($p = 0.002$)	-	-	-	**decrease** ($p = 0.076$)	**increase** ($p = 0.005$)
	6-week Con vs. 2-year Con	-	-	**increase** ($p = 0.007$)	**increase** ($p = 0.002$)	**increase** ($p = 0.002$)	-	-
	6-week HFHFD vs. 2-year HFHFD	-	**increase** ($p = 0.011$)	-	-	-	-	-

Con, control; HFHFD, high fructose high fat diet; Cldn-2, Claudin-2; vs., versus.

Conclusion: Our data showed that high fructose, high-fat diet induced hepatic steatosis and inflammation of small bowel, especially in old rats. This indicates that NAFLD is related with damage of small bowel brush border membrane. The upregulated gene expression of cldn-2 ($p = 0.005$) and the decreased tendency of jejunum villus/crypt ratio ($p = 0.076$) in the aged females rats suggests a potential role of TJP in advancing NAFLD in postmenopausal females.

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PP0296

THE IMPACT OF SMALL INTESTINAL BACTERIAL OVERGROWTH ON THE EFFICACY OF FECAL MICROBIOTA TRANSPLANTATION IN PATIENTS WITH CHRONIC CONSTIPATION

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Introduction: To investigate the impact of small intestinal bacterial overgrowth (SIBO) on the efficacy of Fecal Microbiota Transplantation (FMT) in patients with chronic constipation

Aims & Methods: Inclusion of 218 chronic constipation patients treated with FMT, divided into constipation with SIBO group (SIBO) based on breath test results for SIBO, and constipation without SIBO group (Non-SIBO). The efficacy of the two groups was evaluated using constipation-related scoring scales. At the same time, feces and small intestinal fluid samples were collected from both groups before and after FMT to compare the changes in the intestinal microbiota through 16S rRNA sequencing

Results: In this study, it was found that the clinical efficacy of FMT in the SIBO group was superior to that in the Non-SIBO group. After FMT treatment, both groups showed a significant increase in bowel frequency and improvement in stool characteristics. Abdominal symptoms, rectal symptoms, and defecation symptoms were significantly alleviated (P<0.05), and patients' quality of life was significantly enhanced (P<0.05). After FMT, except for the CAS scale scores, other scale scores showed significant differences between the two groups, the SIBO group scoring significantly better than the Non-SIBO group (P<0.05).

After FMT, there were minor changes in the colonic microbiota but more substantial changes in the small intestinal microbiota. At baseline, the SIBO group had a higher abundance of *Veillonella*, and lower abundances of *Escherichia-Shigella* and *Acinetobacter* compared to the Non-SIBO group.

Characteristics	Overall (N=218)
Age (mean ± standard deviation)	55.62±16.48
Gender (Male/Female)	95/123
SIBO (Positive/Negative)	88/130
Disease Duration (year)	13.46±7.98
Complete Spontaneous Bowel Movements (CSBM) time/week	1.32±0.66
Spontaneous Bowel Movements (CSBM) time/week	1.76±0.59
Wexner Scale	16.27±1.68
Bristol Stool Form Scale (BSFS)	1.30±0.46
PAC-SYM	2.95±1.32

Table 1 General Information of the Patients

Conclusion: Chronic constipation patients with SIBO have a better response to FMT than those without SIBO.

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Disclosure: Nothing to disclose.

PP0297

INCREASING PREVALENCE OF INCIDENTAL ILEITIS FOLLOWING THE COVID-19 PANDEMIC

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Introduction: The incidental finding of terminal ileal inflammation during routine colonoscopy is not uncommon and is of unclear significance. It has been reported among asymptomatic persons undergoing non-diagnostic colonoscopy. There is limited data on long term follow-up and outcomes of such findings.(1)

Aims & Methods: This is a monocentric retrospective observational study spanning from 2018 to 2023, comparing two periods: the first period (P1) before the impact of the Covid-19 pandemic in 2020, and the second period (P2) starting 2020 reflecting the impact the pandemic left on colonoscopies. For each patient, variables were collected from medical records.

Results: Our initial database included 11200 colonoscopies, from which we randomly selected 2394 patients proportionally to the annual activity (mean of 1686/year before 2020 and 1189/year after 2020). Among indications of colonoscopy were screening and follow-up, change in bowel habits, anemia and bleeding. We excluded 287 patients (known history of inflammatory bowel disease and no ileum intubation). Then the sample was divided into P1 (681 patients) and P2 (1483 patients). Characteristics and findings are shown in Table 1. We noted better bowel preparation (77.1 vs 91.7%) and a higher ileum intubation in P2 (79.7 in P1 vs 93.8% in P2).

N=2164		P1 (N1=681)	P2 (N2= 1483)	p
Age (years)		57.5	56.9	NS
Overall endoscopic findings n(%)	Endoscopic aspect of ileitis	0 (0)	71 (5.8)	<0.001
	Abnormal ileal findings on histology	23 (3.4)	169 (11.4)	<0.001
	Diagnosis of Crohn's disease	0 (0)	19 (0.9)	<0.001
Overall histological findings n(%)	Active ileitis	19 (2.8)	147 (9.9)	<0.001
	Focal ileitis	10 (1.5)	89 (6)	<0.001
	Ulcerative ileitis	11 (1.6)	48 (3.2)	0.031
	Erosive ileitis	3 (0.4)	31 (2.1)	0.004
	Architectural distortion	1 (0.1)	16 (1.1)	0.023

Overall ileoscopy findings showed an increased aspect of ileitis on endoscopy (0% in P1 and 5.8% in P2, with p<0.001), correlating with a significant increase in the incidence of histological findings on ileal biopsies (3.4% in P1 vs 11.4% in P2; p<0.001). These histological findings included active, focal and ulcerative ileitis, with more architectural distortion. However, these results corresponded to a less severe ileitis (0.3 vs 2.3%; p=0.001). Nineteen of these patients with positive results of ileitis were diagnosed with Crohn's disease.

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Conclusion: This study showed an increase in endoscopic and histologic ileitis on colonoscopy following the Covid-19 era. This could be the result of the direct virologic effect that the virus has on intestinal inflammation, its indirect effect on immunity triggering the onset of enteritis, or the consequence of the mass vaccination campaign held to protect from the virus. Furthermore, histology showed granulomatous inflammation similar to Crohn's disease, which may lead to misdiagnosis and probably overtreatment.

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Disclosure: Nothing to disclose.

PP0298

EXAMINING LONG-TERM ANTITHROMBOTIC TREATMENT PATTERNS IN PATIENTS WITH SHORT BOWEL SYNDROME POST ACUTE MESENTERIC ISCHEMIA: INSIGHTS FROM AN INTERNATIONAL SURVEY

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Introduction: Long-term antithrombotic therapy is recommended in short bowel syndrome (SBS) caused by acute mesenteric ischemia (AMI). However, evidence-based recommendations on drug choice, duration and monitoring of treatment are lacking. Also, guidelines are based on data on other artery diseases and overlook the altered drug absorption in patients with SBS. To understand current practices, we aimed to assess the long-term antithrombotic management of AMI.

Aims & Methods: An online survey was sent to centers contributing to the chronic intestinal failure database of the HAN-CIF (Home Artificial Nutrition – Chronic Intestinal Failure) special interest group of ESPEN (European Society on Clinical nutrition and Metabolism), treating patients with SBS post-AMI. The survey featured 10 cases varying in postsurgical intestinal anatomy, revascularization and etiology and focused on drug class, administration route, duration and monitoring of antithrombotic therapy.

Results: The survey was completed by 43 centers. In case of idiopathic AMI leading to SBS, anticoagulation (65.1% of all physicians) or antiplatelet (32.6%) therapy were mostly prescribed (Figure 1; jejunocolic anastomosis, idiopathic).

Antiplatelet therapy was prescribed more often to patients with type 3 anatomy (37.2%, largest absorptive capacity) than to other anatomy groups (30.2 and 32.6% in type 1 and 2 respectively). Antiplatelet drugs (39.5%) or combination therapy (18.6%) were more commonly prescribed in case of revascularization compared to no revascularization (32.6% or 2.3% respectively).

Anticoagulants were very common in patients with atrial fibrillation (90.7%), venous thrombosis (95.3%) and antiphospholipid syndrome (83.7%). More antiplatelet drugs (27.9%) or combination therapy (18.6%) were preferred with concomitant peripheral artery disease. In hypovolemic shock, no therapy (41.9%) or anticoagulants (51.2%) were common.

Parenteral drugs were more prescribed to patients with type 1 anatomy (51.2% of all physicians; lowest absorptive capacity) than to patients with type 2 (44.2%) or 3 (20.9%) anatomy.

Antiplatelet drugs were mainly prescribed lifelong (94.7% of this drug class), while anticoagulants were prescribed for 6 (19.4%) to 12 (16.1%) months or lifelong (51.6%), irrespective of anatomy, revascularization or etiology.

Vitamin K antagonists were consistently monitored, whereas aspirin was not. Monitoring of subcutaneous low molecular weight heparins (no monitoring, 36.8% - always monitoring, 42.1% - in specific circumstances, 21.0%) and non-vitamin K oral anticoagulants varied (no monitoring, 50.0%).

Conclusion: In a case-based survey, anticoagulant versus antiplatelet drug choice depended on the anatomy, revascularization and etiology. However, there was a high variability in case of idiopathic AMI. Administration route depended on anatomy. More detailed practice guidelines are needed.

Disclosure: TV has participated in advisory boards, as a consultant and/or as a speaker on behalf of Boehringer Ingelheim, Bayer, Daiichi Sankyo, BMS/Pfizer, Sanofi Aventis, and Leo Pharma.

PP0299

PREDICTIVE FACTORS IMPACTING THE RESULT OF A GLUCOSE BREATH TEST: A RETROSPECTIVE ANALYSIS OF 2373 TESTS

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Introduction: Glucose breath test (GBT) is the most commonly used method for the diagnosis of small intestinal bacterial overgrowth (SIBO). SIBO is a common condition with very unspecific symptoms (bloating, abdominal pain, diarrhea or constipation).

Recently, American and European recommandations have been published to standardize the indications and the conditions to perform these tests.[1,2] However, some discrepancies remain between the European and American recommandations, and many statements were supported by

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low quality of evidence.[1,2] Moreover, several counfounding factors related to the patient's condition may impact the result of the test, leading to a difficult interpretation.

Aims & Methods: This study aimed to assess the factors that could impact the results of a GBT. Therefore, we retrospectively reviewed the results of the GBT performed in our physiology department from 2013 to 2023. The test was performed after ingestion of 75g of glucose, with breath samples collected every 15min during 2 hours. The result was considered positive for hydrogen (H_2) if a rise ≥ 20ppm was observed or positive for methane (CH4) if a rise ≥ 10ppm was observed. Demographic data were collected, such as symptoms and questionnaires during the exam including Rome questionnaire, and the Gastro-Intestinal Quality of Life Index (GIQLI). Severity of symptoms was based on the GIQLI questionnaire, where a higher score represents a better condition. Medical and surgery history, such as drug treatments were then retrospectively collected from medical records, using a computerized research.

Results: A total of 2373 tests from different patients were included. The mean age was 46.0 ± 15.8 years old, with 1727 females (73.7%). The number of positive GBT was 576 tests (24.3%), with 294 tests positive for H_2 only, 214 tests positive for CH_4 only, and 69 tests positive for both. In univariate analysis, a higher age, female sex, and a lower body mass index (BMI) were associated with a positive GBT (Table). A history of diabetes, and a previous gastric surgery (sleeve, or bypass) were associated with a positive test, while proton pump inhibitors (PPI) or opioids intake did not change the outcome (Table). Regarding the symptoms, more severe abdominal bloating and constipation were associated with a positive test (1.28 ± 1.0 vs 1.41 ± 1.1; p=0.02, and 2.67 ± 1.3 vs 2.82 ± 1.2; p=0.03), while abdominal pain and diarrhea were not statistically different (1.57 ± 1.1 vs 1.67 ± 1.1; p=0.07; and 2.23 ± 1.3 vs 2.33 vs 1.4; p=0.14). In multivariate analysis, age (Odds Ratio (OR) = 1.02 [1.01 – 1.03]; p<0.001), BMI (OR=0.93 [0.90 – 0.96]; p<0.001), a previous gastric surgery (OR=9.32 [3.74 – 24.86]; p<0.001) and more severe constipation (OR=0.88 [0.79 – 0.98]; p=0.02) were associated with a positive GBT.

	Negative GBT (n=1797)	Positive GBT (n=576)	p value
Age (years)	44.7 ± 15.5	50.0 ± 16.2	<0.001
Sex (female)	1280 (71.2%)	447 (77.6%)	0.005
BMI (kg/m²)	25.0 ± 5.5	23.9 ± 5.2	<0.001
Diabetes	62 (3.5%)	30 (5.2%)	0.02
Gastric sleeve	8 (0.4%)	14 (2.4%)	<0.001
PPI intake	361 (20.1%)	123 (21.4%)	0.50
Opioids intake	183 (10.2%)	57 (9.9%)	0.91
Irritable bowel syndrome	561 (31.2%)	148 (25.7%)	0.09
Functional dyspepsia	704 (39.2%)	190 (33.0%)	0.08

Conclusion: The result of GBT is more likely to be positive in patients with constipation. A previous gastric surgery is a cofounding factor, leading to a non-interpretable result of the test. A diagnosis of irritable bowel syndrome or functional dyspepsia did not change the result of the test, such as PPI or opioid intake.

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PP0300

THE IMPACT OF GUT MICROBIOTA MODULATION ON THE EVOLUTION OF CELIAC DISEASE

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Introduction: The only effective treatment for celiac disease (CD) is lifelong compliance with a gluten-free diet. *Lactobacilli* and *Bifidobacterium* spp. may play a role in the breakdown of gluten. Ithas been demonstrated that *Lactobacilli* can detoxify gliadin peptides after partial digestion by human proteases. These findings could suggest the use of probiotics as complementary therapy for CD.

Aims & Methods: We designed a study to determine the potential effect of a combination of Lactobacillus (acidophilus HA-122, casei HA-108) and Bifidobacterium bifidum HA-132 on perception of symptoms and on the inflammatory-immunological markers in patients with active CD.

25 adult patients were randomized to receive bid of a probiotic combination of Lactobacilus and Biffidobacter for six weeks (13 pts) va a placebo group (12 pts) while being on gluten-free diet. Laboratory studies included routine blood tests, CD specific antibodies, determination of IL-1b, IL-6 and TNF-alpha. CD serology and duodenal histology were performed at baseline, at the end of treatment and at 30 days after therapy. Histological damage in intestinal biopsy samples was graded according to the Marsh modified classification. At baseline and follow-up visit, patients completed the Gastrointestinal Symptom Rating Scale (GSRS) questionnaire. Histological damage in intestinal biopsy samples was graded according to the Marsh modified classification.

Results: 25 pacients, 13 in the probiotic associated arm, 12 in the placebo arm, with median age of 40, BMI 21.2, 18 with symptomatic CD (equal distribuited) and 7 with sublinical CD (4 in the probiotic arm, 9 in the placebo arm). Baseline serology: IGA tTG 1986.6 (group A) vs 1507.1 (placebo group). Baseline hystology (Marsh classification)- Marsh 3a 1 pacient in probiotic group, 2 in the placebo arm. 5 pts Mars 3b3 in group A and 2 in the placebo arm and 17 pts Marsh 3c (9 pts probiotic, 8 pts placebo). IELs density (mean range) was 44.5 in the first group vs 42.5

After treatment: clinical aleviation of indigestion and diareea (p=0.002), constipation (p=0. 0396), gastro-esophageal reflux (p=0.0493) and abdominal pain (p=0.0378)-superior in the probiotic associated treatment.

Serologic parameters: IgA tTG 0,96 in the probiotic arm vs 1,16 in the placebo (p=0.048), IgA EMA 1,05 vs 0,97 (p=0.028), IgG DGP 0,91 vs 1,12 (p=0.21).

Improvement of the inflamatory status: decrease of IL-1b, IL 6 and TNF-a was superior in the probiotic associated group (p=0.071, p=0.083, p=0.078).

Regarding the histologic response: IELs 0,83 vs 1.07 in the placebo arm (p=0.049).

Conclusion: Tests and the lymphocyte infiltrates in the mucosal biopsy specimen in uur study suggests that prebiotic/probiotic treatment may improve symptoms and serology in celiac disease.

Disclosure: I don t have any conflict of interest to declare.

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PP0301

SLEEP DISTURBANCES IN CELIAC PATIENTS UNDER GLUTEN FREE DIET: A PROSPECTIVE STUDY

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Introduction: Celiac disease (CD) is an immune-mediated enteropathy associated with extraintestinal manifestations. It has been hypothesized that sleep disorders may be more frequent in CD patients.

Aims & Methods: The purpose of this study was to investigate the quality of sleep in CD patients and to assess the relationship between poor sleep quality and biochemical parameters.

We included patients with a proven diagnosis of CD at least twelve months prior to enrollment. Patients not fully adhering to a gluten free diet were excluded. All the participants completed the Sleep Scale from the Medical Outcomes Study (MOS sleep scale) and Sleep Problems Index II (SLP9) was calculated. A high SLP9 scale indicated worse quality of sleep and values >20 indicated significant sleep disturbance. Stress was evaluated by PSS-4 scale. Simultaneously, blood samples from each subject were collected and demographic and clinical data were recorded. Linear logistic regression was used for uni- and multivariate analysis, and b coefficient and 95% confidence intervals (CI) were calculated.

Results: We enrolled 53 patients, mean age 42.75±11.19, 84.9 % female. The average hours of sleep per night was 6.47±1.11. Mean SPL9 was 37.44±20.65. Forty patients (75.5%) showed sleep disturbance. At univariate analysis, high SLP9 values (worse sleep) were inversely related to the number of hours of sleep (b=-0.38, 95% CI -11.8 to -2.2, p=0.005) and to vitamin D blood levels (b=-0.57, 95% CI -2.7 to -0.46, p=0.009). In multivariate analysis only serum levels of vitamin D were inversely related to SLP9 (b=-0.47, 95% CI -12.8 to -0.43, p=0.05).

Conclusion: Our study demonstrates that low serum levels of vitamin D are independently associated with poor sleep quality in celiac patients. These findings need to be confirmed in future experimental and clinical studies.

Disclosure: Nothing to disclose.

PP0302

DIGESTION SUPPLEMENTED WITH COMMERCIAL GLUTENASES: EVALUATION OF THE FATE OF GLUTEN IN PIZZA BY PROTEOMICS AND IMMUNOCHEMISTRY

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Introduction: Celiac disease (CD) is an autoimmune disorder caused by gluten in genetically predisposed subjects. Recently, the market has introduced enzymatic supplements, known as glutenases, that individuals can ingest before a meal. These supplements can include single enzymes or mixtures facilitating the digestion; however, their potency is unclear.

Aims & Methods:

We developed an analytical process for assessing the effect of supplements on gluten digestion. A standardized pizza recipe was created and used as substrate in a novel *in vitro* model of gastroduodenal digestion (INFOGEST)[DL1] [GDA2] . 3 different commercial supplements (S1, S2 and S3) were tested. The digestion products were qualitatively monitored in the gastric and duodenal phases using electrophoresis. Primary amines were used to quantify the production of proteinaceous degradation products. Finally, we assessed the digestion resistance of immune-toxic gluten epitopes by using a commercial G12-based ELISA.

Results: Samples treated with supplements showed a reduced number of electrophoretic gluten-related bands [DL3] [GDA4] [GDA5] after gastric digestion as compared with the control. This suggests a faster gluten digestion in the stomach. The highest content of primary amines was determined in the presence of S1 and S2, supplements formulated to contain proteases, peptidases, and starch-degrading enzymes which are active at various pH levels, assisting gluten degradation by gastroduodenal enzymes. The ELISA assay showed a more rapid degradation of G12-immune-toxic epitopes in the presence of S3-a supplement containing prolyl-oligopeptides, that cleave proline-rich motifs.

Conclusion: The developed approach was effective in assessing the ability of commercial glutenase supplements to reduce the potential immune toxicity of gluten-containing ingredients.

Disclosure: This study was supported by an unrestricted grant from Takeda Pharmaceuticals.

PP0303

THE FREQUENCY OF CELIAC DISEASE AND ITS ASSOCIATION WITH INFERTILITY IN WOMEN REFERRING TO INFERTILITY CLINICS; A CASE-CONTROL STUDY IN SOUTHERN IRAN

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Introduction: Celiac disease (CD) is a multifactorial systemic disease causing enteropathy that can cause a wide spectrum of disorders related to infertility.

Aims & Methods: This case-control study aimed to evaluate the frequency of CD and its association with infertility in women.

This study was conducted on women referred to the Ghadir Mother and Child Hospital in Shiraz. Infertile women were considered as the case group, and fertile women were considered as the control group. Blood samples were taken from participants, and tissue anti-transglutaminase (Anti-TTG Ab) levels were measured. Patients with high levels of Anti-TTG Ab were referred for duodenal biopsy.

Results: One hundred subjects were enrolled in the case group and 200 in the control group. Eight patients in the case group had positive serology, and duodenal biopsy results of four were positive for CD. Among the control group, one had a positive serology, and the duodenal biopsy result was negative (*P* for between-group differences: 0.001 for serology and 0.012 for biopsy results). There was a positive association between high levels of Anti-TTG Ab and infertility (odds ratio=17.30, 95% CI: 2.13-140.39), even after adjusting for age and body mass index (odds ratio =9.92, 95% CI: 1.17-84.21).

Conclusion: The frequency of CD among infertile women was higher than that of fertile women. The increased levels of Anti-TTG Ab were independently associated with infertility.

Disclosure: None.

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PP0304

PATIENT PREFERENCES FOR THE DIAGNOSIS OF COELIAC DISEASE: A DISCRETE CHOICE EXPERIMENT

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Introduction: There is potential for a paradigm shift from a biopsy- to a serology-based diagnosis of coeliac disease in selected adults. However, it remains unknown if this approach would be acceptable to patients. We aimed to explore patients' views and preferences regarding the no-biopsy approach for coeliac disease diagnosis.

Aims & Methods: We developed a discrete choice experiment survey containing 12 different scenarios with 2 possible alternatives (endoscopy & biopsy or serology), to estimate patient preferences. The scenarios were based on 5 attributes: risk of false positive results, risk of missed diagnosis, waiting time to start treatment, risk of complications, discomfort, or pain. Patient preferences and the relative importance of the attributes were estimated using a mixed logit model.

Results: In total, 385 people (70.6% female, 98.2% white) completed the survey. Respondents preferred a serology-based diagnosis over endoscopy and duodenal biopsies (59% vs 41%, β coefficient = 1.88, $p < 0.001$). Diagnostic test accuracy ($p<0.001$), shorter waiting time to start treatment ($p<0.001$), and discomfort levels during the procedure ($p<0.001$) were the most important attributes to respondents. The risk of complications, including perforation and bleeding, did not significantly influence respondents' choices. Respondents with previous endoscopy experience were more willing to undergo endoscopy compared with those who never had one.

Conclusion: The no-biopsy approach to diagnosing coeliac disease is acceptable and preferred by patients over endoscopy and biopsy. Our findings highlight the importance of patient-centred care and shared decision-making in guiding diagnostic strategies for optimal patient outcomes.

Disclosure: Nothing to disclose.

PP0305

EXPLORING THE VIEWS OF PRIMARY AND SECONDARY CARE PHYSICIANS ON THE NO-BIOPSY DIAGNOSIS OF COELIAC DISEASE IN ADULTS: A QUALITATIVE INTERVIEW STUDY

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Introduction: Despite mounting evidence supporting a no-biopsy approach for the diagnosis of coeliac disease in adults, established clinical pathways in primary and secondary care could hinder implementation in clinical practice. The aim of this study was to identify the barriers and facilitators to a no-biopsy diagnostic pathway of coeliac disease in primary and secondary care.

Aims & Methods: We conducted a qualitative study using semi-structured interviews with gastroenterologists and general practitioners (GPs). We used a pre-defined interview topic guide consisting of open-ended questions and prompts to facilitate discussion about the views of primary and secondary care physicians on the no-biopsy diagnosis of coeliac disease in adults. The interviews were conducted in person or virtually, then transcribed verbatim and analysed using reflexive thematic analysis.

Results: We interviewed 24 physicians (12 gastroenterologists and 12 GPs) across England between February and March 2024. Participants (62.5% male, 58.3% from North of England, and 50% non-white) had different levels of experience ranging between 2 and 30 years. Gastroenterologists were more aware of the no-biopsy approach and the evidence supporting it compared with GPs (100% vs. 50%, $p=0.01$). Both groups were supportive of the no-biopsy approach but acknowledged the lack of clear guidelines as a major barrier to implementation in clinical practice. Increased patient satisfaction, shorter waiting times to start treatment, and reduced endoscopy services pressures and costs were perceived as the main advantages of the no-biopsy approach. However, participants had concerns regarding false positive results, missing concurrent pathology, dealing with persistent symptoms after treatment without biopsy confirmation, and the increased workload in primary care.

Conclusion: Primary and secondary physicians are willing to adopt a serology-based diagnosis of coeliac disease in adults. The safe and effective implementation of this no-biopsy approach depends on developing clear evidence-based clinical guidelines and referral pathways, along with comprehensive education for all healthcare providers involved.

Disclosure: Nothing to disclose.

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PP0306

BODY COMPOSITION AND MEDITERRANEAN DIET ADHERENCE OF COELIAC PATIENTS. A PROSPECTIVE, MULTICENTER STUDY (ARCTIC STUDY)

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Introduction: Coeliac disease (CD) is an immune-mediated systemic disorder caused by gluten proteins, affecting approximately 1% of the population. The cardiovascular (CV) and metabolic effects of the disease and the gluten-free diet (GFD) used for its treatment are controversial. Current guidelines suggest that the Mediterranean diet is the most suitable for reducing the risk of CV and metabolic disease, both in terms of primary and secondary prevention. The optimal GFD is based on the Mediterranean diet.

Aims & Methods: Our aim was to assess the body composition and adherence to the Mediterranean diet of CD patients who have been on a GFD for at least 1 year. Body composition was measured with an InBody 770 device and adherence to the Mediterranean diet was assessed with a 14-item questionnaire. Descriptive statistics and Spearman's rank correlation test were applied to analyze our data, provided as mean ± standard deviation.

Results: Data from 105 patients (84 females, 21 males, mean age 36±13 years) were analyzed. Regarding body composition, body weight was 68±17 kg, BMI was 23.9±5.5 kg/m^2, waist-hip ratio was 0.89±0.07, body fat percentage was 27±10%, skeletal muscle mass was 27±7 kg and visceral fat area was 89±50 cm^2 on average. CD patients' Mediterranean diet adherence was low (5.44±1.54 points), due to an inadequate consumption of whole grains (34.3%), fish (22.9%), avocado (13.4%) and red wine (29.5%).

Conclusion: Despite the lack of correlation between body composition parameters and adherence rates, patients with CD have an elevated waist-hip ratio and visceral fat area, as well as insufficient intakes of fibre, fish and red wine. Our data suggest that, in addition to maintaining a GFD, an increased fibre consumption and adherence to a Mediterranean diet are necessary for the prevention of CV and metabolic complications, which should be emphasized during patient follow-up.

Disclosure: Nothing to disclose.

PP0307

ASSOCIATION BETWEEN CELIAC DISEASE SYMPTOM SEVERITY AND QUALITY OF LIFE AMONG PATIENTS IN A US LONGITUDINAL OBSERVATIONAL STUDY

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Introduction: Quality of life (QoL) measures and symptom frequency/severity are common patient-centred outcomes in clinical practice and drug trials. However, the relationship between QoL measures and symptom burden among patients with celiac disease (CeD) is poorly understood. This study evaluated QoL scores in relation to self-reported symptom severity.

Aims & Methods: The Virtual Celiac Symptoms Study (NCT05309330) is a patient survey that followed US adults (≥ 18 years) and adolescents (12–17 years) with CeD currently adhering to a gluten-free diet, for a study period of 12 weeks. Participants self-reported daily symptom severity through the Celiac Disease Symptom Diary 2.1© (CDSD 2.1) and completed the Celiac Disease Quality of Life 20-item questionnaire (CD-QoL) covering a 30-day recall period through a web application. The CDSD 2.1 included five core CeD symptoms (abdominal pain, bloating, diarrhoea, nausea, and tiredness) scored 1 to 5 as none, mild, moderate, severe, and very severe, respectively. The CD-QoL total scores range from 20 to 100, with higher scores corresponding to better QoL. The CDSD 2.1 score was calculated as the daily average symptom severity during the 4 weeks prior to completing the CD-QoL and was categorized as increasing score quartiles; quartile 1 (Q1, reference) represented the lowest possible score and Q4 the highest possible score. In addition, individual CDSD 2.1 symptoms were categorized by severity (none/mild [reference], moderate, and severe/very severe). The mean adjusted percentage difference and 95% confidence interval (95% CI) in CD-QoL scores was estimated by generalized linear regression which included terms for age, sex, race, years since CeD diagnosis, and body mass index.

Results: At Week 7, 404 participants in total had complete information on CD-QoL and CDSD 2.1 scores. The mean (standard deviation [SD]) CD-QoL score was 56.83 (19.21) and the mean (SD) score for CDSD 2.1 was 0.72 (0.56). The CD-QoL scores had a positive linear association with CDSD 2.1 overall scores as well as with individual symptom severity scores. Participants had a 52.7% (95% CI: 39.2%, 67.4%) higher CD-QoL score when comparing those in CDSD 2.1 Q4 (median [min–max] score 1.19 [0.85–3.00]) with those in Q1 (reference; median [min–max] score 0.07 [0.00–0.19]), which was unchanged when tiredness was excluded from the CDSD 2.1 score (53.2% [95% CI: 39.9%, 67.8%]). The positive association between CD-QoL and CDSD 2.1 was mainly driven by abdominal pain, tiredness, and bloating. Comparing participants with severe/very severe symptoms with those with none/mild symptoms, the CD-QoL score was 38.8% (95% CI: 28.2%, 50.2%) higher for abdominal pain, 37.8% (95% CI: 25.5%, 51.3%) higher for tiredness, and 32.2% (95% CI: 21.7%, 43.6%) higher for bloating.

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Conclusion: In patients with CeD assessed via a digital platform, those with a higher symptom severity burden, assessed with CDSD 2.1, had significantly greater (worse) CD-QoL scores, which suggests that the CDSD 2.1 might be a good indicator of clinically active disease among patients with CeD.

Disclosure: MA, NC, SG, RS, and DAL are employees and shareholders of Takeda. LMM was an employee of Takeda at the time of the study. JRM and ES are employees of Analysis Group, Inc., which received research support from Takeda Development Center Americas, Inc. DA has served as a consultant for Takeda Pharmaceuticals and Ironwood Pharmaceuticals. MG is an employee of the Celiac Disease Foundation, which received financial support from Takeda Development Center Americas, Inc. EL serves as a consultant for Takeda Pharmaceuticals and is an author for UpToDate (celiac section) for which he receives royalties.

PP0308

COMORBIDITIES, MEDICATION USE AND PROCEDURES AMONG PATIENTS WITH CELIAC DISEASE BEFORE AND AFTER DIAGNOSIS: A US HEALTH CLAIMS DATABASE STUDY

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Introduction: The healthcare burden among patients with celiac disease (CeD) following a gluten-free diet is still uncertain. The aim of this study was to compare the frequency of comorbidities, medication use and medical procedures among patients with CeD, pre- and post-diagnosis.

Aims & Methods: This case-control study was conducted in US individuals with and without CeD (1:5 ratio), matched for age, sex, region and diagnosis year using MarketScan claims data (2002-2023). CeD diagnosis was defined as ≥1 inpatient or ≥2 outpatient International Classification of Diseases (ICD)-9/-10 diagnostic codes and all individuals had continuous medical/pharmacy enrolment for 2 years pre-/post-index. Odds ratios (ORs) and 95% confidence intervals (CIs) were estimated by logistic regression comparing CeD cases and controls.

Results: The 21-year data coverage period included 39,814 patients with CeD and 198,961 individuals without CeD (matched controls) with similar baseline demographics (mean age 41.6 years [standard deviation 20.1]; 70.5% female). Comorbidities, medication use, and medical procedures were higher among individuals with CeD versus controls at 2 years pre- and post-diagnosis.

Two years pre-diagnosis, the frequency of depression/anxiety and use of antidepressants and anxiolytics was greater for individuals with CeD versus without CeD (OR: 2.0 [95% CI 1.9–2.0]), 1.5 [1.5–1.5] and 1.5 [1.4–1.5], respectively), and remained higher 2 years post-diagnosis (OR: 2.2 [2.1–2.3], 1.6 [1.5–1.6] and 1.6 [1.6–1.7], respectively).

The odds of medical procedures were higher comparing individuals with CeD versus without CeD, with the strongest associations for video capsule endoscopy and oesophagogastroduodenoscopy (OR: 9.6–19.9 pre- and post-diagnosis).

Conclusion: Comorbidities, medication use, and medical procedures were greater in individuals with CeD than matched controls, both pre- and post-diagnosis, indicating a continued burden for patients with CeD as well as the healthcare system.

Disclosure: MA, SM, VG, SG, RS, DAL, IP and NC are employees and shareholders of Takeda. BL has nothing to disclose. FZ served as a speaker for Werfen and a consultant for Takeda.

PP0309

THE PHENOTYPE AND TREATMENT OPTIONS FOR MESENTERIC LYMPH NODE CAVITATING SYNDROME IN COELIAC DISEASE: A CASE SERIES AND SYSTEMATIC REVIEW

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Introduction: There is a paucity of data on mesenteric lymph node cavitating syndrome (MLNCS); a rare complication of coeliac disease, characterized by central necrosis within enlarged mesenteric lymph nodes.

Aims & Methods: A case series of all patients with MLNCS at Sheffield Teaching Hospitals, UK between 2000 and 2022.

A further systematic review using PubMed and Google Scholar for patients with MLNCS and coeliac disease until 2022.

Results: In our case series, 5/6 patients were female (mean age:58 years (SD:14.1 years)). Half of patients presented with weight loss, and all were diagnosed by CT scan. Five patients followed a gluten free diet (GFD) and 3 were still alive 1-6years later, 2 died after 5 and 9 years. One did not follow a GFD and died at 1 year.

From the literature 44 patients were identified with MLNCS (52% female, mean age 52 years (SD:14.0 years)) and 16% had refractory coeliac disease (RCD). Common features were persistent villous atrophy (90%), weight loss (82%) and hyposplenism (80%). Forty followed a strict GFD, and 28 had partial or complete improvement of their symptoms whilst 19 died. Twenty-one were treated with steroids and 12 of them died.

The overall mortality was 44% and associated with cachexia (5), lymphoma (4), and infectious complications (4).

Conclusion: MLNCS was indicative of a poor prognosis and commonly presented with weight loss, hyposplenism, persistent villous atrophy and RCD. An early diagnosis and strict GFD may lead to a better prognosis. The role of immunosuppression is unclear and needs further investigation.

Disclosure: Nothing to disclose.

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PP0310

CELIAC DISEASE: A FORTY-YEAR ANALYSIS IN AN ITALIAN REFERRAL CENTER

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Introduction: Celiac disease (CD) is a gluten-related autoimmune enteropathy affecting human leukocyte antigen (HLA) DQ2 or DQ8 positive subjects.[1] CD onset can occur at any age and with several clinical phenotypes. Modern serological markers have significantly improved the diagnosis of the disease, which affects about 1% of the population worldwide.[1-3] Hence, CD serology helps to detect CD in patients with non-classical symptoms or even in those who are asymptomatic.

Aims & Methods: This study aims to analyze the clinical presentation, serology, histology, associated diseases, complications, and death rate of a large cohort of CD patients diagnosed and followed up over forty years. This is a retrospective study on 1547 CD patients (1170 females and 377 males; mean age at diagnosis: 33.1±24.4 years; F:M ratio = 3.1:1), consecutively diagnosed from January 1980 to December 2020 at our referral center (Department of Medical and Surgical Sciences, St. Orsola-Malpighi Hospital, Bologna, Italy). CD diagnosis was established following positive serological tests and confirmed by duodenal histology. The evaluated serological markers included: total immunoglobulin A (IgA), anti-transglutaminase(TG)-2 and anti-endomysium (EmA), both of IgA type; in cases with IgA deficiency, the immunoglobulin G (IgG) antibody anti-TG2 or against deamidated gluten peptides were considered.[4] Comorbidities and complications were recorded at diagnosis and during follow-up.

Results: The rate of CD diagnoses quadrupled after the year 2000. The most prevalent phenotype was the non-classical CD (63.3%), and the most frequent histotype was Marsh 3C (44.7%). Gastrointestinal manifestations, affecting about half of patients (51%), included diarrhea (24.3%), bloating (28%), and aphthous stomatitis (19.7%). Among the CD-associated disorder, osteopenia was the most common (59.9%) and predominantly affecting females (64.3%). Extraintestinal manifestations were anemia (35.8%, with low levels of ferritin and folic acid malabsorption detectable in 87% of patients), cryptogenic hypertransaminasemia (27.9%), and recurrent miscarriages (11.5%). The prevailing CD-related autoimmune disorders were thyroiditis (26.9%), type 1 diabetes mellitus (2.9%), and dermatitis herpetiformis (1.4%). Six patients had inflammatory bowel disease (2 Crohn's disease and 4 ulcerative colitis). Complications occurred in 1.8% of cases, and the mortality rate was 1.9%.

Conclusion: This forty-year, single-center, large cohort analysis showed that the clinical presentation of CD changed over the years, with an increase of non-classical and subclinical phenotypes. Our study highlighted three main aspects: i) the non-classical and subclinical phenotypes prevailed over the classical forms, according to the recently established criteria; ii) modern serological tests allowed an earlier, faster, and more precise CD diagnosis, even at the subclinical stage, thus contributing to explain the increased diagnostic yield; iii) some CD-associated disorders correlated with gender (e.g., osteopenia which prevailed in women). In conclusion, our data confirmed and broadened current knowledge on CD, and help physicians to recognize new CD presentations, thus avoiding diagnostic delay and inappropriate management.

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Disclosure: Nothing to disclose.

PP0311

IMPACT OF SEROLOGY-BASED CELIAC DISEASE DIAGNOSIS ON PEDIATRIC PATIENTS' OUTCOME

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Introduction: Celiac disease (CD) is an immune-mediated systemic disease, triggered by the ingestion of gluten, which only develops in genetically predisposed individuals. The main characteristics of the disease are the gluten-dependent small intestinal mucosal lesion, villous atrophy and crypt hyperplasia, and the production of the specific serum autoantibodies. The gold standard for diagnosis has been based on the biopsy findings. In 2012, new ESPGHAN diagnostic guidelines removed the necessity of duodenal biopsy in symptomatic children showing a high serum transglutaminase 2 titers (TG2-IgA) together with the presence of endomysial antibodies (EmA) and HLA DQ2 and/or DQ8.

Aims & Methods: Our aim was to assess the clinical outcome of the diagnostic change in children by comparing new serology-based diagnosed patients to the biopsy-based ones.

We conducted a retrospective analysis of data collected from the medical charts of the patients diagnosed during a four-year period, from 2010 to 2014, being a time overlapping the guideline change. From a total of 63 patients, 30 were biopsy-diagnosed and 33 diagnoses were based on serology only. Clinical data and laboratory parameters were compared between the groups at diagnosis and at follow-ups at 1 year, 2 years and at their latest follow-up visit during the gluten-free diet (GFD) treatment, median time 8.7 years for the serology-based group and 10 years for the biopsy-based group.

Results: Our results show clear improvement upon GFD in gastrointestinal symptoms in both groups. However, there was a general trend for better treatment outcomes in the biopsy-based group compared to the serology-based group at 1 and 2 years of follow-up. Body mass index Z score was in the biopsy group at baseline -1.5, at 1 year on GFD -0.2 and at 2 years -0.2. The corresponding results in the serology group was -1.51, -0.8 (p=0-056), and -1.35 (p=0.023). Anemia was observed in the serology group

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at one year as frequently (25%) as at baseline (26.7%) but in the biopsy group improvement from baseline was seen, 27.4% to 7.1 % of patients. At two years, anemia was present in 16.6% of patients. Ferritin values were significantly lower in the serology group at one year (27.5 µg/L vs 59, p=0.040) but not anymore at 2 years. After two years on GFD both TG2-IgA and EmA were often still above the cut-off for positivity in both diagnostic groups (serology group 58.3 % and 50% respectively and in the biopsy group somewhat less, 50% and 28.6% respectively). At long term, no anemia was present anymore. However, 4 children from serology group and 8 from the biopsy group showed antibodies above cut-off even after 8.7 and 10 years of treatment, respectively. Upon long-term follow-up the patients from the biopsy-proven group seemed to be more compliant with the follow-up visits and more adherent to the gluten-free diet.

Conclusion: Both clinical and laboratory results show slow recovery upon gluten-free diet. High percentage of autoantibody positivity still at 2 years on GFD indicates substantial amounts of gluten in the diet. Biopsy-proven patients seemed to have a faster recovery than the serology-diagnosed. Serological diagnosis also appears to negatively influence the adherence and compliance to treatment and follow-up. Novel (drug/vaccine) treatments adjunctive to GFD are an unmet need also in children. Duodenal mucosal recovery results on long-term GFD are warranted.

Disclosure: Nothing to disclose.

PP0312

PREVALENCE OF ANTI-SACCHAROMYCES CEREVISIAE AND ANTI-NEUTROPHIL CYTOPLASMIC ANTIBODIES IN PATIENTS WITH NEWLY DIAGNOSED CELIAC DISEASE

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Introduction: Anti-Saccharomyces cerevisiae antibodies (ASCA) and anti-neutrophil cytoplasmic antibodies (ANCA) are well recognized as serological markers in inflammatory bowel diseases (IBD). With the growing evidence of a bidirectional association between IBD and celiac disease (CD), there has been an increased interest in identifying shared biomarkers for both conditions.

Aims & Methods: We aimed to assess seroprevalence of ANCA, IgA and IgG ASCA, in newly diagnosed CD patients. CD diagnosis was set according to 2013 ACG guideline and 2019 ESsCD guideline. Serum collected at the time of diagnosis for all CD patients was analyzed for IgA and IgG ASCA, and ANCA concentration, using INOVA Quanta Lite® assays. Positivity cut-off was defined as ≥ 25U for ASCA and > 30U for ANCA.

Results: Altogether 82 newly diagnosed CD patients, 82.9% female, were recruited, all of them having atrophic mucosal injury on duodenal biopsy sampling and one being seronegative CD. While none of the CD patients had positive ANCA, seroreactivity against Saccharomyces cerevisiae was seen in 32.93% of cases for IgG ASCA. A lower prevalence for IgA ASCA was recorded, of 4.88%, due to associated IgA deficiency. Moreover, 7.32% of newly diagnosed CD patients had equivocal result for IgG ASCA.

Conclusion: A high prevalence of ASCA, particularly IgG type, was seen in newly diagnosed, biopsy-proven CD patients. Future research should investigate the clinical significance of the seroreactivity to IBD-specific antibodies in CD patients.

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Disclosure: None.

PP0313

SEROLOGICAL DIAGNOSIS OF COELIAC DISEASE - A WEB-BASED SURVEY AMONG NORWEGIAN GASTROENTEROLOGISTS

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Introduction: Serological diagnosis of coeliac disease (CeD) is accepted for pediatric patients in Europe [1]. Both the 2019 European guidelines and the 2023 American guidelines from the American College of Gastroenterology required a diagnostic biopsy in adults [2, 3]. In recent years we have seen several studies addressing the no-biopsy approach in adults [4-6]. Wide-spread implementation of no-biopsy approach demands precise instructions on how the diagnosis can be made and by whom.

Aims & Methods: We aimed at investigation the acceptance among clinicians for a no-biopsy approach, and how they follow current guidelines. A digital survey was sent in November 2023 to members of the Norwegian Society for Gastroenterology. It was sent out once, no reminder, and was open for one month. Replies were collected anonymously. The Society has 452 members including 256 gastroenterologists. Almost all specialists are members of this Society.

Results: We received 72 replies. Of the respondents, 48 were male and 22 female, 1 not-binary and 1 did not answer. Five of them were not clinically active. Fifty-eight of 72 adhered to guidelines and routinely took biopsies both from the bulb and the distal duodenum. Forty-five took 4 biopsies from the distal duodenum, only five of them 2 biopsies. Negative serology was considered insufficient to exclude CeD by 52 respondents. All used IgA-TG2 for diagnosis, 41 also IgG-DGP, whereas 6 used endomysium test (EMA) and 4 also anti-gliadin antibodies. We initially asked if positive serology alone could diagnose CeD and received mixed response; 26 supported the notion, 37 did not and 9 did not know. We thereafter provided information on no-biopsy diagnosis for children. With this information, fourty responders supported serological diagnosis without biopsy, 10 were negative and 21 uncertain. The majority (32) requested 2 strongly positive (more than 10 x ULN) IgA-TG2, whereas 14 considered a single strongly positive test as sufficient. Specialists deemed capable of setting the diagnosis was gastroenterology (48 votes), general practitioner (22 votes), internists (21 votes) any doctor (10 votes). However, half of the respondents considered it not necessary to personally meet the patient to make a diagnosis.

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Conclusion: The no-biopsy approach receives support; but the precise set-up is not clear. If a no-biopsy approach is agreed upon, we should decide on single or repeated tests, on the use of EMA or rely on only IgA-TG2, and if IgG-DGP has a role. The coeliac community should agree on precise guidelines to minimize misdiagnosis and provide the patients with a clear and definite diagnosis.

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Disclosure: Nothing to disclose.

PP0314

VARIABLE PRESENTATIONS OF CELIAC DISEASE & SOCIAL IMPLICATIONS OF CELIAC DISEASE, DATA FROM DEVELOPING COUNTRY

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Introduction: Celiac disease (CD) is an autoimmune enteropathy triggered by ingestion of gluten in genetically susceptible individuals. It has extremely varied clinical presentations. Diagnosis of celiac disease is made by history and clinical presentation compatible with CD, serological screening compatible with CD, histological findings compatible with CD, obvious clinical and serological response to a gluten-free diet, and exclusion of other clinical conditions mimicking CD. Not much is known about presentation of this disease in Pakistani adults.

Aims & Methods: The study is cross-sectional in nature, gathering data from celiac support communities on social media platforms throughout Pakistan. Patients who were diagnosed cases of Celiac
disease on the basis of serology and/or histopathology were interviewed via a structured questionnaire via non probability consecutive sampling technique. Patient's age, gender, age of diagnosis, diagnostic modality, intestinal and extra-intestinal symptoms, gluten free diet status and response and challenges in gluten free diet were recorded while ensuring patient anonymity. Data was entered and statistical analysis was performed via IBM SPSS version 25.

Results: A total of 340 patients were enrolled in present study. Most predominant intestinal symptom was Diarrhea reported by 46.8% of participants followed by Abdominal pain/Bloating in 44.7% (Figure I). Most common Extra-intestinal features were Anemia reported by 51.5% of patients (Figure II). Of note constipation was reported by 21.5% and Stunted growth by 35.3% of patients. Most common utilized diagnostic modality was Serology in 62.9% of patients (Figure III). 86.8% of patients reported symptom resolution with Gluten free diet. This association was statistically significant. Patients who reported Abdominal pain/Bloating were significantly more likely to report improvement in symptoms following Gluten free diet (Figure IV). Challenges in Gluten free diet were availability issues reported by 37.5% followed by financial issues in 25%, social challenges in 19.2% and other issues including compliance and cross contamination by 40% (Figure V)
Pie & Bar Charts for Abstract

Conclusion: Celiac Disease has an extremely Variable Presentation in Pakistani patients. It can present with Classical Diarrhea and Non classical presentation with abdominal pain and bloating as well. Adherence to Gluten free diet was associated with reduction in Symptoms and improvement.
Financial constraints and limited availability of gluten-free products are contributing factors to non-adherence to gluten-free diet. Further Studies are needed to know more about the Variability of Symptoms in underdeveloped Countries.

Disclosure: Nothing to disclose.

PP0315

WORK ABSENTEEISM, QUALITY OF LIFE AND PSYCHOLOGICAL MORBIDITY IN COELIAC PATIENTS ON A GLUTEN-FREE DIET: AN ITALIAN CROSS-SECTIONAL STUDY

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Introduction: The data on the impact of coeliac disease on work loss are still scarce [1-3]. We aimed to evaluate the amount of hours lost at work in adult coeliac patients on a gluten-free diet (GFD), focusing on the reasons behind the work loss (WL), and the relationship with quality of life (QOL) and psychological morbidity.

Aims & Methods: Adult patients with biopsy-proven CD under regular follow-up at two Italian centres were surveyed between October 2022 and January 2024. The survey consisted of specific questionnaires, including the Work Productivity and Activity Impairment questionnaire (WPAI:CD), gastrointestinal symptoms severity score (Visual Analog Scale, VAS), the Coeliac Disease-specific Quality Of Life scale (CD:QOL), the State-Trait Anxiety Inventory scale (STAI-Y1 & STAI-Y2), and the Beck Depression Inventory scale (BDI-II). Amount of hours lost at work in the previous 7 and 30 days were specifically investigated and statistically analysed.

Results: 160 patients on a GFD (121 F, mean age at CD diagnosis 32±14 years, 6% in a GFD for less than 1 year, 63% for more than 5 years, 31% for 1 to 5 years) were enrolled. Eight/160 (6.0%) patients reported having lost working hours in the previous 7 days and 15/132 (11.3%) in the previous 30 days. Reasons for work loss at 30 days included symptoms persistence despite the GFD (40%), attendance of outpatient clinic (40%) and issues with GFD adherence (20%).
The patients had a mean of 3.1 hours of missed work at 7 days and 5.0 hours at 30 days. Focusing on the patients that experienced WL in the previous 30 days, we have found that they were more likely to have on-

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going symptoms despite the GFD (67% vs 93%, p=0.04), scored higher on the survey items regarding depression (median BDI-II score 7.00 vs 12.00, p=0.04), and self-reported a lower QOL (median CD:QOL score 85.00 vs 75.00, p=0.03).

Conclusion: Our clinical data-based study has shown that CD patients can experience a substantial work loss at 30 days, particularly in case of persistence of symptoms despite a GFD, which can have negative impacts on psychological health, as well as overall QOL. This aspect should be considered when trying to develop personalized strategies for the individual follow-ups of coeliac patients.

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Disclosure: Nothing to disclose.

PP0316

A DECADE OF COELIAC DISEASE: A COMPARATIVE STUDY OF THE MALTESE ISLANDS (2013-2023)

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Introduction: Coeliac disease (CeD) is a chronic autoimmune condition affecting the small intestine triggered by gluten.

Aims & Methods: To assess the national management of newly diagnosed CeD patients between 2013 and 2023 in Malta and whether this was carried out in accordance with "European Society for the Study of Coeliac Disease (ESsCD) guideline for CeD and other gluten-related disorders"[1].

The national statistics data of all the newly diagnosed patients with CeD between 2013 and 2023 in the Maltese Islands, were retrospectively analysed. Several factors were considered including patient demographics, serology, histology, imaging and dietitian referral.

Results: 171 patients with newly diagnosed CeD (2013 n=46, 2023 n=125) were recruited. Adults, defined as ≥18 years, accounted for 72.5% (n=124) of the total cohort.

In 2013, 80.4%(n=37) were female while in 2023, 66.4% (n=83) were female (p=0.07), The mean age at diagnosis was 38.5 years (SD19.6) in 2013 and 36.6 years (SD23.4) in 2023. The largest proportion, 17.0%(n=29), were diagnosed in the 31 to 40 age group.

Anti-Tissue Transglutaminase Antibody (anti-TTG IgA-Ab) was taken in 99.4%(n=170), with 92.4%(n=158) having a positive anti-TTG. 4.1%(n=7) had negative anti-TTG IgA-Ab while 2.9%(n=5) had borderline anti-TTG IgA-Ab. Of these 12 patients, 11 patients had biopsies taken with histology confirming the diagnosis of CeD. The other patient was 8 years old with IgA deficiency and had a positive anti-TTG IgG-Ab, with a value of >250 U/mL.

73.1%(n=125) of the total cohort had biopsies taken, with 56.0%(n=70) taken from D2, 3.2%(n=4) taken from D1 and 37.6%(n=47) taken from both D1 and D2. In 3.2%(n=4), the biopsy site was not specified. In 2023, 48.2%(n=41) of patients had D1+D2 biopsies taken, versus only 15%(n=6) in 2013 (p<0.001). Of the remaining 26.9%(n=46) who did not have biopsies taken, 73.9%(n=34) were children.

48%(n=60) of the patients in 2023 were referred to the dietician versus 15.2%(n=7) in 2013 (p<0.001). From the total cohort, only 39.2%(n=67) were referred to the dietician. DEXA scan was carried out in 49.1%(n=84), 40% referred patients in 2023 versus 74% in 2013. 11 patients, 6.4%, required small bowel imaging.

73.7%(n=126) had repeat TTG taken after diagnosis. Only 9 patients, (7.2%), had repeat biopsies taken.

Conclusion: This is the first review of CeD in the Maltese Islands showing a percentage increase over ten years of 171% in diagnosis. The incidence rate was 9.2 new cases per 100,000 in 2013 and 24.3 new cases per 100,000 in 2023. CeD diagnosis was x2.4 times commoner in females. This significant change is also accompanied by the increase in D1 biopsies taken in addition to D2 (p<0.001).

An improvement was noted in the number of dietician referrals carried out from 2013 to 2023. DEXA scan testing was carried out in nearly half the total cohort emphasizing the need to consider this investigation especially in those patients who are at high risk of osteoporosis.

A limitation of this study was the fact that biopsies taken in the private sector were not available for review. Therefore from the total cohort only 7%(n=12) did not have an available biopsy result or review.

A multidisciplinary team approach to the management of CeD is crucial and will help to provide the best outcomes for the patient.

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PP0317

SCREENING FOR COELIAC DISEASE IN INFLAMMATORY BOWEL DISEASE

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Introduction: Coeliac disease (CeD) is an autoimmune condition affecting the small intestine. Research has shown a likely bidirectional association between CeD and Inflammatory Bowel Disease (IBD)[1].

Aims & Methods: Our aim is to retrospectively assess whether patients with IBD are screened for CeD in Malta.

284 patients with a histological diagnosis of IBD were randomly selected from the hospital IBD pool. They were grouped into those receiving biological therapy and those solely on immunomodulators. Data collected included age, sex and whether coeliac serology (TTG – IgA or EMA) and/or duodenal biopsies were taken and whether the results were compatible with a diagnosis of CeD.

Results: From 284 patients (58.1% male, mean age 44.5 years SD=16.4). 58.5% had Crohn's disease, 41.5% had Ulcerative Colitis. 65.1%(n=185) were on biological treatment (Infliximab=21.5%, Adalimumab=22.2%, Vedolizumab=17.6%, Ustekinumab=3.9%).

Out of all patients, 79.9%(n=227) underwent CeD immunology tests either at the time of diagnosis or during follow-up, with no difference in gender (p=0.79) or IBD subtype (p=0.32) whether serology was taken or not. Among them, five patients (2.2%) tested positive for coeliac antibodies. However, only three of these individuals proceeded to have duodenal biopsies performed.

One of these patient's biopsies yielded results consistent with Coeliac disease. All these five patients had repeat TTGs, while four of them had DEXA scan performed. 3 patients with positive CeD serology had small bowel imaging. 33.5% of IBD patients had duodenal biopsies taken during upper GI endoscopy.

Conclusion: It's reassuring to observe that a significant majority (79.9%) of patients with IBD underwent testing for CeD. However, despite consistent follow-up in gastroenterology clinics, one-fifth of IBD patients in our cohort were never tested. Our findings reveal a local incidence of CeD in IBD of 0.4%, which is slightly lower than the reported 0.75% [1].

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This underscores the significance of considering CeD as a potential pathology that may contribute to ongoing symptoms and abnormal blood test results in IBD patients.

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Disclosure: No disclosures or Conflict of Interest.

PP0318

EVALUATION OF GLUTEN-FREE DIET ADHERENCE AND QUALITY OF LIFE IN COELIAC PATIENTS

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Introduction: As there is currently no drug treatment available to cure celiac disease (CD), treatment is based on a strict, lifelong gluten-free diet (GFD). The GFD is very restrictive and patient adherence is not always a guaranty.

Aims & Methods: The aim of our study was to evaluate adherence to the GFD and the impact of the disease on quality of life.

We conducted a cross-sectional study, includoiong celiac patients followed up at the consultation during the period from January to March 2023. Adherence to the GFD was assessed using the validated Biagi questionnaire, and patients' quality of life was assessed using the CD-Qol questionnaire, which we translated into Tunisian dialect.

Results: We included 33 patients followed for celiac disease with confirmed diagnosis assessed by anatomopathological examination according to Marsh classification. The mean age of included patients was 42.2 years. The sex ratio (M/F) was 0.32. The mean course of the disease was 13.3 years. The dietary survey revealed insufficient macronutrient intake and unbalanced macronutrient distribution (67% of patients). Assessment of adherence to the GFD revealed that 44.4% of patients followed the diet strictly, 33.3% followed the GFD non-strictly, and 22.2% did not follow the GFD at all. The most frequent reason for non-adherence to the GFD was economic issues (87.9%). With regard to quality of life, 57.7% of patients had a good quality of life, as evidenced by a high CD-Qol score > 73. Only 12.1% of patients had poor quality of life (low score < 46). Patients with good adherence to RSG had better quality of life (p=0.027).

Conclusion: Our study demonstrates the link between adherence to the RSG and quality of life in coeliac patients, hence the importance of encouraging patients to follow the diet properly.

Disclosure: No conflict of interest.

PP0319

AWARENESS AND KNOWLEDGE OF CELIAC DISEASE AMONG THE HEALTH CARE PROFESSIONALS: A CROSS-SECTIONAL SURVEY FROM THE LARGEST PUBLIC SECTOR HOSPITAL OF KARACHI, PAKISTAN

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Introduction: Celiac disease (CD) is a common chronic illness with various clinical presentations. Despite a rise in prevalence, 95% of patients go undetected, often misdiagnosed or ignored. Inadequate understanding among healthcare professionals contributes to this delay.

Aims & Methods: To evaluate the knowledge of HCPs regarding celiac disease.

Results: Out of 374 participants, 196 (52.4%) were male and 178(47.5%) were female, indicating sample diversity. Furthermore, the split of professional roles demonstrates the vocational diversity of the sample, with nurses comprising the largest group at 147 (39.3%), followed by internees 106(28.3%), house officers 93(24.8%), and medical officers 28(7.49%). The study reveals that a majority of the population perceives the disease as autoimmune, with a higher percentage believing it can be hereditary.

The perception of the illness as lifelong is prevalent, with a high confidence in a cure. A significant proportion acknowledge the potential reversal of symptoms through a strict gluten-free diet, but also associate gluten with sensitivity and allergies.

According to the poll, most people are aware of the symptoms of celiac disease, including chronic diarrhea, bone pain, stomach pain, bloating, exhaustion, weight loss, and skin/nail manifestations.

Conclusion: The study highlights the unsatisfactory knowledge level among HCPs regarding CD, suggesting the need for increased awareness and e-learning activities to improve CD knowledge.

Disclosure: Nothing to disclose.

PP0320

POTENTIAL CELIAC DISEASE IN CHILDREN: HEALTH STATUS ON A LONG-TERM GLUTEN-CONTAINING DIET

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Introduction: Potential celiac disease (PCD) is a clinical condition characterized by the presence of a positive CD-specific serology and a normal intestinal architecture. Asymptomatic PCD patients are generally addressed to continue on a gluten containing diet (GCD), but long-term risks of this approach have never been explored.

Aims & Methods: In the present study we aim at investigating nutritional and autoimmune complications possibly developing overtime in a cohort of asymptomatic PCD children left on a GCD. We compared parameters of growth, nutritional status, and autoimmunity between time of diagnosis and in occasion of last medical check, after a long-term gluten containing diet.

Results: Altogether we collected data from 171 PCD children with a mean follow-up time of 3 years (*range 0.35-15.3 years)*. During follow-up, despite patients did not reduce the amount of daily gluten intake , anti-tissue transglutaminase (anti-TG2) antibodies spontaneously and significantly decreased. Most of parameters analysed did not change over follow-up (height centile, ferritin, albumin., cholesterol, calcium, alkaline phosphatase , parathormone, vitamin D, p>0.05) or even improved (weight and BMI

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centile, haemoglobin, blood iron, HDL, glycaemia and HbA1C, p<0.05), always remaining in the limit of normality. Equally autoantibodies for other concomitant autoimmune disorders did not increase overtime. Similar results were obtained excluding from the analysis patients that stopped producing anti-TG2 and those with a follow up < 3 years.

Conclusion: Our pilot study provides reassuring results on the maintenance of a gluten containing diet in asymptomatic PCD children, even when a long term follow-up is considered.

Disclosure: no conflict of interest to declare.

PP0321

EVIDENCES OF SENESCENCE AND OXIDATIVE STRESS IN THE DUODENAL MUCOSA OF ACTIVE CELIAC DISEASE. AN EXPLORATORY STUDY

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Introduction: The exact molecular pathways contributing to epithelial abnormalities in celiac disease (CeD) remain incompletely understood. Cellular senescence, characterized by cell-cycle arrest and involving oxidative stress (OS), has been associated with the onset of chronic diseases, and attributed to the decline in regenerative capacity and the release of inflammatory agents. However, the potential involvement of senescence and OS in the context of intestinal mucosal damage in CeD remains to be investigated.

Aims & Methods: Aim: To investigate the presence of senescence and OS markers in the duodenal mucosa of both CeD patients and non-CeD controls.

Methods: This was an exploratory study wherein we analyzed duodenal biopsies obtained from 4 adult patients diagnosed with CeD and 5 non-CeD controls. Senescence was analyzed through the expression of the protein p21 and its relevance in terms of cell proliferation was evaluated by the expression of the protein Ki67 (detection with specific primary antibodies and peroxidase immunostaining). Positive cells were semi-quantified in the surface epithelium and crypts according to the percentage of positive cells among the total cells of the analyzed section (no staining reaction = 0; 1-30% stained nuclei = 1; 31–60% = 2; > 60%= 3). Carbonylated proteins were detected using amino-oxyfunctionalized fluorophores. The intensity of fluorescence was quantified in villi and crypts and expressed as relative fluorescence units (RFU) by surface. The statistical analysis was performed by non-parametric tests.

Results: Intestinal biopsies from CeD patients (3 females; median age 26.5 yr; 25%-75% IQR: 21-29) and non-CeD controls (3 females; median age 38 yr; 25%-75% IQR: 33-38) were studied. The percentage of p21-positive cells in the surface epithelium was not different between CeD patients and controls (median [25-75% IQR]): 100% (95-100%) vs 98% (98-100%), respectively (p=0.6).

By contrast, the expression of p21 in the epithelium of crypts was significantly higher in CeD patients vs. controls (45% [40-67.5%] and (0 [0-1]), respectively (p=0.02). Ki-67 expression in the surface epithelium was higher in CeD patients vs. controls (2% [2-3]) and 0% [0–0%]), respectively (p=0.04), while the determinations were similar in the crypts of both groups. Finally, carbonylated proteins were detected with greater intensity in CeD patients compared to controls both in the surface epithelium (5050 RFU [3862-5622]) vs. 2335 RFU [2324-2477]), respectively (p=0.03), and in crypts (2260 RFU [2197-2598]) vs. (1403 RFU 1313-1476]), respectively (p=0.03).

Conclusion: This exploratory study presents the first evidence of senescence and OS in the duodenal mucosa of CeD patients at diagnosis and compared to controls suggesting that these phenomena could be involved in a dysfunctional epithelial regenerative process in CeD. These novel findings offer new perspectives for understanding CeD mucosal damage. However, further confirmation in a larger series of cases is required to validate these findings.

Disclosure: Nothing to disclose.

PP0322

CARBOHYDRASE ACTIVITY IN PROXIMAL AND DISTAL SEGMENTS OF SMALL INTESTINE IN HEALTH, IRRITATED BOWEL SYNDROME AND ENTEROPATHY WITH MEMBRANE DIGESTION DISTURBANCE

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Introduction: Diagnostics of enteropathy with membrane digestion disturbance (EMDD) is based on the evaluation of carbohydrase activity in duodenal mucosa (DM). There is no information about diagnostic value of ileum mucosa (IM) carbohydrases.

Study objective. Assess the diagnostic value of carbohydrases activity in mucous membrane of duodenum and distal ileum in patients with EMDD and irritable bowel syndrome (IBS).

Aims & Methods: The study includes 12 patients with EMDD, 16 patients with IBS and 12 relatively healthy persons (control group). Study design included analysis of food tolerance and evaluation of activity of sucrase (S), glucoamylase (GA), maltase (M) and lactase (L) in mucous membrane of duodenum and ileum based on the method of Dahlquist with N.I.Belostotsky modification. Microsoft Office Excel and STATISTICA 10.0 were used for statistical evaluation of material. Mann-Whitney criterion was used to evaluate the differences between quantitative parameters. The differences were considered to be significant at p <0.05. Control group was used for comparison.

Results: Ratio of carbohydrase activity in duodenum and ileum in control group and patients with IBS and EMDD.

		Glucoamylase	Maltase	Sucrase	Lactase
Control	Duodenum	776 ±163	2664 ±317	175 ±25	83 ±17
	Ileum	714 ±121	2532 ±438	130 ±27	16 ±4
	P	0.763	0.81	0.2	0.001
IBS	Duodenum	590 ±91	2118 ±124	194 ±33	84 ±6
	Ileum	1275 ±158	4142 ±489	191 ±27	14.4 ±6
	P	0.008	0.0004	0.934	0.002
EMDD	Duodenum	164 ±23	859 ±76	69 ±7	27 ±6
	Ileum	858 ±182	3259 ±59.6	143 ±30	10 ±3
	P	0.0014	0.0009	0.03	0.004

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In control group the carbohydrases activity in duodenum mucosa was not significantly statistically different from the one in ileum mucosa.
In patients with IBS glucoamylase and maltase activity parameters in IM were significantly different from activity in DM, which can be interpreted as functional inequality of enzyme activity of various departments of small intestine.
In patients with EMDD GA level in ileum mucosa is 5.2 times higher, maltase level is 3.8 times higher and sucrose level is 2 times higher than duodenum enzyme levels, which demonstrates the possibility of formation of compensatory reaction of adaptive shift type towards distal segments of small intestine, initially described by A.M. Ugolev et al. in 1985
Conclusion: Enzyme activity distribution along small intestine is an adaptor function of its segments. Our study confirmed the need and promising outlook of further investigations in terms of evaluation of proximo-distal gradients of enzyme activity to study compensatory reactions which can be fulfilled by distal segment of small intestine in case of duodenum damage.
The evaluation of carbohydrase activity in ileum mucosa provides additional opportunities for diagnostics of EMDD and its differentiation from IBS, which can be used to improve therapeutic approaches.
Disclosure: Nothing to disclose.

PP0323

WHAT IS THE SIGNIFICANCE OF SYMPTOMS DURING THE GLUCOSE SIBO BREATH TEST IN PATIENTS WITH DISORDERS OF GUT-BRAIN INTERACTION (DGBI)?

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Introduction: Small intestinal bacterial overgrowth (SIBO) is a commonly diagnosed finding in diverse conditions ranging from DGBI to GI motility, connective tissue and neurodegenerative disorders (1). A wide spectrum of signs and symptoms have been attributed to SIBO (1,2). Diagnosis is achieved either by direct culture of duodenal aspirate or indirectly, by demonstrating exhaled fermentation gases with ingested sugars, such as glucose or lactulose (3,4). In the breath tests used for carbohydrate intolerances, the symptoms provoked are at least as important for the diagnostic interpretation as the measured gas concentrations. However, the symptoms generated during SIBO breath tests are neither evaluated nor mentioned in most testing guidelines. SIBO exists in asymptomatic populations and the association with symptoms remains uncertain (1,2).
Aims & Methods: We examined the acute relationship between breath gases and the symptoms generated during glucose breath tests in DGBI. A standardised breath test with glucose (50g in 150ml water) was performed after a 24-hour low-FODMAP diet and >8-hour fast. Exhaled hydrogen and methane concentrations were measured and symptom assessments (composite symptom index score, SI used for intolerance breath tests)(5) were performed every 15 minutes for 2 hours in 790 male (30%) and female patients of average age 40.0 years referred to our gastroenterology practice with DGBI (Rome 4 criteria). Breath gas concentration thresholds of >20ppm and >10ppm above baseline were considered positive tests for hydrogen and methane, respectively.
Results: Overall, 59 (7.5%) and 111 (14.1%) DGBI patients were SIBO positive using the hydrogen and methane thresholds, respectively. The composite SI was positive in 208 (26.3%) of patients. The most common individual symptoms were fullness, nausea, tiredness, headache (each in >30% of patients), followed by bloating, flatulence, abdominal cramps or pain, GORD and problems concentrating (each in 20-30% of patients). A positive SI with a negative hydrogen test was seen in 151 (19.1%) of DGBI patients, in 108 (13.0%) with a negative methane test and in 55 (7.0%) with both negative hydrogen and methane tests. A negative SI occurred in 2.4% of patients if either of the gas tests was positive.
Conclusion: During the glucose SIBO breath test, positive gas results and evoked symptoms are closely associated. However, glucose frequently provokes symptoms in DGBI patients with negative hydrogen and/or methane breath test results. This may be due to a discrepancy between breath and intestinal gases, production of gases beyond hydrogen and methane, or the induction of symptoms by glucose or its metabolites via alternative mechanisms.
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Disclosure: Nothing to disclose.

PP0324

REDUCING THE NUMBER OF BREATH SAMPLES IS A COST-EFFECTIVE STRATEGY TO OPTIMIZE THE LACTOSE BREATH TEST

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Introduction: The lactose breath test (LBT), through the measurement of hydrogen (H_2) and methane (CH_4), is the test of choice in assessing lactose malabsorption (LM). Although direct costs of LBTs are low, indirect costs are high due to the long test duration requiring prolonged involvement of qualified staff and available infrastructures. Sampling frequency represents the principal time-consuming part of the test. The current European guidelines recommend a test duration of 3-5 hours (hrs) and sample intervals of 30-60 minutes (min). In our center, the standard LBT consists of 9 samples collected every 30 min up to 4 hrs after the ingestion of 25 g of lactose.
Aims & Methods: The study aimed to optimize the methodology of LBT either by reducing the test duration by up to 3 hrs (Shortened-test: ST) or reducing the number of breath samples with hourly intervals by up to 4 hrs (Five-sample test: FST). Retrospectively, patients (pts) undergoing LBT at the Gastroenterology Unit from January 2022 to October 2023, were included. Gas peak ≥20 ppm for H_2 and/or ≥10 ppm for CH_4 above baseline was diagnostic for LM. Results of ST and FST were calculated from the standard LBT and compared with them. The delta-ppm (Δ-ppm) and the area under the curve (AUC) were calculated to stratify gas excretion into 3 grades of malabsorption (mild, moderate, severe) according to the percentile distribution. Data were expressed as mean (±DS), and analyzed with Chi-square, Mann-Whitney and Pearson correlation. $p<0.05$ was considered statistically significant. Specificity was always 100% due to the lack of an independent gold-standard test.
Results: Of the 543 pts enrolled (F 71.5%, mean age 43.7 ± 17.6 yrs), 60.4% (328/543) were positive for LM and were significantly older (44.7±17 years) than negative ones (40.5±17.6 years) ($p=0.006$). LBTs were mainly positive for H_2 (80.8% for H_2, 8.2% for CH_4, and 11% for both gasses; $p<0.0001$).

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FST showed higher accuracy than ST. In the ST, the positivity rate was 54.1% (294/543), sensitivity was 89.6% with a false negative rate of 10.4% (34/328) and accuracy was 93.7%. In FST, the positivity rate was 57% (309/543), sensitivity was 94.2% with a false negative rate of 5.8% (19/328) and accuracy was 96.5%.

A statistically positive correlation was found between the Δ-ppm value and the AUC, both for H_2 (r=0.94) and CH_4 (r=0.78). In the FST, according to the percentile stratification of Δ-ppm, 84% (16/19) of negative pts showed mild LM (≤25th percentile) and the remaining 16% (3/19) had Δ-ppm values slightly higher.

Conclusion: Comparing FST and ST, only FST proved to be a cost-effective strategy to optimize LBT without reducing test sensitivity. Extending the interval of breath sampling collection with hourly measurements reduces the indirect costs, by decreasing the involvement of qualified staff and maximizing the procedure by doubling the number of pts who could be tested in a day.

Disclosure: Nothing to disclose.

PP0325

EFFICACY AND QUALITY OF LIFE IMPACT OF INTRAVENOUS FERRIC CARBOXYMALTOSE TREATMENT IN PATIENTS WITH IRON DEFICIENCY ANAEMIA DUE TO IRON MALABSORPTION

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Introduction: Iron malabsorption (FeM) may lead to iron deficiency anaemia (IDA). Its most common causes are celiac disease (CD) due to atrophy of the small intestine mucosa, corpus atrophic gastritis (CAG), and gastric surgery (GS): reduced hydrochloric acid secretion derived from corpus oxyntic mucosa atrophy or surgical resection causes a decreased conversion of Fe(III)ion to Fe(II)ion, the latter being the more easily absorbable form. IDA affects more than two billion people over the world and it can cause troublesome symptoms as asthenia and fatigability. Iron oral supplementation is considered the standard therapy but frequently it leads to side effects reducing long-term compliance. Ferric carboxymaltose (FCM) is an intravenous (iv) preparation shown to be safe and effective in improving the iron status (FeS), but its efficacy in patients with FeM and IDA has not been investigated so far.

Aims & Methods: To evaluate the safety, efficacy and quality of life impact of iv FCM in treating pts with IDA due to FeM. A prospective observational single centre study was conducted on 37 pts with IDA caused by CAG (n=19/51.4%), CD (n=9/24.3%), and GS (n=9/24.3%). Pts with other causes of IDA as GI bleeding were excluded. At baseline (T0), pts were clinically and biochemically evaluated (symptoms, SF12, CBC and FeS: ferritin, iron, transferrin, transferrin saturation (TS)). FCM dosage was calculated according to Ganzoni formula considering Hb values and body weight. If the dosage needed was up to 1000mg, iv infusion was scheduled in a single session, otherwise a 2 nd infusion after 1 wk was planned. After treatment, CBC and FeS were assessed at 4 (T4) and 12 (T12) wks. SF12 was revaluated after 12 wks (cut-off physical component summary (PCS)<44, mental component summary (MCS)<46). IDA was defined as Hb<13 and <12 g/dl for men and women, ferritin<45 ng/dl or TS<15%. Moderate and severe anemia were defined as Hb 8-10 and Hb<8 g/dl. Student's t-test was used to compare CBC and FeS parameters at baseline (T0), 4 weeks (T4) and 12 weeks (T12).

Results: All 37 patients completed 12-wks-follow-up. At T0 Hb was mean±SEM 10.9±0,2g/dl, 18.9% had moderate and 2.7% severe anaemia. In pts treated in a single-session-infusion (64.9%), dosage of FCM was mean±SEM 975±12mg; in pts treated in two sessions (35.1%), FCM dosage was mean±SEM 1400±77mg. One patient reported mild side effects (pruritus, cutaneous erythema) not requiring FCM interruption and reversing after H_2 antagonist treatment within 15 minutes. At T4 and T12, 70.4% and 78.4% of pts recovered from IDA. At T4 mean Hb significantly increased compared to T0 (12.7±0.2 vs 10.8±0.2 g/dl, p<0.0001). At T12 a further increase was observed (13.1±0.2 g/dl, p<0.05). Also, TS increased from 6% at T0 to 20% at T12, p<0.0001. At T4 ferritin levels (ng/dl) significantly increased compared to T0, but slightly decreased at T12 (T0 28.5±11.2 vs T4 188.2±25.7, p<0.0001; T12 124.9±25.7, p=0.10). RDW (%) significantly increased at T4 and significantly decreased at T12(T0 15.4±0.5 vs T4 18.6±0.7, p=0.0003; T12 14.8±0.4, p<0.0001). The proportion of patients with SF12 MCS>46 and PCS>44 scores significantly increased at T12 compared to T0 indicating improved mental and physical health perception after FCM treatment.

Conclusion: In pts with IDA due to FeM, ivFCM is an effective and safe treatment, leading to recovery of IDA in nearly 80% of cases at 12 weeks after treatment and concomitantly to an improved mental and physical health perception. These data show that ivFCM treatment could be considered as a first line treatment for IDA patients with FeM.

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Disclosure: Nothing to disclose.

PP0326

CLINICAL AND HISTOLOGICAL SPECTRUM OF MALABSORPTION SYNDROME: INSIGHTS FROM ENDOSCOPIC EVALUATION AND HISTOLOGICAL ANALYSIS

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Introduction: Malabsorption syndrome encompasses a spectrum of gastrointestinal disorders characterized by impaired absorption of nutrients, leading to various clinical manifestations. Endoscopic and histological evaluations play crucial roles in diagnosing underlying etiologies, including celiac disease.

Aims & Methods: This retrospective study aimed to analyze the clinical presentation, endoscopic findings, and histological yield in patients with malabsorption syndrome, focusing on the subset exhibiting villous atrophy to investigate the prevalence of celiac disease. We conducted a retrospective analysis of 100 patients with malabsorption syndrome who underwent endoscopic evaluation between January 1, 2023, and December 31, 2023. Clinical data, endoscopic findings, and histological results were collected and analyzed.

Results: Among the 100 patients, 60 were males and 40 were females. Common presenting symptoms included fatigue (33 patients), recurrent mouth ulcers (15 patients), bloating (12 patients), diarrhea (24 patients), and abdominal discomfort (16 patients). Endoscopic findings revealed normal mucosa in 34 patients, decreased height of duodenal folds in 24,

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and fissured mucosa in 42. Histological analysis showed increased intraepithelial lymphocytes in 36 patients, crypt hyperplasia with increased intraepithelial lymphocytes in 20, partial villous atrophy in 22, subtotal villous atrophy in 12, and total villous atrophy in 10 patients. Giardia was detected in histological specimens from 4 patients with partial villous atrophy. Interestingly, only 12 patients had positive celiac serology despite a substantial number exhibiting villous atrophy on histological examination.

Conclusion: This study highlights the diverse clinical presentation and histological spectrum of malabsorption syndrome. Despite significant villous atrophy, celiac disease was diagnosed in only a minority of patients, suggesting the need for further investigation into the etiology of villous atrophy in this subset of patients.

Disclosure: Nothing to disclose.

PP0327

ENDOSCOPIC SMALL BOWEL SURVEILLANCE IN PATIENTS WITH SHORT BOWEL SYNDROME IN TREATMENT WITH TEDUGLUTIDE

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Introduction: Concerns regarding colorectal cancer in short bowel syndrome (SBS) patients receiving teduglutide (TED) treatment have led to the recommendation of colonoscopy for screening and surveillance.[1,2] While there are reports of new small bowel polyps in TED patients, data on small bowel cancer risk and surveillance are lacking.[3,4]

This study aims to evaluate the feasibility and effectiveness of using double-balloon enteroscopy (DBE) for surveillance in SBS patients undergoing TED treatment.

Aims & Methods: Data from patients with SBS in TED treatment who underwent DBE within a year of starting TED at a tertiary referral center for enteroscopy between December 2019 and October 2023 were collected. The primary endpoints were technical success, adverse events, and total enteroscopy rates. Secondary endpoints included the incidence of small bowel polyps and residual small bowel length measurement.

Results: Nine patients (7 female, median age 65.5, IQR 58-79) with SBS (22% type 1, 55.6% type 2, 22.2% type 3-SBS) underwent DBE (10 anterograde, 2 retrograde) under procedural sedation. Crohn's disease (33.3%) and post-surgical complications (33.3%) were the most frequent causes of SBS. The median interval from TED prescription to DBE surveillance was 9.1 months (IQR 3.2-32.9). The median depth of maximal insertion was 85 cm (IQR 50-187). No adverse events occurred during all the procedures. The total enteroscopy rate was 75% (80% [8/10] of anterograde DBEs, 50% [1/2] of retrograde DBEs). Panenteric DBE allowed the estimation of small bowel residual length in all cases (2/2) where this important clinical information was unknown and revealed a different residual small bowel length compared to surgery in 44% of patients. No small bowel malignancies were detected during the study period.

Conclusion: DBE surveillance for small bowel malignancy in SBS patients undergoing TED treatment is feasible and effective. Total enteroscopy can be frequently achieved with a single endoscopy in these patients, allowing easy oncologic surveillance.

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Disclosure: Nothing to disclose.

PP0328

UTILITY OF ENDOSCOPIC ISCHEMIC POLYPECTOMY FOR THE TREATMENT OF POLYPS IN THE SMALL INTESTINE OF PEUTZ-JEGHERS SYNDROME

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Introduction: Peutz-Jeghers polyps (PJPs), seen in Peutz-Jeghers syndrome (PJS), are found throughout the gastrointestinal tracts except for the esophagus. Of these, polyps in the small intestine are likely to become numerously developed and enlarged, resulting in intestinal intussusception which requires surgical treatment. Therefore, endoscopic treatment is recommended to prevent intestinal intussusception.

Recently, endoscopic ischemic polypectomy (EIP) as the strangulation of the polyps using endoclips without polypectomy has been a treatment option for PJPs of the small intestine as well as conventional endoscopic mucosal resection (EMR). However, there has been insufficient data on the clinical utility and safety of EIP compared to EMR, and that needs to be clarified.

Aims & Methods: The purpose of this study was to evaluate the utility and safety of EIP for PJP in the small intestine. We retrospectively recruited PJS patients who underwent ballon-assisted enteroscopy at Chiba University Hospital between February 2010 and December 2023. The clinical information such as patient clinical backgrounds, treatment option, number of polyps, maximum diameter of polyps, and incidence of bowel obstruction after treatment were analyzed in patients who underwent endoscopic treatment for PJPs of the small intestine. The safety profile per-treatment was evaluated using the inverse probability weighting (IPW) method regarding the incidence of bleeding and intussusception after treatment.

Results: During the research period, 55 enteroscopies were performed, of which 37 (20 EMR and 17 EIP) endoscopic treatments were performed for a total of 239 PJPs (diameter 10-40 mm) in 10 PJS patients. The pathological evaluation was performed for 15 lesions (diameter 15-40 mm) treated with EMR, and all were diagnosed as PJPs.

While there was no difference in the total time for endoscopic procedures including insertion and treatment between the EMR and EIP groups (182.5 min vs. 179.1 min, P=0.497), the number of treated polyps at one procedure was significantly higher in the EIP group (3.0±2.1 polyps by EMR v.s. 10.5±13.0 polyps by EIP, P=0.015). Post-treatment complications included bleeding in 5 treatments while no intussusception was observed. IPW analysis revealed that EIP was associated with a significantly lower risk of complications (P=0.017, OR=0.186, 95% CI 0.047-0.738) (Table).

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	OR	95%CI	P value
EIP	0.186	0.047 - 0.738	0.017

Table. Weighted logistic regression analysis of the risk of bleeding after treatment for PJP.

Adjusted for gender, age at treatment, presence of intestinal adhesion, scope approach (via-oral or via anal), endoscopist (expert or non-expert), insert length, number of treated polyps, maximum diameter of polyps, and total time as confounding factors.

Conclusion: EIP for PJPs in the small intestine is a safer treatment option for numerous lesions than EMR.

Disclosure: Nothing to disclose.

PP0329

DEEP LEARNING AND CAPSULE ENDOSCOPY: AUTOMATIC PANENDOSCOPIC DETECTION OF PROTRUDING LESIONS

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Introduction: While capsule endoscopy (CE) offers a minimally invasive way to examine the entire digestive tract, current analysis methods are time-consuming and susceptible to errors. Protruding lesions are frequentlyencountered with varying significance throughout the GI tract. This study aims to develop and test a Convolutional Neural Network for panendoscopic automatic detection of protruding lesions on CE exams.

Aims & Methods: A multicentric retrospective study was performed on 1222 CE exams. A total of 191455 images from six different CE device types were evaluated, with 52717 confirmed protruding lesions (epithelial andsubepithelial lesions) after a rigorous triple-validation process. First, we conducted a 5-fold cross validation with exam-split for assessing CNN's robustness. Then, we evaluated CNN's global performance in an exam-split train and test division of data, assessing the latter. Primary outcomes measures were sensitivity, specificity, accuracy, positive predictive value (PPV), negative predictive value (NPV) and area under the conventional receiver operating characteristic curve (AUC-ROC) and the precision-recall curve (AUC-PR).

Results: Regarding test set, the sensitivity were 90.7% and specificity 94.0%. The PPV and NPV were 95.3% and 88.3%, respectively. The global accuracy was 92.1%. The CNN processed 112 frames per second.

Conclusion: This study aims to address the gap in AI-enhanced capsule panendoscopy by investigating the development of the first CNN for the detection of protruding lesions across the GI tract. AI's improvement of CE's diagnostic accuracy, along with the growing interest in minimally invasive procedures, may contribute to increasing access to this diagnostic tool.

Disclosure: Nothing to disclose.

PP0330

DIAGNOSTIC YIELD AND OUTCOMES OF SMALL BOWEL CAPSULE ENDOSCOPY IN PATIENTS WITH SMALL BOWEL BLEEDING RECEIVING ANTITHROMBOTICS

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Introduction: Use of antithrombotics including antiplatelet agents and anticoagulants is associated with upper and lower GI bleeding, thus a higher rate of positive findings is reasonable to be expected when patients receiving these drugs undergo SBCE for the investigation of SBB. However, the results of published studies concerning the impact of antithrombotics on the SBCE diagnostic yield remain conflicting. Furthermore, the clinical outcome of patients under antithrombotics who undergo SBCE for the investigation of SBB is not currently known.

Aims & Methods: We aimed to determine the diagnostic yield and outcome of patients receiving antithrombotic drug therapy subjected to small bowel capsule endoscopy (SBCE) for the investigation of small bowel bleeding (SBB). Multicenter retrospective analysis of collected data from all patients undergoing SBCE for the investigation of SBB from March 2003 to June 2023. The diagnostic yield of SBCE was defined as the detection of positive findings that could explain the cause of the patient's bleeding. Rebleeding was defined as evidence of bleeding within 1 year after the index episode.

Results: During the study period, 8401 patients underwent SBCE for SBB investigation. Bleeding lesions were detected in 1103/2535 (43.5%) patients receiving antithrombotics, compared to 1113/5866 (18.9%) not receiving these drugs (p<0.00001). Following capsule endoscopy, a therapeutic intervention was possible in 390/2216 (17.5%) patients with a bleeding lesion. Rebleeding occured in 927 (36.5%) of patients receiving antithrombotics (36.5%), compared to 795 (13.5%) not receiving these drugs (13.5%, p<0.00001).

Conclusion: Both the diagnostic yield of SBCE and the rebleeding rate were higher in patients with SBB receiving antithrombotics. Therapeutic intervention was possible in a real world setting only in a minority of patients with positive findings.

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Disclosure: Nothing to disclose.

PP0331

EFFICACY OF YOGURT CONTAINING *LACTOBACILLUS PARAGASSERI* IN ALLEVIATING ON SMALL INTESTINAL INJURY INDUCED BY COMBINED TREATMENT OF ASPIRIN AND A POTENT ACID SECRETION INHIBITOR

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Introduction: Our previous study using capsule endoscopy has demonstrated that low-dose aspirin (LDA) frequently induces small intestinal injury, which is shown to be alleviated by some probiotics. However, the mechanism underlying the effects of probiotics remains elusive.

Aims & Methods: Aim: To clarify the mechanism underlying the alleviating effects of probiotics on LDA-induced mucosal injury using gut microbiota analysis.

Methods: Twenty patients in each of the three groups were recruited: control, received no treatment; PPI, treated with either a proton pump inhibitor (PPI) or potassium-competitive acid blocker (P-CAB); and Aspirin+P, treated with aspirin and PPI/P-CAB. Ileal fluid was collected using colonoscopy for ileal microbiota analysis. All patients were treated with *Lactobacillus paragasseri* OLL2716 (LG) for 6 weeks after colonoscopy, and stool samples were collected for faecal microbiota analysis before and after LG treatment. The participants were asked to complete the Gastrointestinal Symptom Rating Scale (GSRS) questionnaire before (at 0 weeks) and after 6 weeks of LG intake.

Results: Faecal microbiota analysis revealed no significant difference in α-diversity among the three groups before and after LG intake. In contrast, β-diversity indices differed significantly between the control and PPI groups as well as between the control and Aspirin+P groups before LG intake. However, no such significant differences between them were found any more after LG intake. The composition of microbiota structures in the PPI and Aspirin+P groups did not differ before or after LG intake. Ileal microbiota analysis revealed no differences among the three groups. The GSRS scores in the control and Aspirin+P groups significantly decreased after LG intake compared to those before LG intake. However, the GSRS scores in the PPI group did not differ significantly before and after LG intake.

Conclusion: The effect of aspirin itself on the microbiota structure and composition was minimal when combined with PPI or P-CAB. The remodelling effect of probiotics on Aspirin+P-induced dysbiosis may be one of the mechanisms underlying the alleviating effects of probiotics on aspirin-induced mucosal injury.

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PP0332

GASTROINTESTINAL ANGIODYSPLASIA RESOLUTION FOLLOWING TRANSCATHETER AORTIC VALVE IMPLANTATION: A PROSPECTIVE COHORT STUDY

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Introduction: Heyde syndrome is the co-occurrence of aortic stenosis and gastrointestinal bleeding secondary to vascular lesions, including angiodysplasias. Several studies have demonstrated cessation of gastrointestinal bleeding after transcatheter aortic valve implantation (TAVI), but the etiology and effects on vascular lesions are largely unknown.

Aims & Methods: We aimed to establish the effects of TAVI on gastrointestinal vascular lesions and identify the factors associated with recovery. In this prospective, single-center cohort study, patients with iron-deficiency anemia on the TAVI waitlist from September 2020 to February 2022 were assessed by capsule endoscopy. Those with vascular lesions were reassessed six months after TAVI. Endoscopic images were anonymized and evaluated by two independent researchers.

Results: Twenty-four patients underwent capsule endoscopy, and vascular lesions were present in 18 (prevalence 75%). TAVI was performed in 15/18 patients with vascular lesions, of which 11 agreed to a second capsule endoscopy. The mean number of vascular lesions across the gastrointestinal tract decreased from 6.4 (SD 5.6) to 2.0 (SD 2.1) (P = .036). The number of vascular lesions decreased in 9/11 (82%) patients, including 6/11 (55%) who no longer had typical angiodysplasias. Cessation of angiodysplasias was less frequent in patients who had multiple valvular heart disease before TAVI (0/3 [0%] vs 6/8 [75%]) and patients with significant paravalvular leakage after TAVI (2/5 [40%] vs 4/6 [67%]).

Conclusion: Angiodysplasias are present in 75% of anemic patients with severe aortic stenosis. TAVI effectively reduces the size and number of angiodysplasias in these patients.

Disclosure: Lia Goltstein, Maxim Rooijakkers, Naomi Thierens, Selene Schoormans, Charles Houdeville, Marlijn Hoeks, and Sanna Rijpma do not have potential conflicts of interest or disclosures to report. Hanneke Beaumont has received speaker fees from Medtronic. Erwin van Geenen has received research funding from Viatris, Boston Scientific, and Olympus and serves as a consultant for MTW-Endoskopie and Microtech. Xavier Dray is cofounder and shareholder of Augmented Endoscopy, has served as a consultant for Norgine and Provepharma, and received speaker fees from Bouchara Recordati, Fujifilm, Medtronic, Norgine, and Sandoz. Niels van Royen has received research funding from Abbott, Philips, Medtronic, and Biotronik, has served as a consultant for RainMed, Castor, and Medtronic, and received speaker fees from Abbott. The Radboudumc, on behalf of Joost Drenth, has received research funding from Gilead to support Hepatitis C elimination in The Netherlands.

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PP0333

IS FIT FIT FOR DETECTION OF SMALL BOWEL PATHOLOGY? A PROSPECTIVE STUDY

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Introduction: The role of faecal immunochemical testing (FIT) as a biomarker to stratify patients with lower gastrointestinal (GI) symptoms is well established. However, its use in patients with iron deficiency anaemia (IDA) is unclear. The aim of this study was to investigate the role of FIT in predicting small bowel (SB) pathology in patients with IDA.

Aims & Methods: This was a single tertiary centre prospective study. Inclusion criteria was adults (>18 years and < 80 years) with IDA who were referred to secondary care for upper and lower gastrointestinal tract (GI) endoscopy. These patients also had a FIT test done in primary care. All participants had a small bowel capsule endoscopy (SBCE) prior to their endoscopic investigations. Participants with subsequent upper or lower gastrointestinal tract malignancy were excluded from the study as this was considered as the most likely cause of IDA. IDA was defined as a Hb < 131g/L for men and < 110 g/L for women with ferritin< 30ug/L and/ or iron levels < 11umol/L.

Results: In total 79 patients were included in the final analysis with a median age of 67yrs (IQR 52-74); haemoglobin 103(IQR 95- 118) and ferritin 11(IQR 6-11). There were 13 patients on NSAID's at the time of their investigations. There were 35 (44%) patients with significant findings on SBCE which was classed as contributing to IDA. These findings included angioectasia in n= 21 (26%) of the patients which was the most common finding. The other findings included erosions= 5 (6.3%); inflammatory strictures= 3 (3.8%); active Crohn's n=1 (1%); visible blood with no clear source n=3 (3.8%); bleeding angioectasia n=1 (1%); and ulcer=1 (1%). A positive FIT (>10) had a sensitivity, specificity, positive predictive value (PPV) and negative predictive value (NPV) of 34.29%, 54.55%, 37.5% and 51.08% respectively.

Conclusion: Patients with IDA have a high diagnostic yield on SBCE and should routinely have SBCE as part of the work up for IDA. There is a pressing need for biomarkers to help triage SBCE referrals, however FIT is not a reliable biomarker for the presence of small bowel pathology in patients with IDA.

Disclosure: Nothing to disclose.

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Poster
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Nutrition

09:00-18:00 / Poster Exhibition

PP0334

USEFULNESS OF LAPAROSCOPIC SLEEVE GASTRECTOMY FOR NAFLD (MASLD)

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Introduction: Since 2006 at our hospital, lifestyle modification for obese patients has been provided through a team approach with a physician, dietician, nurse, physical therapist and psychologist. From 2016, a surgeon and anesthesiologist were added to the team for metabolic/bariatric surgery (laparoscopic sleeve gastrectomy: LSG).

Aims & Methods: The aim of this study is to clarify the characteristics of obese patients with NAFLD for whom LSG was effective. Of 40 obese patients who were received LSG between 2017 and 2021 with follow-up period of more than 1 year, 19 were male and 21 were female, 24 were with diabetes, 16 with other complications. The diagnosis of NAFLD was made by liver imaging for fat deposit and stiffness by Fibroscan, and liver biopsy in some cases.

Results: The mean preoperative BMI was 43, the mean Fib-4 index was 0.98, and the postoperative insulin resistance (CPI) of diabetic patients improved significantly, with 15 patients in complete remission and 6 in incomplete remission after surgery. One year after surgery, the mean weight loss was 18.3%, the Fib-4 index did not decrease significantly, but the mean AST (U/L) improved from 38 to 18, ALT (U/L) from 55 to 15, GGT (U/L) from 78 to 40, CAP (dB/m) from 348 to 279, and liver stiffness (kPa) from 12.1 to 7.6, significantly. In patients who underwent follow-up biopsy, there was a trend toward improvement in liver fibrosis.

Conclusion: Since LSG significantly improves hepatic markers in obese patients with NAFLD, metabolic/bariatric surgery should be positively considered.

Disclosure: Nothing to disclose.

PP0335

HYPERTRIGLYCERIDEMIA IS A DOSE-DEPENDENT RISK FACTOR OF DIABETES MELLITUS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: The prevalence of type 2 diabetes has more than doubled in the past 20 years, and literature suggests that nearly 90% of diabetes cases are preventable if risk factors are eliminated early. One of these potential risk factors is hypertriglyceridemia. Given the high prevalence of hypertriglyceridemia (which can affect nearly one-third of the population), the link between the two conditions is of crucial importance. Therefore, we aimed to investigate the dose-dependent effect of hypertriglyceridemia on the development of diabetes mellitus.

Aims & Methods: We carried out a systematic search in three databases (MEDLINE, Embase, and CENTRAL) on the 9th of November, 2023 with no restriction. In the final analysis, we investigated adult population with different triglyceride levels (exposure) and our outcome of interest was the development of diabetes mellitus. We calculated pooled odds ratios (OR) with 95% confidence intervals (CIs) and assessed the risk of bias with the Quality In Prognosis Studies (QUIPS) tool. The protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO) under the number CRD42023471288.

Results: According to the meta-analysis result of eight cohorts with a total population number of 127431, those who had hypertriglyceridemia had more than a 2-fold higher risk (OR:2.17[CI 1.52-3.10]) of developing diabetes mellitus during the follow-up period than those whose triglyceride levels were below 1.7 mmol/L. Those who had borderline high triglyceride levels at baseline (1.7-2.3 mmol/L) had significantly higher odds of developing diabetes mellitus during follow-up (OR 1.45[CI 1.29-1.63]) according to the data of six cohorts with 36807 patients. Those whose baseline triglyceride levels were higher than 2.3 mmol/L also had significant and even higher odds of the development of diabetes mellitus (OR:2.16 [CI 1.74-2.67]) than patients with normal triglyceride levels, based on the results of 6 cohorts with data on 39068 patients.

Conclusion: Hypertriglyceridemia is a dose-dependent risk factor of diabetes mellitus.

Disclosure: Nothing to disclose.

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PP0336

THE IMPACT OF OBESITY ON DEVELOPMENT OF NON-CARDIA GASTRIC CANCER ACCORDING TO SEX IN SOUTH KOREA: A RETROSPECTIVE COHORT STUDY

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Introduction: Several studies have recently reported an increased risk of gastric cancer (GC) risk in both underweight and obese population in East Asia. However, the impact of underweight or obesity on the development of non-cardia GC is not fully established yet. In addition, GC is a male dominant cancer, and considering the phenotype differences in body composition between males and females, sex differences must be considered in evaluating the association between obesity and the risk of developing non-cardia GC.

Aims & Methods: Therefore, we aimed to evaluate associations between body mass index (BMI) and non-cardia GC risk with consideration of sex. This study was performed based on a retrospective cohort, which consists of Koreans randomly sampled from participants who underwent the National Health Screening programs and is termed National Health Insurance Service–Health Screening Cohort (NHIS-HEALS). The National Health Insurance system provides health checkups biennially for adults aged ≥20 years through the National Health Screening program. This program covers over 70% of total population in Korea, thus the NHIS-HEALS is a nationwide representative database in South Korea. This study includes a total of 341,999 subjects including 4,103 incident non-cardia GC cases identified during a median follow-up period of 10 years after exclusion of incident GC cases identified within 2 years from the follow-up start. We classified subjects into five groups according to BMI (<18.5, 18.5-22.9, 23.0-24.9, 25.0-29.9, ≥30.0kg/m^2) and evaluated the association between BMI level and GC risk by calculating hazard ratio (HR) and corresponding 95% confidence interval (95% CI) based on the Cox proportional hazard regression.

Results: Increased non-cardia GC risks were observed in severely obese (BMI ≥30.0 kg/m^2) males and obese (BMI 25.0-29.9 kg/m^2) females, respectively (HR, 1.40; 95% CI, 1.11-1.75 for severely obese males; HR, 1.25; 95% CI, 1.06-1.46 for obese females).

Age[1]	Sex	Body mass index (kg/m^2)				
		<18.5	18.5-22.9	23.0-24.9	25.0-29.9	≥30.0
		HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)	HR (95% CI)
<50 years	Total	1.53 (0.74-3.16)	1.01 (0.77-1.32)	1.00	0.88 (0.66-1.17)	1.41 (0.77-2.58)
	Males	1.74 (0.75-4.02)	1.06 (0.77-1.46)	1.00	0.94 (0.69-1.30)	1.91 (1.03-3.55)
	Females	1.39 (0.17-11.02)	0.88 (0.38-2.02)	1.00	0.91 (0.34-2.44)	N/A
≥50 years	Total	1.06 (0.85-1.32)	1.01 (0.93-1.10)	1.00	1.13 (1.04-1.23)	1.17 (0.96-1.44)
	Males	1.09 (0.85-1.39)	0.96 (0.87-1.06)	1.00	1.08 (0.99-1.19)	1.33 (1.04-1.70)
	Females	0.97 (0.61-1.54)	1.17 (0.99-1.38)	1.00	1.30 (1.10-1.53)	0.98 (0.68-2.26)

Table. Association between body mass index and non-cardia gastric cancer risk according to sex in Korea.

Abbreviations: HR, Hazard ratio; CI, Confidence interval
1. Age at BMI measurement

In contrast, evidences supporting that underweight (BMI <18.5kg/m^2) is associated with increased non-cardia GC risk was not found overall (HR, 1.08; 95% CI, 0.88-1.33). In males, the increased risk of non-cardiac GC in the severely obese group remained regardless of age at BMI measurement.

However, the increased risk of non-cardia GC risk related to obesity was confirmed only in elderly females, and was not found in females under 50 years of age. In addition, increased non-cardia GC risk in the normal weight group (18.5-22.9 kg/m^2) was found compared to that in overweight group among females over 50 years of age.

Conclusion: Results suggest that elevated risk of non-cardia GC development might be originated from obesity, especially in elderly populations in East Asia regardless of sex. However, this study proposes that being overweight, which is the pre-obesity stage, may rather have a lower risk of developing non-cardia GC than being normal weight in elderly females.

Disclosure: Nothing to disclose.

PP0337

GASTROINTESTINAL ADVERSE EVENTS WITH GLP-1 RAS IN NON-DIABETIC OVERWEIGHT AND OBESE INDIVIDUALS: SYSTEMATIC REVIEW AND DOSE-RESPONSE META-ANALYSIS

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Introduction: Overweight and obesity present global health challenges, often managed through lifestyle changes and medications like glucagon-like peptide-1 receptor agonists (GLP-1 RAs), which can cause gastrointestinal (GI) adverse events. However, research on their effects in non-diabetic populations is lacking.

Aims & Methods: This review addresses this gap by examining the impact of GLP-1 RAs on GI adverse events in non-diabetic overweight or obese individuals. We systematically reviewed the literature by conducting a search using PubMed and EMBASE, adhering to predefined inclusion and exclusion criteria, to assess GI adverse events linked with GLP-1 RAs in non-diabetic overweight or obese populations. We conducted a dose response meta-analysis to evaluate each adverse event associated with different GLP-1 RAs. The quality of the included studies was appraised utilizing the Cochrane Collaboration's assessment tool.

Results: Our systematic review and meta-analysis encompassed thirty-eight articles, incorporating data from a total cohort of 32,882 individuals. Predominant GI adverse effects observed included nausea, vomiting, diarrhea, constipation, and abdominal pain. Across all evaluated GLP-1 RAs, a notable elevation in nausea risk was noted, with orforglipron exhibiting the highest risk, succeeded by exenatide, tirzepatide, semaglutide, and liraglutide. Moreover, liraglutide, orforglipron, semaglutide, and tirzepatide were associated with heightened vomiting risk, while cagrilinitide and exenatide displayed no significant increase. Exenatide, cagrilinitide, and orforglipron demonstrated no significant association with diarrhea risk. Conversely, semaglutide and liraglutide correlated with an elevated risk of constipation, while cagrilinitide and exenatide did not exhibit a significant increase. Notably, certain GLP-1 RAs were linked to a heightened

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risk of abdominal pain, with liraglutide, semaglutide, and tirzepatide displaying significant increases, whereas orforglipron showed no significant elevation in abdominal pain risk.

Conclusion: In conclusion, GLP-1 RAs are associated with an increased risk of GI adverse effects, including nausea, vomiting, diarrhea, constipation, and abdominal pain. Importantly, distinct GLP-1 RAs exhibited varying degrees of risk for these adverse effects, underscoring the importance of tailored risk assessment and management strategies in clinical practice.

Disclosure: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

PP0338

EXPLORING THE IMPACT OF BREASTFEEDING DURATION ON NEURODEVELOPMENT IN FIVE-YEAR-OLD CHILDREN

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Introduction: Breastfeeding, recognized for its essential nutrition and health benefits, potentially influences central nervous system development. This study delves into the nuanced relationship between breastfeeding duration and neurodevelopment in five-year-old children.

Aims & Methods: A cohort of 72 children living in Varna, Bulgaria was followed up from birth to age five, all of them being term-born infants with normal birth weight and devoid of perinatal infections or abnormalities. Comprehensive data on demographics and dietary practices were collected. Neurodevelopmental assessments at five years included motor and speech development, articulation, nonverbal intelligence, and behavior, evaluated using the validated NDT5 tool. Children were stratified into 2 groups based on breastfeeding duration (up to age six months vs. over six months). Statistical analysis was performed using Jamovi 2.0.0.0.

Results: Children in both groups exhibited similar demographic characteristics and BMI distributions. Notably, significant differences emerged between the groups in girls' motor development (U=54, p=0.018) and overall development (U=52, p=0.045). Similarly, boys showed disparities in nonverbal intelligence (U=66.5, p=0.008), as evidenced by both intention-to-treat (ITT) and per-protocol (PP) analyses.

Conclusion: Breastfeeding duration appears to exert a discernible influence on neurodevelopment in five-year-old children. Notably, distinct effects were observed in girls' motor development and boys' nonverbal intelligence. Further longitudinal investigations are warranted to optimize neurodevelopmental outcomes based on breastfeeding duration.

Disclosure: Nothing to disclose.

PP0339

PROBIOTIC INTERVENTION IMPROVES QUALITY OF SLEEP – A RANDOMIZED, DOUBLE-BLINDED, PLACEBO-CONTROLLED TELEMEDICAL STUDY

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Introduction: Poor quality of sleep has detrimental effects on job performance, accident rates and health risks, with a potential link to the gut microbiome composition. First pilot trials have indicated beneficial effects of probiotic modulation of the microbiome on quality of sleep.

Aims & Methods: We conducted a randomized, double-blind, placebo-controlled trial testing the effects of a probiotic intervention on quality of sleep, quality of life and intestinal microbiome.

In total, 130 volunteers with impaired quality of sleep (PSQI>5) were included in the study. They were randomized in a 1:1 ratio to a 28-day intervention with either a multispecies probiotic (Omnibiotic-Stress Repair) or a placebo. Participants completed validated questionnaires to estimate quality of sleep (PSQI), quality of life (SF12, GIQLI) and perceived stress (PSQ20) before, during and after the intervention. Furthermore, stool samples for 16S rRNA gene sequencing were collected before and after the intervention. Ninety-four participants finished the study and could be included in the analysis.

Results: The 50 participants (88.6% female, 41.2±10.6 years old) analysed in the probiotic group and 44 participants (88.0% female, 40.1±10.7 years old) in the placebo group were well comparable at baseline, including the initial PSQI-score (10.1±2.7 vs. 10.5±2.6). After two weeks of intervention, both groups showed a comparable improvement in their PSQI-scores. After four weeks of intervention, the improvement of the PSQI score was significantly more pronounced in the probiotic group compared to the placebo group (6.8±2.9 vs. 7.7±3.1; p=0.036), mainly due to improved sleep efficiency and latency. The intervention did not influence quality of life or perceived stress. Ingested bacteria were partially recovered in the microbiome, causing a slight shift in beta diversity in the probiotic group, while alpha diversity and the overall composition of the microbiome remained unaffected.

Conclusion: In conclusion, the intervention with a multispecies probiotic was able to improve quality of sleep of participants with poor quality of sleep beyond an observable placebo-effect. Participants reported falling asleep more quickly and spending less time awake in bed. This well-powered RCT confirms that the probiotic modulation of the intestinal microbiome can be effective in alleviating sleep disturbances.

Disclosure: VS received research funding and speaker honoraria from Institut Allergosan (Graz, Austria) and Winclove Probiotics (Amsterdam, The Netherlands).

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PP0340

THE IMPACT OF NUTRITIONAL SUPPLEMENTATION ON CLINICAL ACTIVITY, QUALITY OF LIFE AND DISABILITY OF PATIENTS WITH IBD: A BRAZILIAN MULTICENTER LONGITUDINAL STUDY

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Introduction: Inflammatory Bowel Diseases (IBD), including Crohn's Disease (CD) and Ulcerative Colitis (UC), pose significant global health challenges. Nutrient deficiencies and malnutrition are common in both forms of the disease. Dietary supplementation with TGF-β has been suggested to enhance patients' nutritional status, quality of life, and improvement of the symptoms. However, comprehensive evaluations of its clinical impact are needed.

Aims & Methods: This study aims to assess the efficacy of nutritional supplementation in improving clinical outcomes, nutritional status, quality of life, fatigue, disability, and body image among IBD patients.

Methods: Multicenter longitudinal study, enrolling 42 IBD patients in clinical activity and/or at nutritional risk. Patients received the nutritional supplement enriched with TGF-β and underwent assessments before supplementation (M0), at 30 days (M30), and at 60 days (M60) of supplement use. Clinical parameters, disease activity (Harvey-Bradshaw score for CD and partial Mayo score for UC), nutritional evaluation, body composition, presence of fatigue, anxiety and depression, disability, and body image were evaluated. Statistical analyses included descriptive statistics and ANOVA ($p<0.05$).

Results: 42 patients were included: 30 (71%) with CD and 12 (29%) with UC; 48% women, age of 35.1 ± 15.65 years. Among CD patients, 30% were in remission, 16.67% had mild, 43.33% moderate, and 10% severe activity. According to Montreal classification, 36.67% presented ileocolonic disease, 36.67% fistulizing behavior, 26.67% stenosing behavior, 33.33% perianal disease. For UC patients, 41.67% had mild and 58.33% moderate activity; 41.67% presented pancolitis. All patients presented malnutrition or nutritional risk at enrollment. Significant improvements were observed in HBI score comparing M0 (7.97±5.70) vs M30 (4.61±3.16) vs M60 (3.27±1.75), $p=0.0055$. There was also an improvement of the number of bowel movements (5.45±4.65 vs 2.90±2.02 vs 2.65±1.55). Nutritional assessment revealed a significant increase in brachial circumference (24.23±3.68 vs 25.37±3.51 vs 26.03±4.10, $p=0.0182$). Fatigue decreased (40.58±14.07 vs 33.12±15.72 vs 30.38±14.88, $p=0.0293$), as well as disability (51.14±25.34 vs 36±20.89 vs 30.33±22.64, $p=0.0032$). It was observed an improvement of IBD score (135.08±38.88 vs 145.20±48.27 vs 152.71±58.96; $p=0.3737$) and body image (10.78±7.37 vs 9.68±8.42 vs 5.57±6.04; $p=0.1395$), with no significance.

Conclusion: Nutritional supplementation demonstrated notable improvements in clinical and nutritional parameters, as well as reduction in fatigue and disability, underscoring its potential as an adjunctive therapy for IBD management

Disclosure: Nothing to disclose.

PP0341

PREVALENCE OF FOOD ALLERGY AMONG JORDANIAN CHILDREN

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Introduction: Food allergy, characterized by an immune reaction to proteins in food, presents a significant global health challenge, especially among children. It is a growing concern worldwide, impacting over 10% of the general population. Epidemiological studies suggest a rising prevalence of food allergy, with estimates indicating that approximately 1 in 10 adults and 1 in 12 children are affected.

Aims & Methods: Our aim was to assess the prevalence of food allergies among schoolchildren aged 5 to 14 years in central and northern Jordan. We conducted a randomized cross-sectional online questionnaire among schoolchildren aged 5 to 14 in central and northern Jordan. A total of 1629 parents completed the questionnaire, providing information on their children's demographic details, feeding history, food allergy occurrences, as well as atopic history for both the child and their family.

Results: The prevalence of parent-reported food allergy was 11.5% (95% CI = 10-13%). Among these cases, 10.1% (95% CI = 8.6-11.7%) were identified as having convincing food allergies, while 5.7% (95% CI = 4.5-6.8%) were diagnosed by a physician. The most common allergens reported were eggs (17.8%), cow's milk (15.8%), peanuts (14.5%), and strawberries (13.8%). A majority (67.1%) reported having a single allergen, while 15.8% reported two allergens, and 17.2% reported more than two allergens.

Among the participants, Food allergy prevalence was higher among children with coexisting atopic diseases and those with a family history of allergies. Regional variations in prevalence were observed, with urban areas showing higher rates.

Conclusion: This study sheds light on the prevalence of food allergy among schoolchildren in Jordan, emphasizing the importance of considering convincing food allergy alongside parent-reported data. The findings underscore the significant burden of food allergy and its association with other atopic conditions, highlighting the need for comprehensive management strategies and further research to understand underlying factors.

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Disclosure: Nothing to disclose.

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PP0342

SARCOPENIA RELATED FUNCTIONAL STRAINS AND EXPLORE THEIR MECHANISMS BASED ON GERM-FREE ANIMALS

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Introduction: Sarcopenia is a progressive, systemic skeletal muscle disease associated with aging that severely affects the quality of life and overall health of older adults. Emerging research demonstrates a strong association between the onset of sarcopenia and gut microbiota.

Aims & Methods: 16 out of 30 sarcopenia patients who showed significant skeletal muscle improvement after fecal microbiota transplantation (FMT) were selected and validated by metagenomic, cultured and animal experiments.

Results: Our clinical observations show a marked improvement in sarcopenia symptoms among older patients post FMT, leading to a notable rise in muscle mass ($P < 0.05$). To delve deeper into this occurrence, we conducted metagenomic sequencing of patient fecal samples pre- and post-treatment. FMT distinctly modified the intestinal microbiota composition. Utilizing germ-free animal models and culturomics, we identified Roseburia inulinivorans as a potentially beneficial strain for the alleviation of sarcopenia. Further experimental validation displayed that Roseburia inulinivorans escalated MyoG and MyoD expression ($P < 0.05$), augmented muscle mass ($P < 0.01$) and grip strength ($P < 0.01$) in mice by boosting branched-chain amino acid metabolism within the host, diminishing sarcopenia symptoms in the process.

Conclusion: These outcomes not only validate the intestinal microbiota-sarcopenia relationship but also introduce novel insights and strategies for sarcopenia management.

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Disclosure: Nothing to disclose.

PP0343

EXPLORING THE RELATIONSHIP BETWEEN NUTRITIONAL STATUS AND NON-ALCOHOLIC FATTY LIVER DISEASE IN INFLAMMATORY BOWEL DISEASE: A CROSS-SECTIONAL ANALYSIS

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Introduction: Inflammatory Bowel Disease (IBD), encompassing Crohn's Disease (CD) and Ulcerative Colitis (UC), shares chronic and inflammatory characteristics with Non-Alcoholic Fatty Liver Disease (NAFLD). While their precise pathogenesis remains unclear, emerging evidence implicates dysbiosis and metabolic changes. This study aims to investigate the association between IBD and NAFLD, focusing on metabolic syndrome and its nutritional implications

Aims & Methods: Objective: To evaluate the nutritional status and body composition in IBD patients, comparing groups with and without NAFLD, and identifying the associated factors.

Methods: A cross-sectional study was conducted, assessing clinical data, metabolic syndrome presence, nutritional status, and biochemical markers. Patients were categorized into IBD+NAFLD and IBD-only groups based on hepatic ultrasound findings. Statistical analysis included descriptive statistics and logistic regression.

Results: 127 patients were included, comprising 58 CD and 69 UC, with a mean age of 45.56 years**.** Most patients were female and did not engage in physical activity. Patients were mostly in clinical remission and endoscopic activity. 58 patients (45.67%) had hepatic steatosis on ultrasound, forming the IBD+NAFLD group. The IBD+NAFLD group had higher BMI, fat mass, and all circumference values, Table 1, as well as a higher prevalence of hypertension ($p=0.0446$) and dyslipidemia ($p=0.0377$). It is noteworthy that, although most of these patients were classified as overweight by BMI, considering anthropometric data, most IBD+NAFLD patients were classified as eutrophic. 86.21% of patients in the IBD+NAFLD group met the criteria for MAFLD. No difference was observed regarding medication in use. Regarding biochemical tests, the IBD+NAFLD group had higher values of hematocrit ($p=0.0023$), liver enzymes (AST, $p=0.0141$ and ALT, $p<0.0001$), fasting glucose ($p=0.0107$), glycated hemoglobin ($p=0.0146$), basal insulin ($p=0.0003$), and triglycerides ($p=0.0001$), with lower HDL cholesterol ($p=0.0101$)

Conclusion: The study highlights a significant association between nutritional status and NAFLD in IBD patients. Those categorized as eutrophic had a higher prevalence of hepatic steatosis, potentially linked to dietary factors.

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PP0344

WITHDRAWN

PP0345

CAN DIETARY PATTERNS AND DIET QUALITY BE ASSOCIATED WITH DISEASE ACTIVITY?

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Introduction: Inflammatory bowel disease (IBD) is characterised by a relapsing-remitting disease course, known to be influenced by dietary factors. Therefore, this study aims to examine the relationship between dietary patterns and diet quality with disease activity in IBD patients.

Aims & Methods: From a one-year study of IBD patients from two Dutch hospitals, a cross-sectional analysis of the baseline data was conducted. The GINQ-FFQ[1], an IBD-specific food frequency questionnaire was used to assess dietary intake. Consumed foods were categorised into 26 food groups and analysed for dietary patterns using principal component analysis (PCA)[2]. The Dietary Inflammatory Index (DII), a dietary quality score reflecting inflammatory potential, was calculated according to Shivappa et al.[3], using 28 of the 45 components. Patients were categorised by sex, disease type (Crohn's Disease (CD) or ulcerative colitis (UC)), and biochemical disease activity (remission: CRP ≤ 5 mg/L and/or FCP ≤ 150 µg/g). Mann-Withey U and T-tests were performed to assess differences in dietary patterns and diet quality between patient groups.

Results: A total of 177 patients (52% female, 55.4% CD, median age 50 years) correctly filled out the GINQ-FFQ. Three dietary patterns were identified: a western diet, a flexitarian diet, and a convenience diet. The western diet was high in bread, meat and poultry, pastries and biscuits, cold meats, potatoes and root vegetables, sugar and sweets, sweet sauces, and cheese. It was low in nuts and seeds, vegetables, soups, and fruit. The flexitarian diet consisted of many savoury sauces, eggs, vegetables, potatoes and root vegetables, fats and oils, soups, herbs and spices, savoury snacks and crisps, fish, legumes, and grain products and thickeners. Lastly, the convenience diet was high in grain products and thickeners, composite dishes, non-alcoholic drinks, and meat and poultry, but low in fats and oils, cold meats, alcoholic drinks, and cheese. No differences in consumption of these dietary patterns were observed between patients with active (n=40) and remitted disease (n=137) (p=0.779, p=0.087, p=0.212, respectively). Females adhered less to Western (p=0.001) and convenience food patterns (p=0.009) compared to males. UC patients favoured a flexitarian diet more than CD patients (p=0.030). No differences in DII scores were found (p=0.802). Patients in biochemical remission consumed more alcohol (p=0.04) compared to those with active biochemical disease.

Conclusion: This analysis found no direct association between dietary patterns or dietary inflammatory potential and disease activity. However, patients in remission consumed more alcohol, and a trend towards a flexitarian diet was observed in those with active disease. The full study will follow these patients for a year to assess potential changes in diet and disease activity.

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PP0346

IMPACT OF ANAL FISSURE ON QUALITY OF LIFE AND THE EFFECTIVENESS OF HIGH-FIBER DIET IN TREATMENT

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Introduction: Anal fissures are painful benign conditions that significantly impact patients' quality of life (QoL).

Aims & Methods: This study aimed to evaluate the QoL of patients with anal fissures and assess the effectiveness of a high-fiber diet in treatment. A prospective cross-sectional study was conducted over 4 months (September-December 2023) on patients with anal fissures. QoL was assessed using the Hemo/Fiss Burden score and the newly validated REALISE Score. The REALISE Score measures pain severity, post-defecatory pain, analgesic use, bleeding frequency, and impact on QoL.

Results: 62 patients were included, with a female predominance (sex ratio: 1/2). The average age was 54.3 years. Both scores revealed significant impairments in QoL across all domains, including physical discomfort, psychological distress, and sexual dysfunction. A strong correlation was found between the Hemo/Fiss and REALISE scores (r: 0.71; p <10-4), indicating the validity of the REALISE Score for QoL assessment.

Treatment and Effectiveness: All patients were advised to adopt a high-fiber diet, which resulted in healing in 73% of cases within 4 weeks. This improvement is likely due to the softening of stool and facilitation of regular bowel movements, which reduces pain and discomfort associated with anal fissures.

Additionally, the high-fiber diet significantly improved QoL scores compared to those who did not take fiber (p=0.039). Bleeding and pain at defecation were significantly reduced during six weeks of high fiber diet associated to medical treatment compared to those using medical treatment only (p=0.041).

Conclusion: This study demonstrates the significant impact of anal fissures on QoL and highlights the effectiveness of a high-fiber diet in promoting healing and improving QoL. The REALISE Score offers a valuable tool for assessing the severity of anal fissures and monitoring treatment progress.

Disclosure: Nothing to disclose.

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PP0347

NON-CELIAC GLUTEN/WHEAT SENSITIVITY: GUT MICROBIOME AS A DIAGNOSTIC TOOL AND THERAPEUTIC TARGET

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Introduction: Non-celiac gluten/wheat sensitivity (NCG/WS) is a clinical condition in which gluten/wheat ingestion evokes, while withdrawal ameliorates or reverts gastrointestinal (GI) and extra-GI symptoms. The pathogenetic mechanisms of NCG/WS are poorly understood and no diagnostic biomarker is so far available. NCG/WS diagnosis is established by excluding celiac disease (CD) and wheat allergy (WA), evaluating response to a gluten-free diet (GFD) and the effects of gluten reintroduction. Initial evidence suggests a central role played by the gut microbiota in NCG/WS.

Aims & Methods: This study aimed to identify: i) microbiota species valid as biomarkers for rapid and non-invasive NCG/WS diagnosis; ii) the effects of GFD on microbiota in NCG/WS. Subjects with a clinical suspect of either NCG/WS or CD and healthy volunteers (Ctrl) were enrolled while on a free diet. A thorough work-up (including clinical, endoscopic, allergologic and serological evaluations) was applied to diagnose NCG/WS or CD while excluding WA. Instructions for GFD were provided to all participants. Stool samples for gut microbiome were obtained from NCG/WS and CD before (t0) and after 3, 7, 30 and 200 days of GFD (t1, t2, t3, t4). Stool samples from Ctrl were obtained at t0, t1, t2, and t3. The presence and severity of GI (GI Symptom Rating scale) and extra-GI symptoms of NCG/WS were recorded at each time point (t0-t4). NCG/WS diagnosis was confirmed in participants with a 30% improvement in symptom scores in t3 *vs.* t0. Metagenome from stools was shotgun-sequenced. Data were processed and microbiome profiled with MetaPhlAn v.4.1 (Jun.23). Alpha-diversity (Shannon index, SI), beta-diversity and relative abundances were computed. Principal coordinates analysis was performed (Bray-Curtis dissimilarity test) and groups were compared (PERMANOVA test). Differences in SI at t0 among cohorts were assessed (Mann-Whitney U-test). Species present at any time point in at least 10 subjects were assessed. Candidate biomarkers were validated in newly diagnosed NCG/WS and CD cohorts. Species varying relative abundance during the GFD were identified (linear mixed model and Friedman's ANOVA). P-values were combined with the harmonic mean and corrected for multiple testing (Benjamini-Hochberg method, q-value).

Results: Twenty-eight NCG/WS (F, 37±16 yrs), 11 CD (9 F, 32±15 yrs) and 37 Ctrl (25 F, 33±14 yrs) were enrolled. Alpha-diversity at t0 was comparable (p>0.19) in NCG/WS (SI=4.0), CD (SI=3.7) and Ctrl (SI=4.1), whereas beta-diversity was different among groups (p=0.026). A total of 507 species were identified. The candidate biomarker *Clostridiaceae bacterium* (SGB47672) showed lower abundance in NCG/WS *vs.* Ctrl (q<0.2), data confirmed in validation cohorts (172 NCG/WS, 243 CD). Alpha-diversity was preserved (p>0.05) in NCG/WS and CD up to t4 and in Ctrl (t3) in GFD, whereas concerning changes in relative abundance (q<0.2; p<0.05) we identified a reduction of *Bifidobacterium longum* (SGB17248).

Conclusion: We showed that NCG/WS exhibits a peculiar metagenomic signature with *C. bacterium* (SGB47672) being a potential biomarker for NCG/WS diagnosis. Our study suggests that the use of probiotic supplements can help to restore *B. longum* (SGB17248) deficiency during GFD. Beyond integrating GFD treatment, probiotics containing Bifidobacteria can positively contribute by exerting anti-inflammatory and protective effects on the intestinal epithelial barrier in NCG/WS.

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PP0348

DIET AND EARLY-ONSET COLORECTAL CANCER: VALIDATION OF A NOVEL AD HOC DESIGNED FOOD FREQUENCY QUESTIONNAIRE

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Introduction: The incidence of early-onset colorectal cancer (eoCRC), defined as colorectal cancer occurring in young adults under the age of 50, is increasing globally. Questionable eoCRCs' exogenous risk factors are represented by diet and lifestyle, even though a scant literature is available to date. The DEMETRA study is being conducted to assess diet and lifestyle as potential risk factors for eoCRC in various European and non-European countries with populations characterized by large variations in dietary/drinking habits and eoCRC incidence.

Aims & Methods: Therefore, our aims were to (i) develop a unique and shared semi-quantitative food frequency questionnaire (SQFFQ) able to accurately describe dietary and drinking habits of eoCRCs and healthy age-matched controls of different countries at global level that will be involved in the DEMETRA study; (ii) validate the SQFFQ.

We designed an ad-hoc, shared, online SQFFQ to investigate the usual consumption of 329 foods over the past year. These foods were classified in groups and subgroups using the same criteria as the EPIC Italy study [1]. In addition to frequency of consumption, the tool investigates the portions habitually consumed using validated photographs, household measures, and standard units, as well as types of seasoning and methods of cooking and storage. A special software was then developed to analyze responses and link them to food composition tables in order to provide a nutritional

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breakdown of individual and collective diets. The SQFFQ was then validated for repeatability by administering it twice, 3 weeks apart, to a sample of 30 young adults under 50 years (mean age 35.1 y ± sd 7.71; 63,3% females, 33.3% males, 3% preferred not to answer). To evaluate repeatability, the measurement error for each food group was estimated as the percentage change between the estimates of food consumption for the same individual. Afterwards, the agreement between the two measurements for each food group was measured with Cohen's kappa coefficient.

Results: For 40 out of 65 food groups, the difference between the two SFFQs was less than 20%. The greatest percentage differences were found, as expected, for food groups with sporadic consumption (snacks, barbecue sauces, ketchup, mayonnaise) or consumed in minimal quantities (spices). Agreement levels, represented by the calculation of Cohen's kappa coefficient, were as follows: 12 food groups showed fair/sufficient agreement (Cohen's kappa 20-40), 23 foods exhibited good/moderate agreement (Cohen's kappa >40-60), and 22 food groups demonstrated high substantial agreement (Cohen's kappa >60). Insufficient agreement, with Cohen's Kappa values < 0.21, was observed for only three food groups (snacks, sauces, mayonnaises).

Conclusion: The ad hoc designed SQFFQ provides a reasonably repeatable measure of dietary intake and can be used to assess the dietary and drinking habits of volunteers in this age group. Indeed, most staple foods in the Italian diet, including pasta, fruit, vegetables, legumes, eggs, meat, coffee, and tea, are well estimated by the SQFFQ. However, as yet described in literature, challenges persist in estimating the consumption of foods assumed sporadically (such as snacks) or in small quantities such as spices. Definitive conclusions will be drawn after the completion of the external validation, in which the SQFFQ will be validated against the gold standard, represented by the completion of a 4-day food diary followed by a dietary recall.

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Disclosure: Nothing to disclose.

PP0349

PREVALENCE AND IMPACT OF SARCOPENIA IN PATIENTS WITH INFLAMMATORY BOWEL DISEASES: A PROSPECTIVECOHORT STUDY

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Introduction: Sarcopenia is associated with increased morbidity and mortality in IBD. The main objective of thisstudy was to prospectively evaluate sarcopenia by combining morphological and functional criteria inan outpatient IBD population.

Aims & Methods: In a prospective cohort study, we included all IBD patients admitted to the Day Hospital (DH) of ourdepartment between March 1st and 31st, 2023. Patients with short bowel syndrome were notincluded. Muscle mass was assessed using a Bodystat®1500MDD dual-frequency impedance meter,and muscle function was assessed by measuring handgrip strength, Carym® digital dynamometer. Pre-sarcopenia was defined by a handgrip measurement < 16 to 26, and myopenia was defined by anappendicular muscle mass index (AMMI) < 5.5 to 7 kg/m2 or a fat-free mass index (FFMI) < 15 to 17 inwomen or men. Sarcopenia was defined by the association of pre-sarcopenia and myopenia. Patientswere followed-up at 6 months to measure a composite criterion of disease activity combining one ofthe factors: FC > 250µg/g, endoscopic activity, therapeutic change, hospitalization, or IBD-relatedsurgery

Results: In total, 60 patients were included, median age 37 years (IQR 28-54), 55% women, with CD (52%) orUC (48%) diagnosed 9 years (4-17) ago, mostly in remission (67%, n=40). Patients with CD had ileocolic(48%), ileal (35%), or colonic (16%) involvement. Patients with UC had E1 (7%), E2 (41%), or E3 (52%)involvement. A history of digestive surgery linked to IBD was noted in 30% of cases. The biologicsadministered were infliximab (61%),vedolizumab (25%), or ustekinumab (14%). The BMI was 24 (21-27) and serum albumin was 41g/l (39-44). The prevalence of sarcopenia, pre-sarcopenia and myopeniawas 10%, 11% and 20%, respectively. Sarcopenic patients were significantly older (59 vs. 36 years,p=0.01), with older disease (20 vs. 8 years, p=0.003), and associated joint damage (40% vs. 2%,p=0.002) and tended to involve CD more frequently (80 vs. 49%, p=0.36). There was no significantdifference between biologics, BMI or albuminemia in sarcopenic patients, but vitamin D levels tendedto be lower (15 vs. 43, p=0.27). Myopenia was significantly associated with a history of surgery (67 vs.21%, p=0.004), a lower BMI (21 vs. 24, p=0.001) and a more active disease at M6 according to thecomposite criterion (50 vs. 21%, p=0.04 after adjustment for age, sex, and disease activity at baseline). Myopenia indices were strongly correlated with handgrip values(r=0.7, p<0.001)

Conclusion: In a prospective study of Day Hospital IBD patients, sarcopenia was identified by impedancemetry in11% and myopenia in 20% of cases, and associated with an increase in morbidity

Disclosure: No conflict of interest.

PP0350

DIETARY BELIEFS AND HABITS IN INFLAMMATORY BOWEL DISEASES: A CROSS-SECTIONAL STUDY

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Introduction: Inflammatory bowel diseases (IBD) are characterized by recurrent flares and periods of remission. Dietary modifications are commonly made by patients with IBD to manage their symptoms. However, there is no single dietary recommendation for all patients with IBD and little is known about the impact of diet on these patients.

Aims & Methods: This study aimed to identify patients' beliefs, attitudes, and behaviors regarding diet in the management of symptoms and flares in patients with IBD, and to assess the impact on their social life.

Methods: This was a cross-sectional, single-center study involving 50 patients with IBD who were followed up in our department. All patients completed a semi-structured questionnaire.

Results: The majority of patients were male (56%) and had Crohn's disease (84%). Four main themes emerged from the analysis: beliefs about the role of diet in disease onset and flares, attempts to control symptoms through diet, changes in appetite during and outside of flares, and the impact of dietary beliefs on social life. Fifty-six percent (n=28) of patients believed that their diet had triggered their disease, while 84% (n=42) believed that diet could trigger flares. Seventy percent of patients identified specific foods as triggers. Participants described a process of trial and error, identifying foods that worsened their symptoms and others that helped them feel better. Twenty-four percent of patients reported that

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avoiding fiber improved their symptoms during flares. Two patients used diet as their primary treatment for IBD, but 96% of patients considered diet to be complementary to prescribed medication. Eighty-eight percent of respondents reported having received dietary advice at some point. The majority of patients (68%, n = 34) reported that their disease had changed their enjoyment of food. Appetite was lower during flares than during remission.

Only 8% of patients followed a normal diet during flares. Dietary beliefs and behaviors had an impact on social life for 56% of patients. Avoiding eating out was reported by 56% of patients and not sharing meals with others was reported by 44% of patients.

Conclusion: Our findings confirm those of previous studies that patients with IBD perceive that certain foods can trigger their disease or symptoms, and dietary control is one of the ways they choose to manage their IBD symptoms. We also noted that patients used a personalized approach to diet, combining experience and knowledge.

Disclosure: No conflict of interest.

PP0351

INCIDENCE AND RISK PREDICTION SCORE OF REFEEDING SYNDROME IN HOSPITALIZED PATIENTS ADMITTED IN A PUBLIC TERTIARY HOSPITAL: A PROSPECTIVE, ANALYTICAL SINGLE CENTER STUDY

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Introduction: Among critically ill older patients who are hospitalized, malnutrition is prevalent and is also considered a risk factor to develop refeeding syndrome (RFS). There is not enough data on the incidence of refeeding syndrome among them.

This study was conducted to identify the prevalence, severity and risk factors of refeeding syndrome among malnourished hospitalized patients.

Aims & Methods: In this prospective, analytical study conducted in Vicente Sotto Memorial Medical Center in Cebu, Philippines from October 2022 to December 2023, 217 patients were evaluated for malnutrition and risk for developing refeeding syndrome. Malnutrition assessment was conducted using Nutrition Risk Screening 2002 (NRS-2002) through bedside patient interview. The criteria for refeeding syndrome (RFS) were based on the consensus recommendations of the American Society for Parenteral and Enteral Nutrition (ASPEN) for RFS.

Data on electrolyte levels and daily caloric and intake of protein were taken within five days post resumption of nutrition. Significant risk factors in predicting refeeding syndrome were analyzed using Chi-square testing and Pearson rho correlations then subjected to a binary logistic regression to ascertain the effect size.

Results: Among 217 patients, 70 (32%) were at risk for RFS and assessed to have severe nutrition risk with an NRS 2002 score of at least 4 via bedside interview. Refeeding syndrome occurred in 14 patients (20%). Patients with RFS were older (54.71 ± 18.20) than those who were not diagnosed with RFS (50.91 ± 15.77). The total protein (g/day) and total caloric intake of patients who had RFS were significantly lower (41.25g/day ± 12.88 and 884.64 kcal/day ± 559.25, respectively) than those who did not develop RFS (51.20 g/day ± 11.02 and 1275.59 kcal/day ± 295.77, respectively).

Serum albumin (2.59 ± 0.27) and electrolyte levels, particularly, serum phosphorus (1.66 ± 0.29), magnesium (1.96 ± 0.36) and potassium (3.13 ± 0.43) were lower among the RFS patients than the non-RFS group. RFS diagnosis was associated with higher incidence of arrhythmia (50% vs 21%), use of vasopressors (64% vs 32%), ventilatory support (71% vs 23%) and mortality (64.29% vs 21.43%) compared to patients who did not have refeeding syndrome. Predictors of RFS include lower serum magnesium and potassium levels within 5 days of feeding initiation, lower total protein delivered and lower serum albumin level.

Conclusion: Among patients referred to the Section of Gastroenterology and Nutrition Medicine in Vicente Sotto Memorial Medical Center, Cebu City, Philippines, the incidence of refeeding syndrome (RFS) among malnourished patients is 20%. Refeeding syndrome is associated with high morbidity (arrhythmia incidence, use of ventilatory support and vasopressors) and mortality. Parameters such as malnutrition, low total protein intake, low serum electrolyte levels (serum magnesium, serum potassium, serum phosphorus) and low serum albumin were identified to predict the development of refeeding syndrome, with high accuracy, sensitivity, and specificity. However, despite the high morbidity and mortality associated in RFS, most clinicians lack awareness in identifying and managing this syndrome. Clinicians should be able to identify patients at risk to reduce its occurrence.

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Disclosure: Nothing to disclose.

PP0352

A RETROSPECTIVE EXAMINATION OF RISK FACTORS FOR CENTRAL LINE-ASSOCIATED BLOODSTREAM INFECTIONS IN HOME PARENTERAL NUTRITION PATIENTS FROM A QUEENSLAND TERTIARY HOSPITAL

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Introduction: A central line-associated bloodstream infection (CLABSI) is a common and potentially life-threatening complication for patients with intestinal failure (IF) receiving home parenteral nutrition (HPN). In uncomplicated infections, The European Society for Clinical Nutrition and Metabolism (ESPEN) guidelines advocate for central venous access (CVC) salvage for venous access preservation (1).

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Existing knowledge regarding the risk factors of HPN-related CLABSIs have been extrapolated from European and North American studies (2-4), and the effects of tropical climates and its potentially higher support requirements on the incidence of infection has not yet been studied.

Aims & Methods: We sought to analyse the risk factors of developing HPN-related CLABSI, and assess CLABSI management, in a large Australian state with a highly dispersed population. A retrospective observational cohort study was conducted on 34 adult patients receiving HPN via a central venous catheter (CVC) at a Queensland tertiary referral centre, between 2016 and 2023. CLABSI was defined according to the criteria of the Australian Commission on Safety and Quality in Health Care (5). Patient charts were reviewed, and a univariate cox regression analysis model was used to identify predictors of CLABSI in the first CVC during the study period. Kaplan-Meier analysis was employed to build survival curves of time to CLABSI, and log-rank tests analysed survival between patient and CVC characteristics.

Results: Nineteen patients had ≥1 CLABSI(s) episode, accounting for a total of 39 episodes. Patients with ≥1 CLABSI(s) used regular opioids more than those who did not develop CLABSI (p=0.016) but did not differ by age, gender, body mass index, geography, aetiology of intestinal failure or catheter weeks. Fourteen patients (41%, n=14/34) developed a CLABSI in their first CVC. No patient or line characteristics, including geographical remoteness and tropical climate, were found to be predictive of CLABSI in their first CVC. The overall infection rate was 1.02 per 1000 catheter days. Most CLABSIs were caused by Enterobacterales (22%, n=12/55), followed by coagulase-negative Staphylococcus (18%, n=10/55). Administration of empiric antimicrobials within the standardised 3-hour sepsis timeframe was inadequate (50%, n=14/28), with only 25% (n=7/28) providing adequate cover of causative pathogens. The median time to effective antibiotic therapy was 22.7 hours (IQR 4.8-29.8). There were 3 CVC salvages (8%, n=3/39), all of which were successful.

Conclusion: In this cohort of patients, regular opioid use was associated with an increased risk of developing CLABSI which is consistent with international studies (2-4). We also found a low CVC salvage rate in our cohort, representing potential missed opportunities for venous access preservation. Based on our findings, an empiric antimicrobial regime of vancomycin, cefepime and caspofungin for all HPN-related CLABSI cases would be expected to be effective for 84.6%. We recommend that Australian IF units with a similar antimicrobial distribution and resistance pattern adopt this regime.

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Disclosure: Nothing to disclose.

PP0353

EVALUATION OF MALNUTRITION RISK WITH THE MUST SCORE AMONG PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Inflammatory bowel disease patients are at risk of malnutrition due to the nature of the disease, the drugs used, and some dietary restrictions.Several screening tools are available, including the MUST score: the Malnutrition Universal Screening Tool, a simple and an easy-to-use tool.

Aims & Methods: The aim of our study was to assess the frequency of malnutrition risk using MUST score in IBD patients.

We conducted a prospective study that included 142 patients with IBD. All patients with IBD completed a self-administered nutrition screening assessment using the MUST score.Patients with a MUST score of ≥ 2 were concluded as high-risk for malnutrition.

Results: A total of 142 IBD patients were studied over 2 month period. We enrolled a total of 116 (82,1%) patients from outpatient and 26 (18,3%) from inpatient . Risk of malnutrition was identified in 21,1 % of patients according to MUST score. Number of hospitalized patients with malnutrition risk was significantly higher than the outpatient ones (84,6% vs 6.9 %, p<0.01).The frequency of high malnutrition risk was 36,8.% in ulcerative colitis (UC) patients and 15,4% in Crohn's disease (CD) patients (p=0.006). A high MUST score was strongly associated with an elevated Harvey Bradshaw Index score (p<0.01) for patients with Crohn's disease, and with Partial Mayo Index scores (p<0.01) for patients with ulcerative colitis.

Conclusion: In our study, we found that the risk of malnutrition was high in both CD and UC patients, especially inpatients ,and it correlates to the disease activity.Considering the effect of malnutrition on the prognosis of the disease ,IBD patients should be evaluated in terms of malnutrition risk both during their routine follow-ups and during hospitalizations.

Disclosure: Nothing to disclose.

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PP0354

THE IMPACT OF ETRASIMOD ON THE TRANSCRIPTOME IN COLON BIOPSIES USING RNA-SEQ: DATA FROM ELEVATE UC 52 AND ELEVATE UC 12 PHASE 3 CLINICAL TRIALS

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Introduction: Etrasimod is an oral, once-daily (QD), selective sphingosine 1-phosphate $(S1P)_{1,4,5}$ receptor modulator for the treatment of moderately to severely active ulcerative colitis (UC). $S1P_1$ receptor modulation is believed to reversibly sequester lymphocytes[1]; its impact on UC intestinal tissue is unknown.

Aims & Methods: We characterised the impact of etrasimod in inflamed colonic biopsy tissue using ribonucleic acid sequencing (RNA-seq) to examine transcriptional-level impacts on disease, in terms of mucosal barrier inflammatory pathways and immune cell populations. In ELEVATE UC 52 (NCT03945188) and ELEVATE UC 12 (NCT03996369), patients (pts) aged 16–80 years with moderately to severely active UC were randomised 2:1 to etrasimod 2 mg QD or placebo (PBO).

Etrasimod treatment effects were characterised by RNA-seq at Week 12 vs baseline (both trials), and in responders achieving clinical remission vs nonresponders. CytoPro (CytoReason) was used for colon-specific cell type deconvolution analysis and single sample Geneset Enrichment Analysis was used for pathway analysis.

Results: At baseline in ELEVATE UC 52/ELEVATE UC 12, colon biopsy samples were available for 245/166 pts receiving etrasimod and 114/84 pts receiving PBO. There were significant reductions in T cell subsets (CD4+, CD8+, helper [Th]1 and Th17), plasma/mature B cell subsets and innate immune cell subsets (monocytes, macrophages, dendritic cells and natural killer cells) in the colon of pts receiving etrasimod (false discovery rate [FDR] < 0.05]); significant changes were not seen among pts receiving PBO. There were also reductions in inflammatory fibroblasts and increases in epithelial cells in the colon of pts receiving etrasimod but not PBO. The change in immune cell populations for responders vs nonresponders at Week 12 was similar to that for etrasimod Week 12 vs baseline. At Week 12, several inflammatory pathways were reduced in responders vs nonresponders to etrasimod; IL-17 signalling was among the most affected (both trials). Nonimmune pathways related to mucosal healing were significantly increased in Week 12 responders.

Conclusion: Decreases in activated immune cell subset and inflammatory signalling pathway changes in responders to etrasimod suggest reductions in local inflammation may contribute to remission and mucosal healing. RNA-seq with cell type deconvolution and pathway analysis contributes to the understanding of etrasimod's mechanism of action.

References: 1. Scott FL et al. *Br J Pharmacol* 2016; 173: 1778–1792.

Disclosure: BS has received lecture fees from AbbVie, CED Service GmbH, Chiesi, Dr. Falk, Forga Software, Galapagos, IBD Passport, Janssen, Materia Prima, Pfizer Inc and Lilly; consultancy fees from AbbVie, Arena Pharma, Boehringer Ingelheim, BMS, Celgene, CT-Scout, Endpoint Health, Galpagos, Gilead Sciences, Janssen, Landos, Lilly, Pfizer Inc, PredictImmune and PsiCro. RDR is an employee of and holds stock options with Pfizer Inc. CMC, XG and JCW are employees and shareholders of Pfizer Inc. MTA has received consultancy fees from AbbVie, Amgen, Arena Pharmaceuticals, Bristol Myers Squibb, Celsius, Eli Lilly, Gilead, Janssen, Microba Life Sciences, Pfizer, Prometheus, Prometheus Bioscience, Takeda and UCB Biopharma. SV has received lecture/speaker fees from AbbVie, Dr. Falk Pharma, Ferring, Galapagos, Hospira, Janssen, MSD, Takeda and Tillotts; consultancy/advisory fees from AbbVie, AbolerIS Pharma, Alimentiv, Arena , AstraZeneca, Avaxia, BMS, Boehringer Ingelheim, Celgene, CVasThera, Dr. Falk Pharma, Ferring, Galapagos, Genentech/Roche, Gilead, Hospira, Imidomics, Janssen, Johnson & Johnson, Lilly, Materia Prima, MiroBio, Morphic, MrMHealth, MSD, Mundipharma, Pfizer Inc, Prodigest, Progenity, Prometheus, Robarts Clinical Trials, Second Genome, Shire, Surrozen, Takeda, Theravance Biopharma, Tillots Pharma AG and Zealand Pharma; grant/research support from AbbVie, MSD, Pfizer Inc, Galapagos, Janssen and Takeda. ID has received consultancy fees, research grants or honoraria from Abbott, AbbVie, Altman, Arena, Athos, Cambridge Healthcare, Celgene/BMS, Celltrion, Falk Pharma, DSM, Ferring, Food Industries Organisation, Galapagos, Genentech/Roche, Gilead, GSK, Hospital for Sick Children, Integra Holdings, Iterative Scopes, Janssen, Mediahuset, Medscape, Neopharm, Nestle, Peer voice, Pfizer, Rafa Laboratories, Sangamo, Sublimity, Takeda, Wilbio. KMR has received grant/research support from Bristol Myers Squibb. HKK has received consultancy fees from Kinevant Sciences and is an employee/shareholder for Uniquity Bio. MG is an employee and shareholder of Pfizer AG. SD has received lecture/speaker fees from AbbVie, Amgen, Ferring Pharmaceuticals Inc, Gilead, Janssen, Mylan, Pfizer and Takeda; consultancy/advisory fees from AbbVie, Allergan, Amgen, AstraZeneca, Biogen, Boehringer Ingelheim, Celgene, Celltrion, Ferring, Gilead Sciences, Hospira, Janssen, Johnson & Johnson, MSD, Mundipharma, Pfizer Inc, Roche, Sandoz, Takeda, TiGenix, UCB, and Vifor; holds directorship/ownership for Gastroenterology and Endoscopy, IRCCS Ospedale San Raffaele.

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PP0355

CHARACTERISATION OF THE EFFECT OF ETRASIMOD ON PERIPHERAL BLOOD MONONUCLEAR CELLS: DATA FROM ELEVATE UC 52 AND ELEVATE UC 12 PHASE 3 CLINICAL TRIALS

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Introduction: Etrasimod is an oral, once-daily (QD), selective sphingosine 1-phosphate $(S1P)_{1,4,5}$ receptor modulator for the treatment of moderately to severely active ulcerative colitis (UC). S1P receptor modulation reversibly sequesters specific lymphocyte subsets in lymph nodes.[1]

Aims & Methods: This study aimed to better understand immune cell type selectivity of etrasimod using in-depth immunophenotyping of peripheral blood mononuclear cells (PBMC). In ELEVATE UC 52 (NCT03945188) and ELEVATE UC 12 (NCT03996369), patients (pts) aged 16–80 years with moderately to severely active UC were randomised 2:1 to etrasimod 2 mg QD or placebo (PBO). PBMCs were collected and characterised using flow cytometry and epigenetic immune monitoring by Epiontis ID for baseline and Week 12 assessments (both trials).

Outputs were expressed as mean percent change from baseline (%CFB) of cell frequencies. Statistical tests were performed with t-tests at a nominal significance level of 0.05.

Results: In ELEVATE UC 52 and ELEVATE UC 12, 185 (64.0%) and 120 (50.4%) pts receiving etrasimod 2 mg QD, and 98 (68.1%) and 68 (58.6%) receiving PBO, respectively, were included in the biomarker analysis set. Baseline demographics and disease characteristics were similar to the full analysis set.

Using flow cytometry, in both trials, %CFB reductions were observed in total T and B cells with etrasimod (62.24–72.46% decrease) but not PBO. Among CD4+ T cell subsets, the greatest %CFB reductions with etrasimod were for naïve (Tn) cells. Reductions were also seen in central memory (Tcm), helper (Th)2, regulatory, Th1, Th17 and effector memory (Tem) cells (61.00–87.07% decrease) in both trials.

Similarly, CD8+ Tn and Tcm subsets were reduced (69.33–78.48% decrease) with etrasimod; minimal changes were seen in CD8+ Tem cells. For B cell subsets, memory cells and naïve cells were reduced with etrasimod in both trials (62.45–78.03% decrease). In both trials, minimal changes were observed for natural killer cells. With Epiontis ID, etrasimod induced significant reductions in peripheral CD3+, CD4+, GATA3+, IL17A+, FOXP3+, CCR7+ and LRP5A+ cells vs PBO (both trials; Table).

	ELEVATE UC 52		ELEVATE UC 12	
n, mean CFB, % (SD)	PBO (N = 98)	Etrasimod 2 mg QD (N = 185)	PBO (N = 68)	Etrasimod 2 mg QD (N = 120)
CD3+ (T cells)	28, -1.61 (62.62)	64, -45.05 (52.48)*	25, 7.07 (41.22)	47, -32.57 (50.77)***
CD4+ (CD4 T cells)	28, -1.63 (67.21)	62, -65.29 (31.33)***	25, 11.20 (45.41)	47, -49.51 (41.10)***
GATA3+ (Th2 cells)	27, 0.91 (74.49)	58, -62.56 (33.11)***	25, -3.60 (44.35)	45, -43.15 (39.43)***
IL17A+ (Th17 cells)	28, -4.61 (69.25)	60, -59.85 (35.28)***	25, 3.96 (42.72)	46, -34.43 (48.66)**
FOXP3+ (Treg cells)	27, -2.01 (52.99)	59, -52.40 (37.59)***	24, 3.36 (56.09)	44, -41.86 (38.78)**
CD8B+ (CD8 T cells)	28, -6.53 (52.03)	60, -28.65 (70.59)	25, 10.96 (50.60)	47, -11.67 (65.27)
CCR7+ (LN homing cells)	27, -7.18 (48.44)	61, -56.64 (24.04)***	25, 8.35 (46.89)	46, -47.71 (25.33)***
CXCR3+ (Th1, activated T cells)	28, 5.16 (63.25)	56, 1.92 (105.60)	25, 2.16 (41.96)	46, 28.54 (93.48)
LRP5A+ (B cells)	28, -9.45 (54.26)	62, -59.22 (33.40)***	25, 13.94 (64.45)	47, -44.60 (55.28)***
MVD+ (NK cells)	28, 13.72 (50.83)	62, 39.96 (91.54)	25, 17.79 (31.80)	46, 71.31 (89.23)***

*p < 0.05, **p < 0.01 and ***p < 0.001. Significant differences for etrasimod vs placebo are denoted with asterisks. The p-values were determined with a 2-sided t-test of the null hypothesis that the etrasimod-treated population parameter mean is equal to the PBO-treated population parameter mean.
Values shown are for Epiontis ID parameters in PBMC by visit.
%CFB, percent change from baseline; CD, cluster of differentiation; CCR, C-C chemokine receptor; CXCR, C-X-C motif chemokine receptor; GATA, GATA binding protein; FOXP, forkhead box protein; IL, interleukin; LN, lymph node; LRP, low-density lipoprotein receptor-related protein; MVD, mevalonate diphosphate decarboxylase; N, number of patients in cohort; n, number of patients in group; NK, natural killer; PBMC, peripheral blood mononuclear cell; PBO, placebo; QD, once daily; SD, standard deviation; Th, helper T cell; Treg, regulatory T cell; UC, ulcerative colitis.

Table. Epiontis ID PBMC %CFB at Week 12 (biomarker analysis set).

Conclusion: Etrasimod induces selective, pharmacodynamic changes in certain PBMC subsets. Results of cell-type characterisation by flow cytometry and Epiontis ID were complementary.

References: 1. Scott FL et al. Br J Pharmacol 2016; 173: 1778–1792

Disclosure: BS has received lecture fees from AbbVie, CED Service GmbH, Chiesi, Falk, Forga Software, Galapagos, IBD Passport, Janssen, Materia Prima, Pfizer Inc and Lilly; consultancy fees from AbbVie, Arena Pharma, Boehringer Ingelheim, BMS, Celgene, CT-Scout, Endpoint Health, Galpagos, Gilead Sciences, Janssen, Landos, Lilly, Pfizer Inc, PredictImmune and PsiCro. CMC, XG and JCW are employees and shareholders of Pfizer Inc. RDR is an employee of and holds stock options with Pfizer Inc. MTA has received consultancy fees from AbbVie Inc, Amgen, Arena Pharmaceuticals, Bristol Myers Squibb, Celsius, Eli Lilly, Gilead, Janssen, Janssen Biotech Inc, Microba Life Sciences, Pfizer, Prometheus, Prometheus Bioscience, Takeda, and UCB Biopharma. SV has received lecture/speaker fees from AbbVie, Dr. Falk Pharma, Ferring, Galapagos, Hospira, Janssen, MSD, Takeda and Tillotts; consultancy/advisory fees from AbbVie, AbolerIS Pharma, Alimentiv, Arena, AstraZeneca, Avaxia, BMS, Boehringer Ingelheim, Celgene, CVasThera, Dr Falk Pharma, Ferring, Galapagos, Genentech/Roche, Gilead, Hospira, Imidomics, Janssen, Johnson and Johnson, Lilly, Materia Prima, MiroBio, Morphic, MrMHealth, MSD, Mundipharma, Pfizer Inc, Prodigest, Progenity, Prometheus, Robarts Clinical Trials, Second Genome, Shire, Surrozen, Takeda, Theravance Biopharma, Tillots Pharma AG and Zealand Pharma; grant/research support from AbbVie, MSD, Pfizer Inc, Galapagos, Janssen and Takeda. ID has received consultancy fees, research grants or honoraria from Abbott, AbbVie, Altman, Arena, Athos, Cambridge Healthcare, Celgene/BMS, Celltrion, Falk Pharma, DSM, Ferring, Food Industries Organisation, Galapagos, Genentech/Roche, Gilead, GSK, Hospital for Sick Children, Integra Holdings, Iterative Scopes, Janssen, Mediahuset, Medscape, Neopharm, Nestle, Peer voice, Pfizer, Rafa Laboratories, Sangamo, Sublimity, Takeda and Wilbio. KMR has received grant/research support from Bristol Myers Squibb. HKK has received consultancy fees from Kinevant Sciences and is an employee/shareholder for Uniquity Bio. MG is an employee and shareholder of Pfizer AG. SD has received lecture/speaker fees from AbbVie, Amgen, Ferring Pharmaceuticals Inc., Gilead, Janssen, Mylan, Pfizer and Takeda; consultancy/advisory fees from AbbVie, Allergan, Amgen, AstraZeneca, Biogen, Boehringer Ingelheim, Celgene, Celltrion, Ferring, Gilead Sciences, Hospira, Janssen, Johnson & Johnson, MSD, Mundipharma, Pfizer Inc, Roche, Sandoz, Takeda, TiGenix, UCB, and Vifor; holds directorship/ownership for Gastroenterology and Endoscopy, IRCCS Ospedale San Raffaele.

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PP0356

EVALUATION OF FECAL SHORT-CHAIN FATTY ACIDS CONTENT IN PATIENTS WITH ULCERATIV COLITIS.

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Introduction: It is known that disorders of the intestinal microbiota in patients with ulcerativ colitis (UC) lead to increased intestinal permeability, impaired production and intestinal absorption of short-chain fatty acids (SCFA). The ratio of fecal SCFA is an important indicator of the integrity of the intestinal microbial community. The study of the concentrations of SCFA is a very attractive approach to broadening the understanding of the pathogenesis of UC.

Aims & Methods: To study the content of SCFA in the coprofiltrate of patients depending on the degree of severity of the course of UC.

We included in this study 86 patients with UC with a mean age of 43.2 ± 1.8 years. They were divided according to the Mayo index: 7 milds, 64 moderate and 15 severe. The control group consisted of 7 relatively healthy people. The diagnose UC were established according to generally accepted diagnostic standards in gastroenterology. Determination of SCFA was carried out by the method of chromatographic research. Was determined anaerobic index (AI) - the ratio of the sum of propionic and butyric acids concentrations to the concentration of acetic acid. AI values are used as an indicator of the infrastructure of the microbiocenosis - the ratio of anaerobic and aerobic microorganisms.

Results: The total content of SCFA did not show significant changes in patients with mild and moderate severity of UC (only a moderate tendency to decrease by 1.6 times (p>0.05) and 1.8 times (p>0.05), respectively), while in severe severity there was a statistically significant decrease by 3.9 times (p=0.008) compared to control. Also, in severe UC, this indicator had a pronounced tendency to decrease by 2.4 times (p>0.05) and 2.2 times (p>0.05), respectively, compared with mild and moderate severity, respectively. In mild and moderate severity of the disease, only a weak tendency to reduce the content of acetic acid in the coprofiltrate of patients by 1.4 times (p>0.05) and 1.3 times (p>0.05), respectively, and in severe severity - a significant decrease by 1.7 times (p=0.02), compared to the control. The content of propionic acid in the experimental samples of patients did not change significantly, there was only a moderate/weak tendency to its decrease by 1.6 times (p>0.05), 1.3 times (p>0.05) and 1.2 times (p>0.05), respectively, for the three severity levels of UC compared to the control. A significant decrease in the content of butyric acid in the coprofiltrate of patients with moderate and severe UC severity was found by 2.1 times (p=0.049) and 10.4 times (p=0.001), respectively, compared to the control. It is characteristic that in severe UC, the content of butyric acid was significantly reduced by 5.7 times (p=0.034) and 5 times (p=0.022), respectively, compared with mild and moderate severity of this disease.

IA significantly decreased by 1.5 times (p=0.028) and 1.9 times (p=0.033) in moderate and severe UC severity compared to the control group.

Thus, in severe UC severity, there was a marked decrease in the total content of FFAs mainly due to a significant decrease in butyric acid and a moderate decrease in acetic acid.

Conclusion: The obtained results showed that changes in SCFA concentrations in patients with UC differed significantly from the control group, what can lead to deterioration of the course and prognosis of UC. Estimation of the ratio of SCFA fractions with the calculation of the anaerobic index may be useful for the diagnosis of violation of the quantitative and qualitative composition of the intestinal microbiota in patients with UC.

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Disclosure: Nothing to disclose.

PP0357

ACCURATE AND ROBUST COLLAGENOUS COLITIS DETECTION IN HISTOLOGICAL IMAGES USING FEATURES EXTRACTED FROM SEGMENTED AREAS BASED ON SUPERPIXELS TOGETHER WITH DEEP LEARNING

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Introduction: Collagenous colitis (CC) is a form of microscopic colitis (MC), the others being lymphocytic colitis (LC), incomplete CC and incomplete LC (1). It is an inflammatory disease of the large bowel that causes chronic watery diarrhoea, abdominal pain, faecal incontinence, nightly defecation, and weight loss, resulting in significantly impaired quality of life (2). Incidence of MC has increased significantly during the past decades in some countries and the main reason for this increase is thought to be an enhanced disease recognition (3). The diagnosis of MC is challenging as it can only be diagnosed upon histological examination of colonic biopsies taken from normal or near normal appearing mucosa (1).

Histological interpretation of biopsies involves subjective evaluation leading to inter-rater variability discrepancies in diagnosis and treatment plan. Deep learning-based assistance system can objectivise diagnostic key feature selection leading to minimisation of inter-rater variability and improvement of diagnostic accuracy.

Aims & Methods:

The aim of this pilot study was to develop an algorithm for robust segmentation of light microscopy images of histological specimen slides (tissue slides) emphasizing key diagnostic features of CC.

Histological specimens were fixed with formalin, embedded in paraffin, cut into 3 µm sections, and stained with haematoxylin and eosin (H&E) for histological examination. Images were taken using OLYMPUS IX71 light microscope (x20 magnification) equipped with Q IMAGING EXI aqua camera at (1392 x 1040 pix.) resolution and 24-bit pixel encoding. Registered image pixel values were normalized in RGB colour space by histogram alignment using empty tissue-less areas using an algorithm described previously (4).

Superpixel technique using Simple Linear Iterative Clustering algorithm (5) was used for initial segmentation of the images. All 1000 superpixels defined in each image were annotated by the expert, indicating ones containing thickened subepithelial collagen layer as the sought class among the others containing the rest of the tissue indicated as normal class. The

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feed – forward neural network (implemented in MatLab environment) was trained to classify superpixels using their normalized histogram bin-counts of the pixel values as a feature.

Due to significant inequality in the data set, where the number of sought class superpixels was significantly smaller than the number of the remaining ones, the training set was formed by equalising it to the lowest class case's value (18761). Only randomly selected cases of the bigger class were included in the training set.

Results: The proposed algorithm was tested on images from 5 treatment-naive patients with newly diagnosed CC. Patients' histological specimens were rated by an expert pathologist and 50 images were chosen (with changes characteristic of CC and without). The classification training process with extracted data set took 0.13 seconds on personal computer with an AMD Ryzen Threadripper 3970X 32-Core, 3.70 GHz processor and 128GB of RAM memory, epochs for training was set to 30 (after tests the model was not improving after 30 epochs) and data shuffling was set to every-epoch. The algorithm showed 0.807 accuracy, 0.801 sensitivity and 0.813 specificity.

Conclusion: The shown ability of deep learning algorithm to classify histology image segments can assist the diagnostic process of CC emphasizing thickened subepithelial collagen layer – the essential diagnostic feature.

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Disclosure: Nothing to disclose.

PP0358

A MULTI-OMICS ANALYSIS REMINDS THE RELATIONSHIP AMONG GUT MICROBIOTA, BILE ACID METABOLISM, AND THE DISORDER OF INTESTINAL BARRIER FUNCTION IN CROHN'S DISEASE

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Introduction: Crohn's disease (CD) is a complex chronic intestinal inflammatory disease with unknown etiology[1]. Gut microbiota is involved in the bile acid metabolism in the intestine and the regulation of intestinal mucosal immunization to maintain intestinal homeostasis[2]. The histological changes in the intestinal mucosa of CD are closely related to impaired intestinal barrier function[3-5]. Abnormal gut microbiota and metabolic disorders can destroy the normal intestinal barrier function and intestinal homeostasis, thereby continuously aggravating the inflammatory reactions[6,7].

Aims & Methods: To further explore the related factors affecting the incidence and provide essential guidelines for clinical prevention and treatment of CD, the prospective case-control study conducted a multi-omics analysis of gut microbiota, fecal bile acid profiles, serum cytokines and intestinal barrier function in patients with active CD, patients in remission and normal controls.

Results: The study found that CD patients had unique gut microbiota and fecal bile acid profiles compared with normal controls, which were related to the disease activity. The beneficial bacteria represented by Romboutsia were positively correlated with the concentrations of fecal secondary bile acids, while negatively correlated with the concentrations of primary bile acids. When patients were active, their gut-beneficial bacteria decreased, affecting the bile acid metabolism. The variation of CD patients' bile acid profile was increased. The production of secondary bile acids was decreased. The levels of proinflammatory cytokines in peripheral blood were increased, such as FC, ESR, CRP and HBP, with P values all below 0.001. And the CD active group had significantly higher levels of IL-6, IL-8, IL-10 and IL-1β than those in remission, with P values of <0.001, 0.005, 0.04 and 0.044, respectively. The intestinal physical barrier and mucosal barrier were damaged with disease activity of CD.

Conclusion: The study suggested that the occurrence and development of CD might be caused by the interaction between gut microbiota and the metabolic product bile acid, which leads to the consequence of immune response in the body and the damage to the intestinal barrier.

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PP0359

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PP0360

THE CONTACT SYSTEM - AN INFLAMMATORY CONTRIBUTOR IN ULCERATIVE COLITIS

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Introduction: Previous research indicate the contact system (CAS) is associated with ulcerative colitis (UC). The activation of coagulation factor XII leads to the activation of plasma kallikrein. Plasma kallikrein generates bradykinin and cleaved H-kininogen (cHK) by cleaving H-kininogen. Bradykinin induces inflammation.

C1 inhibitor (C1-INH) is the most important regulator of CAS, while activated factor XII and plasma kallikrein are the accelerators. Interestingly, experimental colitis can be induced and alleviated in UC animal models by respectively activating and inhibiting CAS[1-3].

Increased activity of CAS in plasma and colonic tissue was detected in patients with active disease[4, 5].

Human studies are still limited. We, therefore, aimed to characterise CAS in patients with active UC by employing our unique panel of CAS assays.

Aims & Methods: We included adults with active UC at the Section of Gastroenterology, Esbjerg and Grindsted Hospital, Denmark from May 2021 to June 2023. Inclusion criteria were: Patient reported outcome 2 (PRO2) score ≥1, Mayo endoscopic subscore (MES) ≥1 and negative stool culture. The PRO2 considers stool frequency and passage of blood.

Pregnancy and conditions influencing CAS or inflammation was prohibited. Patient were not allowed to use oral contraceptives, non-steroid anti-inflammatory drugs, systemic corticosteroids or anticoagulants. Controls were gender-matched blood donors recruited at the Unit for Thrombosis Research, Esbjerg and Grindsted Hospital, Denmark from 2019 to 2021. In- and exclusion criteria were the Danish blood donation criteria. We used the Montreal classification, MES, Nancy histological index, PRO2, CRP, and fecal calprotectin to describe the UC activity.

Biochemical assays determined the kallikrein generation and plasma concentrations of coagulation factor XII, prekallikrein, H-kininogen, C1-INH and cHK (a surrogate marker of bradykinin generated by plasma kallikrein).

Results: We included 94 patients and 100 controls. Patients were primarily female (n=53) non-smokers (n=91) with a median age of 43 years and limited comorbidity (n=26). Half of the patients had proctitis. Patients displayed mild to moderate disease with medians: MES 2, Nancy histologic index 2, PRO2 4, CRP 2.5 mg/L and fecal calprotectin 515 mg/kg.

Although, patients and controls generated similar amounts of plasma kallikrein, patients still had higher cHK levels in plasma than controls (1.38 [0.21-3.98] µg/mL vs 0.83 [0.02-2.88] µg/mL, p <0.0001). The same applied for C1-INH levels (0.22 [0.14-0.33] g/L vs 0.21 [0.16-0.29] g/L, p =0.0340).

The remaining CAS components were similar in patients and controls. Nor, did CAS correlate to UC extension or severity. Interestingly, cHK remained higher even in patients displaying mild disease (see Table 1).

Patient subgroups	cHK in patients	cHK in controls	p-value
Proctitis, n=50	1.40 [1.13-1.70]	0.83 [0.64-1.16]	<0.0001
Mayo endoscopic subscore 1, n=14	1.35 [1.07-1.61]	0.83 [0.64-1.16]	0.0012
Nancy histological index grade ≤1, n=24	1.29 [0.90-1.56]	0.83 [0.64-1.16]	0.0013
Patient reported outcome 2 score ≤2, n=19	1.33 [0.92-1.61]	0.83 [0.64-1.16]	<0.0001
CRP in plasma ≤5 mg/L, n=63	1.40 [1.13-1.61]	0.83 [0.64-1.16]	<0.0001
Fecal calprotectin ≤250 mg/kg, n=20	1.45 [1.13-1.76]	0.83 [0.64-1.16]	0.0004

Results are expressed as medians [interquartile range 25-75%]. Significance was tested with the Mann-Whitney U test.

Table 1 Cleaved HK in patients with mild disease vs controls.

Conclusion: This is currently the largest study to profile CAS in patients with UC. By employing unique and novel assays, we can conclude that CAS contributes to active UC as verified by increased plasma levels of cHK and C1-INH in patients. Further, cHK might be useful as a serological marker for distinguishing patients with UC from healthy subjects.

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Disclosure: Nothing to disclose.

PP0361

MYENTERIC AND SUBMUCOSAL PLEXITIS AS PREDICTIVE FACTORS FOR POSTOPERATIVE ENDOSCOPIC AND CLINICAL RECCURENCE AFTER SURGERY IN CROHN'S DISEASE

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Introduction: Postoperative recurrence (POR) in Crohn's disease (CD) affects nearly half of patients one year after surgery. The identification of histological features, predictive of POR is relevant. Although it has been suggested that the presence of plexitis is associated with an increased risk of POR, it is not yet common practice to look for plexitis during pathological examination. Therefore, the predictive value of myenteric and submucosal plexitis in POR remains unclear.

Aims & Methods: The aim of our work was to evaluate the histopathological and clinical factors associated to POR after bowel resection in CD patients. We conducted a retrospective study using the Database of the Department of Pathology of Habib Thameur hospital (Tunis). Patients with ileocolonic or ileal resection for CD between 2000 and 2020 were included. The proximal resection margin was analyzed to detect the presence of myenteric and/or submucosal plexitis and assess its grade and the inflammatory cells involved. Plexitis was evaluated based on the appearance of the most severely inflamed ganglion or nerve bundle. Endoscopic recurrence was defined as a Rutgeerts score of ≥ i2 during a colonoscopy performed

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between the 6th and 12th month after anastomotic surgery . Clinical recurrence was defined as the reappearance of disease-related symptoms, confirmed by endoscopy and imaging.

Results: Seventy patients were included in the study. The mean age was 35 ± 10.7 years and the sex ratio was 1.33. Endoscopic recurrence occured in 56% of cases. Clinical POR concerned 50% of patients within 62 ± 49.2 months. The time to onset of clinical recurrence was significantly shorter in case of endoscopic recurence (p=0.05). Age > 30 years at the time of surgery increased the rate of endoscopic POR (p=0.02) and its severity (p=0.015). Endoscopic recurrence was more frequent in cases of a previous bowel resection (p=0.03). An ileal resection > 30 cm was associated with clinical POR (p=0.012). Ileocolic disease location was associated with a significant increase in clinical POR (p=0.04). A macroscopically affected proximal margin was associated with endoscopic (p=0.017) and clinical (p=0.024) recurrence. Histologic activity at the proximal margin was present in 28% of cases and was associated with endoscopic POR (p=0.042). Myenteric plexitis was reported in 49% of cases. Submucosal plexitis was present in 21% of patients.

The presence of myenteric plexitis and its grade did not increase endoscopic (p=0,28) and clinical (p=0,54) POR. However, plasma cell plexitis was associated with clinical POR (p=0.043). The presence of submucosal plexitis and its grade were not significantly related with a higher risk to endoscopic and clinical recurrence (p=0,94, p= 0,31 respectively).

Inflammatory cells involved in submucosal plexitis had no influence on endoscopic and clinical recurrence (p=0,18, p=0,5 respectively). In multivariate analysis, factors associated with endoscopic POR were age > 30 years at the time of surgery (Odds ratio (OR) = 1.069; 95% CI 1.002 to 1.14; p = 0.044) and the presence of histologic activity (OR= 4,969; 95% CI 1,175 to 21,01; p=0,029). Ileal resection > 30 cm was independently associated with clinical POR (OR= 1,043; 95% CI 1,002 to 1,086; p=0,04).

Conclusion: In this present study, histologic activity at proximal margin, age > 30 at surgery and ileal resection > 30 cm were associated with POR. No association was revealed between myenteric/submucosal and POR. Looking for plexitis should be more regular with a common search methodology to better assess its predictive value in POR.

Disclosure: Nothing to disclose.

PP0362

ANALYSIS OF FECAL MICROBIOTA COMPOSITION IN NEWLY DIAGNOSED PATIENTS WITH INFLAMMATORY BOWEL DISEASE BASED ON DISEASE LOCATION

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Introduction: Patients diagnosed with inflammatory bowel disease (IBD) may have damage in different parts of the gut. The development of this disease involves an intricate interaction between a host with genetic susceptibility, environmental factors, an imbalanced immune response, and intestinal microbiota. Several studies have revealed discrepancies in the composition of gut microbiota between IBD patients and healthy individuals, especially concerning microbial diversity and relative abundance of specific bacteria.

Nonetheless, previous research typically involves patients who have undergone immunosuppressive treatment, potentially altering the microbiota and affecting the integrity of study findings.

Aims & Methods: The aim of this study was to elucidate the differences in the microbiota of newly diagnosed patients with Crohn's disease (CD) and ulcerative colitis (UC) compared to healthy controls (HC), based on the location of the affected area. Stool samples were obtained from IBD patients who had not yet initiated treatment for the disease, and from HC. Each sample underwent shotgun metagenomic sequencing, followed by bioinformatic analysis using the Cutadapt, Kraken2, and Bracken programs. Shannon index (α-diversity) and Bray-Curtis dissimilarity (β-diversity) were used as tests of within-species diversity and within-group diversity, respectively.

Results: We included 103 patients with CD, 144 with UC and 49 HC. Patients with CD, which have been classified depending on disease location (ileal, colonic and ileocolonic), showed lower α-diversity compared to HC in each subgroup, showing the lowest index in those with ileocolonic location. In UC patients, divided according to disease extension (proctitis, left-sided colitis and extensive colitis), those with extensive involvement,

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exhibited the lowest α-diversity compared to HC and other groups. β-diversity in CD patients revealed the highest differences between the comparisons ileal vs ileocolonic and colonic vs ileocolonic, suggesting that patients with ileocolonic pattern showed a more different microbiota profile compared to the other groups studied; however, there were no differences in UC patients.

Examination of taxonomic abundance revealed a decrease in *Phocaeicola coprocola* and several species of *Adlercreutzia*, and an increase in *Shigella flexneri*, in CD patients, compared to HC. In UC patients, an increase in *Toxoplasma gondii* and *Gemella morbillorum* was observed in all three extension subgroups compared to HC. However, several species of *Adlercreutzia* and *Alistipes* were decreased in patients with extensive colitis compared to HC.

Conclusion: Our data suggest that newly diagnosed CD and UC patients showed relevant differences in the microbiota depending on the location of the disease, with a reduction in beneficial genus such as *Adlercreutzia* and *Alistipes*, and an increase in pathogenic species.

Disclosure: Nothing to disclose.

PP0363

PHYSICAL ACTIVITY SUPPRESSES THE DELETERIOUS EFFECTS OF HIGH-FAT DIET ON INTESTINAL HOMEOSTASIS THROUGH THE ACTIVATION OF AUTOPHAGY

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Introduction: Crohn's disease (CD) is a chronic inflammatory disorder of the gastrointestinal tract, with its etiology involving genetic, microbial and environmental factors. Dysregulated autophagy, high-fat (HF) diet and abnormal colonization of adherent-invasive *E. coli* (AIEC) have been identified as risk factors for CD. Physical activity (PA) represents a fundamental intervention that offers significant health benefits and has been associated with a reduced risk of developing various diseases, including CD. Although PA has been shown to induce autophagy in different tissues, exerting beneficial effects for the organism, its impact on autophagy in the intestine and its ability to suppress the deleterious effects of HF diet *via* the modulation of autophagy have not been investigated.

Aims & Methods: This study aims to:

(i) explore the influence of PA on the autophagy process in intestinal tissue, and;

(ii) assess the potential of PA to restore intestinal homeostasis and decrease inflammation caused by HF consumption.

Mice deficient for the autophagy-related gene *Atg16l1* specifically in intestinal epithelial cells (*Atg16l1*$^{flox/flox}$*CreVillin*, called by *Atg16l1*$^{\Delta IEC}$ mice) or wild-type mice (WT or *Atg16l1*$^{flox/flox}$) were fed either a control or HF diet for 12 weeks, and then engaged in spontaneous PA on running wheels (active group) or not (sedentary control group) for the same duration. The travelled distance (km), and the speed (m/min) on the wheels were recorded continuously. At the end of the experiment, total fat mass and fat-free mass were measured with echoMRI. The weight of total mesenteric and epididymal adipose tissues were also determined post-mortem. Levels of autophagy markers were assessed by Western blot and qRT-PCR. Intestinal inflammation was assessed by quantification of fecal lipocalin-2 and intestinal pro-inflammatory cytokine levels.

Results: Under HF diet, both *Atg16l1*$^{\Delta IEC}$ and WT mice exhibited substantial weight gain compared to the groups fed the control diet. Spontaneous PA notably attenuated weight gain, particularly in the HF diet-fed groups. Spontaneous PA also diminished the increase in total fat mass and the weight of mesenteric adipose tissue induced by HF consumption, while concurrently augmenting total fat-free mass. HF diet-fed mice exhibited a lower total running distance over the 12 weeks compared to those fed the control diet. Importantly, HF diet consumption led to increased fecal lipocalin-2 level compared to control diet, with a more pronounced effect in *Atg16l1*$^{\Delta IEC}$ *versus* WT mice, and this level was reduced with spontaneous PA. In the absence of PA, HF diet-fed *Atg16l1*$^{\Delta IEC}$ mice exhibited increased *E. coli* abundance compared to other mouse groups, and this was not observed in the active groups.

Compared to control diet, HF diet led to dysregulation of expression levels of several autophagy markers and autophagy flux in intestinal tissues, and this was corrected by PA. In particular, there was a positive correlation between the total travelled distance and levels of several autophagy-related mRNAs.

Conclusion: Spontaneous PA can correct the dysregulation of autophagy and mitigate intestinal inflammation induced by the HF diet. This suggests that combining spontaneous PA with other therapeutic interventions may be a valuable strategy for CD patients with dysregulated autophagy.

Disclosure: Nothing to disclose.

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PP0364

UNREVEALING NEW POTENTIAL KEY MECHANISMS IMPLICATED IN INFLAMMATORY BOWEL DISEASE BY MULTIOMIC APPROACH

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Introduction: Inflammatory bowel diseases (IBD), comprising Crohn´s disease (CD) and ulcerative colitis (UC), are characterised by chronic, relapsing and remitting inflammation of the gastrointestinal tract. Despite significant efforts to understand the pathogenetic mechanisms of IBD, the elucidation of its etiopathology and progression remains incomplete. This study aimed to clarify IBD pathogenesis using a multiomic approach in serum, urine, intestinal tissue and stool samples.

Aims & Methods: We performed a multiomic analysis (transcriptomic, proteomic, metabolomic and metagenomic) in a discovery cohort of 53 patients with active CD, 50 patients with active UC, and 33 healthy controls (HC). Transcriptomic and proteomic analysis of intestinal tissue was performed by RNA-seq and liquid chromatography-mass spectrometry, respectively. Serum and urine samples were used for metabolome study using nuclear magnetic resonance, and metagenomic analysis of stool samples was performed by 16rRNA gene sequencing.

Results: Gene expression profiles showed a detailed transcriptome landscape of intestinal tissue in patients with CD and UC. A total of 4,105 proteins were identified in mucosal biopsies, of which 2,128 were differentially expressed (DE) between CD and HC, 2,715 proteins DE in UC compared with HC, and 1,653 proteins DE between CD and UC. Metabolomic analyses revealed metabolites and pathways related to ketone, amino sugar and methyl histidine metabolism, Warburg effect, and ammonia recycling deregulated in IBD patients. Moreover, metagenomic results indicated that CD and UC differ in their microbial populations according to beta diversity. In comparison to HC, the microbial population in CD was different, and its alpha diversity was significantly lower.

Conclusion: The results showed that some of the identified analytes could play essential roles in the pathogenesis of CD and UC. New potential key mechanisms that could help to understand the information of omics layers that underlies the pathogenesis of IBD have been outlined.

Disclosure: Nothing to disclose.

PP0365

LIVER FIBROSIS IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: A CASE-CONTROL STUDY FROM AN IBD CENTER

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Introduction: Inflammatory Bowel Disease (IBD), comprising Crohn's disease (CD) and ulcerative colitis (UC), is a chronic inflammatory condition affecting the gastrointestinal tract. Although primarily associated with intestinal symptoms, IBD has also been linked to various extraintestinal complications, including liver fibrosis (LF). Additionally, metabolic-associated steatotic liver disease (MASLD), a common liver condition driven by metabolic risk factors such as obesity and insulin resistance, could intersect with IBD, given its systemic nature.

While both LF and MASLD are prevalent among the general population, their incidence and associated risk factors in IBD patients remain underexplored. Understanding these liver-related complications is crucial for developing effective screening and management strategies for patients with IBD.

Aims & Methods: We aimed to assess the prevalence of LF and MASLD in IBD patients compared to matched controls. In this cross-sectional, case-control study, we enrolled consecutive IBD patients from an IBD referral center and matched controls from the general population based on age, sex, type 2 diabetes (T2D) status, and body mass index (BMI) in a 1:1 ratio. Liver fibrosis and MASLD were confirmed through shear-wave elastography (SWE) measurements of stiffness and attenuation coefficient. Potential risk factors were analyzed using multivariate logistic regression.

Results: We included 378 participants, comprising 200 IBD patients (106 with CD and 94 with UC) and 178 controls. There were no statistically significant differences regarding distribution by gender, age, T2D status and BMI. The prevalence of LF (SWE ≥ 6.78 kPa) and advanced LF (SWE ≥ 9.15 kPa) was 42.5% and 16.0%, respectively, in IBD patients (48.1% in CD vs. 36.2% in UC; p=0.088) and 28.1% and 9.0%, respectively, in the control

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group (p<0.05). For UC, history of complicated disease (OR 7.5, CI95% 1.2-46.2 p<0.05) and multiple hospitalizations due to acute disease (OR 5.6, CI95% 1.3-24.8; p<0.05) were significant risk factors. As for CD, previous hospitalizations (OR 9.7, CI95% 1.4-65.8; p<0.05) and heavy smoking habits (OR 4.9, CI95% 2.0-12.4; p<0.001) were the main risk factors.

Extraintestinal manifestations, higher fecal calprotectin level at the time of liver stiffness measurement and latest endoscopic activity were not associated with presence of LF. MASLD was not significantly different between IBD and control group (42.0% vs. 51.7%, respectively; p>0.05). Higher BMI, higher waist circumference, higher abdominal wall thickness and metabolic syndrome were significant risk factors for MASLD (p<0.001).

Conclusion: Our case-control study demonstrates that patients with IBD have a higher prevalence of liver fibrosis compared to matched controls, with significant risk factors including a history of complicated disease, multiple hospitalizations, and heavy smoking habits.

Conversely, the prevalence of MASLD does not significantly differ between IBD patients and controls, suggesting that metabolic factors might play a more substantial role in MASLD risk. Given these findings, it is critical for clinicians to implement routine liver fibrosis surveillance in IBD patients, especially in those with identified risk factors. Proactive management strategies aimed at reducing risk factors, such as smoking cessation and minimizing hospitalizations through effective disease management, could mitigate liver fibrosis risk in this population. These insights emphasize the need for integrated care approaches that address both IBD-related complications and associated liver health.

Disclosure: Nothing to disclose.

PP0366

MIRNA EXPRESSION CHANGES IN COLONIC EPITHELIAL CELLS AND THEIR RELEASED EXTRACELLULAR VESICLES IN RESPONSE TO COMMENSAL BACTERIA IN ULCERATIVE COLITIS

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Introduction: Ulcerative colitis (UC) is an inflammatory bowel disease marked by chronic inflammation of the colon's inner lining. Its pathogenesis involves a multifactorial interplay between genetic and environmental factors, dysregulated immune responses, and alterations in the gut microbiome. MicroRNAs (miRNAs) have emerged as critical regulators of gene expression implicated in UC pathogenesis.

Understanding the differential expression of miRNAs in colonic epithelial cells and their extracellular vesicles (EVs) during bacterial exposure could offer potential insights into the interplay between gut microbiota and epithelial cell-mediated inflammatory responses.

Aims & Methods: The objective of this study was to investigate the expression of UC-associated miRNAs in colonic epithelial cells and their EVs, using UC patients and control subjects-derived colonic epithelial cells co-cultures with *Escherichia coli* and *Phocaeicola vulgatus* bacteria.

Colonic stem cells from healthy controls (HC) (n=10) and patients with active UC (n=10) were used to generate 3D colonic epithelial organoids, which subsequently served as a cell source for constructing polarized monolayers of colonic epithelial cells. These monolayers were co-cultured with commensal *E. coli* or *P. vulgatus* bacteria for 2 hours, washed, replenished with fresh culture medium, and further incubated for an additional 24 hours. EVs from cell-conditioned media were isolated using precipitation method and characterized by ELISA (CD63, HSP70, Apo-A1), Dynamic Light Scattering and cryo-TEM. RNAs from colonic epithelial cells and precipitated EVs were isolated using silica-column based method. MiRNAs expression was evaluated by TaqMan qPCR. Statistical analysis and visualization of the results were performed using R studio.

Results: Although statistically insignificant, but the expression of miR-183-5p and miR-135b-5p in epithelial cells of HC group was tended to decrease after contact with *E. coli* compared to untreated cells (resp. FC=0.84 (p=0.574) and FC=0.87 (p=0.798)), while miR-146a-5p increased (FC=1.37 (p=0.798)). In the EVs samples of HC, the same trends were observed. Interestingly, in the UC patients' cells, after incubation with *E. coli*, the expression of all studied miRNAs - miR-183-5p, miR-135b-5p, miR-146a-5p - tended to increase (resp. FC=1.53 (p=0.730), FC=1.71 (p=0.666), and FC=2.39 (p=0.258)).

Similar expression changes were observed in the EVs of colonic epithelial cells from UC patients following stimulation with *E. coli* (resp. FC=1.32 (p=0.730) (miR-183-5p), FC=1.32 (p=0.666) (miR-135b-5p), and FC=2.07 (p=0.258) (miR-146a-5p)).

After culturing colonic epithelial cells with *P. vulgatus* the expression of all investigated miRNAs, compared to unstimulated cells, remained almost unchanged, both in HC and in UC patients' monolayers. Meanwhile, in both the HC and UC groups, a decrease in miR-183-5p and miR-135b-5p expression was observed in EVs following contact with *P. vulgatus*, compared to control conditions (in the HC group resp. FC=0.59 (p=0.382) and FC=0.51 (p=0.234); in the UC group resp. FC=0.70 (p=0.605) and FC=0.73 (p=0.489)). However, the expression of miR-146a showed a decreasing trend only in the HC group compared to unstimulated cells (FC=0.53 (p=0.161)).

Conclusion: Due to the considerable variation in the response between study subjects, the observed expression changes did not reach statistical significance**.** The experimental study revealed a tendency that the expression of miRNAs associated with UC can be modulated by commensal gut bacteria, particularly *E. coli*, both in colonic epithelial cells and their EVs.

Disclosure: Nothing to disclose.

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PP0367

THE IMPACT OF TOFACITINIB AND BUDESONIDE ON STEMNESS, CELL DEATH, AND CHEMOKINE RELEASE IN IBD PATIENT-DERIVED COLONOIDS

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Introduction: Mucosal healing, defined as the resolution of inflammation and restoration of the intestinal epithelial barrier[1], is believed to cause long-term remission in patients treated for Inflammatory bowel disease (IBD). However, most current therapeutic strategies only alleviate the symptoms and induce endoscopic remission rather than directly addressing histological and molecular healing, and little is known about how IBD drugs may contribute to epithelial repair in the colon[2].

During homeostasis, epithelial renewal is maintained by leucine-rich repeat-containing G-protein coupled receptor 5+ (Lgr5+) intestinal stem cells in the crypts, producing proliferative progenitors that gradually cease to divide as they move towards the surface and differentiate into mature cytokeratin 20 (CK20) expressing colonocytes. Inflammation is a major effector of intestinal mucosal damage in IBD and has been found to affect epithelial stemness, regeneration and cellular functions[3].

Thus, we need more knowledge about how IBD drugs, developed to modulate immune cells, also affect intestinal epithelial cells. Here, we used human colon organoids (colonoids) to recapitulate important features of colonic epithelium in a donor-specific manner[4].

Aims & Methods: Our hypothesis is that mechanisms of action (MOA) of IBD drugs in intestinal epithelium are relevant for homeostasis, and drug classes such as targeted small molecules and broad-spectrum first-line corticosteroids will have distinct MOA and varied effects among patients. Here, we use colonoids from healthy donors and ulcerative colitis patients to elucidate MAO for budesonide and the pan-JAK inhibitor tofacitinib, especially their impacts on stemness in differentiated cells.

The effects of tofacitinib and budesonide treatment during differentiation were examined in uninflamed and TNF+Poly(I:C) stimulated colonoids. Colonoid growth, stemness, cell death, and chemokine release were analyzed by brightfield and confocal imaging, immunohistochemistry, western blot, 40-chemokine plex, and single ELISAs. Scripts were developed for automated size and protein quantifications.

Results: When added during the differentiation process, tofacitinib led to an increase, while budesonide resulted in a decrease in colonoid size. Tofacitinib treatment was associated with increased Ki67 expression in general and specifically in mature CK20-positive cells. We observed large inter-individual differences, but tofacitinib treatment enhanced expression of the stem cell markers LGR5 and EPHB2 receptor in colonoids from most donors. Budesonide treatment did not seem to affect proliferation or stemness consistently. Neither tofacitinib nor budesonide pre-treatment showed definite harmful or protective effects on TNF+Poly(I:C) induced cell death in colonoids, but both drugs attenuated TNF+Poly(I:C) induced chemokine release from colonoids.

Conclusion: Although both budesonide and tofacitinib show anti-inflammatory effects in colonoids, their effects on stemness may differ. Notably, our study represents the first to demonstrate that treatment of colonoids with the pan JAK-inhibitor tofacitinib during differentiation promotes a heightened tendency for proliferation in terminally differentiated colonoids.

Furthermore, our research reveals that tofacitinib enhances the expression of stem cell markers. Our study also demonstrates the utility of human colonoids in investigation of drug effects and inter-individual variations in the intestinal epithelium.

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PP0368

VALERIC ACID: A POTENTIAL THERAPEUTIC TARGET FOR MODULATING INTESTINAL BARRIER FUNCTION IN INFLAMMATORY BOWEL DISEASE

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Introduction: Inflammatory bowel disease (IBD), encompassing Crohn's disease (CD) and ulcerative colitis (UC), is characterized by chronic inflammation of the gastrointestinal tract. Emerging evidence suggest critical role of gut microbial metabolites in IBD progression. Among these, valeric acid (VA), a short-chain fatty acid (SCFA) produced by bacterial fermentation, is known to be decreased in the feces of IBD patients and to exert an anti-inflammatory effect on macrophages. However, it remains underexplored whether VA could alleviate intestinal inflammation and improve the epithelial barrier in IBD.

Aims & Methods: This study aimed to investigate the effects of VA on intestinal epithelial barrier function and delineate its molecular mechanisms in IBD. VA pathway-related gene expression in human colon biopsies was measured by bulk and single-cell RNA sequencing. In addition, we investigated changes in microbial enzymes involved in VA biosynthesis via microbiome sequencing.

Furthermore, we used human colon organoid *in vitro* model to examine the effects of VA on the intestinal epithelial inflammatory response.

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Results: Expression of VA receptors (*Ffar2, Ffar3, Hcar2*), transporters (*Slc5a8, Slc16a1*), and histone deacetylases (*HDACs*) are altered in the colonic mucosa of IBD patients, compared to healthy individuals. The abundance of microbial enzymes responsible for VA synthesis shows significant differences in the microbiota of IBD patients.

Finally, VA treatment decreases the secretion of the inflammatory IL-8 cytokine in a dose-dependent manner and downregulates genes such as *Ffar2* and *Cxcl8* in the colon organoids model.

Conclusion: VA is involved in IBD pathogenesis by affecting the epithelial barrier and modulating immune response.

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Disclosure: Nothing to disclose.

PP0369

SMALL MOLECULE INHIBITORS FOR A4B7 AND IRAK4 SHOW PROMISING EFFECTS IN A PRECLINICAL MODEL OF SPONTANEOUS COLITIS

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Introduction: Current IBD therapy regimens are still frequently inefficient, can exert severe side effects, and loss of treatment response is common. Thus, the development of novel therapeutic approaches is an unmet clinical need – especially for patients featuring a primary non-response, refractory disease and/or severe side effects.

Here we assessed novel small molecule inhibitors targeting a4b7 integrin or IRAK4 regarding their efficacy to relieve colitis in a model of spontaneous intestinal inflammation either as a stand-alone treatment or in combination.

Aims & Methods: IL-10 knockout (IL-10KO) mice, which develop spontaneous immune-driven intestinal inflammation resembling human IBD, were used for all studies. Mice with established disease were fed for three weeks *ad libitum* with control chow or chow containing the inhibitors at two previously defined concentrations: a higher concentration intended to induce optimal treatment response or a lower, sub-optimal concentration that did not induce therapeutic effect as stand-alone therapy.

Results: When administered as stand-alone treatment at the higher dose, a4b7 and IRAK4 inhibitors both significantly reduced colitis severity (reduced endoscopic and histological colitis scores [p<0.001 and p<0.01], reduced disease activity index scores [p<0.001]). This effect was accompanied by a normalization of immune cell populations in the colonic lamina propria.

Particularly, we found reduced numbers of granulocytes, Th1, Th17, and cytotoxic T cell sin a4b7 and IRAK4 treated mice. Furthermore, mRNA and protein levels of inflammatory cytokines (IFNg, IL17, TNFa and IL-1b) were normalized in intestinal tissue of a4b7 and IRAK4 inhibitor-treated mice (p<0.01, each). When administered in lower concentrations, however, the a4b7 and IRAK4 inhibitors were not able to reduce disease activity as a stand-alone treatment. Of note, a combination of the low doses of the a4b7 inhibitor and the IRAK4 inhibitor significantly reduced the disease severity as evidenced by reduced endoscopic and histologic colitis scores [p<0.001, each], reduced disease activity scores [p<0.01] and a normalization of immune cell infiltrates (reduced levels of granulocytes [p<0.001], Th1 cells [p<0.001], Th17 cells [p<0.001] and less IFNg+ CD8 T cells [p<0.001]).

Furthermore, while low doses of anti-IL12/23 p40 antibody were not able to reduce disease severity as a stand-alone treatment, combination of low-dose anti-IL12/23 p40 with the low dose of the a4b7 inhibitor resulted in amelioration of colitis (normalized endoscopic [p<0.001] and histologic [p<0.01] colitis scores, reduced infiltration of granulocytes, Th1, Th17 and IFNg+ CD8 T cells [p<0.01, each], normalization of IFNg, TNFa and IL-1b protein and mRNA levels in the intestine).

Conclusion: Our results demonstrate that small molecule inhibitors for a4b7 and IRAK4 were efficient in reducing colitis in a model of spontaneous immune-cell driven intestinal inflammation. Combination of the a4b7 inhibitor with low doses of the IRAK4 inhibitor or sub-optimal doses of an anti-IL12/IL23 antibody was effective to relieve the disease in the IL-10-KO model. This indicates that combination treatment of a4b7 inhibition with other immune-modulatory drugs, such as IL12/23 neutralization or IRAK4 inhibition might be a promising approach for patients that do not respond to single therapies.

Furthermore, our results indicate that combination therapy is able to induce significant disease-relieving effects when the drugs are administered in low concentrations what could minimize side effects without interfering with treatment response.

Disclosure: Nothing to disclose.

PP0370

TACROLIMUS OPTIMIZES OXIDATIVE METABOLISM OF COLONOCYTES IN EXPERIMENTAL COLITIS

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Introduction: It has emerged that the energy metabolism of colonocytes shapes the gut microbiota, and loss of epithelial hypoxia accompanied by respiration-dependent dysbiosis is observed in experimental colitis[1,2].

The inhibitory effect of tacrolimus on inducible nitric oxide synthase (iNOS), a vital protein in nitrate respiration, was reported in human colonic epithelial cell lines[3].

Our previous study reported that tacrolimus significantly alleviated colitis in mice and induced beneficial alterations of the gut microbiome[4]. However, the mechanism of how tacrolimus improves the gut microbiota in colitis remains to be elucidated.

Aims & Methods: In this study, we aim to explore the effects of tacrolimus on colonocyte metabolism in colitis. Male C57BL/6J mice aged 8 to 10 weeks were employed. Solutions of 2.5% dextran sulfate sodium (DSS) were prepared in the drinking water to induce colitis for 7 consecutive days. Tacrolimus (Tacro) at a dose of 10 mg/kg was administered via gavage, while aminoguanidine hydrochloride (AG), the selective iNOS inhibitor, was given intraperitoneally at a dose of 100 mg/kg per day. The body weight was observed daily. For hypoxia detection, mice were treated with 60 mg/kg of pimonidazole HCl intraperitoneally 1 hour before euthanasia. Colon lengths were measured, and the colon samples were fixed in 4% paraformaldehyde for subsequent hematoxylin and eosin staining and hypoxia staining. Total RNA and protein were extracted from the isolated colonocytes for the measurement of iNOS expression. For the in *vitro*

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study, the human colorectal adenocarcinoma cells (Caco-2) were incubated with 10 ug/mL tacrolimus or 0.1% DMSO for 4 hours, then the oxygen consumption rates were measured on the Seahorse XFe24 Analyzers according to the manufacturer's protocols.

Results: Hypoxia in the colonic epithelium was significantly disrupted in DSS-induced colitis mice, which was partially restored by tacrolimus treatment. The iNOS expression on both mRNA and protein levels was significantly upregulated by DSS and downregulated after tacrolimus administration, while the addition of AG enhanced the effects of tacrolimus.

Compared with the individual tacrolimus intervention, the combined treatment of tacrolimus and AG achieved a better remission in weight loss, colonic shortening, and colonic pathological changes including the disruption of crypt-villus structure, goblet cell loss and immune cell infiltration. In *vitro* measurement of mitochondrial oxidative phosphorylation showed that tacrolimus significantly enhanced oxygen consumption, and promoted ATP production and spare respiratory capacity in Caco-2 cells.

Conclusion: Our results indicated that tacrolimus improved the oxidative metabolism of colonocytes in experimental colitis, which could be an important contributor to the alterations in the gut microbes induced by tacrolimus. The observed effect of tacrolimus in colonocytes may be associated with the inhibition of iNOS.

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PP0371

A NOVEL PEPTIDE CAN SIGNIFICANTLY IMPROVE PRECLINICAL OUTCOMES IN A MURINE DEXTRAN SULFATE SODIUM MODEL OF COLITIS

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Introduction: Most current approaches for therapy for Inflammatory Bowel Disease (IBD) focus on inhibition of excessive inflammation. However, with clinical remission rates for IBD plateauing, new approaches, are needed. The protease, Activated Protein C (APC), can exert a broad range of cytoprotective actions including inhibition of inflammation, stabilization of endothelial and epithelial barriers and suppression of apoptosis. APC shows therapeutic effects in various inflammatory diseases, and its cytoprotective functions often require biased cleavage of protease-activated receptor (PAR)1 and PAR3.

Biparetide is a bivalent PAR1:PAR3-derived peptide formed by covalent linkage of PAR1-residues 47-55 to PAR3-residues 51-65 via a 10-mer oligo-Gly peptide. Biparetide in vitro mimics APC's anti-inflammatory activity.

Aims & Methods: Aim: Our objective was to assess the in vivo therapeutic benefits of intrarectal and intraperitoneal administration of biparetide in a murine preclinical 40kDa dextran sulfate sodium (DSS) model of colitis.

Methods: Eight-week-old female C57Bl/6N mice were given 2% (weight/volume) DSS in drinking water for three 7-day cycles followed by 7 days of regular drinking water to induce colitis. Both biparetide (1, 10 and 50 mg/kg) and scrambled biparetide (peptide with randomized amino acid sequence) were given daily from day 18 to day 41 by concomitant intrarectal (half volume) and intraperitoneal (IP) injection (half volume).

Anti-TNFa was administered IP 15mg/kg 3 times per week from day 18 to day 41. Prednisolone was administered IP (1mg/kg) once daily from day 18 to day 42. Control groups included healthy mice and DSS-treated animals receiving only vehicle. Colitis severity was monitored using a disease activity score (DAI) based on daily evaluation of weight, stool consistency, and presence of blood; at study end, endoscopic scoring was performed. Colon length and weight were measured, and histological analysis on formalin-fixed paraffin sections was performed by a blinded pathologist.

Results: DSS effectively induced chronic colitis. Biparetide dose-dependently reduced body weight loss compared to vehicle-dosed DSS-treated mice from day 24 ($p<0.01$). Biparetide at both 10 and 50 mg/kg significantly improved body weight loss compared to prednisolone ($p<0.01$) and anti-TNF-a from day 31 ($p<0.01$), and it was associated with a dose-dependent, significant reduction in DAI score compared with vehicle-dosed DSS-treated mice, e.g., from day 24 (for 10 mg/kg, $p<0.01$). Biparetide significantly reduced DAI compared with prednisolone ($p<0.01$) and reduced DAI similar to anti-TNF-a. The peptide's protective effects against chronic colitis observed at both 10 and 50 mg/kg were accompanied by improvements in colon length shortening and in colon weight/length ratio ($p<0.01$), with effects comparable to anti-TNF-a. Amelioration of histological and endoscopic parameters of colitis was seen with anti-TNF-a and biparetide (10 and 50mg/kg) treatments.

Conclusion: In this DSS-induced colitis model, biparetide gave dose-dependent improvement in clinical signs and symptoms, reducing body weight loss and improving DAI score. These significant improvements were seen at 10mg/kg and 50mg/kg peptide. Similar improvements were observed during endoscopy and upon histological assessment.

Here we demonstrate proof of concept that biparetide is a novel anti-inflammatory peptide that mimics APC's previously reported in vivo benefits for DSS-induced colitis. Moreover, these data suggest that biparetide merits further evaluation for clinical translation for inflammatory pathologies.

Disclosure: Nothing to disclose.

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PP0372

TRANSCRANIAL DIRECT CURRENT STIMULATION (TDCS) MODULATES ENTERIC NERVOUS SYSTEM (ENS)-CELL EXPRESSION PATTERNS AND INTESTINAL BARRIER IN CHRONIC DSS COLITIS

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Introduction: Chronic visceral pain is a clinical challenge in >50% of Inflammatory Bowel Disease (IBD) cases (Morrison et al., 2013). A randomized controlled trial (RCT) demonstrated that transcranial direct current stimulation (tDCS) can modulate brain activity and effectively alleviate chronic abdominal pain associated with IBD (Neeb et al., 2019).

To elucidate the underlying mechanism of tDCS, we implemented this technique in a mouse model of inflammatory, chronic DSS colitis.

Aims & Methods: The tDCS protocol, including a total of 10 stimulations, was designed based on established human tDCS studies. An anodal direct current (0.5 mA, 20 min) was administered to both primary motor cortices over two separate periods with each period consisting of five consecutive days of stimulation, separated by a two-day break. The study was controlled by including a sham-treatment-group (0.5-minute sham stimulation).

For histology, H&E stained FFPE-sections of colon were assessed by scoring (Koelink et al., 2018). Barrier integrity was analysed by Ussing chamber experiments. To confirm that tDCS was affecting murine activity, mice were tracked in home cages using subcutaneously injected RFID-chips. Myenteric plexus (MP) cells of the murine small intestine, primarily the terminal ileum, were isolated and cultivated for 10 days.

Staining with beta3-tubulin and S100B was performed confirming neuronal cell identity. Bulk-RNA sequencing was conducted using the NovaSeq Xplus system (read length 150 bp, read depth 30 million copies/run).

Results: The distal colon of DSS-treated mice revealed significantly stronger inflammation in the H&E scores compared to control mice (mean±SEM: Ctrl: 0.14±0.08; DSS: 1.52±0.19; MWU: p=0.0005). Only a tendency towards less inflammation was observed in tDCS-treated mice compared to sham-tDCS.

The DSS caused an inflammation-driven thickening of the colon wall, resulting in an increased transmural resistance. This effect was ameliorated by the tDCS-treatment (mean±SEM: DSS+sham-tDCS: 72.2±8.2 $\Omega \cdot cm^2$; DSS+tDCS: 48.1±2.1 $\Omega \cdot cm^2$; MWU: p=0.007).

DSS colitis was associated with a reduced activity, which was partially recovered by tDCS (mean±SEM: DSS+sham-tDCS: 70.7±15.2 relative distance units/day; DSS+tDCS: 205.9±40.4 relative distance units/day; MWU: p=0.029).

Neuronal cell identity was confirmed in isolated murine MP cells by immunostaining (beta3-tubulin, S100B). Bulk-RNAseq revealed a clear separation of tDCS and sham-tDCS treated colitis mice by PCA. Subsequently, 739 differentially expressed genes (DEGs) were identified in tDCS-treated DSS colitis mice compared to a sham-tDCS-treatment (622 upregulated, 117 downregulated genes; log2(fold-change)≥1.5; padj<0.05), *Table 1*.

Analysis of KEGG and GO pathways revealed involvement in several biological processes, comprising neuroactive ligand-receptor interaction ($p=7.94 \cdot 10^{-3}$), gated channel activity ($p=2.4 \cdot 10^{-7}$), substrate-specific channel activity ($p=1.99 \cdot 10^{-7}$), and leukocyte migration ($p=3.99 \cdot 10^{-7}$).

Gene name	Differential expression	Log2(fold-change)	Adjusted p-value
Nos1, Nitric oxide synthase 1	upregulated	$2.68 \cdot 10^{14}$	$5.32 \cdot 10^{-08}$
Fos, FBJ osteosarcoma oncogene	upregulated	$2.21 \cdot 10^{13}$	$4.04 \cdot 10^{-07}$
Ncam2, Neural cell adhesion molecule 2	upregulated	$2.29 \cdot 10^{14}$	$2.66 \cdot 10^{-04}$
Chrnb4, Cholinergic receptor, nicotinic, beta polypeptide 4	upregulated	$1.69 \cdot 10^{14}$	$3.69 \cdot 10^{-04}$
Cend1, Cell cycle exit and neuronal differentiation 1	upregulated	$1.67 \cdot 10^{14}$	$6.71 \cdot 10^{-03}$

Table 1 Selection of neurologically relevant DEGs.

Conclusion: In colitis, transcranial direct current stimulation (tDCS) has the capacity to modulate activity of DSS-treated mice and alters gene expression in MP cells. This reveals another layer of evidence for the impact of central pathways on the gut / brain-gut-axis.

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Disclosure: Nothing to disclose.

PP0373

DOES A TRANSCRIPTIONAL SIGNATURE COMPOSED OF TWELVE METALLOPROTEINASES PREDICT THE RESPONSE TO ANTI-TNF THERAPY IN PATIENTS WITH ULCERATIVE COLITIS?

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Introduction: Failure of anti-TNF therapy contributes to the socio-economic burden of inflammatory bowel disease (IBD). This is particularly relevant when new drugs targeting α4β7 integrin, IL-12/IL-23, Janus kinases or sphingosine-1 are added to the IBD therapeutic armamentarium. Anti-TNFs are the usual first-line biologic agents and, therefore, pre-treatment discrimination of patients who will be refractory to these therapies is important to prevent IBD patients from being exposed to multiple therapeutic failures. Many studies have reported predictive biomarkers for response to anti-TNF therapy.

However, the implementation of easily readable biomarkers in daily clinical practice is an unmet need. Due to the active role of the metalloproteinase (MMP) family in the development of intestinal inflammation, we wanted to isolate robust and reproducible transcriptional markers capable of specifically predicting response to anti-TNF therapy by comprehensive analysis of these proteases.

Aims & Methods: We analysed the expression of twelve MMPs in publicly available microarray and RNAseq datasets from the Gene Expression Omnibus and the European Bioinformatics Institute: GSE12251, GSE16879,

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GSE73661 and E-MTAB-7845 for colon biopsies from ulcerative colitis (UC) patients before anti-TNF or vedolizumab therapy. The R package was used for differential expression analysis and receiver operating characteristic (ROC) calculation. Cytoscape software and Morpheus (broadinstitute.org) were used for network data integration and visualisation.

In addition, we evaluated the expression of these twelve MMPS by RT-qPCR in peripheral blood mononuclear cells and polymorphonuclear cells from a local cohort of UC patients (n=21).

Results: We isolated a colonic transcriptional signature consisting of twelve MMPs that is highly associated with disease activity. Most of the transcripts of this signature were significantly up-regulated in colon biopsies of non-responders to anti-TNFs before treatment in three different microarray datasets, but not in vedolizumab-treated patients. ROC curves of the twelve MMPs showed variability in correlation with anti-TNF response in all datasets analysed.

However, the combination of MMP2, MMP3, MME and PHEX efficiently discriminated patients who failed anti-TNF therapy (ROC AUC = 84%). Interestingly, this four-gene signature was not able to predict the response to vedolizumab (ROC AUC = 63%).

For completeness, we observed that of the twelve MMPs analysed, only MME showed increased expression in peripheral mononuclear cells from patients refractory to anti-TNF therapy, confirming that this MMP could be a feasible biomarker in both colonic tissue and peripheral blood.

Conclusion: Overall, our study reveals a novel and robust colonic transcriptional signature consisting of four MMPs associated with anti-TNF, but not vedolizumab, therapy response.

Furthermore, MME levels in peripheral blood cells emerge as a novel, easily readable predictive biomarker of anti-TNF response. Nevertheless, our results should be confirmed in independent cohorts of UC patients to test their efficacy and sensitivity in the clinical setting.

Disclosure: Nothing to disclose.

PP0374

PATIENTS WITH ULCERATIVE PROCTITIS ON ADVANCED THERAPY EXPERIENCE SIMILAR DISEASE BURDEN AS THOSE WITH LEFT-SIDED COLITIS AND PANCOLITIS

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Introduction: Ulcerative colitis (UC) is a chronic, idiopathic inflammatory bowel disease (IBD), with mucosal inflammation affecting the rectum (proctitis, E1), left side of the colon (E2), or whole colon (pancolitis, E3). Proctitis is considered a mild disease, and often excluded from clinical trials due to heterogeneity in accessing the affected area of the colon.

Proximal extension of the disease is associated with a more severe clinical course. We compared the disease burden of patients with UC according to disease extent.

Aims & Methods: THEFAR is a cross-sectional, observational, multicentre study, including adult patients with UC treated with advanced therapy (AT; biologics or Janus kinase inhibitors). Gastroenterologists across 41 centres included consecutive patients during routine clinic visits.

They completed medical and therapeutic management data, including disease activity (partial Mayo score) using an electronic case report form (eCRF). Patients completed self-reported quality of life (EQ-5D-5L), disability (IBD-Disk), incontinence (Wexner score), bowel urgency (Urgency Numeric Rating Scale, UNRS), and Work Productivity and Activity Impairment UC (WPAI:UC) questionnaires.

Results: 293 patients with UC (mean age 44 ± 16 years; 50% [147/293] male) were enrolled, with 292 self-reported questionnaires returned for analysis; 45 (15%) patients had proctitis (E1), 117 (40%) left-side colitis (E2), and 131 (45%) extensive colitis (E3). Former smokers were more common in E2 (25%) and E3 (28%) than E1 (11%) subgroups (p=0.01). Comorbidities, including chronic rheumatological disease and cardiovascular diseases, were similar across all subgroups (p>0.05). Disease duration was related to the extent of disease: mean duration of 5.3 years (E1) to 8.8 years (E3) (p=0.02). Age at diagnosis was lowest in E3 (33 years) compared to E1 and E2 (38 years each) subgroups (p=0.02). Regardless of subgroup, there was a similar incidence of flares over the past year, presence of extra-intestinal manifestations, and disease activity per partial Mayo score (all p>0.05). There was no proof of difference in current treatment allocation, duration or lines of advanced therapies, or corticosteroid use (p>0.05 for all [except golimumab]) between subgroups. The proportions of hospitalized patients were significantly different in subgroups: E1 (11%), E2 (21%), and E3 (46%) (p<0.0001).

There was no between-group difference for EQ-5D-5L, IBD-Disk, or WPAI:UC scores (all p>0.05). UC-related patient-reported outcomes were similar across subgroups according to both the Wexner score (p>0.05) and UNRS (p>0.05; Table).

Conclusion: In this cohort of patients with UC treated with AT, there was no association between the extent of disease and quality of life, disability, treatment allocation, work impairment, incontinence, or bowel urgency. However, the extent of disease was associated with hospitalizations. Patients with proctitis treated with AT experience similar disease burden and require a comparable level of treatment as those with left-sided colitis and pancolitis.

Demographics and disease characteristics of study participants	E1	E2	E3	p-value
EQ-5D-5L Utility, mean (SD)	0.871 (0.162)	0.893 (0.147)	0.889 (0.174)	0.38
UNRS, mean (SD)	3.5 (3.3)	3.1 (2.9)	2.8 (2.9)	0.46
Wexner Score, mean (SD)	8.4 (4.9)	7.4 (5.1)	8.2 (4.3)	0.35
Total IBD-Disk Score, mean (SD)	36.4 (24.8)	33.5 (25.0)	32.6 (26.1)	0.69
WPAI:UC scores (on people employed*)				
Absenteeism, mean % (SD)	11.1 (29.9)	8.1 (17.6)	7.7 (21.7)	0.52
Work impairment (loss of productivity), mean % (SD)	26.7 (27.1)	22.8 (26.5)	19.4 (26.1)	0.51
Activity impairment, mean % (SD)	27.2 (25.7)	20.1 (23.6)	22.3 (27.0)	0.47

*44 patients in E1, 117 in E2 and 131 in E3 completed the WPAI:UC questionnaire. E1, proctitis, E2; left-sided colitis; IBD, inflammatory bowel disease; UNRS, Urgency Numeric Rating Scale; WPAI-UC, Work Productivity and Activity Impairment-UC; SD, standard deviation. Utility was calculated using the French value set for EQ-5D-5L.

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V.G., I.R., and I.J. are employees and minor shareholders of Eli Lilly and Company Ltd.

N.E. is an employee and minor shareholder of Eli Lilly and Company Ltd and reports previous employment and minor shareholdings with AbbVie and Gliead Science.

R.A. has no conflicting interests to declare.

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PP0375

ML-DERIVED, QUANTITATIVE HISTOLOGIC FEATURES OF THE INFLAMMATORY MICROENVIRONMENT AFTER ADALIMUMAB INDUCTION TREATMENT ALLOW PREDICTION OF RESPONSE IN PATIENTS WITH ULCERATIVE COLITIS

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Introduction: Ulcerative colitis (UC) is a chronic disease characterized by inflammation. Despite the advent of anti-inflammatory biologics, most patients do not achieve a deep response to therapy. Furthermore, predicting response to therapy is challenging. Quantitative histologic features of the inflammatory microenvironment in UC have potential to inform treatment selection and maintenance efficacy once a regimen has started.

To address this hypothesis, we compared such features derived from machine-learning (ML) models with endoscopic response status (Mayo endoscopic score; MES<2) in patients with UC treated with adalimumab (ADA).

Aims & Methods: A retrospective analysis of a phase III clinical trial of ADA (NCT02065622) was performed. A suite of ML models, collectively known as IBD Explore™ (PathAI, Boston[1])* previously trained to quantify tissue regions (e.g., granulation tissue) and cells (e.g., goblet cells) were deployed on 345 hematoxylin and eosin (H&E)-stained whole slide images (WSI) of biopsies from the rectum and sigmoid colon at ADA end-of-induction (week 8). Human interpretable features (HIFs) were extracted from each WSI. Features were then correlated with end-of-maintenance (week 52) region-level endoscopic response status (MES<2) using univariate logistic regression.

Multivariable logistic regression predictive models were also developed on a subset of the samples (277 WSIs). Predictive accuracy was assessed on a held-out test dataset (66 WSIs).

Results: Of the 657 model-derived HIFs assessed at end-of-induction, 251 were associated (p<0.05) with end-of-maintenance region level MES response status after applying Benjamini-Hochberg (BH) false discovery rate correction. Some example HIFs and their associated odds ratio with end-of-maintenance MES response are noted in Table 1.

Feature	Odds ratio	BH p-value
Median density of goblet cells in crypts	2.85	< 1e-10
Median area ratio of goblet cell mucin to epithelium + crypt abscess in crypts	2.50	< 1e-9
Proportion of neutrophils within 20 µm of goblet cells	2.40	< 1e-8
Proportion of inflamed crypts	0.46	< 1e-7
Proportion of plasma cells + intraepithelial lymphocytes in mucosa	0.66	< 1e-3
Area proportion of granulation tissue in mucosa	0.72	0.032
Heterogeneity of crypt area	0.71	0.01

Table 1: Example HIFs taken at end-of-induction and associated odds ratio to end-of-maintenance MES response.

To develop predictive models of region-specific end-of-maintenance endoscopic response, 5-fold cross validation was used. Biopsies from each subject were placed in a single split to minimize data leakage. The selected model (L1 penalized, class balance weighted) achieved a sensitivity and specificity of 0.64 and 0.67, respectively, on WSI not seen during predictive model development.

Conclusion: A quantitative understanding of the histologic changes in the colonic microenvironment shortly after treatment initiation is critical to advance our collective knowledge of UC and may aid in the prediction of a patient's response. Here, quantitative features of goblet cells and goblet cell mucin were especially predictive of endoscopic response from end-of-induction. ML-derived quantitative histology features have the potential to guide therapeutic decisions in UC that leverage mechanistic rationale derived from histology, contributing to precision medicine in immunology.

References: *Gaitan et al., 2023. *Machine learning-based prediction of Geboes score and histologic improvement and remission thresholds in ulcerative colitis.* Digestive Disease Week 2023 Poster Presentation

Disclosure: [1]PathAI Employee and owner of stock options

[2]AbbVie Employee and owner of stock options.

PP0376

THE CAUSAL EFFECTS OF INFLAMMATORY BOWEL DISEASE ON SKIN CARCINOMA: A TWO-SAMPLE MENDELIAN RANDOMIZATION STUDY

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Introduction: Observational studies have indicated that inflammatory bowel disease (IBD) patients have higher incidence of skin carcinoma (SC), including melanoma skin carcinoma (MSC) and nonmelanom skin carcinoma (NMSC) than healthy people. However, whether the correlation is causal remains unclear.

The purpose of this study was to evaluate the causality of IBD on SC using the mendelian randomization (MR) analysis.

Aims & Methods: We performed a two-sample MR analysis with public genome-wide association studies (GWAS) data. Eligible instrumental variables (IVs) were selected according to the three assumptions of MR analysis. The inverse-variance weighted (IVW) method was the main method. Complementary methods included the MR-Egger regression, the Weighted Median, the Weighted Mode and MR pleiotropy residual sum and outlier (MR-PRESSO) methods.

Results: Genetically predicted IBD (IVW OR = 1.07, 95% CI: 1.02-1.13, P = 0.011) and ulcerative colitis (UC; IVW OR = 1.09, 95% CI: 1.03-1.16, P = 0.003) was associated with an increased risk of MSC. Results of complementary methods were consistent with those of the IVW method except Weighted mode. In addition, Crohn's disease (CD; IVW OR = 1.04, 95% CI: 0.99-1.08, P = 0.128) had not a causal effect on MSC. Moreover, IBD (IVW OR = 1.03, 95% CI: 1.00-1.07, P = 0.034) and CD (IVW OR = 1.03, 95% CI: 1.00-1.06, P = 0.045) were associated with a higher risk of NMSC. However, UC (IVW OR = 1.00, 95% CI: 0.97-1.04, P = 0.803) was not associated with NMSC.

Conclusion: Our study revealed genetically predicted associations of IBD on MSC and NMSC in European populations. Additionally, UC increased the risk of MSC, and CD increased the risk of NMSC. But the influence of the use of immunosuppressive agents or biologics cannot be ruled out.

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PP0377

IMPACT OF ENDOMETRIOSIS ON THE EVOLUTION OF INFLAMMATORY BOWEL DISEASES: A MULTICENTER RETROSPECTIVE STUDY

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Introduction: Inflammatory bowel diseases (IBD), including Crohn's disease (CD) and ulcerative colitis (UC), and endometriosis are two nosological frameworks affecting young women. They share a spectrum of abdominal symptoms and partially common pathophysiological mechanisms involving immune system dysregulation. It is described that women with endometriosis have a higher risk of developing IBD. However, the impact of endometriosis on IBD is poorly understood.
The aim of our study was to evaluate the impact of the coexistence of both diseases on the phenotype and evolution of IBD.
Aims & Methods: We conducted a retrospective cohort study in 18 GETAID (Groupe d'Étude Thérapeutique des Affections Inflammatoires du tube Digestif) centers between June 2022 and March 2023. Any patient with both endometriosis and IBD was eligible for inclusion. They were then randomly matched to 1 or 2 patients with IBD without endometriosis, according to the following criteria: type of IBD (CD or UC), year of birth (± 5 years), year of IBD diagnosis (± 4 years), and date of the last consultation (± 2 years). Data from non-exposed patients (IBD alone) were collected from a single center (CHU de Lille).
The evolution of IBD was assessed using a composite severity criterion including intestinal damage (stenosis, fistula, or intra-abdominal abscess) or the need for intestinal resection surgery. The use of immunosuppressants and biologics was also recorded.
Results: In total, 207 patients with both endometriosis and IBD (149 CD and 58 UC) were included and matched to 409 patients with IBD alone. The median age of patients at the diagnosis of IBD was 23 years [IQR 18; 29]. The median follow-up duration for IBD was 10 years [5.75; 17]. For patients with CD, there was no difference between the two groups regarding disease location, disease phenotype, and anoperineal involvement. For patients with UC, there was a difference in disease extent with more rectal involvement in the endometriosis group (29.3% vs 12.6%; p=0.03). Patients exposed to endometriosis had significantly less use of immunosuppressive treatments (UC p < 0.01; CD p < 0.001) and biologics (UC p < 0.01; CD p < 0.001), both anti-TNF (UC p=0.01; CD p < 0.001) and non-anti-TNF (UC p=0.04; CD p=0.001). In the actuarial analysis according to the composite severity criterion, patients exposed to endometriosis had a more favorable evolution of their CD than non-exposed patients (HR=0.68; 95% CI [0.50; 0.92]; p=0.011). In the subgroup of patients with CD who underwent surgery for their endometriosis during follow-up (n=72), a less severe progression was also observed (HR=0.65; 95% CI [9.42; 1]; p=0.05). The luminal surgery rate for CD in these patients was lower (32.2% vs 43.9%; p=0.01) in the "exposed" group.
No difference was observed for anoperineal manifestations (p=0.16) and the use of anoperineal surgery (p=0.82). For the UC group, those with endometriosis seemed to have a less favorable disease course, according to the composite severity criterion (HR=1.73; 95% CI [0.74; 4.00]; p=0.20). However, the rate of colectomy was not significantly higher for exposed patients (19.0% vs 11.5%; p=0.20).
Conclusion: Although patients with endometriosis are at a higher risk of developing IBD, it does not negatively influence the course of IBD. On the contrary, patients with CD appear to have a less severe progression, with significantly less exposure to immunosuppressants and biologics. However, both conditions have a major impact on patients, requiring multidisciplinary follow-up to ensure symptom control and improve their quality of life.
Disclosure: Nothing to disclose.

PP0378

ALTERED EXPRESSION AND ENZYMATIC ACTIVITY OF XANTHINE OXIDASE IN INFLAMMATORY BOWEL DISEASE

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Introduction: Xanthine oxidase (XO) is involved in the last steps of purine metabolism; indeed, it catalyzes the oxidation of hypoxanthine to xanthine and xanthine to uric acid (UA) with the production of superoxide anion. The accumulation of UA has been shown to initiate the inflammatory process through NLRP3 inflammasome and the production of reactive oxygen species (ROS) contributes to inflammation-related tissue damage. In murine models of colitis, XO inhibitors, allopurinol or febuxostat, reduced the expression of proinflammatory cytokines supporting the involvement of XO in the intestinal inflammatory process.
Aims & Methods: This study aims to conduct a thorough investigation of the role XO plays in IBD pathogenesis and to explore the potential of XO inhibition as a therapeutic strategy.
Biopsies from the sigmoid colon of 15 Healthy Controls (HC), 20 Ulcerative Colitis (UC), and 15 Crohn's Disease (CD) patients, as well as from the ileum of 10 HC and 10 CD patients, were collected at the University Hospital of Cagliari. Disease activity was assessed using Mayo and SES-CD scores. XO expression levels were evaluated using Real-Time PCR and validated by GEO database (GSE11223, GSE117993). XO enzymatic activity was

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quantified spectrophotometrically in protein extracts from the biopsies. XO and NLRP3 inflammasome activation pathway, including NLRP3, ASC, and CASPASE1, was investigated using immunohistochemistry (IHC). Biopsies samples were lysed in RIPA buffer and Capillary Western analyses were performed with ProteinSimple Jess System utilizing 1:10 dilution for XO, NLRP3, ASC, and Caspase 1; 1:100 dilution for Beta-actin as primary antibodies. Ex-vivo organ culture were assayed with Intestinal biopsies from UC and CD patients. Briefly, the biopsies were cultivated with or without XO inhibitors, allopurinol and febuxostat, for 24 hours in a 24-well plate with 250 µL RPMI at 37°C with 95% O 2 /5% CO 2 atmosphere. After 24h the biopsies were used for IHC analysis of NLRP3 pathways.

Results: Increased XO mRNA expression was found in UC and CD sigmoid colon mucosa compared to HC in the GEO database. This finding was further confirmed by Real-Time PCR in our patients, with a Relative Quantification (RQ) of 1.93 vs. 0.99 (p=0.004) for UC and CD sigmoid colon, and RQ of 2.72 vs. 1.26 (p=0.004) for CD ileum. XO activity was higher in UC and CD sigmoid colon (3.86 nmol UA/min/mg in UC, 4.82 in CD vs. 1.03 in HC, p<0.001) and CD ileum (8.41 vs. 1.83 in HC, p<0.001). IHC and Western analysis confirmed that XO and the NLRP3 inflammasome pathway are more expressed in IBD patients than in HC, predominantly within epithelial cells. The ex-vivo organ culture of CD patients, and UC patients, showed a reduced UA levels and reduced NLRP3 pathways activation after treatment with XO inhibitors.

Conclusion: XO expression was higher in UC and CD patients than in HC individuals. Alongside this increase in XO activity, there was a concurrent hyperactivation of the NLRP3 inflammasome. The inhibition of XO activity in ex-vivo organ culture resulted in reduced activation of the NLRP3 inflammasome pathway. These findings support the hypothesis that XO plays a critical role in the pathogenesis of IBD and highlight its potential as a therapeutic target.

Disclosure: I declare that there are no conflicts of interest.

PP0379

LIMITED PREDICTIVE VALUE OF THE GUT MICROBIOME AND METABOLOME FOR RESPONSE TO BIOLOGICAL THERAPY IN INFLAMMATORY BOWEL DISEASE

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Introduction:

The complexity of Inflammatory Bowel Diseases (IBD) presents challenges for the management of these diseases. Predicting treatment outcomes remains difficult, leading to suboptimal outcomes and high costs. Emerging evidence suggests the potential of the gut microbiome in predicting response to biologic treatments (1,2). In this prospective study we aimed to predict treatment response to vedolizumab and ustekinumab in 79 IBD patients by integrating clinical data, gut microbiome profiles and fecal metabolites and validating these findings in a replication cohort of 47 IBD patients.

Aims & Methods: Treatment response was defined as either continuation or discontinuation of biologic therapy after six months. We performed whole genome metagenomic shotgun sequencing on the baseline fecal samples to identify microbial and functional profiles. Additionally, abundances of over 1000 metabolites were captured through untargeted metabolomic profiling. Baseline diversity, dissimilarity and differential abundance analyses compared responder and non-responder groups. The prediction tool CoDaCoRe was used to identify predictive log-ratio biomarkers. We tested the identified ratios in an external cohort and attempted to replicate the microbiome-based signals of previous studies. Finally, we used a neural-network framework to model the relationship between metabolites and microbes, comparing identified clusters in responders and non-responders and tested our approach with different definitions of response.

Results: We identified 7 differentially abundant metabolites between responders and non-responders (FDR < 0.05), and no significant differences in bacterial species or pathways. Our prediction analysis indicated only marginal predictive utility of the gut microbiome and fecal metabolites for treatment response, when compared to a clinical model using fecal calprotectin, disease duration and disease activity, among other factors (AUC only clinical features: 0.71±0.13, AUC microbial and clinical features: 0.73±0.12). The main predictive features of these models were the high abundance Phocaeicola vulgatus, Bacteroides uniformis and Alistipes onderdonkii, and low abundance of Ruminococcus gnavus and Faecalibacterium prausnitzii. Testing our identified ratio of 10 species in an external cohort of 47 IBD patients reinforced the lack of predictive power of the microbiome. No replication of previously published predictive signals of the microbiome was observed. Additionally, we identified two metabolite and one microbiome cluster associated with response, and observed that predictors were highly dependent on the definition of response.

Conclusion: While previous studies of similar size have shown that microbial features can predict response to either vedolizumab (VedoNet, 1), or vedolizumab, ustekinumab and anti-TNF (2), our comprehensive study found no significant differences in the gut microbiome at baseline between responders and non-responders among IBD patients treated with ustekinumab or vedolizumab. Microbial features added no predictive power to drug response in our IBD patient cohort or an independent replication cohort. Addition of metabolite features did not improve predictive power. These findings suggest minimal impact of the gut microbiome on treatment outcomes with these medications in IBD patients at this disease stage. Generalizability beyond initial study cohorts is limited, leaving predictors for individualized IBD-medication selection unidentified, but based on this work, likely do not lie in the fecal microbiome.

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Disclosure:

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PP0380

PROTECTIVE EFFECT OF FOOD EMULSIFIERS CARBOXYMETHYLCELLULOSE (CMC) ON MICE COLITIS MODEL IN HIGH-MILK-FAT DIET

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Introduction: Inflammatory bowel disease (IBD) is a complex chronic and relapsing inflammatory disorder of the gastrointestinal tract with unclear pathogenesis. Accumulating evidence suggests food emulsifiers as a risk factor for the development of IBD. It is indicated that food emulsifiers carboxymethylcellulose (CMC) can alter gut microbiota and promote intestinal inflammation in IBD under normal diet. Since high-fat diet is the main life scenario for CMC use, little is known about the effect of CMC on colitis model in different high-fat diet.

Aims & Methods: *C57/BL6* mice (3-5 weeks old) were randomly divided into three high-fat diet (HFD) groups, which only differed by fat composition (40 %), derived from corn oil (HCO, n-6 polyunsaturated fatty acids), olive oil (HOO, monounsaturated fatty acids) and milk fat (HMF, saturated fatty acids), and exposed to CMC (1.0% w/v) in drinking water for 4 weeks before the establishment of dextran sulfate sodium (DSS) induced acute colitis mice model. Colon inflammation severity was evaluated using survival rate, HE staining, colon length, body weight change and DAI score. Ki67 and TUNEL assays were performed to detect proliferation and apoptosis.

Results: Compared with the CMC+DSS group, the HMF+CMC+DSS group markedly increased survival rate (P=0.013), colon length (P=0.027) and body weight (day4, day5, day6 and day7, P<0.05, respectively), and relieved DAI score (day4 and day6, P<0.05, respectively) and histological damage (P<0.0001). The HOO+CMC+DSS group seemly had no significant difference from the CMC+DSS group. In contrast, the HCO+CMC+DSS group developed more aggressive colitis with increased DAI score (day5, P<0.01) and decreased body weight (day1, day2, day5 and day6, P<0.05, respectively). In terms of maintaining homeostasis of the intestinal barrier, the HMF+CMC+DSS group showed the highest Ki67 (P=0.012, P=0.0002 P=0.042, respectively) and TUNEL positive cells (P=0.324, P<0.0001, P<0.0001, respectively) compared with the CMC+DSS, HOO+CMC+DSS or HCO+CMC+DSS group. Between CMC+DSS and HOO+CMC+DSS or HCO+CMC+DSS group, there was no difference in Ki67 positive cells, while TUNEL positive cells significantly decreased (P=0.002, P=0.008, respectively).

Conclusion: The function of CMC on DSS colitis varies in different high-fat. Unexpectedly, CMC protects against the development of high-milk-fat related colitis by maintaining homeostasis of proliferation and apoptosis.

References: The function of CMC on DSS colitis varies in different high-fat. Unexpectedly, CMC protects against the development of high-milk-fat related colitis by maintaining homeostasis of proliferation and apoptosis.

Disclosure: None.

PP0381

A NOVEL MULTIOMIC APPROACH TO UNRAVEL THE MECHANISMS OF ACTION OF BIOLOGICS AND TOFACITINIB IN INFLAMMATORY BOWEL DISEASE

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Introduction: Inflammatory bowel diseases (IBD), which includes Crohn´s disease (CD) and ulcerative colitis (UC), are complex and heterogeneous diseases characterized by a multifactorial etiology. IBD prevalence is increasing worldwide. The availability of diverse treatments with different mechanisms of action have revolutionized the ability to achieve clinical

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remission and endoscopic healing. The aim of this study was to achieve a deeper understanding of the mechanism of action and response to different IBD treatments using a multiomic approach.

Aims & Methods: We analysed the metabolome of serum and urine, metagenome of stool samples and transcriptome and proteome of mucosal biopsies from 53 patients with moderate-to-severe CD and 50 UC patients initiating biologic therapy (anti-TNF, ustekinumab or vedolizumab) or tofacitinib. Patients were assessed at 14 weeks for endoscopic remission and were categorized into responders or non-responders. The methodologies employed in this study were RNA-seq analysis, liquid chromatography-mass spectrometry, nuclear magnetic resonance, and 16S rRNA gene sequencing.

Results: The multiomic analyses revealed substantial alterations in the transcriptome, proteome, metabolome and metagenome of both responders and non-responders IBD patients before and after treatment (14 weeks) to different biologics and tofacitinib. Gene enrichment analysis showed noteworthy differences in all study comparisons, except for patients with CD and UC who did respond to anti-TNF and tofacitinib. Functional enrichment analysis of differentially expressed proteins indicated several processes that could be potentially involved in the mechanisms of action of different treatments. More differences were observed in lipoproteins than in serum and urine metabolites in the different study comparisons. Finally, metagenomic findings indicated changes in the composition of gut bacteria communities among CD patients treated with ustekinumab and UC patients treated with tofacitinib (responders vs non-responders at 14 weeks).

Conclusion: The present study provides novel insights into the mechanisms of action of different IBD treatments from a multiomic approach. Nonetheless, multiomic data integration and validation studies are needed to confirm these findings within a more extensive and independent cohort of patients.

Disclosure: Nothing to disclose.

PP0382

NOVEL BIOMARKERS ASSOCIATED WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Inflammatory bowel disease (IBD) is a chronic multi-factorial disease characterized by inflammation of the gastrointestinal tract. Currently, the mechanisms underlying the disease are far from being elucidated. Thus, the study of proteins involved in its pathogenesis would allow further insights into the molecular mechanisms underlying IBD.

Aims & Methods: The serum and intestinal mucosa (ileum and left colon) proteomic profiles of patients with Crohn´s disease (CD) and ulcerative colitis (UC) and healthy controls (HC) were analyzed through label-free relative quantitative proteomic profiling using nLC MS/MS. Data mining and pattern recognition techniques were performed to identify hidden relationships that are not detectable using classical linear classifiers, and thus identify potential markers able to discriminate between the different study groups. Six candidate biomarkers were selected for further validation using enzyme-linked immunosorbent assay (ELISA) and immunohistochemistry (IHC) in serum and intestinal tissue samples, respectively.

Results: Comparison of serum proteins between UC patients and HC revealed statistically lower levels of protein 1 and 2 and higher levels of protein 3. The comparison of potential biomarkers between CD and HC group showed lower levels of protein 1 and higher levels of protein 3. In addition, protein 4 was differentially upregulated in CD patients compared to UC. Regarding the intestinal biopsies, protein 5 showed higher intensity in the

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IHC staining of ileum from CD patients compared to HC, whereas protein 6 was significantly decreased in left colon and ileum from CD and UC patients compared to HC. These proteins are involved in thyroxine transport, regulation of chemotactic chemokine activity, inflammatory processes, triglyceride homeostasis, oxidative stress and regulation of NF-kappa-B activation in the cytosol.

Conclusion: In this study, we have identified a panel of six potential biomarkers that could help to elucidate mechanisms involved in IBD pathogenesis. Their roles in the pathophysiology need to be clarified in further experiments.

Disclosure: Nothing to disclose.

PP0383

WITHDRAWN

PP0384

WITHDRAWN

PP0385

FETAL THIOPURINE SENSITIVITY IN A MOUSE MODEL BASED ON NUDT15 GENE POLYMORPHISM

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Introduction: Thiopurines are vital therapeutic agents for inflammatory bowel diseases but often cause severe myelosuppression. The NUDT15 polymorphism, particularly p.Arg139Cys (NUDT15R139C), is strongly linked to severe leukocytopenia and is prevalent in Asian populations. Despite thiopurine use during pregnancy, it remains unclear if NUDT15R139C affects fetal thiopurine sensitivity.

Aims & Methods: The human and murine NUDT15 proteins share 89% homology with the preservation of arginine at codon 139 of human NUDT15 in codon 138 of murine Nudt15. We established a knock-in mouse model with a point mutation (c.412C>T) generating p.Arg138Cys (*Nudt15*^{R138C}), mimicking the human phenotype. In the current study, we investigated offspring generated from *Nudt15*$^{+/R138C}$ or *Nudt15*$^{+/+}$ female mice that were mated with *Nudt15*$^{+/R138C}$ male mice and simultaneously administered with mercaptopurine (MP).To evaluate the effects of thiopurine on the fetus, we evaluated the number of hematopoietic stem cells in the fetal liver, which is the center of fetal hematopoiesis.

Results: In administration with the tolerable MP dose (0.5 mg/kg) for adult *Nudt15*$^{+/R138C}$ mice, genotypes of the genotypes of neonatal mice born from breeding between *Nudt15*$^{+/R138}$ male mice and *Nudt15*$^{+/R138}$ female mice were 35 (37.2%) *Nudt15*$^{+/+}$, 59 (62.8%) *Nudt15*$^{+/R138C}$, and 0 *Nudt15*$^{R138C/R138C}$. On embryonic day 14.5, *Nudt15*$^{R138C/R138C}$ fetal mice showed shorter body lengths and fewer hematopoietic stem cells in the fetal liver than *Nudt15*$^{+/+}$ fetal mice. The number of fetal HSCs was not altered in any of the *Nudt15* genotypes without MP treatment. However, it was significantly reduced by MP treatment in *Nudt15*$^{+/R138C}$ fetal mice ($p< 0.01$, mean number 1104 in DMSO and 577 in MP) and in *Nudt15*$^{R138C/R138C}$ fetal mice ($p< 0.01$, mean number 927 in DMSO and 402 in MP), but not in *Nudt15*$^{+/+}$ fetal mice ($p= 0.39$, mean number 1004 in DMSO and 820 in MP). The number of HSCs in *Nudt15*$^{R138C/R138C}$ fetal livers was significantly reduced to less than 50% of that in *Nudt15*$^{+/+}$ fetal livers by MP treatment during gestation ($p= 0.0089$). In administration with the tolerable MP dose (1.0 mg/kg) for *Nudt15*$^{+/+}$ adult mice, genotypes of neonatal mice born from breeding between *Nudt15*$^{+/R138}$ male mice and *Nudt15*$^{++}$/female mice were 23 *Nudt15*$^{+/+}$ and 0 *Nudt15*$^{+/R138C}$. Without MP administration, neonatal mice were born in a Mendelian fashion.

Conclusion: Fetal toxicity may arise if the fetus has more NUDT15 R139C alleles than the mother, despite safe thiopurine doses for the mother. Analysis of the father's NUDT15 genotype may enable safer thiopurine use during pregnancy, emphasizing the importance of considering parental genotypes in therapeutic decisions.

Disclosure: Nothing to disclose.

PP0386

LOSS OF UNCULTURED BACTERIA AND BILE ACID DYSMETABOLISM IN INFLAMMATORY BOWEL DISEASE

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Introduction: The gut microbiome has been implicated in inflammatory bowel disease (IBD), with functional defects leading to disturbances in important classes of metabolites, including bile acids. Bile acids are molecules produced by the liver that help aid in the digestion of dietary fats. While most are reabsorbed in the small intestine, approximately 5% escape into the colon and are metabolized by bacteria in the colon. Bile acids are important signalling molecules and have been implicated in both inflammation and colorectal cancer. More information is needed to understand how dysbiosis modifies bile acid profiles in patients with IBD.

Aims & Methods: This is a bioinformatic analysis of whole metagenomic sequences that searched for bacteria and genes involved in the transformation of primary bile acids into secondary bile acids. The population

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consisted of 215 IBD patients who were previously recruited from France. The faecal samples and metagenome sequences were obtained for each patient. The data was processed through a metagenomic pipeline using HUMAnN 3.0 and MetaPhlAn 3.

Results: Cluster analysis previously revealed 3 distinct patient groups based on their gut microbiome composition, one of which, cluster 3, was strongly associated with bile acid dysmetabolism. *Firmicutes CAG:103* and *Firmicutes CAG:83*, previously uncultured co-abundance groups, most strongly positively correlated with secondary bile acid levels in the dataset. *Firmicutes CAG:103* has previously been identified as a potential dominant secondary bile acid producer in metagenomic surveys. Notably, these bacterial groups were almost completely absent in cluster 3 subjects, implicating them as potential actors in secondary bile acid depletion in IBD patients.

Conclusion: This study suggest loss of *Firmicutes CAG:103* and *Firmicutes CAG:83* could be a driving factor for bile acid dysmetabolism in IBD patients. Further research is underway to validate these findings using metagenome-assembled genomes.

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Lavelle, A., & Sokol, H. (2020). Gut microbiota-derived metabolites as key actors in inflammatory bowel disease. *Nature reviews Gastroenterology & hepatology*, *17*(4), 223-237.

Disclosure: Nothing to disclose.

PP0387

THE TRANSCRIPTION FACTOR HEPATOCYTE NUCLEAR FACTOR 4A IN THE SMALL INTESTINE PROTECTS AGAINST ACUTE INFLAMMATORY STIMULI

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Introduction: Inflammatory bowel disease (IBD), are characterized by chronic inflammation of the intestine, and affects millions of people worldwide. There are two main classifications of IBD: Crohn's disease and ulcerative colitis with different mechanisms and etiology of disease progression. IBD is multifactorial; the incidence of this disease may be associated with a dysregulated immune response, genetics, environment, and interactions with the host microbiome. The gastrointestinal epithelium acts as a barrier against harmful stimuli, and defects in its integrity are a precursor for the development of IBD. Our laboratory has shown that the conditional deletion of the Hepatocyte nuclear factor 4A (HNF4A) in the mouse intestinal epithelium (*Hnf4a)* leads to chronic intestinal inflammation. However, the functional impact of HNF4A on epithelial barrier integrity is still controversial.

Aims & Methods: To evaluate the impact of *Hnf4a* deletion on barrier integrity during acute inflammation.

We used a tamoxifen-inducible Cre^{ER}-loxP system to delete *Hnf4a* in the adult intestinal epithelium. Intestinal permeability was assessed in *Hnf4a*$^{\Delta IEC\text{-}ind}$ and control littermate mice in the collected serum after oral gavage of FITC-dextran. Mice were infected with an invasion-attenuated strain of *Salmonella* Typhimurium (SB103 (invA::strep)). *S.* Typhimurium loads were scored in the feces, liver, and spleen by plating homogenates on LB agar supplemented with streptomycin. Histological assessment was performed by various staining. In addition, we used a model of T-cell-induced enteropathy triggered by a T-cell activating monoclonal anti-CD3 Ab. Control and *Hnf4a*$^{\Delta IEC\text{-}ind}$ mice received an intraperitoneal injection of 50 µg monoclonal antiCD3ε Ab (145-2C11). Mice were assessed for clinical signs of enteropathy and sacrificed 1-, 3-, and five days post-injection. Gene expression levels of selected targets were measured by RT-qPCR. Also, Ki67+ staining was performed in the small intestinal tissue

Results: FITC-dextran assay showed a significant increase of intestinal permeability in uninfected *Hnf4a*$^{\Delta IEC\text{-}ind}$ mice compared to controls. In contrast, an oral infection with an invasion-deficient *Salmonella* Typhimurium strain did not differentially impact tissue bacterial loads or epithelial damage between groups. Likewise, the loss of epithelial HNF4A led to increased Muc2 transcripts expression in the ileum of infected *Hnf4a*$^{\Delta IEC\text{-}ind}$ mice. Histological examinations at four days post-infection showed an elevated number and size of goblet cells in the ileal crypts of *Hnf4a*$^{\Delta IEC\text{-}ind}$ mice. Bacterial localization using the general bacterial probe EUB388-cy3 (red) staining showed that infected *Hnf4a*$^{\Delta IEC\text{-}ind}$ mice tended to enhance barrier mucus layer thickness compared to the controls. On the contrary, acute T cell activation with anti-CD3 mAb induced a significant increase of *Cxcl1* transcripts in the small intestine of *Hnf4a*$^{\Delta IEC\text{-}ind}$ mice. H&E staining showed elongated ileal crypts in *Hnf4a*$^{\Delta IEC\text{-}ind}$ mice compared to controls after anti-CD3 injection. Moreover, five-days after inflammatory stimulus, the ki67+ staining showed a proliferative phenotype located in the hyperplastic crypts of *Hnf4a*$^{\Delta IEC\text{-}ind}$ mice.

Conclusion: Altogether, these results support that HNF4A could play a crucial adaptive role in modulating the small intestine epithelial barrier function and controlling the immune system during acute inflammatory stress.

Disclosure: No conflict of interest in the text.

PP0388

ABLATION OF THE COLONIC EPITHELIAL MUSCARINIC RECEPTOR 3 INDUCES SEX SPECIFIC MODULATION OF INTESTINAL STEM CELLS AND RESPONSE TO INTESTINAL INJURY

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Introduction: Intestinal stem cells (ISCs), located at the intestinal crypt base, are exposed to diverse signals that contribute to their proliferation and differentiation. These signals originate from the stem cell niche, which is comprised of epithelial cell types, enteric neurons, myofibroblasts or endothelial cells. Acetylcholine (ACh) is an important neurotransmitter of the enteric nervous system (ENS), which activates muscarinic (mAChRs) and nicotinic receptors (nAChRs). Mammalian mAChRs consist of five main subtypes (M1–M5) which are coupled to G-proteins. In addition to cells of the ENS, mAChRs are prominently expressed by intestinal epithelial cell (IEC) types including Lgr5-positive ISC. We could previously show that the modulation of mAChRs altered small intestinal differentiation and the cellular activity of Lgr5+ ISCs.

Aims & Methods: We investigated the role of mAChRs in colonic epithelial homeostasis and regeneration by utilizing male and female Vil-Cre×M3R fl/fl mice with epithelial ablation of M3R. The colonic *Lgr5*-expressing ISC

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number was assessed in Vil-Cre×M3R fl/fl and C57bl/6 mice by *in situ* hybridization (ISH). Intestinal inflammation was induced by treating C57bl/6 and Vil-Cre×M3R fl/fl mice with 1% (v/v) DSS in drinking water for 7 days, followed by 5 days of normal drinking water. Epithelial cell proliferation rate and numbers of Lgr5+ ISC were assessed in colonic organoids treated with 250 nM bethanechol (Beth), a muscarinic agonist, for 4 days.

Results: Epithelial deletion of M3R in vivo revealed sex-based differences in colonic *Lgr5*- expressing ISC cell numbers. While male Vil-Cre×M3R fl/fl mice exhibited a severe reduction in colonic *Lgr5*-expressing ISCs, female mice showed the opposite phenotype with a moderate expansion of *Lgr5*-expressing cells. This translated into exacerbated DSS-induced colitis in male Vil-Cre x M3R fl mice as compared to very mild inflammation in female Vil-Cre x M3R fl mice normalized to respective age- and gender matched C56BL WT controls. In vitro, pharmacologic muscarinic agonism using Bethanechol similarly induced sex-specific changes to epithelial proliferation as well as Lgr5-positive ISC counts employing murine colonic organoids. Interestingly, Bethanechol treatment induced increased proliferation and an expansion of Lgr5-positive ISC in vitro in colonoids from female mice, which did not occur in treated organoids from male mice.

Conclusion: In summary, our study highlights the significant role of mAChRs, particularly M3, in ISCs and colonic health. The sex-based disparities observed in Lgr5+ ISC populations and responses to inflammatory stimuli emphasize the potential intricate interplay between mAChRs and sex hormones in intestinal physiology. These findings suggest potential therapeutic targets for intestinal disorders, emphasizing the importance of understanding sex-specific mAChR signaling for personalized treatments.

Disclosure: Nothing to disclose.

PP0389

LONG-TERM FOLLOW UP OF PATIENTS WITH INFLAMMATORY BOWEL DISEASE TREATED FOR CYTOMEGALOVIRUS COLITIS

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Introduction: Inflammatory bowel disease (IBD) describes two chronic inflammatory intestinal diseases: Ulcerative Colitis (UC) and Crohn's Disease (CD). Treatment of these conditions is often through immunosuppression to induce remission then maintenance. Cytomegalovirus (CMV) is a virus commonly found latent in all humans with seroprevalence rates existing between 40-100% depending on geographical location (1). Pathological reactivation of CMV is triggered by inflammation and immunosuppression; both present in the pathogenesis and treatment of IBD respectively (2). Consequentially, CMV reactivation occurs in IBD flare-ups around a rate of 25% (2) and greater than 30% in steroid-refractory patients (1, 3). Whether CMV reactivation is associated with escalating medical therapy, further hospital admissions, and worse clinical outcomes such as toxic megacolon or colectomy when compared to non-CMV associated flare-ups remains controversial. As such, the aims of this study were to follow up patients with IBD following an index episode of CMV colitis and analyse disease progression, relapse of CMV colitis, and clinical outcomes.

Aims & Methods: As such, the aims of this study were to follow up patients with IBD following an index episode of CMV colitis and analyse disease progression, relapse of CMV colitis, and clinical outcomes.

A retrospective study of patients treated at Royal Melbourne Hospital, Parkville, Australia treated between 2018-2023 was completed. Patients were included if they had a diagnosis of IBD, had positive immunohistochemistry for CMV on biopsy, and were treated with Ganciclovir or Valganciclovir for CMV colitis. Data collected included
demographic details and laboratory investigations at time of first episode of CMV colitis. Data regarding treatment for IBD and CMV colitis at time of CMV colitis was also collected. Outcome results were collected at 6-month and 12-month time points after first episode of CMV colitis. Fourteen patients were included in this study; 13 with UC and 1 with CD.

Results: Recurrence of CMV colitis occurred in 21% of cases at 6 months and 29% at 12 months. Additional courses of corticosteroids remained constant at 21% of cases at 6 and 12 months. Forty-three percent of patients had changed their biologic at 6 months and this stayed constant at 12 months. Similarly, 29% of patients had escalated their biologic dose at 6 months which was the same at 12 months. At 6 months, 43% of patients had been re-hospitalised due to IBD or CMV including 2 colectomies. At 12 months, these figures increased to 36% and 3 respectively. Disease remission was only achieved by 7% of patients at 6 months, increasing to 29% at 12 months.

Conclusion: While a descriptive study, our results when compared with previous studies (4) suggest that CMV colitis is a negative prognostic marker in patients with IBD, resulting in higher rates of hospitalisation, failed medical therapy requiring medication change or dose escalation, and colectomy. These risks may be greater at <6 months from index episode of CMV colitis. Further research regarding effective management of CMV colitis is needed to minimise poor outcomes in these patients.

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Disclosure: Nothing to disclose.

PP0390

PREDICTION OF ENDOSCOPIC FEATURES OF DISEASE FROM SINGLE-CELL RNA SEQUENCING DATA IN SIGMOID COLON OF PATIENTS WITH ULCERATIVE COLITIS

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Introduction: Patients suffering from ulcerative colitis (UC) have reduced life qualities and an increased risk of developing cancer. Current treatment options are limited and have a ceiling effect with remission rates of ~35%. A possible reason is the multi-factorial aetiology, which are as-

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sociated with an overactivation of the immune system and a dysbiosis of the gut microbiome. An important step towards a better understanding is the characterization of epithelial lesions. Their severity is classified by the Mayo score system based on their macroscopic appearance. While previous studies analysed cellular changes within the colon in UC patients, a detailed characterization of different UC-associated lesions are still missing, which is the aim of our study in collaboration with Sanofi Pharmaceuticals.

Aims & Methods: We recruit patients with active disease at our endoscopy unit and collect different biopsies from multiple regions within these patients to compare the reproducibility of single cell RNA sequencing (scRNAseq) results of similar appearing lesions within and between patients. For the sample preparation, we performed a cellular enrichment step for immune, epithelial and stromal cells before performing 10x scRNAseq to obtain equal numbers for each cell fraction from every sampled area. Additionally, we collected stool samples and endoscopic brushes to determine the associated microbial communities by 16S rRNA sequencing.

Results: Our cellular isolation strategy allows successful recovery of fibroblast, epithelial and immune cell fractions from each lesion. All expected epithelial cell types as previously described are detected in our dataset as well. Moreover, our experimental setup to obtain comparable numbers of fibroblasts, epithelial and immune cells enables us now to investigate cell-cell communication differences within different Mayo score lesions. For example, CXCL signals increased with the severity of macroscopic lesions and involves multiple cell types.

Additionally, we are in the middle of correlating microbial diversity with the corresponding disease conditions.

Conclusion: Our results provide exciting insights into cell-cell communications at an unprecedented depth, which identifies aberrant signalling pathway that may contribute to treatment failure in the clinics.

Disclosure: We have nothing to disclose.

PP0391

LACK OF DNA METHYLTRANSFERASE DNMT3A IN FIBROBLASTS IMPAIRS CELL GROWTH AND WOUND HEALING

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Introduction: Fibrosis in Crohn´s disease (CD) is a complication of chronic inflammatory processes that may result in impaired gastrointestinal function and bowel strictures. However, exact pathomechanisms leading to intestinal fibrosis in CD are not fully understood. Since the de novo DNA methyltransferase DNMT3A is involved in cellular differentiation processes and genetic variants have been associated with increased risk for CD, we sought to characterize how DNMT3A loss might affect fibroblasts, the key mesenchymal cells involved in fibrosis.

Aims & Methods: We stably deleted Dnmt3a in 3T3 fibroblast cell line using CRISPR-Cas9 technology and subsequently selected GPF-positive clones using FACS sorting. Cell cultures were performed in cell culture plates with or without 0.2% gelatin coat. Reverse transcription quantitative PCR, western blot, and immunofluorescence were used to confirm Dnmt3a gene deletion and characterize wild-type (WT) and knockout (KO) cells. Cell adhesion was evaluated by fluorescence levels of calcein acetoxymethyl ester (CAM)-labelled cells that attached to cell culture plates after 1h of culture and subsequent wash. Collagen lattice assays in presence of proinflammatory and profibrotic stimuli served to assess fibroblast contractibility.

Results: 3T3 fibroblasts lacking Dnmt3a acquire an activated myofibroblast phenotype (increased alpha-SMA expression) that also overexpress tissue remodeling matrix metalloproteinase Mmp3, and the inflammation-induced necroptosis marker RIPK3 when compared to their WT counterparts. In addition, Dnmt3a-depleted 3T3 cells adhere more strongly to cell culture plates. In presence of transforming growth factor (TGF)-beta, cells embedded in a 3D collagen matrix and lacking Dnmt3a build an interconnected cellular network and contract up to a 77.78% (compared to interconnected cells that reach up to 13.73% contraction by WT cells). This contraction was independent of the presence of interleukin (IL)-11. Profibrotic factors such as bleomycin prevent intercellular network formation in both WT and KO cells; hence, incapacitating cells to contract the collagen matrix.

Conclusion: We conclude that Dnmt3a is crucial for fibroblast homeostasis, including cell activation and functionality. Furthermore, TGF-beta induces higher fibroblast contractibility when Dnmt3a is lacking. Thus, additional characterization of Dnmt3a depletion in fibroblasts will deepen the understanding of fibrosis and may point towards new therapeutic approaches for this untreatable complication.

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Disclosure: Nothing to disclose.

PP0392

HIGH SERUM LIPOCALIN-2 EXPRESSION AT THE DIAGNOSIS IN CHILDREN WITH ULCERATIVE COLITIS: A PILOTSTUDY

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Introduction: Lipocalin-2 (LCN2) is a pleiotropic mediator of various inflammatory processes. Upon gastrointestinal damage, bacterial infection, or intestinal inflammation, LCN-2 is strongly upregulated, resulting in high concentrations of this molecule.

Aims & Methods: We determined the serum levels of LCN-2 in a cohort of newly diagnosed UC children, and we compared them with healthy controls.

We conducted a prospective cross-sectional observational study conducted at the Pediatric Gastroenterology Unit of the Umberto I Hospital and at the IBBC – Institute of Biochemistry and Cell Biology – CNR in Rome. All children aged 6-18 years of age, newly diagnosed with UC were consecutively enrolled. Blood withdrawal was conducted upon diagnosis. Human serum LCN-2 (Cat. No. DY1757) was measured using a sandwich enzyme-linked-immunosorbent assay (ELISA) kits (R&D Systems, Minneapolis, MN, USA), according to the protocols provided by the manufacturer. Clinical, demographic, biochemical and endoscopic data were recorded.

Results: Thirty-two children with a new diagnosis of UC [11 (34%) female, mean age 12,7 ± 4] and 38 age- and sex-matched healthy control [21 (55%) females, mean age 11.71 ± 4] were consecutively enrolled. Serum LCN-2

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levels were significantly higher in cases compared to controls (280 ± 152 ng/mL vs 66 ± 55 ng/mL), p < 0,001]. Among UC children, significantly higher LCN2 levels were measured in pancolitis (363.7± 155.2 ng/mL) compared to proctitis, left-sided and extensive colitis (184.9 ± 74.65 ng/mL; P = 0.0003). A significant inverse correlation was observed between LCN-2 and albumin levels at the diagnosis by the Spearman coefficient correlation (r = −0.455; P = 0.03) and a significant direct correlation was observed between LCN-2 and CRP values at the diagnosis (r = 0.44; P <0.05).

Conclusion: Serum LCN-2 levels are significantly higher in children with UC compared to the healthy control group, with the highest levels observed in children with the most extensive disease. Such results as well as and its correlation with the actual disease monitoring tool, make this protein an interesting candidate biomarker.

Disclosure: Nothing to disclose.

PP0393

OCTN1 VARIANTS SHAPE INNATE IMMUNITY AND PREDICT INDIVIDUAL RESPONSE TO VEDOLIZUMAB IN ULCERATIVE COLITIS PATIENTS: AN EXPLORATORY STUDY

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Introduction: Limited evidence is available about predictors of response to specific medications for ulcerative colitis (UC). OCTN1, an organic cation transporter, with its variants, could modulate the inflammatory response and potentially predict individual response to immune-modulatory therapy (anti-TNFα and vedolizumab).

Aims & Methods: We first confirmed the ability of OCTN1 to differentially modulate IL1β secretion under different stimuli in a human leukemia monocytic cell line (THP-1) and in human peripheral blood mononuclear cells. We therefore prospectively enrolled a cohort of UC patients starting either anti-TNFα or vedolizumab, determined their OCTN1 genotype and evaluated their clinical features at baseline and clinical response to therapy after six months, as well as two years therapy persistance.

Results: OCTN1-deficient THP1 cells showed reduced IL-1β secretion in response to peptidoglycan (PGN) and Fusobacterium nucleatum (Fn), whilst primary monocytes bearing OCTN1 503F variant displayed an increased production of IL-1β in response to PGN and live bacteria. We therefore enrolled 90 patients, of which 50 starting anti-TNFα and 40 vedolizumab. At the multivariate analysis, an overall better response to therapy was observed with the mutated 503F variant (OR=16.72, p=0.011) in comparison to wild type (503L) and heterozygous variants; nevertheless, a significant reduction of clinical response after six months was observed in patients with 503F variant treated with vedolizumab (OR=0.02, p=0.04). Furthermore, the same trends were observed the two years steroid free therapy persistence wa taken inot consideration.

Conclusion: Mutated OCNT1 genotype (503F) could favor an increased IL1β secretion from human leukocytes, thus suggesting a worse disease course and lower response to vedolizumab as compared to homozygous 503L and heterozygous.

Disclosure: Nothing to disclose.

PP0394

ENDOTHELIAL AND SUBCLINICAL CARDIAC DYSFUNCTION IN INFLAMMATORY BOWEL DISEASE

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Introduction: Endothelial dysfunction is seen in chronic inflammatory diseases and also associated with inflammatory bowel disease (IBD).(1-3) Similarly, subclinical cardiac dysfunction has also been seen in children with IBD.(4) This has not been studied in adults with IBD.

Aims & Methods: This study aims to assess the prevalence of endothelial and subclinical cardiac dysfunction among patients of IBD.

Adult patients with active IBD, IBD in remission and healthy control (age and sex matched) were recruited. Diagnosis of IBD was based on clinical, endoscopic, and histological findings. Ulcerative colitis (UC) severity was evaluated with Total Mayo Score (MS) and Crohn's Disease (CD) with Crohn's Disease Activity Index (CDAI). All participants underwent investigations to assess disease severity, presence of anemia and dyslipidemia. Conventional 2D and speckled echocardiography was done to assess subclinical cardiac dysfunction. Endothelial dysfunction was detected using endothelium dependent flow mediated dilation (FMD). Vascular doppler ultrasound of brachial artery was done to assess FMD.

Group comparisons of numerical data were done with one-way ANOVA test or Kruskal-Wallis test, depending on the normality of the data. Holm-Sidak test was used as post hoc test for multiple assessment. Proportions were compared using the chi-square test. Statistical significance was determined when the p-value was less than 0.05.

Results: 90 subjects were analyzed in the study (30 with active IBD, 30 IBD in remission and 30 healthy controls). 55 patients had UC and 5 patients had CD. The median duration of disease was 24 (48.3) months and was similar in both active IBD and IBD in remission.

The median FMD was 13.7 %, 22.9% and 23.3% in active IBD, IBD in remission and controls respectively. FMD was significantly lower in patients with active IBD as compared to controls (p=0.025) and IBD in remission (p=0.048). There was no difference in FMD between controls and IBD in remission (p=0.707).There was no correlation between impaired FMD and disease duration (Pearson's r = -0.104, p=0.21). Cardiac evaluation revealed comparable ejection fraction (EF) among different groups. The global longitudinal strain (GLS) was significantly lower in active IBD [-18.0 (5.73)%] (p=0.002) and IBD in remission [-18.0 (3.30)%] (p=0.034) as compared to controls [-20.2 (1.95)%]. GLS was similar in IBD in remission and active IBD (p=0.933). The prevalence of subclinical cardiac dysfunction based on GLS was 4% (4/30) in controls, 14.4% (13/30) in IBD in remission and 15.6% (14/30) in active IBD. The difference among the groups was significant (p=0.011). FMD was not different between subjects with normal and abnormal GLS (p=0.936).

Outcome Median (IQR)	Active IBD	IBD in remission	Controls	p Value	p1	p2	p3
FMD (%)	13.7 (16.3)	22.9 (15.1)	23.3 (6.4)	0.017	0.048	0.025	0.707
GLS	-18.0 (5.73)	-18.0 (3.30)	- 20.2 (1.95)	0.003	0.99	0.002	0.003

Comparison between: p1 –Active IBD and IBD in remission; p2 –active IBD and controls, p3 – IBD in remission and controls

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Conclusion: Endothelial dysfunction was more in active IBD compared to IBD in remission and controls. It was similar in IBD in remission and controls. Subclinical cardiac dysfunction was also more in IBD patients. Endothelial dysfunction did not correlate with disease duration. We also found no correlation between Endothelial Dysfunction and presence of subclinical cardiac dysfunction.

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Disclosure: Nothing to disclose.

PP0395

TRANSCRIPTOMIC ANALYSIS REVEALS DIFFERENTIAL EXPRESSION OF OXIDATIVE STRESS PATHWAYS IN CROHN'S DISEASE PATIENTS

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Introduction: Crohn's disease (CD) poses a clinical challenge due to its unpredictable post-operative recurrence and the necessity for effective therapeutic interventions. The molecular mechanisms governing recurrence are poorly understood, although oxidative stress (OS) has been implicated in CD pathogenesis. However, its specific role in recurrence remains elusive. This study aimed to analyze the gene expression signatures regarding OS and their implications for CD recurrence using transcriptomic analysis.

Aims & Methods: Ileal mucosa samples were obtained from the prospective mutlicentrique cohort of CD patients undergoing surgery. They were collected from ileal resection margins (n = 196) at the time of surgery (M0) and in the neo-terminal ileum (n = 121) 6 months after surgery (M6) while patients were classified as in remission (Rem) or recurrence (Rec) based on the Rutgeerts score. Endoscopic recurrence was defined as a Rutgeerts score ≥i2b. Gene data sets (Affymetrix microarrays) were issued on a princeps paper*. Pathway analysis was conducted to identify significant OS pathways using the Ensemble of Gene Set Enrichment Analyses (EGSEA) method and through unsupervised signature generation (Independent Component Analysis) from M0 and M6 genes dataset. Statistical analyses were carried out to assess differential gene expression and pathway enrichment between Rem and Rec patients at M0 and M6.

Results: No significant genes were differentially expressed between Rem and Rec groups at M6 and M0. However, using EGSEA at M6, of the 1766 significant pathways, 40 were related to OS and predominantly overexpressed in the Rec group, indicating a potential association between OS resolution and pressure relief. At M0, of the 2556 significant pathways, 40 were related to OS and equally distributed between the Rem and Rec groups, and therefore did not predict recurrence at M6. Applying unsupervised signature generation, 140 signatures were identified and 8 were annotated as OS pathways. Theses OS pathways were upregulated at M6 in the Rec group and at M0 in the Rem group.

Conclusion: The study highlights the role of OS in CD progression and post-operative relapse. The significant increase in OS pathways at M6 in relapsed patients and at M0 in patients who remain in remission suggests that resolution of OS is necessary to counteract CD relapse. Further research is needed to delineate the role of oxidative stress in CD recurrence and to identify new therapeutic targets tailored to individual patient profiles.

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Disclosure: There is no conflict of interest regarding this abstract

PP0396

THE E3 UBIQUITIN LIGASE RNF31 MEDIATES THE DEVELOPMENT OF ULCERATIVE COLITIS BY REGULATING NLRP3 INFLAMMASOME ACTIVATION

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Introduction: The NLRP3 inflammasome is a multiprotein complex consisting of the receptor protein NLRP3, the linker protein ASC, and the effector protein caspase-1. Activation of NLRP3 can lead to increased production of proinflammatory cytokines, which disrupt intestinal barrier function and induce the development of ulcerative colitis.

Aims & Methods: Whether RNF31 regulates the expression of the NLRP3 inflammasome is not known. In this study, we examined the expression of RNF31, and NLRP3 in ulcerative colitis tissues by GEO database analysis and immunohistochemical assays, and subsequently determined that RNF31 can regulate NLRP3 inflammasome expression by constructing an RNF31-deficient cell model, and elucidated the role of RNF31 in the development of ulcerative colitis.

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Results: We found that the expression of NLRP3 decreased after the addition of MG132 inhibited the level of ubiquitination. a cycloheximide (CHX)-chase analysis showed that RNF31 was able to make NLRP3 more stable, which was most likely due to the fact that RNF31 changed the level of ubiquitination on NLRP3. Multiple co-immunoprecipitation (CO-IP) assays showed that there was an interaction between RNF31 and NLRP3 and that this interaction increased the level of ubiquitination on NLRP3. And CO-IP experiments using truncated RNF31 plasmids showed that the RBR structural domain interacted with NLRP3, increasing the level of ubiquitination on NLRP3 and making NLRP3 more stable. By constructing a cell model with RNF31 deletion as well as overexpression and using CO-IP experiments we found that the K63 ubiquitination chain plays an important role in stabilizing NLRP3.

Finally, we constructed RNF31-deficient mouse models using adenovirus and showed that the expression of NLRP3 and inflammatory factors IL-18 and IL-1β were reduced in the absence of RNF31.

Conclusion: Our study shows that RNF31 can interact with NLRP3 through the RBR structural domain and increase the level of the K63 ubiquitin chain on NLRP3 thereby making NLRP3 more stable and activating the assembly of the inflammasome, elucidating how RNF31 is involved in the mechanism of colitis development.

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PP0397

EXPLORING BIOMARKERS OF SYSTEMIC OXIDATIVE STRESS AND PLACENTAL INSUFFICIENCY IN PREGNANT WOMEN WITH INFLAMMATORY BOWEL DISEASES

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Introduction: Inflammatory bowel disease (IBD), encompassing Crohn's disease (CD) and ulcerative colitis (UC), often manifests around the fertile age. It is known that disease activity affects pregnancy outcomes. However, little is known about how IBD affects pregnancy outcomes and vice versa. Both IBD and pregnancy complications, particularly those in which placental insufficiency is implicated, are accompanied by oxidative stress. Oxidative stress is defined as an imbalance between reactive oxygen species (ROS) production and antioxidant availability. Free thiols (FT) are circulating biomarkers of oxidative stress since they readily scavenge ROS. Previous research has shown reduced levels of FT in both IBD[1] and pregnancy complications[2]. Classical biomarkers of placental insufficiency include soluble FMS-like tyrosine kinase-1 (sFlt-1) and placental growth factor (PLGF), and their ratio is reflective of maternal pregnancy complications[2]. However, the behavior of these biomarkers in pregnant women with IBD remains unclear.

Aims & Methods: In this study, we aimed to investigate the relationship between FT, sFlt-1, and PLGF and the occurrence of pregnancy complications in patients with IBD, and to assess their dynamics before, during, and shortly after pregnancy.

In this retrospective clinical pilot study, we investigated a cohort of pregnant women with IBD and healthy pregnant controls who were pregnant between 2000 and 2023. Serum samples taken before (IBD n=41), during (IBD n=33, controls n=14), and after (IBD n=25) pregnancy were analyzed for PLGF, sFlt-1 and FT levels. FTs were measured using a colorimetric detection method. sFlt-1 and PLGF were measured using an immunofluorescent assay. Extensive clinical data were collected, including data about pregnancy complications such as pre-eclampsia, fetal growth restriction, and gestational diabetes, and IBD disease parameters, including fecal calprotectin and clinical disease scores.

Results: In total, 40 patients (n=32 CD, n=7 UC, and n=1 IBD-U) with 47 pregnancies and 14 non-IBD control pregnancies were included. Comparing patients with IBD to controls during pregnancy, serum FT levels were significantly reduced in patients, even after adjustment for gestational age (p=0.043). FT levels significantly decreased during pregnancy compared to preconception (p=0.001), suggesting increased systemic oxidative stress during pregnancy. The sFlt-1/PLGF ratio was significantly higher in IBD compared to controls (p=0.011), which was partially due to a difference in gestational age (median (IQR): controls 32.6 weeks (29.4-33.1) vs. 15.4 weeks (7.0-22.7) in IBD). In total, 32 patients (56%) and 3 controls (21%) developed pregnancy complications. The sFlt-1/PLGF ratio was higher in those developing maternal complications, almost reaching significance (p=0.053), while FT showed a trend towards decreased levels in those developing maternal complications (mean (SD): 368.9 μM (38.7) without vs. 349.1 μM (21.9) with maternal pregnancy complications, p=0.082).

Conclusion: This study provides evidence for increased systemic oxidative stress (reflected by lower systemic FT levels) during pregnancy in IBD, when compared to both healthy controls and their own pre-conceptional levels. These results suggest a difference in redox status between pregnant women with IBD compared to those without. Further research is necessary to substantiate these findings and evaluate the utility of these biomarkers in predicting pregnancy complications and guiding clinical decision making in IBD pregnancy.

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Disclosure: Nothing to disclose.

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PP0398

INCIDENCE OF CHRONIC END-STAGE RENAL DISEASE IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: A POPULATION-BASED COHORT STUDY FROM THE FRENCH NATIONAL HEALTH INSURANCE DATABASE

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Introduction: Renal disorders are among the extra-intestinal manifestations of inflammatory bowel disease (IBD), and treatments may also lead to renal failure. However, the incidence of end-stage renal disease (ESRD), defined as the need for renal replacement therapy (dialysis and/or renal transplantation (RT)), is poorly known. The primary objective of our study was to assess the incidence of ESRD in IBD patients, and to compare it with the general health insurance scheme population.

Aims & Methods: We conducted a prognostic cohort study based on data from the French National Health Insurance Database(SNDS). Prevalent cases of IBD affiliated to the general health insurance scheme during the period 2011-2020 were identified on the basis of declarations long-term diseases ALD 24 or the presence of 2 hospital stays for IBD with ICD-10 codes or a single stay with reimbursement of specific treatments. Incident cases of ESRD were identified on the basis of a stay with a RT or a period with dialysis sessions for at least 45 days. Crude incidence rates of ESRD were calculated over the period for the IBD and the non-IBD population, with their 95% confidence intervals.

Results: A total of 280,283 IBD patients were included in the study over the period 2011-2020, 134,567 with ulcerative colitis(UC) and 145,716 with Crohn's disease(CD). The mean age at study inclusion was 39.8 (± 17.1) years. The M/F sex ratio was 0.9. The mean age at incidence of ESRD in the non-IBD population was 65.0 (±17.0) years, the M/F sex ratio cases was 1.7. Among IBD patients, 534 patients began replacement therapy by dialysis or RT, they were younger than non-IBD patients (60.3(±16.7)years, P<0.001).The mean incidence rate of ESRD over the period was 26.4 per 100,000 person-years (IC95% 23.1-29.6), 480 (89.9%) patients entered ESRD by dialysis, and 54 (10.1%) by preemptive RT. In the non-IBD population the mean rate of ESRD was 15.9 per 100,000 person-years (IC95% 14.8-17.1; P<0.0001 compared with IBD): 79,606 (94.5%) patients entered ESRD by dialysis, and 4,634 (5.5%) by RT (as compared with IBD,P<0.001). In subgroup analysis, the mean incidence rate of ESRD was 22.8 per 100,000 person-years (IC95% 18.9-26.7) in UC patients and 29.5 (IC95% 25.3-33.6) in CD patients (P<0,0001). The rates of preemptive RT were 12.4% in UC patients (87.6% dialysis) and 8.6% in CD patients (91.4% dialysis).

Over a median follow-up of 6.8 years, the survival rate after ESRD was 40.8% in ESRD-IBD patients VS 34.2% in ESRD non-IBD patients (P<0.001). This difference remained significant for patients under 65 (p<0.0073).

No significant difference for the risk of re-hospitalization after RT (p=0.84), of retransplantation during the follow-up period in RT patients (p=0.67), or for the risk of hospitalization for severe infection (p=0.97) after RT in patients from the general population compared with IBD patients. The same was true for the type of severe infections after RT (p= 0.068) and for the risk of solid and haematological cancer after RT (p=0.48). However, there was a significant difference in the type of first severe infection after RT(p=0.003), but we are waiting for the results to tell us which one.

Conclusion: The incidence of ESRD is significantly higher in IBD than in the general data base population, confirming the need for regular monitoring of renal function in our patients.

Survival after ESRD onset, is higher in IBD patients.

The high incidence of RT in the IBD population confirms that access to transplantation is not restricted in this
population.

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Disclosure: Nothing to disclose.

PP0399

TISSUE-DIRECTED GENE THERAPY FOR INFLAMMATORY BOWEL DISEASE

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Introduction: There is an unmet need for new therapies to promote mucosal healing and restoration of epithelial barrier function in the inflammatory bowel diseases (IBD). The development of novel approaches to specifically deliver a therapy to the gut has the theoretical advantage of optimizing benefit while limiting side effects. Adenoviral (Ad) gene therapy has been successful in treating cancer and has recently been used in clinical trials as a vaccine for SARS-Cov2. Ad vectors can be modified to target specific tissues and therefore may be more effective vehicles for gene therapy for diseases like IBD than other systemic therapeutics. We propose that gene therapy using adenovirus has promise to achieve these goals in IBD. As a proof of concept, we developed an approach to deliver IL-22 via Ad gene therapy. IL-22 plays an important role in maintaining the epithelial barrier at mucosal sites like the skin, airway, and intestine. Epithelial cells are the primary target of IL-22 signaling, where it activates transcription of genes involved in cell proliferation and promotes antimicrobial functions.

Aims & Methods: We hypothesize that gene therapy with Ad vectors to deliver IL22 directly to the intestine could be an effective therapeutic for IBD by promoting intestinal wound healing and microbial homeostasis. To test this hypothesis, we injected human adenovirus serotype 5 (huAd5) expressing *Il22* (Ad-CMV-Il22) intravenously into 6-week-old C57BL/6 mice.

Results: IL-22 protein levels in the serum, liver, ileum, and colon were increased with Ad-CMV-Il22 compared to the empty vector control (Ad-CMV-GFP). We analyzed transcription of genes downstream of IL-22 signaling and found Ad-CMV-Il22 increased relative expression of *Lcn2* in the liver, ileum, and colon, as well as increased expression of *Mt1* and *Mt2* in the liver and colon. Ad-CMV-Il22 induced increased crypt depth and goblet cell count in the ileum and colon, and increased villus length in the ileum compared to Ad-CMV-GFP. Changes to the stool microbiome included increased abundance of the pro-tolerogenic family Lachnospiraceae-Ruminococcaceae with Ad-CMV-Il22 in both colon and ileum. It is known that huAd5 has liver tropism and gene therapy with hAdv5 has been associated with liver toxicity. We also observed higher levels of IL-22 in the liver than in the serum, ileum, or colon after Ad-CMV-Il22 treatment. To reduce liver tropism, we modified the vector to incorporate myeloid-binding peptide (MBP) on the virus fiber, which re-targets the virus to the intestine and

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vascular endothelium. Incorporation of MBP significantly reduced IL-22 protein levels in the liver by nearly 10-fold. The ratio of IL-22 protein levels in the ileum and colon was 1.35 and 1.81 times higher with the targeted vector compared to untargeted virus. Increased colon and ileal crypt depth (45%) as well as villus length (9%) were seen with Ad-CMV-MBP-Il22 as compared to the untargeted virus.

Conclusion: Together, these data suggest that IL-22 gene therapy in the intestine can promote cell proliferation and microbial homeostasis. In the future, this adenoviral gene therapy could be modified further to increase *Il22* gene delivery specifically to intestinal tissues.

Disclosure: D.T.C. is a consultant for Ocugen, Circero, Asgard, Accession and Tome Biosciences.

PP0400

IMPACT OF PHYSICAL ACTIVITY ON THE QUALITY OF LIFE OF PATIENTS WITH IBD: PRELIMINARY RESULTS OF A PROSPECTIVE STUDY

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Introduction: Chronic inflammatory bowel diseases (IBD) are chronic conditions that profoundly affect the quality of life (QoL), and their evaluation has become essential during IBD follow-up. Regular and tailored physical activity (PA) can contribute to improving QoL; however, engaging in PA may be limited during these diseases.

The aim of this study was to describe the factors associated with PA practice and its impact on QoL.

Aims & Methods: This was a descriptive and analytical prospective study conducted on a sample of 85 patients followed in IBD consultations. We collected data regarding the disease and assessed PA intensity using the International Physical Activity Questionnaire (IPAQ), which distinguishes between three activity levels: light, moderate, and vigorous. We assessed QoL using the Inflammatory Bowel Disease Questionnaire (IBDQ), calculating sub-scores for digestive symptoms, systemic symptoms, emotional disorders, social function, and overall score. Statistical analysis was performed using SPSS 20.0 software. Qualitative variables were compared using the Chi-square test and McNemar's test. Factors associated with PA practice were identified using a binary logistic regression model in univariate and multivariate analysis.

Results: We collected data from 85 patients. The mean age was 35 +/- 3.4 years. The female-to-male ratio was 1.23. Crohn's disease was reported in 65 (76.47%) patients, while 20 (23.53%) had ulcerative colitis. 68 (80%) patients were in remission. Two patients (3.3%) were smokers. The prevalence of joint manifestations was 13.3%. 12 patients (14.1%) had undergone surgery as part of their disease management. 51 (60%) patients were on biologic therapy with a mean treatment initiation of 3.5 +/- 1.2 years; 13 (15.29%) patients were on thiopurines alone, 19 (22.3%) were on salicylates, and 2 (2.35%) were on corticosteroids. Among our patients: 58 (80%) engaged in light PA, 18 (21.1%) in moderate activity, and 9 (10.58%) in vigorous activity. The mean sub-score for digestive symptoms was better in patients engaging in moderate to vigorous activity compared to those engaging in light activity, with respective values of 31.3 +/- 1.16 vs. 29.15 +/- 2.11, $p < 0.001$. The mean sub-score for systemic symptoms was also better in patients engaging in moderate to vigorous activity: 34.6 +/- 3.61 vs. 28.52 +/- 1.21 in cases of light PA, $p = 0.003$. In multivariate analysis, engaging in moderate to vigorous PA was associated with clinical remission (OR 1.06; [1.02; 3.15]; $p = 0.002$) and absence of joint manifestations (OR 2.93; [1.24 - 3.05], $p < 0.001$). Smoking and a history of digestive surgery did not influence PA. Patients engaging in moderate to vigorous activity had a better quality of life than those engaging only in light activity, with mean overall IBDQ scores of 198.3 +/- 21.3 vs. 151.11 +/- 25.3, $p = 0.0003$. The emotional sub-score was better with moderate to vigorous activity: 33.7 +/- 5.1 vs. 31.05 +/- 4.51 in patients with light PA, $p = 0.002$. Similarly, the social function sub-score was better with moderate to vigorous activity: 29.04 +/- 7.16 vs. 26.23 +/- 4.61, $p < 0.001$.

Conclusion: In our study, engaging in moderate to vigorous PA among patients with IBD was associated with better quality of life. Certain factors limit its practice, notably disease flare-ups and joint manifestations. The importance of physical activity should therefore be evaluated during consultations, and patients with IBD should be encouraged to engage in regular and tailored PA according to their disease profile

Disclosure: Nothing to disclose.

PP0401

THROMBOEMBOLIC ACCIDENTS IN IBD: IS THERE A GENDER EFFECT?

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Introduction: Chronic inflammatory bowel disease (IBD) is at high risk of thromboembolic events (TEA), which are a significant cause of morbidity and mortality.

They are attributed to a pre-thrombotic state caused by the inflammatory activity of this disease.

The objective of our study was to evaluate the prevalence and risk factors (FDR) of ATEs in patients with IBD, to determine their clinical and evolutionary particularities while specifying the existence or not of a gender effect.

Aims & Methods: This is a retrospective descriptive and analytical study carried out at the hepato-gastroenterology and proctology department "Médecine B" CHU Ibn Sina Rabat, between January 2001 and July 2023, including all patients who presented during their follow-up with an ATE confirmed by imagery (thoracic angioscanner/ cerebral angio MRI/ doppler ultrasound).

We collected demographic, clinical and paraclinical data.

The Chi-2 test was used to compare categorical variables.

Results: During the study period, among the 1643 IBD cases collected in the department, we recorded 21 cases of IBD complicated by ATE, a prevalence of 1.27%.

The mean age was 42.7 years (23-61 years), with a male predominance of 12 men (57%) and 9 women (43%), with a sex ratio of M/F : 1.3.

Among the 12 men, 7 (58%) had Crohn's disease and 5 (42%) UC, with a mean disease duration of 8 years and 4 months. Among the 9 women, 6 (67%) were being followed for Crohn's disease and 3 (33%) for UC.

The occurrence of a thrombo-embolic complication during hospitalization for a disease exacerbation was more frequent in men than in women (8 (67%) Vs 5 (55%); p=0.3).

Deep vein thrombosis (DVT) was more common in women than in men (6(67%) Vs 1(8%); p= 0.003).

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Cerebral thrombophlebitis was found in 2 men (17%) and 2 women (22%). Budd Chiari syndrome was found in 1 woman (11%).

Pulmonary embolism was found in 3 men (25%), portal thrombosis in 3 men (25%), renal artery thrombosis in 2 men (17%), superior mesenteric vein thrombosis in 1 man (8%). None of the women had these manifestations.

Thrombophilia testing revealed decreased anti-thrombin activity in women, associated with protein C deficiency in 2 patients (22%). In the men, we noted protein C deficiency in 1 patient (8%) and 1 case (8%) of protein S deficiency.

Hyperhomocysteinemia was positive in one patient (11%), with no cases in men.

All patients were treated with anti-coagulants for curative purposes.

The absence of thrombosis and good revascularization was confirmed by follow-up imaging after 6 months of treatment in 5 men (42%) and 3 women (33%); p= 0.4.

A complication in the form of extensive cerebral ischemic necrosis of thrombosed vessels was noted in 1 man (8%).

A single case of death was reported in 1 man (8%) following a severe pulmonary embolism. No cases of death were recorded in women.

Conclusion: Patients with chronic inflammatory bowel disease have a non-negligible risk of developing TEA, particularly during the active phase of the disease, underlining the importance of preventing thromboembolic disease when managing patients during relapses.

Disclosure: Nothing to disclose.

PP0402

CANDIDA YEASTS IN CROHN'S DISEASE: PREVALENCE, ASSOCIATED FACTORS, AND IMPACT ON DISEASE PROGRESSION

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Introduction: Crohn's disease (CD) has a complex etiopathogenesis involving the intestinal fungal microbiota which appears to have a significant impact on the development and progression of the disease.

Aims & Methods: The objectives of our study were to investigate the prevalence of *Candida* yeasts and the factors associated with their presence, as well as their impact on the course of CD. A prospective study involving patients with CD [June 2020 - March 2021] was conducted at the gastroenterology department of HHT in collaboration with IPT. A mycological examination of stool samples was performed to identify *Candida* yeasts.

Clinical and laboratory follow-up was conducted to assess disease progression. Unfavorable evolution was defined in the presence of flare-ups, complications (abscesses, collections, surgical intervention for CD), and therapeutic escalation.

Results: Fifty-two patients with a mean age of 40.7 years [18-67] and a gender ratio (M/F) of 1,59 were included. Yeasts were detected in 71,2% of patients, with *Candida* genus present in 55,8%.

The isolated species were *Candida albicans* (45%), *Candida parapsilosis* (32%), *Candida glabrata* (10%), *Candida tropicalis* (6%), and *Candida krusei* (7%).

Two patients harbored two *Candida* species simultaneously: *C. krusei* and *C. parapsilosis*, and *C. albicans* and *C. parapsilosis*, respectively.

The presence of *Candida* was associated with a history of diabetes (p=0,001), ileal (p=0,018) and ileocolonic (p=0,017) disease locations, and clinical disease activity as assessed by CDAI (p=0,03).

C. albicans was associated with both stenosing and fistulizing phenotypes of CD (p=0,04) and a CRP > five mg/L (p=0,005). *C. glabrata* was associated with ileal localization (p=0,01), and *C. tropicalis* was associated with the occurrence of collections (p=0,049) and therapeutic escalation (p=0,01).

Conclusion: *Candida* genus yeasts are common in CD and are associated with disease activity and an unfavorable disease course. We suggest their systematic investigation during MC, although their treatment is not consensual.

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Disclosure: Nothing to disclose.

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PP0403

CHARACTERISTICS OF LEISHMANIA INFECTION IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Infection by Leishmania spp. in patients diagnosed with inflammatory bowel disease (IBD) is an uncommon infectious complication. The use of anti-TNF-α is well known risk factor for Leishmania infection (LI), particularly in some endemic areas, but only a few case series have been reported in the literature, mainly in the Mediterranean basin.

Aims & Methods: Aim: To evaluate the main demographic, IBD-disease features and risk factors as well as LI disease characteristic and required treatment among IBD patients who develop LI.

Methods: This is a retrospective, observational, multicenter study conducted in twenty-five Spanish hospitals, promoted by the Young Group of GETECCU, including patients diagnosed with IBD and subsequent diagnosis of LI.

Results: We collected a total of 73 patients, 48 of them were men (65.8%) with a mean age of 48 ± 15.06 years. 92% of patients resided in the Mediterranean basin, an endemic area of Leishmania. Only 1 patient was born outside of Spain. Nineteen percent of patients were smokers and 16.4% had some comorbidities.

Fifty patients had Crohn's disease (CD) (68.5%) with ileocolonic involvement (30.1%) and inflammatory behaviour (45.2%) and 42.9% had perianal disease. 94% of CD patients were on clinical remission at the time of LI diagnosis. In addition, 23 patients had ulcerative colitis (UC) (31.5%), being extensive colitis the most frequent extent (78.3%), and all of them were on clinical remission or had mild disease activity. 69 patients (94.5%) were under immunosuppressive and/or biologic treatment during the 6 months prior to the LI. 6.8% were on immunosuppressants monotherapy, 42.5% in combotherapy and 38.4% under biologic monotherapy. Regarding biological treatment, 97% of patients were under AntiTNF and only 2 patients were under treatment with Ustekinumab.

Regarding the characteristics of LI, 62 patients presented cutaneous involvement (85%), 9 visceral (12%) and the rest presented mucocutaneous involvement.

Symptoms associated to LI were fever (9.6%), abdominal pain (4.1%) and constitutional syndrome (8.2%), moreover 24.7% required hospital admission. Regarding the location of the skin lesions, the most frequent were the upper limbs (35.6%) followed by lower limbs (30.1%).

The treatment used for LI was systemic amphotericin in 60.3% of patients and pentavalent antimonials in 35.4%; the remaining patients were treated with other treatments, including miltefosine. LI solved at one month in 33 patients (45.2%) and at one year in 95.3% of patients, and only 11% of patients had recurrent LI.

Treatment was stopped temporarily in 38.4% of patients and definitively in 27.4%, starting Ustekinumab (36.8%), Vedolizumab (15.8%) and antiTNF (10.6%) and the rest of the patients remained without biological treatment after the LI.

Conclusion: The usage of antiTNF is associated to LI in IBD patients, but LI could also happen under inmunossupresants or ustekinumab therapy. Although cutaneous involvement is the most frequent manifestation, 12% of patients have visceral involvement or LI recurrences. This infectious complication should be considered in IBD patients in endemic areas.

Disclosure: No Conflict of Interest.

PP0404

INTERLEUKIN-8 AS A POSSIBLE PREDICTIVE MARKER OF LONG-TERM RESPONSE IN ULCERATIVE COLITIS PATIENTS TREATED WITH VEDOLIZUMAB

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Introduction: Vedolizumab has demonstrated efficacy in moderate-severe Ulcerative Colitis (UC). To date, few tools are available to predict medium- and long-term response to the drug. The aim of the study is to characterise the profile of circulating cytokines associated with the pathophysiology of inflammatory bowel disease (IBD) to identify markers of immune response to vedolizumab in UC patients. As a secondary objective,

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we analysed the possible involvement of baseline characteristics such as smoking and disease duration until drug initiation, previous exposure to other biologics, and baseline faecal calprotectin (FCP) and C-reactive protein (CRP) values in the response to the drug.

Aims & Methods: Retrospective single-centre cohort study including 28 patients diagnosed with UC with inflammatory activity objectified by clinical and biochemical data who started treatment with vedolizumab in the period from 2015 to 2022. Participants consented and authorised the collection of biological samples. Plasma samples were collected prior to drug initiation and response to treatment was assessed at the end of follow-up.

Response to treatment was based on clinical remission (pMAYO < 3 with no subscore >1) together with a biochemical response (reduction in FCP and/or CRP > 50%) at the end of follow-up. In patients where colonoscopy was available, endoscopic response was defined as an endoscopic Mayo of 0 or 1. Cytokines were determined by Magpix® Luminex using a Milliplex® Human High Sensitivity T Cell Magnetic Bead Panel containing 12 proteins.

The cytokines studied were (interferon-gamma, interleukin 10, interleukin 12, interleukin 13, interleukin 17A, interleukin 10, interleukin 2, interleukin 21, interleukin 23, interleukin 7, interleukin 8, TNF-alpha).

Results: Twenty-eight patients with UC (100% pancolitis or left colitis) and S2+S3 94.7% were included in the study. 85.2% were bioexposed (infliximab in 75%). 50% of patients received concomitant corticosteroid therapy at the start of vedolizumab.

Vedolizumab response was seen in 13 patients (46.4%) after a median overall treatment duration in both cohorts of 2.79 years (IQR 7.55). The responder group was on vedolizumab treatment for a median of 5.58 (IQR 6.9) years vs 1.17 (IQR 5.13) years in the non-responder group, p=0.005. There were no baseline differences between responders in pMayo or in FCP and CRP values. Active smoking, disease duration until vedolizumab initiation and previous exposure to other biologics were not associated with response/non-response to vedolizumab treatment.

Vedolizumab responders had significantly lower median interleukin 8 (IL8) levels at baseline than vedolizumab non-responders (median IL8 1.67 pg/mL versus 3.03 pg/mL, p=0.016). No statistically significant differences in baseline concentrations were found in any of the other interleukins analysed.

Conclusion: In our series of UC patients treated with vedolizumab, patients with long-term response to the drug show lower plasma IL8 levels than non-responders at baseline. Therefore, IL8 could serve as a predictive marker of long-term response to vedolizumab in UC patients. These results should be confirmed in prospective studies with larger sample sizes.

Disclosure: Nothing to disclose.

PP0405

CAUSAL ASSOCIATION BETWEEN CIRCULATING IMMUNE CELL COUNTS AND INFLAMMATORY BOWEL DISEASE: A BIDIRECTIONAL TWO-SAMPLE MENDELIAN RANDOMIZATION STUDY

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Introduction: Inflammatory bowel disease (IBD) is a chronic recurrent inflammatory bowel disease, whose cause involves the interaction between genetic and environmental factors. At present, the pathogenesis of IBD has not been fully understood. Although there is a recognized link between immune response and IBD, the causal relationship between circulating immune cell counts and IBD remains controversial.

Aims & Methods: This study aimed to elucidate the causal relationship between genetically predicted circulating immune cell counts and IBD. We conducted a bidirectional two-sample Mendelian randomization (MR) study using aggregated statistics from genome-wide association studies. The causal relationship between five circulating leukocytes cells (monocytes, lymphocytes, eosinophils, basophils and neutrophils) counts and IBD, including Ulcerative colitis (UC) and Crohn's disease (CD) was analyzed. Inverse variance weighting (IVW) was used as the primary MR analysis method, and horizontal pleiotropy test and heterogeneity test were used to ensure the stability of the results.

Results: Our findings indicated that monocytes, lymphocytes, eosinophils, and basophils count were not significantly associated with IBD, however, elevated circulating neutrophils count was significantly associated with higher risk of IBD [odds ratio (OR): 1.002; 95% confidence interval (CI): 1.001–1.003; p = 0.009] and UC [OR: 2.465; 95%CI: 1.236–4.916; p = 0.01].

In addition, we also found that IBD [OR: 12.07; 95% CI: 1.909-76.316; P = 0.008] and UC [OR: 2.465; 95% CI: 1.236 - 4.916; P = 0.01] were significantly associated with higher circulating neutrophils count in reverse MR.

Conclusion: This MR study provides genetic evidence for the causal relationship between the genetically predicted increase in circulating neutrophils count and the risk of IBD (UC and CD). This finding stresses the need for further exploring physiological functions of neutrophils in order to develop effective strategies against IBD.

Disclosure: The authors state that the study was conducted in the absence of any business or financial relationships that could be interpreted as potential conflicts of interest.

PP0406

THE EFFECT OF VEDOLIZUMAB ON THE LEVEL OF ADHESIVE MOLECULES ICAM-1, ICAM-2, SL-SELECTIN IN ULCERATIVE COLITIS

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Introduction: A deeper understanding of the pathogenesis of UC has led to the emergence of new therapeutic strategies. One of them consists of targeted exposure with biological drugs to certain molecules that regulate the interaction between leukocytes and endothelial cells in the inflammatory zone (mainly leukocyte integrins and endothelial cell adhesion molecules of the immunoglobulin superfamily).

Aims & Methods: Purpose of the study: To study the content of adhesion molecules ICAM-1, ICAM-2 and L-selectin in the peripheral blood of patients with ulcerative colitis (UC) and the effect of vedolizumab (Intivio) on their levels.

We examined 12 patients with moderate and severe forms of UC and the ineffectiveness of previous standard therapy. The severity of the UC attack was calculated according to the Mayo index. Antivio® 300 mg was administered to patients intravenously according to the attached instructions. The content of adhesion molecules in the peripheral blood was studied before and 6 weeks after the first injection of vedilizumab. The control group consisted of 15 healthy volunteers. The content of ICAM-1, ICAM-2 and L-selectin in blood serum was determined by multilayer ELISA analysis (Bender med.Systems, Austria). Statistical processing of the obtained results was carried out using SPSS 13.0 software.

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Results: It was found that the initial levels of ICAM-1 (461.9±28.4 ng/ml), ICAM-2 (323.3±18.4 ng/ml) and sL-selectin (1650±12.25 U/ml)) in the peripheral blood of patients with UC were increased (P<0.05 with control), the Mayo index was 9 points. After four infusions of Vedolizumab (week 14), the levels of soluble molecules decreased in 4 patients, with a predominant decrease in sL-selectin (201±9.32 ng/ml, 197±8.54 pg/ml and 1008±15.3 U/ml respectively). The Mayo index was 5 points. In 8 patients, by the 6th week of treatment, the levels of the studied adhesive molecules did not change, although the Mayo index decreased by 3 points.
Conclusion: The initial level of ICAM-1, ICAM-2 and sL-selectin in the peripheral blood of patients with severe forms of UC is increased. The dynamics of the synthesis of adhesive molecules by immunocompetent cells under the influence of the fourth administration of Vedolizumab may predict the possibility of continuing the administration of the drug.
Disclosure: Nothing to disclose.

PP0407

EPIDEMIOLOGY AND THERAPEUTIC MANAGEMENT OF INFLAMMATORY BOWEL DISEASE USING DATA FROM THE FRENCH NATIONAL HEALTH DATA SYSTEM 2014- 2021 – THE EPITHERA STUDY

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Introduction: Inflammatory bowel disease (IBD) prevalence is steadily increasing worldwide, particularly in developed countries. In parallel, the medical treatment of IBD evolved significantly with the marketing approval of advanced therapies over the last decade. The aim of this study was to describe the epidemiology of IBD and to assess treatment patterns in real-life management in France.
Aims & Methods: This was an observational study using the French National Health Databases (SNDS) including all adult patients diagnosed with Crohn disease (CD) or ulcerative colitis (UC) meeting at least one of the following criteria between 2014 and 2021: i) long term disease (LTD) related to IBD ii) at least 2 hospital stays for IBD iii) one hospital stay for IBD and 3 or more IBD treatment deliveries. A look-back period of at least 5 years prior to index date was considered to assess epidemiology and patients' characteristics, and a follow-up period of at least 12 months to describe treatment patterns.
Results: In the study period, 274,694 IBD patients were identified: 139,001 CD patients (including 45,939 (33.0%) incident patients), and 135,693 UC patients (including 53,939 (39.8%) incident patients). CD prevalence varied from 193.5 per 100,000 person-years (py) in 2014 to 234.4 per 100,000 py in 2021, while incidence varied from 12.1 to 9.7 per 100,000 py respectively. UC prevalence and incidence varied respectively from 173.1 to 229.2 per 100,000 py and from 13.7 to 11.8 per 100,000 py.
Prevalent CD patients were 56.7% female, median age was 42 years (IQR 31-56) and median follow-up was 8.0 years (IQR 4.6-8.0). Among patients with at least 1 year of follow up, 42.3% were exposed to at least one biologic: 96.9% received an anti-TNF in first line, 1.6% an anti-interleukin, 1.4% an anti-integrin and 36.4% received a concomitant immunosuppressant. Among incident CD patients (54.6% female, median age of 37 years (IQR 26-53) and median follow-up of 3.9 years (IQR 2.0-5.9)), 50.8% of patients with at least 1 year of follow-up were exposed to at least one biologic. The treatment distribution in first line was similar to prevalent patients.
Prevalent UC patients were 50.8% male, median age was 49 years (IQR 36-62) and median follow-up was 8.0 years (IQR 3.9-8.0). Among patients with at least 1 year of follow up, 20.3% were exposed to at least one biologic: 94.0% received an anti-TNF in first line, 5.3% received an anti-integrin, 0.4% an anti-interleukin, 0.3% a JAK inhibitor (JAKi) and 34.5% received a concomitant immunosuppressant. Among incident UC patients (50.9% male, median age of 43 years (IQR 31-58) and median follow-up of 3.8 years (IQR 1.9-5.8)), 23.5% of patients with at least 1 year of follow-up were exposed to at least one biologic, with comparable treatment distribution in first line as in prevalent patients.
After a switch or a swap from an anti-TNF in first line for CD, the use of anti-TNF in second line tends to decrease (from 92.8% in 2014 to 36.2% in 2020) in favor of anti-integrins (from 2.3% to 16.7%) and anti-interleukins (from 4.9% to 47.1%). The same trend is observed for UC, for anti-TNF (from 93.8% to 30.0%), mostly in favor of anti-integrins (from less than 5.6% to 57.0%).
Conclusion: This study reports an increase in prevalence and a slight decrease in incidence of IBD in France between 2014 and 2021. Treatment patterns evolved with an increased use of advanced therapies and a swap out of class in the second line.
Disclosure: LV received fees from abbvie, amgen, MSD, Ferring, Takeda, Pfizer, Janssen, Lilly, Galapagos, Celltrion, Viatris
LPB received fees from Abbvie, Abivax, Alimentiv, Alma Bio Therapeutics, Amgen, Applied Molecular Transport, Arena, Biogen, BMS, Celltrion, Connect Biopharm, Cytoki Pharma, Enthera, Ferring, Fresenius Kabi, Galapagos, Genentech, Gilead, Gossamer Bio, GSK, Hac Pharma, IAC Image Analysis, Index Pharmaceuticals, Inotrem, Janssen, Lilly, Medac, Mopac, Morphic, MSD
Norgine, Novartis, OM Pharma, ONO Pharma, OSE Immunotherapeuthics, Pandion Therapeuthics, Par' Immune, Pfizer, Prometheus, Protagonist, Roche, Roivant, Samsung, Sandoz, Sanofi, Takeda, Theravance, Thermo Fischer, Tigenix, Tillots, Viatris, Vifor, Vectivbio, Ventyx, Ysopia
JK received fees from Janssen, Takeda, Lilly, Abbvie, Pfizer, Tillots, Amgen, Celltrion, Roche, Gilead, Galapagos
SC, HO, LG, KS, LC works for Janssen-Cilag
PR, GB, TN works for IQVIA.

PP0408

OUTPATIENT CARE FOR CROHN'S DISEASE IN BRAZILIAN HEALTHCARE PUBLIC SYSTEM: FROM COST TO REFUND

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Introduction: Crohn's disease (CD) is characterized by a rare, chronic disease that requires lifelong treatment, representing a burden on healthcare systems. Considering that in Brazil, through the Unified Health System (SUS), patients have access to all treatment, it is important to know how the costs are being distributed in these patients, to better plan they care.
Aims & Methods: The present study aims to evaluate the costs for the hospital and SUS reimbursement for outpatient treatment of patients with CD. This is a retrospective cohort of patients with CD followed between 2006 and 2020 in a Brazilian public tertiary hospital. Patients over 18 years of age and with complete information in their medical records were in-

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cluded. Through medication and exams carried out during follow-up, SUS reimbursement values were calculated using the SIGTAP-SUS table and hospital costs using the table provided by the hospital. The values for the acquisition of medications were provided by the State Health Department. Conversion to dollars was carried out each year, as well as correction for inflation. The statistical difference was evaluated across disease phenotypes according to the Montreal classification using the ANOVA test and the Scheffe post hoc test.

Results: One hundred and ninety-one patients were included, 50.26% being female. The SUS reimbursement cost was $6741.40 ± 2526.87, the hospital cost was $7162.88 ± 2732.30, and adjusted for inflation was $11274.36 ± 5806.85. In post hoc analysis, there was a statistical difference between groups A2 and A3 (age at disease onset between 16 and 40 years and over 40 years) in the reimbursement group, and between the inflammatory and fistulizing phenotypes in the cost and reimbursement groups.

Conclusion: In Brazil, the treatment of CD has lower costs but maintains differences between phenotypic profiles similar to international studies. Showing the difference in cost, reimbursement and inflation, questions are raised about the updating of SIGTAP-SUS tables and the credibility of cost studies and its management planning.

Disclosure: Nothing to disclose.

PP0409

WITHDRAWN

PP0410

INCIDENCE AND NEOPLASTIC RISK ASSOCIATED WITH COLONIC STRICTURE IN PEDIATRIC-ONSET CROHN'S DISEASE: A POPULATION-BASED STUDY

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Introduction: The identification of colonic stricture in patients with Crohn's disease (CD) often leads to surgery due to concerns about associated neoplasia. Nevertheless, information regarding the neoplastic risk associated to colonic stricture is still limited.

Aims & Methods: The objective of this study was to assess both the incidence of colonic stricture in pediatric-onset CD and the associated neoplastic risk at the population-based level. All patients diagnosed with CD before age 17 between 1988 and 2011, and included in a population-based registry, were followed until 2013. Colonic stricture was defined as a narrowing of the digestive lumen observed on endoscopic or radiological examination. Excluded were stricture in the ileocaecal valve, anal canal, or low rectum. Univariate risk factors for colonic stricture were explored using Cox proportional hazards models with time-dependent variables, and a multivariable Cox model analysis was performed.

Results: A total of 1,007 patients diagnosed with CD between 1988 and 2011 were included. Median age at diagnosis was 14.5 years [IQR: 12.0-16.1]. Throughout the follow-up period (median follow-up, 8.8 years [4.6-14.2]), patients were predominantly exposed to 5-aminosalicylates (88.1%), immunosuppressants (70.6%), and biologics (47.9%). Colonic stricture was diagnosed in 52 patients, including 11 at CD diagnosis. The cumulative incidence of colonic stricture in CD was 2.9% [1.8%-4.0%] at 5 years. The overall incidence rate of colonic stricture in CD was 5.4 cases/1000 person-years, 95% CI [4.1-7.1]. In multivariable analysis, an increased risk of colonic stricture was observed during periods of active disease (HR = 2.62, [1.40-4.89], p<0.01).

Conversely, the absence of colonic involvement at diagnosis (HR = 0.18, [0.04, 0.80], p<0.05) or treatment with aminosalicylates (HR = 0.39, [0.19-0.79], p<0.05) was associated with a lower risk of colonic stricture. After a median follow-up of 6.4 [2.4-12.7] years, colonic stricture surgical resection was performed in 29 patients (55.8%). Colon adenocarcinoma was detected in one patient (1.9%) within 6 months after the diagnosis of colonic stricture.

Conclusion: In this population-based study of pediatric-onset Crohn's disease, 52 patients (5%) experienced colonic stricture, with a 2.9% incidence rate observed after 5 years of follow-up. Additionally, one patient (2%) presented with an associated adenocarcinoma.

Disclosure: M.F. has received lecture/consultant fees from AbbVie, Ferring, Tillotts, MSD, Biogen, Amgen, Fresenius, Hospira, Sandoz, Pfizer, Celgene, Gilead, Boehringer, Galapagos, Janssen, and Takeda. C.D. has received consultant fees from AbbVie. N.R. has received lecture/consultant fees from AbbVie and Takeda. The other authors state that they have no competing interests regarding this work to disclose.

PP0411

SPINE AND SACROILIAC JOINT INVOLVEMENT IN NEWLY DIAGNOSED PATIENTS WITH INFLAMMATORY BOWEL DISEASE - SYMPTOMS, CLINICAL AND MAGNETIC RESONANCE IMAGING FINDINGS FROM A POPULATION-BASED COHORT STUDY

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Introduction: Axial spondyloarthritis (axSpA), characterised by chronic inflammation of the sacroiliac joints (SIJs) and spine, is a common extraintestinal manifestation in Inflammatory Bowel Disease (IBD). It is associated with poor outcomes and a substantial impact on quality of life. Thus,

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early detection is important to optimally tailor treatment and suppress the inflammation in all disease domains. However, the prevalence and pattern of axial involvement have not been described in early IBD.

Aims & Methods: We aimed to describe the prevalence and distribution of axial symptoms, clinical findings and MRI-determined inflammatory and structural lesions in the spine and SIJs of patients with newly diagnosed IBD. Patients with newly diagnosed IBD from a prospective population-based inception cohort were consecutively included. Structured rheumatological interview collecting data on peripheral and axial symptoms and clinical evaluation were performed. MRI assessment for spine and SIJ inflammation and structural lesions was based on validated and widely applied scoring systems [1,2]. Further, an expert assessment of the MRIs as +/- indicative of axSpA applying standardized definitions was performed [3,4].

Results: Of 110 included patients (mean age 42 years, 40% male, 40 with Crohn's disease (CD), 70 with ulcerative colitis (UC)), 74 had active IBD. At the time of evaluation, 4 patients received treatment with systemic glucocorticoid and one received TNF-inhibitor. Nine patients were HLA-B27 positive. None were previously diagnosed with SpA.

Back and/or buttock pain was reported by 48(44.9%) patients and inflammatory back pain by 10(9.1%). In 23(21.5%) patients the pain could not be better explained by competing conditions (for example degenerative disc disease, nerve root compression, osteoporosis or fibromyalgia) according to the assessing rheumatologist. The axial symptoms were accompanied by peripheral joint and/or entheseal symptoms in 30(62.5 %) patients and 33(68.8%) had tender and/or swollen joints and/or entheses.

According to the MRI expert, MRIs of 17(16.7%) patients were indicative of axSpA (spine and SIJs evaluated together), while 11(10.8%) of the spine-MRIs and 13(12.7%) of the SIJ-MRIs were indicative of axSpA. Only 10 of the 17 patients with typical axSpA MRI findings had axial symptoms. There was no difference in symptoms or MRI findings between patients with UC and CD, or between patients with active and inactive IBD.

Spine bone marrow oedema (BME) in ≥1 slice was present in 30(30.3%) patients. SIJ BME in ≥1 slice was found in 27(26.5%) patients, erosion was present in 24(23.5%) patients, fat lesions in 12(11.8%) and ankylosis in 3(2.9%). The definition of positive MRI for sacroiliitis was fulfilled in 12 (11.8%) patients of whom six had axial symptoms. The Assessment of Spondyloarthritis International Society (ASAS) classification criteria for axSpA were met in 11 cases.

Conclusion: One out of six patients with newly diagnosed IBD had findings indicative of axSpA on spine and SIJs MRI. Since only 60% of these patients were symptomatic, we conclude that axSpA may be an underdiagnosed early-occurring condition among IBD patients. Our results indicate a low threshold for performing MRI of the spine and SIJs and a need for multidisciplinary collaboration to ensure early detection of axial involvement to enable an optimal therapeutic strategy and prevent future structural damage and disability.

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Disclosure: Nothing to disclose.

PP0412

INNOVATIONS IN HEALTHCARE: AI'S IMPACT ON INFLAMMATORY BOWEL DISEASE AND DIGESTIVE HEALTH

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Introduction: Artificial Intelligence (AI) has significantly changed the landscape of healthcare, particularly in the diagnosis and treatment of Inflammatory Bowel Disease (IBD) and various digestive conditions. Key AI tools like ChatGPT and Google Gemini have the ability to decode endoscopic images, examine various samples, streamline administrative tasks, and aid in the evaluation of medical imagery and device automation. These AI solutions improve the management of digestive diseases by tailoring treatments and predicting potential adverse reactions. The incorporation of AI into healthcare allows patients to make informed decisions through the assessment of personalized treatment options.

Aims & Methods: The aim of the study was to assess the accuracy of two popular chatbots, ChatGPT and Google Gemini, in addressing medical management inquiries. Each chatbot was given a set of questions to answer, with their responses rated on a 1-10 Likert scale, where 1 represented a low score and 10 was high. To ensure fair evaluation, two neutral evaluators examined the responses of each bot. The research sought to highlight the abilities of these chatbots by methodically evaluating their performance in terms of accuracy and reliability. The use of two evaluators and the Likert scale method was instrumental in reducing potential biases, thereby validating the results.

Results: Our research evaluated the effectiveness of ChatGPT and Google Gemini in the realm of medical management. ChatGPT outperformed, attaining a 71% accuracy rate compared to Google BARD's 39% ($p = 0.032$), and surpassed in information reliability with 58% against Google BARD's 29% ($p = 0.021$). This study emphasizes the critical role of accuracy and reliability in the development of chatbots for managing IBS. ChatGPT's impressive performance suggests its potential as a reliable tool, yet it also points to the continuous need for research and improvement in this area.

Conclusion: AI has profoundly transformed healthcare, especially in the diagnosis and management of IBD. ChatGPT and Google Bard stand out as crucial AI instruments for interpreting medical imagery, evaluating samples, and enhancing efficiency. Our study assessed the accuracy and reliability of these chatbots within medical management contexts. ChatGPT outshone Google Bard by achieving 71% accuracy and a 58% reliability rate in information. ChatGPT demonstrates significant promise as a dependable AI tool for IBD care, yet our findings also underline the ongoing need for further research and development in the field.

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Disclosure: Nothing to disclose.

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PP0413

BRIDGING THE GAP: ETHNIC VARIATIONS IN IBS OUTCOMES AFTER BARIATRIC SURGERY

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Introduction: Bariatric surgery emerges as a powerful intervention for obesity, promoting enhanced metabolic health. However, substantial evidence indicates a rise in Irritable Bowel Syndrome (IBS), primarily following such surgical interventions. The effect of race on the occurrence of IBS after surgery has yet to be investigated. Given the widespread occurrence of IBS and its confirmed connection to bariatric surgery, examining the effect of racial disparities is crucial. This study seeks to ascertain whether racial differences impact the probability of an individual developing IBS post-bariatric surgery. Revealing this relationship is vital for improving patient results and optimizing postoperative care.

Aims & Methods: We executed a retrospective study to analyze patients who underwent bariatric surgery at our facility between 2009 and 2022. Data about comorbidities, insurance statuses, surgical techniques, colonoscopy and pathology reports, and initial patient attributes were collected through ICD and CPT codes. Patients who developed IBS post-surgery were monitored until 2023 and categorized by gender, race, and duration post-surgery. We used propensity score matching to balance baseline characteristics. Participants with a previous diagnosis of IBS were excluded from the study. Kaplan-Meier survival analysis and Cox proportional hazards models were leveraged to examine the onset of IBS among different racial groups. Additionally, odds ratios were employed to discern independent variables influencing the study outcomes

Results: Between 2009 and 2022, 960 bariatric surgeries were performed at our institution. Of these, 39 patients, equating to 0.04%, were diagnosed with IBS, typically around 40±9.2 months post-surgery. The subjects' average age was approximately 39±8.7 years, with females constituting 51.2% of the participants. Regarding racial breakdown, 29.2% were African Americans, 33.5% were Whites, 21.8% were Hispanics, and 15.5% were Asians. Notably, White patients manifested symptoms of IBS significantly earlier compared to other racial groups (23±5vs 36±7, 31±8, 29±6) (Breslow: 12.46, p=0.015). Additionally, a heightened risk of IBS was observed among individuals with lower educational attainment (OR: 1.25, P=0.025).

Conclusion: This research aims to bridge the prevailing knowledge void relating to the influence of race on the emergence of IBS following bariatric surgery. Our retrospective analysis of 960 surgeries from 2009 to 2022 highlighted a 0.04% occurrence of IBS, generally identified around 40 months after surgery. Notably, White patients experienced the onset of IBS in considerably less time than other racial groups, with individuals with lower educational levels exhibiting increased risks. These insights emphasize the importance of considering racial and other sociodemographic elements in refining postoperative care

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Disclosure: Nothing to disclose.

PP0414

SHORT-TERM EXPOSURE TO ENVIRONMENTAL AIR POLLUTION AND INFLAMMATORY BOWEL DISEASE FLARES: A TIME-STRATIFIED CASE CROSS-OVER STUDY

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Introduction: Some epidemiological studies suggest that long-term exposure to environmental air pollutants, such as particulate matter (PM), could have a role in inflammatory bowel disease (IBD) incidence and pathogenesis. However, less is known about the potential effect of short-term air pollution exposure. The aim of this study is to investigate whether short-term exposure to environmental air pollution is associated with the occurrence of IBD flares.

Aims & Methods: Anonymized hospital discharge records (HDR) with ICD-9-CM codes 555 (Crohn's disease) and 556 (ulcerative colitis) for years 2002-2009 from all hospitals of the Lombardy region were collected. All HDR of adult patients (18-75 years) resident in Milan were included, except for recurrent hospitalizations within 6 months.

Each record was assigned daily PM ≤10 µm (PM10) concentrations at municipality level by averaging PM10 levels of the day of admission with the levels of the day before (lag01) and of each preceding day up to six days before (lag06).

To evaluate the association between PM10 and IBD-related hospitalizations, a time-stratified case-crossover study design was applied. Conditional logistic regression models, calculating Odds Ratios (OR) and 95% Confidence Intervals (95%CI) per increments of 10 µg/m3 of PM10, were performed.

Results: Overall, 2,176 hospital admissions (males=1,137, Crohn's disease=1,264, ulcerative colitis=912) were included. In the entire population no clear association between PM10 and IBD-related hospitalization was identified, notwithstanding a small risk increase for males (OR: 1.03; 95%CI: 0.99-1.08 at lag04).

When further stratifying for ICD code, an increased risk of hospitalization for "left-sided ulcerative colitis" (ICD code: 556.5) was found in males only, for exposures in all days preceding hospitalization up to lag04 (1.28; 1.05-1.28; 59 cases, 207 controls) (Figure 1). In females, the association was weaker, although positive (1.14; 0.91-1.42; 49 cases, 164 controls). No relevant findings emerged when looking at other ICD codes.

Conclusion: We observed an increased risk of IBD-related hospitalization in male patients with left-sided ulcerative colitis per increasing levels of air pollution in the days preceding hospital admission. Although preliminary, our findings suggest the possible role of sex as an effect modifier of the influence that air pollution might exert on IBD flares.

Further studies are needed to better comprehend the complex framework characterizing the interplay between environmental and individual factors contributing to IBD.

Disclosure: DN received unrestricted research grants from Pfizer, consulting fees Dr. Falk, and lecturer fees from Ferring. MV served as a consultant for Abbvie, MSD, Takeda, Janssen-Cilag, and Celgene. He received lecturer

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fees from Abbvie, Ferring, Takeda, MSD, Janssen-Cilag, and Zambon. MA received consulting fees from Nikkiso Europe, Mundipharma, Janssen, Abbvie, Ferring, Galapagos, and Pfizer. FCa has served as consultant for Mundipharma, AbbVie, MSD, Takeda, Janssen, Roche, and Celgene; received lecture fees from AbbVie, Ferring, Takeda, Allergy Therapeutics, and Janssen; and received unrestricted research grants from Giuliani, SOFAR, MS&D, Takeda, Pfizer, and AbbVie.

PP0415

TESTING MACHINE LEARNING WITH FINNISH REGISTER DATA TO PREDICT NON-RESPONSE TO CONVENTIONAL TREATMENT IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Inflammatory bowel disease (IBD), including both Crohn's disease (CD) and ulcerative colitis (UC), are difficult to diagnose due to a broad symptoms spectrum.1,2

If symptoms remain uncontrolled with conventional therapy (CT), patients are transitioned to biologics. Little is understood about which CD and UC patients will respond to CT and which patients may have a more aggressive disease progression, and therefore should be treated with biologic therapies sooner.

Aims & Methods: We aimed to develop and evaluate machine learning (ML) models to identify non-responders to CT based on patients' demographic characteristics and on data available before their treatment with CT. The study cohort included all Finnish patients ≥18 years of age, with a diagnosis for CD or UC between 1.1.2000–31.12.2019. Data were collected from electronic health records and national healthcare and prescription registers. CT drugs included corticosteroids (CS), immunomodulators and 5-aminosalicylic acids. CT failure outcomes are defined in *Table 1*.

A set of 6,782 predictive features including demographic and clinical characteristics, comorbidities, healthcare visits, procedures, laboratory measurements, and medication were created for three different baseline periods. A Lasso model was run both separately and in combination for each period, splitting the data into training and testing subcorpora ten different times to estimate the error bounds.

Results: After applying specific attrition criteria, the final analysis cohort included 12,818 CD and 30,178 UC patients. In total, 64.2% of CD and 61.1% of UC patients experienced CT failure (*Table 1*).

For CD, CS at baseline and young age (0–18 and 18–30 years) were the highest predictors of CT failure. For UC, the model selected 78 predictive features, the strongest being: baseline CS use, age (0–18, 18–30, and 65+ years), gastrointestinal and schizoaffective disorders, and secondary care visits to rheumatology, general medicine, and emergency.

The key findings of the Lasso model were:

1) The performance of CD-models was slightly better than the UC-models (median test accuracy was 0.65 [CD] and 0.62 [UC]);

2) a slight discrepancy between test and training statistics suggested overfitting, and;

3) most features appeared in all ten splits indicating their relevance.

CT failure outcomes	CD, n	CD, %	UC, n	UC, %	Total, n	Total, %
Prescription for biologics[1]	718	5.6%	828	2.7%	1,546	3,6%
IBD-related surgery	803	6.3%	728	2.4%	1,531	3.6%
90-day gap in CT medication (5-ASA[2], IM[3])	2,339	18.2%	5,325	17.6%	7,664	17.8%
≥2 corticosteroid[4] prescriptions	6,347	49.5%	14,302	47.4%	20,649	48.0%
IBD-related inpatient stay	1,339	10.4%	2,160	7.2%	3,499	8.1%
IBD-related ED visit	193	1.5%	398	1.3%	591	1.4%
Any of the above	8,223	64.2%	18,453	61.1%	26,676	62.0%

[1]adalimumab, vedolizumab, infliximab, golimumab, ustekinumab, and tofacitinib; [2]mesalazine, olsalazine, and sulfasalazine; [3]methotrexate, mercaptopurine, azathioprine, cyclosporine, and metronidazole; [4]prednisolone, hydrocortisone, budesonide, methylprednisolone, prednisone, and dexamethasone. Abbreviations: CD, Crohn's disease; CT, conventional therapy; ED, emergency department; IBD, intestinal bowel disease, IM, immunomodulators; UC, ulcerative colitis; 5-ASA, 5-aminosalicylic acid

Table 1. CT failure outcomes (within one year from CT initiation) for CD, UC, and combined cohorts.

Conclusion: Data quality is critical to build high-performance models. Even though Finnish registries provide access to comprehensive longitudinal data allowing the generation of relatively large analysis cohorts, our ML model demonstrated a weak predictive potential, with low AUCs and accuracies (0.6 range), in line with previous findings.3 Due to scarcity of information on additional clinical variables available in the Finnish IBD registry, their use did not improve prediction of CT failure.

Additional prediction difficulty is introduced by non-standardized data recording practices. Confounding factors and random chance affect complicated outcomes, such as CT failure, making them especially difficult to predict.

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PP0416

IMPACT OF COVID-19 INFECTION ON INFLAMMATORY BOWEL DISEASE COURSE: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Several viral respiratory infections have been implicated in the exacerbation of immune mediated inflammatory diseases (IMID)[1-3]. There has however been limited investigation and conflicted findings of the long-term outcomes in inflammatory bowel disease (IBD) following COVID-19[4]. We therefore performed a systematic review and meta-analysis of the long-term impact of SARS-CoV-2 infection on the disease course of IBD.

Aims & Methods: To understand the impact of early SARS-CoV-2 variants and to be better prepared for future variants, we aimed to carry out a meta-analysis of the available evidence. We searched MEDLINE and EMBASE for studies investigating the risk of adverse IBD-related outcomes (IBD flares, change in disease activity, change in medication, IBD-related hospitalization, and surgery) following a SARS-CoV-2 infection, compared to IBD patients without a SARS-CoV-2 infection. ROBINS-E for observational studies was used to assess the risk of bias in the included studies. We used random effects model (REM) meta-analysis to calculate pooled hazard ratios (HR) for the risk of adverse IBD outcomes following the acute phase of COVID-19 infection. Subgroup meta-analysis was carried out to assess the risk of the outcomes by IBD subtype (Crohn's disease (CD) and ulcerative colitis (UC)). Meta-regression was undertaken to assess whether sex or duration of follow-up were significant predictors of IBD-related outcomes.

Results: A total of 3119 studies were identified of which 5 studies were eligible for inclusion after screening using the pre-defined inclusion criteria. The studies were based on cohorts from Denmark, Italy, the US, and a pan-European consortium including a total of 34,977 SARS-CoV-2 positive IBD patients and 53,270 IBD patients without a recorded SARS-CoV-2 infection.

Two studies were assessed as having a high risk of bias. REM meta-analysis did not show an increased risk of adverse IBD-related outcomes in IBD patients with a recorded SARS-CoV-2 infection compared to those without (HR:1.05, 95% CI:0.75-1.46) and substantial between-study heterogeneity was identified in the included studies (I2=91%, p<0.01).

There was no significant difference in adverse IBD-related outcomes between CD and UC patients (CD HR: 0.91, 95% CI: 0.82-1.02; UC HR: 0.83, CI: 0.75-0.90). Following meta-regression, neither sex nor duration of follow-up were found to be significant predictors of individual study effect size.

Conclusion: In this systematic review and meta-analysis, we find that SARS-CoV-2 infection does not increase the risk of long-term adverse IBD-related outcomes. High-quality studies with longer follow-up periods are required to understand the long-term impact of SARS-CoV-2 infection on IBD outcomes.

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Study	Country	Source Population	Study Period	COVID-19 cohort	Reference cohort	Summary estimates with 95% CI: type of relative estimate	Adverse IBD-outcome	ROBINS-E overall Risk of Bias (RoB)‡
Papa et al., (2022)	Italy	IBD patients from 5 Italian treatment centers	Oct 2020 -Mar 2021 (6)	95	190	0.42 (0.08-2.15): HR	Need for therapy intensification	High RoB
Hadi et al., (2022)	USA	TriNetX (40+ US Health Care Organizations)	Jan 2020 – Mar 2021 (3)	4310	22508	1.33 (1.18-1.51): RR	Inpatient hospitalisation along with a collection of stool studies, intravenous methylprednisolone requirement or flexible sigmoidoscopy	High RoB
Hong et al., (2022)	USA	5 academic medical centers within New York's Crohn's & Colitis Organization	Feb 2020 – Dec 2021 (11)	251	251	0.84 (0.44–1.42): HR	IBD-related hospitalisation/ surgery	Low RoB
Bezzio et al., (2022)	Italy	IG-IBD Italian cohort	Mar 2020 – Dec 2020(7)	219	219	1.48 (0.90-2.42): OR	Change in HBI or PMS	Some concerns
Nørgård et al., (2023)	Denmark	Danish National IBD Cohort	Mar 2020 – Jul 2022(6)	30102	30102	0.85 (0.79-0.91): IRR	Corticosteroid prescription	Some concerns
Amiot et al., (2022)	EU I-CARE cohort†	I-CARE European prospective longitudinal multicenter cohort	Apr 2020 – Dec (9)	233	5224		Change in HBI or SCCAI	Some concerns
Lukin et al., (2021)	USA	I-CARE European prospective longitudinal multicenter cohort	(6 months)	118	118		Change in HBI or PMS	Some concerns

Table.

Abbreviations: HR- Hazard Ratio, RR – Relative Risk, OR – Odds Ratio, IRR – Incidence Rate Ratio, HBI – Harvey Bradshaw Index, SCCAI – Simple Clinical Colitis Activity Index, PMS – Partial Mayo Score.

† Belgium, Denmark, France, Germany, Greece, Hungary, Ireland, Israel, Italy, the Netherlands, Poland, Portugal, Spain, Sweden and the United Kingdom.

‡ The studies are rated as having a high risk of bias, some concerns or low risk of bias.

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PP0417

FATIGUE IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE IN REMISSION (THE IBSEN III STUDY)

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Introduction: Chronic fatigue (CF) is prevalent among patients with ulcerative colitis (UC) and Crohn's disease (CD), but the influence of the disease activity is not established.

Aims & Methods: The primary aim of this study was to evaluate the prevalence and probability of chronic fatigue (CF) in patients with CD and UC one year after diagnosis, stratified by different measures for disease remission (symptomatic, biochemical, endoscopic, and histological). Additionally, we aimed to evaluate factors associated with chronic fatigue among patients in endoscopic/radiological remission.

Patients ≥18 years with CD and UC were recruited from the Inflammatory Bowel Disease South-Eastern Norway (IBSEN III) study, a population-based, observational inception cohort. Fatigue was assessed using Fatigue Questionnaire (FQ), and CF was defined as a dichotomized score of ≥4 with a duration of ≥6 months. Only patients with complete FQ data, without history of colectomy, were eligible for inclusion. Binary logistic regression analyses were used to examine the probability of CF for various indicators of remission, including i) symptomatic remission (Harvey–Bradshaw index score <5 for CD and Simple Clinical Colitis Activity Index score <3 for UC), ii) biochemical remission (Fecal calprotectin ≤ 250 µg/g), iii) endoscopic/radiological remission (Mayo endoscopic score 0 in UC, and endoscopically verified absence of inflammation combined with normal intestinal MRI/CT in CD), and iv) histological remission (normal histology in all mucosal biopsies). Additionally, in patients with endoscopic/radiological remission, associations between CF and selected clinical factors were evaluated using binary logistic regression analysis.

Results: In total, 711/1416 patients with CD or UC diagnosis were included (CD: 33.8%, UC: 66.2%). Among them, 619 (87.1%) underwent colonoscopy, intestinal MRI/CT and/or fecal calprotectin testing. Prevalence of CF stratified by different indicators of remission is presented in table 1. In univariate binary logistic regression analysis regarding measures for disease remission, the probability of CF was only significantly reduced in patients with UC in symptomatic remission (OR: 0.48 (0.30, 0.77), p=0.003). Among patients with CD and UC in endoscopic/radiological remission (n=181), CF was independently associated with current biological treatment (OR: 2.92 (1.002, 8.49), p=0.05) and sleep disturbance (OR: 6.23 (2.03, 19.14), p=0.001) in multivariate logistic regression.

	CD (% (n/n total))	UC (% (n/n total))
Symptomatic remission	33.6 (51/152)	30.0 (85/283)
Biochemical remission	33.3 (25/75)	34.6 (62/179)
Endoscopic/radiological remission	40.4 (21/52)	33.3 (33/99)
Histological remission	36.8 (14/38)	35.0 (28/80)

Abbreviations and table legends: CD: Crohn's disease. UC: Ulcerative colitis. Symptomatic remission: Simple Clinical Colitis Activity Index (SCCAI) score <3 for UC / Harvey–Bradshaw index (HBI) score <5 for CD. Biochemical remission: Fecal calprotectin ≤ 250 µg/g. Endoscopic/radiological remission: Mayo endoscopic score 0 in UC and absence of intestinal disease on endoscopy and/or intestinal MRI/CT in CD. Histological remission: normal histology in all mucosal biopsies.

Table 1: Prevalence of chronic fatigue among CD and UC patients with different measures for remission one year after diagnosis.

Conclusion: The probability of CF one year after diagnosis was not associated with either low fecal calprotectin level, normal endoscopic/radiological examinations, or normal histology. Among patients with UC, but not CD, symptomatic remission was associated with a reduced probability of CF. For patients in endoscopic/radiological remission, CF was independently associated with sleep disturbances and current biological treatment.

Disclosure: Nothing to disclose.

PP0418

ADHERENCE TO MEDITERRANEAN DIET AND DIET QUALITY IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: A SINGLE CENTER, OBSERVATIONAL, CASE-CONTROL STUDY

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Introduction: The nutritional status of patients affected by inflammatory bowel disease (IBD) is often impaired. Although a link between inflammation-related pathophysiologic mechanisms and malnutrition in IBD has been proposed, the role of dietary regimen and adherence to Mediterranean diet, known to have anti-inflammatory properties, is under investigated.

The aim of this study was to assess quality of diet and adherence to Mediterranean diet in a cohort of Sardinian IBD patients.

Aims & Methods: We conducted a case-control study in which IBD patients, both Crohn's disease (CD) and ulcerative colitis (UC), were compared with healthy controls matched for age, sex and BMI at a 1:2 ratio. Diet Quality Index (DQI) and Medi-Lite questionnaires were collected from IBD patients and controls to assess individual diet quality and adherence to Mediterranean diet, respectively. Subgroup analysis by disease phenotypic characteristics and use of biologic or small molecules as proxy of

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disease severity was also performed. Differences between groups were analyzed by descriptive statistics and non-parametric tests were used to assess statistical significance of the observed differences.

Results: Fifty CD and an equal number of UC were matched each with 100 healthy control subjects. Both DQI and Medi-Lite scored significantly lower in IBD patients as compared to controls (DQI: CD 34.5 [IQR 33-37] vs CTRL 40 [IQR 38.5-43] p<0.0001; UC 34.5 [IQR 33-37] vs CTRL 42 [IQR 40-44] p<0.0001; Medi-Lite CD 7.5 [IQR 7-9] vs CTRL 9 [IQR 7-10] p=0.0379; UC 8 [IQR7-10] vs CTRL 9 [IQR 8-10.5] p=0.0046). In CD, DQI was consistently lower than controls independently of disease localization and behavior.

However, Medi-Lite score was significantly lower in patients with stricturing phenotype and ileo-colonic localization. In UC patients, DQI was consistently lower as compared to controls independently of disease extent. Among these patients, Medi-Lite score was significantly lower only in those with extensive colitis.

Finally, DQI was lower in both CD and UC independently of the use of concomitant biologic or small molecules, while Medi-Lite resulted significantly lower in UC but not CD patients undergoing major therapies.

Conclusion: IBD was associated with an overall low quality of diet independently of disease type and phenotype.

However, Medi-Lite score detected a lower adherence to Mediterranean diet in patients affected by ileo-colonic stenosing disease and in UC patients with extensive colitis suggesting that the potential benefit of Mediterranean diet may be deemed by low acceptance in specific IBD patient subgroups.

Disclosure: Nothing to disclose.

PP0419

CURRENT STATUS OF LATENT TUBERCULOSIS INFECTION IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE IN JAPAN

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Introduction: When starting immunosuppressive therapy in patients with inflammatory bowel disease (IBD), screening test for latent tuberculosis infection (LTBI) and subsequent preventive treatment for positive patients are recommended to reduce the risk of tuberculosis reactivation especially in moderate-burden countries like Japan. However, the prevalence of LTBI and TB in patients with IBD and its influence in the selection of immunosuppressive therapy remain uncertain.

Aims & Methods: Our multicenter study group (IBD-Quality team; IBD Q-shu all institutional study team) conducted a retrospective study to clarify the prevalence of LTBI and the difference in therapeutic management according to the LTBI status in Japanese patients with IBD. Electronic clinical records of each institution during the period between July 2010 and June 2020 were reviewed to assess patients' clinical profile and subsequent medication after screening test of LTBI. LTBI was defined as positive when interferon γ release assays (IGRAs) or tuberculin test was positive with normal chest X-ray.

Results: This retrospective study included 10 institutions of IBD-Quality team, where 4257 IBD patients (Crohn's disease: CD 1646, Ulcerative colitis: UC 2611) were treated during the study period. LTBI was diagnosed in 69 patients (1.62%), and their mean ages of the diagnosis of IBD and LTBI were 40.9 ± 2.2 years and 51.8 ± 2.1 years, respectively. Of the 69 patients, 47 patients (68%) were having clinically active IBD, 3 patients (4.4%) had been treated for TB before the diagnosis of IBD, and 5 patients (7.3%) had a family history of TB.

After the diagnosis of LTBI, 57 patients (82.6%) were treated for LTBI (LTBI treatment group), whereas the remaining 12 patients were followed up without treatment of LTBI (non-LTBI treatment group). In LTBI treatment group, patients were treated with anti-TNFα antibody in 40 cases (70.2%), JAK inhibitors in 3 cases (5.2%), ustekinumab in 12 cases, and vedolizumab in 5 cases.

During the observation period, two patients (1 CD and 1 UC) developed TB 9 and 18 months after completing LTBI treatment, and they were treated with anti-TNFα antibody and JAK inhibitor, respectively. On the other hand, patients in non-LTBI treatment group were treated with ustekinumab in 2 cases, vedolizumab in 1 case, and other drugs in 9 cases, however, neither anti-TNFα antibody nor JAK inhibitors were used. No patient developed TB in the non-LTBI treatment group during the observation period.

Conclusion: The prevalence of LTBI was 1.62% (95% CI; 1.28-2.05) and the rate of developing TB during IBD treatment was 0.05%(95% CI; 0.013-1.17). No cases of TB were observed in the LTBI-negative group during the study period. Although rare, TB can develop in patients with IBD treated with immunosuppressive treatment even after LTBI treatment. Careful follow-up with imaging modalities for TB seems to be mandatory, and further research is required to determine the possible influence of immunosuppressive therapies for the development of TB in patients with IBD.

Disclosure: Nothing to disclose.

PP0420

IBD IN THE ELDERLY POPULATION: LONG-TERM DISEASE COURSE, SURGERY RATES AND BIOLOGICAL USE IN CROHN'S DISEASE AND ULCERATIVE COLITIS – A POPULATION-BASED STUDY FROM HUNGARY BETWEEN 1977–2020

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Introduction: There is a scarcity of population-based studies investigating therapeutic strategies and the long-term disease outcomes in inflammatory bowel diseases (IBD) amongst the elderly. This is a population population-based cohort from Veszprem Province, Hungary, with a follow-up of the incidence and disease course for over 40 years.

Aims & Methods: Our aim was to analyze long-term disease course, surgery rates, medication exposures (especially biological therapy) in a prospective population-based cohort of incident, elderly onset Crohn's disease and ulcerative colitis patients.

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Patient inclusion was between January 1,1977 and December 31,2018. Patient follow-up ended December 31, 2020. Elderly-onset IBD was defined if diagnosis was established at more than 60 years of age. Disease outcomes and surgery rates were compared based on time of diagnosis: 'pre-biological era', 1977-2008; vs. 'biological era', 2009-2018. Both in-hospital and outpatient records were collected and comprehensively reviewed at diagnosis and during clinical follow-up.

Results: Data of 840 CD and 1,295 UC patients were analyzed, 58 and 183 of which were elderly-onset IBD patients, respectively. Proportion of patients with colonic (L2) disease location and stenosing behavior (B2) at diagnosis were higher in the elderly population compared to adult-onset CD (p=0.014 and p=0.017, respectively). In UC, the prevalence of extensive colitis was less amongst the elderly (p=0.001).

In time dependent analysis, the probability of biological therapy was 0.0±0% and 3.2±3.2% at 1 and 5 years from diagnosis in elderly CD patients, 7.8±3.3% and 16.6±4.8% among elderly UC patients, respectively. Biological therapy was significantly lower in elderly patients compared to adult-onset disease in CD [pLogRank=0.003], but similar exposure was seen amongst UC patients [pLogRank=0.770].

The cumulative probability of resective surgery in the elderly CD population was 31.1±6.1%, 32.9±6.2%, and 35.3±6.4% after 1, 5 and 10 years. There were no statistically significant differences in resective surgery rates between the elderly and adult-onset cohorts in CD [pLog Rank=0.838]. The cumulative probability of colectomy in elderly UC patients was 2.2±1.1%, 3.4±1.4%, and 3.4±1.4% after 1, 5 and 10 years, respectively. Colectomy rates in the elderly and adult-onset cohorts were similar [pLog Rank=0.435].

The probability of disease behavior progression in CD patients with luminal (B1) behavior into stenosing or penetrating phenotype (B2/B3) was lower among elderly patients, compared to adult-onset CD. [pLogRank=0.015]. Similarly, disease extent progression (E1/2>>E2/3) in elderly UC patients was also lower [pLogRank=0.022].

Logistic regression model showed that male gender (OR 0.22; 95%CI 0.07-0.72; p=0.013) stenosing or penetrating disease (B2/B3) behavior (OR 8.69; 95%CI 2.17-34.85; p=0.001) and ileal or ileo-colonic (L1/L3) location (OR 9.71; 95%CI 2.41-39.13; p=0.001) were predictors of resective surgery in elderly CD patients. Whereas, extensive colitis (E3) (OR 7.17; 95%CI 1.63-31.52; p=0.011), continuous disease activity (OR 11.07; 95%CI 2.28-53.77; p=0.010), and anaemia (OR 10.13; 95%CI 1.97-52.13; p=0.004) were predictors of colectomy in elderly UC.

Conclusion: Elderly patients represent an increasing proportion of the IBD population. Biological therapy use was lower in elderly CD, however no different in elderly UC compared to adult-onset populations. There were no significant differences in resective surgery and colectomy rates between the elderly and adult-onset cohorts, whereas disease phenotype progression was slower in the elderly.

References: .

Disclosure: PLL has been a speaker and/or advisory board member: AbbVie, Amgen, Bristol Myers Squibb, Celltrion, Fresenius Kabi, Gilead, Janssen, Eli Lilly, Merck, Organon, Pendopharm, Pfizer, Roche, Sandoz and Takeda and has received unrestricted research grant: AbbVie, Gilead and Pfizer.

PP0421

IBD HEALTH EQUITY SUMMIT: A CALL TO ACTION

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Introduction: Pfizer's Institute for Translational Equitable Medicine and our Chief Medical Officer, Aida Habtezion, MD, hosted an IBD Health Equity Summit at DDW 2023 in Chicago, Illinois on May 5, 2023 with a group of 22 key opinion leaders in the gastroenterology field. The purpose of the Summit was to uncover health equity research gaps and quality of care gaps that will enable Pfizer and our partners to determine future directions in IBD research and contribute to achieving healthcare equity by providing opportunities to improve patient outcomes, as well as address practice-based disparities.

Aims & Methods: A full day Summit was designed and developed, bringing together thought leaders and experts (representing research, HCPs and patients) to discuss health disparity gaps and possible solutions. The day was split into two parts: a morning session focused on research, with IBD researchers presenting different facets of the research construct with follow-on discussion with the audience. The afternoon was dedicated to clinical care and also included IBD experts addressing Health Equity concerns and opportunities. A follow-on summary report was developed, distributed and serves as the basis for current and future strategies aimed at closing disparity gaps among IBD patients.

Results: Several overarching themes emerged, highlighting IBD as a global affliction, and challenges that transcend geographical borders. Addressing these challenges will require solutions tailored to local and regional contexts that consider factors such as race, ethnicity, socio-economic status, culture, diets, and sexual orientations. It was also recognized that many of the challenges posed by IBD are not unique to this disease, and that valuable insights can be gained by examining other complex disease states to improve the quality of care and address health disparities in patients with IBD.

Conclusion: IBD Research

Key gaps include low clinical trial participation of patients from diverse communities due to lack of healthcare literacy, low historical representation, bias in recruitment, lack of diversity in providers.

Strategies to address these gaps include improving health literacy, workforce development and educating providers, adequate compensation for time, flexibility for participating in trials, and incorporation of digital health tools for follow-ups.

Clinical Care

There are groups of patients in several regions that lack any care, and addressing healthcare disparities in IBD care for these patients cannot be done online but will need to be done on a local and regional level on the ground.

When addressing health inequities, the heterogeneities within underserved populations like Black and Hispanic patients need to be considered.

Gaps in data on the effectiveness of therapies in specific patient populations (ie, response to anti-TNFs in black patients and patients with comorbidities).

Sponsorship is lacking to establish efforts in underserved communities on a global level; There is an opportunity to provide connection for minority physicians to resources.

By identifying gaps and potential strategies to close them, the participants have provided Pfizer and our partners a roadmap to address health disparities among IBD patients worldwide. These data and information have served as the basis for a go-forward plan to support and invest in

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change initiatives to close gaps. Our desire is to enlist additional partners and collaborators through a Call to Action to identify solutions at the local level that can accelerate improved IBD care and outcomes for some of the most vulnerable populations in the world.

Disclosure: Employee of Pfizer Inc.

PP0422

PATIENT AND FAMILY INSIGHTS ON PREDICTION AND PREVENTING INFLAMMATORY BOWEL DISEASE – A WORLDWIDE SURVEY

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Introduction: Significant advances have been made in identifying biomarkers to predict IBD onset. With the use of predictive testing to identify individuals at risk of developing the disease, it would be possible to intervene and try to mitigate that risk. However, the feasibility of such strategy would depend on whether individuals at risk of developing IBD would be willing to undergo predictive testing and preventive interventions.

Aims & Methods: An anonymized electronic survey was created in eight different languages and disseminated by clinicians and patient's associations; patients with IBD, patients with an at-risk first-degree relative (offspring), and their first-degree relatives were invited to provide their feedback.

The survey had four main sections – participants' characterization, knowledge of disease, predictive tests and preventive interventions. Univariate and multivariate logistic regression were used to assess factors associated to acceptance of predictive testing and preventive interventions.

Results: Overall, 2034 participants (73% women, mean age of 43±13 years old) from 66 different countries (80% from Europe), participated. Most participants (52%) were parents with IBD with healthy at-risk children. 51% of individuals with IBD had Crohn's disease, 56% were undergoing treatment with biologics, 28% had a history of surgery, and 61% reported a significant impact on their quality of life caused by disease. 73% recognized IBD as a disease with no defined genetic transmission, but with an increased risk among FDR of affected patients.

Most participants (82%) would be willing to undergo, predictive testing, or to have their children undergo them, mainly with minimally-invasive blood (91%), stool (90%) or saliva tests (78%), or intestinal ultrasound (68%); 46% would accept a colonoscopy.

The main benefits recognized with predictive testing were the possibility of starting preventive interventions to minimize the risk of developing IBD (69%) and to obtain an earlier diagnosis (58%). The main disadvantage was anxiety generated from a positive result (57%).

Overall, 97% of participants were willing to accept preventive interventions. The most accepted were dietary modification (82%), physical exercise (79%) and probiotics (71%). Overall, 40% and 34% would accept oral or intravenous/subcutaneous immunosuppressive treatment, respectively; provided that the risk of minor adverse events was negligible to acceptable (23%), while 44% considered that the risk of major adverse events should be <1% to be acceptable. In multivariate analysis, factors associated with less acceptance of predictive testing were older age ($p<0.001$) and treatment with thiopurine ($p<0.001$).

Factors only associated with less acceptance of preventive interventions were having ulcerative colitis ($p=0.004$) or undetermined colitis ($p<0.001$) and being treated with mesalazine ($p<0.001$). Having a disease with low impact in their/their family members' quality of life were associated with less acceptance of both predictive testing and preventive interventions ($p<0.001$).

Conclusion: The vast majority of parents with children at-risk for IBD, or first-degree relatives at risk for IBD development would accept minimally-invasive predictive testing for IBD. The vast majority expressed acceptance of non-pharmacological interventions, however, more than one-third would be willing to accept immunosuppressive medications, particularly if associated with a low risk of minor and major adverse events. This data provides useful and important information for the development of prediction and prevention strategies.

Disclosure: Nothing to disclose.

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PP0423

FINANCIAL IMPLICATIONS OF FIXED DOSING OF SUBCUTANEOUS INFLIXIMAB IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: In Australian patients receiving intravenous (IV) infliximab (IFX) treatment, 44% of patients do not respond to standard regimen and require dose escalation within a year. The REMSWITCH (6-month evaluation) and REMSWITCH-LT (18-months evaluation) investigated the effectiveness of switching from IV to subcutaneous (SC) IFX in patients with inflammatory bowel disease (IBD) who received standard and dose-escalated IV regimen.

The two studies concluded that a fixed dose of 120 mg Q2W is feasible, safe and well-accepted leading to a low risk of long-term relapse in patients with escalated IV doses except for those on 10mg/kg/4weeks regimen. In this study, we developed a model to estimate the financial implications of switching.

Aims & Methods: Clinical efficacy parameters were derived from the REMSWITCH and REMSWITCH-LT studies and applied to the Australian context. Number of Australian patients who are on IFX, the proportion of patients on dose escalated regimen, and cost parameters were identified from the published literature or publicly available resources. A financial model was developed based on the assumption that a cohort of patients will receive either IV or SC IFX, and the analysis concluded with the cost difference.

Results: The model assumes that a total of 5.6K patients are receiving IV IFX, with 44.2% of them receiving dose escalation and/or shortened interval of IV IFX. The model produced 6 months and 12 months results, suggesting savings between 582 and 882 AUD per patient, respectively, and equates to a yearly societal savings of 5 million AUD.

Conclusion: IFX SC deserves premium over IFX IV, given that the novel formulation offers tangible and intangible benefits that originator Remicade and biosimilars cannot provide. There is a concern regarding the affordability of SC IFX because it is generally priced higher than IV IFX, which has many biosimilars on the market.

However, this study asserts that fixed dosing of 120mg of IFX is cost-saving for society, considering highly prevalent practice of dose escalation and/or shortened intervals of IFX regimen.

Disclosure: Nothing to disclose.

PP0424

INDOOR RADON LEVELS AND ITS ASSOCIATION WITH THE DIAGNOSIS AND EVOLUTION OF INFLAMMATORY BOWEL DISEASES

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Introduction: Environmental factors have only been partially implicated in the aetiopathogenesis of inflammatory bowel disease (IBD). The role of indoor radon, a radioactive gas, has not been formally studied and may play a role in the pathogenesis of the disease.

Aims & Methods: The aim of this study is to analyse the role of individual residential radon exposure on the incidence of IBD in a radon-prone area, also characterised as a high incidence IBD region, and whether this exposure is associated with the development of the disease.

Methods: A case-control study was developed in the health area of Santiago de Compostela (Galicia, Spain) between June 2020 and September 2023, including all consecutive incident cases with IBD. Controls were selected by frequency matching based on sex and age and without IBD. All participants had to have lived in the same home for at least 5 years. All cases were followed up for one year to assess disease progression. Endoscopic index, imaging techniques and analytical parameters were collected. In patients with IBD, hospitalisations related to the disease and the number and severity of flares (measured by Partial Mayo Score and Harvey-Bradshaw) were recorded. Radon levels measured in participants' homes over three months were compared between cases and controls, and between different types of IBD. Each individual was provided with a home radon meter (an RSKS trace detector), which was delivered and read by the Galician Radon Laboratory, a certified radon laboratory. Radon concentrations were categorised into three groups: 0-99 Bq/m3, 100-299 Bq/m3 and >299 Bq/m3. The results are expressed as odds ratios with their 95% CIs obtained by logistic regression.

Results:

RADON LEVELS (Bq/m^3)	IBD PATIENTS (%)	CONTROLS (%)	OR (95%CI)
0-99	34.3	19.8	1
100-299	45.2	55.8	0.5 (0.3-0.8)
>299	20.5	24.4	0.5 (0.3-0.9)

**OR = Odds Ratio; CI = Confidence Interval.*

178 incident cases and 178 controls were included, with a median age of 51 years (IQR 40-59.5). 51.7% were female. 136 IBD patients completed one year of follow-up (78 ulcerative colitis, 52 Crohn's disease and 6 unclassified colitis). The median radon level in the homes of the IBD patients was 144.5 (IQR 83 - 260) and the median radon level in the homes of the controls was 189.5 (IQR 112 - 292.5). After adjustment for age and sex, high radon levels showed a negative association with the development of IBD (taking levels of 0-99 Bq/m3 as reference, OR 100-299 Bq/m3 is 0.5 (95%CI 0.3-0.8) and OR >299 Bq/m3 is 0.5 (95%CI 0.3-0.9). The results are shown in Table 1. Taking into account the different types of IBD, radon levels show

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the same negative association, but without statistical significance. There were no statistically significant differences in radon levels, taking levels of 0-99 Bq/m3 as reference, between hospitalised and non-hospitalised patients (OR 0.8 95%CI 0.3-2.1 for 100-299 Bq/m3, OR 1. 7 95%CI 0.6-5.0 for >299 Bq/m3), nor in patients with a flare compared with patients without a flare during the follow-up year (OR 0.4 95%CI 0.0-4.5 for 100-299 Bq/m3, OR 5.4 95%CI 0.8-34.3 for >299 Bq/m3). Radon levels were also not associated with the number of flares (OR 0.6 95%CI 0.3-1.7 for 100-299 Bq/m3, OR 1.5 95%CI 0.5-4.5 for >299 Bq/m3). Stratification by type of IBD did not alter the results.

Conclusion: High radon levels may be a protective factor against the development of IBD, and indoor radon levels do not appear to be associated with the development of IBD in terms of hospitalisations or flares.

Disclosure: Nothing to disclose.

PP0425

IMPACT OF THE FASTING MONTH OF 'RAMADAN' ON DISEASE ACTIVITY IN EMIRATI PATIENTS WITH INFLAMMATORY BOWEL DISEASE AT A TERTIARY REFERRAL CENTER IN ABU DHABI, UNITED ARAB EMIRATES: RESULTS OF RETROSPECTIVE COHORT STUDY

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Introduction: Current Literature examining how Ramadan affects gastrointestinal disorders indicates that individuals with inflammatory bowel disease (IBD) generally face a low likelihood of worsening symptoms, although older men with ulcerative colitis (UC) are more susceptible to exacerbations during fasting. Moreover, individuals with duodenal ulcers have a heightened risk of experiencing bleeding following fasting during Ramadan (1).

Aims & Methods: Aim: To study the impact of 'Ramadan' on Inflammatory bowel disease (IBD) activity in an Emirati cohort.

Methods: We designed a retrospective cohort study of adults (> 16 years) with IBD seen at our tertiary care IBD referral center in Abu Dhabi, the United Arab Emirates (UAE) for IBD related emergencies, IBD related hospitalizations and in-person clinic visits during Ramadan, the Islamic lunar calendar month of fasting vs. during the preceding month of Shaban in the year 2023. Exposure of interest was the month (30 days) of 'Ramadan' and the unexposed cohort was the month (30 days) of 'Shaban'. Chi-square testing was used to compare emergency department (ED) visits and hospitalizations.

Logistic regression was used to assess the impact of Ramadan vs. Shaban on the risk of active clinical disease during in-person clinic visits. Statistical analysis was performed using R (version 4.0.2) and a p-value of <0.05 was considered statistically significant. IRB approval was in place for this study.

Results: A total of 294 encounters in the cohort (130 in Ramadan vs. 164 in Shaban) During Ramadan, 12 of 130 (9%) IBD encounters were ED visits for an IBD related emergency vs. 4 of 164 (2%) patients during Shaban (X^2 = 5.25, 95%CI 0.06-0.13, P=0.02). During Ramadan, 16 of 130 (12%) IBD encounters led to hospitalization vs. 16 of 164 (10%) during Shaban (X^2=0.26, 95%CI -0.054-0.1, P=0.61). In Ramadan, during 33/77 (43%) in-person clinic visits, patients had active disease vs. 24/106 (23%) in Shaban (X^2=7.59, 95%CI 0.055-0.35, P=0.006). Logistic regression model adjusted for age and gender demonstrated that during Ramadan, the risk (likelihood) of active clinical disease was approximately 3 times higher compared to during Shaban (OR 2.70, 95% CI 1.41-5.25, P= 0.003).

Conclusion: This is the first study examining the effect of the fasting month of 'Ramadan' on disease activity among Emirati patients with IBD. We found that during Ramadan, there was higher likelihood of ED visits, hospitalizations and active clinical disease when compared to during Shaban. This risk was profound and statistically significant regarding ED visits and likelihood of active clinical disease. There may be several factors underlying these significant findings.

For example, in the UAE during Ramadan, there are limited meal and drink options during the day, there occurs a sudden drastic change in diet patterns and there are limited working hours that reduce access to the IBD Clinic. IBD practices in the UAE could try to mitigate the risk of flares during Ramadan by increased dietary and lifestyle counseling of patients in preparation for the month, encourage patients to hydrate adequately and perhaps by increasing access when needed to the IBD clinic during Ramadan.

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Disclosure: Nothing to disclose.

PP0426

INFLAMMATORY BOWEL DISEASE AND CONCOMITANT ENDOMETRIOSIS: FREQUENCY AND CHARACTERISTICS IN A MULTIDISCIPLINARY APPROACH

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Introduction: Inflammatory bowel disease (IBD) and endometriosis are both immuno-mediated chronic inflammatory diseases, sharing some common clinical manifestations (i.e. abdominal pain). A possible association between IBD and endometriosis has been suggested but is currently still undefined.

Aims & Methods: In a multidisciplinary prospective single-center study, we aimed to assess the frequency of undiagnosed endometriosis in IBD patients by searching for symptoms compatible with this condition (dysmenorrhea, dyschezia, dyspareunia, dysuria, heavy menstrual bleeding, HMB).

Secondary end point was to compare symptoms compatible with endometriosis in IBD patients with vs without a subsequent diagnosis of endometriosis. In a prospective study, all female IBD patients in childbearing age with symptoms compatible with endometriosis, as revealed by filling a questionnaire during a scheduled visit at our IBD center, were enrolled. After informed consent, gynecological evaluation and transvaginal ultrasonography (TVUS) was performed by our referral dedicated gynecologist, in order to search for pelvic endometriosis.

Inclusion criteria:

1) female gender;
2) age ≥18 and ≤50 years;
3) well-defined diagnosis of IBD;
4) ≥1 symptom compatible with endometriosis;
5) written informed consent.

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Exclusion criteria:
1) pregnancy;
2) menopause.
Characteristics of endometriosis and adenomyosis were reported according to standard classifications. Data were expressed as median [range].
Results: The study population included 51 IBD patients (age 37 [23-53] years, body mass index 21.6 [17.3-41.6]). In IBD, there were 24 (47.05%) Crohn's Disease (CD) and 27 (52.9%) Ulcerative Colitis (UC) patients.
At enrollment, questionnaires revealed dysmenorrhea in 43 (84.3%), dyschezia in 18 (35.3%), dyspareunia in 31 (60.8%), dysuria in 4 (7.8%), HMB in 33 (64.7%) patients. At TVUS, endometriosis in any form was observed in 35 (68.6%) IBD patients, adenomyosis in 26 (51%).
Both diseases were concomitantly detected in 22 patients (43.1%). In patients showing endometriosis, deep infiltrating endometriosis was diagnosed in 33 (94.3%) IBD patients, and 7 out of these 33 (21.2%) patients showed rectal endometriosis.
Among the 35 patients with endometriosis, IBD included CD in 16 (45.7%) and UC in 19 (54.3%).
The frequency of the tested 5 symptoms compatible with endometriosis were then compared between IBD patients with vs without a subsequent diagnosis of endometriosis. Dysmenorrhea (32 [91.4%] vs 11 [68.75.4%]; p=0.04) and HMB (26 [74.3.%] vs 7 [43.75%]; p=0.04) were more frequent in IBD patients with endometriosis.
Differently, the other considered symptoms were comparable between IBD patients with vs without endometriosis: dyspareunia (24 [68.6%] vs 7 [43.75.%]; p=0.09), dyschezia (11 [31.4%] vs 7 [43.7%]; p=0.39), dysuria (3 [8.6%] vs 1 [6.25%]; p=0.77).
No significant differences were observed between UC and CD patients in terms of characteristics of endometriosis, when excluding a higher frequency of endometriosis involving the uterosacral ligament in UC vs CD (16 [84,2%] vs 6 [36.5%]; p=0.005).
Conclusion: Findings from a prospective multidisciplinary study suggest that endometriosis may be highly underestimated in IBD and that specific symptoms, particularly dysmenorrhea and HMB, should be carefully searched in young patients. In IBD, endometriosis was mostly represented by deep infiltrating endometriosis and a higher frequency of rectal endometriosis, potentially leading to rectal bleeding was observed.
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Disclosure: Nothing to disclose.

PP0427

HIGHER FREQUENCY OF PERIANAL DISEASE AND FAILURE OF BIOLOGICS IN OVERWEIGHT AND OBESE CROHN'S DISEASE PATIENTS: A PROSPECTIVE STUDY

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Introduction: The relationship between obesity and inflammatory bowel disease (IBD) is currently undefined. The clinical outcome of obese and non-obese IBD patients is apparently comparable. However, evidences at this regard are conflictual.
Aims & Methods: Primary aim was to compare, in a prospective study, IBD characteristics between overweight or obese and non-obese consecutive IBD patients. Secondary aim was to compare the frequency of immunomodulator therapies, IBD-related surgery and hospitalizations between overweight or obese and non-obese patients.
In a prospective, single-center study, consecutive IBD patients undergoing scheduled outpatients visit between September 2022 and January 2023 were enrolled. Demoghaphic and clinical data including body mass index (BMI) were recorded. Inclusion criteria: 1)defined diagnosis of IBD; 2)age ≥18 years; 3)regular follow up (≥2 visits/year). Exclusion criteria: 1)BMI <18.5 mk/m^2, 2)incomplete data. Data were expressed as median [range]. Normal distribution of continuous variables tested by the Kolmogonov-Smirnov test. Comparisons assessed by Student-t test, Mann-Whitney u-test and Chi-squared test.
Results: Out of 300 IBD patients assessed, 284 were enrolled (141 [49.6%] Crohn's Disease (CD); 143 [50.4%] UC). The median age of IBD group was 51 [20-87] and male gender was observed in 157 (55.3%) patients. The median IBD duration was 13.5 [1-57] years. Overall, 170 (59.8%) IBD patients were normal-weight and 114 (40.2%) were overweight or obese.
Among the 141 CD patients enrolled, 89 (63.1%) normal-weight patients and 52 (36.9%) overweight or obese patients were observed. Lesions distribution and behavior was comparable between normal-weight and overweight or obese patients (L1: 62 [70%] vs 35 [67.3%], p=0.77; L2: 8 [9%] vs 4 [7.7%], p=0.79; L3: 19 [21%] vs 13 [25%], p=0.61; L4: 7 [7.9%] vs 2 [3.8%], p=0.34; B1: 24 [27%] vs 13 [25%], p=0.79; B2: 44 [49.4%] vs 32 [61.5%], p=0.16; B3: 21 [23.6%] vs 7 [13.5%], p=0.74, respectively). Perianal disease was more frequently observed in overweight or obese vs normal-weight patients (9 [10.1%] vs 12 [23%], p=0.03). Immunosuppressive (ISS) or biologic use was comparable between groups (ISS: 33 [37%] vs 19 [36.5%], p=0.94; biologics: 23 [25.8%] vs 17 [32.7%], p=0.38). Lack of response to biologics was more frequently observed in overweight and obese vs normal-weight patients (0 [0%] vs 4 [23.4%], p=0.01). Loss of response and adverse events were comparable between groups (11 [47.8%] vs 5 [29.4%], p=0.23; 13 [56.5] vs 8 [47.1%], p=0.55).
Among the 143 UC patients enrolled, 81 (56.6%) normal-weight patients and 62 (63.4%) overweight or obese patients were observed. Proctitis and extensive UC were more frequently reported in overweight or obese vs normal-weight patients (E1: 18 [29%] vs 11 [13.6%], p=0.02; E3: 35 [56.4%] vs 31 [38.3%], p=0.03). Left-sided UC were more frequent in normal-weight then in overweight or obese patients (E2: 39 [48.1%] vs 19 [30.6%], p=0.03). ISS use was comparable between groups (14 [17.3%] vs 7 [11.3%], p=0.31) while biologics were more frequent in normal-weight then in overweight or obese patients (15 [18.5%] vs 4 [6.4%], p=0.03). Lack of response, loss of response and adverse events were comparable between groups (6 [40%] vs 1 [25%], p=0.58; 5 [33.3%] vs 1 [25%], p=0.75; 6 [40%] vs 0 [0%], p=0.12).
Conclusion: In the tested population, overweight or obese CD patients showed a higher frequency of perianal disease and refractoriness to biologics than normal weight patients. These preliminary findings in CD suggest that BMI and phenotype may influence responsiveness to biologics.
Disclosure: Nothing to disclose.

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PP0428

INSIGHTS INTO MICROSCOPIC COLITIS: A COMPREHENSIVE ANALYSIS FROM LONDON'S PREMIER TEACHING HOSPITAL IN THE UNITED KINGDOM

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Introduction: Microscopic colitis is a cause of chronic watery, non-bloody diarrhoea diagnosed mainly on histology after taking biopsy from macroscopically normal colonoscopy. Thereby mainly two subtypes can be identified: Lymphocytic and collagenous colitis. So far, no biomarkers for the diagnosis of microscopic colitis have been identified. Here, we looked at, risk factors and treatment outcome of microscopic colitis in a large series of cases.

Aims & Methods: In this study, we analysed all cases of microscopic colitis diagnosed from 2018-2022 in the Northeast London from 5 different teaching hospitals in the UK. 225 cases (121, Lymphocytic, 124 cases are collagenous colitis) were identified and their electronic health record from hospital and from the primary General Practitioner (GP) were manually reviewed.

Results: In this cohort, 32% male, 68% female with ratio of (1:3, P < 0.036) with mean age of 64years. Thirty three percent were active or ex-smoker, Disease symptoms were significantly different compared to non-smoker (p< 0,007). PPI uses were associated with in 81% cases. While NSAIDS and statin were associated in 57% and 49% cases, respectively. We also found that SSRI use was associated in about 31% cases.

Painless watery diarrhoea was the main cause for referral in 90% of cases, while complaint of abdominal pain noted in only 39% cases. Thirty-five (25%) cases noted to have history of weight loss. Forty-eight (48%) patients received treatment with Budesonide, 46% treated with loperamide, 5% with Immunomodulator or Biologics and 1% Bismuth Subsalicylates respectively. Other autoimmune disease was associated in 6% cases and 2% cases were associated with Coeliac disease. in Nineteen (19%) of cases symptoms improved immediately after stopping above associated drugs without any specific therapy.

Twenty-five (25%) percent relapses 6 months after stopping of therapy. Patients present with abdominal pain, and arthralgia were significantly (p<0.001) higher risk of relapsing compared to patients with other symptoms.

Conclusion: In this large series of microscopic colitis, we found that in addition to associated drug use is not only risk, but patient also presents with abdominal pain and arthralgia have higher risk of relapse, after stopping of therapy of microscopic colitis. Further studies are needed to identify biomarker of this frustrating disease which has significant impact on patients of quality of life.

References: nil

Disclosure: Nothing to disclose.

PP0429

SLEEP, ILLNESS PERCEPTION AND QUALITY OF LIFE IN INFLAMMATORY BOWEL DISEASE – SHOULD WE BE CONCERNED?

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Introduction: The management of inflammatory bowel disease (IBD) has traditionally focused on controlling symptoms and reducing or even clearing inflammatory burden. However, the Selecting Therapeutic Targets in Inflammatory Bowel Disease (STRIDE)-II statements have expanded the targets to include health-related quality of life (HRQoL) as a long-term outcome. Despite this, the predictive factors for worse HRQoL have yet to be fully completely determined. Alongside psychosocial status, it has been hypothesized that illness perception may also influence HRQoL. Additionally, a bidirectional relationship between sleep quality and IBD has been observed.

Aims & Methods: We thus aim to investigate the potential association between self-reported sleep quality and HRQoL in patients with IBD and to identify other predictors of HRQoL to so that these factors can be integrated into the evaluation of IBD patients. We conducted a prospective single-center cohort study at a tertiary center. From January 2024 to March 2024, patients previously diagnosed with IBD were invited to complete a comprehensive questionnaire assessing HRQoL using the Short Inflammatory Bowel Disease Questionnaire (SIBDQ), self-reported sleep quality using the Pittsburgh Sleep Quality Index (PSQI), and illness perception using the revised version of the Illness Perception Questionnaire. Demographic, clinical (including history of psychopathology), and symptomatology (evaluated using the Harvey Bradshaw Index (HBI) and Partial Mayo Score), as well as analytical parameters (fecal calprotectin), were assessed at the time of questionnaire completion. Clinical remission was defined as HBI<=3 or Partial Mayo Score<=1. Poor HRQoL was defined as SIBDQ<50 and poor sleep quality defined as PSQI>5. Logistic regression was used to performed both univariate and multivariable analysis. Predictors with p<0.1 in the univariate analysis were included in the multivariable analysis.

Results: A total of 272 patients were included, with 53.3% being male. The median age was 48 years (interquartile range [IQR] 38-58). Ulcerative colitis was present in 184 (68%) patients, while the remaining patients had Crohn's disease. A history of psychopathology was noted in 35 (12.9%) patients. The majority of patients were in clinical remission (245 (90.1%) patients). Poor QoL was reported by 87 (32.7%) and poor sleep quality was reported in 185 (70.1%). Univariate analysis revealed a statistically significant association between female gender, Crohn's disease, absence of clinical remission, self-reported poorer sleep quality, and illness perception (higher scores in four components: timeline cyclical, illness consequences and coherence, and emotional representation of IBD) and worse HRQoL. Multivariable analysis confirmed younger age, absence of clinical remission, worse sleep quality, and illness perception (illness consequences, coherence, and emotional representation of IBD) as independent predictors of poor HRQoL.

Conclusion: HRQoL in IBD is influenced by factors beyond disease activity. Sleep quality emerges as a significant determinant of HRQoL, alongside the patient's illness perception. Therefore, integrating sleep quality assessment into patient clinical evaluation is paramount. Moreover, addressing patients' understanding of their disease, perspective on consequences, and emotional response to the condition is crucial for improving HRQoL.

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PP0430

CAN THE IBD DISK ADEQUATELY ASSESS PSYCHOLOGICAL BURDEN IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE?

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Introduction: Inflammatory bowel disease (IBD) is often diagnosed in early adulthood and can lead to a substantial decline in quality of life. It has been suggested that patients with IBD are at increased risk of depression and anxiety disorders.
The IBD Disk is a validated 10-question, self-administered visual questionnaire that includes characterisation of psychosocial disease burden. Whether scores in these domains correlate with traditional measures, such as the Hospital Anxiety and Depression Scale (HADS), is not known.
Aims & Methods: The aim of this study was to assess the prevalence of anxiety and depression in IBD patients, and the correlation between IBD-Disk and traditional measures.
We conducted a cross-sectional study over 3 months between April and June 2023. IBD patients responded to the IBD-Disk and HADS questionnaires.
HADS Scores are out of 21; with 0–7 normal, 8–10 mild, 11–15 moderate and 16–21 severe.
Results: We enrolled 57 patients (50.9% men) with a median age of 38 years [31-48].
Thirty-eight patients (66.7%) had Crohn's disease (CD). It was ileal in 14 patients (36.8%), ileocolic in 15 (39.5%), and colonic in 9 (23.7%). Ano-perineal involvement was present in 36.8% of cases. Nineteen patients (33.3%) had ulcerative colitis. Pancolic involvement was noted in 13 patients (68.4%).
Thirteen patients were depressed (22.8%) and 16 patients were anxious (28.1%).
In univariate analysis, depression was significantly associated with clinical disease activity (p=0.002), hospitalization at the time of interview (p<0.001), anemia (p=0.01), elevated CRP (p=0.015), hypoalbuminemia (p=0.016) and corticoresistance profile (p=0.01).
Anxiety was significantly associated with clinical activity (p=0.043), hypoalbuminemia (p=0.023) and high CRP (p=0.043). In multivariate analysis, only hospitalization at the time of interview and corticoresistance were independent risk factors for depression (OR=15.169, p<0.001; OR=5.577, p=0.042 respectively). The only independent risk factor for anxiety was clinical disease activity (OR=3.437, p=0.049). A significant correlation was noted between the IBD-Disk global score, the anxiety score and the depression score (respectively r=0.511 and r=0,524, p<0.001).
Conclusion: IBD patients tend to have a higher prevalence of depression and anxiety disorders than the general population. The IBD disk is an easy tool that correlates well with HADS scores. Its utilisation in clinic allows holistic assessment, particularly in patients with active disease.
Disclosure: Nothing to disclose.

PP0431

WITHDRAWN

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PP0432

CLINICOPATHOLOGIC FEATURES AND PROGNOSIS OF COLITIS-ASSOCIATED COLORECTAL CANCER IN ULCERATIVE COLITIS PATIENTS WITH PRIMARY SCLEROSING CHOLANGITIS: A NATIONWIDE COHORT STUDY IN JAPAN

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Introduction: The prevalence of primary sclerosing cholangitis (PSC) among ulcerative colitis (UC) is known to be lower in Eastern Asia, including Japan, than in Western countries. There have been few studies describing the details of ulcerative colitis-associated colorectal cancer (CAC) in UC with PSC (PSC-UC) in Japan so far, owing to its rarity.

Aims & Methods: This retrospective study aimed to evaluate the clinicopathologic features and prognosis of CAC in PSC-UC in the largest nationwide cohort of its kind in Japan. A total of 797 eligible patients diagnosed with CAC from 1983 to 2020 were categorized into those with PSC (PSC-UC group, n = 26) and those without PSC (UC-alone group, n = 771). Clinicopathologic features, 10-year overall survival (OS) and 10-year cancer-specific survival (CSS) were compared between the two groups.

Results: In PSC-UC group, duration of UC was shorter (12.8 vs 17.0 year, p=0.042), and age at CAC diagnosis was younger (47.8 vs 53.4 years old, p=0.042) compared to UC-alone patients. Right-sided colon cancer was significantly more common in PSC-UC group than in UC-alone group (34.6% vs 15.5%, p=0.032). Left-sided colon cancer (38.5%) and rectal cancer (26.9%) accounted for a similar proportion with right-sided colon cancer in PSC-UC group. There were no differences between the two groups in terms of sex, age at UC diagnosis, disease extent of UC, medical therapies for UC, diagnostic opportunity for CAC, histological types, and pathological stages of CAC. PSC-UC group showed a comparable 10-year OS (70.6% vs 83.1%, p=0.072) and 10-year CSS (84.1% vs 87.6%, p=0.50) as UC-alone group.

Conclusion: Our study highlighted the differences and the similarities of CAC between UC with PSC and without PSC patients. In patients with PSC-UC, it may be beneficial to start surveillance colonoscopy earlier because of the shorter duration of UC before the CAC diagnosis.

Disclosure: Nothing to disclose.

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PP0433

PREVALENCE AND TREND OF ANEMIA IN A LARGE COHORT OF CHILDREN WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Anemia is a common extraintestinal manifestation of inflammatory bowel disease (IBD) with an important impact on patients' quality of life. The prevalence is uncertain (6% to 74% depending on the definition and on the subgroup of patients examined). In Italy a prevalence of anemia at the diagnosis of 13.6% among the adult IBD outpatients is reported.

Aims & Methods: We aimed to determine the prevalence of anemia and its characteristics at the diagnosis in children with IBD and to investigate its trend during follow-up.

This was an observational, multicenter cohort study including data of IBD children with anemia at the diagnosis prospectively enrolled in the Italian Society of Pediatric Gastroenterology, Hepatology and Nutrition (SIGENP) IBD registry. Data were collected at the diagnosis and anemia trend was evaluated at 1-year follow-up. Anemia was diagnosed based on the WHO criteria and classified as mild, moderate, and severe accordingly. Based on serum ferritin and CRP level and/or transferrin saturation (TNSat), anemia was defined as iron deficiency anemia (IDA), anemia of chronic disease (ACD) or inflammatory IDA (InflIDA).

Results: Five hundred eighty-nine [295 CD (50%) and 294 UC/IBDU (50%)] out of 1634 children with IBD presented with anemia, resulting in a prevalence of 36%. Anemia rate was higher in CD than in UC (39% vs 33%, p = 0.009) and most patients had moderate anemia (55%). Iron deficiency anemia and InflIDA were diagnosed equally in 42% of children, while only 15.4% presented with ACD. Children with CD showed higher rates of mild anemia (38% vs 33%,p<0.0001) and a higher frequency of ACD compared to UC patients (26.5% vs. 4.5%, p < 0.0001). A severe form of anemia was more common in UC (13% vs 6%, p= 0.001) as well as the IDA (53% vs 31%,p<0.0001). Younger age and lower albumin levels significantly correlated with the severity of the anemia in CD. In UC, females were more likely to experience severe anemia (69% vs 47%, p=0.02) and an extensive disease was prevalent in children with moderate and severe anemia than in those with mild anemia (p=0.01 and p=0.03, respectively).

At 1 year, 99 children (17%) were still anemic. At the multivariable logistic regression analysis, baseline wPCDAI and SES-CD significantly correlated with the risk of the persistence of anemia at 1 year in CD. No variables correlated with the persistence of anemia in UC.

Conclusion: More than one-third of pediatric patients presents with anemia at the diagnosis of IBD, most commonly moderate. A severe anemia is most common in UC compared to CD. IDA (alone or in its form of ACD) is the most common cause of anemia in children with IBD. One out of five is still anemic after 1 year from the diagnosis.

Disclosure: Nothing to disclose.

PP0434

EVALUATION OF THE RELATIONSHIP BETWEEN PAIN, PSYCHOLOGICAL PROFILE AND DYSAUTONOMIA IN NEWLY DIAGNOSED IBD PATIENTS USING THE EXPERIENCE SAMPLING METHOD

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Introduction: Chronic abdominal pain (CAP) occurs in up to 50% of patients with Inflammatory Bowel Disease (IBD), even in remission[1]. Previous studies demonstrate that CAP is associated with fatigue, depression and reduced quality of life in IBD patients[2].

These studies, however, are mostly retrospective questionnaire based which are subject to recall bias.

The Experience Sampling Method (ESM) is a digital patient reported outcome measure (PROM) allowing real-time assessment of current symptoms and reduces recall bias compared to end of day or end of week surveys[3,4]. ESM has not been used in IBD patients to evaluate the relationship between pain, psychological profile and autonomic symptoms.

Aims & Methods: 69 patients with newly diagnosed IBD within the last 12 months were included in the analysis. Patients completed an ESM-based PROM (with a maximum of 10 buzz questionnaires per day) for 7 consecutive days. Within each of the ESM questionnaires, patients reported on current abdominal pain, anxiety, mood, and autonomic symptoms on a scale from 0 (not at all) to 10 (very much). The mean score across the 7 days was calculated for each of the symptoms. Only patients who had completed >25% of the ESM questionnaires (18/70) were included in the analysis.

Results: 63 patients completed >25% of the ESM questionnaires (age range 18-62, 34 male, 29 female) and were included in the analysis. 39 patients (62%) had no abdominal pain (mean pain score 0-1), 15 patients (24%) had mild pain (mean pain score 1-4), whilst 9 patients (14%) had moderate/severe pain (mean pain score 4-10).

The mean scores for low mood (2.35, 95% CI 1.69-3.00) and anxiety (1.57, 95% CI 0.78-2.37) were higher in the mild pain group compared to the mean scores for low mood (1.25, 95% CI 0.75-1.76) and anxiety (1.01, 95% CI 0.59-1.43) in the no pain group (p < 0.01). Likewise, the mean scores for low mood (4.59, 95% CI 2.93-6.25) and anxiety (3.59, 95% CI 1.64-5.53) were higher in the moderate/severe pain group compared to the mild pain group (p < 0.05)

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There was a higher mean score for palpitations and dizziness in both the mild pain and moderate/severe pain groups compared to the no pain group ($p < 0.01$). The mean score for dyspnea was higher in the moderate/severe pain group compared to no pain group ($p < 0.05$), however it was similar between the mild pain and no pain groups ($p = 0.39$).

Finally, the mean score for sweating was similar between the three groups ($p = 0.34$).

A strong positive correlation ($r = 0.84$) between pain intensity and low mood intensity was found ($p < 0.01$).

Similarly, there was also a strong positive correlation ($r = 0.82$) between pain intensity and anxiety intensity ($p < 0.01$).

Conclusion: Newly diagnosed IBD patients experiencing moderate to severe abdominal pain are more likely to have lower mood, higher levels of anxiety as well as more symptoms of dysautonomia.

Further evaluation of the ESM data will provide a greater insight into the course of symptoms as well as potential triggers for the abdominal pain.

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Reuma Margalit Yehuda & Qasim Aziz are joint senior authors.

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Disclosure: Nothing to disclose.

PP0435

FACTORS ASSOCIATED WITH CHRONIC PAIN IN INFLAMMATORY BOWEL DISEASE: A LARGE CROSS-SECTIONAL QUESTIONNAIRE-BASED STUDY

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Introduction: Up to half of inflammatory bowel disease (IBD) patients experience chronic pain despite disease remission. The factors affecting the development of chronic pain are poorly understood and pain-directed therapies are scarce.

Our study aimed to compare the clinical features and psychological characteristics of patients with and without chronic abdominal pain (CAP). The current analysis focuses on patients with IBD in remission and used REDCap to capture electronic data [1].

Aims & Methods: This is a cross-sectional online questionnaire-based study of 2381 IBD patients aged 16 years and above. The Rome IV questionnaire was used to diagnose irritable bowel syndrome (IBS). Remission was defined as Harvey Bradford Index (HBI) < 5 for patients with Crohn's disease (CD) or Simple Colitis Activity Index (SCAI) < 3 for patients with ulcerative colitis (UC) [2]. Participants in remission with at least weekly abdominal pain for at least three months were defined as the cohort with chronic abdominal pain, and the rest were categorised as the pain-free cohort.

The independent samples t-test was used to compare mean Body Mass Index (BMI) and age at diagnosis, and the independent samples Mann Whitney U test to compare the non-parametric scale data of the two cohorts. Spearman's correlation was used to compare the pain intensity and pain interference scores from the Brief Pain Inventory with scale data from the following scales: Big Five Personality Questionnaire (BFPQ), Highly Sensitive Person Scale (HSPS), General Anxiety Disorder-7 (GAD7), Patient Health Questionnaire 9 (PHQ9), and Cognitive and Behavioural Response Questionnaire (CBRQ). Cronbach's alpha was used to assess the reliability of these scales.

Results: 1191 participants in remission were studied. Their ages ranged from 17 to 92, with a mean age of 51, and 52.6% were male. Eighty-eight percent were of White British ethnic origin and 40% were current or previous smokers.

25.9% (308) of those in remission reported CAP, with a mean age of 47 years, and 41.6% of them were males. The mean age at diagnosis of IBD was 35 years in those with CAP in remission, versus 30 in those without CAP (t= -4.548, p <0.001, 95% CI -6.598 to -2.621). In the CAP group, the prevalence of IBS was 73.1% compared to the total prevalence of 18.9% in

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the remission cohort. The prevalence of CD and UC were 72.4% and 27.6% respectively in the CAP in remission cohort, compared to 61.3% and 38.7% in all remission participants. 34.4% of people with CAP in remission had previous surgery.

There was a significant difference between the CAP and non-CAP cohort median scores with p values < 0.001 for neuroticism from the BFPQ, Ease of Excitation (EoE) from the HSPS, GAD7, PHQ9. The CBRQ components of fear avoidance, embarrassment avoidance, symptom focusing, all-or-nothing behaviour score, and avoidance/resting behaviour score also showed differences between the CAP and non-CAP groups (p <0.001, 95 % confidence interval).

Pain intensity and pain interference scores showed positive correlations significant at the 0.01 level with neuroticism, EoE, GAD7, PHQ9, and all components of the CBRQ except catastrophisation score within the CAP in remission group.

Conclusion: Our results demonstrate that an older age at diagnosis, IBS prevalence, Crohn's disease, neurotic personality type, and certain behavioural and psychological profiles are significant features of IBD patients with chronic abdominal pain in remission. Further studies will help to develop management strategies and to establish causality.

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*Dr Ruqaya Idrees and Dr Pavit Luthra are joint first authors on this abstract.

Disclosure: Nothing to disclose.

PP0436

EXPLORING THE RELATIONSHIP BETWEEN HORMONAL CONTRACEPTIVE USE AND ULCERATIVE COLITIS FLARES: A LONGITUDINAL ANALYSIS

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Introduction: Numerous studies have implicated hormonal contraceptive therapy (HCT) as a potential trigger for IBD onset. However, data regarding disease flares in hormonal contraceptive users are limited in ulcerative colitis (UC). This study aims to explore this association.

Aims & Methods: Female patients who were diagnosed with UC after 1st January 2014, and born between 1966-2008 were included. The Andersen-Gill extension of the Cox proportional hazards model was used to account for recurrent events within the same subjects. Two models were employed: the first included each patient corresponding to only one cohort, while in the second one those who initiated HCT after diagnosis, were included twice, one from diagnosis to HCT as non-exposed and as exposed thereafter.

Results: Among 2359 patients in our local database, 545 were women with UC, of whom 132 met the inclusion criteria for analysis. The majority had UC proctitis 57 (43.2%) with 24 (18.2%) were smokers. The median age was 32.5 (23.3-42) years.

Of the cohort, 52 (39.1%) women exposed HCT, with a median exposure duration of 1.5 (0.5-5.0) years. Of these, 33 (25%) were exposed to HCT during follow-up.

Over 518 person-years of follow up, 178 flares and 198 courses of steroids were observed with 56 admissions and one colectomy, differences were observed among the incidence rates between groups Table 1.

Additionally, there were 24 pregnancies documented.

No differences in time to first flare (log-rank test p=0.83). However, the Cox proportional hazard model showed that HCT was associated with higher cumulative risk of flares in both models (HR 1.53, 95%CI 1.05-2.22, p=0.029) and (HR 1.47, 95%CI 1.019-2.11, p=0.039), Figure 1A and 1B. No adverse events associated to HCT occurred during follow-up.

N=132	All N=132	No OCP n=99	HCT n=33	Incidence rate ratio (95%CI)	p
Age at diagnosis, median (IQR)	32.5 (23.3-42)	36 (27-45)	25 (20.5-32)		0.0001
UC extent, n (%)					
E1	57 (43.2)	40 (40.4)	17 (51.5)		0.38
E2	40 (30.3)	33 (33.3)	7 (21.2)		
E3	35 (26.5)	26.3 (26.3)	9 (27.3)		
Time to 1st flare	493 (182-1187)	481 (177-1210)	522 (184-1226)		0.83
Person-year follow-up	517	458	56		N/A
Flare rate (nºevent/person-year follow-up)	167/518=0.34	132/458=0.29	35/56=0.63	0.46 (0.32-0.69)	<0.0001
Hospitalization rate	56/518=0.11	39/458=0.09	17/56=0.3	0.28 (0.16-0.53)	<0.0001
Surgery rate	1/518=0.002	1/458=0.002	0	0.002 (-0.01-0.01)	0.73
Steroid courses rate	198/518=0.38	158/458=0.35	40/56=0.72	0.48 (0.34-0.70)	<0.001

Conclusion: Our study suggests that HCT is associated with an increased cumulative risk of flares, steroid prescription and hospitalization rates in ulcerative colitis patients. Furthermore, no safety signals were observed with the use of HTC in this cohort.

Disclosure: Beatriz Gros has served as consultant to Galapagos and Abbvie and as speaker for Abbvie, Jansen, Takeda, Pfizer and Galapagos.

PP0437

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PP0438

PRIORITIZING PATIENT PERSPECTIVES: IDENTIFYING KEY SYMPTOMS IN CROHN'S DISEASE AND ULCERATIVE COLITIS TO ENHANCE CLINICAL TRIAL DESIGN

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Introduction: Due to the diverse symptoms experienced by patients living with Crohn's Disease (CD) and Ulcerative Colitis (UC), it can be challenging to understand how best to assess the benefit of new therapeutics for these conditions. Identifying priority symptoms that significantly impact patient's well-being and functioning is essential to inform the development of new therapies that truly improve their quality of life. Qualitative interviews with patients are a fundamental step in collecting this critical data, ensuring patient-focused drug development.

This study aimed to identify the most pertinent concepts (signs, symptoms, and impacts) from the patient perspective to provide an in-depth understanding of what aspects of their condition they prioritize.

Aims & Methods: A total of 24 qualitative interviews were conducted with US adult patients with UC or CD. Concept elicitation was conducted remotely using an IRB-approved, semi-structured guide. Participants rated the bothersomeness on a 10-point scale from 0 (no bother) to 10 (greatly bothersome); the average rating of bothersomeness for each symptom was calculated and combined with the total number of incidence reports to assess the saliency of each identified concept.

Data were analyzed in Atlas.ti and then visualized in a graphical format using Microsoft Excel. Salience was assessed for the present time ('at the interview') and at worst.

Results: Twelve UC adults (mean age: 39.5 years, 7 female, 7 white, 4 with only high school education) participated. The average time since UC diagnosis was 5.3 years, with 10 receiving one UC treatment at the time of the interview. Patients with UC most frequently reported rectal bleeding (N=12), fatigue (n=9/12), abdominal pain (n=9/12bowel urgency (n=9/12), and diarrhea (n=8/12).

The symptoms identified as most bothersome in a ranking task were abdominal pain (n=7/12), bowel urgency (n=5/12), rectal bleeding (n=4/12), and diarrhea (n=4/12). In symptom salience tasks 'at its worst', rectal bleeding (bothersomeness, 8.7/10) and bowel urgency (bothersomeness, 9.3/10) were identified as most salient. Most frequently reported UC impacts were diet change (N=12, worrying (n=10/12) and increased need for rest (n=10/12).

Twelve CD adults (mean age: 40.3 years, 8 female, 10 white, 4 with only high school education) participated. The average time since CD diagnosis was 10.3 years, with 11 receiving one CD treatment at the time of the interview. Patients with CD most frequently reported fatigue (n=11/12 bowel urgency (n=9/12), abdominal pain (n=8/12), and diarrhea (n=7/12).

The symptoms identified as one of the most bothersome to patients in a ranking task were fatigue (n=8/12%), frequent bowel movements (n=6/12), diarrhea (n=4/12), bowel urgency (n=4/12) and abdominal pain (n=8/12). In symptom salience tasks 'at its worst', fatigue (bothersomeness, 8.3/10) was identified as most salient. Most frequently reported CD impacts were inability to leave home (n=10/12), low mood (n=10/12) and planning around bathroom use (n=10/12).

Conclusion: The UC and CD patients reported priorities and concerns that may not be adequately assessed by the Mayo score (fatigue, abdominal pain and bowel urgency are not assessed) and CD Activity Index (CDAI) (fatigue, bowel urgency and diarrhea are not assessed) currently used in UC and CD clinical trials, respectively.

This underlines a need to increase patient experience data collection and to investigate whether inflammatory bowel disease (IBD) drugs in development (as well as those already approved) adequately address patients' priorities.

Disclosure: Nothing to disclose.

PP0439

CLOSTRIDIOIDES DIFFICILE INFECTION IN ADMITTED PATIENTS WITH ULCERATIVE COLITIS: ANALYSIS OF 23 YEARS THROUGH PROPENSITY SCORE MATCHING

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Introduction: Clostridioides difficile infection (CDI) represents the leading cause of nosocomial diarrhea in the Western world. Patients with ulcerative colitis (UC) have a higher risk of developing this infection, which has been associated with an increased risk of flare-ups, need for surgery, and mortality. The aim of the present study was to evaluate the impact of this infection in patients hospitalized for a UC flare-up.

Aims & Methods: Retrospective case-control study including all patients hospitalized for a UC flare-up at Reina Sofía Hospital in Córdoba between January 2000 and January 2023. To avoid potential selection biases, Propensity Matching Score (PMS) adjusted for age, gender, duration, and extent of the disease, with a 1:2 ratio was used. Primary objective was to assess the need for treatment intensification (initiation or dose escalation of biologics/small molecules or immunosuppressant). Secondary endpoints included readmission rate, colectomy, mortality, and CDI risk factors identification.

	All patients				PMS		
	Overall N=419	CDI n=46	NO CDI n=373	p	CDI n=46	NO CDI n=92	p
Length of admission, days, median (IQR)	11 (7-16)	11 (8-15)	11 (7-17)	0.79	11 (8-15)	8 (5-15)	0.107
Albumin <3.5gr/L, n (%)	145 (34.6)	17 (37)	128 (34.3)	0.83	17 (37)	31 (33.7)	0.848
Need for blood transfusion, n (%)	45 (10.7)	9 (19.6)	36 (9.7)	0.042	9 (19.6)	10 (10.9)	0.203
Therapy intensification, n (%)							
- During admission	225 (53.7)	19 (41.3)	206(55.7)	0.065	19 (41.3)	46 (50)	0.366
- Within 3 months	59 (14.1)	8 (17.4)	48 (14.6)	0.49	9 (20.9)	14 (15.2)	0.655
- Within 3-12 months	59 (14.1)	11 (23.9)	48 (12.4)	0.035	11 (27.5)	7 (7.6)	0.004
Readmission within 12 months, n (%)	80 (19.1)	17 (37)	63 (16.9)	0.001	17 (37)	16 (17.4)	0.009
Complications during admission, n (%)	66 (15.3)	5 (10.9)	61 (16.4)	0.34	5(10.9)	17 (17.7)	0.39
- Toxic megacolon	26 (6)	3 (6.3)	23 (6.2)		3 (6.3)	7 (7.3)	
- Bowel perforation	5 (1.2)	0	5 (1.3)		0	1 (1)	
- Severe bleeding	7 (1.6)	1 (2.1)	6 (1.6)		1 (2.1)	2 (2.1)	
- Other	26 (6)	1 (2.1)	25 (6.5)		1 (2.1)	7 (7.3)	
Colectomy, n (%)	49 (11.7)	3 (6.3)	46 (12.3)	0.25			
- During admission	14 (3.2)	3 (6.3)	10 (2.7)	0.19	3 (6.3)	11 (12)	0.385
- Within 1st year	11 (2.5)	3 (6.3)	8 (2.1)	0.09	6 (13)	6 (6.6)	0.214
Death within 12 months, n (%)	8 (1.9)	3 (6.5)	5 (1.3)	0.047	3 (6.5)	1 (1.1)	0.108

Table 1. Hospitalisation characteristics, evolution and complications

Results: During the study period, 419 patients were hospitalized for a UC flare-up, of whom 251 (59.9%) were men, with a median age of 41 (IQR 28-55) years and disease duration of 0.89 (0-5.8) years. Of all admissions, 46 (11%) had CDI. After PMS, CDI was associated with increased therapeutic intensification between 3-12 months after discharge (27.5% vs. 11.7%, p = 0.004) and a higher risk of readmission at 12 months (37.5% vs. 13.7%, p = 0.009) but not to risk of colectomy or mortality (Table 1). Infection recurrence was detected in 8 (16.7%) patients.

Classical risk factors for CDI were not found to increase the risk of CDI

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among UC patients, however antiTNF therapy was associated to CDI in both models: full cohort (OR 3.52, 95%CI 1.46-8.49) and PMS (OR 4.63, 95%CI 1.32-16.31).
Chi-square test or Fisher test were used to compare categorical variables. U-Mann Whitney test was used to compare continuous variables.
PROPENSITY SCORE MATCHING ANALYSIS: Cohort matched adjusting for: age, gender, disease duration and disease extension.
Conclusion: CDI in UC was associated with a greater need for therapeutic intensification and a higher readmission rate at one year. There was no evidence of an increased risk of colectomy or mortality in these patients. Classical risk factors for CDI were not identified but patients on antiTNF were at higher risk of CDI.
Disclosure: Beatriz Gros has served as a consultant for Galapagos and Abbvie and as a Speaker for Abbvie, Janssen, Takeda, Pfizer and Galapagos.

PP0440

GENE-ENVIRONMENT INTERACTIONS IN INFLAMMATORY BOWEL DISEASE: A SYSTEMATIC REVIEW OF HUMAN EPIDEMIOLOGIC STUDIES

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Introduction: While genetic susceptibility and environmental exposures are known contributors to the pathophysiology of Inflammatory Bowel Disease (IBD), their complex gene-environment interactions (GXE) remain elusive. This systematic review aims to summarize the current evidence of GXE in IBD.
Aims & Methods: PubMed, EMBASE, Web of Science and Scopus were systematically searched from inception through July 20, 2022 to identify publications examining the interaction effect of genetic variants and environmental factors in IBD. All eligible studies were graded using the STREGA guideline.
Results: 4,305 publications were identified and screened, resulting in 36 eligible studies. 17 studies reported statistically significant interactions. *NOD2*-smoking interaction was most frequently investigated and showed variant-specific effects at rs2066847 regarding the risk of Crohn's disease. Gene-smoking interactions were further identified in other IBD risk genes (*ATG16L1, IL23R,* and *CALM3*), detoxification genes (*GSTP1* and *HMOX1*), smoking-associated genes (*CHRNA3, CHRNA5, PPP1R3C,* and *BDNF*), and the inflammatory cytokine (*IL1B*) through a candidate gene approach. Immunochip-wide interaction analyses yielded 64 smoking interacting variants. Gene-diet interactions were observed across multiple nutritional measures, including fatty acid intake with *CYP4F3* and *FADS2*, serum selenium with *SEPHS1* and *SEPSECS*, potassium intake with *IL21*, alcohol consumption with *IL12B*, heme iron intake with *FCGR2A* and serum vitamin D with *VDR*.
Conclusion: Although previous studies have provided valuable insights into potential roles of GXE in the etiology of IBD, current evidence is sparse and inconclusive due to limited statistical power and lack of replication. Further studies, including genome wide environment interaction studies (GWEIS) and genotype-based nutrition/lifestyle clinical trials, are needed to unravel the missing heritability influenced by environmental exposures, and to construct personalized recommendations of lifestyle/dietary modification based on an individual genetic background.
Disclosure: Nothing to disclose.

PP0441

PREVALENCE AND RISK FACTORS OF GALLSTONE DISEASE IN KOREAN PATIENTS WITH ULCERATIVE COLITIS: A RETROSPECTIVE STUDY

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Introduction: The prevalence of gallstone disease in patients with ulcerative colitis (UC) is higher than in the general population. However, risk factors of gallstone disease in these patients remain unclear. Thus, we investigated the prevalence and risk factors of gallstone disease in Korean patients with UC.
Aims & Methods: Patients diagnosed with UC who underwent abdominal imaging between 1997–2020 were investigated using a well-established referral center-based large volume inflammatory bowel disease cohort. The prevalence and clinical characteristics of patients with gallstone disease were evaluated and compared with those without gallstone disease.
Results: Overall, 2811 patients with UC were enrolled. During the follow-up period (mean 5.7 years), 198 (7.0%) patients were diagnosed with gallstone disease and compared with those without gallstone disease (n = 2613). The proportion of extensive colitis at maximum extent, primary sclerosing cholangitis (PSC), history of cytomegalovirus, corticosteroid use, immunomodulatory use, colectomy, and appendectomy were significantly higher in the gallstone group (all $P < 0.05$). In multivariate analyses, age ≥ 60 years at gallstone evaluation (odds ratio [OR] 1.027; 95% confidence interval [CI] 1.002–1.052; $P = 0.033$), PSC (OR 6.304; 95% CI 3.162–12.565; $P < 0.001$), and history of colectomy (OR 2.494; 95% CI 1.222-5.087; $P = 0.012$) were significant risk factors for gallstone disease in patients with UC.
Conclusion: The prevalence of gallstone disease in Korean patients with UC was 7.0%, and age ≥ 60 years at gallstone evaluation, PSC, and history of colectomy were significant risk factors for UC patients with gallstone disease.
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Disclosure: Nothing to disclose.

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PP0442

TRANSITION OF INFLAMMATORY BOWEL DISEASE PATIENTS FROM PEDIATRIC TO ADULT CARE: A SINGLE-CENTER TEN-YEARS EXPERIENCE

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Introduction: Transition of adolescents with inflammatory bowel disease (IBD) to adult care is a challenge for patients, families, and healthcare providers. Considering that ~ 25% of IBD patients are diagnosed before the age of 16 years, and that these patients are at higher risk for poorer clinical disease outcomes. interest in the transition of care has been increasing considerably in the last decade. To assess the main clinical and epidemiological features of a single centre pedriatic population of IBD patients at diagnosis and at the time of transition to the adult center

Aims & Methods: From June 2014 to June 2023 we performed a multicenter, observational, cross-sectional study that included pediatric patients with IBD enrolled in the only pediatric IBD reference center in Sardinia. Data were obtained from the patients' medical records and from a questionnaire administered at the inclusion visit.

Results: Sixty-five IBD patients were enrolled (UC 40, CD 25, M 32, F 33). Median age at diagnosis was 13 years (IQR 2-18). After a median disease duration of 8 years (IQR: 1-25), no UC patients experienced proximal estension of proctitis or left-sided colitis, and no CD patients experienced new localizations of disease. Fifteen patients developed extraintestinal manifestations. No significant difference was found in median diagnostic delay (DD) between UC [4 months (IQR: 1-84)] and CD patients [4,5 months (IQR: 1-48)]. At the transition visit, overall, twenty-nine patients (44,6%) were exposed to one biologic agent (vs 3% at baseline; p<0.02); 0 patients (0%) were exposed to two biologic agents. 4,6% of patients (3/ 65) underwent surgery

Conclusion: Our results provide important insights into the clinical and epidemiological features of paediatric IBD population, and while waiting for a nation Italian registry, present eligible data on the IBD population in Sardinia

Disclosure: Nothing to disclose.

PP0443

IMPACT OF VITAMIN D LEVEL ON OUTCOMES IN ULCERATIVE COLITIS IN THE EMIRATI PATIENT POPULATION AT A TERTIARY CARE CENTER IN ABU DHABI, UNITED ARAB EMIRATES: RESULTS OF A RETROSPECTIVE COHORT STUDY

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Introduction: Vit D is a known immunomodulator and is essential in maintaining immune balance as well as controlling processes observed in inflammatory disorders like ulcerative colitis (UC) [1,2].

Aims & Methods: Aim: We previously completed a study looking at the effect of vitamin-D on Crohn's disease activity and noted a marginal beneficial effect. In this study, we sought to investigate the association between Vitamin D (VitD) levels and Ulcerative Colitis (UC) activity in the Emirati patient population of the United Arab Emirates (UAE).

Methods: We designed a retrospective cohort study using a sample population of patients (16-85 years) with UC, who were followed for at least one year in the Division of Gastroenterology at our tertiary care center in Abu Dhabi, between January 2021 and October 2023.

Demographics were collected on patients retrospectively identified in the EMR. Eligibility criteria were applied that included patients who had their VitD levels recorded at baseline and whose disease outcomes were measured between 1-12 months post-baseline visit. VitD was documented as a continuous variable and dichotomized into low vs. adequate level categories.

The primary outcome was clinical disease activity measured dichotomously. Summary statistics were calculated. Unadjusted logistic regression analysis was employed to examine the association between VitD level as a continuous variable and when categorized into low vs. adequate levels on clinical disease activity in UC.

Multivariable logistic regression looked at the independent effect of VitD levels by adjusting for covariates of effect size >20%. Statistical analysis was performed using R (version 4.0.2) and a p-value of <0.05 was considered statistically significant. IRB approval was in place for this study.

Results: We identified 89 patients with UC. Of these, 75 met our eligibility criteria, out of which 47 had their Vit D levels recorded at baseline, and 18 were noted to be deficient. 34 (45%) were women. The mean age of the cohort was 36 years. Detailed summary statistics were done. Unadjusted logistic regression analysis revealed that VitD did not appear to impact disease activity.

However, in the multivariable logistic regression analysis, when the model was adjusted for variables with an effect size of greater than 20% (age, BMI, duration of disease, steroid use), a marginal association between VitD as a continuous variable and active clinical disease was observed (OR 0.96 95%CI 0.93-0.99 P=0.039).

Conclusion: This is the first study looking at the impact of VitD on Emirati patients with UC. While there are previous studies in other cohorts that have suggested an association between VitD level and disease activity in UC, it appears that having adequate VitD levels, similar to our experience with Emirati patients with CD has a positive effect. The results of this study highlight the need for IBD practices in the UAE to consider VitD level sampling at baseline and replacement when necessary but discourage the frequent use or overuse of VitD testing in the UC population in the UAE.

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Disclosure: Nothing to disclose.

PP0444

UTILITY OF CANCER GENOME ANALYSIS USING BIOPSY TISSUES FROM COLORECTAL NEOPLASIAS DEVELOPING FROM ULCERATIVE COLITIS MUCOSA

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Introduction: Colorectal neoplasias developing from ulcerative colitis mucosa (CRNUCs) can be divided into ulcerative colitis-associated neoplasias (UCANs) and sporadic neoplasias (SNs); however, it is often difficult to distinguish between UCAN and SN from biopsy.

We previously reported the utility of cancer genomic analysis for differentiating UCAN from SN in post-resection CRNUC samples (Inflamm Bowel Dis. 2021).

Aims & Methods: This study investigates whether genomic analysis can differentiate UCANs from SNs using CRNUC biopsy tissues. In the first step, of the 36 early-stage CRNUCs (high-grade dysplasias and T1 carcinomas) resected at Hiroshima University Hospital between June 1998 and July 2018, and examined for genetic landscapes in the resection specimens, 14 CRNUC samples that underwent biopsy before endoscopic or surgical resection were analyzed.

The genomic landscape of the biopsy and resection specimens was examined for concordance. In the second step, the relationship between the genomic status and pathological diagnosis was investigated using biopsy specimens from 26 CRNUC cases, not included in the first step, which provided sufficient DNA for genomic analysis.

A custom panel of 27 genes, previously identified as mutated in CRNUC studies, was used for deep sequencing analysis of the cancer genome. In the evaluation of biopsy specimens, UCAN diagnosis using p53 immunostaining (IHC) was defined as cases exhibiting strongly positive p53-IHC results, and UCAN diagnosis based on *TP53* mutations was defined as cases with pathogenic mutations. UCAN diagnosis, combining hematoxylin and eosin (HE) staining with p53-IHC or *TP53* mutation analysis, was defined as follows: cases diagnosed with UCAN based solely on HE staining of the biopsy specimen, or cases with strongly positive p53-IHC or a pathogenic *TP53* mutation when HE staining alone was insufficient for a UCAN or non-UCAN diagnosis.

Results: After the first step of analysis, 14 out of the 27 candidate genes were found to be mutated. The concordance rate for pathogenic mutations across all 14 genes between the biopsy and resected specimens was 29% (4/14), whereas that for pathogenic mutations in *TP53* and *KRAS* was 79% (11/14). After the second step of analysis, the accuracy of diagnosing UCAN/non-UCAN using only HE staining of biopsy specimens was 38% (10/26), whereas p53-IHC achieved an accuracy of 73% (19/26, p=0.026), and *TP53* mutation analysis achieved an accuracy of 77% (20/26, p=0.012). The combination of HE pathology with other methods improved UCAN diagnosis. Specifically, combining HE pathology with p53-IHC in biopsy specimens achieved a sensitivity of 73% and an accuracy of 85% (22/26), while combining HE pathology with *TP53* mutation analysis resulted in a sensitivity of 87% and an accuracy of 88% (23/26).

Conclusion: Both p53 immunohistological staining and *TP53* mutation analysis of biopsy tissue were found to be useful for diagnosing UCAN, with *TP53* mutation analysis demonstrating comparable accuracy to p53 immunohistological staining for UCAN diagnosis.

Disclosure: Nothing to disclose.

PP0445

UTILIZING PATIENT-DERIVED COLON ORGANOIDS TO EXPLORE THE EFFECT OF GENETICALLY DETERMINED ERAP2-PROFICIENCY UPON PROINFLAMMATORY STIMULATION

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Introduction: Most immune-mediated disorders display a degree of heritability, with specific genetic variants linked to disease susceptibility. Variants in Endoplasmic Reticulum Aminopeptidase 2 (*ERAP2*) have been associated with numerous inflammatory conditions, including Inflammatory Bowel Disease (IBD). As several species (rodents included) lack the *ERAP2* gene, functional studies have been relatively limited. Several immunological functions of ERAP2 beyond peptide trimming have been proposed, yet most of these findings have been reported in immune cell populations. As the epithelial involvement in IBD pathogenesis is getting increasingly recognized, we set out to explore the impact of epithelial ERAP2 presence during inflammatory encounter using human-derived colonoids as a model system. To investigate the effect of genotype on *ERAP2* expression in different disease states, patient material from a IBD cohort was used to correlate genotype to *ERAP2* expression in inflamed and noninflamed colonic biopsies.

Aims & Methods: Two single nucleotide polymorphisms (SNPs) linked to ERAP2 expression (rs2910686/rs2248374) were analyzed in a cohort of IBD patients. Genotypes for both SNPs were correlated with colonic *ERAP2* expression levels in noninflamed and inflamed biopsies. Based on rs2910686

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genotype, ERAP2-proficient (*n*=4) and ERAP2-deficient (*n*=4) colon organoids were established to study differential expression between proficient and deficient donors after proinflammatory stimulation.

Results: Conditioning on genotype reveals upregulation of *ERAP2* in the inflamed colon of IBD patients. ERAP2 expression proficiency correlates with rs2910686 genotype. IFNγ induces colonic epithelial ERAP2 expression. Western blot confirms that mechanisms independent of the frequently studied SNP rs2248374 can cause ERAP2-deficiency. A total of 586 genes involved in various molecular mechanisms are differentially expressed between ERAP2-proficient and -deficient colonoids upon proinflammatory stimulation, including *AOC1, CPE, ANPEP* and *MEP1A* (catalytic activity), *TNFSF9, MDK, GDF15, ILR6A, LGALS3* and *FLNA* (regulator activity), *SLC40A1* and *SLC5A1* (transmembrane transporter activity), *FGL2, HMCN2,* and *MUC17* (extracellular matrix structural constituents).

Conclusion: Our findings demonstrate that ERAP2 expression is significantly upregulated in the inflamed colon of IBD patients, and expression levels correlates with rs2910686 genotype. Although rs2248374 is commonly used to assess ERAP2 expression proficiency, our findings confirm that mechanisms independent of rs2248374 can lead to ERAP2-deficiency. We further demonstrate that stratifying organoid donors based on genotype is a useful way of exploring the impact of genetic variants. Our data support that ERAP2 play a multifaceted role in modulating the inflammatory response in epithelial cells.

Disclosure: Nothing to disclose.

PP0446

EVALUATION OF DIET-DERIVED MICROBIOME-ASSOCIATED METABOLITES ON GUT INFLAMMATION IN ORGANOID MODELS OF INFLAMMATORY BOWEL DISEASE

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Introduction: Maintaining intestinal homeostasis relies on the interplay between individuals and their unique gut microbiota. Disruptions may result in chronic inflammation and Inflammatory Bowel Disease (IBD) [1]. This study delves into this interplay, emphasizing the characterization of diet-derived microbiome-associated metabolites and prioritizing them based on their effects on epithelial barrier integrity and inflammatory status.

Aims & Methods: Diet-derived microbiome-associated metabolites – epigallocatechin-gallate, gallic acid (GA), protocatechuic acid (PCA), ellagic acid (EA), and urolithins (Uro-A, B, C, D) – were selected through extensive literature research. In the commercial colon cell model (Caco-2) and 3D epithelial organoids from primary adult intestinal-epithelial stem cells, inflammation was induced with TNF-alpha and IFN-gamma (25ng/ml and each) to mimic IBD conditions. The expression of inflammatory cytokines genes (TNF-alpha, CXCL-9, MCP-1, IL-8) and epithelial barrier integrity markers (ZO-1 and OCLN) were evaluated by qRT-PCR. Epithelial barrier integrity was visualized using FITC-dextran assay.

Results: The induction of inflammation significantly increased in key inflammatory markers by an average 7.5-fold change (±1.8SD) in Caco-2 and 3D IBD organoid models. Metabolite analysis in Caco-2 cells showed that GA exhibited a 1.24-fold increase in TNF-alpha (p=0.036) and a -7.22-fold decrease in MCP-1 (p=0.049) gene expressions. EA decreased the expression of TNF-alpha and MCP-1 genes by -1.59-fold (p=0.045) and -3.59-fold (p=0.036) respectively, whereas IL-8 was reduced by -7.62-fold (p=0.01). Uro-A uniquely decreased MCP-1 gene expression by -7.16-fold (p=0.004) but increased IL-8 by 8-fold (p=0.01). PCA demonstrated a statistically significant decrease in all tested inflammatory markers (p<0.05). Analysis of 3D IBD organoids highlighted unique effects in restoring the epithelial barrier and expression of inflammatory marker genes.

Conclusion: Our findings demonstrate the anti-inflammatory properties of gallic acid, ellagic acid, protocatechuic acid, and urolithin-A in both Caco-2 and 3D organoid models. This research elucidates the complex interrelationships within the nutrition-microbiome-metabolome axis and its consequential impact on gut health.

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Disclosure: No conflict of interest to declare.

PP0447

CLINICAL SYMPTOM TRAJECTORIES ARE SUPPORTED BY DIFFERENTIAL REGULATION OF MUCOSAL TRANSCRIPTOMES IN PATIENTS WITH ULCERATIVE COLITIS TREATED WITH FILGOTINIB: A *POST HOC* ANALYSIS OF THE SELECTION STUDY

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Introduction: The SELECTION trial (NCT02914522) demonstrated that filgotinib (FIL), a once-daily, oral, Janus kinase 1 preferential inhibitor, was effective at inducing and maintaining clinical remission in patients with moderately to severely active ulcerative colitis (UC).[1]

A *post hoc* analysis of FIL responders identified five unique patient groups with different trajectories of response modelled from partial Mayo Clinic Score (pMCS) data, ranging from sustained improvement to rebound of symptoms.[2]

Aims & Methods: We investigated the underlying biology associated with these pre-defined symptomatic trajectories using colon biopsies collected during SELECTION; the trial design has been reported previously.[1] Using group-based trajectory modelling, FIL responders (defined as patients treated with FIL who were in clinical remission or with a Mayo Clinic Score response at week 10) were clustered into trajectories based on similar changes in pMCS from baseline to week 58.[2] RNA was extracted from colonic mucosal biopsies collected at baseline and week 10, and RNA

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sequencing was performed as per standard protocols. Trajectory-specific transcriptional changes were assessed at baseline by differentially expressed gene (DEG) analysis.

Key comparisons were:

a) Each trajectory at baseline versus the average of the other trajectories and'

b) The change from baseline to week 10, assessed separately for each trajectory group.

Results: Comparison of DEGs between each trajectory group demonstrated minimal transcriptional differences between the groups at baseline. However, the number of DEGs from baseline to week 10 differed between the trajectory groups (*Table*). Interestingly, trajectories associated with greater symptomatic improvement beyond week 10 had higher numbers of DEGs.

Conclusion: We identified transcriptional differences in the change from baseline to week 10 in DEGs between trajectory groups, aligning with symptomatic improvement beyond week 10 in patients receiving FIL. A qualitative analysis of DEGs is under investigation, for which preliminary results support the potential for personalized medicine in UC, by identifying differences in underlying pathophysiology associated with differential response to targeted therapy. These findings could further help to identify blood biomarkers that predict treatment outcomes in patients with UC on FIL.

	Group A	Group B	Group C	Group D	Group E
n (%)	4683 (8.85%)	4356 (8.23%)	1306 (2.47%)	14 690 (27.76%)	14 677 (27.73%)

Table. Change in number of significantly differentially expressed genes from baseline to week 10 for each trajectory

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SS and MR contributed equally.

MR and AO are employees of Alfasigma S.p.A at the time of abstract submission.

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PP0448

GENOME WIDE ANALYSES UNRAVEL NOVEL INTERACTIONS BETWEEN TOBACCO SMOKING AND GENETIC VARIANTS IN CROHN'S DISEASE AND ULCERATIVE COLITIS

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Introduction: The divergent impact of tobacco smoking in Crohn's disease (CD) and ulcerative colitis (UC) has been widely observed, but it's not completely understood whether the effect of smoking may be modified by genetic makeup.

Our present study aims to systematically identify the genetic modifiers of tobacco smoking in relation to CD and UC disease risk based on genome wide screening.

Aims & Methods: We performed a case-only genome wide association meta-analysis to examine the multiplicative interaction effect between tobacco smoking and single nucleotide polymorphisms (SNPs) in two independent IBD cohorts, the 1000IBD cohort (N=990) and the Dutch Parelsnoer-IBD cohort (N=2540). Imputed Global Screening Array genotype datasets combined with classical HLA alleles were used to compare the genotype frequencies between ever versus never smokers, current versus never smokers, and former versus never smokers in patients with IBD. We further separated entire analyses for CD and UC to characterize interactions where opposite direction was observed.

Functional mapping and annotation of smoking interaction variants were conducted using FUMA, including positional mapping, eQTL mapping and chromatin interaction mapping.

Results: A total of 572 lead SNPs showed consistent evidence of interaction with tobacco smoking between two independent cohorts (meta-analysis P<5e-5, heterogeneity Q test>0.05). Thirty SNPs exhibited opposite associations with smoking behaviors between CD and UC. (Effect size ratio>5 or <0.1) An HLA Class II haplotype (HLA-DRB1*03:01~HLA-DQA-1*05:01~HLA-DQB1*02:01) showed lower frequency in IBD smokers than never smokers.

Functional mapped genes were significantly overrepresented in various immune-response regulation pathways as well as collagen metabolic processes and showed enrichment in specific cell types, such as fetal intestine myeloid cells and midbrain neurotypes. These implicated genes also significantly overlapped with the risk genes for mental disorders, including autism spectrum disorders, schizophrenia, and major depressive disorders.

Conclusion: Our findings indicate that the influence of smoking behaviors on IBD risk depends on individual genetic background. Genetic modifiers of tobacco smoking with opposite directions may explain the differential impact of smoking on the susceptibility to CD and UC. Overrepresentation of these mapped genes in multiple mental disorders reflected complex pleiotropy.

Disclosure: Nothing to disclose.

PP0449

EARLY VEDOLIZUMAB DRUG LEVELS PREDICT ENDOSCOPIC IMPROVEMENT IN BIOLOGICAL-NAIVE PATIENTS: A SINGLE-CENTER EXPERIENCE

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Introduction: Inflammatory bowel diseases (IBD) are chronic gastrointestinal diseases that include ulcerative colitis (UC) and Crohn's disease (CD). IBD is a major public health concern, affecting approximately 6 million people globally (2). Although more common in industrialized countries, the prevalence of IBD is rising in emerging countries as a result of lifestyle changes.

Unfortunately, there is no official data on the occurrence of IBD in Montenegro, either for children or adults. Vedolizumab is a humanized monoclonal antibody that selectively targets α4β7 integrin and was approved for the treatment of adult patients with moderately to severely active UC and CD. The effects of therapeutic drug monitoring or a target serum concentration for vedolizumab are not fully established at this time. It is assumed that higher serum concentrations of vedolizumab lead to a higher probability of clinical response or remission.

Aims & Methods: The aim of the study was to investigate the relationship between vedolizumab serum concentrations (measured at week 6th after beginning of treatment) and clinical and endoscopic responses in IBD patients (evaluated at week 14th). The trial included 22 biologic-naive IBD patients (15 - UC, 7- CD), who received vedolizumab as a first-line treatment. We evaluated endoscopic scores (Mayo Score for UC and Simple Endoscopic Score (SES) for CD) as well as the clinical disease activity scores (Ulcerative Colitis Disease Activity Index (CDAI), Partial Mayo Score (PMS) for UC, and Harvey Bradshaw Index (HBI) for CD). Endoscopic response was defined as a 50% decrease in SES-CD score from the baseline value and Mayo endoscopy subscores of 0 or 1 were considered satisfactory. Clinical remission was defined as an HBI < 5, Mayo Score ≤ 2, and no individual subscore > 1. Clinical response was defined as a decrease in HBI ≥ 3, a decrease in Mayo Score ≥ 30%, and ≥ 3 points from the induction baseline, with a rectal bleeding subscore of 0 or 1.

Results: In the 14th week of therapy, a statistically significant clinical and endoscopic response was achieved, surpassing the overall response of patients with UC and CD (p<0.001). Furthermore, a statistically significant reduction in individual clinical scores (CDAI, HBI, and PMS) was observed in the 14th week of drug application (p=0.004; p=0.026, p=0.005, respectively). The endoscopic response was statistically significant in patients with UC (Mayo endoscopic score, p=0.003) in the 14th week, but not in patients with CD (p=0.176).

The difference in vedolizumab concentration was statistically significant between patients who achieved an endoscopic response in the 14th week and those who did not (p=0.02). Additionally, the difference in drug concentration was close to statistical significance (p=0.088) between patients with UC (28.6±13.6) and CD (21±9.7).

In biologic-naïve IBD patients, a vedolizumab serum concentration of 25.6 mcg/mL or greater at week 6th demonstrated a sensitivity of 88.9% and a specificity of 100% in predicting an endoscopic response at week 14th.

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Conclusion: Early determination of vedolizumab serum concentration in biologic-naive IBD patients, primarily UC, can be a promising predictor of endoscopic improvement in these patients.

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PP0450

THE ULTRASOUND MEASUREMENT OF PSOAS MUSCLE AS AN INDEX OF SARCOPENIA IN A COHORT OF PATIENTS AFFECTED BY INFLAMMATORY BOWEL DISEASE: CORRELATIONS WITH THEIR CLINICAL HISTORY AND VARIABLES

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Introduction: Sarcopenia is a progressive disorder characterized by reduced musculoskeletal mass leading to reduced strength and poor physical performance. Nutritional status is critical to therapeutic success in Crohn's Disease (CD) and Ulcerative Colitis (UC) but it is clear how traditional weight and body mass index (BMI) scores are poorly representative of these.[1,2]

Sarcopenia could better reflect the degree of malnutrition.[3]

In inflammatory bowel disease (IBD) sarcopenia has been associated with refractory to medical treatment, prolonged hospitalization, increased rate of surgery and increased frequency of post-operative complications.[4–6]

Recently the measurement of the psoas muscle (PM) by ultrasound (US) has been proposed as a surrogate of sarcopenia obtaining an high concordance with the reference standard CT scan.[7]

Aims & Methods: Our preliminary prospective study aims to identify correlations between US measurement of PM in a cohort of 57 patients with IBD and their clinical history and variables. The transverse area of the PM in cm^2 (US-TPA) and the same divided the square height in $cm^2/m2$ (US-PMI) were measured in the right iliac fossa using a low frequency probe: General Electric Logiq P8 (Boston, USA) for outpatients whereas Esaote MyLab One/Touch (Genova, Italia) for inpatients.

Results: Normal weight patients have significantly higher US-TPA and US-PMI values (p=0.032 and 0.031) than underweight (BM<18.5 kg/m^2) or overweight (BMI>25 kg/m^2). CD patients have significantly lower US-TPA (p=0.03) than UC patients. Patients with history of fistulizing perianal disease or who have at least two surgeries have significantly lower US-TPA and US-PMI values than the rest of the population (p=0.02). Patients with severe disease activity (Mayo Partial 5 or Harvey-Brashaw Index>8) have lower US-TPA and US-PMI (p = 0.0003 and 0.0004) than those in clinical remission or mild-moderate activity. As well as for patients with disease activity of any degree (Mayo Partial >2 or HBI>5) compared to remission (p= 0.001 and 0.002). With due caution, we have empirically applied the US-PMI cut-offs proposed by Kobayashi[8] in their cirrhotic cohort of 1.89 cm^2/m^2 for males and 1.81 cm^2/m^2 for females: 13 (23%) of our patients were found to be sarcopenic.

Of these, 46% of patients were admitted at the time of inclusion, 50% clinically active and 77% clinically severe. The higher frequency of sarcopenia observed in these groups was statistically significant with OR 5.42 (IC 95% 1.4-21.8) and p=0.017, OR 5.4 (IC 95% 1.4-21.2) and p=0.02 and OR 19.3 (IC 95% 3.2-116) and p=0.0002 respectively. Underweight and overweight subjects (BMI <18.5 kg/m2 or >25 kg/m2) were more frequently sarcopenic according to the same cut-off than normal weighted with OR 3.97 (95% IC 1.08 - 14.53) and p=0.037.

Conclusion: US is a simple and reproducible method of measuring the PM as a surrogate for sarcopenia, an independent nutritional index that could affect therapeutic responses and outcome in IBD. Our study has found a correlation between the so quantified sarcopenia and the disease activity and clearly demonstrates how such a state of malnutrition is linked to both underweight and overweight. The widening of the population will allow us in future to emphasize associations and differences. The impact and predictive role of sarcopenia on IBD outcomes will be investigated. It will be necessary to identify PM US cut-off for IBD, validated and related by nutritional parameters or other quantitative or qualitative methods.

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PP0451

MONITORING STRUCTURAL BOWEL DAMAGE IN PATIENTS WITH CROHN'S DISEASE: CAN IT REVERSED BY BIOLOGICAL TREATMENT?

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Introduction: Crohn's disease (CD) is a chronic inflammatory bowel disease, characterized by a progressive course that leads to structural bowel damage (SBD). There is no clear definition of SBD. The Lémann index has been proposed, but it's too complicated to calculate. However, the presence of detectable complications through imaging techniques, such as strictures, fistulas or abscesses is widely accepted to describe SBD. Their presence is strongly associated with the severity of the disease and predicts the risk of hospitalization and surgery. Therefore, reversing structural bowel damage is an emerging medical goal in the time of biological treatments.

Aims & Methods: The main objective of this study was to assess the ability of biological treatments to reverse structural bowel damage monitored by intestinal ultrasound (IUS) and compare the disease course of patients with CD with inflammatory behaviour vs complicated behaviour (stricturing or penetrating -abscesses and fistulas-).

This was a longitudinal observational study, which included patients with CD who started a biological treatment in our center and that were monitored with IUS. Clinical, laboratory and ultrasound variables were collected at the beginning of biological treatment, after 12 weeks and after one year of treatment in order to study the evolution of the disease.

To achieve our objective, we compared the transmural healing (TH) at 12 weeks and one year after starting biological treatment. TH is defined as intestinal parietal thickness ≤3 mm, with no detectable Doppler signals and absence of complications.

As a secondary objective, we compared the need for medical treatment, hospital admission and surgery during the first year of biological treatment

Results: A total of 137 patients with CD were included in this study. Their characteristics and treatments are shown in Table 1.

Variables		Absolute frequency (%)
Male		67 (48.9%)
Initial bowel damage	Stricturing	31 (22.6%)
	Abscesses	11 (8%)
	Fistula	21 (15.3%)
Montreal Classification	Age	≤16 years: 12 (8.8%) 17-40 years: 79 (57.7%) ≥ 41 years: 46 (33.6%)
	Location	Ileal: 75 (54.7%) Colonic: 19 (13.9%) Ileocolonic: 43 (31.4%) Upper disease: 1 (0.7%)
	Behaviour	Inflammatory: 53 (38.7%) Stricturing: 30 (21.9%) Penetrating: 54 (39.4%)
Indication for initiating biologic treatment	Stenotic complication	25 (18.2%)
	Penetrating complication	22 (16.1%)

Table 1.

At the beginning of biological treatment, 50 patients (36.5%) exhibited a complicated behaviour (strictures, abscesses, and fistulas).

After 12 weeks of treatment, TH was achieved in 22.9% of all evaluated patients. By groups, 8.2% of patients with complicated CD and 31.3% with an inflammatory behavior achieved TH, being the last difference statistically significant (p 0.02). After one year of treatment, TH on IUS was achieved in 46% of all evaluated patients. 31.8% of these patients had a complicated EC and 45.1% presented an inflammatory behavior, without reaching statistical significance (p 0.159).

Furthermore, the group with complicated behavior exhibited a significantly higher need for corticosteroid therapy, hospital admission, and surgery.

Conclusion: Patients with SBD showed a significantly lower rate of TH compared to patients with an inflammatory behaviour after 12 weeks of biological treatment.

Additionally, the need for corticosteroid therapy, hospital admission, and surgery was significantly higher in patients with a complicated CD. However, after one year, the TH rate showed no differences between groups. This suggests that reversing structural bowel damage is possible but that the treatment of these patients is more complex.

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PP0452

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PP0453

TYPE 2 DIABETES AND INFLAMMATORY BOWEL DISEASE: IS THERE A POSSIBLE LINK?

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Introduction: Nowadays, it is widely recognized that the gastrointestinal tract plays an important role in the homeostasis of glycemic metabolism. Some studies suggest that the presence of chronic intestinal inflammation is associated with an over-risk of type 2 diabetes (T2D), due to dysbiosis (1).

However, the impact of diabetes on chronic inflammatory bowel disease (IBD) has been insufficiently studied.

The aim of our study was to compare the clinical and therapeutic features as well as the evolutionary character of IBD according to the presence or not of T2D.

Aims & Methods: We conducted a monocentric retrospective study, including patients with Crohn disease (CD) or ulcerative colitis (UC) IBD, over a 7-year period [2014 - 2021]. Clinico-biological data, disease severity as well as the long-term therapeutic and evolutionary data were collected. Patients were compared according to the presence or absence of diabetes. They were divided into two groups:

- Group 1 (N=21): diabetic patients with IBD
- Group 2 (N=79): non-diabetic patients with IBD.

The statistical study was carried out by using SPSS V23 software.

Results: We enrolled 100 patients with IBD: 50% with CD and 50% with UC. The mean age was 46.5 years [extremes: 11 - 82], with a noticeable male predominance (M/F sex ratio of 2.5). The prevalence of T2D in our population was 21%, essentially affecting patients with UC in two-thirds of cases (N=14) versus seven diabetic patients with CD.

The comparison of these two groups showed no significant difference in terms of sex ratio (p= 0.961), smoking habits (p=0.731), duration of disease progression (p=0.744) or association with extraintestinal manifestations. However, the mean age was higher in cases of diabetes (p=0.017), with a cut-off of 51 years. Among diabetic patients, we observed a higher rate of complications such as acute severe colitis in 38.1% versus 15.2% of non-diabetic patients (p=0.020).

The number and severity of IBD flare-ups were not associated with diabetes. Nevertheless, bacterial or viral superinfection during relapses was significantly more frequent in case of diabetes (38.1% versus 15.2%, p=0.020). It was mainly due to superinfection with Escherichia Coli. Thromboembolic complications also occurred more frequently in diabetic patients (14.3% versus 1.3%, p=0.028). As for the therapeutic component, there was no significant difference between the two groups regarding the use of systemic corticosteroid therapy (p=0.458), immunosuppression (p=0.101) or biotherapy (p=0.803). Nonetheless, we found that side effects of corticosteroids were more important among diabetic patients (21.4% versus 2.3%, p=0.013). The risk of undergoing surgery was similar in both groups (33.3% versus 34.2%, p = 0.944).

In multivariate studies, type 2 diabetes was an independent factor in the occurrence of superinfection during relapses (OR=4.8; 95% CI [1.096 - 21.747]; p=0.038), as well as the side effects of corticosteroid therapy (OR=17.6 95% CI [1.281 - 243.006]; p=0.032) among IBD patients.

Conclusion: In our study, type 2 diabetes seems to have a negative impact by increasing the risk of superinfections and cortico-induced side effects during IBD, justifying increased surveillance in these patients. However, it does not appear to have a significant effect on the prognosis and long-term course of the disease.

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Disclosure: Nothing to disclose.

PP0454

PREDICTIVE VALUE OF THE MOBILE HEALTH INDEX IN DIFFERENT RISK GROUPS IN INFLAMMATORY BOWEL DISEASES: SIX-MONTH OUTCOMES OF A PROSPECTIVE COHORT STUDY

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Introduction: Patients with inflammatory bowel disease (IBD) require long-term, intensive care. To optimise future healthcare delivery, identifying patients suitable for remote monitoring is crucial. Patient-reported outcome measures integrated within eHealth systems, such as the 4-item Mobile Health Index (mHI)[1], offer potential for predicting disease activity. This study explores the predictive value of the mHI within the e-health system DEARhealth[2] at a Dutch university medical centre.

Aims & Methods: This multi-centre prospective cohort study tracked IBD patients within the e-health system. Patients were categorised into high-, intermediate-, and low-risk groups based on current and past biochemical disease activity. High-risk patients had current biochemical disease activity (FCP > 150 µg/g and/or CRP > 5 mg/L), intermediate-risk patients were in remission but experienced a biochemical flare within the last three years, while low-risk patients had been in biochemical remission for at least three years.

A personalised care pathway, considering both risk group and medication usage, dictated the frequency of outpatient visits and lab tests. Over a 24-month period, patients are monitored according to standard of care (biochemical disease activity; endoscopic disease activity) and with mHI scores at both pre-defined time points and patient-initiated moments. These preliminary results focus on assessing the predictive value of the mHI, above a validated cut-off value[1], at baseline for predicting a flare within six months. Sensitivity, specificity, and predictive values are calculated alongside simple logistic regression and Kaplan-Meier analyses.

Results: Out of 319 patients (56.7% Crohn's Disease (CD), median age 43 years, median disease duration 15 years), 89 exhibited biochemical disease activity at baseline and were excluded from the current analysis. Among the 230 patients in remission at baseline, 51 (21.3%) scored above the mHI cut-off point at baseline and 79 (34.3%) experienced a biochemical flare-up within six months of follow-up. The relative risk of experiencing a disease flare in Ulcerative Colitis (UC) patients with a baseline mHI score above the cut-off point was 2.44 (p = 0.04) with sensitivity of 42.1%, specificity of 82.4%, positive predictive value of 40.0%, and negative predictive value of 83.6%. In CD patients, this relative risk was 1.62 (p = 0.39) with sensitivity of 38.5%, specificity of 73.4%, positive predictive value of 14.7%, and a negative predictive value of 90.9%. Logistic regression analysis revealed a significant association between a mHI score above the cut-off point and disease flares in UC patients (OR = 3.394, 95% CI: 1.110-10.330, p = 0.03), whereas in CD patients, such an association was not significant (OR = 1.724, 95% CI: 0.487-5.603, p = 0.37). Similarly, Kaplan-Meier analysis demonstrated a longer flare-free survival in UC patients with a baseline mHI score below the cut-off point (p = 0.03), while no significant difference was observed in CD patients (p = 0.14).

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Conclusion: This study indicates that the mHI could be a valuable tool for predicting disease activity, and particularly quiescent disease, in UC patients. However, this predictive capability was not observed in patients with CD. These results underscore the potential utility of mHI in enhancing the management of IBD patients, with implications for personalised care and resource allocation within healthcare systems.

Further research should explore repeated mHI measurements in combination with biomarkers like calprotectin for a more comprehensive approach.

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PP0455

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PP0456

CLINICAL RECURRENCE AFTER ILEOCOLIC RESECTION FOR CROHN'S DISEASE: INTEREST OF THE REMIND GROUP'S ENDOSCOPIC SCORE

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Introduction: The Rutgeerts score is the most commonly used score for assessing endoscopic recurrence after ileocolic resection in Crohn's disease. However, a significant drawback of this score is that it combines lesions localized to the anastomosis with those in the neo-terminal ileum in the same category, assigning them equal prognostic significance.

Aims & Methods: The aim of this study was to evaluate the performance of the REMIND score in predicting postoperative recurrence of Crohn's disease and compare its performance with the Rutgeerts score.

We conducted a monocentric, retrospective, evaluative study of patients operated for Crohn's disease. All patients had a follow-up colonoscopy within one year of surgery.

The REMIND score was derived from endoscopic details documented in the colonoscopy reports. Clinical relapse was defined by the Crohn's disease related symptoms or complications, such as intra-abdominal collection or occlusion, or subsequent surgery Postoperative. Statistical analysis was performed using SPSS.25 for Windows.

Results: Eighty patients were included, with a mean age of 35.7 years, and a gender ratio H/F=2,2. Median follow-up was 5 years from surgery. Clinical recurrence-free survival was significantly shorter in patients with ileal lesions whatever their severity was I(1,2,3,4)A(x), compared with patients without ileal lesions I(0)A(x): p<0,001 and not significantly different between patients without anastomotic lesions I(x)A(0) and patients with anastomotic lesions whatever their severity was I(x)A(1,2,3): p=0,2.

However, patients with at least semicircumferential anastomotic ulcerations A(2,3) had more anastomotic occlusive complications than patients classified A0 or A1: A0 VS A2 p=0,01 A0 VS A3 p<0,001 A0 VS A1 p=0,9. The I(x) score, compared with the Rutgeerts and A(x) scores, had the best values in terms of discriminative capacity (Area under the curve= 0,83 VS 0,73 VS 0,61), monotonicity (linearity test $\chi 2$ of 26,77 VS 12,36 VS 3,89), and homogeneity (likelihood ratio $\chi 2$ of 33,91 vs 18,45 VS 5,49).

Conclusion: In contrast to the Rutgeerts and A(x) scores, the I(x) score seems to exhibit superior performance in predicting postoperative recurrence. Therefore, the use of a score describing separately ileal and anastomotic lesions might be more appropriate to define postoperative endoscopic recurrence.

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Disclosure: Nothing to disclose.

PP0457

EPITHELIAL AND NOT LAMINA PROPRIA NEUTROPHIL LOCALIZATION AND INTESTINAL BARRIER PROTEIN EXPRESSION ARE INNOVATIVE OUTCOME PREDICTORS IN INFLAMMATORY BOWEL DISEASE

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Introduction: Patients with inflammatory bowel disease (IBD) in clinical and endoscopic remission may still experience disease relapse. Therefore, there is an unmet need for reliable predictors of outcomes to enable a personalized therapeutic approach. Recently, a novel histological score has highlighted the pivotal role of neutrophils in predicting disease outcomes in Ulcerative Colitis (UC)[1]. Furthermore, the impairment of the intestinal barrier plays a key role in forecasting disease outcomes in UC and Crohn's disease (CD). Hence, this study aims to assess the predictive role of neutrophils according to their tissue localization and intestinal barrier protein expression in IBD patients.

Aims & Methods: IBD patients in clinical remission who underwent colonoscopy between January 2020 and June 2022 at two tertiary referral centres were enrolled. Patients with a Mayo Endoscopic Score ≤1 for UC and a Simple endoscopic score of ≤6 for CD were included. The histological activity was assessed using the Robarts Histopathology Index, Nancy Histological Index, and PICaSSO Histological Remission Index. Neutrophil localization in the epithelium and lamina propria was carefully evaluated

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by experienced pathologists. Additionally, the immunohistochemical expression of barrier proteins, such as Claudin-2 and JAM (junctional adhesion molecule)-A, was assessed to evaluate the integrity of the intestinal barrier.

Results: Out of 60 UC and 76 CD patients, 60% were in histological activity. 25.8% of patients developed an adverse outcome within 12 months. Only neutrophils in the epithelium and not in lamina propria could predict adverse outcomes for UC (HR 5.20, *p*=.01) and CD (HR 4.38, *p*=.03) patients in endoscopic remission.

Moreover, epithelial Claudin-2 expression correlated with endoscopic and histological disease activity and predicted outcomes in UC. Combining neutrophils in the epithelium or Claudin-2 expression significantly predicted outcomes in UC patients (HR 3.23, *p*=.035). Similar results were found for JAM-A in CD despite this protein showing less specificity as a barrier predictor of outcome.

	Ulcerative colitis	Crohn's disease
Neutrophils in epithelium		
Study sample	5.20 (1.46 - 18.46; .01)	1.86 (0.72 - 4.81; .20)
Endoscopic remission	2.5 (0.6 - 11.04; .24)	4.38 (1.16 - 16.55; .03)
Endoscopic mild inflammation	Inf (0 - Inf; .999)	0.32 (0.07 - 1.44; .14)
Neutrophils in lamina propria, not epithelium		
Study sample	3.02 (0.85 - 10.72; .09)	1.59 (0.6 - 4.25; .35)
Endoscopic remission	1.75 (0.39 - 7.83; .46)	4.36 (0.94 - 20.21; .06)
Endoscopic mild inflammation	Inf (0 - Inf; .999)	0.32 (0.07 - 1.44; .14)
Claudin-2 expression		
Study sample	2.12 (1.62 - 43.18; .01)	1.28 (0.001 - 1368.4; .94)
Endoscopic remission	2.08 (0.46 - 9.36; .34)	1.45 (0.44 - 4.76; .54)
Endoscopic mild inflammation	1.07 (0.21 - 5.29; .94)	0.27 (0.06 - 1.36; .11)
JAM-A expression		
Study sample	1.20 (0.44 - 3.32; .72)	3.18 (1.22 - 8.30; .02)
Endoscopic remission	1.32 (0.30 - 5.9; .71)	1.98 (0.58 - 6.78; .27)
Endoscopic mild inflammation	1.16 (0.29 - 4.63; .84)	6.88 (1.23 - 38.43; .03)

Conclusion: This multicentre study highlights the potential role of *epithelial* neutrophil localization and Claudin-2 'leaky gut' protein expression as practical clinical tools for predicting outcomes in IBD and potentially guiding patient-tailored therapy.

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Disclosure: Nothing to disclose.

PP0458

THE URGENCY NUMERIC RATING SCALE: PSYCHOMETRIC EVALUATION IN ADULTS WITH CROHN'S DISEASE

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Introduction: Bowel urgency is one of the symptoms that patients with Crohn's disease (CD) frequently experience and consider most important to control. Bowel urgency often has a substantial negative impact on quality of life. The Urgency Numeric Rating Scale (NRS) is a single-item patient-reported outcome assessing severity of bowel urgency in the previous 24 hours on an 11-point scale (0–10), with higher scores indicating more severe bowel urgency. The Urgency NRS was initially developed to assess the urgency to have a bowel movement in ulcerative colitis due to lack of appropriate existing instruments.

Aims & Methods: This work aimed to evaluate the psychometric properties of the Urgency NRS in CD and to estimate thresholds for meaningful change and remission for bowel urgency. Pooled data from VIVID-1 (NCT03926130), a Phase 3, multicentre, randomised, placebo- and active-controlled study of mirikizumab in adults with moderately to severely active CD were analyzed.

The measurement properties (reliability, validity, and responsiveness) of the Urgency NRS were evaluated using weekly average scores at Weeks 0 (Baseline), 4, 12, and 52, which were calculated by averaging the most recent 7 days in the 12 days prior to the visit.

These analyses used other measures administered in VIVID-1, including the Patient Global Rating of Severity (PGRS), Patient Global Impression of Change (PGIC), Inflammatory Bowel Disease Questionnaire (IBDQ), and Crohn's Disease Activity Index (CDAI). PGRS and PGIC were used as primary anchors to establish the Urgency NRS score thresholds for meaningful change and remission of bowel urgency. Data from VIVID-1 treatment arms were pooled for these analyses.

Results: The analysis population comprised 1065 adults (mean age 36.2 years [standard deviation: 13.0]; 55% male, 71% White). The Urgency NRS weekly average score decreased during the trial (from 6.6 at Baseline to 2.9 at Week 52), indicating a reduction in bowel urgency severity. Test-retest reliability was good for patients who were stable between Baseline and Week 4 based on the PGRS (intraclass correlation coefficient [ICC] 0.86) and the PGIC (ICC 0.78).

At Weeks 12 and 52, the Urgency NRS showed moderate correlations ($0.30 \leq r < 0.70$) with the PGRS, IBDQ Bowel Symptom domain, IBDQ total score, and CDAI score; as expected, correlations with clinical and laboratory assessments were weak ($r < 0.30$), demonstrating discriminant validity. The Urgency NRS could differentiate between groups of participants known to be distinct based on PGRS and IBDQ Bowel Symptom domain scores.

Improvement in the Urgency NRS average score from Baseline to Weeks 12 and 52 was statistically significantly different between most PGRS change groups and PGIC groups, as well as groups based on IBDQ Bowel Symptom domain, IBDQ response and remission, and clinical response/remission by CDAI, indicating that the Urgency NRS was able to detect change. Anchor-based analysis indicated that clinically meaningful improvement could be represented by a 3–5-point reduction in the Urgency NRS score and that an Urgency NRS score of ≤2 could represent bowel urgency remission.

Conclusion: These results support the reliability, validity, responsiveness of the Urgency NRS and provide definitions of clinically meaningful improvement (reduction of 3–5 points in the Urgency NRS score) and remission of bowel urgency (Urgency NRS score of ≤2) in adults with moderately to severely active CD. These findings confirm the Urgency NRS as suitable for assessing treatment benefits in patients with CD during clinical trials.

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Farmaceutica, ai4gi, Alcimed, Allergan, Amgen, Aptel, Arena Pharmaceuticals, Asahi Kasei Pharma, Aspen Pharmacare, Astellas, AstraZeneca, Atlantic Pharmaceuticals, Barco, Biocare Medical, Biogen, BL Pharma, Boehringer Ingelheim, Bristol Myers Squibb, BUHLMANN Diagnostics, Calcico Therapeutics, Celgene, Cellerix, Cerimon Pharmaceuticals, ChemoCentryx, Chiesi, Cisbio, Comcast, Coronado Biosciences, Cosmo Pharmaceuticals, Dr. Falk Pharma, Ducentis BioTherapeutics, Dynavax Technologies, Elan Pharma, Eli Lilly and Company, Enterome, Equillium, Ferring Pharmaceuticals, Galapagos NV, Genentech/Roche, Genzyme, Gilead Sciences, GlaxoSmithKline, Glenmark Pharmaceuticals, Grünenthal, GW Pharmaceuticals, Immunocore, Indigo Biosciences, Janssen, Lexicon Pharmaceuticals, Medarex, MedTrix, Merck, Merrimack Pharmaceuticals, Mestag Therapeutics, Millennium Pharmaceuticals, Neovacs, Novartis, Novo Nordisk, NPS Pharmaceuticals/Nycomed, Ocera Therapeutics, OPTIMA Pharma, Origin Pharma, Otsuka, Palau Pharma, Pentax Medical, Pfizer, PharmaVentures, Phesi, Phillips Pharma Group, Procter & Gamble, Pronota, Proximagen, Resolute Pharma, Robarts Clinical Trials, Sandoz, Santarus, Satisfai Health, Sensyne Health, Shire, Sigmoid Pharma, Sorriso Pharmaceuticals, Souffinez, SynDermix, Synthon, Takeda, Theravance Biopharma, TiGenix, Tillotts Pharma AG, Topivert, Trino Therapeutics with Wellcome Trust, TxCell SA, UCB Pharma, Vertex Pharma, VHsquared, Vifor Pharma, Warner Chilcott, and Zeria Pharmaceutical.

PP0459

A PATIENT-REPORTED OUTCOME MEASURE COMPRISING THE STOOL FREQUENCY AND ABDOMINAL PAIN ITEMS FROM THE CROHN'S DISEASE ACTIVITY INDEX: PSYCHOMETRIC EVALUATION IN ADULTS WITH CROHN'S DISEASE

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Introduction: The Crohn's Disease Activity Index (CDAI) is an 8-item disease activity measure with three patient-reported items that cover stool frequency, presence of abdominal pain, and patient's general well-being. Absolute scores of the Stool Frequency (SF) and Abdominal Pain (AP) items, together referred to as the Patient Reported Outcome (PRO), have been used separately in the VIVID-1 study.

Both PRO items have recall periods of 24 hours; the SF item measures the number of liquid or very soft stools per day and the AP item measures abdominal pain severity on a 4-point scale (0=none, 1=mild, 2=moderate, and 3=severe).

Aims & Methods: This study aimed to evaluate the measurement properties of the SF and AP items and estimate thresholds for remission. Pooled data from VIVID-1 (NCT03926130), a Phase 3, multicentre, randomized, placebo- and active-controlled study of mirikizumab in adults with moderately to severely active Crohn's Disease (CD) were analysed.

The measurement properties (reliability, validity, and responsiveness) of the PRO items were evaluated separately using average weekly scores of the SF and AP items at Weeks 0 (Baseline), 4, 12, and 52. Patient Global Rating of Severity (PGRS) and Patient Global Impression of Change (PGIC) were used as primary anchors to estimate the remission thresholds. Exit interviews were conducted in a subgroup of participants to support the remission thresholds.

Results: Data from 1065 participants (mean age: 36.2 years [standard deviation: 13.0], 55% males, 71% White) were analysed. During the trial, average weekly scores decreased for both items: SF from 5.7 (Baseline) to 1.9 (Week 52) and AP from 2.1 (Baseline) to 0.8 (Week 52). Both PRO items demonstrated moderate to good test-retest reliability for participants who were stable between Baseline and Week 4 based on PGRS (intraclass correlation coefficient [ICC] 0.85 for SF and 0.82 for AP) and PGIC (ICC 0.79 for SF and 0.69 for AP). SF had moderate ($0.30 \leq r < 0.70$) correlations with Inflammatory Bowel Disease Questionnaire (IBDQ) item 1–stool frequency (Baseline and Weeks 12 and 52) as well as with IBDQ total score, IBDQ Bowel Symptoms domain score, and PGRS (Weeks 12 and 52 only). AP had moderate to large correlations ($0.30 \leq r \leq 0.90$) with PGRS, IBDQ total score, IBDQ item 13–abdominal pain, 36-item Short Form Health Survey (SF-36) Bodily Pain domain, and EQ-5D-5L Pain/Discomfort (Baseline and Weeks 12 and 52).

As anticipated, both PRO items had weak correlations ($r < 0.30$) with the Simple Endoscopic Score for Crohn's Disease (SES-CD), demonstrating discriminant validity. The PRO items could differentiate between groups of participants known to be distinct based on PGRS (for SF and AP), IBDQ Bowel Symptom domain score (for SF), and EQ-5D-5L Pain/Discomfort (for AP).

The PRO items were able to detect change, as score changes in both items between Baseline and Weeks 12 and 52 were statistically significantly different between most PGRS change groups and PGIC groups. Anchor-based analyses combined with results from the exit interviews suggested that an SF score of ≤3 and an AP score of ≤1 could represent remission.

Conclusion: These psychometric evaluation results support the reliability, construct-validity, and responsiveness of the PRO items when administered to adults with moderately to severely active CD. A combination of an SF score of ≤3 and an AP score of ≤1 could represent clinical remission by PRO.

Disclosure: SS has been a consultant for AbbVie, Amgen, Arena Pharmaceuticals, Bristol Myers Squibb, Boehringer Ingelheim, Celltrion, Dr Falk Pharma, Ferring, Fresenius, Galapagos, Genentech, GSK, Gilead, I-MAB Biopharma, Janssen, Eli Lilly and Company, Merck, Novartis-Sandoz, Pfizer, Protagonist, Takeda, and Theravance. AV, SS, and FD are employees and shareholders of Eli Lilly and Company. XZ is an employee of Syneos Health, which received funding from Eli Lilly and Company in connection with this study. AK, LS, and CC are employees of Evidera, which received funding from Eli Lilly and Company in connection with this study. JL has received personal fees from Johnson & Johnson Consumer Inc; grants, personal fees, and other from Takeda Pharmaceuticals and Pfizer; personal fees and nonfinancial support from AbbVie; grants and personal fees from Janssen Pharmaceuticals, Nestle Health Science; personal fees from Eli Lilly and Company, Samsung Bioepis, UCB, Bristol-Myers Squibb, Bridge Biotherapeutics, Celgene, Merck, Gilead, Arena Parmaceuticals, and Protagonist Therapeutics.

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PP0460

PSYCHOMETRIC EVALUATION OF THE FUNCTIONAL ASSESSMENT OF CHRONIC ILLNESS THERAPY–FATIGUE IN ADULTS WITH MODERATE-TO-SEVERE CROHN'S DISEASE

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Introduction: In Crohn's disease (CD), fatigue is a common symptom that affects ~40% of patients [1] and often has a major impact on their quality of life. The Functional Assessment of Chronic Illness Therapy–Fatigue (FACIT-Fatigue) is a 13-item patient-reported outcome assessing fatigue over the previous week on a 5-point scale. In the FACIT-Fatigue, scores for all items are summed to give a total score between 0 and 52, with a higher score indicating less fatigue. The psychometric properties of the FACIT-Fatigue have been assessed in multiple chronic illnesses, including CD.

Aims & Methods: The objective of this study was to provide further evidence on the measurement properties of the FACIT-Fatigue and to determine thresholds for meaningful improvement in fatigue for patients with moderately to severely active CD. Pooled data from VIVID-1 (NCT03926130), a Phase 3, multicentre, randomised, placebo- and active-controlled study of mirikizumab in adults with moderately to severely active CD were analysed. The measurement properties (reliability, validity, and responsiveness) of the FACIT-Fatigue were evaluated at Weeks 0 (Baseline), 12, and 52. Other assessments completed in the trial were used for these analyses, including the Patient Global Rating of Severity (PGRS), Patient Global Impression of Change (PGIC), Inflammatory Bowel Disease Questionnaire (IBDQ), Crohn's Disease Activity Index (CDAI), and 36-Item Short Form Health Survey (SF-36). PGRS and PGIC were used as primary anchors to estimate the FACIT-Fatigue total score change that would constitute meaningful improvement.

Results: Analyses included 1065 adults (mean age 36.2 years [standard deviation: 13.0], 55% males, 71% White). The mean FACIT-Fatigue total score improved during the study (Baseline: 31.5; Week 12: 37.1; Week 52: 40.0). The measure demonstrated internal consistency across visits (Cronbach's alpha: 0.92–0.94); correlations between individual items and the total score were moderate ($0.30 \leq r < 0.70$) to large ($r \geq 0.70$). Across visits, the FACIT-Fatigue total score showed moderate to large correlations with the PGRS; IBDQ Bowel Function domain, item 2 (fatigue), and total score; CDAI score; and SF-36 Vitality domain score. However, correlations with clinical measures were weak ($r < 0.30$), demonstrating discriminant validity. The FACIT-Fatigue could differentiate between groups of participants known to be distinct based on the PGRS, IBDQ total score, and IBDQ item 2. The FACIT-Fatigue was able to detect change, as FACIT-Fatigue total score improvements from Baseline to Weeks 12 and 52 were statistically significantly different between most PGRS change groups and PGIC groups, as well as between groups based on IBDQ response/remission, IBDQ item 2, SF-36 Vitality domain, and clinical response/remission by CDAI. Anchor-based analyses suggested that a 6–9-point increase in the FACIT-Fatigue total score could represent meaningful improvement.

Conclusion: These findings expand on previous research supporting the internal consistency, reliability, construct validity, and responsiveness of the FACIT-Fatigue. Analyses suggest an improvement range of 6–9 points for FACIT-Fatigue as clinically meaningful in the VIVID-1 trial population of adults with moderately to severely active CD.

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PP0461

CLINICAL USEFULNESS OF IMMUNE PROFILING FOR DIFFERENTIAL DIAGNOSIS BETWEEN CROHNS DISEASE, INTESTINAL TUBERCULOSIS, AND BEHCETS DISEASE

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Introduction: It is important to make a differential diagnosis between inflammatory diseases of the bowel with similar clinical and endoscopic features. Misdiagnosis or delayed diagnosis can affect patients' therapeutic outcomes and prognosis. The profiling of immune cells is helpful for accurately diagnosing inflammatory diseases.

Futhermore, evaluating differences of immune marker expression between Crohn's disease (CD), intestinal Behcet's disease (BD), and intestinal tuberculosis (TB) could be useful in differentiating the diseases.

Aims & Methods: This study aimed to compare the expression levels of immune markers between CD, intestinal BD, and intestinal TB and evaluate the usefulness of immune profiling in differential diagnoses using this advanced imaging system. Biopsy specimens were acquired around ulcerations on the terminal ileum or cecum from five patients with each disease.

Panel 1 included multiplexed immunohistochemistry staining for CD8, CD4, Foxp3, CD20, programmed death-1, and granzyme B. CD56, CD68, CD163, CD11c, and HLA-DR were analyzed in panel 2. Cell density was used to compare the expression levels of immune cells.

Results: The expression of immune cells in panel 1 was not different between diseases. Differences in cytotoxic T cells, helper T cells, and regulatory T cells identified by the double-positivity method were also not significant. However, the expression of $CD68^+$ cells in panel 2 was the highest in intestinal BD among bowel diseases (Table 1).

In the co-expression method, the expression levels of M1 macrophages ($CD56^-CD68^+CD163^-HLA\text{-}DR^-$) and dendritic cells ($CD56^-CD68^-CD163^-CD11c^+HLA\text{-}DR^+$) were significantly different ($P = 0.001$, $P = 0.012$, respectively).

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	Crohn's disease	Intestinal Behcet's disease	Intestinal tuberculosis	p-value
CD56+, median cells/mm² (range)	199 (63-263)	75 (44-335)	160 (14-602)	0.582
CD68+, median cells/mm² (range)	530 (508-840)	1828 (1648-3366)	1224 (508-1404)	<0.001
CD163+, median cells/mm² (range)	63 (17-228)	154 (89-408)	114 (69-167)	0.165
CD11c+, median cells/mm² (range)	362 (20-1160)	19 (0-616)	304 (36-649)	0.252
HLA-DR+, median cells/mm² (range)	258 (193-451)	152 (3-640)	890 (168-1147)	0.208

Table 1. Comparison of immune cell densities in panel 2 between Crohn's disease, intestinal Behcet's disease, and intestinal tuberculosis.

Conclusion: The expression of immune cells, including M1 macrophages and dendritic cells, was different between CD, intestinal BD, and intestinal TB. Immune profiling can be helpful for differential diagnoses of inflammatory bowel diseases.

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PP0462

USEFULNESS OF NOVEL IMAGE-ENHANCED ENDOSCOPY FOR PREDICTING MAINTENANCE OF CLINICAL REMISSION IN ULCERATIVE COLITIS

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Introduction: Clinical remission and mucosal healing have been advocated as treatment goals for patients with ulcerative colitis (UC). Novel molecularly targeted drugs for the treatment of UC have enabled to target not only clinical remission but also endoscopic remission and even histologic remission.

Recently, the usefulness of novel image enhanced endoscopy (IEE) such as texture and color enhancement imaging (TXI) and red dichromatic imaging (RDI) have been reported in endoscopic evaluation of UC. We aimed to verify the usefulness of the novel IEE in prediction of maintenance of clinical remission in UC patients.

Aims & Methods: Patients with UC in clinical remission who underwent colonoscopy with IEE from March 2021 to December 2023 at our hospital were enrolled in this prospective study.

Disease activities in six segments (cecum, ascending colon, transverse colon, descending colon, sigmoid colon, and rectum) were assessed using white-light imaging (WLI), TXI and RDI. Endoscopic evaluation of disease activity using WLI was performed using Mayo endoscopic score (MES) and the Ulcerative Colitis Endoscopic Index of Severity (UCEIS). IEE evaluation was performed using predefined TXI score (0-2: 2 indicates maximum severity) and RDI score (1-4: 4 indicates maximum severity) for UC. MES of ≤1 and UCEIS of ≤1 were defined as endoscopic remission. With IEE, TXI score 0-1 and RDI score 1 were defined as endoscopic remission in accordance with previous reports.

We evaluated the correlation of those scores by Spearman's rank correlation coefficient. Moreover, relationships between endoscopic remission of WLI, TXI and RDI, and maintenance of clinical remission were evaluated by Kaplan-Meier analysis. Relapse was defined as worsening of clinical symptoms that required additional medical therapies for UC.

Results: Total 117 patients with UC were enrolled. Of these, 17 were excluded because adequate images could not be obtained. Therefore, total 100 patients were assessed. The median MES, UCEIS, TXI, and RDI scores were 1 (range 0-3), 1 (range 0-5), 2 (range 1-3) and 2 (range 1-4), respectively. The average follow-up period was 10.1 months, and 8 cases showed clinical relapse during the observation period.

Evaluation of endoscopic images at each colonic site revealed that UCEIS (P<0.001), TXI score (P<0.001), and RDI score (P<0.001) were significantly associated with MES, respectively. In addition, scores at the most inflamed area showed a positive correlation between MES and UCEIS (r=0.91, P<0.001), MES and TXI (r=0.73, P<0.001), and MES and RDI (r=0.77, P<0.001).

In Kaplan-Meier analysis, MES remission (log-rank, P=0.245), UCEIS remission (log-rank, P=0.701), and RDI remission (log-rank, P=0.827) were not significantly associated with subsequent maintenance of clinical remission. On the other hand, TXI remission (log-rank, P=0.045) was significantly associated with subsequent maintenance of clinical remission.

Conclusion: TXI score and RDI score were significantly associated with existing endoscopic score. In particular, TXI remission was significantly associated with subsequent maintenance of clinical remission. Novel IEE may be useful option in prediction of maintenance of remission in UC patients.

Disclosure: Nothing to disclose.

PP0463

PREVALENCE OF SARCOPENIA IN A COHORT OF PATIENTS WITH INFLAMMATORY BOWEL DISEASE: OBSERVATIONAL STUDY IN A TERTIARY CARE CENTRE

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Introduction: Malnutrition is a common condition of patients with inflammatory bowel disease (IBD) and could be associated with sarcopenia i.e. a reduced muscle mass and strength. Recent data suggest that the prevalence of sarcopenia is, respectively, 52% and 37% in Crohn's disease (CD) and ulcerative colitis (UC). It is well demonstrated that sarcopenic IBD patients have two to three times higher risk of lost of response to biotechnology drugs than non-sarcopenic patients.

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Aims & Methods: Consecutive IBD patients accessing the IBD outpatient service were enrolled. Detailed data regarding the characteristics of MC and CU populations, current therapy and disease activity are reported. Each patient was interviewed about the frequency (none, low, high) and the kind (aerobic or anaerobic) of the physical activity they were practicing.

At the enrolment, nutritional assessments were performed by body mass index (BMI) evaluation according with the World Health Organization recommendations, global subjective assessment by the MUST (Malnutrition Universal Screening Tool) index according with the European Society for Clinical Nutrition and Metabolism and the Global Leadership Initiative on Malnutrition [GLIM]. Moreover, the SARC-F questionnaire, recommended by the scientific consensus EWGSOP 2 to identify individuals at risk for sarcopenia was administrated. A SARC-F value ≥ 4 was significant of sarcopenia. Finally, the quality of life was assessed using the inflammatory bowel disease questionnaire (IBDQ). Univariate and multivariate analysis was conducted between BMI, MUST, SARC-F and IBDQ values and patient characteristics (age, current therapy, disease activity and physical activity).

Results: A total of 321 IBD patients (158 CD and 163 UC), 56.07% male, with a mean age of 44.78 years were enrolled. Sixty-seven patients (20.87%) reported a current smoking habit. A mean BMI of 70.37 kg, was calculated. Regarding malnutrition, 199 patients (61.99%) were considered at low risk (MUST=0), 75 subjects (23.36%) at medium risk (MUST=1), 47 (14.64%) at high risk (MUST ≥2). A SARC-F value ≥4 was evaluated in 50 patients (15.58%). A reduction in quality of life, understood as IBDQ < 209, was found in 45 (14.02%) patients. Females and males presented a significantly different prevalence of sarcopenia (chi2: 9.69; p-value: 0.002). Univariate analysis shows a significant protective role of male sex (OR vs. female sex: 0.33; p-value: 0.018) in the risk of sarcopenia; older subjects had a slightly higher sarcopenic risk (OR: 1.04; p-value <0.001). The risk was associated with corticosteroid therapy (OR 1,03, p-value <0.001), with increasing MUST (OR 1,65, p-value <0.001) and with the IBDQ decreases (OR 2,1; p-value<0.001).

Multivariate analysis confirmed an higher incidence of sarcopenia in female subjects of advanced age with active-stage disease (R^2: 0.88, p-value<0.0001), furthermore female patients with active-stage disease who did not engage in physical activity had a greater impairment in quality of life, regardless of age (R^2: 0.75, p-value<0.0001). Aerobic and anaerobic physical activity played a protective role in the onset of sarcopenia regardless of sex and age (R^2: 0.82, p-value<0.0001).

Conclusion: In our cohort of patients, sarcopenia was present in only 18.94%. Male sex was found to be a protective factor for sarcopenia. Disease status is firmly linked to the risk of sarcopenia. Regular physical activity has been shown to have a protective role for good quality of life and it was related with a lower risk of sarcopenia. This study will continue to expand the cohort taken into account.

Disclosure: Nothing to disclose.

PP0464

TRANSMURAL HEALING ON INTESTINAL ULTRASOUND IS ASSOCIATED WITH IMPROVED LONG TERM OUTCOMES IN ULCERATIVE COLITIS

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Introduction: Recently, endoscopic healing (EH) has been recognized as a long term treatment goal for ulcerative colitis (UC)[1]. It has been demonstrated that UC does not often remain limited to the mucosa but affects all layers of the bowel[2]. Intestinal ultrasound (IUS) is a useful tool to assess transmural disease in UC[3].

Aims & Methods: We performed a retrospective study on UC patients in remission and on stable maintenance treatment with EH (endoscopic Mayo score of ≤ 1) who had an IUS performed between 2 weeks before and 6 months after endoscopic evaluation and had clinical follow up data available for at least 6 months. Patients were identified from 4 prospective cohorts (January 1, 2017 up till December 31, 2023). Clinical follow up data were analysed from date of endoscopy till December 31, 2023.

Transmural healing (TH) measured with IUS was defined as bowel wall thickness(BWT) ≤ 3mm and modified Limberg Doppler score of ≤ 1 on all colonic segments except for rectum. Histology data (healing defined as absence of granulocytes) was also obtained.

Primary outcome was occurrence of any UC related event: documented relapse, initiation of corticosteroids, intensification/change of medical therapy, hospitalization and/or surgery. Patients were censored at first event and time to event analyses with Cox regression was used to determine the association between TH, EH, histological healing and UC related event.

Characteristic	Transmural Healing (n=30)	Non Transmural Healing (n=18)	Overall (n=48)	p value
Gender (male, %)	15 (50.0)	8 (44.4)	23 (47.9)	0.71
Duration of disease (years, median, IQR)	8.6 (3.2, 11.7)	7.6 (4.1, 16.7)	7.9 (3.6, 17.1)	0.77
Age at baseline (years, median, IQR)	33 (27, 46)	42 (27, 55)	38 (27, 50)	0.37
C-Reactive Protein (mg/L, median, IQR)	0.6 (0.3, 0.9)	1.3 (0.4, 4.4)	0.7 (0.3, 9.0)	0.22
Number of prior biologics used (%)				
0	12 (40.0)	1 (0)	13 (27.1)	0.09
1	8 (26.7)	9 (50.0)	17 (35.4)	
2	5 (16.7)	5 (27.8)	10 (20.1)	
≥ 3	5 (16.7)	3 (16.7)	8 (16.7)	
Previous anti-TNF use (%)				
Yes	16 (53.3)	14 (77.8)	30 (62.5)	0.09
No	14 (46.7)	4 (22.2)	18 (37.5)	
Current Medications (%)				
5-Aminosalicylate	7 (23.3)	5 (27.8)	12 (25.0)	0.88
Thiopurines / Methotrexate	2 (6.7)	1 (5.6)	3 (6.3)	
Anti-TNF (Infliximab / Adalimumab)	7 (23.3)	4 (22.2)	11 (22.9)	
Vedolizumab	7 (23.3)	0 (0)	7 (14.6)	
Ustekinumab	0 (0)	2 (11.1)	2 (4.2)	
Jak Inhibitors	13 (43.3)	11 (61.1)	24 ((50.0)	
UC Extent (Montreal classification) (%)				
E1 Ulcerative proctitis	2 (6.7)	2 (11.1)	4 (8.3)	0.82
E2 Left-sided UC	12 (40.0)	6 (33.3)	18 (37.5)	
E3 Pancolitis	16 (53.3)	10 (55.6)	26 (54.2)	
Endoscopic Mayo Score (%)				
0	12 (40.0)	6 (33.3)	18 (37.5)	0.64
1	18 (60.0)	12 (66.7)	30 (62.5)	

Table 1.

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Results: There were 48 patients available for the current analysis. Baseline characteristics are shown in Table 1. The mean duration of follow up was 38.1±13.1 months. Primary outcome occurred in 18/48 patients (37.5%; TH=5, no TH=13). All patients had a clinical relapse as the first UC related event. Subsequently, 4 patients were initiated with corticosteroids, 13 patients had intensification/change of medical therapy and 1 patient had a colectomy. The mean time to event was numerically shorter in patients without TH than in those with TH (67.9±46.4 weeks vs 92.4±61.2 weeks; *p=0.370*)

Univariate Cox regression showed that TH (HR 0.17, 95% CI 0.06-0.48), endoscopic Mayo score of 0 (HR 0.52, 95% 0.18-1.48) and histological healing (n = 31, HR 0.96, 95% CI 0.26-3.59) were associated with absence of UC related event. However, in multivariate analysis adjusting for age, gender, duration of disease and number of previous biologics and anti-TNF exposure, only TH remained a significant factor (*p=0.025*) associated with event-free long term outcome. Similar results were observed in analysis involving BWT ≤ 3mm with modified Limberg Doppler score 0.

Conclusion: TH is an independent predictor of favorable long-term outcome for UC patients with EH. This novel finding is currently undergoing validation in larger cohorts.

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PP0465

ANTI-INTEGRIN AVB6 AUTOANTIBODIES PREDICT RESPONSE TO ADVANCED THERAPY IN ULCERATIVE COLITIS

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Introduction: Precision medicine relies on accurate prediction of treatment response. Anti-integrin αvβ6 autoantibodies serve as a serum diagnostic biomarker for ulcerative colitis (UC) and associated with adverse UC-related outcomes.[1]

Aims & Methods: We aimed to investigate the association between anti-αvβ6 autoantibody titers and treatment outcomes of advanced therapies. This study was conducted as a part of the prospective biobank in two centers in Japan and Israel. Patients with active UC who received advanced therapies between June 2017 and February 2023 were prospectively included and followed up for 1 year.

The primary and secondary outcomes were treatment persistence rate for 1 year and clinical remission rate at week 6 in the anti-αvβ6 autoantibody low-titer versus high-titer groups, respectively. Treatment persistency was defined as continuation without switching to another advanced therapy or UC-related surgery, and clinical remission was defined as a two-item patient-reported outcome of ≤ 1 (rectal bleeding = 0 and stool frequency ≤ 1).

The cumulative treatment persistence rate was analyzed using the log-rank test. Cox proportional hazards models adjusted for confounders were used to estimate hazard ratios (HRs) and 95% confidence intervals (CIs) for treatment persistence.

Week 6 remission rate was presented as numbers (%) and differences were analyzed using the Chi square test. Logistic regression models adjusted for confounders were used to estimate odds ratios (ORs) and 95% CIs for Week 6 remission.

Results: A total of 102 patients were enrolled in the cohort (median age, 41 years; male, 55.9%; median disease duration, 78.5 months; proctitis, 1%, left-sided colitis, 12.7%, extensive colitis, 86.3%; prior use of advanced therapies, 56.9%, median modified Mayo score, 7). The median observation period was 10 months (interquartile range [IQR]: 2-12 months), with treatment discontinuation occurring in 49 (48.0%) patients.

The cumulative treatment persistence rate was significantly lower in the high-titer(>128 U/mL) compared to the low-titer (≤128 U/mL) group (p = 0.03).

Multivariate Cox proportional hazards model analysis indicated that anti-αvβ6 autoantibodies were the sole independent predictor of treatment persistence (HR, 2.03 [95% CI, 1.04-3.95], p = 0.04) (Table).

Clinical remission at Week 6 was achieved more frequently in the low-titer group than in the high-titer group (48.8% versus 18.2%, respectively, p = 0.04). Multivariate logistic regression analysis also showed that anti-αvβ6 autoantibody titer (OR, 3.52 [95% CI 1.08-11.47], p = 0.04) was a significant predictor associated with Week 6 remission, in addition to the modified Mayo score (OR, 0.35 [95% CI 0.14-0.90], p = 0.03).

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	Crude analyses		Multivariable adjusted analyses	
	HR (95% CI)	p value	HR (95% CI)	p value
Disease duration				
3 years	1.00 (reference)		1.00 (reference)	
>3 years	0.99 (0.53-1.84)	0.97	0.77 (0.38-1.53)	0.46
Disease extent				
Proctitis or left side colitis	1.00 (reference)		1.00 (reference)	
Extensive colitis	0.60 (0.24-1.52)	0.28	0.85 (0.32-2.29)	0.75
History of advanced therapies use				
Advanced therapy-naïve	1.00 (reference)		1.00 (reference)	
Prior use of advanced therapies	0.70 (0.39-1.27)	0.24	0.71 (0.38-1.34)	0.29
Modified Mayo score				
Moderate	1.00 (reference)		1.00 (reference)	
Severe	0.55 (0.30-1.00)	0.05	0.58 (0.31-1.08)	0.09
CRP (per 1mg/dL increase)	1.00 (0.89-1.15)	0.98	1.02 (0.91-1.21)	0.69
Anti-αvβ6 autoantibody titer				
128 U/mL	1.93 (reference)		1.00 (reference)	
>128 U/mL	1.00 (reference)	0.03	1.00 (reference)	0.04

The multivariable Cox proportional hazards analysis included 5 likely confounders: disease duration, disease extent, history of advanced therapies use, modified Mayo score, and CRP.
Abbreviations: CI, confidence interval; CRP, C-reactive protein; HR, hazard ratio.
The modified Mayo score is the sum of the Mayo stool frequency, rectal bleeding, and endoscopic subscores, with a maximum total score of 9. On the modified Mayo scoring system, a score of 4 to 6 indicates moderate ulcerative colitis, and a score of 7 to 9 severe ulcerative colitis.

Table. Predictors of treatment persistence by Cox proportional hazards model analysis.

Conclusion: Anti-αvβ6 autoantibodies are valuable for predicting the outcomes of advanced therapy in active UC.

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PP0466

INADEQUATE VACCINATION RATES AGAINST HEPATITIS B AMONG PATIENTS WITH INFLAMMATORY BOWEL DISEASE IN GREECE: A SINGLE-CENTER STUDY

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Introduction: Although there is a childhood vaccination program against hepatitis B virus (HBV) in Greece, vaccination rates are thought to be suboptimal both in general population and in inflammatory bowel disease (IBD) patients[1,2].

Wide use of new immunosuppressant and biologic therapy could lead to an oppurtunistic infection. As it is recommended by European Crohn's and Colitis Organisation (ECCO) for prevention of HBV infection, patients should be serologically screened at baseline, before and during an immunosuppressive therapy[3] and be appropriately vaccinated. Vaccination is until now the most effective way in order to protect from HBV offering longtime protection. Lower successful immunization rates are seen after induction of immunosuppressant therapy carrying a higher risk of a current HBV infection and complications . There are limited epidemiological data about successful immunization, among Greek patients with inflammatory bowel disease, prior to immunosuppressant therapy initiation[1].

Aims & Methods: The aim was to detect the vaccination rate against HBV among IBD patients who are going to be treated with immunosuppressant therapy. Retrospectively collected data from medical records of IBD patients in a single referral IBD center between 2017 and 2023 were analyzed. Inclusion criteria were IBD patients prior to immunosuppressant therapy initiation with no natural immunity to HBV, absence of hepatitis B surface antigen (HBsAg), absence of isolated hepatitis B core antibody (anti-HBc) and any status of hepatitis B surface antibody (anti-HBs).

Results: In total 126 patients were included in the study with average age 46 years old (+/- 28 years old). 42.86% (54/126) of them were women and 57.14% (72/126) were men. 52.38% (66/126) had been diagnosed with Crohn's disease and 47.61% (60/126) with ulcerative colitis. In total 40,48% (51/126) had positive anti-HBs, because of successful immunization against HBV and 59.52% (75/126) were seronegative.

Conclusion: Our study confirms inadequate immunization results against hepatitis B among IBD patients in Greece. Intensive care should be given in vaccination protocols and serology testing especially in IBD patients that are going to start immunosuppressive agents. Primary care and IBD specialist should be well informed about serology testing and appropriate vaccination against hepatitis B, carefully advise and strongly recommend vaccination in IBD patients.

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PP0467

NEW BLOOD-BASED TEST, 7-ALPHA-HYDROXY-4-CHOLESTEN-3-ONE (7ALPHAC4), FOR BILE ACID MALABSORPTION IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE OR UNSPECIFIED DIARRHEA

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Introduction: Serum 7AlphaC4 (7-alpha-hydroxy-4-cholesten-3-one) is a diagnostic test for bile acid (BA) malabsorption (BAM) or bile acid diarrhea (BAD) and the only available blood-based test for BAM/BAD. Concentrations >48 ng/mL have a high positive predictive value (PPV) of 82% for BAD and high likelihood of responding to BA sequestrants.[1]
In BAM, less BAs re-enter enterohepatic circulation and more BAs spill into the colon increasing motility and secretion. Less reabsorbed BA means less negative feedback and an ensuing increase in hepatic BA synthesis. As BA synthesis increases, blood levels of 7AlphaC4, a BA precursor, also increase.
This 7AlphaC4 test performed on a random blood collection as a surrogate measure of stool BA aims to simplify the investigation and thereby, improve recognition of BAM/BAD which may often be overlooked in IBD and other patients with more than one possible etiology of diarrhea.
Aims & Methods: We assessed the distribution of numeric serum 7AlphaC4 results (in ng/mL) in 712 patient samples using an in-house developed method utilizing protein precipitation followed by liquid chromatography with tandem mass spectrometry. Reference range of this validated assay is 1.8 – 57 as determined using sera of 120 non-disease individuals.
Results: Analysis of 712 patient samples yielded a mean (± standard deviation) 7AlphaC4 of 36.2 ± 50.1 ng/mL (range, undetectable (<0.5) to 549); 37.6% were low at <15 (associated with a NPV of 85%); 21.3% were high at >48. Most 7AlphaC4 was ordered in the general or urgent medical setting (42.8%), by gastroenterologists (27.8%) or infectious disease specialists (9.3%).
Other testing in order of frequency included: tTG IgA or other celiac tests, fecal calprotectin, fecal pancreatic elastase, stool PCR for 22 GI pathogens, fecal fat, *C. difficile* toxin, *Giardia, H.pylori*, *C.* and stool lactoferrin. Most common International Classification of Diseases (ICD) codes were: diarrhea unspecified, noninfectious gastroenteritis and colitis, abdominal distension, unspecified abdominal pain, irritable bowel syndrome with diarrhea (IBS-D), change in bowel habit, other fatigue, gastroesophageal reflux, functional diarrhea.
ICD codes are not known for all samples. However, of 50 patients with the highest 7AlphaC4 levels (>100), at least 20% had CD (ileal resection status unknown). In 86 patients with known diagnoses, those with CD with small intestine involvement had the highest mean 7AlphaC4 of 83.9 ± 77.1 (n=27) compared to 46.1 ± 83.7 in functional diarrhea (n=20), 37.3 ± 58.9 in unspecified diarrhea (n=191), 31.4 ± 22.3 in IBS-D (n=31), and 22.1 ± 10.8 in UC (n=8).

Serum 7AlphaC4 concentration (ng/mL)	Percentage of total patient results	Interpretation
< 15	37.6%	Low; Associated with 85% NPV for BAM/BAD
15 – 48	41.1%	Normal; Does not help rule in/out BAM/ BAD
> 48	21.3%	High; Associated with 82% PPV for BAM/BAD

Table 1. Distribution of All Serum 7AlphaC4 Results

Conclusion: More than half (58.9%) of all 7AlphaC4 results were low (< 15) or high (>48) and as such, potentially diagnostic, carrying a high NPV or high PPV for bile acids contributing to diarrhea. BAM/BAD may account for 30% of all chronic diarrhea cases, thereby, affecting 1-2% of the general population. Disruptions in BA homeostasis and reabsorption occur more frequently in IBD (up to 40%, with or without ileal resection).
We observed that many of the highest 7AlphaC4 levels occurred in Crohn's patients. In summary, 7AlphaC4 is a new blood-based marker that provides quantitative evidence of the contribution of colonic BA toward diarrhea (and hence, a higher likelihood of responding to BA sequestrants), especially in patients with persistent diarrhea of more than one possible etiology.
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Disclosure: Nothing to disclose.

PP0468

CHARACTERIZATION OF CARDIOVASCULAR RISK, LIVER STEATOSIS AND FIBROSIS IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Recent studies have indicated an increased prevalence of steatosis and liver fibrosis(LF), obesity and cardiovascular diseases among inflammatory Bowel disease(IBD) patients. We aimed to characterize these conditions and risk factors associated.
Aims & Methods: Cross-sectional study on patients with IBD. Pregnancy or known cause of chronic liver disease were excluded. All patients underwent transient elastography, clinical and laboratorial assessment. International Physical Activity Questionnaire(IPAQ) classified activity as low, moderate, or high. Presence of steatosis and fibrosis was defined as Controlled Attenuated Parameter(CAP) measurement of >248dB/m and Liver Stiffness Measurement(LSM) >8kPa, respectively. Fibrosis-4 score (FIB4) was also calculated. For cardiovascular risk (CVR), SCORE2 was calculated according to European Society of Cardiology (if >40 years).
Results: 79 consecutive patients were enrolled [53 Crohn's disease (CD), 24 Ulcerative colitis (UC)]. Baseline characteristics and CVR factors are presented in table 1. Median SCORE2 was 4 (IQR3-10) and 5 (IQR2-10),

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(p=0.83), conferring low cardiovascular risk; 36% of CD and 70% of UC patients had liver steatosis (p=0.04) and 11% of CD and 25% of UC had LF (p=0.13). Median LSM and CAP was 4.7, IQR (4-6) kPa vs 4.6, IQR (3.6-8.6) kPa (p=0.86) and 234, IQR (171-284) vs 217, IQR (179-294) dB/m (p=0.86), for CD and UC respectively. We found a moderate correlation between FIB4 and LSM in CD (r=0.66, p<0.001), but not in UC (r=0.37, p=0.37).
Comparison between patients with and without fibrosis in IBD showed significant differences in terms of age (47, IQR [45-66] vs 36, IQR [29-54]), p=0.01); body mass index (BMI) (31.01, [26.54-37.83] vs 24.61, [20.61-28.05], p=0.03), and history of thromboembolic diseases (23%vs 3%,p=0.007). In a multivariate regression, adjusting for clinically relevant variables, only higher BMI showed higher risk of fibrosis (OR 1.25, CI [1.05-1.48],p=0.01).

	CD	UC	p
Male,%	53%	46%	0.57
Age, median,IQR	55, (31-54)	45,(29-56)	0.42
Low IPAQ,%	87%	71%	0.13
Dyslipidemia,%	11%	8%	0.13
Diabetes,%	4%	4%	0.93
Smoker/Previous smoker,%	44%	65%	0.93
BMI, median,IQR	25.5,(21- 29)	26.53,(21-29)	0.93

Conclusion: In our population, IBD patients had a low cardiovascular risk, and 11% of CD and 25% of UC had liver fibrosis. A potential factor linking both conditions is a higher BMI, was independent risk factor for the development of fibrosis.
Disclosure: Nothing to disclose.

PP0469

ASSESSMENT OF INTESTINAL ULTRASOUND IN THE DIAGNOSIS OF POST-OPERATIVE RECURRENCE IN CROHN'S DISEASE

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Introduction: Ileocolonoscopy and Rutgeerts endoscopic score (RS) remains the gold standard to evaluate post-operative recurrence (POR) in Crohn's disease (CD). Intestinal ultrasound (IUS) is proposed as a non-invasive alternative for the diagnosis of POR, with a sensitivity of 89%, specificity of 86% and area under curve (AUC) of 0.93 (1).
Bowel wall thickness (BWT) alone has a high accuracy in the diagnosis of POR and it is proposed that the association of other parameters such as hyperemia, loss of bowel wall stratification, involvement of mesenteric fat (i-fat) or ultrasound scores, increases sensitivity and specificity in the diagnosis of POR (2).
Aims & Methods: The aim of this study is to assess whether IUS parameters and ultrasound scores correlate with endoscopy in the diagnosis of POR. A unicentric retrospective study was performed. There were included patients with CD with both an ileocolonoscopy and intestinal ultrasound performed for the detection of POR, the time between tests was less than 6 months or there was no therapeutic change between tests.
Endoscopic POR was evaluated with Rutgeers score (RS), RS i0 and i1 was considered no POR and RS ≥i2a was POR. The IUS parameters were: BWT, Limberg Score, bowel wall stratification, inflammation of mesenteric fat, Simple US Score, SUS-CD and IBUS-SAS score.
Results: 129 patients were included, baseline characteristics are in Table 1. In 108 patients (83.7%) the time between test was <6 months and in 21 (16.3%) >6 months without therapeutic changes. 38 (29.7%) were classified as endoscopic POR (RS i0 – i1) and 90 (69.8%) as POR: 28 (21.2%) i2a, 17 (13.3%) i2b and 45 (35.1%) severe (RS i3 – i4). With IUS, 42 patients (32.6%) were classified as no POR and 87 (67.4%) as POR. The mean BWT by IUS in patients without endoscopic POR was 2.6 mm (SD +/- 1.2) versus 4.8 (SD +/- 1.6) in patients with RS ≥i2a (p 0.001). BWT ≥3 had a sensitivity of 86.6%, specificity 78.9% and AUC of 0.83 in the diagnosis of POR. Hyperemia (Limberg ≥1) was present in 75 patients (84.3%) with RS ≥i2a versus in 8 patients (21.1%) without endoscopic POR (p 0.001). 21 patients (23.9%) of patients with RS ≥i2a presented loss of bowel wall stratification versus 0 (0%) with i0-i1 (p 0.001). I-fat was present in 16 patients (12.4%), all of them RS ≥i2a (p 0.02).
The mean results of IUS scores were: Simple Us score 5.6 (SD +/- 2.9), SUS-CD 2.1 (SD +/- 1.8) and IBUS-SAS 31.6 (SD +/- 23.3). In patients without endoscopic POR the mean result of Simple US Score was 3.0 (SD +/- 2.0) versus 6.7 (SD +/- 2.4) in RS ≥i2a (p>0.05). The mean result of SUS-CD without POR was 0.5 (SD +/ 1.0) versus 2.8 (SD +/- 1.5) in POR (p>0.05). In IBUS-SAS the mean result in RS i0-i1 was 13.4 (SD +/- 10.8) versus 39.2 (SD +/- 22.9) in RS ≥i2a (p>0.05). Simple US Score showed an AUC of 0.85 in the diagnosis of POR (95% CI 0.78 – 0.92), SUS-CD of 0.87 (95% CI 0.81 – 0.93) and IBUS-SAS of 0.85 (95% CI 0.77 – 0.92).

Variable	n 129
Men (%)	65 (50.4)
Active smoking (%)	32 (25.4)
Montreal classification (%)	
- L1	59 (45.7)
- L2	2 (1.5)
- L3	60 (46.5)
- L4	8 (6.2)
- B1	19 (14.7)
- B2	49 (38.0)
- B3	61 (47.3)
- p	8 (6.2)
Preventive treatment (%)	
- None	35 (27.1)
- Inmunomodulator	39 (30.2)
- Biological therapy	33 (25.6)
- Inmunomodulator + biological	21 (16.3)
C-reactive protein (mg/L) (mean, SD)	16.3 (127.7)
Fecal calprotectin (µg/g) (mean, SD)	192.1 (391.6)
Bowel wall thickness (mm) (mean, SD)	4.1 (1.8)
Limberg score (0-3) (mean, SD)	1.5 (1.2)
Loss of bowel wall stratification (%)	21 (16.5)

Conclusion: In our experience, IUS parameters such as BWT, Limberg score, loss of wall stratification, i-fat and IUS scores (IBUS-SAS, SUS-CD and Simple US Score), have a high accuracy in the diagnosis of POR.
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Disclosure: Nothing to disclose.

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PP0470

ASSESSMENT OF DISEASE ACTIVITY IN PEDIATRIC INFLAMMATORY BOWEL DISEASE USING INTESTINAL ULTRASOUND: A COMPARATIVE DIAGNOSTIC ANALYSIS

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Introduction: Intestinal ultrasound (IUS) is increasingly used as a point-of-care tool for patients with ulcerative colitis (UC). The non-invasive monitoring of pediatric patients with UC is crucial in managing disease progression and therapeutic intervention. The Milan Ultrasound criteria (MUC), the Intestinal Bowel Ultrasound Segmental Activity Score (IBUS-SAS), and the Civitelli score are promising tools for evaluating disease activity in real-time, offering a potential alternative to invasive procedures like endoscopy.

Aims & Methods: The study aimed to assess the correlation and efficacy of MUC, IBUS-SAS, and the Civitelli score in evaluating disease activity in pediatric UC patients, using endoscopic findings and histological parameters as comparative benchmarks.

A prospective cohort of 16 pediatric UC patients underwent IUS and colonoscopy to assess disease activity using the MUC, IBUS-SAS, and Civitelli scores in relation to endoscopic findings and histological evaluation using the Nancy Index. Endoscopy and ultrasound examinations were performed within a 7-day interval, without any change in therapy between assessments. The diagnostic accuracy was evaluated through receiver operating characteristic (ROC) analysis, considering endoscopy as the reference standard.

Results: The study population consisted of 16 pediatric patients with UC (average age 12 years, 56% male), presenting a range of disease severities and durations. The area under the curve (AUC) for MUC was 0.615, IBUS-SAS was 0.667, and Civitelli was 0.603, indicating moderate diagnostic accuracy. Although no significant differences were observed in ROC curve comparisons among these scores, IBUS-SAS demonstrated the highest accuracy. Regarding histological parameters, the AUC was 0.500 for MUC, 0.579 for Civitelli, and 0.563 for IBUS-SAS.

Conclusion: Although potentially less effective than in adult populations, both MUC, Civitelli score, and IBUS-SAS provide reasonable diagnostic accuracy in the assessment of pediatric UC, compared to traditional endoscopic and histologic evaluations. IBUS-SAS, in particular, may offer superior sensitivity in pediatric contexts due to its focus on vascularization and wall patterns. These features are particularly significant in pediatric cases where the disease is recent and the wall often is not very thickened, yet highly vascularized.

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Disclosure: Nothing to disclose.

PP0471

THE BURDEN OF PSYCHIATRIC MANIFESTATIONS IN INFLAMMATORY BOWEL DISEASES: A SYSTEMATIC REVIEW WITH META-ANALYSIS

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Introduction: Psychiatric manifestations in individuals with inflammatory bowel disease (IBD) have been the subject of extensive research. Still, uncertainties remain regarding its precise magnitude, its impact on intestinal disease, and the whole burden of psychiatric manifestations.

Aims & Methods: This systematic review aims to synthesize available evidence on the prevalence and impact of psychiatric symptoms, including depression, anxiety, bipolar disorders, and schizophrenia in IBD patients.

A systematic search was conducted in the PubMed/MEDLINE, EMBASE and SCOPUS databases to identify relevant studies published between January 2010 and January 2023. Cross-sectional, case-control, cohort, registry and population studies were selected that addressed the prevalence of depression, anxiety, bipolar disorder, and schizophrenia in people with IBD compared to the general population. A meta-analysis was also conducted to assess the strength of the association. Data extraction and quality assessment were performed.

Results: The search yielded a total of 3209 articles. Although 193 met the inclusion criteria, only 26 studies provided complete data. Overall, the prevalence of psychiatric comorbidities was found to be significantly higher in IBD patients compared to the general population. The analysis revealed a moderate but significant association between IBD and depression, with a calculated pooled odds ratio (OR) of 1.42 (p<.0001).

Also, anxiety disorders emerged as a prevalent comorbidity in IBD with a calculated pooled OR of 1.3 (p<.0001). Evidence for the association between bipolar disorders and IBD was limited but significant, with a pooled OR of 1.64 (p<.0001), despite a substantial heterogeneity (I2=94.01%). Only 3 studies examined the association between schizophrenia and IBD, providing widely heterogeneous results, with an inconclusive OR (0.93, p= 0.73).

Conclusion: This systematic review with meta-analysis highlights that psychiatric comorbidities such as depression, anxiety, and bipolar disorders are more prevalent in IBD than in the general population. Depression and anxiety are particularly prevalent. Bipolar disorder is still an under-researched area in the context of IBD. Data on the association between schizophrenia and IBD is sparse and inconclusive and requires further research.

Given the findings of this systematic review, there is a need for a deeper understanding of these associations and early detection in order to develop tailored mental health interventions for patients with IBD.

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Disclosure: Nothing to disclose.

PP0472

WITHDRAWN

PP0473

INFLAMMATORY BOWEL DISEASE IN REMISSION IS ASSOCIATED WITH MINOR COGNITIVE IMPAIRMENT COMPARED TO HEALTHY CONTROLS

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Introduction: Compelling evidence suggests that inflammatory bowel disease (IBD) can affect the brain. IBD can increase the risk of developing mood and behavioural disorders or neurological diseases. However, few studies assessed cognitive function in such patients, yielding inconsistent results.
Aims & Methods: To evaluate minor cognitive impairment in IBD patients compared with age, sex, and education-matched healthy controls (HC). We enrolled adult (18-65 years) IBD patients referred to our tertiary centre. Clinical remission was defined as Harvey-Bradshaw Index <5 for Crohn's disease (CD) and Partial Mayo score <2 for Ulcerative colitis (UC), respectively. Healthy subjects were recruited from friends and relatives of physicians involved in the study. Exclusion criteria were psychiatric or neurologic diseases, language difficulties, sleep disorders, ileostomy or colostomy, severe comorbidities, alcohol abuse, and use of corticosteroid or other medications with known effects on neuropsychological functions in the three months before enrolment. Cognitive function was assessed by using the Montreal Cognitive Assessment (MoCA) test. A MoCA score greater than 26 was considered normal. To rule out the potential impact on cognitive abilities of conditions such as anxiety and depression, we screened patients using the Hospital Anxiety and Depression Scale (HADS) test. A score equal to or greater than 8 on either HADS-D or HADS-A is considered a 'possible case', and above 11 a 'probable' case. Demographic and clinical data were also recorded. The Mann-Whitney rank-sum or t-test and Chi-square or Fisher's exact test were used to compare continuous or categorical variables, respectively.
A univariate analysis was performed to establish whether baseline clinic psychometrics parameters contributed to predicting MoCa score below the lower limit of normal. Multivariate analysis was then performed by logistic regression with the calculation of odds ratio (OR) and a confidence interval (CI) of 95%.
Results: Ninety-four IBD patients [48 CD and 46 UC] and 68 HC were finally considered for statistical analysis. No significant differences in terms of sex (p=0.224), age (p=0.243), and years of schooling (p=0.279) were observed between patients with IBD and HC. A significantly higher percentage of IBD patients had MoCA scores below the lower limit of normal in comparison to HC (57% vs 42%, p=0.037). A similar result was obtained also after excluding patients with probable anxiety and depression as assessed with HADS (42% vs 25%, p=0.025). UC patients had significantly lower mean scores in the memory domain than HC (3.1±1.4 vs 2.5±1.5, p=0.041). Age (p=0.036), years of schooling (p=0.002), and UC (p=0.037) were significantly associated with a MoCA score below the lower limit of normal. At multivariate analysis, age (OR 1.04, 95%CI 1-1.1, p=0.017) and UC (OR 2,2, 95%CI 1.4-5.3, p=0.047) were the only variables that independently predicted a MoCA score below the lower limit of normal.
Conclusion: IBD patients had minor cognitive impairment in a percentage significantly higher than controls as assessed by MoCA. UC patients showed deterioration of the memory domain. Our data support the existence of a gut-brain axis and suggest that colonic rather than small bowel inflammation could be the pathogenetic link for cognitive impairment in IBD. A multidisciplinary approach is required for IBD patients, which should also include cognitive function assessment.
Disclosure: Nothing to disclose.

PP0474

EVALUATION OF THE RESULTS OF A MICROLEARNING PROGRAMME FOR PATIENTS WITH INFLAMMATORY BOWEL DISEASE IN IMPROVING THERAPEUTIC ADHERENCE. THE ADEII PROJECT

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Introduction: Information and communication technologies (ICTs) represent a novel approach to chronic disease management. However, their use in Inflammatory Bowel Disease (IBD) is scarce as no specific tools are available. Our aim was to develop a microlearning programme for IBD patients and to evaluate their outcomes with respect to adherence, medication beliefs and perception of the disease.
Aims & Methods: Prospective cohort study in adult IBD patients (18-70 years) who underwent a microlearning programme over 2 months, through 4-minute videos organised into individualised video pathways that were sent via mobile phone using a Telegram bot. All patients received the main pack of 4 videos, consisting of 2 videos with information about their disease (ulcerative colitis or Crohn's disease), 1 video on adherence and 1 video on corticosteroid treatment. We then add a video for each medication used by the patient in the management of their IBD. Before starting the programme and at 6 months, patients completed the BMQ (Beliefs Medicines Questionnaire) and Brief Illness Perception Questionnaire (BIPQ). We checked the medication possession rate (MPR) at these same intervals, considering a MPR ≥80% as adequate adherence.
Results: Between June and October 2022, 1226 videos were sent to 200 IBD patients: 51.5% female (103), Ulcerative colitis 52.5% (105), median age 46.1 (13.2) years (Table 1). The median of videos per patient was 6 (6-7). Attendance to the ADEII programme ranged from 78% to 90%. Baseline non-adherence in this cohort was 29.8% (59 patients), decreasing significantly to 18.1% (36) after the programme (p=0.02). Likewise, in the BMQ scales, we found significant decreases in the median and IQR of the BMQ scores : Medication Abuse: 8 (7-9) vs 8 (6-9) (p=0.001), BMQ-Medication Harm 10 (8 -12) vs 9.5 (7-11) (p=0.002) and BMQ-Treatment Concern 13.5 (11-16) vs 13 (10-16) (p=0.001), with no change in BMQ-Treatment Need. Regarding illness perception, on the post-intervention BIPQ scale there were significant decreases in P8 Negative Emotions 6 (3-8) vs 5 (3-7) (p=0.003) and P5 Identity 6 (4-8) vs 5 (3-7) (p=0.003), as well as a significant increase in P7 Understanding of illness 7 (5-9) vs 8 (7-9) (p=0.003).

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	Ulcerative colitis (105)	Crohn´s Disease (95)	IBD (200)
Male / Female	59 (56,2%) 46 (43,8%)	38 (40%) 57 (60%)	97 (48,5%) 103 (51,5%)
Age	47 (12,7)	45,1 (13,7)	46,1(13,2)
Studies: Primary school/ Secondary / University	25 (24%) 57 (54,8%) 22 (21,2%)	23 (24,2%) 49 (51,6%) 23 (24,2%)	48 (24,1%) 106 (53,3%) 45 (22,6%)
Years from diagnosis	10,8 (8,2)	15,2 (11,2)	13(10)
UC E1/ E2/ E3	21 / 47 / 36		
CD A1/ A2/ A3 CD: L1/ L2/ L3 CD: B1/ B2/ B3		5 / 77 / 13 52 /13 / 30 49 / 27 / 19	
EIMs	9 (8,6%)	19 (20%)	28 (14%)
Perianal disease	7 (6,7%)	22 (23,2%)	29 (14,5%)
Surgery No One / ≥2	101 (96,2%) 4 (3,8%)	68 (71,6%) 19 (20%) 8 (8,4%)	169 (84,5%) 23 (11,4%) 8 (4%)

Conclusion: The ICT-based ADEII educational programme was widely accepted by patients, demonstrating an improvement in adherence to treatment with a decrease in negative beliefs about medication and negative emotions and representations, improving the understanding of the disease.

Disclosure: Nothing to disclose.

PP0475

HEAD TO HEAD COMPARISON OF A RAPID AND A CONVENTIONAL FECAL CALPROTECTIN KIT AS PREDICTOR OF CLINICAL SCORE IN ULCERATIVE COLITIS AND CROHN'S DISEASE

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Introduction: Measurements of fecal (fe) calprotectin as a surrogate biomarker of gastrointestinal inflammation have great clinical impact in the judgement of clinical activity in inflammatory bowel disease. There is no accepted standardization procedure of ELISA-based calprotectin kits, but several commercial kits use of a combined pioneering polyclonal calprotectin antibody (CA) (1) and a monoclonal CA.

Aims & Methods: To evaluate the accuracy of a new rapid fe-calprotectin kit (RFC) including a polyclonal-based CA to predict clinical activity in comparison to a standard commercial (SCFC) kit in patients with ulcerative colitis (UC) and Crohn's disease (CD) with special focus on clinical remission.

Included were 16 UC and 20 CD patients on ongoing treatment from May 2022 to March 2024 (5ASA, azathioprine, biological therapy). The degree of illness was evaluated at regularly visits every three months using the clinical scoring system- Ulcerative Colitits Clinical Score (UCCS), and Harvey-Bradshaw index (HBI). From the same fe-sample RFC analyses were performed at the Sami Klinikkha, Finnmark Hospital and the SCFC analyses at the University Hospital Tromsø, Norway, both with an upper normal level of 50 µg/g.

Statistical evaluation was done by correlation and limits of agreement analysis; cut-off levels were explored by Receiver Operating Characteristic (ROC) analysis against clinically defined remission

Results: The intra-assay variation of the RFC was 5.3% at a level of 133 µg/g.

Mean calprotectin values (µg/g) with 95 % confidential interval (CI) are shown at various degrees of clinical activities during the observation time (remission, slight and mild), Table 1. In general, the RFC measured lower means and CI levels compared to the SCFC. In clinical remission, the upper RFC CI interval were in proximity to the upper normal level, whereas the corresponding SCFC CI levels were almost three times higher than the upper normal level.

A ROC curve was performed for prediction of clinical remission. Both assays had significant area under the curve (AUC) in both UC and CD patients (p<0.0001). The RFC kit predicted endoscopic remission with 100% specificity at cut-off levels 139 µg/g in both UC and CD. The specificity of the SCFC kit was lower applying the same cut-off levels. Increasing the cut-off to 246 µg/g in the SCFC kit (Youden's J index 0.68 in the UC group and 0.49 in the CD group), the specificity increased to 96.3 and 96.4% in the UC and CD group, respectively.

On the other hand, the test sensitivity decreased (71.4% in UC and 52.2% in the CD patient group).

	Remission UC n=27	Slight UC 10	Moderate UC 8	Remission CD 28	Slight CD 27	Moderate CD 10
RFC	60 (mean) 45-69 (CI)	202 170-232	365 161-569	69 54-84	206 175-236	433 283-583
SCFC	104 69-139	325 187-436	677 143-1209	110 82-138	234 183-286	558 314-801

Table 1.

Conclusion: The rapid fe-calprotectin test is an accurate test and predicts remission in UC and CD patients better than a conventional commercial fe-calprotectin kit

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PP0476

BOWEL URGENCY IN ULCERATIVE COLITIS PATIENTS: PREVALENCE AND ITS ASSOCIATION WITH CLINICAL, ENDOSCOPIC DATA, AND DISABILITY IN A REAL-WORLD SETTING

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Introduction: Bowel Urgency (BU), identified as the sudden need for bowel movements, is increasingly being recognized as an impactful symptom in patients with Ulcerative Colitis (UC), distinct from Stool Frequency (SF) and Rectal Bleeding (RB).

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Aims & Methods: We aimed to assess the association of bowel urgency (BU) with Patient-Reported Outcomes (PRO), including stool frequency (SF) and rectal bleeding (RB), along with endoscopic findings in a real-world setting. Additionally, we investigated its impact on disability and quality of life.

A cross-sectional multicenter study, which consecutively enrolled UC patients aged ≥ 18 years, was conducted between February 2023 and October 2023 using data from the 'IBD-Disk: Italian translation and validation study. Demographic, clinical and endoscopic data were collected. Clinical remission was defined as partial Mayo score ≤ 1 and endoscopic activity as MES ≥ 2. All patients completed the following questionnaire for disability IBD-Disk and IBD Questionnaire-32 (IBDQ-32) for quality of life. BU was assessed using the question about regulating defecation in the past 7 days using a 10-point NRS ranging from 0 (absolutely disagree) to 10 (totally agree) from the IBD-Disk. RB was measured from question 22 of IBDQ and SF from question 1 of IBDQ-32, both using a 7-point NRS ranging from 0 (always) to 7 (never).

Results: 209 UC patients with a median age of 37 (IQR 28-52) were enrolled, 111 (53.1%) being female. Overall, 31 (14.8%) had proctitis, 91 (45.5%) had left colitis and 87 (41.6%) had extended/pancolitis. The median pMAYO score was 4 (IQR=2-6), while the median MES was 2 (IQR=1-2). 75% patients (35.9%) reporting BU were more likely to have moderate clinical (pMAYO=5; IQR=3-6) and endoscopic activity of disease (median MES=2; IQR=1-2) compared to those without urgency (p< .0001). Additionally, BU was associated with a higher median rate of hospitalizations in the previous 12 months (Δ=1; p<0.001). Patients with BU reported a higher degree of RB (33.3%) and SF (50,7%) than patients without BU (p< .0001). Noteworthy, individuals with BU had higher median IBD-Disk total score (29; IQR=9-52) and lower scores for IBDQ-32 (168; IQR=138-196) compared to patients without BU [median IBD-Disk=17 (IQR=5-39); median IBDQ-32=181, IQR=158-202), p<0.001]. In the univariate analysis, BU was associated with a higher IBD-Disk total score (OR=1.04, CI=1.02-1.06; p<0.001) and SF (OR=0.59, CI=0.45-0.78; p=0.001), without being associated with RB (p=0.57) and IBD-Q32 total score (p=0.84).

	BU (n=75)	No-BU (n=134)	OR (CI 95%)	p value
pMAYO score, median (IQR)	5 (3-6)	1 (0-3)		p<0.001
pMAYO >1, n (%)	60 (80,0%)	62 (46,3%)	4,6 (2,4-9,0)	p<0.001
MES, median (IQR)	2 (1-2)	1 (1-2)		p<0.001
MES ≥2, n (%)	53 (70,7%)	54 (40,3%)	3,6 (1,9-6,5)	p<0.001
Relapes in prev 12 months, median (IQR)	1 (0-1)	0 (0-1)		p<0.001
RB, n (%)	25 (33,3%)	9 (6,7%)	6,9 (3,0-15,9)	p<0.001
SF, n (%)	38 (50,7%)	12 (8,9%)	10,4 (4,9-22,0)	p<0.001
IBD Disk, median (IQR)	29 (9-52)	17 (5-39)		p<0.001
IBDQ-32, median (IQR)	168 (138-196)	181 (158-202)		p<0.001

Conclusion: BU can be a surrogate marker for UC clinical and endoscopic activity. Our findings suggest that BU is an independent factor from RB and it is associated with disability. Hence, it can be considered as an additional key PRO in UC. However, a systematic standardized approach for BU definition and assessment is strongly awaited to incorporate it as treat-to-target endpoint.

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PP0477

EXTERNAL VALIDATION OF THE ELIAKIM SCORE IN CAPSULE PANENDOSCOPY: AN EFFECTIVE TOOL FOR DIAGNOSIS AND MONITORING OF CROHN'S DISEASE

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Introduction: Capsule panendoscopy is increasingly used in Crohn's disease (CD), particularly in patients with extensive small bowel and colonic involvement. It offers a non-invasive, one-step evaluation of both segments, making it appealing for disease monitoring and treatment response. However, there is a need for validated scoring systems that objectively assess inflammatory activity in both the small bowel and the colon.

Aims & Methods: The aim of this study was to externally validate the Eliakim Score (Elk-S), developed to assess mucosal inflammation in CD by capsule panendoscopy. Patients referred for PillCam™ Crohn's examination at our institution were selected, and procedure data were collected retrospectively to calculate the Elk-S.

Results: A total of 72 patients were included, 62.5% female, with a mean age of 42±15 years. In 52 (72.2%) individuals, the indication for the procedure was suspected CD. Among them, the diagnosis was rejected in 38 cases, while in 14 cases, the diagnosis was established. There was a significant difference in mean Elk-S between the two groups (5.3±1.5 vs. 21.8±5.3, respectively; p<0.001). Capsule panendoscopy was used to evaluate inflammatory activity in 20 (27.8%) patients with CD. Treatment remained unchanged in 10 cases, while therapy was intensified in 10 patients, and there was a significant difference in mean Elk-S between the two groups (2.2±0.5 vs. 17.1±2, respectively; p<0.001). The Elk-S demonstrated an area under the ROC curve of 0.979 for the diagnosis of CD and 0.875 for predicting therapeutic escalation. An Elk-S ≥ 7 (Table 1) had a Sensitivity, Specificity, Negative Predictive Value, and Positive Predictive Value of 100%, 89.5%, 100%, 77.8%, respectively, for the diagnosis of CD. To determine the need for therapeutic intensification in patients with established CD, the same cut-off value showed a Sensitivity, Specificity, Negative Predictive Value, and Positive Predictive Value of 90%, 70%, 87.5%, 75%, respectively.

Elk-S cutoff: ≥7	Sensitivity	Specificity	NPV	PPV
CD Diagnosis	100%	89.5%	100%	77.8%
Treatment intensification	90%	70%	87.5%	75%

Table 1: Performance of the Elk-S for diagnosing CD and predicting therapeutic escalation.

Conclusion: The Elk-S exhibited high accuracy in establishing the diagnosis of CD and was equally useful in predicting the need for therapeutic intensification in cases of established CD. The present study represents the first external validation of the Elk-S, demonstrating its effectiveness for assessing inflammatory activity via capsule panendoscopy of both the small bowel and colon.

References: Eliakim R, Yablecovitch D, Lahat A, et al. A novel PillCam Crohn's capsule score (Eliakim score) for quantification of mucosal inflammation in Crohn's disease. *United Eur Gastroenterol J*. 2020;8(5):544-551. doi:10.1177/2050640620913368

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PP0478

WITHDRAWN

PP0479

A PREDICTIVE MODEL FOR HISTOLOGICAL RESPONSE TO THERAPY GUIDING THERAPEUTIC STRATEGIES IN ULCERATIVE COLITIS: A POST-HOC DATA ANALYSIS FROM A PHASE-II MIRIKIZUMAB CLINICAL TRIAL

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Introduction: Clinical trials are ongoing to develop novel therapies for ulcerative colitis (UC), aiming to break the existing therapeutic ceiling. Mirikizumab, a monoclonal antibody targeting the p-19 subunit of IL-23, was evaluated in a phase-II clinical trial for the treatment of moderate-to-severe UC. Identifying predictive variables for response to therapy is still an unmet need and would serve as a valuable asset in clinical trials and clinical practice to guide therapeutic strategies.

Aims & Methods: This post-hoc data analysis aimed to identify variables predicting response to mirikizumab in UC patients from a phase II clinical trial. One-hundred and six randomly selected patients from the induction phase were considered: 26 (24.5%) were assigned to receive mirikizumab 50mg IV Q4W, 37 to 200mg IV Q4W (34.9%), 30 to 600mg IV Q4W (28.3%) and 13 to placebo (12.3%). In the maintenance phase, 44 received 200mg SC Q12W (42.3%), 47 received 200mg SC Q4W (45.2%), and 13 received placebo (12.5%). Patients who did not achieve clinical remission (14, 13.2%) were excluded from the analysis related to the maintenance phase.

A stepwise analysis was conducted, initially using a principal coordinate analysis (PCoA) with permutational multivariate analysis of the variance (PERMANOVA), followed by a univariate analysis (Fisher's or Wilcoxon signed-rank test) to select significant variables (p<.05).

Further, a country-nested, three-visit repeated-measured generalized mixed ordinal logistic regression model (GLM) using significant PCoA variables was conducted. For both models the endpoint was histological remission at weeks 12 and 52, defined as Geboes score <2.

Results: At week 12, histological remission was achieved in 34 patients (32.1%) while at week 52 in 42 patients (39.6%). PCoA, PERMANOVA and univariate analysis considered histological remission to be significantly predicted by faecal calprotectin, endoscopic remission (Mayo 0, UCEIS ≤1) and induction therapy at weeks 12 and 52 (p<.05). Also, at week 52, the 32-IBD Questionnaire (systemic symptoms and social function), Physician Global Assessment, partial and total Mayo scores, and maintenance treatment were significant.

Considering these as explanatory variables for the GLM, we confirmed the following as significant predictors of histological remission: faecal calprotectin <250 µg/g (Odds Ratio 0.26; 95% CI 0.12-0.58; p<0.001), endoscopic remission (OR 0.16; 95% CI 0.04-0.57; p=.005), induction treatment with mirikizumab 200mg IV Q4W (OR 0.09; 95% CI 0.02- 0.45; p=.004), 600 mg IV Q4W (OR 0.09; 95% CI 0.01-0.50; p=.006) and maintenance with 200 mg SC Q4W (OR 0.05; 95% CI 0.01-0.30; p=.001). A more severe Physician Global Assessment predicted histological activity (OR 2.29; 95% CI 1.03- 5.11; p=.043).

Conclusion: Our predictive model identified faecal calprotectin, endoscopic remission, and Physician Global Assessment as predictive factors for response to therapy. Notably, our model identified the most effective therapeutic strategies to achieve histological remission during induction and maintenance phases, showing promising ability to guide therapeutic choices in clinical trials and clinical practice.

Additional investigation utilizing a larger clinical dataset would further enhance the ability of this model to identify variables predictive of response to therapy.

Disclosure: Nothing to disclose.

PP0480

CORRELATION BETWEEN PSOASIC INDEX AND DIAPHRAGMATIC INDEX ULTRASOUND IN THE EVALUATION OF SARCOPENIA IN A COHORT OF PATIENTS WITH INFLAMMATORY BOWEL DISEASE: A STUDY IN A TERTIARY REFERRAL CENTER

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Introduction: Sarcopenia is prevalent in patients with inflammatory bowel disease (IBD) and impacts surgical and therapeutic outcomes; thus, effective diagnostic tools are needed to assess muscle mass and function in IBD population. The ileo-psoas muscle (IP) muscle has already been used in the literature for the assessment of sarcopenia, however, as it is a deep muscle and it needs experienced operators for a proper measurement. Therefore, diaphragm muscle (DM) more superficial and easily assessed, was examined to check its correlation with IP.

Aims & Methods: The aim of this work was to evaluated the correlation of positivity SARC-F score with ultrasound (US) measure of the IP and DM. A total of 50 IBD patients with significant suspected sarcopenia with SARC-F score positive ≥4, were enrolled. IP and DM US were detected in the enrolled patients. The transverse diameter of the right IP was evaluated.

We defined "reliable" the measurements of the IP that fulfilled all of the following criteria: all anatomical boundaries could be clearly identified (superior margin of the iliac crest and lumbar vertebrae); the maximum diameter of the psoas between each of the three measurements did not deviate from the other two by more than 5 mm. Moreover, we calculated the US psoas/height ratio as the average of the psoas diameter divided by the patient's height (mm/m). For DM, an extended space between the axillary line and the hemi clavicular line was used, in the diaphragmatic thickness between the hyperechogenic limiting. Diameters during expiration and during inspiration were assessed, then diaphragmatic expansion was evaluated.

We calculated the diaphragm/height ratio (mm/m) as well. The correlation between IP and DM ratio was researched by using the Pearson correlation test. Age and sex were taken into consideration as confounders in a linear regression model, using the psoas/height ratio and these confounders as independent variables and the diaphragm index as dependent variable.

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Results: The mean of the three US IP measured diameters was 30.37 mm, the psoas/height ratio was 17.92 mm/m. The mean maximum inspiration DM diameter was 20.4 mm (± 5 mm), the mean minimum expiration DM diameter was 10.7 mm (± 35 mm). The mean difference between the two diameters was 9.7 mm (± 3,4 mm). The diaphragm/height ratio was 0.59 mm/m (±0.21 mm/m).

A significant correlation was identified between IP and DM ratio, with a Pearson coefficient r = 0.3568 (p-value <0.05). A consistent association was found between the two indices, with diaphragm/height ratio increasing linearly with the psoas/height ratio (OR: 0.018; p-value: 0.008). Multivariate analysis showed that age and sex were not significantly associated with the diaphragm/height ratio (p-value >0.05).

Conclusion: The findings of this study emphasize the potential of muscle US as a reliable diagnostic tool for assessing sarcopenia in IBD patients. IP US is confirmed to correlate with sarcopenia. DM can be used more easily and correlates with IP. This research could take a significant implications for disease management in IBD patients and underscores the importance of further investigations to validate these findings in larger cohorts. This study will continue to expand the cohort under evaluation.

Disclosure: Nothing to disclose.

PP0481

PREVALENCE OF SEXUAL DYSFUNCTION IN INFLAMMATORY BOWEL DISEASE: SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Patients with inflammatory bowel disease (IBD) can face various challenges, including sexual dysfunction, which refers to any physical or psychological perturbation of sexual health. This can manifest through symptoms such as reduced libido, erectile dysfunction, discomfort during intercourse, and other psychosocial issues that negatively impact patients' quality of life. However, uncertainties remain about the magnitude of this problem

Aims & Methods: Therefore, we aimed to assess prevalence of sexual dysfunction in adult patients with IBD. We conducted a systematic review and meta-analysis examining this issue. MEDLINE EMBASE, EMBASE Classic (from inception to 30th March 2023) were searched to identify observational studies that reported prevalence of sexual dysfunction in adult patients with IBD based on validated screening instruments (Female Sexual Function Index (FSFI) and International Index of Erectile Function (IIEF) for women and men respectively). Data were extracted from published study reports, and pooled prevalence, odds ratios (OR), and 95% confidence intervals (CIs) were calculated.

Results: Of 1017 citations evaluated, 18 articles fulfilled eligibility criteria, containing 2,694 patients with IBD recruited from 13 different countries. The pooled prevalence of sexual dysfunction in patients with IBD, was 50.6% (95% CI 40.8% to 60.5%; I2=96.3%). The OR for sexual dysfunction in patients with IBD versus healthy controls was 2.94 (95% CI 1.99 to 4.35, I2=73.4%). When we considered UC or CD separately, the pooled prevalence of sexual dysfunction was 64.8% (95% CI 45.1% to 82.1%; I2=88.8%) in patients with UC, and 58.3% (95% CI 36.0% to 79.0%; I2=95.3%) in patients with CD. When we conducted subgroup analyses based on sex, the pooled prevalence of sexual dysfunction was higher in females with IBD compared with males (62.7% versus 34.0%), with a four-times higher odds of sexual dysfunction. Similar trends were observed when analysing CD and UC patients separately by sex, although limited data were available for this subgroup analysis.

Finally, the pooled prevalence of sexual dysfunction was higher in patients with IBD with active disease compared with inactive disease (75.1% vs 34.2%), with a nine-fold higher odds of sexual dysfunction.

Conclusion: We demonstrated a high prevalence of symptoms of sexual dysfunction in patients with IBD, especially in women with CD. Encouraging gastroenterologists to screen for, and treat, these disorders with a holistic approach might improve quality of life of patients with IBD.

Disclosure: Nothing to disclose.

PP0482

COMBINING ULTRA-HIGH MAGNIFICATION ENDOCYTOSCOPE WITH AUTOMATED SPATIAL MULTISPECTRAL INTESTINAL BARRIER PROTEIN IMAGING FOR ASSESSMENT OF HEALING AND OUTCOMES IN ULCERATIVE COLITIS

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Introduction: Endoscopic and histologic remission are key therapeutic targets in Ulcerative Colitis (UC). Recently, intestinal barrier healing has demonstrated superiority in predicting clinical adverse outcomes. It is promising, especially in patients with primary sclerosing cholangitis (PSC) colitis, which has more aggressive disease behaviour and a higher risk of colorectal cancer.[1]

Hence, novel real-time and in-vivo tools for the intestinal barrier are pivotal.

Aims & Methods: This study aimed to assess intestinal barrier healing in UC using ultra-high magnification endocytoscope (ECS) and automated multispectral imaging. Thirty-four UC patients, 68% with PSC, in clinical remission undergoing surveillance colonoscopy, and 3 healthy controls were prospectively enrolled from two tertiary university referral centres. Ileal, ascending and descending colonic mucosa were evaluated with high-definition, virtual chromoendoscopy, and ECS, with biopsies taken from each segment.

Endoscopic remission was defined as MES ≤1, UCEIS≤1, PICaSSO≤3, while histological remission as RHI≤3 without lamina propria or epithelial neutrophils, NHI ≤1 and PHRI=0. A novel ECS score was applied to assess ileal and colonic intestinal barrier (Table 1).

Furthermore, the barrier was deeply characterised through automated multiplex immunofluorescence (inForm® Akoya Biosciences), quantifying tight junctions (TJs). Specifically, the cell density and mean expression of Claudin-2, Occludin and JAM-A were assessed in all intestinal segments.

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Patients were followed up for a median of twelve months, and adverse outcomes, including flare-ups, hospitalisation, colectomy and changes in medication, were recorded.

ECS Ileum		ECS Colon	
Normal	0	Crypts architecture	
Break		Normal, elongated	1
	1	Irregular	2
		Necrosis	3
Few gaps / drop-out	2	Infiltration of the cell between the crypts	
Multiple gasp / dropout	3	≤ 50%	1
		> 50%	2
Total Score	0 – 3	Distance between crypts	
		Normal (≥ 3 crypts in a VF)	1
		Elongated (< 2 crypts in VF)	1
		Intermediate (2-3 crypts in a VF with infiltrating cells in LP)	2
		Drop-out / necrosis	3
		Visibility of superficial microvessels	
		Not visible	0
		Visible	1
		Total score	3 – 9

Table 1. ECS score for barrier assessment.

Results: The ECS score moderately correlated with endoscopic scores (r=0.48 for MES, r=0.66 for UCEIS and r=0.63 for PICaSSO) and strongly with histological scores (r=0.69 for RHI, r=0.71 for NHI and r=0.84 for PHRI). Seven patients experienced adverse outcomes, with an ECS score <5 in the colon and <3 in the ileum predicting them (HR 2.35; p<0.01 and HR 5.78; p=0.05, respectively). All TJs were less expressed in UC patients than controls, with Claudin-2 and Occludin predominantly expressed in the epithelium and JAM-A mainly in the lamina propria. Notably, Claudin-2 cell density predicted outcomes (3370 vs 1290; p=0.01) and the ECS score in both colon and ileum correlated with Claudin-2 mean expression and cell density (ρ=.15; p=0.02 and ρ=.48, p<0.01, respectively). The combined intestinal barrier assessment by ECS and Claudin-2 cell density and mean expression significantly predicted outcomes in both colon and ileum (log rank test p<0.01).

Conclusion: ECS demonstrated promising ability to assess intestinal barrier in UC, positively correlating with TJ expression. Moreover, the novel barrier assessment through ECS complemented automated multispectral imaging in predicting adverse outcomes. These innovative tools hold promise in clinical practice for assessing barrier healing and guiding therapeutic management in UC and PSC colitis.

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Disclosure: Nothing to disclose.

PP0483

IMPACT OF SICKLE CELL DISEASE ON THE COURSE OF INFLAMMATORY BOWEL DISEASE AND DRUG SAFETY PROFILE: A MULTICENTER STUDY

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Introduction: While patients with Sickle Cell Disease (SCD) often exhibit digestive manifestations, the underlying pathophysiology remains poorly understood. SCD could notably affect the intestinal mucosa through microvascular phenomena. Co-occurrence of SCD and Inflammatory Bowel Diseases (IBD) has been reported in a very limited number of cases.

Our aim was to explore the intersection between SCD and IBD, particularly focusing on the impact of SCD on the natural course of IBD and the safety profile of drugs in this high-risk population.

Aims & Methods: We conducted a multicenter retrospective case-control study involving all adult patients diagnosed with both IBD and concomitant SCD and followed in 3 tertiary university hospitals. Each IBD patient with SCD was matched with up to 4 IBD patients without SCD, based on disease phenotype (UC or CD), sex, date of birth (±3 years), length of disease follow-up (±2 years), and year of diagnosis (±3 years). The primary outcome was a complicated disease course, defined by a composite outcome including intestinal resection, perianal surgery, and/or the need for advanced therapy.

Results: Overall, 125 patients with IBD were studied, including 24 with concomitant SCD and 101 controls. Of the 24 SCD patients, 5 (20.8%) had Crohn's disease (CD), 17 (70.8%) had ulcerative colitis (UC) and 2 (8.3%) had indeterminate colitis. 23/24 (95 %) had colonic involvement. Of note, 8 of 24 (33.3%) SCD patients had concomitant primary sclerosing cholangitis (PSC) compared to 1% of patients in the control group (p<0.001). Survival without a complicated course of the disease was estimated at 58.7 % (IC95[49.6-69.5]) at 5 years and at 38.9% (IC95[29.9-50.4]) at 10 years, for non-SCD patients, and at 63.3% (IC95[45.7-87.6]) and 52.7% (IC95[34.9-79.6]) at 5 and 10 years, respectively, for patients with SCD and IBD (p=0.36, Log-rank test). The survival without the need of advanced therapy was estimated at 66.1% (IC95[57.3-76.2]) at 5 years for non-SCD patient, and at 78.2% (IC95[63-97.2]) in patients with SCD patients (p=0.45). Regarding the safety profile, 26.3% (5/19) of patients with SCD and 13.5 % (14/104) of controls experienced adverse events with advanced therapies (p=0.17, Fisher's exact test). There was one reported vaso-occlusive crisis associated with the use of thiopurines. No death occurred during follow-up.

Conclusion: While patients with SCD and IBD displayed a distinctive phenotype characterized by nearly constant colonic involvement and a high prevalence of associated PSC, the disease natural history of IBD did not differ from that of patients without SCD. Given the notable prevalence of PSC, systematic screening with cholangio-MRI should be considered. Furthermore, no significant safety signal regarding the use of immunomodulator or advanced therapy was raised.

Disclosure: Nothing to disclose.

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PP0484

SEXUAL FUNCTION IN PATIENTS WITH ACTIVE PERIANAL FISTULIZING CROHN'S DISEASE

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Introduction: Sexual function is an important determinant of quality of life (QoL) and may be affected in patients with perianal fistulizing Crohn's disease (pCD). However, the literature on this topic is limited in patients with pCD.

Aims & Methods: Aims: This study aimed to evaluate the sexual function, the correlation between sexual function and quality of life and to identify potential risk factors for SD in patients with active pCD in a national prospective multicenter cohort study.

Methods: Patients with active pCD were identified from a prospective multicenter cohort study in 41 Dutch hospitals. Respondents on the sexual function (FSFI [female patients] or IIEF-5 [male patients]) and HR-QoL questionnaires (CAF-QoL scale and sIBDQ) were included in this study. Primary outcome was the prevalence of SD (FSFI<26.55, IIEF-5<22). Correlation for SD with HR-QoL was assessed with Spearman's correlation test. Potential risk factors for SD were identified using multivariable logistic regression.

Results: 214 patients with active pCD were included. Median age of the study cohort at inclusion was 37 years (IQR 30 – 51) and 52% of the patients were female. The median CD duration and median pCD duration comprised 8 years (3 – 17) and 4 years (2 – 8). The median FSFI and the median IIEF-5 scores were 19 (IQR 3.0 – 27) and 19 (IQR 9 – 24). Overall SD was reported by 69% of the patients and was more prevalent in female patients (75%) as compared to male patients (63%) without reaching significance (p=0.07). A significant higher CAF-QoL score (i.e. decreased pCD-related QoL)(median score 54 points vs. 35 points, p<0.01) and significant lower sIBDQ (i.e. decreased IBD-related QoL)(median score 46 points vs. 52 points, p<0.01) were observed in patients with SD. In addition, SD patients reported significantly more depressive symptoms (49% vs. 27%, p<0.01) as compared to patients with no SD. Higher rates of fatigue were reported in patients with SD, but did not reach significance. A weak correlation of SD with the CAF-QoL scale (r=0.25) and sIBDQ (r=-0.24) was observed. Multivariable analysis identified an older age (aOR 1.0; 95% CI 1.0 – 1.1), a longer pCD duration (aOR 0.9; 95% CI 0.8 – 0.9) and a decreased pCD-related QoL)(aOR 1.0; 95% CI 1.0 – 1.1) were independently associated with SD (**Table 1).**

Variable	aOR	95% CI
Female sex	1.7	0.8 - 3.3
Age at baseline*	1.0	1.0 - 1.1
pCD duration*	0.9	0.8 - 0.9
Low educational level	1.7	0.6 - 5.3
Active luminal disease	1.3	0.7 - 2.7
Seton in situ	0.8	0.4 - 1.6
Ostomy in situ	0.9	0.4 - 2.7
Self-reported depressive symptoms	1.2	0.5 - 2.9
Self-reported fatigue	1.1	0.4 - 2.8
CAF-QoL scale*	1.0	1.0 - 1.1
sIBDQ	0.9	0.9 - 1.1

Table 1. Multivariable logistic regression

Abbreviations. aOR = adjusted odds ratio; CI = confidence interval; pCD = perianal fistulizing Crohn's disease;
CAF-QoL= Crohn Anal Fistula - Quality of Life; sIBD = short Inflammatory Bowel Disease Questionnaire

Conclusion: Sexual dysfunction is highly prevalent in both male and female patients with pCD. A weak correlation between HR-QoL and SD was observed. An older age and a decreased pCD-related QoL were identified as risk factors for SD. Reversely, a longer pCD duration was identified as protective factor. Due to the high prevalence, the sexual function should be evaluated routinely by the treating team. When necessary, a referral to an expert may be considered.

Disclosure: Nothing to disclose.

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PP0485

QUALITY OF CARE IN INFLAMMATORY BOWEL DISEASE THROUGH PATIENTS' EYES: VALIDATION OF THE PORTUGUESE VERSION OF IQCARO-QOC QUESTIONNAIRE

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Introduction: As a chronic disease, patients' perspectives of the quality of care (QoC) provided are increasingly recognized as an essential aspect of inflammatory bowel disease (IBD) management and an essential standard for units that manage patients with IBD[1].

Aims & Methods: Aims:

- Primary: Translation and validation of the Portuguese version of the IQCARO-QoC decalogue[2].
- Secondary: identify factors associated with a perception of better QoC.

Methods: A unicentric cross-sectional study was conducted in an IBD unit in a non-tertiary hospital. All consecutively evaluated IBD patients in the outpatient setting between january and december 2023 were included. Demographic and disease characteristics were collected.

The original version of IQCARO-QoC decalogue was translated and adapted into Portuguese (independent forward and back-translation).

Validity was assessed through correlation between IQCARO-QoC and visual analogue scale (VAS). Fiability was assessed through internal consistency, test-retest and intraclass correlation. Test–retest (4-week repeat) was used for measuring reliability.

Factors associated with QoC were analyzed.

Results: A total of 114 patients were included (51% female; mean age: 44±15years), of whom 63% had Crohn's disease. 69% of patients were receiving biological therapy. Among the patients, 24% were solely in clinical remission and 48% in both clinical and endoscopic remission. 31 patients (27%) perceived their disease to be in remission.

The mean IQCARO score was 9.3±1.9. Construct validity analyses demonstrated moderate-to-strong correlations between the IQCARO-QoC and VAS. The IQCARO-QoC exhibited high internal consistency (Cronbach's $\alpha = 0.703$) and intraclass correlation (intraclass correlation coefficient = 0.622).

In univariate analysis, concerning patients' self-assessment of disease activity, those who considered their disease to be in remission reported higher IQCARO scores (p=0.02). Individuals with higher educational levels tended to perceive higher QoC (p=0.037). Disease duration, previous surgeries, hospitalizations, and number of outpatient appointments did not exhibit a statistically significant impact on QoC perception. Regarding quality of life, as assessed by the SIBDQ[3] with a mean score of 50.7±12.1, a statistically significant positive correlation was observed between QoC and the SIBDQ (r=0.276, p=0.004).

Multivariate analysis revealed that education significantly influenced the perception of QoC provided ($\beta = -0.284$ [-1.490; -0.340], p=0.002).

Relationship between clinical outcomes and IQCARO-QoC	IQCARO-QoC
Treatment (Conventional vs Biological/ Advanced therapies)	p=0.099
Previous intestinal surgery	p=0.296
Hospitalization related to IBD	p=0.346
Clinical remission	p=0.877
Clinical and endoscopic remission	p=0.624
Patient-reported disease activity	p=0.02
Education	p=0.037
Disease duration	r=0.112 (p=0.256)
Quality of life (SIBDQ-PT)	r=0.276 (p=0.004)

Conclusion: The Portuguese version of the IQCARO-QoC is a simple, valid, and reliable tool for assessing the QoC perception among patients with IBD in clinical practice.

Furthermore, these observational data suggest an association between patients' perspectives of QoC and their evaluations of disease activity and quality of life. In our cohort, education emerged as an independent predictor influencing the perception of QoC received.

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Disclosure: Nothing to disclose.

PP0486

TAILORING CROHN'S DISEASE SURVEILLANCE AFTER ILEOCOLIC RESECTION: THE IMPORTANCE OF INFLAMMATORY BIOMARKERS

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Introduction: Recurrence of Crohn's disease (CD) after ileocolic resection is common. However, reliable non-invasive monitoring methods are lacking, for instance, the use of faecal calprotectin (FC) has showed inconsistent results in this subgroup of patients.

Aims & Methods: The aim of this study was to evaluate clinical and biochemical factors associated with post-operative endoscopic recurrence in CD patients.

Retrospective single-center study including patients with CD who underwent ileocolonoscopy for CD surveillance, after ileocolic resection. The patients' clinical status was evaluated through Harvey-Bradshaw index (HBI), with clinical remission considered as ≤ 4 points. Biochemical parameters included erythrocyte sedimentation rate (ESR), c-reactive protein (CRP) and FC.

Clinical and biochemical parameters were evaluated in the 6 months prior to ileocolonoscopy. Post-operative endoscopic recurrence was defined as a Rutgeerts score ≥ i2b.

Results: A total of 81 patients and 121 ileocolonoscopies were included. Most patients were female (55.6%), with a mean age of 48±13 years. Regarding the Montreal classification, 37 patients had ileal disease (45.7%) and 44 ileocolonic disease (54.3%); 9 patients had nonstricturing, nonpenetrating disease (11.1%), 35 stricturing (43.2%) and 37 penetrating disease (45.7%). Nineteen patients had perianal disease (23.5%). The mean time between the surgery and the endoscopic evaluation was 135±79 months. Post-operative endoscopic recurrence was observed in 52 ileocolonoscopies (43.0%).

Symptomatic patients (HBI > 4 points) were 11 times more likely to have post-operative endoscopic recurrence (OR 10.686, 95%CI 2.930-38.974, $p<0.001$). Those with post-operative endoscopic recurrence had a significantly higher FC values (median 189 *vs* 69 µg/g, $p<0.001$) and higher ESR

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values (median 14 *vs* 9 mm, *p*=0.002). In fact, patients with FC values ≥ 150 µg/g and those with ESR values > 12 mm were 4 and 3 times more likely to have post-operative endoscopic recurrence (OR 3.654, 95%CI 1.605-8.316, *p*=0.002 and OR 3.059, 95%CI 1.397-6.696, *p*=0.005, respectively). The presence of post-operative endoscopic recurrence was not associated with CRP values (median 2.9 *vs* 2.9 mg/L, *p*=0.466), and neither with mean time between the surgery and the endoscopic evaluation (mean 126±83 *vs* 142±75 months, *p*=0.263).

Conclusion: The presence of symptoms (HBI > 4 points), FC values ≥ 150 µg/g and ESR values > 12 mm was associated with post-operative endoscopic recurrence in CD. Therefore, the HBI, FC and ESR values can be used to monitor CD patients after ileocolic resection, and prompt endoscopic evaluation should be performed in those with HBI > 4 points, FC values ≥ 150 µg/g and/or ESR values > 12 mm.

Disclosure: Nothing to disclose.

PP0487

ASSESSMENT OF A REMOTE CALPROTECTIN MONITORING TOOL IN IBD: A COMPARATIVE AGREEMENT ANALYSIS OF A CLASSICAL LAB-BASED ASSAY AND A MOBILE PHONE-BASED QUANTITATIVE PATIENT SELF-TEST

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Introduction: Several treatment guidelines and published clinical consensus studies recommend using fecal Calprotectin as a monitoring tool for IBD. Available analysis methods range from well-established laboratory analysis methods to more recently developed rapid methods for remote monitoring and point of care.

Since there is no international reference for calprotectin, an established agreement between different assay platforms is needed to use them interchangeably. Ideally the assays from the same manufacturers should measure fecal calprotectin equally, although this is not always the case.

This study compares fecal calprotectin measurements from two different assay platforms from the same manufacturer.

Aims & Methods: 131 fecal samples from IBD patients were measured using CalproLab ELISA ALP (CALP0170) and CalproSmart Self-Test for remote monitoring of IBD. All fecal samples were extracted and analysed in parallel according to the kit-specific instructions given by the manufacturer, requiring individual sample collection and extraction methods for both assays. 64 samples covered the linear range for the assays and were included in the method comparison analysis.

Results: When comparing the two analysis methods, the slope and intercept values of 0,92 and 4,44 respectively suggest that there is no significant difference between the two methods within their common linear range. This includes all relevant clinical cut-off values for the two methods. Furthermore, the relative difference between the two assays ranges from -1,9% to -6,8% at selected calprotectin values.

Calprotectin values [mg/kg]	Relative difference [%]
77	-1,9%
100	-3,2%
200	-5,5%
250	-5,9%
500	-6,8%

Table 1: The relative difference between CalproSmart and CalproLab ELISA at selected calprotectin values.

Conclusion: CalproSmart and CalproLab ELISA measures fecal calprotectin equally and can be used interchangeably.

Disclosure: All authors are employees at Calpro AS. Experiments performed at Calpro R&D lab.

PP0488

EARLY MONITORING TREATMENT RESPONSE IN ULCERATIVE COLITIS: A MULTICENTER STUDY

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Introduction: Data on using Intestinal Ultrasonography (IUS) for monitoring colonic lesions and treatment response in ulcerative colitis (UC) are still limited and most results are from single-center units usually performed in centers with a special expertise in IUS.

Aims & Methods: Our multicenter study aimed to identify predictive factors and monitor early response after different therapies in UC.

Methods: From July 2022 to Oct. 2023, we conducted a multicenter study at 5 IBD centers. UC pts starting biological therapies or small molecules were included, and underwent colonoscopy (CS), IUS, clinical assessments at baseline and at 12 weeks. All pts received standard of care therapy according to ECCO guidelines.

Exclusion criteria were disease limited to the rectum, active infections, and inability or contraindication to perform CS and/or IUS within the time intervals established by the study. Colonic wall thickening (CWT), vascularization (CWF), Milan Ultrasound Criteria (MUC), Mayo endoscopic score (MES), total/partial Mayo score were evaluated.

Results: Eighty-four UC patients were enrolled and eligible for starting biological therapy or small molecules. Forty-nine out of 84 pts (58.4%) were male, 40 pts (47.6%) had extensive colitis and 44 pts (52.4%) had left colitis, with a disease duration of 20 months (1-336); 53 pts (63%) were steroid-dependent/refractory.

Forty-one pts (48.9%) started anti-TNFs, 14 pts (16.6%) anti-IL12/23 or anti IL 23, 16 pts (19%) vedolizumab, and 13 pts (15.5%) JAKi. After 12 weeks, 39 out of 84 patients (46.4%) had partial Mayo score<2, 31 patients (36.9%) 2-4, 10 patients (11.9%) 5-7, and 4 patients (4.76%) >7.

Forty-three out of 84 patients (51.2%) had MES=0-1, while 41 out of 84 patients (48.8%) had MES=2-3. Fifteen out of 84 patients (17.9%) had ultrasonographic response (defined as CWT<3mm) and 26 out of 84 patients (30.1%) had MUC<6.2 (*1*). Logistic regression was performed to evaluate the relationship between clinical/endoscopic and ultrasonographic

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characteristics and early IUS response at 3 months. Univariable analysis showed that active smoking ([RR], 1.9; 95% CI, 1.2–3.1, p<0.005), appendectomy ([RR], 2.5; 95% CI, 1.9–3.2, p<0.0001), partial Mayo score <2 ([RR], 2.18; 95% CI, 1.5–3, p<0.001) were associated with higher risk of response at 3 months. Inversely, severe endoscopic lesions (MES 3) (RR, 0.3; 95% CI, 0.2–0.6; p<0.0001), presence of lymph-nodes, mesenteric hypertrophy and vascularization (p<0.001, RR 0.36, 0.28, 0.45 respectively) were associated with a lower risk of response at 3 months. CWT was associated with a lower risk of response at 3 months (RR 0.66, 95% CI, 0.53-0.81, p<0.0001) and every increased mm was associated with a 34% reduction of IUS response. MUC was associated with a lower risk of response at 3 months (RR 0.78, 95% CI, 0.7-0.87) and every increased of one point was associated with a 22% reduction of IUS response.

In the multivariable analysis, CWT and MUC were the only predictive factors associated with IUS response at 3 months (RR 0.70, 0.78 respectively, p<0.04). A CWT<5.5 mm showed a 2.5 risk of treatment response at 3 months.

Conclusion: Our multicenter study demonstrated that IUS allows close monitoring of treatment in UC. MUC, and in particular CWT, are valuable parameters to predict early response in UC.

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Disclosure: Nothing to disclose.

PP0489

EVALUATION OF THE IMPACT OF AN IBD NURSE IN AN INFLAMMATORY BOWEL DISEASE CENTER

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Introduction: In some European countries, IBD nurse is not a recognized status. IBD nurses are general care nurses who wants to improve their knowledges and get involved in the « Care and Support » of IBD patients. An increasing number of IBD centers are using the services of IBD nurse. It seems important to evaluate the role and impact of IBD nurse.

Aims & Methods: The aim of this study was to evaluate prospectively the activity and impact of IBD nurse on the management of IBD patients in an IBD center. From March 13th to June 13th, 2023, all patients having interacted with IBD nurse in an expert center, were prospectively recorded. The main points were, the number and characteristics of patients, means and reasons of contact, actions implemented, follow-up and, finally, patient satisfaction assessed using a standardized questionnaire (visual analog scale from 0 to 10).

Results: During three months, 132 patients, 69 women and 63 men (47% Ulcerative Colitis, 52% Crohn's disease and 1% indeterminate colitis), with a median age of 40 years (16 to 87), had contacted the IBD nurse. Contact were made by phone call: n =131 (42%), text messages: n = 25 (8%), e-mail: n = 69 (22%), face-to-face consult: n = 86 (27.5%) and teleconsultion: n = 1 (0.5%).

The reasons for contact were: advices/questions/information: n = 121 (39%), a scheduling/organization appointment issue: n = 60 (19%), care-providing (biological check-up + starting treatment): n = 77 (25%), emergency: n = 22 (7%), request for an appointment/Information with IBD nurse: n = 19 (6%), request for a patient education session (PE): n = 13 (4%).

The 312 contacts had led to 315 actions divided into seven groups:

Action realized	Number of patients (n)	Percentage (%)
Explanation/Reinsurance	166	53%
Implementations or renewing prescription /administrative documents	55	17%
Scheduled visit (doctor, dietician, Nurse, Patient education program, outpatient unit, ...)	40	132%
Nursing care/learning self-care	25	8%
Referral to Patient Education Program	19	6%
Referral to emergency department or another department	7	2%
Patient hospitalization	3	1%

Over this period, IBD nurse was contacted 22 times by 19 patients for a real or perceived emergency. These situations were managed by a multidisciplinary team, and resulted in reinsurance (68%), referral to others medical specialist (18%) or hospitalization (14%).

Overall satisfaction with the IBD nurse intervention has been evaluated at 10/10 for 78% of patients and 9/10 for 11%. To the question: "do you think that the intervention of the IBD nurse improves disease management?": 78.5% agreed at 10/10 and 7% at 9/10. To the question: "do you think contact with the IBD nurse improves access to care: 67.5% agreed at 10/10 and 3.5% at 9/10.

The main patients' advice / reviews to improve care were: to adjust levels of follow-up according to the patient and his/her needs, to strengthen the IBD Nurse coordination team to limited days off, to have explanation sheets and reports of IBD Nurse intervention, to develop more workshops for patients in the Education program.

Conclusion: This pilot study showed the importance of the IBD Nurse activity in an expert center. The patients who benefited from this care were very enthusiastic about it.

Disclosure: This study has been funded by Janssen Pharmaceuticals.

PP0490

METAGENOMIC SEQUENCING IN DE NOVO PEDIATRIC INFLAMMATORY BOWEL DISEASE VERSUS CONTROLS WITH AND WITHOUT GASTROINTESTINAL SYMPTOMS

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Introduction: Inflammatory bowel disease (IBD), including Crohn's disease (CD), ulcerative colitis (UC), and IBD-unclassified (IBD-U), often manifests in adolescence or young adulthood, with a trend towards decreasing age at presentation. Current diagnostic protocols for pediatric IBD involve an invasive colonoscopy under general anesthesia to differentiate between IBD and other conditions, such as functional abdominal pain and irritable bowel syndrome.

Previous microbiome research indicates significant differences in microbial composition of pediatric IBD patients compared to healthy controls (HC), illustrating its potential as non-invasive biomarker for pediatric IBD,

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but an IBD-specific microbial signature has not yet been defined(1). This study aims to profile the microbiome in de novo pediatric IBD patients compared to controls with gastrointestinal symptoms (CGI), and HC.

Aims & Methods: We collected baseline fecal samples and clinical data from a cohort of de novo pediatric IBD patients and CGI. Consecutively, we matched all included patients from this cohort to a HC cohort from the Dutch Microbiome Project, based on age and sex. Metagenomic sequencing was performed to obtain data on bacterial species and functional pathway abundances. Diversity, dissimilarity and differential abundance analyses were conducted for bacterial species and pathways across various group comparisons.

Additionally, prediction analysis using Coda4Microbiome was performed to differentiate IBD from HC, IBD from CGI, and to predict treatment outcomes based on both microbial features and clinical features, including disease activity scores and inflammatory markers.

Results: We included 99 de novo IBD patients (66 CD, 27 UC, 6 IBD-U) and 74 CGI, matched to 341 HC. Microbial composition differed significantly across HC, CGI and IBD groups. Alpha-diversity decreased progressively from HC to CGI (p<0.001), further decreasing in IBD patients (p=0.0056). Disease status significantly explained a part of microbiome composition variance (beta-diversity, p=<0.001, R2=0.045), with distinct clustering patterns between HC, CGI and IBD samples. Differential abundance analysis revealed 116 differential abundant species between HC and IBD, and 30 species between CGI and IBD. We observed an increase in *F. prausnitzii* (p=<0.001) in IBD compared to HC. Prediction models accurately distinguished IBD from HC using microbiome features (microbiome: AUC=0.96±0.02, clinical + microbiome: AUC=0.96±0.02) based on high abundance of *G. pamelaeae* and low abundance of *A. communis*, and IBD from other diseases (microbiome: AUC=0.72±0.10, clinical + microbiome: AUC=0.83±0.09), based on high abundance of *E. coli* and low abundance of a *Clostridia* species. Prediction of response to diet-based induction of remission in CD patients using baseline microbiome features yielded an AUC of 0.67±0.13, suggesting a potential effect of the gut microbiome, possibly limited by our non-responder numbers.

Conclusion: This study confirmed previous findings of profound microbiome differences between de novo pediatric IBD patients and HC. Additionally, we observed distinct microbiome compositions between CGI and IBD. Although clinical and biochemical models are often effective in distinguishing IBD from other diseases, our prediction models demonstrated that microbiome features have predictive power as well, including for predicting responses to nutrition-based treatments. Given the non-invasive nature of fecal sample collection, integrating microbiome features into future clinical diagnostics shows promise for improving diagnostic and therapeutic precision.

References: 1. Fitzgerald, Rachel S et al. "Paediatric Inflammatory Bowel Disease and its Relationship with the Microbiome." Microbial ecology vol. 82,4 (2021): 833-844. doi:10.1007/s00248-021-01697-9

Disclosure: RG received funding by Janssen Pharmaceuticals (for unrelated research projects), received consulting funding from Esox Biologics (for unrelated research projects). RKW has received unrestricted Research Grants from Takeda, Johnson & Johnson, Ferring and Tramedico and speakers fees from Abbvie, MSD and Boston Scientific and has acted as a consultant for Takeda Pharmaceuticals. TGJdM has received an unrestricted grant from Nutricia Benelux Corporation unrelated to this work. NKHdB has served as a speaker for AbbVie and MSD, as consultant and/or principal investigator for TEVA Pharma BV and Takeda and received a (unrestricted) research grant from Dr Falk, TEVA Pharma BV, MLDS and Takeda.

PP0491

IS TRANSMURAL HEALING BETTER THAN MUCOSAL HEALING IN PEDIATRIC CROHN'S DISEASE?

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Introduction: Data on the long-term benefit of transmural healing (TH) are scarce in children.

Aims & Methods: We investigated the outcomes of pediatric Crohn's disease (CD) patients with TH comparing them to those with mucosal healing (MH) and with "no healing".

We conducted a prospective, single-center, observational study conducted at the Pediatric Gastroenterology Unit of Sapienza University of Rome. Children with an established diagnosis of CD and under biological therapy undergoing an MRE and an ileocolonoscopy within a 3-month interval were consecutively enrolled. TH was defined as the complete healing of both the mucosa at endoscopy and of the bowel layers at the MRE. In the presence of a normal colonoscopy but an active MRE children were deemed as MH. The group "no healing" (NH) included children with an active endoscopy, regardless of the MRE classification. At 6, 12, and 24 months number disease flares, CD-related hospitalization, CD-related surgery, need for step-up treatment, and occurrence of complications were collected.

Results: 93 children were included (mean age 11,2±3,8). Twenty-three (25%) were classified as TH, 27 (29%) as MH and 43 (46%) as NH. The risk of any unfavorable outcome was lower in both TH and MH patients compared to NH (Log Rank p<0.0001 and p=0.002). Children with TH were at lower risk of hospitalization (p=0.02), and complications (p=0.007) compared to those with NH.

No difference was found between MH and NH children for the same outcomes. Both TH and MH children were at lower risk of step-up treatment (p<0.0001 and p=0.0002, respectively) compared to NH. TH and MH patients were comparable for all the outcomes evaluated.

Conclusion: Achieving TH in pediatric CD is associated with better outcomes, although the additional benefit over MH alone does not appear significant. However, the persistence of inflammation is associated with worse outcomes in all patients.

Disclosure: Nothing to disclose.

PP0492

A COMPREHENSIVE ASSESSMENT OF FECAL INCONTINENCE, SEXUAL FUNCTION, AND QUALITY OF LIFE IN INFLAMMATORY BOWEL DISEASES

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Introduction: Inflammatory bowel diseases (IBD) are chronic systemic conditions that affect patients' lives in various ways. To date, there is a scarce evidence correlating fecal incontinence, sexual dysfunctions, disease activity, and quality of life (QoL) in these patients.

This study aimed to evaluate the impact of fecal incontinence on sexual functions such as erectile and prostatic function in a cohort of IBD patients; and to analyse the relationship among fecal incontinence, sexual function and QoL.

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Aims & Methods: IBD patients were enrolled from IBD clinic of the Gastroenterology Unit of the University Hospital of Salerno. Demographic characteristics, anthropometric data, and usage of immunosuppressive drugs were collected. Patients underwent a physical examination, and laboratory tests, including C-reactive protein (CRP) and fecal calprotectin (FC). A standardized questionnaire was used to assess gastrointestinal (GI) symptoms such as abdominal pain and fecal incontinence. For the latter, we also used the Wexner score. Crohn's Disease Activity Index (CDAI) and Mayo score as well as FC levels, and endoscopic findings were used to classify IBD patients as in active or in remission phase. Patients with perianal manifestations were excluded. Hormonal values (FSH, LH, Prolactin, 17-betaestradiol, testosterone) were measured. Sexual function was evaluated using the International Index of Erectile Function (IIEF), Male Sexual Health Questionnaire (MSHQ), Ejaculatory Dysfuction (EjD), prostatic dysfunction using the National Institute of Health - Chronic Prostatitis Symptom Index (NIH-CPSI) and International Prostatic Symptoms Score (IPSS). QoL was assessed by FiQol domains (lifestyle, coping behaviour, self-perception and depression, embarrassment).

Results: Eighty-five male IBD patients (47 Crohn's disease [CD] and 38 ulcerative colitis [UC]), were enrolled. Mean age was 43.7 ± 15.9 years; mean BMI was 36.4±25.5 Kg/m^2. Twenty-four CD patients had an active disease and 23 were in remission, while 20 UC patients had an active disease and 18 were in remission. UC patients showed significantly higher fecal incontinence and Wexner scores compared to CD patients (1.0±1.9 vs 0.4±1.2, p=0.05 and 2.84±4.2 vs 2.3±1.1, p=0.01, respectively). Only the Mayo score significantly correlated with Wexner score (Rs=0.578; p<0.001), and fecal incontinence (Rs=0.584; p<0.001).

Sexual function, evaluated by erectile and prostatic scores, nicely correlated each other's (p<0.05 in all cases). However, no association was found between fecal incontinence and all sexual function scores. The FiQol domain self-perception and depression was significantly associated with IIEF (p=0.01).

Conclusion: This study highlighted the prevalence of fecal incontinence is higher in UC than CD patients. However, fecal incontinence did not impact on sexual function, evaluated by erectile and prostatic scores. QoL influenced erectile function.

Disclosure: Nothing to disclose.

PP0493

CAPSULE ENDOSCOPY AND ILEOCOLONOSCOPY FOR ILEAL CROHN'S DISEASE MONITORING, A COMPARATIVE ANALYSIS. PRELIMINARY RESULTS FROM A RETROSPECTIVE STUDY

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Introduction: The terminal ileum stands out as the most frequently affected segment of the gastrointestinal tract in Crohn's disease (CD). However, ileocolonoscopy (IC) may occasionally fail to fully explore this area. Notably, there is a lack of standardized quality criteria for IC in the context of ileal CD. Additionally, assessing inflammation activity only in the terminal ileum may lead to an underestimation of the overall ileal involvement. Capsule endoscopy (CE) offers a complete examination of both the entire terminal ileum and the remaining ileal segment.

Aims & Methods: This study presents the preliminary analysis of a retrospective study conducted at a single center. The primary outcome is the correlation between the total ileal activity, as assessed by CE, and the activity in the terminal ileum, as evaluated by IC.

Furthermore, we explored the correlation between the results of these two examinations specifically concerning the evaluation of the terminal ileum. Patient inclusion criteria permitted a maximum time interval of 3 months between the two examinations.

Activity measurement for CE was graded with the Lewis score of the third segment of the small intestine (Lewis-3S) and the Lewis score of the terminal ileum (Lewis-TI). For IC, the SES-CD index of the terminal ileum was collected.

The study also includes an analysis of the correlation of endoscopic scores with biochemical markers (PCR, fecal calprotectin). Clinical, blood tests, and therapeutic data from medical records were systematically collected.

Results: A total of 18 patients with ileal CD underwent CE and IC. CE revealed terminal ileum activity in 14 out of 18 patients (78%), whereas IC detected activity in only 4 out of 18 patients (22%).

Notably, no significant correlations between CE and IC scores were observed. A moderate to strong correlation was, however, identified between the two CE scores, Lewis-3S and Lewis-TI (ρ 0.72 [0.31-0.94], p=0.001). The median length explored by IC was 15 [15-20] centimeters.

Importantly, no correlation between endoscopic scores and biochemical parameters was observed, and no adverse events were documented.

Correlation SESD terminal ileum and Lewis 3rd segment ρ	0,391 (p>0,05%)
Correlation SESD terminal ileum and Lewis terminal ileum ρ	0,545 (p<0,05%)
Correlation Lewis third tercer segment and Lewis terminal ileum ρ	0,719 (p<0,05)

Table 1: Correlation between analytical data and between the two explorations.

Conclusion: In this preliminary analysis, despite the limited sample size, our findings indicate an underestimation of global ileal activity when measured by IC in comparison to CE examination.

Disclosure: Nothing to disclose.

PP0494

IN HINDSIGHT, DID WE OVERREACT? INSIGHTS ON COVID19 VACCINATION RESPONSE FROM A LARGE, REAL-WORLD IMID-PATIENT COHORT EVALUATED DURING SEQUENTIAL PHASES OF THE PANDEMIC

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Introduction: The BELCOMID study recruited immune mediated inflammatory disease (IMID) patients during the COVID19 pandemic. Patients were evaluated before vaccination, after onset of the vaccination campaign and after onset of booster vaccinations.

Clinical data, Nucleocapsid(N-)antibodies and Spike(S-) antibodies were assessed.

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Aims & Methods: Current subgroup analysis aimed to firstly, explore factors associated with N-/S-antibody response in IMID patients with different COVID19 vaccination status. Five patient groups were identified. G1: no vaccination before the vaccination campaign, G2: complete baseline vaccination, G3: 1 booster, G4: 2 boosters, G5: patients refusing every vaccine even after the booster campaign. We report on influence of IMID treatment and smoking status.

Secondly, we explored influence of SARS-CoV2 infection and vaccination on IMID flare-up.

Results: Demographics of 2165 recruited IMID patients are shown in *Table*.

After baseline vaccination (G2) odds of N-seroconversion were higher in patients treated with combined IMM/TIMT (OR2.52, p=0.023). No other association with N-seroconversion was found.

Mean S-antibody titre increased progressively from G2 to G4. After completed baseline vaccination (G2), odds of S-antibody seroconversion were lower in those treated with TIMT (OR0.28, p=0.0079) particularly rituximab (OR0.0354, p<0.001), IMM (OR0.276, p<0.001), combined IMM/TIMT (OR0.156, p<0.001) and systemic steroids (OR0.183, p<0.001). IMID flare prior to vaccination was associated with lower risk of S-seroconversion (RR0.94, p<0.001). By contrast, mean S-titre was higher for those treated with antiIL23 (mean ratio 2.17, p=0.011) and vedolizumab (mean ratio 1.84, p<0.001) in G2.

After 1 booster (G3), lower odds of S-seroconversion remained with rituximab (OR0.0368, p<0.001), IMM (OR0.222, p=0.0014), combined IMM/TIMT (OR0.15, p<0.001) or systemic steroids (OR0.0637, p<0.001). Vedolizumab was associated with lower odds of being in the lowest S-titer quartile (OR0.206, p<0.001). Odds for S-antibody seroconversion with anti-TNF were higher compared to rituximab both for patients in G2 (OR26.3, p<0.001) and G3 (OR27.5, p<0.001).

Patients who received 2 boosters (G4) all had S-seroconversion and no IMID treatment modality was associated with higher odds of being in the lowest quartile.

After baseline vaccination (G2) smokers had higher risk of being in the lowest S-antibody titer quartile (RR1.45, p=0.0098). This observation persisted after 1 booster (G3)(RR1.42, p=0.0122) but was no longer noticed after 2 boosters (G4).

After start of booster vaccinations, 25.17% of patients had experienced IMID flare-up in the previous months. No significant association between IMID flare-up rate and previous SARS-CoV2 infection was found. There was no increased risk of IMID flare-up with additional booster vaccination. However, patients who received 2 booster vaccines showed a lower RR for IMID flare-up compared to those who only received 1 booster (RR0.70, p=0.0067).

	G1 (N: 2144)	G2 (N: 1532)	G3 (N: 1283)	Group 4 (N: 147)	Group 5 (N: 40)
Mean age (SD)	44.6 (35.5)	46.9 (30.8)	46.6 (32.7)	51.0 (15.7)	39.4 (14.4)
Male/female	1088 (51.0%) /1047 (49.0%)	794 (51.9%) /737 (48.1%)	661 (51.6%) /621 (48.4%)	80 (54.4%) /67 (45.6%)	24 (60.0%) /16 (40%)
Active smoker	369 (17.2%)	261 (17.0%)	201 (15.7%)	24 (16.3%)	12 (30.0%)
TIMT (targeted treatment)	1580 (73.7%)	1232 (80.4%)	1073 (83.6%)	119 (81.0%)	38 (95.0%)
IMM (immunomodulator)	456 (21.3%)	311 (20.3%)	225 (17.5%)	39 (26.5%)	8 (20.0%)
IMM+TIMT combined	263 (12.3%)	197 (12.9%)	152 (11.8%)	24 (16.3%)	8 (20.0%)
Recent systemic steroids	229 (10.7%)	91 (5.94%)	59 (4.60%)	6 (4.08%)	4 (10.0%)

Conclusion: Certain immune-modulating treatment modalities and smoking may blunt serologic COVID19 vaccination responses. However, this is overcome by repeated booster vaccination without increased risk of IMID flare-up.

Disclosure: The BELCOMID study group received research grants from following pharmaceutical companies: Almirall, Roche, Janssen, Pfizer, Eli Lilly, Amgen, Biogen, Mylan, LEO Pharma. These companies had no role in study design, conduct nor reporting of results.

PP0495

IBD-DISK AS A PREDICTIVE TOOL FOR INSOMNIA IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: The IBD-Disk is a validated 10-question, self-administered visual questionnaire used to assess the inflammatory bowel disease (IBD) related disability. Insomnia is common in IBD, and is often associated with a decline in patients' quality of life. The Insomnia Severity Index (ISI) is a validated tool that assesses the severity of insomnia in individuals.

Aims & Methods: The aim of this study was to evaluate the role of IBD Disk in the predilection of sleep disorders.

We conducted a cross-sectional study over 3 months between April and June 2023. IBD patients responded to the IBD-Disk and ISI questionnaires. Insomnia and severe insomnia were defined by a "Sleep" sub-score > 5 and > 7 respectively. The correlation between IBD-Disk and the ISI was analyzed using Pearson's correlation coefficient.

Results: We enrolled 57 patients with a mean age of 40.37 years [20-69] and a M/F sex ratio of 1.03.

Two-thirds of patients had Crohn's disease (CD). It was ileal in 14 patients (24.6%), ileocolic in 15 (15.8%), and colonic in 9 (26.3%). Ano-perineal involvement was present in 38.2% of cases. Nineteen patients (33.3%) had ulcerative colitis. Severe disease activity was noted in 14 patients (24.6%).

According to the ISI, insomnia was present in 24 patients (42.1%), distributed as follows: Mild insomnia in 19.3% of cases, moderate insomnia in 17.5% and severe insomnia in 1.8%. Insomnia was associated only with disease activity (p=0.003). According to the IBD Disk "Sleep" sub-score, insomnia was present in 29.8% of patients, distributed as follows: Moderate insomnia in 12.3% and severe insomnia in 17.5%.

A significant correlation was noted between the ISI, the IBD-Disk global score and the "Sleep" sub-score (respectively r=0.506 and 0.609, p<0.001).

Conclusion: The IBD Disk is a psychometric tool that predicts the presence and severity of insomnia in IBD patients.

Disclosure: Nothing to disclose.

PP0496

VIRTUAL CHROMOSCOPY AND AUTOFLUORESCENCE FOR ASSESSMENT OF INFLAMMATION DEGREE IN IBD PATIENTS

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Introduction: Currently, most IBD clinics are equipped with endoscopic devices with virtual chromoscopy (VC) and autofluorescence imaging (AFI) screening functions.

These functions allow the mucosa and various neoplasms of the gastrointestinal tract to be examined in more detail and allow a presumptive conclusion to be made about the origin and histological nature of the changes detected.

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Aims & Methods: To determine the potential of virtual chromoscopy (VC) and autofluorescence (AFI) for the assessment of inflammation in ulcerative colitis (UC) and Crohn's disease (CD).

151 patients with IBD (51 with CD, 100 with UC) were examined by ileocolonoscopy using the Olympus CF FH260AZL endoscope system EVIS LUCERA ELITE CLV-290SL with narrow-spectrum function and autofluorescence function with biopsy. Similar examinations were performed on 45 controls without bowel disease.

Results: Minimal UC showed the absence of venules in VC($p<0.05$), as well as green staining with purple foci in AFI ($p<0.05$). Moderate inflammation was characterised by visualisation of submucosal venules ($p<0.001$) and violet staining of the mucosa with green foci, which was significantly higher than this colour with minimal activity ($p<0.001$).

Autofluorescence diffuse purple staining occurred significantly more often in patients with severe inflammation than in those with less severe inflammation, in those in remission and in controls ($p<0.001$), and two types of vascular patterns were absent in VC ($p<0.001$).

In CD, the infiltration phase was characterised by a statistically significant higher frequency ($p = 0.033$ and $p = 0.006$) of the visualisation of two types of green staining in the autofluorescence mode in comparable amounts than other colour options.

In the ulcer-fissure phase of VC, we observed predominantly capillary or venous visualization in the same amount ($p < 0.001$). In the scar phase, the frequency of the different types of staining was comparable to that in the remission phase and to that in the control group. In the VC, both types of vascular patterns were detected with a significantly higher frequency ($p = 0.001$).

Conclusion: Based on the results of our study, VC and AFI are additional methods in diagnosing the degree of activity of IBD and can be used in difficult cases to clarify the severity of inflammation in patients with UC and CD.

Disclosure: Nothing to disclose.

PP0497

THE VALUE OF FAECAL CALPROTECTIN TESTING IN PATIENTS PRESENTING WITH LOWER GASTROINTESTINAL SYMPTOMS; A RETROSPECTIVE PRIMARY CARE STUDY

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Introduction: Faecal calprotectin (FC) is of value in differentiating Inflammatory Bowel Disease (IBD) from other causes of lower gastrointestinal (GI) symptoms and is recommended in national guidance. This study aims to examine its role and factors associated with faecal calprotectin requesting in primary care.

Aims & Methods: Retrospective open cohort study using Clinical Practice Research Datalink (CPRD Aurum) of adults aged over 18 years with lower GI symptoms between 2015 and 2019. FC requested 30 days before to 3 months after a lower GI symptom record in primary care were included. Logistic regression analysis explored patient level factors associated with FC testing. Referral pathways and outcomes (IBD diagnoses) of patients with and without FC testing were also examined.

Results: Of 741,190 symptomatic patients, only 18,320 (2.5%) had FC requested in primary care: median age 38.3 (IQR 28.9-50.7); 51.4% female. The rate of FC requesting in symptomatic patients increased from 0.02% in 2010 to 4.4% in 2019 ($p=0.001$).

FC value of <50 ug/g and 50-200 ug/g were predominantly seen in patients with abdominal pain (42.1% and 37.9% respectively), while FC > 200 ug/g was predominantly seen in patients with diarrhoea (45.3%). Factors associated with FC requesting included: increasing age (>70 years adjusted Odds Ratio 0.16 (95% CI 0.15-0.18)); female (0.89 (0.86-0.92); Black (0.61(0.56-0.66) or Asian (0.75 (0.70-0.80) ethnicity; obesity (0.89(0.85-0.93); lower socioeconomic status (0.91(0.86-0.98); comorbidity score >2 (0.73(0.68-0.79); current smoking (0.89(0.85-0.92)); change in bowel habit (7.31(6.90-7.75), rectal bleeding (2.04(1.95-2.13) or diarrhoea (2.88(2.78-2.99) compared to abdominal pain; and ex-smokers (1.06(1.03-1.11).

IBD was diagnosed in 21.5% of patients referred with a FC >200 ug/g, but only 2.9% of referred patients with a FC 50-200 ug/g. The only patients with a significant risk of IBD with FC 50-200ug/g were patients with rectal bleeding (8%), with the risk in all other patients only 1.7%. 27.8% patients with a normal FC test result were still referred to secondary care, of which only 0.4% had IBD.

Conclusion: Despite an increase in FC requests in recent years, only 4.4% of patients with relevant lower GI symptoms had an FC requested in primary care as recommended. Despite a negative FC test, a third of patients were still referred to secondary care, with very few diagnosed with IBD . Patients with an FC between 50 ug/g and 200 ug/g without rectal bleeding have a low risk of IBD and can be safety netted in primary care.

Disclosure: Nothing to disclose.

PP0498

CLUSTERING OF PATIENT-REPORTED OUTCOMES IN INFLAMMATORY BOWEL DISEASES – FINDINGS FROM A DANISH POPULATION-BASED INCEPTION COHORT

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Introduction: Patient-reported outcomes (PROs) are pivotal in assessing treatment efficacy, delivering patient-centered care, and estimating the burden for inflammatory bowel diseases (IBD), including ulcerative colitis (UC) and Crohn's disease (CD).

Aims & Methods: We aimed to investigate PROs at IBD diagnosis, including associated factors.

Since May 1st, 2021, all adult and pediatric patients with incident IBD within a catchment area encompassing 20% of the Danish population were enrolled in the prospective population-based Copenhagen IBD Inception Cohort.[1]

Upon diagnosis, all adult patients diagnosed between May 2021 and May 2023 were asked to report PROs including Short inflammatory Bowel Disease Questionnaires (SIBDQ), IBD-Disability Index (IBD-DI), and Functional Assessment of Chronic Illness Therapy-Fatigue (FACIT-F). Symptoms were evaluated using Memorial Symptom Assessment Scale, IBD Symptom Inventory, and Mayo Assessment of Functional Gastrointestinal

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Disease. Multivariable logistical analysis adjusting for gender, age, and disease phenotype was conducted (adjusted odds ratio, aOR (95% confidence interval)).

Results: The population consisted of 203 and 116 patients with UC and CD (66% and 58% of the original cohort), respectively, who completed the PROs questionnaires within a median of 1.2 (IQR 0.3–3.0) months from diagnosis. The only baseline difference between participants and non-participants was a higher proportion of females with UC in the former group (60% vs. 44%, p<0.01).

In total, 160 (79%) and 99 (85%, *p*=0.18) patients with UC and CD, respectively, reported moderate to severe affection of at least one PRO, while 44% and 64% (*p*<0.01), respectively, reported moderate to severe affection of at least two PROs.

Specifically, a higher proportion of patients with CD reported moderately to severely impaired quality of life (QoL) (96 (82.8%)) compared to UC (141 (69.5%), p<0.01). Of those, 33 (82.5%) and 41 (56.9%), respectively, reported simultaneously moderate to severe fatigue and disability (p=0.48), while 19 (19.8%) and 27 (19.1%, p=0.90), respectively, reported moderate to severe QoL and fatigue but not disability. In addition, among 71 (35.0%) and 52 (44.8%) with UC and CD, respectively, reporting moderate to severe fatigue (p=0.08), all but three patients with UC had moderate to severe QoL, while 57.6% and 63.5%, respectively, reported moderate to severe disability. Accordingly, FACIT-F correlated strongly with SIBDQ in both UC (Pearson correlation coefficient (PCC) 0.76 (0.70-0.82) and CD (PCC 0.84 (0.78-0.89).

The most frequent symptoms of UC were bloody stool (88%), lack of energy (82%), and abdominal pain (78%), whereas those of CD were lack of energy (86%, *p*=0.36) and abdominal pain (84%, *p*=0.17). These symptoms as well as bowel urgency (46% vs. 43%, *p*=0.60) were equally common in UC and CD, while bloody stool was less frequent in CD (54%, p<0.01). Finally, bowel urgency was linked to moderate to severe disability in UC (aOR=4.01 (1.11;14.51)), but not QoL or fatigue, which was confirmed in sensitivity analysis of proctitis.

Conclusion: Our prospective population-based inception cohort study highlights a considerable burden of UC and CD at the time of diagnosis, with moderate to severe impairment of PROs in most patients. In addition, we observed a clustering of impaired PROs in half of the patients, underscoring the interconnectedness of PRO metrics and symptoms. These findings emphasize the importance of tight monitoring of PROs and symptoms in clinical practice, allowing timely interventions.

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PP0499

RAMAN SPECTROSCOPY AS A PROMISING EXPLORATORY MOLECULAR TOOL FOR DIAGNOSIS, PHENOTYPING AND ASSESSMENT OF MUCOSAL HEALING IN INFLAMMATORY BOWEL DISEASE

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Introduction: Diagnosing and assessing Inflammatory Bowel Disease (IBD) requires invasive procedures such as endoscopy and intestinal biopsies. However, reliable and accurate non-invasive tools for assessing and monitoring inflammation in IBD patients are in an exciting phase of early development. Raman spectroscopy (RS), which provides molecular vibrational spectra, can characterise human biofluids and intestinal samples, offering the potential for non-invasive molecular assessment, characterisation and monitoring in IBD.

Aims & Methods: This study aims to explore the potential of RS applied to saliva, blood and intestinal biopsies from IBD patients for molecular characterisation and development of new biomarkers enabling a reliable and accurate non-invasive diagnosis of IBD, phenotyping between Crohn's disease (CD) and Ulcerative Colitis (UC) and predicting mucosal healing (MH). IBD patients undergoing advanced endoscopy, i.e. ultra-high magnification endocytoscopy or probe-based confocal laser endomicroscopy, for disease assessment or surveillance and healthy controls were prospectively enrolled from two university hospitals in Cork, Ireland. Peripheral blood, saliva, and biopsies from the most representative bowel areas were collected. All matching samples were processed through a novel waveguide-enhanced RS technique, enabling sensitive detection of metabolites and other biochemical components.

Additionally, MH from matching biopsies was assessed as RHI≤3 without lamina propria or epithelial neutrophils, NHI ≤1 and PHRI=0. RS data processing was performed using Matlab®. Spectral markers matched for saliva, blood and biopsies and differentiating IBD vs controls, CD vs UC and MH vs active inflammation were identified. A machine learning model based on linear discrimination analysis was used to assess the classification accuracy.

Results: A cohort of 16 IBD patients (median age 39, 11:5 M: F), comprising 9 UC and 7 CD, and 4 healthy controls (median age 43.5, 3:1 M: F) was enrolled, with matching saliva, blood, and biopsies. MH was found in 4 IBD patients. When comparing IBD patients with controls, differences in 3 wavenumbers were observed in all samples, corresponding to decreased tyrosine and lipids signals and increased carotenoids signal in IBD. Similarly, differences in spectral markers were found in CD vs UC, with increased lipids signal in CD and increased signals for carotenoids, nucleic acid, and amide proteins in UC.

Interestingly, 5 matching differences in saliva, plasma and biopsies spectra were found in patients with MH compared to those with active inflammation, including an increase in nucleic acids and amide proteins and a decrease in phosphate, lipids, and carotenoids. The machine learning model showed promising accuracy (above 82%) of RS-aided biofluids analysis in diagnosing IBD and predicting phenotype and MH.

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Conclusion: The combination of spectral markers from saliva and plasma, confirmed at tissue level, showed promise in accurately diagnosing IBD and distinguishing CD from UC. Notably, RS non-invasive biomarkers could predict MH with promising accuracy, identifying metabolites involved in IBD-associated inflammation, particularly lipids, carotenoids, and nucleic acids. Therefore, RS holds potential as a monitoring and predictive tool for inflammation and healing in IBD, with intriguing translational implications. Further studies are required to validate these preliminary findings and shed light on molecular mechanisms.

Disclosure: Nothing to disclose.

PP0500

THE PREDICTIVE VALUE OF INTESTINAL ULTRASOUND FOR TREATMENT RESPONSE IN IBD: A SYSTEMATIC REVIEW

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Introduction: Selecting the right treatment in the early stages of inflammatory bowel disease (IBD) is essential for preventing disease progression and modifying disease course. Intestinal ultrasound (IUS) is a promising tool in predicting early treatment response. We aimed to systematically review current literature on the predictive value of IUS for treatment response.

Aims & Methods: A literature search was performed on the 24th of October and updated on the 26th of March 2024 using PubMed, Medline, Embase, Cochrane Library and Cinahl. We included articles in paediatric and adult IBD patients where IUS was used for the assessment of treatment. Studies were grouped based on timing of IUS after start of treatment. In acute severe ulcerative colitis (ASUC) timing included within 5 days or between 6-14 days. In ulcerative colitis (UC) or Crohn's disease (CD) baseline, 0-3 weeks, 4-8 weeks, 12-16 weeks and >16 weeks were used.

Additional response assessments included clinical or endoscopic disease activity scores and adverse events (e.g. surgery, hospitalization, treatment escalation).

Results: Out of 1739 unique articles, 27 were included with 17 in CD, 6 in UC and 4 in ASUC.

Baseline IUS parameters were not predictive of treatment response in CD, UC or ASUC.

In CD, most studies investigated response to anti-TNF (n=12), though multiple therapies were evaluated (n=5). For anti-TNF, IUS was predictive of response to therapy at all time points from 2 weeks onward. IUS at week 4 to 8 showed conflicting results in studies using multiple treatments. After 12 weeks, IUS predicted clinical and endoscopic outcomes as well as adverse events in all studies regardless of treatment.

Lastly, an IUS several months after starting treatment predicted clinical response, adverse events and endoscopic response in the follow years.

A variety of medical treatments were included in most UC studies (n=3). IUS within 3 weeks of treatment initiation was able to predict both short- and long-term clinical response in UC. Endoscopic response could be predicted by IUS after 6 weeks. The combination of BWT and colour Doppler signal (CDS) measured at 3 and 9 months of therapy predicted endoscopic activity after 9 and 15 months, respectively.

In ASUC, bowel wall thickness (BWT) significantly differed between responders and non-responders of intravenous steroid treatment when measured 1-3 days after treatment initiation.

	IUS time point	Drug*	Clinical outcomes	Adverse outcomes**	Endoscopic outcomes
CD	<3 weeks	Anti-TNF	Yes		Yes
		MT		No	
	4-8 weeks	Anti-TNF	Yes		Yes
		USTE			Yes
		MT	Yes	Inconclusive	Yes
	12-16 weeks	Anti-TNF			Yes
		USTE			Yes
		MT	Yes	Yes	Yes
	>16 weeks	Anti-TNF		Yes	Yes
		MT	Yes		Yes
ASUC	1-5 days	CS	Yes	Yes	
	6-21 days	CS	Yes	Yes	
UC	<3 weeks	CYTA	Yes		
		MT	Yes		No
	4-8 weeks	MT			Yes
	12-16 weeks	CS			Yes
		MT			Yes
	>16 weeks	CS			Yes

** CS=corticosteroids | MT=multiple treatments |CYTA=cytapheresis | USTE=Ustekinumab*

***Hospitalization, surgery, treatment escalation or no discontinuation of corticosteroids*

Conclusion: IUS performed at 1-3 days, 2 weeks and 6 weeks was predictive of treatment response in ASUC, CD and UC, respectively. Further data is needed to characterize IUS response kinetics of the individual therapies.

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PP0501

PHYSIOLOGIC AND ANATOMIC CHARACTERISTICS OF PELVIC FLOOR DISORDERS IN INFLAMMATORY BOWEL DISEASE PATIENTS

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Introduction: With the expansion of treatment options available for inflammatory bowel disease (IBD), the emphasis on refractory functional disorders grows along with new treatment targets, aiming at quality-of-life improvement. Pelvic floor disorders such as fecal incontinence (FI) and obstructive defecation syndrome (ODS) are considered as disabling conditions and are a source of embarrassment and suffering for IBD patients. Reluctance to complain and feeling of incompetence leads to a delay in diagnosis and treatment.

Aims & Methods: Aim: To characterize the etiology and the physiological characteristics of pelvic floor disorders in a cohort of IBD patients as compared with non IBD patients.

Methods: Data was collected from the medical files of IBD patients who went through a workup due to defecation disorder during 2015-2024. All patients completed either an anorectal manometry, trans anal ultrasound (TRUS) or evacuation proctography. Data was compared to that of a control group, that included patients with defecation disorders without IBD, that went through a similar medical workup.

Results: 67 IBD patients were included, 44 patients (66%) with Crohn's disease (CD) and 23 with ulcerative colitis (34%). Perianal involvement was diagnosed in 25% of the CD group. The control group consisted of 140 patients. Gender and age did not differ significantly between the groups. The median number of deliveries in the IBD group was 2.5 (IQR=1) versus 2 births (IQR=1) in the control group (P=0.38). The prevalence of pelvic floor surgery was higher in the control group (36%, VS. 18%, p=0.01).

Pelvic floor dyssynergy was diagnosed in 75% of the patients in the control group compared with 59% of the patients in the IBD group (p=0.03). Rectal hyposensitivity was found in 44% of the patients in the control group compared with 46% of patients in the IBD group (p=0.06), and rectal hypersensitivity was diagnosed in 13% of the patients in the control group versus 33% of the IBD group, p=0.003. (Table 1)

		IBD	Non-IBD	P value
Dyssynergia		34 (59%)	75 (75%)	P=0.03
Anal compliance	normal	12 (25%)	51 (39%)	P=0.07
	low	36 (75%)	79 (61%)	
Balloon explosion		11 (21%)	22 (16%)	P=0.56
	hyposensitive	24 (46%)	58 (44%)	P=0.06
	hypersensitive	17 (33%)	17(13%)	P=0.003
TRUS	Internal sphincter tear	6 (16%)	24 (17%)	0.84
	External sphincter tear	10 (26%)	23 (16%)	0.16
	fistula	4 (11%)	1 (0.7%)	0.0009

Conclusion: Pelvic floor disorders in IBD patients may be associated with higher prevalence of rectal hypersensitivity compared with non-IBD patients. Dyssynergy is common in IBD patients and is associated with FI. Understanding the mechanisms of pelvic floor disorders in IBD patients is essential for choosing the best treatment modalities in this group of.

Disclosure: Nothing to disclose.

PP0502

CPA9-HNE: A NEUTROPHIL-DERIVED FRAGMENT OF CALPROTECTIN MEASURED IN SERUM CAN MONITOR ENDOSCOPIC AND CLINICAL DISEASE ACTIVITY IN ULCERATIVE COLITIS

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Introduction: Ulcerative colitis (UC) is a chronic inflammatory bowel disease (IBD) presenting in remitting ulcerations of the colonic mucosa and submucosa. Mucosal healing is therefore an important treatment target for optimal disease management. While fecal calprotectin is a common biomarker used to monitor mucosal healing in UC, patient compliance is low due to a preference for serological markers over fecal ones.

Aims & Methods: This study aimed at investigating the association of serum calprotectin [CPa9-HNE], a non-invasive neo-epitope biomarker of neutrophil activity, with both clinical and endoscopic disease activity in UC. The cohort included 49 patients with active UC and 50 and healthy controls (HC). Endoscopic and clinical disease activity was assessed using the Ulcerative Colitis Endoscopic Index of Severity (UCEIS) and the full Mayo score. Serum calprotectin [CPa9-HNE] was measured, reflecting a human neutrophil elastase-derived fragment of calprotectin. Spearman's rank correlation was applied. Group differences were evaluated using Kruskal-Wallis with Dunn's test, bonferroni corrected for MCP. Diagnostic and discriminative capabilities were assessed using receiver operating characteristic (ROC) statistics with are under the curve (AUC).

Results: CPa9-HNE was significantly elevated in patients with UC compared to HC (ng/ml [IQR]: 32.4 [29.0–52.3] vs. 26.0 [22.6–29.0], $p<0.0001$) and presented acceptable discriminative capabilities (AUC [95% CI]: 0.77 [0.67–0.87], $p<0.0001$). CPa9-HNE showed a moderate positive correlation to UCEIS and Mayo ($\rho=0.42$; $\rho=0.48$, both $p<0.01$). When patients were stratified according to endoscopic disease activity (UCEIS), CPa9-HNE was significantly elevated in patients with severe disease compared to those with mild disease (ng/ml [IQR]: 55.6 [40.9–66.4] vs. 28.9 [24.6–31.1], $p<0.01$) and could accurately discriminate between the two groups. Similarly, when patients were grouped according to the full Mayo score, CPa9-HNE was significantly elevated in patients with mild vs. severe disease (29.1 [24.3–32.4] vs. 62.9 [55.2–67.5], $p<0.01$) and moderate vs. severe disease (32.0 [29.2–39.2] vs. 62.9 [55.2–67.5], $p<0.01$), and could accurately discriminate between the groups.

Conclusion: CPa9-HNE was elevated in patients with UC compared with healthy subjects. Furthermore, CPa9-HNE accurately discriminated between patients with mild and severe endoscopic disease based on the UCEIS score, as well as between patients with mild, moderate, and severe clinical disease activity according to the full Mayo score. These findings highlight the potential use of CPa9-HNE as a non-invasive tool to monitor both endoscopic and clinical disease activity in UC, with the potential of guiding treatment decisions and better aligning with patient preferences.

Disclosure: MSA, MPE, MAK, ACBJ, and JHM are employees of NB.

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PP0503

ANTI-TNF EXPOSURE IS A RISK FACTOR FOR ENDOSCOPIC ILEAL RECURRENCE IN ILEOSTOMY PATIENTS WITH CROHN'S DISEASE

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Introduction: A small but significant proportion of patients with Crohn's disease (CD) will ultimately require a permanent ileostomy. (1-3) The natural history and treatment success in these patients remain unclear and studies on endoscopic recurrence are lacking.

The aim of this study was to investigate the endoscopic recurrence rate, possible risk factors, and treatment outcomes with current biological agents.

Aims & Methods: We performed a retrospective study of CD patients with a permanent ileostomy in a tertiary referral center. Inclusion criteria were CD adults (≥16 years) and the presence of an ileostomy for at least 12 months. Ileal postoperative recurrence was defined as the presence of ulcerations during ileoscopy.

Traditional risk factors historically associated with postoperative recurrence (age at diagnosis ≤16 years, active smoking, penetrating phenotype, and inflammation at resection margin) were evaluated. (4,5) The probability of developing post-operative ileitis was examined by cox regression and Kaplan Meier analysis.

	All patients (N=166)	Endoscopic surveillance (N=115)
Female, n (%)	105 (63.3)	77 (67.0)
Median age at diagnosis, years (IQR)	22.0 (16.8-31.0)	22.0 (16.0-30.0)
Median disease duration at time of surgery, years (IQR)	7.0 (2.0-15.0)	7.0 (2.0-13.0)
Smoking status at time of surgery, n (%)		
Never	89 (53.6)	68 (59.1)
Former	29 (17.5)	26 (22.6)
Current	36 (21.7)	16 (13.9)
Unknown	12 (7.2)	5 (4.3)
Disease location and behaviour at diagnosis, n (%)		
L1, ileal	13 (7.8)	11 (9.6)
L2, colonic	95 (57.2)	67 (58.3)
L3, ileocolonic	42 (25.3)	30 (26.1)
Unknown location	16 (9.6)	7 (6.1)
B1, inflammatory	71 (42.8)	54 (47)
B2, stricturing	13 (7.8)	7 (6.1)
B3, penetrating	28 (16.9)	20 (17.4)
Unknown behaviour	54 (32.5)	34 (29.6)
Perianal disease at time of surgery, n (%)	74 (44.6)	51 (44.3)
Previous intestinal resection, n (%)	86 (51.8)	55 (47.8)
Previous immunomodulator, n (%)	109 (65.7)	79 (68.7)
Previous biological therapy, n (%)	125 (75.3)	88 (76.5)
Previous anti-TNF	102 (61.4)	71 (61.7)
≥2 prior biologicals	32 (19.3)	23 (20)
≥1 traditional risk factor*, n (%)	138 (83.1)	95 (82.6)
Endoscopic recurrence, n (%)		46 (40)
Endoscopic remission after treatment, n (%)		
Infliximab		4/13 (20.8)
Adalimumab		7/13 (53.8)
Vedolizumab		6/8 (75)
Ustekinumab		4/12 (33.3)

Table 1.

**Defined as age at diagnosis ≤16 years, active smoking, penetrating phenotype, history of IBD-surgery and inflammation at resection margin.*

Results: 166 patients were identified (median age 34 years (IQR 24-46), 63% female) (Table 1). Median follow-up after surgery was 5.5 years (IQR 2-14) and 37% had a loop ileostomy without resection. 115/166 (69%) underwent endoscopy during follow-up. Indications for ileoscopy were symptoms of flare (50%), planning for ostomy closure (14%) and other indications (21%). Endoscopic lesions were observed in 40% of patients, with a cumulative risk at 1-, 3- and 5-years of 13%, 27% and 36%, respectively. Controlling for traditional risk factors, patients with preoperative anti-TNF treatment were at higher risk (HR 2.18, 95% CI 1.11-4.28), with 17%, 41%, and 44% recurrence rates at 1, 3, and 5 years respectively (log rank P = 0.01). Other traditional risk factors for postoperative recurrence were not predictive of endoscopic disease recurrence in this cohort. Ileal recurrence was treated with biological therapy, including infliximab, adalimumab, vedolizumab, or ustekinumab. Patients treated with vedolizumab had the highest rate of remission (6/8 patients) and longest treatment persistence (54 months, IQR 19-102.8) (Table 1).

Conclusion: This retrospective study showed that 40% of ileostomy patients developed endoscopic recurrence. Patients with previous anti-TNF exposure were at higher risk of recurrence. Vedolizumab was the most effective treatment, although this finding needs to be confirmed in larger cohorts.

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PP0504

EARLY INTESTINAL ULTRASOUND IN ULCERATIVE COLITIS PATIENTS TREATED WITH FILGOTINIB: PRELIMINARY DATA FROM THE STEER STUDY

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Introduction: Intestinal ultrasound (IUS), in particular bowel wall thickness (BWT), is accurate in determining endoscopic disease activity in ulcerative colitis (UC). Reduction of BWT six weeks into advanced anti-inflammatory treatment predicted endoscopic response and clinical remission. Filgotinib is a preferential JAK1-inhibitor effective in UC. Currently, there is no data on IUS-kinetics or early prediction of treatment response with filgotinib.

Aims & Methods: In this study, we aimed to identify early IUS predictors of treatment response to filgotinib in patients with moderate to severe UC. This prospective observational study included UC patients with active endoscopic disease (i.e. endoscopic MAYO score (EMS) ≥2) starting with filgotinib. Clinical scores (UCEIS, Lichtiger), biochemical parameters and IUS parameters of the sigmoid were assessed at baseline, after 4 weeks (W4) and at the time of the second endoscopy (between W8 and W23). Endoscopic remission (i.e. EMS=0), endoscopic response (i.e. decrease in EMS ≥1 compared to baseline) and clinical remission (SCCAI ≤2 or Lichtiger score ≤3) were assessed. IUS response (i.e. defined as a reduction in BWT of >25% or >2.0 mm, or >1.0 mm and one colour Doppler signal) was assessed at week 4. All normally distributed data were reported in mean ± standard deviation [SD] and non-normally distributed in median and interquartile range [IQR]. Mann–Whitney U tests were used to compare continuous non-parametric variables, chi square tests for dichotomous variables.

Results: A total of 20 patients were included. 13 patients underwent second endoscopy, 2 refused a follow-up endoscopy and 5 patients were still in follow-up. At baseline, 9 patients (9/20, 45%) had received ≥2 biologicals, 16 patients (16/20, 80%) had previous anti-TNF exposure, and no patients were exposed to other JAK-inhibitors. 8 patients (8/20, 40%) had severe endoscopic disease at baseline (i.e. EMS=3). 5 patients (5/13, 38%) were endoscopic responders at W8-23, and no patient reached endoscopic remission. BWT was significantly lower in the sigmoid colon from W4 onwards for patients reaching endoscopic response (Table 1). Decrease in BWT compared to baseline was statistically greater in endoscopic responders from W4 onwards (Table 1). IUS response was seen at W4 in 6 patients, of which 5 were endoscopic responders (83%, p=0.002). Decrease in BWT from W4 onwards was statistically greater in patients reaching clinical remission at W8-23 (-45% (-55% to -36.6%) vs -9% (-25% to +8%), p=0.014).

Time point	*Endoscopic response (n=5)*	*Endoscopic non-response (n=8)*	*p-value*
Median BWT (IQR) at baseline	4.5 mm (3.2-6.2)	5.1 mm (3.6-6.5)	*p=0.61*
Median BWT (IQR) at W4	1.9 mm (1.8-3.8)	4.7 mm (3.3-5.9)	***p=0.019***
Median BWT (IQR) at W8-23	2.2 mm (1.8-3.7)	5.3 mm (4.3-6.7)	***p=0.008***
Median decrease in BWT (IQR) W4	-42% (-53% to -38%)	-3% (-16% to +9%)	***p=0.003***
Median decrease in BWT (IQR) W8-23	-45% (-51% to -35%)	-10% (-1% to +24%)	***p=0.003***

Conclusion: On IUS, UC patients treated with filgotinib have significantly lower BWT from week 4 onwards compared to baseline, which could predict endoscopic response.

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FV received speaker or honoraria fees from AbbVie, Janssen and Tillotts
ML received consultancy/lecturefees from Abbvie, Bristol Myers Squibb , Eli Lilly, Galapagos, Janssen-Cilag, Johnson & Johnson, Medtronic, Pfizer, Takeda and Tillotts.
RG Received speakers fee from Janssen and Tillotts
GD has served as a consultant and/or received speaker's fees from AbbVie, Alimentiv, AstraZeneca, Boehringer Ingelheim, Bristol Myers Squibb, Celltrion Healthcare, Cytoki Pharma, Eli Lilly, Ferring, Galapagos, Glaxo SmithKline, Gossamer Bio, Immunic, InDex Pharmaceuticals, Janssen, Johnson & Johnson, Kaleido, Pfizer, Progenity, Prometheus Biosciences, Prometheus Laboratories, Protagonist Therapeutics, Takeda, Tillotts, Ventyx Biosciences.
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PP0505

ARTIFICIAL INTELLIGENCE-ASSISTED DIGITAL PATHOLOGY IS COMPARABLE TO EXPERT PATHOLOGISTS AND OUTPERFORMS NON-EXPERTS IN INFLAMMATORY BOWEL DISEASE: A NETWORK META-ANALYSIS

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Introduction: Histological diagnosis and assessment of Inflammatory Bowel Disease (IBD) face significant intra- and inter-observer variability, even among expert pathologists (1). Artificial intelligence (AI)- assisted digital pathology models have been developed to address this challenge (2). However, their comparative accuracy with pathologists remains uncertain.

Aims & Methods: This is the first meta-analysis comparing performance metrics among IBD-related AI models and expert and non-expert pathologists. A systematic search identified papers investigating AI-digital pathology in IBD, published from inception to March 2024. Using the online EPPI-reviewer platform duplicates among PubMed and Scopus were automatically excluded. Consequently, four experienced IBD researchers conducted blind manual screening and excluded studies lacking sufficient data. We computed the pooled diagnostic accuracy metrics, including sensitivity or recall, positive predictive value (PPV) or precision, observed agreement or accuracy, F1 score and Matthew's correlation coefficient (MCC). Data were meta-regressed and network meta-analysed using the "meta" and "netmeta" libraries from R v4.0, respectively (R Foundation for Statistical Computing; Vienna, Austria).

Results: Among 223 studies retrieved after the systematic search strategy, only 23 were selected for full-text evaluation, and 12 were included for final analysis (*Table 1*).

The included studies were published between 2022 to 2024. According to the random effect model, pooled sensitivity, PPV, accuracy, F1 score and MCC of AI models were 0.85 (0.81; 0.88; I^2=94.9%), 0.88 (0.85; 0.90; I^2=91.1%), 0.88 (0.85; 0.90; I^2=96.3%), 0.85 (0.82; 0.88; I^2=96.4%), and 0.71 (0.60; 0.82; I^2=100.0%), respectively. Although high heterogeneity was found, accuracy sub-analysis showed no significant difference between CD vs UC (.88 vs.87) or disease diagnosis vs disease assessment (.89 vs .87). Expert pathologists reached the highest pooled accuracy (.91), fol-

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lowed by AI models (.88) and non-expert pathologists (.82), being significantly different according to the meta-regression (p<.01). Network meta-analysis revealed no significant difference between the performance of AI vs expert pathologists, while AI outperformed non-experts in accuracy (OR 2.68; 95% CI 1.08-.66), F1 score (OR 2.52; 95% CI 1.03-6.16), and MCC (OR 4.53; 95% CI 2.08-9.88).

Studies included		n (%)
Inflammatory Bowel Disease (IBD) phenotype*, n (%)	Ulcerative colitis (UC)	9/12 (75%)
	Crohn's disease (CD)	3/12 (25%)
	Not specified	1/12 (8%)
Aim of the study, n (%)	IBD diagnosis	2 (17%)
	Disease assessment	10 (83%)
Observer, n (%)	Artificial intelligence (AI)	12 (100%)
	Expert pathologist	2 (17%)
	Non-expert pathologist	1 (8%)

Table 1. Studies included in the network meta-analysis.
**Not-mutually exclusive variables.*

Conclusion: AI-assisted digital pathology exhibits promising potential in IBD diagnosis and disease activity assessment, comparable to expert pathologists and superior to non-experts. These findings support the integration of AI models to expedite central reading in clinical trials and hold promise for future application in clinical practice. Nonetheless, standardisation efforts are pivotal to reducing heterogeneity and facilitating widespread AI implementation.

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PP0506

ASSOCIATION OF ULTRASONOGRAPHIC INFLAMMATORY ACTIVITY PARAMETERS WITH FUTURE DRUG FAILURE IN CROHN'S DISEASE PATIENTS DURING MAINTENANCE ANTI-TNF THERAPY

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Introduction: Intestinal ultrasound (IUS) is effective for monitoring treatment efficacy in Crohn's disease (CD). Bowel wall thickness (BWT) is used as a surrogate of inflammatory activity. However, the added value of other IUS parameters ((bowel wall flow (BWF), stratification (BWS), lymph node enlargement, mesenteric fat hypertrophy, and complications)) for monitoring and predicting treatment failure during anti-TNFs' maintenance therapy is still poorly defined and was the goal of this study.

Aims & Methods: This was a retrospective cross-sectional study. The study population included patients treated on infliximab or adalimumab maintenance treatment who had undergone an index intestinal ultrasound examination. We examined the added value of various ultrasonographic activity signs, in addition to bowel wall thickness, for their association with subsequent treatment failure. Treatment failure was defined as the need for CD-related surgery, hospitalization, corticosteroid use, dose escalation, and drug discontinuation.

		Treatment Failure			Drug Discontinuation		
Variable		Adjusted Odds Ratio (aOR)	95% CI	P-Value	Adjusted Odds Ratio (aOR)	95% CI	P-Value
BWT	≥3	3.25	1.30-8.12	0.012	2.3	0.94-5.60	0.068
	≥3.5	2.7	1.07-6.78	0.035	1.78	0.73-4.35	0.209
	≥4	2.47	0.93-6.51	0.068	2.68	1.04-6.89	0.042
High BWF		2.93	1.16-7.39	0.023	1.5	0.63-3.59	0.361
Enlarged lymph nodes		1.41	0.32-6.34	0.651	0.68	0.15-3.07	0.62
Mesenteric fat hypertrophy		4.44	1.15-17.07	0.03	5.29	1.61-17.44	0.006
Disturbed BWS		4.64	0.93-23.20	0.062	2.73	0.71-10.44	0.143
IUS Complications		6.04	0.69-52.59	0.104	4.66	0.80-27.05	0.086

Table 1: Sonographic Characteristics Associated with Treatment Failure and Drug Discontinuation.
Abbreviation: BWT - bowel wall thickness; BWF - bowel wall flow; BWS - bowel wall stratification; IUS - intestinal ultrasound.

Results: The study included 103 CD patients treated with anti-TNF therapy (67% adalimumab and 33% infliximab) who underwent an IUS at a median of 10 months (IQR 6-18) after starting therapy. Of 103 patients, 59.2% experienced treatment failure at a median of 5 months (2-12.5) after the index IUS examination. Among the failure group, bowel thickness at preceding index IUS was significantly higher compared with patients without failure (3.9 vs 3mm, respectively, P= 0.007), BWF was higher (49.2% vs.

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23.8%, P= 0.009), and mesenteric hypertrophy was more prevalent (20.4% vs 7.1%, P=0.006). Logistic regression identified increased BWT and mesenteric hypertrophy as having a significant association with therapy failure.

A sensitivity analysis with the outcome of therapy discontinuation revealed that mesenteric fat hypertrophy was a significant predictive factor along with increased BWT (Table 1).

Conclusion: Our study highlights the significance of BWT, BWF, and mesenteric fat hypertrophy as key sonographic markers associated with anti-TNF treatment failure in Crohn's disease. Integrating these indicators into scoring systems may enhance predictive accuracy, providing a quantitative tool for evaluating disease severity and treatment response.

Disclosure: Nothing to disclose.

PP0507

NON-INVASIVE DIAGNOSTICS OF ULCERATIVE COLITIS REMISSION

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Introduction: Studies using a variety of diagnostic techniques confirm the importance of the gut microbiota in the pathogenesis of ulcerative colitis (UC), as well as its possible influence on the effectiveness of drug therapy and the prognosis of the disease. In different phases of the UC course, structural and quantitative changes in the gut microbiota suggest its use as a non-invasive diagnostic tool and the tool predicting the achievement of clinical and endoscopic remission of the disease.

Aims & Methods: The objective of the study was to determine the taxonomic diversity of the gut microbiota in patients with UC in various phases of the disease and to identify microorganisms that are predictors of remission. The study included 21 patients (age from 20 to 41 years, 13 men and 8 women) in the relapse phase of UC - left-sided UC with mild to moderate activity (Truelov-Witts scale, Mayo index); and 20 patients (age from 26 to 52 years, 10 men and 10 women) in the phase of clinical endoscopic remission. The gut microbiota was assessed by NGS sequencing of the bacterial 16S ribosomal RNA gene structure using the IlluminaMiSeq platform. After the normality test, we performed statistical analysis using the nonparametric Mann-Whitney test, the analysis is presented by median and quartiles.

Results: We predicted clinical and endoscopic remission of UC using a logistic regression model, including microorganisms, the number of which in absolute values changed more significantly. In cases of achieving clinical and endoscopic remission of the disease, we recorded a statistically significant increase in the levels of Streptococcus, NK4A214_group and Pelomonas and a decrease in Enterococcus and Actinomyces. The developed logistic regression model is statistically significant and statistically reliable using evaluation criteria: chi-square (p = 0.002), Cox & Snell's pseudo R-squared is 0.556, Nagelkerk is 0.741, which allows its use in wide gastroenterological practice to make decisions on modifying the course of drug therapy for UC (Table 1).

Genus	Coefficient	Odds ratio	p
Enterococcus	-117,244	$1{,}2{\cdot}10^{-51}$	0,011
Streptococcus	1,117	3,06	0,045
NK4A214_group	2,335	10,33	0,008
Pelomonas	67,529	$2{,}1{\cdot}10^{29}$	0,011
Actinomyces	-55,947	$5{,}0{\cdot}10^{-25}$	0,008
Constant	-0,528	0,59	0,201

Table 1. Results of modeling the gut microbiotope in patients with UC using logistic regression.

Conclusion: The study results provide a prognostic model for non-invasive diagnosis of achieving UC remission. Predictors of achieving clinical and endoscopic remission of UC are high levels of Streptococcus, NK4A214_group and Pelomonas, as well as a decrease in the number of Enterococcus and Actinomyces.

Disclosure: Nothing to disclose.

PP0508

ADHERENCE TO QUALITY PERFORMANCE MEASURES FOR ENDOSCOPY IN IBD: A NATIONWIDE ITALIAN SURVEY

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Introduction: The quality of gastrointestinal endoscopy in inflammatory bowel disease (IBD) patients has been recently identified as a major priority being associated with better outcomes.

Aims & Methods: We aim to assess the adherence of Italian gastroenterologists to current international recommendations on quality performance measures for IBD endoscopy.

From March to July 2023 a total of 178 Italian consultants were invited to participate to an online questionnaire-based survey, which included questions on their demographics, professional characteristics and clinical behaviour as well as structural features of their affiliations. Four clinical domains were analysed: A) the first endoscopy for suspected IBD; B) assessment of endoscopic activity; C) post-operative Crohn's disease; and D) surveillance endoscopy for patients at high risk of colorectal neoplasia. Recommendations from three consensus on quality performance measures for IBD endoscopy (BRIDGE, ECCO, ESGE) were used as reference standards. A p value ≤0.05 was considered statistically significant.

Results: Overall, responders paid low attention to perform ileal biopsies in case of the first colonoscopy for suspected IBD (67% vs standard of care 80%) as well as to the use of the Paris (70%) and mucosal pattern classifications (60%) for lesion characterization of endoscopic visible lesions. Ileal intubation (97%) and colonic biopsies (80%) during the first colonscopy for suspected IBD, adequate tissue samples management (96%), post-operative schedule within 1 year (94%), implementation of surveillance plans (96%) and use of endoscopic index of activity (95%) were highly reported. No significant regional differences were observed, while significantly lower performances were reported for many outcomes by small centers (<100 IBD patients in follow-up and/or no more than 2 physicians for each IBD unit) or physicians spending <50% of their clinical activity for IBD. Young doctors <35 years old had a significantly higher use of modern classifications for lesion characterization (Paris: 89% vs 68%; Kudo:82% vs 56%).

Conclusion: In Italy, the adherence to current standards of care for IBD endoscopy is generally good, with some practices to be improved. There is a need to support the endoscopic activity of small centers and doctors less engaged in IBD.

Disclosure: Nothing to disclose.

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PP0509

SERUM BIOMARKERS OF PROTEOLYTIC TISSUE DESTRUCTION, FORMATION AND MACROPHAGE ACTIVITY CAN DISCERN PATIENTS WITH IBD ACCORDING TO INFLIXIMAB TREATMENT NON-RESPONSE OR RESPONSE

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Introduction: Characterized by chronic inflammation, patients with Inflammatory Bowel Disease (IBD) experience detrimental remodeling of their intestinal extracellular matrix (ECM). Treatment with anti-inflammatory drugs can reduce inflammation, leading to remission and tissue healing. However, adequate monitoring of patients is critical to ensure and maintain treatment response. As potential surrogate markers of ECM remodeling, we investigated blood-based biomarkers of type III and -VII collagen and posttranslational modifications of vimentin in patients with IBD.

Aims & Methods: By quantifying biomarkers of intestinal tissue destruction (C3M), tissue formation of type VII collagen (PRO-C7), and the macrophage activity biomarker, VICM, we aimed to determine the value of these biomarkers for identifying and monitoring response to infliximab (IFX).
160 IBD patients initiating IFX treatment at visit 1 were included, having blood drawn at visits 1, 2, and 3, with treatment response evaluated at visit 3 using the Physicians Global Assessment (PGA) score. The response was defined as a PGA of 0 at Visit 3. Measuring the biomarkers C3M (MMP-9 degraded type III collagen), PRO-C7 (pro-peptide of type VII collagen), and VICM (citrullinated and MMP-degraded vimentin) in the blood, we evaluated their ability to identify responders (n=141) and non-responders (n=19) to IFX induction therapy. The biomarker levels between responders and non-responders were statistically assessed using a stepwise logistical regression model and DeLong et al. receiver operating characteristics.

Results: Using logistical regression, the combination of serum C3M and PRO-C7 measured at visit 1 could discern non-responders and responders with an AUC of 0.717 (p=0.001, sens: 61.1%, and spec: 80.2%). At visit 2, serum VICM provided an AUC of 0.684 (p=0.006, sens: 57.9%, and spec: 78.9%), whereas measuring C3M at visit 3 discerned patients with an AUC of 0.695 (p=0.009, sens: 60.0%, and spec: 74.4%). Finally, we performed a stepwise logistical regression model combining all the biomarkers for a composite score. Measuring PRO-C7 at visit 1, VICM at visit 2, and C3M at visit 3 identified non-responders to IFX treatment with an AUC of 0.797 (p<0.001, sens: 80.0%, and spec: 74.5%).

Conclusion: Quantifying specific fragments of MMP-degraded type III collagen reflecting tissue destruction, the formation of type VII collagen, and the macrophage activity biomarker VICM in patients treated with IFX could, with a moderate to good AUC, identify patients who at visit 3 would be determined a non-responder or responders to IFX treatment based on their PGA. Our data suggest differences in the proteolytic tissue destruction and formation biomarkers, in combination with macrophage activity, can discriminate between responders and non-responders to IFX treatment.

Disclosure: Martin Pehrsson, Marta Alexdottir, Morten Karsdal, Anne-Christine Bay-Jensen, and Joachim Mortensen are employed at Nordic Bioscience A/S
The remaining authors have nothing to declare.

PP0510

BACTERIAL PROFILE OF THE GUT IN PATIENTS WITH ULCERATIVE COLITIS IN DIFFERENT PHASES OF DISEASE PROGRESSION

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Introduction: Numerous studies show that changes in the composition of the gut microbiota constitute an important factor influencing both the development of ulcerative colitis (UC) and the prognosis of the disease, drug metabolism and effectiveness of treatment. The growing global prevalence of UC necessitates continued advances in diagnostics and treatment. Gut microbiota can be considered as a promising dynamic biomarker for non-invasive assessment of the severity of UC and the effectiveness of treatment.

Aims & Methods: The purpose of the research was to study the taxonomic diversity of the gut microbiota in patients with UC at different stages of the disease. The study included 21 patients (ages from 20 to 41, 13 men and 8 women) in the relapse stage of UC - left-sided UC with mild to moderate activity (Truelov-Witts scale, Mayo index); and 20 patients (ages from 26 to 52, 10 men and 10 women) in the stage of clinical endoscopic remission. The study of the coprofiltrate microbiota was carried out by NGS sequencing of the structure of the bacterial 16S ribosomal RNA gene using the Illumina MiSeq platform. Statistical analysis was preceded by testing of the distribution for normality using the Kolmogorov-Smirnov (with Lilliefors significance correction) and Shapiro-Wilk tests. Hypotheses about the distribution corresponding to normal were rejected, and therefore the calculation of descriptive statistics is presented by median and quartiles. Group comparisons were made using the nonparametric Mann-Whitney test.

Results: Statistically significant differences between the relapse and remission groups were identified for 15 genera of microorganisms. In the relapse group, higher levels of Enterococcus, Streptococcus, Granulicatella, Pelomonas, TM7x, Actinomyces, Gemella, Corynebacterium, Bradyrhizobium, and lower levels of NK4A214_group, Pseudoscardovia, Family_XIII_UCG-001, Anaerofilum, Candidatus_Soleaferrea, DNF00809 were found (Table 1).

Genus	Me (Q25%; Q75%)		U Mann-Whitney	p
	Relapse (n=21)	Remission (n=20)		
Enterococcus	0.03 (0.00; 0.26)	0.00 (0.00; 0.00)	24.00	0.012
Streptococcus	1.22 (0.78; 6.27)	0.28 (0.18; 1.15)	25.00	0.035
NK4A214_group	0.07 (0.00; 0.48)	0.87 (0.07; 1.53)	25.50	0.036
Granulicatella	0.08 (0.00; 0.82)	0.00 (0.00; 0.02)	21.00	0.013
Pelomonas	0.08 (0.03; 0.30)	0.01 (0.01; 0.02)	19.50	0.012
TM7x	0.11 (0.01; 0.23)	0.00 (0.00; 0.02)	18.00	0.007
Actinomyces	0.27 (0.07; 0.38)	0.00 (0.00; 0.03)	14.00	0.003
Gemella	0.03 (0.00; 0.37)	0.00 (0.00; 0.01)	21.50	0.013
Pseudoscardovia	0.00 (0.00; 0.01)	0.04 (0.00; 0.14)	25.50	0.028
Family_XIII_UCG-001	0.00 (0.00; 0.01)	0.02 (0.01; 0.03)	25.00	0.025
Corynebacterium	0.01 (0.00; 0.05)	0.00 (0.00; 0.00)	25.00	0.008
Bradyrhizobium	0.01 (0.00; 0.03)	0.00 (0.00; 0.00)	20.00	0.003
Anaerofilum	0.00 (0.00; 0.02)	0.02 (0.01; 0.04)	28.00	0.048
Candidatus_Soleaferrea	0.00 (0.00; 0.00)	0.01 (0.00; 0.01)	24.00	0.009
DNF00809	0.00 (0.00; 0.00)	0.01 (0.00; 0.02)	30.00	0.027

Table 1. Comparison results of the remission and relapse groups of UC (statistically significant differences).

Conclusion: Data obtained demonstrates taxonomic differences at the genus level in patients in the relapse phase and in clinical and endoscopic remission of UC. The identified structural changes in the gut microbiota allow for implementation as a non-invasive method for monitoring the course of progression the disease.

Disclosure: Nothing to disclose.

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PP0511

SYSTEMIC OXIDATIVE STRESS ASSOCIATES WITH CLINICAL AND BIOCHEMICAL DISEASE ACTIVITY IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: A PROSPECTIVE DIAGNOSTIC VALIDATION STUDY

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Introduction: Inflammatory bowel disease (IBD), consisting of Crohn's disease (CD) and ulcerative colitis (UC), are characterized by chronic inflammation of the intestines. This is accompanied by oxidative stress, which plays an important role in the pathophysiology of IBD. Oxidative stress-related biomarkers can serve as potential biomarkers for disease activity monitoring. Systemic free thiols (FT) are found to accurately reflect oxidative stress brought on by inflammation and have shown promise as a biomarker for inflammatory disease activity in IBD.

However, the clinical significance of this marker warrants validation. This prospective diagnostic study aimed to compare FT to other validated biomarkers and investigate the link between FT and endoscopic disease activity in patients with IBD.

Aims & Methods: An interim analysis was conducted in 52 patients. FT were measured using a colorimetric detection method. Clinical disease activity was quantified by the Harvey-Bradshaw Index (HBI) for CD and Simple Clinical Colitis Activity Index (SCCAI) for UC. Biochemical disease activity was determined by measuring blood C-reactive protein [CRP] and fecal calprotectin [fCal] levels. Endoscopic disease activity was quantified with the Mayo endoscopic sub score for UC and the Simple Endoscopic Score (SES-CD) for CD. Serum FT levels were assessed for correlations with disease activity parameters using linear and logistic regression analyses. Discriminative capacity of FT as biomarker was assessed by calculating receiver operating characteristics (ROC) statistics.

Results: Serum FT levels demonstrated a significant inverse correlation with clinical parameters, such as the HBI ($r=-0.53$, $p<0.05$). Additionally, other inflammatory biomarkers, including CRP, exhibited a negative correlation with serum FT levels ($r=-0.43$, $p<0.05$), which were decreased in patients with active IBD. Discrimination between clinically quiescent (HBI<5) and active disease based on serum FT levels was accurate (AUC=0.80, p=0.02) with a value numerically higher than that of fCal (AUC=0.55, p=0.83). Serum FT levels did not significantly differ between biochemically quiescent and active disease based on CRP (AUC=0.69, p=0.20). Both endoscopic scores demonstrated a tendency toward negative associations with serum FT levels.

Patients with moderate and severe disease activity had lower FT levels compared to those with mild disease activity (estimated marginal means [EMM] 497.2 vs 529.5 µM, p=0.43) and 447.1 vs 529.5µM , p=0.06, respectively).

Conclusion: The interim analysis of this study indicates that plasma FT levels inversely correlate with IBD activity and may surpass traditional markers like fCal in detecting disease states. Plasma FT demonstrates potential as a sensitive biomarker for both clinical and endoscopic disease activity in IBD. Further patient inclusion and longitudinal data are essential to validate FT's utility fully. Continued research could establish FT as a key tool for personalized treatment strategies, improving patient outcomes in IBD management.

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PP0512

MICRO VS MACROSCOPY IN PEDIATRIC COLONOSCOPIES: A PROSPECTIVE CLINICAL STUDY

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Introduction: Inflammatory bowel disease (IBD) is a growing cause for concern and often missed or not thought of among pediatric population in developing economies. It is reported that about 25% of IBDs present before the age of 20.

Growing plethora of investigations have not been able to replace the role of direct visualisation of intestinal mucosa in various colorectal issues. Macroscopic findings may or may not help in clinching the diagnosis of IBD, hence "Biopsy" remains the Elixir of diagnosis. In this article, we review and attempt to correlate colonoscopic(macroscopy) and histological (microscopy) findings of colonoscopies in pediatric population, done over a period of 1 year.

Aims & Methods: To compare macroscopic findings versus histopathology findings of colonoscopies in pediatric population in suspected IBD.

To study the correlation of macroscopic findings with histopathologic findings of colonoscopies in pediatric population and to assess the frequency and pattern of microscopic impressions of colonoscopies in our cohort.

A prospective observational study carried out in a tertiary referral centre in children who underwent colonoscopies, aged 1 month to 17 years over a period of 12 months.

All children whose parents consented were included in the study. Inclusion and Exclusion criteria were formulated. Informed consent was obtained from the parents. Clinical details, colonoscopy findings and histopathology reports were. A total of 135 cases were enrolled during our study period. Results were collated using Microsoft Excel ™.

©Ethical clearance was obtained from the institutional ethical committee.

Results: We found that the diagnosis of IBD was much more specific than expected with good sensitivity when data was matched between Macroscopy and Microscopy

	Colonoscopy Finding Marked "Normal"	Colonoscopy -Moderate to Severe inflammation	Histopathology Moderate-Severe inflammation	Sensitivity	Specificity	P value
RECTUM	81%	19%	26%	68.18	94.87	0.028
COLON	86%	14%	27.4%	45.83	96.05	0.019
ILEUM	57%	17%	19%	85.71	94.19	0.021

Conclusion: It is our standard practice to do colonoscopic biopsies from 7 areas (terminal ileum, cecum, ascending colon, transverse colon, descending colon, sigmoid colon, rectum) of the colon to exclude microscopic colitis which may be missed by the endoscopist or marked as 'normal'. Microscopic findings are considered as 'Gold standard' in the diagnosis of IBD and we envisage that it will remain the gold standard in the many next decades to come

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Our data of sensitivity and sensitivity of the microscopic findings inclined the relevance of colonic mucosal biopsies in children with a normal-appearing mucosa and also the need to analyse every segment of the colon to diagnose IBD that may masquerade as "normal" in children which will add burden to the family, society and health care system and more importantly to the child's quality of life.

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Disclosure: Nothing to disclose.

PP0513

METABOLOMIC PROFILING FOR IBD DIAGNOSIS: SEARCH FOR OPTIMAL SETS OF METABOLOMIC IBD MARKERS

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Introduction: Crohn's disease (CD) and ulcerative colitis (UC) are the most common types of IBD. CD and UC are practically indistinguishable at early stages, which makes its diagnostics difficult. Various molecular methods are designed to carry out for global search of the most informative markers for IBD types verification and CD and UC differential diagnosis. Among omics technologies, metabolomics, as one of the promising tools that can contribute to the search and rapid implementation of informative markers. Today, various options are available for the assessment of low molecular weight compounds, such as HPLC-MS/MS, GC-MS/MS, etc. In this study, we combined the search for metabolomic markers using HPLC-MS/MS and HS GC-MS/MS methods for analyzing such groups of metabolites as lipids, volatile metabolites and hydrophilic metabolites in serum and stool samples from Crohn's disease and ulcerative colitis patients.

Aims & Methods: The study was conducted for groups of patients with a verified diagnosis of Crohn's disease and ulcerative colitis. Serum and stool samples were obtained from patients in a hospital. Informed consent for the study was obtained from all patients. The total number of serum and stool samples was 160 samples (CD and UC combined). The samples were provided by the Department of Gastroenterology of the M.F. Vladimirskiy Moscow Regional Research and Clinical Institute, Moscow, Russia., as well as the Department of Gastroenterology of the LOPUKHIN FRCC PCM. Lipid and hydrophilic fractions of metabolites extracted from serum was carried out using HPLC-MS TripleTOF 6600+ System. Stool samples were analyzed using a Shimadzu QP2010 Ultra GC/MS with a Shimadzu HS-20 headspace extractor.

Results: Stool samples analysis shown that isovaleric acid is one of the markers for the IBD and the ratio of valeric and isovaleric acids using the logistic regression method allows us to obtain a risk coefficient for the IBD development. Analysis of the serum hydrophilic fraction demonstrated quantitative difference for the following compounds: Sphingosine, Urocanic Acid, N-Acetylcytidine, 5'-S-Methylthioadenosine, Adenine, -Hydroxyanthranilic acid, Acetylcarnitine, O-Acetyl-L-carnitine, Nicotinamide, Phenylalanine, Isoleucine, Choline chloride, Proline, Cholic acid_CA, Chenodeoxycholic acid, Inosine, Glycoursodeoxycholic acid. Mentioned markers contributed to the IBD verification, while O-Acetyl-L-carnitine contribute to the differentiation between Crohn's disease and ulcerative colitis. When analyzing the lipid fraction, differences were identified for a limited range of lipids, however, such lipids as Cer 37:0;3O|Cer 20:0;2O/17:0;O, PE P-36:2|PE P-18:0_18 :2 and PC 36:2|PC 18:1_18:1 were detected only in CD samples.

Conclusion: The obtained metabolomic differences in the hydrophilic fraction and volatile components make it possible to reliably verify inflammatory bowel diseases, since separate lipid markers and O-Acetyl-L-carnitine contribute to the CD detection.

Disclosure: Nothing to disclose.

PP0514

APPLICATION OF ICCE CRITERIA FOR PATIENTS UNDERGOING SMALL BOWEL CAPSULE ENDOSCOPY FOR SUSPECTED CROHN'S DISEASE AND ITS DIAGNOSTIC YIELD

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Introduction: Small bowel capsule endoscopy (SBCE) is a recommended modality to investigate suspected small bowel Crohn's disease (CD). The Interational Conference on Capsule Endoscopy (ICCE) set forward criteria to optimize patient selection and diagnostic yield. Capsule candidates should meet one criterion from group A (GI symptoms) along with 1 other criterion from group B/C/D (extraintestinal symptoms/ inflammatory markers /abnormal radiology).

Aims & Methods: The aim of this study is to evaluate the diagnostic performance when ICCE criteria are satisfied in those referred for SBCE with suspected CD. We performed a retrospective single-center study of patients referred for SBCE with suspected CD between March 2021 and March 2024. Demographics, clinical and capsule data were assessed from our hospital's database and recorded. Patients with incomplete data were excluded. A positive SBCE was defined as a Lewis Score (LS) ≥135. Patients were grouped according to ICCE criteria: ICCE+ were those who satisfied ≥ 2 ICCE criteria and ICCE- were those who did not fulfil criteria. Where available, histopathological results, MDT outcomes and treatment decisions were analyzed to determine CD diagnosis.

Results: A total of 335 of 433 capsules for suspected Crohn's disease were included (58.5% female, mean age 42 years). In all, 82 capsules (24.5%) had clinically significant enteritis (median LS= 266, range 135-6600). 119 patients (35.5%) satisfied ICCE criteria; ICCE A+B (n=3, 2%), ICCE A+C (n=82, 69%), ICCE A+D (n=19,16%), ICCE x3 (n=15, 13%). Of those, 48 patients (40%) had a positive capsule (average LS = 917.46). As expected ICCE + cases were statistically more likely to have significant enteritis compared with ICCE – cases; 48/119 (40%) versus 34/216 (16%) respectively, OR 3.49, p< 0.0001, 95%CI 2.0892 to 5.8508.

Of note the diagnostic yield was similar for ICCE category B,C &D (p=0.6356). On follow up a histological diagnosis of CD was made in 25 (30%) cases and was similar in both the ICCE + and ICCE – groups, 17/ 48 versus 8/34 (p=0.09).

Conclusion: ICCE criteria in our cohort predicted significant enteritis and CD. Each of the secondary ICCE categories performed similarly. Of interest is the significant rate of enteritis in patients not meeting the criteria, with similar rates of confirmed CD to the ICCE + cohort, suggesting these patients still warrant investigation with capsule and that the clinical features and presentation of small bowel CD is varied.

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Disclosure: Nothing to disclose.

PP0515

THE ACE (ALBUMIN, CRP, ENDOSCOPY) SCORE IN ACUTE ULCERATIVE COLITIS

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Introduction: Patients with acute severe ulcerative colitis (ASUC) have a lifetime risk of an acute flare requiring hospital stay. The diagnosis of acute severe ulcerative colitis is based on Truelove and Witts criteria and the treatment by intravenous steroids showed improved clinical remission and reduced mortality. However, some patients respond poorly to systemic steroids and need a medical treatment by of anti TNF agents or a surgical treatment by colectomy.
The aim of our work is to study the ACE score as a predictive factor of surgery in acute severe ulcerative colitis.
Aims & Methods: We conducted a retrospective study on the Hepato gastro enterology department in IBN Rochd University Hospital of CASABLANCA, including all the admissions of patients with acute severe ulcerative colitis between 2016 and 2023. Demographic (sex, age, smoking habits, age of the diagnosis, prior episodes of severe ulcerative colitis, extent of the disease according to Montreal classification, maintenance medication prior to hospital admission), clinical (lichtiger score at admission and 3 days after IV steroid, laboratory (albumin, CRP) at day 1 and day 3 during hospital stay and endoscopic (Mayo score and UCEIS) characteristics of the patients were collected from medical records.
The primary outcome was to calculate the ACE score and determine the association between a high score and a major risk of surgery.
The score ACE was calculated following this formula: albumin≤30 g/L{score=1}+ CRP ≥50 mg/L{score=1} + increased endoscopic severity {score=1} predicting 78.1% of no responders .
Results: 97 patients were included in the study. The median age was 28 years with interquartile range (IQ) [15, 64]. The sex ratio was 1. 22,7% had a history of smoking. For 53 patients, acute severe ulcerative colitis was inaugural. At day1, the median C-reactive protein (CRP) was 84 mg/dl and the mean of albumin level was 29.9+/-6.28. The evolution at day 5 was marked by the persistence of a high Lichtiger score in 21 patients. 10 were treated with anti TNF alpha and 11 underwent colectomy.
Younger age (<20 years) and inaugural acute severe ulcerative colitis was significantly associated with a higher risk of colectomy (p=0.019; p=0.001). Elevated C-reactive protein (>45) at day 3 (p=0.004), decreased albumin level (<30g/L) (p=0.02) and positive oxford criteria (p<0.01) was also significantly associated with colectomy risk. The ACE (Albumin, CRP, Endoscopic) score was calculated in our patients, and a higher score of 3 was significantly associated to a major risk of colectomy.
Conclusion: Multiple biomarkers have been used to predict a non-response to steroids in the situation of acute severe ulcerative colitis. The most used is Travis score at day 3. The ACE score has been used recently in many studies , and confirm its ability to predict non responders to first line medical therapy. Further research and validation of these scoring systems are warranted to enhance their accuracy and reliability in predicting the need for a second line therapy or surgery.
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Disclosure: Nothing to disclose.

PP0516

THE INFLUENCE OF THE BODY MASS INDEX ON THE COURSE OF INFLAMMATORY BOWEL DISEASES

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Introduction: The role of nutritional status, as indicated by the body mass index (BMI), ininflammatory bowel diseases inflammatory bowel diseases (IBD) is encountered at various levels: etiopathogenic, therapeutic, and prognostic.
Aims & Methods: The aim of this study was to assess the impact of the BMI on the disease course of IBD patients.
Methods: This was an 8-year study period [January 2016-December 2023] of patient records followed for IBD. We collected clinical, biological, radiological, endoscopic, therapeutic, and follow-up data. Patients were divided into three groups based on body mass index (BMI) at diagnosis: group 1: overweight or obese patients (BMI ≥ 25 kg/m^2), group 2: patients with normal BMI (18.5 ≤ BMI < 25 kg/m^2), and group 3: patients with malnutrition (BMI < 18.5 kg/m^2). A comparison of the three groups was performed through univariate logistic regression.
Results: We enrolled 126 patients, with a mean age of 37.3 years [14-75] and a gender ratio (M/F) of 1.5. The median BMI at inclusion was 20.98 kg/m^2 [13-33]. Malnourishment was observed in 24.6% of patients at the disease first diagnosed. Obesity or overweight were noted in 18.25% of patients, while 57.14% had normal BMI. Patients with malnutrition mostly had Crohn's disease (96.77%) with ileal, colonic, and ileocolonic localization in 3, 13, and 14 cases respectively. Comparison of these groups showed no significant differences in terms of age (p = 0.09), sex ratio (p = 0.06), smoking history (p = 0.54), disease duration (p = 0.25), disease phenotype (p = 0.64), perianal (p = 0.67) or upper digestive tract involvements (p = 0.4).
However, a significant association was observed between normal BMI and clinical and biological remission compared to other BMIs (p = 0.013). After a mean follow-up of 56 months [6 - 90], there was no significant difference

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between the three groups regarding the occurrence of CD complications (p = 0.78), the use of immunosuppressive therapies (p = 0.49), biotherapies (p = 0.24), or surgical intervention (p = 0.12).

Conclusion: According to our study, a normal BMI seems to be associated with better control of IBD, highlighting the importance of nutritional status in these patients. This findings need to be confirmed by large-scale studies.

Disclosure: Nothing to disclose.

PP0517

LIVER FIBROSIS PREDICTS RESPONSE TO ADVANCED THERAPY IN INFLAMMATORY BOWEL DISEASE: A NEW PERSONALIZED TREATMENT APPROACH?

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Introduction: Crohn's disease (CD) and ulcerative colitis (UC) are chronic and progressive inflammatory bowel diseases (IBD). It is estimated that up to 30-50% of patients with IBD develop intestinal fibrosis leading to complications such as strictures. Liver fibrosis occurs up to 16.7% of patients with IBD. However, the impact of liver fibrosis on the response to therapy in patients with IBD is unknown.

Aims & Methods: Our aim was to investigate the association between liver fibrosis before starting an advanced therapy and response to treatment in IBD.We conducted a retrospective analysis of data that were prospectively gathered from consecutive patients with a confirmed diagnosis of UC or CD, spanning from November 2021 to December 2023. All patients who started an advanced therapy and had transaminase concentrations, body mass index (BMI), history of diabetes, and platelet counts available were eligible. Several scores of liver fibrosis were measured including AST/ALT ratio (AAR), AAR platelet ratio index (AARPRI), AST to platelet ratio index (APRI), body mass index AST/ALT ratio and diabetes mellitus score (BARD), fibrosis index based on four factors (FIB -4), and modified FIB-4 (m-FIB4).

All available clinical (partial Mayo score for UC and Harvey-Bradshaw index for CD), biochemical (fecal calprotectin, FC), and endoscopic (endoscopic Mayo score for UC and simple endoscopic score for Crohn's Disease (SES-CD) and Rutgeert's score for CD) data at 6 months were collected. A logistic regression multivariate analysis was performed to identify any correlation between scores of liver fibrosis at baseline and response to advanced therapies. Analyses were subjected to correction using the false discovery rate (FDR) method.

Results: In total, 122 patients were included in the analysis (73 UC (59.8%), 65 female (53.3%), median age of 42.8 years). Approximatively half of the patients were biologic naïve (60, 49.2%), while among biologic-experienced subjects 23/62 (37.1%) had been treated with more than one advanced therapy. Patients were treated with ustekinumab (46, 37.8%), infliximab (33, 27.0%), vedolizumab (16, 13.1%), adalimumab (15, 12.3%), upadacitinib (6, 4.9%), tofacitinib (4, 3.3%), golimumab (1, 0.8%), filgotinib (1, 0.8%). A small proportion of patients (16/122, 13.1%, 2 CD and 14 UC) discontinued treatment at 6 months. Clinical remission at 6 months was achieved in approximately half of CD (23/49, 46.9%) and UC patients (39/73, 53.4%). Similarly, half of CD patients (25/49, 51.0%) and one third of UC patients (22/73, 30.1%) had FC normalization (<150µg/g). A quarter of CD patients (4/16, 25.0%) and UC patients (9/28, 32.1%) achieved endoscopic remission at 6 months. Two UC patients (2.7%) were hospitalized. Five patients (4.1%) underwent surgery (2 UC-related and 3 CD-related). No cases of neoplasia or death occurred. In the multivariate analysis, mFIB-4 at baseline was a predictor of clinical activity at 6 months in CD (FDR=0.08), while AARPRI and AAR at baseline predicted clinical activity at 6 months in UC (FDR=0.08 and FDR=0.06, respectively).

Conclusion: Patients with IBD and concomitant liver fibrosis have a lower rate of clinical remission after 6 months of advanced therapy. Scores of liver fibrosis could predict the response to therapy. Further prospective studies are warranted to implement their use in clinical practice.

Disclosure: Nothing to disclose.

PP0518

COMPLETE ENDOSCOPIC HEALING ASSOCIATED WITH BETTER OUTCOMES THAN PARTIAL ENDOSCOPIC HEALING IN PATIENTS WITH CROHN'S DISEASE: RESULTS OF A PROSPECTIVE MULTICENTER STUDY

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Introduction: The STRIDE II guidelines recognize endoscopic healing (EH), defined by an SES-CD score ≤2 or a CDEIS score <3, as one of the main therapeutic targets in Crohn's disease (CD). Nevertheless, complete EH could reduce the risk of long-term complications in CD. The aim of this study was to assess the risk of long-term complications in CD according to the degree of EH achieved.

Aims & Methods: We conducted a prospective multicenter study that included all patients with CD undergoing colonoscopy for EH assessment between September 2019 and September 2022 in three centers and were compared outcomes of patients with complete EH (CDEIS = 0) *vs* partial EH (CDEIS greater than 0 but less than 4). The primary endpoint was therapeutic failure, defined as the need for drug intensification and/or corticosteroid initiation and/or CD-related hospitalization and/or development of a luminal strictures/fistula and/or perianal CD and/or the need for intestinal resection.

Results: A total of 128 patients were included with a median disease duration of 10 (IQR, 4-18) years. Fifty-six (43%) patients had ileocolonic location and 79 (62%) an inflammatory (B1) behavior. Patients had been on treatment for a median of 12 months (8.3-24.3) prior to colonoscopy and 81 (63%) patients had complete EH. After a median follow-up of 22.5 months [IQR, 19.3-23.7], therapeutic failure was observed in 40 patients (31%). The risk of therapeutic failure was significantly higher among patients with partial EH group (25% *vs*. 43% p = 0.036, HR = 1.90; IC95% = [1.02 - 3.53]; p=0.043). The risk of CD-related hospitalizations (2.5% *vs.* 17%), p=0.005, HR= 7.09; IC95% = [1.51 – 33.4]; p=0.013) was significantly higher in patients with partial EH. The risk of intestinal resection, steroid initiation and drug intensification among patients with complete and partial EH, were 0% *vs*. 4.3% (p=0.13), 2.5% *vs.* 11% (p=0.1) and 22% *vs.* 36% (p=0.088), respectively.

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Conclusion: This prospective multicenter study confirms that complete endoscopic healing is associated with better long-term outcomes than partial endoscopic healing in patients with CD, as well as fewer hospitalizations and an overall decreased risk of treatment failure.
Disclosure: Nothing to disclose.

PP0519

A PROSPECTIVE, PILOT, REAL-WORLD STUDY TO ASSESS THE PRESENCE AND SEVERITY OF URGENCY IN PATIENTS WITH CROHN'S ILEITIS

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Introduction: Urgency is not an IBD Patient Reported Outcome (PRO) in the STRIDE II but it has been assessed at baseline and as an outcome measure in recent pivotal RCTs with newer biologicals and targeted synthetic small molecules in both Crohn's disease CD) and ulcerative colitis (UC). However, although it is reasonable to assume that UC patients or CD patients with rectal involvement may complain for severe urgency, urgency has been assessed even in patients with Crohn's ileitis (iCD).
Aims & Methods: The aim of this study was to assess whether urgency is reported by iCD patients with flares or quiescent disease in a Real-World (RW) setting. This prospective, pilot, RW study included CD patients with solely L1 phenotype. L1 location was documented by ileocolonoscopy, histology and, imaging (MRE, US, or capsule endoscopy). Patients with colonic and/or rectal disease, perianal involvement and complicated ileal disease were excluded. Patients were interviewed by office visits during disease flare [HBI>5, CRP <5mg/dl, faecal calprotectin (FC)>250μg/g faecal tissue and SES-CD for terminal ileum >3 if colonoscopy was performed within a month prior to the visit) and during clinical remission (HBI<5, CRP <5mf/dl and FC <200μg/g and ileal SES-CD=0, in a recent ileocolonoscopy). The presence of and severity of urgency was recorded by a Symptom Rating Score of 0-10 (absent to very severe). The presence and severity of abdominal pain, the number and consistency of stools, nocturnal stools, blood and/or mucus in the stools, and the feeling of inadequate evacuation were sought by a structed questionnaire.
Results: During a 15-month period, 85 iCD patients [45 males, median age (range) 33 (18-65) years, median (range) disease duration 9 (0.6-25) years] were enrolled in the study. Patients were receiving a wide range of treatments, including anti-TNF agents (n=35), ustekinumab (N=25), vedolizumab (N=5), azathioprine (n=10), budesonide (n=5) and mesalazine (n=15). Fifty patients had a flare, 65 were in remissionand 30 patients were interviewed sequentially in flare and remission of disease.
All patients in disease flare reported abnormal classical PROs of CD. Moderate to severe urgency [median score 5, range 4-8] was reported by 5/50 (10%) patients with disease flare, 8/65 (12%) patients with disease remission and 2/30 (6.5%) patients who were interviewed during flare and remission of disease. 3/50 patients with urgency during a flare of CD did not report urgency during disease remission. Six of the 8 patients who were interviewed only with iCD in remission and reported urgency had concurrently typical Irritable Bowel Symptoms (IBS, abdominal pain relieved by defecation, alternating diarrhoea/constipation, inadequate bowel emptying).
Conclusion: This prospective, pilot, real-world study shows that the reliability of urgency as classical PRO in patients with iCD is questioned and raises doubts whether urgency should be included in the spectrum of PROs for patients with iCD both in RCTs and in RW studies. Validation of the terminology in different ethnic languages may also pose additional barriers in patient understanding and reporting the symptom in the presence of additional IBS-type symptoms. Large studies are needed to assess if urgency is present in patients with iCD and how reliably is reported especially in multinational RCTs.
Disclosure: Dr GJ Mantzaris: Research grants: AbbVie, Genesis, Merck Sharp & Dohme, Takeda; advisor/speaker: AbbVie, Aenorasis, Dr Falk, Ferring, Hospira, Janssen, Merck Sharp & Dohme, MYLAN, Pfizer, Takeda, Vianex.
Dr A Loukou: Nothing to declare.

PP0520

FAECAL PAI-1 LEVEL IS A NEW TOOL FOR DISEASE MONITORING IN INFLAMMATORY BOWEL DISEASES - RESULTS OF THE LDN-051 PROJECT

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Introduction: Crohn's disease and ulcerative colitis, two forms of inflammatory bowel disease (IBD) require lifelong treatment and patient monitoring. Current predictors of relapse and therapeutic success have limitations, therefore there is a high unmet need to identify novel biomarkers in IBD. Monitoring approaches for IBD are mostly invasive, time-consuming, and expensive. Therefore, a simple, rapid, and non-invasive test, with the ability to differentiate IBD from other gastrointestinal conditions and to monitor disease activity, will have important clinical implications. Our previous mucosal cytokine profiling identified that Plasminogen activator inhibitor type 1 (PAI-1) is expressed in IBD patients with active disease, but not in control.
Aims & Methods: Therefore, we aimed to assess the utility of PAI-1 in monitoring disease activity and therapeutic response of IBD patients in different biological samples.
Serum, biopsy, and faecal samples were collected from more than 160 IBD patients and controls without IBD. ELISA method was used to define the level of PAI-1 in the mucosa, serum, and faecal samples and the serum CRP levels were also determined. For the mucosal gene expression analysis qRT-PCR was used. The correlation and ROC analyses were applied to observe the relationship between the disease activity and PAI-1 level.
Results: The applied screening approach detected significantly higher expression of PAI-1 in active IBD compared to the control. We demonstrated that the serum, mucosal, and faecal PAI-1 concentration was significantly elevated in IBD patients showing clinical and endoscopic activity. Additionally, the initial PAI-1 level decreased significantly upon successful therapy in responder patients, whereas it remained unchanged in non-responders. qRT-PCR analysis of the mucosa showed same results of PAI-1 expression pattern as the ELISA. We also showed that faecal PAI-1 selectively increases in active IBD patients, but not in other organic gastrointestinal diseases. In addition, the stability analysis of faecal PAI-1 showed that the protein was detectable for up to one week at room temperature and 4°C. The serum, mucosal, and faecal PAI-1 levels showed positive correlation to the endoscopic activity. ROC curves demonstrated that faecal

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PAI-1 showed a relatively high power to distinguish between active and control patients. However, the serum CRP level reflected to the disease activity, but no to the therapeutic response. Correlation between the serum CRP and faecal PAI-1 level was not observed.

Conclusion: Our results suggest that serum, mucosal and faecal PAI-1 expression reflects to the disease activity in IBD, which could be used to disease monitoring and possibly therapeutic response. The correlation between the faecal PAI-1 level and the endoscopic activity promotes that it could be used as a novel non-invasive disease specific faecal biomarker in the IBD diagnosis.

Disclosure: LDN-051 is owned by Ladon Therapeutics Ltd.

PP0521

PAIN IN INFLAMMATORY BOWEL DISEASE: A FUNCTION OF DISEASE ACTIVITY OR CENTRAL SENSITIZATION?

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Introduction: Chronic abdominal pain is prevalent in individuals with inflammatory bowel diseases (IBD). However, the determinants and phenotypes of IBD-associated pain are not well-characterized, frustrating effective pain management and mitigation strategies

Aims & Methods: Aims: This is an ongoing descriptive, cross-sectional study aimed at phenotyping the types of pain experienced by patients with IBD, the impact on health-related quality of life (HRQoL) and psychological health, and relationship to disease activity assessed by clinical metrics and endoscopy. Methods: Fifty-eight adult patients diagnosed with IBD were prospectively enrolled at a single academic medical center between October 17, 2022 and November 7, 2023. Patients completed a set of online patientreported outcome measures (PROM)-based standardized and validated questionnaires (Table 2) just prior to a scheduled colonoscopy. Endoscopically active disease was defined as Mayo Endoscopic Score ≥ 1 for ulcerative colitis (UC) and SES-CD ≥ 3 for Crohn's Disease.

Results: A total of 58 patients (28 UC/30 Crohn's, 24 male/34 female, median age 44.5±14.5 years) were included. An initial analysis examined relationships based on endoscopic disease activity. Interestingly, there was no significant difference between those with endoscopically active disease and those without on metrics of pain, HRQoL, or impact on psychological health ($p>0.05$). However, those with endoscopically inactive disease had significantly higher scores on central sensitization (CS) (31.1±11.5 vs. 23±12.1, $p<0.05$). Using a validated score threshold of 40 on the central sensitization inventory (CSI), the cohort was then stratified into two new groups (those with and without evidence of CS).

Overall, 12/58 patients met diagnostic criteria for CS. Table 1 summarizes the baseline characteristics of the two groups. As detailed in Table 2, those with CS had higher scores on abdominal pain (severity and interference, including catastrophizing), visceral hypersensitivity, lower scores on HRQoL questionnaires, and greater scores on depression and anxiety scales ($p<0.05$). There was no significant difference in neuropathic pain ($p=0.85$).

Conclusion: Central sensitization affected at least 20% of this cohort of IBD patients, which is consistent with prior studies. Interestingly, there was a trend towards more central sensitization in those with quiescent disease. CS was associated with worse pain, HRQoL, and psychosocial distress across a broad range of outcomes measured. These data support enhanced attention to contribution of CS to pain in IBD patients, particularly those with persistent pain in remission, as therapies emerge for the treatment of CS.

Disclosure: Nothing to disclose.

PP0522

PREDICTIVE ACCURACY OF FECAL CALPROTECTIN IN ASSESSING CLINICAL ACTIVITY AND DISEASE SEVERITY IN PATIENTS WITH ULCERATIVE COLITIS AND CROHN'S DISEASE

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Introduction: Inflammatory bowel disease is an idiopathic disorder characterized by repeated relapses and remissions. Endoscopy, the gold standard for diagnosis and monitoring, is time consuming, expensive and invasive. Fecal calprotectin has been suggested as a surrogate marker to assess intestinal inflammation and predict disease relapse. Published evidence regarding the utility of fecal calprotectin as an indicator of clinical activity and endoscopic severity in patients with inflammatory bowel disease is not unambiguous.

Aims & Methods: The aim of this study was to determine the use of fecal calprotectin as a predictor of clinical activity and endoscopic severity in patients with inflammatory bowel disease.

Methods: Patients fulfilling selection criteria were recruited for the study. Clinical examination findings, laboratory investigations and fecal calprotectin levels were documented as per proforma. Endoscopic and clinical severity were measured by Simple Endoscopic Score and Crohn's Disease Activity Index for Crohn's disease and by Ulcerative Colitis Endoscopic Index of Severity score and Simple Clinical Colitis Activity Index for Ulcerative Colitis .Lab investigations and calprotectin were repeated after six months and compared with endoscopic and clinical activity indices.

Results: Fecal calprotectin correlated with Crohn's Disease Activity Index both at first visit and follow up (r-0.689, p-0.016)(r-0.425, p-value <0.05). The sensitivity and specificity of calprotectin in detecting active disease and remission at first visit were 93.8% and 80% respectively (AUC-0.869). At follow up, the sensitivity and specificity were 80% and 93.3% respectively (AUC-0.867). In Ulcerative colitis , Simple Clinical Colitis Activity Index score correlated with fecal calprotectin levels at first visit and follow up. (r-0.231 / 0.387, p-value 0.001/0.001). The sensitivity of calprotectin in detecting active disease and remission in ulcerative colitis was 92.6% and specificity was 83.3%. AUC was 0.88. At follow up, the sensitivity of fecal calprotectin in detecting active disease and remission was 89.9% and the specificity was 87.0% ; AUC was 0.879.

Conclusion: Fecal calprotectin levels show significant correlation with clinical and endoscopic activity indices in patients of inflammatory bowel disease. Therefore, calprotectin levels in stool could be used as a surrogate marker for monitoring mucosal status as well as predicting endoscopic remission in these patients.

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Disclosure: Nothing to disclose.

PP0523

FISTULA DRAINAGE ASSESSMENT IS CORRELATED WITH IMPROVEMENT IN QUALITY OF LIFE AFTER 26 WEEKS OF TREATMENT IN PERIANAL FISTULIZING CROHN'S DISEASE

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Introduction: Perianal fistulizing Crohn's disease (pfCD) is a debilitating phenotype that severely impacts quality of life (QoL). pfCD frequently implies a stepwise, combined medical and surgical treatment approach. Treatment goals are ideally individual and include improvement of QoL, however precise outcomes are not determined yet.
Currently, most commonly used outcomes are clinical response/remission, and radiological healing, the latter most relevant to long-term disease remission.
Aims & Methods: The aim of this study was to evaluate QoL in pfCD and compare it with clinical and radiological outcomes after 26 weeks of treatment.
In this prospective exploratory, observational cohort (AMFIBIO), we included patients with pfCD starting on anti-TNF-α therapy, mesenchymal stem cell injection (MST), or hyperbaric oxygen therapy (HBOT) combined with surgical closure. We collected patient-reported IBD-specific (IBDQ-32) and fistula-specific (CAF-QoL) QoL outcomes. MAGNIFI-CD (remission defined as MAGNIFI-CD ≤6), perianal disease activity index (PDAI; response defined as ≤4) and fistula drainage assessment (FDA) were evaluated at baseline and week 26. Paired T tests were used to analyze changes in QoL, clinical and radiological scores. Pearson's correlation coefficients were used to correlate changes between the outcomes.
Results: Twenty-one patients were analyzed at week 26 (Table 1). Mean change between baseline and week 26 in CAF-QoL and IBDQ-32 was 13 (95%CI 3 – 23) and 9 (95%CI -5 – 24) respectively. FDA correlated moderately with improvement in CAF-QoL (ρ=0.59; 95%CI 0.19 – 0.82), and moderately with improvement in IBDQ-32 (ρ=0.49; 95%CI 0.04 – 0.77). Improvement in CAF-QoL (ρ=0.31; 95%CI -0.20 – 0.69) and IBDQ-32 (ρ=0.35; 95%CI -0.16 – 0.72) did not correlate with improvement in PDAI, nor with PDAI response (resp. ρ=0.38; 95%CI -0.009 – 0.71, and ρ=0.40; 95%CI -0.08 – 0.72). No correlation was seen between changes in CAF-QoL and MAGNIFI-CD (ρ=0.08; 95%CI -0.39 – 0.52), or MRI remission (ρ=0.22; 95%CI -0.26 – 0.61).

Age (years) at baseline, mean (95%CI)	**34.7 (28.5 – 40.9)**
Female, n (%)	16 (76)
Caucasian, n (%)	16 (76)
Crohn's disease duration (years), mean (95%CI)	8.8 (3.3 – 14.3)
Perianal disease duration (years), mean (95%CI)	37.1 (3.3 – 10.8)
Perianal fistula classification (Geldof et al.), n (%)	2a: 11 (52) 2b: 8 (38) 2c-ii: 2 (10)
Therapy during study, n (%)	anti-TNF: 10 (48) MST: 6 (29) HBOT: 5 (24)
SES-CD at baseline, median [IQR]	2 [0 – 7.5]
CAF-QOL at baseline, mean (95%CI)	58.1 (51.0 – 65.2)
IBDQ-32 at baseline, mean (95%CI)	145.1 (130.7 – 159.4)

Table 1. Patient and fistula characteristics (n=21)

Conclusion: IBD- and fistula-specific Quality of Life scores improve after therapy at week 26 in patients with pfCD. This improvement correlates with fistula drainage assessment, however it does not correlate with PDAI and MRI outcomes.
Disclosure: LM, KB, JT and RvH have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this abstract. CB has received grants from Boehringer Ingelheim and Roche; speakers honoraria from AbbVie, Janssen, Galapagos, Takeda, and Tillotts. JS is President-elect of the European Society of Gastrointestinal and Abdominal Radiology. GD has served as consultant for AbbVie, Astrazeneca, Alimentiv, Bristol-Myers Squibb, Boehringer Ingelheim, Celltrion, Cosmo, Eli Lilly, Galapagos, GSK, Johnson and Johnson, Takeda, Pfizer Inc, Polpharma, Prometheus Biosciences, Tillotts and Ventyx; received speaker fees from AbbVie, Boehringer Ingelheim, Celltrion, Eli Lilly, Johnson and Johnson, Takeda, Pfizer Inc and Tillotts. KG has received grants from Pfizer Inc, Celltrion and Galapagos; consultancy fees from AbbVie, Arena Pharmaceuticals, Galapagos, Gilead, Immunic Therapeutics, Janssen Pharmaceuticals, Novartis, Pfizer Inc., Samsung Bioepis and Takeda and speaker's honoraria from Celltrion, Ferring, Janssen Pharmaceuticals, Novartis, Pfizer Inc, Samsung Bioepis, Takeda and Tillotts.

PP0524

THE IMPACT OF STRESS ON INFLAMMATORY BOWEL DISEASE PATIENTS' CONDITION AND THEIR QUALITY OF LIFE

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Introduction: It has long been hypothesised and increasingly proven that stress (both acute and chronic) can have a significant impact on inflammatory bowel disease (IBD) patient's condition in terms of relapses and worsening disease.1 It is also understood that patients with IBD are at a higher risk of developing depressive disorders than the healthy population. 2 As such ongoing work into the bidirectional gut brain axis and its association with the inflammatory response in the gut is important in understanding how reduction in stress can assist in managing IBD. 3
Aims & Methods: Our study aim was to assess the impact of stress on our IBD patients' self-reported symptoms, to understand how our IBD patients' managed their stress levels and to assess how their inflammatory bowel disease affects their quality of life.
101 IBD patients attending University Lewisham Hospital completed an anonymous structured paper questionnaire between December 2023 and January 2024. Data was analysed for age, gender, IBD subtype and medications, the impact of stress on their condition, their methods of stress

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reduction as well as the impact of IBD on their social, work and sexual life.
Results: 101 patients completed the survey. 55% were female and the median age was 42 years (IQR 30-58). 45% had ulcerative colitis (UC), 48% had Crohn's disease (CD) and 8% had IBD-unspecified (IBD-U). 5.6% of patients were not on treatment, 29.7% on mesalazine, 10.9% on azathioprine and 75% were on a biologic therapy.
Of the 101 patients surveyed 76.2% (77/101) felt that stress triggers flares in their condition. 26.7% (27/101) felt that stress directly led to their IBD developing, while 23.8% (24/101) felt it was a contributing factor. 40.6% (41/101)felt that controlling stress was more important than their medications. 62.4% (63/101) employed methods of stress reduction.
The most common methods were: exercise/sports/yoga (41.5%). walking (30.7%), reading (26.7%), socialising (22.8%), alcohol (10.9%) smoking (5.9%) and recreational drugs (3.0%).
32.7% (33/101) felt their condition made them exhausted all of the time. 63.4% (64/101) reported avoiding social events due to their IBD. 48.5% (49/101) reported their condition affected their sex life. 25.6% (26/101) felt their relationship with their partner was affected because of their condition.
Over half of patients surveyed (57.4%, 58/101) felt that their condition affected their mental health, but only 24.8% (25/101) felt they had received sufficient NHS mental health support.
19.8% (20/101) of patients felt their condition had stopped them from getting promoted at work, and 47.5% (48/101) attended work whilst in a disease flare for fear of losing their job. 14.9% (15/101) reported feeling discriminated against because of their condition.
Conclusion: Over three quarters of IBD patients surveyed felt that stress is a significant trigger for flares, however surprisingly over a quarter of IBD patients perceive stress as a direct factor for their disease developing. IBD significantly impacts patients social, work, and sexual life with a significant impact seen on mental health. This survey found that although a good proportion of patients used commonly perceived healthy methods of stress reduction, almost 20% of patients relied on unhealthy methods (smoking, alcohol/recreational drugs). This data presents a clear argument for a holistic approach to IBD patient care and integrating psychological support within an IBD service. Further work should follow on the relationship between patient perceived stress levels and disease outcomes and failed treatments.
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Disclosure: Nothing to disclose.

PP0525

NON-ALCOHOLIC FATTY LIVER DISEASE AND LIVER FIBROSIS IN INFLAMMATORY BOWEL DISEASE: A FIBROSCAN-BASED ASSESSMENT

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Introduction: Non-alcoholic fatty liver disease (NAFLD) prevalence is elevated in patients with inflammatory bowel disease (IBD) in Western nations. Both intestinal inflammation and metabolic factors contribute to IBD-associated NAFLD. However, the burden of NAFLD in non-Western populations remains unclear.
Aims & Methods: This study aimed to evaluate the prevalence of NAFLD and liver fibrosis in a cohort of Tunisian IBD patients.
Methods: This cross-sectional, single-center study enrolled 84 consecutive IBD patients followed up at our department from April to December 2023. All participants underwent FibroScan examination for hepatic steatosis and fibrosis assessment using liver stiffness measurement (LSM) and controlled attenuation parameter (CAP). Patients with excessive alcohol consumption or recent steroid use were excluded. Univariate and multivariate analyses were performed. Hepatic fibrosis was defined by transhepatic elastography exceeding 7 kPa.
Results:
A total of 84 patients (37 ulcerative colitis, 47 Crohn's disease) were included. The median age was 39 years, with a median disease duration of 10 years. Body mass index (BMI) distribution was: normal weight (38.9%), underweight (11.1%), overweight (41.3%), and obese (8.7%). Mean CAP values increased progressively with BMI: 143.52 dB/m (underweight), 198.77 dB/m (normal weight), 269.3 dB/m (overweight), and 278.51 dB/m (obese). NAFLD was diagnosed in 28.57% of patients, with 1.19% exhibiting significant fibrosis. Increased BMI (odds ratio [OR] 2.03, 95% confidence interval [CI] 1.1–2.62) and older age at IBD diagnosis (OR 1.34, 95% CI 1–1.61) were significantly associated with NAFLD presence. Patients with CD and UC did not differ significantly in terms of hepatic steatosis.
The median stiffness in IBD patients was normal. Hepatic fibrosis was detected in 7 (8.3%) of all IBD patients, predominantly in Crohn's disease (6/7). Most patients (7.14%) had an F2 grade, 1.2% had an F3 grade and no patient had an F4 grade.
Conclusion: This study demonstrates a 28.57% prevalence of NAFLD in Tunisian IBD patients. Higher BMI and older age at IBD diagnosis were associated with NAFLD. Hepatic fibrosis occurred predominantly in Crohn's disease patients. Further longitudinal studies are warranted to elucidate the impact of IBD on hepatic complications.
Disclosure: Nothing to disclose.

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PP0526

VALIDATION OF THE PORTUGUESE VERSION OF IBD-CONTROL QUESTIONNAIRE AND COMPARISON WITH IBD-DISK

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Introduction: IBD-Control questionnaire and IBD-Disk are two easy-to-use IBD-specific patient-reported outcome measures aimed to measure the impact of IBD in the different health domains. Contrary to IBD-Disk, there is no fully published validated Portuguese version of IBD-Control. Additionally, the two instruments have not yet been compared.

Aims & Methods: The authors aimed to translate and validate the IBD-control in Portugal and to compare it with IBD-Disk. Acceptability was verified in a pilot patient group (n = 10) after the forward-backward translation of the IBD-control (Figure 1). Prospective validation involved completion of the IBD-Control, EQ-5D, short IBDQ, IBD-Disk and disease activity measurement by PRO2. Test-retest (2-week repeat) was used for measuring reliability. We compared user-friendliness (through 3 questions) and the ability to identify good disease control (through ROC analysis) of IBD-Control and IBD-Disk.

Results: Questionnaires were completed by 142 IBD patients (108 Crohn's disease, 34 ulcerative colitis). The mean completion time was 168s. Internal consistency (Cronbach's alpha) was 0.796 for the sub-group of 8 questions (IBD-control-8). Construct validity analyses demonstrated moderate-to-strong correlations of the IBD-control-8 and the other instruments (0.55-0.82) (Table 1).

Test-retest reliability for stable patients (n=54) was high (intraclass correlation coefficient - 0.861). The IBD-control-8 showed good discriminant ability between well-controlled and insufficient/poor-controlled disease (15±2 vs 11±4; p<.001). No significant differences were noted between IBD-Control and IBD-Disk regarding user-friendliness and AUC (0,79 vs 0,76) for identification of good disease control.

Conclusion: The Portuguese IBD-control is a rapid, reliable, and valid instrument for measuring disease control from an IBD patient's perspective, with no significant difference in terms of user-friendliness and measurement ability when compared to IBD-Disk.

Disclosure: Nothing to disclose.

PP0527

COMPARISON BETWEEN THE BOWEL ULTRASOUND SCORE AND INTERNATIONAL BOWEL ULTRASOUND SEGMENTAL ACTIVITY SCORE BASED ON ILEOCOLONOSCOPY IN PATIENTS WITH CROHN'S DISEASE

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Introduction: Intestinal ultrasound (IUS) has proven to be a valuable tool for accurately detecting intestinal damage and characterizing disease behavior in Crohn's disease (CD). IUS scoring systems for CD activity have been developed, however all the studies had methodological flaws. Therefore, studies comparing and validating the existing IUS scores in CD patients are required.

Aims & Methods: In this study, we aimed to compare two main existing scores, Bowel Ultrasound Score (BUSS) and International Bowel Ultrasound Segmental Activity Score (IBUS-SAS) in patients with ileal CD using the simplified endoscopic activity score for CD (SES-CD).

An observational cross-sectional study was performed to evaluate the disease activity in adult CD patients with ileal involvement between November 2019 and February 2024. All patients had undergone ileocolonoscopy and IUS, the time interval between the two procedures was lower than 10 days. Receiver operating characteristic (ROC) curve analyses was performed for BUSS and IBUS-SAS to predict SES-CD ≤2 for CD.

Results: Total 56 adult patients with ileal CD were included in the study, 38 patients (67.9%) were male, the median age was 40.5 (IQR 26-46.75) years and the median duration of the disease was 8.4 (IQR, 4.93-10.29) years. In the endoscopically remission (SES-CD ≤2) group; the median BUSS was 2.63 (IQR 2.12-3.45) and the median IBUS-SAS was 14 (IQR, 11.3-18.4), while in the endoscopically active (SES-CD >2) group; the median BUSS was 5.67 (IQR, 3.47-7.76) and the median IBUS-SAS was 49.7 (IQR, 23.3-72.65). The median BUSS and median IBUS-SAS in the endoscopically remission group were significantly lower than those of the endoscopically active group (p< 0.001, for both).

The cut-off value for BUSS to determine endoscopic remission (SES-CD ≤2) was 3.9 (AUC: 0.839, 95% CI: 0.737-0.941, sensitivity: 92%, specificity: 73%, p < 0.001), while the cut-off value for IBUS-SAS to establish endoscopic remission (SES-CD ≤2) was 24.4 (AUC: 0.871, 95% CI: 0.780-0.962, sensitivity: 100%, specificity: 73%, p < 0.001).

There was a very high positive correlation between BUSS and IBUS-SAS (r=0.914, p<0.001), a high positive correlation between BUSS and SES-CD (r=0.659, p<0.001), a moderate positive correlation between BUSS and CRP (r=0.577, p<0.001). There was a high positive correlation between IBUS-SAS and SES-CD as well as IBUS-SAS and CRP (r=0.728, p<0.001 and r=0.657, p<0.001 respectively).

Conclusion: In this study, we found that the most accurate cut-off value for BUSS was 3.9 and the most accurate cut-off value for IBUS-SAS was 24.4 in CD patients with endoscopic remission (SES-CD ≤2) group. Also, we have confirmed that both BUSS and IBUS-SAS significantly correlate with endoscopic activity in a real-world cohort.

Disclosure: Nothing to disclose.

PP0528

ASSESSING VISCERAL ADIPOSITY WITH CT ENTEROGRAPHY COULD HELP PREDICT RESPONSE TO BIOLOGIC THERAPY IN PATIENTS WITH CROHN'S DISEASE OF THE SMALL BOWEL

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Introduction: Recent studies implicate visceral adiposity in various bad outcomes of Inflammatory Bowel Disease. Patients with moderate to severe Crohn's Disease involving the small bowel are especially difficult to treat. The study aimed to seek any correlation between visceral fat and response to biologic therapy in this group of patients with the use of computed tomography enterography (CTE).

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Aims & Methods: This was a retrospective cohort of patients with Crohn's Disease of the small bowel who underwent baseline assessment with CTE before the start of a new biologic therapy. Repeat imaging assessment with CTE was performed 6 to 12 months post new treatment initiation. Examinations were performed in a 16-sclice CT scanner.

A single cross sectional CT slice at the L3 level was used for the segmentation of visceral and subcutaneous adipose tissue with syngo via dedicated software. We measured the ratio of visceral to subcutaneous adipose tissue (VAT/SAT).

The primary outcome was radiologic response: disease worsening vs complete response, partial response or stable disease. Secondary outcomes were clinical response, treatment persistence, Crohn's related surgeries and hospitalizations.

Results: A total of 114 exams from 62 patients were included in the study. Baseline disease characteristics were: median age 40 years (23-76), median disease duration 15 years (1-40), median BMI 24 (18-46), median body weight 71 Kg (45-160), stricturing disease 72%, previous surgery 56%, smokers 30%.

Most patients received anti-TNF agents: infliximab 40%, adalimumab 40%, ustekinumab 15%, vedolizumab 5%. Visceral fat > 12 was recorded in 65.8% of patients. VAT/SAT was associated with radiologic response (p=0,03), clinical response (p=0,02) and treatment persistence (p=0,03), but not with surgeries or hospitalizations.

Multivariate analysis confirmed the statistical correlation of VAT/SAT with radiologic response. Interestingly enough, no specific type of biologic therapy or disease characteristic showed any association with the primary outcome. Baseline BMI, weight and visceral fat volume also failed to correlate with response to therapy.

Conclusion: The use of visceral to subcutaneous adipose tissue ratio to quantify visceral adiposity helped to predict response to biologic therapy in Crohn's Disease patients with disease of the small bowel. This might confer an added value of enterography to other imaging modalities as for example intestinal ultrasound. Visceral fat seems to play an important role in inflammatory bowel disease and its accurate assessment may aid to risk stratify patients and plan therapy accordingly.

Disclosure: Nothing to disclose.

PP0529

THE INTESTINAL ULTRASOUND SCORE IBUS-SAS SIGNIFICANTLY CORRELATES WITH ENDOSCOPY AND HISTOLOGY IN ULCERATIVE COLITIS

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Introduction: Intestinal ultrasound (IUS) is widely recognised in the monitoring of inflammatory bowel diseases (IBD) and the resolution of bowel wall thickening (transmural healing) is currently investigated as treatment goal in IBD.[1]

The international bowel ultrasound- segmental activity score (IBUS-SAS) was developed to guarantee high inter-observer reliability in Crohn's disease (CD) and data aiming to validate it are promising.[2–4] Regarding IUS in ulcerative colitis studies remain scarce. We aimed to validate IBUS-SAS in UC-patients in comparison to the established Limberg-score.[5]

Aims & Methods: Patients with established diagnosis of UC were eligible for inclusion. The IBUS-SAS in the sigmoid colon was quantified according to the IBUS-groups guidelines.[2] Data regarding clinical (Mayo clinical score [MCS]), endoscopic (Mayo endoscopic score [MES]; ulcerative colitis endoscopic index of severity [UCEIS]), histologic (Nancy index) and biomarker (C-reactive protein [CRP], faecal calprotectin [FC]) activity were registered. The correlations between IBUS-SAS and these parameters were analysed using Spearman's rank test. All p-values <0,05 were considered to be statistically significant.

Results: We enrolled 51 UC patients (31 female), with a median IBUS-SAS (individual factors see table 1) of 41.6 (IQR 24.0; 62.3). The IBUS-SAS was significantly correlated with every individual parameter (p<0.001, r=0.48, 95% CI 0.22–0.67 for MCS; p<0.01, r=0.53, 95% CI 0.25–0.73 for MES; p<0.001, r=0.53, 95% CI 0.24–0.73 for UCEIS; p=0.03, r=0.36, 95% CI 0.02–0.73 for Nancy index; p<0.001, r=0.52, 95% CI 0.26–0.70 for CRP; p=0.03, r=0.61, 95% CI 0.07–0.87 for FC).

The Limberg-score showed a significant correlation to the clinical (p=0.01, r=0.36, 95% CI 0.08–0.59), endoscopic (p=0.001, r=0.50, CI 95% 0.21–0.71 for MES and p=0.001, r=0.50, 95% CI 0.21–0.71 for UCEIS) activity and CRP (p=0.02, p=0.44 95% CI 0.02–0.65), but not for histologic activity (p=0.27) and FC (p=0.14).

Overall, the correlations of the IBUS-SAS with the various disease activity parameters were stronger than their correlation with the Limberg score regarding clinical disease activity (r=0.48 vs. 0.36), the endoscopic picture (r=0.53 vs. 0.50 for MES; r=0.53 vs. 0.50 for UCEIS), histological activity (r=0.36 vs. 0.19) and the biomarkers of inflammation (r=0.52 vs. 0.44 for CRP; r=0.61 vs. 0.43 for FC).

IBUS-SAS features				
Bowel wall thickness (MM)		median 5.0 (IQR 4.0 – 5.9)		
Color Doppler signal	0	n =	20	(39.2 %)
	1	n =	19	(37.3 %)
	2	n =	10	(19.6 %)
	3	n =	2	(3.9 %)
Intestinal fat	0	n =	22	(44.0 %)
	1	n =	8	(16.0 %)
	2	n =	20	(40.0 %)
Bowel wall stratification	0	n =	33	(66.0 %)
	1	n =	10	(20.0 %)
	2	n =	2	(4.0 %)
	3	n =	5	(10.0 %)
IBUS-SAS		median 41.6 (IQR 24.0 – 62.3)		

Table 1: The distribution of the individual IBUS-SAS parameters in this UC-cohort.

Conclusion: We conclude that the IBUS-SAS shows a better correlation to established objective read-outs of disease activity in UC than previously established IUS scores and therefore seems a promising candidate for further validation as a tool for real-world disease monitoring and future clinical trials.

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PP0530

NATURAL HISTORY AND MRI PREDICTIVE FACTORS OF RADIOLOGICAL HEALING IN FISTULIZING PERIANAL CD

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Introduction: Fistulizing perianal Crohn's disease (CD) is a disabling condition affecting up to 30 % of patients. While mucosal healing is associated with a better clinical outcome in luminal CD, the objective of the management of perianal CD is less clear. Radiological parameters are rarely considered as an outcome in perianal CD. Some studies have suggested that a high degree of fibrosis of the fistula tract was associated with a better clinical outcome.
The aim of this study was to describe the natural history of perianal CD including MRI parameters and to identify clinical and radiological factors of perianal CD relapse and radiological healing.
Aims & Methods: All pelvic MRI performed for fistulizing perianal CD in the CHU of Liège, Belgium, between 2018 and 2023 were collected. MRI examinations were performed according to the protocol of the institution, using a 1.5-T [Siemens Aera ou Sola] or 3-T [Siemens Vida] machine, including T2 turbospin- echo, T2 TSE fat-saturated, T1 TSE FS before contrast in the axial plane and T1 TSE FS after gadolinium. The MAGNIFI-CD index and the Van Assche index were evaluated for each patient. The degree of fibrosis of the fistula tract was evaluated by two radiologists, blinded for the clinical history. The first MRI was evaluated in all patients. Patients with perianal flare had only one MRI included in the analysis. The first and the last MRI performed during the study period were analyzed, if available, in patients without perianal flare. Parianal flare was defined by the drainage of an abscess or a stoma for active perianal CD. Radiological remission was defined by a fibrosis of the fistula tract of ^{3}80% or a MAGNIFI CD score of 0.
Results: Seventy-five pelvic MRI from 60 patients were included in the study. The median age of the patients was 37.0 (30.0 - 48.5) years old, 37 (62%) patients were male, the median duration of perianal CD was 36.00 (0.60 - 123.00) months, 19 (31.7%) had a seton at the time of the first MRI, 23 patients (40%) were on anti-TNF. Nineteen (32%) patients, required abscess drainage in the follow-up period. Sixty-three % were on anti-TNF at the time of the drainage. No clinical nor radiological factors at baseline were predictive of perianal flare.
Fifteen patients without perianal flare had a second MRI during the follow-up. Seven patients (46,7%) had radiological healing at the end of the follow-up. One patient (6.7%) had radiological healing at the first MRI but not at the last MRI, 2 patients (13,3%) had radiological healing at both MRI 1 and 2 (13.3%), 5 patients (33,3%) had no radiological healing at MRI 1 but had radiological healing at MRI 2. Longer duration of perianal CD was associated with higher rates of radiological healing at the last MRI whereas high VA hyperintensity T2 (OR=0.13, p=0.099) and hyperintensity on postcontrast T1 (OR=0.11p=0.071) at the first MRI were predictive of absence of mucosal healing during the follow-up.
Conclusion: One third of the patients with perianal CD required abscess drainage whereas about half of the patient had radiological healing at the end of the follow-up. No clinical predictive factors were associated with perianal flare and few radiological factors were predictive of radiological healing. Predicting the outcome of perianal disease remains challenging in clinical practice. Radiological factors are promising and should be validated in a prospective cohort.
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PP0531

ENDOSCOPIC SKIPPING AND PENETRATING COMPLICATIONS IN CROHN'S DISEASE ON TANDEM ILEO-COLONOSCOPY AND CROSS-SECTIONAL IMAGING: A RETROSPECTIVE COHORT STUDY

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Introduction: Crohn's disease (CD) is characterized by discontinuous inflammation. Failure to identify skipping lesions of the terminal ileum (TI) or transmural changes can lead to incorrect management.
Aims & Methods: Eligible adult patients with CD undergoing ileo-colonoscopy and computed tomography enterography (CTE) or magnetic resonance enterography (MRE) within six months. We determined the prevalence of endoscopic skipping (normal ileum on colonoscopy but proximal small bowel inflammation on cross-sectional imaging), skip lesions (discontinuous inflammation along the gastrointestinal tract identified on cross-sectional imaging), and penetrating complications.

Clinical Characteristics	Active Crohn's Disease and Normal TI (n = 81)	Active Crohn's Disease and Abnormal TI (n = 121)	P value
Age, median (range), years	44.00 (15.91)	46.00 (16.55)	P = 0.88
Sex			
Male	34 (41.98)	45 (37.19)	P = 0.50
Female	47 (58.02)	76 (62.81)	
Smoking history, n (%)			
Former smoker	7 (8.64)	13 (10.74)	P = 0.50
Smoker	17 (20.99)	18 (14.88)	
Non-smoker	57 (70.37)	90 (74.38)	
Duration of Crohn's Disease			
≤5 years	35 (43.21)	49 (40.50)	P = 0.74
>5 years	46 (56.79)	71 (58.68)	
Previous Perianal Surgery, n (%)	13 (16.05)	8 (6.61)	P = 0.03
Previous resection, n (%)	30 (37.04)	47 (38.84)	P = 0.80
C-reactive protein, mean (SD), mg/L	4.31 (5.17)	8.00 (10.43)	P = 0.005

Results: Among 202 patients, 45 (22.3%) had endoscopic skipping proximal to TI intubation. Fifty patients (24.5%) had small bowel skip lesions, primarily in the ileum. Strictures were identified in 34 patients (16.8%) through both imaging and ileo-colonoscopy, in 21 patients (10.4%) solely through cross-sectional imaging, and in 3 patients (1.5%) solely through

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ileo-colonoscopy. Approximately 63.8% of stricturing cases would be missed without cross-sectional imaging. Penetrating complications, including abscesses (2.5%) and various fistula types (4.9%), were detected in 15 (7.4%) patients.

Conclusion: Ileo-colonoscopy missed detection of active CD in approximately one-fifth of cases due to more proximal disease location. Stricturing disease would be missed in more than half of cases if cross-sectional imaging was not performed.

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PP0532

THE ROLE OF ULTRASOUND IN EVALUATING RESPONSE TO THERAPY IN ACTIVE ULCERATIVE COLITIS: RESULTS FROM A PROSPECTIVE COHORT STUDY

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Introduction: Intestinal ultrasound is emerging as a non-invasive method of monitoring patients with ulcerative colitis (UC). Our goal was to prospectively evaluate the association between ultrasound and endoscopic response and remission.

Aims & Methods: This is a prospective cohort of patients with left-sided UC (E2) or extensive UC (E3) with active disease (defined by SCCAI≥5), proposed to initiate/optimize therapy. Clinical activity (PRO2/SCCAI), biochemical activity (fecal calprotectin (FCal)/CRP), ultrasound (wall thickening (BWT)/Milan-Ultrasound Criteria (MUC) score) was assessed at week 0 (W0), 2 (W2), 6 (W6), 14 (W14), and 30 (W30), and endoscopic (Mayo endoscopic sub-score (sMayo)) at W0, W14, and W30. Ultrasound response was defined as: 1-reduction in BWT >25%, 2-reduction in BWT >2mm, or 3-decrease in BWT >1mm + decrease in Modified Limberg score ≥1. Endoscopic response was defined as a decrease in sMayo ≥1. Ultrasound remission was defined as MUC <6.2, and endoscopic remission as sMayo = 0.

Results: Twenty patients with UC were included (65% male, median age 54 (28-67) years, 50% E3, 60% on biologic therapy). Clinical, biochemical, ultrasound, and endoscopic activity characterization is found in Table 1. Clinical, biochemical, ultrasound, and endoscopic remission was achieved at W30 in 56%, 57%, 40%, and 33%, respectively. Patients with ultrasound response at W2 or ultrasound remission at W6 showed more endoscopic remission at W14 (p=0.032 and p=0.014, respectively). The presence of ultrasound remission at W14 was associated with endoscopic response at W30 (p=0.011).

Variables, median (IQR)	W0	W2	W6	W14	W30
SCCAI	7 (6-10)	3 (1-5)	3 (0-6)	3 (0-5)	1 (0-6)
FCal (ug/g)	858 (705-2050)	750 (111-2110)	452 (69-1760)	162 (26-1092)	170 (36-580)
BWT (mm)	4.79 (4.13-5.40)	4.30 (3.84-4.91)	3.93 (3.23-4.50)	3.85 (2.65-4.53)	3.80 (2.20-4.70)
MUC	8.4 (7.4-9.6)	7.8 (6.5-8.9)	7.2 (4.9-8.2)	5.7 (3.7-8.0)	7.1 (4.6-8.6)
sMayo	3 (2-3)	-----	-----	2 (1-3)	1 (0-3)

Table 1 - Clinical, biochemical, ultrasound, and endoscopic activity characterization.

Conclusion: The presence of ultrasound response at W2 and ultrasound remission at W6 was associated with endoscopic remission at W14. Additionally, ultrasound remission at W14 was associated with endoscopic response at W30. Ultrasound response in active ulcerative colitis appears to be early, allowing its use as a non-invasive tool for tight control, enabling early optimization of therapy.

Disclosure: Nothing to disclose.

PP0533

BEYOND THE SCOPE: UTILITY OF INTESTINAL ULTRASOUND FOR PREDICTING CLINICAL OUTCOMES IN ULCERATIVE COLITIS IN CLINICAL REMISSION

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Introduction: Intestinal Ultrasound (IUS) is a point-of-care tool that is increasingly utilised in the management of ulcerative colitis (UC) as a marker of active disease and response to medical therapy. IUS has the advantage of being non-invasive, non-irradiating, well tolerated, cost-effective with good reproducibility when compared to ileo-colonoscopy and magnetic resonance enterography.

There is however limited evidence that sonographic remission on IUS predicts UC disease course.

Aims & Methods: This study aimed to examine the relationship of IUS and clinical outcomes in UC patients in clinical remission and determine if presence of sonographic inflammation was associated with medication escalation, corticosteroid use, hospitalisation or surgery.

A retrospective study at The Royal Melbourne Hospital included all UC patients who underwent IUS for IBD between 2017 and 2022 in clinical remission (Simple Colitis Clinical Activity score ≤2) at the time of IUS and followed up for at 6 months. Patients with rectum-confined disease (E1), a stoma, or suboptimal IUS were excluded. Sonographic healing, defined as BWT ≤3 mm without hyperemia, inflammatory fat, or disrupted bowel wall stratification was the combined endpoint.

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Kaplan-Meier analysis was performed to compare medication-escalation-free survival, corticosteroid-free survival, hospitalisation-free survival and surgery-free survival in those with and without sonographic healing. Univariate and multivariate Cox proportional hazard regression analyses were performed to identify predictors of these clinical outcomes.
Results: 2,113 ultrasound reports were assessed for eligibility, and 394 patients had a confirmed UC diagnosis. Of 149 consecutive patients in clinical remission met inclusion criteria (43% male), sonographic inflammation was present in 45%.
During mean follow-up of 634 days, medication escalation occurred in 40%, corticosteroid use in 27%, hospitalisation in 13% and UC related surgery in 2%.
Sonographic inflammation did not emerge as a predictive factor for these outcomes; the only factor predictive of significant outcomes was the baseline Montreal classification (Table 1). Patients with sonographic inflammation on the whole had worse outcomes however.

Variable	Univariate (n=149)			Multivariate (n=149)		
	OR	95% CI	p-value	OR	95% CI	p-value
Gender	1.25	0.64, 2,46	0.5	1.41	0.65, 3.13	0.4
Smoker	1.02	0.63, 1.63	> 0.9	0.95	0.55, 1.61	0.8
Montreal	2.23	1.28, 4.03	0.006	2.43	1.30, 4.78	0.007
Steroid	2.77	1.27, 7.39	0.020	1.93	0.86, 5.20	0.14
Immunomodulator	1.07	0.81, 1.40	0.6	0.84	0.59, 1.17	0.3
Biologic	1.06	0.92, 1.22	0.4	1.02	0.87, 1.20	0.8
Hyperaemia	8.24	1.28, 160	0.057	7.30	0.80, 174	0.12
US Inflammation	0.90	0.46, 1.74	0.8	0.51	0.13, 1.84	0.3

Table 1: Univariate and multivariate predictors of medical escalation in patients with ulcerative colitis. Data is shown as hazard ratio (95% confidence intervals)
OR = Odds Ratio, CI = Confidence Interval

Conclusion: This study is the first to examine the predictive utility of IUS in the setting of clinical remission and interestingly demonstrates that IUS does not appear to predict clinical complications. The role of IUS in UC has been slower to be established compared to Crohn's disease due to concerns about poor rectal views and lower sensitivity to detect mild mucosal inflammation.
This retrospective study challenges the utility of intestinal ultrasound in predicting clinical outcomes in UC patients in clinical remission, highlighting the need for further research to establish its role in guiding real time treatment decisions.
Disclosure: Nothing to disclose.

PP0534

BOWEL PREPARATION OPTIMIZATION OF DOUBLE BALLOON ENTEROSCOPY AND MAGNETIC RESONANCE OF SMALL INTESTINE IN PATIENTS WITH CROHN'S DISEASE IMPROVES PATIENTS' SATISFACTION

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Introduction: Double balloon enteroscopy(DBE) and Magnetic resonance enterography (MRE) are routine examination to detect lesions in the small intestine of Crohn's disease(CD). However, twice bowel preparation within several days reduces patients' satisfaction and compliance.
Aims & Methods: This study combined these two examinations on the same day with only one bowel preparation was required and compared the quality of DBE and MREC with two bowel preparations. This is a retrospective study collected 243 CD patients received both DBE and MREC within three days between April 2023 and April 2024. Patients in OBP group took bowel cleansing medication the night before two examinations. They underwent MREC and DBE in the order the next day: received MRMC in the morning and had DBE in the afternoon. Patients in TBP group underwent twice bowel preparations, MRMC and DBE as scheduled time within three days. Overall sex- and age-adjust analysis was performed.
Results: A total of 123 patients was in one bowel preparation (OBP) group and 120 in twice bowel preparation (TBP) group. 115(95.83%) patients qualified for bowel preparation at both examinations, 1(0.83%) only qualified for bowel preparation at MREC, and 4(3.33%) only qualified for bowel preparation at DBE in OBP group. While in TBP group, 116(94.31%) patients had bowel preparations qualified in both examinations, 2(1.63%) patients had bowel preparations only qualified in MRE, and 5(4.06%) patients had bowel preparations only qualified in DBE(P=0.81).
The average depth of insertion was 255.00±70.52 cm in OBP and 255.59±95.83 cm in TBP for the anterograde approach from the pylorus (P=0.98) and 144.70±75.40cm vs. 155.06±98.35cm for the retrograde procedures beyond the ileocecal valve (P=0.41). Complications of hypoglycemia was observed in 7(5.83%) patients in OBP group and 17(13.82%) patients in TBP group(P=0.037). The questionnaire results showed that 82.50% (99/120) of patients in OBP group while 68.29% (84/123) of patients in TBP group were satisfied with the process of examination(P=0.01). The mean intervention depth of OBP and TBP was 255.00±70.52 cm and 255.59±95.83 cm（P=0.98）, and the mean intervention depth of OBP and TBP was 144.70± 75.40 cm and 155.06± 98.35 cm（P=0.41）. In this paper, we discuss the relationship between the mechanism and the relationship between the mechanism. The results showed that the patients were satisfied with the inspection process of OBP group and the patients were satisfied with the inspection process of OBP group, and the patients were satisfied with the inspection process of TBP group.
Conclusion: One bowel preparation for DBE and MRE has similar effect as split-dose bowel preparation, and increases the compliance and tolerance of patients.
Disclosure: None.

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PP0535

ARTIFICIAL INTELLIGENCE (AI) ASSISTED INTESTINAL ULTRASOUND: DEVELOPMENT AND EVALUATION OF AN AI MODEL FOR BOWEL WALL SEGMENTATION AND MEASUREMENT IN INFLAMMATORY BOWEL DISEASE PATIENTS

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Introduction: Intestinal ultrasound (IUS) is increasingly used to assess inflammatory bowel disease (IBD). It provides physicians with detailed insights into transmural inflammation. Measurements of bowel wall thickness (BWT) based on IUS have proven to be a strong detector of active disease in both ulcerative colitis and Crohn's disease. However, the result of IUS relies heavily on the skill of the operator. To address this problem, we aim to develop an artificial intelligence (AI) model to assist operators in accurately identifying the inner and outer bowel walls, thereby facilitating AI-assisted BWT measurements.

Aims & Methods: We extracted 600 ultrasound frames from 231 ultrasound cine loops (videos) of IBD patients. Each frame was segmented, and the inner and outer bowel walls were labelled by four International Bowel Ultrasound Group (IBUS) certified experts from Denmark, with each expert annotating 150 frames. The annotated frames were split 80/10/10 into training-validation-test sets. There was no overlap in the video between the different splits. An AI model was trained to segment the inner and outer bowel walls in the frames. A postprocessing step was employed to refine the segmentations and perform automated BWT measurement, rejecting any segmentations considered too poor (e.g. a distance measure > 35 pixels).

The model's segmentations were evaluated against the reference expert segmentations using the rejection rate, absolute pixel difference, and Dice Similarity Coefficient (Dice score).

Results: A total of 30 frames were discarded due to poor visibility, leaving 570 annotated frames. In the test set (unseen data), the rejection rate was 4.2%. The mean pixel difference in BWT measurements was 3.712 (SD: 3.979). The AI model achieved a Dice score of 0.42 when compared with expert annotations. These results indicate that the model tended to match the distances measured by experts, albeit with slight shifts in the exact points of measurement.

Conclusion: Our preliminary findings suggest that the AI-assisted IUS model has the potential to aid operators by providing reliable, expert-level approximations of BWT with a low rejection rate. It may become a useful tool in clinical practice for consistent IBD assessments. Further training and refinement of the framework could enhance its accuracy and facilitate its application in real-time diagnostic assessments.

Disclosure: Nothing to disclose.

PP0536

ENHANCING CALPROTECTIN'S PREDICTIVE POWER AS A BIOMARKER OF ENDOSCOPIC ACTIVITY IN ULCERATIVE COLITIS: A MACHINE LEARNING USE CASE

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Introduction:

Ulcerative colitis (UC) is a chronic inflammatory bowel disease characterized by periods of exacerbation and remission, making disease monitoring and management challenging. Endoscopy, the gold standard for assessing disease activity and severity, involves invasive procedures and is associated with patient discomfort and risks. Using machine learning (ML) to combine fecal calprotectin with other clinical or biological tests can significantly enhance the non-invasive prediction of endoscopic disease activity (EDA) in UC.

Aims & Methods: Aim: To prove that by fusing fecal calprotectin with other clinical data into an ML model, the performance of the non-invasive prediction of EDA can be significantly improved. Methods: We conducted a prospective, observational, single-center study encompassing 103 patients diagnosed with UC. We employed multilayer perceptron models as the core ML algorithm for predicting EDA. For the constructed models, we utilized the varImp function from the caret library in R to assess the significance of each variable in predicting the outcome.

Results: Calprotectin as a sole predictor obtained an accuracy of 70% and an area under the curve (AUC) of 0.68. Combining calprotectin with the list of selected predictors that were fed to the MLP models improved accuracy and the AUC. The accuracy of the algorithm on the test set was 85%. Similarly, the AUC increased to 0.93. This is the first study to propose the use of calprotectin as a predictor in an ML model to estimate UC endoscopic disease activity.

Conclusion: The deployment of this ML model can furnish doctors and patients with valuable evaluation of endoscopic disease activity which can be highly beneficial for individuals with UC who need long-term treatment.

Disclosure: Nothing to disclose.

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PP0537

LEVELS OF FATTY ACIDS IN ERYTHROCYTE MEMBRANES AND BLOOD SERUM AS EXACERBATION PREDICTORS IN PATIENTS WITH INFLAMMATORY BOWEL DISEASES

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Introduction: Unclear etiology of IBD, the frequent similarity of their clinical and endoscopic manifestations, lack of a gold standard in diagnosing diseases, which complicates the choice of targeted therapy, urgently require improving the efficiency of currently available diagnostic technologies [1]. Significant efforts are being made to search for biomarkers that can be used to identify patients at high risk of IBD relapse who need more intensive monitoring and therapy in the early stages of these diseases, in remission to prevent long-term irreversible structural damage [2].

Aims & Methods: The purpose of the research was to study the levels of fatty acids (FAs) in red blood cells (rbc) membranes and blood serum in patients with IBD examined under follow-up control as possible predictors of disease exacerbation.

During our study, 44 patients with IBD have been monitored, in the course of that period, the average age was equal to 36.3±5.2. The first examination has been performed in remission of the diseases, the second one 6 to 12 months after the first step, and it has been done in the acute stage of IBD. The composition of the FAs has been studied using gas chromatography-mass spectrometry (*Agilent 7000B (USA).*

Results: Predictors of exacerbation progress in patients with IBD have been found to be reduced levels of such FAs as C20:5 n-3 (p=0.002), C18:3 n-3 (p=0.039), C20:2 n-6 (p=0.002), C16:2 n-6 (p=0.006), C22: 4 n-6 (p=0.008), and on the contrary, increased content of FA C18:0 (p=0.005), SFAs/USFAs (p=0.039), SFAs/PUFAs (p=0.03) in rbc membranes and FA C20:0 in blood serum (p=0.008). The diagnostic model created was able to predict the exacerbation progress with a sensitivity of 87.2%, specificity of 91.3% (AUC 0.936), incorporating the above-described fatty acids with respect to their degree of contribution. We have found correlations of rbc C18:0 with haemoglobin level (r=-0.344, p<0.001), erythrocyte sedimentation rate (r=0.480, p=0.012), fibrinogen level (r=0.352, p=0.004); rbc C18:3n-3 - with level of C-reactive protein (r=-0.472, p=0.023), fibrinogen (r=-0.420, p=0.022), faecal calprotectin (r=-0.391, p=0.044). The content of rbc C16:2n-6 was directly correlated with the level of fecal calprotectin (r=0.407, p=0.028), ratio of SFAs/USFAs was depending on the number of leukocytes (r=0.411, p=0.046). The function of elevated levels of SFAs as predictors of exacerbation of IBD is probably determined by the fact that they potentially regulate inflammatory reactions by acting on the pathways of the PPAR-γ and retinoid X receptors (RXR) [3].

Reduced levels of C18:3n-3 as a predictor of exacerbation may be related to the ability of this FA to reduce the activity of ionised calcium-binding adaptor molecule 1-positive macrophages [4] and to reduce the expression of proinflammatory genes IL-8, COX2 and inducible nitric oxide synthase to reduce intestinal inflammation [5].

The significance of C20:5n-3 for prediction of the active stage of IBD is due to the fact that it inhibits the inflammatory pathways of transforming growth factor beta-1/epidermal growth factor receptor P-EGFR and NF-κB, and it also modulates redox balance and mitigates the progression of UC [6]. Reduced levels of several omega-6 PUFAs may be related to their increased consumption during elongation, desaturation, including in the arachidonic cascade associated with the cynthesis of proinflammatory cytokines [7].

Conclusion: The use of fatty acids of rbc membranes and blood serum is promising and pathogenetically justified, namely as biomarkers for predicting the exacerbation progress in patients with IBD.

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PP0538

CORRELATION OF BOWEL ULTRASONOGRAPHY AND ENDOSCOPY IN CHRONIC INFLAMMATORY BOWEL DISEASE (IBD)

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Introduction: Accurate assessment of disease activity and prediction of treatment response and prognosis are challenges in the management of chronic inflammatory bowel disease (IBD). Due to the invasiveness of colonoscopy, non-invasive methods are of high clinical relevance. Bowel ultrasonography (B-US) is establishing itself as a non-invasive, cost-effective method in daily clinical practice.

Aims & Methods: The aim of this study is to evaluate the correlation between B-US and endoscopy in relation to inflammatory activity and severity of IBD. Patients were retrospectively selected for comparison between B-US and endoscopy.

A total of 473 patients were included, of whom (n=274) had Crohn's disease (CD) and (n=199) had ulcerative colitis (UC). The examination results were divided into four sections (terminal ileum, proximal colon, distal colon and rectum) and systematically analyzed. Inflammatory activity was classified according to intestinal wall thickness in mm:

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none (0): 0-2.9, mild (1): 3.0-4.9, moderate (2): 5.0-7.9, severe (3): >8.0. Endoscopic evaluation was performed according to simplified Mayo and SES-CD criteria.

Results: Based on endoscopy, patients were divided into groups with inflammatory activity (274 CD, 175 UC) and remission (34 CD, 24 UC). Comparison of B-US and endoscopy showed a significant correlation regarding inflammatory activity (r=0.44, p<0.001). The correlation was more significant in UC (r=0.5548, p<0.001) than in CD (r=0.3665), except in the ileum in UC (r=-0.1050, p<0.4822). The best correlation in MC was found in the ileum (r=0.4783, p<0.001) and in the distal colonic segments (r=0.5217, p<0.001), while in UC the best correlation was found in the proximal (r=0.5586, p<0.001) and in the distal colonic segments (r=0.6250, p<0.001). Localization of rectal inflammation was associated with lower diagnostic accuracy. Compared to endoscopy, B-US showed a sensitivity of 82.2%, a specificity of 51.7% and a positive predictive value of 92.4%.

Conclusion: B-US shows reliable diagnostic accuracy compared to endoscopy in assessing inflammatory activity in IBD patients and can be considered as a complementary non-invasive diagnostic tool.

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PP0539

COMPARATIVE EFFICACY AND SAFETY OF ENDOSCOPIC MODALITIES FOR COLORECTAL CANCER SCREENING IN INFLAMMATORY BOWEL DISEASE: A NETWORK META-ANALYSIS USING GRADE METHODOLOGY

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Introduction: Long-standing Inflammatory bowel disease (IBD) increases the risk of colonic neoplasia, necessitating effective screening strategies. Various endoscopic modalities are used for surveillance colonoscopy in IBD patients, yet their comparative effectiveness and safety remain unclear. Previous guidelines have supported using both dye-based and virtual chromoendoscopy as equivalent[1]; however, their additional benefit in the era of high-definition (HD) white light remains controversial.

Aims & Methods: This network meta-analysis (NMA), part of the British Society of Gastroenterology's initiative to update IBD surveillance guidelines, aims to compare the efficacy and safety of different endoscopic techniques to inform clinical practice and guidelines. We searched CENTRAL, ClinicalTrials.gov, Embase via Ovid, MEDLINE via Ovid, WHO ICTPR for randomised controlled trials (RCTs), and reference lists of relevant studies up to February 2024 for RCTs comparing endoscopic modalities for screening colonoscopy in patients with IBD. The primary outcome for efficacy was the detection of any dysplastic lesion (Vienna classification 2-5) per patient and, for safety, serious adverse events. Review authors independently screened and extracted data and assessed the risk of bias using the Cochrane tool. We conducted an NMA under a frequentist framework to estimate the relative effectiveness and safety of the interventions. The GRADE approach was employed to assess the certainty of the evidence for each outcome.

Results: A total of 27 RCTs involving 4,159 participants were included. The network comprised 6 endoscopic modalities: HD white light endoscopy (HD-WLE), HD virtual chromoendoscopy (HD-VCE), HD dye-based chromoendoscopy (HD-DCE), HD-WLE with segmental re-inspection (SR), auto-fluorescence imaging (AFI), and full-spectrum endoscopy (FUSE). There were no RCTs comparing HD to standard definition (SD) WLE/DCE; thus, the SD modalities could not be connected to the main network. HD-DCE may be more effective for detecting dysplasia, with a risk ratio (RR) of 1.42 (95% CI: 1.02-1.98) and low-GRADE certainty compared to HD-WLE. All other modalities had very low-GRADE certainty to draw conclusions about them, with RRs of 0.99 (95% CI: 0.69-1.43) for HD-VCE; 1.35 (95% CI: 0.66-2.77) for HD-WLE with SR; 3.24 (95% CI: 0.66-15.87) for FUSE; and 1.18 (95% CI: 0.55-2.57) for AFI. A similar trend was noted in sensitivity analyses, excluding studies reporting on serrated lesions (RR for HD-DCE: 1.91, 95% CI: 1.36-2.69) and excluding single endoscopist studies (RR for HD-DCE: 1.57, 95% CI: 1.10-2.26) compared to HD-WLE. Similar performances were noted for HD-VCE subtypes and varying concentrations of dye used in HD-DCE. Data on serious adverse events were limited, precluding any meta-analysis. In studies that reported this outcome, 2 serious adverse events were reported out of 2164 patients; however, this outcome had very low-GRADE certainty.

Conclusion: HD DCE is suggested over other modalities for surveillance in IBD, considering its potential to detect a small number of additional patients with dysplastic lesions compared to HD WLE. HD-VCE was not supported as an equivalent alternative to HD-DCE. However, the certainty of this recommendation is low due to the limited evidence. When choosing the surveillance modality, clinicians should also consider patient preferences, resource availability, and individualised risk assessments.

References: 1 Murthy et al. *Gastroenterology* 2021

Disclosure: Nothing to disclose.

PP0540

INTESTINAL ULTRASOUND IDENTIFIES HISTOPATHOLOGICAL APPENDICEAL INFLAMMATION IN ULCERATIVE COLITIS

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Introduction: Growing evidence points to appendectomy as an alternative treatment for ulcerative colitis (UC). Histopathological appendiceal inflammation might be correlated with response to appendectomy. Intestinal ultrasound (IUS) is a non-invasive diagnostic modality used to visualize appendiceal inflammation.

Aims & Methods: We aimed to investigate appendiceal IUS's relationship with histopathology findings in appendix resection specimens. Therapy-refractory UC patients undergoing appendectomy in single center clinical trials were included. IUS parameters included appendix diameter, wall thickness, submucosal layer thickness and submucosal hyperechogenicity. Submucosal layer hyperechogenicity was quantified using the mean aerial grayscale intensity which is a numerical scale 0-255, with zero corresponding to black. To adjust for variations in depth and gain, we measured the Δ grayscale intensity using the difference between the sub-

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mucosa and muscularis propria in a perpendicular fashion. The Robarts Histopathology Index (RHI) was used to measure mucosal histological disease activity in the appendix ranging from 0-33. Appendiceal inflammation was defined as RHI >3, with at least subscores of lamina propria neutrophils or epithelium neutrophils ≥1, or presence of ulcers or erosions.

Results: 37 of 55 (67.3%) included patients had a visualized appendix on IUS. The median RHI was 6 (IQR 0-12) and histopathological appendiceal inflammation was present in 21/37 (57%). Submucosal layer thickness was significantly higher, and the submucosal layer hyperechogenicity was significantly enhanced in microscopically inflamed appendices (Table 1). Submucosal hyperechogenicity predicted histopathological appendiceal inflammation (OR 1.03, 95% CI 1.0-1.05; p=0.035). Submucosal thickness ≥0.75 mm diagnosed histopathological appendiceal inflammation with 66.7% sensitivity, 81.3% specificity and an AUC of 0.69 (p=0.046). Submucosal hyperechogenicity >110.20 diagnosed histopathological appendiceal inflammation with 57.1% sensitivity, 81.3% specificity and an AUC of 0.69 (p=0.046).

IUS characteristic	Total N=37	RHI≤3 N=16	RHI>3 N=21	p-value
Diameter (mm)	4.3 (3.7-5.8)	4.3 (3.5-5.6)	4.4 (3.9-6.0)	0.24
Wall thickness (mm)	1.9 (1.5-2.3)	1.9 (1.4-2.1)	1.9 (1.5-2.4)	0.36
Mucosa (mm)	0.6 (0.4-0.8)	0.7 (0.4-0.8)	0.6 (0.4-0.8)	0.72
Submucosa (mm)	0.7 (0.5-0.9)	0.6 (0.4-0.7)	0.9 (0.6-1.0)	0.04
Muscularis propria (mm)	0.5 (0.3-0.6)	0.5 (0.3-0.6)	0.5 (0.3-0.7)	0.5
Submucosal hyperechogenicity (Δ grayscale value, 0-255)	99.5 (83.4-133.3)	95.4 (76.2-108.1)	113.5 (87.7-151.7)	0.02

Table 1. Ultrasound characteristics of the visible appendices.

Conclusion: IUS is able to identify patients with histopathological appendiceal inflammation who may benefit from appendectomy as treatment for UC.

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PP0541

INTER-RATER AGREEMENT FOR BOWEL WALL COLOUR DOPPLER SIGNAL ASSESSMENT USING INTESTINAL ULTRASOUND IN CROHN'S DISEASE PATIENTS

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Introduction: Intestinal ultrasound (IUS), a non-invasive, point-of-care method, is increasingly emerging as a cornerstone technique for the diagnosis and follow-up of patients with inflammatory bowel disease (IBD). The importance of IUS is even more evident in Crohn's disease (CD) patients where activity of disease doesn't frequently correlate with the symptoms. One of the key assessment parameters in intestinal ultrasound is the Colour Doppler signal (CDS), semiquantified according to the Limberg score (LS).

Although, simple and user friendly, its reliability and reproducibility in CD are not well established. Therefore, we aimed to assess the inter-rater agreement among IUS experienced (E) and IUS non-experienced (NE) group of physicians.

Aims & Methods: The inter-rater agreement among 8 physicians (3 experienced, each with more than 300 procedures, and 5 non-experienced, each with up to 50 procedures) from 3 different IBD centers was assessed. They independently scored 317 de-identified CDS IUS images, including both colon and small intestine views in transverse and longitudinal perspectives, using non-modified LS. All images were recorded using GE Logiq s8 linear probe. CDS was optimized for slow flow with a maximum velocity scale of 5-7 cm/s. Initially, the scale was increased to its maximum, than gradually decreased until noise dissipated. LS was treated as an ordinal variable and Kendall coefficient of concordance was calculated. Additionally Cohen's kappa was used in order to assess agreement between normal (LS 0 and 1) and pathological findings (LS>1).

Results: The test database comprised 36.3% (115) transverse view images and 63.7% (202) longitudinal view images. Overal inter-rater agreement was $W = .73$, $p < .01$, with higher concordance among E group $W = .85$, $p < .01$ compared to NE group $W = .73$, $p < .01$. No difference in inter-rater agreement was found among the three most experienced physicians, regardless of the view modality (transverse vs longitudinal view), $W = .85$, $p < .01$ for both categories. Agreement for the binary outcome (normal vs pathological finding) among most experienced physician was substantial $\kappa = .74$.

Conclusion: Our results showed that assessment of colour doppler signaling in CD patients is reliable even between physicians of varying experience, regardless of probe perspective. Further analyses on cineloops are warranted.

Disclosure: Nothing to disclose.

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PP0542

LONG-TERM, REAL-WORLD TREATMENT PERSISTENCE, MUCOSAL HEALING AND CORTICOSTEROID-FREE CLINICAL REMISSION RATES WITH VEDOLIZUMAB IN PATIENTS WITH ULCERATIVE COLITIS: SYSTEMATIC LITERATURE REVIEW AND META-ANALYSIS

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Introduction: Vedolizumab (VDZ) is approved in Europe and the USA for the treatment of moderately to severely active ulcerative colitis (UC). Real-world evidence is needed to assess the effectiveness and benefit–risk profile of VDZ for UC outside clinical trial settings. A systematic literature review (SLR) and meta-analysis published in 2018 confirmed the real-world effectiveness and positive benefit–risk profile of VDZ treatment for up to 1 year in patients with inflammatory bowel disease.[1] Further real-world evidence is required to support the long-term (> 1 year) effectiveness of VDZ in patients with UC.

Aims & Methods: The aims of the current meta-analysis based on the SLR were to evaluate the rates of treatment persistence at 1 and at 2 years and of mucosal healing and corticosteroid-free clinical remission at 1 year (overall and stratified by biologic-naive versus biologic-experienced patients). Literature searches identified studies published from 2014 to 2022. For the SLR overall, outcomes of interest included clinical remission, mucosal healing, treatment patterns (including treatment persistence) and safety. Data on adults with UC treated with vedolizumab in a real-world setting were extracted for meta-analysis if there were at least 2 studies for each timepoint and population. Weighted proportions and corresponding 95% confidence intervals (CIs) were calculated using the DerSimonian-Laird random-effects model to account for between-study heterogeneity.

Results: Data from 68 studies of 21 774 patients with UC had at least one outcome of interest for analysis. The mean age ranged from 34.0–53.0 years (16 studies), the mean disease duration ranged from 4.9–10.4 years (5 studies) and the mean partial Mayo score ranged from 4.1–6.1 (6 studies). In the 28 and 9 studies that assessed treatment persistence at 1 and at 2 years, the rates of treatment persistence were 72.4% (95% CI 68.7–75.8) and 64.2% (95% CI 55.3–72.2), respectively. Similar treatment persistence rates at 1 year were observed among biologic-naive vs biologic-experienced patients, but at 2 years higher rates were observed for biologic-naive patients vs overall (*Table*). The overall mucosal healing rate at 1 year (assessed in 29 studies) was 42.6% (95% CI 37.0–48.3), and similar rates were observed in biologic-naive vs biologic-experienced patients (*Table*). The corticosteroid-free clinical remission rates at 1 year were 42.7% (95% CI 38.4–47.1; 22 studies) and at 2 years 43.4% (95% CI 27.9–60.2; 5 studies); higher rates were observed among biologic-naive vs biologic-experienced patients at 1 year (*Table*).

	Overall		Biologic-naive		Biologic-experienced	
	Patients[a] (studies)	Rate, % (95% CI)[b]	Patients[a] (studies)	Rate, % (95% CI)[b]	Patients[a] (studies)	Rate, % (95% CI)[b]
1-year VDZ treatment persistence	5642 (28)	72.4 (68.7–75.8)	1666 (7)	74.5 (66.8–80.9)	1368 (5)	73.1 (62.4–81.6)
2-year VDZ treatment persistence	1735 (9)	64.2 (55.3–72.2)	449 (2)	77.6 (70.8–83.2)	–[c]	–[c]
1-year mucosal healing	1851 (29)	42.6 (37.0–48.3)	376 (3)	45.7 (31.4–60.7)	316 (5)	40.9 (30.9–51.8)
1-year corticosteroid-free clinical remission[d]	2797 (22)	42.7 (38.4–47.1)	465 (7)	45.5 (37.9–53.3)	580 (8)	36.6 (30.3–43.5)

[a]Number of patients assessed.
[b]Events per 100 patients.
[c]No meta-analysis because only one study reported this outcome.
[d]Resolution of symptoms or clinical measures of disease activity without the use of corticosteroids.

Table.

Conclusion: This meta-analysis of real-world studies of VDZ demonstrated high rates of long-term treatment persistence, mucosal healing and corticosteroid-free clinical remission in patients with UC, indicating effective disease management for these outcomes. However, the assessment of mucosal healing rates may not have been systematically conducted across all studies.

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References: Schreiber S et al. *J Gastroenterol.* 2018;53:1048–64.

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AA has received advisory board/lecture fees from AbbVie, Amgen, Arena, AstraZeneca, Biogen, Boehringer Ingelheim, Bristol Myers Squibb, Celgene, Celltrion Healthcare, Dr Falk Pharma, Eli Lilly, Ferring, Galapagos, Gilead, Janssen, Lion Health, MSD, Nestlé, Novartis, Pfizer, Protagonist Therapeutics, Roche, Samsung Bioepis, Sandoz, Takeda, Teva Pharmaceuticals and Tillots Pharma.

PB is an employee and stockholder of Takeda Pharmaceuticals International AG, Zurich, Switzerland.

MM is an employee and stockholder of Takeda Pharma AG, Zurich, Switzerland.

ER is an employee of Putnam Associates, which received funding for this study from Takeda Pharmaceuticals International AG, Zurich, Switzerland.

CS is an employee of Putnam Associates, which received funding for this study from Takeda Pharmaceuticals International AG, Zurich, Switzerland.

SS has received personal fees from AbbVie, Amgen, Arena, Biogen, Bristol Myers Squibb, Celgene, Celltrion Healthcare, Dr Falk Pharma, Fresenius, Galapagos/Gilead, Hikma, I-Mab, Janssen, MSD, Mylan, Pfizer, Protagonist Therapeutics, ProventionBio, Takeda, Theravance Biopharma and Ventyx.

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PP0543

CRP/ALBUMIN RATIO: A NEW PREDICTOR OF DISEASE COURSE IN ACUTE SEVERE ULCERATIVE COLITIS

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Introduction: Acute Severe ulcerative colitis (ASUC) is a life-threatening medical emergency of colonic inflammatory bowel disease (IBD). Corticosteroids remain the primary initial therapy, although one-third of patients do not respond to treatment. In this perspective, early identification of steroid non-responsive ASUC patients is the key to an effective management and thus, a better prognosis.

Aims & Methods: The aim of our study was to evaluate the impact of CRP/albumin ratio measured on day 3 on treatment outcomes and disease course in ASUC. We conducted a retrospective descriptive study including patients who presented with ASUC over a 10-year period [January 2010 - December 2019]. The diagnosis of ASUC was established on Truelove and Witts criteria and supported by endoscopic findings. Patients demographics were collected; C-reactive protein (CRP) and albumin levels were recorded at baseline and during admission. Data were entered and analyzed by SPSS software version 26.

Results: We included 54 patients with a sex ratio of M/F=0.71. The mean age at diagnosis of IBD was 41.48 years [12-58]. A family history of IBD was noted in 13% of the patients. Twenty-seven patients had Ulcerative Colitis and 27 patients had Crohn´s disease. ASUC was inaugural of the disease in 35% of the patients. For the diagnosis of ASUC, the median number of bloody stools was 7 and the mean CRP level was 91 mg/L. At day 3 of first-line treatment with corticosteroids, the median number of bloody stools was 1 and the mean CRP level was 49 mg/L. First-line treatment failure was noted in 40% (n=20) of the patients with a mean duration of 8 days [3-20]. The mean level of the CRP/albumin ratio on day 3 was 1.8 [0.8-2.9]. Area under the receiver operating characteristic curve analysis showed that this ratio measured on day 3 was highly predictive of steroid non-responsiveness (AUROC=0.69, p=0.02). The optimal cutoff value was 1.6 (sensitivity, 80%; specificity, 57%). There was no significant linear association (Spearman r=-0.11, p=0.64) between this ratio and the timeframe of corticosteroids failure. A significant association was noted between this ratio and the urge to a rescue medical therapy (p=0.015) without being predictive of response (p=0.327). If on day 3, the ratio was higher than 1.6 and bloody stool frequency was beyond 3, the relative risk of steroid non-response was significantly raised at 3.85 (p<0.001) versus 1.7 if only a ratio above 1.6 was considered. As for longer prognosis, this ratio was higher in patients who underwent colectomy within a year of ASUC, without a significant association (n=13, p=0.07), and no correlation was noted with ASUC recurrence (p=0.23).

Conclusion: Our study showed that CRP/albumin ratio on day 3 is not only an effective predictor of non-response to steroids in ASUC, but also predicts the disease course beyond this severe flare-up.

Disclosure: Nothing to disclose.

PP0544

THE INCIDENCE OF ACTIVE TUBERCULOSIS AND IMPACT OF PROPHYLACTIC TREATMENT AMONG PATIENTS WITH LATENT TUBERCULOSIS, IN THE PHASE 3 ELEVATE UC 52 AND ELEVATE UC 12 TRIALS

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Introduction: Tuberculosis (TB) is a prevalent disease, especially in developing countries.[1] The TB medication isoniazid (INH) has known hepatotoxic and drug–drug interaction (DDI) potential.[2]

With the increasing prevalence of ulcerative colitis (UC) in newly industrialised countries,[3] more patients (pts) are expected to take UC medications concomitantly with INH.

Therefore, the likelihood of DDIs is an important UC treatment consideration. Etrasimod is an oral, once-daily (QD), selective sphingosine 1-phosphate $(S1P)_{1,4,5}$ receptor modulator for the treatment of moderately to severely active UC.

Aims & Methods: This post hoc analysis aimed to evaluate the incidence of active TB in pts receiving etrasimod 2 mg QD or placebo and the impact of concomitant INH on safety outcomes in pts receiving etrasimod 2 mg QD. This analysis pooled data (≤52 weeks) from phase 3 trials, ELEVATE UC 52 (NCT03945188) and ELEVATE UC 12 (NCT03996369). Pts with latent TB infection (LTI; at enrolment or historically) who received/were receiving adequate treatment were allowed to enrol in both studies. We report the incidence of active TB in pts receiving etrasimod 2 mg QD or placebo and in those with an LTI history.

We also report liver enzyme elevations (irrespective of treatment-emergent adverse event [TEAE] classification) in pts taking concomitant etrasimod + INH. Data are summarised descriptively.

Results: Of the 527 and 260 pts receiving etrasimod and placebo, respectively, 17 (3.2%) and 3 (1.2%) had history of LTI. No pts receiving etrasimod experienced active TB and one event of active TB was reported in a pt receiving placebo (without history of LTI).

Overall, 9/20 pts with history of LTI were taking etrasimod + INH per protocol. Most of these pts were White (7/9) and originated from Georgia (2/9), India (2/9), Poland (2/9), Romania (2/9) and South Africa (1/9).

No liver-associated TEAEs were reported during etrasimod + INH use.

One pt had elevated alanine transaminase (ALT; >1 to ≤2 × upper limit of normal [ULN]), aspartate transaminase (AST; >1 to ≤2 × ULN) and γ-glutamyl transferase (GGT; events of >1 to ≤2 and >2 to ≤3 × ULN) while receiving etrasimod + INH concomitantly (Table).

Additionally, post-baseline, two pts had transiently elevated ALT and AST (all >1 to ≤2 × ULN), and two pts had transiently elevated GGT (>1 to ≤2 × ULN) while receiving etrasimod + INH concomitantly (Table).

These were not considered by the investigator to be TEAEs. None of these pts with liver enzyme elevations discontinued treatment or met drug-induced liver injury criteria; all pts continued etrasimod in the open-label extension study (NCT03950232).

The incidence of elevated liver enzymes was generally similar in those receiving etrasimod + INH vs the overall ELEVATE UC cohort.

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Pts, n	Pts receiving etrasimod 2 mg QD (N=527)
With a history of LTI	17
With active TB	0
Liver enzyme elevations recorded any time post-baseline, n	Pts receiving concomitant etrasimod 2 mg QD + INH (N=9)
ALT >1 to ≤2 × ULN	3[a]
ALT >2 × ULN	0
AST >1 to ≤2 × ULN	3[a]
AST >2 × ULN	0
GGT >1 to ≤2 × ULN	3[b]
GGT >2 to ≤3 × ULN	1[b]
GGT >3 × ULN	0

aOne pt receiving etrasimod 2 mg QD was excluded in the count as elevated AST and ALT were experienced after the discontinuation of concomitant etrasimod + INH.
bOne pt had elevation of GGT >1 to ≤2 × ULN and GGT >2 to ≤3 × ULN.

ALT, alanine transaminase; AST, aspartate transaminase; GGT, γ-glutamyl transferase; INH, isoniazid; LTI, latent tuberculosis; n, number of patients with event; N, number of patients in analysis set; pt, patient; QD, once daily; TB, tuberculosis; TEAE, treatment-emergent adverse event; UC, ulcerative colitis; ULN, upper limit of normal.

Table. Incidence of active TB in pts with a history of LTI receiving etrasimod and the proportion of pts receiving etrasimod with liver enzyme elevations (irrespective of TEAE classification) in ELEVATE UC 52 and ELEVATE UC 12 (pooled).

Conclusion: No pts receiving etrasimod experienced active TB, regardless of LTI history, and in the limited population of pts receiving etrasimod + INH, there was no increased risk of liver injury or abnormal liver function.

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PP0545

TO WHAT EXTENT ARE PHYSICIANS CYCLING THROUGH MORE THAN ONE TNFI (TUMOR NECROSIS FACTOR INHIBITOR) IN ULCERATIVE COLITIS (UC) TREATMENT?

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Introduction: With the wider availability of non-TNFi treatments in Ulcerative Colitis (UC), the long-established treatment approach to cycle through two TNFi treatments (such as adalimumab and infliximab) is changing.

Aims & Methods: Objectives: The purpose of this study was to evaluate to what extent TNFi cycling is occurring in the EU5 management of UC, and how this has changed over the last 3 years. More specifically, the data is used to demonstrate which types of patients are likely to cycle through 2 TNFi products, and which are likely to be prescribed a non TNFi after one TNFi treatment choice.

Methodology: In Q3 of 2023, 205 gastroenterologists from France, Italy, Germany, Spain, and the UK contributed 1430 anonymous patient record forms for individuals receiving advanced therapy for UC. All the forms were collected through the Therapy Watch IBD (UC and Crohn's) survey by Research Partnership. The survey has been conducted with similar numbers of gastroenterologists for 3 years at regular intervals. Informed consent from physicians was obtained. Where relevant, statistical comparisons were performed using a pairwise z-test.

Results: UC patients who require an advanced treatment (biologic or JAKi) are most likely to be prescribed a TNFi first-line. From our Q3 23 data, 81% of EU5 UC patients received a TNFi (most commonly adalimumab or infliximab) as first-line. The integrin blocker vedolizumab, IL12/23 inhibitors (such as ustekinumab) and JAK inhibitors are more likely to be prescribed at 2nd or 3rd+ line; at 2nd line, TNFi were prescribed to 37% of patients.

In Q3 23, of the 485 UC patients who were currently on their 2nd line of advanced treatment, 32% had a TNFi in both 1st and 2nd line (i.e. cycling TNFi) and 59% had a TNFi in 1st line and a non TNFi in their 2nd line of treatment. In Q1 21, 41% were cycling TNFi, and 50% were not. This represents a statistically significant reduction in TNFi cycling between Q1 21 and Q3 23 ($p<0.05$).

Certain patient characteristics are more prevalent amongst patients who switch to a non TNFi at 2nd line, compared with those who cycle to a second TNFi, as follows (all represent a statistically significant difference, $p<0.05$):

- More likely to be younger (51% under 40, compared with 41% among TNFi cyclers)
- Less likely to have any comorbidities (44% no comorbidities, compared with 27% among TNFi cyclers)
- More likely to have blood in stool and bowel incontinence (45% compared with 20% among TNFi cyclers)
- More likely to have an increasing rate of UC flares (29% compared with 15% among TNFi cyclers).

Conclusion: Our findings confirm that there has been a statistically significant reduction in TNFi cycling over the past 3 years, with an increase in the proportion of UC patients who are prescribed a non TNFi in 2nd line. TNFis remain the main 1st line advanced treatment choice for UC.

At this point, our data suggests that physicians are making decisions between TNFi cycling and switching directly to a non-TNFi, based on certain patient characteristics or disease status.

References: Data from Therapy Watch IBD survey (UC and Crohn's) conducted by Research Partnership.

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PP0546

EFFECTIVENESS OF USTEKINUMAB FOR PATIENTS WITH MODERATE TO SEVERE ULCERATIVE COLITIS: A MULTI-CENTER REAL-WORLD CANADIAN STUDY (IMPACT STUDY)

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Introduction: We aimed to evaluate the real-world effectiveness and safety of ustekinumab for the treatment of both bio-naïve and bio-exposed ulcerative colitis (UC) patients in a real-world setting.

Aims & Methods: Retrospective, Canadian multicenter cohort study. Standardized data collection was performed on consecutive adult UC outpatients treated with ustekinumab in 4 large IBD centers. The primary outcomes were clinical remission and endoscopic remission. Key secondary outcomes included composite clinical and endoscopic response, safety, and persistence. A multivariable Cox proportional hazards model was used to explore associations between baseline factors and endpoints. For multivariable model selection, backwards elimination was used with a threshold $P < 0.05$.

Results: A total of 198 patients were included, of whom 47.5% were female and 85.9% under 65 years. The majority had severe endoscopic activity at baseline (47.5%, n=87) and were bio-exposed (88.8%, n=176; 33%, n=66 prior failure of 3 or more biologics. Patients were followed for a median of 12.8 (IQR 8.2 to 22.1) months after induction. Clinical remission rates were 43.7% (69/198), 43.8% (56/128), and 62.4% (58/93) at 3, 6 and 12 months respectively. Endoscopic remission rates were 11.3% (7/62), 24.7% (23/93), and 23.4% (11/43) at 3, 6 and 12 months respectively. Bio-naive patients were more likely to achieve clinical and endoscopic remission compared to bio-exposed (HR=1.71, 95% CI: 1.02–2.88, and HR=7.33, 95% CI: 2.15–25.01, respectively). Adverse events were reported in 14.1% (18/198), with colectomies exclusively observed in bio-exposed individuals (8.6%; 17/198).

Conclusion: In a highly refractory and largely bio exposed population more than half of patients with UC treated with ustekinumab achieved clinical remission and one-third achieved endoscopic remission. Prior biologic exposure was independently associated with clinical and endoscopic remission. Safety profile was consistent with interleukin (IL)-12-23 blockade.

Disclosure: VS, AN, KES, AZ, LG have no conflict of interest
NN has been a speaker or advisor for Janssen, Abbvie, Takeda, Pfizer, Merck, Sandoz, Fresenius Kabi, Innomar, Iterative Scopes, Bristol Myers Squibb, Viatris and Ferring
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PP0547

EFFICACY AND SAFETY OF ETRASIMOD IN PATIENTS WITH ULCERATIVE COLITIS STRATIFIED BY BODY MASS INDEX: A POST HOC ANALYSIS FROM THE ELEVATE UC CLINICAL PROGRAMME

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Introduction: A high body mass index (BMI) may negatively impact treatment efficacy in patients with ulcerative colitis (UC) treated with advanced therapies, such as biologics,[1] which is relevant due to the prevalence of obesity and UC increasing substantially over the past decades.[2,3] Etrasimod is an oral, once-daily (QD), selective sphingosine 1-phosphate $(S1P)_{1,4,5}$ receptor modulator for the treatment of moderately to severely active UC.

Aims & Methods: This post hoc analysis of ELEVATE UC 52 and ELEVATE UC 12 investigated the impact of BMI on the efficacy and safety of etrasimod in patients with UC. ELEVATE UC 52 (NCT03945188) and ELEVATE UC 12 (NCT03996369) were phase 3, randomised, placebo-controlled trials.[4] ELEVATE UC 52 had a treat-through design consisting of a 12-week (wk) induction and 40-wk maintenance period; ELEVATE UC 12 comprised a 12-wk induction period. Patients who received ≥1 dose of etrasimod 2 mg QD or placebo were stratified by baseline BMI: <25 kg/m^2, 25–30 kg/m^2 and >30 kg/m^2. Efficacy outcomes (aligned to the predefined outcomes for the ELEVATE UC 52 and ELEVATE UC 12 trials) included the primary endpoint (clinical remission) and selected secondary endpoints at Wk 12

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(data pooled for both trials) and Wk 52 (ELEVATE UC 52 only). Safety outcomes assessed the proportions of treatment-emergent adverse events (TEAEs).

Results: Of the 787 patients enrolled in the ELEVATE UC clinical programme, 443 (56.3%) had baseline BMI <25 kg/m^2, 217 (27.6%) had baseline BMI 25–30 kg/m^2 and 127 (16.1%) had baseline BMI >30 kg/m^2. In the BMI <25 kg/m^2 and BMI 25–30 kg/m^2 subgroups, significantly more patients (p<0.05) receiving etrasimod vs placebo achieved all efficacy endpoints at Wks 12 and 52. In the BMI >30 kg/m^2 subgroup, significantly more patients receiving etrasimod vs placebo achieved clinical remission at Wk 52 (p<0.0001; Wk 12: p=0.05), endoscopic improvement and clinical response at Wks 12 and 52, and corticosteroid-free clinical remission at Wk 52 (all p<0.05; *Table*). The proportions of patients with the most common TEAEs were generally consistent between BMI subgroups receiving etrasimod. Incidence of serious AEs was generally similar across BMI subgroups receiving etrasimod (<25 kg/m^2: 17/308 [5.5%]; 25–30 kg/m^2: 8/137 [5.8%]; >30 kg/m^2: 1/82 [1.2%]).

	BMI <25 kg/m^2			BMI 25–30 kg/m^2			BMI >30 kg/m^2		
	Placebo QD	Etrasimod 2 mg QD	% Difference (95% CI)	Placebo QD	Etrasimod 2 mg QD	% Difference (95% CI)	Placebo QD	Etrasimod 2 mg QD	% Difference (95% CI)
Clinical remission,[b] Wk 12, n (%)	18 (13.3)	86 (27.9)	15.0 (7.4, 22.7)***	8 (10.0)	39 (28.5)	18.6 (8.2, 29.0)***	3 (6.7)	18 (22.0)	12.7 (0.0, 25.4)
Clinical remission,[b] Wk 52, n (%)	7 (10.0)	48 (30.8)	22.7 (12.6, 32.9)***	3 (6.3)	27 (33.3)	26.3 (13.7, 38.9)***	1 (3.9)	19 (36.5)	33.3 (17.3, 49.4)***
Endoscopic improvement,[c] Wk 12, n (%)	28 (20.7)	111 (36.0)	15.2 (6.7, 23.8)***	14 (17.5)	49 (35.8)	18.5 (6.8, 30.1)**	4 (8.9)	26 (31.7)	19.7 (6.0, 33.5)**
Endoscopic improvement,[c] Wk 52, n (%)	11 (15.7)	56 (35.9)	23.2 (12.3, 34.0)***	6 (12.5)	32 (39.5)	25.4 (11.2, 39.6)***	2 (7.7)	25 (48.1)	43.2 (25.5, 60.8)***
Clinical response,[d] Wk 12, n (%)	55 (40.7)	195 (63.3)	22.2 (12.3, 32.1)***	28 (35.0)	88 (64.2)	28.5 (15.2, 41.9)***	17 (37.8)	50 (61.0)	25.4 (7.3, 43.6)**
Clinical response,[d] Wk 52, n (%)	15 (21.4)	72 (46.2)	26.8 (14.6, 38.9)***	13 (27.1)	44 (54.3)	26.4 (10.1, 42.8)**	7 (26.9)	27 (51.9)	24.7 (1.9, 47.5)*
CS-free clinical remission[e] with CS use at baseline, Wk 52, n/N1 (%)	3/24 (12.5)	19/64 (29.7)	19.9 (2.1, 37.6)*	0/9 (0.0)	7/18 (38.9)	40.5 (16.6, 64.4)***	0/9 (0.0)	3/11 (27.3)	38.8 (4.8, 72.9)*

***p<0.001; **p<0.01; *p<0.05.

Population size assessed for each subgroup and time point were: BMI <25 kg/m^2: placebo QD, N=135 (Wk 12) and N=70 (Wk 52) and etrasimod 2 mg QD, N=308 (Wk 12) and N=156 (Wk 52); BMI 25–30 kg/m^2: placebo QD, N=80 (Wk 12) and N=48 (Wk 52) and etrasimod 2 mg QD, N=137 (Wk 12) and N=81 (Wk 52); BMI >30 kg/m^2: placebo QD, N=45 (Wk 12) and N=26 (Wk 52) and etrasimod 2 mg QD, N=82 (Wk 12) and N=52 (Wk 52). Difference (95% CI) and 2-sided p value are based on the Cochran-Mantel-Haenszel method adjusting to reported randomisation stratification of (1) naïve to biologic or Janus kinase inhibitor therapy at study entry, (2) baseline CS use (yes or no), (3) baseline disease activity (MMS: 4–6 or 7–9) and (4) study identifier (ELEVATE UC 52 and ELEVATE UC 12).

[a]Data were pooled at Wk 12 for both trials (ELEVATE UC 52 and ELEVATE UC 12); Wk 52 is ELEVATE UC 52 only.

[b]Clinical remission was defined as SFS=0 (or =1 with a ≥1-point decrease from baseline), RBS=0 and ES ≤1 (excluding friability).

[c]Endoscopic improvement was defined as ES ≤1.

[d]Clinical response was defined as a ≥2-point and ≥30% decrease from baseline in MMS, and a ≥1-point decrease from baseline in RBS or an absolute RBS ≤1.

[e]CS-free clinical remission was defined as the proportion of patients with clinical remission at Wk 52 who had been receiving CS at baseline but had not been receiving CS for ≥12 wks immediately prior to Wk 52.

BMI, body mass index; CI, confidence interval; CS, corticosteroid; ES, endoscopic subscore; MMS, Modified Mayo score; n, number of patients; N, number of patients in the subgroup; N1, number of subjects in each baseline BMI subgroup category, QD, once daily; RBS, rectal bleeding subscore; SFS, stool frequency subscore; UC, ulcerative colitis.

Table. Primary and select secondary efficacy endpoints at Wk 12 in ELEVATE UC 52 and ELEVATE UC 12 (pooled)[a] and Wk 52 in ELEVATE UC 52, stratified by baseline BMI.

Conclusion: In the ELEVATE UC clinical programme, significant improvements in efficacy were observed for patients receiving etrasimod 2 mg QD vs placebo, regardless of baseline BMI. The safety profile of etrasimod was consistent between BMI subgroups and with the overall ELEVATE UC population.

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PP0548

UNDERSTANDING GASTROENTEROLOGIST PREFERENCES AT THE TIME OF TREATMENT ESCALATION TO FIRST-LINE ADVANCED THERAPIES IN ULCERATIVE COLITIS: A DISCREET CHOICE EXPERIMENT IN FIVE EUROPEAN COUNTRIES

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Introduction: As the number of advanced treatment options for moderately to severely active ulcerative colitis (UC) increases, it is necessary to understand the factors driving gastroenterologist (GE) choice when escalating patients from conventional to advanced therapy.

Aims & Methods: We completed a quantitative analysis of GE therapy attribute preferences when choosing to escalate patients to their first advanced UC therapy. We conducted an online, cross-sectional survey using a discrete choice experiment (DCE) design.

Attribute and level selection was informed by a targeted literature search and formative qualitative research with patients and clinicians. Survey responders were practising GEs experienced in treating patients with moderately to severely active UC, recruited from France, Germany, Italy, Spain and the United Kingdom (UK).

Preference weights were estimated using a random parameters logit model for varying levels of seven attributes: time to symptom improvement, probability of remission at one year, difference between probability of remission and CS-free remission, five-year risk of malignancy, annual risk of serious infection, annual risk of major adverse cardiovascular events and mode and frequency of administration.

Relative importance (RI) was calculated using the difference in preference weights between the most and least preferred level of each attribute, scaled from 0 to 100%. An additional survey section was included to understand GE treatment and prescribing practices.

Results: A total of 397 GEs were included (France n = 140; Germany n = 40; Italy n = 40; Spain n = 47; UK n = 130). The most common GE-reported barriers to prescribing advanced therapies were concerns about contraindications and risks/side effects from patients (54.9%) and GEs (39.8%), perceived patient concerns about receiving injections or infusions (35.5%) and concerns about cost or insufficient reimbursement (32.2%). All DCE attributes factored into GE treatment decisions (see Table for RI and preference weights).

The three most impactful attributes were probability of remission at one year (RI 48.4%), followed by 5-year risk of malignancy (RI 11.4%) and time to symptom improvement (RI 11.1%; Table).

	Time to symptom improvement	Probability of remission at one year	Difference between probability of remission and CS-free remission	Five-year risk of malignancy	Annual risk of serious infection	Annual risk of MACE	Mode and frequency of administration
Attribute RI,[a] % (95% CI)[b]	11.1 (8.9, 13.4)	48.4 (45.7, 51.1)	8.0 (6.1, 10.0)	11.4 (9.5, 13.2)	6.7 (4.9, 8.5)	6.8 (5.0, 8.6)	7.5 (5.4, 10.1)
Preference weight level (95% CI)[c]							
Level 1	2 weeks 0.50 (0.31, 0.69)	20% probability -2.11 (-2.42, -1.81)	0% difference 0.27 (0.13, 0.41)	1/1000 patients 0.49 (0.35, 0.62)	1/100 patients 0.27 (0.14, 0.40)	1/1000 patients 0.27 (0.14, 0.40)	Oral pill 1–2 times daily with potential dose change 0.15 (-0.02, 0.33)
Level 2	4 weeks 0.22 (0.09, 0.34)	35% probability 0.26 (0.18, 0.34)	5% difference 0.12 (0.03, 0.21)	3/1000 patients -0.04 (-0.13, 0.05)	3/100 patients 0.02 (-0.07, 0.10)	3/1000 patients 0.01 (-0.08, 0.10)	Oral pill 1–2 times daily with the same dose throughout 0.23 (0.10, 0.35)
Level 3	8 weeks -0.31 (-0.44, -0.18)	45% probability 1.85 (1.65, 2.06)	15% difference -0.39 (-0.48, -0.29)	5/1000 patients -0.45 (-0.54, -0.35)	5/100 patients -0.28 (-0.38, -0.19)	5/1000 patients -0.28 (-0.38, -0.19)	Injection every 1–2 weeks 0.01 (-0.12, 0.14)
Level 4	12 weeks -0.41 (-0.55, -0.27)	N/A	N/A	N/A	N/A	N/A	Infusion every 4–8 weeks -0.39 (-0.52, -0.26)

The DCE model included seven attributes, each with several preference weight levels.

[a]RI is calculated as the difference in preference weights between the most preferred and least preferred level divided by the sum of the differences across all attributes; estimates sum to 100%.

[b]95% CIs that do not include zero indicate a statistically significant RI of an attribute. All seven attributes were statistically significantly important when selecting an advanced therapy. 95% CIs that do not overlap for pairs of attributes indicate a statistically significant difference in importance between attributes. Probability of remission at one year was statistically significantly more important than all other attributes.

[c]Preference weight levels are effects coded; zero indicates the mean effect across all attribute levels.

CI, confidence interval; CS, corticosteroid; DCE, discrete choice experiment; MACE, major adverse cardiovascular events; N, total number of patients; N/A, not applicable; RI, relative importance; UC, ulcerative colitis.

Table. Preference weights and RI of attributes influencing advanced UC therapy choice (N = 397).

Conclusion: All attributes factored into the trade-offs GEs consider when escalating patients with moderately to severely active UC to their first advanced therapy. Whereas risk of side effects was the most stated GE barrier to prescribing advanced therapy, probability of remission outweighed all other DCE attributes.

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PP0549

IMPACT OF OZANIMOD ON TYPE IV COLLAGEN BIOMARKERS AND THE ASSOCIATION WITH OZANIMOD EFFICACY IN PATIENTS WITH MODERATELY TO SEVERELY ACTIVE ULCERATIVE COLITIS: RESULTS FROM THE PHASE 3 TRUE NORTH STUDY

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Introduction: Ozanimod was efficacious and well tolerated for up to 52 weeks in patients with moderately to severely active ulcerative colitis (UC) in the phase 3 True North study (NCT02435992). Type IV collagen, a major component of the intestinal basement membrane, is thought to be involved in the pathogenesis of UC.

Aims & Methods: This analysis assessed the effect of ozanimod on type IV collagen biomarkers and their association with ozanimod efficacy in True North. Patients in Cohort 1 were randomized to ozanimod (n=429) or placebo (n=216) and patients in Cohort 2 received open-label ozanimod (n=367) through Week 10. Ozanimod patients with clinical response at Week 10 were rerandomized to ozanimod (ozanimod/ozanimod, n=230) or placebo (ozanimod/placebo, n=227) through Week 52; placebo patients with clinical response at Week 10 continued placebo through Week 52 (n=69). Levels of biomarkers for type IV collagen formation (type IV collagen propeptide [PRO-C4]), degradation (granzyme B-degraded [C4G] and matrix metalloproteinase-degraded [C4M] type IV collagen), and turnover (PRO-C4/C4M ratio) were assessed at baseline, Week 10, and Week 52. Baseline and Week 10 disease activity scores, including rectal bleeding, stool frequency, Physician Global Assessment (PGA), and endoscopy subscores, and partial and total Mayo scores, were correlated to biomarker levels. Associations between type IV collagen biomarkers and disease activity were assessed using Spearman's *rho* correlation.

	Induction (Week 10)			Maintenance (Week 52)	
	Placebo: Cohort 1	Ozanimod: Cohort 1	Ozanimod: Cohort 2	Ozanimod/ placebo	Ozanimod/ ozanimod
PRO-C4	0.07 (−2.40, 2.60) n=162	−3.38* (−5.05, −1.67) n=346	−2.06 (−3.82, −0.26) n=298	−2.46 (−5.13, 0.29) n=112	−6.71* (−8.91, −4.45) n=157
C4G	0.04 (−4.53, 4.84) n=151	−8.60** (−11.60, −5.50) n=315	−6.41* (−9.52, −3.20) n=282	−3.75 (−9.14, 1.96) n=103	−18.30*** (−22.12, −14.28) n=154
C4M	−1.37 (−5.24, 2.66) n=165	−10.53*** (−13.02, −7.96) n=346	−11.67*** (−14.22, −9.03) n=299	−16.85 (−20.57, −12.96) n=112	−20.40 (−23.50, −17.18) n=157
PRO-C4/ C4M	1.80 (−1.78, 5.50) n=162	8.02** (5.35, 10.76) n=346	10.71*** (7.86, 13.63) n=298	17.32 (12.72, 22.10) n=112	16.88 (12.90, 21.00) n=157

*P<0.05, **P<0.01, ***P<0.001 vs placebo: Cohort 1 for induction or ozanimod/placebo for maintenance.

Table. Adjusted mean percent change from baseline in type IV collagen biomarker levels (95% CI).

Results: At Week 10, significant decreases in PRO-C4, C4G, and C4M and significant increases in PRO-C4/C4M were seen with ozanimod vs placebo (**Table**). C4M decreases and PRO-C4/C4M increases with ozanimod at Week 10 were significantly greater in patients with vs without Week 10 clinical response. Baseline C4M was significantly positively correlated and PRO-C4/C4M was significantly negatively correlated with all baseline disease activity scores except rectal bleeding. In ozanimod-treated patients, changes in PRO-C4 and C4M at Week 10 were significantly positively correlated with changes in all disease activity scores at Week 10 except rectal bleeding and PGA for PRO-C4, while changes in C4G and PRO-C4/C4M at Week 10 were significantly negatively correlated with changes in all disease activity scores at Week 10. Decreases in PRO-C4 and C4G continued through Week 52 and were significantly greater in ozanimod/ozanimod than ozanimod/placebo patients; decreases in C4M and increases in PRO-C4/C4M continued at Week 52 at similar levels in ozanimod/ozanimod and ozanimod/placebo patients (*Table*).

Conclusion: Ozanimod decreased type IV collagen degradation and increased turnover, which are effects associated with disease activity. The T-cell activity marker C4G showed the strongest, but delayed, decrease at Week 52, which is aligned with the mechanism of action of ozanimod. These findings support ozanimod involvement in modulating key molecules in intestinal epithelial barrier function, extracellular matrix remodeling, and T-cell activity in UC.

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PP0550

MIRIKIZUMAB EFFECT ON BOWEL URGENCY RESOLUTION IN A RANDOMISED CONTROLLED PHASE 3 TRIAL OF PARTICIPANTS WITH CROHN'S DISEASE

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Introduction: Bowel urgency (BU), the sudden or immediate need to have a bowel movement, is an impactful symptom of Crohn's disease (CD) that varies in intensity[1]. We report the efficacy of the IL-23 p19 inhibitor, mirikizumab (miri) in improving BU among patients with moderately-to-severely active CD from the phase 3 VIVID-1 trial (NCT03926130). VIVID-1 co-primary and gated endpoints, including comparison with ustekinumab, are already reported[2].

Aims & Methods: Adult patients (N=1065) were randomised to placebo (PBO) or miri 900mg intravenously at weeks (W) 0, 4, and 8, then 300mg subcutaneously every 4 weeks (Q4W) from W12 to W52. At W12, PBO responders continued PBO to W52, non-responders received the blinded miri regimen per protocol. BU was assessed at W12 and W52 using the Urgency Numeric Rating Scale (UNRS)[3].

We evaluated the proportion of patients with baseline (BL) UNRS ≥3 and ≥6 who achieved BU Clinically Meaningful Improvement (CMI; >3 change in UNRS) and BU remission (UNRS ≤2). Treatment comparisons used Cochran-Mantel-Haenszel tests with non-responder imputation for missing values.

Results: Study population: 55.1% male. Mean (SD) age 36.2 (13.0) years. Disease duration 7.4 (7.9) years. At BL, median (Q1, Q3) UNRS 6.9 (5.2, 8.1), 94.5% patients achieved UNRS ≥3 and 66.0% patients achieved ≥6. In patients with BL UNRS ≥3, nominally significantly higher proportions of miri versus PBO patients achieved BU CMI and BU remission at both W12 and W52 (Table).

A nominally significantly greater proportion of miri-treated patients achieved W52 BU CMI-composite and BU remission-composite (Table). Similar results were seen with BL UNRS ≥6.

		W12 PBO	W12 Miri	W12 p-value	W12 95% CI	W52 PBO	W52 Miri	W52 p-value	W52 95% CI
	UNRS[a] CFB (LSM ±SE)	-1.58 (0.168)	-2.44 (0.099)	0.00011	-1.24, -0.48	-1.23 (0.180)	-3.24 (0.106)	<0.0001	-2.42, -1.60
Patients with BL UNRS ≥3	BU CMI-TT	50/186 (26.9)	227/547 (41.5)	0.0004	6.7, 22.0	38/186 (20.4)	301/547 (55.0)	<0.0001	28.0, 42.1
	BU CMI-composite[b]					38/186 (20.4)	251/547 (45.9)	<0.0001	18.5, 32.8
	BU Remission[c]-TT	29/186 (15.6)	134/547 (24.5)	0.01	2.3, 15.0	21/186 (11.3)	209/547 (38.2)	<0.0001	21.0, 33.2
	BU Remission-composite					21/186 (11.3)	180/547 (32.9)	<0.0001	15.7, 27.8
Patients with BL UNRS ≥6	BU CMI-TT	38/137 (27.7)	180/376 (47.9)	0.0001	9.8, 28.2	31/137 (22.6)	229/376 (60.9)	<0.0001	30.1, 47.3
	BU CMI-composite					31/137 (22.6)	188/376 (50.0)	<0.0001	18.4, 36.0
	BU Remission-TT	11/137 (8.0)	70/376 (18.6)	0.007	3.8, 15.8	13/137 (9.5)	126/376 (33.5)	<0.0001	17.1, 31.1
	BU Remission-composite					13/137 (9.5)	111/376 (29.5)	0.0002	13.2, 27.0

PBO, N=199; Miri, N=579. Data are N (%) unless otherwise stated. [a]UNRS scale: 0 (no urgency)-10 (worst possible urgency)[3]. [b]Composite=Clinical response by PRO (2 of the patient-reported items of the CDAI, AP and SF) at W12 and W52 parameter (CMI or Remission). [c]BU remission=no to minimal BU (UNRS≤2).

Treatment comparison: CFB used a mixed model for repeated measures model containing treatment, BL value, visit, stratification factors and interaction of BL value by visit and treatment by visit. Response rates used Cochran-Mantel-Haenszel tests with non-responder imputation. AP=abdominal pain; BL=baseline; CFB=change from BL; CI=confidence interval; CMI=clinically meaningful improvement; LSM=least squares mean; PBO=placebo; PRO=Patient Reported Outcome; SD=standard deviation; SE=standard error; SF=stool frequency; TT=treat-through; UNRS=Urgency Numeric Rating Scale; W=week.

Conclusion: BU is one of the most unrecognised symptoms in patients with CD. Patients enrolled in VIVID-1 had high BU scores at BL. Compared to PBO-treated pts, miri-treated pts achieved nominally significantly higher rates of BU CMI and remission at W12 and W52.

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PP0551

THROUGH THE LENS OF INFLAMMATORY BOWEL DISEASE: IMPACT OF RHEUMATOLOGICAL INVOLVEMENT ON DISEASE COURSE

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Introduction: Inflammatory bowel diseases (IBD) are systemic disorders that may be associated with extraintestinal manifestations in up to 40% of patients. Recognizing the recurrent emergence of these manifestations in IBD, with a particular emphasis on rheumatological aspect, becomes a pivotal focal point for a holistic therapeutic approach.

Aims & Methods: The aim of this study was to evaluate the impact of ankylosing spondylitis in IBD on disease course. We conducted a retrospective and comparative study, including patients followed for IBD over an 8-year period [2014 - 2022]. Clinico-biological data, disease severity, and long-term outcomes (surgery and disease course) were collected. IBD patients with ankylosing spondylitis (group1) and patients without (group2) were compared. Data were entered and analyzed via SPSS 26 software.

Results: One-hundred IBD patients were enrolled with a sex ratio of M/F=2.44 and a mean age of disease diagnose of 37.26 ± 13.46 years. Eighteen patients had ankylosing spondylitis (18%). The two groups were comparable in terms of age at disease onset, gender, smoking status and family history of IBD (p=0.65; p=0.9, p=1 and 0.1 respectively). In Crohn's disease, there was no difference between the two groups regarding the extent of the disease (p=0.71), the presence of upper gastro-intestinal tract involvement (p=0.11) and perineal manifestations (p=0.7). Inflammatory, structuring, and penetrating phenotypes were also comparable between the two groups (p=0.67; p=0.3; p=1 respectively). However, the occurrence of acute intestinal obstruction was significantly more common in ankylosing spondylitis patients (G1:54% vs. G2:10%; p=0.004). As for ulcerative colitis, extensive presentation (G1:28% vs. G2:41%; p=0.66) and the occurrence of disease extension during follow-up was comparable between the two groups (p=0.78). None of the patients with or without rheumatological involvement had any dermatological or ophthalmological involvement related to IBD, but there was a more significant prevalence of hepatic manifestations in ankylosing spondylitis group (G1:16% vs. G2:2%, p=0.01). The mean number of flare-ups was not significantly different between the two groups (G1:3.21±1.47 vs. G2:2.53±1.4; p=0.06). Regarding treatment, recourse to methotrexate (G1:16% vs. G2:2%; p=0.01) and the use of anti-TNF agents (G1:77% vs. G2:22%; p<0.001) was more frequent in group 1, with no difference in the occurrence of adverse events (p=1), the use of corticosteroids (p=0.22) or azathioprine (p=0.1). Surgical treatment was more frequently indicated in ankylosing spondylitis patients (G1:51% vs. G2:30%; p=0.06). Post-operative relapse of Crohn's disease was comparable between the two groups (p=1).

Conclusion: In our study, rheumatological involvement was not likely responsible of a more severe clinical phenotype of IBD.

Disclosure: Nothing to disclose.

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PP0552

COMPARATIVE STUDY OF THE EFFICACY OF TWO BIOLOGIC THERAPY SEQUENCES IN INFLAMMATORY BOWEL DISEASE

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Introduction:

Vedolizumab (VED) and ustekinumab (UST) are valid options for ulcerative colitis (UC) and Crohn's disease (CD) patients who fail anti-tumour necrosis factor (TNF) therapy. However, there are conflicting data on which drug should be preferred as second-line.

Aims & Methods:

UC and CD patients who received VED or UST after anti-TNF failure or intolerance were included, and treatment groups were compared. The primary outcome of clinical remission (CR) was evaluated at 6 and 12 months of therapy, as were the secondary outcomes of corticosteroid-free CR and biochemical remission (BR). Endoscopic remission (ER) was evaluated at 12 months. Predictors of these outcomes were assessed using Cox regression.

Results: A total of 72 patients were included, 21 with UC (14 UST | 7 VED) and 51 with CD (35 UST | 16 VED). There were no significant differences in the various remission rates between both treatment groups, in UC and CD. In CD, multivariate analysis revealed baseline Harvey-Bradshaw index (HBI) as an independent predictor of failed CR (p<0.001, HR 1.42, IC 95% 1.18-1.71) and corticosteroid-free CR (p<0.001, HR 1.43, IC 95% 1.19-1.72) at 12 months. Anti-TNF discontinuation due to primary or secondary failure (vs. adverse event) was an independent predictor of BR at 12 months (p=0.028, HR 2.33, IC 95% 1.09-4.98). Interestingly, subanalysis of 16 patients who advanced to a third-line therapy with the opposite drug revealed a tendency for lower CR rates with VED (40.0%), when compared to UST (72.73%), albeit not statistically significant (p=0.299).

Conclusion:

In this study, clinical, biochemical and endoscopic remission rates in UC and DC were not significantly different between patients treated with VED or UST as second-line therapy. HBI and reason for anti-TNF withdrawal were independent predictors of failed CR and BR, respectively. There may be a tendency for larger differences in CR when comparing VED and UST as third-line agents, with better results for the latter.

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PP0553

ANTI-TUMOUR NECROSIS FACTOR DRUG CONCENTRATION IS NOT ASSOCIATED WITH DISEASE OUTCOMES IN POUCHITIS: A RETROSPECTIVE, INTERNATIONAL, MULTI-CENTRE, CROSS-SECTIONAL STUDY

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Introduction: Therapeutic drug monitoring is an important tool for optimising anti-tumour necrosis factor-a (TNF-a) therapy in inflammatory bowel disease. However, the exposure-response relationship has never been assessed in patients with pouchitis.

Aims & Methods: The aim was to explore the association between anti-TNF-a drug concentration and pouchitis disease activity in patients with a background of ulcerative colitis. A retrospective, international, multicentre, cross-sectional study was conducted in adult patients with pouchitis requiring anti-TNF-a treatment.

Rates of clinical remission (≤1 on a 4-point Physician's Global Assessment scale) and endoscopic remission (lack of friability or ulceration) were calculated, and drug concentrations during maintenance therapy were compared between remission and non-remission cohorts.

Results: Sixty-three patients were included; median age, 48 years (IQR 36-59), male 28 (44%), median time since pouchitis diagnosis, 7 years (IQR 2-13). Patients received infliximab, n=27 (43%), adalimumab, n=29 (46%), or both n=7 (11%). Thirty-two (51%) patients received concomitant immunomodulation.

Median infliximab trough concentrations (mg/ml) were similar between patients in clinical remission (n=21) vs non-remission (n=11), 5.3 vs. 4.4, p=0.73, although patients with moderate-to-severely active endoscopic pouchitis (n=19) had significantly higher trough concentrations than those in endoscopic remission (n=13), 10.6 vs. 3.5, p=0.04. For adalimumab, median drug concentrations did not significantly differ between remission/

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non-remission groups based on clinical (n=18/18), 11.4 vs 7.6, p=0.32, or endoscopic assessment, (n=7/29), 9.0 vs. 7.8, p=0.78. Four patients had positive anti-drug antibodies with undetectable drug concentration.

Conclusion: In a cohort of patients with pouchitis, higher anti-TNF-a drug concentrations were not associated with more clinical or endoscopic remission.

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PP0554

COST-EFFECTIVENESS AND COST-UTILITY OF OPTIMIZED MERCAPTOPURINE TREATMENT VERSUS PLACEBO IN ULCERATIVE COLITIS PATIENTS: THE RANDOMIZED CONTROLLED OPTIC TRIAL

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Introduction: The randomized controlled OPTIC trial reported that optimised mercaptopurine treatment was superior to placebo in achieving clinical, endoscopic and histological outcomes at one year in ulcerative colitis (UC) patients failing 5-aminosalicylates after receiving induction treatment with corticosteroids.

Aims & Methods: We aimed to assess cost-effectiveness and cost-utility of therapeutic drug monitoring (TDM)-guided mercaptopurine treatment in UC. Data were collected during the OPTIC trial to perform a health economic evaluation for mercaptopurine treatment versus placebo in UC patients. The economic evaluation was performed from a health care and societal perspective using a cost-effectiveness and cost-utility analysis with a time horizon of one year. Volumes and costs of in-hospital care, out-of-hospital care, out-of-pocket expenses and productivity loss were assessed. Main outcomes were the costs per patient who achieved combined corticosteroid-free clinical remission and endoscopic improvement at one year and the costs per quality-adjusted life-year (QALY).

Results: In total, 59 patients were randomized to the intervention (n = 29) and control (n = 30) group. Patients on mercaptopurine treatment showed a non-significant gain in QALYs of 0.047 (-0.024 to 0.117) (P=0.184) corrected for baseline health utility. Non-significant differences in costs were €63 (-€1267 to €1434; P=0.927) in favour of placebo from a health care perspective and -€742 (-€3683 to €2016; P=0.644) in favour of mercaptopurine from a societal perspective, including patients' out-of-pocket expenses and employers' productivity loss. The probability of optimised mercaptopurine treatment being cost-effective was 0.80 at a willingness to pay per additional QALY of €20,000.

Conclusion: Optimised mercaptopurine treatment in UC patients failing 5-aminosalicylates is a cost-effective strategy from a societal perspective.

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PP0555

COMPARISON OF REAL-WORLD TREATMENT PERSISTENCE AND SAFETY OF VEDOLIZUMAB AND ANTI-TUMOUR NECROSIS FACTOR-ALPHA TREATMENT IN BIOLOGIC-NAIVE PATIENTS WITH ULCERATIVE COLITIS: SYSTEMATIC LITERATURE REVIEW AND META-ANALYSIS

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Introduction: Vedolizumab (VDZ) is approved in Europe and the USA for the treatment of moderately to severely active ulcerative colitis (UC). A previous systematic literature review (SLR) and meta-analysis published in 2018 demonstrated the real-world effectiveness and positive benefit–risk profile of VDZ treatment for up to 1 year in patients with inflammatory bowel disease.[1] There is limited real-world evidence on the long-term (> 1 year) treatment persistence and safety of VDZ compared with anti-tumour necrosis factor-alpha (TNFα) treatment in biologic-naive patients with UC.

Aims & Methods: The aims of this meta-analysis were to assess the real-world effectiveness and treatment persistence of VDZ in patients with UC. We report an exploratory analysis of treatment persistence and serious adverse events (SAEs) among biologic-naive patients with UC treated with VDZ compared with anti-TNFα treatment at 1 and at 2 years. Literature searches were conducted to identify studies published from 2014 to 2022. For the primary analysis, outcomes of interest included clinical remission, mucosal healing, treatment patterns (including treatment persistence) and safety. Data on adults with UC treated with VDZ or anti-TNFα treatment in a real-world setting were extracted if there were at least 2 studies for each timepoint and population. Weighted proportions and corresponding 95% confidence intervals (CIs) were calculated using the DerSimonian-Laird random-effects model to account for between-study heterogeneity. No formal statistical significance testing between treatments was applied.

Results: Among comparative studies of VDZ and anti-TNFα treatment, 8 studies were identified (10 486 biologic-naive patients with UC). Of these, 5 and 2 studies, respectively, reported the rates of treatment persistence in biologic-naive patients at 1 and at 2 years. The rate of treatment persistence at 1 year was 74.9% with VDZ (95% CI 64.3–83.3) and 59.0% with anti-TNFα treatment (47.2–69.8). Similarly, the rate of treatment persistence at 2 years was 77.6% with VDZ (70.8–83.2) and 60.0% with anti-TNFα treatment (48.3–70.7). Three retrospective cohort studies of 775 biologic-naive patients with UC treated with VDZ (n=465) versus anti-TNFα treatment (n=310) reported SAEs. In the first study, safety outcomes were followed from treatment initiation up to five half-lives after treatment discontinuation; 6.8% treated with VDZ experienced SAEs compared with 19.2% with anti-TNFα treatment.[2] In the second study, the rate of SAEs was similar between patients treated with VDZ (4.5%) and anti-TNFα treatment (5.0%).[3] In the third study that reported SAEs within the first year of treatment; the rates of SAEs were 14.3% and 10.9% among patients who received VDZ and anti-TNFα treatment, respectively.[4]

Conclusion: This analysis of comparative studies in biologic-naive patients with UC suggested that treatment persistence rates were higher with VDZ than with anti-TNFα treatment both at 1 and at 2 years. In addition, these data support a comparable safety profile, with similar rates of SAEs between VDZ and anti-TNFα treatment.

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PP0556

TARGET ACHIEVEMENT AFTER 32 WEEKS OF VEDOLIZUMAB TREATMENT IN PATIENTS WITH MODERATE TO SEVERE ULCERATIVE COLITIS: AN INTERIM ANALYSIS FROM THE VERDICT TRIAL

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Introduction: The primary objective of the VERDICT study, a continuing randomized controlled trial (NCT04259138),[1,2] is to determine the optimal treatment target in moderately-to-severely active ulcerative colitis (UC). Patients are allocated to Group 1 (corticosteroid-free [CSF] symptomatic remission), Group 2 (CSF symptomatic remission + CSF endoscopic remission), and Group 3 (CSF symptomatic remission + CSF endoscopic remission + CSF histologic remission, also termed disease clearance). Patients follow treatment algorithms involving early introduction of vedolizumab (VDZ), without week 10 dosing, nor dose escalation before week 16. Here we report target achievement at weeks 16 and 32 amongst groups.

Aims & Methods: VDZ 300 mg was administered intravenously following a treatment algorithm related to baseline UC treatment until assigned treatment target was reached at week 32, with an escalation step between week 16 and 32 as indicated. CSF symptomatic remission was defined as Mayo rectal bleeding subscore=0, CSF endoscopic remission was defined as Mayo Endoscopic Score [MES]≤1, and CSF histologic remission was defined as Geboes score<2B.0.

Results: As of 29 Feb 2024, 612 patients were enrolled with 154, 198, and 260 patients assigned to target Groups 1, 2, and 3, respectively. Baseline age and disease duration were similar amongst groups, with respective percentages of patients having baseline MES=3 given by 60.4%, 61.6%, and 64.2%, and receipt of concomitant corticosteroids given by 42.9%, 40.9%, and 50.4% (median dose 25.4, 25.3, and 23.7 mg). There were 71.4%, 75.3%, and 83.5% of patients who were bionaïve, respectively. At time of analysis, 63.6%, 71.2%, and 80.8% of randomized patients in the groups had completed the week 32 visit.

	Group 1 Bionaïve N=86	Group 1 Bio-exposed N=12	Group 1 Total N=98	Group 2 Bionaïve N=116	Group 2 Bio-exposed N=25	Group 2 Total N=141	Group 3 Bionaïve N=183	Group 3 Bio-exposed N=27	Group 3 Total N=210
CSF remission, n (%)	72 (83.7)	8 (66.7)	80 (81.6)	74 (63.8)	13 (52.0)	87 (61.7)	101 (55.2)	11 (40.7)	112 (53.3)
Non-CSF remission, n (%)	3 (3.5)	0 (0.0)	3 (3.1)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.5)	0 (0.0)	1 (0.5)
Remission target not achieved, n (%)	11 (12.8)	3 (25.0)	14 (14.3)	40 (34.5)	12 (48.0)	52 (36.9)	81 (44.3)	16 (59.3)	97 (46.2)
Remission target status missing,[b] n (%)	0 (0.0)	1 (8.3)	1 (1.0)	2 (1.7)	0 (0.0)	2 (1.4)	0 (0.0)	0 (0.0)	0 (0.0)

[a]Week 32 completer analysis set includes all patients who reached the week 32 visit. Group 1 is defined as CSF symptomatic remission; Group 2 is defined as CSF symptomatic remission + CSF endoscopic remission; and Group 3 is defined as CSF symptomatic remission + CSF endoscopic remission + CSF histologic remission.

[b]Remission target status missing includes patients who ended the study before the week 32 visit, those who had not reached the week 32 visit, and those who reached the week 32 visit but had no status, as of 29 FEB 2024.

Abbreviation: CSF, corticosteroid-free.

Table 1. Summary of Target Achievement by Week 32[a].

In patients with observed data and a week 16 visit, 64.1% (75/117), 42.2% (73/173), and 40.2% (94/234) in Groups 1, 2, and 3, respectively, achieved CSF remission at week 16. In patients with observed data and a week 32 visit, 81.6% (80/98), 61.7% (87/141), and 53.3% (112/210), respectively, achieved CSF remission by week 32, with higher proportions in bionaïve (83.7%, 63.8%, and 55.2%) compared to bio-exposed (66.7%, 52.0%, and 40.7%) (*Table 1*). In the intention-to-treat population, 48.7%, 36.9%, and 36.2% of patients, respectively, achieved CSF remission at week 16, with higher proportions in patients who were bionaïve (56.4% [62/110], 38.3%

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[57/149], and 38.2% [83/217]) compared to bio-exposed (29.5% [13/44], 32.7% [16/49], and 25.6% [11/43]). By week 32, 51.9%, 43.9%, and 43.1% of the intention-to-treat population achieved CSF remission, with higher proportions in bionaïve (65.5% [72/110], 49.7% [74/149], and 46.5% [101/217]) compared to bio-exposed (18.2% [8/44], 26.5% [13/49], and 25.6% [11/43]). For patients not achieving target at week 16, treatment escalation occurred in 22.7%, 49.5%, and 53.5% of patients; after escalation, proportions achieving CSF remission by week 32 were 41.9% (13/31), 27.0% (20/74), and 19.0% (23/121), respectively.

Conclusion: These interim week 32 results provide evidence of improvement from week 16 to 32 across all endpoints and continue to support the efficacy of VDZ in achieving the stringent endpoint of CSF disease clearance.

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PP0557

LONG-TERM OUTCOMES FOLLOWING WITHDRAWAL OF ANTI-TUMOUR NECROSIS FACTOR TREATMENT IN INFLAMMATORY BOWEL DISEASE PATIENTS IN REMISSION: THE EXIT LONG-TERM STUDY OF GETECCU

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Introduction: The EXIT trial showed there is no difference between inflammatory bowel disease (IBD) patients who were in remission and withdrew anti-TNF therapy, and those who maintained anti-TNF therapy with respect to the percentage of sustained remission at 12 months. However, data of the long-term outcomes are crucial.

Aims & Methods: Aims: To compare long-term risk of relapse in IBD patients in remission who withdrew anti-TNF therapy versus those who maintained anti-TNF therapy, to analyse the response to anti-TNF resumption in those patients who relapsed after anti-TNF withdrawal, and to evaluate the overall success of the anti-TNF resumption strategy.

Methods: This study is an extension of the follow-up from the EXIT trial. Patients included in EXIT trial were IBD patients who have achieved clinical remission for ⩾6 months at standard anti-TNF dosage, with absence of significant endoscopic/radiological lesions, no previous IBD-related surgery, and ongoing immunomodulatory treatment. Patients were randomised either to maintain or to withdraw the anti-TNF therapy[1]. The long-term outcomes of patients in sustained clinical remission who have completed the follow-up of EXIT trial per-protocol were analysed at the end of the EXIT trial. After that, patients were managed at the discretion of their physicians. Patients who were randomised in EXIT trial to continue with the anti-TNF comprised the maintenance arm (MA). For these patients, follow-up continued until either the anti-TNF resumption, or until the initiation of another biologic treatment or until the small molecule therapy (due to relapse or electively) or until the last visit (if no relapse occurred). Patients who were randomised in EXIT trial to withdraw the anti-TNF comprised the withdrawal arm (WA). For these patients, follow-up ended either at the time of relapse, at the withdrawal of anti-TNF treatment (electively, due to relapse, or adverse events), or at the last visit. The success of the anti-TNF resumption strategy was defined as the achievement and maintenance of clinical remission after restarting the anti-TNF therapy due to relapse, without experiencing adverse events or major complications (such as the need for surgery for IBD or the appearance of new perianal fistulas) that led to treatment withdrawal.

Results: 125 patients were included, 63 in MA and 62 in WA. Median follow-up was 12 and 26 months in MA and in WA, respectively (p<0.05). The main characteristics of the study population are detailed in table 1. In MA, the cumulative incidence of relapse was 8% at 12 months, 47% at 24 months, and 65% at 36 months, whereas in the WA was 16% at 12 months, 39% at 24 months and 43% at 36 months of follow-up, (p=0.3). The incidence rate of relapse was 22% (95% CI 14-31) and 19% (95% CI 13-26) per patient-year of follow-up in MA and WA, respectively. In the multivariate analysis, having a fecal calprotectin at baseline of the EXIT trial >250 µg/g was the only variable associated with a higher risk of relapse (HR=3.94, 95% CI=1.92-8.06; p<0.0001). Of the 29 patients of WA who relapsed, 26 patients (90%) resumed the same anti-TNF and 69% of them achieved clinical remission.

At the end the of follow-up, 16 out of 26 patients (62%) who restarted anti-TNF maintained remission, while 10 out of 26 patients (38%) discontinued anti-TNF: 4 due to non-response, 2 due to loss of response, and 4 due to adverse events. The proportion of adverse events was similar in the MA and the WA, 69% and 68%, respectively (p>0.05).

Conclusion: In the long-term analysis of the outcomes of patients included in EXIT trial, the withdrawal of anti-TNF therapy in patients with IBD in remission was not associated with higher risk of relapse. Resumption of anti-TNF after relapse recapture remission in the majority of patients. The anti-TNF withdrawal approach was safe.

Variables	Maintenance arm N=63	Withdrawal arm N=62	P
-Mean age at inclusion, years (range)	41 (18-65)	42 (20-68)	0.5
-Mean age at diagnosis, years (range)	31 (10-58)	32 (12-65)	0.9
-Median time under current anti-TNF (years), IQR	2.8 (1.8-4.9)	2.6 (1.2-5.7)	0.7
-Median time in remission with current anti-TNF at standard dose (months), IQR	23.2 (13-41)	21.9 (9.2-37.4)	0.9
-Median disease duration at inclusion (months), IQR	77.2 (39-163)	100.8 (36.5-188)	0.4
Male gender, n (%)	36 (57)	42 (68)	0.2
Current smoker, n (%)	10 (16)	10 (16%)	0.9
Crohn's disease, n (%)	29 (46)	25 (40)	0.5
-L1, n (%)	9 (31)	7 (28)	
-L2, n (%)	9 (31)	6 (24)	0.7
-L3, n (%)	11 (38)	12 (48)	
-L4, n (%)	1 (3)	1 (3)	0.3
-B1, n (%)	20 (69)	23 (92)	
-B2, n (%)	7 (24)	2 (8)	0.09
-B3, n (%)	2 (7)	0 (0)	
-Perianal, n (%)	2 (6)	2 (6)	1
Ulcerative colitis, n (%)	39 (56)	39 (56)	
-Extensive, n (%)	25 (73)	20 (54)	1
-Left-sided, n (%)	8 (24)	16 (43)	
-Proctitis, n (%)	1 (3)	1 (3)	
Extraintestinal manifestations	13 (21)	8 (13)	0.2
Type of anti-TNF, n (%)			
-Adalimumab, n (%)	9 (14)	9 (15)	0.9
-Infliximab, n (%)	54 (86)	53 (85)	
Indication for the anti-TNF, n (%)			
-Refractoriness to immunomodulators, n (%)	18 (29)	20 (32)	
-Steroid-dependency, n (%)	22 (35)	19 (31)	0.7
-Steroid-refractoriness, n (%)	13 (21)	12 (19)	
-Top-down strategy, n (%)	1 (2)	4 (7)	
-Others, n (%)	9 (14)	7 (11)	
Prior intensification of current anti-TNF, n (%)	11 (18)	9 (15)	0.7
Median C-reactive protein, IQR	0.1 (0.1-0.3)	0.1 (0.1-0.4)	0.1
Faecal calprotectin >250 mg/g, n (%)	4 (7)	8 (15)	0.35

IQR, interquartile range; TNF, tumour necrosis factor

Table 1. Characteristics of the study population at baseline.

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PP0558

THE REAL-WORLD TREATMENT PATTERNS AND CLINICAL OUTCOMES OF 5-AMINOSALICYLIC ACID TREATMENT IN KOREAN PATIENTS WITH ULCERATIVE COLITIS: A POPULATION-BASED RETROSPECTIVE STUDY

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Introduction: This retrospective study analysed the real-world use patterns and clinical outcomes of 5-aminosalicylic acid (5-ASA) treatment in Korean patients with ulcerative colitis (UC) based on the records from national health insurance claims database (NHICD).

Aims & Methods: The records of individuals aged ≥15 years, first diagnosed as having UC between 1 January 2009 and 30 September 2018, were considered eligible for the study if 5-ASA was prescribed for ≥30 days within six months after the diagnosis.

The data were extracted until the events of interest, patient's death, or 31 March 2019, whichever occurred earlier. The events of interest were defined as UC-related hospitalisation, UC-related surgery, and 5-ASA non-response, the latter corresponding to the prescription of steroids, immunosuppressants, and/or biological medications during the follow-up period. The yearly proportions of patients in each average daily dose (ADD) group, i.e., rectal monotherapy (RM), oral low dose (<2 g/day) monotherapy (OML), oral standard-to-high dose (2 g/day to ≥3 g/day) monotherapy, oral low dose combination(oral+rectal) therapy (OCL), and oral standard-to-high dose combination(oral+rectal) therapy (OCSH) group, were presented. Odds ratio for prescribing high dose 5-ASA was calculated from a logistic regression model.

The incidence of the events of interest was expressed per 1000 person-years (PY). Hazard ratios for 5-ASA insufficient-response and UC-related hospitalisation in each ADD group were obtained from a Cox proportional hazard model, with OML as a reference group. Treatment adherence was expressed as a proportion of days covered (PDC) within the first 12 months from the prescription index date.

Results: The analysis included the data of 11,338 patients (mean age 42.15 years, 59.53% of males).

During a median observation period of 1,720 days, a tendency to a decrease in RM proportion was observed, along with an increase in the proportions of oral + rectal combination therapies. The odds of prescribing high-dose 5-ASA increased significantly since 2013.

The proportion of OCSH increased from 18.1% in 2009 to 39.5% in 2018, whereas OML and OCL proportions decreased from 17.5% to 6.6% and from 11.6% to 6.6%, respectively. Incidence rates of 5-ASA insufficient-response and UC-related hospitalization were significantly lower in the RM group than in the OML group, with no significant differences between the OML group and other ADD groups. Mean PDC in overall population was 0.70, ranging from 0.65 to 0.76 across the ADD groups. 82.4% of patients responded to 5-ASA, and Mean PDC in overall population was 0.70, ranging from 0.65 to 0.76 across the ADD groups.

Conclusion: A tendency to a more frequent prescription of standard to high-dose 5-ASA as combination therapy was observed in 2009-2018, along with a decrease in RM prescription frequency, which is consistent with the current guidelines.

Disclosure: Nothing to disclose.

PP0559

MIRIKIZUMAB PHARMACOKINETICS AND EXPOSURE-RESPONSE RELATIONSHIPS IN PATIENTS WITH MODERATELY TO SEVERELY ACTIVE CROHN'S DISEASE IN PHASE 2 AND 3 STUDIES

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Introduction: Mirikizumab (MIRI) is a humanized anti-interleukin-23-p19 monoclonal antibody (mAb) approved for ulcerative colitis (UC) with demonstrated efficacy for Crohn's disease (CD) in Phase 2 (AMAG, NCT02891226) and Phase 3 (VIVID-1, NCT03926130) trials.[1,2] This analysis characterized MIRI pharmacokinetic (PK) properties and exposure-response (ER) relationships in adults with moderately to severely active CD using data from 2 trials.

Aims & Methods: Patients received 12-week MIRI induction (200, 600 or 1000 mg [AMAG] or 900 mg [VIVID-1] intravenously [IV] every 4 weeks [Q4W]), followed by MIRI maintenance (200, 600 or 1000 mg IV [AMAG], or 300 mg administered subcutaneously (SC) [AMAG and VIVID-1] Q4W) up to Week (W)52. PK parameters were estimated by non-linear mixed-effects modeling. Covariate effects on MIRI exposure were evaluated using simulation-based estimations. The exposure efficacy model included data from participants who had model-predicted average drug concentration during multiple dosing. Clinical efficacy at W12 was compared with AMAG graphically and models were conducted on primary endpoints. ER for patients achieving endoscopic response by Simple Endoscopic Score for CD (SES-CD) at W12 and W52 were assessed using logistic regression models.

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Results: MIRI PK was best described by a linear two-compartment model with first-order absorption. Population (PopPK) model-estimated parameters were consistent across AMAG and VIVID-1 (Table). AMAG indicated a near-maximal efficacy for doses between 600 and 1000 mg IV for endoscopic response by SES-CD at W12, suggesting that 900 mg IV in VIVID-1 would produce near-maximal effect for W12 endpoints. Linear relationships between exposure and W12 endoscopic response by SES-CD best described observed data in VIVID-1. In AMAG, model-based analyses of the relationships between exposure and W52 efficacy measures showed no ER trend across MIRI maintenance doses, supporting 300 mg SC Q4W for VIVID-1 Maintenance Period. Linear relationships between exposure and endoscopic response by SES-CD on a logistic scale best described observed data in VIVID-1 at W52. None of the covariates evaluated in PK or ER modelling were considered clinically relevant.

	CL (L/h)	Albumin effect on CL	Q (L/h)	BLBW effect on CL and Q	V_c[c] (L)	V_p[c] (L)	BLBW effect on V_c and V_p	F[d] %	BL BMI effect on F
VIVID-1 Population estimate [95% CI[a], % SEE]	**0.0197**[b] [0.0179 to 0.0230, 4.53]	**-0.0200** [-0.0228 to -0.0173, 7.25]	**0.00921**[b] [0.00529 to 0.0252, 26.6]	**0.268** [0.150 to 0.378, 23.1]	**2.78**[b] [2.71 to 2.82, 0.978]	**1.61**[b] [1.13 to 2.85, 16.5]	**0.445** [0.372 to 0.517, 9.53]	**38.8** [33.8 to 45.8, 5.62]	**-0.0354** [-0.0469 to -0.0263, 13.9]
AMAG Population estimate [95% CI[a], % SEE]	**0.0217**[e] [0.0203 to 0.0306, 6.27]	**-0.754** [-0.976 to -0.598, 12.0]	**0.00884**[f] [0.00605 to 0.202, 42.3]	**0.538** [0.348 to 0.745, 19.7]	**3.31**[f] [3.08 to 3.45, 2.21]	**1.75**[f] [1.49 to 5.67, 18.2]	**0.543** [0.330 to 0.721, 18.1]	**0.334**[g] [0.296 to 0.445, 8.44]	**-0.0318** [-0.0434 to -0.0206, 18.2]

[a]Estimated using bootstrap; [b]CL_i=CL*(BLBW/65.0)$^{0.268}$*(1+Albumin-44.57)*0.02*(BL CRP/7.41)$^{0.0853}$; Q_i= Q*(BLBW/65.0)$^{0.268}$; [c]$V_{c,i}$=V_c*(BLBW/65.0)$^{0.445}$; $V_{p,i}$=V3*BLBW/65.0)$^{0.445}$; [d]Estimate is on the logit parameter for bioavailability; [e]CL_i=CL*(BLBW/71.8)$^{0.538}$*(BL SES-CD/12)$^{0.227}$*(Albumin/44.0)$^{-0.754}$; [f]CL_i=CL*(BLBW/71.8)$^{0.538}$; $V_{c,i}$=V_c*(BLBW/71.8)$^{0.543}$; $V_{p,i}$= V_p*BLBW/71.8)$^{0.543}$; [g]F_i= F*[1+(VL BMI – 24.75)*-0.0318]

BL=baseline; BW=body weight; BMI=body mass index; CI=confidence interval; CL=clearance; CV=geometric coefficient of variation; F=bioavailability; i=individual; L/h=Liters per hour; Q=intercompartmental clearance; SEE=standard error of the estimate; SES-CD=Simple Endoscopic Score for CD; V_c=volume of distribution of the central compartment; V_p=volume of distribution of the peripheral compartment

Conclusion: MIRI PK characteristics in AMAG and VIVID-1 were consistent with UC studies and typical of mAbs.[3] ER analyses showed MIRI induction dose of 900 mg IV at W0, W4, and W8 followed by maintenance dose of 300 mg SC Q4W resulted in near-maximal efficacy in VIVID-1. These findings support MIRI dosing regimen proposed for patients with moderately to severely active CD. PK and ER data suggested that no adjustments based on patient factors are warranted.

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PP0560

EFFECTS OF MIRIKIZUMAB VERSUS PLACEBO ON HISTOLOGIC INFLAMMATION EVALUATED BY COMPREHENSIVE ASSESSMENT IN 5 INTESTINAL SEGMENTS IN A RANDOMIZED CONTROLLED PHASE 3 TRIAL OF PARTICIPANTS WITH CROHN'S DISEASE

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Introduction: Histologic inflammation persists in up to one quarter of patients with Crohn's Disease (CD) despite endoscopic mucosal healing. Responsiveness of histologic inflammation to treatment is an evolving outcome measure in CD. Mirikizumab (MIRI) increased histologic response (H-Res) and remission (H-Rem) relative to placebo (PBO) in the Phase 2 SERENITY trial.[1]

Aims & Methods: We compared the effect of MIRI on H-Res and H-Rem at week (W)12 and W52 to PBO by composite endpoints of response (clinical response by Patient-Reported Outcome [PRO] at W12 and W52 H-Res) and of remission (clinical response by PRO at W12 and W52 H-Rem) at W52. The overall population of patients with moderately to severely active CD, patients without prior biologic failure (non-BF), and patients with prior BF were assessed, using data from the Phase 3 VIVID-1 trial (NCT03926130). Two biopsy specimens from each of 5 intestinal segments (1 ileal and 4 colonic) were obtained from the edge of the ulcers, or the most inflamed mucosa from randomized patients at screening, W12, and W52. Criteria for H-Res were the absence of epithelial neutrophils and epithelial damage, erosions, and ulceration or ≥50% decrease in either the sum of the 5 segments of Robarts Histopathology Index or the Global Histologic Disease Activity Score. H-Rem was defined as complete absence of mucosal neutrophils (in epithelium and lamina propria), and no epithelial damage, erosions, and ulcers; these criteria had to be met in all biopsy specimens. Clinical response by PRO was defined as ≥30% decrease in stool frequency and/or abdominal pain with neither score worse than baseline. H-Res, H-Rem, composite H-Res, and composite H-Rem in all patients were prespecified, non-multiplicity-controlled endpoints. Analyses in non-BF and BF patients were post hoc.

Results: At W12, treatment with MIRI resulted in nominally statistically significantly higher rates of H-Res in all three patient groups analyzed (p<.001) (*Table*). At W12, differences between MIRI vs PBO were nominally significant in achieving H-Rem in all patients (p<.01) and in non-BF and BF patients (both p<.05). For W52 composite H-Res, differences between MIRI vs PBO were nominally statistically significant in all patient groups examined (p<.001). For W52 composite H-Rem, nominally statistically significant differences between MIRI vs PBO were observed in all patients (p<.001), in non-BF patients (p<.05), and in BF patients (p<.001).

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	All patients			Without Prior Biologic failure			With Prior Biologic Failure		
	PBO (N=199)	MIRI (N=579)	Difference (95% CI)	PBO (N=91)	MIRI (N=273)	Difference (95% CI)	PBO (N=88)	MIRI (N=255)	Difference (95% CI)
W12 H-Res	30.7	52.7	22.2 (14.6-29.7)***	37.4	58.6	21.2 (9.7-32.8)***	22.7	47.8	25.1 (14.4-35.8)***
W12 H-Rem	12.1	22.1	10.2 (4.7-15.7)**	14.3	27.5	13.2 (4.3-22.1)*	4.5	12.2	7.6 (1.7-13.5)*
W52 Composite H-Res	16.1	44.4	28.2 (21.6-34.7)***	26.4	46.5	20.1 (9.3-31.0)***	6.8	45.9	39.1 (31.0-47.1)***
W52 Composite H-Rem	8.5	23.3	14.7 (9.5-19.9)***	14.3	24.9	10.6 (1.8-19.5)*	2.3	21.6	19.3 (13.4-25.2)***

Data are %; *p<.05; **p<.01; ***p< .001 vs PBO
"Without prior biologic failure" and "With prior biologic failure" were subgroups of all patients with active histologic disease at baseline. P-values for "All patients" were from the Cochran–Mantel–Haenszel test adjusting for baseline covariates. P-values for the "Without prior biologic failure" and "With prior biologic failure" subgroups were from the Fisher's exact test.
CI=confidence interval; H=histologic; MIRI=mirikizumab; PBO=placebo; Rem=remission; Res=response; W=week

Conclusion: Based upon strict definitions that applied to all 5 intestinal segments, MIRI achieved nominally significantly higher rates of H-Res and H-Rem compared to PBO in all patients at W12 and W52. The statistical difference was more pronounced in the BF population after 1 year of treatment.

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PP0561

EVALUATION OF AN EXERCISE PROGRAM FOR PATIENTS WITH INFLAMMATORY BOWEL DISEASE: A PILOT STUDY

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Introduction: Patients with inflammatory bowel disease (IBD) tend to be less physical active, whilst maintaining an active lifestyle has been associated with enhanced disease control, diminished fatigue, and improved quality of life (1,2,3).

Aims & Methods: This study aims to evaluate the feasibility and potential impact of a specially designed exercise program for IBD patients.

Patients with IBD participated in a 16-week exercise program based on their individual fitness level, between March and June 2023. All outcomes were measured before and after the program. Body composition included, BMI, muscle mass and fat percentage. Physical fitness was based on

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performance of the Fundamental Motor Skills. Effect on quality of life, fatigue and disease control was assessed using validated questionnaires. For statistical analyses, a paired t-test or Wilcoxon matched paired rank test was used.

Results: In total 32 patients were included in the exercise program. Mean age was 50.1 years (SD 12.3), 37.5% were male, and 50% had Crohn's disease (Table 1).

The full program was completed by 84.4% of all patients, and average rating of the program was 8.6 out of 10. After 16 weeks, muscle mass improved compared to baseline (P=0.020) and fat percentage decreased (P=0.003). Physical fitness improved significantly; lower body strength and coordination (P=0.006), flexibility (p=0.002), speed and endurance (P<0.001). Quality of life improved with on average 8 points on the Inflammatory Bowel Disease Questionnaire. The program significantly improved fatigue scores (P=0.013). Disease control showed no significant improvement after the program.

		Overall (n=32)
Mean age, y (SD)		50.1 (12.3)
Male		12 (37.5)
Smoking	Current	3 (9.4)
Physical activity, d (IQR)[a,b]	Light / Moderate / High	4 (2-7) / 2 (1-4) / 1 (0-2)
Diagnosis	Crohn's disease	16 (50)
	Ulcerative Colitis	16 (50)
Median diagnosis age, y (IQR)		34.5 (25-47.3)
Abdominal surgery (Yes)		10 (31.1)
Stoma (Yes)		2 (6.3)

Table 1. Baseline characteristics

Note: data are described as numbers (percentages) unless stated otherwise.

Abbreviations: CD, Crohn's disease; d, days; IQR, interquartile range; n, number; SD, standard deviation; UC, ulcerative colitis; y, year.

[a] Physical activity is based on the patients who filled in the questionnaire (n=25)

[b] Physical activity is categorized in light, moderate and high. Light is walking. Moderate is cycling, exercising at a low intensity level. High exercise is working out or dancing.

Conclusion: This pilot study shows that a specially designed exercise program is feasible and has a positive effect on IBD patients. These findings underline the importance of physical activity in IBD patients and can be used as an initial step towards integrating an exercise program in standard IBD care.

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PP0562

IMPACT OF MIRIKIZUMAB ON CLINICALLY MEANINGFUL IMPROVEMENTS IN FATIGUE AS MEASURED BY FACIT-F IN PATIENTS WITH MODERATELY TO SEVERELY ACTIVE CROHN'S DISEASE

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Introduction: Fatigue is a debilitating multifactorial symptom experienced by many patients with Crohn's disease (CD).[1] Topline efficacy data of mirikizumab (MIRI), an anti-interleukin-23p19 antibody, vs. placebo (PBO) and ustekinumab (UST) in patients with moderately to severely active Crohn's disease in the Phase 3 VIVID-1 study were previously reported.[2-4]

Aims & Methods: This analysis assesses the impact of MIRI vs. PBO in improving fatigue and the correlation of patient variables with fatigue severity at baseline (BL) in the VIVID-1 study. Participants were randomized 6:3:2 to MIRI (900 mg intravenous at Week [W]0, W4, and W8, then 300 mg subcutaneous every 4W from W12-W52), UST or PBO. Fatigue was assessed via the Functional Assessment of Chronic Illness Therapy-Fatigue (FACIT-F) questionnaire (scale 0-52, higher scores indicating less fatigue,[5] meaningful within-patient change identified as a 6–9 point [pt] increase). At W12 we report the proportion of PBO (N=199) and MIRI (N=579) patients achieving FACIT-F ≥6- or ≥9-pt increase from BL. Comparisons with PBO at W52 used composite (C) endpoints: clinical response by Patient-Reported Outcome at W12 (≥30% decrease in stool frequency [SF] or abdominal pain [AP], and neither worse than BL), and FACIT-F increase of either ≥6-pts (C≥6) or ≥9-pts (C≥9) by W52. *p*-Values were calculated via Cochran-Mantel-Haenszel test adjusted by biologic-failed status (yes/no); BL Simple Endoscopic Score for CD (SES-CD) total score (<12, ≥12); either BL SF ≥7 and/or AP ≥2.5 (yes or unknown/no).

The pooled population of patients randomized to PBO or MIRI at BL, excluding those with missing data, with a range of patient variables was analyzed by the random forest method to assess the correlation with fatigue severity. Variables were ranked by importance by the random forest method and Pearson correlation coefficients (*r*) calculated for continuous variables.

Results: At W12, 31.2% (PBO) and 43.0% (MIRI) patients achieved FACIT-F ≥6-pt increase ($p<0.005$) and 22.1% (PBO) and 32.6% (MIRI) patients achieved FACIT-F ≥9-pt increase ($p<0.01$). At W52, 20.1% (PBO) and 34.9% (MIRI) patients met C≥6 ($p<0.0005$) and 15.6% (PBO) and 28.5% (MIRI) patients met C≥9 ($p<0.0005$). A strong association ($r>0.5$) was found between fatigue severity and Inflammatory Bowel Disease Questionnaire (IBDQ), and with the emotional functioning domain of IBDQ and Mental Component Summary (MCS) of Short Form (SF)-36 (Table).

There was a weak association ($r=0.1–0.3$) with AP score, SF score, and Urgency Numeric Rating Scale, and insubstantial association ($r<0.1$) with objective markers of inflammation, including SES-CD, C-reactive protein, and fecal calprotectin.

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Variable	Importance value based on random forest	Pearson correlation coefficient	Association based on Pearson correlation
IBDQ-systemic symptoms score	56.756	0.784	Strong
SF-36 Mental Component Summary	18.816	0.654	Strong
IBDQ-emotional functioning score	12.299	0.686	Strong
SF-36 Physical Component Summary	10.423	0.577	Strong
Inflammatory Bowel Disease-Disability Index score	5.951	-0.567	Strong
IBDQ-social functioning score	3.529	0.602	Strong
IBDQ-bowel symptoms score	1.510	0.538	Strong
Patient Global Rating of Severity score	0.472	-0.307	Moderate
Urgency Numeric Rating Scale	0.342	-0.136	Weak

Table: Correlation of highest ranked BL variables with FACIT-F

Conclusion: MIRI showed a nominally clinically significant improvement in fatigue vs. PBO at W12 and a nominally statistically higher improvement in composite endpoints vs. PBO at W52. At BL, fatigue was not associated with objective inflammatory markers of disease severity or with typical CD symptoms. The observed strong association between fatigue and MCS of SF-36 and the emotional functioning domain of IBDQ implies bidirectional effects of brain-gut interactions that are not well understood.

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PP0563

PSYCHOLOGICAL INTERVENTIONS IMPROVE MENTAL HEALTH-RELATED OUTCOMES IN PATIENTS WITH INFLAMMATORY DIGESTIVE DISEASES: A SYSTEMATIC REVIEW AND META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: Inflammatory digestive diseases can have a major impact on the physical and mental wellbeing of affected patients. Psychosocial factors can alter the behaviour linked to disease and influence treatment outcomes.

Aims & Methods: The aim of this systematic review and meta-analysis was to assess the efficacy and safety of psychological interventions on depression, anxiety, and stress in patients with inflammatory digestive diseases. The study protocol was prospectively registered on PROSPERO (CRD42021282965). The systematic search was conducted on April 1st, 2023 in three medical databases (PubMed, CENTRAL, Embase). Randomized controlled trials (RCTs) that investigated the effects of various psychological interventions on mental and physical health in patients with all types of inflammatory digestive diseases were included. Outcomes were assessed at the end of therapy and at various timepoints during the follow-up period (short-, mid- and long-term follow-up). Owing to the differences in scales and questionnaires, the standardised mean difference (SMD) was used to pool effect sizes, with its corresponding 95% confidence interval (CI). Higgins and Thompson's I^2 statistics were used to assess heterogeneity.

Results: We included 24 RCTs that investigated the effect of psychological interventions on depressive symptoms, totalling 1572 participants. The pooled analysis found that psychological interventions resulted in a moderate reduction of depression compared to controls (SMD -0.46; 95% CI: -0.75; -0.18; I^2 87%). Data from 19 RCTs with 1593 patients also demonstrated a moderate reduction in anxiety symptoms in subjects undergoing psychological interventions (SMD -0.58; 95% CI: -0.94; -0.22; I^2 90%). The pooled analysis of 13 RCTs totalling 772 participants found that psychological therapy was associated with a moderate decrease in stress (SMD -0.53; 95% CI: -1.04; -0.03; I^2 85%). Additional analysis of follow-up data showed that benefits resulting from treatment gradually decrease, showing no significant difference at the end of the long-term follow-up period.

Conclusion: Psychological interventions effectively improve mental health-related outcomes in patients with inflammatory digestive diseases. Since benefits from therapy seem to decrease during the follow-up period, repeated interventions may be necessary.

Disclosure: Nothing to disclose.

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PP0564

HOSPITALIZATION AND EMERGENCY DEPARTMENT VISITS AMONG INDIVIDUALS WITH INFLAMMATORY BOWEL DISEASE TREATED WITH INFLIXIMAB: PRELIMINARY DATA FROM CAN-AIM

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Introduction: Infliximab is used to treat inflammatory bowel disease (IBD) but may be associated with adverse events such as infections. We describe hospitalization and emergency department (ED) visits among IBD patients using infliximab bio-originator (INF-O) or biosimilar (INF-B).

Aims & Methods: We used data from Canadian IBD Research Consortium (CIRC) cohorts contributing to CAN-AIM's biosimilar registry, a pan-Canadian clinical registry of biosimilar/bio-originator users. We included people diagnosed with IBD who initiated infliximab between 2019-2023. Our outcome was self-reported hospitalization or ED visits, any time from INF initiation until outcome, treatment discontinuation or last information available. Categorical variables are presented as absolute numbers (percentage) and continuous as median (interquartile range, IQR). Kaplan-Meier curves with log-rank test and Cox regression model were used for time to event analyses for first hospitalization or ED - adjusted for sex and age at INF initiation.

Results: Of 158 individuals (78% INF-B), 53% had Crohn's and 47% ulcerative colitis. Individuals were 54% male, 85% Caucasian, age at INF initiation at 37 years; 28% were using prednisone at enrolment. 16% were hospitalized at least once (first hospitalization within a median follow-up of 17 months (IQR 7-25) and 25% had at least one ED visit (first ED occurred at a median of 13 months, IQR 5-26). A total of 45 hospitalizations and 61 EDs were recorded. Most ED visits were due to IBD flare (n=33), infections (n=9), and other reasons (n=15). Most hospitalizations were for IBD-related surgery (n=19), IBD flare (n=16), and infections (n=7). Time from treatment initiation to first hospitalization or first ED was similar for INF-B and INF-O. Compared to INF-O, aHR for first hospitalization was 0.64 (95%CI 0.27-1.50) and ED was 0.72 (0.36-1.43) for INF-B.

Conclusion: In this real-world dataset, almost one-third of infliximab users for IBD had hospitalization or ED visits, often due to IBD flare, IBD-related surgery, and infections. We noted no clear differences between biosimilars and bio-originators, though limited sample size precluded precision.

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PP0565

DISEASE CHARACTERISTICS ASSOCIATED WITH BOWEL URGENCY IN CROHN'S DISEASE; RESULTS FROM THE PHASE 3 VIVID-1 TRIAL

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Introduction: Data on bowel urgency (BU) in Crohn's disease (CD) are limited, and associated factors are not well identified. Here we evaluate the association of BU with baseline (BL) disease characteristics in patients with moderately to severely active CD in the randomized, double-blind, double-dummy, treat-through design Phase 3 VIVID-1 trial (NCT03926130).

Aims & Methods: BU was assessed using the average Urgency Numeric Rating Scale (UNRS) (range 0-10) from 7 days prior to the visit. Adult patients as previously described[1], that had an UNRS ≥3 and no missing values at BL were included in this analysis.

The association of UNRS with Crohn's Disease Activity Index (CDAI), abdominal pain (AP), stool frequency (SF), Simple Endoscopic Score for CD (SES-CD) total score, disease location, count of inflamed intestinal segments, rectal involvement, presence of narrowing, presence of perianal fistulae, number of bowel resections and inflammatory biomarkers were analyzed.

Other factors explored included sex, age, age at diagnosis, duration of CD, BL corticosteroid and immunomodulator use, and prior biologic failure. Random forest method (RFM) was applied to rank variables by importance associated with BU score at BL. Pearson correlation coefficients (r-value) were calculated for continuous baseline variables.

Results: At BL, 73.4% (782/1065) patients randomized to placebo (n=141), MIRI (n=433), or ustekinumab (n=208) reported a UNRS ≥3 with a median UNRS of 7.0 Among BL disease characteristics, CDAI total score and AP were the top two important factors based on RFM (CDAI=0.701; AP=0.398) and had moderate positive correlations (CDAI=0.479; AP=0.384) with UNRS (Table). Notably, disease characteristics such as SF (RFM=0.236; r=0.286), SES-CD total score (RFM=0.056; r=0.118), age at diagnosis (RFM=0.037; r=-0.018), age (RFM=0.033; r=-0.022), faecal calprotectin (RFM=0.029; r=0.031), C-reactive protein (RFM=0.028; r=0.050), count of inflamed segments (RFM=0.021; r=0.069), duration of CD (RFM=0.015; r=-0.007), presence of perianal fistula (RFM=0.009; r=0.090), and number of bowel resections (RFM=0.000; r=-0.036) all had weak correlations with

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UNRS. BL corticosteroid (RFM=0.015) and immunomodulator (RFM=0.007) use, disease location (RFM=-0.005), rectal involvement (RFM=0.011), presence of narrowing (RFM=0.007), prior biologic failure (RFM=0.003), and sex (RFM=0.007) had weak correlations with UNRS implied by RFM.

Patient baseline variables	Importance value based on random forest	Pearson correlation coefficient (r)
CDAI Total Score	0.701	0.479
Abdominal pain score	0.398	0.384
Stool frequency score	0.236	0.286
SES-CD total score	0.056	0.118
Age at diagnosis	0.037	-0.018
Age	0.033	-0.022
Faecal calprotectin	0.029	0.031
C-reactive protein	0.028	0.050

Correlation strength defined as: r>0.5=strong, 0.5≥r≥0.3 =moderate, 0.3>r≥0.1 =weak, r<0.1=insubstantial[2]
RFM is used to rank the relative importance of the patient baseline variables to UNRS. The higher values mean more important with a stronger association with UNRS.
Abbreviations: CDAI=Crohn's Disease Activity Index; CI=confidence interval; RFM = random forest method; SES-CD=Simple Endoscopic Score for Crohn's Disease; UNRS = Urgency Numeric Rating Scale

Conclusion: Most patients reported BU ≥3 at BL. The only moderate positive correlations were observed between BU, clinical disease severity (CDAI), and AP. Even though the association of BU with other disease characteristics such as extent and severity of endoscopic inflammation, rectal involvement, presence of active perianal disease, disease location, previous resection, and duration of CD were weak, they potentially interact and contribute together resulting in BU.

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PP0566

EXIT INTERVIEWS EXPLORING PATIENTS' EXPERIENCE OF CHANGES IN THEIR FATIGUE DURING THE PHASE 3 CLINICAL TRIAL OF MIRIKIZUMAB FOR TREATMENT OF MODERATELY TO SEVERELY ACTIVE CROHN'S DISEASE

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Introduction: Fatigue is a common and debilitating symptom of Crohn's disease (CD), associated with functional impairment and reduced quality of life.[1,2]

Aims & Methods: Here we qualitatively explore change in fatigue in participants with moderately to severely active CD who completed the phase 3, randomised, double-blind, placebo- and active-controlled, mirikizumab clinical trial (VIVID-1/NCT03926130). Qualitative, web-assisted, semi-structured, 90-minute exit interviews were conducted in 62 trial participants within 3 weeks of completing the final trial visit (May 2022–July 2023). Clinical trial arms were blinded for this study. A sub-set of participants from either mirikizumab, ustekinumab or placebo arms were asked open-ended questions to explore changes in fatigue (N=31). Interview audio recordings were transcribed verbatim and analysed using a directed content analysis approach in ATLAS.ti v9.

Results: Among the 31 participants who were asked about fatigue (61.3% male [n=19], 90.3% White [n=28]), median age was 42 (n=30; range: 24–75) years and median CD duration was 5.9 (n=30; range: 1.4–54.3) years. Overall, 31.0% (n=9/29) participants had a college degree or higher and 41.4% (n=12/29) participants were employed full-time. Participants were from the United States (n=15), Czech Republic (n=5), Germany (n=5), Poland (n=3), Australia (n=2) and Canada (n=1).

At the start of the trial, participants reported that they had difficulty in performing daily activities (n=7), walking or climbing stairs (n=3), completing outdoor tasks and activities (n=2), concentrating or focusing (n=2) and were unable to feel well-rested from sleep (n=3), lacked energy (n=2), felt exhausted (n=1), dozed off in evening (n=1) and took breaks during the day (n=1) due to fatigue. They also reported socialising less (n=1) and feeling worried (n=3), demotivated (n=2), grumpy (n=1), frustrated (n=1) and depressed (n=1).

Although it was unknown which treatment the participants had received, most participants reported experiencing an improvement in fatigue (74%, n=23/31) during the trial. Seven participants reported experiencing no change in fatigue (23%, n=7/31). Of these, four participants reported experiencing no or little fatigue at the start of the trial, which did not change until their last visit. One participant reported worsening of fatigue (3%, n=1/31) during the trial which impacted their daily activities as they needed to sleep more often during the day.

Among participants who experienced an improvement in fatigue, 11 reported an improved emotional well-being such as feeling more relaxed, confident, independent, and motivated. Overall, 9 participants reported that their physical function improved, and they were able to walk further, had more energy and felt stronger. Eight participants reported being able to function more day-to-day. Improvement in fatigue resulted in better friendships or relationships (n=4), being more successful in work or school (n=3), sleeping better (n=2), napping less (n=2), being able to exercise more (n=2), go out for longer duration (n=2), being able to work (n=1), and gaining more enjoyment in life (n=1). When participants were asked about meaningful improvement in fatigue, 10 out of 13 participants reported a meaningful improvement.

Conclusion: Exit interview results showed that irrespective of the treatment received, most interviewed participants with moderately to severely active CD experienced an improvement in fatigue by end of the trial, resulting in better energy levels, physical function, daily functioning, and emotional well-being.

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PP0567

THE IMPACT OF BODY MASS INDEX ON TREATMENT RESPONSE IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: RESULTS OF THE PROSPECTIVE DUTCH INITIATIVE ON CROHN AND COLITIS (ICC) REGISTRY

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Introduction: To examine the effect of Body Mass Index (BMI) on treatment response in patients with inflammatory bowel disease (IBD).

Aims & Methods: Patients ≥ 16 years old with IBD, a documented baseline BMI and starting thiopurines, intravenous (iv) vedolizumab, subcutaneous (sc) vedolizumab, ustekinumab, ozanimod, filgotinib, or tofacitinib were selected from the Initiative on Crohn and Colitis (ICC) registry in the Netherlands. Patients with a current follow-up shorter than 24 weeks were excluded. Some of the patients in the registry started vedolizumab sc electively whilst in remission after being treated with vedolizumab iv. Therefore, patients with vedolizumab sc were excluded if they were in steroid-free clinical remission (SFCR) (i.e. Simple Clinical Colitis Activity Index (SCCAI) ≤ 2 for ulcerative colitis and Harvey Bradshaw Index (HBI) < 5 for Crohn's disease) at baseline. Patients who discontinued treatment were classified as non-responders at all timepoints after treatment discontinuation. SFCR at baseline, week 12 and week 24 was compared between patients with normal weight (BMI < 25 kg/m^2), overweight (BMI 25-30 kg/m^2) and obesity (BMI ≥ 30 kg/m^2) using Chi-Square tests. SFCR rates over time in the three groups were analysed using generalized linear mixed modelling.

Results: 1101 patients with IBD (631 Crohn's disease, 470 ulcerative colitis) were included, 636 had normal weight, 313 were overweight and 152 were obese. Generalized linear mixed model demonstrated that the course of SFCR over time after initiating treatment was significantly different between the three groups (p = 0.038). Chi-Square tests showed significant differences in SFCR between the normal weight, overweight and obese patients at week 12 (40.1%, 49.8%, 42.9%, respectively, p = 0.027) and week 24 (48.2%, 49.4%, 34.7%, respectively, p = 0.015). No significant difference in SFCR was found between groups at baseline.

Conclusion: Obesity was associated with a lower SFCR rate at week 24. Surprisingly, at week 12, overweight was associated with a higher rate of SFCR. Although overweight patients have shown a faster treatment response, this advantage had vanished at 24 weeks. The week 12 results could possibly be explained by the protective effect of excess bodyweight during illness, which is an hypothesis of the 'obesity paradox', concerning the finding of lower mortality rates in overweight and obese patients with cardiovascular diseases and malignancies[1,2].

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PP0568

PATIENT-REPORTED OUTCOMES WITH USTEKINUMAB FOR CROHN'S DISEASE IN REAL-LIFE CLINICAL PRACTICE IN GREECE: FINAL RESULTS FROM THE PROSPECTIVE 'REASSURE' STUDY

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Introduction: Due to lack of published data on ustekinumab (UST) use in Greece, REASSURE study aimed to generate real-world evidence (RWE) on treatment persistence, and patient-reported outcomes (PROs) in patients with Crohn's Disease (CD).

Aims & Methods: REASSURE was a non-interventional, 24-month prospective study of adult (18-80 years old) CD patients newly initiated on UST as 1st, 2nd, or 3rd biologic therapy per the approved label in Greece. Data were collected by routine assessments, including Harvey-Bradshaw Index (HBI), and PROs at baseline (start of UST) and at 4 follow-up visits at 6-month intervals, with a final visit at Month 24 (±2 months) or at treatment discontinuation (whichever occurred first). PROs included the validated Greek version of the Inflammatory Bowel Disease Questionnaire (IBDQ), the Functional Assessment of Chronic Illness Therapy-Fatigue (FACIT-F) Scale and the Work Productivity and Activity Impairment Questionnaire in Crohn's Disease (WPAI:CD). PROs at 24 months are presented herein.

	Overall (N=169)	Bio-naïve (1L) (N=60)	Bio-experienced (N=109)
Time from CD diagnosis to UST initiation, median (IQR), years	7.0 (1.8, 13.0)	1.3 (0.3, 6.2)	10.1 (5.9, 15.9)
Baseline characteristics, % (n/N)			
Age ≤50 years	62.7 (106/169)	51.7 (31/60)	68.8 (75/109)
Females	50.9 (86/169)	41.7 (25/60)	56.0 (61/109)
Employed (full-time or part-time)	43.4 (72/166)	55.9 (33/59)	36.4 (39/107)
At least 1 comorbidity	55.0 (93/169)	55.0 (33/60)	55.0 (60/109)
Stricturing and/or penetrating disease	43.2 (70/162)	33.3 (19/57)	48.6 (51/105)
Extraintestinal manifestations	36.5 (61/167)	31.7 (19/60)	39.3 (42/107)
Starting UST maintenance dosage: 90 mg SC Q8W	89.9 (152/169)	83.3 (50/60)	93.6 (102/109)
IBDQ at month 24			
Response (change ≥16 points), % (n/N), as observed	56.8 (63/111)	60.5 (26/43)	54.4 (37/68)
Response (change ≥16 points), % (n/N), imputed	44.2 (73/165)	49.2 (29/59)	41.5 (44/106)
Median/mean change from baseline (IQR/SD), as observed	24.0 (5.0, 52.0)[w]	33.0 (39.6)[T]	25.7 (35.6)[T]
Median change from baseline (IQR/SD), imputed	9.0 (0.0, 36.0)[w]	14.0 (-1.0, 47.0)[w]	8.0 (0.0, 36.0)[w]
FACIT-F at month 24			
Significant improvement (≥4 points), % (n/N), as observed	59.3 (67/113)	62.8 (27/43)	57.1 (40/70)
Significant improvement (≥4 points), % (n/N), imputed	47.9 (80/167)	53.3 (32/60)	44.9 (48/107)
Median/mean change from baseline (IQR/SD), as observed	6.0 (0.0, 15.0)[w]	9.1 (14.3)[T]	7.2 (12.1)[T]
Median/mean change from baseline (IQR/SD), imputed	2.0 (0.0, 12.0)[w]	4.0 (0.0, 14.5)[w]	1.0 (0.0, 10.0)[w]
Activity impairment at month 24 (based on WPAI:CD)			
Median change from baseline (IQR), as observed	-20.0 (-40.0, 0.0)[w]	-20.0 (-40.0, 0.0)[w]	-20.0 (-40.0, 0.0)[w]
Patients completing WPAI:CD at both baseline and 24 months	N=113	N=43	N=70
Median change from baseline (IQR), imputed	-10.0 (-40.0, 0.0)[w]	-10.0 (-30.0, 0.0)[w]	-5.0 (-40.0, 0.0)[w]
Patients completing WPAI:CD at baseline	N=166	N=60	N=106
Work productivity loss at month 24 (WPAI:CD)			
CMI (decrease of ≥7 points), % (n/N), as observed	68.4 (26/38)	77.8 (14/18)	60.0 (12/20)
CMI (decrease of ≥7 points), % (n/N), imputed	53.8 (28/52)	63.6 (14/22)	46.7 (14/30)
Median change from baseline (IQR), as observed	-20.0 (-56.8, 0.0)[w]	-20.0 (-60.0, -10.0)[w]	-10.0 (-35.8, 0.0)[w]
Median change from baseline (IQR), imputed	-10.0 (-30.0, 0.0)[w]	-20.0 (-56.9, 0.0)[w]	0.0 (-21.6, 0.0)[w]

Data imputation was performed using last observation carried forward analysis. For change from baseline, median (IQR) or mean (SD) is displayed, depending on p-value of normality (< or ≥0.05, respectively). Abbreviations: 1/2/3L: First/Second/Third Line; CD: Crohn's Disease; FACIT-F: Functional Assessment of Chronic Illness Therapy-Fatigue; IBDQ: Inflammatory Bowel Disease Questionnaire; IQR: Interquartile Range; IV: intravenous; N: number of patients with available data; n, number of patients with variable; ns, not statistically significant; PRO, Patient Reported Outcome; Q8W: every 8 weeks; SC: subcutaneous; SD: Standard Deviation; T: T-test p-value of change <0.01, indicating statistical significance; UST: ustekinumab; W: Wilcoxon signed-rank test p-value of change <0.01, indicating statistical significance; WPAI:CD: Work Productivity and Activity Impairment Questionnaire in Crohn's Disease, CMI: Clinically meaningful improvement.

Table 1: Baseline characteristics and 24-month PROs.

Results: Between Apr/2019 and Feb/2021, 169 eligible patients were enrolled in 19 Gastroenterology Clinics in Greece. The median (IQR) age at baseline was 45.0 (35.0-54.0) years; of the evaluable patients, 17.5% were overweight, 49.0% had tertiary level of education, and 68.5% were ever-smokers. In terms of employment, 43.4% (72/166) of patients were employed, 29.5% (49/166) unemployed (due to CD in 21.6%; 8/37), 16.3% (27/166) retired (due to CD in 24.0%; 6/25), and 10.8% (18/166) reported other status. Among UST treated patients, 35.5% were bio-naïve (BN) and 64.5% were bio-experienced (BE).

Median (IQR) duration of study participation was 23.9 (18.3-24.7) months (median treatment duration: 24.0 months). At 24 months, 72.8% (123/169) [BN: 76.7% (46/60); BE: 70.6% (77/109)] remained on treatment with UST, while 21.9% (37/169) had discontinued UST after a mean (SD) of 10.2 (6.4) months mainly due to lack/loss of response (15/37), safety issues (10/37), or loss to follow-up (6/37). The proportion of evaluable patients in remission (HBI score 0-4) at 24 months was 87.7% (100/114) [BN: 90.7% (39/43); BE: 85.9% (61/71)].

Baseline median IBDQ (170.0), FACIT-F (34.0), WPAI:CD work productivity loss (WPL) (30.0) and activity impairment scores (30.0) significantly improved (P<0.01) at 24 months, with median changes of 24.0, 6.0, -20.0, and -20.0, respectively, among paired observations. Significant improvements were also noted based on imputed data and in both bio-naive and bio-experienced patients. At 24 months, among patients with paired ob-

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servations, 56.8% achieved an IBDQ response (≥16-point increase), 59.3% a significant FACIT-F improvement (≥4-point increase) and 68.4% a clinically meaningful improvement in WPL (≥7-point decrease) from baseline. Further details on baseline characteristics and 24-month PROs are shown in Table 1.

Conclusion: These data provide RWE on the beneficial effect of UST on health-related quality of life, fatigue, and work productivity in both bio-naïve and bio-experienced CD patients in Greece.

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PP0569

EXPLORING THE CHARACTERISTICS OF CROHN'S DISEASE (CD) PATIENTS WHO HAVE BEEN TREATED WITH ADVANCED THERAPY FOR MORE THAN 6 MONTHS

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Introduction: The Crohn's Disease Activity Index (CDAI) is a widely used research tool used to quantify CD patient' symptoms, evaluate the severity of CD and to determine treatment outcomes.

Aims & Methods: The objective of this study was to assess any nuances in treating CD patients with active disease (CDAI score >150), who have remained on the same advanced therapy (biologics/ novel oral molecules) for more than 6 months.

The Ipsos CD Therapy Monitor, a multi-centre online medical chart review study of patients with CD was conducted between July – September 2023 among UK, FR, DE, IT & ES gastroenterologists practicing across hospital and private practices.

Physicians were screened for practice duration and patient volume. Charts of patients prescribed with advanced therapy were included in the analysis.

Results: 208 sampled physicians collectively reported 373 CD patients who were recorded as being on the same advanced therapy for more than 6 months. 286 reported patients were deemed in remission (CDAI score ≤ 150) and 87 patients recorded with active disease (CDAI score >150). Looking at reported CD patient co-morbidities, metabolic conditions were more common in patients with active disease vs those in remission (24% vs 19%)

When looking at extra-intestinal manifestations (EIMs) between the two patient groups, patients with active disease were more likely to experience EIMs vs those in remission.

CD patients	Patients with active disease (CDAI >150)	Patients in remission (CDAI ≤ 150)
% experienced anaemia	21%	11%
% experienced arthritis	9%	7%
% experienced osteoporosis	6%	2%

Table 1: EIMs in reported CD patients (% patients)

A higher proportion of patients with active disease activity were currently receiving non-tumour necrosis factor (TNF) therapy vs those in remission (37% vs 24%) and reported higher steroid usage (% reported patients currently on steroids – 11% vs 3%).

When analysing reasons for prescribing selected advanced therapy between the two patient groups, a greater proportion of patients with active disease had the following reasons specified more frequently by the sampled physicians, vs the patients in remission: 'To achieve steroid-free remission' (51% vs 43%), 'Good safety profile' (40% vs 35%), and 'Low incidence of side-effects' (35% vs 21%).

Conclusion: Comparisons in this study cohort highlighted that active disease in CD patients, as determined by the CDAI score, is associated with a higher number of patient co-morbidities and a greater likelihood of extra-intestinal manifestations; this indicates a more severe or complicated disease burden.

As additional development products with new mechanisms of action become available in gastroenterology, non-TNF therapy usage may grow over time to provide more holistic and individualised treatment strategies for CD patients. Further investigation using comparator cohort is warranted.

References: [1] Use of the Crohn's disease activity index in clinical trials of biological agents, Hugh James Freeman https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2725371/

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PP0570

ROAD TO EXIT STRATEGY IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: READY OR NOT?

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Introduction: Whereas there is an emerging consensus on the optimal approach to initiation of a range of therapies in inflammatory bowel disease (IBD), there remains greater uncertainty about the risks, benefits, and timing of stopping treatment when patients are in stable remission on therapy. When planning an exit strategy for drug withdrawal, the risk of disease relapse must be balanced against the risk of drug-related adverse events and healthcare costs.

Aims & Methods: Our study evaluated the risks of disease relapse associated with withdrawal of the biologic therapy in both ulcerative colitis (UC) and Crohn's disease (CD). In this is a retrospective observational study, the exit strategy was proposed to 60 patients regularly followed in two IBD referral centre from 2015 to 2023. The exit startegy was performed as: (i) discontinuation without de-escalation, (ii) progressive de-escalation and (iii) azathioprine as monotherapy. Data on medical history were collected from electronic health records. Chi-square test was used to compare categorical variables. Logistic regression was used to assess the relationship between disease-related characteristics and the onset of disease relapse after the withdrawal.

Results: 60 IBD patients were enrolled, 46 patients withdrawal their biologic therapy and 14 patients were in a de-escalation stategy. These 46 patients (15 UC and 31 CD) performed discontinuation without de-escalation (65%), progressive de-escalation (26%) and azathioprine as monotherapy (9%). They withdraw anti-TNF (82%) or other biologics (18%). Most of patients (71%) were on first line of biologic therapy. The median follow-up period after drug withdrawal was 29 months. A disease relapse occurred in 14 patiens with a median period of 12 months. A good recapture of previous therapy was observed in 57% of patients. Logistic regression analysis revealed a trend of significance between biologic therapy duration (p 0.05) and absence of flare-up after exit strategy. No correlation was noticed with duration of endoscopic or clinical remission before starting the drug withdrawal.

Conclusion: There are no specific factors related to a successful exit strategy. Biologic drug discontinuation is something best considered on a case-by-case basis and determined by patient preferences and disease features. We need more specific study in order to identify patients who can stop therapy without undue risk of poor outcomes.

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PP0571

THE IMPACT OF DISEASE STATUS AFTER BIOLOGIC INDUCTION THERAPY ON LONG-TERM OUTCOMES IN PERSONS WITH CROHN'S DISEASE

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Introduction: Despite the increase in the therapeutic armamentarium of the treatment of Crohn's disease (CD), historically less than 50% of patients achieve partial response after induction, and less than 20% achieve clinical remission. It remains unknown how response to biologic induction therapy influences disease behavior, treatment patterns, and healthcare utilization in those patients who are maintained on biologic therapy for the treatment of their CD.

Aims & Methods: Aims: To determine the impact of achieving clinical remission after induction therapy for CD on post-induction outcomes.

Methods: A retrospective chart review of all persons with CD, age 18 or above, followed at Mount Sinai Hospital, Canada who were dispensed an approved biologic therapy for IBD between April 1st, 1996, and June 30th, 2021, and completed a complete course of induction was conducted. We stratified patients into those were in clinical remission as per the physician global assessment at completion of biologic induction ("remitters") and those who were not ("non-remitters"). The start of follow-up was the completion date of biologic induction and patients were followed until end of last follow-up date. We extracted data on IBD symptoms, decisions on biologic therapy (such as discontinuation), healthcare utilization such as hospitalizations and emergency department (ED) visits, need for CD surgery, and endoscopic assessments. Endoscopic remission was defined based on a SES-CD score of £ 4 when available, or on the endoscopists' impression of absence of mucosal inflammation.

Results: A total of 160 patients were eligible for data extraction; 60.0% were male, 35.6% and 25.0% had stricturing and penetrating complications respectively, 28.9% had perianal disease and 34.3% had a history of a CD surgery. The index biologic included Infliximab in 24.4% patients, adalimumab in 32.5% patients, vedolizumab in 25.0% patients, and ustekinumab in 11.9% patients. Fifty-five (34.4%) patients were remitters and 105 (65.6%) were non-remitters after completion of biologic induction. Those in clinical remission at induction were less likely to have upper gastrointestinal CD (3.6% vs. 10.5%), have penetrating complications (18.2% vs. 28.6%), and perianal disease (16.4% vs. 35.2%) at time of biologic start.

Remitters were more likely to be in sustained clinical remission throughout follow-up compared to non-remitters (47.2% vs. 27.6%, p=0.01), be in clinical remission at last follow-up date (87.2% vs. 69.5%, p=0.04), and appeared to have a lower likelihood of discontinuing the index biologic (7.2% vs. 16.1%, p=0.06).

Similarly, remitters were less likely to undergo a CD-related intestinal resection during follow-up (3.6% vs. 13.3%, p=0.05).

There appeared to be no significant differences between remitters and non-remitters in proportion of patients being hospitalized for IBD (10.9% vs. 19.0%, p=0.14) or non-IBD (5.4% vs. 7.6%, p=0.62) reasons and visiting the emergency department (7.2% vs. 13.3% p=0.26). Of the 160 patients, 117 (73.1%) underwent a colonoscopy or sigmoidoscopy during follow-up. There were no differences in the rates of ulcer persistence (51.6% vs. 53.5%, p=0.63) or achievement of endoscopic remission (58.9% vs. 48.7%, p=0.76) between remitters and non-remitters at induction.

Conclusion: Achieving clinical remission after biologic induction is associated with sustained clinical remission in follow-up but not endoscopic remission.

Disclosure: Nothing to disclose.

PP0572

IMPACT OF NON-REMISSION IN PATIENTS WITH CROHN'S DISEASE AND ULCERATIVE COLITIS

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Introduction: Current treatments for Crohn's disease (CD) and ulcerative colitis (UC) are associated with loss of treatment response. However, studies assessing the real-world burden of non-response in Canada are scarce and the clinical and humanistic burden remains unknown. We aimed to describe the burden of active disease on symptom prevalence and work impairment by describing patients with CD and UC according to remission status.

Aims & Methods: Data were drawn from the Adelphi Real World Inflammatory Bowel Disease Specific Programme™, a real-world, cross-sectional survey of gastroenterologists (GEs) and their consulting patients with CD and UC in Canada from June 2023 to January 2024. Data included physician-reported patient demographics and clinical characteristics. Descriptive analyses were conducted.

Results: 19 physicians reported data for 144 patients (73 with CD and 71 with UC). Physicians reported 20% of patients not to be in clinical or endoscopic remission (non-remitters), 31% to be in clinical remission only and 49% in both clinical and endoscopic remission. Mean (standard deviation) age was 44.5 (20.5) years among non-remitting patients, 33.5 (13.3) years among those in clinical remission only and 40.4 (17.4) years among patients in those with clinical and endoscopic remission. Among non-remitting patients, 38% were an ex-smoker, versus 14% of clinical remission and 18% of clinical and endoscopic remission patients. The most reported comorbid conditions included anxiety (24% non-remitters, 14% clinical remitters, 18% clinical and endoscopic remitters), depression (14% non-remitters, 0% in clinical remitters, and 14% in clinical and endoscopic remitters) and hypertension (14% non-remitters, 9% clinical remitters, and 14% clinical and endoscopic remitters). Top three current symptoms included fatigue (52% non-remitters, 32% clinical remitters, 15% clinical and endoscopic remitters), abdominal pain (69% non-remitters, 14% clinical remitters, 7% clinical and endoscopic remitters), and non-bloody diarrhea (48% non-remitters, 7% clinical remitters, 6% clinical and endoscopic remitters). Regarding employment, 38% of non-remitters were working full-time versus 48% of clinical remitters, and 58% of clinical and endoscopic remitters.

Conclusion: Patients with CD/UC that had not achieved or maintained remission experienced more symptom burden, comorbid conditions, and were often not in full-time employment than those in disease remission. Notably, non-remitters experienced a high symptom burden, including fatigue, abdominal pain, and diarrhea, which impacted daily living. Patients with both clinical and endoscopic remission experienced fewer symptoms and full-time employment. This highlights the importance of, and significant unmet need for, effective treatment in IBD to manage disease symptoms and decrease inflammation.

Disclosure: Janssen Global Services funded the research and I am an employee of Janssen

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PP0573

5-AMINOSALICYLATES FOR PATIENTS WITH ACTIVE OR QUIESCENT CROHN'S DISEASE: AN UMBRELLA REVIEW OF SYSTEMATIC REVIEWS

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Introduction: 5-aminosalicylates (5-ASAs) are commonly used in patients with Crohn's disease (CD), especially in mild-to-moderate disease. Over half of patients receive 5-ASA during their disease course, with up to 30% continuing long-term therapy for either active (induction of remission) or quiescent (maintenance of remission) disease. However, the evidence regarding their efficacy is mixed, with conflicting findings in systematic reviews (SRs). In preparation for the results of the STATIC trial (NCT 03261206)[1], we conducted an umbrella review to consolidate existing knowledge by identifying, evaluating, and synthesizing findings from published SRs on the use of 5-ASA in patients with active or quiescent CD.

Aims & Methods: Following Cochrane guidelines for "Overviews of Reviews" and adhering to the Preferred Reporting Items for Overviews of Reviews (PRIOR), we systematically searched major databases for relevant SRs published in English until February 28, 2024, summarizing data on 5-ASA use in induction, maintenance, or withdrawal trials of CD. We excluded SRs focusing solely on post-surgical patients or studying the chemoprevention effects of 5-ASA. Two independent reviewers screened articles, extracted data, and assessed quality using AMSTAR 2.

Results: Of 795 screened references, 8 SRs met our inclusion criteria, with the number of included RCTs ranging from 2 to 22 (Table). Most SRs included RCTs of patients with mild-to-moderate ileo-colonic CD. Three SRs focused on induction of remission, two on maintenance of medically induced remission, and three included both, while none investigated withdrawal of therapy. Two SRs were network meta-analyses (NMA) comparing 5-ASAs with other active treatments or placebo; four were Cochrane systematic reviews comparing 5-ASAs with placebo, corticosteroids, or thiopurines (AZA or 6-MP). One SR, published in 2017,used the outdated Jadad scale to assess the risk of bias of RCTs; only Cochrane reviews assessed the level of certainty of evidence. The latest NMA in 2017, including 22 RCTs for induction of remission, suggested that high-dose mesalamine was more effective than placebo, ranking higher than low-dose budesonide and sulfasalazine, though lower than systemic corticosteroid and high-dose budesonide. Conversely, the latest Cochrane review in 2016, including 12 RCTs for maintenance of remission, found no evidence of benefit for oral 5-ASAs over placebo for maintaining medically induced remission. No SRs performed subgroup analysis for duration of remission upon enrollment. Only the NMA conducted a sensitivity analysis to evaluate baseline disease activity, indicating that high-dose mesalamine was superior to placebo when restricted to RCTs with an enrollment CDAI between 150 to 450.

Conclusion: The results of this umbrella review of SRs suggest that high-dose mesalamine may be considered for inducing remission in mild-to-moderate CD. However, evidence does not support the role of 5-ASA for maintaining medically induced remission. Updating SRs with new evidence is imperative to provide guidance on patient subsets who may benefit from 5-ASA use and these results will be updated with the STATIC trial [1].

References: 1. NCT03261206. Stopping Aminosalicylate Therapy in Inactive Crohn's Disease (STATIC) https://classic.clinicaltrials.gov/ct2/show/NCT03261206

Disclosure: Nothing to disclose.

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Review Type (NMA or SR+MA or SR only)	Author, year	Search date (up to)	Searched database	Number of included studies/ study designs	Study Eligibility	Adults or pediatric	Induction trials, maintenance trials, or both	Intervention (5ASA) / dose/ treatment duration/follow-up duration (range)	Comparisons (Control regimens, dose)/ duration	Key conclusions related to 5-ASA
NMA	Coward, 2017	Nov 2015	Cochrane reviews, Medline, EMBASE, CENTRAL	22 RCTs (mesalamine =9, sulfasalazine=4)	RCTs comparing budesonide, sulfasalazine, mesalamine, or systemic corticosteroids with placebo in CD	Adults	Induction	Sulfasalazine: 1 g/15 kg /d to 3 g /d; 6 - 17 wk Mesalamine: 1- 4 g /d; 8-16 wk (above and below 2.4 g/d as high vs low dose)	Placebo, systemic corticosteroids, budesonide	Treatment ranking for induction from best to worst: systemic corticosteroid> high dose budesonide > high dose mesalamine > (low dose budesonide = sulfasalazine)> low dose mesalamine > placebo. Mesalamine at doses above 2.4 g/day was more effective than placebo for the induction of remission in CD, this finding was shown in NMA but not the traditional MA. High dose mesalamine may be an option among patients preferring to avoid steroids. Sulfasalazine was not significantly superior to any therapy including placebo.
SR + MA	Chen, 2017	May 2014	Pubmed/Medline, EMBASE, ISI Web of Knowledge, CINAHL, COCHRANE Database of Systematic Reviews, CBM, CNKI, ChiCTR	Total 8 RCTs, but 6 for CD	RCTs of AZA or 6-MP vs 5-ASA	Adults	Both	AZA: 2-3 mg/kg/d; 1-3 yr 6-MP: 50 mg/d; 2 yr	Mesalamine: 3-4 g/d ; 1-3 yr	Thiopurines (AZA/6-MP) are more effective that 5-ASA for the treatment of CD (clinical relapse rate and remission rate).
SR+MA (Cochrane review)	Akobeng, 2016	Jun 2016	MEDLINE, EMBASE, CENTRAL, Cochrane IBD group specialized register	12 RCTs	5-ASA agents vs either placebo or sulfasalazine in quiescent CD	Both	Maintenance	5-ASA: 1.6 g to 4 g/d; 12-24 mo dose 50 mg/kg/d />6 mo	Placebo	No evidence to suggest that oral 5-ASA preparations are superior to placebo for the maintenance of medically induced remission in patients with CD.
SR+MA (Cochrane review)	Chande, 2016	Oct 2015	MEDLINE, EMBASE and the Cochrane Library	13 RCTs, only two compared vs 5-ASA	RCTs of AZA and 6-MP for induction of remission in active CD (5-ASA as one of the comparative arms). One study 6-MP vs 5-ASA, one study AZA vs sulfasalazine; the two studies were pooled together	Adults	Induction	AZA and 6MP: NR ;NR	Mesalamine: 3 g/d; 30 wk Sulfasalazine: NR; 17 wk	Overall quality of the evidence for both outcomes was low to very low due to sparse data.
SR + MA (cochrane review)	Lim, 2016	Jun 2015	PubMed, EMBASE, MEDLINE, and the cochrane Central Library	20 RCTs	5-ASA drugs (sulfasalazine & mesalamine) vs placebo, corticosteroids, and other 5-ASA (alone or in combo with corticosteroids) in mild to moderate active CD	Adults	Induction	Mesalamine: 1 - 4.5 g /d; 6-16 wk Sulfasalazine: NR; up to 18 wk	Placebo, corticosteroids, other 5-ASA alone or in combination with corticosteroids	Sulfasalazine is modestly effective compared to placebo and inferior to corticosteroids for the treatment of mild to moderate active CD. Olsalazine and low dose mesalamine (1-2g/d) are not superior to placebo. High dose mesalamine (3.2-4g/d) is not more effective than placebo for inducing response or remission. High dose mesalamine (4-4.5g/d) has conflicting results compared to budesonide.
SR + MA (cochrane review)	Chande, 2015	Jun 2016	MEDLINE, EMBASE, and the Cochrane Library	11 RCTs (only two RCTs included mesalamine or sulfasalazine, published in 2000 and in 1979)	RCTs of AZA and 6-MP for maintenance of remission in active CD (5-ASA was one of the comparative arms). One study compared 6-MP to 5-ASA, one study compared AZA to sulfasalazine- the two studies were pooled together in meta-analysis	Adults	Maintenance	6-MP: 1 mg kg/ d Methotrexate: 10 mg/wk; 76 weeks AZA: 1- 2.5 mg/ kg/ d	5-ASA: 3g/d; 76 wk	Only two RCTs; no statistically significant difference in the proportion of patients who maintained remission between AZA (1.0 to 2.5 mg/kg/day) or 6-MP (1.0 mg/day) and mesalamine (3 g/day) or sulfasalazine (0.5 g/15 kg/day) therapy.
NMA	Moja, 2015	Oct 2014	MEDLINE, EMBASE, the Cochrane library and clinicaltrials.gov	25 RCTs (induction =11, maintenance n=14)	Budesonide or mesalamine vs placebo or against each other in active or quiescent CD	Adults	Both	Mesalamine: 1 - 4 g/d; 6 wk for induction, 6 mo for maintenance Budesonide: 3 -18 mg/d; 6 wk for induction, 6 mo for maintenance	Placebo or mesalamine or budesonide in different doses	Budesonide 9mg/d or higher for induction of remission in active mild-to-moderate CD and 6mg/d for maintenance of remission appears to be the best treatment choice. Mesalamine is not effective in induction of remission and preventing relapse in mild-to moderate CD.
SR + MA	Ford, 2011	Dec 2011	MEDLINE, EMBASE, Cochrane central register	23 RCTs (active CD n=7, quiescent CD n=13, in active and quiescent CD=3)	5-ASA drugs (sulfasalazine, mesalamine, balsalazide, olsalazine) vs placebo or no treatment in quiescent or active CD.	Adults	Both	Sulfasalazine: 1 g/15 kg /d to max 5 g/d Mesalamine: 1 - 4 g/d Olsalazine: 2 g /d; 4-12 mo Duration in active CD: 6 wk - 4 mo, in quiescent CD: 6-48 mo	Placebo or no treatment	The role of 5-ASA in inducing remission of active CD and preventing relapse of quiescent CD remains uncertain.

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PP0574

DEMOGRAPHIC FEATURES OF ULCERATIVE COLITIS PATIENTS AS PREDICTORS OF RESPONSE TO ADSORPTIVE GRANULOMONOCYTAPHERESIS AS REMISSION INDUCTION THERAPY: A RETROSPECTIVE ASSESSMENT OF TREATMENT OUTCOMES

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Introduction: Inflammatory bowel disease (IBD) is associated with elevated myeloid lineage leucocytes, which show activation behaviour[1] including the CD14+CD16+DR++ phenotype known as proinflammatory monocytes and a major source of tumor necrosis factor-α.[2] Accordingly selective depletion of myeloid leucocytes by granulomonocytapheresis (GMA) with an Adacolumn is expected to promote remission and enhance drug efficacy. However, hitherto studies in IBD have reported contrasting efficacy outcomes, ranging from an 85%[3] to statistically insignificant.[4] Patients' demographic features should guide to select responder patients.

Aims & Methods: In ulcerative colitis patients, we were interested to identify relevant demographic features, which potentially could indicate a favourable response to GMA. Therefore, in a retrospective setting, we looked at the baseline clinical and endoscopic features of responders and non-responders to adsorptive GMA in 149 consecutive ulcerative colitis patients who had undergone GMA as remission induction therapy, 76 steroid naive, 71 steroid dependent, and 2 steroid refractory. Patients had received up to an 11 GMA sessions over 10 weeks. At entry and week 12, patients were clinically and endoscopically evaluated, allowing each patient to serve as her or his own control. Clinical activity index (CAI) ≤4 was defined as remission. Biopsies from endoscopically detectable inflamed mucosa were processed to see the impact of GMA on leucocytes within the mucosa.

Results: At entry, the average CAI was 12.8, range 10-17. Ninety-five patients (63.8%) had responded to GMA, 53 of 76 steroid naïve (69.7%), 41 of 71 steroid dependent (57.7%), and 1 of the 2 steroid refractory cases. On average remission was sustained for 8.6 months in steroid naïve patients and for 10.4 months in steroid dependent cohort. Mucosal biopsies revealed that infiltrating leucocytes were mostly neutrophils and monocytes. There was a marked reduction of infiltrating leucocytes in the biopsies from the responder patients. Patients with extensive deep UC lesions together with loss of the mucosal tissue at the lesions were non-responders. Patients with the first UC episode were identified as the best responders, followed by steroid naive patients. Further, a short duration of active UC prior to GMA marked a patient as a likely responder.

Conclusion: Patients who responded well to GMA attained a favourable long-term clinical course. GMA was more effective if applied immediately after a relapse than after a lag time. In general, GMA is favoured by patients for its safety profile and for being a non-drug remission induction option. Patients with extensive deep ulcers, with long duration of UC refractory to multiple pharmacologics are unlikely to benefit from GMA. In therapeutic settings, knowing baseline features, which may identify responder patients should help to avoid futile use of medical resources.

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Disclosure: Nothing to disclose.

PP0575

PERSISTENCE OF ADVANCED THERAPY IN PATIENTS WITH ULCERATIVE PROCTITIS – ANALYSIS FROM THE POPULATION-BASED REGISTRY CREDIT

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Introduction: Proctitis, a form of ulcerative colitis, is often highly symptomatic and challenging to manage. Data regarding the outcomes of biologic and small molecule therapy (advanced treatment) for patients with UC proctitis are limited, largely because these patients are usually excluded from randomized controlled trials.

Aims & Methods: Our aim was to evaluate the persistence of advanced treatment in patients with UC proctitis. From the Czech national prospective registry of biologic and innovative therapy of inflammatory bowel disease (CREdIT) which contains 9033 patients with IBD, we identified 200 treatment courses of advanced therapy among 167 patients with UC proctitis and ≥3 months follow-up. In this preliminary analysis, we employed mixed-effects Cox regression models to attempt to identify an association between treatment type and time to treatment discontinuation. An individual, medical centre, and time period were added as random effects into the models. To ensure a balance between the groups inverse probability of treatment weighting (IPTW) was used when comparing persistence between anti-tumor necrosis factor inhibitors (anti-TNF) and vedolizumab. We further analysed corticosteroid-free persistence after 6 months of treatment for those observations where this data were available. Missing data of co-variables were imputed using multiple imputation.

Results: Among 200 treatment courses, 78 were with infliximab (39%), 39 with adalimumab (20%), 59 with vedolizumab (30%), 13 with tofacitinib (7%), three each with filgotinib, golimumab, and ustekinumab, and two with upadacitinib. Most of the treatment courses were given as 1st line (1st 129, 65%; 2nd 46, 23%; >2nd 25, 13%). Concomitant corticosteroids at the start of advanced treatment course were reported in 119 (60%) of cases. Comparison of different treatment types revealed that persistence on vedolizumab was longer than on infliximab in the final adjusted model (Hazard Ratio [HR] 2.33, 95% CI 1.17 - 4.67, p-value=0.017). Moreover, vedolizumab was also more persistent compared with anti-TNF drugs as a group in the IPTW analysis (HR 8.03, 95% CI 2.60 - 24.84, p-value=0.001). Corticosteroid-free persistence at 6 months was found in 31/50 (62%) courses of vedolizumab and 56/97 (58%) courses of anti-TNF treatment. Al-

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though vedolizumab did not show greater corticosteroid-free persistence at 6 months than infliximab (HR 1.50, 95% CI 0.56-3.96, p-value=0.417), IPTW analysis revealed that vedolizumab was significantly more corticosteroid-free persistent compared to anti-TNF drugs as a group (HR 9.55, 95% CI 2.90-31.42, p-value=0.001).

Conclusion: Based on data from real clinical practice, the persistence of advanced treatment in UC proctitis seems to be low. Treatment with vedolizumab appears to be more persistent than anti-TNF therapy. However, efficacy data must also be analysed before vedolizumab can be considered preferable.

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ML: Lectures/congress fee/consultacy: Johnson&Johnson, Celltrion, Abbvie, Takeda, Biogen, Phizer
PD: Lecture fee/consultancy: AbbVie, Pfizer, Takeda, Janssen, Eli-Lilly, Sobi, Celltrion
TV: Lecture fee/consultancy: AbbVie, Takeda
ZS: Lecture fee/consultancy: Takeda
LN: Lecture fee/consultancy: Pfizer, Takeda, Janssen (Johnson and Johnson),Ferring
MB: Lecture fee/consultancy: AbbVie, Pfizer, Takeda, Janssen Cilag, Eli-Lilly, Ferring,
Celltrion, Alfasigma, SOBI, PRO.MED.CS
DD: Lecture fee/consultancy: AbbVie, Pfizer, Takeda, Janssen, Eli-Lilly, Ferring

PP0576

MANAGEMENT OF PERIANAL FISTULIZING CROHN'S DISEASE IS NOT IN KEEPING WITH GUIDELINES – RESULTS FROM A MULTI-CENTRE RETROSPECTIVE COHORT STUDY

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Introduction: Perianal fistulising Crohn's disease (PFCD) affects up to 20% of patients with Crohn's Disease (CD). PFCD can cause pain, purulent discharge and faecal incontinence, leading to significant morbidity and reduction in quality of life. The use of biologic therapy, antibiotics and surgical intervention remain the mainstay of management of perianal fistulas.

Aims & Methods: This study aims to review current practices in the management of PFCD across two large tertiary IBD centres. This retrospective cohort study assessed adult patients with PFCD between two tertiary Australian hospitals between 2006 to 2024. Patients with PFCD were identified using prescribing data. CD phenotype, therapy and fistula-related data were extracted. Biologic choice and therapeutic drug monitoring, antibiotic choice and duration, surgical intervention, and length of stay in hospital were monitored.

Cross-sectional disease activity of perianal fistulizing disease was assessed at the time of data collection using clinical fistula symptom grade and radiological assessment were available. Clinical remission was defined as fistula symptom grade of 1 or less. Radiological remission was defined by healing fistula tracts or no active inflammation seen on MRI. The study was approved by the Central Adelaide Local Health Network Human Research Ethics Committee (HREC19145).

Results:

Gender	**n = 57**
Male, n	**30**
Female, n	**27**
Age, y, median (IQR)	**40 (31-54)**
Active Smoker, n (%)	14 (24.5)
Median length of clinical follow up, months, n (IQR)	46.5 (23.8-72.3)
Antibiotics Prescribed	
Metronidazole, n treated (%)	47 (68.1)
Amoxicillin / Clavulanic acid, n treated (%)	13 (27.7)
Others, n treated (%)	9 (4.2)
Duration of Antibiotics	
Metronidazole, median duration, days (IQR)	5 (5-7)
Amoxicillin / Clavulanic acid, median duration, days (IQR)	5 (5-10)
Others, median duration, days (IQR)	14 (7-14)
Biologic therapy	
Total patients on biologic therapy, n (%)	56 (98.2)
Infliximab, n (%)	43 (75.4)
Adalulimab, n (%)	11 (19.3)
Ustekinumab, n (%)	2 (3.5)
Therapeutic Drug Monitoring (Infliximab)	
Infliximab level performed, n (%)	37 (86%)
Subtherapeutic (Infliximab level <10), n (%)	30 (81.1%)
Active disease and subtherapeutic (Infliximab level <10), n (%)	12 (40%)
Remission status	
Clinical Remission, n (%)	
- Defined as Fistula symptom grade of 1 or less	
Radiological Remission, n (%)	48 (84.2)
- Defined as healing fistula tracts or no active inflammation seen on MRI	33 (57.8)
Subtherapeutic Therapeutic Drug Monitoring (All biologic agents)	
In Remission, n (%)	18 (56.2)
Active Disease, n (%)	14 (43.8)

Table 1. Baseline characteristics.

57 patients (median age, 40 years (31-54); 52.6% male) with PFCD were included in the study and followed over a median duration of 46.5 months (23.8-72.3). There were a total of 157 individual surgical procedures over the period, 2.75 per patient, most of which (94.2%) were examination under anaesthesia. Antibiotic therapy was dispensed for either perianal sepsis or an actively draining fistula in 26 patients, 2.65 number of courses per patient. The most common antibiotic prescribed was Metronidazole (68.1%), followed by Amoxicillin / Clavulanic acid (27.7%). The median duration of antibiotic therapy was 5 days (5-7) for Metronidazole and 5 days (5-10) for Amoxicillin / Clavulanic acid. There were 56 (98.2%) patients on biologic therapy. The most common biologic agent was Infliximab (75.4%), followed by Adalimumab (19.3%) and Ustekinumab (3.5%). At the time of analysis, 48 patients (84.2%) were in clinical remission and 33 patients (57.8%) were in radiographical remission. In patients treated with infliximab, 37/43 (86%) had therapeutic drug monitoring performed. Amongst these, 30 (81.1%) had subtherapeutic levels (defined as Infliximab level < 10), with 12/30 (40%) of patients having active PFCD.

Conclusion: Across two tertiary IBD centres, management of PFCD is variable and not in keeping with current guidelines. Short courses of antibiotic therapy were used, and the majority of patients had subtherapeutic drug levels when tested. Optimising outcomes for patients with PFCD requires awareness of best practice and a multidisciplinary team approach.

Disclosure: Nothing to disclose.

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PP0577

MEDIUM TERM OUTCOMES AND PREDICTIVE FACTORS OF COLECTOMY IN PATIENTS WITH ACUTE SEVERE ULCERATIVE COLITIS

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Introduction: Acute severe ulcerative colitis (ASUC) is a medical emergency associated with a significant mortality rate. The literature amongst medium-term outcomes of ASUC remain limited.

Aims & Methods: This study aims to determine if early medical rescue therapy in patients with ASUC reduces the need for colectomy in the medium-term. This was a retrospective cohort study conducted in an Australian quaternary IBD centre. Adult UC patients admitted between January 2015 and December 2021 that fulfilled the Truelove and Witts' criteria for ASUC were identified and included. Demographic and clinical data was collected including date of colectomy and IBD-related readmission at 6 months. Patients who did not receive Infliximab (IFX) as rescue therapy or following discharge were excluded. Medium term was defined as within 1-3 years. The primary outcome of this study was colectomy (within 1-3 years), secondary outcomes were UC-related hospital re-admissions at 6 months. Kaplan-Meier curves were utilised to investigate colectomy and readmission-free days from the initial flare to the commencement of biologics. Multivariate log binomial models were employed to explore the significant factors related to colectomy and six-month readmission. The study was approved by the Southern Adelaide Local Health Network Ethics Committee LNR/22/SAC/157.

Results:

Baseline Characteristics	Patients (n = 32)
Male / Female, n	17/15
Age, y, median (IQR)	36 (28-48)
Colectomy, n (%)	11 (34.4)
Length of stay in hospital, days, median (IQR)	10 (7.8-13.3)
Flare to IFX commencement, days, median (IQR)	5 (3-7.3)
IFX commencement to colectomy, days, median (IQR)	9 (5-330)
D1 CRP, mg/L, median (IQR)	95 (37.6-157.7)
D3 CRP, mg/L, median (IQR)	29.7 (15.1-98.4)
D1 Albumin, g/L, median (IQR)	29 (23.8-32)
D3 Albumin, g/L, median (IQR)	22 (20.5-27)
Haemoglobin, g/L, median (IQR)	119.5 (110-129.5)
Platelet, 10^9/L, median (IQR)	400 (306-492.75)
Statistical Analysis - Colectomy	Odds Ratio / p-value
Female	6.59 / 0.2
D1 CRP	1.02 / 0.3
D3 CRP	0.98 / 0.3
D1 Albumin	2.09 / 0.074
D3 Albumin	0.24 / 0.05
Statistical Analysis - Re-admission	Odds Ratio / p-value
Female	1.14 / > 0.9
D1 CRP	1.0 / 0.7
D3 CRP	1.0 / > 0.9
D1 Albumin	0.79 / 0.2
D3 Albumin	0.98 / > 0.9

Table 1. Baseline Characteristics

267 patients with UC flares were identified between the shortlisted timeframes. After exclusions, 32 patients remained (median age, 36 years (18-71); 56.3% male). The median time of admission to the commencement of IFX was 5 days (3-7.3). The overall colectomy rate was 34.4% (11/32); of these patients, 3 needed a colectomy during their index presentation and 8 within the subsequent 3 years. The median time from IFX commencement to colectomy was 9 days (5-330). 9 patients (28.1%) had a UC-related readmission within 6 months of discharge. Biochemical results (day 1 CRP, day 1 albumin) and demographics such as age and gender were analysed, however we could not find any statistically significant association with the need for colectomy or UC-related readmission. The overall trend and likelihood of undergoing colectomy and having a UC-related admission rose with longer durations between IFX initiation, however the results were not statistically significant.

Conclusion: The need for colectomy remains high in patients with ASUC. We were unable to demonstrate any statistically significant predictors of colectomy in the medium term due to the small sample size. Future larger prospective studies are warranted to obtain statistical significant findings for our current study.

Disclosure: Nothing to disclose.

PP0578

THE INFLAMMATORY DURATION INDEX: A NOVEL SCORE FOR PREDICTING CORTICOSTEROID RESPONSE IN ACUTE SEVERE ULCERATIVE COLITIS

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Introduction: Acute severe ulcerative colitis (ASUC) represents a life-threatening medical emergency[1]. Patients with ASUC require hospitalization and instant medical treatment, with corticosteroids remaining the primary initial therapy. One-third of ASUC patients are steroid non-responders[2]. Management of ASUC patients should be based on efficient and fast assessment due to the need for further treatment decisions.

Aims & Methods: Our study aimed to create a new ASUC algorithm to predict corticosteroid response on the first day of hospital admission. A new Inflammatory Duration Index (IDI) was created by multiplying the onset duration of symptoms of UC prior to hospital admission and CRP to Albumin Ratio[2,3]. A cross-sectional study included 103 newly diagnosed patients with UC hospitalized at the University Clinical Center of Serbia from 2020 to 2023.

Patients included in the study fulfilled the criteria for ASUC according to Truelove's and Witts' severity index (62 male and 41 female, age ranging from 18 to 75 years). Descriptive statistics, Spearman's correlation, logistic regression, and ROC analysis were used to assess the predictive value of the score.

Results: The Inflammatory Duration Index (IDI) showed a significant negative correlation with glucocorticoid response (Spearman's rho = -0.383, $p < 0.01$). Logistic regression analysis revealed the score's predictive capacity for treatment outcomes. The ROC curve analysis showed an area under the curve (AUC) of 0.722, suggesting a good diagnostic ability to predict the glucocorticoid response. If we were to determine a sensitivity of 91%, this ratio would have to value less than 120 to predict the complete response to glucocorticoids.

Conclusion: The Inflammatory Duration Index (IDI) upon admission can serve as a valuable prognostic tool in UC management. Identifying patients who are less likely to respond to conventional glucocorticoid therapy, facilitates more personalized and effective treatment strategies, but further studies are needed to confirm this.

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Disclosure: Nothing to disclose.

PP0579

TOFACITINIB VERSUS VEDOLIZUMAB AMONG BIO-NAÏVE PATIENTS WITH ULCERATIVE COLITIS: A REAL-WORLD PROPENSITY-WEIGHTED COMPARISON

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Introduction: Over the last decade, treatment options for moderate-to-severe ulcerative colitis (UC) have expanded. However, comparative studies between these agents are limited, especially among biologic naïve patients. We aimed to compare the efficacy, safety and persistence of tofacitinib and vedolizumab as the first advanced treatment for patients with UC.

Aims & Methods: Patients who received either tofacitinib or vedolizumab as their first advanced therapy for UC via the NHS Lothian health board were included. Patients aged 65 years or older were excluded. We used inverse probability of treatment weighting (IPTW). The probability of treatment assignment was calculated via logistic regression using age, gender, IBD duration, Montreal extent, CRP, concomitant corticosteroids, and partial Mayo score at drug commencement. Missing data for these variables were imputed using multivariate imputation by chained equations. Confounder-adjusted survival curves were created using Kaplan-Meier estimates weighted via IPTW. The Pepe and Fleming test was used to compare survival curves.

Results: We included 158 patients of whom 81 (51.2%) received vedolizumab and 77 (48.7%) tofacitinib. Median follow-up for patients on vedolizumab was 3.1 (1.6-4.8) years and for tofacitinib 1.5 (0.34-2.3) years. The cohort was 59.3% male with a median age of 41.1 years (31.5-51.8) and disease duration of 6.21 years (2.56-13.5). At drug commencement there were no differences in steroid prescription, partial Mayo score, CRP or faecal calprotectin.

Vedolizumab persistence was superior to tofacitinib (p=0.005). Primary non-response and secondary loss of response were 9.9% and 17.3% for vedolizumab and 23.4% and 13% for tofacitinib respectively. At week 12, steroid free clinical remission was more frequent in the vedolizumab group (69% vs 51.4%, p=0.030), Table 1.There were no differences in the frequency of adverse events (11 [13.6%] vedolizumab vs 19 [24.7%] tofacitinib, p=0.629). However, adverse events resulting in temporary discontinuation of the drug occurred in 1 (1.2%) vedolizumab versus 11 (14.3%) tofacitinib, p=0.013.

Conclusion: We found that the persistence and tolerability of vedolizumab was superior to tofacitinib in bio-naïve UC, although the rates of clinical and biomarker remission were comparable. These data may help inform positioning of advanced therapy in UC.

Disclosure: Beatriz Gros has served as consultant to Galapagos and Abbvie and as speaker for Abbvie, Jansen, Takeda, Pfizer and Galapagos. Nikolas Plevris has served as a speaker for Janssen, Takeda and Pfizer. Professor Charlie Lees has acted as a consultant to Abbvie, Janssen, Takeda, Pfizer, Galapagos, Bristol Myers Squibb, B.I., Sandoz, Novartis, GSK, Gilead, ViforPharma, Dr Falk, Trellus Health and Iterative Scopes; he has received speaking fees and travel support from Pfizer, Janssen, Abbvie, Galapagos, MSD, Takeda, Shire, Ferring, Hospira and Dr Falk. Gareth-Rhys Jones has served as a speaker for Takeda, Janssen, Abbvie, Fresnius and Ferring.

PP0580

PREDICTIVE FACTORS OF ADVERSE SKIN REACTIONS DURING BIOLOGIC THERAPY FOR INFLAMMATORY BOWEL DISEASES IN CHILDREN - PRELIMINARY DATA FROM CZECH NATIONAL REGISTRY BASED PROSPECTIVE STUDY

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Introduction: No clinical trials have directly compared biologics in children with inflammatory bowel disease (pIBD), with observational studies indicating only modest differences in efficacy [1]. Risk factors for adverse effects of individual preparations are important for treatment personalization. Cutaneous adverse effects of tumour necrosis factor antagonists (anti-TNF) have been widely reported, affecting up to 48% of anti-TNF treated pIBD patients [2]. Identifying patients at risk for developing adverse cutaneous effects may be beneficial for treatment sustainability.

Aims & Methods: The Czech National Prospective Registry of Biologic and Targeted Therapy of Inflammatory Bowel Disease (CREdIT) was used to identify the biologic treatment courses. PIBD patients younger than 19 years were considered eligible if they were treated with anti-TNF, vedolizumab, or ustekinumab. Baseline patient characteristics, including anthropological features, disease classification, and the reason for biologic, were collected in the registry. Data on the family and personal history of skin conditions and atopic diseases were collected at enrolment or, if missing, by systematic telephone interview. The occurrence of cutaneous adverse events, including type (atopic, psoriatic, pustulotic or other), distribution and the quality of life as assessed by validated questionnaires were prospectively collected. Observation was defined as one course of

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biological therapy in one patient. The primary outcome was defined as time to onset or significant worsening of the skin condition during biological therapy. Cox proportional hazard models with mixed effect were designed for potential predictors. Predictors that were statistically significant in the univariate analysis were used to build the multivariate model.
Results: Among the 544 included observations of 423 paediatric patients from 4 centres, 208 were treated with adalimumab (38%), 256 with infliximab (47%), 46 with ustekinumab (8%), and 36 with vedolizumab (7%). Most patients had CD (N=311; 74%) and were also treated with an immunomodulator at the time of initiation of biologic therapy (N=478; 78%). The primary outcome was achieved in 108 observations (atopic [9%], psoriatic [34%], pustulotic [13%]), with most of the children treated with adalimumab (50%), 47% with infliximab and 3% with ustekinumab. Female sex (hazard ratio (HR) 1.84, 95% confidence interval (CI) 1.22–2.78), anti-TNF therapy (HR 6.97, 95% CI 2.15–22.59), and family history of atopic diseases (HR 1.77, 95% CI 1.12–2.79) were identified as predictors of the occurrence or progression of the cutaneous adverse events in univariate analysis. These predictors retained their significance in the multivariate model additionally including diagnosis, concomitant immunomodulator use and line of therapy (Table 1).

Predictor	HR (95% CI)	p-value
Female sex	1.82 (1.18–2.81)	0.01
Family history of atopic diseases	1.82 (1.13–2.91)	0.01
anti-TNF therapy	6.84 (1.98–23.63)	<0.01
Crohn's disease	1.33 (0.75–2.36)	0.43
Biologically naïve	0.79 (0.46–1.37)	0.41
Concomitant immunomodulator	1.48 (0.76–2.88)	0.25

Table 1 – Multivariate Cox proportional hazard model of cutaneous adverse effect predictors.

Conclusion: These preliminary data from the prospective multicentre study suggest that when choosing biologic treatment, the female sex, family history of atopic diseases and, in particular, anti-TNF treatment, may be considered as predictors of cutaneous adverse effects in paediatric IBD patients regarding the occurrence of cutaneous adverse reactions.
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KZ: Consultancy: Nestlé
IC: Nothing to disclose.
TL: Nothing to disclose.
JS: Lectures/congress fees/consultancy: MSD, Abbvie, Takeda, Nutricia, Nestle
EK: Nothing to disclose.
MH: Lectures/congress fees/consultancy: Nutricia, Nestlé
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EV: Workshop/congress fee: Biocodex, Ewopharma
MK: Nothing to disclose.
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PP0581

COMPARISON OF THE EFFICACY AND SAFETY OF CORTICOSTEROIDS IN ELDERLY-ONSET ULCERATIVE COLITIS AND NON-ELDERLY-ONSET ULCERATIVE COLITIS

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Introduction: Clinical practice guidelines recommend corticosteroids use for remission induction therapy in patients with moderate-severe ulcerative colitis (UC); however, there are limited data on their effectiveness and safety in elderly UC patients. Although comparable outcomes in elderly UC patients have been reported, these patients show higher morbidity, and mortality, than younger patients due to the lower malnutrition and immunocompromised state. It has been reported that elderly onset is associated with a lower effectiveness of intravenous corticosteroids with higher risks of surgery and adverse events in UC. Thus, the efficacy and safety of corticosteroids in elderly UC patients remains to be determined.
Aims & Methods: To compare the clinical characteristics in elderly onset UC patients treated with corticosteroid against non-elderly onset patients. We enrolled 170 UC patients presenting at Nippon Medical School Hospital prior to March 2023. The patients were divided into two groups comprising elderly-onset and non-elderly-onset UC patients, and their clinical characteristics were determined according to medical records. We defined elderly-onset UC as UC initially diagnosed in patients at 65 years or older according to WHO criteria.
Results: Clinical characteristics of UC patients were as follows: 12 patients were categorized as elderly-onset UC including 3 males to 9 females, and 158 were non-elderly-onset UC including 91 males to 67 females. The proportion of females was higher in elderly-onset UC patients than in non-elderly onset UC patients. The rate of patients who underwent corticosteroid induction therapy was significantly higher in elderly-onset UC patients; 9 out of 12 (75%) in elderly-onset UC patients and 81 out of 158 (51%) in non-elderly-onset UC patients, although average prednisone doses were comparable. Adverse events were observed in 3 cases (33%) in elderly-onset UC patients: candidiasis in 2 cases and hyperglycemia in 1 case, and no patients required surgery. While adverse events were 21 cases (26%) in non-elderly-onset UC patients: liver disfunction (n=4), followed by hyperglycemia (n=2), depression (n=2); and one each for cytomegalovirus colitis, clostridioides difficile infection, gastric ulcer and so on. Hemoglobin values were significantly lower (P<0.01), and erythrocyte sedimentation ratio was significantly higher (P=0.04) in elderly-onset UC patients. On the other hand, no significant difference was observed between the two groups regarding; pre-treatment WBC counts, serum albumin concentrations, neutrophil-lymphocyte ratio, CRP values, the rate of patients who achieved clinical remission and steroid-free remission, rate of introduction of immunomodulators, rate of use of biologics, number of hospitalizations, and number of relapses.
Conclusion: There was no difference in responsiveness to steroid therapy between non-elderly and elderly UC patients, and no serious adverse events were observed during the period of steroid therapy in both groups. However, consideration should be given to comorbidity in elderly patients.
Disclosure: The authors declare that they have no competing interests.

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PP0582

EXPLORING BIOMARKER OF CORTICOSTEROID THERAPY FOR MODERATELY TO SEVERELY ACTIVE ULCERATIVE COLITIS

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Introduction: Ulcerative colitis (UC) is characterized by chronic immune mediated inflammation of the colon and rectum. Corticosteroid (CS) is a common treatment for patients with moderate to severe active UC, but its effect varies among patients, including resistance, response, and dependence. In addition, the longer CS therapy continues, the greater the concern regarding drug-related adverse effects. New immunomodulatory therapies are being developed for the treatment of UC drug showing promise in the treatment of patients with moderate UC refractory to aminosalicylic acid (5-ASA). Thus, the clinical benefit of being able to predict whether certain patients would benefit from CS therapy prior to administration cannot be overestimated.

Aims & Methods: We compared the clinical characteristics of patients with steroid-responder (SR) and steroid-non-responder (SNR) UC and retrospectively evaluated for biomarkers that could predict the efficacy of CS prior to administration in patients with moderate to severe active UC who failed to respond to 5-ASA.

We enrolled 100 UC patients presenting at Nippon Medical School Hospital with a history of CS administration prior to December 2023. The patients were divided into two groups, SR patients and SNR patients, and their clinical characteristics were determined according to medical records. Eosinophil counts were measured in the high-power field using Direct Fast Scarlet staining.

Results: Clinical characteristics of UC patients were as follows: the median age, 48±17 years; height 167±9.2 cm, and weight, 58±14 kg; including 61 males to 39 females. The most common disease extent type was the total colitis (63%), followed by left-sided colitis (32%), and proctitis (5%). Complete SNR was observed in 15 patients (15%), with remaining 85 patients (85%) classified as SR. There was no significant difference found in Mayo Endoscopic Score, first dose of CS, number of hospitalizations, number of relapses, propotion of patients with total-colitis type, or duration of CS treatment between the SR group and the SNR group. The non-remission rate, defined as failure to achieve mucosal healing, was significantly higher in the SNR group (66%) than in the SR group (36%) (P=0.03). The proportion of patients treated with azathioprine was significantly higher in the SR group (40%) than in the SNR group (39%). The proportion of patients treated with biologics or small molecular compounds was significantly higher in the SNR group (40%) than in the SR group (22%) (P=0.03). We then compared hematological biomarkers in blood samples obtained prior to CS administration between the two groups. There was no significant difference in WBC counts, hemoglobin values, serum albumin concentrations, CRP values, and ESR between the two groups. However, eosinophil profiles (both total count numbers and ratio to white blood cell counts) were significantly higher in the SR group than in the SNR group (P=0.02). Neutrophil-lymphocyte ratio tended to be higher in the SR group than in the SNR group (P=0.06). The number of infiltrated eosinophils in the inflamed rectal mucosal tissue was not significantly different between the two groups. The number of infiltrated eosinophils in the rectal mucosa tended to decrease in the steroid-response groups (p = 0.07). However, there was no difference in the number of infiltrated eosinophils in the steroid-non-response groups.

Conclusion: Both eosinophil in the blood and in the rectal mucosa, and neutrophil-lymphocyte ratio in the blood could be potential biomarkers for steroid responsiveness prior to administration.

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· Mucosal Eosinophil Abundance in Non-Inflamed Colonic Tissue Is Associated with Response to Vedolizumab Induction Therapy in Inflammatory Bowel Disease：J Clin Med Res,2022

Disclosure: Nothing to disclose.

PP0583

PATIENTS WITH INFLAMMATORY BOWEL DISEASES RECEIVING ESCALATED INFLIXIMAB THERAPY ACHIEVE STEROID-FREE CLINICAL AND BIOCHEMICAL REMISSION WITH HIGHER SERUM TROUGH LEVELS: A CROSS-SECTIONAL STUDY AT A GREEK TERTIARY CENTRE

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Introduction: Therapeutic monitoring of infliximab (INF) correlates with improved therapeutic outcomes in inflammatory bowel diseases (IBD).

Aims & Methods: The aim of this cross-sectional study was to assess the relationship between serum INF trough levels (TLs) and positive therapeutic outcomes in patients with ulcerative colitis (UC) and Crohn's disease (CD) in a Greek population. The relationship between adverse events (AEs) and the existence of anti-infliximab antibodies (ADAs) was also investigated. Consecutive adult patients with IBD treated with intravenous INF at maintenance phase in a Greek tertiary centre were included. Investigated therapeutic outcomes included sustained clinical remission (CR), steroid-free clinical remission (SFCR), biomarker remission (CRP <5mg/L, BR) and combined (steroid-free clinical and biochemical) remission (SFCBR). Infliximab TLs and ADAs were assessed using ELISA-based assays.

Results: A total of 77 patients were enrolled (62.3% male, mean age 44 years), 48 (62.3%) with CD, 26 (33.8%) with UC and 3 (3.9%) with unclassified colitis. The mean disease duration was 11.7±8.4 years, while the mean duration of INF infusion was 5.1±4.6 years. Forty-seven (61%) patients had undergone treatment escalation, with a further dose escalation in 7 (9.1%) patients. Concomitant combined therapy was given in 13 (16.8%) patients, 4 (5.2%) of them were receiving azathioprine and 9 (11.7%) corticosteroids. Mean INF-TLs were 7.2±4.9 μg/mL and only correlated with treatment escalation (9.7 vs 3.6 μg/mL, p<0.001). Sixty-eight (88.3%) patients had CR, 62 (80.5%) had SFCR, 48 (62.3%) had BR and 43 (55.8%) had SFCBR. Compared to non-responders, mean INF-TLs were numerically higher, although not significantly different, in CR (7.5 vs 5.5 μg/mL, p=0.254), SFCR (7.3 vs 6.9 μg/mL, p=0.761), BR (7.4 vs 6.9 μg/mL, p=0.707) or SFCBR (7.7 vs 6.6 μg/mL, p=0.349).

In a subgroup analysis, among patients without treatment escalation, 27 (90%) had CR, 25 (83.3%) had SFCR, 22 (73.3%) had BR and 19 (63.3%) had SFCBR. Mean INF-TLs were significantly increased among patients with BR

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(4.2 vs 0.8 µg/mL, p=0.020) and SFCBR (4.3 vs 1.5 µg/mL, p=0.035). In ROC analysis, TLs could significantly predict (p=0.016) with a good accuracy (AUC 0.768, 95% CI: 0.584=0.952) patients with SFCBR, with an optimal TL cut-off at 3.4 µg/mL. Among patients with treatment escalation, 41 (87.2%) had CR, 37 (78.7%) had SFCR, 26 (55.3%) had BR and 24 (51.1%) had SFCBR. In ROC analysis, TLs could significantly predict (p=0.018) with a fair accuracy (AUC 0.653, 95% CI: 0.527=0.755) patients with SFCBR, with an optimal TL cut-off at 11 µg/mL.

Twenty patients had TLs <1 µg/mL, of whom 10 had positive ADAs. Among them, 9 showed combined remission, 4 of whom had positive ADAs. Among 21 (27.3%) patients with reported AEs, 8 (10.4%) had an allergic reaction, which was attributed to INF. Allergic reactions had a tendency of correlation with ADA existence, as 4/5 patients with allergic reaction and TLs <1 µg/mL had positive ADAs, while 4/10 patients with positive ADAs reported allergic reaction, compared to 1/10 patients with negative antibodies (p=0.152).

Conclusion: Infliximab TLs correlate with treatment escalation. Higher TLs appear to result in favourable therapeutic outcomes. It is suggested that patients undergoing treatment escalation could benefit from higher INF-TLs in the achievement of favourable outcomes.

Disclosure: Nothing to disclose.

PP0584

REAL-WORLD EXPERIENCES OF SWITCHING TO SUBCUTANEOUS FORMULATION IN IBD PATIENTS ON MAINTENANCE VEDOLIZUMAB TREATMENT

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Introduction: Usability of subcutaneous vedolizumab (SC VDZ) in inflammatory bowel diseases (IBD, ulcerative colitis [UC], Crohn's disease [CD]) have proved via clinical trials, while real-world data collection is ongoing. Experiences of Central-European population and assessment of patient's perspectives are lacking.

Aims & Methods: Our study aimed to evaluate the real-world efficacy, safety and patient's preferences of SC VDZ maintenance treatment after switching from intravenous (IV) formulation in a Hungarian IBD cohort. In this prospective, multicenter cohort study, IBD patients on maintenance IV VDZ treatment were enrolled, who switched to SC administration. Baseline was the day of the switch, while 52-week follow-up was set. Clinical and demographic data were collected, while serum VDZ level and CRP was measured at baseline and at w52. A non-validated questionary was filled by patients who were on drug at w12 recording patient' satisfaction. Primary outcome was the drug survival rate at one year, while secondary outcomes were change in corticosteroid-free clinical (CSFR, based on physicians' global assessment score PGA<1), and biochemical remission (BR, 5 mg/l^{3}CRP) rates to w52, safety issues and patients' preferences and change in serum drug levels.

Results: In total, 37 IBD patients were followed (15 CD and 22 UC, male/female ratio 45.9 %, median age 42.0 [IQR 34-49] years). In CD group, 66.7% were in CSFR and 60 % in BR, while in UC group 63.6% were in CSFR and 68.2 % in BR (p=0.55 and p=0.61). Overall, 29.7 % of patients ceased SC VDZ treatment after a median of 20 (IQR 6-26) weeks. Cessation was due to secondary response losing in 3/6 and 4/5 patients in CD and UC group. In case of 2 and one patients, SC treatment was terminated due to local reactions. One patient was fear of needle. Severe adverse event did not occur. Factors associated with discontinuation did not identify. The mean serum VDZ level has higher at w52 compared to baseline in CD and UC patients (CD: 8.9 ±6.9 to 30.6 ±17.8 ug/mL, p = 0.047; UC: 13.6±13.8 to 33.6±17.6 ug/mL, p = 0.001), while disease activity was stable. 84 % of the patients who filled the questionnaire were satisfied with injection, while only one patient found it difficult to inject.

Variables	Total (n = 37)	CD (n = 15)	UC (n = 22)	Sig.
Follow-up duration, *weeks, median (IQR)*	52 (26-52)	52 (23-52)	52 (46-52)	0.23
Sex, *male (%)*	17 (45.9)	8 (53.3)	9 (40.9)	0.47
Age at inclusion, *years, median (IQR)*	42.0 (34-49)	38.0 (35-49)	43.5 (34-52)	0.82
Disease duration at inclusion, *weeks, median (IQR)*	11.0 (9-17)	11.0 (9-17)	11.5 (8-16)	0.48
IV VDZ treatment duration, *weeks, median (IQR)*	75.9 (39-106)	86.9 (55-112)	60.3 (32-97)	0.64
Disease activity				
PGA, *mean (± SD)*	0.5 (0.7)	0.4 (0.6)	0.6 (0.8)	0.75
CSFR, n (%)	24 (64.9)	10 (66.7)	14 (63.6)	0.55
CRP, *mg/l, mean (± SD)*	7.0 (6.9)	7.3 (6.2)	6.8 (7.4)	0.825

Conclusion: Transition from IV to SC maintenance VDZ treatment is effective at one year, the overall survival rate is high presumably due to the elevated serum drug levels. Severe safety issue did not raise. Satisfaction of switched patients is high, which may improve treatment adherence and save resources.

Disclosure: K.F. has received speaker's honoraria from AbbVie, Janssen, Ferring, Takeda, and Goodwill Pharma. T.M. has received speaker's honoraria from MSD, AbbVie, Egis, Goodwill Pharma, Takeda, Pfizer, Janssen, Sandoz, MundiPharma, Phytotec, Roche, Fresenius, and Teva. Remaining authors have no conflict of interest to declare.

PP0585

EFFECT OF ANTIBIOTIC TREATMENT ON THE FORMATION OF ANTIBODIES AGAINST ANTI-TNFA IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE: A CASE – CONTROL STUDY

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Introduction: Anti- TNFα are the first advanced therapy in inflammatory bowel disease (IBD). However, up to 50% of patients will lose response during treatment. One of the most common mechanisms is immunogenicity. This is associated with obesity, smoking, monotherapy or the presence of HLA-DQA1*05. The gut microbiota appears to play an important role in pathogenesis and response to treatment. This can be altered by several factors, such as antibiotics. Therefore, antibiotics could (potentially) affect immunogenicity and response to treatment.

Aims & Methods: The aim of our study is to evaluate the effect of antibiotic treatment on the development of immunogenicity to anti-TNFα drugs in patients with IBD.

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This is a retrospective case – control study of patients with IBD from a tertiary hospital treated with anti- TNFα, with data collected from June 2005 to January 2023. A group of patients who has developed antibodies against the drug (ADA) through a non-tolerant test has been compared. Two controls were selected per case, matched by type of IBD, mono/combotherapy, previous biological treatment, and type of anti- TNFα (infliximab/adalimumab).

The epidemiological variables and their association with the prescription of antibiotics and omeprazole 6 months before and 6 months after the start of biological treatment have been analysed.

Results: 100 patients have been included (30 cases / 70 controls): 50% men, 27% smokers, 55% Crohn's disease, 70% infliximab, 61% associated immunosuppression. 60% required intensification. No differences were found in any baseline characteristics between cases and controls. Anti-TNF withdrawal occurred in 19/30 cases (time until withdrawal: 19.5±27.52 months).

The use of antibiotics during de 6 months after the start of biological treatment was associated with developing ADA (76.7% vs. 38.6%; p<0.0001). The use of quinolones (33% vs. 11%; p=0.009) and penicillins (17% vs 4%, p=0.03) were associated with de development of ADA. In the multivariate analysis (adjusted for immunosuppressants, tobacco, obesity, number of biologicals), only the use of antibiotics in the first 6 months was associated with the development of ADA (OR 3.95, p<0.01).

Conclusion: Taking antibiotics during the first months after starting anti-TNFα is associated with an increased risk of ADA. The antibiotics that show the greatest association are quinolones and penicillins.

Disclosure: Nothing to disclose.

PP0586

EFFECT OF OZANIMOD AT THERAPEUTIC AND SUPRATHERAPEUTIC DOSES ON ABSOLUTE LYMPHOCYTE COUNT AND HEART RATE IN HEALTHY ADULTS

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Introduction: Ozanimod (OZA) is an oral selective sphingosine 1-phosphate (S1P) receptor 1 and 5 modulator approved for the treatment of moderately to severely active ulcerative colitis (UC) or relapsing multiple sclerosis. Consistent with the mechanism of action (MOA) of S1P receptor modulators, OZA causes lymphocyte retention in lymphoid organs. S1P receptor modulators also bind to specific receptors in cardiac myocytes, but the associated risk of bradycardia may be mitigated with gradual dose escalation. Evidence from the phase 3 True North study (NCT02435992) suggests that absolute lymphocyte count (ALC) is a pharmacodynamic biomarker but is not predictive or prognostic for clinical response in OZA-treated patients with moderately to severely active UC.

Aims & Methods: We analyzed changes in ALC and heart rate (HR) over 65 d of OZA treatment at the therapeutic dose (0.92 mg) and a supratherapeutic dose (1.84 mg). A phase 1, double-blind, placebo (PBO)-controlled study (NCT04978298) evaluated the pressor effect of oral tyramine in healthy adults receiving OZA. The study included 3 periods; this analysis focuses on Period 2, during which participants (ppts) were randomized to OZA (therapeutic dose group: 7-d dose escalation to 0.92 mg once daily [QD]; supratherapeutic dose group: 10-d dose escalation to 1.84 mg QD) or matched PBO for 65 d (Day 8–72). Safety endpoints included adverse events (AEs), clinically significant changes in laboratory tests and electrocardiograms, and vital signs.

Results: This study included 27 ppts in the OZA 0.92 mg group (mean [SD] age 36.4 [8.9] y, males 55.6%), 26 ppts in the OZA 1.84 mg group (36.3 [8.4] y, 53.8%), and 25 ppts in the PBO group (40.3 [9.1] y, 52.0%). Mean group level decreases from baseline (BL) in ALC ranged from 0.32×10^9/L to 1.18×10^9/L in the OZA 0.92 mg group and 0.12×10^9/L to 1.37×10^9/L in the OZA 1.84 mg group. ALC decreases were similar in OZA groups. Decreases began to plateau after ~14 d of OZA dosing, with mean group level decreases from BL of 0.82×10^9/L in the OZA 0.92 mg group and 1.09×10^9/L in the OZA 1.84 mg group at Day 21. Two ppts discontinued due to treatment-related lymphopenia: 1 event (3.7%) in the OZA 0.92 mg group was moderate in intensity ($0.07–1.02\times10^9$/L, Period 2); 1 event (3.8%) in the OZA 1.84 mg group was considered mild ($0.10–0.88\times10^9$/L, Periods 2 and 3; reported and discontinued in Period 3). Neither ppt had a reported infection or associated AE. Both events of lymphopenia resolved by end of study. ALC reductions in other ppts were not considered clinically significant. HR was slightly lower with OZA 1.84 mg on Days 8–15 and normalized thereafter, but overall, no clinically meaningful trends in mean HR or change from daily BL occurred across groups (*Table*).

	PBO			OZA 0.92 mg			OZA 1.84 mg		
	Mean (min, max) predose (BL) HR (bpm)	Mean (min, max) 6-h postdose HR (bpm)	Mean (min, max) change in HR[a] (bpm)	Mean (min, max) predose (BL) HR (bpm)	Mean (min, max) 6-h postdose HR (bpm)	Mean (min, max) change in HR[a] (bpm)	Mean (min, max) predose (BL) HR (bpm)	Mean (min, max) 6-h postdose HR (bpm)	Mean (min, max) change in HR[a] (bpm)
Day 8	63.2 (46, 80)	69.2 (54, 82)	5.9 (−11, 28)	59.8 (50, 80)	62.5 (51, 83)	2.7 (−8, 23)	62.2 (42, 86)	64.1 (51, 78)	1.9 (−15, 18)
Day 12	63.6 (44, 81)	70.4 (56, 94)	6.9 (−3, 16)	56.4 (48, 78)	60.7 (48, 82)	4.3 (−5, 14)	58.3 (39, 75)	61.0 (43, 79)	2.7 (−11, 16)
Day 15	63.8 (48, 80)	69.1 (54, 104)	5.3 (−6, 24)	57.2 (43, 80)	60.9 (45, 71)	3.7 (−12, 17)	58.1 (38, 73)	58.8 (42, 70)	0.7 (−11, 12)
Day 18	60.7 (41, 81)	68.8 (54, 80)	8.2 (−6, 30)	55.9 (44, 70)	59.7 (43, 79)	3.8 (−9, 14)	57.5 (43, 73)	60.8 (44, 73)	3.3 (−4, 20)
Day 21	62.6 (44, 80)	66.5 (48, 82)	3.9 (−7, 17)	58.7 (47, 75)	63.6 (53, 78)	4.9 (−13, 17)	59.5 (42, 79)	64.7 (50, 84)	5.2 (−7, 24)
Day 42	61.9 (45, 72)	68.1 (52, 82)	6.2 (−7, 16)	58.4 (49, 79)	63.0 (51, 77)	4.5 (−12, 14)	59.7 (46, 72)	66.6 (45, 87)	6.9 (−7, 25)
Day 72	58.4 (41, 75)	65.5 (50, 78)	7.1 (−6, 20)	58.1 (47, 73)	64.7 (53, 80)	6.6 (−2, 15)	59.4 (41, 80)	66.4 (46, 84)	7.0 (−4, 17)

[a]Calculated as the mean change in HR from predose (BL) to 6-h postdose for each day.

Table. Mean predose and 6-h postdose HR and mean change in HR.

Conclusion: In healthy ppts who received OZA at the therapeutic or supratherapeutic dose, ALC decreased as expected given the therapeutic MOA. No clinically significant changes from BL were noted. The ALC decrease was similar across OZA groups and was not dose dependent. No new cardiac safety signals were observed.

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PP0587

THE IMPACT OF CYTOMEGALOVIRUS INFECTION ON DISEASE RELAPSE IN PATIENTS WITH MODERATE-SEVERE ULCERATIVE COLITIS: A RETROSPECTIVE SINGLE CENTRE STUDY

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Introduction: Cytomegalovirus (CMV) infection frequently affects patients with Ulcerative Colitis (UC) and it has been associated with disease relapse and increased severity.

Aims & Methods: Our study aimed to assess clinical and biochemical differences between CMV-positive UC patients with moderate to severe disease relapse and those without CMV infection.

We retrospectively included all patients with UC who were followed for at least one year at the Unit of Gastroenterology of the Azienda Ospedale Università Padova and that were studied for CMV infection at the time of a moderate/severe reactivation (pMayo ≥5 and/or fecal calprotectin ≥ 250 and/or endoscopic Mayo ≥ 2) from May 2019 to October 2023. Patients were considered CMV-positive if CMV antigens were revealed by immunohistochemistry (at least 1 CMV intranuclear inclusion body in routine hematoxylin and eosin-stained sections and/or immunohisto-chemical staining) and/or a positive tissue viral load was detected by qualitative real-time PCR on colonic biopsies. Demographic and clinical features, fecal calprotectin, and previous and ongoing UC therapy were compared between CMV-positive patients and those who tested negative. Finally, the anti-CMV therapy administered was reported. Differences between the two study groups were assessed using the independent-samples t-test, while differences in the categorical analysis were analyzed using the Chi-squared test. A p-value <0.05 was considered significant. STATA 11 was used for statistical analysis.

Results: We retrospectively included 122 UC patients with moderate-to-severe disease (58.2% females, mean age at study 51 ± 17 years, time from diagnosis 11 ± 10 years). 24 patients (19.7%) were positive for CMV infection; no differences in gender, age, and time from diagnosis were observed between CMV-positive and negative patients. Serum quantitative CMV-DNA was tested in 18 patients and > 1000 copies/ml in 4 patients (22.2%) were detected. Approximately half of the patients were treated with immunomodulatory therapy during disease relapse with a similar distribution between CMV-positive and negative patients (p = 0.7). 95.2% of CMV-positive patients had diarrhoea compared to 71.6% of patients without CMV infection (p = 0.02), and abdominal pain was present in 71.4% CMV-positive compared to 42.1% CMV-negative patients (p = 0.015). In contrast, symptoms like fever and bloody diarrhoea had similar rates in the two groups (p=ns). Mean levels of fecal calprotectin were similar between the two groups (p = 0.07). Hospitalization was necessary in 91.7% of CMV-positive patients compared to 30.2% of CMV-negative patients. CMV-positive patients were treated with oral or intravenous steroids and switched to UC biological therapy more frequently than CMV-negative patients (p < 0.05). In parallel, 14/24 CMV-positive patients were treated with Ganciclovir for 3-5 days, and 13/14 were subsequently treated with Valganciclovir for another 2-6 weeks. Ten patients were only treated with oral Valganciclovir.

Conclusion: CMV-positive UC patients required systemic steroid therapy and hospitalization more frequently than non-infected patients, suggesting a more challenging management of the disease. Moreover, symptoms such as abdominal pain and diarrhoea emerged as potential indicators of CMV infection when compared to other clinical presentations.

Disclosure: Nothing to disclose.

PP0588

INFLAMMATORY BIOMARKER REDUCTION AND IMPROVEMENT NORMALIZATION AND IN CLINICAL OUTCOMES WITH RISANKIZUMAB VERSUS USTEKINUMAB IN PATIENTS WITH MODERATE TO SEVERE CROHN'S DISEASE: A POSTHOC ANALYSIS FROM THE PHASE 3 SEQUENCE STUDY

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Introduction: Per STRIDE II recommendations, the normalization of high-sensitivity C-reactive protein (hs-CRP) and fecal calprotectin (FCP) should be an intermediate treatment target for patients (pts) with moderate to severe Crohn's disease (CD).[1] The safety and efficacy of risankizumab (RZB), an interleukin (IL)23 p19 inhibitor, and ustekinumab (UST), an IL12/23p40 inhibitor, in pts with CD have been reported previously.[2–4] Using data from the phase 3 SEQUENCE study, we compared RZB versus UST for the achievement of biomarker reduction and normalization (alone or in combination with disease remission).

Aims & Methods: SEQUENCE was an open-label, multicenter, randomized, efficacy assessor-blinded study that directly compared the efficacy and safety of RZB vs UST in pts with moderate to severe CD and prior anti-TNF failure (NCT04524611). Pts were randomized 1:1 to receive recommended induction and maintenance doses of RZB (N=255) or UST (N=265). A mandatory steroid taper began at week (W)2. Randomization was stratified by baseline (BL) steroid use and number of failed anti-TNFs. In this post hoc analysis, pts with elevated FCP (> 250 mg/kg) or hs-CRP (> 5 mg/L) at BL were assessed for achievement of normalized FCP (≤ 250 mg/kg)) or hs-

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CRP (≤ 5 mg/L) at week W8, week W24, and week W48. At week W24 and W48 week 48, achievement of normalized FCP or hs-CRP in combination with CD activity index (CDAI) clinical remission (CDAI < 150) or with inflammatory bowel disease questionnaire (IBDQ) remission (IBDQ total score ≥ 170), as well as achievement of a 50% reduction in hs-CRP or FCP and CDAI response (reduction of CDAI ≥100 points from BL), were examined. Endpoints were analyzed using non-responder imputation, while incorporating multiple imputation to handle missing data due to COVID-19 and/or geopolitical conflict. All *P* values are nominal.

Results:

Endpoint n/N % (95% CI)	Risanki-zumab Week 8	Ustekinumab Week 8	Response Rate Difference % (95% CI) P value	Risanki-zumab Week 24	Ustekinumab Week 24	Response Rate Difference % (95% CI) P value	Risanki-zumab Week 48	Ustekinumab Week 48	Response Rate Difference % (95% CI) P value
Normalization of hs-CRP[a] in patients with elevated hs-CRP[b] at BL	58/163; 35.6% (28.2, 42.9)	56/171; 32.7% (25.7, 39.8)	2.8% (-7.3, 13.0) *P* = .6	72/163; 44.2% (36.6, 51.9)	48/171; 28.1% (21.3, 34.8)	16.2% (6.0, 26.3) ***P* <.01**	71/163; 43.6% (35.9, 51.2)	38/171; 22.2% (16.0, 28.5)	21.3% (11.5, 31.2) ***P* <.0001**
Normalization of FCP[c] in patients with elevated FCP[d] at BL	49/170; 28.8% (22.0, 35.6)	25/180; 13.9% (8.8, 18.9)	14.9% (6.5, 23.4) ***P* <.001**	65/170; 38.2% (30.9, 45.5)	35/180; 19.4% (13.7, 25.2)	18.8 (9.5, 28.1) ***P* <.0001**	72/170 42.4% (34.9, 49.8)	24/180; 13.3% (8.4, 18.3)	29.0 (20.1, 38.0) ***P* <.0001**
Normalization of hs-CRP[a] or FCP[c] AND CDAI clinical remission In patients with elevated hs-CRP[b] or FCP[d] at BL				84/211; 39.9% (33.3, 46.5)	53/218; 24.3% (18.6, 30.0)	15.6% (6.9, 24.3) ***P* <.001**	97/211 46.0% (39.2, 52.7)	54/218; 24.8% (19.0, 30.5)	21.2% (12.4, 30.0) ***P* <.0001**
Normalization of hs-CRP[a] or FCP[c] AND IBDQ remission in patients with elevated hs-CRP[b] or FCP[d] at BL				57/211 27.0% (21.0, 33.0)	59/218; 27.1% (21.2, 33.0)	-0.1 (-8.5, 8.4) *P* = 1.0	47/211; 22.3% (16.7, 27.9)	34/218; 15.6% (10.8, 20.4)	6.7 (-0.7, 14.1) *P* = 1.0
≥ 50% decrease from BL in hs-CRP or FCP AND CDAI clinical response in patients with elevated hs-CRP[b] or FCP[d] at BL				111/211; 52.6% (45.8, 59.3)	74/218 33.9% (27.7, 40.2)	18.6 (9.4, 27.8) ***P* <.0001**	119/211; 56.4% (49.7, 63.1)	68/218; 31.2% (25.0, 37.3)	25.2 (16.1, 34.3) ***P* <.0001**

BL, baseline; CD, Crohn's disease; CDAI, CD activity index; CI, confidence interval; FCP, fecal calprotectin; hs-CRP, high-sensitivity C-reactive protein; IBDQ, inflammatory bowel disease questionnaire.
CDAI clinical remission: CDAI < 150; IBDQ remission: IBDQ total score ≥ 170.
[a]Normalization of hs-CRP is defined as ≤ 5 mg/L.
[b]Elevated hs-CRP is defined > 5 mg/L.
[c]Normalization of FCP is defined as ≤ 250 mg/kg.
[d]Elevated FCP is defined > 250 mg/kg.
Efficacy population includes all patients randomized to either ustekinumab (i.e., weight-based induction dose IV followed by maintenance dose 90-mg SC), or the selected dosing regimen for risankizumab (600 mg for induction, 360 mg for maintenance), and who received at least one dose of study drug.
95% CI for response rate is the synthetic result based on Student's t-distribution from PROC MIANALYZE procedure if there are missing data due to COVID-19 and/or geopolitical conflict or is based on the normal approximation to the binomial distribution if there are no missing data due to COVID-19 and/or geopolitical conflict.
Response rate difference = (risankizumab - ustekinumab); 95% CI for difference is calculated based on normal approximation to the binomial distribution for the comparison of two treatment groups. P-value is based on Chi-square test. The calculations are based on non-responder imputation incorporating multiple imputation to handle missing data due to COVID-19 and/or geopolitical conflict.
Pooled p-value from Chi-square is calculated following Li et al., *Statistica Sinica*. 1991; 1, 68-92.

A greater proportion of pts receiving RZB vs UST achieved normalization of hs-CRP at W24 weeks 24 and W48, as well as normalization of FCP at weeks W8, W24, and 48 (P <.01 for all; *Table*). Rates of normalization of FCP or hs-CRP plus CDAI clinical remission were also greater at weeks W24 and W48 in pts receiving RZB vs UST, as were rates of CDAI clinical response and a ≥ 50% reduction in FCP or hs-CRP (*P* <.001 for all; *Table*). Rates of normalization of FCP or hs-CRP plus IBDQ remission were similar between pts receiving RZB vs UST (*Table*). The safety profiles of RZB and UST in CD have been previously reported.[2–4]

Conclusion: A greater proportion of pts achieved normalization of biomarkers FCP or hs-CRP with RZB vs UST. Composite endpoints of CDAI clinical remission and biomarker normalization, as well as CDAI clinical response and biomarker reduction (≥ 50%) favored RZB vs UST at weeks 24 and 48.

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EFFICACY OF RISANKIZUMAB VERSUS USTEKINUMAB BY DURATION OF DISEASE IN PATIENTS WITH MODERATE TO SEVERE CROHN'S DISEASE: A POSTHOC ANALYSIS FROM THE PHASE 3 SEQUENCE STUDY

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Introduction: Consistent associations between higher remission rates and shorter disease duration have been reported for advanced therapies in Crohn's disease (CD), but no direct comparisons between newer advanced treatments and efficacy with respect to CD duration have been examined. The efficacy of risankizumab (RZB), an interleukin (IL)23p19 inhibitor, and ustekinumab (UST), an IL12/23p40 inhibitor, in pts with CD have been reported.[1–3]

Herein, using data from the phase 3 SEQUENCE study, the clinical and endoscopic efficacy of RZB vs UST, stratified by baseline (BL) CD duration, is reported.

Aims & Methods: The SEQUENCE open-label, multicenter, randomized, efficacy assessor-blinded study compared the efficacy and safety of RZB vs UST in pts with moderate to severe CD and prior anti-TNF failure (NCT04524611). Pts were randomized 1:1 to receive recommended induction and maintenance doses of RZB (N=255) or UST (N=265). A mandatory steroid taper began at week (W)2. Randomization was stratified by BL steroid use and number of failed anti-TNFs. This post hoc analysis evaluated CDAI clinical remission and endoscopic remission at W24 and W48, and endoscopic response, steroid-free CDAI clinical remission, and steroid-free endoscopic remission at W48, in pts stratified by disease duration (<2 years [Y], 2-5Y, 5-10Y, >10Y) (endpoints defined in *Table* footnote). Missing data due to COVID-19 and/or geopolitical conflict was handled using non-responder imputation while incorporating multiple imputation. *P* values are nominal.

Results: Baseline characteristics were mostly similar across treatments and disease duration subgroups, except total number of pts and proportion of pts failing >1 anti-TNFs (both lower in the <2Y subgroup), and fecal calprotectin (lower with increasing CD duration) (*Table*). With RZB vs UST, a numerically greater proportion of pts achieved all endpoints, irrespective of disease duration (*Table*). A greater proportion of pts in all subgroups achieved endoscopic response with RZB vs UST at W48 (<2Y: *P*<05; 2-5Y, P<001; 5-10Y, *P*<.05, > 10Y, P<.01), with a similar trend for RZB vs UST observed for most, but not all, disease duration subgroups for W24 endoscopic response and W48 endoscopic remission (*Table*).

The numerically highest rates of endoscopic response were observed with RZB in pts with a disease duration ≤2Y, andwhile the numerically greatest treatment differences with RZB vs UST were observed in pts with a disease duration of ≤5Y (*Table*). The safety profiles of RZB and UST in CD were previously reported.[1–3]

BL Demographic and Disease Characteristics	Disease Duration at Baseline (BL)							
	<2 years		2-5 years		>5-10 years		>10 years	
	RZB N=36	UST N=43	RZB N=59	UST N=62	RZB N=67	UST N=71	RZB N=93	UST N=89
Age (years), mean (SD)	36.7 (13.9)	34.2 (13.2)	33.1 (12.6)	36.0 (13.7)	36.3 (13.5)	35.1 (13.8)	43.0 (11.3)	44.3 (12.2)
Sex (male), n (%)	18 (50)	18 (42)	35 (59)	35 (56)	42 (63)	35 (49)	41 (44)	43 (48)
Body mass index (kg/m²), mean (SD)	21.6 (4.2)	22.9 (5.4)	23.1 (5.9)	25.4 (6.6)	24.1 (4.8)	23.3 (5.8)	24.9 (5.9)	26.6 (5.6)
Average daily stool frequency, mean (SD)	5.1 (2.5)	5.0 (2.6)	5.3 (2.4)	5.6 (2.8)	6.0 (3.2)	5.2 (2.4)	5.5 (2.6)	6.3 (2.3)
Average daily abdominal pain, mean (SD)	1.9 (0.5)	2.0 (0.6)	2.0 (0.5)	2.0 (0.5)	2.0 (0.5)	1.9 (0.6)	1.9 (0.5)	1.8 (0.6)
CDAI, mean (SD)	312.5 (58.3)	314.8 (55.3)	318.6 (56.1)	314.6 (63.3)	317.4 (65.1)	304.0 (68.3)	296.8 (62.7)	309.7 (61.3)
SES-CD, n; mean (SD)	n = 36 14.4 (8.2)	n = 43 15.5 (7.7)	n = 59 12.9 (6.7)	n = 62 15.2 (8.3)	n = 67 14.0 (7.4)	n = 71 13.7 (7.4)	n = 93 13.2 (6.6)	n = 89 13.1 (6.4)
Fecal calprotectin (mg/kg), median (IQR)	n = 33 1429 (2877)	n = 41 2116 (2811)	n = 47 1453 (2552)	n = 49 1535 (2079)	n = 53 1016 (2217)	n = 57 1515 (2070)	n = 74 593 (1472)	n = 68 1115 (2306)
hs-CRP (mg/L), n; median (IQR)	n = 35 5 (30)	n = 40 19 (45)	n = 58 10 (22)	n = 60 10 (25)	n = 63 9 (31)	n = 71 8 (15)	n = 90 7 (13)	n = 86 8 (20)
Anti-TNF failure history (> 1), n (%)	2 (6)	5 (12)	11 (19)	11 (18)	19 (28)	14 (20)	27 (29)	31 (35)
Corticosteroid use (yes) - n %)	11 (31)	14 (33)	11 (19)	16 (26)	16 (24)	15 (21)	20 (21)	26 (29)
Crohn's disease location per SES-CD, n (%)								
Ileal only	6 (17)	9 (21)	9 (15)	12 (19)	9 (13)	10 (14)	18 (19)	14 (16)
Colonic only	20 (56)	16 (37)	23 (39)	20 (32)	25 (37)	28 (39)	34 (37)	42 (47)
Ileal-colonic	10 (28)	18 (42)	27 (46)	30 (48)	33 (49)	33 (46)	41 (44)	33 (37)

	RZB N=36	UST N=43	Response Rate Difference % (95% CI) P-value	RZB N=59	UST N=62	Response Rate Difference % (95% CI) P-value	RZB N=67	UST N=71	Response Rate Difference % (95% CI) P-value	RZB N=93	UST N=89	Response Rate Difference % (95% CI) P-value
CDAI clinical remission at week 24, n/N; % (95% CI)	23/36 63.9% (48.2, 79.6)	22/43 51.2% (36.2, 66.1)	12.7 (-8.9, 34.4) P =.3	30/59 50.8% (38.0, 63.6)	26/62 41.9% (29.7, 54.2)	8.9 (-8.9, 26.6) P =.3	40/67 59.7% (48.0, 71.4)	39/71 54.9% (43.4, 66.5)	4.8 (-11.7, 21.3) P =.6	59/93 63.4% (53.7, 73.2)	26/89 29.2% (19.8, 38.7)	34.2 (20.6, 47.8) **P <.0001**
CDAI clinical remission at week 48, n/N; % (95% CI)	23/36 63.9% (48.2, 79.6)	19/43 44.2% (29.3, 59.0)	19.7 (-1.9, 41.3) P =.1	33/59 55.9% (43.3, 68.6)	27/62 43.5% (31.2, 55.9)	12.4 (-5.3, 30.1) P =.2	39/67 58.2% (46.4, 70.0)	31/71 43.7% (32.1, 55.2)	14.5 (-2.0, 31.1) P =.1	60/93 64.5% (54.8, 74.2)	31/89 34.8% (24.9, 44.7)	29.7 (15.8, 43.6) **P <.0001**
Endoscopic response at week 24, n/N; % (95% CI)	19/36 52.8% (36.5, 69.1)	14/43 32.6% (18.6, 46.6)	20.2 (-1.3, 41.7) P =.1	26/59 44.5% (31.7, 57.2)	12/62 19.4% (9.5, 29.2)	25.1 (9.0, 41.2) **P <.01**	33/67 49.3% (37.3, 61.2)	23/71 32.4% (21.5, 43.3)	16.9 (0.7, 33.0) **P <.05**	37/93 39.8% (29.8, 49.7)	21/89 23.6% (14.8, 32.4)	16.2 (2.9, 29.5) **P <.05**
Endoscopic response at week 48, n/N; % (95% CI)	19/36 52.8% (36.5, 69.1)	11/43 25.6% (12.5, 38.6)	27.2 (6.3, 48.1) **P <.05**	24/59 40.7% (28.1, 53.2)	8/62 12.9% (4.6, 21.2)	27.8 (12.7, 42.8) **P <.001**	31/67 46.3% (34.3, 58.2)	18/71 25.4% (15.2, 35.5)	20.9 (5.3, 36.6) **P <.05**	41/93 44.1% (34.0, 54.2)	21/89 23.6% (14.8, 32.4)	20.5 (7.1, 33.9) **P <.01**
Endoscopic remission at week 48, n/N; % (95% CI)	11/36 30.6% (15.5, 45.6)	8/43 18.6% (7.0, 30.2)	12.0 (-7.1, 31.0) P =.2	20/59 33.9% (21.8, 46.0)	6/62 9.7% (2.3, 17.0)	24.2 (10.1, 38.4) **P <.01**	19/67 28.4% (17.6, 39.2)	16/71 22.5% (12.8, 32.3)	5.8 (-8.7, 20.3) P =.4	31/93 33.3% (23.8, 42.9)	13/89 14.6% (7.3, 21.9)	18.7 (6.7, 30.8) **P <.01**
Steroid-free CDAI clinical remission at week 48, n/N; % (95% CI)	23/36 63.9% (48.2, 79.6)	19/43 44.2% (29.3, 59.0)	19.7 (-1.9, 41.3) P =.1	33/59 55.9% (43.3, 68.6)	26/62 41.9% (29.7, 54.2)	14.0 (-3.6, 31.6) P =.1	39/67 58.2% (46.4, 70.0)	31/71 43.7% (32.1, 55.2)	14.5 (-2.0, 31.1) P =.1	60/93 64.5% (54.8, 74.2)	31/89 34.8% (24.9, 44.7)	29.7 (15.8, 43.6) **P <.0001**
Steroid-free endoscopic remission at week 48, n/N; % (95% CI)	11/36 30.6% (15.5, 45.6)	8/43 18.6% (7.0, 30.2)	12.0 (-7.1, 31.0) P =.2	19/59 32.2% (20.3, 44.1)	4/62 6.5% (0.3, 12.6)	25.8 (12.4, 39.2) **P <.001**	19/67 28.4% (17.6, 39.2)	16/71 22.5% (12.8, 32.3)	5.8 (-8.7, 20.3) P =.4	31/93 33.3% (23.8, 42.9)	13/89 14.6% (7.3, 21.9)	18.7 (6.7, 30.8) **P <.01**

BL, baseline; CD, Crohn's disease; CDAI, CD activity index; CI, confidence interval; FCP, fecal calprotectin; hs-CRP, high-sensitivity C-reactive protein; SES-CD, simple endoscopic score for CD; TNF, tumor necrosis factor

CDAI clinical remission - CDAI < 150; endoscopic remission - SES-CD ≤ 4 and ≥ 2 point reduction versus BL and no subscore > 1 in any individual variable, as scored by central reviewer; endoscopic response - decrease in SES-CD > 50% from BL (or for patients with isolated ileal disease and a BL SES-CD of 4, ≥ 2 point reduction from BL), as scored by central reviewer; Steroid-free CDAI clinical remission - CDAI clinical remission without receiving steroids; steroid-free endoscopic remission - endoscopic remission without receiving steroids

Efficacy population includes all patients randomized to either ustekinumab (i.e., weight-based induction dose IV followed by maintenance dose 90-mg SC), or the selected dosing regimen for risankizumab (600 mg for induction, 360 mg for maintenance), and who received at least one dose of study drug.

5% CI for response rate is the synthetic result based on Student's t-distribution from PROC MIANALYZE procedure if there are missing data due to COVID-19 and/or geopolitical conflict or is based on the normal approximation to the binomial distribution if there are no missing data due to COVID-19 and/or geopolitical conflict.

Response rate difference = (risankizumab - ustekinumab); 95% CI for difference is calculated based on normal approximation to the binomial distribution for the comparison of two treatment groups. P-value is based on Chi-square test. The calculations are based on non-responder imputation incorporating multiple imputation to handle missing data due to COVID-19 and/or geopolitical conflict.

Pooled *p*-value from Chi-square is calculated following Li et al., *Statistica Sinica*. 1991; 1, 68-92. All *P*-values are nominal.

Conclusion: At W24 and W48, numerically higher rates of efficacy were observed with RZB vs UST treatment for all endpoints across disease duration subgroups, with notable consistent differences observed for endoscopic response. Endoscopic response rates with RZB were numerically highest in pts with a disease duration of ≤2Y, supporting early intervention for patients with CD.[4]

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PP0590

IMPACT OF EXTRAINTESTINAL MANIFESTATIONS ON DISEASE COURSE IN INFLAMMATORY BOWEL DISEASE

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Introduction: Inflammatory bowel disease (IBD) represents a group of chronic inflammatory disorders of the gastrointestinal tract. While the hallmark features of IBD involve inflammation and damage to the digestive system, the impact of the disease often extends beyond the gut, manifesting as a spectrum of extraintestinal manifestations (EIM), affecting various organs and contributing to the complexity and heterogeneity of IBD.

Aims & Methods: The aim of this study was to assess the impact of EIM in IBD on disease course. We conducted a retrospective, descriptive and comparative study including patients hospitalized for IBD over a 9-year period [January 2011-January 2020]. Patients with less than 2 years of follow-up were excluded. Clinico-biological data, disease severity, and long-term outcomes (surgery and disease course) were collected. Data were entered and analyzed via SPSS 26 software.

Results: We included 225 men and 227 women. We enrolled 341 patients diagnosed with Crohn's disease (CD) and 111 patients with ulcerative colitis (UC). The mean age at diagnosis was 36.15 ± 12.74 years. One-hundred and sixty-five patients (36.4%) had EIM with respectively: rheumatological manifestations (n=144), dermatological manifestations (n=26), ocular manifestations (n=17) and hepatic manifestations (n=8). We noticed a significant association of EIM with male gender (56.3% vs 45.9%; p=0.03). Rheumatological EIM were associated with a significant older age at disease onset (36.7 years vs 32.8 years; p=0.003). No significant association was noted between EIM with neither family history of IBD nor the type of IBD (p=0.33 and p=0.3 respectively). In case of CD, we noted a higher frequency of ileal involvement in patients with EIM, without a significant difference (61.4% vs 51.4%; p=0.07). No association was noted between EIM with neither fistulising behaviour nor with extended ileal disease nor with upper gastrointestinal tract involvement (p=0.1; p=0.5 and p=0.5 respectively). In case of UC, no association was noted between EIM and an extensive disease (p=0.7). As for treatment, EIM had no impact on corticosteroids' use or immunosuppressive treatment (p=0.36 and p= 0.08), but was associated with a significant higher use of anti-TNF therapy (83.4% vs 27.2%; p=0.01). Response to corticosteroids and anti-TNF therapy was similar whether with or without EIM (p= 0.36 and p=0.24 respectively). In the sub-group of ocular EIM, we noted a higher failure of immunosuppressive treatment (70% vs 36.6%; p=0.04). We noted that EIM was associated with a rarer urge to surgical treatment (35.7% vs 44.9%;p=0.05) with a significant longer timeframe to surgery (47.7 months vs 27,2 months; p=0.03). The mean number of years of active disease was significantly higher in patients with EIM (3.3 vs 2.5; p=0.002) with a higher number of flare-ups (p=0.009) and hospitalizations (p<0.001), reflecting a better disease course in patients without EIM.

Conclusion: In our study, EIM were associated with an overall more severe disease course in IBD, thus, emphasizing the importance of comprehensive assessment and management strategies tailored to address both intestinal and extra-intestinal manifestations in the holistic care of IBD patients.

Disclosure: Nothing to disclose.

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PP0591

ADHERENCE TO CORTICOSTEROID RECOMMENDATIONS IN IBD PATIENTS: A CRITICAL EVALUATION

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Introduction: Corticosteroid tapering over a period of 6-8 weeks effectively induces remission in inflammatory bowel disease (IBD). However, prolonged corticosteroid use (lasting more than 3 months) is linked to severe side effects.

Aims & Methods: Detecting and mitigating prolonged use is crucial for optimal care. We conducted a prospective study in 2018 involving 116 IBD patients undergoing biologic therapy at an IBD referral center in Montenegro. We utilized the Steroid Assessment Tool (SAT) as described by Selinger et al., 2017, to assess corticosteroid usage in this cohort.

Results: Of the patients, 61% had ulcerative colitis (UC) and 39% had Crohn's disease (CD). 53% of patients received anti TNF biologics (adalimumab or infliximab), and 47% received anti-integrin therapy (vedolizumab) in the current therapy. In the preceding year 32 patients (28%) had used corticosteroids. Most of those patients (78%) received a single course of steroids, while 19% received two courses, and 3% received three. The duration of steroid use was predominantly three months (78% of patients), with the longest duration being nine months (1 patient). Among steroid users, 81% (26/32) did not develop dependency or relapse after three months, while 19% (6/32) were steroid resistant/dependent.

Conclusion: Our study identified corticosteroid excess in 6.0% of patients. While our findings suggest adherence to corticosteroid recommendations in IBD patients undergoing biologic therapy, there's a need to expand routine monitoring of steroid dependency or excess to all IBD patients, regardless of their biologic therapy history.

Disclosure: Nothing to disclose.

PP0592

EARLY APPLICATION OF TIGHT CONTROL IMPROVES THE RATES OF ENDOSCOPIC AND TRANSMURAL REMISSION IN CROHN'S DIASEASE

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Introduction: Increasing evidence support the use of transmural remission as a treatment target in Crohn's disease (CD), but it is seldom achieved in clinical practice. Tight monitoring of inflammation using fecal calprotectin (FC) with reactive treatment escalation may potentially improve these results.

Aims & Methods: We aim to evaluate if treatment escalation based on FC can improve the rates of transmural remission in CD.

Retrospective cohort study including 256 CD patients with 2 consecutive assessments by MRI-enterography and colonoscopy and with regular monitoring using FC. For each occurrence of an elevated FC (≥ 250 ug/g), we evaluated if a reactive adjustment of medical treatment was performed. The ratio of treatment escalation/elevated FC was correlated with the chances of reaching transmural remission. Early disease was defined as disease duration <18 months without previous exposure to immunomodulators and biologics.

Results: After a median follow-up of 2 years (IQR 1-4), 121 patients (47.3%) reached endoscopic remission and 61 patients (23.8%) reached transmural remission. There was a higher prevalence of early disease (34.4% vs 20.5%, *P*=0.038), non-stricturing non-penetrating phenotype (54.1% vs 37.4%, *P*=0.026) and less biologic use at baseline (13.1% vs 26.7%, *P*=0.037) in patients with transmural remission.

Ratios of escalation ≥50% resulted in higher rates of transmural remission (34.2% vs 15.1%, *P*<0.001). The effect was more pronounced in patients with early disease (50.0% vs 12.0%, *P*=0.003). In multivariate analysis, a treatment escalation ratio ≥50% (OR 3.46, 95%CI 1.67-7.17, *P*=0.001) and early disease intervention (OR 3.24, 95%CI 1.12-9.34, *P*=0.030) were independent predictors of achieving transmural remission.

Conclusion: Tight-monitoring and reactive treatment escalation increased the rates of transmural remission in CD. Intervention in early disease further improved these results.

Disclosure: Nothing to disclose.

PP0593

PROPOSED USTEKINUMAB BIOSIMILAR FYB202: PHARMACOKINETIC EQUIVALENCE DEMONSTRATED IN A RANDOMISED, DOUBLE-BLIND, PARALLEL-GROUP, SINGLE-DOSE TRIAL IN HEALTHY SUBJECTS (RUSTIC)

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Introduction: Ustekinumab is a recombinant human monoclonal IgG1 antibody targeting the interleukins IL-12 and IL-23. Ustekinumab is approved for treatment of moderately to severely active inflammatory bowel disorders, i.e. Crohn's disease and ulcerative colitis, besides its approved use in moderate-to-severe plaque psoriasis and active psoriatic arthritis.

The presented study provides the basis for the marketing authorisation of the proposed ustekinumab biosimilar FYB202, together with an extensive analytical characterisation and the results of an efficacy and safety trial in patients with moderate-to-severe plaque psoriasis.

Aims & Methods: The primary objective of the three-arm trial RUSTIC was A) to prove pharmacokinetic (PK) similarity in terms of $AUC_{0\text{-}inf}$ and C_{max} of the proposed ustekinumab biosimilar FYB202 to both EU-approved (EU-Ref) and US-licensed ustekinumab (US-Ref) and B) to prove PK similarity of both reference drugs. PK similarity was concluded if the 90% confidence interval (CI) of the geometric mean ratios was contained within the acceptance interval of 80.00–125.00% (analysis of covariance [ANCOVA] with body weight as covariate).

Treatments included a single 45-mg subcutaneous injection. The multi-centric clinical trial was conducted at four German trial sites. Safety analyses comprised immunogenicity (anti-drug antibodies [ADAs] and neutralising antibodies [NAbs]), adverse events (AEs), vital signs, body temperature, 12-lead electrocardiograms, and local tolerability (subject- and investigator-assessed).

Healthy male and female subjects (18–55 years, body weight 60–90 kg, body-mass index 20.5–30.0 kg/m^2) were included in the trial and observed for up to 112 days after dosing.

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Results: Overall, 491 subjects were randomised (FYB202: n=164, EU-Ref: n=163, US-Ref: n=164); 486 subjects completed the trial, and 478 subjects were included in the PK analysis set. PK similarity was demonstrated for all three comparisons: all 90% CIs of the respective geometric mean ratios for $AUC_{0\text{-}inf}$ and C_{max} were contained within the acceptance interval of 80.00–125.00% (*Table 1*). After administration of FYB202, fewer subjects had at least one positive ADA result at trial end compared to the reference groups (FYB202: 20%, EU-Ref: 42%, US-Ref: 51%). ADA titre levels were similar between all three groups. The percentage of subjects with at least one positive NAb-reactive result at trial end was slightly lower in the FYB202 group compared to the reference groups (FYB202: 11%, EU-Ref: 15%, US-Ref: 18%). More than half of the subjects had at least one treatment-related AE (FYB202: 53%, EU-Ref: 55%, US-Ref: 57%); no treatment-related serious AEs occurred. Local tolerability at the injection site was comparable in all treatment groups.

Parameter	FYB202 vs. EU-Ref	FYB202 vs. US-Ref	EU-Ref vs. US-Ref
$AUC_{0\text{-}inf}$	102.85 (96.84-109.23)	112.05 (105.24-119.30)	109.00 (102.84-115.53)
C_{max}	97.22 (92.05-102.67)	101.65 (96.19-107.42)	104.59 (99.06-110.44)

Table 1: Point estimates and 90% confidence intervals [%] for the primary pharmacokinetic parameters of ustekinumab (PK analysis set; N=478)
Observations: $AUC_{0\text{-}inf}$: n=318 each comparison; C_{max}: n=319 FYB202 vs. EU-Ref, EU-Ref vs. US-Ref; n=318 FYB202 vs. US-Ref

Conclusion: The RUSTIC trial demonstrated bioequivalent PK characteristics between FYB202 and both EU-approved and US-licensed ustekinumab and for the two reference drugs. No clinically meaningful differences with respect to the overall safety, immunogenicity and local tolerability were observed.

Disclosure: The trial is registered in the EU Clinical Trials Register (EudraCT no. 2022-000832-43).

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PP0594

EUROPEAN AND US PATIENT AND HEALTH CARE PROFESSIONAL PERSPECTIVES ON THE EXPERIENCE OF FATIGUE IN CROHN'S DISEASE AND ULCERATIVE COLITIS: COMMUNICATING NEEDS AND FEATURES OF IBD EXPERIENCES (CONFIDE) SURVEY

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Introduction: Fatigue is a prevalent and debilitating symptom among patients with Crohn's disease (CD) or ulcerative colitis (UC). However, due to its subjective nature and the lack of effective management strategies in CD/UC, fatigue often goes under-recognized and under-treated.[1] The Communicating Needs and Features of IBD Experiences (CONFIDE) study explored patient and healthcare professional (HCP) perspectives on the experiences and impact of CD/UC-related symptoms in Europe (France, Germany, Italy, Spain, and UK), the United States (US) and Japan.

Aims & Methods: Online, quantitative, cross-sectional surveys were conducted separately among patients with moderate-to-severe CD/UC and HCPs who make prescribing decisions for patients. Moderate-to-severe CD/UC were defined using criteria based on previous treatment, steroid use, and/or hospitalization. Descriptive statistics were used here to summarize the European and US data related to fatigue.

Results: A total of 547 European (55.4% male, mean age: 38.0 years) and 215 US (54.9% male, 40.9 years) patients with CD and 556 European (57.4% male, 38.9 years) and 200 US (61.5% male, 40.4 years) patients with UC completed the surveys. The proportions of patients who received advanced therapies at the time of survey completion were 63.4% and 58.1% among European and US patients with CD. Among patients with UC, the percentages were 54.1% and 76.5% of European and US patients.

Overall, 34.2% European and 35.8% US patients with CD, and 20.9% European and 27.5% US patients with UC reported currently experiencing fatigue (in the past month). Similar results were observed among patients receiving advanced therapies: CD (Europe: 34.3%; US: 36.0%); UC (Europe: 19.9%; US: 25.5%). Fatigue was ranked among the top five symptoms with greatest impact on patients by 44.5% European and 52.9% US patients with CD, and 55.2% European and 37.5% US patients with UC. Among patients who reported currently experiencing fatigue, 29.9% European and 36.4% US patients with CD, and 22.4% European and 40.0% US patients with UC discussed it with their HCPs at every appointment. Among patients who did not discuss fatigue with their HCPs at every appointment, at least two-thirds indicated that they would like to discuss it more frequently: CD (Europe: 75.6%; US: 67.3%); UC (Europe: 74.4%; US: 66.7%).

Surveys were completed by 503 European (70.8% male) and 200 US (78.0% male) HCPs, of which 92.5% were gastroenterologists/internal medicine practitioners. A minority of HCPs reported fatigue among the top five symptoms with the greatest impact on patients: CD (Europe: 33.9%; US: 22.0%); UC (Europe: 22.7%; US: 11.5%). More than half of HCPs reported proactively discussing fatigue at a routine appointment with patients: CD (Europe: 69.4%, US: 59.5%); UC (Europe: 64.6%, US: 59.0%). Among HCPs

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not proactively discussing fatigue, the most common reason was that they expected patients to bring it up if it is an issue: CD (Europe: 39.0%, US: 30.9%); UC (Europe: 37.1%, US: 32.9%).

Conclusion: For both patients with CD/UC, fatigue was a common and impactful symptom, despite the use of advanced therapies. A substantial proportion of patients experiencing fatigue who did not discuss it at every HCP appointment, reported a desire to discuss fatigue more frequently. Although more than half of HCPs reported proactive discussion of fatigue with their patients, some HCPs expected patients to bring it up. This indicates a communication gap between patients and HCPs and highlights the underappreciation of the impact of fatigue in CD/UC.

References: 1. Farrell D, et al. (2020) Cochrane Database Syst Rev. 4(4), CD012005.

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PP0595

USE OF FK-ADA BIOSIMILAR (MSB 11022) IN CONTEXT OF BIOLOGIC SWITCH IN INFLAMMATORY BOWEL DISEASES: BASELINE DATA OF THE REAL WORLD IDEA STUDY

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Introduction: Biosimilars offer similar efficacy and safety when compared to their originators, thereby increasing patient access to affordable treatment. FK-ada is an adalimumab biosimilar approved for the treatment of inflammatory bowel diseases (IBD), Crohn's disease (CD) and ulcerative colitis (UC). FK-ada development included 1,234 subjects treated up to 52 weeks (total 420 patients-year). Since first approval, there have been no unexpected safety signals despite more than 115,000 patient-years of exposure. IDEA is an ongoing real-world study in which adult patients with IBD and inflammatory rheumatic diseases (RA, AS, PsA), switching from adalimumab originator, or from adalimumab biosimilar to FK-ada, are followed up to one year. We present here the baseline characteristics of IBD patients, physicians and centres participating to the study.

Aims & Methods: The aim of this study was to identify predictive factors of treatment persistence after switch to FK-ada and to develop a model to predict persistence at 12 months. Data collected include: baseline characteristics of patients, physicians, and the participating centres, reasons for treatment switch, patient clinical status at each follow-up visit including disease activity scores (HBI and Montreal classification), and faecal calprotectin results. Adult patients with either CD or UC were included but those receiving treatment outside of routine clinical care were excluded. Data is presented using descriptive statistics and a logistic regression model will be used to identify possible predictive factors.

Results: IBD patients were recruited in 18 centres (15/18 university or public hospitals, >50% granted incentives to switch to biosimilars, 16/18 secondary and tertiary care level, 12/18 with at least 4 managing physicians in the medical team). All physicians were full time gastroenterologists, (66% over 40 years old, with a median time of 22 years in IBD management and 15 years median time in prescribing biologics). In the past 5 years, 71% have been involved in 6 or more non interventional studies, and 67% in 6 or more clinical trials. 138 IBD patients were recruited (CD: 113, UC: 25 patients were diagnosed with IBD (57% males, median age 43, 90% with secondary or university level of education, median BMI stable at 25.3 [22.2, 28.4] kg/m$^{2)}$. Current smokers were more frequent in CD patients 21.4% vs 8.3 % UC (median 10 pack years). Disease history and current

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status at baseline summarized in *Table 1.* Median fecal calprotectin value: 73.0 [30.0, 299.0] mcg/mg. 54% of patients were switched to FK-ada from the originator, 43% from another adalimumab biosimilar. Main reason for switching was administrative (economic, hospital decision): 88% from originator, 65% from biosimilar; Safety: 1% from originator, 22% from biosimilar.

		CD (N=113)	UC (N=25)	Overall (N=138)
Time since diagnosis (Years)	Median [Q1 - Q3]	11.8 [6.1, 20.1]	7.9 [4.9, 17.4]	11.4 [5.8, 20.1]
Previous surgical procedure	Yes	49/113 (43.4%)	0/24	49/137 (35.8%)
Previous or current non biologic therapy	Yes	75/111 (67.6%)	20/25 (80.0%)	95/136 (69.9%)
Response to steroids	None	26/112 (23.2%)	2/25 (8.0%)	28/137 (20.4%)
Current disease location (Montreal classification)	One location	65/113 (57.5%)	15/25 (60.0%)	80/138 (58.0%)
	>1 location	48/113 (42.5%)	10/25 (40.0%)	58/138 (42.0%)
Current disease activity NA= Not applicable	Clinical remission or mild to moderate (HBI score categorical)	110/113 (97.3%)	NA	NA
	Clinical remission or mild to moderate (Montreal classification)	NA	22/25 (88.0%)	NA

Table 1: Disease history and current status at baseline

Conclusion: FK-ada is a well-known valuable therapeutic option in the treatment of IBD. 52-week IDEA data will support identification of predictive factors of FK-ada treatment persistence after switch.

Disclosure: Pr Laurent Peyrin-Biroulet, Pr Yoram Bouhnik, Pr Stefanie Howaldt, Pr Carlos Taxonera are principal investigators for IDEA study sponsored by Fresenius Kabi.

PP0596

IMPROVED ADHERENCE AND PERSISTENCE IN INFLAMMATORY BOWEL DISEASE PATIENTS TAKING HIGH STRENGTH MESALAZINE TABLETS

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Introduction: While patients at all stages of ulcerative colitis (UC) are affected by non-adherence, those in symptomatic remission are particularly at risk, often taking less than 70% of their prescribed medication[1], with non-adherent patients being more likely to relapse[2]. Therefore, improving adherence to therapy has become one of the most important goals of patient management of chronic diseases. A high dose formulation of mesalazine was introduced in the French market in May 2020, Fivasa® 1600 mg, to reduce the pill burden. However, its impact on treatment compliance has not yet been assessed.

Aims & Methods: This was a retrospective cohort study in French patients using the IQVIA Longitudinal Prescription Data; a patient database based on retail pharmacy claims. An anonymised unique patient ID is assigned to each patient, which enables a longitudinal follow-up of these patients. The French database covers more than 9,600 retail pharmacies (~44% of national coverage). Patients were eligible if they had at least one treatment delivery of oral mesalazine initiated between June 2020 and April 2022, and a pre-observational period of at least 365 days prior to index date. UC and Crohn's disease patients cannot be distinguished in this database.

This study compared the adherence and persistence to oral mesalazine according to the tablet strength (high strength (HS), 1600 mg; standard strength (SS), 800-1000 mg and low strength (LS), 400-500 mg) delivered in the pharmacy.

The primary objective was 1) To compare the adherence to the prescribed dose of oral mesalazine over 1 year between groups of inflammatory bowel disease (IBD) patients in relation to the tablet strength (number of pills per day) provided in retail pharmacy and 2) To compare the persistence to the prescribed dose of oral mesalazine over 1 year between groups of IBD patients in relation to the tablet strength (number of pills per day) provided in retail pharmacy.

Results: The study population is shown in Table 1 and included 9929 patients after exclusion of patients without the pre-observability period, patients <18 years old, patients taking oral granules or patients taking different tablet strengths of mesalazine at index date.

Persistence to medication at 1 month after follow-up was 100%, 73.1% and 64.7% in the HS, SS and LS group, respectively. After 3 months of follow-up, it was 63.7%, 50.9% and 39.7% in the HS, SS and LS group, respectively. After 6 months of follow-up, it was 47.0%, 37.2% and 28.5% in the HS, SS and LS group, respectively. After 1 year of follow-up, it was 24.3%, 17.6% and 11.8% in the HS, SS and LS group, respectively.

Patient adherence to mesalazine tablets was 800/986 (81.1%) patients in the HS group, 5368/6947 (77.3%) patients in the SS group and 1493/1996 (74.8%) patients in the LS group (p<0.001).

At index date	Overall population N=9929	High strength N=986 (9.9%)	Standard Strength N=6947 (70.0%)	Low strength N=1996 (20.1%)
Age, mean (SD)	52.4 (18.6)	46.5 (17.2)	51.3 (18.1)	59.5 (18.6)
Gender, N (%)				
Female	5247 (52.8%)	468 (47.5%)	3693 (53.2%)	1086 (54.4%)
Male	4484 (45.2%)	500 (50.7%)	3107 (44.7%)	877 (43.9%)
Missing data	198 (2.0%)	18 (1.8%)	147 (2.1%)	33 (1.7%)
Number of co-medications, N (%)				
0	3031 (30.5%)	471 (47.8%)	2191 (31.5%)	369 (18.5%)
1	2761 (27.8%)	324 (32.9%)	1987 (28.6%)	450 (22.5%)
2	1663 (16.7%)	119 (12.1%)	1177 (16.9%)	367 (18.4%)
3 or more	2474 (24.9%)	72 (7.3%)	1592 (22.9%)	810 (40.6%)

Table 1: Baseline Characteristics.

Conclusion: These results showed a consistent lower adherence with the increase of pill burden. Both standard and low strength tablets were significantly associated with lower adherence to medication than the high strength tablet group. The impact of pill burden was more prominent on the persistence of medication in patients taking mesalazine and showed significantly/longer persistence in the high strength tablet group.

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Disclosure: HW, ES, OG and RL are employees of Tillotts Pharma AG.

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PP0597

A FOUR-STRAIN PROBIOTIC EXERTS A POSITIVE EFFECT ON SYMPTOMS OCCURING IN IBD IN THE ABSENCE OF INFLAMMATION (IBD/IBS OVERLAP)

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Introduction: Irritable Bowel Syndrome (IBS) is common in patients with inflammatory bowel disease, with a systematic review and meta-analysis of observational studies reporting an overall pooled prevalence of 39% and 35% in patients in clinical remission [1].

This subset of IBD patients report a reduced quality of life and increased anxiety and depression [2].

These patients are usually excluded from randomised clinical trials of treatments for IBS and therefore the evidence base for treatment strategies is small. A multi-strain probiotic (Symprove™, Symprove Ltd, Farnham, United Kingdom) has been shown to lead to improvement in overall symptom severity in patients with IBS and associated with decreased intestinal inflammation in patients with UC, but not in CD and is well tolerated in both populations [3,4].

Aims & Methods: Aim: To assess the efficacy of a liquid, multi-strain probiotic (Symprove) in patients with IBD whose disease is in remission as evidenced by the absence of inflammation but have ongoing IBS symptoms.

Methods: A single-centre double-blind, placebo controlled trial. Baseline demographic and clinical data; IBS symptom severity scores (IBS-SSS), IBD scores, bloods and faecal calprotectin were collected at week 0 and participants will receive either placebo or probiotic. At week 12 we repeated assessment of symptoms, repeat CRP and faecal calprotectin. Participants must be considered in remission from IBD as evidenced by a C-reactive Protein (CRP) <5mg/L and faecal calprotectin (FC) < 250ng/ml at recruitment but with ongoing gastrointestinal symptoms and meeting Rome IV criteria for the diagnosis of IBS.

Primary Outcome: 100-point drop in IBS-SSS post treatment

Secondary Outcomes: Change in IBS-SSS. Negative IBD outcomes including biochemical flare, steroid treatment or escalation of maintenance therapy.

	PLACEBO (N=21)	SYMPROVE (N=40)
MEAN AGE	45 years	41 years
GENDER	71% Female 29% Male	70% Female 30% Male
DIAGNOSIS	38% UC, 62% CD	23% UC, 77% CD
MEDIAN DURATION OF DIAGNOSIS	17.5 years	11 years
PREVIOUS RESECTION	38.1%	20%
STOMA	0	0
BIOLOGIC USE CURRENTLY	57%	70%

Results: 61 patients were enrolled in the study, 40 of whom received active agent and 21 of whom received placebo. The average age of patients was 42 years and 71% of patients were female. 28% (n=17) had Ulcerative Colitis and 72% (n=44) had Crohn's Disease, demographics are outlined in Table 1. Overall 45% of patients receiving the active agent achieved the end point compared to 33% of those receiving placebo (P=0.423). Separating patients with Crohn's Disease from those with UC, 50% of patients with UC receiving placebo achieved the endpoint compared to 44% of those receiving the active agent (p=1.000), however in Crohn's Disease 45% of those receiving the active agent achieved the endpoint compared to 29% of those receiving placebo (p=0.340). 23.8% (n=5) of those receiving placebo experienced a biochemical flare or steroid prescription or Escalation of Therapy compared to 12.5% of those receiving the active agent (p=0.266). In Crohn's Disease the mean IBS-SSS declined by 55 points in the placebo group (SD, -86) compared to 74 points in the active group (SD, -125).

Conclusion: A tendency among patients with Crohn's Disease towards improved IBS-SSS scores as well as fewer adverse IBD outcomes was noted in the probiotic group. Although this did not reach statistical significance this pilot study allows us to adequately power larger studies addressing this question.

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Disclosure: Active agent and Placebo were provided by Symprove Ltd, Farnham, United Kingdom.

PP0598

MANAGEMENT OF COMPLICATIONS IN PATIENTS WITH AN ILEOSTOMY: AN UMBRELLA REVIEW OF SYSTEMATIC REVIEWS FOR THE ENDOTRIAL CONSORTIUM

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Introduction: Ileostomies, whether temporary or permanent, may be required in patients with Crohn's Disease (CD) due to medically refractory disease or penetrating complications. However, standardized clinical care processes for patients with CD and a permanent ileostomy (PI) are lacking. The EndoTrial consortium aims to address this gap by developing pathways for care.

This umbrella review sought to consolidate existing knowledge by identifying, evaluating, and synthesizing findings from published systematic reviews on various aspects of clinical care for ileostomy patients, including managing high stoma output, disease recurrence, peristomal skin care, stoma pouching system and adhesives, behavioral interventions, mental health, and dietary management.

Aims & Methods: Following Cochrane guidelines for "Overviews of Reviews" and adhering to Preferred Reporting Items for Overviews of Reviews (PRIOR), we systematically searched major databases for relevant systematic reviews (SRs) or scoping reviews (ScR) published in English until January 5, 2024. Two independent reviewers screened articles, extracted data, and assessed quality using AMSTAR 2. Data synthesis was conducted narratively.

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Results: Of 1349 screened papers, we selected 24 reviews, including 22 SRs (8 with meta-analysis) and 2 ScRs. None exclusively focused on PI. Furthermore, nine reviews did not mention patients with inflammatory bowel disease (IBD), and only two reviews included patients with high-output ileostomy, highlighting a large evidence gap. The identified reviews covered eleven types of interventions, including ostomy care pathways, clinical management of high-output stoma, peristomal skin care, dietary management, nurse specialists, post-discharge continuous care, telemedicine, patient education, self-management interventions, management of parastomal varices, and chewing gum for postoperative ileus. Notably, none of the reviews exclusively examined stoma pouching system and adhesives, behavioral interventions, or mental health.

Interventions	SR	SR+MA	ScR	Key Findings
Ostomy care pathways	1	0	0	Ostomy care pathways (including preoperative education and counseling, postoperative education and discharge planning, and outpatient home visits and telephone follow-ups), may contribute to patient satisfaction and decrease in both hospital length of stay and hospital readmission rates
Clinical management of high-output stoma	1	1	0	No significant effect of common agents on stoma output. Variation in management strategies and discrepancies in recommendations were noted. Magnesium may benefit hypomagnesemia
Peristomal skin care	1	1	0	Systemic steroids and combination therapies are crucial for peristomal pyoderma gangrenosum, and adjunctive techniques showed no significant benefit in peristomal skin condition compared standard care
Dietary management	1	1	1	Dietary management findings from various reviews and guidelines indicate a range of strategies with inconsistent effects. Low fiber and fat diets seem generally beneficial, while alcohol intake is detrimental. Distal feeding shows promise in reducing complications and improving outcomes. For lower GI surgery, early enteral nutrition reduces hospital stay without added risks
1. Nursing care 2. Post-discharge continuous care	3	2	1	Importance of stoma care nurse specialists. Nursing interventions mitigate stoma-related quality of life issues. Validated HRQOL measurement instruments (QOLS, CO-HQOL-O, Stoma-QOL) identified for research use Continuing care improved health outcomes and care satisfaction for patients with stomas compared with routine care
Telemedicine	1	1	0	Telemedicine reduced the incidence of stoma and peristomal complications. Technologies-based interventions (e.g., sensor-based wearable technologies and computer-based) created a positive effect on the management of an ostomy, provided opportunities for enhancing self-efficacy, self-care, and self-management
1. Patient Education 2. Self-management interventions	4	1	0	Patient education interventions for individuals with stomas, have the potential to positively impact various aspects of patients' lives, including quality of life, psychosocial skills, and self-management. Self-management interventions showed positive effects on self-efficacy and self-reported self-management skills but had mixed outcomes for quality of life and objective measures of self-management. Nurse involvement and post-operative contact improved outcomes, leading to the development of a leaflet with a traffic light system for self-monitoring
Parastomal varices management	1	0	0	TIPS had superior efficacy in preventing recurrent hemorrhage from parastomal varices compared to local measures, surgical or non-operative, which were less effective
Prevention of postoperative ileus	1	1	0	Gum chewing after ostomy was found to significantly reduce the time to pass flatus and stool; it may reduce the time to resolution of postoperative ileus and length of hospital stay

Conclusion: Evidence for best practice interventions to help improve the quality of life for patients living with an ileostomy is limited and heterogeneous. Notably, systematic reviews have largely overlooked interventions for patients with PI. There is an urgent need for standardized clinical care processes and pathways tailored to the unique needs of this patient population, which is a key workstream of the EndoTrial Consortium

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PP0599

A SINGLE CENTER COHORT STUDY ON THE USE OF BIOLOGIC THERAPY IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE AND CANCER

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Introduction: In the years to follow we anticipate a rise in the prevalence of an aging population of inflammatory bowel disease (IBD) patients and a rise in cancer incidence. We are in urgent need of data to help us manage these patients.

We therefore followed a cohort of patients from a referral oncology hospital with prior and active cancer who received biologic therapy to control their IBD.

Aims & Methods: We first recorded data regarding inflammatory bowel disease characteristics: disease type, disease duration, disease severity, previous surgery or hospitalization and received therapy: type and duration of biologic therapy, prior use of immunosuppressants. We also recorded all available data regarding cancer disease: cancer site, stage, virus related cancer, family history of cancer, smoking status and received therapy (curative surgery, chemotherapy, hormone therapy, radiotherapy, immunotherapy). We monitored outcomes of IBD (steroid free clinical remission, IBD-related surgery or hospitalization) and outcomes of cancer (recurrence free survival, progression free survival, overall survival).

A blinded oncologist was asked to retrospectively evaluate all available data regarding cancer prognosis (stage, histology, response to therapy) and to categorize patients in 3 groups: low, intermediate or high risk to recur or progress.

Results: A total of 30 patients were included, 57 % female, median age 60 years, 60% Crohn's Disease, median disease duration 9.5 years, 26% prior IBD surgery, 33% prior azathioprine use, 36% smokers. They were treated with anti-TNF therapy (n=19) anti-integrin therapy (n=8) and anti-IL 12/23 therapy (n=4). The majority (20%) of patients had a history of hematologic

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malignancy (lymphomas, myeloproliferative disorders, leukemias), 13% with a virus related cancer and 66% with curative surgery. Patients were classified retrospectively as 63% low risk, 13% intermediate risk and 10% high risk. Four patients) with active cancer (2 with lung cancer and 2 with myeloproliferative disorders) continue to receive biologic therapy for IBD and oncological therapy with no signs of progression from both diseases. All but 3 patients have remained in steroid free clinical remission: 3 had IBD-related surgery. Median follow up under biologic therapy was 5.5 years: only 2 patients showed cancer recurrence: 1 patient with breast cancer developed bone metastases and finally died, and 1 patient with colon cancer developed bladder cancer and died as well. Both patients were classified as high risk.

In both patients biologic therapy was interrupted and that resulted in administration of high-dose steroids and IBD-related hospitalizations and infections (pneumonia, C Difficile colitis).

Conclusion: In this cohort of cancer patients with IBD treated with biologics and followed up for more than 5 years the risk of cancer recurrence was low and not unexpected in oncology terms.

Biologic therapy was successful in the majority of patients and kept them off surgeries and steroids for their underlying IBD. We need more smart and innovative ways to study this particular clinical scenario in the future.

Disclosure: Nothing to disclose.

PP0600

PHASE 1 BIOMARKER ANALYSES OF GS-1427 A NOVEL ORAL α4β7 INTEGRIN INHIBITOR FOR INFLAMMATORY BOWEL DISEASE DEMONSTRATE α4β7 SELECTIVITY AND ONCE DAILY DOSING

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Introduction: GS-1427 is a novel selective small molecule inhibitor of the integrin receptor α4β7 in clinical development for once daily oral dosing in inflammatory bowel disease (IBD). Inhibition of α4β7 is an established efficacious therapeutic option for the treatment of Ulcerative Colitis (UC) and Crohn's Disease, but current treatment requires intravenous or subcutaneous injection. A Phase 1 first-in-human study evaluated the pharmacokinetics, safety, tolerability, and exploratory biomarkers of GS-1427 in healthy subjects. The biomarker objectives in Phase 1 were to confirm preclinical findings of extended α4β7 receptor occupancy (RO) supporting once daily dosing, selectivity for α4β7 and to inform Phase 2 dose selection.

Aims & Methods: Biomarkers were evaluated before and after single (20mg-1000mg fasted, SAD) and multiple (20-500mg fed, once daily for 14 days, MAD) ascending doses GS-1427.

Using fluorescently labeled MAdCAM-1, α4β7 RO was assessed in fresh whole blood on CD4+CD45RO+β7+ memory T cells; α4β1 RO was evaluated on CD4+CD45RO+α4+β1+β7- memory T cells using fluorescently labeled tripeptide leucine aspartic acid valine (LDV). RO was calculated as a percent using the fluorescence intensity values measured post-dose and pre-dose. Changes in the frequency of β7+CD4+ memory T cells by flow cytometry and *CCR9* gene expression by RNAseq in whole blood served to evaluate downstream functional α4β7 pharmacodynamics (PD); absolute lymphocyte counts (ALC) in fresh whole blood and frequencies of CD34+ Hematopoietic stem cells (HSC) in viably frozen PBMC were measured to estimate PD downstream of α4β1.

Results: α4β7 RO analyses post single and multiple once daily doses of GS-1427 demonstrated sustained and prolonged RO ≥95% confirming pre-clinical findings and supporting a once daily dosing regimen. Duration of α4β7 RO ≥95% post-dose was dose dependent. Doses ≥80mg GS-1427 demonstrated α4β7 RO ≥95% for 24 hours, starting with the first dose on Day 1 and persisted until Day 14.

The highest dose group in the MAD demonstrated extended α4β7 RO ≥95% up to 72 hours in 100% of subjects after the last dose on Day 14. Corroborating α4β7 RO findings, analyses of %β7+CD4+ memory T cells demonstrated dose and time dependent significant increases from baseline compared to placebo, which reached 40% on average in the highest dose group in the MAD. Dose dependent significant increases of *CCR9* mRNA expression levels compared to placebo were observed for doses ≥80mg GS-1427; up to ~2.5-fold increase on average were measured in the higher dose groups in the MAD.

Unlike α4β7 RO, a4β1 RO was partial and transient across the doses tested, and RO levels returned to baseline within ~24 hours post-dose in both SAD and MAD. Individual values of a4β1 RO 24 hours post multiple doses GS-1427 on Day 14 were below the assay's lower limit of quantification. The median maximum a4β1 RO at C_{trough} was estimated to be ~20% in the highest dose group.

Supporting findings for a4β1 RO, percent change of ALC was transient and changes of CD34+ HSC frequencies were highly variable which was expected given the low frequency of this population and were within the natural variation across all GS-1427 doses tested.

Conclusion: In conclusion, the exploratory biomarker results in this Phase 1 study in healthy subjects demonstrating extended α4β7 RO and α4β7 selectivity together with the pharmacokinetics, safety, and tolerability data (abstract in preparation) support further development of GS-1427 with a once daily dosing regimen in a Phase 2 study in IBD.

Disclosure: All authors are employees and shareholders of Gilead Sciences.

PP0601

ADVANCED THERAPY IN PATIENTS WITH ULCERATIVE COLITIS AND MALIGNANT DISEASE: A CASE STUDY

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Introduction: Advanced Therapy (AT) should be carefully considered for patients who have both inflammatory bowel disease and a malignancy. The aim of administering AT is to avoid hospitalization and surgery in patients with otherwise intractable inflammatory bowel disease; however, the AT may result in exacerbation or recurrence of the malignancy. After careful consideration, we have been treating patients with both inflammatory bowel disease and a malignancy with AT. We here report the results of AT in patients with ulcerative colitis (UC) and malignant disease.

Aims & Methods: From January 2012 to December 2023, we treated five patients with UC and malignant disease (excluding colitic cancer) at the Ikoma Gastroenterology and Proctology Clinic. Age, sex, disease duration, disease extent, severity, clinical course, intractable or not, use or not of azathioprine, type of AT, duration of AT, curability of malignant disease, type and duration of AT after onset of malignant disease, whether the onset of malignant disease was diagnosed during AT, duration of follow-up after onset of malignant disease, outcome of UC after onset of malignant disease, and outcome of malignant disease.

Results: The results are shown in the Table. Age: range (median) (years) **:** 45-69(56). Sex:Male/Female: 0/5. Disease duration: range (median) months) : 114–289 (245) . Extent of UC : total colon/left colon: 1/4**.** Severity

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of UC : Severity :severe/moderate : 0/5. Clinical course: chronically persistent/relapsing–remitting: 0/5. Use of azathioprine: yes/no: 0/5.Three patients developed malignant disease during AT (cervical cancer, breast cancer, squamous cell carcinoma of the scalp), the remaining two (breast cancer, thyroid cancer) commencing AT only after the cancer's onset. Four of the patients continued AT, the exception (a patient with squamous cell carcinoma) received adalimumab. After the onset of malignant disease, UC remained in remission in all patients and no new malignant disease developed.

AT	One drug: UST, ADA (2 patients) Two drugs: GLM→TOF (1 patient)
	Three drugs: IFX→UST→TOF→UST (1 patient)
	Five drugs: ADA→GLM→VED→TOF→UST (1 patient)
Duration AT: range (median) (months)	24–83 (41)
Curability of malignancy: yes/no	5/0
Type and duration of AT after onset of malignant disease (months)	UST(30), VED3→TOF24→UST36 total 63 UST 30, GLM1→TOF59 total 60
Malignancy first diagnosed during AT: yes/no	Yes: TOF, GLM, ADA (3 patients)/no (2 patients)
Duration of follow-up after onset of malignant disease: range (median)months	30–378 (52)
Outcome of UC after onset of malignant disease: remission/relapse	5/0
Outcome of malignant disease: exacerbation or recurrence/no exacerbation or recurrence	0/5

(AT:Advanced Therapy, UST:Ustekinumab, ADA:Adalimumab, GLM:Golimumab, TOF:Tofacitinib, IFX:Infliximab, VED:Vedolizumab)

Conclusion: In patients with UC who develop a malignancy before or during AT and achieve cure of that malignancy, continuing AT can prevent progression of the UC without exacerbation or recurrence of the malignant disease.

Disclosure: Nothing to disclose.

PP0602

A NOVEL TREATMENT STRATEGY OPTIMISING MEDICAL AND SURGICAL CARE LEADS TO IMPROVED HEALING OF CROHN'S PERIANAL FISTULAS: A REAL-WORLD PROSPECTIVE COHORT

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Introduction: Despite best standard care, Crohn's perianal fistula healing rates remain low. We herein evaluated the efficacy of a novel treatment strategy optimising medical and surgical care in Crohn's perianal fistulas.

Aims & Methods: A real-world prospective cohort study evaluated the efficacy of a new treatment strategy in adults with Crohn's perianal fistulas at a single tertiary centre. Sixty patients were sequentially included and followed up for a minimum of 12 months. The treatment strategy implemented structured clinician reviews and a novel treatment algorithm centralised around three dynamic stages of care. The stages of care were directed towards patients' clinical disease activity: stage A, active disease, focused on medical and surgical optimisation; stage B, optimised disease with a seton in situ, focused on definitive surgical management; and stage C, healed disease, focused on treatment maintenance. Endpoints included clinical healing, radiologic remission and response on magnetic resonance imaging, and proportion of patients requiring treatment optimisation.

Results: At baseline, 47% were in stage A, 5% stage B, and 48% stage C. Median age was 40 years (IQR: 32-49), 52% were male, and median perianal disease duration was 9 years (IQR: 2-16). After a median of 22 months follow-up (IQR: 16-25), 83% had clinical healing which was significantly higher than baseline inclusion rates (83 vs 48%, P<0.01). For patients in stage A at baseline, 68% achieved clinical healing with estimated rates of 39% (95%CI: 24-60) and 85% (95%CI: 64-97) at 1 and 2 years, respectively. There was no significant difference in follow-up between patients who achieved clinical healing and those with non-healing fistulas.

	Stage A	Stage B	Stage C	Overall
N	28	3	29	60
Clinical follow-up, median months (IQR)	21.8 (15.9, 25.1)	20.9 (16.5, 22.6)	22.6 (15.9, 24.3)	21.8 (15.8, 24.6)
Clinical outcomes, n	28	3	29	60
Non-healing, n (%)	9 (32.1)	0 (0.0)	1 (3.4)	10 (16.7)
Healing, n (%) *	19 (67.9)	3 (100.0)	28 (96.6)	50 (83.3)
Radiologic outcomes, n	26	2	25	53
Radiologic remission, n (%) #	6 (23.1)	1 (50.0)	20 (80.0)	27 (50.9)
Radiologic response, n (%) ‡	18 (69.2)	1 (50.0)	6 (24.0)	25 (47.2)

Table 1: Clinical and radiologic outcomes based on baseline stage of care.

** Clinical healing: absence of draining perianal fistula external openings despite gentle finger compression.*

Radiologic remission: VAI=0, VAI_{infl}=0, MAGNIFI-CD=0, or FS=6.

‡ Radiologic response: VAI reduction >3, MAGNIFI-CD reduction >4, or FS increase ≥1.

[FS = Fibrosis Score; MAGNIFI-CD = Magnetic Resonance Index for Assessing Fistulas in Patients with Crohn's Disease; VAI = Van Assche Index; VAI_{infl} = VAI-inflammatory subscore].

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Overall, radiologic remission was observed in 51% and response in 47%. For patients in stage A at baseline, 23% achieved radiologic remission which was significantly higher than baseline inclusion rates (23 vs 4%, *P*<0.05) and 69% radiologic response (Table 1). Patients who achieved clinical healing had significantly less radiologic disease activity compared to non-healing fistulas. Overall, 53% had biologic dose escalation and 10% switched to an alternative biologic agent with the new treatment strategy. Thirty-seven percent underwent a median of two perianal surgical interventions (IQR: 1-3), 15% had closure or ablative techniques performed, and 15% had seton removal only.

Conclusion: A novel treatment strategy optimising medical and surgical care resulted in high rates of clinical healing and improved radiologic disease activity of Crohn's perianal fistulas. Controlled-matched studies are needed.

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PP0603

EFFECTIVENESS AND SAFETY OF THIOPURINES IN INFLAMMATORY BOWEL DISEASE PATIENTS HARBORING NUDT15 MUTATIONS: A RETROSPECTIVE ANALYSIS

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Introduction: Thiopurines have a well-recognized role as a maintenance therapy in inflammatory bowel disease (IBD). While Nudix hydrolase 15 (NUDT15) polymorphisms are well-known to affect thiopurine metabolism predisposing to leucopenia, especially in Asian population, their appropriate dosing and efficacy are unclear in NUDT15 heterozygous patients.

Aims & Methods: We aimed to study the effectiveness and safety of thiopurine (azathioprine and 6-mercaptopurine) use in NUDT15 mutant patients. We also planned to identify the maximum tolerated dose of the drug among them. We retrospectively analyzed the database of our IBD patients for NUDT15 gene variants (c.415 C>T). Records of those IBD patients who received thiopurine were further studied for dose of the drug at leucopenia, time to develop leucopenia, severity of leucopenia, time to resolution of leucopenia, and maximum tolerated dose of thiopurines. We also compared them with a historical control of IBD patients with wild variants of NUDT15, and TPMT genes.

Results: Of the 1440 patients of IBD tested for NUDT15 polymorphism, 118 (8.2%) patients were detected to have NUDT15 mutation (5 homozygous, 113 heterozygous). Out of these 118 patients, 51 had information available regarding thiopurine use. The rest were excluded for various reasons like incomplete data, those who never were on azathioprine, or were lost to follow-up. Among these 51 patients, 46 were heterozygous for NUDT15 (c.415 C>T) (90.2%), and 5 patients were homozygous (9.2%). In the NUDT15 heterozygous group, the mean age was 39 ± 13 years (30 males). Ulcerative colitis (UC) and crohn's disease (CD) were diagnosed in 38 and 8 patients. Proctitis, left-sided colitis, and extensive colitis were seen in 3, 11, and 19 patients. Among the CD patients, ileal, colonic, and ileocolonic diseases were seen in 1, 1, and 5 patients, respectively, whereas inflammatory and stricturizing phenotypes in 6 and 2 patients, respectively. Twenty-eight patients were on 5-ASA, 14 on corticosteroids, 2 on infliximab, and one patient was on adalimumab. Twenty (43.5%) NUDT15 heterozygous patients developed leucopenia at a mean azathioprine equivalent dose of 81.25 mg (SD 34.3) after 19 weeks (SD 56.9, median 4 weeks). All NUDT15 homozygous patients developed leucopenia at a mean (SD) dose of 65 mg (SD 33.5, median 50 mg) after 1-8 weeks of thiopurine exposure. Concurrent use of 5-aminosalicylate with or without steroids was not significantly different in those with or without leucopenia. Four (8.7%) patients in the heterozygous group and 2 (40%) patients in the homozygous group required hospitalization due to leucopenia. The incidence of cytopenia was significantly higher in the NUDT15 heterozygous group compared to the control group (41.3% vs 7.8%, p <0.001), the maximum tolerated dose of azathioprine was significantly lower in the NUDT15 heterozygous group (mean ± SD 66.23 ± 33.24 mg vs 93.87 ± 27.12 mg, respectively, p =0.02). The mean duration to develop cytopenia was also earlier in the heterozygous group vs control (19.63 ± 58.43 weeks vs 70.5 ± 63.17 weeks, p=0.05). Seven (35%) heterozygous patients could be maintained at a lower dose of thiopurine (71.43 ± 17.25 mg), and in 13 (65%), it has to be permanently stopped. Twenty-five (78% of available data of 32 patients) underwent remission while on thiopurines.

Conclusion: NUDT15 mutants are more likely to develop thiopurine-induced leucopenia. Thiopurines should be avoided in NUDT15 homozygous patients but can be used cautiously at lower dose with frequent monitoring among heterozygous patients.

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PP0604

PHARMACOKINETICS AND TARGET ENGAGEMENT FOLLOWING A SINGLE SUBCUTANEOUS INJECTION OF AN ANTI-TL1A ANTIBODY, TEV-48574, IN CYNOMOLGUS MACAQUES

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Introduction: Tumor necrosis factor-like ligand 1A (TL1A) is a pleiotropic cytokine involved in immune-mediated pathways leading to inflammation and fibrosis. TEV-48574 is a human IgG1 monoclonal antibody that selectively inhibits TL1A binding to the death receptor 3 (DR3) with greater potency compared to the decoy receptor 3 (DcR3) which could preserve the natural homeostatic mechanism.[1] TEV-48574 has previously been shown to reduce fibrosis, inflammation, and mortality in animal models of asthma and IBD.[1] TEV-48574 is in clinical development as a potential treatment for ulcerative colitis and Crohn's disease.

Aims & Methods: The objectives of the study were to characterize the pharmacokinetic properties of TEV-48574 and evaluate target engagement through analysis of free TL1A concentrations post-dose. Biologic-naïve male cynomolgus monkeys (n = 4) were each administered a single 5 mg/kg dose of TEV-48574 by subcutaneous (sc) injection. Serum samples were collected pre-dose and at selected time points up to last day of the study, Day 42. Serum samples were analyzed using ELISA to determine TEV-48574 concentrations and the presence of monkey anti-TEV-48574 antibodies (ADAs). Pharmacokinetic parameters were determined using non-compartmental methods, and free TL1A levels were measured using a ligand binding assay.

Results: Absorption following sc TEV-48574 administration was consistent, median T_{max} was 1 day with a range from 0.25 to 3 days, mean ± SD C_{max} was 42.2 ± 10.8 µg/mL. Mean t½ was 5.55 ± 4.10 days and $AUC_{0-\infty}$ was 473 ± 134 day*µg/mL. ADAs were confirmed on day 14 in serum sample for 1 monkey and on Days 21 and 35 for serum samples from all 4 animals. Administration of TEV-48574 resulted in suppressed free TL1A levels immediately post-dosing. Free TL1A remained below the limit of quantitation until Day 21 and below pre-dose baseline values until Day 42.

Conclusion: TEV-48574 demonstrated an acceptable pharmacokinetic profile with maximum serum concentrations observed within 1 day post-dose of a single sc injection of 5 mg/kg TEV-48574 in 4 male cynomolgus monkeys. Target engagement was shown by a rapid and sustained TEV-48574-mediated suppression of free TL1A in serum, supporting further development of TEV-48574 in patients with ulcerative colitis and Crohn's disease.

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Disclosure: All authors are employees of Teva Global Research and Development. TEV-48574 is being developed in a partnership between Teva and Sanofi.

PP0605

COMPREHENSIVE DISEASE CONTROL IN PATIENTS WITH ULCERATIVE COLITIS: REFINEMENT OF AN INDIVIDUAL, MULTI-COMPONENT ENDPOINT USING DATA FROM THE FILGOTINIB SELECTION TRIAL

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Introduction: Ulcerative colitis (UC) treatment requires a patient-centric efficacy endpoint to assess treatment benefits, enabling a shift towards a personalized approach. Comprehensive disease control (CDC) is an individual multi-component endpoint based on findings from a patient and expert Delphi consensus on remission assessment in UC. CDC was defined as IBDQ remission, partial Mayo Clinic Score (pMCS) remission, biomarker (faecal calprotectin [FCP]) remission and endoscopic improvement.

Aims & Methods: We evaluated different definitions of CDC, and the contribution of different CDC components to the net treatment benefit (NTB) of filgotinib 200 mg (FIL200), an oral, once-daily, Janus kinase 1 inhibitor approved for UC treatment following the SELECTION trial (NCT02914522). CDC was assessed at week 58 (W58) of SELECTION in adult patients with active UC treated with FIL200 or placebo (PBO). The CDC definitions evaluated were: 4-component CDC (CDC4) with FCP <150 µg/g, CDC4-150; CDC4 with FCP <250 µg/g, CDC4-250; 5-component CDC (with histological remission [CDC5]) with FCP <150 µg/g, CDC5-150; and CDC5 with FCP<250 µg/g, CDC5-250. The proportion of patients achieving each combination of components was evaluated. NTB of FIL200 was assessed for each endpoint using generalized pairwise comparisons and is reported here for CDC4-150 (and excluding different components).

Results: At W58, 21.6%, 26.6%, 19.1% and 22.6% of FIL200-treated patients achieved CDC4-150, CDC4-250, CDC5-150 and CDC5-250, respectively. Among patients who did not achieve CDC4-150, 0.5% of patients did not achieve this composite endpoint because they did not achieve pMCS remission; whereas 5.0%, 10.1% and 6.0% did not do so because they did not achieve IBDQ remission, biomarker remission or endoscopic improvement, respectively. Positive overall NTB of FIL200 versus PBO was demonstrated with all CDC definitions, including those with histological remission (CDC5). For CDC4-150 at W58, excluding pMCS remission did not substantially change the NTB (Δ −0.33; *Table*) versus excluding IBDQ or biomarker remission, or endoscopic improvement (Δ 3.87, 6.93 and 3.20, respectively). Similar NTB results were seen for all CDC definitions, with or without histological remission.

Conclusion: CDC is a stringent, patient-level, multi-component endpoint achieved by ~1/5 of FIL200-treated patients at W58. While CDC4-150 could be the optimal operational definition, the addition of histological remission resulted in similar stringency, potentially allowing for the use of dif-

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ferent CDC definitions based on context or resources. Additional NTB of FIL200 was identified in patients when IBDQ or biomarker remission or endoscopic improvement were excluded. Overall, CDC appears to be a suitable endpoint to assess individual comprehensive benefit in patients with UC. Prospective validation is needed.

	W10				W58			
	Number of pairs			NTB (Δ NTB)	Number of pairs			NTB (Δ NTB)
	Favourable	Unfavourable	Neutral		Favourable	Unfavourable	Neutral	
CDC4-150	14905	3616	122932	7.98 (–)	3931	1092	14497	14.47 (–)
CDC4-150 excluding IBDQ remission	3204	1760	117968	9.00 (1.02)	900	146	13451	18.33 (3.87)
CDC4-150 excluding pMCS remission	807	898	121227	7.92 (−0.06)	90	155	14252	14.13 (−0.33)
CDC4-150 excluding biomarker remission	6966	5525	110441	9.00 (1.02)	1760	408	12329	21.40 (6.93)
CDC4-150 excluding endoscopic improvement	7685	2538	112709	11.62 (3.64)	1056	432	13009	17.66 (3.20)

CDC4-150 was defined as simultaneous achievement of IBDQ remission (defined as an IBDQ total score ≥170), pMCS remission (defined as a pMCS ≤2 and no individual rectal bleeding, stool frequency or Physician's Global Assessment subscore >1), biomarker remission (defined as an FCP concentration <150 µg/g) and endoscopic improvement (defined as a Mayo endoscopic subscore of 0 or 1). pMCS is the sum of Physician's Global Assessment, rectal bleeding and stool frequency subscores. Generalized pairwise comparisons were used to compare each patient in the FIL200 group with each patient in the PBO group according to each outcome. The number of favourable pairs is the number of superior outcomes of FIL200 compared with PBO. The number of unfavourable pairs is the number of inferior outcomes of FIL200 relative to PBO. The number of neutral pairs is the number of tied outcomes of FIL200 relative to PBO. NTB for CDC4-150 was calculated as the proportion of favourable outcomes minus the proportion of unfavourable outcomes and represents the net probability that a random patient has a better outcome with FIL200 than with PBO. For example, a 10% NTB indicates a 10% greater probability of a positive outcome with FIL200 than with placebo. The NTB for CDC4-150 with excluded outcomes is the NTB for CDC4-150 achievement without including the given outcome in the definition of CDC4-150. The difference (Δ) in NTB is the additive contribution to the NTB of excluding a given component of CDC4-150 relative to achievement of full CDC4-150.
CDC4-150, four-component comprehensive disease control with faecal calprotectin <150 µg/g; FCP, faecal calprotectin; FIL200, filgotinib 200 mg; IBDQ, Inflammatory Bowel Disease Questionnaire; NTB, net treatment benefit; PBO, placebo; pMCS, partial Mayo Clinic Score; W10, week 10; W58, week 58.

Table. NTB with exclusion of different components of the CDC4-150 endpoint.

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PP0606

EFFECTIVENESS OF A SECOND-LINE ANTI-TNFα AGENT AFTER FIRST ANTI-TNFα FAILURE COMPARED TO A DIFFERENT MECHANISM OF ACTION IN ULCERATIVE COLITIS: CAMBIACU STUDY

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Introduction: Ulcerative colitis (UC) is a chronic and relapsing disease for which the number of biological therapies available are increasing over the years. Data comparing the effectiveness of a second anti-TNFα (aTNF) and the switch to a different mechanism of action (MoA) such as Ustekinumab (UST), Vedolizumab (VDZ), Filgotinib or Tofacibinib (JAK) is lacking.

Aims & Methods: We performed a retrospective and multicenter study (6 Andalusian sites). Patients diagnosed of UC who had failed a first biological therapy with an aTNF and started a second-line biological therapy with another aTNF or a different MoA (UST, VDZ, JAK) were included. Clinical response and remission were assessed at weeks 16 and 52. The aim of our study was to evaluate the effectiveness of a second-line biological therapy with aTNF compared to a different MoA after a first aTNF failure.

Results: 185 UC patients were included. There were no differences in the demographic data except for disease duration (9 years with MoA vs 13 years in aTNF, p=0.02). As first aTNF, infliximab (IFX) was the most common in both groups (49% MoA group, 56% second aTNF), followed by adalimumab (ADA) (48% MoA group, 33% second aTNF). In the second aTNF group, immunomodulators were significatively more used (60% vs 38%, p=0.003). There were no differences in the concomitant use of steroids. 80 patients received a different MoA (37 VDZ, 24 UST, 19 JAK) and 105 a second aTNF (34 IFX, 60 ADA, 11 golimumab). At week 16, 52% of patients were in clinical response (70% with another MoA vs 54% with second aTNF, p=0.07; IFX 65%, ADA 53%, VDZ 68%, UST 71%, JAK 74%, *p*=0.07), being 15.6% more likely to achieve response with another MoA. Clinical remission at week 16 was obtained by 36% (39% with another MoA vs 33% with second aTNF, *p*=0.6; IFX 47%, ADA 30%, VDZ 35%, UST 46%, JAK 37%, *p*=0.1). At week 52, 48% achieved clinical response (51% with another MoA vs 45% with second aTNF, *p*=0.8; IFX 50%, ADA 42%, VDZ 49%, UST 54%, JAK 53%, *p*=0.5) and 39% were in clinical remission (40% with another MoA vs 38% with second aTNF, *p*=0.53; IFX 50%, ADA 32%, VDZ 35%, UST 38%, JAK 53%; *p*=0.06). When the effectiveness is analyzed by first aTNF used, independently of the aTNF, subsequent use of a different MoA achieves better results although not significant compared to a second aTNF. Partial Mayo and calprotectin showed a significant decrease at weeks 16 and 52 in both groups. Hemoglobin and albumin levels remained similar during follow-up in both groups.

Conclusion: Second-line biological therapy is effective regardless of biological therapy used in UC, but switching to a different MoA tend to achieve better results in terms of clinical response and remission compared to a second aTNF.

Disclosure: Nothing to disclose.

PP0607

PREVALENCE OF BILE ACID DIARRHOEA IN PATIENTS WITH ACTIVE MICROSCOPIC COLITIS; A PROSPECTIVE STUDY

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Introduction: Coexistence of bile acid diarrhoea (BAD) and microscopic colitis (MC) has been described (1), and treating this with sequestrants could reduce the need for budesonide. Fasting plasma 7α-hydroxy-4-cholesten-3-one (C4) values >46 ng/mL are specific for BAD but sensitivity is only 47% (37-57%) compared with tauroselcholic [75-Se]
acid (SeHCAT) retention <10% (2). C4 values between 33 and 46 are in a diagnostic grey zone. Budesonide directly affects the bile acid homeostasis making the temporal relation between treatment and testing important. Fortunately, unlike SeHCAT testing, C4 may be sampled immediately when a patient presents with a flare of MC.

Aims & Methods: To assess the prevalence of coexisting BAD in patients with active MC.

Consequtive patients with MC and a flare fulfilling the Hjortswang diarrhoea criteria (≥3 stools or ≥1 watery stool per day) and a negative test for infectious diarrhoea were included. Patients were treated with budesonide 9 mg once daily for 6 weeks. Clinical assessment, Bristol stool charts, and fasting C4 were obtained at baseline, during the last treatment week and 4 weeks after end of treatment. Remission was defined as a daily mean of <1 watery bowel movement and <3 total bowel movements [EudraCT: 2019-002762-12].

Results: 49 patients were included; 26 (53%) with collagenous colitis. Mean age was 60 (range 55-70) years, 22% current smoker, none on NSAIDs. Six (12%; 95%CI: 5–25%) of 49 patients had BAD diagnosed by C4. Another 3 patients had grey zone C4 values. Mean C4 did not change during budesonide treatment, 22 and 23 ng/mL at baseline and at week 6, respectively. SeHCAT was performed outside the study in 26 patients of whom 3 had retention of <10% and 6 of <15%. The mean total number of stools per day declined from baseline 7.9 to 1.9 and watery stools from 3.7 to 0.3 during budesonide treatment. Remission was obtained in 45. Twenty-nine had relapsed four weeks after end of budesonide treatment, 5 of 6 (83%; 95%CI: 36–100%) with MC and BAD, and 24 of 43 (56%; 95%CI: 40–71%) with MC and not BAD.

Conclusion: During an inflammatory flare of MC, 12% (5–25%) had C4-defined BAD. However, C4 only identifies about half of patients with BAD compared with the diagnostic gold standard SeHCAT, meaning about 25% of patients with MC could have coexisting BAD. Screening for BAD may have merit in MC patients with relapse of diarrhoea after budesonide treatment.

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PP0608

RELATIONSHIP BETWEEN INDUCTION SERUM VEDOLIZUMAB CONCENTRATION AND LONG-TERM IBD OUTCOMES : A PROSPECTIVE, MULTICENTRIC, COHORT STUDY (VEDOPREDIRESP)

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Introduction: VEDOPREDIRESP is a phase IV, multicentric, prospective, observational study aiming to evaluate the relationship between induction serum trough concentration and treatment outcomes over 52 weeks in IBD patients treated with intravenous (IV) vedolizumab (VDZ)

Aims & Methods: Real-world data from patients with UC or CD treated with IV vedolizumab (300 mg at W0, W2, W6 and e8w) were collected prospectively in five centres. Clinical activity scores were evaluated (CDAI for CD and pMayo score for UC) at every infusion visits; VDZ trough levels were collected at W2,W6 and W14 ; their results were kept blinded to investigators. VDZ exposure-response relationship was evaluated overall indication by indication and based on baseline characteristics. Three outcomes were defined: (i) group 1: patients in clinical remission under VDZ with no treatment modification until W52, (ii) group 2: patients in clinical remission at W52 with an increased VDZ dose (300mg/4W) and (iii) group 3: patients who stopped VDZ due to loss of response, adverse events or surgery. Clinical remission was defined as CDAI < 150 for CD and pMayo score < 3 for UC.

Results: 115 patients were included (UC, n = 67; CD, n = 48) with a median age of 39 years; (55% men). 87% had prior anti-TNF exposure (46% with two prior anti-TNF). Two resections were performed during the study in CD patients. Severe adverse events were reported in 29% of cases due to loss of response in all cases. 49% of patients were in clinical remission at the end of follow-up (W52) (54% for UC, 47% for CD). 45% of patients had needed a VDZ intensification during follow-up and 50% of these patients were in clinical remission at W52. Mean VDZ serum levels were comparable at W2, W6, and W14 in the three groups (P:NS). These results were similar analyzing Group 1 and 2 all together versus Group 3 (Table 1). The relapse-free survivals were comparable in CD and UC patients (Log-rank test; p=0.45) (Figure 1). In groups 1, 2 and 3, results were similar in CD and UC patients respectively. Finally, mean VDZ serum levels were comparable just before VDZ intensification in patients responders and no responders to optimization. After logistic regression, no factor (clinical caracteristics of patients or disease at inclusion, VDZ levels at each time) was significantly associated Group 3.

Conclusion: In this prospective, multicentric cohort study, serum VDZ concentrations during induction were not predictive of outcomes in both UC and CD patients. So, these measurements are not useful in clinical practice.

Disclosure: TAKEDA

PP0609

WITHDRAWN

PP0610

INSIGHTS FROM THE POLISH REGISTRY OF MICROSCOPIC COLITIS: ANALYSIS OF THE FIRST 165 PATIENTS ACROSS NINE ACADEMIC GASTROENTEROLOGY REFERENCE CENTERS IN POLAND

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Introduction: Microscopic colitis (MC) represents a spectrum of chronic inflammatory bowel diseases (IBD) characterized by chronic watery diarrhea and distinctive histologic features. The data concerning the epidemiology of microscopic colitis in Poland are non-existing or scarce.

Aims & Methods: The Polish Registry of Microscopic Colitis was established to better understand the clinical characteristics, treatment patterns, and outcomes of MC patients in Poland.

We present retrospective and prospective data findings from the initial cohort of 165 patients enrolled in the registry, spanning nine academic gastroenterology centers across Poland. Data collection included demographics, clinical presentation, histopathological features, treatment modalities, and clinical outcomes.

Results: The median age at diagnosis was 57 years, with 36% of patients diagnosed before age 45. The majority (80%) had a prior MC diagnosis, with an average delay of approximately 3 years from symptom onset to diagnosis. MC occurred 3.5 times more frequently in women. Lymphocytic colitis (LC) was the most common subtype (66.7%), followed by collagenous colitis (CC) (24.1%) and incomplete colitis (IC) (9.3%). Women were more predisposed to CC. Normal colonoscopy results were observed in 71.5% of patients. Approximately 48% were active or former smokers, with smokers more likely to have CC. Predominant symptoms included watery diarrhea (86.1%), fecal urgency (66.1%), abdominal pain (58.8%), and bloating (62.4%). Weight loss during exacerbation of symptoms was reported in nearly one-fifth of patients. One-third of MC patients reported abdominal pain consistent with Rome IV criteria for irritable bowel syndrome (IBS) during medical interviews. Additionally, nearly half of the patients had a confirmed Covid-19 infection based on authorized testing. Budesonide MMX demonstrated the highest efficacy (92.4% reported improvement), with other treatment modalities showing varied effectiveness. Three patients with severe MC were successfully treated with adalimumab.

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	Number of patients using treatment	Number of patients reporting improvement after treatment	Number of patients reporting no improvement after treatment	Improvement after treatment (in percentage)
Budesonide MMX	92	85	7	92.4%
Budesonide	23	21	2	91.3%
Loperamide	34	15	19	44.1%
Mesalazine	18	5	13	27.8%
Rifaximine	40	20	20	50.0%
Probiotics	45	15	30	33.3%

Table 1. Polish Microscopic Colitis Registry. The effectiveness of treatment following therapies.

Conclusion: The Polish Registry of Microscopic Colitis provides comprehensive insights into the clinical features and treatment outcomes of MC in Poland. Budesonide MMX emerged as an effective treatment option, while biologic therapy showed promise in refractory cases. Our findings suggest a potential overlap of symptoms between MC and IBS, highlighting the importance of thorough clinical assessment. Our findings inform evidence-based practice and highlight the importance of multidisciplinary collaboration in MC management. The multicenter nature of the Polish MC Registry provides a comprehensive overview of real-world practices and outcomes, facilitating future research and the development of standardized management strategies for this understudied condition.

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PP0611

LOW RATES OF TRANSITION FROM SUBCUTANEOUS TO INTRAVENOUS FORMULATION IN PATIENTS WITH INFLAMMATORY BOWEL DISEASES TREATED WITH VEDOLIZUMAB

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Introduction: Vedolizumab is available in subcutaneous (SC) formulation for maintenance therapy in patients with moderately to severely active ulcerative colitis and Crohn's disease following vedolizumab intravenous (IV) induction therapy. The efficacy and safety of vedolizumab SC maintenance was demonstrated in the pivotal phase 3 VISIBLE clinical studies. Real-world evidence also supports the clinical effectiveness and persistence of vedolizumab SC treatment in patients with IBD; however, some patients transition back from vedolizumab SC to IV maintenance.

Aims & Methods: We aimed to describe current real-world evidence regarding the transition from vedolizumab SC maintenance to IV maintenance treatment. EMBASE and MEDLINE publication databases were searched (date 13 March 2024) for real-world studies, excluding single case reports, that reported vedolizumab-treated patients who transitioned from SC back to IV maintenance. Data on patient proportions and reasons for the SC-IV transition were extracted.

Results: There were 11 publications meeting prespecified search criteria, which included 2 retrospective studies,[1,2] 7 prospective studies,[3-9] 1 case series of 7 patients[10] and 1 report of 31 patients who were transitioned from IV to SC administration during the COVID-19 pandemic.[11] Patient numbers in retrospective and prospective studies ranged from N=19[7] to N=168.[3] Excluding the case series,[10] reported rates of transitioning from SC maintenance back to IV maintenance ranged from 3.2% (1/31) among patients in the UK during the COVID-19 pandemic[11] to 12.9% (12/93) in a 2023 Italian cohort[9], with the overall rate in these 10 publications of 7.4%. The most commonly reported reasons for SC to IV transition were injection site reactions (reported in 1/2[7], 3/5[8], 6/12[9] and 7/7[10] patients who transitioned back to IV maintenance), unspecified AEs (8/8[4] and 9/12[6]), allergic reaction (1/5[2]), fear of needles (3/12[6]) and preferred route of administration (2/5[8] and 2/12[9]). Four of 12 patients who transitioned back to IV in a 2023 Italian cohort did so due to disease flare.[9] Additionally, among 31 patients from the UK who transitioned to SC during the COVID-19 pandemic, 1 transitioned back to IV administration due to disease flare.[11] In a 2023 Italian cohort, 24/168 patients experienced disease relapse following IV to SC transition, 10 of whom transitioned back to IV treatment and 8 then recaptured clinical remission.[3] A 2022 UK study reported that 12/164 patients transitioned from SC back to IV, 4 of whom at an escalated dose, and 1 patient experienced an anaphylactic reaction upon restarting IV.[5] Additionally, a 2023 case series reported that 7 patients experienced allergic manifestations including urticaria, hypotension and lip swelling following the transition from SC to IV, which led to vedolizumab discontinuation in 4 patients.[10]

Conclusion: Real-world data suggest a low rate of patients transitioning back to vedolizumab IV from SC maintenance, and those that transition back to IV administration do so mainly due to injection site and allergic reactions and preferred route of administration, rather than a loss of response. It is not possible to assess from current published data the extent to which infusion reactions occur when transitioning back to IV maintenance or if transitioning back to IV maintenance may be due to a 'nocebo' effect due to more negative patient expectations or perceptions of SC treatment delivery.

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Disclosure: K.B. Russ: Consultant and has served as an advisory board member for Bristol Myers Squibb, Pfizer and Iterative Scopes; C. Agboton, D. Lindner, A.M. Wojtowicz: Employees of and hold stock/stock options in Takeda; A. Afzali: Consultant and has served as an advisory board member for AbbVie, Bristol Myers Squibb, Eli Lilly, Janssen, Pfizer, and Takeda and received speaker's fees from AbbVie, Bristol Myers Squibb, Janssen, Pfizer, and Takeda.

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PP0612

MONITORING THE PSYCHOPATHOLOGICAL PROFILE OF INFLAMMATORY BOWEL DISEASE PATIENTS TREATED WITH BIOLOGICAL AGENTS AND SMALL-MOLECULES

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Introduction: Patients with Inflammatory Bowel Disease (IBD), including Crohn's disease (CD) and ulcerative colitis (UC), often experience anxiety and depression. A recent meta-analysis revealed that 32.1% of IBD patients suffer from anxiety and 25.2% experience depressive symptoms. Furthermore, previous studies have suggested that biological therapy may alter the psychopathological profile of a subset of patients treated for various diseases, including IBD. However, the lack of comprehensive understanding necessitates a more systematic analysis of this association.

Aims & Methods: The aim of this study is to longitudinally monitor the psychopathological profile of a cohort of IBD patients initiating a new biological treatment at our center, and compare the results with a cohort of IBD patients receiving traditional therapy.

In this prospective longitudinal cohort study, we recruited consecutive adult patients affected by CD and UC treated at Fondazione Policlinico Universitario "A. Gemelli", IRCCS, Rome. Enrolled patients were divided into two cohort: those starting a new biological or small molecules therapy (Cohort 1), and those receiving traditional non-biological therapy (Cohort 2). Hetero-administered scales, including Hamilton Depression Rating Scale [Ham-D], Hamilton Anxiety Rating Scale [Ham-A], Young Mania Rating Scale [YMRS], Brief Psychiatric Rating Scale [BPRS], Partial Mayo Score [PMS] and Harvey-Bradshaw Index [HBI], were utilized to assess psychosomatic symptoms and gastrointestinal at the beginning of the new treatment (or at a random time point in the non-biological group) (baseline, T0), one month (T1), three months (T2) and six months (T3) thereafter. Results obtained at each time point were compared between the two patient groups.

Results: Ninety IBD patients were enrolled, 51 in cohort 1 (26 UC and 25 CD) and 39 in cohort 2 (22 UC and 17 CD). At T0, non-significant differences were found between cohort 1 and cohort 2 in mean scores of Ham-D (5.55 SD±6.16 vs 3.18 SD±3.85, p=0.27), Ham-A (5.84 SD±5.66 vs 4.08 SD±4.77, p=0.79), YMRS (1.10 SD±1.80 vs 0.85 SD±1.42, p=0.679), BPRS (22.43 SD±6.65 vs 20.03 SD±6.52, p=0.210). Statistically significant differences were found in the mean PMS and HBI between T0 and T2 in the 32 patients of cohort 1 who completed T2, and in the 20 patients who completed T3, as well as in cohort 2 between T0, T1 and T2 in the 30 patients who completed T2 and between T0 and T2, as well as T0 and T3 in the 11 patients who completed T3. No statistically significant differences were found in any of the psychiatric scales administrated in either cohort at the different time points.

Conclusion: Although a comprehensive understanding of psychosomatic effects of IBD biologic therapy is yet to be defined, our results suggest that they can be safely administered. Close monitoring during the early weeks and months is strongly recommended to assess possible side effects, including psychiatric adverse events.

Disclosure: Nothing to disclose.

PP0613

ACUTE INFUSION REACTIONS DUE TO DIFFERENT BIOLOGIC AGENTS IN A LARGE COHORT OF PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: The therapeutic armamentarium for inflammatory bowel disease (IBD) includes several biological agents: infliximab (IFX), adalimumab (ADA), vedolizumab (VED) and ustekinumab (UST). Persistence in therapy with these drugs is not satisfactory, not only due to the loss of clinical response over time but also due to adverse events. Acute infusion reactions (AIRs) occur during or within 24 hours of the infusion. The intensity of AIRs ranges from mild (flushing, dizziness, headache, itching, rash) to severe (anaphylactic).

Aims & Methods: The primary aim of this study was to compare the prevalence rates of AIRs due to the different biological agents in a large cohort of patients with IBD and to assess risk factors for AIRs.

All reports of AIRs occurring from January 2021 to October 2023 in IBD patients treated with different biological agents (IFX, ADA, VED and UST) in a tertiary IBD center were retrospectively reviewed. The electronic records of the identified patients were checked for the type of IBD, whether Crohn's disease (CD) or ulcerative colitis (UC), type of biological agent, route of administration (intravenous or subcutaneous), concomitant immunosuppressive therapies, atopic status. Furthermore, the type of treatment for the AIRs and the subsequent therapy undertaken for IBD were also recorded.

Results: 22 AIRs were recorded out of 1911 patients (1.1%). Of these, 16/550 in patients on IFX biosimilars (2.9 %), 4/590 (0.7%) in patients on ADA biosimilars, 2/318 (0.6%) in patients on VED and 0/453 (0%) in patients on UST (Table 1).

The difference in the prevalence of AIR rates observed with the different biological agents was statistically significant (p<0.01). 7 of 16 post-IFX AIRs were moderate-severe, with two episodes of anaphylaxis occurring in patients with CD.

The two patients treated with VED experienced only mild symptoms, and no anti-reactive treatment has been used; two mild and two moderate AIRs, respectively, have been reported among patients on ADA. 12 of 22 AIRs occurred during the induction period (from the first to the third administration). 7 of 22 patients had previous allergic reactions.

		Infliximab (n=16)	Adalimumab (n=4)	Vedolizumab (n=2)
Type of IBD, n (%)	Ulcerative colitis	8 (50)	2 (50)	2 (100)
Age, y. Median (range)		34.81 (20-62)	47 (42-50)	62.5 (50-75)
Sex, n (%)	Female	7 (43.8)	2 (50)	0 (0)
Duration of IBD, y. median (range)		6.1 (1-22)	12.2 (7-15)	8.5 (6-11)
AIRs' severity	Mild	9 (56.2)	4 (100)	2 (100)
	Moderate-Severe	7 (43.8)	0 (0)	0 (0)
Infusion number, (when AIR occurred) n (%)	1-3	10 (62.5)	1 (25)	1 (50)
	4-7	5 (31.25)	0 (0)	1 (50)
	>7	1 (6.25)	3 (75)	0 (0)

Table 1.

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Conclusion: Our real-world data on a large population of IBD patients treated with biologic agents confirm that AIRs are rare and occur mainly in patients treated with IFX biosimilars. Most patients who experienced AIRs were atopic. Almost all patients after AIRs changed biological agents within or out of the same biological class. Further studies are necessary to confirm our results and better characterise the profile of patients at greater risk of AIRs.

Disclosure: Nothing to disclose.

PP0614

SHORT- AND LONG-TERM EFFECTIVENESS AND SAFETY OF CYCLOSPORINE THERAPY IN SEVERE ULCERATIVE COLITIS REFRACTORY TO CORTICOSTEROID TREATMENT

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Introduction: For decades, corticosteroids have been the first-line therapy in patients with severe ulcerative colitis (UC), however 30-40% of patients fail to respond and require second-line rescue therapy with cyclosporine (CsA) or infliximab to avoid colectomy.

Aims & Methods: We conducted a single-center, retrospective study including all patients with active severe, steroid-refractory UC and with at least one year follow-up, to assess the short- and long-term effectiveness and safety of CsA therapy. Disease severity was defined according to Mayo disease activity index. Patients with a pMayo score of ≥7 and an eMayo score of ≥2 were enrolled, and received 4 mg/kg/day CsA IV for 5-7 days for second-or third-line rescue therapy between 2001 and 2023. IV CsA therapy was followed by oral CsA in case of clinical response. Colectomy-free survival was assessed using Kaplan-Meier analysis. Multivariable logistic regression analyses were performed to identify the predictive value of various factors for therapeutic response to CsA and colectomy during or after CsA treatment.

Results: Ninety-two steroid-refractory UC patients (54.4% male, mean age: 40.0 ± 14.0 years, median disease duration: 4.0 years) were included with the median follow-up time of 14.0 years (IQR:7-18). Clinical response was achieved in 59/92 (64.1%) patients, and clinical remission was observed in 22/92 (23.9%) patients after the IV phase. During oral CsA therapy (10.4 ± 11.7 months) relapse rate was 42.0%, whereas 33 (40.7%) patients achieved or maintained remission until treatment discontinuation. Remission with oral CsA therapy was significantly higher among those who received concomitant immunomodulator treatment (p=0.004, OR:6.98, 95% CI:1.9-26.4). 20/92 (21.7%) patients underwent colectomy during CsA therapy, while a total of 45 (48.9%) patients required surgery at the end of follow-up.

Non-response to IV CsA was the only predictor of colectomy during CsA therapy (p=0.04. OR: 0.17, 95% CI:0.03-0.94). No predictors for the effectiveness of the IV CsA could be found. Neither the need for colectomy during CsA therapy (p=0.33), nor the rate of remission with oral CsA (p=0.49) was influenced by the prior use of biologics.

Remission with oral CsA was found to be the only predictor of colectomy within 3 years after CsA discontinuation. The probability of colectomy-free survival was 74.7%, 62.6%, 57.1% and 45.6% at 1, 3, 5 and 14 years. In 52 patients (64.1%), biological therapy was needed after an average of 7.2 (± 6.74) months from the start of CsA. Immunomodulators were used in 50 out of 89 patients following CsA discontinuation, as maintenance therapy or as part of combination therapy. Adverse events (AEs) occurred in 53.3% of patients, 57.2% of them required a dose reduction or treatment discontinuation. AEs were mainly reversible, but 1 patient died of opportunistic infections (acute tuberculous pneumonia). The occurrence of AEs was not associated with the use of concomitant immunomodulators (p=0.92). Malignancy was observed in 5 patients during follow-up, mainly melanoma and non-melanoma skin cancer.

Conclusion: CsA is an effective therapeutic option for avoiding colectomy in patients with active severe UC, even in the era of biologicals. Moreover, its effectiveness can be enhanced by concomitant immunomodulator therapy without compromising safety. CsA can be used as a "bridge" until the therapeutic effect of immunomodulators or biological therapy is fully developed. However, due to the high occurrence of AEs, tight patient monitoring during CsA therapy is needed.

Disclosure: Nothing to disclose.

PP0615

WHAT FACTORS ARE ASSOCIATED WITH THE NEED FOR DOSE OPTIMISATION OF USTEKINUMAB IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE (IBD)?

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Introduction: Ustekinumab (USK) is a biological drug that inhibits interleukins 12 and 23. It has been approved for the treatment of inflammatory bowel disease (IBD) as an induction agent, which is administered intravenously, followed by a maintenance subcutaneous therapy at a dose of 90 mg every 8 or 12 weeks. In patients who do not achieve an adequate response with the usual doses, a common clinical practice is to optimise the USK dose.

Aims & Methods: The objective of this study was to evaluate the existence of risk factors associated with the need for this dose optimisation.

Methods: This was a retrospective observational single-centre study which included all adult patients with IBD treated with USK between 2013 and 2023 in the IBD Unit of the Santiago de Compostela Hospital, Spain.

The following variables were evaluated: demographic variables (age, sex, BMI, smoking habit), descriptive variables of the disease (type, extension, behaviour, perianal involvement), biomarkers (calprotectin, CRP) and endoscopic characteristics. Furthermore, intestinal ultrasound was conducted in patients with Crohn's disease (CD).

Dose optimization was defined as a reduction in the time interval between USK doses. The chi-square test was employed for qualitative variables, while quantitative variables were analysed using the Shapiro-Wilk test, Student's t-test, and Mann-Whitney U test.

A multivariate logistic regression analysis was conducted to identify factors associated with the need for therapy optimization. Drug persistence measurements were represented using the Kaplan-Meier method.

Results: A total of 189 patients (50.3% men) with a mean age of 51.9 years (SD 16.7) were included in the study. Crohn's disease (CD) was present in 84.7% of the patients, while the remaining patients had ulcerative colitis (UC). The dosage of ustekinumab (USK) was optimised to 90 mg/4 weeks in 67.2% of all patients (69% in CD and 63% in UC). The univariate analysis revealed that BMI (p = 0.023), disease pattern (p = 0.016), presence of perianal disease (p = 0.011), high calprotectin levels (p = 0.039) and CRP (p

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= 0.006) were significantly associated with the need for therapy optimisation. In the logistic regression analysis, BMI was the only significant risk factor for dose intensification of USK (OR 1.093, IC 95% 1.001-1.194). Conversely, an identified protective factor was an inflammatory pattern (OR 0.234, IC 95% 0.069-0.800). The mean time from the commencement of therapy to the necessity for optimisation was 10 months in patients with CD and 5 months in patients with UC.

Conclusion: In more than half of the patients treated with ustekinumab, dose optimization is required. The necessity for dose optimization of ustekinumab in patients with Crohn's disease may be influenced by a higher body mass index value and the non-inflammatory pattern of the disease.

Disclosure: Nothing to disclose.

PP0616

WHAT IS THE PROFILE OF CROHN'S DISEASE PATIENTS WHO ARE IN THE 3RD OR 4TH LINE OF ADVANCED TREATMENT?

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Introduction: This study looks at the profile of Crohn's patients who have been on at least 3 different advanced treatments, such as tumour necrosis factor inhibitors (TNFi), integrin blockers and IL12/23 inhibitors.

Aims & Methods: Objective: The purpose of this study was to evaluate the profile of Crohn's disease patients who are in the 3rd or 4th line of an advanced treatment and to consider whether patients may be likely to switch to a different treatment.

Methodology: In both Q1 and Q3 2023, ~200 gastroenterologists from France, Italy, Germany, Spain, and the UK contributed ~1200 anonymous patient record forms for individuals receiving advanced therapy for Crohn's Disease. All the forms were collected through the Therapy Watch IBD (UC and Crohn's) online survey by Research Partnership.

The survey has been conducted with similar numbers of gastroenterologists for 3 years at regular intervals. Informed consent from physicians was obtained. Where relevant, statistical comparisons were performed using a pairwise z-test.

In addition, Research Partnership has conducted research with 138 Crohn's disease patients through their Living With Crohn's EU5 survey. Quantitative online interviews were conducted with patients who were currently on an advanced treatment for Crohn's.

Results: Of all the Crohn's disease patient record forms collected in the Therapy Watch survey in 2023, 104 patients were on their 3rd or 4th line of advanced treatment.

The majority of 3rd or 4th line patients were on a non TNFi treatment, such as an integrin blocker or an IL12/ IL23 inhibitor. ~20% of patients were on a TNFi.

The patients were a mix of male and female, at 44% and 56% respectively, and with a wide age range (42% under 40, 30% aged 41-50 and 28% aged over 50). 42% had suffered with Crohn's disease for over 10 years. The majority of patients had an existing co-morbidity, most commonly a CV condition or depression.

At the time of the survey, the current severity of Crohn's was reported by physicians to be: 32% mild, 50% moderate and 18% severe. Approximately a quarter of them also were reported as having flares that are increasing in frequency.

From our Living With Crohn's survey, out of 138 Crohn's patients currently on an advanced therapy*, approximately a quarter stated that their flares had increased in the last 6 months, but they have not requested a different treatment from their physician. The reasons for not requesting a new treatment were mainly: patients feel they can still tolerate the symptoms that they experience and they are scared of potential side effects of a different treatment.

* Living With patients include those in any line of advanced treatment, not only 3rd or 4th line.

Conclusion: Our data suggests that the majority of 3rd and 4th line Crohn's patients have moderate or severe disease, and therefore may be candidates for a switch to a new treatment as these become available for Crohn's. 30% of 3rd and 4th line Crohn's disease patients are classified as mild in severity, and this segment appear to have found a level of efficacy at this stage.

Our patient survey (Living With) has shown that even those patients who suffer increasing frequency of Crohn's symptoms do not feel confident in switching to a different therapy; there is a level of fear of switching.

Disclosure: Nothing to disclose.

PP0617

POSTOPERATIVE COURSE OF CROHN'S DISEASE AND THE EFFICACY OF ENDOSCOPIC DILATION: A SINGLE-CENTER RETROSPECTIVE STUDY

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Introduction: Crohn's disease (CD), characterized by chronic inflammation, frequently results in complications such as strictures, fistulas, and abscess formation, necessitating surgical intervention. However, the significant rates of postoperative recurrence and reoperation underscore the critical importance of effective postoperative management strategies. Recent advancements in therapeutic agents, particularly biologics, have transformed the management of Crohn's disease by not only achieving clinical remission but also promoting mucosal healing. Endoscopic balloon dilation (EBD) has emerged as a promising option for managing strictures, potentially reducing the necessity for surgery in select cases.

Aims & Methods: We therefore undertook a study to assess the postoperative outcomes of patients with Crohn's disease who had previously undergone surgery and those who underwent endoscopic dilatation for stenosis. We performed a retrospective review of patients with Crohn's disease who underwent surgery and identified factors that contributed to their outcomes. We also evaluated patients who underwent endoscopic dilatation for stenosis.

Results: Of the 87 cases with a history of surgery, initial surgical indications (1979-2019) included strictures in 24 cases, fistulas in 15 cases, abscesses in 18 cases, perforations in 18 cases, bleeding in 5 cases, endoscopic complications in 6 cases, and one case of unknown etiology. A comparison of reoperation rates before and after 2008, when maintenance therapy with biologics and EBD became available, showed a significant decrease in reoperation rates after 2008 (p = 0.0081; log-rank test). Multivariate analysis identified biologics and EBD as significant factors associated with reduced reoperation rates (hazard ratios of 0.08 and 0.37, respectively, p-values of 0.000031 and 0.021). Among the cases treated with EBD for strictures (97 cases), an average of 4.3 EBD procedures (431 procedures) were performed, with an average of 2.5 sites dilated per procedure. Before initial dilatation, stricture symptoms included ileus in 35 cases (36.1%), mild symptoms such as abdominal pain or bloating in 32 cases, and no symptoms in 30 cases. Complications observed in the 431 EBD procedures included bleeding requiring prolonged hospitalization in 5 cases, perforation requiring surgery in 4 cases, and local peritonitis in 1 case. During follow-up after the initial EBD, surgery was performed in 22 cases, with

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criticisms being the primary cause in 14 cases and abscess formation, perforation (including those caused by complications), or malignancy in the remainder. Analysis of the cumulative reoperation rate specifically for stricture-related cases treated with EBD showed rates of 10.4% at 5 years and 22.2% at 10 years. Logistic regression analysis identified a stricture length of 2 cm or more as a significant factor associated with cases requiring surgery (hazard ratio of 10.2, p = 0.0036). Kaplan-Meier analysis comparing groups according to whether initial EBD was performed before or after ileus showed no significant differences in the cumulative rates of ileus or reoperation for strictures.

Conclusion: Since 2008, reoperation rates for Crohn's disease have decreased, probably due to maintenance therapy with biologics and EBD. With EBD, it is important to consider that cases with longer strictures may be difficult to manage without surgery, and complications of EBD may require surgery.

Disclosure: Nothing to disclose.

PP0618

SERUM IMMUNE PROTEIN PROFILING AT BASELINE IN CROHN'S DISEASE PREDICTS ENDOSCOPIC RESPONSE TO USTEKINUMAB

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Introduction: Dysregulation of the immune response is central to the pathogenesis of inflammatory bowel disease (IBD), including Crohn's Disease (CD) [1].

Serum inflammatory protein profiling emerges as a potential biomarker for diagnosing and predicting treatment efficacy in IBD [2].

This study aimed to identify serum inflammatory proteins that could predict response to ustekinumab in CD patients before therapy initiation.

Aims & Methods: The study cohort included 20 CD patients, divided equally between those who showed an endoscopic response and those who did not after 24 weeks of ustekinumab treatment. Endoscopic response was defined as a reduction of ≥50% in the Simple Endoscopic Score for Crohn's Disease (SES-CD). Serum protein profiling was performed using the Olink Target 96 Inflammation assay. Univariate t-test analyses with false discovery rate correction were used to differentiate protein expression between responders and non-responders.

Results: Baseline characteristics were similar across both groups, except for age; non-responders tended to be older, with five diagnosed after age 40. Five proteins—CCL11, OPG, CDCP1, SLAMF1, Flt3L, and CXCL10—were significantly overexpressed in non-responders, while IL-22 RA1 was predominantly found in responders. Functional enrichment analyses demonstrated cytokine-mediated signaling pathway as the most notable biological process, chemokine activity as the most significant molecular function, and extracellular region as the major cellular component. KEGG enrichment also highlighted cytokine-cytokine receptor interaction as the most significant pathway.

Conclusion: This study identifies six serum inflammatory proteins that are differentially expressed at baseline and can predict endoscopic response to ustekinumab in CD. These findings suggest that baseline serum protein profiling could serve as a valuable biomarker for forecasting treatment outcomes in IBD.

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Disclosure: Nothing to disclose.

PP0619

VALIDATION OF THE ADMIT-ASC INDEX FOR PREDICTING RESPONSE TO CORTICOSTEROIDS IN ACUTE SEVERE COLITIS

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Introduction: Acute Severe Colitis (ASC) is a medical and surgical emergency that requires prompt and well-coordinated treatment. The initial treatment typically involves intravenous corticosteroids. However, a significant number of patients (around 30-40%) do not respond to corticosteroids and need alternative treatment. Predicting resistance to the first-line treatment early on is a challenge that has not been fully addressed. A new index called ADMIT-ASC (Admission Model for Intensification of Therapy in ASC), which considers biological and endoscopic data, has been developed to predict the failure of steroid treatment.

Our goal is to validate this index's effectiveness in predicting corticosteroid resistance in ASC associated with ulcerative colitis (ASUC).

Aims & Methods: We conducted a retrospective chart review of 210 patients admitted to the gastroenterology department at Sahloul Sousse for ASUC between March 2010 and February 2020. Failure of corticosteroid therapy was defined as the need for second-line medical treatment or colectomy.

The ADMIT-ASC score is an index ranging from 0 to 4, calculated by adding points based on specific parameters at admission: CRP ≥ 100 mg/L (1 point), albumin ≤ 25 g/L (1 point), and endoscopic UCEIS score ≥ 4 (1 point) or ≥ 7 (2 points). The area under the ROC curve (AUROC) was calculated to evaluate the accuracy of this score in predicting the failure of corticosteroid therapy.

Results: Two hundred and seventy patients were included in the study, with an average age of 33.5 years and a male-to-female ratio of 0.68. Of these patients, 62.8% had a history of UC, while ASC was the initial episode in 37.2% of cases. Fifty-two point thirty-eight percent (52.38%) of patients did not respond to intravenous corticosteroid treatment. Among them, 34.3% responded to ciclosporin, 2.4% to infliximab, and 5.2% underwent salvage colectomy. The average ADMIT-ASC score was 2.14 ± 0.99. Patients with scores of 0, 1, 2, 3, and 4 had response rates of 100%, 62.7%, 56%, 43.4%, and 25%, respectively (p=0.01). The failure rate of first-line medical treatment was 52.7% for a score ≥ 2 and 62% for a score ≥ 3 (95% CI). The AUROC of ADMIT-ASC in predicting corticosteroid resistance was 0.628 (p=0.01, 95% CI [0.552; 0.703]).

Conclusion: The ADMIT-ASC index seems to be a reliable predictor of response to corticosteroid therapy in ASC patients, starting from admission. This could help identify patients early on who would benefit from switching to second-line treatment promptly. However, larger prospective studies are needed to confirm these results.

Disclosure: I confirm that I do not have any conflicts of interest regarding this subject.

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PP0620

HISTOLOGICAL AND HISTO-ENDOSCOPIC OUTCOMES WITH ADVANCED TREATMENTS (BIOLOGICS AND SMALL MOLECULES) IN ULCERATIVE COLITIS - A NETWORK METANALYSIS

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Introduction: There is growing awareness of the importance of histological outcomes in ulcerative colitis (UC). However, direct comparisons of histological endpoints are lacking. Thus, we aimed to conduct a network meta-analysis (NMA) to compare the effectiveness of various biologics and small molecules in achieving histological endpoints in UC.

Aims & Methods: We searched four online databases to identify all randomized controlled trials (RCTs) investigating advanced therapies for moderate-to-severe UC that reported histological outcomes. The analysis focused on the proportion of patients achieving histological improvement, histological remission, and histo-endoscopic improvement during induction and maintenance therapy. Using a random-effects frequentist model, we conducted a NMA, presenting the results as odds ratios (OR) with 95% confidence intervals. To evaluate the relative effectiveness of each treatment, we calculated the P-score, which ranges from 0 to 1. We also assessed the risk of bias and evidence certainty using established scoring systems.

Results: Twenty-four RCT were included, comprising 15 different therapies, and enrolling over 8800 patients. Of these trials, 19 provided data on induction therapy, while 10 assessed histological outcomes during maintenance treatment. Outcome definitions were similar among studies. We found that etrasimod 2 mg/day ranked highest in achieving histologic response (P-score 0.98) and remission (P-score 0.90) following induction, while guselkumab ranked first for histoendoscopic improvement. In maintenance trials, upadacitinib 30 mg/day ranked highest in achieving histologic and histo-endoscopic outcomes (P-score 0.88 for both endpoints). Etrasimod 2 mg/day ranked second for histologic remission (P-score 0.70) and histo-endoscopic improvement (P-score 0.73), while mirikizumab 200 mg/month ranked second for histologic response.

Conclusion: Our findings underscore the importance of small molecules in treating moderate-to-severe UC, particularly etrasimod and upadacitinib. Nonetheless, additional head-to-head RCTs are imperative to better inform clinical practice and to address unanswered questions.

Disclosure: Nothing to disclose.

PP0621

NEW OR RECURRENT CANCER OCCURRING IN INFLAMMATORY BOWEL DISEASE PATIENTS TREATED WITH IMMUNOMODULATORS OR BIOLOGICS AFTER CANCER

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Introduction: Whether using immunomodulators (IMM) or biologics after a first cancer in Inflammatory Bowel Disease patients (pts) may increase the risk of new or recurrent cancer is undefined.

Aims & Methods: The primary aim was to assess the frequency of new or recurrent cancer in IBD pts treated with IMM or biologics after a previous diagnosis of cancer. The secondary aim was to evaluate the cancer-related mortality in IBD pts treated with IMM and biologics after the first cancer. In a retrospective, multicentre study, clinical records of all IBD pts treated with IMM or biologics after a previous diagnosis of cancer were reviewed. Inclusion criteria:1) Diagnosis of Crohn's Disease (CD) or Ulcerative Colitis (UC); 2)Cancer history;3)IMM or biologics used after a first cancer;4) IBD and cancer data recorded; 5)Follow-up after cancer ≥4 months (mos). Exclusion criteria:1) IMM used for ≤3 mos. Data considered: IBD features,-cancer type,IBD duration, age at diagnosis of cancer, follow-up after cancer,cancer-related death.Data expressed as median [range].

First cancer	IMM	Biologics	New cancer	Recurrent cancer
Prostatic	-	UST,VDZ	BCC	Prostatic
Hepatocellular	Tacrolimus	VDZ	SCC	-
Colonic	MTX	VDZ	-	Colonic
BCC	-	ADA	BCC	-
BCC	-	ADA	BCC	-
Urinary tract	MTX	VDZ,TOFA	SCC	-
Melanoma	Thiopurines	UST,VDZ	BCC	Melanoma
Testicular	Thiopurines	IFX,ADA,VDZ	Cardiac Myxoma	-

Table. New and recurrent cancers after IMM/biologics

Results: 75 IBD pts (57 CD,18 UC) treated with IMM or biologics after a first diagnosis of cancer were enrolled (n [%]: F 36 [48%]; age 61 [26-89] yrs). CD features included (n=):L1 (37);L2 (2);L3 (16);L4 (8);B1 (17);B2 (30);B3 (10). UC extent included: E1 (2),E2 (8), E3 (8). Cancer in IBD pts treated with IMM/biologics included (n [%]): basal cell carcinoma (BCC) 14 (18.4%), breast 9 (12%), colorectal 9 (12%), melanoma 8 (10.7%), prostatic or testicular 7 (9.3%), urinary tract 7 (9.3%), thyroid 6 (8%), squamous cell carcinoma (SCC) 4 (5.3%), endometrial 2 (2.7%), hematopoietic 4 (5.3%), others 7 (9.3%). In 2 (2.7%) pts, ≥2 cancers occurred. The median time interval between index cancer and ISS/biologics use was 38 mos [0-428]. IMM after cancer were used for a median of 32.5 [4-131] mos, including: methotrexate (MTX) in 13 (17.3%),thiopurines in 8 (10.7%), mycophenolate mofetil, tacrolimus or everolimus in 3 (4%) pts.

Biologics were used after cancer for a median of 36 [3-194] mos:Ustekinumab (UST) in 24 (32%), Vedolizumab (VDZ) in 31 (41.3%), Adalimumab (ADA) in 17 (22.7%), Infliximab (IFX) in 12 (16%), Tofacitinib (Tofa) in 2 (2.7%), Golimumab in 1 (1.3%) pt. Both IMM and biologics were used after a first cancer in 15 (20%),while ≥2 biologics in 16 (%) pts. In a median follow up of 95 [4-509] mos after the first cancer, 8 (10.7%) IBD pts developed a new and/or recurrent cancer after IMM/biologics use.

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In particular, cancer recurred in 3 (4%) pts (1 melanoma,1 prostatic,1 colonic adenocarcinoma), while new cases of cancer occurred in 6 (8%) pts,all 6 represented by non-melanoma skin cancer (NMSC) (both recurrent and new cancer in 1 pt).A not significant trend for a higher frequency of new vs recurrent cancer (p=0.47) was observed.No cancer-related death were observed.

Conclusion: In a retrospective multicentre study, a low frequency of recurrent cancer was observed in IBD pts using IMM or biologics after the first cancer. A not significant trend for a higher frequency of new vs recurrent cancer (p=0.47) was observed, being almost all new cancers represented by NMSC.

Disclosure: Nothing to disclose.

PP0622

REAL WORLD VALIDATION OF A PROGNOSTIC CLINICAL INDEX TO PREDICT STEROID RESPONSE IN ACUTE SEVERE COLITIS: A SINGLE TERTIARY CENTER RETROSPECTIVE STUDY

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Introduction: Acute severe colitis (ASC) is an important cause of morbidity and mortality in Ulcerative Colitis (UC), requiring hospitalisation and, frequently, colectomy. Intravenous corticosteroids (CS) remain the first-line medical treatment. Approximately 40% of episodes of ASC do not respond to this treatment and require rescue therapy with ciclosporin (CIC) or infliximab (IFX)[1]. In the past decades, these treatments and protocol-driven management algorithms led to reduced mortality and colectomy rates. A simple clinical scoring system (the Admission Model for the Intensification of Therapy in Acute Severe Colitis (ADMIT-ASC))[3], based on CRP, albumin, and the Ulcerative Colitis Endoscopic Index of Severity (UCEIS) score, was recently developed to predict the steroid response on admission.

Aims & Methods: We aimed to retrospectively assess the steroid non-response during admission with ASC and the usefulness of the new prognostic clinical index ADMIT-ASC at admission to predict steroid response in a Belgian tertiary center. All patients admitted with ASC between 1st January 2015 and 1st April 2023 were identified from our inflammatory bowel disease database at Erasme University Hospital. Inclusion criteria were ASC patients admitted at the hospital, standard treatment with at least 3 days of intravenous CS and endoscopy evaluation upon admission. ASC was defined according to modified Truelove and Witts' criteria as ≥6 bloody stools/day with one or more markers of systemic disturbance (tachycardia >90 beats/min, temperature >37.8°C, haemoglobin <105 g/L or CRP >30 mg/L). Steroid non-response was defined as the need for medical rescue therapy (CIC of IFX) or colectomy during the admission. ADMIT-ASC index consists of four points: CRP ≥100 mg/L (1 point), albumin ≤25 g/L (1 point) and UCEIS score ≥4 (1 point) or ≥7 (2 points).

Results: Out of 127 patients admitted with UC and treated with intravenous CS during three days, 84 (45% male, median age of 31 years old) were included in the study. Admission demographics and clinical characteristics are described in Table 1.

For 17 patients (20%), the admission was their index presentation of UC. Response to CS was seen in 49 patients (58%). Thirty-three patients (39%) received medical rescue therapy (17/33 IFX and 16/33 CIC). Colectomy rate during admission was 13% after a median duration of 18 days (IQR: 16-54). Admission C reactive protein (CRP) (p=0.036), albumin (p=0.026) and UCEIS scores (p=0.022) were significantly associated with steroid non-response. Patients scoring 0, 1, 2, 3 and 4 had steroid response rates of 90%,75%, 55%, 19% and 0%, respectively. Scoring of ≥3 was 75% (95% CI, 0.43 to 0.95) predictive of steroid failure (OR 5.08, 95%, CI 1.38 to 24.5).

	Steroid response (n=49)	Steroid non-response (n=35)	P value
Median disease duration (years) (IQR)	3.6 (0.3-0.9)	2.1 (0.8-5.5)	0.5
Montreal Classification	E1 : 5/49 (10%) E2 : 20/49 (41%) E3 : 24/49 (49%)	E1 : 0/35 E2 : 12/35 (34%) E3 : 23/35 (66%)	0.084
5-ASA, n (%)	22 (45%)	10 (29%)	0.13
Methotrexate or thiopurine, n (%)	6 (12%)	9 (26%)	0.11
Infliximab, n (%)	4 (8.2%)	8 (23%)	0.058
Vedolizumab, n (%)	7 (14%)	5 (14%)	>0.9
Tofacitinib, n (%)	2 (4.1%)	2 (5.7%)	>0.9
Readmission, n (%)	12 (24%)	10 (29%)	0.7

5-ASA, 5-aminosalicylate
P values from Wilcoxon rank-sum test or Fisher's exact test.

Table I Admission Demographics data (n= 84).

Conclusion: In this tertiary referral cohort study, we confirmed that the ADMIT-ASC index is a simple clinical tool that can be used on the day of admission to predict steroid non-response in ASC with a threshold of ≥3 to identify patients who are unlikely to respond to intravenous CS and enabling timely treatment escalation or surgery.

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CDT has not disclosure.

CM has not disclosure.

AC has received lecture fees from Galapagos, Abbvie, Janssen, celltrion, Pfizer, Takeda.

DF has received lecture fees from Abbvie, Astrazeneca, Dr.Falk Pharma, Ferring, Janssen, Mundipharma, MSD, Pfizer, Takeda, Amgen, and advisory board fees from Abbvie, Janssen,Takeda, Celltrion, Amgen.

CL has received consultancy fees from Janssen, Abbvie, Takeda and Galapago and speaker fees from Sandoz, Janssen, Abbvie.

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PP0623

EVOLVING MANAGEMENT STRATEGIES IN GASTROINTESTINAL IMMUNOTHERAPY RELATED ADVERSE EVENTS

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Introduction: The use of Immune Checkpoint Inhibitors (ICI's) have been demonstrated to significantly improve progression free-survival in a number of malignancies [(1)].

Immune related adverse effects (IRAEs) are, however, an increasingly prevalent unwanted outcome, and now a widely recognised entity, with a number of guideline based management strategies published. Gastrointestinal IRAEs (GIIRAEs), including colitis and hepatitis are among two of the most reported IRAE's with combination ICI use accounting for 15% and 10% of all IRAE's respectively [(2)].

Aims & Methods: We aimed to report the rates and severity of Gastrointestinal Immune Related Adverse Effects (GIIRAE's) for patients who had received combination immunotherapy with nivolumab and ipilimumab, and analyse the investigations, treatment regime and response in a single tertiary referral centre. A retrospective cohort analysis was conducted of all patients receiving ipilimumab and nivolumab for melanoma, renal cell carcinoma and mesothelioma between the dates of 2019 to 2023. The local oncology database and electronic hospital records were reviewed. Baseline demographics, incidence, grade and treatment of GIIRAE's were collected.

Results: A total of 99 patients on combination ICI were identified. 26 patients (26%) had a GIIRAE of whom 15% (15/99) had diarrhoea and 11% (11/99) had hepatitis. 47% (7/15) of patients with diarrhoea had radiological or endoscopic confirmation of colitis (5) or enteritis (2). Of the 5 patients with Grade 3-4 diarrhoea, 3 (60%) required escalation to infliximab, all of which only had mild colitis evident at colonoscopy. 7/11 (64%) of patients had Grade 3-4 hepatitis, with only one patient requiring escalation to steroid sparing agent, despite only 57% (4/7) being commenced on high dose intravenous steroids as recommended by current guidelines. 40% (10/26) of Grade 1-2 IRAEs were managed conservatively. Only 50% (13/26) of all identified cases had involvement of a gastroenterologist.

Conclusion: Management strategies for GIIRAEs have and continue to evolve, and our cohort highlights the heterogeneity in investigation and management practice over the course of four years. Based on our findings, symptom-based assessment of severity for diarrhoea associated IRAEs appears to be a better predictor for the need for escalation of therapy rather than severity of investigation findings given only mild colitis was found on endoscopy for all patients who required biologic therapy. For hepatitis associated IRAEs, it would appear that a less aggressive approach to steroid therapy compared to current published guidelines may be adequate. Further prospective studies with larger cohort would be of value to confirm our observations.

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Disclosure: Nothing to disclose.

PP0624

REAL-WORLD DATA ON THE TREATMENT WITH MIRIKIZUMAB IN A MOSTLY BIOLOGICAL-EXPERIENCED ULCERATIVE COLITIS COHORT

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Introduction: Mirikizumab was approved for the treatment of Ulcerative colitis (UC) in May 2023. However, real-world data especially in biological-experienced patients, is scarce.

Aims & Methods: We aimed to determine treatment efficacy and persistence on Mirikizumab in mostly treatment-experienced UC patients.

All patients, who were treated for UC in the inflammatory bowel disease (IBD) outpatient clinic of the University Hospital Frankfurt from 07/23-03/23 were retrospectively analyzed. Data on clinical (Simple Clinical Colitis Activity Index (SCCAI)) and biochemical disease activity (fecal calprotectin (FC)) as well as patient history were collected. Patients were followed-up for at least 12 weeks.

Results: Twenty-one patients with UC were included. Eighteen patients had already received at least one biological agent prior to starting Mirikizumab. Thirteen patients were exposed to ≥ 2 biological agents and even 6 patients to ≥ 4 biological agents prior to starting Mirikizumab. Two patients received Mirikizumab as a first line treatment. Eight patients were treated with Ustekinumab prior to the therapy with Mirikizumab**.** Ten patients received oral steroids as concomitant treatment. The median SCCAI at the start of induction was 7 (1-18). We report a reduction in clinical disease activity to 2 (0-11) at week 12 based on the preliminary data of the SCCAI. Median SCCAI and FC levels are shown in table 1.

Biochemical disease activity was assessed with FC levels. The median FC level at the start of the induction was 642 (14-2100). At week 12 the median FC level was 253 (21-2100) and 330 (9-2100) at week 12. Intravenous induction therapy was extended in 11 patients, with 10 patients receiving 6 infusions. Four patients were re-induced after having previously received Mirikizumab subcutaneously. One patient discontinued treatment after an extended induction due to a primary non-response. One patient reported new nodulous skin lesions after the treatment with Mirikizumab.

Conclusion: We report preliminary real-world data on the treatment with Mirikizumab. In our interim analysis Mirikizumab proves to be an effective and safe treatment in biological-experienced UC patients.

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PP0625

USTEKINUMAB PROMOTES RADIOLOGICAL FISTULA HEALING IN PERIANAL FISTULIZING CROHN'S DISEASE:RETROSPECTIVE REAL-WORLD ANALYSIS

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Introduction: Perianal fistula is the most common complication of Crohn's disease (CD), affectingapproximately 40% of patients. For patients with perianal abscesses and active draining fistula, seton or fistulotomy should be performed, followed by aggressive medical therapies, including anti-TNF agents. However, a significant proportion of patients are non-responders. There is insufficient evidence to confirm the efficacy of ustekinumab (UST), in promoting fistula closure in perianal fistulizing Crohn's disease (CD) patients.

Aims & Methods: We aimed to evaluate the efficacy of UST in a real-world setting.

The data were retrospectively analyzed. Clinical and endoscopic changes were evaluated. Fistula radiological outcomes were determined using MRI.

Results: A total of 72 patients diagnosed with CD and receiving scheduled UST treatment were enrolled. Of these, 45 patients were excluded due to the absence of perianal fistula based on clinical symptoms and MRI scans, 5 for insufficient follow-up duration, and 5 for incomplete data. A total of 17 eligible patients were finally included, 2 patients did not have follow-up MRI, 65% (n=11) of whom were male, with a mean age of 28.2 ± 2.0 years at diagnosis and a mean disease duration of 3.4 ± 0.6 years. As for the Montreal classification, 62% of the patients were assigned to B1 (non-stricturing, non-penetrating) and 70% to L3 (ileocolonic) phenotypes. Most fistulas were inter-sphincteric (65%). After 58,66+-13,48 weeks of UST therapy, fistula closure was shown in 40% (n=6) of patients. In patients with endoscopical remission at 6 months of UST therapy, we did not detect significant difference in fistula closure (57.1% vs 42.9%, p=0.315).

Conclusion: UST is effective in promoting clinical and radiological fistula remission in patients with CD. Intestinal endoscopical remission may not be correlated with fistula outcome. Further research is needed for determining ustekinumab therapy efficacy for perianal fistula.

Disclosure: Nothing to disclose.

PP0626

WITHDRAWN

PP0627

SHORT AND LONG-TERM OUTCOMES AFTER ACUTE SEVERE ULCERATIVE COLITIS IN ADULTS: A 12 YEAR CANADIAN EXPERIENCE IN THE POST BIOLOGIC ERA

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Introduction: Patients with acute severe ulcerative colitis (ASUC) are at increased risk of colectomy following an episode of acute severe exacerbation. Medical rescue therapy in steroid non-responders has reduced the need for emergency colectomy. However, the impact of biologics on longer-term colectomy risk is unclear.

Aims & Methods: A retrospective cohort study of adult patients aged ≥18 years hospitalized with acute exacerbation of UC requiring hospitalization and managed with intravenous corticosteroids between 2010 and 2022 at two regional hospitals in London, Ontario, Canada. Steroid non-response was defined as requirement of medical rescue therapy or colectomy.

Results: A total of 264 adults hospitalized with ASUC were included in the analysis (male: 51.1%[n=136], mean age at admission: 40.8±17.8 years). The majority had extensive colitis (71.3% [n=169]) at the time of diagnosis. 46.3% (n=118/255) presented within one year of disease onset and 23% (n=61) had ASUC as their first presentation with UC. 25.7%(n=66/257) of patients had prior purine analogues and 23.3% (60/257) had previous biologics prior to ASUC presentation. After admission, 55.7% (n=147) responded to intravenous corticosteroids. Of the steroid non-responders, 37.5% (n=99) patients were managed with infliximab rescue therapy and only one patient received tofacitinib rescue therapy. Median CRP, stool frequency and Lindgren index score were statistically significant between steroid responders and non-responders (Table 1). A higher proportion of steroid non-responders had prior exposure to biologics or tofacitinib compared to steroid responders (31.9%vs16.3%, p<0.01). On multivariate analysis for various factors predicting steroid non-response, assessment by Oxford criteria on day 3 was the only factor that was statistically significant (odds ratio 4.70 (95% CI[1.06-20.8]), p=0.04). Oxford criteria on day 3 had a sensitivity, specificity of 58.6% and70.8%, respectively for predicting steroid non-response. 8% (n=21) required colectomy during index admission. An additional 13% (n=32) underwent colectomy within 12 months of discharge, for which there was no difference between steroid responders and non-responders (12.2% vs 14.7%). The short term (3 months following discharge) and long-term (3-12 months following discharge) colectomy rates following discharge were 8% (20/245) and 5.4% (12/223) respectively. Hospitalization with UC exacerbation following discharge was 17% (38/223).

Conclusion: 8% of patients admitted for ASUC require colectomy during the same admission and additional 13% required colectomy within 12 months of discharge. Despite a high initial response to corticosteroids, long term colectomy rates and re-hospitalization rates remain high.

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Parameter	Overall (n = 264)	Steroid non-responders (n = 117)	Steroid responders (n =147)	P-value
Extent, n/N (%)				
Proctitis	9 (3.8)	5 (4.9)	4 (3.0)	0.71
Left sided colitis	59 (24.9)	26 (25.5)	33 (24.4)	
Extensive colitis	169 (71.3)	71 (69.6)	98 (72.6)	
Day 1				
Stool frequency (median [IQR])	10.0 (7.0 – 15.0)	10.0 (7.0 – 15.0)	10.0 (7.0 – 15.0)	0.71
CRP (mg/L) (median [IQR])	60.1 (20.2 – 122.6)	61.5 (22.4 – 119.9)	56.2 (16.1 – 132.1)	0.54
Albumin (g/L) (mean [SD])	31.2 (6.0)	31.2 (5.5)	31.2 (6.4)	0.94
CRP/Albumin ratio (median [IQR])	2.0 (0.7 – 3.9)	2.1 (0.8 – 3.9)	1.9 (0.6 – 4.4)	0.25
Seo index (mean [SD])	222.1 (28.6)	220.1 (26.7)	223.3 (29.9)	0.64
Lindgren index (mean [SD])	23.3 (11.9)	22.8 (10.3)	23.7 (13.1)	0.59
Day 3				
Stool frequency (median [IQR])	6.0 (3.0 – 8.0)	7.0 (5.0 – 10.0)	4.0 (3.0 – 7.0)	<0.01
CRP (mg/L) (median [IQR])	32.9 (15.7 – 75.2)	43.4 (19.1 – 86.9)	25.4 (9.9 – 62.2)	0.05
Albumin (g/L) (mean [SD])	29.9 (10.2)	29.2 (7.6)	30.6 (12.2)	0.56
CRP/Albumin ratio (median [IQR])	1.3 (0.5 – 2.3)	1.6 (0.6 – 3.2)	0.8 (0.2 – 1.3)	0.25
Lindgren index (mean [SD])	13.7 (8.5)	16.3 (9.0)	11.4 (7.5)	<0.01
Fulfilling Oxford criteria on day 3 (stool frequency >8 or 3-8/day with CRP>45mg/L)	53/123 (43.1)	34/58 (58.6)	19/65 (29.2)	<0.01
Medical rescue therapy, n/N (%)	102/260 (39.2)	102/117 (87.2)	0/143 (0.0)	<0.01
Colectomy in index admission, n/N (%)	21/259 (8.1)	21/116 (18.1)	0/143 (0.0)	<0.01
Hospital admission with exacerbation between 3-12 months of discharge n/N (%)	38/223 (17.0)	13/94 (13.8)	25/129 (19.4)	0.28
Colectomy between 3-12 months of discharge n/N (%)	12/223 (5.4)	6/94 (6.4)	6/129 (4.7)	0.57
Overall colectomy within 12 months of discharge n/N (%)	32/243 (13.0)	14/96 (14.6)	18/147 (12.2)	0.59

Table 1: Baseline characteristics and outcomes in adult patients with ASUC with comparison between steroid responders and non-responders

CRP: C - reactive protein, IQR: interquartile range, SD: standard deviation.

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VS: None

LG: None

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TP: has received Advisory Board/consultant fees from Bristol Myers Squibb, Celltrion, Eli Lilly, Organon and Pfizer recently. He currently serves as an investigator on clinical research studies for Abivax, Alimentiv, Applied Molecular Transport, BristolMyersSquibb, Eli Lilly, Prometheus/Merck and Takeda.

MB: Honoraria: Shire, Allergan, Pfizer, Takeda, Advisory Boards, AbbVie, Takeda, Janssen, Lupin, Gilead, Pfizer, Novo Nordisk, McKesson, Innomar, Clinical Trials: AbbVie, Novo Nordisk, Gilead, Takeda, Boehringer Ingelheim, Janssen, Pfizer, Astra Zeneca, Intercept, Research Publications: Abbvie (2021 – Letter signed)

JG: received speakers fees from AbbVie, Janssen, Takeda, Celltrion, Organon and Ferring

BY: None

MS: None

VJ: has received has received consulting/advisory board fees from AbbVie, Alimentiv Inc, Arena pharmaceuticals, Asahi Kasei Pharma, Asieris, Astra Zeneca, Avoro Capital, Bristol Myers Squibb, Celltrion, Eli Lilly, Endpoint Health, Enthera, Ferring, Flagship Pioneering, Fresenius Kabi, Galapagos, Gilde Healthcare, GlaxoSmithKline, Genentech, Gilead, Innomar, JAMP, Janssen, Merck, Metacrine, Mylan, Pandion, Pendopharm, Pfizer, Protagonist, Prometheus Biosciences, Reistone Biopharma, Roche, Roivant, Sandoz, SCOPE, Second Genome, Sorriso, Takeda, TD Securities, Teva, Topivert, Ventyx, Vividion; speaker's fees from, Abbvie, Ferring, Bristol Myers Squibb, Galapagos, Janssen Pfizer Shire, Takeda, Fresenius Kabi

PP0628

TRANSMURAL REMISSION ASSOCIATES WITH A LOWER RISK OF PHENOTYPE PROGRESSION IN CROHN'S DISEASE

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Introduction: Patients with Crohn's disease (CD) are at risk of progressing from inflammatory to stricturing and penetrating phenotypes. The influence of the type of remission on phenotype progression has not been adequately evaluated.

Aims & Methods: Retrospective cohort study including surgically naïve CD patients with inflammatory or stricturing phenotype evaluated concomitantly by magnetic resonance enterography and colonoscopy. The degree of remission was correlated with the risk of progressing to stricturing and penetrating phenotypes.

Results: 381 CD patients were included, 21.8% with transmural remission, 21.3% with isolated endoscopic remission, 10.5% with isolated radiologic remission, and 46.5% without remission. Patients with transmural remission presented the lowest rates of phenotype progression (1.2%), with a significant difference compared to isolated endoscopic remission (18.3%, $P \leq 0.001$), isolated radiologic remission (17.5%, $P=0.002$), and no remission (36.9%, $P \leq 0.001$). In multivariate regression analysis, transmural remission (OR 0.023 95%CI 0.003-0.171, $P<0.001$), isolated radiologic remission (OR 0.342 95%CI 0.141-0.826, $P=0.017$), and isolated endoscopic remission (OR 0.441 95%CI 0.220-0.881, $P=0.020$) resulted in lower rates of phenotype progression compared to no remission.

Figure 1 presents the Kaplan-Meier estimates of remaining free of penetrating complication (upper curve) and free of stricturing and/or penetrating complication (lower curve) after the baseline assessment in patients with no remission (A), isolated endoscopic remission (B), isolated radiologic remission (C), and transmural remission (D).

Conclusion: The degree of intestinal remission correlates with the risk of phenotype progression. Patients with transmural remission are at the lowest risk of progressing to stricturing and penetrating phenotypes.

Disclosure: Nothing to disclose.

PP0629

REAL-LIFE EFFECTIVENESS AND SAFETY OF TOFACITINIB AND VEDOLIZUMAB AS A SECOND-LINE THERAPY IN ANTI-TNFS EXPERIENCED ULCERATIVE COLITIS PATIENTS: PRELIMINARY RESULTS OF AN IGIBD STUDY (VE2TO-UC)

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Introduction: Drug positioning in ulcerative colitis (UC) patients refractory to anti-tumor necrosis factor (TNF) is still debated. We aimed to compare the real-life effectiveness and safety of tofacitinib and vedolizumab as second-line therapy in anti-TNFs experienced UC patients.

Aims & Methods: In this Italian multicenter cohort study, we retrospectively included consecutive UC adult patients exposed to ≥ 1 anti-TNF agent starting tofacitinib or vedolizumab. Clinical remission (defined as partial Mayo score (PMS) <3 and no subscore >1) and corticosteroid-free clinical remission (CFREM) were assessed at week 26. Patients who discontinued treatment due to a primary or secondary non-response, adverse events, or intolerance were considered as treatment failure and classified as non-responders. The primary outcome was to compare clinical remission in tofacitinib- and vedolizumab-treated patients at week 26. Propensity score with inverse probability of treatment weighting (IPTW) was used to adjust for confounding, including sex, age at drug initiation, age at diagnosis, smoking habit, disease extension, use of immunosuppressant and steroids at baseline, disease activity at baseline by PMS, reason for anti-TNF discontinuation. Multiple imputation was used for missing data. Venous thromboembolism event, herpes zoster reactivation, major cardiovascular adverse event, severe lymphopenia, initiation or escalation of cholesterol-lowering drugs, hospitalization, surgery, dysplasia or colorectal carcinoma, and death were assessed for the safety profile.

Results: Overall, 99 tofacitinib- and 230 vedolizumab-treated patients were included. Baseline characteristics are summarized in Table 1. The groups were comparable except that tofacitinib-treated patients were younger and of pediatric onset, with lower comorbidities and cardiovascular risk factors. Of note, 22 patients were excluded due to a lack of sufficient follow-up. After IPTW analysis, no difference between tofacitinib- and vedolizumab-treated patients in terms of clinical remission 1.23 (0.68 – 2.23) and CFREM 1.27 (0.71 – 2.29) was observed at week 26. Similar results in terms of clinical remission 1.13 (0.60 – 2.14) and CFCR 1.16 (0.62 – 2.20) were obtained when analyzing patients with a baseline PMS ≥ 2 (Figure 1). Safety wasconsistent with known tofacitinib and vedolizumab profiles in ulcerative colitis.

	Tofacitinib (n=99)	Vedolizumab (n=230)	p
Age at baseline, y	36 (28.5 – 46)	46 (33 –57.8)	<0.001
Concomitant steroids	40 (40%)	106 (46%)	0.31
Partial Mayo Score	3 (3 – 6.5)	5 (3 – 6)	0.74
Mayo Endoscopic Subscore			0.56
• 0	1	1	
• 1	6 (8.3%)	8 (5.6%)	
• 2	24 (33%)	58 (41%)	
• 3	41 (57%)	76 (53%)	
Cardiovascular risk factor	13 (13%)	62 (28%)	<0.01

Conclusion: Our preliminary results suggest no difference in terms of clinical remission and CFREM oftofacitinib and vedolizumab as second-line therapy in anti-TNFs experienced UC patients atthe 26th week. Safety was consistent with known tofacitinib and vedolizumab profiles inulcerative colitis.

Disclosure: DN received research grant from Pfizer, consulting fees from Dr. Falk, lecturer fees from Ferring. AO served as an advisory board member for AbbVie, Galapagos, MSD, Janssen, Pfizer, Takeda Pharmaceuticals, and received lecture grants from AbbVie, MSD, Sofar, Chiesi, Janssen, Pfizer, and Takeda Pharmaceuticals. SO has received speaker fees from AbbVie, Takeda, Janssen, Amgen, Galapagos and Norgin MCF received consultancy fees from Roche, Takeda, Jannsen-Cilag, Pfizer, Sandoz, Biogen, Galapagos and research economic support from Abbvie. ES has served as speaker for AbbVie, AGPharma, Alfasigma, EG Stada Group, Fresenius Kabi, Grifols, Janssen, Innovamedica, Malesci, Medtronic, Novartis, Pfizer, Reckitt Benckiser, Sandoz, SILA, Sofar, Takeda and Unifarco; has served as consultant for Alfasigma, Amgen, Biogen, Bristol Myers Squibb, Celltrion, Diadema Farmaceutici, Falk, Fresenius Kabi, Janssen, Merck & Co., Reckitt Benckiser, Regeneron, Sanofi, Shire, SILA, Sofar, Synformulas GmbH, Takeda and Unifarco; and has received research support from Reckitt Benckiser, SILA, Sofar and Unifarco. Grants or contracts from any entity AbbVie, Janssen, Kern, Takeda, Galapagos, Pfizer Consulting fees: Janssen. Payment or honoraria for lectures, presentations, speaker bureaus, manuscript writing or educational events, AbbVie, Janssen, Takeda, Pfizer. Support for attending meetings and/or travel from Ferring and Janssen. Educational activities research projects scientific meetings or advisory boards sponsored by MSD, Ferring and Abbvie. MV served as a consultant for Abbvie, MSD, Takeda, Janssen-Cilag, and Celgene. He received lecturer fees from Abbvie, Ferring, Takeda, MSD, Janssen-Cilag, and Zambon. FC served as consultant to Abbvie, MSD, Takeda, Janssen, Roche, Celgene, Bristol-Meyers Squipp, Galapagos, Gllead, Pfizer, Mundipharma, Galapagos, Biogen, received lecture fees from Abbvie, Ferring, Takeda, Allergy Therapeutics, Janssen, Pfizer, Biogen, and unrestricted research grants from Giuliani, Sofar, MSD, Takeda, Abbvie.

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PP0630

REDUCED COLONOCYTE HYPERPLASIA IN THE PROXIMAL COLON FOLLOWING ORAL ADMINISTRATION OF EMU OIL IN A MOUSE MODEL OF COLITIS-ASSOCIATED COLORECTAL CANCER

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Introduction: Ulcerative colitis, a type of inflammatory bowel disease, is characterised by chronic colonic inflammation often resulting in uncontrolled crypt lengthening and the development of colitis-associated colorectal cancer (CA-CRC). Orally administered Emu Oil, derived from Emu adipose tissue, has demonstrated protective and reparative properties in preclinical models of bowel disease, including CA-CRC. However, its long-term potential to reduce colonocyte proliferation and inhibit CA-CRC progression remains undetermined. Protein Ki-67 has been used widely as a proliferation marker for human tumour cells and may serve as a potential predictive biomarker for cancer progression.

Aims & Methods: We hypothesized that orally administered Emu Oil would reduce colonocyte proliferation, evidenced by decreased Ki-67 expression, in a mouse model of CA-CRC. Female C57BL/6 mice were injected with saline or azoxymethane (7.4mg/kg; intraperitoneal), followed by *ad libitum* access to either water or three dextran sulphate sodium (DSS)/water cycles (each cycle comprising seven days DSS [2%w/v] and 14 days water). Mice (n=9/group) were orally administered either water (80µL) or Emu Oil (low 80µL or high 160µL dose) thrice weekly for nine weeks. Proximal and distal colonic sections were stained with haematoxylin and eosin for crypt depth, cell count and size measurements and stained immunohistochemically to detect and quantify the proliferation marker Ki-67 in upper and lower colonocyte crypt zones. $p<0.05$ was considered statistically significant.

Results: In the proximal colon, numbers of Ki-67-positive cells in the upper and lower crypt regions of CA-CRC mice were increased by up to 100% compared with normal controls; low dose Emu Oil significantly decreased numbers of Ki-67-positive cells by 47% ($p<0.05$) compared with CA-CRC controls. Numbers of Ki-67-positive cells in the distal colon increased only in the upper crypt region in CA-CRC controls compared with normal mice (157%), with no further impact following Emu Oil administration ($p>0.05$). The overall percentage of Ki-67-positive cells remained unchanged in proximal and distal colonic crypts across treatment groups ($p>0.05$). Proximal and distal colonic crypt depth increased in mice with CA-CRC (20% and 37%, respectively; $p<0.01$), with greater distal colonocyte count, compared with normal controls (37%; $p<0.05$). Emu Oil did not impact proximal and distal colonic crypt depth nor cell counts in normal or CA-CRC mice ($p>0.05$). Furthermore, proximal colonocyte count, proximal and distal colonic cell diameter, and number of cells per 100µm remained unchanged among treatment groups ($p>0.05$).

Conclusion: Crypt lengthening in CA-CRC was primarily attributed to crypt cell hyperplasia, determined by increased cell numbers with no change in cell size. Low-dose Emu Oil administration in mice with CA-CRC resulted in attenuation of proximal colonic crypt cell proliferation, with no impact on crypt depth or cell size. Importantly, crypt depth and proliferation in normal mice remained unchanged following Emu Oil treatment, supporting its safety for long-term use to treat intestinal disorders. Future studies would benefit from an assessment of Emu Oil and its impact on expression of the oncogene *Kras*, tumour suppressor genes *p53* and *Apc*, and inflammatory cytokines, including interleukin- 6, tumor necrosis factor-alpha and interferon-gamma, which are implicated in colorectal cancer progression.

Disclosure: Nothing to disclose.

PP0631

EMU OIL-MODULATED FAECAL MICROBIOTA TRANSPLANTATION AMELIORATES CLINICAL INDICATORS OF DISEASE IN A MOUSE MODEL OF CROHN'S-LIKE COLITIS

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Introduction: Crohn's disease (CD), a variant of inflammatory bowel disease, is a complex, debilitating, chronic and relapsing disorder of the gastrointestinal tract. If poorly controlled, patients are at risk of intestinal obstruction and infection due to the formation of strictures, fistulae and narrowing of the intestinal lumen, often requiring surgical intervention. As the precise aetiology remains unclear, current treatments for CD are limited. An altered composition of native gut microbiota may play a role in CD pathogenesis, with recent faecal microbiota transplantation (FMT) studies revealing variable therapeutic efficacy in CD management. Orally administered Emu Oil, derived from Emu adipose tissue, has demonstrated anti-inflammatory and reparative properties in models of intestinal disease.

Aims & Methods: The current study aimed to determine whether Emu Oil-modulated FMT could reduce disease severity and modulate the gut microbiota in a mouse model of trinitrobenzene sulfonic acid (TNBS)-induced Crohn's-like colitis. Female ARC(s) mice were assigned to FMT donor (n=5/group) and recipient (n=10/group) groups. *Donor* mice were orally administered either Water (80µl), Olive Oil (OO) or Emu Oil (160µl) daily (days -7 to 0); fresh faeces were collected (days -1 to 3) and pooled per group. *Recipient* mice were administered TNBS (3mg; day 0) and donor faecal supernatant (120µl; day 3) intrarectally and euthanised on day 6. Daily bodyweight and disease activity index (DAI), colonoscopically-assessed disease severity, organ weights, intestinal permeability (fluorescein isothiocyanate [FITC]-dextran uptake), histologically-assessed colonic damage (disease severity, crypt depth), intestinal barrier function (mucin-secreting cells), acute inflammation (myeloperoxidase [MPO]) and faecal microbiome (16S rRNA amplicon sequencing) were assessed. $p<0.05$ was considered statistically significant.

Results: TNBS induced bodyweight loss in all groups, which normalised to day 0 values on the final day of the experimental period. Daily DAI remained elevated throughout the trial in all groups; only Emu Oil FMT normalised DAI on day 6 compared with day 0 ($p<0.05$). TNBS increased colonoscopically-assessed disease severity scores ($p<0.05$), with no significant

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Emu Oil nor OO FMT effect. Emu Oil FMT decreased liver weights and increased small intestinal weights relative to bodyweight, reduced histologically assessed disease severity and increased colonic mucin counts compared with Water and OO FMT ($p<0.05$). The most abundant genera across all groups were *Muribaculaceae*, *Eisenbergiella* and *Bacteroides*; while alpha and beta diversity, taxonomic profiles, FITC-dextran, crypt depth and MPO activity remained unchanged in all treatment groups ($p>0.05$).

Conclusion: Emu Oil-modulated FMT improved histologically-assessed indicators of intestinal disease and barrier function, compared with Water FMT and OO FMT. Interestingly, Emu Oil-modulated FMT affected liver and small intestinal organs weights, suggesting a systemic effect in this setting of disease. Importantly, key genera across taxonomic profiles were identified, however, the microbiome and remaining disease indicators were largely unchanged between treatments. Overall, Emu Oil-modulated FMT demonstrates therapeutic potential in CD management. Future studies would benefit from a deeper assessment of the gastrointestinal microbiome through long-read whole genome sequencing to further determine the role of the microbiome in EO FMT efficacy compared with conventional FMT.

Disclosure: Nothing to disclose.

PP0632

EVALUATING THE EFFICACY AND SAFETY OF ZASOCITINIB (TAK-279), A HIGHLY SELECTIVE ORAL TYROSINE KINASE 2 INHIBITOR, IN PATIENTS WITH MODERATELY TO SEVERELY ACTIVE CROHN'S DISEASE: STUDY DESIGN FOR A PHASE 2B RANDOMIZED CLINICAL TRIAL

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Introduction: Tyrosine kinase 2 (TYK2) mediates cytokine signalling in immune-mediated inflammatory diseases, including Crohn's disease (CD). Zasocitinib (TAK-279) is a highly selective, oral, allosteric TYK2 inhibitor with efficacy and acceptable safety shown in patients with psoriasis and psoriatic arthritis.[1,2] This phase 2b study (NCT06233461) aims to assess the efficacy and safety of zasocitinib in patients with moderately to severely active CD.

Aims & Methods: This is a multicentre, randomized, double-blind induction, placebo-controlled, dose-ranging, treat-through study in patients (18–75 years) with moderately to severely active ileal, ileocolonic or colonic CD (CD Activity Index [CDAI] score 220–450 and ulcerations characteristic of CD defined by ileocolonoscopy at screening and Simple Endoscopic Score for CD [SES-CD] ≥6 or ≥4 for isolated ileitis). Eligible patients must have inadequate/loss of response or intolerance to ≥1 of oral aminosalicylates, corticosteroids, immunosuppressants, biologics, JAK inhibitors (JAKi) or sphingosine-1-phosphate (S1P) receptor modulators. Patients who have also failed <3 classes of advanced therapies will be included. After screening (30 days), approximately 268 patients will be randomized 1:1:1:1 to receive low-, medium- or high-dose zasocitinib or placebo orally once daily for 12 weeks (induction phase), followed by a 40-week maintenance period and a 4-week safety follow-up. At or after Week 12, patients receiving placebo with disease worsening based on PRO2 score will receive active zasocitinib treatment. Study doses are expected to provide ≥90% TYK2 inhibition over the 24-hour dosing interval, and were associated with efficacy in preclinical models.[3] Randomization will be stratified based on previous biologic, JAKi or S1P receptor modulator therapies, and steroid use at baseline. The number of patients who received previous biologic, JAKi, S1P receptor modulator, anti-interleukin (IL)-23p19 or anti-IL-12/23p40 therapy will be capped.

Primary endpoint analysis will be conducted using a Cochran–Mantel–Haenszel (CMH) test. Binary secondary endpoints will be analysed similarly to primary endpoints. Longitudinal continuous secondary endpoints up to Week 12 will be analysed using a mixed-effects model with repeated measures. Safety data will be presented by treatment group and assessment period.

Results: The study is actively enrolling with the first patient randomized in March 2024. An open-label extension is planned for zasocitinib responders. The primary efficacy endpoint is the achievement of endoscopic response at Week 12 (>50% decrease in SES-CD from baseline [for patients with isolated ileal disease: SES-CD ≤4 or ≥2-point reduction]). Secondary efficacy endpoints at Week 12 include: clinical remission (CDAI <150), clinical response (CDAI reduction from baseline of ≥100), endoscopic remission (SES-CD ≤4 or ≤2 for patients with ileal disease, no subscore >1), patient-reported symptoms and disease-specific health-related quality of life (HRQoL). Safety and tolerability will be assessed by treatment-emergent adverse events (AEs), serious AEs and AEs of special interest throughout the study. Intestinal ultrasound measurements, using the International Bowel Ultrasound Segmental Activity Score (IBUS-SAS), will be assessed at Weeks 4, 12 and 52 in a subset of patients.

Conclusion: Alongside evaluating the efficacy and safety of zasocitinib in patients with moderately to severely active CD, this unique 52-week treat-through study design will assess the impact of treatment on HRQoL, an area of significant unmet need in this population.

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PP0633

DISEASE CLEARANCE IS AN ACHIEVABLE TARGET IN PATIENTS WITH UC: PRELIMINARY RESULTS

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Introduction: Recent delineations by experts from the IOIBD organization have emphasized disease clearance as a comprehensive outcome in ulcerative colitis (UC), incorporating concurrent clinical, endoscopic, and histologic remission [1]. However, real-world data regarding disease clearance following biologic therapy remain scarce.

Aims & Methods: We conducted a retrospective observational cohort study at our academic tertiary care institution in Padova, comprising 119 patients receiving treatment with biologic drugs, totaling 142 treatments. Patient data encompassed demographics, disease phenotype, treatment history, and concomitant medications. Categorical variables underwent comparison using the Chi-Squared exact test, Mann-Whitney U test, and McNemar test, while continuous variables were assessed via either the Student's t-test or the Wilcoxon test, based on their distribution. Propensity score matching was employed to analyze outcomes between patients administered Vedolizumab and Infliximab. Furthermore, conditional logistic regression was utilized to estimate the odds of outcomes. All data analyses were conducted using the JASP 0.18.3 statistical software package, DATAtab: Online Statistics Calculator (DATAtab e.U. Graz, Austria), and XLSTAT 2024 (Assinsoft, Paris, France).

Results: In total, 142 treatments were administered to 119 adult patients diagnosed with UC. Of these patients, 67 (47.1%) presented with pancolitis, 63 (44.4%) had left-sided colitis, and 12 (8.4%) had rectal involvement at the onset of biologic therapy. At initiation, 75 (52.8%) were biologic therapy naïve. The median duration of biologic therapy was 48 months (IQR 35-71). Following Infliximab therapy, disease clearance was achieved by 9 out of 43 patients (20.9%), by 4 out of 47 patients (8.5%) with Vedolizumab, 4 out of 25 patients (16.0%) with Adalimumab, 1 out of 7 patients (14.3%) with Ustekinumab, 4 out of 15 patients (26.7%) with Golimumab, and 1 out of 5 patients (20.0%) with Tofacitinib. Upon applying propensity score matching to the cohorts treated with Infliximab and Vedolizumab, Vedolizumab demonstrated an odds ratio of 0.26 (95% CI 0.04 - 1.56, p = 0.139) compared to Infliximab. Additionally, the log-rank test showed no significant difference in the time to event occurrence among various biologic therapies (p = 0.41).

Conclusion: Attaining disease clearance is an attainable goal for UC patients undergoing biologic therapy. Future data will shed light on whether specific medications facilitate reaching this target more effectively. Continued research endeavors are essential to validate and enhance these findings.

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Disclosure: Nothing to disclose.

PP0634

OPTIMISM MAY OUTWEIGH DISEASE ACTIVITY IN MITIGATING MEDICAL TRAUMA IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE (IBD)

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Introduction: Recent research finds 25%-30% of IBD patients experience chronic post-traumatic stress due to their disease (IBD-PTS) and IBD-PTS is associated with poorer outcomes. Post-traumatic growth (PTG) is defined as "positive psychological changes that occur as a result of one's struggle to cope with trauma or a major life crisis." PTG can reduce the negative impacts of IBD-PTS, which may be moderated by disease activity in IBD.

Aims & Methods: We aim to examine the relationship between remission status, patient expectations regarding if they can achieve and sustain remission, and PTG among patients with IBD. Adults (18+) with IBD were recruited via social media IBD support groups, and flyers posted in gastroenterology clinics and community bathrooms with additional snowball sampling methods. Surveys were completed online via Qualtrics and included demographics, Patient Simple Clinical Colitis Activity Index (P-SCCAI), Post-Traumatic Growth Inventory (PTGI), and PTSD Checklist for DSM-5 (PCL-5). Participants reported if they were in remission or experiencing a flare of symptoms, and their expectations of experiencing a flare in the next 2 months on a 4-Point Likert scale.

Results: 313 adults with IBD: UC (n=140, 44.7%), CD (n=173, 55.3%). Age 18-70 years (M=33, SD=11), 75.1% female. Scores for PTGI, P-SCCAI, and PCL-5 are in Table 1. Participants tended to have active disease and high levels of IBD-PTS. No significant correlation existed between P-SCCAI score and PTG (r= -.09 p=.127).

Those in a current IBD flare (n=133,42.4%) reported lower confidence in the longevity of their remission over the next 2 months compared to participants who were in remission ($X2=114.01$,$p<.001$), with no difference in PTG levels between groups (p=.310). However, participants who demonstrated optimism about sustained remission had higher levels of PTG ($F(3,309)=4.688$,$p=.003$).

Participants who endorsed "I am positive I will experience a flare up in the next two months" had much lower levels of PTG (M=65.91,SD=25.33) than participants who endorsed "I am positive I will not experience another flare up in the next two months" (M=81.52,SD=25.58;p=.004).

Those who endorsed "I am at least 50% positive I will experience another flare up" had lower rates of PTG (M=69.99, SD=22.95) than those who endorsed "I am positive I will not experience another flare up in the next 2 months" (M=65.91,SD=25.33;p=.026). A linear regression found the extent of the differences in the levels of confidence in maintaining remission influenced levels of PTG ($F(3,309)=4.688$ p=.003). Participants who were the least confident in their remission scored 15.61 points lower for PTG than for participants who were the most confident. Participants who were at least 50% sure they would experience another flare in the next two months scored 11.53 points lower for PTG than participants who were the most confident in their remission.

Outcome Measure	Mean (SD)	Maximum Score	Cutoff
PTGI	72.81 (24.36)	105	> 45
PCL-5	45.99 (16.28)	80	> 35
P-SCCAI	5.56 (3.51)	27	> 4

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Conclusion: Our findings suggest IBD patients who expect to achieve disease remission are more likely to have higher levels of PTG than those who are more pessimistic about their disease course, independent of current symptom activity. However, those in an IBD flare are less likely to believe they can attain remission. Due to the unpredictable nature of IBD, patients are chronically exposed to risks for medically traumatic experiences. Patients with confidence in their remission longevity may conceptualize remission as an "end" to the threat of IBD-PTS, thus allowing PTG to occur. Conversely, if remission is seen as a temporary reprieve before the next inevitable flare, it may limit patients' abilities to cultivate PTG. Providers can help patients move towards a PTG mindset by assessing patients' expectations of remission, modeling optimism in a patient's prognosis, and providing evidence for ongoing hope.

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PP0635

A NOVEL MONOCLONAL ANTIBODY DRUG CANDIDATE SPY001 TARGETING INTEGRIN α4β7 FOR THE TREATMENT OF IBD: IN VITRO PROPERTIES AND NON-HUMAN PRIMATE PHARMACOKINETICS AND SAFETY

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Introduction: Antagonism of the interaction between the cellular adhesion integrin α4β7 and endothelial ligand mucosal addressin cell-adhesion molecule-1 (MAdCAM-1) has proven to be relatively safe and effective in the treatment of Crohn's disease (CD) and ulcerative colitis (UC). Additional benefit may be gained from an α4β7 antagonist given via the subcutaneous (SC) route at extended intervals (eg, every 8 to 12 weeks) during maintenance therapy.

Aims & Methods: The humanized monoclonal IgG1 antibody SPY001 binds to the same α4β7 epitope as vedolizumab and includes a YTE modification within the Fc region to increase its serum half-life via increased affinity for the neonatal Fc receptor (FcRn), thereby favoring recycling into the circulation over degradation. The binding affinity of SPY001 to α4β7 was determined by kinetic exclusion assay (KinExA), surface plasmon resonance, and flow cytometry. In vitro functional activity was determined by inhibition of MAdCAM-1-mediated cell adhesion. In vivo half-life extension was evaluated via pharmacokinetic analysis in Tg276 transgenic mice (hemizygous for human FcRn) and in non-human primates (NHP). PK simulations in humans were based on allometric scaling of SPY001 clearance observed in NHP. Safety was assessed via a good laboratory practice toxicity (GLP-tox) study in NHP.

Results: SPY001 binds specifically to α4β7 and not to the related integrins α4β1 and αEβ7. In cell-based binding assays, SPY001 binds α4β7 expressed on the surface of RPMI-8866 cells with an affinity matching that of vedolizumab, but does not bind to Ramos cells expressing α4β1. It demonstrates high affinity for α4β7, with a K_D of <1 nM.
Functionally, SPY001 potently inhibits MAdCAM-1-mediated cell adhesion with an IC_{50} comparable to that of vedolizumab but has no inhibitory activity for VCAM-1-mediated cell adhesion.
In Tg276 mice, the half-life of SPY001 was approximately 12 days compared to approximately 4 days for vedolizumab. In cynomolgus monkeys, the half-life of SPY001 was approximately 22 days following a single SC infusion, substantially longer than the half-life of vedolizumab (~6 days) in the same study. Based on allometric scaling of SPY001 clearance observed in this study, predictive simulations of SPY001 PK in humans suggest that Q8W-Q12W SC dosing will achieve therapeutic exposure in patients across a wide range of body weights and albumin concentrations. In a repeat-dose GLP toxicity study in NHP no adverse or dose limiting toxicities were observed even at the highest dose tested. Both SC and IV administration were well-tolerated and observations related to the administration procedure were similar across doses including controls.

Conclusion: SPY001 is a novel humanized monoclonal IgG1 with an extended half-life relative to that of vedolizumab in Tg276 mice and cynomolgus monkeys. Repeat-dose toxicity studies at doses up to 160 mg/kg showed no SPY001-related adverse events and a safety profile supportive of progression into clinical development. SPY001 offers the potential for effective and safe treatment of CD and UC with the advantage of infrequent SC dosing. Clinical studies are warranted to demonstrate this potential.

Disclosure: EZ, DR, RV, JO, and HS are employees of Paragon Therapeutics.
JF, DN, AS, MR, and OB are employees of Spyre Therapeutics.

PP0636

COMPARISON OF THE EFFICACY AND SAFETY OF GLUCOCORTICOIDS AND BIOLOGICS IN THE TREATMENT OF MODERATE TO SEVERE ULCERATIVE COLITIS

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Introduction: Currently, biologics are widely used in patients with moderate to severe ulcerative colitis (UC), while there is limited research directly comparing the efficacy of biologics and traditional corticosteroid therapy in patients with moderate to severe UC.

Aims & Methods: This study aims to analyze whether the efficacy and safety of biologics (such as infliximab, IFX, or vedolizumab, VDZ) in patients with moderate to severe UC are superior to corticosteroids.A retrospective analysis was conducted on patients diagnosed with moderate to severe UC at the First Affiliated Hospital of Nanchang University from July 2014 to January 2023. Patients were divided into two groups based on prescription treatment: biologics and corticosteroids. A comparison was made between the two groups at 14 weeks for endoscopic remission (ER), clinical remission (CR), clinical response rate, and the difference in survival rates without UC-related events (relapse, discontinuation of medication, surgery, malignancy, death) at 52 weeks. Factors influencing endoscopic remission, clinical remission, and clinical response within 14 weeks in both treatment groups were analyzed. The incidence of adverse events in both treatment groups was also analyzed.

Results: This analysis demonstrates that after IPTW adjustment, the endoscopic remission rate at 14 weeks in the biologics group was significantly higher than in the corticosteroids group (biologics group vs corticosteroids group: OR: 4.14, 95% CI 1.20-15.3; *P*=0.001). However, there were no statistically significant differences between the two groups in terms of

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clinical response rate at 14 weeks (OR: 1.26, 95% CI 0.36-2.05; *P*=0.754) and clinical remission rate (OR: 1.46, 95% CI 0.46-4.75; *P*=0.700).Furthermore, the 52-week survival rate without any UC-related events in the biologics group was significantly higher than in the corticosteroids group (biologics group vs corticosteroids group: HR: 0.26, 95% CI 0.12-0.59;*P*<0.001). In addition, biologic therapy (*P*=0.031), disease severity (*P*=0.019), and baseline high-sensitivity C-reactive protein (CRP) levels (*P*=0.010) were independently associated with endoscopic remission at 14 weeks in UC patients. Baseline high-sensitivity CRP levels were independently associated with clinical remission at 14 weeks in patients with ulcerative colitis (*P*=0.045).Regarding adverse events, there were no statistically significant differences between the two groups, including adverse events (OR: 1.52, 95% CI 0.52-4.50; *P*=0.444) and the incidence of severe adverse events (OR: 0.34, 95% CI 0.03-3.89; *P*=0.364).

Conclusion: Biologic therapy (IFX or VDZ) for severe ulcerative colitis shows higher endoscopic remission rates at 14 weeks and higher survival rates without UC-related events at 52 weeks compared to corticosteroid treatment. Despite baseline imbalances between the two treatment groups, similar conclusions were drawn after adjusting for confounding factors with IPTW. It is hoped that these findings will be further confirmed in prospective randomized trials.

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PP0637

ENDOSCOPIC BALLOON DILATION ASSOCIATED WITH BIOLOGICAL THERAPY IN CROHN'S DISEASE STRICTURES: A SINGLE CENTER EXPERIENCE

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Introduction: Symptomatic luminal stricture in Crohn's Disease (CD) occurs in up to 20% of patients at the diagnosis and in more than 50% during their lifetime. A considerable percentage of patients often require further surgery for post-operative CD recurrence, with a consequent increased burden of short bowel syndrome. In this scenario, combining biological therapy with endoscopic balloon dilation (EBD) could potentially reduce this risk.

Aims & Methods: Our study aimed to evaluate the efficacy of EBD combined with biologic therapy in symptomatic gastrointestinal strictures in CD patients. We conducted a retrospective study including consecutive CD patients who underwent EBD for low gastrointestinal strictures from December 2015 until November 2022 at our tertiary IBD Center. Subsequent subocclusive/occlusive episodes, the need for steroids, and surgery due to CD stenosis were evaluated during a follow-up period of 6, 12 months, and 24 months.

Results: A total of 52 patients treated with EBD with an overall number of 55 strictures were included. The mean age was 46,8 (18-77) years. The median of disease duration was 9 (0-42) years. Among the total of 55 studied strictures: 14 (25%) in patients with active smoking, 21 (38%) with increased CRP values at the time of dilation, 43 (78%) had evidence of moderate or severe endoscopic disease activity with ulcerative lesions at stricture site. Anastomotic strictures accounted for 26 (47%) of all 55 strictures. The remaining 29 (53%) were de novo stricture: 8 (15%) ileal, 11 (20%) at ileocecal valve, 4 (7%) colonic, and 6 (11%) anal . Forty-two patients (76%) were under biologic therapy at the time of EBD and 2 (4%) started after EBD. EBD technical success was reported in 60% of the procedures, and no adverse events were recorded. After 6 months from EBD, at univariate analysis, increased CRP at the time of dilation resulted in a risk factor for the need for steroids (OR 14,63, IC 2,690 to 71,50; p-value = 0,0006, Table 2). Concomitant biological therapy, active smoke, and the presence of ulcerative lesions did not show a significant correlation with considered outcomes. After 12 and 24 months from EBD, concomitant biological therapy resulted in being a protective factor for surgery (OR 0,06; IC 0,005 to 0,502; p-value=0,0047), while subocclusive and occlusive events were less frequent in anastomotic strictures than ileal stenosis (OR 0,07; IC 0,005 to 0,651; p-value=0,0152).

Conclusion: In patients with CD undergoing EBD for intestinal CD stricture, concomitant biological therapy reduces the risk of surgery for CD stenosis recurrence. Further prospective and controlled studies are underway to confirm these observations.

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Disclosure: Antonio Gasbarrini reports personal fees for consultancy for Eisai S.r.l., 3PSolutions, Real Time Meeting, Fondazione Istituto Danone, Sinergie S.r.l. Board MRGE, and Sanofi S.p.A, personal fees for acting as a speaker for Takeda S.p.A, AbbVie, and Sandoz S.p.A, and personal fees for acting on advisory boards
for VSL3 and Eisai. Daniela Pugliese received speaker fees from AbbVie, MSD, Takeda and Janssen, Pfizer; advisory board fees from Pfizer. Franco Scaldaferri i: advisory board for Abbvie, Janssen, MSD, Sanofi and Takeda; All the other authors have no conflict of interest to declare.

PP0638

PHARMACOKINETIC COMPARABILITY OF ABP 654 ADMINISTERED EITHER VIA PREFILLED SYRINGE OR AUTOINJECTOR IN HEALTHY ADULT SUBJECTS: RESULTS FROM A RANDOMIZED, OPEN-LABEL, PARALLEL-GROUP STUDY

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Introduction: ABP 654 is a biosimilar to ustekinumab reference product that is supplied in a prefilled syringe (PFS). The ABP 654 autoinjector (AI) is in development as a single-use, disposable, handheld, latex-free, mechanical injection device that aims to offer additional convenience for patients and healthcare professionals.
Aims & Methods: This clinical study was designed to evaluate the comparability, based on exposure (maximum observed serum concentration [C_{max}] and area under the serum concentration-time curve from time 0 extrapolated to infinity [AUC_{inf}]), of ABP 654 after a single subcutaneous (SC) injection delivered by PFS or AI in healthy male and female adults. The study also assessed additional pharmacokinetics (PK) parameters, safety, tolerability, and immunogenicity.
A total of 156 subjects (78 subjects in the PFS group and 78 subjects in the AI group) were randomized in a 1:1 ratio to receive a single 90 mg SC injection of ABP 654 administered either via PFS or AI. Serum concentrations of ABP 654 were determined using a validated electrochemiluminescent assay. PK variables were calculated from the serum concentration data using noncompartmental methods and actual sampling times. Blood samples for antidrug antibody (ADA) assessments were collected at various times. Samples testing positive for binding ADAs were also tested for neutralizing ADAs. Adverse events (AEs) were graded by severity according to Common Terminology Criteria for Adverse Events (CTCAE) version 4.03.
Results: Mean (± SD) serum ABP 654 concentration-time profiles show a high degree of similarity between the PFS and AI groups with overlapping profiles over the course of the study. PK comparability was established between the PFS and AI groups based on the 90% CIs of the geometric mean ratios (GMR) for the primary PK endpoints of AUC_{inf} and C_{max} being contained within the prespecified margin of 0.8 to 1.25. Exposure was comparable between the 2 groups. Peak concentrations were observed on average about 8 days following injection. After achieving peak ABP 654 concentrations, levels declined in a monophasic manner with a $t_{1/2}$ of about 23 days. The estimated $t_{1/2}$ was comparable for both groups. The mean percentage of AUC_{inf} due to extrapolation from the last quantifiable concentration observed to infinity (AUC%Extrap) was 5.13% and 4.53% for the PFS and AI groups, respectively, confirming the adequacy of the duration of PK sampling for the evaluation of AUC_{inf}. The frequency, type, and severity of AEs were comparable between the groups and consistent with the safety profile of ustekinumab. Through the end of the study, 10 (12.8%) subjects in the PFS group and 16 (20.5%) subjects in the AI group tested positive for the development of binding ADAs. This numerical difference is not considered clinically impactful, based on demonstrated PK comparability and similarity of safety profiles between treatment arms.

Treatment	AUC_{inf} GMR (90% CI)	C_{max} GMR (90% CI)	C_{max} (µg/mL) GM (Geo CV [%])	T_{max} (h) Median (range)	AUC_{last} (h*µg/mL) GM (Geo CV [%])	AUC_{inf} (h*µg/mL) GM (Geo CV [%])	$t_{1/2}$ (h) Mean (SD)	AUC% Extrap (%) Mean (SD)
ABP 654 PFS	1.0069 (0.92, 1.10)	1.0503 (0.97, 1.14)	9.79 (31.7)	191.93 (48.00, 599.62)	9040 (35.1)	9660 (36.6)	575.75 (155.38)	5.13 (3.48)
ABP 654 AI			10.30 (32.8)	191.09 (48.00, 291.92)	9300 (34.8)	9720 (36.8)	556.11 (143.34)	4.53 (2.87)

Table 1. Summary of ABP 654 PFS and ABP 654 AI PK Parameters

Conclusion: PK comparability was established between the PFS and AI groups. Both treatments were safe and well-tolerated. Overall, the results support conclusions of PK comparability and similar safety and immunogenicity profiles between ABP 654 administered via PFS and AI.
Disclosure: V. Chow, D. Mytych, J. Cao, C. Barragan, W. Radziszewski and J. Franklin are Amgen employees and shareholders. P. Winkle has no conflicts.

PP0639

EFFECTIVENESS AND SAFETY OUTCOMES BEYOND FIVE YEARS OF IBD PATIENTS TREATED WITH VEDOLIZUMAB. THE LONG-LIVE IG-IBD STUDY

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Introduction: Long-term real-life data on vedolizumab (VDZ) are scarce. In this study, we aim to evaluate the long-term outcomes of patients (pts) with inflammatory bowel disease (IBD) who received VDZ.
Aims & Methods: The LONG-LIVE is the long-term extension (LTE) of the LIVE study, including IBD pts who had started VDZ from 04 to 06/2017 at 47 Italian centres affiliated with IG-IBD, prospectively followed-up to 06/2019. For this study, data on therapy withdrawal, clinical and endoscopic activity, need for surgery/hospitalizations, deaths, and serious adverse events (SAEs) were retrospectively extracted from 07/2019 up to the last available follow-up (fup).
Results: Seven hundred eighty-two (70% of the original cohort) pts were enrolled in the LTE (394 CD, 388 UC), 316 (40%) of whom had already definitively discontinued VDZ; 43% were female, mean age was 53±16 years, mean disease duration was 17±9 years, 76% of patients were bioexposed to anti-TNFα therapy. Among the remaining 466 still receiving VDZ, 204 (44%) subsequently suspended VDZ permanently and 9 (2%) temporarily: mean VDZ therapy duration was 113±92 weeks and 311±60 weeks among those who suspended and those who continued VDZ, respectively. Mean therapy duration was 177±124 weeks (172±124 weeks for CD and 182±125 weeks for UC, with no significant differences); longer mean persistence was observed in bionaive vs bioexposed CD pts (197±123 vs 182±122 weeks, p<0.05), while no such difference was observed in UC pts (193.1±126 vs 190±122. week, p=ns). Among CD pts who continued VDZ throughout the entire LTE, 74% and 54% were in clinical and endoscopic remission, respectively, at the last fup; among UC pts who continued VDZ throughout the entire LTE, 79% and 50% were in clinical and endoscopic remission, respectively, at the last fup. During the LTE, CD pts had a higher

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risk of IBD-related hospitalizations (Hazard Ration, HR, 1.37, 95% Confidence Interval, CI, 1.04-1.72, p=0.027). Specifically, 153 (20%) pts (83 CD, 70 UC) were hospitalized to receive IBD-related surgery after a mean time of 136±87 weeks from VDZ start (for CD vs UC comparison, HR 1.17, 95% CI 0.85-1.60, p=ns); 85 (11%, 54 CD, 31 UC) pts underwent IBD-related hospitalization for non-surgical reasons (CD vs UC, HR 1.71, 95% CI 1.21-2.62, p=0.015). New or recurrent cancers were reported in 17 pts (the most common localization being 4 breast cancers and 3 gastrointestinal cancers): mean interval between first VDZ administration and cancer diagnosis was 214 (±95) weeks; 20 non-oncological SAEs were recorded in pts receiving VDZ, the most common being infections (n=7), mostly pneumonia (n=4). Finally, 15 deaths were reported (of which 9 occurred in patients still receiving VDZ); the most common causes of death were infections (n=5), cardiovascular diseases (n=5) and cancer (n=3).

Conclusion: About one-third of pts were still receiving VDZ after a fup of more than 6 years, with the majority of them being in clinical remission and more than 50% also having quiescent endoscopic activity at the last observation; bionaive CD patients had longer persistence compared to bioexposed ones. During the LTE, CD pts had higher rates of IBD-related hospitalizations; most hospitalizations were due to the necessity of surgery. SAEs were observed in a minority of patients and predominantly consisted of infections; the rates of cancer diagnosis are seemingly in line with those of the general population. Overall, our study strengthens the notion that vedolizumab is associated with a long-term real-life durability in terms of both effectiveness and safety.

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PP0640

LONG TERM FOLLOW-UP OF PATIENTS WITH ULCERATIVE COLITIS AFTER WITHDRAWL OF INFLIXIMAB FOR DEEP AND STABLE REMISSION

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Introduction: Infliximab (IFX) is the first anti-TNF biologic drug approved for treating moderate to severe ulcerative colitis (UC). While IFX is effective in inducing mucosal healing and enhancing quality of life, concerns persist about the long-term safety and cost-effectiveness of extended immunosuppression. Current guidelines lack clarity on when to discontinue treatment after achieving stable deep remission. Moreover, existing literature is limited to studies characterized by small sample sizes and heterogenous criteria for defining remission. Our study aimed to evaluate the rate of remission maintenance in UC patients who discontinued IFX after achieving stable deep remission, to identify predictors of relapse post-discontinuation, and to examine clinical outcomes in those who experienced a relapse

Aims & Methods: In this single-center, retrospective, observational study, we analyzed UC patients at our tertiary IBD Referral Center who discontinued IFX due to sustained deep remission, defined as the combination of partial Mayo subscore ≤ 1 and endoscopic Mayo score = 0, and who had at least 1 year of follow-up post treatment withrdawl. We assessed the relapse rates at intervals of 6 months, 1 year, 2 years, and beyond 2 years post-IFX discontinuation. For those who relapsed, we recorded the therapeutic strategy and its effectiveness, including the necessity for colectomy. Additionally, we performed univariate and multivariate analyses to identify relapse predictors post-treatment suspension.

Results: From 2006 to 2022, 790 UC patients received IFX treatment at our center; 113 suspended IFX due to stable deep remission with at least 1 year of follow-up and were included in this study. Of these, 41.6% maintained remission over a median of 5.17 years. Flare rates were 14.2%, 23.9%, and 35.5% within 6, 12, and 24 months, respectively, with 58.4% experiencing a flare overall. Multivariate analysis showed that longer treatment duration reduced relapse risk (p=0.001), while steroid use during IFX treatment and early IFX initiation post-diagnosis increased relapse risk (p=0.004 and p=0.043, respectively). Most cases of disease relapsee were successfully managed medically either with a course of steroids, restarting IFX, or initiation of a different type of biologic/small molecule, whereas the rate of colectomy was 7.1%.

Conclusion: This study is one of the largest to date with the longest follow-up that evaluates outcomes post-IFX discontinuation in UC patients in deep remission. Our finding suggest that withdrawal of IFX treatment can be considered in carefully selected cases. In patients who relapse after IFX suspension, medical treatment is often effective, while the rate of colectomy is low. Further studies are warranted to better identify ideal candidates for treatment withdrawal.

Disclosure: Nothing to disclose.

PP0641

FRAILTY IS DYNAMIC OVER TIME AND ASSOCIATED WITH ALL-CAUSE AND ACUTE HOSPITALIZATIONS DURING 18 MONTHS FOLLOW-UP IN OLDER PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Frailty is prevalent among older patients with Inflammatory Bowel Disease (IBD). A geriatric assessment is a method to assess the presence of frailty and its course over time. However, the predictive value of the geriatric assessment and the longitudinal course of frailty have not yet been studied in patients with IBD.

Aims & Methods: The primary aim was to evaluate the course of frailty and to identify factors associated with changes in level of frailty over 18 months follow-up in older patients with Inflammatory Bowel Disease (IBD). The second aim was to study the association of frailty with hospi-

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talizations during follow-up. A prospective multicenter cohort study was conducted in IBD patients (≥ 65 years). At baseline, a geriatric assessment was performed in all 405 patients, assessing impairment in five domains: the somatic domain, (instrumental) activities of daily living, physical capacity, mental and social domain. Follow-up was conducted after 18 months. Part of the patients were seen for a second, follow-up geriatric assessment. The rest of the patients was contacted by phone for a follow-up assessment. Primary outcome was change in the level of frailty, defined as a change in the number of impaired domains at follow-up compared to baseline (an increase in number of impaired domains suggests an increase in level of frailty, a decrease in number of impaired domains suggests a decrease in level of frailty). Secondary outcomes were all-cause, acute and IBD-related hospitalizations during 18 months. Logistic regression was performed to identify factors associated with changes in level of frailty, adjusted for age, sex, type of IBD and educational level. Cox regression analyses were performed to study the association of frailty with adverse outcomes, adjusted for age, sex and biochemical disease activity (C-reactive protein ≥ 10 mg/L and/or fecal calprotectin ≥250 µg/g).

Results: Out of 405 patients (median age: 70 years) with a geriatric assessment at baseline, 160 (39.5%) patients had 2 or 3 impaired domains and 32 patients (7.9%) had 4 or 5 impaired domains. Follow-up was conducted in 356 patients (87.9%), of which 154 (43.3%) patients were seen for a geriatric assessment and 202 patients (56.7%) were contacted by phone. In the geriatric assessment, 32.5% had increased and 26.0% decreased ≥ 1 in number of impaired domains compared to baseline. Use of systemic corticosteroids at or three years prior to baseline associated with an increase in impaired domains (adjusted odds ratio (aOR) 3.00; 95% confidence interval (95% confidence interval (CI)) 1.11-8.08; p-value= 0.030), while initiation of biological therapy associated with a decrease in impaired domains (aOR 4.92, 95%CI 1.41-17.15, p-value=0.013). Over 18 months, all-cause hospitalizations occurred 136 times in 96 patients (23.7%). Of all hospitalizations, 103 (75.7%) were acute occurring in 74 patients and 41 were IBD-related (30.1%) occurring in 28 patients. Frailty was independently associated with both all-cause (adjusted hazard ratio (aHR) 1.28, 95% confidence interval [CI] 1.05-1.56, p-value=0.016) and acute hospitalizations (aHR 1.30, 95% CI 1.04-1.61, p-value=0.020). No association was found for IBD-related hospitalizations.

Conclusion: The level of frailty is dynamic in 58.5% of the older patients with IBD over 18 months. In addition, frailty at baseline is independently associated with all-cause and acute hospitalizations. Future studies should establish how interventions targeting frailty could improve health outcomes.

Disclosure: Nothing to disclose.

PP0642

A RETROSPECTIVE COHORT STUDY TO DEMONSTRATE MAINTENANCE OF REMISSION RATES FOR SUBCUTANEOUS COMPARED TO INTRAVENOUS VEDOLIZUMAB IN THE TREATMENT OF INFLAMMATORY BOWEL DISEASE

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Introduction: Vedolizumab (VDZ) is an anti α4β7 integrin antibody therapy that inhibits migration of T-lymphocytes to the gastrointestinal epithelium through mucosal addressin cell adhesion receptor[1]. As a licensed treatment for inflammatory bowel disease (IBD), it is now available as a subcutaneous (SC) formulation. The efficacy and safety of SC VDZ was comparable to intravenous (IV) therapy in two randomised controlled trials (VISIBLE 1 and 2) in ulcerative colitis (UC)[2] and Crohn disease (CD)[2]. SC biologic preparations offer patients flexibility to self-administer therapy and reduce hospital visits.

Aims & Methods: Objective is to evaluate the efficacy of SC to IV VDZ in IBD and to assess impact on costs.

Patients receiving IV VDZ infusions at a single tertiary centre were assessed for eligibility to switch to SC formulation. Between February 2021 to February 2023, all patients switch and remaining on IV were included. Baseline data was collected with follow-up at 6 and 12 months. The primary outcome was maintenance of clinical remission, defined as HBI<4 for CD and SCCAI<3 for UC at 6 and 12 months. Secondary outcomes included serum CRP, faecal calprotectin (FCP), drug persistence at 6 and 12 months and potential savings calculated using the drug price and bed costs.

Results expressed as rates and odds ratio (OR) were analysed with GraphPad prism using one-way ANOVA, mixed-effect analyses (q value) and chi-squared test. Benjamini-Hochberg false discovery rate (FDR) adjustment was used for P value correction

Results: 246 patients on IV VDZ were identified. 29 were not suitable to switch: 20/29 due to disease phenotype remained in the IV group, 11 patients on escalated dose regimen were excluded from both groups.

Of the 195 included patients, 175 were eligible to switch: 80/175 declined SC treatment due to preferences and continued IV therapy. In the SC switch group (n =95, 54%) median age was 54.5 years, 43% (n=41) were females, 63% (n=60) had UC, 37% (n=35) had CD. The commonest CD phenotype was ileal disease (N=15/35). Compared to the IV group (n =100, 51%) median age was 46 years, 42% (n=42) were females, 63% (n=63) had UC, 37% (n=37) had CD. The commonest CD phenotype was ileocolonic disease (N=11/37).

The primary outcomes are shown in Table 1 which demonstrated statistical significance in maintenance of remission with SC VDZ. The odd for remission with SC VDZ at 6 and 12 months were 1.1 [CI] (p0.7) and 3.6 [CI} (p0.006) at 12 months.

There were no statistically significant differences in CRP and FCP between baseline,6, or 12 months for the IV and SC groups. For CRP, 80% and 75% of patients maintained biochemical remission 6 and 12 months and for FCP 78% at 6 and 81% at 12 months.

Drug persistence for SC at 6 and 12 months were 93% and 87%, respectively. 4% (N=4) patients switched to a different drug due to loss of response, 8% (n=8) switched back to IV VDZ due to adverse events, issues with homecare or symptom control.

Based on this cohort of patients, an expected reduction of £ 72,010 per annum is likely to be achieved.

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	SC		IV	
Baseline	81% (n=77)		77% (n=77)	
6 months	82% (n=78)	q=0.2979 (Baseline vs 6 months)	80 % (n=80)	q=0.894 (Baseline vs 6 months)
12 months	86% (n=82)	q=0.002 (Baseline vs 12 months)	79% (n=79)	q=0.894 (Baseline vs 12 months)

Table 1: Displays baseline, 6-month and 12-month remission rates of VDZ IV and SC in IBD.

Conclusion: Transitioning patients from IV to SC VDZ was associated with low drug discontinuation rate. We observed higher remission rates at 12 compared with 6 months for SC group in comparison to patients who remained on IV VDZ. These findings should be used to encourage uptake of switching strategies.

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Disclosure: None.

PP0643

COMBINING IL-23 BLOCKADE WITH ANTI-α4β7 OR ANTI-TL1A FOR THE TREATMENT OF IBD IS SUPPORTED BY IN VITRO AND MOUSE IBD MODEL EXPERIMENTS

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Introduction: Despite the discovery of advanced treatments for IBD, placebo-adjusted clinical remission rates during induction therapy are typically only 10%-25%, and loss of response is common even among those who initially benefit. Monoclonal antibodies targeting IL-23 or α4β7 are both relatively safe and effective treatments for Crohn's disease (CD) and ulcerative colitis (UC), while antibodies to the cytokine TL1A have shown promise in phase 2 clinical trials. The combined use of targeted biologic agents may improve efficacy by inhibiting multiple pathways while avoiding the risks associated with broad immunosuppression. Pre-clinical studies are needed to assess the potential benefit of combining IL-23 antagonism with α4β7 or TL1A blockade.

Aims & Methods: IL-17 production in response to IL-23 and/or TL1A by human peripheral blood mononuclear cells (PBMC) and mouse splenocytes was assessed in the presence of anti-IL-23 and/or anti-TL1A. The ability of anti-IL-23 and anti-α4β7 as monotherapies or in combination to prevent colitis was assessed by transferring CD4+ CD45RB^{high} T cells into *Rag2*$^{-/-}$ mice. Survival, body weight, disease activity, histology, and colon IL-17 levels were measured.

Results: IL-23 and TL1A had a synergistic effect on IL-17 production by human PBMC and mouse splenocytes, extending previous findings in CD patient-derived lamina propria T cells[1]. Both anti-IL-23 and anti-TL1A inhibited IL-17 production, with the greatest inhibition occurring with combined IL-23 and TL1A blockade.

In the mouse T-cell transfer colitis (TCT) model, the anti-IL-23 and anti-α4β7 combination at high doses resulted in improved body weight relative to either monotherapy. The monotherapies and the combination of anti-IL-23 and anti-α4β7 (both high and low doses) resulted in similar improvements in survival, disease activity index, and histology score. The combination resulted in greater reductions in colonic IL-17 protein than either monotherapy alone.

Conclusion: IL-23 and TL1A have a synergistic effect on promoting IL-17 secretion from human and mouse immune cells. The combination of anti-IL-23 and anti-TL1A suppresses IL-17 secretion more effectively than either agent alone. In the TCT model of IBD, combination therapy with anti-IL-23 and anti-α4β7 improves body weight and reduces colonic IL-17 levels relative to monotherapy with less obvious effect on other disease measures.

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Disclosure: DR, MA, JV, JO, and HS are employees of Paragon Therapeutics.

JF, DN, and AS are employees of Spyre Therapeutics.

PP0644

EXPLORING THE MOLECULAR TARGETS AND MECHANISM OF TARAXASTEROL IN THE TREATMENT OF ULCERATIVE COLITIS BASED ON NETWORK PHARMACOLOGY AND EXPERIMENTAL VALIDATION

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Introduction: Ulcerative colitis (UC) is a chronic, non-specific inflammatory disease of the bowel of unknown etiology. This disease is characterised by high difficulty in cure, high recurrence and cancer rates, which seriously endangers the lives and health of our population. Taraxasterol (TA), a major constituent of the traditional Chinese medicine dandelion, has been reported in several studies to have good anti-inflammatory and antioxidant activities as well as anti-tumour effects.

Aims & Methods: However, the molecular targets and mechanisms of action of TA against UC have not been fully elucidated.The Swiss Target Prediction, PharmMapper databases were used to obtain relevant targets for the drug TA. Targets for UC were obtained from GeneCards, OMIM, PharmGkb, DrugBank databases. Targets of TA anti-UC identified by intersection of drug target and disease target.

The relevant targets of TA anti-UC were uploaded to the String database and the resulting data were imported into Cytoscape 3.9.1 software to construct a protein-protein interaction (PPI) network of drug-disease targets. At the same time, the top 10 hub targets were screened using CytoHubba software.

Gene ontology (GO) and Kyoto Encyclopedia of Genomes (KEGG) pathway enrichment analyses were performed on TA anti-UC targets using R software, and molecular docking was performed by Autodock Vina 1.1.2. The GEO dataset was used to analyze the differential mRNA expression of the TA anti-UC top 10 hub gene in UC patients and the normal population. Lipopolysaccharide (LPS)-induced intestinal epithelial cells (NCM460) were used to construct an in vitro cellular inflammation model and a Dextran Sulfate Sodium Salt (DSS) induced acute mouse colitis model. The pivotal targets and key signaling pathways were also validated by RT-qPCR and Western blotting.

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Results: A total of 129 relevant targets of TA against UC were obtained, of which the top 10 hub targets were GRB2, PTPN11, SRC, PIK3R1, LCK, EGFR, MAPK1, IGF1, ESR1 and JAK2. The PI3K-Akt pathway was significantly associated with TA treatment of UC as identified by KEGG enrichment analysis. GEO dataset analysis demonstrated that EGFR and GRB2 were upregulated in UC patients, while SRC, PIK3R1 and JAK2 were down-regulated. In vitro experiments revealed that TA treatment reversed the expression of EGFR, GRB2, PIK3R1 and JAK2 differential genes.

Furthermore, TA was able to up-regulate the mRNA expression of pro-inflammatory factor IL-10 and decrease the mRNA levels of anti-inflammatory factors IL-6, IL-1β and TNF-α in LPS-induced NCM460 cells. In vivo assay confirmed that TA significantly alleviated DSS-induced colonic inflammation in mice, including reversal of weight loss and shortening of colon length, reduction of DAI score in mice, and improvement of intestinal barrier disruption. In vivo and in vitro experiments have confirmed that TA also inhibits the activation of the PI3K/Akt pathway, thus exerting an anti-inflammatory effect.

Conclusion: TA may play a therapeutic role in UC by inhibiting the PI3K-Akt pathway. EGFR, GRB2, PIK3R1 and JAK2 may be the most relevant potential therapeutic targets for TA in the treatment of UC.

Disclosure: Nothing to disclose.

PP0645

THE VALUE OF TOFACITINIB FOR THE TREATMENT OF MODERATE TO SEVERE ULCERATIVE COLITIS: TWO-YEARS DATA FROM TOFA-UC, A SN-IBD STUDY

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Introduction: Real-world evidence is becoming increasingly important to determine the value of drugs in clinical practice. We have recently reported one-year data on safety and effectiveness of tofacitinib (TOFA) for the treatment of moderate to severe ulcerative colitis (UC) in Sicily, Italy.

Aims & Methods: TOFA-UC is a multicenter, observational, 2-year study performed among the Sicilian Network for Inflammatory Bowel Disease (SN-IBD). All consecutive patients with moderate to severe UC starting TOFA from June 2021 to June 2023 (24 months) were enrolled. The primary outcome was the assessment of safety, in terms of rate of adverse events (AEs). Secondary outcomes included the rates of clinical response and steroid-free clinical remission (based on partial Mayo score) after 8 and 52 weeks, the rate of loss of response, and treatment persistence.

Results: 171 patients were included [median follow-up: 12.0 (IQR 6.0, 19.0) months]. Only 9 patients (5.3%) were naïve to biologics, while most had previous failures to one (39.2%), two (25.1%), three (24.6%), or four (5.8%) biologics.

Thirty adverse events were reported (17.5%; incidence rate: 15.85 per 100 patient-years), with hypercholesterolemia being the most frequent one (13 cases). No herpes zoster reactivation was reported, while one case of pulmonary embolism occurred. At week 8, the rates of clinical response and steroid-free clinical remission were 76.6% and 51.5%, respectively.

At week 52, the rates of clinical response and steroid-free clinical remission were 63.6% and 58.3%, respectively. Forty patients experienced a loss of response after successful induction (38.8%; incidence rate: 108.3 per 100 patient-years).

Sixty-seven patients (39.2%) discontinued TOFA over time, of whom 9 due to AEs, and 58 due to primary non response or loss of response. By multiple regression analysis model, being former smoker (hazard ratio=2.05; 95% CI=1.02-4.13; p=0.043) and a higher partial Mayo score at baseline (hazard ratio=1.20; 95% CI=1.04-1.38; p=0.011) were the variables associated with a higher risk of treatment discontinuation.

Conclusion: After two years of use in clinical practice, our multicenter study confirmed the safety and effectiveness of TOFA for the treatment of patients with moderate to severe UC and multiple previous failures to biological therapies.

Disclosure: Funded by Pfizer. The protocol of this investigator-sponsored study was selected in the context of the "Local Competitive Grant Program RE-CUP 2021 (Global Medical Grants system ID #71844083)", a competition between Italian projects promoted by Pfizer.

PP0646

DESIGN OF A PHASE 2 STUDY TO EVALUATE THE EFFICACY AND SAFETY OF THE HIGHLY SELECTIVE ORAL TYROSINE KINASE 2 INHIBITOR ZASOCITINIB (TAK-279), IN PATIENTS WITH MODERATELY TO SEVERELY ACTIVE ULCERATIVE COLITIS

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Introduction: Tyrosine kinase 2 (TYK2) mediates cytokine signalling pathways involved in immune-mediated inflammatory diseases, including ulcerative colitis (UC). Zasocitinib (TAK-279) is a highly selective, oral, allosteric TYK2 inhibitor that has shown efficacy and acceptable safety in psoriasis and psoriatic arthritis.[1,2]

This phase 2 study (NCT06254950) will assess efficacy and safety of zasocitinib in patients with moderately to severely active UC.

Aims & Methods: This is a multicentre, randomized, placebo-controlled, double-blind induction study in adults (18–75 years) with moderately to severely active UC (modified Mayo score of 5–9 points and endoscopic subscore ≥2) with a documented diagnosis (endoscopic with histology)

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≥30 days before screening. Eligible patients must have inadequate/loss of response or intolerance to ≥1 of oral aminosalicylates, corticosteroids, immunosuppressants, biologics, JAK inhibitors (JAKi), α4 integrin antagonists or sphingosine-1-phosphate (S1P) receptor modulators. Patients who have failed <3 classes of advanced therapies will be included. Approximately 207 patients will be randomized 1:1:1 to receive low- or high-dose zasocitinib or placebo orally once daily for 12 weeks. Study doses were selected to provide ≥90% TYK2 inhibition over the 24-hour dosing interval, and were associated with efficacy in preclinical models.[3] Randomization will be stratified based on previous biologic, JAKi, α4 integrin antagonist or S1P receptor modulator therapies, and steroid use at baseline. The study comprises a 30-day screening period, a 12-week double-blind induction treatment period, a 40-week open-label treatment period and a 4-week safety follow-up. After Week 12, all patients will be reassigned to receive active zasocitinib treatment in the open-label maintenance period. Enrolment of patients who have received previous biologic, JAKi, α4 integrin antagonist, S1P receptor modulator, anti-interleukin (IL)-23p19 or anti-IL-12/23p40 therapy will be capped.

Primary endpoint analysis will be conducted using a Cochran–Mantel–Haenszel (CMH) test. Binary secondary endpoints will also be analysed similarly to primary endpoints. A mixed-effects model with repeated measures will analyse longitudinal continuous secondary endpoints. Safety data will be presented by treatment group and assessment period.

Results: The study is actively enrolling. An open-label extension is planned for zasocitinib responders. The primary efficacy endpoint is clinical remission at Week 12, assessed as the proportion of patients achieving a modified Mayo score ≤2, with stool frequency and endoscopic subscores ≤1 and a rectal bleeding subscore of 0. Secondary efficacy endpoints at Week 12 include proportion of patients achieving clinical response (reduction from baseline in modified Mayo score ≥2 and ≥30%, and a decrease in rectal bleeding subscore ≥1 or absolute rectal bleeding subscore ≤1), symptomatic remission (rectal bleeding subscore 0 and stool frequency subscore ≤1), endoscopic improvement (modified Mayo endoscopic subscore ≤1), endoscopic remission (modified Mayo endoscopic subscore 0), patient-reported symptoms and disease-specific health-related quality of life (HRQoL). Safety and tolerability will be assessed by monitoring treatment-emergent adverse events (AEs), serious AEs and AEs of special interest throughout the study.

Conclusion: Alongside evaluating the efficacy and safety of zasocitinib in patients with moderately to severely active UC, this study will assess the impact of treatment on HRQoL, a significant unmet need in this population.

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PP0647

JAK INHIBITORS AND S1P RECEPTOR MODULATORS FOR THE TREATMENT OF ANTIBIOTIC REFRACTORY CHRONIC POUCHITIS: AN ECCO CONFER MULTICENTRE CASE SERIES

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Introduction: Data regarding effectiveness and safety of JAK inhibitors and S1P receptor modulators in chronic antibiotic refractory pouchitis (CARP) are lacking.[1-4] This ECCO-CONFER project collected cases of CARP treated with JAK inhibitors or S1P receptor modulators.

Aims & Methods: This retrospective multicentre study collected cases across Europe through the ECCO-CONFER project, including patients with ulcerative colitis (UC) diagnosis who underwent proctocolectomy with ileal pouch anal anastomosis (IPAA), treated with JAK inhibitors or S1P receptor modulators for CARP and at least three months of follow up.

Main outcomes were steroid and antibiotics-free clinical response at three months (T3) for each drug, defined as clinical part of modified Pouchitis Disease Activity Index (mPDAI) reduction by ≥1 points, steroid and antibiotics-free clinical response at 1 year (T12) assessment, defined as mPDAI reduction by ≥2 points, steroid and antibiotics-free clinical remission (mPDAI=0) at T3 and T12, endoscopic response at T12 (improvement in PDAI Endoscopic Inflammation subscore reaching <3) and endoscopic remission at T12 (endoscopic PDAI ≤1). Non-response imputation was applied.

Results: Seventeen treatments with small molecules of CARP in 15 patients were collected from nine centres in seven countries. Eight patients were treated with tofacitinib; steroid and antibiotics-free clinical response at T3 was achieved in six patients (75%) and steroid and antibiotics-free clinical remission was achieved in five patients (62.5%). One patient discontinued tofacitinib for loss of response after eight months. Of the patients with at least 12 months of follow-up, steroid and antibiotics-free clinical response was achieved in 50% of patients, steroid and antibiotics-free clinical remission in one patient (16.7%), endoscopic response in 60%, endoscopic remission in 60%.

One patient was treated with filgotinib, neither steroid and antibiotics-free clinical response nor steroid and antibiotics-free clinical remission were achieved. Six treatments with upadacitinib were collected: at T3 steroid and antibiotics-free clinical response was achieved in 33.3% and steroid and antibiotics-free clinical remission was achieved in 16.7% of cases. Two cases were treated with ozanimod. At T3 steroid and antibi-

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otics-free clinical response was achieved in one patient (50%) and steroid and antibiotics-free clinical remission was not achieved. No side effects were reported. Main baseline characteristics are summarized in Table 1.

Age at JAK/S1P introduction Mean years (SD)	38.7 (± 11.2)
Male, n (%)	10 (66.7)
Median years from IPAA (IQR)	50 (4.0-12.0)
Mean years from chronic pouchitis diagnosis (SD)	3.9 (±3.3)
Indication for IPAA, n (%)	Disease refractory to medical treatment, 12 (80.0) Cancer or dysplasia, 3 (20.0)
Target therapies before IPAA, n (%)	
Infliximab	11 (73.4)
Adalimumab	7 (46.7)
Vedolizumab	6 (40.0)
Biologics after IPAA for pouchitis, n (%)	
Infliximab	5 (33.3)
Adalimumab	4 (26.7)
Vedolizumab	9 (60.0)
Ustekinumab	5 (33.3)
Mean score of clinical part of mPDAI, (SD)	3.9 (±1.1)
Mean PDAI endoscopic inflammation subscore, (SD)	4.6 (±0.8)

Conclusion: In a multiple biologic-refractory population, small molecules might be a suitable treatment option for CARP. Based on current limited data, the use of JAK inhibitors should be further investigated.

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Disclosure: Nothing to disclose.

PP0648

SYSTEMATIC REVIEW WITH META-ANALYSIS OF THE EFFECTIVENESS OF SUBCUTANEOUS BIOLOGICS VERSUS INTRAVENOUS BIOLOGICS IN INFLAMMATORY BOWEL DISEASES

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Introduction: Biologic therapies are commonly used for inflammatory bowel disease (IBD) patients. Multiple biologic medicines can now be given both intravenously and subcutaneously. The different administration routes present provide different advantages regarding dose escalation, healthcare resource utilisation, pharmacokinetics, convenience, and safety.[1,2] Comparator effectiveness studies between intravenous and subcutaneous administration are lacking.

Aims & Methods: Our primary outcome was to compare the effectiveness between intravenous (IV) and subcutaneous (SC) biologics in rates of clinical remission. We performed a systematic review and meta-analysis to include all relevant articles from MEDLINE (Ovid), EMBASE, PubMed, and Cochrane central register of controlled trials from January 1st, 1946 to January 28th, 2024. Studies that compared intravenous and subcutaneous administration of the same biologic therapy in IBD patients and reported effectiveness outcomes were included. Eligible studies had to include one of the following outcomes: [a] Clinical remission or response with a formalised clinical scoring system as defined by each study; [b] CRP remission; [c] faecal calprotectin remission. This study was registered on PROSPERO (CRD42023442675).

Results: Twenty studies met the inclusion criteria for the systematic review. There were only two randomised controlled trials and therefore were not analysed. There were eighteen cohort studies. One study was a prospective clinical trial switch from IV to SC vedolizumab. Twelve were prospective cohort studies. Five were retrospective cohort studies. Ten studies compared vedolizumab, seven studies compared infliximab and one study compared ustekinumab. We excluded the ustekinumab study given it was the only study evaluating ustekinumab. Of the 17 included studies, 15 were whole cohort switches from IV to SC formulations, with only 2 studies having both IV and SC arms. Nine vedolizumab cohort studies were meta-analysed for clinical remission and no difference was found in clinical remission rates between IV and SC administration (RR 0.99; 95% CI 0.88, 1.11). Six infliximab cohort studies were meta-analysed for clinical remission and no difference was found in rates of clinical remission between IV and SC administration (RR 0.91; 95% CI 0.77, 1.08). Four studies reported CRP remission rates (1 infliximab study and 3 vedolizumab studies), and the vedolizumab studies were meta-analysed. Seventy one percent (159/224) of the SC group and 71% (207/291) of the IV group were in CRP remission. The pooled RR was 0.88 (95% CI 0.88, 1.07; p=0.09). We observed a high degree of heterogeneity (I^2 = 76%). Four studies, all studying vedolizumab, reported faecal calprotectin remission, and they were meta-analysed. Seventy six percent (158/207) of the SC group and 76% (140/183) of the IV group were in faecal calprotectin remission. The pooled RR was 0.99 (95% CI 0.90, 1.09). We observed no heterogeneity (I^2 = 0%). The quality of the cohort studies was high, with no studies found to be at a high risk of bias.

Conclusion: There was no difference in the effectiveness between subcutaneous and intravenous administration of infliximab and vedolizumab respectively. However, further high-quality studies, particularly head-to-head studies are required to confirm this finding.

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PP0649

TEDUGLUTIDE FOR THE TREATMENT OF SHORT BOWEL SYNDROME IN CROHN'S DISEASE PATIENTS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Severe Crohn's disease (CD) may progress to short bowel syndrome (SBS). Teduglutide stands as a validated treatment for SBS offering promising results for these patients.

Aims & Methods: This study aims to evaluate the performance of teduglutide in the subgroup of patients with SBS secondary to CD. We performed a systematic review in PubMed, Embase and Cochrane databases, searching for studies with patients with SBS and CD treated with teduglutide. The primary outcome was the reduction of parenteral support (PS). The secondary outcome was discontinuation of PS. The odds ratio (OR) was used for binary outcomes with their respective 95% confidence interval (CI). Heterogeneity was assessed using the Cochran Q test and I^2 statistics.

Results: The meta-analyses included 4 studies and 77 patients. All studies evaluated PS reduction and teduglutide demonstrated a good performance (PS reduction in 79.94% of the patients; 95% CI 57.97% to 100.00%; I2=85%). Three studies (61 patients) evaluated PS discontinuation and teduglutide demonstrated a good performance too (PS discontinuation in 28.59% of the patients; 95% CI 17.43% to 39.75%; I2=0%).

Conclusion: Teduglutide is a good therapeutic option for patients with SBS in the subgroup of CD patients, improving PS reduction and discontinuation and leading to better quality of life of these patients.

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Disclosure: Nothing to disclose.

PP0650

EFFICACY OF LOW-DOSE AZATHIOPRINE ASSOCIATED WITH ALLOPURINOL IN INFLAMMATORY BOWEL DISEASE: A SYSTEMATIC REVIEW AND META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: Thiopurines are widely used for treating inflammatory bowel disease, either alone or in combination with anti-tumor necrosis factor alpha therapy. However, intolerance to thiopurines is common, leading to discontinuation of treatment in over 30% of individuals.

Aims & Methods: This study aims to evaluate the effectiveness of combined therapy of allopurinol with low-dose azathioprine concerning therapeutic response and improvement of therapy tolerance. We performed a systematic review in PubMed, Embase, Cochrane and Web of Science databases, searching for randomized studies that evaluated low-dose azathioprine therapy plus allopurinol versus azathioprine monotherapy. The primary outcome was clinical remission at weeks 14-16. The secondary outcomes were 6-TGN, 6-MMP and FC (fecal calprotectin) levels and FC lesser than 50ug/g rate. The odds ratio (OR) was used for binary outcomes with their respective 95% confidence interval (CI). Heterogeneity was assessed using the Cochran Q test and I^2 statistics.

Results: The meta-analyses included 3 studies and 237 patients. There was a significant difference favoring low-dose azathioprine plus allopurinol in clinical remission (OR 2.46; 95% CI 1.43 to 4.25; p=0.001; I2=0%) and in 6-MMP levels (MD -2,329.89; 95% CI -3,869.68 to -790.11; p<0.01; I2=94%). There is no difference in 6-TGN levels (p=0.05) and FC levels (p=0.68).

Conclusion:

Patients using allopurinol plus low-dose azathioprine showed improved therapeutic adherence while maintaining superior clinical remission rates compared to azathioprine monotherapy.

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PP0651

ROUTINE CARE DATA ANALYZED USING EXPLAINABLE ARTIFICIAL INTELLIGENCE IDENTIFY BEST THERAPEUTIC SEQUENCE OF 2ND LINE BIOLOGIC TREATMENTS AFTER A 1ST LINE WITH ANTI-TNF OR VEDOLIZUMAB IN INFLAMMATORY BOWEL DISEASE

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Introduction: Inflammatory bowel disease (IBD) includes Crohn's disease and ulcerative colitis, two conditions characterised by chronic inflammation of the gastrointestinal tract, which can damage the bowel and cause serious disabilities in patients' quality of life. Treatments only have a suspensive effect and the therapeutic sequences are currently based on reimbursement criteria or the prescriber's expertise, in a "trial-and-error" approach rather than on validated scientific evidence.

Routine clinical care data from 1,190 IBD patients collected by a private tertiary care centre dedicated to IBD patients, using the customised Instamed® platform, includes comprehensive information on the patient's treatment history, clinical variables and disease activity scores, in particular the Harvey-Bradshaw Index (HBI), Mayo Partial Score (MPS) and Simple Clinical Colitis Activity Index (SCCAI).

KEM® (Knowledge Extraction and Management) is an explainable Artificial Intelligence (xAI) platform using Formal Concept Analysis (FCA) that systematically extracts and evaluates all relations between variables in a database, thus enabling the identification of patient subgroups with greater likelihood of response to treatment sequences.

Aims & Methods: This study addresses the prevalent clinical challenge of uncertainty in treatment selectionfollowing the initial administration of anti-TNF or vedolizumab given as first-line therapy for IBD, aiming to identify the most effective second-line treatment options.

Data were aggregated into a consolidated database of 1,190 patients and 1,269 variables within the KEM® xAI Platform. Patients with at least 2 treatment lines were selected. To ensure a strong causality between treatment prescription and response, the maximal number of treatment lines was limited to 3. The outcome used was therapeutic response after three treatment lines, evaluated using HBI, PMS and SCCAI. Within 19,230,463 associations between sequence of the first two treatment lines and the response, 70 associations were retained by KEM® with successful treatment sequence with anti-TNFs, vedolizumab or thiopurine as first treatment line.

Results: 285 patients fulfilled the analysis criteria. We identified hypotheses in favor of a second-line biological treatment (ustekinumab, JAKi, 2nd anti-TNF) after failure of a first-line treatment with anti-TNF or vedolizumab. Ustekinumab was 2 times more likely to be associated with a response to treatment. JAKi was 2 times less likely to be associated with a response to treatment. The 2nd anti-TNF was associated with a response to treatment in a specific subgroup. Moreover, successful treatment sequences with patient count above 10 and up to 30, involved thiopurine as first line of treatment, followed by anti-TNFs or thiopurine repeated as second line of treatment.

Conclusion: We show a powerful setup using routine clinical care data from an initial set of 1,190 IBD patients and the KEM® xAI platform to provide actionable hypotheses supporting therapeutic
decisions for disease management. These initial results will be further tested and validated as we accumulate real-life data on new patients and complete the follow-up of patients already included. The flexibility of our platform will enable us to include more therapeutic sequences and to add all the new treatments available in current clinical practice. This will allow us to offer continuous, up-to-date support for difficult medical decisions, enabling patient management to be guided by results observed in real life and leading de facto to precision medicine.

Disclosure: Nothing to disclose.

PP0652

REAL-LIFE USE OF THE CROHN'S DISEASE EXCLUSION DIET (CDED) IN ADULT PATIENTS WITH MILD-TO-MODERATE CROHN'S DISEASE ACTIVITY: AN OPEN-LABEL, RANDOMIZED CONTROLLED TRIAL

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Introduction: The Crohn's Disease Exclusion Diet (CDED) is a dietary approach that exclude specific foods to induce remission in Crohn's disease (CD).

Aims & Methods: We conducted an open-label, randomized controlled trial including adults with mild-to-moderate CD, as determined by the Harvey-Bradshaw Index (HBI). Patients were randomly assigned to either CDED or a control diet (Mediterranean pattern). At enrollment, we collected demographic data and medical histories. We measured weight, body mass index (BMI), HBI, fecal calprotectin, and serum inflammatory markers at baseline, 12 and 24 weeks. Bioelectrical impedance analysis (BIA) evaluated safety for CDED participants. The primary outcome was clinical remission (HBI<5), measured in the intention-to-treat population. This study aims to provide a real-world evidence.

Results: Twenty-four patients were randomized to the CDED diet and 21 to the control group, without significant differences at baseline. Five CDED participants discontinued the diet for intolerance by a median of 5 weeks. After 12 and 24 weeks, CDED patients showed significantly lower HBI and higher remission rates compared to control group (70.8% vs 38.1% at 12 weeks and 79.2% vs 42.9% at 24 weeks; p=0.027 and p<0.0001, respectively). Fibrinogen resulted significantly lower in CDED than control group at week 24. CDED participants reported a median BMI reduction from 25.8 kg/m^2 to 24.4 kg/m^2 at 12 weeks and 24.5 kg/m^2 at 24 weeks (p=0.047), although BIA analysis indicated a reduction in fat mass after 12 and 24 weeks (p<0.001 and p<0.001, respectively) while fat-free mass and body cellular mass significantly increased at 12 weeks (p=0.001 and p=0.042, respectively) and remained stable at 24 weeks (p=0.072 and p=0.120, respectively).

Conclusion: The CDED has proven to be safe and effective for achieving remission in adults with mild-to-moderate CD.

Disclosure: No conflict of interest to disclose.

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PP0653

IMPACT OF THE INTRA-OPERATIVE USE OF CORTICOSTEROIDS AND LOCAL ANESTHETICS ON THE POST-OPERATIVE OUTCOME OF INTESTINAL BOWEL DISEASE PATIENTS UNDERGOING SURGERY FOR THEIR CHRONIC DISEASE

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Introduction: Surgical resections are an integral part of the management of patients with inflammatory bowel diseases (IBD). Typically, for patients with complicated or drug-resistant Crohn's disease (CD), ileocecal resection (ICR) is becoming the treatment of choice. In this inflammatory context, post-operative complications affect 15 to 30% of patients.

The aim of this study was therefore to assess the impact of molecules with anti-inflammatory properties, such as local anesthetics (LA) and/or dexamethasone (DXM), used during the surgical period, on the post-operative outcome of CD patients undergoing surgery for their chronic disease.

Aims & Methods: We conducted a retrospective, monocentric study, including all inflammatory bowel diseases (IBD) patients who underwent major surgery related to their disease. We compared patients who received the combination of LA and DXM (LA+DXM group) versus the remaining patients who received only one of the two or neither (non-LA+DXM group). The primary endpoint was overall morbidity within 30 days post-operatively, assessed according to the Clavien-Dindo classification. The secondary endpoints were minor (mC) and major complications (MC), Comprehensive Complication Index (CCI), length of hospitalization, perioperative kinetics of inflammatory markers and pain assessment, and intra- and postoperative morphine consumption.

Results: Between January 2015 and December 2021, a total of 426 patients were operated and were 219 analyzed. Of these, 114 CD were operated on for scheduled ICR. 49 patients received the combination of LA+DXM during surgery and 65 did not (non-LA+DXM). The percentage of complications were 33% vs 60% in the LA+DXM vs non-LA+DXM groups, respectively. In univariate analysis, overall post-operative morbidity was statistically reduced in the LA+DXM group (OR 0.32 [0.15- 0.70], p = 0.007), and this result was confirmed after logistic regression (OR 0.26 [0.11-0.63], p = 0.003). This reduction in morbidity was also linked to a reduction in mC in the LA+DXM group (OR 0.28 [0.07-0.92], p = 0.005), as well as a clear trend towards a reduction in MC. Median length of hospital stay was 8 days for the LA+DXM group vs. 9 days for the non-LA+DXM group (p = 0.02). Regarding the post-operative inflammatory syndrome, despite a tendency to decrease in the LA+DXM group, there was no statistical difference in thresholds set at D3, D5 and D7. On the other hand, patients in the LA+DXM group received less intraoperative opioids (83.3mg morphine-equivalent vs. 54.76mg for the LA+DXM group, p < 0.01), and the total quantity of opioids titrated in the recovery room was also lower (40.5mg vs. 30mg for the LA+DXM group, p = 0.01).

Conclusion: Intraoperative co-administration of LA+DXM in CD patients undergoing scheduled ileocecal resection for chronic disease significantly reduces overall postoperative morbidity. A controlled study is needed to confirm these promising results.

Disclosure: Nothing to disclose.

PP0654

EFFICACY OF USTEKINUMAB IN PREVENTING POST-OPERATIVE RECURRENCE IN A REAL LIFE COHORT OF PATIENTS WITH CROHN'S DISEASE

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Introduction: Prevention of postoperative recurrence (POR) after ileocolonic resection is one of the most relevant issues in the management of Crohn's disease (CD). The role of ustekinumab (UST) especially in the context of real-life studies is still not entirely clear, often due to small sample sizes, thus requiring further investigation and confirmation (1).

Aims & Methods: Our real life single-centre retrospective study aims to evaluate the effect of UST therapy on prophylaxis of endoscopic recurrence after ileo-colonic resection in CD. In the whole cohort UST was used as first-line therapy for prophylaxis of recurrence. All patients (pts) underwent the first endoscopic evaluation within 6-12 months after surgery to assess postoperative recurrence (POR), using the Rutgeerts score and were subsequently evaluated at 22 (+-3) months with a second endoscopy. Persistence on therapy ~ two years after the surgery was also evaluated.

Results: 27 pts with a mean age of 51 years old were included. 40% of pts had previously been resected and 45% were active smokers. 93% had already been exposed to one or more biological drugs. The most frequent reason for surgery was the presence of stenosing disease. The first post-operative colonoscopy at 6-12 months, showed severe POR (RS score i3 or i4) in 19% of patients, so 11% changed treatment at this time point. Among patients who continued the drug (24 pts), the second endoscopy performed at 22 (+-3) months showed severe POR in 13% of pts, whose treatment was subsequently changed.

At ~ 2 years persistence on therapy in the cohort was 78%, notably endoscopic remission (RS<2) was found in 37%.

No patients discontinued the drug due to adverse effects.

Conclusion: In our real life cohort of complex CD patients, ustekinumab showed high levels of persistence on therapy at the point of time considered, although less than half of the patients achieved endoscopic remission.

References: 1- Tursi A, Mocci G, Picchio M, Elisei W, Maconi G. Letter: ustekinumab for the treatment of post-surgical and refractory Crohn's disease. Aliment Pharmacol Ther. 2021 Apr;53(7):859-860. doi: 10.1111/apt.16281. PMID: 33709395.

Disclosure: None/

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PP0655

MIRIKIZUMAB PHARMACOKINETICS AND EXPOSURE-RESPONSE IN PEDIATRIC PATIENTS WITH MODERATELY TO SEVERELY ACTIVE ULCERATIVE COLITIS: RESULTS FROM THE PHASE 2 SHINE-1 STUDY

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Introduction: Mirikizumab is a p19-directed anti-interleukin-23 antibody approved for treatment of adults with moderately to severely active ulcerative colitis (UC). Here, we report mirikizumab pharmacokinetics and exposure-response in pediatric patients with moderately to severely active UC from the phase-2, multicenter, open-label study SHINE-1 (NCT04004611).

Aims & Methods: Patients aged 2 to <18 years weighing >10kg with a modified Mayo score (MMS) of 4-9 and endoscopic subscore ≥2 received intravenous mirikizumab induction doses (300mg for weight >40kg; 5 or 10mg/kg for weight ≤40kg) every 4 weeks (Q4W) through Week 12. At Week 12, patients achieving MMS clinical response received subcutaneous maintenance doses of mirikizumab 200mg (>40kg), 100mg (>20-≤40kg), or 50mg (≤20kg) Q4W for 36 weeks. Pharmacokinetic parameters were analyzed by nonlinear mixed-effects modeling with fixed allometry for size. Covariate effects on mirikizumab exposure were evaluated using simulation-based estimations. Exposure-response for patients achieving clinical response, remission, and endoscopic response at Weeks 12 and 52 were explored using graphical approaches.

Results: The analysis included 232 samples from 26 patients. Estimated systemic clearance, volume of distribution, and subcutaneous bioavailability were 0.000296 L/hr/kg, 0.069 L/kg, and 49.8%, respectively. Pharmacokinetic model-estimated exposures in all weight groups were similar to those in adult phase-3 studies LUCENT-1 (NCT03518086) and LUCENT-2 (NCT03524092) except for the 10-mg/kg induction dose, which was 2-fold higher (Figure). In the exposure range studied in SHINE-1, exposure-response analysis using post-hoc grouping by average concentration quartile revealed no improvement trend in clinical response, remission, or endoscopic response and showed similar efficacy to that in adults.

Conclusion: Mirikizumab pharmacokinetic exposures and efficacy results at intended phase-3 doses across all pediatric weight groups were similar to those reported in adults. No patient factors other than weight justify dose adjustment. Exposure-matching to the adult exposure data and exposure-response analysis support a phase-3, weight group-divided dosing strategy in pediatric patients.

Disclosure: Y. Otani, L. Chua, W. Komocsar, A. Larkin and X. C. Zhang are employees and shareholders of: Eli Lilly and Company.

PP0656

EFFICACY OF USTEKINUMAB IN THE TREATMENT OF POST-OPERATIVE RECURRENCE IN A REAL LIFE COHORT OF PATIENTS WITH CROHN'S DISEASE

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Introduction: Treatment of post-operative recurrence (POR) after ileocolonic resection is one of the most challenging issues in the management of Crohn's disease (CD). The role of ustekinumab (UST), especially in the context of real-world studies, is still not entirely clear, often due to small sample sizes. Recent studies report encouraging results, with endoscopic remission rates even more than 50% of CD patients, but data are quite heterogeneous (1,2).

Aims & Methods: Our retrospective single-centre real-life study aims to evaluate the effect of UST therapy on the treatment of relapse after ileo-colonic resection in CD. In our cohort, UST was used as second-line therapy in case of recurrence or intolerance of previous therapy.

All patients underwent endoscopic or radiological re-evaluation up to 18 months after the start of therapy. Endoscopic success, defined as a reduction of at least one point in the Rutgeerts score, or radiological success, defined as improvement in the judgement of the radiologist who performed the examination or the multidisciplinary IBD team at our centre, were evaluated. Persistence in therapy ~ two years after surgery was also assessed.

Results: 27 pts with a mean age of 37 years old at surgery were included. 37% of pts had previously been resected and 56% were active smokers. All pts had already been exposed to one or more biological drugs. The most frequent reason for surgery was the presence of stenosing disease. The mean time to start UST therapy since surgery was 21 months.

The reasons for initiation of UST were recurrence of endoscopic and/or radiological disease, or intolerance to previous therapy, Cohort 1 and Cohort 2, respectively.

In Cohort 1 (19 pts) an instrumental re-evaluation was performed after a mean of 14.4 ± 3.3 months following initiation of UST. Endoscopic or radiologic success was reported in 68% 13 of pts, of whom radiologic or endoscopic remission was achieved in 47.3%.

In Cohort 2 (10 pts) an instrumental re-evaluation was performed after a mean of 15.5 ± 3.5 months following initiation of UST, there were no patients who had advanced endoscopic (RS >= 3) and/or severe radiological recurrence of disease at the start of UST in this group. Only one patient showed endoscopic worsening of disease since the start of UST and discontinued treatment. 90% of pts had radiological or endoscopic improvement or stability of disease, of which remission was achieved in 50%.

In the entire cohort (27 pts), persistence on therapy at ~ two years was 74%, and radiological or endoscopic remission in 51%.

Conclusion: In our real life cohort of patients with CD, ustekinumab showed high levels of persistence to therapy at time point considered, with a satisfactory result on endoscopic and/or radiological remission of disease.

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1-Tursi A, Mocci G, Picchio M, et al. Maconi G.Letter: ustekinumab for the treatment of post-surgical and refractory Crohn's disease. Aliment Pharmacol Ther. 2021;53:859–860

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Disclosure: None.

PP0657

THE PLACE OF ENDOSCOPIC DILATATION IN THE MANAGEMENT OF STENOSES IN STENOSING CROHN'S DISEASE.

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Introduction: Crohn's disease (CD) is a chronic inflammatory condition of the digestive tract characterized by a cyclical course, the most frequent and feared complication is the formation of stenoses. The management of these stenoses is often multidisciplinary medico-surgical as well as endoscopic.

Endoscopy plays an essential role in the diagnosis, monitoring, and treatment of these stenoses, particularly through endoscopic dilations. The objective of our work is to study the role of endoscopic dilations during stenosing Crohn's disease in addition to medical treatment.

Aims & Methods: This is a single-center descriptive retrospective study carried out within our department over a period of 6 years: from January 2018 to January 2024, including all patients followed for stenosing Crohn's disease who underwent endoscopic dilation. We collected data related to age, history, location, phenotype, clinical symptoms, endoscopic data as well as evolution.

Results: We included 19 patients followed in our training for stenosing Crohn's disease who benefited from endoscopic dilation out of a total of 109 patients followed for Crohn's disease with a stenosing phenotype (17.4%). The average age was 45 years with extremes ranging from 20 to 71, with a male predominance and a M/F sex ratio: 2.8 (14 M and 5 F).

Three patients were smokers (15.8%), 6 patients underwent anterior ileocecal resection (31.6%) and a total colectomy ATCD in one patient (5.3%). The median time from diagnosis of Crohn's disease to formation of symptomatic stenosis was 9.3 years.

Clinically, abdominal pain was present in 8 patients (42.1%), Koenig syndrome in 7 patients (36.8%), diarrhea in 3 patients (21%), 4 patients (28.5%). had defecatory pain, evacuation constipation in 3 patients (15.8%) and late postprandial vomiting was present in 1 patient (5.3%).

The stenosis was at the last ileal loop in 5 cases (26.3%), 4 (21%) at the rectal level, 1 stenosis respectively at the colonic, anal, esophageal, pyloric and duodenal level (5.3% each) and 5 stenoses at the level of the anastomosis including 4 (21%) at the level of the ileocolic anastomosis and 1 at the level of the ileorectal anastomosis (5.3%).

The average length of the stenoses was 19.2 mm with extremes ranging from 5 to 40 mm. Dilations were carried out using hydrostatic dilation balloons whose caliber varied between 10 and 16.5 mm, 3 (15.8%) patients benefited from dilation to 16.5 mm, 7 (36.8%) at 12mm, 6 (31.6%) at 15mm, and 2 (10.5%) at 11mm.

11 patients benefited (57.9%) from a single dilation session, 7 patients benefited from 2 dilations (36.8%) and 1 patient (5.3%) had recourse to 5 dilations.

Primary success defined by the passage of the endoscope was possible in 15 patients (78.9%). Only one perforation type complication was noted, treated surgically.

The secondary success rate defined by the disappearance of symptoms was in 9 patients (47%). The average follow-up was 31.8 months and the clinical recurrence rate was 37.5% at 6 months. 4 patients (21%) benefited from additional surgery.

Conclusion: Endoscopic dilation is a minimally invasive treatment which constitutes an effective alternative to surgery in the management of stenoses during Crohn's disease; in our study more than half of our patients only had recourse to a single dilation session with a primary success of more than 78.9%, and a disappearance of symptoms at 6 months in almost half of the patients, thus making the use of surgery less frequent.

Disclosure: Nothing to disclose.

PP0658

MANAGEMENT OF ACUTE SEVERE COLITIS IN CROHN'S DISEASE

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Introduction: Acute severe colitis (ASC) is a medical-surgical emergency that can occur at any time during the course of chronic inflammatory bowel disease. It most commonly affects UC.

Aims & Methods: Aims: The aim of our work is to describe the particularities of GCA management during Crohn's disease (CD).

Materials and methods: This is a retrospective descriptive and analytical study spread over a 5-year period from January 2018 to January 2023. All patients who developed GAC during the course of Crohn's disease, whether inaugural or progressive, were included in our study.Data collected included age, sex, Crohn's disease location and phenotype, Lichtiger score before and after coticotherapy, second-line treatment, and use of surgery. SPSS 2.0 software was used for the statistical study. A binary logistic regression model was used for univariate analysis.

Results: Of the 424 IBD cases, 267 were being followed for Crohn's disease, of whom 38 (14.2%) patients presented with GAC. Mean age was 44.21 ±12.7 years (range 19-72 years). The F/H sex ratio was 1.9 (25 women to 13 men).The location of Crohn's disease was pancolic in 28 patients, ileocolic in 9, and ileocolic with high involvement in 1 patient. GCA was inaugural in 10 patients (26.3%), and complicated known Crohn's disease in 28 patients (73.7%). The mean Lichtiger score calculated at D0 was 11.84±1.8. The mean CRP value on admission was 89.8±47. Mean albumin was 27.9±3.6. Emergency abdomino-pelvic CT scans showed no complications such as colectasis, perforation or abscess in our series. All patients received IV methylprednisolone corticosteroids at a dose of 0.8 mg/kg/d, not to exceed 40 mg/d, with close clinico-biological monitoring. 21 patients (55.2%) were corticosensitive. The mean Lichtiger score calculated at D5 in corticosensitive patients was 8.7, with a mean CRP value at D3 of 36.8±12.

In univariate analysis, CRP >45 (on admission) and hypoalbuminemia were predictive factors for failure of IV corticosteroid therapy, with [OR=3.2 [1.3-5.2]; p=0.0005] and [OR = 1.31 [1.015-7,] p=0.002] respectively.Of the 17 (44.8%) cortico-resistant patients, 3 (7.9%) received emergency second-line treatment with Infliximab, and 15 (39.47%) underwent subtotal colectomy with ileostomy. Of these patients, one was scheduled after

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failure of second-line treatment. Good outcome was noted in 36 patients (94.7%). 2 cases of death (5.3%) were reported postoperatively. On discharge, 23 patients (63.9%) were treated with immunosuppressive agents or 13 (36.1%) with biotherapy.

Conclusion: Acute severe colitis can reveal or complicate Crohn's disease at any stage of its evolution. Management must be urgent, and requires medical-surgical coordination. The first-line medical treatment is intravenous corticosteroid therapy. Colectomy should be discussed at every stage of management.

Disclosure: Nothing to disclose.

PP0659

EFFICACY OF DIFFERENT MODALITIES OF FAECAL MICROBIOTA TRANSPLANTATION TO INDUCE CLINICAL REMISSION IN PATIENTS WITH ULCERATIVE COLITIS: A SYSTEMATIC REVIEW AND NETWORK META-ANALYSIS

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Introduction: To compare the efficacy of different modalities of FMT to induce clinical remission in patients with ulcerative colitis (UC).

Aims & Methods: We performed a systematic review and network analysis (random effects model) of randomized controlled trials including at least one arm of FMT in adult patients with active UC.

Primary endpoint, i.e. clinical remission (total Mayo score ≤ 2 with Mayo endoscopic score ≤ 1), was assessed between week 6 and 12. Results are expressed as relative risk with 95% confidence intervals, adjusted for bowel cleansing and pre-FMT antibiotics therapy.

Ranking of FMT modalities for clinical remission was calculated as their surface under the cumulative ranking (SUCRA), which represents the percentage of efficacy achieved by an agent compared with an imaginary agent that is always the best without uncertainty. Heterogeneity among studies was evaluated by τ (global statistical heterogeneity across all comparisons).

Results: Among the 12 selected studies, patients were exclusively bio-naïve in 4 studies (4/12), while between 9 and 32% had prior biologics exposure in the other trials. Risk of bias was low across all domains in 7 studies. Global heterogeneity was negligible ($\tau = 0$). After adjustment on bowel cleansing and pre-FMT antibiotics, oral capsule (RR=7.1[1.8-33.3]), lower GI FMT (RR=4.5[1.7-12.5] and combination of both (RR=12.5[2.1-100]) were more effective than placebo to induce clinical remission contrary to upper GI FMT (RR=1.1[0.2-7.7]) and autologous FMT (RR = 0.8 [0.2-3.8].

The combination of lower GI FMT and oral capsule was significantly more effective than upper GI FMT to induce clinical remission (RR=10.7[1.1-104.2]). Combination of lower GI FMT and oral capsule ranked highest for the induction of clinical remission (SUCRA=0.93), followed by oral capsule (SUCRA=0.82) and lower GI FMT (SUCRA=0.72). Autologous FMT ranked lower than placebo (SUCRA=0.12 vs 0.22). We performed a sensitivity analysis distinguishing single donor FMT and multi-donor FMT (pooled samples from different donors). Combination of lower GI FMT and oral capsule ranked highest for the induction of clinical remission (SUCRA=0.89), followed by single donor lower GI FMT (SUCRA = 0.70), multi-donor (pooled samples) lower GI FMT (SUCRA = 0.65), multi-donor (pooled samples) oral capsule (SUCRA = 0.61), single donor oral capsule (SUCRA = 0.60).

Conclusion: In this network meta-analysis, we confirmed the efficacy of FMT to induce clinical remission in patients with UC. The combination of lower GI and oral capsule FMT seems to be the best modality of FMT for patients with UC while upper GI FMT is not suitable. In clinical trials, autologous FMT should be avoided due to a potential detrimental effect.

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PP0660

THE PREDICTIVE VALUE OF GAMMA-GLOBULIN AND ALBUMIN LEVELS IN ASSESSING ANTI-TNF RESPONSE IN INFLAMMATORY BOWEL DISEASE

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Introduction: Loss of response to anti-TNFα therapy is a significant challenge in the management of inflammatory bowel disease (IBD) patients. Identifying predictive biomarkers for this outcome could aid in treatment optimization.

Aims & Methods: This study aimed to evaluate the predictive value of serum γ-globulin and albumin concentrations for treatment failure to anti-TNFα therapy in patients with IBD.

This was a retrospective analytical study conducted at the Gastroenterology Department of Hedi Chaker University Hospital in Sfax. The study spanned a 16-year period from 2007 to 2023. All patients with inflammatory bowel disease (IBD) treated with anti-TNF therapy were included. Patients using anti-TNF therapy for extra-intestinal manifestations of IBD were excluded.

Epidemiological data, characteristics of IBD, biological and endoscopic parameters, as well as fecal calprotectin levels, were collected. Serum albumin and γ-globulin levels were measured as part of the pre-anti-TNF assessment. Therapeutic evaluation followed the STRIDE-II strategy.

Results: The study included a total of 116 patients, among whom 82 had Crohn's disease (CD) and 34 had ulcerative colitis (UC). The mean age at diagnosis was 30.91 ± 12.227 years for all patients, with slightly lower ages observed in CD patients (30.49 ± 13.137 years) compared to UC patients (31.94 ± 9.792 years).103 patients (88.8%) received infliximab as their first-line anti-TNF therapy while 11.2% patients were initially treated with adalimumab.

The study observed a primary non-response rate of 24.1% (n=28) across all patients with IBD. This rate was significantly higher in patients with UC compared to those with CD (47.1% vs 14.6%; $p < 0.001$).

Secondary loss of response was noted in 47.9% of cases (n=46), with a higher frequency observed among CD patients compared to UC patients (52% vs 33.3%), although this difference was not statistically significant. Regarding specific associations with treatment response, in patients with CD, primary non-response was significantly linked to the presence of hypergammaglobulinemia (>13.5g/l) (p=0.009).

In contrast, for patients with UC, secondary loss of response was notably associated with moderately elevated gammaglobulinemia levels (p=0.039). Notably, there was no significant association observed between albumin levels and treatment response in either patient group.

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Parameter	Total (n=116)	Crohn's (n=82)	UC (n=34)
Age at First Anti-TNF Introduction (years, ± SD)	36.38 ± 13.355	35.96 ± 13.777	37.38 ± 12.417
Gender Male, n (%)	67 (57.8%)	46 (56.1%)	21 (61.8%)
Albuminemia - Number (%)	95 (81.9%)	65 (79.3%)	30 (88.2%)
- Albuminemia (g/l), Mean ± SD	33.484 ± 5.9794	33.269 ± 5.8248	33.95 ± 6.3787
- Hypoalbuminemia (<35 g/l), n (%)	26 (27.4%)	20 (30.8%)	6 (20%)
Gammaglobulinemia - Number (%)	87 (75%)	60 (73.2%)	27 (79.4%)
- Gammaglobulinemia (g/l), Mean ± SD	12.555 ± 3.902	12.48 ± 3.635	12.722 ± 4.5095
- Hypergammaglobulinemia (>13.5 g/l), n (%)	32 (36.8%)	22 (26.7%)	10 (37%)

Table 1: Demographic, Clinical Characteristics, and Biological Parameters of IBD Patients

Conclusion: this study highlights the potential utility of serum biomarkers, particularly gammaglobulinemia, in predicting response to anti-TNFα therapy in IBD patients. Further research in this area could contribute to personalized treatment strategies aimed at improving therapeutic outcomes in this challenging patient population.
Disclosure: No conflict.

PP0661

MEDICAL TREATMENT OF CAPSULE RETENTION IN CROHN'S DISEASE: A SYSTEMATIC REVIEW

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Introduction: Capsule retention is the main adverse event of SBCE. Current guidelines support the use of conservative/medical treatment for asymptomatic patients. However, the appropriate duration of conservative and medical therapy, the optimal regimens, and the timing to perform DAE/surgery in asymptomatic patients is still largely undefined.
Aims & Methods: Through a systematic search of multiple electronic databases until November 2023, studies including CD patients undergoing capsule endoscopy (CE) were identified. Capsule retention was defined according to ICCE criteria.
Results: We included 42 studies, totaling 196 capsule retentions in CD patients. Of these, 50 underwent anti-inflammatory therapy, consisting of corticosteroids (n= 34); corticosteroids + antibiotics (n=1), and corticosteroids + infliximab (n= 15). Steroids allowed for spontaneous passage in 44% (n=22): 3 out of 16 asymptomatic patients; 3 out of 5 symptomatic without complete small bowel occlusion (SBO); 2 in 3 with complete SBO. Combination with infliximab solved the retention in further 6 out of 15 patients. Mean time to resolution was 6,9 days (R 15h-21d), which was reported in 6 patients. Steroid administration route was reported in 2 patients (intravenous) and dosage in 3 patients: 100mg for 3 days, 1.2 mg/kg/day for 17 days, 0.5 mg/kg/day for 17 days, the latter being unsuccessful. Conservative treatment only or after failed medical therapy was used in 19 asymptomatic patients, with spontaneous passage in 6. One study reports retention of a capsule for 8.5 years. Of patients undergoing medical treatment 3 underwent urgent and 15 programed surgeries.
Conclusion: Evidence was of low quality. Treatment duration, regimen and time to spontaneous passage was underreported. We were unable to find predictors of the need to perform DAE/surgery in asymptomatic patients. As capsule retentions are increasingly rarer, the available evidence is unlikely to change.
Disclosure: Nothing to disclose.

PP0662

OUTCOMES OF STEM CELL THERAPY FOR ANO-PERINEAL CROHN'S DISEASE : A PROSPECTIVE STUDY

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Introduction: Perianal fistulas, which may be difficult to treat, are a common complication of Crohn's disease, which is a chronic disorder that is complex and difficult to control. The purpose of our research was to determine whether or not it would be possible and safe to inject hematopoietic mesenchymal stem cells into these patients' bodies, as well as to investigate the effectiveness of the therapy over a period of forty-eight weeks.
Aims & Methods: Between April 2022 and November 2023, we went through the process of conducting a prospective observational research at our hospital. Twenty-three adult patients who were diagnosed with Crohn's disease and perianal fistulas were included in our study. Hematopoietic stem cells were extracted from the bone marrow of every individual patient and delivered at a dosage of 3x10^6 cells. The first half of the cells were injected into the entrance of the internal fistula, while the second half was put into the opening of the exterior fistula. The surgery was followed by the administration of an intravenous stem cell infusion to a certain number of individuals.
Results: 36 percent of patients had achieved full closure of all external fistula openings by the 12th week of therapy, and this percentage improved to 52 percent by the 48th week of treatment. The majority of patients who originally presented with more than two external openings showed considerable improvement when at least one of the holes was closed.
It is important to highlight that throughout the follow-up period, one patient had an anal abscess that required further draining, while all other patients did not develop any abscesses. Furthermore, the level of luminal disease activity, as determined by CDAI, remained low throughout the duration of the follow-up period.
Conclusion: In conclusion, the findings of our research indicate that the use of hematopoietic stem cell injections for the treatment of perianal fistulas in patients with Crohn's disease is not only possible but also risk-free. According to the clinical findings, which are consistent with those published in reference studies, the closure rate at one year is fifty percent. In addition, this therapy has been linked to an improvement in patients' perineal activity levels as well as their quality of life.
Disclosure: Nothing to disclose.

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PP0663

LAMB-INDUCED ALTERATIONS OF THE MACROPHAGE POLARIZATION EXACERBATE INTESTINAL INFLAMMATION

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Introduction: Ulcerative colitis (UC) is a chronic intestinal inflammatory disease mainly involving the mucosa and submucosa. The pathogenesis of UC has not been clearly elucidated and there are limitations of existing treatment measures. Dietary factors have been implicated closely related to the pathogenesis and clinical outcome of UC. Our team in the early clinical trials observed that UC patients had aggravation or recurrence of their disease after taking in lamb. However, the relationship between them is unclear. Does lamb aggravate the intestinal inflammation of UC patients? What is the mechanism if lamb has a negative impact on intestinal inflammation? It is urgent for us to study further.

Aims & Methods: We aimed to clarify the effect of lamb on intestinal inflammation in UC mice and to reveal the mechanism, so as to provide scientific basis for clinical dietary guidance for UC patients.

Mice were given 2.5% DSS (w/v) solution for 5 days and deionized water for 7 days for 1 cycle, repeated 3 cycles to establish the chronic UC model. Take the start of giving DSS solution as day1, corresponding measures of lamb suspensions at a dose of 0.25, 0.50, 1.00 g/mL or sterile water were administered for 31 days. Then the general condition and the severity of colitis were monitored. Colon tissues were obtained to assess the degree of pathological damage by HE staining and detect the levels of key factors of intestinal barrier by RT-qPCR. Spleen tissues were collected to detect the levels of M1/M2 macrophages and related factors by flow cytometry and RT-qPCR.

Results: Our results showed that lamb aggravated intestinal inflammation of UC mice, as indicated by higher weight loss and DAI, shorter colon and more severe pathological damage. Also, the colonic mechanical barrier-associated factors including *ZO-1, Claudin-1, Occludin, E-cadherin* mRNA and colonic mucosal barrier associated-factors such as *Muc1, Muc2, Muc3* mRNA showed significant decrease in lamb-treated colitis mice. Moreover, colitis mice with lamb treatment showed significant increased M1 macrophages and reduced M2 macrophages in the spleen, leading to an increased the ratio of M1/M2 macrophage. Meanwhile, the M1 macrophage-associated inflammatory factors gene *TNF-α, IL-6* mRNA increased and M2 macrophage-associated inflammatory factors gene *IL-10, TGF-β* mRNA decreased significantly in lamb-treated colitis mice.

Conclusion: Our results show that lamb aggravates the intestinal inflammation and damages the mucosal barrier of colitis mice, whose mechanism is related to the promotion of macrophages to M1 cells and inhibition M2 cells differentiation, thereby aggravating the imbalance of M1/M2 macrophages.

Disclosure: Nothing to disclose.

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TRANSITION FROM INTRAVENOUS TO SUBCUTANEOUS INFLIXIMAB: EFFECTIVENESS IN A COHORT OF PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Inflammatory bowel disease (IBD), ulcerative colitis (UC) and Crohn's disease (CD), is a group of chronic inflammatory disorders. Biologics are indicated for patients failing conventional maintenance therapy with moderate to severe activity. Different routes of administration, intravenous (IV), subcutaneous (SC) or oral have been approved. For chronic diseases such as IBD, some patients may prefer self-administered SC dosing to IV dosing as a less time-intensive and more convenient treatment option.

Aims & Methods: This study aims to evaluate evaluate the success of infliximab subcutaneous administration after switching from intravenous administration in IBD patients. The transition from IV to SC administration was proposed to 60 IBD patients in clinical remission. Data on medical history and laboratory tests were collected from electronic health records. Moreover, patients filled a paper questionnaire 8 weeks after the switch focused on the level of satisfaction and onset of adverse event (AE).

Results: Overall 60 patients were enrolled for switching from IV to SC therapy. Most of patients (>80%) did not experience any difficulties in handling the device and respected the correct time-table of administration. The switch was a complete success in 50 out of 60 patients (83.3%). Ten patients stopped subcutaneous formulation: five of these switched back to intravenous formulation for adverse event or patient's preference, four swap or swop to other therapy and one withdraw from biologic. No predictor of loss of response was found at multivariate analysis. Faecal calprotectin and C-reactive protein values significantly improve after the switch at 3, 6 and 12 months (p= 0.03, p= 0.004, p = 0.01 and p= 0.06, p= 0.001, p= 0.007 respectively). About 16.7 % of patients had at least one side effect after switching. Local pain and swelling at site of injection were the most common AEs. The vast majority of AE were mild and lasted only a few days. No serious AEs were reported and no patient was hospitalized.

Conclusion: Effectiveness of switching from intravenous to subcutaneous infliximab administration in IBD patients is confirmed in our real world cohort of patients.

Disclosure: Nothing to disclose.

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PP0665

ADVANCED THERAPIES, THE BEST COMBINATIONS IN REFRACTORY IBD PATIENTS

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Introduction: The combination of two biological agents or a biological agent with novel small molecules, known as advanced combination therapy (ACT), for inflammatory bowel disease (IBD) has shown promising results in phase II studies as well as in real world studies. We describe the real-world experience of 2 Swiss IBD referral centers.

Aims & Methods: A chart review of all IBD patients on ACT from June 2020 to December 2023 was performed in Bern University hospital and the Crohn and Colitis Center Lausanne (Gastro-entérologie Beaulieu SA). Data on disease duration, extension, previous or concomitant treatments as well as efficacy and safety were collected prospectively.

Results:

Disease	Combination of advanced therapy (number of patients)	Partial or complete Clinical or biological response	Adverse effects, Infections
Ulcerative colitis (N=11)	Vedolizumab / Tofacitinib (5)	2/5 (40%)	none
	Vedolizumab / Anti-TNFs (3)	2/3 (50%)	none
	Vedolizumab /Cyclosporin or Tacrolimus (2) Ustekinumab / Tacrolimus (1)	1/2 (50%) 1/1 (100%)	none
Crohn's disease (N= 17)	ustekinumab (5)/ Anti-TNFs(infliximab (1), adalimumab (4))	5/ 5 (100%)	Eczema, tumefaction left parotide
	ustekinumab (4) / JAK inhibitors Upadacitinib,(3) , Tofacitinib (1)	4/4 (100%)	Lymphoma, mycosis, otitis media, lymphopenia, CMV ileitis
	Vedolizumab / Infliximab (1) Vedolizumab / Tacrolimus	0/1 (0%) 1/1 (100%)	none
	Vedolizumab / Upadacitinib (3)	3/3 (100%)	Acnea, Headache, nausea
	Anti-TNF(adalimumab (2), cimzia (1) / Upadacitinib (3)	2/3 (66%)	Ophthalmic zona
Total		21/28 (75%)	11 events (14%)

A total of 25 IBD patients (48% women, 10 ulcerative colitis, 15 Crohn's disease) received 28 ACT. The indications were extended and refractory luminal disease and/or extraintestinal manifestations. The 25 patients were all steroid dependent or refractory and half of them were either refractory or intolerant to at least two anti-TNF alpha agents.

These treatments were started after a median disease duration of 5 years (range 1-31). The mean duration of ACT was 10 months (range 1-40). Most frequent ACT combinations were vedolizumab or ustekinumab combined with either JAK inhibitors or TNF- alpha inhibitors.

A partial or complete response was observed in 21/28 therapies (75%) with a mean decrease of CRP of 9,4 mg/l (range - 23 to + 59). Among responders, a median decrease of calprotectin by 871 mcg/g (range 0 – 4002; N=8, skewed) was observed.

Concerning safety, 3 infections (ophthalmic herpes zoster, otitis media, cutaneous fungal infection) and 1 case of Non-Hodgkin Lymphoma, were reported. 8 minor adverse events (left parotid gland, tumefaction, eczema, 2 patients with lymphopenia, acne nausea and headaches) were reported, with mostly small molecules implicated. One patient had a successful pregnancy and delivery on combined adalimumab and ustekinumab ACT.

Conclusion: Vedolizumab or ustekinumab combined with either JAK or TNF- alpha inhibitors were the most frequent combinations of ACT in this cohort of severe and steroid-dependent/refractory IBD patients. 75% of patients achieved a clinical /biological response. Small molecules appear to be associated with of lymphopenia, and minor infections. There was one case of Non-Hodgkin Lymphoma while receiving ustekinumab and tofacitinib ACT in the context of previous long exposure to adalimumab.

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PP0666

EFFECTIVENESS AND SAFETY OF ADALIMUMAB BIOSIMILAR IN BIO"NAIVE" PATIENTS WITH INFLAMMATORY BOWEL DISEASE: THE GALICIAN EXPERIENCE

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Introduction: Biological drugs has revolutionized the management of inflammatory bowel disease (IBD). However, its use has led to a significant increase in cost that has strained health systems. Biosimilars will reduce cost related to treatment, allowing access to a greater proportion of patients. However, data on the efficacy and safety of the various adalimumab (ADA) biosimilars in IBD patients "naive" to biologics remains scarce.

Aims & Methods: We performed an observational cohort study in 8 Spanish hospitals located in Galicia. All patients with IBD naïve to biologics who started ADA biosimilar between November 2018 and January 2022 for control of their disease were enrolled. The primary endpoints were efficacy defined as clinical remission at week 8 and long term efficacy defined as persistence on drug treatment at the end of follow-up. The other co-primary endpoint was to determine the incidence of adverse events that lead to suspension or discontinuation of the drug or hospitalization of the patient.

Results: 383 patients (95 CU; 283 CD) with a median age of 47.0 (34.0-59.3) years and a median disease duration of 3.4 (0.8-10.0) years before starting treatment were included. After induction (at 8 weeks) 63.8 % of patients were in clinical remission and 85.3 % achieved clinical response. At this time of follow-up, 14 patients (4.9 %) discontinued treatment mainly due to primary non-response (PNR) (57.1 %) and adverse events (42.9

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%). In total, 114 patients discontinued treatment during follow-up. The main reasons for stopping therapy were adverse events (26.3 %), PNR (30.7 %) and secondary loss of response (43.0 %). Clinical remission was maintained in 78.4 % of patients, after a median follow-up of 18 (12-24) months. Drug persistence shown by Kaplan-Meier survival curves were significantly higher in CD patients (p = 0.012) and in the group of patients treated with immunomodulators (IMM) (p = 0.001). Additionally, the rates of clinical remission and clinical response were higher in CD patients and in IMM-treated patients (Table 1). Adalimumab levels were also measured during follow-up, showing a median of 10.4 (6.7-13.0) µg/mL. Significantly, drug persistence was higher in patients with post-induction ADA levels ≥ 7 µg/mL (p = 0.002). Regarding treatment safety in this population, adverse events that required treatment suspension or discontinuation were reported in 30 patients.

	EC (N = 383)	UC (N = 95)	P	IMM (N = 113)	Non-IMM (N = 252)	P
Clinical remission, n (%)	69.6	63.2	0.243	72.5	65.5	0.161
Clinical response, n (%)	80.6	70.5	0.041	80.2	77.4	0.532
Biochemical remission, n (%)	58.3	58.9	0.912	62.6	57.9	0.378
PNR, n (%)	7.8	13.7	0.086	6.9	10.3	0.267
SNR, n (%)	12.7	13.7	0.809	9.9	14.3	0.225

Table 1. Comparison of the clinical effectiveness at the end of follow-up between EC vs. UC patients and IMM vs. non-IMM treated patients.

Conclusion: ADA biosimilars in usual clinical practice are safe and effective in inducing and maintaining remission in Galician patients with IBD. Drug persistence was significantly higher in patients with CD, treated with IMM and with post-induction ADA levels ≥ 7 µg/mL.
Disclosure: Nothing to disclose.

PP0667

GUIDELINES SURROUNDING USE OF BIOSIMILARS IN GASTROENTEROLOGY: PERCEPTIONS FROM CLINICAL SOCIETIES

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Introduction: Since their introduction, biologics have drastically improved management of clinical symptoms in patients with inflammatory bowel disease (IBD); however, historically, biologics have been very expensive medications. Biosimilars provide similar clinical efficacy and safety to reference product (RP) biologics at a reduced cost, which has helped to alleviate undue strain on the healthcare system and improve patient access to biologics treatment. Despite these advantages, uptake remains slow in some regions, which may be due to limited awareness among patients and clinicians. This review sought to understand the recommendations for biosimilar use from government agencies and clinical societies, as they can be an important source for education
Aims & Methods: A literature search was conducted in MEDLINE and Google Scholar focusing on the use of biosimilars in gastroenterology. Search terms included biosimilar and gastroenterology, in combination with position, guidance, recommendation, workshop, or consortium. The literature search was restricted from 2006 (time of biosimilar approval) to present.
Results: Following screening there were 20 position statements/guidelines identified. The use of biosimilars was discussed among gastroenterology societies in Belgium (n = 2), Brazil (n = 1), Canada (n = 2), Italy (n = 2), Korea (n = 2), Mexico (n = 1), Poland (n = 2), Spain (n = 1), and the USA (n = 1) with regional representation from Europe (eg, ECCO) and Asia (n =2). Five were published before 2015 and 15 were published in 2015 or later. Although government agencies such as the European Medicines Agency and Food & Drug Administration published positive recommendations for the use of biosimilars, statements from key clinical societies from Canada, Europe, Italy, Poland, and Spain expressed concerns regarding the use of biosimilars for patients with IBD in the early 2010s. The main objection was the extrapolation of biosimilar indications to IBD from clinical studies in patients with rheumatoid arthritis, as statements suggested that clinical trials of biosimilars specific to IBD were necessary before their use. Since then, the efficacy and safety data from real-world observational studies, and the acceptance of indication extrapolation have contributed to positive recommendations from most clinical societies regarding biosimilar use. For example, 2014 guidelines published in Poland was not supportive of biosimilars; however, updated guidelines in 2019 noted that approved biosimilars were not significantly different from originator medicines. Most clinical societies approve biosimilar use for patients with IBD naïve to the molecule as well as those with stable disease, switching from the RP medicine to the biosimilar. One notable exception to this trend is a 2020 Canadian position statement recommending against nonmedical switching from originator to biosimilar in patients with stable IBD. However, several Canadian provinces have since adopted policies that support nonmedical switching, suggesting that medical practices and future Canadian position statements may change.
Conclusion: As real-world efficacy and safety studies continue to demonstrate the clinical value of biosimilars, perceptions surrounding their use have greatly improved among several key clinical societies globally. Notably, most societies expressed the importance of patient consent when deciding to switch to biosimilars. Thus, services that provide education to patients to increase awareness of and confidence in biosimilars are critical to ensure their use and should be optimized to promote expanded patient access.
Disclosure: Kunal Shastri is an employee of Fresenius Kabi, a manufacturer of biosimilars. Kerise Clarke, Margaret Ainslie-Garcia, and Nicole Ferko are employees of EVERSANA who receives consulting fees from Fresenius Kabi.

PP0668

ANTI-JAK EFFICACY IN A REAL WORLD COHORT OF ULCERATIVE COLITIS PATIENTS

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Introduction: Ulcerative colitis (UC) is a chronic inflammatory disease requiring sustained therapy for both induction and maintenance of remission. Biologics radically changed the natural history of UC patients and are still the cornerstone of treatment, but they have some limitations. The introduction of oral Janus kinase inhibitors (JAKi), recently approved for UC management, presents a novel treatment option. Data are needed to confirm their efficacy in a real world setting.
Aims & Methods: In this retrospective, observational study at the IBD Unit in Bologna, we enrolled consecutive UC patients treated with the JAKi tofacitinib (TOFA) and upadacitinib (UPA). We collected demographic and disease-specific data, assessing clinical activity via the partial Mayo (pMayo) score, biomarker status through C-reactive protein levels, and endoscopic activity via the Mayo endoscopic score at baseline and at 2, 6, and 12 months. Concomitant use of steroids were registered at every time point. We measured clinical response (≤3-point reduction in pMayo or dis-

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continuation of steroids) at 2 months, clinical remission rates (pMayo < 2) and endoscopic remission (Mayo score of 0) at each time point. Need for surgery was also recorded.

Results: A total of 50 patients were treated with JAKi between December 2021 an June 2023 (39 TOFA and 11 UPA). Males were 59.57% and median age was 35.5 years. Nobody were bio-naïve and 80.86% were exposed to ≥ 2 biologics/small molecules prior to JAKi initiation. Concomitant steroids were present in 76% of cases at baseline. Clinical response at T2 was obtained in 65.30% of cases. Clinical remission rate was 40% at T2, 32.6% at T6 and 24.4% at T12. Endoscopy was performed at T2 in 24 patients, at T6 in 26 and at T12 in 10 patients; endoscopic remission was observed in 5/24, 9/26 and 4/10, respectively. Treatment optimization was adopted in 52% of cases. A total of 5 patients (10%) underwent colectomy during follow up period. No serious adverse events were observed.

Conclusion: In a cohort of multirefractory UC patients, JAKi demonstrated substantial efficacy in inducing clinical and endoscopic remission, though there was a decline in effectiveness during the maintenance phase. The rate of colectomy remained relatively low, underscoring the potential utility of JAKi in this challenging patient population.

Disclosure: Nothing to disclose.

PP0669

RISK OF ACUTE INFLIXIMAB INFUSION-RELATED REACTIONS IN IBD PATIENTS WITH IMMUNOLOGICAL HYPERSENSITIVITY BACKGROUND

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Introduction: Acute infusion-related reactions (IRRs) are common adverse events associated with infliximab, affecting up to 23% of patients. Their etiology involves cytokine release, complement activation, and mast cell degranulation. Consequently, individuals with a history of immunological hypersensitivity reactions are often perceived to be at high risk due to their exaggerated immune responses, and are frequently prescribed premedication.

Aims & Methods: We aimed to determine if patients with immunological hypersensitivity background, receiving infliximab for inflammatory bowel disease (IBD), have an increased risk of IRRs. This retrospective cohort study included all patients diagnosed with IBD who underwent intravenous (IV) infliximab at our centre over a 13-year period. All patients received standard infliximab 2-hour infusion protocol, preceded by premedication with 200mg of IV hydrocortisone. IRRs were defined as any type of mild to severe reaction occurring during infusion or within 2 hours of its completion. Patients were considered to have an allergic predisposition if they had atopic diathesis (such as allergic rhinitis, asthma or atopic dermatitis) or a documented history of allergy to other medications.

Results: Ninety-five patients, with a mean age of 35 ± 13 years at the initiation of infliximab therapy, were included in the study, with females accounting for 52.6%. The majority of patients were diagnosed with Crohn's disease (n=76, 80%), while the remaining 19 patients (20%) had ulcerative colitis. A history of allergy was reported in 20 patients (21.1%) within the cohort.

Over the study period, a total of 258 person-years (PY) of exposure to infliximab were recorded, with a median of 12 infusions per patient (interquartile range: 5-33). In most cases, infliximab therapy was combined with immunomodulators (n=63, 66.3%).

Fourteen IRRs occurred during the follow-up period, resulting in an overall incidence rate of 49.5 per 1000 PY. Patients with and without a history of immunological hypersensitivity showed comparable incidence rates of IRRs (32.5 per 1000 PY and 57.8 per 1000 PY, respectively), as well as similar cumulative incidence (15% and 22.7%, respectively). No statistically significant difference was observed between the two groups (p=0.446).

Conclusion: Having an immunological hypersensitivity background does not appear to increase the risk of developing acute infliximab infusion-related reactions. Consequently, it does not warrant specific premedication or additional precautionary measures.

Disclosure: Nothing to disclose.

PP1248

UNVEILING THE IMPACT ON IN UTERO EXPOSURE TO BIOLOGIC TREATMENTS FOR INFLAMMATORY BOWEL DISEASE (IBD) ON CHILDREN'S PSYCHOMOTOR DEVELOPMENT: INSIGHTS FROM THE DUMBO REGISTRY OF GETECCU

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Introduction: Women with inflammatory bowel disease (IBD) are highly recommended to maintain treatment during pregnancy because discontinuation has been associated with flare-ups, which are linked to a worse prognosis for the mother and the fetus. It is widely known that biologic drugs cross the placenta, but most studies focused on the impact on the risk of infection. Data regarding the impact on psychomotor development are very limited, particularly on non anti-TNF biologics.

Aims & Methods: Our aim was to evaluate the impact of the exposure of biologics in utero on the psychomotor development of children during the first year of life.

Data from children included in the DUMBO registry with complete AQS-3 available up to 12 months of age were analysed. DUMBO is a prospective, observational, and multicentre registry endorsed by GETECCU, which enrols pregnant women with IBD throughout 5 years in 60 centres in Spain. This subanalysis to evaluate the neurodevelopment of the babies, follows up mothers and babies since the pregnancy diagnosis until delivery during each trimester and then, every 3 months during the first 12 months of babies´ life. Normal psychomotor development was defined by ASQ-3 scores above the lower limit of normality (referral zone) in all domains. Serious adverse events (SAE) were defined in accordance with the ICH Topic E 2 A Clinical Safety Data Management.

Results: 352 children born to 343 mothers were included (9 pair of twins).134 children (38%) had been exposed to biologics in utero; from them, 50 (37%) had been exposed to adalimumab, 44 (32%) to infliximab, 3 (2.2%) to certolizumab, 1 (0.7%) to golimumab, 28 (20%) to ustekinumab, and 10 (7.5%) to vedolizumab. To assess psychomotor development, 2073 ASQ-3 questionnaires were analyzed. No differences were observed in any of the evaluations or domains throughout the first year, except at 12 months, where exposed babies were more likely to achieve normal results in the domain of gross motor skills.

In the univariate analysis between babies with normal development at 12 months and those who did not we only found differences in: Exposure to biologic drugs, which was higher in those who presented a normal ASQ-3. The type of maternal IBD, that babies from mothers with UC had lower likelihood of normal psychomotor development, and prematurity, serious adverse events, and hospitalizations, all of which were more frequent in those who presented abnormal ASQ-3 results. In the multivariate analysis, to have been born to a mother with CD (vs. UC) was associated with higher likelihood (OR=2, 95%CI=1.1-3.9), while to be premature was associated with lower likelihood (OR=0.3, 95%CI=0.1-0.6) of having ASQ-3 scores above the limit of normality in all domains at 12 months of age.

Conclusion: In the multicenter, prospective DUMBO registry, the exposure to biologics for the treatment of IBD in utero (including anti-TNF and non-anti-TNF agents) did not impair the psychomotor development of the children. Our findings support the safety of these drugs and reinforce the recommendation to continue them during pregnancy.

Disclosure: Nothing to disclose.

PP0670

ISCHEMIC ULCERS DO NOT SIGNIFICANTLY IMPACT THE ENDOSCOPIC RECURRENCE SCORE IN CROHN'S DISEASE: A MULTICENTER COMPARATIVE STUDY AFTER RIGHT-SIDE RESECTIONS FOR CROHN'S DISEASE AND COLORECTAL CANCER

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Introduction: Post-operative recurrence (POR) is common in patients with Crohn's disease (CD) who have undergone surgery. Endoscopy, mainly based on Rutgeerts' score, is crucial for the diagnosis of POR, showing also high prognostic value. More recently, several studies have suggested that the current endoscopic score system could be affected by the ischemic (not inflammatory) nature of some anastomotic ulcers, particularly at the level of the mucosal inverted stapled line, so overestimating the prevalence of the "real" anastomotic POR.

However, these studies, based on the comparison between CD patients and subjects undergone right-side resection for colon cancer (CRC), included a very small population.

Aims & Methods: Starting from these considerations, we performed a retrospective study aimed to directly compare the prevalence of anastomotic POR in CD with that of patients operated of right-side resection for CRC in a wide population. A retrospective studies was conducted at 4 Italian Institutions (3 with high volume of IBD admissions; 2 with high volume CRC admissions). All patients performed endoscopy within 8 months from surgery. We selected all consecutive CD and CC patients undergone a surgical resection with an anisoperistaltic stapled L-L anastomosis. The diagnosis of POR of CD was made in accordance with the modified Rutgeerts' score (i0, i1, i2A, i2B, 3, 4). The global prevalence of PSR (Rutgeerts's score ≥2) was recorded. Furthermore, we directly compared the prevalence of the isolated anastomotic ulcers of CD (Rutgeerts i2A) to that detected in CC patients.

Results: At the end of the study, we enrolled 221 CD patients and 72 CRC subjects. POR was recorded in 127 CD patients (57%). According to the endoscopic grade system, POR was scored as i0: 56 pts (25%), i1: 38 pts (17%); i2A: 42 pts (19%), i2B: 26 pts (12%), i3: 20 pts (9%), i4: 39 pts (18%). In the CRC group, endoscopy detected anastomotic ulcers in 2 out of 72 patients (3%); 1 case (1.4%) of CRC early recurrence was observed. When

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comparing the prevalence of isolated anastomotic ulcerations in CD (i2A) *vs* those in CRC patients, the prevalence of ulcerations was significantly higher in CD patients (19% vs 3%; p<0.01; O.R. 6.3).

Conclusion: Our study confirms the high prevalence of early recurrence in CD patients, so confirming the role of early (within 6-8 months) post-operative endoscopy in CD setting. The prevalence of anastomotic (ischemic) ulcers in CRC patients is very low and, as a consequence, this issue should not represent a real diagnostic matter if shifted into the POR setting of CD. Prospective, multicentre and directly comparative studies are needed to confirm our results.

Disclosure: Nothing to disclose.

PP0671

POST-OPERATIVE BMI IN PATIENTS WITH CROHN'S DISEASE UNDERGOING ILEO-COLONIC RESECTION IS DETERMINED BY BASELINE RATHER THAN PERI-OPERATIVE BMI

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Introduction: Patients diagnosed with Crohn's disease (CD) often undergo ileo-colonic resection (ICR) due to disease activity and complications. Post-operative weight gain is common, secondary to reduction in inflammatory burden, resolution of strictures and improved nutrition, yet may result in overweight and obesity (oBMI). Recent studies have shown that obesity in CD may lead to disease exacerbation and decreased response to medical treatment.

Aims & Methods: In this retrospective, case control, single tertiary center study, we aimed to identify risk factors for post-ICR oBMI and to explore the association of post-ICR overweight and obesity with disease recurrence in CD patients. All adult patients with CD who underwent ICR between 1/1/2010 to 31/12/2022 were included. Data collected from hospital records included patients and disease characteristics, post-ICR adverse events and endoscopic recurrence (Rutgeerts score), medical treatment before and after ICR. BMI was recorded at 6-12 months prior to disease exacerbation leading to surgery (baseline-BMI), at ICR and at 12-months post-ICR. Patients with (1) protective or end ileostomy, (2) non-CD exacerbation related surgery (e.g. neoplasia, ischemia), (3) lack of in-hospital IBD clinic follow up, and (4) incomplete BMI records, were excluded. The cohort was divided into two groups by post-ICR BMI [non-overweight BMI (nBMI)<25 kg/m^2 or oBMI≥25 kg/m^2]. Logistic regression model was used and two-sided p-value <0.05 was considered significant.

Results: A total of 94 patients met the inclusion criteria and were included in the study. At 12 months post-ICR, 32 (33.7%) patients had oBMI (mean BMI 28.1±2.8 kg/m^2) and 62 (66.3%) had nBMI (mean BMI 21.5±2.1 kg/m^2). Groups were comparable in baseline characteristics [Table 1]. Mean baseline-BMI and ICR-BMI were higher in oBMI group compared to nBMI group (26.6±3.3 kg/m^2 vs. 21.4±2.9 kg/m^2 at baseline and 24.2±4.0 kg/m^2 vs. 19.6±2.6 kg/m^2 at ICR, p<0.001 for both). oBMI patients had a higher rate of ileocolonic (L3) disease (vs. ileal (L1) disease) compared to the nBMI group (62.5% vs 32.3%, p=0.005). No differences were observed in post-operative complications and re-hospitalizations within 12 months of follow up. The rate of endoscopic disease recurrence was comparable within 18 months of ICR (Rutgeerts≥i2b 30.8% for oBMI vs 43.1% for nBMI, p=NS). A logistic regression analysis adjusted to age, gender, and disease location, demonstrated that baseline-BMI, but not ICR-BMI was associated with 12 months post-ICR-oBMI (baseline-BMI: OR 1.699, 95%CI 1.242-2.326, p<0.001; ICR-BMI: OR 1.208, 95%CI 0.932-1.565, p=0.153).

	Post-ICR-nBMI <25kg/m^2 (n=62)	Post-ICR-oBMI ≥25kg/m^2 (n=32)	p-value
Body Mass Index (kg/m^2), mean±SD			
At baseline	21.4±2.9	26.6±3.3	**<0.001**
At time of ICR	19.6±2.6	24.2±4.0	**<0.001**
12 months post-ICR	21.5±2.1	28.1±2.8	**<0.001**
Male gender, n [%]	36 [58.1]	22 [68.8]	0.313
Age at CD diagnosis, n [%]			
<17 years	7 [11.3]	5 [15.6]	
17-40 years	46 [74.2]	16 [65.6]	0.682
>40 years	9 [14.15]	6 [18.8]	
CD location, n [%]			
Ileal	42 [67.7]	12 [37.5]	**0.005**
Ileocolonic	20 [32.3]	20 [62.5]	
Disease behavior, n [%]			
Inflammatory	1 [1.6]	1 [3.1]	
Stricturing	26 [41.9]	9 [28.1]	0.400
Penetrating	35 [56.5]	22 [68.8]	
Perianal disease, n [%]	18 [29.0]	9 [28.1]	0.927
Smoking, n [%]			
Never	32 [51.6]	16 [50.0]	
Former	12 [19.4]	11 [34.4]	0.173
Active	18 [29.0]	5 [15.6]	
Disease duration at ICR (years), mean±SD	10.2±8.1	10.5±12.5	0.871
Biologic treatment experienced, n [%]	44 [71.0]	21 [65.6]	0.595

Table 1: Patients' characteristics

Abbreviations: nBMI – non-overweight body mass index; oBMI – overweight/obese body mass index; SD – standard deviation; ICR – ileo-colonic resection; CD – Crohn's disease; n – number

Conclusion: BMI≥25 kg/m^2 prior to CD exacerbation is associated with post-ICR overweight and obesity. Though at short term follow-up, no increase in postoperative complications and endoscopic disease recurrence were observed, longer follow-up may be needed to assess possible effects of post-ICR overweight and obesity on CD's course.

Disclosure: Nothing to disclose.

PP0672

PRELIMINARY REPORT ON SHORT-TERM POST-OPERATIVE OUTCOMES OF ROBOTIC VERSUS LAPAROSCOPIC APPROACH FOR RESTORATIVE PROCTO-COLECTOMY WITH ILEAL-POUCH-ANAL ANASTOMOSIS IN ULCERATIVE COLITIS: A COMPARATIVE STUDY

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Introduction: Restorative proctocolectomy with ileal pouch–anal anastomosis (IPAA) performed in two or three stages has become the standard surgical treatment in patients with ulcerative colitis (UC) in which the medical management fails. It is a technically complex operation associated with significant morbidity that can be performed with a laparoscopic or robotic approach. Robotic colorectal surgery is gaining popularity because it overcomes many of the mechanical and optical disadvantages of laparoscopy, although limited data are available regarding its safety, efficacy, and costs.

Aims & Methods: This study compares the postoperative short-terms surgical outcomes of the two surgical approaches. Robotic and laparoscopic restorative proctocolectomy procedures with IPAA consecutively per-

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formed between June 2015 and May 2023 were retrospectively collected. Inclusion criteria was UC not responders to medical therapy. Exclusion criteria was conversion to open surgery.
Primary analyzed outcome were complications rate in the first month (30 days) according to Clavien Dindo Score and length of hospital stay (days). Statistical analysis was performed with a dedicated software (SPSS).
Results: A total of 54 patients with UC refractory to medical therapy were included, with 31 laparoscopic procedure and 23 robotic procedures.
Demographic data were comparable between two groups in terms of age (48.7 vs 45.7 years; p= 0.62), BMI (22,0 vs 23,7 Kg/m2; p= 1.311) and ASA score (p=0,72).Short-term complications defined as Clavien Dindo scores 2-3, (compared with score 1) were 13% in the robotic group and 31% in the laparoscopic group (RR= 0.27, 95% conf. interval. 0.11-1.06).
In particular, no high-grade Clavien-Dindo complications occurred in the robotic group while 3 events (grade 3) occurred in the laparoscopic group. No surgical intervention was needed in both groups.
Length of hospital stay was a mean of 7.3 days in lap-group and 6.3 days in rob-group with no statistical significativity (p=0,09).
Conclusion: This preliminary study concludes that robot-assisted surgery allows a reduction of short-term complications compared to laparoscopic approach. Moreover, robotic surgery allows a reduction of length of hospital stay although not statistically significant. Limits of the study were the retrospective design and the low numerosity of the sample, although comparable with the experiences already published.
Disclosure: Nothing to disclose.

PP0673

THE ENHANCED RECOVERY AFTER SURGERY PROTOCOL TAKE ITS PLACE IN INFLAMMATORY BOWEL DISEASE SURGERY

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Introduction: The enhanced recovery after surgery (ERAS) protocol is an evidence-based standardised multimodal programme formulated to reduce surgical stress and optimise recovery after surgery. The ERAS represents the best care practice for patients undergoing colorectal surgery. To ensure high compliance with active ERAS elements, patients must be educated to actively participate in the perioperative care pathway. ERAS protocols were originally designed to treat patients with CRC, only few studies have assessed this protocol in IBD patients.
Aims & Methods: The aim of this study was to investigate the adherence to ERAS in IBD patients. This retrospective study enrolled adult patients who underwent elective colorectal surgery and the ERAS protocol at our institute between May 2020 and December 2022. The patients were divided into IBD and non-IBD group. Data on medical history were collected from electronic health records.Preoperative, intraoperative, and postoperative data were collected for each procedure using a self-reported mobile application. Adherence was calculated for at least 75% of active ERAS items. Chi-square test was used to compare categorical variables.
Results: 619 patients, 492 non-IBD and 127 IBD, were enrolled. The overall adherence to ERAS protocol items was similar between the two groups (84% vs 82%, p 0.2). Adherence was evaluated during the three ERAS phases in the groups: preoperative (92% vs 90%, p 0.11), hospitalization (63% vs 62%, p 0.8), post-discharge period (96% vs 93%, p 0.2). Moreover, focusing on the preoperative ERAS items adhrence, there were significant difference between non-IBD and IBD group in nutritional optimization (84% vs 67%, p <0.001) and smart communication (93% vs 89%, p 0.002). No difference were reported in the patient's prehabilitation adherence (74 % vs 71%, p 0.4).
Conclusion: The feasibility of ERAS protocol was good in colorectal surgery as well as in IBD surgery. We need to focus on a tailored nutritional optimization in order to improve adherence and compliance among IBD patient.
Disclosure: Nothing to disclose.

PP0674

ANALYSIS OF COLECTOMY RATES AND ADVANCED THERAPY PRESCRIBING IN ULCERATIVE COLITIS OVER 2 DECADES IN LOTHIAN, SCOTLAND

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Introduction: The impact of increased advanced therapy prescribing on colectomy rates in ulcerative colitis (UC) to date are conflicting. We have previously demonstrated a decrease in the rates of colectomy over time from 2005 to 2018.[1] The influence of the expanding number of available therapies for UC on colectomy rates is unknown.
Aims & Methods: We aimed to describe prescribing trends of all available advanced therapies for UC in the Lothian IBD cohort up until the end of 2023 along with the corresponding colectomy rates. We identified all patients who started an advanced therapy for UC, and all patients who underwent colectomy for UC in the Lothian population from January 1st, 2005, to December 31st, 2023. Incidence and prevalence data of UC in Lothian was obtained from the rigorously validated Lothian IBD Registry.[2] Prescribing and colectomy rates are presented as a proportion of the prevalent population. Linear and segmental regression analyses were used to identify the annual percentage change and temporal trends in colectomy rates.
Results: From 2005 to 2023, 509 patients with ulcerative colitis underwent colectomy. The median age at colectomy was 42 years old (IQR 29-56). The median time from diagnosis to colectomy was 2.6 years (IQR 0.4-9.7). Colectomy rates per 100 UC patients were 1.47 in 2005, 1.53 in 2010, 0.68 in 2015, and 0.22 in 2023. 68% of colectomies (346/509) were performed for acute severe UC, of which 34% were within 90 days of diagnosis. 3.5% (18/509) of colectomies were for cancer (n=11) or dysplasia (n=7).

Year	06-07	08-09	10-11	12-13	14-15	16-17	18-19	20-21	22-23
No. of 1st line starts	19	7	20	6	39	67	114	131	199
2nd lines starts	0	1	1	1	7	35	43	83	94
3rd lines starts	0	0	0	0	1	3	14	37	66
4th lines starts	0	0	0	0	0	1	4	17	32
Total no. of new prescriptions of ATs	19	8	21	7	47	106	175	268	391
UC prevalence*	2382	2813	3107	3369	3677	3907	3948	4078	4115
No. of colectomies	71	68	82	63	66	48	28	21	30
Emergency colectomies	49 (69%)	47 (69%)	51 (62%)	39 (62%)	44 (67%)	36 (75%)	18 (64%)	16 (76%)	22 (73%)

*AT, advanced therapies. *Prevalence is at the start of the latter year. Prevalence from 2006-2019 is actual prevalence, with 2020-2023 predicted prevalence.*

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Over the same time period 615 patients received at least one advanced therapy for UC. The median age at first advanced therapy was 40 years old (IQR 28-54). New advanced therapy prescriptions per 100 UC patients increased from 0.67 in 2005, to 0.37 in 2010, 1.01 in 2015, 3.21 in 2020, and 5.15 in 2023. Persistence rates on first line therapy were highest with vedolizumab, followed by infliximab. By the end of 2023, 43% (n=265), 20% (n=121), 9% (n=54), and 3% (n=21), of the 615 patients had received a second, third, fourth, and fifth line of advanced therapy respectively.

Conclusion: The incidence of colectomy for ulcerative colitis has substantially decreased over time and correlates with an increased use, and number of available advanced therapies. However, correlation cannot ascribe causation.

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PP0675

EVALUATION OF THE EFFECTIVENESS AND SAFETY OF MESENCHYMAL STEM CELL TREATMENT IN FISTULIZING CROHN'S DISEASE – INTERNATIONAL, MULTICENTER, RETROSPECTIVE COHORT STUDY

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Introduction: Approximately 10-15% of patients with Crohn's Disease (CD) experience penetrating complications during disease course, which significantly burdens patients' lives and impacts the healthcare system. Efficacy and safety of adipose-derived mesenchymal stem cell (MSC) transplantation (darvastrocel) in treating CD complicated by complex perianal fistulas are contradicting and real-world data are lacking.

Aims & Methods: Our study aimed to assess the effectiveness and safety of MSC therapy in managing perianal Crohn's disease. This was a retrospective, multicenter cohort study enrolling CD patients with perianal fistulas. Baseline is defined as the day of MSC treatment while the follow-up is set at 6 to 12 months. Primary outcome was perianal clinical remission at w26 (defined by closure of all external openings), while secondary outcomes were clinical response rate, perianal clinical activity (defined by Perianal Disease Activity Index [PDAI]), and adverse events. Demographic, clinical, and surgical data were recorded at baseline, w12, w26, and at w52. Predictors of treatment failure were analyzed by logistic regression models.

Variables	CD (n = 69)
Follow-up duration, weeks, median (IQR)	52 (48 - 53)
Sex, male (%)	36 (52.2)
Age at inclusion, years, median (IQR)	36 (31 - 44)
Disease duration at inclusion, weeks, median (IQR)	11 (7 - 18)
Disease activity	
CDAI, mean (± SD)	61,2 (53)
PDAI, mean (± SD)	6,9 (3,4)
CRP, mg/l, mean (± SD)	3,8 (5,2)
Number of fistulas, mean (± SD)	2 (1)

Table 1. Baseline demographic and clinical characteristics of enrolled patients.

Results: Overall, data of 69 patients (male/female ratio was 0.52 and median age was 36 [IQR 31-44]) were analyzed with a median follow-up duration of 52 [IQR 48-53] weeks. Perianal clinical remission was achieved in 49.3% until w26 and 44.9% (intention-to-treat) and 53.44% (per protocol) until w52, while baseline PDAI (OR 0.77, p = 0.043) score was associated with treatment failure. Characteristics of perianal fistulas were not linked

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to primary outcome. Clinical response rates were 69.6% at w26 and 78.3% (intention-to-treat) and 74.1% (per protocol) at w52. PDAI scores were decreased to w52 (5.2 [±3.1] to 3.2 [±2.8], p < 0.001). Worsening of perianal symptoms were not recorded, while improvement was reported in 78.8%, 81.1%, and 85.7% of cases at w12, w26, and w52. Total adverse events occurred in 8.7%, which were needed surgical reintervention. Furthermore, emerging stenosis, anal incontinence and severe infection were not registered.

Conclusion: Efficacy data are consisted with the registration ADMIRE study, more than 80% of patients benefit from the treatment, however the appearance of spontaneous closure is high in the literature. The safety profile is reassuring, and patients' satisfaction is high.

Disclosure: Nothing to disclose.

PP0676

REAL-LIFE EFFECTIVENESS OF ALLOGENEIC EXPANDED ADIPOSE-DERIVED MESENCHYMAL STEM CELL THERAPY FOR PERIANAL FISTULIZING REFRACTORY CROHN'S DISEASE

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Introduction: Perianal fistulizing Crohn's disease (CD) is frequent (in up to one quarter of patients within two decades of the diagnosis) (1) and represents a chronic, highly debilitating and difficult-to-treat condition. Whereas the efficacy and safety of darvastrocel (a suspension of expanded human allogeneic, adipose-derived mesenchymal stem cells) have been demonstrated in a phase 3 randomized double-blind placebo-controlled multicenter trial (ADMIRE-CD 1), a second randomized trial (ADMIRE CD 2) was recently reported negative. Real life data about the long-term efficacy of darvadstrocel for treating perianal fistulas in Crohn's disease (CD) remain scarce.

Aims & Methods: Aim: to report the effectiveness and safety of darvadstrocel therapy to close perianal fistula in a real-life cohort of CD patients. Methods: We retrospectively collected data from all CD patients with draining perianal fistulas treated with a single local injection of darvadstrocel. All patients with nonactive or mildly active luminal CD with complex anoperineal fistula who were treated with darvadstrocel were enrolled. Darvadstrocel was locally injected following the standardized procedure of ADMIRE trial. All patients underwent removal of seton, fistula curettage, transanal closure of internal opening(s) by suture followed by local injection of 120 million of mesenchymal stem cells in outpatient surgery.

The primary endpoint was clinical remission (closure of all external openings with no discharge at pressure) at week 24. Combined remission (defined as clinically plus MRI-assessed fistula closure) were also assessed at week 24 and 52.Incidence and time to fistula relapse assessed by Kaplan Meier estimates or new fistula occurrence as well as need for anal surgery were recorded. Short and long-term adverse events were also collected.

Results: A total of 36 refractory CD patients (19 Male, mean age 38.5 years) from 2 referral French IBD centers were included with a median follow up of 16 months. Patients had CD for a mean (±SD) of 10.5 ± 8.5 years and all had received previous immunosuppressants in the past 6 months and were also receiving concomitant immunosuppressive medications. One quarter of them had a history of diverting stomas. The vast majority of patients had a transphincteric fistula tract (83 %) and only one internal opening (80 %). Clinical remission was achieved in 17 out of 36 patients (47.2 %) and combined remission in 15 out of 36 patients (41.6 %) at week 24. At week 52, clinical and combined remission was achieved in 17 out of 36 patients (47.2 %) and in 15 out of 33 evaluable patients (45.4 %), respectively. The probabilities of relapse-free survival was 87.1 %, 74.2 % and 55.6 % at week 24, 40 and 60, In responders at week 12, fistula relapse occurred in 50 % of patients with a median time of 34 weeks (10-71) and no new fistula was clinically and radiologically evidenced. We failed to detect any predictors of response. No new safety issues, including changes in anal continence, tumorigenity or ectopic tissue formation were recorded throughout the study period.

Conclusion: This real-life based study shares additional experience on the effectiveness and safety of a single local injection of darvadstrocel for treating complex perianal fistula in a particularly severe and refractory cohort of patients with CD. Clinical remission was achieved in 47.2 % of patients at week 24 and combined remission in one more than 4 out of 10 patients at 52 weeks. Long-term larger studies are needed to better identify predictors of sustained fistula healing using darvadstrocel in perianal fistulizing CD.

Disclosure: Nothing to disclose.

PP0677

WHAT ARE THE INDICATIONS FOR SURGERY IN CROHN'S DISEASE IN THE ERA OF BIOLOGICS?

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Introduction: Despite significant progress in medical therapy, surgery remains a common treatment for Crohn's Disease (CD), particularly in cases with complications such as intestinal obstruction, fistulas, abscesses, or severe bleeding. A substantial proportion of patients require surgery within the first two decades after diagnosis, even with the widespread use of biological therapies.

Aims & Methods: This study is a retrospective, descriptive analysis spanning 8 years, from January 2015 to March 2024, conducted in a gastroenterology department of a university hospital. It includes all patients with Crohn's Disease who underwent surgery during this period. Data were collected from patient records, including demographics, disease characteristics, surgical indications, and treatment history.

This study evaluates the prevalence of surgical interventions and the primary indications for surgery in the era of biological treatments for Crohn's Disease.

Results: The study cohort comprised 292 patients diagnosed with Crohn's Disease, of whom 50 (17.12%) required surgical intervention. The median age at diagnosis among these surgical patients was 30 years, with a range from 9 to 64 years. The mean age at the time of surgery was 32 years (range: 16-64 years). Among the surgical cohort, 28 patients (54%) were male and 22 patients (46%) were female. Additionally, 7 of the 50 surgical patients (14%) had a prior history of appendectomy.

Regarding disease location, 16 patients (32%) had localized ileum involvement, while 42 patients (84%) had ileo-colic involvement. Notably, 13 of the surgical patients (26%) had perineal disease. Concerning the frequency of surgery, 38 patients (76%) underwent a single surgical procedure, 8 patients (16%) had two surgeries, 2 patients (4%) had three surgeries, and 2 patients (4%) required four surgical interventions.

Of the total surgical procedures, 18 (36%) were conducted on an urgent

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basis, indicating a significant proportion of acute complications. Regarding medication use prior to surgery, 6 patients (12%) were on corticosteroids, 21 patients (42%) were receiving immunosuppressants, and 5 patients (10%) were undergoing biological therapy.

The primary indications for surgery in this cohort were stenosis, accounting for 35 cases (67.3%), intestinal perforation representing 7 cases (13.5%), fistulation comprising 7 cases (13.5%), and abscess with 1 case (1.9%).

Conclusion: Despite the increasing use of biological therapies, surgery remains necessary for many patients with Crohn's Disease, with 17.12% of the study cohort requiring surgical intervention. The most common indications for surgery were stenosis, perforation, and fistulation, demonstrating the continued need for surgical management in patients with complex disease profiles. The relatively high rate of urgent surgeries (36%) emphasizes the need for timely and effective treatment strategies. The use of corticosteroids and immunosuppressants prior to surgery indicates that a multi-faceted approach to treatment is crucial. These findings highlight the ongoing importance of surgery in the care of Crohn's Disease and underscore the need for continued research to improve patient outcomes.

Disclosure: Nothing to disclose.

PP0678

ASSESSMENT OF ADHERENCE TO POSTOPERATIVE ILEOCOLONIC RESECTION GUIDELINES AND IMPACT ON EARLY ENDOSCOPIC RECURRENCE IN CROHN'S DISEASE: AN AUSTRALIAN RETROSPECTIVE STUDY

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Introduction: Recent studies have demonstrated the efficacy of ileocolonic resection in the treatment of ileal Crohn's disease (CD). Stratification based on risk of recurrence is current standard of care, with international guidelines recommending prophylaxis to high-risk patients. Existing data indicate suboptimal adherence to these guidelines, potentially leading to increased healthcare costs. This study aimed to evaluate (1) guideline adherence in the postoperative setting, and (2) the impact of guideline driven prophylaxis on post-operative endoscopic recurrence (POR).

Aims & Methods: We performed a retrospective cohort study of patients who underwent ileal or ileocolonic resection and anastomosis for CD between 2017 and 2021 in four Australian centres. Adherence to three specific guideline aspects was assessed: (1) colonoscopy within 12 months, (2) documentation of a Rutgeert's score at colonoscopy, and (3) prophylactic therapy prescription in high-risk patients. High-risk patients were defined as those with one or more of the following risk factors for recurrence: postoperative smoking, prior CD-related intestinal resection, presence of penetrating or perianal disease, or presence of granulomas in the resection specimen. Early POR (defined as a Rutgeert's score of ≥ i2 within 12 months of surgery) was assessed; if this was not documented, it was inferred from the report text and available images.

Results: A total of 180 patients were analysed, with a median age of 46 years, and of whom 55% were female. Disease was confined to the ileum in 59% of patients, and 18% had additional perianal involvement. Disease behaviour was stricturing in 66% and penetrating in 30%. 32% of the cohort had undergone prior ileocolonic resection, and 25% were smokers.122 (68%) underwent colonoscopy within 12 months, and 162 (90%) had a colonoscopy during the follow-up period (median 240 days, IQR: 174 – 402 days). A Rutgeert's score was documented in 89/162 (55%). Of the 142 (79%) patients with at least one risk factor for recurrence, 98 (69%) received prophylactic therapy. Early POR occurred in 54 of 122 patients (44%) who underwent timely colonoscopy (high-risk group 43/94, 46%; low-risk group 11/28, 39%, p = 0.55). Prophylactic therapy in the high-risk group did not reduce early POR (28/62, 45% vs 15/32, 47%, p = 0.87).

Colonoscopy within 6-12 months	122 (68)
Rutgeert's score documentation	89 (55)
Prophylaxis in high-risk group	98 (69)
Prophylaxis in low-risk group	20 (53)
Early post-operative recurrence (POR)	
High-risk group (all)	43 (46)*
High-risk group, prophylaxis prescribed	28 (45)
Low-risk group	11 (39)*
* no difference in POR between high-risk and low-risk groups (p = 0.55)	

Table 1. Selected Results n (%).

Conclusion: Adherence to guidelines in the management of postoperative Crohn's is suboptimal, particularly in the documentation of a Rutgeert's score. Irrespective of prophylactic therapy prescription in our cohort, early POR remains high, and guideline driven prophylaxis did not significantly reduce this rate. Additionally, recurrence was still high in patients without traditional risk factors for recurrence. This invites further investigation into alternative risk-stratification and treatment strategies to reduce POR in Crohn's disease.

Disclosure: Nothing to disclose.

PP0679

FIRST SURGERY IS DELAYED IN DIFFICULT-TO-TREAT PATIENTS WITH CROHN'S DISEASE

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Introduction: The majority of patients with Crohn's disease (CD) undergo surgery at some point of their disease evolution. Timing of surgical procedure is currently at the centre of discussion, with the concept of surgery first showing promising results with regards to short-term quality of life and cost-effectiveness but it is unclear whether this approach can have long-term modifying effect on disease behaviour. With the enlargement of CD therapeutic armamentarium during the past years, the surgical procedure is often being delayed with unclear benefit for the patient.

Aims & Methods: The aim of this study was to determine whether the time to first surgery differ between patients with uncomplicated disease course compared with difficult-to-treat patients. All patients with CD followed at one tertiary IBD centre by March 2024 were identified through automated medical record system and patients with the history of IBD-related surgery were further analysed. Basic demographics, disease phenotype, surgical procedures and biologic use were retrieved from medical records. Surgical procedure of interest for this study purpose was defined as bowel resection of any localisation, stricturoplasty, proctectomy, gastro-entero-anastomosis. Patients with more than one surgical procedure and/or two or more biologics were categorized as difficult-to-treat; patients with one surgical procedure and no or one biologic were considered as patients

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with uncomplicated disease course. The differences in basic demographics, disease phenotype and time to surgery between the two groups were analysed statistically.

Results: In total 152 patients with a history of CD-related surgery were included; mean age 40 years (range 19-81), 57 (38%) men, mean duration of the disease 14,9 years (range 1-42), the majority of patients had stenosing or stenosing and penetrating disease behaviour (54% and 33%, respectively) and ileal or colonic localisation (53% and 38%, respectively). Thirty-nine patients (26%) had history of two or more surgeries.

Seventy-seven patients (51%) were considered as difficult-to-treat, there were no differences in disease localisation and behaviour between these patients and patients with uncomplicated disease course. Time to surgery was significantly longer in the difficult-to-treat group (6.5 ± SD 5.6 yrs) compared with the patients with uncomplicated disease course (3.8 ± SD 5.5 yrs); p=0.003. Among patients with uncomplicated disease course, significantly higher proportion of patients have been operated during the first year after diagnosis compared with difficult-to-treat patients (38 pts, 51% vs. 18 pts, 23%; p=0.000488).

Conclusion: Difficult-to-treat patients with Crohn's disease have significantly longer time to surgery than patients with uncomplicated disease course. Considering the finding that these two groups do not differ in disease phenotype as risk factor for difficult-to-treat disease, delayed first surgery might be an independent factor for complicated disease course in these patients at long term.

Disclosure: Nothing to disclose.

PP0680

SECRET-IBD: PRELIMINARY RESULTS OF A MULTICENTRE RETROSPECTIVE OBSERVATIONAL COHORT STUDY ON ONCOLOGICAL OUTCOMES OF PATIENTS WITH INFLAMMATORY BOWEL DISEASE WHO HAVE UNDERGONE SEGMENTAL COLONIC RESECTIONS FOR DYSPLASIA AND CANCER

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Introduction: The current international guidelines on the management of dysplasia in Inflammatory Bowel Disease (IBD) recommend proctocolectomy (PC) in cases of endoscopically unresectable dysplasia, multifocal dysplasia and colorectal cancer (CRC), and to consider it in cases of Low Grade (LGD) or Indeterminate dysplasia (IND). However, population-based studies have demonstrated that segmental colectomies (SC) are occurring in 40-70% of patients with IBD-associated CRC. Data on their outcomes is scarce, with only a handful of published articles, as evidenced by a recent systematic review.

Aims & Methods: Our aim is to contribute to this limited body of evidence from English IBD centres. Data of all adult IBD patients undergoing colorectal resections in 2004-2019 for dysplasia and CRC were collected from three centres. Primary outcome was oncological and secondary outcome was post-operative. Descriptive statistics were applied to compare outcome between SC (including subtotal colectomy) and proctocolectomy (PC).

Results: Of 131 patients identified, 45 underwent SC, and 86 PC. No statistically significant difference between SC and PC were identified for: gender, age at IBD diagnosis and subsequent surgery, comorbidity status (including Primary Sclerosing Cholangitis), indication for surgery (CRC vs dysplasia), and IBD subtype (CD vs UC) (p>0.05). Patients undergoing PC had more extensive and active colitis than those undergoing SC, as well as earlier stage and multifocal cancer. Median follow up time was 86 months (range 0 – 204), and 90 months (range 17 – 211), respectively. Unexpected cancer was found in 41/131 (31%), with no significant difference between SC and PC cohorts (28% vs 33%, p=0.581), however only 9/131 (7%) was at a site different to the original neoplasia. Metachronous cancer was found in 1/45 (2%) on completion proctectomy at 125 months post-index extended right hemicolectomy in a patient having completion for High Grade Dysplasia (HGD, SC cohort). Completion surgery occurred in 11/45 (25%) of SC, and only 4/11 were for a neoplastic indication (2/11 LGD, and 2/11 HGD). No significant difference in Overall Survival was observed between SC and PC groups (81% vs 94% at 5 years (p=0.376)).

Conclusion: This study failed to identify oncological advantages to proctocolectomy in patients with IBD neoplasia undergoing resection. Although there is potential for type II error with this small cohort, the findings are consistent with those of the published literature and recent systematic review. This study highlights the need for larger prospective studies.

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Disclosure: Nothing to disclose.

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PP0681

IMPACT OF SURGERY ON THE PSYCHOLOGICAL PROFILE OF INFLAMMATORY BOWEL DISEASE PATIENTS: A PROSPECTIVE STUDY

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Introduction: Surgery represents an important part of the therapeutic armamentarium for the treatment of IBD, and has been associated with variable effects on the psychological profile of patients with inflammatory bowel disease. However, surgical treatment is often distressingly perceived by these patients as the failure of their therapies, and can be a trigger for anguish, anxiety or depression.

Aims & Methods: The aim of this study is to describe the psychological impact of surgery on operated IBD patients.

This is a prospective descriptive monocentric study, conducted at the Department of Hepato-Gastro-Enterology and Proctology "Medicine B" CHU Ibn Sina Rabat, which collected all patients followed for IBD (Crohn's + UC) and who were operated between August 2021 and August 2023, the psychological impact and apprehension regarding surgery will be addressed in this study. Patients of both sexes who agreed to answer the questionnaire were included, and patients who refused to answer the questionnaire and complicated patients who stayed in intensive care were excluded.

Results: 43 operated patients were included in our study, including 29 (67.4%) in the emergency setting and 14 (32.6%) scheduled. Mean age was 43 ± 16.17 years. The sex ratio was 1.26 (F/H=24/19). 35 patients (81.4%) were being followed for Crohn's disease and 8 patients (18.6%) for ulcerative colitis (UC), of whom 11 patients (25.5%) had a history of previous resection, and 1 patient (2.3%) was being followed for depression.

The surgical indications were GCA in 12 cases (27.9%), stenosis in 18 cases (41.8%), perforation in 1 case (2.3%), abscess in 8 cases (18.6%), and fistula in 4 cases (9.3%).

Our patients underwent ileo-caecal resection in 22 cases (51.2%), hemicolectomy in 6 patients (14%), coloproctectomy in 3 patients (7%), sub-total colectomy in 11 patients (25.6%), of whom 18 patients (41.9%) had an ostomy.

On the announcement of a possible surgical procedure, the thymic symptoms and feelings expressed by patients can be classified as follows: preoperative anxiety in 18 patients (41.9%), fear of death in 7 patients (16.3%), while 24 patients (55.8%) were indifferent.

The main anxiety-generating concern was postoperative pain in 7 patients (14%), potential complications in 7 patients (16.3%), failure of the procedure in 6 patients (14%), fear of dying in 7 patients (16.3%), and 25 patients (58.1%) expressed no anxiety.

The postoperative psychological experience was unremarkable in 37 patients (86%), 3 patients (7%) reported transient postoperative sadness and 3 patients (7%) developed postoperative depression, prompting consultation with a psychiatrist.

Conclusion: The impact of surgery in IBD patients not only concerns the body, but also has an effect on psychological well-being.

This study found that the pre-operative period induces a significant burden of anxiety in these patients, and depression in some.

Disclosure: Nothing to disclose.

PP0682

COLECTOMY INDICATIONS DURING ULCERATIVE COLITIS: A TWENTY-YEAR RETROSPECTIVE FROM A UNIVERSITY HOSPITAL CENTER

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Introduction: Over the past two decades, advancements in medical and surgical therapy have significantly impacted hemorrhagic rectocolitis management. However, it is still not clear what happens in three situations when a colectomy is done during ulcerative colitis (UC): severe acute colitis, an illness that does not get better with medicine, and precancerous lesions.

Aims & Methods: The objective of this study was to evaluate the changes in the reasons and methods for using the UC during the last two decades. This study is an observational retrospective at a university hospital center, examining the progress of operational indicators in patients monitored for colorectal hyperplasia and who underwent segmental or complete colectomy between 2003 and 2023. The primary aim is to compare changes in patient characteristics, compare surgical methods (laparotomy and laparoscopy), and determine the frequencies of early and late post-operative problems based on the Clavien Dindo classification.

Results: The study analyzed 286 patients, with 57% being males and a median age of 40 years. The majority were pancholic and 35% were left-handed. Between 2003 and 2013, 87 patients underwent surgery for refractory chronic colitis (RCH) and 199 patients for severe acute colitis or (pre-) neoplastic lesions. The preoperative clinical features showed similarities between the two groups, particularly in terms of the duration of the course and severity. Over time, there has been a decline in colectomies for refractory chronic colitis, but an increase in colectomies for severe acute colitis or (pre-)neoplastic lesions. The utilization of laparotomy decreased from 52.3% to 18.6%, while the use of laparoscopy increased from 47.7% to 81.4%. An increase in the rate of mechanical ieo-anal anastomosis was detected.

There was no significant difference in the death rate 30 days following colectomy. The incidence of severe early problems following colectomy fell from 12.6% to 5.5%, while the average number of late complications reduced from 1.6 to 0.8. The rate of late recurrence of operation decreased from 38.6% to 19.5%. However, the incidence of infectious complications increased from 27% to 45.7%.

When comparing the two periods, there were notable disparities in the average number of biotherapies received, preoperative exposure to infliximab, and subcutaneous anti-TNFα. The utilization of infliximab considerably increased after 2013 in cases of severe acute colitis.

Conclusion: The study confirms changes in colectomy criteria for chronic ulcerative colitis over the past two decades, with a decrease in treatment-resistant cases and an increase in severe acute colitis or dysplasia/cancer cases, while laparoscopy usage has increased, reducing morbidity despite biotherapies usage.

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PP0683

USEFULNESS OF UNDERWATER EMR FOR SPORADIC NEOPLASM IN ULCERATIVE COLITIS

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Introduction: According to Western guidelines, endoscopic resection (ER) is acceptable for neoplastic lesions with clear boundaries, even within areas affected by ulcerative colitis (UC), and this has increased the opportunities for endoscopic resection However, this procedure is also difficult due to advanced fibrosis of mucosa associated with previous inflammation within areas affected by UC. Underwater endoscopic mucosal resection (UEMR)has become popular in recent years and has been useful for lesions with advanced fibrosis, such as lesions on scars after endoscopic resection. However, there are no reports on lesions on fibrotic mucosa caused by previous inflammation within areas affected by UC.

Aims & Methods: In this study, we verified the safety and usefulness of UEMR for lesions diagnosed as sporadic neoplasms (SN) before ER, among neoplastic lesions that developed within areas affected by UC. We enrolled 35 lesions in 26 patients that underwent UEMR for neoplastic lesions 20 mm or smaller in diameter, which had developed within areas affected by UC, from December 2021 to February 2023 at Keio University Hospital. The primary endpoint was R0 resection rate, while the secondary endpoints were en bloc resection rate, procedure time. and adverse events. Before ER, chromoendoscopy with indigo carmine-dye spraying was performed to confirm the absence of flat type dysplasia around the lesion, then the patients diagnosed with SN were recruited into the study. UEMR was performed on all cases with background mucosa in remission (Mayo Endoscopic Subscore 0 or 1).

Results: The disease types of the subjects consisted of pancolitis in 22 patients (84.6%) and left-sided colitis type in four patients (15.4%), the mean tumor diameter of resected lesions was 11.1 mm, and background mucosa consisted of normal mucosa (regenerated mucosa after inflammation) in nine lesions (25.7%), atrophic mucosa in 20 lesions (57.1%), and atrophic mucosa with ulcer scars in six lesions (17.2%). A total of 34 lesions (97.1%) underwent en bloc resection; of them, 29 lesions (85.3%) underwent R0 resection, while only one (2.9%) underwent piecemeal resection. Median procedure time was 8 minutes. The final pathological diagnosis was SN for all cases: two cases (5.7%) of tubular adenocarcinoma, 21 cases (60%) of tubular adenoma, and 12 cases of sessile serrated lesions (34.3%). In terms of procedural accidents, there were no cases of intraoperative perforation, delayed perforation, post polypectomy coagulation syndrome, and delayed bleeding. Endoscopies performed 1 year later revealed no local recurrence in all cases.

Conclusion: UEMR may be safe and effective endoscopic resection method for SN within areas affected by UC.

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Disclosure: Nothing to disclose.

PP0684

META-ANALYSIS: REAL-WORLD EFFECTIVENESS AND SAFETY OF DARVADSTROCEL FOR COMPLEX PERIANAL FISTULIZING CROHN'S DISEASE

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Introduction: Local injection of darvadstrocel, a suspension of expanded adipose-derived allogenic mesenchymal stem cells with anti-inflammatory and immunomodulatory properties, represents a novel effective treatment for complex perianal fistulas in patients with Crohn's disease (CD).

Aims & Methods: The aim of this systematic review was to summarize reported evidence on the real-world outcomes of darvadstrocel for complex perianal fistulas in CD and to conduct a meta-analysis of effectiveness and safety data.

A comprehensive search was led in electronic databases PubMed, Web of Science, EMBASE, Scopus and ScienceDirect until 15 April 2024 to identify observational studies evaluating darvadstrocel for complex fistulizing CD. Joanna Briggs Institute (JBI) Critical Appraisal tool was used to assess the study quality and the risk of bias. Random-effects meta-analysis model was used to calculate the pooled rates of clinical and safety outcomes. Heterogeneity across studies was assessed with the I^2 statistic according to Cochrane Handbook.

The primary endpoint was clinical remission (closure of all treated external openings) at months 6 and 12. Adverse events (AE) were reported as incidence rates (IRs) per 100 patients-year (PY) of exposure.

Results: Of the 72 citations identified, 12 studies with 595 patients were eligible for inclusion. According to JBI criteria, 7 and 5 studies showed a low or moderate risk of bias, respectively. The pooled rate of patients achieving clinical remission was 68.1% at month 6 (95% CI 63.4-72.7) and 77.2% (95% CI 70.1-83.8) at month 12 (Table).

Median time to achieve fistula closure was 6 weeks (95% CI 3.8-9.5). Combined remission defined as clinical remission and absence of collections > 2 cm confirmed by magnetic resonance imaging was reported in 60.6% and in 69.7% of patients at month 6 and month 12, respectively. Clinical response was observed in 79.9% and in 81.7% of patients at month 6 and month 12, respectively.

The rate of patients with treatment failure, defined as no clinical remission, at last follow-up (mean 18.7 month; SD 9.9) was 34.5% (95% CI 25.7-43.9). Linear regression analysis indicated that failure rate was independent of follow-up time (R^2 = 0.001). For all effectiveness outcomes, between-study heterogeneity might not be important (Table).

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Pooled IR per 100 PY of AEs and serious AEs were 19.6 and 3.2, respectively. IR per 100 PY of new perianal abscesses and reoperations were 16.9 and 7.1, respectively. For safety analyses, between-study heterogeneity ranged between moderate and substantial (Table).

Effectiveness outcomes	Pooled rate	95% CI	Nº studies	Patients	Heterogeneity (I^2)	Risk of heterogeneity
Month 6 clinical remission	68.1%	(63.4-72.7)	7	424	0% (P = 0.78)	Not be important
Month 12 clinical remission	77.2%	(70.1-83.8)	7	191	0% (P = 0.99)	Not be important
Month 6 combined remission	60.6%	(51.0-69.9)	4	111	0% (P = 0.84)	Not be important
Month 12 combined remission	69.7%	(47.6-88.6)	2	25	0% (P = 0.87)	Not be important
Month 6 clinical response	79.9%	(73.1-86.0)	4	370	31% (P = 0.22)	Not be important
Month 12 clinical response	81.7%	(75.2-87.6)	4	177	0% (P = 0.43)	Not be important
Failure at last follow-up	34.5%	(25.7-43.9)	7	191	35 % (P = 0.16)	Not be important
Safety outcomes	Incidence rate/100 PY	95% CI	Nº studies	Total PY	Heterogeneity (I^2)	Risk of heterogeneity
AEs	19.6	(10.1-29.1)	7	1245	71 % (P < 0.01)	Substantial
SAEs	3.2	(0.6-5.8)	8	1279	59 % (P = 0.02)	Moderate
Perianal abscesses	16.9	(7.6-26.2)	8	1267	69 % (P < 0.01)	Substantial
Reoperation	7.1	(2.2-12.1)	7	222	44 % (P = 0.09)	Moderate

Table. Pooled rates for effectiveness outcomes and pooled incidence rates per 100 patients-year (PY) for safety outcomes.

Conclusion: Real-world evidence supports the efficacy and safety of darvadstrocel for complex perianal fistulas in CD. Studies reported high fistula healing rates that can be sustained long-term in most patients, with high homogeneity across studies. Darvadstrocel showed a favourable safety profile.

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PP0685

COMPARATIVE EFFECTIVENESS OF VANCOMYCIN AND METRONIDAZOLE ON EVENT-FREE SURVIVAL AFTER INITIAL INFECTION IN PATIENTS WITH *CLOSTRIDIOIDES DIFFICILE* - A GERMAN MULTICENTRE COHORT STUDY (IBIS)

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Introduction: The European Society of Clinical Microbiology and Infectious Diseases (ESCMID) published an update of its guideline in 2021, in which vancomycin or fidaxomicin replaced metronidazole as the preferred first-line treatment option for non-severe *Clostridioides difficile* infection (CDI). This decision was controversially discussed since outcome data in this particular patient group remains limited. The IBIS study aimed to close this evidence gap.

Aims & Methods: The IBIS multicentre cohort enrolled patients with an index episode of CDI between August 2017 and September 2020. The primary endpoint was event-free survival (EFS), defined as response to treatment with metronidazole or vancomycin within ten days of initiation, absence of recurrence and death from any cause up to 90 days post-treatment. A Cox proportional hazards model with inverse probability of treatment weighting was used to investigate the comparative effectiveness of this outcome. Additionally, subgroup analyses were performed based on severe and non-severe infections.

Results: Of the 489 patients included, 118 (24%) received initial treatment with metronidazole and 371 (76%) with vancomycin. Of these, 78/118 (66.1%) and 247/371 (66.6%), respectively, responded to treatment within ten days, neither developed a recurrence nor died within 90 days and thus achieved the outcome of EFS. In the subgroup of non-severe infections, 74/293 patients (25.3%) received metronidazole, and 219/293 (74.7%) received vancomycin. Of these, 33/74 (44.6%) metronidazole patients and 150/219 (68.5%) vancomycin patients survived event-free. The Cox proportional hazards model revealed differences in EFS for the overall population and both sub-groups (reference metronidazole: all severity levels: hazard ratio [HR] 0.46, [95% Cl 0.33-0.65]; non-severe: HR 0.39; [95% Cl 0.24-0.60]; severe: HR 0.52; [95% Cl 0.28-0.95]).

Conclusion: Our analysis confirms current changes in guidelines, as it supports the superiority of vancomycin compared to metronidazole across all severity levels.

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PP0686

COMPARATIVE ANALYSIS OF GUT MICROBIOTA CHANGES, DIARRHEA FREQUENCY, AND PROBIOTIC SENSITIVITY DURING HELICOBACTER PYLORI ERADICATION IN ELDERLY AND YOUNGER ADULTS

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Introduction: Antibiotic-associated diarrhea frequently arises due to disruptions in the intestinal microbiome, often becoming severe in elderly individuals. Such instances of diarrhea are also notably prevalent during *Helicobacter pylori* eradication, affecting approximately 26.6% of patients [1]. The composition and diversity of gut microbiota vary with age, suggesting that the frequency of diarrhea associated with antibiotic disruption of the gut flora, as well as sensitivity to probiotics, may differ across age groups.

Aims & Methods: Aims: This study aims to examine the differences in the frequency of diarrhea and disruption of the microbiome during *Helicobacter pylori* eradication between elderly and younger Japanese adults, and to evaluate the effectiveness of the antibiotic-resistant probiotic *Enterococcus faecium 129 BIO 3B-R* in mitigating these effects.

Methods: Seventy-six Japanese adults undergoing *Helicobacter pylori* eradication treatment were enrolled in the study and categorized into two groups: those aged 70 and above (elderly) and those under 70 (younger adults). The diversity of the gut microbiota prior to treatment initiation was assessed using meta 16S sequencing to evaluate differences between the two age groups. Participants were randomly assigned to receive either a resistant lactic acid bacteria preparation (*Enterococcus faecium 129 BIO 3B-R*) or a placebo in a blind manner, and the incidence of diarrhea and changes in microbiota composition were analyzed.

Results: Results: Meta 16S rDNA sequencing analysis revealed distinct microbiota compositions between younger and elderly adults, with significantly higher alpha diversity observed in the elderly. LEfSe analysis showed a higher abundance of the *Lactobacillus* and *Blautia* genera and a lower presence of *Bammaproteobacteria* class in the elderly group. The frequency of antibiotic-induced dysbiosis-related diarrhea was compared between younger and elderly adults, showing a higher tendency in the elderly, although there was no significant difference in the extent of diversity reduction between the age groups. Administration of the resistant lactic acid bacteria preparation, *Enterococcus faecium 129 BIO 3B-R*, resulted in a lower frequency of diarrhea and a tendency to preserve alpha diversity in the microbiota of the elderly compared to the placebo group.

Conclusion: Elderly individuals, with their distinct gut microbiota composition and diversity compared to younger adults, may have different susceptibilities to antibiotic-induced diarrhea. Furthermore, incorporating resistant lactic acid bacteria as a supplementary treatment could be advantageous for managing diarrhea and stabilizing the gut flora in the elderly.

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PP0687

THE SAFETY OF FECAL MICROBIOTA TRANSPLANTATION FOR GASTROINTESTINAL DISEASES: LONG-TERM FOLLOW-UP OF 11025 CASES IN CHINA

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Introduction: Fecal microbiota transplantation (FMT) has been successfully used in a series of diseases in recent years. Most studies have focused on short-term risks, which are mild and generally associated with delivery methods. There have been no reports of FMT safety in the treatment of multiple diseases and long follow-up periods.

Aims & Methods: We performed a retrospective analysis of 11025 patients who underwent FMT in the Tenth People's Hospital of Tongji University between January 2012 and December 2023. We included patients with various gastrointestinal diseases and patients with extraintestinal diseases that involved intestinal dysfunction.

A questionnaire was administered to the included patients during follow-up (2 weeks, 3 months, 1 year, and 5 years after FMT) via telephone and health portal software Haodaifu app).

Results: Most adverse events during short-term follow-up (2 weeks) were related to the method of delivery. The main adverse events in the nasogastric tube group were tube-related respiratory difficulty (40.6%,include sore throat 14.1%, nasal congestion 8.7%, nasal irritation due to nasogastric intubation 17.8%) and nausea/vomiting (12.6%). In the oral capsules group, nausea and vomiting (8.9%) occurred during oral administration because the capsules are associated with a fecal smell and taste, both of which are difficult to remove completely. For colonoscopy and enema routes, the main adverse events were diarrhea (10.7%, 9.9%) and perianal discomfort (15.3%, 8.4%), respectively.

Two cases in the colonoscopy group and one case in the enema group were complicated with colon perforation, but remission with conservative treatment. The most frequently reported adverse events were mild or moderate in severity and transient. Those symptoms associated with nasogastric tube administration disappeared after tube removal at the completion of treatment. Several other new symptoms were reported during short-term follow-up, although none occurred in >1% of patients. The most common symptom reported was bloating/flatulence. Although more adverse events were reported in the nasogastric tube group, most were mild.

Requirements	Descriptions
Informed consent	Informed consent must be obtained from patients and family members before treatment
Safety of the transplanted bacterial solution	Optimize the quality control of bacterial solution preparation and storage. Ensure that the bacterial solution is used within 1 week.
Assessment of patient suitability for FMT	Ensure normal vital signs, as well as the absence of fever, severe infection, sepsis, SIRS, and other inflammatory diseases. Confirm that the patient can tolerate full oral nutrition or enteral nutrition.
Transplantation route safety	The establishment of transplantation pathways must comply with the requirements of the Chinese guidelines*, including transplantation management, safety assessment, patient tolerance, and individualized selection.
Emergency procedures for suspected adverse events during transplantation	During the transplantation process, if adverse events are suspected, the transplantation should be paused. The cause of the event should be identified; the transplantation can be resumed when the adverse event is resolved.

Table 1.Tongji FMT Requirements.

Additionally, patients are more comfortable with the upper gastrointestinal route because they could use it for enteral nutrition support or gastrointestinal imaging according to their medical needs. Based on our clinical

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experience, we proposed the "Tongji FMT Requirements"and implemented them in our clinic (beginning in 2019) to standardize the indications and timing of FMT treatment and to improve FMT safety.

After the implementation of these requirements for 2 years, we compared safety data collected before and after implementation; The application of the requirements has effectively reduced adverse events. The short-term (mainly FMT-related diseases) incidence rate significantly decreased from 32.6% in 2017–2019 to 17.2% in 2019–2021.

Conclusion: FMT appears safe during both short-term and long-term follow-up. The patient group that received FMT did not exhibit an increased incidence of new diseases, and the risk of infection transmission appears to be low. Considering the widespread use of FMT, there is a need for the establishment of a national clinical and patient-reported FMT safety collection system in future.

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Disclosure: Nothing to disclose.

PP0688

VALIDATION OF *CLOSTRIDIOIDES DIFFICILE* QPCR METHODS FOR COMPARISON BETWEEN BACTERIAL LOAD AND CONCENTRATION OF TOXIN GENES

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Introduction: Adverse outcomes of *Clostridioides difficile* infection (CDI) are derived from toxin presence, thus the gold standard has focused on cell cytotoxin assays (1,2). However, quantitative polymerase chain reaction's (qPCR) cycle threshold values linked to genomic load may be significantly associated with disease outcome, aiding in the distinction between causative pathogens, coinfections and asymptomatic carriage (3). Genes encoding toxins A/B (*tcdA*/*tcdB*) are usually targeted by PCR diagnostics (3). While most virulent strains produce them simultaneously, both alone can also cause CDI (2,4,5). These genes are located in the genome within the Pathogenicity loci (*PaLoc*), which can be horizontally transferred (4,6). Recent studies have also discovered extrachromosomal molecules carrying *tcdB* (7). This suggests that the number of cells does not necessarily correlate to the number of toxin genes.

Aims & Methods: The aim was to develop a reliable qPCR method to quantify *C. difficile* toxin A/B genes to compare them to bacterial load by targeting a house-keeping gene (*adK*). Standard curves were validated according to official guidelines and reference publications (efficiency E=90-110%; slope S=(-3.1)-(-3.6); linearity R^2≥0.98) (8–11), testing 10 replicates by 9 consecutive 10-fold dilutions starting from copies/mL of 3 quantified dsDNA gBlock stocks. The linear range was validated running 5 dilution points (to copies/mL) in triplicate. The last detected dilution inside the linear range with a coefficient of variation ≤25% was considered limit of quantification. Precision, repeatability and reproducibility were evaluated. Commercially supplied *C. difficile* isolates and retrospective clinical samples confirmed to be *C. difficile* positive were quantified.

Results: All standard curve assays developed showed acceptable qPCR efficiency and qPCR linearity (E=96-103%; S=(-3.2)-(-3.4); R^2≥0.99). Four commercially supplied *C. difficile* isolates and 14 clinical samples were tested with all 3 validated methods. For each sample, concentrations resulted in the same order of magnitude but different total numbers for all assays (adK to copies/mL; tcdA to copies/mL; tcdB to copies/mL), except for three samples that presented one order of magnitude less tcdA than adK. In general, adK gene copies resulted in higher numbers than the toxins, with *tcdB* more concentrated than *tcdA*.

Conclusion: Three reliable qPCR standard curve quantification methods were successfully developed and validated to allow for the detection of toxigenic *C. difficile* and quantification of bacterial load. Results indicate that the number of copies for each gene is not the same within one single sample. This raises doubts regarding CDI, such as the possibility of harbouring both toxigenic and nontoxigenic strains (*PaLoc* negative) within one single patient, which would explain the higher concentration of *adK* gene copies in samples. Since multiple copies of the *PaLoc* have not been described, a higher concentration of *tcdB* compared to *tcdA* could suggest either the existence of some tcdA-tcdB+ organisms or of multiple copies of tcdB+ extrachromosomal molecules, the percentage of which might have been underestimated. Further studies would be required to elucidate the relevance and implications that these findings might have to CDI diagnosis and treatment.

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PP0689

HUMAN MICROBIOTA DIVERSITY IN TISSUE SAMPLES OF UPPER AND LOWER GI TRACT

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Introduction: The microbiota of human gastrointestinal tract is a complex and dynamic population of microorganisms and plays a major role in the maintenance of gut health and disease conditions. The microbial population, structure and function vary considerably throughout the upper and lower gut, but specific population and its diversity for each segment of GI tract is still not well described.

Aims & Methods: Aim: The retrospective evaluation of microorganism population and its diversity separately for upper and lower GI tract by cultivation of tissue samples that were obtained at routine endoscopy of 271 patients indicated to examination for different reasons.

Methods: Biopsies were taken using sterilized standard endoscopic biopsy forceps. Two pieces of tissue were obtained without the removal of forceps from the duodenum, from the antrum of the stomach and in same way from different segments of large intestine. Samples were placed into a 10% glucose solution for microbial cultivation.

Results: Overall, 271 biopsy specimens from the upper and lower GI tract were cultivated, specifically in 235 patients samples were obtained from antrum of stomach and duodenum, in 127 patients from large intestine and in 36 patients from both parts of GI tract. The presence of bacteria in upper GI tract was found in 70,21% and was negative in 29,79% of cases, while in lower GI tract bacteria was found in 93,70% and was negative 6,30% of cases. Tissue samples from both parts of GI tract was obtained from 36 patients and the same genera was found in 44,44%. In upper GI tract *E. coli* was present in 15,74%, followed by *Enterobacter cloacae* in 10,64%, *Pseudomonas aeruginosa* in 8,51%, *Enterococcus species* in 7,66%, *Klebsiella pneumonie subs pneumoniae* in 5,11%, *Candida albicans* in 4,26%, *Helicobacter pylori* and *Klebsiella pneumoniae* in 2,98%, *Proteus mirabilis* in 2,55%, *Stenotrophomonas maltofilia* in 1,70%, *Neisseria subflava* and *E. coli ESBL po*sitive in 0,85% and 0,43% belongs to *Klebsiella pneumoniae ESBL positive, Streptococcus salivarius, Acinetobacter baumani, Staphylococcus warneri, Enterococcus faecalis, Streptococcus maltofilia, Enterococcus hirae* and *Candida complex*. In lower GI tracts *E. coli* was present in 64,57%, followed by *Klebsiella pneumoniae* in 8,66%, *Candida albicans, Proteus mirabilis* and *Enterobacter cloacae in* 3,15%, *Enterococcus species and Klebsiella pneumoniae subs pneumoniae* in 2,36%, *E. coli ESBL positive, Enterococcus faecalis* and *Pseudomonas aeruginosa* in 1,57%, *Candida non albicans* and *Citrobacter koseri* in 0,79%.

The diversity of upper and lower GI tract was measured by Shannon diversity index, and it was 2.48 for upper GI tract and 1.29 for lower GI tract.

Conclusion: Most of studies which reveals fundamental differences between microbiota of upper and lower GI tract is based on examination of luminal content. We are presenting our results of the diversity of microbiota based on cultivation of biopsy samples.

Disclosure: Nothing to disclose.

PP0690

MICROBIOME AWARENESS AND BEHAVIORS AMONG PATIENTS: RESULTS FROM THE INTERNATIONAL MICROBIOTA OBSERVATORY IN 11 COUNTRIES

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Introduction: The International Microbiota Observatory aimed to measure awareness, knowledge and behaviors about the microbiome in general populations worldwide and in patients.

Aims & Methods: The survey was conducted among 7,500 individuals aged 18 years old and over in 11 countries (France, Spain, Portugal, Poland, Finland, Morocco, United States of America, Brazil, Mexico, China and Vietnam). Automated selection from the Ipsos online Panel ensured representative samples of general population fit the quotas method based on gender, age, employment status, and country regions. Participants were asked to fill out an online questionnaire with single- or multiple-choice questions. Demographics, microbiome awareness and knowledge, behaviors, information provided by healthcare professionals were documented and analyzed using descriptive statistics. Our focus is a self-reported patients' sub-population that included 4,509 individuals with respiratory (n=888), cardiovascular (n=1,177), digestive (n=1,036), genitourinary and kidney (n=529), neurological (315), autoimmune (n=716), osteoarticular (n=1040), psychiatric diseases (n=652), diabetes (n=745), cancer (n=270) and allergies (n=2,097).

Results: 4,509 people with various chronic diseases from 11 countries completed the questionnaire (52% female, mean 47,7 years). 70% of these patients have heard of the term "microbiome", with 24% knowing exactly what it is. However, patients' awareness varies according to the disease they have: people with diabetes are less aware (65%; (p<0.001) and people with digestive diseases are more aware (79%; (p<0.001) of this term compared to all patients. For patients with cardiovascular disease, awareness is not statistically different compared to all patients (68%). In their daily lives, 61% of patients have changed their behaviors to keep their microbiome as balanced and functional as possible. This proportion varies according to the disease they have: patients with cardiovascular diseases are less likely to change their behaviors (57%; p=0.001) while patients with digestive diseases are more likely to change them (73%; p<0.001). And among patients with diabetes, the proportion who have changed their behaviors (59%) is not statistically different compared to all patients. Only 48% of patients had received from a healthcare professional an explanation of what a microbiome is, its role and function. But, compared to all the patients, people with cardiovascular disease (43%; p<0.001) were the patients the least likely to have received this piece of information and those with digestive diseases were more likely to have received it (61%, p<0.001). Finally, 78% of patients would trust healthcare professionals as the first source of relevant and trustworthy information.

Conclusion: This survey provides insights into the need for additional education in patients. Healthcare professionals could play a crucial role in providing information about the role and functions of the microbiome, more precisely tailored to each chronic disease that patients suffer from.

References: https://www.biocodexmicrobiotainstitute.com/en/international-microbiota-observatory

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PP0691

GUT MICROBIOME IN AUTISTIC CHILDREN: THE EFFECT OF OXYTOCIN ON THE GUT MICROBIOME AND THE IMPACT OF LIFESTYLE VARIABLES

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Introduction: Gut microbiome has been suggested to play a role in the etiology of autism. Specifically, recent studies have pointed toward altered gut microbiome compositions, resulting in overgrowth of particular bacteria that have been linked to autism symptom severity, increased repetitive behavior, and gastrointestinal problems. However, how variations in the gut microbiome relate to autistic behavioral difficulties remains sparse. Recent autism research points towards alterations in the oxytocinergic system in autistic individuals (1), which have been implicated to form a major mediator of the gut microbiome and his impact on behavior.

However, there have been no studies to date directly addressing whether stimulation of the oxytocinergic system could modulate gut microbiome compositions in autistic children. Furthermore, it is largely unknown how potentially confounding factors may impact the results of this intervention.

Aims & Methods: In the current study we analysed gut microbiome from a previous clinical trial (2). We examined diagnosis-related differences in gut microbiome compositions in autistic versus non-autistic children (8-12 years), and the effects of a 4-week intranasal oxytocin administration on gut microbiome compositions, taking into account confounding variables.

This double-blind randomized placebo-controlled trial consisted of a double-blind phase followed by a 4-week single-blind phase during which all participants received intranasal oxytocin. Effects of oxytocin (12 IU, twice daily) on gut microbiome compositions and behavioral difficulties were assessed immediately after the administration period and 4 weeks after the last administration. Eighty autistic children and 40 typically developing peers were recruited. Note that the collection of stool samples were optional, resulting in 52 autistic children and 27 non-autistic children included in the gut microbiome part of this study.

Behavioral difficulties were assessed using the Social Responsiveness Scale, the Repetitive Behavior Scale-Revised, and the child- and parent-reported version of the Screen for Anxiety Related Emotional Disorders. Gut microbiome was analysed by 16S rRNA sequencing.

Results: No difference in gut microbiome major confounder transit time (Bristol Stool Scale) was observed between autistic and non-autistic children. Diagnosis-related difference analyses in alpha-diversity revealed a significant lower observed richness (Z = 2.476, p = .013) and a higher microbial evenness (Z = -2.377; p = .017) in autistic vs non-autistic children after Benjamini-Hochberg correction (BH-correction). Moreover, no significant differences in (CLR-transformed) abundances of bacterial genera were found between the autistic and non-autistic children after BH-correction.

However, examining the overall associations with behavioral difficulties, a significant positive association (q-value (p-value after BH-correction) <.25) was evident between *Romboutsia* and parent-reported anxiety-like behavior (Coeff. = 0.048, q = .06).

Ongoing analyses will unveil initial findings on oxytocin's effect on the gut microbiome in association to clinical responses at UEG 2024.

Conclusion: Gut microbiome differs modestly between autistic and non-autistic children. The anticipated outcomes of the intervention will allow to advance our understanding of the intricate interplay between the oxytocinergic system and the gut microbiome, offering a foundation for the development of targeted therapeutic interventions.

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Disclosure: Nothing to disclose.

PP0692

STUDY OF THE PREBIOTIC POTENTIAL OF A PREPARATION USED IN FUNCTIONAL GI SYMPTOMS, MARSHMALLOW ROOT EXTRACT, USING THE HIGH THROUGHPUT EX VIVO TECHNOLOGY SIFR

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Introduction: The extract of the marshmallow (*Althaea officinalis* L.) root is used in gastrointestinal complaints involving irritations of the gastrointestinal mucosa [1]. While the underlying mucosa protective and anti-inflammatory effects have been studied thoroughly over the last years, data on ist effect on the gut microbiota are missing.

Aims & Methods: Given that a considerable part of the constituents of the extract can be assumed to be not fully absorbed in the small intestine, its influence on the gut microbiota in the large intestine is worth to be studied.

The impact of marshmallow root extract (STW 42) on the gut microbiome was investigated using the high-throughput, *ex vivo* SIFR® technology (Cyptobiotix, Ghent, Belgium). Considering 6 subjects individually, this allowed to cover the broad spectrum of microbial composition that occurs *in vivo*, in line with the so-called enterotypes, thus ensuring representative findings. Simulating a single intake, Althaea radix (AR) was dosed at 1.875 g/d and inulin (IN; used as positive control) was dosed at equivalent of 2 g/day.

Results: AR impacted microbial metabolite production as was shown by significantly increased total SCFA and significantly decreased pH. While the extent of the effects of IN was larger, nevertheless, AR significantly increased health-related propionate/butyrate generation, suggesting a health-promoting potential of AR. AR showed a specific effect on butyrate production, which suggests a product-specific impact on the human adult gut microbiome.

In contrast to the major interpersonal differences at family and phylum, potent treatment effects were noted at higher phylogenetic resolution (OTU level): AR exerted effects including the consistent stimulation of an OTU related to *Ruminiclostridium cellulolyticum.*

Conclusion: Overall, the SIFR® study demonstrated the prebiotic potential

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of marshmallow root extract (AR) based on the impact on microbial metabolite production and microbial composition. This adds to the evidence supporting its use in functional GI symptoms.

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PP0693

ESTABLISHING A STOOL BANK FOR FECAL MICROBIOTA TRANSPLANTATION: A PILOT PROJECT IN CATALONIA

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Introduction: Faecal microbiota transplantation (FMT) involves the transfer of microbiota from healthy donors to patients with the aim of correcting dysbiosis and restoring intestinal diversity. FMT has been shown to be successful in the treatment of recurrent or refractory *Clostridioides difficile* infection. The main limiting factor to carry out FMT is access to treatments due to the absence of intestinal microbiota banks (IMB), a process that requires a complex structure.

Aims & Methods:

• Define and standardize procedures for the creation of an IMB.
• Fabrication of viable FMT samples.

The project was carried out at the Hospital Clinic of Barcelona. Processes were defined for recruitment donors, collection and processing samples for their later use in clinical practice. A multidisciplinary team was made up with gastroenterologists, microbiologists, infectologists and two clinical nurses.

Results: From May to December 2021, through a questionnaire distributed to the donation unit of the Blood and Tissue Bank (BTB) of Hospital Clinic, 207 potential donors were identified. 52.7% (109/207) of them were considered suitable, but after the a first screening including clinical visit and microbiological analytical studies carried out, only 18.3% (20/109) of them were able to start the donation process. During this period, 79 donations were collected which enabled the processing of 155 stool samples for FMT. After 2 months of quarantine and the second screening, only 73% of the samples (114/155) were ready to be released for clinical use. Two donors were responsible for 40.5% of the donations, while 30% (6/20) did not contribute any samples during the donation period. If we consider suitable donors those who did not present exclusion criteria in any phase of the established screening, the final number was only 9.1% (10/109).

Conclusion: 1. Correct organization of all processes is essential for the establishment of a IMB.
2. Loyalty of committed donors is essential.
3. This was the first step of a collaborative project currently ongoing at the Clinic Hospital and the Bellvitge Hospital, supported by the Department of Health in Catalonia.

Disclosure: Nothing to disclose.

PP0694

THE IMPACT OF QUALITY IMPROVEMENT PROGRAM ON CLOSTRIDIOIDES DIFFICILE INFECTION: A QUASI-EXPERIMENTAL STUDY

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Introduction: Clostridioides difficile infection (CDI) is a major cause of nosocomial diarrhea. This study aimed to implement a quality improvement program (QIP) to expedite proper CDI treatment, including discontinuing laxatives and causative antibiotics.

Aims & Methods: Stool test results positive for CDI were automatically sent via text message to the quality improvement team (QIT), specialists in CDI management. The QIT played an advisory role in this treatment. The outcome of this study was the competency of CDI treatment within 24 h of stool test reporting.

Competency was investigated using three different models:

Model 1, initiation of CDI treatment within 24 h of positive stool test report; Model 2, Model 1 criteria met with no concurrent laxative use; and Model 3, Model 2 criteria met with no concurrent causative antibiotics. Competency rates were compared between pre- and post-intervention periods (1 year each). Analyses were performed for all patients with CDI and inpatients with CDI.

Results: In total, 340 patients with CDI (pre-intervention, n=144; post-intervention, n=196) were included in this study. The numbers of inpatients with CDI were 129 and 181, respectively. For the full patient analysis, the rates of competency for Model 1 (P=0.002), Model 2 (P<0.001), and Model 3 (P<0.001) in the post-intervention group were higher to those in the pre-intervention group. In the inpatient analysis, the competency rate was also higher in the post-intervention period in all three models.

Conclusion: QIP enhanced the quality of CDI treatment in terms of prompt treatment and discontinuation of concomitant laxative and causative antibiotics.

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PP0695

DISPARITIES OF CONSUMPTION AND EXPENDITURE ON FIDAXOMICIN AND VANCOMYCIN FOR CLOSTRIDIOIDES DIFFICILE INFECTION (2012 TO 2023): A LONGITUDINAL STUDY OF 43 COUNTRIES

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Introduction: While the latest treatment guidelines recommended fidaxomicin and oral vancomycin as first-line treatments for *C. difficile* infections, it is imperative to understand the changes in the use of these medications at the global level, given the disparities on disease prevalence and medication availability.

Aims & Methods: We aim to assess the multinational trends in the consumption and expenditures of fidaxomicin and oral vancomycin for *C. difficile*. This longitudinal trend study used the IQVIA-MIDAS global sales database to analyse data from 2012 to 2023 for fidaxomicin and oral vancomycin in 43 countries. Consumption was quantified in a defined daily dose per year per 100,000 inhabitants (DDD/100K). Expenditure was calculated in dollars per capita per year at manufacture level price, standardised to 2023 US dollars. Changes in consumption and expenditure were estimated using compound annual growth rates (CAGR).

Results: Global consumption of fidaxomicin increased rapidly from 21.68 DDD/100K in 2012 to 126.68 DDD/100K in 2023, with a consumption CAGR of 17.42% and an expenditure CAGR of 14.96%. Oral vancomycin also displayed a slower upward consumption trend from 71.61 to 120.82 DDD/100K between 2012 and 2023 with a CAGR of 4.87 %, but a decreasing trend in expenditure CAGR (-15.16%). Consumption of fidaxomicin and oral vancomycin in high-income countries were substantially higher than in countries of upper- and lower-middle income. At contient level, consumption and expenditure of fidaxomicin was highest in North America, which exceeded the cumulative total of all other continents. However, the most increase in expenditure was seen in Africa continent and lowest in North America.

	Consumption (DDD/100K)			Expenditure (US$ per capita)		
	2012	2023	CAGR	2012	2023	CAGR
Fidaxomicin (All, n=34)	21.68	126.86	17.42	0.07	0.34	14.96
High income countries (=31)	24.42	143.86	17.49	0.00	0.39	15.05
Upper middle income (n=2)	0.00	2.72	168.23*	0.00	<0.01	149.43*
Lower middle income (n=1)	0.00	0.00	104.18**	0.00	0.00	99.74**
Oral Vancomycin (All, n=29)	71.61	120.82	4.87	0.20	0.03	-15.16
High income countries (n=22)	203.16	350.79	5.09	0.61	0.10	-14.93
Upper middle income (n=5)	0.11	2.52	37.34	<0.01	<0.01	32.05
Lower middle income (n=2)	1.21	5.70	15.08	<0.01	<0.01	39.62

*2017-2023; **2018-2021

Conclusion: Consumption of fidaxomicin and oral vancomycin continues to increase multi-nationally, with high-income countries and North America consistently being the largest consumers. However, there are marked disparities on increase in expenditure, which was more marked in countries with middle income and African continent. This disparity could jeopardize the optimal delivery of effective medication for patients with *C. diffile* infection in these regions.

Disclosure: Nothing to disclose.

PP0696

DEEP LEARNING-BASED PREDICTION OF *CLOSTRIOIDES DIFFICILE* INFECTION CAUSED BY ANTIBIOTICS USING LONGITUDINAL ELECTRONIC HEALTH RECORDS

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Introduction: *Clostrioides difficile* infection (CDI) is a major cause of antibiotic-associated diarrhea and colitis. It is recognized as one of the most significant hospital-acquired infections. Although CDI can develop severe complications and spores of *Clostrioides difficile* can be transmitted by the fecal-oral route, CDI is occasionally overlooked in clinical settings.

Aims & Methods: Thus, it is necessary to monitor high CDI risk groups, particularly those undergoing antibiotic treatment, to prevent complications and spread. We developed and validated a deep learning-based model to predict the occurrence of CDI within 28 days after starting antibiotic treatment using longitudinal electronic health records. For each patient, timelines of vital signs and laboratory tests with a 35-day monitoring period and a patient information vector consisting of age, sex, comorbidities, and medications were constructed.

Results: When prediction performance of our model was evaluated, the area under the receiver operating characteristic curve was 0.952 (95% CI: 0.932-0.973) in internal validation and 0.972 (95% CI: 0.968-0.975) in external validation. Platelet count and body temperature emerged as the most important features. The risk score, the output value of the model, exhibited a consistent increase in the CDI group, while the risk score in the non-CDI group either maintained its initial value or decreased.

Conclusion: Using our CDI prediction model, high-risk patients requiring symptom monitoring can be identified. This could help reduce the underdiagnosis of CDI, thereby decreasing transmission and preventing complications

Disclosure: Nothing to declare

PP0697

EFFICACY AND SAFETY OF FECAL MICROBIOTA TRANSPLANTANTION (FMT) IN RECURRENT *C.DIFFICILE* INFECTION– RESULTS OF THE ONE-CENTER RETROSPECTIVE STUDY

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Introduction: Many studies have indicated a high effectiveness of fecal microbiota transplantation (FMT) in treatment of recurrent or refractory *Clostridioides difficile* infection (rCDI). Nevertheless, there is still lack of data from subpopulations regarding application of FMT and factors related to its response.

Aims & Methods: The aim of the study was to analyse the data and evaluate retrospectively the efficacy and safety of FMT for recurrent CDI. While working on the task we analysed retrospectively data of characteristics and outcomes of hospitalised patients from tertiary center who received FMT from 2015 to 2022. In all cases FMT was performed through the nasoenteric tube placed during gastroscopy. Frozen donor faces were used for

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FMT from unrelated healthy donors. To avoid effect of the single donor, FMT from 2-3 unrelated donors was used. Three days prior FMT vancomycin was applied and preparation with macrogol was administered according to European guidelines. Good response to 1 FMT was considered as improvement, more than 1 FMT as suboptimal response.

Results: FMT was performed for total of 98 patients, out of which 75 patients with recurrent CDI (23 patients needed 2 FMT, 6 - 3FMT and 2 -5FMT). The average age of patients was 68 years. 56 % (42) were women. 41 (54.7%) of patients previously used antibiotics, 2 (2.6%) steroids and 4 (5.3%) proton pump inhibitors (PPI). None of them was under immunosuppressive therapy. After the first FMT procedure clinical improvement was observed in 52 patients (69.3%). What is more 23 out of 30 patients initially non-responding patients had a clinical resolution after a second FMT. Whereas, the remaining 6 patients needed 3 FMT and 1 needed 5 FMT. Though, patients who required more than 1 FMT comorbidity had any cardiovascular disease (CV) such as cardiac insufficiency, diabetes mellitus (DM) or inflammatory bowel disease (IBD). Odds of suboptimal effect of therapy (need of more than 1 FMT) was associated with prior use of metronidazole then vancomycin or fidaxomicin surprisingly. According to our data there was only one irrelevant adverse event documented (abdominal pain). There was no death attributed to CDI in the studied population.

Conclusion: In our study, FMT was effective and safe treatment of recurrent CDI in our population. Nevertheless, at the same time we have identified risk factors of suboptimal response to FMT in our cohort such as CV disease, DM and IBD, as well as use of metronidazole.

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Disclosure: Nothing to disclose.

PP0698

PSEUDOMEMBRANOUS COLITIS AS A RISK FACTOR FOR RECURRENT *CLOSTRIDIOIDES DIFFICILE* INFECTION

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Introduction: Clostridioides difficile infection (CDI) is one of the most common causes of nosocomial diarrhoea with high morbidity and mortality. Recurrent and refractory courses in particular represent a special challenge in the management of these patients. The identification of risk factors that lead to recurrent or complicated courses could improve management, e.g. through the use of new and effective therapies in patients with CDI.

Aims & Methods: The aim of this study was to investigate the relationship between pseudomembranous colitis (PMC) and the course of CDI.

In a retrospective, monocentric cohort study of a tertiary centre over a period of 10 years, patients with CDI-related hospitalisations were examined. The study included 336 patients with available information on PMC and CDI history. The patients were systematically evaluated according to concomitant diseases and co-medication. In addition, the presence of ribotype 027 (RT027) in the sub-cohort was investigated using PCR genotyping.

Results: Of the 336 patients included in the study, 210 (62.5%) had an endoscopic evaluation. Of these 210 patients with CDI, 102 (48.57%) had PMC and 108 (51.43%) had no PMC. Patients with endoscopically detected PMC were more likely to have recurrences (OR 3.7 (95%CI 2.05-6.75). The presence of PMC was associated with older age (71.4 ± 13.41 years PMC+ vs. 61.78 ± 19.42 years PMC-). In contrast, patients with chronic inflammatory bowel disease and patients under immunosuppression were less likely to have PMC (4.90% PMC+ vs. 18.52% PMC- and 16.67% PMC+ vs. 31.48% PMC-, respectively). In patients with PMC, the RT027 ribotype was detected more frequently (44.74% PMC+ vs. 20.00% PMC-). Gender, the other comorbidities analyzed and the use of PPIs played no role in the development of PMC.

Conclusion: Endoscopic detection of PMC is associated with an increased risk of recurrent CDI. Implementation of early assessment of PMC in patients with CDI could potentially lead to improved risk assessment for recurrence.

Disclosure: Nothing to disclose.

PP0699

THE PREDICTIVE FACTORS CONTRIBUTING TO THE RECURRENCE OF *CLOSTRIDIUM DIFFICILE* INFECTION

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Introduction: *Clostridium difficile* infection (CDI) presents a complex challenge in healthcare, particularly within hospitals and long-term care facilities where the use of antibiotics is usually unavoidable. Understanding the factors that contribute to CDI recurrence is crucial for effective treatment and management.

Aims & Methods: This study aimed to assess possible risk factors contributing to the recurrence of CDI. We performed a retrospective analysis of 107 adults diagnosed with CDI who were managed in the tertiary care center during 2018-2020, focusing on variables including demographics, comorbidities, clinical features, history of prior antibiotic and glucocorticoid treatments.

Results: Overall, this study included 63 (58.9%) females, with a median age for all patients of 77 years (IQR 68–82). The cohort comprised a diverse array of comorbid conditions: inflammatory bowel disease (3.7%), diabetes mellitus (10.3%), chronic kidney disease (CKD) (15.9%), with 4 (23.5%) patients requiring haemodialysis, and oncological diseases (21.5%). Obesity/overweight was noted in 28 (26.2%) patients, and a history of stroke was present in 14 (13.1%). Nursing care was required for 41 (38.3%) individuals, while dementia symptoms were reported in 31 (29 %) cases.

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There was a common history of prior hospitalization (65.4%), glucocorticoids were used by 14 (13.1%) patients, proton pump inhibitors (PPI) by 23 (21.5%) subjects, and 18 (16.8%) patients had surgeries within 6 months before hospitalization. Antibiotics were administered to 48.6% of patients before the onset of CDI.

During hospitalization, 4 (3.7%) patients died: 3 of whom had oncological history, 2 of them worsened after chemotherapy, and other 2 patients experienced progressive heart failure. Comparative analysis between patients with a first-time CDI diagnosis (n = 39), who all remained recurrence-free over a 3-year follow-up, and those with recurrences (n = 64), revealed that recurrence was more common in females (73.3% vs. 46.5%, p=0.006), while no significant age differences were observed. Neither surgical history (p = 0.974) nor PPI usage (p = 0.232) demonstrated a significant influence on CDI recurrence. Moreover, diabetes, oncological history, and CKD did not significantly impact CDI recurrence rates (p = 0.405, p = 0.078, and p = 0.974) either. In contrast, a significant correlation was found between dementia and increased CDI recurrence: 83.3% of patients with dementia experienced recurrence versus 53.4% without dementia (p = 0.004). While patients with a history of stroke and those requiring nursing care had higher rates of recurrence, these trends were not statistically significant (p = 0.533 and p = 0.065, respectively).

Conclusion: While many clinical and demographic factors were assessed, only female sex and dementia showed a significant influence on CDI recurrence. These findings suggest that among the array of potential risk factors, cognitive impairment presents a unique challenge in the management and prevention of CDI recurrence.

Disclosure: Nothing to disclose.

PP0700

UTILITY OF MYCOBACTERIA GROWTH INDICATOR TUBE (MGIT) CULTURE AND XPERT MTB RIF ASSAY IN DIAGNOSING ABDOMINAL TUBERCULOSIS (ATB)

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Introduction: Diagnosis of abdominal tuberculosis relies on microbiological tests to detect Mycobacterium tuberculosis (MTB) and histopathology. Newer microbiological tests like MGIT (culture test for MTB) and Xpert MTB Rif assay (PCR test for MTB) have an established role in diagnosing pulmonary tuberculosis (TB). This study aimed at assessing the diagnostic accuracy of these tests in abdominal tuberculosis (ATB).

Aims & Methods: We aimed to study the utility of MGIT culture and gene Xpert MTB Rif assay in diagnosing Abdominal Tuberculosis (ATB).

We conducted a retrospective study among patients diagnosed with Abdominal Tuberculosis (ATB) in 2018. The clinical and investigation details were recorded on a data sheet. This included the nature and duration of symptoms and the site of the lesion. The sample of extra-intestinal lesions was acquired under radiological guidance and sent for histopathology, MGIT culture, and Xpert test. The intestinal lesions were detected on colonoscopy or enteroscopy, and the sample was sent for histopathology and microbiological tests - MGIT & Xpert. The final diagnosis of ATB was done using composite reference standard based on the presence of histopathology findings or positive culture or the presence of diagnosed extra-intestinal tuberculosis or response to anti-tuberculosis treatment (ATT). Diagnostic yield of MGIT and Xpert tests were calculated.

Results: A total of 68 patients were studied. Forty patients had intestinal tuberculosis (ITB) and 28 patients had extra-intestinal Abdominal tuberculosis (EIATB).

In patients with ITB, mean age was 37.1±13.4 years, and 30 (70%) were males. The median duration of symptoms was four months (IQR, 2-11.3 months). Abdominal pain (80%) was the commonest symptom. Fever was noted in 37.5% of cases and diarrhea in 17.5%. MGIT culture was positive in 5(12.5%) cases. Xpert test was positive in two of the five cases with positive MGIT culture report. Another patient had Xpert positive, but MGIT culture was negative. The sensitivity of MGIT culture was 12.5%, and Xpert was 7.5%. The combined sensitivity of the two tests was 15%.

In patients with EIATB, mean age was 47.8±13.6 years and 16(57.1%) were males. Fever was present in 60.7% of cases, and abdominal pain was present in 35.7% of cases. Tissue sample for diagnosis of TB was obtained from omentum in 16(57.1%) intra-abdominal lymph node in 7(25%) and other sites in 5(17.9%) patients. Histopathology supported a diagnosis of TB in 67.9% of cases. Xpert test was positive in eight (28.6%) cases. MGIT culture was positive in 3(10.7%), of which two also had a positive Xpert test. The combined sensitivity of the two tests was 32.1%. Six of the seven intra-abdominal lymph node tissue had a positive Xpert test, while none of the omental tissue samples showed a positive result for this test.

Overall, XPERT was positive in 13 cases with a sensitivity of 19.1%. MGIT was positive in 8 cases with a sensitivity of 11.7%. The combined sensitivity of both tests was 28.6%.

Conclusion: Newer microbiological tests (MGIT, Xpert) help in the diagnosis of about one-sixth of cases with ITB and in about one-third of cases of EIATB. The faster availability of results compared to conventional culture is the key advantage, but due to the paucibacillary nature of ATB, its yield is quite low.

Disclosure: Nothing to disclose.

PP0701

ENDOSCOPICALLY-DELIVERED PURASTAT FOR THE TREATMENT OF HAEMORRHAGIC RADIATION PROCTOPATHY IS SAFE, FEASIBLE AND WELL TOLERATED: RESULTS FROM A RANDOMISED FEASIBILITY STUDY

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Introduction: Severe bleeding from radiation proctopathy (RP) affects around 6% of people treated with radiotherapy for pelvic cancer[1]. Purastat, a haemostatic agent licensed to treat bleeding from blood vessels in the gut, has been shown to have haemostatic benefit during surgery and endoscopic procedures[2-6]. Early case series data for patients with RP have shown improvements in rectal bleeding and endoscopic appearance with no adverse effects[7], however the effects compared to treatment as usual (TAU, sucralfate enemas) have not previously been studied.

Aims & Methods: In this first clinical trial of Purastat for RP, we investigated the feasibility, acceptability and safety outcomes in comparison to standard treatment (ISRCTN NCT04918758).

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Adults with significant bleeding from endoscopically confirmed RP recruited from 3 centres (Manchester University NHS Foundation Trust, Royal Liverpool University Hospital, St Mark's Hospital) were randomised to Purastat (5mls at baseline, week 4 and week 8) or TAU (sucralfate enemas 2g BD for 8 weeks starting at baseline visit).

Significant bleeding was defined as bleeding on an at least weekly basis. Patients with other causes of rectal bleeding, or who had previously been treated with sucralfate enemas or Purastat, were excluded.

Clinical and endoscopic assessments were made at baseline, week 4, week 8 and week 20. Episodes of rectal bleeding were assessed using a patient reported diary 7 days before and after each assessment. Endoscopic grade was recorded at each assessment using the Zinicola score, Rectal Telangectasia Density Score (RTDS) and Vienna Rectoscopy Score. Participant satisfaction was assessed using Client Satisfaction Questionnaire-8 (CSQ8).

Results: 143 patients were screened over 25 months. 26 patients were consented (18%) and 25 started the study. 13 were randomised to the Purastat group. 20 participants completed the study. 5 participants (20%) withdrew (4 TAU, 1 Purastat).

Baseline demographic and clinical outcome data are shown in Table 1.

There were no adverse events relating to Purastat or TAU.

There was an improvement in rectal bleeding episodes, endoscopic grade and number of participants with anaemia in both groups. There is a possible trend towards better outcomes in the Purastat group, but as this is a small feasibility study it is not powered to detected differences between groups.

All participants reported being "very satisfied" or "mostly satisfied" overall (CSQ8).

		Purastat (n=13)	**Sucralfate** (n=12)
Age (years) [median (range)]		69 (56 to 77)	74.5 (46 to 81)
Gender	**Female**	2 (15%)	3 (25%)
	Male	11 (85%)	9 (75%)
Tumour site [n (%)]	**Bladder**	1 (8%)	1 (8%)
	Cervix	0	2 (17%)
	Prostate	10 (77%)	8 (67%)
	Uterine	0	1 (8%)
	Vaginal	2 (15%)	0
Anti-coagulant/ Anti-platelet therapy [n (%)]	**Warfarin**	0	1 (8%)
	Apixaban	1 (8%)	0
	Aspirin	0	2 (17%)
Previous treatments for RP [n (%)]	**Radiofrequency Ablation**	0	1 (8%)
	Argon Plasma Coagulation	2 (15%)	0
	Rectal Steroids	0	3 (25%)
Total no. of patient reported bleeding episodes [median (IQR);range]	**Prior to Baseline**	12 (5, 21); 0 to 34 (n=13)	9 (4, 12); 1 to 50 (n=11)
	Post Week 8	6 (0, 11); 0 to 49 (n=10)	5 (3, 12); 1 to 35 (n=7)
	Prior to Week 20	6 (3, 13); 0 to 23 (n=11)	8 (4.5, 11); 2 to 26 (n=8)
Modal endoscopic grade [Zinicola score; RTDS; Vienna telangectasia density score]	**Baseline**	C; 3; 3	B,C; 3; 3
	Week 8	A; 2; 2,3	B; 2; 2,3
	Week 20	A; 2; 2	B; 2,3; 2
No. with anaemia [n (%)]	**Baseline**	7 (54%, n=13)	3 (30%, n=10)
	Week 8	6 (50%, n=12)	1 (13%, n=8)
	Week 20	4 (17%, n=12)	2 (29%, n=7)
Haemoglobin concentration (g/L) [mean (range)]	**Baseline**	126.3 (80-154)	134.4 (112-159)
	Week 8	128.5 (88-154)	136.8 (126-154)
	Week 20	128.7 (85-153)	141.9 (125-165)

Table 1: Baseline demographic and clinical outcome data

Conclusion: This study shows that the study process and intervention is safe and acceptable. Clinical and endoscopic improvement was seen in both groups. A definitive national, multicentred RCT is required to test the clinical and cost effectiveness of Purastat compared with TAU.

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PP0702

CHARACTERISTICS OF POST-EMR COLORECTAL BLEEDING IN THE ELDERLY: FOCUSING ON THE NEED FOR ENDOSCOPIC HEMOSTASIS

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Introduction: Bleeding after colorectal endoscopic mucosal resection (EMR) is a relatively frequent complication. Some patients require hemostasis, and some patients can be monitored. Appropriate judgment of which cases are amenable to follow-up is beneficial in avoiding unnecessary invasion, especially in the elderly who are highly complicated. In this study, we aimed to clarify the characteristics of these cases while focusing on age-related factors.

Aims & Methods: Of 2099 colorectal EMR patients with 5744 lesions performed at our hospital between January 2007 and December 2019, 59 patients (M:F=43:16) and 154 lesions in the suspected delayed bleeding (DB) group with hematochezia (including small amounts of blood) or a decrease in Hb of 2 g/dL or more the day after EMR were included in the analysis. The DB group was divided into hemostasis group (H group; endoscopic hemostasis was performed) and no hemostasis group (no-H group; no endoscopy or endoscopy without hemostasis was performed), and the characteristics of patients who could be without hemostasis were identified by comparing patient and lesion's backgrounds. Elderly defined as 75 years and older.

Results: 17 patients (28.8%) in the suspected DB group were elderly, and the mean age of the elderly/nonelderly was 78 (75-92)/ 63 (40-74). Of these, (7 of 17 and 5 of 42 were on antiplatelet drugs, significantly more in the elderly (82.3% vs. 33.3%, $p < 0.05$). The H/ no-H group was 39/20. The 20 patients in the no-H group consisted of 8 elderly and 12 non-elderly. The elderly were significantly more amenable to no hemostasis than the non-elderly (47.1% vs 28.5%. $p < 0.05$). Mean lesion diameter (mm) and polypoid (Is, Isp, Ip) were 10 (4-30) /6 (1-50), 36 (78.2%)/33 (82.5%), respec-

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tively, with no significant difference. In univariate analysis, the factors that tended to contribute to no-hemostasis were antiplatelet medication (Odds Ratio (OR) 3.66, 95% Confidence Interval (CI) 0.98-13.62, p=0.053) and 75 years and older (OR 3.17, CI 0.979-10.264, p=0.054).

Conclusion: Many cases of suspected DB in the elderly are amenable to follow-up. It was suggested that antiplatelet drug use leads to small amount of hematochezia that no hemostasis was required. This characteristic should be taken into account when making decisions regarding appropriate endoscopic hemostasis after suspected DB in the elderly.

Disclosure: Nothing to disclose.

PP0703

SIGNIFICANCE OF HB DECREASE IN DELAYED BLEEDING AFTER COLORECTAL ESD

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Introduction: Delayed bleeding (DB) is a major complication of colorectal endoscopic submucosal dissection (ESD). There are several definitions of DB, and one of the main ones is "cases requiring hemostasis with hematochezia or hemoglobin (Hb) decrease ≥2 g/dL as compared to the last preoperative level" [1]. However, it is not clear whether or not cases with Hb decrease actually requires endoscopic hemostasis, and this is sometimes difficult to determine in actual clinical practice.

Aims & Methods: We defined DB as a case requiring hemostasis with hematochezia or a Hb decrease ≥2 g/dL after colorectal ESD. The purpose of this study was to clarify the significance of Hb decrease in DB after colorectal ESD. Of the 595 patients (615 tumors) who underwent ESD or hybrid ESD for colorectal tumors at our hospital between June 2009 and April 2022, 145 patients (145 tumors) with suspected DB (suspected DB; with hematochezia or a Hb decrease ≥2 g/dL) were analyzed. We compared the hemostasis group (H group; hemostasis performed) and the no-hemostasis group (no-H group; no endoscopy performed or endoscopy performed but no hemostasis performed after hematochezia or Hb decrease) in a retrospective study.

Results: Thirty-two patients with 32 tumors were categorized into the H group, while 113 patients with 113 tumors were in the no-H group. Hematochezia was significantly higher in the H group than in the no-H group (100% vs 83.2%, p = 0.008, Fisher's exact test).

On the other hand, the incidence of Hb decrease on postoperative day (POD) 1 was significantly higher in the no-H group than in the H group (26.5% vs 0%, p <0.001, Fisher's exact test). There were 7 cases in the 145 cases who had both hematochezia and a Hb decrease on POD 1, and every 7 cases were in the no-H group. Among the H group, only 8 cases (25%) had a Hb decrease ≥2 g/dL on the day of hemostasis compared to the preoperative level.

On multivariate logistic regression analysis, date from ESD to first hematochezia ≤Day3 (OR 3.57, 95%CI 1.46–8.74; p = 0.005) was identified as the significant factor contributing to the no-H group, followed by bleeding duration ≤1 day (OR 4.58, 95%CI 1.45–14.43; p = 0.009).

Conclusion: The duration and the first date of hematochezia seemed to be important in determining whether to perform hemostasis when DB was suspected. On the other hand, there were no cases in which a Hb decrease ≥2 g/dL on POD 1 required hemostasis. This suggests that a Hb decrease may be a low priority factor in DB after colorectal ESD.

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Disclosure: Nothing to disclose.

PP0704

UNRAVELING THE CLOTTING CONNECTION: D-DIMER LEVELS AS PREDICTORS OF MESENTERIC ISCHEMIA

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Introduction: This study investigates the potential link between elevated D-dimer levels and the development of mesenteric ischemia, a severe vascular condition. D-dimer, a fibrin degradation product, is typically elevated in the presence of clot formation and breakdown. Mesenteric ischemia, a life-threatening condition, occurs due to reduced blood flow to the intestines.

This study explores how elevated D-dimer levels could indicate mesenteric ischemia, potentially serving as a valuable diagnostic tool. Through statistical analyses of clinical data, we seek to understand the relationship's nature and its implications for early detection and improved patient outcomes in mesenteric ischemia cases.

Aims & Methods: Our retrospective study, spanning 2009-2023, included all Mesenteric ischemic patients from the Nationwide Inpatient Sample (NIS) database. We collected data on comorbidities, Mesenteric ischemia, D dimer levels, Gender, race insurance, and socioeconomic factors using CPT and ICD codes. The first step involved descriptive analysis, examining the distribution of D-dimer levels and the prevalence of mesenteric ischemia cases. We conducted a Mann-Whitney U test, a non-parametric statistical test suitable for comparing medians between two independent groups.

Spearman's rank correlation coefficient was used to assess the strength and direction of the association between D-dimer levels and mesenteric ischemia. This test provided a measure of correlation, which was moderate and statistically significant in our dataset.

Results: A total of 7854 patients diagnosed with Mesenteric ischemia were included in the study. 4398 (54%) were female. The average age in the study is 58±7 years. The average BMI in the study group was 31± 10. Among the races, The Mann-Whitney U test (10.5 p<0.01) showed a higher D dimer level was associated with Mesenteric ischemia Similarly, the Spearman correlation coefficient of approximately 0.260 (p<0.05) demonstrates a moderate positive correlation between D-dimer levels and mesenteric ischemia, further supporting this association.

Conclusion: The results of our study reveal a noteworthy association between elevated D-dimer levels and mesenteric ischemia. The Mann-Whitney U test indicated significantly higher D-dimer levels in patients with mesenteric ischemia, reinforcing the potential of D-dimer as a marker for this condition.

Furthermore, the Spearman correlation's moderate positive coefficient strengthens the evidence for this link. These findings suggest that monitoring D-dimer levels could aid in the early identification of mesenteric ischemia, potentially improving patient outcomes through timely intervention.

However, the study highlights the need for cautious interpretation and consideration of other influencing factors in clinical settings.

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Disclosure: Nothing to disclose.

PP0705

CLINICAL AND ENDOSCOPIC FEATURES OF SOLITARY RECTAL ULCER SYNDROME PRESENTED TO TERTIARY CARE HOSPITAL: MULTICENTER STUDY IN KARACHI, PAKISTAN

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Introduction: Solitary rectal ulcer syndrome (SRUS) is a chronic benign disorder characterized by the presence of an abnormality of the rectum in persons who have a long history of straining during defecation. It is a misnomer as only 3rd of the patients have solitary ulcer, many have no ulcer.
Aims & Methods: This was a retrospective observational longitudinal study, conducted in the department of Gastroenterology at Jinnah Post graduate Medical Center and National Medical Center, Karachi, from 1st March 2017 to till to date . All adult patients of either gender diagnosed with SRUS on basis of clinical history and endoscopic features were included in the study. The data obtained was analyzed on the statistical software SPSS version 26.Descriptive statistics were obtained by frequencies and percentages.
Results: There were total 85 patients in our study out of which 49(58%) were female and 36(42%) were male. Mean age was 35 ±17 years. Most common presentation was per rectal bleed 68(80%) ,Constipation 58 (68.57%) and followed by abdominal pain 32(37.9%).The most common predisposing risk factor was 85 (100%) self-digitation, 51(60%) prolong stay in toilet and rectal prolapse 5 (7.1%). Endoscopically most common finding was of single ulcer 80(95.7%), followed by proctitis 59(70%) and multiple ulcers 12(14.3%) and Histology revealed mucosal ulceration 75(88%) extension of muscular fibers 65 (77%), fibromuscular obliteration of lamina propria 59(70%) and crypt distortion 54 (64.3%). Associated psychological condition among patients were obsessive compulsive disorder 51 (60%) and anxiety disorder 46(52%). Most of the patients were treated through lifestyle modification and biofeedback. Only 4 patients were referred to surgical intervention.
Conclusion: SRUS is a rare disorder with variable clinical presentation. In this study patients commonly presented with bleeding p/r followed by abdominal pain. However, common predisposing factor was self -digitation, in this regard awareness is needed in our region.
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PP0706

PROSPECTIVE, MULTI-CENTER, COMPARISON OF RISK SCORES FOR THE PREDICTION OF SAFE DISCHARGE FROM THE EMERGENCY DEPARTMENT IN PATIENTS WITH LOWER GASTROINTESTINAL BLEEDING

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Introduction: Lower gastrointestinal bleeding (LGIB) is usually self-limited, although serious bleeding requiring blood transfusion, surgery, and death can occur in a minority of individuals. Risk scores can differentiate between these patients and identify who can be safely discharged from the emergency department (ED), although which score should be used is unknown.
Aims & Methods: We conducted a prospective cohort study at 8 hospitals across Canada of patients presenting with hematochezia to the ED. Individuals 18 years or older were eligible regardless if they were admitted or discharged from the ED. LGIB risk scores, informed by a prior systematic review and meta-analysis[1], were calculated at presentation for the Oakland, Strate, and BLEED scores.
Two commonly used upper gastrointestinal bleeding risk scores were also included: the Glasgow-Blatchford score (GBS) and the clinical Rockall score (cRockall). All subjects were followed for 28 days. The primary outcome was safe discharge, defined as the absence of rebleeding, readmission, need for blood transfusion, endoscopic hemostasis, radiologic hemostasis, surgery, and death at 28 days.
The performance of each score was evaluated for its overall discriminative ability (c-statistic) in predicting safe discharge. Calibrations were assessed using calibration belt plots as well as grouped calibration plots. The c-statistics of LGIB and UGIB risk scores were separately compared and adjusted for multiplicity with a focus on comparison to the Oakland score. Sensitivities and specificities were determined for each score.
Results: Between Sep 2019 and Feb 2024, a total of 529 subjects were screened, of whom 344 were enrolled. The mean (SD) age was 59.5 years (19.6), 48.5% were female, and 27.4% had a Charlson Comorbidity Index score ≥3. Overall, 39.7% of patients were admitted to hospital. At 28 days

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follow-up, 4.6% were readmitted, 10.7% needed blood transfusions, 6.2% had endoscopic hemostasis, 0.3% had radiological hemostasis, 3.3% had surgery, and 1.3% died.

Overall, 79.3% had a composite endpoint of safe discharge within 28 days. Among LGIB risk scores, the AUC (95% CI) was 0.71 (0.63-0.78) for the Oakland score, 0.67 (95% CI 0.57-0.75) for the Strate score, and 0.61 (95% CI 0.53-0.68) for the BLEED score. For UGIB risk scores, the AUC (95% CI) was 0.76 (0.67-0.84) for the GBS and 0.62 (0.54-0.70) for the cRockall score. The sensitivity and specificity for each score were as follows: 23.1% and 92.1% for the Oakland score; 16.7% and 91.7% for the Strate score; 47.4% and 77.8% for the BLEED score; 50% and 88.1% for the GBS; 25.8% and 88.8% for the cRockall, respectively.

Conclusion: The Oakland score can predict safe discharge with a high degree of specificity, but clinical judgment should still be used. If a single risk score for both UGIB and LGIB is desired, then the GBS may be a good alternative although further validation is required.

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Disclosure: HS has been on advisory boards or consulted to Pendopharm, Amgen Canada, Abbvie Canada, Innomar, Sandoz Canada, Takeda Canada, and Guardant Health, Inc. None of this is related to the current work. All other co-authors have no relevant disclosures related to this study.

PP0707

HAEMOCER PLUS IN THE TREATMENT AND PREVENTION OF LOWER GI POST-RESECTIONAL BLEEDING: PROSPECTIVE MULTICENTER REGISTRY

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Introduction: GI bleeding associated to endoscopic procedure is defined as clinical evidence of bleeding and a drop in hemoglobin of ≥ 2g/dL on the day of the procedure (early bleeding) or up to 14 days after the procedure (delayed bleeding). GI bleeding is a common complication of endoscopic procedures, such as endoscopic mucosal resection (EMR) and endoscopic sub mucosal dissection (ESD). Endoscopic treatments encompass injection therapy (epinephrine, sclerosing agents), mechanical therapy (hemoclipsplacement), and thermaltherapy (monopolar and bipolar coagulation, argonplasma coagulation, or heat probe).

New endoscopic hemostasis modalities (topical hemostatic agents) are emerging as possible alternative endotherapies for primary hemostasis when bleeding is refractory or not amenable to standard endoscopic hemostasis therapies.

Aims & Methods: We aim to establish a multicenter, observational registry to collect data related to the use of HaemoCer PLUS for the primary prevention of delayed bleeding linked to endoscopic resectional procedures of the lower GI tract. Primary aim: prevention of delayed bleeding after colonic EMR or ESD larger than 30 mm. Secondary aim: evaluation of possible adverse events (AEs) related to the application of the powder.

We enrolled all adults patients undergoing colorectal ESD or EMR for lesions bigger than 30 mm where HaemoCer PLUS has been used will be included.

Results: Preliminary results of first forthytwo enrolled patients: 23 male; mean age 75.2 years old (51-88). Mean diameter of lesions was 63 mm (35-120 mm). During procedure only four major bleeding occurred treated with hemostatic forceps. No mechanical therapy was applied. In all patients hemostatic powder was applied when there was no visible bleeding. Four patients underwent procedure with ongoing antiplatelet therapy.

We encountered only two bleeding, in day 1 post-procedure. Of these, one patient who undergone procedure with ongoing antiplatelet and anticoagulant therapies, due to cardiovascular and hepatic comorbidities; the other had no ongoing or past therapy.

Conclusion: Even if this is a preliminary results report, this hemostatic powder seems to be a valuable additional therapy to prevent bleeding in patients with large colonic endoscopic resection with ongoing antiplatelet therapy. Combined antiplatelet and anticoagulant therapy, conjunct to several comorbidities still remain a difficult to manage situation in daily practice.

Disclosure: Nothing to disclose.

PP0708

RECURRENT ISCHEMIC COLITIS: WHAT RISK FACTORS?

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Introduction: Ischemic colitis (IC) is the most common form of intestinal ischemia, arising from inadequate blood circulation in the colon. Recurrent ischemic colitis has not been extensively studied.

Aims & Methods: Our study aimed to identify the main risk factors for recurrent ischemic colitis.

An observational, cross-sectional, and descriptive study included all patients diagnosed with IC at our Hospital Center from January 2011 to December 2021. Clinical records of identified patients were analyzed, collecting demographic, clinical, imaging, endoscopic, and histological data, as well as hospitalization duration and mortality. Patients were divided into two groups - recurrent IC and non-recurrent IC - for characterization and statistical analysis.

Results: A total of 142 patients with IC were identified, with recurrence observed in 11 patients (7.7%) during the study period. Smoking was more prevalent in the recurrent IC group compared to the non-recurrent IC group (45.5% vs. 13%; p=0.014). Constipation was more frequent in the recurrent IC group (40.0% vs. 4.7%; p=0.003). Laparotomy was more common in patients with recurrent IC compared to those with non-recurrent IC (27.3% vs. 6.1%; p=0.041). The use of clopidogrel was more prevalent in the recurrent IC group (36.4% vs. 10%; p=0.029). Mean hemoglobin levels on hospital admission were lower in the recurrent IC group (11.07±3.67; 12.99±2.27; p=0.012). There was a higher need for blood transfusion in the recurrent IC group compared to the non-recurrent IC group (36.4%; 6.9%; p=0.01). Other findings did not show an association with recurrent IC.

Conclusion: Recurrent IC occurred in 7.7% of patients in the study. Smoking, constipation, laparotomy, and the use of clopidogrel were identified as possible risk factors for recurrent IC.

Disclosure: Nothing to disclose.

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PP0709

BRAZILIAN TREND IN ANAL CANCER MORTALITY - 2012 TO 2021

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Introduction: Anal cancer accounts for 2% of malignant neoplasms of the gastrointestinal tract. Most cases are squamous cell carcinoma, and the global incidence is increasing alongside the rise in Human Papillomavirus (HPV) incidence.

Individuals infected with HPV, particularly those co-infected with human immunodeficiency virus (HIV), as well as smokers, individuals with more than ten sexual partners, and those who maintain receptive anal intercourse are at increased risk of developing this cancer.

Aims & Methods: Our objective was to analyze the trends in anal cancer mortality rates in Brazil from 2012 to 2021, with a focus on gender and age. This descriptive study of anal cancer deaths (ICD-10 code C21) in Brazil utilized data from the Mortality Information System of the Unified Health System (DATASUS)

Results: In Brazil, from 2012 to 2021, there were a total of 5,955 deaths attributed to anal cancer, with 3,600 deaths in women and 2,355 deaths in men, and most of these deaths occurred after the age of 50. During this period, anal cancer mortality rates increased in both sexes. In 2012, the mortality rate was 0.22 deaths per 100,000 women and 0.13 deaths per 100,000 men, rising to 0.55 deaths per 100,000 women and 0.48 deaths per 100,000 men by 2021.

Furthermore, the average annual percent change (AAPC) in anal cancer mortality rates was higher in males than females across all age groups except for ages 30-39.

In males aged 40 and over, the AAPC of anal cancer mortality rates increased progressively with age, while in females, it remained stable across different age groups.

Notably, among individuals over 80 years old, the AAPC was 11% per year among women and 44% per year among men. As a result, the mortality ratio between women and men decreased from 1.88 (2012-2016) to 1.41 (2017-2021), reaching its lowest value of 1.14 in 2021.

Conclusion: Despite its rarity, anal cancer warrants recognition due to its rising mortality rates.

While initially, mortality rates from anal cancer were significantly higher among women, the accelerated increase in mortality rates among men has led to similar mortality rates between genders by the year 2021.

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PP0710

INCIDENCE OF POST-COLONOSCOPY COLORECTAL CARCINOMA AT THE UNIVERSITY OF SZEGED – RETROSPECTIVE COHORT STUDY

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Introduction: Colonoscopy and polypectomy reduce the incidence of colorectal carcinoma (CRC), however it does not entirely prevent the emergence of tumours. Post-colonoscopy CRC (PCCRC) is defined when a tumour is detected in a short follow-up period after an index colonoscopy, where no malignancies were found.

This period is hard to define however, as the World Endoscopy Organisation (WEO) does not have a unified decision on this subject, the most widespread period is a five-year long.

Aims & Methods: Our aims were to define the incidence of PCCRC at the University of Szeged, and to identify factors that could influence its detection. Patients diagnosed with CRC between 01.01.2020. and 31.12.2023 were involved. For PCCRC, we used 5-, 3- and 1-year cut-off point. We also investigated the indicators of quality colonoscopy as a predictor and assessed the adherence to guidelines regarding the management of the involved patients.

Results: Overall, 3900 CRCs were diagnosed during the study period, the ratio of PCCRC was 3.82% (N=149), 3.08 (N=120), and 1.13% (N=44) with 5-year, 3-year, and 1-year cut-off, respectively. Regarding PCCRC etiology, 64.5% (N=111) of the cases were missed lesions, 9.4% (N=14) were due to incomplete polypectomy and 16.11% (N=24) were de novo lesions. 40% (N=60) was localized on the right side of the colon, 33% (N=49) was on the left, 30% (N=45) in the rectum. Duplex tumours were detected in 5 cases. From the indicators of quality colonoscopy, coecal intubation was documented in 79% (N=117) of the cases, and adequate bowel preparation was achieved in 76% (N=113), both of which does not reach the required minimum value set by the guidelines. There was no substantial difference between the ratios of CRCs and missed lesion PCCRCs regarding lymph node involvement (grade III) and distant metastases (grade IV) (27.24% vs 23.42% and 26.30% vs 22.52%). Adherence to the adequate follow-up strategy stated by the guidelines currently in effect regarding the results of the index colonoscopy was 60.40% (N=90), negligence of the patients was responsible in 32.20% (N=19) in the non-adherent cases.

Conclusion: The incidence of PCCRC in our institution correlates with the data available worldwide. 75% of PCCRCs are missed lesions, and every third examination was carried out with insufficient bowel preparation and/or without coecal intubation and with inadequate follow-up strategy. The incidence of PCCRC can be lowered with quality colonoscopy and strict adherence to guidelines.

Disclosure: Nothing to disclose.

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PP0711

MONITORING OUTCOMES OF COLORECTAL CANCER SCREENING BY ETHNICITY COULD INFORM ALTERNATIVE STRATEGIES TO IMPROVE SCREENING EFFICACY: THE EXAMPLE OF THE NEW ZEALAND NATIONAL BOWEL SCREENING PROGRAM

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Introduction: After pilot studies, New Zealand implemented a national colorectal cancer (CRC) screening program in 2017 with a graduate roll out by district health board. The program invites individuals aged 60 to 74 years to perform a fecal immunochemical test (FIT), using a positivity cut-off of 40 microgram Hemoglobin per gram feces. While health disparities exist between Māori and non-Māori populations across various health domains, there has been no systematic assessment of these inequities in the context of CRC screening.

Aims & Methods: This study aimed to assess the differences on key performance indicators of the National Bowel Screening Program between Māori and non-Māori populations in New Zealand and inform policy to improve equity. In this population-based study, we used data from the national population register of invitees for CRC screening between 2017 and 2023, which contained outcomes of up to four screening rounds of the National Bowel Cancer Screening Program in New Zealand. Key performance indicators were the FIT and follow-up colonoscopy participation rate, FIT positivity rate, positive predictive value (PPV) and detection rate of advanced neoplasia (AN; CRC and advanced adenomas). Multivariate logistic regression analyses were used to assess the association between ethnicity and key performance indicators, adjusted for confounding factors such as age, sex and invitation round.

Results: A total of 1,171,205 individuals were invited for CRC screening between 2017 and 2023 for up to four screening rounds. Of these, 675,274 (57.7%) individuals participated (Table 1).

		Participation FIT, % (OR; 95%CI)	FIT positivity rate, % (OR; 95%CI)	Participation follow-up colonoscopy, % (OR; 95%CI)	PPV AN, % (OR; 95%CI)	DR AN, % (OR; 95%CI)
Total		57.7%	4.4%	81.0%	34.0%	1.20%
Ethnicity	European	62.0%	4.2%	81.2%	36.2%	1.23%
	Maori	48.0% (0.60; 0.60-0.61)	6.0% (1.52; 1.46-1.57)	80.4% (0.93; 0.84-1.13)	27.5% (0.67; 0.61-0.73)	1.33% (1.13; 1.04-1.22)
	Pacific	37.2% (0.37; 0.37-0.38)	5.0% (1.27; 1.19-1.35)	80.3% (0.97; 0.83-1.13)	22.0% (0.54; 0.45-0.63)	0.89% (0.78; 0.67-0.90)
	Asian	45.2% (0.51; 0.51-0.52)	3.9% (0.98; 0.94-1.03)	81.1% (1.02; 0.91-1.15)	26.4% (0.67; 0.60-0.76)	0.84% (0.73; 0.67-0.81)
	Missing	27.3% (0.21; 0.19-0.23)	11.5% (2.31; 1.79-2.97)	56.5% (0.27; 0.17-0.44)	41.0% (1.01; 0.53-1.91)	2.66% (1.46; 0.89-2.40)

Abbreviations: FIT, faecal immunochemical test; OR, odds ratio; CI, confidence interval; AN, advanced neoplasia = colorectal cancer + advanced adenomas; DR, detection rate.

Table 1 Multivariate logistic regression analysis: association between ethnicity and key performance indicators in the National Bowel Screening Program in New Zealand between 2017-2023 adjusted for sex-, age- and screening round.

Compared to European descent, the Māori, Pacific and Asian populations were less likely to participate in FIT screening, with odds ratios (OR) of 0.60, 0.37 and 0.51 respectively, while the Māori ((OR: 1.52) and Pacific (OR: 1.27) were more likely to have a positive FIT result compared to the European descent population. Participation to follow-up colonoscopy was not significantly different by ethnicity. Asian, Māori and Pacific people had a significantly lower PPV for AN compared to European descent (OR 0.67, 0.67, 0.54 respectively). The detection rate for AN was significantly higher for Māori individuals compared to the European descent population (OR 1.13), while being significantly lower for Pacific and Asian (0.78 and 0.73, respectively).

Conclusion: CRC screening outcomes in New Zealand differ significantly by ethnicity, in particular for the Māori and Pacific people, with Māori exhibiting the highest yield of CRC screening. These findings highlight the importance of increasing participation in this population subgroup. Given the disparities in screening yield, risk-based screening strategies tailored to ethnicity could also be considered.

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Disclosure: Nothing to disclose.

PP0712

WITHDRAWN

PP0713

SHORT-TERM COMBINED IMPACT OF COVID-19 PANDEMIC AND POPULATION-BASED SCREENING ON THE INCIDENCE AND CLINICAL CHARACTERISTICS OF COLORECTAL CANCER – RETROSPECTIVE COHORT STUDY

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Introduction: In Hungary, the population-based colorectal carcinoma (CRC) screening programme was introduced at the end of 2018, shortly before the start of the COVID-19 pandemic.

Aims & Methods: The aim of our retrospective cohort study was to evaluate the impact of two presumably opposing causal factors, COVID-19 pandemic and population-based screening, on the incidence and clinical characteristics of CRC.

All patients with newly diagnosed CRC from 2014 to 2023 were included and categorized into pre-pandemic (confirmed between 2014-2019), pandemic (confirmed in 2020-2021) and post-pandemic (confirmed in 2022-2023) groups.

The extended pre-pandemic period was required to assess the impact of the screening programme, further dividing it into screening-affected (2015, 2016, and 2019) and unaffected (2014, 2017, 2018) subgroups.

Results: A total of 2803 patients were diagnosed with CRC during the study period. The annual number of CRC cases was significantly lower in the pandemic subgroup compared to the pre-pandemic screened and post-pandemic subgroups (254/year, 295/year and 295/year, p=0.009). In addition, in the pandemic subgroup, the incidence rates of CRCs detected at localized stage were significantly lower compared to the other three

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groups (36.61%, 50.85%, 46.35% and 49.49%, p<0.001).
Statistically significant difference in the rate of T1 tumors was only observed between the pandemic and the post-pandemic subgroups (7.09% and 12.20%, p=0.026). In the pandemic period, the annual number of colonoscopies decreased by 64.31% compared to the pre-pandemic period overall, however, the post-pandemic period (n=4574) saw a modest 8.70% increase relative to the pre-pandemic period, offering only partial compensation for the shortage experienced during the pandemic.
Furthermore, the annual number of colonoscopies in the extended pre-pandemic subgroup with screening was higher by 20.06% compared to the pre-pandemic subgroup without screening (4591 examinations/year vs 3824 examinations/year) and it was almost the same as the post-pandemic subgroup (4574 examinations/year).
Substantially fewer examinations were required to detect a CRC in the pandemic group compared to the pre-pandemic overall and post-pandemic groups (10.67, 15.56, 13.97 and 15.50 examinations, respectively), but within the extended pre-pandemic group, there was no relevant difference between the two subgroups in the mean number of colonoscopies required to detect a CRC (15.56 vs. 13.97).
Conclusion: Our study confirmed the negative impact of COVID-19 pandemic on the CRC epidemiology. The newly introduced population-based CRC screening has only been able to partially compensate this negative effect during pandemic period, however, was moderately compensated by during the pandemic period, but in the post-pandemic period it caused a favorable change in the clinical features of CRC. Our study demonstrates both the effectiveness of the screening programme and that by introducing non-invasive pre-screening of patients (e.g. iFOBT) in a pandemic situation, it can reduce its negative public health impact.
Disclosure: Nothing to disclose.

PP0714

WITHDRAWN

PP0715

WITHDRAWN

PP0716

WITHDRAWN

PP0717

PEPTIDE TYROSINE TYROSINE IS INCREASED IN INDIVIDUALS WITH BILE ACID DIARRHOEA

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Introduction: Bile acid diarrhoea (BAD) is a socially debilitating disease characterized by chronic and watery diarrhoea due to excess bile reaching the colon resulting in increased water secretion and motility. Since bile acids may stimulate the release of gut hormones, including peptide tyrosine tyrosine (PYY) implicated in diarrhoeal pathophysiology, we investigated fasting as well as postprandial plasma levels of PYY in BAD and healthy controls.
Aims & Methods: Total PYY concentrations were measured in plasma samples from 12 individuals with colesevelam-treated idiopathic or post-cholecystectomy BAD and 12 healthy controls matched on age, sex and body mass index undergoing two 4-hour liquid mixed meal tests on separate days with either the bile acid sequestrant colesevelam (1,875 mg) or placebo performed in a randomized and double-blinded order. Fasting PYY concentrations, area under the curve (AUC) and baseline-subtracted AUC (bsAUC), respectively, were evaluated.
Results: When comparing the two groups on placebo days, individuals with BAD exhibited higher mean fasting PYY concentrations (95.8 [SEM 12.6] pg/ml vs. 51.5 [12.6] pg/ml, *P*=0.0095) and AUC (27.1 [3.0] ng/ml × min vs. 16.5 [2.4] ng/ml × min, *P*=0.0087) whereas bsAUC was similar between groups (4.1 [1.9] ng/ml × min vs. 4.1 [1.6] ng/ml × min, *P*=0.99). Compared to placebo, colesevelam did not significantly affect PYY levels postprandially in any of the groups.
Conclusion: Here, we demonstrate elevated fasting PYY levels, but normal postprandial PYY responses in individuals with BAD compared to carefully matched healthy controls supporting the notion that bile acids in the gut lumen stimulates the secretion of PYY in individuals with BAD. Whether and how elevated PYY levels contribute to or might counteract BAD pathophysiology remains uncertain.
Disclosure: Filip K. Knop is currently employed by Novo Nordisk A/S, Søborg, Denmark. The remaining authors report no potential conflicts of interest relevant to this abstract.

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PP0718

DELTEX E3 UBIQUITIN LIGASE 2 ASSOCIATED WITH DNA DAMAGE REPAIR IN COLORECTAL CANCER

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Introduction: Different E3 ligases recognize specific substrates and thus can participate in many different cellular processes, such as proliferation, apoptosis, DNA damage repair, immunity, and metabolism, thus playing different roles in the development and progression of tumors. Deltex E3 ubiquitin ligase 2 (DTX2) has been recognized as an E3 ubiquitination ligase and has two main transcripts. However, the role and mechanism of DTX2 in colorectal cancer(CRC) is unclear.

Aims & Methods: The tissue of CRC patients and paired normal tissue (3 cm away from the tumor) were subjected to a quantitative real-time polymerase chain reaction (qRT-PCR) and Western blot (WB). Immunohistochemistry (IHC) was also used to detect the expression of DTX2 proteins in CRC tissues. CRC cell lines HCT-116, SW620, RKO, HCT15, SW480, 174T and human colonic epithelial cells (HCoEpiC) were cultured, and the expression levels of DTX2 mRNA and protein in these cells were compared. The potential substrate proteins of DTX2 in CRC cells were investigated by IP-MS experiment.

Full-length human DTX2 and short human DTX2 were amplified from total RNA by RT-PCR. Full-length and short transcript overexpression plasmids and empty vectors were constructed and transfected into SW480 cells. WB was used to detect the expression level of DTX2 and proteins associated with the DNA damage repair (DDR) pathway.

Results: The results of qRT-PCR and WB showed that the expression level of DTX2 mRNA and protein in CRC tissues was significantly higher than in normal tissues ($p < 0.05$). In addition, the IHC showed that DTX2 was mainly localized in CRC tissue cells. qRT-PCR and WB confirmed the expression of DTX2 mRNA in CRC cells was higher than HCoEpiC. PARP1 was recognized as a key substrate protein of DTX2 in colorectal cancer cells through an IP-MS experiment.

In addition, the expression of DTX2 was positively correlated with the activation of DDR-related pathways ($P<0.05$). Compared with the control and full-length groups, overexpression of the short DTX2 transcript significantly promoted the DDR in SW480.

Conclusion: In this study, we report that DTX2 is highly expressed in colorectal cancer and associated with PARP1. In addition, the short transcript of DTX2 is implicated in DNA damage repair in colorectal cancer.

Disclosure: Nothing to disclose.

PP0719

WITHDRAWN

PP0720

ATPASE ACTIVITY OF THE SUBCELLULAR FRACTION OF COLORECTAL CANCER SAMPLES UNDER THE ACTION OF BILE ACID

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Introduction: The development of colorectal cancer is related to genetic mutations, lifestyle, and diet and eating habits. Fat consumption correlates with increased levels of bile acids in the intestinal lumen, where they stimulate the growth and activity of bacteria that convert primary bile acids into secondary bile acids that have pronounced tumorigenic activity. We assumed that one of the mechanisms of the tumor genesis of bile acids may be a violation of the trans-membrane transport of ions in the intestinal mucosa.

Aims & Methods: Therefore, the aim of this work was to study the effect of taurolithocholic acid 3-sulfate (TLC-S) on the ATPase activity of colorectal cancer samples.The investigation was conducted on colorectal cancer samples of 10 patients. Adjacent samples of healthy tissue served as controls. All patients signed an informed consent to participate in this research. ATPase activity was measured spectrophotometric by the content of P_i in the incubation medium.

Results: We found that the addition of TLC-S to the incubation medium of the subcellular fraction of samples of colorectal cancer patients caused the increase in the specific activity of the Na^+, K^+-ATP-ase by 1.5 times ($P\leq0.05$; n=10) vs to the control (without TLC-S). In healthy samples the activity of this pump did not change under the influence of TLC-S. However, the addition of TLC-S to the incubation medium of healthy samples of the intestinal mucosa caused a statistically significant decrease in the activity of the basal Mg^{2+}-ATPase of the subcellular fraction by 3.6 times vs to the control ($P\leq0.05$; n=10), but in cancer tissue samples the activity of basal Mg^{2+}-ATPase did not change under the influence of TLC-S.

Conclusion: We found that TLC-S causes an increase in the specific activity of the Na^+,K^+-ATPase in colorectal cancer samples, and in normal tissue TLC-S reduces the basal Mg^{2+}-ATP-ase activity

Disclosure: Nothing to disclose.

PP0721

WITHDRAWN

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PP0722

CHEMO(RADIO)THERAPY FOR RECTAL ADENOCARCINOMA AND THE RISK OF MULTIPLE PRIMARY GASTROINTESTINAL SQUAMOUS CELL NEOPLASMS

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Introduction: Rectal cancer represent 30% of colorectal cancer with 46,220 new cases estimated in the United States in 2024. About 98% of rectal cancer patients have rectal adenocarcinoma. Gastrointestinal (GI) squamous cell neoplasm (SCN) is very rare with no enough studies to estimate its incidence and clinical features.

Thus, our aim was to assess the risk of developing second primary GI SCN following chemo(radio)therapy after rectal adenocarcinoma.

Aims & Methods: We used the Surveillance Epidemiology and End Results (SEER) database to obtain the data of 145,789 patients with first primary rectal adenocarcinoma.

We used MP-SIR session to calculate the standardized incidence ratios (SIR) ratio as (observed/Expected) with a latency exclusion period of 0 months, Excess risk (ER) was calculated per 10.000 and 95% Confidence Interval (CI). The results were significant if P> 0.05.

Results: Out of 145.789 patients with rectal adenocarcinoma, 4333 patients developed second primary SCN. Among who developed second primary GI SCN, 21.9% received chemotherapy with an O/E of 4.07 (P<0.05, CI: 4.50 - 3.67, ER=74.53) while 3.25% received radiotherapy with an O/E of 2.23 (P < 0.05, CI: 2.60- CI: 1.90, ER=42.59).

The group who had no systemic therapy had an O/E of 2.83 (P< 0.05, 95%CI: 2.93–2.74, ER=57.27). The group who received combined chemo-radiotherapy had a 2-fold increased risk of developing second primary GI SCN (O/E =2.34, P< 0.05, CI:2.44- 2.24, ER=34.88).

There was a significant increase in the 0-11 months interval in the four groups. The group who received chemotherapy O/E was 11.73 (P < 0.05, 95 % CI: 13.27– 10.32, ER=233.48) and the group who received radiotherapy had increased O/E to 5.16 (P<0.05, ER=150.85, 95%CI: 6.51–4.03).

Conclusion: Chemotherapy increases the risk of multiple primary GI SCN after the first primary rectal adenocarcinoma, especially in the first year after diagnosis. Chemo(radio)therapy have a significantly high risk second primary GI SCN with different potential and variations across different intervals. Thus, more attention to be given for the screening of second primary GI malignancies after chemo(radio)therapy administration.

Disclosure: Nothing to disclose.

PP0723

UNVEILING THE LANDSCAPE OF SERRATED POLYPS: INSIGHTS FROM A LARGE RETROSPECTIVE COHORT STUDY

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Introduction: Colorectal carcinoma (CRC) screening strategies traditionally focused on conventional adenomas, but reports have highlighted the significance of serrated polyps, particularly sessile serrated polyps (SSP), in CRC pathogenesis.

This study aimed to assess the prevalence, characteristics, and clinical implications of serrated polyps and hyperplastic polyps in our practice.

Aims & Methods: A retrospective analysis covering a decade (2015-2023) of colonoscopy data from our Gastroenterology department was conducted. Inclusion criteria comprised individuals 18 years or older undergoing initial colonoscopies for multiple indications, excluding those with prior colectomies, personal CRC history, or other abdominal tumors. Detailed demographic, procedural, endoscopic, and pathological data were collected and analyzed. Polyp endoscopic characteristics were thoroughly documented, including size, location, type, and pathologic data thoroughly reviewed to identify polyp type and grade of dysplasia.

Results: Among 22,175 colonoscopies reviewed, 5,741 polypectomies with complete histologic and endoscopic data were included. Adenomas comprised the majority, with 3,297 tubular adenomas, 885 tubulo-villous, and 137 villous adenomas. Hyperplastic polyps totaled 1,343, while only 74 were recognized as serrated, primarily located in the proximal colon (90%). Serrated polyps were predominantly sessile (71.6%), flat (24.4%), and pedunculated (4%) in endoscopic report. Reviewing pathology reports revealed 14.6% of serrated polyps exhibiting high-grade dysplasia. Notably, 18% of hyperplastic polyps were proximal to the sigmoid colon, and upon pathological revision, 15.7% were reclassified as serrated polyps.

Conclusion: The prevalence and clinical insights provided shed light on the diagnostic challenges and underscore the importance of accurate identification and management of SSP lesions. Establishing standardized detection metrics for serrated polyps is imperative for enhancing CRC screening efficacy and patient outcomes.

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Disclosure: Nothing to disclose.

PP0724

WITHDRAWN

PP0725

IMPACT OF ARTIFICIAL INTELLIGENCE ON COLONOSCOPIES AT HOSPITAL SÍRIO-LIBANÊS: A RETROSPECTIVE ANALYSIS.

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Introduction: According to the latest statistics, the global estimate for the year 2020 pointed to over 1.9 million new cases of colorectal cancer (CRC) (10.0%), making it the third most incident tumor among all cancers. Among men, there were approximately 1.1 million new cases, with an estimated risk of 23.40 cases per 100,000 men. Among women, there were 865 thousand new cases, making it the second most frequent tumor, with an incidence rate of 16.20 cases per 100,000 women. The highest incidence rates of CRC in men were observed in central, northern, and southern Europe. Among women, the highest incidences were observed in Oceania and northern Europe.

In terms of mortality in Brazil, in 2020, there were 20,245 deaths from CRC (9.56 per 100,000). Among men, there were 9,889 deaths, corresponding to 9.55 deaths per 100,000 men. Among women, there were 10,356 deaths, corresponding to 9.57 deaths per 100,000 women.

Colonoscopy is considered the gold standard for CRC screening as it allows for both diagnosis and resection of lesions. Observational studies

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have shown a reduction of up to 77% in the incidence and 29 to 37% in mortality from CRC compared to the group of patients not undergoing prior colonoscopy.

Following successful validation by the Food and Drug Administration (FDA) in 2021, the Computer-Aided Detection (CADe) system was introduced into clinical practice. Subsequently, it became possible to assess its potential benefit in increasing the adenoma detection rate (ADR) and the rate of adenomas per colonoscopy (APC).

On the other hand, some potential drawbacks were identified, such as the downgrading of the endoscopist's skills or an increase in examination time.

Aims & Methods: The aim was to evaluate the adenoma detection rate (ADR) in a group of patients undergoing colonoscopies with AI assistance and in a group of patients undergoing the procedure without AI assistance at Hospital Sírio-Libanês, Bela Vista/SP unit.

The sample size calculation was based on a comparison test of two proportions. At a 5% significance level, minimum power of 80%, considering the values obtained from a previous similar study, 254 individuals were required in each group, totaling n=508.

In the first group, examinations were performed using the Fujifilm processor model VP-7000, with a colonoscope model EC-760ZP-V/M. In this group, throughout the examination period, the CADe system remained activated and operational, allowing AI assistance in lesion detection. It was at the discretion of the colonoscopist whether to resect the lesion after the alarm emitted by the CADe.

In the second group, the examinations were performed using the Olympus processor model CV-190, with a colonoscope model CF-HQ190L/I. In this group, no AI assistance was utilized.

Results: The ADR in the entire sample was 37.0%, with a rate of 36.6% among patients whose examinations were performed without the use of AI and 37.4% among those in which the examination was conducted with the use of AI, without statistical difference. The ADR was also stratified by age groups within each gender, showing no significant association with the use of AI in the examination. Within the female gender, a higher ADR was observed in women aged 60 or older compared to those under 60, but without statistical significance (p=0.451).

Conclusion: In this study, after a retrospective analysis of colonoscopies conducted at Hospital Sírio-Libanês, Bela Vista/SP unit, the ADR of examinations performed with AI did not show statistically significant superiority compared to examinations performed without AI.

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Disclosure: Nothing to disclose.

PP0726

A MULTICENTRE EVALUATION OF FIT LEVELS IN COLORECTAL CANCER PATIENTS

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Introduction: NICE recommend FIT (faecal immunochemical test) in almost all suspected colorectal cancer (CRC) cases. FIT-based prediction models perform well [1], but it is unclear how levels change along the adenoma-carcinoma sequence and how lesion type and location affect results.

We previously reported retrospective FIT data from CRC patients at a single site [2]. We have re-evaluated this two years later and, utilising our regional trainee network, have included five more sites.

Aims & Methods: Our aim was to evaluate FIT results at various time points and in different tumour types/locations.

This was a multicentre, retrospective evaluation involving 6 hospital sites. All patients diagnosed with a malignant lesion of colon, rectum or anus in a 6-month period were identified using cancer registries. Data were collected from medical records.

Results: 856 CRC patients were included. Where histology was available, 90% were adenocarcinoma. Referrals were 85% symptomatic, 14% BCSP and 1% surveillance. 492 patients had ≥1 non-BCSP FIT prior to diagnosis. 530 individual results were recorded with a mean of 178 days pre-diagnosis.

Overall, 11% were <10µg/g, however the proportion increased as time from diagnosis increased (≤30 days 5.2%; 31-90 days 7.4%; 91-180 days 22.2%; 181-365 days 21.7%; >366 days 47.2%). When limited to ≤90 days, 6.3% were <10µg/g. This reduced to 5.3% in proven adenocarcinoma. The proportion varied according to lesion location (Table 1).

Location	FIT results	
	Total	<10µg/g
Right colon	113	14 (12.4%)
Left colon	124	2 (1.6%)
Rectum	151	8 (5.3%)

Table 1: Results by lesion location.

Conclusion: Time since FIT is relevant clinically and in research. FITs >90 days old may not be suitable for guiding clinical decisions, however our sample for this time point was small. Prospective, sequential testing would more accurately study the change in FIT along the adenoma-carcinoma sequence. Studies assessing the value of repeat testing at different time points would help guide clinicians. The highest proportion of results <10µg/g was in right colon lesions, consistent with existing literature. However, we identified more false negatives in rectal lesions compared to left colon and recommend further research assessing this.

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Disclosure: Nothing to disclose.

PP0727

STUDY OF INTESTINAL DIFFERENTIATION MARKERS (CDX2 AND SATB2) IN DIGESTIVE NEUROENDOCRINE NEOPLAMS

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Introduction: Digestive neuroendocrine tumors (DNETs) are rare and ubiquitous neoplasm of the digestive tract, with an incidence that is increasing significantly in recent years. Special AT-rich sequence-binding protein 2 (SATB2) and the homeotic transcription factor (CDX2) are markers of intestinal differentiation that may be involved in the in the topographic diagnosis of DNET and thus orientate towards the site of the primary NET.

Aims & Methods: The aim of our work was to evaluate the immunohistochemical expression of CDX2 and SATB2 in DNET according to the topography and the histoprognostic factors. We conducted a longitudinal and retrospective study including 39 DNETs. Epidemiological and anatomopathological data according to the 2019 WHO classification were studied. The study of CDX2 and SATB2 was performed by immunohistochemical method. Positivity was considered in the presence of a nuclear immunoreactivity greater than 10%.

Results: The mean age of patients diagnosed with DNETs was 51.4 years with a sex ratio M/F of 0.77. The predilection sites were the small intestine and the appendix, in 33.3% of cases each. CDX2 was expressed in 64% of DNETs. It was positive in 90-100% of small intestine, appendix and colon neuroendocrine neoplasms. SATB2 was expressed in 60% of DNETs. It was found in 100% of appendicular and colorectal neuroendocrine tumors. A weak staining was observed in 69% of small intestine neuroendocrine tumors. Gastric and pancreatic tumors pancreatic tumors did not express CDX2 nor SATB2. This variation in expression was statistically significant ($p<0.001$). Grade G1 tumors (67%) were in 70% CDX2+/SATB2+. Grade G2/G3 tumors (33%) were in 68% of cases SATB2- without significant difference. The pT1/pT2 DNET (n=19) were CDX2+/SATB2+ in 68%. The pT3/pT4 tumors (n=17) were CDX2- in 35% and SATB2- in 47% without significant difference. Lymph node and distant metastases were found in 28% and 14% of cases respectively. No statistically significant correlation was noted between CDX2 or SATB2 and the presence of lymph node metastases. However, the loss of SATB2 was significantly associated with the metastatic potential ($p<0.001$).

Conclusion: CDX2 and SATB2 are highly sensitive and relatively specific markers of the digestive tract. They would be useful in the topographic approach of neuroendocrine tumors of unknown origin.

Disclosure: Nothing to disclose.

PP0728

SCREENING COLONOSCOPY IN GREECE: DISCOURAGING PUBLIC AWARENESS

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Introduction: Colorectal cancer incidence is increasing worldwide. Nowadays it is the third most diagnosed cancer[1]. Screening colonoscopy is the gold standard for prevention and often treatment, both economically and efficiently. Hellenic Society of Gastroenterology recommends that screening colonoscopy should starts by 50 years old for average risk population[1]. There are limited data about participation in screening colonoscopy in general population in Greece and the existing ones shows discouraging participation rates[2,3].

Aims & Methods: The aim of this study was to determinate general population's knowledge about screening colonoscopy in Greece. We created a digital, anonymous questionnaire at Google Forms platform and transferred it via social media in general population that lives in Greece.

Results: In total 711 individuals answered the survey. 65.8% of them(468/711) were woman. 61.5% (437/711) were under 40 years old, 19.7% (140/711) were between 40 and 50 years old and 18.3% (130/711) were between 50 and 75 years old. 40.2% (286/711) did not know if there is an appropriate age for the first screening colonoscopy. 39% (277/711) answered that they thought an appropriate age is at 50 years old, 26.7% (190/711) at 45 years old and 8.2% (58/711) answered that they thought there is not an appropriate age for a screening colonoscopy. Regarding the performance of screening colonoscopy 363 out of 711 (51.1%) are willing to perform it in the future, 20.1% (143/711) answered that they do not know or do not answer the question,11% (78/711) are not willing to perform it and 2.1% (15/711) are not willing to perform it, because they think it is not necessary for preventing colorectal cancer.

Conclusion: Our study confirms that there is lack of information in general population and unwillingness to screen for colorectal cancer in Greece. General population tends to be uninformed and unwilling about screening colonoscopy[1]. Despite the effectiveness of colonoscopy there is not an organized screening program in Greece. Individuals are usually self-referred for screening and usually later than the appropriate age and after symptoms occure[1]. Primary health care providers should be well informed and correctly advise individuals about colorectal cancer screening and its privilege, options, and alarm symptoms[4].

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Disclosure: Nothing to disclose.

PP0729

WITHDRAWN

PP0730

VIDEO ASSESSMENT OF LARGE NON-PEDUNCULATED COLORECTAL POLYPS SIMILARLY PREDICTS HARD OUTCOMES IN THE ASSESSMENT OF SUBMUCOSAL INVASIVE CANCER RISK COMPARED TO LIVE ASSESSMENT

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Introduction: Cancer in large non-pedunculated colorectal polyps (LNPCPs) is poorly discriminated outside expert centres. It is unknown whether video- is comparable to live-assessment for determining submucosal invasion (SMI) in LNPCPs. Video-assessment has multiple advantages including collaborative discussion of imaging findings at multi-disciplinary meetings avoiding extra assessment procedures for patients and incorrect decision-making. We compared live LNPCP assessment with analysis of high-quality endoscopic videos produced by an expert.

Aims & Methods: Blinded live assessments of LNPCPs from consecutive, consenting patients were conducted by 15 endoscopists in a single center. After the live assessment, an expert recorded a standardized video of each LNPCP. The remaining endoscopists [who had not assessed the LNPCP live] assessed these videos using an online survey. Endoscopists were classified as experienced (>50 EMRs ever and >1000 colonoscopies) or inexperienced. Parameters such as Blink Impression(BI) [1], 6-Blink Features(BF) – namely fold deformation, extra redness, depression, chicken skin mucosa, ulceration and spontaneous bleeding - presence of a Demarcated Area(DA), Vascular Pattern(VP) of any DA, size, location, Paris classification, and morphology were recorded to form the endoscopist's impression of cancer presence in the LNPCP. The results were later compared to expert consensus including an external validator. The primary endpoint was to evaluate the accuracy of endoscopists' final impression of cancer and choice of correct treatment between live and video assessments.

Results: 64 live assessments (12 endoscopists) & 96 video assessments (9 endoscopists) of 18 LNPCPs were performed. 5 (41.7%) (live) vs 4 (44.4%) (video) endoscopists were inexperienced. The comparison of live versus video assessments revealed no significant difference in the final impression of cancer presence by the endoscopist (sens 80.0% vs 100%, spec 87.0% vs 85.5%), BI (sens 85.7% vs 92.3%, spec 88.6% vs 80.7%) and the presence of ≥2BF (sens 71.4% vs 84.6%, spec 81.8% vs 68.7%) vs histology (P=.77, .34 & .21 respectively). Further the ability of DA & JNET to predict SMI did not differ nor did the rate of correct treatment determination (71.9% live vs 60.4% video, P=.14). Versus expert consensus the determination of ≥2BF was equally accurate live (82.4%) vs video (76.0%,P=.38). Conversely the exact number of BF vs experts (live 60.4% vs video 37.5%,P=.01), the presence of ≥3 features (location, size, morphology or Paris classification) of COVERT (not visible on the surface) SMI live (81.3%) vs video (65.6%, P=.03) and the correct outcome of a structured algorithm[2] (live 96.9%, video 81.3%,P=.002) were determined differently between the groups.

	LIVE			Video			P
	Sens	Spec	Acc	Sens	Spec	Acc	
Blink pre	85.7% [59.8%-100%]	88.6 % [79.3% - 98.0%]	88.2% [79.4% - 97.1%]	92.3% [77.8% - 100%]	80.7% [72.2% - 89.2%]	82.3% [74.7% - 89.9%]	0.345
Blink (vs SMI, ≥2 BF)	71.4% [378.0%-100%]	81.8% [70.4% - 93.2%]	80.4% [69.5% - 91.3%]	84.6% [65.0% - 100%]	68.7% [58.7% - 78.7%]	70.8% [61.7% - 79.9%]	0.207
Blink (vs expert, ≥2 BF)	70.0% [41.6%-98.4%]	85.4% [74.6% - 96.2%]	82.4% [71.9% - 92.8%]	63.5% [50.4% - 76.6%]	90.9% [82.4% - 99.4%]	76.0% [67.5% - 84.6%]	0.377
OVERT DA	100.0% [100% - 100%]	75.9% [64.5% - 87.3%]	79.7% [69.8% - 89.5%]	100.0% [100% - 100%]	77.1% [68.1% - 86.2%]	80.2% [72.2%-88.2%]	0.920
OVERT VascPattern JNET	100.0% [100% - 100%]	70.4% [58.2%-82.6%]	75.0% [64.4% - 85.6%]	100.0% [100% - 100%]	69.9% [60.0% - 79.8%]	74.0% [65.2% - 82.7%]	0.887
Cancer present	80.0% [55.2%-100%]	87.0% [78.1%-96.0%]	85.9% [77.4% - 94.5%]	100.0% [100% - 100%]	85.5% [78.0% - 93.1%]	87.5 [80.9% - 94.1%]	0.777

Conclusion: Final impression of the presence of cancer and determination of correct treatment in LNPCPs are predicted equally well from blinded video assessment versus live assessment in a group of similarly experienced endoscopists.

This finding could minimise the need for patients to undergo a repeat procedure to assess their LNPCP prior to resection relying instead upon expert assessment of a video.

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Disclosure: Nothing to disclose.

PP0731

QUALITY INDICATORS AND RESULTS OF COLONOSCOPY EXAMINATIONS IN INITIAL FIVE-YEAR PERIOD OF HUNGARIAN POPULATION-BASED COLORECTAL CANCER SCREENING PROGRAM

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Introduction: The Hungarian population-based colorectal cancer (CRC) screening program (EFOP-1.8.1-VEKOP-15-2016-00001) was launched in 2019 among asymptomatic individuals between the ages of 50 and 70 with average risk of CRC.

Aims & Methods: Aim of the study was to retrospectively assess the quality indicators and results of screening colonoscopies performed during the initial five-year period of screening program (from 2019 to 2023). Our

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non-interventional, observational cohort study retrospectively analyzed clinical records of screening colonoscopies which were prospectively collected in the registry of National Public Health Institute. The quality indicators and results of examinations was determined based on cecal intubation rate, polyp, adenoma and cancer detection rates (PDR, ADR, and CDR), proportion of photo-documentation and sedation.

Results: Total of 27,021 colonoscopies were performed during the initial five-year period of CRC screening program. Cecal intubation rate was 92.54%, from which 66.77% of cases were confirmed by cecal image documentations. The causes of incomplete colonoscopies were inadequate bowel cleansing (n=172, 11.48%), anatomical reasons (n=369, 24.63%) and malignant colonic obstruction (n=178, 11.88%), and only in 191 cases (12.75%), the examination had discontinued due to patients' intolerance. Total of 17,981 patients (66.54%) of participants received some kind of sedation. The result of colonoscopy was non-negative in 80.83% of cases. At least one polyp was found (PDR) in 56.56% (n=15,282) of participants among which the largest polyp was located in rectosigmoid colon in 64.92% of cases and the polyp size was higher than 1 cm in 34.69% of cases. Morphological features of polyps could not be assessed due to inappropriate colonoscopy reports in registry. A total of 36,829 polyps were found, of which 30,494 were removed during the registered examinations, becoming polyp-free 82.15% of patients (polypectomy per colonoscopy 1,13). The ADR was 33.53%, where the most relevant lesion histological types were tubular, tubulovillous, villous and serrated in 67.46%, 22.83%, 3.60% and 2.92% respectively. CRC was found in 5.51% of participants from which the most cancers were located in the sigmoid colon (n=578, 38,82%) and rectum (n=458, 30,76%). The data of registry to description of polyp morphology, usage of advanced imaging assessment, polyp retrieval rate and the image documentation were often missing or inappropriate.

Conclusion: Cecal intubation, polyp and adenoma detection rates of screening colonoscopies of Hungarian CRC screening program are satisfactory according to guideline of European Society of Gastrointestinal Endoscopy, however, the description of polyp morphology, usage of advanced imaging assessment, polyp retrieval rate and the image documentation are inappropriate yet, therefore the evaluation was not possible. The aim for the future is to improve the quality of data input to the register.

Disclosure: Nothing to disclose.

PP0732

PATIENT SATISFACTION MEASUREMENT OF SCREENING COLONOSCOPIES IN THE HUNGARIAN POPULATION-BASED SCREENING PROGRAM - NATIONWIDE, CROSS-SECTIONAL SURVEY STUDY

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Introduction: In colorectal cancer (CRC) screening programs utilizing colonoscopy, patient satisfaction plays a pivotal role in persistent adherence, thus influencing the program's long-term efficacy. Our study aimed to evaluate patient satisfaction following endoscopic procedures within the Hungarian population-based CRC screening program (EFOP-1.8.1-VEKOP-15-2016-00001) on a national scale.

Aims & Methods: Our study aimed to evaluate patient satisfaction following endoscopic procedures within the Hungarian population-based CRC screening program (EFOP-1.8.1-VEKOP-15-2016-00001) on a national scale. The survey evaluated data from the anonymous, standardized self-administered questionnaire devised and distributed by the Hungarian National Centre for Public Health and Pharmacy between January 2020 and November 2023. The questionnaire predominantly comprised closed-ended questions. Patient satisfaction was assessed quantitatively by a 5-point scale. Categorical variables were reported as event rates and relative frequencies, and continuous variables as means with standard deviation. Fisher's exact test was used to analyze categorical data.

Results: A total of 4083 questionnaires were evaluated, with a slight male predominance (55.71%) and with 41.96% of patients under 61 years of age. Questionnaire completed by 19.76% of colonoscopy patients (0.46% in 2020, 33.69% in 2021, 27.70% in 2022 and 22.71% in 2023). Respondents were evenly split between county centers/large cities, small towns and villages, but Budapest was under-represented (31.55%, 32.92%, 32.32% and 2.94% respectively). There was considerable variation in the willingness of endoscopic units to cooperate, so that no data at all were available from 4 counties. Sedation was used in 60.40% of patients (analgesia 55.57%, general anesthesia 4.82%), but 493 patients did not answer this question. Patients rated the quality of the colonoscopy units, competence of endoscopy staff and the overall satisfaction with exam execution as 4.68±0.58, 4.89±0.42 and 4.47±0.77 respectively. They were equally satisfied with the communication of the doctor (4.87±0.41) and the staff (4.87±0.44). Only 1.67% of patients reported that they would prefer to be switched to another doctor in case of a future screening, and the question remained unanswered in 5.44% of cases. Lack of rest room was complained by 9.09% of patients, where it was available, it was rated 4.68±0.67 points, however 7.67% of patients didn't respond to this question. In 33.70% of the cases, colonoscopy was considered to be substantially unpleasant and painful, giving an overall score of 3.85±1.06. Procedure-related pain and discomfort were rated as significantly less severe in men ($p < 0.001$) and in patients over 61 years of age ($p=0.001$), as well as in patients who received detailed information from the endoscopist immediately before the endoscopy ($p<0.001$) and who received sedation ($p<0.001$). The detailed information provided by the endoscopist ($p<0.001$), the sedation ($p<0.001$) and the mild pain experienced during the examination ($p<0.001$) had a positive influence on patients' overall satisfaction with the colonoscopy.

Conclusion: This is the first large-volume patient satisfaction survey among patients invited to CRC screening program in Hungary. Positive patient feedback indicates high quality of care. Overall satisfaction can be further improved by modulation of procedure-related pain, rate of sedation and quality of postprocedural rest, as well as the quality of information delivered by endoscopists prior to procedures.

Disclosure: Nothing to disclose.

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PP0733

ENDOSCOPIC PREDICTORS OF LYMPH NODE METASTASIS IN T1B COLORECTAL CANCER IN THE ABSENCE OF PATHOLOGIC RISK FEATURES: A SINGLE-CENTER STUDY OF 305 LESIONS

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Introduction: There are insufficient data to guide clinical decision-making regarding the implementation of additional surgery following endoscopic therapy in patients with deep submucosal invasive colorectal cancer (T1b CRC) and no additional risk factors for lymph node metastasis (LNM). This investigation sought to identify factors associated with LNM in cases of T1b CRC without other risk factors.

Aims & Methods: Consecutive cases with pathologically proven T1b CRC were retrospectively collected from 2014 to 2023 at a single tertiary hospital in Korea. These patients received either surgical or endoscopic treatment. In patients who received endoscopic treatment alone, no LNM was deemed to exist if there was no recurrence after at least 3 years of follow-up. Excluded were cases with other risk factors, including poor differentiation, lymphovascular invasion, perineural invasion. The cases were compared based on the presence of LNM.

Results: A total of 305 cases were enrolled and LNM was observed in 28 cases (9.18%). In univariate analysis, differentiation, sm3 invasion, tumor growth pattern, and peri-tumoral white spots were related with LNM. Multivariate analysis identified presence of non-polypoid growth parts (odd ratio [OR] 6.39; 95% CI 2.76-14.79), and peri-tumoral white spots (OR 2.85; 95% CI 1.22-6.64) as significant predictive factors for LNM.

Factor	OR	95% CI	P-value
Peri-tumoral white spots	2.85	1.22-6.64	0.015
Non-polypoid growth	6.39	2.76-14.79	0.000

Table. Multivariate analysis

Conclusion: Endoscopic findings including presence of non-polypoid growth and peri-tumoral white spots were related with increased risk of LNM in T1b CRC in the absence of other pathologic risk factors. These findings may be useful in determining further treatment strategy for patients with Tb1 CRC received endoscopic resection.

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Chihiro Yasue, Akiko Chino, et al. Pathological risk factors and predictive endoscopic factors for lymph node metastasis of T1 colorectal cancer: a single-center study of 846 lesions. J Gastroenterol(2019) 54: 708-717

Disclosure: Nothing to disclose.

PP0734

IS MODERATELY DIFFERENTIATED ADENOCARCINOMA A RISK FACTOR FOR LYMPH NODE METASTASIS IN T1 COLORECTAL CANCER? A SINGLE-CENTER STUDY OF 881 LESIONS

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Introduction: Several guidelines, including those from the US, Europe, and Japan, recognize “tumor differentiation“ as a risk factor for lymph node metastasis (LNM) in T1 colorectal cancer. Surgical resection with lymph node dissection is recommended if poorly differentiated adenocarcinoma, signet-ring cell carcinoma, or mucinous carcinoma are observed in histological findings from endoscopically resected specimens. Recent reports suggest that moderately differentiated adenocarcinoma (tub2) may also be a risk factor for LNM.

Aims & Methods: This study aims to investigate whether tub2 is a risk factor for LNM. We initially evaluated 1,742 patients with T1 colorectal cancer who underwent surgical resection with lymph node dissection at our institution from April 2001 to April 2024. Lesions were assessed based on the worst grade of differentiation. Exclusions were made for patients with (a) metachronous or synchronous advanced cancer, (b) invasion deeper than the muscularis propria, (c) Lynch syndrome, familial adenomatous polyposis, or ulcerative colitis, (d) preoperative chemotherapy or radiotherapy, (e) missing data, or (f) lesions without surgical resection. In addition, after excluding 177 patients with poorly differentiated, mucinous, or signet-ring cell adenocarcinoma, 881 patients were enrolled for analysis. We investigated the sensitivity, specificity, and area under the receiver operating characteristics (ROC) curve of tub2 for predicting LNM, alongside correlations between pathological factors and LNM using multivariate analysis.

Results: LNM was detected in 9.9% (87/881) of patients. The sensitivity, specificity, and area under the ROC curve for tub2 were 78.2%, 42.2%, and 0.60, respectively. In the univariate analysis, tub2 was associated with LNM ($p<0.001$). Multivariate analysis identified lymphatic invasion, vascular invasion, and tub2 as significant risk factors for LNM, with odds ratios of 7.49, 1.97, and 1.95 ($p < 0.01$, < 0.01, and $= 0.02$, respectively).

Conclusion: Moderately differentiated adenocarcinoma (tub2) is a risk factor for LNM in T1 colorectal cancer.

Disclosure: Conflicts of interest: None.

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PP0735

USEFULNESS OF COMPUTER-AIDED DIAGNOSIS (CADX) BASED ON NON-MAGNIFIED WHITE LIGHT IMAGES IN THE DIAGNOSIS OF DEEP INVASIVE CANCER IN T1 COLORECTAL CANCER

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Introduction: The optical diagnosis of T1 colorectal cancer (CRC) with deep invasion (≥T1b CRC) is crucial in determining subsequent treatment. Recently, computer-aided diagnosis (CADx) has become available for the diagnosis of deep invasive cancer, but until now, it must be combined with image enhanced endoscopy (IEE) or endocytoscopy. We have developed a CADx system to diagnose deep invasive cancer using non-magnified white light imaging (WLI).

Aims & Methods: This study aimed to compare the diagnostic accuracy of the CADx in T1 CRC diagnosis with that of non-expert endoscopists and expert endoscopists. A total of 365 images of 37 T1 CRCs (16 T1a CRCs and 21 T1b CRCs) from prospectively enrolled consecutive patients were included in the study. Images were independently diagnosed by two non-expert endoscopists and two expert endoscopists; sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and accuracy of T1b CRC diagnosis by the CAD system were calculated and compared with the endoscopists diagnostic accuracies.

Results: The sensitivity, specificity, PPV, NPV, and accuracy of CAD were 70.2%, 60.4%, 84.2%, 49.3%, and 70.7%, respectively, while those of non-experts were 39.9%, 80.6%, 82.7%, 36.6%, and 51.9%, respectively. The accuracy rate was significantly higher in the CAD group ($p < 0.01$). On the other hand, those of experts were 47.9%, 87.7%, 79.2%, 63.2%, and 67.7%, respectively, which were comparable to the diagnostic accuracy of CAD. The diagnostic time per image was significantly shorter for CAD than for endoscopists (0.49 sec. for CAD, 7.72 sec. for non-expert, and 5.77 sec. for expert, $p < 0.01$, $p < 0.01$, comparison with CAD).

Conclusion: Non-magnified WLI-based CADx is useful for non-expert endoscopists in diagnosing T1 CRC with deep invasion.

Disclosure: This work was supported by JSPS KAKENHI Grant Number 21K07885 and JFE (The Japanese Foundation for Research and Promotion of Endoscopy) Grant.

PP0736

SENSITIVITY OF THE AMINOINDEX CANCER SCREENING FOR EARLY COLORECTAL NEOPLASTIC LESIONS; A PROSPECTIVE COHORT STUDY

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Introduction: Screening colonoscopy reduces colorectal cancer (CRC) mortality. However, several studies have reported a lower participation rate of colonoscopy. Now it is important to provide less invasive test for CRC screening.

AminoIndex® Cancer Risk Screening (AICS®) is an examination that measures plasma amino acid concentrations and estimates the risk of existing CRC based on the difference in amino acid balance in patients with CRC and healthy individuals. AICS is calculated from 0.0 to 10.0 with the higher value indicating a higher the risk of existing CRC.

We aimed to conduct a prospective cohort study to evaluate the accuracy of AICS® in detecting the risk probability for early colorectal lesions, such as advanced adenoma (AA), intramucosal neoplasia (Tis), or T1 CRC.

Aims & Methods: Consecutive patients who underwent colonoscopy between 06/02/2017 and 31/03/2021at the National Cancer Center Central Hospital and endoscopically diagnosed adenoma larger than 20 mm, intramucosal neoplasia (Tis), or submucosal invasive carcinoma (T1) were included as a selection criterion. Patients with endoscopically suspected intramucosal neoplasia or high-grade dysplasia by preoperative colonoscopy, but finally diagnosed with adenoma were also included in the study. We excluded the following patients;

1) patients with polyposis such as familial colorectal adenomatosis or inflammatory bowel disease,
2) patients with concurrent cancer of other organs,
3) patients with obvious bleeding,
4) patients with severe defecation disorder, and;
5) patients with stenosis of colon or intestine.

We measured plasma amino acid concentrations calculated the AICS score. FIT was also measured at the same time. The lesions were endoscopically resected using EMR or ESD technique. The sensitivity of AICS, FIT, and the combination of FIT and AICS were examined according to the pathological diagnosis.

Results: Two hundred and seventeen patients were included in the primary analysis, and 208 patients excluding 9 patients who did not perform FIT were included in the secondary analysis.

The median age of the 217 patients was 71 years old (range: 32- 91). The mean lesion size was 28.9 mm±12.9. Prevalence of AA, Tis, T1, and advanced lesion was, 63, 118, 30, and 6, respectively. Six lesions were suspected to be T1 carcinoma on preoperative endoscopy, following pathological examination revealed advanced cancer after endoscopic resection.

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The sensitivity of AICS for the advanced neoplasia (AN), AA, intramucosal neoplasia (Tis), submucosal invasive carcinoma, and advanced CRC was 28.1%, 30.9%, 25.4%, 23.3%, and 33.3%, respectively.
In the secondary endpoint, the sensitivity of FIT for AN, AA, intramucosal neoplasia (Tis), submucosal invasive carcinoma, and advanced CRC was 47.0%, 35.3%, 49.1%, 46.7%, and 100%, respectively. The sensitivity of the combination of FIT and AICS for AN, AA, intramucosal neoplasia (Tis), submucosal invasive carcinoma, and advanced CRC was 61.1%, 52.9%, 64.0%, 60.0%, 100%, respectively.
Conclusion: Although it was a reasonable value compared to the development data, AICS® has insufficient sensitivity for lesions requiring endoscopic intervention. It could be clinically applicable as a minimally invasive examination to stratify the risk of existing CRC in combination with FIT for colorectal screening.
Disclosure: The primary investigator (Hiroyuki Takamaru) had received research funding from Ajinomoto Co. Ltd.

PP0737

BACK-TO-BACK COLON CAPSULE ENDOSCOPY AND COLONOSCOPY IN BOTH SCREENING AND SYMPTOMATIC POPULATION

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Introduction: Colon capsule endoscopy (CCE) has been demonstrated to be comparable to colonoscopy and better than CT colonography (CTC) for the detection of colonic pathology. Challenges include poor completion rates and high re-investigation rates.
Aims & Methods: The aim of this study was to assess the role of CCE with faecal immunochemical testing (FIT) as a non-invasive, diagnostic tool in symptomatic and screening populations attending Connolly Hospital.
Prospective participants were identified from both the central endoscopy waiting list at Connolly Hospital and from our national colorectal screening programme, Bowel Screen. All participants were due to attend Connolly Hospital for a symptomatic or surveillance colonoscopy during the time period of the study.
Consenting participants underwent CCE the day prior to their scheduled colonoscopy. CCEs were read by two expert readers. Participants were sent FIT kits to complete prior to CCE. Primary outcomes were CCE polyp detection rates (PDR). The potential benefit of using FIT as a triage tool to identify low-risk individuals appropriate for CCE over colonoscopy was explored.
Results: 80 back-to-back procedures were completed January 2023-February 2024 (M: F 50:30, mean age 59.6, range 35-78) in a single centre. The most common indications for colonoscopy were surveillance 41.5%, PR bleeding 21% and diarrhoea/ IBD assessment 15%.
CCE detected a total of 108 polyps in 45/80 (56%) participants versus 93 polyps in 39/80 (48%) participants at colonoscopy (p= 0.34). When only complete CCE are included, the polyp detection rate increased to 31/45 (68%), p< 0.05. 79% supplied FITs prior to CCE. CCE PDR for all polyps when FIT < 50 ng/ml was 19/33 (57%) versus (20/27) 74% with FIT > 50 ng/ml (p=0.18). There is trend towards significance. A lower FIT cut-off, closer to limit of detection, not available to this study, may have more significant results.
Only one participant had a cancer diagnosed at colonoscopy. It was located on the ileocaecal valve. The cancer was not detected by CCE despite the capsule passing through this region. This, to our knowledge, is only the second missed cancer by CCE documented.
ESGE 2012 guidelines recommend follow up colonoscopy for one polyp measuring 6mm or more, or if three polyps of any size are detected. 50/80 (62.5%) CCE examinations would have required follow up if they were not part of a study and scheduled for next day colonoscopy. Even when looking at only those CCE examinations that were complete, 17/45 (38%) required further investigation. Reasons for re-investigation included incomplete examination 32/49 (65%), for polypectomy 16/49 (33%) and for biopsy in 1/49 (2%).
False negative CCE rates for polyps greater than 6mm range from 2-15.5% in the literature. In keeping with this, the false negative CCE for all polyps in this study was 8.75%, and 6.25% for polyps greater than 6mm.
There were no adverse events reported. There were technical issues reported in 16% of CCE. These included the capsule not turning on until it was beyond the caecum, participants removing the belt intermittently and one interference which was believed to be secondary to a topical glucose monitor.
Conclusion: The increasing demand on colonoscopy waiting lists raises concerns regarding the ability of health services to deliver traditional endoscopy in a timely fashion. Alternative modalities are necessary and CCE is a viable diagnostic alternative. There is likely a role for FIT testing as a triage tool prior to CCE to reduce re-investigation rates.
Disclosure: Nothing to disclose.

PP0738

CLINICAL FEATURES OF DEPRESSED TYPE COLORECTAL NEOPLASMS - USEFULNESS OF MAGNIFYING ENDOSCOPY AND ENDOCYTOSCOPY

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Introduction: In the carcinogenic pathway of colorectal cancer, the first concept is the “adenoma-carcinoma sequence“ and the second concept is the “ serrated pathway “. In these concepts, adenomas and serrated lesions gradually progress to cancer. The third concept is the “de novo pathway,“ in which cancer develops without adenomas or serrated lesions and is also characterized by high malignancy and rapid progression.
We have endoscopically characterized depressed colorectal cancers considered to be of the “de novo pathway“ and validated the usefulness of pit pattern and EC classification.
Aims & Methods: A total of 50119 colorectal neoplasms were resected endoscopically or surgically in our institution from April 2001 to December 2023. They were subjects of this study. Advanced cancers were excluded. Of these, 1633 lesions were submucosally invasive (T1) carcinomas, 6802 lesions were classified as high-grade dysplasia and 38952 lesions as low-grade dysplasia. Morphologically, they were divided into 3 types: depressed, flat, and protruded type comprising 1.1 (%), 45.0 (%),and 54.0 (%), respectively. Specifically, the distinctive features of depressed type neoplasms concerning their pit pattern and EC classification were examined.
Results: Most of the flat (91.5%) and protruded type (94.9%) lesions showed type IIIL or IV pit pattern, which are common findings in adenomas. On the other hand, most of the depressed type lesions (92.3%) classified as type IIIs, VI or VN pit patterns, are typically seen in carcinomas. As for endocytoscopy, most of flat and protruded type lesions were identified as EC2 corresponding to adenomas. In contrast, relatively high rates of depressed type lesions were categorized as EC3a (40.3%) and EC3b (52.3%), consistent with invasive carcinomas.

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The proportion of T1 carcinomas in depressed type lesions was 60.0%, whereas in flat and protruded type lesions it was 2.7% and 2.6%, respectively. Additionally, 11.6% of depressed type lesions less than 5mm in diameter were T1 carcinomas, whereas flat and protruded type lesions were 0.01% and 0%, respectively.

Conclusion: This study revealed that depressed lesions were classified as type IIIs, VI or VN pit patterns on magnifying endoscopy and as type EC3a or EC3b on endocytoscopy. Lesions with these features or pit patterns tend to invade the submucosal layer even if they are small size. Despite their high potential for malignancy, depressed lesions are often ignored. However, it is critical to recognize their features and accurately diagnose lesions based on morphology, pit pattern, and EC classification.

Disclosure: Nothing to disclose.

PP0739

CLINICAL VALUE OF FECAL MICRORNA COMBINATION FOR PRECANCEROUS AND CANCEROUS COLORECTAL LESIONS DETECTION

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Introduction: Colorectal cancer (CRC) screening programs can improve patient prognosis and survival rate. However, the Fecal Immunochemical Test based screening has low sensitivity for precancerous lesions and is suboptimal for CRC detection, leading to a high false-negative rate. This limitation increases the number of undiagnosed patients, decreases the overall screening effectiveness, and increases the urgency for the development of new tools. In the past few decades, several molecules have been investigated as potential biomarkers to enhance CRC screening, among them emerged microRNAs (miRs). MiRs play an essential role in the genetic events associated with carcinogenesis. In CRC context, these small molecules have a dysregulated expression that is easily detected in biological samples, such as stool, making them a promising tool for CRC screening.

Aims & Methods: In this respect, we aimed to identify miRs profiles in fecal samples that could potentially serve as biomarkers to improve the early detection of CRC. MiR expression was evaluated using RT-qPCR in stool samples from 96 individuals who underwent colonoscopy with different clinical outcomes, namely, individuals without lesions found on colonoscopy, low-grade dysplasia lesions, high-grade dysplasia lesions, and patients diagnosed with CRC.

Results: Our results showed altered expression of seven miRs in fecal samples from different clinical groups. Of these, miR-451a showed significantly decreased expression in the high-grade dysplasia group, with good performance in distinguishing between this clinical group and individuals without lesions found on colonoscopy (AUC, 0.706; sensitivity, 63%). Moreover, miR-199a-5p, described for the first time in stool, miR-21-5p, miR-4516, and miR-92a-3p were upregulated in advanced lesions, specifically in CRC and high-grade dysplasia lesions groups. In addition, a panel of three fecal miRs showed high discrimination power between the high-grade dysplasia lesion group and individuals without lesions found in the colonoscopy group (AUC, 0.828; sensitivity, 88%). Also, a panel consisting of two miRs showed moderate performance in identifying the CRC group (AUC, 0.724; sensitivity, 83%).

On the ther hand, the use of these two panels in combination improved the performance for CRC detection (sensitivity of 88%) and high-grade dysplasia lesion identification (sensitivity of 92%). Furthermore, regarding the detection of advanced lesions, these panels achieved a sensitivity of 90%.

Conclusion: These findings suggest that the combined use of these miR panels can identify patients with CRC and individuals with high-grade dysplasia lesions and has the potential to be used clinically in combination with the Fecal Immunochemcial Test.

Disclosure: The results provided in the abstract are unpublished data, although we have prepared a manuscript that has not yet been submitted. However, these findings will be presented in oral communication at the Porto Cancer Meeting, a Portuguese congress in I3s, Porto.

PP0740

COLORECTAL CARCINOMA IN YOUNG PATIENTS, SINGLE CENTRE STUDY FROM KARACHI, PAKISTAN

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Introduction: Colorectal cancer (CRC) is rising among young adults. It is cancer of colon, appendix and rectum. According to GLOBOCON 2018, It is the 4th most common cancer in the world and the third leading cause of cancer mortality. The expected rise of the CRC rate to 60% by year of 2030 globally, will result in 2.2 million new cases and 1.1 million deaths annually. CRC is Heterogeneous,70% cases of CRCs are sporadic in onset, 10-30% are familial and 5-7% are due to hereditary diseases.

Aims & Methods: To evaluate the clinicopathological features of Colorectal Cancer (CRC) among young adults under the age of 30).

This was cross-sectional Retrospective study conducted at the Gastroenterology and Oncology Department of JPMC, Karachi from Jan. 2018 till Dec. 2022. All patients with CRC under age 30 were included in the study. Diagnosis was made on Colonoscopy and histopathology confirming CRC. The data obtained was analyzed on the statistical software SPSS version 23. Descriptive statistics were obtained by frequencies and percentages.

Results: In this study total 566 patients of CRC and Young patients 117 were analyzed. The majority, 67 (57.3%) patients were male and 50 (42.7%) were female. Most commonly present in Urdu speaking 47(40%) followed by Sindhi 40 (34%) and Pathan and Panjabi 13(11%). Smoking has a strong association,13(9%) were active smokers followed by Naswar 8 (6%).Most common area was from Orangi Town 8(6%) followed by Korangi 3(4%) Colonoscopy revealed polypoidal growth in 79 (67%) of patients followed by ulcerated lesions in 23 (19.7%). The most common site of CRC was Rectum affecting 48 (41%) followed by Sigmoid colon 46 (39.3%) and the least common site was Ascending colon 7(2.6%). Moderately differentiated Adenocarcinoma was the most common histological type found in 96 (82.5%) and 17 (10%) having signet ring type. The common clinical presentation was abdominal pain 64 (53%) constipation 45 (38.5%) and Bleeding per Rectum 42 (36%).

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Conclusion: CRC is the leading cause of mortality. It is on rising pattern among young adults, most commonly involving the left side of the colon. We need more studies to make guidelines for screening and surveillance of CRC in our population.
Disclosure: Nothing to disclose.

PP0741

IMPACT OF SEDATION ON POST COLONOSCOPY COLORECTAL CANCER (PCCRC) DEATHS

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Introduction: Use of sedation is associated with a positive effect on endoscopists ADR and a higher proportion of colonoscopies with a withdrawal time > 10. However, the impact of sedation on the risk of post colonoscopy colorectal cancer (PCCRC)deaths remains unclear.
Aims & Methods: Retrospective cohort study within a large population based screening program database in Austria, the association of sedation in screening colonoscopy with post-colonoscopy colorectal cancer death was analysed. Logistic regression was used to estimate the effect of sedation on PCCRC-death.
Results: 407.598 screening colonoscopies were analyzed. In 371.132 (91,1%) sedation was used. 46.8% of sedated patients were men compared to 68,2% in unsedated patients (HR 1.94 (1.60, 2.35), p<0,001). Coecum was reached in 97.3% of sedated and 95.5% of unsedated individuals. (HR 4.22 (3.06, 5.82) p<0,001).
Cumulative incidence of PCCRC in sedated individuals was 0.05 (95% CI 0.05-0.5) after 4 years, 0.07% (0.07-0.07) after 6 years, 0.11% (0.11-0.11) after 8 years, 0.14 (0.14-0.14) after 10 years, 0.18% (0.18-0.18) after 12 years. In unsedated patient, cumulative incidence was 0.1% (0-1-0-1) after 4 years, 0.15% (0.15-0.15) after 6 years, 0.22% (0.21-0.23) after 8 years, 0.25% (0.24-0.26%) after 10 years, 0.27% (0.26-0.28) after 12 years. Hazard ratio was 1.31 (0.94-1.83, p=0.1) for sedation in PCCRC-deaths.
Conclusion: Incidence of PCCRC-death was higher in unsedated patients, however this effect was not significant and probably an effect of lower cecal intubation rate within unsedated patients.
Disclosure: Nothing to disclose.

PP0742

COLONOSCOPY IN ELDERLY VS YOUNG SUBJECTS: WHAT INDICATIONS ARE ASSOCIATED WITH HIGHER RISK LESIONS OF COLORECTAL NEOPLASIA?

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Introduction: The indications for colonoscopy are clearly defined and its main objective is to look for neoplasia or lesions at high risk of colorectal neoplasia, even if benign lesions are much more frequent. Common indications for young adults < 50 years include rectal bleeding and diarrhea, while for subjects over 50 years, the indications may be larger given the risk of colorectal cancer, despite the recent increase in neoplastic lesions in young adults.
Aims & Methods: A retrospective descriptive and analytical study, including 1518 patients, was carried out over 3 and a half years. The patients were divided into 2 groups: group A for patients aged less than 50 years, and group B for patients aged greater than 50 years. We compared epidemiological, clinical, and endoscopic data between the two groups. Patients known to have inflammatory bowel disease (IBD) were excluded from our study. The statistical analysis was carried out by the JAMOVI software.
Our study aims to compare the indications and results of colonoscopy between those under and over 50 years old and to determine the indications associated with a risk of neoplastic lesion.
Results: 65.1% of patients were over 50 years old, and 34.9% were under 50 years old. The average age was 37.5 ± 9 years in group A and 64.4 ± 9 years in group B. The female gender predominated among those under 50 years old (55.3%), while the male gender predominated among those over 50 years old (56.3%) (p<0.001).
History of colorectal cancer predominated among those over 50 years old at 10.5% (vs 4.4%; p<0.001). The indications for colonoscopy were dominated by constipation in patients in group B at 29.1% (vs 22%; p=0.01) and by diarrhea in patients in group A at 33.3% (vs 19. 3%; p<0.001).
Colonoscopy was pathological in 49.1% of patients in group B vs 32.9% of patients in group A (p<0.001). It was dominated by colitis or rectocolitis (47.9%) in subjects under 50 years old (vs 11.7%; p<0.001), by polyps (61.5% vs 41.3%; p<0.001) and colorectal neoplasia (16% vs 5%; p=0.002) in older patients.
A higher risk of neoplastic lesions was associated with rectal bleeding and constipation (p < 0.001) in subjects > 50 years old, and with constipation in subjects < 50 years old (p = 0.007).
Conclusion: The presence of neoplastic lesions was strongly associated with the presence of constipation in subjects < 50 years old and with rectal bleeding and constipation in subjects > 50 years old. Early screening with colonoscopy might be beneficial for young individuals with these higher-risk lesions of colorectal cancer.
Disclosure: Nothing to disclose.

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PP0743

ANTICOAGULATION IS ASSOCIATED WITH EARLIER GASTROINTESTINAL CANCER DETECTION: A RETROSPECTIVE STUDY

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Introduction: A significant amount of the population, especially the elderly, are receiving antithrombotic drugs. There is a significant association between antithrombotic therapy (AT) and gastrointestinal bleeding that leads to gastrointestinal cancer detection.

Aims & Methods: The aim of our study is to investigate the possibility that the administration of AT contributes, through the early onset of symptoms, to the early diagnosis of digestive neoplasms. Patients of the department's gastroenterology and oncology units who were followed for gastric or colon neoplasm were analyzed retrospectively. Exclusion criteria included screening colonoscopy, and patients with idiopathic inflammatory bowel diseases or neuroendocrine neoplasms. AT was defined as anticoagulant or antiplatelet therapy.

Results: 152 patients were included, with a mean age of 69.6 ± 10 years, of which 89 were men (58.6%). The majority (84.2%) had colorectal cancer. Forty-nine patients were diagnosed with a bleeding episode (32.2%) and 42 in the context of anemia investigation (27.6%). Fifty-five patients were receiving AT (36.2%). The main reason for receiving AT was the presence of coronary artery disease. Most patients were diagnosed with cancer stage III (45.4%) and II (30.3%). A statistically significant association was observed between the use of AT and an earlier cancer stage at diagnosis (p=0.001). 54.5% of patients receiving AT were diagnosed at an early stage (I or II), compared to 30.9% of patients not receiving AT. Patients diagnosed after a bleeding episode or onset of anemia while receiving AT had an earlier stage of disease at diagnosis than the rest of the sample (p=0.002). 62.5% of these patients were diagnosed at an early stage (I or II), compared to 32.1 % of patients in other groups.

Cancer Stage	AT	No AT
I	8 (14.5%)	6 (10.9%)
II	22 (40%)	24 (24.7%)
III	22 (40%)	47 (48.5%)
IV	3 (5.5%)	20 (20.6%)
Total	55	97

Table: Cancer stage at diagnosis according to AT use
AT: antithrombotic therapy

Conclusion: Antithrombotic drugs may promote an earlier diagnosis of neoplasms of the digestive tract due to the earlier appearance of bleeding, diffuse or occult. In this subset of patients, a bleeding event, barring significant short term morbidity, may facilitate earlier diagnosis and treatment of the underlying disease..

Disclosure: Nothing to disclose.

PP0744

EFFICACY OF HEMOSTATIC POWDER (UI-EWD) IN REDUCING COAGULATION SYNDROME IN PATIENTS WITH COLONIC NEOPLASMS REMOVED BY ENDOSCOPIC SUBMUCOSAL DISSECTION (ESD)

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Introduction: Endoscopic submucosal dissection (ESD) is a widely used and effective treatment modality for colonic neoplasm, such as lateral spreading tumors or early colorectal cancer. However, ESD can cause several procedure-related complications. Coagulation syndrome (CS) is one of the common complications described as localized abdominal pain, fever, and leukocytosis after ESD temporarily.

Aims & Methods: This study aims to evaluate the effect of reducing the incidence of CS by administering UI-EWD after ESD and the practicability and safety of applying UI-EWD to post-ESD ulcer sites. CS occurs more frequently when a tumor size ≥20 mm and a lesion located in the left side colon. This study was conducted retrospectively on 79 consecutive patients were treated with ESD between November 2022 and January 2024. We evaluated the rate of CS and the rate of bleeding within 30 days. CS was defined as localized abdominal pain and fever at the ESD site that occurred within 3 days of ESD without perforation.

Results: The total incidence of CS in enrolled patients with CS high risk is large colorectal tumor is 21.5% (14/65). The incidence of CS after ESD was statistically significantly lower in patients who applied UI-EWD. (0% (0/14) vs 27.5% (14/51), p=0.029) The rebleeding rate within 30 days was not observed in the UI-EWD application group, but there was no statistically significant difference. (0% (0/14) vs 9.8% (5/51), p=0.347) No UI-EWD-related adverse events were observed.

Conclusion: Less CS was observed in the UI-EWD group. It was suggested that the application of UI-EWD after colon ESD in CS high-risk group has a good association with a decrease in CS occurrence. To obtain significant information on the performance of UI-EWD in a decrease of CS, a well-designed, large-scale, randomized control study is needed.

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Disclosure: Nothing to disclose.

PP0745

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PP0746

IMPACT OF COLORECTAL SCREENING ON THE MANAGEMENT OF T1 COLORECTAL CANCERS

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Introduction: The incidence of early-stage colorectal cancer (CRC) rises with the introduction of colorectal screening programs; there is a shift towards less invasive treatment in screen-detected compared to non-screen-detected CRCs.

Aims & Methods: This retrospective, single-center cohort study aimed to assess the impact of colorectal screening on the incidence of submucosal invasive (pT1) CRC and compare treatment characteristics of screen-detected and non-screen-detected lesions between 2010 and 2023.

Screen-detected pT1 CRC (involving lesions detected with the pilot CRC screening in Csongrád-Csanád county in 2015, and those with the Hungarian population-based screening program [EFOP-1.8.1-VEKOP-15-2016-00001] initiated in 2019) lesion, patient and treatment characteristics were compared to pT1 lesions detected outside the screening program in average-risk patients aged 50-70 years during the screening period (2015-2016 and 2019-2023). The impact of screening on pT1 CRC incidence was assessed with investigating new pT1 CRC cases throughout the screening period (2015-2016 and 2019-2023) compared to non-screening periods (2010-2014 and 2017-2018).

Results: During the 14-year study period, a total of 182 pT1 CRCs were detected in average-risk patients aged 50-70 years. 81 of these lesions (44.5%) were detected during non-screening periods. During the screening period, 44 lesions were screen-detected (24.2%, 16 of these detected during the pilot screening) and 57 lesions were non-screen-detected (31.3%). Mean pT1 incidence was higher in the screening than in the non-screening periods (14.7 vs. 11.6 lesions/year, p<0.01); with the highest incidence during the pilot screening (2015-2016; 20 lesions/year), and the lowest during the pandemic (2020-2021, 9 lesions/year).

Patient and lesion characteristics were similar in screen-detected and non-screen-detected lesions (mean age: 63.1 vs 63.5 years, male ratio: 62.8% vs 64.3%, follow-up period: 3.85 vs 3.70 years; location: 31.8% vs 33.3% rectal, morphology: 34.1% vs 42.1% pedunculated, mean lesion size: 21.0 vs 24.7 mm).

No difference in histologic prognostic factors were found between the two groups. Treatment choices were similar for both screen-detected, and non-screen-detected lesions: local excision in 65.9% vs 72.2% (incl. ER in 63.6% vs 63.2%, and minimally invasive local surgical resection in 2.3% vs 7.0%), and surgical oncologic resection in 34.1% vs 29.8%, respectively; p=0.167. Screen-detected colon cancers were more likely to be treated with surgical oncologic resection (43.3% vs 28.9%, p=0.218), while screen-detected rectal cancers were preferred to be treated with local excision (85.7% vs 68.4% [ER: 78.6% vs 47.4%], p=0.416) compared to non-screen-detected lesions.

During the follow-up, adverse outcomes (local recurrence and/or dissemination) occurred in two screen-detected, and in one non-screen-detected patient, p=0.797. Lymph node involvement rate was similar in those undergoing surgical oncologic resection in screen-detected, and non-screen-detected patients (6.7% vs 5.9%, respectively; p=0.927).

Conclusion: The incidence of submucosally invasive CRC rises with the implementation of colorectal screening program. There is no difference in the management and outcomes of screen-detected and non-screen-detected pT1 CRCs, but local excision, especially endoscopic resection is preferred for screen-detected rectal lesions.

Disclosure: Nothing to disclose.

PP0747

NEOADJUVANT THERAPY IN RECTAL CANCER - ONE YEAR FOLLOW-UP RESULTS OF STANDARD VERSUS TOTAL NEOADJUVANT STRATEGIES

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Introduction: Locally advanced rectal cancer (LARC) poses a significantly challenge in clinical management, involving multimodal treatment approach.

Among innovative strategies, Total Neoadjuvant Therapy (TNT) has emerged, delivering all planned chemotherapy before surgery.

Aims & Methods: Our aim was to evaluate the real-world application and efficacy of TNT and to compare it with non-TNT strategy.

This retrospective study compared 107 locally advanced rectal adenocarcinoma patients treated with Total Neoadjuvant Therapy (TNT) from 2022 and traditional chemoradiotherapy (CRT) from 2020-2021. Evaluation included primary endpoints of efficacy, tolerability, and completion rates.

Results: Among 107 patients (54.2% male, mean age 62.48 years), non-TNT and TNT follow-ups were 26.66 and 8.19 months, respectively. No differences in gender(p=0.163), staging (p=0.707), or location (p=0.727) were noted. TNT patients received more short-course radiotherapy (42.5% vs1.5%, p<0.001).

Clinical responses favored TNT (p= 0.030) with no significant differences in pathological responses, recurrence rates, or survival. TNT exhibited higher chemotherapy completion (p = 0.007) and lower adverse events (p<0.001). Post-surgery events showed no significant differences (p=0.470).

Conclusion: In conclusion, our data add to the body of literature favoring the TNT treatment strategy for locally advanced rectal cancer, aiming to achieve higher complete response rates with less adverse events.

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Disclosure: Nothing to disclose.

PP0748

WATCH AND WAIT STRATEGY IN RECTAL CANCER – FOLLOW-UP RESULTS FROM A TERTIARY ONCOLOGY CENTRE

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Introduction: The treatment of rectal cancer (RC) has been evolving in recent decades, aiming at improving organ preservation. This work evaluates the results of the Watch & Wait (W&W) strategy in a tertiary cancer centre.
Aims & Methods: Patients diagnosed with adenocarcinoma of the rectum up to 6cm from the anal margin, cM0, >cT1N0, with diagnosis and first multidisciplinary consultation between June 2014 and June 2020 were included. All patients received neoadjuvant radiotherapy ± chemotherapy and underwent reassessment at 8-12 weeks. Patients with complete clinical response (cCR), defined by proctological examination, fibrosigmoidoscopy and magnetic resonance imaging, initiated surveillance in the W&W protocol and the other were offered surgery. We present the follow-up of patients until December 2023.
Results: The study included 137 patients, 56% male, mean age 63.7 ±15 years. Stage at diagnosis: III (75%), II (13%) and I (12%). After neoadjuvant treatment, 18 patients (13.6%) had a complete response. Stage I at diagnosis was associated with a higher probability of achieving cCR (p=0.008). All patients without CRC (113) underwent surgery: 50.4% abdominoperineal resection and 40.7% anterior resection of the rectum. The operative specimen revealed N+ disease in 23.7% of patients and complete pathological response in 14. Median follow-up time of 44 months ± 26 (3.6 years). In the W&W group, 6/18 patients presented regrowth, between 3 and 35 months (mean at 15.8 months), 4 were treated with surgery (2 refused), 1 of 4 had local recurrence after surgery). None of the patients had distant disease. At the end of follow-up, 15 patients were alive without disease, 2 alive with disease, and 1 died from disease.
Of the remaining 113 patients: 28 (26.4%) patients showed distant recurrence and 3 patients local and distant relapse. At the end of follow-up, 70 (62%) of patients (62 had no disease, 7 had disease, 1 was unknown) and 11.4% had death attributed to disease.
Comparing the two approaches, there is no difference between local recurrence, but there is a higher probability of distant recurrence by the traditional approach (p<0.009) as well as death (p=0.006).
When comparing patients under WW approach with patients with complete pathological response after traditional approach, there was only a higher probability of local recurrence in WW (0.011), with no statistically significant differences in distant recurrence or mortality.
Conclusion: In this cohort, with a longer follow-up period, the W&W approach continues to be a safe oncological approach, as described internationally.
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Disclosure: Nothing to disclose.

PP0749

MANAGEMENT OF COLORECTAL CANCER IN PATIENTS WITH LIVER CIRRHOSIS - RETROSPECTIVE STUDY IN A TERTIARY CENTER

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Introduction: The actual screening of colorectal cancer (CRC) includes people of 45 years old and patients with alcoholic liver cirrhosis (LC) have an elevated risk of developing colon cancer. Curative treatment of CRC in patients with an existing diagnosis of LC can be a challenge, requiring major surgical resection and may include liver-toxic chemotherapy. Postoperative morbidity and mortality are elevated in patients with LC. For improved survival, optimization of liver function is more important than adjuvant chemotherapy.
Aims & Methods: Our study included 149 hospitalized LC patients Classes Child- Pugh A, B and C, meanage 56 years old, during April 2023-March 2024, of which 13 cases presented lower bleeding, iron deficiency anemia, intestinal transit disorders, abdominal pain, and familial history of CRC. Clinical and paraclinical diagnostic of CRC in LC patients were made by digital rectal examination, hematological samples, rectosigmoidoscopy, total colonoscopy, abdominal ultrasound, computed tomography and magnetic resonance imaging.
Results: Biological tests indicated anemia, trombocitopenia, increased ESR, biochemical liver tests increased, occult bleeding (FIT), and the increase of the tumor markers CA19-9, AFP, CEA that is particularly useful for monitoring treated CRC. Flexible rectosigmoidoscopy detected six cases of rectal cancer, three cases of rectosigmoidian tumors, and total colonoscopy was the “gold standard“ imaging investigation that detected two cases of descending colon cancer, one case of hepatic flexure tumor and one patient with ascending colon cancer. Of patients with LC Class Child Pugh A, 6 cases were operated by laparotomy, being referred to Oncology Institute with favorable outcomes, and the rest of seven patients with LC Class Child Pugh B and C had advanced CRC, being treated by chemotherapy. Abdominal ultrasound visualizes liver metastases, abdominal lymph nodes and CT, MRI were the main methods used for staging CRC before surgical and oncological treatment.
Conclusion: Colonoscopy is the exploration of choice in CRC detection with 97% sensitivity, providing histopathological examination by tumor biopsy and also diagnosis of synchronous cancer. Early diagnosis of CRC is essential to prevent complications and increase the survival rate in LC. Men with LC are predominantly affected of CRC. Abdominal ultrasound, CT or MRI are the main imaging investigations recommended in CRC staging before the surgical and oncological treatment. Laparotomy should be preferred in patients with LC. Survival rate is strongly associated with the improvements in liver function. Echoendoscopy, immunoscintigraphy with monoclonal antibodies can be useful in CRC and genetic testing of tumor DNA markers in stool can be recommended in families with genetic syndromes (Turcot, FAP).
Disclosure: Nothing to disclose.

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PP0750

OPTIMIZING ENDOSCOPIC MANAGEMENT OF MALIGNANT COLORECTAL POLYPS: INVESTIGATING THE FACTORS HINDERING CURATIVE RESECTION

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Introduction: Endoscopic recognition of malignant colorectal polyps is challenging, as most do not exhibit obvious signs of submucosal invasion, despite advancements in optical diagnostic techniques. While endoscopic resection might achieve cure, comprehensive data on cure rates and predictive factors are lacking.

Aims & Methods: This study aimed to elucidate the endoscopic predictors associated with achieving cure for malignant polyps. We conducted a retrospective analysis of electronic medical records for patients undergoing colonoscopy with endoscopic mucosal resection at our institution. Patients with confirmed colorectal adenocarcinoma classified as stage T1 were included and those with biopsied and tattooed polyps referred for surgery were excluded. Data regarding polyp size, morphology, location, and indirect signs of submucosal invasion (non-lifting sign; chicken-skin sign; edge retraction; depressed areas; fold convergence; ulceration) were collected. Endoscopic cure was defined by negative margins (>1mm) and favorable histological prognosis factors.

Results: A total of 165 patients were included, 57.6% males, with a mean age of 65 years. A total of 170 malignant polyps were resected with a global cure rate of 33.5%. Pedunculated polyps accounted for 37.1% and exhibited a significant association with endoscopic cure (p<0.001, OR 3.4, 95%CI 1.74-6.6) compared to nonpedunculated polyps (Paris classification 0-Is, any 0-II). Most pedunculated polyps were localized in the left colon (87.3%), and 50.8% were resected with cure. Interestingly, neither a large (p=0.532) nor a short stalk (p=0.672) nor polyp size (p=0.308) impacted the likelihood of cure. In contrast, the cure rate for nonpedunculated polyps was 23.4%. The distribution of nonpedunculated polyps was 38.3% in the rectum, 43.9% in the left colon, and 17.8% in the right colon. A "piecemeal" endoscopic resection (p<0.001), the presence of indirect signs of submucosal invasion (p<0.001) and size (cure: 19.8±7.4mm, no cure 26.4±12mm, p=0.01) were associated with non-curative resections. However, on multivariate analysis, only *en-bloc* resection emerged as an independent factor associated with cure (P=0.02, OR 5.6, 95% CI 1.3-24).

Conclusion: Our data underscores endoscopic resection viability for potentially malignant pediculated polyps, irrespective of size or stalk type. However, sessile/flat polyps present challenges, particularly when large or with endoscopic signs of submucosal invasion. In these scenarios, *en-bloc* removal using advanced techniques might contribute to better outcomes.

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Disclosure: Nothing to disclose.

PP0751

ENDOSCOPIC SUBMUCOSAL DISSECTION OF ANORECTAL JUNCTION: RETROSPECTIVE REVIEW

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Introduction: Endoscopic submucosal dissection (ESD) of anorectal junction lesions (ARJL) presents technical challenges. This location may be associated to increased risk of adverse events, such as anal stricture, fecal incontinence, and chronic pelvic pain. We aim to identify the safety and effectiveness of the ESD for ARJL.

Aims & Methods: We conducted a retrospective cohort study on consecutive patients undergoing ESD for rectal submucosal lesions between 2018 and 2023 at two centers. Patients without follow-up were excluded. Lesions were categorized as ARJL or proximal rectal lesions (RL). Complications were assessed starting from 12 hours post-procedure.

Results: Our study included 61 patients (54% male; mean age 71±9 years). Forty patients had RL and twenty-one had ARJL. 89% of the patients had an R0 resection. 90% had en bloc resection. On histology 62% of the lesions were low-grade dysplasia adenoma, 16% were high-grade dysplasia adenoma, 7% had intramucosal adenocarcinoma and 15% had adenocarcinoma with submucosal invasion (T1), of which 33% had deep submucosal involvement (sm1), all RL. 3% were submitted to surgery. Mean procedure times were 95±63 and 128±117 minutes (p=0.21) with mean sizes of 36.72±16.31 mm and 36.94±25.16 mm (p=0.96), for RL and ARJL respectively. Median follow-up time was 18 months (IQR 7-39). In total, only five patients developed complications during follow-up: From the ESD of ARJL, 1 patient developed an asymptomatic stenosis and from the ESD of RL, one also developed an asymptomatic stenosis, two patients developed post-polypectomy syndrome, one had delayed bleeding and one had chronic pelvic pain.

Conclusion: Our study identified a low incidence of adverse effects associated with ESD of the ARJL, being categorized as mild, indicating favorable tolerability and safety of these interventions. Limitations include the retrospective design and small sample size. Further research is needed to validate these findings and to compare ESD and transanal surgery outcomes.

Disclosure: Nothing to disclose.

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PP0752

ESD VS EFTR IN SMALL PT1 LESIONS' RESECTION, A COST-EFFECTIVENESS ANALYSIS

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Introduction: Endoscopic submucosal dissection (ESD) and Endoscopic Full-thickness Resection (EFTR) represent treatments of choice in the endoscopic resection of small gastrointestinal (GI) lesions with superficial submucosal invasion. It is often challenging to assess the best curative option in daily practice, since ESD is thought to be more technical demanding, and requires a longer learning curve.

Aims & Methods: We conducted a monocentric retrospective study, analyzing all GI pT1 lesions < 35 mm resected by either ESD or EFTR in our Unit from 2016 and 2023. All procedures were performed in a Day Hospital setting. As primary endpoints we assessed en bloc resection, R0 resection (clear deep margins) and curative resection (no further need for surgery) rate. As secondary endpoints we analyzed mean procedural time, cost and adverse events that occurred within or after both procedures. Chi-square test, Fisher Test and T-test were performed for categorical variables.

Results: 40 ESD and 37 EFTR were enrolled. En bloc resection was achieved in, respectively, 87.5 % (35/40) and 97% (36/37), without reaching statistical difference (RR 0.9, p= 0.74), likewise R0 resection (30/40, 75% vs 36/37, 97%, RR 0.77, p=0.44) and curative resection (22/40, 55% vs 25/37, 67%, RR 0.82, p= 0.57). Adverse events analysis showed no statistical difference between the two cohorts (4/40, 12% vs 3/37, 8%, RR 1.5, p= 0.79). Mean procedural time appeared significantly lower in EFTR, (69.85 ± 31.94 min vs 48 ± 16.9 min, p=0.0002), whilst mean procedural cost per hospitalization day was lower in ESD than EFTR (2746 € ± 1003 vs 3458 ± 746 €).

Conclusion: ESD and EFTR turned out to be equally successful in terms of resection outcomes in pT1 small lesions. Nevertheless, ESD appeared to be more cost-effective in our study, and therefore should be confirmed as gold standard in daily practice. However, EFTR could allow easier and faster en bloc resection in most challenging procedures, representing a valid alternative in less experienced settings.

Disclosure: Nothing to disclose.

PP0753

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PP0755

LONG-TERM RESULTS OF THE ENDOSCOPIC RESECTION OF APPENDICEAL LESIONS

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Introduction: Endoscopic resection of lesions involving the appendiceal orifice is an alternative to surgery. Long term outcomes have not been determined.

Aims & Methods: To determine the long-term efficacy and safety of endoscopic resection of appendiceal lesions.

Retrospective analysis of the endoscopic resection of lesions with appendiceal involvement was performed in seven centers in Spain between January 2016 and July 2022. The technical success rate (complete macroscopic resection at index procedure), recurrence rate and short- and long-term complications were evaluated, including delayed appendicitis with a clinical follow-up of at least 12 months.

The intra-appendicular extension of the lesion was described as superficial (visible appendiceal margin) or deep (if it was not possible to determine the appendiceal margin of the lesion before resection) and the circumferential extension was described according to the affected percentage.

Results: 98 lesions with appendiceal involvement were treated, 73 had a clinical follow-up >12 months with a median follow-up of 26 months ([IQR]16-37 months). There were 35 lesions with deep intra-appendicular involvement and 63 with superficial appendiceal involvement, 64 with a circumferential involvement ≥50%. The median size of the lesions was 20mm([IQR]10-30mm) and 19(19%) had previous manipulation. Underwater endoscopic mucosal resection (UEMR) was performed in 53% of the lesions (24% by underwater cap-suction pseudopolyp formation, CAP-UEMR) with a technical success of 93% (49% en bloc and 51% piecemeal).

Among these cases, there were 7 incomplete resection, with 3 patients undergoing surgery and the remaining 4 undergoing endoscopic treatment during a subsequent session.

Within the entire cohort, there were 3 cases of malignant histology: one low-risk pT1 which underwent surgery resulting in no residual cancer and negative lymph nodes, another low-risk pT1a (which chose follow-up and experienced no recurrence), and a third with suspected incomplete resection and pT2 cancer, necessitating surgery.

There were 6 intraprocedural bleeding and 2 delayed bleeding, all managed endoscopically. 45 patients completed at least 12 months of endoscopic follow-up (median 18, [IQR] 8-29months), showing recurrence in 12

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cases (6 lesions with deep intra-appendicular extension), 8 (67%) treated endoscopically. There was 1 case of intraprocedural perforation, managed endoscopically.

Surgery was needed during follow-up in 7 additional cases [recurrence=3; pT2 cancer =1; pT1a =1; appendicitis=2 (at 11 months and 56 months post-endoscopic resection)], in total 10 surgeries were needed in the entire cohort.

INTRA-APPENDICULAR EXTENSION	Superficial (n=63)	Deep (n=35)	p-value
Age years(mean+/-SD)	67+/-8.8	71+/-10.6	0.040
Female, n(%)	30(48%)	12(34%)	0.201
Antithrombotic agents	20(32%)	13(37%)	0.588
Lesion size, median mm(median[IQR])	15(8-20)	30(19-45)	<0.001
Circumferential extension			<0.001
<50%	29(46%)	5(14%)	
50-90%	28(44%)	17(49%)	
100%	6(10%)	13(37%)	
Previous manipulation	12(19%)	7(20%)	0.516*
Complete macroscopic resection	60(97%)	31(88%)	0.065*
En bloc resection	39(62%)	9(26%)	<0.001
Clinical follow -up,months(median[IQR])	25(15-34)	27(17-49)	0.154

Table 1- Baseline characteristics of patients and resected lesions
**Fisher's exact test applied.*

Conclusion: The endoscopic management of lesions involving the appendix may indeed prove effective and safe after following a minimum follow-up of 12 months, offering a viable alternative to surgery. It is important to note that delayed-onset acute appendicitis, though rare can be observed beyond 11 months post-procedure, although a casual mechanism cannot be proven.

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Disclosure: Dr Hugo Uchima collaborates as proctor with Erbe Spain and Olympus Iberia. Has received congress registration from Casen Recordati.

PP0756

STOMA-FREE SURVIVAL AFTER PALLIATIVE COLONIC STENTING

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Introduction: Self-expandable metal stents (SEMS) are the preferred treatment of malignant large bowel obstruction in those individuals who are not likely to have a curative resection (ESGE recommendations 2020). The differentiation between palliative and surgical candidacy may not be possible at presentation. We wished to measure the stoma-free prognosis of palliative patients, who do not have a surgical resection.

Aims & Methods: We analysed all patients undergoing SEMS at our centre over 10 years, collecting data on stoma formation, significant events and survival. Palliative stenting was defined in those who never underwent a resection of their primary obstructing cancer. All patients who had a technically successful SEMS were followed (a stent was placed across the stricture) so cases where there was early clinical failure were included. In our centre, we opt to attempt SEMS in all areas of the colon including the right colon.

Results: 171 patients with malignant large bowel obstruction (mean age 72.6, 57% male, 31% right-sided with obstruction beyond the splenic flexure) were referred for SEMS and no resection of the primary cancer was undertaken (this was 80% of total patients referred for SEMS). In 31% of patients, a stated plan for curative or palliative intent was amended in later decision-making. Where a stent was successfully deployed (150 patients, 88% technical success), 96% of patients avoided a stoma at any time in their palliation and median stoma-free survival was 283 days (9.4 months). Palliative chemotherapy was administered in 38.4% of cases. In those 21 cases where stenting was unsuccessful, 15 had surgery with a stoma formed and 6 were palliated. No procedural perforations occurred. Re-stenting of the primary cancer was required in 15 cases out of 150 (10%) at a median time of 6.5 months. Long term survival showed 33% of patients lived at least 1 year, 16% of patients survived more than 2 years, and 3.6% more than 3 years. Four patients (2.7%) were found to have incidental stent migration on CT to assess response to chemotherapy, but no further intervention (re-stenting or surgery) was required. Three patients may have died prematurely from COVID. Factors associated with significantly lower rate of re-intervention were obstruction in the proximal colon and older age.

Conclusion: In palliative SEMS management of malignant bowel obstruction, 96% of patients can avoid lifetime stoma surgery and can expect a median survival of 283 days.

Disclosure: Nothing to disclose.

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PP0757

SURGERY AFTER ENDOSCOPIC REMOVAL OF A MALIGNANT POLYP: IS RESIDUAL DISEASE A REAL ISSUE?

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Introduction: The implementation of colorectal cancer screening programmes has been associated with an increase in the incidence of malignant polyps removed endoscopically. In those without endoscopic cure criteria, surgery can be performed to improve oncological outcomes.

Aims & Methods: Evaluate factors associated with the presence of residual malignant disease in the surgical specimen after endoscopic removal of a malignant polyp without endoscopic cure.

Retrospective, cohort study including patients with a malignant polyp endoscopically removed without endoscopic cure, who underwent surgery. The absence of endoscopic cure was defined as positive vertical resection margins (< 1 mm), poorly differentiated grade, depth of submucosal invasion > 1 mm, presence of lymphatic or vascular invasion, or presence of tumour budding. Patients with a synchronous neoplastic lesion and those with a malignant polyp without complete endoscopic removal were excluded.

Results: Included 74 consecutive patients submitted to surgery after the endoscopic removal of a malignant polyp without endoscopic cure. Most patients were male (62.2%), with a mean age of 64 ± 10 years. Regarding the endoscopic mucosal resection (EMR), 45 polyps were removed in one fragment (60.8%) and the other 29 were submitted to piecemeal EMR (39.2%). In 15 cases, snare tip spray coagulation (STSC) was performed at the margins of mucosal defects after EMR, 6 in normal-appearing margins (8.1%) and 9 in margins suspected of residual adenomatous tissue (12.2%). Considering the post-operative outcomes, 7 patients required surgical re-intervention (9.5%), 1 was admitted in the intensive care unit (1.4%), 1 patient died (1.4%), and 2 patients required re-hospitalization in the following 6 months (2.7%).

The presence of residual malignant disease in the surgical specimen occurred in 9 patients (12.2%), 7 had intramural disease (9.5%) and 2 had nodal metastasis (2.7%). Patients with polyps with a flat component were 5 times more likely to have residual disease in the surgical specimen (p=0.043). The polyp size and its location in the right colon were not associated with the presence of residual disease in the surgical specimen (p=0.140 and p=0.654, respectively). Patients with polyps submitted to piecemeal EMR were 7 times more likely to have residual disease in the surgical specimen (p=0.024), with 2 patients with residual disease after EMR in one fragment (4.4%) and 7 patients with residual disease from the piecemeal EMR group (24.1%). Those in which STSC was performed were also more likely to have residual disease in the surgical specimen (p<0.001). No statistically significant differences were found between the presence of positive margins, lymphatic or vascular invasion or well or moderately differentiated histologic grade, and the presence of residual disease in the surgical specimen (p=1.000, p=0.675 and p=0.686, respectively). Tumour budding was described in 3 patients, none of which had residual disease in the surgical specimen.

Conclusion: A minority of patients had residual malignant disease in the surgical specimen, with a higher risk for those with polyps with a flat component or that were submitted to piecemeal EMR. In fact, with piecemeal EMR almost one quarter of patients had residual disease e less than 5% of those with EMR in one fragment had residual disease in the surgical specimen, highlighting the need for a careful evaluation of the polyp before endoscopic removal and the importance of removal in one fragment, in case of suspected submucosal invasion.

Disclosure: Nothing to disclose.

PP0758

WITHDRAWN

PP0759

CLINICAL OUTCOMES OF ENDOSCOPIC RESECTION FOR T1 COLORECTAL CARCINOMA - SURVEILLANCE OR SURGICAL TREATMENT?

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Introduction: Complete endoscopic resection and accurate histological evaluation for T1 colorectal cancer (CRC) is critical to determine subsequent treatment. Endoscopic full-thickness resection (EFTR) and endoscopic submucosal dissection (ESD) are potential curative methods for treatment of T1 CRC.

Aims & Methods: Primary aims were R0 resection rate, histopathological risk-assessment, and curative resections. Secondary aims were short-term outcomes of patients with T1 CRC after endoscopic and surgical resection.

Consecutive EFTR and ESD procedures for T1 CRC at one tertiary endoscopic centre, prospectively recorded in National CRC Registry between 2015 and 2023, were retrospectively analysed. High risk factors included positive resection margin, low-grade differentiation (G3, G4), deep submucosal invasion (DSI) and lymphovascular invasion (LVI).

Results: We analysed 178 procedures: 114 (64 %) ESD and 64 (36 %) EFTR. From all resected lesions, 24 (13 %) were T1 carcinomas, 10 (6 %) subepithelial lesions (GIST/NET), 9 (5 %) T2 carcinomas, 2 (1%) T3 carcinomas, 58 (33 %) intramucosal carcinomas and 75 (42 %) low-grade adenomas. From 24 T1 CRC (54 % men, average diameter of lesion 25 mm) were 15 (63 %) lesions resected by EFTR, 7 (29 %) lesions resected by ESD, and 2 (8 %) lesions resected by hybrid technique ESD/EFTR.

Overall R0 resection was achieved in 18 (75 %) lesions. Seven (29 %) T1 CRC had superficial submucosal invasion (≤ 1000μm), 17 (71 %) T1 CRC had deep submucosal invasion, including 1 lesion with lymphovascular invasion. Curative resection rate for resected T1 CRC was 29 % and 71 % after excluding deep submucosal invasion as a risk factor.

Additional oncologic surgery was performed in 9 (38 %) patients, with residual cancer in 1 patient. Endoscopic follow-up was available in 22 (92 %) patients with a median of 3 years and no residual cancer were observed.

Conclusion: EFTR and ESD techniques are effective methods for treatment T1 CRC. Appropriate surveillance without additional surgery may be considered as another option in selected cases after endoscopic resection of T1 CRC.

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Disclosure: Nothing to disclose.

PP0760

THE RELATIONSHIP BETWEEN PRETHERAPEUTIC MALNUTRITION ACCORDING TO GLIM CRITERIA AND CLINICAL OUTCOMES IN PATIENTS WITH COLORECTAL CANCER

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Introduction: Malnutrition is a highly prevalent condition in patients with cancer that has a negative impact on clinical outcomes. The GLIM criteria for malnutrition aim to standardize its diagnosis. In colorectal cancer, poor nutritional status is associated with increased morbidity and mortality and reduce tolerance to treatment.
Aims & Methods: We aimed to assess the prevalence of malnutrition base on the GLIM criteria and to correlate this with patient outcomes and survival in colorectal cancer.
Patients with colorectal cancer diagnosed from January 2015 to December 2022 were included. All of them were screened for malnutrition using the GLIM criteria.
The relationships between GLIM-malnutrition and suvival were analyzed.
Results: Sixty patients were included. The mean age was 54 years [35 - 82] with a sex ratio of 1.8 (M/F). Tumors were located in the right colon in 20 patients (33%), the left colon in 22 patients (36%), and the rectum in 15 patients (25%). According to TNM classification, patients were predominantly classified in stages II and III. In terms of treatment, 20 patients received neoadjuvant chemotherapy (30%), 15 patients received neoadjuvant radiotherapy (25%), and 45 patients underwent surgery (75%). Malnutrition, according to GLIM criteria, was present in 40% of patients. It was classified as mild, moderate, and severe in 40%, 45%, and 15% of cases, respectively.
Regarding survival, the difference between the two Kaplan-Meier curves of patients fulfilling GLIM malnutrition criteria versus patients without malnutrition was statistically significant (Log-rank test p = 0.001), [HR = 4.35 (95% CI = 2 - 9)], with a mean survival period of 18 ± 4.7 versus 32 ± 5.2 months, respectively (p = 0.028). Mild malnutrition was associated with better survival compared to moderate and severe malnutrition: 20 ± 2.3 versus 17 ± 3.2 months, respectively (p = 0.003).
Conclusion: Pretherapeutic screening for malnutrition seems to be mandatory due to its negative prognostic impact in patients with colorectal cancer, allowing for better pre-therapeutic stratification.
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Disclosure: Nothing to disclose.

PP0761

SURGERY AFTER RESECTION OF MALIGNANT POLYPS: BETTER SAFE THAN SORRY?

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Introduction: After resection of a malignant colorectal polyp, patients may be proposed for surgical treatment or surveillance, depending on the presence of cure criteria.
Aims & Methods: The aim of this study was to compare the detection of metachronous lesions in patients with malignant polyps undergoing surgery versus surveillance.
Retrospective cohort study including patients with endoscopic removal of malignant polyps. Patients were divided into 2 groups, according to the strategy defined in the oncology group consultation: surveillance (group 1) and surgery (group 2). The diagnosis of metachronous lesions during the follow-up period was the main outcome studied. A metachronous lesion was defined as the detection of adenoma/adenocarcinoma, 6 or more months after index colonoscopy.
Results: 77 patients were included, 55.8% in group 1, 44.2% in group 2. Mean age was 65.2±10.8 years, 45.5% were women.
In group 1, significantly more patients developed metachronous lesions compared to group 2 (66.7% vs 33.3%, p=0.036). The mean time to diagnosis of metachronous lesion was significantly shorter in group 1 (18.6±2.8 vs 36.0±5.6 months, p=0.004).
In group 1, 32.1% of patients developed metachronous lesions in the same colonic segment as the endoscopic resection of the malignant polyp.
In histological characterization, the two groups did not present statistically significant differences: low-grade dysplasia in 92.9% (group 1) and 85.7% (group 2); high-grade dysplasia in 7.1% (group 1) and 14.3% (group 2), p=0.457. No adenocarcinomas were detected during the follow-up period.
Conclusion: Surgical approach was associated with a lower incidence of metachronous lesions, suggesting that a significant proportion of the risk of metachronous lesion occurs in the segment of the index lesion.
Disclosure: Nothing to disclose.

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PP0762

THE EFFECTS OF SMALL MOLECULE AXIN-STABILIZER (KYA1797K) IN MONOTHERAPY AND COMBINED TREATMENT WITH CANNABINOID AGONIST (WIN 55,212-2) IN COLORECTAL CANCER CELL LINES

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Introduction: Colorectal cancer (CRC) is one of the most common neoplasms world widely. G-protein coupled receptors (GPCR) affect crucial elements of oncogenesis, such as responses to various endo- and exogenous stimuli (e.g. medications), angiogenesis and forming metastases. The magnitude of GPCR signaling is terminated by regulators of G-protein signaling (RGS), which typically possess GTPase activity. AXIN is an atypical RGS without GTPase function. Its main role is stabilization of β-catenin destruction complex in which it cooperates with adenomatous polyposis coli (APC) and glycogen synthase kinase3 (GSK3). Through this complex, AXIN antagonizes prooncogenic canonical Wnt signaling pathway that depends on β-catenin. On the other hand, recently attention has been paid to the potential tumor-suppressive properties of cannabinoids (CB) which are ligands of Gαi/o-type GPCR – CB1 and CB2.

Aims & Methods: In this project we aim to test the therapeutic potential of simultaneous targeting of AXIN and CB receptors with KYA1797K and WIN 55,212-2 in CRC cell lines (SW480, Caco-2) and non-cancerous enterocytes CCD 841 CoN as comparator. We hypothesize that KYA1797K is effective in monotherapy and in combination with cannabinoids in CRC. Moreover, we expect that stabilization of AXIN does not to weaken the effect of CB receptor ligands.

Cytotoxicity of KYA1797K (1; 10; 100, 300 μM) and WIN 55,212-2 (1; 10; 100 μM) in monotherapies and combined treatment (KYA1797K(10μM)+WIN 55,212-2(10μM) [K+W10] and KYA1797K(100μM)+WIN 55,212-2(10μM)) was assessed using MTT test. Relative genes' expression was evaluated using quantitative real-time PCR and Western Blot. Statistical analysis included Shapiro-Wilk test, one-way ANOVA with post-hoc Dunnett's and Tukey's tests.

Results: Caco-2 cells were insensitive to KYA1797K in monotherapy excluding concentration of 300 μM, for which we observed significant cytotoxicity. Treatment with WIN 55,212-2 in concentration of at least 10 μM in both mono- and combined therapy significantly reduced viability of this cell line. Similar profile of changes was observed in SW480 line, however the strength of observed effect and sensitivity to KYA1797K were higher. In CCD 841 CoN cells significant additive effect of KYA1797K and WIN 55,212-2 was observed for 100 μM with no changes observed for 10 μM. Moreover, we found that treatment with K+W10 causes significant cytotoxicity in cancerous SW480 cells but not in normal enterocytes CCD 841 CoN. Moreover, K+W10 treatment resulted in increased expression of AXIN 1 and 2 in each cell line except AXIN 2 in Caco-2 line, in which opposite effect was observed. The combined treatment decreased CB1 receptors' expression in all examined cell lines, whereas opposing tendency was observed for CB2. Furthermore, K+W10 treatment led to increased expression of RGS2, 11, APC and GSK-3. At the protein level K+W10 treatment surprisingly tended to decrease β-catenin relative expression. CB2 receptors were significantly more expressed in SW480 than in CCD 841 CoN that was consistent with cytotoxicity results.

Conclusion: AXIN stabilization is a promising new approach in CRC treatment. AXIN stabilization does not weaken CB1 and 2-dependent effects. The combined treatment of AXIN stabilizer and CB agonist may be especially important as life-prolonging therapy for patients with advanced CRC.

Disclosure: Nothing to disclose.

PP0763

INVESTIGATION OF THE IN VITRO CYTOTOXIC EFFECT OF PALLADIUM(II) COMPLEXES WITH 2-AMINOTHIAZOLE DERIVATIVES ON MOUSE AND HUMAN COLORECTAL CARCINOMA CELL LINES

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Introduction: Over the last four decades, metal-based antitumor medications, notably cisplatin, have been extensively utilized in chemotherapy, despite notable side effects. Colorectal cancer, a significant contributor to cancer-related mortality worldwide, has prompted extensive research into innovative treatment methods. Thiazoles and their derivatives constitute integral components of certain cytostatic agents, garnering considerable attention in studies exploring their biological and pharmacological properties.

Aims & Methods: The primary objective of this research is to examine the in vitro cytotoxic effect of newly synthesized palladium(II) complexes with 2-aminothiazole derivatives on colorectal carcinoma cell lines. The cytotoxic effects of newly synthesized palladium(II) complexes (C1, C2) and corresponding ligands (L1, L2) were determined using the MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide) assay against both mouse (CT26) and human (HCT116) colorectal carcinoma cell lines. Non-tumor cells, specifically mouse mesenchymal stem cells (mMSCs), were employed as controls.

Results: The tested complexes and corresponding ligands reduce the viability of tumor cells in a dose-dependent manner. The C1 complex demonstrated strong cytotoxic activity against all tested tumor cell lines, with the HCT116 cell line being more sensitive to its toxic effects. All tested complexes and corresponding ligands reduced the viability of non-tumor mMSC cells, with mMSC cells being particularly sensitive to the cytotoxic effect of the C2 complex. The IC50 (half maximal inhibitory concentration) value obtained for the C1 complex was higher compared to the gold standard, cisplatin. The C1 complex exhibits the highest cytotoxic capacity towards tumor cells, followed by the highest IC50 values for control.

Conclusion: Palladium(II) complexes with 2-aminothiazole derivatives demonstrate direct antitumor activity in vitro. The C1 complex exhibits a high level of cytotoxicity. Further research is necessary to indicate the potential use of these compounds in experimental tumor therapy.

Disclosure: yes.

PP0764

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PP0765

RISK OF PERFORATION OF SELF-EXPANDABLE METAL STENTS WHO RECEIVED BEVACIZUMAB COMPARED TO STENT ALONE OR NON-BEVACIZUMAB CHEMOTHERAPY IN PATIENTS WITH ADVANCED COLORECTAL CANCER: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Bevacizumab is an anti-angiogenic monoclonal antibody targeting vascular endothelial growth factor used to treat advanced colorectal cancer (CRC). Self-expandable metal stents (SEMS) are employed in cases of advanced obstructive CRC but are linked to adverse events like bleeding, stent migration, and bowel perforation. To date, it is unclear whether bevacizumab-based chemotherapy increases the risk of perforation in patients with SEMS. Therefore, we conducted a systematic review and meta-analysis to evaluate the risk of bevacizumab-related perforation in these patients.

Aims & Methods: We conducted a comprehensive literature search from major databases until December 2023. We identified all potential studies that evaluated the risk of colonic perforation in patients with bevacizumab-based chemotherapy after SEMS insertion. We compared the risk of colonic perforation in patients who received bevacizumab-based chemotherapy with patients who received chemotherapy without bevacizumab or best supportive care. We calculated the odds ratio (OR) and 95% confidence interval (CI) using a M-H random-effects model and I-square method.

Results: Twelve studies, including 1,599 patients, met our inclusion criteria and were included in the meta-analysis. Of 1,559 SEMS patients, 320 received bevacizumab-based chemotherapy, and 1,274 received chemotherapy without bevacizumab or best supportive care. A total of 106 perforations have been reported. The risk ratio for colonic perforation in bevacizumab-based chemotherapy was 2.78 (95% CI: 1.21-6.38, p=0.02; I2=57%) when compared to non-bevacizumab-based therapy. In subgroup analysis, bevacizumab-based chemotherapy showed an increased risk of perforation when compared with chemotherapy without bevacizumab (OR, 3.71; 95% CI: 1.43-93.5; p=0.007; I2=0), but no statistically significant difference when compared with SEMS alone (OR, 3.75; 95% CI: 0.64-22.08; p=0.14; I2=62%).

Conclusion: Bevacizumab-based chemotherapy appears to increase the risk of perforation in advanced CRC patients after SEMS compared with non-bevacizumab-based therapy. These results were especially pronounced in chemotherapy without bevacizumab than in SEMS alone.

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Disclosure: Nothing to disclose.

PP0766

ENDOSCOPIC SUBMUCOSAL DISSECTION AS THE PROCEDURE OF CHOICE FOR RESECTION OF LARGE COLORECTAL NEOPLASMS: AN INTERNATIONAL, MULTICENTER STUDY

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Introduction: Piecemeal endoscopic mucosal resection (EMR) vs endoscopic submucosal dissection (ESD) has been a long standing debate in the endoscopic community worldwide with multiple studies done to evaluate the preferred method to manage large colorectal neoplasms. Piecemeal EMR with thermal ablation of the margin has been strongly advocated and reported by certain groups to be a simple, inexpensive and effective technique and this has been preferred widely due to its perceived ease of performance and shorter learning curve.[1]

However, close, repeated surveillance colonoscopy is recommended even when the lesion is thought to have been completely resected[2] as residual tumors can cause interval cancers.[3]

Some groups, mainly from east Asian countries, have recommended en bloc resection by ESD due to significantly lower risk of local recurrence and precise pathological diagnosis.[4]

But its usage is limited due to concerns about risk of serious adverse events. With improvement in training and devices, colorectal ESD is now being performed by various centers globally. This study aims to report the outcomes of an international, multicenter study on colorectal ESD.

Aims & Methods: Patients who underwent colorectal ESD for lesions greater than 20mm between January 2022 to April 2024 from centers in USA, Japan, and India respectively were included in this study. Recurrent lesions after endoscopic therapy, neoplasms identified in patients with ulcerative colitis, and non-epithelial lesions were included.

Primary outcomes were en bloc and R0 resection rate, delayed adverse events (bleeding/perforation), and local recurrence rate. Secondary outcomes were procedure time, lesion size, and pathology. Univariate and multivariate analysis were done to identify the risk factors for primary outcomes.

Results: 309 patients (median size, 40(27.5-60)mm; location rectum 52.4%, left colon 26.2%, right colon 21.4%) were involved in this study. The en bloc resection rate was 96.1% and R0 resection rate was 90.3%. 3 patients experienced delayed bleeding which was managed endoscopically. 2 patients underwent surgery for delayed perforation. 6 patients had a local recurrence. 67(23%) patients had intramucosal carcinoma on pathology.

Univariate analysis showed lesser procedure time (median(IQR) 145(80-236.25) vs 270(180-300) minutes, *P*= 0.005) and smaller lesion size (median(IQR) 40(26-56.25) vs 60(40-90) mm, *P*= 0.004) were significant factors for successful en bloc resection. However, lesser procedure time (median(IQR) 74(29.5-97.5) vs 150(82.5-240) minutes, *P*= 0.034) led to an

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increased delayed adverse event rate. Multivariate analysis using Cox regression showed en bloc resection (odds ratio 0.038; 95% confidence interval 0.003-0.421; *P*= 0.008) was an independent factor preventing local recurrence.

Factors		Variables		P value
En bloc resection rate		En bloc resection (n=297)	Piecemeal resection (n=12)	
	Procedure Time (median (IQR) minutes)	145 (80 – 236.25)	270 (180 – 300)	0.005
	Lesion Size (median (IQR) mm)	40 (26 – 56.25)	60 (40 – 90)	0.004
Recurrence rate		Recurrence (n=6)	No recurrence (n=257)	
	En Bloc resection n (%)	4 (66.7)	246 (98)	0.009
Adverse events		Delayed adverse events (n=5)	No delayed adverse events (n=304)	
	Procedure Time (median (IQR) minutes)	74 (29.5 – 97.5)	150 (82.5 – 240)	0.034

Conclusion: This international, multicenter study with data from three countries shows favorable outcomes for colorectal ESD for large lesions. The high en bloc and R0 resection rate, and low adverse event rate show increased global adaptation to this procedure. Reduced local recurrence is an added benefit. Data from a more centers globally will be valuable to establish ESD as the standard for large colorectal neoplasms.

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Disclosure: Nothing to disclose.

PP0767

REAL-WORLD DATA ON PROGNOSIS AND RISK FACTORS FOR LYMPH NODE METASTASIS AND RECURRENCE IN T1 COLORECTAL CANCER TREATED WITH ENDOSCOPIC OR SURGICAL RESECTION: A MULTICENTER RETROSPECTIVE STUDY

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Introduction: According to the European Society of Gastrointestinal Endoscopy (ESGE) guidelines, the curative criteria of endoscopic resection (ER) for T1 colorectal cancer (CRC) are well/moderately differentiated adenocarcinoma or papillary carcinoma, no lympho-vascular invasion, submucosal invasion depth <1000μm, and budding grade (BD) 1. Although an additional surgery is recommended for non-curative ER, the incidence of lymph node metastasis (LNM) is less than 12%. Many reports on lymph node metastasis and recurrence in T1 CRC originate from academic centers, with limited reports of real-world data.

Aims & Methods: The aim of this study was to clarify clinical outcomes in patients with T1 CRC. A total of 686 consecutive patients (Male:447 Female:239) were enrolled from April 2009 to August 2019 at 10 institutions. Exclusion criteria included evidence of familial adenomatous polyposis, hereditary nonpolyposis colorectal cancer, or inflammatory bowel disease; presence of active, malignant diseases in any other organs; presence of synchronous or metachronous advanced CRC. Risk factors of LNM and recurrence, and survival were analyzed.

Results: The mean age of patients was 69.3±10.2 years old. The mean tumor size was 20.3±12.4mm. The location of the lesion was 439(64%) and 247(36%) in the colon and rectum, respectively. The histopathological types included 681 cases of differentiated type and 5 cases of undifferentiated type. A number of patients with ER without additional SR (ER alone) were 176(curative resection 86, non-curative resection 90), those with SR (SR alone) were 510, and those with ER and additional SR (ER + additional SR) were 116.

(i) Risk factor of LNM: LNM was found in 13(11.2%) patients with ER + additional SR and in 40(10.6%) patients with SR alone. In multivariate analysis, positive lymphatic invasion and tumor size were significant independent risk factor for LNM (OR 5.9, 95% CI [2.2-16], p<0.01, OR 1.04, 95%CI［1.00-1.07］p=0.028, respectively).

(ii) Recurrence: During the mean observation period of 45.5±34.3 months, recurrence was found in 27(3.9%) patients. No recurrence was found in the ER curative resection group. Although 43% (90/206) patients with ER non-curative resection did not receive additional SR, there was no signif-

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icant difference in recurrence rate between ER alone with non-curative resection and ER + additional SR groups (5.6% (5/90) vs. 2.6% (3/116), respectively; p=0.3). In the SR alone group, recurrence was found in 4.8% (19/394). The significant independent risk factor for recurrence was tumor size (OR 1.12, 95% CI [1.05-1.2], p<0.01).

(iii) Survival: There were 50 deaths (7.3%, 76.2±6.6 years old) during the observation period of 45.5±34.3 months, but CRC-related death was found in only 6 cases (0.8%); one case was ER alone with non-curative resection and the other cases were SR alone.

Conclusion: This study suggests that the possibility of expanding the indication for ER of T1 CRC, because positive lymphatic invasion and tumor size were the only risk factor for LNM and better long-term outcomes were shown despite including significant number of patients with ER alone with non-curative resection.

Fifty cases of death showed a higher average age and fewer deaths from CRC. Therefore, even in cases of non-curative resection for T1 CRC after ER, avoiding additional SR may be considered due to advanced age or comorbidities.

Disclosure: Nothing to disclose.

PP0768

FEASIBILITY, SAFETY, AND OUTCOMES OF LGI ENDOSCOPIC SUBMUCOSAL DISSECTION FROM UK ESD REGISTRY: A MULTICENTRE PROSPECTIVE STUDY ON WESTERN POPULATION

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Introduction: The role of EMR in large colonic polyp is already well established whereas the practice and outcomes of ESD are variable depending on the volume and experience, especially in the west.

Aims & Methods: We aim to evaluate practice and outcome of colonic ESD and hence, a multicentre national registry was set up in the UK. This is a prospective observational study for patients undergoing ESD for early GI neoplasia. All technical and outcomes data was prospectively collected onto REDCap.

Results: There were 568 colorectal ESDs from 6 large tertiary referral centres in the UK. Mean age of the patients was 67.3 years (range 37 - 90 years) with male preponderance (341 males vs. 227 females). Majority were in the rectum: 300, 52.8% while 126 (22.2%) in the left colon and 142 (25%) were in the proximal colon. Polyp size ranges from 10 to 210 mm (mean 47 mm). 318 (56%) were flat polyp.

Of 568 ESDs, 128 (22.6%) were hybrid ESDs where more than 80% of submucosal dissection was completed by using a knife before using a snare for completion. Technical success was achieved in 97.7% (555 cases) with en-bloc rate of 83.1%. For Rectal ESDs, 12.0% were hybrid ESDs with overall en-bloc rate of 89.5% whereas, for right colonic ESDs, 38.7% of them were hybrid ESDs with en-bloc resection rate of 71.7%.

Overall, there were 22 (4%) intraprocedural perforation (7 proximal and 15 distal colon) and 7 (1.3%) intraprocedural bleeding (3 proximal and 4 distal colon) which were all managed endoscopically. There were 3 (0.5%) delayed perforation (2 in caecum and 1 in rectum) and out of which, 1 required surgery and 2 were managed conservatively. There were 15 (2.7%) delayed bleeding (6 proximal and 9 distal colon) and 4 of them required blood transfusion while others were managed conservatively/endoscopically. 5 were readmitted with post polypectomy syndrome. Overall, ESD related complication rate was 8.6% (12.0% in right colon and 6.6% in left colon) as shown in Table 1.

Complication rates	All location (%) n= 568	Right sided (%) n= 142	Left sided (%) n= 426
Delayed bleeding	15 (2.7)	6 (4.2)	9 (2.1)
Intraprocedural perforation	22 (3.9)	7 (4.9)	15 (3.5)
Delayed perforation	3 (0.5)	2 (1.4)	1 (0.2)
Post-polypectomy syndrome	5 (0.9)	2 (1.4)	3 (0.7)
Overall (%)	8.6%	12.0%	6.6%

Table 1. Complication rates of colonic ESD.

Post resection histology showed 508 adenomatous neoplasia, 18 sessile serrated lesions (SSLs) with dysplasia, 4 SSLs without dysplasia, 3 hyperplastic polyps, 6 neuroendocrine tumours and 1 squamous cell cancer and 1 melanoma.

Post resection ESD of these adenomatous lesions showed majority (263, 62.2%) was low grade dysplasia with lateral margin R0 rate of 73%. 130 (30.7%) were high grade dysplasia with R0 rate of 88.3%. There were 60 (14.2%) submucosally invasive cancers with deep margin R0 rate of 81.4%.

Conclusion: Our data demonstrates that colonic ESD is feasible and the majority of the lesions were in rectum where ESD has high en-bloc resection rate. It is safe with low complication rate but we observed higher complication rates in right colonic ESD compared to the left. Overall, lateral margin R0 rate of LGD is suboptimal which could be related to very close resection margin which needs improvement.

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PP0769

EXPLORING UNWARRANTED CLINICAL VARIATION IN THE MANAGEMENT OF EARLY RECTAL CANCER. FINDINGS FROM AN AUSTRALIAN HOSPITAL NETWORK

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Introduction: Radical resection in the rectum is associated with significant morbidity and poses a mortality risk, particularly in elderly or comorbid patients. Early or T1 rectal cancer is often curable and can be managed with local endoscopic resection techniques with reduced morbidity and negligible mortality. To establish a pathway for appropriate treatment of these early lesions through the multidisciplinary team (MDT), an early rectal cancer working party (ERCWP) was formed to evaluate management trends and outcomes within our hospital network. The ERCWP was multi-speciality comprising of gastroenterologists, colorectal surgeons and radiation oncologists.

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Aims & Methods: A retrospective review was performed across four institutions that make up our network to identify all patients with a new diagnosis of localised (T1-T4,N0,M0) rectal or recto-sigmoid cancer between January 2015 and August 2023. Patients with confirmed adenocarcinoma staged as T1N0M0 or T2N0M0 were included in the analysis. Available electronic medical records from the time of index resection and all encounters for the following 12 months were reviewed. Patients were excluded if their treatment was completed at an outside institution. Patient outcomes were stratified by location of tumour (rectum / recto-sigmoid) and mode of resection (Endoscopic resection, Local surgical excision or Radical surgical resection). Major adverse events included any one of the following: ostomy, deep incisional site infection, abscess, wound dehiscence, pneumonia, sepsis, septic shock, acute renal failure, deep venous thrombosis, pulmonary embolus, stroke/cerebrovascular accident, myocardial infarction, ventilator >48 hours, still in hospital >30 days, cardiac arrest, reoperation, readmission and/or mortality.
Results: A total of 111 localized (T1/T2,N0) and 153 locally advanced (T3/T4,N0) lesions were newly diagnosed during the study period. 95/111 patients with localized disease were included (M=62/95, 65.3% F=33/95, 34.7%). Mean age was 67.8 years in the endoscopy group and 67.0 years in the radical surgical resection group. Mortality was 2.2% in the group of patients who had surgery (1/46). There was no mortality in patients who received endoscopic resection. Major adverse events were recorded in 16/46 (34.8%) of patients who underwent surgery. There were no major adverse events in patients who received endoscopic resection. Mean length of stay was 1.47 days in patients who had endoscopic resection and 10.2 days in patients who had radical resection. Mean length of stay was prolonged where there was a treatment related major adverse event (14.8 days, mean difference 9.8 days, p=<0.001). 78.5% (51/65) of the cohort with T1 cancer was low risk (well or moderately well differentiated and no lymphovascular invasion). Of the 16 patients with T1 rectal cancer who underwent definitive surgery, 15 (93.8%) were suitable for cure by en-bloc endoscopic excision. Following establishment of the ERCWP in 2020, there have been no cases of radical resection in the rectum for T1 cancer. For T1 cancer in the rectum, major adverse events were 5/34 (14.7%) pre ERCWP formation and 1/13 (7.7%) post ERCWP formation (p=<0.001).
Conclusion: Surgery carries a high risk of morbidity in patients with rectal cancer. Endoscopic resection carries a favourable risk profile and is currently being underutilised. Formation of a local expert group focusing on outcomes in T1 cancer can reduce unwarranted variation and similar groups should be formed at local institutions to optimise patient outcomes.
Disclosure: Nothing to disclose.

PP0770

ENDOSCOPIC SURVEILLANCE OF COLORECTAL CANCER: THE ROLE OF INTENSIFIED ENDOSCOPIC CONTROL

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Introduction: Endoscopic surveillance after surgery for colorectal cancer (CRC) has the main objectives of early identification of metachronous lesions and neoplastic recurrence. This study aimed to evaluate the possible benefit of intensified endoscopic surveillance compared to standard recommendations.
Aims & Methods: In this retrospective cohort study, all patients who underwent follow-up colonoscopy after CRC surgery over a period of 6 months were included. Patients with endoscopically treated CRC and patients with postoperative complications were excluded. The recommended follow-up period was considered the 1st year, the 4th year, and the 9th year after surgery and a time period of +/- 6 months to the recommended date was considered. Statistical analysis: Chi-square test; t-student test; Mann-Whitney test.
Results: 102 patients (38 women) were included, aged 72±11 years. Patients were divided into two groups, those who followed the recommended protocol (55.9%; n=57) and those who underwent intensified surveillance (44.1%; n=45). No statistically significant differences were identified between the two groups regarding demographics, location of the neoplasm or type of surgery performed. There was only a higher adenoma detection in the intensive surveillance subgroup in the 1st year after surgery (p=0.042), with no differences in the remaining endoscopic controls. None of the advanced adenomas identified in this group required earlier endoscopic surveillance. No advanced adenomas were identified in additional colonoscopies carried out outside the recommended period. The adenoma detection rate didn't show statistically significant differences between the two groups. An excess of 156 colonoscopies were performed over a total of 361 years of follow-up.
Conclusion: Intensified endoscopic surveillance after CRC surgery does not appear to confer benefit and is a significant burden to endoscopy units, in addition to unnecessary risks and discomfort for patients. Adherence to international recommendations should be promoted among referring physicians and a strict colonoscopy vetting process should be enforced.
Disclosure: Nothing to disclose.

PP0771

THE INFLUENCE OF ANTHOCYANINS BLACK RICE BRAN EXTRACT (ABRIBE) ON MRNA EXPRESSION OF THE ABCA1 GENE AND COLORECTAL CANCER CELL PROLIFERATION IN THE HUMAN HT-29 COLORECTAL CANCER CELL LINE

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Introduction: Anthocyanins decrease the proliferation of colorectal cancer cell and cholesterol regulation via the adenosine triphosphate-binding cassette subfamily A member 1 (ABCA1) protein, which is underexpressed in colon cancer.[1,2,3] Anthocyanins, which are found in Indonesian native black rice bran, can also promote the outflow of cholesterol from cancer cell.
Aims & Methods: This study aims to explore the influence of Anthocyanins Black Rice Bran Extract (ABRiBE) on mRNA expression of the ABCA1 gene

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and colorectal cancer cell proliferation in the human colorectal cancer cell line, HT-29. ABRiBE is thought to suppress the proliferation of HT-29 cell while increasing the expression of the ABCA1 gene. Anthocyanins derived from black rice bran powder using ethanol acidified with citric acid were applied in this experimental work on HT-29 cells. The MTS assay used 12 extract concentrations and control groups to determine the inhibitory concentration (IC_{50}) of 50% cell growth inhibition for 24 and 48 hours. The quantitative Polymerase Chain Reaction (qPCR) method was used to examine the expression of the ABCA1 gene in groups higher than the IC_{50} value and a control group.

Results: The proliferation of HT-29 cells was lowered by ABRiBE, which had an IC_{50} of 1.57 and 1.35 mg/mL for incubation of 24 and 48 hours with values of r=0.97, p=0.001 and r=0.95, p=0.001, respectively. The expression of the ABCA1 gene was enhanced by 1.16 and 2.32 fold when extract supplementating with 2 doses beyond the IC_{50} values of 1.7 and 2.0 mg/mL and a control group.

Gene Name	mRNA Relative Expression (fold)	Primary Sequence		Amplicon Size (base pair/ bp)
		Forward	Reverse	
Human HRPT1	1 fold	CCAGTACCAAGACCAGTTTCTC	GGTGGTAGCTGCAGGAATTAT	93 bp
Human ABCA1				
1.7 mg/ mL	1.16 fold	GGTGGTGTTCTTCCTCATTACT	CCGCCTCACATCTTCATCTT	112 bp
2.0 mg/ mL	2.32 fold			

Table 1. The mRNA relative expression of the ABCA1 and HRPT1 (reference) genes as well as their primer pair sequence and amplicon sizes

Conclusion: The ABRiBE decreased the proliferation of the HT-29 cells while increasing the expression of the ABCA1 gene.

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Disclosure: The authors declare no conflict of interest.

PP0772

PLATELETS TO LYMPHOCYTES RATIO: A PROGNOSTIC INFLAMMATION-BASED FACTOR IN COLORECTAL CANCER

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Introduction: Colorectal cancer remains a significant public health concern, with its incidence steadily rising worldwide. Despite notable strides in treatment modalities, predicting disease progression and tailoring therapeutic approaches remains a challenge. In this context, identifying reliable and simple prognostic factors is crucial for optimizing disease outcomes.

Aims & Methods: The aim of this study was to investigate the predictive and prognostic value of platelet-to-lymphocyte ratio (PLR). We conducted a retrospective and descriptive study including patients diagnosed with primary colon or rectal cancer, over an 8-year period [January 2014-June 2022]. Patient demographics, inflammatory markers, tumor characteristics, treatment and follow-up data were collected. Baseline PLR was calculated for each patient. Data were entered and analyzed by SPSS software version 26.

Results: Eighty-four patients were included, with a mean age at disease diagnosis of 60 ± 12 years and a sex ratio M/F=1.89. The median level of PLR was 169.80 [123.1-269.06]. Smoker patient have significantly higher PLR (p=0.01) as well as for patients with older age at diagnosis (Pearson p=0.001, r=0.35).

No significant association was noted of PLR neither with family history of colorectal cancer (p= 0.47), nor gender (p=0.69), nor tumor localization (p=0.15). For biological data, anemia and hypoalbuminemia were significantly associated with a higher level of PLR (p=0.01 and p=0.001 respectively). Regarding treatment, neither primary surgery, nor adjuvant chemotherapy were associated with PLR (p=0.36 and p= 0.97 respectively).

However, increased PLR was significantly associated with overall survival (p=0.001). Area under the receiver operating characteristic curve analysis showed that higher PLR was predictive of a shorter progression-free survival (AUROC = 0.706). The optimal cutoff value was 163.12 (sensitivity, 80%; specificity, 59%).

A total of 48 (57%) patients had a PLR above this threshold. These patients presented more frequently with bowel obstruction (23% vs 6%, p=0.03). They also had a higher carcinoembryonic antigen rates (p=0.007). Advanced TNM stage 3 or 4 was more common in patients with PLR above this threshold and thus, neoadjuvant chemotherapy was more indicated in this group (29% vs 11%; p=0.04). Overall, disease recurrence was similar for PLR above and below 163.12 (p=0.12).

Conclusion: Our study demonstrated that PLR has a good predictive value of progression-free survival connected with clinicopathological parameters in patients with colorectal cancer. It is a valuable tool for a simple prognostic assessment during colorectal cancer, contributing to a more personalized treatment decisions.

Disclosure: Nothing to disclose.

PP0773

WITHDRAWN

PP0774

EVALUATION OF THE EFFECTIVENESS OF INCREASED RADIATION DOSE IN NEOADJUVANT THERAPY FOR POTENTIALLY RESECTABLE RECTAL CANCER

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Introduction: Radiation therapy as part of neoadjuvant treatment for rectal cancer aims to improve local disease control with increased response rates and the possibility of sphincter preservation. The addition of the "watch and wait" (W&W) strategy has increased the efficacy and safety associated with neoadjuvant radiotherapy.

Despite the use of low doses (between 40-50.4 Gy), most studies report complete pathological response rates of 10-27%. However, there is still a lack of studies with higher doses to evaluate effectiveness in terms of clinical and pathological response, versus the toxicity associated with increased radiation dose.

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Aims & Methods: The aim was to evaluate the efficacy and tolerance of neoadjuvant therapy for potentially resectable rectal cancer with higher radiation doses.

A retrospective observational preliminary study included 41 individuals with potentially resectable rectal cancer (T2-T4, N0-1, M0), treated with capecitabine (825mg/m2 every 12 hours, 5 days a week) and 25 sessions of 54 Gy + 50 Gy to pelvic lymph nodes, considering patients on W&W. Partial response was defined as downstaging of at least 1 level in T and/or N. Complete clinical response was defined as reevaluation by MRI, endoscopy with biopsies, and digital rectal examination without evidence of residual disease. Short- and long-term complications of increased radiation dose were also evaluated.

Results: Out of 41 individuals with potentially resectable disease observed during the study, downstaging was observed in 78% (n=32), with 29% (n=12) showing complete response (clinical and/or pathological). Of the patients with complete clinical response (n=12), 8 underwent total mesorectal excision due to unpassable stenosis affecting adequate visualization of the rectal mucosa (n=1), presence of rectal ulcer with negative biopsies (n=6), and concurrent ulcerative colitis (n=1). All these patients showed complete pathological response. Four patients entered the W&W protocol, 3 of them with residual ulcer and negative biopsies, without recurrence at present (4 years follow-up). Radiation therapy complications were observed in 7% (n=3) of patients, respectively due to rectal stenosis, radiation proctitis, and radiation cystitis.

Conclusion: Our study found higher rates of downstaging and complete response compared to most studies under similar conditions and with lower radiotherapy doses, with a relatively low frequency of adverse effects.

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PP0775

BLUEBERRY INGESTION IMPROVES GASTROINTESTINAL SYMPTOMS AND QUALITY LIFE AND MODULATES PROTECTIVE TIGHT JUNCTION PROTEINS IN PATIENTS WITH DISORDERS OF GUT-BRAIN INTERACTION

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Introduction: Changes in intestinal microbiota, stress and inflammation are factors known to be relevant in the pathogenesis of pain and further symptoms in disorders of gut-brain interaction (DGBI), previously designated as functional gastrointestinal disorders (1). The same factors are also implicated in modulating cellular tight junction integrity and thereby also blood-brain and intestinal barrier permeabilities (1,2). Dietary nutrients, especially polyphenols in fruit and vegetables, can beneficially affect cellular tight junction integrity, modulate the gut microbiome and reduce pro-inflammatory effects (3).

Aims & Methods: The effects of blueberries on, 1. GI symptoms (Gastrointestinal Clinical Rating Scale: GSRS), life functioning and quality of life (Outcome Questionnaire OQ45.2), 2. circulating tight junction proteins especially relevant for the integrity of the intestinal and blood-brain barriers and, 3. stool consistency (Bristol stool scale: BSS) were prospectively examined in DGBI. Successive patients with irritable bowel syndrome or functional dyspepsia (Rome 4) received freeze-dried blueberries (equivalent to 180g of fresh blueberries) or sugar and energy-matched placebo for 6 weeks in a double-blind, randomized, cross-over study, with a washout period of 2 to 4 weeks between treatments. Serum concentrations of claudin-3, claudin-5 and zonula occludens-1 (ZO-1) were measured by ELISA and the GSRS, OQ45.2 and BSS scores were collected at baseline and after 6 weeks. The changes in protein concentrations and in GSRS, OQ45.2 and BSS scores from baseline were correlated.

	Blueberry			Placebo		
	Baseline	Week 6	Change (95%CI)	Baseline	Week 6	Change (95%CI)
Questionaries						
Mean GSRS ± SD	37.7 ± 13.4	33.0 ± 12.9	-4.7 (-8.4 to 1.0) *	38.2 ± 11.3	36.6 ± 12.6	-1.7 (-5.2 to 1.9)
Median BSS (IQR)	3.5 (1.5)	3.5 (1.5)	0 (0 to 1.0)	4.0 (1.5)	4.0 (2.0)	0 (-0.7 to 0.5)
Median OQ-45.2 (IQR)	62.0 (15.0)	61.0 (17)	-3.0 (-5.0 to 0) **	63.0 (21)	66.0 (16)	0 (-3.0 to 3.0)
Serum tight junction protein concentrations						
Median claudin-3 (IQR) (pg/ml)	76.3 (16.3)	77.2 (17.4)	2.1 (1.0 to 4.3) ***	74.5 (16.4)	72.1 (18)	0 (-1.5 to 1.2)
Median claudin-5 (IQR) (pg/ml)	38.5 (15.7)	40.3 (14.9)	1.2 (-1.1 to 4.3)	39.8 (20.2)	40.3 (15.8)	0.1 (-1.6 to 2.1)
Median Z0-1 (IQR) (pg/ml)	81.3 (25.2)	81.3 (12.5)	-1.1 (-3.9 to 1.0)	81.4 (21.3)	80.0 (14.1)	-1.7 (-6.6 to 1.5)

*Table. Outcome measures at baseline and after 6 weeks of treatment with blueberry or placebo. Within-group differences (baseline vs. week 6): *p=0.0129, **p=0.0143,***p=0.00340**

Results: In 43 patients (37 females, mean age±SD 31±12yr) blueberry treatment for 6 weeks increased claudin-3 serum concentrations and improved GSRS and OQ45.2 scores significantly (see Table). Increased serum claudin-5 concentrations correlated with improved GSRS and OQ45.2 scores (Spearman's r= -0.38, p=0.016; r= -0.36, p=0.024, resp.), while increased ZO-1 levels correlated with improved OQ45.2 scores (r= -0.32,

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p=0.046). There were neither significant serum concentration increases in tight junction proteins, changes in clinical outcomes nor correlations between these outcomes during placebo treatment. The median BSS was not significantly changed during either treatment, although constipation was relieved more frequently by blueberries.

Conclusion: Blueberries given for 6 weeks decreased symptoms and improved measures of quality of life and life functioning scores in patients with DGBI. Concomitantly, increased serum concentrations of claudin-3, a tight junction protein important in the regulation of intestinal and other barrier tight junction integrity, were observed. Increases in circulating tight junction protein levels relevant in tightening blood-brain (claudin-5) and other (ZO-1) barrier tight junctions correlated with the improvement in symptoms.*This study was supported by a grant from the US Highbush Blueberry Council.*

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PP0776

THE INADEQUACY OF MALABSORPTION AS A PREDICTOR OF CARBOHYDRATE INDUCED SYMPTOMS: EXPLORING SYMPTOM BURDEN AFTER LACTOSE CONSUMPTION

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Introduction: A substantial portion of individuals diagnosed with lactose malabsorption fail to experience significant relief from a lactose-free dietary regimen (1).

The 'adult Carbohydrate Perception Questionnaire' (aCPQ) is a simple, test-specific questionnaire for the assessment of gastrointestinal symptoms following a carbohydrate challenge (2). The aCPQ has undergone rigorous validation, it has excellent psychometric properties, and its use has been recommended to replace missing or unvalidated symptom assessments in carbohydrate breath testing (3).

Aims & Methods: Our aim was to assess symptom burden after lactose ingestion by means of the aCPQ with respect to the presence or absence of malabsorption. 753 patients with functional gastrointestinal disorders received a lactose (50 g) intolerance/H_2-breath test (180 min duration), utilizing the aCPQ for test specific, validated symptom assessment and an electrochemical hydrogen detector for quantification of end-expiratory breath hydrogen. Malabsorption (M+) was defined as an increase in end-expiratory H_2 by >20 ppm over baseline (as opposed to M-). The presence of relevant symptoms after lactose ingestion (S+) was defined as an increase of symptom scores by >20 cm in one of the five VAS scales over baseline (as opposed to S-). The maximal rise of symptom severity over baseline was determined (maximal possible rise: 100 for each symptom) and considered the individual symptom burden; the sum of individual symptom burdens was considered the total symptom burden (maximum possible sum of 500).

Results: 308 patients tested M+, 445 M-, and 417 tested S+ vs. 336 S-. 80 patients (26.0%) testing M+ had no clinically relevant symptoms (S-) following the lactose challenge; in contrast, 189 patients (45.3%) testing M- had relevant symptoms after the lactose challenge. Table 1 shows the individual and total symptom burden based on the diagnosis of lactose intolerance/H2 breath test: patients with malabsorption who had no clinically relevant lactose-induced symptoms (S-/M+) reported slightly higher intensities of flatulence and diarrhea as compared to S-/M-. Malabsorbers with symptoms had higher symptom burdens as compared to absorbers except for nausea, which was not statistically different between S+/M+ and S+/M-.

mm VAS:	S-/M-	S-/M+	P-value S-/M- vs. S-/M+	S+/M-	S+/M+	P-value S+/M- vs. S+/M+
Abdominal Pain	2.7	3.2	0.4	16.4	32.0	<0.001
Nausea	1.9	2.6	0.2	18.2	16.9	0.5
Bloating	2.8	3.5	0.7	19.9	32.0	<0.001
Flatulence	2.3	3.8	0.02	16.4	34.4	<0.001
Diarrhea	0.8	1.6	0.02	9.4	33.2	<0.001
Total Burden	10.5	14.7	0.02	80.3	148.5	<0.001

Table: Symptom burden after lactose ingestion

Conclusion: Malabsorption may be just one of the factors that lead to the onset of symptoms following lactose consumption. This is crucial regarding both the diagnosis and therapy of lactose intolerance. The aCPQ is the only reliable and valid, test specific instrument to evaluate symptoms triggered by the consumption of carbohydrates. It can be used in conjunction with a breath test, which is necessary if lactose malabsorption is to be detected.

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PP0777

SYSTEMIC ADVANCED GLYCATION END-PRODUCTS ARE LOWER IN FRUCTOSE INTOLERANT PATIENTS WITH DISORDERS OF GUT-BRAIN INTERACTION (DGBI)

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Introduction: Fructose and fructans/fructooligosaccharides are implicated in the production of advanced glycation end products (AGE), which cause potent oxidative stress and inflammatory responses related to ageing, inflammatory and degenerative diseases (1-3).

It has recently been shown that fructose intolerance is unrelated to fructose malabsorption and more probably to differences in fructose metabolism by the intestinal microbiome (4).

Individual species of the intestinal microbiota can differentially either produce or metabolise AGEs, as well as ferment fructose and fructans (2). DGBI are characterised by a high prevalence of fructose and fructan intolerance, pro-inflammatory markers and changes in gut microbiota (5).

The relationships between fructose intolerance, the intestinal microbiome and AGEs are unclear.

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Aims & Methods: We examined the association between skin AGEs by autofluorescence (SAGE) as a validated, reproducible and non-invasive marker of metabolically long-acting AGEs (6), exhaled breath hydrogen and symptoms following fructose provocation during breath tests in 1188 patients with DGBI (Rome 4 criteria, 30% male, mean age 40.9 years, mean body mass index 23.6 kg/m^2) referred to our Gastroenterology practice. A standardised breath test with fructose 35g was performed after a 24-hour low-FODMAP diet and >8-hour fast. Exhaled hydrogen concentrations and gastrointestinal symptoms (composite symptom index score, 7) were collected every 30 minutes for 5 hours. SAGE was performed with the AGE Reader® (Diagnoptics Technologies B.V., Groningen, NL) before the breath test. Lower SAGE values reflect a lower systemic pro-inflammatory AGE load.

Results: Overall, the SAGE values and hydrogen breath concentrations correlated inversely (Rho= -0.09, p=0.0022, Spearman's rank test). Extensive fructose fermenters, defined by an increase in breath gas concentrations above baseline of >20ppm of hydrogen after fructose ingestion, showed lower SAGE values than poor metabolisers (1.85±0.56 vs. 1.92±0.55; p= 0.0221).

Furthermore, patients with an increased symptom index following fructose and thereby classified as fructose intolerant (7), had significantly lower SAGE scores than fructose tolerant patients (1.84±0.55 vs. 1.96±0.57; p=0.0004).

Conclusion: Patients with DGBI and pronounced symptoms and gas production during fructose breath tests appear to have lower systemic AGEs. Metabolism by the intestinal microbiota plays an important role in both fructose intolerance and in modulating the levels of long-acting AGEs. This association may be of crucial relevance in gastrointestinal and extra-gastrointestinal dysfunction in DGBI and requires further investigation. Potential confounders, such as dietary factors, require careful consideration.

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PP0778

LACTOCOCCUS LACTIS CNCM I-5388 AS A GLUTAMATE DECARBOXYLASE (GAD)-ENZYME POSTBIOTIC ALLEVIATES VISCERAL PAIN IN RATS THROUGH GABA PRODUCTION.

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Introduction: Irritable Bowel Syndrome (IBS) is a chronic disorder of the gut-brain interaction characterized by visceral hypersensitivity. There is growing evidence underscoring the beneficial impact of probiotics and postbiotics with psychobiotics properties in relieving stress-related gastrointestinal disorders in humans especially in patients suffering from IBS. We have previously shown that high GABA-producer *L. lactis* CNCM I-5388 exerts antinociceptive properties after 5- and 10-days oral treatment in an IBS-like rat model. We demonstrated, using both mutant *L. lactis* Δ*gadB* strains and $GABA_B$ receptor antagonist that these antinociceptive properties were dependent on the GAD-enzymatic activity of *L. lactis* and mediated by host's $GABA_B$ receptors. In addition, we have shown that CNCM I-5388 displayed a high cell-bound GAD activity and survival *in vitro* under HCl-mediated acidic stress.

Aims & Methods: In this work, we were interested in studying the GABA-producer *L. lactis* CNCM I-5388 as an active-enzyme postbiotic to relieve visceral hypersensitivity mediated by a psychological stress.

L.lactis CNCM I-5388 was inactivated by ethanol (EtOH) while maintaining its GAD activity. Female Wistar rats were treated orally for 1, 5 or 10 days by the EtOH-treated nonviable *L.lactis* CNCM I-5388 form or the viable *L. lactis* CNCM I-5388 or vehicle. Visceral sensitivity in response to colorectal distension was measured by electromyography before and after partial restraint stress (PRS). GABA levels were quantified by HPLC both in the stomach collected from rats and in the gastric compartment of the TIM-1 human gut model in fed state.

Results: A daily treatment for 5 and 10 days by *L. lactis* CNCM I-5388 both in its viable and nonviable forms counteracted visceral hypersensitivity-induced by PRS in response to colorectal distension. However, after a single administration only viable *L. lactis* CNCM I-5388 tended to reduce visceral hypersensitivity. After 5-days treatment, the production of GABA within the stomach was enhanced in stressed rats treated with the viable or nonviable *L. lactis* CNCM I-5388 form. This increase was confirmed by using the TIM-1 human gut model.

Conclusion: Our results highlight that a postbiotic with an active-GAD of *L. lactis* CNCM I-5388, similarly to its viable psychobiotic form, exerts visceral antinociceptive properties in an IBS-like rat model. These effects are associated with GABA production in the stomach where the low pH promotes GAD activity. From a clinical point of view, this "active GAD-bag" concept offers a new therapeutic perspective in the management of visceral pain in IBS patients with a superior safety profile compared with GABA-producing pyschobiotics that could acquire antibiotic resistance and virulence factors in chronic use.

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PP0779

THE GUT VASCULAR BARRIER IS INVOLVED IN SYMPTOM DEVELOPMENT IN IRRITABLE BOWEL SYNDROME

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Introduction: Irritable bowel syndrome (IBS) is a multifactorial disorder characterized by recurrent abdominal pain associated with defecation or a change in bowel habits. The pathophysiology of IBS involves alterations in the epithelial barrier, but data on the intestinal vascular barrier are lacking.

Aims & Methods: The aims of this study were to investigate the morpho-functional integrity of the vascular barrier and to correlate barrier dysfunction with IBS symptoms. Forty-eight healthy subjects (AC) and 144 patients with IBS (61 IBS-D, 51 IBS-C and 32 IBS-M) were enrolled in the study. Rome IV criteria, Bowel disease questionnaire, Short Form 12, Hospital Anxiety Depression Scale were used to phenotype patients. Immunohistochemistry, western blot and electron microscopy were used to characterize the vascular barrier. Vascular permeability was evaluated by assessing two markers: plasmalemma vesicle–associated protein-1 (PV1) and vascular endothelial cadherin (VEC). A translational model based on HUVEC was used to evaluate permeability changes induced by mediators spontaneously released by mucosal biopsies, and to highlight the underlying molecular mechanisms. Correlation analyses between experimental and clinical data were performed.

Results: Mucosal vessel density was significantly increased by 1.5 fold in the colonic mucosa of IBS patients compared to AC ($P<0.01$). PV1 expression was increased ($P<0.01$) while VEC expression was decreased in the colonic mucosa of IBS patients compared to AC ($P<0.05$). Compared to AC, IBS mucosal mediators increased permeability in the HUVEC monolayer through the activation of protease-activated receptor (PAR)-2 and histone deacetylase (HDAC)11, resulting in VEC downregulation. PV1 expression significantly correlated with anxiety and depression scores and health-related quality of life reported by IBS patients.

Conclusion: The vascular barrier is impaired in IBS and contributes to clinical manifestations

Disclosure: Nothing to disclose.

PP0780

EVALUATION OF FOOD INTOLERANCE AND FOOD AVOIDANCE IN IRRITABLE BOWEL SYNDROME PATIENTS

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Introduction: Many irritable bowel syndrome (IBS) patients report gastrointestinal (GI) symptoms due to various foods, resulting in food avoidance. However, a detailed description of trigger foods and related GI symptoms and avoidance behaviour is lacking. Therefore, we evaluated the extent and nature of food intolerance and avoidance in IBS patients, and its relation to GI symptoms and psychological comorbidities.

Aims & Methods: Food intolerance and avoidance behaviour were evaluated in 124 IBS patients and 113 healthy controls (HC) using a questionnaire with 257 food products of 13 food groups. Per product, participants were asked if they experience symptoms (*i.e.* nausea, constipation, diarrhoea, bloating, abdominal pain) after consumption, and if they consumed this product despite these symptoms. Furthermore, IBS symptom severity and psychological factors were assessed.

Results: IBS patients reported a higher number of food products with perceived intolerance than HC (median of 18.0 (25th-75th percentile 10.0-33.5) vs 1.0 (0.0-8.0), $p<0.001$). A wide variety of trigger foods was reported, with gas-producing foods and fatty/cream-based dairy products most frequently reported by both groups. Predominant symptoms were abdominal pain and bloating. The number of avoided foods was higher in IBS (15.0 (8.0-27.0) vs 1.0 (0.0-7.0), $p<0.001$). Food avoidance was not significantly related to symptom type, but was associated with an IBS diagnosis, female sex, and higher screening scores for somatisation and Avoidant/Restrictive Food Intake Disorder.

Conclusion: Self-reported food intolerance and avoidance is higher in IBS than HC, with a wide variety of trigger foods. Food avoidance was significantly related to psychological factors, but not to symptom type. This warrants individual dietary guidance, as well as attention for psychological factors.

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PP0781

DEPRESSION, ANXIETY, AND VISCERAL SENSITIVITY IN PATIENTS WITH DIARRHEAL IRRITABLE BOWEL SYNDROME TREATED WITH MULTI-STRAIN PROBIOTICS: A RANDOMIZED STUDY

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Introduction: Irritable bowel syndrome (IBS) is considered a disorder of the altered microbiota-gut-brain axis, where gut dysbiosis and impaired central nervous system processing play important roles. The latter, in the form of depression, anxiety, and gastrointestinal-specific anxiety (GSA), frequently coexist and exacerbate IBS. Probiotics might be a promising therapeutic approach to treat IBS and coexisting depression, anxiety, and GSA.

Aims & Methods: The aim of the trial was to compare the impact of standard care combined with multi-strain probiotics with standard care alone on depression, anxiety, and visceral sensitivity in patients with diarrheal IBS (IBS-D). It was a 12-week randomized trial that included 108 patients with IBS-D that were divided into 2 groups: a control group (n = 51) receiving only standard care (diet, peppermint oil) and an experimental group (n = 57) receiving a combination of standard care (diet, peppermint oil) and the study probiotic (based on Lactobacillus, Streptococcus, Bifidobacterium, and Saccharomyces boulardii strains). Patient Health Questionnaire-9 (PHQ-9) and Beck's Depression Inventory (BDI) were used to assess depression; the Hospital Anxiety and Depression Scale (HADS) was used to assess anxiety and depression; and the Visceral Sensitivity Index (VSI) was used to assess GSA.

Results: Baseline data were not different between the study groups (p> 0.05). After 12 weeks of treatment, PHQ-9, BDI, and anxiety subscale of HADS were not different between the study groups (p> 0.05), while the central tendency of the depression subscale of HADS in the control group was higher (3 (2-4)) than in the experimental group (3 (2-3)), p = 0.046. VSI was higher in the control group (31±5) compared to the experimental group (29±5.7), p = 0.032, after 12 weeks of trial.

Conclusion: In patients with IBS, additional prescribing of a multi-strain probiotic led to a decrease in depression according to HADS, as well as a reduction in GSA and visceral sensitivity assessed by VSI compared to IBS standard care alone.

Disclosure: Nothing to disclose.

PP0782

GUT PERMEABILITY AND ITS ROLE IN PARKINSON'S DISEASE: PRELIMINARY RESULTS FROM A PILOT STUDY

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Introduction: Parkinson's disease (PD) is a neurodegenerative disorder characterized by both motor and non-motor symptoms, such as gastrointestinal symptoms. Emerging evidence suggests that gut permeability and inflammation play a role in PD pathophysiology via the gut-brain axis. According to Braak's theory, the pathological degeneration associated with PD may originate from the gut and subsequently develop in the brain through the vagus nerve.

Aims & Methods: The aims were to assess the relationship between gut permeability and neuroinflammation in a cohort of patients with PD and REM behavior disorder (RBD), a condition that often precedes the onset of PD. The study prospectively included 28 PD patients and 18 RBD patients both complaining of GI symptoms. No serious organic diseases or therapies potentially affecting GI motility were reported. Relevant demographic and clinical data were collected for all patients. There was the inclusion of 14 healthy controls (HCs). A serum ELISA assay of tumor necrosis factor (TNF), caspase-1, and lipopolysaccharide-binding protein (LBP) was performed in all participants to investigate systemic inflammatory status and gut permeability, respectively.

In a total of 10 PD patients, 4 RBD patients and 9 HCs, multiple colic biopsies were obtained to assay tissue expression of inflammation markers by both western blot (WB) (i.e. caspase-1 and procaspase-1) and immunofluorescence (IF) (i.e. ASC and CD68). Data were expressed as mean ± SD, and ANOVA and Dunnett's tests were used to compare the three groups.

Results: All three groups were comparable in age and sex. Serum assays showed significantly increased LBP, in patients with PD (35.0 ± 0.25; p = 0.002) and RBD (32.2 ± 1.1; p = 0.009) compared with HCs. TNF (0.1 ± 0.02; p = 0.03) and caspase-1 (16.9 ± 9.4; p = 0.04) were significantly raised only in the PD group in comparison with HCs (Table 1).

WB analysis showed significant differences in the expression of procaspase-1 (157.3 ± 17.8; p = 0.02) and caspase-1 (168.3 ± 23.3; p = 0.03) only in PD patients compared with HCs. IF showed a qualitative increase in both CD68+ macrophage infiltration and ASC expression in PD and RBD groups compared with HCs.

	PD	RBD	HCs	P-value
LBP (ng/ml)	35.0 ± 0.25	32.2 ± 1.1	23.1 ± 2.1	0.009 (RBD vs HCs) 0.002 (PD vs HCs)
Caspasi-1 (ng/ml)	16.9 ± 9.4	11.4 ± 2.1	9.9 ± 2.7	0.04 (PD vs HCs)
TNF (pg/ml)	0.1 ± 0.02	0.05 ± 0.02	0.04 ± 0.02	0.03 (PD vs HCs)

Table 1. Serum ELISA assay of TNF, caspase-1, and LBP from HCs and in patients with RBD and PD.

Abbreviations: TNF = tumor necrosis factor; LBP = lipopolysaccharide-binding protein; HCs = healthy controls; RBD = REM behavior disorder; PD = Parkinson's disease.

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Conclusion: Our results show a bowel activation of NLRP3 inflammasome in PD, as demonstrated both by the levels of procaspase-1 and caspase-1 on WB analysis and by the increase of CD68+ macrophages and ASC in the lamina propria of colonic mucosa.
Only a qualitative increase in NLRP3 activation at IF could be observed in RBD patients. However, findings from colonic biopsies, if related to elevated serum levels of TNF and caspase-1 in PD patients, and LBP in PD and RBD patients, suggest that bowel inflammation, sustained by increased gut permeability, may represent an early stage in the develop of RBD and PD.
Disclosure: Nothing to disclose.

PP0783

REGULATOR OF G PROTEIN SIGNALING (RGS) 6 BLUNTS THE EFFECT OF CANNABINOIDS IN VITRO AND IN VIVO IN MOUSE MODELS OF INTESTINAL MOTILITY

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Introduction: RGS proteins promote silencing of the GPCR responses, e.g. cannabinoid (CB) receptors. In our previous work we showed that RGS6 is highly expressed in the gut and RGS6-/- animals exhibit increased intestinal motility in physiological conditions[1].
Activation of CB receptors cause several well-recognized effects in the GI tract, including smooth muscle relaxation, reduced motility and analgesia. However, little is known about how the effect of those compounds is modulated by RGS6.
Aims & Methods: The aim of this study was to elucidate the function of RGS6 in physiology and pathophysiology of the GI tract. METHODS: We used intestinal Caco-2 cells (WT and RGS6-/-) to measure the effect of CB agonist WIN 55,212-2 on the level of intracellular, second messenger cAMP and phosphorylation of extracellular signal-regulated kinase (ERK).
Then in the in vivo conditions we used an RGS6-/- mice to evaluate the effect of this protein on intestinal motility (stress-induced hypermotility) in the course of DSS-induced inflammation in in naïve mice as well in the presence of WIN 55,212-2.
Results: WIN 55,212-2 (1 μM) caused significant reduction of both basal and forskolin-stimulated intracellular cAMP in WT and RGS6-/- Caco-2 cells. This effect was dose-dependent and was significantly stronger in the absence of RGS6. Phosphorylation of ERK was significantly increased in RGS-/- Caco-2 cells as compared to WT and stimulation with WIN 55,212-2 (1 μM) diminished this effect in the dose-dependent manner. Previously we observed a significant increase in RGS6 expression in the mouse colon samples during intestinal inflammation[1].
However, we did not see any difference between anti-inflammatory effect of WIN 55,212-2 (1 mg/kg i.p.) in WT vs. RGS6-/- animals in the model of DSS-induced colitis. On the other hand, we observed that in RGS6-/- mice with colitis WIN 55,212-2 (1 mg/kg i.p.) alleviates stress-induced motility significantly stronger than in WT animals.
Conclusion: Serving as a central control point in GPCR signaling cascades, RGS proteins hold great promises as targets for the drug development. Here we present RGS6 protein as a novel candidate for drug targeting in the GI tract.
Our study suggests that loss and perhaps also pharmacological blockade of RGS6 might be used to increase the potency of cannabinoids in the gut.

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Disclosure: Nothing to disclose.

PP0784

LOW LEVELS OF FECAL PRIMARY BILE ACIDS ARE ASSOCIATED WITH ALTERED MICROBIOTA AND COLONIC IMMUNE GENE EXPRESSION IN IRRITABLE BOWEL SYNDROME

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Introduction: Irritable bowel syndrome (IBS) is a common disorder rooted in a disruption of the gut-brain-axis, characterized by abdominal pain and altered bowel habits.
Altered gut microbiota is reported in IBS, with microbial metabolites, including bile acids, recognized to be potentially involved in the generation of IBS symptoms.
Aims & Methods: This study aims to clarify the potential link between the gut microbiota composition, fecal bile acid levels, colonic gene expression, and IBS symptoms. In patients with IBS (n = 152), and healthy controls without gastrointestinal symptoms (n = 32), we measured fecal bile acids using targeted metabolomics.
We profiled the fecal microbiota by whole-genome metagenomics, and analyzed colonic gene expression patterns by RNA sequencing transcriptomics of mucosal biopsies.
All individuals completed self-reported questionnaires on GI symptom severity and patterns (GSRS-IBS, Bristol Stool Form Scale-based stool diary).
Spearman's correlation with Benjamini-Hochberg multiple testing adjustment was used for correlation analyses, permutational ANOVA for analysis of gut microbial composition on Bray-Curtis dissimilarity, and the Wald test for analysis of differences in colonic gene expression. Differences in bile acid levels were tested with Kruskal-Wallis and post hoc Conover-Iman tests, with Benjamini-Hochberg adjustment.
Results: The levels of the hepatically synthesized primary bile acids, chenodeoxycholic acid and cholic acid, were lower in IBS-C (constipation-predominant IBS) compared to IBS-D (diarrhea-predominant IBS, $p = 2.31e^{-7}$), IBS-M (mixed type IBS, $p = 0.002$) and healthy controls ($p = 2.57e^{-6}$, Table 1). In IBS patients, levels of primary bile acids correlated with perceived symptom severity of constipation (negatively, rho = -0.3, p = 0.0002) and diarrhea (positively, rho = 0.26, p = 0.00106).
When dividing IBS patients into quartiles by primary bile acid levels, 95 genes in the lowest quartile showed altered colonic expression. Gene-set enrichment analysis revealed enrichment of immune response-related genes, including those involved in chemokine and cytokine activity. Permutational ANOVA tests revealed altered gut microbiota composition in the two lowest quartiles of primary bile acid levels compared to healthy controls (p = 0.0001, p = 0.0003).

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Group	Primary bile acid levels (pmol/mg)
Healthy controls (n = 32)	173.8 (67.3 - 717.4)
IBS-C (n = 57)	25.8 (8.5 - 88.1)
IBS-D (n = 59)	129.4 (63.9 - 1008.8)
IBS-M (n = 36)	86.5 (41.1 - 249.5)

Table 1. Median (interquartile range) of fecal primary bile acid levels in healthy controls and IBS patients.

Conclusion: Alterations in fecal primary bile acid levels were associated with stool consistency in IBS, and lower levels of primary bile acids were linked with altered gut microbiota composition and expression of colonic immune-related genes, suggesting involvement of gut microbiota-bile acid-host crosstalk in symptom generation in IBS.
Disclosure: Nothing to disclose.

PP0785

WITHDRAWN

PP0786

DIAGNOSTIC SCORE OF DYSSYNERGIC DEFECATION IN PATIENTS WITH REFRACTORY CONSTIPATION

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Introduction: Dyssynergic defecation (DD) is identified in approximately 30% of patients presenting with chronic constipation. Patients with DD usually poorly responded to standard laxative and biofeedback can be offered as the specific treatment. The diagnostic of DD relies on physiological tests including anorectal manometry (ARM) or defecography, which is not universally accessible. As there are lack of single predictive symptom for DD, hence, we aimed to establish a diagnostic score to help with DD diagnosis.
Aims & Methods: This is a retrospective study conducted at Siriraj Hospital. Derivation cohort included 402 refractory constipation patients who were referred to our center for ARM from 2017-2023. While validation cohort of 100 patients were recruited during 2014-2016. Clinical characteristics and results of physiological tests [ARM, balloon expulsion test (BET), and colonic transit time (CTT)] were collected.
The interpretation of ARM was conducted in accordance with the London classification. Univariable and multivariable logistic regression analyses were utilized and indicative features of DD were identified. The scoring system was formulated from beta-coefficients and simplified into an integer-point format. The diagnostic performance of this score was then assessed using both cohorts.
Results: The derivation cohort had DD prevalence of 52.49%. 82.09% of this cohort were female and the average age of 52.39 years. Logistic regression analysis identified significant associations between DD and the following clinical characters: duration of symptoms less than two years (OR 1.88; 95% CI=1.02-3.49), had three or more bowel movements per week (OR 4.67; 95% CI=2.47-8.86), straining (OR 7.76; 95% CI=3.57-16.90), sensation of incomplete evacuation (OR 1.95; 95% CI=1.02-3.73), sensation of anorectal blockage (OR 3.71; 95% CI=1.75-7.88), and the use of manual maneuvers to facilitate defecation (OR 2.55; 95% CI=1.37-4.78). An 11-point diagnostic scoring system was developed: 3 points are assigned for straining during defecation; 2 points each for frequent bowel movements, the sensation of anorectal obstruction, and the use of manual maneuvers; 1 point each for a symptom duration of less than two years and the sensation of incomplete evacuation. The diagnostic performance of the score was robust, with an area under the receiver operating characteristic curve (AUROC) of 0.91 (95% CI, 0.88–0.93) in the derivation cohort and 0.88 (95% CI, 0.81–0.95) in the validation cohort. The optimal cutoff was determined to be ≥ 5 points, resulting in a sensitivity of 93.4% (95% CI = 89.1-96.3), a specificity of 72.8% (95% CI = 65.9-79.0), a positive predictive value (PPV) of 79.1% (95% CI = 73.5-84.0), and a negative predictive value (NPV) of 90.8% (95% CI = 85.1-94.9). The score also exhibited excellent calibration performance.

Characteristic	Univariable Analysis		Multivariable Analysis			Score
	Crude OR	p-value	Adjusted OR	p-value	Regression coefficient	
Demographic features						
Age > 35	1.47 (0.85-2.52)	0.165				
Female Sex	1.08 (0.65-1.81)	0.753				
Clinical features						
Duration ≤ 2years	3.72 (2.39-5.79)	<0.001	1.88 (1.02-3.49)	0.044	0.63	1
BM > 3 times/week	10.45 (6.57-16.60)	<0.001	4.67 (2.47-8.86)	<0.001	1.54	2
BSFS > 3	1.65 (1.02-2.66)	0.039	1.31 (0.64-2.66)	0.458		
Straining	17.28 (9.62-31.02)	<0.001	7.76 (3.57-16.90)	<0.001	2.05	3
Incomplete evacuation	7.38 (4.74-11.49)	<0.001	1.95 (1.02-3.73)	0.044	0.67	1
Digital evacuation	7.13 (4.55-11.18)	<0.001	2.55 (1.37-4.78)	0.003	0.95	2
Sensation of blockage	10.55 (5.74-19.38)	<0.001	3.71 (1.75-7.88)	0.001	1.31	2
Present of urge	10.79 (4.99-23.35)	<0.001	2.53 (0.88-7.29)	0.085		
Proctalgia	3.22 (1.26-8.22)	0.014	1.48 (0.43-5.07)	0.529		
Previous anorectal surgery	9.89 (2.28-42.92)	0.002	2.19 (0.39-12.35)	0.374		
Spinal cord disease	2.76 (0.55-13.87)	0.216				
Psychiatric disease	11.38 (0.61-3.16)	0.439				
Medication						
Using of rectal suppository	0.93 (0.53-1.62)	0.793				
Endoscopic features						
SRUS	1.75 (0.73-4.22)	0.213				
Anal fissure	3.72 (0.78-17.76)	0.099	3.03 (0.28-32.73)	0.362		
Hemorrhoid	1.37 (0.77-2.46)	0.286				

Univariable and multivariable analysis of clinical predictors for defecation disorder

Conclusion: We proposed the first diagnostic score for DD diagnosis. As its simplicity and excellent diagnostic performance, this can be served as a practical tool for all physicians to facilitate early identification of DD. Further diagnostic confirmation by standard physiological tests and quicken triaging patients for biofeedback therapy.
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Disclosure: Nothing to disclose.

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PP0787

NEW PATHOPHYSIOLOGICAL MECHANISMS OF FUNCTIONAL CONSTIPATION EXAMINED FROM THE PERSPECTIVE OF THE BRAIN GUT AXIS BASED ON RS-FMRI

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Introduction: Functional constipation (FCon), a highly prevalent functional gastrointestinal disorder (FGID), often accompanies by mental and psychological disorders. Previous neuroimaging studies have demonstrated brain functional and structural alterations in FCon patients. However, little is known about whether and how regional homogeneity (ReHo) is altered in FCon patients. Moreover, the potential genetic mechanisms associated with these alterations remain largely unknown.

Aims & Methods: A total of 73 patients with FCon and 68 healthy controls were recruited, and we conducted ReHo comparison to identify the abnormal spontaneous brain activities in FCon patients. Based on Allen Human Brain Atlas, we further employed partial least squares regression analysis to explore gene expression profiles associated with ReHo alterations in FCon patients.

Results: Compared with healthy controls, FCon patients demonstrated significantly decreased ReHo in both bilateral caudate nucleus, putamen, anterior insula, thalamus and right middle cingulate cortex, supplementary motor area, and increased ReHo in the bilateral orbitofrontal cortex. Genes related to central nervous system development, fatty acid metabolism, and immunity were spatially correlated with abnormal ReHo patterns.

	FCon (N = 73) (Mean ± SD)	HC (N = 68) (Mean ± SD)	FCon vs. HC P value
Age (yrs)	51.14 ± 18.03	53.20±12.24	0.3487
Gender	20M/53F	23M/45F	0.408
BMI (Kg/M2)	21.96 ±4.10	24.78 ± 3.198	< 0.0001
Depression (SDS)	45.54 ± 11.16	37.46 ± 8.15	< 0.0001
Anxiety (SAS)	40.51 ±11.05	35.44± 7.52	0.0019
PAC-QOL	45.16 ± 19.07	N/A	N/A
Wexner constipation score	11.71 ± 3.76	N/A	N/A

Table 1. The demographic and clinical characteristics of FCon and HC.

Conclusion: Our findings showed significant ReHo alterations in FCon patients, and these changes may be related to complex polygenetic and poly-pathway mechanisms, which provides a new perspective to understand the pathophysiology of FCon.

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Disclosure: Nothing to disclose.

PP0788

DOCUMENTATION OF BOWEL, BLADDER AND SEXUAL DYSFUNCTION SYMPTOMS IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Bowel, bladder and sexual dysfunction symptoms are common in patients with Inflammatory Bowel Disease (IBD). Despite significantly impacting on quality of life, few patients proactively seek help for these symptoms. For this reason clinician awareness of these functional symptoms in IBD patients is vital to identifying patients who may benefit from targeted treatment.

Aims & Methods: This study aimed to describe how well clinicians identified bowel, bladder and sexual dysfunction symptoms in IBD patients in the outpatient setting. Clinic letters from consecutive patients attending specialist IBD clinics at a tertiary hospital in the United Kingdom, from 1 January 2024, were reviewed retrospectively for documentation of bowel, bladder and sexual dysfunction symptoms. Patients with unclassified IBD, microscopic colitis, stomas and J-pouches were excluded. Data on patient demographics, IBD disease phenotype and activity were also collected.

Results:

One hundred and twenty patients were included, of which 60/120 (50%) were male, 76/120 (63%) had ulcerative colitis (UC), and 44/120 (37%) had Crohn's disease (CD), with median age 45 (range 18-88). Faecal urgency (FU) was the most commonly documented bowel symptom, documented in 26/120 (22%) patients, with 14/120 (12%) documenting the presence of FU and 12/120 (10%) documenting no FU. Faecal incontinence and constipation were each documented in 5/120 (4%) patients, and anorectal pain was documented in 3/120 (2.5%) patients. Urinary symptoms were only documented in one instance: urinary urgency in 1/120 (0.8%) patients. There was no documentation of urinary frequency or voiding dysfunction. Sexual dysfunction symptoms were not documented in any of the 120 letters, nor was impact of bowel symptoms on quality of life. FU was more frequently documented in patients with UC (11/76[14%]) compared to CD (3/44[7%]), whilst all cases of anorectal pain were documented in CD patients 3/44(7%), with no documentation of anorectal pain in any patients with UC. In patients with CD, documentation of FU was similar in remission (Harvey Bradshaw-Index (HBI) <4) and active disease (HBI≥ 4) groups (2/44 vs. 1/44, p=n.s.).

Conclusion: Bowel, bladder and sexual dysfunction symptoms are infrequently documented in patients with IBD. Failure to recognise these symptoms is likely to impair access to appropriate treatment. Introduction of a standardised reporting tool and improving IBD clinician awareness may facilitate best care for this patient group.

Disclosure: Nothing to disclose.

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PP0789

STUDY OF FUNCTIONAL CONSTIPATION: IS ANORECTAL MANOMETRY ENOUGH?

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Introduction: Constipation is a common anorectal symptom that causes significant changes in the patients' quality of life. Anorectal manometry (ARM) and magnetic resonance (MR) defecography are important tools in diagnosing disorders associated with constipation, which are mainly structural and/or functional. Pelvic floor dyssynergia is diagnosed in the presence of alterations in 2 of the following: defecation kinetics in ARM, balloon expulsion test (BET) and/or MR defecography.

Aims & Methods: Compare the findings in ARM and MR defecography, in patients with constipation.

Retrospective cohort study including patients who fulfilled the Roma IV criteria for functional constipation, who were submitted to ARM and MR defecography. The London classification was used to describe the findings in ARM, establishing 3 groups of patients: Group 1 with those who had normal anorectal coordination and normal BET, Group 2 had altered anorectal coordination and altered BET, and Group 3 with those who had inconclusive findings: altered anorectal coordination and normal BET or normal anorectal coordination with abnormal BET. The structural alterations considered significant in MR defecography were those described as moderate to severe, which lead to defecation obstruction.

Results: A total of 51 patients were included, most were female (98.0%), with a mean age of 55 ± 12 years. Regarding the ARM findings, 16 patients were in Group 1 (31.4%), 11 in Group 2 (21.6%) and 24 in Group 3 (47.1%). Forty-three patients had alterations in MR defecography (84.3%), with 42 patients having at least one structural alteration and the most common being rectocele (72.5%) and descending perineum syndrome (66.7%). Pelvic floor dyssynergia was diagnosed by MR defecography in 9 patients (17.6%).

In the group of patients with normal anorectal coordination in ARM and normal BET (Group 1: n=16), 14 had significant structural alterations in MR defecography (87.5%) and 2 patients had no alterations (12.5%). In the group of patients with altered anorectal coordination in ARM and altered BET (Group 2: n=11), 8 also had structural alterations in MR defecography (72.7%). In the group of patients with inconclusive findings in ARM (Group 3: n= 24), 20 had structural alterations (83.3%) and 3 patients had pelvic floor dyssynergia diagnosed by MR defecography (12.5%), 2 of whom had both alterations (8.3%). In this group, 3 patients had no alterations in MR defecography (12.5%).

Patients with pelvic floor dyssynergia, with or without structural alterations, had a significantly lower rectoanal pressure differential than those with solely structural alterations (-25.0 ± 27.1 vs 5.7 ± 30.0 mmHg, p=0.004). There were no statistically significant differences in the values of anal pressure at rest and voluntary contraction pressure between patients with pelvic floor dyssynergia and those with solely structural alterations (p=0.350 and p=0.833), as well as in the values of residual anal pressure, intrarectal pressure and percentage of relaxation during defecation (p=0.267, p=0.257 and p=0.348).

Conclusion: Most patients had alterations in MR defecography, regardless of the ARM findings. Therefore, it is important to interpret the results of the ARM in conjunction with an imaging test capable of identifying structural alterations that may condition defecation in patients with functional constipation. In addition, those with pelvic floor dyssynergia, with or without associated structural alterations, had a lower rectoanal pressure differential than those with solely structural alterations.

Disclosure: Nothing to disclose.

PP0790

FECAL CONSISTENCY AS A MARKER OF GOOD RESPONSE IN FUNCTIONAL BOWEL DISORDERS TREATMENT AND THE ROLE OF SOLUBLE FIBERS

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Introduction: Functional bowel disorders are among the most frequent gastrointestinal complaints. As a first step in clinical therapy, current guidelines are clear in indicating the least invasive, reversible and low morbidity treatments. In this context, soluble dietary fibers are included on some constipation and fecal incontinence guidelines.

A simple way to assess normal intestinal transit is through fecal consistency, commonly demonstrated by the Bristol Stool Scale (BSS). Hard stools (BSS types 1 and 2) can often indicate slowed intestinal transit. Similarly, individuals with normal stools (BSS 3 and 4) tend to have a lower incidence of gastrointestinal symptoms.

Aims & Methods: Objective: To evaluate the clinical response to the use of partially hydrolyzed guar gum and inulin, according to fecal consistency, as the first approach in the treatment of functional intestinal disorders.

Methods: Prospective study of 648 patients with functional bowel disorders (functional constipation, functional diarrhea, irritable bowel syndrome and anal incontinence) consecutively treated on an outpatient basis by the same researchers.

Patients were asked to take one measuring spoon (5g) of a mixture composed of partially hydrolyzed guar gum (60%) and inulin (40%) added to water, tea or juice, two times a day for a 30-day period, and to drink it within the first minute.

If BSS was 1 to 2, patients were told to mix the measuring spoon to 250 - 300 mL of water, tea, or juice. If BSS was 5 to 7, to mix it to 50 mL, and finally, if BSS was 3 to 4, to mix the measuring spoon to 200 mL.

Mean BSS, global improvement percentage, complication rate, Constipation Scoring System (CSS) and Cleveland Clinic Incontinence Score (CCIS) were applied by the authors to the patients during the first appointment and during the 30-day follow-up appointment.

All patients accepted to participate by signing the informed consent form.

Results: Of the 648 patients, 61.9% were female, with a mean age of 48.1 years. All patients informed no difficulty in complying with the soluble fiber protocol.

Initial mean CSS was 12,0 (+-4,8) and CCIS was 2 (+-1). Mean BSS of constipation patients was 2 (+- 1) of functional diarrhea and fecal incontinence was 5 (+-1) and to irritable bowel syndrome was 5 (+-1), respectively.

By the time of the 30-day follow-up appointment, 71.7% of the patients had BSS types 3 and 4, associated with 63.1% (+- 18.4) of overall improvement, while in the patients who presented the other types of fecal consistency (28.3%), the overall improvement was only 21.8% (+-14.3).Mean CSS was 6,7 (+-3,9) and CCIS was 1 (+1), $p < 0,01$.

Conclusion: Achieving normal fecal consistency is associated with significant overall improvement. Oral soluble fibers use according to fecal consistency are a valid tool in treating functional bowel disorders.

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PP0791

LIVING WITH FAECAL INCONTINENCE: A QUALITATIVE INVESTIGATION OF PATIENT EXPERIENCES AND PREFERRED OUTCOMES THROUGH SEMI-STRUCTURED INTERVIEWS

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Introduction: Faecal incontinence (FI) is a prevalent and debilitating anorectal problem which can cause embarrassment, anxiety, social isolation and diminishes quality of life (QoL). Currently, there is no optimal treatment option for FI, consequently, treatments primarily focus on symptom reduction and improving QoL. In order to gain a clearer understanding of the challenge's patients encounter, it is essential to actively listen to patients' perspectives and experiences, as well as the specific outcomes they aim to achieve from treatment. This study aims to explore how FI impacts patients' lives and identify important treatment outcomes as part of the development of a Core Outcome Set (COS).
Aims & Methods: Patients with FI were recruited from surgery and gastroenterology outpatient clinics in the Netherlands. Semi-structured interviews were performed, audio recorded, transcribed per verbatim and coded. Thematic analysis was performed to identify (sub)themes and categories relevant to the patients.
Results: Twelve interviews were conducted before saturation was reached (75% female, 25% male, mean age 63, range 39-83 year). Four main themes emerged 'Physical symptoms', 'Impact on daily life', 'Emotional impact' and 'Coping'. Patients expressed how FI severely limits daily activities and emotional wellbeing. Patients' treatment priorities centred on resuming normal activities rather than solely on (physical) symptom reduction.
Conclusion: The impact of FI extends far beyond uncontrolled loss of faeces, affecting psychological, emotional, and social wellbeing. Patients prioritise outcomes focussed on reclaiming normalcy and independence rather than focusing on physical symptoms alone. Integrating these patient-centered outcomes in future studies could enhance treatment satisfaction and patient-perceived treatment success.
Furthermore, outcomes identified through this study will be combined with other outcomes identified through literature review, forming a comprehensive list for a Delphi survey with relevant stakeholders to establish a consensus on a final list of outcomes for a COS for FI.
Disclosure: Nothing to disclose.

PP0792

TOWARDS A CORE OUTCOME SET FOR FAECAL INCONTINENCE – A SYSTEMATIC REVIEW OF OUTCOMES REPORTED IN LITERATURE

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Introduction: Faecal incontinence (FI) is a prevalent issue which can have a detrimental impact on the patient's quality of life. Current international guidelines lack strong treatment recommendations due to few studies in the field, in combination with the heterogeneity in outcome reporting. To address this, a Core Outcome Set (COS) is proposed to standardize outcome reporting in FI studies, facilitating meta-analyses and enhancing therapeutic recommendations.
This systematic review aims to identify which outcomes have been used in FI intervention studies so far in adult patients as a starting phase for the development of a future COS for FI.
Aims & Methods: Medline, Embase and the Cochrane library were searched to identify all outcomes reported in comparative effectiveness trials assessing one or more treatment option in adult patients suffering from FI. The outcomes were categorized according to the Core Outcome Measurement in Effectiveness Trials (COMET) taxonomy to standardize outcome terminology, assess completeness, and inform subsequent steps in COS development.
Results: A total of 109 studies were included, which revealed 51 unique outcomes which were classified into 38 domains within four core areas. On average four outcomes were reported per study (range 1-11). The most

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commonly reported outcomes were 'Severity of FI' (83%), 'Quality of life' (64%), 'Number of FI episodes' (40%), 'Anorectal motor function' (39%), and 'Frequency of bowel movements' (16%).

Conclusion: This systematic review offers an overview of outcomes reported in FI studies, highlighting the heterogeneity between studies. This heterogeneity emphasises the need for standardising outcome reporting which can be established through the creation of a COS. Outcomes identified through this study will be combined with other outcomes identified through patients interviews, forming a comprehensive list for a Delphi survey with relevant stakeholders to establish a consensus on a final list of outcomes for a COS for FI.

Disclosure: DK and SB have received grant support from UEG and ESCP for the development of a guideline for faecal incontinence. DK has received a grant from ZonMw for the study of the clinical efficacy of anal plugs in faecal incontinence. DK has received research funding from Rome Foundation, MLDS, Horizon 2020, Horizon Europe, Allergan, Grunenthal, and speaker's free from Dr Falk (paid to host institute), all outside of submitted work. The other authors declare that they have no conflict of interest.

PP0793

ANORECTAL PAIN IS A PREDICTOR OF DYSSYNERGIC DEFECATION IN PATIENTS WITH REFRACTORY FUNCTIONAL CONSTIPATION

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Introduction: Functional Defecation Disorders (FDD) are a frequent cause of refractory functional constipation (FC). The diagnosis of FDD requires anorectal manometry (ARM) and a balloon expulsion test (BET), which are not easily available outside referral centers. FDD often do not respond to conventional treatments, while they respond to biofeedback.

Aims & Methods: The aim of our study was to evaluate if pain reported by patients during defecation or elicited during digital rectal examination (DRE) predicted a FDD diagnosis in a cohort of refractory FC patients.

Two-hundred and thirty-eight FC (Rome III Criteria) patients failing a 30-day conservative treatment trial (increased fiber and fluids, tailored laxatives) were enrolled in the study.

The presence of weekly anorectal pain during defecation and the elicited pain/tenderness consequent to a stretching of the levator ani muscle during DRE (DRE-pain) had been evaluated before and after the macrogol trial. Possible causes of organic anorectal pain were excluded.

All patients underwent:

- ARM+BET;
- barium defecography, if ARM and BET were discordant;
- second DRE with evaluation of DRE-pain.

OR and accuracy were calculated for both pain during defecation and DRE-pain in predicting a Rome III FDD.

Results: One hundred and twenty patients (50.42%) showed features of FDD: 82 dyssynergic defecation (DD) and thirty-eight inadequate defecatory propulsion (IDP). Thirty-three structural outlet obstructions (SOO) were identified using barium defecography.

Pain during defecation was strongly associated with DD, with an OR of 7.85 (p<0.001) and an accuracy of 75.2% in diagnosing a FDD (Tab.1).

DRE-pain was strongly associated with DD, with an OR of 11.78 (p<0.001) and an accuracy of 77.3% in diagnosing DD (Tab.1). Furthermore, DRE-pain showed a very good reproducibility between T0 and T30 (Cohen's K=0.87).

The combination of pain during defecation and DRE-pain showed an even stronger association to DD compared to single items, with an OR of 20.84 and an accuracy of 82.8% in diagnosing DD (p<0.001, Tab.1).

Neither pain during defecation nor DRE-pain were predictive of IDP.

		Rome III DD		p-value	OR (95% CI)	Sens.	Spec.	PPV	NPV	Acc.
		No	Yes							
Weekly anorectal pain during defecation	No	121 (77.6%)	24 (29.3%)	<0.001	7.85 (4.2 – 14.68)	70.7%	77.6%	62.4%	83.4%	75.2%
	Yes	35 (22.4%)	58 (70.7%)							
DRE-pain	No	96 (76.8%)	16 (21.92%)	<0.001	11.78 (5.9 – 23.58)	78.1%	76.8%	66.3%	85.7%	77.3%
	Yes	29 (23.2%)	57 (78.08%)							
DRE-pain + Weekly anorectal pain during defecation	No	110 (88%)	19 (26.03%)	<0.001	20.84 (9.83 - 44.17)	73.9%	88%	78.3%	85.3%	82.8%
	Yes	15 (12%)	54 (73.97%)							

Table 1 - Association and diagnostic devaluation of reported pain during defecation and DRE-pain

Abbreviations: DD=dyssynergic defecation; OR=odds ratio; 95% CI=95% confidence interval; Sens.=sensitivity; Spec.=specificity; PPV=positive predictive value; NPV=negative predictive value; Acc.=accuracy; DRE-pain= pain/tenderness consequent to a traction on the levator ani muscles during digital rectal examination

Conclusion: Anorectal pain during defecation or elicited during DRE were shown to be predictive of a DD diagnosis in FC patients. An early referral to anorectal physiology testing may be considered in patients presenting these clinical features, leading to an improvement in their management.

Disclosure: Nothing to disclose.

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PP0794

EVALUATION OF A MODIFIED LONDON PROTOCOL INVOLVING ONLY TWO 10-SECOND PUSHES TO DIAGNOSE DEFECATION DISORDERS WITH HIGH-DEFINITION ANORECTAL MANOMETRY

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Introduction: Anorectal manometry (ARM) is one of the required investigations to diagnose defecation disorders according to Rome IV criteria. The International Anorectal Physiology Working Group (IAPWG) introduced into clinical practice a standardized sequence of manoeuvres to perform ARM (London Protocol) (Carrington EV, 2020). IAPWG Protocol suggested the need of performing three push attempts, each of 15s duration, to assess recto-anal coordination, with a consensus <75% between experts.

Aims & Methods: The aim of this study was to assess the diagnostic accuracy, compared to IAPWG recommendations, of simplified ARM protocols featuring:

- only two push attempts (each of 15 seconds);
- three pushes attempts (each of 10s);
- two pushes attempts (each of 10s)

High-Definition ARM (HD-ARM) traces from 150 patients (118 F, mean aged 54 years) who underwent ARM for faecal incontinence (n=37) or constipation (n=105) were retrospectively reviewed. All traces were performed according to the IAPWG Protocol.

Firstly, we evaluated if the mean rectal and anal pressures were different among the three pushes. Subsequently, ARM traces were re-evaluated considering the following modifications to the IAPWG protocol:

- considering only the first two pushes (as if the third had never been performed);
- considering the first ten seconds of each push (as if each push featured only ten seconds);
- considering the first ten seconds of the first two pushes (as if the third had never been performed and the first two featured only ten seconds).

In all three cases, the best push (defined as the most qualitatively normal) among the available pushes was chosen.

New diagnoses were made based on these modified protocols and compared with the standard to evaluate whether these modifications could affect the final diagnosis.

Results: According to the IAPWG London Protocol, the qualitative best pushes were the first, the second or the third in 58, 45 and 47 patients, respectively. No statistically significative differences were found in anal and rectal pressures among the three pushes.

No statistically significative differences were found in anal and rectal pressures comparing the three modified protocols with the IAPWG's one.

Regarding the diagnostic accuracy, all the proposed modified London Protocol showed an almost perfect concordance compared with the standard of practice (K=0.97, Tab.1).

This result was confirmed for the constipated and incontinent patients' subgroup.

		No disorders of rectoanal co-ordination	Normal expulsion with abnormal manometric pattern of rectoanal co-ordination	Abnormal expulsion with normal manometric pattern of rectoanal co-ordination	Abnormal expulsion with poor propulsion	Absormal expulsion with dyssyn-ergia	Abnormal expulsion with poor propulsion and dyssyn-ergia	Comparison of proportions (Kappa Cohen agreement)
All patients N=150	3 Push 15 sec	5	18	16	25	44	42	-
	2 Push 15 sec	5	18	14	25	45	43	k=0.974
	3 Push 10 sec	5	18	15	25	45	42	k=0.974
	2 Push 10 sec	4	19	15	25	45	42	k=0.966
Constpated patients N=105	3 Push 15 sec	3	15	10	14	30	33	-
	2 Push 15 sec	3	15	9	14	30	34	k=0.975
	3 Push 10 sec	3	15	9	14	30	34	k=0.975
	2 Push 10 sec	3	15	9	14	30	34	k=0.975

Table 1 - Cohen's K concordance according to London Protocol among all patients and in constipated patients

Conclusion: Based on our results, it seems that two pushes of ten second each may be enough to reach an accurate manometric diagnosis of defecation disorders.

Performing only 2 push manoeuvres of 10s each may reduce the examination length of 55s, reducing patient's discomfort and, likely, improving HD-ARM's acceptability.

Disclosure: Nothing to disclose.

PP0795

SHORT-COURSE BIOFEDBACK THERAPY MAY BE USEFUL IN THE MANAGEMENT OF FUNCTIONAL DEFECATORY DISORDER IN RESOURCE-POOR SETTINGS- A CENTRE PROSPECTIVE OBSERVATIONAL STUDY BASED ON THE LONDON CLASSIFICATION OF ANORECTAL DYSFUNCTION

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Introduction: Dyssynergic defecation constitutes more than half the cases of refractory chronic constipation, and biofeedback therapy is the first-line treatment in the management of functional defecatory disorder (FDD). The studies done by Rao et al[1] and Chiarioni G et al[2]. showed response rates of 79% and 80%, respectively, with the standard protocol of Biofeedback therapy (BFT). However, the need for six to eight sessions of biofeedback therapy once every two weeks can pose economic and logistic challenges to patients.

Aims & Methods: Our study aims to determine whether a short biofeedback course helps manage dyssynergic defecation, especially in a resource-poor setting. We also wanted to know the factors predicting the response to biofeedback therapy.

It was a single-center, prospective observational study. Consecutive patients of age more than 16 years who fulfilled Rome IV criteria for functional defecatory disorder were recruited in the study. Patients with structural and metabolic causes of constipation were excluded. They were classified according to Rao's type of dyssynergia and London classification of anorectal dysfunction. All patients underwent one biofeedback therapy session per day for three consecutive days. Each patient was asked the PAC-SYM questionnaire before initiating biofeedback therapy, and the total score was calculated. The same questionnaire was used after three

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months, either in person or via a telephone call. Patients with a reduction in average score by 0.6 points or more at three months were considered to have responded to the biofeedback therapy[3].

Results: A total of 70 patients were included in the study. Most patients were males (70%), and the median duration of constipation was 5 (2-10) years. The median PAC- SYM questionnaire score at baseline was 1.33 (1.08- 1.58). Most patients belonged to Rao's Type 1 dyssynergia (91.4%), followed by Type 3 dyssynergia (7.1%). When patients were classified according to the London classification, Anal hypotension with normal contractility was noted in 12.8% patients and anal normotension with hypocontractility was noted in 4.2% patients. Abnormal expulsion with dyssynergia was noted in 78.5% of patients, and abnormal expulsion with poor propulsion and dyssynergia was noted in 21.4%. None of the patients had absent Recto- anal areflexia, abnormal expulsion with normal manometric pattern of recto-anal coordination, abnormal expulsion with poor propulsion, normal expulsion with abnormal manometric pattern of recto-anal coordination, or rectal hyposensitivity.

The study also showed that 52.8% of patients (95% CI 40.5%- 65.9%) responded to biofeedback therapy at the end of three months. However none of the factors like age, sex, duration of illness, and types of anorectal dyssynergia were statistically significant between BFT responders and non-responders (Table-1).

Variables	Responded to BFT	Not responded to BFT	p value
Patients (n, %) (95% CI)	37 (52.8%) (40.5%- 64.9%)	33 (47.1%) (35.1%- 59.4%)	
Age (years)	31 (26-43)	39 (27- 54)	0.12
Sex (%)			0.79
Male	25 (51.0%)	24 (48.9%)	
Female	12 (57.1%)	9 (42.8%)	
Duration of illness (years)	5 (3-10)	4 (2-10)	0.30
Rao's type of Dyssynergia (n,%)			0.38
Type 1	32 (50%)	32 (50%)	
Type 2	1 (100%)	0 (0%)	
Type 3	4 (80%)	1 (20%)	
Type 4	0 (0%)	0 (0%)	
Anal hypotension with normal contractility (n,%)	5 (55.6%)	4 (44.4%)	1.00
Abnormal expulsion with dyssynergia (n,%)	28 (50.9%)	27 (49.0%)	0.39
Abnormal expulsion with poor propulsion and dyssynergia (n,%)	9 (60%)	6 (40%)	0.57

Conclusion: Standard course biofeedback therapy (6- 8 sessions fortnightly) is superior to short course biofeedback therapy in managing dyssynergic defecation. However short course of biofeedback therapy is helpful in the managing dyssynergic defecation in resource-poor setting.

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Disclosure: Nothing to disclose.

PP0796

IMPACT OF PUBLISHING A NATIONAL POSITION PAPER ON THE PERFORMANCE AND INTERPRETATION OF ANORECTAL MANOMETRY: ADDRESSING THE GAP BETWEEN INTERNATIONAL RECOMMENDATIONS AND CLINICAL PRACTICE

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Introduction: In 2019, an international anorectal physiology working group published a consensus paper ('London') with recommendations for the performance of anorectal manometry (ARM) and balloon expulsion testing (BET)[1]. However, little is known on the implementation of these recommendations in the clinical setting.

Aims & Methods: We aimed to (1) evaluate, on a national level, current practice in all aspects of ARM and BET performance and interpretation, (2) identify gaps compared to both updated international recommendations and between testing centers, and if present, (3) jointly draft a national position paper, advocate for its instillation, and examine the impact of the publication on the identified gaps.

Between October and November 2020, gastroenterologists from 20/20 (100%) centres who perform and/or interpret ARM in Israel completed a 61- item online survey encompassing all stages of ARM and BET testing (preparation, testing protocol and reporting).

Results: Major inconsistencies between centres were demonstrated (table 1, baseline survey), especially in reporting, including:

(1) the use of London classification for description of anorectal dysfunction

(2) lack of normal values for reference

(3) selective reporting of anal squeeze pressure and anal slow waves, and;

(4) report template and language.

High concordance (>90%) between centres in most stages of preparation and testing was noted, consistent with international recommendations, with the exception of testing for anal squeeze endurance, cough, rectoanal inhibitory reflex (RAIR) and BET methodology.

	Baseline survey N=20	Follow up survey N=22	P value for difference
Squeeze endurance testing- n (%)	17 (85%)	20 (91%)	NS
Cough/ Valsalva testing - n (%)	15 (75%)	20 (91%)	NS
RAIR 50 ml, repeat to 100 ml if needed - n (%)	5 (25%)	10 (45%)	NS
BET - on toilet and dedicated expulsion balloon - n (%)	2 (10%)	6 (27%)	NS
BET Balloon fill: water - n (%)	5 (25%)	10 (45%)	NS
Volume of BET balloon: 50 ml - n (%)	15 (75%)	22 (100%)	NS
London classification on report - n (%)	13 (65%)	22 (100%)	<0.01
Anal squeeze pressure on report: absolute and increment - n (%)	6 (30%)	13 (59%)	0.07
Anal slow waves on report- n (%)	4 (21%)	4 (18%)	NS

Table 1. Changes in performing anorectal manometry and balloon expulsion testing following local position paper publication and hands-on educational course.

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Following these results, an ARM and BET position paper was endorsed by the Israeli Neurogastroenterology Society (October 2021), followed by a free-of-charge hands-on educational course focused on proper performance of tests (Feb 2022) and publication of the final version of the position paper (Jan 2023)[2].

One year following publication, in a repeat survey of centres (22/24; 4 new centres), almost all responders (19/22;86%) stated they have changed their clinical practice following publication of the national position paper, and 21/22 (95%) reported now using the published normal values.

Compared to the baseline survey, the following improved (table 1, follow up survey): (1) reporting according to the London classification, and (2) uniform volume of balloon used for BET.

High concordance (>90%) was also demonstrated now for performing squeeze endurance and cough, as well as reporting templates, but not for RAIR or for the use of a dedicated evacuation balloon or sitting position for BET.

Conclusion: The current study highlights major gaps in ARM and BET testing between consensus papers and clinical practice on a national level. Publication of a national position paper along with an educational curriculum can address existing gaps.

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Disclosure: Nothing to disclose.

PP0797

DISORDERS OF GUT BRAIN INTERACTIONS FOLLOWING SALMONELLA GASTROENTERITIS: A 30-YEARS LONGITUDINAL AND PROSPECTIVE STUDY

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Introduction: Salmonella Enteritidis, a Gram-negative bacterium, poses a significant threat to gastro-intestinal health, often leading to widespread outbreaks. Acute gastroenteritis increases the likelihood of developing a disorder of gut brain interaction (DGBI), such as irritable bowel syndrome (IBS) and functional dyspepsia (FD).

This study investigates the long-term risk of DGBI development 30 years post-Salmonella exposure and explores additional predictive factors.

Aims & Methods: In 1994 a Salmonella Enteritidis outbreak involved 36 schools in Bologna, 1811 individuals tested positive for this bacterium in stool cultures. Following this event, a prospective cohort study was initiated after 30 years.

300 subjects were randomly selected and assessed for DGBI prevalence using Rome III and Rome IV questionnaires, persistent gastrointestinal symptoms via the Gastrointestinal Symptoms Rating Scale (GSRS), anxiety and depression levels with the Hospital Anxiety and Depression Rating Scale (HADS), and quality of life using the SF12 questionnaire (12 item short form).

The exposed group comprised 150 people who contracted S. Enteritis infection in 1994, while the control cohort consisted of individuals, matched for sex, age and geographic origin, who had not experienced a bacterial gastroenteritis. The DGBI course was estimated by comparing our results with a previous follow-up carried out in 2009.

Results: Questionnaires were completed by 175 subjects (58,3%, 88 cases and 87 controls). Among the exposed subjects 39.8% developed IBS and 34.1% FD compared to 6.9% IBS and 4.6% FD in the control group (p=<0.001). Multivariate logistic regression in the study population revealed predictive factors for IBS: S. Enteritidis infection [Odd Ratio (OR) 7.056, 95% CI 2.684-18.554, p=<0.001] and anxiety level evaluated in 2024 (OR 1.244, 95% CI 1.105-1.400, p=<0.001).

Similarly, predictors for FD included S. Enteritidis infection (OR 10.218, 95% CI 1.259-82.925, p=0.03), anxiety level assessed in 2009 (OR 1.212; 95% CI 1.058-1.390, p=0.006), and Mental component summary (MCS) score (OR 0.047; 0.002-0.985; p=0.049).

Notably, anxiety level evaluated in 2024 was predictive of IBS development in the exposed group, while anxiety level in 2009, along with MCS score, was predictive for FD. We found an increased IBS and decreased FD occurrence in 2024 compared to 2009 follow-up.

Exposed individuals exhibited a statistically significative higher rates and severity rates of gastrointestinal symptoms in the following areas: abdominal pain, hunger pains, diarrhea, loose stools, urgency, rumbling bloating, constipation, hard stools and incomplete emptying. Psychiatric comorbidities resulted more common in exposed group than in non-exposed one, in particular anxiety evaluated in 2024 (31.8% vs 10.3%, p=0.002). The exposed group showed lower scores in all SF12 subcategories compared to the control group.

	OVERALL (n = 175)	CASES (n = 88)	CONTROLS (n= 87)	P
ROME IV				
IBS, n (%)	41 (23.4)	35 (39.8)	6 (6.9)	<0.001
FD, n (%)	34 (19.4)	30 (34.1)	4 (4.6)	<0.001
PDS, n (%)	21 (12.0)	17 (19.3)	4 (4.6)	0.003
EPS, n (%)	19 (10.9)	19 (21.6)	0	<0.001
OVERLAP PDS-EPS, n (%)	6 (3.4)	16 (18.1)	1 (1.2)	<0.001
HADS				
Depression, Mean (SD)	3.68 ± 2.83	3.92 ± 3.01	3.43 ± 2.63	0.433
Anxiety, Mean (SD)	5.33 ± 3.42	6.19 ± 3.92	4.47 ± 2.63	0.001

Conclusion: This study, following a 1994 foodborne culture-proven S. Enteritis outbreak, highlights that a bacterial gastroenteritis is a major risk factor for IBS and for FD even after 30 years from the infection.

Disclosure: Nothing to disclose.

PP0798

TESTING FOR INTOLERANCE TO FRUCTANS, A LONG-ACTING FODMAP, IN 600 PATIENTS WITH DISORDERS OF GUT-BRAIN INTERACTION (DGBI)

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Introduction: Fructans or fructooligosaccharides are included in the spectrum of FODMAP (fermented) sugars, which are relevant in the induction of symptoms in patients with disorders of gut-brain interaction (DGBI) and in non-coeliac gluten sensitivity (1).

Fructans have been shown to induce visceral hypersensitivity, an increase in mucosal mast cells, and changes in the colonic mucosal barrier in mice, and hypersensitivity associated with an altered gut microbiome in humans (2-5).

Breath tests are used in the assessment of intolerances to several FODMAPs, but the test is poorly standardised for fructans (6).

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Aims & Methods: A standardised breath test with fructans (10g Orafti®HSI) was performed afer a 24-hour low-FODMAP diet and >8-hour fast. Exhaled hydrogen and methane concentrations were measured every 30 minutes until 5h, and synchronised symptom assessments performed for 24h (composite symptom index score, SI, 7) in 600 successive male (33%) and female patients of average age 41 years referred to our gastroenterology practice with DGBI (irritable bowel syndrome and/or functional dyspepsia, Rome 4 criteria).

Breath gas concentration thresholds of >20ppm and >8ppm above baseline were used for hydrogen and methane, respectively.

Results: The SI was positive in 61.2% (n=367) of patients within 24 hours of fructan ingestion. The SI peaked in the first 3 hours in 17.5% (n=105) and in the first 5 hours in 42.5% (n=255) of patients. Suprathreshold increases in hydrogen and methane concentrations within 5 hours were seen in 62.7% and 14.9% of patients, respectively. Discordant SI and breath gas results were obtained in 35.1% (hydrogen) and 57.8% (methane) of patients. A negative SI with a *supra*threshold hydrogen or methane result occurred in 18.2% and 5.2%, respectively.

A positive SI with a *sub*threshold hydrogen or methane result occurred in 16.9% and 52.6%, respectively. Suprathreshold concentrations in methane and not in hydrogen were apparent in 4.5%. Main symptoms during tests were excessive flatulence (58.8%), bloating (50.3%), abdominal fullness (49.3%), abdominal cramps (29.0%), nausea (26.7%), GORD (26.7%), diarrhoea (9%) and extra-GI symptoms (16.2%).

Conclusion: Standardised oral provocation with fructans resulted in familiar clinical symptoms in a majority of patients with DGBI, suggesting a high prevalence of fructan intolerance. Due to the slow fermentation of intestinal fructans and the associated delayed symptoms, prolonged data collection over at least 12, preferably 24 hours is recommended. The test duration of 3 hours currently commonly used in test centres is inadequate. Symptom and breath gas concentration results are often divergent, as is known from other FODMAP breath tests, and validation against clinical outcome results is required. Fructans are of interest in DGBI as the current literature suggests they have both protective as well as detrimental effects on intestinal physiology, which may be dependent on the resident microbiota.

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Disclosure: Nothing to disclose.

PP0799

CHARACTERISTICS OF PATIENTS WITH CARBOHYDRATE INTOLERANCE IN DISORDERS OF GUT-BRAIN INTERACTION

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Introduction: Patients with Disorders of Gut-Brain Interaction (DGBI) often experience meal-related symptoms, potentially due to carbohydrate intolerances like fructose and/or lactose. While breath tests are common for exploring carbohydrate malabsorption, assessing intolerances, defined by the onset of gastrointestinal (GI) symptoms after a defined load of carbohydrates, is valuable. The adult carbohydrate perception questionnaire (aCPQ) can be used for detecting carbohydrate intolerances (1).

Aims & Methods: We aimed to determine lactose and fructose intolerance prevalence in DGBI and characterize these patients. DGBI patients who underwent fructose (25g) and lactose (25g) breath tests alongside the aCPQ in our tertiary center between May 2022 and December 2023 were included. Carbohydrate intolerance was a 20 mm increase in any of the five Visual Analog Scales (VAS) scores (for abdominal pain, nausea, diarrhea, bloating, and flatulence) in aCPQ. An increase from baseline of ≥20 ppm in exhaled H2 or ≥10 ppm of CH4 (≥20 ppm for fructose) indicated malabsorption. Patients completed questionnaires on irritable bowel syndrome (IBS) symptom severity (IBS-SSS), somatic symptoms (PHQ-15), quality of life (QOL)(GIQLI), anxiety, and depression (HAD).

Results: We included 314 DGBI patients, among them, 124 were lactose intolerant, 138 were fructose intolerant, and 77 were intolerant to both carbohydrates. Lactose intolerant patients were younger (42± 15.9 vs 46 ± 15.3; p=0.022), and more likely to have lactose malabsorption and fructose intolerance, while fructose intolerant patients were more likely to have fructose malabsorption, lactose intolerance, and higher anxiety/depression levels. Both intolerant groups were more likely to be female (81.5% vs 61.6%; p<0.0001 for lactose, 77.5% vs 63.1%; p=0.006 for fructose), with 2 DGBI or more (63.1% vs 40.7%; p<0.0001 for lactose, 57.4% vs 43.5%; p=0.021 for fructose), to have a reduced QOL, higher levels of GI symptoms, and somatic symptoms (table).

Variable	Positive Lactose Intolerance (n=124)	Negative Lactose Intolerance (n=190)	P-value	Positive Fructose Intolerance (n=138)	Negative Fructose Intolerance (n=176)	P-value
Positive Lactose Malabsorption	49 (39.5)	29 (15.3)	<0.0001	38 (27.5)	40 (22.7)	0.328
Positive Fructose Malabsorption	36 (29)	45 (23.7)	0.290	49 (35.5)	32 (18.1)	<0.0001
Positive Fructose Intolerance	77 (62.1)	61 (32)	<0.0001			
Positive lactose intolerance				77 (55.8)	47 (26.7)	<0.0001
GIQLI	86 ± 20.85	95 ± 19.49	<0.0001	85.1 ± 19.96	96.4 ±19.45	<0.0001
IBS-SSS	252 ± 100.87	207 ± 101.53	<0.0001	255.6 ± 98.48	201.2 ± 101.06	<0.0001
HAD-Anxiety	7.9 ± 4.11	7.8 ± 4.03	0.970	8.4 ± 3.97	7.4 ± 4.08	0.032
HAD-Depression	5.7 ± 3.8	5.2 ± 3.6	0.233	6 ± 3.74	4.9 ± 3.63	0.012
PHQ-15	13 ± 5.51	10 ± 4.97	0.0001	13 ± 5.34	10 ± 4.98	<0.0001

Table: Comparison of DGBI patients with and without lactose and fructose intolerance.

Values are n (%), or mean ± SD.

GIQLI, Gastrointestinal Quality of Life Index; IBS-SSS, irritable bowel syndrome symptom severity score; HADS, Hospital Anxiety and Depression Scale; PHQ-15, patient health questionnaire 15.

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Logistic regressions showed independent associations of female sex, fructose intolerance, and lactose malabsorption with the lactose intolerance variable, and IBS symptom severity, lactose intolerance, and fructose malabsorption with the fructose intolerance variable.

Conclusion: DGBI patients with lactose and fructose intolerance exhibit severe IBS symptoms and reduced QOL. Interestingly, intolerance to one carbohydrate is associated with malabsorption of this specific carbohydrate and the coexistence of intolerance to the other carbohydrate.

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Disclosure: CM has served as a consultant/advisory board member for KyowaKirin, Norgine, Biocodex, Mayoly Spindler, Tillots, Ipsen, Nestlé HealthScience and Viatris.

FM has served as a consultant/speaker for Laborie, Medtronic, Dr Falk Pharma.

GG has served as a consultant/speaker for Laborie, Medtronic, Kyowa Kirin, Lilly and Enterra medical.

PP0800

THE EFFECT OF MICROBIOTA ON ZONULIN EXPRESSION IN FECES IN PATIENTS WITH FUNCTIONAL BOWEL DISEASES

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Introduction: violations of the "gut-brain" axis, changes in intestinal permeability, as well as changes in the qualitative and quantitative composition of the intestinal microbiota.

Zonulin is a protein of intercellular tight contacts capable of reversibly affecting intestinal permeability. It is well known that the main provocateurs of zonulin production are gluten and bacteria. Taking into account the research data indicating differences in the microbial composition in patients with functional intestinal diseases and healthy volunteers, it can also be assumed that their zonulin levels change, as well as its decrease after treatment of such patients with probiotics.

Aims & Methods: To evaluate the relationship between probiotic intake and changes in zonulin levels in patients with functional bowel diseases.

We determined zonulin levels in 108 patients with functional bowel diseases, such as functional constipation (FC), functional diarrhea (FD), irritable bowel syndrome with diarrhea (IBS-D) and constipation (IBS-C), before and 4 weeks after taking a multi-strain probiotic containing Lactobacillus acidophilus 1.25 x 10^9 SFU, Bifidobacterium lactis - 1.25 x 10^9 SFU, Lactobacillus paracasei - 1.25 x 10^9 SFU, Lactobacillus rhamnosus - 1.25 x 10^9 SFU.

Zonulin concentration in feces was determined by enzyme immunoassay (Human insulin ELISA kit zonulin, 96 (BlueGene Biotech Shanghai (PRC)).

Zonulin levels were assessed as follows:

- within the reference values – less than 83.5 ng/ml
- elevated levels – from 83.5 to 110 ng/ml
- high level – more than 110 ng/ml).

Results: According to the data obtained during our study, the level of zonulin in feces ranged from 13.6 to 458.5 ng/ml, and the median values were 113.5 (98.6-122.9; 95% CI) ng/ml. Zonulin was detected within the reference values in 30 (28%) patients, an elevated level in 24 (22%) patients, and a high level in 54 (50%). When comparing for each of the 4 functional diseases, it was noted that patients with IBS-D were statistically significantly more likely to have an increase in zonulin levels above the reference, compared with IBS-C - 86.0% vs 55.6% (p=0.019; according to the Fisher criterion) and compared with FC - 86.0% vs 58.0% (p=0.033; according to the Fisher criterion). In patients with FD, there was also a statistically significant difference in the frequency of increased zonulin in feces: 81.0% vs 55.6% (p=0.013; according to Fisher's criterion) between FD and IBS-C and 81% vs 58.0% between FD and FC (p=0.014; according to Fisher's criterion). When comparing IBS-C and FC, as well as IBS-D and FD, there was no statistically significant difference.

Against the background of probiotic therapy for 4 weeks, zonulin levels ranged from 13.6 to 173.7 ng/ml, the median was 82.3(73.9-93.9) ng/ml. In patients with IBS-D, as well as in FD, there was a statistically significant decrease in the level of zonulin in feces: the median values decreased from 121.7 to 67.8 ng/ml (p=0.027) in IBS-D and from 130.7 to 81.1 ng/ml (p=0.039) in FD. In patients with constipation syndrome, a decrease in indicators was also noted, but the results were not statistically significant: in IBS-C, zonulin levels decreased from 89.4 to 79.1 ng/ml (p=0.089), and in FC from 107.6 to 92.8 ng/ml (p=0.077).

Conclusion: Thus, the data of the conducted study show that functional intestinal diseases are associated with an increase in the level of zonulin in feces. The highest values were observed in patients with diarrheal syndrome. Probiotic therapy for 4 weeks had a positive effect on the level of zonulin in the stool.

Disclosure: Nothing to disclose.

PP0801

COLONIC VOLUMES IN IRRITABLE BOWEL SYNDROME AND HEALTHY INDIVIDUALS; THE ASSOCIATIONS WITH SYMPTOMS AND GUT PHYSIOLOGY

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Introduction: The pathophysiological mechanisms of irritable bowel syndrome (IBS) are complex and incompletely known. Studies investigating the role of colonic volumes and their relation with symptoms and measurements of gut physiology in IBS are sparse.

Aims & Methods: The aim of this study was to explore the associations between colonic volume and symptoms and measurements of gut physiology in IBS and healthy individuals. We included 18 IBS patients (4 IBS-diarrhea, 5 IBS-constipation, 9 IBS-mixed or IBS-unspecified) and 19 healthy individuals. Total and segmental colonic volumes were determined by magnetic resonance imaging (MRI). The GSRS-IBS was used to measure gastrointestinal symptoms and a stool diary (Bristol Stool Form Scale) was completed 14 days before the MRI.

Oro-anal transit time (radiopaque markers) and rectal sensitivity (rectal barostat, including pressure thresholds for first perception, defecation, and discomfort) were measured.

Results: The total colonic volume did not differ between IBS patients and healthy individuals (736 ml [528-903] vs 597 [498-939] (median [IQR]);

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p=0.44). In the IBS group, there were trends towards inverse associations between total colonic volume and the perceived severity of diarrhea (r_s= -0.43; p=0.08), as well as the volume of the descending colon and stool frequency (r_s= -0.48; p=0.07). Other GI symptoms and stool consistency were not associated with colonic volumes. When investigating the physiological parameters in IBS, the total colonic volume was positively associated with oro-anal transit time (r_s=0.54; p=0.02) and negatively associated with rectal pressure thresholds for sensations (first perception: r_s= -0.51; p=0.04, defecation: r_s= -0.44; p=0.09), Table 1. A similar pattern was seen when investigating individual colonic segments, with tendencies towards positive associations between segmental colonic volumes and oro-anal transit time, and negative associations with rectal sensory thresholds. The associations in healthy individuals between colonic volumes and oro-anal transit time and rectal sensitivity, respectively, were overall weaker (all p>0.05).

	Total colonic volume (ml)	
	IBS	Healthy individuals
Oro-anal transit time (days)	0.54*	0.37
Rectal barostat		
First perception (mmHg)	- 0.51*	- 0.21
Defecation (mmHg)	- 0.44	- 0.14
Discomfort (mmHg)	- 0.15	- 0.02

Table 1. Spearman's correlation between total colonic volume measured with magnetic resonance imaging and oro-anal transit time (radiopaque markers) and rectal barostat.
** p≤0.05*

Conclusion: In this pilot study, we have shown that larger colonic volumes, measured with MRI, seem to be associated with slower oro-anal transit and rectal hypersensitivity in IBS. This suggests a potential role for measuring colonic volumes in IBS pathophysiology studies, but larger studies are needed to better appreciate subtype differences and their relevance for symptoms.
Disclosure: Nothing to disclose.

PP0802

DIFFERENCES IN SYMPTOMS AND GUT PHYSIOLOGY BETWEEN IRRITABLE BOWEL SYNDROME SUBGROUPS IDENTIFIED BY MIXTURE MODEL ANALYSIS

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Introduction: Irritable bowel syndrome (IBS) is subtyped based on bowel habits. However, an alternative based on combined gastrointestinal (GI), non-GI somatic and psychological symptom profiles has been proposed[1,2].
Aims & Methods: This study aimed to validate the novel more comprehensive way to subgroup IBS and investigate differences in symptoms and gut physiology in these groups. In total, 760 IBS patients (Rome III and IV) from six study cohorts were included. Subgroups were based on Gaussian mixture model analysis using 13 single-item GI symptom scores (GSRS-IBS), stool frequency and consistency (Bristol Stool Form Scale), 11 single-item non-GI somatic symptom scores (PHQ-12, excluding menstruation problems), and anxiety and depression scores (HADS). Differences between subgroups were investigated for fatigue (MFI), GI-specific anxiety (VSI), IBS-specific quality of life (IBS-QOL), and measurements of gut physiology, including oro-anal transit time (OATT) assessed by radiopaque markers, visceral sensitivity assessed by a rectal barostat procedure and lactulose nutrient challenge test (LNCT), and breath hydrogen and methane production during LNCT as markers for microbial fermentation.
Results: Six subgroups based on GI and non-GI symptom profiles were identified: diarrhea with low comorbidities (n=67), constipation with low comorbidities (n=120), overall mild severity (n=166), diarrhea with high comorbidities (n=231), constipation with high comorbidities (n=89), and mixed GI symptoms with high comorbidities (n=87). The groups with high comorbidities had higher levels of fatigue and GI-specific anxiety, and lower QOL compared to the low comorbidities groups, Table 1. OATT was longer in constipation with low comorbidities vs diarrhea with low comorbidities (1.7 [1.0-2.7] vs 0.9 [0.5-1.3] days (median [IQR]), p<0.01), but there was no difference in OATT comparing constipation with high comorbidities vs. diarrhea with high comorbidities. Visceral sensitivity and postprandial hydrogen and methane production did not differ between groups, but with a tendency of methane production to be highest in constipation with high comorbidities.

	Diarrhea, Low comorbidities	Constipation, Low comorbidities	Overall mild severity	Diarrhea, High comorbidities	Constipation, High comorbidities	Mixed GI, High comorbidities	p-value
High fatigue (general fatigue score ≥13)	61 %	52 %	53 %	69 %	79 %	87 %	<0.001
GI-specific anxiety (VSI), Median (IQR)	45 (36, 54)	41 (31, 53)	40 (27, 51)	45 (32, 56)	47 (34, 58)	53 (41, 62)	<0.001
Quality of life							
IBS-QOL (total) Median (IQR)	70 (62, 79)	65 (56, 79)	71 (65, 83)	65 (54, 75)	55 (47, 65)	56 (38, 68)	<0.001
Gut physiology							
Oro-anal transit time (days), Median (IQR)	0.9 (0.5, 1.3)	1.7 (1.0, 2.7)	1.1 (0.8, 2.0)	1.3 (0.9, 2.0)	1.3 (0.9, 1.9)	1.2 (0.9, 2.2)	<0.01
Rectal barostat (discomfort Z-score*), Median (IQR)	0.22 (- 0.22, 0.44)	- 0.24 (- 0.79, 0.97)	- 0.24 (- 0.75, 0.47)	- 0.16 (- 0.75, 0.37)	- 0.24 (- 0.75, 0.27)	- 0.24 (- 0.79, 0.09)	0.20
Exhaled hydrogen (LNCT, AUC), Median (IQR)	5415 (3233, 12473)	7388 (5344, 10009)	6758 (3278, 9825)	5708 (4035, 9938)	7155 (3600, 8715)	6210 (3986, 11505)	0.73
Exhaled methane (LNCT, AUC), Median (IQR)	1050 (525, 2220)	1403 (701, 3270)	1320 (743, 2460)	1170 (683, 4073)	2280 (1050, 5535)	1140 (529, 2513)	0.09

Table 1. Symptoms, quality of life and gut physiological parameters.
** Z-scores of rectal barostat discomfort thresholds (mmHg) based on different protocols AUC, area under the curve; IQR, interquartile range; LNCT, lactulose nutrient challenge test; VSI, visceral sensitivity index*

Conclusion: We have confirmed previously reported IBS subgroups based on combinations of GI and non-GI symptom profiles. Demonstrated associations with fatigue, GI-specific anxiety, and oro-anal transit time, suggest different pathophysiological mechanisms in these novel subgroups. Our results suggest that the presence or absence of non-GI symptoms could be important when planning IBS treatment strategies.
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Disclosure: Nothing to disclose.

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PP0803

HETEROGENEITY OF IRRITABLE BOWEL SYNDROME IN CHILDREN: AN ACCENT ON THE TRIGGER FACTOR

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Introduction: In accordance to Rome IV Criteria irritable bowel syndrome (IBS) is classified into four subtypes: IBS with predominant constipation (IBS-C), IBS with predominant diarrhea (IBS-D), with mixed bowel habits (IBS-M) or IBS, unsubtyped. Some authors distinguish a post-infectious subtype of IBS, in which an acute episode of infectious gastroenteritis is a trigger of the disorder.

In all the other cases, mostly psycho-emotional stress is the trigger factor, with this variant being defined as stress-associated subtype of IBS.

Aims & Methods: Aim: to study the peculiarities of the formation and course of IBS in children, depending on the trigger factor. A total of 54 patients aged 6-12 years with a verified diagnosis of IBS according to Rome criteria IV have been examined. The trigger factor of the onset of IBS was established by the interrogation of patients and their parents. The levels of trait anxiety and somatization were assessed by Children's Form of the Manifest Anxiety Scale (CMAS) questionnaire and Somatoform Symptom Screening (SOMS) test respectively.

The *Catechol-O-methyltransferase* (*COMT)* gene val158met polymorphism was estimated by the polymerase chain reaction. Enzyme immunoassay Ridascreen Calprotectin (R-Biopharm AG, Germany) was used for the quantitative determination of calprotectin in stool samples.

Results: We have diagnosed post-infectious subtype of IBS in 30 children. In this group we have found significantly higher concentration of fecal calprotectin (60.0 (35.6; 129.0) vs 31.2 (21.1; 58.7 μg/g)), which is consistent with pathogenesis of this subtype of IBS. On the contrary, we have discovered that psycho-emotional stress was the trigger factor of the onset of IBS in 22 patients (stress-associated subtype of IBS). Children with stress-associated IBS had higher levels of trait anxiety (25.1±5.7 vs. 18.7±6.4) and somatization (11.1 ± 4.0 vs. 8.6 ± 3.1) in comparison to patients with post-infectious subtype of the disorder. Furthermore, the analysis of the *COMT* gene val158met polymorphism revealed, that almost the half patients (48%) with stress-associated IBS had 472 AA (Met/Met) genotype. Met/Met carriers have genetically determined low enzymatic activity of COMT, which may lead to decreased elimination of catecholamines and is associated with chronic stress. In contrast, only 31% of children with post-infectious subtype of IBS were found to have 472 AA (Met/Met) genotype.

Conclusion: Our study suggests that not only predominant bowel habit, but also trigger factor in the development of IBS determine the heterogeneity of the disorder. This approach may be used for classification of IBS in children, as it dictates treatment strategy.

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PP0804

THE LINK BETWEEN ALEXITHYMIA, PSYCHOLOGICAL COMORBIDITIES AND THE SEVERITY OF IRRITABLE BOWEL SYNDROME

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Introduction: Irritable bowel syndrome (IBS) is a disorder of gut-brain interactions that combines abdominal pain and transit disorders in the Rome IV classification. The severity of IBS varies from patient to patient and is influenced by psychological comorbidities such as anxiety and depression. Alexithymia, also called emotional blindness, is the difficulty in identifying and expressing one's own emotions and feelings or those of others, and is present in 2-20% of healthy subjects. It is frequently found in different types of chronic pain disorders and has also been described in IBS (1). The association between alexithymia and pain intensity varies according to the studies and the type of pain, however. The aim of our study was to assess the prevalence of alexithymia in patients with IBS and to look for links between this trait and both the clinical phenotype of the disorder and the levels of anxiety and depression of the patients.

Aims & Methods: This study was carried out, after approval from a local ethics committee, between January and August 2023 in consecutive patients suffering from IBS according to Rome IV criteria assessed in a multidisciplinary day hospital setting (workup by a neurogastroenterologist, a dietician and a psychologist). Clinical characteristics (age, gender, subtype of IBS, disease duration and severity (IBS-SSS)), levels of anxiety-depression (HADS), quality of life (EQ-5D) and alexithymia (TAS-20, a score composed of 20 questions rated from 1 to 5, maximum score of 100, high-level cutoff >60) were recorded.

Results: We included 76 IBS patients (67.1% female; mean age 48.3 ± 16.9 years; BMI 22.1 ± 4.0 kg/m^2). The median duration of the disorder was 13 years (IQR 6.7 - 26.4). All subtypes were represented (IBS-D 27.6%, IBS-C 48.7% and IBS-M 17.1%). The majority of patients had a moderate-severe disorder, with a mean IBS-SSS score of 318 ± 107 points (minimal, moderate and severe forms in 1.3%, 34.7% and 58.7% respectively, and remission in 5.3%). Quality of life was impaired, with a median EQ-5D health score of 50 (IQR 35.0-69.5). The HADS-A subscore was 10.8 ± 4.4 points, with 59.7% of patients with definite anxiety (>11), while the HADS-D subscore was 6.2 ± 3.6 points, with 16.4% of patients with definite depression (>11). The total HAD score was 17.0 ± 6.7 points. The severity of IBS was correlated with depression (rho 0.35, p=0.005) and inversely correlated with quality of life (rho -0.28, p=0.02) scores. The mean alexithymia score was 58.7 ± 9.8 points, with 45.4% of patients having a high level of alexithymia (TAS-20 >60). The level of alexithymia was not different between men and women (58.1 ± 9.5 vs 59.8 ± 10.2; p=0.48) and was not influenced by the duration of the disorder (p=0.92). Alexithymia correlated well with

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anxiety (rho 0.33, p=0.007), depression (rho 0.36, p=0.003) and quality of life (rho -0.31, p=0.016) scores but not with IBS severity (rho 0.17, p=0.18). However, the level of IBS severity was greater in the case of high levels of alexithymia (IBS-SSS of 359.2 ± 73.1 vs 298.4 ± 107.7 points for a TAS-20 > 60 vs ≤ 60; p=0.009).

Conclusion: In this series of patients with moderate-severe IBS and all subtypes, a high level of alexithymia was found in almost one out of two patients. Alexithymia was linked to the level of anxiety-depression and the deterioration in quality of life. Patients with a high level of alexithymia had more severe IBS. The prognostic value of alexithymia for the response to treatments such as hypnotherapy and cognitive behavioural therapies deserves to be evaluated in patients with IBS.

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PP0805

PSYCHIATRIC DIAGNOSES AND DISEASE BURDEN IN PATIENTS WITH IRRITABLE BOWEL SYNDROME (IBS)

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Introduction: Psychiatric comorbidity, most commonly measured as severity of anxiety and depression symptoms, has been found to be associated with IBS symptom severity. However, available studies have rarely assessed the importance of formal psychiatric diagnoses on disease burden in IBS. In this analysis from a pan-European project (DISCOvERIE) focusing on the relevance of comorbid mental (anxiety, depression) and/or somatic (fibromyalgia, chronic fatigue syndrome) diagnoses in patients with IBS, we aimed to assess whether presence of psychiatric diagnoses according to the DSM-5 criteria has an impact on overall symptom severity in IBS.

Aims & Methods: We included patients with IBS according to the Rome IV criteria from nine different European centers. Exclusion criteria included the presence of severe psychiatric disease (i.e. bipolar disorder, schizophrenia, autism spectrum disorder, schizoaffective disorder, or organic psychiatric disorder demanding psychopharmacotherapy, psychiatric day-care or inpatient treatment). The patients underwent The Mini International Neuropsychiatric Interview (MINI) to diagnose major psychiatric diagnoses (a total of 17). Fibromyalgia and chronic fatigue were diagnosed by current diagnostic criteria. They also filled in questionnaires assessing Severity of IBS symptoms (IBS-SSS), depression (PHQ-9), anxiety (GAD-7), somatic symptoms (PHQ-15), and fibromyalgia symptoms (FIQ). Severity of symptoms were compared between groups with increasing number of psychiatric diagnoses using One-Way Between-Groups ANOVA with linear trend analysis.

Results: In total, 617 patients with IBS (median age 44 years, 80% females) were included. Among the IBS patients, 269 (44%) had none of the mental (anxiety, depression) or somatic (fibromyalgia, chronic fatigue syndrome) comorbidities (median age 44 y, 75% females), whereas 174 (28%) had comorbid mental disorder only (median age 38 y, 82% females), 49 (8%) had comorbid somatic disorder only (median age 50 y, 86% females) and 124 patients (20%) had mental and somatic comorbidities (median age 47 y, 86% females). The most common psychiatric diagnoses among the IBS patients were depression (n=284), generalized anxiety disorder (n=145) and panic disorder (n=38) followed by agoraphobia (n=23) and social anxiety (n=19). Participants were divided into four groups according to the number of psychiatric diagnoses (0, 1, 2, and ≥3). The severity of IBS, depression, anxiety, somatic, and fibromyalgia symptoms all increased gradually with increasing number of psychiatric diagnoses. The effect sizes (partial eta squared) were small for IBS symptoms, medium for somatic symptoms and large for the other severity measures (Table 1).

	No. of psychiatric diagnoses (MINI)				Partial eta squared	P-value
	0 (n=355)	1 (n=83)	2 (n=84)	≥3 (n=95)		
IBS symptom severity (IBS-SSS)	230 [170–310]	267 [210–339]	254 [184–348]	282 [232–351]	0.017	0.002
Depression (PHQ-9)	4 [1–8]	8 [4–13]	10 [3–15]	13 [8–17]	0.170	<0.001
Generalized anxiety (GAD-7)	4 [1–7]	5 [3–9]	8 [5–14]	11 [8–15]	0.227	<0.001
Somatic symptoms (PHQ-15)	10 [7–14]	13 [10–16]	13 [8–19]	15.5 [12–21]	0.097	<0.001
Fibromyalgia symptoms (FIQ)	16 [6–29]	31 [17–45]	37 [19–55]	45 [27–62]	0.154	<0.001

Table 1: Symptom severity in groups with increasing number of psychiatric diagnoses

Scores presented as medians with interquartile range.

Conclusion: This multicenter pan-European study shows that psychiatric diagnoses are common among IBS patients and that the overall disease burden in IBS is associated with psychiatric comorbidity. These findings highlight the importance of taking psychiatric comorbidity into consideration and to acknowledge the relevance of gut-brain axis interactions when managing patients with IBS.

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PP0806

AGREEMENT RATE OF ACUTE FOOD INDUCED MUCOSAL ALTERATIONS VISUALIZED USING CONFOCAL LASER ENDOMICROSCOPY IN IRRITABLE BOWEL SYNDROME PATIENTS

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Introduction: Confocal laser endomicroscopy (CLE) enables real-time microscopic visualization of the duodenal mucosa during upper endoscopy and has shown acute food-triggered disruption of the epithelial barrier in the duodenum of irritable bowel syndrome (IBS) patients. While this technique offers novel scientific insights, image interpretation is subjective and agreement rates between observers are unknown.

Aims & Methods: We aimed to assess the agreement rate between assessors regarding acute food induced mucosal changes visualized by CLE in IBS. To this end we performed a multicentric, international post-hoc evaluation of CLE recordings. CLE sequences from three centers were blindly assessed. Five investigators from the three participating centers blindly assessed all videos both in real-time and in slow-motion in a random order, including some duplicates.

Mucosa was considered altered if fluorescein leakage AND "cell shedding" were simultaneously present. Outcomes of the global assessment were assessed using the binary outcome "altered" vs. "unaltered". Agreement was quantified using Fleiss' kappa for >2 assessors and Cohens' kappa for 2 assessors – both are reported as κ.

Results: A total of 124 video sequences (94572 frames, 197 minutes of real-time recording) were reviewed. Intraindividual agreement rate was between 90 and 100% (κ=0.74-1.00). Overall interobserver agreement rate for all assessors and all videos was 86% (κ=0.69). Agreement rates were not higher when videos were assessed in slow-motion (81%, κ=0.58). Among the 3 centers, agreement between all blinded assessors and real-time assessment by the endoscopist ranged from 78% (κ=0.55) to 86% (κ=0.67). A total of 63 videos were judged the same by all assessors (100% agreement) – here agreement rates with the endoscopist were 97% (κ=0.91). Using these videos as reference, a cutoff of 2 altered spots in a recording of 5 spots showed a sensitivity and specificity of 100% for the presence or absence of alterations.

Conclusion: We report here for the first time the interobserver agreement rates of acute food induced mucosal alterations as visualized on CLE. The agreement rates were substantial overall – both among blinded assessors and between blinded assessors and real-time judgement by the endoscopist. Slow-motion revision did not increase agreement rates. A cutoff of 2/5 spots was ideal for the detection of acute mucosal alterations.

Disclosure: Nothing to disclose.

PP0807

IMPROVED LIPID AND GLUCOSE PROFILE IN IBS PATIENTS AFTER A DIETARY INTERVENTION WITH SSRD OR LOW FODMAP

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Introduction: Irritable bowel syndrome (IBS) is the most common disorder of gut-brain interaction (DGBI) and associated with other pain syndromes, lifetime stress, and psychological disorders. During the last years, associations have also been described between IBS and overweight and the metabolic syndrome. Low content of fermentable oligo-, di-, and monosaccharides and polyols (FODMAP) and a starch- and sucrose-reduced diet (SSRD) reduce IBS symptoms (1,2). However, concerns have been raised whether a diet may lead to malnutrition when adhered to for a longer time.

Aims & Methods: The aim of the study was to randomize IBS patients according to Rome IV with a total score of IBS-SSS > 175 to either SSRD or low FODMAP and measure nutritional status over time. 155 patients, 131 women (84.5%), age 42 (32-55) years, weight 69.2 (63.0-82.9) kg, BMI 25 (23-28) kg/m^2 were included. 77 received SSRD (5 dropouts) and 78 low FODMAP (6 dropouts) for 4 weeks, with a second follow-up at week 24. Rome IV, IBS-severity scoring system (IBS-SSS) and visual analog scale-IBS were completed at baseline, and after 4 and 24 weeks. Nutritional status, lipid profile, and glycosylated hemoglobin A1c (HbA1c) were analyzed in blood according to clinical routines at the laboratory. Differences were calculated by Wilxon Signed Ranks with values at week 4 and 24 in comparison with baseline. P<0.05 was considered statistically significant.

Results: There were no differences in baseline values between the groups. All gastrointestinal and extraintestinal symptoms were improved by both diets (p<0.001 in most variables). The weight reduction was most pronounced in SSRD. At week 4, folate was increased (p=0.003), and iron decreased (p=0.001), in the SSRD group, whereas vitamin D was increased in both groups (p=0.004 vs. p=0.017) and HbA1c was decreased (p=0.010 vs. p=0.009). The only negative nutritional measurement at week 24 was lower cobalamin levels in both groups (p=0.009 vs. p=0.002) and lower ferritin in the low FODMAP group (p=0.007). Cholesterol, low-density lipoprotein (LDL) cholesterol, and non-high-density lipoprotein (non-HDL) cholesterol were decreased throughout the study in the SSRD group, but only after 4 weeks in the low FODMAP group (Table).

	SSRD N = 77		Low FODMAP N = 78	
P-Cholesterol (3.3-6.9 mmol/L)		P-value		P-value
Baseline	5.0 (4.4-5.6)		4.7 (4.4-5.8)	
4 weeks	4.8 (4.0-5.4)	0.007	4.9 (4.2-5.5)	0.003
24 weeks	4.8 (4.3-5.4)	0.009	4.8 (4.2-5.7)	0.166
P-LDL (1.4-4.7 mmol/L)				
Baseline	3.2 (2.7-3.8)		3.1 (2.7-3.8)	
4 weeks	3.0 (2.4-3.7)	0.004	3.0 (2.6-3.6)	0.036
24 weeks	3.0 (2.5-3.6)	0.017	3.0 (2.5-3.8)	0.088
P-non-HDL (1.9-5.1 mmol/L)				
Baseline	3.4 (2.8-4.0)		3.3 (2.8-4.1)	
4 weeks	3.3 (2.6-3.8)	0.015	3.2 (2.7-3.9)	0.002
24 weeks	3.2 (2.6-4.0)	0.008	3.3 (2.8-4.2)	0.164

Conclusion: Both SSRD and low FODMAP decrease gastrointestinal and extraintestinal symptoms. Although some nutritional analyses showed lower values after the intervention, other showed improved values. Metabolic risk factors such as weight, lipid values and HbA1c were improved by

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the diets, especially SSRD. Thus, dietary interventions in IBS do not only improve symptoms but also improve lipid and glycemic profiles without any major impact on nutritional status when the IBS patients are followed for 6 months. Reducing carbohydrate intake appears to positively impact bowel symptoms and metabolism in IBS patients, irrespective of the specific carbohydrates targeted by the corresponding diet. Future studies are warranted to identify the individual saccharides most likely to exert therapeutic effects, eventually in relation to their breakdown capacity in individual patients.

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PP0808

A STARCH- AND SUCROSE-REDUCED DIET HAS SIMILAR EFFICIENCY AS LOW FODMAP IN IBS. A RANDOMIZED NON-INFERIORITY STUDY

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Introduction: A diet with low content of fermentable oligo-, di-, and monosaccharides and polyols (FODMAP) are established treatment for irritable bowel syndrome (IBS), with well-documented efficiency. Recently, a starch- and sucrose-reduced diet (SSRD) has shown similar promising effects.

Aims & Methods: This randomized, non-inferiority study aimed to test SSRD against low FODMAP and compare the responder rates to a 4-week dietary intervention of either diet.

The 155 IBS patients were randomized to SSRD (n=77) or low FODMAP (n=78) for 4 weeks, with a follow-up 20 weeks later. The SSRD advice means removal of sucrose and reduction of starch-containing food such as cereals, some vegetables (e.g., potatoes), rice, and pasta, with maintained gluten and lactose. Low FODMAP means reduction of fermentable saccharides and polyols. The questionnaires Rome IV, IBS-severity scoring system (IBS-SSS), and visual analog scale for IBS (VAS-IBS) were completed at baseline and after 2, 4, and 24 weeks. Weight, height, waist circumference, and blood pressures were measured at week 2, 4, and 24. Comparisons were made within the groups and the differences between the changes in the two groups were compared.

Results: The patients were categorized according to the Rome IV criteria as constipation-predominated IBS (IBS-C) (n=26, 16.8%), diarrhea-predominated IBS (IBS-D) (n=44, 28.4%), mixed IBS (IBS-M) (n=54, 34.8%), unspecified IBS (IBS-U) (n=7, 4.5%), and unspecific functional bowel disorder (FBD) (n=24, 15.5%) with weekly abdominal pain but weak association between the pain and bowel habits. There were no differences between the groups at baseline. In the SSRD group, 72 (93.5%) completed the study at week 4 and 53 (68.8%) at week 24, with corresponding figures 72 (92.3%) and 49 (62.8%) in low FODMAP. The responding rate (RR=ΔTotal IBS-SSS ≥ −50) was non-inferior in SSRD compared with low FODMAP at week 2 (84.7% vs. 81.4%); p=0.658), week 4 (84.7% vs. 84.7%; p=1.00), and week 24 (52.8% vs. 68.0%; p=0.159). All gastrointestinal and extraintestinal symptoms were equally improved (p<0.001 in most variables) after the 4-week intervention. This remained at week 24 for all specific abdominal symptoms, total IBS-SSS, total extraintestinal IBS-SSS, and most specific extraintestinal symptoms. The decrease in total IBS-SSS correlated at all time points with the decrease of total extraintestinal symptoms (p<0.001) and improved psychological well-being (p<0.001). At week 4, 26.4% in both groups were considered healthy with <75 in total IBS-SSS. At week 24, around 40-50% had abominal pain 2-3 times a month, or less, or did have <75 in total IBS-SSS. SSRD rendered more pronounced reductions in weight (p=0.006), body mass index (BMI) (p=0.005), and sugar craving (p=0.05) compared with low FODMAP, whereas waist circumference and blood pressure were equally decreased in both groups. Weight and BMI were regained at follow-up. The decrease in total IBS-SSS at week 24 correlated with the decrease of weight (p=.010) and BMI (p=0.013), and the responders at week 24 in the SSRD group still had lowered weight (-0.7 (-2.5-0.1) vs. 0.2 (-0.7-2.2) kg; p=0.005) and BMI (-0.25 (-0.85-0.03) vs. 0.07 (-0.35-0.77) kg/m^2; p=0.009) compared with baseline in contrast to non-responders. Those who had tested both diets preferred SSRD (p=0.032).

Conclusion: The SSRD effect on gastrointestinal and extraintestinal IBS symptoms were not inferior to the low FODMAP effect. Reductions of weight, BMI, and sugar craving were most pronounced following SSRD. Thus, dietary intervention in IBS do not only affect symptoms but also anthropometric parameters.

Disclosure: Nothing to disclose.

PP0809

ONDANSETRON IN DIARRHOEA PREDOMINANT IRRITABLE BOWEL SYNDROME AND FUNCTIONAL DIARRHOEA: OUTCOMES FROM A REAL-LIFE CLINICAL EXPERIENCE

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Introduction: 5-HT3 receptor antagonists are a second line treatment for irritable bowel syndrome with predominant diarrhoea (IBS-D), but there is no evidence on their efficacy for functional diarrhoea (FDr) (Savarino, 2022). Randomized controlled trials show that ondansetron seems to be effective in treating IBS symptoms (Garsed 2014, Gunn 2023), but real-life data are lacking.

Aims & Methods: Our aim was to study the clinical response to ondansetron in patients with IBS-D or FDr in a real-life setting. A retrospective analysis on consecutive patients (December 2014 – February 2024) meeting criteria for IBS-D or FDr according to Rome III criteria was conducted in a single Neurogastroenterology outpatient clinic. Only patients prescribed with ondansetron were included. A standardized template was used to

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collect clinical data and patient-reported response to treatment. Ondansetron was prescribed at initial dosage of 4mg 1 tablet every other day with the possibility of increasing the dosage up to 6 tablets/daily. Clinical response was defined as (i) an improvement in the number of bowel movements per day vs baseline and (ii) firmer stool consistency (Bristol stool form types 3 and 4, 'normal').

Results: Ninety-four patients (female 68.1%, mean age 45 ±15 years old) were included. Among them, 77 (82%) were diagnosed with IBS-D and 17 (18%) with FDr. The mean follow-up duration was 19.1 months (SD±16.71). Seventy patients (74.5%) achieved a clinical response to ondansetron. Response was maintained with ondansetron at a mean dosage of 2.5 tablets/daily (SD±1.5) for a mean follow-up duration of 23.5±17 months. There was no significant difference between the number of IBS-D and FDr patients who responded to ondansetron (table 1). Patients with IBS-D reported a significantly higher improvement in abdominal pain vs baseline (p<0.01, table 1). Twenty-four patients (25.5%) did not respond to ondansetron. Adverse events were significantly higher in non-responders compared to responders (p=0.02, table 1).

Main characteristics at baseline	Responders, n (%)	Non-responders, n (%)	p-value*
IBS-D FDr	55 (71.4) 15 (88.2)	22 (28.6) 2 (11.7)	0.22
Bristol Scale 5 Bristol Scale 6-7	24 (34.3) 46 (65.7)	8 (33.3) 16 (66.7)	1
Vomit No vomit	14 (20) 56 (80)	1 (4.2) 23 (95.8)	0.07
Urgency No urgency	29 (41.3) 41 (58.7)	8 (33.3) 16 (66.7)	0.63
Opioids No opioids	14 (20) 56 (80)	7 (29.2) 17 (70.8)	0.39
Neuromodulators No neuromodulators	22 (31.4) 48 (68.6)	10 (41.6) 14 (58.4)	0.45
Main characteristics post-treatment	Responders, n (%)	Non-responders, n (%)	p-value*
Improvement in pain in IBS-D No improvement in pain in IBS-D	40 (72.7) 15 (27.3)	4 (18.2) 18 (81.8)	<0.01
Adverse events No adverse events	11 (15.7) 59 (84.3)	10 (41.7) 14 (58.3)	0.02

Table 1. Main characteristics of responders and non-responders at baseline and post-treatment

**Fisher's exact test*

Abbreviations: FDr= functional diarrhoea, IBS-D= irritable bowel syndrome with diarrhoea

Conclusion: This is the first real-life study assessing the response to ondansetron in IBS-D or FDr patients showing long-term reduction in the frequency of bowel movements, improvement in faecal consistency, and reduction in abdominal pain. Our data suggest that ondansetron can be used in FDr patients.

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PP0810

SYSTEMATIC REVIEW AND META-ANALYSIS: PLANT BASED THERAPY FOR IRRITABLE BOWEL SYNDROME

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Introduction: Irritable bowel syndrome (IBS) is characterized by a combination of abdominal pain, bloating, and stool irregularity [1]. Many patients are dissatisfied with the pharmacological treatment of IBS and turn to "complementary and alternative medicine" (CAM). In particular, plant-based preparations to treat abdominal symptoms are popular [2-4].

Aims & Methods: A systematic review and meta-analysis to present the current evidence on phytotherapy for IBS was performed.

A literature search was done using 3 electronic databases (Medline via PubMed, Embase via Ovid, and Web of Science Core Collection) from inception to April 2023. A total of 7894 citations were found. Studies were excluded if they met the following exclusion criteria: (i)were homeopathic medicines, (ii) unpublished or only partially published studies, (iii) patients under 18 years of age, (iv) studies with fewer than ten participants, and (v) publication languages other than German or English. Efficacy was assessed using dichotomous ratings of change in IBS symptoms. To pool data a random effects model with 95% confidence intervals was used. I2 statistics were used to test for heterogeneity of included studies. Scores were 0-24% no-, 25-49% low-, 50-75% moderate- and >75% high heterogeneity.

Results: Through the screening process, 56 eligible studies were identified (6054 patients).

Meta-analysis was performed for fibre (8 studies), peppermint oil (7 studies), aloe vera (3 studies), and Chinese herbal medicine (7 studies). At least one randomized controlled trial (RCT) was performed for the following plant-based products: Ginger, Kiwi fruit, St John's Wort and Aniseed oil; however, there was insufficient data to perform meta-analysis. For 192 of 387 patients fibre was helpful; whereas, placebo intake was beneficial for 127 of 346 patients (RR=1.22; 95% CI: 1.00-1.49). For 144 of 311 patients peppermint oil improved symptoms; whereas, in the control group, 97 of 357 patients experienced a beneficial effect (RR=1.63; 95% CI: 1.33-2.00). For 62 of 155 patients aloe vera relieved symptoms; whereas, the control intervention was helpful for 51 of 144 patients (RR=1.25; 95% CI: 0.76-2.06). In 403 of 731 patients, Chinese Herbal Medicine therapy had a

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positive effect; whereas, in the placebo group, 245 of 723 patients found the therapy helpful (RR=1.56; 95% CI: 1.27-1.92. The heterogeneity in the studies ranged from none to moderate. Many studies had some concern of bias in the Risk of Bias 2 (RoB 2) assessment (42/53).

Conclusion: This analysis supports the use of selected plant-based preparations in IBS management. Peppermint oil and Chinese Herbal Medicine preparations demonstrated significant efficacy compared to placebo. Dietary fiber improved stool frequency but had borderline positive effects on IBS scores. The quality of many studies was limited. Additional well-powered RCTs are needed to establish the places of phytotherapy in the management of functional gastrointestinal symptoms.

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PP0811

INVESTIGATION OF INTESTINAL PERMEABILITY AND MUCUS LAYER IN DIFFERENT COLONIC SEGMENTS IN PATIENTS WITH IRRITABLE BOWEL SYNDROME AND EFFECT OF A MUCOPROTECTANT FORMULATION

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Introduction: Irritable bowel syndrome (IBS) is one of the most frequent gastrointestinal disorders with high impact on quality of life and national health systems. Diarrhea predominant IBS (IBS-D) is characterized by abdominal pain and chronic diarrhea. Elevated intestinal permeability is a pathophysiological element of the disease pathogenesis. Mucus layer is a novel target of IBS research.

Aims & Methods: Our aims were to examine the permeability and mucus alterations in IBS-D patients' different colonic segments and the effect of formulation of xyloglucan (XG), pea protein (PP), grape seed extract (GS) and xylo-oligosaccharides (XOS) on pain and intestinal permeability in IBS-D subjects.

Methods: 14 subjects (IBS-D patients and control) were involved. After initial colonoscopy with biopsy sampling XG+PP+GS+XOS was administered to IBS-D patients for 12 weeks. Stool count and pain scores were documented. After treatment, colonoscopy was repeated. Permeability of biopsy samples was measured in Ussing-chambers. Adherent mucus layer thickness and Muc-2 expression were evaluated by histology/immunohistochemistry.

Results: XG+PP+GS+XOS treatment significantly reduced diarrhea and abdominal pain. Intestinal permeability in IBS-D patients was elevated compared to controls, XG+PP+GS+XOS restored permeability in ascendent colon. PAS-stained mucus layer was significantly thinner in IBS-D compared to controls. XG+PP+GS+XOS therapy caused a further reduction in mucus thickness. After XG+PP+GS+XOS treatment, proportion of Muc-2 positive cells was reduced compared to controls.

Conclusion: The study shows the physiological variance regarding to the adherent mucus thickness and mucus secreting cell count in the different colon segments.

Our study verifies the segmental difference in permeability alteration and shows adherent mucus thickness reduction in IBS-D. These results show the segmentally different pathophysiology in IBS-D. Mucus thickness alteration can be a new therapeutic target for the treatment. In fact, treatment with XG PP+GS+XOS significantly improved abdominal pain, and diarrhea. Moreover, it was able to restore the permeability in IBS-D patients to a physiological state. Based on this study, XG+PP+GS+XOS can be considered as an effective therapy for IBS-D symptoms.

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PP0812

ANTIDEPRESSANTS IN IRRITABLE BOWEL SYNDROME – A SYSTEMATIC REVIEW AND META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: Irritable bowel syndrome (IBS) is among the most prevalent gastrointestinal conditions. However, its therapeutic management poses challenges, with many current recommendations resting on a low level of evidence.

Aims & Methods: Our objective is to conduct a systematic review with meta-analysis of randomized, double-blind, placebo-controlled trials to assess the efficacy of antidepressants in treating IBS.

Systematic review with meta-analysis performed according to PRISMA guidelines and registered on the PROSPERO platform (CRD42024502427). PubMed, EMBASE, and The Cochrane Library platforms were searched from inception to January 2024. Only randomized, double-blind, placebo-controlled trials, conducted in patients with IBS patients, with a minimum treatment period of 4 weeks, were included. The quality of evidence was assessed using the revised version of the Cochrane tool (RoB 2). A random-effects model was used. Heterogeneity was assessed by the I^2 statistics. Funnel plots and the Egger test were used to assess publication bias. Statistical analysis was performed using STATA v16.

Results: The search strategy identified 1340 studies, of which 16 were included, totaling 1428 patients. Among these studies, 733 patients were assigned to the antidepressant group and 698 to the placebo group. In the intervention group, 405 (55.3%) patients reported improvement, compared to only 227 (32.5%) patients in the placebo group. The meta-analysis showed that the use of antidepressants in patients with IBS provides a benefit in overall symptom improvement (OR=3.02; 95% CI 2.16-4.2) compared to placebo.

A subgroup analysis of 9 studies, comprising 1008 patients, assessed the efficacy of tricyclic antidepressants (TCAs) in the same setting and revealed similar results (OR=3.39, 95% CI 2.24-5.12), as did the sub-analysis of 6 studies (including 363 patients) evaluating the efficacy of selective serotonin reuptake inhibitors (SSRIs) (OR= OR=2.39, 95% CI 1.14-5.01). The

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efficacy also extends to patients refractory to first-line measures (6 studies including 713 patients) (OR=2.96, 95% CI 1.67-5.25) and to patients without psychopathology (8 studies including 471 patients) (OR=2.92, 95% CI 1.6-5.31). Concerning symptoms, a sub analysis also confirmed the efficacy of antidepressants specifically both in the improvement of abdominal pain (OR=4.33, 95% CI 2.1-8.93) and in abdominal distension (OR=2.68, 95% CI 1.42-5.04).

Only in the sub-analysis assessing the efficacy of SSRIs and of the improvement of abdominal pain, I^2 was slightly >60%, indicating substantial heterogeneity, but <75%. Causes of heterogeneity were searched. Publication bias was detected, and potential sources were identified. Sub-analysis without the studies identified as sources of this bias revealed similar results with lower heterogeneity.

Conclusion: Antidepressants, including both TCAs and SSRIs, demonstrate efficacy in patients with IBS, presenting a validated treatment option supported by evidence. Moreover, these medications can be beneficial to patients resistant to initial treatments and those lacking psychopathological symptoms.

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PP0813

EFFECT OF PEDIOCOCCUS ACIDILACTICI CECT7483, LACTOPLANTIBACILLUS PLANTARUM CECT7484 AND LACTOPLANTIBACILLUS PLANTARUM CECT7485: ON PERMEABILITY ALTERATIONS INDUCED BY IRRITABLE BOWEL SYNDROME MEDIATORS: A TRANSCRIPTOMIC AND METABOLOMIC STUDY

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Introduction: The pathophysiology of irritable bowel syndrome (IBS) is complex and involves several factors including intestinal permeability alterations. The soluble mediators spontaneously released by the colonic mucosa (SUP) of IBS patients increased paracellular permeability in the *in vitro* model based on Caco-2 cells (1).

Aims & Methods: The aims of this study were to evaluate the effect of probiotic blend consisting of Pediococcus acidilactici (CECT7483) and Lactobacillus plantarum (CECT7484 and CECT7485) on Caco-2 permeability and trascriptomic profile and to correlate these results with the metabolites produced by the probiotic blend.

11 IBS-D, 10 IBS-M and 7 asymptomatic control (AC) were enrolled. IBS patients were diagnosed according to ROMEIV criteria. All subjects enrolled in the study were 18-65 years old and without relevant organic disease or dietary intolerance. IBS/AC SUP were incubated on Caco-2 monolayer with or without the probiotic blend (10^4 and 10^6). Permeability changes were evaluated by assessing the flow of dextran-FITC to the Caco-2 monolayer. At the end of permeability experiments RNA was extracted and used for library preparation for RNAseq analysis. Gene expression was validated by real-time qPCR. Untargeted metabolomics approach was used.

Results: Both doses of probiotic bland significantly recovered permeability changes induce by IBS-SUP. RNAseq results, validated by qPCR showed that the co-incubation of the higher dose of probiotic bland and IBS-SUP compare to IBS-SUP alone induced a significant increase in the expression of Cytochrome P450 Family 1 Subfamily A Member 1 *(CYP1A1)* ($P<0.05$) and a decrease in the expression of HtrA Serine Peptidase 1 (*HTRA1)* gene. On the basis that *CYP1A1* expression is induced by the activation of the aryl hydrocarbon receptor (AhR), we evaluated the production of indoles, natural ligand of AhR, by the strains contained in the probiotic blend. We found that the Indole-3-lactic acid was the most produced one.

Conclusion: This study demonstrated the molecular mechanisms underlying the effect of the probiotic blend containing *Pedicococcus acidilactici* CECT7483, *Lactobacillus plantarum* CECT7484, and *Lactobacillus plantarum* CECT7485 on an *in vitro* model of intestinal permeability.

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PP0814

THE TREATMENT PREFERENCES OF PHYSICIANS REGARDING IRRITABLE BOWEL SYNDROME: A DISCRETE CHOICE EXPERIMENT

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Introduction: Irritable Bowel Syndrome (IBS) is a highly prevalent, chronic disorder of the Gut-brain interaction which forces increased demands of healthcare facilitations to manage these symptoms. Several treatment options are available, all with comparable clinical efficacy, such as pharmacotherapy, dietary treatment and psychotherapy. Therefore, the guidance of a physician during consultation is essential for effective decision-making. In our previous study, the most-preferred treatment among IBS patients was dietary intervention, subsequently pharmacotherapy and psychotherapy. The preferences of physicians remain unclear, but could be of major importance to give insight in the management strategy of different physicians and to compare with patients' perspective.

Aims & Methods: The aim of this study was to examine these treatment preferences among different types of physicians and evaluate if treatment wishes differ between subgroups of physicians.

Physicians from the Dutch Association of Specialists for Gastroenterology & Hepatology were invited to participate. A labeled discrete choice experiment (DCE) survey was filled in, containing twelve scenarios with each three alternatives (medication, diet, psychotherapy). Each scenario was accompanied by a patient profile differing in predominant symptom and anxiety symptoms. Five attributes were included in the treatment scenario: effectiveness, time until response, cessation of response following discontinuation of therapy, side effects and treatment burden. The preference weights were analyzed using a mixed logit model.

Results: In total, 52 physicians completed the discrete choice experiment (58% female). The majority of these respondents were gastroenterologists (51.9%), followed by gastroenterology trainees (34.7%). A small amount was general practitioner, pediatric gastroenterologist or gastroenterology nurse practitioner. The average work experience with treating IBS patients was 8.5 years (SD 3.5). The discrete choice experiment showed that the most-preferred treatment was pharmacotherapy, which was chosen in 39.71% of the cases. Dietary intervention was chosen 195 times (31.26%) and psychotherapy in 29.02% of the cases. Effectiveness, treatment burden and mild side effect were reported to be important treatment attributes.

Subgroup analysis regarding gender showed that male physicians preferred pharmacotherapy, while female physicians tend to prefer dietary treatment. Important attributes in female physicians were effectiveness, treatment burden, time to response, mild side effects and time until recurrence regarding psychotherapy. Male physicians reported effectiveness, time until recurrence regarding medication and diet and mild side effects, to be important in IBS treatment.

Conclusion: Pharmacotherapy was the most-preferred IBS therapy among physicians. Revealing physicians' preferences regarding the treatment of IBS raises awareness of the important treatment factors which can contribute to improved shared decision-making during consultation.

Disclosure: Nothing to disclose.

PP0815

TRANSIENT COLONIZATION WITH *BLASTOCYSTIS* SPP. AFTER TRANSMISSION VIA FECAL MICROBIOTA TRANSPLANTATION

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Introduction: The pathogenicity of *Blastocystis spp.* is still debated. Guidelines for donor feces screening differ in their advice to screen for *Blastocystis spp.,* but when tested, its presence is a common reason for exclusion.[1] *Blastocystis spp*. are correlated to increased bacterial alpha-diversity[2] and distinct bacterial groups[3] and therefore its presence may indicate favorable efficacy of fecal microbiota transplantation (FMT). The latest European consensus report no longer advices to reject feces donors that test positive for *Blastocystis spp*.[4] Only one paper has been published on human transmission of *Blastocystis spp*. via frozen FMT.[5]

Aims & Methods: We aimed to investigate the long-term effects of the application of Blastocystis-positive FMT, prepared with fresh feces.

In a FMT trial on FMT with refractory Irritable Bowel Syndrome (IBS) adolescents (age 16 – 20 years) received two administrations (at baseline and after 6 weeks) of fresh allogeneic FMT via nasoduodenal tube prepared with fecal samples from a *Blastocystis spp*. positive feces donor (NCT03074227). *Blastocystis spp*. presence and subtyping were determined using microscopy, PCR and Sanger sequencing.

Results: Three recipients received FMT from one donor colonized with *Blastocystis* subtype 3 (ST3). Two recipients were negative for *Blastocystis spp.* and one recipient carried ST2 at baseline prior to FMT. After FMT, the two prior-negative recipients tested positive for identical ST3 as the donor after FMT at 12 weeks, but lost carriership of this subtype at week 24 and week 48. The ST2 colonized recipient remained colonized with ST2 and did not acquire ST3 after FMT. Transient adverse events did occur, but did not differ from patients treated with *Blastocystis spp.*-negative FMT. No FMT-related serious adverse events occurred.

	Recipient	Baseline	Before FMT 1	Donor FMT 1	Before FMT 2	Donor FMT 2			
		Week -2	Week 0	Week 0	Week 6	Week 6	Week 12	Week 24	Week 48
1	Woman, 18 y	negative	negative	ST3	ST 3	+, no seq	ST3	negative	negative
2	Woman, 20 y	negative	negative	ST3	+, no seq	ST3	ST 3	ST3	ST3
3	Woman, 16 y	ST2	ST2	ST3	+, no seq	ST3	ST2	ST2	ST2

Table 1: Sanger PCR sequencing results of frozen feces samples from three adolescents with IBS and one healthy donor.

Samples were collected before, during and after the FMT treatment schedule with follow-up unto week 48. Light blue cells indicate a different sequence order of Blastocystis ST3 than the Blastocystis ST3 indicated in the dark blue cells. Grey cells indicate columns representing donor samples.
Abbreviations: FMT; fecal microbiota transplantation, ST; subtype, seq; sequence, y; years.

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Conclusion: We present the first long-term results of *Blastocystis* spp. transmission by fresh FMT in three cases. Transient colonization with Blastocystis spp. did occur without the occurrence of FMT-related serious adverse events.

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PP0816

MELATONIN FOR FUNCTIONAL ABDOMINAL PAIN: A SYSTEMATIC REVIEW AND META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: Functional abdominal pain is a common disorder for which pharmacological strategies are still inadequate. Melatonin has many direct and indirect effects on the gastrointestinal tract and seems to have an important influence on the brain-gut axis.

Aims & Methods: We aimed to assess the efficacy of melatonin versus placebo for treating patients with functional abdominal pain, performing subgroup analyses by age range, dosage of melatonin, and follow-up time. We systematically searched PubMed, Embase, and Cochrane Library databases until March 31, 2024. The primary outcome was the improvement of abdominal pain severity. The secondary outcomes were reduction in abdominal pain frequency and improvement in quality of life. We analyzed continuous outcomes using the standardized mean difference (SMD) with 95% confidence interval (CI). We used random-effects model, and statistical significance was regarded at p-value <0.05. Between-study heterogeneity was assessed using Cochran's Q test and I^2 statistics. Statistical analyses were performed using R statistical software. We assessed the certainty of evidence for each outcome using the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) tool.

Results: Five randomized controlled trials (including 236 patients) were included. The summary of findings for the primary and secondary outcomes is shown in **Table 1**. Melatonin was associated with greater improvements in abdominal pain severity (SMD 1.18; 95% CI 0.75, 1.61), abdominal pain frequency (SMD 1.19; 95% CI 0.62, 1.75), and quality of life (SMD 0.47; 95% CI 0.05, 0.88) compared to placebo. The subgroup analysis of adult patients showed that melatonin had a significantly greater decline in pain severity (SMD 1.13; 95% CI 0.57,1.69) and pain frequency than placebo (SMD 1.30; 95% CI 0.51,2.09). In the subgroup analysis based on the lowest dosage of melatonin (3 mg), the intervention showed a greater decline in pain severity scores compared to placebo (SMD 1.14; 95% CI 0.58,1.69). When analyzing only studies with longer follow-up (≥8 weeks), melatonin improved pain severity (SMD 0.56; 95% CI 0.04,1.07), pain frequency (SMD 0.89; 95% CI 0.04,1.76), and quality of life (SMD 0.56; 95% CI 0.003,1.11) compared to placebo.

Outcome	Pooled effect. SMD [95% CI]	p-value	I^2	Certainty of evidence
Abdominal pain severity	1.18 [0.75 - 1.61]	< 0.001	0%	High
Abdominal pain frequency	1.19 [0.62 – 1.75]	<0.001	60%	Moderate
Quality of life	0.47 [0.05 – 0.88]	0.027	0%	High

Table 1. Results from the pooled analyses.
CI, confidence interval; SMD, standardized mean difference.

Conclusion: In patients with gastrointestinal functional disorders, melatonin was associated with a statistically significant decline in abdominal pain severity and frequency when compared to placebo, as well as greater improvement in quality of life. The subgroup analyses suggest that the improved outcome can be achieved even with low melatonin dosage, may be sustained over a longer follow-up time, and is effective in adult patients.

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Disclosure: Nothing to disclose.

PP0817

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PP0818

PREDICTIVE CT FINDINGS TO BETTER RISK-STRATIFY RECURRENCES IN THE OUTPATIENT MANAGEMENT OF UNCOMPLICATED ACUTE DIVERTICULITIS

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Introduction: Among the different clinical phenotypes of diverticular disease, acute diverticulitis (AD) and its recurrences represent the most significant economic burden for healthcare systems. As indicated by current guidelines, AD care programs have been organized for outpatient management of acute uncomplicated diverticulitis to contain costs. This approach resulted in safe and economically efficient but to improve the quality of patient care, it would be useful to establish the risks of early and late AD recurrence.

Aims & Methods: The aim of the study was to identify possible CT findings predictive of AD recurrence to better risk-stratify the management of outpatients after the 1st episode. Over the past year, 66 patients referred to the Emergency Department of Policlinico Umberto I Hospital (Rome) for uncomplicated AD, defined as modified Hinchey stages 1a and 1b, were included. 33 of these patients were those enrolled in the outpatient AD care program with 1 year-follow- up without recurrence (NR-pts) and 33 patients, matched per gender and age, were patients admitted for a recurrent AD (R-pts) within 12 months from the previous 1st access for AD to the Emergency Department. 1st episode CT images were analyzed for known parameters of disease severity by two radiologists and compared with $\chi 2$ and Student t-tests. Data are expressed as mean±SD. Univariate and multivariate Cox regression models and Hazard Ratio (HR) were used to identify risk factors predictive of AD recurrence.

Results: No statistically significant differences were found regarding gender and age distribution between NR-pts (F 36%, ages 58.0±12.2) and R-pts (F 39%, ages 61.0±14.0). Among the analyzed CT parameters, at univariate analysis, R-pts showed significantly greater parietal thickness compared to NR-pts (16.0±4.2mm vs 11.5±2.6 mm, p <0.001) and grade 4 peridiverticular inflammation (40% R-pts vs 12% NR-pts, p=0.014), with HR of 1.25 (1.15-1.36, p<0.001) and 3.44 (1.50-18.71 p=0.004) respectively. At multivariate analysis, only parietal thickness was confirmed as a significant risk factor, with Kaplan-Meyer plots showing a 6-fold increase in recurrence probability with a wall increase >15 mm. For early recurrence within 90 days from first discharge, further risk factors were a longer inflamed segment length (9.0±2.3 mm < 90 days vs 8.0±2.3 mm >90 days; p<0.001) and the onset Hinchey stage, recurrence being more frequent in Hinchey Ib than Hinchey Ia (54% vs 16%, p=0.011).

Conclusion: The use of CT predictive factors could represent useful tools in clinical practice to tailor a personalized follow-up and appropriate treatment for outpatient management of acute uncomplicated diverticulitis.

Disclosure: Nothing to disclose.

PP0819

A PROSPECTIVE STUDY ON THE INTERRELATIONSHIP BETWEEN CLINICAL, MANOMETRIC, AND IMAGING FINDINGS IN FAECAL INCONTINENCE

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Introduction: Faecal incontinence (FI) is prevalent, being associated with a high biopsychosocial burden and considerable health costs. The pathophysiology of FI is varied and often multifactorial, representing a diagnostic and therapeutic challenge. Anorectal manometry is the first recommended diagnostic method, although its utility is debatable due to overlapping findings between healthy individuals and patients.

Aims & Methods: The aim of this study is to establish a relationship between FI clinical presentation, manometric findings, and imaging.

We prospectively included all patients undergoing high-resolution anorectal manometry, according to the London protocol, at a tertiary centre, who agreed to respond to a questionnaire to characterize their FI, between August 2021 and March 2023. Clinical records were consulted, in order to access demographic data and past imaging reports. Statistical analysis was performed using IBM SPSS Statistics version 26®.

Results: Forty-seven patients were included with a mean age of 63.5±11.6 years, 87.2% women, of whom 73.2% had an obstetric history. Depressive/anxious symptoms were described in 59.6%, previous perineal surgery in 38.3%, and concomitant urinary incontinence in 40.0%. Mean FI duration was 5.4±5.2 years, with 51.1% reporting daily complaints and a bowel movement median of 2.0 (IQR 3.0). Males tended to have more bowel movements than females (3.5 (IQR 5.0) vs 2.0 (IQR 2.0); *p*-value 0.03). FI for flatus and faeces was present in 57.4%, for faeces alone in 34.0%, and for flatus alone in 8.5%. Urgency FI was verified in 37.0%, passive FI in 37.0%, and mixed FI in 26.1%. Mean Wexner score was 10±5. Rectal examination revealed hypotonicity in 15.9%. Mean resting pressure on anorectal manometry was 55.9±22.5 mmHg, dyssynergy was identified in 16.7% and altered balloon expulsion in 19.5%. The mean value of first sensation, first urge to defecate, and maximum tolerated volume were within the normal range. Imaging was performed in 19 patients (40.4%), with external anal sphincter injury in 15.8% and internal sphincter injury in 10.5%. Mean thickness of the internal anal sphincter (IAS) was 2.3±1.0 cm. Patients with hypotonicity in rectal examination had lower anal resting pressure (35.7±16.0 mmHg vs 60.3±22.4 mmHg; *p*-value 0.09) and lower IAS thickness (1.6±0.8 mm vs 2.6±0.9 mm; *p*-value 0.07), although not statistically significant. Patients with depressive/anxious symptoms showed lower maximum incremental pressure (56.6±34.0 mmHg vs 81.2±46.2 mmHg; *p*-value 0.045), as well as females when compared to males (60.0±34.9 mmHg vs 108.7±53.2 mmHg; *p*-value 0.005). No other statistically significant results were found, especially correlating clinical findings with manometry or imaging.

Conclusion: Our study showed, as previously replicated, that clinical, manometric and imaging findings have low correlation in the assessment of FI, reflecting its pathophysiological complexity.

Disclosure: Nothing to disclose.

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PP0820

UNRAVELLING SOLITARY RECTAL ULCER SYNDROME: INSIGHTS FROM A CASE SERIES

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Introduction: Solitary rectal ulcer syndrome (SRUS), often mistakenly classified as a rectal motility disorder, arises from defecatory irregularities and presents with diverse clinical manifestations. This prospective study conducted at a leading medical center aims to elucidate the spectrum of clinical presentations, potential risk factors, colonoscopic findings, histopathological characteristics, and therapeutic approaches in individuals affected by SRUS.

Aims & Methods: The prospective analysis included all patients diagnosed with SRUS at the Advanced Gastroenterology Institute (AGI) - The GASTROCITI, Ludhiana, over 48 months from March 2020 to February 2024. Cases were identified using the comprehensive database of the Endoscopy Department. Data retrieval from medical records provided insights into clinical symptoms, endoscopic observations, and histopathological features.

Results: From approximately 7000 diagnostic sigmoidoscopies & colonoscopies, a meticulous identification of 48 patients with SRUS was achieved. The average age was 48.5 years (±18.5), with 75% (36) being males. Rectal bleeding was reported by 77% (37) of patients, 64.5% (31) experienced constipation, and 50% (24) described bowel movement difficulties, with an average symptom duration of 13.7 months.

Associated factors included constipation (75%), digital rectal manipulation (43.75%), high-pressure water jet usage (25%), prior rectal surgeries (10.5 %), and rectal prolapse (10.5%). Endoscopic findings revealed solitary ulcers in 64.5% of cases, multiple ulcers in 27 %, and polypoidal appearance in 10.5%. Two patients presented with a proliferative mass-like anomaly (pseudotumor) and two had a circumferential rectal stricture.

The finding of difficult-to-take endoscopic biopsies due to underlying submucosal fibrosis showed a robust positive predictive value (PPV) exceeding 90% for SRUS diagnosis. Histopathological analysis revealed surface ulceration (90%), lamina propria fibrosis (70%), architectural distortions (60%), and increased muscle hypertrophy with mucin production (50%). Conservative interventions were effective in all cases, without the need for surgical intervention.

Conclusion: SRUS, often misinterpreted as a rectal motility disorder, presents with diverse manifestations. Management strategies including high-dose laxatives, avoidance of digital evacuation and high-pressure water jet usage, adoption of a fiber-rich diet, and localized application of mesalamine/sucralfate and biofeedback therapy have shown efficacy. This study highlights the complex nature of SRUS and emphasizes the importance of a comprehensive approach to diagnosis and management.

References:

Disclosure: Nothing to disclose.

PP0821

INSIGHTS INTO RESULTS OF GLOBAL AUDIT FOR FECAL INCONTINENCE ON BEHALF OF THE EUROPEAN SOCIETY OF COLOPROCTOLOGY (ESCP) COLLABORATING GROUP

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Introduction: Faecal incontinence (FI) is a common problem, affecting about 9% of Western populations. However, only a few patients seek consultation and the number of such patients reaching hospital care is unknown. Current treatment pathways are poorly evidence based and are thought to vary greatly between different countries.

Aims & Methods: Background: The aim of this study was to evaluate the incidence of fecal incontinence (FI) in patients attending colorectal clinics in a global setting, treatment choices and accessibility to diagnostic and treatment tools.

Methods: An international, multi-center snapshot audit was performed in an 8-week period between January 9, 2023 and March 28, 2023. All eligible cross-sectional data was recorded. Data was entered and saved in a REDCap secured database. Two physician- and center-level surveys were filled for current practice evaluation.

Results: 1,813 outpatients with FI from 110 hospitals, from 35 countries were included in this audit. Majority of patients with FI were females, parous, and in age range 50-70. Patients most commonly waited 1 to 3 years until seen in outpatient clinic. Etiology skewed toward anal injury and rectal prolapse. Stress incontinence was diagnosed more common than urge incontinence (36% vs 23% respectively). Majority of patients received previous treatment (n=222 (61%)), which mostly included pelvic floor physiotherapy (n=149), bowel retraining with biofeedback (n=114), and nurse-led continence support (n=90). Sacral nerve stimulation was the most commonly performed procedure (n=114/395) for FI in outpatient clinics, followed by sphincterocplastty (n=87/395). Endoanal ultrasound (n=206/1,226) as well as anorectal manometry (n=185/1,1226) are the basic diagnostic tools used to diagnose FI in outpatient settings. More than a half (56%) of clincs have access to pelvic floor MDT. The majority of physicians offer surgical treatment (colostomy or sphincter repair) for FI.

Conclusion: Patients FI wait 1-3 years until attending colorectal clinics. Anal injury and rectal prolapse are the main causes of FI. FI treatment consists of conservative and surgical treatment, and there has not been a single best treatment option.

Disclosure: Nothing to disclose.

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PP0822

HIGH ANAL HR-HPV PREVALENCE AND ABNORMAL ANAL CYTOLOGY IN MEN HAVING SEX WITH MEN WHO USE PREP: ARE THEY A DISTINCT RISK GROUP FOR ANAL CANCER?

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Introduction: Persistent High-Risk Human Papillomavirus (HR-HPV) infection can lead to an anal high-grade squamous intraepithelial lesion (HSIL), the precursor lesion to anal squamous cell carcinoma (SCC). Anal cytology and HR-HPV testing are strategies used for anal SCC screening. Anal HR-HPV prevalence in HIV-negative MSM and in MSM living with HIV (MSM-HIV) is 41% and 74%, respectively. The use of Pre-Exposure Prophylaxis (PrEP) in HIV-negative MSM to prevent HIV contraction created a new Sexual Minority Group among MSM, i.e. MSM using PrEP (MSM-PrEP). The aim of this study was to determine the prevalence of anal HR-HPV infection and abnormal anal cytology in MSM-PrEP and MSM-HIV.

Aims & Methods: MSM-PrEP and MSM-HIV were enrolled in this mono-centric study during consultations at the S-clinic and the HIV-reference centre respectively. Patient characteristics, sexual behavior and demographics were collected using a questionnaire, completed on the day of the anal swab testing.

Results: We enrolled 150 MSM-PrEP and 152 MSM-HIV. Quality of anal swabs was sufficient in respectively 96.7% (n=145) and 96.1% (n=146). HR-HPV prevalence in MSM-PrEP was comparable with HR-HPV prevalence in MSM-HIV; respectively 74.5% (n=108) and 76.6% (n=111) tested positive for at least one HR-HPV (p=0.78). No significant difference in abnormal cytology was seen; 53.1% (n=77) of the MSM-PrEP and 57.2%(n=83) of MSM-HIV had either ASCUS, ASC-H, LSIL or HSIL (p=0.56).

Conclusion: The risk of HR-HPV infection and abnormal anal cytology in MSM-PrEP is comparable to that in MSM-HIV, both of which are higher than observed in HIV-negative MSM not using PrEP. This heightened prevalence of HPV in MSM-PrEP poses an increased risk for HSIL. Recent guidelines on anal screening from International Anal Neoplasia Society recommend screening for anal cancer in both MSM-HIV and HIV-negative MSM, reflecting respective anal cancer incidence rates of 70-100 per 100,000 person-years and 18-34 per 100,000 person-years. Our findings suggest that MSM-PrEP constitutes a distinct risk group for anal SCC positioned between MSM-HIV and HIV-negative MSM not using PrEP.

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Disclosure: Nothing to disclose.

PP0823

PREVALENCE OF FUNCTIONAL CONSTIPATION AND ASSOCIATED FACTORS IN THAI INFANTS AND TODDLERS: A MULTI-CENTER CROSS-SECTIONAL STUDY

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Introduction: The global prevalence of pediatric functional constipation (FC) varies from 1 to 30% due to the heterogeneity of the studies in terms of target population sampling, method of data acquisition, diagnostic criteria, among others. Studies conducted by Osatakul *et al.* and Piriyanon *et al.* in Thai young children, utilizing Rome III criteria, indicated prevalence rates of 1.6% and 25%, respectively. However, there is currently no study on the prevalence of FC in Thai young children using the Rome IV criteria, mainly because no standardized questionnaire exists for this age group.

Aims & Methods: The Thai version of the Rome-IV Diagnostic Questionnaire for Pediatric Functional Gastrointestinal Disorder (R4PDQ)-Toddler was developed then authorized by the Rome foundation. Guardians of children aged 6-48 months from six hospitals in five regions of Thailand from August 2023 to February 2024 were surveyed with the questionnaire composed of demographic data, the R4PDQ-Toddler for defecation, and the associated factors of FC.

Results: With a 98.1% response rate, a total of 1,874 guardians participated. The prevalence of FC was 16.1% (95% CI 14.5%-17.8%). Most presenting symptoms were painful or hard bowel movements (81.5%), large diameter stools (77.5%) and excessive stool retention (72.2%). Independent associated factors were age >1 year [aOR=3.36 (95%CI 2.29-4.94)], first-degree relative history of constipation [aOR=2.48 (95%CI 1.85-3.35, fiber supplementation [aOR=7.85 (95%CI 3.56-17.30)], and stressful life events in family [aOR=1.41 (95%CI 1.03-1.92)].

Conclusion: The prevalence of FC in Thai young children has been increased. Older age, family history of constipation, stressful life events in the family and supplementary fibers were associated factors of FC, warranting further evaluation.

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PP0824

LIVING WITH FAECAL INCONTINENCE: A SYSTEMATIC REVIEW AND META-ETHNOGRAPHY

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Introduction: Faecal incontinence (FI) affects up to 8.3% of the population and may have physical, psychological and social consequences (1). Sadly, patient experiences remain under-explored, and most symptom measures are based on clinical outcomes rather than key priorities or concerns of affected individuals (2).

Aims & Methods: This study aimed to describe the lived experience of FI through review of published qualitative studies. A systematic literature search of studies of patients with FI published between 2003-2023 was conducted across Medline, Embase, CINAHL and PsycINFO. Studies were included if they provided qualitative data on patient experiences of living with FI. Studies in children and those principally focused on experiences following stoma formation were excluded. Methodological quality was described using the Critical Appraisal Skills Programme (CASP) tool for qualitative studies. An inductive approach by 2 reviewers enabled identification of key concepts. Collaboration with a third reviewer then generated higher-order interpretations. Finally, a meta- ethnographic method was utilized to construct a comprehensive thematic framework of lived experiences of FI.

Results: Of the 3394 studies identified, 29 studies were included in this synthesis. All studies were judged to be of good quality. Six major themes were identified (figure 1): quality of life, adaptation, protection, wider environment, perception and healthcare experiences. The theme 'Quality of Life' reflected the impact of FI on mental health, physical health, social well-being and functional ability. 'Adaptation' encompassed behavioural changes, dietary modification, and adjustments to daily routine to manage FI. The theme 'Protection' comprised risk management strategies and symptom concealment. 'Wider Environment' described navigation of the professional and public environment. 'Perception' reflected the impact of FI on self-perception, as well as the perception of other people's attitudes towards FI. Finally, 'Healthcare Experiences' reflected patient experiences of accessing care and interactions with clinical care teams.

Conclusion: This qualitative synthesis offers a novel patient-centric framework of the lived experiences of FI and provides insight into the breadth of factors contributing to the burden of this condition. Many factors identified in this review are not represented in current outcome measures, highlighting the need for new ways of holistically assessing and measuring the impact of FI on patient wellbeing.

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Disclosure: Nothing to disclose.

PP0825

EFFICACY OF PRANIC TREATMENT IN THE MANAGEMENT OF CHRONIC PAIN, ASSOCIATED MEDICAL CONDITIONS AND PSYCHOLOGICAL SYMPTOMS IN CHILDREN AND ADOLESCENTS

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Introduction: Chronic-pain and anxiety with increased school absences are common problems in children and adolescents. The majority of abdominal pain cases fulfil the Rome IV criteria for functional abdominal pain disorders (FAPDs).

Aims & Methods: We aim to evaluate the efficacy of Pranic treatment in the management of chronic-pain, non-pain-clinical-symptoms associated with medical diagnosis, psychological-symptoms and school attendance. A prospective, proof-of-concept single group study within Child Health, King's College Hospital, London, UK.

Eligibility criteria

- Chronic-pain
- Age range 5-21 years old
- Paediatric consultants' purposive selection of patients with difficult-to –manage chronic-pain
- Patients underwent 8-weekly treatment sessions and review Week-12,16,20
- Participant's psychological and medical diagnosis determined the corresponding Pranic Psychotherapy and Pranic Healing protocols.
- Pain-intensity score 0-10 (PVAS), Pain-frequency (PF), Non-pain-clinical-symptoms (NPCS) School attendance (SAA), variables measured with metric ordered questionnaires
- Psychological symptoms variables measured with binary categorical questionnaires.

Results:

- 100 participants underwent treatment, July 2022 – November 2023.
- M: F ratio 37:63. Median age 13.5 years.

At Baseline:

- 77% participant's medical diagnosis included Functional-abdominal –pain-disorders.
- 62% had experienced chronic-pain for 3 years or more.
- 98% non- zero PVAS score (median 8 = intense dreadful, horrible pain)
- 85% persistent non-pain-clinical-symptoms e. g chronic diarrhoea, chronic constipation, cyclical vomiting (median= 3= routine life disturbed)
- 78% reported school attendance had been regularly affected (median =2= school attendance regularly affected)
- 98% reported low mood and poor concentration, 83% sleep-difficulties, 79% fatigue
- 89% reported 'yes' to routine life being affected by symptoms.

Pranic Treatment Outcomes:

- 93/100 completed treatment 8-weekly treatment sessions
- 78/100 completed week-12, 16, 20 review.
- Pranic treatment proved beneficial to, Pain-Intensity and Pain-Frequency and School Attendance (PVAS (P-value<0.001)) (PF (P-value<0.001)) (SAA (P-value<0.001))
- Pranic Healing Protocols proved beneficial for Non- pain –clinical-symptoms (NPCS (P-value<0.001))
- Pranic Psychotherapy Protocols proved beneficial for psychological symptoms (Psychological symptoms (P-value<0.001)).

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- The Kendall's coefficient for concordance for PVAS=0.904, PF=0.865, NPCS=0.703 and SAA=0.759 shows agreement in Patient's response to treatment.
- Benefits were sustained over 3 months follow-up period
  - 78 participants' completing final review at the week -20 time point patient outcomes:
  - 69/78 (88%) of symptomatic participants reported PVAS= 0= asymptomatic
  - 66/68 (97%) of symptomatic patients NPCS = 0 = asymptomatic
  - 63/77 (82%) of participants with school attendance affected reported SAA score =0 = school attendance no longer affected by symptoms
  - 68/78 (87.2%) and 67/78 (85.9%) symptomatic participants reported resolution of low mood and poor concentration respectively.
  - 71/78 (91%) and 72/78 (92.3%) symptomatic participants reported resolution of fatigue and sleep difficulties respectively.
  - 68/78 (87.2%) of participants with routine life affected, regained the ability to participate in routine life activities.

Conclusion:

- Pranic treatment is capable of simultaneous improvements in psychological and physical symptoms.
- Resulting in clear improvements in chronic-pain and school attendance.
- Pranic treatment has an after-treatment effect with continuing improvements in behaviour and physical function.

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Disclosure: Nothing to disclose.

PP0826

DOES BOTULINUM TOXIN STILL HAVE A ROLE TO PLAY IN THE TREATMENT OF CHRONIC ANAL FISSURE?

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Introduction: Chronic anal fissure is a frequent cause of referral to the proctology clinic and is associated with significant morbidity. In cases refractory to conservative measures, botulinum toxin (BT) is an alternative to a surgical approach, without the risk of permanent damage to the internal anal sphincter.

Aims & Methods: The aim of this study was to assess the efficacy of BT in anal fissure healing and to identify predictors of non-response. This was a retrospective, single-centre, observational study. Patients with chronic anal fissure who underwent TB between 2019 and 2023. Patients with a follow-up of less than 3 months or inflammatory bowel disease were excluded. On clinical reassessment, patients were categorised as complete, partial or refractory.

Results: Of the 118 patients selected, 80 were included. The average age was 49.1 ± 14.9 years, with 73.8% (n=59) of the patients female. The overall TB response rate was 62.0%. More than one TB session was carried out on 40.0% (n=32) of the patients, with an average of 1.5±0.6 per patient, with an interval between sessions of 3.4 ± 5.8 months. On reassessment after the first application, 30.0% (n=24) showed a complete response, 56.3% (n=45) a partial response and 13.8% (n=11) were refractory. Of the partial or refractory responders, after completing the treatment cycle, 56.5% (n=13) showed a complete response, 26.1 (n=6) a partial response and 17.4% (n=4) were refractory. During a median follow-up of 10.0 [3-48] months, 7 patients became asymptomatic and 2 patients had a recurrence of the cleft. 21 patients (26.3%) were referred for surgery. The duration of symptoms was an independent predictor of TB treatment failure and the need for surgery (OR 4.895; p=0.049).

Conclusion: TB has a satisfactory response rate in the treatment of patients with chronic anal fissure, making it possible to avoid surgery in patients, specially with a short duration of symptoms.

Disclosure: Nothing to disclose.

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PP0827

EFFICACY AND SAFETY OF OFFICE-BASED PROCEDURES FOR HEMORRHOIDAL DISEASE IN PATIENTS WITH INFLAMMATORY BOWEL DISEASE

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Introduction: Hemorrhoidal disease affects 2-7% of patients with inflammatory bowel disease (IBD) (1,2), but treatment recommendations are limited due to apparent higher morbidity in this subset of patients (3,4). Currently, there is limited data about hemorrhoidal disease treatment in IBD, there is even less information regarding non-surgical treatment.

Aims & Methods: The authors aim to evaluate the safety and efficacy of office-based procedures in hemorrhoidal disease in patients with IBD. This is a retrospective and Portuguese multicenter study (7 centers) that included patients with IBD who underwent office-based procedures for hemorrhoidal disease between July 2013 and July 2021. Data relating to IBD and hemorrhoidal disease were analyzed namely success, relapse, and complications.

Success was defined as a significant improvement in symptoms after office-based procedures, without the need for additional treatment. Relapse was defined as a worsening of symptoms, requiring new office-based procedures/surgery.

Results: 129 patients were included, 70% with ulcerative colitis; in Crohn's disease, 5.4% had a history of perianal disease. Only 62% of patients had endoscopic remission and 18% were on biologics. Regarding Goligher's classification, the majority had grade II (48%). 77 patients underwent rubber band ligation (RBL), 44 patients underwent polidocanol injection and 8 patients underwent both techniques, in a total of 304 procedures (median of 2(2) sessions/patient).

The treatment was effective in 88% of patients, of which 14% relapsed, with an average follow-up time of 55.3±34.5 months. Of 304 procedures, 17 (5.6%) procedures were complicated, in a total of 16 patients (12.4%), 2 (0.65%) required invasive treatment, without cases of pelvic sepsis. Success, relapse, and complications were not associated with IBD (type, medication, or remission at the time of treatment) (p>0.05). Success was greater in patients with Goligher grades I and II (p=0.037).

Complications were higher in patients undergoing RBL (p=0.030) and in patients with Goligher grade III (p=0.031).

Conclusion: This is the first study about office-based procedures for hemorrhoidal disease in IBD, showing similar efficacy to the general population, with a low rate of complications, and should therefore be recommended in these patients.

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Disclosure: Nothing to disclose.

PP0828

UNDERWATER ENDOSCOPIC MUCOSAL RESECTION OF RECTAL NEUROENDOCRINE TUMORS – A CASE SERIES OF AN INNOVATIVE ENDOSCOPIC APPROACH

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Introduction: Neuroendocrine neoplasms are a group of tumors originating from neuroendocrine cells and peptidergic neurons. Rectal neuroendocrine tumors (NETs) are the second most common location for gastrointestinal NETs, albeit representing only 2% of all rectal neoplasms. Most of these are nonfunctional and asymptomatic although pain, blood loss and perianal discomfort can occur in larger lesions.

Endoscopically, rectal NETs typically appear as small and well-delimited submucosal lesions, with a yellowish color. The risk of lymph node metastasis is related to the tumor size, therefore, the current guidelines consider endoscopic resection as an option for rectal NETs smaller than 2 centimeters.

The classical options for lesion removal include endoscopic mucosal resection (EMR) and endoscopic submucosal dissection (ESD). However, EMR has a higher risk of incomplete resection, whereas ESD is more invasive and has an increased risk of complications (namely bleeding and perforation).

In this setting, underwater endoscopic mucosal resection (UEMR) appeared as an alternative that facilitates the complete removal of a lesion through the filling of the intestinal cavity with water. Previous Eastern studies suggest a higher complete resection rate with this endoscopic technique when compared to conventional EMR.

Aims & Methods: The present study aimed to evaluate the outcomes of patients submitted to UEMR of rectal NETs in a reference center.

Results: A total of 5 patients underwent UEMR for rectal NETs (Table 1). 3 patients had a tumor from the proximal rectum and 2 patients from the distal rectum. The tumor size ranged between 4 and 8 millimeters.

After marking the lesion borders, the entire rectal cavity was filled with saline and UEMR was performed. The lesion scar was closed with endoscopic clips in all cases. In a single case, delayed rectal bleeding was observed 3 days after procedure, solved with application of an additional endoscopic clip, without the need of inpatient management. No other complications were reported.

A R0 resection was observed in 80% of the cases. In a single case a positive basolateral margin was observed. However, after performance of a second UEMR in the scar (where no endoscopic evidence of residual lesion was found), there was absence of lesion in the histopathological sample.

All the tumors had a G1 grade in the WHO 2019 Classification for Neuroendocrine Tumors, with a Ki-67 index ranging between 1 and 3%.

In all the cases there was absence of residual lesion or recurrence in the follow up endoscopy.

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Patient	Tumor size, millimeters	Resection margins (R)	Tumor grade (G)	Ki67 (%)	Recurrence
♂, 79	5	R0	G1	3	No
♀, 64	8	R0	G1	2	No
♂, 33	4	R0	G1	3	No
♀, 46	8	R0	G1	3	No
♂, 74	8	R1	G1	1	No

Conclusion: This case series showcases UEMR as an effective alternative to conventional EMR, achieving high satisfactory complete resection rates, while having a low rate of procedure related complications. Therefore, UEMR should be contemplated when considering an endoscopic removal for rectal neuroendocrine neoplasms.

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PP0829

BENIGN PROCTOLOGICAL PATHOLOGIES AND QUALITY OF LIFE: AN UNDERESTIMATED SOURCE OF DISABILITY

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Introduction: The aim of our study is to highlight the impact of these pathologies on quality of life using the BURDEN haemo/fiss score validated by the French Society of Colo-proctology (SNFCP).

Aims & Methods: This is a prospective cross-sectional study over a 3-month period (January, February and March 2024) including our patients with benign proctological pathology (hemorrhoidal pathology/anal fissure) examined in our proctology consultation.

Quality of life was assessed using the BURDEN haemo/fiss score, which comprises 23 questions covering four domains exploring physical and psychological problems, discomfort during defecation and impact on sexual life. Pain was rated using the VAS (Visual Analogue Scale) from 0 to 10.

Results: We enrolled 112 patients with a mean age of 48 ±15 years. Among our patients, 60.7% were women with a sex ratio of 0.64.

Hemorrhoidal pathology topped the list with a percentage of 68.8%, while patients with anal fissure presented 42.9%. Two conditions coexisted in 31.3% of cases (N=15).

The mean Hemo/Fiss score was 60.08±14.29 [extremes 28-93]. The mean score for each dimension of the score was 59.84±23.58 for physical disorders, 55.91±24.71 for psychological disorders, 76.86±21.37 for difficulty with defecation and 47.71±20.14 for sexual disorders.

All pathologies had a negative impact on quality of life, irrespective of sex and type of underlying proctological pathology ($p>0.05$), with significant concordance between the VAS and the BURDEN haemo/fiss score ($p<0.001$).

Forty-eight of our patients (42.9%) had unfavorable socioeconomic conditions, making it impossible to obtain the full prescribed treatment, which led to non-improvement and/or early recurrence of symptoms. Work stoppage was noted in 36.6% of our patients (N=41).

Conclusion: Our study clearly shows the deterioration in quality of life in patients with benign proctological pathologies, with work stoppage in a considerable number of our patients, often young, regardless of gender or the specific type of underlying pathology. This finding prompts us to pay particular attention to these conditions in all their dimensions.

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Disclosure: Nothing to disclose.

PP0830

ARTIFICIAL INTELLIGENCE AND HIGH-RESOLUTION ANOSCOPY: DEVELOPMENT OF AN INTEROPERABLE ARTIFICIAL INTELLIGENCE MODEL FOR THE DETECTION AND DIFFERENTIATION OF ANAL SQUAMOUS CELL CARCINOMA PRECURSORS – A MULTICENTRIC STUDY

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Introduction: High-resolution anoscopy (HRA) plays a central role in the detection and treatment of precursors of anal squamous cell carcinoma (ASCC). The visual characterization of high-grade intraepithelial lesions (HSIL) is hampered by a limited interobserver agreement. Artificial intelligence (AI) algorithms have shown high levels of efficiency in detecting and differentiating HSIL from low-grade squamous intraepithelial lesions (LSIL) in HRA images. Nevertheless, these studies have been conducted using a digital videoproctoscope, whose use is not predicted in the standards of the International Anal Neoplasia Society (IANS). Our aim was to develop a deep learning system for automatic detection and differentiation of HSIL versus LSIL using HRA images from both conventional and digital proctoscopes.

Aims & Methods: A convolutional neural network (CNN) was developed based on 151 HRA exams from 137 patients undergoing HRA at two large volume centres (GH Paris Saint-Joseph, Paris, France and Emílio Ribas Infecciology Institute, São Paulo, Brazil), using conventional HRA (KLP 200 LED®, Kolplast) and digital HRA (THD Proctostation®, THD SpA). A total of 58,498 images were included, 29,212 images containing HSIL and 29,286 LSIL. Partial subanalyses were performed to evaluate the performance of the CNN in the subset of images with no staining (n=2,820), acetic acid (n=27,191), lugol (n=10,011), and after treatment of the anal canal (n=11,047). For these subanalyses, training, and testing datasets comprised 90% and 10% of each class's total number of images. The sensitivity, specificity, accuracy, positive and negative predictive values, and area under the curve (AUC) were calculated.

Results: The overall accuracy of the CNN in distinguishing HSIL from LSIL during the testing stage was 98.6%. The algorithm had an overall sensitivity and specificity of 97.3% and 99.1%, respectively. The overall AUC was 0.99. For staining with acetic acid, HSIL was differentiated from LSIL with

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an overall accuracy of 96.4%, while for lugol and after therapeutic manipulation these values were 96.6% and 99.3%, respectively. The AUC ranged between 0.98 and 1.00 in these particular subgroups.

Conclusion: The introduction of AI algorithms to HRA may enhance the early diagnosis of ASCC precursors, and this system showed to perform adequately across conventional and digital HRA interfaces. Improving the detection of ASCC precursors may improve the treatment outcomes and prognosis. The developed AI model is the first model capable of performing adequately in different clinical sub settings and in different systems available on the market.

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Disclosure: Nothing to disclose.

PP0831

MILLIGAN AND MORGAN IN THE MANAGEMENT OF HEMORRHOIDAL DISEASE: PROCTOLOGY UNIT EXPERIENCE

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Introduction: Hemorrhoidal disease (HD) is the most common pathology in proctology, with an estimated prevalence of between 4.4% and 39%. It can be managed in a variety of ways: hygienic-dietary measures, medical, instrumental or surgical treatment. The latter is considered especially after failure of medical and instrumental treatment.

Aims & Methods: Aims : The aim of our study is to report our experience with Milligan and Morgan technique in the management of hemorrhoidal disease.

Materials and methods: This was a monocentric retrospective descriptive-analytical study conducted over a 12-year period: from January 2011 to January 2023. The data studied were age, sex, clinical manifestations, stage of HD on proctological examination and therapeutic data. All patients who underwent Milligan and Morgan hemorrhoidectomy (MMH) and presented with stage IV and III hemorrhoids (after failure of instrumental treatment) were included in our study. Excluded were asymptomatic patients, patients with surgical contraindications and other stages treated by non-surgical means. The statistical study was carried out using IBM-SPSS 20.0 software. The chi-2 test was used to compare categorical variables.

Results: 468 patients were enrolled during the study period. The mean age of this population was 41 years (22-71). The sex ratio M/F was 2,1. Clinical expression was dominated by rectal discharge in 246 patients (74%), proctalgia in 132 patients (28%), anal pruritus in 119 patients (25%) and anal oozing in 56 patients (12%). Proctological examination revealed stage IV haemorrhoids in 201 patients (43%) and stage III haemorrhoids in 267 patients (57%). 25 patients (5.3%) had an associated anal fissure. Anemia was present in 341 patients (72.8%), with a mean hemoglobin value of 7.8 g/dl. Transfusion was required in 65 patients (14%) who presented with severe anemia of less than 7g/dl. Colonoscopy was indicated in 87 patients (18.5%) over the age of 45 with rectal bleeding, and in 42 patients (9%) with a history of digestive cancer. Milligan and Morgan's indication was stage III hemorrhoids with severe anemia in 65 patients (14%), after failure of infrared photocoagulation in 194 patients (41.4%); after failure of HAL-RAR treatment n=8 (1.7%) and stage IV hemorrhoids in 201 patients (43%). The procedure performed was a hemorrhoidectomy.An open pedicle MMH type was performed. Patients with an associated anal fissure (5.3%) underwent the Bellan procedure in addition to hemorrhoidectomy. Immediate post-operative complications included post-operative pain in all our patients, urinary retention in 31 patients (6.6%), significant post-operative bleeding in 2 patients (0.5%) treated by electrocoagulation in the OR, and minimal post-operative bleeding in 117 patients (25%) treated by compression bandaging. The latter was reported in 43 patients (36.7%) with stage III hemorrhoids and 74 patients (63.2%) with stage IV hemorrhoids. This difference was statistically significant ($p<0.001$). Late complications included delayed healing in 9 patients (1.9%), 4 of whom were diabetic, anal stenosis in 5 patients (1%), requiring sphincterotomy combined with anoplasty in 3 patients, and finger dilatation in 2 patients. Anal stenosis was reported in 4 patients who did not undergo postoperative digital rectal examination and 1 patient who did. This difference is statistically significant $p=0.03$.

Conclusion:

MMH remains the gold standard for the treatment of hemorrhoidal disease. The results are satisfactory, especially in the long term. Post-hemorrhoidectomy anal stenosis remains the most serious complication.

Disclosure: Nothing to disclose.

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PP0832

EXPERT'S PERCEPTION OF CHAT-GPT BASED ANSWERS TO COMMON PATIENT'S QUERIES IN GASTROINTESTINAL ENDOSCOPY: UNVEILING THE IMPACT OF AN AI CONVERSATIONAL TOOL

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Introduction: Chat Generative Pre-Trained Transformer (Chat-GPT) by OpenAI, released in November 2022, is an artificial intelligence (AI) language model that was designed to generate human-like text based on inputs provided. Given the high potential of a pre-trained technology, it has recently gained attention among healthcare professionals for its capabilities of communicating information. This technology has, indeed, proven to be effective in providing information to patients regarding specific queries and doubts in common gastrointestinal diseases, ensuring high-quality answers. Gastrointestinal (GI) endoscopy is perceived by patients as uncomfortable and worrying, despite the advancements regarding instrumentation and sedation.

Aims & Methods: We aimed to assess the effectiveness of Chat-GPT in answering common patients' queries on GI endoscopy in three domains (reliability, accuracy, and comprehensibility). Two healthcare providers selected the 18 most common patient queries regarding GI endoscopy. Questions, concerning pre-procedural, intraprocedural or post-procedural queries, were administered to Chat-GPT 4.0 with the following prelude "As a physician, how would you answer to a patient asking…". Afterwards, ten healthcare professionals, including eight physicians and two nurses, along with two non-healthcare professionals were asked to evaluate the AI-based answers. All 12 participants had expertise in GI endoscopy and were asked to rank all 18 answers according to a Likert scale for each of the following features: reliability (1 to 6 points), accuracy and comprehensibility (1 to 3 points). The one-way non-parametric ANOVA (Kruskal-Wallis test) was applied to test the distribution of respondents' ranks.

Results: The mean overall reliability was 5.2±1.7 (Figure 1), with questions 3 and 8 receiving the highest score (5.7±0,7 and 5.7±0.7, respectively), and questions 13 and 15 ranking the lowest (4.6±0.9 and 4.8±0.4, respectively). The mean overall accuracy was 2.7± 0.4, with most of the questions ranking more than 2,5, except for questions 1, 13 and 14, which all had 2.5±0.5 scores. Finally, the mean overall comprehensibility was 2.9± 0.2: regarding this specific feature, question 18 scored had the lower mean rank (2.5±0.5). The mean ranks with standard deviations for each feature of all 18 answers are reported in Table 1.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Reliability	5.3 ±0.5	5.3 ±0.6	5.7 ±0.7	5.5 ±0.5	4.9 ±0.9	4.8 ±1.2	5.4 ±0.7	5.7 ±0.7	5.1 ±0.7	5.3 ±0.8	5.5 ±0.5	5.4 ±0.8	4.6 ±0.9	4.9 ±1.0	4.8 ±0.4	5.4 ±0.7	5.4 ±0.7	5.0 ±06
Accuracy	2.5 ±0.5	2.6 ±0.5	2.8 ±0.5	3.0 ±0.0	2.6 ±0.5	2.7 ±0.5	2.8 ±0.4	2.9 ±0.3	2.7 ±0.5	2.9 ±0.5	2.8 ±0.5	2.8 ±0.5	2.5 ±0.5	2.5 ±0.5	2.8 ±0.4	2.8 ±0.5	2.8 ±0.4	2.8 ±0.4
Comprehensibility	2.9 ±0.3	2.9 ±0.3	2.9 ±0.3	3.0 ±0.0	2.8 ±0.5	2.8 ±0.6	2.9 ±0.3	2.9 ±0.3	2.9 ±0.3	2.9 ±0.3	2.8 ±0.4	2.9 ±0.3	2.8 ±0.4	2.8 ±0.5	2.9 ±0.3	2.8 ±0.4	2.9 ±0.3	2.5 ±0.5

All participants have expressed a positive opinion regarding the potential use of the answers provided by the AI, except for answer number 13. The score distribution was not similar for all the 12 participants in each feature, reliability (p<0.001), accuracy (p=0.005) and comprehensibility (p=0.006).

Conclusion: The findings of our study show that Chat-GPT can deliver reliable, accurate and easy-to-understand responses to a broad range of common queries from patients undergoing GI endoscopy. The use of AI-based conversational programs, like Chat-GPT, holds promise as a valid option for physicians to support daily clinical practice, also within the setting of GI endoscopy.

Disclosure: Nothing to disclose.

PP0833

OPIOID ENTEROPATHY AND PANCREATOBILIOPATHY: FIRST DESCRIPTION OF A NEW ENTITY RESULTING FROM LONG-TERM OPIUM ADDICTION

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Introduction: Opium addiction is common in north-western India. Opioid-induced bowel dysfunction (OIBD) and narcotic bowel syndrome are well-described gastrointestinal disorders in patients with opioid addiction. We describe patients with prolonged opium addiction presenting with ulceroconstrictive disease of the bowel and pancreatobiliary abnormalities.

Aims & Methods: We prospectively collected data on patients using opium for prolonged durations and presenting with gastrointestinal symptoms. Opioid enteropathy (OE) was diagnosed in patients with evidence of ulcerocontrictive disease of the bowel and opioid pancreatobiliopathy (OPB) was diagnosed in patients with dilated common bile duct (CBD) and/or pancreatic duct (PD) (double-duct sign), with all other causes excluded by careful evaluation.

Results: Of 148 patients [mean age 57.3 ± 11.6 years; 145/148 (98%) males] with opioid addiction and gastrointestinal complaints, 59 (39.9%), 53 (35.8%), 18 (12.2%) & 18 (12.2%) patients were diagnosed with OIBD, isolated OE, isolated OPB, and combined OE+OPB, respectively. Mean duration of opioid consumption was 18.4±10.9 years. Of 148 patients, 48 (32.4%) consumed raw opium (yields 9.5-14.2% morphine), 89(60.1%) consumed opium husk (yields 0.1-0.3% of morphine), and 4(2.7%) consumed tramadol tablets. Median duration of symptoms was 12(3-60) months. Common presenting symptoms were pain abdomen (n=116, 78.4%), constipation (n=93, 62.8%), vomiting (n=29, 19.6%), anorexia (n=79, 53.4%), weight loss (n=74, 50%) and easy fatigability (n=64, 43.2%). Evaluation in patients with OE revealed anemia (69%) and hypoalbuminemia (74.6%) with a mean Hemoglobin of 9.5g/dL. Endoscopy revealed pyloric and duodenal strictures in 9.9% & 8.5% of the patients, respectively. Colonoscopy revealed multiple web-like focal circumferential partial strictures with superficial ulceration in terminal ileum in 8.5% of patients. CECT enterography showed multiple short-segment jejunal and ileal strictures in 73.2% of patients, while duodenal/pyloric strictures were seen in 8.5% & 9.9%, respectively. Histopathology from the stricture site (18 patients) showed only ischemic changes. Patients with OPB (n=36) revealed dilated CBD (mean diameter: 11.6 mm) (91.7%) and MPD (mean diameter: 5.7mm) (69.4%) up to papilla with normal LFTs and imaging, including endoscopic ultrasound. Conservative management with opioid deaddiction, iron supplements, rifaximin, and laxatives was done in all patients, with endoscopic balloon dilatation of pyloric stricture needed in six patients. Only 1/3 rd of the patients agreed and could undergo successful opioid de-addiction.

Conclusion: Long-term, high-dose opium addiction can lead to multifo-

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cal ulceration and stricture formation in the GI tract, and pancreatobiliary dilatation. This entity should be recognized as an important differential of Crohn's disease in areas with a high prevalence of opium addiction. Further, these patients with double-duct sign are often subjected to evaluation and surgery, suspecting periampullary malignancies. Knowledge about this spectrum of disorders may prevent unnecessary diagnostic and surgical interventions.

Disclosure: Nothing to disclose.

PP0834

A SYSTEMATIC REVIEW AND META-ANALYSIS OF TEST PERFORMANCE IN RANDOMIZED CONTROLLED TRIALS FOR COLONOSCOPY SCREENING FOR COLORECTAL CANCER

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Introduction: The modality for colorectal cancer (CRC) screening has transferred from guaiac fecal occult blood testing (gFOBT) to fecal immunochemical testing (FIT), but its sensitivity remains lower. While total colonoscopy (TCS) has been anticipated as a new modality for CRC screening, conclusive results based on randomized controlled trials (RCTs) are still pending. Test performance RCTs have been utilized to predict the effectiveness of new modalities for CRC screening. They were adopted for the evaluation of cancer screening using a stepwise approach in the World Endoscopic Evaluation (Young et al. Cancer 2016).

Aims & Methods: A systematic review of test performance RCTs for CRC screening was conducted. Candidate articles were searched from inception to October 14, 2023, using Ovid MEDLINE, Cochrane Central Registry, Embase, and Ichushi-web. We compared the test performance of TCS screening, referring to the following modalities: gFOBT, FIT, flexible sigmoidoscopy (FS), and CT colonography (CTC). Single and multiple rounds were searched. Test performance indicators included participation rates in CRC screening, advanced neoplasia (AN) detection rates, and CRC detection rates. The primary results were evaluated using an intention-to-screen (ITS) based network meta-analysis, and a per-protocol analysis was also performed.

Results: Seventeen RCTs (13 from Europe, two from the US, and one each from Israel and China), involving a total of 341,123 healthy screening invitees, were included. The median participation rates for the programs based on FIT, FS, and TCS were 50.4 % (28.6-94.0%), 32.3% (7.9-52.1%), and 21.5% (10.0-42.5%), respectively. In the network meta-analysis, participation rates for TCS screening were significantly lower than the programs based on FS (RR=0.58; CI: 0.37-0.89), FIT (RR=0.45; CI: 0.31-0.65), and gFOBT (RR=0.45; CI: 0.30-0.69). Detection rates for AN were consistently higher for TCS than for FIT (RR= 2.18, 95%CI:1.46-3.26) but not statistically significant for FS (RR=1.06, 95CI: 0.65-1.73). Similarly, TCS had higher CRC detection rates compared to FIT (RR= 1.80, 95%CI:1.04-3.12) and FS (RR= 1.52, 95%CI:0.87-2.64). While AN and CRC detection rates for TCS were higher than those of CTC, the differences were not statistically significant (AN: RR= 1.02, 95%CI:0.65-1.73, CRC: RR= 0.73 95%CI: 0.32-1.64). The follow-up period of single-round RCTs was limited until diagnosis, but the screening frequency varied depending on the follow-up duration, which could affect effectiveness. Screening intervals were typically two years for FIT, five years for FS, and ten years for CS. One quasi-experimental study combined the following arms from different RCTs with extended follow-up: one-time CS, one-time FS, and four rounds of FIT (Grobbee et al. Clin Gastroenterol Hepatol 2020). The participation rate was higher for FIT (73%) compared to FS (31%) and CS (24%). Cumulative AN and CRC detection rates were higher for FIT (AN = 4.5%, 95%CI:4.2-4.9, CRC =0.5% 95%CI: 0.4–0.6), but nearly equivalent for CS (AN = 2.2%, 95%CI:1.8-2.6, CRC =0.2% 95%CI: 0.1–0.3) and FS (AN = 2.3%, 95%CI:2.0-2.7, CRC =0.2% 95%CI: 0.1–0.3).

Conclusion: The single-round test performance of colonoscopy screening might be superior to that of other modalities. To confirm the priority of TCS, it is insufficient to draw conclusions based on the results of a single-round performance, given variations in screening intervals and frequencies among other modalities. Cumulative effects among different modalities should be compared within the same follow-up period.

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Disclosure: Nothing to disclose.

PP0835

A SINGLE-CENTRE ANALYSIS OF POST-COLONOSCOPY COLORECTAL CANCER

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Introduction: Post colonoscopy colorectal cancer (PCCRC) is colorectal cancer diagnosed after a normal colonoscopy with no cancer found. In England, the rate of PCCRCs has been decreasing but remains around 6.5% for colonoscopies performed from 2011-2013[1]. English study suggested that 80% of PCCRCs are avoidable[2]. We aim to analyse PCCRC cases using the local data and determine the demographics, nature of cancer and the cause of PCCRC.

Aims & Methods: The PCCRCS are identified following the linkage of Hospital Episode Statistics, Bowel cancer screening programme (BCSP) and cancer registry datasets. As new colorectal cancers are registered with the National Cancer Registration and Analysis Service (NCRAS), PCCRC cases will be identified. Each colorectal cancer (ICD-10 code C18-20) provisionally registered within the Hospital Episode Statistics dataset will be identified and determined whether a colonoscopy was performed within 6- 48 months before the diagnosis which failed to identify the cancer.

This audit includes patients diagnosed with PCCRC 6-48 months after a colonoscopy was performed in a single medical centre in England.

Once the cases had been confirmed, we reviewed the clinical, pathology and radiological findings of each case. We also looked at interval cancer which the World Endoscopy Organisation (WEO) defines as one diagnosed before the next recommended interval. Important to note that all interval cancers are PCCRCs but not all PCCRCs are interval cancer.

Results: 39 cases were included and 29 were confirmed to be true PCCRC which fulfilled the above criteria. 7 did not meet the criteria. (2 index colonoscopies performed at other trusts, 3 missing data, 2 other types of cancers such as Neuroendocrine tumour and appendiceal tumour, 1 has <6 months to cancer diagnosis from index colonoscopy, 2 flexible sigmoidoscopy was performed as index colonoscopy to investigate per rectal bleed-

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ing but subsequent cancer was found in ascending colons).

52% (n=15) of patients were male with a median age of cancer diagnosis at 74 years old (range 22-84). In terms of PCCRC, 24% (n=7) presented via the National Bowel Screening Programme (BCSP), 14%(n=4) presented with per rectal bleeding, 7% (n=2) presented with abnormal imaging and the same proportion of patients presented due to anaemia, non-specific change in bowel habit, weight loss, diarrhoea, abdominal pain and surveillance for family history of gene mutation. 3% (n=1) were diagnosed after presenting for polypectomy and 3% (n=1) presented with intestinal obstruction.

In terms of investigations that led to PCCRC, 45% (n=13) were via 2-weeks-wait colonoscopy, 24% (n=7) were via National Bowel Screening Programme, 17% (n=5) were diagnosed by Computed Tomography (CT) imaging, 7% (n=2) via flexible sigmoidoscopy and 7% (n=2) was via surgically resected sample. Upon diagnosis of PCCRC, 7% (n=2) tested positive for Lynch syndrome, 24% (n=7) was tested but no marker was found and 69% (n=20) was not tested. In terms of size of PCCRC, 66% (n=19) had tumour size >2cm. 66% (n=17) of tumour size was determined by histopathological sample, 21 (n=6) via endoscopy, 3% (n=1) via CT imaging and 17% (n=6) was not known.

In terms of PCCRC staging, 48% (n=14) was Dukes' A staging, 21% (n=6) was Dukes' B, 7% (n=2) was Dukes' C and 24% (n=7) was Dukes' D. 83% (n=24) was planned for curative treatment and 17% (n=5) was planned for palliative treatment. 52% (n=16) had surgical resection only, 17% (n=29) had surgical resection and chemotherapy, 3% (n=11) had surgical resection and radiotherapy, 3% (n=1) had surgical resection, chemotherapy and radiotherapy, 7% (n=2) had chemotherapy and radiotherapy, 3% (n=1) had chemotherapy only, 3% (n=1) had endoscopic resection only and 7% (n=2) had no treatment.

In terms of the type of interval cancer, 14% (n=4) was non-interval A, 7% (n=2) was non-interval B, 66% (n=19) was non-interval C and 21% (n=4) was interval cancer. Important to note was 66% (n=19) had no further follow-up or test arranged due to adequate examination on the index endoscopy. 90% of index endoscopies report no difficulties. 75% (n=22) was due to clinician factors, 10% (n=3) was due to administrative errors such as booking delay and 10% (n=3) was due to patient factors such as patient's decision and intercurrent illness.

Conclusion: The majority of the PCCRC had treatment with curative intent. Most PCCRC cases are due to clinician factors and measures needed to improve this. The high percentage of non-interval type C PCCRC would be explained by either a fast-growing cancer or a missed lesion since up to 90% of the index procedures reported no difficulty.

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Disclosure: Nothing to disclose.

PP0836

IMPACT OF ELECTRONIC PATIENT RECORDS AND THE INTRODUCTION OF A CLINICAL NURSE SPECIALIST ENDOSCOPY ON ADENOMA DETECTION RATES

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Introduction: Colorectal cancer screening, particularly through colonoscopy, is a pivotal element of preventive healthcare. Understanding the multifaceted factors influencing Adenoma Detection Rates (ADR) is crucial for optimizing the effectiveness of screening programs. This study aims to provide a detailed analysis of the impact of electronic patient records, clinical interventions, and other variables on ADR trends in a gastroenterology unit.

Aims & Methods: Aim: This study aims to investigate the factors contributing to the increase in ADR over time through the implementation of a comprehensive colonoscopy Quality Indicator recording system.

Methods: This study investigates the trends in Adenoma Detection Rates (ADR) and associated factors in a gastroenterology unit over a four-year period. The primary outcome is the evolution of ADR, with secondary outcomes including the relationship between ADR and withdrawal time, endoscopists' experience, and the Boston Bowel Preparation Scale (BBPS). The introduction of electronic patient records in 2019 establishes the baseline for data analysis.

In 2022, a full-time clinical nurse specialist in endoscopy began actively monitoring key indicators, such as ADR, BBPS, and withdrawal time. These metrics were regularly communicated to endoscopists through a PowerBi dashboard. As part of clinical interventions, a patient preparation video was introduced to enhance communication and standardize information for patients undergoing colonoscopies.

Results: We included patients aged 50 or older, with a total of 3340 colonoscopies performed between January 2019 and September 2023. 8000 colonoscopies are performed per year, but we only include the adenoma screening indications.

The analysis revealed a significant increase in ADR in 2020, coinciding with the introduction of electronic patient records. This marked rise, at 13.5%, suggests a positive influence of improved record-keeping on diagnostic accuracy. However, subsequent years showed a more modest increase of 0.4%, indicating a potential plateau effect. In 2022, the introduction of a full-time clinical nurse specialist in endoscopy corresponded with changes in key indicators.

The BBPS showed a significant decrease in 2023 compared to 2022 (p=0.03), while a reverse trend was observed between 2020 and 2021 (p = 0.03). Endoscopist's experience (p= 0,2470), and BBPS scores (p= 0,1652) were no significant predictors of the increased ADR. Adjusted for the other variables included in the model. These findings underscore the complex interplay of various factors influencing outcomes.

Conclusion: The dynamic nature of ADR observed in this study prompts a closer examination of the long-term effects of electronic patient records on diagnostic accuracy. The initial surge in ADR following their introduction suggests a positive and immediate impact on procedural outcomes. However, the subsequent stabilization of ADR raises questions about

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the sustainability of this effect over time. The introduction of a full-time clinical nurse specialist in endoscopy in 2022 may have contributed to the observed changes in BBPS. This emphasizes the potential benefits of multidisciplinary collaboration in optimizing colonoscopy outcomes. The observed decrease in BBPS in 2023, coupled with the sustained ADR levels, suggests a need for ongoing assessment of clinical interventions. The patient preparation video, introduced as part of enhanced communication strategies, may have contributed to these changes. Future research should delve deeper into the nuanced impact of such interventions on procedural outcomes and patient experiences.

Disclosure: Nothing to disclose.

PP0837

ESTABLISHING A BRIDGE BETWEEN PATIENTS UNDERGOING GASTROSCOPY AND COLONOSCOPY: A RETROSPECTIVE CROSS-SECTIONAL STUDY

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Introduction: It remains unknown whether gastric histopathology is associated with the occurrence of colorectal diseases.

Aims & Methods: Aims：We aimed to clarify the association between gastric histopathology and colorectal diseases, especially colorectal polyps (CP). On this basis, we established risk prediction models for different colorectal diseases.

Methods: A retrospective cross-sectional study was conducted on 2540 patients who underwent gastroscopy and colonoscopy simultaneously in The Affiliated Wuxi People's Hospital of Nanjing Medical University from January 1, 2021 to August 31, 2023. Basic demographic data, previous medical history, clinical symptoms, important laboratory indicators, and gastric histopathology classification were collected for univariate and multivariate analysis. The nomogram model was constructed with R language while its calibration was tested by Hosmer-Lemeshow, its discrimination was evaluated by Area Under the ROC Curve (AUC), and its clinical efficacy was evaluated by Decision Curve Analysis (DCA).

Results: 1: According to the results of colonoscopy, patients were divided into Normal group (N=601); Non-adenomatous polyp group (N=576); Adenomatous polyp group (N=1116); Inflammatory bowel disease group (N=149) and Colorectal cancer group (N=98). The average age of the five groups was (55.04±13.95) years, (58.74±11.58) years, (61.83±10.19) years, (50.48±16.64) years, and (66.10±10.80) years, respectively. The CRC group had the largest proportion of males, accounting for 59.2% (n=58).

2: The results of gastroscopy were divided into: Normal; NAG (non-atrophic gastritis); AG or IM (atrophic gastritis/intestinal metaplasia); Gastric polyps; Esophageal cancer; and SC-patho (pathological gastric cancer). The association between indicators and colonoscopy results was analyzed to establish risk prediction models for different colorectal diseases. We found that gastric histopathology was significantly correlated with NAP and AP; for clinical symptoms, abdominal pain was significantly correlated with colorectal diseases, and the contribution ranking was: NAP>IBD>CRC>AP; hematochezia was significantly correlated with IBD; for clinical indicators, anemia was significantly correlated with AP, and CEA was significantly correlated with CRC; for previous medical history, previous colonoscopy examination was significantly correlated with colorectal diseases, and the contribution ranking was: NAP>CRC>AP>IBD; history of hypertension and rheumatism disease were significantly correlated with NAP; history of chronic kidney disease and rheumatism disease were significantly correlated with AP; history of chronic kidney disease and gastrointestinal tumor were significantly correlated with CRC; history of hypertension and thyroid nodules were significantly correlated with IBD; for demographic indicators, age was significantly correlated with NAP, AP, and CRC; while BMI was significantly correlated with IDB.

3: Using R language to establish the nomogram prediction model, we could find that: all the prediction models had good discrimination, and the AUC of the CRC model reached 0.885. Through the calibration curve, we could find that all prediction models has strong consistency, among which the AP model has the highest consistency and the smallest absolute error value of 0.007.

Conclusion: The nomogram prediction model has good discrimination and calibration, and has good clinical value for predicting the occurrence probability of different colorectal diseases.

Disclosure: Nothing to disclose.

PP0838

DELINEATION OF RECURRENCE AMONG NON-EXPERT ENDOSCOPISTS IS LESS ACCURATE THAN EXPERTS BUT IS READILY TRAINABLE BY A SHORT LEARNING INTERVENTION

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Introduction: Endoscopic mucosal resection (EMR) is a safe and effective method to resect large (>=20mm) non-pedunculated colorectal polyps (LNPCPs). Incorrect decision making regarding the endoscopic resection scar after EMR risks unrecognized residual or recurrent adenoma (RRA) propagating into post-colonoscopy colorectal cancer. Despite expert series suggesting that scar recognition and interrogation are well performed with a high negative predictive value(2), the performance of non-experts remains unknown.

Aims & Methods: We aimed to investigate the diagnostic performance of endoscopic imaging in detecting RRA at an endoscopic resection scar amongst non-experts and the impact of a learning intervention.

Methods: A survey containing 15 paired High-Definition White Light (HD-WL) and virtual chromoendoscopy (VC) images, a learning intervention on detection of RRA (LT), followed by 15 different HD-WL & VC images was circulated to the GIEQs foundation mailing list. Images were selected from a prospective database of surveillance after EMR. For each image the same questions were asked: Is this a resection scar? Is there evidence of RRA? What is your confidence? Demographic data collected included participant number of EMRs performed, current grade and years of experience. Comparisons were made to expert opinion derived at a consensus meeting of the senior authors regarding the appearances of the images using the approach set out in Desomer et al. 2017(2).

Results: 82 participants completed the survey (response rate: 4.1%) resulting in 2,460 observations. 15 (18.3%) participants were experienced (performed >100 EMRs).

The overall accuracy of RRA detection pre-LT was 75.3% (95%CI 73.5-77.2) vs post-LT: 79.5% (95%CI 77.2-82.0, p =.006). Specificity significantly im-

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proved post-LT (50.0% [95%CI 44.1-55.4] pre vs 94.0% [95%CI 90.0-98.0] post, p<.001).

Both accuracy and specificity were significantly higher amongst experienced endoscopists (acc: 83.6% [95%CI 79.4-87.7], spec: 67.3% [95% CI 50.2-84.3]) versus inexperienced endoscopists (acc: 74.0% [95% CI 72.1-75.8], P<.001, spec: 46.7% [95% CI 41.1-52.3], P<.003) pre-LT. Specificity improved in both groups post-LT (experienced 67.3% [95%CI 50.2-84.3] vs 97.6% [95% 100-92.7], P<.001 and inexperienced 46.7% [95% CI 41.1-52.3] vs 93.5% [95% CI 88.3-98.7], P<.001). Furthermore the difference in specificity disappeared post-LT (spec experienced 97.6% [95% CI 100-92.7] vs inexperienced 93.5% (95% CI 88.3-98.7] p = .440).

	PRE learning intervention				POST learning intervention				P
	Experienced*	Inexperienced*	P*	Overall***	Experienced**	Inexperienced**	P**	Overall***	Overall PRE- vs POST -LT
Accuracy (95% CI)	83.6% (79.4-87.7)	74.0% (72.1-75.8)	<.001	75.3% (73.5-77.2)	82.6% (77.6-87.7)	78.7% (75.9-81.7)	0.220	79.5% (77.2-82.0)	.006
Sensitivity (95% CI)	73.3% (64.9-81.7)	71.7% (68.6-74.8)	.650	72.0% (69.1-74.8)	75.0% (67.2-82.2)	69.8% (65.6-74.1)	0.300	70.9% (67.5-74.5)	.649
Specificity (95% CI)	67.3% (50.2-84.3)	46.7% (41.1-52.3)	.003	50.0% (44.1-55.4)	97.6% (100-92.7)	93.5% (88.3-98.7)	0.440	94.0% (90.0-98.0)	<.001
PPV (95% CI)	85.4% (78.1-92.8)	68.3% (65.2-71.3)	<.001	70.9% (67.8-74.0)	99.3% (100.0-97.7)	97.8% (96.4-99.3)	0.338	98.0% (99.1-98.0)	<.001
PPV (95% CI)	85.4% (78.1-92.8)	68.3% (65.2-71.3)	<.001	81.3% (80.0-83.0)	99.3% (100.0-97.7)	97.8% (96.4-99.3)	0.338	65.1% (62.1-68.1)	<.001

RRA smaller than 10 mm was more likely to be missed (<10mm: 77.5%, <5mm: 58.9% of missed detections). RRA occupying ≤20% of the scar accounted for two-thirds of missed detections (66.4%). RRA located on the edge of a scar (versus centrally) accounted for the vast majority of missed RRA (88.1%).

Conclusion: The accuracy of RRA detection in non-experts is low compared to previously published expert series, even amongst endoscopists performing EMR regularly. Low accuracy predominantly resulted from low specificity which was mitigated by the training intervention. These results suggest that the potential negative effects of missing RRA at an endoscopic resection scar (propagation between surveillance procedures and the potential for development of PCCRC) can be mitigated by training.

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Disclosure: Nothing to disclose.

PP0839

CLINICAL OUTCOMES OF COLORECTAL NEOPLASM WITH INCOMPLETE ENDOSCOPIC SUBMUCOSAL DISSECTION

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Introduction: A positive resection margin after endoscopic submucosal dissection (ESD) of a colorectal neoplasm is associated with an increased risk of recurrence. However, it may be caused by cauterization or tangential specimen cutting. We aimed to identify the clinical significance of positive resection margins in colorectal neoplasms after ESD.

Aims & Methods: We reviewed 1010 patients who underwent colorectal ESD at two hospitals between 2015 and 2020. The inclusion criteria were that the patient 1) had an *en bloc* resection, 2) had at least one follow-up colonoscopy. The recurrence rates and presence of residual tumors in the surgical specimens were evaluated.

Results: A total of 632 patients were enrolled. Among them, 557 patients had complete resections, and 75 had incomplete resections with positive resection margins in vertical and/or horizontal margin. The rate of additional surgery after ESD and recurrence was significantly higher in patients with incomplete resection compared to patients with complete resection. However, in patients with non-invasive colorectal neoplasm, all 411 patients, including those with complete and incomplete resection, had no recurrence. Among 221 patients with invasive lesions, 39 underwent additional surgery due to non-curative resection. Positive vertical margin and lymphovascular invasion were associated with remnant tumors in the surgical specimen. Lymphovascular invasion was associated with lymph node metastasis. However, no remnant tumor nor lymph node metastases were found in patients with only one unfavorable histological factor, including 1) positive vertical margin, 2) a submucosal invasion depth > 1000 μm, 3) lymphovascular invasion, 4) poorly differentiated cancer, and 5) high tumor-budding grade.

Conclusion: A positive resection margin after *en bloc* colorectal ESD for noninvasive lesions was not associated with recurrence. In T1 colorectal cancer with a positive vertical resection margin, salvage surgery can be considered in selected patients with additional risk factors.

Disclosure: Nothing to disclose.

PP0840

FACTORS ASSOCIATED WITH TECHNICAL DIFFICULTY IN COLORECTAL ENDOSCOPIC SUBMUCOSAL DISSECTION: A HONAM ASSOCIATION FOR THE STUDY OF INTESTINAL DISEASE (HASID) MULTICENTER STUDY

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Introduction: Colorectal endoscopic submucosal dissection (ESD) is promising but challenging procedure. It is not widely performed due to its technical difficulty. We aimed to find the predictive factors associated with technical difficulty in colorectal ESD before procedure.

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Aims & Methods: Clinical data from patients who underwent ESD for colorectal tumor in five hospitals in Honam province of South Korea between 2015 to 2020 were reviewed retrospectively. Technically difficult colorectal ESD procedure was defined in three points. Long procedure time (longer than 60 minutes), occurrence of perforation and failure of *en bloc* resection. Factors associated with technically difficult ESD were included as main outcome measure.

Results: 1446 patients were identified and their data were analyzed. Median procedure time was 30.0 minutes and median long axis of the tumor was 20.1 mm. Technically difficult procedures including long procedure time were 231 cases (16.0%), perforation occurred in 34 cases (2.3%) and *en bloc* resection was done in 1292 cases (89.3%).

Tumor size larger than 35 mm (odd ratio [OR] 1.474, p=0.047), central depression or ulceration in the lesion (OR 1.474, p=0.013), previous endoscopic mucosal resection or polypectomy procedure (OR 2.428, p=0.020) were associated with technically difficult ESD. Descending colon located tumor (OR 5.355, p<0.001), use of IT knife (OR 4.157, p=0.003) were associated with perforation.

Conclusion: Recognizing factors associated with technically difficult ESD can help planning the ESD procedure beforehand.

Disclosure: Nothing to disclose.

PP0841

ANALYZING LEARNING CURVES OF EXPERT COLONOSCOPISTS IN SEDATION-FREE WATER EXCHANGE COLONOSCOPY

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Introduction: Water exchange, replacing air insufflation during colonoscopy's insertion phase, has become popular for its effectiveness in enhancing endoscopic quality. Its adoption by novices, who show improved performance, raises questions about its benefits for experienced colonoscopists.

This study examines the adaptation process of three seasoned colonoscopists to water exchange, evaluating if they achieve similar improvements, to understand its broader utility.

Aims & Methods: In this study, three expert colonoscopists, each with over 10,000 air insufflation colonoscopies, first conducted 100 standard procedures. They then switched to the water exchange technique for 100 more procedures across four groups. Key outcomes—cecal intubation rates, learning curves, pain scores, polyp detection, complications, abdominal compression, position changes, and bowel preparation quality—were compared between methods to evaluate the water exchange's effectiveness and patient experience.

Results: Cecal intubation success rates and median times for water exchange groups were over 90% and below 6 minutes, matching results from traditional air insufflation methods. No notable differences were found in intubation times between endoscopists A and B, but endoscopist C showed significant time variations, especially in the third quartile, indicating a performance shift (P<0.001).

A slight, non-significant decrease in abdominal compression was observed for endoscopist A, while endoscopist B's transition to water exchange significantly reduced compression needs (37% vs. 21%; P<0.05), showcasing its advantages. No changes were evident for endoscopist C. Position change demands differed significantly between methods for endoscopists B and C, hinting at endoscopist C's preference for air insufflation (P=0.001). In this study, the pain scores in the water exchange group were significantly lower than those in the air insufflation group (P=0.008) for only one expert. For the other experts, although the number of patients with a pain score of 10 decreased and the number of patients with a pain score of 0 increased in the water exchange group, no significant difference was observed. This suggests that for seasoned professionals, the water method primarily played a significant role in reducing severe abdominal pain. For patients with mild pain, the effect was unclear.

An analysis of the learning curve, considering cecal intubation success and time, found no correlation with the number of water exchange procedures performed, indicating all endoscopists quickly adapted to the method without a marked learning curve, maintaining success rates above 90%.

Conclusion: The study found the water method effectively reduced severe abdominal pain, with its impact on mild pain less clear. Experts showed proficiency with this technique from the start, achieving 95% success rates in cecal intubation, indicating quick adaptation. Despite this, traditional colonoscopy's known drawbacks weren't significantly highlighted. However, experts familiar with conventional techniques found the water method more challenging and saw no major advantages for colorectal screening.

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Disclosure: Nothing to disclose.

PP0842

PREVALENCE AND PREDICTIVE FACTORS OF COLORECTAL SESSILE SERRATED LESIONS IN YOUNGER INDIVIDUALS

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Introduction: Sessile serrated lesions (SSLs) are obscured lesions predominantly at right sided colon and associated with interval colorectal cancer. However, the prevalence and risk factors among younger individuals remain unclear.

Aims & Methods: This retrospective study enrolled individuals who underwent index colonoscopy. The primary outcomes were SSL prevalence between younger (<50 years) and older (≥50 years) groups, whereas the secondary outcomes included clinically significant serrated polyps (CSSPs). Multivariable logistic regression was employed to identify predictors. The aim of this study is to investigate the prevalence of SSL of younger age people in the single center cohort. Individuals who visited outpatient services were mostly symptomatic, were eligible for screening colonoscopy, or had abnormal fecal/tumor marker test results.

Additionally, asymptomatic individuals who underwent health checkups at their own expense or who adhered to national labor law requirements were included. The exclusion criteria were as follows:

(1) instances of duplicated cases due to prior colonoscopy examinations,

(2) cases with inadequate bowel preparation,

(3) history of hospitalization or colonoscopy as part of an emergency room visit,

(4) cases of unsuccessful cecal intubation or flexible sigmoidoscopy, and;

(5) suspicious or confirmed inflammatory bowel disease. Notably, indi-

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viduals who had previously undergone colonoscopy at other hospitals, as confirmed by their electronic medical records or chart review, were also excluded.

Results: Of the 9854 eligible individuals, 4712 (47.8%) were categorized into the younger age group. Individuals in the younger age group exhibited a lower prevalence of adenomas (22.6% vs. 46.2%, *P* < 0.001) and right-sided adenomas (11.2% vs. 27.2%, *P* < 0.001) compared with their older counterparts. However, both groups exhibited a similar prevalence of SSLs (7.2% vs. 6.5%, *P* = 0.157) and CSSP (10.3% vs. 10.3%, *P* = 0.956). Multivariable analysis results revealed that age = 40–49 years (odds ratio [OR] = 1.81, 95% confidence interval [CI] = 1.01–3.23), longer withdrawal time (OR = 1.18, 95% CI = 1.15 – 1.21 per minute increment), and endoscopist performance (OR = 3.35, 95% CI = 2.44 – 4.58) were independent predictors of SSL detection in the younger age group. No significant correlation was observed between adenoma and SSL detection rates among endoscopists (P = 0.083).

	Younger age group (< 50-years-old)	Elder age group (≥ 50-years-old)	P value
Cases (%)	4712 (47.8)	5142 (52.1)	Not applicable
Symptomatic cases (%)	1985 (42.1)	2763 (53.7)	< 0.001*
Sex (female, %)	2425 (51.5)	2649 (51.5)	< 0.001*
Family history of CRC (%)	318 (6.7)	164 (3.2)	< 0.001*
Adenoma (%)	1063 (22.6)	2375 (46.2)	< 0.001*
Advanced adenoma (%)	188 (4.0)	728 (14.2)	< 0.001*
SSL (%)	341 (7.2)	335 (6.5)	0.157
CSSP (%)	485 (10.3)	531 (10.3)	0.956
CRC (%)	19 (0.4)	72 (1.4)	< 0.001*

Conclusion: SSLs are not uncommon among younger individuals. Moreover, diligent effort and expertise are of paramount importance in SSL detection. Future studies should explore the clinical significance of SSLs in younger age individuals.

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Disclosure: This study is recently accepted by Endoscopy (2024/02). We hope this research can be considered for the congress with its clinical importance and impact.

PP0843

CAN 5-MINUTE COLONOSCOPY WITHDRAWAL TIME BE THE BENCHMARK BASED ON EXPERIENCED ENDOSCOPISTS AND EXCELLENT BOWEL PREPARATION? A MULTICENTER, RANDOMIZED CONTROLLED, BACK-TO-BACK TRIAL IN CHINA

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Introduction: Although current quality indicators of colonoscopy recommend 6 minutes as the minimum standard for withdrawal time (WT), long-term implementation of the 6-minute WT results in low colonoscopy turnaround efficiency and it don't take into account the effects of endoscopists experience and the quality of bowel preparation of colonoscopy .Whether it is still necessary to adhere to the 6-minute WT standard under the condition of experienced endoscopists and excellent bowel preparation quality is worthy of further discussion. Therefore, this study based experienced endoscopists and excellent bowel preparation (Boston Bowel Preparation Scale score, BBPS≥8) is aimed to compare the effects of 5-minute WT versus 6-minute WT on adenoma detection rate (ADR) and adenoma miss rate (AMR).

Aims & Methods: Aims：This study based experienced endoscopists and excellent bowel preparation (Boston Bowel Preparation Scale score, BBPS≥8) is aimed to compare the effect of 5-minute WT versus 6-minute WT on adenoma detection rate (ADR) and adenoma miss rate (AMR).

Methods：This multicenter, randomized controlled trial involving 298 patients was conducted from June 2023 to February 2024 in 5 centers, including Wuxi People's Hospital, Jiangsu Province People's hospital, Xuzhou First People's Hospital, The Second Affiliated Hospital of Soochow University, Taizhou People's Hospital. Colonoscope was inserted and withdrew twice on each patient and all procedures for the same patient were done by the same endoscopist. Participants were randomly divided into the 5-minute WT group (the first WT-5-minute and the second WT-6-minute, n =161) and the 6-minute WT group (the first WT-6-minute and the second WT-6-minute, n=137), and a timer was used to adjust the withdrawal speed. The primary outcome was ADR and AMR.

Results: The baseline characteristics of the two groups were similar (Table 1) and colonoscopy characteristics of the patients in the two groups regarding quality of bowel preparation, cecal intubation rate, cecal intubation time, cecal intubation distance, number of abdominal compression, number of position change were comparable (Table 2). Overall ADR(44.5% vs 41.6%, *P* = 0.613), ADR in the first (37.2% vs 36.6%, *P* = 0.918) and second withdrawal (15.3% vs 14.3%, *P* = 0.800) were not significantly increased in the 6-minute WT group compared with the 5-minute WT group. Likewise, the overall advanced adenoma detection (AADR) (8.0% vs 6.8%, *P* = 0.694), AADR in the first (7.3% vs 6.8%, *P* = 0.875) and second withdrawal (1.5% vs 0.6%, *P* = 0.888) did not differ statistically. Further analysis showed that AMR (23.5% vs 24.6%, *P* = 0.826) and advanced adenoma miss rate (AAMR) (6.7% vs13.3%, *P* = 0.543) were lower in the 5-minute WT group than in the 6-minute WT group, less adenomas and advanced adenomas were missed in the 5-minute WT group, although not statistically significant (Table 3).

Conclusion: Shortening WT from 6 to 5 minutes didn't reduce ADR based experienced endoscopists and excellent BBPS, a 5-minute WT benchmark was acceptable with experienced endoscopists and excellent bowel preparation quality of colonoscopy.

Disclosure: Nothing to disclose.

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PP0844

MULTICENTER PROSPECTIVE FEASIBILITY STUDY OF COMPLIANCE, SAFETY, AND ACCEPTANCE OF COLON CAPSULE ENDOSCOPY (CCE) AND SMALL BOWEL CAPSULE ENDOSCOPY (SBCE) IN THE OUT-OF-CLINIC SETTING IN JAPAN (HOMECAM-J STUDY)

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Introduction: The U.S. FDA has granted the PillCam™ SB3 @HOME program which allows patients to undergo SBCE at home. This program for SBCE remote use, however, is not cleared in the other countries, neither is a telehealth option of CCE worldwide.

Aims & Methods: We aimed to confirm compliance, safety, and acceptance of CCE and SBCE in the out-of-clinic setting in Japan. We enrolled 30 subjects who had not undergone CE (24 for CCE and 6 for SBCE) at 6 gastroenterological centers between June 2023 and February 2024. All examinees were provided with detailed instructions on how to interact with data-recorder automatic signaling, and for CCE examinees, low residue diet and a bowel preparation set including magnesium citrate, sodium picosulfate, PEG solution plus ascorbic acid, and castor oil. We prospectively collected data regarding baseline characteristics, compliance with procedures, CE results, acceptability with a subjectively categorized questionnaire survey, and adverse events using REDCap system (jRCTs042220163).

Results: Of 24 CCE examinees (male/female: 11/13, median age of 51 years, 23 for health check-up and 1 for positive fecal immunological test), 16 and 8 underwent CCE at home and at the workplace, respectively. Compliance rates of intake of low residue diet, recorder-capsule self-pairing, ingestion of CCE, intake of a bowel preparation set, return of equipment were all 100%. During the procedure, 12 examinees (50%) called the physician 18 times about when to finish (66.7%), how to take preparation (16.7%), and how to operate recorder (11.1%). Capsule excretion rate within examination time was 83% (20/24). In the 4 examinees who did not excrete the capsule, last images showed the rectum (n=1), transverse colon (n=1), ascending colon (n=1), and cecum (n=1). Median examination time between capsule ingestion and removal of sensor belt was 227 (80-1233) min; 208.5 (80-320) min and 697 (240-1233) min in examinees with or without capsule excretion, respectively (P=0.0118). The overall adequate endoscopic cleansing rate was 83%. Abnormalities were detected at 37.5% (9/24); colorectal polyps (25.0%), diverticula (8.3%), suspicious small bowel subepithelial tumor (4.2%). There was only one adverse event including intermittent abdominal pain. The proportions of examinees who reported "quite easy", "fairly easy", "moderate", "fairly hard", "quite hard", and wished for CCE as the next examination were 17%, 25%, 25%, 29%, 4%, and 67%, respectively.

Of 6 SBCE examinees (male/female: 4/2, median age of 39 years, 2 for obscure GI bleeding and 1 for chronic diarrhea, 1 for ileitis, 1 for abdominal pain, 1 for health check-up), 4 and 2 underwent SBCE at home and at the workplace, respectively. Compliance rates of recorder-capsule self-pairing, ingestion of SBCE, return of equipment were all 100%. During the procedure, 3 examinees (50%) called the physician 4 times about when to finish (75.0%) and how to operate recorder (25.0%). Total enteroscopy rate within examination time was 100% (6/6). Median examination time between capsule ingestion and removal of sensor belt was 198 (158-372) min. The overall adequate endoscopic observation rate was 100%. Abnormalities were detected at 16.7% (1/6); duodenal polyp (n=1). There were no adverse events. The proportions of examinees who reported "quite easy", "fairly easy", and "moderate" were 33%, 33%, and 33%, respectively.

Conclusion: CCE and SBCE with all procedures in an out-of-clinic setting were feasible, safe, and comparable to those in a hospital setting, but remote consultations would be needed during the procedures.

Disclosure: Nothing to disclose.

PP0845

COMPUTER-AIDED DIAGNOSIS FOR THE RESECT AND DISCARD STRATEGY FOR COLORECTAL POLYPS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: The Resect and Discard strategy allows endoscopists to replace post-polypectomy pathology with real-time prediction of polyp histology during colonoscopy (optical diagnosis). The present systematic review and meta-analysis investigated how implementation of computer-aided diagnosis (CADx) for polyp pathology affected this optical diagnosis procedure.

Aims & Methods: We conducted a search of MEDLINE, EMBASE, and Scopus databases, covering studies published from inception to February 20, 2024. We included histologically verified accuracy diagnostic studies that evaluated the real-time optical diagnosis performance of CADx systems for polyps ≤5mm in the entire colon. The aim was to estimate the potential benefits and harms associated with integrating CADx assistance into the

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endoscopist optical diagnosis, both before and after its implementation. The primary outcomes were the proportion of polyps that would have avoided pathological assessment (main benefit) and the proportion of polyps incorrectly predicted (main harm) due to false-positive or -negative optical diagnosis. Only diagnosis with a high-level of confidence were considered.

Results: A total of 11 studies with 7,400 polyps were eligible for analysis, providing stand-alone performance of CADx with sensitivity and specificity in identifying adenomatous histology of 87% (95% confidence interval [CI]: 0.82, 0.91) and 75% (95% CI: 0.68, 0.82), respectively. Three studies (4,086 polyps) were included to analyze the primary benefit/harm outcome measures. There was no significant difference between the CADx-unassisted and CADx-assisted strategies for the proportion of polyps that would have avoided pathological assessment (90% vs. 90%; risk ratio: 1.01; 95% CI: 0.99, 1.04; I^2=53.49%; moderate certainty evidence). CADx-assisted strategy reduced the proportion of incorrectly predicted polyps as compared with CADx-unassisted (13% vs. 12%; risk ratio: 0.88; 95% CI: 0.79, 0.98; I^2=0%; moderate certainty evidence) mainly due to a decrease in false-positive diagnosis.

Conclusion: CADx implementation did not result in clinically relevant benefits for the Resect-and-Discard strategy.

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PP0846

UNDERWATER ENDOSCOPIC MUCOSAL RESECTION AND HOT-SNARE POLYPECTOMY IN COLONOSCOPY

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Introduction: Conventional endoscopic mucosal resection (CEMR) and underwater endoscopic mucosal resection (UEMR) are techniques for the removal of colorectal lesions.

Aims & Methods: The our study aim is to compare the efficacy and safety of UEMR versus CEMR in patients with lesions under 30mm in diameter.

The study included 124 patients with colorectal lesions with a diameter < 20mm, 20-30mm, type 0-Ip (hot-snare polypectomy), 0-Is, 0-IIa or 0-IIa+Is according to the Paris Classification. Cases were divided into two groups: Group I 64/124(51,61%) CEMR cases (0-Is, 0-IIa, 0-IIa+Is - < 20mm 43,75%, 20-30mm 20,31%; type 0-Ip - 35,93%) and Group II 60/124(48,38%) UEMR patients (0-Is, 0-IIa, 0-IIa+Is - < 20mm 45%, 20-30mm 23,33%; type 0-Ip - 31,66%). Cases with deep submucosal invasion were not included. Optical evaluation with chromoscopy and NBI was performed by expert endoscopist, followed by endoscopic resection.

The outcomes were the lesion recurrence rate, en-bloc resection, R0 resection rates, adverse events. Data were analysed using Chi-sq, descriptive statistic by SPSS version 26.0.

Results: The recurrence rate was lower for II Group (UEMR) (1/14; 7,14%) in cases of 20-30mm lesions size (I Group (CEMR) - 2/13; 15,38%), and in 0-Ip (hot-snare polypectomy) cases (II Group (UEMR) - 1/19; 5,26%; I Group (CEMR) - 3/23; 13,04%); en-bloc resection rates was higher in II Group (UEMR) (13/14; 92,85%) in cases of 20-30mm lesions size. The R0 resection and adverse events showed the same tendency, only for polyps between 20 and 30 mm. There were no differences in the overall recurrence rate between groups. Underwater endoscopic mucosal resection was faster and easier to perform than CEMR.

Conclusion: Underwater endoscopic mucosal resection is safety method and could be considered for treatment of colorectal lesions with low adverse events rates.

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Disclosure: Nothing to disclose.

PP0847

ENDOSCOPIC CHARACTERISTICS DIFFERENTIATING SESSILE SERRATED LESIONS AND MICROVESICULAR HYPERPLASTIC POLYPS FROM GOBLET CELL RICH HYPERPLASTIC POLYPS: A CROSS-SECTIONAL STUDY

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Introduction: Differentiating between colorectal serrated polyps (SPs) on endoscopy can be challenging since sessile serrated lesions (SSLs) and hyperplastic polyps (HPs) tend to have a similar appearance[1-4]. Furthermore, the endoscopic differences between the two subtypes of HPs, namely microvesicular HP (MVHP) and goblet cell-rich HP (GCHP), are not well understood[5,6]. Therefore, we compared the endoscopic features of SSL, MVHP, and GCHP.

Aims & Methods: This retrospective, cross-sectional study was conducted at the Toyoshima Endoscopy Clinic. We examined the polyp size, location, Paris classification type, mucus cap, indistinct border, expanded crypt opening, varicose microvascular vessels (VMV), and Japan NBI expert team (JNET) classification type. Multivariable analysis was performed to determine the endoscopic finding that predicted SP histology using a binomial logistic regression model. We constructed endoscopic microvesicular goblet (MG) score for differentiating SPs using the endoscopic findings associated with SP histology in the multivariable analysis, and calculated the area under the receiver operating characteristic curve (AUC) and diagnostic accuracy.

Results: A total of 1366 SPs were eligible for this study, including 321 SSLs, 718 MVHPs, and 327 GCHPs. The mean polyp size was 6.24 mm. SPs in the proximal colon accounted for 70.7%. In the comparison between MVHP and GCHP, a mucus cap (partial regression coefficient 1.44, 95% confidence interval [CI] 1.06-1.82), expanded crypt opening (1.92, 1.44-2.40), and VMV (1.04, 0.58-1.49) were more often found in MVHP in contrast to GCHP (Table). comparison of SSL and MVHP showed that SSLs were located in the proximal colon (0.70, 0.30-1.10), were larger (0.21, 0.16-0.25), and had mucus cap (0.54, 0.16-0.93) compared to MVHPs. On comparing the SSL plus MVHP group and GCHP group, larger size (0.12, 0.06-0.19), a mucus cap (1.76, 1.39-2.14), expanded crypt opening (2.04, 1.57-2.52), VMV (1.21, 0.75-1.68), and JNET type 2A (1.17, 0.73-1.61) were more often observed in SSL + MVHP group compared with GCHP. Mucus cap, expanded crypt opening, and VMV were each assigned 1 point and the sum of

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the points was defined as the endoscopic MG score. In the examination of MVHPs and GCHPs, the AUC of the endoscopic MG score for predicting MVHP was 0.78 (95% CI 0.76-0.81) and the optimal cut-off value was 1; the sensitivity, specificity, and positive predictive value (PPV) were 74.7%, 75.5%, 87.0%, respectively. In the assessment of SSLs, MVHPs, and GCHPs, the AUC of the endoscopic MG score for predicting SSL plus MVHP group was 0.82 (0.79-0.84) and the optimal cut-off value was 1; the sensitivity, specificity, and PPV were 79.4%, 75.5%, and 91.2%, respectively.

	Partial regression coefficient	95% confidence interval	Degree of freedom	P value
Polyp size	0.04	-0.03, 0.11	1	0.23
Mucus cap	1.44	1.06, 1.82	1	<0.001
Indistinct border	0.07	-0.32, 0.46	1	0.73
Expanded crypt opening	1.92	1.44, 2.40	1	<0.001
VMV	1.04	0.58, 1.49	1	<0.001

Table. Multivariable analysis for comparison of endoscopic appearance between MVHP and GCHP.

Conclusion: MVHP and GCHP, while belonging to the same HP category, exhibited different endoscopic appearances such as mucus cap, expanded crypt opening, and VMV. Conversely, SSL and MVHP despite belonging to different histopathological categories, showed similar endoscopic appearances except for mucus cap. Moreover, SSL plus MVHP showed distinct endoscopic appearances compared to GCHP group. Considering MVHP and GCHP as separate entities can aid in the endoscopic diagnosis of SPs.

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PP0848

AWITHDRAWN

PP0849

DIAGNOSTIC PERFORMANCE OF JAPAN NBI EXPERT TEAM (JNET) CLASSIFICATION AND CONECCT CLASSIFICATION FOR LARGE COLORECTAL LATERALLY SPREADING LESIONS TREATED BY ENDOSCOPIC SUBMUCOSAL DISSECTION: A LARGE WESTERN COHORT STUDY

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Introduction: Endoscopic characterization aims to predict the histology of a lesion and thus the risk of cancer. This characterization step is essential, especially in the case of large superficial colorectal lesions, as the choice of resection technique directly depends on it. The JNET (Japan NBI Expert Team) classification is the classification proposed by European and Japanese guidelines to select the optimal therapeutic approach between EMR (endoscopic mucosal resection), ESD (endoscopic submucosal dissection), or surgery, but has not been evaluated in large laterally spreading lesions (LSTs) (>20 mm). We therefore evaluated the diagnostic performance of the JNET classification for predicting the risk of submucosal invasive cancer (SMIC) in LSTs >20 mm.

Aims & Methods: Single-center, prospective, observational study including all patients referred for endoscopic resection of lesions >20 mm, between 2017 and 2022. All LST samples with SMIC underwent a secondary pathological examination to establish the correlation between "covert" carcinoma (SMICs categorized as JNET 2A) and "buried" carcinoma (existence of SMICs more than 0.5 mm from the surface). The primary outcome was the diagnostic accuracy of the JNET classification for histological prediction of LSTs >20 mm. The secondary objective was to assess the combined performance of the JNET classification and pejorative macroscopic (macronodule >1 cm, Paris 0-IIc area, JNET 2B area, Non-Granular LST) criteria also called the CONECCT classification. We also attempted to determine the characteristics and risk factors of LSTs with covert or buried carcinoma, and to identify potential associations between the two.

Results: 972 lesions were removed by en-bloc ESD, of which 110 (11.3%) of these lesions contained SMIC. The median lesion size was 57 mm. Of the 110 submucosal cancers, 33 had covert carcinoma and 35 were classified as buried carcinoma. The sensitivity and specificity for predicting high-grade dysplasia (HGD) to SMIC were 46% and 75% for JNET 2B and 87% and 38% for CONECCT IIC, respectively; and for predicting superficial SMIC were 67% and 67% for JNET 2B and 94% and 27% for CONECCT IIC, respectively. The false negative rates (FNR) (probability of missing SMIC) of JNET 2B and CONECCT IIC were 31% (31/100) and 3% (3/100) for predicting SMIC, respectively; and 33% (16/48) and 6% (3/48) for predicting superficial SMIC, respectively. JNET 2B exhibited significantly higher specificity than CONECCT IIC for predicting HGD to superficial SMIC (75% vs. 38%; p< 0.001) and for predicting isolated superficial cancer (67% vs. 27%; p< 0.0001). And CONECCT IIC was significantly more sensitive than JNET 2B for predicting HGD to superficial SMIC (46% vs. 87%; p< 0.001) and for predicting isolated superficial cancer (67% vs. 97%; p< 0.0001). No association was found between covert carcinoma and buried carcinoma (p 0.889).

Conclusion: The JNET classification has low diagnostic accuracy for predicting the risk of submucosal cancer for large superficial colorectal lesions and for making decisions on *en bloc* endoscopic treatment using

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ESD. If a selective ESD and piecemeal EMR strategy is used based on the submucosal cancer risk, the CONECCT classification is preferred to prevent under-treatment of SMIC.

Disclosure: Nothing to disclose.

PP0850

PARIS-IS NODULES ≥10MM COMMONLY OCCUR IN LARGE GRANULAR RECTAL LESIONS AND ARE A RISK FACTOR FOR COVERT SUPERFICIAL SUBMUCOSAL INVASIVE CANCER

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Introduction: Cancer in large non pedunculated colorectal polyps (LNPCPs) may go unrecognised if it is not visible on the surface (hidden or COVERT cancer). Increasing size, non-granular morphology, rectosigmoid location and combined Paris classification have been identified as risk factors for COVERT cancer in LNPCPs.[1]

It is unknown whether the size of the dominant nodule (largest Is component) in LNPCPs is correlated with risk of COVERT cancer or depth of submucosal invasion (SMI).

Aims & Methods: The endoscopic appearance and histopathology of LNPCPs ≥20 mm was analysed from a prospectively collected multicenter database of polyps referred for endoscopic resection between November 2017 and November 2023. Visible (OVERT) cancers and all serrated polyps were removed. LNPCPs with large Is components (Paris 0-Is and 0-IIa+Is polyps with Is component ≥10mm as defined by expert consensus and retrospective image review) were compared to LNPCPs without large Is component. The primary endpoint was the correlation between large Is component and presence and depth of SMI. Secondary end points were correlation between a large Is component and other risk factors for COVERT disease (location in the colon, size and morphology).

Results: 3470 LNPCPs without evidence of OVERT cancer were included, of which 265 (7.6%) contained a large Is component and 3205 (92.4%) did not.

LNPCPs with large Is component were more likely to contain SMI (4.9% [95% confidence interval (CI):2.3% - 7.5%] vs. 2.1% [95% CI: 1.6% - 2.6%], P<0.01). Subgroup analysis showed that LNPCPs with a large Is component were more likely to contain superficially invasive cancer (large Is: 2.3% [95% CI: 0.5% - 4.1%] vs. no large Is: 0.7% [95% CI: 0.4% - 1.0%], P=0.01). There was no significant difference in presence of deep SMI between LNPCPs with a large nodular component vs. LNPCPs without large Is component (large Is: 2.6% [95% CI: 0.7% - 4.6%] vs. no large Is: 1.4% [95% CI: 1.0% - 1.8%], P=0.17).

LNPCPs with large Is component were larger (≥40mm; 57.0% [95% CI: 51.0% - 62.9%] vs. 37.8% [95% CI: 36.1% - 39.5%], P<0.01), more often granular (86.8% [95% CI: 82.7% - 90.9%] vs. 58.0% [95% CI: 56.3% - 59.7%], P<0.01) and found in the rectum (34.0% [95% CI: 28.3% - 39.7%] vs. 16.4% [95% CI: 15.1% - 17.7%], P<0.01) vs. LNPCPs without large Is component (Table 1).

	No Large Is Component [95% CI]	Large Is Component [95% CI]	P-Value
Number of included LNPCPs	3205	265	-
Size ≥ 40mm	37.8% [36.1% - 39.5%]	57.0% [51.0% - 62.9%]	<0.01
Granular	58.0% [56.3% - 59.7%]	86.8% [82.7% - 90.9%]	<0.01
Rectal location	16.4% [15.1% - 17.7%]	34.0% [28.3% - 39.7%]	<0.01
SMI (all)	2.1% [1.6% - 2.6%]	4.9% [2.3% - 7.5%]	<0.01
Superficially invasive	0.7% [0.4% - 1.0%]	2.3% [0.5% - 4.1%]	0.01
Deeply invasive	1.4% [1.0% - 1.8%]	2.6% [0.7% - 4.6%]	0.17

Table 1: Comparison of endoscopic features and histopathology between large colonic laterally spreading lesions without and with at least one large nodular (Is, size of the nodule ≥10mm) component. LNPCPs, large non pedunculated colorectal polyps. SMI, submucosal invasion. CI, confidence interval.

Conclusion: LNPCPs with a Is component ≥10mm are at higher risk of containing COVERT cancer which, if present, is commonly superficially invasive. These results have implications for choice of resection technique on detection of large rectal LNPCPs with nodules ≥10mm and stratification of these patients to endoscopic en-bloc resection.

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Disclosure: Nothing to disclose.

PP0851

PROSPECTIVE EVALUATION OF THE USE OF ARTIFICIAL INTELLIGENCE-ASSISTED MONITORING OF EFFECTIVE WITHDRAWAL TIME VERSUS STANDARD WITHDRAWAL TIME ON ADENOMA DETECTION RATE

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Introduction: Effective withdrawal time (EWT), as measured by EndoScreener QC (Wision A.I.), is a novel artificial intelligence (AI)-derived colonoscopy quality metric, which measures the actual duration of clear images obtained during scope withdrawal from cecum to rectum. We have previously demonstrated the superiority of EWT to standard withdrawal time (SWT) on adenoma detection rate (ADR) in a retrospective study.(1) This prospective study determines the association between EWT and colonic lesion detection rates.

Aims & Methods: This was a prospective evaluation of the use of real-time determination of EWT, as measured by the EndoScreener QC, of endoscopists who were blinded to the output of the system. Patients were recruited in our hospital from Nov 2023 to Feb 2024. All examination was performed with high-definition endoscopes (EVIS-EXERA 290 video system, Olympus Optical, Tokyo, Japan) under white light. In all examination, colonoscope was first advanced to the cecum as confirmed by identification of the appendiceal orifice and ileocecal valve or by intubation of the ileum. After cecal intubation, the colonoscopy was slowly withdrawn. The AI derived (AI) real time effective mucosal examination monitor system (EndoScreen QC) was initiated during scope withdrawal, starting from cecum to anus. The SWT was defined as total withdrawal time from cecum to rectum after excluding polypectomy and biopsy time. All detected polyps were removed during scope withdrawal only. The size (measured with

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biopsy forceps), location and morphology of each polyp was recorded by an independent observer. The bowel preparation quality was graded according to the Boston Bowel Preparation Scale. The primary outcome was the correlation between EWT and ADR, which was determined by the Pearson correlation coefficient (r); and compared with SWT. Other outcomes included correlation between EWT and polyp detection rate (PDR) or serrated lesion detection rate (SDR).

Results: In the prospective analysis, 185 colonoscopies performed by 9 endoscopists were analyzed. The mean age was 67.4 year (SD: 10.5) and 57.3% (n=106) were male. 83.9% (n=154) were diagnostic colonoscopy. The median Boston Bowel Preparation score (BBPS) was 7. The mean ADR was 49.2% (95% CI: 41.6-56.2, n=91), mean EWT was 4.4 (95% CI: 4.1-4.6) minutes and mean SWT was 7.1 (95% CI: 6.7-7.5) minutes. The correlation between EWT and ADR was significantly higher (r=0.98, 95%CI: 0.96-1.00) than between SWT and ADR (r=0.80, 95%CI: 0.53-0.98, p<0.01). For PDR, the correlation between EWT and PDR was also significantly higher than between SWT and PDR (r=0.97, 95%CI: 0.95-1.00 vs r=0.73, 95%CI: 0.30-0.98, p=0.02). However, there was no significant difference between EWT and SWT on SDR [r=0.92, 95%CI: 0.72-1.00 vs r=0.93, 95%CI: 0.86-0.98, p=0.89].

Number	185
Mean age in years (SD)	67.4 (10.5)
Male sex, % (n)	106 (57.3)
Indications:	
Screening, % (n)	31 (16.8)
Diagnostic, % (n)	154 (83.9)
Bowel preparation in BBPS, median (range)	7 (6-9)
Mean standard withdrawal time in mins (95%CI)	7.1 (6.7-7.5)
Mean EWT in mins (95%CI)	4.4 (4.1-4.6)
ADR %(n)	49.2 (91)
PDR %(n)	58.9 (109)
SDR % (n)	20.0 (37)
Mean number of adenoma (SD)	1.1 (0.1)
Mean number of polyp (SD)	1.6 (0.1)
Mean number of serrated lesion (SD)	0.4 (0.1)
r between SWT and ADR (95%CI)	0.80 (0.53-0.98)
r between EWT and ADR (95%CI)	0.98 (0.96-1.00)
r between SWT and PDR (95%CI)	0.73 (0.30-0.98)
r between EWT and PDR (95%CI)	0.97 (0.95-1.00)
r between SWT and SDR (95%CI)	0.93 (0.86-0.98)
r between EWT and SDR (95%CI)	0.92 (0.72-1.00)

Table 1: Baseline characteristics and outcomes of the study.

Conclusion: The novel AI-derived EWT demonstrated better correlation with ADR and PDR than SWT in this prospective study. Further studies should evaluate the role of applying real-time monitoring EWT on endoscopy training and quality improvement of endoscopists with low ADR.

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PP0852

ENDOSCOPIC FREE-HAND SUTURING IS SAFE AND FEASIBLE FOR DEFECT CLOSURE IN THE UPPER GASTROINTESTINAL TRACT AND COLORECTUM: THE LARGEST RETROSPECTIVE ANALYSIS TO DATE

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Introduction: The management of post-interventional and iatrogenic defects in the gastrointestinal tract presents significant challenges in endoscopic practice, often carrying substantial risks for patients. Various closure techniques have been developed, yet consistently effective outcomes remain elusive in diverse scenarios. Endoscopic free-hand suturing (EFHS) using a needle holder is a new technique that has demonstrated potential for effectively addressing these defects.

In this study, we present one of the largest cohorts to date, analyzing the safety and feasibility of endoscopic free-hand suturing across multiple scenarios and locations.

Aims & Methods: This study aimed to investigate the feasibility and safety of EFHS and the attainability of sustained closure following various procedures: endoscopic submucosal dissection (ESD), perforation, per-oral endoscopic myotomy (POEM), per-oral endoscopic myotomy for Zenker's diverticulum (Z-POEM), submucosal tunneling endoscopic resection (STER), and endoscopic submucosal excavation (ESE).

ESD, or the respective above-mentioned interventions, were followed by EFHS, utilizing a gastroscope or colonoscope and a barbed absorbable wire (V-loc 180, Covidien, Dublin, Ireland). Needle steering involved a distal cap attachment and a needle holder (SutuArt; Olympus, Hamburg, Germany). Suturing was performed in a zigzag pattern at 5-10 mm intervals. For safety, the needle was concealed, and post-suturing, the remaining suture and needle were cut with a loop cutter (Olympus, Hamburg, Germany) for removal.

Results: In this retrospective single-center study, 30 post-interventional defects in 26 patients (female *n* = 10, male *n* = 16) and performed EFHS were retrospectively followed. Technical success, clinical success after the procedure, and severe EFHS-related adverse events were evaluated.

EFHS was applied for defect closure after the following interventions and/or complications: colonic endoscopic submucosal dissection (*n* =10; ceacum *n* =4, ascending *n* = 1, transverse *n* = 2, sigmoid *n* = 3), rectal endoscopic submucosal dissection (*n* = 9), gastric subepithelial lesion removal (*n* = 2; STER *n* = 1, ESE *n* = 1) duodenal perforation (*n* = 1), colonic perforation (*n* = 1), peroral endoscopic myotomy (POEM) (*n* = 1), peroral endoscopic myotomy for the treatment of Zenker's diverticulum (Z-POEM) (*n* = 2).

The median of the resected area was 1450 mm^2 (range 500 – 3600 mm^2).

In none of the 26 patients were severe procedure-related events observed, specifically, there were no instances of bleeding or perforation.

Endoscopic follow-up was conducted in 15 of 30 EFHS closures, demonstrating adequate closure in 13 cases, in 2 cases in the rectum, the mucosa was partially dehiscent, although showing re-epithelialization. The median time between the intervention and endoscopic follow-up was 15 days (range 1 - 69 days). The median duration of hospital stay after endoscopic intervention was 2 days (range 1 – 5 days).

Conclusion: Endoscopic free-hand suturing is safe and feasible within the gastrointestinal tract, including esophagus, stomach, duodenum, colon and rectum. Endoscopic free-hand suturing has the potential to enhance the safety of endoscopic interventions and reduce hospital stay duration.

Disclosure: Nothing to disclose.

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PP0853

A NOVEL TOOL PERFORMS SIMILARLY TO JNET AMONGST WESTERN EXPERTS IN THE DETECTION OF CANCER WITHIN LARGE NON-PEDUNCULATED COLORECTAL POLYPS: EXPERT VALIDATION OF CRIS

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Introduction: The JNET (Japan NBI Expert Team) classification can be used to predict large non-pedunculated colorectal polyp (LNPCP) histology and presence/depth of submucosal invasion (SMI). Japanese endoscopist accuracies >84% have not been replicated in the West; for example, in a recent study of European experts the accuracy of JNET was 55%. Furthermore, European guidelines suggest both low-grade dysplasia (LGD) and high-grade dysplasia (HGD) can be treated using endoscopic mucosal resection (EMR) thus there is no clinical consequence to interpreting HGD as JNET 2A (excluded in the original interpretation of JNET).

Aims & Methods: We aimed to validate a novel tool CRIS (Colorectal Regular-Irregular Score) by comparing it with JNET for LNPCP histology prediction amongst Western experts.

32 images of flat LNPCPs under narrow-band imaging were consecutively obtained with matching histopathology data. Each image was rated (anonymous online survey) using JNET and then all the images were rated again (order randomised) using CRIS, a tool which simply asks whether the most disordered vascular pattern visible in the image is 'regular(=CRIS-R), regularly-irregular(=CRIS-RI) or completely-irregular(=CRIS-I)'. No further explanation of JNET or CRIS was offered. Accuracy of JNET and CRIS vs. histopathology was analysed using the original JNET interpretation (where JNET 1/2A and CRIS-R = LGD) and a clinically relevant approach (where JNET1/2A and CRIS-R can also = HGD). JNET2B & CRIS-RI were matched to HGD & superficial SMI and JNET3 & CRIS-I to deep SMI. Correct treatment was defined as pEMR for LGD or HGD, en-bloc EMR/ESD for HGD or superficial SMI and surgery for deep SMI.

Results: 12 experts (median 150 colorectal EMRs and 13 ESDs/year) rated 32 images.

The overall accuracy of JNET vs. histopathology using the original JNET interpretation was 69.7% (95%CI) 46.0-89.3%] and CRIS 69.6% (95%CI 45.3-88.7%). Expert agreement was similar between JNET (91.3% (95%CI) 82.8-98.3%) and CRIS (88.5% [95%CI 78.2-97.3%]) with K=0.64 (moderate) for JNET and K=0.57 (moderate) for CRIS.

The accuracy of JNET 2B was 51.7% (95%CI 45.9-57.4%) and JNET 3 85.8% (95%CI 81.2-89.3%). Comparably CRIS-RI and CRIS-I showed accuracies of 51% (95%CI 45.2-56.7%) and 85.1% (95%CI 80.4-88.7%) respectively. Correct treatment was determined in 144/288 (50.0%) observations using JNET and 144 (50.0%) using CRIS. Using the clinically relevant approach the accuracy of JNET was 73.9% (95%CI 46.0-89.3%) vs. CRIS 72.3% (95%CI 45.3-88.7%).

The accuracy, sensitivity and specificity of JNET 1/2A changed by +13.2%, -8.0% and +38.0% and CRIS-R by 8.3%, -8.2% and 35.5%. Correct treatment determination increased by 29.5% to 79.5% for JNET and by 27.1% to 77.1% for CRIS.

	ORIGINAL JNET INTERPRETATION								CLINICALLY RELEVANT INTERPRETATION							
Imaging	JNET 1/2A	CRIS-R	JNET 2B	CRIS-RI	JNET 3	CRIS-I	JNET Overall	CRIS Overall	JNET 1/2A	CRIS-R	JNET 2B	CRIS-RI	JNET 3	CRIS-I	JNET Overall	CRIS Overall
Histology	LGD		HGD, superficial SMI		Deep SMI				LGD, HGD		HGD, superficial SMI		Deep SMI			
Accuracy % (95%CI)	62.5% (56.7 - 67.8%)	63.9% (58.0 – 69.0)	51.7% (45.9 - 57.4)	51.0% (45.2 - 56.7)	85.8% (81.2 - 89.3)	85.1% (80.4 - 88.7)	69.7% (46.0- 89.3)	69.6% (45.3- 88.7)	75.7% (70.4 - 80.2)	72.2% (66.7 - 77.0)	no change				73.9% (46.0- 89.3)	72.3% (45.3- 88.7)
Sensitivity % (95%CI)	77.4% (67.3 - 85.0)	72.6% (62.0 – 81.0)	27.6% (21.1 - 35.0)	28.2% (21.7 - 35.7)	75.0% (61.2 - 85.0)	81.2% (68.0 - 89.8)	53.8% (21.2- 85.1)	54.3% (21.7- 89.8)	69.4% (63.0 - 75.2)	64.4% (57.7 - 70.4)					58.4% (21.2- 85.1)	56.4% (21.7- 89.8)
Specificity % (95% CI)	56.4% (49.5 - 62.9)	60.3% (53.0 – 67.0)	80.3% (72.7 - 86.1)	78.0% (70.2 - 84.2%)	87.9% (83.1 - 91.4)	85.8% (80.8- 89.6)	76.6% (49.5- 91.5)	76.2% (53.4- 89.7)	94.4% (86.5 - 97.8)	95.8% (88.4 - 98.5)					87.4% (72.7- 97.8)	85.8% (70.2- 98.6)
Correct treatment n (%) /288	42.2% (34.3- 50.4	43.0% (34.7- 51.5)	62.3% (49.8- 73.4)	60.3% (48.1- 71.3)	55.4% (42.5- 67.5)	53.4% (41.4- 65.0)	50.0% (44.2- 55.7)	50.0% (44.2- 55.7)	97.4% (93.0- 99.1)	97.9% (93.4- 99.4)					79.5% (74.2- 83.9)	77.1% (71.7- 81.7)

Conclusion: Both JNET and CRIS, a novel score named and described by 3 self-explanatory words, can be used by Western experts to accurately determine LNPCP histology.This suggests JNET may be able to be simplified for non-experts. Importantly the accuracy of both JNET and CRIS can be increased by framing their interpretation versus. histology in a clinically relevant fashion. This has significant implications for signposting patients to correct treatment after detection of LNPCPs especially amongst non-experts.

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PP0854

ENDOSCOPIC CALCIUM-ELECTROPORATION FOR PALLIATIVE THERAPY IN ADVANCED COLORECTAL CANCER: FIRST CASE SERIES FROM THE UNITED KINGDOM

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Introduction: Colorectal cancer(CRC) is the commonest gastrointestinal cancer and the 2nd commonest cause of cancer-related mortality. Approximately 20% are metastatic and/or inoperable at diagnosis, usually requiring palliative treatment which may not be feasible in frail patients. Endoscopic Calcium-electroporation is a less aggressive alternative which uses controlled electric pulses to enhance cellular calcium influx and induce cell death via apoptosis/necrosis as well as pyroptosis.

Aims & Methods: The aim of this study is to determine efficacy and safety of Endoscopic electroporation using calcium in the palliative treatment of advanced CRC in a major UK hospital.

Consenting frail patients with locally symptomatic, inoperable sigmoid colon and rectal cancer were included. Diagnosis and staging were according to standard guidelines while frailty and/or risk was assessed using performance status(PFS), Charlson co-morbidity index(CCI) as well as American Society for Anesthesiology(ASA) score. Calcium Electroporation was administered via out-patient flexible endoscopy at intervals of two to three months with clinical and/or radiological reviews preceding each therapeutic session. Outcomes assessed include symptom relief, quality of life (QoL), survival rates and adverse events (AE).

Results: Fourteen sessions of Endoscopic Calcium-Electroporation were administered to 7 patients over 15months with an average age of 89.8years(85-93years). Luminal symptoms reported included rectal bleeding(N=5), constipation(N=3) and pain(N=2). Five out of seven patients had metastatic disease and none received chemotherapy or other oncologic treatment. The PFS and ASA for all subjects were >3 and 5 respectively while the average CCI was 15(0% estimated 10-year survival).

Symptomatic relief and improved QoL was reported in 71.4% of patients, usually after the 1st treatment session. The average survival rate was 85.7% with a mean life span of 8months (4-15months). No AE was recorded. One patient (who received a single session of treatment) in this cohort died after 11months from progression of the metastatic disease.

Conclusion: Endoscopic Calcium electroporation is a safe and effective palliative option in frail patients with advanced colorectal cancer. It should be considered in patients who are unfit for other conventional oncological therapies.

Disclosure: Nothing to disclose.

PP0855

SNARE TIP SOFT COAGULATION VS ARGON PLASMA COAGULATION AFTER LARGE NON-PEDUNCULATED COLORECTAL POLYP RESECTION: A META-ANALYSIS OF RANDOMIZED TRIALS

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Introduction: Thermal ablative methods (such as argon plasma coagulation (APC) and soft tip snare coagulation (STSC) are commonly used to treat polyp margins.

Aims & Methods: We aim to appraise the current literature and compare clinical outcomes between patients with treated (with APC vs STSC) and non-treated EMR margins.

We searched major databases from inception until November 2023 for RCTs comparing EMR of large non-pedunculated colorectal polyps with and without treated margins. Pooled data was analyzed for the primary outcome of recurrence at first screening colonoscopy, as well as adverse events. Analysis was performed using a random effects model and data was reported using 95% confidence intervals (CI).

Results: A total of five RCTs were found which included 1020 polyps (577 in treatment and 443 in control groups). Three studies included treatment with STSC and three studies used APC as the modality for margin ablation. Of included patients, 53% were female and average age was similar between treatment and control groups (65.9 vs 66.1 years). Overall, 71% of lesions were proximal to the splenic flexure. Mean follow up to first colonoscopy and average polyp size were comparable (6.3 vs 6.2 months; 28.2 mm vs 28.0 mm, respectively). Pooled analysis showed that margin ablation was associated with significantly lower rates of recurrence (OR = 0.267, 95% CI 0.18 - 0.4, P< 0.001) with low heterogeneity between studies (I^2=0%, P = 0.47). Pooled analysis showed no significant difference between STSC and APC in terms of recurrence (OR = 0.6, 95% CI 0.27 - 1.7, I^2=0%, P=0.3) or adverse events (OR = 0.67, 95% CI 0.3 - 1.6, I^2 13%, P = 0.46).

Conclusion: Our study shows that ablation of EMR margins is very effective at preventing recurrence at first surveillance colonoscopy. We found no difference between STSC or APC in terms of polyp recurrence or adverse outcomes.

Disclosure: Nothing to disclose.

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PP0856

COMPARATIVE EFFICACY OF DIFFERENT BOWEL PREPARATIONS FOR COLONOSCOPY: A NETWORK META-ANALYSIS

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Introduction: The quality of a colonoscopy is heavily reliant on the effectiveness of bowel cleansing, which can impact the diagnostic accuracy and identification of adenomas. Various bowel cleansing products have been suggested, but their comparative effectiveness remains uncertain.

Aims & Methods: This systematic review and network meta-analysis aims to compare the performance of different bowel preparations for colonoscopy. MEDLINE, Embase, Scopus, and the Cochrane Library were systematically searched for randomized controlled trials (RCTs) comparing the efficacy of different bowel preparations administered as split or same-day regimen. The primary outcome was cleansing success (CS) defined as a total BBPS score of ≥6. The secondary outcome was ADR.

Results: On network meta-analysis for CS (22 RCTs, 7179 patients, 14 bowel preparations assessed), 2L PEG+simethicone (RR=1.25 [95%CI=1.13-1.37]), 2L PEG+lactulose (RR=1.22 [95%CI=1.10-1.38]) and 1L PEG+ascorbate (ASC) (RR=1.03 [95%CI=1.01-1.06]) were significantly superior to 2L PEG+ASC. Overall, 2L PEG+lactulose resulted as the best product (SUCRA 0.94), followed by 2L PEG+simethicone (SUCRA 0.93), whereas 2L PEG+ASC showed the poorest performance. On network meta-analysis for ADR (17 RCTs, 6639 patients, 11 bowel preparations assessed), only 2L PEG+simethicone (RR=1.60 [95%CI=1.05-2.43]) resulted significantly superior to 2L PEG + ASC.

Conclusion: 2L PEG+simethicone, 2L PEG+lactulose, and 1L PEG+ASC seemed to provide high rates of cleansing success, albeit only 2L PEG+simethicone was associated with a significantly higher ADR. Consequently, these products should be preferred for bowel preparation of colonoscopy. Further randomized studies with adequate sample sizes are needed for a more accurate comparison of these products on ADR.

Disclosure: Nothing to disclose.

PP0857

TRENDS OF THE POLYP, ADENOMA, AND SESSILE SERRATED DETECTION RATE BY DIFFERENT AGE GROUPES IN INDIVIDUALS UNDERGOING COLONOSCOPY 2021–2023

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Introduction: Colonoscopy plays a key role in colorectal cancer (CRC) screening. It is effective in reducing morbidity and mortality from CRC. To optimize quality of colonoscopy, performance measures (PM) have been identified and published by major international societies. While adenoma detection rate (ADR) is an imperative quality measure the sessile serrated lesion detection rate (SSLDR) is not currently a quality indicator for screening colonoscopy.

This study aimed to determine the trend of polyp detection rate (PDR), ADR, and SSLDR in participants in different age groups.

Aims & Methods: We performed a retrospective observational study of patients who underwent colonoscopy at Olymed, Kyiv, Ukraine from April 2021 through December 2023. All data were extracted and collected anonymously from the colonoscopy database and pathology reports and further analyzed. We divided patients by the different age groups and calculated PDR, ADR, and SSLDR for each group. PDR, ADR, and SSLDR were defined as the number of colonoscopies with at least one polyp, adenoma, and sessile serrated lesion detected respectively, divided by the total number of complete colonoscopies performed.

Results: This study was performed initially on 2676 patients who underwent colonoscopy from April 2021 through December 2023. Colonoscopies were mainly performed with colonoscopes with magnification and artificial intelligence assistance. After the exclusion of colonoscopies where examinations weren't extended to the cecum or Boston Bowel Preparation Scale < 6 the final study dataset consisted of 2402 colonoscopies. The patients were divided into 6 main groups and considering the discussion about lowering the screening age for CRC 2 additional groups were added. Results of PDR, ADR, and SSLDR for each group are shown in Table 1.

Age	Patients n	PDR	ADR	SSLDR
18-29	170	24.7%	7.7%	10.0%
30-39	422	43.6%	22.5%	18.0%
40-49	594	63.3%	40.0%	27.3%
50-59	605	75.9%	59.2%	27.9%
60-69	432	84.7%	70.6%	28.0%
≥70	179	86.0%	79.9%	18.4%
45-49	323	66.8%	44.9%	26.7%
≥50	1216	80.9%	66.3%	26.6%

Table 1.

Conclusion: This study reveals a notable trend of the PDR, ADR, and SSLDR in different age groups during colonoscopy. While ADR and PDR show an increase with patient age, a similar pattern is not observed with SSLDR. This finding underscores the importance of recognizing SSLs, particularly in younger adults, as they appear to be more common than previously assumed. Further investigation into the role of SSLs in the development of CRC, particularly among younger adults, is warranted. Additionally, there is a need to establish the ideal SSDR metric and define a target for endoscopists to meet.

By advancing our understanding of SSLs and their implications for CRC risk, we can improve screening and prevention strategies to reduce the burden of colorectal cancer in the population.

Disclosure: Nothing to disclose.

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PP0858

DURATION-DEPENDENT RISK OF HYPOXEMIA IN COLONOSCOPY PROCEDURES

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Introduction: Colonoscopy is a critical diagnostic and therapeutic procedure in gastroenterology. However, it carries risks, including hypoxemia, which can impact patient safety. Understanding the factors that contribute to the incidence of severe hypoxemia, specifically the role of procedure duration, is essential for improving patient outcomes.

Aims & Methods: This study aims to elucidate the relationship between the length of colonoscopy procedures and the occurrence of severe hypoxemia. We conducted a retrospective cohort study at Sheba Medical Center, Israel, including 21,524 adult patients who underwent colonoscopy from January 2020 to January 2024. The study focused on the incidence of severe hypoxemia, defined as a drop in oxygen saturation below 90%. Sedation protocols, involving a combination of Fentanyl, Midazolam, and Propofol, were personalized based on the endoscopist's discretion. Data were collected from electronic health records, covering patient demographics, clinical scores, sedation and procedure details, and outcomes. Statistical analyses, including logistic regression, were used to examine the association between procedure duration and hypoxemia, adjusting for various patient and procedural factors.

Results: The study found a significant correlation between colonoscopy duration and the risk of hypoxemia. Short procedures (under 20 minutes) had a 17.3% incidence of hypoxemia, which increased to 36.1% for procedures lasting over 60 minutes. Age and procedure duration were significant predictors of hypoxemia, with older patients and longer procedures associated with higher risks. The analysis highlighted the need for careful sedation management, especially in longer procedures.

	Duration <20 min	Duration 20-40 min	Duration 40-60 min	Duration ≥60 min	p-value
Age (years)	Median 56.0, IQR (42.4 - 68.1), Mean 54.1, SD ± 17.8	Median 62.7, IQR (50.9 - 71.9), Mean 59.8, SD ± 16.0	Median 68.6, IQR (59.4 - 74.8), Mean 66.1, SD ± 12.9	Median 67.0, IQR (56.5 - 74.9), Mean 64.4, SD ± 14.9	<0.001
Sex (n, %)	4814, (54.1%)	5446, (52.8%)	1001, (57.8%)	343, (59.2%)	<0.001
Inpatient (n, %)	797, (9.0%)	832, (8.1%)	177, (10.2%)	61, (10.5%)	<0.001
ASA Score	Median 2.0, IQR (2.0 - 2.0), Mean 1.9, SD ± 0.6	Median 2.0, IQR (2.0 - 2.0), Mean 2.0, SD ± 0.5	Median 2.0, IQR (2.0 - 2.0), Mean 2.1, SD ± 0.5	Median 2.0, IQR (2.0 - 2.0), Mean 2.1, SD ± 0.6	<0.001
Mallampati Score	Median 2.0, IQR (2.0 - 2.0), Mean 1.8, SD ± 0.4	Median 2.0, IQR (2.0 - 2.0), Mean 1.8, SD ± 0.4	Median 2.0, IQR (2.0 - 2.0), Mean 1.8, SD ± 0.4	Median 2.0, IQR (2.0 - 2.0), Mean 1.9, SD ± 0.4	<0.001
Midazolam (mg)	Median 2.0, IQR (2.0 - 2.0), Mean 1.9, SD ± 0.6	Median 2.0, IQR (2.0 - 2.0), Mean 1.9, SD ± 0.5	Median 2.0, IQR (2.0 - 2.0), Mean 1.9, SD ± 0.5	Median 2.0, IQR (2.0 - 2.0), Mean 1.9, SD ± 0.6	<0.001
Propofol (mg)	Median 200.0, IQR (100.0 - 200.0), Mean 185.9, SD ± 95.8	Median 200.0, IQR (160.0 - 300.0), Mean 229.9, SD ± 118.2	Median 300.0, IQR (200.0 - 400.0), Mean 300.6, SD ± 163.8	Median 300.0, IQR (200.0 - 500.0), Mean 380.3, SD ± 259.1	<0.001
Fentanyl (mcg)	Median 50.0, IQR (0.0 - 50.0), Mean 38.9, SD ± 28.2	Median 50.0, IQR (50.0 - 50.0), Mean 44.0, SD ± 26.4	Median 50.0, IQR (50.0 - 50.0), Mean 47.4, SD ± 30.0	Median 50.0, IQR (50.0 - 50.0), Mean 45.8, SD ± 30.2	<0.001
SatO2 < 90%	1540, (17.3%)	2494, (24.2%)	562, (32.4%)	209, (36.1%)	<0.001

Table 1- patient characteristics and outcomes based on the colonoscopy duration.

Conclusion: The duration of colonoscopy procedures significantly impacts the risk of severe hypoxemia, with longer durations associated with higher risks. This study underscores the importance of optimizing procedural efficiency and tailoring sedation protocols to individual patient risk profiles to enhance the safety of colonoscopy. Further research is needed to develop strategies that minimize procedure duration without compromising the quality of care, thereby reducing the risk of hypoxemia and improving patient safety.

Disclosure: Nothing to disclose.

PP0859

ASSESSING THE IMPACT OF BOWEL PREPARATION QUALITY ON COLORECTAL CANCER SCREENING EFFECTIVENESS: A STUDY WITH UNEXPECTED RESULTS

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Introduction: The diagnostic accuracy of colorectal cancer (CRC) screening via colonoscopy is influenced by the quality of bowel preparation. The adenoma detection rate is another well-known quality measure for CRC screening colonoscopies. Counting the total number of adenomas identified during each colonoscopy can provide a more comprehensive assessment of the procedure's effectiveness. Adenomas that are 1 cm or larger, exhibit high-grade dysplasia, or have a villous architecture are considered advanced adenomas and carry a higher risk of progressing to adenocarcinoma.

Aims & Methods: The aims of our study were to assess whether there was an association between the adequacy of bowel preparation and both adenoma detection rate and the advanced adenoma detection rate, and to investigate whether there was a correlation between the quality of preparation and the number of adenomas detected per colonoscopy. We present a retrospective, single-center study of complete screening colonoscopies performed on patients with 50 years old or more, over 18 months (from April 2022 to September 2023), and their respective histological reports. Adequate preparation was considered if all colonic segments scored 2 or more points on the *Boston Bowel Preparation Scale* (BBPS). Statistical analysis was conducted using SPSS v29.0.

Results: 672 total colonoscopies were analyzed. The average age was 64.5 ± 8.0 years, and there was no predominance between genders. BBPS scores ranged from 0 to 9. The majority of patients had adequate bowel preparation (74%, n = 502), with BBPS scores of 6, 7, 8, and 9 in 37.7%, 7.0%, 22.9%, and 7% of cases, respectively. The adenoma detection rate was 41.2% and the detection rate of advanced adenomas was 16.4%. The number of adenomas detected per colonoscopy ranged from 0 to 11 adenomas. No association was found between having adequate preparation and having at least 1 adenoma (p>0.05) or having at least 1 advanced adenoma (p>0.05). There was no statistically significant difference in the average number of adenomas per colonoscopy between patients with and without adequate preparation (p>0.05). There was no correlation between the BBPS score and the number of adenomas per colonoscopy (p>0.05).

Conclusion: The detection rates of adenomas and advanced adenomas were not associated with the adequacy of bowel preparation. No correlation was found between the quality of preparation and the number of adenomas per colonoscopy. Despite the need to repeat another colonoscopy with adequate BBPS for optimal observation of the mucosa, concluding screening procedures in patients with inadequate bowel preparation has still proven to be useful. As long as it is considered safe, endoscopists should be encouraged to conclude all screening procedures.

Disclosure: Nothing to disclose.

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PP0860

ENHANCING THE ADENOMA DETECTION RATE IN COLORECTAL CANCER SCREENING: ADENOMAS PER COLONOSCOPY AS A QUALITY METRIC

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Introduction: Adenomas per Colonoscopy (APC) might serve as an adjuvant quality metric to the Adenoma Detection Rate (ADR). This indicator is calculated by dividing the total number of detected adenomas by the number of colonoscopies performed. It motivates the operator to maximize adenoma identification and removal during screening procedures and allows a more comprehensive assessment of the colonoscopy's quality.

Aims & Methods: The aims of this study were to assess the association between the ADR and the APC metrics in colorectal cancer screening and to explore factors influenciating each of these indicators. This retrospective, single-center study analyzed total screening colonoscopies conducted on 50 years old or more patients, over a period of 18 months (from April 2022 to September 2023) and their corresponding histological reports. Colonoscopies with at least one colonic segment with less than 2 points on the *Boston Bowel Preparation Scale* (BBPs) were excluded. Statistical analysis was performed using SPSS v29.0.

Results: 502 total colonoscopies were included. ADR was 41.7% and APC was 76.5%. The number of adenomas per colonoscopy ranged from 0 to 11 adenomas. There was a positive statistically significant association between the ADR and the APC (β=0.6, $p<0.001$). No statistically significant difference was found in the average age between patients with and without adenomas ($p>0.05$), nor a correlation between age and the number of APC ($p>0.05$). Men were 2.1 times more likely to have at least one adenoma ($p<0.05$) and the average number of adenomas per colonoscopy was statistically significant higher in men ($p<0.05$). The BBPS score was not associated with the detection of at least one adenoma or to the total number of adenomas per colonoscopy ($p>0.05$). The average withdrawal time was higher in patients with adenomas ($p<0.05$), and there was a positive, statistically significant, moderate correlation between withdrawal time and the number of APC ($r=0.5$; $p<0.001$). No association was found between the colonoscopy schedule, endoscopist fatigue or experience and the presence of at least one adenoma ($p>0.05$). The average number of adenomas per colonoscopy was also not statistically significantly different among the previously mentioned groups ($p>0.05$).

Conclusion: Our study confirmed a significant association between the ADR and the APC, encouraging the use of APC as a complementary metric in colorectal cancer screening. Influencing factors were similar between the two indicators. While gender and withdrawal time significantly influenced adenoma detection and the number of adenomas detected per colonoscopy, with men and longer procedures showing higher ADR and APC rates, other factors such as age, bowel preparation quality, and procedural variables showed no significant impact.

Disclosure: Nothing to disclose.

PP0861

CAN ARTIFICIAL INTELLIGENCE PREDICT HYPERPLASTIC POLYP HISTOLOGICAL SUBCLASSES BASED ON ENDOSCOPIC APPEARANCE?

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Introduction: Recent insights in polyp molecular biology have suggested that hyperplastic polyps (HP) are closely related to sessile-serrated lesions (SSL). Their hallmark serrated crypts closely resemble those found in SSL. Furthermore, similar to SSL, specific subtypes of HP can harbor BRAFV600E-mutations. These findings, amongst others, have raised the hypothesis that some HP may have an intrinsic malignant potential through the serrated neoplasia pathway. However in everyday clinical practice, no routine HP subdifferentiation is performed.

Aims & Methods: In this single-center trial, we aimed to develop a deep learning algorithm capable of differentiating between the subclasses of hyperplastic polyps, namely microvesicular hyperplastic polyps (MVHP) and goblet cell-rich hyperplastic polyps (GCHP), based on their endoscopic appearance. We retrospectively selected 198 hyperplastic polyps in 127 patients. All polyps had high quality endoscopic video available in white light endoscopy and virtual chromoendoscopy (i-scan) using Pentax EC38-i20c and EC34-i10 endoscopes. We reviewed all histological slides with an expert pathologist and updated their pathology diagnosis using the MVHP/GCHP subdifferentiation. Of the total 198 HP, 144 were GCHP, 26 were MVHP and 28 were mixed GCHP/MVHP. We selected the polyps without mixed constitution for development of a deep learning algorithm. These 170 polyps were randomly split into 70% for training, 10% for validation and 20% for testing. Sequences of still images of the polyps were manually annotated and subsequently used to train a deep learning algorithm. The imbalance in the training dataset was adjusted by randomly oversampling the MVHP.

Results: We trained a deep learning algorithm using a convolutional neural network on 170 HP (144 GCHP and 26 MVHP). This model can differentiate between MVHP and GCHP based on still endoscopic images with an AUC of 0.743, corresponding to a sensitivity of 70% for MVHP. However due to the low prevalence of MVHP (14%) compared to GCHP (86%) the positive predictive value for MVHP is only 24.1%. The specificity (for GCHP) is 64.2%

	Predicted - MVHP (+)	Predicted - GCHP (-)
True - MVHP (+)	14	6
True - GCHP (-)	44	79

Conclusion: Our preliminary results show that a deep learning algorithm is capable of detecting MVHP with a good sensitivity (70%) based on their endoscopic appearance. However, due to the low prevalence of MVHP compared to GCHP, the positive predictive value for MVHP is only 24.1%. Moreover, the current algorithm has some difficulties correctly identifying the GCHP, as is shown by the rather low specificity of 64.2%. We plan to further expand the dataset to mitigate these issues, however this pilot study serves as a proof of concept that artificial intelligence is capable of predicting hyperplastic polyp subclasses based on endoscopic appearance.

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PP0862

FEW-SHOT LEARNING FOR CLASSIFICATION OF COLORECTAL NEUROENDOCRINE TUMORS AND POLYPS ON ENDOSCOPIC IMAGES

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Introduction: Colorectal neuroendocrine tumors (NETs) and polyps have similar endoscopic manifestations and are easily misidentified. Due to the low incidence of NETs, it is difficult to obtain adequate images to fit deep-learning models. This study introduced a few-shot learning (FSL) model for the ternary classification of NETs and polyps on endoscopic images.

Aims & Methods: Three classes of endoscopic images (56 serrated lesions and polyps [SLPs], 86 traditional adenomas and 53 NETs) were collected from three datasets. First, three images from each class were randomly selected for Support Set (SS), while others for Query Set (QS). Second, ResNet50 V2 was pretrained based on ImageNet and then on esophageal endoscopic images Third, imaging eigenvectors were extracted based on a dual-pretrained model. Fourth, the Euclidean distances to the nine SS images were calculated for each QS image. Finally, a k-nearest neighbor classifier was used for classification based on Euclidean distances. The proposed FSL model was tested in three rounds.

Results: The performance of the FSL model (macro-area under the curve [AUC], 0.731; macro F1-score, 0.674; Matthews Correlation Coefficient [MCC], 0.526; Kappa, 0.523) was better than that of the traditional ResNet50 V2, junior endoscopist, and senior endoscopist. When identifying NETs, the proposed FSL model achieved the highest accuracy 0.823, sensitivity 0.653, precision 0.673 and F1-score 0.659, whereas the senior endoscopist achieved the highest specificity 0.904.

Conclusion: This study presented a FSL method based on transfer learning and metric learning to distinguish between SLPs, adenoma, and NETs on endoscopic images. This suggests that FSL might be an effective method for computer-aided diagnosis of rare diseases.

Disclosure: Nothing to disclose.

PP0863

IS THERE AN IMPACT OF DEEP SEDATION WITH PROPOFOL ON ADENOMA DETECTION RATE, POLYP DETECTION RATE, AND CECAL INTUBATION RATE IN ELDERLY PATIENTS?

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Introduction: Early detection of colon adenomas and polyps lowers the risk of colorectal cancer development. Quality indicators for screening or diagnostic colonoscopy encompass the adenoma detection rate (ADR), polyp detection rate (PDR), and cecal intubation rate (CIR). Previous research on ADR, PDR, and CIR in groups of patients who were deeply sedated with propofol versus those who underwent colonoscopy without sedation is undetermined.

Aims & Methods: The objective of this study was to analyze the effects of deep sedation versus no sedation on ADR, PDR, and CIR in a population of elderly patients.

This was a retrospective cohort study that included adult patients over the age of 60, presenting for a first screening or diagnostic colonoscopy performed in the period from January 2017 to September 2023 at our clinical center. The exclusion criteria were emergency colonoscopies, inflammatory bowel diseases, intent therapeutic colonoscopies, and bowel preparation quality (based on the Boston Bowel Preparation Score, BBPS) of less than 6. Participants' characteristics were assessed using descriptive statistics. Normal distribution was assessed using Kolmogorov-Smirnov and Shapiro-Wilk tests. Continuous variables were analyzed using Mann–Whitney test and categorical variables using Chi-square or Fisher exact test. Binary logistic regression was conducted to identify significant predictors of the outcomes. Two-tailed tests were conducted, and statistical significance was considered when $p < 0.05$.

Results: Among 2034 patients (46.4% female; mean age: 70 years) deep sedation was performed in 622 patients (30.6%). Total PDR was 51.65%, ADR was 33.3%, and CIR was 94.25%. Patients' characteristics are shown in table 1.

After adjusting for confounders (age, gender, BMI, BBPS for PDR/ADR and age, gender, BMI, BBPS, previous abdominal/pelvic operation, diverticulosis for CIR) no significant difference was shown in PDR, ADR and CIR between no-sedation and deep sedation groups. Higher BMI (B = .041, p < .01) and male gender (B = .36, p = .02) were independent predictors for ADR.

	No-sedation (N=1412)	Deep sedation with propofol (N=622)
Age*	70 (65-76)	68 (64-73)
Female gender *	655 (46,4%)	359 (57,7%)
BMI	27 (24,3 - 30,5)	27,1 (23,7 - 29,9)
BBPS	9 (6-9)	9 (6-9)
PDR	731 (51,8%)	320 (51,5%)
ADR	472 (33,5%)	202 (32,5%)
CIR	1316 (93,2%)	593 (95,3%)
Abdominal/pelvic operation	488 (34,6%)	186 (29,9)
Diverticulosis *	523 (37,1%)	272 (43,8%)

*Table 1; Patient s' characteristics, BMI; Body mass index, BBPS; Boston bowel preparation scale, PDR; Polyp detection rate, ADR; Adenoma detection rate, CIR; Cecal intubation rate, *gender $\chi2(3, N=2034)=27,37$, $p<0.01$, *diverticulosis $\chi2(1, N=2031)=8,14$, $p<0.01$, *age $U=367989$, $z=-5,78$; $p<0,01$, PDR: $\chi2(1, N=2033)=0,01$, $p=.92$, ADR: $\chi2(1, N=2031)=0,17$, $p=.68$*

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Conclusion: In our mixed cohort of elderly patients undergoing screening and diagnostic colonoscopies, propofol-induced deep sedation did not enhance ADR, PDR, or CIR. Further research is needed to determine the impact of deep sedation on colonoscopy quality parameters in elderly patients.

Disclosure: Nothing to disclose.

PP0864

THE IMPORTANCE OF KNOWING WHAT IS INSIDE THE BOX: A CASE STUDY OF WHAT MIGHT GO WRONG IN ARTIFICIAL INTELLIGENCE TRAINING IN ENDOSCOPY

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Introduction: The use of artificial intelligence (AI) in colonoscopy has already been shown enable a higher polyp detection and characterization. New applications are also being developed at an increasingly high pace, but there are potential pitfalls that might appear when developing new AI methods.

Aims & Methods: The aim of this case study is to show potential pitfalls that can arise during the training of new AI models, most importantly on smaller datasets. We aim to illustrate the hidden pitfalls that might arise during development, aside from known problems such as dataset imbalance and overfitting.

This case study is based on the development of an AI network for differentiating between hyperplastic polyp (HP) subclasses using colonoscopy imagery. The original primary outcome was to determine the feasibility of such a setup, while secondary outcome was to determine the effectiveness of such a model. Finally, the tertiary outcome was to use explainable AI techniques on the trained model to see which elements were picked up during training to correctly identify both HP subclasses. The specific technique used is the SHAP technique [1], which highlights the areas in an image used for the classification.

We retrospectively collected images of 198 polyps with histological assessment of hyperplastic polyps from different colonoscopy procedures. These polyps in these images were manually segmented and the histological slices were re-examined for further subclassification into goblet cell-rich hyperplastic polyps (GCHP) and microvesicular hyperplastic polyps (MVHP).

The dataset consists of 144 GCHP, 28 MVHP and 28 mixed MVHP/GCHP polyps, of which only the unmixed polyps were included in this study. The polyps were cropped out of the image with a margin of 50% to take the mucosa immediately surrounding the polyp into account. The total dataset was randomly split into training (70%), validation (10%) and test set (20%). Finally, the imbalanced training and validation datasets were balanced by randomly oversampling the images of MVHP.

Results: We obtained good results for this model, with sensitivity of 90% and 69,9% for MVHP and GCHP respectively and the area under the curve (AUC) was 86,6%. These results can be successfully presented and published, establishing our original primary and secondary objectives. The problem is that when checking the model with explainable AI techniques to achieve our tertiary objective, we noticed a problem. For MVHP, the features in the image that were used, were usually either be the pit pattern or the existence of mucus. The images that were classified as GCHP seemed to be decided using the surrounding mucosa. The problem is that around 40% of the images classified as GCHP seemed to be selected using the black edges of the scope recording.

It seems that during training there were only black edges visible in training samples of GCHP, making the model assume that all samples with black edges are GCHP, which is obviously a wrong assumption. A simple solution such as randomly adding black edges in all samples solves this problem, correcting the result to a sensitivity of 70% and 64,2% for MVHP and GCHP respectively and with an AUC of 74,3%, which is slightly lower, but does not have the built-in problem.

Conclusion: In preliminary stages of AI development, it is always important to verify achieved results if possible. A model will try to capture features using all the available training data, but uncorrelated features might exist unintentionally. A good verification method is to open the black box-like AI model using explainable AI techniques.

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Disclosure: Nothing to disclose.

PP0865

IMPACT OF A COMBINATION REGIMEN OF JIAWEI DACHENGQI DECOCTION AND SHUTAIQING ON INTESTINAL CLEANNESS AND POLYP DETECTION RATE

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Introduction: The objective of this study was to compare and analyze the effects of a combination of "*Jiawei Dachengqi* decoction and *Shutaiqing*" and "*Shutaiqing*" regimens on intestinal preparation, intestinal cleanness, and polyp detection rate in a randomized controlled study.

Aims & Methods: The study included 1,000 patients who came to the hospital for electronic colonoscopy and were randomly divided into two equal groups. Patients in the experimental group received a combination regimen of "*Jiawei Dachengqi* decoction and *Shutaiqing*" while patients in the control group received only "*Shutaiqing*". We compared the intestinal preparation and rates of polyp detection in the two groups.

Results: The quality of intestinal preparation (U = 71254.000, P < 0.001), pain assessment (U = 53340.000, P < 0.001), and adverse event assessment (U = 56902.000, P < 0.001) were significantly better than those of the control group. The rate of polyp detection in the experimental group (298/500 cases, 59.60%) was significantly higher than that in the control group (148/500 cases, 29.60%) (χ^2=91.062, P < 0.001). Of those, the overall detection rates of inflammatory polyps, tubular adenomas, and tubular adenomas with low-grade intraepithelial neoplasia were significantly higher than those in the control group (P < 0.05). The polyp detection rate also significantly differed with different degrees of intestinal cleanness (χ^2 = 153.153, P < 0.001). Compared with the control group, the rates of polyp detection in participants with intestinal cleanness of 8–9 points (χ^2 = 9.191, P = 0.002) and 6–7 points (χ^2 = 15.723, P < 0.001) were significantly higher.

Conclusion: The "*Jiawei Dachengqi* decoction combined with *Shutaiqing*" regimen was effective in reducing the dose of polyethylene glycol, improving the quality of intestinal preparation, effectively lowering the adverse

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events and sensory evaluation of pain in participants, as well as enhancing the rate of polyp detection.

Disclosure: Nothing to disclose.

PP0866

CURRENT REALITIES OF SCREENING COLONOSCOPY IN UKRAINE: CHALLENGES AND OPPORTUNITIES

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Introduction: Colonoscopy plays a key role in colorectal cancer (CRC) screening. It is effective in reducing morbidity and mortality from CRC, which continues to be diagnosed at advanced stages in Ukraine.

This research evaluates the challenges faced by endoscopists in Ukraine and suggests solutions to improve screening colonoscopy in Ukraine.

Aims & Methods: Semi-structured interviews were conducted with Ukrainian endoscopists regarding screening colonoscopy. Interview questions were first pilot-tested by a Ukrainian endoscopist and refined by a qualitative expert. Sampling continued until data saturation was achieved.

The interviews were recorded and transcribed. Rapid qualitative analysis was conducted. Two researchers independently coded the data and identified common themes. Interviews were conducted in Ukrainian and translated into English.

Results: Six endoscopists were interviewed. Six themes were developed: quality of endoscopy training; equipment and consumables; performance measures recording; cooperation with National Health Services of Ukraine (NHSU); collaboration with specialists in related fields and recommendations for improving quality of colonoscopy in Ukraine.

Most endoscopists cite poor quality of endoscopy training, insufficient provision of equipment and consumables, problems with collaboration between specialists. The top recommendations they cite for improvement of screening colonoscopy include equipment upgrade and consumables provision, the importance of structured training and coordination between related specialists. The most representative quotes for each theme are shown in Table 1.

Conclusion: These results summarize challenges and suggest solutions to improve screening colonoscopy in Ukraine, emphasizing the importance of structured training, equipment and consumables provision, co-operation with specialists in related fields, and the development of funding programs.

These results are important in establishing an effective country specific screening program.

Disclosure: Nothing to disclose.

Theme	Subtheme	Representative quote	Representative quote
Quality of endoscopy training		"The training methods have already changed: now they learn through videos, on mannequins with artificial intestines. Now, it's feasible to shift the approach of training young professionals. Establishing training centers with a modern curriculum: starting from diagnostics and progressing to advanced procedures." - Endoscopist 1.	„This is a very painful issue. I've done half one colonoscopy and zero gastroscopies. It's like I've gone through specialization, but I can't do anything." - Endoscopist 5. „It was a disappointment both in terms of the training provided and in terms of how the work was executed." - Endoscopist 6.
Equipment and consumables	Equipment	"When it comes to equipment, I'd say I've been quite fortunate." - Endoscopist 6. "We are waiting for new equipment to be provided." - Endoscopist 2.	" The major challenge lies in having just one colonoscope. Everything is done through our own efforts, aiming for the department not to resemble the Soviet times." - Endoscopist 5. "Something Happened With The Internet Connection: The Electricity Went Out." - Endoscopist 6.
	Consumables	"We have all the necessary equipment for colon examinations." - Endoscopist 1. "I switched to disposable instrumentation a long time ago and always discuss this with the patient." - Endoscopist 4.	"When it comes to consumables, compromises sometimes have to be made." - Endoscopist 6.
	Equipment processing	"There are no difficulties." - Endoscopist 4. "We have processing machines." - Endoscopist 1.	"We have manual processing. It takes a lot of time. We hope for an update on the processing equipment." - Endoscopist 2.
Performance measures recording			" I do not document. We have such a large flow of patients that we lack the strength and motivation to handle it. The motivation to begin documentation would arise if it was mandatory for everyone." Endoscopist 4.
Relationship with National Health Services of Ukraine (NHSU)		"Yes, we are contracted. Everything is fine because there is a registrar who takes care of all this." - Endoscopist 5.	"Payments do not cover everything. „Do more - pay less." And it should be: „Pay more and do the right thing."" - Endoscopist 1. „If we had adequate funding, most of the problems would be solved. There is also a question about coding - we need to put the name in order" - Endoscopist 4.
Relationship with specialists in related fields		" Our doctors are pure love. There is understanding and feedback. Pathologists are wonderful. There is a team of anesthesiologists working with us." - Endoscopist 2. "As for the pathologists, until we started going to each other and seeing - there were problems. Working together - there are no questions." - Endoscopist 1. "Surgeons have confidence in me, I have confidence in what they do." - Endoscopist 6.	"The main problem is primary health care physicians who do not understand the purpose of CRC screening." - Endoscopist 1. "A complete disaster with histology." - Endoscopist 5. "Until now, for surgeons, endoscopists have remained the ones who have to go in and see if there is something there or not." - Endoscopist 5
Recommendations for improving colonoscopy quality in Ukraine.		"For development at the European level, coordinated actions of the state and associations, funding, and the development of funding programs are needed. If everyone acts separately, it won't be nationwide." - Endoscopist 1.	"For the development of any field of medicine, the doctor's motivation is important". - Endoscopist 6. "New equipment, training of doctors, promotion of endoscopy are needed." - Endoscopist 3.

Table 1.

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PP0867

PREDICTING COLONIC PREPARATION: EVALUATING THE LAST BOWEL MOVEMENT APPEARANCE, BODY MASS INDEX, DIABETES, AND DIET DURATION

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Introduction: Adequate colonic preparation is crucial for successful colonoscopy, ensuring optimal visualization of the colon and accurate diagnosis of potential abnormalities. Various factors have been implicated in influencing bowel preparation quality, including patient characteristics, dietary habits, and the chosen preparation regimen.

Aims & Methods: This study investigates factors influencing colonic preparation quality for colonoscopy, focusing on the last bowel movement appearance, body mass index (BMI), diabetes mellitus, diet duration, and age.

We prospectively analyzed data from 158 patients undergoing colonoscopy between August and September 2023. Bowel preparation quality was assessed using a pre-endoscopy visual aid based on the color of the patient's last bowel movement an hour before the procedure. Additionally, we compared the 1-day vs. 4-day bowel preparation regimens.

Results:

The pre-endoscopy visual aid effectively predicted adequate bowel cleansing. Patients with yellow and clear stools like urine (YCSU) or light orange and almost clear stools (LOAC) had significantly higher rates of successful bowel preparation (98.4%) compared to those with darker or murkier stools.

Increasing BMI values did not correlate with higher proportions of inadequate bowel preparation. Obese patients achieved successful bowel preparation at a similar rate (81.48%) to non-obese patients (76%).

Diabetic patients had significantly lower successful bowel preparation rates (55.5%) compared to non-diabetic patients (80.3%). However, cecal intubation rates were similar between both groups.

Diet duration significantly impacted colonic preparation. Patients on a 4-day regimen achieved a higher rate of successful preparation compared to those on a 1-day regimen (p=0.042).

Age was significantly associated with non-prepared colonoscopies (p=0.033). Patients with a BBPS score below 6 were more likely to be older than 70 years old.

Conclusion: This study highlights the effectiveness of the pre-endoscopy visual aid in predicting bowel preparation adequacy. While diabetes adversely affects bowel preparation success, BMI and age above 70 years old are also associated with higher rates of inadequate preparation. Extending diet duration from 1 to 4 days significantly improves colonic preparation quality.

Disclosure: Nothing to disclose.

PP0868

EFFICACY AND SAFETY OF A NOVEL FULLY SYNTHETIC AND SELF-ASSEMBLING PEPTIDE SOLUTION FOR SUBMUCOSAL INJECTION IN ENDOSCOPIC MUCOSAL RESECTION OF COLORECTAL TUMORS: A MULTICENTER RETROSPECTIVE OBSERVATIONAL STUDY

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Introduction: Submucosal injection solutions create a mucosal cushion, which provides sufficient separation between the muscle layers and resection sites. It decreases the risk of perforation, without interfering with histological evaluation. In Japan, several submucosal injection solutions, including normal saline, hyaluronate solution, and sodium alginate, have been used during endoscopic resection procedures. A novel fully synthetic and self-assembling peptide solution, PuraLift (3-DMatrix, Tokyo, Japan), is a non-biologic preparation that self-assembles to form a gel composed of nanofibers upon contact with a neutral pH or the response to inorganic salts such Na^+ and K^+.

Aims & Methods: In the present study, we aimed to evaluate the efficacy and safety of PuraLift in endoscopic mucosal resections (EMRs) of colorectal neoplasms. This study included 70 patients with 110 neoplastic lesions, measuring 5–20 mm in the colorectal mucosa, from five referral hospitals in Japan, between January and May 2023, who underwent EMR with PuraLift. This study was conducted by board-certified fellows of the Japan Gastroenterological Endoscopy Society (48.2%) and the others. The primary outcome measures were included complete resection rate (en-bloc resection with a histopathologically negative resection margin) and maintenance of mucosal lesion lifting during endoscopic resection (calculated as the number of additional injections required due to the loss of mucosal lesion lifting). The secondary outcome was safety, evaluated by analyzing adverse events during the study period.

Results: The median age and sex ratio (male/female) of the patients were 67 (38–86) years and 43/27, respectively, and 14 patients (20%) were treated with antithrombotic agents. Among the lesions, 29 (26.4 %) were laterally spreading tumors, and the remaining were protruding lesions. The mean tumor size was 9.45 ± 3.26 mm (range: 5–20 mm).

Only two patients required an additional injection due to the loss of mucosal lesion lifting, and the mean injection volume was 2.01 (±2.46) mL. En-bloc resection was performed for 109 lesions (99.1%). Eight lesions (7.3%) showed inconclusive horizontal margins on histopathological evaluation. No cases of bleeding, perforation, or other accidental intraoperative event was observed. Evaluation of the ease of EMR (injection and snaring) with PuraLift revealed easy or normal results for 99.1% (109/110) of the lesions.

Conclusion: This novel submucosal injection material enabled sufficient lifting of the colorectal intramucosal lesion during EMR, reducing the need for additional injections and total amount of local injections. Furthermore, this solution contains the same ingredients as the hemostatic material PuraStat (3-DMatrix); therefore, it may have a perioperative hemostatic effect and promote wound healing, contributing to the reduction of complications.

Disclosure: Nothing to disclose.

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PP0869

EVALUATION OF CECAL WITHDRAWAL TIMER IN SCREENING COLONOSCOPY TESTS - A POTENTIAL TOOL FOR ENHANCING WITHDRAWAL TIME AND ADENOMA DETECTION RATE

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Introduction: Effective colorectal cancer prevention relies on the quality of screening colonoscopy examinations. Cecal withdrawal time is significantly associated with adenoma detection rate (ADR), which is an established quality indicator for screening colonoscopy.

Therefore, we introduced a real-time cecal withdrawal timer (CWT) that alerts the physician of the elapsed time since cecal intubation.

Aims & Methods: An observational, retrospective, single-center study to evaluate CWT in screening colonoscopy tests. The CWT is activated by a vocal command when the physician achieves cecal intubation. Following activation, the elapsed time is audio-announced every minute until the end of the procedure.

The ADR of colonoscopies with and without the CWT was evaluated following CWT implementation from January 2023 to February 2024. Moreover, the ADR of the entire department in this period was compared with the year before.

Results: Of 1,557 colonoscopies between January 2023 and February 2024, 908 (58.3%) were performed with the CWT with a median withdrawal time of 8.75 (IQR: 6.9-12.0) minutes.

During this period, the CWT usage was greater than 50% in 11 out of 14 months. Considering only months with CWT usage over 50%, the ADR of colonoscopies with CWT was 28.3% compared to 19.2% in those without ($p<0.001$).

Among physicians who performed at least 20 tests with and 20 tests without the CWT, 6 out of 8 demonstrated improved ADR with the cecal timer. Notably, the 2 physicians who did not improve ADR had baseline ADR exceeding 30%.

The overall ADR of the department between January 2023 and February 2024 was 25.5%, compared to 26.2% in 2022 ($p=0.704$).

Conclusion: Initial findings of the CWT program suggest its potential in enhancing screening colonoscopy quality indicators, although long-term prospective follow-up is warranted.

Disclosure: Nothing to declare.

PP0870

EFFICACY AND SAFETY OF CONSCIOUS SEDATION WITH MIDAZOLAM CONTINUOUS INFUSION FOR ESD PROCEDURES: A SINGLE CENTRE PROSPECTIVE STUDY

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Introduction: To date sedation protocol for colorectal endoscopic submucosal dissection (ESD) has not been standardized. (1)

In a recent Italian survey 48,2% of responders perform colorectal ESD with anaesthesiologist assisted deep sedation. (2)

A randomized controlled trial showed an higher incidence (37.9%) of procedure discontinuation in upper GI ESD when performed with midazolam administration.(3)

The aim of this study was to analyse the efficacy and safety of colorectal ESD performed under endoscopist assisted conscious sedation with midazolam continuous infusion (MCI).

Aims & Methods: Consecutive patients who underwent standard and hybrid ESD for colorectal lesions in our institution from April 2017 to September 2023 were prospectively analysed.

In all procedures, a bolus of 0.05mg/kg followed by 0.07mg/kg/h continuous infusion of midazolam was administered with any further boluses and additional drugs (meperidine and scopolamine butyl bromide) according to the patient's level of sedation and pain control.

The main outcome was the incidence of discontinuation of the procedure due to a poor response to sedation.

The second outcome was the incidence and type of sedation side effects.

Treatment outcomes, including adverse events were evaluated for each lesion.

Additional drugs administration and past medical history were collected.

Results: 101 ESDs were performed in the period of analysis.

In 5 out of 101 procedures (4.9%) was registered a discontinuation due to poor response to MCI sedation and was requested a conversion to anaesthesiologist assisted deep sedation to complete the procedure.

N. 96 cases were considered for safety analysis (M/F 49/47, median age 70yrs [IQR 14.25]; mean BMI 25.2 [SD: 2.92]).

Sedation drug doses are listed in Table 1.

Lesions had a mean size of 37.1mm [range 10-80], being mostly located in the rectum (96.9%).

En bloc rate was 85,4% , while R0 rate was 92,7%.

The incidence of sedation side effects was 13,5% (13/96): in 12 patients occurred bradycardia, while in 1 patient hypotension.

All those side effects were conservatively managed with medical therapy without occurrence of cerebrovascular or cardiac ischemic events.

In the univariate analysis, sedation side effect occurrence was associated with additional intraprocedural scopolamine butyl bromide administration and patient history of respiratory disease ($p<0.05$).

Among early adverse events, in 3 patients occurred superficial muscular damage, which was managed endoscopically, while in one patient occurred perforation, which required surgery.

Regarding late adverse events, in 5 patients was reported a late bleeding, which was treated endoscopically with success, whereas in 6 patients occurred post-polypectomy syndrome and in 4 patients occurred urinary retention, both resolved with medical therapy

Induction + Continuos infusion		Mg (mean)
Midazolam		19
Additional bolus	N(%)	Mg (mean)
Midazolam	91 (95)	2.5
Scopolamine	37 (38,5)	6.04
Meperidine	80 (83,3)	66.56

Table 1. Sedation drug doses

Conclusion: The use of MCI is a safe and effective sedation option for colorectal ESD procedures, with a low rate of procedure discontinuation.

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Disclosure: V. Cennamo: Member of the scientific board of Olympus Italia; Consultant for Novità Medicali and Euromedical. All other authors disclosed no financial relationships.

PP0871

PREDICTIVE FACTORS OF INCOMPLETE SEDATION-FREE COLONOSCOPY

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Introduction: Colonoscopy is the gold standard of an optimal exploration of the colon. Complete colonoscopy stands as the cornerstone for detecting precancerous lesions and colorectal screening.

Aims & Methods: The aim of our study was to identify factors limiting the completeness of the colonoscopy. We retrospectively colonoscopy reports performed between August 2023 and April 2024. Colonoscopies performed under general anesthesia were excluded from our analysis. Demographics, clinical and endoscopic data were collected. We investigated predictive factors of incomplete examinations.

Results: In total, 400 performed colonoscopies were included. The mean age of the patients was 56.5 years ± 14.22 (ranging from 16 to 91 years) and a male-to-female ratio of 2. History of diabetes was present in 24% of patients, hypertension in 38% of patients, and 7.8% of patients had chronic inflammatory bowel disease. The predominant indications for colonoscopy were constipation in 29% of cases, digestive bleeding in 16.1% of cases, diarrhea in 11% of cases, anemia in 9% of cases, screening in 4.8% of cases, and anal fistula in 3.8% of cases. The cecum was intubated in 381 colonoscopies (79.8% of cases). Interruption of the endoscopic examination were attributed to various causes, including inadequate preparation (45.5%), poor tolerance (32.5%) and irreducible loops (13%). In univariate analysis, anemia (p=0.033), history of benign prostatic hyperplasia, tolerance (p=0.05), and poor preparation (assessed by the Boston Bowel Preparation Scale) (p<0.0001) were identified as associated factors to incomplete colonoscopy. Demographic characteristics, comorbidities, irritable bowel syndrome, body mass index, abdominal hernia, history of anterior abdominal surgery were not associated with incomplete examination. On multivariate analysis, poor tolerance (p<0.0001, OR= 2.45 95%IC: 1.667-3.672), colonic preparation (p<0.0001, OR= 4.04 95%IC: 2.384-6.845) were predictive factors of incomplete colonoscopies.

Conclusion: This study confirms the importance of addressing limiting factors of a complete colonoscopy to avoid rescheduling the procedure and to prevent its psychological impact and delay in management and economic burden. Adequate analgesia and improved preparation are necessary for successful cecal intubation.

Disclosure: Nothing to disclose.

PP0872

THE EFFECT OF SMARTPHONE APPLICATION ON THE QUALITY OF BOWEL PREPARATION FOR COLONOSCOPY

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Introduction:

Cancer is the second leading cause of death globally. Colon cancer is the fifth most prevalent type of cancer. Screening, particularly colonoscopy, reduces incidence by 40% and mortality by 50%. The process of bowel preparation before a colonoscopy is complex and requires the patient's compliance.

Enhanced education using a smartphone application has been shown to improve bowel preparation quality. However, heterogeneity was observed among the studies.

Aims & Methods: AIMS: To determine if providing instructions through an application enhances bowel preparation quality in Thai patients undergoing elective colonoscopy.

METHODS: From July 2022 to January 2024, a prospective, endoscopist-blinded, randomized, controlled trial was conducted. Consecutive patients aged > 18 years old who had convenient access to the Line application were randomly assigned in a 1:1 ratio to be educated via the smartphone application (APP group) or the standard education (control group). All patients received the same purgative regimen.

The primary outcome was the Boston Bowel Preparation Scales (BBPS). Secondary outcomes included adenoma detection and cecal intubation rates, cecal intubation time, withdrawal time, and patient satisfaction.

Results: The trial aims to recruit 488 patients. A total of 272 patients (APP group, n = 131; control group, n =141) were preliminary analyzed. The mean age was 61.5 years old, and 29% were male.

The mean BBPS was comparable between the APP and control groups (8.30 ± 0.99 vs 8.51 ± 1.19, P = 0.173). Both groups achieved adequate bowel preparation (BBPS>6) in 99.2% of cases.

All secondary outcomes corresponded with the primary outcome. In the subgroup analysis among the elderly, levels of education, income, and waiting times for colonoscopies, BBPS was not impacted by the providing methods. No adverse events related to education were observed.

Outcome	Control group (n = 141)		Intervention group (n = 131)		p-value
	n	(%)	n	(%)	
BBPS, mean ± SD	8.51	± 0.99	8.30	± 1.19	0.173
Adequate bowel preparation, BBPS ≥ 6	140	(99.2)	130	(99.2)	1.000
NPO time to colonoscopy (hour), mean ± SD	4.73	± 1.2	4.87	± 1.1	0.292
Withdrawal time (min), IQR	9	(6.3 – 12.0)	8	(6.4 – 10.4)	0.240
Polyp detection rate	75	(53.2)	64	(48.9)	0.475
Adenoma detection rate	54	(38.3)	43	(32.8)	0.346
Cancer	1	(0.7)	1	(0.8)	1.000
Satisfaction, IQR	4.2	(3.6 – 4.8)	4	(3.5 – 4.5)	0.084

Conclusion: Both standard and enhanced education using the smartphone application resulted in excellent bowel preparation. Eligible patients included in this study might influence this outcome. However, using the smartphone application was well-received and could potentially replace standard education for bowel preparation.

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Disclosure: Nothing to disclose.

PP0873

ASSESSING COLONOSCOPY ADHERENCE TO GUIDELINES: MITIGATING INAPPROPRIATE PROCEDURES AND MINIMIZING ENVIRONMENTAL IMPACT

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Introduction: In line with the European Society of Gastrointestinal Endoscopy (ESGE) and European Society of Gastroenterology and Endoscopy Nurses and Associates (ESGENA) position statement on reducing the environmental footprint of gastrointestinal (GI) endoscopy, reducing the current rate of unnecessary procedures is the most effective action to mitigate greenhouse gas (GHG) emission in GI endoscopy.[1]

The study assesses colonoscopies for adherence to BSG guidelines and evaluates associated environmental burdens, considering that a standard colonoscopy produces 6.71 Kg of CO2.[2]

Aims & Methods: It is a service evaluation of colonoscopy practice at our Endoscopy unit. The procedure requisitions were retrospectively analyzed for all colonoscopies done between January and June 2022 and their appropriateness was evaluated based on BSG/NICE guidelines. This activity included assessing non-invasive test reports, if any, recommended before colonoscopy. Repeat colonoscopies due to inadequate bowel preparation were included too.

Inappropriate indication for colonoscopy	Count	Percentage
Abdominal pain and bloating (no FIT/ FCP)	26	19%
Alternating bowel habits + Normal FCP/FIT	47	34%
Loose stools + Normal FCP/FIT	30	21%
Others	26	19%
Polyp surveillance at inappropriate intervals	3	2%
Unexplained weight loss+ Normal FCP/FIT	7	5%
Total	139	

Table 1.

Results: Out of 938 colonoscopies done in six months, 14.8% (n= 139) were inappropriate as per the clinical indication. Table 1 provides a summary of inappropriate indications for colonoscopy and their corresponding frequencies. Just over 7.9% (n= 75) of colonoscopies could have been avoided by performing less invasive tests like FIT or faecal calprotectin (FCP). About 4.2% (n=40) of the colonoscopies were repeated due to inadequate bowel preparation.

Conclusion: Just over 14.8% of colonoscopies were performed inappropriately. In 6 months, our unit contributed to an additional 933kg of CO2 (estimated annual carbon print 3.74 tons). Every unit should implement robust vetting to reduce carbon costs secondary to inappropriate endoscopy.

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Disclosure: Nothing to disclose.

PP0874

NEW ENDOSCOPIC PLATFORM TO FACILITATE ENDOSCOPIC SUBMUCOSAL DISSECTION

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Introduction: Endoscopic submucosal dissection (ESD) is gaining popularity around the word. However, ESD remains a technically difficult, labor-intensive and time-consuming procedure.

Aims & Methods: Study aim: To describe first clinical experience with recently developed new endoscopic platform to facilitate ESD.

Methods: C[1] endoscopic platform (Lumendi LLC, Westport, CT, USA) is a flexible overtube with a balloon on its distal end and attached instrument channel. The C[1] was preloaded on top of any colonoscope and inserted into colon till pre-cancerous or early cancerous lesion is reached.

After circumferential incision around the lesion was completed with a commercially available ESD knife, a specially designed flexible grasper (Lumendi LLC, Westport CT, USA) was advanced through the instrument channel.

The separated edge of the lesion was grasped with the grasper to facilitate ESD. After completion of the ESD, C[1] was used a conduit to deliver Overstitch endoscopic suturing device mounted on a double channel upper endoscope (GiF 2T180, Olympus, Center Valley, PA, USA) to the site of lesion removal to perform sutured closure of the post-ESD defect.

Results: C[1] platform was used in 7 patients (3 females, 4 males, age 62.1±10.1 years) with large (35.0±13.5 mm) sessile colonic polyps. Most lesions were located in anatomically difficult positions (splenic and hepatic flexure, appendix and ileocecal valve). On-demand inflation of the C[1] balloon facilitated advancement of the colonoscope through the colon and shortening and straightening of the colonic turns.

None of the patients required external pressure or a change of patient position and in all cases the colonoscope was successfully advanced to the cecum. When the lesion was reached, the balloon was distended providing stabilization of the colonoscope during ESD. Flexible grasper provided multidirectional dynamic retraction of the lesion making endoscopic dissection simple and fast (total ESD time 38.7±16.5 minutes with mean dissection speed 24.8 mm^2/minute).

After completion of dissection, the lesion was retrieved through C[1] and then Overstitch suturing device was easily delivered to the ESD site. Suturing closure of post-ESD defects was successfully achieved in all patients. There were no device related adverse events during the study.

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Conclusion: Newly developed C[1] endoscopic platform with a flexible grasper significantly facilitates colonic ESD.
Disclosure: Nothing to disclose.

PP0875

ENDOSCOPIC CLOSURE TECHNIQUE PERFORMANCE FOR IATROGENIC PERFORATION: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Perforation is the most feared endoscopy-related adverse event. Various closure techniques are now available including through-the-scope clips (TTSC), over-the-scope clips (OTSC) and over-the-scope suturing (OTSS).[1,2] Given the lack of comparative performance estimates, we sought to systematically appraise TTSC, OTSC and OTSS performance for iatrogenic perforation.
Aims & Methods: MEDLINE and EMBASE were independently searched between 06 / 1947 – 01 / 2024 for citations evaluating TTSC, OTSC or OTSS performance. Technical success was defined as complete apposition of the borders of the defect with visual confirmation. Clinical success was defined as technical success with lack of further intervention. Pooled estimates using random-effects modelling were generated stratified by closure modality, location, perforation etiology, and time (year of publication).
Results: Thirty seven citations assessing 795 perforations were included. Pooled technical success and clinical success were 91.5% (95% confidence interval [CI] 87.9-94.1%; I^2 = 26%) and 86.3% (95% CI 81.9-89.8%; I^2 = 39%). Technical success of TTSC, OTSC and OTSS were 89.3% (95% CI 82.6-93.6%), 90.7% (95% CI 86.0-93.9%) and 94.9% (95% CI 83.8-98.5%; p=0.53). Clinical success of TTSC, OTSC and OTSS was 85.3% (95% CI 77.9-90.6%), 86.2% (95% CI 80.4-90.5%) and 94.9% (95% CI 83.8-98.5%) (p=0.24). Clinical success was significantly higher for tissue resection-related perforations (p = 0.02) and for perforations in the stomach (p < 0.04); no difference in clinical success was identified when stratifying by closure modality and time (all p > 0.05).
Conclusion: TTSC, OTSC and OTSS are effective for iatrogenic perforation management; specifically, tissue-resection related perforations and perforations in the stomach. No difference in performance was identified based on closure modality or time.
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Neal Shahidi: Speaker Honorarium: Boston Scientific, Pharmascience.
All other authors have no conflicts of interest to disclose.

PP0876

PREVALENCE OF ENDOSCOPICALLY CURABLE LOW-RISK CANCER AMONGST LARGE (≥20MM) NON-PEDUNCULATED POLYPS IN THE RIGHT COLON

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Introduction: Endoscopic submucosal dissection (ESD) is increasingly promoted as the optimal treatment strategy for all large non-pedunculated colorectal polyps (LNPCP). The effect of a universal en bloc strategy on oncological outcomes for the treatment of LNPCP in the right colon is unknown. We evaluated this in a large Western population.
Aims & Methods: We retrospectively analysed prospectively collected data from major tertiary centres where patients were referred for endoscopic resection (ER) of LNPCP (N=5084). Patients found to have cancer after ER and those referred directly to surgery were included. The primary outcome was to determine the proportion of right colon LNPCP with low-risk cancer.
Results: Over 180 months until June 2023, 3294 sporadic right colon LNPCP in 2956 patients were referred for ER at 7 sites (median size 30mm [IQR 15]). 63 (2.1%) patients were referred directly to surgery and cancer was proven in 56 (88.9%). 2851/2956 (96.4%) LNPCP underwent ER (median size 35mm [IQR 20]) of which 75 (2.6%) were cancers. The overall prevalence of cancer in the right colon was 4.4% (131/2956). Detailed histopathological analysis was possible in 115/131 (88%) cancers (71 after ER, 44 direct to surgery, Table 1). 23/2940 (0.78%) sporadic right colon LNPCP were low-risk cancers.

	Direct to Surgery (n=44)	EMR Performed (n=71)	p
Depth of SM invasion (Kikuchi) N (%)			<0.001
SM1	4 (9.1)	22 (30.9)	
SM2/3	8 (18.2)	39 (54.9)	
Invasion beyond SM	32 (72.7)	10 (14.1)	
High tumour grade N (%)	5 (11.4)	9 (12.7)	1
LVI N (%)	8 (18.2)	12 (16.9)	1
High grade TB N (%)	5 (20.8)	10 (14.9)	0.68
Any high-risk histology (high tumour grade, LVI, high grade TB, deep SM invasion) N (%)	40 (90.9)	52 (73.2)	0.04

Table 1: Final Oncologic Outcomes; comparison between histopathologic variables of cases referred directly to surgery versus following endoscopic resection.

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Conclusion: The proportion of right colon LNPCP referred for ER containing low-risk cancer amenable to endoscopic cure was <1%, in a large, multicentre Western cohort. A universal ESD strategy for the management of right colon LNPCP is unlikely to yield improved patient outcomes given the minimal impact on oncological outcomes, significant resource demands and increased adverse events.

Disclosure: Nothing to disclose.

PP0877

PROSPECTIVE VIDEO-BASED STUDY ASSESSING EFFECT OF COMPUTER-ASSISTED OPTICAL DIAGNOSIS ON DISTINGUISHING SESSILE SERRATED FROM HYPERPLASTIC AND ADENOMATOUS COLORECTAL POLYPS

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Introduction: Computer-Aided Diagnosis (CADx) systems have become available to assist endoscopists with optical polyp diagnosis. However, current CADx systems have not been trained to recognize sessile serrated lesions (SSLs) and currently classify SSLs incorrectly either as hyperplastic or adenomatous. We aimed to assess the effect of current CADx systems on diagnostic accuracy of SSL recognition among a diverse group of endoscopists.

Aims & Methods: We conducted a prospective video-based study including 17 endoscopists (gastroenterologists and surgeons, experts, and trainees). Each endoscopist reviewed the same video and image series of 115 polyps (33% SL, 33% Hyperplastic Polyps (HP), 34% tubular adenoma (TA)). Histopathological diagnosis of the polyps was used as the gold standard for comparison. Presentation of polyps (SSL, TA, HP) was randomized so that 50% of the polyps seen were presented with displaying a CADx diagnosis for the polyp and 50% were presented without displaying a CADx diagnosis.

Results: A total of 1955 optical polyp diagnoses were performed. The optical diagnostic accuracy amongst various endoscopists to recognize SSLs was: expert gastroenterologists (52.3%, 95% Confidence Interval (CI) 46.3-58.3), trainee gastroenterologists (38.2%, CI 27.2-49.1), expert surgeons (22.1%, CI 16.2-28.0) and trainee surgeons (14.9%, CI 8.4-21.5). Further details are provided in Table 1. CADx identified 52.6% of SSL as HP and 47.4% as TA. CADx identified 60.5% of HP correctly and 82.1% of TA correctly. The optical diagnostic accuracy for HP and TA improved among endoscopists when a correct CADx diagnosis was displayed to the endoscopists compared to when no CADx information was displayed (HP: 63.7% vs 38.4% / TA: 81.5% vs 70.5%; p=0.05). The optical diagnostic accuracy of the endoscopists worsened when the displayed CADx diagnosis was incorrect (HP: 24% / TA: 25%). The diagnostic accuracy of the endoscopists for SSL did not differ significantly when the CADx diagnosis, which is currently untrained to diagnose SSL and can only classify polyps as hyperplastic or adenomatous, was displayed (37.2% vs 33.1%, p=0.05).

	Sessile Serrated Lesion	Hyperplastic Polyp	Adenoma
CADx Alone	HP: 52.6 (36.8-68.5) TA: 47.4 (31.5-63.2)	Sn 60.5 (49.5-71.5) Sp 82.1 (70.0-94.1) PPV 86.8 (77.7-95.9) NPV 51.6 (39.2-64.1)	Sn 82.1 (70.0-94.1) Sp 60.5 (49.5-71.5) PPV 51.6 (39.2-64.1) NPV 86.8 (77.7-95.9)
CADx diagnosis NOT displayed All Endoscopists (n=17)	Sn 37.2 (31.9-42.4) Sp 78.6 (75.5-81.8) PPV 46.2 (40.1-52.2) NPV 71.7 (68.4-75.0)	Sn 38.4 (33.1-43.7) Sp 79.8 (76.8-82.9) PPV 48.4 (42.3-54.6) NPV 72.4 (69.2-75.7)	Sn 70.5 (65.6-75.4) Sp 64.7 (61.0-68.4) PPV 50.6 (46.1-55.2) NPV 81.0 (77.6-84.4)
CADx Displayed Correct Diagnosis All Endoscopists (n=17)	Sn 33.1 (28.0-38.3) Sp 79.4 (76.3-82.5) PPV 44.2 (38.0-50.5) NPV 70.6 (67.3-73.9)	Sn 63.7 (57.4-70.0) Sp 78.7 (74.9-82.5) PPV 60.2 (53.9-66.4) NPV 81.1 (77.4-84.8)	Sn 81.5 (76.9-86.2) Sp 86.5 (83.2-89.9) PPV 80.7 (76.0-85.3) NPV 87.2 (83.9-90.5)
CADx Displayed Incorrect Diagnosis All Endoscopists (n=17)		Sn 24.0 (15.6-32.4) Sp 77.4 (71.7-83.0) PPV 33.3 (22.4-44.2) NPV 68.3 (62.4-74.2)	Sn 25.0 (14.0-36.0) Sp 45.2 (39.1-51.4) PPV 9.8 (5.1-14.5) NPV 71.7 (64.7-78.7)
Breakdown of the performance by endoscopists category **GE:** Gastroenterologist Expert (n=7) **GT:** Gastroenterologist Trainee (n=2) **SE:** Surgery Expert (n=5) **ST:** Surgery Trainee (n=3)			
Overall	GE: 52.3 (46.3-58.3) GT: 38.2 (27.2-49.1) SE: 22.1 (16.2-28.0) ST: 14.9 (8.4-21.5)	GE: 35.0 (29.2-40.7) GT: 61.8 (50.9-72.8) SE: 53.7 (46.6-60.8) ST: 42.1 (33.0-51.2)	GE: 74.7 (69.6-79.9) GT: 64.1 (53.5-74.7) SE: 66.2 (59.5-72.8) ST: 74.4 (66.4-82.3)
CADx diagnosis NOT displayed	GE: 54.1 (45.7-62.6) GT: 39.5 (23.9-55.0) SE: 23.2 (14.7-31.6) ST: 19.3 (9.1-29.5)	GE: 27.8 (20.2-35.4) GT: 57.9 (42.2-73.6) SE: 53.7 (43.7-63.7) ST: 24.6 (13.4-35.7)	GE: 72.8 (65.3-80.3) GT: 66.7 (51.9-81.5) SE: 70.4 (61.4-79.4) ST: 69.5 (57.7-81.2)
CADx Displayed Correct Diagnosis	GE: 50.4 (41.9-58.9) GT: 36.8 (21.5-52.2) SE: 21.1 (12.9-29.3) ST: 10.5 (2.6-18.5)	GE: 50.6 (40.2-60.9) GT: 84.6 (70.7-98.5) SE: 67.2 (55.9-78.4) ST: 73.2 (59.6-86.7)	GE: 85.0 (78.4-91.5) GT: 75.0 (60.0-90.0) SE: 73.4 (63.7-83.2) ST: 91.5 (83.5-99.5)
CADx Displayed Incorrect Diagnosis		GE: 25.0 (12.2-37.8) GT: 25.0 (0.5-49.5) SE: 21.4 (6.2-36.6) ST: 25.0 (3.8-46.2)	GE: 37.5 (18.1-56.9) GT: 0.0 (0.0-0.0) SE: 16.7 (0.0-33.9) ST: 27.3 (1.0-53.6)

Table 1. CADx and endoscopists performance in optical diagnosis of SSL, HP and adenomas depending on the polyp histology in percentages of correct optical diagnoses with 95% confidence intervals. For SSL, CADx is not able to provide correct diagnosis; thus is presented as CADx shown as either HP or TA)

Sn: sensitivity; Sp: specificity; PPV: positive predictive value; NPV: negative predictive value

Conclusion: Optical polyp diagnosis that includes SSL, TA, and HP recognition remains challenging. Current CADx systems when displaying correct diagnostic outputs improve diagnostic accuracy of the endoscopists. CADx systems not trained for SSL recognition do not significantly affect SSL recognition among endoscopists even if an incorrect diagnostic output (TA or HP) is displayed. Development of improved CADx systems with SSL recognition and higher diagnostic accuracy is warranted.

Disclosure: Dr Jeffrey Marks: Consultant for Boston Scientific and Steris Endoscopy.

PP0878

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PP0879

COMPARATIVE ANALYSIS OF PERFORMANCE SCORES ON A COLONOSCOPY SIMULATOR (MIKOTO): A NON-INFERIORITY STUDY WITH UK NATIONAL ENDOSCOPY DATABASE KEY PERFORMANCE INDICATORS

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Introduction: Gastrointestinal endoscopy training relies increasingly on simulation tools, especially in foundation years, to teach skills like scope handling and tip control. However, access to simulators offering accurate, real-time feedback is limited. Existing simulators also lack a performance score aligned with UK National Endoscopy Database (NED) Key Performance Indicators (KPIs) such as Caecal intubation rate (CIR), Polyp detection rate (PDR), and Colonoscopy comfort score (CCS).

The MIKOTO colonoscopy training simulator aims to replicate diagnostic colonoscopy conditions and provide real-time feedback based on a simulator score derived primarily from patient comfort during the procedure.

Aims & Methods: This study aimed to validate the comparability of real-time performance scores from the MIKOTO (colonoscopy training simulator) with endoscopists' NED KPIs. Conducted in a major UK Joint Advisory Group (JAG) accredited centre, the non-inferiority study involved 20 endoscopists of varying experience levels, categorized into Novice, Less Competent, Competent, and Expert groups based on their lifetime colonoscopy numbers. Endoscopists provided their NED KPIs before using the MIKOTO simulator. They then performed three full colonoscopies on the simulator, with caecal intubation time and MIKOTO scores collected after each attempt. Statistical analyses were performed to compare the results.

Results: The mean lifetime colonoscopies of the novices, less competent, competent and expert endoscopists were 31, 255, 593 and 2890 respectively. The median CIR for novices, less competent, competent and expert endoscopists were 31%, 92%, 97% and 99% respectively while the PDR across the groups were 7%,20% 30% and 45% respectively. In addition, the CCS for the various groups were 37.5%, 10%, 4% and 0.8%. The MIKOTO scores for novices, less competent, competent and expert endoscopists were 47, 69, 84 and 81 respectively. There is statistically significant correlations were observed between CCS, CIR, and MIKOTO scores across all groups (Table 1). No difference was found between Competent and Expert endoscopists in MIKOTO scores and CIR. Overall, higher MIKOTO scores were associated with better NED KPIs (Fig 1).

NED KPI	Novice (MSS=47)	Less competent (MSS=69)	Competent (MSS=84)	Expert (MSS=81)	P-value
PDR (%)	7	20	30	45	0.008
CIR (%)	31	92	97	99	0.566
*CCS (%)	37.5	10	4	0.8	0.003

Table 1: Comparison between NED-KPIs and MIKOTO simulator scores for different competency levels.

*NED= National Endoscopy Database, KPI= Key performance Indicator, PDR= Polyp Detection rate, CIT= Caecal Intubation time, CIR= Caecal intubation rate, *CCS= Colonoscopy Comfort Score (% of patient where comfort level is recorded as moderate or severe discomfort), MSS= Mikoto Simulator Score*

Conclusion: The MIKOTO simulator score closely aligns with NED KPIs for colonoscopy, potentially serving as a real-time indicator of endoscopic competence, and a valuable tool for early endoscopy training. Larger, multi-institutional studies are needed to assess the integration of the MIKOTO simulator into UK endoscopy training for skill development and assessment.

Disclosure: No Conflict of interest

PP0880

NON GRANULAR AND GRANULAR MIXED LATERALLY SPREADING TUMOR, WHERE SUBMUCOSAL INVASION LIES BENEATH

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Introduction: Endoscopic diagnosis and resection strategy of colorectal laterally spreading tumors (LSTs) is a relevant topic of debate. Current guidelines suggest that LSTs can be treated with endoscopic mucosal resection (EMR) provided that superficial submucosal invasion (SMI) stigmata are ruled out. By contrast, the treatment of choice for high risk SMI lesions should be en bloc resection with endoscopic submucosal dissection (ESD).

Aims & Methods: Since prediction of SMI risk is still challenging, in this study we evaluated which endoscopic features associated with presence of SMI in LSTs.

In this monocentric retrospective study, we included all patients with resected LSTs between January 2018 and November 2023. Lesions were classified according to morphology into granular (G), granular mixed (GM) and non granular (NG) phenotype. Paris Classification, Kudo pit Pattern, location and lesion diameter were evaluated as predictors of SMI. Univariate and multivariate logistic regression analysis was performed to assess predictors of SMI.

Results: Overall, 459 lesions were analyzed: n=172 NG-LST (37,5%), n=172 GM-LST (37,5%) and n=115 G-LST (25%). Superficial SMI was detected in 35 NG-LSTs (20,3%) , 21 GM-LSTs (12,2%) and 4 G-LSTs (3,5%). In the NG-LST group the prevalence of SMI was non-significantly higher in distal compared to proximal colon (24,5 % vs 17% , OR=1.59 [0.70-3.62] p= 0.27). According to Paris classification, the risk of SMI was significantly higher in depressed NG-LSTs (Paris IIc- IIa + IIc) compared to non-depressed lesions (31 % vs. 12%, OR=3.62 [1.59-8.26]; p=0.002).

In GM-LSTs cohort, a not significantly higher SMI prevalence was reported comparing distal with proximal lesions (12,5% vs 5,9%, OR=2,26 [0.61-8.30] p=0.22), whereas rectal lesions showed significant higher risk than proximal lesions (22,9% vs 5,9 %, OR=5.87 [1.94-17.73]; p= 0.002). Kudo pit pattern Vi lesions in comparison to Kudo pit pattern IIIs/IIIL/IV lesions showed a significantly higher SMI prevalence both in NG-LSTs (32% vs. 2%, OR=36.57 [4.80-278.60]; p< 0.001) and in GM-LSTs (73% vs 5%, OR=40.36 [11.96-136.18]; p <0.001). Besides, in GM-LST group a lesion diameter ≥4 cm was associated with higher SMI prevalence only in rectal lesions (40% vs 5%, OR 11.33 [1.33-96.81]; p=0.03).

At the multivariate analysis, independent predictors of SMI presence were Kudo pit pattern Vi (OR 35.02 [4.31-284.85] p<0,001) in NG-LSTs, and Kudo pit pattern Vi (OR 74.15 [14.08-390.43], p<0.001) and rectal lesions (OR 13.45 [2.00- 90.25], p=0,007) in GM-LSTs.

Conclusion: Our study confirmed that NG-LST lesions deliver a higher overall risk of SMI than GM-LSTs and G-LSTs, independently on the location. GM-LSTs carried an intermediate risk of SMI and, considering only rectal lesions, size was an important risk factor for SMI. Furthermore, the Kudo pit pattern was an independent risk factor for SMI in both groups. As a consequence, our study stresses that a proper optical diagnosis is para-

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mount in order to stratify SMI risk and choose the appropriate treatment for these lesions.

Disclosure: Nothing to disclose.

PP0881

ENDOSCOPIC RESECTION OF CIRCUMFERENTIAL AND NEAR-CIRCUMFERENTIAL LARGE NON-PEDUNCULATED COLORECTAL POLYPS – A WESTERN PARADIGM SHIFT

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Introduction: Endoscopic resection (ER) of large non-pedunculated colorectal polyps (LNPCP) has advanced significantly, pushing boundaries to treat lesions of any size. However, a dearth of published data exists on the extremes of ER practice, particularly for large lesions creating circumferential or near-circumferential mucosal defects. While Japanese studies, focusing on ESD-resected LNPCP, show conflicting outcomes, two prominent Western series lean towards EMR-resected lesions. The Western landscape is evolving, with a notable rise in ESD for LNPCP.

Our study delivers fresh insights, presenting outcomes from ER with ≥75% circumferential mucosal defects in a leading Western centre that predominantly favours ESD for such lesions. This signifies a transformative trend in ER methodology, underlining the efficacy of ESD in addressing intricate colorectal lesions, thereby shaping evolving standards.

Aims & Methods: 1477 ERs of LNPCP ≥20mm were performed and grouped according to the extent of the mucosal defect and comparisons made between those with <75% and ≥75% defects. Independent predictors of stenosis were identified.

Results: 137 patients had ER leaving ≥75% circumference defect, most located at or distal to the rectosigmoid, with ≥90% defects in 29 and 100% in 14. 31 (22.6%) were resected using EMR and 106 (77.3%) using ESD. En bloc resection was achieved in 64%, increasing to 88% in the last 4 years of the study where ESD was increasingly employed routinely. Overall complications were more likely in these patients (11.7% versus 3.4%, p<0.001), although there was no difference in the incidence of clinically significant perforation (p=0.41) or post-procedure bleeding (p=0.45).

101 patients had undergone surveillance. Stenosis developed in only 2 of 71 (2.8%) patients with defects ≥75% circumference but 10 of 18 (55.6%) patients with defects ≥90% and 10 of 12 patients (83.3%) with 100% circumferential defects. ≥90% circumference defect was an independent risk factor for stenosis (p<0.001).

All patients with stenosis were managed successfully with either expectant management if asymptomatic or balloon dilatation. None have required surgical intervention for stenosis. Recurrence was more likely in those with ≥75% defect (OR 7.9, 95% CI 3.8-16.4, p<0.001) but was acceptable at an incidence of 15% at SC1

Conclusion: This is the largest western series to date reporting on outcomes of ER of LNPCP leaving near or fully circumferential defects, and the first to include significant numbers of ER performed by ESD. ER of LNPCP resulting in defects ≥75% of the luminal circumference is challenging but highly effective in avoiding surgery when performed in an expert centre. En bloc resection can be achieved in most cases, reducing recurrence. The only independent predictor of stenosis is ≥90% circumference defect but some patients improve with expectant management therefore pre-emptive intervention may not be warranted.

Disclosure: Nothing to disclose.

PP0882

THE YIELD OF COLONOSCOPY IN THE EVALUATION OF CONSTIPATION: AN AGE-BASED ANALYSIS OF OUTCOME

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Introduction: Chronic constipation is a prevalent gastrointestinal complaint with increasing incidence among the aging population. Examining colonoscopy performance in chronic constipation across age groups, this study aims to comprehensively assess diagnostic yield and compare results with average-risk controls.

Aims & Methods: A retrospective analysis was conducted on 50,578 colonoscopy procedures performed over 12 years, including 5,478 constipated patients. An average-risk control group (n=4,100) was included. Data extracted from electronic medical records covered demographics, operational aspects, and colonoscopy findings. The primary outcome measures included the diagnosis rate of colorectal cancer (CRC), polyp detection rate (PDR) and inflammatory bowel disease (IBD) diagnoses in constipated patients versus controls, with age-based and multivariate analyses.

Results: Constipated patients exhibited lower rates of adequate bowel preparation (92.7% vs. 85.3%; p < 0.001) and a lower cecal intubation rate. No significant variances between CRC and PDR were observed between constipated and controls, except for a potential of a slightly elevated CRC risk in constipated patients older than 80 (2.50% vs. 0% in controls; p = 0.07). However, multivariate analysis demonstrated that constipation did not confer an increased risk for CRC or polyp detection across all age groups.

Notably, younger constipated patients exhibited a higher rate of IBD diagnoses (1.7% vs. 0.1% in controls; p < 0.001).

Conclusion: Conclusion: Constipation did not confer an increased risk for CRC or polyps among all age groups, except for a potential signal of elevated CRC risk in patients older than 80, and was associated with higher rates of IBD in younger patients.

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Disclosure: Nothing to disclose.

PP0883

COLON CAPSULE ENDOSCOPY: WHAT FACTORS PREDICT AN INCOMPLETE EXAMINATION?

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Introduction: The most important factors for a successful colon capsule endoscopy (CCE) study are the quality of bowel preparation and the capsule excretion during battery life. Incomplete colonoscopy is one of the main indications for CCE.

Aims & Methods: The aim of this study was to analyze clinical and demographic factors for complete CCE after incomplete colonoscopy.

A retrospective single-center study was conducted, including patients consecutively submitted to CCE with previous incomplete colonoscopy. Complete examination was defined as visualization of hemorrhoidal plexuses during battery time. Demographic (gender and age) and clinical data (diabetes mellitus, hypertension, smoking, BMI, hypothyroidism, constipation, depression, opioid therapy, psychotropic medication and previous abdominal surgery) were collected.

Results: 197 patients were analyzed, with a mean age of 67 years, 71.6% were women. The main reasons for incomplete colonoscopy were impassable angulation (54.8%) and irreducible loop (40.1%).

CCE was incomplete in 32.5% (n=64) of cases. Bowel preparation was inadequate in 30.5% (n=60).

Constipation was a risk factor for incomplete CCE study (OR 2.962; p=0.021, 95% CI 1.178-7.264), as well as obesity (OR 4.407; p=0.020, 95% CI 1.263-15.373).

The clinical indication for the examination did not show a statistically significant relationship with the rate of incomplete examinations (p=0.603). There were no statistically significant differences between the percentage of incomplete CCE in patients with incomplete colonoscopy due to impassable angulation versus irreducible loop (33.3% vs 33.3%, p=1.000).

Conclusion: Constipation and obesity are factors associated with a higher risk of incomplete CCE study. In these patients, intensification or alteration of the bowel preparation protocols used may be considered.

Disclosure: Nothing to disclose.

PP0884

CLINICAL IMPLEMENTATION OF RESECT AND DISCARD USING COMPUTER-ASSISTED OPTICAL POLYP DIAGNOSIS (CADX): A PILOT STUDY

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Introduction: Histopathological assessment remains the gold standard for evaluating diminutive colorectal polyps, despite the potential of optical diagnosis (OD) to reduce colonoscopy-associated costs (1). However, implementing OD strategies remains limited in clinical practice (2). This study aimed to evaluate patient acceptance of truly replacing pathology evaluation of diminutive polyps with resect and discard (RD) and diagnose and leave (DL) strategies using computer-aided polyp diagnosis (CADx).

Aims & Methods: In this prospective clinical pilot study at the Montreal University Hospital Center, 102 patients undergoing colonoscopy were invited to participate in the study. Patients received information on CADx, RD, DL and were informed that if they agree to participate in the study, we will diagnose all polyps using CADx during the colonoscopy, but polyps will be discarded or left in place and no pathology exams will be performed. For consenting patients, blue light imaging (BLI)-assisted CADx was used and endoscopists made the final histology determination and rated their confidence level (high vs. low) in the OD. Endoscopists also determined if high-risk features were present before using RD or DL strategies. Then all diminutive polyps optically diagnosed with high confidence as adenomas were resected and discarded and all diminutive rectosigmoid polyps optically diagnosed as hyperplastic with high confidence were left in situ. All polyps >5mm, with disagreement between endoscopists and CADx OD, low-confidence OD, or high-risk features (JNET 2b or higher) were sent for pathology evaluation. All cases were video recorded, and a quality review of the diagnoses performed with two expert endoscopists (DR and HP).

Results: Among the 102 patients approached, 97 (95%) accepted participating and utilization of CADx instead of pathology. Three stated that they did not want to participate in any research, and two patients stated that they did not trust OD or CADx as the reason for refusal. Overall, 164 diminutive polyps were removed in 97 patients (mean age=67.2 years), of which 36% were in the rectosigmoid and 64% in the proximal colon. CADx-assisted OD was performed on 97% of the polyps. The proportion of polyps with a low-confidence OD or disagreement between endoscopists' and CADx-based OD was 19/159 (11.9%). Overall, 98/159 (61.6%) polyps underwent the RD strategy, of which 4 polyps were discarded by the endoscopist's discretion despite disagreement between endoscopists and CADx OD. One of these 4 polyps had to undergo the DL strategy but was RD because of serrated features. One polyp in the proximal colon was resected and retrieved because of serrated features. 40 polyps were DL in the rectosigmoid, of which 1 polyp was neoplastic based on CADx and endoscopist's OD.

Conclusion: We observed a very high patient acceptance for truly replacing diminutive polyp histopathology with CADx-assisted OD. CADx-based OD was used to RD 98/159 (61.6%) or DL 40/159 (25.2%) diminutive polyps found in the cohort.

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Disclosure: Dr Rex is a consultant for Olympus Corporation, Boston Scientific, Aries Pharmaceutical, Braintree Laboratories, Lumendi Ltd., Norgine, Endokey, GI Supply,
Medtronic, and Acacia Pharmaceuticals; the recipient of research support from EndoAid, Olympus Corporation, Medivators, Erbe USA Inc, and Braintree Laboratories; and a shareholder in Satisfai Health. Dr Pohl has received research funding from Steris, Cosmo, and is an advisory board member of InterVenn. Dr von Renteln has received research funding from ERBE Elektromedizin GmbH, Ventage, Pendopharm, Fujifilm, and Pentax and has received consultant or speaker fees from Boston Scientific Inc, ERBE Elektromedizin GmbH, and Pendopharm. The remaining authors disclose no conflicts.

PP0885

INFLUENCE OF AGE ON COLORECTAL POLYP CHARACTERISTICS AND FACTORS ASSOCIATED WITH DYSPLASIA: A BI-CENTER RETROSPECTIVE-PROSPECTIVE STUDY

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Introduction: Colorectal polyps are benign tumours detected by endoscopy, with a higher prevalence in older people. The adenoma-carcinoma sequence is well established. Therefore, systematic screening is important to remove them at an early stage.
Aims & Methods: This is a retroprospective study including all polyps found during 1393 colonoscopies performed in 2 private university hospitals. According to age, 3 groups were defined: Group 1 (less than 45 years), Group 2 (between 45 and 65 years) and Group 3 (more than 65 years). Data were collected using an evaluation form and analyzed using Jamovi (version 2.4). The aim of our study is to investigate the effect of age on colorectal polyp characteristics and factors associated with the occurrence of dysplasia.
Results: Out of 1484 colonoscopies, we included 187 patients (12.6%) with 347 polyps. The mean age was 53.3 years [25 - 87 years], with a male predominance at 59% (male-to-female sex ratio: 1.4).
The 347 diagnosed polyps were distributed by age: 44 in Group 1 (12.8%), 171 in Group 2 (49.1%), and 132 (38%) in Group 3.
The mean number of polyps was higher in Groups 2 and 3 (respectively 2.16 and 2.06 polyps per patient). The mean number of adenomas was higher in Group 2 and 3 (72% and 74%) versus 54% in group 1.
The majority of polyps found were sessile (80%) with mean size of 6.7mm. all groups combined, the most common histological type was tubular adenomas (48%) followed by hyperplasic polyps (22%) with high rate of low-grade dysplasia (56-66%), and less than 10% f High-grade dysplasia.
Statistical analysis revealed a statistically significant correlation between age and the number of polyps (R=0.14; p < 0.01) as well as their size (R=0.13; p = 0.013).
Low-grade dysplasia was statistically associated with polyp size (p < 0.001; Z = 4.5; OR 1.14 CI (1.08-1.2)), as well as certain histological types: tubular adenomas (p < 0.001 Z 4.8 OR 195) and tubulovillous adenomas (p < 0.001 Z 3.4 OR 45).
However, no statistically significant correlation was found between the occurrence of dysplasia and age, number, or macroscopic appearance of the polyps.
Conclusion: Polyps and age are risk factors for colorectal cancer. In our series, advanced age was statistically associated with an increase in the number and size of polyps.
Additionally, a correlation was found between the histological type of polyps and low-grade dysplasia. Our results support the recommendations to commence colorectal screening from age 45 to treat polyps at an early stage.
Disclosure: Nothing to disclose.

PP0886

COMPLEXITY OF ENDOSCOPIC RESECTION FOR APPENDICEAL ORIFICE LESIONS SUGGESTS CONSIDERATION OF BOTH SURGICAL AND ENDOSCOPIC OPTIONS

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Introduction: Large non-pedunculated colorectal polyps (LNPCP) and laterally spreading tumours (LST) >10mm involving the appendix orifice (AO) are notoriously challenging to treat with endoscopic resection (ER). Few studies report outcomes of conventional ER to treat these lesions. The optimal therapy and expected outcomes remain unclear.
Aims & Methods: We report the outcome after EMR and ESD of lesions involving the AO from a large series of ER performed at a tertiary referral centre. Patients undergoing ER at a tertiary referral centre between January 2011 and September 2023 were included. Involvement of the AO was defined as any LNPCP or LST encroaching within 5mm of the orifice.
These were classified as encroaching the AO, involving part of the circumference of the AO and involving the full circumference of the AO. Techniques and outcomes of ER of lesions involving the AO were compared with other colorectal locations.
Results: ER was performed for 1596 LNPCP and LSTs during the study period. 40 lesions involved the AO: 10 encroached within 5mm of the AO, 14 involved part of the circumference and 16 involved the full circumference. 25 (63%) were resected by EMR, 8 by ESD (20%) and 7 by Hybrid ESD (18%). Initial successful ER was similar for lesions involving the AO and other colonic LNPCP (95.0% *versus* 97.6%, p=0.27).
Clinically significant perforation occurred in 3 patients with lesions involving the AO (2 with partial circumference involvement and 1 with complete circumference involvement) and was more frequent after ER of these lesions than others (7.5% *versus* 1.2%, p<0.001). 2 of these patients required surgery. The incidence of post procedure bleeding was similar (p=0.44).
One patient developed appendicitis, more than 12 months after resection.
Recurrent/residual adenoma at first surveillance was significantly more likely after ER of AO lesions than others (22.2% *versus* 4.8%, p<0.001). Significantly more patients with lesions involving the AO ultimately required

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surgery for any reason than patient with other colorectal lesions (12.5% *versus* 3.9%, p=0.02). This was significantly more likely for AO lesions involving the complete or partial circumference (18.8% *versus* 14.3% partial circumference involvement *versus* 0% encroaching AO *versus* 3.9% other locations, p=0.004).

Conclusion: These results demonstrate the challenging nature of ER for lesions involving the AO. Safe and effective ER without recurrence is achieved for many patients, including those with lesions involving the complete circumference, but significant complications, recurrence and surgery are more common with these lesions. Given there are relatively low risk minimal access surgical options to treat these lesions, these data will help inform detailed discussions with patients regarding the risks and benefits of ER (which differ from LNPCP elsewhere in the colon).

Disclosure: Nothing to disclose.

PP0887

ASSOCIATION OF SUCCESSFUL COLONOSCOPY RATE AND QUALITY OF BOWEL PREPARATION BY TWO DIFFERENT BOWEL PREPARATION AGENTS

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Introduction: Intubation of the terminal ileum (TI) is the gold standard for completion of colonoscopy.

Our aim was to assess if this objective was achieved in our centre between endoscopists using various bowel preparations.

Poly ethylene glycol 4000 Vs Magnesium sulphate combined with potassium and sodium sulphates were the two commonly used bowel preparations. Magnesium containing preparations must be used with caution and avoided in patients with renal insufficiency, cardiac disease or pre-existing electrolyte abnormality and patients on diuretic therapy.

The Boston Bowel Preparation Scale was developed to limit inter-observer variability in the rating of bowel preparation quality, while preserving the ability to distinguish various degrees of bowel cleanliness and is mainly designed for adults. There is no dedicated bowel preparation score available for paediatric colonoscopies, BBPS is being used for critical comparison in our prospective cohort.

Aims & Methods: Objectives: To compare the intubation rate of TI between two endoscopists with quality of bowel preparation of colonoscopies in a tertiary health care centre and to assess the quality of bowel preparation of individual colonoscopies performed vis-a-vis, Boston bowel preparation scale as a prospective study over a period of 1 year.

Material and methods: A prospective study in a tertiary referral centre in children who underwent colonoscopies, aged 1 month to 17 years over a period of 1 year.

All children whose parents consented were included in the study. Results were collated using Microsoft Excel ™.A total of 135 cases were enrolled during our study period.

Results:

1. TI intubation rate in PEG-4000 + bisacodyl based laxative formula is 87%
2. TI intubation rate in Magnesium/sodium/potassium based saline laxative formula is 79%
3. No side effects were encountered/ observed in either of the cohort.

Conclusion: In our study albeit the fact that our endoscopist chose different bowel preparations based on their anecdotal evidence based experience, we found that the bowel preparation was adequate.

There was no standardised paediatric bowel preparation globally and we note that every hospital or department or clinician have devised their own bowel preparation based either on personal experience or current available scientific literature.

We also note that we do not have a BBPS for paediatrics which will of immense value for research evaluation and revalidation of bowel preps.

	BBPS (mean)	Sex (m:f)	Mean Age	Sensitivity	Specificity	P value	TII rate
Endoscopist # 1 PEG 4000+ Stimulant laxative	2.37	3:2	8.96	93%	82%	0.015	87%
Endoscopist # 2 Magnesium preparation	2.28	3:2	9.32	86%	77%	0.043	79%

© Ethical clearance was obtained from the institutional ethical committee.

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Disclosure: Nothing to disclose.

PP0888

PREVALENCE AND OUTCOMES OF SERRATED POLYPOSIS SYNDROME IN GASTROENTEROLOGY PRACTICES: A PRELIMINARY ANALYSIS

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Introduction: Serrated polyposis syndrome (SPS) is the most common polyposis syndrome found in approximately 1 in 125 screening colonoscopies and is associated with an increased risk of colorectal cancer (CRC).[1,2] Diagnosing SPS can be challenging due to cumulative polyp tracking and physician awareness.[3]

Serrated polyposis syndrome prevalence and patient outcomes, outside of established SPS cohorts, remains unclear.

Aims & Methods: We sought to explore the prevalence of SPS within academic gastroenterology practices and patient outcomes. Between 2013-2018, using an established histopathology database, patients who underwent colonoscopy with 1 of 11 academic gastroenterologists and with at least 1 serrated lesion underwent complete chart review. All patients were screened for SPS based on 2019 WHO criteria.[3]

Demographic, procedural and patient outcome variables were reviewed for all patients meeting SPS diagnostic criteria. Interim analysis (2013-2018) was conducted on the available dataset (2013-2023).

Results: From 2013-2018, 1800 patients with at least 1 serrated polyp underwent complete chart review; of which 83 (4.6%) met criteria for SPS (58 WHO I, 10 WHO II, 15 WHO I/II). Median age at first colonoscopy was

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58 years (IQR 50-68 years) and 58 (69.6%) were female. Eighteen patients (21.7%) were referred with fecal immunohistochemistry test (FIT) positivity. Only twenty-eight (33.7%) patients were diagnosed by the clinician.
The median number of colonoscopies to diagnosis was 3 (IQR 2-4) over a median of 2 years (0-6 years). On index colonoscopy, patients with SPS had a median of 3 (IQR 2-5) serrated polyps with median largest size of 10mm (IQR 5-15), whereas patients without SPS had a median of 1 (IQR 1-2) serrated polyps with median largest size of 6mm (IQR 4-8) (both p<0.001).
CRC was found in 6 (7.2%) patients, either before or at diagnosis of SPS. Twelve (14.5%) patients underwent surgical resection; 5 (6%) for CRC and 7 (8%) for polyp burden. 71 (85.5%) patients with SPS were managed endoscopically.
Conclusion: In academic gastroenterology practices, the prevalence of SPS in patients with at least 1 serrated polyp was 4.6%; with only 33.7% receiving a formal diagnosis. This highlights the need for increased awareness and strategies for cumulative polyp burden tracking.
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Disclosure: Nothing to disclose.

PP0889

ASSESSMENT AND REPORTING OF LARGE NON PEDUNCULATED COLORECTAL POLYPS OUTSIDE OF SPECIALIST CENTRES IS INCONSISTENT AND LEADS TO UNDERESTIMATION OF SIZE AND UNDER-DIAGNOSIS OF INVASIVE CANCER

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Introduction: Endoscopic resection (ER) is the preferred treatment for large non-pedunculated colorectal polyps (LNPCP) in many centres.
Despite standardized classification systems, discrepancies exist between the endoscopic assessment by experts in tertiary ER referral centres and endoscopists in peripheral settings. This significantly impacts planning for ER, leading to duplicated procedures and increased burdens on patients and healthcare.
Aims & Methods: Our aim was to compare the final morphology, surface characteristics, histological diagnosis, and lesion size of LNPCPs referred to a tertiary center for consideration of ER with the exact descriptions provided during the initial referral.
We included patients with LNPCP referred for ER at a tertiary referral centre between June 2022 and September 2023.
Clinical referrals were assessed for morphological classification (Paris/Japanese classifications), surface characteristics (pit pattern and/or vascular pattern), estimated lesion size, and preliminary endoscopic diagnosis.
We compared the endoscopic diagnosis and size with our unit's advanced endoscopic assessment. The final confirmed size of the lesion after ER was also recorded for comparison.
Results: A total of 328 LNPCP were assessed and resected following referral during the study period. The location of LNPCP was the most consistently reported feature by referring endoscopists (98.2%).
Standardised morphology according to Paris/Japanese classifications was reported in only 133 cases (40.7%). Pit pattern evaluation was reported in 109 cases (33,2%), while vascular pattern evaluation was reported in 66 cases (20,1%).
A size estimate was provided in 276 cases (84.2%). However, 51 cases (18.6%) underestimated the size by ≥20mm compared to the final LNPCP size after ER.
Regarding suspected diagnoses of invasive cancer, 50 cases (15.3%) were suspected, of which 15 had a final diagnosis of invasive cancer. In 277 cases (84.7%) with no suspicion of invasive cancer, 15 had a final diagnosis of invasive cancer, resulting in a sensitivity of 50% and specificity of 88% for a final diagnosis of invasive cancer.
After advanced endoscopic assessment, a diagnosis of likely invasive cancer was made for 41 LNPCP, resulting in a sensitivity of 73% and specificity of 93% for a final diagnosis of invasive cancer. This sensitivity is however most likely an underestimation, due to the number of lesions that were previously manipulated (biopsies, attempted resection), complicating surface characterization.
Conclusion: Despite broad agreement on standardised reporting and classifications for endoscopic diagnosis, our study reveals significant deficiencies in the assessment of LNPCP outside of specialist centres. The majority of referred LNPCP lacked a morphological classification and detailed interrogation of surface characteristics.
These shortcomings, coupled with the limited sensitivity for diagnosing potential invasive cancer, hinder the ability to plan advanced resection without subjecting patients to a further procedure for lesion assessment, with its attendant patient and institutional burdens.
Moving forward, efforts should be made to improve the quality and consistency of LNPCP assessment in non-specialist settings. This may involve enhanced training and education for endoscopists, as well as the development of standardized protocols for LNPCP evaluation and referral. Further research is warranted to explore optimal strategies for improving diagnostic accuracy and streamlining patient management in this context.
Disclosure: Nothing to disclose.

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PP0890

COLD VS HOT SNARE ENDOSCOPIC MUCOSAL RESECTION FOR LARGE (≥15MM) FLAT NON-PEDUNCULATED COLORECTAL POLYPS: A RANDOMIZED CONTROLLED TRIAL

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Introduction: Endoscopic mucosal resection (EMR) is standard of care for the management of large (≥20mm) non-pedunculated colonic polyps (LNPCPs). Its efficacy and cost effectiveness were limited by recurrence. Margin thermal ablation (MTA) to the post-EMR defect, has greatly mitigated this problem. However, electrocautery-related complications of delayed bleeding and perforation incur significant morbidity.

Cold snare polypectomy is effective and safe for the resection of small polyps with negligible recurrence rates. Application of cold snare resection to LNPCPs, while an attractive alternative, lacks high-quality evidence. We conducted a randomized trial to compare the efficacy and safety of cold EMR (C-EMR) to conventional (H-EMR).

Aims & Methods: A prospective single-centre randomized controlled trial was conducted at an Australian tertiary referral centre for flat, 15-50mm adenomatous LNPCPs referred for endoscopic resection. Prior to resection, eligible LNPCPs were randomly assigned to EMR using a dedicated cold snare or conventional H-EMR with margin thermal ablation (control). First surveillance colonoscopy (SC1) was conducted 6 months following resection. Scars were evaluated with high-definition white-light, narrow-band imaging and biopsy. The primary outcome was endoscopically visible and/or histologically confirmed recurrence at SC1. Secondary outcomes were clinically significant post-EMR bleeding (CSPEB), delayed perforation and technical success.

Results: 931 LNPCPs were referred for endoscopic resection between November 2019-September 2023. 177 LNPCPs in 177 patients were randomized to either the C-EMR arm (n=87) or H-EMR arm (n=90). Treatment groups were equivalent with no significant difference in lesion size or technical success (86/87 (98.9%) vs 90/90 (100%); p=0.31). One C-EMR required conversion to H-EMR due to unexpected submucosal fibrosis. Following exclusions, 146 LNPCPs undertook SC1 (76 C-EMR, 70 H-EMR). 5 patients await surveillance. On intention to treat and per protocol analysis, recurrence was significantly greater in C-EMR (16/87, 18.4% vs 1/90, 1.1%; p<0.001 and 16/76, 21.1% vs 1/70, 1.4%; p<0.001 respectively). Significant deep mural injury (27/90 (30%) vs 0; p<0.001) and delayed perforation (1/90 (1.1%) vs 0; p=0.32) only occurred in the H-EMR group. CSPEB and intraprocedural bleeding were significantly greater in the H-EMR arm (7/90 (7.8%) vs 1/87 (1.1%); p = 0.034 and 12/90 (13.3%) vs C-EMR arm 1/87 (1.1%); p=0.002 respectively).

Conclusion: Compared to conventional H-EMR, C-EMR for flat, adenomatous LNPCPs, demonstrates superior safety with equivalent technical success. However, endoscopic recurrence is significantly greater for cold snare resection and is currently a limitation of the technique.

Disclosure: Michael Bourke: Research Support: Olympus Medical, Cook Medical, Boston Scientific

The remaining authors have no financial, professional, or personal conflicts of interest to disclose.

PP0891

IMPACT OF MARGIN THERMAL ABLATION ON RECURRENCE AFTER ENDOSCOPIC MUCOSAL RESECTION OF LARGE (>/-40MM) COLORECTAL LESIONS

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Introduction: Endoscopic mucosal resection (EMR) of large (>/-20mm) non-pedunculated colorectal polyps (LNPCP) </- 40mm is associated with an increased risk of recurrent or residual adenoma (RRA). Margin Thermal Ablation to the EMR defect margin using snare-tip soft coagulation (MTA) reduces the rate of RRA. We sought to analyse its effect in a large prospective cohort of LNPCP >/-40mm.

Aims & Methods: We retrospectively analysed a prospectively collected cohort of consecutive patients who underwent EMR in an expert tissue resection centre. MTA technique was developed in 2012. Enrolment for the MTA randomised control occurred between July 2013-May 2016 and became standard practice from mid-2017. The cohort was divided into three phases: 'pre-MTA'; April 2008- June 2012, 'MTA adoption'; July 2012-June 2017 and 'standardised MTA'; July 2017-April 2023. The primary outcome was to compare the RRA rate between the three time periods.

Results: Over 15 years until April 2023, 973 LNPCP >/-40mm in 973 patients underwent EMR (median lesion size 50mm, IQR 20mm) of which 893 (92%) were successful. 824 patients (median lesion size 50mm, IQR 20) underwent SC1 (median of 6 months [IQR 4]). The rates of RRA in the pre-MTA, MTA-adoption and standardised-MTA phases were 20.3% (n=45/221), 19.1% (n=59/309) and 4.4% (n=13/294) respectively (p=<0.001). The rates of complete MTA in the pre-MTA, MTA adoption and standardised-MTA phases were 0, 38.2% (n=118/309) and 93.9% (n=276/294). 18 (6.1%) of cases within the standardised-MTA did not receive MTA due to difficult access (n=7), concern for stricture formation in fully circumferential LNPCP (n=9) or sedation-related issues (n=2). The rate of RRA in this group was 22.2% (n=4/18).

	Pre-MTA N=221	MTA-adoption N=309	Standardised-MTA N=294	p
Age, years median (IQR)	69 (16)	69 (15)	67 (15)	0.06
Lesion size, mm median (IQR)	50 (20)	50 (20)	50 (20)	0.17
Location n (%)				
• Right*	123 (55.7)	165 (53.4)	163 (55.4)	0.85
• Left	98	144	131	
RRA n (%)	45 (20.3)	59 (19.1)	13 (4.4)	<0.001
Complete MTA n (%)	0	118 (38.2)	276 (93.9)	<0.001
RRA in MTA Cohort n (%)	NA	3 (2.5)	9 (3.3)	<0.001
RRA in no MTA Cohort n (%)	45 (20.3)	56 (29.3)	4 (22.2)	0.11

*The right colon was defined as proximal to the splenic flexure

Conclusion: MTA post-EMR of LNPCP >/-40mm (median size 50mm) significantly reduced the rate of RRA in a large, prospective cohort. When MTA was not applied, RRA was similar across the three time periods (p=0.1), indicating that the main impact on RRA reduction was the use of MTA with snare-tip soft coagulation, rather than a factor of EMR technique improvement over time.

Disclosure: Nothing to disclose.

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PP0892

ROUTINE APPLICATION OF THE SYDNEY DEEP MURAL INJURY CLASSIFICATION SYSTEM ELIMINATES DELAYED PERFORATION

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Introduction: Delayed perforation remains the most dreaded complication following endoscopic mucosal resection (EMR) of large (>/-20mm) non-pedunculated colorectal polyps (LNPCP) and is reported with a frequency of 1.1-3.6% (1,2). It is associated with prolonged hospitalisation and high rates of surgery and stoma formation. The Sydney Deep Mural Injury (DMI) Classification System (3) was first published in July 2016 and we sought to evaluate its subsequent impact on the frequency of delayed perforation

Aims & Methods: A prospective cohort of consecutive patients who underwent colonic EMR in an expert tissue resection centre was evaluated. Following EMR, the standard practice included methodical defect assessment, DMI classification and clip closure of DMI Type 2 or greater. DMI type 3 or greater was defined as significant DMI. Clip closure was performed using through-the-scope clips, targeting the area of DMI rather than complete defect closure.

Clinical follow-up by way of a structured telephone interview was conducted at two weeks to screen for adverse events. Routine endoscopic surveillance was performed 6 months post EMR. Data collected included patient and lesion characteristics, the rate of DMI by sub-type, rates of clip closure and the rate of delayed perforation.

Results: Between July 2016 and June 2023 (84 months), 1069 LNPCP underwent EMR (median size 35mm, IQR 20). The rates of DMI, clip closure and perforation by DMI subtype are shown in Table 1. Two (2/1069 [0.19%]) delayed perforations were recorded.

Case 1 resulted after EMR of a 50mm LNPCP involving the full circumference of the ileocaecal valve; the fat component within the defect created difficulty in accurately assessing the defect for DMI.

The patient was managed conservatively. Case 2 involved a 40mm LNPCP in the ascending colon. No significant DMI was identified after EMR. Clip placement to prevent delayed bleeding was performed with the colon fully distended, creating a shearing effect on the muscularis propria.

The patient experienced post-procedural pain and cross-sectional imaging revealed gas and contrast leak. Right hemicolectomy with primary anastomosis was performed. One clip prong had penetrated beyond the muscularis propria.

Of the cases with significant DMI, 96% (n=47/49) were discharged on the same day. Successful endoscopic closure was achieved in 48 (98%). One case of DMI type III was not closed as the defect was located in the distal rectum, distal to the peritoneal reflection; there were no adverse events.

DMI Type	N (%)	Clip Closure n (%)	Delayed perforation n (%)
0	778 (72.8)	147 (18.9)	2 (0.25)
1	60 (5.6)	7 (11.6)	0
2	182 (17.0)	165 (90.7)	0
3	32 (3.0)	31 (96.9)	0
4	17 (1.6)	17(100)	0

Table 1: Rates of intraprocedural and delayed perforation by Sydney Deep Mural Injury (DMI) sub-type in 1069 LNPCP which underwent conventional EMR between July 2016 and June 2023

Conclusion: Meticulous defect interrogation, accurate DMI classification and targeted clip closure of significant DMI resulted in a low rate (<0.2%) of delayed perforation in an expert tissue resection centre. Universal application of the Sydney DMI Classification system greatly attenuates the risk of delayed perforation.

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PP0893

RISK OF SUBMUCOSAL INVASIVE CANCER IN NON-GRANULAR LATERALLY SPREADING TUMORS (LST-NGS) IN A WESTERN TERTIARY REFERRAL CENTRE EXPERIENCE

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Introduction: Laterally spreading colorectal tumors (LSTs) are defined as ≥10 mm non polypoid lesions with lateral growth, representing between 1% and 6% of all colorectal lesions.

The clinical relevance of LSTs is strictly associated with the risk of covert (ie, without clear signs of invasion) submucosal invasive cancer (SMIC) due to its implications on both the endoscopic approach and the need of subsequent surgery.

Aims & Methods: According to European guidelines, Non-Granular Laterally Spreading Tumors (LSTNGs) require en-bloc resection due to the increased risk of covert submucosal Invasive cancer (SMIC). The overall reported SMIC of LST-NGs is 11.7 %, varying from near 5% for flat elevated (FE) to near 30% for pseudodepressed (PD) subtypes.

However, this risk stratification is mainly based on data from Asian cohorts, while Western data on LST-NGs management are heterogeneous and poor.

Thus, we aimed to evaluate risk stratification of SMIC in a Western cohort of LSTNGs according to endoscopical features.

All LST-NGs evaluated for resection in our academic tertiary referral centre from 2016 to 2022 and prospectively collected in a dedicated database (comprehending clinical, endoscopical and histological characteristics) were analysed. Univariate and multivariate analysis were conducted to assess predictors of covert SMIC.

Results: We included 197 LST-NGs, 149 (75.6%) FE and 48 (24.4%) PD. Mean age of patients (F: 39%) was 69.1±8.8 (yrs±SD). Mean lesions size was 22.4±8.6 (mm±SD).

The LSTs location was 149 (75.6%) in Right colon, 33 (16.8%) in Left colon and 15 (7.6%) in the Rectum. Vascular pattern was JNET 2A in 162 (82.2%), JNET 2B in 35 (17.8%), while pit pattern was invasive (Kudo Vi) in 30 (15.2%) and non-invasive (Kudo II-III) in 167 (84.8%).

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SMIC was found in 22 (11.2%) of LST-NGs, 17 (77.3%) were PD and 5 (22.7%) were FE. Of these, 8 (36.4%) underwent surgery upfront and 14 (63.6%) were treated endoscopically, with an en-bloc resection in 85.7% of cases. Statistically significant predictive factors of SMIC were found to be: size >20 mm [OR (95%CI): 4.55 (1.62; 16.2)], Rectal location [OR (95%CI): 6.92 (2.09; 21.8)], PD subtype [OR (95%CI): 15.8 (5.77; 51.0)], Kudo Vi [OR (95%CI): 42.4 (14.4; 147)] and JNET 2B [OR (95%CI): 29.7 (10.4; 99.4)].

At multivariate analysis, Kudo Vi was the only independent predictor of SMIC.

Conclusion: Our data confirm that in LST-NGs size >20 mm, Rectal location, PD morphology, Kudo Vi and JNET2B are strongly associated with increased risk of SMIC, comparable to what shown in Eastern series. Furthermore, these data underline how an accurate optical diagnosis is crucial for establishing the best therapeutic approach.

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PP0894

ENDOSCOPIC FULL THICKNESS RESECTION IS A VIABLE ALTERNATIVE TO ENDOSCOPIC SUBMUCOSAL DISSECTION (ESD) FOR HIGH-RISK COLONIC LESIONS

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Introduction: Endoscopic submucosal dissection (ESD) is the primary resection modality recommended for high-risk colonic lesions to ensure an R0 resection. Endoscopic full-thickness resection (EFTR) is a viable alternative for complete resection of high-risk lesions. However there is little available data on endoscopic cure, size limitations and long-term outcomes following EFTR.

Aims & Methods: We aimed to evaluate long-term outcomes of patients with lesions containing high-grade dysplasia (HGD) and invasive cancer, resected by EFTR.

Consecutive non-lifting lesions referred to a single tertiary center with suspected localised advanced pathology for consideration of EFTR were prospectively studied. Multiple data points were recorded including technical success, EFTR histopathological data, complications, and follow-up surveillance data.

Results: Over a 4 year period to July 2023, 39 lesions were referred of which 33 (85%) underwent EFTR.

18/33 (55%) lesions contained cancer and 15/33 (45%) lesions contained HGD only. The median size of cancer was 7mm [Interquartile range (IQR) 4-10mm] contained within a median lesion size of 30mm (IQR 21-31mm). The median depth of invasion was 2550µm (IQR 1625-3075µm) and the deepest resected layer contained sub-serosal fat. Of the lesions containing cancer removed by FTR, 17/18 (94%) had an R0 resection. The case without an R0 resection had deep margin involvement, resected to the sub-serosa (depth 4000µm).

At 6 month [Interquartile range (IQR) 5-7 months] surveillance colonoscopy, there was 1/23 (4%) case with recurrence, successfully re-resected with FTR, with cure confirmed on second surveillance colonoscopy. Of the patients with cancer who were referred for surgical resection post-FTR, no (0/7, 0%) cases had residual cancer within the surgical specimen.

Conclusion: E-FTR is safe and effective for the resection of localised colorectal cancer. EFTR offers a viable alternative to ESD for lesions with advanced pathology. Compared to ESD, it has a shorter learning curve, is less resource intensive and has similar curative outcomes. Resection options need to be individualised and clearly outlined to the patient.

Disclosure: No conflicts.

PP0895

STRUCTURED ENDOSCOPIC TIP CONTROL TRAINING – AN EX-VIVO MODEL OF SNARE TIP SOFT COAGULATION IMPROVES PARTICIPANT TIP CONTROL (HAM-2 STUDY)

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Introduction: There are no objective performance metrics for basic endoscopic skills to guide training and career trajectory. The fidelity of endoscopic tip manipulation is a key skill akin to hand steadiness in surgery. Two recent studies using ex-vivo[1] and in-vivo[2] models of snare tip soft coagulation (STSC) applied to a polypectomy defect margin reliably stratified endoscopists by interventional profile, number, and complexity of polypectomies performed.

Aims & Methods: Tip-control of endoscopists with varied expertise was assessed using a STSC model containing 3D-printed figures constructed from 1-4 perfect intersecting circles (radius=17.2mm) (shape 1=1 circle, shape 4=4 circles) printed onto the centre of a single piece of cooked ham. The study consisted of two assessments (=all 4 shapes, maximum 5 minutes/shape), 1 initial, two sessions of training on the model, and a second final after the training. Videos of assessments were rated via a web application for accuracy and speed (=correct hits/second) by 6 blinded raters in a random order using a web application. Each video was rated three times by three different raters. Accuracy and speed were determined and

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mutated into a composite-score (TIP-score) using Z-score normalization (score-mean/sd), added together with the addition of 5 to the result removing negative values.

Results: 13 endoscopists participated (1 expert, 7 endoscopy fellows, 5 trainees) performing 8 procedures each. 312 assessments of 104 STSC videos were performed. Overall median speed [pre 0.25 vs. post 0.34 correct hits/sec; P<0.001] and TIP-score [pre 4.6 vs. post 5.4; P<0.001] were significantly increased after training but accuracy was not [pre 93.8% vs. post 94.7; P=0.1].

Fellows had an overall accuracy of 94.2% (vs. 97.6% expert [P <0.001], vs. 94.1% trainees [P <0.31]), a speed of 0.29 correct hits/sec (vs. 0.55 expert [P <0.001], vs. 0.27 trainees [P <0.001]), and a TIP-score of 5.05 (vs. 7.1% expert [P <0.001], vs. 4.9% trainees [P <0.02]). The expert showed worse accuracy (P=.006), and no change in speed (P=.1) or TIP-score (P=.9) post-training.

Fellows and trainees showed no change in accuracy (P=.06, .10) but significantly increased speed (both P<.001) and TIP-score (both P<.001) post-training. Compared to trainees and fellows the expert was more accurate than fellows and trainees pre-training (P<.001) but not post- (P=.40).

Expert speed and TIP-scores were significantly superior to trainees (all P<.001) and fellows (all P<.001) in both pre- and post-training. Fellows were significantly superior to trainees in speed pre+post (P=.03 & <.001) and TIP-score only post-training (P=.05). The expert showed a significant decrease in accuracy pre- vs post- readings indicating that performance can vary in a single individual at different time points.

	PRE	POST		PRE	POST		PRE	post	
	ACCURACY % (IQR)	ACCURACY % (IQR)	P	SPEED correct hits/s (IQR)	SPEED correct hits/s (IQR)	P	TIP-score (IQR)	TIP-score (IQR)	P
Overall	93.8 (6.4)	94.7 (4.9)	.10	.25 (.10)	.34 (.16)	<.001	4.6 (1.7)	5.4 (1.7)	<.001
Expert	99.1 (1.9)	95.4 (3.1)	.006	.53 (.14)	.56 (.29)	.10	7.3 (1.3)	7.1 (1.6)	.90
Fellow	93.7 (6.5)	94.6 (5.8)	.06	.26 (.08)	.36 (.18)	<.001	4.7 (1.3)	5.4 (1.8)	<.001
Trainee	92.9 (5.1)	95.2 (5.5)	.10	.23 (.08)	.30 (.08)	<.001	4.3 (1.3)	5.2 (1.1)	<.001

Conclusion: An ex-vivo model to assess tip-control was found to stratify endoscopists by expertise. Furthermore, performing two structured training sessions using the model led to a significant increase in the tip-control of fellows, and trainees but not experts. These results have implications for standardising the assessment of endoscopic competency, benchmarking progress during training, and allowing expedited acquisition of complex endoscopic skills in an ex vivo setting.

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PP0896

USE OF HIGH-RESOLUTION COLONOSCOPY TO ACCURATELY MONITOR DISEASE PROGRESSION IN A MURINE MODEL OF COLITIS-ASSOCIATED COLORECTAL CANCER

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Introduction: Colonoscopy is a gold standard technique to survey and diagnose various diseases of the colon and rectum, including inflammatory bowel diseases (IBD) and colorectal cancer (CRC). Numerous mouse models of IBD and CRC have been developed to understand disease pathogenesis, monitor response to novel therapies and for pre-operative assessment, though often require euthanasia at various timepoints for monitoring. The azoxymethane (AOM)-dextran sulphate sodium (DSS) model of colitis-associated CRC accelerates the progression of tumour development, inducing colonic tumours that share the histopathological characteristics of human CRC.

Aims & Methods: Herein, we review techniques for monitoring colitis and tumour progression in AOM/DSS-induced colitis associated-CRC in mice. Female C57BL/6 mice were injected with AOM (7.4mg/kg; intraperitoneal) followed by *ad libitum* access to three DSS (7 days)/water (14 days) cycles. Colitis and tumour progression were monitored by colonoscopy on days 20, 41 and 62 using the Coloview miniendoscope system (Karl Storz, Tuttlingen, Germany), which includes a rigid straight forward colonoscope (1.9mm diameter) with operating sheath, air pump for insufflation, light source, camera and video monitor. The colonoscope was advanced to the splenic flexure at approximately 4cm and videos recorded upon withdrawal. Videos were assessed for colitis severity including colon thickening, vasculature pattern, fibrin, mucosal surface granularity and stool consistency, and colonic tumours counted. On the day of euthanasia (day 63), a total colectomy was performed and longitudinally opened to visualise and photograph tumours (Canon, Tokyo, Japan). Tumour number and size were determined using Olympus Soft Imaging Solutions software (Tokyo, Japan).

Results: In healthy control mice, there was an unobstructed circumferential view of the colonic mucosal surface which consistently appeared translucent (thin), shiny and smooth, without evidence of prominent vasculature or disrupted patterns; visualised stool was formed and mobile. In AOM/DSS mice, signs of mucosal inflammation were present from day 20, including less translucency hence mucosal thickening, loss of vasculature pattern, occasional bleeding, adherence of loose stool and evidence of small tumour development (approximately 10 per mouse). By day 40, mice demonstrated mucosal surface granularity with complete loss of translucency, fibrin development, bleeding and diarrhoea, often impairing circumferential view; this required intermittent and careful colonic clearance with saline. On day 62, visualisation was largely impaired in the setting of ongoing diarrhoea and near-impassable tumours, prompting careful navigation around tumours and colonic clearance using saline. On the day of euthanasia, direct visualisation of longitudinally opened colons

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revealed tumours concentrated in the rectum and distal colon; tumour numbers directly correlated with colonoscopic assessment, highlighting that colonic tumours were confined to the anatomical location distal to the splenic flexure.

Conclusion: High-resolution colonoscopy represents an important, accurate and cost-effective technique for repeated monitoring of colitis activity and CRC development in mice. Advanced assessment can also be employed using temporal biopsies to analyse immunological and molecular parameters. Moreover, this technique lends itself to the evaluation of other colorectal diseases, including diverticular and fistulating disease, to promote the development of potential new therapies.

Disclosure: Nothing to disclose.

PP0897

EVALUATION OF A TRAINING PLATFORM USING AI-GENERATED IMAGES OF COLORECTAL POLYPS

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Introduction: To prevent colorectal cancer, it is crucial to accurately identify and classify precancerous colonic lesions. Previous studies have shown that the classification of precancerous lesions still depends on the individual examiner. In this study, we attempted to create realistic AI-generated images for training physicians in the assessment of colorectal lesions. Artificially generated realistic images make it independent of patient data, allowing to specifically create various precancerous lesions and practice their classification.

Aims & Methods: We trained an AI using 25 million endoscopic images, that can generate images of different precancerous colorectal lesions. To show that the AI is capable of producing realistic-looking images, we designed a study in which we presented artificial polyps and real images of polyps to eleven physicians and asked them to discriminate between them. They were also asked about their level of confidence for the assignment. We used 20 real and 20 AI generated images. We included five experts and six beginners in this study. Beginners were defined as endoscopists who have performed less than 100 examinations or have been practicing for less than 2 years.

Results: Real images were identified as such with an accuracy of 79%, while only 66% of AI-generated images were recognized as artificially generated. The p-value is 0.0034. The physicians had a higher confidence when choosing an answer for real images (63%) and a lower for fake images (56%). Especially the AI-generated images that clearly showed mucosal vessels were classified as real images by many physicians.

Conclusion: In this study we were able to generate synthetic images with a high degree of authenticity. We have identified features that make images appear particularly realistic and we will continue to train the AI with these specific characteristics to achieve even higher quality. The images that were identified as most realistic will be used in further projects to classify precancerous lesions.

Disclosure: Nothing to disclose.

PP0898

WITHDRAWN

PP0899

ENDOSTYLE: ENDOSCOPIC IMAGE STYLE TRANSFER TO CUSTOMIZE DATA FOR ARTIFICIAL INTELLIGENCE MODELS

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Introduction: The use of artificial intelligence (AI) systems in gastrointestinal endoscopy has become increasingly common. However, these systems are typically trained using data sourced exclusively from a single type of processor, limiting their use. In addition, the availability of expertly annotated images is limited in the medical field, which reduces the variability of the data and consequently the performance of AI systems.

Aims & Methods: The objective of this study is to develop and validate a novel AI-based system (EndoStyle) capable of generating multiple endoscopic images from a single source, simulating images captured by different processors from various manufacturers (Olympus EZ1500, Olympus CV-190, Olympus CV-170, Pentax EPK-i7000, and Storz-s1). Leveraging the StarGAN-v2 architecture, our AI model was fine-tuned using 239 875 images of 5 different endoscopic processors. Data was split into 90% for training and 10% for validation. The Fréchet Inception Distance (FID) was used as a metric to assess the quality of images created by the generative model. To further evaluate the performance of the developed AI model, 10 experienced endoscopists initially visualized 28 ten-second colonoscopy sequences in the testing phase. Subsequently, they were presented with three images and tasked with identifying those taken during the same colonoscopy procedure they viewed in the video. These images comprised images from another processor modified to resemble the processor of the video sequence (EndoStyle group), images from the same video sequence (positive control group), and images from distinct processors (negative control group).

Results: EndoStyle obtained an FID score of 38.5 during the validation phase (FID < 50 is considered excellent). In the testing phase, endoscopists selected in 60.1 % (220/366) that the Endostyle group image belonged to the displayed video. The positive and negative control groups were selected in 86.8 % (243/280) and in 4.6 % (9/185), respectively. The proportion of selected images in the Endostyle group was significantly different to the proportion of the positive and negative control group (both $p<0.05$).

Conclusion: This study demonstrates the potential of EndoStyle to mimic the visual characteristics of different endoscopic processors. The results suggest that EndoStyle successfully replicated the visual attributes of the targeted processor, as confirmed by endoscopists' selections. EndoStyle allows customizing the style of data to train AI models for a particular system. In addition, it could function as a data augmentation method to increase the variability, and hence the performance, of new AI models.

Disclosure: Nothing to disclose.

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PP0900

AI-BASED PIPELINE FOR POLYP DETECTION AND IMAGE RETRIEVAL FROM COLONOSCOPY EXAMINATIONS

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Introduction: In colonoscopy reports, the total number of polyps discovered and the characteristics of the polyp images are essential data in determining follow-up timing and treatment. Using artificial intelligence (AI) to differentiate individual colorectal polyps may potentially provide the needed information and reduce workload during examinations, especially in cases with multiple polyps. However, there is currently no widely adopted solution to accomplish this goal.

Aims & Methods: In this study, we built an AI-based pipeline to discriminate frames with polyps from colonoscopy videos. The pipeline consists of a Neural Network to compute features of each image and a hierarchical density-based spatial clustering method to group the features. Through this pipeline, we intend to count the number of polyps appearing in colonoscopy videos. The AI model was trained on 398 full-length colonoscopy videos with frames containing polyps and was validated on 44 videos. These videos were recorded in 7 different endoscopy centers from November 2021 to November 2022. The pipeline was tested in 50 colonoscopies, which contained a total of 83 different polyps. Two evaluation metrics are used: the error in the number of polyps, to quantify the disparity between the actual and predicted numbers of polyps; and precision, to evaluate the accuracy of retrieving individual polyp images.

Results: Testing on frames with visible polyps obtained from 50 colonoscopy examinations demonstrated that the pipeline achieved an average error in the number of polyps of 1.3. And the average precision was 0.66, representing the probability of retrieving the corresponding individual polyp images. Among these examinations, there are 14 examinations with more than one polyp, which have an average error in the number of polyps of 1.1 and an average precision of 0.55.

Conclusion: We have developed a pipeline that utilizes artificial intelligence techniques to identify individual polyps and retrieve representative polyp images. Despite the pipeline encountering challenges in cases with only one polyp, it still holds potential. Overall, this approach provides a promising direction for retrieving individual polyp images.

Disclosure: Nothing to disclose.

PP0901

WITHDRAWN

PP0902

IMPACT OF BOWEL PREPARATION QUALITY ON COLONOSCOPY FINDINGS AND COLORECTAL CANCER DEATHS IN A NATION-WIDE COLORECTAL CANCER SCREENING PROGRAM

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Introduction: Adequate bowel preparation is paramount for a high-quality screening colonoscopy. Despite the importance of adequate bowel preparation, there is a lack of large studies that associated the degree of bowel preparation with long-term colorectal cancer outcomes in screening patients.

Aims & Methods: In a large population-based screening program database in Austria, quality of bowel preparation was estimated according to the Aronchick scale by the endoscopist (excellent, good, fair, poor and inadequate bowel preparation). We used logistic regression to assess the influence of bowel preparation on the detection of different polyp types and the inter-physician variation in bowel preparation scoring. Time to event analyses were performed to investigate the association of bowel preparation with post-colonoscopy colorectal cancer (PCCRC) death.

Results: 335466 colonoscopies between 01/2012 and follow-up until 12/2022 were eligible for the analyses. As compared to excellent bowel preparation, adenoma detection was not significantly lower for good bowel preparation (OR 1.01, 95% CI 0.9971 - 1.0329, p = 0.1023), however, adenoma detection was significantly lower in fair bowel preparation (OR 0.97, 95% CI 0.9408 - 0.9939, p = 0.0166). Individuals who had fair or lower bowel preparation at screening colonoscopy had significantly higher hazards for PCCRC death (HR for fair bowel preparation 2.56, 95% CI 1.67 - 3.94, p <0.001).

Conclusion: Fair bowel preparation on the Aronchick scale was not only associated with a lower adenoma detection probability, but also with increased risk for PCCRC death. Efforts should be made to increase bowel cleansing above *fair* scores.

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All other authors have no conflict of interest to declare.

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PP0903

COMPARATIVE OPTICAL DIAGNOSIS PERFORMANCE OF AUTONOMOUS COMPUTER-AIDED DIAGNOSIS (CADX) VERSUS HUMAN-ASSISTED BY CADX. A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Computer-aided diagnosis (CADx) allows the real-time prediction of polyp histology during colonoscopy based on their appearance. This systematic review aims to compare the diagnostic performance of two future CADx utilization: the autonomous CADx strategy (CADx-alone) and the endoscopist with CADx assistance strategy (CADx-assisted).

Aims & Methods: We searched MEDLINE, EMBASE, and Scopus databases published before 20 March 2024. We screened for histologically-verified accuracy diagnostic studies that compared the real-time optical diagnosis performance of CADx-alone strategy versus CADx-assisted strategy for polyps ≤5mm. Only high confidence optical diagnosis were condsidered for the CADx-assisted optical diagnosis. A random effect model was used to calculated the pooled sensitivity, specificity, negative predictive value, positive predictive value and accuracy. Two separate analyses were performed: one on polyps in the whole colon and another restricted to rectosigmoid polyps.

Results: The analysis was based on seven studies. The entire colon, included 5,765 polyps for CADx-alone and 5,009 polyps for CADx-assisted; CADx-alone showed a significant reduction in sensitivity (88% vs. 93%; RR: 0.95; 95% CI: 0.92, 0.99; I^2: 79.46%), negative predictive value (NPV) (85% vs. 90%; RR: 0.94; 95% CI: 0.91, 0.98; I^2: 74.96%), PPV (81% vs. 83%; RR: 0.98; 95% CI: 0.96, 1.00; I^2: 0%) and overall accuracy (84% vs. 86%; RR: 0.96; 95% CI: 0.95, 0.98; I^2: 0%). However, there was no significant difference in specificity (74% vs. 75%; RR: 0.98; 95% CI: 0.94, 1.02; I^2: 55.78%). When the analysis was restricted to rectosigmoid polyps, which included 2,515 polyps evaluated by CADx-alone and 2,153 by CADx-assisted, CADx-alone showed significantly lower specificity (83% vs. 85%; RR: 0.97; 95% CI: 0.95, 1.00; I^2: 26.36%), PPV (73% vs. 79%; RR: 0.94; 95% CI: 0.89, 1.00; I^2: 15.50%) and overall accuracy (86% vs. 88%; RR: 0.97; 95% CI: 0.95, 0.98; I^2: 0%). No significant difference was observed in sensitivity (87% vs. 92%; RR: 0.96; 95% CI: 0.92, 1.01; I^2: 33.60%) and NPV (94% vs. 95%; RR: 0.99; 95% CI: 0.97, 1.00; I^2: 0%)

Conclusion: Our findings show that autonomous CADx failed to reach the CADx-assisted sensitivity, NPV, PPV and accuracy in the whole colon. When restricted to rectosigmoid polyps, autonomous CADx failed to reach specificity, PPV and accuracy. These findings underscore the necessity for further enhancements in the autonomous CADx diagnostic performance and the need for further studies that investigate the human-machine interaction.

Disclosure: Cesare Hassan: Fujifilm Co. (consultancy); Medtronic Co. (consultancy), Yuichi Mori: Olympus Corp (consultancy, speaking honorarium, equipment loan); Cybernet System (ownership interest), Alessandro Repici: Fujifilm Co. (consultancy); Olympus Corp (consultancy); Medtronic Co. (consultancy).

PP0904

REASONS FOR INCOMPLETE COLONOSCOPIES DURING SCREENING COLONOSCOPY

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Introduction: Screening colonoscopy is effective in preventing colorectal cancer by removing precancerous lesions. Incomplete examinations are associated with a higher risk of post-colonoscopy colorectal cancer. Therefore, the minimum standard for cecal intubation rate is >90%, and the target standard >95%. However, little is known about the distribution of reasons for incomplete colonoscopy in colorectal cancer screening programs.

Aims & Methods: Based on data from the Austrian quality assurance program for screening colonoscopy, we analyzed the reasons for incomplete colonoscopies. We further investigated whether the use of sedation and the adenoma detection rate (ADR) are associated with lower rates of complete examination.

Results: 489,800 screening colonoscopies were analyzed within the Austrian quality assurance program. Of these, 2.9% (n=14,325) were incomplete. 59,7% of those patients were female and 40.3% male. 85.4% of patients with an incomplete vs. 91.2% with a complete examination were sedated (p<0.001). The reasons for not reaching the cecum were insufficient bowel preparation in 23.3% (n=3335), pain in 23% (n=3291), stenosis in 11.7% (n=1674), complications in 1.6% (n=227), and other factors in 40.5% (n=5798). 69% of incomplete colonoscopies were performed by endoscopists with an ADR lower than 25% and 31% with an ADR over 25%. 54.6% of complete examinations were done by endoscopists with an ADR lower than 25% and 45.4% with an ADR over 25% (p<0.001).

Conclusion: The target standard of the cecal intubation rate has been achieved (97.1%). Incomplete colonoscopies were more often in female (84.4%) compared to male (40.3%), performed in unsedated patients and by endoscopists with an ADR <25%. The most common reason for not reaching the cecum was categorized as other factors (40.5%), this category is not well defined and requires further investigation.

Disclosure: Nothing to disclose.

PP0905

CORRELATION BETWEEN CONECCT CLASSIFICATION AND HISTOLOGY OF COLONIC POLYPS: PRELIMINARY RESULTS OF A PROSPECTIVE STUDY

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Introduction: The CONECCT classification groups together all the predictive criteria from various classifications, enabling us both to predict the histology of polyps and to propose an appropriate therapeutic strategy. The use of this classification requires specific training for endoscopists.

Aims & Methods: This is a prospective descriptive and analytical study, spanning a period of 1 year between April 2023 and March 2024, on 68 colonic polyps identified in 50 colonoscopies performed for various indications.The CONNECT classification includes: CONECCT IH: hyperplastic

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polyp, CONECCT IS: sessile scalloped lesion with or without dysplasia, CONECCT IIA: adenoma with low or high grade dysplasia, CONECCT IIC: low- or high-grade dysplastic adenoma or intramucosal or submucosal adenocarcinoma < 1000 microns, CONECCT IIC+ : intramucosal or submucosal adenocarcinoma of any depth, CONECCT III : deep invasive adenocarcinoma > 1000 microns in the submucosa.

The aim of our study is to evaluate the concordance between endoscopic prediction using the CONECCT classification and the definitive histology of lesions.

Results: 50 patients with 68 consecutive lesions were included in our study, the mean age of our patients was 59.29 with extremes ranging from 30 to 70 years. Our series was characterized by a male predominance of 73.5%, i.e. a sex ratio (M/F) of 0.73. The mean polyp size was 6.4 mm (3 - 20 mm), with the left colon being the most frequent site (29.4% of cases).All polyps were classified according to CONECCT, including 50 IIa, 10 IH, 4IIc, and 4 Is. Concordance between CONECCT and definitive histology was 70.5%. Correct classification was achieved for 80% of hyperplastic polyps, 75% of scalloped polyps, 68% of low-grade adenomatous polyps and 75% of high-grade adenomatous polyps.

Conclusion: The correlation between histology and CONECCT classification remains insufficient, and further technical and training resources will be required to improve polyp characterization.

Disclosure: Nothing to disclose.

PP0906

DOUBLE BALLOON ENDOSCOPIC PLATFORM FACILITATES PERFORMANCE AND ENABLES COMPLETION OF DIFFICULT COLONOSCOPY

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Introduction: Screening colonoscopy has become the gold standard in discovery and removal of colonic polyps and prevention of colon cancer. However, in 13.1-25.8% of patients colonoscopy was not completed and cecal intubation was not achieved.[1-3] DiLumen® (Lumendi LLC, Westport CT, USA) double balloon endoscopic platform was developed to facilitate endoscopic interventions in gastrointestinal tract.

Aims & Methods: Aim: To evaluate use of DiLumen® to facilitate difficult colonoscopy.

Methods: DiLumen® is a flexible overtube with 2 balloons on its distal end. One (Aft) balloon is rigidly attached to the overtube and a second (For) balloon can be advanced forward or pulled back using 2 push rods moved by a slider located on the handle of DiLumen®. The assembled platform is pre-loaded on top of any colonoscope and introduced into the rectum. The colonoscope-DiLumen® assembly is advanced through sigmoid into descending colon where Aft-balloon is distended to fix the overtube inside the colonic segment. Then DiLumen® is pulled back to straighten and to shorten the colon. The colonoscope is advanced forward till transverse colon is reached. The Aft balloon is collapsed and DiLumen® is advanced into transverse colon over the colonoscope. The Aft balloon is distended inside the transverse colon and DiLumen® is pulled back now shortening and straightening the left colon. This process is repeated till the cecum is reached.

Results: We reviewed use of DiLumen® to facilitate colonoscopy in 53 patients (25 males, 28 females, mean age 68.0±10.1 years). Prior to referral for DiLumen®-assisted procedure, one colonoscopy attempt was unsuccessful in 31 patients, 19 patients had 2 unsuccessful colonoscopies, 2 patients had 3 unsuccessful colonoscopies and 1 patient had 4 unsuccessful colonoscopy attempts. The most common reasons for failure were long, redundant, torturous colons. DiLumen® effectively straightened and shortened the colon markedly facilitating advancement of the colonoscope and allowing successful examination of the entire colon with cecal intubation in all 53 patients (technical success 100%). Mean time to reach cecum was 13.1±6.8 minutes and total procedure time was 30.6±15.2 minutes. Normal colonic mucosa was seen in 13 patients (24.5%). Majority (75.5%) of patients had significant pathology found: only 1 polyp was found in 14 patients (26.4%), 2 polyps were found in 10 patients (18.9%), 3 and more (up to 9) polyps were found in 16 patients (30.2%). The polyps ranged in size from 3 to 80 mm and were found in all colonic segments from rectum to ileocecal valve.

Conclusion: Endoscopic double balloon interventional platform markedly facilitated colonic straightening and shortening and allowed successful completion of colonoscopy with removal of previously undetected colonic polyps in patients with difficult (long, redundant and torturous) colon and unsuccessful previous colonoscopies.

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Disclosure: Nothing to disclose.

PP0907

ACCURACY OF WHITE LIGHT ENDOSCOPY VERSUS NARROW BAND IMAGING IN SCAR ASSESSMENT AFTER COLORECTAL MUCOSAL RESECTION: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction:

Endoscopic mucosal resection has been associated with lesion recurrence in 16%-27% of cases. Guidelines recommend endoscopic follow-up but it is unclear if endoscopic assessment has to include biopsies.

Aims & Methods: This study aims to evaluate the recurrence prediction of white light endoscopy (WLE) and narrow band imaging (NBI) in the assessment of colorectal mucosal resection scar. We performed a systematic review in PubMed, Embase, Cochrane and Web of Science databases, searching for studies comparing WLE and NBI. Statistical analysis included data pooling and heterogeneity testing. Quality of the included studies was assessed by Quality Assessment of Diagnostic Accuracy Studies 2 (QUADAS-2).

Results: The meta-analysis included 5 studies and 798 mucosal resections. The pooled sensitivity and specificity and AUC for WLE were 80.9% (95% CI 71.8% to 87.5%), 95.6% (95% CI 92.0% to 97.6%) and 0.917 (95%

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CI 0.885 to 0.972), respectively. The pooled sensitivity and specificity and AUC for NBI were 92.1% (95% CI 87.3 to 95.2%), 94.1% (95% CI 91.9% to 95.7%) and 0.935 (95% CI 0.772 to 0.972), respectively. There was no difference in AUC of WLE and NBI (p=0.633).

Conclusion: Our study demonstrates a high sensitivity, specificity and accuracy for endoscopic diagnosis of recurrent lesions after colorectal mucosal resection, specially with NBI. In all cases of low confidence assessment, biopsy is required.

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Disclosure: Nothing to disclose.

PP0908

ADENOMA DETECTION BY CHROMOENDOSCOPY WITH INDIGO CARMINE VERSUS ENDOCUFF: A RANDOMIZED CLINICAL TRIAL

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Introduction: The adenoma detection rate (ADR) is the percentage of screening colonoscopies that detect at least one conventional adenoma verified by pathology. International guidelines recommend a minimum ADR of 25% (1,2), for every 1% increase in ADR reducing 3% and 5% the incidence of colorectal cancer and mortality, respectively (3).

Increasing ADR under any context is highly desirable, so the primary outcome of our study is to compare the ADR using Endocuff versus indigo carmine staining.

Aims & Methods: A prospective, randomized, single-blind clinical trial was carried out on patients aged 45-85 years undergoing screening colonoscopy in a private tertiary hospital in Mexico City from December 2016 to September 2022. Patients were randomized 1:1 for Endocuff vision or indigo carmine group arms. The primary outcome was global ADR; secondary outcomes were ADR by segment, polyp detection rate (PDR) and serrated lesions detection rate (SDR).

Results: A total of 324 patients were analyzed, 163 in the Endocuff vision arm and 161 in the indigo carmine arm. The ADR was 46% (95% CI 54-38) in the Endocuff group and 52% (95% CI 60-44) in the indigo carmine group without significant difference (p=0.27), with a higher ADR in the right colon for the use of indigo carmine (p=0.06).

In the subgroup analysis, there was a higher PDR in the indigo carmine group (69 vs 79%, p=0.03) but no significant difference between groups in advanced adenomas or SDR (p=0.23)

	Endocuff vision (n=163)	Indigo carmine (n=161)	p-value
PPC, mean (SD)	2.1 (2.8)	3.4 (3.8)	0.0005
PDR, n (%)	112 (68.7)	127 (78.9)	0.03
APC, mean (SD)	0.8 (1.2)	1.4 (2.1)	0.002
ADR, n (%)	75 (46.0)	84 (52.2)	0.27
Right ADR, n (%)	40 (24.5)	55 (34.2)	0.06
Left ADR, n (%)	38 (23.3)	48 (29.8)	0.18
Advanced adenomas (10mm, villous features, high-grade dysplasia), n (%)	25 (15.3)	35 (21.7)	0.34
Carcinoma, n (%)	3 (1.8)	1 (0.6)	0.32
SDR, n (%)	4 (2.4)	8 (4.9)	0.23

Table 1. Outcomes results

APC, adenomas detected per colonoscopy; ADR, adenoma detection rate; PPC, polyps detected per colonoscopy; PDR, polyp detection rate; SDR, serrated lesions detection rate.

Conclusion: Any technique to improve ADR is useful and favorable over conventional colonoscopy, without a difference in the use of indigo carmine or Endocuff. In Mexico, ADR can be achieved as high as 46% to 52% with an added tool.

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Disclosure: Nothing to disclose.

PP0909

A NOVEL COMPUTER-AIDED DIAGNOSTIC SYSTEM BASED ON THE YOLOV8 ALGORITHM FOR DETECTING CRUCIAL QUALITY INDICATORS IN COLONOSCOPY

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Introduction: To improve the effectiveness of screening colonoscopies, it is important to maintain a high level of colonoscopy quality. We aimed to develop an artificial intelligence (AI) system to monitor key quality indicators for colonoscopy.

Aims & Methods: We implemented and tested the YOLOv8 model in various configurations (ranging from nano to extra-large) across multiple datasets, including five public datasets: Kavasir-SEG, which consists of 5597 polyp images, and the Severance Hospital colonoscopy database, which comprises 6316 images including polyps, the ileocecal valve, appendix orifice, and fecal materials. The model focused on detecting polyps and key anatomical landmarks such as the ileocecal valve and appen-

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dix orifice. Performance was evaluated based on the accuracy of detecting these features. Additionally, we investigated a novel path prediction method using a calibration plate and stereo matching.

Results: The integration of the YOLOv8 model into colonoscopy procedures has significantly improved the detection capabilities for colonic polyps, essential anatomical landmarks, and fecal materials indicative of bowel preparation quality. The precision and mAP50 for each metric were as follows: 0.90 and 0.85 for Polyp Detection, 0.99 and 0.94 for the Appendix Orifice, 0.90 and 0.82 for the Ileocecal Valve, 0.61 and 0.58 for Fecal Material, and 0.75 and 0.77 for Four-Class Detection, as detailed in Table 1. Additionally, a novel path prediction method has demonstrated potential for accurately measuring the colon path in short-term frames.

Category	Precision	Recall	mAP50	mAP50-95	Fitness
Polyp Detection	0.8993	0.8064	0.8531	0.6517	0.6719
Appendix orifice	0.9926	0.8292	0.9494	0.8559	0.8653
Ileocecal valve	0.9019	0.7716	0.8296	0.6242	0.6447
fecal material	0.6186	0.5408	0.5841	0.2996	0.3280
Four-Class Detection	0.7586	0.7618	0.7681	0.6435	0.6559

Table 1. Results of AI system for major quality indicators detection.

Conclusion: We developed a colonoscopy quality indicator monitoring system utilizing the YOLOv8 model. Further research is required to refine and validate the model's performance.

Disclosure: Nothing to disclose.

PP0910

ENDOSCOPIC INTERMUSCULAR DISSECTION (EID) AS FURTHER DEVELOPMENT OF ENDOSCOPIC SUBMUCOSAL DISSECTION (ESD) AND NEW PERSPECTIVE FOR THE ENDOSCOPIC RESECTION OF T1 RECTAL CANCERS AND HIGH-RISK LESIONS IN THE RECTUM - FIRST SINGLE-CENTER EXPERIENCE IN 8 CASES

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Introduction: In contrast to mucosal rectal T1 adeno-CA with an R0 rate of up to 92% at centers, the R0 RR in studies drops to 62-64% in case of deep sm infiltration.[1,2] Traditionally total mesorectal excision (TME) follows as completion surgery. Deep sm infiltration as isolated risk factor excluding L+, V+, G3 and Bd2-3 seems associated with a low risk of LN metastases of only 2.6%. [3] MRI and even EUS often lead to a significant T and N overstaging.[4] A min.-invasive diagn.-ther. resection procedure providing a clear histo-pathologic staging without potentially destroying parts of the meso-rectum as it has been reported for full thickness resection techniques would be desirable for potential later surgical resection (TME).[5] Endoscopic Intermuscular Dissection (EID), similar to PAEM uses a new flexible endoscopic dissection technique with separation of the circumferential and longitudinal muscular layer providing a specimen ideally including the entire mucosa, submucosa and the inner circumferential muscle layer.[6,7] The new technique achieved in a first prospective MC study including 67 patients a technical success rate of 96% and a histologic R0 resection rate of 85%.[5] In the following we report on our first experiences using EID as referral center for ESD.

Aims & Methods: To evaluate EID in lesions with biopsy proven ADC of the rectum without endoscopic, EUS or clinical signs of local infiltration. Rectal polyps with biopsy proven HGIEN and/or suspicious for local malignant infiltration. Additionally 2 patients were included who had undergone neo-adjuvant RT/Ch therapy 24 and 26 months before refusing definitive surgery (TME). Both presented with local flat polyps with histopathologically SSA (1) / hyperplastic polyp with focal LGIEN adenoma (2) for diagn.-therapeutic resection. After endoscopic, EUS, MRI, abd/th CT-staging and rectal digital examination without suspicion of local malignant infiltration pts were offered to prospectively undergo EID after extensive informed consent within a prospective protocol. All procedures were carried out under gen. anesthesia by an exp. interv. endoscopist with >1000 ESDs. Post-Op all pts. were hospitalized and closely monitored.

Results: From 11-2023 to 4-2024, 8 EIDs in 8 pts (3 f/ 5 m) were carried out (mean 63 y; 34-83y). 6/8 lesions lower rectum, 1/8 middle, 1/8 upper rectum. 'En bloc' resection was achieved in 100%. Histopathology showed 3x ADC G2, 3 x HGIEN and in both cases with neodajuvant RT/Ch LGIEN adenoma. R0 resection was achieved in 2 of the 3 ADCs, with deep sm T1 infiltration in both but without any other risk factors. Tumor board recommendation was close clinical & imaging FU. In the 3rd ADC tumor infiltration reached the inner muscular layer (T2) and the pat. underwent uneventful TME 10 d after EID. All other cases were follow clinically. Mean length of hosp. stay was 7.1 days. Serious major complications did not occur. All pts. are closely followed with uneventful FU.

Conclusion: EID seems a new perspective for the endosc. resection of T1 ADC and high-risk polyps in the rectum. Two cases after potentially curative neo-adjuvant treatment for rectal cancer and refusal of TME show EID as potential diagnostic tool but have to be regard with caution due to a remaining risk of scattered tumor cells in the mesorectum with TME remaining the gold standard. More experience in multicenter prospective studies with matched pair analysis or a comparison to surgical local procedures (TAMIS etc.) is necessary for the final assessment of EID as new promising endoscopic technique.

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PP0911

INTERVENTIONAL CLOSURE FORM COMPLEX FISTULA AND ANASTOMOTIC INSUFFICIENCIES IN THE LOWER GI TRACT USING ENDOSCOPIC SUTURING

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Introduction: Treatment of complex fistula and residual anastomotic insufficiencies remain a challenge even in an interdisciplinary setting. Endoscopic standard techniques for closure like clips, OTSC or SEMS are limited to special indications. Endoscopic suturing may offer an approach to close complex fistula orifices.

Aims & Methods: Endoscopic suturing was used for interventional closure of complex insufficiencies in the lower GI-tract. 5 consecutive patients were treated using standard Endosuturing (SutuArt, Olympus) or modified suture strategy using an additional working channel (ACEES). In three patients residual insufficiencies in the rectum were visible after complex vacuum therapy. In one patient a recto-vaginal fistula was clinical apparent. In one patient an iatrogenic perforation was occurred within dilation therapy for anastomotic stenosis. Due to size or location of the defects a standard closure technique was either not applicable or failed in a prior attempt.

Results: Technical success, defined as macroscopic defect closure, was achievable in 5 of 5 patients. Mean time for intervention was 57(±14)min. For 3 patients bimanual work using additional working channel enhanced endoscopic suturing (ACEES) was performed. In endoscopic surveillance permanent closure was achieved in 4 of 5 patients.

Conclusion: Endoscopic suturing is a newly available technique. Beyond its basic indication for defect closure after ESD this procedure, also in combination with bimanual suturing (ACEES), may offer an excellent opportunity for treatment of complex fistula.

Disclosure: Nothing to disclose.

PP0912

DIAGNOSTIC POTENTIAL OF FECAL IMMUNOCHEMICAL TESTING OUTSIDE THE NATIONAL COLORECTAL CANCER SCREENING PROGRAM IN DENMARK: A NATIONWIDE COHORT STUDY

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Introduction: Colorectal cancer (CRC) and other bowel diseases is a significant health concern globally. In several countries, the fecal immunochemical test (FIT) is a cornerstone in CRC screening for early detection of cancer in asymptomatic individuals. Moreover, the utility of FIT extends beyond this, as it is increasingly recommended for evaluating symptomatic patients as part of the clinical pathway to render possible or rule out serious bowel pathology. Therefore, we aimed to quantify the findings of serious bowel conditions following a non-screening FIT test in the Danish population.

Aims & Methods: This study investigated use of FIT beyond the national CRC screening program in Denmark. We aimed (1) to assess utilization patterns of FIT in contexts other than screening and (2) to examine ensuing diagnoses. Our goal was to quantify the findings of CRC and other bowel diseases in symptomatic patients, thereby assessing its broader potential. This cohort study was conducted within the Danish universal public healthcare system, delivered by five administrative regions. National CRC screening using FIT was initiated in 2014, but utilization of FIT for symptomatic patients varies between regions. The cut-off for a positive test is ≥10 µg haemoglobin/g faeces for symptomatic populations.

The study used data from the Danish Civil Registration System, the Danish National Patient Registry covering all Danish hospitals, and the Clinical Laboratory Information Register. The cohort included all patients who underwent FIT between 2014 and 2021 outside the Danish CRC screening program.

We employed descriptive statistics to examine FIT administration and demographic distributions across the Danish regions, as well as fecal haemoglobin levels among tested individuals. In addition, we estimated the cumulative incidence of CRC and inflammatory bowel disease (IBD) diagnoses following FIT, considering death as a competing risk and evaluated the proportion of diagnoses for CRC and IBD during the year following FIT.

Results: We identified 19,787 individuals who underwent FIT outside the national CRC screening program in 2014 to 2021. Of these, 4,283 (22%) individuals had a positive FIT result. Among tested individuals, 10,807 (55%) were female. Median age at testing was 64 years. Use of FIT varied across the regions, with 36% of the study cohort being from the Central Denmark Region, and 1.6% being from the Region of Southern Denmark.

Positive FIT cases had a shorter median time to CRC diagnosis (28 days)

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and any diagnosis (36 days) compared to negative FIT cases (711 and 58 days, respectively). Within one year of the FIT, 370 (1.9%) individuals were diagnosed with CRC, with higher occurrence for FIT positive individuals (7.9%) than for FIT negative individuals (0.2%) (risk difference: 7.7 percentage points [95% CI: -3.2, 11.1]).

Among individuals with a positive FIT result, 83 (1.9%) were diagnosed with IBD within one year, compared to 51 (0.3%) individuals with a negative FIT (risk difference: 1.6 percentage points [95% CI: -5.8, 7.3]).

Conclusion: We found that FIT used outside the national CRC screening program had over 20% positivity. A positive test indicated higher risk of CRC and IBD. Differences in diagnoses are expected due to varied follow-up procedures after positive and negative FIT results. A negative test, while not definitive, suggests a potential to rule out disease and could reduce unnecessary investigations when assessing symptomatic patients.

Disclosure: Nothing to disclose.

PP0913

INVESTIGATION OF THE EFFICACY OF TEXTURE AND COLOR ENHANCEMENT IMAGING (TXI) ON IMPROVEMENT OF COLORECTAL POLYP DETECTION

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Introduction: Identification and subsequent removal of colorectal polyps has been demonstrated to deduce the mortality of colorectal cancer. However, approximately 25% of colorectal polyps can be missed under conventional endoscopic examination using white light imaging (WLI). Texture and color enhancement imaging (TXI) is a new modality developed to improve the detection of polyps, which enhancing the color tone and mucosal structure of lesions using an image color tone algorithm. We conducted a crossover prospective randomized observational study comparing the detection rate of colorectal polyps between WLI group and TXI group.

Aims & Methods: A total of 700 patients who underwent lower gastrointestinal endoscopy at Yokohama Sakae Kyosai Hospital between June 2022 and January 2023 were included in this study. After reaching the cecum, polyps were identified while the scope was withdrawn to the hepatic flexure in a randomly assigned observation mode. The scope was reinserted into the cecum and polyps were identified using the other observation mode. After evaluation of size and morphology of colorectal polyps, they were removed endoscopically and pathological evaluation was performed. Primary outcome was polyp detection rate. We also evaluated the characteristics of missed lesions. Additionally, multivariate logistic regression analysis was performed to reveal the factors affecting the polyp detection rate.

Results: There were no significant differences in age, sex, bowel cleanliness, or cecum intubation rate between TXI and WLI groups. Except in cases of inadequate bowel preparation and discontinuation, 336 and 338 patients were included in TXI and WLI groups, respectively. Although the total number of polyps were not differed between TXI and WLI groups (226 vs 210, p=0.48), the polyp detection rate was significantly higher in TXI group (37.5% vs 28.4%, p=0.01). The adenoma detection rate was also higher in TXI group (31.8% vs 24.9%, p=0.03). Additionally, the polyp miss rate was significantly lower in the TXI group (19.9% vs 32.4%, p=0.003). No difference in polyp miss rate between the two groups for lesion sizes <5mm and 10mm>, but for lesions 5-9mm, the TXI group had a significantly lower polyp miss rate (12.3% vs 38.2%, p=0.001).

Multivariate logistic regression analysis revealed that TXI observation (OR 1.51, 95% CI 1.09-2.09, p=0.01) and older age (OR 1.71, 95% CI 1.18-2.48, p=0.004) were independent significant factors affection the colorectal polyp detection.

Prevalence and number of lesions according to study group	TXI (n = 336)	WLI (n = 338)	P value
Detected polyps, n	226	210	0.48
First inspection			
Patient with polyps, n (%)	126 (37.5%)	96 (28.4%)	0.014
Patient with adenomas, n (%)	107 (31.8%)	81 (24.9%)	0.026
The polyp miss rate	19.9%(45/226)	32.4%(68/210)	0.003
Subgroup analysis of missed lesions	Polyp miss rate (PMR)		
Polyp size	TXI	WLI	P value
5-9mm	12.3% (8/65)	38.2% (29/76)	0.001

Conclusion: TXI may improve the identification of colorectal polyps, especially for small polyps.

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Disclosure: Nothing to disclose.

PP0914

QUANTIFYING OBSERVATIONAL SKILLS FOR COLONOSCOPY USING ARTIFICIAL INTELLIGENCE

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Introduction: Recently, artificial intelligence-based computer-aided detection (CADe) for colonoscopy has garnered attention due to its effectiveness in increasing the adenoma detection rate (ADR). However, theoretically, CADe systems cannot reduce exposure errors for polyps obscured by objects. In this regard, computer-aided quality improvement (CAQ) which expects to induce the meticulous mucosal inspection and leading to a reduction of blind spot area. In this study, we developed and evaluated CAQ for colonoscopy.

Aims & Methods: This was a single-center, retrospective study conducted from April 1, 2022 to October 31, 2023. We randomly extracted videos of the entire colonoscopies and created 18,922 training images. Quadrant images were made from one image and three endoscopists annotated the following scores; i) (overall) quality of image(not evaluable 0 or evaluable 1), ii) appropriate distance between the colon mucosa and the endoscope(Scored on a scale of 0 to 8), iii) the amount of residue in the colon,(Scored on a scale of 0 to 8) and iv) presence of blind area in the field of view (e.g., fold or flexure) (Scored on a scale of 0 to 4). We conducted supervised machine learning using annotated images. The CAQ deployed multi-head deep convolutional neural network model which can output multiclass prediction from one image. Consequently, the CAQ system outputs aforementioned score from colonoscopy video. To validate its performance, we randomly selected videos from the cecum to the hepatic flexure performed by 19 endoscopists and calculated CAQ model scores.

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We compared CAQ scores between endoscopists with low and high ADR (ADR cutoff is 35%) and with and without the history of detecting early depressed-type colorectal cancer by endoscopists.

Results: 62 videos were included as validation samples. Among the 19 endoscopists, 11 had a low ADR and 8 had a high ADR. The results showed a significant difference in the rate of evaluable images between endoscopists with low and high ADR (median 70% vs. 77%, p = 0.003, t-test). There was also a significant difference in the evaluation score of the distance (median 3.8 vs. 4.2, p = 0.003, t-test). A significant difference was also obtained in the score for the amount of residue in the colon(median 5.3 vs. 5.7, p = 0.01, t-test). However, there was no significant difference in the scores evaluating the presence of blind areas(median 1.01 vs. 1.10, p = 0.13, t-test)

Among the 19 endoscopists, 5 endoscopists had a history of detecting early depressed-type colorectal cancer, while 14 did not. The results showed a significant difference was observed in the rate of evaluable images between endoscopists with and without a history of detecting early depressed-type colorectal cancer(median 80% vs. 71%, p < 0.001, t-test) . Additionally, there was a difference in the evaluation scores for the distance (median 4.5 vs. 3.8, p < 0.001, t-test). There was also a difference in the scores assessing the presence or absence of blind spots (median 1.2 vs. 1.0, p = 0.006, t-test). Moreover, a significant difference was found in the evaluation scores for the amount of residue in the colon(median 5.9 vs. 5.3, p < 0.001, t-test) .

Conclusion: The CAQ system developed in this study was able to quantify the skill of endoscopists. Such CAQ is expected to standardize the observation technique of colonoscopy and may lead to elucidation of detection techniques for significant lesions such as early depressed-type colorectal cancer.

Disclosure: Nothing to disclose.

PP0915

CLINICOPATHOLOGICAL FEATURE OF DEPRESSED-TYPE COLORECTAL NEOPLASM

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Introduction: Several development concepts have been proposed for colorectal neoplasms in the carcinogenesis of colorectal cancer. One is “adenoma-carcinoma sequence”. In this concept, polypoid adenoma develops into early polypoid cancer and finally develops to advanced cancer. One is “serrated-pathway”. In this concept, serrated lesion develops into early cancer. The other is “de novo” pathway. In this concept, early cancers develop without preexisting adenoma. We investigated clinicopathological characteristics of depressed-type colorectal carcinomas considered as “de novo” pathway.

Aims & Methods: A total of 50119 colorectal neoplasms excluding advanced cancers were resected endoscopically or surgically in our center from April 2001 to December 2023 at Showa University Northern Yokohama Hospital. Of these, 1633 lesions were T1 cancers. According to the developmental morphology classification, they were divided into 3 types: 323 lesions (20.1%) were depressed-type, 602 lesions (36.9%) were flat-type and 703 lesions (43.0%) were protruded-type. We analyzed the pathological difference and recurrence of these types.

Results: The average tumor size were 15mm, 32mm, and 18mm in depressed-type, flat-type and protruded-type lesions, respectively. The rates of vessel invasion were 42%, 21% and 21%, that of poorly differentiated or mucinous adenocarcinoma were 14%, 10% and 15% in depressed-type, flat-type and protruded-type lesions, respectively. The rates of these pathological factors were significantly higher in depressed-type lesions than other types.

On the other hand, the rates of adenomatous component were 4%, 54% and 52%, and the rates of polypoid growth were 13%, 56% and 88%, respectively. These rates were significantly lower in depressed-type lesions than other types, suggesting that they emerge directly from normal epithelium without going through the adenomatous stage. The rate of distant metastasis or recurrence was observed in 0.8% (13/1633).

Conclusion: Depressed-type colorectal carcinomas had significantly lower adenomatous components and smaller size than flat-types or protruded-types, and they also contained worth pathological features than other types. So, depressed colorectal carcinoma can be the important lesions for clinical diagnosis. Therefore, during routine colonoscopy, we should give a greater attention to the morphological features of the depressed-type colorectal carcinomas.

Disclosure: Nothing to disclose.

PP0916

ENDOSONOGRAPHY-GUIDED COLOENTEROSTOMY: A SAFE AND EFFECTIVE ALTERNATIVE FOR PALLIATIVE MANAGEMENT OF MALIGNANT SMALL-BOWEL ILEUS – PRELIMINARY RESULTS

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Introduction: Some malignant diseases such as ovarian carcinoma, neuroendocrine neoplasias or gastrointestinal cancers tend to manifest with peritoneal carcinomatosis. The accompanying small-bowel ileus (SBI) is often decisive for the prognosis even in an otherwise controlled oncologic setting while posing a significant burden on the patients quality of life (QoL). Conventional procedures of palliative care such as bypass surgery or gastric tube placement deliver unsatisfying results and are not well-tolerated by patients.

A new method for improving the quality of life and possibly survival may therefore be to restore the passage through the endosonographic (EUS) placement of lumen-apposing metal stents (LAMSs) for creation of a colo-enterostomy (CE), without opening the abdominal cavity.

Aims & Methods: Aim of this ongoing multicenter intervention study is a comprehensive evaluation of feasibility and clinical value of CE in patients with malignant SBI. Biometric, technical and clinical data, symptoms, periinterventional management, QoL (EORTC_QLQ_C30), adverse events as well as the ability to tolerate oncologic treatment and enteral nutrition are recorded longitudinally at predefined time points (d-1, d+2, d+10, d+34).

Results: We report on 10 patients that underwent EUS-guided CE so far. Eight of these patients (80%) were female and 4/10 (40%) had an intestinal neuroendocrine tumor, 4/10 (40%) an ovarian carcinoma and 2/10 (20%) colorectal cancer. The intervention was technically successful in 10/10 (100%) and clinically successful 9/10 (90%) patients. In one patient, only a second technically successful CE on day 3 could induce clinical symptom relief. Subsequently, all patients experienced considerable

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improvement of ileus-related symptoms (e.g. vomiting, stool frequency, abdominal pain) and an increase in QoL to an extent, were 10/10 (100%) were able to tolerate enteral nutrition and 6/10 (60%) could be re-exposed to systemic oncologic treatment. All 4 patients (100%) depending on morphine based pain medication prior to SBI could be reintroduced to their previous regimen after the intervention.

Conclusion: EUS-guided CE for treatment of malignant SBI is feasible and safe. The intervention boasts a high technical and clinical success rate with immediate symptom relief and improved QoL. A relevant number of patients was enabled to be reintroduced to systemic oncologic treatment and morphine based pain medication. Larger cohorts and long-term data are needed to evaluate sustainability as well as possible impact on survival and to form the foundation for prospective trials.

Disclosure: Alexander Meining: Ovesco Consultant
Alexander Weich: Novartis Pharma Gmbh Consulant.

PP0917

ENDOSCOPIC ULTRASOUND-GUIDED ENTERO-COLOSTOMY WITH LUMEN APPOSING METAL STENT AS A RESCUE TREATMENT FOR MALIGNANT INTESTINAL OCCLUSIONS: A MULTICENTRE STUDY

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Introduction: Surgery is the first-choice treatment for malignant intestinal obstruction (MIO), however many patients are deemed inoperable. Endoscopic ultrasound guided enterocolostomy (EUS-EC) with lumen apposing metal stents (LAMS) could represent a new treatment option.

Aims & Methods: The primary aim of the present multicentre retrospective study was to assess the technical success rate of EUS-EC in patients with malignant intestinal occlusion unfit for surgery. Secondary aims of the study included the clinical success rate, the safety of the procedure, the time of re-feeding, the duration of hospital stay and the clinical outcome.

All consecutive patients undergoing EUS-EC for MBO from November 2021 to September 2023 were retrospectively enrolled at four tertiary referral European centres. All cases were discussed in a multidisciplinary meeting.

Inclusion criteria were:

1) malignant small bowel/colonic obstruction;
2) CT-scan showing ileal or colonic MIO with dilated intestinal loops <7 days from procedure;
3) multidisciplinary meeting discussion;
4) unfit for surgery;
5) unsafe colonic stent positioning.

Exclusion criteria were:

1) acute MIO requiring emergency treatment;
2) unavailability of CT scan <7 days;
3)rectal malignancy.

Demographic, clinical and procedural data including the clinical outcome and adverse events were recorded. Technical success was defined as ability to deploy the LAMS between the colon and the colonic or small bowel target, clinical success as resolution of MIO for at least 2 weeks from the procedure and adverse events defined and classified according to the AGREE classification. Data were expressed as median [range]. Comparisons were made by using the $\chi 2$ test, Student t-test, or Mann-Whitney u-test.

Results: Twelve patients were enrolled (58.3% female, median age 72.5 [42-85] years). The most frequent cause of MIO was colonic adenocarcinoma in 9 (75%) patients. The majority (11 [91.7%]) of patients had a Stage IV disease and the median malignancy duration was 13 [1-108] years. Site of MIO was ileum and right colon in 3 (25%) patients and left colon in 6 (50%). in In 83.3% of the procedures the LAMS used was the Hot AXIOS, in 16.7% the Hot SPAXUS. In 7 (58.3%) patients a EUS-guided ileo-colostomy was performed while in 5 (41.7%) a colo-colostomy. Technical success was achieved in all procedures (100%). No LAMS misdeployment or procedural adverse events were observed. Post-procedural complications were recorded in 3 (25%) patients, however direct correlation with the procedure could not be ascertained. Clinical success was achieved in 10 (83.3%) patients. The median time of oral diet resumption was 1 [1-2] day. The median post-procedure hospital stay was 9 [1-20] days and overall median survival was 47.5 [2-270] days.

When comparing EUS-guided ileo-colostomy vs colo-colostomy no difference were observed in terms of technical (7/7 [100%] vs 5/5 [100%], p=1) or clinical (5/7 [71.4%] vs 5/5 [100%], p=0.6) success or adverse events (3/7 [42.8%] vs 0/5 [0%], p=0.31). Time of re-feeding, duration of hospital stay and survival were also comparable between these 2 subgroups of patients (p=0.06, p=0.78, p=0.2, respectively).

Conclusion: EUS-EC is a feasible technique and could be considered as a valid alternative to standard approaches for MIO in highly selected patients.

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Disclosure: Nothing to disclose.

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PP0918

EFFECTS OF OPIUM TINCTURE ON GASTROINTESTINAL FUNCTION AND MOTILITY IN HEALTHY VOLUNTEERS: A MAGNETIC RESONANCE IMAGING STUDY

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Introduction: Opium tincture in oral solution has been used for antidiarrheal treatment for centuries, and it is often used to treat chronic idiopathic diarrhea as a secondary choice after loperamide and bulking agents. However, the underlying mechanisms of opium tincture on gastrointestinal function are not evident and have not been investigated in clinical trials. This study aimed to investigate the effects of opium tincture on gastrointestinal motility, intestinal volumes, and water content of different gut segments assessed by magnetic resonance imaging (MRI).

Aims & Methods: Twenty healthy volunteers were included in a randomized, placebo-controlled, cross-over study with 9-day treatment (30 drops per day of opium tincture corresponding to 15 mg of oral morphine or placebo). MRI was performed on day 1 (before treatment) and day 9 (during treatment) using a pan-alimentary scan protocol. Images were obtained during fasting and after a 285-kcal granola bar meal. MRI measurements included assessments of gastric volume, gastric emptying half-time, gastric motility, small bowel volume, small bowel water content, small bowel motility, colon volume, colon water content, and whole gut transit. Data was analyzed using repeated measure mixed models and adjusted using baseline data as a covariate and a Bonferroni post hoc correction.

Results: The median age of the included subjects was 24 years (range 20-72 years), and 10 subjects were female. Opium tincture delayed gastric half-emptying time as compared to placebo (mean [95% confidence intervals] values: 38.5 [35.3-41.6] min vs. 32.8 [29.9-35.8] min, p=0.004). Opium tincture increased gastric meal volumes 15 and 30 minutes after the meal as compared to placebo (ps<0.018). Gastric contraction amplitude and frequency were not affected by treatment. Small bowel endpoints were not different between treatments (all p>0.05). Ascending-, transverse-, and descending colon volumes were increased after opium tincture as compared to placebo (ascending: 226 [197-253] ml vs 168 [140-195] ml, p=0.003; transverse: 208 [180-236] ml vs. 158 [131-185] ml, p=0.017; descending: 155 [125-184] ml vs. 110 [83-138] ml, p=0.045). T1-relaxation time (a proxy for water content) of the descending colon chyme was decreased during opium tincture as compared to placebo, indicating dryer feces (517 [425-609] ms vs. 690 [595-786] ms, p=0.030). Whole gut transit time was delayed during opium tincture as compared to placebo (p=0.027).

Conclusion: Opium tincture as compared to placebo induces changes in gastric and colonic motility and volumes as measured using a pan-alimentary MRI framework in a population of healthy volunteers. Based on the current results, opium tincture seems to affect the gastrointestinal tract in ways that are relevant for treating diarrhea. An improved understanding of how opioids affect gut functions may lead to a better understanding of and optimized management of diarrhea.

Disclosure: Nothing to disclose.

PP0919

DAY-TO-DAY PHYSIOLOGICAL VARIATION IN GASTROINTESTINAL MOTILITY AND VOLUMES USING A PAN-ALIMENTARY MAGNETIC RESONANCE IMAGING PROTOCOL

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Introduction: Gastrointestinal symptoms can originate from different segments of the gastrointestinal tract, however, symptoms seldom overlap with clinical findings. Magnetic resonance imaging (MRI) has lately shown promising results when investigating volume and motility disturbances in diseases like diabetes mellitus and Crohn's disease. MRI offers the possibility of examining the full gastrointestinal tract in one examination; however, to our knowledge, intrasubject physiological variability has not been investigated in detail.

The current study aimed to describe day-to-day intrasubject physiological variability in MRI-assessed segmental volumes and motility of the gastrointestinal tract in healthy volunteers.

Aims & Methods: Forty healthy volunteers were investigated with MRI at fasting and 0, 15, and 45 minutes after a 285-kcal meal. MRI measurements included gastric emptying half-time, small bowel motility, whole gut transit capsules, and volume assessments of the stomach, small bowel, and colon. The MRI protocol was repeated after a mean period of 10.5 weeks (a minimum of five weeks and a maximum of 24 weeks). Assessments of variability were performed using the intraclass correlation coefficient and [95% confidence intervals].

Results: Gastric emptying half-time was moderately reproducible with an ICC of 0.56 [0.28-0.75]. Gastric meal volumes showed overall poor reproducibility with the highest ICC at the fasting scan with an ICC of 0.35 [0.03-0.61], while gastric gas volumes showed poor to moderate reproducibility with the highest ICC of 0.56 [0.29-0.75] at the 0-minute scan. Small bowel motility showed the highest reproducibility at the fasting scan with an ICC of 0.60 [0.34-0.77] and the lowest reproducibility at the 45-minute scan with an ICC of 0.13 [-0.37-0.56]. Small bowel volumes at fasting were moderately reproducible with an ICC of 0.64 [0.40-0.79], while colon volume had an ICC of 0.45 [0.15-0.67]. Whole gut transit capsules showed poor reproducibility with an ICC of 0.30 [-0.15-0.66].

	N	Fasting scan (ICC)	0-minute scan (ICC)	15-minute scan (ICC)	45-minute scan (ICC)
Gastric half emptying time (t50)	36	0.56 [0.28-0.75]			
Gastric meal volume (meal ingested after fasting scan)	37	0.35 [0.03-0.61]	0.24 [-0.06-0.51]	0.31 [-0.1-0.57]	0.16 [0.45-0.62]
Gastric gas volume	37	0.43 [0.13-0.66]	0.56 [0.29-0.75]	0.51 [0.23-0.71]	0.45 [0.16-0.67]
Small bowel motility	37	0.60 [0.34-0.77]	0.25 [-0.08-0.53]	0.38 [0.07-0.63]	0.13 [-0.37-0.56]
Small bowel volume	37	0.64 [0.40-0.79]	0.77 [0.60-0.88]	0.65 [0.42-0.80]	0.68 [0.32-0.87]
Colon volume	37	0.45 [0.15-0.67]			
Whole gut transit	18	0.30 [-0.15-0.66]			

Table: Overview of day-to-day reliability on MRI-assessed endpoints in fasting and following a granola bar meal (285 kcal and 400 ml of water) using the intraclass correlation coefficient (ICC) and [95% confidence intervals].

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Conclusion: Intrasubject physiological variation in MRI-assessed segmental volumes and motility of the gastrointestinal tract was high, and most comparisons resulted in indications of poor to moderate variability when interpreting the ICC values. When designing clinical studies or performing clinical patient examinations that include MRI-based assessments of the gastrointestinal tract, the normal physiological variation must be considered.

Disclosure: Nothing to disclose.

PP0920

SELF-SUPERVISED LEARNING FOR CLASSIFYING BOWEL OBSTRUCTION ON UPRIGHT ABDOMINAL RADIOGRAPHY: A RETROSPECTIVE COHORT STUDY

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Introduction: Accurate labeling of obstruction site on upright abdominal radiograph is a challenging task. The lack of ground truth leads to poor performance on supervised learning models. To address this issue, self-supervised learning (SSL) is proposed to classify normal, small bowel obstruction (SBO), and large bowel obstruction (LBO) radiographs using a few confirmed samples.

Aims & Methods: A total of 1500 confirmed and 1922 unlabeled radiographs were categorized based on the ground truth. The SSL model was firstly trained on the unlabeled radiographs, and then fine-tuned on the confirmed radiographs. ResNet50 and VGG16 were used for the embedded base encoders, whose weights and parameters were adjusted during training process. Furthermore, it was tested on an independent dataset, compared with supervised learning models and human interpreters. Finally, the t-SNE and Grad-CAM were used to visualize the model's interpretation.

Results: The proposed VGG16-backboned SSL model showed superior classification performance (accuracy 0.847, F1-score 0.846, MCC 0.873, Kappa 0.770), followed by the ResNet50-backboned model and senior radiologist. When recognizing SBO and LBO radiographs, the SSL model reached the best accuracy (0.907, 0.907) and F1-score (0.865, 0.860). Additionally, the SSL model exhibited better cluster performance through t-SNE and demonstrated outstanding lesion recognition ability through Grad-CAM.

Conclusion: The proposed SSL model exhibited outstanding performance in classifying bowel obstruction on upright abdominal radiograph, using unlabeled images and limited ground truth. Our study supports the feasibility of SSL model for computer-aided diagnosis of diseases that are difficult and time-consuming for labeling.

Disclosure: Nothing to disclose.

PP0921

STENTING AS BRIDGE TO SURGERY VERSUS EMERGENCY SURGERY FOR ACUTE LEFT COLONIC MALIGNANT OBSTRUCTION

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Introduction: Emergency surgery (ES) is associated with higher morbidity and mortality for acute left colonic malignant obstruction. Since the first application of self-expandable metallic stent (SEMS), the technique has been popularly applied as "bridge to surgery (BTS)". The short-term clinical outcomes and long-term oncologic prognosis remain controversial.

Aims & Methods: This retrospective study aimed to compare the performance for managing acute left colonic malignant obstruction by ES or BTS. The medical records were collected from May 2015 to December 2019 in two tertiary hospitals in Eastern China. Patients were followed up after surgery every 6 months for 3 years and once a year afterwards. Clinical success was defined as resumption of intestinal function for 30 days without morbidity or mortality after surgical procedure. All demographics, intervention procedures and follow-up were analyzed for short-term and long-term outcomes. Risk factors for long-term survival were discussed as well.

Results: A total of 136 patients were eligible for the study, 67 in BTS group and 69 in ES group. No significant differences concerning demographics were found, including age, sex, body-mass index (BMI), symptoms, coexisting disorders, laboratory tests, preoperative radiotherapy or chemotherapy, previous history of abdominal surgery, cancer location, obstructive degree and TNM stage. Technical success of SEMS insertion was achieved in 64 patients (95.5%). The interval from SEMS placement to elective surgery was 16 days. Minor perforation was detected on 4 patients during the elective surgery, while no symptomatic complaint was recorded.

BTS group showed better short-term performance, including less intraoperative decompressive intervention (85.1% vs. 69.6%, p=0.045), more complete decompression (64.2% vs. 27.5%, p<0.001), more primary anastomosis (65.7% vs. 17.4%, p<0.001), less temporary and permanent stoma (34.3% vs. 82.6%, 13.4% vs. 58.7%, both p<0.001), less intensive care unit requirement (10.4% vs. 27.5%, p=0.011), less intraoperative complications (3.0% vs. 13.0%, p=0.031) and reduced 30-day mortality (0 vs. 8.7%, p=0.028). Longer hospitalization was observed in the BTS group (21.7±13.3 days vs. 14.9±7.6 days, p<0.001). Postoperative hospitalization and total cost were comparable between the two groups. No significant difference was observed about clinical success (73.1% vs. 63.8%, p=0.240), overall mortality (53.7% vs. 64.4%, p=0.247) or long-term survival (52.0 months vs. 35.0 months, p=0.121). TNM stage Ⅳ, permanent stoma and postoperative complication were the independent risk factors for long-term survival.

	Patients under BTS n=67	Patients under ES n=69	T/χ2	p value
Intraoperative complication, no. (%)	2 (3.0%)	9 (13.0%)	4.626	0.031
ICU requirement, no. (%)	7 (10.4%)	19 (27.5%)	6.420	0.011
Temporary stoma, no. (%)	23(34.3%)	57(82.6%)	32.713	0.001
Permanent stoma, no. (%)	9(13.4%)	37/63(58.7%)	29.138	0.001
Postoperative hospitalization, day (mean±SD)	12.4±11.0	11.1±5.8	0.825	0.411
Total cost, ¥ (mean±SD)	63360.0±37302.9	56315.2±31224.4	1.180	0.240
Clinical success, no. (%)	49(73.1%)	44(63.8%)	1.379	0.240
30-day mortality, no. (%)	0	6 (8.7%)	/	0.028
Overall mortality, no. (%)	29/54 (53.7%)	38/59 (64.4%)	1.338	0.247

Table.1 Operation related characteristics.

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Conclusion: SEMS served as a safe and effective bridge to elective surgery for acute left colonic malignant obstruction, without adverse effect on long-term survival. It significantly improved short-term outcomes by reducing perioperative complication, 30-day mortality and stoma. More prospective studies are necessary for further evaluating the clinical values of SEMS, and weighing its benefits against potential complications.

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Disclosure: Nothing to disclose.

PP0922

ENDOSCOPIC RETROGRADE APPENDICITIS THERAPY FOR MANAGEMENT OF CHRONIC FECALITH APPENDICITIS

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Introduction: Chronic appendicitis is a condition with chronic abdominal pain or mild attacks of appendicitis, seriously affecting the patient's quality of life. Endoscopic retrograde appendicitis therapy (ERAT) has emerged as a promising, non-invasive treatment for acute uncomplicated appendicitis.

Aims & Methods: We aim to assess the safety and efficacy of ERAT for chronic fecalith appendicitis.

We retrospectively reviewed the medical records of consecutive patients who underwent ERAT for chronic fecalith appendicitis at Zhongshan Hospital, Fudan University, Shanghai, China between December 2017 and June 2023.

Clinicopathological characteristics, procedure-related parameters, AEs, and follow-up outcomes were analyzed.

Results: A total of 60 patients were included. ERAT intubation was successfully performed in all patients (100%). The median procedure time was 15 minutes (IQR, 12-25 minutes).

The postoperative abdominal pain scores were significantly reduced (P<0.0001). Short-term adverse events included 5 cases of abdominal pain (8.3%) and 2 cases of fever (3.3%).

Long-term adverse events included recurrent abdominal pain in 3 cases (5.0%) and recurrent appendicitis in 2 cases (3.3%). The median time to recurrence was 5 months (IQR, 2-12 months).

Logistic regression analysis revealed appendiceal stenosis (OR=25.000, 95% CI, 1.114-561.281, *P*=0.043) and distorted appendix lumen (OR=12.500, 95% CI, 1.373-113.806, *P*=0.025) as significant risk factors for long-term adverse events.

Conclusion: ERAT may be a safe, effective and minimally invasive alternative approach for chronic fecalith appendicitis. Appendix lumen stenosis and distortion are risk factors of recurrence. Further large-scale prospective studies are necessary to assess the efficacy and safety of ERAT compared with antibiotic therapy and surgery.

Disclosure: Nothing to disclose.

PP0923

THE PREOPERATIVE AUTOLOGOUS BLOOD TATTOOING FOR LOCALIZATION IN LAPAROSCOPIC COLORECTAL SURGERY : A SAFETY AND EFFICACY COMPARISON STUDY

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Introduction: There are several preoperative localization methods for laparoscopic colorectal surgery. ICG (indocyanine green) injection has recently been in the limelight for preoperative localization. However, although the incidence is low, ICG injections may cause complications due to allergic reactions. In this study, we tried to check the feasibility and safety of autologous blood tattooing as an alternative to ICG.

Aims & Methods: From august in 2017 to July 2023, a total of 42 patients were preoperative endoscopic localization. Through preoperative endoscopy, 33 patients underwent autologous blood tattooing and 9 patients underwent ICG tattooing. The surgeon checked visibility and other complication such as abscess or spillage at the start of surgery. The characteristics of patients, outcomes, and complications were collected retrospectively.

Results: Patients characteristics showed no significant difference between the two groups. As for the location of localization, sigmoid colon was the most common in both groups. The two groups each had one imprecise localization. In autologous blood tattooing patient, the omentum interfered with detection, and in ICG tattooing patient, blurring was severe. In addition, in patients with autologous blood tattooing, detection was not possible in 2 patients due to adhesion and technical failure. In the case of complications, there was only one spillage in the ICG group, which resulted in blurring.

Conclusion: Preoperative autologous blood tattooing is a simple and useful technique because it has high visibility, and no complication. This method has the advantage that it does not require additional agents or facilities and can be performed quickly and accurately through the patient's blood. However, unlike ICG tatooing, there is a disadvantage that it is difficult to secure a field of view when there is an obstruction structure such as an omentum. In the future, further study of large size will be needed.

Disclosure: Nothing to disclose.

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PP0924

PREOPERATIVE PREDICTION OF POSTOPERATIVE COMPLICATIONS FOLLOWING MINIMALLY INVASIVE TOTAL MESORECTAL EXCISION UTILIZING ARTIFICIAL INTELLIGENCE

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Introduction: Recent investigations have highlighted specific dimensions within the bony pelvis as potential indicators of total mesorectal excision (TME) outcomes. However, manual pelvimetry is labor-intensive.

Aims & Methods: This study endeavors to develop an automated pelvimetry system employing artificial intelligence (AI) to facilitate the preoperative identification of high-risk patients. Specifically, our research aims to construct and validate an AI-based automated pelvimetry tool and assess the prognostic utility of pelvimetric measurements in predicting postoperative complications necessitating reintervention (Clavien Dindo classification ≥ 3) in patients undergoing minimally invasive TME. This study is a multicenter retrospective cohort investigation incorporating clinical and imaging data from 1,700 patients treated at eight experienced TME centers. Leveraging a Deep-Learning network, our algorithm locates multiple anatomical landmarks to quantify pelvic inlet, pelvic outlet, and depth distances accurately. The accuracy of the developed algorithm is compared against inter-observer variability observed in manually conducted pelvimetry. The prognostic significance of pelvimetry for (post-)operative outcomes is evaluated through multivariate regression analyses.

Results: The AI-driven landmark localization achieved a mean margin of error of 5 millimeters across various pelvic dimensions, while manual pelvimetry demonstrated an inter-observer variation of 2.5 millimeters. Analyses of pelvimetric measurements and postoperative complications identified pelvic depth as an independent predictor of such complications ($p < 0.001$).

Conclusion: An artificial intelligence algorithm could successfully be used for accurate automated pelvic dimension determination. These pelvimetric measurements hold promise as a valuable decision support tool for surgeons in preoperatively assessing the risk of postoperative complications following minimally invasive TME.

Disclosure: Nothing to disclose.

PP0925

COLD VERSUS HOT SNARE POLYPECTOMY AND COLD FORCEPS FOR SMALL COLORECTAL POLYPS: A SYSTEMATIC REVIEW AND NETWORK META-ANALYSIS

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Introduction: Endoscopic removal of colorectal polyps is a cornerstone of colorectal cancer prevention.[1]

Optimal techniques for removing small polyps (≤10mm) remain scarce,[2–4] with cold snare polypectomy (CSP), hot snare polypectomy (HSP), and cold forceps (CF) being the main options.[1–4]

Aims & Methods: This systematic review and network meta-analysis aims to compare the efficacy of these techniques for removing small colorectal polyps and determine the optimal approach.

We searched Web of Science, PubMed, Cochrane CENTRAL, and Scopus from inception until March 2024 for relevant randomized controlled trials and observational studies comparing CSP, HSP, and CF (including sub-techniques) for small polyp removal.

Records were screened for eligible studies, and all relevant outcomes were pooled as a risk ratio (RR) with the corresponding 95% confidence intervals (CI) in a network meta-analysis (NMA) model, using R software.

Results: Our findings from 13 studies (2767 patients) showed insignificant differences among HSP versus CSP in terms of complete resection rate (CRR) and incomplete resection rate (IRR) [RR 2.00, (95% CI, 0.53 to 7.51); P = 0.74] and [RR 1.24, (95% CI, 0.20 to 7.85); P = 0.43], respectively. In addition, no significant variance was detected between the CSP and CF in terms of complete histological polyp eradication (CHPE), incomplete histological polyp eradication (IHPE), and visual polyp eradication (VPE) [RR 0.97, (95% CI, 0.87 to 1.08); P = 0.1], [RR 1.36, (95% CI, 0.57 to 3.26); P = 0.9], and [RR 0.75, (95% CI, 0.62 to 0.91); P = 0.07], respectively.

Conclusion: In conclusion, our results revealed no significant difference between any of the techniques in terms of their success rate in completely removing the polyp. These findings suggest that all three methods might be equally effective for removing small polyps.

However, we recommend further studies with more participants, and longer follow-up periods might be needed to definitively determine the best approach for all situations. This could include considering factors like specific polyp characteristics or patient preferences.

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Disclosure: Nothing to disclose.

PP0926

THE ROLE OF LASER FILAC™ IN THE MANAGEMENT OF ANAL FISTULAS: EXPERIENCE FROM MEDICINE SERVICE B

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Introduction: "Complex" anal fistulas are characterized by a tract that cannot be laid open by fistulotomy without risking significant incontinence due to the depth of the tract and/or the patient's condition. Over the past 20 years, several techniques known as "sphincter-saving" have been proposed to address this surgical challenge. Among these techniques, FiLaC™ is one of the most recent. It involves closing the fistula tract by coagulation. The aim of our work is to evaluate the results of this technique in terms of healing of anal fistulas through a series of cases.
Aims & Methods: This is a descriptive prospective study conducted between May 2023 and March 2024 involving 14 patients who underwent treatment for anal fistula using the FiLaC™ technique. All patients underwent an initial surgical intervention to identify the fistula tract, place a loose seton, drain any abscess, and/or lay open any diverticula. The type of fistula tract was determined by the proctologist during this first operation. In cases of doubt about the quality of drainage, multi-operated fistula, and/or Crohn's disease, an MRI was performed. The procedure was performed under spinal anesthesia, and all patients received perioperative antibiotic prophylaxis. We used a 1.2 mm diameter radial fiber powered by the Leonardo Duel© diode laser platform. The primary outcome measure was fistula healing, clinically defined by closure of the internal and external openings and absence of pain and discharge.
Results: Fourteen patients were included, with a mean age of 45.1 +/- 2.1 years. Among our patients, 12 (85%) had cryptogenic fistulas, 1 case had a fistula as part of ano-perineal manifestations of Crohn's disease (CD), and one anal fistula was secondary to perianal tuberculosis. The fistula was classified as Parks 1 in 14.2% of cases, Parks 2 in 57.14% of cases, Parks 3 in 14.2% of cases, and Parks 4 in 14.2% of cases. The internal opening was tight in 64.28% of patients and gaping in 35.72%. The mean operative time before laser closure was 2.07. All patients had a seton drain placed. The mean time for drain removal was 124.90 seconds, and the mean delivered energy was 799.88 Joules. No cases of postoperative complications were reported. The mean follow-up was 2.21 months. Healing was achieved in 78.5% of cases, while in 21.5% of patients, closure of the internal and/or external opening was not achieved, and surgical revision was proposed. No cases of recurrence were reported.
Conclusion: The treatment of complex fistulas still presents a challenge: achieving fistula healing while preserving anal continence. Laser coagulation appears as a promising technique that can meet this challenge.
Disclosure: Nothing to disclose.

PP0927

IS FISSURECTOMY WITH ANOPLASTY BETTER THAN FISSURECTOMY ALONE?

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Introduction: Anal fissure is a common reason for consultation in proctology. Surgery is proposed in cases of chronic, complicated, or recurrent anal fissure. Fissurectomy can be performed alone (FS) or combined with anoplasty (F+A). The benefit of the latter compared to fissurectomy alone has never been demonstrated. The objective of the study was to compare the effectiveness and tolerance of the two techniques.
Aims & Methods: This was a retrospective descriptive and analytical study conducted over a period of 18 years from January 2005 to August 2023, involving 2070 patients surgically treated for anal fissure. We included all patients without a proctological surgical history who were operated on for a single, non-infected idiopathic anal fissure. We evaluated postoperative pain, early postoperative complications (bleeding, dysuria, constipation, superinfection), and healing time. Statistical analysis was performed using SPSS 20.0 software. Qualitative variables were compared using the Chi-square test. Predictive factors for healing were identified using a binary logistic regression model in univariate and multivariate analysis.
Results: The overall prevalence of anal fissures compared to other reasons for proctology consultation was 55.9%. The median age of the patients was 40 [21-69], with a female-to-male ratio of 0.85. The anal fissure was posterior in 1493 (72.1%) patients and anterior in 577 (27.9%) patients. Other proctological pathologies (anal fistula, hemorrhoidal disease) were associated in 358 cases (17.3%). FS was performed in 1304 patients (62.9%) and F+A in 766 patients (37%). Sphincterotomy was associated in 357 patients (17.24%). The median postoperative follow-up was 2 [1-3.9] months. Continence was respected in all patients. The time to disappearance of postoperative pain was the same for both procedures: 1.1 +/- 0.5 months vs. 1.0 +/- 1.2 months, p=0.51, respectively for FS and F+A. The overall rate of early complications was 5.29%, with them being more frequent in cases of F+A: 7.1% vs. 2.3%, p=0.031. The healing rate was similar in both procedures: 92.9% vs. 97.7%, p=0.31. The median healing time was shorter in patients who underwent F+A: 1.5 +/- 0.2 months vs. 2.8 +/- 0.4 months, p = 0.0003. In multivariate analysis, independent risk factors associated with delayed healing were age ≥ 40 years [OR = 3.1; 95% CI [1.1 to 14.1] p = 0.004) and a history of diabetes [OR 6.5, 95% CI [1.7 to 43.2] p <0.001. The postoperative recurrence rate was similar in both procedures: 2.1% in the case of FS vs. 3.2% in the case of F+A, p=0.1.
Conclusion: Fissurectomy remains an effective procedure in the management of anal fissures. Our study showed that fissurectomy with anoplasty has the advantage of accelerating healing. However, at present, no technique is consensual, hence the importance of multicenter randomized studies evaluating both surgical techniques.
Disclosure: Nothing to disclose.

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PP0928

PERIOPERATIVE MANAGEMENT FOR PATIENTS WITH HEMOPHILIA A UNDERGOING ENDOSCOPIC MUCOSAL RESECTION TO CONTROL POSTOPERATIVE DELAYED HEMORRHAGE

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Introduction: Endoscopic mucosal resection (EMR) are popularly used to remove gastrointestinal polyps. Successful treatment will be hampered by postoperative delayed bleeding. Due to the deficiency of coagulation factors, the risk of hemorrhage increases for patients with hemophilia A. It is necessary to receive the infusion of clotting factor Ⅷ (FⅧ) during perioperative period. The infusion duration depends on operation trauma, while no accepted guidelines are reported for EMR.

Aims & Methods: We aim to describe perioperative management strategy for patients with hemophilia A undergoing EMR for controlling postoperative delayed hemorrhage. This study is designed as an observational study at First Affiliated Hospital of Soochow University and approved by the committee ethics (approval number: 2023-280). A total of six patients were included into analysis, including 2 gastric polyps and 4 colorectal polyps. The concentration of FⅧ was monitored during perioperative period. The strategy of replace treatment was timely adjusted. Demographics and perioperative characteristics were recorded, including age, sex, body-mass index (BMI), the location and size of polyps, the testing results of FⅧ, the infusion strategy and prognosis.

Results: The concentration of FⅧ on admission ranged from 0.4%-26.2%. FⅧ was infused preoperatively to increase the concentration to more than 20.0%, except for the patient whose initial FⅧ concentration was 26.2%. No intraoperative hemorrhage was observed. The first three patients received the infusion of FⅧ for just one day postoperatively, however, delayed bleeding attacked them. The longest interval from operation to hemorrhage was 15 days. The hemorrhage could be successfully controlled upon the infusion of FⅧ. In contrast, the other three patients received appropriate infusion of FⅧ during postoperative seven days (Table.1) to maintain the concentration over 20.0%. The total infusion dose of FVIII for each patient all exceeded 1500U. Eventually, no postoperative delayed bleeding was observed.

Patient No.	Age	Gender	BMI (kg/m²)	Polyp location	FⅧ concentration at admission	Preoperative FⅧ infusion strategy	FⅧ concentration on preoperative day	Postoperative FⅧ infusion strategy	Postoperative delayed bleeding
1	29	Male	21.1	Stomach	0.4%	2000U qd*day1	28.6%	1000U qd*day1	Yes
2	43	Male	24.3	Stomach	12.6%	200U qd*day1-4	39.5%	1200U qd*day1	Yes
3	36	Male	24.6	Rectum	4.0%	1200U qd*day1	21.4%	1400U qd*day1	Yes
4	53	Male	17.4	Colon	19.5%	1000U qd*day1	43.0%	600U q12h*day1-2, 400U q12h*day3-4, 800U qod*day5-7	No
5	28	Female	21.8	Rectum	26.2%	No infusion	26.2%	400U qd*day3-6	No
6	58	Male	22.6	Colon	7.4%	1800U qd*day1	30.4%	800U qd*day1-3, 600U qd*day4-6	No

Table.1 Demographics and management strategy.

Conclusion: In spite of the super minimal invasiveness of endoscopic surgery, it is critical for patients with hemophilia A to receive enough FⅧ infusion in peri-EMR management. The concentration of FⅧ should be increased to more than 20% before EMR, no matter which infusion dose to be chosen. The duration of operation and the size of surgical wound are needed to be reduced. It is crucial to keep infusion of FⅧ and maintain its concentration over 20% for at least 7 days postoperatively.

Furthermore, resumption of FⅧ might be beneficial to the patients at high risk for delayed bleeding, including the aged (>60 years), larger lesions (>10mm) and rectal location. This is the first report about perioperative management when performing EMR on patients with haemophilia A. More researches are necessary for verifying the clinical benefits against potential complications.

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Disclosure: Nothing to disclose.

PP0929

EFFECT OF THE NUTRACEUTICAL "MICODIGEST 2.0" ON THE COMPLICATION RATE OF COLORECTAL CANCER SURGERY WITH CURATIVE INTENT: A RANDOMISED, DOUBLE-BLIND CLINICAL TRIAL (NCT04821258)

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Introduction: Fungal polysaccharides have the potential to reduce complications associated with surgery. This is due to their role in modulating the microbiota and their anti-inflammatory activity. Therefore, the nutraceutical MICODIGEST 2.0, composed of 9 fungal extracts, could be a valuable therapeutic resource to be considered in reducing the complications associated with colorectal surgery.

Aims & Methods: A randomised, double-blind clinical trial was conducted on patients with colorectal cancer (CRC) who were candidates for surgical treatment with curative intent[1]. Patients were randomised to take the nutraceutical MICODIGEST 2.0 or a placebo, and were followed for 4-6 weeks until the time of surgery. At baseline and at the end of follow-up, analytical determinations were made for inflammatory parameters and assessments of nutritional status and quality of life. Adverse effects (AE) were also collected during the follow-up period. Post-surgical complications were collected at hospital discharge and assessed using the Clavien-Dindo (CD) classification.

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Results: A total of 46 patients were included in the study, with a median age of 67 (62.3-72.8) years and a percentage of women of 50.0 %. A total of 26.1 % of patients had some type of postoperative complication. The majority of these complications were minor (84.4 %), classified as grade I (50 %) or grade II (50 %) according to the CD classification. The remaining complications were more severe (16.6 %), and were classified as grade III according to the CD classification. There were no significant differences between the two groups in terms of clinical characteristics, nutritional status, quality of life, or analytical determinations. The percentage of post-surgical complications was similar between patients treated with nutraceutical and those treated with placebo (25.9% vs. 26.3%, p=0.9). Adverse events were reported in a total of 14 patients (30.4%) during follow-up, of which only 7.1% were serious. However, the percentage of adverse events was not significantly different between the nutraceutical and placebo groups (33.3% vs. 41.1%, p=0.5). A comparison between patients with postoperative complications and those without revealed no significant difference. Similarly, no difference was found between data collected at baseline and at the end of the study.
Conclusion: The nutraceutical MICODIGEST 2.0 has not demonstrated a beneficial effect in reducing complications associated with surgery for CRC treated with curative intent. However, these results are limited to the sample size obtained and should be validated in larger cohorts of patients.
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Disclosure: Nothing to disclose.

PP0930

FIRST EXPERIENCES WITH THE USE OF VACSTENT FOR TREATMENT OF ANASTOMOTIC LEAKAGE IN THE LOWER GI TRACT

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Introduction: Anastomotic leakage is a serious complication following colorectal surgery and is associated with early and long-term morbidity and mortality. Various endoscopic and surgical approaches have been established for management. A new therapeutic option involves the placement of a fully covered VACStent for the lower GI tract (36x80 mm, MICRO-Tech, Düsseldorf, Germany), which offers advantages such as completely covering the defect, continous drainage, and facilitating stool passage simultaneously. Here we report our initial experiences with the VACStent in the lower GI tract for treatment of anastomotic leakage.
Aims & Methods: The fully covered VACStent (36x80 mm) was placed endoscopically under direct visualization. A continuous negative pressure between -75 and -125 mmHg was applied. The VACStent was flushed daily with 20 ml of saline to prevent adhesions of the open-pored sponge of the stent. Six hours before the stent extraction/exchange, the negative pressure was discontinued. The integrity of the leak closure was confirmed visually and via fluoroscopy.
Results: Between February and April 2024, we treated four patients with anastomotic leakage (rectal leakage n=2, sigmoid leakage n=2) using the VACStent. The leakage was detected on the 7th ± 4 postoperative day. The leak was successfully sealed with the VACStent, and one to two exchanges were performed. The leak closure occurred after 17 ± 3 days. Inflammatory parameters such as CRP decreased by around 85 ± 13%. Three patients had a stoma before the VACStent was placed. The distance of the leak from the anal canal averaged 10 ± 9 cm. Patients with leaks involving the anal canal required additional pain reliever. One patient experienced point-shaped erosion bleeding in the anal canal, which was effectively treated with APC.
Conclusion: The lower GI VACStent represents a novel promising and effective endoscopic treatment option for colonic anastomotic defects.
Disclosure: Nothing to disclose.

PP0931

WITHDRAWN

PP0932

LOW ANTERIOR RESECTION SYNDROME – A NEVERENDING CHALLENGE. WHAT IS THE ROLE OF THE ENTEROSORBENTS?

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Introduction: Low anterior resection syndrome (LARS) occurs after transabdominal rectal surgical removal, most related to colorectal tumors.
It has a significant impact on quality of life, so that some patients prefer a permanent stoma. Main symptoms are diarrhea, fecal incontinence, increased defecation frequency, disturbing fecal urgency and perception of incomplete evacuation.
Treatment is still challenging, and many protocols were developed so far.
Enterosorbents are oral intestinal adsorbents, based on organic mineral with adsorption capacity towards some toxic and harmful substances in the gut, e.g., endotoxins and enterotoxins, physically binding them to its surface, reducing stool frequency and improving its consistency. Until the present day, little is known about their effect on LARS patients.
Aims & Methods: Fifteen consecutive adult patients were referred to treatment with the enterosorbent methylsilicic acid due to poor response to optimal clinical protocol to LARS.
Patients were asked to take one measuring spoon (5g) of 70% methylsilicic acid hydrogel added to 50 mL of water two times a day for a 30-day period.
Mean BSS, global improvement percentage, defecatory frequency, Constipation Scoring System (CSS) and Cleveland Clinic Incontinence Score (CCIS) were applied by the author to the patients during the first appointment and during the 30-day follow-up appointment.
All patients accepted to participate by signing the informed consent form, in compliance to the respective Ethical Committee authorization.
Results: Nine patients were female and average age was 32 years (+- 7 years). All patients informed no difficulty in complying with the enterosorbent protocol.
Initial mean defecatory frequency was 10 (+-3). Mean CSS was 9 (+-5) and CCIS was 15 (+-5). Mean BSS was 5 (+- 2).
By the time of the 30-day follow-up appointment, mean defecatory frequency was 5 (+-2). Mean CSS was 5 (+-3), CCIS was 5 (+-3), and mean BSS was 4 (+-1), $p < 0,01$. Global improvement percentage was 52,3% (+- 10,2) with no side effects.
Conclusion: The enterosorbent methylsilicic acid hydrogel is a safe and valid tool in treating low anterior resection syndrome patients and may be suitable as a new therapeutic option.
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Disclosure: Nothing to disclose.

PP0933

EFFICACY AND SAFETY OF MULTI-LOOP TRACTION DEVICE-ASSISTED COLORECTAL ENDOSCOPIC SUBMUCOSAL DISSECTION: A MULTICENTER RANDOMIZED CONTROLLED CLINICAL TRIAL

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Introduction: Colorectal endoscopic submucosal dissection (C-ESD) has been established as a standard treatment for colorectal neoplasms which need to be resected in an en bloc fashion. However, the dissection of thin submucosa under inadequate visualization is still technically demanding. A traction device is a key to this problem.
Aims & Methods: We recently developed a novel traction device (multi-loop traction device, MLTD; Boston Scientific Co. Ltd., Tokyo, Japan). The present study focuses on evaluating the effectiveness and safety of MLTD during C-ESD. A multicenter randomized controlled trial was conducted involving patients aged 20 years or older with colorectal lesions (≥ 20 mm) suspected of non-invasive carcinoma. Participants were randomly assigned in 1:1 ratio to undergo ESD with MLTD (MLTD-ESD group) or without any traction device (control group). In both groups, the operators were given the option to convert the treatment to the other group if endoscopist found it difficult to continue treatment under the conditions given in each group for more than 10 minutes. The primary endpoint was dissection speed, with key secondary endpoints being technical success rate and adverse events. The technical success rate was defined as the proportion of cases in which the protocol treatment was completed in each group.
Results: A total of 108 participants were randomized into the MLTD-ESD group (n = 53) and the control group (n = 55). Median dissection speeds were not significantly different between the MLTD-ESD group and the control group (14.8 [IQR: 8.9-23.9] vs. 13.3 [IQR: 8.9-18.8] mm^2/min; P = 0.30). However, the technical success rate was significantly higher in the MLTD-ESD group (96.2%) compared to the control group (71.0%).
There were two cases in which treatment was converted in the MLTD-ESD group, and 16 cases in which treatment was converted in the control group. There was no significant difference between the MLTD-ESD group and the control group in the en bloc resection rate (98.1% vs. 100%, P = 0.31), R0 resection rate (94.3% vs. 91.0%, P = 0.46) and the incidence of adverse events (7.5% vs. 9.1%, P = 0.77).
Two intraoperative perforations, one delayed bleeding, and one delayed perforation occurred in the MLTD-ESD group. All patients recovered with nonsurgical management.
The subgroup analysis among expert endoscopists with over 200 clinical ESD experiences showed a faster dissection speed in the MLTD-ESD group compared to the control group (21.6 [IQR: 15.5-28.8] vs. 14.4 [IQR: 9.9-21.2] mm^2/min; P = 0.009).
Conclusion: The use of MLTD did not lead to a faster dissection speed during C-ESD in this multicenter randomized controlled trial. However, it was associated with a higher technical success rate, and faster ESD procedures among expert endoscopists.
To better define its role in C-ESD, further research with a more targeted approach is warranted.
Disclosure: Kazuki Sumiyama is named as an inventor in patent licensing of the "Multi-Loop Traction Device."

PP0934

OUTCOMES AND LEARNING CURVE OF ENDOSCOPIC SUBMUCOSAL DISSECTIONS OF COLON LESIONS IN A WESTERN COUNTRY

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Introduction: Endoscopic submucosal dissection (ESD) is used for en bloc resection of selected superficial lesions of the colon and rectum. ESD of colon lesions is particularly challenging, mostly limited to few reference centers in Western countries, with small series in the literature.
Aims & Methods: Assessment of outcomes of ESD of colon lesions and the relationship between endoscopist's experience.
Retrospective cohort study, from prospective collected data, including all consecutive patients submitted to ESD in the colon, in a tertiary Western center. The patients were divided into 3 groups, each made up of 63 patients, in chronological order of the procedure. The procedure was always performed by the same endoscopist.
Results: A total of 189 ESD in the colon were performed in 185 patients (60.8% male; median age 69), with an overall rate of en bloc, R0 and curative resections of 94.18%, 78.84% and 74.07%, respectively. There were 3 perfurations (1,6%) all solved endoscopically, without any death or complications requiring surgical approach.
With increasing experience, a higher percentage of R0 and curative resections was found, with statistical significance (p-value = 0.0004 and 0.0006, respectively). In the first group, R0 was reached in 63.49%, whereas in the second group R0 was attained in 80.85% and in the third group in 92.06%. In terms of en bloc resection, there was no significant difference between the three groups (p-value = 0,067). Regarding perforations, there was no difference between the 3 groups (1 case in each).

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Conclusion: In this real life study from a Western center, ESD for colonic lesions was very safe and with high rates of en bloc resections. However, a learning curve was observed with a significant increase in the rate of complete and curative resections over time.
Disclosure: Nothing to disclose.

PP0935

INNOVATIVE INTERVENTIONS: COMPARING ENDOSCOPIC DIRECT APPENDICITIS THERAPY WITH ENDOSCOPIC RETROGRADE APPENDICITIS THERAPY FOR UNCOMPLICATED APPENDICITIS

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Introduction: Endoscopic retrograde appendicitis therapy (ERAT) offers an appendix-preserving approach, yet visual and specificity challenges persist. In contrast, endoscopic direct appendicitis therapy (EDAT) provides direct visual observation with diagnostic and therapeutic capabilities. This study aimed to assess the efficacy and feasibility of EDAT compared to ERAT in uncomplicated appendicitis.
Aims & Methods: We retrospectively reviewed of patients diagnosed with suspected uncomplicated appendicitis undergoing ERAT or EDAT from Jan 2021 to Mar 2024.
The primary outcome was to evaluate the intervention success of two groups.
The secondary outcomes included hospitalization, procedure time, stent placement, post-procedure hospitalization period, pain relief, recurrence, and endoscopic direct-view characteristics were compared.
Results: Sixty-nine patients were enrolled, among which 34 patients underwent ERAT and 35 patients underwent EDAT. EDAT showed significantly higher intervention success (100% and 85.3%, respectively, $p = 0.018$), and shorter hospitalization stay ($p = 0.003$), and post-procedure hospitalization stay ($p < 0.001$) compared to ERAT group. With comparable procedure time ($p = 0.915$) and recurrence ($p = 0.164$), significant fewer proportions of stent placements ($p < 0.001$), guidewire assists ($p < 0.001$), and post-procedure antibiotic administrations ($p = 0.036$) associated in EDAT group compared to ERAT group.
Conclusion: EDAT has demonstrated to be an effective and feasible approach, offering both diagnostic and therapeutic capabilities for uncomplicated appendicitis compared to ERAT.
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Disclosure: Nothing to disclose.

PP0936

ACCURACY OF PRE-RESECTION OPTICAL EVALUATION FOR SERRATED AND ADENOMATOUS LARGE NON-PEDUNCULATED COLORECTAL POLYPS

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Introduction: Optical evaluation differentiating between adenomatous and serrated histopathology is the first step in management of large non-pedunculated colorectal polyps (LNPCP) as it informs appropriate endoscopic resection technique selection. Performance of optical evaluation is modest for diminutive polyps but has not been reported for LNPCP.[1]
Aims & Methods: We sought to evaluate the performance of optical evaluation to differentiate between adenomatous and serrated histopathology for LNPCP. Consecutive patients > 18 years of age who underwent endoscopic resection for a LNPCP were enrolled in a prospective multi-center observation cohort study (ClinicalTrials.gov NCT05402696). Pre-resection optical evaluation was performed using high-definition white-light and narrow-band imaging (NBI) with or without near-focus. The Japanese NBI Expert Team (JNET) classification was used to differentiate between serrated (JNET I) vs. adenomatous (JNET IIA, IIB) LNPCPs. Traditional serrated adenomas (TSA) and cancers were excluded from analysis. Sensitivity, specificity, and accuracy were used to evaluate optical evaluation performance.
Results: Between June 2022 to April 2024, 440 patients underwent 476 procedures for 558 LNPCP. Median LNPCP size was 30 mm (IQR 20-40) and 404 (72.4%) LNPCP were proximal to the splenic flexure. Histopathology of all lesions included 365 (65.4%) adenomas, 137 (24.6%) serrated lesions, 34 (6.1%) cancers, and 22 (3.9%) other lesions, which included 6 (1.1%) TSA. Adenomatous histopathology was found predominantly in LNPCP with Paris 0-IIA+IS (107/136, 78.7%) and Paris 0-IS (70/82, 85.4%) morphology. Paris 0-IIA lesions were both adenomatous (184/328, 56.1%) and serrated (125/328, 38.1%). Of 137 serrated LNPCP, 133 (97.1%) were predicted as serrated, with sensitivity, specificity, and accuracy of 97.08% (95% CI 92.69-99.20), 99.45% (95% CI 98.03-99.93), and 98.80% (95% CI 97.42-99.56), respectively. Of 365 adenomatous LNPCP, 350 (96.0%) were predicted as adenomatous, with sensitivity, specificity, and accuracy of 95.89% (95% CI 93.31- 97.68), 97.08% (95% CI 92.69- 99.20), and 96.22% (95% CI 94.15- 97.71), respectively.

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	Serrated Histopathology, N (%)	Adenomatous Histopathology, N (%)
All lesions	137	365
Predicted Serrated	133 (97.1)	2 (0.5)
Predicted Adenomatous	4 (2.9)	350 (95.9)
Predicted Cancer	0 (0.0)	13 (3.6)
Sensitivity	97.08% (95% CI 92.69-99.20)	95.89% (95% CI 93.31- 97.68)
Specificity	99.45% (95% CI 98.03-99.93)	97.08% (95% CI 92.69- 99.20)
Accuracy	98.80% (95% CI 97.42-99.56)	96.22% (95% CI 94.15- 97.71)

Table 1: Optical Performance.

Conclusion: For LNPCP, optical evaluation had greater than 95% accuracy to differentiate between serrated and adenomatous lesions, forming the basis of a selective resection algorithm.

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Disclosure: Nothing to disclose.

PP0937

REAL WORLD EFFICACY OF A COLONIC SELECTIVE RESECTION ALGORITHM FOR LARGE NON-PEDUNCULATED COLON POLYPS

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Introduction: Endoscopic resection is the primary treatment strategy for most large (≥20 mm) non-pedunculated colonic polyps (LNPCP). A selective resection algorithm with endoscopic mucosal resection (EMR) and endoscopic submucosal dissection (ESD) has been shown to be cost effective and optimizes oncologic outcomes in the rectum.[1,2] However, an algorithm for the colon which incorporates cold snare resection (CSR) has not been described.

Aims & Methods: We sought to evaluate the performance of a colon-specific selective resection algorithm for LNPCP. Consecutive patients > 18 years of age who underwent endoscopic resection for a LNPCP were enrolled in a prospective multi-center observation cohort study (ClinicalTrials.gov NCT05402696). Modality selection was determined by optical evaluation using the Paris and the Japan NBI Expert Team classifications: 1) JNET I – CSR; 2) JNET IIA – EMR; 3) JNET IIB or Paris 0-IIC morphology – en bloc resection (EMR/ESD); 4) JNET III – referral to surgery/multi-disciplinary team review. Algorithm performance was evaluated by technical success (all neoplastic tissue removed at index procedure), procedure-related adverse events, referral to surgery, and recurrence at first surveillance colonoscopy (SC1).

Results: From June 2022 to April 2024, 382 patients underwent 402 procedures for 480 LNPCP. Median age was 67 years (IQR 62-74 years) and 174 (45.5%) were female. Median lesion size was 30 mm (IQR 20-40 mm) and 386 (80.4%) were located in the proximal colon. Cancer was present in 16 (3.3%) LNPCP. Based on optical evaluation, LNPCP were resected by CSR (25.8%), EMR (72.5%), and ESD (1.7%). Out of protocol deviations occurred in 23 (4.8%) LNPCP. Technical success was 97.0%, 100.0% and 100.0% for EMR, CSR, and ESD, respectively (p=0.172). Procedure duration was significantly shorter for CSR (median 10 min; IQR 5-15 min) compared to EMR (15 min; IQR 10-30 min) and ESD (55 min; IQR 38-90 min) (p<0.001). Perforation was seen in 11 (3.3%), 1 (14.3%) and 0 (0%) patients undergoing EMR, ESD and CSR, respectively (p=0.030). There were no significant differences between resection modalities in their frequencies of clinically significant post-endoscopic resection bleeding (5.2%) or delayed perforation (0.0%). 26 (5.4%) LNPCP were referred to surgery. Of 194 lesions completing SC1, there were 4 recurrences (0.8%), comprised of 1 (0.8%) and 3 (0.9%) in the CSR and EMR group, respectively (p=1.000); all of which were treated endoscopically.

	N (%)	CSR, N (%)	EMR, N (%)	ESD, (N %)	P-value
Lesions attempted	480	124	348	8	
	Serrated	114 (91.9)	20 (5.7)[NS1]	0 (0.0)	<0.001
	Adenomatous	8 (6.5)	302 (86.8)	6 (75.0)	
	Cancer	0 (0.0)	14 (4.0)	2 (25.0)	
	Other	2 (1.6)	12 (3.4)[NS2]	0 (0.0)	
Duration in minutes (median, IQR)		10 (5-15)	15 (10-30)	55 (38-90)	<0.001
Technical success		124 (100.0)	338 (97.1)	8 (100.0)	0.172
Recurrence		1 (0.8)	3 (0.9)	0 (0.0)	1.000

Table 1: LNPCP Characteristics and Outcomes

Conclusion: A colon-specific selective resection algorithm for LNPCP using optical evaluation to select between CSR, EMR, and ESD has high efficacy, efficiency, and safety.

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Disclosure: Nothing to disclose.

PP0938

LONG-TERM EFFICACY OF TRANSANAL DOPPLER-GUIDED HEMORRHOIDAL ARTERY LIGATION WITH MUCOPEXY (HAL-RAR) IN HEMORRHOIDAL DISEASE

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Introduction: Doppler-guided artery ligation with Mucopexy, or HAL-RAR, is a minimally invasive surgical technique that allows for better postoperative comfort. An anoscope linked with a light source and a doppler probe allows the proctologist to locate the submucosal branches of the superior rectal artery and to ligate them via a transfixing parietal approach. Mucosal pexis is performed afterward: the mucosa of the prolapsed hemorrhoidal pedicles is vertically sutured, consequently leading to a plication of the prolapsed muco-hemorrhoidal tissue and its reposition inside the canal anal.

Nevertheless, long-term recurrence rate of internal hemorrhoidal disease after this procedure has been poorly studied. The aim of our study is to evaluate the long-term recurrence rate of hemorrhoidal disease post Doppler-guided artery ligation (HAL-RAR) with Mucopexy in our center

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Aims & Methods: The aim of our study is to evaluate the long-term recurrence rate of hemorrhoidal disease post Doppler-guided artery ligation (HAL-RAR) with Mucopexy in our center.

Materials and Methods:

An ambispective observational and analytical study was conducted over a 5-year period, from January 2019 to January 2024. Surgical indications included grade II and III internal hemorrhoids that resisted to medical treatment and small localized non-circumferential grade IV hemorrhoids. Patients with external hemorrhoids, bulky grade IV hemorrhoids or pelvic floor disorders were excluded. All data were analyzed using Jamovi version 2.0.0.0 software. Patient characteristics were described as number and percentage for qualitative items and median (interquartile range [IQR]) or mean (± standard deviation) for quantitative ones. A p value of ≤0.05 was deemed to be significant. Our primary objective was to evaluate the rate of recurrence more than 24 months after surgery. The secondary objectives were to determine the clinical features of relapsing patients.

Results: 114 patients were included in this study. The mean age was 46.5 ± 9.6 years. The sex ratio was 6 (16f, 98H). Preoperatively, 69% (n=79) of the patients reported proctalgia, 53% (n=60) post-defecation rectal pain and 58% (n=66) prolapse.

We staged the internal hemorrhoidal disease according to the Goligher classification, identifying 32 with grade II hemorrhoids, 61 with grade III hemorrhoids and 21 grade IV hemorrhoids. At the end of a median follow-up of 4 years (7-2), 9 patients (7.9%) presented a recurrence, made in 6 cases of a muco-hemorrhoidal prolapse and in 3 cases of post-defecation rectal bleeding. None of our patients presented with anal stenosis or incontinence during the follow-up period. In our study, there was no statistically significant relationship between the recurrence and the preoperative clinical symptomatology (p=0.21) nor with grade of hemorrhoids (p=0.17). Overall, 94% of the patients included were satisfied with this technique. 4 of the patients who relapsed benefited from Milligan-Morgan hemorrhoidectomy. However, they claimed that they were more satisfied following the HAL-RAR procedure

Conclusion: transanal doppler-guided hemorrhoidal artery ligation with mucopexy (HAL-RAR) is a safe, quick, easy to perform technique and can be considered an effective long-term treatment for internal hemorrhoidal disease.

Disclosure: Nothing to disclose.

PP0939

OUTCOMES OF ENDOSCOPIC INTERMUSCULAR DISSECTION FOR RECTAL NEUROENDOCRINE NEOPLASMS: A MULTI-CENTER RETROSPECTIVE STUDY

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Introduction: Endoscopic resection is the therapeutic standard for localized rectal neuroendocrine tumors (NETs), yet the innovative endoscopic intermuscular dissection (EID) technique remains under-examined. This study endeavors to assess the clinical outcomes and efficacy of EID in the management of rectal NETs.

Aims & Methods: This retrospective analysis compiled patient data from three esteemed institutions: South China Hospital of Shenzhen University, Shenzhen Hospital of Southern Medical University, and Ningbo Medical Center Lihuili Hospital. Subjects with rectal NETs treated via EID from November 2017 to November 2023 were evaluated. Primary endpoints included technical and clinical success rates, while secondary endpoints involved procedural duration, postoperative hospital stay, adverse event occurrence, and longitudinal follow-up outcomes.

Results: A total of 18 patients (mean age 46.0; 72.2% male), with an average lesion size of 9.1 mm (SD 2.8), were enrolled, and 44.44% presented lesions over 10 mm in diameter. All patients (100%) reported technical and clinical success. Median operative time was 55 minutes (IQR: 43-67 minutes). Complete resection, validated by histopathological examination indicating G1 classification with clear margins, was accomplished universally. Median postoperative hospitalization was 5.6±1.0 days, devoid of complications such as hemorrhage, perforation, infection, or pain. Over an average of 28.8 months of follow-up (SD 22.9 months), there was no evidence of residual or recurrent disease, nor distant metastasis.

Conclusion: EID is substantiated as a viable and effective technique for rectal NET management. The observed outcomes suggest that EID could be a significant addition to the therapeutic modalities for rectal NETs, meriting further scrutiny in future research.

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PP0940

FUNCTIONAL OUTCOMES AND QUALITY OF LIFE FOLLOWING OPEN VERSUS LAPAROSCOPIC VERSUS ROBOT-ASSISTED VERSUS TRANSANAL TOTAL MESORECTAL EXCISION IN RECTAL CANCER PATIENTS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: The standard surgical treatment for rectal cancer is total mesorectal excision (TME), which may negatively affect patients' functional outcomes and quality of life (QoL). However, it is unclear how different TME techniques may impact patients' functional outcomes and QoL.

Aims & Methods: This systematic review and meta-analysis evaluated functional outcomes of urinary, sexual, and fecal functioning as well as QoL after open, laparoscopic (L-TME), robot-assisted (R-TME), and transanal total mesorectal excision (TaTME). A systematic review and meta-analysis, based on the Preferred Reporting Items for Systematic Reviews and Meta-Analysis statements, were conducted (PROSPERO: CRD42021240851). A literature review was performed (sources: PubMed, Medline, Embase, Scopus, Web of Science, and Cochrane Library databases; end-of-search date: September 1, 2023), and a quality assessment was performed using the Methodological index for non-randomized studies. A random-effects model was used to pool the data for the meta-analyses.

Results: Nineteen studies were included, reporting on 2495 patients (88 open TME, 1171 L-TME, 995 R-TME, and 241 TaTME). Quantitative analyses comparing L-TME vs. R-TME showed no significant differences regarding urinary and sexual functioning, except for urinary function at three months post-surgery, which favored R-TME (SMD [CI] -0.15 [-0.24 – -0.06], p=0.02; n=401). Qualitative analyses identified most studies did not find significant differences in urinary, sexual, and fecal functioning and QoL between different techniques.

Conclusion: This systematic review and meta-analysis suggests no significant differences in patients' urinary, sexual, and fecal functioning and their QoL between various TME techniques. However, further high-quality, randomized-controlled, and prospective cohort studies are needed to confirm these findings.

Disclosure: Nothing to disclose.

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PP0941

COX-2 MEDIATES FERROPTOSIS THROUGH REGULATING NRF2 IN HEPATOCYTES

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Introduction: Ferroptosis is an iron-dependent form of regulated cell death (RCD) characterized by lipid peroxidation and excessive oxidation (1). Cyclooxygenase-2 (COX-2), as a pivotal inducible sensor of inflammation, has been regarded as a remarkable marker in ferroptosis. However, the regulation of COX-2 on hepatic ferroptosis is unknown.

Aims & Methods: The aim of this study is to investigate whether and how COX-2 modulate ferroptosis during liver fibrosis.

Thioacetamide (TAA)-induced liver fibrosis models were constructed, and mice were divided into *Cox-2*$^{+/+}$ NS, *Cox-2*$^{-/-}$ NS, *Cox-2*$^{+/+}$ TAA and *Cox-2*$^{-/-}$ TAA groups. Primary mice hepatocytes (PMHs) were isolated. Mitochondrial morphology, iron metabolism, lipid peroxidation, and major anti-oxidation element GPX4 and Nrf2 were examined.

Results: Our data showed that ferroptosis existed in liver fibrosis induced by TAA. COX-2 knockout markedly attenuated liver ferroptosis through recovery of mitochondrial morphology, reduction of hepatic iron (Cox-2+/+ TAA vs. Cox-2-/- TAA: 0.48±0.12 mg/gram protein vs. 0.22±0.11 mg/gram protein, p=0.017), alleviation of lipid peroxidation mainly for polyunsaturated fatty acids (PUFAs) as well as remission of oxidation stress, the content of GSH dropped from 3.63±0.16 μmol/gprot to 2.24±0.24 μmol/gprot ($p<0.001$) with the decrease of GPX4 expression ($p<0.05$). Moreover, experiments in vitro also confirmed that inhibition of COX-2 significantly alleviated PMHs' ferroptosis through activating Nrf2.

Conclusion: Inhibition of COX-2 can effectively alleviate ferroptosis, representing the first demonstration of the functional role of COX-2 to regulate ferroptosis by Nrf2 in liver fibrosis.

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Disclosure: Nothing to disclose.

PP0942

THE ROLE OF OSM/OSMRB SIGNALLING AXIS IN THE DEVELOPMENT OF MASLD-RELATED HEPATOCELLULAR CARCINOMA

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Introduction: Oncostatin M (OSM) is a pleiotropic cytokine of the IL-6 family which can bind two different heterodimeric receptors: Glycoprotein 130 (GP130)/ Leukemia inhibitory factor receptor beta (LIFRβ) and GP130/OSM receptor beta (OSMRβ) that is the receptor able to mediate the most relevant biological effects of OSM in mice. OSM was found to be involved in different aspects related to tumor development and progression in several cancers including hepatocellular carcinoma (HCC). High levels of OSM were found in cirrhotic patients with different etiology carrying HCC.

In particular, patients with HCC developed on a metabolic dysfunction-associated steatotic liver disease (MASLD) background showed increased OSM serum levels that correlate with clinical parameters and disease outcome.

Aims & Methods: The aim of the study is to explore the role of OSM as a cytokine able to modulate the tumor microenvironment and affect crucial events of MASLD-related HCC development and progression taking advantage of:

a) cohort of MASLD patients with HCC and related clinical data to correlate OSM levels with HCC incidence and severity;

b) advanced molecular techniques to study gene expression changes on human THP1 macrophage cell lines and liver cancer cell lines (HepG2 cells);

c) animal models to investigate the role of OSM/OSMRβ axis in the progression of MASLD-related HCC, in particular mice genetically manipulated to delete OSMRβ in hepatocytes (OSMRβ $^{-/-}$ mice) as well as control WT littermates, submitted to the experimental DEN/CDAA MASLD-related protocol of liver carcinogenesis.

Results: Data obtained indicate that liver tumor masses in OSMRβ $^{-/-}$ mice differ significantly when compared to those observed in WT mice, including: I) a significant reduction in the angiogenic switch; II) an impaired proliferation indexes; III) a marked decreases in the recruitment/function of TAMs and Tregs lymphocytes. A similar immunosuppressive trend was observed in a small cohort of human patients with MASLD exhibiting fibrosis or HCC. Multiplex immunoassay analyses revealed a significant serum

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increase of a subset of cytokines that may be related to the tumor immunosuppressive microenvironment (TIME) (like IL1β, CCL2, IL8, CXCL13), and transcript levels of these cytokines correlate with OSM expression in MASLD patients. Of interest, higher serum levels of CCL15, a cytokine involved in favoring a TIME, were found to be associated with a worse prognosis in the cohort of MASLD-related HCC patients examined. CCL15 expression was increased in MASLD-related HCC compared to F3/F4 MASLD patients, and correlated with OSM expression.

Moreover, spatial transcriptomics analysis confirmed that CCL15 is mainly expressed by malignant hepatocytes in HCC patients of different etiologies. Furthermore, the transcript levels of CCL15 were increased in mice subjected to the DEN/CDAA carcinogenic protocol compared to mice fed a CSAA control diet or CDAA diet, and were drastic reduced in the DEN/CDAA OSMRβ$^{-/-}$ mice.

Conclusion: These results suggest a possible crucial role of OSM in the progression of MASLD, contributing to the formation of an immunosuppressive tumor microenvironment that would facilitate tumor evasion mechanisms by the immune system.

Disclosure: Nothing to disclose.

PP0943

TARGETING OF INFLAMMATORY MONOCYTES VIA CCL2/CCR2 SIGNALING AS A THERAPEUTIC STRATEGY AGAINST AUTOIMMUNE HEPATITIS BY USING MACROPHAGE-DERIVED EXTRACELLULAR VESICLES

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Introduction: Autoimmune hepatitis is a serious chronic liver disease with immune disorders, histological lesions and liver dysfunction, with a gradually increasing prevalence. Yet the cellular and molecular mechanisms of immune dysregulation in AIH are poorly understood.

Aims & Methods: Using proteomic analysis, we comprehensively profiled the differentially expressed proteins and signaling pathways of liver during human AIH. Then, monocytes and macrophage from blood and livers of AIH patients and controls were analyzed. The recruitment and polarization of monocyte-derived macrophages in AIH and the mechanism of CCL2/CCR2 axis activation were investigated by Concanavalin induced experimental AIH (EAH). The CCL2/CCR2 axis was blocked by CCL2 neutralizing antibody and CCR2 antagonist to determine its effect on AIH mice. Finally, M2 macrophage-derived extracellular vesicles (M2-EVs) were isolated and extracted as drug delivery tool of CCR2 antagonist, and its therapeutic effect on AIH mice was determined.

Results: Proteomic analysis took expression ratio (FC) > 1.5 times and P<0.05 as screening criteria, and a total of 1028 proteins were identified as increased or decreased. KEEG analysis suggested that differential expressed proteins were mainly associated with metabolic processes. The expression of mononuclear macrophage system marker proteins CD14, S100A9, CD163, CD68 and CD11b increased in the liver tissue of AIH patients as revealed by proteomic analysis and immunohistochemistry.

The proportion of classical monocytes in peripheral blood of AIH was increased, which was positively correlated with the levels of ALT and AST of AIH patients. The co-localization analysis of liver tissue suggested that CCL2 originated from Kupffer cells (KC), and the expression of CCR2 increased after circulating monocytes infiltrated liver. The expression of M1 marker in AIH liver tissue increased. AIH mouse models suggest mobilization of inflammatory monocytes on the bone marrow-liver axis and spleen-liver axis. Blocking-up of CCL2/CCR2 axis with CCL2 neutralizing antibody or CCR2 antagonist, respectively, alleviated liver injury in AIH mice, while recombinant CCL2 injection increased recruitment of inflammatory monocytes with bone marrow-liver axis and spleen-liver axis to liver, aggravating liver injury. CCR2 antagonist-M2-EV can target circulating mononuclear cells and activated mononuclear macrophages in the liver, respectively, to reduce liver injury in AIH mice.

Conclusion: AIH mediates the recruitment of inflammatory monocytes from bone marrow and spleen to liver through the CCL2/CCR2 axis, which can be inhibited by different methods to reduce liver inflammatory injury in AIH mice. M2- EVs delivers CCR2 antagonists targeting activated pro-inflammatory monocytes and hepato-splenic mononuclear macrophage system in the circulating pool, indicating that CCR2 antagonists-EVs could be a potential agent for liver and monocyte targeted therapy for AIH.

Disclosure: N/A

PP0944

PROTECTIVE EFFECT OF PUNICALAGIN ON ACUTE LIVER INJURY INDUCED BY CARBON TETRACHLORIDE IS ASSOCIATED WITH MICROPHAGE POLARIZATION

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Introduction: Acute liver injury (ALI) is a common diseases and globally important public health issue. Without properly treatment, ALI rapidly progresses to acute liver failure, which seriously threaten the life of patients. Hepatic macrophages show extreme heterogeneity under both physiological and pathological conditions. The different polarization states of macrophages are closely related to the repair of ALI. Punicalagin (PUN), the most abundant ellagic tannin in pomegranate, is a polyphenolic active substance, exhibiting antioxidant and anti-inflammatory properties to facilitate tissue repair. However, the precise mechanism of PUN on ALI remains unclear.

Aims & Methods: The aim of present study was to explore the protective effect of PUN on ALI and underling mechanism and provide a new strategy and theoretical basement for ALI of PUN treatment. The ALI mouse model was induced by intraperitoneal injection of Carbon tetrachloride (CCL_4) and then treated with PUN. The liver function and pathological damage were detected by biochemical test and Hematoxylin-Eosin (H&E) staining. Enzyme-linked immunosorbent assay (ELISA) was used to detect the expression of pro-inflammatory and anti-inflammatory factors both in peripheral blood The liver non-parenchymal cells was isolated through collagenase perfusion in situ combined with density gradient centrifugation. The total number of macrophages and the proportion of M1/M2 macrophages in liver tissues were detected by multi-color flow cytometry and multiple immunofluorescence staining. The total and phosphorylated expression of nuclear factor-kappa B (NF-κB), STAT1 and STAT3 in liver tissue were detected by Western blot (WB).

Results: PUN intervention had a protective effect on CCL4-induced ALI mice, as indicated by reducing liver index, attenuating the hydropic degeneration of hepatocytes and areas of liver injury and decreasing the levels of alanine aminotransferase (ALT) and aspartate aminotransferase (AST). Meanwhile, PUN down-regulated the concentrations of pro-inflammatory factors IL-6 and TNF-α and up-regulated the expression of anti-inflammatory factor IL-10 in liver tissue and plasma of ALI mice by inhibiting activation of NF-κB pathway. Multi-color flow cytometry and immunofluorescence staining showed that the total number of $CD45^+F4/80^+$ macrophages in CCL_4-induced ALI mice liver tissue was significantly increased compared

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with those in the control group. Monocyte-derived macrophages (MDMs, $CD45^+F4/80^+CD11b^+$) and M1 macrophages ($CD45^+CD86^+$) were significantly increased in the CCL_4-induced ALI mice liver tissue. Compared with ALI model mice, PUN treatment significantly reduced the total number of $CD45^+F4/80^+$ macrophages, the proportion of MDMs subsets and pro-inflammatory M1 macrophages, accompanied by increasing the proportion of anti-inflammatory M2 macrophages.

The results of multi-color immunofluorescence were consistent with those of flow cytometry. Western Blot showed that the levels of both phosphorylated STAT1 and STAT3 protein was remarked increased in the liver tissue of ALI mice, while PUN treatment inhibited the phosphorylation of STAT1 protein and promoted the phosphorylation of STAT3 protein.

Conclusion: The protective effect of PUN on acute liver injury induced by carbon tetrachloride is associated with microphage polarization.

Disclosure: Nothing to disclose.

PP0945

INFLAMMATORY T CELL RESPONSE INDUCED BY LSECTIN DOWNREGULATION IN CIRRHOSIS CONTRIBUTES TO ALTERATIONS OF THE LIVER-BRAIN AXIS IN AN AGED MODEL OF CIRRHOSIS

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Introduction: During chronic liver disease (CLD) the loss of the innate receptor C-type lectin LSECtin, expressed mainly by liver sinusoidal endothelial cells (LSECs), is associated to the dysregulation of hepatic immune response and Th17 expansion in the liver. CLD is also accompanied by intestinal dysbiosis, both promoting systemic inflammation and metabolic alterations that affect neurocognitive function through the liver-brain axis. As aging may worsen neurological damage and CLD and its complications are more prevalent in the aging population, we assessed the relationship of LSECtin with liver and neurocognitive damage in an aged model of cirrhosis.

Aims & Methods: CLD was induced by oral administration of carbon tetrachloride (CCl_4) for 12 weeks (w) in young mice (6 w old) and aged mice (40 w old). Liver damage was assessed through histology and molecular biology. Levels of ammonia in different tissues were examined, and a panel of behavioural assays was conducted to assess neurological impairment. Leukocyte populations from brain and liver were studied by flow cytometry (FC).

Results: Hepatic damage was more severe following the induction of CLD in aged mice compared to young ones, exhibiting significantly elevated levels of ALT and AST as well as higher levels of histological markers of fibrosis and leukocyte recruitment. This damage was accompanied by both, a loss of endothelial cells ($CD31^+$ in FC) and a decrease in LSECtin expression, especially in old cirrhotic livers.

Interestingly, we found a greater increase of Th17 (1.81 ± 0.58 vs 3.39 ± 0.63 % of CD4 cells, p = 0.003) and CD8 (12.90 ± 2.62 vs 19.22 ± 5.21 % of CD3 cells, p = 0.042) cells in the livers of aged cirrhotic mice as well. Additionally, ammonia concentrations were also higher in cirrhotic old mice than in young ones, in both serum (2.47 ± 0.74 vs 5.18 ± 2.36 nmol/uL, p = 0.015) and the brain (4.65 ± 0.41 vs 5.27 ± 0.86 nmol/uL, p = 0.114). Cirrhotic animals showed motor impairment and a tendency towards anxiety compared to healthy controls. Aged cirrhotic animals exhibited a significantly higher severity index in motor function than the young ones (p = 0.023).

Interestingly, only ammonia levels in the brain correlated with results in the open field test (r: -0.544, p = 0.003) and motor evaluation test (r: 0.444, p = 0.018), suggesting a direct relationship between brain ammonia and neurological impairment. Interestingly, we have found higher levels of CD8 cells (12.90 ± 2.62 vs 19.22 ± 5.21 % of CD3 cells, p = 0.042) in old cirrhotic mice' brains compared to young ones.

Conclusion: Aged animals exhibit lower hepatic levels of LSECtin along with more severe liver damage during the experimental cirrhosis model, with higher levels of ammonia and CD8 cells in the brain, correlating with worse neurological function. Immune imbalance in the liver favoured by LSECtin downregulation might modulate immune populations in the brain, contributing to neurocognitive damage. Young cirrhotic animals show similar neurocognitive impairment than healthy aged animals, appointing to a direct contribution of liver damage to neurological impairment during experimental cirrhosis.

Disclosure: Nothing to disclose.

PP0946

WITHDRAWN

PP0947

DIETARY SELENIUM INTAKE IN RELATION TO NON-ALCOHOLIC FATTY LIVER DISEASE ASSESSED BY FATTY LIVER INDEX AND HEPATIC STEATOSIS INDEX; A CROSS-SECTIONAL STUDY ON THE BASELINE DATA OF PROSPECTIVE PERSIAN KAVAR COHORT STUDY

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Introduction: Non-alcoholic fatty liver disease (NAFLD) encompasses a variety of hepatic disorders, including steatosis, steatohepatitis, fibrosis, cirrhosis, and hepatocellular carcinoma [1].

The global prevalence of NAFLD is 25.24%, with the highest rates related to the Middle East and South America [2].

This disorder is also becoming one of the most common causes of liver transplantation [3].

There is limited and conflicting evidence on the association between selenium and NAFLD.

Aims & Methods: Therefore, the present population-based cross-sectional study aimed to explore the relationship between dietary selenium intake and the risk of NAFLD.

A total of 3026 subjects from the PERSIAN (Prospective Epidemiological Research Studies in IrAN) Kavar cohort study were included in the analysis. The daily selenium intake was evaluated using a semi-quantitative food frequency questionnaire, and energy-adjusted quintiles of selenium intake (μg/day) were calculated. NAFLD was defined as the fatty liver index (FLI) ≥ 60 or the hepatic steatosis index (HSI) > 36. The association between dietary selenium intake and NAFLD was evaluated using logistic regression analysis.

Results: The prevalence rates of NAFLD were 56.4% and 51.9%, based on the FLI and HSI markers, respectively. The odds ratios (ORs) for FLI-defined NAFLD were 1.31 (95% confidence interval (CI): 1.01–1.70) and 1.50 (95% CI: 1.13–1.99) for the fourth and fifth quintiles of selenium intake, respectively, after adjustment for sociodemographic variables, smok-

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ing status, alcohol drinking, physical activity, and dietary factors (*P* trend = 0.002). There was also a similar association between selenium intakes and HSI-defined NAFLD (OR = 1.34 (95% CI: 1.03–1.75) for the fourth quintile and OR = 1.50 (95% CI: 1.12–2.01) for the fifth quintile of selenium intake) (*P* trend = 0.006).

Conclusion: In this large sample study, we observed a weak positive association between dietary selenium intake and NAFLD risk.

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Disclosure: None.

PP0948

EVALUATION OF INSULIN RESISTANCE AND LIPID-RELATED INDICES TO PREDICT THE PRESENCE OF NAFLD IN OBESE ADOLESCENTS

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Introduction: Non-alcoholic fatty liver disease (NAFLD) is an increasingly common health issue in obese children and adolescents. NAFLD does not occur in every obese patient and in some cases, ultrasonography is inadequate for diagnosis. In this study, BMI SDS, WC SDS, triponderal mass index (TPMI), HbA1c, HOMA-IR, TG/HDL, LAP index, triglyceride-glucose (TyG) index and aminotransferase index, used in the diagnosis of NAFLD in obese adolescents,were examined together and their diagnostic values for clinical use were compared.

Aims & Methods: Non-alcoholic fatty liver disease (NAFLD) is an increasingly common health issue in obese children and adolescents. NAFLD does not occur in every obese patient and in some cases, ultrasonography is inadequate for diagnosis. In this study, BMI SDS, WC SDS, triponderal mass index (TPMI), HbA1c, HOMA-IR, TG/HDL, LAP index, triglyceride-glucose (TyG) index and aminotransferase index, used in the diagnosis of NAFLD in obese adolescents,were examined together and their diagnostic values for clinical use were compared.

Results: In the comparison between NAFLD(+) and NAFLD(-) obese adolescent groups, HbA1c, HOMA-IR, aminotransferase index and TyG index were higher in the NAFLD (+) group (p=0.012;p=0.001; p=0.012; p=0.002, respectively). There was a positive correlation between liver fat percentage and HOMA-IR,TyG index,aminotransferase index and TG/HDL. According to regression analysis, male gender and HOMA-IR elevation were determined to be significant risk factors for NAFLD)

Conclusion: Even if no NAFLD is detected on USG imaging, evaluation of HOMA-IR and TyG index levels, especially in high-risk patients, will support the diagnosis of NAFLD. In obese adolescents with a HOMA-IR value of ≥ 4.55and a TyG index value of ≥ 8.38, evaluation should be made with liver MRI, even if no NAFLD is detected on ultrasound imaging. A basic chart with ALT, HOMA-IR, TyG, and gender with a diagnostic accuracy of 80%, positive predictive value of82.5% and sensitivity of 80% may assist in diagnosis of NAFLD in obese adolescents.

Disclosure: None.

PP0949

MYELOID-MAS DEFICIENCY MODULATES THE GUT-LIVER AXIS IN MASLD MICE: AN UNFORESEEN PROTECTIVE MECHANISM

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Introduction: The molecular mechanisms underlying lipid metabolism in metabolic dysfunction-associated steatotic liver disease (MASLD) regulated by the gut-liver axis remain poorly elucidated. Previous studies have shown that the important effector protein Mas in the RAS non-classical pathway ACE2/Ang(1-7)/Mas axis has multiple target effects such as regulating glucose metabolism and lipid metabolism, but the specific mechanism is unknown.[1-5]

This study investigated the role of Mas by examining serum bile acid (BA) profile and fecal microbiota composition.

Aims & Methods: Serum and feces samples from MASLD patients and mouse models were analyzed using metabolomics and 16S rDNA. Mas was in vivo evaluated using mice with systemic (*Mas1*$^{-/-}$) and specific deletion of *Mas1* in myeloid cells (*LysM*^{cre}*Mas1*$^{f/f}$), intestinal epithelial cells (*Vil1*^{cre}-*Mas1*$^{f/f}$), hepatocytes (*Alb*^{cre}*Mas1*$^{f/f}$), and endothelial cells (*Cdh5*^{cre}*Mas1*$^{f/f}$), after being fed with a methionine/choline-deficient(MCD) diet or high fat diet. Mouse bone marrowderived macrophages and intestinal organoids were prepared for the in vitro experiments.

Results: The relative abundance of serum BA and fecal microbiota showed similar changes in both patients and mouse models with MASLD. In *Mas1*$^{-/-}$-MCD mice, the reduced chenodeoxycholic acid (CDCA) levels observed in patients with MASLD were effectively restored, along with an notable increase in beneficial gut microbiota like Akkermansia and Bifidobacterium. These changes were also associated with a significantly improved disease phenotype and enhanced intestinal barrier function. The enriched CDCA inhibits lipid accumulation by activating hepatic and intestinal farnesoid X receptor (FXR) signaling. At the same time, we determined that myeloid cells are the primary effector cells of Mas function through cell-specific knockout of *Mas1* in myeloid cells.

Considering the metabolic reprogramming of colonic epithelial cells facilitated by polyamines derived from macrophages,[6] our findings highlight the crucial role of Mas in augmenting monocyte polyamine synthesis and promoting intestinal cell proliferation through the Mas-pS6-Arg1 pathway.

Conclusion: Myeloid-Mas deficiency improves the mouse MASLD phenotype by promoting polyamines synthesis, enhancing gut barrier function, increasing

beneficial gut microbiota, and enriching CDCA to activate hepatic and intestinal FXR signaling, thereby reducing lipid accumulation.

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Disclosure: Nothing to disclose.

PP0950

ROLES OF MODIFIABLE RISK FACTORS IN MEDIATING THE CAUSAL EFFECT OF LIVER CIRRHOSIS ON COLORECTAL CANCER: A TWO-STEP MENDELIAN RANDOMIZATION STUDY

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Introduction: This study aims to investigate the causal estimates of the correlation between liver cirrhosis and colorectal cancer (CRC) and assess the mediation effect of modifiable risk factors direct to clinical treatments.
Aims & Methods: We performed a two-step mendelian randomization (MR) analysis using publicly accessible summary-level data on liver cirrhosis and modifiable risk factors from databases except for the FinnGen, as well as CRC from the FinnGen. Single-nucleotide polymorphisms (SNPs) significantly related to exposures were selected as instrumental variables. A random-effects inverse variance weighted (IVW) method was used in the primary analysis and sensitivity analyses were complemented to ensure the reliability of causal associations. Gene set enrichment analysis (GSEA) was performed to explore the potential biological mechanisms of the causal associations.
Results: In the IVW analysis, liver cirrhosis was causally associated with CRC without adjusting for confounding factors (OR = 1.193, 95%CI = 1.008, 1.411, *p* =0.040). Multivariate MR analysis showed that body mass index, whole body fat mass, waist to hip ratio (WHR) and alcohol consumption were independent risk factors of CRC (*p* values < 0.05). The estimate of the mediator, WHR, is 0.013 and the standard error is 0.006. WHR explained 7.45% of the total effect of cirrhosis on CRC risk. GSEA analysis identified two intersection genes which worked both in liver cirrhosis and CRC tissues, namely SERPINE1 and ARKND1. The potential biological mechanisms might include epithelial–mesenchymal transition, coagulation, inflammatory response, and *etc*.

Exposure	Outcome	Method	nSNPs	Beta	Standard error	OR (95% CI)	p	Mediation effect (%)
Cirrhosis	Waist-to-hip ratio	IVW	3	0.044	0.009	1.045 (1.026, 1.064)	0.000*	
Cirrhosis	body mass index	Wald ratio	1	-0.041	0.010	0.960 (0.941, 0.980)	0.000*	
Cirrhosis	Alcoholic drinks per week	IVW	4	0.001	0.006	1.001 (0.988, 1.013)	0.914	
Cirrhosis	Whole body fat mass	IVW	4	0.007	0.022	1.007 (0.964, 1.052)	0.745	
Waist-to-hip ratio	Colorectal cancer	Multivariate IVW	228	0.302	0.107	1.352 (1.096, 1.667)	0.005*	0.013 (7.45%)
body mass index	Colorectal cancer	Multivariate IVW	462	0.006	0.090	1.006 (0.843, 1.200)	0.950	
Alcoholic drinks per week	Colorectal cancer	Multivariate IVW	33	-0.040	0.387	0.961 (0.450, 2.050)	0.918	
Whole body fat mass	Colorectal cancer	Multivariate IVW	396	0.045	0.090	1.046 (0.876, 1.250)	0.617	

Table 1. The mediator effect of modified risk factors on the causal association between liver cirrhosis and colorectal cancer.

Conclusion: Genetically predicted liver cirrhosis had potentially risk causal effect on CRC with the mediation by WHR. SERPINE1 and ARKND1 might be candidate targets for preventing CRC in patients with liver cirrhosis.
Disclosure: Nothing to disclose.

PP0951

WITHDRAWN

PP0952

WITHDRAWN

PP0953

PROGRESSION OF METABOLIC DYSFUNCTION-ASSOCIATED STEATOTIC LIVER DISEASE AMONG PATIENTS WITH TYPE 2 DIABETES MELLITUS AND NORMAL SERUM AMINOTRANSFERASE LEVELS: A 5-YEAR PROSPECTIVE COHORT

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Introduction: The incidence and severity of metabolic dysfunction-associated steatotic liver disease (MASLD) are notably higher in individuals diagnosed with type 2 diabetes mellitus (T2DM). However, studies evaluating the progression of MASLD in patients with T2DM and normal serum aminotransferase (ALT) levels are limited.
Aims & Methods: This 5-year prospective cohort study aimed to evaluate the progression of MASLD in T2DM patients with normal ALT. People with type 2 diabetes who had MASLD (defined by controlled attenuation parameter (CAP) >250 dB/min) and ALT levels that stayed normal (defined by ALT ≤40 IU/L for at least two occasions spanning at least six months) at Rajavithi Hospital, Bangkok, from November 2016 to 2017 were included. We excluded patients with viral hepatitis and alcohol consumption exceeding moderate levels. Transient elastography (TE) and CAP measurements were performed at baseline and ≥5 years of follow-up. The cutoffs to define significant and advanced fibrosis were TE ≥8 kPa and TE ≥12 kPa, respectively. All patients were instructed to reduce their weight and abstain from herbal remedies; medications for comorbidities were prescribed without limitations, subject to the physician's assessment. (most patients were under endocrinologist care.)
Results: The study included 84 patients, 57% of whom were female and had a mean age of 58.5 years. Of these, 62% were obese, with a mean BMI of 28.6 kg/m2. At baseline, the median TE was 6.92±3.35 kPa; 57 (68%) had no significant fibrosis, 22 (26%) had significant fibrosis, and 5 (6%) had advanced fibrosis, and after a 5-year follow-up, the median TE was 6.54±3.57 kPa; 69 (82%) had no significant fibrosis, 15 (18%) had significant fibrosis, and 2 (2%) had advanced fibrosis. During the follow-up, fibrosis decreased in 45 (54%), remained stable in 6 (7%), and increased in 33 (39%) patients, while a small proportion of patients experienced changes in the stages of fibrosis: 1.2% progressed from significant to advanced fibrosis; 17.4% regressed from advanced to significant fibrosis or transitioned from significant to no fibrosis; 12.8% remained stable with significant and advanced fibrosis; and the majority (66.3%) remained without significant fibrosis.
A significant factor associated with fibrosis progression was alcohol consumption (p=0.013), while the use of angiotensin receptor blockers (p=0.042) was associated with fibrosis regression. Significant factors associated with steatosis progression included high baseline BMI (p=0.013), an increase in BMI ≥ 5% (p=0.04), high fasting plasma glucose, and LDL

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cholesterol levels, while a decrease in BMI ≥ 5% (p=0.002) the use of calcium channel blockers (p=0.038) and coffee consumption >3 cups per day (p=0.031) were associated with steatosis regression. ALT elevation during the follow-up occurred in a small proportion of patients, and it was not significantly associated with fibrosis or steatosis progression.

Conclusion: The majority of MASLD patients with T2DM and normal ALT maintained a relatively stable liver disease during 5 years of follow-up. Obesity, weight gain, alcohol consumption, and inadequate metabolic control were all associated with the progression of MASLD, whereas weight loss, coffee consumption, angiotensin receptor blockers, and calcium channel blockers were all associated with the improvement of MASLD.

Disclosure: I have Nothing to disclose.

PP0954

INTRADUCTAL PAPILLARY MUCINOUS NEOPLASM IN POLYCYSTIC LIVER DISEASE: A RETROSPECTIVE STUDY

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Introduction: Intraductal Papillary Mucinous Neoplasm (IPMN) is the most frequent type of pancreatic cystic neoplasm found in the general population. Due to the advances in imaging technologies, the rate of detection of pancreatic cystic neoplasms has been increasing in the past years. A large-scale single-center cohort study conducted in Seoul revealed a prevalence of 1.7% in asymptomatic healthy individuals [1].

Polycystic liver disease (PLD) is a rare disorder characterized by the presence of at least 10 parenchymal cysts in the liver. It may be inherited in an autosomal dominant fashion and, depending on the most prominent phenotype, PLD can be an isolated manifestation in autosomal dominant polycystic liver disease (ADPLD) or it can be an extrarenal manifestation in autosomal dominant polycystic kidney disease (ADPKD) [2,3].

In ADPKD, cysts can be detected also in the pancreas, seminal vesicles and arachnoid membrane [4]. In literature there are isolated reports of association between Intraductal Papillary Mucinous Neoplasm and Autosomal Dominant Polycystic Kidney Disease [5,6].

Aims & Methods: A retrospective observational study was conducted to include all PLD patients followed at CEMAD (Centro Malattie Apparato Digerente) Fondazione Policlinico Universitario A. Gemelli to determine the prevalence of IPMNs in PLD patients (isolated or associated to ADPKD) using abdominal resonance imaging (MRIs) and/or abdominal CT scan, in order to evaluate whether IPMN can be a not acknowledged manifestation of PLD.

Results: Abdominal MRIs and/or abdominal CT scans were examined for 92 PLD patients (56 ADPKD and 36 isolated PLD). IPMNs were detected in 8 patients (8.7% of total), 4 with isolated PLD (11.7% of total isolated PLD patients) and 4 with PLD in ADPKD (7% of total ADPKD patients). As for the IPMN site, body-tail localization of the IPMN was the most frequent detected (50% of total IPMNs).

Conclusion: A high prevalence of IPMN in PLD patients was found, which seems to be higher than prevalence found in the general population (8.7% vs 1.7%). Moreover, IPMN appeared to be more frequent in isolated PLD than in PLD associated with ADPKD (11.1% vs 7%). Our results suggest we should consider the possibility of IPMN being another extrahepatic manifestation of PLD. Of course, further studies in a larger population of patients are needed in order to confirm our findings.

	TOTAL	ISOLATED PLD	PLD IN ADPKD
N	92	36	56
Mean Age (years)	59.7	59.7	59.6
IPMN: N°	8 (8.7%)	4 (11.1%)	4 (7%)
IPMN site: head;	1 (12.5%);	0;	1 (25%);
body-tail;	4 (50%);	1 (25%);	3 (75%);
multifocal	3 (37.5%)	3 (75%)	0

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Disclosure: Nothing to disclose.

PP0955

CARDIOVASCULAR DISEASE BURDEN IN PATIENTS WITH METABOLIC DYSFUNCTION-ASSOCIATED STEATOHEPATITIS VERSUS MATCHED-CONTROL PATIENTS: DATA FROM THE UNCOVER-MASH LONGITUDINAL COHORT STUDY

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Introduction: Metabolic dysfunction-associated steatohepatitis (MASH) is linked to increased risk of cardiovascular (CV) disease (CVD), but many aspects of CVD burden in patients with MASH remain unknown. Particularly, data on the clinical burden of CVD and specific CV events in patients with non-cirrhotic MASH vs those without non-cirrhotic MASH are limited.

Aims & Methods: The aim of this real-world study was to assess CVD burden in patients with and without non-cirrhotic MASH. Patients with MASH were identified using the International Classification of Diseases code for non-alcoholic steatohepatitis (K75.81) from October 2015–June 2022, using data from a federated network (TriNetX). The index date was defined as the date of first MASH diagnosis. Exclusion criteria were viral hepatitis, alcohol use disorder, chronic liver diseases other than MASH or metabolic

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dysfunction-associated steatotic liver disease, cirrhosis, decompensated MASH, hepatocellular carcinoma and human immunodeficiency virus. Patients with MASH were matched 1:1 to patients without MASH based on demographics (sex, age, year of start/end of insurance enrolment, and insurance scheme).

Outcomes were CVD prevalence (assessed up to 1 year prior to index) and risk of CV events(s) during follow-up (index date to end of enrolment, death or study end) amongst patients with no history of respective CV event(s) at baseline. For CV risk, cumulative incidence was plotted using Aalen-Johansen curves after adjusting for competing risk of non-CV death. Incidence rate (IR) was calculated per 100 person-years (PY) and hazard ratios (HR) were calculated using Cox proportional hazard models (crude and adjusted for CV risk factors [age, sex, type 2 diabetes (T2D), chronic kidney disease (CKD), obesity, hyperlipidaemia and hypertension]).

Results: In total, 9,642 patients (4,821 with MASH; 4,821 without MASH) were included. In both groups, mean (standard deviation) age was 50.8 (13.2) years and 58.5% were women. Compared with those without MASH, more patients with MASH had hypertension (58% vs 31%), hyperlipidaemia (57% vs 30%), obesity (45% vs 17%), T2D (41% vs 14%) and CKD (6% vs 3%) at baseline.

Prevalence of any CVD was higher in patients with MASH than without MASH (23% vs 12%), as was risk of any CV event (IR: 11.1 vs 4.5 per 100 PY; crude HR [95% confidence interval]: 2.72 [2.30, 3.22], p < 0.0001; adjusted HR: 2.27 [1.95, 2.66], p < 0.0001). Similar results were seen for individual CV events, namely ischaemic heart disease (IR: 5.3 vs 2.3), cerebrovascular disease (IR: 4.4 vs 2.1), atherosclerosis (IR: 3.9 vs 1.8), and heart failure (IR: 3.2 vs 1.5).

Conclusion: CVD prevalence at baseline and risk of CV events during follow-up were significantly higher in patients with non-cirrhotic MASH compared with matched controls without MASH in a real-world cohort. These data provide additional evidence of the association between MASH and CVD and complement a previous analysis of the unCoVer-MASH study.

Disclosure: EB has acted as a consultant for AstraZeneca, Boehringer Ingelheim, Bristol Myers Squibb, Gilead Sciences, Intercept Pharmaceuticals, Inventiva SA, Merck Sharp & Dohme (UK) limited, Novo Nordisk and Pfizer.

AM has no conflicts of interest.

KKM, ASC, KG, NK and ASP are employees of Novo Nordisk A/S and own employee shares/stocks.

KEC has acted as a consultant for Theratechnologies and Novo Nordisk, and has received grants from Bristol Myers Squibb and Novartis.

PP0956

SARCOPENIA IN LIVER TRANSPLANT RECIPIENTS: AN EMERGING CHALLENGE OF LONG-TERM CARE.

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Introduction: Sarcopenia has been described in 10-70% of patients with cirrhosis and it negatively impacts the outcomes of liver transplant (LT). Less information is known about changes in body composition after LT, especially at long-term follow-up.

Aims & Methods: This study aims to assess the prevalence of sarcopenia and its possible associated factors in a cohort of LT recipients with a long-term follow-up.LT recipients were enrolled from the Liver Transplant Follow-up Center of the University Hospital of Salerno. Demographic characteristics, anthropometric data, data about LT, and prevalence of comorbidities were collected.

Patients underwent a physical examination, laboratory tests (including immunosuppressive drug levels), the Hand Grip Test for the evaluation of the manual grip force (FPM) using a calibrated hand-held dynamometer (DynX digital dynamometer - Akern), and a bioelectrical impedance analysis (BIVA), to assess body composition using the device "BIA 101" (Akern, Florence).

The considered cut-off FPM values were<27 kg for males and <16 kg for females. Appendicular lean mass (ASM) values lower than 20 kg in males and 15 kg in females and ASMM/ h^2 ratio lower than 7 kg/m^2 in males and 5.5 kg/m^2 in females were considered diagnostic for sarcopenia.

Parameter	B	S.E.	Wald	Sig.	Exp(B)	95,0% CI per EXP(B) Lower Limit	95,0% CI per EXP(B) Upper Limit
Years of follow-up	.076	.075	1.019	.313	1.079	.931	1.250
BMI (Kg/m^2)	-1.066	.480	4.924	.026	.344	.134	.883
Age (years)	-.021	.058	.126	.723	.980	.874	1.098
Type of immunosuppression	-1.308	1.408	.863	.353	.270	.017	4.272
Etiology of LT	-.547	.462	1.403	.236	.579	.234	1.430
Costant	28.558	13.162	4.708	.030	3E+012		

Table 1. Linear regression model with the presence of sarcopenia at BIVA as dependent variable, and age, years of follow-up, BMI, type of immunosuppressive therapy (calcineurin inhibitors/mTOR inhibitors), and disease etiology at LT as covariates.

Legend: S.E.= Standard Error; BMI= body mass index; LT=liver transplant

Results: 81 LT recipients (57 males, 70.4%) were included. Mean age was 64.7±10.3 years, mean body mass index (BMI) was 27.8±4.3Kg/m^2; 25/81 (30.9%) were obese. Seventeen patients (21%) underwent LT for Hepatitis B Virus Cirrhosis, 21 (25.9%) for Hepatitis C Virus Cirrhosis, 11 (13.6%) for alcoholic cirrhosis, 5 (6.1%) for NASH cirrhosis, the remaining for other causes (autoimmune diseases, cryptogenic). The mean follow-up after LT was 14.0 ±7.2 years. Thirty-five/81 (43.2%) LT recipients showed pathological FPM values and 18/81 (22.2%) showed BIVA values diagnostic for sarcopenia.

At logistic regression, the presence of sarcopenia at BIVA was significantly associated with low BMI whereas no significant associations were found with other covariates: age, years of follow-up, BMI, type of immunosuppressive therapy (calcineurin inhibitors/mTOR inhibitors), disease etiolo-

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gy at LT. (Table 1).

Conclusion: Sarcopenia affects a significant percentage of LT patients, also at long-term follow-up. A low BMI seems to be associated with sarcopenia, whereas no correlation was found with the type of immunosuppression and the etiology of the disease at LT.

Disclosure: Nothing to disclose.

PP0957

HYPERHOMOCYSTEINEMIA IS ASSOCIATED TO MASLD INDEPENDENTLY FROM ATHEROSCLEROSIS

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Introduction: Homocysteine (Hcy) levels are elevated in a variety of conditions, including cardiovascular diseases (CVD), diabetes, hyperinsulinemia, and metabolic-associated steatotic liver disease (MASLD). Although the relationship between MASLD and cardiovascular diseases (CVD) is epidemiologically well established (1), the pathogenetic process linking such conditions has not been completely clarified.

Aims & Methods: In this cross-sectional study, we analyzed Hcy levels in a population of 901 control and metabolic outpatients, considering the presence of MASLD and other dysmetabolic conditions, aiming to detect and deepen the putative role of Hcy in the relationship between liver steatosis and CVD. Hyperhomocysteinemia (HHcy) was defined for Hcy levels ≥15 µmol/l. Abdominal and carotid ultrasounds and comparison of Hcy levels were performed in all subjects.

Results: HHcy was identified in 140 subjects. Comparisons between subjects with Hcy<15 and Hcy>15 showed that the latter had significantly increased waist circumference (WC) (p=0.00001). HDL-c (p=0.00071) and 25-OH-Vitamin D (p=0.00002) levels were significantly lower, and triglycerides were significantly raised (p=0.04979) in subjects with HHcy compared to patients with Hcy<15.

We evaluated Hcy levels in subjects with and without such conditions, finding that Hcy did not differ between patients with and without carotid plaques (p=0.2182), but was different in presence of liver steatosis (p=0.00192) *(Table 1)*.

Subjects with HHcy had a strong association with liver steatosis (chi-square p=0.00001), whereas we did not find the same association in presence of atherosclerotic plaques (chi-square p=0.08388). ORs analysis showed that subjects with HHcy subjects displayed an increased risk of fatty liver, with an OR of 2.1 (95% CI 1.3-3.2, p=0.00001), whilst the risk of being diagnosed with atherosclerotic plaques, though slightly increased (OR 1.2, 95% CI 0.8-1.7), was not significant (p=0.09938).

We calculated chi-square and ORs for HHcy association with increased BMI and each criterion of the IDF definition of MetS. In the overall population, regardless of the liver steatosis diagnosis, the cardiometabolic risk conditions significantly associated with HHcy were impaired fasting glycemia (FPG) (OR 2.1, 95% CI 1.4-3.1, p=0.00001), hypertriglyceridemia (OR 1.8, 95% CI 1.2-2.6, p=0.00028) and arterial hypertension (AH) (OR 1.7, 95% CI 1.4-3.1, p=0.00088).

In subjects with MASLD, the same analysis highlighted higher ORs, with hypertriglyceridemia showing the highest OR (3.3, 95% CI 2.6-3.8, p=0.00001), followed by AH (OR 2.6, 95% CI 1.9-2.9, p=0.00011), FPG (OR 2.5, 95% CI 2.1-2.8, p=0.00083) and WC (OR 2.4, 95% CI 1.6-2.8, p=0.00099), whilst the BMI criterion was not associated with HHcy.

	NO Plaque N=313 (128M:185F)	Plaque N=343 (199M:144F)	p-value
Hcy	11.6±6.3	12.1±8.2	0.2182

	NO Steatosis N=321 (147M:174F)	Steatosis N=510 (282M:228F)	p-value
Hcy	12.6±8.3	18.3±9.2	0.00192**

Table 1. Comparison of homocysteine (Hcy) levels in patients with and without atherosclerotic plaque and liver steatosis.

Conclusion: Our data suggest an intriguing scenario where the reciprocal correlation between liver steatosis and HHcy may worsen underlying metabolic dysfunctions leading to CVD, and further exacerbate the key pathological features of MASLD. Addressing these interconnected risk factors could represent a crucial strategy for mitigating CVD burden in the increasingly prevalent population of patients with liver steatosis.

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Disclosure: Nothing to disclose.

PP0958

WITHDRAWN

PP0959

RECURRENT ACUTE LIVER FAILURE (RALF) IN CHILDREN – UNRAVELING THE MYSTERY

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Introduction: Pediatric Acute Liver Failure (PALF) presents a considerable threat to children with a high risk of mortality, despite its rarity. However, the exact cause remains unknown in about half of the cases, emphasizing the need for continued investigation and advancements in our understanding of the condition. Recurrent Acute Liver Failure (RALF) represents an even more uncommon phenomenon, defying diagnosis despite thorough examination. The advent of Whole Exome Sequencing offers hope in uncovering its elusive genetic origins, potentially shedding light on specific genetic disorders underlying this condition.

Aims & Methods: This study aimed to report spectrum of Recurrent Acute Liver Failure (RALF) cases and role of Whole Exome Sequencing (WES) in establishing a cause for RALF, during January 2023 – January 2024 in a tertiary care institute.

Results: We report 4 cases of Recurrent Acute Liver Failure (RALF). First case was a 5 year old 1st born female child of 3rd degree consanguineous marriage who presented with RALF during febrile episodes at 10 months, 18 months and 5 years. Child had low GGT (19 IU/L) cholestasis initially. Etiological workup for Wilson, infective, metabolic was normal. Provisionally, Fever Associated Infantile Liver Failure (ILFS) was thought. But, Upper GI endoscopy revealed esophageal varices confirming portal hypertension, and liver biopsy revealed- acute on chronic hepatitis with cirrhosis, confirming chronic liver disease, which was in contrast to complete liver function resolution seen in ILFS.

Change in gait and expressive language ability was lost from 3 years of age. The child had rigidity, increased deep tendon reflexes, tremor and ataxia on examination. Genetic analysis revealed homozygous (autosomal recessive) likely pathogenic variant of SCYL1 gene mutation leading to Spinocerebellar Ataxia 21.

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The second child also presented with RALF at 8 months, 1year and 1.5 years during febrile episodes, with low GGT Cholestasis. WES revealed pathogenic variant of SCYL1 gene mutation leading to Spinocerebellar Ataxia 21. Child died at 1.5years of age.

Third child is a 10 year old girl presented with multiple fever-related RALF since 9 years of age and was asymptomatic in between episodes. As the cause was indeterminate after extensive evaluation, WES was done which revealed novel compound heterozygote mutation in RINT1 gene in exon 8 and 12.

The fourth child is a sibling of third case who succumbed with the first acute liver failure following a febrile episode at 12 years of age.

Conclusion: Understanding the genetic basis of PALF is crucial for accurate diagnosis and management. Genetic testing, including targeted gene sequencing or whole-exome sequencing, can help identify these mutations and provide insights into the underlying causes of PALF in affected individuals. This information is essential for personalized treatment approaches and genetic counseling for families. In addition to metabolic and mitochondrial diseases, mutations in genes related to both cytosolic and mitochondrial pathways—such as NBAS, TRMU, IRAS, MARS, RINT1, and SCYL1—have been implicated in pediatric acute liver failure (PALF).

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Disclosure: Nothing to disclose.

PP0960

FIBROSIS AND INFLAMMATION MARKERS IN PEDIATRIC METABOLIC-ASSOCIATED STEATOTIC LIVER DISEASE

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Introduction: The non-invasive identification of liver fibrosis related to metabolic-associated steatotic liver disease is crucial for risk stratification in children [1]. Non-invasive tests, including serum collagen-derived and inflammatory markers, may represent a good surrogate for risk stratification [2].

Aims & Methods: The aim of our study was to investigate the differences in fibrosis and inflammation markers in children with metabolic-associated steatotic liver disease (MASLD) depending on the liver fibrosis presence.

94 patients aged 9 to 17 years (average age 12.15 ± 2.51 years) were included in the study. According to body mass index, transient elastography (Fibroscan®502 touch, Echosens, France) data patients were divided into 4 groups: 1 group consisted of 27 obese patients with MASLD and liver fibrosis ≥F1 (METAVIR), 2 group - 35 obese patients with MASLD without fibrosis, 3 group - 18 obese patients without MASLD, 4 group (control) - 14 patients with normal weight without MASLD. Serum content of hydroxyproline free (HPf) and hydroxyproline protein-bound (HPp/b) according to Osadchuk and glycosaminoglycans (GAG) according to Rimington were evaluated. Serum IL-10 and IL-6 levels were evaluated by an ELISA kit (Wuhan Fine Biotech Co., Ltd, China).

Results: The levels of serum HPp/b were elevated in 100% of 1 group children compared to the control group as well as in 86.9% and 53.8% of 2 and 3 group children respectively (p<0,05). HPp/b /HPf ratio was increased in children of 1 group by 1.4 times (p<0.05), 2 group - by 1.4 times (p<0.05), 3 group - by 1.3 times (p<0.05) in comparison with the control group. Correlation of HPp/b /HPf ratio with CAP (r=0,307; p<0,05), liver stiffness (r=0,323; p<0,05) was found. Serum GAG content in 1 group of children was 1.3 times (p<0.05) higher as well as IL-6 /IL-10 ratio was 5,4 times higher in comparison with the control group (p<0.05).

Conclusion: Thus, an imbalance between the collagen synthesis and degradation in obese MASLD children was revealed combined with a shift towards pro-inflammatory cytokine production, which might promote fatty liver fibrosis development.

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Disclosure: Nothing to disclose.

PP0961

FROM HEPATIC STEATOSIS TO METABOLIC DYSFUNCTION–ASSOCIATED STEATOTIC LIVER DISEASE (MASLD): A HOSPITAL COHORT OF 450 PATIENTS

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Introduction: Metabolic dysfunction–associated steatotic liver disease (MASLD), formerly known as non-alcoholic fatty liver disease (NAFLD), is now recognized as the leading cause of chronic liver disease. Diagnosis can be made by simple tests available in primary care setting, particularly abdominal ultrasound and FIB4 index. More accurate assessment of fibrosis and specialized follow-up may be necessary to prevent progression to cirrhosis or hepatocellular carcinoma.

Aims & Methods: The aim of our study was to define the prevalence of hepatic steatosis and MASLD in a hospital cohort, to analyze the factors associated with MASLD in our population, and to select patients who need more accurate fibrosis assessment and closer follow up. A prospective descriptive and analytical study was conducted over a 7 months period. After consenting to participate to the study, patients consulting our department for gastroenterological or proctological complaints were included, excluding those followed up for known chronic liver disease.

Epidemiologic and clinical data were collected, followed by abdominal ultrasonography. We first divided our population into 2 groups according to the presence of steatosis on ultrasound. FIB4 index were then calculated for patients with steatosis. Statistical analysis was performed with IBM SPSS Statistics 20.0 software.

Results: Among the 450 patients included, the prevalence of hepatic steatosis was 24.88 %. The mean age was 50.6 (16 - 98 years). Older age (p = 0.005), obesity (p = 0.02) and diabetes (p < 0.001) were significantly asso-

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ciated with the presence of steatosis. In the steatotic population, 37.5% were diabetic (88.09% of type 2 diabetes), and 20.5% hypertensive. All these patients were overweight or obese, with a mean BMI of 29.01 kg/m^2 and a mean waist circumference of 92.7 cm. Seven point forteen percent were drinking alcohol, with an average intake of 25 g/week. Twenty-two patients (29.34%) were found to have a Fib4 > 1.3.

Comparing these two steatotic subgroups, we found a significant difference in diabetes and obesity, which were significantly more prevalent in the Fib4>1.3 group (p<0.001 and p=0.02, respectively).

Conclusion: Our study confirms the high prevalence of hepatic steatosis, especially MASLD, which becomes a major public health problem. Advanced age, diabetes and obesity were the main risk factors associated with MASLD in our study. These data highlight the importance of systematic screening for fibrosis in these populations, as well as specialized follow-up.

Disclosure: Nothing to disclose.

PP0962

THE GUT MICROBIOME IN OBESE CHILDREN WITH METABOLIC-ASSOCIATED STEATOTIC LIVER DISEASE DEPENDING ON THE GALLBLADDER FUNCTION

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Introduction: Numerous studies suggested a pivotal role of gut microbiota and its metabolites in the development and progression of metabolic-associated steatotic liver disease (MASLD) [1]. So, modification of the gut microbiota and/or the metabolites could be beneficial in children with MASLD [2].

Aims & Methods: The aim of our study was to determine the gut microbiome composition in obese children with metabolic-associated steatotic liver disease (MASLD) depending on the contractile function of the gallbladder.

A survey included 73 children aged 10 to 17 years (mean age – 12.15 ± 2.51 years). According to body mass index, transient elastography (Fibroscan®502 touch, Echosens, France), and ultrasound data patients were divided into 3 groups: 1 group – 35 obese MASLD children with gallbladder hypokinesia, 2 group – 30 obese MASLD children with gallbladder normokinesia, 3 group (control) – 8 healthy children with normal weight and gallbladder normokinesia. Contractile gallbladder function was assessed by ultrasound examination after physiological nutritional load.

Qualitative and quantitative intestinal microbiome composition were investigated by bacterial culture method using ten-fold dilutions (10^{-1} - 10^{-9}) on a standard set of selective and differential diagnostic culture media for the aerobic and anaerobic microbes isolation. Fecal short-chain fatty acid (SCFA) content was evaluated with gas chromatography (Chromatec-Crystal-5000).

Results: An increase in the fecal butyric acid concentration in 2.0 times (p<0.05) was observed in MASLD children with gallbladder hypokinesia compared to the control group. Also, an increase in the fecal acetic acid concentration in 6.9 times (p<0.05) in 1 group children and in 6.8 times (p<0.01) in 2 group children was found compared to the control group.

However, a significant decrease in the fecal propionic acid content in 2.4 times (p<0.001) in MASLD children with gallbladder hypokinesia and in 3.6 times (p<0.001) in MASLD children with gallbladder normokinesia was revealed compared to the control group.

Due to the increased content of fecal acetic and butyric acids, the total content of SCFA (Σ (C2-C4)) was increased in 2.3 times in 1 group children compared to the control group (p<0.05).

Patients with decompensated dysbiosis predominated in 1 group children (37.1% of patients). In 2 group children the subcompensated dysbiosis was more common (36.7% of patients). The concentration of *Lactobacillus* and *Enterococcus* in 1 group patients was significantly lower than in 2 group patients: in 1.9 times (p<0.05) and in 1.4 times (p<0.05), respectively.

Conclusion: Impaired contractile function of the gallbladder in MASLD children is associated with a sharp decrease in the number of major symbionts of the intestinal microbiota as well as increased production of fecal acetic and butyric SCFA.

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Disclosure: Nothing to disclose.

PP0963

PREVALENCE OF MASLD AND ITS ASSOCIATION WITH THE CONSUMPTION OF SUGAR SWEETENED BEVERAGES AMONG A NON-COMMUNICABLE DISEASE COHORT IN RURAL TANZANIA

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Introduction: The global health community has increasingly shifted its focus toward non-communicable diseases (NCDs), recognizing them as a critical public health emergency, particularly pronounced in low and lower middle-income countries (LMICs)[1]. Metabolic dysfunction associated steatotic liver disease (MASLD) is the most common liver disease with a worldwide prevalence of 30%[2], and represent the leading cause of liver-related morbidity and mortality. The prevalence, associated factors, and consequences of MASLD in the African continent, particularly in SSA, remain poorly understood. Indeed, no SSA studies were included in the latest major literature review and meta-analysis on the global epidemiology of MASLD[2].

Tanzania, like many other SSA countries, is undergoing a demographic and epidemiological transition, leading to a marked increase in the burden of NCDs [3].

Aims & Methods: The primary aim of this study was to estimate the prevalence of MASLD among an NCDs population in rural Tanzania. Additionally, secondary objective was to investigate whether the consumption of sugary drinks is associated with MASLD.

This study entailed a retrospective analysis of a prospectively collected cohort of consecutive patients attending the NCDs outpatient clinic of Tosamaganga Hospital, from August 2022 to March 2023. Detailed socio-

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demographic, clinical, laboratory, and abdominal ultrasound data were collected for all participants. Additionally, the Beverage Intake Questionnaire (BEVQ-15) was administered to evaluate their everyday drinking behaviors; and for each item, the average daily intake in fluid ounces (fl-oz), and calories (kcal) were calculated.

Results: Of 183 patients enrolled in the cohort, 128/183 (70%) were female. The mean age was 58±13.7 years and mean BMI was 28±6.67 Kg/m^2. Hypertension affected 141 individuals (77.0%), diabetes affected 77 (42.1%), and both conditions co-occurred in 35 patients (19.1%). Steatotic liver disease was found in 88/183 (48%) patients; with 39/88 (44.3%) of moderate or severe steatosis. MASLD was identified in all steatotic patients. Overall, on average the participants drank a mean of 12.6 fl-oz of SSB (throght sodas). MASLD patients drank an average of 22.4 fl-oz of SSB per day. Multivariable analysis revealed that BMI and daily SSB consumption (fl-oz) exhibited significant associations with MASLD, with respective OR of 1.16 (95% CI: 1.06 – 1.28, p = 0.002) and 1.19 (95% CI: 1.12 – 1.27, p < 0.001).

	Univariable model OR (CI 95%)	P-value	Maultivariable model OR (CI 95%)	P-value
Gender male	0.50 (0.26 – 0.96)	0.039	1.28 (0.42 – 3.84)	0.663
Age (years)	1.02 (0.99 – 1.04)	0.087	0.92 (0.94 – 1.03)	0.422
BMI (Kg/m^2)	1.24 (1.16 – 1.33)	<0.001	1.16 (1.06 – 1.28)	0.002
Hypertension (%)	2.53 (1.22 – 5.27)	0.013	1.01 (0.24 – 4.14)	0.994
Hemoglobin (g/dL)	1.40 (1.13 – 1.72)	0.002	1.51 (1.07 – 2.15)	0.020
Sartans	3.58 (1.81 – 7.10)	<0.001	1.63 (0.52 – 5.09)	0.404
Insulin	0.27 (0.08 – 0.63)	0.005	1.63 (0.28 – 9.39)	0.585
Water, daily (fl-oz)	0.99 (0.98 – 1.00)	0.018	0.99 (0.98 – 1.00)	0.150
Sugar-sweetened beverages, daily (fl-oz)	1.19 (1.13 – 1.26)	<0.001	1.19 (1.12 – 1.27)	< 0.001

Conclusion: This study unveils a considerable prevalence of MASLD among the rural Tanzanian NCDs population, affecting nearly half (48%) of all patients studied. Timely identification of MASLD, facilitated by straightforward diagnostic means, holds pivotal importance for patient management and the dissemination of vital lifestyle modification guidance. Furthermore, our study, one of the pioneering efforts to demonstrate a significant association between SSB consumption and MASLD in an African context, underscores the imperative to heighten awareness regarding the risks posed by SSB consumption among NCDs patients and potentially the wider population.

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Disclosure: Nothing to disclose.

PP0964

CLINICAL AND BIOLOGIC DIFFERENCE BETWEEN THE LEAN PHENOTYPE AND THE NON LEAN PHENOTYPE IN NAFLD

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Introduction: NAFLD has been linked to metabolic syndrome and obesity. Nowadays more lean patients, who do not necessarily have a metabolic syndrome are diagnosed with NAFLD.

Aims & Methods: We aim to determine the clinical and biological characteristics of patients diagnosed with NAFLD, and to compare patients with a lean phenotype to those overweight and obese. A retrospective and descriptive study was conducted in Mongi Slim hospital. Clinical and biological data were retrieved after consulting patient records, in addition to results of transient elastography. Lean phenotype was defined by having a body mass index of less than 25 kg/m^2.

Results: A total of 78 patients were included in this study, with a mean age of 55 ±10.3 years [26-77 years] and a gender ratio F/M of 1.5. Lean NAFLD was found in 23.1%. Type 2 diabetes, dyslipidemia and hypertension were found in 38.5%, 33.3% and 41% of patients respectively. Significant fibrosis was found in 16.7% of patients. Comparatively to overweight and obese patients, prevalence and diabetes was significantly lower in lean patients (45% vs. 16.7%; p=0.03). These patients also had significantly less abdominal fat than the fore mentioned groupe (waist circonference: 102.8 ±11.3 cm [83-140cm] vs. 83.16 ±6.5 cm [75-95cm]; p<0.001). Although hypertension and dyslipidemia were more prevalent in obese and overweight patients, difference between the two groups was not statistically significant (hypertension: 46.7% vs. 22.2%; p=0.06; Dyslipidemia: 38.3% vs. 16.7% ; p=0.08). Advanced fibrosis was present in comparable amount in the two groups of patients (p=0.1). In terms of prediction of fibrosis, the performance of the fib-4 and NFS score seem to be superior in the lean NAFLD group (p<0,001 respectively).

Conclusion: In this study, the prevalence of metabolic syndrome was greater in overweight and obese patients compared to lean patient all diagnosed with NAFLD. there seems to be no difference in the two groups in terms of prevalence of advanced fibrosis.

Disclosure: Nothing to disclose.

PP0965

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PP0966

LONG-TERM ADMINISTRATION OF PEMAFIBRATE PLUS SODIUM-GLUCOSE COTRANSPORTER-2 INHIBITOR IN PATIENTS WITH DYSLIPIDEMIA AND TYPE 2 DIABETES COMPLICATED BY METABOLIC DYSFUNCTION-ASSOCIATED FATTY LIVER DISEASE

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Introduction: Metabolic dysfunction-associated fatty liver disease (MAFLD) has become the most common chronic liver disease in recent years. However, medications to improve MAFLD has no standard. Several potential agents have been widely investigated for the treatment of MAFLD. Pemafibrate, a selective peroxisome proliferator-activated receptor-α modulator, has been reported to improve dyslipidemia. Sodium-glucose cotransporter-2 inhibitor (SGLT2i) is an oral antidiabetes drug that promotes glucose excretion into the urine. These medications have potential therapeutic effects on MAFLD.

Aims & Methods: This retrospective study aimed to evaluate the effects of long-term administration of pemafibrate plus SGLT2i in patients with dyslipidemia and type 2 diabetes complicated by MAFLD. Transient elastography (FiboScan; Echosens, Paris, France) was used for liver stiffness measurement (LSM) and assessment of the controlled attenuation parameter (CAP). In total, 594 patients with dyslipidemia and type 2 diabetes complicated by MAFLD started pemafibrate plus SGLT2i treatment between 2019 and 2020 at our hospital. Pre- and post-treatment transient elastography data were compared.

Results: Following 3 years of pemafibrate plus SGLT2i treatment, the median LSM significantly decreased from 10.1 kPa to 6.4 kPa ($P < 0.001$) and the median CAP significantly decreased from 321 dB/m to 219 dB/m ($P < 0.001$), in spite of no significant reductions of the studied patients' weight. Moreover, serum parameters, including hemoglobin A1c, alanine aminotransferase, and gamma-glutamyl transferase, significantly decreased over the study period.

Conclusion: The findings of our study confirm that long-term administration of pemafibrate plus SGLT2i can improve hepatic fibrosis by ameliorating hepatic steatosis and inflammation in patients with dyslipidemia and type 2 diabetes complicated by MAFLD.

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Disclosure: Nothing to disclose.

PP0967

DRUG-INDUCED LIVER INJURY CAUSED BY METAMIZOLE: IDENTIFICATION OF A PATHOGNOMONIC INJURY PATTERN

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Introduction: Drug-induced liver injury (DILI) caused by metamizole has gained increasing attention in recent years[1, 2]. However, causality assessment remains a challenge, in particular in patients with multiple drug intake.

Aims & Methods: We further characterized metamizole DILI cases with the goal to identify typical patterns that can help adjudication. To this end, the data of 88 patients with metamizole intake from our prospective study on acute liver injury with potential drug-related cause were analysed. DILI diagnosis and causality assessment was based on clinical, laboratory and histological features as well as expert adjudication[3]. Alternative causes of liver injury were identified or ruled out by a thorough hepatological work-up.

Results: DILI was associated with metamizole in 63 of 324 DILI patients from our cohort (prevalence 19.4%). Out of the remaining 25 patients with metamizole intake, nine patients suffered from non-metamizole DILI and 16 patients from an alternative cause for liver injury (mainly autoimmune hepatitis [n=10] and biliary obstruction [n=4]). A distinctive phenotype was observed in 45 of the 63 metamizole DILI cases, which was characterized by a relatively long latency from the start of drug intake until recognition of liver injury (median: 51 days), high transaminases at onset (aspartate aminotransferase [AST] 36.9xULN, alanine aminotransferase [ALT] 42.5xULN) and a delayed peak of bilirubin with a median of 7 days after DILI recognition. Patients with this phenotype had higher rates of jaundice (100.0% vs. 62.8%, p<0.001), coagulopathy (73.7% vs. 20.9%, p<0.001) and acute liver failure (40.0% vs. 7.0%, p<0.001) when compared to metamizole DILI cases without this clinical picture, non-metamizole DILI and non-DILI controls, however remission rates did not differ significantly when compared to the controls (71.1% vs. 76.7%, p=0.481) or to the recovery rates described in larger registry studies on DILI in general[4]. Overall, adverse outcome as defined by death or liver transplantation was observed in nine patients with metamizole DILI (14.3%). On multivariate analysis, AST and INR at DILI recognition were the only parameters independently associated with this endpoint. Regarding prediction of an adverse outcome INR had the highest discrimination (c-statistic 0.89, 95% confidence interval [CI]: 0.74-0.99) and with this performed better than Hy's law, transaminases, bilirubin or the model for end-stage liver disease (MELD) score at baseline in terms of prognosis. INR at a cut-off of ≥1.95 showed a sensitivity and specificity as high as 77.8% and 93.5% for an adverse outcome.

	Metamizole signature pattern (n=40)	Control patients† (n=43)	p
AST at onset (xULN)	36.9 (2.5-115.3)	12.7 (1.2-213.8)	<0.001*
ALT at onset (xULN)	42.5 (3.9-138.5)	15.1 (0.8-223.0)	<0.001*
TBIL at onset (xULN)	9.8 (0.2-24.3)	1.4 (0.2 -22.3)	<0.001*
Time to peak ALT (days)	1 (0-18)	1 (0-50)	0.467
Time to peak TBIL (days)	7 (0-22)	3 (0-82)	0.029*
Coagulopathy	33 (73.7 %)	9 (20.9 %)	<0.001*
Acute liver failure	18 (40.0 %)	3 (7.0 %)	<0.001*
Remission	32 (71.1 %)	33 (76.7 %)	0.481
Time to remission (weeks)	10 (3-168)	14 (1-87)	0.209

†Control cases are comprised of other metamizole DILI (n=18), non-metamizole DILI (n=9) and non-DILI cases with metamizole intake (n=16).

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Conclusion: Patients with DILI due to metamizole can present with a specific pattern which can help to identify metamizole as the causative agent and avoid erroneous association of liver injury with innocent bystander medication. Outcome, however, is not associated with this clinical picture and should rather be predicted by AST and especially INR at onset.

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PP0968

THROMBOXANE A_2 RECEPTOR SIGNALING IN MACROPHAGES PROMOTES LIVER REPAIR AFTER ACETAMINOPHEN-INDUCED LIVER INJURY BY INCREASING ACCUMULATION OF REPARATIVE MACROPHAGES AND PRODUCTION OF HGF

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Introduction: Acetaminophen (APAP) overdose causes severe acute liver failure. Impaired liver repair and regeneration after APAP hepatotoxicity leads to failed recovery and mortality. Accumulating evidence indicates that macrophages play an important role in liver repair after APAP-induced liver injury; however, underlying mechanisms of involvement of macrophages remain unknown.

Aims & Methods: We examined the role of endogenous thromboxane A_2 (TXA_2) in macrophages in liver repair after APAP-induced liver injury.

APAP (300 mg/kg, ip) was administered to macrophage-specific thromboxane prostanoid receptor (TP) deficient mice ($TP^{\triangle mac}$) and control mice (Cont).

Results: Compared with Cont, $TP^{\triangle mac}$ exhibited severe liver injury as indicated by increased levels of ALT, necrotic area, and pro-inflammatory cytokines (TNF-α, IL-1β, and IL-6) and decreased expression of hepatocyte growth factor (HGF) and PCNA, a marker of hepatocyte proliferation at 48 h after APAP treatment. CD68-positive macrophages less accumulated in the livers from $TP^{\triangle mac}$, accompanied by reduced expressions of chemokines such as CCL2, CCL7 and CCL9. Flow cytometry analysis revealed that the numbers of pro-inflammatory macrophages (M1) in $TP^{\triangle mac}$ were higher than Cont, while the numbers of reparative macrophages (M2) in $TP^{\triangle mac}$ were lower than Cont. In cultured bone marrow-derived macrophages from $TP^{\triangle mac}$, M1-related gene expressions were increased and M2-related gene expressions were decreased. HGF treatment in $TP^{\triangle mac}$ attenuated liver injury and promoted liver repair after APAP-induced liver injury.

Conclusion: TP receptor signaling in macrophages promoted liver repair after APAP-induced liver injury by accumulation of reparative macrophages and production of HGF in the livers.

Disclosure: Nothing to disclose.

PP0969

GLUCAGON-LIKE PEPTIDE RECEPTOR 1 ANALOGUE COULD REDUCE ALCOHOL INTAKE IN PEOPLE WITH ALCOHOL USE DISORDER: A SYSTEMATIC REVIEW

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Introduction: Alcohol use disorder (AUD) remains a significant global health concern. Despite the availability of various pharmacological and behavioural interventions, alcohol-related mortality is rising.

Aims & Methods: This systematic review aimed to critically evaluate the existing literature on the role of glucagon-like peptide receptor 1 analogues (GLP-1 RAs) in people with AUD.

Electronic search was conducted on Ovid Medline, EMBASE, Cochrane Central and PsycINFO from inception to March 2024. Grey literature was searched. The primary outcome of the review was to explore the impact of GLP-1 RAs on self-reported alcohol intake in people with AUD. The extracted data were organized and synthesized to identify key findings and trends evaluating the potential impact of GLP-1 RAs on alcohol consumption in AUD. Narrative synthesis was done to provide a coherent and detailed narrative of the evidence, highlighting significant findings, relationships, and implications. PROSPERO (CRD42024531982).

Results: Seven studies comprising 88,163 participants were included; of these, 38,746 (43.9%) were treated with GLP-1 RAs. Pooled mean age was 49.6 years (SD=10.5), 56.9% (n=50,151/88157) were male, and if the study provided ethnicity 87.5 % (n=426/487) were white. An RCT comparing exenatide with placebo found no significant difference in self-reported alcohol measures at 26 weeks (percentage point reduction: heavy drink days: 6.0, 95% CI -7.4 to 19.4, p=0.37; total alcohol consumption: -42.0 g/30 days, 95% CI -507.7 to 423.7, p=0.86; days without alcohol use: -10.5, 95% CI -2.6 to 23.4, p=0.11). But in participants (n=30) with BMI > 30 kg/m^2, exenatide compared with placebo was associated with a significant reduction in heavy drinking days (percentage point reduction 23.6, 95% CI, –44.4 to –2.7, P=0.034) and in total alcohol consumed (-1,205 g, 95% CI, –2,206 to –204, P= 0.026). Another RCT showed dulaglutide users compared to placebo were 29% more likely to reduce weekly alcohol consumption (relative effect size: 0.71, p=0.04). In an observational study, authors reported with

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liraglutide (n=42) 21.4% stopped and 78.6% reduced drinking (p<0.05). A prospective cohort control study showed people on semaglutide or tirzepatide (n=153) compared to no medication significantly reduced number of alcoholic drinks consumed and binge drinking (p<0.05). In a case series (n=6) semaglutide use was associated with a 9.5-point reduction in alcohol use disorder identification test score. A population-based study from Denmark (n=87,676) reported that at 90 days people on GLP-1 RAs compared to one on dipeptidyl peptidase-4 inhibitors (DPP-4) were less likely to experience alcohol-related events (HR 0. 46, 95% CI 0.24 to 0.86) and effect remained positive long term (356-4619 days: HR 0.72, 95% CI 0.60–0.86). Two qualitative studies examining Reddit social media posts from people on GLP-1 RAs noted over two-thirds addressed a reduction in alcohol consumption. Functional brain imaging suggested significant reductions in cue reactivity were observed in the ventral striatum [F(1,31) = 4.744, P = 0.037, partial η2 = 0.133], dorsal striatum [F(1,31) = 6.124, P = 0.019, partial η2 = 0.165], and putamen [F(1,31) = 4.730, P = 0.037, partial η2 = 0.132] after 26 weeks of exenatide treatment compared with placebo (Table 1). Two studies were of high quality, two medium, and three low.

Conclusion: GLP-1 RAs show promise in reducing alcohol consumption and improving outcomes in individuals with AUD, although heterogeneous study findings warrant further research to establish true effectiveness and safety of GLP-1 RAs in AUD.

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Disclosure: Nothing to disclose.

PP0970

ETIOLOGY, PROGNOSIS AND CLINICAL COURSE OF TOXIC HEPATITIS: A SINGLE CENTER EXPERIENCE

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Introduction: Toxic hepatitis is a significant hepatic condition that can occur due to drugs, fungi, herbal agents and supplements, sometimes progressing fulminantly and resulting in transplantation or mortality. As it is a diagnosis of exclusion, it can be challenging to establish and the etiological factors vary societally. In our study, we investigated the etiology, epidemiology, factors related to prognosis, and clinical outcomes of patients diagnosed and treated for toxic hepatitis.

Aims & Methods: The demographic characteristics, etiological factors, laboratory and imaging results, biopsy findings, and information about the drugs used in the treatment of 254 patients hospitalized and treated for toxic hepatitis between February 2019 and June 2022 at Ankara Bilkent City Hospital were retrospectively examined. After evaluating the results according to the RUCAM scale, patients interpreted as possible and highly probable were included in the study, and they were classified based on severity as mild, moderate, severe or fatal.

Results: Of the patients included in the study, 60.6% were female, and 30.3% of these patients presented to the hospital with jaundice symptoms. The mean age of the patients was 43.9 years. Hepatocellular damage was observed in 68.5% of the developed liver toxicities. The most common etiological factors were antibiotics and NSAIDs in sequence. When antibiotic drugs were evaluated separately, amoxicillin-clavulanate (9.8%) was considered the most common antibiotic causing toxic hepatitis, and dexketoprofen (9.1%) was identified as the most common NSAID. A total of 45 patients (18.1%) were associated with herbal and dietary supplements, while 18 patients (7.1%) were related to mushroom consumption in the etiology. Of the patients included in the study, 56% had a moderate or severe hepatitis course, 3 (1.18%) patients progressed to liver transplantation, and 6 (2.5%) patients died, including one post-transplant. The severity of toxic hepatitis was mild in 109 participants (42.9%), moderate in 89 patients (35.0%), severe in 48 patients (18.9%), and fatal in 8 patients (3.1%). In 5.9%, there was worsening of the disease (transition from mild or moderate severity to severe or fatal groups) after admission. In our study, 14.3% of patients in the severe-fatal group died due to acute liver failure or underwent liver transplantation. Both groups exhibited a dominant pattern of hepatocellular-type damage, and there was no difference in the evaluation of etiological factors between the groups. When biochemical parameters were evaluated, elevation in ALT, AST, LDH, initial and peak total bilirubin, direct bilirubin, initial and peak INR, prolonged prothrombin time, and low albumin were significant in both the severe patient group and the group admitted to intensive care. Liver biopsy was performed on 60 of the 254 patients included in the study.

Conclusion: In our study, the median age, gender distribution, dominant injury type, and the most commonly detected agents in etiology (NSAIDs and antibiotics) and dominant were consistent with the literature. Factors such as the consumption of Toxic hepatitis is a disease that can be challenging to diagnose and treat. Given that our center receives a high number of patient admissions, accepts patients from surrounding areas, and serves as a transplant center, we believe that the data we have obtained can closely approximate current and real data. We also believe that our study on these 254 patients will contribute to the literature in terms of etiological, epidemiological, and prognostic insights.

Disclosure: Nothing to disclose.

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PP0971

DRUG-INDUCED HEPATITIS: ETIOLOGICAL PROFILE AND THERAPEUTIC MANAGEMENT

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Introduction: Drug-induced hepatotoxicity is one of the main forms of iatrogenic pathology, which is often under-diagnosed. Numerous drugs are implicated and can cause hepatic damage reproducing the whole spectrum of non-iatrogenic liver pathology, ranging from a simple transient disturbance of the liver balance to fulminant, life-threatening hepatitis.

Aims & Methods: The aim of our work is to study the etiological profile and therapeutic management of drug-induced hepatitis.

This is a retrospective descriptive monocentric study from January 2010 to August 2023 of patients with drug-induced hepatitis in the Hepato-Gastro-Enterology and Proctology Department “Medicine B”.

Diagnosis was based on the elimination of infectious, immunological and toxic causes of hepatitis. For all patients, viral serologies for hepatotropic viruses and a pharmacovigilance investigation were carried out before concluding that the cause was drug-induced.

Results: 194 patients were included in our study, with a mean age of 44.2 years, ranging from 23 to 84 years. Female predominance was observed, with 110 women (56.7%) versus 84 men (43.2%), so, a F/H sex ratio of 1.3.

In 59 cases (30.4%), liver damage was discovered by chance during a biological check-up. In the other cases, clinical manifestations were dominated by jaundice in 112 patients (57.7%), followed by asthenia in 42 patients (21.6%), pruritus in 35 cases (18%) and fever in 17 cases (8.7%). Liver function tests showed cytolysis in 97 cases (50%), mixed disease in 74 patients (38.1%), and cholestasis in 23 patients (11.8%). The time to onset of clinical or biological signs of hepatotoxicity ranged from 03 days to 09 months. Severe acute hepatitis occurred in 70 cases (36%), including 48 cases (24.7%) of fulminant hepatitis. Viral serologies were negative in all cases. Liver biopsy was consistent with drug-induced hepatitis in 35 cases (18%). The main drugs implicated were anti-tuberculosis drugs in 110 cases (56.7%), followed by immunosuppressants, particularly azathioprine in 44 cases (22.6%), then paracetamol in 23 cases (11.8%), antidepressants in 6 cases (3%), and non-steroidal anti-inflammatory drugs (NSAIDs) in 3 cases (1.5%). Other drugs were less frequently associated with liver dysfunction, such as: allopurinol, fluconazole (1 case: 0.5%), synthetic antithyroid drugs (1 case: 0. 5%), antiepileptics (Carbamazepine) (1 case: 0.5%), hypolipidemics (1 case: 0.5%), ciprofloxacin (1 case: 0.5%), anti TNF (Adalimumab) (1 case: 0.5%), and anti-angoretics (1 case: 0.5%).

Discontinuation of the treatment in question was considered in all patients. Patients with fulminant acute hepatitis were managed in intensive care, and N-acetylcysteine was introduced in patients admitted for paracetamol-induced hepatitis. The outcome was favorable in 160 patients (82.4%), while 34 patients (17.5%) died.

Conclusion: Drug-induced hepatitis is a delicate condition, given its polymorphic presentation and the diversity of potentially hepatotoxic drugs prescribed. Diagnosis is based on the elimination of other possible causes of liver dysfunction, a suggestive or compatible chronology between the administration of the drug and the onset of hepatitis, and a favorable evolution after discontinuation of the drug. Prevention requires compliance with prescribing rules, early detection of signs of therapeutic intolerance and rigorous monitoring of progress under treatment.

Disclosure: Nothing to disclose.

PP0972

ELEVATED SERUM FERRITIN LEVELS AS A PREDICTIVE FACTOR OF SEVERITY AND COMPLICATIONS IN CIRRHOTIC PATIENTS

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Introduction: Cirrhosis is a chronic liver disorder characterized by fibrosis, leading to liver dysfunction and portal hypertension. Ferritin has emerged as a simple biomarker associated with prognosis in cirrhotic patients. Our study aimed to determine the presence of a significant relationship between serum ferritin levels and the incidence of complications.

Aims & Methods: In this retrospective descriptive and analytic study, serum ferritin levels were assessed in adult cirrhotic patients upon admission to our Hepato-Gastroenterology department at a University Hospital between March 2017 and March 2023. Subsequently, patients were evaluated for the occurrence of complications within the following year. Serum ferritin levels normal range were of 22 to 275 ng/ml for men and 4.63 to 204 ng/ml for women.

Results: Our study comprised 140 cirrhotic patients with varied etiologies, including 72 females and 68 males, with an average age of 58 ± 14.88 years (range: 21 to 90). These patients presented with varying disease stages and diverse etiologies.

Among patients with elevated ferritin levels, 86.4% experienced complication within the year following the measurement, compared to only 43.8% of patients with normal ferritin levels. The most commonly observed complication was hepatic encephalopathy, followed by variceal bleeding, and ascites. Our findings suggest a significant association between serum ferritin levels and the occurrence of complications within the subsequent year ($p < 0.001$).

These patients were distributed across different disease stages: 50 had Child-Pugh score A, 66 had Child-Pugh score B, and 24 had Child-Pugh score C. We used the Anova test to compare the average ferritin levels among the three groups. we found a significant difference between ferritin values in the 3 groups: A (88.27 ± 152), B (528 ± 1209), C (105 ± 128) ($p: 0.027$). And by using post Hoc tests, we found that this difference was significant between group A and B ($p: 0.04$), but non significant difference between A and C, and B and C.

Conclusion: In conclusion, our study reveals a significant association between elevated serum ferritin levels and increased incidence of complications in cirrhotic patients, highlighting ferritin as a potential prognostic marker for disease progression.

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Disclosure: Nothing to disclose.

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PP0973

DIFFERENTIAL EXPRESSION OF P53 PROTEIN ISOFORMS AND THEIR ROLE IN THE PATHOGENESIS OF SPONTANEOUS BACTERIAL PERITONITIS

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Introduction: Translocation of intestinal bacteria into the mesenteric lymph nodes and ascites plays a central role in the development of spontaneous bacterial peritonitis (SBP), a severe complication of liver cirrhosis. Immunosuppression, bacterial dysbiosis and increased intestinal permeability favor bacterial translocation. We have shown that the mucus layer in the intestine of patients with liver cirrhosis is thinner, supporting direct contact of bacteria with epithelial cells. This leads to degradation of cell-junctions (Haderer et al., Gut 2022), tissue damage and cellular stress. p53 is an important regulator of cellular stress responses. Depending on its specific isoforms, p53 can perform different functions. We investigate the association between bacterially induced cellular stress and the expression of different p53 isoforms using a novel luciferase-based reporter system (Truong et al. Nat Cell Biol 2021).

Aims & Methods: To quantify p53 isoform expression, we integrated exon-specific reporter systems (EXSISERS) into TP53 exons 2, 4, and 7. This allows distinction among three primary isoform groups (full-length p53, Δ40p53, and oncogenic Δ133p53 and Δ160p53) in HCT116 colon carcinoma cells. Using split proteins facilitates the removal of luciferase from the resultant p53 polypeptide, thereby maintaining the integrity of the p53 protein isoforms. This method enables the determination of the cellular ratio of p53 protein isoforms. HCT116 EXSISERS cells were co-cultured with a reference strain of *Escherichia coli* (*E. coli*) O6:Hnt, as well as clinical isolates of *E. coli* Ont:H7, Ont:H41, *Proteus mirabilis*, and *Klebsiella pneumonia* obtained from ascites of patients with SBP, for up to 2 hours. Isoform expression was quantified by measuring luciferase activities at intervals of 15 minutes to 2 hours using specific substrates. In addition, qPCR analyses were performed. Further, p53 isoform expression was evaluated after co-incubation with the cytokine IL-8 (mimicking inflammatory conditions) or bacterial components (LPS, MDP, CpG).

Results: Co-incubation with bacteria at MOI 5 resulted in upregulation of the cell-death inducing △40p53 isoform expression within 15 minutes. Simultaneously, we observed a gradual reduction of △133p53 and △160p53, known for their anti-apoptotic and pro-oncogenic properties, over a period of 2 hours.

Thus, after bacterial co-culture isoforms inducing cell death or cell cycle arrest were expressed at a constant level, while oncogenic isoforms were downregulated. Interestingly, stimulation with CpG motifs of bacterial DNA interacting with TLR9, resulted in an upregulation of △40p53 and a reduced expression of △133p53 and △160p53 similar to that observed after bacterial co-incubation. In contrast, bacterial co-culture of EXSISERS cells with IL-8, LPS or MDP resulted in a downregulation of Δ40p53 isoform expression after 15 minutes.

Conclusion: p53 plays a central role in SBP pathogenesis. Direct contact with bacteria induces the pro-apoptotic isoform △40p53 to eliminate damaged cells, while constant translation of full-length p53 triggers a cell cycle arrest allowing epithelial cells to adapt to bacterial stress. Thus, in SBP, there is a delicate balance between cell survival and cell death in the intestinal epithelium. While △40p53 remained unaffected by IL-8, CpG motifs of bacterial DNA induced △40p53 upregulation.

Further studies will decipher the signaling pathways and interactions leading to Δ40p53 activation of in order to develop targeted therapeutic approaches to protect the epithelial barrier.

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Disclosure: Nothing to disclose.

PP0974

TRANSITION FROM 2D TO 3D – INVESTIGATING BACTERIAL EFFECTS ON INTESTINAL BARRIER INTEGRITY IN SPONTANEOUS BACTERIAL PERITONITIS (SBP) USING SPHEROID AND ORGANOID MODELS

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Introduction: The use of 3D cell culture models, including spheroids and organoids, has proven to be a valuable method to approximate *in vivo* conditions in basic research. Our recent work on the pathogenesis of spontaneous bacterial peritonitis (SBP) – a life-threatening complication of liver cirrhosis with a 1-year-mortality of 66% – shows that bacteria induce the degradation of important cell-cell contact proteins upon direct contact via various processes.

The resulting destabilization of the epithelial barrier enables bacterial translocation and thus represents a mechanism in the development of SBP (Haderer et al., Gut 2022).

To evaluate these processes under more physiological conditions, we investigate bacterial-induced effects on cell-cell contact proteins in a 3D spheroid model and subsequently in a patient-based organoid model.

Aims & Methods: To establish the method and create a reference, spheroids were generated from colon carcinoma cell line HCT116 with a concentration of 70,000 cells/ml. Spheroid development and formation of cell-cell contacts were assessed by fluorescence microscopy. Subsequently, bacterial-induced effects on the tight junction proteins Occludin, Claudin4 and Zonulin1 as well as on the adherens junction protein E-Cadherin were investigated after up to 12 hours of bacterial co-culture using qPCR, fluorescence microscopy and Western blot analysis. Bacterial co-culture was performed with the reference strain *Escherichia coli* (*E.coli*) O6:Hnt and the *E. coli* strain Ont:H41, which was isolated from the ascites of a patient with SBP.

In addition, organoids were generated from intestinal biopsies of patients with liver cirrhosis and co-cultured with both bacterial strains. Bacterial-induced effects on cell-cell-contact proteins were investigated using Western blot analysis.

Results: HCT116 cells formed formed spheroids composed of viable cells with well-defined cell-cell contacts. Co-culture with *E. coli* for up to 12 hours led to a significant degradation of Occludin (up to 60% reduction), E-Cadherin (up to 85% reduction) and Zonulin1 (up to 95% reduction), whereas Claudin4 levels remained mostly constant. Co-culture with heat-inactivated bacteria did not affect cell-cell contacts. The intensity of these effects was dependent on the respective protein, the bacterial strain used, as well as the multiplicity of infection (MOI) and the duration of the co-culture. Degradation of Occludin could also be visualized by fluorescence microscopy.

We were then able to transfer this experimental approach to an organoid setting. Bacterial-induced degradation of the epithelial barrier was confirmed in organoids, showing a clearly visible reduction of E-Cadherin, Occludin and Zonulin1 after 8 hours of bacterial co-culture. The intensity of the effects varied depending on the bacterial strain, with the patient strain showing a slightly higher level of aggressiveness.

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Conclusion: Understanding the pathophysiological changes of the intestinal epithelial barrier in patients with liver cirrhosis is essential for advancing therapeutic strategies for SBP. In this context, a representative *ex vivo* model is crucial to draw reliable conclusions.

Our work shows that relevant mechanisms of SBP, such as bacterial-induced degradation of cell-cell contacts, can be successfully modeled in spheroids and organoids. Thus, our study provides an approach for further approximation to *in vivo* conditions – bridging the gap between basic research and clinical utility.

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Disclosure: Nothing to disclose.

PP0975

FRIALTY CORRELATES WITH GREATER ODDS OF ENCEPHALOPATHY, ASCITES, & SPONTANEOUS BACTERIAL PERITONITIS, A STUDY FROM PAKISTAN

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Introduction: Frailty is associated with morbidity and mortality in advanced cirrhosis, However, the information on the correlation between frailty and complications in decompensated cirrhosis is scarce. We aim to evaluate the impact of Frailty on risk of Hepatic encephalopathy (HE), Ascites, Spontaneous bacterial peritonitis (SBP) & impact of multidomain implementation on these outcomes.

Aims & Methods: We prospectively evaluated the impact of Frailty on the prevalence of ascites, HE, and SBP among cirrhotic adults. Frailty was defined by the Liver Frailty Index (LFI)and a level of > 4.4 was defined as frial. Development of complications (worsening ascites, HEor SBP) & unplanned hospitalization were recorded.

In this study one arm of frail continued with their normal diet and was taken as control and the other arm(intervention)was given training of taking 5 meals/day, containing appropriate protein and fat contents. Additional branched-chain amino acids were also given to these patients, Physical exercise (both cardiac and resistance) was advised. they were evaluated for a period of 6 months for the complication rates. Between-group comparisons used chi-squared methods and adjusted multivariate logistic regression.

Results: 179 patients were included(MELD 13.2±3.4, Child-Pugh A/B/C 34.9%/35.1%/30%), and 74.3% were frail. Compared to the robust group, pre-frail and frail patients had significantly higher cumulative 1-year probabilities of worsening ascites (0% *vs.* 8.5% *vs.* 18.4%, $p = .009$), and spontaneous bacterial peritonitis(0% *vs.* 13.5% *vs.* 34.2%, $p < .001$), encephalopathy (0% vs 28.9% vs 71.1%,p =0.02 and unplanned hospitalizations (100% *vs.* 90.8% *vs.* 80.4%, $p = .014$).

Two models of multivariable Cox regression analysis were done and adjusted with MELD-Na and Child-Pugh C, frailty was associated with developing decompensation (HR 3.01, $p = .04$; and 2.98, $p = .04$, respectively) and unplanned hospitalization (HR 2.46, $p = .02$; and 2.39, $p = .03$, respectively). Control arm with frailty were significantly more likely to have ascites, HE, and SBP.(Control arm vs. interventional arm: ascites, 66% vs. 93%; HE, 81% vs. 49%; SBP, 4% vs. 16%, p < 0.001 for all).

Conclusion: Frailty is associated with higher odds of cirrhosis complications including ascites, HE, and SBP, which was observed across all liver disease etiologies.

With the implementation and mitigating the knowledge regarding PROPER DIET AND EXERCISE, WE CAN improve the condition of patients and can avoid complications.

Disclosure: Nothing to disclose.

PP0976

ENDOSCOPIC ULTRASOUND ASSESSMENT OF PORTAL HYPERTENSION VERSUS TRANSJUGULAR MEASUREMENTS

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Introduction: Endoscopic ultrasound (EUS) has emerged as a new tool in assessing portal hypertension (PH). EUS portal pressure measurement (EUS-PPG) was described in few studies using a compact manometer, but the use of a pressure transducer connected to a pressure measurement monitor was previously described in a very small number of patients. The comparative assessment of portal pressure measured by EUS and transjugular route has not been done before. Our study aimed to compare the value of EUS-PPG and transjugular measurements (HVP, PPG) in patients with portal hypertension.

Aims & Methods: Patients with signs of PH on transabdominal imaging and need for haemodynamic and/or endoscopic assessment were included. Patients with platelet count <50000/µL and INR>2.5 were excluded. EUS was performed using a 22 G fine needle aspiration attached to a central venous pressure measurement monitor, and it evaluated the presence of gastrointestinal varices, portal vein (PV) diameter and pressure, median hepatic vein/inferior vena cava pressure. All patients had HVPG measurements through the transjugular approach under fluoroscopic guidance, and portal pressure was assessed in patients scheduled for postosystemic shunt (TIPS). The results obtained by the two methods were statistically analysed.

Results: We prospectively enrolled (January to December 2023) 19 patients of 50±14 years old, male to female ratio=2,1:1). The aetiology of PH was porto-sinusoidal vascular disease (n=3, 15%), alcohol cirrhosis (n=15,75%), viral cirrhosis (n=1,5%) and metabolic cirrhosis (n=1, 5%). The patients with cirrhosis had Child-Pugh class B or C in 50% of cases. The mean INR was 1.41and the mean platelet level was 122.000/mcL

EUS-PPG was technically successful in 18 patients (94%), with one failure attributed to obesity and important perihepatic ascites which limited the needle deep advancement. The mean EUS-PPG was 13.15 ± 5.48 mmHg. 73% of patients had oesophageal and/or gastric varices present. The mean PV diameter was 12.8 ± 2.7 mm.

HVPG measurements were technically performed in 18 cases (94%), the one exception was related to anatomical factors impeding hepatic vein catheterisation. Mean HVPG was 14.7 ± 5.26 mmHg and it was similar to EUS-PPG (p = 0.09), with strong Pearson correlation coefficient (r=0.88).

In patients scheduled for TIPS (n=10), mean PPG was 16±4,15 mmHg and it correlated with EUS-PPG (r=0,81).

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No difference was observed between the patients performing measurements during TIPS under general anaesthesia and the group assessed under mild sedation with midazolam. No adverse effects were noted.
Conclusion: The measurements of PH by transjugular route or EUS are comparable. EUS-PPG measurement with 22G needle proved to be an accurate and safe technique. Further investigation is required to determine its indications in patients requiring endoscopic assessment of portal hypertension
Disclosure: Nothing to disclose.

PP0977

THE MEAN PLATELET VOLUME (MPV) AS A PREDICTOR FACTOR OF SEVERITY IN CIRRHOTIC PATIENTS

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Introduction: Mean platelet volume (MPV) is a routinely measured, non-invasive, and inexpensive parameter of complete blood count. Recent research suggests that MPV has the potential to serve as a valuable prognostic indicator in clinical practice. The aim of our study is to assess the relationship between MPV and the severity of cirrhosis.
Aims & Methods: we included in a cross-sectional study, 140 cirrhotic patients of various grades of severity and various etiologies, followed in our university Hepato-Gastroenterology department. MPV was assessed in all patients, with a normal range set between 7 and 9 fl. Our aim was to evaluate the relationship between MPV values and cirrhosis severity, as determined by Child-Pugh scores and the Model for End-Stage Liver Disease (MELD).
Results: Our study involved 140 cirrhotic patients with diverse etiologies and stages, including 72 females and 68 males, with an average age of 58 ± 14.88 years (range: 21 to 90). Among patients classified as Child A, only 52% had a high MPV, while in groups B and C, higher rates of MPV were noted, reaching 100% and 95.8% respectively. A significant relationship between MPV levels and Child stage was noted ($P < 0.001$).
These patients were distributed across different disease stages: 50 had Child-Pugh score A, 66 had Child-Pugh score B, and 24 had Child-Pugh score C. We used the ANOVAs test to compare the average MPV levels among the three groups. We found a significant difference between MPV values in the 3 groups: A (9.94 ± 1.2), B (11.56 ± 1.19), C (15.27 ± 3.51) ($p < 0.001$). And by using post Hoc tests, we found that this difference was significant between group A and B ($p < 0.001$), A and C ($p < 0.001$), and B and C ($p < 0.001$). Additionally, a significant positive correlation was observed between MPV and MELD score (r: 0.7; $p < 0.001$) using Pearson's correlation coefficient.
Conclusion: MPV can serve as a valuable and inexpensive biomarker in cirrhotic patients for assessing the severity and prognosis of the disease. However, further researches are needed to explore the full potential and utility of MPV in clinical practice.
Disclosure: Nothing to disclose.

PP0978

OUTCOMES AND PROGNOSTIC FACTORS OF CIRRHOTIC PATIENTS WITH ACUTE VARICEAL BLEEDING AND CONCURRENT HEPATOCELLULAR CARCINOMA: A CASE-CONTROL STUDY

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Introduction: The treatment outcomes and risk factors for the prognosis of acute variceal bleeding (AVB) in hepatocellular carcinoma (HCC) patients remain unclear. Hence, we assessed the clinical outcomes and prognostic factors of these patients.
Aims & Methods: To evaluate the treatment outcomes and risk factors of cirrhotic patients with hepatocellular carcinoma (HCC) and acute variceal bleeding (AVB).
This is a retrospective case-control study. Consecutive patients with HCC and AVB admitted between 2016 and 2022 were included. For each patient with HCC, a patient without HCC was matched by age, sex, and Child-Pugh class. Follow-up was continued until death and transplantation.
Results: A total of 620 patients were included. The five-day treatment failure and 6-week mortality rates of the whole cohort were 18.7% and 16.6%, respectively. Patients with HCC had higher 6-week mortality, respectively (24.2% vs. 9.0%, $P<0.001$).
Among AVB patients with HCC, the Barcelona Clinic Liver Cancer (BCLC) stage (C-D vs. 0-B) (HR, 14.409; 95% CI, 5.758-36.055; $P<0.001$), and Child-Pugh score (HR, 1.239, 95% CI, 1.121-1.370; $P<0.001$) were independent risk factors of 6-week mortality in multivariate analysis.
Furthermore, the risk of 6-week mortality was 60.2% among patients with a Child-Pugh score ≥ 9 and BCLC stage C–D, much higher than patients with low Child-Pugh score (<9) and earlier BCLC stage (0-B) ($P<0.0001$).
Conclusion: Among patients with cirrhosis and AVB, patients with HCC had a worse outcome than patients without. The severity of liver disease and the stage of HCC are the main determinants of mortality in HCC patients.
Disclosure: Nothing to disclose.

PP0979

THE SEVERITY OF PORTAL VEIN THROMBOSIS IS ASSOCIATED WITH THE PROGNOSIS OF CIRRHOTIC PATIENTS WITH ACUTE VARICEAL BLEEDING

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Introduction: The clinical implications of portal vein thrombosis (PVT) on the prognosis of acute variceal bleeding (AVB) in cirrhosis remain under debate. Most studies have not been specifically designed to assess the severity of PVT.
Aims & Methods: This study aimed to evaluate the severity of PVT and its impact on outcomes in patients with cirrhosis and AVB. From January 2016 to December 2016, a total of 1541 consecutive cirrhotic patients admitted for AVB, with (n=432; 28.1%) or without PVT (n=1109; 71.9%), were

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included retrospectively. The degree of luminal obstruction was used to classify the grades of PVT according to Baveno VII criteria. The primary endpoint was 6-week mortality.

Results: During the follow-up period, 179 patients (11.6%) died within 6 weeks. Patients with PVT had a higher risk of 6-week mortality compared to those without PVT (18.3% vs 8.9%; HR 2.12, 95% CI 1.52–2.94, p <0.001). Among them, those with malignant main PVT had a significantly elevated risk (39.3% vs 9.2%; HR 4.86, 95% CI 2.74–8.62, p <0.001). In the non-malignant PVT cohort, only completely occlusive main PVT was associated with increased 6-week mortality (completely occlusive 26.1% vs cavernous transformation 9.7% vs partially occlusive 6.5% vs minimally occlusive 5.9% vs without PVT 9.2%, p = 0.023). These results were consistent across various relevant subgroups.

Conclusion: Our study demonstrates that PVT is linked to a poorer outcome in cirrhotic patients with AVB, particularly in cases of malignant main PVT and completely occlusive main PVT.

Disclosure: None of the authors have any conflict of interest.

PP0980

EFFECT OF VOLUME OVERLOAD MEASURED BY VEXUS SYSTEM ON THE PROGRESSION TO HEPATORENAL SYNDROME AND VARIATION IN MELD NA SCALE IN PATIENTS WITH HEPATIC CIRRHOSIS DEVELOPING ACUTE KIDNEY INJURY

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Introduction: Acute kidney injury is a complication in patients with decompensated cirrhosis (26%). Hepatorenal syndrome has high morbidity and mortality (90%), characterized by renal functional circulatory changes. Abdominal ultrasound, as a medical tool, in the VexUS protocol assessing intravascular congestion (Inferior vena cava, suprahepatic vein, portal vein, and renal vein), could be routinely used in cirrhotic patients with acute kidney injury to objectify the degree of venous congestion and improve decision-making, changing the treatment paradigm with albumin.

Aims & Methods: Prospective, observational, and analytical cohort study in hospitalized hepatic cirrhosis patients with acute kidney injury due to acute decompensation, receiving intravenous albumin at a dose of 1 g/kg/day for 48 hours with subsequent assessment of response. Before albumin administration, abdominal ultrasound (VExUS) was performed to determine the venous congestion status (Period: July 2022 to December 2023). No sample calculation was performed, as all patients meeting selection criteria were included. Sociodemographic, clinical, and biochemical variables were recorded.

Results: Out of 36 patients, 16 (44.4%) responded to albumin therapy, while 20 (55.6%) did not respond. Analyzing according to the VExUS score, 12 patients with VExUS 0 had a response rate of 54.5%, and 10 did not respond (45.5%). When classifying patients based on the presence or absence of venous congestion (VExUS = 0 vs. VExUS 1 or 2), it was observed that in patients with venous congestion, 4 responded (28.6%) and 10 did not respond (71.4%), with a decrease in the p-value without reaching statistical significance.

A logistic regression analysis revealed that both the VExUS score and serum creatinine at admission are predictors of non-response to albumin. The presence of congestion (VExUS I or II) and a 1 mg/dL increase in creatinine were associated with a 5 and 10 times higher risk of non-response, respectively. Regarding the MELD Na score, the 14 deceased patients had a median of 32.5, while the 22 survivors had a median of 23.77.

Conclusion: Non-invasive diagnostic tools like ultrasound (VexUs), widely used due to availability, prove useful in determining the volume status of cirrhotic patients with acute kidney injury and hepatorenal syndrome. This helps establish the prognosis of response to a volume expander like albumin or the utility of other targeted therapeutic measures in patients with evidence of volume overload and/or abdominal hypertension data.

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Disclosure: Nothing to disclose.

PP0981

GASTROINTESTINAL BLEEDING DUE TO PORTAL HYPERTENSION IN CIRRHOSIS: PROGNOSTIC VALUE OF THE CAGIB SCORE

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Introduction: Gastrointestinal variceal bleeding is a serious complication of cirrhosis. Currently, no score has been validated to assess the prognosis of a bleeding episode in cirrhotic patients. The aim of our study was to evaluate the performance of the CAGIB score for recurrence and mortality following upper gastrointestinal bleeding in cirrhotic patients.

Aims & Methods: We conducted a retrospective study of cirrhotic patients hospitalized for upper gastrointestinal bleeding due to portal hypertension [2012-2022]. We calculated the scores CHILD-PUGH, MELD, MELD-Na, MELD 3.0, ALBI, ROCKALL, GBS, NLR and CAGIB, and compared their performance in terms of early recurrence (5-days) , short-term recurrence (6-weeks) and early mortality (6-weeks) through the study of ROC curves (SPSS software, P significant if <0.05).

Results: We included 105 patients with a mean age of 63 years and a sex ratio F/M=1.05. The origin of the bleeding was oesophageal varices rupture in 94.28% of cases and gastric varices rupture in 17.14%. Endoscopic treatment was performed in 96 patients, 90 of whom underwent band ligation of esophageal varices. Short-term recurrence was noted in 20% of patients. Early recurrence was noted in 3.8% of patients. Fourteen cases of early death were recorded. The CAGIB score was the best predictor of early recurrence (AUC=0.995, Sensibility=83.3%, Spécificity=99.6%, Odds Ratio=47, p<0.001) but it has a poor prognostic value for short-term re-

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currence (p=0,08). For early mortality, the CAGIB score performed better than the other scores (AUC=0.891) with a Sensibility=85.7% and Specificity=83.5% for predicting mortality from a cut-off ≥-4.648 with an OR=18.
Conclusion: The CAGIB score had the best predictive value for early recurrence and mortality following an episode of upper gastrointestinal bleeding due to portal hypertension in cirrhosis. We therefore propose it as an easy-to-calculate and reproducible tool, for a cut-off >-4.648.
Disclosure: Nothing to disclose.

PP0982

ACUTE KIDNEY INJURY IN CIRRHOTIC PATIENTS EXPOSED TO IODINATED CONTRAST: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Iodinated contrast (IC) is one of the leading causes of acute kidney injury (AKI). Nephrotoxicity of IC in cirrhosis is a debated issue due to low-quality evidence.
Aims & Methods: This study aims to evaluate the association between exposition to IC in cirrhotic patients and the development of AKI. We performed a systematic review in PubMed, Embase and Cochrane databases, searching for studies conducted in cirrhotic patients exposed to IC. The primary outcomes were: 1) AKI development in cirrhotic patients exposed or not to IC; and 2) AKI development in cirrhotic and health patients exposed to IC. The odds ratio (OR) was used for binary outcomes with their respective 95% confidence interval (CI). Heterogeneity was assessed using the Cochran Q test and I^2 statistics.
Results: The meta-analyses included 4 studies and 1,055 patients. Three studies (911 patients) evaluated cirrhotic patients exposed or not exposed to IC. There was a significant increase in AKI in cirrhotic patients exposed to IC (OR 3.10; 95% CI 1.45 to 6.61, p=0.003; I2=0%). Three studies (654 patients) evaluated cirrhotic and health patients exposed to IC. There was no difference in AKI comparing both groups (OR 1.61; 95% CI 0.63 to 4.16, p=0.322; I2=0%).
Conclusion: Patients with cirrhosis exposed to IC showed higher incidence of AKI compared with those not exposed, but when compared with health individuals also exposed to IC the incidence of AKI seems to be the same.
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Disclosure: Nothing to disclose.

PP0983

PARA-VESSELS AROUND CARDIA SIGNIFICANTLY AFFECT THE EFFICACY OF ENDOSCOPIC MANAGEMENT OF GASTROESOPHAGEAL VARICES

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Introduction: Acute variceal bleeding is a life-threatening complication of portal hypertension, and it has a high mortality rate, responsible for almost one-third of cirrhosis-related deaths[1]. Collaterals develop between the portal vein tributaries and systemic veins under the sustained portal high pressure[2]. Gastroesophageal varices are varices in direct continuity from the esophagus to the stomach, draining into the systemic circulation via the gastroesophageal venous system and/or the gastrophrenic venous system[3]. Para-gastric and para-esophageal varices are also important collaterals for reducing portal pressure and may influence the choice and effectiveness of treatment. Endoscopic gastric variceal tissue adhesive treatment followed by the endoscopic esophageal variceal ligation is the preferred treatment for the gastroesophageal varices[4]. However, some patients still experience re-bleeding after endoscopic treatment, even after the combination of drug and surgical therapy[5]. Clinical risk factors affecting the prognosis of gastroesophageal varices still remain to be analyzed.
Aims & Methods: The aim of this study was to investigate the effect of esophageal and gastric extraluminal vessels on the efficacy of endoscopic prevention of esophageal and gastric variceal bleeding. The clinical data of patients with cirrhosis-related esophageal and gastric varices who underwent endoscopic ligation and/or tissue adhesive treatment and hepatic venous-portal gradient (HVPG) measurement to prevent rebleeding between January 2020 and December 2020 were analyzed. According to the esophageal and gastric extraluminal blood vessels on portal vascular CT, patients were divided into no combined para-vessels group and with combined para-vessels group. The patients were followed up for re-bleeding and survival after endoscopic treatment for hematiostasis.
Results: A total of 69 patients were analyzed, 27 patients were combined with para-gastric or para-esophageal varices and 42 were not. Baseline characters including age, sex, hepatitis B, diabetes, hypertension and portal thrombosis were comparable between the two groups. Hemoglobin, prothrombin time and albumin levels were also comparable after the hospital admission, so was the HVPG (Median 14 (9-20) mmHg, p=0.9951). All patients received similar endoscopic treatment for the secondary prophylaxis of gastroesophageal variceal rebleeding. The median follow-up was 745 days. A total of 27 patients experienced rebleeding and 1 patient underwent liver transplantation. Kaplan-Meier analysis showed that the 3-year rebleeding after endoscopic treatment was significantly higher in patients with para-vessels than in patients without (74.33% vs 32.79%, p=0.005).
After further multivariate analysis including hepatitis B and albumin levels, the combination of para-vessels was found to be an independent predictor of rebleeding after endoscopic treatment in patients with gastroesophageal variceal bleeding (HR 2.56, 95% CI 1.149-5.705, p=0.042).

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Conclusion: The presence of para-gastric and para-esophageal varices is an independent predictor of rebleeding after endoscopic treatment for gastroesophageal variceal bleeding. As a result, portal vascular CT is required to assess para-gastric and para-esophageal varices for patients with gastroesophageal variceal bleeding undergoing endoscopic treatment to prevent rebleeding. For patients with large para-varices, other treatment options, such as interventional transjugular portosystemic shunt are highly recommended for a better prognosis.

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Disclosure: Nothing to disclose.

PP0984

ACUTE VARICEAL BLEEDING DURING THE SARS-COV-2 PANDEMIC

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Introduction: Acute variceal hemorrhage (AVH) is a dreaded complication of liver cirrhosis and continues to be associated with high mortality despite significant medical knowledge and advances in recent years. In addition, the severe acute respiratory syndrome coronavirus-2 (SARS-COV-2) pandemic has led to a temporary reduction in the diagnosis, surveillance and treatment of chronic liver diseases and their complications due to the necessary protective measures.

Aims & Methods: This study aimed to analyze the impact of the SARS-COV-2 pandemic on the course of disease in patients with AVH. In this retrospective multi-center study of patients with AVH due to cirrhosis disease courses during to two different time periods (01.03.2016-28.02.2020, and 01.03.2020-28.02.2022, termed COVID period) were compared regarding patients baseline characteristics, management, post-bleeding infections, rebleeding and mortality.

Results: For the COVID and pre-COVID period 96 and 151 patients were analyzed, respectively. During the COVID period the amount of alcohol-associated liver disease or active alcohol consumption did not differ between cohorts (58.2% vs. 47.9%; p=0.147 and 44.2% vs 39.1%; p= 0.496). There was a trend towards a higher median Model for End-Stage Liver Disease (MELD) (16 vs. 19; p=0.06) while the Child-Pugh Score was comparable (10 vs. 10; p=0.304). Intubation rates during endoscopy (16.1% vs. 16.0%; p= 1.000) as well as active bleeding rates (59.6% vs. 50.0%; p=0.150) did not differ but transjugular intrahepatic portosystemic shunt placement (11.3% vs. 23.2%; p=0.019) was performed more frequently during the pandemic.

Additionally, during the pandemic there was a trend to a higher incidence of post-bleeding infections in patients without present infection at the time of AVH, who received antibiotic prophylaxis (21.6% vs. 35.4%; p=0.068). A Coronavirus-infection was present in only one patient. The 6-week recurrent bleeding and mortality rates did not differ between groups (22.3% vs. 22.8%; p=1.000 and 30.1% vs. 32.2%; p=0.767).

Conclusion: Despite an observed trend to a higher incidence of post-bleeding infections and a lower MELD score mortality in patients with AVH was not increased during the SARS-COV-2 pandemic.

Disclosure: Nothing to disclose.

PP0985

THERAPEUTIC PLASMA EXCHANGE IN PATIENTS WITH ACUTE-ON-CHRONIC LIVER FAILURE

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Introduction: Acute-on-chronic liver failure (ACLF) is a severe form of acute decompensation of chronic liver disease with a high short-term mortality rate. It is characterized by excessive systemic inflammatory response and extrahepatic multiorgan failures (1).

Previously published data have shown that therapeutic plasma exchange (TPE) decreases mortality rate in patients with ACLF reducing cytokine levels, serum bilirubin, and coagulopathy (2,3).

Aims & Methods: This retrospective study was conducted at the University Clinical Center of Serbia. All patients diagnosed with ACLF according to the EASL criteria and treated with TPE from January 2021 to January 2024 were included. (1)

The main indication for providing TPE was serum bilirubin level >12 mg/dl with no improvement to standard medical treatment (SMT). The following scores were calculated at the hospital admission: Model for end-stage liver disease (MELD), Model for end-stage liver disease-Sodium (MELD-Na), Chronic Liver Failure Consortium- acute-on-chronic liver failure (CLIF-C ACLF). All patients were followed for 90 days. MELD and MELD-Na scores were calculated as well on the day 28.

The aims of this study were: assessment of 28-day and 90-day survival in patients with ACLF, the impact of TPE on MELD, and MELD-Na score.

Results: The study included 40 patients in whom SMT did not lead to improvement, of which 32 (80%) were male and 8 (20%) were female with a median age of 45.5 (39.75-58.5) years. The most frequent cause of liver cirrhosis was alcohol (n=28; 70%), followed by Wilson disease (n=4; 10%), metabolic dysfunction-associated fatty liver disease (MAFLD) (n=3; 7,5%), autoimmune hepatitis (n=2; 5%), primary biliary cholangitis (n=2; %%), AIH/ PSC syndrome overlap (n=1; 2.5%). Eighteen patients (45%) were previously diagnosed with acute decompensation (Table 1).

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The average number of performed TPEs was 6±12.73. In alcohol-related liver disease (ALD) cirrhosis, alcoholic hepatitis was the main cause of ACLF in 22 (78.5%) patients. Twenty-two patients (55%) were diagnosed with Community-acquired infections. At admission, the average MELD and MELD-Na scores were 28.5±6.43 and 31±6.21. The mean MELD and MELD-Na scores in surviving patients after 28 days were 24.3 ±7.78 and 25.52±7.58, respectively. The average CLIF C-ACLF score at admission was 47±8.07. ACLF grade 1 was observed in 20 patients (50%), ACLF grade 2 in 18, (45%), and ACLF grade 3 in 2 (5%). The 28-day mortality in this study group was 37.5% (n=15), with 25% (n=5) in ACLF grade 1, 44.4%(n=8) in ACLF grade 2, and 100% (n=2) in ACLF grade 3. Ninety-day mortality was 62.5% (n=25). Regarding the ACLF grade, the following 90-day mortality was observed: grade 1-45%(n=9), grade 2-77.8%(n=14), and grade 3-100 (n=2). With the use of TPE in addition to SMT, a statistically significant decrease in MELD and MELD-Na as well was observed after 28 days (p <0.01).

Variables	Total number (n=40)
Gender m/f	32(80%)/8(20%)
Age[a]	45.5 (39.75-58.5)
Etiology of liver cirrhosis:	
Alcohol	28 (70%)
Wilson disease	4 (10%)
MAFLD	3 (7.5%)
Autoimmune hepatitis	2(5%)
PBC	2 (5%)
PSC/AIH	1 (2.5%)
Previous acute decompensation	18 (45%)
ACLF grade I/II/III	20(50%)/18 (45%)/2 (5%)
28-day/90 day mortality	15(37.5%)/25 (62.5%)

Table 1. Baseline demographic and clinical characteristics.

Conclusion: TPE may be a beneficial treatment option for patients who do not respond to SMT, but in our study, short-term mortality is still high. New prospective randomized studies with a larger number of subjects are needed to examine this therapeutic modality accurately.

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Disclosure: There is no conflict of interest.

PP0986

EFFECTS OF PROBIOTICS USE IN PATIENTS WITH LIVER CIRRHOSIS AND HEPATIC ENCEPHALOPATHY: A SYSTEMATIC REVIEW AND NETWORK META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: Bacterial translocation exacerbates circulatory dysfunction present in patients with liver cirrhosis. Probiotics can reduce bacterial translocation and consequently improve the outcomes in this population.

Aims & Methods: This study aims to evaluate the efficacy of probiotics in the cirrhotic patients. We performed a systematic review in PubMed, Embase and Cochrane Central database, searching for studies analyzing the effects of probiotics use in overt hepatic encephalopathy incidence in cirrhotic patients. A Bayesian network meta-analysis was conducted to estimate the relative effects between treatments using risk ratios (RR) and to rank each one according to the Surface Under the Cumulative Ranking Curve (SUCRA).

Results: We included eight studies, comprising 975 patients. The assessment of SUCRA values indicated that lactulose plus probiotics (0.95) emerged as the most effective prevention for overt hepatic encephalopathy. Moreover, lactulose plus probiotics demonstrated superior efficacy compared to lactulose (0.54), probiotics only (0.51), LOLA (0.48), placebo (0.24), and no therapy (0.21). The risk of overt hepatic encephalopathy was low for lactulose plus probiotics (0.000025; credible interval [CrI] 1.3x1016, 1.5), lactulose only (RR 0.64; CrI 0.09, 4.0), probiotics (RR 0.69; CrI 0.17, 2.9), LOLA (RR 0.64; CrI 0.04, 8.9) and no therapy (RR 1.1; CrI 0.22, 7.0).

Conclusion: In our network meta-analysis, the best results in the treatment of the patients with cirrhosis and hepatic encephalopathy was seen with the association of lactulose plus probiotics. Both drugs have good tolerability and are widely disponible, so we suggest considering this association.

Disclosure: Nothing to disclose.

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PP0987

VALIDATION OF BAVENO VII CRITERIA: RECOMPENSATION OF DECOMPENSATED VIRAL CIRRHOSIS

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Introduction: Managing decompensated cirrhosis is a significant challenge due to high morbidity and mortality rates. The recently proposed Baveno VII criteria provide a standardized definition to assess recompensation. In this study, we validated the Baveno VII definition of recompensation and aimed to evaluate recompensation factors in patients with decompensated viral cirrhosis.

Aims & Methods: This is a retrospective descriptive study over a 10-year period between 2012 and 2022, enrolling 99 patients from the hepato-gastroenterology department of Mohamed Taher Maamouri Hospital in Nabeul who had been treated for decompensated viral cirrhosis B or C and achieved sustained virological response with a follow-up of at least 12 months.

Results: A total of 99 patients with decompensated viral cirrhosis (B or C), who had received treatment and achieved sustained virological response, were included in the study. The mean age was 62 ± 11 years, and 56 patients were female. The most common mode of decompensation was ascites, observed in 81.8% of the patients. The mean MELD score was 15 ± 4, and the majority (59.7%) were classified as Child-Pugh class B. Overall, 45.5% of the decompensated patients achieved recompensation during the study period. The recompensation rate was higher in patients with hepatitis B-related cirrhosis (51.1%) compared to those with hepatitis C-related cirrhosis (40.4%).

According to the results of logistic regression, four independent factors associated with recompensation were found: albumin (OR = 2.033, IC95% [1,461-2,827], p<0.001), diabetes (OR =0.003, IC95%[0.000-0.045], p<0.001), Child Pugh score (OR=0.162, IC95% [0.057-0.459], p=0.001), INR (OR =0.017, IC95%[0.000-0.939].

Conclusion: This study identified key factors associated with recompensation in patients with viral decompensated cirrhosis, including serum albumin levels, INR values, history of diabetes, and severity of liver disease. The knowledge of recompensated indicators may help support the development of different prevention strategies to reduce the incidence of acute decompensation.

Disclosure: Nothing to disclose.

PP0988

ACCURACY OF NEUTROPHIL GELATINASE-ASSOCIATED LIPOCALIN TO DIAGNOSIS ACUTE TUBULAR NECROSIS IN CIRRHOSIS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction:

Renal impairment is a common complication in cirrhosis. Urinary neutrophil gelatinase-associated lipocalin (NGAL) has shown promise in diagnosis of the different types of acute kidney injury (AKI) in cirrhosis.

Aims & Methods:

This study aims to evaluate the accuracy of NGAL in the differential diagnosis of acute tubular necrosis (ATN) in cirrhosis. We performed a systematic review in PubMed, Embase and Cochrane databases, searching for studies analyzing NGAL as a biomarker of ATN. Statistical analysis included data pooling and heterogeneity testing. Quality of the included studies was assessed by Quality Assessment of Diagnostic Accuracy Studies 2 (QUADAS-2).

Results: The meta-analyses included 3 studies and 317 patients. The NGAL cut-off evaluated were similar between the three studies (delta 143 to 244µg/g). The pooled sensitivity and specificity of NGAL in predicting ATN in cirrhosis were 78.7% (95% CI 68.4% to 86.4%) and 80.8% (95% CI 55.5% to 93.4%). The AUC was 0.866 (95% CI 0.646 to 0.955).

Conclusion: In patients with cirrhosis, urinary NGAL is a good predictor of ATN and can be used as an additional tool to the differential diagnosis of acute kidney injury.

Disclosure: Nothing to disclose.

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PP0989

A 5-YEAR LONGITUDINAL STUDY ON REPEATED VARICEAL LIGATION IN ACUTE ESOPHAGEAL VARICEAL BLEEDING

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Introduction: Acute esophageal variceal bleeding (EVB) typically requires initial treatment involving pharmacotherapy and endoscopy, particularly endoscopic variceal ligation (EVL). In cases of rebleeding, transjugular intrahepatic portosystemic shunt (TIPS) is recommended, but in regions with limited TIPS access, repeated endoscopic therapy is common. This study evaluates the outcomes of EVL in managing rebleeding and mortality in resource-constrained settings.

Aims & Methods: Patients experiencing acute EVB and undergoing EVL were followed for up to five years or until death. Treatment involved vasoactive drugs and EVL until variceal eradication was achieved. Primary endpoints included bleeding control effectiveness and the need for a second EVL session to manage rebleeding. Rebleeding and mortality rates at 6 weeks and 5 years were analyzed. Logistic regression analysis identified risk factors associated with rebleeding and mortality.

Results: Among 118 patients with acute EVB undergoing EVL, no treatment failures occurred within five days. Eighty-three percent were successfully controlled with one EVL session, while 17% required at least two sessions for recurrent bleeding. In the rebleeding group, no patients rebled within five days after the first EVL, but 75% rebled within 6 weeks. Second EVL was performed for these patients, with a failure rate of 25%. Recurrent EVBs were ultimately controlled by other endoscopic therapy. There was no significant difference in 6-week mortality between rebleeding (15%) and non-rebleeding (18%) groups (p=0.72). Factors associated with 6-week mortality included age, hepatocellular carcinoma, post-bleeding beta-blocker use, creatinine, bilirubin, albumin, Child-Pugh class, and MELD score (Table). Beta-blocker use was associated with preventing recurrent EVB within six months (OR 0.053, 95% CI: 0.004-0.687, p = 0.025). Over five years, the rebleeding group showed a higher mortality trend (90%) compared to the non-rebleeding group (68.8%), though not statistically significant (p=0.06).

	OR (95%CI)	p
Age	1.04 (1.00-1.08)	0.04
HCC	4.05 (1.45-11.37)	0.01
Beta-blockers used after EVB	0.09 (0.03-0.29)	<0.001
Total bilirubin	1.1 (1.02-1.19)	0.02
Creatinine	1.83 (1.11-3.01)	0.02
Albumin	0.27 (0.10-0.75)	0.01
Child-Pugh class	2.50 (1.14-5.50)	0.02
MELD score	1.12 (1.04-1.20)	0.001

Conclusion: In resource-limited settings, multiple attempts of EVL demonstrate success in controlling rebleeding. Long-term trends indicate higher mortality in rebleeding cases, although not statistically significant.

Disclosure: The authors declare that they have no conflict of interest.

PP0990

IDENTIFICATION OF POTENTIAL EPIGENETIC MECHANISMS REGULATING LSECTIN EXPRESSION IN HEPATIC ANTIGEN PRESENTING CELLS

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Introduction: LSECtin/*Clec4g* is a C-type lectin that recognizes pathogen patterns, and a T-cell ligand expressed by Kupffer Cells (KCs) and Liver Sinusoidal Endothelial Cells (LSECs). Its expression decreases during chronic liver disease progression, however, the mechanisms that regulate this process have not been fully characterised.

Aims & Methods: Given the relevance of epigenetic control in gene expression regulation during liver disease progression, our main aim was to characterise the epigenetic regulation of LSECtin regarding DNA methylation and Histone Posttranslational Modifications (HPTMs), main mechanisms of epigenetic gene regulation. For this purpose, immortalised KCs and LSECs lines were cultured and stimulated with selected epigenetic drugs: OTS186935 (Suv39h2 inhibitor), UNC0642 (G9a/Glp inhibitor), UNC1999 (EZH2/1 inhibitor), 5-Aza-2'-deoxycitidine (DNMT inhibitor) with or without IL-4, an LSECtin inductor. Simultaneously, probe-based methylation array and ChIP-Seq experiments were conducted in primary KCs and LSECs isolated from both control and an experimental chronic liver disease mice model, generated by a 12-week protocol of CCl_4 oral gavage.

Results: Different significant changes were observed in specific genomic loci, both in LSECs and Kupffer cells, these last particularly susceptible to methylation changes after CCl_4 treatment. We didn't observe significant changes at general methylation levels in KCs and LSECs methylome from untreated vs CCl_4-treated mice. Beta-values of specific probes for *Clec4g* gene body and promoter region resulted to be statistically significantly different, especially in Kupffer cells. ChIP-Seq of KCs and LSECs from both untreated and CCl*4*-treated mice allowed the study of peak profile after immunoprecipitation with H3K9me3 and H3K27me3 antibodies. Both populations showed a differential peak profile at genomic level, but no differences at *Clec4g* gene body or proximal promoter. Pharmacological inhibition of histone-lysine N-methyltransferase enzymes for 48 h of treatment showed a downregulation in LSECtin expression. Consequently, LSECtin expression could be retrieved by IL-4+drug treatment, highlighting results obtained in KCs. Nevertheless, Ezh2/1 inhibition did not follow this trend, moderately increasing LSECtin expression after the treatment. In addition, LSECtin expression was upregulated by DNA passive demethylation due to DNMT inhibition, with a higher increase in Aza+IL-4 treated cells, thus supporting the potential role of DNA methylation in Clec4g regulation.

Conclusion: LSECtin expression upregulation by DNA passive demethylation together with methylation array analysis strongly points out DNA methylation as a regulatory mechanism of LSECtin expression. On the other hand, LSECtin expression modulation by epigenetic drugs, altogether with the absence of a specific peak profile at *Clec4g* suggests a more complex, indirect effect of HPTMs in the protein regulation.

Disclosure: Nothing to disclose.

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PP0991

SCREENING FOR VIRAL HEPATITIS IN ALL INPATIENTS OF A UNIVERSITY INTERNAL MEDICINE DEPARTMENT

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Introduction: The use of effective vaccines and drugs has led the global community to set a goal of shrinking the prevalence of viral hepatitis by 2030. Most efforts are focused on screening large segments of the population that are unaware that they have these diseases.

Aims & Methods: The aim of our study was to screen a large number of hospitalised patients for viral hepatitis. Patients hospitalised between 1/2021 and 9/2023 at the First Propaedeutic Department of Internal Medicine of the Aristotle University of Thessaloniki were prospectively analysed. Patients with known pre-admission viral hepatitis or patients investigated for liver disease were omitted. Patients underwent testing for viral hepatitis B and C markers.

Results: A total of 3914 patients were screened. The mean age was 69.8 ± 16.9 years. 112 (2.9%) patients had positive surface antigen, and 1281 (32.8%) patients had evidence of prior HBV infection (Anti-Hbc+), of which most (952, 74.3%) had concurrent positive Anti-Hbs antibodies. The presence of positive surface antigen was more common in patients 41-65 years old (4.1%) compared to patients > 65 years old (2.4%) and patients ≤ 40 years old (2.7%). Patients born in a foreign country constituted 21.2% of patients with positive surface antigen, with the vast majority being in the 41-65 age group (58.3%). Also, the presence of prior infection had a significant correlation with patient age (p < 0. 001), with patients >65 years of age having a much higher prevalence (39.9%) compared to patients 41-65 years (20.9%) and patients ≤ 40 years (9.5%). HBV DNA testing was performed in 65 patients (58% of positives), with 60 patients showing positive HBV DNA (92.3%). Of these, 13 had chronic HBV infection, and 47 had chronic HBV hepatitis. Finally, 28 patients (59.6%) received treatment. Regarding screening for hepatitis C, 102 patients (2.7%) showed anti-HCV (+), with 53 patients (52%) undergoing HCV RNA testing. The presence of positive anti-HCV antibodies was more common in patients ≤ 40 years (9.1%), compared to patients 41-64 years (3.5%) and patients > 65 years (1.6%); 29 patients showed positive HCV RNA (54.7%), with 13 patients eventually receiving treatment (44.8%).

Conclusion: The study confirms the high prevalence of viral hepatitis in our country. Additionally, a significant proportion of patients with previous hepatitis B infection is recorded, a fact that is of considerable importance in the era of widespread use of immunosuppressive therapies. The increased prevalence of HBV infection in patients aged 41-65 years is probably explained by the presence of many patients of foreign origin. Finally, an important factor regarding the low treatment initiation rates is probably the performance status of patients hospitalised in internal medicine departments in our country, with high rates of morbidity and both inpatient and outpatient mortality. To conclude, hospitalised patients are a population in which screening for viral hepatitis can be easily and effectively performed.

Disclosure: Nothing to disclose.

PP0992

WITHDRAWN

PP0993

EXTENDING HCV ELIMINATION EFFORTS TO UNDERSERVED POPULATIONS - A VIENNESE APPROACH FOR HCV CARE AMONG HOMELESS PEOPLE WITHOUT HEALTH INSURANCE

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Introduction: Despite the increased risk for hepatitis C virus (HCV) infection among people experiencing homelessness (PEH), tailored HCV elimination efforts are currently lacking as this population is often not covered by health insurance.

We present the first Austrian HCV elimination project for PEH and people without health insurance (PHI).

Aims & Methods: "neunerhaus", a low-threshold facility providing health care, support by social workers, and housing for >5.000 PEH and PHI per year in Vienna, served as a platform for this project. From March 2020, all PEH and PHI using one of the services offered by neunerhaus had the opportunity to receive serological antiHCV screening directly on-site. AntiHCV(+) blood samples were automatically tested for HCV-RNA via PCR (reflex testing). Through the establishment of an on-site hepatology service at neunerhaus, HCV-viremic individuals received treatment with direct-acting antiviral agents (DAA).

For the duration of DAA therapy, housing (PEH) was provided by neunerhaus while DAA treatment (PHI) was kindly provided by Abbvie and Gilead. Through social workers as well as at the neunerhaus housing facilities, PEH and PHI received support in DAA adherence.

Results: Serologic antiHCV screening was performed in 601 persons between March 2020 and July 2023, among whom 116 (19.3%) showed antiHCV(+). Negative HCV-RNA PCR results indicated HCV clearance in 24/116 (20.7%), while 92 (79.3%) individuals remained HCV viremic (HCV-RNA+). Among the 92 viremic patients, 36 (39.1%) already started DAA and 21/36 (58.3%) finished treatment. Sustained virologic response (SVR) was documented in 17/21 (81.0%) patients and in 4 (19.0%) post-treatment follow-up remains pending. No treatment interruptions were recorded among the study population.

Conclusion: Despite COVID19-related restrictions affecting most of the study period, a relevant number of underprivileged PEH and PHI underwent HCV screening through this project. HCV viremia was found in 15.3% of the overall cohort, underlining the need for specific HCV elimination strategies for PEH and PHI as an underserved population.

Furthermore, our data demonstrate the effectiveness of streamlined DAA therapy for PEH and PHI, provided that safe housing for the duration of therapy, support through social workers, and low-barrier health care are available.

Future HCV elimination strategies should include underserved populations such as PEH and PHI in whom HCV prevalence remains alarmingly high despite the availability of DAA treatment.

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PP0994

HCV MICROELIMINATION THROUGH INDIVIDUALIZED INTERVALS FOR DIRECTLY OBSERVED THERAPY - A STREAMLINED APPROACH TO HCV CARE FOR PWID

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Introduction: Treating hepatitis C virus (HCV) infection among people who inject drugs (PWID) on opioid agonist therapy (OAT) according to the concept of directly observed therapy (DOT) with direct-acting antiviral agents (DAA) is a highly effective approach to HCV microelimination. The principle behind DOT is DAA prescription via a low-threshold facility (LTF) and supervised OAT/DAA ingestion at a pharmacy or the LTI. In individuals with good adherence to OAT, the intervals for OAT/DAA dispensation may be extended from daily to once weekly.

However, the effect of these alterations in OAT/DAA dispensation schedules on sustained virologic response (SVR) rates remains unclear. We present an individualized approach to DAA DOT for PWID that was used throughout the COVID19 pandemic and thereby represents one of only few HCV elimination programs that were successfully continued during the pandemic.

Aims & Methods: Two groups of Viennese PWID were compared regarding DOT distribution intervals, SVR rates, and socioeconomic characteristics: The "COVID period" was defined as 15.3.2020-2.5.2023 and comprised 239 PWID starting DAA treatment within that time frame. The "pre-COVID period" was defined as 26.9.2014-14.3.2020 and consisted of 441 PWID who began DAA therapy during that period.

Results: Daily DOT was the most frequent OAT/DAA distribution schedule pre-COVID (348 PWID, 78.9%) while 19 (4.3%) were on 2-3 times weekly and 74 (16.8%) on once weekly distribution. By comparison, during COVID, only 75 (31.4%) PWID were on daily DOT, 60 (25.1%) underwent 2-3 times weekly and 104 (43.5%) once weekly dispensation regimens (p<0.001). SVR rates, however, were comparably high between the two treatment periods (167/168 [99.4%] COVID vs. 402/405 [99.3%] pre-COVID according to modified intention-to-treat analysis; p=0.849) and no significant difference concerning missed visits (59 [0.8%] COVID vs. 84 [0.4%] pre-COVID; p=0.239) was recorded.

During COVID, socioeconomic status among the study population was equal or worse as compared to pre-COVID (unemployment: 83.7% vs. 67.3%, p=0.967; homelessness: 38.5% vs. 35.1%, p=0.227; harmful alcohol use: 29.3% vs. 10.7%, p<0.001; ongoing intravenous drug use: 64.0% vs. 57.8%, p=0.616). No treatment interruptions or COVID19-related deaths were observed among the study population.

Conclusion: Individualized DOT featuring extended OAT/DAA distribution intervals resulted in similarly high SVR rates as traditional daily DOT. Hence, the effectiveness of DOT seems to primarily result from the streamlined approach to HCV care comprising low-barrier access to healthcare and DAA treatment via an LTI and is not mainly linked to strict supervision.

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PP0995

EMERGING TRENDS: RISING INCIDENCE OF ACUTE HEPATITIS A AMONG ADULTS AND CLINICAL CHARACTERISTICS

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Introduction: For decades, Hepatitis A virus (HAV) had been a leading cause of acute hepatitis among children and was less prevalent among adults. However, recently a paradigm shift has been observed in the epidemiology of HAV evident by cases of acute hepatitis due to HAV among adults. There has been a global surge in the incidence of acute hepatitis A among adults as shown in Korean and Brazilian studies.[1,2] The objective of our study is to estimate the frequency of HAV among adult patients presented with acute viral hepatitis at Aga Khan University Hospital (AKUH), Karachi. To compare the clinical and laboratory characteristics of acute viral hepatitis due to HAV and Hepatitis E virus (HEV) infection among adults.

Aims & Methods: This was a retrospective cross sectional study conducted at AKUH from 2019 till 2023. All patients above 18 years age, diagnosed with acute viral hepatitis secondary to hepatotropic viruses were included. Non hepatotropic virus related acute hepatitis, autoimmune, alcoholic and other metabolic causes were excluded. Data regarding clinical and laboratory parameters was collected by reviewing patient medical records. Yearly trend of occurrence was noted, frequencies of HAV, HEV, HBV (Hepatitis B virus) and HCV (Hepatitis C virus) were recorded, comparative analysis was done between HAV and HEV using spss.

Results: A total of 396 patients were found to have acute hepatitis during our study duration. 234 patients were diagnosed with acute HAV infection, 157 patients were found to have acute HEV infection. Additionally, HBV infection was identified as the cause in three patients, whereas HCV infection was found in only two cases of acute hepatitis. The mean age of patients with HAV was 28±8 while those with HEV was 30±8 with a p value of 0.017 showing that older adults were affected with HEV infection. Higher levels of total bilirubin were seen in HEV infection while higher levels Alanine transaminase (ALT) were seen in HAV infection. (p value<0.005). Increased frequency of acute liver failure (ALF), coagulopathy and mortality was observed in HEV. 10 patients with HEV required ICU while only 4 patients with HAV required ICU (p value=0.038). 13 patients with HEV were found to have underlying chronic liver disease (CLD) while only one patient with HAV was found found to have CLD (p value=0.000) . Yearly

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trends showed increasing occurrence of HAV infection among adults. (Table) Prolonged hospital stay was observed among HEV patients with a p value <0.005.

Year	HAV	HEV
2019	30	33
2020	27	28
2021	31	40
2022	61	19
2023	85	37
Total bilirubin (mean)	6.2 mg/dL	13.3 mg/dL (p 0.000)
ALT (mean)	3274 IU	1760 IU (p 0.000)
ALF	8	24 (p 0.000)
Mortality	2	11 (p 0.001)

Conclusion: HAV seems to be the rising cause of acute hepatitis among adults. This signifies the need of improved sanitation, hygiene and a robust vaccination programme against HAV among adult population. Although it is less severe than acute HEV infection which causes more morbidity and mortality. More epidemiological studies are needed to establish this increase in HAV among adults.

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PP0996

WITHDRAWN

PP0997

NEONATAL SCLEROSING CHOLANGITIS- PHENOTYPE GENOTYPE OBSERVATIONS OF NOVEL BIALLELIC VARIANTS AND SINGLE EXON DELETIONS IN DCDC2 GENE IN INDIAN CHILDREN – EXPANDING HORIZON

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Introduction: Neonatal Sclerosing Cholangitis (NSC) is associated with progressive biliary fibrosis that often requires liver transplantation in childhood. Children present early in life with jaundice, hepatosplenomegaly, hyperbilirubinemia and high serum gamma glutamyl transferase activity (GGT) or intracranial bleed. DCDC2 pathogenic variants have been reported to be associated with renal-hepatic ciliopathy with nephronophthisis, deafness, dyslexia and central nervous system impairment.

Aims & Methods: The aim of our study was to present the phenotypic genotypic observations in five Indian children with novel biallelic variants and single exon deletions in DCDC2 gene. Prospective data of children diagnosed with NSC between March 2023 to February 2024 in a tertiary care institute in India were analyzed.

Results: We report five cases of NSC with mutation in DCDC2 gene of which patient four and five were siblings. Mean age at first presentation was 51 days. All children had cholestatic jaundice with high Gamma-Glutamyl Transferase (GGT). Two children presented with seizures due to intracranial bleed at their second month of life, and other three with jaundice and hepatosplenomegaly. Three children had variceal bleed within 3 years and endotherapy had been done. None of them had renal disease or deafness. Liver biopsy in two children showed portal fibrosis and cholestasis. Magnetic Resonance Imaging of one child showed focal narrowing and dilatations of bile duct while others had minimal intrahepatic biliary radical dilatation. On Next Generation Sequencing, four different novel variants were identified. Two novel variants with frame shift variant in exon7, c.840_841delinsT p.Lys280AsnfsTer4 (depth -181X), and splice site variant c.759+1_759+2dup in Intron6 were classified as likely pathogenic, as these variants were predicted to cause loss of function. Child with splice site variant had motor developmental delay, which improved with age. Homozygous single exon deletions in DCDC2 in exon2 [chr6:(?_24357674-24358132_?)del] and exon5 [chr6:(?_24291148-24291316_?)del], which were reported previously in patients with reading disability, language impairment were identified in three children. As these two children were 3years and 9months respectively at last follow up, reading and language impairment could not be definitely established.

Sl.no	Gender	Age at Presentation	Parental Consanguinity	Symptoms at onset	Pruritis	Variceal Bleed	Highest Bilirubin (mg/dl)	Highest GGT (IU/L)	Variant Nomenclature	Zygosity	Location	Change	Extrahepatic manifestations	Growth Retardation	Follow up
1	Female	1 month	No	Neonatal cholestasis	Yes	Yes, 3 years	8.4	1179	c.840 _841 delinsT p.Lys 280Asnfs Ter 4 (depth -181X)	Homozygous	Exon 7	Frameshift variant	No	No	Alive till 13yrs Good synthetic function
2	Female	55 days	Yes	Cholestatic jaundice, subdural hematoma	Occasional pruritis	Yes, 3 years	9.9	130	c.759+1 _759 +2dup [Depth -92X]	Homozygous	Intron 6	Splice site variant	Motor developmental delay -Improved after 2 yrs of age	No	Alive till 11 years Good synthetic function
3	Male	2 months	Yes	intracranial bleed, neonatal cholestasis	No	Yes, 2.5 years	6.3	1997	chr6: (?_24 3576 74_ 2435 8132_?) del	Homozygous	Exon 2	Deletion	No	No	Alive till 3.5 years Good synthetic function
4	Male	50days	Yes	Neonatal cholestasis	No	No	4.6	600	[chr6: (?_24 2911 48-2429 1316_?) del]	Homozygous	Exon 5	Deletion	Growth retardation Developmental delay	Yes	Died at 2.5years (not of liver failure)
5	Female	2 months	Yes	Neonatal cholestasis	No	No	7.6	1287	[chr6: (?_24 2911 48-2429 1316_?) del]	Homozygous	Exon 5	Deletion	Severe growth retardation	Yes	Alive till 1 year 4 months Good synthetic function

Conclusion: Till date only thirty-eight cases have been reported. Novel homozygous spice site, frameshift variants and homozygous single exon deletions of DCDC2 reported here widens the spectrum of DCDC2 in NSC.

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Disclosure: Nothing to disclose.

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PP0998

DIAGNOSIS AND SCREENING OF LOW PHOSPHOLIPID-ASSOCIATED CHOLELITHIASIS SYNDROME (LPAC): CHALLENGES AND PERSPECTIVES IN A MOROCCAN POPULATION

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Introduction: Low Phospholipid-Associated Cholelithiasis (LPAC) syndrome is a rare disorder caused by mutations in the ABCB4 phospholipid transporter, found in biliary canaliculi. These mutations lead to a deficiency in phospholipids and the subsequent precipitation of cholesterol crystals within the small intrahepatic bile ducts. LPAC syndrome can be diagnosed following the emergence of biliary symptoms or through familial screening. The objective of our study is to characterize the epidemiological, clinical, and paraclinical data of patients monitored for LPAC syndrome in our institution and to assess therapeutic responses and outcomes.

Aims & Methods: We collected data from all patients diagnosed with LPAC syndrome at our institution. Diagnosis was established based on the presence of at least two of the following criteria: (1) age at onset of biliary symptoms ≤40 years; (2) presence of hyperechoic intrahepatic foci ("comet tails" or micro-spots); (3) recurrence of biliary symptoms post-cholecystectomy.

Results: 10 patients were included. The Average age at diagnosis was 33,6 years, with symptoms typically beginning around age 30. The sex ratio was 2/8 (M/F).a family history of biliary lithiasis and cholecystectomy before the age of 40 were found in 70% of cases . None of the female patients had a history of pregnancy-related cholestasis. 70% of the patients had undergone cholecystectomy prior to their LPAC diagnosis, with recurrence of biliary symptoms before age 40 in all these patients. Clinical presentations upon admission included symptomatic gallstone disease in 5 cases, acute cholangitis due to main bile duct stones in 4 cases, and acute lithiasic pancreatitis in 1 case. Hepatobiliary ultrasound was performed in all patients, revealing comet tail signs in 50% of cases, hyperechoic intrahepatic spots in 20% of cases, and intrahepatic stones in 50% of cases. Biochemical tests showed hyperbilirubinemia in 30% of cases, elevated Gamma-GT and alkaline phosphatase in 80% of cases, and hepatocellular damage in 40% of cases. A mutation in the ABCB4 gene was found in 2-patients.8 patients benefit from An Ursodeoxycholic Acid (UDCA) dosage averaging 11mg/kg, experiencing significant clinical and biological improvements.

Conclusion: LPAC syndrome is a rare pathology in our series, predominantly affecting young, non-overweight women with personal and family history of early biliary stones occurrence . Treatment with Ursodeoxycholic Acid (UDCA) controlled symptoms in the majority of cases.

Disclosure: Nothing to disclose.

PP0999

SELADELPAR IN PATIENTS WITH PRIMARY BILIARY CHOLANGITIS: A SYSTEMATIC REVIEW AND META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: Peroxisome proliferator-activated receptor delta (PPARδ) agonists, such as seladelpar, emerged as potential therapies for PBC and other cholestatic diseases. PPARδs are nuclear receptors widely expressed in hepatocytes, cholangiocytes, Kupffer cells, and hepatic stellate cells. Activation of this receptor may modify the progression of PBC, particularly through its bile acid retention mechanism and antifibrotic and anti-inflammatory effects, inducing biochemical and clinical improvement in PBC patients.

Aims & Methods: We aimed to compare the efficacy and safety of seladelpar *vs* placebo for treating patients with primary biliary cholangitis (PBC) and inadequate response or intolerance to ursodeoxycholic acid (UDCA). We systematically searched MEDLINE, Embase, and Cochrane databases from inception to March 12, 2024. Risk ratios (RRs) and mean differences (MDs) with 95% confidence intervals (CIs) were employed for dichotomous and continuous outcomes, respectively, using a random-effects model. We used Review Manager 5.4.1 for statistical analyses.

Results: We included 3 randomized clinical trials (395 patients). The results of the pooled analyses are shown in *Table 1*.

Outcome	Pooled effect estimate (95%CI)	P-value	I^2, %	Certainty of evidence
Composite biochemical response (week 4)	RR 6.67 (4.00 to 11.13)	<0.001	0	⊕⊕⊕⊕
Composite biochemical response (week 12)	RR 6.07 (3.71 to 9.9)	<0.001	0	⊕⊕⊕⊕
Least-squares percent change of ALP (week 4)	MD -34.28 (-39.56 to -29.00)	<0.001	29	⊕⊕⊕
Least-squares percent change of ALP (week 12)	MD -40.76 (-47.95 to -33.58)	<0.001	67	⊕⊕⊕
Alkaline phosphatase normalization (week 12)	RR 18.57 (3.81 to 90.42)	<0.001	0	⊕⊕⊕
Least-squares mean change of pruritus in overall population (week 12)	MD -0.64 (-1.06 to -0.22)	0.003	0	⊕⊕⊕⊕
Least-squares mean change of pruritus in patients with baseline NRS pruritus score ≥4 (week 12)	MD -1.24 (-2.03 to -0.45)	0.002	0	⊕⊕⊕
Any adverse event	RR 0.99 (0.90 to 1.10)	0.87	1	⊕⊕⊕⊕
Any serious adverse event	RR 0.88 (0.32 to 2.44)	0.81	0	⊕⊕⊕⊕

Table 1. Results from the pooled analyses.
⊕⊕⊕⊕, high-quality evidence; ⊕⊕⊕, moderate-quality evidence; 95%CI, 95% confidence interval; MD, mean difference; NRS, numerical rating scale; RR, risk ratio.

Compared with placebo, seladelpar increased the probability of achieving composite biochemical response both at week 4 (RR 6.7; 95% CI 4.0 to 11.1) and at week 12 (RR 6.1; 95% CI 3.7 to 9.9), decreased alkaline phosphatase levels both at week 4 (MD -34.3%; 95% CI -39.6 to -29.0%) and at week 12 (MD -40.8%; 95% CI -47.9 to -33.6), increased the probability of achieving alkaline phosphatase normalization at week 12 (RR 18.6; 95% CI 3.8 to 90.4), and decreased pruritus scores at week 12 in both the overall population (MD -0.6; 95% CI -1.1 to -0.2) and the population with baseline

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pruritus score ≥4 (MD -1.2; 95% CI -2.0 to -0.4). Additionally, there was a non-significant difference between seladelpar and placebo in the risk of any adverse event (RR 1.0; 95% CI 0.9 to 1.1), any serious adverse event (RR 0.9; 95% CI 0.3 to 2.4), or any adverse event leading to treatment discontinuation (RR 0.7; 95% CI 0.2 to 2.0).

Conclusion: Seladelpar may be an effective and safe treatment option for patients with PBC and inadequate response or intolerance to UDCA. However, new large RCTs assessing long-term efficacy and safety outcomes are necessary to better evaluate its place in PBC management.

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PP1000

CAROLI DISEASE AND SYNDROME: ABOUT 8 CASES

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Introduction: Caroli disease (CD) is a rare congenital disease defined by the presence of segmental dilations in the intrahepatic bile ducts. It is called Caroli syndrome (CS) when it is associated with liver fibrosis or cirrhosis. It is often diffuse, rarely localized to a single lobe of the liver.

The aim of this study was to describe the epidemiological, clinical and therapeutic aspects as well as the observed complications associated with this pathology

Aims & Methods: We collected patients followed for CD or CS in the gastroenterology department of the Tahar Maamouri University Hospital in Nabeul from January 2008 to December 2018. We retrospectively collected epidemiological, clinical, radiological, therapeutic and evolutionary data.

Results: We collected 8 patients: 4 men and 4 women whose average age was 64 years (38 -79 years). The circumstances of discovery were an acute cholangitis in two patients and biliary colic in six. The average time between the onset of symptoms and definitive diagnosis was 10 months with extremes ranging from 1 month to 10 years. A morphological assessment (abdominal ultrasound, abdominal CT or Biliary MRI) was carried out in all patients. It showed dilation of the intrahepatic bile ducts (IHBD) in all cases. It was diffuse in four cases and localized to the left in 4 cases. It was associated with intrahepatic stones in 50% of patients (N=4). Gallstone lithiasis was associated in 38% of cases (N=3). Polycystic kidney disease was observed in 25% of patients (N=2). All patients had normal renal function.

Two patients had signs of portal hypertension consistent with a Caroli syndrome. In our series, complications were observed in 50% of cases such as: iterative attacks of acute cholangitis in two patients, acute pancreatitis due to a biliary lithiasis in one patient and a case of liver abscess in one patient.

Episodes of cholangitis were controlled by appropriate antibiotic therapy and ursodeoxycholic acid was prescribed in 3 patients at a dose of 15 mg/kg/day for intrahepatic lithiasis. Surgical treatment was carried out in 4 patients, consisting of a left hepatectomy in 3 patients with localized disease and a choledocotomy with hepaticojejunal anastomosis in one patient for lithiasis of the main bile duct. Endoscopic sphincterotomy was performed in 2 patients: one for residual lithiasis of the common bile duct in and one as sequential treatment. The median follow-up was 84 months, two deaths were observed: one secondary to hepatocellular failure and the other secondary to uncontrolled septic shock. In our series there was no case of cholangiocarcinoma.

Conclusion: The diagnosis of CD is often late. It should be considered in the presence of abscesses or recurrent cholangitis associated with the dilatation of the bile ducts. Hepatectomy of localized forms protects against septic complications and degeneration. Liver transplantation remains to be developed to treat patients with a diffuse form.

Disclosure: Nothing to disclose.

PP1247

CHALLENGES IN MANAGEMENT OF PRIMARY BILIARY CIRRHOSIS IN UNIVERSITY HOSPITAL LEWISHAM, A 2 -YEAR AUDIT

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Introduction: Primary Biliary cholangitis (PBC) is a cholestatic condition caused by autoimmune damage of the biliary channels and potentially Precipitating biliary liver cirrhosis and ESLD. The National PBC audit highlighted the nationwide variability in diagnosing and managing PBC (1). The BSG guideline-recommended standard dosing of UDCA (13–15 mg/kg/ day) and prompt liver transplant assessment when indicated (2). Additionally, it endorsed early detection of cirrhosis with related screening to establish and treat hepatocellular carcinoma and oesophageal varices early and it supports adequate control of symptoms such as fatigue or itching which can be challenging (1,2).

Aims & Methods: We obtained medical records from our Electronic Prescription and Medication Administration (EPMA) team for all PBC patients on UDCA between May 2021 and May 2023. A total of 141 patients were identified; 45 and 12 patients were excluded as duplicate and incomplete records, respectively. Only 84 patients were included in this retrospective analysis all with confirmed PBC as defined by the BSG and BASL. Mostly Females (84%) with a mean age of 56.5 years old (35:78). Data were collected from clinic letters and medical records regarding UDCA dose per body weight, symptoms control, refer to liver transplantation, FibroScan and Variceal/HCC screening when liver cirrhosis is a concern.

Results: Our audit revealed that only 78% of the patients were on ideal dosing of UDCA as per documented body weight, additionally, no clear symptoms scoring (fatigue, itching, joint pain, etc) was used. However,

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in the 19 patients with established biliary cirrhosis on fibroScan, there was good compliance with HCC screening (84%) and oesophageal varices screening (78.9%).

Additionally, only 7 patients were referred for liver transplant assessment or discussed with the tertiary center (see Table 1 for details).

Parameter	result
UDCA dose (13–15 mg/kg/ day)	66/84 (78%)
Associated conditions at diagnosis like Ulcerative colitis	4.7%
Symptoms Burden (fatigue and itching score)	Poor documentation and inconsistency throughout records.
FibroScan for liver cirrhosis (>13 Kpa)	19/84 established LC (22.6%)
Refer for liver transplantation assessment or discussions	7/84 (8.3%)
HCC screening (AFP & US)	17/19 (84%)
Variceal screening (Gastroscopy)	15/19 (78.9%)

Study Results (Table 1): - (UDCA, ursodeoxycholic acid - LC, liver cirrhosis – HCC, Hepatocellular carcinoma, AFP, alpha-fetoprotein- US, ultrasound)

Conclusion: More compliance with the recommended dose of UDCA in our patient is needed to achieve clinical outcomes and normalization of liver functions. there was great compliance with variceal and HCC screens in cirrhotic patients However we also advised prompt early referrals for liver transplant assessment. Furthermore, documentation and scoring of extrahepatic symptoms should be improved.

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Disclosure: Nothing to disclose.

PP1001

IS MYCOPHENOLATE MOFETIL EFFECTIVE FOR PATIENTS WITH AUTOIMMUNE HEPATITIS WHO FAILED STANDARD THERAPY?: AN UPDATED SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Standard treatment of autoimmune hepatitis (AIH) typically includes a combination of azathioprine and corticosteroids. However, approximately 20% of patients do not respond well to this conventional therapy. In this scenario, mycophenolate mofetil (MMF) may be used as an alternative treatment for patients with intolerance or non-response to first-line options.

Aims & Methods: This study aims to evaluate the performance of MMF in intolerant or non-responders patients with AIH previously treated with conventional therapy. We performed a systematic review in PubMed, Embase, and Cochrane databases. We analyzed overall remission, remission at AZA intolerance, and remission at AZA non-response. We also compared the outcome of overall remission in patients with AZA intolerance vs. AZA non-response. Statistical analyses were performed using R language 4.3.1.

Results: 21 studies and 646 patients were included. The pooled overall remission rate was 62.21 (95% CI 54.31, 70.10). We found an overall remission rate of 66.74 (95% CI 57.48, 76.00) in patients intolerant to AZA, and 27.80 (95% CI 11.56, 44.04) in AZA non-responders. The comparison of overall remission in both groups favored the intolerance group (OR 4.98; 95% CI 1.69, 14.72; p = 0.004).

Conclusion: MMF is effective in inducing remission in patients with AIH who failed the first-line treatment. The overall effect is more prominent in patients who failed conventional therapy due to intolerance. Non-response to AZA may be correlated to suboptimal response to MMF as well.

Disclosure: Nothing to disclose.

PP1002

MYCOPHENOLATE MOFETIL IS MORE EFFECTIVE THAN AZATHIOPRINE AS A FIRST-LINE AGENT TO INDUCE COMPLETE BIOCHEMICAL REMISSION IN PATIENTS WITH AUTOIMMUNE HEPATITIS: RESULTS FROM A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Mycophenolate mofetil (MMF) is considered an alternative treatment for patients with autoimmune hepatitis (AIH) who have insufficient response to azathioprine (AZA). However, it is hypothesized that MMF may be more effective than AZA to induce remission in treatment-naive patients as well.

Aims & Methods: This study aims to compare the effectiveness of MMF and AZA in inducing remission in patients with treatment-naive AIH. PubMed, Embase, and Cochrane were systematically searched for randomized controlled trials and observational studies. The primary outcome was complete biochemical remission (CBR) at the end of the follow-up period. The secondary outcome was non-response rate. Heterogeneity was examined using I^2 statistics, and a random-effects model was employed. Review manager 5.4 was used for statistical analyses.

Results: The meta-analysis included 3 studies (1 RCT and 2 observational) and 265 patients (180 in the MMF group vs. 85 in the AZA group). CBR was significantly higher in the MMF group (OR 3.81; 95% CI 1.98 to 7.36; p < 0.0001; I2 = 0%). There was no significant difference between groups regarding the non-response rate (OR 0.46; 95% CI 0.14 to 1.52; p = 0.20; I2 = 53%).

Conclusion: In patients with treatment-naive AIH, MMF is more effective in inducing CBR compared to conventional treatment with AZA. Larger RCTs are needed to consolidate the evidence of MMF's impact on AIH treatment as a first-line agent and to provide concise insights for guiding clinical decisions.

Disclosure: Nothing to disclose.

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PP1003

PORTAL HYPERTENSION AFTER LIVER TRANSPLANT: AN ITALIAN-SINGLE CENTER EXPERIENCE

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Introduction: Post liver transplantation portal hypertension (post LT-PH) is an emerging late complication after LT and a challenging disorder.

Aims & Methods: The aim of this study is to describe the prevalence, risk factors, clinical consequences (oesophageal varices, ascites, portal vein thrombosis, hydrothorax, hepatic encephalopathy) and the mortality rate of post LT-PH. We retrospectively enrolled 358 patients receiving a liver transplant at the University Hospital Policlinico Umberto I, between 2000 and 2018 to ensure a follow-up period of at least 5 years. For the statistical analysis Chi squared test, Fisher Test and T test were used.

Results: In our cohort 54 (15%) patients developed post LT-PH. The average time between LT and the occurrence of clinical evidence of PH was 6 ± 3.7 years: specifically 5.4 ± 3.4 years in patients with cirrhosis recurrence, and 3.4 ± 4.5 years for all non-cirrhotic portal hypertension causes. Patients with post LT-PH showed significant differences from the other LT patients regarding the immunosuppression regimen, with a more frequent use of cyclosporine (27.7 vs 1.3% p> 0.0001). Etiology of portal hypertension was cirrhotic in 41 (76%) patients and non-cirrhotic in 12 (24%) patients. The most frequent etiologies of cirrhosis recurrence were HCV or HBV, biliary complications, chronic rejection, de novo autoimmune hepatitis. Among the non-cirrhotic portal hypertension, we identified 3 cases of Porto Sinusoidal Vascular Disorder (PSVD), 1 case of Budd-Chiari Syndrome de novo and several cases of pre or post-hepatic anastomotic stenosis. The most frequent complications of post LT-PH were gastroesophageal varices, ascites, and portal vein thrombosis. Four patients underwent TIPS placement for the management of gastrointestinal bleeding. Moreover, as expected, more patients with cirrhotic post LT-PH died for liver failure (61%) while in the non cirrhotic group only in 3 (25%) cases death was liver related (p value 0.0471).

Conclusion: PH may develop during the follow up of liver transplant patients. This event has been mainly reported due to cirrhosis recurrence, especially for viral etiology, and numbers will decrease following the introduction of efficacious antiviral therapies. Despite this, it is important to consider other causes of post LT-PH. In our cohort 3 cases of PSVD have been highlighted, which is a rare and sparsely described event in the literature among transplant patients. Gastroesophageal varices and ascites were the most frequent complications of portal hypertension. Close clinical and instrumental follow-up is suggested for early diagnosis and management.

Disclosure: Nothing to disclose.

PP1004

ENDOSCOPIC ULTRASOUND-GUIDED FINE-NEEDLE BIOPSY IN A PORCINE LIVER MODEL: A COMPARATIVE STUDY BETWEEN FRANSEEN AND NOVEL TRICORE NEEDLES

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Introduction: Endoscopic ultrasound-guided fine-needle biopsy (EUS-FNB) is a reliable and minimally invasive method for obtaining tissue samples from organs like the pancreas, adrenal glands, and liver [1]. However, challenges in obtaining sufficient tissue samples persist due to limitations of needles, including design and materials [2]. To overcome these limitations, various FNB needles with different tip design and materials, such as Franseen and fork-tip needles, have been developed [3]. Nevertheless, ongoing enhancement of FNB needles remains crucial for increasing the diagnostic yield and efficacy of EUS-FNB.

Aims & Methods: This study aimed to compare the efficacy of newly designed Tricore needles with Franseen needles in a porcine liver model under EUS guidance. A total of four male Yorkshire pigs were randomly divided into two groups: the Franseen group and the Tricore group. Nine EUS-FNB procedures were performed on each pig using three FNB needles. All pigs were euthanized immediately after the final procedure. The maximum resistance force during the puncture and the angle of the FNB needle tip was measured to compare puncture performance of two FNB needles. Visibility under ultrasound, blood contamination, and tissue acquisition time were also evaluated immediately after the EUS-FNB procedure. The sufficiency of the specimen for diagnosis was assessed by measuring the total specimen area, number of fragments, fragment areas, and number of complete portal tracts (CPTs).

Results: Procedural outcomes were successful with no complications observed, yielding specimens with a 100% (36/36) success rate. The Franseen needle (5.97 ± 0.53 N) exhibited significantly higher resistance compared to the Tricore needle (3.83 ± 0.84 N, $P < 0.001$). The both needle tips were observed considerable bending pre- and post-procedure ($P < 0.01$); however, the Franseen needle (84.58 ± 39.69°) became more significantly bent than the Tricore needle (32.82 ± 14.17°, $P < 0.001$). The visibility under ultrasound of the Tricore needle was significantly higher than that of the Franseen group ($P < 0.01$). While blood contamination was similar between groups ($P = 0.926$), the Franseen group (59.90 ± 19.15 s) required significantly longer acquisition time than the Tricore group (48.94 ± 9.26 s, $P < 0.05$). Histological findings revealed significantly larger total specimen area ($P < 0.05$) and fragment area ($P < 0.001$) in the Tricore group compared to the Franseen group. Additionally, the Tricore group exhibited significantly higher number of CPTs ($P < 0.05$).

Conclusion: In conclusion, our study demonstrated that the newly designed Tricore EUS-FNB needle exhibits superior puncture ability and ultrasound visibility, while acquiring liver tissue specimens of greater quantity and quality compared to the commercially available Franseen needle. These findings suggest that the Tricore needle may offer a promising alternative for EUS-FNB procedures. Although further clinical validation is needed to confirm these findings, this novel needle design holds potential for improving the diagnostic yield and efficacy of EUS-FNB in clinical practice.

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Disclosure: Nothing to disclose.

PP1005

CONCORDANCE OF NON-INVASIVE TESTS FOR THE ASSESSMENT OF LIVER FIBROSIS

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Introduction: Liver fibrosis consists in the abnormal accumulation of extracellular matrix proteins due to chronic inflammation. Liver biopsy has been considered the gold-standard for diagnosing and staging fibrosis, which is vital to establish the prognosis and guide therapeutic management. Nevertheless, it is an invasive technique with certain drawbacks which must not be neglected. Given this scenario, new radiologic and serum-based non invasive diagnostic tools have arised to accurately predict liver stiffness.
Aims & Methods: The aim of the study is to evaluate the concordance between different non-invasive tests to assess liver fibrosis including transient elastography (TE), point shear wave speed measurement (SWM) and serum biomarkers (APRI and FIB-4). Moreover, a validation analysis for SWM was performed. It is a single-center, descriptive, observational and retrospective study. Eligible patients had a known history of chronic liver disease, with a maximum time interval between TE and SWM of 6 months. Subsequently, APRI and FIB-4 were calculated. A concordance and validation analysis was conducted in patients with alcoholic liver disease (ALD), metabolic dysfunction associated steatotic liver disease (MASLD) and combined (MetALD).
Results: A total of 267 patients (68 % male, mean age: 59 years) with chronic liver disease regarding multiple aetiologies were included; MASLD (28.5%) was the most prevalent subgroup. The global concordance for fibrosis staging between ET and SWM was adequate with an intraclass correlation coefficient (ICC) of 0.684. Stratified analysis demonstrated a solid concordance for ALD with an ICC 0.7; and moderate for MASLD and MetALD with an ICC 0.53 and 0.48 respectively. However, the overall concordance between ET/SWM and serum biomarkers (APRI, FIB-4) was poor with an ICC < 0.4. To evaluate the diagnostic performance of SWM receiver operating characteristic (ROC) curves were obtained. The global validation analysis was favourable with an area under the curve (AUC) of 0.92.
Conclusion: TE and SWM demonstrated an adequate overall concordance when assessing liver fibrosis, especially in ALD. SWM global validation analysis was favourable making it a reliable non-invasive technique for the evaluation of liver stiffness as a surrogate quantitative biomarker.
Disclosure: Nothing to disclose.

PP1006

ADHERENCE TO BAVENO VII RECOMMENDATIONS IN PORTAL HYPERTENSION SCREENING: WHERE DO WE STAND?

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Introduction: Since the advent of the BAVENO VII consensus, Transient elastography (Fibroscan) has taken on a key role in the assessment of clinically significant portal hypertension (CSPH) and the prediction of decompensation and death. A liver stiffness measurement (LSM) ≤ 15 kPa associated with platelets ≥ 150 G/L excludes CSPH (NPV > 90%), while a LSM > 25 kPa is very much in favor of CSPH (PPV > 90%). CSPH remains possible in non-obese patients with LSM between 20 and 25 kPa with platelets < 150 G/L or LSM between 15 and 20 kPa with platelets < 110 G/L) but with a 40% risk of error.
Aims & Methods: We conducted this study to evaluate the reliability of transient elastography (TE) and platelet count in the diagnosis of CSPH, and to determine the extent of adherence to the consensus recommendations in our center.
This is a monocentric and retrospective study made in the gastroenterology and hepatology department of Mohamed Taher Maamouri Hospital in Nabeul, Tunisia, in which we collated reports of Fibroscans performed from March 2023 to March 2024.
We excluded patients with etiologies other than viral, alcoholic, or non-obese non-alcoholic steatohepatitis (NASH).
Platelet count and upper gastrointestinal endoscopy (EGD) data (if performed) were collected.
Data were entered and analyzed using SPSS version 27 software, with a significance level of 0.05.
Results: One hundred and fifty-nine Fibroscans were performed, of which 127 were included.
Women accounted for 64.6%, with a M/F sex ratio of 0.55. Mean age was 55.13 years [20-83].
The indications for Fibroscan were distributed as follows: 53.5% (68 patients) for non-obese NASH, 33.1% (42 patients) for HBV, 11.8% (15 patients) for HCV and only 2 cases for chronic ethylism.
The mean LSM value was 6.45 kPa [2-41.5], with thresholds distributed as follows; LSM <5 kPa in 70 patients (55.1%), LSM between 5 and 10 kPa in 44 patients (34.6%), LSM between 10 and 15 kPa in seven patients (5.5%), LSM between 15 and 20 kPa in two patients (1.6%), LSM between 20 and 25 kPa in a single patient, and LSM >25 kPa in three patients (2.4%).
The mean platelet count was 220 G/L [20-440], with a count > 150 G/L in 111 patients (87.4%), a count between 110 and 150 G/L in 13 patients (10.2%) and a count < 110 G/L in 3 patients.
Of all patients, 24 FOGDs were performed. Signs of Portal hypertension (PH) were noted in only 5 cases, represented by grade I esophageal varices. In analytical study, hepatic elastography and platelet count were significantly correlated with the presence of endoscopic PH (p=0.009). Indeed, among patients with signs of endoscopic PH, 60% had a LSM > 25 kPa, 20% had a LSM between 15 and 20 kPa with a platelet count < 110 G/L and only 20% had a LSM < 10 kPa.
Furthermore, the consensus recommendations for the indication or not of EGD were not respected in our center (p=0.004); of all endoscopies performed 87.5% (21 EGD) were not indicated, and we noted one case where EGD was indicated but not performed.
Conclusion: Our study highlighted the recommendations of the latest BAVENO VII consensus regarding the reliability of transient elastography and platelet count in the evaluation of CSPH (p=0.009). This must be taken

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into consideration by practitioners in order to implement adequate preventive measures and to rationalize the indication of EGD which remains an inconvenient procedure for our patients.

Disclosure: Nothing to disclose.

PP1007

A DEEP LEARNING-BASED LINEAR SYSTEM FOR LIVER STATION IN ENDOSCOPIC ULTRASOUND

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Introduction: Endoscopic ultrasound (EUS) has made significant strides in recent decades, whose current application in liver is mainly concentrated in liver parenchymal diseases, vascular diseases and portal hypertension. Incorporating EUS into the hepatic examination process is advantageous, which requires the endoscopist to have a good grasp of the anatomy of the liver. However, the learning curve for EUS-guided operations is well known to be steep. Thus, in the evaluation of EUS liver, it is necessary to have a system assisting the endoscopist to complete the scan at each station. Based on the anatomical markers in the liver, we construct a system which can perform a comprehensive inspection of the liver segments through the scanning imaging of 8 stations.

Aims & Methods: EUS of the liver can be done from the stations as follows: Station 1: Left hepatic vein (LHV); Station 2: Umbilical potion of portal vein (PV); Station 3: PV trunk and common bile duct (CBD); Station 4: Middle hepatic vein (MHV) and venous ligament; Station 5: Inferior vena cava (IVC) and right hepatic vein (RHV); Station 6: Gallbladder (GB) and left and right branches of PV; Station 7: Anterior and posterior branches of right portal vein; Station 8: Right kidney.

The system was internally validated in images, and subsequently compared with the performance of EUS endoscopists. We retrospectively collected 2743 images from Hospital A for internal validation, and an independent dataset containing 340 images for human-machine competitions. To filter out the white light images, we apply a DCNN1 model primarily. The output ultrasound images are then fed into DCNN2. In this model, they are classified into standard and non-standard categories. And the DCNN3 is then activated with standard images. DCNN3 identifies certain features in the image and groups them into eight stations. We use accuracy (defined as the number of correctly classified images divided by the total number of images) to evaluate the performance of the model. In addition, a T-test with a significance level of 0.05 was performed to assess significant differences between endoscopists and the model.

Results: The accuracy of DCNN1 in white light image and EUS image classification reached 100% [200/200, 95%CI]. The accuracy of DCNN2 in standard and non-standard image classification reached 81.24%. For the classification of standard station images, the overall accuracy of DCNN3 on the internal image verification set was 92.52% [2572/2743, 95%CI]. In the man-machine competition, the system correctly classified the liver standard stations with an accuracy of 88.53% [301/340, 95%CI]. The average accuracy of the two specialists and ten endoscopists was 84.56% [575/680, 95%CI] and 74.99% [2548/3398, 95%CI], respectively. The P-value for this analysis was less than 0.0001, indicating a statistically significant difference between endoscopists and the model.

Conclusion: In this study, we have constructed an intelligence-assisted linear EUS system based on deep learning algorithms. The system allows for real-time localization of liver standard sub-stations and provides valuable guidance to endoscopists during procedures. The system's effectiveness has been extensively validated through internal validations using images and. Moreover, in a competition with experienced endoscopists, our system has demonstrated superior accuracy, comparable to that of EUS specialists. The result shows that the system is expected to overcome the difficulties of ultrasonographic interpretation among trainees in liver ultrasound endoscopy.

Disclosure: Nothing to disclose.

PP1008

WITHDRAWN

PP1009

EXPRESSION OF DEIODINASES-3 IN HCC AS NEGATIVE PROGNOSTIC FACTOR

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Introduction: The complex signalling pathways of thyroid hormones (TH) has been recently involved in the liver diseases pathogenesis. Particularly, it has been proven that the expression of deiodinases type-1 and 3 (D1-3) changes during liver injury, but their role is still poorly understood in hepatocarcinogenesis process.

Aims & Methods: We aimed to evaluate the role of TH signalling pathway in liver carcinogenesis. In this prospective study, 49 patients underwent liver surgery for HCC in cirrhosis (31 cases) or for other non-neoplastic indication in non-cirrhotic context (18 controls). We evaluated genes and protein expression of the main TH metabolism factors (D1, Monocarboxylate transporter-MCT8, Thyroid receptors-TR alpha and beta, Kruppel-like factor-KLF-9 and TH-responsive SPOT-14 gene), with RT-PCR and Western blot analysis in HCC, cirrhotic tissue and healthy liver samples.

Results: RT-PCR analysis showed a progressive statistically significant decrease of D1 (p=0.003), MCT-8 (p=0.001), and TRβ (p=0.04) mRNA expression from healthy liver to HCC. The expression of KLF9, involved in cell differentiation and proliferation, decreased accordingly (p=0.01). Similarly, the expression of SPOT14 decreased in relation to reduced TH activation (p=0.009). Western Blot analysis showed a decreased expression of D1 protein in all cirrhotic samples (p=0.01), while D3 increased in 50% of HCC (p=0.02). Among HCC patients, D3 expression was associated with poor tumor differentiation compared to D3 negative HCC subjects (p=0.007). Furthermore, at Kaplan-Meyer analysis, overall survival of D3 positive HCC was significantly lower than D3 negative HCC (p=0.04).

Conclusion: These preliminary data showed that D3 expression could define a more severe phenotype of HCC with poor tumor differentiation and reduced survival. Moreover, the signalling pathway of TH is dysregulated in HCC, with intrahepatic hypothyroidism. However, these results need to be confirmed in large cohorts.

Disclosure: Nothing to disclose.

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PP1010

WITHDRAWN

PP1011

HEPATOCELLULAR CARCINOMA DETECTED IN HEPATOMA SURVEILLANCE CLINIC HAVE BETTER OUTCOME THAN THOSE DIAGNOSED SYMPTOMATICALLY: ABERDEEN ROYAL INFIRMARY EXPERIENCE

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Introduction: HCC is well known complication of liver cirrhosis. The current standard of care for patients with cirrhosis involves systematic 6 monthly screening with ultrasound and serum alpha-fetoprotein testing. (1) The Aberdeen Hepatoma surveillance clinic has been instituted since 2005 and currently 707 patients under screening program. It is a one-stop nurse led clinic wherein the ultrasound scan, blood testing and clinical review is done in a single visit, with sequential visits planned every 6 months.

Aims & Methods: In this study we aimed to identify outcome of HCC detected in hepatoma clinic compared to those identified symptomatically. Data collected retrospectively of patients referred to HCC MDT at Aberdeen Royal infirmary over period of 5 years. Diagnosis of HCC was made based on liver ultrasound and serum AFP as per guidelines

Results: 130 patients were identified with HCC through both pathways. 17 patients were detected as part of the surveillance programme and the rest were directly referred to the HPB MDT through the symptomatic pathway. The predominant subjects were males 102 (78.4%) and the cause of liver cirrhosis was as follows: MASLD 39%, ARLD 29%, Chronic HCV 11%, HBV 4%, Haemochromatosis 3%, PBC 2%, autoimmune hepatitis and PSC 1%. The mean age of patients detected in the surveillance group was 65 years ± 6.14 SD as opposed to those in the symptomatic group 71 years ± 9.99 SD. AFP was normal in 50% of surveillance group. The BCLC staging in the surveillance group was stage 0-A (20%), stage B 45%, In the symptomatic group patients were in stage 0-A 7%, Stage B 22%, and stage C, D 55% respectively.

Curative options including liver transplant and resection was possible in 20 % of surveillance patients as opposed to 7% in symptomatic patients. Palliative options like RFA, TACE and Sorafenib was offered in 60% of surveillance patients and 42% of symptomatic patients. In 20 % of surveillance patients no treatment could be offered as opposed to 55% symptomatic patients.

Conclusion: The mean age of the patients detected in the surveillance clinic was 5 years less than the symptomatic group suggesting that the programme was picking patients early. More curative and palliative procedures could be offered to this subgroup for this reason. The BCLC staging confirmed that most patients in the surveillance programme were in the early stages as opposed to the symptomatic group. Despite having an entrenched surveillance programme most patients were presenting symptomatically, suggesting that more efforts are needed to identify patients with cirrhosis who are at an added risk for developing HCC.

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Disclosure: Nothing to disclose.

PP1012

LIVER FUNCTION IS A PREDICTOR OF SURVIVAL IN PATIENTS WITH HEPATOCELLULAR CARCINOMA IN BEST SUPPORTIVE CARE

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Introduction: The prognosis of patients with hepatocellular carcinoma (HCC) is very variable, and the relative contribution of tumour burden and liver dysfunction to survival is uncertain. Median overall survival (OS) of patients managed with best supportive care is around 3-6 months, although longer values may be observed in clinical practice. Aim of this study was to identify the factors linked to tumour or liver dysfunction associated with survival in patients with HCC treated with BSC.

Aims & Methods: We retrospectively evaluated the clinical characteristics of 1414 patients who had an indication for BSC recorded in the Ita.Li.Ca database between 2000 and 2020. We analysed both patient and tumour characteristics to identify predictors of OS.

Results: Median age was 69y and 76% of patients were male. Aetiology included chronic viral infection (68.3%), alcohol use disorder (30.9%) and non-alcoholic steatohepatitis (8.3%). 67.4% of patients had a performance status 0-1 and 41.4% were in Child-Pugh B class. Median MELD was 13.60% of patients had a multifocal HCC with a median number of 3 lesions and a median size of 33mm. 533 patients had vascular invasion. Median alpha-fetoprotein was 49.25 ng/ml. 111 patients were classified as BCLC-A, 148 as BCLC-B, 791 as BCLC-C and 325 as BCLC-D (12 unknown). No differences in terms of OS were observed considering the aetiology of liver disease or the presence of cirrhosis. Obesity ($p<0.001$), hypercholesterolemia ($p=0.036$) and hypertriglyceridemia ($p=0.034$) were associated with lower OS. Absence of symptoms (6 vs 10 months, $p<0.001$), lack of vascular invasion (9.1 vs 5.03, $p<0.001$), and absence of metastasis (8.167 vs 4.733, $p<0.001$) were associated with a better OS. Survival in BCLC-A patients was longer than in stages B or C. Survival progressively declined according to severity of liver function using three different scores (CPS, ALBI, pALBI, $p<0.001$). Women tended to survive longer (23 vs 19 months, $p=0.053$). Comparing patients surviving more or less than 12 months (398 vs 1016), age, size of lesions, albumin, bilirubin, alpha-fetoprotein, and MELD were significantly different. At Cox univariate analysis presence of cirrhosis, number and size of lesions, vascular invasion, metastasis, ALBI and pALBI grades, MELD and CPS were significantly associated with OS. Using different models to avoid collinearity ALBI, pALBI and CPS maintained an independent prognostic role on OS.

Conclusion: In a large series of patients with HCC in BSC, parameters of liver function are strongly associated with survival.

Disclosure: Nothing to disclose.

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PP1013

RELIABLE PERFORMANCE OF MALBI GRADE-BASED RISK MODELS FOR PREDICTING THE PROGNOSIS OF PATIENTS WITH HEPATOCELLULAR CARCINOMA RECEIVING ATEZOLIZUMAB PLUS BEVACIZUMAB: COMPARATIVE ANALYSIS OF 13 RISK MODELS

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Introduction: This study aimed to compare the prognostic performance of the risk models for patients with hepatocellular carcinoma (HCC) receiving atezolizumab and bevacizumab (Atez/Bev) as first-line treatment.

Aims & Methods: A total of 449 patients were included in this retrospective multicenter study. We compared the prognostic performance of 13 risk models for the 12- and 18-month survival status using area under the curve (AUC), net reclassification improvement (NRI), and integrated discrimination improvement (IDI) analysis. We also constructed a calibration plot to assess the fitness of each model.

Results: Regarding the analysis of the 12-month survival status, none of the risk models demonstrated AUC values higher than the modified albumin-bilirubin (mALBI) grade. In the NRI analysis, only the IMmunotherapy with AFP, BCLC staging, mALBI, and DCP evaluation (IMABALI-De score) exhibited a statistically significant improvement compared with the mALBI grade (p=0.009).

While the modified albumin-bilirubin grade (mALF) score and prognostic nutritional index (PNI) did not exhibit significant differences compared to the mALBI grade (p=0.3 and 0.2, respectively), the remaining risk models were inferior to the mALBI grade. In the IDI analysis, none of the risk models showed a significant improvement compared with the mALBI grade. The calibration plot of the PNI was unsatisfactory. The results for the 18-month survival status were consistent with those for the 12-month survival status.

Conclusion: The mALBI grade, as well as the IMABALI-De and mALF scores (both of which are risk models based on mALBI grade), exhibited reliable performance in predicting the prognosis of HCC patients.

Disclosure: Takeshi Hatanaka received lecture fees from Eisai. Atsushi Hiraoka received lecture fees from Eli Lilly, AstraZeneca, and Chugai. Toshifumi Tada received lecture fees from AbbVie, Eisai, and Chugai. Hidenori Toyoda received lecture fees from Eisai, Chugai, Takeda, Terumo, AbbVie, Gilead, Fujifilm WAKO, and Abott. Hidekatsu Kuroda received lecture fee from Eisai. The other authors declare no conflicts of interest associated with this study.

PP1014

HEPATOCELLULAR CARCINOMA AND IMMUNOTHERAPY – OUR THREE YEARS EXPERIENCE

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Introduction: Hepatocellular carcinoma (HCC) is the most common primary malignant liver cancer globally and ranks as the third leading cause of cancer-related mortality. Treatment with immune checkpoint inhibitors (most commonly atezolizumab in combination with bevacizumab) is the most effective first-line systemic therapy for patients with advanced HCC. Since it is a new method of treatment, there is little clinical data on the use of this combination of drugs in practice (outside of prospective studies).

This study aims to present the clinical outcomes of patients with HCC who underwent immunotherapy at our center over the past three years.

Aims & Methods: In this retrospective observational study, we included patients treated with the atezolizumab and bevacizumab combination between March 2020 and September 2023. Demographic information, general physical assesments, disease etiology, disease staging, liver cirrhosis status, portal hypertension and laboratory values (aminotransferases and total bilirubin) were collected. Adverse events were analyzed according to the Common Terminology Criteria (CTCAE). Descriptive statistics and survival analysis were performed using Statistical Package for the Social Sciences (SPSS) version 22.

Results: 53 HCC patients received atezolizumab and bevacizumab in studied time period, 8 (15.1%) females and 45 (84.9%) males. The median age at the initiation of immunotherapy was 68 years (range 41 – 84 years). Thirteen patients (24.5%) had esophageal varices prior to immunotherapy, with three (5.6%) requiring ligation. The majority of patients had liver cirrhosis A 27 (50.9%) according to Child Pugh criteria, a minority B (9.4%), and 21 patients (39.6%) were without cirrhosis.

The primary causes of the disease were alcohol (51.6%), metabolic dysfunction-associated steatotic liver disease (MASLD) (16.1%), hepatitis C virus (12.9%), hepatitis B virus (6.5%), and hemochromatosis (3.2%). Stage C HCC, according to The Barcelona Clinic Liver Cancer, World Health Organization (WHO) performance status 0, and absence of macrovascular invasion and extrahepatic spread were predominant at treatment initiation. Grade I or II adverse events occurred in almost all patients (98.1%).

Severe adverse events were present in 36 patients (67.9%). Due to severe side effects, 8 (15.1%) patients discontinued treatment, 2 (3.8%) died due to treatment related complications (encephalitis and bleeding from esophageal varices). Treatment was modified in 21 (39.6%) patients due to adverse events. The estimated median survival was 15.7 months.

Conclusion: The majority of patients with unresectable HCC, treated with a combination of atezolizumab and bevacizumab immunotherapy, experienced adverse events, primarily grades I-II. The immunotherapy combination proved to be more effective in patients with viral etiology of liver cirrhosis. The estimated median survival was 15.7 months. This combination of immunotherapy has been demonstrated to be a safe and effective treatment method. Currently, there is limited data regarding the utiliza-

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tion of this treatment in standard clinical practice and our data aligns reasonably well with the findings of the registration study.

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Disclosure: Nothing to disclose.

PP1015

EFFECT OF NEUTROPHIL-TO-LYMPHOCYTE RATIO IN TREATMENT WITH ATEZOLIZUMAB-BEVACIZUMAB IN CIRRHOTIC AND NON-CIRRHOTIC PATIENTS WITH HEPATOCELLULAR CARCINOMA

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Introduction: The majority of hepatocellular carcinoma (HCC) patients has cirrhosis. The increased incidence of HCC in non-alcoholic steatohepatitis (NASH) and on the other hand the decrease of incidence of viral hepatitis, have as a result increased incidence of HCC in non-cirrhotic patients.

There are limited data for response differentiation in cirrhotic and non-cirrhotic patients regarding treatment with atezolizumab-bevacizumab[1]. Elevated neutrophil-to-lymphocyte ratio (NLR) seems to be in favor of worst prognostic for immunotherapy.

Aims & Methods: The aim of this study was to determinate NLR in HCC for cirrhotic and non-cirrhotic patients as a biomarker for effective treatment. 57 patients with Barcelona Clinic Liver Cancer(BCLC) B/C HCC that received treatment with combination atezolizumab-bevacizumab in our referral center through the last three years were included.45/57 were men, mean age was 66.3 years old and mean body mass index (BMI) was 27.23, 33 viral related HCC, 19 with diabetes,29 ALBI-I,41 BCLC-C,16 BCLC-B,22 macrovascular infiltration (MVI),23 extrahepatic disease, 31 had received prior therapy.

Patients with NLR>3 categorized as NLR-H patients and the rest of them as NLR-L. Patients divided into 4 groups, depending on the presence of cirrhosis or not and NLR value, as follows: Group A (non-cirrhotic, NLR-L/N=13), group B (non-cirrhotic, NLR-H/N=14), group C (cirrhotic, NLR-L/N=11), group D (cirrhotic, NLR-H/N=19). The 4 groups were compared for their baseline characteristics and survival (OS and PFS) and multivariate analysis were performed. Next, cirrhotic and non-cirrhotic patients were examined separately and compared for baseline characteristics and survival based on baseline NLR values and receipt of prior therapy.

Results: The groups were comparable in baseline characteristics except for liver disease etiology (p<0.001), ALBI score (p=0.014), MVI(p<0.001), BCLC(p=0.014) and prior treatment (p=0.009). Median OS was 30 months(m),10m,12m,5m and median PFS was 14m,4m,8m,2m for groups A,B,C,D respectively (p<0.001). In multivariate analysis, a statistically significant difference regarding worst survival was found in group D patients compared to group A (p=0.005, HR=0.1), group C (p=0.032, HR=0.333) and marginally group B (p=0.055, HR=0.321). In analysis of cirrhotic patients alone based on NLR, patients were comparable for all baseline characteristics. Cirrhotic patients with NLR-L had better OS(12m vs.5m,p=0.002) and PFS(8m vs. 2m, p=0.028) compared to cirrhotic patients with NLR-H. NLR was an independent determinant of OS (p=0.015,HR=3.480). In analysis of non-cirrhotic patients based on NLR, patients were comparable except for prior treatment. Patients with NLR-L had significantly longer OS (30m vs 10m,p=0.006) and PFS (15m vs 4m,p=0.01). In multivariate analysis,only the presence of prior treatment and not NLR influenced survival. Patients were re-divided according to the presence or absence of prior therapy into 2 groups. Previously treated non-cirrhotic patients had better OS (30m vs 8m, p<0.001) and PFS (24m vs 4m, p<0.001). In the multivariate analysis, the presence of prior treatment was an independent factor determining survival in the non-cirrhotic patients.

Conclusion: Cirrhotic patients with high baseline NLR values show the worst survival among the subgroups. Baseline NLR is an independent determinant of OS in cirrhotic patients, but not PFS, while in non-cirrhotic patients receiving prior therapy is the only factor that independently determines both OS and PFS. The results need to be verified in larger patient cohorts.

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Disclosure: Nothing to disclose.

PP1016

ENDOSCOPIC ULTRASOUND-GUIDED LIVER BIOPSY USING A 22-G FINE-NEEDLE BIOPSY NEEDLE

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Introduction: Liver biopsy is a crucial diagnostic procedure for assessing hepatic pathologies, providing valuable insights into the etiology, severity, and prognosis of liver diseases. In recent years, Endoscopic Ultrasound-Guided Liver Biopsy (EUS-LB) using 22G- fine needle biopsy (FNB) needle has emerged as a promising alternative, leveraging the superior imaging capabilities of endoscopic ultrasound (EUS) to guide biopsy with increased precision and safety. The 22G- FNB needle features a crown tip with three symmetrical cutting edges that enable swift core tissue procurement for histological analysis. The 22G- FNB needle provide greater than 95% tissue adequacy and an excellent diagnostic accuracy.

Aims & Methods: The aim of our study is to evaluate the diagnostic utility and safety of a 22G-FNB needle using the slow -pull technique for EUS-LB of parenchymal and focal liver lesions.

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We performed a retrospective analysis of prospective data basefrom January 2021 to March 2024. Each patient underwent anendoscopic ultrasound by using linear scope (Fujifilm EG-580UT) and fine needle biopsy using 22G - tip needle with the slow-pull aspiration technique:

- For 33 patients we used a 22G Trident™ - tip FNB needle (Micro-Tech Endoscopy);
- For 26 patients we used a 22G Acquire- tip FNB needle (Boston Scientific Corp.);
- For 4 patients we used a 22G SonoTip Top Gain - tip FNB needle (MediGlobe);

The median number of needle passes was approximately 3 (between2 and 5).

Results: We identified 63 patients who underwent EUS-LB of the liver during the study period(28 men and 35 women): 47 (74.6%)of them were with malignant and 16 (25.4%)–with benign disease. The medium age of the patients was 56.9 years. The specimen was considered adequate in all 63 patients.

We classified the EUS-FNB of liver diseases into EUS-FNB of parenchymal and focal liver lesions.

- EUS-FNB of parenchymal liver disease were performed to 13 (20.6%) of the patients: cirrhosis was found in 2 (3.17%), nonalcoholic fatty liver disease – in 5 (7.9%) and cholestasis in 6 (9.5%).
- We performed EUS-FNB of focal liver lesions in 50 (79.36%) of the patients.

Metastasis were found in 39 (61,9%): pancreatic ductal adenocarcinoma was detected in 27 (42.8%), pancreatic acinar cell carcinoma – in 1 (1.58%), pancreatic neuroendocrine carcinoma – in 5 (7.9%), lung adenocarcinoma – in 3 (4.7%), ductal adenocarcinoma of breast – in 2 (3.17%) , colorectal adenocarcinoma – in 1 (1.58%) and Ewing sarcoma – in 1 (1.58%).

Primary liver cancer - hepatocellular cancer was diagnosed in 4 (6.34%) patients.

Cholangiocarcinoma was found in 3 (4.7%) patients.

There were 3 (4.7%) patients with negative histology for tumor cells in the specimen: in 2 (3.17%) cases liver abscess was proved and 1 (1.58%) hepatic cyst.

No adverse events were observed during the first 48h and after a 30 days follow-up.

There was no death during the study period.

Conclusion: Our study illustrates that EUS-guided liver biopsy using a 22 G- FNB needle provides adequate specimen for histologic analysis, lesser postprocedural recovery time and it is a safe, highly effective alternative to other methods of liver biopsy. The use of real time imaging guidance with doppler also helps to reduce the adverse events.

Disclosure: Nothing to disclose.

PP1017

CLINICAL CHARACTERISTICS AND PROGNOSIS OF DE NOVO HEPATOCELLULAR CARCINOMA AFTER TIPS PLACEMENT: A RETROSPECTIVE COHORT STUDY

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Introduction: Portal hypertension often accompanies hepatocellular carcinoma (HCC) in cirrhosis. Transjugular intrahepatic portosystemic shunt (TIPS) serves as an effective intervention for portal hypertension, while HCC can exacerbates it.

Aims & Methods: To investigate the prognosis of post-TIPS HCC patients and factors influencing it. Retrospective analysis of post-TIPS HCC patients, assessing overall survival (OS), progression-free survival (PFS), and objective response rate (ORR) using Kaplan-Meier, logistic regression, and Cox model.

Results: A total of 68 patients were included (April 2015 to October 2023). Forty-seven percent died during follow-up, with liver decompensation as a major cause (27.9% vs 10.3%, P=0.18). Median OS was 33 months, median PFS was 24 months, and ORR was 80%. The 2-year OS rates for BCLC stage 0, A, B, and C were 85.6%, 54.6%, 25%, and 0%, respectively. Tumor number (HR 1.58, P=0.01) and moderate to severe ascites (HR 5.16, P=0.003; HR 5.52, P=0.01) were OS risk factors, while mild to moderate ascites affected PFS (HR 3.19, P=0.021; HR 6.20, P=0.002). Age (HR 0.91, P=0.031) and ascites severity (HR 0.06, P=0.022; HR 0.025, P=0.022) influenced ORR.

Conclusion: Liver decompensation is a primary mortality cause in post-TIPS HCC. Ascites presence affects outcomes, underscoring post-TIPS ascites management importance.

Disclosure: Nothing to disclose.

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PP1018

TREATMENT OF HEPATOCELLULAR CARCINOMA ACCORDING TO MULTIPARAMETRIC THERAPEUTIC HIERARCHY APPROACH: A PROSPECTIVE MULTICENTER VALIDATION STUDY

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Introduction: Treatment strategy of hepatocellular carcinoma (HCC) is a complex decision-making process that considers several variables in clinical practice. A "multiparametric therapeutic approach" has been proposed (*Lancet Oncol* 2023): the feasibility of all treatments is evaluated systematically in a hierarchical order, to match each patient with the optimal therapy.

Aims & Methods: The aim of the study is to validate the reliability of the new multiparametric decisional framework (MPDF) and the impact of the different variables included in the framework on the final treatment decision in clinical practice. This is a prospective ongoing study with a first monocentric phase (A) and a second multicentric validation phase. Primary endpoints are: 1) adherence to MPDF for treatment allocation, 2) identification of the relative weight of variables involved in this MPDF for treatment decision, while secondary endpoints are overall survival and radiological response after treatment. The study includes all consecutive patients with HCC managed in each center from September 1st, 2023. The disease is staged according to Barcelona Clinic Liver Cancer (BCLC) classification, while patient's characterization is conducted by Charlson Comorbidity Index (CCI), Liver Frailty Index (LFI), Child-Pugh (CPT) and ALBI, MELD and skeletal muscle mass index by CT. Each patient is allocated to treatment according to MPDF, BCLC 2022 was the benchmark.

Results: As of November 1st, 2023, 48 patients were recruited into phase A of the study [median age 70 (52-87) years, 81% male, 37% HCV, 27% first treatment for HCC]. Thirteen (27%) patients were classified BCLC 0, 17 (35%) A, 6 (12.5%) B, 6 (12.5%) C and 6 (12.5%) D. The median CCI was 8 (4-13), the median LFI was 4.15 (2.99-4.81). Thirty-nine (81%) patients were in CPT class A, 7 (15%) in B and 2 (4%) in C, while 32 (67%) were classified ALBI grade 1 and 16 (33%) ALBI grade 2, and the median MELD score was 8.5 (6-22). Forty-four (92%) patients received treatment through a multidisciplinary team (MDT). According to the MPDF, 9 (19%) patients were allocated to liver transplant, 5 (10%) to laparoscopic surgical resection, 10 (21%) to microwave thermal ablation (MWTA), 7 (15%) to transarterial treatments (4 TACE and 3 TARE), 3 (6%) to other locoregional therapies (2 MWTA+TACE and 1 electrochemotherapy), 6 (13%) to systemic treatment and 5 (10%) to palliative care only. In 4 cases, the MDT decided for reassessment at 3 months. The treatment allocation was coherent to the BCLC 2022 in 69% BCLC 0, 82% BCLC A, 83% BCLC B, 67% BCLC C and 100% BCLC D.

Conclusion: Preliminary data of our ongoing prospective study suggest that personalized approach for HCC treatment by MPDF is feasible in clinical practice and might help ensuring a customized approach in a standardized framework. Updated data will be presented at the meeting.

Disclosure: Nothing to disclose.

PP1019

THE IMPACT OF METFORMIN ON THE OVERALL SURVIVAL OF DIABETIC PATIENTS WITH ADVANCED HEPATOCELLULAR CARCINOMA ON IMMUNOTHERAPY: A META-ANALYSIS

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Introduction: Non-alcoholic steatohepatitis (NASH) is a common known risk factor for developing hepatocellular carcinoma (HCC). Many patients with NASH are given metformin, a widely prescribed medication for type 2 diabetes and obesity. Its primary mode of action involves the activation of AMP-activated protein kinase (AMPK), a central regulator of cellular energy homeostasis. Beyond its anti-hyperglycemic effects, it also exhibits pleiotropic properties that extend to modulating various cellular processes, including inflammation, metabolism, and immune response. In this similar mechanism, growing evidence indicates that metformin also has anti-cancer properties. It has shown significant synergistic effects in combination with standard chemotherapy and immune check-point inhibitors (ICI). However, its effect on advanced HCC treated with ICI (e.g. atezolizumab + bevacizumab, pembrolizumab, durvalumab) is not well understood.

Aims & Methods: This meta-analysis aims to determine the effect of metformin in the overall survival of diabetic patients with advanced HCC treated with ICIs. Major electronic databases and grey literature sources were searched up to January 2023 for studies assessing the effect of metformin in the survival of diabetic HCC patients on immunotherapy.

Results: The hazard ratios for the overall survival of metformin (experimental) vs non-metformin (control) groups were compared from the studies done by Kang 2022, Lin 2022 and Ramini 2023. Resulting I2 of 0% (p value = 0.54) implies that heterogeneity does not exist. Resulting pooled hazard ratio of 1.08, with a narrow 95% confidence interval 0.85 to 1.38, is not significant (Z = 0.65, p = 0.51), implying no significant difference between the two groups. The forest plot also shows that the diamond market intersects the 1 axis or the line of no effect, suggesting that the hazard ratio is not significant. However, eyeballing the said forest plot, all 3 studies tend toward the non-metformin or control group which signifies a worse prognosis with metformin on HCC patients on ICIs.

Conclusion: The use of metformin has no significant difference in the overall survival in diabetic HCC patients receiving immunotherapy, although trends are leading towards worse prognosis. Hence, physicians must take into consideration the diabetes status of HCC patients on immunotherapy due to its impact on their clinical outcomes and negative prognostic effect of metformin use.

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Disclosure: No conflict of interest.

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PP1020

A MODIFIED CHARLSON COMORBIDITY INDEX TO IMPROVE MANAGEMENT OF PATIENTS WITH HEPATOCELLULAR CARCINOMA: A STEP TOWARDS PRECISION MEDICINE

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Introduction: Patients with hepatocellular carcinoma (HCC) frequently have comorbidities that limit access to treatments and might increase mortality. The Charlson comorbidity index (CCI) assigns a numeric score to 17 diseases according to their effect on mortality. In patients with cirrhosis and HCC, these variables might be excluded from the CCI (modified CCI, mCCI).

Aims & Methods: Aim of the study was to evaluate the performance of CCI and mCCI for HCC patients' characterization.

We used data from our retrospective monocentric study of patients with first HCC diagnosis. We evaluated the performance of CCI and mCCI in dissecting our population and predicting overall survival (OS). As secondary end-point, the impact of mCCI on adherence to Barcelona Clinic Liver Cancer (BCLC) staging and treatment allocation was evaluated.

Results: The study included 385 patients (289 males, 68 years median age, 78% Child Pugh (CPT) A, 66% viral etiology) with HCC (271 BCLC 0/A, 64 B, 40 C and 10 D). According to CCI, 94% of our patients were "high-risk" (CCI≥5), 5.7% "intermediate-risk" (CCI 3-4) and 0.3% "low-risk" (CCI 0-2). These cutoffs were applied to mCCI: 21% "high-risk", 48% "intermediate-risk", and 31% "low-risk". Patients' mCCI correlated with CCI (Kendall's τ=0.47; p-value< 0.0001). Overall, patients belonging to a "high" or "intermediate-risk" mCCI classes had a shorter OS as compared to "low-risk" patients (median OS 37 vs 49 vs 67 months, p-value=0.007). This was confirmed in BCLC 0/A (p-value=0.002). Moreover, "high-risk" mCCI class, worse CPT class, presence of varices, BCLC stage and AFP>200 ng/mL were independently associated to mortality [HR1.36 (95%CI 1.09-1.70, p-value=0.006; HR1.62 (95%CI 1.14-2.30), p-value=0.007; HR1.58 (95%CI 1.11-2.24), p-value=0.01; HR1.96 (95%CI 1.66-2.31) p-value<0.001; HR1.80 (1.17-2.79) p-value=0.008]. mCCI was also an independent predictor of mortality in BCLC 0/A, with higher tumor burden and worse CPT. No differences were observed in adherence to BCLC2022 algorithm and access to treatment among the mCCI risk classes.

Conclusion: mCCI better stratified patients with HCC as compared to CCI and independently predicted mortality. Implementation of a multiparametric therapeutic approach to treatment allocation including evaluation of comorbidities and frailty is warranted

Disclosure: Nothing to disclose.

PP1021

EVALUATION OF RISK FACTORS AND PROGNOSIS OF HEPATOCELLULAR CARCINOMA RECURRENCE AFTER LIVER TRANSPLANTATION

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Introduction:

Transplantation techniques and immunosuppression treatments have been increasing notably in the past years, particularly when considering liver transplant (LT). LT is the preferred treatment for early-stage hepatocellular carcinoma (HCC). Despite the use of the Milan criteria, recurrence remains a major concern, with an estimated incidence ranging from 5% to 20%. Therefore, our objective is to assess potential risk factors, including clinical, laboratory and pathological parameters, as well as prognosis of HCC recurrence in LT recipients.

Aims & Methods:

Retrospective evaluation of medical records from a Brazilian tertiary hospital, including adults who were submitted to LT due to HCC, during the period of 01/01/2002 up to 31/12/2021. Patients who had a LT for other causes, who presented a follow-up less than 1 year or with incidental detection of HCC in the liver explants were excluded. Data collection included demographic information, immunosuppressive treatments, clinical, radiological and histhopathological findings related to the HCC, as well as characteristics associated with the post-transplant follow-up and tumoral recurrence, including death. Primary outcome was recurrence of HCC, whereas secondary outcomes were survival and adopted treatments after cancer recurrence, as well as time elapsed until HCC recrudescence.

Results: 150 patients were included (males 95 [63.3%], median age 60 years-old [IQR 56-66]), 130 (86.6%) were classified as Milan-in. The main cirrhosis etiologies were viral (88%) and metALD (10%). Half of patients presented an A CHILD score (50%) and a median MELDNa score of 12 (IQR 9-14). Main locoregional therapies included transarterial chemoembolization (TACE) and percutaneous ethanol injection (PEI), accounting for 48 (32%) and 45 (30%) respectively. 47 (31.33%) patients underwent two or more locoregional therapy modalities, and downstaging was achieved in 20 patients (13.3%). Explant analysis with microvascular invasion and Milan-out were detected in 18 (12%) and 30 (20.1%) individuals, respectively. HCC recurrence occurred in 21/150 patients (14%), while non-HCC neoplasias in 27/150 (18.1%) after LT. The median time to recurrence was 30 months (IQR 20-43), and systemic therapy to HCC with sorafenib was used in 12/21 patients (57.14%). Among those who presented recurrence, 19/21 died (90.4%), and prognosis after recurrence was dismal (median survival of 14.5 months, IQR 3.25-33.5). Evaluation of possible associations between HCC recurrence and clinical, laboratorial, imaging and histopathological parameters was performed, but none showed impact on HCC recurrence after LT. However, there was a trend to an association between recurrence and CHILD score (p = 0.06) and smoking (p = 0.08).

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Also, when evaluating patients who presented early recurrence (less than 24 months after LT), a trend of an association with the number of involved liver segments was found (p = 0.07).

Conclusion: Identifying patients at risk of HCC recurrence after LT remains as a great challenge. There are some parameters previously suggested as possible predictors, but, in accordance to literature findings, our study didn't identify significant associations.

Nonetheless, the possible association between early recurrence and the number of affected liver segments hasn't been studied yet, and this is the first study that suggests an association.

Thus, evaluating the number of liver segments affected by HCC might help physicians to identify patients with higher risk of recurrence, as well as giving another possible parameter to investigate in further studies.

Disclosure: Nothing to disclose.

PP1022

HEPATOCYTE NUCLEAR FACTOR 1 ALPHA AS RISK FACTOR FOR HEPATOCELLULAR CARCINOMA DEVELOPMENT AMONG EGYPTIAN PATIENTS WITH AND WITHOUT DIABETES MELLITUS

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Introduction: HNF1A gene variants have been reported to be involved in developing mature onset diabetes mellitus (DM). Many studies reported the role of DM as a risk factor for hepatocellular carcinoma (HCC) development. To date, it has not been reported whether HNF1A gene variants are associated with the risk of DM in cirrhotic patients and their subsequent HCC.

Aims & Methods: Aim: To evaluate the HNF1A (the hepatocyte nuclear factor 1 homeobox A) genetic variants as a cofactor with DM for HCC development in HCV-infected patients.

Methods: This study was conducted on 140 subjects; 30 had HCC without DM, 30 HCC with DM, and 40 patients had DM with no HCV infection or had HCC; in addition, 80 healthy volunteers with matched ages and genders were enrolled in the study as a control group. Liver function tests, hepatitis viral markers, alpha-fetoprotein (AFP), fasting sugar and HBA1c and HNF1A (rs2464196 and rs1169310) using real-time polymerase chain reaction (PCR) were done for all participants.

	DM (n = 40)		HCC (n = 30)		HCC+DM (n = 30)		Control (n = 40)		χ^2	P
	No.	%	No.	%	No.	%	No.	%		
HNF SNP2 (rs1169310)										
CC	10	25.0	10	33.3	6	20.0	7	17.5		
CT	13	32.5	10	33.3	13	43.3	17	42.5	3.389	0.759
TT	17	42.5	10	33.3	11	36.7	16	40.0		
$^{HW}\chi^2$ (p)	0.037*		0.068		0.552		0.508			
Allele										
C	33	41.3	30	50.0	25	41.7	31	38.8	1.910	0.591
	47	58.8	30	50.0	35	58.3	49	61.3		

Table 1: Comparison between the studied groups according to HNF SNP2 (rs1169310) genotypes and allele frequencies.

Results: The frequency of HNF1A rs2464196 (AA) genotype in patient groups (DM, HCC, HCC+DM) was significantly higher compared to the control group (P=0.006, P=0.018, P<0.001respectively). The combined dominant model (AA + GA) of rs 2464196 was significantly higher than the (GG) genotype in patient groups (DM, HCC, HCC+DM) than the control group. In addition, the frequency of the AA genotype is more prevalent in HCC+DM (73%) compared to the group of DM or HCC patients.

In contrast, the HNF1A rs 1169310 (TT, TC or CC genotypes) showed no significant difference among the four studied groups and their T or C allele distributions.

Conclusion: This finding suggested that the HNF1A rs2464196 (AA) genotype could be associated with DM and may raise the possibility of HCC development among HCV-infected patients who harbour this genotype more than (GG).

On the contrary, the HNF1A rs 1169310 polymorphism was of no significance as a risk factor in the current study. Large scale studies are recommended on other variants of HNF1A to clarify the role of this gene in HCC development.

Disclosure: Nothing to disclose.

PP1023

ACCURACY OF RADIOMICS IN PREDICTING MACROTRABECULAR HEPATOCELLULAR CARCINOMA: A SYSTEMATIC REVIEW AND DIAGNOSTIC META-ANALYSIS

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Introduction: Hepatocellular carcinoma (HCC) is the second-most deadly cancer. Pathological examination is the gold standard for subtype differentiation. Patients with macrotrabecular HCC (MTHCC) have more aggressive lesions, higher recurrence rates and poorer survival. Radiomics can be seen as a digital biopsy that allows a characterization of tumor heterogeneity and phenotypes.

Aims & Methods: This study aims to evaluate the diagnostic accuracy of radiomics in predicting MTHCC subtype, considering that this could change the approach of the patient.

We performed a systematic review in PubMed, Embase and Cochrane databases, searching for studies evaluating the diagnostic accuracy of radiomics in predicting the MTHCC subtype. Statistical analysis included data pooling and heterogeneity testing.

Results: The meta-analysis included 6 studies and 529 patients. Data of the training cohort were not considered.

When internal and external validation cohorts were available, both were considered. The pooled sensitivity and specificity and AUC of radiomics for predicting MTHCC were 75.3% (95% CI 66.4% to 82.5%), 71.8% (95% CI 60.6% to 80.8%) and 0.792 (95% CI 0.697 to 0.833).

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Conclusion: Our study shows that radiomics does not have enough diagnostic accuracy to predict MTHCC with a high degree of certainty. However, this could be an additional resource to improve radiological diagnosis. Improvements in this field are necessary once the diagnosis of this HCC subtype can change the management of the patient.
Disclosure: Nothing to disclose.

PP1024

ANNEXINA2 AS A MARKER OF HEPATOCELLULAR CARCINOMA DETECTION AMONG PATIENTS WITH HEPATITIS C VIRUS-RELATED LIVER CIRRHOSIS AFTER HCV TREATMENT USING DIRECTLY ACTING ANTIVIRALS

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Introduction: Hepatocellular carcinoma (HCC) is considered the sixth most prevalent malignant tumor in the world, and it ranks first in Egypt. Early detection of HCC is of great value, as early stages can be treated with curative modalities. Current surveillance programs depend on ultrasonography with or without alpha-fetoprotein (AFP) measurement, but the ultrasonography is operator-dependent and has many limitations, while AFP is neither adequately sensitive nor specific. Serum Annexin A2 has the potential to be a promising biomarker for the early detection of HCC.
Aims & Methods: The aim of the present study is to evaluate the potential contribution of serum annexin A2 in the detection of hepatocellular carcinoma. This case-control study included four groups:
Group A: Early hepatocellular carcinoma; BCLC stages 0 and A.
Group B: Advanced hepatocellular carcinoma; BCLC Stage C.
Group C: liver cirrhosis without HCC.
Group D: healthy subjects as a control group.
Chronic hepatitis C was the cause of cirrhosis in the three groups (A, B, and C). All of them attained sustained virological response (SVR) after receiving treatment with direct-acting antiviral medications (DAAs). Subjects with other liver cirrhosis causes, those with active hepatitis C viremia, those with combined hepatitis B virus infection, and those with any other malignancy were excluded.
Assuming a common standard deviation (14.6) using the F test (one-way ANOVA power analysis), at significance level.05, a sample size of 100 participants (25 per group) was needed to achieve 80% power to detect an overall difference of at least .35 ng/mL in the mean of serum annexin A2 between the studied 4 groups.
Both serum AFP and annexin A2 were measured in all included subjects.
Results: One hundred subjects were included in the study, 25 in each group. There were no statistically significant differences between groups regarding age and sex. The 3 diseased groups were not statistically significantly different regarding the modified Child-Pugh scores (median was 5), ALBI grading, MELD scores, platelet counts, INR, albumin, and total serum bilirubin levels.
By ROC curve analysis, at a cut-off point >12 (ng/ml), AFP was a statistically significant discriminator between HCC patients (group A+B) and patients with liver cirrhosis (group C), with an area under the ROC curve (AUC) = 0.670 (95% CI 0.550 to 0.789) (*p =.017), sensitivity* of 60.00%, specificity of 76.00%, positive predictive value (PPV) of 83.30%, and negative predictive value (NPV) of 48.70%, but at the same cut-off point, AFP is not a statistically significant discriminator between patients with early HCC (group A) and patients with liver cirrhosis (group C) with an AUC of 0.619 (95% CI 0.450 to 0.783) (*p =.148*), with a sensitivity of 52.00%, a specificity of 76.00%, a PPV of 68.40%, and a NPV of 61.30%.
Serum annexin A2 levels showed a significant difference between the advanced HCC and the early HCC groups compared to the group of healthy individuals (group D) (*p*<.001). Annexin A2 levels were also significantly higher between the early HCC and advanced HCC groups when compared to the cirrhosis-only group (p =.005 *and p*<.001, respectively).
Moreover, annexinA2 levels were significantly higher in the advanced HCC group when compared to the early HCC group (*p =.003*). Annexin A2 was also higher in the cirrhosis-only group than in the healthy group (*p =.002*).
By ROC curve analysis, at a cut-off point >7.7 (ng/ml), serum annexin A2 was a statistically excellent discriminator between HCC patients (group A+B) and patients with liver cirrhosis (group C) with an AUC of 0.962 (95% CI 0.910 to 1.000) (*p*<0001), with a sensitivity of 98.00%, a specificity of 88.00%, a PPV of 94.2%, and a NPV of 95.7%.
At the same cut-off point, annexinA2 is a statistically significant discriminator between patients with early HCC (group A) and those with liver cirrhosis (group C), with an AUC of 0.934 (95% CI 0.851 to 1.000) (*p >.001*), a sensitivity of 96.00%, a specificity of 88.00%, a PPV of 88.90%, and a NPV of 95.7%.
Conclusion: Serum annexin A2 is a promising biomarker that can improve the sensitivity of HCC detection among patients with liver cirrhosis after HCV eradication using DAAs.
Disclosure: Nothing to disclose.

PP1025

PRESENTING COMPLAINTS IN 250 PATIENTS WITH CONGENITAL PORTOSYSTEMIC SHUNTS

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Introduction: Congenital portosystemic shunts (CPSS) are associated with an array of clinical signs and symptoms of varying severity in both adults and children. Yet, symptoms are often missed and ascribed to other etiologies. Understanding presenting complaints will increase awareness of this rare disorder.
To this end the International Registry of Congenital Portosystemic shunts (IRCPSS) was created. Here we report an up-to-date description of the patients enrolled in the IRPCSS focusing on presenting complaints.
Aims & Methods: Descriptive analysis of the ESPGHAN sponsored IRCPSS described in (1)

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Results: 256 patients were enrolled - 125 females (49%). Type of CPSS was available for 212: 88 (41.5%) presented with intrahepatic (IHS) and 124 (58.5%) with extrahepatic (EHS) shunts. Clinical data were available for 71 with IHS and 106 with EHS. 51 (72 %) of IHS were asymptomatic. Of the 20 (28%) symptomatic, median age at diagnosis was 0.59 years (14 days-19.9 years). Presenting complaint included liver in 14, cardiopulmonary in 6, neurological in 2, other in 7. Of 56 IHS closed, 21 (37.5%) closed spontaneously before age 1. 29 (52%) were closed using interventional radiology. Among 106 patients with EHS, 72 (68%) were asymptomatic. Among symptomatic patients, median age at diagnosis was 7.49 years (14 days-24.9 years). Presenting complaints included liver in 15, cardiopulmonary in 13, neurological in 13, and other in 17. 74 patients had closed shunts at last follow-up; they were closed at a median age of 2.7 years (18 days- 25.4 years). Spontaneous closure was reported in 2 patients, otherwise closure was performed by interventional radiology in 25 (34%), or surgery in 40 (54%) and 7 undergoing LT.

Conclusion: Most shunts were asymptomatic at diagnosis. Among symptomatic patients, the median age at diagnosis was higher in EHS than IHS. IHS and EHS were equally likely to be closed, with IHS more likely to close spontaneously within the first year of life.

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Disclosure: Nothing to disclose.

PP1026

SPLEEN ELASTOGRAPHY IN NON-CIRRHOTIC CHRONIC EXTRA-HEPATIC PORTAL VEIN THROMBOSIS PATIENTS ALLOWS FOR PERSONALIZED SCREENING FOR ESOPHAGEAL AND GASTRIC VARICES

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Introduction: In patients with cirrhosis, a transient elastography value of spleen stiffness (SS) <40 kPa can be used to identify those with low probability of high-risk varices and in whom endoscopy may be avoided. We aimed to evaluate if SS <40KPa may be useful to select patients with non-cirrhotic chronic extrahepatic portal vein thrombosis (CEPVT) in whom upper gastrointestinal endoscopy (UGE) to screen for high risk esophageal varices (EV) or gastric varices(GV) may be avoided.

Aims & Methods: We performed a retrospective study including consecutive patients with non-cirrhotic CEPVT, with data from SS and upper gastrointestinal endoscopy (UGE) within a 2 years interval. We evaluated 42 patients and finally excluded 20 (No SS measurement -10, previous splenectomy - 7 and no UGE in the previous 2 years). We compared two groups of patients: Group A - Those with high risk EV/GV or those who previously underwent variceal endoscopic band ligation (EBL) and group B – patients without EV/small EV or GV. The performance of SS <40kPa as indicator of probability of high risk EV was evaluated.

Qualitative data-analysis was performed with Chi-squared test. Quantitative data with non-normal distribution was analyzed with Mann-Whitney test. A p value < 0.05 was considered significant.

Results: We included 22 patients, with a median age 44(3-77) years with, 55% (12) being females. From these, 50% (11) were included in group A with 2 of them having had a previous EV bleeding episode. Comparing the 2 groups, the median SS value was 75kPa (IQR=0) in group A and 35 kPa (IQR=2.5) in group B (p=0.009).

All of the patients in group A had SS value >40KPa and thus would not have been excluded from UGE.

In group B, 45% (n=5) of patients had SS value >40KPa would have undergone UGE despite not having high risk EV/GV. In this cohort the cut-off showed a sensitivity of 100% and a negative predictive value of 100% despite a low specificity (55%).

Conclusion: In our cohort, a SS <40kPa in patients with non-cirrhotic CEPVT had a high negative predictive value and can be useful to select those patients in whom endoscopy can be avoided.

Disclosure: Nothing to disclose.

PP1027

SPLEEN STIFFNESS IS CORRELATED WITH HEPATIC VENOUS PRESSURE GRADIENT AND CAN PREDICT TRANSJUGULAR INTRAHEPATIC PORTOSYSTEMIC SHUNT DYSFUNCTION: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Hepatic venous pressure gradient (HVPG) measurement is invasive. Non-invasive methods for evaluating the severity of portal hypertension in patients with transjugular intrahepatic portosystemic shunt (TIPS) are warranted.

Aims & Methods: This study aims to evaluate the efficacy of spleen stiffness in predicting HVPG and TIPS dysfunction. We performed a systematic review in PubMed, Embase and Cochrane databases, searching for studies comparing spleen stiffness measurements pre-TIPS and post-TIPS. The primary outcome was spleen stiffness. The secondary outcome was spleen stiffness in the context of TIPS dysfunction. The MD was used for continuous outcomes with their respective 95% confidence interval (CI). Heterogeneity was assessed using the Cochran Q test and I^2 statistics.

Results: The meta-analyses included 13 studies and 521 patients. There was a significant difference in spleen stiffness reduction after TIPS compared to baseline, measured by SSW m/s (MD -0.49; 95% CI -0.55 to -0.43; p<0.01; I^2=0%) and by SSM kPa (MD -11.19; 95% CI -13.32 to -9.07; p<0.01; I^2=0%). TIPS dysfunction was associated with increased spleen stiffness values (MD 0.35; 95% CI 0.24 to 0.45; p<0.01; I^2=0%) compared to normal tips function similarly to those found in pre-TIPS patients.

Conclusion: Our study showed that spleen stiffness is a non-invasive measure to evaluate HPVG reduction after TIPS placement. Spleen stiffness may also be useful in monitoring TIPS dysfunction.

Disclosure: Nothing to disclose.

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PP1028

INTEREST OF ANTICOAGULANTS IN THE EVOLUTION OF CHRONIC PORTAL THROMBOSIS IN A NON-CIRRHOTIC PATIENT

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Introduction: Cruoric thrombus in the portal vein represents the most frequent cause of infrahepatic portal hypertension. It most often complicates cirrhosis, but when it occurs in the absence of cirrhosis, it is favored by a pro-thrombotic state in 2/3 of patients. It can be total or partial, localized or extended to the spleno-mesenteric trunk. The absence of repermeabilization leads to the development of a collateral venous network known as portal cavernoma (or chronic PVT). Management is essentially based on treatment of the cause, and in the absence of contraindication, anticoagulation is recommended to prevent recurrence or extension of thrombosis. The aim of this study is to report our department's experience of the benefits of anticoagulation in cases of chronic portal thrombosis in a non-cirrhotic patient.

Aims & Methods: This is a retrospective, descriptive study over a 6-year period from 2018 to 2024, involving all patients with chronic portal thrombosis without cirrhosis on anticoagulation for more than 6 months. Curative-dose anticoagulation with vitamin K antagonsit and rivaroxaban 20mg/d was administered in patients with a major pro-thrombotic risk factor, and anticoagulation with rivaroxaban dosed at 15 mg/d in a low risk.

Results: Out of 58 patients followed for chronic portal vein thrombosis, 17 were put on anticoagulants: vitamin k antagonist (VKA) and direct oral anticoagulants (DOACs), with a prevalence of 29.31%. The mean age was 45.6 years (31-72), with a sex ratio of 0.4 M/F .Tow patient's groups were identified: Patients with thrombosis at high risk of recurrence, requiring lifelong anticoagulantion this group included 10 patients for 58.8% : the dominant etiology is myeloprolierative syndrome in 60% of patients (n=6),followed by anti-phospholipid antibody syndrome in 40% (n=4). Treatment in these patients was based on curative anticoagulation with 3 patients on rivaroxaban 20mg/d, and 7 patients on adapted VKA with an INR target of between (2-3). In the second group at low risk of recurrence, receiving rivaroxaban 15mg/d, involving 41.1% of our patients (n=7). The etiologies were as follows: isolated protein C or S deficiency in 42.8% (n=3), celiac disease in 28.5% (n=2), hydatid cyst surgery in 14.2% (n=1), no etiology found in 14.2% of patients (n=1).The evolution was marked by the absence of thrombosis extension assessed by imaging after a minimum of 6 months' anticoagulant treatment in 58.8% of cases (n=10), a partial repermeabilization in 29.4% (n=5), And complete repermeabilization was observed in 11.7% (n=2).

Conclusion: Anticoagulant treatment proved effective in preventing recurrence and thrombotic extension in all patients in our study, as assessed by imaging, which, although not systematically recommended in the follow-up of portal cavernoma, enabled us to confirm non-extension of the thrombosis in 58.8% of cases, and partial repermeabilization in 29.4% and complete repermeabilization in 11.7% of patients , Hence the importance of early anticoagulation with etiological management.

Disclosure: Nothing to disclose.

PP1029

FACTORS PREDICTING SURVIVAL IN PATIENTS DIAGNOSED WITH HEPATOCELLULAR CARCINOMA (HCC) IN FAR NORTH QUEENSLAND, AUSTRALIA

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Introduction: Far-North Queensland a region of 380,000 km^2 in tropical Australia has a population of approximately 290,000, of whom 17% identify as Aboriginal and/or Torres Straits Islander Australians. The region has 3 of the 7 most disadvantaged local government areas in Australia . Performing HCC surveillance in dispersed, disadvantaged populations across vast distances is challenging, as many other health conditions compete for finite health resources. Over 80% of Aboriginal Australians in this region have the less oncogenic HBV/D genotype infection.

Aims & Methods: A retrospective cohort study of individuals diagnosed with HCC in Far North Queensland between Jan 2014 and Sep 2021. The study's primary outcome was overall survival.

Results: There were 169 individuals diagnosed with HCC during the study period, their mean (95 % confidence interval (CI)) age was 65 (64 - 67). First Nations Australians were not over-represented (15 % of the cohort). The most common risk factors for liver disease were alcohol use disorder (66%), chronic hepatitis C virus (HCV) (47 %) and metabolic dysfunction (24 %). HBV-related HCC was less common (10 %) and there were no reported cases in Aboriginal Australians. The mean (95 % CI)) survival in the cohort was 421 (304 - 570) days. In multivariate analysis, death was more common in individuals with Child-Pugh C cirrhosis (hazard ratio (HR) (95 % CI)): 3.63 (2.20 - 5.99), $p < 0.0001$) and individuals with HBV infection (HR (95 % CI): 2.17 (1.25 - 3.77), $p = 0.006$). Individuals with chronic HCV infection were less likely to die (HR (95 % CI): 0.46 (0.31 - 0.69), $p < 0.0001$). Only 24 % of the cohort was participating in HCC surveillance at time of diagnosis despite most patients having guideline-recommended indications for screening; individuals participating in HCC surveillance were less likely to die (HR (95 % CI): 0.54 (0.33 - 0.90), $p = 0.02$).

Conclusion: HCC surveillance is associated with improved survival, but uptake in the region is suboptimal and contributes to the very poor outcomes. The absence of HBV-related HCC in Aboriginal individuals despite a high community prevalence of HBV infection suggests that HBV genotype might be employed to inform optimal HCC surveillance strategies in this remote region of Australia.

Disclosure: None.

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PP1030

HEPATOCELLULAR CARCINOMA SCREENING IN CIRRHOSIS: EXPERIENCES OF A PERIPHERAL IRISH HOSPITAL

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Introduction: 2018 European Association for the Study of the Liver (EASL) guidelines suggest a six-month radiological surveillance interval for cirrhotic patients with either Child-Pugh stage A and B, or C awaiting liver transplantation, to reduce hepatocellular carcinoma (HCC) associated mortality.

Aims & Methods: We aimed to audit our compliance with HCC screening guidelines.

Our hepatology clinic, liver elastography and radiology databases were cross referenced to yield 91 patients attending with cirrhosis over the years 2013 to 2022.

Results: 69% (n=63) were male. Median age at diagnosis was 58 years (IQR: 50-64.75). 84% (n=76) and 12% (n=11) had alcohol and non-alcoholic steatohepatitis (NASH) as a documented aetiology. Median MELD3 was 13 (IQR: 9-17). 14% (n=13) had died at the time of writing. Median age at death was 62 (IQR 52-66).

34% (n=31) and 46% (n=42) had radiological features of splenomegaly and hepatic steatosis respectively. 56% (n=51) had evidence of varices with 67% (n=34) of those taking non-selective beta blockers (NSBBs). 20% (n=18) had prior hepatic encephalopathies and 67% (n=12) of those were taking rifaximin. 33% (n=30) had prior GI bleeding. 71% (n=65) had prior ascites with 75% (n=49) prescribed diuresis. 4% (n=4) had prior documented hepatorenal syndrome. 10% (n=9) had prior spontaneous bacterial peritonitis (SBP) with 44% (n=4) taking secondary prophylaxis. 2% (n=2) had prior trans-jugular intrahepatic portosystemic shunts (TIPSS). 3% (n=3) had prior portal vein thrombosis.

5% (n=5) were diagnosed with HCC. Median age at time of HCC diagnosis was 74 years (IQR: 66-74). Median alpha-fetoprotein (AFP) at time of diagnosis was 10.9 (IQR: 5.4-756).

Patients with Child Pugh C cirrhosis (N=24, 18%) were excluded from the analysis.

167 surveillance studies on those with no more than Child Pugh B cirrhosis were analysed over the ten-year period. The median number of months between surveillance imaging was 9.77 months (IQR: 5.9-17.8). 40% of surveillance studies were carried out at an interval of greater than 12 months. The average number of months between imaging pre- and post-COVID were 12.8 and 8.26 months respectively (95% CIs: 10.9-14.8, 6.4-10.1).

60% (n=3) of HCC diagnoses were made in the absence of a known history of cirrhosis. Two patients with Child-Pugh C cirrhosis had HCC diagnosed. One was a male in his 60s who had a diagnosis made on ultrasound 10.5 months after prior surveillance at another site. Another male in his late 70s had a HCC diagnosis made 35 months after a prior ultrasound identified cirrhosis, though his performance status made surveillance inappropriate.

Conclusion: Alcohol was the most prevalent cause of cirrhosis in our cohort. 5% of our cohort were diagnosed with HCC, with none as a result of failures to follow surveillance guidelines. Surveillance intervals post March 2020 fell significantly ($p < 0.01$). This may be attributable to an increase in radiology resources in the latter part of the decade.

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Disclosure: Nothing to disclose.

PP1031

POPULATION-BASED HEPATITIS C SCREENING IS FEASIBLE IN COUNTRY WITH MODERATE SEROPREVALENCE: 21-MONTH RESULTS OF THE PROGRAM IN LITHUANIA AND SCENARIOS FOR ACHIEVING WHO ELIMINATION TARGETS

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Introduction: Lithuanian HCV screening program started on May, 2022. State health authorities pay general practitioners (GPs) a special promoting fee for performing serological tests for HCV antibodies: 1. For the population born in 1945-1994 (once per life) and 2. For people who inject drugs (PWID) or are HIV-infected (annual HCV testing). Such an initiative is the first in Central and Eastern Europe.

Aims & Methods: This study aimed to evaluate the results of the first 21-month of the program and different scenarios to achieve WHO 2030 targets. GPs invited people to perform a serum blood test for HCV antibodies. Anti-HCV positive patients were referred to a gastroenterologist or infectologist for HCV RNA testing, and if result was positive, direct-acting antiviral (DAA) therapy was prescribed. Information about patients was obtained from the database of the National Health Insurance Fund. The Markov disease progression model elaborated by the CDA Foundation was used to assess HCV elimination progress. The data from the 2022-2023 screening were used as inputs. Three scenarios were developed: the 'Base' scenario - return to pre-screening program level in 2023 and 2 scenarios with different extents of treatment.

Results: At the beginning of 2022, about 1.8 million people born in 1945-1994 lived in Lithuania. Between May 5, 2022 and January 31, 2024 from this population 1145121 people (63,6%) were tested for HCV antibodies. Positive test results were found in 1.23% of cases. In the risk group, 9604 PWID and HIV+ people were screened, of whom 30,1% were seropositive. All seropositive patients were referred to specialists. Viremia was detected in 58 %. During 21-month period 4811 patients were treated with DAA. The number of treated patients during 21 months of program increased 2,8 times in comparison to previous 21 month before program (Aug 2020 - Apr 2022). Treatment delay for 48% of HCV patients was related to organizational problems (consultation waiting line, some treatment restrictions, delay in drugs supply).

Solving organizational challenges, by the end of 2024 we expect to perform HCV antibody testing in 80% of the target population and treatment up to 70% of RNA-positive individuals, further aiming to cure all of infected patients by 2030. The Markov disease progression model to assess HCV elimination progress showed the following scenarios: Scenario 1: if the

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same number of patients are treated as before the screening, the WHO targets will not be reached. Scenario 2: treating 70% of infected patients will meet most but not all WHO targets. Scenario 3: by treating all infected patients by 2030, the WHO target will be met by saving 150 lives and preventing 90 new cases of decompensated cirrhosis and 120 cases of hepatocellular carcinoma.

Conclusion: Our data show that in European country with moderate HCV seroprevalence a population-based screening program can be feasible, screening near 64 % (more than 1,1 million) of the population born between 1945 and 1994 during the first 21 months of the program. We hope that WHO 2030 goals can be achievable in Lithuania.

Disclosure: Nothing to disclose.

PP1032

PREVALENCE OF SEXUAL DYSFUNCTION IN PATIENTS WITH LIVER DISEASE

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Introduction: Chronic liver disease, particularly cirrhosis, leads to significant morphological and psychological endocrine changes, which can have an impact on patients' sexual health. Although sexual health is an important parameter of quality of life, it remains little studied in hepatology.

Aims & Methods: The aim of our work is to study the prevalence and degree of sexual dysfunction in male patients with chronic liver disease, and the associated risk factors.

This is a monocentric, comparative, descriptive and analytical study conducted during the year 2023.We included male patients followed for chronic hepatopathies (group 1 (G1)) as well as a control group of subjects followed in gastroenterology consultation presenting no hepatic pathology (group 2(G2)). Exclusion criteria were the existence of overt hepatic encephalopathy, cognitive disorders or active alcohol abuse, limiting the reliability of the survey.

Sexual health was assessed using the IIEF (International Index of Erectile Function) questionnaire, which covers five domains: erection, orgasm, desire, satisfaction with intercourse and overall satisfaction. Two software packages were used for statistical analysis: Jamovi version 2.3.28 and Microsoft Excel.

Results: Our study included 120 patients, 60 followed for chronic liver disease, and 60 without liver pathology (control group).

Mean age was 44.6 years (21-73 years).

In the group of patients with hepatopathies, we found:

34 (65%) cirrhotics, 24 (40%) of whom had a history of decompensation. 18 (30%) HBV, 11(18%) HCV, 5 (8%) NAFLD, 5 (8%) of undetermined cause, 5 (8%) portal hypertension, 4 (6.6%) portal thrombosis, 12 (19.8%) others.

Patients in the liver disease group had more erectile dysfunction, orgasmic dysfunction, disorders of desire, disorders of sexual satisfaction and disorders of overall satisfaction compared with the control group, with respectively 40/60 (66. 6%) vs 16/60 (26.6%) *p 0.003*; 35/60 (58.3%) vs 14/60 (23.3%) *p 0.012*; 41/60 (68.3%) vs 10/60 (16.6) *p <.001*; 38/60 (63.3%) vs 18/60 (30%) *p <.001*; and 45/60 (75%), vs 17/60 (28.3%) *p <.001*.

After statistical analysis, cirrhosis was the only statistically significant risk factor for sexual dysfunction in our study (OR 7.9 ;p 0.012).

Moreover, 100% of patients said they had never been asked about their sexual health during a consultation.

Conclusion: Sexual dysfunction is common in patients followed for liver disease, with cirrhosis as the main risk factor. Sexual dysfunction should be systematically screened, with a multidisciplinary approach to the management of these patients.

Disclosure: Nothing to disclose.

PP1033

FACTORS INFLUENCING THE QUALITY OF LIFE IN CIRRHOTIC PATIENTS

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Introduction: Cirrhosis is a chronic liver disease that, due to its natural progression and complications, affects the quality of life (QOL). Tunisian data related to the QOL of cirrhotic patients are limited. Our objectives were to evaluate the QOL of cirrhotic patients and identify factors influencing it.

Aims & Methods: We conducted a cross-sectional study, including cirrhotic patients followed in our outpatient clinic. QOL was assessed using the following questionnaires: Chronic liver disease questionnaire (CLDQ) (QOL impairment threshold defined at five), 36-item short form (SF-36), The Hospital anxiety and depression scale , the Clinical frailty scale , the fatigue severity scale , the Insomnia severity index , and the Work productivity and activity impairment index .

Patient data regarding medical history, cirrhosis characteristics, and socio-economic conditions were collected.

Statistical analysis was performed using the SPSS software (significance threshold p=0.05).

Results: One hundred (56 women and 44 men) were included, with an average age of 60.62 years. The cirrhosis was due to viral hepatitis in 55% of cases with a CHILD-PUGH score classified as B or C (49 cases). Thirty-four patients had a history of upper gastro-intestinal bleeding. Thirty patients presented with ascites.

The mean SF-36 was 54.06, and the mean CLDQ was 4.7. Twenty-five patients experienced anxiety, and 35 had depression, 22 suffered from clinically patent insomnia.

In multivariate analysis . The factors identified as influencing the SF-36 were the education level (p=0.014), the CHILD-PUGH score (p=0.016), the presence of esophageal varices (p=0.016) or ascites (p=0.012), anemia (p=0.016), and low serum albumin levels (p=0.014).

The factors influencing CLDQ were the CHILD-PUGH score (p=0.020), the presence of ascites (p=0.012), anemia (p=O.025), and low serum albumin levels (p=0.018).

A CLDQ below five was associated with a CHILD-PUGH score B or C (p=0.012), the presence of ascites (p=0.011), a portal vein thrombosis (=0.048), and low serum albumin levels (p=0.010).

Conclusion: The results of our study highlight the importance of assessing QOL during cirrhosis and developing comprehensive care for our patients

Disclosure: Nothing to disclose.

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Biliary

09:00-18:00 / Poster Exhibition

PP1034

THE CONCENTRATIONS OF TNFRSF14 AND LIGHT IN THE SERA AND BILE OF THE PATIENTS WITH PRIMARY SCLRELOSING CHOLANGITIS

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Introduction: Primary sclerosing cholangitis (PSC) is an idiopathic cholestatic disease. Recent genome-wide studies (GWAS) have identified about 30 disease susceptible genes to PSC.

We investigated the expression profiles of these genes in the liver biopsy samples of the patients with PSC, and revealed that the expression of TNFRSF14, which is a one of the risk genes for PSC, was upregulated in the biliary epithelial cell (BEC) of PSC compared to that in normal liver tissue and primary biliary cholangitis (PBC) liver tissue. TNFRSF14 is a member of the tumor necrosis factor receptor superfamily associated with inflammation and fibrosis. We also clarified that the expression of LIGHT, an active ligand for TNFRSF14, was elevated in the BEC of PSC[1)].

These molecules are mainly expressed on lymphocytes, and in other inflammatory diseases, many reports have mentioned to the associations between these molecules in the sera and the disease progression.

Aims & Methods: Aims: In this study, we aimed to evaluate the concentrations of these molecules in the sera and bile of PSC.

Methods: The concentrations of soluble TNFRSF14 and LIGHT in the sera and bile were measured by enzyme-linked immunosorbent assay. The samples were collected from the following cases:

• Serum: 73 patients with PSC, 35 patients with PBC and 53 healthy controls.

• Bile: 15 patients with PSC, 12 patients with malignant biliary obstruction (MBO), 19 patients with biliary lithiasis (BL).

The bile samples were obtained during endoscopic retrograde cholangiopancreatography. The median values of the serum (s) and bile (b) concentrations of each group were evaluated by Mann-Whittney U test.

Results: sTNFRSF14 in the PSC group was significantly higher than those in the healthy controls (6.301 v.s. 4,247 pg/mL, $P < 0.001$) and the PBC group (4,987 pg/mL, $P < 0.001$).

sLIGHT in the PSC group was 87 pg/mL, which was higher than those in the healthy controls (61 pg/mL, $P = 0.012$), and was comparable to those in the PBC group (106 pg/mL, $P = 0.142$).

bTNFRSF14 in the PSC group was 2,005 pg/mL, showing no significant difference compared to those in the MBO (1,794 pg/mL, $P = 0.516$) and the BL group (1,987 pg/mL, $P = 0.504$).

bLIGHT in the PSC group was 523 pg/mL, remarkably higher than those in the MBO (82 pg/mL, $P < 0.001$) and the BL group (268 pg/mL, $P = 0.024$).

Conclusion: sTNFRSF14 in the PSC group was higher than those in the healthy control and the PBC group, and this result was equivalent to the previous report about the upregulated expression in the liver biopsy samples. While soluble LIGHT was secreted non-specifically in the sera of cholestatic liver disease, the elevated bile concentration of LIGHT can be involved in in the unique biology of PSC. Further research is needed to elucidate the involvement of these two molecules in the pathogenesis of PSC.

References: 1. Digestive Liver Disease 2024 Feb;56(2):305-311.

Disclosure: Nothing to disclose.

PP1035

THE EFFECT OF URSODEOXYCHOLIC ACID (UDCA) ON SYMPTOMS AND QUALITY OF LIFE IN PATIENTS WITH BILIARY SLUDGE AND EARLY STAGE OF CHOLELITHIASIS IN ROUTINE PRACTICE: MULTICENTER OBSERVATIONAL STUDY RESULTS

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Introduction: As known, bIliary sludge has mostly asipmtomatic course, thus lack of data for biliary sludge diagnostics and treatment in Russia is observed.

Aims & Methods: Evaluation of the effectiveness and safety of therapy with UDCA in patients with the biliary sludge and early stage of cholelithiasis in routine practice.

Multicenter observational study (10 sites in total) has been conducted. Patients with the biliary sludge and/or early stage of cholelithiasis were enrolled. The main inclusion criteria were ultrasound (US) criteria (biliary sludge or low density stones less than 1 cm), history of uncomplicated cholelithiasis. Patients were given UDCA (10 mg/kg/day).

Other medication were also given to address their abdominal symptoms (include pain that does not meet the criteria for biliary colic) in accordance with approved standards.The duration of the observation was 180 days.

Primary outcome was to evaluate the effect of complex therapy with UDCA on the patients': quality of life, pain syndrome (include appearance biliary colic), changes in severity of other symptoms. The secondary outcomes were dynamics of US characteristics of the biliary system and the safety of UDCA.

Results: 1056 patients were included in the study: with biliary sludge – 948 (89.8%), stones – 270 (25.6%), of which sludge plus stones – 162 (15.3%). At baseline, 40.2% patients had mild or medium abdominal pain, 44% - abdominal discomfort, 35.5% - sensitivity to abdominal palpation. The most common complaint was bitter taste in mouth (529 patients, 50.1%). Statistically significant increase of the quality of life and decrease the frequency of abdominal pain on 60 days and 180 days of treatment was shown.

There was a statistically significant decrease in the frequency of several symptoms such as: nausea, bitter taste in mouth, heartburn, feeling of fullness after eating, bloating, flatulence at the end of treatment compared to the baseline (p<0,001).

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Parameter	Base	60 days	p(0-60 d)	Visit 7	p(0-180 d)
SF-36, physical, scores	49.9	53.5	<0.001	54.9	<0.001
SF-36, mental, scores	46.9	50.7	<0.001	52.5	<0.001
No pain, %	59.8	95.2	<0.001	99.3	<0.001

Results of repeated US showed the positive effect complex therapy with UDCA. The frequency of biliary sludge had decreased from 89.8% to 12%, the frequency of gallbladder stones - from 25.6% to 9.8%, simultaneous of biliary sludge and gallbladder stones - from 15,3% to 0.3%. There were no patients with enlargement of the gallbladder (against 8.5% at baseline).
In the vast majority of patients these parameters returned to normal level (ALT in 94.9% of cases, AST in 95.5%, direct bilirubin in 94.8%, total bilirubin in 92.2%, alkaline phosphatase in 95.0%). Among patients with normal values during treatment were recorded increase ALT (5 cases), total bilirubin (6), GGT (4), alkaline phosphatase (2).
High safety of treatment with UDCA was recorded: 4 adverse events associated with taking UDCA, no serious adverse events. Compliance in the vast majority of patients (fluctuations 83.3-90.1% of patients during the study) was high.
Conclusion: The results of this practice study have shown the effectiveness of complex therapy with UDCA in patients with cholelithiasis and biliary sludge. Complex therapy with UDCA improved patients' quality of life, reduced the frequency of mild and moderate abdominal pain, prevents biliary colic. Results demonstrated that UDCA was highly safe. Thus, the UDCA may be extensively used in patients with biliary sludge & early stage of cholelithiasis and may prevent excessive surgery intervention.
Disclosure: Current study was conducted with the financial support of Binnopharm Group

PP1036

FACTORS ASSOCIATED WITH SPONTANEOUS MIGRATION OF STONES FOR COMMON BILE DUCT

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Introduction: Common bile duct (CBD) stones can spontaneously migrate through the duodenal papilla. In this case, ERCP may be unnecessary and a significant rate of complications may be avoided..
Aims & Methods: Aims: The aim of our study is to evaluate the factors associated with spontaneous migration of CBD stones.
Methods:This is a retrospective descriptive and analytical study including 547 patients who underwent ERCP for CBD stones between January 2019 and September 2023.
All patients underwent MRCP for the diagnosis of CBD stones.
MRCP was considered positive if stones were present in CBD.
A positive ERCP was defined as the presence of stones in the VBP on cholangiography.
Patients were classified into 2 groups:
- Group A (n=309): positive MRCP and positive ERCP.
- Group B (n=265): positive MRCP but negative ERCP.
Statistical analysis was performed using JAMOVI software.
Results: The mean age in group A was 59.1 +/- 13.8 years and in group B 56.9 +/- 13.2 years, with no statistically significant difference (p=0.056).
The sex ratio (M/F) in group A was 0.7 and in group B 0.54, with a significant female predominance in both groups (p=0.03).
There was no statistically significant difference between the 2 groups in CBD diameter or presence of a periampullary diverticulum.
Comparing group A versus group B, patients with small stone diameter (P=0.001), a single stone (p=0,001) and distal stones (P=0.04) tended to pass their stones spontaneously.
Conclusion: In our study, the factors associated with spontaneous migration of CBD stones were a small stone diameter, a single stone and distal stones.
Disclosure: Nothing to disclose.

PP1037

EARLY PREDICTORS OF SUCCESSFUL KASAI PORTOENTEROSTOMY

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Introduction: This study was done to identify factors associated with successful Kasai Portoenterostomy (KPE)
Aims & Methods: Retrospective analysis of 49 infants who underwent KPE between January 2018 to October 2023. Successful KPE was defined as total bilirubin < 1 mg/dl by 6 months of age. None of the patients received post KPE steroids and none were tested for concomitant CMV.
Results: The median age at KPE was 71 (59-83) days, with 21 (42.9%) less than 65 days. Median follow up was of 11 (6-36) months with native liver survival in 27 (55.1%).Thirty six (73.5%) were type III biliary atresia and 2 (4%) were associated with other congenital anomalies. Successful KPE was in 22 (44.9%). Age less than 65 days at the time of KPE was significantly associated with successful KPE (42.9% vs 57.1%, OR 5, 95% CI 1.472 -16.988, p=0.009). Weight (p=0.932), total bilirubin (p=0.907), direct bilirubin (p=0.667), AST (p=0.837), ALT (p=0.937), Total protein (p=0.051), albumin (p=0.201) and INR (p=0.220) at the time of surgery did not affect outcome. Ishak fibrosis grade (p=0.267) and inflammation grade (p=0.743) were not associated with outcome. Post surgery, lower total bilirubin (7.95 mg/dl vs 10.1 mg/dl, MD 0.07 to 4.66, p=0.043) and direct bilirubin (4.92 mg/dl vs 6.5 mg/dl, MD 0.16 to 3.03, p=0.03) by post operative day 7 were associated with a successful outcome. Parameters like ascites for more than 7 days (p=0.071), albumin (p=0.431), AST (p=0.091) , ALT (p=0.108) and INR (p=0.241) at 7 days post KPE did not affect the outcome. Seventeen patients suffered from complications post KPE with sepsis in 17, bowel perforation in 3 and biliary ascites in 2. Presence of complications post KPE did not affect the outcome (p=0.825). Twenty (40.8%) patients suffered from a median of 1 (1-3) cholangitis episodes. However, cholangitis was not associated with outcome (p=0.374). On logistic regression analysis age less than 65 days (OR 8.024, 95% CI 1.865 to 34.530, p=0.005) and lower total bilirubin at day 7 post KPE (OR 0.789, 95% CI 0.642 to 0.970, p=0.028) were found to be significantly associated with successful KPE increasing the correctness of classification of model from 55.1% to 73.5%
Conclusion: There is a 8 folds increase in successful outcome if KPE is done at less than 65 days of age. Bilirubin at post operative day 7 is an early predictor of successful KPE.
Disclosure: Nothing to disclose.

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PP1038

COMPARISON OF SHORT-COURSE VERSUS LONG-COURSE ANTIMICROBIAL TREATMENT FOR ACUTE CHOLANGITIS WITH GRAM-POSITIVE BACTEREMIA AFTER ERCP

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Introduction: There is no clear duration of antimicrobial treatment of acute cholangitis complicated by Gram-positive coccus (GPC) bacteremia. The American guideline[1] suggests 4-7 days of antimicrobial treatment after appropriate source control, irrespective of the organism or presence of bacteremia, while the Japanese guideline[2] suggests 14 days when GPC bacteremia is present.

However, data to support these recommendations are limited and deviation from real-world practice is proposed, therefore, studies for adequate treatment duration are warranted.

Aims & Methods: We conducted a single-center, retrospective study of adult patients with acute cholangitis with GPC bacteremia treated with ERCP (Endoscopic Retrograde Cholangiopancreatography) from July 2003 to December 2023. We analyzed two groups according to the duration of effective antimicrobial treatment: short-course (SCT, <14 days) and long-course (LCT, ≧14 days). We constructed logistic regression models for mortality and a composite outcome, including mortality, recurrence, and infection of the same organism related to cholangitis. Malignant biliary strictures were excluded.

Results: A total of 45 patients with acute cholangitis complicated by GPC bacteremia underwent ERCP in our hospital, 22 in the SCT group and 23 in the LCT group. The median duration of antimicrobial treatment was 9 days (IQR 5.5-12.5) in the SCT group and 16 days (IQR 14.5-19.5) in the LCT group, with a significant difference between the two groups ($p<0.05$).

There were no significant differences in age, gender, severity, cause of cholangitis, the complication rate of pancreatitis and cholecystitis, heart valve disease, community acquisition, number of positive blood cultures, rate of echocardiography during treatment, intravascular devices or time to biliary drainage between the two groups.

Common bile duct stones were the most common cause with 82.2% of all patients, followed by diverticulum and stent obstruction with 8.9% and 6.7%, respectively.

Regarding the causative organisms, mixed infections were found in 71.1%, and *Enterococcus spp.* was the most common strain with 66.7%, of which *E. faecalis* with 23.3%, *E. faecium* with 16.7%, and followed by *Streptococcus spp.* with 31.1% of all patients, with no significant differences between the two groups.

Regarding antibiotics, penicillins were most commonly used with 73.3%, followed by cephalosporins with 66.7%, and glycopeptides with 26.7% in all patients. Penicillins (91.3% vs. 54.5%, $p<0.05$) and glycopeptides (43.4% vs. 9.1%, $p<0.05$) were used more frequently in the LCT group.

The 30-day mortality and reinfection with the same organisms within 3 months were 0.0% in both groups, and the length of hospital stay was significantly shorter in the SCT group (median 14.0 vs 10.5 days, $p < 0.05$).

The rate of cholangitis recurrence within 3 months was 5% (1/20) in the SCT group and 9.5% (2/22) in the LCT group, with no significant difference.

	Shorter (<14 days) n=22 (%)	Longer (≧14 days) n=23 (%)	p-value
30-day mortality, n(%)	0 (0%)	0 (0%)	
Recurrence of cholangitis, n(%) Yes No	 1 (5.0%) 19 (95.0%)	 2 (9.1%) 20 (90.9%)	0.715
Reinfection due to the same organism within 3 months, n(%) Yes No	 0 20 (100%)	 0 22 (100%)	0.968
Length of hospital stay (median days, IQR)	10.5 (7-13)	14.0 (10-18.5)	0.032

Conclusion: For acute cholangitis complicated by GPC bacteremia, antimicrobial treatment for less than 14 days might be considered after appropriate biliary drainage.

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Disclosure: Nothing to disclose.

PP1039

PREDICTING FACTORS OF ACUTE GANGRENOUS CHOLECYSTITIS IN PATIENTS TREATED WITH LAPAROSCOPIC CHOLECYSTECTOMY: A SINGLE CENTER RETROSPECTIVE STUDY

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Introduction: Acute gangrenous cholecystitis, a severe complication of cholecystitis, carries a high risks of morbidity and mortality. Despite its severity, preoperative diagnosis is challenging, often only confirmed during surgery, highlighting the need for effective pre-operative predictors to facilitate timely surgical intervention. Therefore, we analyzed various factors that could predict acute gangrenous cholecystitis in patients undergoing laparoscopic cholecystectomy.

Aims & Methods: We conducted a retrospective review of patients who underwent laparoscopic cholecystectomy at Bundang Jasaeng General Hospital between January 2018 and December 2021. From this group, we included only those patients who proceeded directly to surgery without any radiologic interventional treatments. Prior to surgery, all patients underwent an ultrasonographic examination of the gallbladder.

Results: Our study investigated 314 patients diagnosed with acute cholecystitis, of which 37 (11.8%) were identified with acute gangrenous cholecystitis.

Analysis revealed that patients with gangrenous cholecystitis were significantly older, more likely to be male, and had a higher prevalence of chronic kidney disease compared to those with uncomplicated cholecystitis. Laboratory results indicated higher WBC, C-reactive protein, and delta neutrophil index (DNI) levels in gangrenous cases, with a DNI level of 0.5% or higher marking a notable increase in risk.

Ultrasound findings showed a higher incidence of sonographic Murphy's signs (9% vs. 21.6%, p=0.039), thickened gallbladder wall (3.1 ± 0.9 mm vs 5.2 ± 0.9 mm, $p < 0.001$) and defects (0.4% vs. 5.6%, p=0.035) in gangrenous cholecystitis.

Univariate and multivariate analyses highlighted that a thickened gallbladder wall remained a significant predictor of gangrenous cholecystitis ($p<0.001$).

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	Univariate analysis				Multivariable analysis			
	OR	95% CI		p value	OR	95% CI		p value
		Lower	Upper			Lower	Upper	
Thickened GB wall (≥ 4 mm)	85.64	24.95	538.84	< 0.001	79.37	20.39	579.91	< 0.001

Conclusion: Patients diagnosed with gangrenous cholecystitis were typically older, more frequently male, and had a higher prevalence of chronic kidney disease. The results of this study suggest that thickened GB wall (≥ 4 mm) is an effective predicator of gangrenous cholecystitis; therefore, urgent management is recommended for these patients due to the high morbidity and mortality associated with the condition.

Disclosure: Nothing to disclose.

PP1040

CHARACTERISTICS AND METABOLIC POTENTIAL OF BILIARY MICROBIOTA IN PATIENTS WITH GIANT COMMON BILE DUCT STONES

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) is an effective minimally invasive approach for the treatment of choledocholithiasis, while successful extraction is hampered by large diameter of stones. Emerging studies have revealed the close correlation between biliary microbiota and common bile duct stones (CBDS).

Aims & Methods: We aimed to investigate the community characteristics and metabolic functions of biliary microbiota in patients with giant CBDS. The present study was designed as a case-control study, which was approved by the ethics committee of First Affiliated Hospital of Soochow University (approval number: 2020-082). Consecutive patients were prospectively enrolled from February 2023 to October 2023. The inclusion criteria were as follows: (1) CBDS were confirmed by at least one imaging examination; (2) no antibiotics use within three months. The patients who had a history of surgery involved in upper gastrointestinal tract or hepatobiliary system were excluded. The CBDS would be recognized as a giant stone if its diameter was larger than 15mm and the diameter ratio of stone against CBD was larger than 1.0. Bile samples were collected through ERCP. The microbiota was analyzed using 16S rRNA sequencing. Metabolic functions were predicted by PICRUSTs 2.0 calculation based on MetaCyc database. Bile acids were tested and identified using ultra performance liquid chromatography-tandem mass spectrometry.

Results: A total of 26 patients were successfully included into final analysis, 8 in giant stone (GS) group and 18 in control group. No significant difference concerning demographics was found between the two groups (Table.1). Distinct biliary microbial composition was identified in patients with giant CBDS, with a significantly higher abundance of Firmicutes at phylum level. The unique composition at genus level mainly consisted of Enterococcus, Citrobacter, Lactobacillus, Pyramidobacter, Bifidobacterium and Shewanella. Pyramidobacter was exclusively found in GS group, along with the absence of Robinsoniella and Coprococcus. The contents of free bile acids were significantly higher in GS group, including cholic acid (98.39μmol/mL vs. 26.15μmol/mL, p=0.035), chenodesoxycholic acid (54.69μmol/mL vs. 5.86μmol/mL, p=0.022) and ursodeoxycholic acid (2.70μmol/mL vs. 0.17μmol/mL, p=0.047). Decreasing tendency of conjugated bile acids were also observed. Metabolic pathways concerning cholelithiasis were abundant in GS group, including geranylgeranyl diphosphate biosynthesis, gluconeogenesis, glycolysis and L-methionine biosynthesis.

	GS group n=8	Control group n=18	p value
Age, years (mean±SD)	83.50±6.28	66.67±14.48	0.001
Gender, male/female	6/2	13/5	1.000
BMI, kg/m^2 (mean±SD)	20.43±3.06	22.54±2.11	0.052
Stone no., single/multiple	2/6	8/10	0.420
TBIL, μmol/L, (median)	24.55	20.10	0.429
DBIL, μmol/L, (median)	17.10	8.80	0.198
γ-GGT, U/L, (median)	444.80	255.00	0.574
ALP, U/L, (median)	276.20	123.65	0.110
CA199, U/mL, (median)	15.47	9.22	0.261

Table.1 Patient demographics and preoperative characteristics.

Conclusion: This study demonstrated the community structure and metabolic potential of biliary microbiota in patients with giant CBDS. The unique biliary microbial composition holds valuable predictive potential for clinical conditions. These findings provide new insights into the etiology of giant CBDS from the perspective of biliary microbiota.

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Disclosure: Nothing to disclose.

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PP1041

APPLICATION OF EUS OR MRCP PRIOR TO ERCP IN PATIENTS WITH SUSPECTED CHOLEDOCHOLITHIASIS IN CLINICAL PRACTICE

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Introduction: Patients with symptomatic cholelithiasis can be stratified according to the 2019 European Society for Gastrointestinal Endoscopy (ESGE) guideline into low-, intermediate- and high-likelihood categories for the presence of choledocholithiasis. For the intermediate group, endoscopic ultrasound (EUS) or magnetic resonance cholangiopancreatography (MRCP) is recommended to assess whether an endoscopic retrograde cholangiopancreatography (ERCP) is necessary prior to cholecystectomy. The aim of the study is to investigate adherence to the guideline for diagnostic and treatment strategy for cholelithiasis in daily clinical practice.

Aims & Methods: Multicenter, retrospective cross-sectional observational study of the diagnostic pathway of patients with suspicion of choledocholithiasis between 2019 and 2021. Patients were stratified according to the ESGE guideline.

Results: A total of 305 patients were included in the analysis and stratified into low (17%), intermediate (40%) and high (43%) likelihood of choledocholithiasis. In these three categories, 182 (60%) patients underwent ERCP. Adherence to the ESGE guideline recommendation to perform additional imaging was 59.7% overall and was the highest in the intermediate group (83.6%) ($p < 0.001$). In 195 patients who underwent additional imaging, 55 ERCPs (28.2%) could be avoided.

Conclusion: This study shows that stratification according to the ESGE guidelines is useful to reduce the number of unnecessary diagnostic imaging procedures and ERCPs in patients with a suspicion of choledocholithiasis. It can be worthwhile to perform EUS prior the ERCP in the same session.

Disclosure: Mike J.P. de Jong, Megan M.L. Engels, Christina J. Sperna Weiland, Tanya M. Bisseling, Robin Krol and Foke van Delft have no conflict of interest or financial ties to disclose. Jeanin E. van Hooft reports lecture fees from Cook Medical, dr. Falk Pharma, Boston Scientific, AbbVie, and consultancy fees from Olympus. Erwin M. van Geenen reports research grants from Olympus, Boston Scientific, and MTW Endoskopie. Peter D. Siersema reports research grants from Pentax and Fujifilm. All outside the submitted work.

PP1042

PERCUTANEOUS CHOLECYSTOSTOMY IN SEVERE ACUTE CHOLECYSTITIS, COULD BE USED AS A DEFINITIVE TREATMENT IN UNFIT PATIENT

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Introduction: Although percutaneous cholecystostomy (PC) is generally accepted as a bridge to definitive therapy for acute cholecystitis (AC), which remains cholecystectomy, some patients did not undergo cholecystectomy, mainly due to contraindications to surgery. Here, we aimed to investigate the predictors of recurrence and the outcome after PC.

Aims & Methods: This is a retrospective study from a single general hospital at Tunbridge Wells, United Kingdom. One hundred twenty-six patients who presented with AC grade 3 and were initially managed with PC were included. In addition, the proportion of patients who did not undergo subsequent laparoscopic cholecystectomy (LC) and their characteristics were analyzed.

Results: The mean age of the study cohort was 72 (36-98) years, and the median length of drain insertion was 39.5 days. The majority (52%) presented with severe AC grade 3 with failed medical treatment to control the disease, while 7% had an emphysematous gallbladder. Eighty percent of patients did not develop any further attacks of AC after PC removal. The most common comorbidity was hypertension (35%). The mean age-adjusted Charlson comorbidity score was 3.72. Thirty-six percent (45/126) of the study cohort underwent LC, while the remaining patients did not receive any surgical intervention. Nine percent were deemed unfit for surgery. Forty-one patients (33%) were managed conservatively as they did not have a further attack of cholecystitis after PC removal or had a mild attack managed with antibiotics. In addition, 22% experienced procedural complications, including a blocked stent, pain, and cellulitis around the tube. The 30-day mortality rate of patients who did not undergo LC was 5%. Predictors of interval cholecystectomy were younger age, calculus cholecystitis, low Charslson index score, and uncomplicated and shorter length of hospital stay with PC.

Conclusion: Most severe AC patients treated initially with PC did not undergo subsequent LC. Therefore, PC in high-surgical-risk patients with AC could be a definitive treatment.

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Disclosure: Nothing to disclose.

PP1043

URSODEOXYCHOLIC ACID IS NOT ASSOCIATED WITH REDUCED RECURRENCE OF PRIMARY COMMON BILE DUCT STONES

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Introduction: Recurrence of common bile duct (CBD) stones after initial treatment is estimated to occur in 5%-20% of cases[1], requiring additional interventions and is associated with adverse patient outcomes. Previous small scale studies evaluated the role of UDCA in the prevention of CBD stone[2,3]. We aimed to investigate whether UDCA is effective in preventing de-novo CBD stone formation.
Aims & Methods: This was a cohort analysis. We analyzed the electronic records of patients diagnosed with cholangitis or choledocholithiasis who underwent endoscopic retrograde cholangiopancreatography (ERCP) with evidence of complete common bile duct (CBD) stone removal. Due to the lack of consistent evidence, recommendations for using ursodeoxycholic acid (UDCA) to prevent stone recurrence varied among endoscopists. All patients were contacted and interviewed after providing informed consent. The follow-up period was defined as the time from the index ERCP to either evidence of CBD stone recurrence or the date of the patient interview. The primary outcome was a documented case of CBD stone recurrence. We compared baseline variables and outcomes between patients who were recommended to receive UDCA following ERCP and those who did not receive a UDCA recommendation. We performed univariate survival analysis using the log rank score to compare survival distributions between groups.
Results: Ninety-eight patients were included in the analysis and contacted. Of these, 50 patients received a recommendation for UDCA treatment, and 30 patients took UDCA. Additionally, 48 patients did not receive any recommendation for UDCA, contributing to a total of 68 patients who did not take UDCA. Comparison of baseline characteristics, index ERCP findings and outcomes between groups is presented in *Table 1*. There were no significant differences in patient ages and rate of major comorbidities. Most patients undergone cholecystectomy more than 24 months prior to the index ERCP. Previous sphincterectomy was more common in the UDCA treatment group (70% vs 17.6%; p<0.01). No marked differences were noted in regard to other index ERCP findings between the two groups including papillary anatomy, number, size, type of stones and technique of stone removal (*Table 1*). No significant difference in stone recurrence was between patients who took and did not take UDCA (recurrence in 8/30 patients [26.7%] and 22/68 [32.4%] respectively, P=0.75). There was no significant difference in survival without recurrence between the two groups (p=0.84).
Conclusion: Our findings do not suggest efficacy of UDCA in prevention of de-novo CBD stone recurrence after cholecystectomy and complete CBD clearance.
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Disclosure: The authors report no conflict of interests

PP1044

MAGNOLOL INHIBITS THE PROLIFERATION OF CHOLANGIOCARCINOMA THROUGH THE JAK2/STAT3 SIGNALING PATHWAY IN VIVO AND IN VITRO

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Introduction: Cholangiocarcinoma (CCA), a highly aggressive solid tumor with a five-year survival rate of only 5-15%, presents a significant challenge in terms of treatment efficacy. Consequently, there is a pressing demand for the identification of a safe and efficacious therapeutic approach for CCA. Magnolol, a compound derived from the Chinese herbal medicine Magnolia officinalis, has garnered attention for its potential anti-cancer properties. Nonetheless, the precise mechanism underlying the interaction between magnolol and CCA remains ambiguous.
Aims & Methods: Aims: The aim of this research is to investigate the inhibitory effect of magnolol on the proliferation of cholangiocarcinoma and elucidate the underlying mechanism.
Methods: At the cellular level, a series of experiments were conducted utilizing CCA cell lines, including assessments such as cell viability assay, EdU assay, colony formation assay, Transwell assay, and wound-healing assay, to investigate the impact of magnolol on CCA. Furthermore, a zebrafish model of cholangiocarcinoma was developed to examine the effects of magnolol on proliferation, angiogenesis, metastasis, and apoptosis. Additionally, a subcutaneous xenograft tumor model was established in nude mice to explore the effects of magnolol *in vivo*. Lastly, bioinformatics analysis, molecular docking, and western blot were performed to investigate the anticancer mechanism of magnolol.
Results: This study, utilized cellular-level experiments to demonstrate the inhibitory effects of magnolol on the proliferation, migration, and invasion of CCA in a time- and dose-dependent manner. Additionally, a zebrafish model was established to confirm the efficacy of magnolol in suppressing tumor growth, angiogenesis, and metastasis, as well as inducing apoptosis in CCA cells. Moreover, magnolol demonstrates its anti-tumor efficacy at a lower concentration compared to gemcitabine. Furthermore, our study utilizing a subcutaneous xenograft tumor model illustrates that magnolol effectively inhibits the advancement of CCA with minimal toxicity. Mechanistically, magnolol was found to downregulate the JAK2/STAT3 signaling pathway.
Conclusion: Overall, our research verified that magnolol had promising anti-tumor effects by targeting the JAK2/STAT3 pathway, which laid a solid foundation for CCA management.
Disclosure: The authors declare that there are no conflicts of interest.

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PP1045

STENT-DIRECTED IRREVERSIBLE ELECTROPORATION FOR NOVEL MINIMALLY INVASIVE ENDOBILIARY NON-THERMAL ABLATION

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Introduction: Minimally invasive irreversible electroporation (IRE) has gained increasing attention for the treatment of endobiliary tumors.[1-3] However, the clinical application of IRE in hollow organs has been limited due to the lack of suitable electrode designs that can ensure sufficient electrode-tissue contact.[4] Recently, a novel stent-directed electrode (SDE) made of chemically polished nickel-titanium (Ni-Ti) alloy wires was developed for localized endobiliary IRE.

Aims & Methods: The aim of this study was to investigate feasibility and efficacy of localized IRE using a SDE in both *in vitro* and *in vivo* models. Ni-Ti wires were chemically polished to improve surface roughness and electrical conductivity. The SDE was fabricated using the polished wires, and its mechanical properties were characterized. The cytotoxicity and cell viability after IRE treatment were investigated using a colon cancer cell line. The optimal IRE parameter for the *in vivo* study was determined through triphenyl tetrazolium chloride-stained potato model (field strength: 500 V/cm, pulse width: 100 μs, pulse interval: 2,000 μs, number of pulses: 32). Subsequently, the safety and efficacy of endobiliary IRE using the SDE were evaluated in a porcine common bile duct model.

Results: The chemical polishing process significantly improved the surface roughness, current density, and electrical conductivity of the polished Ni-Ti wires while maintaining their mechanical properties compared with the control wire ($p < 0.001$). The polished Ni-Ti wire electrode effectively delivered electrical energy to cancer cells, cytotoxicity and cell viability were proportionally decreased according to the electric field strength increased ($p < 0.001$). IRE-treated potato tissue clearly revealed irreversibly and reversibly electroporated areas containing dead cells in electrical strength-dependent manner. *In vivo* studies showed that the SDE could be successfully delivered to the target site and achieve uniform and localized ablation of the ductal mucosa without causing severe complications. Histological analyses showed significantly increased apoptosis ($p < 0.001$), heat shock protein expression ($p < 0.01$), and inflammatory cell infiltration ($p < 0.05$) in the IRE-treated tissues compared to the sham control group.

Conclusion: The novel SDE made of chemically polished Ni-Ti wires demonstrated excellent performance for endobiliary IRE in terms of surface characteristics, electrical properties, and ablation efficacy. The *in vivo* experiments confirmed the safety and efficacy of the stent-directed IRE using the SE for the treatment of endobiliary tumors in a porcine model. These results suggest that this minimally invasive technique may provide a promising treatment option for endobiliary tumors of the gastrointestinal and urinary tracts. Additional preclinical and clinical studies are necessary to assess the long-term safety and efficacy of this technique for various clinical applications.

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Disclosure: Nothing to disclose.

PP1046

IMAGE-ENHANCED ENDOSCOPIC IMAGES FOR CATEGORIZING GALLBLADDER LESIONS IN RESECTED POST-CHOLECYSTECTOMY SPECIMENS: A PRELIMINARY STUDY

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Introduction: The diagnosis of gallbladder (GB) lesions depends on imaging findings, and the definitive diagnosis is still challenging. Meanwhile, the advancement of cholangioscopy has enabled targeted biopsy under direct vision, leading to the improved diagnosis of bile duct lesions.
We have reported cholangioscopy-guided targeted biopsy in the GB and characteristic endoscopic images of GB lesions by image-enhanced endoscopy (IEE)[1-3].
In the near future, biliary endoscopy with IEE may expectedly work for diagnosing GB lesions.

Aims & Methods: This study aimed to investigate the characteristics endoscopic IEE findings of each disease in the lumen of the GB in resected GB specimens. 50 patients who underwent cholecystectomy between November 2020 and July 2022 were included.
Immediately after cholecystectomy, resected GB specimens was incised contralaterally to the liver bed, and the GB lumen was observed with white light and Narrow Band Imaging (NBI) magnification under saline flooding whenever possible using a magnifying gastroscope (GIF-H290Z or XZ1200; Olympus). The obtained endoscopic images were contrasted with the pathological findings (Ethics Committee Approval No. R2-22, 20210507002).

Results: The preoperative diagnosis of 50 patients included 6 cases of GB cancer, 29 of cholelithiasis, 8 of GB polyps, 2 of adenomyosis of the GB, 1 of xanthogranulomatous cholecystitis, and 4 of gastric cancer. The lumen of the GB removed during gastric cancer surgery appeared to be normal, with a regular arrangement of cylindrical epithelium on endoscopic image and a uniform vascular pattern.
Cancerous areas in patients with early-stage cancer exhibited irregular papillary surface, multi-caliber and irregular distribution of vasculature, while cancerous areas in patients with advanced cancer showed non-structure and irregular dilated and tortuous vasculature. In other cases where chronic inflammation was suggested, the vessels were slightly dilated, but not irregularly dilated or tortuous.

Conclusion: Endoscopic IEE images in resected GB specimens were mainly characterized to distinguish the stage of GB cancers. This trial may contribute to the establishment of classification for GB lesions under direct vision. We also present a typical GB cancer diagnosed by cholangioscopic biopsy in clinical practice.

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PP1047

THE IMPACT OF ENDOSCOPIC BILIARY DRAINAGE ON THE NUTRITIONAL STATUS OF PATIENTS WITH MALIGNANT PERIHILAR OR PERIAMPULLARY BILIARY STRICTURES

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Introduction: Malignant biliary obstruction is the cause of numerous complications affecting nutritional status. Biliary drainage with stenting is the recommended palliative treatment for these patients. This study assesses the impact of biliary drainage on nutritional status by comparing patients before and after the procedure.
Aims & Methods: The aim of this study was to evaluate, through a retrospective analysis, the variation in patients nutritional status before and after biliary drainage wether in perihilar or periampullary strictures.
We studied 200 patients with malignant biliary stenosis who underwent endoscopic biliary drainage.
Demographic, anthropometric and clinical data were collected before and after drainage in two stages (1 month and 3 months). Statistical analysis was performed using SPSS26 software.

	BEFORE DRAINAGE	1 MONTH AFTER DRAINAGE		3 MONTHS AFTER DRAINAGE	
	Mean (SD) or %	Mean (SD) or %	P value	Mean (SD) or %	P value
WHO score					
-STAGE 0	4.1%	14.4%		25%	
- STAGE 1	43.2%	53.2%	0.000	53.6%	0.000
- STAGE 2	52.1%	24.3%		17.9%	
- STAGE 3	0.6%	3.6%		8.1%	
WEIGHT (Kg)	56.95 kg (±9.83)	57 kg (±10.11)	0.000	59.27 kg (±10.14)	0.000
BMI Kg/m^2	19.63 kg/ m^2 (±3.42)	19.66kg/m^2 (±3.40)	0.000	20.39 kg/m^2 (±3.46)	0.000
NRI					
Normal nutrition	0 (0%)	12 (6.84%)		25 (17.24%)	
Moderate undernutrition	70 (35%)	86 (48.86%)	0.000	79 (54.48%)	0.000
Severe undernutrition	130 (65%)	78 (44.31%)		41 (35.17%)	
ALBUMIN (ALB)	29 g/l (±4.31)	32 g/l (±5.48)	0.000	34.52 g/l (±4.5)	0.000
TOTAL BILIRUBIN (TB)	200.72mg/l (±105.94)	32.82mg/l (±34.3)	0.000	15.03mg/l (±23.96)	0.000
C-REACTIVE PROTEIN (CRP)	61.73 mg/l	57.1mg/l	0.89	56mg/l	0.80

Table.

Results: The results showed a significant increase in mean weight, reaching 59.27kg after 3 months (p<0.05). Mean BMI also increased significantly, (20.39kg/m^2 after 3 months (p<0.05)). NRI (Nutritional risk index) decreased significantly after drainage. 35.17% still had severe undernutrition after 3 months (p<0.05). Mean ALB (Albumin) increased significantly, (34.52g/l after 3 months (p<0.05)). Mean TB (Total Bilirubin) also decreased significantly, (15.03mg/l after 3 months (p<0.05)).
However, the decrease in CRP was not statistically significant. The WHO performance score showed significant improvement, with 25% of patients presenting with Stage 0 after 3 months. Pruritus and jaundice were absent in 100% of patients after 3 months, and vomiting absent in 90% of patients after 3 months (p<0.05). When comparing the perihilar and periampullary stenosis groups, no significant differences were observed for any of the parameters studied.
Conclusion: The study suggest that endoscopic biliary drainage significantly improves the nutritional and clinical status of patients with malignant biliary strictures, independently of the location of the obstruction.
Disclosure: No conflict of interest.

PP1048

ACUTE UPPER GASTROINTESTINAL BLEEDING IN PATIENTS WITH BILIARY TRACT CANCER

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Introduction: Upper gastrointestinal bleeding (UGIB) is associated with an unfavorable prognosis in cancer patients. In biliary tract cancers (BTCs), little is known about the occurrence of UGIB and its prognostic relevance for the course of the disease.
Aims & Methods: The aim of this study was to analyze the prevalence of UGIB in BTC patients, compare therapeutic approaches for efficacy, and identify parameters associated with survival and recurrence rates. This retrospective observational study included all patients with BTCs and UGIB, that were treated in our tertiary care hospital between 2010 and 2023.
Results: A total of 1033 patients with BTC were analyzed. 41 (4.0%) of these patients had a total of 60 bleeding events. At the time of the bleeding event, 30.8% of patients displayed symptoms of shock, and 67.5% received red blood cell transfusions.
The most common causes of bleeding were tumor infiltration (24.4%), variceal bleeding (19.5%), or ulcer bleeding (14.6%). At the time of presentation, 56.4% of patients had stage IV cancer. Most patients had perihilar (39.0%) or intrahepatic (34.1%) cholangiocarcinoma.
At index bleeding, endoscopic treatment was performed in 41,5% of cases, while conservative measures (53,7%) or interventional radiological procedures (4.9%) were used in the remaining cases. In cases of bleeding caused by tumor infiltration, endoscopic hemostasis was usually performed using hemostatic powder (18.8%) or clip application (12.5%).
Admission to an intermediate or intensive care unit (odds ratio 6.7; 95% CI 1.5 – 33.9) as well as symptoms of shock at the time of the bleeding event (odds ratio 2.6; 95% CI 0.5 – 15.6) were risk factors of rebleeding within 14 days.
In patients with UGIB due to tumor infiltration, bleeding recurred after endoscopic therapy in 33.3% of cases. With conservative therapy, the recurrence rate was 28.6%. Survival of Patients with UGIB due to tumor infiltration was significantly shorter compared with patients having non-tumor-associated UGIB (median survival 0.93 vs. 6.5 months, log-rank test p = 0.018).

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Conclusion: UGIB is a rare but serious complication in patients with BTC. Patients with major bleeding episodes should be closely monitored for short-term rebleeding. The occurrence of UGIB due to tumor infiltration is associated with adverse outcomes and significantly shortened survival time.

Disclosure: Nothing to disclose.

PP1049

CONTRIBUTION OF BILIARY MRI IN THE DIAGNOSIS OF EXTRAHEPATIC CHOLANGIOCARCINOMA

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Introduction: Cholangiocarcinoma defines a group of tumors with a poor prognosis arising from the epithelial cells of the bile ducts. They can occur anywhere along the biliary tree. The diagnosis of cholangiocarcinoma represents a real challenge, given the difficulties of accessing a cytological diagnosis. Anatomopathological diagnosis remains the "gold standard", but in certain situations, biliary-MRI may suffice to make a positive diagnosis.

Aims & Methods: The aim of our work is to study the contribution of biliary-MRI to the diagnosis of extrahepatic cholangiocarcinoma.

This is a retrospective descriptive monocentric study extending over a 02-year period (January 2021-August 2023) collecting cases of extrahepatic cholangiocarcinoma hospitalized in the Hepato-Gastro-Enterology and Proctology Department "Medicine B".

We only included patients whose diagnosis was evoked by the radiological appearance of biliary MRI.

Results: 27 patients were included in our study, with a mean age of 65.5 ± 10 years. Male predominance was observed, with 16 men (59.2%) and 11 women (40.7%), so, a M/F sex ratio of 1.45.

The circumstances of discovery were dominated by altered general condition, which was reported in 18 patients (66.6%), jaundice in 16 (59.2%), and angiocholitis in 11 (40.7%). Biologically, median total bilirubin was 203 [129- 264], median CEA 6 [3- 120], and median CA 19-9 976 [60- 1933].

The tumor was distal in 6 patients (22.2%), and hilar in 21 (77.7%), including 13 cases (48.1%) classified as bismuth IV, 3 cases (11.1%) as bismuth III; bismuth II in 3 (11.1%), and bismuth I in one (3.7%). The process was infiltrative in 25 cases (92.5%), budding in 2 cases (7.4%), and stenosing in 19 cases (70.3%). In all our patients, the cholangiocarcinoma presented as an irregular process, enhanced late and progressively after contrast injection, with diffusion restriction, and with different signal intensities: Hyposignal T1 in 19 cases (70.3%), iso signal T1 in 07 cases (25.9%), hyper signal T2 in 14 cases (51.8%), intermediate signal T2 in 11 patients (40.7%), and iso signal T2 in 2 patients (7.4%). Liver metastases were noted in 9 patients: bilobar in 8 (29.6%) and unilobar in one (3.7%).

The diagnosis of cholangiocarcinoma was confirmed histologically in 11 patients (40.7%). Only 3 patients (11.1%) underwent surgery, while treatment was palliative in 25 patients (92.5%).

Conclusion: The definitive diagnosis of cholangiocarcinoma remains histological, but Biliary-MRI plays an essential role. Our study showed that late and progressive enhancement with T1 hyposignal and T2 hypersignal were the most frequent cholangiographic sequences in the diagnosis of extrahepatic cholangiocarcinoma.

Disclosure: Nothing to disclose.

PP1050

SHORT VERSUS LONG COURSE ANTIBIOTICS AFTER ERCP IN PATIENTS WITH ACUTE CHOLANGITIS IN A REFERRAL HOSPITAL

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Introduction: Acute cholangitis is an important cause of mortality and morbidity, especially in elderly patients and those with multiple comorbidities. However, the appropriate duration of antibiotic therapy after biliary drainage remains controversial. The aim of this study is to evaluate the efficacy of a short versus a long course of antibiotics after biliary drainage in patients with acute cholangitis.

Aims & Methods: Retrospective analytical cohort study of patients with acute cholangitis treated at a fourth level hospital in Bogotá, Colombia, from 2014 to 2022. Short antibiotic therapy was defined as ≤ 4 days after successful post-ERCP drainage. The primary outcome was a composite of in-hospital mortality, ICU admission, and hospital readmission 30 days after discharge.

A univariate and multivariate logistic regression model was performed to assess the association between duration of antibiotic therapy and the primary composite outcome.

	Univariate analysis		Multivariate analysis	
Variable	OR (CI 95%)	p value	OR (CI 95%)	p value
SCT	1.73 (0.76-3.89)	0.18	-	-
Oncological pathology	1.51 (0.47-4.87)	0.48	1.77 (0.96-3.29)	0.06
Previous antibiotic use*	0.66 (0.15 -2.87)	0.57	-	-
Previous hospitalization*	0.86 (0.19-3.51)	0.84	-	-
Etiology			-	-
Non-tumorous	Reference	Ref	-	-
Tumor	0.80 (0.25-2.6)	0.70		
Severity (Tokyo score)				
I	Reference	Ref	2.70 (0.98-9.54)	0.07
II	2.72 (0.96-9.77)	0.08	32.07 (11.84 113.16)	<0.001
III	30.67 (10.57-113.97)	<0.001		
Bacteremia	1.34 (0.68-2.59)	0.38	-	-
Carbapenem resistance	5.21 (1.24-23.66)	0.02	4.07 (1.02-16.96)	0.04

Table. Univariate and multivariate analysis of clinical factors associated with mortality, ICU admission or hospital readmission at 30 days in patients with acute cholangitis undergoing ERCP

**In the three months prior to the index hospitalization. SCT: Short course therapy; OR: Odds Ratio, 95% CI: 95% Confidence Interval.*

Results: 317 patients were included. E. coli was the most commonly isolated microorganism regardless of severity, and 54% proved 100% antimicrobial susceptibility. 59 patients received short therapy and 258 received long therapy. There were no statistically significant differences in the primary outcome (p=1), but a longer hospital stay was observed in the long-term treatment group (p<0.001). The presence of Tokyo III severity and carbapenem resistance were risk factors for the primary outcome (OR 32.07; 95% CI 11.84-113.16, p<0.001) and (OR 4.07; 95% CI 1.02-16.96, p=0.04) respectively.

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Conclusion: Shorter courses of antibiotics after ERCP drainage can be considered a therapeutic option for patients with acute cholangitis. However, the methodological limitations of this study should be considered.

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Disclosure: Nothing to disclose.

PP1051

SINGLE-SESSION ERCP WITH COLONOSCOPE AND DUODENOSCOPE IN PATIENTS WITH SURGICALLY ALTERED ANATOMY: A FEASIBILITY STUDY

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Introduction: ERCP in patients with surgically altered anatomy (SAA) is one of the most challenging endoscopic procedures. For patients in whom cannulation cannot be achieved with a colonoscope, we placed a tube with the tip near the papilla/anastomosis, and the end out of patient's mouth; a duodenoscope was then inserted following tube placement to finish ERCP in the same session.

Aims & Methods: The aim of the current study was the evaluation of feasibility and safety of two-step ERCP (Figure 1) for biliopancreatic interventions in patients with SAA. Patients with SAA receiving two-step ERCP at a single endoscopic reference center were retrospectively enrolled between January 2016 and June 2023.

Results: 58 patients (20 female, 38 male), median age 58 years (range 43-88) and biliary (n=45) and pancreatic (n=13) indications for two-step-ERCP were enrolled. Types of surgical reconstruction: Roux-enY n=47, Billroth II with Braun anastomosis n=11.

Technical success rates for intubation, cannulation and interventions were 94.8%, 85.5% and 100%, respectively. Overall two-step-ERCP-success rate was 77.6%. One major complication occurred (1.7%; intestinal perforation during intubation).

Conclusion: The current study showed for the first-time feasibility and safety of two-step-ERCP with colonoscope and duodenoscope in SAA patients at an expert center. This data justifies further evaluation of this new technique preferably in a prospective multicenter trial.

Disclosure: Nothing to disclose.

PP1052

UNFORTUNATE GENDER DISPARITIES IN PATIENTS WITH SPHINCTER OF ODDI DYSFUNCTION (SOD) UNDERGOING ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY (ERCP): A DECADE LONG STUDY

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Introduction: The sphincter of Oddi (SOD) is a valve composed of smooth muscle which helps regulate the flow of pancreaticobiliary secretions into the duodenum. SOD can involve the biliary sphincter, pancreatic sphincter, or both. Several studies have shown that patients with SOD have higher complications when undergoing endoscopic retrograde cholangiopancreatography (ERCP).

This study aimed to identify risk factors that contribute to worsened in-hospital outcomes in patients with SOD undergoing ERCP as well as to identify high-risk populations. We sought to identify gender differences between these two populations because gender disparities in health care can be linked to worse health outcomes.

Aims & Methods: The National Inpatient Sample (NIS) is an extensive, publicly available all-payer inpatient care database in the USA. We distinguished between adult patients (> 18 years old) undergoing ERCP with SOD and those without SOD from the years 2010-2020.

Multivariate logistic regression and linear regression analyses using the Kernel Method with propensity matching were employed to account for confounders. STATA 17 MP was used to compute the odds ratios of in-hospital mortality, average length of hospital stay (LOS), and hospital charges (TOTHC). Weights were used to generate nationally and regionally representative estimates.

Results: 3,100 of the 2,176,350 patients undergoing ERCP (2010-2020) had SOD, of which a total of 2,564 female patients (82.68%) were found. Female patients had a significantly higher mortality than males (15.55% Female vs. 3.11% Male, p<0.01) as well as a higher overall mortality rate (OR 2.5, CI 2.1-2.9, p<0.0001).

Additional gender disparities were found after adjusting for age, race, BMI, hospital region, hospital bed size, socioeconomic status, teaching status (teach vs. non-teach), and hospital location (urban vs. rural): female patients had higher LOS (+5.5 days, CI 4.5-6.5 days, p<0.0001) and TOTHC (+$51,856, CI $47,533-$56,179, p<0.0001) compared to male patients. Analysis of post-procedural complications is noted in Table 1.

Conclusion: Significant disparities between male and female patients were observed in this retrospective nationwide cohort of over 2.1 million patients. Female patients with SOD undergoing ERCP had higher overall mortality, mortality rate, and increased LOS and TOTHC when compared to males.

Our study identified risk factors and post-procedural complications such as aspiration, perforation, sepsis, etc (outlined in Table 1) in patients with SOD undergoing ERCP that lead to worse overall outcomes. To help minimize risks and lessen the healthcare burden, the underlying causes of gender differences should be further investigated through randomized controlled trials.

Disclosure: Nothing to disclose.

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PP1053

THE IMPACT OF COVID-19 PANDEMIC ON ERCP IN THE AOSTA VALLEY TERTIARY CARE MEDICAL CENTER

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) is an essential technique for pancreato-biliary diseases. Pancreatico-biliary infection occurs after 0.4–0.8% of endoscopic biliary procedures, with cholangitis and sepsis occurring in up to 3% of cases. These episodes must always be taken seriously because fatality has been reported in 8–20% of such cases.

During the pandemic from coronavirus disease 19, recommendations by the European Society of Gastrointestinal Endoscopy (ESGE) suggested postponing elective procedures, use of triage, use of personal protective equipment and creation of differentiated in-hospital pathways, in order to protect both patients and endoscopy unit personnel.

Currently, it is not know whether, as a result of the change in these procedures, the overall mortality and the infection's rate related to ERCP has been modified in the post-COVID era.

Aims & Methods: In this study, we retrospectively compared ERCP activity, post-ERCP infection rate and 30 days-mortality in our tertiary center before and after the adoption of new criteria for patient's selection and protective equipment implementation, specifically from October 2016 to December 2019 (before COVID) and from January 2020 to March 2023 (after COVID).

Results: During the 39 months before COVID, we performed 417 ERCP for 285 patients, while in the same amount of time after COVID, we executed 257 consecutive ERCP for 191 patients. The indications for performing ERCP were made for obstructive cholangitis caused by stones, by pancreatic or bile ducts neoplasms, by malignant lymph nodes or for post-surgical complications.

Positive blood cultures done within 1 week from ERCP were 13 among 417 exams before COVID and 9 among 257 ERCP after COVID, with septicemia related to ERCP only in 2 procedures (0.5%) before COVID and another 2 (0.8%) after COVID. Cultures resulted positive for Klebsiella pneumoniae and Candida albicans in the first group, while in the group after COVID, patient's blood cultures were positive for Enterococcus faecium and Klebsiella oxytoca.

Overall mortality at 30-days after ERCP accounted for 16 patients (5.6%) during 39 months considered before COVID and 9 patients (4.7%) after COVID. In the first group, the fatal complications were related to ERCP in 2 cases (0.7% of patients), due to the perforation of the digestive tract, while in the second group, 1 patient (0.5%) had a septic shock from Klebsiella oxytoca after ERCP.

Conclusion: In conclusion, our retrospective analysis of a real life cohort of consecutive patients undergoing ERCP in a single tertiary center, demonstrated that the shortages of medical resources caused a significant drop in the ERCP activity, notably about 38%.

Ultimately, the delay due to the COVID pandemic, the rational use of antibiotics and the intensification of personal protective equipment did not change significantly the rate of adverse events, like post-ERCP septicemia (0.5-0.8% before and after COVID respectively), or the 30-days mortality rate after ERCP (0.7%-0.5% before and after COVID respectively).

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PP1054

IN-HOSPITAL OUTCOMES OF PATIENTS WITH COVID-19 UNDERGOING ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY (ERCP)

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Introduction: The novel coronavirus emerged in 2019 and led to a worldwide pandemic. COVID-19 had a catastrophic effect on the world's population, resulting in millions of deaths. Our study aimed to evaluate and determine the outcomes of patients who had an underlying diagnosis of COVID-19 and to assess overall mortality, length of stay (LOS), and total hospital charges (TOTHC) associated with patients undergoing ERCP who had a concomitant diagnosis of COVID-19.

Aims & Methods: We identified patients 18 years and older with COVID-19 undergoing ERCP versus those undergoing ERCP only from the National Inpatient Sample (NIS) from 2020. The NIS is a large publicly available all-payer inpatient care database in the USA. Using STATA 17 MP, multivariate regression analysis was used to calculate the odds ratios of in-hospital mortality, the average LOS, and TOTHC.

Results: We identified 1,685,745 patients with COVID-19, of which 161,985 underwent ERCP. The analysis revealed that mortality was increased (OR 1.41, p<0.001, CI: 1.24 - 1.69), LOS was increased (+2.59 days, p<0.0001, CI: 1.17 - 4.01), and TOTHC were increased (+ $56,220.52, p<0.0001, CI: $28,340.61- $84,100.44) for patients undergoing ERCP with COVID 19 compared to patients undergoing ERCP without COVID-19.

The presence of acute kidney injury (AKI) (OR 6.49, p<0.0001, CI: 6.18-6.81), Stroke (CVA) (OR 1.75, p<0.035, CI: 1.04- 2.94), Non-ST-elevation myocardial infarction (NSTEMI) (OR 2.58, p<0.0001, CI: 2.41- 2.76), coronary artery disease (CAD) (OR 1.47, p<0.001, CI: 1.182 -1.83), and hemodialysis (HD) (OR 4.37, p<0.0001, CI: 4.02- 4.75) worsened mortality in patients with COVID-19 undergoing ERCP.

The presence of AKI (+ 8.56 days, p<0.0001, CI: 8.20- 8.92), venous thrombo-embolism (+8.82 days, p<0.0001, CI: 6.83- 10.82), CVA (+3.92 days,

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p<0.003, CI:1.35 - 6.48), and HD (+5.72 days, p<0.0001, CI: 5.32 - 6.11) led to an increase in LOS in patients with COVID-19 undergoing ERCP.
The presence of hypertension ($1,577.52, p<0.021, CI: $241.04- $2,914.00), AKI ($187,274.00, p<0.0001, CI: $175,994.50 - $198,553.50), VTE ($103,434.30, p<0.0001, CI: $70,698.35 - $136,170.30), cerebrovascular accident (CVA) ($50,131.06, p<0.034, CI: $3,697.798- $96,564.33), NSTEMI ($ 38,293.22 , p<0.0001, CI: $31,289.62 - $45,296.81), obesity ($11,439.68, p<0.0001, CI: $9,247.63 - $13,631.74) and HD ($135,782.10, p<0.0001, CI: $124,296.70- $147,267.60) led to increased TOTHC in patients with COVID-19 undergoing ERCP.
Conclusion: Compared to patients without COVID-19, concomitant COVID-19 in patients undergoing ERCP led to an increased overall mortality, total hospital charges, and length of stay. We also identified positive predictors of mortality, as well as increased LOS and TOTHC in this patient population.
Disclosure: Nothing to disclose.

PP1055

SAFETY AND EFFICACY OF ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY IN CIRRHOTIC PATIENTS WITH OBSTRUCTIVE JAUNDICE

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) has become the main treatment modality for many pancreatobiliary disorders. There are limited prospective studies regarding safety of ERCP in cirrhotic patients. We evaluated the outcome of ERCP in a group of cirrhotic patients in comparison to non cirrhotic group.
Aims & Methods: the aim of the study was to investigate complications and success of ERCP in cirrhotic patients. In this prospective (cohort) comparative study, 54 cirrhotic patients undergoing therapeutic ERCP were compared to 65 non cirrhotics as control group. Basic patient pre endoscopic, interventional data and 30 days follow up data were collected and analysed.
Results: One hundred and nineteen consecutive patients were prospectively enrolled in this study. The most common indication for ERCP in both groups was choledocholithiasis. (70.4% in cirrhotics vs 53.8% in non cirrhotics). There was no significant difference in the frequency of overall complications between cirrhotics and noncirrhotics (11.1% vs 12.3 respectively, P=0.84). Incidence of individual complications in cirrhotics versus non cirrhotics were as follows :Post-ERCP pancreatitis (PEP) (0% vs 4.6%, P=0.250), bleeding (3.7 vs 3.1%, p=0.375), cholangitis (7.4% vs 3.1%, p=0.4) , perforation (0% vs 1.5%). All were non significant.
Regarding 30 days mortality among our patients, it was higher in the cirrhotic group but did not reach the level of statistical significance (11.5% in cirrhotics versus 4.6% in non cirrhotics; P=0.18). Clinical success was insignificantly higher in the non cirrhotic group (87.7% vs 79.6%, p=0. 478). On multivariable analysis, the independent predictors of mortality among our patients were previous history of variceal bleeding, presence of splenomegaly and ascites and occurrence of post ERCP cholangitis. There was significant delay in improvement of serum bilirubin level in the cirrhotic group.
Conclusion: ERCP is considered safe and effective (as regard clinical success) in cirrhotic patients with results comparable to non cirrhotics, however advanced status of liver disease leads to delay in improvement of serum bilirubin in cirrhotic patients and this delay should not be considered as failure of ERCP procedure.
Disclosure: Nothing to disclose.

PP1056

MORPHOLOGY OF THE PAPILLA CAN PREDICT PROCEDURAL SAFETY AND EFFICACY OF ERCP - A RETROSPECTIVE COHORT STUDY

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Introduction: Endoscopic Retrograde Cholangiopancreatography (ERCP) is the most commonly used therapeutic procedure for pancreaticobiliary disorders. Despite the essential role of bile duct cannulation in procedural safety and success, research on this topic is still limited. Certain studies suggest that the anatomy of the papilla affects its efficacy and safety.
Aims & Methods: We aimed to assess the influence of papilla morphology on ERCP outcomes and adverse events.
Patients with a naive papilla scheduled for ERCP were reviewed in a single tertiary care center between September 2022 and January 2023. The papilla was classified into one of four types using the Haralddson classification system. Data on the presence of a periampullary diverticulum (PAD) was also assessed. The indications for ERCP, the rate of post-ERCP adverse events (pancreatitis, bleeding, and perforation), the rate of failed cannulation, and the need and type for rescue cannulations were collected. For the effect size measure, event rates or risk ratios were calculated. For categorical data, the χ^2 test, and for the test of subgroup (papilla type) differences, a “Cochrane Q“ test was used.
Results: A total of 231 patients with naive papillae were included. We found a statistically significant difference in the event rate for cannulation failure between the different papilla types. The highest rate was observed in the case of type II (“small”) papilla (14%), followed by type I (“regular”) (13%) and type III (“protruding or bulging”) (8%). The event rate was lowest in the case of type IV (“creased or ridged”) papilla (6%). The presence of PAD was associated with a higher risk of cannulation failure (RR: 2.64; CI: 1.22 – 5.39).
Moreover, a statistically significant difference in the rate of post-ERCP pancreatitis was observed between the papilla types. The event rate was the highest in type III (“protruding or bulging”) and IV (“creased or ridged”) papilla (13%). The lowest rate rate was observed in type I (“regular”) and II (“small”) papillae, 6%. There was no significant difference between the post-ERCP bleeding and perforation rate in the different papilla types, likely due to the low event number. The need and the types of the various rescue cannulation techniques also significantly differed between the papilla types.
Conclusion: Certain papilla morphologies are associated with a higher rate of cannulation failure and PEP. The presence of PAD is also associated with an increased rate of cannulation failure. Moreover, we can conclude that the preferred rescue cannulation technique differs between the papilla types.
Disclosure: Nothing to disclose.

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PP1057

NATIVE PAPILLA CANNULATION IS NOT ASSOCIATED WITH INCREASED ADVERSE EVENTS OR TECHNICAL FAILURE RATES AMONG TRAINEES

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Introduction: The deep selective cannulation of the bile duct or pancreatic duct has been described as the cornerstone of a successful endoscopic retrograde cholangiopancreatography (ERCP), and perhaps the most challenging aspect of the procedure for trainees. A selective cannulation rate of >80–90% of native papilla cases is usually considered as a good surrogate marker of trainee competence. We aimed to assess whether the presence of a native papilla influences differences in ERCP outcomes and adverse events between trainees and attending endoscopists.

Aims & Methods: We conducted a retrospective study of 1384 patients who underwent ERCP in a single large tertiary insititution between January 2019 and July 2023. Clinical data were prospectively collected during routine ERCP procedures. Demographics, native papilla rates, ASGE-grading system of procedure complexity, cholangioscopy, procedure and fluoroscopy time, adverse events, and technical success rates were compared between ERCP with and without trainee involvement. Technical success was defined as successful cannulation and achievement of therapeutic goal. Categorical variables were compared using Pearson's chi-square test and continuous variables using a standard t-test. Multivariate regressions were performed to adjust for sex, age, clinical setting, and ASGE complexity.

Results: A total number of 903 (65.3%) procedures had a trainee involved and 481 (34.7%) did not have a trainee involved. The mean patient age was 62 years, 53% were male. The proportion of difficult cases (ASGE Grade ≥3) was in 17.3% with trainees involved vs. 15.4% without trainees involved (p = 0.504). The proportion of native papilla was 44.2% with trainee involvement vs. 42% without trainee involvement (p = 0.434). The native papilla was successfully cannulated in 94% with trainees vs. 93.6% without trainees involved (p = 0.839).

Cholangioscopy was used in 11.3% of cases with trainees involved vs. 9.2% without trainees involved (p = 0.218). Trainee participation added 4.8 mins to average procedure time (aOR: 1.04, p = 0.001) and 0.6 mins to fluoroscopy time (aOR: 0.93, p = 0.159). No significant difference was noted between trainees and attending endoscopist in the odds of ERCP adverse events (aOR: 1.9, p = 0.27) or native papilla cannulation/technical success (aOR: 3.11, p = 0.062) for native papilla cases, but procedure time remained longer among trainees (aOR 1.03, p = 0.008)

Conclusion: Native papilla cannulation success > 90% was achieved in this study with trainee participation. Native papilla cannulation performed by trainees was associated with increased procedure time but was not associated with a significant difference in ERCP technical success rates or adverse events.

Variable	aOR (95% CI)	P-value[b]
Procedure time [a]	1.031 (1.008-1.054)	0.0075
Fluoroscopy time [a]	0.944 (0.849-1.051)	0.2941
Cholangioscopy	0.681 (0.282-1.646)	0.3935
Native papilla cannulation/ ERCP technical success	3.107 (0.946-10.21)	0.0618
ERCP adverse events	1.895 (0.608-5.905)	0.2701

Table. Adjusted odds ratios for ERCP outcomes with and without trainee involvement for native papilla only

Abbreviations: aOR, adjusted odds ratio; CI, confidence interval; ERCP, endoscopic retrograde cholangiopancreatography.

a. Based on increasing time intervals

b. Adjusting for sex, age, and clinical setting.

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Disclosure: Nothing to disclose.

PP1058

IMPACT OF THE ASGE GRADING SYSTEM OF ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY (ERCP) ON ERCP OUTCOMES IN A TERTIARY REFERRAL CENTER

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Introduction: The American Society for Gastrointestinal Endoscopy (ASGE) grading system for endoscopic retrograde cholangiopancreatography (ERCP) has been shown to predict success and adverse events rates. We compared ERCP outcomes and adverse events between complex and non-complex procedures using the ASGE-grading system. Our study also sought to assess the impact of the ASGE-grading system on ERCP outcomes with and without trainee involvement.

Aims & Methods: We conducted a retrospective study of 1392 patients who underwent ERCP procedures in a single large tertiary insititution between January 2019 and July 2023. Clinical data were prospectively collected during routine ERCP procedures. Demographics, cholangioscopy, procedure and fluoroscopy time, adverse events, and technical success rates were compared between complex (ASGE Grade ≥3) and non-complex (ASGE Grade <3) ERCP procedures as well as with and without trainee involvement. Categorical variables were compared using Pearson's chi-square test and continuous variables using a standard t-test. Multivariate regressions were performed to adjust for sex, age, clinical setting, and native papilla.

Results: Approximately 83.3% of ERCPs were ASGE Grade <3 and 16.7% were ASGE Grade ≥3. The mean patient age was 62 years, 53% were male. The ampulla was native in 45.3% of ASGE Grade <3 vs. 34.1% of ASGE Grade ≥3 (P=0.002). Cholangioscopy was needed in 1% of ASGE Grade <3 vs. 58.6% of ASGE Grade ≥3 (aOR: 72.2, P <0.001). Procedure and fluroscopy time were longer by 25.4 mins (aOR: 1.04, P < 0.001) and 5 mins (aOR: 1.24, P < 0.001) respectively for ASGE Grade ≥3 . Adverse events (aOR: 0.44, P= 0.274) and technical success (aOR: 1.00, P= 0.983) were similar in both groups.

No significant difference was noted between trainees and attending endoscopist in the odds of ERCP adverse events or technical success for ASGE Grade <3 or ASGE Grade ≥3 respectively.

Conclusion: ERCPs with ASGE Grade ≥3 is associated with increased procedure and fluoroscopy time, and displayed increased odds of requiring cholangioscopy. ERCP technical success and adverse events did not significantly differ with trainee participation regardless of procedural complexity.

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Variable	aOR (95% CI)	P-value[b]
Procedure time [a]	1.040 (1.021- 1.059)	< 0.001
Fluoroscopy time [a]	1.236 (1.137- 1.343)	< 0.001
Cholangioscopy	72.2 (34.3- 151.6)	< 0.001
ERCP technical success	1.00	0.983
ERCP adverse events	0.438 (0.1-1.9)	0.274

Table. ERCP outcomes for complex (ASGE Grade ≥ 3) vs. non-complex procedures (ASGE Grade <3). N=1392.

Abbreviations: aOR, adjusted odds ratio; CI, confidence interval; ERCP, endoscopic retrograde cholangiopancreatography.

a. Based on increasing time intervals

b. Adjusting for native papilla

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Disclosure: Nothing to disclose.

PP1059

EFFICACY OF DETACHABLE BILIARY PLASTIC STENTS WITHOUT FLANGES IN PREVENTING POST-ERCP CHOLANGITIS IN PATIENTS WITH COMMON BILE DUCT STONES: A COMPARATIVE STUDY

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Introduction: ERCP is effective for common bile duct stone treatment but carries post-op risks like pancreatitis and cholangitis. Biliary drainage post-surgery is vital, with methods like ENBD and ERBD being common. ENBD allows monitoring but has drawbacks like infection and inconvenience. ERBD mitigates these but can cause complications. A new solution, Detachable biliary plastic stents without flanges, uses soft materials for self-expulsion, reducing the need for further procedures and patient burden.

Aims & Methods: This study aimed to investigate the efficacy and safety of using detachable biliary plastic stents without flanges for the prevention of cholangitis following endoscopic retrograde cholangiopancreatography (ERCP) in patients with common bile duct stones.

The study cohort comprised patients diagnosed with common bile duct stones and treated at the First People's Hospital affiliated with West Lake University School of Medicine between January 2019 and January 2024. Participants were randomly assigned to either the group receiving detachable biliary plastic stents without flanges or the nasobiliary drainage group. In the former group, the stents were deployed for biliary drainage following the clearance of stones during ERCP. The latter group received nasobiliary drains for similar purposes. Comprehensive clinical data were gathered, encompassing preoperative and postoperative details, including laboratory investigations, endoscopic procedures, complications, hospital stay, and comfort levels assessed using the Visual Analogue Scale (VAS). Additionally, the incidence of cholangitis, the rate of spontaneous drain removal, and stent detachment were compared between the two groups.

Results: 1. The study included 820 patients who underwent ERCP for common bile duct stones. After accounting for 10 patients lost to follow-up, a total of 810 patients were analyzed, with equal distribution across both study groups. The incidence of postoperative cholangitis was 2.0% in the stent group and 3.0% in the nasobiliary drainage group, showing no significant difference (P=0.365). Pancreatitis occurred in 2.7% of patients in the stent group and 4.2% in the drainage group, also not statistically significant (P=0.248). No instances of perforation or bleeding were reported in either group.

2. The median postoperative hospital stay was significantly shorter in the stent group at 2.0 days (range: 1.0 to 4.0d) compared to 3.0 days (range: 2.0 to 5.0d) in the nasobiliary drainage group (P<0.001).

3. The VAS scores for postoperative comfort were significantly lower in the stent group with a mean of 2.86±1.08 versus 5.13±2.19 in the nasobiliary drainage group (P<0.001).

4. All 405 patients in the stent group returned for follow-up one month postoperatively. Stent detachment was confirmed in 373 patients (92.1%). Among the 32 patients with retained stents, two developed cholangitis, and two reported recurrent abdominal pain, resulting in a complication rate of 1.0%. In the nasobiliary drainage group, 10 patients (2.5%) spontaneously removed their drains, with an average age of 72.2±8.3 years.

Conclusion: The deployment of detachable biliary plastic stents without flanges following ERCP for common bile duct stones is as effective as traditional nasobiliary drainage in preventing postoperative cholangitis. This approach offers enhanced patient comfort, reduced hospital stay, and a high rate of spontaneous stent detachment, demonstrating its safety and efficacy in clinical practice.

Disclosure: The authors declare that there are no conflicts of interest.

PP1060

PREVALENCE, CLINICAL FEATURES AND ROLE OF ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY (ERCP) IN BILIARY ASCARIASIS MANAGEMENT

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Introduction: Hepatobiliary and pancreatic ascariasis (HPA) is caused by entry of the nematode, Ascaris lumbricoides (A. lumbricoides), from the duodenum in to the biliary and pancreatic ductal lumen[1]. Patient present with wide spectrum of symptoms of biliary and pancreatic disease.i.e 1.biliary colic

2. acute cholangitis.3.acalculous cholecystitis.4 hepatic abscess

5. acute pancreatitis 6. recurrent pyogenic cholangitis. Ascariasis is prevalent worldwide with an overall prevalence of 25%. An estimated 1.4 billion people are infected.

Aims & Methods: To evaluate the prevalance,clinical features, technical success, complications, and role of ERCP in management of biliary ascariasis.The results of all the procedures were evaluated retrospectively and the prevalence of biliary parasitosis was evaluated.Technical success of the procedure was defined as achieving the diagnostic information or succeeding with endoscopic therapy.

Results: A total of 6070 patients were treated with ERCP for different indications in our departments

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All ERCPs were analyzed during a 6-year period from jan 2018 to dec 2023.29 patients were identified with biliary ascariasis on the basis of ERCP results.There were 7 men and 22 women, with a mean age of 28.5 years ranging from 16 to 55 years. Most of the patients were from Karachi (25, 86.2%). Remaining patients were from Hyderabad (2, 6.89%) and Sukkur (2, 6.89%). abdominal pain was the predominant symptom in all patients with duration ranging from 1 week up to 3 years. Bilirubin was raised in 16 out of 29 (55.17%) patients with mean bilirubin of 7.7 mg/dl and range of 2.16 to 26.7 mg/dl.ALP was raised in 27 out of 29 patients (93.1%) with range of 140 u/L to 2670 u/l (normal upper limit 120u/l and mean of 448.36 u/l.ultrasound abdomen showed worm in 12 out of 29 patients (41.3%) and Dilated cbd was seen in 25 out of 29 patients (86.21%) With mean of 1.1cm .4 patients had dilated IHD as well. On ERCP Worm were seen as linear filling defect in common bile duct (CBD) in 15 (51.72%) cases and CBD along with intrahepatic ducts in 3 cases (10.34%).11(37.93%) patients had worm coming out of ampullary orifice and was removed by a grasper. Out of 29 patients, 5 had stone in CBD and three had pus indicating cholangitis.Only 2 patients developed minor bleeding which was controlled without any intervention.

In all of the cases we successfully and safely removed all the worms endoscopically without having any surgical outcome.The worms were removed by balloon trawl.All the patients received mebendazole following the endoscopic therapy and were dewormed. There was no procedure-related mortality in our patients.

Conclusion: ERCP is a contemporary procedure; our results are the depiction of the fact that ERCP should be utilized for the treatment of biliary ascariasis as a prime procedure.

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Disclosure: Nothing to disclose.

PP1061

PERIAMPULLARY DIVERTICULUM INTERFERES ERCP PROCEDURES AND PROMOTES THE RECURRENCE OF COMMON BILE DUCT STONES

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Introduction: Periampullary diverticulum (PAD) is ordinarily asymptomatic, which is inspected occasionally during gastroscopy and endoscopic retrograde cholangiopancreatography (ERCP). Previous studies demonstrated inconsistent effect of PAD on ERCP procedures and choledocholithiasis recurrence. Thus, this study was aimed to investigate the impact of different types of PAD on the technical success, post-ERCP complications and the safety of different stone extraction methods, as well as the association of PAD with the recurrence of common bile duct stones (CBDS).

Aims & Methods: The ERCP database (March 2019 to March 2021) in West China Hospital of Sichuan University was retrospectively reviewed. Inclusion criteria were: 1) presence of original duodenal papilla; 2) patients with CBDS. However, patients with the following characteristics were excluded: 1) ERCP for other reasons; 2) absence of original papilla; 3) duodenal ampullary mass; 4) anatomical abnormalities caused by surgery; 5) under 18 years old; 6) incomplete clinical data.

In this study, PAD is classified into three types based on the location of the duodenal papilla within the diverticulum (type I), the edge of the diverticulum (type II), and the outside of the diverticulum (type III). The successful cannulation, procedures of ERCP and post-ERCP complications were compared between the PAD group and the non-PAD group, as well as among the different types of PAD groups. The patients with successful stone removal were followed up every 6 months until December 2023 to determine whether PAD was associated with the recurrence of CBDS.

Results: A total of 830 CBDS cases with original duodenal papilla were eventually enrolled, including 239 cases with PAD and 591 cases without PAD. The successful cannulation rate in the PAD group was lower than that in the non-PAD group (96.7% vs. 99.2%, P = 0.013). According to the different types of PAD, the successful cannulation rate of type I PAD was lower compared with non-PAD group (76.9% vs. 99.2%, P<0.001). Additionally, PAD was associated with a wider bile duct diameter (1.55 ± 0.58 vs. 1.37 ± 0.50 mm, P<0.001), larger stone size (1.21 ± 0.68 vs. 0.97 ± 0.54 mm, P<0.001) and a higher rate of endoscopic sphincterotomy (EST) combined with endoscopic papillary balloon dilation (EPBD) (29.0% vs. 15.7%, P<0.001) and mechanical lithotripsy (41.6% vs. 26.5%, P<0.001) during the ERCP procedure. But PAD did not affect the incidence of post-ERCP complications.

Of note, PAD group had a higher recurrence rate (18.8% vs. 7.3%, P<0.001) and shorter recurrence-free time (39.9 ± 13.3 vs. 42.8 ± 9.8 months, P = 0.005) compared with non-PAD group during follow-up period. Based on multivariate analysis, the presence of PAD (OR = 3.265, 95%CI: 1.854-5.751) was an independent risk factor for the recurrence of CBDS, after adjusting for age, gender and other factors.

Conclusion: PAD, especially type I PAD, decreased successful cannulation rate and increased the complexity of stone extraction of ERCP in CBDS. However, PAD did not affect the incidence of post-ERCP complications in CBDS, nor did it affect the choice of stone removal methods. PAD was an independent risk factor for the recurrence of CBDS.

Disclosure: Nothing to disclose.

PP1062

IMPACT OF PERIAMPULLARY DIVERTICULUM ON THE OUTCOME OF ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY

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Introduction: Periampullary diverticulum (PAD) is associated with a higher incidence of biliary stone formation and common bile duct (CBD) obstruction. Endoscopic retrograde cholangiopancreatography (ERCP) is an effective therapeutic procedure in case of biliary obstruction, however; the presence of PAD and its different types could affect ERCP outcome [1-3].

Aims & Methods: Aim: To study the impact of periampullary diverticulum on the outcome of ERCP regarding the cannulation technique, the procedure time, the success rate and the complications.

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Method: A Retrospective study included all patients subjected to ERCP procedures at Qena university hospitals, South Valley University, Qena, Egypt from February 2019 to August 2023. ERCP Will be performed under general anaesthesia with endotracheal intubation according to our centre policy by an expert ERCP endoscopist using Pentax duodenoscopy (ED-3490TK, Pentax Incorporation, Tokyo, Japan). Papilla of vater will be examined endoscopically before starting cannulation trials to evaluate presence of diverticulum, type, size and its relation to the papilla. Data were analyzed using Statistics Package for Social Sciences (SPSS) version 26, p-value < 0.05 was considered significant.

Results: A Total of 686 patients subjected to ERCP procedure for different indications were included in this study, males were 273 patients (39.8%), the mean age was 53.7±17 years, the overall procedure success rate was 91% and the overall complications rate was 6%. Almost all cannulation techniques were used in our cases including wire-guided cannulation (WGC), transpapillary fistulotomy (TPF), double-guidewire technique (DGT), transpapillary papillotomy (TPP), and transpancreatic septotomy with pancreatic stent insertion (TPS). PAD was detected during ERCP procedure in 61 patients (9%). There was statistically significant higher age (65.3 vs 52.6, p< 0.001) and statistically insignificant higher frequency of males (41% versus 39.7%, $p_=$ 0.8) in patients with PAD compared to patients without PAD. During ERCP procedures, presence of PAD was associated with higher number of cannulation attempts (3.5 versus 2.5, p< 0.005), longer time required for cannulation (8 vs 5.1 minutes, p< 0.01), and amazingly; higher rate of cannulation success (100% vs 98%, $p_=$ 0.2), and longer total procedure time (50.4 ± 21.9 minutes vs 47.01 ± 24.04, $p_=$ 0.2). There was no significant predilection of certain cannulation technique in PAD patients; higher WGC: 68% vs 67%, higher TPS (1.8% vs 0.5%), equal chance for WGC: 24.6% vs 24.6% and less chance for TPP: 0.7 vs 0%. The overall procedure success rate was lower in PAD patients (85.2% vs 91.2%, $p_=$ 0.1). Post-ERCP pain was noticed more frequently in PAD patients (4.9% vs 3.4%, $p_=$ 0.5) while the overall complication rate with lower (1.6% vs 6.4%, $p_=$ 0.1).

Conclusion: We concluded that PAD is increasingly detected during ERCP procedure in elderly patients, and despite it requires higher number of cannulation attempts and longer time for cannulation, it has no negative impact on the overall success rate or complication rate of ERCP.

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Disclosure: Nothing to disclose.

PP1063

ENTEROSCOPY-ASSISTED ERCP IN PATIENTS WITH SURGICALLY ALTERED ANATOMY: RESULTS OF AN INTERNATIONAL, MULTICENTRE, PROSPECTIVE REGISTRY (SAMISEN) USING MOTORIZED SPIRAL ENTEROSCOPY

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Introduction: Motorized Spiral Enteroscopy (MSE) was introduced into clinical practice and initial trials at expert centres proved it to be safe and effective for deep enteroscopy even in patients with surgically altered anatomy.

Aims & Methods: Prospective study of efficacy and safety of MSE to perform biliary endoscopic retrograde cholangiopancreatography (ERCP) in patients with surgically altered anatomy at 5 European centers. Pancreatic indications were excluded. Primary objective: procedural success and serious adverse event (SAE) rate.

Secondary objectives: diagnostic and therapeutic yield, procedural time, and patient radiation exposure. The study was prematurely terminated because of MSE withdrawal from the market in July 2023 by the manufacturer (and sponsor of the study). Trial registration: NCT05129449 (clinicaltrials.gov)

Results: 89 patients were enrolled, 1 was excluded because of pancreatic indication: 51 (58%) males and 37 (42%) females with mean age of 58 years (range 24-92) and mean bodyweight of 74 kg (range 44-178). Surgically altered anatomy was based on Billroth II reconstruction (29.5%) or Roux-en-Y reconstruction (70.5%), either with naive papilla (39%) or hepaticojejunostomy (61%). Biliary ERCP indications were mainly anastomotic stricture treatment and/or biliary stone removal. 57% of the procedures were performed under general anesthesia with intubation, the remaining under deep sedation. Technical success was defined by the combination of successful enteroscopy followed by cannulation of the bile duct and endoscopic therapy. Enteroscopy was possible in 65/88 patients (74%) in a mean time of 29 min (range 4-104). Bile duct cannulation with cholangiography was possible in 54/88 (61%) and therapeutic ERCP was technically successful in 48/88 (54%). In Billroth II reconstructions technical success was achieved in 13/26 (50%) compared to 35/62 (57%, p=0.0574 Chi square) in Roux-en-Y reconstructions. Technical success was achieved in 17/34 (50%) in patients with intact naive papilla compared to 31/54 (57%, p=0.4968 Chi square) in hepaticojejunostomy. In 21 patients with bariatric Roux-en-Y gastric bypass surgery technical success was 12/21 (57%). In July 2023 the study was prematurely terminated because of MSE withdrawal from the market by the manufacturer because of safety issues. Overall, in 12/88 (14%) patients adverse events (AE) were recorded, of which 6 (7%) were considered serious. Only 1 was related to MSE enteroscopy, 4 to the ERCP procedure and 1 related to the use of a duodenoscope that replaced MSE after reaching the bile duct. MSE enteroscopy-related SAE was a perforation of the proximal oesophagus during MSE insertion. The patient underwent surgical repair and recovered. ERCP-related SAE were cholangitis in 2, sphincterotomy bleeding in 1 and suspicion of air embolism in 1. There was 1 intestinal perforation occurring during duodenoscope insertion that replaced MSE to facilitate ERCP. There was no procedure-related mortality.

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Conclusion: This prospective multicentre study was prematurely discontinued due to MSE withdrawal from the market by the manufacturer because of safety issues. In total 88 patients underwent enteroscopy-assisted biliary ERCP in different types of surgically altered anatomy, resulting in an overall technical success rate of 54%. This result is lower as compared to literature data based on balloon-assisted enteroscopy. SAE rate was 7% with only 1 SAE attributable to the use of MSE (oesophageal perforation). MSE allows enteroscopy-assisted ERCP in patients with surgically altered anatomy, but other techniques may be preferred.
Disclosure: The study was sponsored by Olympus Europe (Hamburg).

PP1064

SINGLE OPERATOR CHOLANGIOSCOPY IN THE MANAGEMENT OF PROXIMALLY MIGRATED BILIARY STENTS – SINGLE CENTER EXPERIENCE

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Introduction: Endoscopic biliary stenting is a well–established therapy in the management of hepatobiliary diseases. However, migration is reported in 5%-10% of cases. Management of proximally migrated stents using standard techniques(guidewire cannulation, snare, forceps, Schohenra stent retriever) can be technically challenging sometimes. Cholangioscopy and retrieval under direct visualization could be a useful tool in such difficult cases.
Aims & Methods: To evaluate the application of single-operator cholangioscopy in the management of proximally migrated biliary stents in terms of efficacy and safety.
Methods:We performed a retrospective analysis of the data of a prospective group of patients. The cohort included consecutive 100 cases of proximally migrated stents from January 2018 through March 2024. We evaluated the retrieval technique, success rate, and adverse events.
Results: Patients were divided into two groups. In the first collective of patients- cholangioscopy was not performed n=62/62%/. We found a technical success rate – defined as successful stent extraction in 69,35% of cases/n=43/. In 19 of the cases removal was not achieved, followed by placement of new plastic stent n=19/30.65%/.In the second group, n=38/38%/ cholangioscopy with single operator cholangioscope was the retrieval technique with a technical success rate of 100%.
There was a statistical difference between the two groups in terms of technical success in favor of the group with cholangioscopy - Pearson chi-square $\chi 2 = 21.534$, $p < 0.001$.
Cholangioscopy was performed either after the failure of standard techniques or as the first-choice tool for stent removal. In five cases we performed pancreatoscopy for extraction of migrated pancreatic stents with only one case of postprocedure pancreatitis.
We experienced adverse events/AE/ in 17 cases as pancreatitis was the most common complication n=11/64,71% of all AE/, followed by cholangitis n=5/29,41% of all AE/ and one case of bleeding n=1/5,88%/. In the first group, the rate of adverse events was 17,74%/n=11/ against 13,95%/n=6/ for the second group.
There was no significant statistical difference in the rate of AE between the two groups. The mean procedure duration was 34,35 minutes for the first group and 42,24 minutes for the second with a statistically significant difference p=0.029 in favor of the first group.
However, it is important to note that in some cases cholangioscopy was performed after the failure of standard techniques which by ilself prolonged the procedure time.

Type of AE	Group 1	Group 2
Pancreatitis	N=7/63,64% of AE for the group; 41,18% of all AE; 63,64% of all cases of pancreatitis/	N=4/66,67% of AE for the group;23,53% of all AE;33,36% of all cases of pancreatitis/
Cholangitis	N=3/27,27% of AE for the group; 17,65% of all AE;60% of all cases of cholangitis/	N=2/33,33% of AE for the group; 11,76% of all AE;40% of cases of cholangitis/
Bleeding	N=1/9,09% of AE of the group;5,88% of all AE;100% of cases of bleeding/	NA

Conclusion: Cholangioscopy is a safe and effective technique for the extraction of migrated biliary stents with a high rate of efficacy and an acceptable rate of adverse events. It could be proposed as a first choice in cases where failure of standard techniques is suspected/persistent stenosis, residual bile duct stones/.
Disclosure: Nothing to disclose.

PP1065

CHARACTERIZING LARGE COMMON BILE DUCT STONES: DEFINING SIZE AND CLINICAL IMPLICATIONS

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Introduction: The precise definition of a large stone is ambiguous, with sizes often ranging from 10 to 15 mm. Authors refer to a stone as “large“ if its size exceeds the diameter of the CBD by more than 2mm (ratio of stone size to CBD diameter > 1).
Aims & Methods: Our study aimed to determine the threshold at which the diameter of the stone leads to endoscopic treatment failure.This research is a retrospective analysis that encompassed all patients who had endoscopic retrograde cholangiopancreatography (ERCP) for the treatment of common bile duct stones from January 2019 to December 2022. The examination of the ROC curve helped identify the threshold at which the diameter of the stone can predict the failure of the ERCP.
Results: We enrolled a total of 181 patients, with a mean age of 64 years (ranging from 22 to 103 years), and a male-to-female sex ratio of 0.41. The primary reasons for doing ERCP were the presence of remaining or recurring gallstones (69%, n = 129) or the need for sequential therapy (18%, n = 33). The efficacy of the first therapy was 61.5%. The mean stone size was 12.5mm, with a range of 3-40mm. Upon examining the ROC curve, it was determined that a diameter of 12 mm provided the highest accurate prediction of endoscopic treatment failure, with a sensitivity of 74% and a specificity of 73%. Univariate analysis revealed that the presence of a stone bigger than 12 mm was a significant predictor of failure of routine endoscopic therapy (p = 0.001). The average ratio of stone size to diameter of the CBD was 0.74 [0.27-1.67]. In our series, we found no significant association between this factor and ERCP failure (p = 0.276).
Conclusion: Within our dataset, we observed that stones larger than 12 mm were indicative of a higher likelihood of ERCP failure. There was no significant correlation between the ratio of stone size to the diameter of the common bile duct (CBD) and the failure of endoscopic therapy.
Disclosure: Nothing to disclose.

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PP1066

CLINICAL PROFILE, MICROBIOLOGICAL SPECTRUM IN BILE AND PATIENT OUTCOMES IN CASES OF ACUTE BACTERIAL CHOLANGITIS

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Introduction: Acute bacterial cholangitis is a medical emergency, requiring prompt antibiotic administration and biliary decompression. Normally, bile is devoid of microbes. Obstruction
of the biliary system leads to bile stagnation, which predisposes to bacterial proliferation. Bacteria enter the bile duct from duodenum. Also, the increased bile duct pressure in the
obstructed biliary system leads to systemic dissemination of bacteria via cholangio-venous reflux leading to septicemia. Indiscriminate and widespread use of antibiotics over the years has altered the antibiotic sensitivity pattern of micro-organisms and the emergence of multidrug resistant bacteria pose a therapeutic challenge which needs further study.

Aims & Methods: We aimed to analyze the clinico-microbiological profile and outcomes of patients with acute bacterial cholangitis, who underwent endoscopic retrograde cholangiopancreatgraphy (ERCP) at a tertiary care hospital. Data on consecutive patients (n=116) with acute cholangitis who underwent ERCP, from January 2019 to August 2020, was retrospectively extracted. The demographic, clinical, etiological and biochemical profile was recorded. Also, bile culture reports, and antibiotic sensitivity pattern of organisms was recorded.

Diagnosis of cholangitis was based on presence of fever and/or leukocytosis, in patients with biliary obstruction (cholestatic liver function tests, biliary dilatation on imaging). Organ failure was present if patients had altered sensorium, hypotension (systolic blood pressure <90 mm Hg), renal failure (serum creatinine >1.5 mg/dL), hypoxia (blood oxygen saturation <90%).

Results: The mean age of our study cohort was 47.56 +/- 13.58 years and 64.7% were females. Patients presented with abdominal pain (89.6%), jaundice (84.4%) and fever (56.8%) with 41.4% patients exhibiting Charcot's triad and 1 having Reynolds' pentad.

30 (25.8%) patients had organ failure with renal failure being the most common. 5 patients had multiple organ failure, all of which died during hospitalization. 60.3% patients had a benign cause of biliary obstruction; choledocholithiasis was the most common benign cause. Other benign causes included post cholecystectomy bile duct stricture, chronic pancreatitis, portal biliopathy. The most common malignant cause was carcinoma gall bladder (43.5%) followed by peri-ampullary carcinoma (30.4%), carcinoma head of pancreas (13%).

Positive bile culture was seen in 95 (81.9%) patients; of which 17 (17.9%) had polymicrobial growth in bile. 92.9% organisms were gram negative bacteria (GNB). Escherichia coli (45.13%) was the most common GNB followed by Pseudomonas (29.2%). 8 patients had gram positive organisms on culture (Enterococcus, 7.1%).

Most common Multi-drug resistant organisms isolated were carbapenem-resistant Enterobacteriaceae. Escherichia coli and Klebsiella were most sensitive to amikacin, netilmicin and tigecycline. Pseudomonas had greatest sensitivity to amikacin, netilmicin and colistin.

80 (69%) patients were naïve, whereas the remaining 36 patients had previously undergone ERCP. 10.3 % patients died during hospital stay. Median duration of hospital stay was 6 (3-60) days.

Old age, lower serum albumin, higher blood urea and serum creatinine, septic shock and malignant etiology of biliary obstruction were associated with mortality on univariate analysis. On multivariate analysis, low serum albumin and septic shock were associated with higher in-hospital mortality.

Conclusion: Indiscriminate antibiotic use has resulted in increasing prevalence of multi-drug resistant organisms, with emergence of carbapenem resistant organisms posing a major therapeutic challenge.

Disclosure: Nothing to disclose.

PP1067

POST-LIVER TRANSPLANT ANASTOMOTIC STRICTURES – PREDICTORS FOR SUCCESSFUL ENDOSCOPIC REMODELLING AND NEED FOR SURGICAL INTERVENTION

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Introduction: Biliary strictures are a recognised complication of liver transplant. The criteria for proceeding with Endoscopic Retrograde Cholangiography [ERC] remains a subject of contention.

This study seeks to explore management strategies of anastomotic strictures, compare rates of successful remodelling and complication rates.

Aims & Methods: All patients with a post-liver transplant stricture who underwent ERC in a tertiary centre between 2016 and 2023 were included. Categorical divisions were made between patients with a single anastomotic stricture and multi-focal or hilar strictures.

Time for stricture development, pre and post-ERC liver serum markers, frequency of procedures performed, percentage successfully remodelled by ERC, percentage referred for surgical intervention and associated adverse events were retrospectively recorded.

Results: During the specified period, 107 patients with prior liver transplant (70 male, 37 female, median age 55.5 (IQR 15)) with a biliary stricture were referred for ERC. 88/107 had a single anastomotic stricture whereas 19 had multi-focal or hilar strictures. Median time from transplant to radiological discovery of biliary stricture was 5 months with 71% occurring within 12 months.

Patients with a diagnosis of a single anastomotic stricture (n=88) underwent a total of 237 total ERCs (mean 2.6 per patient). There was a significant improvement in liver function profile and bilirubin after ERC (unpaired t-test - p<0.01).

The number of remodelled strictures was 64/88 (72.7%). There were 32 ERC associated complications (13.5%); 10 cholangitis, 9 pancreatitis, 13 other. Ten patients were referred for surgical intervention (11.4%).

Patients with multi-focal / hilar strictures (n=19) underwent a total of 45 ERCs (mean 2.4 per patient). There was a non-significant improvement in bilirubin (P=0.096) and ALP (P= 0.184) after ERC and a significant improvement in ALT (P<0.02). The number of remodelled strictures was 7/19 (36.8%). There were 8 ERC associated complications (17.8%), 7 cholangitis and 1 pancreatitis. Five patients were referred for surgical intervention (26.3%).

Single anastomotic strictures, when compared to multi-focal strictures, were associated with a higher re-modelling rate (x^2, p= 0.003) and reduced need for surgical intervention (x^2, p = 0.089).

Conclusion: ERC is efficacious in managing single anastomotic strictures but significantly less effective in multi-focal proximal strictures. A significant complication rate in this cohort needs to be taken into consideration when planning intervention.

However, the majority of these patients can be managed via ERC without the need for surgical intervention.

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Disclosure: Nothing to disclose.

PP1068

SAFETY OF EARLY DISCHARGE POST ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY (ERCP): A SINGLE CENTRE EXPERIENCE IN SINGAPORE

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Introduction: Early discharge post ERCP can potentially reduce strain on the tight bed occupancy status in tertiary hospitals.
Aims & Methods: The aim of this study is to evaluate the safety of early discharge (<24 hours) for patients who had undergone ERCP in a tertiary hospital in Singapore.
This is a retrospective observational study which reviewed patients who were admitted in the TTSH Medical Ambulatory Center (MAC) and discharged within 16 hours post ERCP. To assess safety of this service, electronic medical records of patients admitted between January 2022 and February 2024 were reviewed. Complications post ERCP, predisposing factors, patient demographics were reviewed.
Results: There were 400 patients admitted to MAC for early discharge after ERCP during this period. The overall complication rate was 5.8% (23 patients). Post-ERCP pancreatitis was diagnosed in 1.8% (7 patients) of the admissions, whereas those who developed post-ERCP cholangitis accounted for 3.5% (14 patients). 1 patient developed non-specific chills with no fever nor abdominal pain (not conclusive for either cholangitis or pancreatitis) 2 days post discharge from MAC. 1 patient had post ERCP cholangiopancreatitis (0.3%). No other patients presented with complications related to ERCP after discharge.

	Post ERCP Cholangitis	Post ERCP Pancreatitis	Post ERCP Cholangiopancreatitis	Others
Number	14	7	1	1
Age (years)	Median:72 Youngest: 32 Oldest: 80	Median:72 Youngest: 56 Oldest: 85	63	74
Gender distribution	Male: 7 Female: 7	Male: 1 Female: 6	Female: 1	Male: 1
Risk factors	Diabetes Mellitus, Concurrent cholangioscopy, Complex pancreatico-biliary anatomy	Complex pancreatico-biliary anatomy, Female gender, Difficult cannulation	Complex pancreatico-biliary anatomy	Not Applicable
Onset post ERCP (hours)				
1-6	2	3	5 hours (1 patient)	unclear
7-11	5	3		
12-16	7	1		

Out of the 23 patients with post ERCP complications, 5 (21.7%) patients developed within 1-6 hours (3 pancreatitis, 2 cholangitis). 8 (34.8%) patients (3 pancreatitis, 5 cholangitis) developed within 7-11 hours, whereas 8 (34.8%) patients (1 pancreatitis,7 cholangitis) developed within 12-16 hours post ERCP. 1 patient with cholangiopancreatitis presented within 6 hours.
Of the 14 patients with post ERCP cholangitis, 7 were females and 7 were males. The median age (years) for this group was 72, youngest being 32 and eldest 80. 6 patients in this group had undergone a concurrent spyglass cholangioscopy during ERCP. All of them received prophylactic antibiotics prior to cholangioscopy. It was also noted that there were 6 patients in this group of post ERCP cholangitis with Diabetes Mellitus
Out of 7 patients with post ERCP pancreatitis, 6 were females and 1 was male. The median age (years) for this group was 72, youngest being 56 and eldest being 85. All but 1 patient in this group had sphincterotomy performed. All patients in this group received 100 milligrams of rectal diclofenac for prophylactically. Only 1 patient had documented difficult cannulation.
Conclusion: From this study, it could be concluded that it was safe to allow early discharge for patients who had undergone ERCP, if there is a clear monitoring protocol and discharge plans. The onset of post ERCP pancreatitis tend to be earlier compared to those who developed post ERCP cholangitis. There is female predominance in the post ERCP pancreatitis group. The patients who develop post ERCP pancreatitis tend to be older. Complex pancreatico-biliary anatomy is a significant risk for post ERCP complications. As there were 94.2% of the patients who remained well after ERCP, they could be potentially discharged even earlier (<10 hours). Further research could focus on early identification of patients who are likely going to develop post ERCP complications to facilitate early discharge, thus relieving tight bed occupancy status.
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Disclosure: There is no conflict of interest to declare.

PP1069

OTHER WAYS OF TRANSITION – PANCREATICOBILIARY ENDOSCOPY IN THE PEDIATRIC POPULATION. SINGLE, TERTIARY-CARE CENTER EXPERIENCE

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Introduction: Endoscopic interventions in children require specialized training and expertise due to unique anatomy and physiology challenges. Pancreaticobiliary endoscopies represent highlighted procedures in this population due to the often-unusual indications and the inherent high risk of complications.
Aims & Methods: The current work demonstrates our experience in pancreaticobiliary endoscopy in the pediatric population. We retrospectively analyzed data on all patients <18 years old who underwent endoscopic retrograde cholangiopancreatography (ERCP) or endoscopic ultrasound (EUS) from 2021 to the present days at our department. We collected demographical data, indications, procedural findings, interventions, and complications.

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Results: Overall, 19 ERCPs and 7 EUS procedures were performed on 24 children with regular endoscopes. 15 were males, and the mean age was 10.7 years (±4.7). Indications for ERCP included bile duct stones, benign stricture, pancreatic divisum and disconnected duct syndrome. Successful intervention was carried out in 18 cases, while there was one case of unsuccessful deep pancreatic duct cannulation due to abnormality of the duct. There was one case of perforation (Stapfer II), which was effectively managed endoscopically by placing a fully covered self-expanding metallic stent. 1 case of mild post-ERCP acute pancreatitis occurred following ERCP. EUS was performed for chronic pancreatitis, pancreatic cystic neoplasia, acute necrotizing pancreatitis, biliary compression and polycystic pancreas. 3 tissue acquisitions were done and one EUS-guided transluminal drainage was established. There were no complications.
Conclusion: To our knowledge, this is the most comprehensive domestic dataset of pancreaticobiliary endoscopies in children. Our findings imply that ERCP and EUS are effective diagnostic and treatment modalities in pediatric patients with acceptable safety profiles.
Disclosure: Nothing to disclose.

PP1070

SAFETY AND EFFICACY OF SPYGLASS™ CHOLANGIOPANCREATOSCOPY IN ROUTINE CLINICAL PRACTICE AT A TERTIARY HOSPITAL IN RIO DE JANEIRO

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Introduction: This study examined the efficacy and safety of cholangiopancreatoscopy using the SpyGlass™ system in routine clinical practice.
Aims & Methods: Clinical data from endoscopic retrograde cholangiopancreatography (ERCP) performed at a regional hospital from January 2021 to January 2024 were retrieved from an electronic database and reviewed. All patients undergoing SpyGlass™ Cholangiopancreatoscopy were included. Patient demographics, procedure indications, technical success rates, clinical success rates, and complication rates were analyzed. A subanalysis of clinical outcomes was performed comparing legacy SpyGlass (fiber optic) and other digital systems.
Results: Out of 2,550 ERCP procedures performed, 97 patients underwent 100 cholangiopancreatoscopy procedures. Clinical indications were: difficult common bile duct (CBD) stones (39.2%, n = 38), indeterminate CBD stricture (50.5%, n = 49), indeterminate pancreatic duct stricture (8.2%, n = 8), and proximally migrated CBD stent (2.1%, n = 2). Complete stone extraction was achieved in 36 (94.7%) out of 38 patients. Among patients with strictures, the correct diagnosis of malignancy based on image visualization was achieved in all 11 cases. Sensitivity and specificity of SpyBite™' biopsies were 91.8% (95% confidence interval (CI] 48.2%-97.7%) and 100.0% (95% CI 15.8%-100.0%), respectively. Proximally migrated CBD stent was successfully extracted. Complications included pancreatitis (1.03%, n = 1), suspected sealed perforation after laser lithotripsy treated with a biliary stent (1.03%, n = 1), and cholangitis (5.15%, n - 5).
Conclusion: SpyGlass™ cholangiopancreatoscopy is a safe and effective tool in routine clinical practice.
Disclosure: Nothing to disclose.

PP1071

DIGITAL SINGLE-OPERATOR CHOLANGIOSCOPY IN THE DIAGNOSIS AND TREATMENT OF BILIO-PANCREATIC DISEASES: A SINGLE-CENTRE EXPERIENCE

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Introduction: Single-operator cholangioscopy with SpyGlass™ DS Direct Visualization System (Boston Scientific Corp.) allows easier and more effective management of bilio-pancreatic diseases, with a relatively low rate of adverse events. Its main applications are the treatment of difficult biliary lithiasis and the diagnosis of indeterminate biliary strictures, although data confirming its efficacy are still limited. In the present study, we evaluated efficacy and safety of single-operator digital cholangioscopy in a secondary digestive endoscopy centre.
Aims & Methods: We retrospectively evaluated 113 consecutive patients undergoing SpyGlass™ DS cholangioscopy between February 2016 and September 2022 at our centre. Demographic and procedural characteristics and post-procedural outcomes were collected for the purpose of statistical analysis.
Results: A total of 126 SpyGlass™ DS procedures were performed on 113 patients (females 69, 61.1% - males 44, 38.9%) with a mean age of 73.4 ± SD 17 years. 92% of the patients underwent only one procedure, while the remaining 8% required 2 to 4 consecutive procedures. Indications were treatment of difficult biliary stones (85/126, 67.5%), diagnosis of indeterminate biliary strictures (30/126, 23.8%) and others (11/126, 8.7%). Holmium laser lithotripsy was the most frequently used (76/85, 89.4%). Complete clearance was achieved in almost all cases (80/85, 94.1%) with an average number of sessions of 1.05. The complete clearance rate at the first session was 93.75%.
40 diagnostic procedures were performed to characterise indeterminate biliary strictures, since diagnostic cholangioscopy was required for incidental biliary stricture in 10 patients who initially underwent lithotripsy. In 36/40 patients (90%) a stricture was confirmed and among these 72.2% required targeted biopsies with SpyBite™ forceps (21/36, 58.3% malignant biliary strictures - 15/36, 41.7% benign biliary strictures). The sensitivity, specificity and accuracy values at follow-up were 90%, 93% and 92% for the visual impression at cholangioscopic imaging and 86%, 100% and 88% for the histological examination of the targeted biopsy. A total of two cases of cholangioscopy-related adverse events were reported (2/126, 1.59%): one mild pancreatitis and one mild cholangitis.
Conclusion: The data obtained from our experience confirm the therapeutic and diagnostic value of digital single-operator cholangioscopy in the management of bilio-pancreatic diseases. Cholangioscopy-guided lithotripsy has been shown to have a high rate of complete clearance already at the first session. Regarding the management of indeterminate biliary stenosis, sensitivity of visual impression was superior to that of targeted biopsy; conversely, specificity of histological examination by targeted biopsy appeared superior. Prospective and multicentre studies would be advisable to consolidate the role of cholangioscopy in the management of bilio-pancreatic pathology.
Disclosure: Nothing to disclose.

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PP1072

RETRIEVAL OF PROXIMALLY MIGRATED PLASTIC STENTS IN ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY: INSIGHTS FROM A SINGLE-CENTER STUDY

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Introduction: Plastic stent placement during endoscopic retrograde cholangiopancreatography (ERCP) is a well-established therapy for a wide variety of obstructive biliary and pancreatic disorders. While proximal or distal stent migration is uncommon, with migration rates ranging from 4.1% to 13.5%, it can result in serious complications. Endoscopic retrieval of proximally migrated stents is intricate and may present a technical challenge to the endoscopist.

Aims & Methods: Aims: This study aims to ascertain the proximal stent migration rate, complications associated, risk factors, success rate, and the endoscopic techniques applied for retrieval.

Materials and Methods: We conducted a retrospective cohort study including 546 consecutive ERPC procedures performed at a tertiary center between January 2022 and March 2023. Data regarding indications for stenting, presentation of migration, stent length and diameter, endoscopic retrieval techniques, and success rate were collected.

Results: Overall, a total of 429 plastic stents were inserted (372 biliary and 57 pancreatic) in 245 patients. Proximal migration occurred in 2.8% of cases (9 biliary and 3 pancreatic), all of which were asymptomatic. Stent length <10cm (p=0.08) was identified as a risk factor for proximal migration. The success rate of endoscopic retrieval was 100%. Stent extraction was accomplished on the first attempt in 9 cases (75%), with three cases (25%) requiring re-attempt. Retrieval techniques employed included the Soehendra stent retriever (n=5, 41.7%), forceps (n=3, 25%), snare (n=3, 25%), and balloon catheter (n=1, 8.3%). Cholangiopancreatoscopy-assisted retrieval was necessary in two cases as a rescue therapy. Notably, there were no complications or procedure-related mortality.

Conclusion: ERCP is an effective and safe approach for the retrieval of proximally migrated plastic stents, however, it requires experience with different techniques and endoscopic accessories. In challenging scenarios, cholangiopancreatoscopy-assisted stent retrieval may offer a viable solution when conventional methods are unsuccessful. Our cohort analysis revealed that stent length <10cm was potentially associated with an increased risk of proximal migration.

Disclosure: Nothing to disclose.

PP1073

THE INFLUENCE OF PERIAMPULLARY DIVERTICULUM ON THE ERCP OUTCOMES: A TERTIARY CENTER EXPERIENCE

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Introduction: Patients with periampullary diverticulum (PAD) are often found accidentally and may increase the complication risk of endoscopic retrograde cholangiopancreatography (ERCP) or may decrease success cannulation rates.

Aims & Methods: We aimed to compare ERCP procedures and outcomes in naive patients with or without PAD. All ERCP procedure and outcomes data of 3072 patients between January 2021 and December 2023 referred to our tertiary center were retrospectively analyzed. ERCP procedures were divided into two groups based on the presence or absence of PAD.

Results: A total of 929 naive patients underwent ERCP, of whom 135 (14.5%) patients with PAD and 794 (85.5%) patients without PAD. Median age of the patients with PAD was significantly higher than the patients without PAD (73 years vs. 60 years, p<0.001), but the two groups had a similar regarding the genders.

The median ERCP procedure time (20 min vs. 18 min, P=0.410), cannulation time (8 min vs. 7 min, P=0.491) did not differ in patients with and without PAD respectively. Successful cannulation rate in the group with PAD was 134 (99.3%), and in the group without PAD was 791 (99.6%), there was no difference between the two groups (P=0.467). One of the indications of ERCP procedure, choledocholithiasis was a more common reason in patients with PAD than in those without PAD [116 (85.9%) vs. 551 (69.4%), p<0.001]. Indications of other ERCP procedures in patients without PAD, malignant biliary strictures [93 (11.7%) vs. 6 (4.4%), P=0.011] and biliary leak [94 (11.8%) vs. 6 (4.4%), P=0.010] were more frequent reasons than in the patients with PAD. There were no significant differences between the patients with PAD and without PAD in term of all ERCP-related complications [6 (4.5%) vs. 73 (9.2%) respectively, P=0.751].

Our results showed notable differences in cannulation techniques between the two groups. Rate of the selective cannulation technique in the patients with PAD was higher than the patients without PAD [102 (75.6%) vs. 506 (63.7%) respectively, P=0.008]. Needle knife fistulotomy in patients without PAD was a higher rate of using technique than in the patients with PAD [139 (17.5%) vs. 8 (5.9%) respectively, P=0.001]. In the Li-Tanaka classification, the distribution of the PAD types identified 9.6% type 1 diverticula, 20.7% type 2A diverticula, 35.6% type 2B diverticula, 18.5% type 3 diverticula, 9.6% type 4A diverticula and 5.9% type 4B diverticula. Li-Tanaka type 2B (35.6%) was the most common type of the diverticula. There was no significant difference seen in any parameter in the PAD subgroups.

Conclusion: Our study revealed that ages of the patients with PAD was significantly higher than the patients without PAD. In the presented study, the overall success rates and total durations of the ERCP procedures were unaffected by the presence of PAD. The overall cannulation rates and complications following the procedures were unaffected by the type of the diverticulum.

The periampullary diverticulum does not affect the success and duration of cannulation, while it may affect the type of cannulation.

Disclosure: Nothing to disclose.

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PP1074

WITHDRAWN

PP1075

A COMPARISON OF ERCP OUTCOMES OF THE NONAGENARIAN PATIENTS AND YOUNGER ADULT PATIENTS

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Introduction: There are few studies evaluating the safety of endoscopic retrograde cholangiopancreatography (ERCP) in patients aged 90 years and above (nonagenarian). This study's main goal was to compare the technical success and complications of ERCP in patients over 90 years or younger patient group.

Aims & Methods: 929 naive patients who underwent ERCP at the tertiary referral center between January 2021 and December 2023 were retrospectively analyzed. The patient's medical records were reviewed for comorbidities, periampullary diverticulum, cannulation time, cannulation method, technical success, procedure duration, and ERCP-related complications.

	Total (n= 929)	Age ≥90 (n=34, 3.7%)	Age <90 (n=895, 96.3%)	P-value
Gender, female, n (%)	539 (58)	25 (73.5)	514 (57.4)	0.062
Age, years	63 (47-74)	91 (90-93.25)	61 (47-73)	-
ERCP procedure time, min	18 (12-28)	18.5 (11.75-23.25)	18 (12-28)	0.660
Total cannulation time, min	7 (3-15)	8 (2.75-15.5)	7 (3-15)	0.824
Indications of ERCP procedure, n (%)				
Choledocholithiasis	667 (71.8)	28 (82.4)	639 (71.4)	
Malignant biliary strictures	99 (10.7)	3 (8.8)	96 (10.7)	0.142
Benign biliary strictures	44 (4.7)	-	44 (4.9)	
Biliary leak	100 (10.8)	1 (2.9)	99 (11.1)	
Others	19 (2)	2 (5.9)	17 (1.9)	
Complications, n (%)				
Post ERCP Pancreatitis	42 (4.5)	1 (2.9)	41 (4.6)	
Perforation	4 (0.4)	-	4 (0.4)	0.970
Bleeding	27 (2.9)	2 (5.9)	25 (2.8)	
Cholangitis	3 (0.3)	-	3 (0.3)	
Others (Cardiopulmonary)	2 (0.2)	-	2 (0.2)	
Past surgery history, n (%)				
History of cholecystectomy	241 (25.9)	10 (29.4)	231 (25.8)	0.638
Operation Billroth I and II	4 (0.4)	1 (2.9)	3 (0.3)	0.139
Cannulation techniques, n (%)				
Selective	608 (65.4)	27 (79.4)	581 (64.9)	
Precut	45 (4.8)	-	45 (5)	0.440
Needle-knife fistulotomy	147 (15.8)	4 (11.8)	143 (16)	
Double guide wire (DGW)	63 (6.8)	3 (8.8)	60 (6.7)	
Trans-pancreatic biliary sphincterotomy	12 (1.3)	-	12 (1.3)	
Periampullary Diverticulum type, n (%)				
1	13 (1.4)	1 (2.9)	12 (1.3)	
2A	28 (3)	1 (2.9)	27 (3)	
2B	48 (5.2)	4 (11.8)	44 (4.9)	0.472
3	25 (2.7)	-	25 (2.8)	
4A	13 (1.4)	1 (2.9)	12 (1.3)	
4B	8 (0.9)	-	8 (0.9)	

Results: A total of 929 naive patients underwent ERCP; 895 (96.3%) patients were < 90 years old and 34 (3.7%) patients were≥ 90 years old. The median ERCP procedure time (18.5 min vs. 18 min, P=0.660), and cannulation time (8 min vs. 7 min, P=0.824) did not differ in patients with two groups respectively.

The successful cannulation rate in the nonagenarian group was 34 (100%), and in the younger group was 891 (99.6%), there was no significant difference between the two groups (P=1).

The main indication of the ERCP procedure was choledocholithiasis in both groups. There was no significant difference in the ERCP-related complications between the two groups. The main cannulation technique was selective sphincterotomy in both groups.

There was no significant difference in cannulation techniques between the two groups. Patients belonging to the nonagenarian group had a higher frequency of periampullary diverticulum (20.6%) compared to the younger group (14.3%).

Type 2B diverticula was the most seen in both groups (11.8% vs. 4.9%). However, there was no significant difference in the presence of periampullary diverticulum between the two groups. The number of chronic concomitant diseases (cardiovascular diseases, chronic lung diseases, chronic renal diseases) was significantly higher in the nonagenarian group [13 (38.2%) vs. 114 (12.7%), P<0.001].

Conclusion: In this study, we showed that ERCP was a safe and successful procedure in patients with nonagenarians. Although comorbidities were substantially higher, ERCP was a successful procedure for nonagenarian patients because the rates of complications are similar to those of younger adult patients.

Disclosure: Nothing to disclose.

PP1076

WITHDRAWN

PP1077

WITHDRAWN

PP1078

THE RESULTS OF USING NONSTEROIDAL ANTI-INFLAMMATORY DRUGS (NSAID-S) PROPHYLAXIS IN POST-ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY (ERCP) PANCREATITIS - A REAL-LIFE DATA FROM A SINGLE TERTIARY CENTRE

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Introduction: ERCP is a commonly used endoscopic procedure for diagnosing and treating disorders of pancreaticobiliary system. NSAIDs have been proven to minimize the risk of post-ERCP pancreatitis, which occurs in approximately 8% of average-risk procedures (1,2).

This study aimed to analyze the occurrence of post-ERCP pancreatitis in patients in whom ERCP was performed for the first time, before and after the use of NSAIDs in routine clinical practice.

Aims & Methods: Between January 2012 and December 2014 and between January 2019 and December 2021, 621 and 664 patients, respectively, were consecutively evaluated. In all those patients therapeutic ERCP procedure with biliary sphyncteroctomy has been done for the first time and the post-procedure complications were consecutively evaluated. NSAIDs were administered in a form of rectal suppositories at least an hour before the procedure.

Post-ERCP pancreatitis was defined as an appearance of pain with elevated amylase concentrations at least 3x above the upper limit of normal for more than 24 hours after the procedure. Chi-Square test and Mann-Whit-

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ney U tests were used to compare discrete and continuous variables where p<0.05 was considered statistically significant.

Results: A total of 1285 ERCP-naïve patients were analyzed. Patients were split into two groups, one receiving and the other not receiving NSAID prophylaxis (667 vs 618 patients, respectively). Patients who received NSAID prophylaxis showed a significantly lower incidence of post-ERCP pancreatitis (2.9% vs. 7.8%, p<0.05) and transient hyperamylasemia (9.3% vs. 28.2%, p<0.05), as well (Table 1).

Conclusion: Real-life experience from a single tertiary center demonstrated that including NSAID prophylaxis into standard clinical practice resulted in a considerable reduction in post-ERCP pancreatitis and hyperamylasemia.

Patients		667	618
Age (years mean ± SD)		68 ± 15	67 ± 15
Male/Female (n)		328/339	249/369
	Choledocholithiasis	399 (59.8%)	436(70.5%)
	Cholangitis	145 (21.7%)	75(12%)
	Malignant biliary obstruction	120 (17.8%)	101(16.4%)
	Acute pancreatitis	55 (8.2%)	57 (9.2%)
	Miscellaneous	58 (8.7%)	33 (5.3%)
Post ERCP pancreatitis n (%)		20 (3.1%)	48 (7.8%) *
Hyperamylasemia n (%)		46 (6.9%)	174 (28%)*

Table 1. NSAID non-NSAID

*Chi-Square test and Mann-Whitney U tests, *p<0.05*

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Disclosure: no conflict of interest

PP1079

MULTIDRUG RESISTANCE PATTERNS IN ACUTE CHOLANGITIS: A PROSPECTIVE MULTICENTER STUDY ON ANTIMICROBIAL RESISTANCE (ARISE)

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Introduction: The "Antimicrobial Resistance in Acute Cholangitis" (ARISE) study is a prospective investigation focused on the antimicrobial resistance patterns in patients with acute cholangitis undergoing ERCP. Conducted across multiple tertiary gastroenterology centers in Romania, this study aims to identify, analyze, and mitigate the challenges posed by multidrug-resistant (MDR) organisms in managing acute cholangitis.

Aims & Methods: Data collected included microbial cultures from bile samples, patient demographics, clinical outcomes, and antibiotic resistance profiles. From the study's onset in January 2024, 138 eligible patients were identified as having positive bile cultures. Among these patients, the predominant symptoms included abdominal pain (82.6%), jaundice (71%), and fever or shaking chills (36.2%). The diagnosis of acute cholangitis was established using TG18 criteria.

Results: Bile cultures were positive in 71.0% of cases. The most frequently isolated bacteria were Escherichia coli, found in 23.2% of positive cultures, followed closely by Enterococcus spp., present in 22.5% of cases. Klebsiella pneumoniae was also notably prevalent, identified in 7.2% of the bile samples. Ciprofloxacin showed the highest resistance rate among the tested antibiotics, with a resistance rate of 23.9%. Ampicillin and gentamicin displayed significant resistance rates at 10.9% and 18.8%, respectively. Piperacillin/tazobactam and trimethoprim/sulfamethoxazole each showed a resistance rate of 15.2%.

The study's data on tigecycline and linezolid showed lower resistance rates, with tigecycline maintaining a sensitivity rate of 20.3% and linezolid at 19.6% in cultures where testing was applicable. In comparison, their resistance rates were only 0.7% for each.

Multidrug resistance was identified in a significant fraction of isolates, with 15.2% of samples. A concerning finding was the presence of MDR strains of Escherichia coli, which exhibited resistance rates to ampicillin (21.0% sensitivity, 10.9% resistance) and amoxicillin/clavulanic acid (17.4% sensitivity, 14.5% resistance).

Conclusion: We observed a concerning prevalence of MDR organisms in acute cholangitis cases, particularly against antibiotics such as ampicillin, gentamicin, and ciprofloxacin, with a high incidence of MDR Escherichia coli strains.

Disclosure: Nothing to disclose.

PP1080

STUBBORN STRICTURES! A RETROSPECTIVE ANALYSIS OF SOEHENDRA STENT RETRIEVER IN MANAGEMENT OF BILIARY AND PANCREATIC STRICTURES

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Introduction: Biliary strictures are defined as narrowing in the intrahepatic or extra hepatic biliary system, resulting in upstream dilatation, stagnant bile flow, ultimately causing biliary obstruction. Patients may present with a variety of symptoms including jaundice, pruritis, weight loss, along with development of recurrent cholangitis, stone formation and liver abscesses. This results in immediate and long term morbidity and mortality. It therefore warrants immediate investigation and treatment. In ERCP, endoscopic techniques exist that allow strictures to be traversed. These include wire guided catheter techniques, balloon dilators and stent placement. When traditional techniques fail, Soehendra stent retriever can be used.

Aims & Methods: Patients with evidence of benign/malignant biliary or pancreatic stricture were retrospectively analyzed. The pancreatic or biliary strictures allowed passage of 0.035 inch guidewire, without allowing passage of >5.5fr cannultomes, 5.8fr balloon dilators, or sohendra stricture dilator. After successful biliary cannulation and guide wire access, a SSR would be passed through the wire to the distal end of the stricture.

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The SSR handle would be rotated clockwise, allowing the SSR to cross the stricture. Once the stricture is dilated, a stent would then be placed.
Results: A total of 72 patients were analyzed. Of these, 38 (52%) were female and 35 (48%) were male. The average age was 44.8 years. Predominant symptoms included jaundice in 36 (50%) patients, and abdominal pain in 35 (48.6%). A total of 59 (81.9) ERCPs were for biliary strictures, and 13 (18%) were for pancreatic strictures.
The predominant findings prior to biliary ERCP included evidence of CBD stricture in 31(52%) cases, unspecified obstructive jaundice in 30 (50.8%) cases, peri-ampullary growth in 13 (22%) patients, and cbd stones in 11 (18%) patients. The main indication for pancreatic ERCP was Chronic Pancreatitis in 100% cases. Out of the 72 patients, 2 (2.7%) developed brief hemobilia during ERCP controlled spontaneously. No patient was demonstrated to have major bleed. No evidence of duct injury, evident by dye extravastion, or perforation was observed. A stent could not be passed in 7 cases (9.7%) despite SSR traversing stricture.
Conclusion: SSR is a useful device for crossing biliary and pancreatic stricture that otherwise could not be negotiated with established techniques. It is safe, and tolerable for strictures defying conventional wire guided techniques
Disclosure: Nothing to disclose.

PP1081

FEASIBILITY OF THREE DIMENSIONAL CHOLANGIOGRAM RECONSTRUCTION FOR COMPLEX BILIARY STRICTURES UNDERGOING ENDOSCOPIC RETROGRADE CHOLANGIOGRAPHY (ERC). -A PILOT STUDY

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Introduction: Endoscopic Retrograde Cholangiopancreatography (ERCP) is pivotal in managing complex biliary strictures due to various causes like Cholangiocarcinoma (CCA), Primary Sclerosing Cholangitis (PSC), and mass-related intrahepatic strictures. Traditional ERCP relies on two-dimensional (2D) X-ray imaging, which often presents challenges in accurately localizing and selectively cannulating specific intrahepatic ducts. This study aims to assess the feasibility and effectiveness of three-dimensional (3D) fluoroscopic ERCP in diagnosing and treating complex biliary strictures.
Aims & Methods: We conducted a prospective pilot study, approved by the Institutional Review Board. Adult patients scheduled for ERCP for complex biliary strictures were enrolled. Exclusions included inability or unwillingness to consent, existing percutaneous biliary drainage, or unfeasible selective trans-papillary biliary cannulation.
Performance of 3D ERCP Patients were positioned on a 3D fluoroscopic table. ERCP followed standard protocols, with high-volume contrast injections opacifying bile ducts. After achieving full biliary tree opacification, a balloon was inflated in the distal bile duct in preparation for 3D imaging. To minimize radiation exposure to staff, the endoscope was fixed in a holder, and the room was evacuated before 3D image acquisition. These images were reconstructed to produce detailed 3D visuals of the target area, aiding real-time therapeutic maneuvers like selective intrahepatic duct cannulation.
Data Collection and Analysis Data included continuous variables (mean ± standard deviation) and categorical variables (percentages). Radiation exposure was quantified separately for ERCP and 3D reconstruction, using thermoluminescent dosimeters. Results were reported as Effective Surface Dose (ESD) in mGY and Dose Area Product in mGY.cm^2.
Results: Eighteen patients (median age 66.6) underwent 3D ERCP. 40% had a malignant biliary structure etiology, with 50% showing both intrahepatic and extrahepatic bile duct involvement. 3D reconstruction facilitated successful selective wire cannulation in 80% of cases, previously unachievable with conventional ERCP. One case provided crucial imaging for liver resection planning. The additional radiation dose was 16.44 mGY.cm^2.
Conclusion: This pilot study proves the feasibility of 3D ERCP in managing complex biliary strictures. This first-of-its-kind study highlights the utility of 3D reconstruction in ERCP. While significantly aiding selective intrahepatic duct cannulation and surgical planning for cholangiocarcinoma resections, it doubles the procedure's radiation exposure, potentially raising cancer risk to 1 in 1000. However the dose needed is significantly lower the IR guided biliary interventions.Further larger studies are required to evaluate the clinical efficacy of 3D ERCP.
Disclosure: Nothing to disclose.

PP1082

OPTIMAL TIMING FOR A SECOND ERCP AFTER FAILURE OF INITIAL BILIARY CANNULATION FOLLOWING PRECUT SPHINCTEROTOMY IN DIFFICULT BILIARY CANNULATION: A RANDOMIZED CLINICAL TRIAL

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Introduction: Precut sphincterotomy is a technique usually employed for difficult biliary cannulation during endoscopic retrograde cholangiopancreatography (ERCP), it increases the success of deep biliary cannulation, but the method fails at the initial ERCP in 5-12% of cases. Although other invasive strategies are often used to access the bile duct, a second ERCP may be effective and safe.
Aims & Methods: We evaluated the efficacy, safety, and factors related to a second ERCP after failed cannulation using a precut sphincterotomy.
Method: In this prospective, monocenter, randomized, clinical trial, patients who were referred for therapeutic biliary ERCP and difficult biliary cannulation with failed cannulation after precut, a second ERCP was performed. They were randomized to early repeat ERCP within 4 days (Group A) or after 7 days after precut (Group B). Efficacy was based on the cannulation rate of the second ERCP, and safety was assessed in terms of adverse events.
Results: We identified 238 patients with failed cannulation after precut, and a second ERCP was performed in 154(64.7%). Pancreatitis post ERCP developed in 5 of the 77 patients (6.4%) in Group A and 11 of the 77 (14.8%) in Group B (odds ratio [OR] 0.55; 95% confidence interval [CI] 0.16–0.78). The incidence of PEP was significantly lower in the early precut group (5/77, 6.4%) than in the delayed precut subgroup (11/77 [14.8%]; OR 0.42, 95%CI 0.17–1.07). There were no differences in biliary cannulation success rates, bleeding, perforation, and cholangitis.
Conclusion:
A second ERCP after failure of initial biliary cannulation following precut appears to be safe and effective. early precut can significantly reduce the incidence of PEP. Repeated biliary cannulation attempts are a real risk factor for this complication.
Disclosure: Nothing to disclose.

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PP1083

EFFICACY AND SAFETY OF EENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY (ERCP) IN PEDIATRIC POPULATION

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Introduction: Data regarding diagnostic and/or threupatic endoscopic retrograde cholangiopancreatography (ERCP) in pediatric population is limited. ERCP is very rarely necessary in this population and most of ERCP procedures are usually performed by adult endoscopists. We evaluated the efficacy and safety of ERCP in pediatric patients.

Aims & Methods: We retrospectively evaluated data of pediatric patients who underwent ERCP during the period of 2013 and 2024 from two adult gastroenterology centers (Harran University and Gaziantep University Gastroenterology)

Results: We included 80 patients (40%, male) with median age of 13 years (range: 3-17) at the time of ERCP procedure. ERCP was performed in 73 patients (91 %) due to biliary disorders while in 7 patients (9%), pancreatic diseases were the indication. ERCP was performed under sedation (propofol ±midazolam ±ketamine). The most common reason was choledocholithiasis (n=35, 44%) followed by hydatic cyst complications (n=22, 27%), and post-liver transplant complications (n=6, 7%).

Overall, the procedure was successful in 79 (99%) patients while in only one patient common bile duct could not be cannulated due to tumor invasion. Post ERCP pancreatitis developed in 5 (6.2%) patients and was mild in all. No major adverse event was observed.

Conclusion: Our data suggest that ERCP is a safe and effective method in the management of pancreatobiliary diseases in pediatric patients.

References: none

Disclosure: Nothing to disclose.

PP1084

FREEHAND PRECUT- A SAVIOR FOR DIFFICULT ERCP- RETROSPECTIVE ANALYSIS OF 3000 ERCPS AT TERTIARY CARE CENTER IN INDIA

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Introduction: With the advent of EUS guided biliary interventions, there is a lot of hype created that ERCP will cease to be performed. The most important and limiting factor for ERCP is successful bile duct cannulation. We have studied a large series of our cases and tried to study the impact of properly performed precut on success of bile duct cannulation.

Aims & Methods: Aim: To analyze the efficacy of freehand precut for selective cannulation in difficult ERCP and its role in improving technical success rate, obviating need for EUS guided / other rendezvous techniques.

Methods: Retrospective analysis of 3000 patients undergoing ERCP from 2017 till date for varied etiologies. No. of patients required precut accessotomy were noted and technical success rate was measured.

Results: No. of patients analyzed- 3000. Selective biliary cannulation- 2452 (81.74%). Difficult cannulations requiring freehand precut accessotomy- 548 (18.26%). Technical success after precut achieving biliary access- 539 (98.36%). Failure of biliary cannulation even after precut- 09 (1.64%). Most common etiology in precut casesstone disease 360 (65.70%) followed by malignant lesions 118 (21.53%), benign strictures 28 (5.11%), post cholecystectomy biliary leak 28 (5.11%), post traumatic biliary leak 10 (1.82%), post hydatid surgery biliary leak 02 (0.36%), post central hepatectomy biliary leak 02 (0.36%).

Most common cause for failure in precut cases was extensive edema and deformed anatomy. Of all failed cases 3 were large stones who were subjected to surgery and 6 were malignant lesions who were subjected to EUS guided biliary drainage.

Conclusion: Free hand precut technique, done correctly, is a savior in difficult cannulation cases and it increases the success rate of selective biliary cannulation significantly avoiding the need for other costlier techniques such as EUS guided / Rendezvous techniques.

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Disclosure: Nothing to disclose.

PP1085

WIRE GUIDED CANNULATION

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) has evolved from a diagnostic to a therapeutic procedure for pancreatobiliary pathologies. Successful cannulation of pancreatobiliary ducts especially deep biliary cannulation (DBC) through the ampulla is of prime importance and involves a spectrum of different techniques (1). Safe technique of DBC cannulationis a necessary prerequisite to most biliary interventionists and is still under constant evolution, with an aim to decrease the incidence of complications related to it (2). Initial technique of DBC was a conventional contrast-assisted cannulation (CC), it has been replaced by a less traumatic and preferred wire-guided cannulation technique (WGC) (4). Previous technique of CC had an approximate success rate of 65% (5). Repeated unsuccessful attempts to cannulate with CC, may result in trauma to the papilla; intramural injection and inadvertent pancreatic duct injections, increasing the risk of post-ERCP pancreatitis (PEP) (6). WGC technique was first introduced by Siegel (7) in 1987, aiming to increase the success rate of biliary cannulation and a possible decrease in the rate of PEP.

Pancreatitis is the commonest complication of ERCP, with rates ranging from 1% to 40%, but figures of 2% to 6% are typical in most large prospective studies (8). Multivariate analysis in most studies have revealed following factors involved in PEP; suspected sphincter of Oddi dysfunction, young age, history of PEP, difficult or failed cannulation, pancreatic duct injection, pancreatic sphincterotomy (especially minor papilla), balloon dilatation of intact biliary sphincter and precut sphincterotomy (9); widespread prophylactic administration of these agents may significantly reduce the incidence of PEP, resulting in major clinical and economic benefit (8). Mechanical approaches are pancreatic duct (PD) stent placement and WGC; PD stent has a definite role in prevention but the utility of WGC is still debatable (10). Hypothesis behind the preventive use of WGC is to reduce the mechanical trauma to the papilla or pancreatic sphincter,

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thus avoiding an increase in hydrostatic pressure associated with contrast injection (8) ERCP in this part of the world is under evolutionary phase with few tertiary care centers and this is the first study of its type from our center. Our aim was to analyze the success and complications with the wire-guide technique.

Aims & Methods: To evaluate the clinical features, laboratory values, success of cannulation with the wire-guided and complications related to wire-guided cannulation (WGC) technique, in patients who underwent endoscopic retrograde cholangiopancreatography (ERCP).

Results: This analysis was performed on patients records registered in our wards, between January 2016 to may 2023. A total of 5400 patients underwent 7000 ERCP procedures for a diagnostic or therapeutic purpose; 1527 patients were included. 1480 ERCPs (96.9%) were performed for therapeutic purpose and 47 (3.1%) for diagnostic purpose. There were 631 (41.3%) men and 896 (58.7%) women with a mean age of 47.3 ± 0.3. Technical success was achieved in all the cases (100%); our first procedure successful cannulation rate was 92.3% in 1410 patients, with a repeat procedure and success in all the remaining 117 (7.7%) patients. There was no procedure-related mortality in our patients. ERCP affected management in 97.3% (1486/1527 procedures).

Conclusion: WGC technique is helpful and efficacious technique and may increase the success rate of biliary or pancreatic duct cannulation.

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Disclosure: Nothing to disclose.

PP1086

WITHDRAWN

PP1087

PREDICTORS OF RECURRENT BILE DUCT STONE AFTER CLEARANCE BY ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) is now the gold standard for the treatment of bile duct stones. However, recurrence stones is frequent after their complete removal by endoscopic treatment.

Aims & Methods: The present study is retrospective, descriptive and analytic, including all patients who underwent ERCP for common bile duct (CBD) stones between January 2008 and January 2024.

Recurrent CBD stone was defined as the finding of a stone at least 6 months after the initial ERCP in which complete stone extraction was performed.

Factors associated with recurrent CBD stones were studied by logistic regression analysis.

Statistical analysis was performed using Jamovi software.

The aim of our study is to identify predictors of recurrent CBD stones after ERCP.

Results: Among 1151 patients who underwent ERCP for lithiasis pathology, 88 patients or 7.7% had recurrent CBD stone.

Mean age of the patients was 64.5 +/-13.4 years, with extremes ranging from 22 to 94 years. Sex ratio (M/F) was 0.69, with a female predominance of 59.1%.

20.5% of patients presented with acute angiocholitis and 6.8% with acute pancreatitis.

A periampullary diverticulum was found in 5.7% of patients (n=5).

In univariate analysis, the risk factors for recurrent CBD stone were: older age (OR :37; IC95% : [10,7- 127,2] ; p<0,001), female gender (OR :9,9 ;IC95% : [4,49-22,04] ; p<0,001), CBD diameter > 15 mm (OR : 7,36 ;IC95% : [5,51-9,8] ; p<0,001), presence of angiocholitis (OR : 2.06 ;IC95% : [1,12-3,8] ; p=0,02), number of CBD stones (OR :8,61; IC95% : [6,51-11,39] ; p=0,018), size of CBD stones (OR :2,51 ;IC95% : [1,43-4,39] ; p=0,001), while enlargement EST (OR :0,18; IC95% : [0,08-0,44] ; p<0,001) was a protective factor.

After multivariate analysis, and adjusting for the factors studied, only older age (OR:1.96; CI95%: [1.94-2.93]; p=0.001), number of CBD stones (OR: 1.99; CI95%: [1,2-3,4] ; p=0,037) and size of CBD stones (OR :4,06; IC95% : [3,51-6,72] ; p=0,003) were risk factors, and enlargement EST (OR :0,31; IC95% : [0,11-0,85] ; p=0,024) was protective factor.

Conclusion: In our study, the predictive factors for recurrent CBD stones were advanced age, the presence of impaction and a large stone.

Disclosure: Nothing to disclose.

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PP1088

DIAGNOSTIC PERFORMANCE OF ENDOSCOPIC ULTRASOUND-GUIDED FINE NEEDLE BIOPSY IN MALIGNANT LESIONS OF THE HEPATIC HILUM: A RETROSPECTIVE ANALYSIS FROM A TERTIARY REFERRAL CENTRE.

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Introduction: Diagnosing hepatic hilar malignancies is challenging due to the complex anatomical location and intricate nature of these lesions, necessitating advanced imaging modalities and specialised diagnostic approaches. To date, limited and dated studies with small sample sizes have focused on the diagnostic utility of endoscopic ultrasound fine-needle aspiration (EUS-FNA) in suspected malignancy of the hepatic hilum. However, only a few case series have reported similar findings for EUS fine-needle biopsy (EUS-FNB).

Aims & Methods: To address this gap, our study aims at assessing the diagnostic performance and safety profile of EUS-FNB a first-line tool in the diagnostic approach for suspected hepatic hilum malignancies. A single-centre, retrospective analysis of all the patients with biliary strictures and lesions involving the hepatic hilum who underwent EUS-FNB from October 2014 to April 2023 was conducted. All included patients were monitored after EUS-FNB via clinical follow-up for a minimum of 6 months or until death (follow-up <6 months). A descriptive statistical analysis for all the variables was performed. Sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and accuracy were finally calculated.

Variables	Patients (n=83)
Prior upper GI surgery, n (%)	5 (6.0)
Pancreaticoduodenectomy	1 (1.2)
Roux-en-Y gastric bypass	2 (2.4)
Distal gastrectomy (Billroth I reconstruction)	1 (1.2)
Ivor Lewis esophagectomy	1 (1.2)
Lesion size, mean ± SD	29.10 ± 13-66
Biopsy needle diameter, n (%)	
19G	2 (2.4)
20G	1 (1.2)
22G	76 (91.6)
25G	4 (4.8)
Passes, n (%)	
2	13 (15.7)
3	53 (63.9)
4	17 (20.4)
Presence of biliary stent, n (%)	19 (22.9)
Sample adequacy, n (%)	77 (92.8)
Repeated EUS-FNB, n (%)	9 (10.8)
Final histological results, n (%)	
CCA	63 (75.9)
HCC	3 (3.6)
Metastasis	5 (6.0)
NEN	2 (2.4)
Multiple myeloma	1 (1.2)
Undefined dysplasia	1 (1.2)
Non-malignant	8 (9.7)

Table 1. Characteristics of the study population and EUS-FNB procedures. Legend: CCA: cholangiocarcinoma; EUS-FNB: endoscopic ultrasound-guided fine needle biopsy, GI: gastrointestinal; HCC: hepatocellular carcinoma; NEN: neuroendocrine neoplasm; SD: standard deviation.

Results: A total of 83 consecutive subjects (M:F 51:32, mean age 69.01±10.10) with a radiological suspicion of malignant lesion of the hepatic hilum were included in the analysis (details in Table 1). Sensitivity, specificity, PPV, and NPV were 88.00% (95%CI: 78.44-94.36), 100.00% (95%CI: 63.06-100.00), 100.00% (95%CI: 94.56-100.00), and 47.06% (95%CI: 32.50-62.13), respectively. Overall accuracy was 89.16% (95%CI: 80.41-94.92).

The repetition of EUS-FNB after the first non-diagnostic one demonstrated increased sensitivity (92.00%, 95%CI: 83.40-97.01%) and overall accuracy (92.77%, 95%CI: 84.93-97.30%). Importantly, histological results were not affected by prior surgery of the upper gastrointestinal tract (n=5, p=0.083), or the presence of a biliary stent in place (n=19, p=0.728). The reported post-procedural adverse events were cholangitis (n=3, 3.6%) and mild pancreatitis (n=1, 1.2%).

Conclusion: This study emphasises the efficacy and safety of EUS-FNB as a first-line diagnostic tool for lesions of the hepatic hilum. Further prospective studies are warranted to determine the diagnostic performance of EUS-FNB for hilar lesions in comparison with ERCP-guided biopsies or brushing.

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Disclosure: Nothing to disclose.

PP1089

OPTIMIZING EUS-GUIDED CHOLEDOCHODUODENOSTOMY WITH LUMEN-APPOSING METAL STENTS FOR PRIMARY DRAINAGE OF MALIGNANT DISTAL BILIARY OBSTRUCTION (SCORPION-IIP): A PROSPECTIVE PILOT STUDY

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Introduction: EUS-guided choledochoduodenostomy (EUS-CDS) with a lumen-apposing metal stent (LAMS) is an alternative for ERCP in patients with a malignant distal biliary obstruction (MBO).(1,2) The main drawback of EUS-CDS using LAMS is the high rate of stent dysfunction which leads to cholangitis and reinterventions.(3,4)

Presumably, the short length and perpendicular angle of the LAMS to the bile duct contribute to the risk of stent dysfunction.(5) Therefore, the aim was to investigate whether placement of a fully covered self-expandable metal stent (FCSEMS) through the LAMS, thereby changing the axis of biliary drainage towards the descending duodenum, will decrease the risk of stent dysfunction while maintaining high technical success and low adverse event rates.

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Aims & Methods: We performed a prospective pilot study in patients with proven MBO and a bile duct diameter of at least 12mm, requiring biliary drainage, excluding patients with gastric outlet obstruction. Patients underwent biliary drainage with (as first procedure) EUS-CDS using a 6 or 8 mm LAMS with a 6 cm by 8 or 10 mm FCSEMS placed through the LAMS.
Primary outcome was stent dysfunction, defined as recurrent jaundice after initial clinical success, ongoing jaundice in combination with persistent dilatation of the bile ducts, or cholangitis. Secondary outcomes were technical success, clinical success, and adverse events (AEs).
Results: Overall, 27 consecutive patients eligible for EUS-CDS were enrolled. The placement of a LAMS was successful in 24/27 patients (89%), and placement of FCSEMS through the LAMS was successful in 20/24 patients (83%), in the remaining 4 patients a coaxial double pigtail stent was placed. In 2 patients there was persistent cholestasis requiring stent revision (10%), leading to a clinical success rate of 90%. No patients developed stent dysfunction after initial clinical success.
Periprocedural AEs occurred in 3 patients (11%) due to LAMS maldeployment which was solved intraprocedurally in all patients. In 1 patient this led to biliary peritonitis and fluid collections requiring percutaneous drainage, the other 2 patients recovered uneventfully.
Two patients experienced cholecystitis within 30 days after the procedure (10%), one patient who also had concomitant renal failure subsequently deceased.
The other patient recovered after antibiotics and percutaneous drainage. Two other patients deceased within 30 days which was unrelated to the procedure.

Median age, y (IQR)	**69 (62.5-77.5)**
Male sex, n (%)	14 (51.9)
Type of tumor, n (%)	
· Pancreatic ductal adenocarcinoma	27 (100)
Tumor stage, n (%)	
· Resectable	16 (59.3)
· Locally advanced	5 (18.5)
· Metastatic	6 (22.2)
Median serum total bilirubin, µmol/L (IQR)	224 (182-336.5)
Median diameter common bile duct, mm (IQR)	16 (16-20)

Table. Baseline characteristics in 27 patients undergoing EUS-CDS.

Conclusion: This study showed a stent dysfunction rate of 10% following technically successful EUS-CDS with placement of a FCSEMS through the LAMS. Improving the LAMS design may reduce the rate of stent dysfunction by improving the direction of bile flow through the stent towards the descending duodenum.
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Disclosure: Jeska A. Fritzsche, Marc G. Besselink, Olivier R. Busch and Mattheus C.B. Wielenga have no conflicts of interest or financial ties to disclose. Paul Fockens performed as a consultant for Olympus and Cook Endoscopy. Freek Daams reports research grants from Medtronic, and received speaker 's fees from Medtronic, and proctoring fees from Intuitive. Johanna W. Wilmink reports research grants from Servier, Merck, Nordic and Astra Zeneca. Rogier P. Voermans reports research grants from Boston Scientific and Prion Medical, performed as a consultant for Boston Scientific, and received speaker's fees from Mylan and Zambon. Roy L.J. van Wanrooij performed as a consultant for Boston Scientific. All outside the submitted work.

PP1090

ENDOSCOPIC ULTRASOUND-GUIDED TECHNIQUES FOR BILIARY DRAINAGE - COMPARATIVE ANALYSIS OF SINGLE CENTER DATA

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Introduction: Endoscopic retrograde cholangiopancreatography /ERCP/ is the standard technique in the treatment of unresectable malignant biliary obstruction/MBO/, unfortunately, a failure rate of 3% up to 12 % and a rate of complications up to 6,85% have been reported in the literature. Endoscopic ultrasound-guided biliary drainage/EUS – BD/ is a feasible alternative for biliary drainage, applicable especially with surgically altered anatomy. With several techniques available, hepaticogastrostomy /EUS - HGS/ and choledochoduodenostomy/EUS - CDS/ are most commonly performed amongst them.
Aims & Methods: To compare the most commonly used techniques for endoscopic ultrasound-guided biliary drainage in terms of efficacy and safety.
Methods:Retrospective analysis of a prospective database of patients undergoing EUS -BD in a single tertiary referral center. EUS–BD was performed after the failure of ERCP or as a first choice of drainage procedure in patients with altered anatomy. 102 patients were included in the EUS-HGS group 94,12%(n=96) with malignant biliary obstruction and 5.12%(n=6) with benign underlying disease. Of 24 patients included in the EUS–CDS group 95,83% (n=23) had underlying malignancy against one patient with benign disease.
Results: Considering the application of EUS – BD as a first choice of drainage procedure for patients with surgically altered anatomy we had 46 patients in the HGS group (45.1%) against 16.67%(n=4) for the CDS group. In 56/44,44%/ cases EUS BD was performed during the same session after unsuccessful cannulation. Evaluating results in terms of technical success rate, defined as successful stent placement – we found a 98,04 % success rate for the EUS-HGS group and 95,83% in the EUS–CDS group. We defined clinical success as a drop in the bilirubin levels by one-half in the first week following the procedure. The results between the two groups were not significantly different - 80.46% for the HGS against 80,95% for the CBS group. Concerning adverse events during the procedure, we found a ratio of 7.84% for the patients undergone HGS and 4,76% for EUS – CDS with no significant difference. The most common AE found was proximal stent misdeployment/n=3/. All of the events were managed either conservatively or during the same session with the placement of fully covered SEMS. Concerning early AE /during the first 7 days / we experienced 14,71% early

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adverse events in the HGS group/n=15/ and 16,67% /n=4/ in the CDS group with most common event postprocedure infection with no significant difference.

Conclusion: The two most common transluminal techniques for EUS biliary drainage - HGS and CDS are a promising tool with comparable technical and clinical success rates, adverse events, and overall survival. Both procedures are valid alternatives after failed ERCP or as a first step especially in patients with surgically altered anatomy.

Disclosure: Nothing to disclose.

PP1091

WITHDRAWN

PP1092

TRANS-DUODENAL GALLBLADDER EUS-GUIDED DRAINAGE: A SINGLE CENTER EXPERIENCE

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Introduction: Endoscopic ultrasound-guided gallbladder drainage (EUS-GBD) is becoming an important and effective alternative to percutaneous drainage (PTD) for patients not suitable for surgery. It is also an acceptable option for biliary drainage in oncologic patient, if retrograde drainage failed. Placing of a lumen apposing metal stent (LAMS) for EUS-GBD allows the removal of gallbladder stones too, so becoming a possible alternative to surgery in very selected cases.

Aims & Methods: The study aims to describe our endoscopic experience and management in patients unfit for surgery and in those oncologically advanced. Retrospectively analyzed cases with EUS-GBD, from December 2019 to September 2023, in a single peripheral hospital (Hopital Riviera-Chablais), using 3 types of LAMS (10x10mm, 15x10mm, 16x20mm), delivered with an electrocautery-enhanced catheter.

Results: EUS-GBD was performed in 23 patients (15:8=M:F) with median age of 83 years old (52-89). 13 patients (56.2%) had advanced cancer (6 pancreatic metastatic cancer, 5 cholangiocarcinoma, 1 breast metastatic cancer, 1 metastatic pulmonary NET) and 10 patients were affected by symptomatic lithiasic cholecystitis and they were not considered fit for surgery (whose 2 already received a PTD). EUS-GBD was attempted through the duodenum in all cases but one, previously treated with PTD and that was drained through the stomach.

In one case, the stent was mis-deployed through the duodenum (without entering the gallbladder lumen), so requiring the placement of an over-the-scope-clip and a different biliary drainage for biliary obstruction in an oncological patient. A 15x10 mm LAMS was used in 14 cases, 10x10 mm LAMS was used in 6 cases (including the failed one), and 16x20 mm in 2 cases. A double pigtail 5 Fr 7cm was placed through all 22 successfully placed LAMS. 3 patients underwent electrohydraulic-lithotripsy (EHL) through the LAMS after 4-8 weeks. In 5 patients LAMS was then removed.

Generally, except one complication, it was not described short term or long-term adverse events. 6/23 (26%) patients died for oncological disease progression and for other causes not procedure related.

Conclusion: EUS-GBD represents a promising therapeutic approach in patients with high risk of mortality. Considering the high rate of clinical and technical success as well as the possibility of managing endoscopically potential complications, this approach could be also proposed in peripheral centers and not only in advanced endoscopic ones.

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Disclosure: Nothing to disclose.

PP1093

WITHDRAWN

PP1094

SAFETY OF ENDOSCOPIC ULTRASOUND-GUIDED CHOLEDOCHO-DUODENOSTOMY IN PATIENTS WITH MALIGNANT OBSTRUCTION AND ASCITES

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Introduction: Ascites of any etiology presents a major problem in patients with malignant biliary obstruction (MBO) and failed ERCP, where trans-mural drainage is indicated, but presumed to be unsafe due to the risk of bile leak. We planned to evaluate the safety of Endoscopic Ultrasound-guided choledocho-duodenostomy (EUS-CDS) in our patients with ascites.

Aims & Methods: We palnned to evaluate the safety of endoscopic ultrasound-guided choledocho-duodenostomy (EUS-CDS) in malignant biliary obstruction (MBO) and ascites in a retrospective study. Prospectively maintained records of consecutive patients with MBO, with or without ascites, and undergoing EUS-CDS for biliary drainage, were retrospectively analysed. Technical success was defined as successful stent deployment, while clinical success was defined as biliary decompression (biochemical and radiological), and relief in cholangitis. Complications such as bile leak, fever and abdominal pain were recorded, and the data were compared between patients with and without ascites.

Results: Between November 2019 and March 2024, 60 patients underwent EUS-CDS for MBO. Of these, 38 (65%) were males and the mean age was 61.49 years. At the time of the procedure, 18 patients had ascites (30%) had ascites, of which 11 had Grade II and 7 had Grade III ascites. The most common malignancies wer pancreatic cancer (28), peri-ampullary carcinoma (12) and distal cholangiocarcinoma (9).

Other lesions seen were duodenal cancer (5), malignant lymph node metastases causing MBO (4) and gall bladder cancer (2). Technical success was achieved in 100% procedures, and clinical success was also achieved in all

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the patients. In the group with ascites, 3 patients had bile leak, which was managed conervatively with percutaneous trans-hepatic biliary drainage (PTBD). One patint had fever resposnsive to antibiotics; the other two had mild abdominal pain responding to analgesics. In the group with without ascites, two patients had bile leak, one had mild pain, and the other two had self-limited fever.

Conclusion: EUS-guided drainage is generally presumed to be contra-indicated in the presence of ascites [1].

EUS-hepatico-gastrostomy has been shown to be safe in the presence of ascites [2].

Our study shows that EUS-CDS is safe & effective in MBO with ascites. Further, larger studies are needed to evaluate its long-term safety and the effect on the survival and quality of life of the patients.

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Disclosure: Nothing to disclose.

PP1095

ENHANCING MRCP PRECISION: A COMPREHENSIVE ANALYSIS OF THE IMPACT OF INTRAMUSCULAR GLUCAGON ADMINISTRATION ON BILIARY VISUALIZATION

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Introduction: Magnetic resonance cholangiopancreatography (MRCP) has continuously evolved to enhance visualization capabilities. However, diagnosing biliary ductal system pathology, particularly early Primary sclerosing cholangitis (PSC), remains challenging.

Aims & Methods: This study investigates the influence of intramuscular glucagon (IMG) administration on final image quality and pancreato-biliary ductal system diameter in MRCP.

Forty patients (57.5% female; average age 34.45 ± 8.2) referred for Magnetic Resonance Enterography (MRE) underwent MRCP before and 8-12 minutes after IMG administration. Coronal T2-weighted fast spin-echo high-resolution 3D MRCP images were analyzed quantitatively and qualitatively by two independent MRI specialists. Quantitative assessments included transverse diameter measurements of 5 predefined biliary ductal (CBD, LHD, RHD…) and 3 pancreatic duct segments (head, body, tail). Qualitative evaluation used a five-point Likert-type scale (1=perfect visualization; 5=not visible) for the respective predefined segments. Interobserver variation was assessed using the Intraclass Correlation Coefficient (ICC).

Results: Following IMG administration, the diameters of all corresponding biliary and pancreatic segments significantly increased, with consistently strong interobserver agreement demonstrated pre and post IMG administration. Moreover, in qualitative analysis, post-IMG administration scores indicated a significant decrease ($P < 0.01$) in visualization scores, signifying improved visualization at all corresponding points for both radiologists compared to the pre-glucagon administration scores. The ICC scores pre and post IMG administration demonstrated moderate to strong agreement.

Conclusion: IMG administration improves MRCP imaging parameters by increasing ductal diameters and enhancing biliary tree visualization, underscoring its potential to aid in the detection of subtle or early pathological changes.

Disclosure: Nothing to disclose.

PP1096

WITHDRAWN

PP1097

EFFICACY AND SAFETY OF SEQUENTIAL CUTTING OF NASOBILIARY TUBES AS AN ALTERNATIVE TO STENT PLACEMENT AFTER NASOBILIARY DRAINAGE FOR THE TREATMENT OF HILAR MALIGNANT BILIARY STRICTURE

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Introduction: Hilar malignant biliary stricture, caused by malignant tumors, often leads to late diagnosis due to subtle initial symptoms, resulting in poor outcomes. Effective endoscopic interventions significantly improve quality of life and survival, yet often lead to post-operative complications like cholangitis.

Aims & Methods: This study aims to assess the effectiveness and safety of utilizing sequential nasobiliary tube cutting as an alternative to stent placement post-nasobiliary drainage in patients, specifically in mitigating post-operative cholangitis. Additionally, to identify risk factors associated with the development of post-operative cholangitis.

Patients diagnosed at Hangzhou First People's Hospital from January 2015 to January 2023 were divided into nasobiliary tube cutting group and conventional stent group. The former underwent nasobiliary drainage followed by tube cutting when stable, transitioning from external to internal drainage, while the latter received standard stent placement. Statistical analyses compared postoperative cholangitis incidence, drainage success, interventions, complications, and hospital stay, further examining risk factors for postoperative cholangitis.

Results: 1. This study encompassed a total of 208 patients, with 103 in the nasobiliary tube cutting group and 105 in the conventional stent group. Postoperative cholangitis occurred in 40 patients, with 12 cases (11.7%) in the nasobiliary tube cutting group and 28 cases (26.7%) in the conventional stent group, showing a statistically significant difference between the two groups (P=0.006). The number of patients whose total bilirubin levels decreased by more than 50% within 7 days postoperatively was 50 (48.5%) in the nasobiliary tube cutting group and 29 (27.6%) in the conventional stent group, a statistically significant difference favoring the nasobiliary tube cutting group (P=0.002). There was no statistically significant difference in the overall drainage success rate between the two groups (P=0.207). Besides postoperative cholangitis, there were no statistically significant differences in other postoperative complications between the two groups (P>0.05). The postoperative hospital stay was

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longer for patients in the nasobiliary tube cutting group than those in the conventional stent group, with median durations of 10.0 (7.0, 14.0) days versus 7.0 (5.5, 12.5) days, respectively, a difference that was statistically significant (*P*=0.024).

2. Based on whether postoperative cholangitis occurred, the patients were further categorized into a postoperative cholangitis group (n=40) and a non-postoperative cholangitis group (n=168). Univariate analysis revealed significant differences between the postoperative cholangitis group and the non-postoperative cholangitis group in terms of gender (male), Bismuth-Corlette classification (Type IV), number of ERCP procedures (≥2 times), and re-intervention (*P*<0.05). Multivariate logistic regression analysis identified Bismuth-Corlette classification (Type IV) (OR=3.207, 95%CI: 1.253-8.210, *P*=0.015) as an independent risk factor for developing postoperative cholangitis.

Conclusion: For patients with hilar malignant biliary stricture, sequential cutting of nasobiliary tubes as an alternative to stent placement after nasobiliary drainage can reduce the incidence of postoperative cholangitis and achieve early successful drainage, but it results in a prolonged postoperative hospital stay. Bismuth-Corlette classification (Type IV) is an independent risk factor for the development of postoperative cholangitis.

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PP1098

LONG-TERM RESULTS ON TREATMENT OF HEPATICOJEJUNOSTOMY ANASTOMOTIC STRICTURE WITH FCSEMS BY SINGLE BALLOON ENTEROSTOMY & EUS

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Introduction: Hepaticojejunostomy anastomotic stricture (HJAS) is one of the common adverse events after bile duct reconstruction surgery, which is reported to occur in 3–13%. Endoscopic approach using BE-ERCP has become the first choice due to less invasiveness, and long-term resolution rate is reported 50-90%.

Aims & Methods: In order to treat symptomatic hepatico-jejunostomy anastomosis stenosis (HJAS), we firstly perform balloon dilation or plastic stent or fully covered self-expandable metallic stent (FCSEMS) by single balloon ERCP (SBE-ERCP) and secondary insert FCSEMS by EUS-BD, in cases of severe stenosis or non-canulation.

Then we have investigated the efficacy of FCSEMS for HJAS. From January 2015 to December 2022, we have performed 698 procedures of SBE-ERCP in 348 cases and inserted FCSEMS in 37 cases with symptomatic HJAS (only benign cases, excluded malignancy). We have planned to deploy FCSEMS for 6 months and removed it after that. Then we investigated resolution rate of HJAS at removing FCSEMS, complication, and long-term resolution rate of HJAS after FCSEMS.

Results: Subjects were Male 26: Female 11 and Median age was 70 y/o. Surgical operation were 33 cases in B-Ⅱ+PPPD/ 4 cases in R-Y. Reason of deploying FCSEMS by SBE-ERCP were unresolved stenosis by balloon dilation or plastic stent (PS) in 29 cases. And FCSEMS for severe stenosis in 6 cases were inserted by direct EUS-Choledocho-jejunostomy (EUS-CJS) and undetectable HJA in 2 cases by Rendezvous EUS-HGS. Additionally, side by side PS were inserted in 32 cases.

Finally total subjects were 36, except for 1 case not reached to 6 months. Although 1 FCSEMS migrated into intrahepatic bile duct, 21 FCSEMS were removed endoscopically and remaining 15 FCSEMS were spontaneously dislocated after resolving HJAS. Resolution rate of HJAS at removing FCSEMS were 92% (33/36). Remained 3 cases were treated by additional deploying FCSEMS for 141 days in average.

In median 19.0 (0.4-69.1) months followed up period, complications were non-obstructive cholangitis in 8 cases which were treated conservatively and intrahepatic stone in 2 cases which were treated by SBE.

Resolution rate of HJAS at 1, 2, 3 years was 100%, 100%, and 100%. Recurrent HJAS occurred in 1 case at 46 months and treated by re-FCSEMS for 189 days.

Conclusion: Treatment of HJAS by FCSEMS were effective in long term period. SBE, EUS-CJS, and EUS-HGS were complimentary each other.

Disclosure: Nothing to disclose.

PP1099

SAFETY AND EFFICACY OF KAFFES INTRADUCTAL SELF-EXPANDING METAL STENTS IN THE MANAGEMENT OF POST-LIVER TRANSPLANT ANASTOMOTIC STRICTURES

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Introduction: Endoscopic management is the first-line therapy for post-liver-transplant anastomotic strictures. Although the optimal duration of treatment with plastic stents has been reported to be 8-12 months, data on safety and duration for metal stents in this setting is scarce. Due to limited access to endoscopic retrograde cholangiopancreatography (ERCP) during the COVID-19 pandemic in our centre, there was a change in practice towards increased usage and length-of-stay of the Kaffes biliary intraductal self-expanding stent in patients with suitable anatomy. This was mainly due to the theoretical benefit of Kaffes stents allowing for longer indwelling periods compared to the traditional plastic stents.

Aims & Methods: To compare the safety and efficacy profile of different stenting durations using Kaffes stents.

Adult liver transplant recipients aged 18 years and above who underwent ERCP were retrospectively identified during a 10-year period through a database query. Unplanned admissions post-Kaffes stent insertion were identified manually through electronic and scanned medical records. The main outcome was the incidence of complications when stents were left

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indwelling for 3 months *vs* 6 months. Stent efficacy was calculated *via* rates of stricture recurrence between patients that had stenting courses for ≤ 120 d or > 120 d.

Results: During the study period, a total of 66 ERCPs with Kaffes insertion were performed in 54 patients throughout their stenting course. In 33 ERCPs, the stent was removed or exchanged on a 3-month interval. No pancreatitis, perforations or deaths occurred. Minor post-ERCP complications were similar between the 3-month (abdominal pain and intraductal migration) and 6-month (abdominal pain, septic shower and embedded stent) groups - 6.1% *vs* 9.1% respectively, *P* = 0.40. All strictures resolved at the end of the stenting course, but the stenting course was variable from 3 to 22 months. The recurrence rate for stenting courses lasting for up to 120 d was 71.4% and 21.4% for stenting courses of 121 d or over (*P* = 0.03). There were 28 patients that were treated with a single ERCP with Kaffes, 21 with removal after 120 d and 7 within 120 d. There was a significant improvement in stricture recurrence when the Kaffes was removed after 120 d when a single ERCP was used for the entire stenting course (71.0% *vs* 10.0%, *P* = 0.01).

Conclusion: Utilising a single Kaffes intraductal fully-covered metal stent for at least 4 months is safe and efficacious for the management of post-transplant anastomotic strictures.

Disclosure: Nothing to disclose.

PP1100

WHAT IS THE BEST ULTRASOUND-GUIDED BILIARY DRAINAGE STRATEGY IN CASE OF GASTRIC OUTLET OBSTRUCTION? A MULTICENTER RETROSPECTIVE STUDY

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Introduction: Pancreatic cancers are dramatically increasing. At an advanced stage, pancreatic cancers can cause biliary obstruction (BO) and gastric outlet obstruction (GOO). The coexistence of BO and GOO is not uncommon and can lead to the inability to access the papilla resulting in ERCP failure or impossibility. The development of therapeutic endoscopic ultrasonography (EUS) has created new treatment options for double stenosis management. Optimising nutritional status by appropriate GOO management and an efficient biliary drainage are an important part of the oncological treatment of these patients.

This study aimed to compare the EUS-biliary drainage efficiencies in case of GOO in order to find the best therapeutic strategy.

Aims & Methods: We conducted an international multicenter retrospective study in 10 centers. All patients who underwent EUS-guided choledochoduodenostomy (EUS-CDS) or hepaticogastrostomy (EUS-HGS) between 2017 and 2023 for biliary obstruction that occurred simultaneously or after the management of duodenal obstruction were eligible for inclusion.

Patients with previous biliary drainage or a post-procedural follow-up shorter than 30 days (unless death occurs before) were excluded. Patients were followed until death or if they underwent pancreaticoduodenectomy.

Primary outcome was the rate of biliary stent obstruction requiring a new intervention. Secondary outcomes included technical and clinical success (defined as a reduction of bilirubin levels by at least 50 % at day 15 or normalization at day 30) and procedure-related adverse events.

Results: 204 patients from 10 centers have been included. The mean age was 70 years [62-78]. The mean bilirubinemia was 162.8 µmol/L (+/- 124.61). 65 % suffered from pancreatic cancer and more than half had a metastatic disease. 12.3 % had ascite. 79.9 % of patients underwent management for simultaneous biliary and gastric outlet obstruction.

The biliary drainage technical success was 96 % and the clinical success was 88 %. 33.3 % were treated by EUS-CDS + enteral stenting (ES) combination (A), 4.4 % by EUS-CDS + EUS-guided gastroenterostomy (EUS-GE) (B), 42.6 % by EUS-HGS + ES (C) and 19.6 % by EUS-HGS + EUS-GE (D).

The median follow-up was 95 days but 84.8 % of the patients died at the end of the follow-up. No difference was observed between the groups in term of technical success and clinical success (respectively *p=0.359* and *p=0.899*).

The post procedural severe adverse events (AGREE ≥ 3) rate was 9.3 % with a significative difference between combinations (A = 4.3 % ; B = 0 % ; C = 15.1 % ; D = 7.5 % ; *p = 0.022*), occurring more frequently in patients treated with EUS-HGS.

The biliary stent obstruction rate during follow-up was 16.2% with a significative difference between groups (A = 16.4 % ; B = 44.4 % ; C = 21.5 % ; D = 5.1 % ; *p = 0.023* %), in favour of EUS-HGS + EUS-GE combination.

Conclusion: Patients suffering from combined BO and GOO have a poor prognostic. The combination of EUS-HGS + EUS-GE appears to be the most effective in reducing the rate of biliary stent dysfunction. However, this efficacy comes at the cost of severe adverse events, suggesting that these fragile patients should be managed in a centre of adequate expertise.

Disclosure: Nothing to disclose.

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PP1101

EUS-CDS IS ASSOCIATED WITH LESS COMPLICATIONS COMPARED TO PTBD AFTER A FAILED ERCP IN PATIENTS WITH A DISTAL MALIGNANT COMMON BILE DUCT OBSTRUCTION: PROSPECTIVE OUTCOMES OF A DUTCH REGISTRY

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Introduction: Endoscopic retrograde cholangiopancreatography (ERCP) with placement of a self-expanding metal stent (SEMS) is the gold standard for drainage in case of malignant obstruction of the distal common bile duct (CBD). Nevertheless, 12.5% of these procedures fail. Percutaneous transhepatic biliary drainage (PTBD) is the most frequently used alternative, despite a reported 30-day mortality of 15-20% and morbidity of up to 60%. EUS-guided choledochoduodenostomy (EUS-CDS) with a lumen apposing metal stent (LAMS) is an emerging alternative drainage modality. Data on the long-term outcomes of these alternative techniques are limited.

Therefore, this Dutch registry aimed to provide insight in the chosen drainage modality after a failed ERCP-procedure and evaluated the outcomes prospectively.

Aims & Methods: This multicenter investigator-initiated registry included consecutive patients with malignant obstruction of the distal CBD after an unsuccessful ERCP over an inclusion period of 18 months. Patients were treated by either EUS-CDS or PTBD based by discretion of the endoscopist. Follow-up duration was 180 days. Primary endpoints were procedure-related morbidity and mortality within 90 days post-procedure.

Results: In total, 55 patients were included in the registry. EUS-CDS was performed in 44 patients (technical success 98%) and PTBD in 11 patients (technical success 100%). Median post-procedural hospital stay was four days shorter in the EUS-CDS group compared to the PTBD group (1 day (IQR 1-2 days) vs. 5 days (IQR 2-6 days), respectively; p=0.003). Median bilirubin reduction after 30 days was 117.5 µmol/L (IQR 50.3-210.0 µmol/L) in the EUS-CDS group and 102 µmol/L (IQR 83-191 µmol/L) in the PTBD group (p=1.00). Furthermore, LAMS was in place for a median of 85.5 days (IQR 28.5-180 days), while the PTBD needed revision after 21 days (IQR 7-32 days; p=0.001). Procedure-related morbidity after 90 days was 54.5% in the EUS-CDS group and 100% in the PTBD group (p=0.004). In the EUS-CDS group, 37.5% of the patients developed infectious complications and 48.5% non-infectious complications compared to 77.8%, and 72.7%, respectively in the PTBD-group.

Additionally, mortality rates within 90 days were higher in the PTBD group (63.6% vs 22.7%; p=0.024).

Conclusion: EUS-CDS is associated with less complications compared to PTBD after a failed ERCP in patients with malignant distal CBD obstruction, as it results in a shorter post-procedural hospital admission and lower morbidity and mortality rates. Although, a randomized controlled trial is needed to confirm this superiority.

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PP1102

WITHDRAWN

PP1103

SUPRA-PAPILLARY VERSUS TRANS-PAPILLARY STENTING FOR MALIGNANT BILIARY OBSTRUCTION: AN UPDATED META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: Malignant biliary obstruction (MBO) encompasses a variety of malignancies arising from the pancreaticobiliary system. The relatively common occurrence of duodenobiliary reflux (DBR) in patients with transpapillary stent placement for palliative management raises the importance of the development of alternative methods to prevent adverse outcomes.

Aims & Methods: We aimed to compare supra-papillary to trans-papillary stenting in patients with MBO for stent dysfunction and adverse events. PubMed, Embase, and Cochrane were systematically searched for randomized controlled trials (RCTs). The primary outcome was stent occlusion. Secondary outcomes included stent migration, occurrence of cholangitis or pancreatitis, cumulative duration of stent patency in days, and patient survival in days. Statistical analyses were conducted using R (version 4.3.1) and a random-effects model was employed.

Results: We found 3 RCTs with 161 patients (80 in the supra-papilla stent group vs. 81 in the trans-papilla group). Supra-papillary stent placement was associated with significantly lower rates of stent occlusion (OR 0.42;

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95% CI 0.20 to 0.87; p = 0.019), and longer cumulative duration of stent patency (MD 77.59; 95% CI 23.84 to 131.34; p < 0.01). The Trial Sequential Analysis for this outcome showed that this study has surpassed the required information size (n = 158) to ascertain the statistical difference for this outcome.

The analyses of stent migration (p = 0.471) occurrence of cholangitis (p = 0.525), and patient survival (p = 0.05) revealed no significant difference between both groups.

Conclusion: We found a significant improvement in stent occlusion rates and cumulative duration of stent patency favoring the supra-papilla group, with no difference for stent migration, occurrence of cholangitis, and patient survival for this comparison.

Disclosure: Nothing to disclose.

PP1104

LONG-TERM OUTCOMES OF BILIARY STENTING IN PATIENTS WITH UNRESECTABLE BILIOPANCREATIC MALIGNANCIES

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Introduction: The prevalence of biliopancreatic malignancies is steadily increasing. Most patients are inoperable due to the typically advanced stage of the disease at the time of diagnosis and require biliary drainage. Despite numerous large-scale studies, the identification of predictive factors for survival in these cancers to improve their prognosis remains a challenge.

The aim of our study was to evaluate the survival of patients who underwent biliary stenting as part of the management of unresectable biliopancreatic malignancies and to assess the factors influencing the survival rate.

Aims & Methods: We conducted an observational cohort study between January 1, 2021, and March 28, 2024. All patients who underwent biliary drainage for obstructive jaundice due to biliopancreatic cancer (pancreatic head adenocarcinoma, peri-hilar cholangiocarcinoma, gallbladder cancer, ampulloma) were included. Contraindications to palliative chemotherapy were poor general condition (ECOG/WHO performance status 3 or more) and hyperbilirubinemia due to inadequate biliary drainage.

Results: 82 patients were included in our study. The mean age in our series was 63.4 +/- 12 years. The sex ratio was 1. The median survival was 350 days (95% CI 268 – 514). 49.4% had pancreatic head adenocarcinoma, 30.6% had cholangiocarcinomas, 10% had ampullomas, and 10% had gallbladder cancer.

The median survival for pancreatic head adenocarcinoma was 450 days vs. 562 days for ampullomas, 356 days for cholangiocarcinomas, and 301 days for gallbladder cancer.

The type of cancer did not have a statistically significant difference on median survival (p = 0.95).

15% of patients had failed ERCP and underwent percutaneous biliary drainage. They had a 2.5 times higher risk of death (RR = 2.57, 95% CI 1.26-5.25, p = 0.01). Indeed, the median survival without death in cases of percutaneous drainage was 39 days vs. 375 in cases of successful ERCP (p = 0.007).

65% of patients were able to undergo palliative chemotherapy after stenting. The median survival with chemotherapy was 562 days vs. 30 days for patients who could not initiate it (p < 0.001). The latter had a 10 times higher risk of death in the month following stent placement (RR = 10.7, 95% CI 5.23-21.9, p < 0.001).

In our series, the presence of pancreatic head adenocarcinoma was associated with a higher probability of receiving chemotherapy (p = 0.04). However, it did not have a significant impact on median survival.

In multivariate analysis, only palliative chemotherapy seemed to improve the survival median (p < 0.001), while recurrence of jaundice and stent occlusion were associated with death (p = 0.03).

Conclusion: According to the results of our study, the prognosis of unresectable biliopancreatic cancers is poor but appears to be improved by a successful biliary stenting. Patients in whom biliary stent placement was a bridging procedure that allows chemotherapy seem to have higher survival rates. The generalization of endoscopic ultrasound-guided biliary drainage as an alternative to biliary percutaneous drainage could improve the survival rate in patients with failed ERCP.

Disclosure: Nothing to disclose.

PP1105

BRAIDED-TYPE STENT VERSUS LASER-CUT-TYPE STENT FOR PATIENTS WITH DISTAL MALIGNANT BILIARY OBSTRUCTION: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: There are two types of covered self-expandable metallic stents (CSEMS): laser-cut (L) and braided (B). The B-type has a strong axial force and can cause problems such as recurrent obstructive jaundice and cholangitis. The L-type is less stressful for the bile duct.

Aims & Methods: This study aims to compare CSEMS B-type versus L-type in patients with unresectable distal malignant biliary obstruction. We performed a systematic review in PubMed, Embase and Cochrane databases, searching for studies comparing both types of CSEMS in this population. The primary outcome was stent occlusion.

The secondary outcomes were tumor ingrowth, tumor overgrowth, stent migration, pancreatitis and cholecystitis. The odds ratio (OR) was used for binary outcomes with their respective 95% confidence interval (CI). Heterogeneity was assessed using the Cochran Q test and I^2 statistics.

Results: The meta-analyses included 4 studies and 430 patients. There was no significant difference for any outcome: stent occlusion (OR 1.28; 95% CI 0.73 to 2.23; p=0.388; I^2=0%), tumor ingrowth (OR 1.12; 95% CI 0.46 to 2.76; p=0.800; I^2=0%), tumor overgrowth (OR 1.23; 95% CI 0.36 to 4.25; p=0.740; I^2=0%), stent migration (OR 1.60; 95% CI 0.28 to 9.20; p=0.600; I^2=73%), pancreatitis (OR 1.28; 95% CI 0.53 to 2.90; p=0.559; I^2=0%) and cholecystitis (OR 0.73; 95% CI 0.26 to 2.06; p=0.553; I^2=0%).

Conclusion: In patients with distal malignant biliary obstruction, no significant difference was observed between CSEMS B-type and L-type.

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Disclosure: Nothing to disclose.

PP1106

12-MM VERSUS 10-MM METALLIC STENTS FOR DISTAL BILIARY OBSTRUCTIONS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Endoscopic drainage of malignant biliary obstruction using metal stents is widely used to treat obstructive jaundice. However, the occurrence of stent occlusion or migration raises the hypothesis using stents with larger diameters may be associated with better outcomes.
Aims & Methods: PubMed, Embase, and Cochrane were systematically searched for randomized controlled trials (RCTs) and observational studies comparing 10-mm vs. 12-mm metallic stents for recurrent unresectable biliary obstruction. The primary outcome was stent obstruction. Secondary outcomes included overall adverse events and stent migration. Statistical analyses were performed using R software version 4.3.1.
The odds ratio (OR) was used for binary outcomes, and the mean difference (MD) for continuous outcomes with their respective 95% confidence interval (CI). Heterogeneity was assessed using the Cochran Q test and I^2 statistics.
Results: We included six studies comprising 592 patients who underwent stent placement. Of those, 328 patients received a 10mm and 264 patients a 12mm stent. 12mm stent is associated with a lower incidence of stent obstruction (OR 0.48; 95% CI 0.27 to 0.88; p=0.02; I^2=53%). However, no statistically significant differences between groups were found for stent migration (OR 0.68; 95% CI 0.39 to 1.19; p=0.18; I^2=0%) and incidence of adverse events (OR 0.84; 95% CI 0.39 to 1.82;p= 0.66; I^2=46%).
Conclusion: 12-mm metallic stents significantly reduce the incidence of stent obstruction compared to 10-mm stents in managing recurrent unresectable biliary obstruction. However, our analysis did not demonstrate statistically significant differences between the two stent sizes regarding stent migration and overall adverse events.
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Disclosure: Nothing to disclose.

PP1107

ENDOSCOPIC BILIARY DRAINAGE IN THE PALLIATIVE TREATMENT OF KLATSKIN TUMORS: OUTCOMES AND FACTORS ASSOCIATED WITH SUCCESS OR FAILURE

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Introduction: Klatskin's tumor is a cholangiocarcinoma that develops from the right or left bile ducts and the upper part of the main bile duct. They are usually diagnosed at an advanced, inoperable stage, and have an extremely poor prognosis.
Biliary drainage is proposed in palliative situation and carries a high risk of infectious complications.
The aim of our work is to report the results of endoscopic biliary drainage as well as the factors associated with its success or failure.
Aims & Methods: This is a retrospective and analytical study of 75 patients, conducted between 2012 and 2024, including all patients admitted with Klatskin's tumor and for whom endoscopic drainage was indicated.
Factors associated with the success or failure of endoscopic treatment were studied by logistic regression analysis.
Statistical analysis was performed using SPSS version 22.0 software.
Results: The average age of our patients was 62.67 +/-12 years with extremes ranging from 31 years to 93 years. Our series was characterized by a male predominance of 68% or a sex ratio of 2.12.
Cholangiocarcinoma was classified as bismuth IV in 50.6% of patients, bismuth IIIa in 30% of patients, bismuth IIIb in 13% of patients and bismuth II in 6% of patients.
Sixteen percent of patients had liver metastases.
Endoscopic drainage was successfully performed in 81.3% of patients by plastic prosthesis in 32% of cases (n=24), by a metal prosthesis in 45.2% (n=34) and by nasobiliary drain in 4.1% (n=3).
Forty-seven percent of patients (n=35) had dilatation of the stenosis prior to prosthesis placement.
Causes of stenting failure were primarily related to failure of papilla catheterization, failure to pass the guidewire through the stenosis, or duodenal invasion by the tumor.
In multivariate analysis and adjusting for age, gender, Bismuth tumor type, presence of metastases and endoscopic dilatation of the stenosis,

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only the presence of metastases, endoscopic dilatation of the stenosis and Bismuth tumor type modified the outcome.

The Bismuth classification affect the success rate.

Indeed, endoscopic dilatation of the stenosis prior to stenting increases the success rate fourfold. Prosthesis increases the success rate by a factor of 4 [OR=4; p=0.01], whereas the presence of metastases decreases this rate by 65% [OR=0.35; p<0.001].

However, tumors classified as Bismuth IV [OR=8; p<0.001] or Bismuth IIIa [OR=5; p=0.004] were associated with a risk of endoscopic treatment failure.

Conclusion: Our study suggests that the presence of metastatic hilar cholangiocarcinoma classified as Bismuth IV or Bismuth IIIa appear to be associated with failure of endoscopic biliary drainage of Klatskin's tumors, whereas endoscopic dilatation prior to prosthesis placement appears to be associated with success.

Disclosure: Nothing to disclose.

PP1108

STUDY OF OUTCOME AND SUCCESS RATE IN PERCUTANEOUS TRANSHEPATIC BILIARY STENT FOR MALIGNANT BILIARY OBSTRUCTION IN SRINAGARIND HOSPITAL, KHON KAEN UNIVERSITY

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Introduction: Malignant biliary obstruction (MBO) is likely to be inoperative at first presentation. Palliative biliary drainage is essential to increase overall survival. Percutaneous transhepatic biliary stent (PTBS) is one of the most effective and safe procedure of biliary drainage in malignant biliary obstruction.

However, there is no previous study in outcome, success rate and complications of PTBS insertion for biliary drainage of MBO in Thai population.

Aims & Methods: Aims: to determine outcome, success rate and complications of PTBS insertion in MBO in tertiary care medical center in Thailand. Methods: A retrospective cohort study was performed on 41 patients with MBO underwent PTBS insertion from 2015 to 2021. Technical data, success rate, and outcome data including survival analysis, occlusion rate and complications were analysed.

Results: Of 41 patients, PTBS insertion had technical success 82.93% and functional success 70.59%. Median survival time of patients was 95 days (IQR 50-147 days). Occlusion rate occurred in only 22.6% of patients and all of them underwent percutaneous transhepatic biliary drainage (PTBD) as alternative palliative biliary drainage. Mean occlusion free survival time was 77.14 days (SD 51.68 days).

Majority of patients (79.4%) did not have PTBS-related complications. The most common complication was acute cholangitis that found only in 6 patients (17.6%). Median complication free survival was 91 days (IQR 60-155 days).

For the group of patients with functional success after PTBS insertion, the cumulative survival rates in 30- day,120-day, and 390-day and cumulative occlusion-free survival rates were statistically significantly better than those with functional failure (p=0.030 and p=0.007).

Demographic data	Patients
Age(year), mean (SD)	64.02 (9.21)
Male (%)	20 (48.78)
Etiologies of MBO Cholanhiocarcinoma (%) -Hilar type (%) -Distal type (%)	 33 (80.49) 32 (96.97) 1 (3.03)
Pancreatic cancer (%)	3 (7.32)
Ampullary cancer (%)	3 (7.32)
Survival data after PTBS insertion	Days
Overall survival, median (IQR)	95 (50-147)
Occlusion free survival, mean (SD)	77.14 (51.58)
Complication free survival, median (IQR)	91 (60-155)

Conclusion: Outcomes and success rate of PTBS insertion in Srinagarind hospital were 82.93% in technical success and 70.59% in functional success. Functional success of PTBS insertion was the most important issue that related with better cumulative survival and occlusion-free survival.

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Disclosure: Nothing to disclose.

PP1109

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PP1110

INNOVATIVE APPLICATION OF TIP-SPLIT DRAINAGE TUBE IN LAPAROSCOPIC HEPATOBILIARY SURGERY: A PARADIGM SHIFT FOR BILIARY FISTULA PREVENTION

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Introduction: Cholecystectomy is a common procedure in hepatobiliary surgery, with iatrogenic injury often leading to biliary fistula. Laparoscopic subtotal cholecystectomy (LSC) is employed when achieving a critical view of safety is challenging, but it comes with a high complication rate, including bile leaks. Laparoscopic common bile duct exploration (LCBDE) with laparoscopic cholecystectomy (LC) is utilized for cholecystolithiasis complicated with common bile duct (CBD) stones. Handling the CBD incision during LCBDE involves two approaches: indwelling T-tube drainage and primary suture.

Aims & Methods: Our research explores routine extra-drainage with tip-split drainage tubes (TSDT) for LCBDE+LC and LSC to reduce postoperative biliary fistulas. Early inspection of the cystohepatic triangle determines the feasibility of achieving a critical view of safety. If unattainable, LSC is considered, with attempts at standard laparoscopic cholecystectomy. TSDT drainage is decided based on the identification of bile flow from the cystic duct orifice. For cases without bile flow, TSDT drainage alone is used. If bile flow is identified, a purse-string suture of the cystic duct is performed, followed by TSDT placement. The LCBDE approach involves a longitudinal incision matching the stone's diameter. Choledochoscope exploration is completed with absorbable sutures or T-tube placement based on inflammation or the likelihood of residual stones. The TSDT is then placed.

Results: Our study, spanning 350 cases over five years, showed a high TSDT success rate, with no postoperative ERCP or PTCD interventions required. A learning curve of 1 to 2 cases was observed. Out of 350 patients, 30 underwent LSC and 320 LCBDE, resulting in 17 postoperative biliary fistula cases. Twelve were Grade A with no change in clinical management, and five were Grade B requiring minimal adjustments. All Grade A cases resolved within a week, while Grade B cases lasted over a week. Follow-up continued until bile leakage disappeared before catheter removal, with no progression to Grade C or relaparotomy requirements.

Conclusion: Our innovative TSDT technique effectively reduces postoperative bile leaks after LSC and LCBDE. Although not previously reported, TSDT appears superior to routine extra-drainage, being less technically demanding and less risky. General surgeons can likely adopt TSDT widely for improved hepatobiliary surgery outcomes.

Disclosure: Nothing to disclose.

PP1111

POST-CHOLECYSTECTOMY GUT MICROBIOME CHANGES AND CLINICAL IMPACT: A SYSTEMATIC REVIEW WITH NARRATIVE SYNTHESIS

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Introduction: Cholecystectomy (CCE) is commonly performed for treatment of gallstone disease. However, CCE may result in post-CCE syndrome (PCS) and diarrhea (PCD) in a subgroup of patients. In addition, epidemiological association of CCE with colorectal cancer (CRC) and metabolic-associated fatty liver disease (MAFLD) has raised concerns. CCE is expected to affect the enterohepatic circulation and result in gut microbiome changes. Such changes may account for clinical symptoms and risk for CRC and MAFLD. However, the spectrum and type of gut microbiome changes post-CCE has not been fully quantified and results have been inconsistent.

Aims & Methods: This systematic review aimed to clarify the effect of CCE on gut microbiome composition and the clinical impact. A systematic search was conducted on PubMed, Web of Science and Scopus in March 2024, combining keywords such as "cholecystectomy" or "post-cholecystectomy" with "gut microbiome" or "gut dysbiosis". Reference and citation searches were conducted on included articles. Original articles irrespective of study design that described gut microbiome changes post-CCE were included. Articles were independently reviewed by two authors and data were extracted and synthesized using narrative review. The quality of studies was assessed using the Newcastle-Ottawa scale.

Results: A total of 1373 articles (after excluding duplicates) were initially selected, and after review of the titles and abstracts, 14 eligible studies were identified and the full text reviewed. There were 3 prospective cohort and 11 cross-sectional/ case control cohort studies, of which 13 examined the bacteria microbiome using 16S rRNA gene sequencing, and 1 cross-sectional study examined the fungal microbiome using ITS1 rDNA gene sequencing. Changes in microbiome diversity indices and relative abundances were inconsistent. Among 3 prospective cohort studies, an increase in phylum Bacteroidetes was observed in a study with post-CCE interval more than 3 months, but not in the study with 3-month post-CCE interval. One study reported no change at the phylum level, but an effect for the class Clostridia and the orders Clostridiales and Coriobacteriales within 3 days of CCE. An increase in phylum Bacteroidetes post-CCE was also seen in 3 other cross-sectional studies. Five studies attempted to correlate post-CCE microbiome changes with symptoms. A decreased Firmicutes/ Bacteroidetes (F/B) ratio was observed in patients with PCS/ PCD in 3 out of 5 studies. Although changes in relative abundances of taxa correlated with symptoms, these were inconsistent, with an increase in genus Prevotella being the only common finding in 2 out of these 5 studies. Other significant observations post-CCE included a decrease of the potentially beneficial genus Faecalibacterium and an increase in the opportunistic pathogen Escherichia/Shigella, as well as an attenuated aging-associated intestinal microbiota. The single cross-sectional study that examined post-CCE fungal microbiome changes found that the relative abundance of Candida glabrata and Aspergillus were enriched, and Candida albicans was depleted.

Conclusion: CCE is associated with significant changes in gut microbiome diversity and relative abundances of taxa. Such changes have been correlated with clinical symptoms. These microbiome changes may potentially explain the risk of CRC and MAFLD noted in epidemiological studies.

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Discordant results between studies may relate to differences in patient populations and study methodologies.

Disclosure: Nothing to disclose.

PP1112

ANTIBIOTIC PROPHYLAXIS IN EMERGENCY CHOLECYSTECTOMY FOR MILD TO MODERATE ACUTE CHOLECYSTITIS: A SYSTEMATIC REVIEW AND META-ANALYSIS OF RANDOMIZED CONTROLLED TRIALS

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Introduction: Emergency cholecystectomy is the mainstay in treating acute cholecystitis (AC). In actual practice, perioperative prophylactic antibiotics are used to prevent postoperative infectious complications (PIC), but their effectiveness lacks evidence. We aim to investigate the efficacy of prophylactic antibiotics in emergency cholecystectomy.

Aims & Methods: We searched PubMed, Embase, Cochrane CENTRAL, Web of Science (WOS), and Scopus up to June 14, 2023. We included randomized controlled trials (RCTs) that involved patients diagnosed with mild to moderate AC according to Tokyo guidelines who were undergoing emergency cholecystectomy and were administered preoperative and/or postoperative antibiotics as an intervention group and compared to a placebo group. For dichotomous data, we applied the risk ratio (RR) and the 95% confidence interval (CI), while for continuous data, we used the mean difference (MD) and 95% CI.

Results: We included seven RCTs encompassing a collective sample size of 1747 patients. Our analysis showed no significant differences regarding total PIC (RR= 0.84 with 95% CI (0.63, 1.12), P=0.23), surgical site infection (RR= 0.79 with 95% CI (0.56, 1.12), P=0.19), distant infections (RR= 1.01 with 95% CI (0.55, 1.88), P=0.97), non-infectious complications (RR= 0.84 with 95% CI (0.64, 1.11), P=0.22), mortality (RR= 0.34 with 95% CI (0.04, 3.23), P=0.35), and readmission (RR= 0.69 with 95% CI (0.43, 1.11), P=0.13).

Conclusion: Perioperative antibiotics in patients with mild to moderate acute cholecystitis did not show a significant reduction of postoperative infectious complications after emergency cholecystectomy. (PROSPERO registration number: CRD42023438755).

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Disclosure: Nothing to disclose.

PP1113

SINGLE-DIGITAL CHOLANGIOSCOPY-ASSOCIATED ADVERSE EVENTS IN SLOVAKIA – RETROSPECTIVE ANALYSIS

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Introduction: Single-operator cholangioscopy (SOC) has become an important diagnostic utility principally indicated for the diagnosis of indeterminate strictures of the biliary system and for the treatment of difficult biliary or pancreatic stone(s). However, SOC carries additional risks compared to standard endoscopic retrograde cholangiopancreatography.

Aims & Methods: We performed a retrospective analysis of all SOCs between 2015-2023 in three tertiary centres in Slovakia. The primary outcome was the frequency and severity of adverse events (AEs) after SOC according to the AGREE classification. The operating characteristics of the SOC to detect malignant lesions and treat lithiasis with electrohydraulic (EHL) or laser lithotripsy (LL) were also analysed.

Results: A total of 182 SOCs (180 peroral, 2 percutaneous) were performed in 149 patients (pts) [48,9 % female; mean age 59,8 years (range: 20 – 91)]. The main indications for SOC were determination of biliary strictures [92 (51%)] and removal of stones from the common bile duct or the intrahepatic duct stones [82 (45%)] and pancreatic stone(s) [8 (4%)], whereas in 54% procedures EHL was used for the treatment of stones. Overall, AEs occurred in 14% (25/182); cholangitis 7% (Grade II), pancreatitis 6% (Grade I) and bleeding 1% (Grade II) according to AGREE. AEs occurred more frequently after diagnostic (14/25) compared to therapeutic procedures (11/25). There was no difference in overall AEs between EHL and LL, however, LL was associated with a higher single session disruption and shorter procedure time compared to EHL. Indeterminate stricture was evaluated in 56 non- PCS pts, while PSC stenosis was evaluated in 18 pts. A woman underwent simultaneously peroral and percutaneous SOC to disrupt 60 mm of stone in the same session. Tissue sampling was performed in 74 procedures, diagnostic accuracy for visual and histological assessment achieved 97.8% vs 83.2% respectively.

Conclusion: (1) SOC provides a high diagnostic yield in patients with undetermined biliary stenosis;

(2) SOC guided stone lithotripsy was effective in most patients and (3) The most frequent complication was cholangitis in 7% of patients.

Additionally, this is the first Slovak analysis of the efficacy and safety of SOC in clinical practice.

Disclosure:

Nothing to disclose.

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PP1114

PREDICTORS OF FAILURE OF ENDOSCOPIC RETROGRADE CHOLANGIOGRAPHY IN CLEARING BILE DUCT STONE ON THE INITIAL PROCEDURE.

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Introduction: Evaluate the predictive factors of failure rate of clearing the biliary system from stones at the initial ERCP.

Aims & Methods: This is a retrospective descriptive and analytical study from September 2004 to September 2023.

All patients with bile duct stones, who underwent ERCP with endoscopic sphincterotomy and stone extraction by standard techniques (extraction balloon or Dormia basket) were included in this study.

Results: A total of 1080 ERCPs were performed to extract biliary duct stones .

The mean age was 58.9 +/- 14.4 years and 59.5% were females.

The mean diameter of the Bile duct was 13.4+/- 4.31 mm. The presence of a biliary stricture in 6.3%.

The primary vacuity rate was 75.1%. Supplementary techniques were used in 22.7% of cases.

On univariate analysis, the predictive factors for failure after use of standard techniques were: age(OR: 0.9;CI95%:-0.03- -0.01;p<0.001); gender(OR:0.7;CI95%:-0.5--0.02;p=0.036); previous cholecystectomy (OR: 1.3,CI95%:0.02-0.6;p=0.035);presence of angiocholitis(OR: 0.4; CI95%:-1.1- -0.4;p<0.001); dilatation of the CBD >15mm (OR:0,8,CI95%:-0.2-0.1;p<0.001), biliary stricture (OR:0.3;CI95%:-1.7--0.7;p<0.001) and the presence of impacted and/or large stone (OR:0.1;CI95%:-2--1;p<0.001).

On multivariate analysis, only the presence of angiocholitis (OR:1.9;-CI95%: 0.2-1;p=0.001), impacted and/or large stone (OR:2.5;CI95%:0.5-1.3;p<0.001), dilatation of the CBD >15mm (OR:0.88;CI95%:-0.17--0.07;p<0.001), and biliary stricture (OR:2.9;CI95%:0.4-1.7;p=0.002) were significantly associated with failure of the primary CBD vacuity rate.

The overall vacuity rate after using the supplementary techniques was 92.4%.

Conclusion: In our study, predictive factors for failure of the primary CBD vacuity rate were the presence of angiocholitis, impacted and/or large stone, dilatation of the CBD (>15mm) and biliary stricture.

Disclosure: Nothing to disclose.

PP1115

ENDOBILIARY RADIOFREQUENCY ABLATION USING A NOVEL BIPOLAR STENT-BASED ELECTRODE FOR MALIGNANT BILIARY OBSTRUCTION: PRECLINICAL STUDY IN PORCINE MODEL

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Introduction: Malignant biliary obstruction (MBO) poses significant therapeutic challenges, particularly in patients ineligible for surgery [1]. Endobiliary radiofrequency ablation (RFA) has emerged as a promising palliative therapy, aiming to enhance stent patency and patient survival [2]. However, conventional catheter-based electrodes may encounter limitations, such as incomplete tumor ablation due to insufficient contact with the inner wall of the bile duct [3].

Aims & Methods: This study aimed to validate the feasibility of the stent-based electrode (SBE)-RFA system with polarity-switching and evaluate its efficacy in achieving circumferential ablation lesions in the porcine bile duct. A bipolar SBE consisting of two electrodes fabricated by braiding nitinol wires was designed and characterized along with a custom RFA generator capable of a polarity-switching system. *Ex vivo* studies using porcine livers were performed to evaluate the ablation ranges under various ablation protocols combining output power and ablation times.

Subsequently, an animal study involving nine male Yorkshire pigs was performed, employing selected protocols based on *ex vivo* findings. The RFA procedure was conducted in the porcine common bile duct (CBD) under fluoroscopic guidance, with pre- and post-procedural assessments including cholangiography, endoscopic examination, and histological analysis.

Results: The SBE with custom RFA generator was demonstrated that temperature feedback and power output were accurate and showed high reproducibility. The SBE was successfully positioned at the correct location during the procedure through fluoroscopy and ultrasound guidance, and complete contact with the bile duct wall was confirmed. The ablation ranges of the porcine liver increased with output power and time, showing proportional growth and an even ablation area between electrodes. Endobiliary RFA in the porcine CBD was technically successful without ductal perforation, resulting in luminal narrowing proportional to output power and ablation time. Mucosal damage, including hemorrhagic erosion and erythema, was observed, intensifying with higher RFA-related parameters. Histological examination revealed increased inflammatory cell infiltration, hemorrhage, and cellular necrosis in correlation with output power and time. Additionally, the ablation area in the porcine CBD was formed in an even circumferential shape, and this also proportionally increased as output power and time increased.

Conclusion: In conclusion, this study demonstrates the technical feasibility and efficacy of endobiliary RFA utilizing a novel bipolar SBE-RFA system with polarity-switching system. This system exhibited controlled and predictable ablation ranges in the porcine CBD, offering potential as a palliative therapeutic option for unresectable MBO. Although further studies in obstructive animal models and long-term efficacy assessments are necessary, these findings suggest that promising prospects form the clinical application of the bipolar SBE-RFA system in managing MBO.

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PP1116

FORGING NEW PATHS: GROUNDBREAKING INSIGHTS FROM MEXICO'S FIRST COMPARATIVE STUDY ON ENDOSCOPIC ULTRASOUND-GUIDED BILIODIGESTIVE DERIVATIONS

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Introduction: Biliodigestive diversions are surgical procedures aimed at bypassing obstructions or restoring the flow of bile from the liver to the digestive tract. These procedures are commonly performed in patients with biliary tract obstructions caused by conditions such as tumors, strictures, or stones. By creating an alternate pathway for bile drainage, biliodigestive diversions help alleviate symptoms and prevent complications associated with biliary obstruction, improving the patient's quality of life and overall prognosis.
There are various techniques for biliodigestive diversions, including hepaticojejunostomy, hepaticogastrostomy, and choledochoduodenostomy, each tailored to the specific needs and anatomical considerations of the patient.
Aims & Methods: Aims: Compare the efficacy and safety of biliodigestive diversion procedures, hepaticogastrostomy and choledochoduodenostomy, in patients with biliary tract obstructions.
Methods: Restrospective comparative study in patients with biliary tract obstructions undergoing either hepaticogastrostomy or choledochoduodenostomy at Centro Medico Nacional 20 de Noviembre will be included in the study from 2018- 2024.
Results: We evaluated 65 patients in total, median age 65 years, 44% were female, we had 3 biliodigestive procedures: Coledocoduthdenostomy 50%, Hepaticogastrostomy 28.1% and Rendez-vous 12.5%, the type of prothesis were 10x8 30%, 8x8 10%, Hot Axius in 5%, 10x6 16%, technical success reported was 87.5% and clinical success was 89.1, Five patients experienced complications, with peritonitis being the most frequent, followed by bleeding.
These findings underscore the importance of vigilant post-procedural monitoring and prompt management of complications to ensure optimal patient outcomes.
Conclusion: These results are encouraging and suggest that both choledochoduodenostomy and hepaticogastrostomy are effective procedures for addressing biliary obstructions in this patient cohort.
Additional Considerations that we want to mention, It's important to note that both technical and clinical success are important indicators of procedure efficacy and safety and that further analysis would be helpful to explore any association between the type of procedure performed and long-term outcomes, such as patency duration and patient quality of life.Overall, these results provide valuable insights into the practice of endoscopic ultrasound-guided biliodigestive diversion in Mexico and may serve as a basis for future research and clinical practice in this field.
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PP1117

ADVERSE EVENTS ASSOCIATED WITH ENDOSCOPIC ULTRASOUND-GUIDED BILIARY DRAINAGE

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Introduction: Endoscopic ultrasound-guided biliary drainage (EUS – BD) has become widely accepted over the past several years for complicated biliary drainage when conventional endoscopic retrograde cholagiopancreatography is not successful or feasible due to variety of constrains. The success rate of the procedure is between 90% an 97% and the complication rate is less than 10%.
Aims & Methods: The aim of the study is to evaluate the safety and efficacy of EUS-BD performed in our unit while assessing the risk factors associated with the procedure.
We performed retrospective analysis of a prospective data base from March 2020 to October 2023. The patients underwent EUS – BD using linear scope (Fujifilm EG – 580UT). Several procedures are possible. The procedure was tailored to individual patient's anatomy.
Results: We identified 140 patients who underwent EUS-BD during the study period (men- 75, women- 65) – 130 (92.8%) with malignant and 10 (7.1%) with benign disease. Gender distribution shows higher rate of post-procedural complications in men (29,7% vs 21,5%), nevertheless there is no statistically significant correlation between gender and complication rate. The medium age of the patients was 66.3 years (38 years – 92 years). No statistically significant correlation was found between age, presence of arterial hypertension, diabetes mellitus and adverse events.
We analyzed the different therapeutic modalities including EUS - CDS (n = 20), EUS - HGS (n = 88), antegrade stenting (n = 12) and rendezvous (n = 13), hepaticojejunostomy (n = 5), cholecystogastrostomy and -duodenostomy (n = 2). We found no statistically significant correlation between the type of technical modality and the incidence of peri- and post-procedural complications.

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The technical success was 96.6% (n = 137). Mean duration of the procedure was 42.15 min. (10 min. – 170min.). Mean hospital stay was 5 days (2 days – 10 days). Adverse events were detected in 38 patients (27.1%). The various complications that have been reported were classified into periprocedural and postprocedural:

- Periprocedural adverse events were detected in 7 (5%) patients: 2 (1.4%) with bleeding, 2 (1.4%) with stent migration, 3 (2,1%) with loss of access.
- Postprocedural complications are deteced in 29 (20,7%) and we divided into:

1. Early (up to 7 days after procedure) – n = 21 (15%) of them 13 (9,3%) with cholangitis, 3 (2,1%) with abdominal pain, 1 (0,7%) with bleeding, 2 (1,4%) with pancreatitis, 1 (0,7%) with pneumoperitoneum and 1 (0,7%) patient with biloma.
2. Late (a week after the procedure) – n = 8 (5,7%) with cholangitis.

- There were 2 deaths related to the procedure (1.4%) in patients with advanced oncologic disease.

All cases were managed conservatively and did not require surgery, intensive care or interventional radiology. Three (2,1%) of the periprocedural complications were solved by performing PTBD immediately after EUS-BD by the same team.

Conclusion: EUS-BD is effective tool for biliary drainage both in naïve patients and those with altered biliary anatomy. Along with the possibility of emergency PTBD, it has favourable technical and clinical success rate and safety profile and therefore the potential to shift the current paradigms that define management of complicated biliary obstruction.

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PP1118

THE IMPACT OF CHOLANGIOSCOPE DIAMETER VARIATION ON CHOLANGIOSCOPY OUTCOMES: A RETROSPECTIVE COMPARATIVE STUDY

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Introduction: Digital single-operator cholangioscopy (DSOC) stands as an invaluable diagnostic and therapeutic instrument, offering direct visualization of the biliary ducts. The use of cholangioscopes with higher diameter may raise concerns regarding patient discomfort and safety.

Smaller diameter cholangioscopes could offer enhanced maneuverability and decreased risk of adverse events, improving procedural success rates.

Aims & Methods: The aim of this study is to compare the impact of cholangioscope diameter variation on cholangioscopy outcomes, with a focus on procedure success rates. A retrospective, comparative study performed in patients who underwent either diagnostic or therapeutic DSOC from January 2021 to August 2023 were retrospectively collected. A p <0.05 was considered statistically significant.

Results: A total of 128 procedures were performed in 119 patients. Median age was 62.5 (44 – 71) and 46.1% were female. 52.3% of procedures were performed with the 9.3F cholangioscope, 53 (41.4%) were performed with the 11.1F, and 6.2% procedures were performed with the 7F cholangioscope. The 9.3F cholangioscope was used mainly for diagnostic procedures (61.2%) when compared to the 11.1F (39.6%) (*P*= .01). The most common diagnostic indication was indeterminate biliary stricture (9.3F: 31.3% vs 11.1F: 13.2%) (*P*= .01).

Most lesions were in the common bile duct (CBD). Two procedures performed with the 9.3F scope switched to a lower caliber scope (7F) to surpass the lesions, while 4 procedures performed with 11.1F required the application of a 9.3F to surpass the lesions and finish the procedures (P=.176). The 9.3F allowed the evaluation of the intrahepatic ducts in 6 procedures. More therapeutic procedures were performed with the 11.1F cholangioscope (60.4%), while 38.8% with the 9.3F. The main indication was lithotripsy.

Most procedures with a stone size >20mm were performed with the 11.1F (61%) (P=.01), with a total removal of stones of 64.5%. Technical success was higher in the 9.3F (97.0% vs 90.6%), clinical success was higher in the 11.1F (96.2% vs 94.0%).

All cases performed with the 7F cholangioscope were diagnostic, 7/8 procedures with 7F allow a complete evaluation of the intrahepatic ducts, 1/8 had a complete post-surgical CBD stenosis, and achieved a 100% technical and clinical success rate.

Conclusion: Cholangioscope diameter should be tailored to the specific diagnostic or therapeutic needs. While the 9.3F cholangioscope is more suitable for diagnostic procedures and has a higher technical success rate, the 11.1F is preferable for therapeutic interventions, particularly for larger biliary stones where higher caliber lithotripter is required. The 7F cholangioscope presents an effective alternative for complete intrahepatic duct evaluations.

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Disclosure: Carlos Robles-Medranda is a key opinion leader and consultant for Pentax Medical, Steris, Medtronic, Motus, Micro-tech, G-Tech Medical Supply, CREO Medical, EndoSound, and mdconsgroup. All other authors have nothing to declare.

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Poster
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Pancreas

09:00-18:00 / Poster Exhibition

PP1119

ISLET AUTOANTIBODY POSITIVITY ACCORDING TO GLUCOSE TOLERANCE AND AGE IN CYSTIC FIBROSIS: SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: With chronic inflammation, recurrent infections, and dysbiosis, people with CF (pwCF) are predisposed to autoimmunity. However, the role of autoimmune-mediated pancreatic beta-cell damage is yet unclear in the development of cystic fibrosis-related abnormal glucose tolerance (CF-AGT).

Aims & Methods: We aimed to assess the prevalence of various islet autoantibodies (AABs) according to glucose tolerance and age and the risk associated with islet AAB positivity for developing CF-AGT. A systematic search across four databases (CENTRAL, Embase, PubMed, Web of Science) identified studies reporting on AABs against glutamic acid decarboxylase (GADA), insulin (IAA), islet cell (ICA), insulinoma-associated protein 2 (IA-2A), and zinc transporter 8 (ZnT8A) in pwCF (CRD42020155846). Prevalence and odds ratios (OR) were calculated with 95% confidence intervals (CI) using a random effects model with subgroup analysis by glucose tolerance and age.

Results: Analysis of 20 eligible studies (2229 pwCF) revealed an overall prevalence of islet AAB positivity at 4% (CI: 2-9%) and multiple positivity at 1% (CI: 0-11%). IAA exhibited the highest prevalence at 6% (CI: 3-14%), followed by GADA at 5% (CI: 2-11%). IA-2A and ICA had prevalence rates of 2% (CI: 1-7%) and 1% (CI: 0-9%), respectively. Islet AAB prevalence trended higher in children than adults and in CFRD vs. non-CFRD. Notably, CFRD showed significantly higher odds for multiple (OR 2.71, CI: 1.49-4.93) and any AAB positivity (OR 2.82, CI: 1.22-6.51) compared to the non-CFRD group. Among the AABs, GADA exhibited the strongest CFRD association (OR 4.63, CI: 3.42-6.28), with also significant associations for ICA and IA-2A (OR 3.57, CI: 1.05-12.18 and OR 2.36, CI: 1.29-4.34).

Conclusion: While islet AAB prevalence remains low in pwCF overall, higher AAB prevalence was detected in childhood, and islet AAB positivity was associated with CFRD, emphasizing the need for early screening. Timely intervention in these high-risk groups is essential to manage early-onset CF- AGT and prevent potential complications.

Disclosure: Nothing to disclose.

PP1120

EXPLORING LONG-TERM VIABILITY AND GENE EXPRESSION DYNAMICS IN MOUSE PANCREATIC DUCTAL ORGANOIDS

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Introduction: Organoid models simulate organ function and structure, offering a physiologically relevant platform for studying organ development, disease mechanisms, and drug responses. Mouse pancreatic ductal organoids serve as valuable models for pancreatic research, yet their long-term viability remains uncertain. Simultaneously, pancreatic cancer, predominantly pancreatic ductal adenocarcinoma (PDAC), poses significant clinical challenges globally. Despite its prevalence and lethality, the genetic alterations driving PDAC progression remain largely elusive. Thus, understanding the gene expression changes in PDAC is crucial for therapeutic advancements.

Aims & Methods: Our study aimed to sustain mouse pancreatic ductal organoids long-term and explore associated gene expression changes, focusing on cancer-related markers. Ductal organoid cultures from wild-type FVB/N mice (N=3) were propagated for 40 passages (p40). RNA was isolated at various time points (p3/5/10/20/30/40) for transcriptomic analysis, with p3 serving as the starting time point. Passaging frequency and organoid morphology were monitored. Transcriptomic data underwent Ingenuity Pathway Analysis (IPA) to elucidate changes over passages in the organoids.

Results: The passaging frequency subsequently declined with time. The morphology of the organoids underwent noticeable changes starting from p20, characterized by the emergence of outgrowths surrounding the domes and the presence of adhesions in the organoids. The preliminary findings from the transcriptome data indicate that the markers associated with ductal cell fate and brush border development exhibit the greatest transcripts per million (TPM) values at p10. The expression of *Cftr*, *Ano1*, and *Adcy8*, which play a crucial role in maintaining the balance of ions in ductal cells, exhibited a notable decline starting at p20. Similarly, starting from p20, the TPM values of cancer-related markers exhibited an unregulated escalation, and there was a notable alteration in the expression of genes that govern the G1/S checkpoint regulation in the cell cycle.

Conclusion: Our research shows that mouse organoid cultures are sustainable for studying ductal cell functionality up to p10. Additionally, we observed potential signs of spontaneous carcinogenesis making suggesting that this model could be utilized for *in vitro* tumorigenesis studies. These findings have the potential to enhance comprehension of the mechanisms involved in the progression of PDAC.

Disclosure: Nothing to disclose.

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PP1121

THE EFFECTS OF CFTR MODULATOR THERAPY ON ENDOCRINE PANCREATIC FUNCTION IN PATIENTS WITH CYSTIC FIBROSIS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: CFTR modulators represent a groundbreaking therapeutic approach in cystic fibrosis (CF) management, targeting the underlying cause of the disease. These therapies have the potential to directly influence pancreatic beta-cell function and pathways involved in insulin secretion, thereby improving the glycemic status of people with CF (pwCF).

Aims & Methods: We aimed to evaluate the effect of CFTR modulator therapy on endocrine pancreatic function in patients with CF. A systematic search of three databases (PubMed, Embase, and CENTRAL) was conducted up to March 6, 2024, to identify studies examining the impact of CFTR modulators on glucose metabolism in pwCF comparing variables before and after CFTR modulator therapy.

Outcomes of interest included measures from oral glucose tolerance tests (OGTT), continuous glucose monitoring (CGM), and glycated hemoglobin (HbA1c). Meta-analysis was initiated using random-effects models to estimate pooled mean differences (MD) and 95% confidence intervals (CIs). Subgroup analyses were created based on CFTR modulators and length of therapy.

Covariate regression analyses were made to assess the relationship between therapy length and pooled MD. A p-value less than 0.05 was considered to be statistically significant. The study protocol was registered in PROSPERO with registration number CRD42024516198.

Results: A total of 1081 records were identified, 73 of which proved to be eligible for inclusion. Pooled analysis of CFTR modulators did not demonstrate a significant effect on 60-minute and 120-minute plasma glucose levels (Table 1). However, elexacaftor/tezacaftor/ivacaftor therapy resulted in significantly reduced plasma glucose levels at 120 minutes (MD: -1.80 mmol/L; 95% CI -2.60, -1.00). Average glucose, glycemic variability, and % of time below 3.9 mmol/L interstitial glucose level were comparable before and after therapy. Meta-regression revealed no significant effect of length of therapy on 60- and 120-minute plasma glucose levels (p=0.512 and p=0.631 respectively).

Outcome	N° of included studies (N° of patients)	Result of the meta-analysis (mean difference with 95% CI)	Statistical heterogeneity
OGTT 60' plasma glucose	11 (253)	0.34 mmol/L (95% CI -0.54, 1.21)	I^2= 79% (95% CI 62, 88), p<0.001
OGTT 60' plasma glucose (lumacaftor/ivacaftor subgroup)	4 (72)	1.41 mmol/L (95% CI -0.60, 3.42)	I^2= 80% (95% CI 46, 92), p=0.002
OGTT 60' plasma glucose (elexacaftor/ tezacaftor/ivacaftor subgroup)	6 (133)	-0.05 mmol/L (95% CI -1.28, 1.18)	I^2= 81% (95% CI 58, 91), p<0.001
OGTT 120' plasma glucose	15 (338)	-0.62 mmol/L (95% CI -1.29, 0.04)	I^2= 80% (95% CI 67, 87), p<0.001
OGTT 120' plasma glucose (lumacaftor/ivacaftor subgroup)	5 (116)	-0.73 mmol/L (95% CI -1.90, 0.44)	I^2= 68% (95% CI 18, 88), p=0.013
OGTT 120' plasma glucose (elexacaftor/ tezacaftor/ivacaftor subgroup)	3 (86)	-1.80 mmol/L (95% CI -2.60, -1.00)	I^2= 0% (95% CI 0, 90), p=0.750
Average glucose	5 (65)	-0.16 mmol/L (95% CI -0.49, 0.17)	I^2= 0% (95% CI 0, 79), p=0.428
Glycemic variability (standard deviation of glucose level)	3 (30)	-0.12 mmol/L (95% CI -0.30, 0.06)	I^2= 0% (95% CI 0, 90), p=0.859
% of time below 3.9 mmol/L glucose level	4 (41)	-0.12 % (95% CI -0.60, 0.37)	I^2= 0% (95% CI 0, 85), p=0.460

Conclusion: Our results revealed an improvement in the 120-minute OGTT plasma glucose level in the elexacaftor/tezacaftor/ivacaftor subgroup. The absence of significant changes in other analyses may indicate that these medications could potentially slow or halt the deterioration of glucose metabolism. Further prospective studies are required to overcome the limitation of a relatively small patient number and to validate our findings.

Disclosure: Nothing to disclose.

PP1122

CLINICAL PERFORMANCE OF AN AUTOMATED TURBIDIMETRIC IMMUNOASSAY TO QUANTIFY FECAL PANCREATIC ELASTASE

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Introduction: Pancreatic exocrine insufficiency (PEI) is defined as a reduction of pancreatic exocrine activity in the intestine at a level that prevents normal digestion(1). PEI leads to malnutrition-related complications and a reduced quality of life. Faecal pancreatic elastase is an ideal biomarker to assess pancreatic function in patients experiencing PEI. Due to its high stability, faecal pancreatic elastase is concentrated during intestinal passage, thus reflecting the total overall pancreatic secretion.

Aims & Methods: The aim of this work was to evaluate the clinical performance of a particle-enhanced turbidimetric immunoassay (PETIA) that allows automated quantification of pancreatic elastase in faecal extracts on clinical chemistry analyzers. Hundred and eighty-nine samples from 76 controls and 113 diagnosed patients were used to compare the turbidimetric immunoassay and a reference ELISA assay performance. For both assays, sensitivity and specificity were determined using a cutoff of 200 µg/g. Moreover, for samples within the measuring range for both assays, a kappa value and the overall agreement were assessed.

Results: A better clinical performance was obtained by the turbidimetric immunoassay compared to the reference ELISA assay. The turbidimetric immunoassay revealed a sensitivity and specificity of 78% and 74%, respectively, while the reference ELISA assay revealed a sensitivity and specificity of 74% and 68%, respectively. The hundred and twelve samples used to perform a method comparison showed a relative bias at the cutoff (200 µg/g) of -5.72% and an overall analytical agreement between the turbidimetric immunoassay and the reference ELISA assay of 85.7%. Cohen's kappa agreement showed a substantial agreement of 0.70.

Conclusion: The turbidimetric immunoassay showed good diagnostic performance and a strong agreement with the reference ELISA assay. The particle-enhanced turbidimetric immunoassay is an accurate, non-invasive, automated test and an ideal assay for large-scale determination of pancreatic elastase levels in fecal extracts.

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Disclosure: "All authors are employees of BÜHLMANN Laboratories AG"

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PP1123

DIABETES-ASSOCIATED REGULATORY SEQUENCES REVEAL NOVEL PATHWAYS IN PANCREATIC DEVELOPMENT AND DISEASE

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Introduction: Rare early onset cases of diabetes stem from single gene mutations, whereas frequent gene variants can also contribute to multifactorial diabetes, indicating shared disease mechanisms. Our recent study has identified *ONECUT1* as a gene implicated in both monogenic and multifactorial diabetes, establishing it as crucial target for investigating fundamental signaling pathways in diabetes pathology [1,2]. However, numerous diabetic characteristics remain unexplained by typical genetic alterations, highlighting the underestimated significance of the non-coding genome. We have identified a neonatal diabetes patient carrying a heterozygous *ONECUT1* truncating variant. Remarkably, the patient exhibited a phenotype akin to individuals with homozygous *ONECUT1* truncating mutations. Upon further examination, we detected an additional heterozygous deletion in the non-coding genome, leading us to hypothesize that it could impact the regulation of the remaining *ONECUT1* allele. Consequently, the dysregulation of the ONECUT1-dependent transcriptional machinery crucial for proper beta-cell development occurs, ultimately leading to the severe diabetic phenotype observed in the patient.

Aims & Methods: Following deletion of specific genomic regions in human embryonic stem cells, we initiated their differentiation into the pancreatic endocrine lineage and meticulously characterized the developmental stages.

Results: Importantly, we successfully replicated the patient's diabetic phenotype, elucidating deficiencies in both pancreatic development and beta-cell functionality. The differentiation of pancreatic progenitors was notably hindered by the loss of the regulatory region associated with nearly complete ablation of *ONECUT1*. Moreover, the stem cell-derived islets exhibited a reduced number of beta-cells as well as impaired insulin secretion, even in heterozygous knockouts.

Conclusion: Altogether, our ongoing comprehensive mechanistic characterization provides insights into the precise role of the identified regulatory element in pancreatic beta-cell development and diabetes pathophysiology, emphasizing the significance of the non-coding genome. By unveiling unknown molecular pathomechanisms linked to alterations in regulatory sequences, we contribute to a deeper understanding of the pancreatic transcriptional network and establish the foundation for personalized diabetes treatment.

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Disclosure: Nothing to disclose.

PP1124

SAFETY AND EFFECTIVENESS OF CHOLECYSTECTOMY IN PREGNANT WOMEN WITH MILD BILIARY PANCREATITIS. PRELIMINARY RESULTS OF THE BORN STUDY

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Introduction: Acute biliary pancreatitis in pregnant patients poses a rare yet significant challenge for clinicians. While early cholecystectomy is a recommended approach after mild biliary pancreatitis in the general population to prevent recurrence and mitigate gallstone-related complications, specific guidelines for managing acute pancreatitis (AP) in pregnant patients are lacking.

Aims & Methods: This study aimed to investigate the safety and effectiveness of cholecystectomy during pregnancy following an episode of mild acute biliary pancreatitis. An international survey was conducted across seventeen centers worldwide in 2023, using both Excel and Redcap plat-

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forms. The survey encompassed questions related to demographic information, clinical presentation, management strategies, timing of cholecystectomy, approaches to the procedure, complications, and outcomes (including recurrence of AP, other gallstone-related complications, and maternal and fetal outcomes). Descriptive statistics were calculated for continuous variables, while categorical values were represented as case numbers and percentages. Statistical significance was determined using a t-test and the Mann–Whitney U and Mood's test.

Results: The study included pregnant women who underwent cholecystectomy during or after pregnancy. According to our international survey, 22% (11/50) of patients had cholecystectomy during pregnancy. Patients who underwent cholecystectomy had a lower rate of readmission due to recurrent AP episodes or other gallstone-related complications (0% and 28%; 0/11 and 11/39). There were no significant differences in cholecystectomy-related complications (9.1% and 12%; 1/11 and 3/26). The incidence of preterm birth was 11% and 8.1% (1/9 and 3/37), while fetal loss rates were 18% (2/11) both fetal losses occurred when surgery was performed in the first trimester and 0% (0/39).

Conclusion: Cholecystectomy effectively prevents the recurrence of AP and other biliary complications, with no observed differences in intra- and postoperative complications during and after pregnancy. Thus, cholecystectomy should be performed in mild biliary pancreatitis in the second and third trimesters of pregnancy. Further studies are needed to evaluate the safety of cholecystectomy in the first trimester.

Disclosure: Nothing to disclose.

PP1125

RISK FACTORS FOR POST-ACUTE PANCREATITIS DIABETES MELLITUS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Approximately 40% of patients develop diabetes mellitus (DM) after acute pancreatitis (AP). Even a single episode of mild AP confers a two-fold increase in DM risk compared to the general population. Nonetheless, information on risk factors is limited.

Aims & Methods: We aimed to identify risk factors for developing PD or DM following AP.

We conducted a systematic search of three databases Up until September 4, 2023, to get direct, within-study comparisons of risk factors on the incidence of PD and DM in patients with AP. If it was not possible to distinguish the PD and DM incidence rates, we presented it as a composite outcome: PD/DM. The random-effects model was used in the meta-analysis to compute pooled odds ratios (OR) with 95% confidence intervals (CI).

Results: The 61 eligible studies reported on 79 risk factors. The meta-analysis included 50 studies, 76,797 participants and 34 risk factor and outcome pairs. We found that the odds of developing DM was significantly higher if the AP was severe or moderately severe (OR: 2.11; CI: 1.30-3.41), the etiology was alcoholic (OR: 1.82; CI: 1.09–3.04), organ failure occurred (OR: 3.19; CI: 0.55–18.64), the patient had recurrent AP (OR: 1.89; CI: 0.95–3.77), liver cirrhosis (OR: 2.48; CI: 0.18–34.25), chronic kidney disease (OR: 2.10; CI: 1.85–2.38), dyslipidemia (OR: 1.82; CI: 0.68–4.84) or was obese (OR: 1.85; CI: 1.43–2.38). A higher probability of developing PD/DM was linked to severe and moderately severe AP (OR: 4.32; CI: 1.76–10.60), pancreatic necrosis (OR: 5.53; CI: 1.59–19.21) and hypertriglyceridemic AP etiology (OR: 3.27; CI: 0.17–63.91).

Conclusion: A number of factors increase the risk of developing PD or DM following AP, including severe and moderately severe AP, alcoholic and hypertriglyceridemic etiologies, pancreatic necrosis, organ failure, recurrent acute pancreatitis and comorbidities of obesity, chronic kidney disease, liver disease, and dyslipidemia.

Disclosure: Nothing to disclose.

PP1126

NUTRITIONAL STATUS OF ACUTE PANCREATITIS PATIENTS

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Introduction: Malnutrition has been shown to worsen the course and outcome of many diseases. However, the Malnutrition Universal Screening Tool (MUST), an internationally accepted nutritional score, has not been widely studied in acute pancreatitis, despite the known fact that the dietary habits of patients with pancreatitis of different etiologies vary.

Aims & Methods: Our objective was to examine the MUST score in pancreatitis.

All patients were assessed for MUST on admission at Semmelweis University Institute of Pancreatic Diseases between 04.10.2021 and 21.02.2024. The MUST score was compared with the etiology and outcome of the disease.

Results: Of the 1148 patients admitted to the unit 994 patients had complete data on MUST score and severity and 992 patients had data on etiology.

Most patients were at low risk of malnutrition (MUST 0) 76,3% (n=760), 10,7% (n=107) had a moderate risk (MUST 1) and 12,9% (n=129) had a high risk (MUST score greater than 1). In patients with a MUST score over two alcoholic etiology was the most common (44,4%), while in patients at low risk of malnutrition biliary etiology was predominant (51,3%). Overweight and obese patients (BMI 25+) typically had biliary AP (47,9%, 61,3%), while normal and underweight patients most often had alcoholic AP (46,4%, 41,3%).

The largest proportion of obese patients (BMI 30+) were seen in hypertriglyceridemic AP (72,2%) compared to other etiologies. The proportion of non-mild AP severity was similar in patients with MUST scores of 0, 1, and 2 (32,7%, 35,5%, and 30,1%, respectively), rising to 50% in patients with MUST scores over two. The largest proportion of severe cases (19,4%) was seen in patients with MUST scores over two.

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Conclusion: Patients with higher BMI and lower malnutrition risk tend to have more often biliary etiology, while higher MUST score and lower BMI is characteristic of alcoholic AP. Malnutrition is a risk factor for severe AP.
Disclosure: Nothing to disclose.

PP1127

IDENTIFYING PREDICTORS FOR THE EVOLUTION OF ACUTE PANCREATITIS INTO RECURRENT AND CHRONIC FORMS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: The transition from acute pancreatitis (AP) to recurrent (RAP) and chronic (CP) pancreatitis poses significant clinical challenges.
Aims & Methods: This systematic review and meta-analysis aimed to identify risk factors associated with this progression. Following the protocol registration on PROSPERO (CRD42022368931), we searched Medline, Embase, and Cochrane databases up to December 19th, 2023, for studies examining risk factors for the evolution of AP into RAP or CP. Pooled odds ratios (OR) with 95% confidence intervals (CI) were calculated using the random effects model. Heterogeneity was evaluated using the I^2 statistic. The risk of bias assessment was performed using the Quality in Prognostic Studies (QUIPS) tool.
Results: A total of 124 articles were included in the meta–analysis, and several risk factors were identified for the progression of AP into RAP and CP. We found the following risk factors of AP recurrence: male gender, smoking, alcoholic etiology, hypertriglyceridemia, diabetes mellitus, pseudocyst, etc. The pooled OR for the male gender was 1.49 (95% CI: 1.31-1.62, I^2=25%), for smoking was 1.47 (95% CI: 1.22-1.71, I^2=57%), for alcoholic etiology was 1.78 (95% CI: 1.45-2.14, I^2=78%), for hypertriglyceridemia was 2.48 (95% CI: 2.16-2.83, I^2=9%), for diabetes mellitus was 1.51 (95% CI: 1.21-1.75, I^2=0%), for pseudocyst was 2.23 (95% CI: 1.61-3.12, I^2=0%). We also found risk factors of RAP progression into CP. The risk of bias was moderate in the majority of the included studies.
Conclusion: These findings suggest that addressing modifiable risk factors might be key in preventing the progression of pancreatitis
Disclosure: Nothing to disclose.

PP1128

PREDICTING ACUTE PANCREATITIS SEVERITY WITH CT-DERIVED BODY COMPOSITION: A RETROSPECTIVE STUDY

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Introduction: Accurately predicting the severity of acute pancreatitis (AP) is a challenge in clinical practice. Recent advancements in artificial intelligence and neural network computing enable the use of CT scans for body composition analysis in addition to their diagnostic role. However, the prognostic application of various CT metrics has been inconsistent, resulting in conflicting data regarding the most reliable indicators of disease severity.
Aims & Methods: Our study aimed to evaluate the key CT body composition metrics for predicting severity in patients with acute pancreatitis (AP). We conducted a retrospective analysis of 279 AP patients from the University of Pécs, Hungary, all of whom underwent a CT scan within the first 24 hours of admission to the Department of Gastroenterology.
Using the TotalSegmentator tool[1] of the 3D Slicer software, we measured the areas of visceral and subcutaneous adipose tissues (VAT and SAT) and skeletal muscle (SMA) at the level of the third lumbar vertebra (L3), with the results in square centimeters.
These area measurements were then normalized for patient height by dividing them by the square of the patient's height to produce indices similar to the Body Mass Index (BMI), namely the VATI (Visceral Adipose Tissue Index), SATI (Subcutaneous Adipose Tissue Index), and SMI (Skeletal Muscle Index).
We also quantified the average muscle radio-density in Hounsfield Units (HU), indicative of fatty muscle infiltration - associated with lower density values. We explored these metrics as potential predictors of severe AP, as defined by the Revised Atlanta Criteria[2].
To assess predictive accuracy, we created receiver operating characteristic (ROC) curves and computed the area under the curve (AUC) for both genders separately due to inherent differences in normal values. The actual clinical progression of AP in these patients provided the reference standard. All statistical analyses were done using IBM SPSS Statistics software; preliminary results are available below.
Results: The data encompassed 174 male and 105 female patients, with an AP severity distribution of 139 mild, 116 moderately severe, and 24 severe cases. The AUC values for various CT metrics, differentiated by gender, are displayed in the subsequent table.

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CT metric	Metric type	Value associated with severe AP	AUC for Females	AUC for Males
SMA	area	Low	0.57	0.59
SMI	index	Inconsistent	0.58	0.54
SAT	area	High	0.55	0.68
SATI	index	High	0.57	0.69
VAT	area	High	0.66	0.63
VATI	index	High	0.72	0.65
Muscle radio-density	average CT density	Low	0.64	0.72
Visceral / Subcutaneous fat area	ratio	Inconsistent	0.6	0.57
Fat / Muscle area	ratio	High	0.66	0.74

Conclusion: CT-derived body composition metrics demonstrate substantial prognostic value in identifying severe AP. Index-based metrics demonstrated slightly improved accuracy over simple area measurements. Elevated levels of subcutaneous and visceral fat emerged as predictors of severe AP, as evidenced by the SAT, SATI, VAT, and VATI metrics.

Furthermore, a decreased muscle radio-density was a better predictor of severity than the skeletal muscle area (SMA) or skeletal muscle index (SMI) alone. The combination of significant fat accumulation and skeletal muscle loss - often referred to as sarcopenic obesity - was also predictive of more severe AP. Determining the precise cutoff values for these metrics to optimize predictive accuracy continues to be a key objective of our ongoing research.

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Disclosure: Nothing to disclose.

PP1129

INVESTIGATING THE PREVALENCE OF ANXIETY AND DEPRESSION IN PATIENTS WITH PANCREATITIS

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Introduction: The bidirectional interplay of the brain-gut axis alongside manifestations of anxiety and depression has already been demonstrated. Despite this fact, routine mental health assessments and interventions are not well-established in the treatment of acute (AP) and chronic pancreatitis (CP), potentially worsening clinical outcomes.

Our objective is to achieve a comprehensive understanding of the mind-body connection within the context of pancreatitis by assessing the prevalence of depression and anxiety in patients with AP and CP.

Aims & Methods: Our protocol was registered on PROSPERO (CRD42023481739). We conducted a systematic search on 10 November 2023 in three databases (MEDLINE, Embase, and CENTRAL) to identify studies reporting the occurrence of depressive and anxiety-related symptoms in pancreatitis. The pooled proportion rate with 95% confidence intervals (CI) was calculated using a random-effects model.

Results: The prevalence of depressive symptoms was 16% in AP (CI 0.06; 0.39), while it was 34% in CP (CI 0.27; 0.43). The prevalence of anxiety symptoms in AP and CP was 15% (CI 0.07; 0.32) and 35% (CI 0.26; 0.45), respectively.

Conclusion: Anxiety and depression are significantly more frequently observed among individuals with pancreatitis compared to the general population. These conditions occur twice as often in CP compared to AP. Thus, we underscore the significance of psychological interventions in this patient population.

Disclosure: Nothing to disclose.

PP1130

HYPERCALCEMIA CAUSES MORE SEVERE PANCREATITIS

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Introduction: Acute pancreatitis (AP) is one of the most common gastroenterological diseases requiring admission to the hospital. The main etiological factors of AP are gallstones and excessive alcohol consumption, but rarer causes, such as hypercalcemia, should keep on mind. Only few cases are reported in the literature, and the severity of hypercalcaemia-induced acute pancreatitis is not known.

Aims & Methods: We aim to assess the occurrence and severity of hypercalcaemia-induced AP and compare it with the clinical characteristics of AP caused by other etiological factors. We collected patients from the Hungarian Acute Pancreatitis Registry, Hungarian GOULASH study and the Europe-wide EASY study, who had AP and had a serum calcium level above 2.6 mmol/l. Patients were divided into three groups: hypercalcemia-induced AP, where no other etiology than hypercalcemia were present, mixed etiology group, where both hypercalcemia and other etiologies were revealed and the control group with normal serum calcium level. Propensity score matching adjusted by age, sex, diabetes, lipid disorder, previous pancreatic disease was made and several outcome parameters were analyzed during the statistical analysis. Hypercalcemic cases were divided into 3 subgroups based on the level of serum calcium: mild (2.6< >3.0 mmol/l), moderate (3.0 < >3.5 mmol/l) or severe hypercalcemia (>3.5 mmol/l).

Results: 1.02% of our AP patients (19/1867) had clear hypercalcemic etiology and 3.75 % (70/1867) of the patients had mixed etiology, 1778 patients were in the control group. There was no difference in the mean ages of the patients in the three groups (57.3± 16.3 and 56.1 ± 20.1 vs. 55.1 ±

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14.6 years). Severe AP was significantly more frequent in patients with hypercalcemia-induced AP compared to the control group [21.05% (4/19) vs 6.13% (109/1778); odds ratio: 4.08 (95% CI, 1.08-15.38)]. Renal failure was more frequent in hypercalcemic etiology compared to AP patients with normal serum calcium level [27.27% (3/11) vs 5.35% (95/1776); odds ratio: 6.63, (95% CI, 1.36-32.35)].

There was no significant difference in CRP, amylase, lipase, WBC, creatinine, length of hospital stay, mortality, intensive care admission, the presence of pancreatic necrosis, pseudocysts, and other organ failure between the hypercalcemic or mixed etiology and the control groups.

There was no significant difference in the severity of AP between the mixed etiology and normal serum calcium groups. The outcome of acute pancreatitis was not influenced by the severity of hypercalcemia within the hypercalcemic group.

Conclusion: Hypercalcemia-induced AP has a higher risk for developing into severe AP and renal failure compared to other etiology.

Disclosure: Nothing to disclose.

PP1131

CLINICAL AND LABORATORY FACTORS ASSOCIATED TO ANTIBIOTIC USE IN ACUTE PANCREATITIS

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Introduction: Infectious complications are a significant concern in Acute Pancreatitis (AP), contributing to increased morbidity and mortality. Antibiotics are crucial in their management and should be initiated when infection is suspected. However, their overuse can lead to complications and antimicrobial resistance.

Aims & Methods: The aim of this retrospective study is to evaluate antibiotic utilization in real-world practice within a Spanish cohort and identify clinical and laboratory factors associated with their use. Patients admitted to our tertiary center for acute pancreatitis were included. Clinical and laboratory parameters routinely used in acute pancreatitis were collected upon admission and after 48 hours. Any infections (mainly infected necrosis, acute cholangitis following Tokyo 2018 criteria, pneumonia, and urinary tract infections) were documented, and patients were followed until discharge or death.

Poor outcomes, such as severe acute pancreatitis based on the Atlanta 2012 classification and death, were assessed in patients who received antibiotics and those who did not.

Results: 211 patients were recruited from July 2017 to December 2017. 111 (52.6%) were women, and in 170 (61.8%) the etiology was biliary. Antibiotic therapy was used in 77 (36.5%) patients, 21 had severe acute pancreatitis (9.95%) and 11 died (5.21%).

Comparison between patients to whom antibiotics were prescribed and those who did not are shown on Table 1. Variables included in the logistic regression were SIRS on admission, PCR > 150mg/dL after 48h, PCT >1 ng/ml on admission and BT> 3mg/dL on admission with the presence of SIRS and BT >3 mg/dl at admission proving to be significant in the end.

N (%)	No antibiotics 134 (63,5%)	Need of antibiotics 77 (36,5%)	P value
Leucocytes, mean (sd)	12.642 (5.317)	14.390 (5.536)	< 0,005
Procalcitonin > 1 ng/ml, n (%)	13 (11,6)	18 (26,5)	< 0.01
Total Bilirrubin , mean (sd)	1.68 (1.84)	2.74 (2.57)	0,001
C reactive protein after 48h, mean (sd)	128 (112)	173 (129)	0,022
Temperature (Celsius), mean (sd)	36,2 (0,5)	36,3 (0,607)	n.s
SIRS, n (%)	13 (9.7)	29 (37.6)	<0.001
Extrapancreatic infection, n (%)	4 (1,9)	25 (32.4)	<0.001
Severe Acute Pancreatitis, n (%)	6 (4.4)	15 (19)	0.001
Death, n (%)	2 (1.4)	9 (11.6)	0.001

Table 1. Patient Characteristics: Comparison between patients with Indicated antibiotic therapy and those without antibiotic therapy.

Conclusion: Antibiotic therapy is routinely used in AP due to infected necrosis, acute cholangitis, or extrapancreatic infections. However, the early inflammatory response typical in AP makes it troublesome to distinguish whether sepsis is occurring due to an extrapancreatic infection, leading to potential overuse of antibiotics. In our sample, antibiotics were used in 36.5% of patients, and those developed SAP and death more often. Among clinical and laboratory factors related to antibiotic use, only total bilirubin >3mg/dL and the presence of SIRS on admission were independently related to the need for antibiotics. However, it is possible that some patients with SIRS and high bilirubin on admission do not truly require antibiotics, as these changes can also be due to AP. Although procalcitonin has been positioned as a potential marker of infection and can guide the need for antibiotics, future research could explore other factors to better select patients without overt infection for empiric antibiotic therapy.

Disclosure: Nothing to disclose.

PP1132

A LARGE COHORT STUDY REGARDING THE HYPERTRIGLYCERIDEMIA INDUCED ACUTE PANCREATITIS

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Introduction: This study aims to examine the extent of notable variations in recurrence rates, admissions to the intensive care unit (ICU), length of hospital stay (LoS), morphological characteristics, disease severity, and patient age between hypertriglyceridemia-induced acute pancreatitis (HTG-AP) and other causes of acute pancreatitis (OAP).

Aims & Methods: A retrospective, single-center cohort study was conducted using data from the BUC-API registry, which includes 1,855 consecutive cases of acute pancreatitis. We aim to seek if there are differences between cases with HTG-AP and OAP regarding demographical and clinical data.

Results: Our analysis reveals a modest association between HTG-AP and increased recurrence rates ($\chi^2(1)=6.9$, $p<0.01$, Cramer's $V=0.07$). Patients with HTG-AP had a median age of 44.5 years and experienced longer ICU stays (median=7.0 days) compared to those with OAP (median=3.0 days). Furthermore, patients with HTG-AP were more likely to develop acute peripancreatic fluid collections, require ICU admission, experience severe disease, and be treated in a gastroenterology ward ($p=0.02$, $p<0.01$, respectively).

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		HTG-AP (n=58)	OAP (n=1514)	p-value
Severity	Mild	19 (32.8%)	796 (52.6%)	p<0.01
ICU	Yes	10 (17.2%)	125 (8.3%)	p=0.02
Ward of care	Gastroenterology	42 (72.4%)	744 (49.1%)	p<0.01
Recurrence or first attack	Recurrence	18 (31.0%)	265 (17.5%)	p<0.01
Morphology	APFC	9 (15.5%)	110 (7.3%)	p=0.02
	Normal pancreas	1 (1.7%)	241 (15.9%)	
Age (years)	Median	44.5 (IQR=9)	57.0 (IQR=25)	p<0.01
Length of stay (days)	Median	8.3 (IQR=9.3)	7.0 (IQR=5.0)	p=0.01
Length of stay ICU (days)	Median	7.0 (IQR=22.5)	3.0 (IQR=4.0)	p<0.01

Table 1 – Comparative analysis. Only findings with p-value <=0.01 presented.

Conclusion: HTG-AP is frequently linked with intensive care needs. The association between HTG-AP and recurrence may be linked to suboptimal adherence to lipid-lowering therapy. The typical profile of an HTG-AP patient is a middle-aged, mainly male individual with a history of acute pancreatitis, necessitating gastroenterology ward care, facing longer ICU and hospital stays, at a higher risk for severe acute pancreatitis (SAP), and with an increased likelihood of developing acute peripancreatic fluid collection (APFC).

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Disclosure: Nothing to disclose.

PP1133

RISK FACTORS OF PANCREATIC EXOCRINE INSUFFICIENCY ASSOCIATED AFTER ACUTE PANCREATITIS – A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: 96-98% of patients survive acute pancreatitis (AP), but in the post-AP phase, patients are at high risk of morbidity and mortality. It has been shown that pancreatic exocrine insufficiency can occur after AP.

Aims & Methods: We aimed to investigate the rate, risk factors, and the effect of pancreatic enzyme replacement therapy (PERT) for PEI in patients with AP. The study protocol was registered on PROSPERO (CRD42024516403). We conducted a systematic search on 5 February 2024 in three databases (PubMed, EMBASE, and CENTRAL). We included in the analysis studies reporting on risk factors and PERT efficacy in PEI associated with AP. Pooled proportion and odds ratios (OR) with a 95% confidence interval (CI) were calculated using a random effect model.

Results: Fifty-six articles were included in our meta-analysis. Among 17896 patients with AP, the pooled proportion of PEI was 20 % ([CI]: 15–26). The odds of PEI development were two-fold higher in severe acute pancreatitis (OR=2.44, [CI]: 0.98–6.11) compared with mild or moderate AP and three times higher after necrosectomy compared to patients without necrosectomy (OR=3.26, [CI]: 1.6–6.64). There was no significant difference between the odds of PEI among AP patients with necrosis (OR=1.43, [CI]: 0.32–6.45) and those without necrosis. However, the odds of PEI were twice as high in patients with infected necrosis (OR=2.13, [CI]: 0.93–4.87) than in patients with sterile necrosis. Pancreatic enzyme replacement therapy was prescribed for 22% ([CI]: 16–30) of patients after AP.

Conclusion: One in five patients with AP develops PEI. The rate is higher in patients with severe AP, infected necrosis, or who need necrosectomy. Patients with AP should be screened for PEI to allow timely PERT administration.

Disclosure: Nothing to disclose.

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PP1134

UTILITY OF DYNAMIC ASSESSMENT OF FACTORS RELATED TO SEVERE ACUTE PANCREATITIS

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Introduction: Acute pancreatitis (AP) carries an overall mortality rate of 4.2%, but in cases of severe acute pancreatitis (SAP), mortality dramatically increases to over 50%. Despite the use of several scoring systems to predict SAP in the early stages of the disease, their clinical value is limited as their performance does not significantly influence decision-making due to their low post-test probability. Although early prediction upon admission would be ideal, the development of a marker that accurately predicts severity at this stage seems unlikely. Therefore, a dynamic assessment based on early identification of changes in clinical and laboratory parameters within the first 48 hours could prove valuable in detecting unfavorable outcomes.

Aims & Methods: The aim of our study was to evaluate the clinical utility of changes in routine clinical and laboratory parameters within the first 48 hours of admission in a Spanish cohort, based on likelihood ratios and post-test probability of severe acute pancreatitis (SAP). Patients admitted to our tertiary hospital with a diagnosis of acute pancreatitis were included. Demographic information, comorbidities, and the Bedside Index for Severity in Acute Pancreatitis (BISAP) score were collected upon admission, while vital signs, routine laboratory parameters typically used in acute pancreatitis, and the Systemic Inflammatory Response Syndrome (SIRS) criteria were collected upon admission and after 48 hours. Patients were followed until death or discharge and categorized into mild to moderately severe and severe acute pancreatitis groups.

Results: 227 patients were included in the study. The mean age was 65.3 years (standard deviation 18.9), and 115 patients were women (50.7%). Among them, 22 (10.7%) developed severe acute pancreatitis, and 13 (5.7%) died. BISAP≥2, rising BUN, increasing heart rate over 20 beats per minute, and persistent SIRS were identified as independent risk factors for severe acute pancreatitis. The positive likelihood ratio and post-test probability of SAP are shown in Table 1. When combining the parameters independently related to SAP, we found that BISAP ≥2 combined with persistent SIRS after 48 hours and any rise in BUN combined with persistent SIRS showed the highest post-test probabilities (82% and 80%, respectively).

Predictor	+ LR (CI 95%)	+ Post-test probability of SAP
Rise in heart rate > 20 bpm	3.27 (1.44-7.43)	26% (13-44%)
Rising BUN	3.57 (2.52- 5.04)	28% (21-35%)
Rising Hematocrit	1.59 (0.75-3.36)	15% (8-28%)
BISAP ≥ 2	2.81 (2.16-3.66)	23% (19-28%)
SIRS on admission	2.80 (1.74-4.48)	23% (16-32%)
SIRS after 48h	12 (5.11-26)	56% (36-74%)
BISAP ≥ 2 and rising BUN	8.22 (4.80-14)	47% (34-60%)
BISAP ≥ 2 and SIRS after 48h	42 (9.62-181)	82% (51-95%)

Table 1. Positive likelihood ratio of clinical and laboratory parameters related to SAP with a pre-test probability of 10.7%.

Conclusion: Monitoring the evolution of the disease in the first hours is crucial, especially in patients who develop organ failure. For those who do not, dynamic assessment of known risk factors for SAP can be valuable in clinical practice. In our sample, the combination of BISAP and dynamic assessment of BUN, heart rate, and SIRS raised the probability of SAP up to 80%.

Ideally, assessing the specific incidence of SAP in multicenter cohorts within a specific area and the impact of these and other factors that aid in monitoring patient evolution would provide valuable insights.

Disclosure: Nothing to disclose.

PP1135

ALTERED ENDOCRINE FUNCTION IN A MOUSE MODEL OF CHRONIC PANCREATITIS

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Introduction: It is well known that chronic pancreatitis increases the risk of diabetes development, however, the underlying mechanism is not completely clear. Therefore, our aim in this study was to investigate the function of the endocrine pancreas in a mouse model of CP, with particular regard to the expression of the cystic fibrosis conductance regulator (CFTR) chloride channel, which presumably plays role in beta cell insulin secretion.

Aims & Methods: CP was induced in FVB/N mice by i.p. administration of cerulein. Following completion of disease development, we performed i.p. glucose tolerance tests (GTT) and ELISA. Freshly dissected pancreatic tissues were either fixed in paraformaldehyde for pathology or digested enzymatically for isolation of islets. Islets were dispersed and the cells were stained for CFTR, insulin and glucagon and imaged with direct stochastic optical reconstruction microscopy.

Results: GTT after four weeks of disease development shows decreased glucose tolerance and decreased, elongated insulin response to glucose load. The expression of CFTR in beta-cells is decreased after two weeks of disease development, but exceeds the control in the third week, proceeding the increased glucose intolerance. Inflammation alters islet cell composition by increasing α-cell mass and decreasing β-cell mass.

Conclusion: Regeneration processes in CP resulted in increased expression of CFTR in β-cells that seems to correlate with increased basal insulin secretion and decreased glucose tolerance. This study was supported by the CFRD-SRC Grant (SRC007) and the National Research, Development and Innovation Office (SNN134497).

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PP1136

CLINICAL, RADIOLOGICAL AND ENDOSCOPIC ULTRASOUND FINDINGS IN GROOVE PANCREATITIS: A SINGLE-CENTER RETROSPECTIVE STUDY

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Introduction: Groove pancreatitis is a rare form of focal disease that affects the groove area. As this form of pancreatitis can be mistaken for malignancy, it should be considered in patients with mass lesions in the head of the pancreas or duodenal stenosis to avoid unnecessary surgical procedures. The aim of the study was to document the clinical, radiological and endoscopic characteristics and treatment outcomes of patients with this disease.

Aims & Methods: This retrospective multicenter observational study included all patients diagnosed with one or more imaging criteria suggestive of groove pancreatitis at the participating centers. Patients with confirmed malignant findings on fine-needle aspiration/biopsy were excluded. All patients were followed up in their own centers and evaluated retrospectively.

Results: Of the 30 patients initially included with imaging criteria suggestive of groove pancreatitis, 9 patients (30%) were excluded due to malignant fine-needle aspiration results by endoscopic ultrasound or biopsy. The average age of the 21 patients included was 49 ± 10.6 years, with a male predominance of 71%. There was a history of smoking in 66.7% and alcohol consumption in 76.2% of the patients. The main endoscopic finding was gastric outlet obstruction in 16 patients (76,2%). There was thickening of the duodenal wall in 9 (42.8%), 5 (23.8%) and 16 (76.2%) patients on computed tomography, magnetic resonance imaging and endoscopic ultrasound, respectively. In addition, pancreatic head enlargement/mass was observed in 10 (47.6%), 8 (38,1%) and 12 (57,1%) patients, and duodenal wall cysts in 5 (23.8%), 1 (4.8%) and 11 (52.4%) patients, respectively. Conservative and endoscopic treatment achieved favorable results in more than 90% of patients.

Conclusion: Groove pancreatitis should be considered in any case with duodenal stenosis, duodenal wall cysts or thickening of the groove area. Various imaging modalities, including computed tomography, endoscopic ultrasound and magnetic resonance imaging, play a valuable role in characterizing groove pancreatitis. However, endoscopic fine-needle aspiration or biopsy should be considered in all cases to diagnose groove pancreatitis and exclude malignancy, which can have similar results.

Disclosure: Nothing to disclose.

PP1137

ENHANCING PANCREATIC CARE - EXPERIENCES OF A SPECIALIZED UNIT FOR BENIGN PANCREATIC DISORDERS

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Introduction: Benign pancreatic disorders pose significant challenges in diagnosis, treatment, and management due to their complex nature and diverse etiology. As the prevalence of pancreatic diseases continues to rise, the demand for specialized care and expertise becomes imperative. Thus the Section of Interdisciplinary Pancreatology (SIP) has been launched in 2023 at Ulm University Hospital to manage a broad spectrum of pancreatic disorders, encompassing cystic pancreatic lesions, autoimmune pancreatitis and IgG4-related disease (IgG4-RD), and all types of chronic pancreatitis.

Here, we will present insights from two years of experience within this unit, focusing on the disease spectrum, the impact of a multidisciplinary board structure, and the direct advantages of Patient-Reported Outcome Measures (PROMs) in enhancing patient care.

Aims & Methods: To basically show the structure of an interdisciplinary section of pancreatoloy and emphasise resulting advantages and chances in pancreas care. Moreover, descriptive analyses from PROMs, impact on decision making and paths will be shown.

Results: The structured implementation of the SIP created a level of interdisciplinarity that has not been achieved before. Thus collaborative efforts resulted in relevant changes of daily-patient care and patient paths. This is assessed and further shown by the integration of PROMs that are continuously recorded. By systematically collecting patient-reported data on symptoms, functional status, and treatment outcomes, the division gains valuable insights into the patient experience and treatment efficacy and allows patient centered therapy management. Specifically by the integration of a dedicated multidisciplinary board to review cases of benign pancreatic disorders, consensus-driven decision-making optimized diagnostic accuracy and therapeutic planning, fostering individualized approaches. One of these successful approaches that has been established are high numbers of hybrid procedures in robotic enucleation of pancreatic cysts.

Conclusion: In conclusion, our two-years experience in a division of interdisciplinary pancreatology underscores its efficacy in addressing the diverse array of pancreatic disorders. Through collaborative decision-making, regular case reviews, and the integration of PROMs, the SIP exemplifies a patient-centric approach to pancreatic care that prioritizes clinical excellence, interdisciplinary collaboration, and continuous quality improvement.

Disclosure: Nothing to disclose.

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PP1138

QUALITY OF MANAGEMENT AND ADHERENCE TO UNITED EUROPEAN GASTROENTEROLOGY GUIDELINE ON IMMUNOGLOBULIN-G4-RELATED DIGESTIVE DISEASE

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Introduction: United European Gastroenterology (UEG) Guideline on IgG4-related digestive disease provides evidence-based recommendations for the diagnosis and management of immunoglobulin G4 (IgG4)-related digestive disease in adults and children. IgG4-related digestive disease can be diagnosed only with a comprehensive work-up that includes histology, organ morphology at imaging, serology, search for other organ involvement, and response to glucocorticoid treatment. Indications for treatment are symptomatic patients with obstructive jaundice, abdominal pain, posterior pancreatic pain, and involvement of extra-pancreatic digestive organs. Treatment with glucocorticoids should be weight-based and initiated at a dose of 0.6–0.8 mg/kg body weight/day orally for 1 month to induce remission and then be tapered within two additional months. Response to initial treatment should be assessed at week 2–4 with clinical, biochemical, and morphological markers.

Aims & Methods: The aim of this study is to evaluate the adherence to diagnostic, therapeutic and management recommendations of the IgG4 guideline across Europe. The invitation to participate in the study was published at *My Connect* UEG platform. Questions in the survey were related to the organ involvement, diagnosis, treatment and follow up of newly diagnosed patients with IgG4-related digestive diseases in the three-year period (January 2021 - December 2023) after the publication of the guideline in 2020. We have received data from 14 European centers (12 countries).

Results: One hundred ninety-nine patients (76% males) were included in this multicentric study.

IgG4-related digestive disease was diagnosed in 54, 75, and 70 patients during 2021, 2022 and 2023 respectively.

Most of the patients had concomitant pancreatic and biliary tree involvement (46%), followed by pancreatic (34%), biliary tree (19%), esophageal (1.5%) or bowel (0.5%) involvement. Among autoimmune pancreatitis (AIP) patients, 76% received a definite diagnosis.

The combination of CT and MRI was employed in 63% of the patients, whereas 13% received only either CT or MRI.

Endoscopic ultrasound was performed in 56% of the patients.

Pancreatic exocrine and endocrine function at baseline were controlled in 78% and 76% of the patients, respectively.

Most of the patients (72%) received induction treatment with glucocorticoids (GC) according to UEG guidelines, whereas 8%, 7%, and 13% received a different GC scheme, a different drug, or no treatment, respectively.

Eleven patients underwent surgery due to suspected cancer.

Follow up assessment was performed according to guidelines (between 2-4 weeks from treatment start) in 48% of the patients and after 4 weeks in 38% of the patients, but 7% did not receive any follow-up imaging.

Among the 37 patients with biliary tract and liver involvement, 51% underwent MRI and CT, while EUS was used in 43% of the patients.

Treatment and imaging follow up mirrored UEG guidelines in 78% and 57% of the patients, respectively.

Five patients underwent surgery due to suspected cancer.

Three cases of IgG4-RD of the esophagus and one case of IgG4-related bowel disease were detected.

Conclusion: IgG4-related digestive disease was diagnosed in pancreas and the bile duct (concomitant or isolated) in most of the patients. IgG4-related disease in other digestive organs is very rare. Adherence to the European guidelines is high in the context of diagnosis and the first line treatment. Surgery was performed in 8% of all cases due to the suspected cancer. There are still differences across Europe regarding the follow up of patients.

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PP1139

PRESENCE OF TYPE 2 DIABETES MELLITUS IS ASSOCIATED WITH INCREASED SHEAR-WAVE VELOCITY IN PANCREATIC PARENCHYMA OF BRANCH-DUCT INTRADUCTAL PAPILLARY MUCINOUS NEOPLASM: A PROPENSITY SCORE MATCHED ANALYSIS

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Introduction: In branch-duct intraductal papillary mucinous neoplasm (BD-IPMN), pancreatic ductal adenocarcinoma (PDAC) develops in the parenchyma of the pancreas. However, it is unclear why PDAC occurs from sites beyond the cysts. Endoscopic ultrasound guided shear-wave elastography (EUS-SWE) is a new imaging modality for the quantification of tissue fibrosis as hardness. In this study, we investigated the clinical features of pancreatic parenchyma in BD-IPMN patients.

Aims & Methods: We selected 138 patients in the BD-IPMN group and 47 patients in the normal group who underwent EUS-SWE from April 2020 to March 2022. We used propensity score matching with age in a 2:1 ratio (76 patients in the BD-IPMN group and 38 in the normal group). Elastic velocity measured by EUS-SWE (Vs, m/s) in the pancreatic body at least 5 measurements were examined. Each Vs had a reliability measurement (VsN, %) and >50% was considered reliable. ALOKA ARIETTA 850 (FUJIFILM, Tokyo, Japan) and GF-UCT260 (Olympus, Tokyo, Japan) were used.

Results: The patients were 43.3% males with a mean age of 65.8 years in the BD-IPMN groups and 50.0% with 65.5 in the normal group with no significant difference in clinical background. Type 2 diabetes mellitus (T2DM) was observed in 25.0% of the BD-IPMN group and 21.1% of the normal group, and a history of alcohol consumption (>60g/day) was 3.0% in the BD-IPMN group. The Vs values were 1.95±0.71m/s in the BD-IPMN group and 1.64±0.46m/s in the normal group, which were significantly higher in the BD-IPMN group (p=0.02). EUS findings (Hyperechoic foci and strands: 42.1%, Hyperechoic duct wall: 22.4%, Lobularity: 5.3%, Dilated side branch: 18.4%) were observed in the BD-IPMN group. In multivariate analysis, the presence of T2DM and EUS findings (more than one) was an independent factor to increase the Vs value (p=0.04, <0.001). Secondly, the normal group and the BD-IPMN group were divided into 4 groups according to the presence or absence of T2DM. The Vs value was 2.47m/s in the BD-IPMN with T2DM group which was higher compared to the absence of T2DM (1.77m/s, p<0.001), and there was no significant difference in the normal group regardless of T2DM (1.63 and 1.67m/s). There was no correlation between age and Vs value.

Conclusion: In the BD-IPMN patients, the stiffness of pancreatic parenchyma was significantly higher than the normal pancreas. Furthermore, the coexistence of T2DM was a risk factor that increased pancreatic stiffness. The results suggest the presence of inflammation and fibrosis, and long-term surveillance may find a relationship with carcinogenesis.

Disclosure: Nothing to disclose.

PP1140

CYSTIC TUMOURS OF THE PANCREAS: THE VALUE OF ENDOSCOPIC ULTRASOUND

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Introduction: Cystic pancreatic tumors are rare tumours that are increasingly being discovered incidentally due in part to the improved performance of pancreatic imaging modalities (ultrasound, multidetector CT scan, MRI) and advancements in knowledge regarding the characteristics of these lesions. They are represented by serous cystadenomas (SC), mucinous cystadenomas (MC), intraductal papillary mucinous neoplasms (IPMN), and solid pseudopapillary tumors of the pancreas (SPTP). The aim is to study the epidemiological, clinical, biological, and etiological aspects of these pancreatic tumors.

Aims & Methods: This single-centre study was carried out from September 2015 to August 2023 and included all patients with cystic tumours of the pancreas. False pancreatic cysts were excluded from the study. Epidemiological, clinical, biological and echo-endoscopic data were collected from endoscopic-ultrasounds registers. All endoscopic-ultrasounds were performed under Propofol sedation using a Pentax-type radial and/or linear video echo-endoscope, and cytoponctions were performed using 19 G echotip needles for biochemical analysis of the cystic fluid and/or anatomopathological study.

Results: Out of a total of 649 biliopancreatic endoscopic ultrasounds performed, 53 patients were included, representing a prevalence of 8.16%. The mean age was 62.3 years (29-82 years) with a clear female predominance (Sex Ratio = 0.65). Discovery circumstances were incidental findings in 21 cases (39%), epigastric pain in 17 cases (32%), acute pancreatitis in 6 cases (11.3%), jaundice in 6 cases (11.3%), and abdominal pain in 3 cases (5.6%). Biochemically, cholestasis was found in 12 cases (22.6%), cytolysis in 5 cases (9.4%), and elevated Ca19-9 in 6 cases (11.3%). Endoscopic ultrasound revealed the location, number, and size of the tumor: 21 cases (39.6%) were located in the pancreatic head, 13 cases (24.5%) in the body, 9 cases (17%) in the tail, and 10 cases (18.9%) were multifocal, with an average tumor size of 28.5 mm (1.9-76 mm). Communication with pancreatic ducts was observed in 35 cases (66%), and Wirsung duct dilation in 16 cases (30.1%). Cytology was performed in 27 cases (51%). The endoscopic ultrasound appearance combined with cystic fluid analysis favored IPMN in 42 cases (79.2%), mucinous cystadenoma in 5 cases (9.4%), serous cystadenoma in 3 cases (5.7%), and SPTP in 3 cases (5.7%).

Conclusion: Cystic pancreatic tumors are rare neoplasms predominantly affecting women in their sixties, often discovered incidentally. They predominantly occur in the pancreatic head and are mainly comprised of IPMN followed by mucinous cystadenomas, with serous cystadenomas and SPTPs being less common.

Disclosure: no links of interest.

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PP1141

APPLICATION OF A DIGITAL TOOL (ICYST) IN THE MANAGEMENT OF PATIENTS WITH PANCREATIC CYSTIC LESIONS: USEFUL OR FUTILE?

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Introduction: Pancreatic cystic lesions (PCLs) have an increasing incidence and their management should be guided by recent guidelines. Currently, there is an electronic tool available, called iCyst, which can assist physicians in determining patient follow-up. The aim of this study was to retrospectively determine, using the iCyst application, whether guidelines were correctly followed at the time of diagnosis and, if not, whether this had a negative impact on patients.

Aims & Methods: Patients with PCLs followed in the outpatient clinic of the Gastroenterology Department of a large tertiary center between January 2022 and December 2023 were included.

Results: One hundred patients were evaluated (female sex - 60%; mean age - 64.2 ± 12.5 years; mean follow-up time - 4.3 ± 2.8 years). Comparing the suggestions of iCyst with what was implemented, it was found that the guidelines were respected to the following extents: European - 73%; AGA - 71%; IAP - 58%. In 9% of cases, none of the guidelines were applied. Regarding follow-up, a significant increase (>5 mm) in lesion size was observed in 11% of cases. Of the 9 patients in whom no guideline was followed, only 1 had dimensional growth, which was an IPMN with high-risk features and surgical indication; however, this patient refused any intervention and subsequently died from a non-pancreatic cause. In the remaining patients with lesion growth, in 6 cases, the plan followed was similar to that proposed by all three guidelines simultaneously, in 3 to that proposed by the EU and AGA guidelines, and in one case, the IAP guideline was followed. There was no statistically significant association between evolution and the implemented guideline.

Conclusion: The results show that, in the vast majority of evaluated patients, the plan proposed by at least one of the three international recommendations was respected, especially the European. It is noteworthy that there seems to be no correlation between strict adherence to these guidelines and the final outcome for the patient.

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PP1142

SURGICAL TACTICS FOR THE TREATMENT OF IPMN, WHAT'S NEW

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Introduction: Until recently, all cystic tumours were treated surgically, including intraductal papillary mucinous neoplasms of the branch type (BD-IPMN). Currently, most patients with BD-IPMN are being followed up.

Aims & Methods: The aims of this study was to estimate the results of treatment of patients with IPMN.

A total of 228 patients with cystic pancreatic tumours who were treated at our department between 2012 and 2023 were included in the analysis. IPMN was diagnosed in 78 (34.2%) patients. All patients with IPMN were divided into two groups. The primary group included 56 patients who underwent surgery between 2018 and 2023 in accordance with the proposed algorithms.

The comparison group comprised 22 patients who underwent surgery between 2012 and 2017. The two groups demonstrated comparable demographic characteristics, including age, gender distribution, and pathological structure.

In the diagnostic algorithm, we always perform MRI with the DWI and with magnetic resonance cholangiopancreatography. We also perform endoscopic ultrasound with fine-needle aspiration if required.

Until 2018, all types of IPMN were operated. Since 2018, we have revised our tactics. Indications for surgery were considered to be the presence of central and mixed type of IPMN or branch duct of IPMN with the presence of HRS and WF.

Of the 22 patients in the control group, central type IPMN was diagnosed in 10 patients (adenocarcinoma was diagnosed in 9, 1 patient was unresectable at the time of diagnosis), mixed type in 4 (2 patients had malignant tumours, 1 patient was unresectable at the time of diagnosis, 1 patient did not undergo surgery), and branch type in 8 (4 patients had malignant tumours, 2 patients - benign and in 2 patients we did not perform surgery). 17 patients with IPMN underwent surgery. Pancreatoduodenectomy (PDE) was performed in 9 patients, total pancreatectomy (TPE) in 1, distal pancreatectomy (DPE) in 7. Five patients were not operated: 2 patients had unresectable tumours diagnosed preoperatively, 2 patients with BD-IPMN were offered observation, 1 patient with mixed type IPMN refused surgery.

IPMN was diagnosed in 56 (40.6%) patients in the main group. Adenocarcinoma on the background of IPMN was diagnosed in 21 (37.5%) patients, all patients with central or mixed type of IPMN, of which PDE was performed in 12 patients, DPE with splenectomy in 8 patients, TPE - in 1 patient. IPMN with mild or severe dysplasia was diagnosed in 20 (35.7%) patients. Of these, PDE was performed in 13 patients, TPE in 1 patient and DPE in 6 patients. In 15 (26.8%) patients BD-IPMN without HGD and WF was diagnosed.

Results: In the comparison group, 17 (77.3%) patients underwent resection. Complications occurred in 6 (35.3%) patients, 3 of which were pancreatogenic complications (17.6%). In the main group, 41 (73.2%) patients underwent surgery. Complications occurred in 11 patients (26.8%). There was no mortality among these patients in either group. Of the 2 patients in the comparison group who were followed up, 1 patient had no progression of the disease, 1 patient was lost to follow-up, the outcome is unknown. Of the 15 patients in the main group with BD IPMN who were followed, none had disease progression.

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Conclusion: The treatment of IPMN of the pancreas requires an individualized approach. In select patients, surveillance of presumed BD-IPMN can be safely discontinued. When choosing a wait-and-see approach in patients with BD-IPMN, it is imperative that the diagnosis not raise doubts.
Disclosure: Nothing to disclose.

PP1143

HYPERTHERMIA IMPROVES GEMCITABINE SENSITIVITY OF PANCREATIC CANCER CELLS BY SUPPRESSING THE EPHRIN A4/B-CATENIN AXIS AND ACTIVATING DEOXYCYTIDINE KINASE

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Introduction: Previously,our investigations have underscored the potential of hyperthermia to improve the therapeutic efficacy of gemcitabine(-GEM) in pancreatic cancer(PC). Nonetheless,the precise underlying mechanisms remain elusive.
Aims & Methods: We aimed to explored the underlying mechanisms of how hyperthermia enhances GEM sensitivity of PC cells. We engineered two GEM-resistant PC cell lines(BxPC-3/GEM and PANC-1/GEM) and treated them with GEM alongside hyperthermia. The impact of hyperthermia on the therapeutic potency of GEM was ascertained through 3-(4, 5-dimethylthiazol-2-yl)-2, 5-diphenyltetrazolium bromide (MTT) assay, assessment of the concentration of its active metabolite difluorodeoxycytidine triphosphate(dFdCTP), and evaluation of deoxycytidine kinase (dCK) activity. Lentivirus-mediated dCK silencing was further employed to validate its involvement in mediating the GEM-sensitizing effect of hyperthermia. The mechanism underlying hyperthermia-mediated dCK activation was explored using bioinformatics analyses. The interplay between hyperthermia and the ephrin A4 (EFNA4)/β-catenin/dCK axis was investigated, and their roles in GEM resistance was further explored via the establishment of xenograft tumor models in nude mice.
Results: Hyperthermia restored dCK expression in GEM-resistant cell lines, concurrently enhancing GEM sensitivity and fostering DNA damage and cell death. These observed effects were negated by dCK silencing. Regarding the mechanism, hyperthermia activated dCK by downregulating EFNA4 expression and mitigating β-catenin activation. Overexpression of EFNA4 activated the β-catenin
while suppressing dCK, thus diminishing cellular GEM sensitivity—a phenomenon remediated by the β-catenin antagonist MSAB. Consistently, in vivo, hyperthermia augmented the therapeutic
efficacy of GEM on xenograft tumors through modulation of the ephrin A4/β-catenin/dCK axis.
Conclusion: This study delineates the role of hyperthermia in enhancing GEM sensitivity of PC cells, primarily mediated through the suppression of the EFNA4/β-catenin axis and activation of dCK.
Disclosure: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

PP1144

FORMONONETIN COMBINED WITH SALVIANOLIC ACID B INHIBITS THE PROGRESSION OF PANCREATIC CANCER THROUGH DOWN-REGULATING MBOAT2

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Introduction: Pancreatic cancer (PC) presents as a highly aggressive malignancy characterized by limited treatment modalities and a poor prognosis. Its prevalence is steadily increasing, with projections indicating that it will rank as the second most common cause of cancer-related mortality by the year 2030, reflecting a substantial escalation in the global burden [1].
Due to the early metastasis of PC and the limited efficacy of current treatment for advanced stages, the overall five-year survival rate remains below 9% [2-3]. Therefore, the development of novel therapeutic approaches is crucial.
In recent years, there has been a surge of interest in natural products due to their perceived potential as anti-cancer agents. Our initial findings suggest that the co-administratin of Formononetin (FMN, a compound derived from Astragalus mongholicus) combined with Salvianolic Acid B (SalB, one of the major water-soluble components of Salvia miltiorrhiza), referred to as "FcS" in subsequent sections of this study, effectively suppresses the proliferation, metastasis, and promotes apoptosis of Panc-1 cells in a zebrafish embryo xenograft model. While FMN and SalB individually inhibited the PC cell viability, their combined treatment exhibited a stronger and significant suppressive effect. However, the mechanism of this combined treatment suppressing the malignant progression of PC remains unclear.
Aims & Methods: This study aims to assess the potential therapeutic effectiveness of FcS therapy for PC, with a focus on investigating the impact of this treatment strategy on PC cell survival, proliferation, and migration capabilities through a series of experiments conducted *in vitro* and *in vivo*. Additionally, our study investigated the expression of MBOAT2 in PC and its correlation with prognosis and tumor invasion.
The proliferation of PC cells was evaluated through the utilization of CCK-8 and EdU assay *in vitro* experiments to investigate the effects of FcS. Wound healing assay and transwell assay were conducted to assess the impact of FcS on the migration and invasion of PC cells. Through molecular docking and analysis of the Cancer Genome Atlas (TCGA) database, the potential regulatory factor MBOAT2 was identified. A Mboat2 knockdown cell line was established and treated with FcS to investigate the interaction between MBOAT2 and FcS. *In vivo* experiments were conducted using the LSL-Kras (G12D/+), LSL-Trp53 (R172H/+), and Pdx-1-Cre (KPC) transgenic mouse model, with tumor weight serving as a measure of the effects of FcS. Immunohistochemistry was utilized to assess protein expression levels from mice tumor tissues.
Results: Experiments conducted on three PC cell lines revealed that treatment with FcS resulted in reduced cell viability, inhibited proliferation, and decreased cell migration and invasion.
Furthermore, analysis of TCGA data was utilized to investigate the impact of FcS on MBOAT2 expression in pancreatic cancer cells, demonstrating a down-regulation of MBOAT2 and subsequent reductions in viability, proliferation, and invasion.

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Additionally, the study identified the JAK/STAT signaling pathway as a potential regulatory mechanism. *In vivo* administration of FcS also exhibited tumor growth suppression in a KPC transgenic mouse model.

Conclusion: FcS treatment may be effective in suppressing the growth and invasion of PC cells through the downregulation of MBOAT2. The unique contribution of this research lies in the therapeutic impact of combining natural compounds and identifying MBOAT2 as a potential target, along with the involvement of the JAK/STAT signaling pathway.

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PP1145

ANALYSIS OF PERIPHERAL BLOOD STEM AND PROGENITOR MARKERS MRNA EXPRESSION IN PANCREATIC NEOPLASTIC AND INFLAMMATORY DISEASES

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Introduction: The development of pancreatic malignancies is related to metabolic and genetic alterations. Genetic alterations of both progenitor genes and stem cells genes and markers were described in pancreatic neoplastic and inflammatory disorders.

In our study we comprehensively asses the expression of mRNA of both stem cells and progenitor pancreatic genes in peripherial blood of patients with pancreatic malignancy in different clinical stages,neuroendocrine tumors, and pancreatic inflammatory disorders (acute(AP) and chronic pancreatitis(CP)), to identify the tumor markers and underlying mechanisms of tumor progression.

Aims & Methods: Our aim was to assess the peripheral blood mononuclear cells (PBMNCs) messenger RNA (mRNA) expression profile of stem cell genes (*OCT4, NANOG*) and pancreatic progenitor genes (*CK19, HES1, INS, PDX1*) in selected pancreatic diseases: neoplastic (cancer, neuroendocrine tumors) and inflammatory (acute and chronic pancreatitis).

The 49 patients with pancreatic neoplastic diseases (37 with cancer, 12 with neuroendocrine tumors) and 48 patients with pancreatitis (28 with acute, 23 with chronic) and 34 patients from gender and age-matched control group hospitalized in tertiary center were enrolled into the study. Venous blood was collected and RNA was isolated from PBMNCs. The mRNA expression of 6 stem cell and pancreatic progenitor genes: *OCT4* (POU class 5 homeobox 1), *NANOG, CK19 (keratin 19), HES1 (hes family bHLH transcription factor 1), INS (insulin), PDX1 (*pancreatic and duodenal homeobox 1) was assessed by real-time quantitative PCR. The results were analyzed statistically and their associations with available clinical, biochemical and morphological data (WBC, ERY, PLT, CRP, Ca 19.9, weight, height and BMI) were studied with Kruskall Wallis test, Mann-Whitney U test and Spearman rank correlation coefficient.

Results: We found that the expression of gene characteristic for early stem cells *NANOG* (p=0.03) and the expression of genes encoding insulin and cytokeratin 19 (*INS* p=0.02,*CK19* p=0.005) was increased in pancreatic cancer patients. However, it was not significantly different in other neoplastic (neuroendocrine) and inflammatory conditions.

Moreover, we found that *OCT4*, *HES1*, *PDX1* expression is significantly decreased in PBMNCs of patients with pancreatitis (both acute and chronic). There were no significant differences in genetic expression between the patients with different stages of pancreatic cancer and patients with acute pancreatitis with different severity (mild vs moderate/severe). What is more, *CK19* overexpression correlated with inflammatory markers (both CRP and WBC) in patients with pancreatic cancer.

Conclusion: PBMNCs mRNA expression of selected stem and progenitor genes (*NANOG, INS, CK19*) may serve as a marker of pancreatic cancer and correlates with coexisting inflammation.

Disclosure: Nothing to disclose.

PP1146

MULTIPLE SEQUENCING AND MULTI-MARKER STAINING REVEALS THE MOLECULAR SUBTYPES AND MICROENVIRONMENT FEATURES OF PANCREATIC DUCTAL ADENOCARCINOMA IN THE CHINESE POPULATION

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Introduction: The prognosis of Pancreatic ductal adenocarcinoma (PDAC) remains poor with a 5-year overall survival of around 10%. In previous whole-genome studies of PDAC in the West, four common mutations have been identified, including KRAS, TP53, SMAD4, and CDKN2A. Due to limited research on Asian populations, we utilized multi-omics sequencing and immune microenvironment evaluation using multiple biomarkers to present the tumor and microenvironment characteristics of PDAC in the Chinese population.

Aims & Methods: The current study obtained 95 primary treatment-naïve PDAC samples from Chinese patients who underwent curative surgery. RNA-sequencing was carried out on all 95 samples, with paired normal sample sequencing for 27 cases, and WES-sequencing for 73 patients. Additionally, to assess the microenvironment features, various markers including immune cell markers, immune checkpoint markers, stroma markers, and markers related to tumor proliferation and subtype were examined using immunohistochemistry (IHC).

Furthermore, Olink sequencing was performed on serum samples from a subset of the cohort (25.26%).

Results: KRAS mutations were found in about 76.7% (56/73) of the samples, with most mutations occurring on codon 12 of exon 2, including G12D (30%), G12V (22%), Q61H (12%), and G12R (7%).

Additionally, oncogenic fusions such as MET and BRAF, as well as EGFR gain, were identified in KRAS wild-type PDAC samples. Tumors with KRAS wild-type displayed higher levels of CD3, CD4, and CD8 T cells, and lower levels of CD68 and CD206 macrophages. Additionally, the KRAS wild-type group exhibited a greater number of PD1-positive samples.

Pathological staining can yield results consistent with RNA sequencing to categorize PDAC into tumor subtypes (basal-like, classical, and double positive) and stromal subtypes (deserted, active, and hybrid) with prognostic significance. It is noteworthy that PDAC tissues with both basal-like

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and classical subtype components (double positive) contain more tertiary lymphoid structures (TLS) and B cells, suggesting potential responsiveness to immunotherapy.
Conclusion: Chinese individuals have a higher proportion of KRAS wild-type PDAC patients, which represents a potentially actionable and immunologically active tumor subtype. Simplified tumor and stromal subtypes based on pathological staining methods are more clinically practical and provide prognostic and therapeutic implications.
Disclosure: Nothing to disclose.

PP1147

PSYCHOLOGICAL BURDEN IN A PANCREATIC CANCER SURVEILLANCE PROGRAM BASED ON "INDIVIDUALS AT RISK"

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Introduction: The prognosis for pancreatic cancer is poor, it is an aggressive cancer and comes with few or unspecific symptoms. Diagnosis at an early stage is vital for survival and chance for curative treatment. Surveillance programs for high-risk individuals, so called, "individuals at risk" are therefore of utter importance. However, there is limited data on the psychological burden within participants of these programs.
Aims & Methods: The aim was to investigate the psychological burden for participants in a pancreatic cancer surveillance program. Additionally, this study aimed to explore whether psychological burden was related to the individual's risk-level for pancreatic cancer.
Results: In total, 278 patients answered the questionnaire. Results indicated high cancer worry among respondents with high scores on CWS (mean=16.45). A majority of 69.3% had scores indicating high cancer worry (cut-off >= 14). Anxiety was not equally high among respondents (mean STAI-T = 35.13; STAI-S=35.9). Female gender and younger age were significantly correlated with higher cancer worry and anxiety (Chi2-tests and linear regression analysis). Outcomes in coping and perceived health were similar to normative populations.
Conclusion: Cancer worry is particularly high within participants, among these women and younger individuals have more cancer worry and anxiety. No correlation was found between risk-level and psychological burden.
Disclosure: None.

PP1148

PREDICTIVE FACTORS OF ACTUAL FIVE-YEAR RECURRENCE-FREE SURVIVAL AFTER UPFRONT SURGERY FOR RESECTABLE PANCREATIC CANCER. -EXPLORATION OF PATIENTS WHO DID NOT REQUIRE NEOADJUVANT TREATMENT-

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Introduction: Neoadjuvant treatment for resectable pancreatic cancer (R-PC) is expected to decrease the postoperative recurrence and prolong the survival. However, It remains unclear whether NAC is necessary for all patients.
Aims & Methods: The present study investigated the prognostic factors associated with actual five-year recurrence-free survival (RFS) after upfront surgery for R-PC in patients who were deemed not to require neoadjuvant treatment. Between 2007 and 2016, 316 patients who underwent pancreatectomy for radiologically R-PC were retrospectively reviewed to evaluate the predictors of actual 5-year RFS. Predictors were identified using logistic regression analysis of preoperative evaluable factors. The cutoff values for continuous variables were determined based on a minimum P-value approach (model 1) or the value that maximized the rate of five-year RFS survivors (model 2).
Results: Fifty-one patients (16.1%) achieved a 5-year RFS. A tumor size ≤23 mm, absence of serosal invasion on computed tomography (CT), and neutrophil-to-lymphocyte ratio (NLR) <1.0, were significantly associated with the 5-year RFS in model 1. A prognostic nutritional index (PNI) ≥58 and the absence of serosal invasion and extrapancreatic nerve plexus invasion on CT were significantly associated with 5-year RFS in model 2. Only six (11.8%, model 1) and four (7.8%, model 2) patients had all three prognostic factors, and their 5-year RFS rates were 83.3% and 100%, respectively.
Conclusion: A modest number of patients who underwent upfront surgery achieved 5-year RFS, but only approximately 10% of them could be identified preoperatively. Based on these results, almost all R-PC patients are forced to undergo NAT in daily practice.
Disclosure: Nothing to disclose.

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PP1149

PREDICTING PANCREATIC CANCER FROM ARTIFICIAL INTELLIGENCE-BASED PANCREATIC VOLUME CHANGES ON CT IMAGING OVER TIME

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Introduction: A previous study, which found that focal parenchymal atrophy of the pancreas is an indirect sign of small pancreatic cancer (PC), documented the changes in the pancreas until cancer onset.［1-3］ However, the pancreas atrophies and changes with age, so issues remain regarding the criteria for determining pancreatic atrophy as a sign of PC. The application of artificial intelligence (AI) technology to quantify changes in the pancreas over time and predict the onset of PC and high-risk groups for PC from objective indicators has the potential to facilitate early detection of PC.

Aims & Methods: Based on the rate of change in the pancreatic volume on computed tomography (CT) using AI technology, the present study predicted PC onset. Forty patients in whom PC was histopathologically diagnosed at Komagome Hospital between 2015 and 2023 and who had a history of three or more follow-up CT examinations over five years before PC onset (the PC group) were analyzed. Forty patients who underwent CT periodically for malignant skin disease but had no pancreatic disease were enrolled into a non-PC group. Patients with pancreatic carcinoma in situ (pancreatic intraepithelial neoplasm), an intraductal papillary mucinous neoplasm or chronic pancreatitis with multiple calcifications were excluded. Changes in the pancreatic volume in the groups over time were then compared using CT images taken one year or earlier before PC onset. The pancreatic volume was calculated using SYNAPSE VINCENT cloud (Fuji Film, Japan), an AI image application program. Age and pancreas volume were combined to create a scatter diagram, and the rate of pancreatic volume change (the temporal pancreatic volume change coefficient) was calculated from the regression line and compared.

Results: The lesion site in PC group included the head (n=17), body (n=7), and tail (n=16). The median maximum tumor diameter was 31 mm (interquartile range: 27-37). A comparison of the two groups found a cStage classification of IB, IIA, IIB, III, and IV in eight, ten, three, eight, and 11 patients, respectively. The median age was 72.5 and 72.5 years in the respective group (P=0.182). The male: female ratio was 23:17 and 24:16 (P=0.820), the median observation period was 83.5 and 107 months (P<0.05); the mean number of examinations was 10.7 and 18; the total number of CT examinations was 430 and 745; the mean total pancreatic volume was 74.4 and 78.15 ml (P<0.01); and the mean pancreatic volume change coefficient was -0.141 and -0.036 (P=0.01), respectively. The rate of decrease in the pancreatic volume was faster in the PC group with increasing age, the average being 8.46 ml over five years in the PC group compared to 2.16 ml in the non-PC group. The cutoff value for the coefficient of pancreatic volume change over time was set at -0.125. The sensitivity was 50%, the specificity was 90%, the positive predictive value was 83%, and the AUC value was 0.67.

Conclusion: In comparison with the age-related changes in the pancreas, the rate of decrease in the pancreatic volume was faster in the PC group, suggesting that PC onset may be predicted using images obtained one year or earlier before PC onset. These findings further suggested that identifying patients with a high risk of PC may be possible by creating a pancreatic volume histogram using AI technology based on age at the time of CT, intervals between past examinations, and the rate of decrease in the pancreatic volume.

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Disclosure: Nothing to disclose.

PP1150

THE EFFECTIVENESS OF CHECK SCORE SHEET FOR THE EARLY DETECTION OF PANCREATIC CANCER IN GASTROENTEROLOGY CLINIC

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Introduction: The early-stage pancreatic cancer (e-PC; stage I/II) detection rate is very low at approximately 25%. The role of general physician is very important for e-PC detection.

Aims & Methods: The aim of this study was to evaluate the effectiveness of the check score sheet used clinical findings to detect e-PC in gastroenterology clinic.

Patients of clinic with a score of ≥ 2 points on clinical findings were enrolled: symptoms of abdominal pain/back pain and/or body weight loss (1 point), high amylase, lipase and/or trypsin (1 point), high carbohydrate antigen 19-9 (CA19-9) (1 point), new-onset or worsened diabetes (2 point), family history of pancreatic cancer (2 points) and ultrasonography (US) findings including direct (e.g., a solid pancreatic tumor) and/or indirect findings (e.g., dilatation of a pancreatic diameter of ≥ 2 mm and/or cystic lesions) (2 points). Abdominal CT and/or EUS (Endoscopic ultrasonography) were performed for all these enrolled patients, and we evaluated about pancreatic disease and pancreatic cancer detection rates of this check score sheet.

Results: Between July 2017 and March 2024, 621 patients (mean age 67.6 years, female 391 [63%]), were enrolled in clinic. The 458 pancreatic cystic disease (IPMN and SCN) cases (73.7%), the 28 chronic pancreatitis disease cases (4.5%), the 18 PC patients (2.9%) and the 10 pancreatic solid disease (SPN and NEM) (1.6%) were detected by abdominal CT and/or EUS. The 6 patients with e-PC were detected. The mean clinical finding score of the PC group (3.2 ± 1.4) was significantly higher than that of the overall non-PC group (2.2 ± 0.6) (p < 0.05). US findings were significantly more frequent in the PC group than in the non-PC group (p <0.05). Although the sensitivity of direct findings by US was 42.3%, that of indirect-findings was 91.7% in the PC group.

Conclusion: Many pancreatic diseases, which may have potential of pancreatic cancer in future, were detected by this simple check score of ≥ 2 points on clinical findings. And in clinic pancreatic cancers were detected efficiently. Indirect findings by US may play an important role in improving the e-PC detection rate.

Disclosure: Nothing to disclose.

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PP1151

IMPACT OF TISSUE HETEROGENEITY OF PANCREATIC CANCER ON ENDOSCOPIC ULTRASOUND GUIDED SHEAR-WAVE MEASUREMENTS

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Introduction: Endoscopic ultrasound guided Shear wave elastography (EUS-SWE) is a noninvasive technique to evaluate
organ fibrosis as hardness. Velocity of SWE (Vs, m/s) enables to quantify, and the measured value is
also evaluated with confidence index: VsN (Percentage of the net amount of effective of Vs, %). EUS-SWE has been reported as a useful diagnostic modality for chronic pancreatitis (S Shintani et al *JMU* 2023). As for pancreatic cancer, previous reports using transabdominal ultrasound have reported only that Vs of pancreatic cancer is elevated, but there are no reports of VsN or using EUS devices. In this study, we examined the characteristics of pancreatic cancer using EUS-SWE.

Aims & Methods: The subjects were patients who underwent EUS-SWE at Shiga University Hospital from April 2019 to December 2022. The retrospective study was conducted with 121 patients (pancreatic cancer, chronic pancreatitis, normal pancreas). EUS-SWE was performed at least 5 times, and the median value was measured. Vs and VsN were measured at central site of tumor cases, and pancreatic body with chronic pancreatitis and normal cases. Histogram analysis (Complexity) at the SWE measurement site was also performed to evaluate tissue heterogeneity. ALOKA ARIETTA 850 (FIJIFILM, Tokyo, Japan) and GF-UCT260 (Olympus Co., Tokyo, Japan) was used.

Results: The subjects were 69 cases of pancreatic cancer (39 males, age: 68.9±10.6 years), 10 cases of chronic pancreatitis (6 males, age: 74.3±5.5 years), and 42 normal cases (22 males, age: 57.5±11.1 years) The EUS-SWE results (pancreatic cancer / chronic pancreatitis / normal) were Success rate (%): 87.2 / 93.5 / 98.1, Vs (m/s): 2.27±0.61 / 2.6±0.67 / 1.68±0.49, VsN (%): 22.6±15.9 / 76.2±21.2 / 79.6±20.6. Complexity: 30.1±12.3 / 23.9±4.8 / 13.5±10.6 indicate that heterogeneity within the tissue is higher in the order of pancreatic cancer > chronic pancreatitis > normal group. VsN and Complexity were inversely correlated, and the higher the tissue heterogeneity, the lower the VsN ($r=-0.45$, $p<0.001$, Spearman correlation test).

Conclusion: Vs was increased in pancreatic cancer compared to normal pancreas. However, VsN was more likely to be decreased in pancreatic cancer, which associated with higher tissue heterogeneity. EUS-SWE for pancreatic cancer should consider tissue heterogeneity.

References: none

Disclosure: Nothing to disclose.

PP1152

EXOCRINE PANCREATIC INSUFFICIENCY AND ADVANCED PANCREATIC CANCER - A NATIONAL REGISTRY ANALYSIS

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Introduction: Diagnostic algorithms and treatment of exocrine pancreatic insufficiency (PEI) in patients after pancreatic surgery have been sufficiently established. Comparable systematically collected data in patients with advanced pancreatic cancer (PDAC) is largely missing. In line, national European guideline recommendations for diagnosis and therapy remain controversial.

Aims & Methods: A registry based analysis of prospectively collected data with regard to PEI was done in patients with advanced PDAC. Data collection was based on EORTC questionnaires (QlQ-c30 and Pan26) and supplemented with fecal Elastase-1 measurements and specific treatment information.

Results: In a period over 5 years more than 100 patients could be included in the analysis at several specialised centres. Compared to other GI cancers (e.g. colon) potential PEI associated symptoms were stated more frequently. In this real-world setting fecal Elastase-1 measurement was not done routinely by centers and surprisingly carried out only in few cases. Frequency of pancreatic enzyme replacement therapy in PDAC patients was below the reported incidence in the literature of PEI in this cohort.

Conclusion: Awareness of PEI in advanced pancreatic cancer seems underrepresented. This survey illustrates the need for further prospective in-depth evaluation.

Disclosure: Nothing to disclose.

PP1153

PREOPERATIVE LYMPH NODE METASTASIS AS A PROGNOSTIC FACTOR IN PATIENTS WITH OPERABLE PANCREATIC CANCER

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Introduction: Pancreatic cancer is a fatal disease with a poor prognosis. Even after curative surgery, the overall survival (OS) is low, owing to the high frequency of relapses or surgical complications. Neoadjuvant chemotherapy (NAC) has recently been reported to improve OS; however, early postoperative death has been reported in some cases. Conversely, chemotherapy for unresectable pancreatic cancer is progressing; the median OS has been extended to approximately 15 months. Therefore, it is necessary that we carefully consider the possibility of surgery including NAC. There are many reports based on postoperative staging; however, there are few reports about preoperative prognostic factors in pancreatic cancer. This study aimed to evaluate these factors in patients with operable pancreatic cancer.

Aims & Methods: Between April 2010 and March 2023, 284 patients were diagnosed with resectable (R) or borderline resectable (BR) pancreatic cancer at the Sendai Kousei Hospital. Sixteen patients who were diagnosed with unresectable cancer after NAC were excluded. The preoperative clinical factors and OS were retrospectively evaluated. Survival data were compared using the log-rank test. P-value <0.05 was considered statistically significant. Clinical factors included in the univariate analy-

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ses were as follows: age (<75 years vs. ≧75 years), sex, region (head vs. body and tail), resectability (R vs. BR), CA19-9 (<37.0 U/ml vs. ≧37.0 U/ml), preoperative chemotherapy (upfront surgery vs. NAC), and staging (evaluated using computed tomography). Clinical staging was determined using preoperative TNM classification based on the General Rules for the Study of Pancreatic Cancer in Japan (8th Edition). Multivariate analysis was performed using a Cox proportional hazards regression model.

Results: The median follow-up period for OS analysis was 30.6 months. Among the clinical parameters, lymph node metastasis (clinical N1; cN1), tumor extent (cT3 or T4), and CA19-9 (≧37.0 U/ml) were significant predictors for OS in the univariate analysis. The median OS of the upfront surgery group (n=214) and the NAC group (n=54) were 28.7 and 39.7 month, respectively, showing an increase of approximately 11 months, which was not significant (p=0.33). Multivariate analysis revealed that only lymph node metastasis (cN1) remained an independent OS predictor (Hazard Ratio, 2.004; 95%CI 1.453-2.875; p<0.001). Median OS was 39.7 and 12.9 months in the cN0 (n=204) and cN1 groups (n=64) (p<0.001), respectively. The survival rate at three years were 51.8% and 21.3% in the cN0 and cN1 groups, respectively.

Conclusion: Preoperative lymph node metastasis was the strongest prognostic factor in patients with resectable or borderline resectable pancreatic cancer.

Disclosure: Nothing to disclose.

PP1154

FEASIBILITY OF ENDOSCOPIC ULTRASOUND-GUIDED FINE NEEDLE BIOPSY FOR GENERATING PATIENT-DERIVED ORGANOIDS OF PANCREATIC DUCTAL ADENOCARCINOMA: PRELIMINARY RESULTS FROM A SINGLE-CENTRE PROSPECTIVE DATABASE

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Introduction: Pancreatic ductal adenocarcinoma (PDAC) is notorious for its aggressive nature and grim prognosis, underscoring the urgent need for personalized pre-clinical models to develop targeted therapies. Patient-derived organoids (PDOs) are three-dimensional models of human tumours that have emerged as promising research tools.

Aims & Methods: Our study aimed to assess the feasibility of endoscopic ultrasound-guided fine needle biopsy (EUS-FNB) for the generation of PDAC-PDOs. A single-centre prospective database was established to collect data from suspected PDAC patients undergoing EUS-FNB for histological confirmation between May and October 2023. All biopsies were performed using a 22G-needle. Two to three biopsies were taken for histological examination before performing a single-pass FNB to establish PDO cultures. The primary outcome was the successful propagation of PDAC-PDOs for at least three passages (P3). Secondary outcomes included sample adequacy, diagnostic rate, and safety.

Results: A total of 22 patients (M:F 13:9, median age 66.5 years, range 55-86) were enrolled. One case was subsequently diagnosed as pancreatic lymphoma and excluded. PDAC-PDOs were successfully established in 9 out of 21 PDAC cases, resulting in a success rate of 43% (please see Table 1 for further details on singular included patients). Sample adequacy and diagnostic rates were both achieved in 100% of subjects. The presence of a biliary stent during EUS-FNB did not impact the success rate of PDAC-PDO generation (n=1 in the generated PDAC-PDOs group vs n=3 in the non-generated PDAC-PDOs group; p>0.05). No adverse events related to the procedure occurred.

Patient	Gender	Age	Lesion Size (cm)	Pancreatic Location	Sample Adequacy	Adverse Events
1	Male	62	3.5	Head	Yes	No
2	Female	85	3.0	Head	Yes	No
3	Male	61	1.2	Head	Yes	No
4	Male	68	3.5	Tail	Yes	No
5	Female	59	4.0	Head	Yes	No
6	Male	56	4.0	Body	Yes	No
7	Female	64	4.5	Tail	Yes	No
8	Female	60	5.0	Head	Yes	No
9	Female	58	4.0	Neck	Yes	No

Conclusion: This study demonstrates that PDAC-PDOs can be successfully generated through EUS-FNB sampling with a single-pass fine needle biopsy, without compromising the safety profile of the procedure.

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Disclosure: Nothing to disclose.

PP1155

ATTITUDES TOWARD GENETIC TESTING, FAMILY PLANNING AND PREIMPLANTATION GENETIC TESTING IN FAMILIES WITH A GERMLINE *CDKN2A* PATHOGENIC VARIANT

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Introduction: Individuals with a germline *CDKN2A* pathogenic variant (PV) have an increased lifetime risk of up to 70% for melanoma and up to 20% for pancreatic cancer, and are recommended to undergo skin and pancreatic cancer surveillance. Little is known about the psychosocial issues associated with this hereditary cancer syndrome.

Aims & Methods: This cross-sectional study included confirmed *CDKN2A* PV carriers and those with a 50% risk of the PV (at-risk carriers) under cancer surveillance. Individuals ≥18 years were identified from the skin and pancreatic cancer surveillance programs at the Leiden University Medical Center and invited for a one-time questionnaire with study-specific questions concerning genetic testing, family planning, and preimplantation genetic testing (PGT).

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Results: A total of 537 individuals were screened for eligibility, of whom 208 of 366 (57%) confirmed carriers (56% female, median age 54 years [IQR 46-63]) and 39 of 171 (23%) at-risk carriers (59% female, median age of 26 years [IQR 22-32]) participated in the study. Primary motivations for genetic testing were to gain control over their personal and children's cancer risk, as well as increasing cancer surveillance practices. In contrast, concerns about obtaining a mortgage and life insurance were frequently cited as reasons for postponing genetic testing.

Family planning decisions remained largely unaffected in both confirmed and at-risk carriers; however, the majority of confirmed carriers were still unaware of their familial or personal cancer risk when starting a family. More than 60% of the participants were unfamiliar with PGT and only a minority (19% of confirmed carriers and 10% of at-risk carriers) would be open to considering PGT as a reproductive option (Table 1).

	Confirmed carriers (n = 208)	At-risk carriers (n = 39)
Have you heard of PGT before? [a]		
No	134 (66%)	24 (62%)
Yes	68 (34%)	15 (39%)
Would you have preferred to receive counseling about PGT? [a]		
Yes	43 (21%)	4 (10%)
I don't know	64 (32%)	15 (36%)
No	95 (47%)	21 (54%)
Would you (have) consider(ed) using PGT for family planning? [b]		
Yes	38 (19%)	4 (10%)
Maybe	66 (33%)	25 (64%)
No	95 (47%)	10 (26%)

Table 1. Attitudes toward PGT

PGT, preimplantation genetic testing; PV, pathogenic variant. [a] 6 confirmed carriers missing. [b] 9 confirmed carriers missing.

Conclusion: This study found different attitudes toward genetic testing, family planning, and PGT among individuals affected by the *CDKN2A* PV. Understanding these different attitudes can help clinicians to address the complexities surrounding these issues, especially for younger individuals facing difficult decisions about the timing of genetic testing, family planning, and the potential use of reproductive assisted options.

Disclosure: Nothing to disclose.

PP1156

INSIGHTS INTO EARLY ONSET PANCREATIC CANCER: CLINICAL AND EPIDEMIOLOGICAL ANALYSIS

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Introduction: The reported prevalence of early onset pancreatic cancer (EOPC), defined in this context as pancreatic cancer diagnosed in individuals under the age of 50, ranges from 4% to 18%. Its incidence has shown a significant increase compared to later onset pancreatic cancer (LOEP). However, there remains limited understanding regarding the extent of this escalation and which subgroups are predominantly affected. This study aims to characterize the clinical and epidemiological characteristics of EOPC in comparison to LOPC.

Aims & Methods: We conducted a retrospective analysis of consecutive pancreatic cancer (PC) patients with definitive histopathological diagnoses, treated at our institution between January 2018 and December 2023. This study aims to characterize the clinical and epidemiological characteristics of EOPC in comparison to LOPC.

Results: Among 223 patients with pancreatic cancer (PC), 37 (16.6%) had EOPC. The mean age of these patients was 37 ± 7 years (range: 18-50). No significant difference in terms of sex was found, with a male predominance in both groups (54.1% in EOPC vs. 55.9% in LOPC). Regarding risk factors, no significant difference was observed between the two groups for smoking, alcohol consumption, obesity, and diabetes. However, a significant increase in family history of pancreatic cancer was noted in the EOPC group (10.8% vs. 1.6%, OR: 3.7 [95% CI: 1.8 – 7.6], p=0.003). No significant difference was found between the two groups regarding tumor stage; cancer was resectable in 27% of cases in the EOPC group compared to 16.1% in the LOPC group (p=0.240). Regarding nutritional status, according to the Nutritional Risk Index (NRI), younger patients had a lower risk of severe malnutrition (18.9% vs. 39.8%, OR: 0.4 [95% confidence interval: 0.2 -0.9], p=0.01), and the mean albumin level was significantly higher in the EOPC group (36 ± 4 vs. 33 ± 5 g/L, p=0.016).

Conclusion: Our study not only highlights the substantial prevalence of EOPC, but also underscores the presence of distinct clinical and epidemiological differences compared to LOPC. Notable findings include a heightened familial predisposition, disparities in nutritional status. These insights emphasize the imperative for tailored screening and management strategies to effectively address the nuances of EOPC.

Disclosure: Nothing to disclose.

PP1157

CANNABINOIDS IN PANCREATIC CANCER PALLIATIVE CARE: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Pancreas adenocarcinoma is the 6th leading cause of cancer-related death worldwide. Most patients are diagnosed with advanced disease, which is associated with distressing symptoms such as pain, fatigue, or anorexia. Cannabinoids are a new class of drugs for the palliation of cancer-associated symptoms; however, they lack clear indication.

Aims & Methods: Our aim is to investigate the safety and efficacy of cannabinoids for symptom control in advanced cancer patients.The study protocol was registered on PROSPERO (CRD42023479375), and a systematic search was conducted using three main databases (PubMed, Embase, and CENTRAL).

Interventional and observational articles, where cancer patients were administered any type of cannabinoid for symptom control, including pain, constipation, nausea and insomnia, were eligible for analysis. The change from baseline in symptom intensity was assessed, for which weighted means and pooled proportions were calculated with a 95% confidence interval (CI) and a random-effects model.

Results: After title-abstract and full-text selection, 96 articles were found to be eligible. Pain, constipation, nausea, and insomnia improvement were assessed using different types of symptom rating scales (0-absence

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of symptom, highest score-worst symptom intensity). The preliminary results show that cannabinoids do not improve pain compared to placebo, the mean change from baseline was 0.17 (-0.53; 0.87 95% CI); nor constipation, mean change from baseline of 0.05 (-0.65; 0.75 95% CI). Cannabinoids modestly improve sleep quality, with a mean change from baseline of -0.14 (-0.85;0.58 95% CI), and nausea, the mean difference from baseline -0.06 (-0.42; 0.29 95%CI), however not clinically significant.

Conclusion: Cannabinoids could be beneficial for advanced pancreatic cancer patients. However, the effects may vary significantly depending on the type of cannabinoid, dosage, and form of administration. These preliminary results need to be followed by further analysis based on subgroups, so proper indication/contraindication of the drugs can be suggested.

Disclosure: Nothing to disclose.

PP1158

PREVENTION OF POSTOPERATIVE COMPLICATIONS IN PATIENTS WITH PANCREATIC CANCER FOLLOWING PANCREATODUODENECTOMY

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Introduction: Despite improvements in the perioperative care of patients with pancreatic cancer the incidence of postoperative complications following pancreatic resections remains high.

Aims & Methods: The objective of this study was to develop surgical techniques to prevent postoperative complications following pancreatoduodenectomy. A retrospective analysis was conducted on the surgical outcomes of 456 patients with resectable pancreatic and periampullary cancer who underwent pancreatoduodenectomy between 2015 and October 2023. Since November 2018, new surgical tactics have been implemented to prevent the occurrence of postoperative pancreatic fistula (POPF) and other postoperative complications. Preliminary data indicates that the development of postoperative complications after pancreatoduodenectomy correlates with postoperative pancreatic fistula, the presence of sarcopenia has been demonstrated to significantly affect the number of clinically relevant postoperative pancreatic fistula and other postoperative complications.

Since that time, we have used new approaches: our modification of the pancreatic fistula risk score, which assesses pancreatic density according to the degree of pancreatic fibrosis. Depending on the risk, we used our surgical tactics. In cases with a high risk of postoperative pancreatic fistula, during pancreaticojejunostomy we used external pancreatic stent. In patients with intermediate risk we performed invaginated duct to mucosa pancreaticojejunostomy. In low-risk patients, the technique of pancreaticojejunostomous anastomosis was considered by the operating surgeon. We performed early removal of abdominal drains in patients with low risk of pancreatic fistula. Prior to surgery, the presence of sarcopenia was assessed, and nutritional support was provided to patients with sarcopenia. During the period from November 2018 to October 2023, proposed mitigation strategies were applied to 227 patients in the main group. The comparison group consisted of 229 patients who underwent surgery without the use of the developed surgical strategies between January 2015 and October 2018.

Results: The groups were comparable in terms of sex, age, laboratory results, disease stage, American Society of Anesthesiologists score, and comorbidities. In the main group, 26 patients (11.4%) experienced POPF grade B or C. In the comparison group, 64 out of 229 patients (27.9%) experienced a postoperative pancreatic fistula of grade B or C, which is significantly higher than the main group (c^2= 19.5, p = 0.0001). Furthermore, the main group had a significantly lower level of postoperative complications compared to the comparison group. A total of 69 patients (30.1%) in the main group experienced complications, while 94 patients (41.0%) in the comparison group did so (c^2=5.6, p=0.02). The mortality rate in the main group was reduced from 2.2% to 1.3% due to the developed tactics (c^2=0.49, p=0.5).

Conclusion: The surgical tactics developed to prevent postoperative pancreatic fistula after pancreatoduodenectomy were found to significantly reduce its occurrence from 27.9% to 11.4%, postoperative complications from 41.0% to 30.1%, and mortality from 2.2% to 1.3% (c^2= 0.49, p = 0.5).

Disclosure: Nothing to disclose.

PP1159

THE ROLE OF CHEMOTHERAPY IN THE MANAGEMENT OF PANCREATIC MUCINOUS ADENOCARCINOMA: A RETROSPECTIVE COHORT

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Introduction: Mucinous adenocarcinoma (MAC) is a mucin-producing tumor. In MAC, about >50% of the tumor is composed of extracellular mucin with clear malignant epithelial cells in clumps and layers. Pancreatic MAC represents only 3.5% of all pancreatic tumors and 5.1% of all MAC cases [1]. Few studies with Poor evidence investigated the survival of pancreatic MAC, especially for patients treated with chemotherapy.

Aims & Methods: In this study, we aimed to assess the impact of chemotherapy on the survival outcome in pancreatic MAC. We conducted a retrospective cohort study Using the Surveillance Epidemiology and End Results (SEER) database. We extracted the data of 4837 patients diagnosed with pancreatic MAC from 2000 to 2020. patients were sub-stratified according to the treatment modality into Surgical and Non-surgical groups. We conducted a survival analysis by Kaplan–Meier and Log-rank test using Jamovi v2.3.26 software. Cox regression analysis was conducted to investigate the association between other risk factors and survival.

Results: The median age for pancreatic MAC was 68 years (IQR: 59-76) [2] [3] with a 5-year Relative survival of 8.3%. The 5-year Relative survival in the group treated Surgically was significantly improved compared to the Non-surgical group (35% and 1.7%; P<0.001). In the Group that had surgical intervention, systemic Chemotherapy showed no significant survival benefit compared to who had no chemotherapy (33.1% and 37.3%; P=0.15). In the group treated medically, Chemotherapy showed a significant survival benefit compared to the group who received no chemotherapy with a 2-year relative survival of 11.6% and 4.2% while the 5-year relative survival was 1.4% and 2.1% (P<0.001) [4] [5]. COX-regression model showed males had better survival outcomes compared to females (HR=0.88, 95%CI: 0.83-0.93, P<0.001) while African Americans had worse survival compared to Caucasians (HR=1.15, 95%CI: 1.05-1.27, P=0.003) [6] [7][8][9].

Conclusion: Pancreatic MAC is an extremely aggressive tumor with poor survival outcome. Surgery had a significant survival benefit of 33% compared to medical treatment only which highlights surgery as the modality of choice for pancreatic MAC. Chemotherapy improved the survival of patients who were not treated surgically. Further evaluation of each patient is needed to confirm if the patient is qualified for the surgical intervention and chemotherapy. Males and Caucasians seem to have better survival compared to females and African Americans.

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Disclosure: Nothing to disclose.

PP1160

THE RISK OF MULTIPLE PRIMARY GASTROINTESTINAL TUMORS FOLLOWING PRIMARY PANCREATIC CANCER

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Introduction: Pancreatic cancer is the seventh leading cause of deaths worldwide representing 4.7% of all cancers. Cancer head of pancreas is the most common (45.95%) while 13.2% arise in the tail and 11.35% in the body.

The association between the three parts of pancreas(head, body and tail) and the risk of multiple primary gastrointestinal (GI) malignancies were poorly discussed in the literature.

The aim of this study is to assess the risk of multiple primary GI malignancies following primary pancreatic cancer with further stratification by the primary site as head, body and tail.

Aims & Methods: We used Surveillance, Epidemiology, End result (SEER) database to extract data of pancreatic cancer patients diagnosed from 2000 to 2020 .we further stratified them by site (head, body, tail). We used MP- SIR session to assess the Excess Absolute Risk (EAR) per 10,000. We calculated SIR as observed/expected (O/E) with 95% confidence interval and P-value was considered significant if P<0.05.

Results: Out of 221800 patients with pancreatic cancer ,1284 patients had GI second primary malignancy (SPM) with O/E of 2.28 (P<0.05, 95% CI: 2.15-2.41, EAR=35.96). In cancer head of pancreas, we observed 578 had GI SPM with O/E of 2.03 (P<0.05, 95% CI: 1.87-2.21, EAR=29.43) with significant increased risk of small intestine SPM in the first year after diagnosis (O/E 23.44 ,P<0.05,95% CI: 18.64-29.10, EAR=17.11). In cancer body of pancreas, 125 had GI SPM with O/E of 2.12 (P <0.05,EAR=29.54) with increased risk of stomach SPM in the 0-11 months interval. In cancer tail of pancreas, 187 patients had GI SPM (O/E of 2.31 P<0.05, EAR=31.93) with increased risk of other sites of pancreas and the stomach SPM.

Conclusion: Pancreatic cancer has very poor prognosis and significantly increased risk to develop second primary GI malignancies by two folds especially in small intestine, stomach, colon and liver. Cancer head of pancreas has the highest risk of multiple primary GI tumors compared to the other sites especially for small intestine in the first year after diagnosis. These results would take part in understanding the potential risk of GI SPM for better management plan and outcome and encourage to pay more attention for early GI malignancies screening after the diagnosis of pancreatic cancer.

Disclosure: Nothing to disclose.

PP1161

PANCREATIC CANCER BIOMARKERS SUCH AS INSULIN-LIKE GROWTH FACTOR BINDING PROTEIN-3, MACROPHAGE MIGRATION INHIBITORY FACTOR AND ADRENOMEDULLIN IN THE DIFFERENTIATION OF CHRONIC PANCREATITIS FROM PANCREATIC DUCTAL ADENOCARCINOMA

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Introduction: Insulin-like growth factor binding protein-3 (IGFBP-3), macrophage migration inhibitory factor (MIF) and adrenomedullin (ADM) are connected with pancreatic ductal adenocarcinoma (PDAC) development.

Aims & Methods: Evaluation of serum levels of IGFBP-3, MIF and ADM in patients with chronic pancreatitis (CP), PDAC and control group and their potential role for differentiation of CP and PDAC.The study included 70 CP, 73 PDAC, and 23 healthy subjects as a control group. Serum concentrations of IGFBP-3, MIF, and ADM were measured with ELISA (Corgenix UK Ltd, R&D Systems). Diagnostic efficiency of the biomarkers was assessed by means of AUROC calculation. The Cambridge and TNM scales were used.

Results: Mean IGFBP-3 serum level was significantly higher in CP patients vs. controls (3.98 ± 2.44 ng/ml vs. 1.40 ± 1.45 ng/ml, p<0.05). Mean IGFBP-3 level was also higher in CP than in PDAC group (3.98 ± 2.44 ng/ml vs. 2.02 ± 1.69 ng/ml, p<0.05). Mean MIF serum level was also significantly higher in CP patients vs. control group (8589.50 ± 3607.78 pg/ml vs. 7365.76 ± 5009.52 pg/ml, p<0.05). Mean MIF level was also higher in CP vs. PDAC patients (8589.50 ± 3607.78 pg/ml vs. 8160.2 ± 4840.32 pg/ml, p>0.05), but the difference was not statistically significant. Mean ADM serum level was significantly lower in CP vs. controls (76.72 ± 135.02 ng/ml vs. 177.12 ± 348.99 ng/ml, p<0.05), but it did not differ between CP and PDAC patients (76.72 ± 135.02 ng/ml vs. 152.72 ± 285.43 ng/ml, p>0.05). Mean CA19-9 serum concentration was 50.98 +/- 153.44 U/ml in CP vs. 439.33 +/- 441.24 U/ml in PDAC; p<0.0000). Mean MIF and ADM levels did not show significant differences between CP and PDAC groups. AUROCs for MIF, IGFBP-3 and ADM ranged below 0.75, which is lower than the AUROC for CA19-9 (0.8703; 0.8120 within 95% CI). CA19-9 & IGFBP-3 taken together showed AUROC equal to 0.8676 (0.8094 within 95% CI). When additional variable IGFBP-2 was added AUROC raised to 0.8912 (0.8378 within 95% CI). When data on CA19-9, IGFBP-2 and IGFBP-3 were considered together with patients' age and sex – AUROC was 0.9535 (0.9227 within 95% CI). Male sex and advanced age raised the probability of PDAC. When examining markers' differentiation according to the degree of advancement of CP and PDAC, it was found that CA19-9 better differentiates groups at an

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early stage of disease development (1 & 2 Cambridge in CP vs. I & II TNM in PDAC), AUROC 0.9259 vs. 0,8755. The combination of CA 19-9, IGFBP-2, IGFBP-3, age and sex gave better results in the advanced group, AUROC 0.9591 vs. 0.9708.

Conclusion: Panel of IGFBP-3, MIF, ADM and CA19-9 may be useful in PDAC and CP differentiation. Further studies are necessary to confirm those findings.

Disclosure: Nothing to disclose.

PP1162

WITHDRAWN

PP1163

MACHINE LEARNING PREDICTION METHODS FOR PANCREATIC DUCTAL ADENOCARCINOMA PROGNOSIS ASSESSMENT

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Introduction: Pancreatic Ductal Adenocarcinoma (PDAC) is a lethal cancer accounting for over 90% of pancreatic cancer cases and it aggressively develops metastases. It is currently the 4th leading cause of all cancer deaths in both males and females, with expectations to become the 2nd most by 2030. PDAC prognosis is grim with most being unresectable at the time of diagnosis.

Aims & Methods: Our objective was to develop machine learning algorithms based on clinicopathologic characteristics and biomarker levels to determine a prognostication model for PDAC assessment. A total of 76 patients diagnosed by EUS-FNB sampling with PDAC were included in the study. Clinical, paraclinical, and specific biomarker expressions were collected. We considered the following: alcohol consumption, smoking, diabetes type II, CRP, albumin, neutrophil/lymphocyte, thrombocyte/lymphocyte, CRP/albumin, CEA, CA 19.9, CA125, plasma levels of mir107 and CD63 immunohistochemistry expression. The endpoint was to compare T1-T2 with T3-T4 stages by using various algorithms.

We calculated error rates for Latent Dirichlet Allocation (LDA), Quadratic Discriminant Analysis (QDA), and Logistic Regression (LR). To evaluate the model and check how well it performs on unseen data, we took a validation set and checked the model's performance on it. Then modeled K-Nearest Neighbors (K-NN), where we demonstrated the validation set performance check.

Results: The 4 models had high accuracy ranging from 0.77 to 1, the best model was QDA and the worst was K-NN. LDA obtained an accuracy of 85% showing that the Neutrophil/Lymphocyte ratio will most influence a T1+T2 stage, and the PRC/Albumin and Thrombocyte/Lymphocyte ratio will lead to a T3+T4 stage. LR obtained an accuracy of 83%. Platelet/Lymphocyte ratio, PRC/Albumin, and CD63 indicators can be used for high prediction of T3+T4 stage, and Neutrophil/Lymphocyte ratio, PRC, and mir107 are suitable indicators for T1+T2 stage prediction.

Conclusion: Machine-learning methods based on clinicopathologic and biomarkers may serve as an auxiliary method to diagnose and predict PDAC prognosis.

Disclosure: Nothing to disclose.

PP1164

THE IMPACT OF EUS ON THE DIAGNOSIS, STAGING, HISTOLOGICAL PROOF, AND TREATMENT OF COMPLICATIONS OF SOLID PANCREATIC TUMORS: A SINGLE-CENTER SERIES

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Introduction: The sensitivity of endoscopic ultrasound (EUS) in detecting pancreatic tumors exceeds 90% and can detect tumors with a diameter of less than 2 cm. EUS plays an important role in the management of patients with pancreatic cancer, providing precise diagnosis, determining local-regional extension and resectability assessment, conducting biopsies, and managing complications.

The objective of our study was to specify the impact of EUS on the diagnosis, staging, histological proof, and treatment of complications of solid pancreatic tumors.

Aims & Methods: A retrospective study was conducted over 5 years (2019-2024), including all patients admitted with clinical suspicion of solid pancreatic tumors and who experienced EUS.

Results: Out of the 115 patients admitted with clinical suspicion of pancreatic tumor, 86 (74.8%) had undergone EUS. The mean age was 68.2 years, ranging from 30 to 90 years, with a clear male predominance (sex ratio 1.1). Biological assessment showed cholangitis grade I and grade II in 35.6% (n=31) of cases, and grade III in 4 patients (4.6%).

EUS revealed a pancreatic mass in 98.9% (n=85) of cases, with 75.5% (n=64) in the head, 14% (n=12) in the uncinate process, 7% (n=6) in the body, and 3.5% (n=3) in the neck. The size was over 3 cm in 59 cases (69.4%), between 2-3 cm in 23 cases (27.1%), and less than 2 cm in 3 cases (3.5%). double duct dilation was shown in 57 cases (66.3%), isolated bile duct dilation in 31 cases (36%), and isolated Wirsung duct dilation in 4 cases (4.6%). Pancreatic atrophy was observed in 31.4% (n=27) of cases.

EUS allowed the determination of vascular invasion in 78 cases (90.7%), invasion of the portal trunk (TP) and superior mesenteric vein (SMV) was present in 43 cases (55.1%), with invasion over >180 in 28 cases (35.9%) and <180 in 15 cases (19.2%), thrombosis in 3 cases (3.5%), and absence of invasion in 15 cases (19.2%).

Invasion of the celiac trunk (CT) and superior mesenteric artery (SMA) was present in 31 cases (39.9%), with invasion over >180 in 22 cases (28.2%) and <180 in 9 cases (11.6%), thrombosis in 2 cases (2.3%), and absence of invasion in 24 cases (30.7%). Invasion of the common hepatic artery (CHA) was present in 6 cases (7.7%), with invasion over >180 in 5 cases (83.3%) and <180 in one case (16.7%), and absence of invasion in 6 cases (7.7%).

Locoregional lymph node extension was observed in 12 cases (15.4%), invasion of the common bile duct (CBD) in one case (1.3%), duodenal invasion in 3 cases (3.9%), secondary hepatic localization in 7 cases (9%), and peritoneal in 2 cases (2.6%).

Cytopuncture under EUS was performed in 70 cases (89.7%), the results of which were conclusive in 65.4% (n=51) of cases and inconclusive in 24.6% (n=19). The histological type was adenocarcinoma in 48 cases (94.1%), clear cell carcinoma, squamous cell carcinoma, and neuroendocrine tumor in one case (1.9%).

Therapeutically, EUS allowed the treatment of pancreatic abdominal pain resistant to medical treatment by celiac plexus block in 2 cases, neurolysis in one case, and also performing anterograde hepatic gastric biliary drainage after the failure of retrograde drainage due to duodenal strictures in 5 cases.

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Conclusion: EUS reveals pancreatic tumors in 98.9% of cases, with its main importance lying in locoregional staging, where vascular invasion was correctly determined in 90.7% of cases, and lymph node extension in 15.4%. The value of EUS lies in determining histological type (65.4%), as well as therapeutic aspects such as pain management and biliary drainage.
Disclosure: Nothing to disclose.

PP1165

METABOLIC IMPLICATIONS OF PANCREATIC STEATOSIS: INSIGHTS AND CORRELATION ANALYSIS WITH MRI FINDINGS

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Introduction: Pancreatic steatosis (PS) frequently coexists with metabolic dysregulation, including insulin resistance, type 2 diabetes, and obesity. Magnetic resonance imaging (MRI) can be useful in quantifying intrapancreatic fat through the three-dimensional volumetric multi-echo gradient echo sequence (IDEAL-Q) to obtain pancreatic proton density fat fraction (P-PDFF).
Aims & Methods: Our study aims to establish a correlation between P-PDFF and metabolic indices. We enrolled 26 patients with increased parenchymal echogenicity in the pancreatic body on ultrasound evaluation and subsequently subjected them to MRI for quantification of intrapancreatic fat. The mean age of the patients was 56.26 years, including 16 women and 10 men, with an average BMI of 29.33+/- 8.38 kg/m2. Patients underwent standard oral glucose tolerance test (OGTT) with 75g glucose load. We calculated the area under the curve of glucose (AUC-Glu) and insulin (AUC-Ins) responses during OGTT.
Further, we evaluated beta-cell function through Disposition Index and Insulinogenic Index and insulin sensitivity through Matsuda Index and Homeostasis Model Assessment Index (HOMA-IR). MRI was performed in order to calculate PDFF within three pancreatic body regions of interest (ROIs). Linear correlations were conducted between mean PDFF and metabolic parameters, including the Matsuda Index, HOMA-IR, Insulinogenic Index, Disposition Index, and BMI.
Results: In the cohort of patients, 73.1% presented normal glucose tolerance (NGT), while the remaining 26.9% displayed impaired glucose tolerance (IGT).
Diabetic patients were excluded from the analysis. Stratification of patients was conducted according to the median of P-PDFF. Comparative analysis between these stratified groups revealed no statistically significant differences in Matsuda Index, HOMA-IR, Insulinogenic Index, or Disposition Index. Linear correlations between the mean PDFF and the analyzed metabolic parameters revealed no statistically significant associations, except for BMI ($p < 0.05$).
Conclusion: Our findings suggest there is no significant correlation between PDFF measured pancreatic steatosis, with glucose levels, insulin secretion and beta cell function. However, PDFF was significantly related to BMI suggesting that overweight and obesity is significantly related to pancreatic fat accumulation. Our study seem to suggest that pancreatic fat accumulation does not impact on endocrine pancreatic function.
Disclosure: Nothing to disclose.

PP1166

MORPHOLOGY OF THE PAPILLA CAN PREDICT PROCEDURAL SAFETY AND EFFICACY OF ERCP - A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Endoscopic Retrograde Cholangiopancreatography (ERCP) is the most commonly used therapeutic procedure for pancreaticobiliary disorders. However, how to best achieve safe and effective bile duct cannulation is still debated. Endoscopists doing ERCP routinely recognize the differences in the macroscopic appearance of the major papilla. This has led to a conception that certain appearances of the papilla are more difficult to cannulate and, therefore, more prone to adverse events.
Aims & Methods: We aimed to quantify the influence of papilla morphology on ERCP outcomes.
PROSPERO registration number: CRD42022360894. We systematically searched three medical databases from inception in September 2022. Studies detailing the cannulation or the rate of adverse events in the context of papilla morphology were included. For the primary classification of the major papilla, the Haraldsson system was used, differentiating four morphological types. Data on studies using other classification systems were also collected. A pooled event rate with a 95% confidence interval (CI) was used for the effect size measure. For the test of subgroup differences, a "Cochrane Q" test was used. The risk of bias assessment was performed using the Joanna Briggs Institute Critical Appraisal tool for studies reporting prevalence.
Results: A total of 17 studies were eligible, and 14 were included in the quantitative synthesis. In studies using the Haraldsson classification, the rate of difficult cannulation was the lowest in type I ("regular") papilla (26%; CI: 18–37), followed by type III ("protruding or bulging") (35%; CI: 25–48) and type II ("small") papilla (39%; CI: 28–52). The highest rate was observed in the case of type IV ("creased or ridged") papilla (41%; CI: 28–55). For post-ERCP pancreatitis, the event rate was the highest in type II ("small") papilla (11%; CI: 8–15) and the lowest in type I ("regular") (6%; CI: 5–8) and III ("protruding or bulging") papilla (6%; CI: 4–8). There was no statistically significant difference in the cannulation failure and post-ERCP bleeding event rates between the different papilla types. Similar trends were observed in the analyses including other classification systems comparable to the Haraldsson's. Most studies carried a low risk of bias.
Conclusion: In conclusion, other types are associated with a higher rate of difficult cannulation compared to the regular papilla type. The small papilla is associated with a higher rate of post-ERCP pancreatitis.
Disclosure: Nothing to disclose.

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PP1167

EXPLORING THE ROLE OF ECHO-ENDOSCOPY IN UNRAVELING ACUTE PANCREATITIS ETIOLOGY: A RETROSPECTIVE STUDY

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Introduction: Acute pancreatitis (AP) can pose challenges in diagnosing its underlying cause, with 10-30% of cases remaining elusive despite thorough clinical, biological, and morphological assessments. Echo-endoscopy (EUS) is particularly effective in exploring the bilio-pancreatic region. Our study aims to assess the role of EUS in elucidating the etiology of unexplained AP.

Aims & Methods: We performed a retrospective descriptive study at a university hospital's gastroenterology department between January 2018 and December 2023. It involved 118 patients diagnosed with AP, whose etiology was either suspected or remained undetermined after an initial radiobiological assessment (US, CT scan and MRI). These patients underwent EUS as part of their diagnostic evaluation.

Results: The average age of our patients was 55 ± 16 years, with a predominance of females (73.7%) and a male-to-female sex ratio of 0.35. 32.2% of patients had undergone cholecystectomy. 94.1% were admitted for their first episode of AP, while 5.9% had experienced at least one previous episode. Jaundice was observed in 29.7% of cases, and liver tests abnormalities were noted in 38.5%. Following initial radiological evaluation (ultrasonography, CT scan, MRCP), 53.4% of patients had biliary duct dilatation without visible obstruction, 33.9% had no dilatation, and 12.7% had diagnostic uncertainty. The diagnostic yield of EUS was 83%: micro lithiasis of the main bile duct accounted for 75.4% of cases, tumor-related causes for 4.2%, IPMN in 2 cases, biliopancreatic junction anomaly in 1 case, and Pancreas Divisum in 1 case. EUS findings were normal in 17% of cases.

Conclusion: EUS shows promise in elucidating the etiology of unexplained AP, boasting a high diagnostic yield in exploring the bilio-pancreatic region. Integrating EUS into diagnostic protocols could enhance the management of patients with AP, particularly in cases where initial investigations are inconclusive.

Disclosure: Nothing to disclose.

PP1168

ENDOSCOPIC PAPILLECTOMY FOR LATERALLY SPREADING AMPULLARY LESIONS

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Introduction: Endoscopic papillectomy (EP) is a standard endoscopic treatment for ampullary lesions (AL), most of which are small and limited to the papillary mound. Laterally spreading lesions (LSL) of the papilla Vateri represent an rare subtype of AL, characterized by extension beyond the ampullary mound and extensive involvement of the surrounding duodenum. In this study, we analyzed the outcomes of EP in matched cohorts, comparing cases of LSL with non-LSL of the ampulla.

Aims & Methods: The ESAP study encompassed 1422 endoscopic papillectomies (EPs). We employed propensity-score matching using the nearest-neighbor method using age, gender, comorbidities and histologic subtype as cofactors. The main outcomes assessed were complete resection (R0), technical success, complications, and recurrence.

Results: Propensity-score-based matching identified 232 patients (116 non-LSL and 116 LSL) with comparable baseline characteristics. LSL sizes were significantly larger compared to non-LSL (median 27.0mm vs. 16.5mm, $p<0.001$). Therefore, en bloc resection rates were significantly higher in the non-LSL group (71.6% vs. 34.5%, $p<0.001$). After first intervention, the R0-rate was significantly lower in the LSL group (54.3% vs. 69.0%, $p=0.002$). Following repeated interventions (such as EP, RFA, APC), the technical success was comparable in both groups at 82.8% ($p=1.00$). There were significantly more recurrences in the LSL group (23.8% vs. 8.6%, $p=0.013$). Complication rate exhibited no significant variance between the two groups.

Conclusion: LSL can be safely managed endoscopically, though repeated interventions are necessary for outcomes comparable to non-LSL. The higher risk of recurrence in LSL necessitates a vigilant surveillance strategy.

Disclosure: Honoraria from FUJIFILM

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PP1169

POST-ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY PANCREATITIS WAS AN INDEPENDENT RISK FACTOR OF PANCREATIC CANCER DEVELOPMENT IN PATIENTS WITH TYPE 2 DIABETES

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Introduction: Both acute pancreatitis and diabetes mellitus (DM) are known risk factors for pancreatic cancer (PC) development. However, whether post-endoscopic retrograde cholangiopancreatography (ERCP) pancreatitis (PEP) is an independent risk factor for PC is under-investigated.

Aims & Methods: We aimed to investigate whether PEP was associated with a higher risk of PC after adjusting for known risk factors of PC among type 2 diabetes (T2D) patients.

All newly diagnosed T2D patients aged ≥18 years between January 1993 and December 2020 from Hong Kong's electronic healthcare database were studied. DM was identified by (i) two abnormal HbA1c ≥6.5% or fasting glucose ≥7 mmol/L test results, (ii) anti-DM medications use, or (iii) International Classification of Diseases Ninth Revision (ICD-9) code 250. Exclusion criteria included history of PC, pancreatic neuroendocrine tumor, serous cystadenoma, and chronic pancreatitis. Index date was defined as three years after T2D diagnosis in order to avoid reverse causality, as new onset diabetes (NOD) may signify underlying pre-existing PC. Primary outcome was PC development. Primary exposure of interest was PEP, which was identified by an amylase or lipase level at least three times the upper limit of normal more than 24 hours and within 7 days after ERCP.

We divided the cohort into three groups:

(i) those who had never underwent ERCP before index date [ERCP(-)],

(ii) history of ERCP without PEP [ERCP(+)/PEP(-)], and;

(iii) history of PEP [ERCP(+)/PEP(+)].

The comparison of ERCP(+)/PEP(+) group with ERCP(+)/PEP(-) group serves to address indication bias (and hence higher PC risk) necessitating performance of ERCP. Covariates were age, sex, lifestyle factors, metabolic disorders, time-weighted mean HbA1c, cardiovascular diseases, other anti-DM medications, non-aspirin non-steroidal anti-inflammatory drugs, statins, and episode of history of acute pancreatitis. Multivariable Cox regression models were used to calculate adjusted hazard ratios (aHRs) of PC with exposure of interest.

	No. of patients	No. of events	Person-years*	Crude incidence rate	HR (95% CI)†	p-value
Whole cohort						
ERCP(-)	751,184	1,491	6,776,558	0.22	Ref	-
ERCP(+)/PEP(-)	20,990	1,024	228,958	4.47	16.8 (15.5 – 18.3)	<0.001
ERCP(+)/PEP(+)	375	12	2,521	4.76	28.0 (15.7 – 50.2)	<0.001
ERCP-cohort						
ERCP(+)/PEP(-)	20,990	1,024	228,958	4.47	Ref	-
ERCP(+)/PEP(+)	375	12	2,521	4.76	1.94 (1.08 – 3.48)	0.025

* Person 1,0000 person-years
† Adjusted for covariates mentioned above
Abbreviations: ERCP, endoscopic retrograde cholangiopancreatography; PEP, post-ERCP pancreatitis; HR, hazard ratio; Ref, reference; 95% CI, 95% confidence interval

Table 1. Association between risk of pancreatic cancer and post-ERCP pancreatitis.

Results: We identified 772,549 T2D patients (mean [SD] age:60.6 [12.4] years;50.3% male), of whom 751,184 (97.2%) was ERCP(-) group , 20,990 (2.7%) ERCP(+)/PEP(-) group, and 375 (0.05%) ERCP(+)/PEP(+) group. The mean (SD) duration of follow-up was 9.1 (+/-5.9) years, and 2,128 (0.3%) incident PC cases were recorded with an incidence rate of 3.0 per 10000 person-years. The PC crude incidence rate was 0.22, 4.47 and 4.76 per 10,000 person-years among ERCP(-), ERCP(+)/PEP(-) and ERCP(+)/PEP(+) groups, respectively. Compared with ERCP(-), ERCP(+)/PEP(-) and ERCP(+)/PEP(+) were associated with a higher risk of PC (aHR:16.8, 95%CI:15.5-18.3; aHR:28.0, 95%CI:15.7-50.2) (*Table 1*). Compared with ERCP(+)/PEP(-), the aHR of PC was 1.94 (95%CI:1.08-3.48) with ERCP(+)/PEP(+).

Conclusion: In a large population of T2D patients, PEP was an independent risk factor of PC development. Our study findings further consolidate the necessity of minimizing the occurrence of PEP to not just prevent short-term complications of acute pancreatitis but also PC development.

Disclosure: Nothing to disclose.

PP1170

DELAY RETROGRADE CHOLANGIOPANCREATOGRAPHY (ERCP) IN PATIENTS WITH MILD BILIARY ETIOLOGY ACUTE PANCREATITIS AND EVIDENCE OF COMMON BILE DUCT STONE NEGATIVELY INFLUENCE REFEEDING TIME, TOTAL LENGTH OF HOSPITALIZATION AND RELATED COMPLICATIONS

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Introduction: Acute pancreatitis (AP) is one of the most common gastrointestinal disorders with biliary etiology (BE) being the most common cause in Italy. Recent guidelines suggest medical therapy over an early retrograde cholangiopancreatography (ERCP) in patients with AP and evidence of biliary obstruction (BO) unless acute cholangitis (AC). Although in AP the refeeding shall be offered as soon as clinically tolerated, in clinical practice in patients with BO, it is usually postponed after ERCP. This study aims to evaluate whether delayed ERCP influences early refeeding, total length of hospital stay and hospitalization-related complications in patients with BE-AP and BO.

Aims & Methods: Single center retrospective study including patients with AP admitted to our unit from January 2020 to March 2024. Refeeding was evaluated as oral refeeding in mild AP and oral or enteral refeeding via enteral-feeding tube in moderate to severe AP.

Results: Overall, 109 patients were included: with a median age of 68 (19-98), 57% men. Biliary etiology in 57%, 12% alcoholic, 3% idiopathic and 28% other causes. Regarding the severity of AP, 60% of patients had mild, 28% moderate, and 12% severe pancreatitis. Mild AP had a median length of stay of 5 days (1-22), moderate/severe AP, 14 days (3-32). Patients with BE AP had a similar median length of hospital stay compared to those with non-biliary etiology (NBE), 7 days (BE 2-27) vs 6 days (NBE 1-32). Overall, 21% of patients with BE-AP without cholangitis had BO on radiological examinations and were scheduled for a not urgent ERCP. BE patients with mild AP and BO had a statistically significant difference in terms of median

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length of hospital stay (10 days, 3-22), and median fasting days (7 days, 4-18) compared to those without BO [5 days (2-15) and 3 days (1-14), respectively] (p<0.05). The median time from admission to ERCP was 7 days (4-17). Furthermore, the percentage of hospitalization-related complications (vascular, renal, pulmonary, and infective) was, as expected, directly proportional to the length of hospital stay.

Conclusion: In a real-world setting, delayed refeeding was observed in patients with mild BE-AP and evidence of BO with no indication to an early ERCP. This probably reflect the concerns of clinicians regarding clinical and biochemical worsening after refeeding in presence of BO. ERCP and refeeding delay also negatively influence the length of hospitalization and related complications.

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Disclosure: Nothing to disclose.

PP1171

COMPARATIVE DIAGNOSTIC PERFORMANCE OF ENDOSCOPIC ULTRASOUND-GUIDED FINE NEEDLE ASPIRATION (EUS-FNA) VERSUS ENDOSCOPIC ULTRASOUND-GUIDED FINE NEEDLE BIOPSY (EUS-FNB) FOR TISSUE SAMPLING OF SOLID PANCREATIC AND NON-PANCREATIC LESIONS WITHOUT ROSE: A PROSPECTIVE MULTICENTER STUDY

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Introduction: Endoscopic ultrasound-guided tissue acquisition, including both fine needle aspiration (EUS-FNA) and fine needle biopsy (EUS-FNB), has been widely performed to obtain samples from pancreatic and non-pancreatic lesions.[1,2] The impact of the diagnostic yield between FNA and FNB remains unclear.

Aims & Methods: We conducted this study to compare the diagnostic performance and accuracy of the 22-gauge FNA needles with the 22-gauge FNB needles in sampling solid pancreatic and non-pancreatic lesions. This is a prospective multicenter study conducted on 465 cases presented with solid pancreatic or non-pancreatic lesions.

Results: Patients were 275 male and 190 females with a mean age of 59 years. Three hundred twenty-seven patients had solid pancreatic lesions, while 138 had non-pancreatic lesions; 245 cases underwent EUS-FNA, and the remaining 211 cases underwent EUS-FNB. The presence of intact tissue core and sample adequacy were significantly higher in the FNB cases in solid pancreatic and non-pancreatic lesions. Blood contamination was significantly more in cell blocks and smears of EUS-FNA compared to that of EUS-FNB in solid pancreatic and non-pancreatic lesions. Based on histologic assessment of cell block only, EUS-FNB had more diagnostic accuracy (99%) than FNA (61%) (P-value<0.005). However, cytological diagnosis by smears only showed no significant difference. The combined cytological and histological evaluation had 100% sensitivity, specificity, and accuracy.

Conclusion: EUS-FNA and EUS-FNB have comparable accuracy in diagnosing solid pancreatic and non-pancreatic lesions without ROSE. EUS-FNB is superior to EUS-FNA in acquiring intact tissue core and adequate samples with little blood contamination. Based on histological assessment (cell block/tissue core) only, EUS-FNA has less accuracy than EUS-FNB in diagnosing solid pancreatic lesions.

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Disclosure: Nothing to disclose.

PP1172

DIAGNOSIS OF PANCREATIC SOLID LESIONS BY ULTRASOUND ENDOSCOPY (EUS) FINE-NEEDLE BIOPSY WITH FRANSEEN NEEDLES: MACROSCOPIC ON-SITE (MOSE) VERSUS RAPID ON-SITE (ROSE) EVALUATION. IS THE PATHOLOGIST NECESSARY IN THE ENDOSCOPY SUITE?

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Introduction: Biopsy samples obtained with fine-needle biopsy (FNB-EUS) using Franseen needles by EUS with endoscopist evaluation (Macroscopic OnSite Evaluation: MOSE) improves the diagnostic accuracy of pancreatic solid lesions. The achievement of macroscopic visible core (MVC) >4mm has a high diagnostic accuracy with MOSE.

Aims & Methods: Aim: To compare diagnostic accuracy of Rapid On-Site Evaluation (ROSE) and MOSE by endoscopist in pancreatic solid lesions. Methods: Prospective bicentric study, consecutive recruitment of patients with pancreatic solid lesions in 2 general hospitals (MOSE was performed in hospital A and ROSE in hospital B). Franseen needles were used (Adquire, Boston Scientific). We evaluated: 1) diagnostic achievement and accuracy (including sensitivity, specificity, positive predictive value and negative predictive value) in solid pancreatic lesions; 2) MVC >4mm acquisition; and 3) number of needle passes needed with MOSE vs ROSE.

Results: A total of 132 patients were included (33 women, 68 men, median age 68,2±1,025). MOSE was performed in 90 (68.2%) and ROSE in 42 (31.8%); in 110 (83.3%) malign and 22 (16.7%) benign pancreatic solid lesions. Correct diagnosis by EUS-FNB was achieved in 89.4% patients (with

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MOSE 88.8%; and ROSE 92.9%; p=0.54). Sensitivity was 100%, specificity 89%, PPV 81.3% and NPV 100% (with MOSE: sensitivity 100%, specificity 87.5%, PPV 88.8% and NPV 100%; with ROSE: sensitivity 100%, specificity 92.1%, PPV 92.6% and NPV 100%). MVC >4mm was obtained in 72.7% (71.1% in MOSE, 76.2% in ROSE, p=0.67). There were no statistica ly significant differences in diagnosis achievement between MVC >4mm and <4mm (88.6% in MOSE and 90.6% in ROSE, p=0.74). No differences in number of passes necessary to achieve diagnosis was found between MOSE and ROSE.

Conclusion: Evaluation of biopsy samples of pancreatic solid lesions obtained by FNB-EUS with Franseen needles by endoscopist (MOSE) has a diagnostic accuracy similar to ROSE, making unnecessary the presence of a pathologist in the endoscopy suite.

Disclosure: Nothing to disclose.

PP1173

EUS-GUIDED PANCREATIC DUCT DRAINAGE: ENDOSCOPIC AND CLINICAL PRACTICE EXPERIENCE

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Introduction: EUS-guided pancreatic duct drainage (EUS-PDD) is an endoscopic procedure which is indicated in symptomatic patients with evidence of pancreatic duct obstruction after failure or impossibility to perform transpapillary drainage or EWSL in the contest of chronic pancreatitis or altered anatomy causes (e.g. Whipple or Roux-en-Y surgery). Nowadays, it is a challenging procedure performed by few endoscopist. Clinical and technical success rate is still lacking and heterogenous, therefore acquisition of new data and experience is necessary.

Aims & Methods: This study aims to describe our initial endoscopic experience evaluating also some clinical aspects. This is a retrospective observational single-center study including patients (pts) undergone EUS-PDD from March 2018 to July 2023 using plastic stent (7Fr 10cm), delivered with an electrocautery-enhanced catheter, after failure or impossibility to transpapillary drainage. It was evaluated exocrine pancreatic function by fecal elastase (FE-1) and nutritional markers (prealbumin and magnesium).

Results: Total of 15 pts with median age of 72 years old. 13 pts (86.6%) had a chronic pancreatitis diagnosis with radiological evidence duct pancreatic alterations (5 stenosis, 3 lithiasis, 4 both, 1 duct rupture); 2 pts had a post-surgical stenosis. All pts were symptomatic: 6 pts (40%) shows multiple episodes of acute pancreatitis and 9 (60%) had abdominal pain non responder to medical therapy. Pancreatic duct had median dilatation of 7 mm (4-15.6 mm). Every transmural access was performed through gastric wall with single plastic stent (7Fr 10cm) placement. Technical success was obtained in 11 pts (73%) of which 2 pts had early stent migration; in 4 pts (26.6%) EUS-PDD failed. No other adverse events. 13 (86.6%) had a condition of pancreatic exocrine insufficiency: FE-1 between 200 µg/g and 100 µg/g in 8 pts and FE-1 < 100 µg/g in 5 pts under pancreatic enzyme replacement therapy. Magnesium median value was 0.74 mmol/L (0.5-0.8 mmol/L) and prealbumin median value was 0.22 g/L (0.11-0.26 g/L), both at the low normal serum level. After 4 weeks, it was placed a second plastic stent in 8 pts and during clinical follow-up (3 and 6 months), pts had no pain and described improvement of general quality of life and nutritional markers.

Conclusion: EUS-guided pancreatic duct drainage remains a challenging endoscopic procedure indicated only for selected cases and performed by expert endoscopists. However, it seems to have a reasonable clinical success rate but we need further data to improve endoscopic experience and clinical awareness.

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PP1174

ABSOLUTE BETTER THAN PERCENTAGE

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Introduction: For symptomatic walled off necrosis(WON) drainage, initial assessment of WON is done by percentage of debris and accordingly LAMS, irrigation and necrosectomy(DEN) are performed. This percentage assessment may go wrong once the fluid is drained off. Percentage of debris may not adequately represent the necrosectomy sessions or time. More necrosectomy procedures, require longer endoscopy time, multiple accessories, intravenous sedation and higher costs.

Aims & Methods: This study is aimed at finding the relation between debris amount and necrosectomy sessions.Consecutive symptomatic WON patients who underwent drainage and necrosectomy are enrolled for this study. Collections without definite wall or irregular shaped collections, trans duodenal drainages are excluded. Percentage of debris in WON is assessed. Solid component of collection is measured directly (length, width, sector, segment). Volume of debris is calculated using mathematical calculations for ellipsoid segment calculations. Every patient underwent CT abdomen 24-48hrs after drainage to assess stent position, any complications and amount of resolution of collection. In difficult to define cases by EUS, CT done after WON drainage is used for calculating the amount of debris.

Results: Five patients of symptomatic WON who underwent drainage and necrosectomy are included in this study. Three male and two female. Average debris percent is 28 % highest being 50 %. Necrosectomy is done in all cases following the step up approach. Mean volume of debris is 182.2 cm^3. Mean number of necrosectomy sessions are 2.5. On an average, 50-80cc of debris is removed per hour. More amount of debris requires more number of hours of necrosectomy or more sessions of necrosectomy as depicted in the table 1.

Serial no	Etiology	Debris %	Size of collection (cm)	Necrosectomy sessions	Volume of debris
1	Biliary	30%	15.2 x 8 x 4.6	3	232
2	Biliary	30%	8 x 6.2 x 5.1	2	183
3	Alcoholic	20%	6.3 x 4.2 x 3.7	2	178
4	Idiopathic	50%	10.5 x 7.3 x 4	3	220
5	Hypertriglyceridemia	10%	12 x 7 x 3.5	1	98

Conclusion: In the presence of fluid, debris in WON may be under or overestimated if assessed by percentage. Absolute volume of debris is a better measure to quantify than percentage for deciding and planning on necrosectomy. More volume of debris requires more duration or number of necrosectomy sessions.

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Disclosure: Nothing to disclose.

PP1175

DIAGNOSTIC VALUE OF EE-FNB IN PATIENTS AT RISK OF PANCREATIC CANCER WITHOUT OBSTRUCTIVE JAUNDICE IN FAMILIES WITH BREAST CANCER AND WITH FOCAL LESION ON COMPUTERIZED TOMOGRAPHY/ MAGNETIC RESONANCE

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Introduction: An increased risk of pancreatic cancer has been reported in families with breast cancer history carrying BRCA1 and BRCA2 mutations. Pancreatic cancer is associated with familial and hereditary cancer syndromes. Multigene panel testing for pancreatic cancer detected germline variants BRCA1/2, PALB2, ATM, TP53, MLH1, STK11/LKB1, APC, CDKN2A, and SPINK1/PRSS1 as high-risk genes. Active pancreatic surveillance using computed tomography (CT), magnetic resonance imaging (MRI), and echoendoscopic (EE) are recommended for high-risk individuals who have a family history of pancreatic cancer or harbor these pathogenic germline variants to improve the detection rate and the prognosis of pancreatic cancer. We evaluated the performance characteristics of echoendoscopically guided fine needle biopsy (EE-FNB) in this subset of patients.
Aims & Methods: This is a retrospective analysis of a prospective database total of 99 families with ≥1 case of breast cancer tested for mutations in BRCA1 and/or BRCA2. The eligible patients were the ones that underwent EE-FNB at a tertiary hospital with an Oncology center due a focal pancreatic lesion observed on CT/MRI, from January 2022 to January 2024. Patients were excluded if (1) they had obstructive jaundice or (2) the lesion appeared cystic on CT/MRI. The main outcome measures were (1) prevalence of the association between BRCA 2 positive and pancreatic cancer (2) performance characteristics of EE-FNB for identification of malignancy.
Results: In the 99 patients in the study, a focal pancreatic lesion was identified with pancreatic adenocarcinoma by EE-FNB. The final diagnosis included 20 patients with breast cancer and BRCA2 positive undergoing oncological treatment or follow-up, with a mean age of 32.8 +/- 2.2 years. EE was performed in an average of 19 +/- 11.3 days. EE-FNB showed an accuracy of 100% for the diagnosis of malignant neoplasia, with a sensitivity of 100%, specificity of 100%.
Conclusion: EE-FNB diagnostic yield was substantial in established high-risk mutation carriers, is highly accurate in patients with focal pancreatic lesion on CT/MRI, without obstructive jaundice. EE-FNB can potentially be used as a definitive diagnostic test in the management of these patients.
Disclosure: Nothing to disclose.

PP1176

EUS-GUIDED LUMEN APPOSING METAL STENTS (LAMS) WITH AND WITHOUT ELECTRO-CAUTERY FOR DRAINAGE OF WALLED-OFF PANCREATIC NECROSIS

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Introduction: EUS-guided drainage using lumen-apposing metal stents (LAMS) has become standard treatment of symptomatic walled-off necrosis (WON). The development of these fully covered self-expanding metal stents with special design of the flanges and large diameter, improved the drainage of debris and facilitated further direct endoscopic necrosectomy. Delivery systems with integrated electro-cautery (EC) enable direct access and stent deployment whilst the conventional stent insertion technique requires several steps including access using a needle or cystotome, wire insertion and enlargement of the created tract before the stent placement.
Aims & Methods: In this single center study, we aimed to compare the practicality of EUS-guided procedures and their outcome between conventional LAMS (cLAMS) and those with integrated EC (LAMS-EC).
From our prospectively maintained database, we included consecutive patients undergoing EUS-guided LAMS insertion with or without integrated electro-cautery for drainage of walled-off necrosis. Infection of the walled-off pancreatic necrosis, biliary compression with cholestasis or gastric outlet obstruction were accepted indications for intervention. Data on procedure time, sedation, hospital stay, technical and clinical success as well as adverse events were analysed.
Results: We analysed 107 consecutive procedures of transmural EUS-stenting for drainage of walled-off necrosis. 39 cLAMS and 68 LAMS-EC were placed successfully. Mean size of the walled-off necrosis was 12.9 ± 4.2 cm with mean 30% ± 15% solid necrosis.
Procedure times were shorter with LAMS-EC (18 .0 ± 6.6 min vs. 39.7 ± 8.3 min; p<0.05). All EUS-guided procedures with cLAMS drainage were performed under propofol sedation administered by an anaesthetic team. In the LAMS-EC group, 36 patients tolerated the procedure under iv sedation (fentanyl/midazolam) in left lateral position and 17 interventions were performed as daycase procedures. None of the patients developed sedation related adverse events such as aspiration.
Adverse events, technical success and clinical outcome did not differ between both groups.
Conclusion: The findings of this study suggest that cLAMS and EC-LAMS achieve similar rates of technical and clinical success without difference in the rate of adverse events. Compared to cLAMS, deployment of EC-LAMS

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is faster and technically less demanding. This might allow to perform the procedure under conventional intravenous sedation and as daycase procedure in carefully selected patients.

Disclosure: Nothing to disclose.

PP1177

FACTORS AFFECTING THE DIAGNOSTIC ACCURACY OF EUS-FNA/B FOR PANCREATIC AND PERIPANCREATIC LESIONS: A TWO-CENTER EXPERIENCE

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Introduction: EUS-FNA and EUS-FNB have become the standard of care for the diagnosis of pancreatic and peripancreatic tumors, and several factors are associated with its diagnostic accuracy. The aim of this study is to evaluate factors that are associated with the diagnostic accuracy of EUS-FNA/B in two referal centers

Aims & Methods: Data from patients with pancreatic and peripancreatic lesions who underwent EUS-FNA/B from 2017 to 2023 were collected retrospectively. The associations between mass location, mass size, needle type, number of needle passes, the presence of a biliary stent and the cytology result were analyzed.

Results: A total of 438 patients underwent 440 EUS-FNA/B procedures. The overall accuracy of EUS-FNA/B was 90,6% (398/440). Mass location on the head of the pancreas was associated with a higher diagnostic accuracy ($p<0.05$). Two or more needle passes were statistically significantly correlated with higher diagnostic accuracy ($p<0.05$) compared to single-pass EUS-FNA/B. Mass size, needle type and presence of a biliary stent did not reach statistical significance for the diagnostic yield of the cytology. A subgroup of 109 patients (24,7%) had lesions in the corpus or the tail of the pancreas. In this subgroup, tumor size greater than 3 cm was statistically significant correlated with a diagnostic cytology result ($p<0.05$).

Conclusion: The results of this study suggest that 2 or more needle passes and lesions in the head of the pancreas are associated with higher diagnostic accuracy of EUS-FNA/B in two referal centers.

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PP1178

COMPARISON OF DIAGNOSTIC PERFORMANCE OF EUS-FNA GUIDED TISSUE ACQUISITION VERSUS EUS-FNB FOR SOLID PANCREATIC MASS WITHOUT ROSE: RETROSPECTIVE STUDY

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Introduction: Endoscopic ultrasound-guided (EUS) tissue acquisition is an established diagnostic procedure for solid pancreatic mass. However, the diagnostic yield between fine needle aspiration (FNA) and fine needle biopsy (FNB) remains unclear. The aim of the study was to evaluate and compare the diagnostic yields between FNA and FNB using puncture by conventional needle aspiration (PAA) and Franseen-tip needles of the same size, 22G, in patients retrospectively. All procedures were performed without a cytologist on site. The diagnostic yield and average number of needle passes between FNB and FNA were then compared.

Aims & Methods: All cases of EUS-PAA by FNA or FNB for solid pancreatic mass between January 2017 and October 2020 in a tertiary care hospital were retrospectively reviewed. All procedures were performed without a cytologist on site. The diagnostic yield and the average number of needle passes between FNB and FNA were then compared.

Results: 91 patients (FNA n = 45; BNF, n = 46) with solid pancreatic mass detected by cross-sectional imaging underwent EUS-PAA. The average age was 62.3 ± 12.8 years, 55 (60.4%) were female. Age, sex, mass location, tumor size, and disease stage on imaging studies were not significantly different between the two groups. Diagnostic yield was higher in EUS-FNB (96%) than in EUS-FNA (91%). The average number of needle passes was clearly lower in FNB than in FNA (2.8 vs. 3.8, $p < 0.001$). The total procedure time was shorter in BNF (24 min) than in FNAC (31 min). The rate of adverse events between FNB and FNA was not significantly different.

Conclusion: The diagnostic yield of solid pancreatic mass was higher in FNB with a Franseen-tip needle than in FNA with a conventional needle in a center where ROSE is not available, without serious adverse events. Additionally, FNB had fewer needle passes and a shorter total procedure time.

Disclosure: Nothing to disclose.

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PP1179

THE FINAL RESULTS OF PANCREATIC FIBROSIS NON-INVASIVE ASSESSMENT BY MULTIDETECTOR COMPUTED TOMOGRAPHY

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Introduction: Pancreatic fibrosis is a part of the pathogenesis in most pancreatic disorders and plays a crucial role in chronic pancreatitis development. Therefore, pancreatic fibrosis early diagnosis gives an invaluable opportunity to be a part of the strategy for preventing disease progression. Thus, it makes non-invasive pancreatic fibrosis evaluation techniques research perspective.

Aims & Methods: Aim was to clarify relationships between pancreatic fibrosis grade and results of multidetector computed tomography (MDCT) post-processing parameters. Methods. Results of contrast-enhanced 128-row MDCT from 148 patients (mean age 57.4±13.8) who underwent surgery from April 2022 to October 2023 due to pancreatic tumors and advanced chronic pancreatitis were analyzed.

We calculated contrast enhancement ratio (CER) between venous phase and non-contrast MDCT, and normalized CER (NCER) during the pancreatic phase and venous phase, also we measured pancreas density during MDCT examination. Pancreatic fibrosis morphological grade was assessed according to Klöppel & Maillet's scoring system [1] and scaled into three grades: mild, moderate and severe pancreatic fibrosis.

Results: We received a significant (p=0.007) growth of CER in patients with severe pancreatic fibrosis (mean CER=1.12±0.6) compared with CER in patients with mild pancreatic fibrosis (mean CER=0.79±0.3). Unenhanced pancreas density (UPD) value was significantly (p=0.001) lower in patients with severe pancreatic fibrosis (31.5±11.8 HU) than in patients with mild pancreatic fibrosis (39.8±8.5 HU). Mean CER and UPD value were significantly (p=0.001) different in patients with morphological signs of inflammation (1.13±0.7 and 33.5±11.4, respectively), than in patients without these signs (0.79±0.3 and 39.1±8.7, respectively). CER and UPD value showed a statistically significant correlation with morphologically inflammation signs presence (Spearmen's rho 0.28 and -0.28, p=0.001, respectively) and pancreatic fibrosis grade (Spearmen's rho 0.26, p=0.001 and -0.32, p=0.0001).

There were no significant influences of pancreatic fibrosis on NCER values in pancreatic phase and venous phase, which we had shown in publications of our preliminary results [2]. We suppose the main reason of this fact is study groups enlargement as well as measurement technique improvement, led to results clarification.

Conclusion: MDCT pancreatic tissue enhancement indicator, as CER, significantly increased in patients with severe pancreatic fibrosis as well as in those who had morphological signs of inflammation. UPD value can also provide additional information due to its decrease as pancreatic fibrosis progresses. Consequently, it could be useful for noninvasive pancreatic fibrosis diagnosis and prognosis.

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PP1180

MINIMAL-INVASIVE METHODS CONTROL PAIN WITH FEWER SIDE EFFECTS COMPARED TO OPIOID TREATMENT IN PATIENTS WITH UNRESECTABLE PANCREATIC CANCER – SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Pancreatic cancer (PC) is the fourth-leading cause of cancer-related death, with over 80% of patients having unresectable disease upon diagnosis. Despite opioid treatment, more than 70% of this population is experiencing chronic pain during the disease, along with a plethora of side effects. In the past decades, interventional radiological procedures have gained a significant role in pain management.

Aims & Methods: Our study aims to identify the most effective approach for managing pain in individuals with unresectable pancreatic cancer. We registered the study on PROSPERO (CRD42023477094) and systematically searched five major scientific databases on the 29th of October, 2023. Only randomised controlled trials (RCTs) reporting on adult patients diagnosed with unresectable PC and cancer-related pain undergoing medical or interventional treatment were considered eligible. The main intervention groups were endoscopic ultrasound-guided celiac plexus neurolysis (EUS-CPN), percutaneous celiac plexus neurolysis (P-CPN), high-intensity focused ultrasound (HIFU), and pharmacological therapy (opioids). Our

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outcomes of interest were pain, survival, and adverse events. Pooled means and proportions were calculated with 95% confidence intervals (CIs), and a random-effects model was applied.

Results: After title-abstract and full-text article selection, 1644 patients of 21 RCTs were found eligible. The pain was assessed using the 11-point Visual Analogue Scale from randomisation at 4, 8 and 12 weeks to determine its dynamic over time. Pain values at baseline were 6.09 (CI: 4.24-7.94) for EUS-CPN, 6.67 (CI: 5.20-8.14) for P-CPN, 6.57 (CI: 4.98-8.15) for HIFU, and 6.47 (CI: 5.02-7.93) for opioids. At one month, the overall pain score was 1.53 (CI: 0.23- 2.82) for EUS-CPN, 2.21 (CI: 1.43-2.91) for P-CPN and 2.68 (CI: 1.94-3.42) for opioids. At two and three months after randomisation, there was a slow but continuous increase in pain scores for all treatment groups. The median survival time in days was 126.22 (CI: 87.21-165.31) for pharmacological treatment and 86.71 (CI: 62.45-136.88) for P-CPN. Patients receiving only opioids developed more adverse events compared to P-CPN, the most common being nausea, vomiting, and constipation. EUS-CPN had the lowest rate of adverse events.

Conclusion: Our study shows that minimal-invasive pain treatments provide pain control with fewer adverse events associated, with EUS-CPN having the most promising results. However, further RCTs with higher sample sizes are needed to confirm our findings.

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Disclosure: Nothing to disclose.

PP1181

EFFICACY AND SAFETY OF EUS-GUIDED DRAINAGE IN POST-PANCREATECTOMY VS. POST-PANCREATITIS FLUID COLLECTIONS WITHOUT NECROSIS

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Introduction: The etiology and natural history of fluid collections complicate pancreatic resection differ from those associated with pancreatitis. Understanding whether one group benefits more from EUS-guided drainage could facilitate the development of condition-specific treatment protocols.

Aims & Methods: We aimed to compare the efficacy and safety of EUS-guided drainage of fluid collections following pancreatic resection and post-pancreatitis pseudocysts.

This study evaluates the outcomes of patients who underwent EUS-guided drainage for pancreatic fluid collection without necrosis complicating pancreatic resection (PR) at a high-volume tertiary university center from 2018 to 2022. These patients were compared to a matched cohort with post-pancreatitis (PP) fluid collections without solid components, using a 1:1 ratio for confounding factors.

In both conditions, treatment was standardized to involve the implantation of 10 French double-pigtail stents as the therapeutic modality. Primary outcomes included the duration of hospital stay and the total number of interventions required for treatment success. This encompassed the initial intervention, stent removals and replacements, and any additional procedures, such as managing complications during a 12-month follow-up.

Early clinical success of EUS-guided drainage was defined as:

1. a reduction of the collection to <50% within 3 months, as confirmed by follow-up imaging, and;
2. clinical improvement.

Late clinical success was determined by the complete resolution of the lesion within 12 months.

Secondary endpoints were the duration of ICU stay, mortality, and adverse events, defined according to the lexicon of the American Society of Gastrointestinal Endoscopy[1].

Results: In the final analysis, 54 patients (70.4% male, mean age 57.7 ± 13.9) were included, 27 in each group. The most common cause within the PP group was alcohol-induced pancreatitis (40.7%), followed by biliary (18.5%) and idiopathic (14.8%) etiology. In the PR- group most patients have had a pancreatic tail resection (55.6%), followed by pylorus-preserving pancreaticoduodenectomy (22.2%) and total or subtotal pancreatic resection (22.2%).

No significant differences were observed between the cohorts in relation to potential confounding variables, including age, gender, comorbidities, presence of fluid infection, number and type of stents utilized, time interval to initial intervention, and pre-drainage size of fluid collections. The number of reinterventions and clinical success after EUS-guided drainage were similar in both groups.

While the overall hospital stay was not significantly different between the groups, the PP patients showed a significantly longer need for ICU services compared to the post-surgery group (11.9 vs. 1.7 days, p=0.03). The complications rate tends to be higher in the post-pancreatitis group but did not reach significance due to the low number of events that occurred (29.6 vs. 7.4%, p=0.08).

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	Fluid collection		
Factors	Post- pancreatectomy	Post- pancreatitis	p-value
Age (years)	57.9 (52.8-63.2)	52.6 (48.1-57.1)	0.11
Male	20 (74.1%)	18 (66.7%)	0.77
BMI (kg/m2)	25.1 (22.5-27.7)	24.4 (21.9-26.9)	0.67
Comorbidities	16 (59%)	20 (74.1%)	0.37
Cardiac	3 (11%)	7 (25.9%)	0.29
Pulmonary	2 (7%)	7 (25.9%)	0.13
Size (mm)	73.7 (60.4- 86.9)	66.5 (54.5-78.6)	0.41
Infection	11 (40.7%)	11 (40.7%)	0.78
Number of Stents	1 (1-5)	1 (1-5)	0.98
Time to intervention (days)	30 (7-100)	24 (2-90)	0.47
Outcomes			
Re-Interventions	3.0 (2.3-3.7)	3.5 (2.5-4.4)	0.44
Hospital stay	24.1 (18.2-29.9)	30.6 (11.6-49.7)	0.47
ICU stay	1.7 (0.5-2.9)	11.9 (1.5-22.3)	0.03
Early outcome	22 (81.5%)	21 (77.8%)	0.99
Late outcome	27 (100%)	27 (100%)	
Complications	2 (7.4%)	8 (29.6%)	0.08
Mortality	1 (3.7%)	3 (11.1%)	0.60

Table.

Conclusion: Despite comparable clinical outcomes after EUS-guided drainage in both groups, patients with post-pancreatitis fluid collection were associated with greater ICU utilization and tended to experience higher complication rates at initial intervention.

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Disclosure: Nothing to disclose.

PP1182

OPTIMIZING PANCREATIC WON MANAGEMENT: THE QNI CLASSIFICATION APPROACH FOR ENDOSCOPIC DRAINAGE

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Introduction: Pancreatic walled-off necrosis (WON) presents a significant challenge in acute pancreatitis management, often necessitating endoscopic intervention for drainage and necrosectomy. However, the optimal selection of stent type for endoscopic drainage remains debatable. This study aimed to evaluate the utility of the QNI classification in guiding stent selection for WON drainage.

Aims & Methods: We conducted a retrospective analysis of patients who underwent endoscopic drainage and necrosectomy for symptomatic pancreatic WON at a tertiary referral center between January 2022 and September 2023.

Patients were divided into two groups based on the QNI classification:

Group 1 consisted of patients who had WON affecting 1 quadrant or 2 quadrants with a necrosis of less than 30%.

Group 2 consisted of patients whose WON had affected 3 quadrants, 2 quadrants with 30% necrosis, or 1 quadrant with more than 60% necrosis at the time of the index procedure.

A WON-based decision was made by the endoscopist regarding the use of LAMS or plastic stents. Necrosectomy was performed in cases of infection following WON drainage. Resolution was considered in clinically asymptomatic patients with WON < 30 mm at the 3-month imaging follow-up.

Results: Forty-eight patients were included in the study, with a mean age of 55.7 (±12.28) and a male-to-female ratio of 2.4:1, 20 patients in Group 1 and 28 patients in Group 2. Except for infection, which occurred more frequently in Group 2, the indications for drainage (pain, weight loss) were similar between the two groups (28% vs 5%, p=0.03). LAMS were used more frequently in group 2 than in group 1 (64% vs 40%, p=0.001). The mean number of endoscopic procedures in Group 1 was 1.7 and in Group 2 it was 3. The interval between endoscopic procedures in Group 1 and Group 2 was 20.3 days and 12.4 days, respectively (p=0.05). Clinical resolution at 3 months showed no significant difference between groups, with rates of 90% vs 85% (p=0.87), and the complication rates was 10% vs 14.2% (p=NS). Percutaneous drainage was done in 0% vs 7.14% cases, and surgery was required in 0% vs 7.1% of cases.

Conclusion: As a result of the newly proposed QNI classification system, patients can be guided in the selection of LAMS placement as well as for multidisciplinary management. Severe QNI scores may need more aggressive interventions, yet outcomes are comparable to those with milder disease.

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Disclosure: Nothing to disclose.

PP1183

EFFICACY AND SAFETY OF ENDOSCOPIC ULTRASOUND-GUIDED RADIOFREQUENCY ABLATION OF PANCREATIC TUMORS - INITIAL EXPERIENCE OF SINGLE CENTER

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Introduction: Surgery is the mainstay therapeutic strategy for pancreatic tumors. Although effective, the rates of peri-operative morbidity and mortality remain a concerning issue. This poses the need for developing less invasive, safe but effective techniques. Endoscopic ultrasound-guided radiofrequency ablation/EUS RFA/ is a novel tool for targeted local destruction, applicable for pancreatic neuroendocrine tumors /PanNEN/, locally advanced adenocarcinoma, and cystic pancreatic neoplasms.

Aims & Methods: To evaluate EUS RFA of pancreatic tumors in terms of efficacy and safety. Primary endpoint was clinical success rate defined as a complete response after the procedure and during follow-up. The secondary endpoint was the assessment of adverse events/AE/ associated with the procedure.

Methods: Retrospective analysis of a prospective database including all consecutive patients who were treated with EUS RFA for the period August 2022 – December 2023 in a single tertiary center. We evaluated electronic patient records and gathered data on patients' baseline characteristics, procedure characteristics, and outcomes. The follow-up of the group included imaging studies - computed tomography and/or contrast-enhanced EUS, done on the third and the sixth month post-procedure.

Results: Ten sessions of RFA in seven patients were performed. Male to female ratio was 4:3, with a mean age of 65.42 years. The procedures were performed with EUS–guided RFA 19 G Needle. The group included five patients with neuroendocrine tumors - two with non-functional pNEN –

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G1 grade of differentiation and three with insulinoma; one patient with metastasis from renal cell carcinoma, and one patient with pancreatic ductal adenocarcinoma. The mean size of the lesions was 23.5714 ±4.755 (±20.17%)mm. Technical success defined as successful puncture of the lesion and application of alternating current was achieved in all cases. Clinical success defined as the disappearance of lesions on imaging studies at the 3rd and 6th month was achieved in 85,71 % of the patients. In three of the patients, we performed a second session on the third month during follow-up due to signs of vital tissue. In the 6thmonth during follow-up in two of them, we found a complete response. There was no correlation between response rate and functional status of neuroendocrine tumors as well as no association of procedure outcome and location of lesions. In terms of efficacy, we found a negative correlation between lesion size and response rate. We observed the resolution of symptoms due to hormonal hypersecretion in the cases of insulinoma in the first 24 postprocedural hours with no events of hypoglycemia. No early or late adverse events were reported in the observed group.

Patient	Histology	Size of lesion	Number of sessions	Number of passes during the session	Response	Adverse events
1	NF PanNET ki-67-2% G1	43/27mm	2	4	Partial response after 2 sessions	None
2	NF PanNET ki-67-2% G1	22/15mm	1	3	Complete response after one session	None
3	FNET - insulinoma G1	20mm	2	4	Clinical response after one session;complete response after second session	None
4	Metastasis from renal cell carcinoma	18/15mm	2	3	complete response after two sessions	None
5	Locally advances adenocarcinoma	38mm	1	3	Complete response after one session	None
6	NET Insulinoma G1	17/13mm	1	3	complete response after one session	None
7	NET Insulinoma G1	7mm	1	2	Complete response after one session	None

Conclusion: EUS RFA is a highly effective and safe mini-invasive technique for treating pancreatic tumors, especially PanNENs.It could be offered to selected patients as an alternative to surgical treatment, providing targeted selective local tissue destruction. The procedure can be applied as an adjunctive tool in the palliative treatment of locally advanced ductal adenocarcinoma but more data are needed.

Disclosure: Nothing to disclose.

PP1184

ADVANCEMENTS IN ENDOSCOPIC PANCREATIC NECROSECTOMY: INSIGHTS FROM THE LAMS ERA

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Introduction: Walled-off pancreatic necrosis (WOPN) stands as a grave complication of acute pancreatitis, posing a substantial risk of sepsis-induced multiple organ failure when infected. Traditional surgical debridement, though effective, often entails elevated rates of adverse events and mortality. However, the emergence of endoscopic ultrasound (EUS)-guided cysto-gastrostomy employing LAMS presents a promising alternative for WOPN treatment, offering advantages in managing necrotic collections via endoscopic necrosectomy. This study aims to evaluate the technical and clinical success as primary objectives, alongside assessing adverse events and recurrence rates as secondary objectives.

Aims & Methods: We conducted a retrospective analysis of consecutive patients diagnosed with infected walled-off pancreatic necrosis (WOPN) and treated over a 48-month period from March 2020 to February 2024. The treatment protocol involved the placement of a novel LAMS – HOT AXIOS stent, utilizing an electrocautery-enhanced delivery system. This was followed by direct endoscopic necrosectomy (DEN), performed immediately post-stent deployment and repeated at intervals of 3 to 7 days until complete resolution was attained. All procedures were performed by expert clinicians under anesthesia with meticulous management of the necrotic cavity.

Results: 31 patients diagnosed with infected WOPN underwent EUS assessment for LAMS placement. 5 patients were excluded: 3 due to cavity distance from gastric wall and 2 due to large amount of solid necrotic tissue (>30%). 26 patients underwent LAMS placement with average age 34.5 years. Drainage indications included sepsis with infected collection(53.8%), pain(23%), gastric outlet obstruction(11.5%), and a combination of these (11.5%). For primary LAMS insertion,100% technical success rate was achieved. Technical success rates for complete endoscopic necrosectomy were 30.8% & 69.2% at the end of 7 days & 14 days respectively. In cases of failed endoscopic necrosectomy, persistently infected necrosis, clinical deterioration, or complications, a step-up approach was pursued, leading to ultimately a surgical intervention in 23% (6 patients). Complications during endoscopic necrosectomy included transient self-limiting bleeding in 6 patients (23%), aspiration pneumonitis in 3 patients (11.5%), and spurting bleed from necrotic bed necessitating intra-procedural clip placement in one patient. Average procedure duration was 55 minutes with an average of 3 sessions required per patient. Clinical success, defined as pure endoscopic management without the need for surgical or radiological intervention, was achieved in 11 patients (42.3%). 6 patients (23%) required angioembolization, 6 patients (23%) required surgical intervention for incomplete drainage/sepsis and 3 patients(11.5%) required percutaneous catheter drain placement. Average hospital stay was 19 days. Patients who underwent successful endoscopic necrosectomy with LAMS were subjected to MRCP to identify disconnected ducts before LAMS removal, which occurred after an average duration of 18 days. Following stent removal, 11 patients (42.3%) required double pigtail plastic stent (5 Fr x 3 cm) placement due to disconnected ducts. Patients were followed up for an average of 60 days with no recurrences observed during this period.

Conclusion: Adoption of specialized LAMS and subsequent direct endoscopic necrosectomy produced commendable results in managing infected WOPNs. This approach showcased remarkable technical success and favorable clinical outcomes when executed according to established protocols in expert hands.

Disclosure: Nothing to disclose.

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PP1185
THE ECONOMIC IMPACT OF USTEKINUMAB IN THE TREATMENT OF INFLAMMATORY BOWEL DISEASE: A TARGETED LITERATURE REVIEW

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Introduction: Ustekinumab, an interleukin (IL) 12/23 inhibitor is a biologic used to treat inflammatory bowel disease (IBD), including Crohn's disease (CD) and ulcerative colitis (UC). Clinical and real-world evidence has demonstrated its ability to provide a clinically meaningful response and remission among patients; however, the economic impact of its use may not be favourable from the payer perspective. The objective of this targeted literature review was to understand the economic impact of ustekinumab reference product in the treatment of patients with IBD.

Aims & Methods: A literature search was conducted in MEDLINE for economic publications that evaluate the economic impact of ustekinumab or include ustekinumab within their analyses. Search terms included "budget impact model", "budget impact analysis", "cost-utility", "cost-effect", "ulcerative colitis", "Crohn's disease" and "ustekinumab" from database inception until Feb 8th, 2024. Only studies published in English were eligible for inclusion.

Results: A total of 30 hits were screened for inclusion, yielding 12 relevant studies that discussed economic implications of ustekinumab. Five articles were conducted from the US payer perspective, two from the Japanese payer perspective, and one each from the Canadian, Swedish, Polish, French, and German perspectives. The economic models included budget impact models (n = 2), cost-effectiveness models (n = 8), and cost per responder/remitter analyses (n = 2). Five studies used a CD patient population, six studies focused on UC, and one study included a mixed population of CD and UC. In CD studies, two found ustekinumab was cost-effective: these studies were conducted from the perspective of Polish and Swedish payers in CD patients who failed anti-TNF therapy. Both studies used a Markov based model to assess cost-effectiveness of treatments. In the remaining CD studies, ustekinumab was associated with the highest cost per responder/remitter (Japanese payer perspective) and was dominated by other treatments in cost-effectiveness analyses (US payer perspective). The studies conducted in the UC patient population also found that, although clinically effective, ustekinumab was dominated (based on the incremental cost effectiveness ratio [ICER]) by other biologics included in the analyses (eg, vedolizumab and infliximab). Similar to the CD patient population, the cost per responder/remitter was highest for ustekinumab compared to five other therapies including tofacitinib and golimumab (US payer perspective). However, in two UC studies, ustekinumab was observed to be more effective than the most cost-effective treatment option (from the Japanese and US payer perspectives), highlighting that its price negatively affected its ICER. Moreover, cost-effectiveness analyses conducted by the German and Canadian perspectives, demonstrated that ustekinumab was not a cost-effective therapy at the current willingness to pay threshold (0% probability). Note, that no studies tested the possible cost-effectiveness of an ustekinumab biosimilar which would positively influence the cost associated with ustekinumab treatment.

Conclusion: This literature review demonstrated that, despite its high efficacy, ustekinumab is among the highest costing biologics available for the treatment of IBD with low probabilities of cost-effectiveness across several studies.

As the economic burden for IBD is increasing globally, cost-saving therapeutic options such as biosimilars may help to provide savings that can be redistributed in a healthcare system or provide funds to expand patient access.

Disclosure: Kunal Shastri is an employee of Fresenius Kabi a manufacturer of biosimilars. Kerise Clarke, Margaret Ainslie-Garcia, Nicole Ferko are employees of EVERSANA a company who receives consulting fees from Fresenius Kabi.

PP1186
WHY DO MANY PEOPLE WITH DISORDERS OF GUT-BRAIN INTERACTION (DGBI) NOT SEEK MEDICAL CARE? ANSWERS FROM A NEW ASSESSMENT INSTRUMENT

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Introduction: A large proportion of people with DGBI in the general population, including irritable bowel syndrome (IBS) and functional dyspepsia (FD), do not seek healthcare (HC) for their condition, but their reasons are poorly understood. We aimed to develop a questionnaire to assess different reasons an individual may not consult a doctor for current gastrointestinal (GI) symptoms.

Aims & Methods: The Reasons for Not Seeking Healthcare (RNSH) questionnaire was created from content derived from 8 focus groups with 23 DGBI patients in the United States and Australia and guided by author experience. Participants rated how important each of 21 items is in causing them not to see a doctor for current (within the last 3 months) GI symptoms on a 4-point Likert scale (Not at all, A Little, Moderately, Very).

Adults in 9 countries (Canada, China, Germany, Japan, Mexico, Poland, Romania, Spain, United States) completed an online survey including the RNSH, the Rome IV Diagnostic questions for IBS and FD, and questions about medical history, HC utilization, and demographics.

Respondents who experienced gastroduodenal or bowel symptoms in the past 3 months but had not sought care for those symptoms were included in Exploratory Principal Component Factor Analysis (PCFA) with Promax

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rotation evaluating the structure and potential sub-scales of RNSH. Items for each factor were summed and transformed into equal 0-100 sub-scales. Reliability of the RNSH was also assessed.

Results: A total of 17,663 adults completed the online survey and 12,666 (80%) of those had GI symptoms in the last 3 months but had not sought care for them: 51% female; mean age 45.5 years (SD=16.3); 5.7% had Rome IV IBS and 10.7% had FD. Of these, 73.7% had never sought care for digestive problems. The RNSH questionnaire demonstrated very good internal consistency (Cronbach α= 0.88) and split-half reliability (Guttman λ^2= 0.89). PCFA yielded 3 factors; 4 items were removed due to poor loading. Reasons for not seeking care include: "negative HC expectations", "prefer non-medical alternatives", and "prefer a wait-and-see approach". The 3 factors collectively accounted for 48% of the variance (Adjusted R^2) in RNSH score. The strongest factor keeping people from seeking HC for GI symptoms (highest standardized mean subscale score) was adopting a "wait and see" approach (Table).

People meeting Rome IV criteria for IBS or FD were markedly more likely to avoid seeking care due to negative HC expectations than those who did not meet Rome criteria (p<.001) with a medium effect size (d=0.48) and to prefer using non-medical alternatives (p<.001) with a small effect size (d=0.29). No differences existed between the two groups for preferring a wait-and-see approach (p=.087).

				Mean Adjusted Score [95% CI]		
Subscale	Items	Eigenvalue	Variance	All (N=12,666)	No IBS or FD (N=11,038)	IBS or FD (N=1,628)
Negative HC Expectations	8	6.46	19.6%	29.97 [29.6-30.4]	28.61 [28.2-29.0]	39.21 [38.1-40.3]
Prefer Non-medical Alternatives	5	2.21	14.7%	27.84 [27.4-28.2]	27.00 [26.6-27.4]	33.51 [32.4-34.6]
Wait and See Approach	4	1.41	13.6%	51.26 [50.7-51.6]	50.99 [50.5-51.5]	52.26 [51.1-53.4]

Table. Factor Structure with Mean Adjusted Scores for the Three Sub-Scales of the RNSH.

Conclusion: This study identified, and showed the relative importance of, three distinct factors that keep people from seeing a doctor for their current digestive symptoms. Specifically, people with IBS or FD may avoid seeking HC based on negative expectations for such care or prefer to use alternative methods such as diet modification or stress reduction techniques. The RNSH may be useful for identifying and helping to overcome specific barriers to timely DGBI treatment that may lead to poorer outcomes.

Disclosure: Nothing to disclose.

PP1187

HEALTHCARE RESOURCE UTILIZATION AND COSTS IN CROHN'S DISEASE AND ULCERATIVE COLITIS PATIENTS INITIATING USTEKINUMAB TREATMENT VIA JANSSEN'S BIOADVANCE® PROGRAM

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Introduction: There is limited research on HealthCare Resource Utilization (HCRU) and costs among patients receiving ustekinumab (UST) treatment. This study aims to evaluate HCRU and costs in patients with Crohn's Disease (CD) and Ulcerative Colitis (UC) who received UST treatment through Janssen's BIOADVANCE® patient support program (PSP) in Ontario, Canada.

Aims & Methods: The primary objective of this study is to compare annual HCRU and costs (excluding medication expenses) during a period of one year before initiating UST treatment and up to four years post-initiation.

This retrospective observational study analyzed data from a consecutive cohort of Ontario CD and UC patients who initiated UST treatment through Janssen's BIOADVANCE® PSP.

Patients were indexed based on their first recorded UST prescription fill, spanning from January 1, 2017, to July 31, 2021, for CD patients, and from February 1, 2020, to July 31, 2021, for UC patients.

Quasi-identifiers (date of birth, gender, postal code, indication) were used for probabilistic linkage of patients in BIOADVANCE® PSP to those in Ontario Crohn's and Colitis Cohort (OCCC) from the Institute for Clinical Evaluative Sciences (ICES). A follow-up period for up to four years post-index date was used to assess all-cause and IBD-specific HCRU and costs. A 12-month lookback period was utilized to establish baseline demographic and clinical profiles and baseline HCRU and costs.

A mixed gamma regression model was employed to compare costs before and after UST initiation. IBD-specific HCRU and costs included general practitioner visits, specialist visits, emergency department visits, inpatient hospitalization, and same-day surgeries. IBD-specific HCRU and costs were identified based on the IBD-specific diagnosis code as the most responsible diagnosis.

Results: A probabilistic linkage of 88.1% for CD cases and UC cases was achieved. A total of 2,279 patients with CD and 626 patients with UC were included. UST was the initial biologic therapy for 35.72% and 22.68% of patients with CD and UC, respectively. The average time since diagnosis was 10.72 years for CD and 8.07 years for UC. The mean age was 45.6 ± 17.2 years for UC patients and 45.7 ± 16.2 years for CD patients. UC showed a male-female ratio of 56% to 44%, whereas for CD, it was 46% to 54%.

IBD-specific HCRU and costs significantly decreased post-UST initiation in Years 1, 2, and 3 for both CD patients and UC patients, compared to baseline (Table).

IBD Specific Costs ($ CAD) Mean +/- (SD)		Year 1	p-value	Year 2	p-value	Year 3	p-value
Crohn's Disease	(n)	n=2,279		n=1,659		n=1,057	
	Baseline (BL)	2,754 ± 6,025	0.0057	2,851 ± 6,361	<0.0001	2,861 ± 6,428	<0.0001
	Follow-up (FU)	2,206 ± 7,550		1,644 ± 5,300		1,417 ± 5,194	
Ulcerative Colitis	(n)	n=626		n=317		n=49	
	Baseline (BL)	3,373 ± 6,745	0.0203	3,910 ± 8,051	<0.0001	6,277 ± 11,499	0.0037
	Follow-up (FU)	2,535 ± 7,865		1,491 ± 4,808		1,335 ± 4,121	

Table.

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Specifically in Year 2, inpatient hospitalization cost was reduced by 43.7% for CD (BL $1,642 vs. FU $925, p= 0.0001) and 67.3% for UC (BL $2,315 vs. FU $756, p= 0.0011). Total physician visit cost was reduced by 33.6% for CD (BL $728 vs. FL $483, p<0.0001) and 45.8% for UC (BL $858 vs. FL $465, p= <0.0001).

Conclusion: Initiation of UST through the BIOADVANCE® PSP was related to substantial reductions in IBD-specific HCRU and costs among patients with CD and UC in Ontario, Canada.

These findings underscore the potential benefits of UST as an advanced therapy in IBD treatment, in managing CD and UC, and support its role in optimizing patient outcomes while reducing the burden on healthcare resources.

References: Data source:

1. Janssen BIOADVANCE® Database (January 1, 2016 – March 31, 2023) and Ontario Crohn's and Colitis Cohort (OCCC) data in Institute for Clinical Evaluative Sciences (ICES).

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PP1188

TELEREGULATION IN GASTROENTEROLOGY AND THE BOTTLENECK OF SPECIALIZED HEALTH CARE

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Introduction: The increase in population and the presence of bottlenecks in specialized health care access demonstrate the importance of developing measures for better management. Telemedicine was widely established during the COVID-19 pandemic, and in this context the implementation of teleregulation aims to bring greater resoluteness to health systems.

Aims & Methods: The main objective of the present study is to evaluate the results of teleregulation in gastroenterology in the referrals in a large Brazilian city.

The secondary objectives were to evaluate agreement between the primary care physician in requesting exams or specialist referral and evaluate the number of incorrect referrals and patients who should have been prioritized, to suggest points for continuing medical education.

Methodology: This is a retrospective cross-sectional study of primary health care teleregulations requests for gastroenterology carried out in the city of Curitiba-PR-Brazil by the Gastroenterology team at the Clinical Hospital Complex of the Federal University of Paraná. Teleregulation data were collected between the months of October 2022 and June 2023 in patients over 18 years old. Patient demographic data was collected, as well as the reason for requesting and its outcome. Requests for reassessment were excluded.

Results: Of the 3000 teleregulations screened in the period, 71.1% were included, of which 68.17% were from female patients, with an average age of 54.32 +/- 16.19 years. The reasons for referral and the outcome are shown in table 1.

Incorrect requests were 6.61%, and the main reasons were acute conditions, exams already requested and patient awaiting appointment, asymptomatic patients, out of criteria for screening colonoscopy, incomplete information and without sufficient basic tests for specialized clinical reasoning. Among all patients, 14.95% of cases were prioritized.

Reasons for referral	n 2133 (%)	Outcome	n (%)
Requesting exams	1368 (64,13)	Gastroenterologists agreed	1212 (88,6)
		Gastroenterologists did not agree	156 (11,40)
Discuss conduct	568 (26,63)	Patients maintained in primary health care	130 (22,89)
		Additional information requested	47 (8,27)
		Additional exams requested	304 (53,52)
		Referred to specialist	87 (15,32)
Request a referral to a specialist	197 (9,24)	Referred to specialist	129 (65,48)
		Patients maintained in primary health care	68 (34,52)

Table 1. Reasons for referrals and outcomes.

Conclusion: The present study highlights that terregulation represents an important tool in health management, being able to bring resolution in 89.9% of cases.

Disclosure: Nothing to disclose.

PP1189

WITHDRAWN

PP1190

A COST ANALYSIS OF PROPOFOL SEDATION FOR OUTPATIENT ENDOSCOPY IN THE ITALIAN HEALTHCARE SYSTEM

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Introduction: In recent years, propofol has become the sedative of choice for gastrointestinal endoscopy. Despite large cohort studies demonstrating its safety and effectiveness, the adoption of Non-Anesthesiologist Administration of Propofol (NAAP) in clinical practice still faces numerous challenges, both in Europe and globally. Consequently, propofol-based sedation remains largely in the hands of anesthesia professionals.

Aims & Methods: This study was conducted in two phases. Initially, a preliminary analysis was carried out to compare adverse event (AE) rates between NAAP and Anesthesiologist Administration of Propofol (AAP) at a tertiary academic center (San Raffaele Hospital, Milan, Italy). In the second phase, potential cost savings from implementing NAAP over AAP at the hospital, local, and national levels were examined.

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Propensity score matching (PSM), based on age, sex, body mass index (BMI), and ASA Physical Status Classification System score, was used to compare the two cohorts. A 3-year budget impact model (BIM) was implemented based on the annual volume of outpatient endoscopy procedures performed at San Raffaele Hospital and estimates of the Italian adult population by Italian National Institute of Statistics (ISTAT).

Results: From May 2019 to October 2021, 2,721 patients who underwent esophagogastroduodenoscopy (EGD) (2,439 NAAP and 282 AAP) and 2,748 who underwent colonoscopy (2,491 NAAP and 257 AAP) in outpatient setting were enrolled.

Only diagnostic procedures were included. After PSM, the EGD cohort included 282 patients under NAAP and 188 under AAP, while the colonoscopy cohort consisted of 256 patients under NAAP and 174 under AAP.

No significant differences in AE rates were observed between the groups in both the EGD (overall: NAAP 1.1% vs AAP 0.8%, p=0.815; after PSM: NAAP 0.4% vs. AAP 1.0%, p=0.452) and colonoscopy cohorts (overall: NAAP 1.7% vs AAP 3.5%, p=0.202; after PSM: NAAP 3.5% vs. AAP 0.6%, p=0.249). All AEs were managed without leading to major or long-term sequelae.

According to the BIM, a complete implementation of NAAP instead of AAP for patients classified as ASA 1 and 2 would lead to financial savings of €851,649 at our hospital, €8.3 million for the Local Health Authority of Milan, €19.8 million for the Lombardy region, and €124.7 million for the Italian National Health System over the triennium 2023-2025.

Conclusion: NAAP is a safe sedation modality for outpatient gastrointestinal endoscopy, with AE rates comparable to those of AAP. Its widespread implementation, at the expense of AAP, could result in savings exceeding €120 million for the Italian National Health System over the triennium 2023-2025.

References: NAAP is a safe sedation modality with AE rates comparable to AAP. The widespread implementation of NAAP, rather than reliance on anesthesiologist-led propofol sedation, could result in savings exceeding €120 million for the Italian National Health System over the triennium 2023-2025.

Disclosure: Nothing to disclose.

PP1191

PRE- AND DURING COVID-19 ANALYSIS OF EARLY AND LATE-ONSET COLORECTAL CANCER CASES AMONG AFRICAN AMERICANS: DOES RACE MATTER?

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Introduction: The incidence of early-onset colorectal cancer (EOCRC, <45 years) is on the rise among adults, including African Americans (AA).

Aims & Methods: Aim: To investigate differences between EOCRC and late-onset colorectal cancer (LOCRC, >45 years) in AA patients and understand the trend patterns in the last decade, particularly during the COVID era. Methods: We conducted a retrospective review of patient records at Howard University Hospital from 2011 to 2022. Our analysis included demographics, clinical presentations, pathology reports, and colonoscopy records. A three-year interval analyses were performed, with a specific focus on the COVID-19 period (2020-2022) and compared it with preceding years to discern temporal trends. Our cohort consisted of 23 EOCRC individuals, and 192 LOCRC.

Results: The majority (80%) of our cohort consisted of AA. A high proportion of males were diagnosed with colorectal cancer (CRC) during the COVID-19 period (2020-2022), with rates of 87.5% in EOCRC versus 75% in LOCRC, in contrast to a nearly equal gender distribution in preceding years. The incidence of CRC fluctuated over the study period, with a higher observed incidence between 2011-2013 that gradually declined until 2019, followed by a notable spike during the COVID era (2020-2022), particularly in EOCRC patients probably due to an increase in smoking, alcohol consumption, and obesity in both LOCRC and EOCRC groups.

The predominant initial symptom for EOCRC patients persisted as abdominal pain. However, our temporal analysis uncovered a noteworthy rise in the incidence of altered bowel habits (75% vs. 50%) and hematochezia (75% vs. 52%), especially in the recent COVID era in EOCRC patients compared to LOCRC patients. Importantly, these shifts were noted to be the primary indications for colonoscopy among EOCRC patients (p=0.044). In the COVID era, EOCRC patients exhibited a predominant left-side occurrence (p=0.040), unlike the balanced distribution in 2017-2019. Overall, LOCRC patients mostly had cancers on their right side. Analysis of cancer stage revealed a significant shift in EOCRC patients presenting in stage 4 during 2011-2016 to stage 2 in the COVID era (p=0.042), suggesting increased awareness or a more symptomatic presentation that led to early diagnostic evaluation. There was no statistically significant difference in cancer type between the groups, with adenocarcinoma being the most common. due to decreased screening.

Conclusion: In our study, we found a notable increase in CRC during the COVID-19 era, especially among young AA males. Shifts in lifestyle and clinical presentations highlight the necessity for addressing the healthcare strategies in this group. EOCRC patients showed left-sided cancer predominance and a shift from stage 4 to stage 2, suggesting potential improvements in awareness and early detection.

Disclosure: None.

PP1192

UNVEILING COLON CLEANSING QUALITY: A FACE-OFF BETWEEN ARTIFICIAL INTELLIGENCE AND CONVENTIONAL METHODS USING BOWL IMAGE FEATURES

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Introduction: The quality of colonic preparation remains a critical benchmark in ensuring the efficacy of colonoscopy procedures (1). Despite concerted efforts and elaborate protocols, inadequate preparation persists in up to 20% of colonoscopies (2).

To address this challenge, there is a need for innovative solutions to assess bowel preparation providing more reliable and standardized results before the procedure.

Aims & Methods: The aim of this study is to assess and compare the predictive accuracy of models predicated on artificial neural network frameworks with those established via traditional logistic regression analysis, to predict the efficacy of the bowel preparation using bowl images characteristics. This prospective study included 512 bowl images from 100 patients taken after the second dose of a split-dose preparation (using high volume while adhering to a three day low fiber diet), between April 2023 and October 2023 in Hotel Dieu de France, a tertiary hospital center in Beirut. The Boston bowel preparation scale (BBPS) was divided into two cat-

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egories: Adequate (≥ 8) and Inadequate (< 8). Manual cropping techniques were employed to refine accuracy and extract images' features using snipping tool. Then, we specified the corresponding mean scores of the image in RGB (Red, Green, Blue) and HSV (Hue, Saturation, Value). Two statistical analysis strategies were considered: the artificial neural network analysis, and the logistic regression analysis.

Results: The artificial neural network for predicting the BBPS from the bowl image characteristics showed an AUROC= 0.797, and an accuracy of 90.4% for the training group (forming 68.2% of the sample), and 92.8% for the testing group (representing 31.8% of the sample). For the traditional statistical analysis, a threshold was defined for each image feature using the area under the ROC curve (AUC). Each feature was categorized into four quartiles. The binary logistic regression showed a significant correlation between blue (threshold= 67) and green colors (threshold= 98), the saturation (threshold= 0.63064), the first quartile of the hue (threshold ≤ 0.073), and the last quartile of the value (threshold ≥ 0.527). Then, a binary logistic regression formula was established: -1.23 * First quartile of the hue (threshold ≤0.073) - 1.7 * Last quartile of the Value (threshold ≥ 0.527) - 1.321 * Saturation (threshold= 0.63064) + 0.688 * Blue (threshold= 67) + 1.08 * Green (threshold=98) + 2.932. The AUROC of the new formula was equal to 0.787 and revealed a threshold= 1.55. The classification based on the new formula (BBPS as dependent factor) showed an accuracy of 88.1%.

Conclusion: The characteristics of the bowl images serve as predictive indicators of colonic preparation quality. A model utilizing artificial neural network architecture yields higher accuracy in prediction than one derived from conventional statistical methods. This method may allow the adjustment of the preparation strategy before undergoing colonoscopy to improve the outcome of the procedure.

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Disclosure: Nothing to disclose.

PP1193

ENDOSCOPISTS' KNOWLEDGE, PERCEPTIONS, AND ATTITUDE TOWARDS THE USE OF ARTIFICIAL INTELLIGENCE IN ENDOSCOPY - A SYSTEMATIC REVIEW

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Introduction: Artificial Intelligence (AI) is rapidly evolving in the field of gastrointestinal (GI) endoscopy. However, the prevailing perceptions and attitudes of endoscopists towards AI remain poorly understood.

Aims & Methods: This review aims to provide a summary on the current views and perspectives covering the incorporation of AI into GI endoscopy, as well as to identify any obstacles hindering its adoption. Electronic databases including MEDLINE, PubMed, and Scopus were searched from inception to 25th March 2024 to identify qualitative studies, such as questionnaires or surveys, conducted on endoscopists regarding the use of AI in GI endoscopy. The Joanna Briggs Institute (JBI) critical appraisal checklist was used to assess study quality. Qualitative synthesis of included studies was conducted by employing thematic analysis classified into the following five domains:

1) Knowledge and attitudes towards AI,
2) Perception on the ability of AI to improve accuracy and quality of endoscopy,
3) Impact of AI on endoscopists,
4) Impact of AI on patients and;
5) Barriers towards the implementation of AI.

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and was registered with the International Platform of Registered Systematic Review and Meta-analysis Protocols (Registration ID: INPALSY202430064).

Results: Ten studies were included in this systematic review comprising of 1587 endoscopists across Europe (32.6%), North America (42.8%), and Asia (24.6%). Domain 1: Most endoscopists (69–100%) reported having basic knowledge of AI. Furthermore, the majority (79.5–87.5%) expressed interest and optimism regarding the use of AI in endoscopy. Domain 2: Most respondents (62.5–97%) supported the notion that AI would positively impact endoscopic performance and quality. Domain 3: There were mixed perceptions regarding the extent of operator dependence on AI (6.2–62.8% agreement), and whether AI would prolong procedural time (21–81.3% in agreement).

Most endoscopists (71–100%) disagreed that AI would replace them at work. Domain 4: A substantial proportion of respondents (81.3%) felt that AI would lead to an increase in the quality of patient care. Only a minority of endoscopists (6.2–24%) felt that AI could improve the patient-physician relationship and provide empathetic care. Domain 5: A significant proportion of endoscopists (75.2–91%) identified costs as a potential barrier to the implementation of AI in endoscopy. Opinions on ethical concerns and the lack of regulation varied (12.5–100% and 35–88%, respectively), with many endoscopists feeling that there was need for clear guidelines and regulations when integrating AI into clinical practice.

Included Studies	Country	Sample size	Domains covered:	1) Awareness and knowledge	2) Improving endoscopic performance	3) Impacts on Endoscopists	4) Impacts on patients	5) Barriers to implementation
Wadhwa et al. (2020)	**USA**	**n = 124**		**X**	**X**	**X**	**-**	**X**
Kochhar et al. (2021)	USA	n = 165		X	X	-	X	X
Leenhardt et al. (2021)	UK	n = 333		X	X	X	X	X
Van der Zander et al. (2022)	Netherlands	n = 80		X	X	X	X	X
Tian et al. (2022)	China	n = 210		X	X	X	X	X
Tham et al. (2022)	Singapore	n = 16		X	X	X	X	X
Kader et al. (2022)	UK	n = 104		X	X	-	-	X
Nehme et al. (2023)	USA	n = 16		X	X	X	-	X
Goh et al. (2023)	Singapore	n = 165		X	X	X	X	X
Leggett et al. (2024)	USA	n = 374		X	X	X	X	X

Table 1. Study characteristics of included studies

Conclusion: Overall, there is a positive attitude towards the use of AI in endoscopy. However, concerns regarding impact on clinical practice, costs and medicolegal considerations remain. The establishment of robust regulatory frameworks is imperative to ensure successful integration of AI into clinical practice.

Disclosure: Nothing to disclose.

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PP1194

A PRE-TRAINING MODEL FOR INTESTINAL METAPLASIA RECOGNITION IN THE GASTRIC CORPUS: PRELIMINARY DATA ANALYSIS

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Introduction: Gastric cancer (GC) is a significant healthcare concern and the recognition of high-risk patients is crucial. The common mechanism for the progression of GC is the Correa cascade, which outlines a multi-step progression from chronic gastritis through atrophic gastritis, intestinal metaplasia (IM), dysplasia and finally GC. The detection of IM in the stomach is not always easy, especially when it is focal. While electronic chromoendoscopy has demonstrated high diagnostic accuracy for IM, its limited availability and the variable expertise of endoscopists pose challenges. Additionally, the potential of artificial intelligence (AI) to aid endoscopists has obtained significant attention.

Aims & Methods: This study aims to develop a deep-learning system to assist in IM diagnosis based on endoscopic image patches. A retrospective dataset of endoscopic images from gastroscopies conducted at Sant'Andrea Hospital, La Sapienza University of Rome between January 2020 and December 2023 was collected. All the gastroscopies were performed using both White Light Imaging (WLI) and Blue Light Imaging (BLI) assessment and targeted biopsies or biopsies according to the updated Sydney system protocol were conducted. Only BLI-assessed gastric corpus images were analyzed for the preliminary machine learning model development data. We tested the most common CNN architecture for image classification (VGG, ResNet, Inception). We used a dataset of coarsely segmented images to extract patches and we cast the problem to a classification task over patches. From each segmented image we extracted 20 patches (with possible overlap). We trained the model to classify patches (positive vs negative). Then we used a voting scheme over patches to classify an entire image.

Results: Best results obtained with ResNet50 architecture without Pre-training. The dataset included 1384 high-resolution corpus BLI images from 300 patients with 721 images classified as GIM-negative and 663 as GIM-positive. Then we obtained a dataset of 10 569 patches that we split into training-set (8264 patches), validation-set (1202 patches), and test-set (1103 patches). On the patches test-set we obtained a specificity of 76%, and a sensitivity of 72%. The classification of the entire image was obtained through a learnable voting scheme, in which each patch expressed its vote (positive or negative). We used the validation set to optimize the following hyperparameters: the decision threshold and the number of positive patches that made the entire image classified as positive. With the best configuration we obtained a specificity of 70% and a sensitivity of 100%, but of course we could use a different voting scheme to trade specificity with sensitivity at need.

Conclusion: The study addresses common challenges in medical scenarios, such as limited data availability and the need to work with a specific dataset, without access to pre-trained models, the difficulty of storing images and the poor quality of routine endoscopic images. Despite constraints, the proposed approach delivers promising results suitable for preliminary data screening, showing a high specificity and sensitivity. Furthermore, using the voting scheme, the best configuration showed a sensitivity of 100% that could permit to aid the endoscopist in the diagnosis of gastric intestinal metaplasia with a zero percentage of false negatives.

Disclosure: Nothing to disclose.

PP1195

USABILITY OF A SMARTPHONE APPLICATION TO SUPPORT PERIOPERATIVE MANAGEMENT OF ENDOSCOPIC SUBMUCOSAL DISSECTION FOR EARLY GASTRIC NEOPLASM

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Introduction: Endoscopic submucosal dissection (ESD) for early gastric neoplasm in Japan is generally performed in the hospitalization to deal with perioperative complications. Although early detection of patients' symptoms is possible during hospitalization, it is difficult to perceive the symptoms during the period of home care after discharge from the hospital.

To solve this problem, we developed a smartphone application (app) to support perioperative management of ESD for early gastric neoplasm that collects data on the patient's own condition and shares this data with medical institutions.

Aims & Methods: The aim of this study was to evaluate the usability of the app during the perioperative period of ESD for early gastric neoplasm. This study was a single center prospective observational study. Patients who own their smartphone and underwent ESD for early gastric neoplasm at our hospital from August 2023 to February 2024 were included.

Exclusion criteria were as follows;

i) patients with severe dementia,

ii) patients who use implantable medical electrical devices such as pacemakers.

Patient downloaded the app to their smartphones prior to ESD and entered their vital signs and symptoms according to the medical questionnaire (including the daily defecation state and the contents of the meals consumed, and these images were also uploaded as needed by the patients) via the app in the morning and at night each day from the day before ESD to the day one month after ESD.

One week after discharge, patients had an online medical examination with a physician using the app. The app included an informational website with ESD procedures, preparation for hospitalization, and precautions for life after discharge, and the patient could check this website at any time.

The primary endpoint was the usage the app that was calculated by: the number of times the patient answered the medical questionnaire / total number of times the questionnaire was delivered. The secondary endpoint was the patients' usability of the app based on a questionnaire survey.

Results: Fifteen patients were enrolled during the period. The median age was 65 years (range; 46-75 years), male / female = 9 / 6. The tumor location was upper / middle / lower = 7 / 6 / 2, and the median tumor size was 14mm (range; 4-75 mm). The median hospital stay was 7 days (range; 6-9 days).

For the primary endpoint, the usage rate of the app was 88.8% (90.9% in the morning and 86.7% at night). All patients had no severe adverse events after discharge. For the usability through the questionnaire survey, all patients found it easy to input their dairy data, and 13 patients (86.7%) answered that daily input into the app has improved their own health awareness. Regarding the online medical examination, 14 patients (93.3%) reported that they had no problem conducting the online connection, and 12 patients (80.0%) reported that they felt more relieved by

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talking to a physician. Fourteen patients (93.3%) indicated that the informational website helped them to be more attentive to their daily life.
Conclusion: Using the app during the perioperative period of ESD for early gastric neoplasm, it was possible to collect data of patients simply. It was expected that the app for perioperative management will improve patients' health awareness and be useful to understanding the status of ESD perioperative patients after discharge from the hospital.
Disclosure: Nothing to disclose.

PP1196

EFFICACY OF A STOOL STATE CHECK APPLICATION USING ARTIFICIAL INTELLIGENCE DURING BOWEL PREPARATION BEFORE COLONOSCOPY: A MULTICENTER PROSPECTIVE STUDY (SCAN STUDY)

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Introduction: In Japan, the stool state during bowel preparation (BP) is one of indicators for starting a total colonoscopy (TCS). For one reason, we have the concept that the clearer the stool state during BP for TCS is, the less residue there is in the colon. During BP, the patients or medical staffs assess patients' stool state using a visual scale. However, the visual scale lacks objectivity and is judged differently by those who assess it. In addition, checking the stool may increase their stress and workload. To address these issues, we developed a smartphone application (app) allows an artificial intelligence (AI)-based evaluation of the patient's stool state during BP and the viewer to enable the real-time sharing of patients' stool images and the app's evaluation with medical staffs.
Aims & Methods: The aim of this study was to assess the feasibility of performing TCS with appropriate quality using the app and the viewer. This study was a prospective observational study conducted from November 2022 to December 2023 involving patients scheduled for TCS at 10 domestic facilities in Japan. Inclusion criteria were as follows; i) patient age between 20 and 70 years old, ii) patient who own their smartphone. Exclusion criteria were as follows; i) patient with known entire or sub-entire circumferential colorectal cancer, ii) patient after colorectal resection, iii) patient with inflammatory bowel disease, and iv) patient with severe dementia. The procedures of BP followed the usual practice of each facility. The patients downloaded the app to their smartphone and captured their stool during BP, while medical staffs reviewed the app's evaluation either through the app or the viewer. The app displayed the evaluation results as follows; 1 star: solid to muddy stools, 2 stars: cloudy watery stools, 3 stars: clear watery stools. TCS was performed when the app displayed 3 stars. The primary endpoint was the proportion of patients with Boston Bowel Preparation Score (BBPS) ≥6. BBPS was evaluated by the endoscopist who performed the TCS. Secondary endpoint included adenoma detection rate (ADR), cecal intubation rate (CIR), withdrawal time for negative TCS, and average BBPS. Additionally, we evaluated patient's satisfaction with the app and medical staff's workload burden with the app and the viewer usage via questionnaire surveys. We set the threshold primary endpoint of 90% and the expected value of 95% based on European guidelines. The required sample size was 316, using the binomial distribution with a one-sided alpha of 2.5% and power of 90%. To account for ineligible patients, the target number of patients was set at 340.
Results: A total of 343 patients enrolled in the study. Finally, 326 patients were included in the analysis, comprising 188 males and 138 females, with a median age of 57 years. The primary endpoint was 99.1% (95%CI: 97.3-99.8%). ADR and CIR were 46.9% and 99.7%, respectively. The median withdrawal time for negative TCS was 9 minutes and 45 seconds. The average BBPS was 8.5. In the questionnaire survey, 295 patients (90.5%) indicated they would like to use the app next TCS. Moreover, 67 medical staffs answered the questionnaire and 60 out of 67 (89.6%) reported a reduction in workload burden.
Conclusion: The proportion of patients with a BBPS ≥6 in the TCS using the app exceeded the set expected value in addition to the threshold value. The app and the viewer are expected to provide TCS with enough quality and also allow real-time monitoring of BP progress, leading to a reduction in burden on both patients and medical staffs.
Disclosure: Nothing to disclose.

PP1197

REDUCING OUTPATIENT VISITS IN FIT-BASED CRC SCREENING PROGRAMS - FEASIBILITY OF A DIGITAL INTAKE TOOL

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Introduction: Currently, all screenees of the Dutch colorectal cancer (CRC) screening program with a positive faecal immunochemical test (FIT) undergo face-to-face counselling at an outpatient clinic. In order to reduce the burden for screenees and healthcare systems, we have developed a Digital Intake Tool (DIT) for the Dutch CRC screening population. A web-application that facilitates the entire pre-colonoscopy counselling process digitally, eliminating the need for a physician. In this study we evaluated whether the DIT could replace in-person counselling.
Aims & Methods: A prospective non-inferiority trial was conducted in seven centres, including academic and regional hospitals, and endoscopy centres. All participants received DIT-counselling. The DIT guided partic-

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ipants through the counselling process, which included spoken animated videos for patient education, a medical questionnaire for health assessment, and the collection of pre-procedural informed consent. For screenees in who no additional information was required, colonoscopy was scheduled without an in-person visit.

The primary outcome was the proportion of participants achieving adequate bowel preparation, defined as a Boston Bowel reparation Scale (BBPS) score of ≥ 6. BBPS scores were compared to patients that received in-person counselling. To assess whether the DIT could replace the 'human touch' of in-person counselling, we evaluated patient-reported outcomes related to anxiety, psychological distress and satisfaction as secondary outcomes.

Results: Between October 2021 and September 2023, 1000 (58.8%) out of 1690 approached patients were included. Median age was 64.0 years with 45% being female. DIT-counselling achieved a BBPS of ≥ 6 in 96.5% of the participants, compared to 97.6% after in-person counselling, demonstrating non-inferiority (-1.1%, 95% CI -2.64 – 0.444, non-inferiority margin -8%). A significant reduction in outpatient visits was established, with only 3.4% requiring face-to-face consultations, and 40.9% of the participants could be scheduled for colonoscopy solely via DIT-counselling. After digital counselling, a significant decrease in distress was reported on a 11-point Likert scale (Z = -6.136 p<0.001). Also anxiety levels, measured by the STAI-6, decreased (Z = -3.210 p =0.001). Nearly all participants (98.9%), were satisfied with the information provided, mean score of 8.5 (SD 1.0) and 88.9% would choose the DIT again, while 5.3% preferred face-to-face counselling.

Conclusion: This study showed that digital counselling can replace in-person counselling for FIT-positive screenees of the CRC screening program. An adequate bowel preparation was achieved in >96% of the participants, meeting the 90% requirement of the CRC screening program. A substantial reduction in in-person outpatient visits was established, and patients reported highly positive experiences after digital counselling. Implementing this home-based approach will decrease the burden for patients and healthcare systems. Patients can receive care at the comfort of their own home at their own time, reducing travel and carbon emissions, while healthcare systems increase their outpatient capacity.

Disclosure: This study received a PPP-allowance provided by Health-Holland, Top Sector Life Sciences & Health, to stimulate public-private partnerships.

PP1198

DOMAIN-SPECIFIC DATA AUGMENTATION IMPROVES ROBUSTNESS OF DEEP LEARNING SYSTEMS IN ENDOSCOPY

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Introduction: Artificial intelligence (AI) systems in endoscopy are predominantly developed in expert centers using standardized, high-quality imaging. However, performance of these systems may decline due to heterogeneous imaging conditions found in daily clinical practice. A primary source of this variability is the diversity in post-processing enhancement settings used across different endoscopy units. These enhancement settings may improve image quality parameters such as color, contrast and sharpness. While often subtle to the human eye, these alterations can significantly impact the accuracy of AI predictions. A potential solution is to simulate the heterogeneity of these enhancement settings during the AI systems training phase using domain-specific data augmentation methods.

Aims & Methods: We aimed to evaluate the negative impact of post-processing enhancement settings on AI performance and explore domain-specific data augmentation as a method to mitigate performance loss.

We investigated two commonly used AI systems in endoscopy: a computer-aided detection (CADe) system for Barrett's neoplasia[1] and a computer-aided diagnosis (CADx) system for the optical diagnosis of colorectal polyps[2]. First, both systems were trained using their original datasets, which comprised limited variability of enhancement settings. These CAD systems were then tested across a wide range of test sets. Each test set comprised the same images but were displayed with a different enhancement setting. This represented the image heterogeneity of daily clinical practice. Finally, both CAD systems were retrained using domain-specific data augmentation. This involved the use of a proprietary software tool to precisely simulate the variability of the complete spectrum of available enhancement settings during training. The array of test sets was then re-evaluated by these updated CAD systems.

Results: The CADe and CADx system trained using their original training set, displayed substantial performance variability across the different test sets. Depending on the test sets enhancement setting, the CADe system displayed sensitivity scores between 83% and 92%, a 9% difference. Specificity scores ranged between 84% and 91%, a 7% difference. Similarly, the CADx system reached differences of up to 7% (78-85%) for sensitivity and 18% (45-63%) for specificity. After retraining both CAD systems using domain-specific data augmentation, sensitivity and specificity variability was reduced to 2% (89-91%) and 1% (90-91%) for CADe and 2% (78-80%) and 8% (55-63%) for CADx.

Conclusion: The performance of current endoscopic AI systems can vary substantially depending on the post-processing enhancement setting of its input data. Domain-specific data augmentation can effectively reduce this variability to ensure consistent performance across different endoscopy units.

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Disclosure: Nothing to disclose.

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PP1199

INTRODUCING INNOVATIVE LOW-CODE/NO-CODE APPLICATIONS IN ENDOSCOPY: AN ERCP AND EUS APP FOR STREAMLINING TRIAGE, ENHANCING SAFETY, AND BUILDING REGIONAL CONNECTIVITY FOR A HIGH VOLUME, LARGE TERTIARY REFERRAL HOSPITAL

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Introduction: Low-code/no-code (LCNC) application development remains underutilised as a clinician-driven solution for daily hospital tasks. Existing paper-based systems used for triage and booking of endoscopy referrals are inefficient, prone to documentation errors, and contribute to paper wastage.

Aims & Methods: Our objective was to create an online referral pathway and real-time tracking board for Endoscopic Retrograde Cholangiopancreatography (ERCP) and Endoscopic Ultrasound (EUS) to replace the current paper-based system at a large tertiary referral hospital.

Using Microsoft's Power Platform™, PowerApps™, and Power Automate™, gastroenterology clinicians developed an online referral system, bookings and management board, and automated notifications centre for inpatient and outpatient ERCP and EUS referrals. After 2 months of usability testing, the system went live in May 2023, results were monitored to April 2024.

Results: 1,392 requests for ERCP and EUS were monitored using the online clinician-built board, 762 (55%) originating from the new online referral pathway, 405 (29%) were manually uploaded from existing paper requests and 225 (16%) re-bookings facilitated by a re-request feature in the app. Of the total requests, 1030 (74%) were tracked to completion of the procedure, with 219 pending procedures and 103 awaiting future surveillance, while 32 were declined and 8 on hold. Of the online referrals, 170 triggered high suspicion of cancer notifications, with 90% prioritised to first procedure dates within 2 weeks or less. Online referrals additionally had 100% documentation of patient co-morbidities, and anti-platelet / anti-coagulant history through compulsory user input for safer triage, this compared to 44% of the original paper-based requests. On average 187 automated email and push notifications per month informed referrers of confirmed booking dates and delayed or completed procedures, as patients moved through the board. 21 of 165 referring doctors responded to a survey representing six referring hospitals, 100% found the online referral system very or somewhat easy to fill out, 95% found the automated messages helpful, and one third booked a follow up clinic after receiving a completed notification. 100% of responders recommending continuing with the system.

Conclusion: Clinician-built LCNC applications are underutilised as solutions to common tasks in healthcare. Our ERCP and EUS referral pathway and patient tracking board has enhanced clinical decision making and efficiency of triage in a high-volume large referral hospital, while providing co-ordinated input and feedback to referrers and other endoscopy staff. The boards real-time interface provides a full patient journey overview and has additional layers of safety checks, essential for these interventional procedures. The online referral pathway board and app were created using standard business Microsoft 365™ licencing and is available on Microsoft Teams mobile or desktop for accessible and secure use and can be integrated into other hospital systems.

Disclosure: Nil

PP1200

THE USE-AI STUDY: A MULTICENTER, SINGLE BLIND PROOF-OF-CONCEPT STUDY ON THE USE OF AI IN GASTROINTESTINAL ENDOSCOPY

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Introduction: The use of artificial intelligence (AI) systems for decision support has recently gained great popularity in the medicine.[1] However, recent studies have shown possible limitations emerging from the interaction between humans and AI. Among these, the term "automation bias (AB)" refers to human errors deriving from over-relying on AI's false alarms such that the human changes a correct initial interpretation.[2] AB is poorly investigated in the medical field and was never tested in the gastroenterological setting.

Aims & Methods: The aim of the study was to investigate the impact of AI on human diagnostic performance in gastrointestinal endoscopy, including AB.

Ninety high-quality endoscopic videoclips (16 seconds) extracted from small bowel (SB) capsule endoscopy (N=60) and proctosigmoidoscopy (N=30) were edited with the addition of a mock AI system designed with diagnostic accuracy, sensitivity and specificity of 80%, 86% and 75%. The ground truth was settled on the agreement of 9 established experts about the presence of SB lesions with significant hemorrhagic potential according to the Saurin classification (30 videoclips), inflammatory lesions relevant for the diagnosis of SB Crohn's disease according to the Lewis score (30 videoclips) and assessing the Ulcerative Colitis (UC) Endoscopic Index of Severity (30 videos). The videoclips were randomly divided into three sessions and submitted around Europe to fully trained endoscopist, trainees and endoscopic technicians. Participants were instructed to the accuracy measures of AI annotations and scored each videoclip first without, then with AI support. Participants' performances with and without AI were calculated. Subgroup analyses were made after stratification by participants' experience level in each endoscopic setting and test performances.

Results: A total of 12.660 videoclips (both with and without AI) were visualized in 422 sessions (182, 118 and 122 participants completed the first, second and third session, respectively). A strong and significant difference between the respondents' performance (N=281) when they were unaided by AI (average accuracy (M)=0.78, [0.77, 0.79] and when they were supported by the AI (M=0.81, [0.80, 0.82], paired t-test, t(276)=11.2, p<0.001) was found. Size effect and the evidence in favour of a significant positive effect of AI were large, 0.67 (Cohen's D) and very large (Bayes Factor>100), respectively. Interestingly, no significant difference between trainees and specialists both in regard to their performance, either unaided, M=0.77 [0.74, 0.79] vs M=0.78, [0.77, 0.79] or aided M=0.80 [0.78, 0.82] vs M=0.81 [0.80, 0.82] and their accuracy improvement (M=0.03, [0.02, 0.05] vs M=0.03, [0.02, 0.04] (unaided t-value=-0.89, p=0.37; aided t-value=-0.69,

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p=0.49, difference t-value=0.54, p=0.59) were found. A strong leveller effect emerged, with low-performers improving their accuracy much more than top-performers (average improvement 5%, [0.04, 0.06] vs 1%, [0.00, 0.02], paired T test, t=6.53, p<0.001). Nine percent of the final decisions involved disregarding correct AI advice in favour of incorrect human judgment, while AB occurred in 2% of cases.

Conclusion: In this study, the application of AI to assess SB bleeding, SB inflammatory lesions, and UC activity brings significant improvement in diagnostic accuracy, with the benefit appearing to be greater in the low-performers. However, in rare cases, AI may induce diagnostic errors due to AB. Future studies directly evaluating the impact of AI in real-life clinical settings are needed to confirm these data.

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Disclosure: Nothing to disclose.

PP1201

A NEXT STEP FOR AI DRIVEN PANENDOSCOPY - DEVELOPMENT OF CONVOLUTIONAL NEURAL NETWORK FOR EVALUATION OF SMALL BOWEL AND COLON CLEANSING DURING DEVICE ASSISTED ENTEROSCOPY

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Introduction: Device-assisted enteroscopy (DAE) offers comprehensive examination of the gastrointestinal tract, but its efficacy is dependent on adequate bowel preparation. While prior research explored AI models for colonoscopy cleansing, there is a lack of studies evaluating both small bowel and colon cleansing during DAE.

Aims & Methods: The aim of this study was to develop and validate a convolutional neural network (CNN) capable of automatically classifying bowel cleansing levels during DAE, encompassing both small bowel and colon regions. We retrospectively analyzed 28 procedures (single balloon, double-balloon and motorized spiral enteroscopy) from a high-volume center. Bowel preparation was graded as excellent (≥90% visible mucosa), satisfactory (50-90%), or unsatisfactory (<50%). A dataset of 88,623 images (training: 90%, testing: 10%) was used, covering both small bowel and colon areas. The CNN's performance was evaluated against expert consensus using sensitivity, specificity, accuracy, and AUC-ROC.

Results: We included 51948 frames that were considered excellent, 28820 frames that were considered satisfactory and 7855 frames that were considered unsatisfactory. The CNN demonstrated the following performance metrics: excellent cleansing (sensitivity: 97.8%, specificity: 80.3%, accuracy: 90.6%, AUC-ROC: 0.95), satisfactory cleansing (sensitivity: 81.8%, specificity: 97.9%, accuracy: 92.7%, AUC-ROC: 0.95), and unsatisfactory cleansing (sensitivity: 68.7%, specificity: 99.5%, accuracy: 96.8%, AUC-ROC: 0.96).

Conclusion: This study presents the first globally developed CNN for accurate classification of bowel cleansing throughout DAE, encompassing both small bowel and colon. This advancement represents a necessary step for optimizing diagnostic accuracy and therapeutic efficacy in gastrointestinal procedures, namely DAE, underscoring the pivotal role of innovative technologies in advancing medical practice

Disclosure: Nothing to disclose.

PP1202

PERFORMANCE OF LARGE LANGUAGE MODELS IN INTERPRETATION AND SUMMARIZATION OF ENDOSCOPY REPORTS: A COMPARATIVE ANALYSIS

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Introduction: Large language models (LLMs) offer new opportunities for the transformation of healthcare. One potential application of these models is to serve as consulting tools for patients.

Aims & Methods: The study aimed to assess the performance of different LLMs for the interpretation and summarization of endoscopy reports. This study was conducted in March 2024, and involved three mainstream LLMs (Claude 3 opus, Gemini pro, and GPT-4) which were tasked with interpreting and summarizing six fictitious endoscopy reports based on common clinical scenarios. Three gastroenterologists used Likert scales to blindly score the content generated by the LLMs, evaluating their accuracy, completeness, comprehensibility, or conciseness. Inter-rater reliability analysis and correlation analysis were performed on the scoring dimensions. The interpretation of the reports was also evaluated for readability and safety.

Results: The acceptable proportions for the interpretations in terms of accuracy, completeness, and comprehensibility were 100%, 94.4%, and 100%. For the impressions, the acceptable proportions for accuracy, completeness, and conciseness were 94.4%, 72.2%, and 100%, respectively. Interpretations from Claude 3 opus ($p<0.001$) and GPT-4 ($p<0.0001$) were significantly more legible than fictitious reports, while interpretations from Gemini pro showed no differences. For score based on Likert scales, the overall mean interpretation score for Claude 3 opus ($p=0.025$) and GPT-4 ($p=0.027$) was significantly higher than that from Gemini pro. Regarding impressions, Claude 3 opus ($p=0.011$) and GPT-4 ($p=0.009$) also achieved significantly higher overall mean score than Gemini pro.

Conclusion: LLMs excel in interpreting and summarizing endoscopy reports, but inter-model differences exist. Further optimization and development of specialized LLMs for endoscopic text processing applications may be warranted.

Disclosure: Nothing to disclose.

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PP1203

FAST-TRACKING ERCP LEARNING WITH THE BOŠKOSKI-COSTAMAGNA TRAINER: RESULTS OF A MULTICENTER RANDOMIZED CLINICAL TRIAL

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Introduction: Increased interest in ERCP simulator-based education is driven by the need for efficient and safe endoscopy training in a dedicated environment.

This study assessed whether a structured training program including initial training with the Boškoski-Costamagna ERCP Trainer (BCT) offers benefits to conventional training.

Aims & Methods: This was a one-year prospective, multicenter, parallel-arm, randomized controlled trial involving 16 novice ERCP trainees randomly assigned to study (SG) and control (CG) groups. Both underwent hands-on ERCP training. Additionally, SG had simulator-based training during the first three months. Performance was evaluated with TEESAT score in all ERCPs with any degree of trainee involvement until each trainee completed a minimum of 60 ERCPs. The primary outcome was overall trainee competence rate; secondary outcomes included successful biliary cannulation and adverse event (AE) rates. Learning curves were generated cumulatively over the training period. Both outcomes and the analysis of specific ERCP steps predictive of overall competence were assessed using mixed-effect logistic regression models (MELR). The most effective predictive model of overall competence was identified using a stepwise backward procedure. MELR was employed to investigate potential confounding factors.

Results: 1,106 ERCPs were included (SG=562,CG=544). SG trainees demonstrated higher success in native biliary cannulation (SG=52%, CG=42%, p<0.001), achieved faster biliary cannulation [SG=3(6)min, CG=5(8)min, p<0.001], and showed a faster evolution on overall performance, native biliary cannulation, and sphincterotomy. Successful biliary cannulation [OR=10.10, 95% CI (5.15-19.83)] and sphincterotomy [OR=7.71, 95% CI (3.77-15.76)] were the most important predictors for overall competence. The multivariate random-effects model demonstrated an almost significant association between simulator-based training [OR=3.24, 95% CI (1.01-10.39), p=0.049] and overall competence, accounting for biliary cannulation, sphincterotomy, and stone removal. No confounding factors were identified.

Conclusion: This study suggests that early simulator training with BCT accelerates the learning curve of trainees in performing key steps of the ERCP procedure and improves overall competence as assessed by dedicated tools during ERCP training.

Disclosure: Nothing to disclose.

PP1204

AN EFFECTIVE STRATEGY FOR THE IMPLEMENTATION OF SCREENING FOR HEREDITARY COLORECTAL CANCER IN CLINICAL PRACTICE. THE ROMANIAN EXPERIENCE

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Introduction: Early recognition and risk-targeted surveillance of patients with hereditary colorectal cancer (HCRC) and familial CRC (FCRC) reduce CRC incidence, improving in turn the effectiveness of population-based organised screening programmes. However, HCRC/FCRC often remain under-recognized and suboptimally managed and opportunities for CRC prevention are often missed. In 2020, pilot screening programmes for CRC have been started in Romania. In this context, a screening initiative for high-risk groups was started at national level, as an adjunct to population-based screening.

Here, we present our experience in setting up an interdisciplinary team as a first step in implementing a screening programme for HCRC/ FCRC with coverage across the country.

Aims & Methods: We used a stepwise approach: first, an evidence-based educational intervention was provided in order to create a national network of clinical specialists; next, implementation of this knowledge in clinical practice and setting-up of a HCRC registry. Between Nov 2022-Jan 2023, we trained clinicians from university hospitals and cancer centers covering the main administrative regions of the country, using an evidence-based online curriculum (weekly, 12 weeks, in total +/-22 hours) for achieving mastery standards (excellence for all) on the recognition and management of HCRC/FCRC. The curriculum was created by an interna-

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tional panel of GI experts and education experts. The training consisted of pre-class preparation (online study of clinical guidelines, instructional videos), virtual classroom (live active learning session with case-based quizzes and targeted feedback) and post-class tutoring sessions (case-based quizzes with instructor/ peer feedback). Pretest and posttest(s) scores were measured. The main outcomes were the proportion of participants reaching the minimum passing standard (MPS) and the learning effect size. Wilcoxon signed-rank test was used to compare the differences between test scores and Cohen's d coefficients for effect size. Secondary outcome was the course satisfaction.

Results: We enrolled 20 gastroenterologists and 3 non-gastroenterologists (median 37.8 (28-58) years, 61% female, from academic (82.6%) and cancer (17.4%) centers. None received previous training in HCRC/FCRC. At baseline, none of the participants reached the MPS. At the final post-test, all participants reached/ exceeded the MPS. Median test scores significantly increased from pretest to initial posttest and final posttest: 55 (45-62)% vs 84 (82-86)% vs 91 (87-97)%, (P<0.0001). The educational sessions produced a very large learning effect size compared to pretest scores (Cohen's d=3.91, 95%CI=2.33-5.50). All participants expressed very high satisfaction about course participation. Immediately after the course, they implemented the acquired knowledge into practice, set up local interdisciplinary teams and continued training of their peers/ students. A 2-step approach to genetic testing was adopted in clinical practice: (universal) IHC/ MSI and in case of a dMMR tumor, targeted germline testing.

Conclusion: We launched a national screening initiative for HCRC/ FCRC as an adjunct to population-based CRC screening. Evidence-based education to achieve mastery standards on HCRC recognition was the first critical step in creating an expert working group to coordinate the performance of clinicogenetic diagnosis and targeted risk-reducing interventions. Efforts are currently underway for achieving full rollout of the training program across the country and ultimately set-up a hereditary cancer registry.

Disclosure: Nothing to disclose.

PP1205

A COMPREHENSIVE STUDY ON GASTROENTEROLOGISTS' BELIEFS, ATTITUDES, AND KNOWLEDGE OF PHYSICAL ACTIVITY AND EXERCISE

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Introduction: Physical inactivity is currently recognized as a global epidemic that necessitates collective action, and affects approximately 33% of the global population (1,2).

Physical inactivity is known to be a significant risk factor for several gastrointestinal diseases such as colon cancer, metabolic dysfunction-associated steatotic liver disease, and inflammatory bowel diseases (3-5).

Aims & Methods: The aim of this study was to determine the beliefs, attitudes, and knowledge levels of gastroenterologists on Physical Activity (PA), and exercise. A survey was created by researchers to assess the beliefs, attitudes, and knowledge levels of gastroenterologists regarding PA, exercise, and healthy nutrition. Exploratory Factor Analysis (EFA) was conducted to simplify the survey interpretation by reducing the number of items into sub-dimensions and assessing construct validity. The internal consistency of each factor was assessed using Cronbach's alpha. Latent class analysis was conducted to categorize gastroenterologists' responses to items, identify positive and negative attitudes classes towards each factor.

Results: The study involved 143 gastroenterologists who were residents (30%), clinicians (29%), or academicians (41%). A majority of the gastroenterologists (78%) thought they had knowledge about physical activity and exercise. Following the provided brief information, this rate increased to 93%. Based on the participants' responses, EFA identified five factors in the questionnaire: i. effectiveness of PA and exercise; ii. necessity or harm of PA and exercise; iii. barriers to recommend exercise; iv. success in behavior change in patients; v. tendency to consult a physiotherapist or dietician. The responses to the five factors determined by EFA were categorized into two classes as positive and negative attitudes by latent class analysis (Table 1).The items showed high or satisfactory levels of internal consistency overall and for specific factors, ranging between 0.559 and 0.842. Although, 46% of the participants believed in the effectiveness of PA and exercise, 24% of them believed that PA was unnecessary or harmful. Three out of four gastroenterologists reported difficulty in prescribing exercise for patients with comorbidities. Additionally, they reported a rate of 81% for consulting their patients with a physiotherapist or a dietitian and a rate of 88% for achieving successful modification of patients' diet or exercise program. However, the reported rates for regularly or consistently seeking advice from a dietitian and physiotherapist were significantly different (respectively 36% and 76%; p<0.001).

Factors	Positive attitudes N (%)	Negative attitudes N (%)
Factor 1 Effectiveness of PA and exercise	66 (46)	77 (54)
Factor 2 Necessity or harm of PA and exercise	34 (24)	109 (76)
Factor 3 Barriers to recommend exercise	109 (76)	34 (24)
Factor 4 Success in behavior change in patients	126 (88)	17 (12)
Factor 5 Tendency to consult a physiotherapist or dietician	116 (81)	27 (19)

Conclusion: The gastroenterologists have an awareness on their lack of knowledge about lifestyle modifications, and also are willing to consult with physiotherapists, especially for the patients with comorbidities. Participation of gastroenterologists in training seminars on PA counseling and strengthening multidisciplinary collaboration may contribute to the improvement of health services.

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PP1206

EXPLORING CONTRIBUTORY FACTORS IN ENDOSCOPY-RELATED PATIENT SAFETY INCIDENTS INVOLVING DEATH AND SEVERE HARM OUTCOMES: A CONTENT ANALYSIS OF PATIENT SAFETY INCIDENT REPORTS FROM ENGLAND AND WALES USING A HUMAN FACTORS FRAMEWORK

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Introduction: Patient safety incidents (PSIs) occur across healthcare and evaluating these events presents an opportunity to learn, improve work systems and prevent future harm. Patient harm related to the care of patients due to undergo endoscopy, during endoscopy or during post-endoscopy care are devastating events calling for careful systems-based analyses to explore contributory factors to improve safety. The Systems Engineering Initiative for Patient Safety (SEIPS) model provides a framework for analysing PSIs to identify contributory factors (C.Fs) across the wider healthcare system, and has been used to analyse incidents and improve patient safety across a range of settings, although it has not been used previously for incidents related to endoscopy care (1,2). A large cohort of endoscopy PSIs across England and Wales were analysed using SEIPS, focussing on cases with patient fatality or severe harm outcomes.

Aims & Methods: This study aims to identify C.Fs to endoscopy-related PSIs involving death and severe harm outcomes, reported to a national incident database across England and Wales.

PSI reports involving endoscopy-related care between January 2017 and January 2022 were obtained from the National Reporting and Learning System (NRLS) and analysed using content analysis, enriched by the SEIPS framework. Descriptive data was also reported.

Results: Of 6,971 reported endoscopy PSIs, 125 involved death and 305 involved severe harm. After exclusion of unrelated PSIs,368 (5.3%) were analysed, spanning pre-procedure (126;34%), peri-procedure (104;28%) and post-procedure (138;38%) care. Follow-up Issues (115;31%) and known complications (109;30%) were the most frequent incident types. Content analysis was performed on 176 PSIs (48% of included PSIs with severe harm and death outcomes) with sufficient narrative data. Using SEIPS, C.Fs related to "People and Teams" were the most common (see table 1), identified 106 times (36%) across 77 (44%) incidents, with the most common subtype being 'Communication', identified 70 times across 57 incidents (32%). Organisational factors were identified 88 times (29%) across 78 (44%) incidents, with the most common being 'Resource Availability'. Task factors were identified 61 times (20%) across 52 (30%) incidents, with the most common being 'Documentation'. Technology and Tools factors were identified 23 times (8%) across 23 (13%) incidents, with the most common being 'Referral System Failures'. External environment factors were identified 13 times across 13 (4%) incidents, with the most common being 'Pandemic Disruption'. Physical Environment factors were identified 8 times (3%) across 7 (4%) incidents, with the most common being 'Out-of-hours Environment'.

SEIPS Contributory factors	References across all incidents, n (%)	Incidents, n (%)
People and Teams	106 (36%)	77 (44%)
Organisation	88 (29%)	78 (44%)
Task factors	61 (20%)	52 (30%)
Technology and tools	23 (8%)	23 (13%)
External environment	13 (4%)	13 (4%)
Physical environment	8 (2%)	7 (4%)

Conclusion: This study highlights the importance of applying a systems approach to understand the complex interplay of C.Fs in endoscopy-related PSIs resulting in severe harm or death. The SEIPS framework revealed that factors related to people and teams, organization, and tasks were the most common contributors to these incidents. Addressing these factors through targeted interventions could potentially mitigate future incidents and improve patient safety in endoscopy services.

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PP1207

THE EFFICACY OF AN E-LEARNING SYSTEM TO IMPROVE PATHOLOGICAL DIAGNOSIS OF GASTRIC CANCER: GLOBAL RANDOMIZED CONTROLLED TRIAL

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Introduction: Even if early gastric cancer is detected endoscopically, if a pathological diagnosis of cancer is not made based on biopsy tissue, there is a risk that curative endoscopic resection will not be performed, and the cancer will be left untreated until it becomes an advanced cancer.

The intramucosal component of early gastric cancer collected by biopsy often shows no signs of invasion and has low atypia, making it often difficult to determine whether it is a neoplasia or a non-neoplasia.

To solve these problems, we developed an e-learning system that improves the pathological diagnosis of gastric biopsies and evaluated its effectiveness using an Internet connection.

Aims & Methods: The design of this study was a randomized controlled trial. All tests and e-learning were conducted using the Internet. After receiving the pre-test (60 questions on biopsy tissue photos), participants were randomly assigned to either an e-learning group (E-group) or a non-e-learning group (NE-group), and only members of the E-group had access to the e-learning system.

Two months later after the pre-test, both two groups received the post-test (the same 60 questions as the pre-test). The primary endpoint was the difference in the rate of improvement in test results between the two groups.

Results: 215 pathologists from 36 countries who were assessed for eligibility were included in this study, with 106 assigned to the E-group and 109 to the NE-group. Using the test score (88.3 points) of the top 30% (34/109) of the NE-group as the cutoff value, high-ranking participants (more than

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88.3 points) were 57 among 106 participants in the E-group. There were significantly more high-ranking participants in the E-group than in the NE-group (p=0.00048).

Conclusion: This global research demonstrated the efficacy of the developed e-learning system in improving the histopathological diagnosis of gastric cancer.

Disclosure: Nothing to disclose.

PP1208

SOCIAL MEDIA SENTIMENTS AMONG MEDICAL PROFESSIONALS: MELD 3.0 COMPARED TO MELD-NA ON TWITTER

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Introduction: Our study investigates healthcare professionals' sentiments, particularly among physicians, regarding the adoption of MELD 3.0 for assessing short-term survival in end-stage liver disease patients. The Model for End-Stage Liver Disease (MELD) has been a reliable measure for short-term survival. Its current version, MELDNa, includes crucial parameters such as international normalized ratio, serum bilirubin, creatinine, and sodium, vital for organ allocation in liver transplants. MELD 3.0, introduced in June 2023, enhances predictive accuracy by incorporating additional variables, including sex and serum albumin, and addresses gender disparities on the transplant waitlist.

Aims & Methods: We used the Twitter application programming interface (API), Hypertext preprocessor (PAP), and Response capture interface (RAI) to collect the response across Twitter. we conducted a Chi-Square Test for Independence to assess whether there was a significant association between the sentiment labels across the datasets. This test helped determine if the sentiments in one dataset were dependent on those in the other. Additionally, we calculated Cohen's Kappa, a measure of inter-rater agreement, to quantify the level of agreement between the datasets, considering the probability of random agreement. Lastly, we used trend analysis, visualizing sentiment distributions over time, to observe sentiment changes and patterns across different years.

Results: Between December 2019 and November 2023, we collected a total of 74,684 responses. Out of these, there were 220,980 positive, 11,710 negative, and 11,154 neutral responses for MELD 3.0. For MELD-NA, we observed 11,724 positive, 10,832 negative, and 7,164 neutral responses. Comparing MELD 3.0 to MELD-NA, we found that MELD 3.0 exhibited significantly more positive sentiments (Chi-Square: 29.68, $p < 0.01$). Additionally, there was a 15.85% sentiment agreement for MELD 3.0, with a Cohen Kappa coefficient of 0.249.

These results indicate that there is a statistically significant association between the sentiment labels in the two datasets, and there is relatively high agreement in sentiment for MELD 3.0.

Conclusion: In this study, we explored healthcare professionals' views on adopting MELD 3.0 for evaluating end-stage liver disease patients' short-term survival. MELD has been a reliable measure, with MELDNa incorporating essential parameters for organ allocation in liver transplants. MELD 3.0, introduced in June 2023, enhances accuracy and addresses gender disparities.

Using Twitter data, we found that MELD 3.0 garnered significantly more positive sentiments compared to MELD-NA. There was also a notable level of sentiment agreement, highlighting the potential acceptance and value of MELD 3.0 among healthcare professionals.

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Disclosure: Nothing to disclose.

PP1209

A GLIMPSE INTO TRAINING IN BASIC DIGESTIVE ENDOSCOPY: A NATIONAL CROSS-SECTIONAL E-SURVEY FROM THE ITALIAN ASSOCIATION OF YOUNG GASTROENTEROLOGISTS AND ENDOSCOPISTS

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Introduction: Gastrointestinal (GI) endoscopy demands complex cognitive and practical skills. Yet, the landscape of training in this field remains heterogeneous and often lacks standardisation, with an absence of data specific to training practices within Italy.

Aims & Methods: This study aims at elucidating the current practices and perceptions of basic GI endoscopy training among Italian trainees. A national cross-sectional e-survey was conducted between November 2023 and April 2024. Participants were recruited from the Italian Association of Young Gastroenterologists and Endoscopists (AGGEI). The e-survey included questions regarding demographics, training experiences, perceived proficiency levels, and opinions on current training programs and resources. Descriptive and inferential statistical analysis was finally performed to identify trends and disparities based on gender.

Results: A total of 108 (M:F 63:45, 0 non-binary) GI endoscopy trainees out of 788 (13.7%) AGGEI members responded to the e-survey. Among them, 63 (58.3%) were under 30 years old with a majority (73,67.6%) working in university hospitals, being all gastroenterology residents. 49(45.4%) and 41(38%) started endoscopy during their second or third year of residency, respectively. 44(40.7%) spent 30-40 hours per week in the endoscopy room. A significant portion (61,56.6%) lacked a designated endoscopy mentor, while those who had one reported daily meetings (29,55.8%). Regarding the consistency of teaching among different instructors or mentors, 50(46.3%) perceived variability, 39(36.1%) moderate consistency, and only 11(10.2%) very high consistency, though most respondents (44,40.7%) preferred a standardised approach. 96(88.9%) responders attended a GI endoscopy observership before performing procedures, with 48(44.4%) finding this beneficial. Colonoscopy practice typically began concurrently with esophagogastroduodenoscopies (EGDs), with over half starting at least 20 procedures per week for both EGDs and colonoscopies. Only 16(14.8%) completed a core curriculum in more than 75% of the procedures. On a 5-point scale, confidence in cognitive skills was rated 4 by 42(48.9%) and 5 by 15(13.9%), while confidence in motor skills was rated 4 by 37(34.3%) and 5 by 21(19.4%). Most programs did not include

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simulators (85,78.7%), though 27(39.7%) participants considered the simulation-based training helpful, particularly for beginners. Competence in endoscopy was primarily assessed based on trainers' subjective opinion (47,43.5%) or a minimum number of completed procedures (35,32.4%), with only 8(7.4%) using skill assessment tools. 45(41.7%) responders received training in performing urgent endoscopy procedures, while 26(24.1%) reported that advanced procedures were not included in their training.

Overall, the endoscopy training programs were rated as excellent by 19(17.6%), very good by 37(34.3%), and good by 46(42.6%), with only 6(5.6%) rating them as fair. Additionally, no statistically significant differences between males and females were observed for any of the variables considered.

Conclusion: This national e-survey highlights the heterogeneous scenario of basic GI endoscopy training among young gastroenterologists and endoscopists in Italy. The findings underscore the need for enhanced standardisation and structured training pathways to ensure consistent competency levels across trainees. Investment in simulation-based training and the development of standardised curricula may serve to address existing gaps and optimise the quality of endoscopic training programs.

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PP1210

EQUITY IN HEALTHCARE: EXPLORING RACIAL AND ECONOMIC DISPARITIES IN COLONOSCOPY PREPARATION

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Introduction: The impact of race, socio-economic status, and insurance on colonoscopy preparation is a critical aspect of understanding healthcare disparities. Colonoscopy, essential for colorectal cancer detection and prevention, demonstrates varied effectiveness across different demographics.

This study aims to explore how these factors influence the quality of colon prep, a vital step in the procedure. By examining the interplay between race, socioeconomic background, and insurance coverage, we can uncover potential disparities in patient outcomes. Such insights are crucial for healthcare providers and policymakers to ensure equitable access to effective screenings, thereby enhancing the success rates of colonoscopies in diverse populations.

Aims & Methods: Colonoscopy records from 2010-2023 were retrospectively collected. Data included colonoscopy report, Gender, Race, Poor Colon Prep quality, socioeconomic status, and Insurance status. Chi-squared tests were conducted to identify statistically significant associations between poor colon prep outcomes and categorical variables: race, socioeconomic status, and insurance status. A Logistic Regression analysis was carried out to assess the combined impact of multiple factors on poor colon prep. The results were interpreted based on p-values, coefficients, and confidence intervals obtained from statistical tests. The significance level was set at 0.05 for all tests.

Results: A total of 3330 who underwent colonoscopy between 2010-2023 were included in the study group, of which 2135 (64%) were female. The racial distributions were 807 (24.2%) Hispanics, 1665 (48%) white, 501 (15%) Blacks, 232 (6%) Asians, 129 (3%) Others. Race significantly impacts poor colon prep outcomes (Chi-sq. 40.78, p-value <0.01). Low Socioeconomic status is also significantly linked to poor prep (Chi-sq. ≈ 20.86, p-value ≈ 0.00011). Patients with no Insurance were statistically associated with poor colon prep (Chi-sq. ≈ 15.48, p-value ≈ 0.000083). Logistic regression, suggested insurance status as potentially influential (coefficient ≈ -0.2140, p-value ≈ 0.068). The model's pseudo-R-squared value was 0.003725.

These findings stress the importance of considering demographic and socio-economic factors in colonoscopy patient management, acknowledging study limitations like dataset scope and model explanatory power.

Conclusion: Our study on race and colonoscopy preparation reveals significant disparities in colon prep quality among racial groups influenced by socio-economic status and insurance coverage. The study finds race, low socio-economic status, and lack of insurance significantly affect colon prep outcomes. While acknowledging its limitations, this study emphasizes the need for healthcare professionals and policymakers to consider these demographic factors, aiming for equitable colorectal cancer screening and prevention.

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Disclosure: Nothing to disclose.

PP1211

EXAMINING THE INTERPLAY BETWEEN RACE, GENDER, AND GASTRIC INTESTINAL METAPLASIA IN CANCER DEVELOPMENT

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Introduction: The impact of race and gender on the development of gastric cancer among patients with gastric intestinal metaplasia is a complex and multifaceted issue. While gastric intestinal metaplasia is a precursor to gastric cancer, studies suggest that race and gender may play significant roles in influencing the progression of this condition to malignancy. Understanding the interplay between these factors is essential for developing targeted prevention and intervention strategies to reduce the burden of gastric cancer within diverse populations.

Aims & Methods: Our retrospective study, spanning 2009-2023, included all Gastric intestinal metaplasia patients. We collected data on comorbidities, Gastric cancer incidence, Gender, race insurance, and socioeconomic factors CPT and ICD codes. we employed logistic regression and Chi-square testing to analyze the impact of race and gender on the incidence of gastric cancer. Logistic regression was used to ascertain the statistical significance and effect size of race and gender on the likelihood of gastric cancer.

The logistic regression model's findings were further explored through a Chi-square test of independence, which evaluated the association between race and gastric cancer across different racial groups. The results were visually represented through a bar graph, illustrating the predicted probabilities of gastric cancer by race, enhanced with the model's overall p-value for the race variable.

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Results: A total of 378 patients diagnosed with gastric intestinal metaplasia were included in the study, of which 72 patients developed gastric cancer in 14 years. 38 (51.3%) patients were female. Among the races 24(33%) Asians, 16(21.6%) Hispanics, 14(19%) Whites, 20 (27.7%) African American. Logistic regression revealed that race has a statistically significant association with the incidence of gastric cancer ($p < 0.001$), with the odds of gastric cancer increased in Asians (Coefficient: 0.68, $p<0.05$). In contrast, gender did not show a significant impact ($p = 0.847$). Further examination using a Chi-square test confirmed a significant association between race and gastric cancer (Chi-square = 37.83, $p < 0.001$).

Conclusion: In conclusion, our study underscores the intricate relationship between race and the development of gastric cancer in patients with gastric intestinal metaplasia. The findings reveal a significant association between race, particularly among Asians, and an increased likelihood of gastric cancer. However, gender did not demonstrate a significant impact in our study cohort. These results emphasize the importance of tailored interventions and screenings for at-risk populations, considering the role of race in gastric cancer progression. Further research is needed to delve deeper into the underlying mechanisms of these disparities and to develop more targeted prevention strategies.

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Disclosure: Nothing to disclose.

PP1212

WHAT DO PATIENTS WITH INFLAMMATORY BOWEL DISEASE KNOW ABOUT THEIR PATHOLOGY?

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Introduction: Shared decision-making is defended as an ideal model in the management of patients with chronic pathology, namely inflammatory bowel disease (IBD).

The objective of this study was to identify knowledge gaps in IBD population, as well as correlate with patient demographic and sociocultural factors, to implement strategies aimed at improving care provision and patient empowerment.

Aims & Methods: Application of a survey to patients with IBD under biological therapy in a Day Hospital, from March 4th to 22nd 2024, and subsequent descriptive analysis.

Results: Out of the 102 patients with IBD (65% Crohn's Disease, 31% Ulcerative Colitis), 53% were men and 47% women, the majority of whom are Portuguese up to 50 years of age, with at least secondary education. The majority obtains information through health professionals and the Internet. The self-assessment of knowledge was higher than the demonstrated knowledge (90% consider themselves to be informed). The majority prefers to obtain oral information from health professionals or information leaflets (although only 50% had received information leaflets prior to starting biologics).

The population over 50 years old, with a lower level of education and diagnosed more than 2 years ago, tends to have important knowledge gaps. 46% are unaware what medication can worsen the disease and 24% are unaware of the impact of tobacco. 56% of patients believe that a restrictive diet should be preferred, with 59% never being evaluated by a nutritionist, although 63% have already discussed this topic with their doctor. 85% of women are unaware of the implications of IBD on fertility, pregnancy and breastfeeding, particularly the population under 30 years of age and with a lower level of education.

Conclusion: The application of a survey across IBD population under biological therapy allowed to detect important gaps in the transmission of information adapted to each patient and identify key points to be optimized during the medical appointment or in dedicated Nursing and Nutrition appointments.

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PP1213

IMPACT OF AN ONLINE MICROBIOTA EDUCATION PROGRAM ON HEALTHCARE PROFESSIONALS' KNOWLEDGE

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Introduction: While there is increasing awareness of the importance of microbiota education for healthcare professionals (HCPs), there is variability in the extent of coverage and accessibility of educational resources across different settings and regions.

Aims & Methods: We aimed to evaluate the effectiveness of an online educational program with the learning objectives to understand the critical role of gut microbiota (GM) in health and disease, recognize its significance across the lifespan, and apply evidence-based approaches to clinical practice.

We implemented an enduring on-demand continuing medical education (CME) program consisting of 6 independent 1-hour courses. It is accessible on an EACCME®-accredited CME platform and is delivered by expert faculty in a video micro-learning format, available in 7 languages.

After each course, learners completed an evaluation survey and a 10-question multiple-choice test. AI-based analysis was conducted on survey responses to identify learners' intention to change. A pre- vs post-assessment model, using 3 multiple-choice questions per course, was employed to evaluate educational effectiveness, with statistical analysis conducted using a paired samples t-test and Cohen's d with correction to estimate effect size. Data collection spanned from the program launch in July 2021 through April 2024.

Results: 5,216 learners from 118 countries enrolled in the program, primarily physicians (led by gastroenterologists, and followed by family medicine and general practice), with nurses also represented.

Evaluation survey ratings (n=1195) indicated high levels of satisfaction, with scores of 4.5/5 for 'Fulfillment of expected learning outcomes,' 4.6/5 for 'Usefulness for professional activity,' and 4.6/5 for 'Scientifically based and balanced information.

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91% of participants agreed to implement the learnt information in their practice. We identified the following common themes reflecting intentions to change after completing the courses:
Patient education (use of probiotics, treatment expectations, importance of GM for health)
Enhanced management of gastrointestinal conditions such as irritable bowel syndrome
More cautious use of antibiotics
Greater emphasis on maternal and child health (breastfeeding, use of probiotics)
More active probiotics incorporation in daily clinical practice
More careful probiotic selection (strain specificity, individual patient needs)
Across the program, participants achieved an average of 77% correct responses in the multiple-choice test, with 70.41% correctly answering at least 80% of questions. Analysis of pre- and post-course assessments (n=224; collected Feb-April 2024), showed that the average percentage of correct answers increased from 54.8% at the baseline (pre-course tests) to 73.0% in the post-course tests ($p < 0.001$), and revealed a moderate educational impact, with a Cohen's d of 0.59. The percentage of learners who correctly answered all three questions in a course increased from an average of 19.9% at the baseline to 42.1% in the final assessment.
Conclusion: The comprehensive online microlearning-based program saw significant participation from a global audience, attracting not only gastroenterologists but also a diverse range of HCPs. Our data revealed significant improvements in clinicians' knowledge, and intentions to change patient care practices. These findings support the need to update and extend such CME activities to ensure that HCPs are well-equipped to understand and address the role of microbiota in patient care.
Disclosure: Nothing to disclose.

PP1214

WHAT MOTIVATES MEDICAL INTERNS TO PURSUE IN GASTROENTEROLOGY? A MULTICENTRIC CROSS-SECTIONAL STUDY

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Introduction: Gastroenterology (GE) is a medical discipline that studies all diseases of the digestive system. It comprises several areas of study and offers a large variety of sub-specializations: proctology, interventional endoscopy or even digestive oncology. The decisive elements in the choice of this specialty vary among interns. The aim of our study is to determine the various factors influencing the choice of gastroenterology as a specialty among medical interns.
Aims & Methods: This is a cross-sectional, multicentric, survey-type study, conducted from September 1 to 30, 2023, involving 7 national medical faculties. All medical interns, regardless of sex or age, participated in our study.
An electronic survey with 29 items was created using Google Forms®. It was divided into two parts: the first part contained 13 questions on socio-demographic characteristics and exposure to GE during the academic curriculum, while the second part concerned the factors motivating the choice or not of GE as a specialty.
A statistical analysis using the Fisher's exact test was performed. P values of <0.05 were considered statistically significant. Data were analyzed using Jamovi (Version 2.3).
Results: A total of 329 responses were collected: the majority were women 208 (63.2) with a F/M sex ratio of 1.7 and a median age of 27 [25; 28] years. The majority had already completed a GE rotation during their training course n=205 (62.3), of which 90 (44.1) had found it "satisfactory". More than half of the participants n= 178 (54.1) had already considered GE as a future specialty, and this choice was mostly made during GE lectures given in medical school n= 66 (36.9). Among those who had never considered GE, the main reasons were lack of interest in the specialty, the fact that it is a medical specialty, lack of information about GE, potential remuneration, and dissatisfaction with GE rotation during the training course (59.2%, 28.9%, 13.9%, 10.4% and 10% respectively).
Among those who finally opted for GE, the majority were women (70.2% vs. 29.8%, p=0.004). Satisfaction with the GE rotation during the training course had an impact on the final choice: "satisfactory" vs. "mediocre" (43.8% vs. 6.3% p=0.009).
Multivariate analysis showed that early interest in fields such as endoscopy (OR= 4.46, CI95%: 1.44-13.78, p=0.009) or proctology (OR= 4.26, CI95%: 1.01-18.01, p=0.05) increased the chance of choosing GE.
Conclusion: This study showed that there is a gender disparity in the choice of GE as a specialty. The fact that it is a medical discipline may lead men to choose another path. Our study also highlighted the importance of GE training courses during the medical curriculum to motivate interns to join our specialty, as well as the need to spotlight certain areas such as endoscopy or proctology.
Disclosure: Nothing to disclose.

PP1215

ASSESSING ENDOSCOPIC TRAINING IN GASTROENTEROLOGY FELLOWSHIPS: A MULTICENTRIC CROSS SECTIONAL PILOT STUDY

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Introduction: Gastroenterology is a medical specialty that includes several areas of study. Endoscopy occupies a central role in professional practice, whether in hospitals or in the private sector. The quality of initial training and the different needs of trainees are key to the quality of care delivered to the population. This study aims to assess endoscopic training experience during fellowship.
Aims & Methods: This is a cross-sectional, multicenter study conducted on gastroenterology residents practicing endoscopy and in the final phase of their curriculum (6th, 7th and 8th semesters). Postgraduated doctors have been excluded from the study. An electronic survey made by Google Forms and was sent nationwide on January 9, 2024. The survey consisted of 42 questions divided into 3 parts:
The first part focused on the participants' socio-demographic data and their endoscopic experience, the second part focused on the quality of training of core endoscopic skills (cognitive and motor skills) during fellowship where a 5-point Likert scale was used to express residents' satisfaction with their skills, ranging from 1="very dissatisfied", 2="dissatisfied", 3="neither satisfied nor dissatisfied", 4="satisfied" and 5="very satisfied".
The third part included questions in which participants expressed their needs and expectations for improving their training in endoscopy.
A statistical analysis using the Student-t test and the Fisher's exact test were performed. P values of <0.05 were considered statistically significant. Data were analyzed using Jamovi (Version 2.3).

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Results: We were able to collect data from 56 respondents. Most were female residents n=40 (71.4) with a mean age of 29.2 ± 2.5 years. The majority of fellows started practicing EGD in their 4th semester (91.1%), and colonoscopy in their 6th semester (57%). On average, the fellows are supervised by 5 senior doctors, including 2 who practice interventional endoscopy.

Furthermore, 37.5% of participants report performing approximately more than 100 EGDs, 14.2% more than 100 colonoscopies, 80.4% report that they have performed less than 20 endoscopic variceal ligation and 14.3% have performed more than 10 polypectomies. An average rating score of 3.34 (1= very dissatisfied; 5= very satisfied) concerning the quality of training and feedback on cognitive aspects of endoscopy was found and we reported greatest emphasis for Procedure indication/contraindication and least on Capsule endoscopy interpretation.

The quality of training in technical aspects of endoscopy was rated higher (average rating score of 3.5). We reported greatest emphasis for Withdrawal/mucosal inspection and least on Esophageal dilation.

A significant association between motor skills and cognitive skills was found with a p<0.001.

Finally, to improve their training, 87.5% of respondents considered simulation to be important/very important, 98.2% of respondents intend to complete their endoscopy training at the end of their curriculum, and 87.5% are prepared to finance such a project (median: 1000 USD [1000;3000]).

Conclusion: Despite a significant improvement in training sites and technical facilities, training in endoscopy in healthcare institutions does not yet provide all the cognitive and technical skills required to enable fellows to be fully autonomous.

With the development of endoscopic procedures, it seems essential for fellows to be able to perform more standard endoscopic procedures such as polypectomy and mucosectomy by the end of their curriculum. This may involve introducing regular endoscopy conferences and simulation training as a part of the curriculum.

Disclosure: Nothing to disclose.

PP1216

ASSESSING THE IMPACT OF A STRUCTURED CAPSULE ENDOSCOPY TRAINING PROGRAM USING A NEW VALIDATED ASSESSMENT TOOL

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Introduction: There is scarce evidence regarding the impact of capsule endoscopy (CE) training programs.

Aims & Methods: We aimed to develop and validate a simple capsule endoscopy (CE) training assessment tool, the Capsule Endoscopy Training Assessment (CETA), and prospectively use it to analyze the learning progression achieved by participants in our CE training program.

Over a 3-year period, all participants in our CE training program completed pretraining and posttraining CETA, ranging between 0-100%, and encompassing theoretical questions and interpretation of segmented CE videos. We compared the mean differences in overall, theoretical, and practical pretraining-posttraining CETA and assessed the influence of previous endoscopic experience (upper gastrointestinal endoscopy [UGE], colonoscopy, device-assisted enteroscopy [DAE] and CE) using generalized linear models.

Results: Fifty-seven participants were included. After training, there was a significant increase in participants' overall (mean difference, 26.3; 95% confidence interval [CI], 20.70 to 31.83), theoretical (mean difference, 27.2; 95% CI, 19.81 to 34.57), and practical (mean difference, 25.9; 95% CI, 20.09 to 31.63) CETA components.

Compared to those without experience, participants with previous experience with these techniques had higher overall pretraining CETA (UGE, mean difference, 15.8; 95% CI, 5.79 to 25.75; colonoscopy, mean difference, 12.5; 95% CI, 2.35 to 22.65; DAE, mean difference, 14.2; 95% CI, 3.92 to 24.54, CE, mean difference, 16.3; 95% CI, 8.72 to 23.93), lower mean difference between overall pretraining and posttraining CETA (UGE, rate ratio, 0.76; 95% CI, 0.63 to 0.91; colonoscopy, (rate ratio, 0.80; 95% CI, 0.67 to 0.9; DAE, (rate ratio, 0.84; 95% CI, 0.73 to 0.97; CE, rate ratio, 0.81; 95% CI, 0.72 to 0.92, respectively) but not significantly different overall posttraining CETA.

Conclusion: CETA is a valid and useful tool in assessing the learning progression achieved by participants after CE training program. We demonstrated a significant improvement in participants' CETA after training, being the least experienced participants in endoscopic procedures who benefited the most from CE training.

Disclosure: Nothing to disclose.

PP1217

UNVEILING CHATGPT'S LIMITATIONS IN EUROPEAN BOARD OF GASTROENTEROLOGY AND HEPATOLOGY EXAM: INCONSISTENCIES AND CHALLENGES IN COMPLEX MEDICAL REASONING

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Introduction: The integration of artificial intelligence (AI) in medicine has the potential to revolutionize healthcare. Exams, such as the European Board of Gastroenterology and Hepatology (EBEGH) exam, are currently the gold standard for assessing medical competency and provide credentialing across several EU countries. This study evaluates the performance of ChatGPT in the high-stakes EBEGH exam by assessing its strengths and limitations in managing complex medical content, focusing on its ability to handle various question types, cognitive task demands, and subspecialty domains.

Aims & Methods: Utilizing a dataset of 200 EBEGH study questions, we analyzed ChatGPT4's performance across question complexity, cognitive task demands, subspecialty domains, and patient-centeredness. The dataset ensured the same number of questions and a balanced representation of subspecialties and focused on textual questions, excluding visual questions. Each question was asked to the model 5 times, with the most frequent answer selected as the final response.

A suitable prompt for the task was used for all questions. Logistic regression, ANOVA, and chi-square tests investigated the impact of these factors on the AI's likelihood of correct responses, performance variance, and categorical data associations. Krippendorff's Alpha evaluated ChatGPT4's precision in replicating responses across multiple iterations. The study questions were categorized based on cognitive task complexity using

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Bloom's taxonomy, and the evidence base supporting each question was copied from the question bank reference. The model's performance was further analyzed in relation to the interdisciplinary nature of the questions and procedural knowledge requirements.

Results: ChatGPT correctly answered a mean of 67.5% (n=135, range 121-142) of the questions, with performance significantly influenced by question complexity. Low complexity cases significantly increased the odds of correct responses by 3.87 times, (p=0.001), while medium complexity cases showed a similar trend by 2.82 times (p=0.013). Non-patient-centered questions significantly decreased the odds of correct answers (OR=0.30, p=0.030). The model demonstrated inconsistency in replicating responses (Krippendorff's Alpha=0.00219).

Questions requiring higher cognitive tasks, such as analysis and evaluation, posed significant challenges for the model (p<0.01). The AI's performance was not significantly influenced by the interdisciplinary nature of the questions or the requirement for procedural knowledge.

Conclusion: While ChatGPT effectively handles simpler queries, its performance in complex, specialized medical testing scenarios, such as EBEGH exams, is currently limited. The AI's performance varies across subspecialties, question types, and cognitive task demands, and it demonstrates inconsistency in replicating responses.

These findings underscore the need for a cautious and nuanced approach to integrating AI in high-stakes medical examinations, particularly in domains requiring deep understanding and patient-centered considerations. The EBEGH exam's ability to effectively challenge ChatGPT demonstrates its value as a robust tool for evaluating medical knowledge and expertise.

Disclosure: Nothing to disclose.

PP1218

AN INNOVATIVE INTERPROFESSIONAL APPROACH TO THERAPEUTIC DRUG MONITORING OF CARBAPENEMS IMPROVES QUALITY OF CARE IN THE TREATMENT OF PATIENTS WITH ACUTE-ON-CHRONIC LIVER FAILURE ON THE INTENSIVE CARE UNIT

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Introduction: Acute-on-chronic liver failure (ACLF) is a severe complication in patients with liver cirrhosis, often escalating rapidly and requiring urgent interventions. Meropenem, an important antibiotic for treating severe infections, necessitates accurate dosing to be effective and safe.

Aims & Methods: In this study aims the impact of implementing therapeutic drug monitoring (TDM) for meropenem in ACLF patients in an intensive care unit (ICU) with an innovative interprofessional approach was assessed.

An interprofessional team of physicians, hospital pharmacists, and staff nurses collaborated to optimize meropenem therapy through TDM. Meropenem was administered via continuous infusion following an initial loading dose, with serum levels monitored twice weekly using high-performance liquid chromatography (HPLC). The evaluation metrics included serum meropenem concentrations, compliance with the interprofessional team's recommendations, and total consumption of the antibiotic.

Results: The initial TDM results showed a mean meropenem serum concentration of 20.9 ± 9.6 mg/L. Of note, only 16.0% of the patients had serum concentrations within the designated target range, while 84.0% exceeded it. Follow-up measurements in Week 2 and Week 3 showed improved targeting, with average concentrations of 15.2 ± 5.7 mg/L and 11.9 ± 2.3 mg/L, respectively. By Week 2, 41.7% of patients were within the target range, and by Week 3, this increased to 50%. The advice of the interprofessional team was fully implemented.

Furthermore, there was a notable reduction in meropenem usage: from 42.1 RDD/100 patient days in the preceding period to 37.9 RDD/100 patient days during the study, marking a 10.0% decrease in application density.

Conclusion: The introduction of an interprofessional TDM approach for meropenem in patients with ACLF significantly improved dosing of carbapenems. Our innovative interprofessional approach brought serum levels within desired ranges more effectively and managed to reduce overall meropenem consumption.

The study underscores the value of innovative collaborative, interprofessional efforts in optimizing drug therapy, particularly in complex ICU settings, enhancing patient safety and treatment efficacy.

Disclosure: Nothing to disclose.

PP1219

THE IMPACT OF GASTROINTESTINAL ENDOSCOPIES ON MUSCULOSKELETAL DISORDERS

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Introduction: Musculoskeletal (MSK) injuries are frequent occupational hazards amongst endoscopists, partly attributed to the physically demanding nature and prolonged durations of endoscopic procedures. Understanding the prevalence and patterns of these injuries is crucial to improve the endoscopist's occupational health by implementing appropriate preventive strategies.

Aims & Methods: Aim: The primary aim of this study was to determine the prevalence and distribution of MSK injuries among endoscopists. Endoscopists from across the world were surveyed regarding the types and frequencies of procedures they perform to evaluate the prevalence of MSK injuries among endoscopists with varying procedural type and frequency. The secondary aim was to investigate whether additional factors, such as exercise habits during adulthood and adolescence and the endoscopy environment, influence the frequency of reported MSK injuries.

Method: An international questionnaire-based study was conducted among endoscopists practising in numerous healthcare centres. Data gathered included the frequency and localisation of injuries, type of endoscopic procedures carried out, access to occupational health (OH) and ergonomics of the endoscopy theatre.

Additionally, the study explored the potential impact of past and present exercise practice during adulthood and adolescence on the prevalence of MSK injuries.

Results: 174 endoscopists participated in this study, from 25 countries. The most commonly reported MSK complaints were back, neck, shoulder and hand pain (Table 1).

Endoscopists' experience was as follows; 32.2% -0-5 years, 46% -5-20 years and 21.8% -20 years of experience. Only 35.1% of participants had formal training on scope handling at the start of training.

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	Percentage of endoscopists reporting pain
Back Pain	23.6%
Neck Pain	19.5%
Shoulder Pain	16.0%
Hand Pain	14.9%
Thumb Pain	13.2%
Hip Pain	7.4%
Elbow Pain	6.7%
Finger Pain	5.1%

Table 1: Pain in endoscopists.

Participants reported to use 1, 2, 3 and 4 monitors respectively during endoscopy sessions: 37.9%, 56.0%, 4.6% and 0.6%. 75.9 % of participants ensure that their screen is at eye level during endoscopies, whilst 75.3% ensure that the monitor is the right distance away.

23% of participants stated that they have had a formal diagnosis of an occupation related injury or have been reviewed by a physiotherapist / orthopaedic surgeon because of an endoscopy related injury. In contrast, only 13.2% of patients have reported their occupational injury to OH services.

51.1% of participants have had MSK injuries that they think are attributable to current endoscopy practices. However, only 13.2% have sought help from the OH team. 74% of these felt they were supported by the OH team.

10.9% of participants have been diagnosed with osteoarthritis or an inflammatory arthropathy, unrelated to work.

Conclusion: This study provides a valuable insight into the relationship between endoscopic procedures and the occurrence of MSK injuries among endoscopists, offering opportunities for prevention to safeguard the health of endoscopists and to encourage them to seek help for such injuries.

Disclosure: The authors declare they have no conflict of interest.

PP1220

TOWARDS GREENER ENDOSCOPY: IMPACT OF A SUSTAINABILITY INTERVENTION ON ENDOSCOPIC WASTE CARBON FOOTPRINT

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Introduction: Climate change is responsible for increasing global health consequences and healthcare-related expenses. Despite efforts to address these issues, the healthcare sector remains a significant contributor to the world's carbon footprint due to increasing greenhouse gas emissions, compromising its own sustainability. Particularly, endoscopy stands out as the second highest waste generator per clinical procedure, largely attributed to its resource-intensive processes and extensive use of single-use devices.

This study aimed to assess the waste carbon footprint generated at a high-volume tertiary endoscopy center both before and after implementing a targeted sustainability intervention. Additionally, we assessed the intervention's impact on the daily workflow of the endoscopy staff.

Aims & Methods: A 3-stage prospective interventional study was conducted from December 2023 until March 2024. Pre-intervention: four-week observational audit, during which daily endoscopic waste (landfill, regulated medical waste (RMW), and recycled paper and plastic) was weighed in both endoscopy rooms and recovery areas. Intervention: following data collection, a dedicated training session was conducted for the endoscopy staff.

The session included a presentation of the collected data and outlined key sustainability strategies aimed at enhancing waste segregation.

Additionally, recycling streams were established with placement of dedicated bins in assessment areas. Post-intervention: four-week period during which daily endoscopic waste was weighed in both endoscopy rooms and recovery areas.

Weighing was conducted using an engineer-calibrated scale. For the purpose of this study, waste carbon footprint calculations were based on the equivalence of 1kg of landfill waste to 1kgCO_{2e} and 1kg of RMW to 3kgCO_{2e}. Upon conclusion of the study, an anonymous survey was distributed to all staff members to assess the impact of the study on the daily work routine.

Results: Total waste carbon footprint was reduced by 28.2%. In endoscopy rooms, waste carbon footprint decreased 20.4% (pre- vs. post-intervention: 1472.8kgCO_{2e} vs. 1173.7kgCO_{2e}, p=0.552). RMW was reduced by 24.4% (pre- vs. post-intervention: 462.8kg vs. 336.3kg, p=0.020), whereas landfill waste (pre- vs. post-intervention: 72.2 vs. 151.1, p<0.001) and recycling waste increased (paper – pre- vs. post-intervention: 0 vs. 21, p<0.001; plastic – pre- vs. post-intervention: 0 vs. 33.6, p<0.001). Mean endoscopy load was similar (pre- vs. post-intervention: 517 vs. 541, p=0.48).

In recovery areas, waste carbon footprint decreased 58.3% (pre- vs. post-intervention: 386.7kgCO_{2e} vs. 161.1kgCO_{2e}, p=0.002). RMW was reduced by 42.1% (pre- vs. post-intervention: 109.4kg vs. 28kg, p<0.001), whereas landfill waste (pre- vs. post-intervention: 58.4 vs. 77.2, p=0.040) and recycling waste increased (paper – pre- vs. post-intervention: 0 vs. 6, p<0.001; plastic – pre- vs. post-intervention: 0 vs. 9.9, p<0.001).

All staff members agreed the project did not disrupt their daily work routine, emerging as a valuable asset for promoting further sustainable practices.

Conclusion: Our study demonstrated that reducing RMW and implementing daily recycling in endoscopy rooms and recovery areas are attainable objectives that do not hinder endoscopy productivity. As a result, a yearly reduction of 2104kgCO_{2e} could be achieved in our endoscopy unit.

Additionally, this study highlighted the effectiveness of educational initiatives in raising awareness towards more sustainable endoscopy practices.

Disclosure: Nothing to disclose.

PP1221

A SCALABLE, REPRODUCIBLE APPROACH TO CASE-BASED TRAINING FOR PRE-ENDOSCOPIC MANAGEMENT OF ANTICOAGULANTS

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Introduction: Guidelines reflect the latest, evidence based consensus for our clinical routine. The existing recommendations are numerous and subject to frequent change. Current learning methods are predominantly non-interactive. Although case-based learning is recognized as an cost effective educational tool [1, 2], it is often constrained by the limited number of available up-to-date case examples. To our knowledge, there is no reproducible and easily maintainable source for case-based training on guideline content.

Aims & Methods: We aimed to develop a versatile framework for the procedural generation of cases that represent a set of guideline recommendations in a structured format.

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For this prototype, we evaluated recommendations for the pre-endoscopic assessment of risks for major bleeding or thromboembolic events based on the German guideline for quality in gastrointestinal endoscopy [3] (Chapters 3.2.2 to 3.2.5) to identify necessary information for database population (e.g., interventions with high or low bleeding risks). A list of case templates was then created, each consisting of a selected guideline excerpt along with a set of predefined properties inherited by the resulting cases. Undefined parameters were generated using random values reflecting unremarkable findings to introduce diversity.

The platform, including the database definition, was developed as a Python web application using the Django framework. Pre-defined database content such as entries for diseases, classifications, interventions, and guideline excerpts are stored in easily extendable or modifiable ".yaml" files.

Results: We established a structured database which stores the data required to determine risk of thrombembolic event. This includes pre-existing diseases (e.g., chronic kidney disease (CKD), atrial fibrillation) with corresponding classifications (e.g., CKD Stage 1 - 5), laboratory values (e.g., hemoglobin, eGFR), and relevant previous events (e.g., stroke, coronary in-stent stenosis).

Furthermore, the database contains data to determine bleeding risk including different interventions (e.g., polypectomy, EMR, ESD, eFTRD, ultrasound guided biopsies as well as patient factors like liver cirrhosis) and lesions. Lesions like polyps may also be specified using morphology (Paris or lateral spreading type classification), location (organ and organ part), and size.

The resulting Python-based Django web application can repeatably generate lists of different cases representing the same set of predefined guideline recommendations. Moreover, future changes can be easily adapted by modifying the implemented rules or guideline texts.

Based on the desired focus - for example apixaban intake and different stages of CKD - the user may define a list of case templates which represent those aspects. Additionally, the user adds the desired questions for the trainee including corresponding guideline citations. Those cases may then be presented to trainees. In our prototype, the trainee was instructed to assess the risk of bleeding and thromboembolic events. The trainee was also asked to determine the pausing of each drug before the procedure if it is necessary. After completing a case, we present the relevant guideline excerpts, including exact citations, to the trainee.

Conclusion: Our platform can be adapted to other domains and future guideline updates can be incorporated with minimal effort, making this tool suitable for diverse case-based learning or for evaluating knowledge in a consistent, benchmark-like manner.

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Disclosure: Nothing to disclose.

PP1222

ENHANCING HEALTHCARE OUTCOMES THROUGH INTERPROFESSIONAL TRAINING WARDS: A COMPREHENSIVE STUDY OF PATIENT CARE, COST EFFICIENCY, AND EDUCATIONAL BENEFITS AT THE A-STAR REGENSBURG

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Introduction: Interprofessional teamwork is essential in modern healthcare, with interprofessional training wards established since 1996 to enhance professional collaboration and communication. Despite their importance in optimizing patient care and training, research into patient outcomes in these wards is limited and often restricted to patient satisfaction. This study expands existing research by comparing patient outcomes in the interprofessional training ward Regensburg (A-STAR) and conventional wards, hypothesizing that interprofessional training wards offer no disadvantage to patient outcomes.

Aims & Methods: This comprehensive study, which delved into various aspects of interprofessional training wards, was conducted at the University Hospital Regensburg from October 2019 to December 2023. To ensure a robust analysis, we assessed patient outcomes, cost efficiency, patient and family satisfaction, and gainage of interprofessional competence in the interprofessional training ward A-STAR. Our data collection methods were diverse, including digital and paper-based anonymous standardized questionnaires (Collaboration and Satisfaction About Care Decisions (CSACD), Family Satisfaction in the Intensive Care Unit—(FS-ICU)), and information from the clinical data system.

Results: The study analyzed outcomes from 1,482 A-STAR and 5,752 conventional ward patients, revealing that the clinical outcomes of A-STAR patients, such as discharges against medical advice, complication-driven readmissions, ICU transfers, and mortality rates were similar to those in conventional wards. A-STAR was also more cost-effective, generating higher profits per case due to increased revenues and reduced material costs. A-STAR received high satisfaction ratings from patients and family members regarding treatment and decision-making. Moreover, medical students and nursing apprentices reported significant improvements in their ability to make interprofessional medical decisions after their training periods (medical students: mean difference -8.23, p=0.004; nursing apprentices: mean difference -9.5, p=0.003).

Conclusion: Interprofessional training wards like A-STAR ensure comparable patient care to conventional wards and enhance the educational experience for healthcare students. Our findings demonstrate that these wards are cost-effective and beneficial for patient and family satisfaction. This underscores the importance of integrating interprofessional training wards into medical curricula, as they significantly improve interprofessional collaboration and decision-making skills, which are crucial for patient safety and fostering a positive team atmosphere. In light of these compelling results, the study strongly advocates for the expansion of such training models in healthcare settings to better equip future professionals for the complexities of modern medical practice.

Disclosure: Nothing to disclose.

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PP1223

TESTOSTERONE AND ESTRADIOL'S PROTECTIVE INFLUENCE ON PRIMARY LIVER CANCER

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Introduction: Primary liver cancer (PLC) poses a significant global health threat, with sex hormones implicated in its pathogenesis. Testosterone and estradiol's influence on PLC risk remains debated. Mendelian randomization (MR) offers a robust method to assess causal relationships.

Aims & Methods: Aims: Investigate the causal relationship between testosterone and estradiol levels and the risk of primary liver cancer (PLC) using Mendelian randomization (MR) analysis, while assessing potential mediators such as tumor necrosis factor receptor superfamily member 9 (TNFRS9) and urokinase-type plasminogen activator (uPA).

Methods: We conducted MR analyses using genome-wide association study (GWAS) data to investigate the causal impact of testosterone and estradiol on PLC risk. Tumor necrosis factor receptor superfamily member 9(TNFRS9) and urokinase-type plasminogen activator(uPA) were explored as potential mediators.

Results: Our MR analysis revealed a protective effect of testosterone and estradiol against PLC. Testosterone was negatively associated with PLC risk (OR = 0.9987, P = 0.0316), with TNFRS9 mediating 9.1% of this effect. Estradiol exhibited a negative correlation with PLC risk (OR = 0.9985, P = 0.0125), with uPA mediating 8% of this effect.

Conclusion: Contrary to traditional beliefs, both testosterone and estradiol demonstrate protective effects against PLC. TNFRS9 and uPA pathways mediate a significant portion of these effects, providing novel insights into the sex hormone-PLC relationship and potential therapeutic targets. Further research should validate these findings across diverse populations.

Disclosure: Nothing to disclose.

PP1224

OCCUPATIONAL STRAIN AND REPRODUCTIVE HEALTH: A COMPARATIVE STUDY OF PREGNANCY OUTCOMES IN FEMALE VISCERAL MEDICINE DOCTORS

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Introduction: Recent research has highlighted significant health risks associated with pregnancy among female surgeons in USA. They reported increased risks of infertility and pregnancy complications among female surgeons, attributed to work schedules and delayed childbearing due to professional demands (1).

Aims & Methods: This joint study by the Young Gastroenterology (JUGA) and Surgical Working Group of Young Surgeons (CAJC) of the German Society for Gastroenterology, Digestive and Metabolic Diseases (DGVS) and the German Society for General and Visceral Surgery (DGAV) focuses on the prevalence and factors of major pregnancy complications among female visceral medicine doctors in Germany and compares their experiences with a non-doctor control group from lawyers and teachers.

Utilizing an anonymous online survey, responses from 352 doctors detailing 757 pregnancies and 92 non-doctors with 172 pregnancies were collected and analyzed.

Generalized Linear Mixed Models (GLMM) were employed to assess risk factors linked with pregnancy complications, with particular attention to work conditions and use of assisted reproductive technologies.

Results: The study revealed no differences in maternal age at first live birth, number of births as well as the overall frequency and number of pregnancy complications between female visceral medicine doctors and non-doctors. However, visceral medicine doctors experienced higher rates of pregnancy loss (23.1% vs. 13.7%, p = 0.008) and intrauterine growth restriction (3.0% vs. 0.0%; p = 0.025). Work hours, especially close to end of pregnancy and during nights and weekends, were significantly longer for doctors ($p \leq 0.001$) and associated with pregnancy complications.

Additionally, a higher proportion of doctors expected pregnancy to have more impact on their career compared to non-doctors (90.3% vs. 70.1%; p = 0.001), with both groups commonly aligning family planning with career goals.

Conclusion: These results reveal significant differences in pregnancy-related complications, with a higher prevalence among medical professionals, likely influenced by working hours and times.

Further analysis is warranted to better understand the underlying factors contributing to these disparities and to inform targeted interventions.

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Disclosure: Nothing to disclose.

PP1225

THE NATIONAL BOWEL CANCER SCREENING PROGRAM AND ACCESSIBILITY FOR PEOPLE WITH DISABILITY: A QUALITATIVE INVESTIGATION OF LIVED EXPERIENCE WITHIN THE AUSTRALIAN CONTEXT

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Introduction: The societal benefits of population-based bowel cancer screening are well documented in both an Australian and global context. Australia's National Bowel Cancer Screening Program (NBCSP) has helped reduce rates of bowel cancer morbidity and mortality since its inception in 2006.(1)
Despite the reported successes of the NBCSP, there are limitations in participation data for sub-populations, making it difficult to demonstrate equitable program impacts across the diverse Australian population. One of these groups is people with disability.
While available data is limited, that which is available shows that people with severe or profound activity limitations participating in the NBCSP show higher screening positivity rates (13% vs 7%), a lower diagnostics assessment follow-up rate (43% vs 64%) and experience a longer median time between positive screen and diagnostic assessment (63 days vs 49 days), demonstrating further disparities and accessibility issues. (2)
Aims & Methods: This study aimed to create knowledge gains on barriers and enablers to accessing the NBCSP for people with disability. To our knowledge this is the first Australian study to collect data on support needs for people with disability (and their carers) when accessing the NBCSP. Multiple qualitative data collection methods were adopted, including arts-based methods such as visual experience, storytelling and photo-elicitation to provide rich and empathic insight into the experience of completing an iFOBT for people with disability. This process was facilitated throughout online in-depth interviews. Participants were recruited through Queenslanders with a Disability Network.
Results: 15 participants with diverse communication, mobility and self-care abilities participated in the online in-depth interviews. Preliminary analysis of data findings indicates several key system level barriers preventing equal access to participating in the NBCSP. Kit packaging, instruction format and complexity level, size of collection tubes for faecal sample and the need for additional carer support to complete the kit was listed as key barriers across the participant cohort.
While barriers differed significantly between disability sub-populations, several synergies remained around the need for simplification of instructions, improved kit packaging and design and the need for prompts or helpline information to facilitate engagement with a carer to support ease of iFOBT completion. Photos submitted by participants about their experience accessing the kit highlighted that although everyone in Australia receives the kit the same way (in the mail) not everyone completes the kit in the same way, with many people needing supports and assistance to navigate kit completion successfully.
Conclusion: This study provided novel, critically needed information and empathetic insights into the lived experience for people with a disability. Results highlighted the need for priority population consideration and consumer consultation to improve the design of the iFOBT, with particular focus on the packaging/kit instructions, with a call for easy read formats, inclusive video guides and references to support carer engagement. Many additional barriers exist that impact the ability for diverse population groups to participate in the program and without reasonable adjustments and navigation inclusions in Program design, disparities between and across population groups will continue to exist. Findings will be shared and used to shape policy and practice impacting system level barriers to equal program access for diverse populations across Australia.
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Disclosure: Nothing to disclose.

PP1226

EXPLORING THE AGE SPECTRUM OF REAL-WORLD PATIENTS WITH INFLAMMATORY BOWEL DISEASE: A MACHINE LEARNING APPROACH

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Introduction: Ulcerative colitis (UC) is a chronic, gastroenterological disease, heterogenic in presentation[1]. Little is known regarding where specific clinical characteristic heterogeneity lies in the context of age groups.
Aims & Methods: We aimed to investigate whether naturally occurring age groups existed among adult patients with UC based on symptomatology and health-related quality of life (HRQoL) and to assess differences in the symptomatology and flaring status of these age groups. Data were drawn from the Adelphi Disease Specific Programme™ for UC, a cross-sectional survey of gastroenterologists (GIs) and UC patients in France, Germany, Italy, Spain, the United Kingdom (UK), the United States (US) and Japan from Jan 2020–Mar 2021. GIs reported data for their next seven consulting patients, including demographics, clinical characteristics, and components of the Mayo score, allowing full Mayo scores to be calculated for all patients. Patients voluntarily reported on HRQoL via the Short Inflammatory Bowel Disease Questionnaire (SIBDQ). 10-fold cross-validation was used to identify age groupings by adjusting knot placement in linear splines of age in regression models with outcomes of patients' Mayo scores and SIBDQ scores. Patients' clinical characteristics across the age groups were compared using bivariate analysis.
Results: Overall, 3052 patients with UC were included (55.8% male, mean [standard deviation; SD] BMI 24.1 [3.9], mean [SD] disease duration 5.1 [5.9] yrs), of which 960 had known SIBDQ scores (52.1% male, mean [SD] BMI 23.5 [3.5], mean [SD] disease duration 3.1 [6.2] yrs).
Age groupings based on Mayo scores were 18-28 yrs (n=651), 29-60 yrs (n=2114), 61+ yrs (n=287). Current flaring status was significantly differ-

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ent across groups with 19.2% of patients in the 18-28 yrs group flaring (p<0.01). Of symptoms with the greatest difference in prevalence between age groups, prevalence of rectal bleeding, abdominal pain, bloody diarrhoea, bowel movement urgency, and anaemia differed, although the results were not always significant.

Age groupings based on SIBDQ scores were 18-37 yrs (n=506), 38-40 yrs (n=83), 41-50 yrs (n=173), and 50+ yrs (n=198). Flaring differed significantly across age groups. Among symptoms with the greatest difference in prevalence between age groups, prevalence of flatulence, fatigue/tiredness, abdominal pain, abdominal distension, and anaemia differed, although not always significantly.

Mayo Derived Age Categories		18-28 Yrs (n=651)	29-60 Yrs (n=2114)	61+ Yrs (n=287)	P value
	Sex, male, n (%)	345 (53.0)	1189 (56.2)	169 (58.9)	0.19
	BMI, mean (SD)	22.67 (3.1)	24.26 (3.9)	26.00 (5.1)	<0.01
	Time since diagnosis (yrs)	n=609	n=1874	n=249	
	Mean (SD)	2.53 (2.7)	5.36 (5.7)	9.13 (9.5)	<0.01
	UC Disease Location, n (%)	n=592	n=1888	n=251	
	Ulcerative proctitis	93 (15.7)	311 (16.5)	38 (15.1)	
	Proctosigmoiditis	119 (20.1)	383 (20.3)	47 (18.7)	0.74
	Left-sided colitis	171 (28.9)	594 (31.5)	82 (32.7)	
	Pan-ulcerative (total) colitis	209 (35.3)	600 (31.8)	84 (33.5)	
	Symptoms with greatest difference in prevalence (top five), n (%)	n=640	n=2100	n=285	
	Rectal bleeding	91 (14.2)	222 (10.6)	22 (7.7)	<0.01
	Abdominal pain	167 (26.1)	545 (26.0)	57 (20.0)	0.08
	Diarrhoea (bloody)	118 (18.4)	321 (15.3)	36 (12.6)	0.05
	Bowel movement urgency	148 (23.1)	435 (20.7)	52 (18.3)	0.21
	Anaemia	146 (22.8)	442 (21.1)	77 (27.0)	0.06
	Current flaring status, n (%)	n=610	n=1950	n=250	
	Currently flaring	117 (19.2)	267 (13.7)	37 (14.8)	<0.01

SIBDQ Derived Age Categories		18-37 (n=506)	38-40 (n=83)	41-50 (n=173)	50+ (n=198)	P value
	Sex, male, n (%)	260 (51.4)	43 (51.8)	84 (48.6)	113 (57.1)	0.40
	BMI, mean (SD)	22.85 (3.1)	23.48 (4.1)	23.48 (3.3)	24.94 (4.0)	<0.01
	Time since diagnosis (yrs)	n=470	n=75	n=154	n=172	
	Mean (SD)	3.15 (3.3)	6.13 (5.6)	6.94 (6.1)	9.01 (9.6)	<0.01
	UC Disease Location, n (%)	n=450	n=70	n=152	n=179	
	Ulcerative proctitis	71 (15.8)	12 (17.1)	25 (16.5)	26 (14.5)	
	Proctosigmoiditis	97 (21.6)	14 (20.0)	30 (19.7)	42 (23.5)	0.58
	Left-sided colitis	146 (32.4)	30 (42.9)	61 (40.1)	62 (34.6)	
	Pan-ulcerative (total) colitis	136 (30.2)	14 (20.0)	36 (23.7)	49 (27.4)	
	Symptoms with greatest difference in prevalence (top five), n (%)	n=505	n=83	n=173	n=196	
	Flatulence	48 (9.5)	17 (20.5)	23 (13.3)	26 (13.3)	0.03
	Fatigue/ tiredness	117 (23.2)	12 (14.5)	36 (20.8)	37 (18.9)	0.26
	Abdominal pain	132 (26.1)	29 (34.9)	56 (32.4)	51 (26.0)	0.18
	Abdominal distension	78 (15.5)	7 (8.4)	26 (15.0)	22 (11.2)	0.22
	Anemia	105 (20.8)	16 (19.3)	37 (21.4)	51 (26.0)	0.46
	Current flaring status, n (%)	n=478	n=73	n=157	n=170	
	Currently flaring	66 (13.8)	6 (8.2)	24 (15.3)	29 (17.0)	0.32

BMI – body mass index; SD – standard deviation; UC – ulcerative colitis; Yrs – Years
Missing data not imputed; n numbers given where sample size varies (patients with known data)

Table 1. Patient demographics and clinical characteristics.

Conclusion: Naturally occurring age groups were identified by exploring disease activity measures and HRQoL outcomes. Further analyses found differences in flaring status and symptomatology across age groups. This highlights that age should be considered when treating patients with UC as patients of different ages are likely to present and respond differently. While further research is warranted, aspects of treatments in clinical trials, such as safety and efficacy, should be tested within different age cohorts, and publicising individual age data would aid in enabling the analysis of drug disposition and safety in patients with UC.

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Disclosure: Nothing to disclose.

PP1227

WITHDRAWN

PP1228

ENDOSCOPY-RELATED HAND INJURIES AND USABILITY OF ENDOSCOPES IN JAPANESE FEMALE ENDOSCOPISTS

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Introduction: Recently, much attention has been given to the well-being of health care workers, but endoscopists are no exception. Endoscopy examinations require complex and repetitive maneuvers that can put a strain on the musculoskeletal system. Female endoscopists are greatly disadvantaged in muscle strength and hand size compared to male endoscopists when operating a one-size-fits-all endoscope. Although there have been some reports, especially from Western countries, on endoscopy-related injuries (ERI),there have been a few reports on ERI from Japan.[1-3] A survey was conducted to examine ERI with a focus on hand disorders and operability of endoscope handles for Japanese female endoscopists.

Aims & Methods: Our aim was to examine ERI with a focus on hand disorders and operability of endoscope handles for Japanese female endoscopists from a survey.

A survey was designed in the Hokkaido chapter of the Women Endoscopists Career Support Committee in the Japan Gastroenterological Endoscopy Society (JGES). An anonymous electronic questionnaire was distributed to endoscopists who have been performing endoscopy in Japan. Participants accessed the Google form web site during the period from July to December in 2023.

A 25-item survey with multiple choice answers was created. Twelve items were dedicated to obtaining non-identifiable demographic data and practice-related information such as the number of endoscopies performed per week performed by each participant. Six-items focused on ERI including current and past injuries and the duration and severity of injuries. The other 7-items were about the usability of the current endoscope handle and practice of ergonomic endoscopy. ERI were defined as past or present endoscopy-related muscle and joint disorders including pain, discomfort, movement restriction and numbness.

Results: Data for 352 endoscopists were analyzed. The incidences of ERI were 55.6% (150/270) in men and 61.0% (50/82) in women and the most common site of ERI was the hands including fingers, thumbs and wrists followed by the lower back and then the neck (incidence of 28.1% in hands, 26.4% in lower back and 24.1% in neck). ERI in hands occurred in 35.4% of the women and increased with age. There were significant differences in age and endoscopy career between female endoscopists with and those without ERI in the hands, and there were no relations to glove size, number of endoscopy procedures and usability of the endoscopy handle. Regardless of the presence or absence of hand injuries, many fe-

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male endoscopists were not satisfied with the shape and usability of endoscopes (75.6%).Especially for female with small hands (gloves size 6 or smaller), the current endoscope handle is so large and they cannot reach the left and right angle knobs with their fingers.

Conclusion: In conclusion, ERI in hands were related to long endoscopy career and aging in female endoscopists. Improved usability of endoscopes and ergonomic endoscopy are important for female endoscopists to continue clinical practice.

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Disclosure: Nothing to disclose.

PP1229

GENDER DISPARITIES IN GASTROENTEROLOGY: PARENTHOOD'S IMPACT ON CAREER AND TRAINING

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Introduction: Career trajectories in gastroenterology can be influenced by parenthood, with potential disparities between genders in academic progression and workload management. This abstract presents findings from a survey by the Young Gastroenterology section of the German Society for Gastroenterology, Digestive and Metabolic Diseases (DGVS), exploring the impact of parenthood on academic achievements and training within the field.

Aims & Methods: This study was conducted through an anonymous online survey distributed via email to DGVS members. The survey was accessible from April 4, 2023, to June 19, 2023, and was conducted exclusively in German. It employed a combination of single and multiple-choice questions, with all questions being skippable to ensure participant comfort and engagement.

Depending on respondents' answers, query logic branched to gather comprehensive data on various aspects of parenthood and career progression in gastroenterology. The survey aimed to identify differences between male and female gastroenterologists with children in terms of academic positions, training delays, and work hours.

Results: The survey garnered 811 responses, distributed among 173 women without children, 206 mothers, 179 men without children, and 251 fathers. Significant gender disparities were found: Most notably, female physicians reported that childbearing often postponed their specialist training ($p < 0.001$), whereas men indicated that their training was seldom delayed. Employment data revealed that women with children typically worked fewer hours, whereas the majority of men, regardless of parental status, worked over 50 hours per week. Fatherhood predominantly led to full-time employment of at least 40 hours per week, suggesting that parenthood induces part-time work mainly among women. No childless men reported work hour reduction as a cause for delayed specialist status, in stark contrast to over half of the mothers citing this as a significant reason for their postponed completion of training. Post-childbirth work adaptations were markedly different: while men's work hours generally remained unchanged across all specialties, women were particularly less active in endoscopy after the birth of their first child (58%). Women with children held professorships less frequently than their childless or male counterparts ($p < 0.001$). Men with children were less likely to lack an academic degree.

Conclusion: The survey indicates that parenthood has a differential impact on the professional development of male and female gastroenterologists. Mothers are more likely to experience delays in training and are less represented in high academic positions. In contrast, fatherhood correlates with sustained or increased work hours, reflecting a gendered divide in work-life balance within gastroenterology. These findings highlight the need for structural changes in the medical profession to support equitable career progression for gastroenterologists, regardless of gender and parental status.

Disclosure: Nothing to disclose.

PP1230

COVID-19 INFECTION AND ILLNESS HISTORY IS LINKED TO HIGHER RATES OF IRRITABLE BOWEL SYNDROME (IBS) AND FUNCTIONAL DYSPEPSIA (FD) WORLDWIDE

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Introduction: Multiple recent studies suggest that COVID-19 (COVID) infection may increase risk of developing Disorders of Gut-Brain Interaction (DGBI). However, the extent of this effect on a population level, and how directly it is related to COVID illness, remains poorly understood.

Aims & Methods: We aimed to test in a multi-national sample the a priori hypotheses that IBS and FD prevalence rates (1) have increased after the COVID-19 pandemic, and are higher in individuals (2) who were more ill with their COVID infection, (3) who have long COVID, and (4) who were not vaccinated for COVID. A population-based Internet survey with pre-defined demographic quotas to obtain equivalent sex and age distribution

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across countries was conducted nationwide in 9 countries (Canada, China, Germany, Japan, Mexico, Poland, Romania, Spain, United States) in late 2023, and included demographic questions, Rome IV IBS and FD diagnostic questions, and questions about history of COVID infection and illness. Group prevalence was compared with Chi-Square tests to test the 4 hypotheses. For hypothesis 1, the total sample was compared to a pre-pandemic Rome Foundation Global Epidemiology Study (RFGES; Sperber et al. Gastroenterology 2021;160(1):99-114) sample from the same 9 countries, surveyed with comparable methodology in 2017-2018, using weighting to equalize country sample sizes.

Results: The survey was completed by 17,663 participants (50.2% females; mean age 46.3 years, range 18-96). People in the current sample had higher prevalence of both Rome IV IBS [% and 95% CI: 7.9% (7.5, 8.3) vs. 3.8% (3.5, 4.1); p<0.0001] and FD [12.7% (12.2, 13.2) vs. 7.0% (6.6, 7.4); p<0.0001] compared to the pre-COVID RFGES sample (N=19,669, 50.2% females, mean age 45.4 years, range 18-95) for the same countries. The post- vs. pre-pandemic prevalence of FD and IBS was numerically higher in all countries, and in all age and sex subgroups. In the current survey, people who reported having had test-confirmed COVID (41.3%) had higher IBS [9.4% (8.8, 10.1) vs. 6.8% (6.3, 7.2); p<0.0001] and FD [14.6% (13.8, 15.4) vs. 11.3% (10.7, 11.9); p<0.0001] prevalence than those without a COVID test-positive history. Those who were moderately or very ill during COVID infection had higher prevalence of IBS [12.0% (10.9, 13.0) vs. 6.8% (6.0, 7.7); p<0.0001] and FD [17.6% (16.4, 18.9) vs. 11.4% (10.4, 12.4); p<0.0001] compared to those with mild or no symptoms, and even higher if they had new-onset GI symptoms during their COVID illness (15.1% IBS, 21.3% FD). People with current lingering symptoms from past COVID infection ("long COVID") had higher IBS [16.3% (14.4, 18.3) vs. 7.8% (7.1, 8.5); p<0.0001] and FD [22.9% (20.7, 25.1) vs. 12.6% (11.8, 13.5); p<0.0001] prevalence than fully recovered individuals. People who had not been vaccinated for COVID (19.4% of the sample) had higher prevalence of FD [14.3% (13.1, 15.4) vs. 12.3% (11.7, 12.8); p<0.0001] but not IBS [8.4% (7.4, 9.4) vs. 7.7% (7.3, 8.2); p=0.18] compared to the vaccinated.

Conclusion: Our findings indicate that both IBS and FD prevalence rates are significantly higher now among adults worldwide than before the COVID-19 pandemic. Greater illness severity during COVID infection and current long COVID status show striking associations with elevated prevalence of these DGBI. Although retrospective infection/illness self-report cannot prove causation, this pattern of findings strongly implicates COVID-related illness as a potent driver of subsequent vulnerability to having IBS and FD, leading to an increased population prevalence of these disorders.

Disclosure: Nothing to disclose.

PP1231

WITHDRAWN

PP1232

TREATMENT OF LIVER INJURY WITH URSODEOXYCHOLIC ACID IN PATIENTS WITH COVID-19

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Introduction: Liver damage is often observed in COVID-19 patients, especially in severe cases. It may be caused by a systemic inflammatory response, cytokine storm, immune dysregulation, hypoxemia, multiorgan dysfunction syndrome , and the use of COVID-19 drugs (1).

Abnormal liver function tests were observed in up to 52% of COVID-19 patients (2).

Ursodeoxycholic acid (UDCA) inhibits the production of pro-inflammatory cytokines involved in the cytokine storm development. Thus, it may be beneficial in restoring liver function in patients with COVID-19 and hepatic injury

Aims & Methods: This non-interventional prospective study aimed to evaluate efficacy and safety of UDCA in adult patients with signs of liver damage and COVID-19 infection.

140 patients (mean age 67,43 ± 14,40; 67 males) with laboratory-confirmed COVID-19 and hepatic injury were enrolled, 93% of them had moderate COVID-19. Hepatic injury referred to elevated level of at least one of the following parameters: alanine aminotransferase (ALT), aspartate aminotransferase (AST), gamma-glutamyl transferase (GGT), alkaline phosphatase (ALP) or total bilirubin. UDCA was prescribed to 70 patients for up to 2 months. The control group comprised of 70 patients with COVID-19 and hepatic injury who were not prescribed UDCA

Results: At baseline, both ALT and AST levels were elevated above the upper limit of normal (ULN) dualfold in 25% of patients of the total population, by three-fold in 20%. Isolated elevation of ALT by more than 2-3 times above the ULN was observed in 31% of patients, isolated elevation of AST – in 8%. Hepatocellular liver injury was diagnosed in 62.85% of patients in the UDCA group and 57.14% in the control. Mean ± standard deviation (SD) ALT levels at baseline were 79.44 ± 81.43 and 66.98 ± 32.34 IU/L in the UDCA and control groups, respectively (not significant, NS). On day 10, the ALT levels were 45.75 ± 39.17 and 69.72 ± 64.99 IU/L (p = 0.002 between the groups), on day 60 – 17.39 ± 7.61 and 21.59 ± 9.14 IU/L (p = 0.011). Mean AST levels at baseline were 82.57 ± 97.12 and 62.24 ± 50.77 IU/L in the UDCA and control groups, respectively (NS), a significant difference between the groups was noted on day 60: 17.39 ± 7.42 and 22.68 ± 7.62 IU/L respectively (p < 0.001).

A significantly lower ALP and total bilirubin levels were observed in the UDCA group compared to control on day 60 (ALP: 57.49 ± 27.91 vs. 74.09 ± 21.85 IU/L, p < 0.001; total bilirubin: 10.32 ± 3.60 vs. 15.03 ± 10.34 μmol/L, p < 0.001). GGT levels were similar in both groups on all visits.

At baseline, hepatomegaly was observed onsonography in 48,57% and 51,43% (NS) of patients in the UDCA and control groups, respectively. By day 60 hepatomegaly remained in only 21,43% patients in the UDCA group but in 48,57% in the control group (p < 0.001 between the group). Biliary dyskinesia was initially observed in 24,29% and 27,14% of patients in the UDCA and control groups respectively, by day 60 it persisted in 1,43% and 15,71% of patients, respectively (p = 0.007).

Positive dynamics on ultrasound and normalization of liver size during treatment were observed in 60.0% of patients in the UDCA group and 7.1% in the control group, while negative dynamics were observed in 0% and 25.7% of patients, respectively.

No changes were seen in 27.1% and 50.0% of patients, respectively.

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Conclusion: Oral administration of UDCA to patients with COVID-19 and liver damage demonstrated a faster normalization of liver function, as evidenced by the reduction in elevated levels of ALT, AST, ALP, and total bilirubin, as well as the restoration of liver size.

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PP1233

PREVALENCE AND RISK FACTORS OF LIVER INJURY AMONG HOSPITALIZED PATIENTS WITH SEVERE OR CRITICAL COURSE OF COVID-19 INFECTION

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Introduction: Several studies have found that the course of infection by COVID-19 virus is often associated with varying degrees of hepatic damage, but the occurrence and its risk factors is yet to be mapped.

The aim of the study was first, to determine the prevalence of liver injury in patients hospitalized for COVID19 infection.

Second, to compare the groups with severe and critical course of COVID19.

Third, to identify independent risk factors of several types of liver injury in the group of critically ill patients.

Aims & Methods: During several waves of the pandemic, we collected data about cases hospitalized for COVID-19 pneumonia in two groups: severe course, group 1 and critical course with mechanical ventilation, group 2. We recorded basic demographic parameters, comorbidities and their chronic treatment, clinical status and laboratory findings at admission, administered treatment and evolution of hepatic tests during hospitalization, maximum levels of ALT, ALP, and bilirubin.

We defined liver injury according to the Common terminology criteria for adverse events v.5.0: ALT > 5 times the upper limit of normal (ULN, cytolytic damage 2nd degree), ALT > 3 times and bilirubin > 2 times ULN (Hy's law), ALP > 5 times ULN (cholestatic damage 2nd degree). In the group 2, we also identified independent risk factors for individual adverse events.

Results: 345 patients had available data, 58. 3 % were men, with a median age of 66 years, a the Charlson comorbidity index of 4. 0, BMI of 28.4, with an initial CRP of 97 mg/l, an oxygen saturation of 91. 5 %, and leukocytes 7.7 x10.9/l. There were 201 patients in group 1 and 144 patients in group 2. Group 2 differed significantly (p < 0.001) from group 1 in median age (61.5 vs. 71 years), proportion of men (70 vs. 49.8%), comorbidity index (2 vs. 6), initial CRP (117 vs. 86.8 mg/l), by entry ALT (0.75 vs. 0.49) and AST (1.0 vs. 0.77ukat/l), but not by a history of liver disease or cirrhosis (11.1 vs. 15.5 % and 2.1 vs. 3.0 %). In group 2, all defined events of liver injury were significantly more frequent (P<0.001): elevation of ALT > 5xULN (39.4 vs. 5%), Hy's law (30.3 vs. 3%), ALP > 5xULN (26.5 vs. 2%).

In group 2, the independent risk factors for ALT elevation > 5xULN were: entry AST (OR 1.75), aminoglycoside antibiotics (OR=4.28), methylprednisolone (OR=2.7), tigacycline (OR=3.06) and mean erythrocyte volume (OR=0.92). Hy's law risk factors were: history of hepatopathy (OR=3.3), aminoglycoside antibiotics (OR=4.04), caspofungin (OR=3.43) and linezolid (OR=3.07). Risk factors for ALP > 5xULN were: methylprednisolone (OR=4.1), level of fibrinogen on admission (OR=1.45), number of types of antibiotics administered (OR=1.41) and albumin on admission (OR=0.86).

Conclusion: Adverse events of liver injury among critically ill patients with COVID-19 were 8 to 12 times more frequent compared to patients with a severe course. The risk factors were mainly the initial state of the liver (history of hepatopathy, AST, albumin), methylprednisolone and selected antimicrobial drugs.

Disclosure: Nothing to disclose.

PP1234

LIVER FIBROSIS INDEX FIB4 AND COVID-19: A SERIES OF 330 PATIENTS IN AN INTENSIVE CARE UNIT

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Introduction: Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) is recognized as the cause of a severe multisystem disease that may require intensive care unit (ICU) treatment. Because of its known hepatic tropism, this virus may be responsible for fatal liver damage. The aim of our study was to evaluate the association between elevated fib4 and the incidence of death in patients hospitalized for SARS-CoV-2 infection in an ICU.

Aims & Methods: We conducted a retrospective descriptive and analytical study including patients hospitalized for COVID19 between 2020 and 2021 in an intensive care unit. A description of the collected clinical and biological variables was first performed with calculation of the FIB4 index. We then performed an analytical study using SPSS software to determine the association between high FIB4 score and infection severity, estimated by mortality and length of hospital stay.

Results: A total of 330 patients were enrolled, with a clear male predominance of 64.2% and a sex ratio of 1.7. The mean age was 62.5 years. Diabetes and hypertension, at 45.5% and 38.2%, respectively, were the most common comorbidities. Sixteen point one percent of patients were smokers and 9.4% were overweight or obese. Overall, 68.2% of patients had cytolysis on admission and 13.3% had thrombocytopenia. Twenty seven point three percent of patients had a fib4>3.25. Median hospital stay was 7.5 days (1-100). Fib4 > 3.25 was significantly associated with death (p=0.006), as were age over 60 years (p=0.025), the presence of pre-existing lung disease (p=0.054), AST levels (p=0.001) and ALT levels (p=0.008). In contrast, fib4 > 3.25 was not significantly associated with longer hospital stay. On multivariate analysis, fib4 > 3.25 was significantly associated with the occurrence of death (odds ratio, 2.109; 95% CI, 1.0054 to 4.43), whereas age (p=0.09), platelet count (p=0.075), ASAT level (p=0.8) and ALAT level (p=0.503) showed no significant difference between the 2 groups.

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Conclusion: According to our results, fib4>3.25 was significantly associated with a higher mortality in IUC patients with COVID19.
Disclosure: Nothing to disclose.

PP1235

HOSPITALIZATION OUTCOMES AND LONG-TERM GASTROINTESTINAL SYMPTOMS FLUCTUATION OF IRRITABLE BOWEL SYNDROME OR FUNCTIONAL DYSPEPSIA PATIENTS POSITIVE AFTER COVID-19

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Introduction: Exposure to COVID-19 has been shown previously associated with a higher risk for long-term gastrointestinal symptoms and disorders of gut-brain interaction (DGBI) occurrence, including irritable bowel syndrome (IBS) (1). However, it is still unclear whether SARS-CoV-2's effects on the gastrointestinal tract are more severe in patients already diagnosed with a DGBI and symptoms fluctuations in these patients over time after COVID-19 exposure.
Aims & Methods: To explore hospitalization outcomes and gastrointestinal (GI) symptoms fluctuation over a 1-year follow-up of COVID-19 patients with IBS or functional dyspepsia (FD) at enrollment compared to COVID-19 patients without any previous GI symptoms/disorders/diseases/surgeries.
In this retrospective multicenter cohort study, patient charts were reviewed for demographic data, medical history including GI chronic diseases, and hospitalization outcomes. All participants completed the Gastrointestinal Symptom Rating Scale (GSRS) over 4 time points: at COVID-19 onset, at 1, 6, and 12 months, and the Hospital Anxiety and Depression Scale (HADS) over 2 time points: at 6 and 12 months. The cohort was divided into two groups: (1) IBS/FD group (2) control group.
Results: A total of 599 patients were eligible for the study (n=61 IBS/FD group, n=538 control group). There were no differences between the groups regarding hospitalization outcomes. There was a significant decrease in both IBS/FD and control groups in depression [6.17±4.45 at 6 months and 3.17±4.14 at 12 months for IBS/FD group (p=0.007) and 3.55±4.02 at 6 months and 3.05±3.74 at 12 months for the control groups (p=0.019), respectively] and anxiety levels over time [5.94±4.32 at 6 months and 4.10±4.58 at 12 months for IBS/FD group (p=0.019) and 3.58±3.71 at 6 months and 3.26±3.65 at 12 months for the control groups (p<0.001), respectively]. The pooled Odds Ratios (OR) of the effect of IBS/FD over 4 time points ranged from 1.93 for diarrhea to 5.10 for increased distention, and were statistically significant for all GI symptoms, even after adjustments for gender, depression, and anxiety. While there was a significant improvement over time in the severity of abdominal pain (p=0.008), hunger pains (p=0.011), nausea (p=0.001), diarrhea (p=0.005) and loose stool (p=0.007), there was also a significant deterioration in constipation (p=0.002), hard stool (p<0.001) and feeling of incomplete evacuation (p=0.007) among the IBS/FD group.
Conclusion: Gastrointestinal symptoms within and between the control group and the IBS/FD group changed over time after COVID-19 onset. Patients with IBS or FD at enrollment showed increased complaints of constipation, hard stool, and incomplete evacuation after COVID-19 exposure.
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Disclosure: Nothing to disclose.

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PP1236

SEROLOGICAL RESPONSE TO VACCINATION AGAINST SARS-COV2 IN PATIENTS WITH CROHN'S DISEASE

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Introduction: Inflammatory bowel disease physiopathology involves various factors including the host immune system and management often requires immunosuppressive therapy posing, consequently, challenges in the context of vaccination. The advent of vaccines against SARS-CoV-2 a major role in the global flight against the pandemic, however questions regarding efficacy in specific patient populations particularly those with underlying immune-mediated conditions such as Crohn's disease are raised.

Aims & Methods: The aim of our study was to assess the humoral response in patients with CD under conventional immunosuppressive or biotherapies and to compare with healthy controls. We conducted a single-center descriptive study involving CD patients who had received at least one dose of the SARS-CoV-2 vaccine between January and June 2022. Additionally, healthy volunteers (HV) who had received at least one vaccine dose were included. To evaluate the humoral response, we assessed quantitative Anti-SARS-CoV-2 antibodies (using IgM and IgG assays) with the VIDAS® method for all participants.

Results: In total, we collected a total of 58 blood samples (28 patients with CD and 30 HV), the median age was 48.39 years ± 12.14. Regarding medical treatment of CD, 12 patients were on biotherapies: Infliximab (n=9), Adalimumab (n=2) and Ustekunimab (n=1). Twenty-one patients had conventional immunosuppressive: azathioprine (n=19), methotrexate (n=1), corticosteroids > 10 mg/d (n=2). The majority of the patients had received at least 2 doses of the vaccine with a median number of doses in both groups was 2 IC95% [1-4]. There was no statistically significant difference regarding the number or the type of the vaccine between the 2 groups. CD patients had a lower mean level of Ig G against SARS-Cov2 antibodies (20.34 IU +/- 15.57) comparing to healthy controls (32.97+/-10.18) with statically significant difference (p= 0.001). Regarding the type of treatment, there was no significant difference of the level of antibodies between conventional immunosuppressive drugs and biotherapies In CD patients.

Conclusion: This study highlights an altered immune response in patients with Crohn's disease, reflected in a reduced humoral response compared to healthy controls. Assessing the serological response to SARS-CoV2 vaccination in this vulnerable population is of paramount importance for optimizing vaccination strategies and protecting the public.

Disclosure: Nothing to disclose.

PP1237

MANAGEMENT OF CODE STATUS IN PATIENTS UNDERGOING ENDOSCOPIC PROCEDURES - SURVEY ON THE CURRENT CLINICAL PRACTICES AND BELIEFS OF PORTUGUESE GASTROENTEROLOGISTS

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Introduction: Cardiorespiratory arrest is a recognized complication of endoscopic procedures, particularly when sedation is administered. Hospitalized patients, especially those with a prescribed Do Not Resuscitate (DNR) order, are more likely to have severe underlying illness, and consequently a higher risk of cardiorespiratory complications. Additionally, the request for endoscopic exams, especially urgent ones, in this group of patients is becoming increasingly common.

Aims & Methods: The aim of this study was to asses the current clinical practices, as well as the beliefs of gastroenterologists in Portugal regarding the management of the code status in hospitalized patients with prescribed DNR undergoing endoscopic procedures.

Self-reported data was obtained through a voluntary and anonymous survey of gastroenterologists and gastroenterology trainees in Portugal.

Results: One hundred and forty five gastroenterologists, of which 26 (17.9%) were trainees, responded to the survey, with an average age of 42 years (27-83 years), and 59.3% female. The majority are aware of the patient's code status prior to the endoscopic procedure (73.1%, n=106). However, DNR is rarely discussed (<25% of cases) with the patient/legal representative (66.9%, n=97) or attending physician (58.6%, n=85). In patients with prescribed DNR, the rate of reversal prior to the procedure is rarely performed (reversal in all or almost all in 8,3%, n=12). Most respondents are unaware of the procedures necessary for DNR prescription (81.3%, n=118), as well as existing national/international recommendations on DNR reversal (96.5%, n=140). Regarding personal beliefs, the majority consider that in these patients DNR should be reversed to an intermediate option that includes limited resuscitation maneuvers according to the patient's preference (62.1%, n=90). They also believe that the urgency of a procedure should not influence the decision to reverse DNR (n=89, 61.4%), and that if DNR is reversed, it should be done by the attending physician who initially prescribed it (91.7%, n=133). Finally, respondents are divided when asked if they would perform an endoscopic procedure on a patient with prescribed DNR that has not been previously reversed: 49.7% answered yes (n=72), 22.1% no (n=32), and 28.3% have no formed opinion (n=41). There is a consensus on the need for recommendations on this topic for gastroenterologists (97.9%, n=142).

Conclusion: The majority of Gastroenterologists rarely reverse DNR before endoscopic procedures, however, it is mostly advocated that limited resuscitation maneuvers should be offered according to patient preference. Most are unaware of the existence of institutional, national, or international policies regarding DNR. These findings suggest considerable variation in knowledge and approach to hospitalized patients with DNR, highlighting the need for specific recommendations for Gastroenterology in this area.

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Disclosure: Nothing to disclose.

PP1238

WORK-LIFE CONFLICT AND ITS HEALTH EFFECTS ON MOROCCAN GASTROENTEROLOGISTS

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Introduction: Gastroenterologists are exposed to various health-related risks in their professional roles,
Because they have various roles including endoscopists, clinicians, and in some cases, academic professionals. Our study aims to assess the work-life balance model and the prevalence of work-related physical and mental symptoms, as well as the effects of work-life balance on the health of Moroccan gastroenterologists.
Aims & Methods: A total of 152 gastroenterologists from various cities in the country participated in an anonymous electronic questionnaire regarding their daily activities and symptoms. A work-life balance ratio was calculated by dividing the number of working hours (per week) by the number of hours spent outside of work (including activities and sleep). Musculoskeletal, gastrointestinal, and mental symptoms were recorded. The Maslach Burnout Inventory (MBI) was used to measure the burnout score.
Data was entered and analyzed using the JAMOVI 2.2 software for Mac.
Results: 152 Moroccan gastroenterologists were included in the study, with 72.4% being female and 27.6% male, resulting in a sex ratio of 0.38.
The median age of the respondents was 31 years (range: 26 to 64 years), with 61.8% being married, of which 70.2% were married to physicians.
The mean number of working hours per week was 50.2 ± 12.7, with a median number of endoscopies per week of 6.5 (range: 2 to 12). 86.8% reported musculoskeletal disorders, and 71.1% reported digestive disorders.
Regarding mental symptoms, 23.7% reported anxiety, and 46.1% reported depression, with 21.1% experiencing both simultaneously.
The median work-life balance ratio was 0.42 (range: 0.32-0.55).
In univariate analysis, this ratio was not correlated with sex or musculoskeletal and digestive symptoms but was correlated with age ($p < 0.001$) and mental symptoms ($p < 0.001$ for depression and $p = 0.007$ for anxiety). However, in multivariate analysis by adjusting the parameters studied, namely age, sex, presence of confirmed depression, and confirmed anxiety, only age emerged as an associated factor (Beta = -0.24 + (-0.46; 0.013)).
Using the MBI scale, 72.4% of the respondents were found to be experiencing burnout. Burnout was not correlated with sex but was associated with the work-life balance ratio ($p = 0.04$). It is noteworthy that two of the three categories of this scale, namely depersonalization and personal accomplishment, were associated with the age of the respondents ($p = 0.002$ and $p = 0.003$). Indeed, a moderate negative correlation was found between age and depersonalization ($r = -0.32$), and a moderate positive correlation was found between age and personal accomplishment ($r = 0.31$).
Conclusion: Moroccan gastroenterologists suffer from musculoskeletal, digestive, and mental symptoms, and are prone to burnout likely due to long and laboting work. Our study found that younger individuals worked more and were more likely to experience burnout, particularly with decreased empathy towards patients and diminished personal accomplishment.
Disclosure: No conflict of interested was noted.

PP1239

THE USE OF GRIP STRENGTH MEASUREMENT IN HIGH-VOLUME ENDOSCOPISTS AND NON-INTERVENTIONAL GASTROENTEROLOGISTS - A NEW APPROACH FOR THE ASSESSMENT AND PREVENTION OF MUSCULOSKELETAL HEALTH DISORDER IN GASTROINTESTINAL ENDOSCOPY?

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Introduction: Up to 80% of practicing GI endoscopists are affected by musculoskeletal health disorder with significant influence on personal health, absence from work and reduction of endoscopic workload (1). It is estimated that overuse injury is a main mechanism of health impairment, but methods to prevent musculoskeletal disorder in GI endoscopists are elusive. Since hand grip strength measurement has been identified as an important indicator of musculoskeletal health and physical fitness (2), the further evaluation of hand grip strength in working endoscopists in comparison to non-endoscopists might bring further evidence to prevent musculoskeletal health disorder in GI endoscopists.
Aims & Methods: In this prospective single-center pilot study, measurement of hand grip strength was conducted on 20 workdays by a high-volume GI endoscopist and a gastroenterologist who did not perform endoscopic procedures before and after work at University Hospital Ulm. For hand grip strength measurement, JAMAR© Plus Hand Dynamometer was used according to the Southampton protocol (3). Three serial measurements were performed with each hand to obtain an average grip strength at each measurement point. Statistical analysis was performed using SPSS Statistics 29 (IBM, USA). Mann-Whitney U test, students t-test, univariate analysis of variation (ANOVA) and Kruskal-Wallis test were performed whenever applicable.
Results: Average hand grip strength of the high-volume endoscopist was not significant higher than the hand grip strength of the non-endoscopic working gastroenterologist (left hand: 102.2 lbs vs. 93.3 lbs, $p=0.26$; right hand: 114.0 lbs vs. 108.2 lbs, $p=0.44$). In comparison, hand grip strength was significant higher after work in non-endoscopist subject for both hands (left: 89.2 lbs vs. 95.4 lbs, $p=0.025$; right: 95.9 lbs vs. 101.4 lbs, $p=0.035$) while the participating endoscopist had a significant lower grip strength after work (left: 107.4 lbs vs. 95.2 lbs, $p<0.001$; right: 106.8 lbs vs. 95.6 lbs, $p<0.001$). In both subjects, the day of the week (beginning vs. end of the week) had no effect on hand grip strength.
Conclusion: As hand grip strength is associated with health performance status, significant difference can be observed between endoscopists and non-endoscopists in the impairment of hand grip strength within a workday. To prevent musculoskeletal health issues in endoscopists, timing and effect of health interventions could be implemented by evidence from hand grip strength measurement. Therefore, further observation studies are mandatory to develop methods to prevent musculoskeletal health disorder especially for high-volume endoscopists.
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Disclosure: Nothing to disclose.

PP1240

A QUESTIONNAIRE SURVEY-BASED ANALYSIS FOR LYNCH SYNDROME SCREENING AND SURVEILLANCE TRENDS AMONG GASTROENTEROLOGISTS IN JAPAN

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Introduction: Recently, in Japan, insurance coverage for microsatellite instability (MSI) testing as a companion diagnostic agent for immune-checkpoint inhibitors and comprehensive cancer genome profiling has highlighted the growing importance of Lynch syndrome (LS) awareness among gastroenterologists.

Aims & Methods: Our aim was to assess the shifts in knowledge regarding LS and practice trends among Japanese gastroenterologists. We conducted a questionnaire survey on physicians' knowledge and practice trends regarding LS in their facilities, collaborating with the councilors of the Japanese Society of Gastroenterology in 2020 and 2022. The knowledge survey on LS encompassed an understanding of the genetic background of LS, primary and secondary screening methods for LS, and spillover effects of comprehensive cancer genome profiling tests on LS practice. Meanwhile, the practice trends survey on LS examined information on the LS screening methods utilized, the number of patients screened for LS, and LS surveillance methods. Each questionnaire item was compared between the first and second surveys to determine the proportion of physicians who chose each option based on the total number of physicians who completed the questionnaire. The responses of physicians who participated in both surveys were analyzed to assess changes in their responses to each question.

Results: The knowledge survey was completed by 146 in the first survey and 228 doctors in the second survey. Besides, the practice trend survey was completed by 362 in the first survey and 195 doctors, respectively, in the second survey. The second survey revealed a significant increase in several key indicators compared to the first survey. These included the number of physicians opting for universal tumor screening as the primary screening method for LS (26/195 vs. 20/354) , the proportion of physicians conducting the BRAF V600E test for chemotherapy selection (100/195 vs. 137/354), the number of newly diagnosed patients with LS in the last 3 years per physician (1.64±4.15 vs. 0.96±3.27), and the number of patients with LS currently under surveillance per physician (2.00±5.09 vs. 1.15±4.03). Among physicians who participated in both surveys, the knowledge about primary and secondary screening methods, the proportion offering BRAF V600E testing in colorectal cancer tissue and performing screening tests for endometrial cancer, and the number of patients with LS currently under surveillance per physician were significantly higher in the second survey than in the first survey. Physicians who initiating or strengthening the inspection of primary and secondary screening for LS between the first and second surveys had a significantly higher proportions of increasing currently surviving patients with LS compared with those who continued or discontinued screening tests. (13/33 vs. 13/100, 11/33 vs. 13/100)

Conclusion: In conclusion, there has been a notable increase in both the number of patients diagnosed with LS and the number of patients screened for LS in Japan in recent years. This increase can be attributed to an increase in the number of physicians performing primary screening, such as universal tumor screening, and secondary screening utilizing MSI and immunohistochemistry for mismatch repair protein tests.

Disclosure: This study was supported by a research grant from The Japanese Society of Gastroenterology.

PP1241

RESULTS AND LEARNING CURVE AFTER IMPLEMENTATION OF ENDOSCOPIC SUBMUCOSAL DISSECTION IN STOMACH AND RECTUM – PROSPECTIVE ANALYSIS FROM TWO WESTERN ENDOSCOPISTS

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Introduction: Endoscopic submucosal dissection (ESD) enables the *en bloc* ressection of superficial lesions of the gastrointestinal tract. Initially developed in the East, the implementation of the technique has been more difficult in the West, mainly due to the lower incidence of gastric neoplasia.

Aims & Methods: In our study, we aimed to evaluate the clinical outcomes since the introduction of the ESD technique in two distinct centers and describe the learning curve of 2 endoscopists that performed the procedures. This is a prospective analysis composed of patients that underwent rectal and gastric ESD between 2019 and 2024. We evaluated *en bloc* and R0 resection rates and recurrence rates. Complications were defined as adverse events related to the procedure that required reintervention or prolongation of the hospital stay. The area of the resected lesion was calculated in cm^2 (assuming an eliptical shape) and the speed of dissection in cm^2/hour.

Using cumulative sum analysis, we identified the procedure in which a significative increase in the speed of dissection was observed and compared outcomes in the two time periods defined by this cut-off.

Results: We included a total sample of 100 patients (59% male, age 69.9±5.4 years).

Gastric ESD was conducted in 56 patients (57% male, age 59.3±2.9 years). *En bloc* ressection rate was 98.2% (n=55) and R0 resection was 91.1% (n=51). 10.7% of the procedures had complications (1 perforation, 2 lesions of the muscular layer and 3 post-procedural bleedings). The mean area of the resected specimens was 7.8 ± 5.0 cm^2. The mean dissection speed was 8.2 ± 5.2 cm^2/hour. The mean hospital stay was 1.3 ± 0.3 days.

There was a significant increase in proficiency by the procedure number 27. When comparing the two groups, there was a tendency for an increase in R0 resection (88.9% vs 93.0%, p=0.294) and a decrease in the recurrence rate (12% vs 5%, p=0.194) in the later time period. There was a significant reduction in the complication rate (18.5% vs 3.4%, p<0.001).

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Rectal ESD was performed in 44 patients (61.4% male, age 71.5±3.1 years). *En bloc* resection was 97.7% (n=43) and R0 resection was 86.4% (n=38). The complication rate was 38.6% (9 post-procedural bleedings; 8 post-coagulation syndromes), none of which required surgical treatment. The mean area of the resected specimens was 13.4 ± 4.6 cm^2 and mean dissection speed was 6.5 ± 4.1 cm^2/hour. The mean hospital stay was 2.2 ± 0.7 days.

There was a significant increase in proficiency by the procedure number 33. The R0 resection rate was similar between groups (85%vs82%, p =0.34), with a tendency to a decrease in complication rate (36%vs27%, p=0.29). A total of 9 adenocarcinomas were resected, with endoscopic cure in 88.9% cases (n=8).

Conclusion: We believe that the evaluation of the results of this intervention allows not only to assure that the quality criteria are being met but also to aid in the creation of learning programs for trainees. The learning curve associated with gastric ESD was steeper than that of rectal ESD. It appears that rectal ESD requires a larger number of procedures until the quality criteria proposed by the European Society of Gastrointestinal Endoscopy are met.

Disclosure: No conflict of interest.

PP1242

ENDOSCOPIC MANAGEMENT OF GASTROINTESTINAL SUBEPITHELIAL LESIONS IN A SPANISH TERTIARY HOSPITAL

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Introduction: Subepithelial lesions (SELs) are a common incidental finding in the gastrointestinal tract during endoscopy. Endoscopic ultrasonography (EUS) is the best tool to characterize SELs features and in several cases it allows anatomopathological diagnosis after fine-needle biopsy (FNB) or aspiration (FNA) (1).

The indication and technique for endoscopic resection of SELs depend on the type, size, depth of invasion and location (1).

Superficial SELs can be resected using endoscopic submucosal dissection (ESD)(2) and others can even be removed by endoscopic mucosal resection (EMR), such as neuroendocrine tumors (NETs) ≤ 10 mm (3).

Aims & Methods: The aim of this study is to assess the endoscopic management of SEL. A unicentric retrospective study was performed. There were collected the LSE endoscopically removed in an advanced endoscopy center between 2017 and 2024. The resection technique was at the discretion of the endoscopist. Those LSE that required surgical treatment were excluded.

Results: 50 lesions were included, the main characteristics are described in table 1.

In bloc resection (R0) was achieved in 45 SELs (90%). There were 5 cases (10%) of fragmentated resection (R1), 2 (40%) resected by ESD, 1 hybrid (20%) and 2 (40%) with hot snare. The mean size of R1 lesions was 17.2 mm (SD +/- 14.0). The histology and location were: 3 rectal NETs (60%) and 2 gastric fibroid polyps (40%).

The histopathological result of 13 SELs (26%) revealed focal contact with the deep margin. 12 (92.3%) were NETs located: 8 (66.7%) in rectum, 3 (25%) in duodenum and 1 (8.3%) gastric.

USE was performed in 21 cases (42%). In 8 LSE (38.1%) FNA was performed, and only in 3 (37.5%) was the histopathological sample representative: 1 GIST (33.3%) and 2 NETs (66.7%). 18 (85.7%) of them were located in the muscularis mucosa or submucosa. 2 LSE (9.5%) involved the muscular layer: 1 GIST (50%) and 1 fibroid polyp (50%). Both cases were resected using ESD and a perforation was documented during the procedure in each. There were 10 (18%) complications registered: 7 bleeding (70%), 2 perforations (20%) and 1 gastrostomy (10%) to retrieve the lesion. 7 of them (70%) required endoscopic treatment, 2 (20%) surgical treatment, and in 1 case (10%) no treatment was needed. ESD presented a higher tendency of complication rates compared to the other techniques.

In 20 patients (40%) a revision endoscopy was performed, only 1 (5%) recurrence of a gastric NET was registered one year after the resection.

Variable	N 50
Men (%)	27 (54)
Age (mean, SD)	64 (14)
Gastrointestinal localization (%):	
Gastric	- 20 (40)
Duodenum	- 6 (12)
Colon	- 3 (6)
Rectum	- 21 (42)
Size (mm) (mean, SD)	14.16 (13.1)
Type of LSE (%):	
NET	33 (66)
GIST	1 (2)
Leiomioma	3 (6)
Lipoma	6 (12)
Fibroid polyp	7 (14)
Resection technique (%):	
ESD	13 (26)
RME	4 (8)
L-RME	18 (36)
Endoloop	3 (6)
Hybrid	4 (8)
Hot snare polypectomy	7 (14)
Cold snare polypectomy	1 (2)
In bloc resection (R0) (%)	45 (90)
USE performed (%)	21 (42%)
Gastric	13 (61.2)
Duodenum	4 (19.1)
Rectum	4 (19.1)
Complications (%)	10 (20)
Hemorrhage	7 (70)
Perforation	2 (20)
Other	1 (10)

Conclusion: In our experience the most frequent SELs are gastric NETs and the main resection technique is L-EMR which achieves high rates of R0 with low complications. ESD is the second most used technique with a higher tendency for complication rates. The risk of recurrence after resection appears to be low.

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PP1243

SINGLE-CENTER EXPERIENCE OF MANAGMENT OF PERFORATION AFTER ENDOSCOPICAL SUBMUCOSAL DISSECTION (ESD) USING OVER-THE-SCOPE CLIP (OTSC)

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Introduction: The Endoscopical Submucosal Dissection (ESD) is an advanced resection technique for removing "en bloc" gastrointestinal (GI) lesions, bigger than 20 mm and with macroscopic features of no deep infiltration (limited to the submucosal). It is a complex procedure and, therefore, with risk of complications (early and/or delayed), first of all bleeding, then perforation.

Aims & Methods: Our aim is to confirm the pivotal role of Over-The-Scope Clip (OTSC) for menagement of perforation occurred during the ESD. Our Centre started the ESD techinque from March 2023 by an expert endoscopist, with a collegue in training, and we have included in a database the information of all procedures carried out from that period to September 2023.

Results: Were included 34 adult patient (M/F: 19/15) with an age mean of 69 years. The mean lesions size was of 31 mm ± 20 for the upper-GI, 25 mm ± 15 for the lower-GI. Histological findings were reported in the tab. 1. All procedures were performed under general anesthetic care. Lesions were succesfully resected "en bloc" in all patients (100%) by ESD procedure using minimal CO2 insufflation. The overall intraprocedure perforation occured in 3 out of 34 (8,8%): 1 out of 18 was located in the upper-GI tract (fundus); 2 out of 16 lesions were located in the lower-GI tract (1 in the ascending colon and 1 in the discending colon, respectively).

According to clinical and medical stability of the patients, all the parietal defect were treated successfully, at the end of the complete ESD, by positioning OTSC, allowing a full-thickness closure of the GI tract defect, avoiding emergency surgery. The R0 rate was of 94.1%, with two cases that underwent to surgery beacause of "pT2" at the final histological report.

Conclusion: OTSC is a known safe and efficacy device for closing perforation occured during ESD, in the course of the same endosocpic procedure. More and larger studies are needed to further validate these findings.

Disclosure: Nothing to disclose.

PP1244

USE OF A DYNAMIC SELF-ADJUSTING PROSTHESIS (P.A.D.) IN INGUINAL HERNIA REPAIR: OUR EXPERIENCE IN 200 PATIENTS

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Introduction: P.A.D. is a therapeutic solution for inguinal hernias, featuring a double polypropylene mesh anchored to the inguinal canal. Its adaptable design accommodates morphological variations during positional changes. A trapezoidal area, determined through patient studies, guides the shape and placement of the prostheses. The lower prosthesis fits this area and is anchored to the internal oblique muscle, while the upper prosthesis is secured to the pubic tubercle and inguinal ligament. Dynamic adjustments between the prostheses ensure continuous support and prevent weak areas. The procedure aims for well-stretched, tension-free prostheses parallel to the inguinal floor, minimizing risks associated with traditional techniques.

Aims & Methods: Between September 01, 2022, and September 30, 2023, our General Surgery department at the Santa Maria della Pietà dei Religiosi Camilliani Hospital in Casoria (Naples) conducted 300 inguinal hernioplasties on male patients. Prior to surgery, all patients underwent laboratory and radiological examinations. A standardized inguinal hernioplasty procedure using P.A.D. was performed for all patients. The procedure involved a 5-6 cm incision between the iliac spine and the pubic tubercle, followed by dissection of skin, subcutis, and Scarpa's fascia. The aponeurosis of the external oblique muscle was identified and opened, allowing access to the external inguinal ring. The ilioinguinal and iliohypogastric nerves were identified, and the cremaster muscle was dissected to access the hernial sac and spermatic cord. The type of inguinal hernia was determined, and appropriate steps were taken to address it, including reduction of the hernia sac and plication of the transversalis fascia if necessary. Pre-hernial lipomas were resected, and additional exploration was conducted to reduce the risk of recurrence. The P.A.D. was affixed using a standardized procedure, followed by closure of the external oblique muscle, subcutis, and skin. Patients were able to walk within 3-4 hours post-surgery and were discharged within 6-8 hours or the following day for afternoon surgeries. Follow-up evaluations were conducted up to 6 months post-surgery to monitor for complications such as hematomas, seromas, pain, inflammation, infection, and recurrence.

Results: Out of 200 recruited, a total of 120 patients underwent open groin right hernia repair with a right:left ratio of 1.3:1. Their age ranged from 0 to 88 years with a mean age of 33.94 ± 23.40 years. Among the patients, 60.2% of the hernias occurred in ages below 50 years; 3.6% had bilateral hernias. Post-operative complication rate was 2.3%, hematoma resulted as the most common one (0.65%). Relapses in 6 months were 0.5%.

Relapses	0.5%
Seromas	0.4%
Hematomas	0.65%
Wound infections	0.5%
Pain over 4 days	2.5%
Persistent neuralgia beyond 7 days	0.15%

Conclusion: The inguinal hernioplasty technique with P.A.D. is currently standardized. We have found the technique to be easy to replicate and easy to perform, even in cases of large hernias. Under no circumstances did we have to cut out and modify the size and shape of the pre-shaped P.A.D. which therefore adapted well to the anatomy of the IC. Postoperative complications, both short and long term, were limited, similar to the results found in scientific literature. This suggests that the P.A.D. inguinal hernioplasty technique is a valid alternative as an open onlay inguinal hernioplasty technique.

Disclosure: Nothing to disclose.

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CORRELATION BETWEEN BOWEL SOUNDS VERSUS CLINICAL AND ENDOSCOPIC DATA : AI-POWERED ANALYSIS

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Introduction: Bowel sound analysis holds promise as a non-invasive, cost-effective alternative to conventional diagnostic methods. However, standardized recording and interpretation methods are lacking. Our study aims to address this gap by investigating the correlations between bowel sounds and clinical/endoscopic data in patients.

Aims & Methods: We conducted a single-center, observational, cross-sectional study at the Mongi Slim Hospital's Gastroenterology Department. The study included patients scheduled for colonoscopy who had followed a residue-free diet for at least three days and ingested a bowel prep the day before the examination. We collected the patients' medical and surgical history, their digestive complaints, and performed a bariatric assessment. Bowel sounds were then recorded for 5 minutes using a modified version of the *Littman® CORE Digital stethoscope* and were then analyzed using AI programs to extract acoustic features. Afterwards, these features were compared with clinical and endoscopic data to establish correlations.

Results: We included 95 patients in our study (mean age = 56.32 years, sex ratio M/F = 1.02). Colonoscopy was performed in 83 patients (87.37%) and was total in 62 patients (74.70%). At least one endoscopic anomaly was present in 47 patients (54.63%). The most common finding was polyps in 27 patients, representing 57.45% of patients with an endoscopic anomaly. We labeled 33,837 bowel sounds from the 95 recordings of the included patients, with an average of 356.18 bowel sound labels per recording. The mean ratio of bowel sound duration to recording duration was 22.75%, corresponding to 68 seconds of bowel sounds out of the 5 minutes recorded.

In the analytical study, we found that gender was correlated with the RMS and Envelope features, with p-values of 0.021 and 0.032, respectively, suggesting that men tend to have an increased overall amplitude of the signal, while women exhibit a greater variability of amplitude over time. Difficulty passing gas was also correlated with RMS and Envelope with p-values of 0.039 and 0.032, respectively, implying that patients having trouble passing gas produce sounds with lower average amplitude and increased variability over the course of the signal.

Difficulty passing gas was also correlated with Spectral Contrast with a p-value of 0.036. This means that in these patients, the spectral energy is evenly distributed across frequencies and there are no major frequency variations over the course of the signal.

The presence of an endoscopic anomaly was also correlated with RMS and Envelope with p-values of 0.027 and 0.024, respectively, suggesting that in the presence of an endoscopic anomaly, the average amplitude of the signal would be lower, and the amplitude of the sounds would be more variable over the course of the signal.

We also found that the presence of polyp(s) was correlated with RMS and Envelope with p-values of 0.041 and 0.042, respectively, indicating that in the presence of polyp(s), the average signal amplitude is lower, and the amplitude of the sounds is more variable over the course of the signal.

Conclusion: Although conducted under experimental conditions, since we used a controlled diet and bowel prep, this study demonstrated promising correlation between bowel sounds and both clinical data and endoscopic findings in patients, particularly the amplitude of bowel sounds and its variations over time. Further studies are needed to verify these correlations under more real-life conditions.

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Disclosure: This study was conducted along with DigeHealth, a swiss start-up working on developing a non invasive monitoring system for bowel disorders.

PP1246

THE IMPACT OF CFTR MODULATOR THERAPY ON LOWER AIRWAY MICROBIOTA IN PEOPLE WITH CYSTIC FIBROSIS: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: In people with cystic fibrosis (CF), CFTR modulator therapy has resulted in marked improvements in pulmonary exacerbations, lung function, body weight, and quality of life. However, it remains unclear how restoring CFTR function modifies existing airway infection and inflammation. Understanding the dynamics of airway microbiome changes in response to modulator therapy plays an essential role in developing strategies for managing airway infections.

Aims & Methods: Our aim was to conduct a systematic review and meta-analysis of all available studies.

We conducted a systematic search on the 26th January, 2024 in MEDLINE, Embase, Web of Science, Scopus and CENTRAL. We performed a random-effects meta-analysis of any provided data pertaining to lower airway microbiota composition before versus after the initiation of CFTR modulator therapy. We calculated mean difference (MD) and standardized MD (SMD) with 95% confidence intervals (CI) and visualized the results on forest plots. Chi-squared tests were performed and I^2 values calculated to assess statistical heterogeneity.

Results: 30 studies were eligible for inclusion, with most prospectively examining people with CF before and 4-12 months after the initiation of CFTR modulator therapy. The initiation of CFTR modulator therapy was associated with an increase in lower airway alpha-diversity (Shannon index; SMD: +0.39 (95% CI: +0.20 - +0.59), p<0.0001), evenness (Pielou index;

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SMD: +0.84 (95% CI: +0.55 - +1.12), p<0.00001) as well as a reduction in the relative abundance of classical CF lung pathogen species *Pseudomonas* (MD: -7.1% (95% CI: -1.2% - -13.1%), p=0.02) and *Staphylococcus* (MD: -4.3% (95% CI: -2.0% - -6.6%), p=0.0003).

Conclusion: Our results demonstrate a clear and statistically significant association between the initiation of CFTR modulator therapy and the stabilization of lower airway microbiota in terms of increased biodiversity, more balanced distribution of microbial species and reduction in the abundance of potentially pathogenic species, all of which contribute to a lower risk of respiratory tract infections.

Disclosure: Nothing to disclose.

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